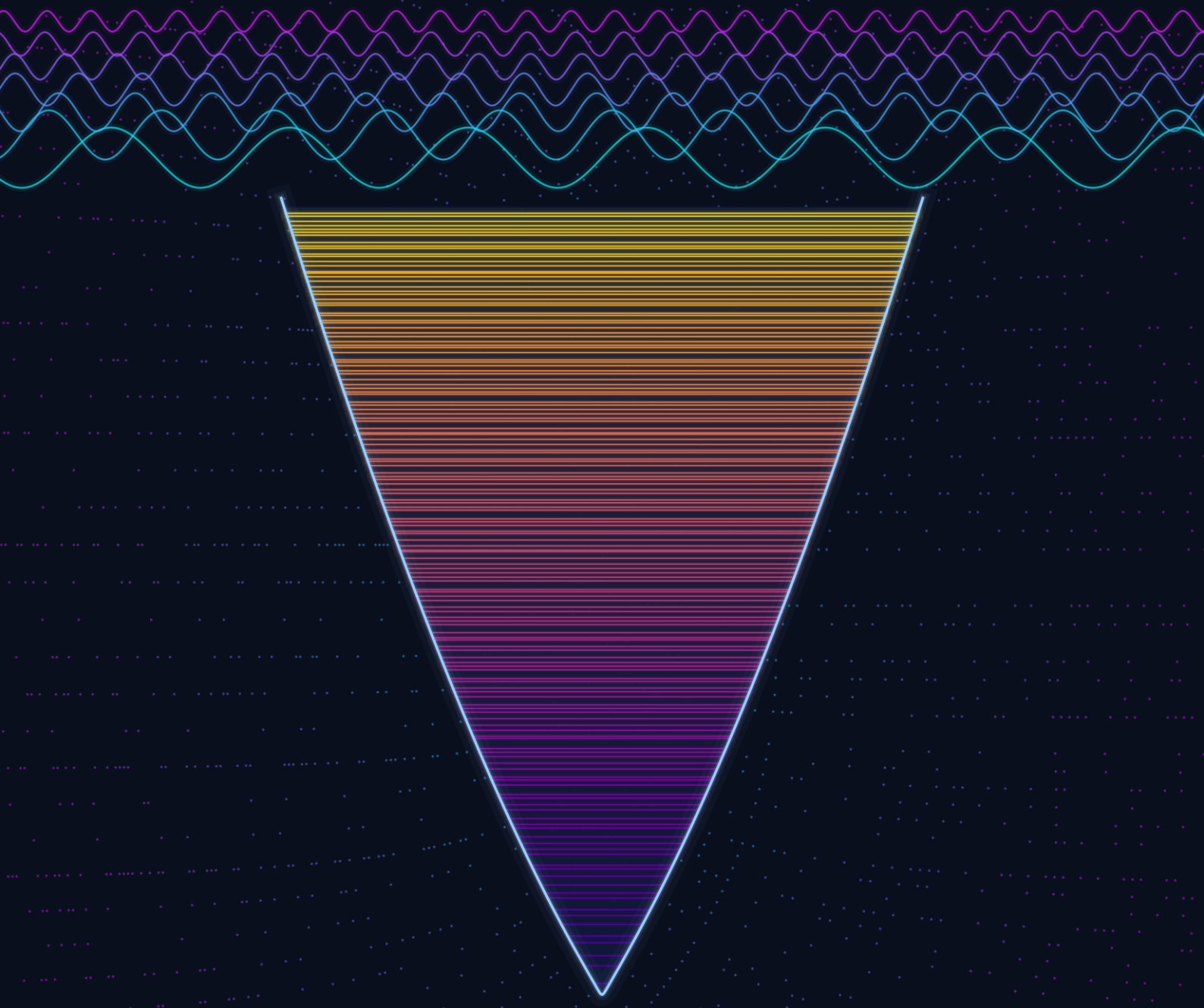
Potentials, Primes and Prediction

A First Course on the Riemann Zeros,
Quantum Spectra and Learning from Data



Directed by Dharmaj Soni
Written with Claude (Anthropic)

About this book

This book takes a reader who has completed a first-year university physics course to the point where they can read, line by line, the research paper "Fitted potentials on the Wu–Sprung background memorize the Riemann zeros while the explicit formula predicts them" [1], and rerun its key computations. Along the way it is a self-contained course in complex numbers, Fourier analysis, linear algebra, one-dimensional quantum mechanics, the Riemann zeta function, random matrices, numerical methods, and the statistics of prediction.

How this book was made

The book was planned and directed by Dharmaj Soni, the author of the paper. Its chapters were drafted by the large language models Claude Opus 5.5 and reviewed by Claude Fable 5.2 (Anthropic), following a written specification (the "book bible", included in the source repository). Every numerical statement in a worked example or answer is recomputed by a verification script that is part of the source, and every figure is generated by a script in the repository. AI-written text can contain errors that such checks do not catch; corrections are welcome through the repository.

Pictures and facts

Every chapter opens with a picture computed from that chapter's own mathematics (the scripts are in the repository folder `tex/figs/art`); none is a stock image. The "Did you know?" boxes, roughly one every ten pages, were drafted by Claude and checked by the coordinating editor; the source behind each fact is recorded in `bible/did_you_know.json`.

Navigating the PDF

Every entry in the Contents, every cross-reference and every citation is a link. Each chapter's opening page has a link back to the Contents, and the PDF bookmarks panel lists all parts, chapters and sections.

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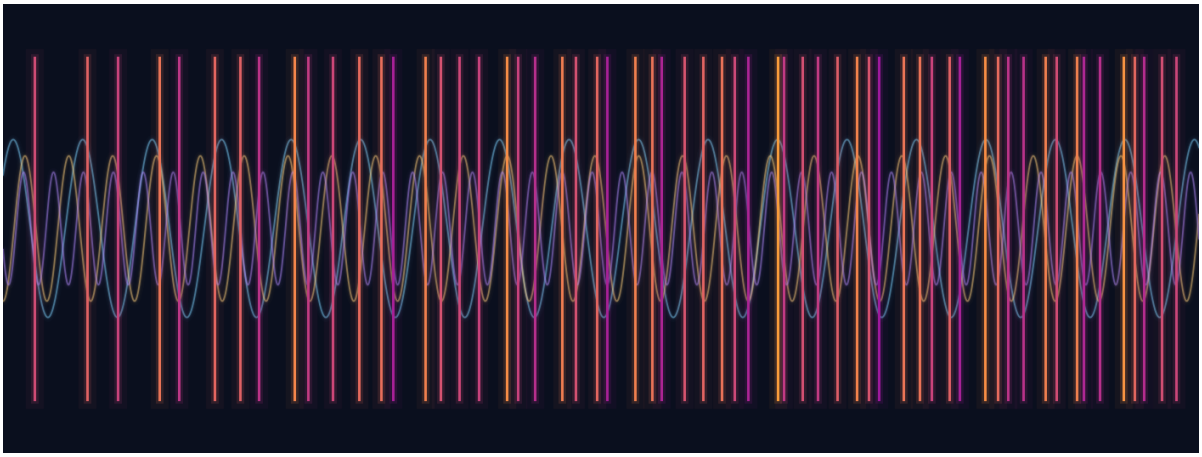
Part I

Mathematical Foundations



The heights of the first 60 Riemann zeros, drawn as energy levels; colour shows how far each sits from its smooth position.

Chapter 1

The Question: Can a Fitted Potential Learn the Riemann Zeros?

↑ Contents

What this chapter is about

Here is a list of numbers: 14.134725, 21.022040, 25.010858, 30.424876, 32.935062, and it goes on forever. Mathematicians have studied these numbers for more than a century and a half. They are built, in a way we will make precise later in the book, out of the prime numbers. Physicists have stared at the same list and seen something familiar: the numbers are spread out the way the allowed energies of a quantum system are spread out. That observation has fed a hope, still unproved, that somewhere there is a quantum system whose energy levels are exactly these numbers.

One way to chase that hope is purely empirical. You take a quantum particle trapped in a valley of potential energy, and you adjust the shape of the valley until its allowed energies match the list. If you can make them match very well, have you found something? Or have you only bent a flexible curve until it passed through your data points? This book is about how to tell the difference, and about one research paper [1] that asked exactly this question and answered it with a held-out test.

In this chapter we meet the numbers, see roughly where they come from, see why they look like energy levels, and learn in plain language what it means to fit a model, to memorize, and to predict. We finish with the paper's result in one page and a road map of the book. Where this shows up in the paper: the abstract, Fig. 1, and all of Sec. I (the Introduction and its four strands of prior work, Sec. I A).

What you need

- First-year physics: potential energy curves and turning points (a ball rolling in a valley turns around where its kinetic energy reaches zero).
- First-year physics: standing waves on a string with fixed ends and their harmonics.
- Calculus basics: logarithms (always the natural logarithm log in this book), exponentials, and the value $\pi = 3.14159\ldots$
- Arithmetic averages (the mean of a list of numbers) and squares and square roots.
- No earlier chapter is assumed. Everything else is introduced here in words and developed properly later.

1.1 A list of mysterious numbers

Let us start with the data, the way an experimentalist would. Table 1.1 lists the first ten numbers of our list. We call the n-th number t_n, so $t_1 = 14.134725$, $t_2 = 21.022040$, and so on. The index n is a counting label: $n = 1, 2, 3, \ldots$. These ten values are the first running example of the book (RE1), and you will see them again and again.

Table 1.1: The first ten heights t_n of the nontrivial Riemann zeros (running example RE1), rounded to six decimal places, and the spacing $t_{n+1} - t_n$ to the next one.

n	t_n	$t_{n+1} - t_n$
1	14.134725	6.887315
2	21.022040	3.988818
3	25.010858	5.414018
4	30.424876	2.510186
5	32.935062	4.651116
6	37.586178	3.332541
7	40.918719	2.408354
8	43.327073	4.678078
9	48.005151	1.768681
10	49.773832	

What do you notice? The numbers increase. They are not equally spaced: the gaps jump around between about 1.8 and 6.9. There is no obvious formula like "add 4 each time". And yet they are not completely haphazard either: among these ten no gap is extremely small (small gaps are observed numerically to be rare at all heights computed, Section 1.3), and on average the numbers come a little closer together as you go up. The paper you will learn to read uses the first 10^4 of them for its explicit-formula oracle and its spacing statistics (paper Table V, Fig. 2) and at most the first 1000 in its fitted potentials; its zero number 1000 is $t_{1000} = 1419.42248\ldots$ (paper Sec. II A).

These numbers are called the *heights of the nontrivial zeros of the Riemann zeta function*. We will explain each word of that phrase in Section 1.2. For now, treat them as measured data: a long, precise list that someone has handed you.

Sample Problem 1.1: Reading the spacings of the first ten zeros

Using Table 1.1, find the average spacing between consecutive zeros among the first ten, and the largest and smallest spacings.
Step 1: There are ten zeros, so there are nine spacings between consecutive ones: $t_2 - t_1$,

$t_3 - t_2$, up to $t_{10} - t_9$.
Step 2: The sum of the nine spacings telescopes: every intermediate t_n appears once with a plus sign and once with a minus sign, so

$$(t_2 - t_1) + (t_3 - t_2) + \cdots + (t_{10} - t_9) = t_{10} - t_1 = 49.773832 - 14.134725 = 35.639107.$$

Step 3: The average spacing is the sum divided by the number of spacings:

$$\frac{35.639107}{9} = 3.959901.$$

Step 4: Reading down the last column of the table, the largest spacing is $t_2 - t_1 = 6.887315$ and the smallest is $t_{10} - t_9 = 1.768681$.
Answer:

$$\boxed{\text{average spacing} = 3.959901, \quad \text{largest} = 6.887315, \quad \text{smallest} = 1.768681}$$

1.1.1 Counting the zeros: a staircase

A second way to look at a list of increasing numbers is to count. Define $N(T)$, read "N of T", as the number of zeros whose height t satisfies $0 < t \le T$:

$$N(T) = (\text{number of } n \text{ with } t_n \le T). \tag{1.1}$$

Below $T = 14.134725$ there are no zeros, so $N(T) = 0$ there. At $T = t_1$ the count jumps to 1 and stays at 1 until $T = t_2$, where it jumps to 2, and so on. Plotted against T, $N(T)$ is a staircase with a step of height 1 at each zero (Figure 1.1).

Here is a remarkable fact, which we take on trust now and derive in Chapter 17. The staircase hugs a smooth curve,

$$N_{\text{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{1.2}$$

which is Eq. (1) of the paper. The subscript "sm" stands for "smooth". The staircase wobbles around this curve, sometimes a little above it, sometimes a little below, but it never wanders far. The wobble is the interesting part. It is where the primes hide.

Sample Problem 1.2: The staircase against the smooth curve

Find $N(50)$ and $N(100)$, and compare with $N_{\text{sm}}(50)$ and $N_{\text{sm}}(100)$. You are told that the eleventh zero is $t_{11} = 52.970321$, the twenty-ninth is $t_{29} = 98.831194$ and the thirtieth is $t_{30} = 101.317851$.
Step 1: count. From Table 1.1, $t_{10} = 49.773832 \le 50$ and $t_{11} = 52.970321 > 50$, so exactly ten zeros lie at or below 50: $N(50) = 10$. Similarly $t_{29} \le 100 < t_{30}$ gives $N(100) = 29$.
Step 2: the ratio $t/2\pi$. For $t = 50$, $t/2\pi = 50/6.2831853 = 7.957747$. For $t = 100$, $t/2\pi = 15.915494$.
Step 3: the logarithms. $\log 7.957747 = 2.074146$ and $\log 15.915494 = 2.767293$.
Step 4: assemble Equation (1.2) for $t = 50$.

$$N_{\text{sm}}(50) = 7.957747\times 2.074146 - 7.957747 + 0.875 = 16.505528 - 7.957747 + 0.875 = 9.422781.$$

Step 5: assemble for $t = 100$.

$$N_{\text{sm}}(100) = 15.915494\times 2.767293 - 15.915494 + 0.875 = 44.042844 - 15.915494 + 0.875 = 29.002350.$$

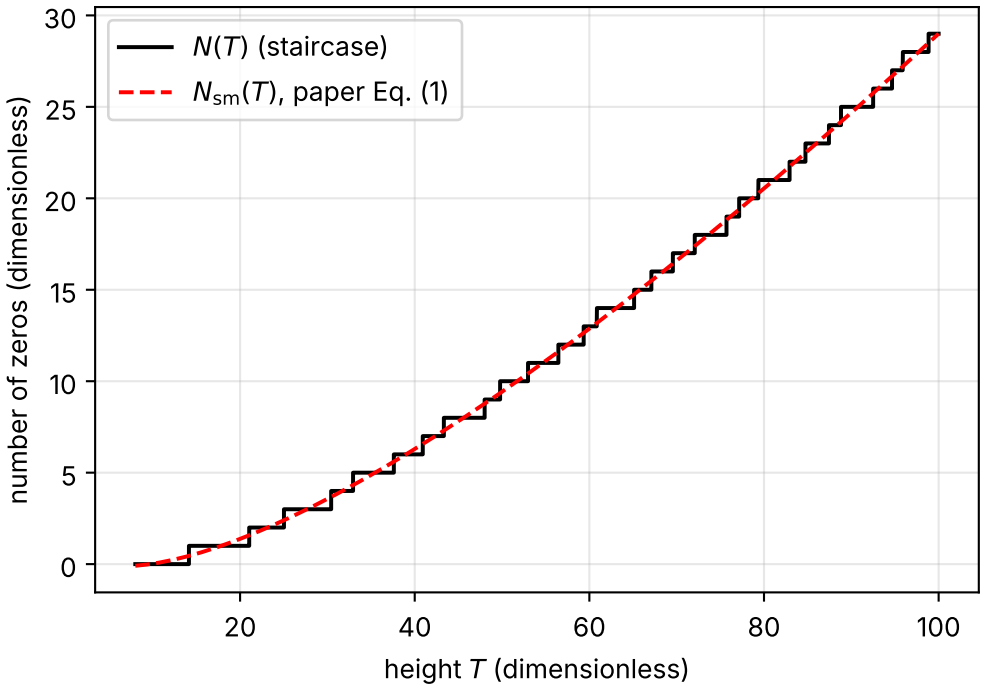


Figure 1.1: The counting staircase $N(T)$ of Equation (1.1), which steps up by 1 at each zero height t_n, and the smooth curve $N_{\mathrm{sm}}(T)$ of Equation (1.2) (paper Eq. (1)) for $8 \le T \le 100$. The staircase oscillates around the smooth curve; in this range the departure stays below one step (checked numerically in this chapter's code).

Answer:

$$\boxed{N(50) = 10,\ N_{\mathrm{sm}}(50) \approx 9.423; \qquad N(100) = 29,\ N_{\mathrm{sm}}(100) \approx 29.002}$$

At $T = 50$ the staircase sits about 0.58 steps above the smooth curve; at $T = 100$ it sits almost exactly on it. The difference $N(T) - N_{\mathrm{sm}}(T)$ is the wobble.

1.1.2 The mean spacing and the fluctuations

Because the staircase follows N_{sm}, the smooth curve tells you how densely the zeros are packed near any height. If the curve climbs by one unit over a stretch of length ℓ in t, then on average one zero sits in that stretch, so the average spacing there is ℓ. The slope of the smooth curve is

$$N'_{\mathrm{sm}}(t) = \frac{1}{2\pi} \log \frac{t}{2\pi} + \frac{t}{2\pi} \cdot \frac{1}{t} - \frac{1}{2\pi} = \frac{1}{2\pi} \log \frac{t}{2\pi}$$

(the product rule on the first term of Equation (1.2), using $\frac{\mathrm{d}}{\mathrm{d}t} \log \frac{t}{2\pi} = \frac{1}{t}$, produces a $+\frac{1}{2\pi}$ that cancels the $-\frac{1}{2\pi}$ from the second term). So one unit of climb takes a stretch $\ell \approx 1/N'_{\mathrm{sm}}(t)$, provided the slope changes little over that stretch. This gives the *local mean spacing*; near height t it is approximately

$$\text{mean spacing near } t \approx \frac{2\pi}{\log(t/2\pi)}. \tag{1.3}$$

The paper writes this spacing as gap_n for the region near zero n. Since $\log(t/2\pi)$ grows as t grows, the spacing shrinks slowly: the zeros crowd together as you go up, which matches what you noticed in Table 1.1.

The paper also uses the *smooth zeros* s_n: the positions where the smooth curve passes through the half-integers, $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$. You can think of s_n as "where zero n would be if there were no wobble". The paper defines the spacing near zero n precisely as $\mathrm{gap}_n = \mathrm{d}s_n/\mathrm{d}n$, the rate at which the smooth zero moves as n increases (treating n as a continuous variable). Differentiating $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ with respect to n by the chain rule gives $N'_{\mathrm{sm}}(s_n)\,\mathrm{d}s_n/\mathrm{d}n = 1$, so $\mathrm{gap}_n = 1/N'_{\mathrm{sm}}(s_n) = 2\pi/\log(s_n/2\pi)$: exactly Equation (1.3) evaluated at $t = s_n$ (Chapter 17 returns to this). The difference $t_n - s_n$, divided by the local mean spacing, is called the *fluctuation* of zero n, measured in mean spacings. Typical fluctuations are a fraction of one spacing. All the predictions in the paper are predictions of these fluctuations. (For the fitted potentials the paper measures fluctuations relative to the levels of the unperturbed potential, which track s_n closely; see Section 1.5.)

Sample Problem 1.3: Measuring an error in mean spacings

The paper reports $t_{500} = 811.1843588465$ (paper Appendix A, key anchors). (a) Estimate the local mean spacing near zero 500 from Equation (1.3). (b) Express an error of 0.5 in t at that height in units of mean spacing. (c) Compare the spacing with the average spacing of the first ten zeros from the Sample Problem "Reading the spacings of the first ten zeros".

Step 1: $t/2\pi = 811.1843588/6.2831853 = 129.10399$.
Step 2: $\log 129.10399 = 4.860618$.
Step 3: Mean spacing $\approx 6.2831853/4.860618 = 1.292672$.
Step 4: An error of 0.5 in t is $0.5/1.292672 = 0.3868$ mean spacings.
Step 5: The ratio of spacings is $3.959901/1.292672 = 3.063$.
Answer:

$$\boxed{\text{spacing} \approx 1.293, \quad 0.5 \text{ in } t = 0.387 \text{ spacings}, \quad \text{low zeros are } 3.06\times \text{ farther apart}}$$

The same error in t is a much bigger mistake near zero 500 than near zero 5, because the zeros are packed about three times more tightly. That is why the paper quotes every error in mean spacings.

Sample Problem 1.4: Fluctuations of the first two zeros

Find the smooth zeros s_1 and s_2, the local mean spacings there from Equation (1.3), and the fluctuations $(t_n - s_n)/\mathrm{gap}_n$ of the first two zeros.

Step 1: rewrite the condition. With $u = t/2\pi$, Equation (1.2) reads $N_{\mathrm{sm}} = u\log u - u + \frac{7}{8}$. The condition $N_{\mathrm{sm}}(s_1) = 1 - \frac{1}{2} = \frac{1}{2}$ becomes $u\log u - u = \frac{1}{2} - \frac{7}{8} = -0.375$. Call the left side $f(u)$.

Step 2: bracket by trial. $f(2.30) = 2.30 \times 0.832909 - 2.30 = -0.3843$, below -0.375; $f(2.32) = 2.32 \times 0.841567 - 2.32 = -0.3676$, above it. The function f increases for $u > 1$, because its derivative is $\log u + 1 - 1 = \log u > 0$ (product rule: $\frac{\mathrm{d}}{\mathrm{d}u}(u\log u) = \log u + 1$). Since f is increasing there, it crosses -0.375 exactly once, and the crossing lies between 2.30 and 2.32.

Step 3: refine. $f(2.311) = 2.311 \times 0.8376803 - 2.311 = -0.375121$, short of -0.375 by 0.000121. Near $u = 2.311$ the slope of f is $\log u = 0.838$, so u must grow by a further $0.000121/0.838 = 0.000144$, giving $u = 2.311144$. (Chapter 4 turns this step into Newton's method.)

Step 4: s_1 and its spacing. $s_1 = 2\pi u = 6.2831853 \times 2.311144 = 14.5213$. The mean spacing is $2\pi/\log 2.311144 = 6.2831853/0.837743 = 7.500$.

Step 5: fluctuation of zero 1. $(14.134725 - 14.5213)/7.500 = -0.3866/7.500 = -0.0515$.

Step 6: zero 2. Now $f(u) = \frac{3}{2} - \frac{7}{8} = 0.625$. Trials: $f(3.28) = 3.28 \times 1.187843 - 3.28 = 0.6161$ (below), $f(3.29) = 3.29 \times 1.190887 - 3.29 = 0.6280$ (above), $f(3.2875) = 3.2875 \times 1.1901274 - 3.2875 = 0.625044$, above 0.625 by 0.000044. With slope $\log u = 1.190$, reduce u by $0.000044/1.190 = 0.000037$, giving $u = 3.287463$, which we round to 3.28746. Then $s_2 = 6.2831853 \times 3.28746 = 20.6557$, the spacing is $6.2831853/\log 3.28746 = 6.2831853/1.19012 = 5.2795$, and the fluctuation is $(21.022040 - 20.6557)/5.2795 = 0.3663/5.2795 = 0.0694$.
Answer:

$$\boxed{s_1 = 14.5213,\ \mathrm{fluct}_1 = -0.052; \qquad s_2 = 20.6557,\ \mathrm{fluct}_2 = +0.069}$$

The first zero sits a little below its smooth position, the second a little above; both displacements are a small fraction of one spacing. The library call `qmnt.fluct(qmnt.zeros(2), np.array([1, 2]))` returns the same two numbers.

Pitfall

Do not compare raw errors in t across different heights. An error of 0.5 is about one eighth of a spacing near $t = 30$ but nearly two fifths of a spacing near $t = 811$. The paper's rms values (Tables I–V) are all in units of the local mean spacing, so that a number like 0.263 means the same thing at every height.

```
import numpy as np
import qmnt

t = qmnt.zeros(10)          # first ten zero heights t_1..t_10
print(t)

gaps = t[1:] - t[:-1]       # nine spacings
print("mean spacing:", gaps.mean())
print("largest, smallest:", gaps.max(), gaps.min())

T = 100.0
print("N(100) =", np.sum(qmnt.zeros(40) <= T))
print("Nsm(100) =", qmnt.Nsm(T))
```

Checkpoint 1.1

Two zeros have fluctuations of $+0.2$ mean spacings and -0.2 mean spacings. One sits near $t = 30$ and the other near $t = 800$. Which of the two is displaced farther from its smooth position, measured in raw t units?

Answer: The one near $t = 30$. Both are displaced by 0.2 of their local spacing, but the spacing near $t = 30$ (about 4) is about three times the spacing near $t = 800$ (about 1.3), so the raw displacement is about three times larger.

Key Ideas

- The data are an infinite increasing list $t_1 = 14.134725, t_2 = 21.022040, \ldots$ (RE1).
- The counting staircase $N(T)$ follows the smooth curve $N_{\mathrm{sm}}(T)$ of Equation (1.2) (paper Eq. (1)) and wobbles around it.

- The local mean spacing, about $2\pi/\log(t/2\pi)$, shrinks slowly with height.
- The fluctuation of zero n is its displacement from the smooth zero s_n, measured in mean spacings. Every prediction in the paper is a prediction of these fluctuations.

1.2 Where the numbers come from: the primes

A *prime* is a whole number greater than 1 whose only whole-number divisors are 1 and itself: $2, 3, 5, 7, 11, 13, \ldots$. Every whole number greater than 1 can be written as a product of primes in essentially one way ($12 = 2 \times 2 \times 3$), which is why primes are called the atoms of arithmetic. Chapter 15 proves these facts.

1.2.1 A function built from primes

In 1859 Bernhard Riemann published a short paper [2] about how the primes are distributed. His main tool was a function now called the *Riemann zeta function*, written $\zeta(s)$ ("zeta of s"). You feed it a number s and it returns a number. For s larger than 1 it can be written as an infinite sum, $\zeta(s) = 1 + 2^{-s} + 3^{-s} + 4^{-s} + \cdots$, and Euler had found earlier that the same function equals an infinite product with one factor for each prime. That product is the bridge between ζ and the primes. We build ζ properly in Chapter 16, after learning about infinite sums and products in Chapter 3.

The input s is allowed to be a *complex number*. You may have heard of i, a number whose square is -1. A complex number has the form $\sigma + it$ with ordinary (real) numbers σ and t: think of it as a point in a plane, with σ the horizontal coordinate and t the vertical one. Chapter 2 teaches all you need about complex numbers. For now, "complex number" means "a point in a plane".

1.2.2 Zeros and the critical line

A *zero* of a function is an input where the function's output equals zero. The zeta function has some zeros that are easy to understand, called trivial zeros. The rest, the *nontrivial zeros*, are the mysterious ones. Every nontrivial zero that has ever been computed lies on one vertical line in the plane, the line where the horizontal coordinate equals one half. So the n-th nontrivial zero (counting upward from the horizontal axis) is written

$$\rho_n = \tfrac{1}{2} + i\,t_n, \tag{1.4}$$

where ρ is the Greek letter rho and t_n is the height of the zero above the horizontal axis. Our list of mysterious numbers is the list of these heights. (Zeros also occur in mirror-image pairs below the axis; we only need the ones above.)

The statement that *all* nontrivial zeros lie on that line is the **Riemann hypothesis**. It is one of the most famous unsolved problems in mathematics. It is a *conjecture*: it has been checked numerically for an enormous number of zeros (Chapter 16 gives the reported extent of these checks) but it has not been proved. The zeros used in the paper lie far below the height up to which all zeros have been verified numerically to lie on the line (the range described in Chapter 16), so for our purposes Equation (1.4) is a fact about the data, not an assumption; the paper itself simply writes $\rho_n = \frac{1}{2} + it_n$. This finite check has no bearing on whether the Riemann hypothesis itself is true.

1.2.3 Primes as musical notes

Now the key connection, which Chapter 18 develops carefully. The wobble of the staircase around the smooth curve can be written, approximately, as a sum of waves, one family of waves

for each prime p. Approximately, and in a sense that Chapter 18 shows to be heuristic rather than exact, the wave for the prime p oscillates as you move up in t with an angular frequency equal to $\log p$, and its amplitude shrinks like $1/\sqrt{p}$. Small primes give the loudest notes. Higher "overtones" with frequencies $2\log p$, $3\log p$, and so on are quieter still. In the paper's Eq. (4) the m-th overtone of prime p has amplitude $1/(\pi m p^{m/2})$ and an overall minus sign, and only $m \le 3$ is kept; Chapter 18 explains where each factor comes from.

The loudest note comes from $p = 2$. The second running example of the book (RE2) is the prime-2 wave,

$$-\frac{1}{\pi}\left[\frac{\sin(t\log 2)}{2^{1/2}} + \frac{\sin(2t\log 2)}{2\cdot 2^{2/2}} + \frac{\sin(3t\log 2)}{3\cdot 2^{3/2}}\right], \tag{1.5}$$

which is the $p = 2$ part of the paper's Eq. (4). You know from first-year physics that $\sin(\omega t)$ repeats when ωt advances by 2π, so its period is $2\pi/\omega$. The first term has $\omega = \log 2 = 0.693147$, so its period in t is $2\pi/\log 2 = 9.0647$. The other two terms have periods half and a third of that, so they also repeat every 9.0647; the whole bracket in Equation (1.5) repeats with period 9.0647 (Figure 1.2).

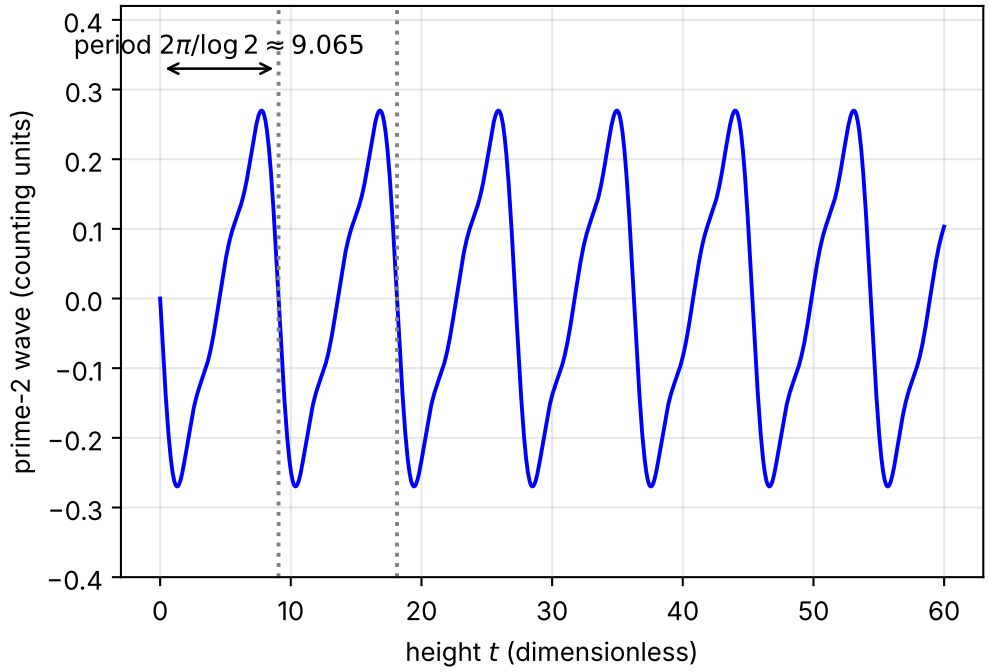


Figure 1.2: The prime-2 wave of Equation (1.5) (running example RE2), plotted for $0 \le t \le 60$. Its period in t is $2\pi/\log 2 \approx 9.065$ (two periods marked). In the paper's continuation probe (Sec. III) a fitted potential is trained on a window of this wave and asked to continue it.

Adding the waves for $p = 2, 3, 5, 7, 11, 13$ gives a formula with no adjustable numbers in it at all. You solve "smooth curve plus waves $= n - \frac{1}{2}$" for t and get a prediction of the n-th zero. The paper calls this the *arithmetic oracle* (paper Sec. II D, Eq. (4)). It is not new mathematics: it is a truncation of the classical explicit formula connecting zeros and primes [3, 4]. But it predicts the zeros remarkably well, and it will serve as the yardstick that tells us how much predictable structure exists.

Sample Problem 1.5: How many zeros fit in one prime-2 period?

Near zero 500 the mean spacing is 1.292672 (Sample Problem "Measuring an error in mean spacings"). How many zeros, on average, lie within one period of the prime-2 wave

there? What is the corresponding number near $t = 50$, where Equation (1.3) gives a mean spacing of 3.029288?
Step 1: The period of the wave in t is $2\pi/\log 2 = 6.2831853/0.6931472 = 9.064720$.
Step 2: The number of zeros per period near zero 500 is the period divided by the spacing: $9.064720/1.292672 = 7.012$.
Step 3: Near $t = 50$: $9.064720/3.029288 = 2.992$.
Answer:

$$\boxed{\text{about 7.0 zeros per period near zero 500; about 3.0 near } t = 50}$$

So even the slowest prime wave wiggles on a scale of a handful of zeros. The paper makes the same point in Sec. V: the $p = 2$ period of about 9 in t is only about 6–7 levels at the heights it studies.

Taken on trust

What this chapter takes on trust. The list of zeros (computed with the mpmath library, Chapter 16); the smooth counting curve Equation (1.2) and the mean-spacing formula Equation (1.3) (derived in Chapter 17); the description of the wobble as a sum of prime waves (Chapter 18, where we also explain that this sum is a heuristic approximation whose exact rigorous form is more delicate). Nothing in this chapter depends on any of these beyond their use as illustrations. Classic accounts are Edwards [5] and Titchmarsh [6].

Checkpoint 1.2

A friend says: "The Riemann hypothesis was proved when computers found that billions of zeros lie on the line." What is wrong with this statement?

Answer: Checking finitely many zeros, however many, does not prove a statement about infinitely many. The Riemann hypothesis remains an open conjecture; the numerical checks are evidence, not proof.

Key Ideas

- The numbers t_n are heights of nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of the Riemann zeta function, a function built from the primes.
- The Riemann hypothesis (all nontrivial zeros on the line $\frac{1}{2} + it$) is open; all zeros used in the paper lie in the range where the line has been verified numerically.
- The wobble of the zeros is approximately a sum of waves with frequencies $m \log p$ in t. The $p = 2$ wave (RE2) has period $2\pi/\log 2 \approx 9.065$.
- A truncated version of this sum, with no adjustable parameters, is the paper's arithmetic oracle (Eq. (4)).

1.3 Do the zeros look like energy levels?

★ Did you know? ***A Fateful Teatime***

In 1972 the number theorist Hugh Montgomery visited the Institute for Advanced Study in Princeton. Over afternoon tea, Sarvadaman Chowla introduced him to the physicist Freeman Dyson. Montgomery described his formula for how pairs of zeta zeros are spaced, and Dyson recognized it at once. It was the pair correlation of eigenvalues of large random complex Hermitian matrices, the kind of model physicists had built for the energy levels of heavy nuclei.

1.3.1 Allowed frequencies and allowed energies

You already know a physical system with a discrete list of allowed values: a string fixed at both ends. It can vibrate only at frequencies $f_1, 2f_1, 3f_1, \ldots$, the harmonics [7]. A plucked guitar string has a "spectrum" of equally spaced frequencies.

Quantum mechanics generalizes this. A particle trapped in a valley of potential energy $V(x)$ can only have certain energies $E_1 < E_2 < E_3 < \cdots$, called its *energy levels* or its *spectrum*. The reason is the same as for the string: the particle is described by a wave, and only waves that "fit" in the valley are allowed. You will learn this from scratch in Chapters 10 and 11. The shape of the valley decides the levels. A box with vertical walls gives levels whose spacing grows with energy. A parabolic valley (a spring) gives equally spaced levels. For valleys of power-law shape, $V \propto |x|^a$, walls rising more slowly than a parabola ($a < 2$) give levels whose spacing shrinks as energy grows, just like the zeros (Chapter 11 shows this; it is a semiclassical result, accurate for high levels).

1.3.2 Level repulsion

There is more to the resemblance than crowding. Look at Figure 1.3. The top row shows the first 29 zeros as tick marks; the middle row shows 29 points thrown down independently at random over the same range; the bottom row shows 29 equally spaced ticks, like harmonics. The zeros are irregular, but not in the way the random points are irregular. Random points often land almost on top of each other. The zeros almost never do: small gaps are rare (a numerical observation, made over all the zeros computed). Physicists call this *level repulsion*, and it is the hallmark of the energy levels of complicated quantum systems.

This is not just an impression. Odlyzko's extensive computations [8] show numerically that the statistics of the gaps match those of the eigenvalues of large random matrices of a particular type, the Gaussian unitary ensemble (GUE). (A matrix is a square table of numbers; its eigenvalues are a list of characteristic numbers attached to it, which for the matrices of quantum mechanics are the energy levels; a random matrix has entries drawn at random. Chapters 8 and 20 explain these ideas.) On the theoretical side, a theorem of Montgomery [9], proved assuming the Riemann hypothesis, shows that the pair correlation of the zeros (how often two zeros lie a given distance apart) agrees with the GUE prediction, but only when tested with a restricted class of test functions. The full agreement, and in particular the agreement of the nearest-neighbor gap distribution, is a numerical observation, not a theorem. Fig. 2 of the paper shows this match for the first 10^4 zeros. Chapter 20 explains what these words mean and derives the curve plotted there.

1.3.3 The Hilbert–Pólya hope

Put these hints together and you arrive at an idea attributed to Hilbert and Pólya: perhaps the t_n are exactly the energy levels of some quantum system. In mathematical language, perhaps

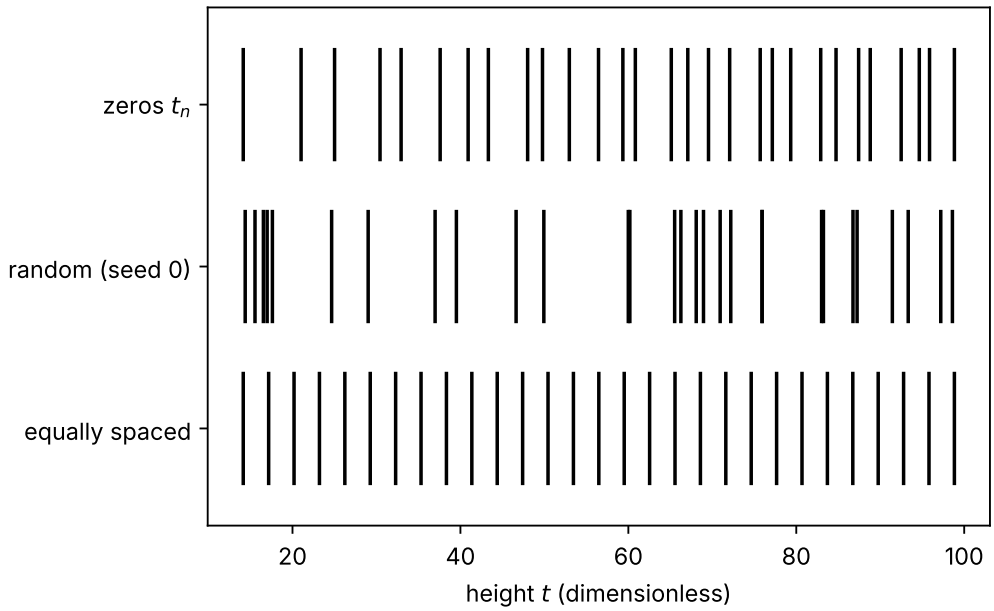


Figure 1.3: Top: the first 29 zero heights $t_1, \ldots, t_{29}$ (all zeros with $t \leq 100$) as tick marks. Middle: 29 points placed independently at random over the same range (fixed seed), for comparison. Bottom: 29 equally spaced ticks spanning the same range, like the harmonics of a string. The zeros are irregular but avoid close pairs, unlike the random points.

they are the spectrum of a *self-adjoint operator*. You will meet operators in Chapter 10; the important property of "self-adjoint" is that it forces every energy level to be an ordinary real number. If someone found a self-adjoint operator whose spectrum is exactly the set of heights t_n (each nontrivial zero being written as $\frac{1}{2} + it_n$), the realness of its energies would force every t_n to be real, and that would put every zero on the line: the Riemann hypothesis would follow. No such operator is known; the idea is a conjecture. A second hint is that the wobble formula looks like the semiclassical formulas physicists use for chaotic quantum systems, with primes playing the role of closed classical orbits [10, 11]. Chapter 21 tells this story.

Pitfall

The Hilbert–Pólya idea is a *conjecture*, a hope. No such operator is known. The GUE statistics are numerical observations and partial theorems; they show the zeros *behave statistically* like energy levels of a certain class of systems, not that a specific system exists. Never read "the zeros look like energy levels" as "the zeros are energy levels".

1.3.4 Two programs: analytic and empirical

Researchers have pursued the hope in two ways. The *analytic program* tries to write down an operator from first principles and then show its spectrum is the list of zeros. The paper does not test this program and makes no claim against it (paper Sec. I and Sec. VI).

The *empirical program* works backwards from the data. Wu and Sprung [12] built a smooth one-dimensional valley whose energy levels reproduce the *smooth* part of the zeros, the positions s_n. Chapter 22 derives their construction; for now, picture a symmetric valley with its bottom at energy $V_0 = 3.10073\pi$, rising slowly on both sides, inside a box with hard walls at $x = \pm L$ (the paper's first scale uses $L = 44.883$; running example RE3). Then they added small fitted wiggles to the valley to reproduce the wobble too. Successors refined this [13, 14, 15]. The implicit logic, in the paper's words, is that "a potential reproducing many zeros accurately has

captured something about the putative zeta operator" (paper Sec. I).

1.3.5 Each level feels the valley near its turning points

One piece of first-year physics will help you understand the paper's results. A ball rolling in a valley with total energy E turns around at the points where the valley's height equals E: the turning points. A quantum particle in level n is not a ball, but it behaves similarly in one way that matters here: its wave is most sensitive to the shape of the valley near its classical turning points (a semiclassical, approximate statement, made precise in Chapter 12). Low levels turn near the bottom of the valley, close to $x = 0$; high levels turn far up the walls. So, roughly, *position in the valley maps to level index.* If you change the valley near some position x, you mainly move the levels that turn near x. Chapter 12 makes this precise.

Checkpoint 1.3

You raise the walls of the valley slightly, only near $x = \pm 25$, leaving the rest unchanged. Which levels move most: the lowest levels, the levels whose turning points are near $x = \pm 25$, or the highest levels?

Answer: The levels whose classical turning points lie near $x = \pm 25$. Their waves are most sensitive to the valley there. Levels that turn well inside never reach that region strongly, and levels that turn far outside are affected too, but less: by the classical analogy the particle moves fast there and spends little time in that region (Chapter 12 makes this quantitative).

Key Ideas

- A quantum particle in a potential valley has a discrete spectrum of energy levels, as a string has harmonics.
- The zeros crowd together and show level repulsion, matching GUE statistics (paper Fig. 2); this is evidence of resemblance, not of a specific operator.
- The Hilbert–Pólya idea (zeros as the spectrum of a self-adjoint operator) is open.
- The empirical program fits a potential, the Wu–Sprung background plus fitted wiggles, to the zeros.
- Each level is most sensitive to the valley near its turning points, so position in the valley maps to level index.

1.4 What it means to fit a model

1.4.1 The ruler and the points

Suppose you hang masses on a spring and record the stretch. You get five points on graph paper. You lay a ruler across them and draw a line that passes close to all of them. That line is a *model*: a rule that turns an input (mass) into an output (stretch). The line has two adjustable numbers, its slope and its intercept. Adjustable numbers in a model are called *parameters*, and choosing them to match data is called *fitting*.

The differences between the data and the model at the data points are the *residuals*; in this chapter we also call them errors. To summarize how well a model fits, we use the root mean square (rms) of the errors $e_1, \dots, e_K$, where K is simply how many errors there are:

$$\text{rms} = \sqrt{\frac{e_1^2 + e_2^2 + \cdots + e_K^2}{K}}. \tag{1.6}$$

Read the name backwards to compute it: square each error, take the mean, take the square root. Squaring makes positive and negative errors count alike. The rms has the same units as the errors. (It is not the same as the standard deviation, which measures spread about the mean of a list; Chapter 19 explains the difference.)

Sample Problem 1.6: Fitting with a ruler

A student records five points (x, y): $(0, 1.0)$, $(1, 2.9)$, $(2, 5.2)$, $(3, 6.9)$, $(4, 9.1)$. She draws the straight line through the first and last points. (a) Find its slope and intercept. (b) Find the residuals and their rms. (c) Predict y at $x = 5$. Later she measures $y = 11.0$ at $x = 5$; what is the prediction error?

Step 1: slope. The line passes through $(0, 1.0)$ and $(4, 9.1)$, so its slope is $(9.1 - 1.0)/(4 - 0) = 8.1/4 = 2.025$.

Step 2: intercept. At $x = 0$ the line passes through $y = 1.0$, so the intercept is 1.0 and the line is $y = 1.0 + 2.025x$.

Step 3: model values. At $x = 1, 2, 3$ the line gives 3.025, 5.050, 7.075; at $x = 0$ and 4 it reproduces the data exactly.

Step 4: residuals (data minus model). At $x = 0, 1, 2, 3, 4$: 0, $2.9 - 3.025 = -0.125$, $5.2 - 5.050 = +0.150$, $6.9 - 7.075 = -0.175$, 0.

Step 5: rms. The squares are 0, 0.015625, 0.0225, 0.030625, 0. Their sum is 0.06875. Divide by $K = 5$: 0.01375. The square root is 0.11726.

Step 6: prediction. At $x = 5$: $1.0 + 2.025 \times 5 = 11.125$. The error against the later measurement is $11.125 - 11.0 = 0.125$.

Answer:

$$y = 1.0 + 2.025x, \quad \text{in-sample rms} = 0.117, \quad \text{prediction at } x = 5 : 11.125 \text{ (error } 0.125)$$

The prediction error is about the same size as the in-sample residuals. That is the signature of a model that has captured the trend.

1.4.2 More knobs: a curve through every point

Now suppose instead of a ruler you use a flexible curve with as many knobs as there are data points. With five points you can always find a polynomial of degree 4, $y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$, which has five parameters, that passes *exactly* through all five. Its in-sample rms is zero. Is it a better model?

Sample Problem 1.7: The memorizing polynomial

For the five points of the Sample Problem "Fitting with a ruler", the degree-4 polynomial through all of them is unique. (Two such polynomials would differ by a polynomial of degree at most 4 that vanishes at all five points; a nonzero polynomial of degree at most 4 has at most four roots, a fact from algebra taken on trust here and revisited with polynomial fitting in Chapter 9; so the difference is zero.) Predict its value at $x = 5$ without finding its coefficients, using the fact that the fourth differences of a degree-4 polynomial at equally spaced points are constant.

Step 1: first differences (each value minus the previous one): $2.9 - 1.0 = 1.9$, $5.2 - 2.9 = 2.3$, $6.9 - 5.2 = 1.7$, $9.1 - 6.9 = 2.2$.

Step 2: second differences: $2.3 - 1.9 = 0.4$, $1.7 - 2.3 = -0.6$, $2.2 - 1.7 = 0.5$.

Step 3: third differences: $-0.6 - 0.4 = -1.0$, $0.5 - (-0.6) = 1.1$.

Step 4: fourth difference: $1.1 - (-1.0) = 2.1$. For a degree-4 polynomial this stays 2.1 forever. (Why: for a power x^k, the binomial expansion gives $(x + 1)^k - x^k =$

kx^{k-1} + (terms with lower powers of x), so taking differences at points spaced by 1 turns a polynomial of degree k into one of degree $k-1$. Four differencings turn a degree-4 polynomial into a polynomial of degree 0, a constant. For example, the values of x^4 at $x = 0, 1, 2, 3, 4$ are $0, 1, 16, 81, 256$; their first differences are $1, 15, 65, 175$, the second $14, 50, 110$, the third $36, 60$, and the fourth 24, and every further fourth difference is also 24. Problem 23 asks you to confirm the prediction numerically.)

Step 5: extend the table one column to the right. New third difference: $1.1 + 2.1 = 3.2$. New second difference: $0.5 + 3.2 = 3.7$. New first difference: $2.2 + 3.7 = 5.9$. New value: $9.1 + 5.9 = 15.0$.

Answer:

polynomial predicts $y(5) = 15.0$; error 4.0 against the measured 11.0

Its in-sample error is zero, yet its prediction error is 32 times that of the ruler line ($4.0/0.125 = 32$). The polynomial bent itself to the noise in the five points. Figure 1.4 shows the picture.

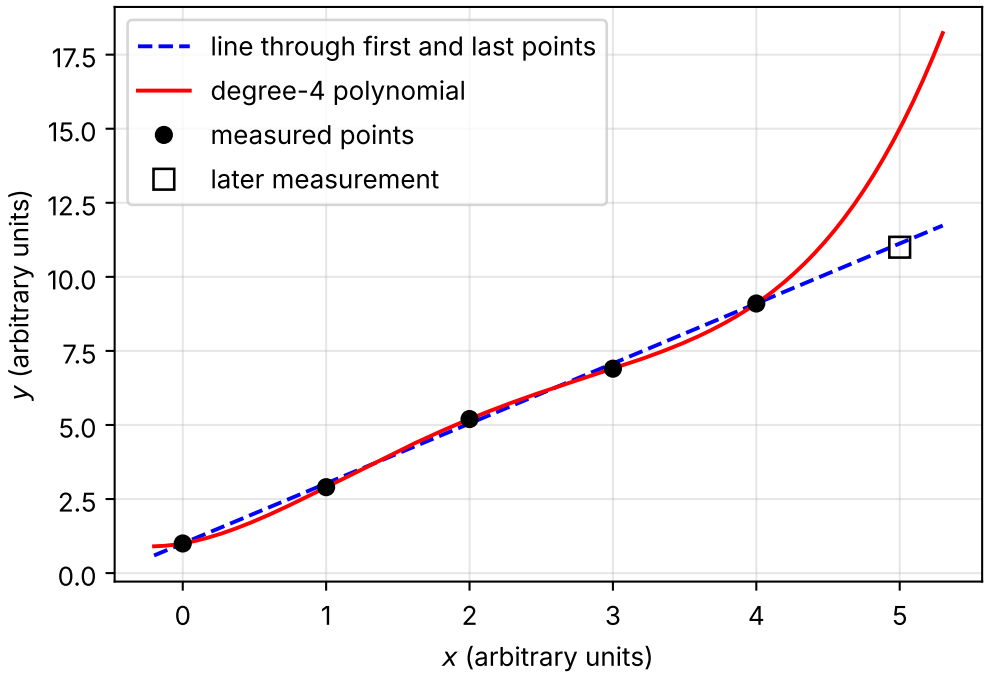


Figure 1.4: Five measured points (dots) fitted by the straight line through the first and last points (dashed; in-sample rms 0.117) and by the degree-4 polynomial through all five (solid; in-sample rms 0). The later measurement at $x = 5$ (square, $y = 11.0$) is predicted to within 0.125 by the line and missed by 4.0 by the polynomial, which predicts 15.0.

This is the simplest picture of memorization, and you will see a fuller version (running example RE4, a polynomial fitted to noisy samples of $\sin t$) in Chapters 9 and 25. The lesson: *when a model has enough knobs to pass through every data point, a perfect fit tells you nothing about whether it has learned the rule.*

1.4.3 Fitting a potential

The paper fits something much more elaborate than a line, but the idea is the same. The model is the Wu–Sprung valley plus a correction $\delta V(x)$ ("delta V") built from M smooth wiggles, each multiplied by an adjustable coefficient. The default wiggles are cosines across the box. The M coefficients are the knobs. For each setting of the knobs, a computer finds the energy levels of the corrected valley (Chapter 23), and an optimizer adjusts the knobs to make the levels match the zeros (Chapter 24). The paper uses M between 200 and 900, fitted to a few hundred zeros. With 900 knobs and 500 zeros there are more knobs than data, a situation even more permissive than the degree-4 polynomial through five points, where knobs and data were equal in number. (Because the levels depend on the knobs in a nonlinear way, counting knobs alone does not guarantee an exact fit; in practice the paper's fits come very close, Finding 1 of Section 1.6.)

Checkpoint 1.4

A model with 400 adjustable coefficients reproduces 400 data points with an rms error of 10^{-4}. A colleague says this proves the model has found the law behind the data. What single additional test would you ask for?

Answer: Ask how well it predicts data points it was not fitted to (held-out data), compared with a simple baseline. With as many knobs as data points, a near-perfect in-sample fit is expected whether or not the model has found the law.

Key Ideas

- A model is a rule with adjustable parameters; fitting chooses them to match data; residuals are data minus model.
- The rms error Equation (1.6) summarizes the size of a list of errors.
- A model with as many parameters as data points can fit exactly while predicting badly.
- The paper's model is the Wu–Sprung potential plus a correction with M fitted coefficients.

1.5 Memorizing versus predicting

1.5.1 Two students

Two students prepare for an exam. The first obtains last year's paper with its answers and memorizes them. The second works through the course and learns how to solve the problems. On last year's paper, both score 100%. The difference only appears on a new exam.

Sample Problem 1.8: The memorizing student

A multiple-choice exam has 20 questions, each with 4 options. Student A has memorized the answers to last year's 20 questions and nothing else. This year's exam reuses 4 of last year's questions; on the other 16, student A guesses at random. Student B understands the material and answers 85% of any set of questions correctly. Find each student's expected score on this year's exam, and on last year's exam.

Step 1: last year's exam. A knows all 20 answers: score 20/20, that is 100%. B gets 85% of 20, which is 17, so 85%.

Step 2: A on the repeated questions. A answers all 4 correctly.

Step 3: A on the new questions. A random guess among 4 options is right with probability 1/4, so on average A gets $16 \times \frac{1}{4} = 4$ of the 16 right.
Step 4: A's expected total: $4 + 4 = 8$ out of 20, which is 40%.
Step 5: B's expected total: $0.85 \times 20 = 17$ out of 20, that is 85%, the same as last year.
Answer:

Last year: A 100%, B 85%.	This year: A 40%, B 85%.

On the old paper A looks better than B. Only the new paper reveals which student learned the rule. And A's new score should be compared with what pure guessing would give: $20 \times \frac{1}{4} = 5$ out of 20, that is 25%. A beats guessing only because of the 4 repeated questions. Those repeats are a *leak* of training data into the test. A clean held-out test contains none of the training questions; on such a test A would expect exactly $20 \times \frac{1}{4} = 5$ out of 20, the guessing (null) score, while B would still expect 17. The clean case is the one the paper sets up for the zeros.

Every element of this story has a name. Last year's paper is the *training data*: the data the model is fitted to. This year's new questions are the *held-out data* (also called test data, or *out-of-sample*, OOS, data): data deliberately kept away from the fitting and used only to score the model. The score on training data is the *in-sample* score. Pure guessing is a *null model*: the simplest possible prediction, against which any claim of skill must be measured.

1.5.2 Training and held-out zeros

In the paper the training data are the first n_{fit} zeros, for example zeros 1–400. The held-out data are the next ones, for example zeros 401–500, which is the paper's principal holdout and the fifth running example of the book (RE5). The model is fitted to zeros 1–400 only; then its predicted levels 401–500 are compared with the true zeros. Neither the fitted potential's prediction nor that of the explicit formula (which has nothing fitted at all) uses any zero past the boundary (paper Fig. 1 caption).

What is the null model here? It is "no fluctuation": predict that each zero sits exactly at its smooth position s_n, or, for the potential, that its levels are those of the unperturbed Wu–Sprung valley. The paper finds that these two nulls give the same error on zeros 401–500, an rms of 0.263 mean spacings (paper Sec. II C). The two are different objects (an eigenvalue spectrum and the solution of a counting equation); it is their numerical coincidence on this holdout (both 0.2631) that lets the paper use one baseline. In other words, the typical fluctuation of those zeros is about a quarter of a spacing, and a model must beat 0.263 to show any skill.

1.5.3 Two scores: rms and correlation

The paper reports two numbers for each prediction. The first is the rms error Equation (1.6) in mean spacings. The second is the *correlation* between predicted and true fluctuations, a number between -1 and $+1$ defined precisely in Chapter 19. For now: correlation $+1$ means the prediction goes up exactly where the truth goes up and down where it goes down (the shapes agree perfectly, whatever the size); 0 means no relation; -1 means the shapes are mirror images. The two scores measure different things, as the next example shows.

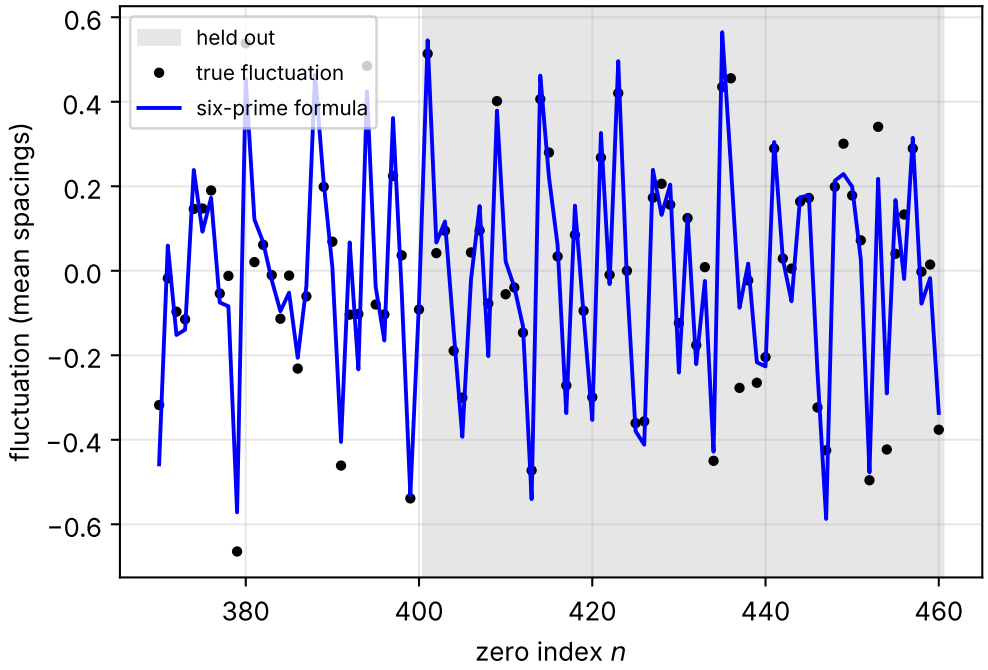


Figure 1.5: True fluctuations (dots, in mean spacings) of zeros 370–460, and the prediction of the six-prime explicit formula of paper Eq. (4) (line), which has no fitted parameters. The shaded region is held out: in the paper's experiment, models are fitted only to zeros 1–400. The paper reports held-out rms 0.074 and correlation +0.96 for this formula on zeros 401–500 (paper Table V). The plotted window 370–460 matches paper Fig. 1; the quoted scores refer to the whole holdout 401–500. This figure omits the fitted operator shown in paper Fig. 1.

Sample Problem 1.9: Why both rms and correlation are reported

The true fluctuations of five zeros are $0.2, -0.3, 0.1, 0.4, -0.4$ (in mean spacings). Compare three predictions: (a) the null, predicting 0 for every zero; (b) a prediction equal to half the truth, $0.1, -0.15, 0.05, 0.2, -0.2$; (c) a prediction equal to minus half the truth. Give each one's rms error and say what its correlation with the truth is.

Step 1: the null. Its errors are the truth itself. Squares: $0.04, 0.09, 0.01, 0.16, 0.16$; sum 0.46; mean 0.092; rms $\sqrt{0.092} = 0.3033$. Its correlation is undefined: correlation is defined only for predictions that vary from zero to zero, and a constant prediction has no shape. So no correlation is reported for the null; it carries no information about the shape.

Step 2: half the truth. Each error is truth minus prediction $= 0.5\times$ truth. Squaring multiplies every square by 0.25, so the mean square is $0.25 \times 0.092 = 0.023$ and the rms is $0.5 \times 0.3033 = 0.1517$. The prediction has exactly the shape of the truth, so the correlation is $+1$.

Step 3: minus half the truth. Each error is truth $-(-0.5\times$ truth$) = 1.5\times$ truth. The rms is $1.5 \times 0.3033 = 0.4550$. The shape is the mirror image, so the correlation is -1.

Answer:

$$\boxed{\text{rms: null } 0.303, \quad \text{(b) } 0.152 \text{ (corr} + 1), \quad \text{(c) } 0.455 \text{ (corr} - 1)}$$

A prediction with the right shape but the wrong size can still have a sizeable rms; a prediction with the wrong shape can be worse than the null. That is why the paper always gives the pair "rms / corr", and why, in this book, we call a result skillful only if the rms is below the null *and* the correlation is reproducibly positive, which is how the paper reads its own tables.

Checkpoint 1.5

A fitted model's held-out rms is 0.30 where the null is 0.263, and its correlation with the truth is -0.02. Has it learned anything that transfers to the held-out zeros?

Answer: No. It does worse than predicting no fluctuation at all, and its output has no relation to the shape of the truth. Whatever it learned in-sample does not transfer.

Key Ideas

- Training data are used for fitting; held-out (out-of-sample) data are used only for scoring.
- Every claim of skill is relative to a null model; for the zeros the null is "no fluctuation", with rms 0.263 on zeros 401–500 (RE5).
- rms measures size of error; correlation measures agreement of shape. Both are needed.
- Memorizing gives a perfect training score and no held-out skill; learning the rule gives similar scores on both.

1.6 The paper's result in one page

We can now say in plain words what the paper [1] did and found. Its title is the summary: "Fitted potentials on the Wu–Sprung background memorize the Riemann zeros while the explicit formula predicts them."

The question. The empirical Hilbert–Pólya program judges a fitted potential by how

well its levels match the zeros it was fitted to. The paper applies the test that program had lacked: does the fitted potential predict zeros it was not shown? The scope is deliberately narrow: fitted corrections on the Wu–Sprung background, in three families of wiggles (cosines; Chebyshev polynomials, a standard family of polynomials; and local "hat" functions, little triangular bumps; all defined in Chapter 24), with one fitting method, in two box sizes that the paper calls scale 1 ($L = 44.883$, first 500 levels scored) and scale 2 ($L = 59.957$, first 1000 levels). Other backgrounds and other families are not tested, and analytic operator proposals are not touched (paper Sec. I).

Finding 1: fitting is easy. With a few hundred coefficients the fitted potential reproduces its training zeros very closely; on all 500 zeros of the first scale it reaches rms below 10^{-4} spacings at $M = 900$ (paper Sec. III).

Finding 2: transfer is nil. In every configuration examined (two scales, several splits, three bases, a range of M, regularization, that is a penalty that discourages large coefficients (Chapter 9), and optimizer restarts), the fitted potentials show no reproducible held-out skill: their held-out rms never improves on the null, and no reproducible positive correlation with the truth appears (paper Tables I–III). One Chebyshev row is marginal, and the paper reports it as such: rms 0.291 with correlation +0.18, whose bootstrap intervals include the null and reach down to zero; a small positive correlation for that row is not excluded (paper Table II). For the principal case (fit zeros 1–400 with $M = 400$ cosines, score zeros 401–500) the in-sample rms is 0.137 but the held-out rms is 0.369, worse than the null 0.263, with correlation −0.07 (paper Table I).

Finding 3: the fit does as well on fakes as on the real zeros. In the principal configuration (scale 1, $M = 400$, zeros 1–400, ten fake draws of each kind), refitted to shuffled versions of the fluctuations, to phase-randomized fakes [16] and to amplitude-adjusted fakes [17] (the last two keep the overall wiggliness, the power spectrum, but have no arithmetic content), the fit reaches the same in-sample accuracy as on the real zeros (paper Sec. III). Its in-sample success measures its capacity, not its understanding.

Finding 4: it did not continue even one prime wave. Trained on levels 1–400 with a target that is nothing but the prime-2 wave (RE2), it reproduces the wave in-sample (rms 0.056, correlation +0.95) and, under the fitting protocol used, did not continue it on levels 401–500 (held-out correlation +0.07, rms worse than predicting zero fluctuation) (paper Sec. III, "Arithmetic-continuation probe"). The paper calls this "the memorization mechanism exhibited in isolation".

Finding 5: the arithmetic predicts. The explicit formula truncated to six primes, with zero fitted parameters, scores rms 0.074 and correlation +0.96 on zeros 401–500; with 25 primes, 0.043 and +0.99; the skill hardly changes up to zero 10^4 (paper Table V, Fig. 1).

Finding 6: the obstacle is the parametrization, not the parameter count. A generic 900-coefficient sum of sinusoids also fails to beat the null, but a fitted model told only that the fluctuation is a sum of a *few* sinusoids finds, from 400 zeros, frequencies at logarithms of primes and prime powers, and predicts zeros 401–500 at 0.034 with correlation +0.99 (paper Sec. IV, Table IV). That the zeros' wobble is concentrated at those frequencies is classical [18, 19]; the paper uses it as a control, not as a new result.

A side lesson. A separate investigation of complex-valued (non-Hermitian) corrections ("Hermitian" is the property that guarantees real energies, Chapter 10; these corrections lack it, Chapter 14) found an apparently meaningful in-sample "floor" that turned out to be manufactured by an approximation used outside its range of validity and by a numerically dead parameter feeding on round-off noise; exact computation dissolved it (paper Sec. V).

The proposal. Any future claim that a fitted operator "captures" the zeros should come with held-out scores against its own null, including a split within one height range; scoring of the actual operator; evidence of robustness to regularization and restarts; and refits to shuffled and phase-randomized targets (paper Sec. VI).

The paper keeps its explanation of *why* the fits fail modest: it offers a heuristic reading (the potential addresses levels locally through their turning points, while the fluctuation is a sum of long-range waves in t) and states that deriving the memorization result is an open problem.

Sample Problem 1.10: Reading headline numbers against the null

Using the paper's numbers, express (a) the operator's held-out rms for the principal case and (b) the six-prime oracle's held-out rms on the same holdout as multiples of the null 0.263.
Step 1: Operator, paper Table I (scale 1, fit zeros 1–400, $M = 400$): $0.369/0.263 = 1.403$.
Step 2: Six-prime oracle, paper Table V (holdout 401–500): $0.074/0.263 = 0.281$.
Step 3: Interpret. The operator's error is about 40% larger than predicting no fluctuation. The oracle's error is about 28% of the null, a reduction of $1 - 0.281 = 0.719$, about 72%.
Answer:

operator: $1.40 \times$ null (worse); six-prime oracle: $0.28 \times$ null (72% better)

Checkpoint 1.6

The arithmetic oracle "is not a fair machine-learning competitor" (paper Sec. IV). Why not, and what is its role then?

Answer: It is handed the correct functional form (waves at frequencies $m \log p$) rather than learning anything. Its role is informational: it proves that transferable, predictable structure about unseen zeros exists, and it measures how much of it simple arithmetic captures.

Key Ideas

- The paper tests fitted Wu–Sprung potentials by held-out prediction. They fit their training zeros closely and predict nothing beyond the null.
- The fits succeed equally on shuffled and phase-randomized fakes, and did not, under the fitting protocol used, continue a single prime wave.
- The truncated explicit formula, with zero fitted parameters, predicts held-out zeros with correlation up to $+0.99$ through zero 10^4.
- A fitted model with only a sparsity prior in frequency (the assumption that only a few frequencies are present) recovers the prime frequencies; the obstacle, for the families and fitting protocol tested, is the potential as parametrization rather than the number of parameters.
- The explanation of the failure is heuristic; the scope is limited to the families tested.

1.7 A road map of the book

★ Did you know? *Eight Pages That Changed Mathematics*

Riemann's famous 1859 paper, *On the Number of Primes Less Than a Given Magnitude*, was written on his election to the Berlin Academy and runs only about eight pages. It was his only paper on number theory. Yet it introduced the zeta function on the complex plane, the explicit formula linking zeros to primes, and the Riemann hypothesis itself.

To read the paper line by line you need tools from several fields. The book is organized in six parts (Figure 1.6).

Part I, Mathematical foundations (Chapters 2–9): complex numbers and the exponential; series and infinite products; approximation, asymptotics and root finding; Fourier series; the Fourier transform and the periodogram; vectors and matrices; eigenvalues; least squares, the singular value decomposition and regularization.

Part II, Quantum mechanics of a particle on a line (Chapters 10–14): wavefunctions and the Schrödinger equation; bound states; the semiclassical (WKB) approximation and level counting; perturbation theory; complex potentials and PT symmetry.

Part III, Primes and the zeta function (Chapters 15–18): the primes; the zeta function; counting the zeros; the explicit formula.

Part IV, Spectra, randomness and operators (Chapters 19–22): probability and statistics; level statistics and random matrices; the Hilbert–Pólya dream; the Wu–Sprung construction.

Part V, Computation, fitting and generalization (Chapters 23–27): solving the Schrödinger equation on a computer; fitting a potential to a spectrum; memorization and generalization; uncertainty and controls; learning frequencies from data.

Part VI, The paper, section by section (Chapters 28–32): five chapters that teach no new mathematics but read the paper in order, rerun its key computations at small scale, and weigh its strengths and limits.

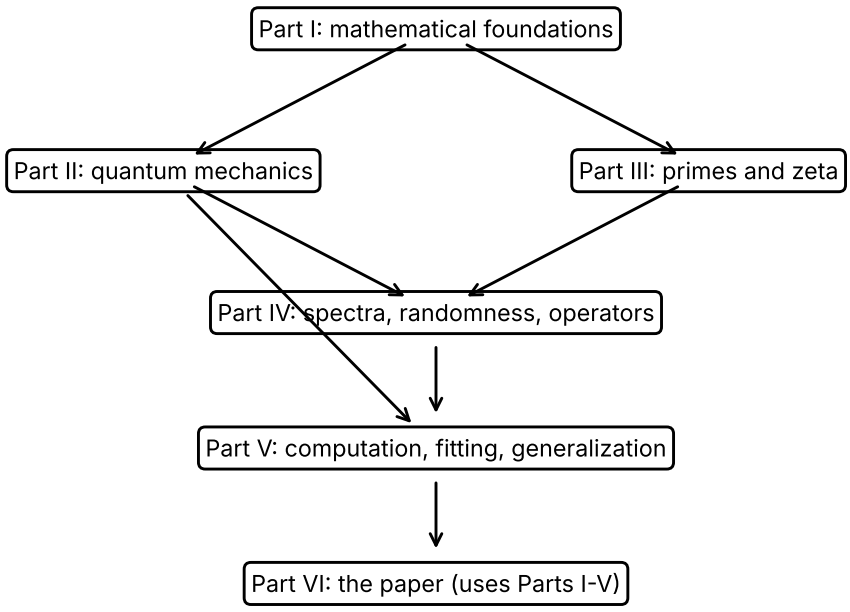


Figure 1.6: Dependency diagram of the six parts of the book. An arrow from one part to another means the second uses tools from the first. Part I underlies everything; Parts II and III can be read in either order; Part VI uses all earlier parts.

If you are mainly interested in the physics, Parts I and II, then Chapters 17, 21 and 22, give the quantum side. If you are mainly interested in the data science, Chapters 9, 19 and Part V give the fitting and testing side. To read the paper fully you need all of it.

Key Ideas

- Six parts: mathematics, quantum mechanics, number theory, spectra and statistics, computation and fitting, and the paper itself.
- Part I underlies everything; Part VI uses everything.

1.8 How to use this book

Sample Problems are fully worked examples, with every step numbered and the answer boxed. Cover the solution and try each one first.

Checkpoints are quick questions to test understanding; the answer follows immediately.

Pitfalls warn of common mistakes. **Beyond** boxes say exactly what is being taken on trust and where the proof can be found.

Status labels. Mathematics and science mix statements of very different strength. We label them consistently: a *theorem* is proved (in the book or, if the proof is beyond us, in a cited source, and we say which); a *heuristic* or *approximate* statement holds under stated conditions; a *conjecture* is open (the Riemann hypothesis and the Hilbert–Pólya idea are conjectures); a *numerical observation* is something seen in computations, which may or may not hold beyond them.

Problems at the end of each chapter are graded: level 1 problems check that you can use the ideas directly; level 2 problems require combining ideas; level 3 problems are harder or open-ended. Several problems in every chapter use the paper's own numbers and methods, and several are computational.

Code. Computational problems use Python with the numpy library and the book's companion library `qmnt`, which computes zeros, smooth zeros, the explicit-formula oracle, and the potentials and spectra of the paper. Appendix A teaches all the Python you need, starting from nothing. The listing in Section 1.1 is a typical example: a few lines that compute something you can check by hand.

Reproducibility. The paper's code and data are archived publicly [20]. In Part VI you will rerun parts of the paper's pipeline at small scale. The habit we want you to form is the one the paper itself argues for: before believing a number, see how it was computed and whether it can be recomputed.

Key Ideas

- Work the Sample Problems yourself before reading the solutions; use Checkpoints to check understanding.
- Watch the status of every statement: theorem, heuristic, conjecture, or numerical observation.
- Appendix A teaches the Python used throughout.

1.9 Paper connection

This chapter prepares you to read the opening of the paper [1]. Here is what each part corresponds to, using the R1 numbering.

Title and abstract. The abstract contains the whole argument in compressed form. You can now decode most of its phrases: "empirical Hilbert–Pólya constructions" (Section 1.3); "fit quantum Hamiltonians to the nontrivial Riemann zeros" (Section 1.4; a Hamiltonian is the energy operator of a quantum system, Chapter 10); "out-of-sample prediction" (Section 1.5); "up to

900 fitted coefficients, in cosine, Chebyshev, and local piecewise-linear bases" (the three families of wiggles); "relative to the unfitted potential" (the null); "shuffled and phase-randomized surrogate targets" (Finding 3 of Section 1.6); "a pure prime-2 explicit-formula oscillation" (RE2, Equation (1.5)); "with zero fitted parameters predicts held-out zeros with correlation up to +0.99 through zero 10^4" (Finding 5). Chapter 28 unpacks the abstract sentence by sentence.

Sec. I, first paragraph. The definition $\rho_n = \frac{1}{2} + it_n$ is our Equation (1.4). The phrase "the spectrum of a self-adjoint operator" is the Hilbert–Pólya idea (Section 1.3). The two strands of circumstantial evidence (GUE statistics [9, 8] and the trace-formula analogy [10, 11]), together with the continuing stream of explicit operator proposals, are the subjects of Chapters 20 and 21.

Sec. I, second paragraph. The empirical program of Wu and Sprung [12] and successors, and its "implicit logic", are summarized in Section 1.3. The scope statement (only fluctuation potentials on the Wu–Sprung background, three bases; no claim against analytic constructions; "capture" means predictive transfer measured by rms and correlation relative to the smooth baseline) is the honest boundary of every result. Notice also the sentence "The novelty claimed is likewise not the explicit formula": the paper claims novelty only for the controlled out-of-sample comparison.

Sec. I, third paragraph, and Fig. 1. The summary of findings is our Section 1.6. Fig. 1 shows zeros 370–460 after a fit on zeros 1–400 at scale 1 with $M = 400$; past the boundary the fitted operator's held-out correlation is -0.07 and the six-prime formula's is $+0.96$. Our Figure 1.5 reproduces the true fluctuations and the six-prime prediction (not the operator, which needs Chapter 24).

Sec. I A, Relation to prior work. Its four strands are: (a) inverse-spectral fits [12, 13, 14, 15], which the paper says have been evaluated mainly by agreement with their own training zeros, a statement it supports by a citation-graph audit recorded in Appendix A; (b) explicit formulas for the n-th zero [3, 4], including Connes's prime-only construction [21], all on the arithmetic side of the comparison; (c) machine learning on zeros and arithmetic data; (d) operator proposals, which are not the target. Chapters 21, 22 and 28 return to these. The Lambert-W function mentioned in strand (b) is introduced in Chapter 4.

Numbers you have already met. Paper Eq. (1) is our Equation (1.2); "All errors below are quoted in units of the local mean spacing $\mathrm{gap}_n = ds_n/dn$" (paper Sec. II A) is the convention of the Sample Problem "Measuring an error in mean spacings". The null 0.263 on zeros 401–500 comes from paper Sec. II C and Tables I and V. The headline rows used in the Sample Problem "Reading headline numbers against the null" are from paper Table I (operator, $0.369/-0.07$) and Table V (six primes, $0.074/+0.96$). The prime-2 period of about 9 in t and "6–7 levels" appear in paper Sec. V. Our Figure 1.3 illustrates qualitatively, with 29 zeros, the level repulsion that paper Fig. 2 shows quantitatively as a spacing histogram of the first 10^4 zeros; Chapter 20 explains that histogram.

Sec. II F, Use of artificial intelligence. The paper discloses that its computations, analysis and drafting were carried out with substantial assistance from a large language model, and that every reported number is regenerated by an archived scripted pipeline [20]. We note this neutrally here; Chapters 28 and 32 discuss why such disclosures and reproducibility records matter.

1.10 Summary

- The data are the heights t_n of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of the Riemann zeta function, Equation (1.4); the first ten (RE1) are in Table 1.1.

- The counting staircase $N(T)$, Equation (1.1), follows the smooth curve $N_{\mathrm{sm}}(T)$, Equation (1.2) (paper Eq. (1)); for example $N(100) = 29$ and $N_{\mathrm{sm}}(100) \approx 29.00$.

- The local mean spacing is about $2\pi/\log(t/2\pi)$, Equation (1.3): about 1.29 near zero 500. Errors are measured in these units.
- The wobble of the zeros is approximately a sum of prime waves with frequencies $m \log p$; the $p = 2$ wave (RE2, Equation (1.5)) has period $2\pi/\log 2 \approx 9.065$.
- The zeros resemble quantum energy levels (crowding, level repulsion, GUE statistics). The Riemann hypothesis and the Hilbert–Pólya idea are open.
- Fitting chooses parameters to match data; the rms Equation (1.6) summarizes errors. A model with as many parameters as data can fit perfectly and predict badly (Figure 1.4).
- Skill is measured on held-out data, against a null model, with both rms and correlation. For zeros 401–500 (RE5) the null rms is 0.263.
- The paper's result: fitted Wu–Sprung potentials memorize (held-out rms 0.369, correlation -0.07 in the principal case) while the six-prime explicit formula predicts (0.074, $+0.96$) with no fitted parameters; a sparsity prior in frequency suffices to learn the arithmetic structure.

1.11 Problems

1. • In your own words, explain the difference between a model that memorizes its data and one that predicts. Give an everyday example of each that is not in this chapter.

2. • Using Table 1.1, compute the spacing between the ninth and tenth zeros directly from their heights, and say whether it is larger or smaller than the average spacing among the first ten zeros.

3. • Near $t = 50$ the mean spacing from Equation (1.3) is 3.029288. A prediction misses a zero near $t = 50$ by 0.6 in t. Express the error in mean spacings.

4. • Which of the following statements are correct? (a) The Riemann hypothesis has been proved. (b) Every zero used in the paper has been verified to lie on the line $\frac{1}{2} + it$. (c) A self-adjoint operator whose spectrum is the t_n is known. (d) The gap statistics of the zeros match those of GUE random matrices numerically.

5. •• A student records the points $(0, 2.0)$, $(1, 3.4)$, $(2, 5.1)$, $(3, 6.4)$ and draws the line through the first and last. Find its slope, the residuals at $x = 1$ and $x = 2$, and its prediction at $x = 4$.

6. •• From paper Table V, the null on zeros 401–500 is 0.263, the six-prime oracle scores 0.074, and the 25-prime oracle 0.043. From paper Table IV, the sparse frequency model C2 scores 0.034. Express each of the three as a percentage reduction from the null.

7. •• From paper Table I (scale 1, fit 1–400), the unregularized $M = 400$ row has OOS rms 0.369 and the ridge $\mu = 0.3$ row has 0.265; the null is 0.263. By what percentage is each worse than the null? What does the second number suggest regularization does?

8. •• A student memorizes the answers to 30 past questions. A new exam has 25 multiple-choice questions with 5 options each, of which 5 are past questions. Find the student's expected percentage score if she guesses on the rest, and compare it with the score from guessing everything.

9. • The paper's largest cosine fit on the first scale uses $M = 900$ coefficients for 500 zeros. How many coefficients per zero is that? How many coefficients per training zero does the principal $M = 400$, zeros 1–400 fit use? How many fitted parameters does the six-prime oracle have?

10. •• The errors of a prediction on four zeros are $0.3, -0.1, 0.0, -0.2$ mean spacings. Find the rms error.

11. •• Find the periods in t of the second ($m = 2$) and third ($m = 3$) terms of the prime-2 wave Equation (1.5), and explain why the sum of all three terms has period $2\pi/\log 2$.

12. ••• Suppose the true fluctuations have rms R and a prediction equals a times the truth for some number a. (a) Show the rms error is $|1 - a|\ R$. (b) Evaluate it for $a = 0, 0.5, 2, -1$ when $R = 0.263$. (c) For which a is the correlation $+1$? Which of these predictions beat the null? Comment on a prediction with perfect shape and rms equal to the null.

13. •• Using `qmnt` or numpy, compute the average spacing of the first ten zeros and of the first 100 zeros. How does the answer change, and why?

14. •• Using `qmnt` or numpy, count the zeros with $t \le 50$ and with $t \le 100$, and evaluate N_{sm} at 50 and 100. Find the wobble $N(T) - N_{\mathrm{sm}}(T)$ at both heights.

15. ••• Using `qmnt`, compute (a) the rms of the fluctuations of zeros 401–500 (the null), and (b) the rms and correlation of the six-prime oracle on the same zeros, using `qmnt.oracle_zeros` and `qmnt.score`. Compare with paper Table V.

16. •• Paper Fig. 1 shows zeros 370–460 after a fit on zeros 1–400. How many training zeros and how many held-out zeros are shown? Why does the figure state that neither prediction uses any zero past the boundary?

17. •• In the paper's interpolation split (Sec. III) the potential is fitted to zeros 1–200 and 301–500 and scored on 201–300. (a) How many training zeros are there? (b) The gap error is 2.64 spacings and the null on the gap is 0.255. What is the ratio? (c) Why is an error far above the null worse news than an error equal to the null?

18. ••• Using the idea that each level is most sensitive to the valley near its turning points (Section 1.3), explain why an optimizer fitting levels 1–200 and 301–500 might place a large correction in the region where levels 201–300 turn, and why this costs it nothing in-sample. Then read the paragraph "Interpolation split" of paper Sec. III and compare.

19. •• In the arithmetic-continuation probe (paper Sec. III) the fit reproduces the prime-2 wave in-sample with correlation $+0.95$ and continues it with held-out correlation $+0.07$. Explain in words what each number shows, and why the paper calls this experiment "the memorization mechanism exhibited in isolation".

20. • Near $t = 50$ the mean spacing is 3.029288. How many zeros lie, on average, within one period of the prime-2 wave there?

21. ••• Design a test you would require before believing a claim that a fitted model "captures" the zeros. List at least four ingredients, then compare your list with the four requirements (i)–(iv) proposed in paper Sec. VI.

22. •• A weather forecaster claims skill because her forecast of tomorrow's temperature has an rms error of 3 degrees. What null model would you compare it with? Why is the claim meaningless without it? Relate your answer to the null 0.263 of the paper.

23. ••• Using numpy, fit a degree-4 polynomial through the five points of the Sample Problem "Fitting with a ruler" with `np.polyfit` and evaluate it at $x = 5$ and $x = 6$. Compare with the straight line of that Sample Problem at $x = 6$. Also confirm the value at $x = 6$ by extending the difference table of the Sample Problem "The memorizing polynomial" one more column.

24. •• A prediction misses a zero by 0.5 in t. Express this error in mean spacings if the zero is near $t = 50$ (spacing 3.029288) and if it is near zero 500 (spacing 1.292672). Which is the bigger mistake, and why do the paper's tables use mean spacings?

25. ••• In the paper's permutation and surrogate tests (Sec. III) the in-sample rms of the fit is 0.134 ± 0.010 (shuffled), 0.141 ± 0.006 (phase-randomized) and 0.138 ± 0.005 (amplitude-adjusted), against 0.137 for the true zeros and 0.244 for the null. What does this tell you about what in-sample accuracy measures? Relate it to the memorizing student who could memorize a fake answer key just as easily as the real one.

1.12 Answers to selected problems

1. A memorizing model reproduces the data it was fitted to but has no rule that extends to new cases; a predicting model has captured a rule that gives similar accuracy on new data. Example: someone who memorizes a city's bus timetable for one route (memorizing) versus someone who knows the buses run every 12 minutes from 6 a.m. (a rule that predicts departures on any day). The test is always new data.

2. $t_{10} - t_9 = 49.773832 - 48.005151 = 1.768681$, smaller than the average 3.959901; it is in fact the smallest of the nine spacings.

3. $0.6/3.029288 = 0.198$ mean spacings.

4. (a) Incorrect: it is open. (b) Correct: the zeros used lie far below the height to which all zeros have been verified numerically to lie on the line (Chapter 16); the paper itself simply writes $\rho_n = \frac{1}{2} + it_n$. (c) Incorrect: no such operator is known. (d) Correct, as a numerical observation (paper Fig. 2; Odlyzko's computations). Montgomery's theorem, proved assuming the Riemann hypothesis, concerns pair correlation for a restricted class of test functions; it does not establish the gap distribution.

5. Slope $(6.4 - 2.0)/3 = 1.4667$; line $y = 2.0 + 1.4667x$. At $x = 1$ the line gives 3.4667, residual $3.4 - 3.4667 = -0.0667$. At $x = 2$ it gives 4.9333, residual $5.1 - 4.9333 = +0.1667$. Prediction at $x = 4$: $2.0 + 4 \times 1.4667 = 7.867$.

6. Reduction $= 1 - (\text{score}/0.263)$. Six primes: $1 - 0.074/0.263 = 0.719$, so 71.9%. 25 primes: $1 - 0.043/0.263 = 0.837$, so 83.7%. C2: $1 - 0.034/0.263 = 0.871$, so 87.1%.

7. $0.369/0.263 = 1.403$: 40.3% worse. $0.265/0.263 = 1.0076$: 0.8% worse. Strong regularization pulls the fit back toward the null; it removes the damage but does not create skill (paper Sec. III: regularization moves the fits "from 'memorize more' to 'do nothing' and never creates skill").

8. Expected correct: $5 + 20 \times \frac{1}{5} = 5 + 4 = 9$ out of 25, that is 36%. Guessing everything: $25 \times \frac{1}{5} = 5$ out of 25, that is 20%.

9. $900/500 = 1.8$ coefficients per zero. $400/400 = 1$ coefficient per training zero. The oracle has zero fitted parameters.

10. Squares $0.09, 0.01, 0, 0.04$; sum 0.14; mean 0.035; rms $\sqrt{0.035} = 0.187$.

11. The $m = 2$ term has $\omega = 2\log 2$, period $2\pi/(2\log 2) = 4.532$; the $m = 3$ term has period $2\pi/(3\log 2) = 3.022$. The first term repeats after 9.065; after the same interval the second term has completed exactly two periods and the third exactly three, so all three return to their starting values together. No shorter interval works for the first term, so the sum has period 9.065.

12. (a) The error at each zero is truth $-\,a\times$ truth $=(1-a)\times$ truth. Each squared error is $(1-a)^2$ times the squared truth, so the mean square is $(1-a)^2R^2$ and its square root is $|1-a|\,R$. (b) $a=0$: 0.263 (the null); $a=0.5$: 0.1315; $a=2$: 0.263; $a=-1$: 0.526. (c) Correlation is $+1$ for every positive a, that is $a=0.5$ and $a=2$. Only $a=0.5$ beats the null (any $0<a<2$ would). The prediction $a=2$ has perfect shape yet rms equal to the null: rms alone would call it useless, correlation alone would call it perfect. Each score sees only half the story.

13. First ten: 3.959901 (Sample Problem "Reading the spacings of the first ten zeros"). First 100: $(t_{100}-t_1)/99=(236.524230-14.134725)/99=2.246359$. The average spacing is smaller because the zeros crowd together with height, as Equation (1.3) predicts. Code:

```
import qmnt
t = qmnt.zeros(100)
print((t[9] - t[0]) / 9, (t[-1] - t[0]) / 99)
```

14. $N(50)=10$, $N_{\rm sm}(50)=9.423$, wobble $+0.577$. $N(100)=29$, $N_{\rm sm}(100)=29.002$, wobble -0.002. Code:

```
import numpy as np
import qmnt
t = qmnt.zeros(40)
for T in (50.0, 100.0):
    N = np.sum(t <= T)
    print(T, N, qmnt.Nsm(T), N - qmnt.Nsm(T))
```

15. (a) About 0.263, matching the null of paper Table V (0.2631 in paper Appendix A anchors). (b) About 0.074 and $+0.96$, matching paper Table V. Code:

```
import numpy as np
import qmnt
n = np.arange(401, 501)                  # holdout indices
t = qmnt.zeros(500)
print(qmnt.rms(qmnt.fluct(t[n-1], n)))   # null rms
pred = qmnt.oracle_zeros(n, qmnt.first_primes(6))
print(qmnt.score(pred, n))               # (rms, corr)
```

16. Training zeros shown: 370–400, that is $400-370+1=31$. Held-out zeros shown: 401–460, that is 60. The statement guarantees that the predictions in the held-out region are genuine predictions: if any held-out zero had been used, a good score could be partly copying.

17. (a) $200+200=400$. (b) $2.64/0.255=10.4$, roughly ten times the null as the paper says. (c) An error equal to the null means the model added nothing; an error far above it means the model actively did harm, displacing levels that would have been closer to the truth had it done nothing.

18. The training levels 1–200 turn inside, and levels 301–500 turn outside, the band where levels 201–300 turn. A correction concentrated in that band moves mainly levels 201–300, which are not in the fit, so it changes the training error very little; an optimizer that only sees training errors has no reason to keep it small, and any structure there that helps a training level even slightly is kept. The paper finds exactly this: the correction is largest in the band of turning points of levels 201–300 ($x=22.5$–28.5), the predicted gap levels are displaced by -6 to $+2$ spacings, and ridge regularization removes the damage.

19. The in-sample correlation $+0.95$ shows that the potential can reproduce the wave on the levels it was trained on; the held-out correlation $+0.07$ shows that it has no tendency to keep oscillating beyond them. Since the target is a single known wave whose correct continuation is certain, the failure cannot be blamed on the complicated mixture of primes in the real data: it isolates the inability of the family, as fitted, to express "continue oscillating".

20. $9.064720/3.029288=2.99$, about 3 zeros.

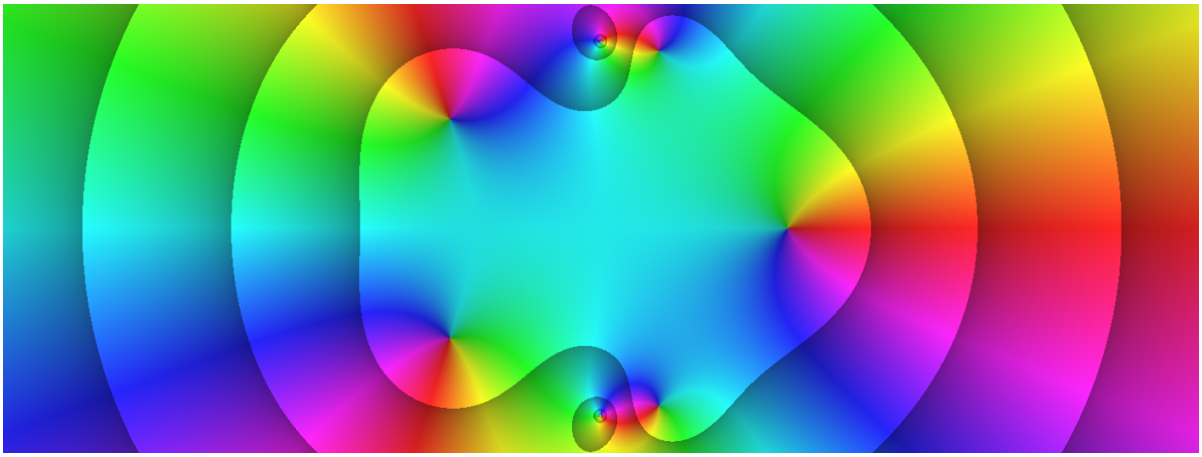
21. A reasonable list: held-out scoring against the model's own null, with both rms and correlation; a split within the same height range as well as one above it; scoring the model's actual spectrum, not an approximation of it; checking robustness to regularization and optimizer restarts; refitting to shuffled and phase-randomized targets to see whether the in-sample fit distinguishes real from fake. These match the paper's (i)–(iv) in Sec. VI.

22. Compare with a null such as "tomorrow will be the same as today" or "tomorrow will be the long-term average for this date". If the null already has an rms error of 3 degrees, the forecast has no skill. Likewise, a held-out rms of 0.3 for the zeros means nothing until compared with the null 0.263, which it fails to beat.

23. `np.polyfit(x, y, 4)` followed by `np.polyval` gives 15.0 at $x = 5$ and 29.9 at $x = 6$. The line gives $1.0 + 2.025 \times 6 = 13.15$ at $x = 6$. Difference table: new third difference $3.2 + 2.1 = 5.3$; new second difference $3.7 + 5.3 = 9.0$; new first difference $5.9 + 9.0 = 14.9$; new value $15.0 + 14.9 = 29.9$. The polynomial's departure from any sensible trend grows rapidly with distance from the data: extrapolating a memorizing model is especially dangerous.

24. Near $t = 50$: $0.5/3.029288 = 0.165$ spacings. Near zero 500: $0.5/1.292672 = 0.387$ spacings. The second is the bigger mistake relative to how tightly the zeros are packed: it is more than a third of the way to the neighboring zero. Mean-spacing units make errors comparable across heights.

25. The fit reaches essentially the same in-sample accuracy on fakes with no arithmetic content as on the true zeros, and all are far below the null 0.244. So in-sample accuracy measures the model's capacity to fit any sequence of this size, not that it has captured the structure of the zeros. A memorizing student could learn a fake answer key as easily as the real one; her perfect score on old papers says nothing about which key was right.



Domain colouring of $(z^5-1)/(z^2+1)$: hue is the argument, the rings are levels of the modulus.

Chapter 2

Complex Numbers and the Exponential

↑ Contents

What this chapter is about

The numbers at the center of this book are not ordinary real numbers. The n-th nontrivial zero of the Riemann zeta function is written $\rho_n = \frac{1}{2} + it_n$, where i is a number whose square is -1. The list $t_1 = 14.134725$, $t_2 = 21.022040, \ldots$ that you met in Chapter 1 is the list of *imaginary parts* of these zeros. The formula that predicts the zeros so well in the paper contains terms like $\sin(mt \log p)/(m\, p^{m/2})$. As you will see in this chapter, each such term is, up to the factor $-1/m$, the imaginary part of a complex power p^{-ms} evaluated at $s = \frac{1}{2} + it$: an arrow that turns around a circle as t increases. To read even the first sentence of the paper you need complex numbers. To understand where its prime-number formula comes from you need the complex exponential.

This chapter builds that language from the ground up. We define i and the arithmetic of complex numbers, draw them as points and arrows in a plane, and describe them by a length and an angle. Then we derive Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, the single most useful identity in this book. With it, rotations, oscillations, logarithms and powers like p^{-s} all become simple. We end with functions of a complex variable, their zeros, and the way the angle ("argument") of a product splits into a sum of angles. That last fact is how the primes enter the counting of zeros later in the book.

Where this shows up in the paper: the notation $\rho_n = \frac{1}{2} + it_n$ in the first sentence of Sec. I; the sine terms and the factor $p^{-m/2}$ in Eq. (4); the frequencies $\log p^m$ of Sec. IV and of the fitted controls in Sec. II D; the period $2\pi/\log 2 \approx 9$ quoted in Sec. V; the complex eigenvalues of the non-Hermitian operators $H + iA$ in Secs. II E and V; and the "phase-randomized" surrogates of Sec. III.

What you need

- Chapter 1: the list of zeros $t_1, t_2, \dots$ (running example RE1) and the outline of the paper's question.
- First-year physics and calculus: trigonometric functions and the addition formulas $\cos(a+b) = \cos a \cos b - \sin a \sin b$ and $\sin(a+b) = \sin a \cos b + \cos a \sin b$; $\cos^2\theta + \sin^2\theta = 1$; cos is even and sin is odd.
- Derivatives of e^x, $\sin x$, $\cos x$; the natural logarithm $\log x$ as the inverse of e^x (in this book log always means the natural logarithm).
- Two-dimensional vectors: components, magnitude, the angle a vector makes with the x axis.
- Uniform circular motion and simple harmonic motion (SHM) as its projection; angular frequency ω and period $2\pi/\omega$.
- The finite geometric sum $1 + r + r^2 + \cdots + r^{n-1} = (r^n - 1)/(r - 1)$ for $r \neq 1$.

2.1 Why a new kind of number?

★ Did you know? *Ten Split Into Two*

In his 1545 book *Ars Magna*, Gerolamo Cardano asked for two numbers that add to 10 and multiply to 40. The answer, $5 + \sqrt{-15}$ and $5 - \sqrt{-15}$, struck him as subtle but useless. Yet his cubic formula sometimes passed through square roots of negative numbers even when the final roots were real. In 1572 Rafael Bombelli showed how to calculate with them consistently.

Try to solve $x^2 + 1 = 0$. You need a number whose square is -1. No real number works: the square of a positive number is positive, the square of a negative number is positive, and $0^2 = 0$. So within the real numbers this equation has no solution at all.

Mathematicians did not invent a new number just to be stubborn about such equations. In the sixteenth century, Italian algebraists found that their formulas for solving cubic equations sometimes passed *through* square roots of negative numbers on the way to perfectly ordinary real answers. The square roots of negatives were a useful intermediate tool even when nobody believed in them. Over the following centuries it became clear that the extended number system is consistent, and that it is the natural language for anything that rotates or oscillates.

That last point is why physicists care. An oscillation $\cos \omega t$ is the shadow of a point going around a circle (you met this in first-year physics: SHM is the projection of uniform circular motion). Complex numbers let us work with the whole circle instead of its shadow, and the circle is much easier to compute with. Alternating-current circuits, waves, and all of quantum mechanics (Chapter 10) are written in complex numbers for this reason.

For us there is a further reason. The Riemann zeta function, from which our mysterious list of numbers comes, is a function of a *complex* variable s. Its zeros are complex numbers. And the primes enter through quantities p^{-s}, which on the vertical line where all the computed zeros sit are arrows of fixed length turning at a rate set by $\log p$. By the end of this chapter you will be able to compute such arrows, add them, and take their angles.

Key Ideas

- Some equations, like $x^2 + 1 = 0$, have no real solutions. We extend the number system by adding a number i with $i^2 = -1$.
- Complex numbers are the natural language of rotation and oscillation.
- The zeta zeros $\rho_n = \frac{1}{2} + it_n$ are complex numbers, and the primes enter the theory

through complex powers p^{-s}.

2.2 The imaginary unit and complex arithmetic

2.2.1 Definitions

We introduce a new number, the **imaginary unit** i, defined by the single property

$$i^2 = -1. \tag{2.1}$$

A **complex number** is an expression

$$z = a + bi, \qquad a, b \text{ real numbers.} \tag{2.2}$$

The real number a is called the **real part** of z, written $\operatorname{Re} z = a$. The real number b is called the **imaginary part** of z, written $\operatorname{Im} z = b$. We also write $a + ib$ for $a + bi$; the order of the factors b and i does not matter. A complex number with $b = 0$ is just the real number a, so every real number is also a complex number. A complex number with $a = 0$, such as $3i$, is called **purely imaginary**.

Two complex numbers are **equal** when both their real parts are equal and their imaginary parts are equal:

$$a + bi = c + di \quad \Longleftrightarrow \quad a = c \text{ and } b = d. \tag{2.3}$$

So one complex equation is really *two* real equations. Keep this in mind; it will matter when we look for zeros of complex functions.

Pitfall

The imaginary part of $z = a+bi$ is the *real* number b, not bi. For example $\operatorname{Im}(3-4i) = -4$. When the paper writes $\rho_n = \frac{1}{2} + it_n$ and calls t_n the "imaginary part", t_n is an ordinary real number such as 14.134725.

2.2.2 Addition, subtraction, multiplication

We add and subtract complex numbers part by part:

$$\begin{aligned} (a + bi) + (c + di) &= (a + c) + (b + d)i, \\ (a + bi) - (c + di) &= (a - c) + (b - d)i. \end{aligned} \tag{2.4}$$

To multiply, we use the ordinary distributive law of algebra, treating i as a symbol, and then replace i^2 by -1 using Eq. (2.1):

$$\begin{aligned} (a + bi)(c + di) &= ac + a(di) + (bi)c + (bi)(di) \\ &= ac + adi + bci + bd\,i^2 \\ &= ac + adi + bci - bd \quad (\text{using } i^2 = -1) \\ &= (ac - bd) + (ad + bc)i. \end{aligned} \tag{2.5}$$

Look at the final line: if you swap (a, b) with (c, d), the real part $ac - bd$ and the imaginary part $ad + bc$ do not change. So complex multiplication is commutative, $zw = wz$. The distributive law $z(w + u) = zw + zu$ also holds. With $z = a + bi$, $w = c + di$, $u = e + fi$, Eq. (2.4) gives $w + u = (c + e) + (d + f)i$, so by Eq. (2.5)

$$\begin{aligned} z(w + u) &= [a(c + e) - b(d + f)] + [a(d + f) + b(c + e)]i \\ &= [(ac - bd) + (ae - bf)] + [(ad + bc) + (af + be)]i, \end{aligned}$$

which by Eqs. (2.5) and (2.4) is $zw + zu$. The associative law $(zw)u = z(wu)$ holds too: expanding both sides with Eq. (2.5) produces the same real products in each part. Problem 9 asks you to carry out this expansion in general and for a concrete case. The upshot is that you may manipulate complex numbers with all the familiar rules of algebra, remembering only that $i^2 = -1$.

2.2.3 Powers of i

The powers of i repeat with period 4:

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = i^2 \cdot i^2 = (-1)(-1) = 1, \tag{2.6}$$

and then $i^5 = i^4 \cdot i = i$, $i^6 = -1$, and so on. To find i^n, divide n by 4 and keep the remainder. Also $i^{-1} = 1/i$; since $i \cdot (-i) = -i^2 = 1$, we have $i^{-1} = -i$. This cycle of four will be the engine of Euler's formula in Section 2.5.

Sample Problem 2.1: Multiplying complex numbers

Compute zw for $z = 3 + 2i$ and $w = 1 - 4i$, and compute i^{27}.
Step 1: Expand the product: $(3 + 2i)(1 - 4i) = 3 \cdot 1 + 3 \cdot (-4i) + 2i \cdot 1 + 2i \cdot (-4i) = 3 - 12i + 2i - 8i^2$.
Step 2: Replace i^2 by -1: $-8i^2 = +8$. So the product is $3 + 8 + (-12 + 2)i = 11 - 10i$.
Step 3: Check with Eq. (2.5), $a = 3$, $b = 2$, $c = 1$, $d = -4$: real part $ac - bd = 3 - (-8) = 11$; imaginary part $ad + bc = -12 + 2 = -10$. Agreed.
Step 4: For i^{27}: $27 = 4 \cdot 6 + 3$, so $i^{27} = (i^4)^6 \, i^3 = 1 \cdot (-i)$.

$$\boxed{(3 + 2i)(1 - 4i) = 11 - 10i, \qquad i^{27} = -i}$$

Pitfall

Do not use the rule $\sqrt{x}\sqrt{y} = \sqrt{xy}$ with negative x, y. If we agree that $\sqrt{-1}$ means i, then $\sqrt{-1}\sqrt{-1} = i^2 = -1$, but $\sqrt{(-1)(-1)} = \sqrt{1} = 1$. The rule is true only for nonnegative real numbers. In this book we avoid writing square roots of negative numbers except in the quadratic formula, where we say exactly which root we mean.

Checkpoint 2.1

Is i a real number? What is $i^2 \cdot i^2$?

Answer: No: no real number squares to -1. And $i^2 \cdot i^2 = (-1)(-1) = 1$, which is i^4.

2.2.4 Complex numbers in Python

Python has complex numbers built in. It follows the engineering convention of writing the imaginary unit as `j`, so $3+2i$ is typed `3 + 2j`. (Appendix A explains how to run such snippets.)

```
z = 3 + 2j            # Python writes i as j
w = 1 - 4j
print(z + w)          # (4-2j)
print(z * w)          # (11-10j)
print(z.real, z.imag) # 3.0 2.0  (both are real numbers)
print(1j**2)          # (-1+0j)
```

Key Ideas

- The imaginary unit satisfies $i^2 = -1$. A complex number is $z = a + bi$ with real part $\operatorname{Re} z = a$ and imaginary part $\operatorname{Im} z = b$, both real.
- Two complex numbers are equal when both parts agree, so one complex equation is two real equations.
- Add part by part; multiply by expanding and using $i^2 = -1$: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, Eq. (2.5).
- Powers of i cycle: $1, i, -1, -i, 1, \ldots$.

2.3 The complex plane: modulus and conjugate

2.3.1 Points and arrows

A complex number $z = a + bi$ is fixed by two real numbers (a, b). So we can draw it as the point with coordinates (a, b) in a plane, or as the arrow (two-dimensional vector) from the origin to that point. This picture is called the **complex plane** (or Argand diagram). The horizontal axis is the **real axis**; the vertical axis is the **imaginary axis**. Figure 2.1 shows several examples.

The picture makes addition familiar. By Eq. (2.4), adding complex numbers adds their components, which is exactly how you add vectors in first-year physics: place the arrows head to tail (the parallelogram rule).

2.3.2 The complex conjugate

The **complex conjugate** of $z = a + bi$ is

$$\bar{z} = a - bi. \tag{2.7}$$

(Some books write z^*; the paper does not need either symbol, but later chapters on quantum mechanics use both.) Geometrically, $\bar{z}$ is the reflection of z in the real axis. Reflecting twice returns the original point: $\bar{\bar{z}} = \overline{a - bi} = a + bi = z$. Three identities follow directly by writing out the parts:

$$z + \bar{z} = (a + bi) + (a - bi) = 2a = 2\operatorname{Re} z, \tag{2.8}$$

$$z - \bar{z} = (a + bi) - (a - bi) = 2bi = 2i \operatorname{Im} z,$$

$$z\bar{z} = (a + bi)(a - bi) = a^2 - abi + abi - b^2 i^2 = a^2 + b^2. \tag{2.9}$$

The last line is important: $z\bar{z}$ is always a real number, and it is never negative.

Conjugation respects arithmetic. With $z = a + bi$ and $w = c + di$, Eq. (2.5) gives $zw = (ac - bd) + (ad + bc)i$, so $\overline{zw} = (ac - bd) - (ad + bc)i$. On the other hand

$$\bar{z}\,\bar{w} = (a - bi)(c - di) = ac - adi - bci + bd\,i^2 = (ac - bd) - (ad + bc)i.$$

The two agree, so

$$\overline{zw} = \bar{z}\,\bar{w}, \qquad \text{and similarly} \qquad \overline{z + w} = \bar{z} + \bar{w}. \tag{2.10}$$

2.3.3 The modulus

The **modulus** (or absolute value) of $z = a + bi$ is the length of its arrow,

$$|z| = \sqrt{a^2 + b^2}. \tag{2.11}$$

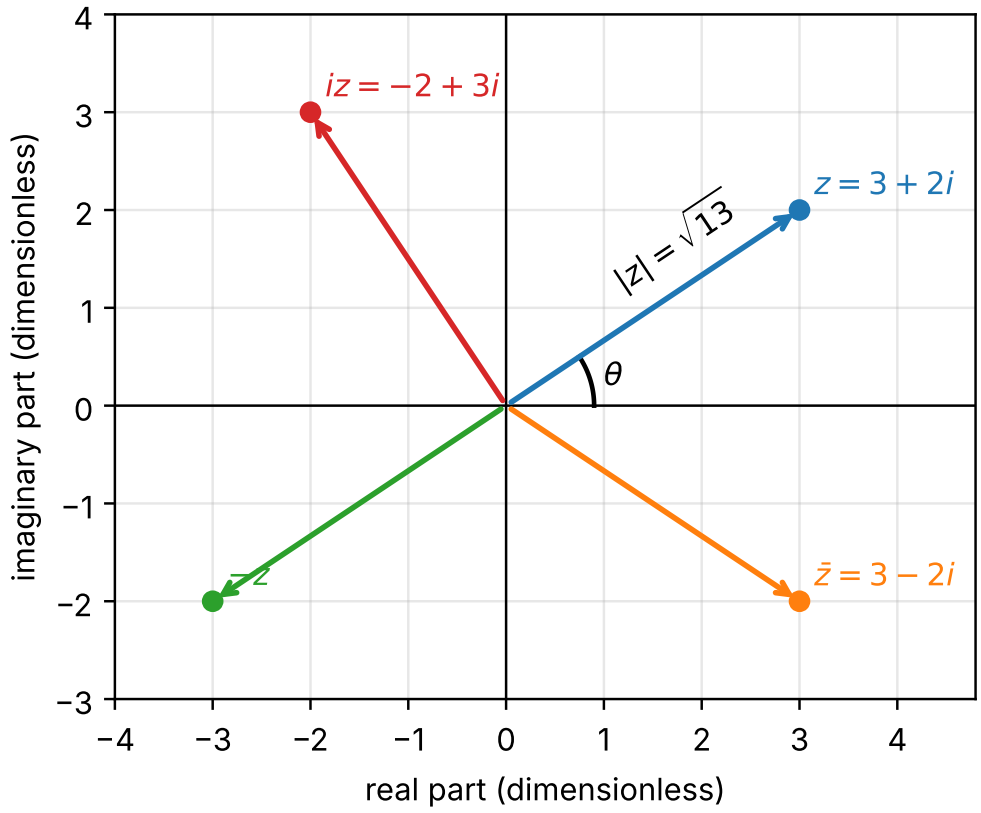


Figure 2.1: The complex plane. The number $z = 3 + 2i$ is the point $(3, 2)$, or the arrow from the origin to it. Its length is the modulus $|z| = \sqrt{13}$ and its angle from the positive real axis is the argument $\theta = \arg z$. The conjugate $\bar{z} = 3 - 2i$ is the mirror image in the real axis; $-z$ is the arrow reversed; and $iz = -2 + 3i$ is z rotated by 90° counterclockwise.

By Pythagoras this is the distance from the origin to the point (a, b); it is the magnitude of the vector (a, b) from first-year physics. For a real number ($b = 0$) it reduces to the ordinary absolute value $\sqrt{a^2} = |a|$. Comparing with Eq. (2.9),

$$z\bar{z} = |z|^2 . \tag{2.12}$$

Also $|z| = 0$ only if $a^2 + b^2 = 0$, which for real a, b forces $a = b = 0$, i.e. $z = 0$. The distance between two complex numbers z and w in the plane is $|z - w|$, the length of the arrow from w to z.

2.3.4 Division

To divide by a nonzero complex number z, multiply the top and bottom by $\bar{z}$ (this multiplies by $\bar{z}/\bar{z} = 1$, so it changes nothing), and use Eq. (2.12) to make the bottom real:

$$\frac{w}{z} = \frac{w\,\bar{z}}{z\,\bar{z}} = \frac{w\,\bar{z}}{|z|^2}. \tag{2.13}$$

In particular $1/z = \bar{z}/|z|^2$. Dividing by the real number $|z|^2$ just divides both parts.

Sample Problem 2.2: Dividing complex numbers

Compute $(3 + 2i)/(1 - 4i)$ in the form $a + bi$, and check the answer.
Step 1: The denominator is $z = 1 - 4i$, so $\bar{z} = 1 + 4i$ and $|z|^2 = 1^2 + 4^2 = 17$.
Step 2: Multiply the numerator by $\bar{z}$: $(3+2i)(1+4i) = 3+12i+2i+8i^2 = 3-8+14i = -5+14i$.
Step 3: Divide by 17: $\dfrac{-5+14i}{17} = -\dfrac{5}{17} + \dfrac{14}{17}i \approx -0.294118 + 0.823529\,i$.
Step 4: Check by multiplying back: $\dfrac{(-5+14i)(1-4i)}{17} = \dfrac{-5+20i+14i-56i^2}{17} = \dfrac{51+34i}{17} = 3 + 2i$. Correct.

$$\boxed{\frac{3+2i}{1-4i} = \frac{-5+14i}{17} \approx -0.294118 + 0.823529\,i}$$

2.3.5 The modulus of a product, and the triangle inequality

Two inequalities and identities about moduli will be used throughout the book.

Modulus of a product. For any complex z, w,

$$|zw| = |z|\,|w| . \tag{2.14}$$

Proof: by Eq. (2.12) and Eq. (2.10),

$$|zw|^2 = (zw)\overline{(zw)} = z\,w\,\bar{z}\,\bar{w} = (z\bar{z})(w\bar{w}) = |z|^2\,|w|^2 .$$

In the first equality we used Eq. (2.12) for the complex number zw: by definition $|zw|$ is the nonnegative square root of $(zw)\overline{(zw)}$. Both sides of Eq. (2.14) are nonnegative, so taking square roots gives the result. Dividing, $|w/z| = |w|\,/\,|z|$ for $z \neq 0$.

Triangle inequality. For any complex z, w,

$$|z + w| \leq |z| + |w| . \tag{2.15}$$

Geometrically: one side of a triangle is no longer than the other two together. Proof in steps:

$$\begin{aligned}|z+w|^2 &= (z+w)(\bar{z}+\bar{w}) \quad \text{(Eqs. (2.12), (2.10))}\\ &= z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w}\\ &= |z|^2 + |w|^2 + (z\bar{w} + \overline{z\bar{w}}) \quad \text{(since } \overline{z\bar{w}} = \bar{z}\,\overline{\bar{w}} = \bar{z}w \text{ by Eq. (2.10) and } \overline{\bar{w}} = w)\\ &= |z|^2 + |w|^2 + 2\,\mathrm{Re}(z\bar{w}) \quad \text{(Eq. (2.8))}.\end{aligned}$$

Now for any complex $u = c + di$, $\mathrm{Re}\,u = c \le \sqrt{c^2+d^2} = |u|$. With $u = z\bar{w}$ and $|z\bar{w}| = |z|\,|\bar{w}| = |z|\,|w|$ (Eq. (2.14), and $|\bar{w}| = |w|$ because changing the sign of the imaginary part does not change $c^2 + d^2$),

$$|z+w|^2 \le |z|^2 + |w|^2 + 2\,|z|\,|w| = (|z| + |w|)^2.$$

Taking nonnegative square roots proves Eq. (2.15).

2.3.6 Complex numbers cannot be put in order

Real numbers can be compared: $2 < 3$. Complex numbers cannot, at least not in any way that cooperates with multiplication. Suppose there were an ordering in which, as for real numbers, the product of two positive numbers is positive and exactly one of z, $-z$ is positive for $z \neq 0$. If i were positive, then $i \cdot i = -1$ would be positive, and so would $(-1)(-1) = 1$; but then both 1 and -1 are positive, a contradiction. If instead $-i$ were positive, then $(-i)(-i) = i^2 = -1$ would be positive, the same contradiction. So the phrase "the larger of two complex numbers" has no meaning. This matters later: the eigenvalues of the paper's non-Hermitian operators (Sec. II E) can be complex, so they have no natural "increasing order" of the kind real energy levels have. In any case, order alone would not tell which perturbed state came from which unperturbed one. The paper identifies states by overlap (Sec. II E (iii); Chapter 14).

Checkpoint 2.2

What is the conjugate of a real number? What does conjugation do to the modulus?

Answer: A real number $a = a + 0i$ is its own conjugate, since $a - 0i = a$. Conjugation leaves the modulus unchanged: $|a - bi| = \sqrt{a^2 + (-b)^2} = |a + bi|$ (a reflection does not change length).

Key Ideas

- A complex number $a + bi$ is the point (or arrow) (a, b) in the complex plane; addition is vector addition.
- The conjugate $\bar{z} = a - bi$ is the reflection in the real axis; $\overline{zw} = \bar{z}\bar{w}$.
- The modulus $|z| = \sqrt{a^2+b^2}$ is the length of the arrow; $z\bar{z} = |z|^2$; $|zw| = |z|\,|w|$; $|z+w| \le |z| + |w|$.
- Divide by multiplying top and bottom by the conjugate: $w/z = w\bar{z}/|z|^2$.
- Complex numbers have no ordering compatible with multiplication.

2.4 Polar form and the argument

2.4.1 Length and angle

An arrow in the plane can be described by its components (a, b), or by its length r and the angle θ it makes with the positive real axis (measured counterclockwise, in radians). From the

right triangle in Figure 2.1,

$$a = r\cos\theta, \qquad b = r\sin\theta, \qquad \text{so} \qquad z = r(\cos\theta + i\sin\theta). \tag{2.16}$$

This is the **polar form**. The length $r = |z|$ is the modulus. The angle θ is called the **argument** of z, written $\arg z$. Physicists also call it the **phase** of z.

The argument is not unique. Adding 2π to θ turns the arrow by a full circle and brings it back, so θ, $\theta + 2\pi$, $\theta - 2\pi$, and in general $\theta + 2\pi k$ for any integer k all describe the same z. When we need a single definite value we use the **principal argument**, written $\operatorname{Arg} z$, which is the value in the interval $(-\pi, \pi]$. The argument of $z = 0$ is not defined, because the zero arrow points nowhere.

2.4.2 Computing the argument without quadrant errors

From Eq. (2.16), $\tan\theta = b/a$ when $a \neq 0$. But the inverse tangent arctan only returns angles in $(-\pi/2, \pi/2)$, the right half of the plane, while $-1 - i$ and $1 + i$ have the same ratio $b/a = 1$ and point in opposite directions. The correct rule for the principal argument of $a + bi \neq 0$ is

$$\operatorname{Arg}(a + bi) = \begin{cases} \arctan(b/a) & a > 0, \\ \arctan(b/a) + \pi & a < 0,\ b \geq 0, \\ \arctan(b/a) - \pi & a < 0,\ b < 0, \\ \pi/2 & a = 0,\ b > 0, \\ -\pi/2 & a = 0,\ b < 0. \end{cases} \tag{2.17}$$

Computers provide this as the two-argument function `atan2(b, a)`; in numpy, `np.angle(z)` returns $\operatorname{Arg} z$ directly.

Pitfall

Never compute an argument as $\arctan(b/a)$ alone. For $z = -1 - i$ that gives $\arctan 1 = \pi/4$, which points into the first quadrant. The correct principal argument (third line of Eq. (2.17)) is $\pi/4 - \pi = -3\pi/4$. Always ask which quadrant the point lies in.

Sample Problem 2.3: Modulus and argument of the first zeta zero

The first nontrivial zero of the zeta function is $\rho_1 = \frac{1}{2} + it_1$ with $t_1 = 14.134725$ (running example RE1). Find $|\rho_1|$ and $\operatorname{Arg}\rho_1$ in radians and degrees.

Step 1: Real part $a = 0.5$, imaginary part $b = 14.134725$.

Step 2: Modulus: $|\rho_1|^2 = 0.5^2 + 14.134725^2 = 0.25 + 199.790451 = 200.040451$, so $|\rho_1| = \sqrt{200.040451} \approx 14.1436$.

Step 3: Since $a > 0$, the first line of Eq. (2.17) applies: $\operatorname{Arg}\rho_1 = \arctan(14.134725/0.5) = \arctan(28.26945)$. It is easier to use $\arctan(b/a) = \pi/2 - \arctan(a/b)$ for $a, b > 0$ (the two acute angles of a right triangle add to $\pi/2$): $\arctan(0.5/14.134725) = \arctan(0.0353739) \approx 0.0353592$, so $\operatorname{Arg}\rho_1 \approx 1.5707963 - 0.0353592 = 1.53544$.

Step 4: In degrees: $1.53544 \times 180/\pi \approx 87.974^\circ$.

$$\boxed{|\rho_1| \approx 14.1436, \qquad \operatorname{Arg}\rho_1 \approx 1.53544 \text{ rad} \approx 87.974^\circ}$$

The zero lies very close to the imaginary axis direction, because its imaginary part is much larger than its real part $\frac{1}{2}$. All the zeros used in the paper sit on the vertical line $\operatorname{Re} s = \frac{1}{2}$ (Figure 2.7 in Section 2.11).

2.4.3 Multiplying in polar form: moduli multiply, arguments add

Here is the reason polar form is so useful. Take $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. Multiply, using Eq. (2.5) with $a = \cos\theta_1$, $b = \sin\theta_1$, $c = \cos\theta_2$, $d = \sin\theta_2$:

$$\begin{aligned} z_1 z_2 &= r_1 r_2\big[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2)\big] \\ &= r_1 r_2\big[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big] \quad \text{(addition formulas)}. \end{aligned} \tag{2.18}$$

So the modulus of a product is the product of the moduli (as already found in Eq. (2.14)) and *an argument of the product is the sum of the arguments*:

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2 \quad \text{(up to a multiple of } 2\pi). \tag{2.19}$$

We abbreviate "up to adding an integer multiple of 2π" as "mod 2π". The phrase "up to a multiple of 2π" matters: if $\operatorname{Arg} z_1 = 3$ and $\operatorname{Arg} z_2 = 1$, the sum 4 is outside $(-\pi, \pi]$, and the principal argument of the product is $4 - 2\pi$.

For quotients, since $z_1 = (z_1/z_2)z_2$, Eq. (2.19) gives $\arg(z_1/z_2) = \arg z_1 - \arg z_2$, and $|z_1/z_2| = r_1/r_2$. For the conjugate, $\bar{z} = r(\cos\theta - i\sin\theta) = r(\cos(-\theta) + i\sin(-\theta))$ (cosine even, sine odd), so $\arg\bar{z} = -\arg z$.

2.4.4 De Moivre's formula

Applying Eq. (2.18) repeatedly to equal factors gives, for every positive integer n,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta. \tag{2.20}$$

Proof by induction. For $n = 1$ there is nothing to prove. Suppose it holds for some n. Then

$$(\cos\theta + i\sin\theta)^{n+1} = (\cos n\theta + i\sin n\theta)(\cos\theta + i\sin\theta) = \cos(n\theta+\theta) + i\sin(n\theta+\theta),$$

using Eq. (2.18) with $r_1 = r_2 = 1$. That is the formula for $n+1$, so it holds for all $n \geq 1$. Hence $z^n = r^n(\cos n\theta + i\sin n\theta)$: raising to the n-th power raises the modulus to the n-th power and multiplies the argument by n.

Sample Problem 2.4: A power the easy way

Compute $(1+i)^8$.
Step 1: Polar form: $|1+i| = \sqrt{1+1} = \sqrt{2}$, and since $a = b = 1 > 0$, $\operatorname{Arg}(1+i) = \arctan 1 = \pi/4$.
Step 2: By De Moivre, $(1+i)^8 = (\sqrt{2})^8[\cos(8\cdot\pi/4) + i\sin(8\cdot\pi/4)] = 16(\cos 2\pi + i\sin 2\pi) = 16(1+0i)$.
Step 3: Check another way: $(1+i)^2 = 1 + 2i + i^2 = 2i$, so $(1+i)^8 = (2i)^4 = 16i^4 = 16$.

$$\boxed{(1+i)^8 = 16}$$

Checkpoint 2.3

Suppose $\operatorname{Arg} z = 3$ and $\operatorname{Arg} w = 1$ (radians). What is $\operatorname{Arg}(zw)$?

Answer: The arguments add to 4, which exceeds π. Subtract 2π: $\operatorname{Arg}(zw) = 4 - 2\pi \approx -2.2832$.

Key Ideas

- Polar form: $z = r(\cos\theta + i\sin\theta)$ with $r = |z|$ and $\theta = \arg z$; the argument is defined up to multiples of 2π, and the principal value $\operatorname{Arg} z$ lies in $(-\pi, \pi]$.
- Compute arguments with the quadrant-aware rule Eq. (2.17) (`atan2`, `np.angle`).
- In a product, moduli multiply and arguments add (mod 2π), Eqs. (2.18)–(2.19).
- De Moivre: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

2.5 Euler's formula

2.5.1 A function that behaves like an exponential

Define, for real θ, the complex-valued function

$$E(\theta) = \cos\theta + i\sin\theta.$$

Equation (2.18) with $r_1 = r_2 = 1$ says

$$E(\theta_1)\,E(\theta_2) = E(\theta_1 + \theta_2), \qquad E(0) = 1. \tag{2.21}$$

That is exactly the law obeyed by exponentials: $e^a e^b = e^{a+b}$ and $e^0 = 1$. The derivative of a complex-valued function of a real variable is defined part by part: if $F(\theta) = u(\theta) + iv(\theta)$ with u, v real functions, then by definition $\mathrm{d}F/\mathrm{d}\theta = \mathrm{d}u/\mathrm{d}\theta + i\,\mathrm{d}v/\mathrm{d}\theta$. The sum rule follows at once. The product rule $(FG)' = F'G + FG'$ also carries over: write both factors in parts, multiply them out with Eq. (2.5), and apply the real product rule to each of the four real products. Now differentiate E with respect to θ in this sense:

$$\frac{\mathrm{d}E}{\mathrm{d}\theta} = -\sin\theta + i\cos\theta = i(\cos\theta + i\sin\theta) = iE(\theta),$$

where in the middle step we used $i \cdot i\sin\theta = -\sin\theta$. Compare $\frac{\mathrm{d}}{\mathrm{d}\theta}e^{k\theta} = k\,e^{k\theta}$. Everything suggests that $E(\theta)$ "is" $e^{i\theta}$, an exponential with $k = i$.

But what could $e^{i\theta}$ mean? "e multiplied by itself $i\theta$ times" makes no sense. We need a definition of e^z for complex z, and then we can *prove* that $e^{i\theta} = E(\theta)$. The definition comes from power series.

2.5.2 Power series from derivatives

Suppose a function $f(x)$ can be written as an infinite sum of powers,

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots = \sum_{n=0}^{\infty} a_n x^n. \tag{2.22}$$

(An infinite sum means the limit of the finite sums $\sum_{n=0}^{N} a_n x^n$ as $N \to \infty$; Chapter 3 treats such limits carefully.) How must the coefficients a_n be chosen? Set $x = 0$: every term except the first vanishes, so $a_0 = f(0)$. Differentiate once term by term,

$$f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots,$$

and set $x = 0$: $a_1 = f'(0)$. Differentiate again, $f''(x) = 2a_2 + 3\cdot 2\,a_3 x + \cdots$, so $a_2 = f''(0)/2$. In general, the n-th derivative of x^k is $k(k-1)\cdots(k-n+1)x^{k-n}$ when $k \geq n$ and zero when

$k < n$. At $x = 0$ the terms with $k > n$ still contain a positive power of x and vanish, so only $k = n$ survives, contributing $n(n-1)\cdots 1 \cdot a_n = n!\, a_n$. Hence

$$a_n = \frac{f^{(n)}(0)}{n!}, \tag{2.23}$$

where $f^{(n)}$ is the n-th derivative and $n! = 1 \cdot 2 \cdots n$ (with $0! = 1$). The series with these coefficients is the **Taylor series** of f about 0.

Now apply this to three functions whose derivatives you know.

Exponential. Every derivative of e^x is e^x, which equals 1 at $x = 0$. So $a_n = 1/n!$:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}. \tag{2.24}$$

Cosine. The derivatives of $\cos x$ cycle through $\cos x, -\sin x, -\cos x, \sin x, \cos x, \ldots$, whose values at 0 are $1, 0, -1, 0, 1, \ldots$. So only even powers appear, with alternating signs:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots. \tag{2.25}$$

Sine. The derivatives of $\sin x$ cycle through $\sin x, \cos x, -\sin x, -\cos x, \ldots$, with values $0, 1, 0, -1, \ldots$ at 0. So only odd powers appear:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots. \tag{2.26}$$

Taken on trust

We have shown what the coefficients must be *if* the function equals a power series and *if* we may differentiate the series term by term. Two facts are taken on trust here: (i) the series (2.24)–(2.26) converge for every real x, and their sums equal e^x, $\cos x$ and $\sin x$; (ii) term-by-term differentiation of a power series is allowed inside its interval of convergence. Convergence is studied in Chapter 3; complete proofs are in Riley, Hobson and Bence [22] (chapter on series and limits).

A quick numerical check: summing Eq. (2.24) at $x = 1$ up to $n = 5$ gives $1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} \approx 2.716667$, already close to $e \approx 2.718282$. The factorials in the denominators grow so fast that the terms eventually shrink for any x.

2.5.3 Defining the exponential of a complex number

Equation (2.24) only involves adding, multiplying and dividing by real numbers, and we can do all of these with complex numbers. So we *define*, for any complex z,

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots. \tag{2.27}$$

For real z this is the familiar exponential, so the definition extends it rather than replacing it.

2.5.4 Derivation of Euler's formula

Put $z = i\theta$ with θ real in Eq. (2.27). Each term is $(i\theta)^n/n! = i^n\theta^n/n!$, and the powers i^n cycle as in Eq. (2.6). Writing out the first eight terms:

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{i^2\theta^2}{2!} + \frac{i^3\theta^3}{3!} + \frac{i^4\theta^4}{4!} + \frac{i^5\theta^5}{5!} + \frac{i^6\theta^6}{6!} + \frac{i^7\theta^7}{7!} + \cdots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} - \frac{\theta^6}{6!} - i\frac{\theta^7}{7!} + \cdots. \end{aligned}$$

For even $n = 2k$, $i^{2k} = (i^2)^k = (-1)^k$ is real; for odd $n = 2k+1$, $i^{2k+1} = (-1)^k i$ is imaginary. Collect the even-n (real) terms and the odd-n (imaginary) terms separately:

$$\begin{aligned} e^{i\theta} &= \Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots\Big) + i\Big(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots\Big) \\ &= \cos\theta + i\sin\theta \quad \text{(Eqs. (2.25) and (2.26)).} \end{aligned} \tag{2.28}$$

This is **Euler's formula**. Equation (2.28) is a theorem, given the definition (2.27) and the convergence facts below.

Taken on trust

Splitting an infinite sum into two sub-sums (here, even and odd terms) is a rearrangement. For some infinite series rearranging changes the answer (conditionally convergent series, met in Chapter 3, can behave this way). It is safe when the series converges *absolutely*, meaning that the sum of the moduli $\sum_n |z^n/n!| = \sum_n |z|^n/n! = e^{|z|}$ is finite, which it is for every complex z. We take on trust that (i) the series (2.27) converges absolutely for every complex z, and (ii) absolutely convergent series may be rearranged freely. Absolute convergence is defined in Chapter 3; both facts are proved in [22].

Sample Problem 2.5: Euler's formula by brute force

Evaluate the partial sums $S_N = \sum_{n=0}^{N} (i\theta)^n/n!$ for $\theta = \pi/3$ and compare with $\cos(\pi/3) + i\sin(\pi/3)$.

Step 1: Exact target: $\cos(\pi/3) = \frac{1}{2}$, $\sin(\pi/3) = \sqrt{3}/2 \approx 0.866025$, so $e^{i\pi/3} = 0.5 + 0.866025\,i$.

Step 2: With $\theta = \pi/3 \approx 1.047198$, the terms are 1, $1.047198\,i$, $-\theta^2/2 = -0.548311$, $-i\theta^3/6 = -0.191397\,i$, $\theta^4/24 = 0.050108$, $i\theta^5/120 = 0.010494\,i$, $-\theta^6/720 = -0.001832$, $-i\theta^7/5040 = -0.000274\,i$, $\theta^8/40320 = 0.000036$.

Step 3: Add them cumulatively (see the table below). Each even term changes only the real part and each odd term only the imaginary part, exactly as in the derivation.

Partial sums of the exponential series for $e^{i\pi/3}$.

N	$\mathrm{Re}\, S_N$	$\mathrm{Im}\, S_N$
0	1.00000	0.00000
1	1.00000	1.04720
2	0.45169	1.04720
3	0.45169	0.85580
4	0.50180	0.85580
5	0.50180	0.86630
6	0.49996	0.86630
7	0.49996	0.86602
8	0.50000	0.86602
exact	0.50000	0.86603

Step 4: By $N = 8$ the partial sum agrees with the exact value to about 10^{-5}.

$$\boxed{e^{i\pi/3} = \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\,i \approx 0.5 + 0.866025\,i}$$

```
import numpy as np
from math import factorial
```

```
theta = np.pi / 3
S = 0
for n in range(9):
    S = S + (1j * theta)**n / factorial(n)
    print(n, round(S.real, 5), round(S.imag, 5))
print(np.exp(1j * theta))   # numpy's built-in complex exponential
```

Figure 2.2 shows the partial sums for a larger angle, $\theta = \pi$. The early terms are large ($\pi^2/2 \approx 4.9$), so the partial sums swing widely, but each term is perpendicular to the previous one (multiplying by $i\theta/n$ turns by 90°) and the lengths eventually shrink, so the path spirals in to the point -1.

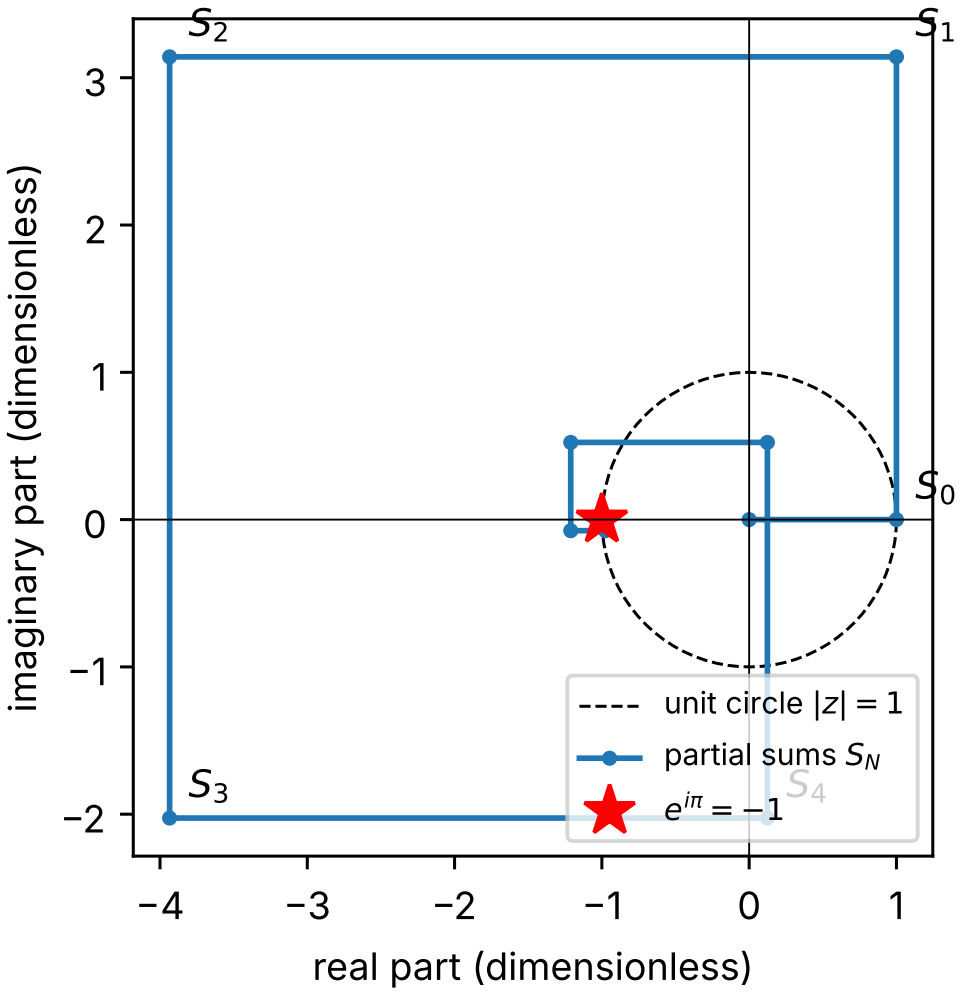


Figure 2.2: Partial sums $S_N = \sum_{n=0}^{N} (i\pi)^n/n!$ of the exponential series for $e^{i\pi}$, plotted in the complex plane for $N = 0, \ldots, 14$. Consecutive steps are at right angles; the path spirals into $e^{i\pi} = -1$ (star). The dashed curve is the unit circle $|z| = 1$, on which every $e^{i\theta}$ with real θ lies.

2.5.5 Consequences

Special values. From Eq. (2.28):

$$e^{i\pi/2} = i, \qquad e^{i\pi} = -1, \qquad e^{2\pi i} = 1, \qquad e^{i(\theta+2\pi)} = e^{i\theta}. \tag{2.29}$$

The second is the famous $e^{i\pi} + 1 = 0$. The last says $e^{i\theta}$ is periodic in θ with period 2π.

The unit circle. For real θ, $\left|e^{i\theta}\right| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$. So $e^{i\theta}$ is always on the circle of radius 1, at angle θ. Its conjugate is $\overline{e^{i\theta}} = \cos\theta - i\sin\theta = \cos(-\theta) + i\sin(-\theta) = e^{-i\theta}$.

Exponential polar form. Equation (2.16) becomes

$$z = r\,e^{i\theta}, \qquad r = |z|\,, \ \theta = \arg z. \tag{2.30}$$

The product rule Eq. (2.18) now reads $r_1 e^{i\theta_1}\, r_2 e^{i\theta_2} = r_1 r_2\, e^{i(\theta_1+\theta_2)}$: the old trigonometric addition formulas have become the law of exponents.

Exponentials of general complex numbers. The exponent law holds for all complex numbers:

Taken on trust

For all complex z and w, $e^z e^w = e^{z+w}$, where both sides are defined by the series (2.27). The proof multiplies the two absolutely convergent series term by term (the "Cauchy product") and uses the binomial theorem; see [22]. For purely imaginary z, w we proved it above, as Eq. (2.21).

With $z = x$ real and $w = iy$ purely imaginary,

$$e^{x+iy} = e^x e^{iy} = e^x(\cos y + i\sin y). \tag{2.31}$$

So $\left|e^{x+iy}\right| = e^x$ (using Eq. (2.14) and $\left|e^{iy}\right| = 1$) and $\arg e^{x+iy} = y$. The real part of the exponent sets the size; the imaginary part sets the angle. Since $e^x > 0$ for every real x, e^z is never zero.

Derivative. We found $\frac{\mathrm{d}}{\mathrm{d}\theta}e^{i\theta} = ie^{i\theta}$. More generally, for real t and a complex constant $k = \alpha + i\omega$, Eq. (2.31) gives $e^{kt} = e^{\alpha t}(\cos\omega t + i\sin\omega t)$, with real part $u(t) = e^{\alpha t}\cos\omega t$ and imaginary part $v(t) = e^{\alpha t}\sin\omega t$. By the real product rule,

$$\begin{aligned} u'(t) &= \alpha e^{\alpha t}\cos\omega t - \omega e^{\alpha t}\sin\omega t, \\ v'(t) &= \alpha e^{\alpha t}\sin\omega t + \omega e^{\alpha t}\cos\omega t. \end{aligned}$$

On the other hand, by Eq. (2.5),

$$\begin{aligned} k\,e^{kt} &= (\alpha + i\omega)\,e^{\alpha t}(\cos\omega t + i\sin\omega t) \\ &= e^{\alpha t}\big[(\alpha\cos\omega t - \omega\sin\omega t) + i(\alpha\sin\omega t + \omega\cos\omega t)\big], \end{aligned}$$

whose real part is u' and whose imaginary part is v'. Hence $\frac{\mathrm{d}}{\mathrm{d}t}e^{kt} = u' + iv' = k\,e^{kt}$ (Problem 27 explores why this characterizes the exponential).

Checkpoint 2.4

What is $\left|e^{i\theta}\right|$ for real θ? What is $\left|e^{2+3i}\right|$?

Answer: $\left|e^{i\theta}\right| = 1$. By Eq. (2.31), $\left|e^{2+3i}\right| = e^2 \approx 7.389$; the imaginary part 3 only sets the angle.

Key Ideas

- Taylor coefficients are $a_n = f^{(n)}(0)/n!$; this gives the series for e^x, $\cos x$, $\sin x$, Eqs. (2.24)–(2.26).
- The complex exponential is defined by the same series, Eq. (2.27).
- Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, Eq. (2.28), follows by splitting the series into even and odd terms (allowed by absolute convergence, taken on trust).
- $e^{x+iy} = e^x(\cos y + i\sin y)$: size e^x, angle y. Polar form is $z = re^{i\theta}$; e^z is never zero.

2.6 The exponential as rotation: phasors and oscillations

★ Did you know? *Phasors Power the Grid*

Charles Proteus Steinmetz, an engineer later at General Electric, showed in 1893 that alternating-current circuits become simple algebra if each sinusoidal voltage and current is represented by a complex number. Derivatives turn into multiplication by $i\omega$. His phasor method spread rapidly, and engineers still use it to analyze AC power networks, though they write j instead of i.

2.6.1 Multiplying by $e^{i\alpha}$ rotates

Take any $z = re^{i\theta}$ and a real angle α. Then $e^{i\alpha}z = re^{i(\theta+\alpha)}$: same length, angle increased by α. *Multiplying by $e^{i\alpha}$ rotates the arrow counterclockwise by α.* In components, with $z = x + iy$,

$$e^{i\alpha}(x+iy) = (\cos\alpha + i\sin\alpha)(x+iy) = (x\cos\alpha - y\sin\alpha) + i(x\sin\alpha + y\cos\alpha), \tag{2.32}$$

which you may recognize as the formula for rotating a 2D vector (x, y) by angle α. Multiplying by $i = e^{i\pi/2}$ rotates by 90°; that is why iz in Figure 2.1 is perpendicular to z. Multiplying by $-1 = e^{i\pi}$ rotates by 180°.

2.6.2 Uniform circular motion and SHM

Let t be time, $R > 0$ a radius and ω an angular frequency (radians per second). The complex number

$$z(t) = R\,e^{i\omega t} \tag{2.33}$$

has constant modulus R and argument ωt increasing at a steady rate, so it moves around a circle of radius R counterclockwise, completing one turn when $\omega t = 2\pi$, i.e. with period $2\pi/\omega$. Its real part is $x(t) = R\cos\omega t$, simple harmonic motion: SHM is the projection of uniform circular motion, now in one line of algebra. Differentiating, the velocity is $\mathrm{d}z/\mathrm{d}t = i\omega Re^{i\omega t}$: length ωR and direction rotated by 90° from the position (tangent to the circle). The acceleration is $(i\omega)^2 Re^{i\omega t} = -\omega^2 z$: length $\omega^2 R$, pointing toward the center. You recover the centripetal acceleration of first-year physics.

2.6.3 Cosine and sine from exponentials

Write Euler's formula for θ and for $-\theta$, using that cosine is even and sine is odd:

$$e^{i\theta} = \cos\theta + i\sin\theta, \qquad e^{-i\theta} = \cos\theta - i\sin\theta.$$

Adding the two lines and dividing by 2, or subtracting them and dividing by $2i$:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \tag{2.34}$$

Equivalently, $\cos\theta = \operatorname{Re} e^{i\theta}$ and $\sin\theta = \operatorname{Im} e^{i\theta}$. A real cosine is the sum of two arrows turning in opposite directions; their imaginary parts cancel and their real parts add. Formulas (2.34) turn every trigonometric identity into algebra with exponentials, as the samples below show.

2.6.4 Phasors

A real sinusoid with amplitude $A > 0$, angular frequency ω and phase constant ϕ can be written

$$A\cos(\omega t + \phi) = \operatorname{Re}\left[(Ae^{i\phi})\,e^{i\omega t}\right]. \tag{2.35}$$

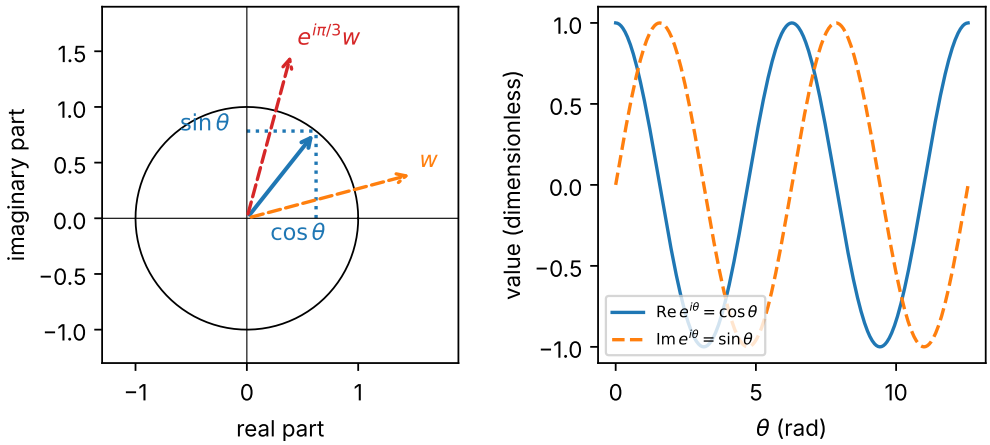


Figure 2.3: Left: the point $e^{i\theta}$ on the unit circle (here $\theta = 0.9$ rad); its horizontal and vertical projections are $\cos\theta$ and $\sin\theta$. The dashed arrow $w = 1.5 + 0.4i$ is rotated by 60° when multiplied by $e^{i\pi/3}$. Right: as θ increases, the real and imaginary parts of $e^{i\theta}$ trace out $\cos\theta$ and $\sin\theta$ (angle θ in radians).

Check: $Ae^{i\phi}e^{i\omega t} = Ae^{i(\omega t+\phi)}$, whose real part is $A\cos(\omega t + \phi)$. The complex number $Ae^{i\phi}$ is the **complex amplitude** or **phasor**: its modulus is the amplitude and its argument is the phase. Because $\mathrm{Re}(u + v) = \mathrm{Re}\, u + \mathrm{Re}\, v$, adding sinusoids of the same frequency is the same as adding their phasors.

Sample Problem 2.6: Adding two sinusoids of the same frequency

Write $f(t) = 3\cos\omega t + 4\sin\omega t$ as a single cosine $A\cos(\omega t + \phi)$.
Step 1: Phasor of $3\cos\omega t$: by Eq. (2.35) with $A = 3, \phi = 0$, it is 3.
Step 2: Phasor of $4\sin\omega t$: since $\sin\omega t = \cos(\omega t - \pi/2)$, it is $4e^{-i\pi/2} = -4i$. (Check: $\mathrm{Re}(-4i\, e^{i\omega t}) = \mathrm{Re}(-4i\cos\omega t + 4\sin\omega t) = 4\sin\omega t$.)
Step 3: Add the phasors: $3 - 4i$. Its modulus is $\sqrt{9+16} = 5$; since its real part is positive, its principal argument is $\arctan(-4/3) \approx -0.927295$.
Step 4: So $f(t) = \mathrm{Re}[(3 - 4i)e^{i\omega t}] = 5\cos(\omega t - 0.927295)$.

$$\boxed{3\cos\omega t + 4\sin\omega t = 5\cos(\omega t - 0.927295)}$$

Sample Problem 2.7: Two prime frequencies: beats

Show that $\sin(t\log 2) + \sin(t\log 3) = 2\sin(\frac{t\log 6}{2})\cos(\frac{t\log(3/2)}{2})$, and find the period of the slow envelope.
Step 1: General identity. Let A, B be real and set $P = (A + B)/2$, $Q = (A - B)/2$, so $A = P + Q$ and $B = P - Q$. Then $e^{iA} + e^{iB} = e^{iP}(e^{iQ} + e^{-iQ}) = 2e^{iP}\cos Q$ and likewise $e^{-iA} + e^{-iB} = 2e^{-iP}\cos Q$, using Eq. (2.34).
Step 2: By Eq. (2.34),

$$\sin A + \sin B = \frac{(e^{iA} + e^{iB}) - (e^{-iA} + e^{-iB})}{2i} = \frac{2e^{iP}\cos Q - 2e^{-iP}\cos Q}{2i} = 2\cos Q\,\frac{e^{iP} - e^{-iP}}{2i} = 2\sin P\cos Q$$

Step 3: Put $A = t\log 2$, $B = t\log 3$. Then $P = t(\log 2 + \log 3)/2 = t\log 6/2$ and $Q = t(\log 2 - \log 3)/2 = -t\log(3/2)/2$. Cosine is even, so $\cos Q = \cos(t\log(3/2)/2)$.
Step 4: The fast factor oscillates with angular frequency $\log 6/2 \approx 0.895880$; the envelope $2\cos(\cdot)$ has angular frequency $\log(3/2)/2 \approx 0.202733$. The loudness $|2\cos Q|$ repeats

when Q changes by π, i.e. after $\Delta t = \pi/0.202733 = 2\pi/\log(3/2) \approx 15.50$.

$$\boxed{\sin(t\log 2) + \sin(t\log 3) = 2\sin\!\left(\tfrac{t\log 6}{2}\right)\cos\!\left(\tfrac{t\log(3/2)}{2}\right), \quad \text{beat period} \approx 15.50}$$

Unlike beats between two frequencies in a whole-number ratio, this sum never repeats exactly. The ratio $\log 2/\log 3$ is irrational: if $\log 2/\log 3 = b/a$ with positive integers a, b, then $a\log 2 = b\log 3$, so $2^a = 3^b$, which is impossible because 2^a is even and 3^b is odd. Sums of sinusoids with irrationally related frequencies are "almost periodic"; Chapter 5 shows that they never repeat. The paper's Eq. (4) is such a sum.

Checkpoint 2.5

Geometrically, what does multiplying a complex number by i do? By $-i$?

Answer: Multiplying by $i = e^{i\pi/2}$ rotates the arrow by 90° counterclockwise; multiplying by $-i = e^{-i\pi/2}$ rotates it by 90° clockwise. The length is unchanged in both cases.

Key Ideas

- Multiplying by $e^{i\alpha}$ rotates by α without changing length; $Re^{i\omega t}$ is uniform circular motion with angular frequency ω and period $2\pi/\omega$.
- $\cos\theta = (e^{i\theta} + e^{-i\theta})/2$ and $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$, Eq. (2.34).
- A sinusoid $A\cos(\omega t + \phi)$ is the real part of the phasor $Ae^{i\phi}$ times $e^{i\omega t}$; same-frequency sinusoids add by adding phasors.
- Trigonometric identities become exponent algebra.

2.7 The complex logarithm and complex powers

2.7.1 Undoing the exponential

For a positive real number x, $\log x$ is the unique real y with $e^y = x$. For a nonzero complex z, we look for complex $w = u + iv$ with $e^w = z$. Write $z = |z|\,e^{i\theta}$ with $\theta = \operatorname{Arg} z$. By Eq. (2.31), $e^w = e^u e^{iv}$. For $e^u e^{iv} = |z|\,e^{i\theta}$, the moduli must agree, $e^u = |z|$, so $u = \log|z|$ (an ordinary real logarithm, since $|z| > 0$). Then the unit-length factors must agree: $e^{iv} = e^{i\theta}$.

Lemma. For real v, θ: $e^{iv} = e^{i\theta}$ if and only if $v = \theta + 2\pi k$ for some integer k. Proof: multiply both sides by $e^{-i\theta}$ (Eq. (2.21)): the condition becomes $e^{i(v-\theta)} = 1$, that is $\cos(v-\theta) = 1$ and $\sin(v-\theta) = 0$ (Eq. (2.3)). The cosine equals 1 exactly at integer multiples of 2π, where the sine is indeed 0.

So the solutions are

$$\log z = \log|z| + i(\operatorname{Arg} z + 2\pi k), \qquad k = 0, \pm 1, \pm 2, \ldots. \tag{2.36}$$

A nonzero complex number has infinitely many logarithms, differing by integer multiples of $2\pi i$. This is the logarithmic face of the fact that the argument is defined only up to 2π. The choice $k = 0$ gives the **principal logarithm**,

$$\operatorname{Log} z = \log|z| + i\operatorname{Arg} z, \qquad -\pi < \operatorname{Im}\operatorname{Log} z \le \pi. \tag{2.37}$$

For positive real z it is the ordinary logarithm (since $\operatorname{Arg} z = 0$). And $\log 0$ is not defined at all, because e^w is never zero.

Branches, informally. Picking one value of k in a consistent way over a region of the plane is called choosing a *branch* of the logarithm. The principal branch has a flaw: as z crosses the negative real axis from above to below, $\operatorname{Arg} z$ jumps from nearly π to nearly $-\pi$, so $\operatorname{Log} z$ jumps by $2\pi i$. The negative real axis is called the *branch cut* of the principal logarithm. Away from it, Log changes continuously. When we follow a logarithm along a path that crosses the cut, we may instead switch k to keep the logarithm continuous; Section 2.10 does exactly that for arguments.

Pitfall

The rule $\log(zw) = \log z + \log w$ holds only up to a multiple of $2\pi i$. With principal values it can fail: $\operatorname{Log}(-1) = i\pi$, so $\operatorname{Log}(-1) + \operatorname{Log}(-1) = 2\pi i$, but $\operatorname{Log}((-1)(-1)) = \operatorname{Log} 1 = 0$. The two sides differ by $2\pi i$.

2.7.2 Complex powers of positive numbers

For a positive real number a and any complex exponent s, we define

$$a^s = e^{s \log a}, \tag{2.38}$$

where $\log a$ is the ordinary real logarithm. No ambiguity arises because a is a positive real number with a single real logarithm. This definition agrees with the familiar one for real s (since $e^{s \log a} = (e^{\log a})^s = a^s$ for real s), and by the exponent law it obeys $a^s a^w = e^{s \log a + w \log a} = a^{s+w}$. For two positive numbers a, b it also obeys $(ab)^s = e^{s \log(ab)} = e^{s \log a + s \log b} = a^s b^s$.

For a complex base, say i^i, we would have to choose a logarithm of the base, and the answer depends on the branch. We will only ever need positive integer bases: the integers k in $\sum k^{-s}$ and the primes p in p^{-s}.

Sample Problem 2.8: Logarithms and a surprising power

Find $\operatorname{Log}(-2)$, $\operatorname{Log}(1+i)$, and the principal value of i^i defined as $e^{i \operatorname{Log} i}$.
Step 1: -2 has modulus 2 and principal argument π, so $\operatorname{Log}(-2) = \log 2 + i\pi \approx 0.693147 + 3.141593\, i$.
Step 2: $1 + i$ has modulus $\sqrt{2}$ and argument $\pi/4$, so $\operatorname{Log}(1+i) = \log\sqrt{2} + i\pi/4 = \frac{1}{2}\log 2 + i\pi/4 \approx 0.346574 + 0.785398\, i$.
Step 3: i has modulus 1 and argument $\pi/2$, so $\operatorname{Log} i = 0 + i\pi/2$. Then $i \operatorname{Log} i = i \cdot i\pi/2 = -\pi/2$, a real number, and $i^i = e^{-\pi/2} \approx 0.207880$.

$$\boxed{\operatorname{Log}(-2) = \log 2 + i\pi, \quad \operatorname{Log}(1+i) = \tfrac{1}{2}\log 2 + \tfrac{i\pi}{4}, \quad i^i = e^{-\pi/2} \approx 0.207880}$$

With a different branch, $\log i = i(\pi/2 + 2\pi k)$ and $i^i = e^{-\pi/2 - 2\pi k}$: infinitely many real values. This is why we restrict complex powers to positive real bases.

Key Ideas

- A nonzero z has infinitely many logarithms $\log|z| + i(\operatorname{Arg} z + 2\pi k)$; the principal one has imaginary part in $(-\pi, \pi]$ and jumps by $2\pi i$ across the negative real axis.
- $\log(zw) = \log z + \log w$ only up to multiples of $2\pi i$.
- For positive real a and complex s, $a^s = e^{s \log a}$ is unambiguous and obeys the usual exponent laws.

2.8 Prime powers on the critical line

2.8.1 The arrow p^{-s}

The zeta function and its relatives are built from quantities k^{-s}, with k a positive integer and s a complex variable. Following the universal convention of number theory, we write the real and imaginary parts of s as

$$s = \sigma + it, \qquad \sigma = \operatorname{Re} s, \quad t = \operatorname{Im} s.$$

Here σ is the Greek letter sigma. Note that t is *not* time in this context: it is a position along the vertical direction of the complex s-plane, the "height". All the nontrivial zeros used in this book have been verified numerically to lie on the line $\sigma = \frac{1}{2}$, with heights $t = t_n$; whether every nontrivial zero does is the Riemann hypothesis, which is open (Chapter 16). (Zeta also has so-called trivial zeros on the negative real axis, Chapter 16; they play no role here.)

Apply the definition (2.38) with $a = k$ and exponent $-s$:

$$\begin{aligned} k^{-s} &= e^{-s\log k} \\ &= e^{-(\sigma+it)\log k} \\ &= e^{-\sigma\log k}\, e^{-it\log k} \quad \text{(exponent law, Eq. (2.31))} \\ &= k^{-\sigma}\, e^{-it\log k} \quad \text{(since } e^{-\sigma\log k} = k^{-\sigma} \text{ for real } \sigma\text{).} \end{aligned} \tag{2.39}$$

This is the most important formula of the chapter for what follows. It splits k^{-s} into a real, positive size $k^{-\sigma}$ and a unit-length rotating factor. So

$$\left|k^{-s}\right| = k^{-\sigma}, \qquad \arg k^{-s} = -t\log k \pmod{2\pi}. \tag{2.40}$$

The size depends only on σ, the angle only on t. In real and imaginary parts,

$$k^{-s} = k^{-\sigma}\cos(t\log k) - i\,k^{-\sigma}\sin(t\log k). \tag{2.41}$$

2.8.2 On the critical line

On the vertical line $\sigma = \frac{1}{2}$, which number theorists call the **critical line** (Chapter 16 explains the name), Eq. (2.40) gives for a prime p

$$\left|p^{-s}\right| = p^{-1/2} = \frac{1}{\sqrt{p}} \qquad (\sigma = \tfrac{1}{2}). \tag{2.42}$$

As t increases, the arrow p^{-s} keeps its length $1/\sqrt{p}$ and its angle $-t\log p$ decreases steadily: the arrow turns *clockwise* around a circle of radius $1/\sqrt{p}$. Its angular rate, in radians per unit of t, is $\log p$. In the language of Section 2.6, p^{-s} rotates with **angular frequency** $\log p$ **in the variable** t, and its period in t is

$$\text{period} = \frac{2\pi}{\log p}. \tag{2.43}$$

Larger primes give smaller circles and faster rotation (Table 2.1 and Figure 2.4).

2.8.3 Powers p^{-ms} and harmonics

Raise p^{-s} to the m-th power ($m = 1, 2, 3, \dots$). Applying the exponent law $e^z e^w = e^{z+w}$ $m-1$ times gives $(p^{-s})^m = (e^{-s\log p})^m = e^{-ms\log p}$, and since $m\log p = \log p^m$ this equals $e^{-s\log p^m} = (p^m)^{-s}$, which we write p^{-ms}. Then Eq. (2.39) with $k = p^m$, together with $(p^m)^{-\sigma} = p^{-m\sigma}$, gives

$$p^{-ms} = (p^m)^{-s} = p^{-m\sigma}\, e^{-imt\log p}. \tag{2.44}$$

Table 2.1: The arrow p^{-s} on the critical line $\sigma = \frac{1}{2}$ for the first six primes: radius $1/\sqrt{p}$, angular frequency $\log p$ in t, and period $2\pi/\log p$.

p	$1/\sqrt{p}$	$\log p$	$2\pi/\log p$
2	0.7071	0.6931	9.0647
3	0.5774	1.0986	5.7192
5	0.4472	1.6094	3.9040
7	0.3780	1.9459	3.2289
11	0.3015	2.3979	2.6203
13	0.2774	2.5649	2.4496

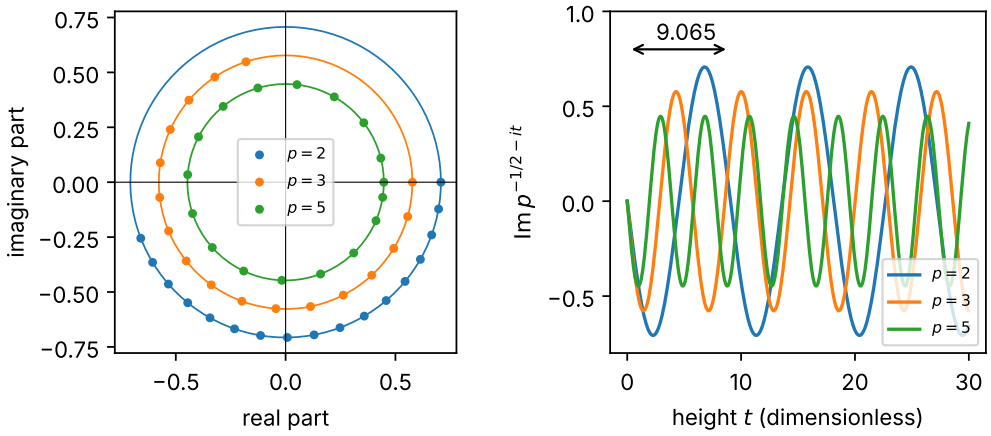


Figure 2.4: Left: as t runs from 0 to 4 in steps of 0.25, the arrow $p^{-1/2-it}$ moves clockwise around a circle of radius $1/\sqrt{p}$, for $p = 2, 3, 5$; dots mark the steps, and larger primes turn faster. Right: the imaginary parts $\operatorname{Im} p^{-1/2-it} = -p^{-1/2}\sin(t\log p)$ as functions of the height t (dimensionless). The $p = 2$ curve has period $2\pi/\log 2 \approx 9.065$.

On the critical line its length is $p^{-m/2}$ and its angular frequency in t is $m\log p = \log p^m$: the m-th harmonic of the prime-p rotation. Taking the imaginary part, with Eq. (2.41),

$$\operatorname{Im}\left[p^{-ms}\right] = -p^{-m/2}\sin(mt\log p), \qquad \text{so} \qquad \frac{\sin(mt\log p)}{m\,p^{m/2}} = -\frac{1}{m}\operatorname{Im}\left[p^{-ms}\right] \quad (\sigma = \tfrac{1}{2}). \tag{2.45}$$

The left-hand form is exactly the shape of each term in the paper's Eq. (4). In particular our running example RE2, the prime-2 wave, is

$$-\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 2)}{m\,2^{m/2}} = \frac{1}{\pi}\sum_{m=1}^{3}\frac{1}{m}\operatorname{Im}\left[2^{-m(1/2+it)}\right]. \tag{2.46}$$

Its three frequencies $\log 2$, $2\log 2 = \log 4$, $3\log 2 = \log 8$ are whole-number multiples of $\log 2$, so RE2 repeats with the period of the slowest one, $2\pi/\log 2 \approx 9.0647$ (Problem 18). Where the factor $-1/\pi$ and the sum over primes come from is the story of Chapters 17 and 18.

Sample Problem 2.9: The arrow 2^{-s} at the first zero

Compute $2^{-\rho_1}$ for $\rho_1 = \frac{1}{2} + 14.134725\,i$, giving its modulus, its principal argument and its real and imaginary parts.
Step 1: Modulus, Eq. (2.42): $|2^{-\rho_1}| = 2^{-1/2} \approx 0.70711$.
Step 2: Argument, Eq. (2.40): $-t_1\log 2 = -14.134725 \times 0.693147 \approx -9.79744$ rad.
Step 3: Reduce to $(-\pi, \pi]$ by adding $4\pi \approx 12.56637$: $-9.79744 + 12.56637 = 2.76893$ rad.
Step 4: Real and imaginary parts: $\cos(2.76893) \approx -0.931360$ and $\sin(2.76893) \approx 0.364101$, so $2^{-\rho_1} \approx 0.70711(-0.931360 + 0.364101\,i)$.

$$\boxed{2^{-\rho_1} \approx -0.65857 + 0.25746\,i, \qquad \left|2^{-\rho_1}\right| \approx 0.70711, \qquad \operatorname{Arg} 2^{-\rho_1} \approx 2.76893}$$

```
import numpy as np
t1 = 14.134725141734695      # height of the first zeta zero
s = 0.5 + 1j * t1
for p in [2, 3, 5, 7, 11, 13]:
    w = p**(-s)              # Python computes exp(-s*log p)
    print(p, abs(w), np.angle(w), 2 * np.pi / np.log(p))
```

Checkpoint 2.6

On the critical line, what happens to 3^{-s} when t increases by $2\pi/\log 3 \approx 5.7192$? What happens to its modulus as t changes?

Answer: Its angle $-t\log 3$ decreases by exactly 2π, so 3^{-s} returns to the same value. Its modulus stays $1/\sqrt{3} \approx 0.5774$ for every t.

Key Ideas

- With $s = \sigma + it$, $k^{-s} = k^{-\sigma}e^{-it\log k}$: size $k^{-\sigma}$, angle $-t\log k$, Eqs. (2.39)–(2.40).
- On the critical line $\sigma = \frac{1}{2}$, p^{-s} is an arrow of length $1/\sqrt{p}$ turning clockwise with angular frequency $\log p$ in t, period $2\pi/\log p$.
- p^{-ms} has length $p^{-m/2}$ and frequency $m\log p = \log p^m$; $\sin(mt\log p)/(mp^{m/2}) = -\operatorname{Im}[p^{-ms}]/m$, the building block of the paper's Eq. (4).

2.9 Functions of a complex variable and their zeros

2.9.1 What a complex function is

A **function of a complex variable** assigns to each complex number z (in some region) a complex number $f(z)$. Examples: $f(z) = z^2$, $f(z) = 1/z$ (for $z \neq 0$), $f(z) = e^z$, and $f(s) = 1 - 2^{-s}$. The zeta function is another, defined in Chapter 16.

You cannot draw the graph of such a function in the usual way: input and output each need two real numbers, so the graph would live in four dimensions. Instead we use one of two tricks. (i) Split into two real functions of two real variables: writing $z = x + iy$, $f(z) = u(x, y) + iv(x, y)$ with $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$. (ii) Plot the modulus $|f|$ (a "landscape" over the plane) and, if needed, the argument.

For $f(z) = z^2$ with $z = x + iy$: $z^2 = x^2 + 2ixy + i^2y^2 = (x^2 - y^2) + i(2xy)$, so $u = x^2 - y^2$ and $v = 2xy$.

2.9.2 Zeros

A **zero** of f is a point z_0 with $f(z_0) = 0$. By Eq. (2.3), this means *both* real equations

$$u(x_0, y_0) = 0 \qquad \text{and} \qquad v(x_0, y_0) = 0 \tag{2.47}$$

hold at once. Equivalently, $|f(z_0)| = 0$, since $|f| = \sqrt{u^2 + v^2}$ vanishes only when u and v both vanish. Each equation alone typically holds along a curve in the plane; the zeros are where two such curves cross. Two equations in two unknowns usually have isolated solutions, so we expect the zeros of a complex function to be separate points that can be listed one by one, $\rho_1, \rho_2, \ldots$, rather than lines or regions. This counting of equations is only a heuristic. The function $f(z) = \operatorname{Re} z$ assigns a complex number to each z, yet it vanishes on the whole imaginary axis: its imaginary part v is 0 everywhere, so one of the two equations imposes no condition at all.

Taken on trust

For *analytic* functions, meaning those with a complex derivative $f'(z) = \lim_{h\to 0}[f(z + h) - f(z)]/h$ as the complex number h approaches 0 from every direction in the plane, the zeros of a function that is not identically zero are isolated points. Polynomials, e^z, $1 - 2^{-s}$ and the zeta function (away from its pole at $s = 1$, Chapter 16) are analytic; $\operatorname{Re} z$ is not. We take this theorem on trust; it is proved in the complex-variables chapters of [22].

For $f(z) = z^2$: $v = 2xy = 0$ requires $x = 0$ or $y = 0$. If $y = 0$, then $u = x^2 = 0$ forces $x = 0$. If $x = 0$, then $u = -y^2 = 0$ forces $y = 0$. So the only zero is $z = 0$: the curves $u = 0$ (the diagonal lines $y = \pm x$) and $v = 0$ (the two axes) cross only at the origin.

2.9.3 Zeros of polynomials

The quadratic formula for complex numbers. Take $az^2 + bz + c = 0$ with complex a, b, c and $a \neq 0$, and write $D = b^2 - 4ac$ for the discriminant. Completing the square uses only the commutative, associative and distributive laws, which hold for complex numbers (Section 2.2):

$$\begin{aligned} az^2 + bz + c &= a\Big[z^2 + \frac{b}{a}z + \frac{c}{a}\Big] \\ &= a\Big[\Big(z + \frac{b}{2a}\Big)^2 - \frac{b^2}{4a^2} + \frac{c}{a}\Big] \quad \Big(\text{since } (z + \tfrac{b}{2a})^2 = z^2 + \tfrac{b}{a}z + \tfrac{b^2}{4a^2}\Big) \\ &= a\Big[\Big(z + \frac{b}{2a}\Big)^2 - \frac{D}{4a^2}\Big] \quad \Big(\text{since } -\tfrac{b^2}{4a^2} + \tfrac{4ac}{4a^2} = -\tfrac{D}{4a^2}\Big). \end{aligned}$$

Let w be a complex number with $w^2 = D$. (One always exists: if $D = re^{i\theta}$, take $w = \sqrt{r}\, e^{i\theta/2}$, whose square is $re^{i\theta}$ by Eq. (2.18).) Then $D/(4a^2) = (w/2a)^2$, and with $u = z + b/(2a)$ the bracket is $u^2 - (w/2a)^2 = (u - w/2a)(u + w/2a)$. Since $a \neq 0$, the equation holds exactly when this product is zero. A product of complex numbers is zero only if a factor is zero: by Eq. (2.14), $|u_1 u_2| = |u_1|\,|u_2|$ vanishes only if $|u_1| = 0$ or $|u_2| = 0$. So $u = \pm w/(2a)$, that is

$$z = \frac{-b \pm w}{2a}, \qquad w^2 = D.$$

This is the familiar formula, with the square root of D meaning either complex number whose square is D (the $\pm$ covers the other one). For real coefficients and a negative discriminant $D < 0$, the two square roots of D are $\pm i\sqrt{|D|}$ (check: $(i\sqrt{|D|})^2 = -|D| = D$).

Conjugate pairs. If a polynomial $q(z) = c_0 + c_1 z + \cdots + c_d z^d$ has *real* coefficients and $q(z_0) = 0$, then $q(\bar{z}_0) = 0$ too. Proof: by Eq. (2.10), conjugation passes through sums and products, and $\bar{c}_k = c_k$ for real c_k, so $\overline{q(z_0)} = \bar{c}_0 + \bar{c}_1 \bar{z}_0 + \cdots + \bar{c}_d \bar{z}_0^{\,d} = q(\bar{z}_0)$. Since $q(z_0) = 0$, $q(\bar{z}_0) = \bar{0} = 0$. So the non-real roots of a real polynomial come in mirror-image pairs.

Taken on trust

The *fundamental theorem of algebra*: every polynomial of degree $d \geq 1$ with complex coefficients has exactly d complex roots, counted with multiplicity. Complex numbers are therefore "enough" for polynomial equations: we never need to invent another kind of number. The proof uses methods beyond this book; see [22]. We will use the theorem in Chapter 8, where eigenvalues are roots of a polynomial.

Sample Problem 2.10: A quadratic whose roots are the first zero and its mirror image

Solve $z^2 - z + (\frac{1}{4} + t_1^2) = 0$ with $t_1 = 14.134725$.
Step 1: Coefficients: $a = 1$, $b = -1$, $c = \frac{1}{4} + t_1^2 \approx 200.0405$.
Step 2: Discriminant: $D = b^2 - 4ac = 1 - 4(\frac{1}{4} + t_1^2) = 1 - 1 - 4t_1^2 = -4t_1^2 < 0$.
Step 3: The square roots of D are $\pm i\sqrt{4t_1^2} = \pm 2it_1$.
Step 4: Quadratic formula: $z = \dfrac{1 \pm 2it_1}{2} = \frac{1}{2} \pm it_1$.

$$\boxed{z = \tfrac{1}{2} + 14.134725\, i \quad \text{and} \quad z = \tfrac{1}{2} - 14.134725\, i}$$

The roots are ρ_1 and its conjugate $\bar{\rho}_1$, as the conjugate-pair theorem requires. The zeta zeros also come in conjugate pairs (Chapter 16); by convention we list only those with $t > 0$.

2.9.4 The zeros of $1 - 2^{-s}$

Our first example built from a prime is $f(s) = 1 - 2^{-s}$. This factor reappears in Chapter 16. First a lemma.

Lemma. For complex w: $e^w = 1$ if and only if $w = 2\pi i k$ for some integer k. Proof: write $w = u + iv$. Then $e^w = e^u e^{iv}$ has modulus e^u, which equals 1 only if $u = 0$. Then $e^{iv} = 1 = e^{i0}$, which by the lemma of Section 2.7 means $v = 2\pi k$.

Sample Problem 2.11: Zeros built from the prime 2

Find all zeros of $f(s) = 1 - 2^{-s}$, first with the lemma and then by checking that real and imaginary parts both vanish.
Step 1: $f(s) = 0$ means $2^{-s} = 1$, i.e. $e^{-s \log 2} = 1$ (Eq. (2.38)).

Step 2: By the lemma, $-s\log 2 = 2\pi ik$, so $s = -2\pi ik/\log 2$. As k runs over all integers, so does $-k$; hence the zeros are $s = 2\pi ik/\log 2$, $k = 0, \pm 1, \pm 2, \ldots$.
Step 3: Cross-check with real and imaginary parts. With $s = \sigma + it$, Eq. (2.41) gives $f = [1 - 2^{-\sigma}\cos(t\log 2)] + i[2^{-\sigma}\sin(t\log 2)]$. The imaginary part vanishes only if $\sin(t\log 2) = 0$ (since $2^{-\sigma} > 0$), i.e. $t\log 2 = \pi j$ for an integer j. Then $\cos(t\log 2) = (-1)^j$, and the real part is $1 - (-1)^j 2^{-\sigma}$. For odd j this is $1 + 2^{-\sigma} > 0$, never zero. For even $j = 2k$ it is $1 - 2^{-\sigma}$, zero only at $\sigma = 0$. So $\sigma = 0$ and $t = 2\pi k/\log 2$, as before.
Step 4: The zeros lie on the imaginary axis, equally spaced by $2\pi/\log 2 \approx 9.0647$: at $0, \pm 9.0647i, \pm 18.1294i, \ldots$

$$\boxed{1 - 2^{-s} = 0 \iff s = \frac{2\pi ik}{\log 2},\ k \in \{0, \pm 1, \pm 2, \ldots\}}$$

The spacing is the period of the prime-2 rotation, Table 2.1. None of these zeros is on the critical line; on that line $|2^{-s}| = 1/\sqrt{2} < 1$, so 2^{-s} can never equal 1.

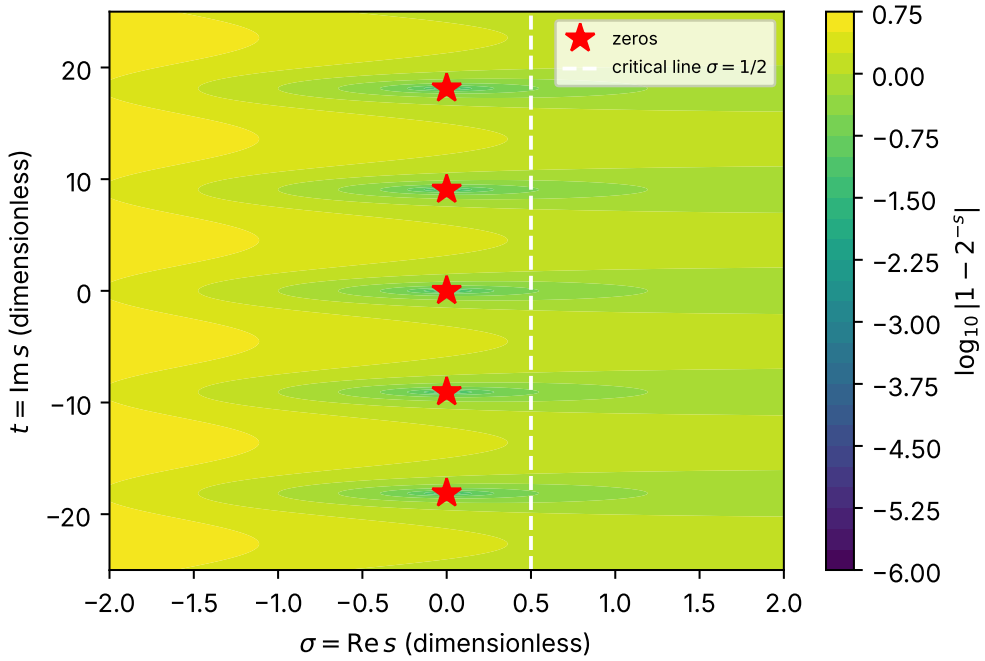


Figure 2.5: The landscape of $\log_{10}|1 - 2^{-s}|$ over the complex s-plane ($\sigma = \mathrm{Re}\, s$ horizontal, $t = \mathrm{Im}\, s$ vertical, both dimensionless). Dark wells are zeros: they sit on the imaginary axis at $t = 2\pi k/\log 2$, spaced by 9.065 (stars). The dashed line is the critical line $\sigma = \frac{1}{2}$, on which $|1 - 2^{-s}| \geq 1 - 1/\sqrt{2}$ and the function never vanishes.

Checkpoint 2.7

Why does e^z have no zeros anywhere in the complex plane?

Answer: By Eq. (2.31), $|e^{x+iy}| = e^x$, and $e^x > 0$ for every real x. A number of positive modulus is not zero.

Key Ideas

- A complex function $f(z) = u + iv$ can be studied through two real functions u, v or through $|f|$ and $\arg f$.
- A zero requires $u = 0$ and $v = 0$ simultaneously (equivalently $|f| = 0$); zeros are typically isolated points.
- Real polynomials have non-real roots in conjugate pairs; every polynomial of degree d has d complex roots (taken on trust).
- $e^w = 1$ iff $w = 2\pi ik$; hence $1 - 2^{-s}$ vanishes exactly at $s = 2\pi ik/\log 2$.

2.10 The argument of a function along a line

2.10.1 Arguments of products add

Equation (2.19) extends to any number of factors by applying it repeatedly: if $F = f_1 f_2 \cdots f_K$, then

$$\arg F = \arg f_1 + \arg f_2 + \cdots + \arg f_K \pmod{2\pi}, \tag{2.48}$$

and for a reciprocal, $\arg(1/f) = -\arg f$. A product of many factors has an angle that is a *sum* of many angles. This simple fact is how the primes get into the counting of the zeta zeros: Chapter 16 writes zeta as an infinite product with one factor per prime (the Euler product), each factor being $1/(1-p^{-s})$. Since $\arg(1/f) = -\arg f$, the argument of zeta becomes minus a sum over primes, $\arg\zeta(s) = -\sum_p \arg(1-p^{-s}) \pmod{2\pi}$. This holds for $\operatorname{Re} s > 1$, where the infinite product converges (Chapter 16). On the critical line the product does not converge, so the corresponding formula there is only formal (heuristic), as Chapter 18 discusses honestly.

Sample Problem 2.12: Argument of a product, and of a prime factor

(a) Find $\operatorname{Arg}[(1+i)(\sqrt{3}+i)]$ two ways. (b) Find $\operatorname{Arg}(1-2^{-s})$ on the critical line at the height t^* where $t^* \log 2 = \pi/2$.

Step 1: (a) $\operatorname{Arg}(1+i) = \pi/4$ and $\operatorname{Arg}(\sqrt{3}+i) = \arctan(1/\sqrt{3}) = \pi/6$. Their sum $5\pi/12 \approx 1.308997$ lies in $(-\pi, \pi]$, so it is the principal argument of the product.

Step 2: Directly: $(1+i)(\sqrt{3}+i) = \sqrt{3} + i + i\sqrt{3} + i^2 = (\sqrt{3}-1) + i(\sqrt{3}+1) \approx 0.732051 + 2.732051\,i$. Then $\arctan(2.732051/0.732051) = \arctan(3.732051)$. To recognize this angle, divide the addition formulas for $\sin(A+B)$ and $\cos(A+B)$ and then divide top and bottom by $\cos A \cos B$: $\tan(A+B) = (\tan A + \tan B)/(1 - \tan A \tan B)$. With $A = 45^\circ$ ($\tan A = 1$) and $B = 30^\circ$ ($\tan B = 1/\sqrt{3}$), $\tan 75^\circ = \frac{1+1/\sqrt{3}}{1-1/\sqrt{3}} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = \frac{(\sqrt{3}+1)^2}{3-1} = \frac{4+2\sqrt{3}}{2} = 2 + \sqrt{3} \approx 3.732051$. Since arctan returns angles in $(-90^\circ, 90^\circ)$, the argument is $75^\circ = 5\pi/12$. Agreed.

Step 3: (b) $t^* = \pi/(2\log 2) \approx 2.266180$. On the critical line, $2^{-s} = 2^{-1/2} e^{-it^* \log 2} = 2^{-1/2} e^{-i\pi/2} = -i/\sqrt{2}$.

Step 4: So $1 - 2^{-s} = 1 + i/\sqrt{2}$, with positive real part, and $\operatorname{Arg}(1-2^{-s}) = \arctan(1/\sqrt{2}) \approx 0.615480$.

$$\boxed{\text{(a) } \tfrac{5\pi}{12} \approx 1.308997; \qquad \text{(b) } \operatorname{Arg}(1-2^{-s})\big|_{t^*} = \arctan\tfrac{1}{\sqrt{2}} \approx 0.615480}$$

2.10.2 Following the argument continuously

Suppose $f(t)$ is a complex number that depends continuously on a real parameter t running over an interval (for instance, a function evaluated along the critical line, $f(\frac{1}{2}+it)$) and is never zero along the way. A **continuous argument** of f is a continuous real function $\alpha(t)$ with $f(t) = |f(t)|\, e^{i\alpha(t)}$ for all t. Three facts:

It exists. We take this on trust, with the following sketch.

Taken on trust

Two calculus theorems are used without proof (see [22]): a continuous function on a closed interval attains a minimum, and it is uniformly continuous there. Since $|f(t)|$ is continuous and never zero on the closed interval $[t_a, t_b]$, it has a positive minimum μ. By uniform continuity we can cut $[t_a, t_b]$ into finitely many pieces so short that on each piece $|f(t) - f(t_j)| < \mu$, where t_j is the left end of the piece. Then $|f(t)/f(t_j) - 1| < \mu/|f(t_j)| \le 1$, so $f(t)/f(t_j)$ lies in the disc of radius 1 about the point 1, which is inside the right half-plane. There Arg is continuous, so $\operatorname{Arg} f(t_j) + \operatorname{Arg}[f(t)/f(t_j)]$ is a continuous argument of f on that piece (Eq. (2.19)). At the end of each piece, the next piece's argument differs from it by a multiple of 2π; subtracting that constant joins the pieces into one continuous argument.

It is unique up to a constant multiple of 2π. If α_1 and α_2 are both continuous arguments, their difference is continuous and, at every t, an integer multiple of 2π (by the lemma of Section 2.7). Recall the intermediate value theorem of calculus: if g is continuous on an interval $[t_a, t_b]$, then g takes every value between $g(t_a)$ and $g(t_b)$. Let $g = \alpha_1 - \alpha_2$, defined on the interval over which t runs. If g took two different values $2\pi j < 2\pi j'$ at two points, it would also take every value between them, including $2\pi j + \pi$, which is not a multiple of 2π. That is impossible, so g is constant. Fixing α at one starting point fixes it everywhere.

For products it adds exactly. If α and β are continuous arguments of f and g, then $fg = |f|\,|g|\, e^{i(\alpha+\beta)}$ and $\alpha + \beta$ is continuous, so $\alpha + \beta$ is a continuous argument of fg. With continuous arguments there is no "mod 2π" left in Eq. (2.48).

The principal argument Arg is continuous as long as $f(t)$ avoids the negative real axis (Section 2.7). When the path does cross that axis, Arg jumps by 2π, and we must add or subtract 2π to remove the jump; numpy's `np.unwrap` does this. If the path circles around the origin, a continuous argument can grow without bound; counting such windings is the idea behind counting zeros in Chapter 17 (see also Problem 28).

A continuous argument of $1-2^{-s}$ on the critical line. Let $x = 2^{-1/2-it}$, so $|x| = 1/\sqrt{2}$. Then $\operatorname{Re}(1-x) = 1 - \operatorname{Re} x \ge 1 - |x| = 1 - 1/\sqrt{2} > 0$. So $1 - 2^{-s}$ stays in the right half-plane for all real t, never touches the negative real axis, and its principal argument is continuous and lies in $(-\pi/2, \pi/2)$. No unwrapping is needed.

2.10.3 A preview: the phase of a prime factor is the prime-2 wave

Here is a numerical observation that previews the central mechanism of Chapter 18. Plot $-\frac{1}{\pi}\operatorname{Arg}(1 - 2^{-1/2-it})$ against t, and on the same axes plot RE2, the three-term prime-2 wave of Eq. (2.46). Figure 2.6 shows that the two curves are nearly the same; a numerical computation (the code listing at the end of this subsection) finds that they differ by less than 0.05 for all t (they are both periodic with period $2\pi/\log 2$, so checking one period suffices). At t^* of the last Sample Problem, for instance, the argument gives $-0.615480/\pi \approx -0.195913$ while RE2 gives $-\frac{1}{\pi}\left(\frac{1}{\sqrt{2}} - \frac{1}{3\cdot 2^{3/2}}\right) \approx -0.187566$ (the $m = 2$ term vanishes because $\sin\pi = 0$).

The status of this observation: in this chapter it is a numerical observation. It becomes a theorem once we take one fact on trust.

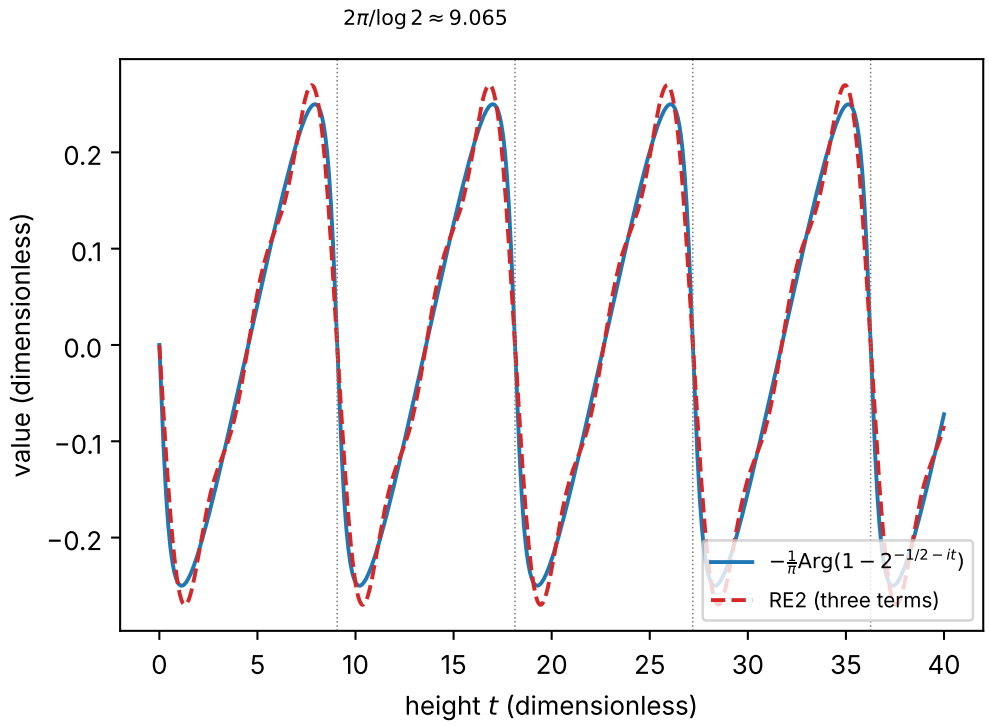


Figure 2.6: Solid: minus the phase of the prime-2 factor, $-\frac{1}{\pi}\operatorname{Arg}(1-2^{-1/2-it})$, against the height t. Dashed: the three-term prime-2 wave RE2, $-\frac{1}{\pi}\sum_{m=1}^{3}\sin(mt\log 2)/(m2^{m/2})$. They agree to within 0.05; both repeat with period $2\pi/\log 2 \approx 9.065$ (marked).

Taken on trust

For a complex number x with $|x| < 1$, the series $\mathcal{S}(x) = -(x + x^2/2 + x^3/3 + \cdots)$ converges, and $e^{\mathcal{S}(x)} = 1 - x$. For real x in $(-1, 1)$ this is the Taylor series $\log(1-x) = -(x + x^2/2 + x^3/3 + \cdots)$ derived in Chapter 3. The extension to complex x, including the identity $e^{\mathcal{S}(x)} = 1 - x$, is proved in the complex-variables chapters of [22].

Given this, $\mathcal{S}(x)$ is one of the logarithms of $1 - x$ in Eq. (2.36), and $\operatorname{Im}\mathcal{S}(x)$ is one of the arguments of $1 - x$. Which one? Apply the triangle inequality, Eq. (2.15), repeatedly to a partial sum, using $|x^m| = |x|^m$ (Eq. (2.14)):

$$\left|x + \frac{x^2}{2} + \cdots + \frac{x^N}{N}\right| \le |x| + \frac{|x|^2}{2} + \cdots + \frac{|x|^N}{N}.$$

The right side is a partial sum of the real series for $-\log(1-|x|)$; its terms are positive, so it never exceeds $-\log(1-|x|)$. Letting $N \to \infty$, and using $|\operatorname{Im} u| \le |u|$ for any complex u, we get $|\operatorname{Im}\mathcal{S}| \le |\mathcal{S}| \le -\log(1-|x|)$. For $|x| = 1/\sqrt{2}$ this is $-\log(1-1/\sqrt{2}) \approx 1.228 < \pi$. Arguments of the same number differ by multiples of 2π, so only one of them lies in $(-\pi, \pi)$, and that one is the principal argument. Hence $\operatorname{Im}\mathcal{S}(x) = \operatorname{Arg}(1-x)$. With $x = 2^{-1/2-it}$, Eq. (2.45) gives $\operatorname{Im} x^m = -2^{-m/2}\sin(mt\log 2)$, so $\operatorname{Arg}(1-2^{-s}) = \sum_{m\ge 1}\sin(mt\log 2)/(m2^{m/2})$ exactly, and RE2 keeps the first three terms. Chapter 18 then does the same for every prime factor of zeta, which is where the paper's Eq. (4) comes from (with an honest discussion of what is rigorous and what is not).

```
import numpy as np, qmnt
t = qmnt.zeros(10)                    # RE1: heights of the first ten zeros
rho = 0.5 + 1j * t
```

```
print(np.abs(rho), np.angle(rho))  # moduli and principal arguments
tt = np.linspace(0, 40, 4001)
x = 2.0**(-0.5 - 1j * tt)
phase = -np.angle(1 - x) / np.pi
re2 = -sum(np.sin(m * tt * np.log(2)) / (m * 2**(m / 2))
           for m in (1, 2, 3)) / np.pi
print(np.max(np.abs(phase - re2)))  # below 0.05
```

Key Ideas

- The argument of a product is the sum of the arguments (mod 2π), Eq. (2.48).
- Along a path that avoids zeros, a continuous argument exists, is unique up to a constant $2\pi k$, and adds exactly over products.
- On the critical line $1 - 2^{-s}$ stays in the right half-plane, so its principal argument is continuous; numerically, $-\frac{1}{\pi}\operatorname{Arg}(1 - 2^{-s})$ matches the prime-2 wave RE2 to within 0.05 (proved later via the series for $\log(1 - x)$).

2.11 Paper connection

★ Did you know? *Who Named i?*

The symbol i for $\sqrt{-1}$ is due to Leonhard Euler, who used it in a memoir written in 1777 and published in 1794, after his death. Gauss later adopted it, which helped make it standard. Euler also introduced the letter e, popularized π, and found $e^{ix} = \cos x + i \sin x$. Every zeta zero in this book is written $\frac{1}{2} + it_n$ in his notation.

The zeros are complex numbers (Sec. I, first sentence; Sec. II A). The paper opens: "The Hilbert–Pólya conjecture proposes that the imaginary parts t_n of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of the Riemann zeta function are the spectrum of a self-adjoint operator" [1]. You can now parse every symbol: ρ_n is a complex number with real part $\frac{1}{2}$ and imaginary part t_n, a real number. Sec. II A reports $t_1 = 14.134725141734695$ and $t_{1000} = 1419.42248094599$; these are imaginary parts, as in our Sample Problem on $|\rho_1|$. Writing every zero with real part $\frac{1}{2}$ places them all on the vertical line $\sigma = \frac{1}{2}$ of the complex plane (Figure 2.7). That *every* nontrivial zero lies on this line is the Riemann hypothesis, which is open (Chapter 16); the paper only uses zeros for which this has been verified numerically (a numerical observation, not a proof). From Sec. II on, the paper works only with the real list t_n; the complex structure is hidden in that choice, and in the conjugate-pair symmetry that lets one keep only $t_n > 0$.

The sine terms of Eq. (4). The paper's arithmetic oracle, Eq. (4) in Sec. II D, is

$$N_{\text{fl}}(t) = -\frac{1}{\pi} \sum_{p \in P} \sum_{m=1}^{3} \frac{\sin(mt \log p)}{m\, p^{m/2}}.$$

By Eq. (2.45), every term is $-\operatorname{Im}[p^{-ms}]/m$ evaluated at $s = \frac{1}{2} + it$. The factor $p^{-m/2}$ is the length of the arrow p^{-ms} on the critical line, Eq. (2.42): it arises because the sum is evaluated at $s = \frac{1}{2} + it$, that is at $\sigma = \frac{1}{2}$, the real part shared by every zero ρ_n that the paper uses, and there $|p^{-ms}| = p^{-m\sigma} = 1/\sqrt{p^m}$. The argument $mt \log p$ is the arrow's angle (up to sign), so $m \log p$ is an angular frequency in the variable t. So Eq. (4) is a sum of rotating arrows, one family per prime, projected onto the imaginary axis. Chapters 17 and 18 derive why this sum approximates the fluctuating part of the zero count.

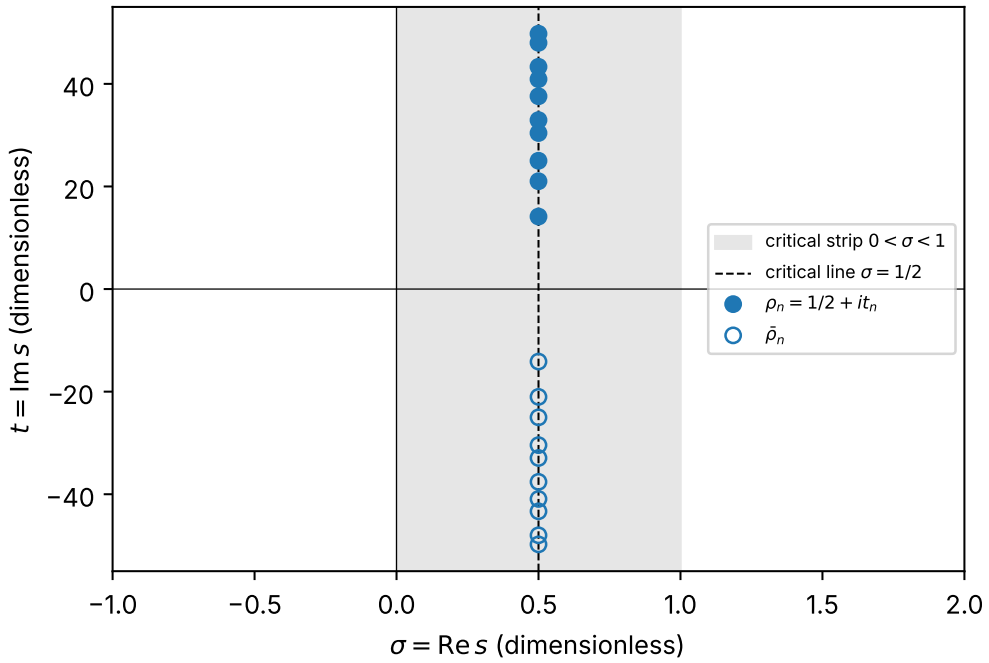


Figure 2.7: The first ten nontrivial zeta zeros $\rho_n = \frac{1}{2} + it_n$ (RE1, filled circles) and their conjugates $\bar{\rho}_n = \frac{1}{2} - it_n$ (open circles) in the complex s-plane. The shaded band $0 < \sigma < 1$ is the critical strip and the dashed line is the critical line $\sigma = \frac{1}{2}$ (Chapter 16). Note the very different scales of the two axes: the zeros are tall and thin in this picture, which is why $\operatorname{Arg} \rho_1$ is close to 90°.

Frequencies $\log p^m$ (Secs. II D and IV; Table IV; App. A steps 20, 23). The paper's abstract says that the sparse control C2 "recovers frequencies at $\log p^m$". Sec. IV states this more carefully: every selected frequency "lies within half the Fourier resolution of the training window of the logarithm of a prime power". In our language $\log p^m = m \log p$ is the angular frequency of p^{-ms} (Eq. (2.44)). The smallest is $\log 2 \approx 0.693$, period 9.0647 in t; the controls search the range $0.05 < \omega < 8$ (Sec. II D). Note that $\omega = 8$ corresponds to the prime power $q = e^8 \approx 2981$, which is consistent with the cutoff $q \le 3000$ used in the alignment check of App. A step 23. In the 800-zero C2 fit, the matched prime powers include $4 = 2^2$ and $8 = 2^3$, the $m = 2, 3$ harmonics of the prime 2 (Sec. IV lists "those up to 113 except 32, 64, and 81").

The prime-2 wave and its period (Sec. III continuation probe; Sec. V). The arithmetic-continuation probe of Sec. III replaces the targets by "a synthetic fluctuation consisting solely of the p=2 term of Eq. (4)": our RE2, Eq. (2.46). It is a periodic signal with period $2\pi/\log 2 \approx 9.0647$ in t (Problem 18). Sec. V quotes this period: "The dominant p=2 oscillation, period $2\pi/\log 2 \approx 9$ in t, is only ∼6–7 levels at these heights." Problem 30 checks that statement using the mean spacing derived in Chapter 17: near height $t = 650$, one period of the prime-2 wave spans about 6.7 mean spacings. Our Figure 2.6 also shows that RE2 is, up to a tail smaller than 0.05, minus $1/\pi$ times the phase of the single factor $1 - 2^{-s}$: the probe asks a fitted potential to continue the phase of one rotating arrow, and the paper finds that it cannot (held-out correlation +0.07, Sec. III).

Complex eigenvalues (Sec. II E; Sec. V; App. A step 11). The non-Hermitian classes of Sec. II E perturb $H \to H + iA$, an operator containing i explicitly; the absorptive class models a term $-i\Gamma/2$. Such operators can have complex eigenvalues (the PT-odd class has a real spectrum to leading order, Sec. II E), and App. A step 11 uses "dense complex `eig`" to

compute them. The imaginary part of an eigenvalue $E - i\Gamma/2$ controls decay: in quantum mechanics a state with energy E evolves with the factor e^{-iEt} (Chapter 10, with t now time), and by Eq. (2.31) $\left|e^{-i(E-i\Gamma/2)t}\right| = e^{-\Gamma t/2}$ (Problem 24). When Sec. V reports that "58% of its widths have the wrong sign", it refers to the sign of this imaginary part. Complex eigenvalues have no natural "increasing order" (Section 2.3), and in any case order does not tell which perturbed state came from which unperturbed one; the paper identifies states "by overlap" (Sec. II E (iii)), and Chapter 14 explains the procedure.

Phases and surrogates (Sec. III; App. A step 18). The "phase-randomized surrogate" [16] of Sec. III is described as "power spectrum kept exactly, no arithmetic content". In Chapter 6 you will see that a sequence can be written as a sum of complex exponentials with complex coefficients; the surrogate keeps the modulus of each coefficient and replaces its argument by a random angle ("uniform phases" in App. A step 18). Modulus and argument, the two halves of polar form, are exactly the quantities being kept and scrambled.

Fig. 1. The curves of Fig. 1 (fluctuations of zeros 370–460) are real numbers measured in mean spacings, but the six-prime prediction plotted there is built from six families of rotating arrows p^{-ms} (presumably the first six primes, $p \le 13$, Table 2.1; the paper says only "six-prime"), projected onto the imaginary axis.

2.12 Summary

- The imaginary unit satisfies $i^2 = -1$ (Eq. (2.1)); a complex number is $z = a + bi$ with real part a and imaginary part b, both real. Multiplication: Eq. (2.5).
- In the complex plane z is the point (a, b). Conjugate $\bar{z} = a - bi$ (Eq. (2.7)), modulus $|z| = \sqrt{a^2 + b^2}$ (Eq. (2.11)), $z\bar{z} = |z|^2$ (Eq. (2.12)), division by Eq. (2.13), $|zw| = |z|\,|w|$ (Eq. (2.14)), triangle inequality Eq. (2.15).
- Polar form $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ (Eqs. (2.16), (2.30)); principal argument by Eq. (2.17); moduli multiply and arguments add (Eqs. (2.18), (2.19)); De Moivre Eq. (2.20).
- Taylor coefficients $a_n = f^{(n)}(0)/n!$ (Eq. (2.23)); complex exponential defined by the series Eq. (2.27); Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ (Eq. (2.28)); $e^{x+iy} = e^x(\cos y + i\sin y)$ (Eq. (2.31)).
- Multiplication by $e^{i\alpha}$ is rotation (Eq. (2.32)); $\cos\theta = (e^{i\theta}+e^{-i\theta})/2$, $\sin\theta = (e^{i\theta}-e^{-i\theta})/(2i)$ (Eq. (2.34)); phasors (Eq. (2.35)).
- Logarithm $\log z = \log|z| + i(\operatorname{Arg} z + 2\pi k)$ (Eq. (2.36)), principal branch Eq. (2.37); powers of positive numbers $a^s = e^{s\log a}$ (Eq. (2.38)).
- With $s = \sigma + it$: $k^{-s} = k^{-\sigma}e^{-it\log k}$ (Eq. (2.39)), $|k^{-s}| = k^{-\sigma}$ (Eq. (2.40)); on the critical line p^{-s} is an arrow of length $1/\sqrt{p}$ rotating with angular frequency $\log p$ in t, period $2\pi/\log p$ (Eqs. (2.42), (2.43)).
- $\sin(mt\log p)/(mp^{m/2}) = -\operatorname{Im}[p^{-ms}]/m$ on the critical line (Eq. (2.45)); RE2 in this form is Eq. (2.46).
- Zeros of a complex function need both real and imaginary parts to vanish (Eq. (2.47)); real polynomials have conjugate-pair roots; $1 - 2^{-s} = 0$ iff $s = 2\pi ik/\log 2$.
- Arguments of products add (Eq. (2.48)); continuous arguments along a path are unique up to $2\pi k$.

2.13 Problems

1. • Let $z = 2 - 3i$ and $w = -1 + 5i$. Compute $z + w$, zw and z/w in the form $a + bi$.

2. • Simplify i^{7}, i^{10}, i^{-1} and i^{2027}.

3. • Find the modulus and principal argument of $-\sqrt{3} + i$, -4, $-2i$ and $3 - 3i$. Sketch each in the complex plane.

4. • Write $2e^{i\pi/4}$, $5e^{-i\pi/2}$ and $e^{3\pi i}$ in the form $a + bi$. Then write $-1 - i$ in the form $re^{i\theta}$ with θ the principal argument.

5. • Starting from $z = a + bi$, show that $z + \bar{z} = 2\,\mathrm{Re}\, z$, $z - \bar{z} = 2i\,\mathrm{Im}\, z$ and $z\bar{z} = |z|^2$. Use these to show that z is real if and only if $z = \bar{z}$.

6. • The second zeta zero is $\rho_2 = \frac{1}{2} + it_2$ with $t_2 = 21.022040$ (RE1). Compute $|\rho_2|$ and $\mathrm{Arg}\,\rho_2$ (in radians and degrees). What is $\bar{\rho}_2$?

7. • Evaluate $e^{i\pi}$, $e^{i\pi/2}$, $e^{2\pi i}$ and $e^{-i\pi/4}$ in the form $a + bi$.

8. • For $z = 5 + 12i$, find $\bar{z}$, $|z|$, $1/z$ in the form $a + bi$, and $|1/z|$.

9. •• Using components $z = a + bi$, $w = c + di$ (not polar form), prove that $|zw| = |z|\,|w|$ and $\overline{zw} = \bar{z}\bar{w}$. Then, writing $u = e + fi$, prove the associative law $(zw)u = z(wu)$ by expanding both sides, and check it for $z = 1 + i$, $w = 2 - i$, $u = 3i$ by computing both sides.

10. •• (a) Prove the reverse triangle inequality $|z| - |w| \le |z - w|$. (Hint: apply Eq. (2.15) to $z = (z - w) + w$.) (b) Use it to show that for $\sigma = \mathrm{Re}\, s > 0$, $|1 - 2^{-s}| \ge 1 - 2^{-\sigma} > 0$, so $1 - 2^{-s}$ has no zeros in the right half-plane. Compare with Figure 2.5.

11. •• Expand $(\cos\theta + i\sin\theta)^3$ with the binomial theorem and compare with De Moivre's formula to derive $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$.

12. •• Using Eq. (2.34), prove $\cos a\cos b = \frac{1}{2}[\cos(a-b) + \cos(a+b)]$ and $\sin a\sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$.

13. •• Use Eq. (2.34) to write $\cos^2(\omega t)$ as a constant plus a cosine, and hence show that its average over one period $2\pi/\omega$ is $\frac{1}{2}$. (This is why the rms of a sinusoid of amplitude A is $A/\sqrt{2}$.)

14. •• (a) Find all three solutions of $z^3 = 1$ by writing $z = re^{i\theta}$ and $1 = e^{2\pi ik}$. (b) Find all four solutions of $z^4 = -16$. Sketch both sets; what shape do they form?

15. •• Find the principal values of $\mathrm{Log}(-e)$, $\mathrm{Log}(2i)$, $i^i = e^{i\,\mathrm{Log}\, i}$ and $2^i = e^{i\log 2}$. Give numerical values to five decimals.

16. •• (Paper.) The six-prime oracle of the paper's Fig. 1 and Table V uses six primes; the natural reading is the first six, $2, 3, 5, 7, 11, 13$. For each, compute the radius of the circle traced by p^{-s} on the critical line and its period in t. Which prime's arrow has the largest amplitude? Which turns fastest? Compare your table with Table 2.1.

17. •• (Paper.) Show that on the critical line $\sin(mt\log p)/(m\,p^{m/2}) = -\,\mathrm{Im}[p^{-ms}]/m$. Hence rewrite the paper's Eq. (4) as $N_{\mathrm{fl}}(t) = \frac{1}{\pi}\sum_{p\in P}\sum_{m=1}^{3}\frac{1}{m}\,\mathrm{Im}\big[p^{-m(1/2+it)}\big]$.

18. •• (Paper.) For the prime-2 wave RE2, list the angular frequencies of its three terms, write each as $\log q$ for an integer q, and give each term's period in t. Explain why the sum is periodic and find its period. (The paper's abstract says that the sparse control C2 recovers

frequencies at $\log p^m$; Sec. IV says that each selected frequency lies within half the Fourier resolution of the training window of the logarithm of a prime power.)

19. •• Find all zeros of (a) $f(s) = 1 - 3^{-s}$ and (b) $g(s) = 1 + 2^{-s}$. Where do they lie, and how far apart are neighboring zeros?

20. •• For $f(z) = z^2 - 1$ with $z = x + iy$, find $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$. Sketch the curves $u = 0$ and $v = 0$ and locate their intersections. Check that these are the zeros you expect.

21. •• Prove that if a polynomial with real coefficients has a root z_0, then $\bar{z}_0$ is also a root. Apply this to $z^2 - z + (\tfrac{1}{4} + t_n^2)$ for any real t_n.

22. •• (Computational.) Using qmnt or numpy, form $\rho_n = \tfrac{1}{2} + it_n$ for the ten zeros of RE1 (`qmnt.zeros(10)`) and compute $2^{-\rho_n}$ for each. Print the modulus and the principal argument. Explain why every modulus is the same, and give the argument for $n = 1$ and $n = 2$.

23. •• (Computational.) Using numpy, plot $-\tfrac{1}{\pi}\operatorname{Arg}(1 - 2^{-1/2-it})$ and RE2 on $0 \le t \le 40$ and compute the largest difference between them. Show that each omitted term $\tfrac{1}{\pi}\sin(mt\log 2)/(m2^{m/2})$ with $m \ge 4$ has size at most $1/(\pi m 2^{m/2})$, sum these bounds numerically for $m = 4, \dots, 200$, and compare with your observed maximum.

24. •• (Paper.) In quantum mechanics a state of complex energy $\mathcal{E} = E - i\Gamma/2$ (E, Γ real) carries the time factor $e^{-i\mathcal{E}t}$, t being time. Show that $\left|e^{-i\mathcal{E}t}\right| = e^{-\Gamma t/2}$ and that the probability $\left|e^{-i\mathcal{E}t}\right|^2$ decays as $e^{-\Gamma t}$. For $E = 100$ and $\Gamma = 1$ (units with $\hbar = 1$), when has the modulus fallen to one half? The probability? Why does Sec. V of the paper call $\Gamma < 0$ the "wrong sign"?

25. •• (Paper, computational.) Show that for any zero on the critical line, $1 - \rho_n = \bar{\rho}_n$, and hence $\rho_n(1 - \rho_n) = \tfrac{1}{4} + t_n^2$ is real and positive. Using `qmnt.zeros(10)`, verify numerically that $\rho_n(1 - \rho_n) - (\tfrac{1}{4} + t_n^2)$ vanishes to rounding error for $n = 1, \dots, 10$, and report the value of $\rho_1(1 - \rho_1)$. (The symmetry $s \leftrightarrow 1 - s$ appears in the functional equation of Chapter 16.)

26. •• A particle moves on a circle of radius $R = 0.5$ m at angular frequency $\omega = 4$ rad/s, with position $z(t) = Re^{i\omega t}$ in the complex plane (meters). Find its velocity and acceleration as complex numbers, and their magnitudes. Show that the acceleration points toward the center.

27. ••• Let $E(\theta) = \cos\theta + i\sin\theta$. (a) Show that $\overline{E(\theta)}$ satisfies $\frac{\mathrm{d}}{\mathrm{d}\theta}\overline{E} = -i\overline{E}$. (b) Suppose $F(\theta)$ is any differentiable complex function with $F' = iF$ and $F(0) = 1$. By differentiating $F\overline{E}$, show that $F\overline{E} = 1$, and deduce $F = E$. This gives a second proof that the only "exponential with rate i" is $\cos\theta + i\sin\theta$.

28. ••• Use the finite geometric sum to show $\sum_{k=0}^{n-1} e^{ik\theta} = \dfrac{e^{in\theta} - 1}{e^{i\theta} - 1}$ for $e^{i\theta} \neq 1$. By factoring $e^{in\theta/2}$ from the numerator and $e^{i\theta/2}$ from the denominator, derive $\sum_{k=0}^{n-1}\cos k\theta = \dfrac{\sin(n\theta/2)\cos((n-1)\theta/2)}{\sin(\theta/2)}$. Check numerically for $n = 5$, $\theta = 0.7$. (This sum reappears in Chapter 6.)

29. ••• Let $f(t) = 1 - re^{-it}$ with $r > 0$ fixed and t running from 0 to 2π. (a) Show that $f(t)$ traces a circle of radius r centered at 1, clockwise. (b) For $r < 1$, show that the continuous argument of f returns to its starting value at $t = 2\pi$. (c) For $r > 1$, show that it changes by -2π. (d) Using numpy's `np.unwrap`, confirm (b) and (c) numerically for $r = 1/\sqrt{2}$ and $r = 2$. Relate (b) to the absence of branch jumps in $\operatorname{Arg}(1 - 2^{-1/2-it})$.

30. • • • (Paper.) Take as given (it is derived in Chapter 17) that the local mean spacing of the zeros at height t is $2\pi/\log(t/2\pi)$. How many mean spacings does one period $2\pi/\log 2$ of the prime-2 wave span at $t = 300$ and at $t = 650$? Compare with the paper's statement in Sec. V that the $p = 2$ oscillation is "only ∼6–7 levels at these heights", and explain in words why the number of levels per period grows with height.

2.14 Answers to selected problems

1. $z + w = 1 + 2i$. $zw = -2 + 10i + 3i - 15i^2 = 13 + 13i$. $z/w = \dfrac{(2-3i)(-1-5i)}{1+25} = \dfrac{-2-10i+3i+15i^2}{26} = \dfrac{-17-7i}{26} \approx -0.653846 - 0.269231\,i$.

2. $7 = 4 + 3$, so $i^7 = i^3 = -i$. $10 = 8 + 2$, so $i^{10} = -1$. $i^{-1} = -i$ (since $i(-i) = 1$). $2027 = 4 \cdot 506 + 3$, so $i^{2027} = -i$.

3. $-\sqrt{3} + i$: modulus 2, $a < 0, b > 0$, so Arg $= \arctan(-1/\sqrt{3}) + \pi = -\pi/6 + \pi = 5\pi/6$. -4: modulus 4, argument π. $-2i$: modulus 2, argument $-\pi/2$. $3 - 3i$: modulus $3\sqrt{2} \approx 4.2426$, argument $-\pi/4$.

4. $2e^{i\pi/4} = 2(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}) = \sqrt{2} + i\sqrt{2}$. $5e^{-i\pi/2} = -5i$. $e^{3\pi i} = \cos 3\pi + i \sin 3\pi = -1$. $-1 - i = \sqrt{2}e^{-3i\pi/4}$.

5. Eqs. (2.8)–(2.9) written out. If $z = \bar{z}$, then $z - \bar{z} = 2i \operatorname{Im} z = 0$, so $\operatorname{Im} z = 0$ and z is real; conversely a real z has $\bar{z} = z$.

6. $|\rho_2|^2 = 0.25 + 21.022040^2 = 0.25 + 441.926166 = 442.176166$, so $|\rho_2| \approx 21.0280$. Arg $\rho_2 = \pi/2 - \arctan(0.5/21.022040) \approx 1.570796 - 0.023780 = 1.547016$ rad $\approx 88.64°$. $\bar{\rho}_2 = \frac{1}{2} - 21.022040\,i$.

7. $e^{i\pi} = -1$; $e^{i\pi/2} = i$; $e^{2\pi i} = 1$; $e^{-i\pi/4} = \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \approx 0.70711 - 0.70711\,i$.

8. $\bar{z} = 5 - 12i$. $|z| = \sqrt{25 + 144} = \sqrt{169} = 13$. By Eq. (2.13), $1/z = \bar{z}/|z|^2 = (5-12i)/169 \approx 0.029586 - 0.071006\,i$. By Eq. (2.14), $|1/z| = 1/|z| = 1/13 \approx 0.076923$.

9. $zw = (ac-bd)+(ad+bc)i$, so $|zw|^2 = (ac-bd)^2+(ad+bc)^2 = a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2 = (a^2+b^2)(c^2+d^2)$; take square roots. For the conjugate, see the computation before Eq. (2.10). Associativity in general, with $u = e + fi$: $(zw)u$ has real part $(ac-bd)e - (ad+bc)f = ace - bde - adf - bcf$ and imaginary part $(ac-bd)f + (ad+bc)e = acf - bdf + ade + bce$. Next, $wu = (ce-df)+(cf+de)i$, so $z(wu)$ has real part $a(ce-df) - b(cf+de) = ace - adf - bcf - bde$ and imaginary part $a(cf+de) + b(ce-df) = acf + ade + bce - bdf$. The same four products appear in each part, so the two sides are equal. Numerical case: $zw = (1+i)(2-i) = 2 - i + 2i - i^2 = 3 + i$, $(zw)u = (3+i)3i = 9i + 3i^2 = -3 + 9i$; $wu = (2-i)3i = 6i + 3 = 3 + 6i$, $z(wu) = (1+i)(3+6i) = 3 + 6i + 3i + 6i^2 = -3 + 9i$. Equal.

11. Binomial expansion: $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta + 3i^2\cos\theta\sin^2\theta + i^3\sin^3\theta = (\cos^3\theta - 3\cos\theta\sin^2\theta) + i(3\cos^2\theta\sin\theta - \sin^3\theta)$. By De Moivre this equals $\cos 3\theta + i\sin 3\theta$; equate parts (Eq. (2.3)). Then use $\sin^2\theta = 1 - \cos^2\theta$: real part $\cos^3\theta - 3\cos\theta(1-\cos^2\theta) = 4\cos^3\theta - 3\cos\theta$; with $\cos^2\theta = 1 - \sin^2\theta$: imaginary part $3(1-\sin^2\theta)\sin\theta - \sin^3\theta = 3\sin\theta - 4\sin^3\theta$.

13. $\cos^2(\omega t) = (\frac{e^{i\omega t}+e^{-i\omega t}}{2})^2 = \frac{e^{2i\omega t}+2+e^{-2i\omega t}}{4} = \frac{1}{2} + \frac{1}{2}\cos(2\omega t)$. Over one period $2\pi/\omega$, $\cos(2\omega t)$ completes two full cycles and averages to zero, so the average is $\frac{1}{2}$. The rms of $A\cos\omega t$ is $\sqrt{A^2/2} = A/\sqrt{2}$.

15. $\operatorname{Log}(-e) = \log e + i\pi = 1 + i\pi \approx 1 + 3.14159\,i$. $\operatorname{Log}(2i) = \log 2 + i\pi/2 \approx 0.69315 + 1.57080\,i$. $i^i = e^{-\pi/2} \approx 0.20788$. $2^i = \cos(\log 2) + i\sin(\log 2) \approx 0.76924 + 0.63896\,i$.

17. By Eq. (2.44) with $\sigma = \frac{1}{2}$, $p^{-ms} = p^{-m/2}[\cos(mt\log p) - i\sin(mt\log p)]$, so $\operatorname{Im}[p^{-ms}] = -p^{-m/2}\sin(mt\log p)$; divide by $-m$. Substituting into Eq. (4): $-\frac{1}{\pi}\sum \frac{\sin(mt\log p)}{mp^{m/2}} = -\frac{1}{\pi}\sum(-\frac{1}{m}\operatorname{Im}[p^{-ms}]) = \frac{1}{\pi}\sum\frac{1}{m}\operatorname{Im}[p^{-ms}]$.

19. (a) $3^{-s} = 1 \iff -s\log 3 = 2\pi ik$, so $s = 2\pi ik/\log 3$: on the imaginary axis, spacing $2\pi/\log 3 \approx 5.7192$. (b) $2^{-s} = -1 = e^{i\pi(2k+1)}$ (using the lemma of Section 2.7: $e^w = -1$ iff

$w = i\pi + 2\pi ik$, from $e^{w-i\pi} = 1$), so $-s\log 2 = i\pi(2k+1)$ and $s = -i\pi(2k+1)/\log 2$: on the imaginary axis at odd multiples of $\pi/\log 2 \approx 4.5324$, spacing 9.0647, interleaved halfway between the zeros of $1 - 2^{-s}$.

21. See the conjugate-pair proof in Section 2.9. The polynomial $z^2 - z + (\frac{1}{4} + t_n^2)$ has real coefficients and root $\frac{1}{2} + it_n$ (as in the Sample Problem), so $\frac{1}{2} - it_n$ is also a root.

23. Each term has $|\sin| \le 1$, so its size is at most $1/(\pi m 2^{m/2})$. The sum of these bounds for $m \ge 4$ is about 0.0487, so the two curves can never differ by more than 0.0487; the observed maximum on a fine grid is below this bound. (Exact identity behind the figure: the argument after Figure 2.6, which uses the fact stated in the `beyond` box there.)

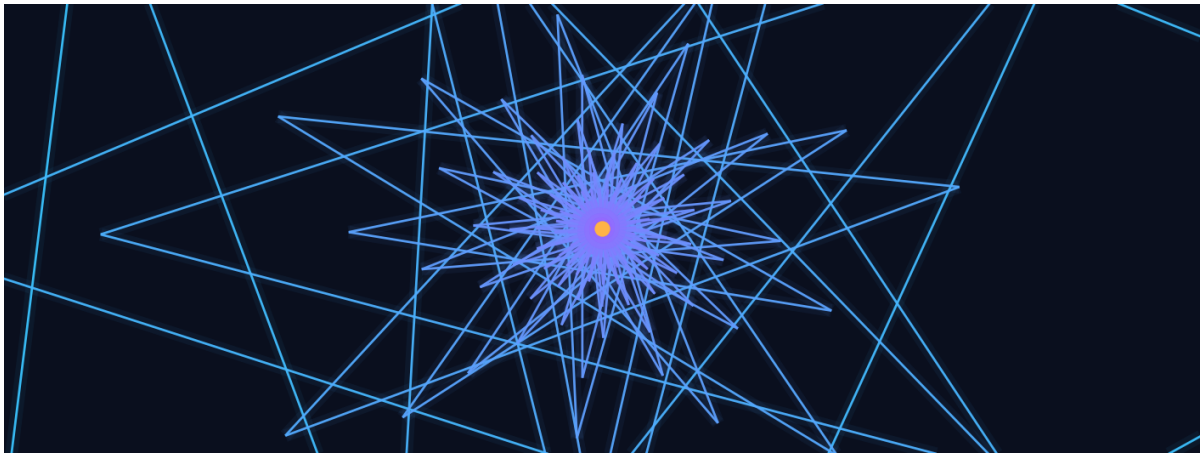
25. On the critical line $\rho_n = \frac{1}{2} + it_n$, so $1 - \rho_n = \frac{1}{2} - it_n = \bar{\rho}_n$. Hence $\rho_n(1-\rho_n) = \rho_n\bar{\rho}_n = |\rho_n|^2 = \frac{1}{4} + t_n^2$ by Eq. (2.12), which is real and positive. Numerically the difference $\rho_n(1-\rho_n) - (\frac{1}{4}+t_n^2)$ is at the level of rounding error for all ten zeros, and $\rho_1(1-\rho_1) \approx 200.040451$ (compare $|\rho_1|^2$ in the Sample Problem on ρ_1).

26. $v = \mathrm{d}z/\mathrm{d}t = i\omega R e^{i\omega t}$, magnitude $\omega R = 2$ m/s, direction 90° ahead of the position (tangent). $a = \mathrm{d}v/\mathrm{d}t = (i\omega)^2 R e^{i\omega t} = -\omega^2 z$, magnitude $\omega^2 R = 8$ m/s^2; the factor -1 means it points from z back toward the origin, the center of the circle.

27. (a) $\overline{E} = \cos\theta - i\sin\theta$, derivative $-\sin\theta - i\cos\theta = -i(\cos\theta - i\sin\theta) = -i\overline{E}$. (b) $(F\overline{E})' = F'\overline{E} + F\overline{E}' = iF\overline{E} - iF\overline{E} = 0$, so $F\overline{E}$ is constant, equal to its value $F(0)\overline{E(0)} = 1$ at $\theta = 0$. Since $E\overline{E} = |E|^2 = 1$, multiply $F\overline{E} = 1$ by E: $F = E$.

29. (a) $|f(t) - 1| = |-re^{-it}| = r$, so f is on the circle of radius r about 1; the angle of $-re^{-it} = re^{i(\pi - t)}$ is $\pi - t$, decreasing, so the motion is clockwise. (b) For $r < 1$ the circle lies in the half-plane $\mathrm{Re}\, f \ge 1 - r > 0$; the principal argument is continuous there and is a continuous argument, so it returns to $\mathrm{Arg}\, f(0)$ at $t = 2\pi$. (c) For $r > 1$ the circle encloses the origin. As $f(t)$ goes once clockwise around it, the arrow from the origin to $f(t)$ makes one full clockwise turn, so the continuous argument changes by -2π. (d) `np.unwrap(np.angle(f))` ends where it started for $r = 1/\sqrt{2}$, and 2π lower for $r = 2$. With $r = 1/\sqrt{2}$ and $t\log 2$ in place of t, this is $1 - 2^{-1/2-it}$, which is why its argument needs no unwrapping.

30. At $t = 300$: $\log(300/2\pi) \approx 3.8659$, mean spacing ≈ 1.6253, and $9.0647/1.6253 \approx 5.58$ spacings per period. At $t = 650$: $\log(650/2\pi) \approx 4.6391$, spacing ≈ 1.3544, giving ≈ 6.69. The paper's "∼6–7 levels" matches the upper part of the fitted range: the 400th zero sits at $t_{400} \approx 680$ (from `qmnt.zeros(400)`), where one period spans $(2\pi/\log 2)/(2\pi/\log(t_{400}/2\pi)) = \log(t_{400}/2\pi)/\log 2 \approx 6.8$ spacings. The period in t is fixed, while the zeros crowd together as t grows (the spacing shrinks like $1/\log t$), so more levels fit into one period.



Partial sums of the alternating series $\sum(-1)^{k+1}k^{-s}$ *at the first zero,* $s = \frac{1}{2} + 14.1347\ldots i$*, spiral into* 0 *(the gold dot).*

Chapter 3

Sequences, Series and Infinite Products

↑Contents

What this chapter is about

Drop a ball onto a hard floor. It bounces back to half its height, falls again, bounces to half of that, and so on forever. It makes infinitely many bounces, yet you can hear that it comes to rest after a few seconds. The total time is a sum of infinitely many positive pieces, and that sum is a finite number. When does adding infinitely many numbers give a finite answer? How close do you get if you stop after a few terms? And what goes wrong when the answer is not finite, or when it depends on the order in which you add?

These questions are not academic for us. The Riemann zeta function (Chapter 16) is defined by an infinite sum $\sum_k k^{-s}$. The Euler product (Chapter 16) writes the same function as an infinite product over the primes. The explicit formula (Chapter 18) expresses the fluctuations of the zeros as an infinite double sum over primes p and powers m, and nobody can add infinitely many terms on a computer, so every practical use of it is a *truncation*: a finite piece of an infinite sum. This chapter gives you the tools to decide when such a sum exists, how to bound the error of stopping early, how products turn into sums through the logarithm, and when the order of the terms matters.

Where this shows up in the paper: the arithmetic oracle of Eq. (4) of the paper [1] is a finite double sum over a prime set P and powers $m \le 3$; Table V shows what happens as more primes are kept; Eq. (5) is a sum over unperturbed states that the paper truncates at 652 states, and Sec. V A states a sum rule that rests on rearranging a finite double sum. By the end of the chapter you will be able to say precisely which of these truncations are controlled by theorems and which are justified only by numerical evidence.

What you need

- From Chapter 2: complex numbers $z = x + iy$, modulus $|z|$, conjugate $\bar{z}$, Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, complex powers $a^s = e^{s\log a}$ for real $a > 0$, and in particular $k^{-s} = k^{-\sigma}e^{-it\log k}$ with $|k^{-s}| = k^{-\sigma}$ when $s = \sigma + it$; the logarithm of a complex number and its branches (informally).
- From first-year physics and calculus: limits, derivatives, definite integrals, the fundamental theorem of calculus, substitution, the functions e^x and $\log x$ (always the natural logarithm) and their properties; free-fall kinematics ($h = \frac{1}{2}gt^2$).
- From Chapter 1: the list of zeros $t_1 = 14.134725,\ t_2 = 21.022040, \ldots$ (running example RE1) and the idea of a truncated prime sum.

3.1 Sequences and their limits

3.1.1 What a sequence is

A **sequence** is an infinite ordered list of numbers $a_1, a_2, a_3, \ldots$, one number a_n for each positive integer n. We write it (a_n) or simply a_n. The numbers may be real or complex. Some sequences you already know:

- $a_n = 1/n$: $1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4}, \ldots$
- $a_n = (-1)^n$: $-1,\ 1,\ -1,\ 1, \ldots$
- $a_n = r^n$ for a fixed number r: $r,\ r^2,\ r^3, \ldots$
- $a_n = t_n$, the heights of the zeta zeros (RE1): $14.134725,\ 21.022040,\ 25.010858, \ldots$ This sequence increases without bound. (This is a proved fact: there are infinitely many zeros and their heights are unbounded; Chapter 17 counts them.)

Sometimes it is convenient to start at $n = 0$ or $n = 2$; the starting index is part of the definition and will always be stated.

3.1.2 Convergence

Intuitively, a_n "tends to a" if the terms eventually stay as close to a as you like. Here is the precise version.

Definition (limit of a sequence). A sequence (a_n) **converges** to the number a, written $a_n \to a$ or $\lim_{n\to\infty} a_n = a$, if for every tolerance $\varepsilon > 0$ there is an index N such that

$$|a_n - a| < \varepsilon \qquad \text{for all } n > N. \tag{3.1}$$

A sequence that does not converge to any number **diverges**. If for every bound B there is an N with $a_n > B$ for all $n > N$, we say the sequence **diverges to** $+\infty$ and write $a_n \to \infty$.

Think of ε as the resolution of your measuring instrument. Convergence means: whatever resolution you choose, after some point in the list the terms are indistinguishable from a.

Sample Problem 3.1: How far out must you go?

For $a_n = 1/n^2$ the limit is $a = 0$. Find an N such that $|a_n - 0| < 10^{-3}$ for all $n > N$.
Step 1: We need $1/n^2 < 10^{-3}$, which (both sides positive) is equivalent to $n^2 > 1000$.
Step 2: Since $n > 0$, this is equivalent to $n > \sqrt{1000} = 31.62\ldots$
Step 3: So every integer $n \geq 32$ works, and we may take $N = 31$.

$$\boxed{N = 31 \text{ (all } n \geq 32 \text{ satisfy } 1/n^2 < 10^{-3})}$$

For a smaller tolerance ε the same steps give N at least $1/\sqrt{\varepsilon}$, which exists for every $\varepsilon > 0$. That is exactly what Eq. (3.1) demands.

The limit laws you used for functions in calculus hold for sequences too: if $a_n \to a$ and $b_n \to b$ then $a_n + b_n \to a + b$, $a_n b_n \to ab$, and $a_n/b_n \to a/b$ when $b \neq 0$ and every $b_n \neq 0$ (so that each quotient is defined). If f is a continuous function, $a_n \to a$ implies $f(a_n) \to f(a)$. These are proved exactly like the corresponding limit laws for functions in calculus (the proofs replace "x close to x_0" by "n large"); we use them freely.

3.1.3 The most important sequence: powers r^n

Theorem. If $|r| < 1$ (real or complex r), then $r^n \to 0$. If $|r| > 1$, then $|r^n| \to \infty$.

Proof. If $r = 0$ every term is 0. Otherwise $|r^n - 0| = |r|^n = e^{n \log |r|}$. When $|r| < 1$, $\log |r| < 0$. Given ε with $0 < \varepsilon < 1$, the condition $e^{n \log |r|} < \varepsilon$ is equivalent to $n \log |r| < \log \varepsilon$ (take logs; log is increasing), which is equivalent to

$$n > \frac{\log \varepsilon}{\log |r|} \qquad \text{(dividing by the negative number } \log |r| \text{ reverses the inequality).}$$

So $N = \log \varepsilon / \log |r|$ works. (For $\varepsilon \geq 1$ any N works, since $|r|^n < 1$.) When $|r| > 1$, $\log |r| > 0$ and $e^{n \log |r|}$ exceeds any bound B once $n > \log B / \log |r|$. □

The borderline cases behave differently: $r = 1$ gives the constant sequence 1 (converges to 1), while $r = -1$ gives $-1, 1, -1, \ldots$, which jumps forever and diverges.

3.1.4 Complex sequences

A complex sequence $z_n = x_n + iy_n$ converges to $z = x + iy$ if $|z_n - z| \to 0$, that is, if Eq. (3.1) holds with the modulus in place of the absolute value. Geometrically: the points z_n in the complex plane (Chapter 2) eventually enter and stay inside every disk centered on z.

Theorem. $z_n \to z$ if and only if $x_n \to x$ and $y_n \to y$.

Proof. For any complex number $w = u + iv$ we have $|u| \leq |w|$ and $|v| \leq |w|$ (because $|w| = \sqrt{u^2 + v^2}$), and also $|w| \leq |u| + |v|$ (square both sides: $u^2 + v^2 \leq u^2 + v^2 + 2|u||v|$). Apply these with $w = z_n - z$. If $|z_n - z| < \varepsilon$ then $|x_n - x| < \varepsilon$ and $|y_n - y| < \varepsilon$. Conversely, if $|x_n - x| < \varepsilon/2$ and $|y_n - y| < \varepsilon/2$ for $n > N$, then $|z_n - z| \leq |x_n - x| + |y_n - y| < \varepsilon$. □

Two examples set the scene for the rest of the book.

Example 1. $z_n = (i/2)^n$. Its modulus is $|i/2|^n = 2^{-n} \to 0$, so $z_n \to 0$. The points spiral inward: each step rotates by 90° (multiplication by i) and halves the distance.

Example 2. $z_n = e^{in\theta}$ with $\theta = \log 2$. Every point has $|z_n| = 1$: the sequence walks around the unit circle by a fixed angle $\log 2 \approx 0.693$ radians per step. It does not converge. To prove it, note first that in a convergent sequence consecutive terms get close: if $z_n \to z$ then

$$|z_{n+1} - z_n| \leq |z_{n+1} - z| + |z - z_n| \to 0.$$

Here, however,

$$|z_{n+1} - z_n| = \left|e^{in\theta}\right| \left|e^{i\theta} - 1\right| = \left|e^{i\theta} - 1\right| = 2\,|\sin(\theta/2)|\,,$$

a fixed positive number. The last equality comes from

$$\left|e^{i\theta} - 1\right|^2 = (\cos\theta - 1)^2 + \sin^2\theta = \cos^2\theta - 2\cos\theta + 1 + \sin^2\theta = 2 - 2\cos\theta = 4\sin^2(\theta/2),$$

where the third step uses $\cos^2\theta + \sin^2\theta = 1$ and the last step uses the half-angle identity $\cos\theta = 1 - 2\sin^2(\theta/2)$, so that $2 - 2\cos\theta = 2 \cdot 2\sin^2(\theta/2)$. Taking the square root gives

$2\,|\sin(\theta/2)|$, which is positive because $0 < \theta/2 < \pi$ for $\theta = \log 2$. So the differences do not tend to zero, and the sequence diverges. The phase $e^{-it\log p}$ that rotates at angular frequency $\log p$ in t (Chapter 2) is exactly this kind of object, and it never settles down.

3.1.5 Monotone sequences

Many of our proofs need a way to show that a limit exists without knowing its value in advance. The tool is the following property of the real numbers.

Taken on trust

Monotone convergence (taken on trust). If a real sequence is increasing ($a_{n+1} \geq a_n$ for all n) and bounded above ($a_n \leq B$ for all n), then it converges, and its limit is at most B. Likewise a decreasing sequence that is bounded below converges.
Status: this is equivalent to the completeness of the real number line (the line has no "holes"). Proving it requires a construction of the real numbers, which is part of a first course in real analysis and is beyond this book. We use it as a basic fact about real numbers.

Checkpoint 3.1

Does $a_n = \sin(n)/n$ converge? If so, to what?

Answer: Yes, to 0. Since $|\sin n| \leq 1$, $|a_n| \leq 1/n$, and $1/n < \varepsilon$ for all $n > 1/\varepsilon$. The wild oscillation of $\sin n$ does not matter because its amplitude is squeezed by $1/n$.

Key Ideas

- A sequence converges to a if, for every tolerance ε, all terms beyond some index lie within ε of a (Eq. (3.1)).
- $r^n \to 0$ when $|r| < 1$; $|r^n| \to \infty$ when $|r| > 1$.
- A complex sequence converges exactly when its real and imaginary parts both converge.
- $e^{in\theta}$ circles forever and does not converge (unless $e^{i\theta} = 1$).
- Increasing and bounded above implies convergent (monotone convergence, taken on trust).

3.2 Series, partial sums and the geometric series

★ Did you know? *Archimedes Sums Forever*

In the third century BC, Archimedes found the area of a parabolic segment by filling it with triangles. Each new layer of triangles had one quarter the area of the layer before, so he needed the sum $1 + \frac{1}{4} + \frac{1}{16} + \cdots$. He had no modern theory of limits, but a careful argument showed him that the total is exactly $\frac{4}{3}$. It is one of the earliest infinite geometric series ever summed.

3.2.1 Infinite sums defined through partial sums

An expression $\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \cdots$ is called an **infinite series**. You cannot literally perform infinitely many additions, so we define its value through finite sums. The n-th **partial**

sum is

$$S_n = \sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n.$$

Definition. The series **converges** to S if the sequence of partial sums converges, $S_n \to S$. Then we write $\sum_{k=1}^{\infty} a_k = S$ and call S the **sum**. Otherwise the series **diverges**.

The difference $S - S_n = \sum_{k=n+1}^{\infty} a_k$ is the **tail** after n terms. It is the error you make by **truncating** the series at n terms. Much of this chapter is about bounding tails.

Theorem (term test). If $\sum a_k$ converges, then $a_k \to 0$.

Proof. For $n \geq 2$, $a_n = S_n - S_{n-1}$. Both S_n and S_{n-1} tend to S, so by the limit laws $a_n \to S - S = 0$. □

Pitfall

The converse is false. Terms tending to zero do *not* guarantee convergence. The harmonic series $\sum 1/k$ has terms tending to zero and diverges (Section 3.4). The term test can only prove divergence: if $a_k \not\to 0$, the series diverges.

3.2.2 The geometric series

The **geometric series** with ratio r (real or complex) is $\sum_{k=0}^{\infty} r^k = 1 + r + r^2 + \cdots$. Its partial sums can be computed exactly. Let

$$S_n = \sum_{k=0}^{n-1} r^k = 1 + r + r^2 + \cdots + r^{n-1} \qquad (n \text{ terms}).$$

Multiply by r:

$$rS_n = r + r^2 + \cdots + r^{n-1} + r^n.$$

Subtract the second line from the first. Every power $r^1, \ldots, r^{n-1}$ appears in both lines and cancels, leaving

$$S_n - rS_n = 1 - r^n.$$

For $r \neq 1$ divide by $1 - r$:

$$S_n = \sum_{k=0}^{n-1} r^k = \frac{1 - r^n}{1 - r} \qquad (r \neq 1). \tag{3.2}$$

Nothing in this derivation used that r is real, so Eq. (3.2) holds for complex r too.

Now take $n \to \infty$. If $|r| < 1$, then $r^n \to 0$ (Section 3.1), so

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r} \qquad (|r| < 1), \tag{3.3}$$

and the tail after n terms is

$$\frac{1}{1 - r} - S_n = \frac{1}{1 - r} - \frac{1 - r^n}{1 - r} = \frac{r^n}{1 - r}. \tag{3.4}$$

If $|r| \geq 1$, then $\left|r^k\right| = |r|^k \geq 1$ for every k, the terms do not tend to zero, and by the term test the series diverges. (For $r = 1$ the partial sums are $S_n = n \to \infty$.)

For example, with $r = \frac{1}{2}$: $S_5 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = 1.9375$, and Eq. (3.2) gives $(1 - 2^{-5})/(1/2) = 2 - 2^{-4} = 1.9375$. The sum is 2.

The geometric series has a long history. In the third century BC Archimedes found the area of a parabolic segment by filling it with layers of triangles. Each layer had one quarter the area

of the layer before, so the total area is a multiple of $1+\frac{1}{4}+\frac{1}{16}+\cdots$. By Eq. (3.3) this sum equals $1/(1-\frac{1}{4})=\frac{4}{3}$. Archimedes reached this value by a careful exhaustion argument, without a modern theory of limits.

A complex example: with $r=i/2$, Eq. (3.3) gives

$$\sum_{k=0}^{\infty}\left(\frac{i}{2}\right)^{k}=\frac{1}{1-i/2}=\frac{1+i/2}{(1-i/2)(1+i/2)}=\frac{1+i/2}{1+1/4}=0.8+0.4\,i,$$

where we multiplied numerator and denominator by the conjugate $1+i/2$ (Chapter 2).

Sample Problem 3.2: A geometric series with ratio $2^{-1/2}$

In the explicit formula the terms belonging to the prime $p=2$ have amplitudes proportional to $2^{-m/2}=(2^{-1/2})^{m}$ (running example RE2). Find $\sum_{k=0}^{\infty}(2^{-1/2})^{k}$ exactly, and find the smallest number of terms n for which the tail is below 0.01.

Step 1: $r=2^{-1/2}=1/\sqrt{2}=0.707107$, and $|r|<1$, so Eq. (3.3) applies: the sum is $1/(1-1/\sqrt{2})$.

Step 2: Simplify. Multiply numerator and denominator by $\sqrt{2}$: $1/(1-1/\sqrt{2})=\sqrt{2}/(\sqrt{2}-1)$. Multiply numerator and denominator by $\sqrt{2}+1$: $\sqrt{2}(\sqrt{2}+1)/((\sqrt{2}-1)(\sqrt{2}+1))=(2+\sqrt{2})/(2-1)=2+\sqrt{2}=3.414214$.

Step 3: By Eq. (3.4) the tail after n terms is $r^{n}/(1-r)$, with $1-r=0.292893$. We need $r^{n}<0.01\times 0.292893=0.00292893$.

Step 4: Take logs: $n\log r<\log 0.00292893$, that is $n\times(-0.346574)<-5.83317$, so $n>5.83317/0.346574=16.83$.

Step 5: Check the two integers around the threshold: $n=16$ gives tail $2^{-8}/0.292893=0.0133$ (too big); $n=17$ gives $2^{-8.5}/0.292893=0.0094$ (small enough).

$$\boxed{\sum_{k=0}^{\infty}2^{-k/2}=2+\sqrt{2}=3.414214,\qquad n=17\text{ terms}}$$

Sample Problem 3.3: Infinitely many bounces in finite time

A ball is dropped from height $h=1.00$ m. After each bounce it rises to a fraction $r=0.5$ of its previous maximum height. Ignore air resistance and the duration of each contact, and take $g=9.8\ \mathrm{m/s^{2}}$. Find the total distance traveled and the total time until it comes to rest.

Step 1 (distance): The first fall is h. After bounce k ($k=1,2,\dots$) the ball rises to $r^{k}h$ and falls back the same distance, covering $2r^{k}h$. Total distance

$$D=h+2h\sum_{k=1}^{\infty}r^{k}=h+2h\,\frac{r}{1-r},$$

using $\sum_{k=1}^{\infty}r^{k}=r\sum_{k=0}^{\infty}r^{k}=r/(1-r)$. With $h=1$ m, $r=0.5$: $D=1+2\times 0.5/0.5=3$ m.

Step 2 (time of one fall): From $h=\frac{1}{2}gt^{2}$ (first-year kinematics), falling a height H takes $\sqrt{2H/g}$. Rising to H takes the same time by symmetry of free flight.

Step 3 (sum of times): The first fall takes $\tau=\sqrt{2h/g}$. The up-and-down flight after bounce k takes $2\sqrt{2r^{k}h/g}=2\tau(\sqrt{r})^{k}$. The total time is

$$T_{\mathrm{tot}}=\tau+2\tau\sum_{k=1}^{\infty}(\sqrt{r})^{k}=\tau\left(1+\frac{2\sqrt{r}}{1-\sqrt{r}}\right),$$

a geometric series with ratio $\sqrt{r}$, which is less than 1.

Step 4 (numbers): $\tau = \sqrt{2/9.8}$ s $= 0.451754$ s; $\sqrt{r} = 0.707107$ (the same ratio as in the previous Sample Problem); $1 + 2(0.707107)/0.292893 = 5.828427$. So $T_{\text{tot}} = 0.451754 \times 5.828427$ s.

$$\boxed{D = 3.00 \text{ m}, \qquad T_{\text{tot}} = 2.633 \text{ s}}$$

Infinitely many bounces fit into 2.633 s because the flight times shrink geometrically.

Figure 3.1 previews four series we meet in this chapter: the geometric series of this section, $\sum 1/k^2$ (Section 3.5), the harmonic series (Section 3.4) and the alternating harmonic series (Section 3.7). Return to it as each one is introduced.

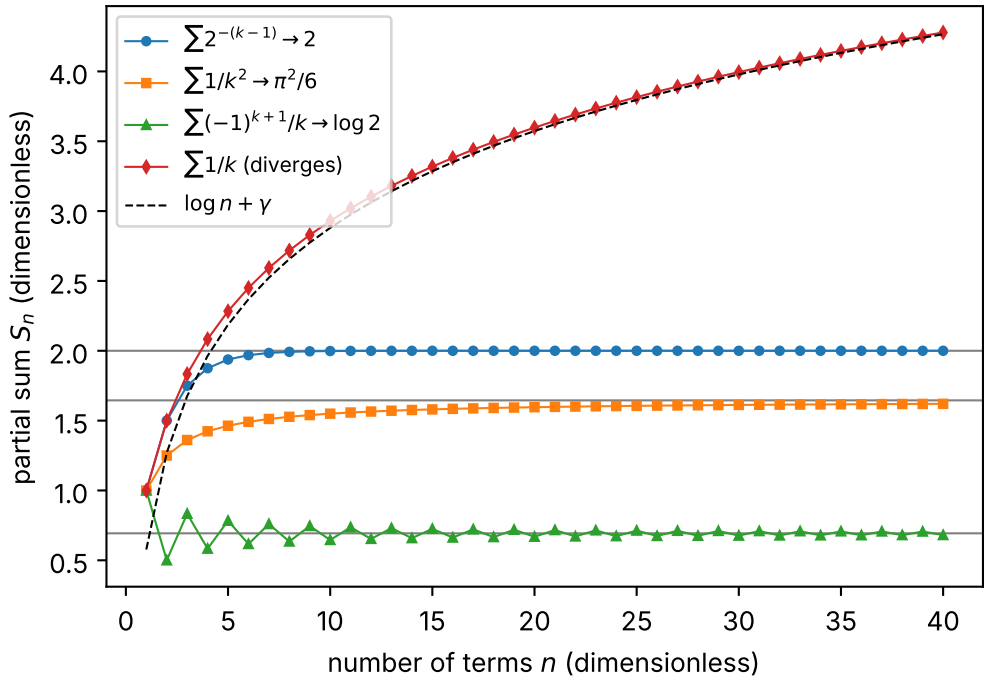


Figure 3.1: Partial sums S_n of four series for $n = 1$ to 40. The geometric series $\sum_{k\geq 1} 2^{-(k-1)}$ approaches 2 very quickly; $\sum 1/k^2$ approaches $\pi^2/6 = 1.644934$ slowly (tail about $1/n$); the alternating harmonic series zigzags around $\log 2 = 0.693147$; the harmonic series $\sum 1/k$ grows without bound, following $\log n + \gamma$ with $\gamma = 0.577216$. Horizontal gray lines mark the limits.

```
import numpy as np
# partial sums of the geometric series with ratio r
r = 2**-0.5
k = np.arange(0, 30)
S = np.cumsum(r**k)            # S[n-1] = sum of first n terms
print(S[16], 1/(1 - r))        # 17 terms vs exact sum 2+sqrt(2)
print(1/(1 - r) - S[16])       # tail after 17 terms, below 0.01
```

Checkpoint 3.2

For which complex numbers r does $\sum_{k=0}^{\infty} r^k$ converge? Does it converge for $r = e^{i\theta}/2$?

Answer: Exactly for $|r| < 1$. For $r = e^{i\theta}/2$, $|r| = \frac{1}{2}$, so it converges, to $1/(1 - e^{i\theta}/2)$, for every real θ.

Key Ideas

- A series converges when its sequence of partial sums S_n converges; the tail $S - S_n$ is the truncation error.
- If a series converges, its terms tend to zero; the converse is false.
- $\sum_{k=0}^{n-1} r^k = (1 - r^n)/(1 - r)$ exactly (Eq. (3.2)); for $|r| < 1$ the sum is $1/(1 - r)$ and the tail is $r^n/(1 - r)$ (Eqs. (3.3), (3.4)). This holds for complex r.

3.3 Telescoping sums

Some sums collapse like a pocket telescope. Suppose each term is a difference of consecutive members of another sequence, $a_k = b_k - b_{k+1}$. Then

$$\sum_{k=1}^{n} (b_k - b_{k+1}) = (b_1 - b_2) + (b_2 - b_3) + \cdots + (b_n - b_{n+1}) = b_1 - b_{n+1}, \tag{3.5}$$

because each of $b_2, \ldots, b_n$ appears once with a plus sign and once with a minus sign. Therefore the series $\sum_{k=1}^{\infty}(b_k - b_{k+1})$ converges if and only if b_n converges, and then its sum is $b_1 - \lim_{n\to\infty} b_n$.

Example 1. Since

$$\frac{1}{k} - \frac{1}{k+1} = \frac{(k+1) - k}{k(k+1)} = \frac{1}{k(k+1)},$$

Eq. (3.5) with $b_k = 1/k$ gives $\sum_{k=1}^{n} \frac{1}{k(k+1)} = 1 - \frac{1}{n+1} \to 1$. So $\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = 1$.

Example 2. With $b_k = -\log k$ we have $b_k - b_{k+1} = \log(k+1) - \log k = \log \frac{k+1}{k} = \log\left(1 + \frac{1}{k}\right)$. Hence

$$\sum_{k=1}^{n} \log\left(1 + \frac{1}{k}\right) = -\log 1 + \log(n+1) = \log(n+1) \to \infty.$$

The terms $\log(1 + 1/k)$ tend to 0, yet the series diverges: a second warning that the term test is not a convergence test.

Sample Problem 3.4: A telescoping sum with a gap

Find $\sum_{k=2}^{\infty} \frac{1}{k^2 - 1}$.

Step 1 (partial fractions): $k^2 - 1 = (k - 1)(k + 1)$, and

$$\frac{1}{k-1} - \frac{1}{k+1} = \frac{(k+1) - (k-1)}{(k-1)(k+1)} = \frac{2}{k^2 - 1},$$

so $\frac{1}{k^2 - 1} = \frac{1}{2}\left(\frac{1}{k-1} - \frac{1}{k+1}\right)$.

Step 2 (write out the partial sum):

$$\sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k+1}\right) = \left[\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n-1}\right] - \left[\frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n+1}\right].$$

Step 3 (cancel): The terms $1/3, \ldots, 1/(n-1)$ appear in both brackets. What survives is $1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}$.

Step 4 (limit): So $\sum_{k=2}^{n} \frac{1}{k^2-1} = \frac{1}{2}\left(\frac{3}{2} - \frac{1}{n} - \frac{1}{n+1}\right) \to \frac{3}{4}$. For instance at $n = 10$ the partial sum is $\frac{1}{2}(1.5 - 0.1 - 0.090909) = 0.654545$, still far from the limit.

$$\boxed{\sum_{k=2}^{\infty} \frac{1}{k^2-1} = \frac{3}{4}}$$

Checkpoint 3.3

The terms of $\sum_{k=1}^{\infty} \log(1+1/k)$ tend to zero. Does the series converge?

Answer: No. It telescopes to $\log(n+1)$, which grows without bound.

Key Ideas

- $\sum_{k=1}^{n}(b_k - b_{k+1}) = b_1 - b_{n+1}$ (Eq. (3.5)); the series converges exactly when b_n does.
- Partial fractions turn many rational sums into telescoping ones.

3.4 The harmonic series diverges

The **harmonic series** is $\sum_{k=1}^{\infty} \frac{1}{k}$, and its partial sums are the **harmonic numbers** $H_n = \sum_{k=1}^{n} \frac{1}{k}$. The name comes from music: a string fixed at both ends and vibrating in its k-th harmonic has a wavelength equal to $1/k$ of the fundamental wavelength (standing waves, HRW [7]), so the relative wavelengths $1, \frac{1}{2}, \frac{1}{3}, \dots$ are the terms of the series.

3.4.1 Oresme's proof

Theorem. The harmonic series diverges: $H_n \to \infty$.

Proof. Group the terms of H_{2^K} into blocks:

$$H_{2^K} = 1 + \underbrace{\tfrac{1}{2}}_{\text{block } 1} + \underbrace{\left(\tfrac{1}{3}+\tfrac{1}{4}\right)}_{\text{block } 2} + \underbrace{\left(\tfrac{1}{5}+\cdots+\tfrac{1}{8}\right)}_{\text{block } 3} + \cdots + \underbrace{\left(\tfrac{1}{2^{K-1}+1}+\cdots+\tfrac{1}{2^K}\right)}_{\text{block } K}.$$

Block j runs from $1/(2^{j-1}+1)$ to $1/2^j$, so it contains $2^j - 2^{j-1} = 2^{j-1}$ terms, and each of them is at least the smallest one, $1/2^j$. Hence block j is at least $2^{j-1} \times 1/2^j = \frac{1}{2}$. Adding the K blocks and the leading 1,

$$H_{2^K} \geq 1 + \frac{K}{2}. \tag{3.6}$$

Given any bound B, choose an integer $K > 2(B-1)$. Then $H_{2^K} > B$, and since H_n increases with n (all terms are positive), $H_n > B$ for all $n \geq 2^K$. So $H_n \to \infty$. $\square$

For example $H_8 = 1 + \frac{1}{2} + \cdots + \frac{1}{8} = 2.717857$, while Eq. (3.6) with $K = 3$ guarantees only $H_8 \geq 2.5$.

3.4.2 How fast does H_n grow?

Compare the sum with an integral. On the interval $k \leq x \leq k+1$ we have $1/x \leq 1/k$, so

$$\int_{k}^{k+1} \frac{\mathrm{d}x}{x} \leq \frac{1}{k}.$$

Summing over $k = 1, \dots, n$, the integrals join into one: $\int_1^{n+1} \mathrm{d}x/x = \log(n+1) \leq H_n$. Similarly, on $k-1 \leq x \leq k$ we have $1/x \geq 1/k$, so $1/k \leq \int_{k-1}^{k} \mathrm{d}x/x$; summing over $k = 2, \dots, n$ gives $H_n - 1 \leq \int_1^n \mathrm{d}x/x = \log n$. Together,

$$\log(n+1) \leq H_n \leq 1 + \log n. \tag{3.7}$$

So H_n grows like $\log n$: without bound, but extremely slowly. At $n = 10$: $\log 11 = 2.397895 \le H_{10} = 2.928968 \le 1 + \log 10 = 3.302585$.

3.4.3 The Euler–Mascheroni constant

Let $g_n = H_n - \log n$. Two facts:

(i) g_n decreases. Compute

$$g_{n+1} - g_n = \frac{1}{n+1} - [\log(n+1) - \log n] = \frac{1}{n+1} - \int_n^{n+1} \frac{\mathrm{d}x}{x}.$$

On $n \le x < n+1$ we have $1/x > 1/(n+1)$, so the integral exceeds $\int_n^{n+1} \frac{\mathrm{d}x}{n+1} = \frac{1}{n+1}$. Hence $g_{n+1} - g_n < 0$.

(ii) g_n is bounded below by 0. By Eq. (3.7), $g_n \ge \log(n+1) - \log n > 0$.

A decreasing sequence bounded below converges (monotone convergence, Section 3.1). Its limit is the **Euler–Mascheroni constant**

$$\gamma = \lim_{n\to\infty} (H_n - \log n) = 0.5772156649\ldots \tag{3.8}$$

(The digits are a standard constant; our proof shows only that $0 \le \gamma \le g_1 = 1$.) At $n = 10$, $g_{10} = 2.928968 - 2.302585 = 0.626383$, still decreasing toward γ.

Taken on trust

A sharper form of H_n (proved; proof beyond this book). The Euler–Maclaurin summation formula, a refined comparison of sums with integrals, gives

$$H_n = \log n + \gamma + \frac{1}{2n} + \mathcal{O}(n^{-2}),$$

where $\mathcal{O}(n^{-2})$ denotes a remainder whose size is at most a constant times n^{-2} for all large n (big-O notation, treated in Chapter 4). We do not prove it here; see Riley, Hobson and Bence [22] or Apostol [23]. We use it only to estimate numbers, never inside a proof.

Sample Problem 3.5: When does H_n first exceed 10?

Estimate the smallest n with $H_n > 10$, first from Oresme's bound and then from γ.

Step 1 (Oresme): Eq. (3.6) gives $H_{2^K} \ge 1 + K/2$. Setting $1 + K/2 = 10$ gives $K = 18$. The inequality is in fact strict for $K \ge 2$, because block 2 is $\frac{1}{3} + \frac{1}{4} > \frac{1}{2}$. So $H_{2^{18}} > 10$, and $n = 2^{18} = 262144$ certainly works. This is a guarantee, not an estimate.

Step 2 (with γ): Using $H_n \approx \log n + \gamma$, solve $\log n + \gamma = 10$: $n \approx e^{10-\gamma} = e^{9.4227843} \approx 12367$.

Step 3 (exact check): Summing directly on a computer (a short numpy loop of the kind used in Problem 19) gives $H_{12366} < 10 < H_{12367}$.

$$\boxed{n = 12367}$$

Oresme's bound is about 21 times too pessimistic, but it is a proof; the γ estimate is accurate because, by the sharper form quoted in the box above, the error of $H_n \approx \log n + \gamma$ is about $1/(2n)$, only 4×10^{-5} here.

The same estimate shows how slowly the harmonic series diverges. To pass 100 you need about $e^{100-\gamma} \approx 1.5 \times 10^{43}$ terms, far more than any computer could add. Oresme's grouping argument, made in the fourteenth century, proves divergence. Direct summation, by contrast,

never gets far enough to show it. A truncated sum can look convergent long before it reveals its true nature.

Key Ideas

- The harmonic series diverges (Oresme, Eq. (3.6)), although its terms tend to zero.
- $\log(n+1) \le H_n \le 1 + \log n$ (Eq. (3.7)); $H_n - \log n \to \gamma = 0.5772\ldots$ (Eq. (3.8)).
- Comparing a sum with an integral, rectangle by rectangle, gives two-sided bounds. This idea is the integral test of the next section.

3.5 Series of positive terms: comparison and the integral test

When all terms are nonnegative, the partial sums can only increase. Monotone convergence then says: *a series of nonnegative terms converges if and only if its partial sums are bounded.* (If they are bounded, they converge; if not, they exceed every bound and the series diverges to $+\infty$.) This single fact powers the two tests below.

3.5.1 The comparison test

Theorem (comparison). Suppose $0 \le a_k \le b_k$ for all k. If $\sum b_k$ converges, then $\sum a_k$ converges and $\sum a_k \le \sum b_k$. If $\sum a_k$ diverges, then $\sum b_k$ diverges.

Proof. Let $B = \sum_{k=1}^{\infty} b_k$. Since the b_k are nonnegative, every partial sum of the b's is at most B. Then $\sum_{k=1}^{n} a_k \le \sum_{k=1}^{n} b_k \le B$, so the partial sums of $\sum a_k$ are bounded, hence converge, and the limit is at most B. The second statement is the first one read backwards: if $\sum b_k$ converged, $\sum a_k$ would too. $\square$

Finitely many terms never affect convergence (they change every partial sum beyond some point by the same fixed amount), so it is enough that $0 \le a_k \le b_k$ for all k beyond some K.

Example. For $k \ge 2$, $k^2 \ge k(k-1)$, so $1/k^2 \le 1/(k(k-1))$. The comparison series $\sum_{k=2}^{\infty} \frac{1}{k(k-1)} = \sum_{j=1}^{\infty} \frac{1}{j(j+1)} = 1$ (substitute $j = k-1$ and use Section 3.3). Therefore $\sum_{k=1}^{\infty} \frac{1}{k^2} \le 1 + 1 = 2$: the series converges.

Example. $1/\sqrt{k} \ge 1/k$ for $k \ge 1$, and $\sum 1/k$ diverges, so $\sum 1/\sqrt{k}$ diverges.

3.5.2 The integral test

We write $\int_a^{\infty} f(x)\,\mathrm{d}x$ for $\lim_{K\to\infty} \int_a^{K} f(x)\,\mathrm{d}x$ when that limit exists (an **improper integral**).

Theorem (integral test). Let f be continuous, positive and decreasing on $[1, \infty)$, and let $a_k = f(k)$. Then for every $n \ge 1$

$$\int_1^{n+1} f(x)\,\mathrm{d}x \;\le\; \sum_{k=1}^{n} f(k) \;\le\; f(1) + \int_1^{n} f(x)\,\mathrm{d}x, \tag{3.9}$$

and the series $\sum_{k=1}^{\infty} f(k)$ converges if and only if the integrals $\int_1^n f(x)\,\mathrm{d}x$ stay bounded as $n \to \infty$. Moreover, when the series converges, its tail after n terms satisfies

$$\int_{n+1}^{\infty} f(x)\,\mathrm{d}x \;\le\; \sum_{k=n+1}^{\infty} f(k) \;\le\; \int_n^{\infty} f(x)\,\mathrm{d}x. \tag{3.10}$$

Proof. Because f decreases, on the interval $[k, k+1]$ we have $f(x) \le f(k)$, so $\int_k^{k+1} f\,\mathrm{d}x \le f(k)$: the rectangle of height $f(k)$ and width 1 lies above the curve (Figure 3.2, left). Summing $k = 1, \ldots, n$ gives the left inequality of Eq. (3.9). On $[k-1, k]$ we have $f(x) \ge f(k)$, so

$f(k) \le \int_{k-1}^{k} f\,\mathrm{d}x$: the rectangle lies below the curve (Figure 3.2, right). Summing $k = 2, \dots, n$ gives $\sum_{k=2}^{n} f(k) \le \int_1^n f\,\mathrm{d}x$; adding $f(1)$ gives the right inequality.

If $\int_1^n f \le I$ for all n, the right inequality bounds the partial sums by $f(1) + I$, so the series converges. If the integrals are unbounded (they increase with n because $f > 0$), the left inequality shows the partial sums are unbounded, so the series diverges.

For the tail, sum the same rectangle inequalities over $k = n+1, \dots, K$ and let $K \to \infty$: $\int_k^{k+1} f \le f(k)$ summed gives $\int_{n+1}^{K+1} f \le \sum_{n+1}^{K} f(k)$, and $f(k) \le \int_{k-1}^{k} f$ summed gives $\sum_{n+1}^{K} f(k) \le \int_n^K f$. Taking $K \to \infty$ in both gives Eq. (3.10). The improper integrals in it exist. For positive f the integrals $\int_a^K f\,\mathrm{d}x$ increase with K. When the series converges with sum S, they are bounded, because the left inequality of Eq. (3.9) gives $\int_1^{K+1} f\,\mathrm{d}x \le S$. Monotone convergence then applies. □

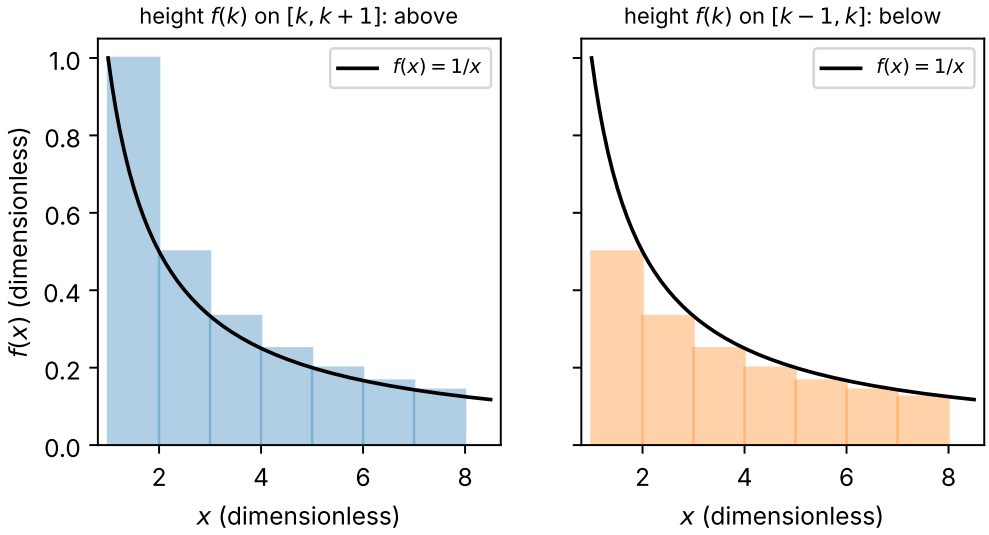


Figure 3.2: The integral test for $f(x) = 1/x$. Left: rectangles of height $f(k)$ on $[k, k+1]$ lie above the curve, so $\sum_{k=1}^{7} f(k) \ge \int_1^8 f\,\mathrm{d}x$. Right: rectangles of height $f(k)$ on $[k-1, k]$ lie below the curve, so $\sum_{k=2}^{8} f(k) \le \int_1^8 f\,\mathrm{d}x$. Squeezing the sum between two integrals gives Eqs. (3.9) and (3.10).

3.5.3 The p-series $\sum k^{-s}$ for real s

Series of the form $\sum k^{-s}$ are traditionally called **p-series**, because the exponent is often written p. We write s instead and keep p for primes.

Theorem. For real s, the series $\sum_{k=1}^{\infty} k^{-s}$ converges if and only if $s > 1$.

Proof. If $s \le 0$, then $k^{-s} = k^{|s|} \ge 1$ for all k, the terms do not tend to zero, and the series diverges. If $s > 0$, $f(x) = x^{-s}$ is positive, continuous and decreasing on $[1, \infty)$ (its derivative $-sx^{-s-1}$ is negative), so the integral test applies. For $s \ne 1$,

$$\int_1^n x^{-s}\,\mathrm{d}x = \left[\frac{x^{1-s}}{1-s}\right]_1^n = \frac{n^{1-s} - 1}{1-s},$$

and for $s = 1$ the integral is $\log n$.

- If $s > 1$, then $1 - s < 0$, $n^{1-s} \to 0$, and the integral tends to $\frac{-1}{1-s} = \frac{1}{s-1}$: bounded. The series converges.
- If $0 < s < 1$, then $1 - s > 0$, $n^{1-s} \to \infty$: unbounded. The series diverges.

- If $s = 1$, $\log n \to \infty$: the harmonic series diverges (as already shown).

□

For $s > 1$ the sum is denoted

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s} \qquad (s > 1), \tag{3.11}$$

the Riemann zeta function on the real axis; Chapter 16 studies it in depth. Eq. (3.10) gives the tail after n terms:

$$\frac{(n+1)^{1-s}}{s-1} \le \zeta(s) - \sum_{k=1}^{n} k^{-s} \le \frac{n^{1-s}}{s-1}.$$

Sample Problem 3.6: Bracketing $\zeta(2)$

Compute $S_{10} = \sum_{k=1}^{10} 1/k^2$ and use Eq. (3.10) to bracket $\zeta(2)$. Compare with the known value $\zeta(2) = \pi^2/6$ (Euler's result; it follows from Parseval's theorem applied to a suitable Fourier series, the tool of Chapter 5, and here we take it as a known value).
Step 1: $S_{10} = 1 + 0.25 + 0.111111 + 0.0625 + 0.04 + 0.027778 + 0.020408 + 0.015625 + 0.012346 + 0.01 = 1.549768$.
Step 2: With $f(x) = x^{-2}$, $\int_a^\infty x^{-2}\,\mathrm{d}x = 1/a$. So the tail lies between $1/11 = 0.090909$ and $1/10 = 0.1$.
Step 3: Therefore $1.549768 + 0.090909 \le \zeta(2) \le 1.549768 + 0.1$.

$$\boxed{1.640677 \le \zeta(2) \le 1.649768}$$

The true value $\pi^2/6 = 1.644934$ lies inside. The midpoint of the bracket, 1.645222, is off by only 0.00029, much better than S_{10} itself (off by 0.095). Averaging works so well because the two integral bounds err in opposite directions: the lower bound misses the true tail by the small curved slivers between the rectangles and the curve on one side, the upper bound overshoots by similar slivers on the other side, and for a slowly varying f these nearly cancel in the average.

Sample Problem 3.7: How fast does $\sum k^{-1/2}$ diverge?

On the critical line $\mathrm{Re}\, s = \frac{1}{2}$ (Chapter 16) the terms of $\sum k^{-s}$ have modulus $k^{-1/2}$. Bound $\sum_{k=1}^{1000} k^{-1/2}$.
Step 1: $f(x) = x^{-1/2}$ is positive and decreasing, with $\int_a^b x^{-1/2}\,\mathrm{d}x = 2(\sqrt{b} - \sqrt{a})$.
Step 2: Eq. (3.9) with $n = 1000$: lower bound $2(\sqrt{1001} - 1) = 2(31.638584 - 1) = 61.277168$; upper bound $1 + 2(\sqrt{1000} - 1) = 1 + 2(30.622777) = 62.245553$.

$$\boxed{61.277 \le \sum_{k=1}^{1000} k^{-1/2} \le 62.246}$$

(Direct summation gives 61.801.) The sum grows like $2\sqrt{n}$: by $n = 10^6$ it is near 2000. A sum of moduli this large tells you that cancellation between terms, not the size of the terms, would have to do all the work on the critical line.

Checkpoint 3.4

Does $\sum_{k=1}^{\infty} k^{-1.5}$ converge? Bound the tail after 100 terms.

Answer: Yes, since $1.5 > 1$. By Eq. (3.10) the tail is at most $\int_{100}^{\infty} x^{-1.5}\,\mathrm{d}x = 2/\sqrt{100} = 0.2$. Convergence is slow: the tail decays only like $2/\sqrt{n}$.

Key Ideas

- A series of nonnegative terms converges if and only if its partial sums are bounded.
- Comparison: smaller than a convergent series converges; larger than a divergent series diverges.
- Integral test: for positive decreasing f, $\sum f(k)$ and $\int f$ converge or diverge together, and integrals bracket the tail (Eqs. (3.9), (3.10)).
- $\sum k^{-s}$ converges exactly for $s > 1$; the tail after n terms is about $n^{1-s}/(s-1)$.

3.6 Absolute convergence and complex series

3.6.1 Definition and the main theorem

A series $\sum a_k$ (real or complex terms) **converges absolutely** if the series of moduli $\sum |a_k|$ converges. A series that converges but not absolutely is said to **converge conditionally**.

We need the triangle inequality for complex numbers, $|z+w| \le |z| + |w|$. Here is a short proof. Using $|u|^2 = u\bar{u}$,

$$\begin{aligned} |z+w|^2 &= (z+w)(\bar{z}+\bar{w}) = |z|^2 + |w|^2 + z\bar{w} + \bar{z}w \\ &= |z|^2 + |w|^2 + 2\,\mathrm{Re}(z\bar{w}) \qquad (\text{since } \bar{z}w = \overline{z\bar{w}}) \\ &\le |z|^2 + |w|^2 + 2\,|z\bar{w}| \qquad (\mathrm{Re}\,u \le |u|) \\ &= (|z| + |w|)^2 \qquad (|z\bar{w}| = |z|\,|w|). \end{aligned}$$

Taking square roots gives the inequality. By induction it extends to $\left|\sum_{k=1}^{n} a_k\right| \le \sum_{k=1}^{n} |a_k|$.

Theorem. An absolutely convergent series converges, and

$$\left|\sum_{k=1}^{\infty} a_k\right| \le \sum_{k=1}^{\infty} |a_k|\,. \tag{3.12}$$

Proof for real terms. Let $b_k = a_k + |a_k|$. Then $0 \le b_k \le 2\,|a_k|$ (if $a_k \ge 0$, $b_k = 2a_k$; if $a_k < 0$, $b_k = 0$). Since $\sum 2\,|a_k|$ converges, so does $\sum b_k$ by comparison. Now $a_k = b_k - |a_k|$, so the partial sums of $\sum a_k$ are differences of the partial sums of two convergent series, and therefore converge.

Proof for complex terms. Write $a_k = x_k + iy_k$. Since $|x_k| \le |a_k|$ and $|y_k| \le |a_k|$, comparison shows that $\sum x_k$ and $\sum y_k$ converge absolutely, hence converge by the real case. The complex partial sums then converge because their real and imaginary parts do (Section 3.1).

The inequality. For each n, the finite triangle inequality gives $|S_n| \le \sum_{k=1}^{n} |a_k| \le \sum_{k=1}^{\infty} |a_k|$. Let $n \to \infty$; $|S_n| \to |S|$ by continuity of the modulus. $\square$

The practical value of Eq. (3.12): applied to the tail, it says

$$\left|\sum_{k=n+1}^{\infty} a_k\right| \le \sum_{k=n+1}^{\infty} |a_k|\,,$$

so the truncation error of an absolutely convergent series is controlled by the moduli alone, whatever the phases do.

3.6.2 Rearranging an absolutely convergent series

A **rearrangement** of $\sum a_k$ is a series $\sum_j a_{q(j)}$ in which the reordering q lists every positive integer exactly once in some new order (so $q(1), q(2), \dots$ is a shuffled copy of $1, 2, \dots$). For finite sums the order never matters. For infinite sums it can matter (Section 3.7), but not when the series converges absolutely.

Theorem. If $\sum a_k$ converges absolutely to S, every rearrangement converges absolutely to the same S.

Proof. First suppose all $a_k \geq 0$. Take any partial sum $T_J = \sum_{j=1}^{J} a_{q(j)}$ of the rearranged series. It uses finitely many terms of the original series, all with index at most $K = \max(q(1), \dots, q(J))$, so $T_J \leq S_K \leq S$ (adding the missing nonnegative terms only increases the sum). So the rearranged partial sums are bounded by S, increase, and converge to some $T \leq S$. The original series is in turn a rearrangement of the new one, so the same argument gives $S \leq T$. Hence $T = S$.

For real terms of either sign, write $a_k = b_k - |a_k|$ with $b_k = a_k + |a_k| \geq 0$ as before; both $\sum b_k$ and $\sum |a_k|$ are nonnegative convergent series, each rearrangement of them has the same sum, so the rearrangement of $\sum a_k$ has sum $\sum b_k - \sum |a_k| = S$. For complex terms apply this to real and imaginary parts. □

3.6.3 The Dirichlet series $\sum k^{-s}$ for complex s

A series of the form $\sum_k c_k k^{-s}$ is called a **Dirichlet series**; here all $c_k = 1$.

Recall from Chapter 2 that for a positive integer k and a complex number $s = \sigma + it$ (σ, t real),

$$k^{-s} = e^{-s \log k} = e^{-\sigma \log k}\, e^{-it \log k} = k^{-\sigma} e^{-it \log k},$$

so that

$$\left|k^{-s}\right| = k^{-\sigma}. \tag{3.13}$$

The modulus depends only on the real part σ; the imaginary part t only rotates the phase, at angular frequency $\log k$ in t.

Theorem. For $\operatorname{Re} s = \sigma > 1$, the series $\sum_{k=1}^{\infty} k^{-s}$ converges absolutely, and its sum $\zeta(s)$ satisfies, for every $n \geq 1$,

$$\left|\zeta(s) - \sum_{k=1}^{n} k^{-s}\right| \leq \frac{n^{1-\sigma}}{\sigma - 1}. \tag{3.14}$$

Proof. By Eq. (3.13), $\sum |k^{-s}| = \sum k^{-\sigma}$, which converges for $\sigma > 1$ (Section 3.5). The tail bound is Eq. (3.12) applied to the tail, followed by Eq. (3.10) with $f(x) = x^{-\sigma}$. □

Sample Problem 3.8: Truncating $\zeta(s)$ off the real axis

Take $s = 2 + i\,t_1$ with $t_1 = 14.134725$ (RE1). How many terms guarantee that the partial sum is within 0.01 of $\zeta(s)$?
Step 1: Here $\sigma = 2$. Eq. (3.14) gives $|\zeta(s) - S_n| \leq n^{1-2}/(2-1) = 1/n$.
Step 2: $1/n \leq 0.01$ requires $n \geq 100$.

$$\boxed{n = 100 \text{ terms suffice}}$$

Notice that the bound does not depend on t at all. Whether t is 14 or 10^6, absolute convergence controls the error through the moduli k^{-2} alone. (Computing the 100-term sum and comparing it with `mpmath.zeta` confirms that the actual error at $n = 100$ is below 0.01.)

3.6.4 What happens on the critical line

At $\sigma = \frac{1}{2}$ the moduli are $k^{-1/2}$, whose sum diverges (the p-series theorem with $s = \frac{1}{2}$; see also the Sample Problem "How fast does $\sum k^{-1/2}$ diverge?" in Section 3.5). So the series is not absolutely convergent there. Does it at least converge conditionally? The terms do tend to zero, so the term test is silent. The answer is no.

Taken on trust

Partial sums of $\sum k^{-s}$ for $0 < \sigma < 1$ (proved; proof beyond this book). Fix $s = \sigma + it$ with $0 < \sigma < 1$. Then, as $n \to \infty$,

$$\sum_{k=1}^{n} k^{-s} = \zeta(s) + \frac{n^{1-s}}{1-s} + \frac{1}{2}n^{-s} + \epsilon_n, \qquad \epsilon_n \to 0,$$

where $\zeta(s)$ is the analytically continued zeta function of Chapter 16, and $|\epsilon_n|$ is small compared with $\frac{1}{2}n^{-\sigma}$ once n is large compared with $|t|$. The proof uses Euler–Maclaurin summation; see Titchmarsh [6], in the discussion of approximate formulas for $\zeta(s)$ in the critical strip, or Edwards [5]. Since $|n^{1-s}| = n^{1-\sigma} \to \infty$, the partial sums do not converge: the series diverges everywhere in $0 < \sigma < 1$, including the critical line.

Figure 3.3 shows both regimes. At $s = 2 + it_1$ the partial sums wind briefly and settle on a point. At the first zero $s = \rho_1 = \frac{1}{2} + it_1$, where $\zeta(\rho_1) = 0$, the beyond box predicts $S_n \approx n^{1-s}/(1-s)$: a spiral around the origin with radius $\sqrt{n}/\,|1-s|$ growing without bound. More precisely, because $\zeta(\rho_1) = 0$, the box gives $S_n - \frac{n^{1-s}}{1-s} = \frac{1}{2}n^{-s} + \epsilon_n$, whose modulus should be close to $\left|\frac{1}{2}n^{-s}\right| = \frac{1}{2}n^{-1/2}$. Numerically, at $n = 1000$ the radius is $\sqrt{1000}/14.1436 = 2.236$, and $|S_{1000} - 1000^{1-s}/(1-s)|$ is about 0.016, matching $\frac{1}{2} \cdot 1000^{-1/2} = 0.0158$ (a numerical observation; the listing below computes it).

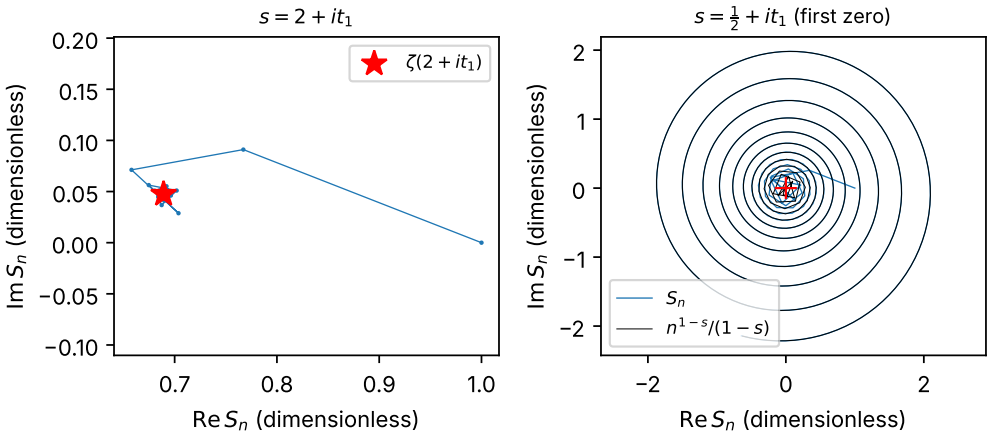


Figure 3.3: Partial sums $S_n = \sum_{k=1}^{n} k^{-s}$ in the complex plane. Left: $s = 2 + i\,t_1$ with $t_1 = 14.134725$; the series converges absolutely and the partial sums settle on $\zeta(s)$ (star). Right: $s = \frac{1}{2} + i\,t_1$, the first nontrivial zero; the partial sums spiral outward around the origin, following the curve $n^{1-s}/(1-s)$ (thin line), and never converge, although $\zeta(\frac{1}{2} + it_1) = 0$.

```
import numpy as np, qmnt
t1 = qmnt.zeros(1)[0]
s = 0.5 + 1j*t1                          # first zero rho_1
k = np.arange(1, 1001)
S = np.cumsum(np.exp(-s*np.log(k)))      # partial sums of k^(-s)
```

```
main = k**(1 - s) / (1 - s)           # growing spiral term
print(abs(S[-1]), np.sqrt(1000)/abs(1 - s))
print(abs(S[-1] - main[-1]))          # about 0.016
```

Checkpoint 3.5

Is $\sum_{k=1}^{\infty} k^{-s}$ absolutely convergent at $s = 1.5 + 100i$? At $s = 0.5 + 100i$?

Answer: At $s = 1.5 + 100i$ yes: the moduli are $k^{-1.5}$, and $1.5 > 1$; the value of t is irrelevant. At $s = 0.5 + 100i$ no: the moduli $k^{-0.5}$ sum to infinity, and by the beyond box the series does not converge at all.

Key Ideas

- Absolute convergence ($\sum |a_k| < \infty$) implies convergence, with $|\sum a_k| \le \sum |a_k|$ (Eq. (3.12)); the tail is bounded by the moduli alone.
- Absolutely convergent series may be rearranged freely without changing the sum.
- $|k^{-s}| = k^{-\sigma}$, so $\sum k^{-s}$ converges absolutely for $\operatorname{Re} s > 1$, with tail at most $n^{1-\sigma}/(\sigma - 1)$ (Eq. (3.14)), independent of t.
- On the critical line the Dirichlet series diverges; its partial sums spiral outward. The zeros of ζ cannot be found by summing $\sum k^{-s}$ there.

3.7 Conditional convergence and alternating series

★ Did you know? *Rearrange to Any Answer*

Bernhard Riemann proved that a conditionally convergent series can be rearranged to add up to *any* real number you choose, or even to diverge. The alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots$ sums to $\log 2$. Reorder the terms as two positives, then one negative, and the sum becomes $\frac{3}{2} \log 2$. The result is in his 1854 habilitation thesis on trigonometric series, published only after his death.

3.7.1 The alternating series test

When terms alternate in sign, cancellation can make a series converge even though the moduli sum to infinity.

Theorem (alternating series test). Let $a_1 \ge a_2 \ge a_3 \ge \cdots \ge 0$ with $a_k \to 0$. Then $\sum_{k=1}^{\infty} (-1)^{k+1} a_k = a_1 - a_2 + a_3 - \cdots$ converges to a sum S, and for every n

$$|S - S_n| \le a_{n+1}. \tag{3.15}$$

Proof. Look at even and odd partial sums separately.

- $S_{2n+2} - S_{2n} = a_{2n+1} - a_{2n+2} \ge 0$: the even partial sums increase.
- $S_{2n+1} - S_{2n-1} = -a_{2n} + a_{2n+1} \le 0$: the odd partial sums decrease.
- $S_{2n} = S_{2n-1} - a_{2n} \le S_{2n-1}$: each even sum lies below the preceding odd sum.

Combining these, $S_2 \le S_4 \le \cdots \le S_{2n} \le S_{2n-1} \le \cdots \le S_3 \le S_1$. The even sums increase and are bounded above by S_1, so they converge to some E; the odd sums decrease and are

bounded below by S_2, so they converge to some O (monotone convergence). Their difference is $O - E = \lim(S_{2n-1} - S_{2n}) = \lim a_{2n} = 0$, so $O = E = S$. Given ε, choose N so that both the even and the odd sums beyond N lie within ε of S; then all partial sums beyond N do, so $S_n \to S$.

Finally, S is at least every even partial sum and at most every odd one. Consecutive partial sums S_n, S_{n+1} are one even and one odd, so S lies between them, and $|S - S_n| \le |S_{n+1} - S_n| = a_{n+1}$. $\square$

The **alternating harmonic series** $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ satisfies the hypotheses with $a_k = 1/k$, so it converges. It does not converge absolutely (the moduli form the harmonic series), so it converges *conditionally*. Its sum is $\log 2$, as we prove in Section 3.8.

Sample Problem 3.9: Computing $\log 2$ slowly

Use the alternating harmonic series (sum $\log 2 = 0.693147$, proved in Section 3.8). (a) Compute S_4 and check Eq. (3.15). (b) How many terms guarantee an error at most 10^{-3}?
Step 1: $S_4 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{12-6+4-3}{12} = \frac{7}{12} = 0.583333$.
Step 2: The actual error is $0.693147 - 0.583333 = 0.109814$, and the bound is $a_5 = \frac{1}{5} = 0.2$. The bound holds.
Step 3: We need $a_{n+1} = 1/(n+1) \le 10^{-3}$, i.e. $n + 1 \ge 1000$.

$$\boxed{S_4 = \tfrac{7}{12}, \quad \text{error } 0.1098 \le 0.2; \qquad n = 999 \text{ terms}}$$

Conditionally convergent series often converge painfully slowly. Problem 11 shows a far faster route to $\log 2$.

3.7.2 Order matters: a rearrangement with half the sum

Rearrange the alternating harmonic series so that each positive (odd-denominator) term is followed by two negative (even-denominator) terms:

$$R = 1 - \tfrac{1}{2} - \tfrac{1}{4} + \tfrac{1}{3} - \tfrac{1}{6} - \tfrac{1}{8} + \tfrac{1}{5} - \tfrac{1}{10} - \tfrac{1}{12} + \cdots$$

Every term of the original series appears exactly once: the positive terms are all $1/(2j-1)$; the negative terms are $1/(4j-2)$ (even numbers not divisible by 4) and $1/(4j)$ (multiples of 4), which together are all even denominators. The j-th group of three is

$$\frac{1}{2j-1} - \frac{1}{2(2j-1)} - \frac{1}{4j}.$$

Combine the first two terms: $\frac{1}{2j-1} - \frac{1}{2(2j-1)} = \frac{1}{2(2j-1)} = \frac{1}{4j-2}$. So the group equals

$$\frac{1}{4j-2} - \frac{1}{4j} = \frac{1}{2}\left(\frac{1}{2j-1} - \frac{1}{2j}\right).$$

The bracket is exactly the j-th pair of the original alternating series. Summing J groups therefore gives $R_{3J} = \frac{1}{2} S_{2J}$, half of the original partial sum with $2J$ terms. As $J \to \infty$, $R_{3J} \to \frac{1}{2}\log 2$. The partial sums $R_{3J+1} = R_{3J} + \frac{1}{2J+1}$ and $R_{3J+2} = R_{3J+1} - \frac{1}{4J+2}$ differ from R_{3J} by at most $\frac{1}{2J+1} \to 0$, so the whole rearranged series converges, to

$$R = \tfrac{1}{2}\log 2 = 0.346574.$$

Same terms, different order, half the sum. Figure 3.4 shows both sequences of partial sums.

Taken on trust

Riemann's rearrangement theorem (proved; proof beyond this book). If a real series converges conditionally, then for every real number C there is a rearrangement that converges to C, and there are rearrangements that diverge to $+\infty$ or $-\infty$. The idea of the proof is to add positive terms until you pass C, then negative terms until you fall below it, and so on; it is given in standard real analysis texts. We will not need the theorem, only its moral, which our explicit example already proves: for a conditionally convergent series, the order of summation is part of the definition of the sum. Historically, Riemann gave this result in his 1854 habilitation thesis on trigonometric series, which was published after his death. Problem 25 shows a rearrangement of the alternating harmonic series (two positive terms, then one negative) whose sum is $\frac{3}{2}\log 2$.

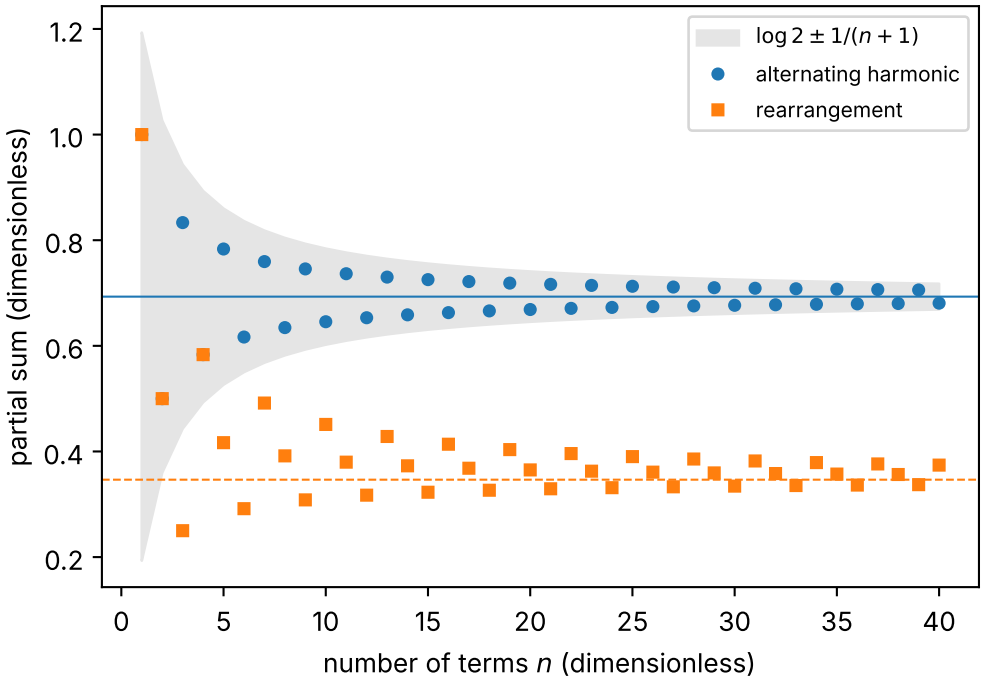


Figure 3.4: Partial sums of the alternating harmonic series (circles), which zigzag around $\log 2$ with error at most the next term $1/(n+1)$ (shaded band), and of its rearrangement $1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \cdots$ (squares), which uses exactly the same terms in a different order and converges to $\frac{1}{2}\log 2$.

Pitfall

Never reorder, regroup in a way that changes which terms are paired, or "sum by a different cutoff" a series that is not absolutely convergent and expect the same answer. For an absolutely convergent series any order is fine (Section 3.6); for a conditionally convergent one, the answer depends on the order. Sums over primes on the critical line are not absolutely convergent, so they fall outside the safe first kind (Section 3.10).

Checkpoint 3.6

You may rearrange $\sum (-1)^{k+1}/k^2$ freely, but not $\sum (-1)^{k+1}/k$. Why?

Answer: The first converges absolutely, since $\sum 1/k^2$ converges, so every rearrangement has the same sum. The second converges only conditionally, and we exhibited a rearrangement with half the sum.

Key Ideas

- Alternating series with decreasing terms tending to zero converge, and the error is at most the first omitted term (Eq. (3.15)).
- The alternating harmonic series converges conditionally to $\log 2$.
- Rearranging a conditionally convergent series can change its sum: the order of summation matters.

3.8 The ratio test, power series and the logarithm series

3.8.1 The ratio test

The geometric series is the yardstick for many others. If successive terms shrink by a factor eventually smaller than some fixed $r < 1$, the series is dominated by a geometric one.

Theorem (ratio test). Suppose $a_k \neq 0$ and $|a_{k+1}/a_k| \to L$. If $L < 1$ the series $\sum a_k$ converges absolutely; if $L > 1$ it diverges; if $L = 1$ the test gives no information.

Proof. Case $L < 1$: choose $r = (L+1)/2$, so $L < r < 1$. Since the ratios tend to L, there is a K with $|a_{k+1}/a_k| < r$ for all $k \geq K$. Then $|a_{K+1}| < r\,|a_K|$, $|a_{K+2}| < r\,|a_{K+1}| < r^2\,|a_K|$, and by induction $|a_{K+j}| < r^j\,|a_K|$. The series $\sum_j |a_K|\,r^j$ is geometric with ratio $r < 1$ and converges, so by comparison $\sum_{k \geq K} |a_k|$ converges; the finitely many terms before K do not matter.

Case $L > 1$: there is a K with $|a_{k+1}/a_k| > 1$ for $k \geq K$, so $|a_k| \geq |a_K| > 0$ for all $k \geq K$. The terms do not tend to zero, and the series diverges.

Case $L = 1$: for the p-series, $|a_{k+1}/a_k| = (k/(k+1))^s \to 1$ for every real s, yet the series converges for $s > 1$ and diverges for $s \leq 1$. So $L = 1$ decides nothing. □

3.8.2 Power series and the radius of convergence

A **power series** in a variable z (real or complex) is $\sum_{m=0}^{\infty} c_m z^m$. Applying the ratio test to $a_m = c_m z^m$,

$$\left|\frac{c_{m+1}z^{m+1}}{c_m z^m}\right| = |z|\left|\frac{c_{m+1}}{c_m}\right|.$$

Assume all $c_m \neq 0$. If $|c_m/c_{m+1}| \to R$ with $0 < R < \infty$, this ratio tends to $|z|/R$. If $R = \infty$ the ratio tends to 0 for every z. If $R = 0$ it tends to infinity for every $z \neq 0$, and the series converges only at $z = 0$. So the series converges absolutely for $|z| < R$ and diverges for $|z| > R$. The number R is the **radius of convergence**: in the complex plane the series converges inside the disk of radius R and diverges outside it. On the circle $|z| = R$ anything can happen and each point must be examined separately. (A radius of convergence exists for every power series, even when the ratio limit does not; we only need the ratio version.)

Examples:

- Geometric, $c_m = 1$: $R = 1$.
- Exponential, $c_m = 1/m!$ (Chapter 2): $|c_m/c_{m+1}| = (m+1)!/m! = m+1 \to \infty$, so $R = \infty$; e^z converges for every z.
- $c_m = 1/m$ $(m \geq 1)$: $|c_m/c_{m+1}| = (m+1)/m \to 1$, so $R = 1$. This is the logarithm series, next.

3.8.3 The series for $-\log(1-x)$

We derive the Taylor series of $\log(1-x)$ with complete control of the error, starting from the finite geometric sum. For $u \neq 1$, Eq. (3.2) says $\sum_{m=0}^{n-1} u^m = (1-u^n)/(1-u)$. Rearranged:

$$\frac{1}{1-u} = \sum_{m=0}^{n-1} u^m + \frac{u^n}{1-u}.$$

This is an exact identity, not an approximation. Integrate both sides over u from 0 to x, for real $x < 1$ (so that $1-u > 0$ throughout).

The left side: substitute $w = 1-u$, $\mathrm{d}w = -\mathrm{d}u$, with w running from 1 to $1-x$:

$$\int_0^x \frac{\mathrm{d}u}{1-u} = -\int_1^{1-x} \frac{\mathrm{d}w}{w} = -\log(1-x).$$

On the right, $\int_0^x u^m \,\mathrm{d}u = x^{m+1}/(m+1)$. Relabel $m+1 \to m$ in the sum:

$$-\log(1-x) = \sum_{m=1}^{n} \frac{x^m}{m} + R_n(x), \qquad R_n(x) = \int_0^x \frac{u^n}{1-u}\,\mathrm{d}u. \tag{3.16}$$

Now bound the remainder.

Case $0 \le x < 1$. For $0 \le u \le x$, $1-u \ge 1-x > 0$, so $0 \le u^n/(1-u) \le u^n/(1-x)$, and

$$0 \le R_n(x) \le \frac{1}{1-x}\int_0^x u^n \,\mathrm{d}u = \frac{x^{n+1}}{(n+1)(1-x)} \to 0.$$

Case $-1 \le x < 0$. Substitute $u = -v$ ($\mathrm{d}u = -\mathrm{d}v$; v runs from 0 to $|x|$):

$$R_n(x) = \int_0^{|x|} \frac{(-v)^n}{1+v}(-\mathrm{d}v) = (-1)^{n+1}\int_0^{|x|} \frac{v^n}{1+v}\,\mathrm{d}v.$$

Since $1/(1+v) \le 1$ for $v \ge 0$, $|R_n(x)| \le \int_0^{|x|} v^n \,\mathrm{d}v = |x|^{n+1}/(n+1) \to 0$, even at $x = -1$.

In both cases $R_n \to 0$, so the partial sums converge to the left side:

$$-\log(1-x) = \sum_{m=1}^{\infty} \frac{x^m}{m} = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots \qquad (-1 \le x < 1). \tag{3.17}$$

At $x = -1$ this reads $-\log 2 = -1 + \frac{1}{2} - \frac{1}{3} + \cdots$, that is,

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots, \tag{3.18}$$

the promised value of the alternating harmonic series. At $x = 1$ both sides fail: $\log 0$ is undefined and the harmonic series diverges. The radius of convergence is 1, consistent with the ratio computation above. Figure 3.5 shows how the truncations approach $-\log(1-x)$ inside the interval and fail near $x = 1$.

Sample Problem 3.10: $\log 1.5$ from three terms

Use Eq. (3.16) with three terms to approximate $\log 1.5$ and bound the error.
Step 1 (choose x): $-\log(1-x) = \log\frac{1}{1-x}$. We want $\frac{1}{1-x} = 1.5$, i.e. $1-x = \frac{2}{3}$, $x = \frac{1}{3}$.
Step 2 (three terms): $\frac{1}{3} + \frac{(1/3)^2}{2} + \frac{(1/3)^3}{3} = 0.333333 + 0.055556 + 0.012346 = 0.401235$.
Step 3 (error bound, case $0 \le x < 1$, $n = 3$): $R_3 \le \frac{x^4}{4(1-x)} = \frac{1/81}{4\cdot\frac{2}{3}} = \frac{0.0123457}{2.666667} = 0.004630$.

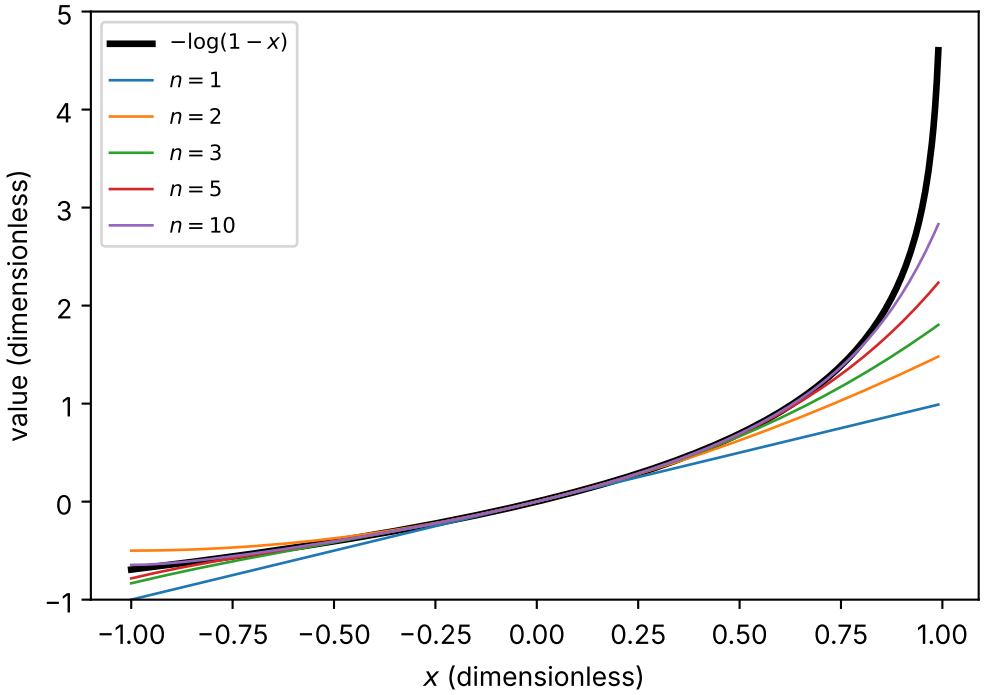


Figure 3.5: The function $-\log(1-x)$ (thick line) and its truncations $\sum_{m=1}^{n} x^m/m$ for $n = 1, 2, 3, 5, 10$. Near $x = 0$ a few terms suffice; as $x \to 1$ the function blows up while every truncation stays finite, so more and more terms are needed. The radius of convergence is 1.

Step 4 (compare): The true value is $\log 1.5 = 0.405465$; the actual error is 0.004230, positive and below the bound, as the analysis predicts.

$$\log 1.5 \approx 0.401235, \quad 0 < \text{error} \le 0.00463$$

3.8.4 Complex arguments

For complex z with $|z| < 1$, the series $\sum_{m\ge1} z^m/m$ converges absolutely, by comparison with the geometric series $\sum |z|^m$ (since $|z^m/m| \le |z|^m$). What is it the logarithm of?

Taken on trust

The complex logarithm series (proved; proof beyond this book). For complex z with $|z| < 1$, the sum $L(z) = \sum_{m=1}^{\infty} z^m/m$ satisfies $e^{L(z)} = \frac{1}{1-z}$. Thus $L(z)$ is a logarithm of $1/(1-z)$; it is the principal one (Chapter 2), with imaginary part between $-\pi/2$ and $\pi/2$ because $1 - z$ lies in the right half-plane. The proof differentiates the power series term by term, which is legitimate inside the radius of convergence; see Riley, Hobson and Bence [22]. A direct numerical computation, summing 200 terms at $z = \frac{1}{2}e^{i}$, confirms the identity to within 10^{-12}.

The case we will use again and again: $z = p^{-s}$ with p a prime and $s = \sigma + it$. By Eq. (3.13), $|p^{-s}| = p^{-\sigma}$, which is below 1 whenever $\sigma > 0$. So

$$-\log(1-p^{-s}) = \sum_{m=1}^{\infty} \frac{p^{-ms}}{m} \qquad (\operatorname{Re} s > 0),$$

with the series converging absolutely. This is the step that turns the Euler product into the explicit formula (Chapter 18).

3.8.5 Running example RE2: truncating the powers of a prime

The prime-2 wave of the paper's continuation probe (RE2) is

$$f_2(t) = -\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 2)}{m\,2^{m/2}}.$$

The explicit formula, before truncation in m, would contain all $m \geq 1$. Call that version $f_2^{\infty}(t)$. Does it exist, and how much does stopping at $m = 3$ change it?

Sample Problem 3.11: How much is lost by keeping only $m \leq 3$?

Show that $f_2^{\infty}(t)$ converges absolutely for every real t and bound $|f_2^{\infty}(t) - f_2(t)|$.
Step 1 (absolute convergence): $|\sin(mt\log 2)| \leq 1$, so the m-th term has modulus at most $\frac{1}{\pi}\frac{2^{-m/2}}{m}$. The series $\sum_m 2^{-m/2}/m$ is Eq. (3.17) at $x = 2^{-1/2} = 0.707107 < 1$, so it converges. By comparison, $f_2^{\infty}(t)$ converges absolutely for every t.
Step 2 (exact tail of the moduli): By Eq. (3.17), $\sum_{m=1}^{\infty}\frac{2^{-m/2}}{m} = -\log(1-2^{-1/2}) = -\log 0.292893 = 1.227947$. The first three terms are $0.707107 + \frac{0.5}{2} + \frac{0.353553}{3} = 0.707107 + 0.25 + 0.117851 = 1.074958$.
Step 3 (bound): By Eq. (3.12) applied to the tail,

$$|f_2^{\infty}(t) - f_2(t)| \leq \frac{1}{\pi}\sum_{m=4}^{\infty}\frac{2^{-m/2}}{m} = \frac{1.227947 - 1.074958}{\pi} = \frac{0.152989}{\pi}.$$

Step 4 (an alternative, cruder bound, for comparison): Using $1/m \leq 1/4$ for $m \geq 4$ and Eq. (3.3): $\frac{1}{4\pi}\sum_{m\geq 4} 2^{-m/2} = \frac{1}{4\pi}\frac{2^{-2}}{1-2^{-1/2}} = 0.067924$. This is also a valid upper bound, but it is larger than the Step 3 value $0.152989/\pi = 0.048698$, so the Step 3 bound is the tighter one and is the one we keep.

$$\boxed{|f_2^{\infty}(t) - f_2(t)| \leq 0.048698 \quad \text{for all } t} \tag{3.19}$$

For scale: the amplitude of the leading term is $2^{-1/2}/\pi = 0.225079$. The neglected powers can shift the wave by at most about a fifth of its leading amplitude ($0.0487/0.2251 = 0.216$), and that worst case requires all neglected sines to peak together. Figure 3.6 compares the truncations.

Checkpoint 3.7

For which real x does $\sum_{m=1}^{\infty} x^m/m$ converge? For which complex z does it converge absolutely?

Answer: For real x: exactly $-1 \leq x < 1$ (diverges at $x = 1$ as the harmonic series, diverges for $|x| > 1$ by the ratio test). It converges absolutely for complex $|z| < 1$; at $x = -1$ it converges only conditionally.

Key Ideas

- Ratio test: $|a_{k+1}/a_k| \to L < 1$ gives absolute convergence, $L > 1$ divergence, $L = 1$ no information.
- A power series converges absolutely inside its radius of convergence $R = \lim |c_m/c_{m+1}|$

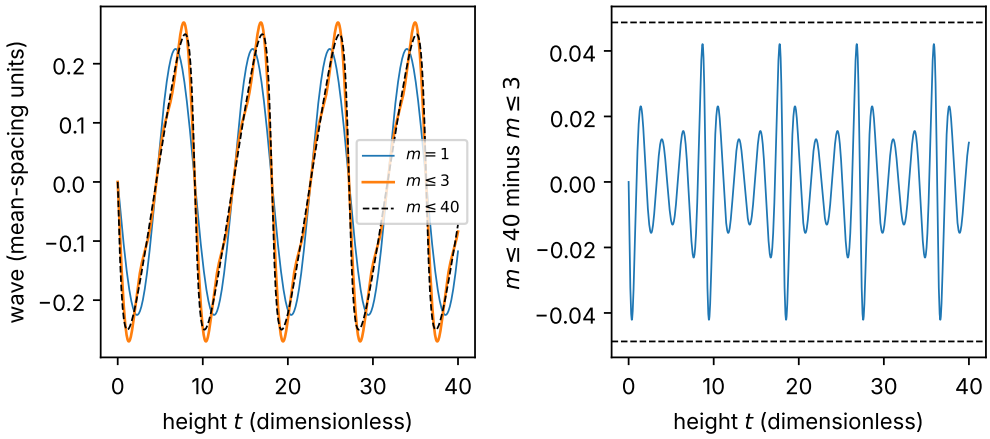


Figure 3.6: Running example RE2. Left: the prime-2 wave $-\frac{1}{\pi}\sum_m \sin(mt\log 2)/(m2^{m/2})$ truncated at $m=1$, at $m\le 3$ (the paper's choice), and at $m\le 40$ (effectively the full absolutely convergent series), for $0\le t\le 40$. The pattern repeats with period $2\pi/\log 2\approx 9.065$. Right: the difference between the $m\le 40$ and $m\le 3$ versions stays inside the guaranteed bound ± 0.0487 of Eq. (3.19) (dashed).

and diverges outside.

- $-\log(1-x)=\sum_{m\ge 1}x^m/m$ for $-1\le x<1$, derived with an explicit remainder (Eqs. (3.16), (3.17)); at $x=-1$ it gives $\log 2$.
- For $|z|<1$ the same series is the principal logarithm of $1/(1-z)$; in particular $-\log(1-p^{-s})=\sum_m p^{-ms}/m$ for $\operatorname{Re} s>0$.
- Truncating the prime-2 wave at $m\le 3$ changes it by at most 0.0487 (Eq. (3.19)).

3.9 Infinite products

3.9.1 Definition

An **infinite product** $\prod_{k=1}^{\infty}u_k=u_1u_2u_3\cdots$ is defined, like a series, through its partial products

$$P_n=\prod_{k=1}^{n}u_k=u_1u_2\cdots u_n. \tag{3.20}$$

Definition. The product **converges** if $P_n\to P$ with P a finite number *different from zero*. If $P_n\to 0$ while no factor is zero, we say the product **diverges to zero**.

Why exclude zero? Two reasons. First, for a finite product, $P=0$ happens only if some factor is zero; we want the same for convergent infinite products, so that a product formula for a function tells us where it vanishes (this is how Chapter 16 proves that $\zeta(s)\neq 0$ for $\operatorname{Re} s>1$). Second, we want logarithms to turn convergent products into convergent sums, and $\log 0$ does not exist.

Writing $u_k=1+a_k$, a convergent product must have $a_k\to 0$: indeed $u_n=P_n/P_{n-1}\to P/P=1$ (the division is allowed because $P\neq 0$). This is the product version of the term test.

3.9.2 Three telescoping examples

(i) Diverges to infinity. $\prod_{k=1}^{n}\left(1+\frac{1}{k}\right)=\prod_{k=1}^{n}\frac{k+1}{k}=\frac{2}{1}\cdot\frac{3}{2}\cdots\frac{n+1}{n}=n+1\to\infty$. (Each numerator cancels the next denominator.)

(ii) Diverges to zero. $\prod_{k=2}^{n}\left(1-\frac{1}{k}\right)=\prod_{k=2}^{n}\frac{k-1}{k}=\frac{1}{2}\cdot\frac{2}{3}\cdots\frac{n-1}{n}=\frac{1}{n}\to 0$. No factor is zero, yet the partial products vanish in the limit. By our definition this product diverges.

(iii) Converges. See the Sample Problem.

Sample Problem 3.12: A product that converges to one half

Evaluate $\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right)$ and its limit.

Step 1 (factor): $1-\frac{1}{k^2}=\frac{k^2-1}{k^2}=\frac{(k-1)(k+1)}{k\cdot k}=\frac{k-1}{k}\cdot\frac{k+1}{k}$.

Step 2 (split the product): $\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right)=\left[\prod_{k=2}^{n}\frac{k-1}{k}\right]\left[\prod_{k=2}^{n}\frac{k+1}{k}\right]$ (finite products may be reordered).

Step 3 (telescope each): The first is $\frac{1}{n}$ (example (ii)). The second is $\frac{3}{2}\cdot\frac{4}{3}\cdots\frac{n+1}{n}=\frac{n+1}{2}$.

Step 4: So $P_n=\frac{1}{n}\cdot\frac{n+1}{2}=\frac{n+1}{2n}\to\frac{1}{2}$. At $n=10$, $P_{10}=\frac{11}{20}=0.55$.

$$\boxed{\prod_{k=2}^{\infty}\left(1-\frac{1}{k^2}\right)=\frac{1}{2}}$$

Notice that the two pieces individually diverge (to 0 and to ∞), but their product converges. Splitting an infinite product into divergent pieces is dangerous; here we split only the *finite* partial products, which is always safe.

3.9.3 From products to sums: the logarithm

For a finite product of positive numbers, $\log(u_1u_2\cdots u_n)=\log u_1+\log u_2+\cdots+\log u_n$ (apply $\log(ab)=\log a+\log b$ repeatedly). The infinite version:

Theorem. Let $u_k>0$. The product $\prod u_k$ converges (to a positive P) if and only if the series $\sum\log u_k$ converges, and then

$$\log\prod_{k=1}^{\infty}u_k=\sum_{k=1}^{\infty}\log u_k. \tag{3.21}$$

The product diverges to zero exactly when $\sum_{k\le n}\log u_k\to-\infty$.

Proof. Let $L_n=\sum_{k=1}^{n}\log u_k=\log P_n$, so $P_n=e^{L_n}$. If $L_n\to L$, then $P_n=e^{L_n}\to e^{L}>0$ because e^x is continuous; so the product converges to e^L and $\log P=L$. Conversely, if $P_n\to P>0$, then $L_n=\log P_n\to\log P$ because log is continuous at the positive number P. Finally $P_n=e^{L_n}\to 0$ if and only if $L_n\to-\infty$. □

Example (ii) in this language: $\sum_{k=2}^{n}\log(1-1/k)=\log(1/n)=-\log n\to-\infty$.

Pitfall

For complex numbers, $\log(zw)=\log z+\log w$ can fail for the principal logarithm, because the arguments may add up beyond π and the principal value wraps around (Chapter 2). Example: $z=w=e^{2\pi i/3}$. Principal values: $\log z+\log w=\frac{4\pi i}{3}$, but $zw=e^{4\pi i/3}=e^{-2\pi i/3}$ has principal logarithm $-\frac{2\pi i}{3}$. They differ by $2\pi i$. Eq. (3.21) is safe as stated for positive factors; for complex factors one must keep track of the branch, a point Chapter 18 handles when it takes $\log\zeta(s)$.

3.9.4 When does $\prod(1+a_k)$ converge?

Theorem. Let $a_k\ge 0$. Then $\prod_{k=1}^{\infty}(1+a_k)$ converges if and only if $\sum_{k=1}^{\infty}a_k$ converges. Moreover, with $A=\sum a_k$,

$$1+A\le\prod_{k=1}^{\infty}(1+a_k)\le e^{A}. \tag{3.22}$$

Proof. Each factor is at least 1, so the partial products P_n increase. Let $A_n = \sum_{k=1}^{n} a_k$.

Upper bound. For $a \geq 0$, $1 + a \leq e^a$: the function $g(a) = e^a - 1 - a$ has $g(0) = 0$ and $g'(a) = e^a - 1 \geq 0$ for $a \geq 0$, so $g(a) \geq 0$. Multiplying these inequalities (all quantities positive), $P_n \leq e^{a_1} \cdots e^{a_n} = e^{A_n}$.

Lower bound. $P_n \geq 1 + A_n$, by induction: $P_1 = 1 + a_1$; if $P_n \geq 1 + A_n$ then

$$P_{n+1} = P_n(1 + a_{n+1}) \geq (1 + A_n)(1 + a_{n+1}) = 1 + A_n + a_{n+1} + A_n a_{n+1} \geq 1 + A_{n+1},$$

since $A_n a_{n+1} \geq 0$.

If $\sum a_k$ converges, $P_n \leq e^{A_n} \leq e^{A}$: increasing and bounded, so P_n converges to a limit at least $P_1 \geq 1$, hence nonzero. If $\sum a_k$ diverges, $P_n \geq 1 + A_n \to \infty$. Taking $n \to \infty$ in the two bounds gives Eq. (3.22). □

Example (i) is the divergent case: $a_k = 1/k$, harmonic. By contrast $\prod(1 + 1/k^2)$ converges, and Eq. (3.22) brackets it between $1 + \pi^2/6$ and $e^{\pi^2/6}$ (Problem 13).

Taken on trust

Complex factors (proved; proof beyond this book). If a_k are complex and $\sum |a_k|$ converges, then $\prod(1+a_k)$ converges in the sense above unless some factor $1+a_k$ is exactly zero; in that case the limit is 0, and only then. Such a product can also be rearranged without changing its value. Proofs are in Apostol [23], in the chapter on Euler products, and in Titchmarsh [6].

3.9.5 A first look at the Euler product

List the primes in increasing order, $p_1 = 2,\ p_2 = 3,\ p_3 = 5, \ldots$ (Chapter 15). For real $s > 1$ consider

$$\prod_{p} \frac{1}{1 - p^{-s}} = \frac{1}{1 - 2^{-s}} \cdot \frac{1}{1 - 3^{-s}} \cdot \frac{1}{1 - 5^{-s}} \cdots \tag{3.23}$$

It converges. Write each factor as $1 + a_p$:

$$\frac{1}{1 - p^{-s}} = \frac{p^s}{p^s - 1} = 1 + \frac{1}{p^s - 1}, \qquad a_p = \frac{1}{p^s - 1} > 0.$$

Since $p \geq 2$ and $s > 1$, $p^s > 2$, so $p^s/2 > 1$ and therefore $p^s - 1 > p^s - p^s/2 = p^s/2$. Hence $a_p < 2p^{-s}$. The sum $\sum_p p^{-s}$ over primes is at most $\sum_k k^{-s} = \zeta(s)$ over all integers (its terms are a subset of nonnegative terms), so it converges; by comparison $\sum_p a_p$ converges, and by the theorem the product converges to a number at least 1.

Euler discovered that its value is $\zeta(s)$. The proof needs the unique factorization of integers into primes (Chapter 15) and is given in Chapter 16; here we only look at the numbers for $s = 2$, where $\zeta(2) = \pi^2/6 = 1.644934$. Each factor is $\frac{1}{1-p^{-2}} = \frac{p^2}{p^2-1}$: $\frac{4}{3},\ \frac{9}{8},\ \frac{25}{24},\ \frac{49}{48},\ \frac{121}{120},\ \frac{169}{168}, \ldots$

Sample Problem 3.13: The Euler product at $s = 2$ over primes up to 13

Write $\Pi(x) = \prod_{p \leq x} p^2/(p^2 - 1)$ for the partial Euler product over the primes up to x. Compute $\Pi(x)$ for $x = 2, 3, 5, 7, 11, 13$ and compare each value with $\pi^2/6$.
Step 1: $\Pi(2) = \frac{4}{3} = 1.333333$.
Step 2: $\Pi(3) = \frac{4}{3} \cdot \frac{9}{8} = \frac{36}{24} = 1.5$.
Step 3: $\Pi(5) = 1.5 \cdot \frac{25}{24} = \frac{25}{16} = 1.5625$.
Step 4: $\Pi(7) = \frac{25}{16} \cdot \frac{49}{48} = \frac{1225}{768} = 1.595052$.
Step 5: $\Pi(11) = 1.595052 \cdot \frac{121}{120} = 1.608344$; $\Pi(13) = 1.608344 \cdot \frac{169}{168} = 1.617918$.
Step 6: Divide by $\pi^2/6$ (Table 3.1).

$$\prod_{p\le 13}\frac{p^2}{p^2-1} = 1.617918 = 0.9836\times\frac{\pi^2}{6}$$

Six primes already give 98.4% of $\zeta(2)$. Figure 3.7 (right) continues to 100 primes.

Table 3.1: Partial Euler products $\Pi(x)=\prod_{p\le x}(1-p^{-2})^{-1}$ and their ratio to $\zeta(2)=\pi^2/6=1.644934$.

largest prime x	partial product $\Pi(x)$	ratio to $\pi^2/6$
2	1.333333	0.8106
3	1.500000	0.9119
5	1.562500	0.9499
7	1.595052	0.9697
11	1.608344	0.9778
13	1.617918	0.9836

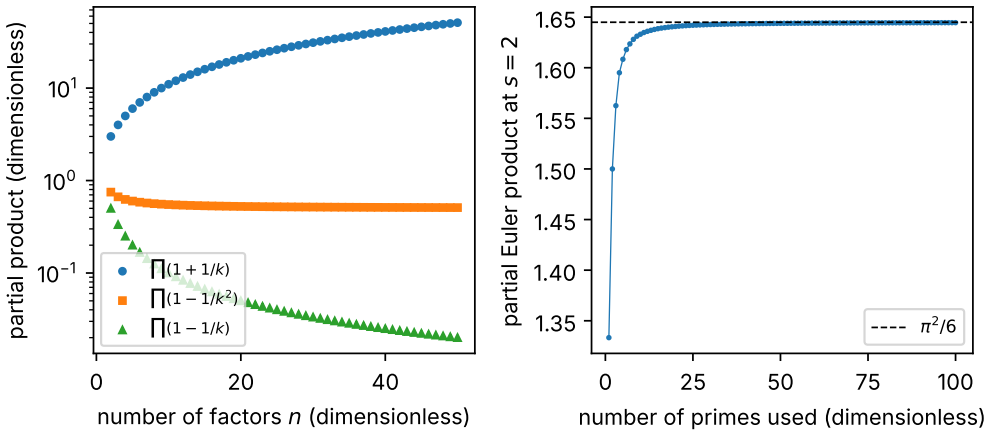


Figure 3.7: Left: partial products of $\prod(1+1/k)$ (grows like $n+1$), $\prod(1-1/k^2)$ (converges to $\frac{1}{2}$) and $\prod(1-1/k)$ (diverges to zero like $1/n$), for n up to 50, on a logarithmic vertical axis. Right: the partial Euler product $\prod_{p\le x}(1-p^{-2})^{-1}$ against the number of primes used, approaching $\zeta(2)=\pi^2/6$ (dashed line).

Taking the logarithm of the partial Euler product and expanding each factor with Eq. (3.17), using p^{-s} (which lies in $(0,\frac{1}{2}]$) in place of the variable of that equation:

$$\log\prod_{p\le x}\frac{1}{1-p^{-s}} = \sum_{p\le x}-\log(1-p^{-s}) = \sum_{p\le x}\sum_{m=1}^{\infty}\frac{p^{-ms}}{m}.$$

Letting $x\to\infty$ produces a *double series* over primes p and powers m. All its terms are positive here, so we may sum them in any order; the precise statement is in the box below. Chapter 18 repeats this computation for complex s and then moves toward the critical line, where this freedom is lost.

Taken on trust

Double series (proved; proof beyond this book). If $b_{pm} \geq 0$, or more generally if $\sum_p \sum_m |b_{pm}| < \infty$, then the double series $\sum_{p,m} b_{pm}$ has the same value whether you sum over m first, over p first, or along any enumeration of the pairs (p, m). The proof extends the rearrangement argument of Section 3.6; see Apostol [23] or Riley, Hobson and Bence [22].

Checkpoint 3.8

The partial products of $\prod_{k\geq 2}(1 - 1/k)$ tend to a limit. Does the product converge?

Answer: No. The limit is 0 while no factor is zero, so by definition the product diverges to zero. Equivalently, $\sum \log(1 - 1/k) = -\infty$.

Key Ideas

- An infinite product converges if its partial products tend to a finite *nonzero* limit (Eq. (3.20)); a convergent product needs factors tending to 1.
- For positive factors, $\log \prod u_k = \sum \log u_k$, and the product converges iff the sum of logs does (Eq. (3.21)).
- For $a_k \geq 0$: $\prod(1 + a_k)$ converges iff $\sum a_k$ does, and $1 + A \leq \prod \leq e^A$ (Eq. (3.22)).
- The Euler product $\prod_p (1 - p^{-s})^{-1}$ converges for real $s > 1$; numerically it approaches $\zeta(2)$ at $s = 2$ (Table 3.1).

3.10 Truncation: when is a finite sum a good stand-in?

★ **Did you know?** ***The Slowest Infinity***

The harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \cdots$ diverges, as Nicole Oresme showed in the fourteenth century by grouping its terms. But it diverges incredibly slowly, because its partial sums grow like $\log N + 0.5772\ldots$. To pass 100 you need roughly 1.5×10^{43} terms. A truncated sum can look convergent long before it reveals its true nature.

Every numerical use of an infinite series is a truncation. We now collect what the chapter has taught about the error of truncating, because the distinction matters for the paper.

Case 1: absolutely convergent series. The truncation error is bounded by the tail of the moduli, $|S - S_n| \leq \sum_{k>n} |a_k|$ (Eq. (3.12)). That tail can be bounded in advance by a geometric series (Eq. (3.4)) or an integral (Eq. (3.10)). The order of the terms does not matter. You know, before computing anything, how many terms you need. Examples: $\zeta(s)$ for $\operatorname{Re} s > 1$ (Eq. (3.14)); the sum over m for one prime on the critical line (Eq. (3.19)).

Case 2: conditionally convergent series. The sum exists but depends on the order of the terms (Section 3.7). A truncation is meaningful only together with the order in which terms are added. For alternating series there is still an error bound (Eq. (3.15)), but it is a special structure, not a general guarantee.

Case 3: divergent series. The "infinite sum" does not exist, and adding more terms need not improve anything. Example: $\sum k^{-s}$ on the critical line (Figure 3.3). A finite piece of such a series can still be useful, but only if some *other* argument says so: a smoothing that makes the sum converge, a rigorous theorem about a modified version, or numerical evidence. Until such an argument is supplied, the truncation is a heuristic.

Table 3.2: Three kinds of infinite sums and what truncation means for each.

kind	example	tail bound?	order matters?
absolutely convergent	$\sum k^{-2-it}$	yes, from moduli	no
conditionally convergent	$\sum(-1)^{k+1}/k$	only special cases	yes
divergent	$\sum k^{-1/2-it}$	no sum to approach	meaningless

The sum over primes on the critical line. The explicit formula (Chapter 18) contains, on the line $s = \frac{1}{2} + it$, the double sum

$$-\frac{1}{\pi}\sum_{p}\sum_{m=1}^{\infty}\frac{\sin(mt\log p)}{m\,p^{m/2}}.$$

Which case is it? Look at the sum of moduli bounds, power by power.

- $m \geq 3$: the moduli are at most $\frac{p^{-m/2}}{m\pi}$, and their total over all primes and all $m \geq 3$ is finite (Problem 27 bounds it by 0.584). This part is absolutely convergent: Case 1.
- $m = 1$ and $m = 2$: the moduli are at most the *amplitudes* $\frac{p^{-1/2}}{\pi}$ and $\frac{p^{-1}}{2\pi}$ (we drop the factor $|\sin(mt\log p)| \leq 1$). The sums of these amplitudes over primes diverge (first box below). So the comparison argument that worked for $m \geq 3$ gives no tail bound here. In fact, for $t \neq 0$ the sum of the actual moduli, $\frac{1}{\pi}\sum_p |\sin(t\log p)|\ p^{-1/2}$, also diverges, because $|\sin(t\log p)|$ is not small for most primes. Proving this needs the distribution of the primes, so we take it on trust in the second box below. Hence for $t \neq 0$ the double sum is not absolutely convergent. Whether, and in what order of summation, it converges at all on the critical line is not something the tools of this chapter can settle; Chapter 18 treats it as a formal, heuristic object. What we can say is that for $t \neq 0$ it is not absolutely convergent, so truncating it to a finite prime set comes with no tail bound from moduli, and it must be handled like Case 3: a truncation justified only by a separate argument.

Taken on trust

The sum of reciprocals of the primes diverges (proved; proof beyond this chapter). $\sum_p \frac{1}{p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \cdots = \infty$. This is due to Euler; a proof, and the finer statement that $\sum_{p\leq x} 1/p$ equals $\log\log x$ plus a quantity that stays bounded as $x \to \infty$, is in Hardy and Wright [24] and Apostol [23]. Since $p^{-1/2} \geq p^{-1}$, it follows by comparison that $\sum_p p^{-1/2}$ diverges as well.

Taken on trust

The actual moduli also have infinite sum (proved; proof beyond this book). For every real $t \neq 0$, $\sum_p |\sin(t\log p)|\ p^{-1/2} = \infty$, and likewise $\sum_p |\sin(2t\log p)|\ p^{-1} = \infty$. The idea is that the angles $t\log p$ are spread around the circle, so a fixed fraction of the primes have $|\sin(t\log p)|$ bounded away from zero. Making this precise uses the prime number theorem (Chapter 15) in a form that controls oscillating sums over primes; see

Davenport [25] or Titchmarsh [6] for the tools. At $t = 0$ every term vanishes, which is why the restriction $t \neq 0$ is needed.

That the truncated prime sum nevertheless predicts the zeros remarkably well is a fact, but it is established in the paper by numerical tests, and in the literature by rigorous theorems about smoothed versions of the formula (Chapter 18), not by the convergence theory of this chapter.

Key Ideas

- Truncation error is under control in advance only for absolutely convergent series (Table 3.2).
- In the critical-line prime sum, the $m \geq 3$ part converges absolutely. For $m = 1$ and $m = 2$ the amplitude sums diverge (because $\sum_p 1/p$ does), so comparison gives no tail bound. For $t \neq 0$ the actual moduli also have infinite sum (taken on trust), so these parts are not absolutely convergent.
- A finite piece of a non-absolutely-convergent sum needs a separate justification: a theorem about a modified sum or numerical evidence.

3.11 Paper connection

Here is where this chapter's tools meet the paper [1], revision R1.

The zeros cannot be computed by summing the defining series (Sec. II A). The paper writes the zeros as $\rho_n = \frac{1}{2} + it_n$ and records $t_1 = 14.134725141734695$. At $s = \rho_n$ the Dirichlet series $\sum k^{-s}$ is not absolutely convergent and in fact diverges (Section 3.6, Figure 3.3). So the statement "$\zeta(\rho_n) = 0$" refers to the analytically continued function of Chapter 16, and the zeros themselves are computed by specialized methods (`mpmath.zetazero`, App. A step 1), not by adding terms of $\sum k^{-s}$.

Eq. (4) is a finite double sum. The arithmetic oracle of the paper (Sec. II D, Eq. (4)) is

$$N_{\mathfrak{fl}}(t) = -\frac{1}{\pi} \sum_{p \in P} \sum_{m=1}^{3} \frac{\sin(mt \log p)}{m\, p^{m/2}},$$

a sum of $3\,|P|$ sinusoids. It is truncated in two ways.

- *In m:* App. A step 8 states $m \leq 3$. For each prime the full m-series converges absolutely on the critical line, and Eq. (3.19) bounds what is dropped: at most 0.0487 for $p = 2$, and at most 0.0169 for $p = 3$ (Problem 15). These bounds are theorems.
- *In p:* the prime set P is finite (one prime, 3, 6 or 25 primes in Table V; 300 primes, "900 terms", in row A of Table IV). The full sum over all primes is not absolutely convergent for $t \neq 0$ (Section 3.10), so there is no tail bound from this chapter and the effect of adding primes is a matter of evidence.

The evidence (Table V and Table IV). On the principal holdout, zeros 401–500 (running example RE5, null rms 0.263), Table V reports rms errors (in mean spacings) of 0.198 with {2}, 0.116 with 3 primes, 0.074 with 6 primes and 0.043 with 25 primes, with correlations +0.66, +0.90, +0.96, +0.99. Table IV, row A, gives 0.018 with 300 primes, and Sec. IV states that the

oracle "improves monotonically" as the budget grows. Read with this chapter's eyes, these are *numerical observations* about a truncation of a non-absolutely-convergent sum. They show that the truncated sum is useful at these heights; they are not a proof that the sum converges, and the paper does not claim one (it calls Eq. (4) a truncation with "no novelty" and attributes its theory to Refs. [3, 4]). You can reproduce the Table V row yourself:

```
import numpy as np, qmnt
n = np.arange(401, 501)                      # holdout RE5
# the null predicts zero fluctuation, so its corr is undefined (nan)
print("null", qmnt.score(qmnt.smooth_zeros(n), n))
for k in [1, 3, 6, 25]:                      # prime budgets of Table V
    P = qmnt.first_primes(k)
    tp = qmnt.oracle_zeros(n, P, mmax=3)     # solve Nsm + Nfl = n - 1/2
    rms, corr = qmnt.score(tp, n)
    print(k, round(rms, 3), round(corr, 2))
```

The continuation probe uses the RE2 wave (Sec. III, App. A step 15). The paper's arithmetic-continuation probe replaces the true targets by the $p = 2$, $m \le 3$ part of Eq. (4), which is exactly $f_2(t)$ of Section 3.8, a finite sum of three sinusoids with period $2\pi/\log 2 \approx 9.065$. Our bound Eq. (3.19) tells you that choosing $m \le 3$ rather than all m changes this target by at most 0.0487 at any height. So a probe built from the full m-series would have used a target differing from the paper's by at most 0.0487. The choice $m \le 3$ is therefore immaterial to what the probe tests (in-sample rms 0.056, held-out correlation +0.07), namely whether the fit can continue an oscillation at all (Chapter 5).

The state sum of Eq. (5) is a truncated sum (Sec. II E). The second-order shift in Eq. (5),

$$\delta E_n = -\sum_{m \neq n} \frac{|A_{nm}|^2}{E_n - E_m},$$

runs over unperturbed states m (here m labels states, not prime powers; A is the non-Hermitian operator of Chapter 14. It is not the sum $A = \sum a_k$ of Section 3.9, nor the sensitivity matrix A_{nk} of the paper's Sec. II C; the paper overloads the letter.). The paper sums "over all 652 basis states with $E \le 1000$" and notes that "truncating at 500 states measurably biases the top levels". A plausible reading (ours, not the paper's): if the matrix elements $|A_{nm}|^2$ of nearby states are of comparable size, the terms are largest when E_m is close to E_n, so for the top levels the first omitted states are close in energy and contribute terms that are not small. A truncation is only as good as the size of what it leaves out, and for the top levels the leftover is not negligible. Chapters 13 and 14 develop the formula.

The sum rule rests on reordering a finite double sum (Sec. V and Sec. V A). Sec. V A states that $\sum_n \delta E_n = 0$ "exactly when summed over a complete truncated basis, since the contribution of each pair (n, m) is antisymmetric". Here is the general fact. Let b_{nm} be numbers, for n, m in a finite index set, with $b_{mn} = -b_{nm}$. Let

$$T = \sum_n \sum_{m \neq n} b_{nm}. \tag{3.24}$$

Rename the summation letters ($n \leftrightarrow m$); this changes nothing, so $T = \sum_m \sum_{n \neq m} b_{mn}$. A finite double sum may be summed in either order, so $T = \sum_n \sum_{m \neq n} b_{mn} = \sum_n \sum_{m \neq n} (-b_{nm}) = -T$. Hence $T = 0$. With $b_{nm} = -|A_{nm}|^2/(E_n - E_m)$, antisymmetry holds because swapping n and m flips the sign of the denominator and leaves $|A_{nm}|^2 = |A_{mn}|^2$ unchanged (true for a real

multiplication operator, as Chapter 14 shows). The word "complete" matters: if the sum over m is cut off for some n but the partner terms are kept for others, pairs are broken and the cancellation fails; for an infinite basis, the reordering step would additionally need absolute convergence (Section 3.6). The paper reports the sum verified to 4×10^{-15} in Sec. V, and to 3.6×10^{-15} (odd class) and 1.4×10^{-14} (even class) in Sec. V A: zero up to floating-point round-off (Chapter 4).

3.12 Summary

- A sequence converges to a if its terms eventually stay within any tolerance of a (Eq. (3.1)); $r^n \to 0$ for $|r| < 1$; complex sequences converge iff real and imaginary parts do. Monotone bounded sequences converge (taken on trust).

- A series converges when its partial sums converge; terms tending to zero is necessary, not sufficient.

- Geometric series: $\sum_{k=0}^{n-1} r^k = (1-r^n)/(1-r)$; sum $1/(1-r)$ and tail $r^n/(1-r)$ for $|r| < 1$ (Eqs. (3.2)–(3.4)).

- Telescoping: $\sum_{k=1}^{n}(b_k - b_{k+1}) = b_1 - b_{n+1}$ (Eq. (3.5)).

- The harmonic series diverges (Eq. (3.6)); $\log(n+1) \le H_n \le 1 + \log n$ and $H_n - \log n \to \gamma$ (Eqs. (3.7), (3.8)).

- Comparison and integral tests for nonnegative series; integral bounds on tails (Eqs. (3.9), (3.10)); $\sum k^{-s}$ converges iff $s > 1$ (Eq. (3.11)).

- Absolute convergence implies convergence, $|\sum a_k| \le \sum |a_k|$, and free rearrangement (Eq. (3.12)). Since $|k^{-s}| = k^{-\sigma}$ (Eq. (3.13)), $\sum k^{-s}$ converges absolutely for $\operatorname{Re} s > 1$ with tail at most $n^{1-\sigma}/(\sigma - 1)$ (Eq. (3.14)); it diverges on the critical line.

- Alternating series converge with error at most the first omitted term (Eq. (3.15)); conditionally convergent series can change their sum under rearrangement.

- Ratio test; radius of convergence; $-\log(1-x) = \sum x^m/m$ for $-1 \le x < 1$ with explicit remainder, giving $\log 2$ (Eqs. (3.16)–(3.18)); $-\log(1-p^{-s}) = \sum_m p^{-ms}/m$ for $\operatorname{Re} s > 0$.

- Truncating the prime-2 wave at $m \le 3$ costs at most 0.0487 (Eq. (3.19)).

- Infinite products converge to nonzero limits; logs turn them into sums (Eqs. (3.20), (3.21)); $\prod(1 + a_k)$ with $a_k \ge 0$ converges iff $\sum a_k$ does, with $1 + A \le \prod \le e^A$ (Eq. (3.22)); the Euler product converges for $s > 1$ (Eq. (3.23)).

- Truncations have controlled error only for absolutely convergent series (Table 3.2); the critical-line prime sum is not absolutely convergent for $t \neq 0$ (its $m = 1, 2$ amplitude sums diverge, and, as taken on trust, so do its actual moduli), so the success of Eq. (4) of the paper is a numerical observation.

- Antisymmetric pair contributions cancel in a finite double sum (Eq. (3.24)), the basis of the paper's sum rule.

3.13 Problems

1. • Compute the partial sum $S_4 = \sum_{k=0}^{3} (1/3)^k$ as a fraction, and the sum $\sum_{k=0}^{\infty} (1/3)^k$.

2. • Decide whether each series converges, naming the test you use: (a) $\sum_{k\ge 1} k^{-1.01}$; (b) $\sum_{k\ge 1} k^{-1/2}$; (c) $\sum_{k\ge 1} \frac{k}{k^2+1}$; (d) $\sum_{k\ge 1} 3^{-k}$.

3. • Use partial fractions and telescoping to find $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)}$ and its limit as $n \to \infty$.

4. • For $s = \frac{1}{2} + 21.022040\,i$ (the second zero, RE1), find $|3^{-s}|$ and the angular frequency in t at which the phase of 3^{-s} rotates.

5. • True or false? (a) If $a_k \to 0$ then $\sum a_k$ converges. (b) If $\sum a_k$ converges then $a_k \to 0$. (c) An absolutely convergent series converges. (d) A convergent series converges absolutely.

6. •• Show that $\sum_{k=1}^{\infty} 1/k^2 \le \frac{7}{4}$ by comparing $1/k^2$ with $1/(k(k-1))$ only for $k \ge 3$.

7. •• Compute $S_5 = \sum_{k=1}^{5} k^{-3}$ and use Eq. (3.10) to bracket $\zeta(3)$.

8. •• Show that $H_n - \log(n+1)$ is increasing in n, and deduce that $\gamma \ge 1 - \log 2$.

9. •• How many terms of $\sum_{k\ge 1} (-1)^{k+1}/k^2$ guarantee an error at most 10^{-4}? Using the fact that this series is absolutely convergent, show that its sum is $\frac{1}{2}\zeta(2) = \pi^2/12$.

10. •• Find the radius of convergence of (a) $\sum m^2 x^m$; (b) $\sum x^m/m!$; (c) $\sum m!\, x^m$; (d) $\sum 2^{-m} x^m$.

11. •• Compute $\log 2$ from Eq. (3.17) at $x = \frac{1}{2}$ (note $-\log\frac{1}{2} = \log 2$). How many terms does the remainder bound require for an error below 10^{-6}? Compare with the number of terms the alternating series of Eq. (3.18) needs for the same guarantee.

12. •• Show that $\sum_{k=2}^{n} \log(1 - 1/k) = -\log n$ and explain, using Eq. (3.21), why $\prod_{k\ge 2}(1 - 1/k)$ diverges to zero. Relate this to the divergence of the harmonic series.

13. •• Show that $\prod_{k=1}^{\infty}(1 + 1/k^2)$ converges and use Eq. (3.22) to bound it. (Its exact value is $\sinh(\pi)/\pi$; check that it lies within your bounds.)

14. •• For real θ, find $\sum_{k=0}^{\infty} (e^{i\theta}/2)^k$ and, by taking the real part, show $\sum_{k=0}^{\infty} \frac{\cos k\theta}{2^k} = \frac{4 - 2\cos\theta}{5 - 4\cos\theta}$.

15. •• (Paper.) Eq. (4) of the paper keeps $m \le 3$ for each prime. Bound the neglected part $\frac{1}{\pi}\sum_{m\ge 4} \frac{p^{-m/2}}{m}$ for $p = 3$ exactly (using Eq. (3.17)) and with the crude geometric bound $\frac{1}{4\pi}\frac{p^{-2}}{1 - p^{-1/2}}$. Compare with $p = 2$ (Eq. (3.19)).

16. •• (Paper.) Prove the pair-cancellation identity of Eq. (3.24) carefully for the index set $\{1, 2, 3\}$ by writing all six terms. Then explain why Sec. V A of the paper insists on a "complete truncated basis".

17. •• (Paper.) From Table V of the paper, compute the ratio of the oracle rms to the null rms on zeros 401–500 for each prime budget. Does the decrease with more primes prove that the prime sum converges on the critical line? Explain using Section 3.10.

18. ••• Use the integral test to prove $\sum_{k=1}^{n} k^{-1/2} \ge 2\sqrt{n+1} - 2$. Deduce that on the critical line the moduli of the terms of $\sum k^{-s}$ have infinite sum, and estimate the n at which that sum of moduli first exceeds 100.

19. •• Using numpy, compute H_n for $n = 10^6$ and compare it with $\log n + \gamma + \frac{1}{2n}$.

20. •• Using numpy, compute $\zeta(2) - S_n$ for $n = 10^2, 10^3, \ldots, 10^6$ and show that $n(\zeta(2) - S_n)$ approaches 1. Explain this from Eq. (3.10).

21. •• Using qmnt (the listing in Section 3.11), compute the oracle rms and correlation on zeros 401–500 for the first 1, 3, 6 and 25 primes, and compare with Table V of the paper.

22. •• Using qmnt and numpy, compute the partial Euler products $\Pi(x) = \prod_{p \le x} (1 - p^{-2})^{-1}$ for x up to 1000 and plot their ratio to $\pi^2/6$ against x. Estimate the relative deficit at $x = 100$.

23. •• Using qmnt and numpy, compute $|S_n|$ for $S_n = \sum_{k=1}^{n} k^{-\rho_1}$ ($\rho_1 = \frac{1}{2} + it_1$) at $n = 100$, 1000, 10^4, and compare with $\sqrt{n}/\,|1 - \rho_1|$.

24. ••• Let $0 \le a_k \le \frac{1}{2}$. Show $-\log(1-a) \le 2a$ for $0 \le a \le \frac{1}{2}$, and deduce that if $\sum a_k$ converges then $\prod (1 - a_k)$ converges to a nonzero limit.

25. ••• Consider the rearrangement of the alternating harmonic series with two positive terms followed by one negative: $1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \cdots$. Using $H_n = \log n + \gamma + \epsilon_n$ with $\epsilon_n \to 0$, show that it converges to $\frac{3}{2}\log 2$.

26. •• (Physics.) In the bouncing-ball Sample Problem, what rebound fraction r makes the total time exactly three times the first fall time τ? Is the total distance finite for every $r < 1$?

27. ••• (Paper.) Show that the $m \ge 3$ part of the critical-line prime sum converges absolutely: prove $\frac{1}{\pi}\sum_p \sum_{m \ge 3} \frac{p^{-m/2}}{m} \le \frac{1}{3\pi} \cdot \frac{1}{1 - 2^{-1/2}} (\zeta(\frac{3}{2}) - 1)$ and evaluate the bound (use $\zeta(3/2) = 2.612375$). Why does this method give no bound for the $m = 1$ and $m = 2$ parts?

28. •• (Paper.) Eq. (5) of the paper sums over 652 states, and the paper notes that truncating at 500 states biases the top levels. Assuming the matrix elements $|A_{nm}|^2$ are of similar size for states near each other, explain why the bias is largest for the levels closest to the cutoff and smallest for low levels.

29. • (Paper.) How many sinusoidal terms does Eq. (4) of the paper contain for each prime budget of Table V (1, 3, 6, 25 primes)? Check that 300 primes give the "900 terms" of row A in Table IV.

30. ••• Use the integral test to show that $\sum_{k \ge 2} \frac{1}{k (\log k)^q}$ converges if and only if $q > 1$. Why is this series a warning about the ratio test?

3.14 Answers to selected problems

1. By Eq. (3.2), $S_4 = \frac{1 - (1/3)^4}{1 - 1/3} = \frac{80/81}{2/3} = \frac{40}{27} = 1.481481$. The sum is $\frac{1}{1 - 1/3} = \frac{3}{2}$.

2. (a) Converges: p-series with $s = 1.01 > 1$. (b) Diverges: p-series with $s = \frac{1}{2} \le 1$. (c) Diverges: for $k \ge 1$, $k^2 + 1 \le 2k^2$, so $\frac{k}{k^2+1} \ge \frac{k}{2k^2} = \frac{1}{2k}$, and $\sum \frac{1}{2k}$ diverges; use comparison. (d) Converges: geometric with ratio $\frac{1}{3}$; the sum is $\frac{1/3}{1 - 1/3} = \frac{1}{2}$.

3. $\frac{1}{2k-1} - \frac{1}{2k+1} = \frac{2}{(2k-1)(2k+1)}$, so the term is $\frac{1}{2}(\frac{1}{2k-1} - \frac{1}{2k+1})$. With $b_k = \frac{1}{2k-1}$ (so $b_{k+1} = \frac{1}{2k+1}$), Eq. (3.5) gives $\sum_{k=1}^{n} = \frac{1}{2}(1 - \frac{1}{2n+1}) = \frac{n}{2n+1} \to \frac{1}{2}$.

4. By Eq. (3.13), $|3^{-s}| = 3^{-1/2} = 0.577350$, independent of t. The phase is $e^{-it \log 3}$, rotating at angular frequency $\log 3 = 1.098612$ per unit of t.

5. (a) False: the harmonic series. (b) True: the term test. (c) True: Section 3.6. (d) False: the alternating harmonic series converges but not absolutely.

7. $S_5 = 1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \frac{1}{125} = 1.185662$. With $\int_a^\infty x^{-3} \mathrm{d}x = \frac{1}{2a^2}$, the tail lies between $\frac{1}{72} = 0.013889$ and $\frac{1}{50} = 0.02$. So $1.199551 \le \zeta(3) \le 1.205662$. The known value $\zeta(3) = 1.202057$ lies inside.

9. The terms $1/k^2$ decrease to zero, so Eq. (3.15) applies: we need $1/(n+1)^2 \le 10^{-4}$, i.e. $n + 1 \ge 100$, so $n = 99$ terms. For the sum: since $\sum 1/k^2$ converges, we may reorder freely.

Then $\sum(-1)^{k+1}/k^2 = \sum_k 1/k^2 - 2\sum_{k\text{ even}} 1/k^2$, and $\sum_{j\ge1} 1/(2j)^2 = \frac{1}{4}\zeta(2)$. So the sum is $\zeta(2) - \frac{1}{2}\zeta(2) = \frac{1}{2}\zeta(2) = \pi^2/12 = 0.822467$.

11. With $x = \frac{1}{2}$, the remainder bound for $0 \le x < 1$ is $\frac{x^{n+1}}{(n+1)(1-x)} = \frac{2^{-n}}{n+1}$. For $n = 15$ this is 1.9×10^{-6}; for $n = 16$ it is $9.0 \times 10^{-7} < 10^{-6}$. So 16 terms suffice. The alternating series needs $\frac{1}{n+1} \le 10^{-6}$, i.e. $n = 999999$ terms. Choosing x well inside the radius of convergence makes the geometric factor x^n do the work.

13. With $a_k = 1/k^2 \ge 0$ and $\sum a_k = \zeta(2) = \pi^2/6$ finite, the product converges. Eq. (3.22): $1 + \pi^2/6 = 2.644934 \le \prod \le e^{\pi^2/6} = 5.180668$. The exact value $\sinh(\pi)/\pi = 3.676078$ lies inside.

15. For $p = 3$: $3^{-1/2} = 0.577350$; Eq. (3.17) gives $\sum_{m\ge1} \frac{3^{-m/2}}{m} = -\log(1 - 0.577350) = 0.861212$; the first three terms are $0.577350 + \frac{1/3}{2} + \frac{3^{-3/2}}{3} = 0.808167$. The neglected part is $\frac{0.053045}{\pi} = 0.016885$. The crude bound is $\frac{1}{4\pi}\frac{1/9}{1-0.577350} = 0.020920$. Both are smaller than the $p = 2$ value 0.048698: the neglected powers matter least for large primes, since the bound shrinks roughly like p^{-2}.

17. Ratios: $0.198/0.263 = 0.753$; $0.116/0.263 = 0.441$; $0.074/0.263 = 0.281$; $0.043/0.263 = 0.163$. No. The decrease is a numerical observation at these heights. The critical-line prime sum is not absolutely convergent (for $t \ne 0$ its $m = 1$ and $m = 2$ parts have divergent sums of moduli; even their amplitude sums diverge, so comparison gives no tail bound), so convergence theory gives no tail bound, and a decreasing error over four budgets cannot by itself establish that an infinite sum exists. The paper's claim is correspondingly about predictive performance of the truncations, not about convergence.

19. Summing with numpy gives $H_{10^6} = 14.392727$, and $\log 10^6 + \gamma + \frac{1}{2\cdot10^6} = 13.815511 + 0.577216 + 0.0000005 = 14.392727$. They agree to about 10^{-13}; the next correction is $-\frac{1}{12n^2} \approx -8 \times 10^{-14}$.

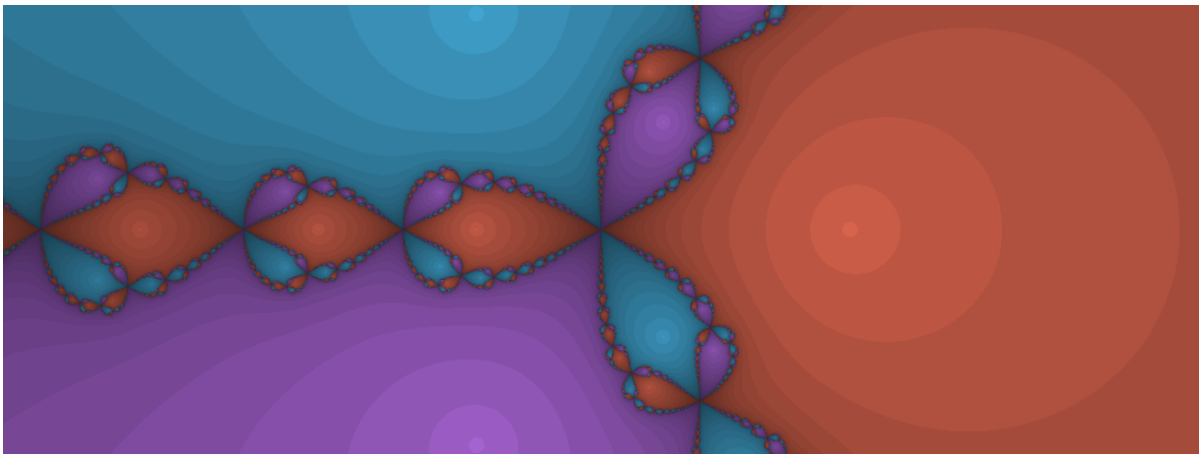
21. The listing reproduces Table V's first row: null 0.263; {2}: 0.198/ + 0.66; 3 primes: 0.116/ + 0.90; 6 primes: 0.074/ + 0.96; 25 primes: 0.043/ + 0.99 (rms in mean spacings / correlation).

23. With $|1 - \rho_1| = \sqrt{0.25 + t_1^2} = 14.1436$, $\sqrt{n}/\,|1 - \rho_1| = 0.707$, 2.236, 7.070 for $n = 100$, 1000, 10^4. The computed $|S_n|$ agree with these to within about $0.5/\sqrt{n}$ (that is, 0.05, 0.016, 0.005), the size of the correction term $\frac{1}{2}n^{-s}$ in the beyond box of Section 3.6. The partial sums grow without bound: the series diverges at the zero.

25. After n blocks (two positive terms, one negative) the partial sum is $\sum_{k=1}^{2n}\frac{1}{2k-1} - \sum_{k=1}^{n}\frac{1}{2k}$. The odd reciprocals up to $4n-1$ are $H_{4n} - \frac{1}{2}H_{2n}$ (all reciprocals up to $4n$ minus the even ones), and $\sum_{k=1}^{n}\frac{1}{2k} = \frac{1}{2}H_n$. So the block partial sum is $H_{4n} - \frac{1}{2}H_{2n} - \frac{1}{2}H_n = \log 4n - \frac{1}{2}\log 2n - \frac{1}{2}\log n + (\gamma - \frac{1}{2}\gamma - \frac{1}{2}\gamma) + (\epsilon_{4n} - \frac{1}{2}\epsilon_{2n} - \frac{1}{2}\epsilon_n)$. The logs combine to $\log 4 - \frac{1}{2}\log 2 = \frac{3}{2}\log 2$, the γ's cancel and the ϵ's tend to zero. Partial sums inside a block differ by terms tending to zero, so the series converges to $\frac{3}{2}\log 2 = 1.039721$.

27. For each prime and $m \ge 3$, $\frac{1}{m} \le \frac{1}{3}$, and $\sum_{m\ge3} p^{-m/2} = \frac{p^{-3/2}}{1-p^{-1/2}} \le \frac{p^{-3/2}}{1-2^{-1/2}}$ (geometric series; $p \ge 2$ makes the denominator smallest at $p = 2$). Summing over primes and enlarging to all integers $k \ge 2$: $\sum_p p^{-3/2} \le \sum_{k\ge2} k^{-3/2} = \zeta(\frac{3}{2}) - 1 = 1.612375$. So the total is at most $\frac{1}{3\pi} \times 3.414214 \times 1.612375 = 0.584$. For $m = 1$ and $m = 2$ the same method bounds the moduli by the amplitude sums $\frac{1}{\pi}\sum_p p^{-1/2}$ and $\frac{1}{2\pi}\sum_p p^{-1}$. Both are infinite because $\sum_p 1/p$ diverges (first beyond box of Section 3.10), so the method yields no finite bound uniform in t. The sums of the actual moduli also diverge for $t \ne 0$; this is true but needs more than this chapter (second beyond box of Section 3.10).

29. Each prime contributes 3 terms ($m = 1, 2, 3$): {2} gives 3, 3 primes give 9, 6 primes give 18, 25 primes give 75 terms. 300 primes give $3 \times 300 = 900$ terms, matching row A of Table IV.



Newton's method for $z^3 = 1$: each starting point is coloured by the root it reaches; the boundaries are fractal.

Chapter 4

Approximation, Asymptotics and Root Finding

↑ Contents

What this chapter is about

Almost nothing in the paper can be computed with pencil and paper alone. The smooth zeros s_n are defined as the solutions of an equation, $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$, that has no solution in terms of logarithms and powers. The Wu–Sprung potential is defined backwards: the paper gives the position x as a function of the height V, and the computer has to turn this around to get V as a function of x. The truncated explicit formula predicts a zero by solving yet another equation. Every one of these steps is *root finding*, and the paper names the algorithm it uses: Brent's method.

Before you can find a root you need to know roughly where it is, and before you can trust a number you need to know how large its error is. So this chapter has three threads. The first is *asymptotics*: the language of big-O, little-o and $\sim$, which lets you say precisely that one quantity grows like another, and the tools (sums compared with integrals, Stirling's formula for factorials and the Gamma function) that produce such statements. The second is *root finding*: inverting monotone functions, bisection with a guaranteed error bound, Newton's method with its very fast convergence, and Brent's method, which combines the safety of the first with the speed of the second. The third is *trust*: absolute and relative errors, significant digits, and the floating-point numbers that computers actually use, including the surprising fact that a quantity that is exactly zero on paper can come out of a computer as a number of size 10^{-7}.

Where this shows up in the paper: solving Eq. (1) for the smooth zeros (Sec. II A and Appendix A step 2), inverting Eq. (2) (Sec. II B, Appendix A step 3), solving the oracle's level equation with Brent's method bracketed by ± 3 around the smooth zero (Appendix A step 8), the Lambert-W form of the smooth zeros mentioned in Sec. I A, the root ambiguity flagged in Table IV, and the float32 noise artifact that manufactured a spurious result in Sec. V.

What you need

- Chapter 2: the Taylor series of e^x; complex numbers, modulus and argument, the logarithm of a complex number and its branches (the last three are needed only for one box on the complex Gamma function).
- Chapter 3: limits of sequences, partial sums, the geometric series, the Taylor series of $\log(1-x)$, the integral test.
- First-year calculus: derivatives (chain and product rules), integrals, integration by parts, substitution, the tangent line to a graph, the Taylor series of $\sin x$ and $\cos x$.
- Throughout, log means the natural logarithm, as in the paper.

4.1 Errors, relative errors and significant digits

Every number a computer gives you is an approximation. Even the first Riemann zero, which the paper quotes as $t_1 = 14.134725141734695$ (Sec. II A), is a rounded version of an infinitely long decimal. Before we can talk about how good an approximation is, we need words for the size of the mistake.

4.1.1 Absolute and relative error

Let x be a true value and $\hat{x}$ an approximation to it.

Definition. The *absolute error* is $\hat{x} - x$ (sometimes its magnitude $|\hat{x} - x|$ is meant). The *relative error* is

$$\epsilon_{\text{rel}} = \frac{\hat{x} - x}{x} \qquad (x \neq 0). \tag{4.1}$$

The absolute error carries the units of x; the relative error is a pure number. Which one matters depends on the question. If you measure the length of a football field to within 1 cm you have done very well; if you measure the thickness of a coin to within 1 cm you have learned nothing. The absolute errors are the same; the relative errors differ by a factor of about 50 000 (about 10^{-4} for a 100 m field against about 5 for a 2 mm coin).

Errors in natural units. Physicists often measure an error in units of some natural scale of the problem. For the zeta zeros the natural scale is the *local mean spacing*: the typical distance between neighboring zeros near a given height. The paper writes it gap_n and quotes every error in these units ("all errors below are quoted in units of the local mean spacing", Sec. II A). We derive the formula $\text{gap}_n = 2\pi/\log(s_n/2\pi)$ in Section 4.5; for now, all you need is that near the first zero the mean spacing is about 7.500, and near the thousandth zero it is about 1.159. An error of 0.01 in t is therefore a much larger fraction of a spacing at the thousandth zero than at the first.

4.1.2 Significant digits

Definition. A number written in decimal has d *significant digits* if it is written with d digits starting from the first nonzero one. When a true value x is rounded to d significant digits, the rounding changes it by at most half a unit in the last place kept, and the relative error then satisfies (proof immediately below)

$$|\epsilon_{\text{rel}}| \leq 5 \times 10^{-d}. \tag{4.2}$$

Here is why. Write $x = m \times 10^q$ with $1 \leq m < 10$ and q an integer. Keeping d significant digits means keeping m to $d-1$ decimal places, so the rounding error in m is at most

$\frac{1}{2} \times 10^{-(d-1)} = 5 \times 10^{-d}$. The absolute error in x is therefore at most $5 \times 10^{-d} \times 10^{q}$, and dividing by $|x| = m \times 10^{q} \geq 10^{q}$ gives Equation (4.2).

Conversely, we say loosely that an approximation is "good to d significant digits" when its relative error is about 10^{-d} or smaller.

Sample Problem 4.1: How wrong is $t_1 \approx 14.13$?

The first zero is $t_1 = 14.134725141734695$ (paper, Sec. II A). You round it to 14.13. Find the absolute error, the relative error, and the error in units of the local mean spacing $\mathrm{gap}_1 = 7.500$ (derived in Section 4.5).
Step 1: Absolute error: $14.13 - 14.134725 = -0.004725$, so $|\hat{x} - x| = 0.004725$.
Step 2: Relative error: $0.004725/14.134725 = 3.34 \times 10^{-4}$. This is below the bound 5×10^{-4} of Equation (4.2) for $d = 4$ digits, as it must be.
Step 3: In mean spacings: $0.004725/7.500 = 6.30 \times 10^{-4}$.

$$\boxed{|\Delta t| = 0.004725, \quad |\epsilon_{\mathrm{rel}}| = 3.34 \times 10^{-4}, \quad 6.30 \times 10^{-4} \text{ spacings}}$$

Even a crude four-digit value of t_1 is accurate to better than a thousandth of a spacing. The paper's quoted rms errors (such as 0.263 spacings) are hundreds of times larger, so they are not rounding effects.

Checkpoint 4.1

A friend reports an error of 0.001 in a computed zero and says "that is negligible". What extra information do you need before agreeing?

Answer: You need a scale. An absolute error of 0.001 is negligible compared with the mean spacing (7.5 near t_1, about 1.16 near t_{1000}) but would be enormous if the quantity were itself of size 10^{-4}. Ask for the relative error or the error in mean spacings.

Key Ideas

- Absolute error $\hat{x} - x$ has units; relative error $(\hat{x} - x)/x$ is dimensionless, Equation (4.1).
- Rounding to d significant digits gives a relative error at most 5×10^{-d}, Equation (4.2).
- The paper measures errors in units of the local mean spacing gap_n, a natural scale that shrinks with height.

4.2 Orders of growth: $\mathcal{O}$, o and $\sim$

★ Did you know? *Big O Is Older Than Computers*

The symbol $\mathcal{O}$ was introduced by Paul Bachmann in an 1894 book on analytic number theory. Edmund Landau made it famous in his 1909 treatise on the distribution of primes, which is why $\mathcal{O}$ and o are called Landau symbols. Only in 1976 did Donald Knuth standardize Ω and Θ for computer science. Asymptotic notation was born in the study of primes.

The smooth counting function of the paper, Eq. (1),

$$N_{\mathrm{sm}}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{4.3}$$

has three terms. Which one matters most for large t? How fast does the whole thing grow? To answer such questions crisply, mathematicians and physicists use three symbols.

4.2.1 Definitions

Let $f(t)$ and $g(t)$ be functions defined for large t, with $g(t) \neq 0$ there.

Big-O. We write $f(t) = \mathcal{O}(g(t))$ as $t \to \infty$ if there are constants $C > 0$ and t_0 such that

$$|f(t)| \leq C\,|g(t)| \qquad \text{for all } t \geq t_0. \tag{4.4}$$

In words: f grows no faster than a fixed multiple of g.

Little-o. We write $f(t) = o(g(t))$ as $t \to \infty$ if

$$\lim_{t\to\infty} \frac{f(t)}{g(t)} = 0. \tag{4.5}$$

In words: f becomes negligible compared with g.

Asymptotic equivalence. We write $f(t) \sim g(t)$ as $t \to \infty$ if

$$\lim_{t\to\infty} \frac{f(t)}{g(t)} = 1. \tag{4.6}$$

Equivalently, $f(t) = g(t) + o(g(t))$: the relative difference goes to zero.

The same symbols are used as $x \to 0$ with "for all $t \geq t_0$" replaced by "for all $0 < |x| \leq x_0$". For example, from the Taylor series $\cos x = 1 - x^2/2 + x^4/24 - \cdots$ we have $1 - \cos x = \frac{1}{2}x^2 + \mathcal{O}(x^4)$ as $x \to 0$.

4.2.2 Examples and rules

- $5t^2 + 3t = \mathcal{O}(t^2)$: for $t \geq 1$, $3t \leq 3t^2$, so $|5t^2 + 3t| \leq 8t^2$; take $C = 8$, $t_0 = 1$.
- $5t^2 + 3t \sim 5t^2$, since $(5t^2 + 3t)/(5t^2) = 1 + 3/(5t) \to 1$.
- $\sin t = \mathcal{O}(1)$, because $|\sin t| \leq 1$.
- $t^2 + t \sim t^2$ but the difference $(t^2 + t) - t^2 = t$ grows without bound. *Asymptotic equivalence controls the ratio, not the difference.*

Three rules follow directly from the definitions. If $f_1 = \mathcal{O}(g)$ and $f_2 = \mathcal{O}(g)$ then $f_1 + f_2 = \mathcal{O}(g)$ (add the two inequalities and use the larger t_0; the constant becomes $C_1 + C_2$). If $f_1 = \mathcal{O}(g_1)$ and $f_2 = \mathcal{O}(g_2)$ then $f_1 f_2 = \mathcal{O}(g_1 g_2)$ (multiply the inequalities; the constant becomes $C_1 C_2$). And $o(g)$ implies $\mathcal{O}(g)$ (if the ratio tends to zero it is eventually below 1).

Pitfall

The equals sign in $f = \mathcal{O}(g)$ is a one-way statement. We write $t = \mathcal{O}(t^2)$ and $t^2 = \mathcal{O}(t^2)$, but you may not conclude $t = t^2$. Read "$= \mathcal{O}(g)$" as "is a function bounded by a multiple of g".

4.2.3 Logarithms lose to every power

The single most useful fact about growth rates in this book is:

$$\log t = o(t^a) \quad \text{as } t \to \infty, \text{ for every } a > 0. \tag{4.7}$$

Proof. Substitute $t = e^u$, so $u = \log t \to \infty$ as $t \to \infty$ and $t^a = e^{au}$. We need $u/e^{au} \to 0$. The Taylor series $e^y = 1 + y + y^2/2 + \cdots$ (Chapter 2) has only positive terms when $y > 0$, so $e^y > y^2/2$. With $y = au$:

$$\frac{u}{e^{au}} < \frac{u}{(au)^2/2} = \frac{2}{a^2 u} \longrightarrow 0.$$

□

Two relatives follow in the same way. First, for $y > 0$ every term of the Taylor series of e^y is positive, so $e^y > y^{k+1}/(k+1)!$ for any whole number $k \geq 0$; hence $y^k/e^y < (k+1)!/y \to 0$ as $y \to \infty$. This says that an exponential beats any power, $t^k = o(e^t)$. Second, with $t = e^u$ and $y = au$, $(\log t)^k/t^a = u^k/e^{au} = a^{-k}\, y^k/e^y \to 0$, so a power beats any power of a logarithm, $(\log t)^k = o(t^a)$.

Pitfall

"Eventually" can be a long time. The ratio $\log t/t^{0.1}$ tends to zero, yet it first *increases*: with $u = \log t$ the ratio is $ue^{-0.1u}$, whose derivative $e^{-0.1u}(1 - 0.1u)$ vanishes at $u = 10$. The maximum, at $t = e^{10} \approx 2.20 \times 10^4$, is $10/e = 3.68$. Even at $t = 10^{30}$ the ratio is still 0.069. Asymptotic statements say nothing about the size of the constants or the place where the limit "kicks in".

4.2.4 Leading-order behavior

The *leading-order behavior* of f is the simplest function g with $f \sim g$. To find it, identify the term that dominates all others and check that the others are little-o of it.

Sample Problem 4.2: Leading order of the smooth counting function

Find the leading-order behavior of $N_{\text{sm}}(t)$ in Equation (4.3), write the remainder with $\mathcal{O}$, and test how good the leading term is at the thousandth zero, $t_{1000} = 1419.42248094599$ (paper, Sec. II A).

Step 1: Expand the logarithm: $\log(t/2\pi) = \log t - \log 2\pi$. Then

$$N_{\text{sm}}(t) = \frac{t}{2\pi}\log t - \frac{t}{2\pi}(\log 2\pi + 1) + \frac{7}{8}.$$

Step 2: Compare each term with $\frac{t}{2\pi}\log t$. The second term divided by the first is $-(\log 2\pi + 1)/\log t \to 0$. The third divided by the first is $\frac{7}{8} \cdot 2\pi/(t \log t) \to 0$. So both are little-o of the first, and

$$N_{\text{sm}}(t) \sim \frac{t}{2\pi}\log t, \qquad N_{\text{sm}}(t) = \frac{t}{2\pi}\log t + \mathcal{O}(t).$$

(The remainder $-\frac{t}{2\pi}(\log 2\pi + 1) + \frac{7}{8}$ is bounded by a constant times t for $t \geq 1$.)

Step 3: Evaluate at t_{1000}. Then $t/2\pi = 225.908$, $\log(t/2\pi) = 5.4201$, and Equation (4.3) gives $N_{\text{sm}}(t_{1000}) = 999.42$. The leading term gives $225.908 \times \log(1419.42) = 225.908 \times 7.2580 = 1639.6$.

Step 4: The ratio is $1639.6/999.42$:

$$\boxed{\left.\frac{(t/2\pi)\log t}{N_{\text{sm}}(t)}\right|_{t=t_{1000}} = 1.64}$$

So at the heights used in the paper, the "leading term" overestimates by 64%. The ratio does tend to 1, but only as fast as $1/\log t$ shrinks, which is very slowly. This is why the paper always works with the full Equation (4.3), never with its leading term.

Checkpoint 4.2

True or false: if $f \sim g$ then $f - g \to 0$.

Answer: False. $t^2 + t \sim t^2$, but the difference is t, which grows. Only the ratio tends to 1.

4.2.5 Asymptotic expansions

Often we can do better than the leading term. A sequence of "gauge functions" $\varphi_0(t), \varphi_1(t), \dots$ with $\varphi_{k+1} = o(\varphi_k)$ (for example $1, 1/t, 1/t^2, \dots$) defines an *asymptotic expansion*

$$f(t) \approx \sum_k a_k \varphi_k(t)$$

when, for every K, the error after K terms is $\mathcal{O}(\varphi_K)$: $f(t) - \sum_{k=0}^{K-1} a_k \varphi_k(t) = \mathcal{O}(\varphi_K(t))$. The key point is that this is a statement about each *fixed* truncation as $t \to \infty$, not about the infinite sum. Stirling's series in Section 4.4 is the classic example.

Key Ideas

- $f = \mathcal{O}(g)$: bounded by a multiple of g; $f = o(g)$: negligible relative to g; $f \sim g$: ratio tends to 1 (Equations (4.4) to (4.6)).
- Logarithms grow more slowly than any positive power, Equation (4.7), but the crossover can be very late.
- $N_{\mathrm{sm}}(t) \sim \frac{t}{2\pi} \log t$, yet at t_{1000} the leading term is off by 64%: logarithmic corrections converge slowly.

4.3 Comparing sums with integrals

Sums are hard; integrals are easy. A sum of values of a smooth monotone function is squeezed between two integrals, and this squeeze is the source of many asymptotic formulas, including Stirling's.

4.3.1 The squeeze

Theorem (comparison of sums and integrals). Let f be positive and decreasing on $[1, \infty)$. Then for every integer $n \geq 1$,

$$\int_1^{n+1} f(x)\,\mathrm{d}x \;\leq\; \sum_{k=1}^{n} f(k) \;\leq\; f(1) + \int_1^{n} f(x)\,\mathrm{d}x. \tag{4.8}$$

Proof. On the interval $[k, k+1]$ a decreasing function satisfies $f(k+1) \leq f(x) \leq f(k)$. Integrating over this unit-length interval,

$$f(k+1) \leq \int_k^{k+1} f(x)\,\mathrm{d}x \leq f(k).$$

Sum the right inequality over $k = 1, \dots, n$: $\int_1^{n+1} f\,\mathrm{d}x \le \sum_{k=1}^{n} f(k)$, the left half of Equation (4.8). Sum the left inequality over $k = 1, \dots, n-1$: $\sum_{k=2}^{n} f(k) \le \int_1^n f\,\mathrm{d}x$; add $f(1)$ to both sides to get the right half. □

Geometrically (Figure 4.1), the sum is the total area of unit-width rectangles of heights $f(1), f(2), \dots$, which sit partly above and partly below the curve.

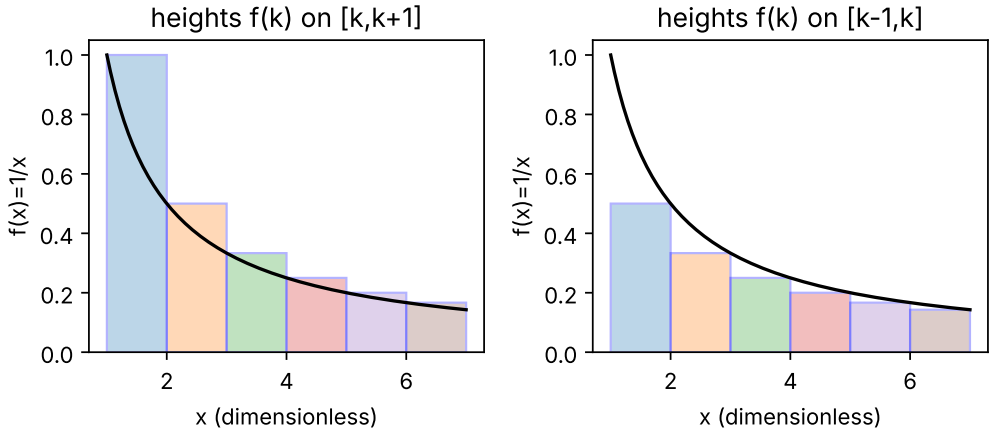


Figure 4.1: Comparing a sum with an integral for the decreasing function $f(x) = 1/x$. Left: rectangles of height $f(k)$ on $[k, k+1]$ lie above the curve, so $\sum_{k=1}^{n} f(k) \ge \int_1^{n+1} f\,\mathrm{d}x$. Right: rectangles of height $f(k)$ on $[k-1, k]$ lie below the curve, so $\sum_{k=2}^{n} f(k) \le \int_1^n f\,\mathrm{d}x$. The axes are dimensionless.

For an *increasing* positive function the inequalities reverse. On $[k, k+1]$ we now have $f(k) \le f(x) \le f(k+1)$, and the same summing gives

$$\int_1^n f(x)\,\mathrm{d}x \;\le\; \sum_{k=2}^{n} f(k) \;\le\; \int_1^n f(x)\,\mathrm{d}x + f(n) - f(1) \quad (f \text{ increasing}). \tag{4.9}$$

In detail: integrating $f(k) \le f(x) \le f(k+1)$ over the unit interval $[k, k+1]$ gives $f(k) \le \int_k^{k+1} f\,\mathrm{d}x \le f(k+1)$. Summing the right inequality over $k = 1, \dots, n-1$ gives $\int_1^n f\,\mathrm{d}x \le \sum_{k=2}^{n} f(k)$, the left half of Equation (4.9). Summing the left inequality over $k = 1, \dots, n-1$ gives $\sum_{k=1}^{n-1} f(k) \le \int_1^n f\,\mathrm{d}x$; adding $f(n) - f(1)$ to both sides and using $\sum_{k=1}^{n-1} f(k) + f(n) - f(1) = \sum_{k=2}^{n} f(k)$ gives the right half.

4.3.2 The harmonic numbers and Euler's constant

The *harmonic number* is $H_n = \sum_{k=1}^{n} 1/k$. In Chapter 3 you saw that $H_n \to \infty$. How fast? Apply Equation (4.8) with $f(x) = 1/x$, using $\int_1^X \mathrm{d}x/x = \log X$:

$$\log(n+1) \le H_n \le 1 + \log n. \tag{4.10}$$

Dividing by $\log n$ and letting $n \to \infty$ (note $\log(n+1)/\log n \to 1$) squeezes $H_n/\log n \to 1$: $H_n \sim \log n$.

The difference $\gamma_n = H_n - \log n$ actually converges (Problem 26 asks you to prove it). Its limit is *Euler's constant* $\gamma = 0.5772156649\ldots$ (a standard constant).

Taken on trust

Refining H_n. Euler's summation formula expresses a sum $\sum_{k=1}^{n} f(k)$ as the integral $\int_1^n f\,\mathrm{d}x$ plus correction terms built from f and its derivatives at the endpoints. Applied to $f(x) = 1/x$ it gives the expansion Equation (4.11) below. We take it on trust; the

proof is in [23].

The expansion, taken on trust as stated in the box, reads

$$H_n = \log n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + \mathcal{O}(n^{-4}). \tag{4.11}$$

Sample Problem 4.3: Squeezing H_{10}

Compute H_{10} exactly, the bounds of Equation (4.10), and the approximation Equation (4.11).
Step 1: $H_{10} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{10} = 2.928968$.
Step 2: Bounds: $\log 11 = 2.397895$ and $1 + \log 10 = 3.302585$. Indeed $2.397895 \le 2.928968 \le 3.302585$.
Step 3: $H_{10} - \log 10 = 2.928968 - 2.302585 = 0.626383$, still noticeably above $\gamma = 0.577216$.
Step 4: With the correction terms: $2.302585 + 0.577216 + 0.05 - 0.000833 = 2.928968$, matching H_{10} to all six decimals.

$$\boxed{2.397895 \le H_{10} = 2.928968 \le 3.302585}$$

The crude squeeze pins H_{10} to within about ± 0.45; the asymptotic expansion, even at $n = 10$, is accurate to about 10^{-6}.

4.3.3 The sum of logarithms

The logarithm of a factorial is a sum: $\log n! = \log 1 + \log 2 + \cdots + \log n = \sum_{k=2}^{n} \log k$ (the $k = 1$ term is zero). Since $\log x$ is increasing, Equation (4.9) applies. By integration by parts,

$$\int_1^n \log x \, \mathrm{d}x = \Big[x \log x\Big]_1^n - \int_1^n x \cdot \frac{1}{x} \, \mathrm{d}x = n \log n - (n-1) = n \log n - n + 1.$$

With $f(n) - f(1) = \log n - 0$, Equation (4.9) becomes

$$n \log n - n + 1 \;\le\; \log n! \;\le\; n \log n - n + 1 + \log n. \tag{4.12}$$

So $\log n! = n \log n - n + \mathcal{O}(\log n)$, and since $\log n = o(n \log n)$, $\log n! \sim n \log n$. For $n = 10$ the bounds read $14.025851 \le 15.104413 \le 16.328436$.

Key Ideas

- For monotone f, $\sum f(k)$ lies between two integrals, Equations (4.8) and (4.9).
- $H_n = \log n + \gamma + \mathcal{O}(1/n)$ with Euler's constant $\gamma = 0.5772\ldots$
- $\log n! = n \log n - n + \mathcal{O}(\log n)$, Equation (4.12).

4.4 Stirling's formula and the Gamma function

The bounds Equation (4.12) leave an uncertainty of $\log n$. We now remove it almost completely. The result, Stirling's formula, is the tool from which Chapter 17 derives the paper's Eq. (1), including the curious constant $\frac{7}{8}$.

4.4.1 Pinning down $\log n!$

Define

$$d_n = \log n! - \left(n + \tfrac{1}{2}\right)\log n + n. \tag{4.13}$$

We will show that d_n decreases, stays bounded below, and therefore converges, and we will bound how fast.

Step 1: the difference of consecutive terms. Using $\log(n+1)! = \log n! + \log(n+1)$,

$$d_n - d_{n+1} = \log n! - \left(n + \tfrac{1}{2}\right)\log n + n - \log n! - \log(n+1) + \left(n + \tfrac{3}{2}\right)\log(n+1) - (n+1).$$

Collecting terms: the $\log n!$ cancel; $-\log(n+1) + (n+\frac{3}{2})\log(n+1) = (n+\frac{1}{2})\log(n+1)$; and $n - (n+1) = -1$. Hence

$$d_n - d_{n+1} = \left(n + \tfrac{1}{2}\right)\log\frac{n+1}{n} - 1. \tag{4.14}$$

Step 2: a convenient variable. Let $x = 1/(2n+1)$. Then

$$\frac{1+x}{1-x} = \frac{(2n+1)+1}{(2n+1)-1} = \frac{2n+2}{2n} = \frac{n+1}{n}, \qquad n + \tfrac{1}{2} = \frac{2n+1}{2} = \frac{1}{2x}.$$

Step 3: a series. From Chapter 3, for $|x| < 1$, $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$ and, replacing x by $-x$, $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$. Subtracting, the even powers cancel and the odd ones double:

$$\log\frac{1+x}{1-x} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right).$$

Step 4: substitute. Putting Steps 2 and 3 into Equation (4.14),

$$d_n - d_{n+1} = \frac{1}{2x}\cdot 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right) - 1 = \frac{x^2}{3} + \frac{x^4}{5} + \frac{x^6}{7} + \cdots.$$

Step 5: lower bound. Every term is positive, so $d_n - d_{n+1} > 0$: the sequence d_n is decreasing.

Step 6: upper bound. Replace each denominator $5, 7, \ldots$ by 3 (making each term larger, strictly so from the second term on) and sum the geometric series:

$$d_n - d_{n+1} < \frac{x^2}{3}(1 + x^2 + x^4 + \cdots) = \frac{x^2}{3(1-x^2)} = \frac{1}{3((2n+1)^2 - 1)}.$$

Since $(2n+1)^2 - 1 = 4n^2 + 4n = 4n(n+1)$,

$$d_n - d_{n+1} < \frac{1}{12n(n+1)} = \frac{1}{12}\left(\frac{1}{n} - \frac{1}{n+1}\right).$$

Step 7: a second monotone sequence. Rearranging Step 6, $\left(d_n - \frac{1}{12n}\right) - \left(d_{n+1} - \frac{1}{12(n+1)}\right) < 0$: the sequence $d_n - \frac{1}{12n}$ is *increasing*.

Taken on trust

Monotone convergence. A decreasing sequence that is bounded below converges to a limit, and an increasing sequence that is bounded above converges to a limit. This is a consequence of the completeness of the real numbers and is proved in any course on real analysis; we take it on trust here and again in Section 4.7.

Step 8: conclude. For every n, $d_n - \frac{1}{12n} \geq d_1 - \frac{1}{12}$. From Equation (4.13), $d_1 = \log 1 - \frac{3}{2}\log 1 + 1 = 1$, so $d_n > d_n - \frac{1}{12n} \geq \frac{11}{12}$. A decreasing sequence that is bounded below converges (monotone convergence, box above); call the limit C. Because d_n decreases to C, $d_n > C$; because $d_n - \frac{1}{12n}$ increases to the same limit C, $d_n - \frac{1}{12n} < C$. Therefore

$$C < d_n < C + \frac{1}{12n}. \tag{4.15}$$

Taken on trust

The value of the constant. The limit is $C = \frac{1}{2}\log 2\pi = 0.9189385\ldots$. Finding it requires an extra ingredient, usually Wallis's product $\frac{\pi}{2} = \prod_{k\geq 1} \frac{4k^2}{4k^2-1}$ or the Gaussian integral $\int_{-\infty}^{\infty} e^{-y^2}\,dy = \sqrt{\pi}$. We take the value on trust; derivations are in [22] and [23].

Substituting $C = \frac{1}{2}\log 2\pi$ into Equations (4.13) and (4.15) gives *Stirling's formula*:

$$\log n! = n\log n - n + \tfrac{1}{2}\log(2\pi n) + \frac{\vartheta_n}{12n}, \qquad 0 < \vartheta_n < 1. \tag{4.16}$$

(Here $(n+\frac{1}{2})\log n - n + \frac{1}{2}\log 2\pi = n\log n - n + \frac{1}{2}\log n + \frac{1}{2}\log 2\pi$, and $\frac{1}{2}\log n + \frac{1}{2}\log 2\pi = \frac{1}{2}\log(2\pi n)$.) Exponentiating, $n! \approx \sqrt{2\pi n}\,(n/e)^n$ with relative error about $1/(12n)$.

Taken on trust

The Stirling series. Our proof pinned the remainder between 0 and $1/(12n)$. A sharper tool, the Euler–Maclaurin summation formula (which expresses a sum $\sum f(k)$ as the integral $\int f\,dx$ plus correction terms involving derivatives of f at the endpoints), refines this into the expansion Equation (4.17) below, with next term $-1/(360n^3)$ and an $\mathcal{O}(n^{-5})$ remainder. We take this expansion on trust; the derivation is in [22] and [23].

The start of the *Stirling series*, taken on trust as stated in the box, is

$$\log n! = n\log n - n + \tfrac{1}{2}\log(2\pi n) + \frac{1}{12n} - \frac{1}{360n^3} + \mathcal{O}(n^{-5}). \tag{4.17}$$

The numerical check $d_{10} = 0.927269$ sits, as Equation (4.15) demands, between $C = 0.918939$ and $C + \frac{1}{120} = 0.927272$.

Sample Problem 4.4: Stirling at $n = 10$ and $n = 100$

Compare $\log n!$ with three successive truncations of Equation (4.17).
Step 1: Exact values: $\log 10! = \log 3628800 = 15.104413$ and $\log 100! = 363.739376$.
Step 2: The truncations are (a) $n\log n - n$, (b) (a) $+\frac{1}{2}\log(2\pi n)$, (c) (b) $+\frac{1}{12n}$. For $n = 10$: (a) $= 23.025851 - 10 = 13.025851$; $\frac{1}{2}\log(20\pi) = 2.070231$, so (b) $= 15.096082$; (c) $= 15.096082 + 0.008333 = 15.104415$.
Step 3: For $n = 100$: (a) $= 360.517019$, (b) $= 363.738542$, (c) $= 363.739376$.
Step 4: Tabulate the errors (exact minus approximation):

truncation	error, $n = 10$	error, $n = 100$
(a) $n\log n - n$	2.078562	3.222357
(b) $+\frac{1}{2}\log(2\pi n)$	0.008331	0.000833
(c) $+\frac{1}{12n}$	-2.77×10^{-6}	-2.78×10^{-9}

$$\boxed{\text{error of (b)} \approx \tfrac{1}{12n}, \qquad \text{error of (c)} \approx -\tfrac{1}{360n^3}}$$

After each added term, the remaining error is roughly the size of the next omitted term, exactly as an asymptotic expansion should behave. The leading-order formula (a) alone is useless for the value (its error even grows), although it gives the correct ratio in the limit.

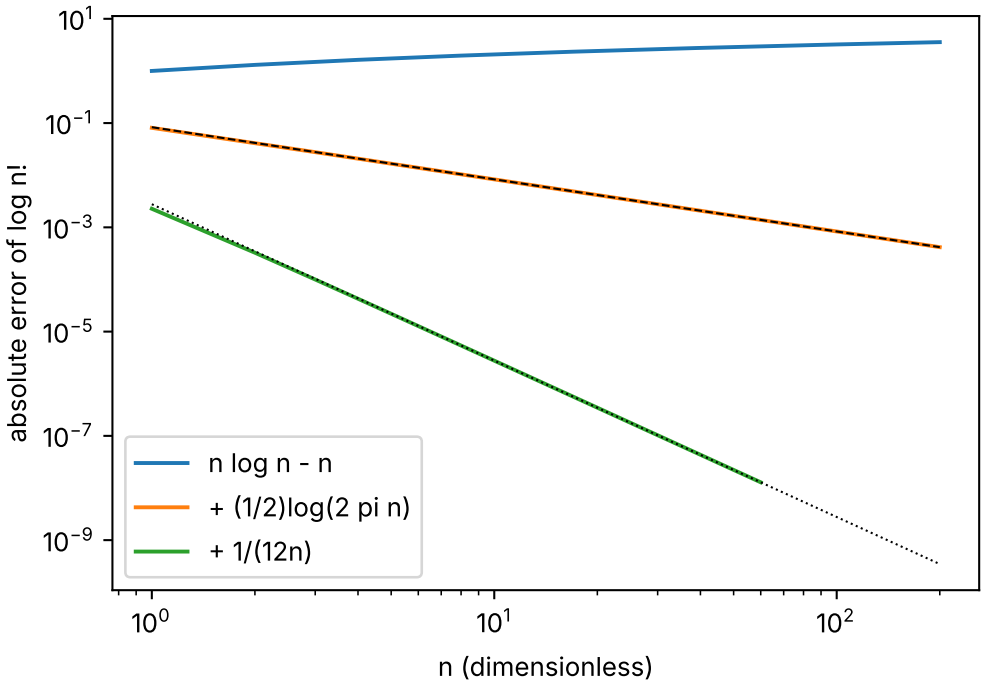


Figure 4.2: Absolute error of three truncations of Stirling's series for $\log n!$, for $1 \le n \le 200$ (both axes logarithmic, dimensionless). The leading term $n \log n - n$ has an error growing like $\frac{1}{2}\log(2\pi n)$; adding $\frac{1}{2}\log(2\pi n)$ leaves an error close to $1/(12n)$ (dashed); adding $1/(12n)$ leaves an error close to $1/(360n^3)$ (dotted).

Taken on trust

Asymptotic, not convergent. If you continue the Stirling series Equation (4.17) forever, it diverges for every n: the coefficients eventually grow factorially, faster than any geometric sequence c^k, so for every fixed n the terms eventually increase. Each fixed truncation is still an excellent approximation for large n, and the error is smallest if you stop near the smallest term. This is a proved theorem whose proof is beyond this book; see [5] and [22].

4.4.2 The Gamma function

Factorials are defined only for whole numbers. Chapter 17 needs a "factorial" of a complex number, and the Gamma function provides it.

Definition. For real $x > 0$,

$$\Gamma(x) = \int_0^\infty u^{x-1} e^{-u}\, \mathrm{d}u. \tag{4.18}$$

The integral converges: near $u = 0$ the integrand behaves like u^{x-1}, whose integral $\int_0^1 u^{x-1}\mathrm{d}u = 1/x$ is finite for $x > 0$; for large u, pick an integer $k \ge x + 2$. For $u > 0$ every term of the Taylor series of e^u is positive, so $e^u > u^k/k!$. Hence $u^{x+1}e^{-u} < k!\, u^{x+1-k} \le k!/u$ for $u \ge 1$,

which tends to 0. The integrand $u^{x-1}e^{-u} = u^{-2}\cdot u^{x+1}e^{-u}$ is therefore below u^{-2} once u is large enough, and $\int_1^\infty u^{-2}\,\mathrm{d}u = 1$ is finite. (This is the precise meaning of "exponentials beat powers" used below.)

The factorial property. First, $\Gamma(1) = \int_0^\infty e^{-u}\mathrm{d}u = 1$. Next, integrate by parts with "$f = u^x$, $\mathrm{d}g = e^{-u}\mathrm{d}u$":

$$\Gamma(x+1) = \int_0^\infty u^x e^{-u}\mathrm{d}u = \Big[-u^x e^{-u}\Big]_0^\infty + \int_0^\infty x u^{x-1}e^{-u}\mathrm{d}u.$$

The bracket vanishes at both ends (at 0 because $u^x \to 0$ for $x > 0$; at ∞ because exponentials beat powers), leaving

$$\Gamma(x+1) = x\,\Gamma(x). \tag{4.19}$$

Applying this repeatedly, $\Gamma(n+1) = n\Gamma(n) = n(n-1)\Gamma(n-1) = \cdots = n!\,\Gamma(1) = n!$. So Γ is a smooth function that passes through the factorials, shifted by one: $\Gamma(n+1) = n!$. A half-integer value: substituting $u = y^2$ in Equation (4.18) gives $\Gamma(\frac{1}{2}) = 2\int_0^\infty e^{-y^2}\mathrm{d}y = \sqrt{\pi} = 1.772454$, using the Gaussian integral (taken on trust, as in the box above).

Stirling for the Gamma function. Since $\log\Gamma(n) = \log n! - \log n$, Equation (4.17) gives $\log\Gamma(n) = (n-\frac{1}{2})\log n - n + \frac{1}{2}\log 2\pi + \frac{1}{12n} - \cdots$. That the same expansion holds for real x, not only for integers, is the content of the following box.

Taken on trust

Stirling for real arguments. The expansion Equation (4.20) below holds as $x \to \infty$ through all real values, not only through the integers where we derived it. This is a proved result whose proof (by the Euler–Maclaurin summation formula applied to $\log\Gamma$) is beyond this book; see [22]. We take it on trust.

Taken on trust as stated in the box,

$$\log\Gamma(x) = \Big(x-\tfrac{1}{2}\Big)\log x - x + \tfrac{1}{2}\log 2\pi + \frac{1}{12x} - \frac{1}{360x^3} + \mathcal{O}(x^{-5}), \quad x \to \infty. \tag{4.20}$$

For example at $x = 5$, $\log\Gamma(5) = \log 24 = 3.178054$; the formula through $1/(12x)$ gives 3.178076, and with the $-1/(360x^3)$ term it gives 3.178054.

Taken on trust

Stirling for complex arguments. The Gamma function extends to complex z (except $z = 0, -1, -2, \ldots$), and Equation (4.20) remains true with x replaced by complex z:

$$\log\Gamma(z) = \Big(z-\tfrac{1}{2}\Big)\log z - z + \tfrac{1}{2}\log 2\pi + \frac{1}{12z} + \mathcal{O}(|z|^{-3}),$$

as $|z| \to \infty$ in any sector $|\arg z| \le \pi - \delta$ with $\delta > 0$ fixed. Here $\log z$ is the principal branch (Chapter 2), and $\log\Gamma(z)$ is the branch that is real on the positive real axis and continuous away from the negative real axis. Proofs are in [5] and [6]. We use this only as a tool.

Preview: where the $\frac{7}{8}$ comes from. Chapter 17 introduces the Riemann–Siegel theta function $\theta(t) = \arg\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ and uses the complex Stirling formula with $z = \frac{1}{4} + \frac{it}{2}$ to derive

$$\theta(t) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \mathcal{O}(t^{-3}). \tag{4.21}$$

That chapter also shows the counting function is $N(t) = \theta(t)/\pi + 1 + S(t)$. Dropping S and the $1/(48t)$ term, $\theta/\pi + 1 = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} - \frac{1}{8} + 1$, which is exactly Equation (4.3) with $-\frac{1}{8} + 1 = \frac{7}{8}$. You can check Equation (4.21) numerically with `mpmath.siegeltheta`: at $t = 100$ it returns 87.97216, while the first three terms give 87.97196, a difference of 2.08×10^{-4}, which is $1/(48t)$ to within 10^{-8}.

Checkpoint 4.3

Why does Stirling's formula Equation (4.16) contain $\frac{1}{2}\log(2\pi n)$ while the crude bounds Equation (4.12) only bracket the error between 1 and $1 + \log n$?

Answer: The bounds treat each rectangle as entirely above or below the curve. For a slowly curving function the true sum is close to the average of the two, which accounts for the $\frac{1}{2}\log n$ (half of $\log n$). The remaining constant $\frac{1}{2}\log 2\pi$ comes from the exact limit C, which crude bounds cannot see.

Key Ideas

- Stirling: $\log n! = n\log n - n + \frac{1}{2}\log(2\pi n) + \vartheta_n/(12n)$ with $0 < \vartheta_n < 1$, Equation (4.16); the bound on ϑ_n was proved here, the constant $\frac{1}{2}\log 2\pi$ taken on trust.
- $\Gamma(x) = \int_0^\infty u^{x-1}e^{-u}\mathrm{d}u$, $\Gamma(x+1) = x\Gamma(x)$, $\Gamma(n+1) = n!$.
- Stirling extends to $\log\Gamma$ for real and (beyond this book) complex arguments; this is the origin of the $\frac{7}{8}$ in the paper's Eq. (1).

4.5 Inverting a monotone function

The paper defines the smooth zeros implicitly: s_n is the number with $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ (Sec. II A). In the same way, Eq. (2) gives x as a function of V, and the solver needs V as a function of x. Both are inversions of a monotone function.

4.5.1 Existence, uniqueness and the derivative of the inverse

Taken on trust

The intermediate value theorem. If f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then there is at least one $c \in [a, b]$ with $f(c) = y$. In particular, if $f(a)$ and $f(b)$ have opposite signs, f has a root in $[a, b]$. This is proved in any course on real analysis; we take it on trust.

Theorem (inverse of a strictly increasing function). Let f be continuous and strictly increasing on an interval I. Then for every y such that $f(a) < y < f(b)$ for some points $a < b$ of I, there is exactly one $x \in I$ with $f(x) = y$. We write $x = f^{-1}(y)$.

Proof. Existence: with such $a < b$, apply the intermediate value theorem to f on $[a, b]$. Uniqueness: if $x_1 < x_2$ then $f(x_1) < f(x_2)$, so two different points cannot give the same value. □

Derivative of the inverse. Write $g = f^{-1}$, so $g(f(x)) = x$ for all $x \in I$. If f is differentiable with $f'(x) \neq 0$, then g is differentiable at $y = f(x)$ (a fact from calculus that we accept), and differentiating $g(f(x)) = x$ with the chain rule gives $g'(f(x))\,f'(x) = 1$, that is,

$$\frac{\mathrm{d}x}{\mathrm{d}y} = \frac{1}{\mathrm{d}y/\mathrm{d}x}. \tag{4.22}$$

4.5.2 The smooth counting function is invertible for $t > 2\pi$

Differentiate Equation (4.3) term by term. By the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\frac{t}{2\pi}\log\frac{t}{2\pi}\Big] = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{1}{2\pi}.$$

(The derivative of $\log(t/2\pi) = \log t - \log 2\pi$ is $1/t$.) The second term contributes $-1/(2\pi)$ and the constant contributes 0. The $\pm 1/(2\pi)$ cancel:

$$N'_{\mathrm{sm}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}, \qquad N''_{\mathrm{sm}}(t) = \frac{1}{2\pi t}. \tag{4.23}$$

So N'_{sm} is negative for $t < 2\pi$, zero at $t = 2\pi$, and positive for $t > 2\pi$; and $N''_{\mathrm{sm}} > 0$ for all $t > 0$ (N_{sm} is *convex*). Convexity in this sense means that the graph lies above every tangent line: by Taylor's theorem with remainder (box in Section 4.7), $f(r) = f(x) + f'(x)(r-x) + \frac{1}{2}f''(\xi)(r-x)^2 \geq f(x) + f'(x)(r-x)$ whenever $f'' > 0$. The minimum value is $N_{\mathrm{sm}}(2\pi) = 1\cdot 0 - 1 + \frac{7}{8} = -\frac{1}{8}$. As $t \to 0^+$, $t\log t \to 0$ (substitute $t = e^{-u}$ and use exponentials beat powers), so $N_{\mathrm{sm}} \to \frac{7}{8}$. As $t \to \infty$, $N_{\mathrm{sm}} \to \infty$.

By the theorem, on $(2\pi, \infty)$ the function N_{sm} takes every value above $-\frac{1}{8}$ exactly once. (Given $y > -\frac{1}{8}$, continuity lets us take a slightly above 2π with $N_{\mathrm{sm}}(a) < y$, because $N_{\mathrm{sm}}(2\pi) = -\frac{1}{8} < y$, and b large with $N_{\mathrm{sm}}(b) > y$, because $N_{\mathrm{sm}} \to \infty$.) Every $n - \frac{1}{2}$ with $n \geq 1$ is at least $\frac{1}{2} > -\frac{1}{8}$, so each smooth zero $s_n > 2\pi$ exists and is unique. Figure 4.3 shows the picture.

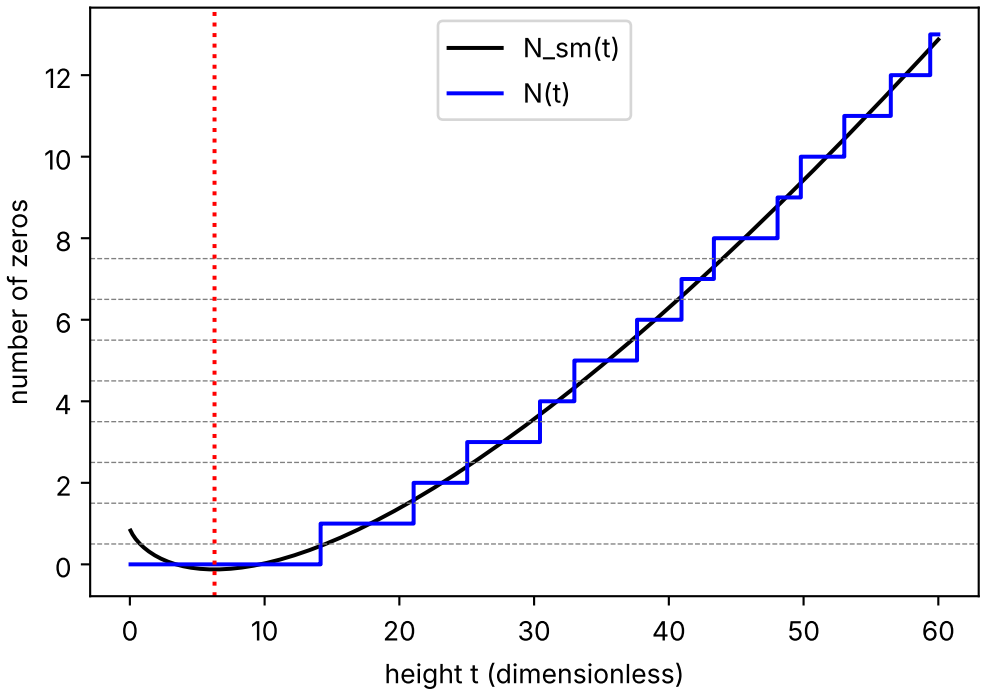


Figure 4.3: The smooth counting function $N_{\mathrm{sm}}(t)$ of the paper's Eq. (1) (solid) and the true zero-counting staircase $N(t)$ (steps at the zeros t_n), for $0 < t \leq 60$. Dashed horizontal lines are the levels $n - \frac{1}{2}$; the smooth zeros s_n are where N_{sm} crosses them for $t > 2\pi$. N_{sm} has its minimum $-\frac{1}{8}$ at $t = 2\pi$ and tends to $\frac{7}{8}$ as $t \to 0$, so the level $\frac{1}{2}$ is crossed twice: at $s_1 = 14.521$ and at a spurious point $t = 0.756$. The horizontal axis is the height t (dimensionless); the vertical axis counts zeros.

Pitfall

Because $N_{\mathrm{sm}}(t) \to \frac{7}{8}$ as $t \to 0$, the equation $N_{\mathrm{sm}}(t) = \frac{1}{2}$ has a *second* root, near $t = 0.756$, on the decreasing branch $t < 2\pi$. It has nothing to do with the zeta zeros. A root finder started carelessly at small t can return it. Always restrict the search to $t > 2\pi$.

4.5.3 The mean spacing

Treat n as a continuous variable and differentiate $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ with respect to n: by the chain rule $N'_{\mathrm{sm}}(s_n)\,\frac{\mathrm{d}s_n}{\mathrm{d}n} = 1$, which is Equation (4.22). With Equation (4.23),

$$\mathrm{gap}_n = \frac{\mathrm{d}s_n}{\mathrm{d}n} = \frac{1}{N'_{\mathrm{sm}}(s_n)} = \frac{2\pi}{\log(s_n/2\pi)}. \tag{4.24}$$

This is the "local mean spacing" in which the paper measures all errors. It shrinks slowly with height (Chapter 17 develops this in detail).

4.5.4 The Lambert W function

Some inverses are important enough to have names. The *Lambert W function* is the inverse of $w \mapsto we^w$:

$$W(z)\,e^{W(z)} = z. \tag{4.25}$$

The function $f(w) = we^w$ has $f'(w) = (1+w)e^w$, which is negative for $w < -1$ and positive for $w > -1$, so f has a minimum $f(-1) = -1/e$. For $w < 0$, $f(w) < 0$; for $w \ge 0$, f increases from 0 to ∞. Consequently:

- for $z \ge 0$ there is exactly one real solution $W(z) \ge 0$ (for example $W(0) = 0$, $W(e) = 1$);
- for $-1/e < z < 0$ there are two real solutions: the *principal branch* $W_0(z) \in (-1, 0)$ and the *lower branch* $W_{-1}(z) < -1$;
- for $z < -1/e$ there is no real solution.

In Python, `scipy.special.lambertw(z, k)` returns branch k ($k = 0$ by default, or $k = -1$); take the real part of its complex output.

Solving $x \log x = y$. Put $x = e^u$. Then $x \log x = ue^u = y$, so $u = W(y)$ and $x = e^{W(y)}$. From Equation (4.25), $e^{W(y)} = y/W(y)$, so

$$x \log x = y \quad \Longleftrightarrow \quad x = \frac{y}{W(y)}. \tag{4.26}$$

The smooth zeros in closed form. Combine the first two terms of Equation (4.3): $\frac{t}{2\pi}(\log\frac{t}{2\pi} - 1) = \frac{t}{2\pi}\log\frac{t}{2\pi e}$, because $\log e = 1$. So $N_{\mathrm{sm}}(s) = n - \frac{1}{2}$ becomes

$$\frac{s}{2\pi}\log\frac{s}{2\pi e} = n - \frac{1}{2} - \frac{7}{8} = n - \frac{11}{8}.$$

Let $v = s/(2\pi e)$, so $s/(2\pi) = ev$. Then $ev \log v = n - \frac{11}{8}$, i.e. $v \log v = y_n$ with $y_n = (n - \frac{11}{8})/e$. By Equation (4.26), $v = y_n/W(y_n)$, and $s_n = 2\pi e\, v = 2\pi e\, y_n/W(y_n)$:

$$s_n = \frac{2\pi\,(n - \frac{11}{8})}{W((n - \frac{11}{8})/e)}. \tag{4.27}$$

Moreover $v = e^{W(y_n)}$ means $\log(s_n/2\pi) = \log(ev) = 1 + W(y_n)$, so Equation (4.24) becomes

$$\mathrm{gap}_n = \frac{2\pi}{1 + W((n - \frac{11}{8})/e)}. \tag{4.28}$$

This is the "Lambert-W asymptotic of the counting function" that the paper mentions in Sec. I A in connection with França and LeClair [3].

Sample Problem 4.5: The first smooth zero and its impostor

Use Equation (4.27) for $n = 1$ and explain which branch of W to use.
Step 1: $n - \frac{11}{8} = -\frac{3}{8}$, so $y_1 = -\frac{3}{8e} = -0.137954$. This lies in $(-1/e, 0) = (-0.3679, 0)$, so there are two real values of W.
Step 2: Principal branch: $W_0(-0.137954) = -0.16226$. Then $s = 2\pi(-0.375)/(-0.16226) = 2.356194/0.16226 = 14.521$.
Step 3: Lower branch: $W_{-1}(-0.137954) = -3.118$, giving $s = 2.356194/3.118 = 0.756$.
Step 4: Decide. By Equation (4.28)'s derivation, $\log(s/2\pi) = 1 + W$. The physical branch has $s > 2\pi$, i.e. $1 + W > 0$, i.e. $W > -1$: the principal branch. The lower branch gives $s < 2\pi$, the spurious root of the pitfall above.

$$\boxed{s_1 = 14.521\ (W_0), \qquad \text{spurious root } 0.756\ (W_{-1})}$$

The mean spacing there is $\mathrm{gap}_1 = 2\pi/(1 - 0.16226) = 7.500$. For $n \geq 2$, $y_n > 0$ and only one real branch exists, so there is no ambiguity.

Sample Problem 4.6: The thousandth zero in mean spacings

The paper gives $t_{1000} = 1419.42248094599$ (Sec. II A). Find s_{1000}, gap_{1000} and the fluctuation $(t_{1000} - s_{1000})/\mathrm{gap}_{1000}$.
Step 1: $y = (1000 - \frac{11}{8})/e = 998.625/2.718282 = 367.374$. Solving $we^w = 367.374$ gives $W = 4.4202$.
Step 2: By Equation (4.27), $s_{1000} = 2\pi \times 998.625/4.4202 = 6274.546/4.4202 = 1419.52$. Here W must be carried to five significant digits: with the rounded value $W = 4.420$ the same division gives 1419.58, and Step 4 would come out as -0.14 instead of -0.08. The fluctuation $t - s$ is a small difference of two large numbers, so it magnifies small relative errors in s (the cancellation effect of Section 4.9).
Step 3: By Equation (4.28), $\mathrm{gap}_{1000} = 2\pi/5.420 = 1.159$.
Step 4: $(1419.422 - 1419.52)/1.159 = -0.08$.

$$\boxed{s_{1000} = 1419.52, \quad \mathrm{gap}_{1000} = 1.159, \quad \text{fluctuation} \approx -0.08 \text{ spacings}}$$

The true zero sits about a twelfth of a spacing below its smooth position. The paper's null rms of 0.263 (Sec. II C) is the typical size of such fluctuations on zeros 401–500.

How good is the leading order? For large y, $W(y) \sim \log y$ (take logarithms of Equation (4.25): $\log W + W = \log y$, and $\log W = o(W)$ by Equation (4.7)). Hence $s_n \sim 2\pi n/\log n$ and $\mathrm{gap}_n \sim 2\pi/\log n$. But at $n = 1000$, $2\pi n/\log n = 909.6$, far from $s_{1000} = 1419.52$, and $2\pi/\log n = 0.9096$, far from 1.159. Figure 4.4 shows how slowly these ratios approach 1. The lesson of Section 4.2 repeats: with logarithms, use the exact inverse, not its leading term.

Numerical inversion in general. Most monotone functions have no named inverse. Eq. (2) of the paper, which gives the Wu–Sprung position $x(V)$, is such a case; the paper says it is "inverted pointwise by bracketed root finding" (Sec. II B). The general recipe for computing

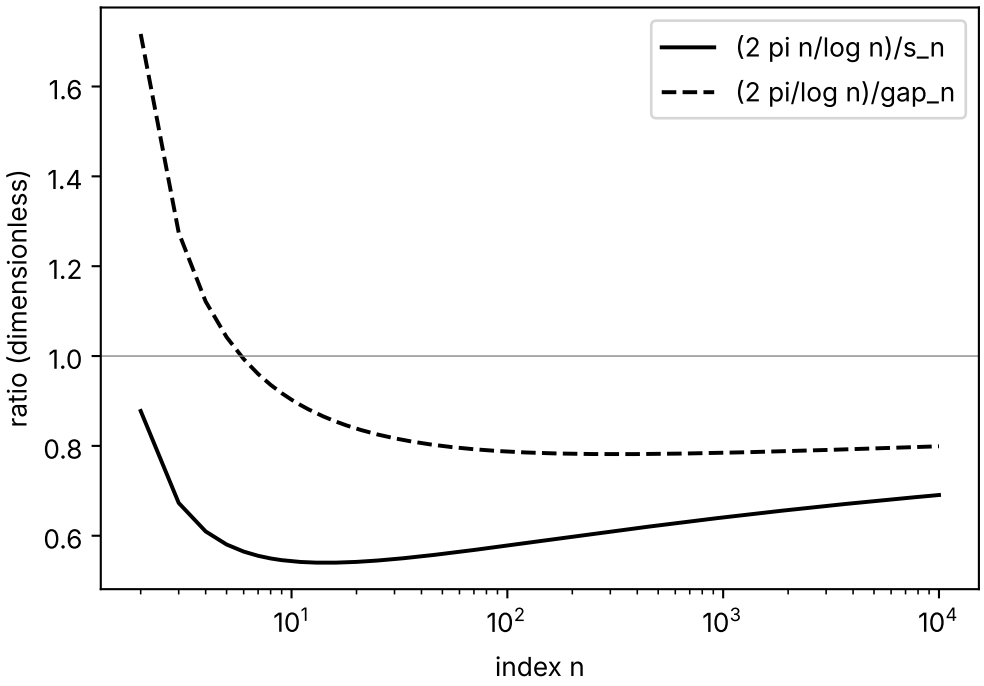


Figure 4.4: Ratios of leading-order approximations to exact values for the smooth zeros, $2 \le n \le 10^4$ (logarithmic horizontal axis, dimensionless). Solid: $(2\pi n/\log n)/s_n$; dashed: $(2\pi/\log n)/\mathrm{gap}_n$. Both tend to 1 as $n \to \infty$, but at $n = 10^4$ they are still far from it. The exact values come from the Lambert-W formulas Equations (4.27) and (4.28).

$f^{-1}(y)$ is: (i) find an interval $[a, b]$ with $f(a) < y < f(b)$; (ii) solve $f(x) - y = 0$ in it with a root finder. The rest of this chapter is about step (ii).

Checkpoint 4.4

The Lambert-W formula gives every smooth zero in one line. Why might a program still use a root finder for s_n?

Answer: Because a root finder works for any continuous function with a sign change, including the Wu–Sprung inversion and the level equations $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ used later (N_{fl} is an oscillating prime sum, Chapter 18), which need not even be monotone and have no closed form. Using one tool for all of them is simpler and lets you cross-check against the closed form where it exists.

Key Ideas

- A continuous strictly increasing function has a unique inverse; $\mathrm{d}x/\mathrm{d}y = 1/(\mathrm{d}y/\mathrm{d}x)$, Equation (4.22).
- $N'_{\mathrm{sm}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$, so N_{sm} is increasing for $t > 2\pi$ and each s_n is unique there; $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$, Equation (4.24).
- $s_n = 2\pi(n - \frac{11}{8})/W((n - \frac{11}{8})/e)$ with the principal branch, Equation (4.27); the other branch gives a spurious root.

4.6 Root finding I: bisection

★ Did you know? *A Theorem Behind Bisection*

Bisection works because a continuous function that changes sign must cross zero somewhere in between. That sounds obvious, but Bernard Bolzano gave one of the first rigorous proofs only in 1817. Each bisection step halves the interval, so ten steps shrink it by a factor $2^{10} = 1024$, gaining about three decimal digits. The method is slow, but once a sign change is bracketed it never fails.

A *root* of a function f is a number r with $f(r) = 0$. To compute s_n we find the root of $f(s) = N_{\mathrm{sm}}(s) - (n - \frac{1}{2})$. The simplest reliable method is to trap the root in an interval and keep halving it.

4.6.1 The algorithm

Suppose f is continuous on $[a, b]$ and $f(a)$, $f(b)$ have opposite signs; we call $[a, b]$ a *bracket*. By the intermediate value theorem the bracket contains a root.

1. Compute the midpoint $m = (a + b)/2$ and $f(m)$.
2. If $f(a)$ and $f(m)$ have opposite signs (or $f(m) = 0$), the new bracket is $[a, m]$; otherwise it is $[m, b]$.
3. Repeat until the bracket is short enough; return its midpoint.

In step 2, if $f(a)f(m) > 0$ then $f(m)$ has the sign of $f(a)$, which is opposite to the sign of $f(b)$, so $[m, b]$ is a bracket. Either way the new interval is a bracket.

4.6.2 A guaranteed error bound

Let $[a_k, b_k]$ be the bracket after k halvings, with $[a_0, b_0] = [a, b]$. Each step halves the width, so $b_k - a_k = (b - a)/2^k$. The root r lies in $[a_k, b_k]$, and the midpoint $m_k = (a_k + b_k)/2$ is at most half the width from any point of the bracket:

$$|m_k - r| \le \frac{b - a}{2^{k+1}}. \tag{4.29}$$

To guarantee $|m_k - r| \le \delta$ we need $2^{k+1} \ge (b - a)/\delta$, that is,

$$k \ge \log_2 \frac{b - a}{\delta} - 1. \tag{4.30}$$

Each step costs one function evaluation and gains one binary digit, about 0.3 decimal digits. The bound holds whatever f looks like, provided it is continuous: that is the great virtue of bisection.

Sample Problem 4.7: Bisection for the first smooth zero

Find s_1 by bisection on $f(s) = N_{\mathrm{sm}}(s) - \frac{1}{2}$, starting from $[10, 20]$. Do five steps and determine how many steps would guarantee an error below 10^{-10}.
Step 1: Check the bracket: $f(10) = -0.4769$ and $f(20) = 0.8775$ have opposite signs. Both points exceed $2\pi \approx 6.28$, where f is increasing, so the bracket contains exactly one root.
Step 2: Iterate. Each row shows the midpoint, the sign of f there, and the new bracket:

k	midpoint m	$f(m)$	new bracket
1	15	$+0.0651$	$[10, 15]$
2	12.5	-0.2460	$[12.5, 15]$
3	13.75	-0.0995	$[13.75, 15]$
4	14.375	-0.0194	$[14.375, 15]$
5	14.6875	$+0.0223$	$[14.375, 14.6875]$

Step 3: After five steps the width is $10/2^5 = 0.3125$; the midpoint 14.53125 has guaranteed error at most 0.15625 by Equation (4.29). The actual error, using the converged root $s_1 = 14.5213$ (found with Brent's method, Section 4.7), is about 0.0099.
Step 4: For $\delta = 10^{-10}$, Equation (4.30) requires $k \geq \log_2(10^{11}) - 1 = 36.54 - 1 = 35.54$, so $k = 36$.

$$\boxed{s_1 \in [14.375,\ 14.6875] \text{ after 5 steps;} \quad 36 \text{ steps for } 10^{-10}}$$

```python
import numpy as np

def Nsm(t):                          # paper Eq. (1)
    u = t / (2 * np.pi)
    return u * np.log(u) - u + 7 / 8

def bisect(f, a, b, tol):
    fa, fb = f(a), f(b)
    if fa * fb > 0:
        raise ValueError("f(a) and f(b) have the same sign")
    while (b - a) / 2 > tol:
        m = (a + b) / 2
        fm = f(m)
        if fa * fm <= 0:             # sign change in [a, m]
            b, fb = m, fm
        else:                        # sign change in [m, b]
            a, fa = m, fm
    return (a + b) / 2

s1 = bisect(lambda s: Nsm(s) - 0.5, 10.0, 20.0, 1e-10)
print(s1)                            # 14.5213...
```

Checkpoint 4.5

You start bisection with a bracket of width 1 and do 10 steps. What is the largest possible error of the midpoint?

Answer: By Equation (4.29), $1/2^{11} = 4.9 \times 10^{-4}$.

Key Ideas

- A bracket $[a, b]$ with $f(a)f(b) < 0$ contains a root (for continuous f).
- Bisection halves the bracket each step: error at most $(b - a)/2^{k+1}$ after k steps, Equation (4.29).
- It always works but gains only one bit per evaluation.

4.7 Root finding II: Newton, secant and Brent

Bisection uses only the *signs* of f. If we also use its *values*, and perhaps its slope, we can converge much faster.

4.7.1 Newton's method

Near a point x_k, a smooth function is close to its tangent line, $f(x) \approx f(x_k) + f'(x_k)(x - x_k)$. Set this to zero and solve for x; call the answer x_{k+1}:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \tag{4.31}$$

This is *Newton's method*. Geometrically, follow the tangent at $(x_k, f(x_k))$ down to where it crosses the axis.

Taken on trust

Taylor's theorem with remainder. If f has a continuous second derivative between x and r, then $f(r) = f(x) + f'(x)(r - x) + \frac{1}{2}f''(\xi)(r - x)^2$ for some ξ between x and r. This is a standard calculus result (it is the Taylor series of Chapter 3 stopped after two terms, with an exact formula for what is left); we take it on trust.

Quadratic convergence. Let r be a root with $f'(r) \neq 0$, and let $e_k = x_k - r$ be the error. Apply Taylor's theorem at $x = x_k$, using $f(r) = 0$:

$$0 = f(x_k) + f'(x_k)(r - x_k) + \tfrac{1}{2}f''(\xi_k)(r - x_k)^2.$$

Divide by $f'(x_k)$ and rearrange:

$$x_k - \frac{f(x_k)}{f'(x_k)} - r = \frac{f''(\xi_k)}{2f'(x_k)}(x_k - r)^2.$$

The left side is $x_{k+1} - r = e_{k+1}$ by Equation (4.31). Hence

$$e_{k+1} = \frac{f''(\xi_k)}{2f'(x_k)}\,e_k^2 \;\approx\; \frac{f''(r)}{2f'(r)}\,e_k^2 \quad \text{when } x_k \text{ is close to } r. \tag{4.32}$$

The error is *squared* at each step: once you have a few correct digits, the number of correct digits roughly doubles per iteration.

Sample Problem 4.8: Newton's method for $\sqrt{2}$ and for s_1

(a) Find $\sqrt{2}$ as the positive root of $f(x) = x^2 - 2$, starting at $x_0 = 1$. (b) Find s_1 starting at $x_0 = 14$.
Step 1: For (a), $f'(x) = 2x$, so Equation (4.31) gives $x_{k+1} = x_k - (x_k^2 - 2)/(2x_k) = \frac{1}{2}(x_k + 2/x_k)$.
Step 2: Iterate and record $e_k = x_k - \sqrt{2}$, with $\sqrt{2} = 1.41421356237$:

k	x_k	e_k
1	1.5	8.58×10^{-2}
2	1.4166667	2.45×10^{-3}
3	1.4142157	2.12×10^{-6}
4	1.41421356237	1.59×10^{-12}

Step 3: Compare with Equation (4.32): $f''/(2f') = 2/(4\sqrt{2}) = 1/(2\sqrt{2})$, so $e_{k+1} \approx e_k^2/2.828$. From $e_2 = 2.45 \times 10^{-3}$ this predicts $e_3 \approx 2.13 \times 10^{-6}$, and from e_3 it predicts $e_4 \approx 1.59 \times 10^{-12}$, in agreement with the table.
Step 4: For (b), $f(s) = N_{\mathrm{sm}}(s) - \frac{1}{2}$ and $f'(s) = \frac{1}{2\pi} \log \frac{s}{2\pi}$ by Equation (4.23). At $x_0 = 14$: $f(14) = -0.0680$ and $f'(14) = 0.1275$, so $x_1 = 14 + 0.0680/0.1275 = 14.5333$.
Step 5: The next iterate x_2 agrees with the converged root $s_1 = 14.5213$ to better than 2×10^{-5}, and x_3 to better than 10^{-10}.

$\sqrt{2}$: 12 correct digits in 4 steps; $s_1 = 14.5213$ in 2 to 3 steps

Compare: bisection needed 36 steps for 10^{-10}.

4.7.2 When Newton fails, and when it cannot

Newton's method has no bracket and no guarantee. If $f'(x_k)$ is small, the tangent is nearly flat and the step is huge. A classic failure is $f(x) = \arctan x$, whose only root is 0. Here $f'(x) = 1/(1+x^2)$, so $x_{k+1} = x_k - (1+x_k^2)\arctan x_k$. From $x_0 = 1.5$: $x_1 = -1.694$, $x_2 = 2.321$, and the iterates fly off with growing magnitude (Figure 4.5). From $x_0 = 1$ the iteration converges.

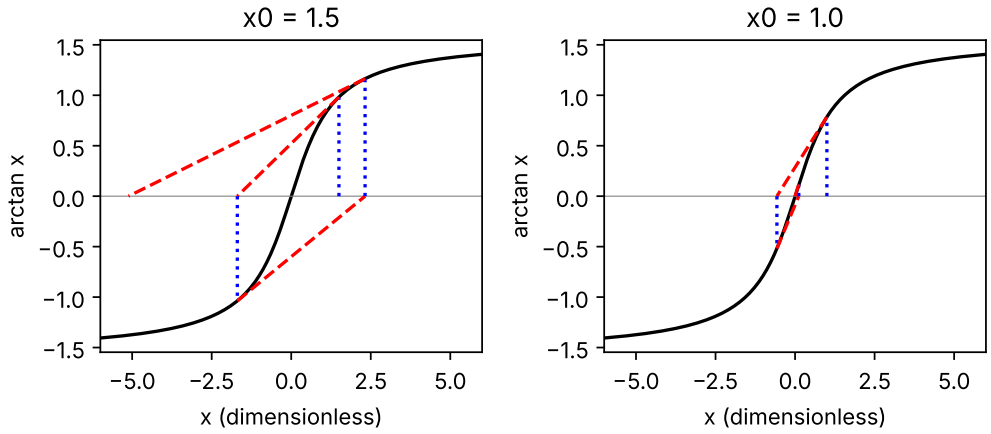


Figure 4.5: Newton's method on $f(x) = \arctan x$ (dimensionless). Each dashed segment is a tangent line followed from $(x_k, f(x_k))$ to the axis. Left: from $x_0 = 1.5$ the iterates -1.694, $2.321, \ldots$ alternate in sign and grow; the method diverges. Right: from $x_0 = 1$ the iterates converge rapidly to the root 0.

For our problem, though, convexity rescues Newton completely.

Theorem (Newton on an increasing convex function). Let f be continuously differentiable, with $f' > 0$ and f convex on an interval I (f lies above all its tangent lines), and let $r \in I$ be a root. If $x_k \in I$, then $x_{k+1} \geq r$. If moreover $x_k > r$, then $r \leq x_{k+1} < x_k$. Hence, as long as the iterates stay in I, from the first step on they decrease monotonically to r.

Proof. Convexity at x_k, evaluated at r: $f(r) \geq f(x_k) + f'(x_k)(r - x_k)$. With $f(r) = 0$ this is $0 \geq f(x_k) - f'(x_k)(x_k - r)$, so $f(x_k)/f'(x_k) \leq x_k - r$, i.e. $x_{k+1} = x_k - f(x_k)/f'(x_k) \geq r$. If $x_k > r$, then $f(x_k) > 0$ (increasing f), so $x_{k+1} < x_k$. A decreasing sequence bounded below by r converges (monotone convergence, box in Section 4.4) to some ℓ with $r \leq \ell \leq x_1$, so $\ell \in I$ (an interval containing r and x_1 contains every point between them). Since f and f' are continuous

and $f'(\ell) > 0$, the right side of Equation (4.31) converges to $\ell - f(\ell)/f'(\ell)$, while the left side converges to ℓ. Hence $\ell = \ell - f(\ell)/f'(\ell)$, so $f(\ell) = 0$, and because f is increasing, $\ell = r$. □

By Equation (4.23), N_{sm} is increasing and convex on $(2\pi, \infty)$. Take $I = (2\pi, \infty)$ and $r = s_n$. If $x_0 \in I$, the theorem gives $x_1 \geq s_n > 2\pi$, so x_1 again lies in I; repeating the argument, every iterate stays in I, and the theorem applies at every step. So Newton's method for any smooth zero, started anywhere above 2π, lands at or to the right of s_n after one step and then marches down to it. (Problem 23 shows that the first step from near 2π can overshoot enormously, but it overshoots to the right, which is safe.)

4.7.3 The secant method

Newton needs f'. If the derivative is unavailable, replace it by the slope of the chord through the last two points:

$$x_{k+1} = x_k - f(x_k)\,\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}. \tag{4.33}$$

This is the *secant method*. Its error satisfies $e_{k+1} \approx C e_k e_{k-1}$, which leads to $|e_{k+1}| \approx \text{const} \times |e_k|^{1.618}$; the exponent is the golden ratio $(1 + \sqrt{5})/2$ (Problem 18 derives this). That is slower than Newton per step, but each step needs only one new function evaluation.

Inverse quadratic interpolation. With three points (a, f_a), (b, f_b), (c, f_c) with distinct f-values, fit x as a quadratic function of y through them and evaluate at $y = 0$. To build that quadratic, define

$$\ell_a(y) = \frac{(y - f_b)(y - f_c)}{(f_a - f_b)(f_a - f_c)},$$

and ℓ_b, ℓ_c in the same way with the letters cycled. Each ℓ is a quadratic in y that equals 1 at its own f-value and 0 at the other two, so $x(y) = a\,\ell_a(y) + b\,\ell_b(y) + c\,\ell_c(y)$ passes through all three points (this is the *Lagrange form*). Setting $y = 0$ gives

$$x_{\text{new}} = \frac{a\,f_b f_c}{(f_a - f_b)(f_a - f_c)} + \frac{b\,f_a f_c}{(f_b - f_a)(f_b - f_c)} + \frac{c\,f_a f_b}{(f_c - f_a)(f_c - f_b)}.$$

(Each fraction is the Lagrange basis polynomial in y evaluated at $y = 0$: it equals 1 when y is the matching f-value and 0 at the other two.) Fitting $x(y)$ instead of $y(x)$ avoids solving a quadratic equation.

4.7.4 Brent's method: safety plus speed

The fast methods can wander off; bisection is slow but cannot fail. *Brent's method* [26] combines them. It keeps a bracket $[b, c]$ with $f(b)f(c) < 0$ at all times, with b the best estimate so far, plus the previous iterate a. At each step:

1. It proposes a fast step: inverse quadratic interpolation through a, b, c if their f-values are distinct, otherwise a secant step through a, b.

2. It accepts the fast step only if the new point lies well inside the bracket (between $(3c+b)/4$ and b) and the step is shrinking quickly enough (less than half the step before last). Otherwise it takes a bisection step instead.

3. It evaluates f at the new point and updates the three points so that the bracket property is preserved.

The result never does much worse than bisection (Brent [26] shows, and we quote without proof, that the number of evaluations never exceeds roughly $(\log_2(w_0/\delta))^2$, where w_0 is the width of the

initial bracket and δ the requested tolerance) and on smooth functions converges superlinearly like the secant method [26, 27]. We use it as a reliable black box: `scipy.optimize.brentq(f, a, b)`. Its default tolerances are an absolute 2×10^{-12} and a relative 8.88×10^{-16} (four times the double-precision machine epsilon of Section 4.9). Figure 4.6 compares all four methods on the first smooth zero.

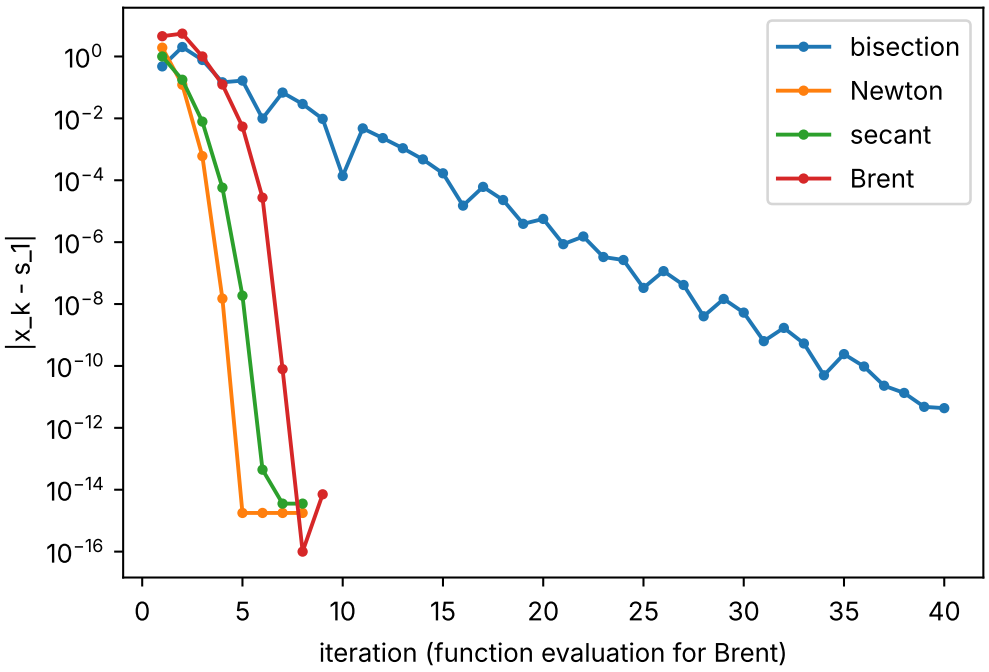


Figure 4.6: Absolute error $|x_k - s_1|$ (logarithmic) against iteration number for four root finders solving $N_{\mathrm{sm}}(s) = \frac{1}{2}$. Bisection and Brent start from the bracket $[10, 20]$; the secant method from the points 10 and 20; Newton from $x_0 = 10$. For Brent every function evaluation is plotted. Bisection gains a fixed factor of 2 per step (a straight line); Newton, secant and Brent bend downward (superlinear convergence). Errors below 10^{-16} are plotted at 10^{-16}.

```
import numpy as np
from scipy.optimize import brentq

def Nsm(t):
    u = t / (2 * np.pi)
    return u * np.log(u) - u + 7 / 8

def newton(f, fprime, x, steps):
    for k in range(steps):
        x = x - f(x) / fprime(x)
    return x

n = 1
f  = lambda s: Nsm(s) - (n - 0.5)
fp = lambda s: np.log(s / (2 * np.pi)) / (2 * np.pi)

print(newton(f, fp, 14.0, 4))        # Newton
print(brentq(f, 10.0, 20.0))         # Brent on a bracket
```

Checkpoint 4.6

Newton's error is 10^{-3} at some step and $f''/(2f') \approx 0.04$ near the root. Roughly what is the error after the next step? After two more?

Answer: By Equation (4.32), $0.04 \times (10^{-3})^2 = 4 \times 10^{-8}$, then $0.04 \times (4 \times 10^{-8})^2 \approx 6 \times 10^{-17}$, which is below double-precision resolution.

Key Ideas

- Newton: $x_{k+1} = x_k - f(x_k)/f'(x_k)$, error squared each step, Equation (4.32); can diverge in general.
- For an increasing convex f (such as N_{sm} on $t > 2\pi$) Newton converges monotonically after one step.
- Secant and inverse quadratic interpolation avoid derivatives; Brent's method combines them with bisection to keep a guaranteed bracket.

4.8 Bracketing failures and bracket expansion

Bisection and Brent need a bracket. Getting one is often the hardest part.

4.8.1 How brackets fail

- **No sign change, but roots.** If $[a, b]$ contains an even number of roots, $f(a)$ and $f(b)$ have the same sign. For example $f(t) = N_{\text{sm}}(t) - \frac{1}{2}$ is positive at both $t = 0.5$ ($f = 0.094$) and $t = 20$ ($f = 0.8775$) although it vanishes twice in between, at 0.756 and 14.521. A root of even multiplicity, like $x = 0$ for x^2, never produces a sign change at all.
- **No sign change, no root.** The obvious case: the root is outside the interval.
- **Sign change without a root.** If f is discontinuous, the intermediate value theorem does not apply. Bisection on $1/x$ over $[-1, 2]$ produces midpoints $0.5, -0.25, 0.125, \ldots$ that converge to 0, which is a pole, not a root. (On $[-1, 1]$ it would fail at once: the first midpoint is exactly 0, where $1/x$ cannot be evaluated.)
- **Several roots inside.** A bracket with a sign change may contain three (or any odd number of) roots; the method returns one of them, and which one depends on details of the algorithm.

4.8.2 Bracket expansion

When a root is expected near a known guess, a standard remedy is to start from a small interval around the guess and widen it until the signs differ; for an increasing function the root found this way is the only one. The paper does exactly this for the oracle: Brent "bracketed ±3 around the smooth zero (bracket expanded on sign failure)" (Appendix A step 8). For the oracle a root near s_n is expected, but monotonicity is not guaranteed (see the discussion of root ambiguity below). The paper does not state the units of the ±3; the companion library `qmnt` uses ±3 mean spacings, and so do we.

```
import numpy as np
from scipy.optimize import brentq
import qmnt
```

```
def solve_level(F, n, width=3.0, grow=1.5, maxtries=20):
    """Root of Nsm(t) + F(t) = n - 1/2 near the smooth zero s_n."""
    s = qmnt.smooth_zeros(np.array([n]))[0]
    g = qmnt.mean_gap(np.array([n]))[0]
    h = lambda t: qmnt.Nsm(t) + F(t) - (n - 0.5)
    for k in range(maxtries):
        a = max(s - width * g, 2 * np.pi)    # never cross t = 2 pi
        b = s + width * g
        if h(a) * h(b) < 0:
            return brentq(h, a, b)
        width *= grow                        # widen and try again
    raise RuntimeError("no sign change found")

# with F = 0 this reproduces the smooth zero itself
print(solve_level(lambda t: 0.0, 1))        # 14.5213...
```

Sample Problem 4.9: A safe bracket for s_1

Find a bracket for $f(t) = N_{\text{sm}}(t) - \frac{1}{2}$ that is guaranteed to contain s_1 and nothing else.
Step 1: The naive choice $[0.5, 20]$ fails: $f(0.5) = 0.094 > 0$ and $f(20) = 0.8775 > 0$.
Step 2: Use the structure from Section 4.5: on $[2\pi, \infty)$, f is strictly increasing. At the left end, $f(2\pi) = -\frac{1}{8} - \frac{1}{2} = -0.625 < 0$.
Step 3: At the right end, $f(20) = 0.8775 > 0$.

$$\boxed{[2\pi,\ 20] \text{ brackets exactly one root, } s_1 = 14.521}$$

The lesson: use what you know about the function (here, monotonicity on a half-line) to choose brackets. Blind brackets can trap the wrong root or none.

4.8.3 Root ambiguity

Adding an oscillating term to N_{sm} can destroy monotonicity. The level equation of the paper's oracle is $N_{\text{sm}}(t) + N_{\text{fl}}(t) = n - \frac{1}{2}$ (Sec. II D), where N_{fl} is a sum of sines (Chapter 18). The left side is increasing as long as $|N'_{\text{fl}}(t)| < N'_{\text{sm}}(t)$. If the oscillation is steep enough to violate this, the left side can wiggle back down and the equation can have three roots near one level. Then "the" predicted zero is ambiguous, and different brackets give different answers. The same can happen for the paper's fitted controls, whose level equation is $N_{\text{sm}}(t) + F(t) = n - \frac{1}{2}$ with a fitted sum of sinusoids F: Table IV flags the control-B row with 900 fitted coefficients as root-ambiguous (Sec. IV). Problem 30 explores the monotonicity condition for the oracle, and Chapter 27 treats the ambiguity in detail.

Checkpoint 4.7

Why can bisection not find the root of $f(x) = x^2$?

Answer: $x^2 \geq 0$ everywhere, so no interval gives a sign change; bisection needs $f(a)f(b) < 0$. A double root touches the axis without crossing it.

Key Ideas

- A bracket needs a sign change of a *continuous* function; even numbers of roots, double roots and poles defeat it.
- Use known monotonicity to choose brackets; expand them systematically when the

sign test fails (paper, Appendix A step 8).

- Non-monotone level equations can have several roots: a root finder then returns one of them, not "the" answer.

4.9 Floating-point numbers

A computer cannot store π or even 0.1 exactly. It stores a nearby *floating-point number*, and every arithmetic operation rounds its result. Usually this is harmless. Sometimes, as in Sec. V of the paper, it is not.

4.9.1 Binary scientific notation

A floating-point number is stored as

$$x = \pm\,(1.b_1 b_2 \ldots b_p)_2 \times 2^e, \tag{4.34}$$

where each b_i is a binary digit (0 or 1), $(1.b_1 \ldots b_p)_2 = 1 + b_1/2 + b_2/4 + \cdots + b_p/2^p$, and e is an integer exponent. The leading 1 is not stored (it is implied), so there are $p+1$ significant binary digits. The two formats that matter here are summarized in Table 4.1.

Table 4.1: The two IEEE floating-point formats used in scientific computing. "Digits" is $(p+1)\log_{10} 2$, the number of decimal digits the significand can hold.

format	bits	exponent bits	p	machine epsilon 2^{-p}	digits	largest value
double (float64)	64	11	52	2.22×10^{-16}	15.95	1.80×10^{308}
single (float32)	32	8	23	1.19×10^{-7}	7.22	3.40×10^{38}

Some numbers are exact: $10 = (1.01)_2 \times 2^3$. Others are not: $0.1 = (1.100110011\ldots)_2 \times 2^{-4}$ repeats forever in binary, just as $1/3$ repeats in decimal, so it must be rounded.

Machine epsilon. The *machine epsilon* ε is the gap between 1 and the next larger floating-point number. From Equation (4.34), the next number after $(1.00\ldots0)_2$ is $(1.00\ldots01)_2 = 1 + 2^{-p}$, so $\varepsilon = 2^{-p}$: 2.22×10^{-16} in double and 1.19×10^{-7} in float32.

The rounding model. Rounding to the nearest floating-point number moves x by at most half the local spacing. Near 2^e the spacing is $\varepsilon\, 2^e$, and $|x| \geq 2^e$, so the relative error is at most $\varepsilon/2$:

$$\mathrm{fl}(x) = x(1+\delta), \qquad |\delta| \leq \frac{\varepsilon}{2}. \tag{4.35}$$

The same holds for each $+$, $-$, $\times$, $\div$: the computer returns the exact result of the operation on the stored numbers, rounded by Equation (4.35). Figure 4.7 shows the relative rounding error of float32 across several powers of 2.

4.9.2 Consequences

Small additions vanish. In double precision, $1 + 10^{-16}$ rounds to exactly 1, since $10^{-16} < \varepsilon/2 = 1.11 \times 10^{-16}$. In float32, $1 + 10^{-8}$ equals 1. And in float32 the integer $16777217 = 2^{24} + 1$ cannot be stored: it rounds to 16777216.

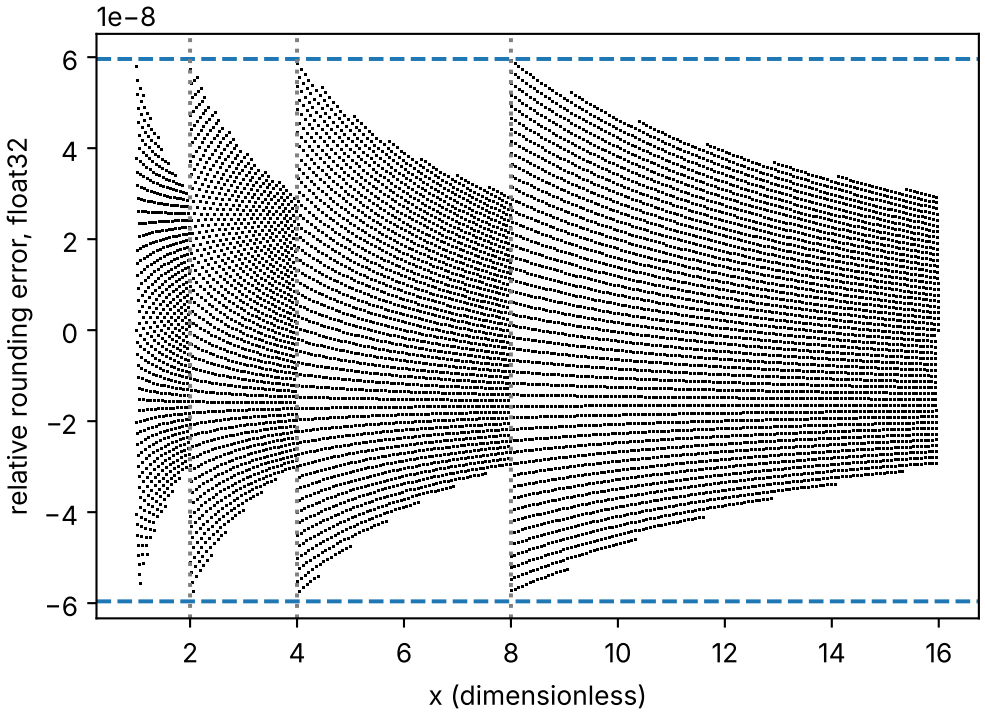


Figure 4.7: Relative error $(\mathrm{fl}_{32}(x) - x)/x$ of rounding to float32, for 10^4 values $1 \le x \le 16$ (dimensionless). The dashed lines are $\pm\varepsilon_{32}/2 = \pm 5.96 \times 10^{-8}$. Within each interval $[2^e, 2^{e+1})$ the absolute spacing is constant, so the relative error envelope shrinks from $\varepsilon_{32}/2$ to $\varepsilon_{32}/4$ and jumps back at each power of 2 (vertical dotted lines).

Accumulation. Rounding errors add up. Summing 0.1 ten times gives 0.9999999999999999 in double and 1.0000001192092896 in float32. Neither is wrong by more than a few ε, but neither is 1.

Catastrophic cancellation. Subtracting nearly equal numbers keeps their absolute rounding errors (about $\varepsilon\,|a|$) but makes the result small, so the *relative* error of the difference is about $\varepsilon\,|a|\,/\,|a-b|$, which can be huge.

Sample Problem 4.10: A harmless rounding error and a catastrophic one

(a) Compute $\sum_{k=1}^{10} 0.1$ in double precision. (b) Compute $(1-\cos x)/x^2$ at $x = 10^{-8}$, whose true value is very close to $\frac{1}{2}$.
Step 1: For (a), each addition rounds by Equation (4.35), and 0.1 itself is stored slightly wrong. The result is 0.9999999999999999, which differs from 1 by 1.1×10^{-16}, half an ε: a harmless error, but enough to make the test `total == 1.0` fail.
Step 2: For (b), $\cos(10^{-8}) = 1 - 5 \times 10^{-17} + \ldots$. The doubles just below 1 are spaced by $\varepsilon/2 = 1.11 \times 10^{-16}$, so the nearest double to $1 - 5 \times 10^{-17}$ is 1 itself. The computer gets $\cos x = 1$, then $1 - \cos x = 0$, and the answer 0.
Step 3: Repair (b) by removing the subtraction analytically: $1 - \cos x = 2\sin^2(x/2)$, so $(1-\cos x)/x^2 = 2\sin^2(x/2)/x^2$, which the computer evaluates as 0.5.

(a) 0.9999999999999999, (b) naive 0, rewritten 0.5

Rule of thumb: never subtract nearly equal floating-point numbers if algebra lets you avoid it, and never test computed numbers with `==`.

4.9.3 When "exactly zero" becomes noise

Many quantities in physics vanish exactly for a structural reason. For example, certain lists of numbers (eigenvectors, Chapter 8) have dot product exactly zero, and certain sums cancel term by term in pairs (sum rules, Chapter 14). On a computer, such a quantity is computed from rounded ingredients and comes out not as 0 but as a small number whose size is set by ε times the size of the terms that cancel.

A clean example. On a grid of N points, the vectors $\psi_n(j) = \sqrt{2/(N+1)}\,\sin\!\big(nj\pi/(N+1)\big)$, $j = 1, \ldots, N$, are exactly orthonormal: $\sum_j \psi_n(j)\psi_m(j) = 0$ for $n \neq m$ (you will prove this in Chapter 8; they are the eigenvectors of the discrete box). Compute the dot product of ψ_3 and ψ_7 with $N = 999$ by adding the terms in order. In double precision the result is below 10^{-14}. Store the vectors in float32 and repeat: the result is again small but nonzero, and at least a thousand times larger than the double-precision result; its size is set by $\varepsilon_{32} \approx 1.2 \times 10^{-7}$ times the size of the individual terms. (The exact digits depend on how the platform rounds the sines, so we do not quote them; the listing below prints them.) Nothing is wrong with either computation; the second simply has fewer digits.

The danger comes when such noise is fed to something that can amplify it.

Sample Problem 4.11: A parameter that should do nothing

A model predicts a correction $p = c\,\eta$ to a quantity, where c is a fitted coefficient and η is a sensitivity that is *analytically zero* (the parameter has no physical effect). Because of float32 rounding the computed sensitivity is $\eta = 10^{-7}$. A least-squares fit asks for the c that makes p equal a residual $r = 0.01$. What does the fit do?
Step 1: Minimizing $(r - c\eta)^2$ over c: setting the derivative $-2\eta(r - c\eta)$ to zero gives $c = r/\eta$.

Step 2: With the numbers: $c = 0.01/10^{-7} = 10^5$.
Step 3: Interpret. The fit "explains" the residual perfectly by driving a meaningless parameter to an enormous value. If the noise had come out as $\eta = -3 \times 10^{-8}$ instead, the same fit would give $c = -3.3 \times 10^5$. The improvement is real in the computer and meaningless in physics.
Step 4: The repair is to set the analytically zero sensitivity to exactly 0. Then $p = 0$ for every c, the fit cannot use the parameter at all, and the coefficient stays where it started (Chapter 9 explains how such "null directions" are handled systematically).

$$c = r/\eta = 10^5 : \text{ a huge coefficient harvesting noise}$$

This is, in miniature, what happened in Sec. V of the paper: the optimizer drove an exactly-null coefficient to $c_0 \approx 2.2 \times 10^6$ by exploiting float32 noise in analytically zero tensor entries (sensitivity coefficients of the second-order model, Chapter 14). Chapter 31 dissects the full story.

```
import numpy as np

total = 0.0
for k in range(10):
    total += 0.1
print(total, total == 1.0)            # 0.9999999999999999 False

x = 1e-8
print((1 - np.cos(x)) / x**2)         # 0.0   (cancellation)
print(2 * np.sin(x / 2)**2 / x**2)    # 0.5   (rewritten)

N = 999
j = np.arange(1, N + 1)
def psi(n, dtype):
    v = np.sqrt(2 / (N + 1)) * np.sin(n * j * np.pi / (N + 1))
    return v.astype(dtype)
for dt in (np.float64, np.float32):
    prod = psi(3, dt) * psi(7, dt)
    # cumsum adds the terms in order, as in the text; np.sum adds
    # pairwise, which hides part of the rounding
    print(dt.__name__, np.cumsum(prod)[-1])   # "zero", in each precision

print(np.finfo(np.float64).eps, np.finfo(np.float32).eps)
```

Pitfall

Never decide that a computed quantity "is zero" by comparing with `0.0`, and never decide it "is not zero" because it prints as 3×10^{-8}. Compare it with ε times the size of the terms that produced it. If theory says a quantity is exactly zero, set it to zero in the code rather than computing it.

Checkpoint 4.8

In float32, is $1 + 10^{-8}$ different from 1? In double?

Answer: In float32 it equals 1, because $10^{-8} < \varepsilon_{32}/2 = 5.96 \times 10^{-8}$. In double it is different from 1, because $10^{-8} \gg \varepsilon_{64}/2 = 1.11 \times 10^{-16}$.

Key Ideas

- Floating-point numbers have about 16 (double) or 7 (float32) significant decimal digits; machine epsilon is 2.2×10^{-16} and 1.2×10^{-7}.
- Each operation rounds with relative error at most $\varepsilon/2$, Equation (4.35); subtraction of nearly equal numbers loses relative accuracy.
- Analytically zero quantities come out as noise of size $\varepsilon\times$(term size); an optimizer can amplify that noise into large, meaningless parameters.

4.10 Paper connection

★ Did you know? *Newton's Favorite Cubic*

Isaac Newton illustrated his root-finding method around 1669 on the equation $y^3-2y-5=0$, and Joseph Raphson published a streamlined version in 1690. Near a simple root the method roughly doubles the number of correct digits at each step. Much earlier, Heron of Alexandria described an iteration for square roots that is exactly Newton's method applied to $x^2 - a = 0$.

Here is where each tool of this chapter appears in the paper [1].

Smooth zeros (Sec. II A, Eq. (1), Appendix A step 2). The paper defines s_n by $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ and computes it with "Brent on $N_{\mathrm{sm}}(t) = n - \frac{1}{2}$". Section 4.5 showed that this equation has exactly one solution with $s_n > 2\pi$ because N_{sm} is increasing there, that for $n = 1$ it has a second, spurious solution near 0.756, and that the closed form Equation (4.27) agrees with the root-finder solution. The mean spacing is computed in Appendix A step 2 as `np.gradient`(smooth zeros), a finite-difference approximation to the derivative $\mathrm{d}s_n/\mathrm{d}n$ of Equation (4.24). For interior indices $20 \le n \le 999$ the two agree to better than 0.1%; at the very first index the one-sided difference $s_2 - s_1 = 6.13$ differs noticeably from the derivative $\mathrm{gap}_1 = 7.50$. Every rms in the paper is divided by these spacings.

The Lambert-W form (Sec. I A, paragraph b). The paper describes França and LeClair's equation for the n-th zero as having a smooth part that "is the Lambert-W asymptotic of the counting function" [3, 4]. Our Equations (4.27) and (4.28) are that smooth part, derived in five lines from Eq. (1).

Precision of the data (Sec. II A). The zeros are computed "at 20 significant digits" up to $n = 1000$ and "at 15 digits" beyond, and the paper says "both precisions exceed every use below by many orders". With Section 4.1 you can verify this: the 15-digit value quoted for t_{1000} has rounding error at most 5×10^{-12}, about 4×10^{-12} mean spacings (Problem 22), and the paper actually computed zeros up to 1000 at 20 digits. The smallest errors the paper quotes are of order 10^{-4} spacings (the in-sample rms below 10^{-4} at $M = 900$, Sec. III), still more than seven orders of magnitude larger.

Inverting the Wu–Sprung potential (Sec. II B, Eq. (2), Appendix A step 3). Eq. (2) gives $x(V)$, an increasing function for $V \ge V_0$ with $V_0 = 3.10073\pi$. The solver needs $V(x)$; the paper inverts Eq. (2) "pointwise by bracketed root finding" (Sec. II B), "by Brent inversion of Eq. (2) on a dense reference grid, cubic-spline interpolated" (Appendix A step 3). The box half-widths $L = x(1000) = 44.883$ and $L = x(1500) = 59.957$ are values of the forward function. Chapter 22 derives Eq. (2).

The oracle's level equation (Sec. II D, Appendix A step 8). Predicted zeros solve $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ by "Brent bracketed ±3 around the smooth zero (bracket expanded on sign failure)" (the paper gives no units for the ±3; see Section 4.8). This is the bracket-expansion strategy of Section 4.8. In Sec. IV the paper reports that the fitted-amplitude control B with 900 coefficients is "root-ambiguous (the level equation has several roots for about a third of the held-out levels)" and that "the score varies by at most 0.012 with the choice of root"; this is the non-monotonic situation of Section 4.8, flagged with a dagger in Table IV.

Richardson extrapolation as an asymptotic expansion (Sec. II B, Appendix A steps 3 and 5). The finite-difference levels behave like $E(h) = E^* + ah^2 + \mathcal{O}(h^4)$ as the grid spacing $h \to 0$ (Chapter 23 derives this). Halving h gives $E(h/2) = E^* + ah^2/4 + \mathcal{O}(h^4)$, and $4E(h/2) - E(h) = 3E^* + \mathcal{O}(h^4)$, so $(4E(h/2) - E(h))/3$ cancels the h^2 term. This is the paper's E^{Rich}, built from the "Richardson pair 12000/24000" grids. (The paper writes this combination as $(4E_{2h} - E_h)/3$. The cancellation above works only if the factor 4 multiplies the levels on the *finer* grid of the pair, so in the paper's formula E_{2h} must be read as the finer-grid levels, our $E(h/2)$. Chapter 23 returns to this notation.) Problem 28 practices the idea on a simpler function.

The float32 artifact (Sec. II E safeguard (i), Sec. V, Appendix A step 9). The constant mode $\phi_0 = 1$ of the absorptive-even class has, analytically, zero second-order real shift. Its tensor rows were at first "left as float32 noise", and the optimizer "had driven the exactly-null constant mode to $c_0 \approx 2.2 \times 10^6$, harvesting in-sample improvement from float32 noise in analytically-zero tensor entries" (Sec. V). Before repair the floor was 0.2322, spuriously close to the PT-odd value 0.2321; after the rows were "zeroed analytically" (Appendix A step 9) it rose to 0.2358. The sample "A parameter that should do nothing" in Section 4.9 is this mechanism in its simplest form; Problem 20 estimates the size of c_0 in a quadratic version. Chapter 31 gives the full account.

Asymptotic thinking in the claims. The paper's statement that the oracle predicts "essentially as well at $n = 10^4$ as at $n = 400$" (Sec. IV), and its use of heights where the mean spacing is still near 1, both rest on the slow, logarithmic growth of Sections 4.2 and 4.5: between zeros 401 and 10000 the spacing changes only from about 1.3 to about 0.85. (The following is our own remark, not a claim made in the paper: over that range $\log(s_n/2\pi) = 2\pi/\mathrm{gap}_n$ grows only from about 4.7 to about 7.4, so "nearly height-independent" refers to a modest stretch of the logarithmic scale.)

Key Ideas

- The paper's smooth zeros, potential inversion and oracle predictions are all bracketed root finding.
- Lambert W gives the smooth zeros in closed form; the paper cites this form in Sec. I A.
- The spurious result of Sec. V began as float32 noise in quantities that are exactly zero on paper.

4.11 Summary

- Absolute error $\hat{x} - x$, relative error $(\hat{x} - x)/x$ (Equation (4.1)); rounding to d significant digits gives relative error $\leq 5 \times 10^{-d}$ (Equation (4.2)). The paper measures errors in mean spacings gap_n.

- $\mathcal{O}$, o, $\sim$ (Equations (4.4) to (4.6)); $\log t = o(t^a)$ for $a > 0$ (Equation (4.7)); $N_{\mathrm{sm}}(t) \sim \frac{t}{2\pi}\log t$ but converges slowly.
- Sums of monotone functions lie between integrals (Equations (4.8) and (4.9)); $H_n = \log n + \gamma + \mathcal{O}(1/n)$; $\log n! = n\log n - n + \mathcal{O}(\log n)$ (Equation (4.12)).
- Stirling: $\log n! = n\log n - n + \frac{1}{2}\log(2\pi n) + \vartheta_n/(12n)$, $0 < \vartheta_n < 1$ (Equation (4.16)); Stirling series Equation (4.17); Gamma function Equations (4.18) and (4.19); $\log\Gamma$ asymptotics Equation (4.20), extended to complex arguments on trust; preview of $\theta(t)$ Equation (4.21).
- Inverse functions: existence for strictly monotone continuous functions; $\mathrm{d}x/\mathrm{d}y = 1/(\mathrm{d}y/\mathrm{d}x)$ (Equation (4.22)); N'_{sm}, N''_{sm} (Equation (4.23)); $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$ (Equation (4.24)).
- Lambert W (Equation (4.25)); $s_n = 2\pi(n - \frac{11}{8})/W((n - \frac{11}{8})/e)$ and $\mathrm{gap}_n = 2\pi/(1 + W)$ (Equations (4.27) and (4.28)).
- Bisection error bound $(b - a)/2^{k+1}$ (Equations (4.29) and (4.30)); Newton's method and quadratic convergence (Equations (4.31) and (4.32)); guaranteed monotone convergence for increasing convex f; secant method (Equation (4.33)); Brent's method as bracketed fast steps.
- Brackets fail for even root counts, double roots, poles; expand brackets using known monotonicity; non-monotone level equations are root-ambiguous.
- Floating point (Equation (4.34), Table 4.1): $\varepsilon_{64} = 2.2 \times 10^{-16}$, $\varepsilon_{32} = 1.2 \times 10^{-7}$; rounding model Equation (4.35); cancellation; analytic zeros become noise that fits can amplify.

4.12 Problems

1. • The approximation $\pi \approx 22/7$ is famous. Find its absolute error, its relative error, and the number of significant digits to which it agrees with $\pi = 3.14159265\ldots$.

2. • Which of the following are true as $t \to \infty$? (a) $5t^2 + 3t = \mathcal{O}(t^2)$; (b) $\log t = o(t^{0.1})$; (c) $t\log t \sim t$; (d) $\sin t = \mathcal{O}(1)$; (e) $e^{-t} = o(t^{-100})$.

3. • Compute the ratio $\left[\frac{t}{2\pi}\log t\right]/N_{\mathrm{sm}}(t)$ at $t = 10^6$ and compare with its value 1.64 at t_{1000} (Section 4.2). What does this say about using the leading-order term?

4. •• Use Equation (4.10) to bound H_{100}, and compare with the exact value.

5. •• Use the comparison of sums and integrals for the increasing function $f(x) = x^2$ to show $\frac{n^3}{3} \le \sum_{k=1}^{n} k^2 \le \frac{(n+1)^3 - 1}{3}$, and deduce $\sum_{k=1}^{n} k^2 \sim n^3/3$.

6. • Use Stirling's series Equation (4.17) through the $1/(12n)$ term to compute $\log 20!$, and find the relative error of the simpler approximation $20! \approx \sqrt{40\pi}\,(20/e)^{20}$.

7. •• (a) Show from Equation (4.19) that $\Gamma(3) = 2$. (b) Using $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, compute $\Gamma(\frac{5}{2})$.

8. •• Apply two steps of Newton's method to $f(x) = x^3 - 10$ starting from $x_0 = 2$. Compare with $10^{1/3}$.

9. •• *(Paper.)* The oracle solver of Appendix A step 8 starts from a bracket ± 3 around the smooth zero; read the ± 3 in mean spacings, as `qmnt` does, so the bracket is $[s_n - 3\mathrm{gap}_n,\ s_n + 3\mathrm{gap}_n]$. If bisection were used instead of Brent, how many steps would guarantee an error below 10^{-12} mean spacings?

10. •• Show that the integer $2^{24}+1 = 16777217$ is not representable in float32, and find the float32 value it is stored as. Using numpy, confirm it.

11. •• Compute $\sqrt{x+1}-\sqrt{x}$ at $x = 10^{12}$ in double precision directly, and again as $1/(\sqrt{x+1}+\sqrt{x})$. Find the relative error of the direct method and explain it with Equation (4.35).

12. •• *(Paper.)* Appendix A of the paper lists the anchor $t_{500} = 811.1843588465$. Find s_{500}, gap_{500} and the fluctuation $(t_{500}-s_{500})/\mathrm{gap}_{500}$.

13. •• (a) Show that $x\log x = 1$ has solution $x = 1/W(1)$. (b) Compute $W(1)$ by Newton's method on $g(w) = we^{w}-1$ from $w_0 = 0.5$: give w_1 and the converged value.

14. •• *(Computational.)* Using `qmnt.smooth_zeros` and `scipy.special.lambertw`, verify that Equation (4.27) reproduces the smooth zeros for $1 \le n \le 10^4$. What is the largest discrepancy?

15. ••• Let $a > 0$ and $f(x) = x^2 - a$. (a) Use the convexity theorem of Section 4.7 to show that Newton's method converges to $\sqrt{a}$ from every $x_0 > 0$. (b) Show that $e_{k+1} = e_k^2/(2x_k)$ exactly, where $e_k = x_k - \sqrt{a}$.

16. •• *(Computational, paper.)* Using `qmnt.zeros` and `qmnt.fluct`, compute the rms of the fluctuations $(t_n - s_n)/\mathrm{gap}_n$ for $401 \le n \le 500$. Compare with the paper's null 0.263 on zeros 401–500 (Sec. II C, Table V).

17. •• For each of the following, say whether $[a,b]$ is a valid bisection bracket and whether bisection would return a root: (a) $f(x) = (x-1)(x-2)$ on $[0,3]$; (b) $f(x) = x^2$ on $[-1,1]$; (c) $f(x) = 1/x$ on $[-1,2]$; (d) $f(x) = (x-1)(x-2)(x-3)$ on $[0,4]$.

18. ••• The secant method Equation (4.33) has errors that satisfy $e_{k+1} \approx C e_k e_{k-1}$ near a simple root (you may assume this). Suppose $|e_{k+1}| \approx K\,|e_k|^{q}$ for some constants K, q. Show that q must satisfy $q^2 = q+1$, so $q = (1+\sqrt{5})/2 \approx 1.618$.

19. •• *(Computational.)* Reproduce the float32 orthogonality experiment of Section 4.9 for $N = 999$, $n = 3$, $m = 7$. Then, in the "parameter that should do nothing" sample, find c if the computed sensitivity is $\eta = -3\times10^{-8}$ and the residual is $r = 0.01$.

20. ••• *(Paper.)* In a second-order model the contribution of a parameter is quadratic, $\delta E = \eta\, c_0^2$, where η is an analytically zero tensor entry computed with rounding noise. Suppose (illustratively; these are not the paper's values) $\eta = 2\times10^{-14}$ and the optimizer wants $\delta E = 0.1$. What c_0 does it choose? Compare with the paper's $c_0 \approx 2.2\times10^{6}$ (Sec. V) and explain why the fix of Appendix A step 9 removes the effect.

21. •• Show that $\mathrm{gap}_n \sim 2\pi/\log n$. Compare $\mathrm{gap}_{1000} = 1.159$ with $2\pi/\log 1000$.

22. • *(Paper.)* The paper quotes $t_{1000} = 1419.42248094599$. How many significant digits is this? Bound the rounding error of this value in units of $\mathrm{gap}_{1000} = 1.159$, and relate your answer to the paper's statement in Sec. II A.

23. •• Apply one Newton step for s_1 (i.e. for $f(s) = N_{\mathrm{sm}}(s) - \tfrac12$) starting from $x_0 = 7$, just above 2π. Where does it land? Explain, using the convexity theorem, why the iteration still converges.

24. ••• *(Computational, paper.)* Using `qmnt.oracle_zeros` with the six primes $2,3,5,7,11,13$ and `qmnt.score`, compute the held-out rms and correlation on zeros 401–500. Compare with Table V. Then write your own version with the bracket-expansion listing of Section 4.8 and `qmnt.Nfl`, and check that it gives the same predicted zeros.

25. •• *(Computational.)* Evaluate `mpmath.siegeltheta(100)` and the three-term approximation $\frac{t}{2}\log\frac{t}{2\pi}-\frac{t}{2}-\frac{\pi}{8}$ at $t=100$. Show that their difference is close to $1/(48t)$, as Equation (4.21) predicts.

26. •• Let $\gamma_n = H_n - \log n$. Show that γ_n is decreasing and positive, and conclude that it converges. (Hint: use $\int_n^{n+1} \mathrm{d}x/x > 1/(n+1)$ and Equation (4.10).) Compute γ_{10}.

27. ••• The equation $v(\log v - 1) = n$ (the leading part of $N_{\mathrm{sm}}(2\pi v) = n$) can be solved by the fixed-point iteration $v_{k+1} = n/(\log v_k - 1)$. (a) For $n = 1000$ and $v_0 = 1000$, compute $v_1, \ldots, v_4$. (b) Show that near the solution v^* the error is multiplied by approximately $-1/(\log v^* - 1)$ at each step, and evaluate this factor.

28. •• The symmetric difference quotient $D(h) = (e^h - e^{-h})/(2h)$ approximates $\frac{\mathrm{d}}{\mathrm{d}x}e^x\big|_{x=0} = 1$ with $D(h) = 1 + h^2/6 + h^4/120 + \mathcal{O}(h^6)$. (a) Derive this expansion from the Taylor series of e^h. (b) Compute $D(0.1)$, $D(0.05)$ and the Richardson combination $(4D(0.05) - D(0.1))/3$. (c) Predict the error of the combination from the expansion and check it.

29. •• Keep only the $p = 2$, $m = 1$ term of the paper's Eq. (4): $N_{\mathrm{fl}}(t) = -\frac{1}{\pi\sqrt{2}}\sin(t\log 2)$. Apply one Newton step to $g(t) = N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) - \frac{1}{2}$ starting at $t_0 = 14.5214 \approx s_1$, and then find the converged root. Compare with s_1 and with $t_1 = 14.1347$.

30. ••• *(Paper.)* The level equation $N_{\mathrm{sm}}(t)+N_{\mathrm{fl}}(t) = n-\frac{1}{2}$ with the six-prime N_{fl} of Eq. (4) ($m \le 3$) is certainly monotone where $|N'_{\mathrm{fl}}(t)| < N'_{\mathrm{sm}}(t)$. (a) Show that $|N'_{\mathrm{fl}}(t)| \le \frac{1}{\pi}\sum_p \sum_{m=1}^{3} \frac{\log p}{p^{m/2}}$ and evaluate the bound. (b) Compare it with $N'_{\mathrm{sm}}(800)$. (c) Does your result prove that the oracle's level equation is root-ambiguous? Relate to the root-ambiguous row of Table IV.

31. •• *(Computational, paper.)* Using `qmnt.wu_sprung_x` and `qmnt.wu_sprung_V`, compute $x(1000)$ and $x(1500)$, check that `wu_sprung_V` inverts `wu_sprung_x` at these points, and check numerically that $x(V)$ is increasing on $V_0 < V \le 1500$. Compare with the paper's L values (Sec. II B).

4.13 Answers to selected problems

Problem 1. $22/7 = 3.142857$. Absolute error $3.142857 - 3.141593 = 1.264 \times 10^{-3}$. Relative error $1.264 \times 10^{-3}/3.141593 = 4.02 \times 10^{-4}$. The first three digits 3.14 agree; the fourth (2 versus 1) does not, so $22/7$ agrees with π to 3 significant digits. Its relative error 4.0×10^{-4} is just under the 4-digit bound 5×10^{-4} of Equation (4.2), but that bound is only a necessary condition: rounded to four digits, $22/7$ gives 3.143 while π gives 3.142.

Problem 2. (a) True: $5t^2 + 3t \le 8t^2$ for $t \ge 1$. (b) True by Equation (4.7) with $a = 0.1$. (c) False: $(t\log t)/t = \log t \to \infty$. (d) True: $|\sin t| \le 1$. (e) True: $e^{-t}/t^{-100} = t^{100}/e^t \to 0$ since exponentials beat powers.

Problem 3. At $t = 10^6$, $\log(t/2\pi) = 13.815511 - 1.837877 = 11.977634$, so $N_{\mathrm{sm}} \approx \frac{t}{2\pi}(11.977634 - 1)$ (the $\frac{7}{8}$ is negligible here) while $\frac{t}{2\pi}\log t = \frac{t}{2\pi} \times 13.815511$. The ratio is 1.26. Even a million units up, the leading term is 26% too large; it should never be used where accuracy matters.

Problem 5. For increasing f, the rectangles of height $f(k)$ on $[k-1, k]$ lie above the curve and those on $[k, k+1]$ lie below it: $\int_{k-1}^{k} x^2\mathrm{d}x \le k^2 \le \int_{k}^{k+1} x^2\mathrm{d}x$. Summing over $k = 1, \ldots, n$: $\int_0^n x^2\mathrm{d}x = \frac{n^3}{3} \le \sum k^2 \le \int_1^{n+1} x^2\mathrm{d}x = \frac{(n+1)^3-1}{3}$. Dividing by $n^3/3$, both bounds tend to 1, so $\sum k^2 \sim n^3/3$. (Exactly, $\sum k^2 = n(n+1)(2n+1)/6$.)

Problem 6. $\log 20! = 20.5\log 20 - 20 + \frac{1}{2}\log 2\pi + \frac{1}{240} = 61.412512 - 20 + 0.918939 + 0.004167 = 42.335617$, against the exact 42.335616 (difference $-3.5 \times 10^{-7} \approx -1/(360 \cdot 20^3)$). The plain Stirling approximation omits $1/(12n)$ and higher terms, so its ratio to 20! is about $e^{-1/240}$; the relative error is about 0.42%.

Problem 7. (a) $\Gamma(2) = 1 \cdot \Gamma(1) = 1$, $\Gamma(3) = 2\Gamma(2) = 2$. (b) $\Gamma(\frac{3}{2}) = \frac{1}{2}\Gamma(\frac{1}{2}) = \frac{\sqrt{\pi}}{2}$ and $\Gamma(\frac{5}{2}) = \frac{3}{2}\Gamma(\frac{3}{2}) = \frac{3\sqrt{\pi}}{4} = 1.329340$.

Problem 8. $x_{k+1} = x_k - (x_k^3 - 10)/(3x_k^2)$. $x_1 = 2 - (8-10)/12 = 2.166667$. Then $x_1^3 = 10.171$, so $x_2 = 2.166667 - (10.171296 - 10)/(3 \times 2.166667^2) = 2.166667 - 0.171296/14.083333 = 2.154504$. The exact value is $10^{1/3} = 2.154435$, so after two steps the error is 6.9×10^{-5}.

Problem 9. The bracket has width $6\mathrm{gap}_n$, and we want $\delta = 10^{-12}\mathrm{gap}_n$. By Equation (4.30), $k \geq \log_2(6 \times 10^{12}) - 1 = 42.45 - 1 = 41.45$, so $k = 42$ steps are needed. Brent normally needs far fewer than this.

Problem 11. The true value is $1/(\sqrt{x+1} + \sqrt{x}) = 4.99999999999875 \times 10^{-7}$, and the rewritten form returns exactly this. In the direct form, $\sqrt{10^{12}+1} = 10^6 + 5 \times 10^{-7}$ must be stored as a double near 10^6. Doubles near 10^6 are spaced by $2^{-33} \approx 1.16 \times 10^{-10}$, so by Equation (4.35) the stored value can be wrong by up to 5.8×10^{-11}. The subtraction of 10^6 is then exact, so it keeps this absolute error while the result is only 5×10^{-7}. The relative error of the direct form is therefore up to about 1.2×10^{-4}. Running the code confirms an error of this order, which means roughly 12 of the 16 digits are lost to cancellation.

Problem 13. (a) Put $x = e^u$ as in Equation (4.26). Then $ue^u = 1$, so $u = W(1)$, and $x = e^{W(1)} = 1/W(1)$ (using $W(1)e^{W(1)} = 1$). (b) The derivative is $g'(w) = (1+w)e^w$. At $w_0 = 0.5$: $g = 0.5 \times 1.648721 - 1 = -0.175639$ and $g' = 1.5 \times 1.648721 = 2.473082$, so $w_1 = 0.5 + 0.071020 = 0.571020$. Further steps converge to $W(1) = 0.567143$, which gives $x = 1.763223$.

Problem 15. (a) On $(0, \infty)$ we have $f'(x) = 2x > 0$ and $f'' = 2 > 0$, so f is increasing and convex there, with root $\sqrt{a}$. From any $x_0 > 0$ the first step gives $x_1 = \frac{1}{2}(x_0 + a/x_0) > 0$, so x_1 stays in the interval, and the theorem gives $x_1 \geq \sqrt{a}$. From then on the iterates decrease monotonically to $\sqrt{a}$. (b) Write the Newton step and subtract $\sqrt{a}$: $x_{k+1} - \sqrt{a} = \dfrac{x_k^2 + a}{2x_k} - \sqrt{a} = \dfrac{x_k^2 - 2\sqrt{a}\,x_k + a}{2x_k} = \dfrac{(x_k - \sqrt{a})^2}{2x_k}$. This is $e_{k+1} = e_k^2/(2x_k)$ exactly.

Problem 17. (a) Not a bracket: $f(0) = f(3) = 2 > 0$, even though the interval contains two roots. (b) Not a bracket: $x^2 \geq 0$ everywhere, so there is no sign change at the double root. (c) There is a sign change, $f(-1) = -1$ and $f(2) = \frac{1}{2}$, but f is not continuous. Bisection converges to the pole $x = 0$, which is not a root. (d) A valid bracket, since $f(0) = -6$ and $f(4) = 6$, containing three roots. The first midpoint is 2, which is itself a root, so bisection returns 2 immediately.

Problem 19. In double precision the computed dot product is below 10^{-14}. In float32 it is at least a thousand times larger, at a level set by ε_{32} times the size of the terms (record the value your machine prints). For the second part, minimizing $(r - c\eta)^2$ gives $c = r/\eta = 0.01/(-3 \times 10^{-8}) = -3.33 \times 10^5$. The sign and size of c are set entirely by the noise.

Problem 21. Since $s_n \sim 2\pi n/\log n$, we have $\log(s_n/2\pi) = \log n - \log\log n + o(1)$. Dividing by $\log n$ and using $\log\log n = o(\log n)$ (Equation (4.7)) gives $\log(s_n/2\pi) \sim \log n$, so $\mathrm{gap}_n \sim 2\pi/\log n$. At $n = 1000$, $2\pi/\log 1000 = 0.9096$, while the exact value is $\mathrm{gap}_{1000} = 1.159$, a ratio of 1.27. For comparison, at $n = 10^4$ the exact spacing is $\mathrm{gap}_{10^4} = 0.854$, against the leading-order value $2\pi/\log 10^4 = 0.682$.

Problem 22. The value has 15 significant digits: four before the decimal point and eleven after it. Rounding error is at most half a unit in the last place, 5×10^{-12}, which is 4.3×10^{-12} spacings. The smallest errors the paper quotes are of order 10^{-4} spacings (the in-sample rms below 10^{-4} at $M = 900$, Sec. III), more than seven orders of magnitude larger, so its statement that the precisions "exceed every use below by many orders" (Sec. II A) is confirmed. (The paper in fact computed t_{1000} at 20 digits.)

Problem 23. At $x_0 = 7$: $f(7) = -0.61873$ and $f'(7) = \frac{1}{2\pi}\log(7/2\pi) = 0.017194$. The step is $0.61873/0.017194 = 35.985$, so $x_1 \approx 43.0$. The tangent at x_0 is nearly flat, so the step is huge. However, it overshoots to the right of s_1. By the convexity theorem, every later iterate stays at

or above s_1 and decreases monotonically to $s_1 = 14.5213$.

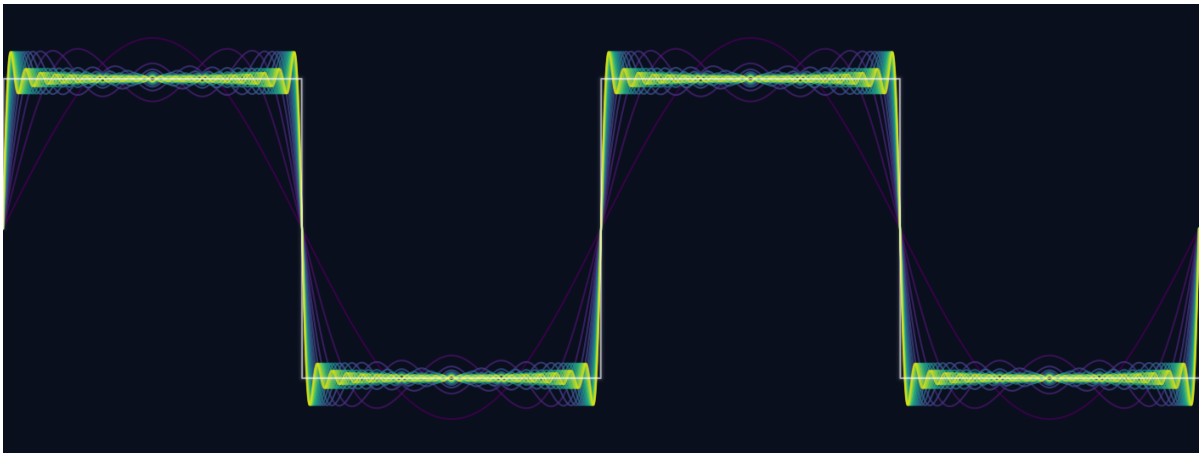
Problem 25. `mpmath.siegeltheta(100)` returns 87.97216, and the three-term formula gives 87.97196. The difference is 2.083×10^{-4}, compared with $1/(48 \times 100) = 2.083 \times 10^{-4}$. They agree to within 10^{-8}; the next term of Equation (4.21) is of order 10^{-9}.

Problem 27. (a) $v_1 = 1000/5.907755 = 169.27$, $v_2 = 242.04$, $v_3 = 222.76$, $v_4 = 226.96$. The iterates oscillate about the solution $v^* \approx 226.2$. (b) Write $g(v) = n/(\log v - 1)$. Then $g'(v) = -n/(v(\log v - 1)^2)$. At the fixed point, $n/(\log v^* - 1) = v^*$, so $g'(v^*) = -1/(\log v^* - 1) \approx -0.226$. Each step therefore multiplies the error by about -0.23, which explains both the oscillation and the steady but linear convergence.

Problem 29. At $t_0 = 14.5214$: $N_{\mathrm{fl}} = +0.1346$, $N'_{\mathrm{fl}} = +0.1251$ and $N'_{\mathrm{sm}} = 0.1333$. Hence $g = 0.1346$ and $g' = 0.2584$, and one Newton step gives $14.5214 - 0.5207 = 14.0007$. Iterating further converges to about 14.021. Starting from t_0, which is 0.387 above $t_1 = 14.1347$, the single prime-2 term moves the prediction to within about 0.11 of t_1.

Problem 30. (a) By the chain rule, the derivative of the term $-\frac{1}{\pi}\frac{\sin(mt\log p)}{mp^{m/2}}$ is $-\frac{1}{\pi}\frac{m\log p\,\cos(mt\log p)}{mp^{m/2}} = -\frac{1}{\pi}\frac{\log p\,\cos(mt\log p)}{p^{m/2}}$ (the factor m cancels). The magnitude of a sum is at most the sum of the magnitudes, and $|\cos| \le 1$, so $|N'_{\mathrm{fl}}(t)| \le \frac{1}{\pi}\sum_p \sum_{m=1}^{3} \frac{\log p}{p^{m/2}}$. For $p \in \{2,3,5,7,11,13\}$ the double sum is 6.568, so the bound is 2.091. (b) $N'_{\mathrm{sm}}(800) = \frac{1}{2\pi}\log\frac{800}{2\pi} = 0.771$. (c) The bound exceeds the slope, so this argument does *not* guarantee monotonicity. But the bound assumes all eighteen cosines take their worst values simultaneously, so it does not prove that the equation is root-ambiguous either. The root-ambiguous row of Table IV is not an oracle row: it is control B with 150 primes and fitted amplitudes. The paper attributes that row's degradation to the underdetermined fit (900 coefficients from 400 zeros). Fitted amplitudes are not bounded by $1/(mp^{m/2})$, so F' can exceed N'_{sm}. Note that the unfitted 900-term oracle A of Table IV is not flagged.

Problem 31. `qmnt.wu_sprung_x` gives $x(1000) = 44.883$ and $x(1500) = 59.957$, the paper's L values for scales 1 and 2 (Sec. II B). Feeding these back into `qmnt.wu_sprung_V` returns 1000 and 1500 to within 10^{-4}, so the bracketed root finder undoes the forward map. At the bottom, $x(V_0) = 0$, because both $\sqrt{V - V_0}$ and $\log\frac{\sqrt{V}+\sqrt{V-V_0}}{\sqrt{V}-\sqrt{V-V_0}} = \log 1$ vanish there. On 2000 equally spaced values of V from just above V_0 to 1500, every successive difference of $x(V)$ is positive. This is a numerical check of monotonicity on a grid, not a proof; Chapter 22 discusses the shape of $x(V)$.



Fourier partial sums of a square wave with 1, 3, 5, …, 39 terms; the overshoot near each jump never disappears (Gibbs).

Chapter 5

Oscillations, Superposition and Fourier Series

↑ Contents

What this chapter is about

When an orchestra tunes, two instruments playing almost the same note produce a slow throbbing, the *beats* you studied in first-year physics. When you pluck a guitar string, the sound you hear is a mixture of the string's harmonics, each a pure sinusoid. These two everyday facts point to one of the most useful ideas in all of science: complicated oscillations can be built by adding simple sinusoids, and, under mild conditions, every repeating shape can be taken apart into sinusoids again. The recipe for taking a shape apart is the *Fourier series*, and the property that makes the recipe work is *orthogonality*.

This chapter builds that machinery from the ground up. We start with a single sinusoid and its three numbers (amplitude, angular frequency, phase), add sinusoids together, derive the orthogonality integrals, and use them to derive the Fourier coefficients, their complex form, and Parseval's theorem, which turns the "size" of a function into a sum of squared coefficients. We then specialize to even functions on an interval $[-L, L]$, where only cosines are needed. Finally we meet sums of sinusoids whose frequencies have an irrational ratio, such as $\sin(t\log 2)+\sin(t\log 3)$. Such sums never repeat, and we prove that no finite window of them can be extended periodically into the right answer.

Where this shows up in the paper [1]: the fitted correction to the potential is a cosine series $\delta V(x)=\sum_{k=0}^{M-1} c_k\cos(k\pi x/L)$ on $[-L,L]$ (paper Sec. II C); the arithmetic predictor of paper Eq. (4) is a sum of sinusoids with frequencies $m\log p$, which is exactly the kind of never-repeating sum studied at the end of this chapter; and the paper's "arithmetic-continuation probe" (Sec. III) asks a fitted model to continue the single prime-2 wave, whose period in t is $2\pi/\log 2\approx 9.065$.

What you need

- Chapter 2: complex numbers, Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, $\cos\theta = (e^{i\theta} + e^{-i\theta})/2$, $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$, complex conjugate, modulus.
- Chapter 3: partial sums, convergence of series, the geometric series $\sum_{j=0}^{K-1} z^j = (1 - z^K)/(1 - z)$ for $z \neq 1$.
- Chapter 4: approximating a sum by an integral; the idea of a limit as a parameter grows.
- First-year physics: simple harmonic motion, waves on a string, standing waves and harmonics, superposition, beats.
- Calculus: derivatives, integrals, substitution, integration by parts, the trigonometric addition formulas $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$ and $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$.

Throughout the book log means the natural logarithm. So $\log 2 = 0.693147\ldots$ and $\log 3 = 1.098612\ldots$.

5.1 Sinusoids: amplitude, angular frequency and phase

A mass on a spring, the voltage across an AC outlet, and the height of a point on a vibrating string all trace the same curve in time: a sinusoid. In this book the independent variable will often *not* be time. In the paper it is usually the height t of a zeta zero, a pure number with no units, or the position x in a potential well. Nothing in the mathematics cares what the variable means, so we write t and speak of "angular frequency in t".

5.1.1 Definition

A **sinusoid** is a function of the form

$$f(t) = A\cos(\omega t + \varphi), \tag{5.1}$$

with three constants:

- the **amplitude** $A \geq 0$, the largest value of $|f|$;
- the **angular frequency** $\omega > 0$, measured in radians per unit of t;
- the **phase constant** φ, measured in radians, which slides the curve along the t axis.

The quantity $\omega t + \varphi$ is called the **phase** at t. Because cosine and sine differ only by a shift, $\sin\theta = \cos(\theta - \pi/2)$, a sine is also a sinusoid: $A\sin(\omega t) = A\cos(\omega t - \pi/2)$.

Period. A function f is **periodic with period** $\tau > 0$ if $f(t + \tau) = f(t)$ for every t. For the sinusoid,

$$A\cos(\omega(t + \tau) + \varphi) = A\cos(\omega t + \varphi + \omega\tau). \tag{5.2}$$

This equals $A\cos(\omega t + \varphi)$ for every t exactly when the extra phase $\omega\tau$ is a whole multiple of 2π (cosine repeats only after a full turn of 2π). The smallest positive such τ is

$$\tau = \frac{2\pi}{\omega}. \tag{5.3}$$

The number of cycles per unit t is the (ordinary) frequency $\omega/2\pi = 1/\tau$. In this chapter we use τ for a period; the paper uses T for a training window and P for a set of primes, so we keep those letters free.

Pitfall

Do not confuse ω (radians per unit t) with $\omega/2\pi$ (cycles per unit t). The paper always quotes *angular* frequencies: the prime-2 term $\sin(t\log 2)$ has $\omega = \log 2 = 0.6931$, so its period is $2\pi/0.6931 = 9.065$, not $1/0.6931$.

5.1.2 Two equivalent forms

Expanding Equation (5.1) with the addition formula $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$, taking $\alpha = \omega t$ and $\beta = \varphi$:

$$A\cos(\omega t + \varphi) = A\cos\varphi\,\cos\omega t - A\sin\varphi\,\sin\omega t. \tag{5.4}$$

So every sinusoid can be written as

$$a\cos\omega t + b\sin\omega t, \qquad a = A\cos\varphi, \quad b = -A\sin\varphi. \tag{5.5}$$

Going back the other way, square and add the two relations: $a^2 + b^2 = A^2(\cos^2\varphi + \sin^2\varphi) = A^2$, so

$$A = \sqrt{a^2+b^2}, \qquad \cos\varphi = \frac{a}{A}, \quad \sin\varphi = -\frac{b}{A}. \tag{5.6}$$

Both $\cos\varphi$ and $\sin\varphi$ are needed to fix φ uniquely in $(-\pi, \pi]$; knowing only $\tan\varphi = -b/a$ leaves a two-fold ambiguity.

5.1.3 The complex representation

Recall from Chapter 2 Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, so that $\cos\theta = \mathrm{Re}\; e^{i\theta}$. Then

$$A\cos(\omega t + \varphi) = \mathrm{Re}\left[A e^{i(\omega t+\varphi)}\right] = \mathrm{Re}\left[Z\, e^{i\omega t}\right], \qquad Z = Ae^{i\varphi}. \tag{5.7}$$

The complex number Z (the **complex amplitude**, or phasor) stores both the amplitude ($|Z| = A$) and the phase constant ($\arg Z = \varphi$). As t increases, $Ze^{i\omega t}$ rotates around the origin of the complex plane at angular speed ω, and the sinusoid is its shadow on the real axis. Using Equation (5.5), $Z = A\cos\varphi + iA\sin\varphi = a - ib$.

Sample Problem 5.1: Amplitude and phase of a sum of a cosine and a sine

Write $f(t) = 3\cos\omega t + 4\sin\omega t$ in the form $A\cos(\omega t + \varphi)$.
Step 1: identify a and b. Comparing with Equation (5.5), $a = 3$ and $b = 4$.
Step 2: amplitude. From Equation (5.6), $A = \sqrt{3^2+4^2} = \sqrt{25} = 5$.
Step 3: phase. $\cos\varphi = 3/5 = 0.6$ and $\sin\varphi = -4/5 = -0.8$. The angle with these values lies in the fourth quadrant: $\varphi = -\arctan(4/3) = -0.9273$ rad (-53.13°).
Step 4: check at $t = 0$. $5\cos(-0.9273) = 5\times 0.6 = 3$, matching $f(0) = 3$.

$$\boxed{3\cos\omega t + 4\sin\omega t = 5\cos(\omega t - 0.9273)}$$

Sample Problem 5.2: The three sinusoids of the prime-2 wave (RE2)

The book's running example RE2 is

$$f_2(t) = -\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(m\,t\log 2)}{m\,2^{m/2}}. \tag{5.8}$$

Find the amplitude and period of each term, and the period of the sum.

Step 1: amplitudes. Term m is $-A_m \sin(mt \log 2)$ with $A_m = 1/(\pi m 2^{m/2})$:

$$A_1 = \frac{1}{\pi\sqrt{2}} = 0.2251, \qquad A_2 = \frac{1}{4\pi} = 0.0796, \qquad A_3 = \frac{1}{3\pi\, 2^{3/2}} = 0.0375.$$

Step 2: angular frequencies. $\omega_m = m \log 2$: 0.6931, 1.3863, 2.0794.
Step 3: periods. From Equation (5.3), $\tau_m = 2\pi/(m \log 2)$:

$$\tau_1 = 9.0647, \qquad \tau_2 = 4.5324, \qquad \tau_3 = 3.0216.$$

Step 4: period of the sum. Shifting t by $\tau_1 = 2\pi/\log 2$ changes the phase of term m by $m \log 2 \cdot \tau_1 = 2\pi m$, a whole number of turns, so every term returns to its value. Hence the sum is periodic with

$$\boxed{\tau = \frac{2\pi}{\log 2} = 9.065}$$

Terms whose frequencies are whole multiples of one **fundamental** frequency are called **harmonics**, exactly as for a guitar string.

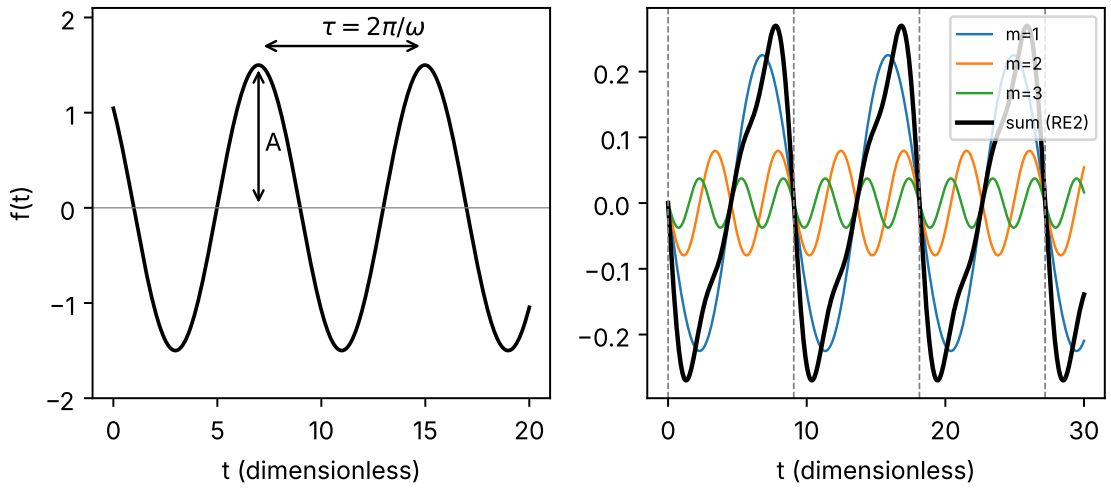


Figure 5.1: Left: anatomy of a sinusoid $A\cos(\omega t + \varphi)$ with $A = 1.5$, $\omega = 2\pi/8$ and $\varphi = 0.8$; the vertical arrow is the amplitude and the horizontal arrow one period $\tau = 2\pi/\omega = 8$ between successive maxima. Right: the three harmonics of the prime-2 wave RE2, Equation (5.8), and their sum (black), which repeats every $2\pi/\log 2 \approx 9.065$ in t (dashed lines). Both axes are dimensionless.

Checkpoint 5.1

If you double the angular frequency of a sinusoid, what happens to its period? What happens to its amplitude?

Answer: The period $2\pi/\omega$ halves. The amplitude is unchanged: A and ω are independent constants.

Key Ideas

- A sinusoid $A\cos(\omega t+\varphi)$ has amplitude A, angular frequency ω, phase constant φ and period $\tau = 2\pi/\omega$.
- Equivalently $a\cos\omega t + b\sin\omega t$ with $A = \sqrt{a^2+b^2}$; or $\mathrm{Re}[Ze^{i\omega t}]$ with complex amplitude $Z = Ae^{i\varphi} = a - ib$.
- Sinusoids with frequencies $\omega, 2\omega, 3\omega, \ldots$ are harmonics; their sum repeats with the fundamental period $2\pi/\omega$. RE2 is such a sum, with period $2\pi/\log 2 = 9.065$.

5.2 Superposition and beats

★ Did you know? *Tuning by Ear*

Play tones of 440 Hz and 442 Hz together and you hear a single pitch that swells and fades twice per second. The beat frequency is the difference of the two frequencies, here 2 Hz. Piano tuners and guitarists exploit this, adjusting a string until the beats slow down and vanish. Superposition turns a tiny frequency mismatch into something your ear detects easily.

On a stretched string, two small waves add: the displacement is the sum of the separate displacements. This **superposition principle** holds whenever the governing equation is linear, and it is why sums of sinusoids are everywhere.

5.2.1 Same frequency: still a sinusoid

Adding two sinusoids of the same ω gives another sinusoid of that ω. Using Equation (5.7),

$$\mathrm{Re}[Z_1e^{i\omega t}] + \mathrm{Re}[Z_2e^{i\omega t}] = \mathrm{Re}[(Z_1+Z_2)e^{i\omega t}], \tag{5.9}$$

because the real part of a sum is the sum of the real parts. So the complex amplitudes simply add like vectors in the plane. Sample Problem "Amplitude and phase" was an instance: $3\cos\omega t$ has $Z_1 = 3$, $4\sin\omega t$ has $Z_2 = -4i$, and $|Z_1+Z_2| = |3-4i| = 5$.

5.2.2 Different frequencies: beats

Now add two sinusoids of equal amplitude and different angular frequencies $\omega_1 < \omega_2$. We need the sum-to-product identity. Start from the addition formulas

$$\sin(u+v) = \sin u\cos v + \cos u\sin v, \tag{5.10}$$

$$\sin(u-v) = \sin u\cos v - \cos u\sin v. \tag{5.11}$$

Adding them,

$$\sin(u+v) + \sin(u-v) = 2\sin u\cos v. \tag{5.12}$$

Set $u+v=\alpha$ and $u-v=\beta$, that is $u=(\alpha+\beta)/2$ and $v=(\alpha-\beta)/2$:

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}. \tag{5.13}$$

With $\alpha=\omega_2 t$, $\beta=\omega_1 t$, and writing $\bar\omega = (\omega_1+\omega_2)/2$ and $\Delta\omega = \omega_2-\omega_1$,

$$\sin\omega_1 t + \sin\omega_2 t = \underbrace{2\cos\left(\frac{\Delta\omega}{2}t\right)}_{\text{slow envelope}}\ \underbrace{\sin(\bar\omega t)}_{\text{fast carrier}}. \tag{5.14}$$

When $\Delta\omega \ll \bar{\omega}$, the result looks like a fast oscillation at the average frequency $\bar{\omega}$ whose amplitude $2|\cos(\Delta\omega\, t/2)|$ swells and fades slowly. For these equal amplitudes the loudness envelope $|\cos(\Delta\omega\, t/2)|$ vanishes when $\Delta\omega\, t/2 = \pi/2 + j\pi$ for integer j, so successive silences are separated by

$$\tau_{\text{beat}} = \frac{2\pi}{\Delta\omega}. \tag{5.15}$$

Complete silences need equal amplitudes. With amplitudes $A_1 \neq A_2$, add the two complex amplitudes as in the previous subsection: their relative angle $\Delta\omega\, t$ turns steadily, so the length of their sum swings between $|A_1 - A_2|$ and $A_1 + A_2$ and never reaches zero. The beat period is still $2\pi/\Delta\omega$. In terms of ordinary frequency this is the familiar first-year result: the beat frequency is the difference $f_2 - f_1$. Two tuning forks at 440 Hz and 442 Hz beat twice per second.

Pitfall

The envelope function $\cos(\Delta\omega\, t/2)$ itself has period $4\pi/\Delta\omega$, twice τ_{beat}. The beat period is half that because the ear (and any amplitude measurement) responds to $|\cos|$, which repeats twice as often as cos.

Sample Problem 5.3: Beats between the primes 11 and 13

Paper Eq. (4) contains the terms $\sin(t\log 11)$ and $\sin(t\log 13)$ (with different amplitudes, but take them equal here). Find the carrier frequency, the carrier period, the beat period, and the number of carrier cycles per beat.

Step 1: frequencies. $\omega_1 = \log 11 = 2.397895$, $\omega_2 = \log 13 = 2.564949$.

Step 2: average and difference. $\bar{\omega} = (\log 11 + \log 13)/2 = \frac{1}{2}\log 143 = 2.481422$. $\Delta\omega = \log 13 - \log 11 = \log(13/11) = 0.167054$.

Step 3: carrier period. $2\pi/\bar{\omega} = 2.5321$.

Step 4: beat period. From Equation (5.15), $\tau_{\text{beat}} = 2\pi/0.167054 = 37.61$.

Step 5: cycles per beat. $\tau_{\text{beat}}/(2\pi/\bar{\omega}) = \bar{\omega}/\Delta\omega = 14.85$.

$$\boxed{\bar{\omega} = 2.4814, \quad \tau_{\text{beat}} = 37.61, \quad \text{about } 14.85 \text{ carrier cycles per beat}}$$

Checkpoint 5.2

In the sample problem above, how many silences (zeros of the envelope) occur for $0 < t < 100$?

Answer: Silences occur at $t = (2j+1)\pi/\Delta\omega = (2j+1) \times 18.81$, that is at $t \approx 18.8$, 56.4, 94.0; the next is at 131.6. So three.

Key Ideas

- Superposition: for linear systems, oscillations add.
- Same frequency: complex amplitudes add; the sum is one sinusoid.
- Two equal-amplitude sinusoids with close frequencies produce beats, Equation (5.14), with beat period $2\pi/\Delta\omega$.

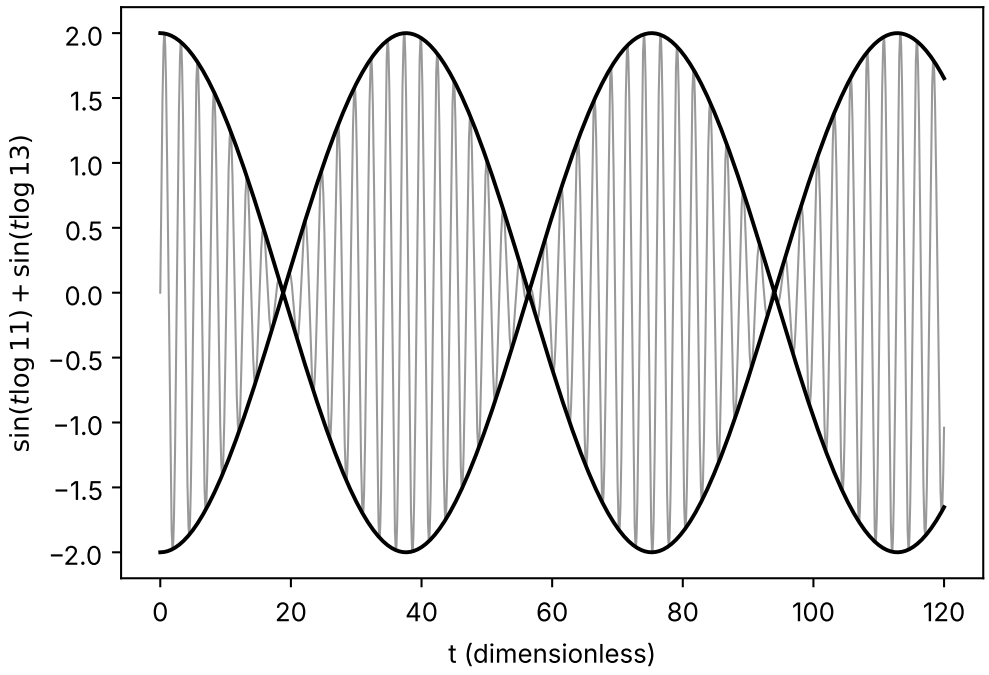


Figure 5.2: Beats: $\sin(t\log 11)+\sin(t\log 13)$ (gray) and its envelope $\pm 2|\cos(t\log(13/11)/2)|$ (black). Silences recur every $\tau_{\text{beat}} = 2\pi/\log(13/11) \approx 37.6$ in t. Both axes are dimensionless.

5.3 Orthogonality of sines and cosines

In 3D you find the x component of a vector $\vec{v}$ by taking a dot product with the unit vector: $v_x = \vec{v}\cdot\hat{x}$. This works because $\hat{x},\hat{y},\hat{z}$ are *orthogonal*: $\hat{x}\cdot\hat{y}=0$, and so on. Remarkably, sines and cosines of different whole-number frequencies behave in exactly this way, provided we replace the dot product by an integral of the product over one period. Chapter 7 develops this "functions as vectors" picture in general; here we only need the integrals.

5.3.1 Product-to-sum identities

From the addition formulas,

$$\cos(\alpha-\beta)+\cos(\alpha+\beta) = (\cos\alpha\cos\beta+\sin\alpha\sin\beta)+(\cos\alpha\cos\beta-\sin\alpha\sin\beta) = 2\cos\alpha\cos\beta, \tag{5.16}$$

$$\cos(\alpha-\beta)-\cos(\alpha+\beta) = 2\sin\alpha\sin\beta, \tag{5.17}$$

$$\sin(\alpha+\beta)+\sin(\alpha-\beta) = 2\sin\alpha\cos\beta. \tag{5.18}$$

Hence

$$\cos\alpha\cos\beta = \tfrac{1}{2}[\cos(\alpha-\beta)+\cos(\alpha+\beta)], \tag{5.19}$$

$$\sin\alpha\sin\beta = \tfrac{1}{2}[\cos(\alpha-\beta)-\cos(\alpha+\beta)], \tag{5.20}$$

$$\sin\alpha\cos\beta = \tfrac{1}{2}[\sin(\alpha+\beta)+\sin(\alpha-\beta)]. \tag{5.21}$$

5.3.2 Two basic integrals

Fix a half-width $L>0$ and integrate over the symmetric interval $[-L,L]$, which has length $2L$. For an integer n, define

$$I_c(n) = \int_{-L}^{L}\cos\frac{n\pi x}{L}\,\mathrm{d}x, \qquad I_s(n) = \int_{-L}^{L}\sin\frac{n\pi x}{L}\,\mathrm{d}x. \tag{5.22}$$

For $n = 0$ the cosine is 1 and $I_c(0) = 2L$. For $n \neq 0$, an antiderivative of $\cos(n\pi x/L)$ is $(L/n\pi)\sin(n\pi x/L)$, so

$$I_c(n) = \frac{L}{n\pi}\left[\sin(n\pi) - \sin(-n\pi)\right] = \frac{2L}{n\pi}\sin(n\pi) = 0, \tag{5.23}$$

because $\sin(n\pi) = 0$ for every integer n. Similarly, for $n \neq 0$, an antiderivative of $\sin(n\pi x/L)$ is $-(L/n\pi)\cos(n\pi x/L)$, so

$$I_s(n) = -\frac{L}{n\pi}\left[\cos(n\pi) - \cos(-n\pi)\right] = 0, \tag{5.24}$$

because cosine is even, $\cos(-n\pi) = \cos(n\pi)$. For $n = 0$, $I_s(0) = \int 0\,\mathrm{d}x = 0$. Summary: $I_s(n) = 0$ always, and $I_c(n) = 2L$ if $n = 0$, otherwise 0.

5.3.3 The orthogonality relations

Let j, k be non-negative integers. Using Equation (5.19) with $\alpha = j\pi x/L$, $\beta = k\pi x/L$,

$$\int_{-L}^{L} \cos\frac{j\pi x}{L}\cos\frac{k\pi x}{L}\,\mathrm{d}x = \tfrac{1}{2}\left[I_c(j-k) + I_c(j+k)\right]. \tag{5.25}$$

Three cases:

- $j \neq k$: then $j - k \neq 0$ and $j + k \geq 1$, so both terms vanish: the integral is 0.
- $j = k \geq 1$: $I_c(0) = 2L$ and $I_c(2k) = 0$, so the integral is $\frac{1}{2}(2L) = L$.
- $j = k = 0$: both terms are $I_c(0) = 2L$, so the integral is $2L$.

In the same way Equations (5.20) and (5.21) give the sine-sine and sine-cosine integrals. Introducing the **Kronecker delta** δ_{jk}, equal to 1 if $j = k$ and 0 otherwise, the results are collected in Table 5.1.

Table 5.1: Orthogonality relations on $[-L, L]$ for non-negative integers j, k.

integral over $[-L, L]$	value
$\int \cos\frac{j\pi x}{L}\cos\frac{k\pi x}{L}\,\mathrm{d}x$	$L\,\delta_{jk}$ for $j, k \geq 1$; $2L$ for $j = k = 0$
$\int \sin\frac{j\pi x}{L}\sin\frac{k\pi x}{L}\,\mathrm{d}x$	$L\,\delta_{jk}$ for $j, k \geq 1$
$\int \sin\frac{j\pi x}{L}\cos\frac{k\pi x}{L}\,\mathrm{d}x$	0 for all j, k

For the sine-sine row with $j, k \geq 1$: Equation (5.20) gives $\frac{1}{2}[I_c(j-k) - I_c(j+k)]$, which is 0 for $j \neq k$ and $\frac{1}{2}(2L - 0) = L$ for $j = k$. For the sine-cosine row, Equation (5.21) gives $\frac{1}{2}[I_s(j+k) + I_s(j-k)] = 0$.

Dividing the $j = k \geq 1$ case by the interval length $2L$ gives a fact worth remembering: the **average of $\cos^2$ or $\sin^2$ over a whole number of periods is $\frac{1}{2}$.**

Sample Problem 5.4: Orthogonality on the paper's box

Take $L = 44.883$, the half-width of the paper's scale-1 box (RE3). Evaluate (a) $\int_{-L}^{L}\cos(2\pi x/L)\cos(3\pi x/L)\,\mathrm{d}x$, (b) $\int_{-L}^{L}\cos^2(2\pi x/L)\,\mathrm{d}x$, (c) $\int_{-L}^{L} 1\cdot 1\,\mathrm{d}x$.

Step 1 (a). $j = 2 \neq k = 3$, so by Table 5.1 the integral is 0. Directly: $\frac{1}{2}[I_c(-1) + I_c(5)] = \frac{1}{2}[0 + 0] = 0$.

Step 2 (b). $j = k = 2 \geq 1$: the integral is $L = 44.883$.

Step 3 (c). $j = k = 0$: the integral is $2L = 89.766$.

0,	44.883,	89.766

A numerical quadrature (CHECKS) confirms all three.

Checkpoint 5.3

Without computing anything, what is $\int_{-L}^{L} \sin(3\pi x/L)\cos(3\pi x/L)\,dx$?

Answer: Zero: it is the sine-cosine row of Table 5.1. (Also $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$, whose integral over whole periods vanishes.)

Key Ideas

- On $[-L, L]$ the functions 1, $\cos(k\pi x/L)$, $\sin(k\pi x/L)$ $(k = 1, 2, \dots)$ are mutually orthogonal: the integral of the product of any two different ones is zero.
- The "squared length" of $\cos(k\pi x/L)$ or $\sin(k\pi x/L)$ is L; that of the constant 1 is $2L$.
- The average of $\cos^2$ or $\sin^2$ over whole periods is $\frac{1}{2}$.

5.4 Fourier series

A guitar string of length ℓ fixed at both ends vibrates in standing-wave modes $\sin(k\pi x/\ell)$, $k = 1, 2, 3, \dots$. Pluck it into some triangular shape and the subsequent motion is a superposition of these modes. Joseph Fourier's bold claim (1807) was that essentially any shape can be written as such a superposition. Orthogonality tells us how to find the recipe.

5.4.1 Definition

Let f be periodic with period $2L$, so $f(x + 2L) = f(x)$. Its **Fourier series** is

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \left[a_k \cos\frac{k\pi x}{L} + b_k \sin\frac{k\pi x}{L} \right]. \tag{5.26}$$

Each term has period $2L/k$, which fits exactly k times into $2L$. So each term repeats after $2L$, and the whole right-hand side has period $2L$, consistent with the left. A function defined only on $[-L, L]$ can always be treated as one period of its **periodic extension**, obtained by copying the graph on $[-L, L]$ into $[L, 3L]$, $[3L, 5L]$, and so on.

5.4.2 Deriving the coefficients

Suppose Equation (5.26) holds. Pick an integer $n \geq 1$, multiply both sides by $\cos(n\pi x/L)$, and integrate over $[-L, L]$, integrating the right-hand side term by term (see the box below for when

this is allowed):

$$\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\mathrm{d}x = \frac{a_0}{2}\int_{-L}^{L}\cos\frac{n\pi x}{L}\mathrm{d}x + \sum_{k\geq 1} a_k \int_{-L}^{L}\cos\frac{k\pi x}{L}\cos\frac{n\pi x}{L}\mathrm{d}x$$

$$+ \sum_{k\geq 1} b_k \int_{-L}^{L}\sin\frac{k\pi x}{L}\cos\frac{n\pi x}{L}\mathrm{d}x \tag{5.27}$$

$$= \frac{a_0}{2}\cdot 0 + \sum_{k\geq 1} a_k\, L\,\delta_{kn} + \sum_{k\geq 1} b_k \cdot 0 \quad \text{(Table 5.1)} \tag{5.28}$$

$$= a_n L. \tag{5.29}$$

Every term except one is killed by orthogonality. So

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,\mathrm{d}x. \tag{5.30}$$

For $n = 0$, integrate Equation (5.26) directly (multiply by $\cos 0 = 1$): the constant term gives $(a_0/2)(2L) = a_0 L$ and every other term integrates to zero, so $a_0 = (1/L)\int f\,\mathrm{d}x$. This is Equation (5.30) with $n = 0$; that is the reason for writing the constant term as $a_0/2$. Multiplying instead by $\sin(n\pi x/L)$ and using the sine rows of Table 5.1,

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,\mathrm{d}x. \tag{5.31}$$

Note that $a_0/2 = (1/2L)\int f\,\mathrm{d}x$ is the **average value** of f over a period.

Taken on trust

What we take on trust: convergence. The derivation assumed that the series converges to f and may be integrated term by term. The standard theorem (Dirichlet's theorem) states: if f has period $2L$ and, on one period, f and f' are continuous except for finitely many jump discontinuities, then the Fourier series converges at every x; its sum is $f(x)$ wherever f is continuous and is the midpoint $\frac{1}{2}[f(x^-) + f(x^+)]$ at a jump. For such f the coefficient formulas Equations (5.30) and (5.31) are valid. Riley, Hobson and Bence [22] state these (Dirichlet) conditions in their chapter on Fourier series but do not prove the theorem; a proof, via the so-called Dirichlet kernel, is found in texts on Fourier analysis or real analysis. Every function expanded in a Fourier series in this chapter satisfies these conditions.

Sample Problem 5.5: The square wave

Find the Fourier series of the square wave of period 2π: $f(x) = -1$ for $-\pi < x < 0$ and $f(x) = +1$ for $0 < x < \pi$.

Step 1: set $L = \pi$. Then $k\pi x/L = kx$.

Step 2: cosine coefficients. $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,\mathrm{d}x$. Split at 0 and substitute $x \to -x$ in the left half: $\int_{-\pi}^{0}(-1)\cos nx\,\mathrm{d}x = -\int_0^{\pi}\cos nx\,\mathrm{d}x$, which cancels the right half $\int_0^{\pi}\cos nx\,\mathrm{d}x$. So $a_n = 0$ for all n (including $n = 0$: the average is zero).

Step 3: sine coefficients. The same substitution gives $\int_{-\pi}^{0}(-1)\sin nx\,\mathrm{d}x = \int_0^{\pi}\sin nx\,\mathrm{d}x$ (two sign changes), so

$$b_n = \frac{2}{\pi}\int_0^{\pi}\sin nx\,\mathrm{d}x = \frac{2}{\pi}\left[-\frac{\cos nx}{n}\right]_0^{\pi} = \frac{2}{n\pi}\left(1-\cos n\pi\right) = \frac{2}{n\pi}\left(1-(-1)^n\right).$$

So $b_n = 4/(n\pi)$ for odd n and 0 for even n.

Step 4: the series.

$$\boxed{f(x) = \frac{4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right)}$$

The first coefficient is $b_1 = 4/\pi = 1.2732$: the best-fitting single sine (best in the mean-square sense, as shown in Section 5.6) is *taller* than the square wave.

Step 5: a bonus. At $x = \pi/2$, $f = 1$ and $\sin(n\pi/2) = 1, -1, 1, \ldots$ for $n = 1, 3, 5, \ldots$, so $1 = \frac{4}{\pi}(1 - \frac{1}{3} + \frac{1}{5} - \cdots)$, Leibniz's series $\pi/4 = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$. At the jump $x = 0$ every term vanishes, giving 0, the midpoint of -1 and $+1$, as Dirichlet's theorem promises.

```
import numpy as np
from scipy.integrate import quad

# Fourier sine coefficients of the square wave, by numerical integration
def f(x):
    return np.sign(x)            # -1 on (-pi,0), +1 on (0,pi)

for n in range(1, 6):
    bn = quad(lambda x: f(x)*np.sin(n*x), -np.pi, np.pi, points=[0])[0]/np.pi
    exact = 4/(n*np.pi) if n % 2 == 1 else 0.0
    print(n, round(bn, 6), round(exact, 6))
```

5.4.3 The Gibbs phenomenon

Truncate the square-wave series after K nonzero terms:

$$S_K(x) = \frac{4}{\pi}\sum_{j=1}^{K}\frac{\sin((2j-1)x)}{2j-1}. \tag{5.32}$$

Figure 5.3 shows that near the jump S_K overshoots the value 1, and the overshoot does not shrink to zero as K grows; it settles to a fixed size and only gets narrower. This is the **Gibbs phenomenon**. We can compute the size of the overshoot exactly.

Locating the first peak. Differentiate term by term (a finite sum, so this is allowed):

$$S_K'(x) = \frac{4}{\pi}\sum_{j=1}^{K}\cos((2j-1)x) = \frac{4}{\pi}\,\mathrm{Re}\sum_{j=1}^{K}e^{i(2j-1)x}. \tag{5.33}$$

The sum is geometric (Chapter 3) with first term e^{ix} and ratio e^{2ix}:

$$\sum_{j=1}^{K}e^{i(2j-1)x} = e^{ix}\,\frac{1-e^{2iKx}}{1-e^{2ix}}. \tag{5.34}$$

Divide numerator and denominator of $e^{ix}/(1-e^{2ix})$ by e^{ix}: it equals $1/(e^{-ix}-e^{ix}) = 1/(-2i\sin x) = i/(2\sin x)$, using $e^{ix}-e^{-ix} = 2i\sin x$ and $1/(-i) = i$. Therefore

$$\sum_{j=1}^{K}e^{i(2j-1)x} = \frac{i\,(1-\cos 2Kx - i\sin 2Kx)}{2\sin x} = \frac{\sin 2Kx + i(1-\cos 2Kx)}{2\sin x}, \tag{5.35}$$

whose real part is $\sin(2Kx)/(2\sin x)$. Hence, for $0 < x < \pi$,

$$S_K'(x) = \frac{2}{\pi}\,\frac{\sin 2Kx}{\sin x}. \tag{5.36}$$

This is positive for $0 < x < \pi/(2K)$ (there $\sin 2Kx > 0$ and $\sin x > 0$) and changes sign at $x^* = \pi/(2K)$. So S_K rises from 0 at $x = 0$ to a local maximum at x^*. This is the *first* peak to the right of the jump, and it is the one whose height we compute. Figure 5.3 shows that it is also the tallest, the later peaks (at the other sign changes of $\sin 2Kx$) being lower. We do not prove that last statement, and nothing below depends on it.

Height of the peak. At x^*, the j-th term has argument $u_j = (2j-1)\pi/(2K)$. Since $1/(2j-1) = (\pi/2K)/u_j$,

$$S_K(x^*) = \frac{4}{\pi} \cdot \frac{\pi}{2K} \sum_{j=1}^{K} \frac{\sin u_j}{u_j} = \frac{2}{\pi} \sum_{j=1}^{K} \frac{\sin u_j}{u_j}\, \Delta u, \qquad \Delta u = \frac{\pi}{K}. \tag{5.37}$$

The points u_j are the midpoints of K equal subintervals of $[0, \pi]$ (subinterval j is $[(j-1)\pi/K,\, j\pi/K]$, whose midpoint is $(j - \tfrac{1}{2})\pi/K = u_j$). So the sum is a Riemann sum (the midpoint rule) for an integral over $[0, \pi]$. The integrand $\sin u/u$ is continuous on $[0, \pi]$ once we give it its limiting value 1 at $u = 0$, and for a continuous function the sum of (value at one point of each subinterval) $\times$ (subinterval width) tends to the integral as the widths tend to zero; that is the definition of the definite integral. Hence, as $K \to \infty$,

$$S_K(x^*) \to \frac{2}{\pi} \int_0^{\pi} \frac{\sin u}{u}\, \mathrm{d}u = \frac{2}{\pi}\, \mathrm{Si}(\pi) = 1.17898, \tag{5.38}$$

where $\mathrm{Si}(z) = \int_0^z (\sin u/u)\, \mathrm{d}u$ is the *sine integral*, a tabulated function with $\mathrm{Si}(\pi) = 1.851937$ (computed in CHECKS with `scipy.special.sici`). In the limit $K \to \infty$ the overshoot above 1 is 0.17898, which is 8.95% of the total jump 2, about 9%. For finite K the first peak is higher (for $K = 1$ it is $4/\pi = 1.273$, an overshoot of 13.7% of the jump), so the overshoot approaches about 9% but never decays to zero, however many terms you keep. For $K = 50$, the finite sum gives $S_{50}(x^*) = 1.1790$.

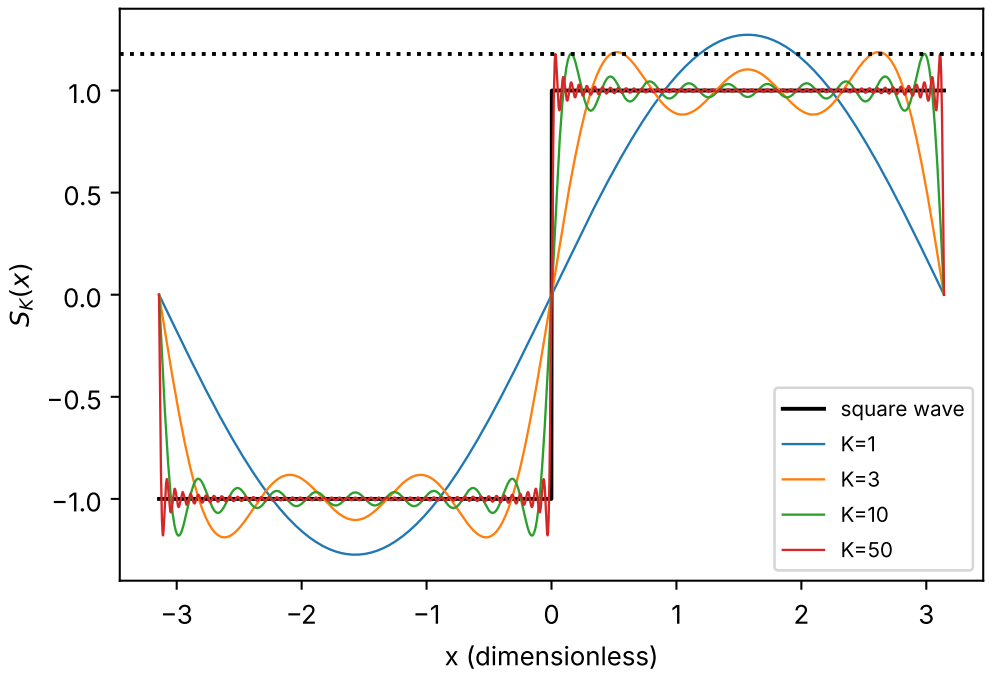


Figure 5.3: Partial sums S_K of the square-wave series with $K = 1, 3, 10, 50$ terms. The overshoot near the jump at $x = 0$ tends to $(2/\pi)\mathrm{Si}(\pi) = 1.179$ (dotted line), about 9% of the jump, and moves closer to the jump as K grows. Axes dimensionless.

Pitfall

Adding more terms does *not* remove the Gibbs overshoot. It squeezes it into a narrower region (width $\propto 1/K$), so the series still converges at each fixed $x \neq 0$, but the overshoot at the first peak next to the jump approaches about 9% of the jump instead of zero. Jumps are expensive for Fourier series; we will see that the paper's cosine basis avoids jumps at the walls for even functions.

Key Ideas

- A $2L$-periodic function has the Fourier series Equation (5.26) with $a_n = \frac{1}{L}\int f\cos\frac{n\pi x}{L}$, $b_n = \frac{1}{L}\int f\sin\frac{n\pi x}{L}$, derived by orthogonality.
- The series converges to f where f is continuous and to the midpoint at jumps (Dirichlet's theorem, taken on trust).
- Near a jump, the overshoot of the partial sums tends to about 9% of the jump as $K \to \infty$ and never decays to zero (Gibbs phenomenon), Equation (5.38).

5.5 The complex form

Sines and cosines come in pairs; complex exponentials package each pair into one function and make the algebra shorter. They will be the natural language of the Fourier transform in Chapter 6.

5.5.1 Orthogonality of complex exponentials

For integers k and l consider $\int_{-L}^{L} e^{ik\pi x/L}\,\overline{e^{il\pi x/L}}\,\mathrm{d}x$, where the overbar is the complex conjugate (Chapter 2); for complex functions the "dot product" conjugates the second factor. The reason for the conjugate: with it, the product of a function with itself is $\int f\bar{f}\,\mathrm{d}x = \int |f|^2\mathrm{d}x \geq 0$, a genuine squared length; without it, $\int_{-L}^{L} e^{i\pi x/L}e^{i\pi x/L}\mathrm{d}x$ would be 0 (by the computation below with $k-l$ replaced by 2) even though the function is never zero. To integrate a complex-valued function of the real variable x, integrate its real and imaginary parts separately: $\int(u+iv)\,\mathrm{d}x = \int u\,\mathrm{d}x + i\int v\,\mathrm{d}x$. For a real constant $\alpha \neq 0$ this gives

$$\int e^{i\alpha x}\mathrm{d}x = \int \cos\alpha x\,\mathrm{d}x + i\int \sin\alpha x\,\mathrm{d}x = \frac{\sin\alpha x}{\alpha} - i\frac{\cos\alpha x}{\alpha} = \frac{e^{i\alpha x}}{i\alpha},$$

where the last step multiplies $\cos\alpha x + i\sin\alpha x$ by $1/i = -i$. So complex exponentials integrate exactly like real ones. Since $\overline{e^{i\theta}} = e^{-i\theta}$, the integrand is $e^{i(k-l)\pi x/L}$. If $k = l$ it is 1 and the integral is $2L$. If $k \neq l$,

$$\int_{-L}^{L} e^{i(k-l)\pi x/L}\,\mathrm{d}x = \frac{L}{i(k-l)\pi}\left[e^{i(k-l)\pi} - e^{-i(k-l)\pi}\right] = \frac{L}{i(k-l)\pi}\,2i\sin((k-l)\pi) = 0. \tag{5.39}$$

So

$$\int_{-L}^{L} e^{ik\pi x/L}e^{-il\pi x/L}\,\mathrm{d}x = 2L\,\delta_{kl}. \tag{5.40}$$

5.5.2 The complex Fourier series

The complex Fourier series of a $2L$-periodic function is

$$f(x) = \sum_{k=-\infty}^{\infty} \hat{f}_k\, e^{ik\pi x/L}, \qquad \hat{f}_k = \frac{1}{2L}\int_{-L}^{L} f(x)\, e^{-ik\pi x/L}\,\mathrm{d}x. \tag{5.41}$$

The coefficient formula follows exactly as before: multiply the series by $e^{-in\pi x/L}$, integrate over $[-L, L]$ term by term, and use Equation (5.40); only $k = n$ survives, giving $2L\hat{f}_n$. Notice the sum runs over *negative* k too.

Relation to the real form. Substitute $\cos\theta = (e^{i\theta} + e^{-i\theta})/2$ and $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$ into one term of Equation (5.26), with $\theta = k\pi x/L$:

$$a_k \cos\theta + b_k \sin\theta = \frac{a_k}{2}(e^{i\theta} + e^{-i\theta}) + \frac{b_k}{2i}(e^{i\theta} - e^{-i\theta}) \tag{5.42}$$

$$= \frac{a_k - ib_k}{2} e^{i\theta} + \frac{a_k + ib_k}{2} e^{-i\theta} \quad (\text{using } 1/i = -i). \tag{5.43}$$

Comparing with Equation (5.41),

$$\hat{f}_0 = \frac{a_0}{2}, \qquad \hat{f}_k = \frac{a_k - ib_k}{2}, \qquad \hat{f}_{-k} = \frac{a_k + ib_k}{2} \qquad (k \geq 1). \tag{5.44}$$

For a real function, a_k and b_k are real, so $\hat{f}_{-k} = \overline{\hat{f}_k}$. You can also see this directly: conjugating the integral in Equation (5.41) with f real flips the sign of the exponent and turns $\hat{f}_k$ into $\hat{f}_{-k}$.

Sample Problem 5.6: RE2 in complex form

Write the prime-2 wave Equation (5.8) as a complex Fourier series.
Step 1: choose the period. RE2 has period $2\pi/\log 2$, so set $2L = 2\pi/\log 2$. Then $k\pi/L = k\log 2$, and term m of RE2 is harmonic $k = m$.
Step 2: read off real coefficients. Term m is $-A_m \sin(mt\log 2)$, so $a_m = 0$ and $b_m = -A_m$ for $m = 1, 2, 3$; all other coefficients vanish.
Step 3: apply Equation (5.44). $\hat{f}_m = (0 - i(-A_m))/2 = iA_m/2$ and $\hat{f}_{-m} = -iA_m/2$.
Step 4: numbers. Using A_m from the earlier sample problem:

$$\boxed{\hat{f}_{\pm 1} = \pm 0.1125\, i, \quad \hat{f}_{\pm 2} = \pm 0.0398\, i, \quad \hat{f}_{\pm 3} = \pm 0.0188\, i}$$

and all other $\hat{f}_k = 0$. Six nonzero complex coefficients describe the whole wave for all t. Check: $\hat{f}_{-m} = \overline{\hat{f}_m}$, as it must be for a real function.

Key Ideas

- $f(x) = \sum_k \hat{f}_k e^{ik\pi x/L}$ with $\hat{f}_k = \frac{1}{2L}\int f e^{-ik\pi x/L} dx$, by the orthogonality Equation (5.40).
- $\hat{f}_k = (a_k - ib_k)/2$; for real f, $\hat{f}_{-k} = \overline{\hat{f}_k}$.

5.6 Parseval's theorem and best approximation

5.6.1 Parseval's theorem

How "big" is a function? A natural measure, the one behind every rms in the paper, is its **mean square** over a period, $\langle |f|^2 \rangle = \frac{1}{2L}\int_{-L}^{L} |f|^2 \, dx$ (for real f, $|f|^2 = f^2$ and we write simply $\langle f^2 \rangle$), and its square root, the root mean square (rms). Parseval's theorem says the mean square is the sum of the squared moduli of the Fourier coefficients, exactly as the squared length of a 3D vector is $v_x^2 + v_y^2 + v_z^2$.

Derivation. Write $|f|^2 = f\,\bar{f}$ and insert the complex series for f and for $\bar{f}$ (for $\bar{f}$, conjugate each term, and use a second summation index l):

$$\frac{1}{2L}\int_{-L}^{L}|f|^2\mathrm{d}x = \frac{1}{2L}\int_{-L}^{L}\Big(\sum_k \hat{f}_k e^{ik\pi x/L}\Big)\Big(\sum_l \overline{\hat{f}_l}\, e^{-il\pi x/L}\Big)\mathrm{d}x \tag{5.45}$$

$$= \sum_k\sum_l \hat{f}_k\overline{\hat{f}_l}\,\frac{1}{2L}\int_{-L}^{L} e^{i(k-l)\pi x/L}\mathrm{d}x \tag{5.46}$$

$$= \sum_k\sum_l \hat{f}_k\overline{\hat{f}_l}\,\delta_{kl} \qquad \text{(Equation (5.40))} \tag{5.47}$$

$$= \sum_k |\hat{f}_k|^2. \tag{5.48}$$

In real form, Equation (5.44) gives $|\hat{f}_0|^2 = a_0^2/4$ and $|\hat{f}_k|^2 = |\hat{f}_{-k}|^2 = (a_k^2+b_k^2)/4$, so the pair $\pm k$ contributes $(a_k^2+b_k^2)/2$:

$$\frac{1}{2L}\int_{-L}^{L} f^2\,\mathrm{d}x = \frac{a_0^2}{4} + \frac{1}{2}\sum_{k=1}^{\infty}\left(a_k^2+b_k^2\right). \tag{5.49}$$

Taken on trust

The interchange of the double sum with the integral needs justification. Parseval's theorem holds for every f with $\int_{-L}^{L}|f|^2\mathrm{d}x$ finite, including functions with jumps such as the square wave; Riley, Hobson and Bence [22] give the formal term-by-term derivation shown above; the rigorous proof rests on the completeness of the trigonometric system (mean-square convergence of Fourier series) and is found in texts on Fourier analysis.

A single sinusoid $A\cos(\omega t+\varphi)$ has $a^2+b^2 = A^2$, so its mean square is $A^2/2$ and its rms is $A/\sqrt{2}$, the familiar rms of an AC voltage. Parseval says that for a sum of harmonics, the mean squares simply add: there are no cross terms, because of orthogonality.

Sample Problem 5.7: Parseval for the square wave

Use Parseval on the square wave to evaluate $\sum_{\text{odd }k} 1/k^2$.
Step 1: left side. $f^2 = 1$ everywhere (except at isolated points, which do not affect integrals), so the mean square is 1.
Step 2: right side. $a_k = 0$, $b_k = 4/(k\pi)$ for odd k: $\frac{1}{2}\sum_{\text{odd}} 16/(k^2\pi^2) = \frac{8}{\pi^2}\sum_{\text{odd}}\frac{1}{k^2}$.
Step 3: equate.

$$\boxed{1+\frac{1}{9}+\frac{1}{25}+\cdots = \frac{\pi^2}{8} = 1.2337}$$

Sample Problem 5.8: The rms of the prime-2 wave

Find the rms of RE2 over a period.
Step 1: Parseval in real form. RE2 has $b_m = -A_m$, so the mean square is $\frac{1}{2}\sum_{m=1}^{3} A_m^2$.
Step 2: squares. $A_m^2 = 1/(\pi^2 m^2 2^m)$: $\sum_m 1/(m^2 2^m) = \frac{1}{2}+\frac{1}{16}+\frac{1}{72} = 0.576389$.
Step 3: mean square. $0.576389/(2\pi^2) = 0.029200$.
Step 4: rms.

$$\boxed{\text{rms of RE2} = \sqrt{0.029200} = 0.1709}$$

Most of it comes from $m=1$: $A_1/\sqrt{2} = 0.1592$ on its own. We return to this number in the Paper connection.

5.6.2 Truncated Fourier series are the best approximations

Suppose you may use only finitely many harmonics. Which coefficients should you use? We show that the Fourier coefficients themselves are the best choice in the mean-square sense. We do it for even functions and cosines (the case used in the paper); the general case is identical.

Let f be even on $[-L, L]$ and write its cosine series in the convention the paper uses,

$$f(x) = \sum_{k=0}^{\infty} c_k \cos\frac{k\pi x}{L}, \qquad c_0 = \frac{1}{2L}\int_{-L}^{L} f\,\mathrm{d}x, \quad c_k = \frac{1}{L}\int_{-L}^{L} f\cos\frac{k\pi x}{L}\mathrm{d}x \ (k \geq 1), \tag{5.50}$$

so $c_0 = a_0/2$ and $c_k = a_k$. Approximate f by $g(x) = \sum_{k=0}^{M-1} d_k \cos(k\pi x/L)$ with arbitrary numbers d_k, and measure the error by $\mathcal{E} = \frac{1}{2L}\int_{-L}^{L}(f-g)^2\mathrm{d}x$ (we write $\mathcal{E}$, not E, because this book reserves E for energies). Expand the square:

$$\mathcal{E} = \frac{1}{2L}\int f^2\mathrm{d}x - 2\sum_{k<M} d_k \frac{1}{2L}\int f\cos\frac{k\pi x}{L}\mathrm{d}x + \sum_{j,k<M} d_j d_k \frac{1}{2L}\int \cos\frac{j\pi x}{L}\cos\frac{k\pi x}{L}\mathrm{d}x. \tag{5.51}$$

By Table 5.1, the last average is $\nu_k\delta_{jk}$ with $\nu_0 = 1$ and $\nu_k = \frac{1}{2}$ for $k \geq 1$. By Equation (5.50), the middle average is $\nu_k c_k$ (for $k \geq 1$ it is $c_k/2$; for $k = 0$ it is c_0). Hence

$$\mathcal{E} = \langle f^2\rangle + \sum_{k<M}\nu_k\left(d_k^2 - 2d_k c_k\right) = \langle f^2\rangle + \sum_{k<M}\nu_k(d_k - c_k)^2 - \sum_{k<M}\nu_k c_k^2, \tag{5.52}$$

where we completed the square, $d^2 - 2dc = (d-c)^2 - c^2$. Only the middle sum depends on the d_k, and it is a sum of non-negative terms, smallest (zero) when $d_k = c_k$. So:

- the truncated Fourier series is the best mean-square approximation with M cosines;
- its error is $\mathcal{E}_{\min} = \langle f^2\rangle - \sum_{k<M}\nu_k c_k^2 \geq 0$ (Bessel's inequality), which tends to zero as $M \to \infty$ by Parseval.

This is a first example of *least squares* (Chapter 9): with an orthogonal basis, the least-squares coefficients are simply projections. The paper also determines cosine coefficients by least squares, but for a different target, the energy levels, so its c_k are not the projections of any known function (Chapter 24).

Checkpoint 5.4

Does the rms of RE2 depend on where you start the period, or on the phases of the three terms?

Answer: No. The mean over any full period is the same, and Parseval uses only $|\hat{f}_k|^2$, which ignores phases.

Key Ideas

- Parseval: $\langle|f|^2\rangle = \sum_k |\hat{f}_k|^2 = \frac{a_0^2}{4} + \frac{1}{2}\sum(a_k^2 + b_k^2)$; mean squares of orthogonal pieces add.
- A sinusoid of amplitude A has rms $A/\sqrt{2}$. RE2 has rms 0.1709.
- Truncated Fourier series minimize the mean-square error among all combinations of the same harmonics, Equation (5.52).

5.7 Even and odd functions; the cosine basis on $[-L, L]$

★ Did you know? *Fourier Faced Skeptics*

Joseph Fourier presented his theory of heat flow, built on expanding functions in sines and cosines, to the Paris Academy in 1807. Examiners, including Lagrange, doubted that arbitrary functions, even ones with corners or jumps, could be written as such series. Fourier won an Academy prize for the work a few years later, and his book *Théorie analytique de la chaleur* finally appeared in 1822.

5.7.1 Definitions and rules

A function is **even** if $f(-x) = f(x)$ for all x (mirror-symmetric about $x = 0$: cos, x^2, $|x|$) and **odd** if $f(-x) = -f(x)$ (antisymmetric: sin, x, x^3). Most functions are neither, but every function splits uniquely into an even and an odd part:

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{f_{\mathrm{e}}(x)} + \underbrace{\frac{f(x) - f(-x)}{2}}_{f_{\mathrm{o}}(x)}. \tag{5.53}$$

(Replace x by $-x$ in f_{e}: the two terms swap, so f_{e} is even; in f_{o} they swap and change sign, so f_{o} is odd. Uniqueness: if also $f = g_{\mathrm{e}} + g_{\mathrm{o}}$ with g_{e} even and g_{o} odd, then $f(-x) = g_{\mathrm{e}}(x) - g_{\mathrm{o}}(x)$, so $f(x) + f(-x) = 2g_{\mathrm{e}}(x)$ and $f(x) - f(-x) = 2g_{\mathrm{o}}(x)$, which says $g_{\mathrm{e}} = f_{\mathrm{e}}$ and $g_{\mathrm{o}} = f_{\mathrm{o}}$.)

Products obey sign rules like multiplying ± 1. If f, g are even, $(fg)(-x) = f(x)g(x)$: even. If both are odd, $(fg)(-x) = (-f(x))(-g(x)) = f(x)g(x)$: even. If f is even and g odd, $(fg)(-x) = f(x)(-g(x))$: odd.

Integrals over symmetric intervals. If h is odd, split $\int_{-L}^{L} h = \int_{-L}^{0} h + \int_{0}^{L} h$ and in the first piece substitute $x = -y$ ($\mathrm{d}x = -\mathrm{d}y$):

$$\int_{-L}^{0} h(x)\,\mathrm{d}x = \int_{L}^{0} h(-y)(-\mathrm{d}y) = \int_{0}^{L} h(-y)\,\mathrm{d}y = -\int_{0}^{L} h(y)\,\mathrm{d}y, \tag{5.54}$$

so $\int_{-L}^{L} h\,\mathrm{d}x = 0$. If h is even, the same substitution gives $+\int_{0}^{L} h$, so $\int_{-L}^{L} h = 2\int_{0}^{L} h$.

5.7.2 Cosine and sine series

If f is even, $f(x)\sin(k\pi x/L)$ is odd, so every $b_k = 0$; and $f\cos(k\pi x/L)$ is even, so $a_k = \frac{2}{L}\int_0^L f\cos(k\pi x/L)\mathrm{d}x$. An even function has a **pure cosine series**. If f is odd, all $a_k = 0$ and it has a **pure sine series** (the square wave of Section 5.4 was odd, and indeed had only sines).

5.7.3 The cosine basis on $[-L, L]$

Why care about evenness? In Chapter 11 you will see that a potential symmetric under $x \to -x$ (as the Wu–Sprung potential is) has states of definite parity, and that a correction which keeps the symmetry must itself be even. So the natural expansion of an even correction on the box $[-L, L]$ is the **cosine basis**

$$\phi_k(x) = \cos\frac{k\pi x}{L}, \qquad k = 0, 1, \ldots, M-1, \tag{5.55}$$

with coefficients given by Equation (5.50). The basis function ϕ_k has wavelength $2L/k$ and wavenumber $k\pi/L$ (radians per unit x); $\phi_0 = 1$ is the constant.

Pitfall

Conventions for the constant term differ. In Equation (5.26) the constant is $a_0/2$ with $a_0 = \frac{1}{L}\int f$. In the paper's form $\sum_k c_k \cos(k\pi x/L)$ the constant is c_0 itself, so $c_0 = \frac{1}{2L}\int f$, the average. Mixing the two conventions doubles or halves the constant.

No jumps at the walls. The cosine series of f on $[-L, L]$ represents the $2L$-periodic extension of f. Just to the right of $x = L$ the extension equals $f(-L + \varepsilon)$, which tends to $f(-L)$; just to the left it is $f(L)$. For an even function $f(-L) = f(L)$, so the extension has *no jump* at the walls and there is no Gibbs overshoot there. There is generally a *kink*: the slope just left of L is $f'(L)$ and just right of it is $f'(-L) = -f'(L)$.

How fast do the coefficients fall? Integrate by parts twice. With $q = k\pi/L$ and $k \geq 1$,

$$c_k = \frac{2}{L}\int_0^L f\cos qx\,\mathrm{d}x = \frac{2}{L}\left[\frac{f\sin qx}{q}\right]_0^L - \frac{2}{Lq}\int_0^L f'\sin qx\,\mathrm{d}x \tag{5.56}$$

$$= -\frac{2}{Lq}\left(\left[-\frac{f'\cos qx}{q}\right]_0^L + \frac{1}{q}\int_0^L f''\cos qx\,\mathrm{d}x\right) \quad (\sin k\pi = \sin 0 = 0) \tag{5.57}$$

$$= \frac{2L}{k^2\pi^2}\left[(-1)^k f'(L) - f'(0^+)\right] - \frac{2L}{k^2\pi^2}\int_0^L f''\cos qx\,\mathrm{d}x, \tag{5.58}$$

using $\cos(k\pi) = (-1)^k$ and $2/(Lq^2) = 2L/(k^2\pi^2)$. When f is smooth, the remaining integral itself shrinks as k grows (integrate by parts again), so a kink makes c_k fall like $1/k^2$, while a jump would make coefficients fall only like $1/k$ (Problem 25). Smooth features need few cosines; sharp features need many.

Sample Problem 5.9: The cosine series of $|x|$ on the paper's box

Expand $f(x) = |x|$ on $[-L, L]$ with $L = 44.883$, and find the rms error of keeping only $k = 0, 1, 2, 3$ ($M = 4$).

Step 1: constant. $c_0 = \frac{1}{2L}\int_{-L}^{L}|x|\mathrm{d}x = \frac{1}{2L}\cdot 2\cdot\frac{L^2}{2} = \frac{L}{2} = 22.4415$.

Step 2: integrate by parts. For $k \geq 1$, $c_k = \frac{2}{L}\int_0^L x\cos qx\,\mathrm{d}x$ and

$$\int_0^L x\cos qx\,\mathrm{d}x = \left[\frac{x\sin qx}{q}\right]_0^L - \int_0^L \frac{\sin qx}{q}\mathrm{d}x = 0 + \left[\frac{\cos qx}{q^2}\right]_0^L = \frac{(-1)^k - 1}{q^2}.$$

Step 3: coefficients. $c_k = \frac{2}{L}\cdot\frac{L^2((-1)^k-1)}{k^2\pi^2}$, which is $-\dfrac{4L}{k^2\pi^2}$ for odd k and 0 for even $k \geq 2$. Numerically $c_1 = -18.190$, $c_3 = -2.0212$. (Compare Equation (5.58): $f'(L) = 1$ and $f'(0^+) = 1$ give exactly this, with zero remainder since $f'' = 0$.)

Step 4: error. By Equation (5.52), $\mathcal{E}_{\min} = \langle f^2\rangle - c_0^2 - \frac{1}{2}(c_1^2 + c_3^2)$ with $\langle f^2\rangle = \frac{1}{2L}\int x^2\mathrm{d}x = L^2/3$:

$$\mathcal{E}_{\min} = L^2\left[\frac{1}{3} - \frac{1}{4} - \frac{8}{\pi^4}\left(1 + \frac{1}{81}\right)\right] = 1.9155\times 10^{-4}L^2.$$

Step 5: rms error.

$$\boxed{c_0 = 22.44,\ c_1 = -18.19,\ c_3 = -2.021; \quad \text{rms error} = \sqrt{\mathcal{E}_{\min}} = 0.621}$$

Four cosines (two nonzero beyond the constant) capture 99.94% of the mean square of $|x|$; the rms error is 2.4% of the rms of $|x|$ itself ($L/\sqrt{3} = 25.91$).

Sample Problem 5.10: The cosine series of x^2 and the Basel sum

Expand $f(x) = x^2$ on $[-L, L]$ and use it to evaluate $\sum_{k\geq 1} 1/k^2$.

Step 1: constant. $c_0 = \frac{1}{2L}\int_{-L}^{L} x^2 \mathrm{d}x = \frac{1}{2L}\cdot\frac{2L^3}{3} = \frac{L^2}{3}$.

Step 2: first integration by parts. $\int_0^L x^2\cos qx\,\mathrm{d}x = \left[\frac{x^2\sin qx}{q}\right]_0^L - \frac{2}{q}\int_0^L x\sin qx\,\mathrm{d}x = -\frac{2}{q}\int_0^L x\sin qx\,\mathrm{d}x$.

Step 3: second integration by parts. $\int_0^L x\sin qx\,\mathrm{d}x = \left[-\frac{x\cos qx}{q}\right]_0^L + \int_0^L \frac{\cos qx}{q}\mathrm{d}x = -\frac{L(-1)^k}{q} + 0$.

Step 4: coefficients. $\int_0^L x^2\cos qx\,\mathrm{d}x = \frac{2L(-1)^k}{q^2}$, so $c_k = \frac{2}{L}\cdot\frac{2L(-1)^k}{q^2} = \frac{4L^2(-1)^k}{k^2\pi^2}$.

Step 5: evaluate at $x = L$. The periodic extension is continuous there, so the series converges to L^2. With $\cos(k\pi) = (-1)^k$, each term is $4L^2/(k^2\pi^2)$:

$$L^2 = \frac{L^2}{3} + \frac{4L^2}{\pi^2}\sum_{k\geq 1}\frac{1}{k^2} \quad\Longrightarrow\quad \boxed{\sum_{k\geq 1}\frac{1}{k^2} = \frac{\pi^2}{6} = 1.6449}$$

This is the Basel sum $\zeta(2)$, our first value of the Riemann zeta function (Chapter 16). At $x = 0$ instead, $0 = L^2/3 + \frac{4L^2}{\pi^2}\sum(-1)^k/k^2$, so $1 - \frac{1}{4} + \frac{1}{9} - \cdots = \pi^2/12 = 0.8225$.

How many cosines does a narrow feature need? Take an even target made of two narrow top-hat bumps: $g(x) = 1$ where $||x| - x_0| < w/2$ and $g(x) = 0$ elsewhere, with $w/2 < x_0 < L - w/2$ so that both bumps lie inside the box. For $k \geq 1$ put $q = k\pi/L$. Since $g\cos qx$ is even, only the right bump needs integrating:

$$c_k = \frac{2}{L}\int_{x_0-w/2}^{x_0+w/2}\cos qx\,\mathrm{d}x = \frac{2}{Lq}\Big[\sin q(x_0 + \tfrac{w}{2}) - \sin q(x_0 - \tfrac{w}{2})\Big] = \frac{4}{Lq}\cos(qx_0)\sin\frac{qw}{2}, \quad (5.59)$$

where the last step subtracts the two addition formulas, $\sin(u+v) - \sin(u-v) = 2\cos u\sin v$. Writing $z_k = qw/2 = k\pi w/(2L)$, this is $c_k = \frac{2w}{L}\cos(qx_0)\,\frac{\sin z_k}{z_k}$. The factor $\cos(qx_0)$ oscillates with k and records where the bumps sit. The envelope $\sin z_k/z_k$ stays close to 1 while $z_k \ll 1$ and first reaches zero at $z_k = \pi$, that is at $k = 2L/w$, where the wavelength $2L/k$ has shrunk to w. Example: $w = 6$ on the scale-1 box $L = 44.883$ gives $z_k = 0.2100\,k$; the envelope is 0.826 at $k = 5$, 0.411 at $k = 10$ and zero at $k = 2L/w = 14.96$. So every cosine up to $k \approx 2L/w$ is needed at nearly full strength; truncating earlier smears the bump out. Beyond that the envelope is at most $1/z_k$, the slow $1/k$ tail caused by the top hat's jumps; smoother bumps have smaller tails, by the integration-by-parts argument of Equation (5.58). The heuristic rule: *a feature of width w needs cosines of wavelength down to about w, that is k up to order $2L/w$.*

```
import numpy as np
from scipy.integrate import trapezoid

L = 44.883                          # scale-1 half-width (RE3)
x = np.linspace(-L, L, 20001)
f = np.abs(x)                       # an even target
M = 4
c = np.empty(M)
c[0] = trapezoid(f, x)/(2*L)        # paper convention: c_0 is the average
for k in range(1, M):
    c[k] = trapezoid(f*np.cos(k*np.pi*x/L), x)/L
g = sum(c[k]*np.cos(k*np.pi*x/L) for k in range(M))
print(c)                            # approx [22.44, -18.19, 0, -2.02]
print(np.sqrt(trapezoid((f - g)**2, x)/(2*L)))   # approx 0.621
```

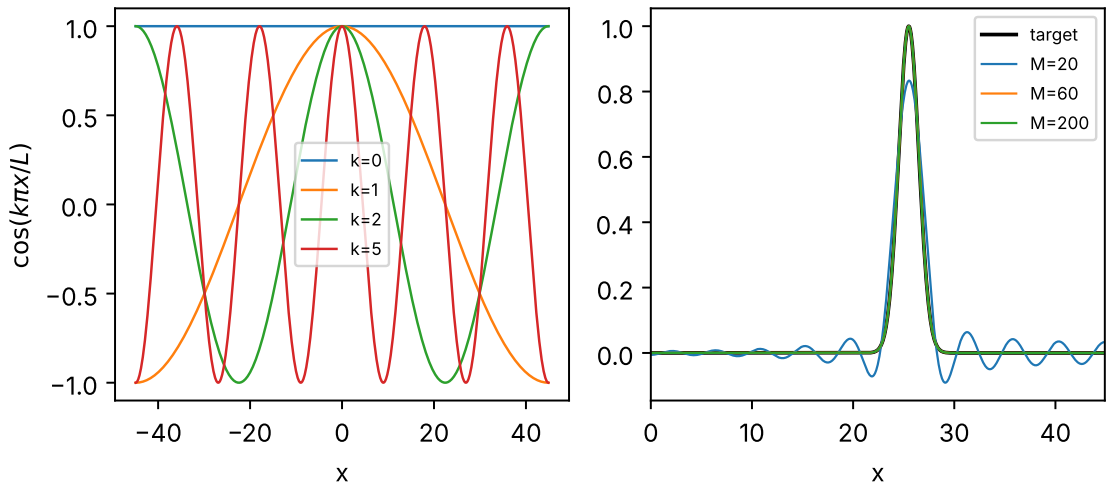


Figure 5.4: Left: cosine basis functions $\cos(k\pi x/L)$ on the paper's scale-1 box, $L = 44.883$, for $k = 0, 1, 2, 5$. Each is even and has slope zero at $x = 0$. Right: an even target made of two narrow bumps centered at $|x| = 25.5$ (black) and its truncated cosine series with $M = 20$, 60 and 200 terms. Building a narrow localized feature requires cosines whose wavelength $2L/k$ is comparable to its width. x is in units with $\hbar = 2m = 1$.

Checkpoint 5.5

Is the $2L$-periodic extension of an even function on $[-L, L]$ continuous at $x = \pm L$? Is the extension of the odd function $f(x) = x$?

Answer: Even: yes, because $f(-L) = f(L)$. Odd $f(x) = x$: no, it jumps from L to $-L$, a jump of $2L$, so its sine series shows Gibbs overshoot at the walls and coefficients falling like $1/k$.

Key Ideas

- Even $\times$ even and odd $\times$ odd are even; even $\times$ odd is odd. Odd functions integrate to zero over $[-L, L]$.
- Even functions have pure cosine series; odd functions have pure sine series.
- The cosine basis $\cos(k\pi x/L)$ on $[-L, L]$, with $c_0 = \frac{1}{2L}\int f$ and $c_k = \frac{1}{L}\int f \cos\frac{k\pi x}{L}$, has no wall discontinuity for even f; kinks give $1/k^2$ decay.
- Heuristic (top-hat example, Equation (5.59)): a feature of width w needs cosines of wavelength down to about w, that is k up to order $2L/w$.

5.8 Quasi-periodic sums: frequencies that never line up

Everything so far concerned functions that repeat. Now add two sinusoids whose periods do not fit together, and something genuinely new happens.

5.8.1 When is a sum of two sinusoids periodic?

Two angular frequencies $\omega_1, \omega_2 > 0$ are called **commensurate** if their ratio is rational, $\omega_2/\omega_1 = l/j$ with positive integers j, l, and **incommensurate** otherwise.

Theorem (proved). *A two-frequency sum is periodic exactly when the frequencies are commensurate.* Let $f(t) = \sin(\omega_1 t) + \sin(\omega_2 t)$ with $0 < \omega_1 < \omega_2$. Then f is periodic if and only if ω_2/ω_1 is rational (a ratio of two integers; $\mathbb{Q}$ denotes the set of rational numbers).

Proof.

Step 1 (rational ⇒ periodic). If $\omega_2/\omega_1 = l/j$, set $\tau = 2\pi j/\omega_1$. Then $\omega_1\tau = 2\pi j$ and $\omega_2\tau = 2\pi j\,\omega_2/\omega_1 = 2\pi l$, whole numbers of turns, so both terms, and hence f, satisfy $f(t+\tau) = f(t)$.

Step 2 (periodic ⇒ rational): set up. Suppose $f(t+\tau) = f(t)$ for all t with some $\tau > 0$. Differentiating this identity twice with respect to t gives $f''(t+\tau) = f''(t)$.

Step 3: isolate one frequency. $f''(t) = -\omega_1^2 \sin\omega_1 t - \omega_2^2 \sin\omega_2 t$, so

$$g(t) \equiv \omega_2^2 f(t) + f''(t) = (\omega_2^2 - \omega_1^2)\sin\omega_1 t.$$

Since f and f'' both have period τ, so does g. Because $\omega_2^2 - \omega_1^2 \neq 0$, $\sin(\omega_1(t+\tau)) = \sin(\omega_1 t)$ for all t.

Step 4: conclude for ω_1. Choose t with $\omega_1 t = \pi/2$: then $\sin(\pi/2 + \omega_1\tau) = 1$, which requires $\omega_1\tau = 2\pi j$ for a positive integer j (positive because $\tau > 0$).

Step 5: the other frequency. The same steps with $\omega_1^2 f + f'' = (\omega_1^2 - \omega_2^2)\sin\omega_2 t$ give $\omega_2\tau = 2\pi l$.

Step 6: divide. $\omega_2/\omega_1 = l/j$, rational.

$$\boxed{\sin\omega_1 t + \sin\omega_2 t \text{ is periodic } \iff \omega_2/\omega_1 \in \mathbb{Q}}$$

This completes the proof. □

Amplitudes and phases. The proof is unchanged in substance for $f(t) = A_1\sin(\omega_1 t + \varphi_1) + A_2\sin(\omega_2 t + \varphi_2)$ with $A_1, A_2 \neq 0$. Step 1: a shift $\tau = 2\pi j/\omega_1$ still adds $2\pi j$ and $2\pi l$ to the two phases, whatever φ_1, φ_2 are. Step 3 becomes $\omega_2^2 f + f'' = (\omega_2^2 - \omega_1^2)A_1\sin(\omega_1 t + \varphi_1)$; dividing by the nonzero constant $(\omega_2^2 - \omega_1^2)A_1$ shows that $\sin(\omega_1 t + \varphi_1)$ has period τ. Step 4: choose t with $\omega_1 t + \varphi_1 = \pi/2$; then $\sin(\pi/2 + \omega_1\tau) = 1$ forces $\omega_1\tau = 2\pi j$ as before. Step 5 swaps the indices.

Many frequencies. Let $F(t) = \sum_{j=1}^{J} A_j\sin(\omega_j t + \varphi_j)$ with distinct positive frequencies ω_j and nonzero amplitudes A_j, and suppose $F(t+\tau) = F(t)$ for all t, with $\tau > 0$. For each j define the operation $D_j g = \omega_j^2 g + g''$. Differentiating $g(t+\tau) = g(t)$ shows that derivatives of a τ-periodic function are τ-periodic, so D_j turns τ-periodic functions into τ-periodic functions. On a single term, $D_j\sin(\omega_i t + \varphi_i) = (\omega_j^2 - \omega_i^2)\sin(\omega_i t + \varphi_i)$: D_j kills term j and multiplies every other term by a constant. Fix i and apply D_j once for each $j \neq i$. Every term except term i is killed, and what remains is $C_i A_i\sin(\omega_i t + \varphi_i)$, where $C_i = \prod_{j\neq i}(\omega_j^2 - \omega_i^2)$ (the product of these factors over all $j \neq i$) is nonzero; this remainder is τ-periodic. Exactly as in Step 4, $\omega_i\tau = 2\pi l_i$ for a positive integer l_i. Hence every ratio $\omega_i/\omega_1 = l_i/l_1$ is rational. Equivalently: *if two frequencies of a finite sum of sinusoids (with nonzero amplitudes) have an irrational ratio, the sum is not periodic.*

5.8.2 The logarithms of primes are incommensurate

Is $\log 3/\log 2$ rational? Suppose $\log 3/\log 2 = a/b$ with positive integers a, b. Then $b\log 3 = a\log 2$, so $\log(3^b) = \log(2^a)$, and since log is one-to-one, $3^b = 2^a$. But 2^a is even and 3^b is odd. Contradiction. Therefore

$$\frac{\log 3}{\log 2} = 1.5849625\ldots \quad \text{is irrational,} \tag{5.60}$$

and by the theorem, $\sin(t\log 2) + \sin(t\log 3)$ is **not periodic**: it never exactly repeats itself.

The same argument works for powers of two different primes $p \neq q$: $\log p^m / \log q^{m'} = a/b$ would mean $p^{mb} = q^{m'a}$, two different factorizations of one integer into primes. Ruling that out for general primes needs the uniqueness of prime factorization, proved in Chapter 15; for 2 against an odd prime, the parity argument above is enough. Powers of the *same* prime behave differently: $\log p^m / \log p^{m'} = m/m'$ is rational, so for example $\log 4 = 2\log 2$ and $\log 8 = 3\log 2$ are commensurate; they are harmonics of $\log p$.

5.8.3 It never repeats, but it comes close

Let $f(t) = \sin(t\log 2) + \sin(t\log 3)$. Its largest conceivable value is 2, reached only if both sines equal 1 at once: $t\log 2 = \pi/2 + 2\pi j$ and $t\log 3 = \pi/2 + 2\pi l$. Dividing, $\log 3/\log 2 = (1+4l)/(1+4j)$, rational, which is impossible. So f *never equals* 2. Yet it comes as close as you like.

Sample Problem 5.11: Getting close to the maximum

Find a t in $[0, 200]$ where $f(t) = \sin(t\log 2) + \sin(t\log 3)$ exceeds 1.999.
Step 1: put the first sine at its peak. Take $t = (\pi/2 + 2\pi j)/\log 2$; then $\sin(t\log 2) = 1$.
Step 2: phase of the second sine. Its phase is $t\log 3 = (\pi/2 + 2\pi j)\rho$ with $\rho = \log 3/\log 2 = 1.58496$. We want this to be close to $\pi/2$ modulo 2π, that is, after discarding whole turns of 2π.
Step 3: scan j. We need the second phase within $\arccos(0.999) = 0.0447$ rad of $\pi/2$ (modulo 2π), so that the second sine exceeds 0.999. Computing this mismatch for $j = 0, 1, 2, \ldots$ in turn (a short loop, rerun in CHECKS) gives, for example, 0.62 rad at $j = 3$ and 0.45 rad at $j = 5$; the first j with mismatch below 0.0447 is $j = 10$. For $j = 10$: $t = 20.5\pi/\log 2 = 92.913$, and $t\log 3 = 102.0759$. Subtracting $16 \times 2\pi = 100.5310$ leaves 1.5449, which is 0.0259 below $\pi/2 = 1.5708$.
Step 4: value. $\sin(1.5449) = \cos(0.0259) = 0.99967$, so

$$\boxed{f(92.913) = 1.99967}$$

Near-periods. Similarly, f never satisfies $f(t+\tau) = f(t)$ exactly, but it nearly does for special τ. Take τ equal to a periods of the $\log 2$ term, $\tau = 2\pi a/\log 2$, so the first sine returns exactly. The second sine has advanced by $\tau\log 3/(2\pi) = a\rho$ cycles; if $a\rho$ is close to a whole number, it nearly returns too. Writing ε for the distance of $a\rho$ from the nearest integer (in cycles), the shift of the second term changes its phase by $\delta = 2\pi\varepsilon$ (modulo 2π). To compare $\sin(\beta+\delta)$ with $\sin\beta$, replace β by $-\beta$ in Equation (5.13), using $\sin(-\beta) = -\sin\beta$ and $\cos(-u) = \cos u$: $\sin\alpha - \sin\beta = 2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$. Now set $\alpha = \beta + \delta$:

$$\sin(\beta+\delta) - \sin\beta = 2\cos\Big(\beta + \tfrac{\delta}{2}\Big)\sin\tfrac{\delta}{2}, \tag{5.61}$$

whose largest absolute value over β is $2|\sin(\delta/2)| = 2|\sin(\pi\varepsilon)|$. So

$$\max_t |f(t+\tau) - f(t)| = 2|\sin(\pi\varepsilon)| \qquad (\tau = 2\pi a/\log 2). \tag{5.62}$$

Table 5.2 lists the best a up to 53. For a general shift τ, not of this special form, we measure the mismatch by $D(\tau) = \sup_t |f(t+\tau) - f(t)|$. Here sup (the **supremum**, or least upper bound) is the smallest number that the quantity never exceeds as t varies; unlike a maximum, it need not be attained at any t.

The row $a = 12$ is a near-coincidence musicians know: $12\log 3 \approx 19\log 2$ means $3^{12} \approx 2^{19}$ (531441 against 524288, a ratio of 1.01364 called the Pythagorean comma). That is why twelve

Table 5.2: Near-periods of $f(t) = \sin(t\log 2) + \sin(t\log 3)$ built from a periods of the $\log 2$ term. ε is the distance of $a\log 3/\log 2$ from the nearest integer; the last column is Equation (5.62).

a	$\tau = 2\pi a/\log 2$	$a\log 3/\log 2$	ε	$\max_t \lvert f(t+\tau) - f(t)\rvert$
1	9.065	1.58496	0.41504	1.929
2	18.129	3.16993	0.16993	1.018
5	45.324	7.92481	0.07519	0.468
12	108.777	19.01955	0.01955	0.1228
41	371.654	64.98346	0.01654	0.1039
53	480.430	84.00301	0.00301	0.0189

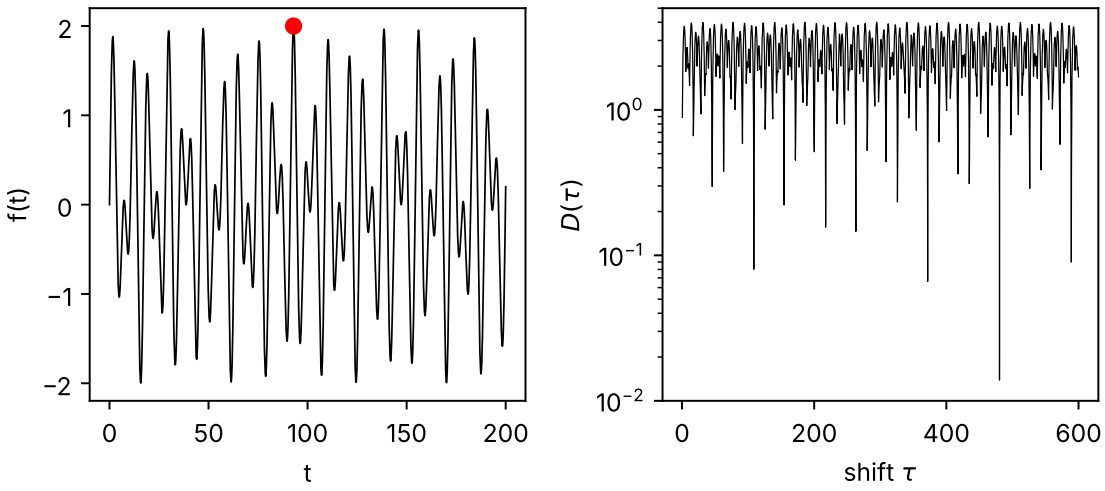


Figure 5.5: Left: the quasi-periodic sum $f(t) = \sin(t\log 2)+\sin(t\log 3)$; the dot marks $t = 92.913$ where $f = 1.99967$, close to (but below) the never-attained value 2. Right: the largest mismatch $D(\tau) = \sup_t \lvert f(t+\tau) - f(t)\rvert$ as a function of the shift τ (logarithmic scale). It dips near $\tau = 108.8$ and 480.4 (Table 5.2) but never reaches zero. All axes dimensionless.

equal semitones nearly, but not exactly, produce perfect fifths. The row $a = 53$ says $3^{53} \approx 2^{84}$, within a factor 1.0021.

Taken on trust

Kronecker's theorem and almost periodic functions. Two facts used informally here are proved elsewhere. (1) Kronecker's theorem: if ω_1/ω_2 is irrational, the pair of phases $(\omega_1 t, \omega_2 t)$ modulo 2π comes arbitrarily close to every pair of angles as t ranges over the reals. Consequently $\sup_t f = 2$ (approached, never reached), because both phases can be brought, at the same t, as close as we like to $\pi/2$. Likewise, for a general shift τ write $f(t+\tau) - f(t) = [\sin(\alpha+\delta_2) - \sin\alpha] + [\sin(\beta+\delta_3) - \sin\beta]$ with $\alpha = t\log 2$, $\beta = t\log 3$, $\delta_2 = \tau\log 2$, $\delta_3 = \tau\log 3$. By the difference identity used for Equation (5.62), the two brackets have largest absolute values $2|\sin(\delta_2/2)|$ and $2|\sin(\delta_3/2)|$, and Kronecker's theorem lets α and β approach, at the same t, angles where both brackets are near their maxima with the same sign. So the supremum of the mismatch (approached, in general not attained) is $D(\tau) = 2|\sin(\tau\log 2/2)| + 2|\sin(\tau\log 3/2)|$, the formula plotted in Figure 5.5. A proof is in Hardy and Wright [24], chapter XXIII. (2) H. Bohr's theory of *almost periodic functions*: a finite sum of sinusoids with any real frequencies has, for every $\epsilon > 0$, "ϵ-almost-periods" τ with $|f(t+\tau) - f(t)| < \epsilon$ for all t, and such τ occur with bounded gaps along the whole real line. Functions of this kind are called almost periodic or quasi-periodic. Bohr's theory, including the existence of ϵ-almost-periods with bounded gaps, is developed with proofs in his monograph *Almost Periodic Functions* and in later texts on harmonic analysis. We take both results on trust; we use only the conclusions stated.

5.8.4 Long-time averages and Parseval for quasi-periodic sums

Over a period, orthogonality killed cross terms exactly. A quasi-periodic sum has no period, but cross terms still die in the long run. Define the average over a window of length T starting at 0: $\langle g\rangle_T = \frac{1}{T}\int_0^T g\,\mathrm{d}t$. For $\omega_1 \neq \omega_2$, by Equation (5.20),

$$\langle \sin\omega_1 t \sin\omega_2 t\rangle_T = \frac{1}{2T}\int_0^T [\cos(\omega_1-\omega_2)t - \cos(\omega_1+\omega_2)t]\,\mathrm{d}t \tag{5.63}$$

$$= \frac{1}{2T}\left[\frac{\sin(\omega_1-\omega_2)T}{\omega_1-\omega_2} - \frac{\sin(\omega_1+\omega_2)T}{\omega_1+\omega_2}\right], \tag{5.64}$$

and since $|\sin| \le 1$,

$$|\langle \sin\omega_1 t \sin\omega_2 t\rangle_T| \le \frac{1}{2T}\left(\frac{1}{|\omega_1-\omega_2|} + \frac{1}{\omega_1+\omega_2}\right) \xrightarrow{T\to\infty} 0. \tag{5.65}$$

With $\omega_1 = \omega_2 = \omega$, Equation (5.20) gives $\langle \sin^2\omega t\rangle_T = \frac{1}{2} - \frac{\sin 2\omega T}{4\omega T} \to \frac{1}{2}$. Phase constants change nothing. By Equation (5.20), $\sin(\omega_1 t+\varphi_1)\sin(\omega_2 t+\varphi_2) = \frac{1}{2}[\cos((\omega_1-\omega_2)t+\varphi_1-\varphi_2) - \cos((\omega_1+\omega_2)t+\varphi_1+\varphi_2)]$, and for $\Omega \neq 0$, $\frac{1}{T}\int_0^T \cos(\Omega t+\psi)\,\mathrm{d}t = \frac{\sin(\Omega T+\psi)-\sin\psi}{\Omega T}$, which is at most $2/(|\Omega|T)$ in size. So the cross-term average is bounded by $\frac{1}{T}(\frac{1}{|\omega_1-\omega_2|} + \frac{1}{\omega_1+\omega_2})$ and again tends to zero; and $\sin^2(\omega t+\varphi) = \frac{1}{2} - \frac{1}{2}\cos(2\omega t+2\varphi)$ has average within $1/(2\omega T)$ of $\frac{1}{2}$. Now take a finite sum $F(t) = \sum_j A_j \sin(\omega_j t+\varphi_j)$ with *distinct* frequencies and expand its square: $F^2 = \sum_j A_j^2 \sin^2(\omega_j t+\varphi_j) + \sum_{i\neq j} A_i A_j \sin(\omega_i t+\varphi_i)\sin(\omega_j t+\varphi_j)$. The square terms average to $A_j^2/2$ and the cross terms to zero as $T\to\infty$, so

$$\lim_{T\to\infty}\langle F^2\rangle_T = \sum_j \frac{A_j^2}{2}, \tag{5.66}$$

Parseval's theorem for quasi-periodic sums. The bound Equation (5.65) carries a lesson for later: two frequencies only "look orthogonal" once T is large compared with $1/|\omega_1 - \omega_2|$. Telling close frequencies apart needs a long window. Chapter 6 turns this into the resolution $\Delta\omega \approx 2\pi/T$.

Checkpoint 5.6

Is $\sin(t\log 4) + \sin(t\log 8)$ periodic? If so, find its period.

Answer: Yes: $\log 4 = 2\log 2$ and $\log 8 = 3\log 2$, ratio $3/2$. Both are harmonics of $\log 2$, so the period is $2\pi/\log 2 = 9.065$.

Key Ideas

- $\sin\omega_1 t + \sin\omega_2 t$ is periodic if and only if ω_2/ω_1 is rational (proved).
- $\log 3/\log 2$ is irrational (parity proof); logarithms of powers of *different primes* are incommensurate (unique factorization, Chapter 15), while powers of the same prime are harmonics of $\log p$.
- A finite sum of sinusoids with nonzero amplitudes is periodic only if all its frequency ratios are rational (proved, many-frequency version).
- Quasi-periodic sums never repeat exactly but have near-periods, Table 5.2.
- Long-time averages kill cross terms between distinct frequencies, so mean squares still add, Equation (5.66), once $T \gg 1/|\Delta\omega|$.

5.9 Why a finite window cannot capture an almost-periodic function

Suppose you know a function only on a window $[0, T]$ and you expand it in a Fourier series on that window. What does the series predict outside?

5.9.1 A Fourier series on a window predicts its own periodic extension

On the window $[0, T]$ (length T, so "$2L$" $= T$), the complex Fourier series is

$$F(t) = \sum_k \hat{f}_k e^{2\pi ikt/T}, \qquad \hat{f}_k = \frac{1}{T}\int_0^T f(t)e^{-2\pi ikt/T}\,\mathrm{d}t. \tag{5.67}$$

(The derivation is that of Equation (5.41) with the interval $[0, T]$; orthogonality holds on any interval of length equal to the period.) Each term satisfies $e^{2\pi ik(t+T)/T} = e^{2\pi ikt/T}e^{2\pi ik} = e^{2\pi ikt/T}$, so $F(t+T) = F(t)$. Inside the window, $F = f$ (where f is continuous). Outside, F is forced to repeat the window: for $T < t < 2T$, $F(t) = F(t-T) = f(t-T)$. So the prediction error on the next window is

$$F(t) - f(t) = f(t-T) - f(t), \qquad T < t < 2T. \tag{5.68}$$

This vanishes for all such t only if f repeats with period T there.

Consequence. If f is quasi-periodic, like $\sin(t\log 2) + \sin(t\log 3)$, it has no period at all (Section 5.8), so *for every window length* T, the window's Fourier series mispredicts somewhere outside the window. The size of the error is set by Equation (5.62) and Figure 5.5: it is small only for special near-periods and is never zero. Choosing T to fit a whole number of $\log 2$ periods cannot also fit a whole number of $\log 3$ periods, because that would make $\log 3/\log 2$ rational.

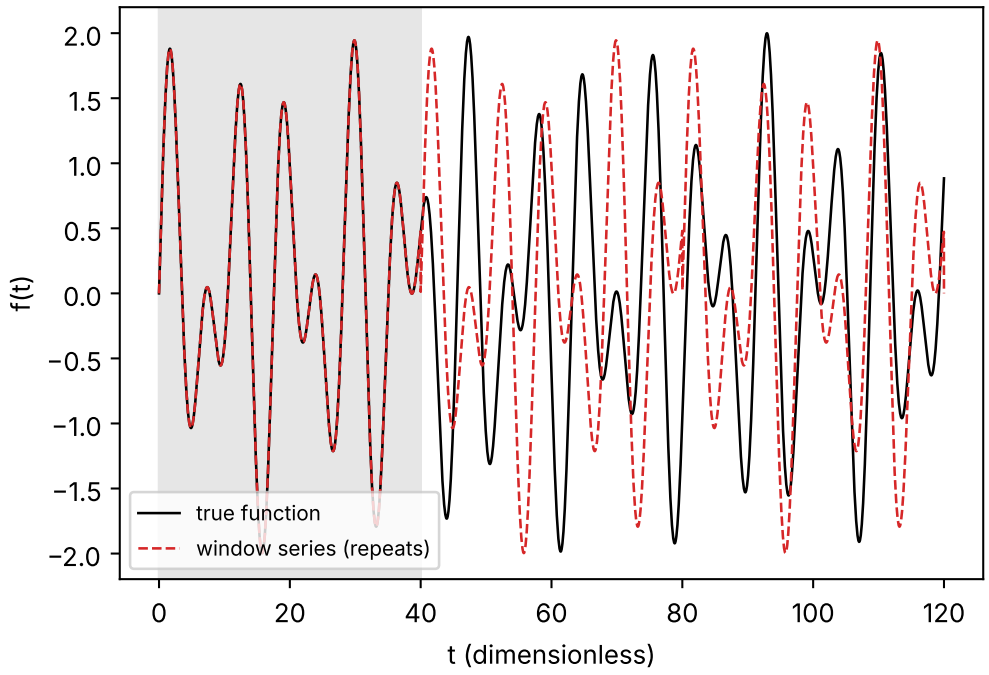


Figure 5.6: The quasi-periodic function $\sin(t\log 2)+\sin(t\log 3)$ (solid) and the Fourier series of its window $[0,40]$ (shaded), which inside the window reproduces it and outside merely repeats the window (dashed). The repetition jumps at $t=40$ and 80 and does not continue the true function. Axes dimensionless.

5.9.2 Even one sinusoid spreads over many harmonics

The problem appears even for a single frequency that does not fit the window. Take $f(t)=e^{i\omega t}$ on $[0,T]$ and write $\omega_k=2\pi k/T$ for the window's harmonic frequencies. From Equation (5.67),

$$\hat{f}_k=\frac{1}{T}\int_0^T e^{i(\omega-\omega_k)t}\mathrm{d}t=\frac{e^{i(\omega-\omega_k)T}-1}{i(\omega-\omega_k)T}. \tag{5.69}$$

Let $\theta=(\omega-\omega_k)T/2$. Factor $e^{i\theta}$ out of the numerator: $e^{2i\theta}-1=e^{i\theta}(e^{i\theta}-e^{-i\theta})=e^{i\theta}\,2i\sin\theta$. So

$$\hat{f}_k=e^{i\theta}\,\frac{\sin\theta}{\theta},\qquad |\hat{f}_k|=\left|\frac{\sin\theta}{\theta}\right|,\qquad \theta=\pi\left(\frac{\omega T}{2\pi}-k\right). \tag{5.70}$$

If the window holds a whole number k_0 of cycles ($\omega T/2\pi=k_0$), then $\theta=\pi(k_0-k)$, so $\hat{f}_{k_0}=1$ and every other $\hat{f}_k=0$: one harmonic. Otherwise *every* $\hat{f}_k$ is nonzero, falling off only like $1/|k|$, the signature of the jump that the periodic extension creates at the window edge. The function $\sin\theta/\theta$ (the "sinc") is the subject of the next chapter.

Sample Problem 5.12: A window that does not fit the prime-2 frequency

For $f(t)=e^{it\log 2}$ on a window of length $T=40$, find the number of cycles in the window and $|\hat{f}_4|$, $|\hat{f}_5|$.

Step 1: cycles. $\omega T/2\pi=40\times 0.693147/6.283185=4.4127$, not a whole number.

Step 2: $k=4$. $\theta=\pi(4.4127-4)=1.2966$, $|\hat{f}_4|=\sin(1.2966)/1.2966=0.742$.

Step 3: $k=5$. $\theta=\pi(4.4127-5)=-1.8450$, $|\hat{f}_5|=\sin(1.8450)/1.8450=0.522$.

$$\boxed{|\hat{f}_4|=0.742,\qquad |\hat{f}_5|=0.522}$$

A single pure frequency is smeared over the window's harmonics. Choosing $T = 2\pi \cdot 4/\log 2 = 36.26$ instead would put exactly four cycles in the window and give a single harmonic.

5.9.3 Representing is not continuing

Here is the lesson in plain words. A flexible enough set of building blocks can *represent* any function on a window as closely as you like. That says nothing about whether the blocks *continue* the function correctly outside the window. The window Fourier series reproduces the window perfectly and then repeats it, because "repeat with period T" is the only continuation its vocabulary can express. To continue $\sin(t \log 2)$ correctly, a model needs to contain the frequency $\log 2$ itself, not an imitation of one window of it. We will meet this distinction again as *memorization versus generalization* (Chapter 25) and as *sparse frequency learning* (Chapter 27).

Key Ideas

- The Fourier series of a window $[0, T]$ equals the function inside and its T-periodic repetition outside, Equation (5.68).
- A quasi-periodic function is not T-periodic for any T, so no window's periodic extension continues it correctly.
- A sinusoid whose frequency does not fit the window spreads over all harmonics with weights $|\sin\theta/\theta|$, Equation (5.70).
- Fitting a window (representation) and continuing beyond it (prediction) are different abilities.

5.10 Paper connection

5.10.1 The cosine basis of δV (paper Sec. II C)

The Hermitian model of the paper [1] (Sec. II C) perturbs the even Wu–Sprung potential by

$$\delta V(x) = \sum_{k=0}^{M-1} c_k \cos\frac{k\pi x}{L}, \tag{5.71}$$

exactly the basis Equation (5.55) with the constant convention of Equation (5.50). You now know why it is cosines: the background is even (Sec. II B, "with V even"), so an even correction needs no sines (Section 5.7), and the periodic extension of an even function has no jump at the walls $\pm L$. The half-widths are $L = 44.883$ at scale 1 and $L = 59.957$ at scale 2 (Sec. II B), and Table I uses M from 200 to 900.

What scales can such a basis resolve? The shortest wavelength present is $2L/(M-1)$: for $M = 400$ at scale 1 it is $2 \times 44.883/399 = 0.225$ (wavenumber $399\pi/L = 27.93$); for $M = 900$ at scale 2 it is 0.1334 (Problem 14). The paper's first paragraph of Sec. III states a basis-coverage law, "the cosine family becomes effectively complete for joint level shifts once $M\pi/L > 2\sqrt{E}$". Chapter 24 derives this heuristic; here we only do arithmetic with it. For level 500 at scale 1, with $E \approx t_{500} = 811.18$ (App. A key anchors), it requires $M > 2\sqrt{811.18} \times 44.883/\pi = 813.8$, consistent with the paper's report that $M = 900$ reproduces all 500 zeros to rms $< 10^{-4}$ spacings. For $M = 400$, $M\pi/L = 28.0$, so the law covers only levels with $2\sqrt{E} < 28.0$, that is E below about 196.

The interpolation split (Sec. III, Fig. 4) shows the superposition principle at work: the fitted cosine series builds a localized structure, "rms $\delta V = 40$" in the band $x = 22.5$–28.5 "against

15 inside it and 13 outside it, with a maximum $|\delta V| = 199$". A correction that is largest in a band of width 6 (the classical turning points of the held-out levels 201–300, Chapter 12) while remaining nonzero elsewhere (rms 15 and 13) is a superposition of many cosines, as in Figure 5.4 and the top-hat example of Section 5.7 (for $w = 6$ at scale 1 the main lobe extends to $k \approx 2L/w = 14.96$); the highest $M = 400$ cosine fits about 27 wavelengths into that band (Problem 27).

The even/odd rules also explain a detail of Sec. II E. The PT-odd class uses the *odd* basis $\sin((k+1)\pi x/L)$ and the paper notes $A_{nn} = 0$. In an even potential each $|\psi_n|^2$ is even (Chapter 11), so $A_{nn} = \int |\psi_n|^2\,(\text{odd})\,\mathrm{d}x$ integrates an odd function over $[-L, L]$ and vanishes by Section 5.7. The absorptive-even class uses the cosines, whose $k = 0$ member is the constant $\phi_0 = 1$; its special "null mode" role is the subject of Chapter 14.

5.10.2 Paper Eq. (4) is an almost-periodic sum

The arithmetic oracle, paper Eq. (4),

$$N_{\text{fl}}(t) = -\frac{1}{\pi} \sum_{p \in P} \sum_{m=1}^{3} \frac{\sin(mt \log p)}{m\, p^{m/2}}, \tag{5.72}$$

is a finite sum of sinusoids in t with amplitudes $1/(\pi m p^{m/2})$ and angular frequencies $m \log p = \log p^m$. (Here P is the paper's set of primes, not a period.) Within one prime the three frequencies are harmonics of $\log p$, so a one-prime oracle is periodic with period $2\pi/\log p$. Across different primes the frequencies are incommensurate (Section 5.8); for example $\log 3/\log 2$ is irrational. With two or more primes, N_{fl} is a finite sum of sinusoids with distinct frequencies and nonzero amplitudes, two of which have an irrational ratio, so by the many-frequency version of the periodicity theorem (Section 5.8) it is not periodic: it is quasi-periodic and never repeats. The six-prime oracle of Fig. 1 uses the six smallest primes, $P = \{2, 3, 5, 7, 11, 13\}$ (the paper says only "six-prime"; this is the natural reading, it is what `qmnt.first_primes(6)` returns, and with it `qmnt` reproduces the six-prime entry 0.074 / +0.96 of Table V on zeros 401–500), and has $6 \times 3 = 18$ distinct frequencies (Problem 26). Chapter 18 explains where Eq. (4) comes from; this chapter explains what kind of function it is. Because it is a sum of sinusoids with *known* frequencies, it is defined for every t, and it continues automatically past any training window; nothing in it needs to be learned from the zeros. That is the property the fitted potentials, as parametrized and optimized in the paper, did not show.

The paper's Discussion (Sec. VI) puts its heuristic explanation of the failure in exactly these terms: "the quantity being fitted is, by the explicit formula, an almost-periodic function of t that is sparse in frequency, whereas a potential addresses the levels through their turning points, locally in energy." The paper labels this reading heuristic, and so do we.

5.10.3 The prime-2 wave in level index

RE2 is the $p = 2$ part of Eq. (4). Its period in t is $2\pi/\log 2 = 9.065$. The paper scores levels by their index n, and consecutive levels near zero n are about one local mean spacing $\text{gap}_n = 2\pi/\log(s_n/2\pi)$ apart (Chapter 17), where s_n is the smooth zero. So one period of RE2 spans $9.065/\text{gap}_n$ levels. Solving $N_{\text{sm}}(s_n) = n - \frac{1}{2}$ numerically gives $s_{400} = 679.9$, $\text{gap}_{400} = 1.341$ and $9.065/1.341 = 6.76$ levels per period; at $n = 500$, $\text{gap}_{500} = 1.293$ and 7.01 levels per period. So one period is about 7 levels, in line with the paper's approximate remark in Sec. V that the $p = 2$ oscillation "is only ∼6–7 *levels* at these heights"; our value 7.01 at $n = 500$ sits at the upper edge of that rounded range. Figure 5.7 shows RE2 sampled at the smooth zeros: perfectly periodic in t, it looks only roughly regular in n, because the mean spacing shrinks slowly with height.

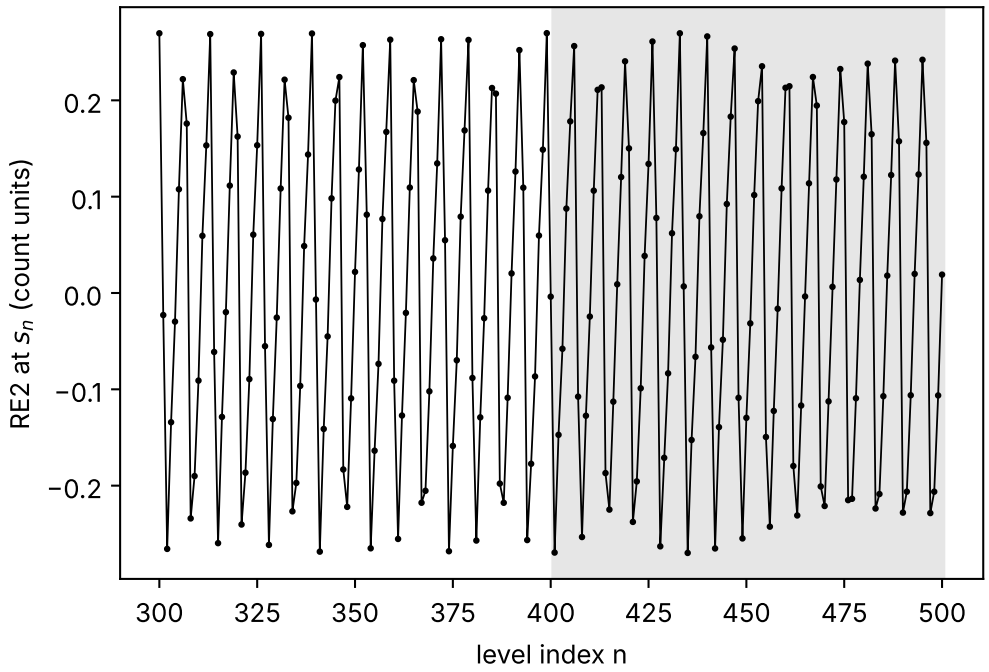


Figure 5.7: The prime-2 wave RE2 (the $p = 2$, $m \leq 3$ part of paper Eq. (4)) evaluated at the smooth zeros s_n, $n = 300$–500. One period (9.065 in t) spans about 6.4 levels at $n = 300$ and 7.0 at $n = 500$. The shaded region $n = 401$–500 is the held-out window of the paper's continuation probe (Sec. III), whose target is the negative of this sequence (App. A step 15). Vertical axis: count units, equivalently mean spacings.

```
import numpy as np
from scipy.optimize import brentq

def Nsm(t):                          # paper Eq. (1)
    return t/(2*np.pi)*np.log(t/(2*np.pi)) - t/(2*np.pi) + 7/8

def smooth_zero(n):                  # s_n: Nsm(s_n) = n - 1/2
    return brentq(lambda t: Nsm(t) - (n - 0.5), 2*np.pi, 1e5)

def re2(t):                          # RE2: the p = 2 part of Eq. (4)
    m = np.arange(1, 4)
    return -np.sum(np.sin(np.outer(t, m*np.log(2)))/(m*2**(m/2)), axis=1)/np.pi

for n in (400, 500):
    s = smooth_zero(n)
    gap = 2*np.pi/np.log(s/(2*np.pi))
    print(n, s, gap, (2*np.pi/np.log(2))/gap)    # levels per period

s = np.array([smooth_zero(n) for n in range(401, 501)])
print("rms of RE2 on 401-500:", np.sqrt(np.mean(re2(s)**2)))
```

5.10.4 The arithmetic-continuation probe (paper Sec. III, App. A step 15)

The probe replaces the true targets by the synthetic fluctuation $-N_{\mathrm{fl}}^{p=2}(s_n)$ in mean spacings (App. A step 15 defines targets $T_n = E_n[\text{smooth}] - N_{\mathrm{fl}}^{p=2}(s_n)\,\mathrm{gap}_n$; this T_n is a target level, not a window length). In other words, the target fluctuation is $-$RE2 sampled at the smooth zeros,

the negative of the sequence in Figure 5.7: a target level sits below its smooth position when the count fluctuation is positive. The sign affects neither the rms nor the period. The fitted potential is trained on levels 1–400 and asked to continue on 401–500.

Parseval tells us what "doing nothing" costs. The long-run rms of RE2 is 0.1709 (Sample Problem "The rms of the prime-2 wave"). The paper's probe null, the rms of predicting zero fluctuation on 401–500, is 0.1708 (App. A key anchors). The two agree closely, as they should: levels 401–500 span about 130 in t, roughly 14 periods, enough for the cross terms in Equation (5.65) to be small. (CHECKS recomputes the sample rms on 401–500 and confirms it lies within 0.003 of both.)

The results (Sec. III): in-sample rms 0.056 and correlation +0.95; held-out correlation +0.07 and held-out rms 0.1958, worse than the null 0.1708 (App. A key anchors). In the paper's words, "The family can paint one window of $\sin(t \log 2)$ onto its levels but cannot express 'continue oscillating': the arithmetic frequency is not in its vocabulary." Section 5.9 gives you a precise model of the phrase "not in its vocabulary": a window Fourier series also paints its window perfectly and then continues it wrongly, because its only built-in continuation is periodic repetition. This is an analogy, not the paper's mechanism: the potential's cosines live in x, not in t, and the paper describes its own explanation as heuristic. What the analogy captures is that representing a window and continuing a frequency are different abilities. Notice also that RE2 is exactly periodic in t; the probe shows that the failure does not need quasi-periodicity, since "a single known arithmetic frequency, learned from a finite window, already defeats the family's extrapolation" (Sec. III).

5.10.5 A Fourier-series view of the dense control C1

The control C1 (Sec. II D, Table IV, App. A step 20) fits $F(t) = \sum_j [a_j \sin \omega_j t + b_j \cos \omega_j t]$ on the uniform grid $\omega_j = 8j/450$, 450 frequencies and 900 coefficients. Every ω_j is a whole multiple of $8/450$, so by the argument of Section 5.1 the model is *exactly periodic* with period $2\pi \times 450/8 = 353.43$ in t: it is a truncated Fourier series in disguise. Its training zeros run from $t_1 = 14.13$ up to about $t_{400} \approx 680$, and the held-out levels 401–500 sit at heights from about 681 to 811, so the model's values there are copies of its values at heights about 328 to 458, inside the training range (Problem 15). This is our observation about the form of the model, not a result reported in the paper; the paper reports only that C1 "stays within 0.005 of the null" (rms 0.258, correlation +0.28 with interval $[+0.08, +0.46]$) and concludes that "capacity without a suitable prior does not generalize". It illustrates Section 5.9 in a model the paper actually ran: a generic periodic vocabulary, however large, continues by repetition.

By contrast, C2 (Sec. II D, Sec. IV) searches for a few frequencies one at a time and finds them at $\log p^m$; that is the subject of Chapter 27, built on the periodogram of Chapter 6.

5.11 Summary

★ Did you know? *The Overshoot That Won't Quit*

Approximate a square wave by a finite Fourier series and you always see an overshoot near each jump, about 9% of the jump size, however many terms you keep. It is called the Gibbs phenomenon after J. Willard Gibbs, who discussed it in *Nature* in 1899, but Henry Wilbraham had described it in 1848. Albert Michelson's mechanical harmonic analyzer helped bring it to attention.

- A sinusoid $A\cos(\omega t+\varphi) = a\cos\omega t + b\sin\omega t = \mathrm{Re}[Ze^{i\omega t}]$ has period $2\pi/\omega$, Equations (5.1), (5.3), (5.5) and (5.7). RE2 is a sum of three harmonics of $\log 2$ with period $2\pi/\log 2 = 9.065$.

- Two sinusoids of nearby frequencies beat with period $2\pi/\Delta\omega$, Equations (5.14) and (5.15).
- Sines and cosines of whole-number frequencies on $[-L, L]$ are orthogonal (Table 5.1); so are complex exponentials, Equation (5.40).
- Fourier series and coefficients: Equations (5.26), (5.30) and (5.31); complex form Equations (5.41) and (5.44). Convergence (Dirichlet) is taken on trust.
- Gibbs: the overshoot of partial sums at a jump tends to about 9% of the jump as the number of terms grows, never to zero, Equation (5.38).
- Parseval: mean squares of orthogonal pieces add, Equations (5.48) and (5.49). The rms of RE2 is 0.1709. Truncated Fourier series are the best mean-square approximations, Equation (5.52).
- Even functions have pure cosine series. The paper's basis is $\cos(k\pi x/L)$ on $[-L, L]$ with c_0 equal to the average, Equations (5.50) and (5.55). Kinks give $1/k^2$ decay of the coefficients, Equation (5.58); jumps give $1/k$.
- A two-frequency sum is periodic if and only if the frequency ratio is rational (proved). $\log 3/\log 2$ is irrational, so sums over several primes, like paper Eq. (4), are quasi-periodic: they have near-periods (Table 5.2, Equation (5.62)) but no period.
- Long-time averages give Parseval for quasi-periodic sums, Equations (5.65) and (5.66); separating frequencies needs $T \gg 1/|\Delta\omega|$.
- A window Fourier series continues its window only by repetition, Equation (5.68); an off-grid frequency spreads as $|\sin\theta/\theta|$, Equation (5.70). Representation is not continuation: the lesson of the paper's continuation probe (Sec. III).

5.12 Problems

1. • Find the angular frequency and the period of $\sin(t\log 3)$, the $p = 3$, $m = 1$ term of paper Eq. (4).

2. • Write $5\cos\omega t + 12\sin\omega t$ in the form $A\cos(\omega t + \varphi)$.

3. • The terms $\sin(t\log 5)$ and $\sin(t\log 7)$ are added with equal amplitudes. Find the beat period.

4. • Evaluate $\int_{-L}^{L}\cos^2(5\pi x/L)\,\mathrm{d}x$ for the scale-2 half-width $L = 59.957$, using Table 5.1.

5. • Classify as even, odd or neither: x^3, $x\cos x$, $|x|\sin x$, $x\sin x$, e^x. For which of them is $\int_{-L}^{L} f\,\mathrm{d}x = 0$ guaranteed?

6. •• Use complex amplitudes to write $2\cos\omega t + 2\cos(\omega t + \pi/3)$ as a single sinusoid.

7. •• Find the Fourier series of the sawtooth $f(x) = x$ on $(-\pi, \pi)$, extended with period 2π.

8. •• Apply Parseval's theorem Equation (5.49) to the sawtooth of Problem 7 to obtain $\sum_{k\geq 1} 1/k^2$.

9. •• Apply Parseval's theorem to the cosine series of $|x|$ (Sample Problem on $|x|$) to evaluate $\sum_{k\ \text{odd}} 1/k^4$.

10. •• Show that over a window of length T that is not a whole number of periods, $\langle\cos^2\omega t\rangle_T = \frac{1}{2} + \sin(2\omega T)/(4\omega T)$. Evaluate it for $\omega = \log 2$, $T = 10$.

11. •• What fraction of the mean square of RE2 comes from its $m = 1$ term?

12. •• Verify the Pythagorean comma: compute $3^{12}/2^{19}$ and relate it to the row $a = 12$ of Table 5.2.

13. •• Prove that $\log 5/\log 3$ is irrational without using unique factorization in general.

14. •• (Paper.) Find the shortest wavelength $2L/(M-1)$ in the cosine basis for $M = 900$ at scale 2 ($L = 59.957$).

15. •• (Paper.) Show that the dense control C1 (Sec. II D, App. A step 20), with frequencies $\omega_j = 8j/450$, is exactly periodic. Find the period and the training heights whose values it reuses on held-out zeros 401–500.

16. •• (Paper.) Using the coverage law of Sec. III and $E \approx t_{1000} = 1419.42$ (Sec. II A), what M would be needed to cover level 1000 at scale 2? Compare with the largest M in Table I.

17. •• (Computational.) Using numpy, compute the complex Fourier coefficients $\hat{f}_k$, $k = -4, \dots, 4$, of RE2 over one period $2\pi/\log 2$ by numerical integration, and compare with the Sample Problem "RE2 in complex form".

18. •• (Computational, paper.) Using `qmnt.mean_gap`, find how many levels one period of the $p = 3$ wave $\sin(t \log 3)$ spans near $n = 400$.

19. ••• (Paper.) Use Equation (5.66) to find the long-time rms of the six-prime oracle N_{fl} (paper Eq. (4) with $P = \{2, 3, 5, 7, 11, 13\}$). Why can you add the mean squares of all 18 terms?

20. ••• (Computational.) Using numpy, evaluate the Gibbs peak $S_K(\pi/2K)$ for $K = 5, 10, 50, 500$ and show that it approaches 1.17898.

21. •• Find the mean-square error of approximating the square wave by its first term $(4/\pi)\sin x$.

22. •• Show that the cosine series of a constant function $f = V_0$ on $[-L, L]$ has $c_0 = V_0$ and all other $c_k = 0$.

23. •• Evaluate the near-period entry for $a = 5$ in Table 5.2 from Equation (5.62).

24. •• (Paper.) Using the parity rules, show that $A_{nn} = \int |\psi_n|^2 \sin((k+1)\pi x/L)\,\mathrm{d}x = 0$ whenever $|\psi_n|^2$ is even, as stated for the PT-odd class in paper Sec. II E.

25. •• Show by one integration by parts that a function with a jump has Fourier coefficients falling like $1/k$. Use $f(x) = x$ on $(-L, L)$ (odd, jump $2L$ at the walls) as the example.

26. • (Paper.) How many distinct frequencies does the six-prime oracle of paper Fig. 1 contain, and what is the largest?

27. • (Paper.) How many wavelengths of the highest $M = 400$ cosine ($k = 399$, scale 1) fit into the band $x = 22.5$–28.5 of paper Fig. 4?

28. •• (Computational, paper.) Using `qmnt.smooth_zeros` and `qmnt.mean_gap`, compute the rms of RE2 sampled at s_n for $n = 401$–500 and compare with the probe null 0.1708 (paper App. A key anchors).

29. ••• Take $f(t) = e^{i\omega t}$ on a window $[0, T]$ holding $k_0 + \frac{1}{2}$ cycles. Find $|\hat{f}_{k_0}|$ and $|\hat{f}_{k_0+1}|$, and explain why $\sum_k |\hat{f}_k|^2 = 1$.

30. ●●● (Computational.) Using numpy, build the window Fourier series of $\sin(t\log 2)+\sin(t\log 3)$ on $[0,40]$ with harmonics $|k|\le 40$, and compute its rms error on $[0,40]$ and on $[40,80]$. Interpret with Equation (5.68).

31. ●● Show that RE2 has zero average over a period, and that therefore its c_0-type constant term vanishes.

32. ●●● (Paper.) Explain, using Section 5.9, why a model that is periodic in t with period shorter than the full data range can never continue paper Eq. (4) with two or more primes exactly, however many coefficients it has.

5.13 Answers to selected problems

1. $\omega=\log 3=1.0986$; $\tau=2\pi/\log 3=5.719$.

2. $a=5$, $b=12$, $A=\sqrt{169}=13$; $\cos\varphi=5/13$, $\sin\varphi=-12/13$, so $\varphi=-\arctan(12/5)=-1.176$ rad: $13\cos(\omega t-1.176)$.

3. $\Delta\omega=\log 7-\log 5=\log 1.4=0.33647$; $\tau_{\text{beat}}=2\pi/\Delta\omega=18.674$.

4. The $j=k=5\ge 1$ case of Table 5.1: the integral is $L=59.957$.

5. x^3 odd; $x\cos x$ odd (odd times even); $|x|\sin x$ odd; $x\sin x$ even (odd times odd); e^x neither ($e^{-x}\ne\pm e^x$). The three odd ones integrate to zero over $[-L,L]$.

7. f is odd, so $a_n=0$. $b_n=\frac{2}{\pi}\int_0^\pi x\sin nx\,dx$. By parts, $\int_0^\pi x\sin nx\,dx=[-x\cos nx/n]_0^\pi+\int_0^\pi\cos nx/n\,dx=-\pi(-1)^n/n+0$. So $b_n=2(-1)^{n+1}/n$ and $x=2(\sin x-\frac12\sin 2x+\frac13\sin 3x-\cdots)$.

9. Mean square of $|x|$ is $L^2/3$; Parseval gives $L^2/3=c_0^2+\frac12\sum c_k^2=L^2/4+\frac12\sum_{\text{odd}}16L^2/(k^4\pi^4)$. So $1/12=(8/\pi^4)\sum_{\text{odd}}k^{-4}$ and $\sum_{\text{odd}}1/k^4=\pi^4/96=1.01468$.

11. Mean squares are proportional to $1/(m^2 2^m)$: $\frac12$, $\frac{1}{16}$, $\frac{1}{72}$, total 0.576389. The $m=1$ fraction is $0.5/0.576389=0.8675$, about 87%.

13. If $\log 5/\log 3=a/b$ with positive integers, then $5^b=3^a$. The right side is divisible by 3; the left is not, because $5\equiv 2$ and powers of 5 leave remainders $2,1,2,1,\ldots$ on division by 3, never 0. Contradiction.

15. Every ω_j is an integer multiple of $8/450$, so by the harmonic argument of Section 5.1 $F(t+\tau)=F(t)$ with $\tau=2\pi\cdot 450/8=353.43$. The held-out heights $t_{401}\approx 681$ to $t_{500}=811.18$ map to $t-\tau\approx 328$ to 457.8, inside the training range 14.1–680. (Our observation about the model form, not a claim of the paper.)

17. The numerical integrals give $\hat f_{\pm1}=\pm0.1125i$, $\hat f_{\pm2}=\pm0.0398i$, $\hat f_{\pm3}=\pm0.0188i$, and $\hat f_0=\hat f_{\pm4}=0$ to rounding: the window is exactly one period, so there is no leakage.

19. The 18 frequencies $m\log p$ are distinct (different prime powers, Chapter 15), so by Equation (5.65) every cross term averages to zero and Equation (5.66) applies with $A=1/(\pi m p^{m/2})$: $\langle N_{\text{fl}}^2\rangle=\frac{1}{2\pi^2}\sum_p\sum_{m\le 3}\frac{1}{m^2p^m}=\frac{1.472299}{2\pi^2}=0.074588$, rms 0.2731. (This is a long-run average in t of the oracle function itself, in count units.)

21. By Equation (5.52) for sines, $\mathcal{E}=\langle f^2\rangle-\frac12 b_1^2=1-\frac12(16/\pi^2)=1-8/\pi^2=0.1894$.

23. $5\log 3/\log 2=7.92481$, nearest integer 8, $\varepsilon=0.07519$; $2\sin(\pi\times 0.07519)=2\sin(0.23621)=0.468$.

25. For odd f, $b_k=\frac{2}{L}\int_0^L f\sin qx\,dx$ with $q=k\pi/L$; by parts this is $\frac{2}{L}[-f\cos qx/q]_0^L+\frac{2}{Lq}\int_0^L f'\cos qx\,dx$. The boundary term is $-2f(L)(-1)^k/(k\pi)$, which does not shrink faster than $1/k$ unless $f(L)=0$ (no jump). For $f=x$: $f(L)=L$, $f'=1$, the integral term vanishes and $b_k=2L(-1)^{k+1}/(k\pi)$, falling like $1/k$.

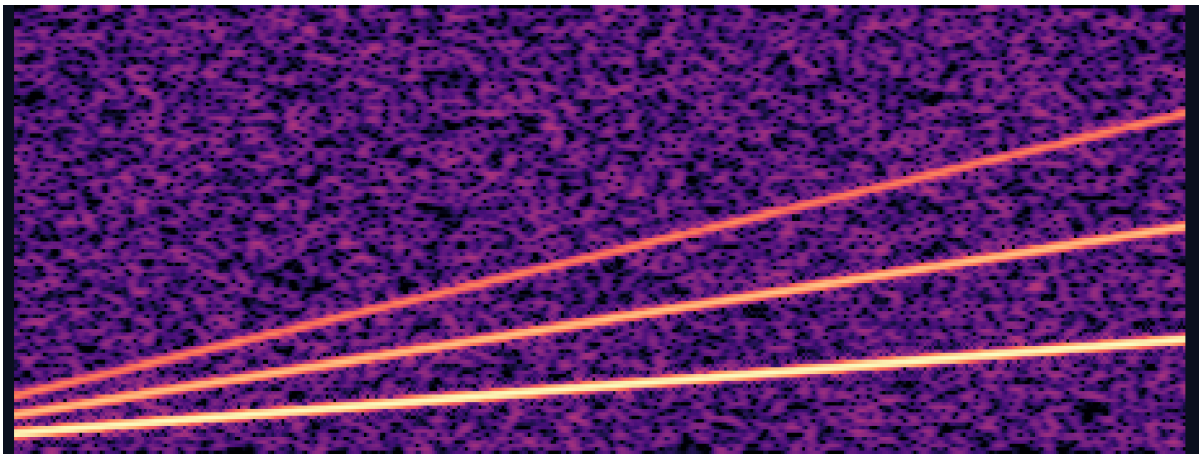
26. 6 primes $\times$ 3 values of $m=18$ distinct frequencies (distinct prime powers have distinct logarithms). The largest is $3\log 13=\log 2197=7.695$.

27. Wavelength $2L/399=0.2250$; $6/0.2250=26.7$, about 27 wavelengths.

29. $\theta=\pi(k_0+\frac12-k)=\pm\pi/2$ for $k=k_0,k_0+1$, so $|\hat f|=|\sin(\pi/2)/(\pi/2)|=2/\pi=0.6366$ for both. Parseval Equation (5.48) on the window: $\langle|e^{i\omega t}|^2\rangle=1=\sum_k|\hat f_k|^2$, so the weight leaked

to all other harmonics is $1 - 2(4/\pi^2) = 0.189$.

31. Each term $\sin(mt \log 2)$ integrates to zero over the period $2\pi/\log 2$, which contains exactly m of its periods (I_s in Section 5.3). So the average, which is the constant coefficient, is zero.



A spectrogram: a signal whose frequency glides upward, with two harmonics, seen through short sliding windows.

Chapter 6

The Fourier Transform, Sampling and the Periodogram

↑Contents

What this chapter is about

In Chapter 5 you learned to break a *periodic* function into sines and cosines. Most signals you will ever analyze are not periodic, and you only ever see them over a finite stretch. A single drum hit, the ring-down of a tuning fork and the fluctuating part of the Riemann zeros are all like this. The zeros are a particularly awkward case. We know a finite list of heights $t_1, t_2, \ldots, t_{10000}$; they are spaced irregularly; and their fluctuation is a sum of oscillations whose frequencies, $\log 2$, $\log 3$, $\log 4$, ..., have no common period, so the pattern never repeats exactly.

This chapter builds the tools for asking "which frequencies are present?" about such data. We let the period of a Fourier series grow without limit to obtain the *Fourier transform*. We then work out what a *finite* observation window does to a spectrum: every sharp frequency is smeared into a "sinc" shape of width about $2\pi/T$, where T is the window length, and energy leaks into neighboring frequencies. We treat sampling, both uniform (where aliasing appears) and irregular (as for the zeros), the discrete Fourier transform and its fast algorithm, the periodogram and how to find its peaks, the autocorrelation of a sequence and its link to the power spectrum (the Wiener–Khinchin theorem), and finally *phase randomization*, which manufactures fake data with exactly the same power spectrum as the real data.

Where this shows up in the paper: the fitted sinusoidal controls C1 and C2 of Sec. II D (a uniform frequency grid; a periodogram peak search on $0.05 < \omega < 8$), the phrase "within half the Fourier resolution" of Sec. IV, the shuffled and phase-randomized surrogate targets of Sec. III, the remark in Sec. IV that "the Fourier transform of the zeros is concentrated at logarithms of prime powers", and steps 18, 20 and 23 of Appendix A.

What you need

- Complex numbers, the complex exponential $e^{i\theta} = \cos\theta + i\sin\theta$, modulus, argument, conjugate and the triangle inequality $|z_1 + z_2| \le |z_1| + |z_2|$ (Chapter 2).
- Partial sums and the finite geometric series $\sum_{j=0}^{N-1} r^j = (1 - r^N)/(1 - r)$ for $r \neq 1$ (Chapter 3).
- Big-$\mathcal{O}$ notation and comparing sums with integrals (Chapter 4).
- Fourier series in real and complex form, orthogonality of $e^{ik\omega t}$ over a period, Parseval's relation, and almost periodic sums such as $\sin(t\log 2) + \sin(t\log 3)$ (Chapter 5).
- First-year physics: angular frequency, period, superposition of waves, beats; single-variable calculus including integration by parts and substitution.

6.1 From Fourier series to the Fourier transform

A plucked guitar string vibrates periodically, and Chapter 5 showed that a periodic motion is a sum of harmonics at multiples of one fundamental frequency. Now think of a single clap of the hands. It happens once. It has no period, yet your ear still hears it as "high" or "low"; it has a frequency content. We want a definition of frequency content that does not need a period.

The idea is simple: treat a non-periodic function as a periodic one whose period is so long that the next repetition never arrives.

6.1.1 Letting the period grow

Recall from Chapter 5 the complex Fourier series. If $f(t)$ has period T, then

$$f(t) = \sum_{k=-\infty}^{\infty} c_k\, e^{i\omega_k t}, \qquad \omega_k = \frac{2\pi k}{T}, \tag{6.1}$$

with coefficients

$$c_k = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\, e^{-i\omega_k t}\,\mathrm{d}t. \tag{6.2}$$

Here t is the independent variable (time for a clap; for the zeros, the height t of a zero $\frac{1}{2} + it$, where the line of complex numbers with real part $\frac{1}{2}$ is called the critical line, Chapter 16), k is an integer labeling the harmonic, and ω_k is the angular frequency of harmonic k, in radians per unit of t.

Now let f be defined on the whole real line, not periodic, and small when $|t|$ is large. Pick a long interval $[-T/2, T/2]$. The series (6.1) with coefficients (6.2) only "sees" the values of f inside this interval and reproduces f there (it reproduces outside it the periodic repetition of that piece, which we will not use). Define, for *every* real ω, not only the harmonics,

$$F_T(\omega) = \int_{-T/2}^{T/2} f(t)\, e^{-i\omega t}\,\mathrm{d}t. \tag{6.3}$$

Comparing with (6.2), $c_k = F_T(\omega_k)/T$. The harmonics are spaced by

$$\Delta\omega = \omega_{k+1} - \omega_k = \frac{2\pi}{T}, \qquad \text{so} \qquad \frac{1}{T} = \frac{\Delta\omega}{2\pi}. \tag{6.4}$$

Substituting into (6.1), for $|t| < T/2$,

$$\begin{aligned} f(t) &= \sum_k \frac{1}{T} F_T(\omega_k)\, e^{i\omega_k t} \\ &= \frac{1}{2\pi}\sum_k F_T(\omega_k)\, e^{i\omega_k t}\,\Delta\omega. \end{aligned} \tag{6.5}$$

The last line is a Riemann sum: values of the function $F_T(\omega)e^{i\omega t}$ at equally spaced points ω_k, each multiplied by the spacing $\Delta\omega$. As $T \to \infty$ the spacing shrinks to zero, the Riemann sum turns into an integral over ω, and F_T turns into the integral over the whole line. This is a heuristic argument (two limits are taken at once and we have not checked that they may be interchanged), but it tells us what to define.

$$\boxed{\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-i\omega t}\, \mathrm{d}t} \tag{6.6}$$

is the **Fourier transform** of f, and the limit of (6.5) suggests the **inverse transform**

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega)\, e^{i\omega t}\, \mathrm{d}\omega. \tag{6.7}$$

The integral over the whole line means $\lim_{A\to\infty, B\to\infty} \int_{-A}^{B}$, an "improper" integral. We call $\hat{f}(\omega)$ the *spectrum* of f; its modulus $\left|\hat{f}(\omega)\right|$ is the *amplitude spectrum* and its argument is the *phase spectrum*. If t were time in seconds, ω would be in rad/s. In this book t is usually a height on the critical line (a pure number), so ω is in radians per unit of t.

Taken on trust

Taken on trust: the Fourier inversion theorem. If $\int_{-\infty}^{\infty} |f(t)|\, \mathrm{d}t$ is finite and f is piecewise smooth, then (6.7) holds at every point where f is continuous (at a jump it gives the midpoint of the jump). Our argument above is a motivation, not a proof. A proof at the level of a mathematical-methods course is in [22].

6.1.2 When does the transform exist?

Because $|e^{-i\omega t}| = 1$ for real ωt, the size of the integrand in (6.6) is $|f(t)|$. The integral of a complex-valued function $g = u + iv$ of the real variable t means $\int u\, \mathrm{d}t + i \int v\, \mathrm{d}t$. It is the limit of the Riemann sums $\sum_j g(t_j)\Delta t$, because the real and imaginary parts converge separately. By the triangle inequality $|\sum z_j| \le \sum |z_j|$ of Chapter 2, each Riemann sum satisfies $\left|\sum_j g(t_j)\Delta t\right| \le \sum_j |g(t_j)|\, \Delta t$. The left side tends to $|\int g\, \mathrm{d}t|$, since the modulus is continuous, and the right side tends to $\int |g|\, \mathrm{d}t$. An inequality between two sequences survives in the limit, so $|\int g\, \mathrm{d}t| \le \int |g|\, \mathrm{d}t$. Hence

$$\left|\hat{f}(\omega)\right| \le \int_{-\infty}^{\infty} |f(t)|\, \mathrm{d}t. \tag{6.8}$$

So if f is *absolutely integrable* (the right side is finite), $\hat{f}(\omega)$ exists and is bounded for every ω. A pure sinusoid $\sin(\omega_0 t)$ is *not* absolutely integrable: the area under $|\sin(\omega_0 t)|$ grows without limit. Its transform does not exist as an ordinary integral. That is not a disaster; it is the first hint that, for oscillations that go on forever, we should look through a finite window (Section 6.2).

Sample Problem 6.1: The transform of a rectangular pulse

Find $\hat{f}(\omega)$ for the "box" $f(t) = 1$ for $|t| \le T/2$ and $f(t) = 0$ otherwise. Evaluate it for $T = 4$ at $\omega = 1$, and find where it first vanishes.

Step 1: Set up. The integrand vanishes outside $[-T/2, T/2]$, so

$$\hat{f}(\omega) = \int_{-T/2}^{T/2} e^{-i\omega t}\, \mathrm{d}t.$$

Step 2: Integrate (for $\omega \neq 0$). An antiderivative of $e^{-i\omega t}$ is $e^{-i\omega t}/(-i\omega)$:

$$\hat{f}(\omega) = \frac{e^{-i\omega T/2} - e^{i\omega T/2}}{-i\omega}.$$

Step 3: Simplify. From $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$ (Chapter 2), the numerator is $-2i\sin(\omega T/2)$. Dividing by $-i\omega$,

$$\hat{f}(\omega) = \frac{2\sin(\omega T/2)}{\omega}.$$

At $\omega = 0$ the integral is directly $\int_{-T/2}^{T/2} 1\,\mathrm{d}t = T$, which is also the limit of the formula since $\sin x \approx x$ for small x.
Step 4: Numbers. With $T = 4$ and $\omega = 1$: $\hat{f}(1) = 2\sin 2 = 2 \times 0.909297 = 1.8186$. The formula vanishes where $\sin(\omega T/2) = 0$ with $\omega \neq 0$, i.e. $\omega T/2 = \pm\pi, \pm 2\pi, \ldots$; the first zero is at $\omega = 2\pi/T = \pi/2 = 1.5708$.

$$\boxed{\hat{f}(\omega) = \frac{2\sin(\omega T/2)}{\omega}, \quad \hat{f}(1) = 1.8186 \text{ for } T = 4, \quad \text{first zero at } \omega = 2\pi/T.}$$

Notice the trade: a wider box (larger T) has its first zero at a smaller ω, so its spectrum is *narrower*. Long in t means narrow in ω.

Sample Problem 6.2: A decaying signal and its linewidth

Find the transform of $f(t) = e^{-a|t|}$ with $a > 0$ (a signal that dies away symmetrically). For $a = 0.5$ find $\hat{f}(0)$ and the full width of $\hat{f}$ at half its maximum.
Step 1: Split the integral at $t = 0$, where $|t|$ changes formula:

$$\hat{f}(\omega) = \int_{-\infty}^{0} e^{at}e^{-i\omega t}\,\mathrm{d}t + \int_{0}^{\infty} e^{-at}e^{-i\omega t}\,\mathrm{d}t.$$

Step 2: First piece. The integrand is $e^{(a-i\omega)t}$, with antiderivative $e^{(a-i\omega)t}/(a - i\omega)$. At $t = 0$ it equals $1/(a - i\omega)$; as $t \to -\infty$ its modulus $e^{at} \to 0$. So the first piece is $1/(a - i\omega)$.
Step 3: Second piece. The integrand is $e^{-(a+i\omega)t}$, antiderivative $-e^{-(a+i\omega)t}/(a + i\omega)$, which tends to 0 as $t \to \infty$ and equals $-1/(a + i\omega)$ at $t = 0$. So the second piece is $0 - (-1/(a + i\omega)) = 1/(a + i\omega)$.
Step 4: Add over a common denominator $(a - i\omega)(a + i\omega) = a^2 + \omega^2$:

$$\hat{f}(\omega) = \frac{(a + i\omega) + (a - i\omega)}{a^2 + \omega^2} = \frac{2a}{a^2 + \omega^2}.$$

Step 5: Numbers. The maximum is at $\omega = 0$: $\hat{f}(0) = 2/a = 4$. Half of it, $1/a$, is reached when $a^2 + \omega^2 = 2a^2$, i.e. $\omega = \pm a$. The full width at half maximum is $2a = 1$.

$$\boxed{\hat{f}(\omega) = \frac{2a}{a^2 + \omega^2}; \quad a = 0.5: \ \hat{f}(0) = 4, \ \text{FWHM} = 2a = 1.}$$

This bell shape is called a Lorentzian. Define the *lifetime* of the signal as $\tau = 1/a$: the time in which $|f|$ falls by a factor e. The spectral width is FWHM $= 2a$. So FWHM $\times\, \tau = 2a \cdot (1/a) = 2$, a pure number that does not depend on a. A short-lived signal has a broad spectrum. You meet the same rule in physics as the "linewidth" of a damped oscillator: the faster it rings

down, the broader its resonance.

6.1.3 Five properties we will use

Each property follows in one or two lines from the definition (6.6).

(P1) Linearity. $\widehat{(\alpha f + \beta g)} = \alpha \hat{f} + \beta \hat{g}$ for constants α, β, because the integral of a sum is the sum of the integrals.

(P2) Real functions have symmetric spectra. If f is real, take the complex conjugate of (6.6). Conjugation passes through the integral and leaves f unchanged, and $\overline{e^{-i\omega t}} = e^{i\omega t}$:

$$\overline{\hat{f}(\omega)} = \int f(t) e^{i\omega t}\, \mathrm{d}t = \hat{f}(-\omega). \tag{6.9}$$

So $\left|\hat{f}(-\omega)\right| = \left|\hat{f}(\omega)\right|$: for real data the amplitude spectrum at negative frequency repeats the one at positive frequency, and we only need $\omega \geq 0$.

(P3) Shift. Let $g(t) = f(t - t_0)$. Substitute $u = t - t_0$, $\mathrm{d}t = \mathrm{d}u$:

$$\hat{g}(\omega) = \int f(u) e^{-i\omega(u+t_0)}\, \mathrm{d}u = e^{-i\omega t_0}\, \hat{f}(\omega). \tag{6.10}$$

Since $\left|e^{-i\omega t_0}\right| = 1$, shifting a signal changes only the phase of its spectrum, never the amplitude.

(P4) Modulation. Let $g(t) = e^{i\omega_0 t} f(t)$. Then

$$\hat{g}(\omega) = \int f(t) e^{-i(\omega - \omega_0)t}\, \mathrm{d}t = \hat{f}(\omega - \omega_0). \tag{6.11}$$

Multiplying by a complex oscillation slides the whole spectrum along the ω axis by ω_0.

(P5) Scaling. Let $g(t) = f(t/b)$ with $b > 0$ (a stretched copy). Substitute $u = t/b$, $\mathrm{d}t = b\,\mathrm{d}u$:

$$\hat{g}(\omega) = \int f(u) e^{-i\omega b u}\, b\, \mathrm{d}u = b\, \hat{f}(b\omega). \tag{6.12}$$

Stretching by b in t squeezes the spectrum by b in ω. This is the general form of the trade-off we saw in both Sample Problems.

Checkpoint 6.1

A real signal has $\left|\hat{f}(-2)\right| = 3$. What is $\left|\hat{f}(2)\right|$? If the signal is delayed by $t_0 = 5$, what is the new value of $\left|\hat{f}(2)\right|$?

Answer: Both are 3. By (P2) the amplitude spectrum of a real function is even; by (P3) a delay multiplies $\hat{f}$ by $e^{-i\omega t_0}$, which has modulus 1.

Key Ideas

- The Fourier transform $\hat{f}(\omega) = \int f(t) e^{-i\omega t}\, \mathrm{d}t$, Eq. (6.6), is the limit of Fourier series as the period grows; the inverse is Eq. (6.7) (proof beyond this book).
- It exists when f is absolutely integrable, Eq. (6.8); an everlasting sinusoid is not, which is why we will look through finite windows.
- A box of width T has transform $2\sin(\omega T/2)/\omega$, first zero at $2\pi/T$; $e^{-a|t|}$ has the Lorentzian $2a/(a^2 + \omega^2)$. Long in t means narrow in ω.
- Real f: $\hat{f}(-\omega) = \overline{\hat{f}(\omega)}$. Shifts change only phases; multiplying by $e^{i\omega_0 t}$ shifts the spectrum by ω_0.

6.2 Finite windows: the sinc, resolution and leakage

You never observe a signal forever. You switch on your recorder, record for a time T, and switch it off. For the zeros, you have computed them up to some height and no further. Mathematically, the data you actually have is the true signal multiplied by a *window* that equals 1 while you observe and 0 otherwise. This section works out exactly what that multiplication does to a spectrum.

6.2.1 The transform of a finite piece of a sinusoid

Take the complex oscillation $e^{i\omega_0 t}$ and observe it for $|t| \le T/2$:

$$g(t) = \begin{cases} e^{i\omega_0 t}, & |t| \le T/2, \\ 0, & \text{otherwise.} \end{cases}$$

This is the box of the first Sample Problem multiplied by $e^{i\omega_0 t}$. By the modulation property (6.11), its transform is the box transform shifted to ω_0:

$$\hat{g}(\omega) = \frac{2\sin\big((\omega-\omega_0)T/2\big)}{\omega-\omega_0} = T\,\mathrm{sinc}\left(\frac{(\omega-\omega_0)T}{2}\right), \qquad \mathrm{sinc}(x) = \frac{\sin x}{x},\ \mathrm{sinc}(0)=1. \tag{6.13}$$

(The second form follows by multiplying and dividing by $T/2$.) If the window were $[0, T]$ instead of $[-T/2, T/2]$, the shift property (6.10) says only the phase would change; $|\hat{g}|$ would be identical.

Read off the features of (6.13), shown in Figure 6.1:

- **Peak.** At $\omega = \omega_0$ the value is T: it grows in proportion to how long you watch.
- **Zeros.** $\hat{g} = 0$ when $(\omega - \omega_0)T/2 = \pm\pi, \pm 2\pi, \ldots$, i.e. at $\omega = \omega_0 \pm 2\pi k/T$, $k = 1, 2, \ldots$
- **Main lobe.** The central hump runs from $\omega_0 - 2\pi/T$ to $\omega_0 + 2\pi/T$, full width $4\pi/T$.
- **Sidelobes.** Between the zeros the sinc oscillates with slowly decreasing size, since $|\mathrm{sinc}(x)| \le 1/|x|$.

A perfectly sharp frequency (the ideal spectrum of $e^{i\omega_0 t}$ is a single spike at ω_0) has been smeared into a peak of width of order $2\pi/T$, with ripples on both sides. The smearing is not an error of computation; it is the honest statement of what a finite window can tell you.

6.2.2 The largest sidelobe

Where are the sidelobe maxima? Differentiate with the quotient rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{\sin x}{x} = \frac{x\cos x - \sin x}{x^2}. \tag{6.14}$$

This vanishes when $x\cos x = \sin x$, i.e. $\tan x = x$. Besides $x = 0$, the first positive solution lies between π and $3\pi/2$; solving numerically (bisection or Brent, Chapter 4) gives $x = 4.4934$, where $|\mathrm{sinc}\, x| = 0.2172$. So the tallest ripple of a finite-window sinusoid is about 22% of the main peak in amplitude. Engineers quote this as $20\log_{10} 0.2172 = -13.3$ decibels.

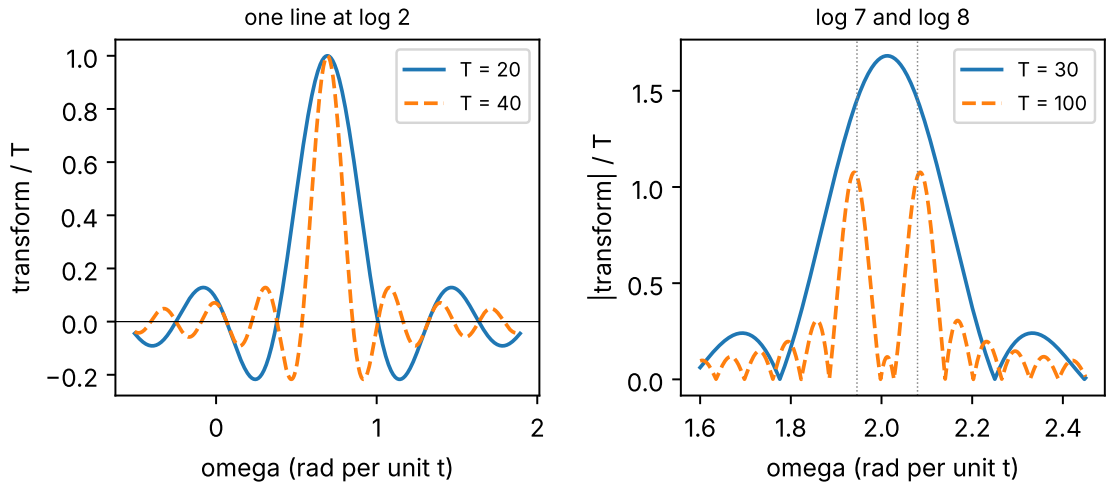


Figure 6.1: Left: the transform (6.13) of $e^{i\omega_0 t}$ observed for a window of length T, divided by T, with $\omega_0 = \log 2$. Doubling T from 20 to 40 halves the width of the main lobe; the zeros sit at $\omega_0 \pm 2\pi k/T$. Right: two equal oscillations at $\log 7 = 1.946$ and $\log 8 = 2.079$, separated by 0.134. With $T = 30$ ($2\pi/T = 0.209$) they merge into one peak; with $T = 100$ ($2\pi/T = 0.063$) they are resolved.

6.2.3 Real sinusoids

A real oscillation $A\sin(\omega_0 t)$ is a sum of two complex ones (Chapter 2):

$$A\sin(\omega_0 t) = \frac{A}{2i}e^{i\omega_0 t} - \frac{A}{2i}e^{-i\omega_0 t}.$$

By linearity (P1) and (6.13), its windowed transform is

$$\frac{AT}{2i}\left[\operatorname{sinc}\left(\frac{(\omega-\omega_0)T}{2}\right) - \operatorname{sinc}\left(\frac{(\omega+\omega_0)T}{2}\right)\right]. \tag{6.15}$$

There are two peaks, at $+\omega_0$ and $-\omega_0$. At $\omega = \omega_0$ the first sinc equals 1 and the second is evaluated at $x = \omega_0 T$, where $|\operatorname{sinc}| \le 1/(\omega_0 T)$. *If* $\omega_0 T \gg 1$ (many cycles fit in the window), that second term is negligible and

$$|\hat{g}(\omega_0)| \approx \frac{AT}{2}. \tag{6.16}$$

For the prime-2 wave of RE2 with T in the hundreds, $\omega_0 T = T\log 2$ is several hundred, so this condition is comfortably met.

6.2.4 Resolution: when can two frequencies be told apart?

Suppose the signal contains two oscillations of equal strength at ω_1 and ω_2. Each produces its own sinc peak. If they are far apart compared with the main-lobe half-width, you see two peaks. If they are much closer, the two humps merge into one. A conventional dividing line, the same one used for the resolving power of a telescope (the Rayleigh criterion in your physics course), is that the peak of one falls on the first zero of the other:

$$\boxed{|\omega_2 - \omega_1| \gtrsim \frac{2\pi}{T} \quad \text{(resolution of a window of length } T\text{)}.} \tag{6.17}$$

The quantity $2\pi/T$ is called the **Fourier resolution** of the window. The "$\gtrsim$" reminds you that the exact factor depends on the criterion chosen; what does not depend on any convention is the scaling $1/T$.

There is a second, more physical way to see (6.17). Over the window, the phase of $e^{i\omega_2 t}$ relative to $e^{i\omega_1 t}$ changes by $(\omega_2-\omega_1)T$. In Chapter 5 you met this as beats: the two oscillations drift in and out of step with beat period $2\pi/|\omega_2-\omega_1|$. Unless the window contains about one full beat, you never see them drift apart. We can make this quantitative. For real θ,

$$\left|e^{i\theta}-1\right| = \left|e^{i\theta/2}\right|\,\left|e^{i\theta/2}-e^{-i\theta/2}\right| = 2\,|\sin(\theta/2)| \le |\theta|\,,$$

using $|\sin y| \le |y|$. With $\theta=(\omega_2-\omega_1)t$,

$$\left|e^{i\omega_2 t}-e^{i\omega_1 t}\right| = \left|e^{i\omega_1 t}\right|\,\left|e^{i(\omega_2-\omega_1)t}-1\right| \le |\omega_2-\omega_1|\;|t| \le |\omega_2-\omega_1|\,\frac{T}{2} \tag{6.18}$$

for every t in the window $|t|\le T/2$. If $|\omega_2-\omega_1|\,T \ll 1$, the two functions differ by a tiny amount everywhere in the window: no analysis of the windowed data, Fourier or otherwise, can tell them apart. The same inequality has a sobering consequence for prediction that Chapter 27 develops: a frequency known only to within $\delta\omega$ gives a phase error $\delta\omega\,\Delta t$ at a distance Δt outside the window, which grows the further you extrapolate.

Sample Problem 6.3: Resolving prime-power frequencies

The fluctuation of the zeros contains oscillations at angular frequencies $\log q$ for prime powers q (Chapter 18). (a) What window length T is needed to resolve $\log 8$ from $\log 9$? (b) The paper's operator fits use zeros 1–400, which span heights from $t_1 = 14.13$ to $t_{400} \approx 680$. Can that window resolve $\log 8$ from $\log 9$? Can it resolve $\log 127$ from $\log 128$?

Step 1: Separations. $\log 9 - \log 8 = \log(9/8) = 0.1178$, and $\log 128 - \log 127 = \log(128/127) = 0.00784$.

Step 2: Required T from (6.17): $T \gtrsim 2\pi/0.1178 = 53.3$ for the first pair and $T \gtrsim 2\pi/0.00784 = 801$ for the second.

Step 3: The paper's window. We take the window length to be the distance between the first and last sample, $T = t_{400} - t_1 \approx 666$. (The paper does not spell out whether its T is $t_{400}-t_1$ or t_{400}; the two differ by 2%, which changes nothing below.) Then $2\pi/T \approx 0.0094$.

Step 4: Compare. $0.1178 \gg 0.0094$, so $\log 8$ and $\log 9$ are resolved with room to spare. $0.00784 < 0.0094$, so $\log 127$ and $\log 128$ are not: within this window they look like a single frequency.

$T \gtrsim 53.3$ for 8/9; $\quad T \gtrsim 801$ for 127/128; $\quad$ with $T \approx 666$: 8/9 yes, 127/128 no.

6.2.5 Lines and almost periodic functions

For a function that oscillates forever, (6.13) suggests dividing by T: the peak value T becomes 1 and the width $\to 0$ as $T\to\infty$. This leads to the *mean Fourier coefficient*

$$a(\omega) = \lim_{T\to\infty}\frac{1}{T}\int_0^T f(t)\,e^{-i\omega t}\,\mathrm{d}t, \tag{6.19}$$

which is the right spectral notion for the almost periodic sums of Chapter 5 (the theory of such functions is due to Harald Bohr).

Let $f(t)=\sum_{j=1}^{J} b_j e^{i\lambda_j t}$ be a finite sum with distinct real frequencies λ_j and complex amplitudes b_j. By linearity, (6.19) needs only $\frac{1}{T}\int_0^T e^{i(\lambda_j-\omega)t}\,\mathrm{d}t$. If $\lambda_j=\omega$ the integrand is 1 and the result is 1. If $\lambda_j\neq\omega$,

$$\frac{1}{T}\int_0^T e^{i(\lambda_j-\omega)t}\,\mathrm{d}t = \frac{e^{i(\lambda_j-\omega)T}-1}{i(\lambda_j-\omega)T}, \qquad |\cdot| \le \frac{2}{|\lambda_j-\omega|\,T} \to 0,$$

because $\left|e^{i\theta}-1\right| \le 2$. Therefore

$$a(\omega) = \begin{cases} b_j, & \omega = \lambda_j, \\ 0, & \omega \text{ not among the } \lambda_j. \end{cases} \tag{6.20}$$

The spectrum of an almost periodic sum consists of isolated *lines*, whether or not the frequencies share a period. For a real term $A\sin(\lambda t) = \frac{A}{2i}e^{i\lambda t} - \frac{A}{2i}e^{-i\lambda t}$ the lines sit at $\pm\lambda$ with $|a| = |A|/2$.

Applied to RE2, the prime-2 wave $-\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 2)}{m2^{m/2}}$: the amplitudes $|A_m| = 1/(\pi m 2^{m/2})$ are 0.2251, 0.0796 and 0.0375 for $m = 1, 2, 3$, so the lines at $m\log 2$ have $|a| = 0.1125$, 0.0398 and 0.0188. The longest period is $2\pi/\log 2 \approx 9.065$ in t.

Applied to the paper's Eq. (4): $N_{\text{fl}}(t) = -\frac{1}{\pi}\sum_{p}\sum_{m=1}^{3}\frac{\sin(mt\log p)}{mp^{m/2}}$, with p running over a set of primes (an integer greater than 1 divisible only by 1 and itself) and log the natural logarithm. Its spectrum is a set of lines at $\omega = \pm m\log p = \pm\log(p^m)$ with $|a| = 1/(2\pi m p^{m/2})$. These frequencies are all different: $m\log p = k\log q$ would mean $p^m = q^k$, which by the unique factorization of integers (Chapter 15) forces $p = q$ and $m = k$. Through a window of length T, each line becomes a sinc of height $\approx T\,|a|$ and width $\sim 2\pi/T$.

6.2.6 Leakage and tapered windows

Look again at Figure 6.1: the ripples of a strong line spill into frequencies where nothing is actually present. This is **spectral leakage**. A weak line sitting under a sidelobe of a strong one can be hidden, or a sidelobe can be mistaken for a weak line. The culprit is the abrupt switch-on and switch-off of the box: sharp edges in t need many frequencies to build.

A *tapered* window rises and falls smoothly. The most common is the **Hann window**, on $|t| \le T/2$,

$$w(t) = \tfrac{1}{2}\left(1 + \cos(2\pi t/T)\right) = \tfrac{1}{2} + \tfrac{1}{4}e^{i2\pi t/T} + \tfrac{1}{4}e^{-i2\pi t/T}, \tag{6.21}$$

which is 1 at the center and falls to 0 at the edges. Its transform follows from linearity and modulation applied to the box transform $B(\omega) = 2\sin(\omega T/2)/\omega$:

$$\hat{w}(\omega) = \tfrac{1}{2}B(\omega) + \tfrac{1}{4}B\Big(\omega - \tfrac{2\pi}{T}\Big) + \tfrac{1}{4}B\Big(\omega + \tfrac{2\pi}{T}\Big). \tag{6.22}$$

Measure frequency in *bins*, $\delta = \omega T/(2\pi)$, so one bin is one Fourier resolution. Then $B = T\,s(\delta)$ with $s(\delta) = \sin(\pi\delta)/(\pi\delta)$, and

$$\frac{\hat{w}}{T} = \tfrac{1}{2}s(\delta) + \tfrac{1}{4}s(\delta - 1) + \tfrac{1}{4}s(\delta + 1). \tag{6.23}$$

We can simplify. Since $\sin(\pi\delta \pm \pi) = -\sin(\pi\delta)$, $s(\delta \pm 1) = -\sin(\pi\delta)/(\pi(\delta \pm 1))$. Factor out $\sin(\pi\delta)/\pi$:

$$\begin{aligned} \frac{\hat{w}}{T} &= \frac{\sin\pi\delta}{\pi}\left[\frac{1}{2\delta} - \frac{1}{4(\delta-1)} - \frac{1}{4(\delta+1)}\right] \\ &= \frac{\sin\pi\delta}{\pi}\left[\frac{1}{2\delta} - \frac{2\delta}{4(\delta^2-1)}\right] \quad \text{(common denominator of last two)} \\ &= \frac{\sin\pi\delta}{\pi}\cdot\frac{(\delta^2-1)-\delta^2}{2\delta(\delta^2-1)} \\ &= -\frac{\sin\pi\delta}{2\pi\delta(\delta^2-1)}. \end{aligned} \tag{6.24}$$

Three consequences, plotted in Figure 6.2:

- **Peak.** As $\delta \to 0$, $\sin(\pi\delta)/(\pi\delta) \to 1$ and the formula gives $-1/(2\cdot(-1)) = \frac{1}{2}$: the peak is $T/2$ (the Hann window has half the area of the box).

- **Wider main lobe.** At $\delta = \pm 1$ the numerator and the factor $\delta^2 - 1$ both vanish, and the limit is $\frac{1}{4}$, not 0 (Problem 10). The first zeros are at $\delta = \pm 2$: the main lobe is twice as wide as for the box.

- **Much lower sidelobes.** For large $|\delta|$ the size falls like $1/|\delta|^3$ instead of $1/|\delta|$. The highest Hann sidelobe, divided by the Hann peak value $\frac{1}{2}$, is 0.0267 (-31.5 dB). The corresponding box figure is 0.217, divided by the box peak value 1.

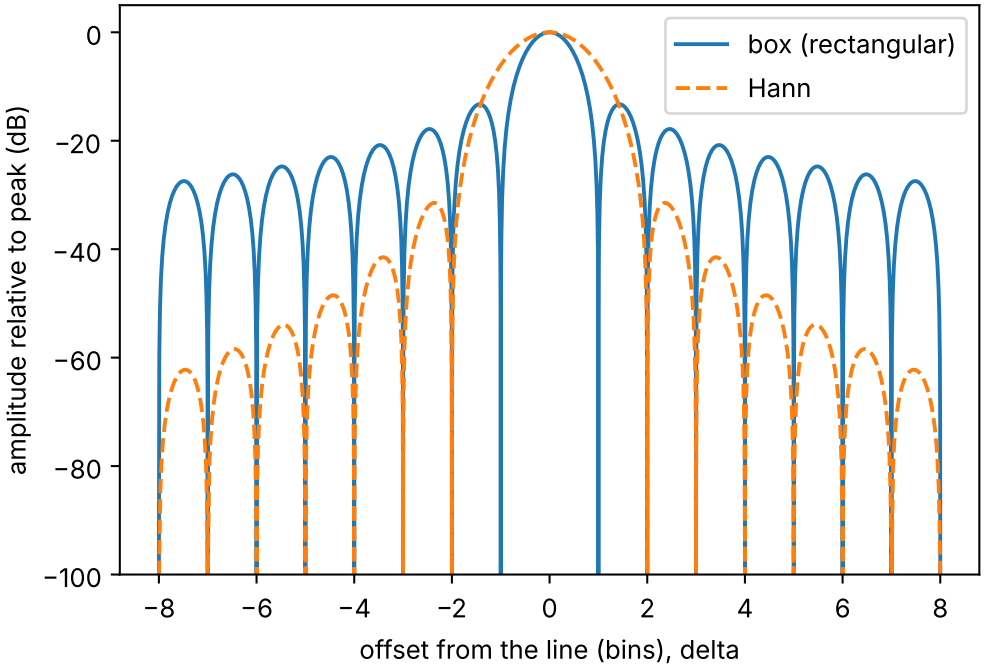


Figure 6.2: Leakage: the amplitude spectrum of a single line seen through a rectangular window (box) and through a Hann window, each normalized to its own peak and shown in decibels ($20 \log_{10}$ of the amplitude ratio). The horizontal axis is the offset from the line in bins, $\delta = (\omega - \omega_0)T/2\pi$. The Hann main lobe is twice as wide, but its sidelobes are far lower and decay much faster.

Sample Problem 6.4: How much of a peak do you see if you miss it by half a bin?

You compute a windowed spectrum only at frequencies spaced one bin ($2\pi/T$) apart, and a line happens to lie exactly halfway between two of them. What fraction of the true peak amplitude do you see with the box and with the Hann window?

Step 1: Box. The offset is $\delta = \frac{1}{2}$, so the ratio to the peak is $s(\frac{1}{2}) = \sin(\pi/2)/(\pi/2) = 2/\pi = 0.6366$.

Step 2: Hann. From (6.24) at $\delta = \frac{1}{2}$: $-\sin(\pi/2)/(2\pi \cdot \frac{1}{2} \cdot (\frac{1}{4} - 1)) = 1/(0.75\pi) = 0.4244$. Divide by the Hann peak $\frac{1}{2}$: $8/(3\pi) = 0.8488$.

$$\boxed{\text{box: } 2/\pi = 0.637; \qquad \text{Hann: } 8/(3\pi) = 0.849.}$$

The loss is called "scalloping". The cure, whichever window you use, is to evaluate the spectrum on a grid finer than one bin; Section 6.5 returns to this.

Pitfall

A tapered window does *not* improve resolution. Only a longer T does. Tapering trades a wider main lobe (worse at separating two close lines of equal strength) for lower sidelobes (better at seeing a weak line far from a strong one). Choose the window for the question you are asking.

Checkpoint 6.2

You double the length T of the window through which you observe $e^{i\omega_0 t}$. What happens to the height of the peak of $|\hat{g}|$, to the width of its main lobe, and to the height of the peak of $|\hat{g}|/T$?

Answer: The peak height T doubles, the main-lobe width $4\pi/T$ halves, and $|\hat{g}|/T$ still peaks at 1.

Key Ideas

- A finite window of length T turns each line at ω_0 into $T\,\mathrm{sinc}((\omega-\omega_0)T/2)$, Eq. (6.13): main lobe from $\omega_0 - 2\pi/T$ to $\omega_0 + 2\pi/T$, largest sidelobe 0.217 of the peak.
- Two frequencies are resolved when $|\Delta\omega| \gtrsim 2\pi/T$, the Fourier resolution, Eq. (6.17); when $|\Delta\omega|\,T \ll 1$ they are indistinguishable in the window, Eq. (6.18).
- An almost periodic sum has a line spectrum, Eq. (6.20); the explicit-formula fluctuation has lines at $\log p^m$ of size $1/(2\pi m p^{m/2})$.
- Leakage comes from the window's sidelobes; the Hann window, Eq. (6.24), lowers them at the price of a doubled main lobe.

6.3 Sampling: uniform, irregular and aliasing

★ Did you know? *Why CDs Use 44.1 kHz*

Human hearing tops out near 20 kHz. To capture a signal without aliasing you must sample faster than twice its highest frequency, so audio CDs sample 44,100 times per second, just above 2×20 kHz. Aliasing shows up in films too. A spinning wagon wheel filmed at 24 frames per second can appear to turn slowly backward.

A computer never holds a function $f(t)$ for all t; it holds a list of values $y_j = f(t_j)$ at sample points t_j. For an oscilloscope the t_j are equally spaced. For the Riemann zeros the "samples" are the zeros themselves, and they are not equally spaced. Both cases matter to us.

6.3.1 Uniform samples and aliasing

Let $t_j = j\,\Delta t$ for $j = 0, 1, \ldots, N-1$, so the samples span a window of length about $T = N\Delta t$. The natural approximation to the windowed transform is the Riemann sum

$$\int_0^T f(t) e^{-i\omega t}\,\mathrm{d}t \approx \Delta t \sum_{j=0}^{N-1} f(t_j)\, e^{-i\omega t_j}. \tag{6.25}$$

Something new happens on the right side. Replace ω by $\omega + 2\pi/\Delta t$:

$$e^{-i(\omega+2\pi/\Delta t)j\Delta t} = e^{-i\omega j\Delta t}\, e^{-2\pi i j} = e^{-i\omega j\Delta t},$$

since $e^{-2\pi ij} = 1$ for every integer j. So the sampled sum is periodic in ω with period $2\pi/\Delta t$. Equivalently, the two oscillations $e^{i\omega t}$ and $e^{i(\omega+2\pi/\Delta t)t}$ have *identical samples*: from the samples alone they cannot be distinguished. This is **aliasing**. For real signals, (P2) adds a mirror symmetry, so ω is also confused with $2\pi/\Delta t - \omega$. All distinct information lives in

$$0 \le \omega \le \omega_{\mathrm{N}} = \frac{\pi}{\Delta t}, \tag{6.26}$$

where ω_{N} is the **Nyquist frequency**: half the sampling rate, or "two samples per cycle". A component above ω_{N} does not disappear; it masquerades as one below. You have seen this in films where a fast-spinning wagon wheel appears to turn slowly backwards: the camera samples at a fixed rate.

Taken on trust

Taken on trust: the sampling theorem. If $\hat{f}(\omega) = 0$ for all $|\omega| > \omega_{\mathrm{N}}$ (a "band-limited" signal), then f is determined at every t by its uniform samples at spacing $\Delta t = \pi/\omega_{\mathrm{N}}$. This is the Shannon (or Nyquist–Shannon) sampling theorem; a practical discussion is in [27]. We will not need it, only the aliasing identity above, which we have proved.

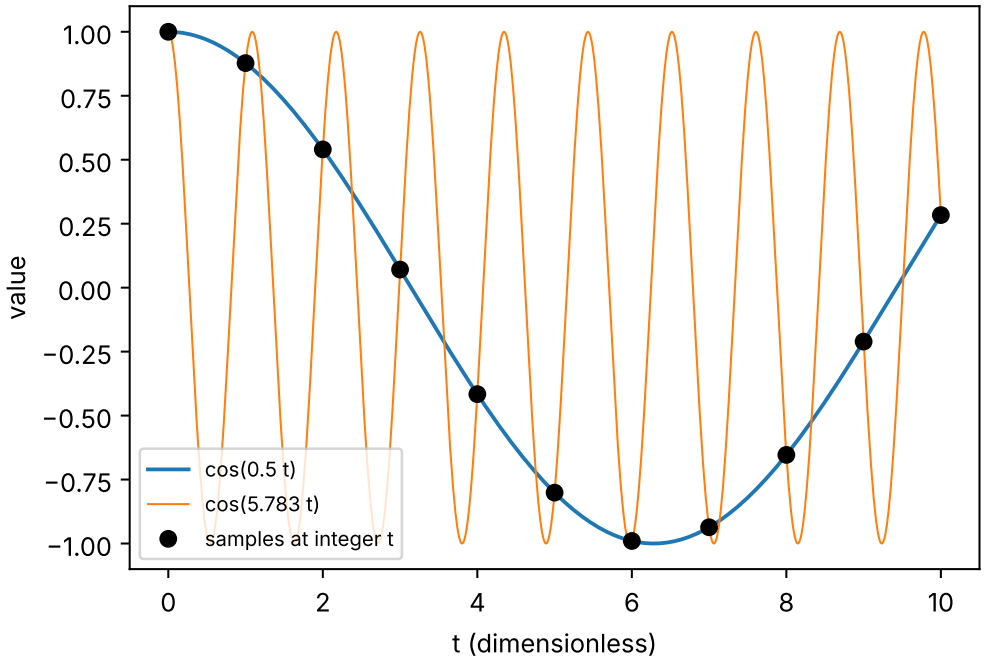


Figure 6.3: Aliasing. The slow oscillation $\cos(0.5t)$ and the fast one $\cos((2\pi - 0.5)t) = \cos(5.783t)$ pass through exactly the same values at the integer sample times $t = 0, 1, 2, \ldots$ (dots). With $\Delta t = 1$ the Nyquist frequency is π, and 5.783 lies above it.

Sample Problem 6.5: An alias, and the "Nyquist frequency" of the zeros

(a) With $\Delta t = 1$, show that $\cos(0.5t)$ and $\cos(5.7832\,t)$ have the same samples. (b) Near zero number $n = 400$, the zeros are spaced on average by the mean gap $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$, where s_n is the smooth zero of Chapter 17 (the solution of $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$, paper Eq. (1)). If the zeros were equally spaced at this gap, what would their Nyquist frequency be?

Step 1: The alias. $5.7832 = 2\pi - 0.5$ to the digits shown. At $t = j$, $\cos((2\pi - 0.5)j) = \cos(2\pi j)\cos(0.5j) + \sin(2\pi j)\sin(0.5j) = \cos(0.5j)$, using $\cos(2\pi j) = 1$ and $\sin(2\pi j) = 0$.
Step 2: Smooth zero and gap. Solving $N_{\mathrm{sm}}(s) = 399.5$ numerically gives $s_{400} = 679.9$; then $\log(s_{400}/2\pi) = 4.684$ and $\mathrm{gap}_{400} = 2\pi/4.684 = 1.341$.
Step 3: Equivalent Nyquist frequency. $\omega_{\mathrm{N}} = \pi/\mathrm{gap} = \frac{1}{2}\log(s_{400}/2\pi) = 2.342$.

$$\boxed{\cos(5.7832\,j) = \cos(0.5\,j); \qquad \pi/\mathrm{gap}_{400} = \tfrac{1}{2}\log(s_{400}/2\pi) \approx 2.34.}$$

The neat form $\pi/\mathrm{gap}_n = \frac{1}{2}\log(s_n/2\pi)$ shows that the zeros sample more densely as you go higher, so this "equivalent Nyquist frequency" slowly rises with height. But the zeros are *not* equally spaced, and that changes the story, as we now see.

6.3.2 Irregular samples and the spectral window

When the t_j are irregular, no FFT shortcut applies (Section 6.4), but nothing stops us from computing the sum directly:

$$F(\omega) = \sum_{j=1}^{N} y_j\, e^{-i\omega t_j}. \tag{6.27}$$

This is sometimes called a nonuniform discrete Fourier transform. Its cost is N complex exponentials per frequency; for $N = 1000$ zeros and 10^4 trial frequencies that is 10^7 operations, a fraction of a second on a laptop. For real data y_j, $F(-\omega) = \overline{F(\omega)}$ by the same argument as (P2).

What does (6.27) do to a single oscillation? Put $y_j = b\,e^{i\omega_0 t_j}$:

$$F(\omega) = b\sum_j e^{-i(\omega-\omega_0)t_j} = b\,W(\omega-\omega_0), \qquad W(\omega) = \sum_{j=1}^{N} e^{-i\omega t_j}. \tag{6.28}$$

The function W is the **spectral window** of the sampling pattern: it plays for sampled data the role the sinc played for the continuous window. It depends only on *where* you sampled, not on the data. Always $W(0) = N$.

Uniform case. With $t_j = j\Delta t$, $j = 0, \ldots, N-1$, W is a geometric series with ratio $r = e^{-i\omega\Delta t}$ (Chapter 3). For $r \neq 1$,

$$W(\omega) = \frac{1 - r^N}{1 - r} = \frac{1 - e^{-iN\omega\Delta t}}{1 - e^{-i\omega\Delta t}}.$$

Write $1 - e^{-i\theta} = e^{-i\theta/2}(e^{i\theta/2} - e^{-i\theta/2}) = 2i\,e^{-i\theta/2}\sin(\theta/2)$ for numerator ($\theta = N\omega\Delta t$) and denominator ($\theta = \omega\Delta t$). The factors $2i$ cancel, the exponentials have modulus 1, and

$$|W(\omega)| = \left|\frac{\sin(N\omega\Delta t/2)}{\sin(\omega\Delta t/2)}\right|. \tag{6.29}$$

This ratio of sines is called the **Dirichlet kernel**. For small $\omega\Delta t$, $\sin(\omega\Delta t/2) \approx \omega\Delta t/2$ and (6.29) becomes $|2\sin(\omega T/2)/\omega|\,/\Delta t$ with $T = N\Delta t$: the box transform, as the Riemann sum (6.25) promised. But the denominator also vanishes at $\omega = 2\pi k/\Delta t$, where $|W|$ climbs back to N: those repeated full-height peaks *are* aliasing.

Irregular case. For $|W(\Omega)|$ to equal N again at some $\Omega \neq 0$, all the unit arrows $e^{-i\Omega t_j}$ would have to point the same way (only then is the triangle inequality an equality). So all the phases Ωt_j would have to agree up to multiples of 2π; that is, every difference $\Omega(t_j - t_l)$ would have to be a multiple of 2π. Generic irregular points do not allow this. So exact aliasing disappears. What remains is a "floor" of partial cancellations. Its typical size follows from a

heuristic count. Expanding the modulus squared, $|W(\Omega)|^2 = \sum_j \sum_l e^{-i\Omega(t_j - t_l)}$. The N terms with $j = l$ each equal 1. For points scattered at random, the remaining $N(N-1)$ terms are unit arrows pointing in all directions, and they largely cancel. If their total were negligible, we would have $|W|^2 \approx N$, that is, $|W| \approx \sqrt{N}$. This is a heuristic, not a proof; Chapter 19 supplies the language of averages needed to make it precise. It shows that the floor lies well below the central peak N for large N. In principle, then, irregular samples can reveal frequencies above the "equivalent Nyquist" value $\pi/\overline{\Delta t}$. Section 6.9 describes a result of the paper that is compatible with this (frequencies recovered above the equivalent Nyquist value), though the mechanism is not established.

Two warnings specific to the zeros. First, irregular sampling also *weights* the data: regions with more samples count more in (6.27). The zeros become denser with height; their average number per unit t is $\frac{1}{2\pi}\log\frac{t}{2\pi}$ (we derive this in Section 6.8), about 0.13 at t_1 and about 0.75 at t_{400}. Sampling at the zeros therefore acts like a window that rises across the data range. Second, the zeros are not random points at all: their spectral window W has sharp spikes at the logarithms of prime powers (Section 6.8). A sampling pattern with its own structure can imprint that structure on any spectrum computed from it.

Checkpoint 6.3

You sample a signal every $\Delta t = 0.1$. What is the Nyquist frequency? A component at $\omega = 40$ is present; at what frequency in $[0, \omega_{\mathrm{N}}]$ does it appear?

Answer: $\omega_{\mathrm{N}} = \pi/0.1 = 31.4$. The alias period is $2\pi/\Delta t = 62.83$, so 40 is confused with $62.83 - 40 = 22.83$, which lies below the Nyquist frequency.

Key Ideas

- Uniform samples at spacing Δt cannot distinguish ω from $\omega + 2\pi/\Delta t$ (aliasing); real signals are fully described on $0 \le \omega \le \pi/\Delta t$, Eq. (6.26).
- For irregular samples compute the direct sum $F(\omega) = \sum y_j e^{-i\omega t_j}$, Eq. (6.27); a sampled oscillation produces the spectral window $W(\omega - \omega_0)$, Eq. (6.28).
- The uniform spectral window is the Dirichlet kernel (6.29), with full-height aliases; irregular sampling removes exact aliases but weights the data by sample density, and for the zeros the window itself has structure.

6.4 The discrete Fourier transform and the FFT

For uniformly sampled data there is a special set of frequencies at which the sum (6.27) has beautiful properties and can be computed astonishingly fast.

6.4.1 Definition and inverse

Given N numbers $x_0, x_1, \ldots, x_{N-1}$ (real or complex), their **discrete Fourier transform** (DFT) is the list of N numbers

$$X_k = \sum_{j=0}^{N-1} x_j \, e^{-2\pi i j k/N}, \qquad k = 0, 1, \ldots, N-1. \tag{6.30}$$

If $x_j = f(j\Delta t)$, then X_k is the sum (6.27) evaluated at

$$\omega_k = \frac{2\pi k}{N\Delta t} = k\,\frac{2\pi}{T}, \qquad T = N\Delta t, \tag{6.31}$$

because $\omega_k t_j = 2\pi kj/N$. So the DFT samples the spectrum exactly once per Fourier resolution, one value per bin. Indices $k > N/2$ are aliases of negative frequencies: $e^{-2\pi ijk/N} = e^{-2\pi ij(k-N)/N}$ since $e^{2\pi ij} = 1$.

The key fact is an orthogonality relation, a discrete cousin of the orthogonality of $e^{ik\omega t}$ in Chapter 5. For an integer m,

$$\sum_{k=0}^{N-1} e^{2\pi ikm/N} = \begin{cases} N, & m \text{ a multiple of } N, \\ 0, & \text{otherwise.} \end{cases} \tag{6.32}$$

Proof. If m is a multiple of N, every term is $e^{2\pi i \times \text{integer}} = 1$, and there are N terms. Otherwise the ratio $r = e^{2\pi im/N} \neq 1$, and the geometric series gives $(1 - r^N)/(1 - r)$ with $r^N = e^{2\pi im} = 1$, so the sum is 0. □

Inverse DFT. Multiply (6.30) by $e^{2\pi ilk/N}$, for a fixed index l, and sum over k:

$$\begin{aligned} \sum_{k=0}^{N-1} X_k e^{2\pi ilk/N} &= \sum_k \sum_j x_j e^{-2\pi ijk/N} e^{2\pi ilk/N} \\ &= \sum_j x_j \sum_k e^{2\pi ik(l-j)/N} \quad \text{(swap finite sums)} \\ &= \sum_j x_j \, N\,[j = l] \quad \text{(by (6.32); } l - j \text{ is a multiple of } N \text{ only if } j = l) \\ &= N x_l. \end{aligned}$$

Here $[j = l]$ is 1 if $j = l$ and 0 otherwise. Hence

$$x_l = \frac{1}{N} \sum_{k=0}^{N-1} X_k \, e^{2\pi ilk/N}. \tag{6.33}$$

Unlike the continuous case, this inversion is fully proved, with no conditions: N numbers in, N numbers out, nothing lost.

Real data. If all x_j are real, conjugating (6.30) gives $\overline{X_k} = \sum x_j e^{2\pi ijk/N} = \sum x_j e^{-2\pi ij(N-k)/N} = X_{N-k}$, using $e^{-2\pi ijN/N} = 1$. So

$$X_{N-k} = \overline{X_k} \qquad (x \text{ real}), \tag{6.34}$$

and in particular $X_0 = \sum x_j$ is real, and if N is even, $X_{N/2} = \sum x_j(-1)^j$ is real. Only about half the coefficients are independent; `numpy.fft.rfft` returns exactly those, $X_0, \ldots, X_{\lfloor N/2 \rfloor}$.

Parseval for the DFT. We prove in Section 6.6 that

$$\sum_{j=0}^{N-1} |x_j|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X_k|^2, \tag{6.35}$$

the discrete form of Parseval's relation of Chapter 5.

Sample Problem 6.6: A DFT by hand

Compute the DFT of $x = (2, 1, 0, 1)$ and verify Parseval's relation (6.35).
Step 1: The phases. With $N = 4$, $e^{-2\pi ijk/4} = (e^{-i\pi/2})^{jk} = (-i)^{jk}$. The powers of $-i$ are $1, -i, -1, i, 1, \ldots$

Step 2: Each coefficient.

$$\begin{aligned} X_0 &= 2+1+0+1 = 4, \\ X_1 &= 2+1(-i)+0+1(-i)^3 = 2-i+i = 2, \\ X_2 &= 2+1(-i)^2+0+1(-i)^6 = 2-1-1 = 0, \\ X_3 &= 2+1(-i)^3+0+1(-i)^9 = 2+i-i = 2. \end{aligned}$$

Note $X_3 = \overline{X_1}$, as (6.34) requires.
Step 3: Parseval. Left side: $4+1+0+1 = 6$. Right side: $(16+4+0+4)/4 = 6$.
Step 4: Interpret. $x_j = 1+\cos(\pi j/2)$, a constant plus one cosine that completes one cycle in four samples; the DFT finds the constant in X_0 and the cosine in X_1 and its mirror X_3.

$$\boxed{X = (4,2,0,2), \qquad \sum |x_j|^2 = 6 = \tfrac{1}{4}\sum |X_k|^2 .}$$

```
import numpy as np

x = np.array([2.0, 1.0, 0.0, 1.0])
N = len(x)
idx = np.arange(N)                      # sample indices 0..N-1
# the DFT straight from the definition; 2j is Python for 2i
X_def = np.array([np.sum(x*np.exp(-2j*np.pi*idx*k/N)) for k in range(N)])
X_fft = np.fft.fft(x)                   # numpy uses the same sign convention
print(np.round(X_def, 12))              # [4, 2, 0, 2]
print(np.allclose(X_def, X_fft))        # True
print(np.sum(x**2), np.sum(np.abs(X_fft)**2)/N)   # 6.0 6.0 (Parseval)
print(np.fft.rfft(x))                   # the independent half: X_0, X_1, X_2
```

Pitfall

Different books and libraries put the $1/N$ in different places (in front of the forward sum, the inverse sum, or $1/\sqrt{N}$ on both) and some use $e^{+2\pi ijk/N}$ in the forward direction. `numpy.fft.fft` uses exactly Eq. (6.30) with no factor in front, and `numpy.fft.ifft` includes the $1/N$. Peak *positions* never depend on the convention; peak *heights* do, so always check before comparing numbers.

6.4.2 The fast Fourier transform

Computing all N coefficients of (6.30) directly takes N multiplications each, N^2 in total. The **fast Fourier transform** (FFT) does the same job in about $N \log_2 N$ operations when N is a power of 2. Here is the whole idea.

Write $w = e^{-2\pi i/N}$, so $X_k = \sum_j x_j w^{jk}$. Split the sum into even $j = 2l$ and odd $j = 2l+1$ indices, $l = 0, \ldots, N/2-1$:

$$\begin{aligned} X_k &= \sum_l x_{2l} w^{2lk} + \sum_l x_{2l+1} w^{(2l+1)k} \\ &= \sum_l x_{2l} (w^2)^{lk} + w^k \sum_l x_{2l+1} (w^2)^{lk}. \end{aligned} \tag{6.36}$$

Now $w^2 = e^{-2\pi i/(N/2)}$ is exactly the "w" of a DFT of length $N/2$. So $X_k = E_k + w^k O_k$, where E_k and O_k are the length-$N/2$ DFTs of the even-indexed and odd-indexed samples. These have

period $N/2$ in k, and $w^{N/2} = e^{-i\pi} = -1$, so for $k = 0, \dots, N/2 - 1$,

$$X_k = E_k + w^k O_k, \qquad X_{k+N/2} = E_k - w^k O_k. \tag{6.37}$$

Two half-size DFTs plus N simple combinations give the full DFT. Apply the same split to each half, and so on down to DFTs of length 1 (which are just $X_0 = x_0$). If $C(N)$ is the operation count, $C(N) = 2C(N/2) + N$ and $C(1) = 0$. Divide by N: $C(N)/N = C(N/2)/(N/2) + 1$. With $N = 2^q$, applying this q times gives $C(N)/N = C(1)/1 + q = q$, so

$$C(N) = N \log_2 N. \tag{6.38}$$

(Algorithms of the same speed exist for N that are not powers of 2; libraries handle this for you.)

Sample Problem 6.7: Bins, Nyquist and the FFT saving

You record $N = 1024$ uniform samples at $\Delta t = 0.5$. Find the bin spacing of the DFT, the Nyquist frequency, and the ratio of the direct cost N^2 to the FFT cost $N \log_2 N$.
Step 1: Window. $T = N\Delta t = 512$.
Step 2: Bin spacing. $\omega_{k+1} - \omega_k = 2\pi/T = 2\pi/512 = 0.01227$.
Step 3: Nyquist. $\omega_{\mathrm{N}} = \pi/\Delta t = 2\pi = 6.283$; equivalently the bin $k = N/2 = 512$.
Step 4: Saving. $N^2/(N \log_2 N) = N/\log_2 N = 1024/10 = 102.4$.

$$\boxed{\Delta\omega = 0.01227, \quad \omega_{\mathrm{N}} = 6.283, \quad \text{FFT about 102 times faster.}}$$

Pitfall

The FFT needs *uniform* sampling. The Riemann zeros are irregular in t, so a spectrum of the zeros in the frequency ω of t must be computed as the direct sum (6.27). But a sequence *indexed by* n, such as the fluctuation d_n of the n-th zero, is uniform in n, and the FFT applies to it perfectly well. Keep track of which variable your frequencies refer to: radians per unit of t, or radians per level.

Key Ideas

- The DFT (6.30) samples the spectrum once per bin $2\pi/T$; its inverse (6.33) follows from the orthogonality (6.32).
- For real data $X_{N-k} = \overline{X_k}$; only $X_0, \dots, X_{\lfloor N/2 \rfloor}$ are independent.
- The FFT splits even and odd samples, Eq. (6.37), and costs $N \log_2 N$ instead of N^2; it requires uniform sampling.

6.5 The periodogram and finding peaks

We now have the tools to answer, with real data, the question "which frequencies are present, and how strongly?"

6.5.1 Definition

For samples y_j at points t_j, $j = 1, \dots, N$ (uniform or not), the **periodogram** is

$$P(\omega) = \frac{1}{N}\left|\sum_{j=1}^{N} y_j\, e^{-i\omega t_j}\right|^2 = \frac{|F(\omega)|^2}{N}. \tag{6.39}$$

For uniform samples at the DFT frequencies it is $P_k = |X_k|^2/N$. It is a function of ω that you can evaluate on any grid you like. Because $P(-\omega) = P(\omega)$ for real data, we plot only $\omega > 0$. Before computing it, subtract the mean of the y_j: otherwise the line at $\omega = 0$, of height $N\bar{y}^2$, leaks into everything near $\omega = 0$.

6.5.2 How tall is a line, and how tall is the noise?

A line. Let $y_j = A\sin(\omega_0 t_j) = \frac{A}{2i}(e^{i\omega_0 t_j} - e^{-i\omega_0 t_j})$. Then

$$F(\omega_0) = \frac{A}{2i}\Big(\sum_j 1 - \sum_j e^{-2i\omega_0 t_j}\Big) = \frac{A}{2i}(N - W(2\omega_0)), \tag{6.40}$$

with W the spectral window (6.28). *If* $|W(2\omega_0)| \ll N$ (the analogue of $\omega_0 T \gg 1$ for a continuous window), then $|F(\omega_0)| \approx NA/2$ and

$$P(\omega_0) \approx \frac{NA^2}{4}. \tag{6.41}$$

The line height grows in proportion to the number of samples.

Noise. Now let $y_j = \varepsilon_j$ be "noise": values that are independent of one another, average zero, and have average square σ^2. (Chapter 19 defines these notions properly; here we use them informally.) Expanding the modulus squared,

$$|F(\omega)|^2 = \sum_j \sum_l \varepsilon_j \varepsilon_l \, e^{-i\omega(t_j - t_l)}. \tag{6.42}$$

Average over many realizations of the noise. The terms with $j \neq l$ average to zero, because ε_j and ε_l are independent with average zero. The N terms with $j = l$ each average to σ^2, and their phase factor is $e^0 = 1$. So the average of $|F|^2$ is $N\sigma^2$, and the average periodogram is

$$\langle P(\omega) \rangle = \sigma^2, \tag{6.43}$$

independent of N and of ω ("white" noise). This is a heuristic result: a single realization fluctuates about this average by an amount of the same order. Putting (6.41) and (6.43) together, the line stands above the noise by a factor

$$\frac{P(\omega_0)}{\langle P \rangle} \approx \frac{NA^2}{4\sigma^2}, \tag{6.44}$$

which grows with N. Longer data make lines stand out, however weak.

Sample Problem 6.8: Detecting a weak oscillation

A sinusoid of amplitude $A = 0.1$ is buried in noise whose root-mean-square size $\sigma = 0.3$ (the square root of the average of the squared noise values) is three times the sinusoid's amplitude. With $N = 400$ samples, how much does its periodogram peak rise above the average noise level? From the peak, how would you estimate A?
Step 1: Line height from (6.41): $NA^2/4 = 400 \times 0.01/4 = 1$.
Step 2: Noise level from (6.43): $\sigma^2 = 0.09$.
Step 3: Ratio: $1/0.09 = 11.1$.
Step 4: Amplitude estimate. Since $|F(\omega_0)| \approx NA/2$, invert: $\hat{A} = 2\,|F(\hat{\omega})|/N = 2\sqrt{NP(\hat{\omega})}/N$, where $\hat{\omega}$ is the location of the peak. The phase of $F(\hat{\omega})$ estimates the phase of the oscillation.

$$\boxed{P(\omega_0)/\langle P\rangle \approx NA^2/(4\sigma^2) = 11.1; \qquad \hat{A} = 2\,|F(\hat{\omega})|\,/N.}$$

Even though each sample is dominated by noise, the oscillation is plain to see in the periodogram. That is the power of adding up N samples with the right phases: the signal adds coherently (like N), the noise incoherently (like $\sqrt{N}$ in $|F|$).

6.5.3 Finding the peaks

A practical peak search on a periodogram has three ingredients.

1. **A search range.** Exclude ω so small that, within the window, the oscillation is indistinguishable from a slow trend (fewer than a few cycles in T), and stop at the highest frequency you care about.

2. **A coarse grid.** Evaluate P on a grid with step a fraction of the resolution $2\pi/T$, and take the largest value (or all local maxima above a threshold). One step per bin is not enough: Sample Problem "How much of a peak do you see if you miss it by half a bin?" showed you could see only 64% of the amplitude.

3. **Refinement.** Re-evaluate P on a fine grid around the coarse maximum and take the maximum there (or fit a parabola through the top three points).

Sample Problem 6.9: Why a step of a quarter of the resolution?

The paper's sparse search (App. A step 20) uses a coarse step of a quarter of $2\pi/T$. Show that with this step the coarse grid never sees less than 97% of a line's peak amplitude (rectangular window, a single isolated line).
Step 1: Worst offset. The step is $\frac{1}{4}$ bin. A line is at most half a step, $\frac{1}{8}$ bin, from the nearest grid point.
Step 2: Main-lobe shape in bins. The amplitude relative to the peak is $s(\delta) = \sin(\pi\delta)/(\pi\delta)$ (Section 6.2).
Step 3: Evaluate. $s(\frac{1}{8}) = \sin(\pi/8)/(\pi/8) = 0.382683/0.392699 = 0.9745$. In power (amplitude squared) that is 0.9496.

$$\boxed{\text{at least } 97.4\% \text{ of the amplitude } (95.0\% \text{ of the power}).}$$

So the coarse grid reliably lands in the right main lobe, and the refinement then pins down the peak.

6.5.4 A first example: the prime-2 wave

Figure 6.4 shows the periodogram of RE2 computed two ways over the same range of t as zeros 1–400. On uniform samples, the three lines at $\log 2$, $2\log 2$, $3\log 2$ appear with heights close to the prediction $NA_m^2/4$ of (6.41). Sampled at the zeros themselves, the lines are in the same places but the heights differ somewhat, because the sampling is denser at the top of the range and, as Section 6.8 explains, the zeros are not a neutral set of sample points.

```
import numpy as np
import qmnt

t = qmnt.zeros(400)                          # t_1 ... t_400
n = np.arange(1, 401)
u = t/(2*np.pi)
```

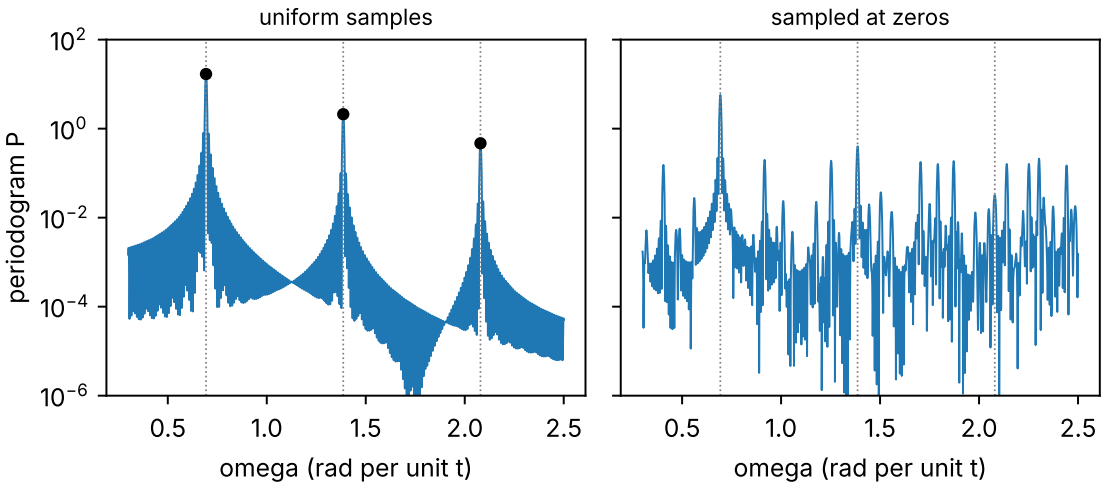


Figure 6.4: Periodogram (6.39) of the prime-2 wave RE2, $-\frac{1}{\pi}\sum_{m=1}^{3}\sin(mt\log 2)/(m2^{m/2})$, on the range $t_1 \le t \le t_{400}$ of zeros 1–400 (logarithmic vertical scale). Left: uniform samples at spacing 0.5; dots mark the prediction $NA_m^2/4$. Right: the same function sampled only at the 400 zeros. Vertical lines: $m\log 2$, $m = 1, 2, 3$.

```
y = n - 0.5 - (u*np.log(u) - u + 7/8)      # y_n = n - 1/2 - Nsm(t_n)
y = y - y.mean()                           # remove the omega = 0 line

def P(w):                                  # periodogram at one frequency
    return abs(np.sum(y*np.exp(-1j*w*t)))**2 / len(t)

T = t[-1] - t[0]                           # window length
step = 0.25*2*np.pi/T                      # a quarter of the resolution
w = np.arange(0.3, 3.0, step)              # coarse grid
Pw = np.array([P(v) for v in w])
k = np.argmax(Pw)                          # coarse peak
wf = np.linspace(w[k]-step, w[k]+step, 201)   # refine on 201 points
Pf = np.array([P(v) for v in wf])
w_peak = wf[np.argmax(Pf)]
print(w_peak, np.log(2), 2*np.pi/T)        # peak lies within 2*pi/T of log 2
```

6.5.5 The periodogram of the zeros

Now the real thing. Define, as in the paper's Sec. II D, the quantity

$$y_n = n - \tfrac{1}{2} - N_{\mathrm{sm}}(t_n), \tag{6.45}$$

where N_{sm} is the smooth counting function of paper Eq. (1) (Chapter 17). The counting function $N(t)$ jumps from $n-1$ to n at t_n; $n-\frac{1}{2}$ is the midpoint of the jump, so y_n is the fluctuating part $S = N - N_{\mathrm{sm}}$ read off at the n-th zero. Chapter 18 explains why S is approximated by the prime sum N_{fl} of paper Eq. (4). If that is right, the periodogram of y_n sampled at the t_n should show lines at $\log p^m$. Figure 6.5 shows it does. The tallest peak in $0.3 < \omega < 3$ sits at $\log 2$, within the resolution $2\pi/T$ of this 1000-zero window. Within $2\pi/T$ of each of $\log 3$, $\log 5$ and $\log 7$ there is a peak more than three times the median level of the periodogram over this range. The dashed lines mark further prime powers for comparison.

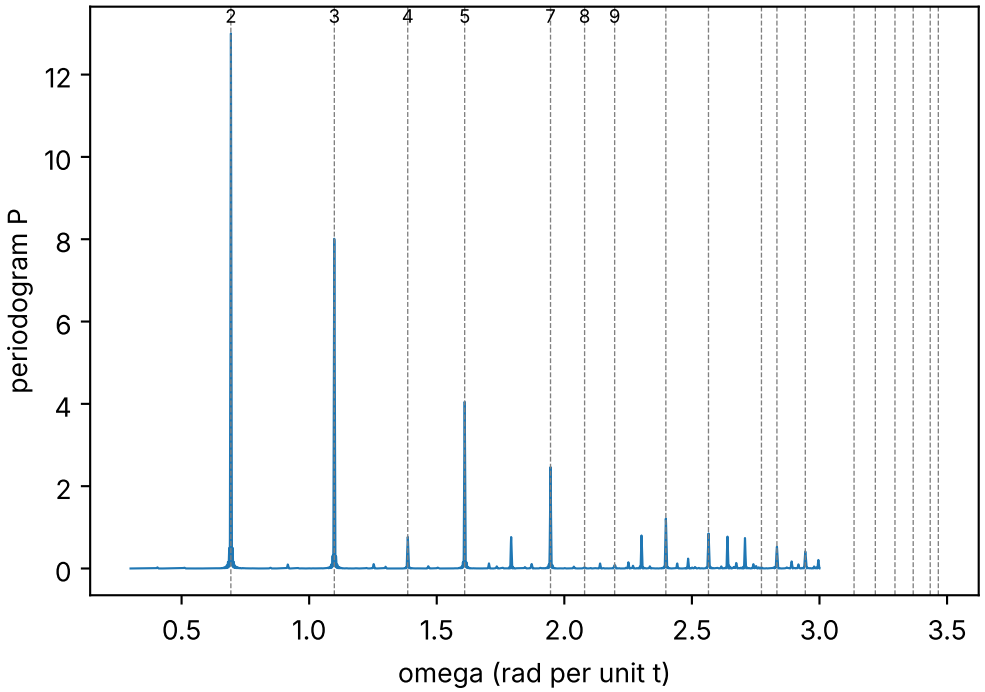


Figure 6.5: Periodogram of the fluctuation $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ of the first 1000 zeros, computed as the direct sum (6.39) at the irregular points t_n (mean subtracted). Dashed vertical lines mark $\log q$ for prime powers $q \le 36$; the first few are labeled. The tallest peak is at $\log 2$.

Checkpoint 6.4

The line heights $NA^2/4$ in (6.41) grow with N, but the noise level (6.43) does not. Does that mean that with enough data you can resolve $\log 127$ from $\log 128$ even though $T < 801$?

Answer: No. More samples in the same window raise the peaks above the noise, but resolution depends only on the window length T, Eq. (6.17). Two lines closer than about $2\pi/T$ merge into one peak no matter how many samples you take.

Key Ideas

- The periodogram $P(\omega) = \left|\sum y_j e^{-i\omega t_j}\right|^2 / N$, Eq. (6.39), works for any sampling; subtract the mean first.
- A line of amplitude A reaches $\approx NA^2/4$; white noise sits at σ^2; the ratio $NA^2/(4\sigma^2)$ grows with N, Eq. (6.44). The amplitude estimate is $2\,|F|\,/N$.
- Search on a coarse grid finer than one bin (a quarter bin loses at most 2.6% of amplitude), then refine.
- The periodogram of the zeros' fluctuation has its peaks at $\log p^m$, the tallest at $\log 2$.

6.6 Power spectrum, autocorrelation and the Wiener–Khinchin theorem

★ Did you know? ***Einstein Got There First***

The Wiener-Khinchin theorem says that the power spectrum of a fluctuating signal is the Fourier transform of its autocorrelation. Norbert Wiener proved it in 1930 and Aleksandr Khinchin in 1934. Yet Albert Einstein had stated the essential idea in a short 1914 paper on analyzing irregularly fluctuating observations. His note went largely unnoticed for decades.

The periodogram keeps the *size* $|X_k|$ of each Fourier coefficient and throws away its *phase*. What, exactly, does it still know about the sequence? The answer is: its autocorrelation, and nothing else. The list of squared moduli $|X_k|^2$ is called the **power spectrum** of the sequence; equivalently one can use the periodogram $P_k = |X_k|^2/N$. The name "power" comes from Parseval's relation: these numbers, divided by N, add up to the sum of squares of the data.

6.6.1 Autocorrelation

The **autocorrelation** measures how much a sequence resembles a shifted copy of itself. For a real sequence $x_0, \dots, x_{N-1}$ and an integer lag m, define the *ordinary* (lagged-product) autocorrelation

$$c_m = \sum_{j=0}^{N-1-m} x_j\, x_{j+m}, \qquad m = 0, 1, \dots, N-1, \tag{6.46}$$

and the *circular* autocorrelation, in which the sequence is wrapped around on itself,

$$r_m = \sum_{j=0}^{N-1} x_j\, x_{(j+m) \bmod N}. \tag{6.47}$$

Here $a \bmod N$ is the remainder of a on division by N. (The letters are reused: c_m and r_m here are not the Fourier coefficients c_k of Section 6.1, nor the ratio r of a geometric series.) For lags m small compared with N, the two differ only by the m wrap-around terms. In data analysis one usually subtracts the mean first and divides by c_0, to get the normalized autocorrelation c_m/c_0, which equals 1 at lag 0. A positive value at lag m means values m steps apart tend to have the same sign; a negative value means they tend to have opposite signs. The fluctuations of the zeros have negative autocorrelation at lags 1–3, as the paper reports in Sec. III; Figure 6.6 shows this for zeros 1–400.

6.6.2 The theorem

Wiener–Khinchin theorem (discrete, proved here). For a real sequence x with DFT X, the DFT of the circular autocorrelation is the squared modulus of the DFT:

$$R_k = \sum_{m=0}^{N-1} r_m\, e^{-2\pi i m k/N} = |X_k|^2 . \tag{6.48}$$

Proof. Write $w = e^{-2\pi i/N}$. Substitute (6.47):

$$R_k = \sum_{m=0}^{N-1} \sum_{l=0}^{N-1} x_l\, x_{(l+m) \bmod N}\, w^{mk}.$$

Fix l and replace the summation variable m by $j = (l+m) \bmod N$. As m runs over $0, \dots, N-1$, so does j, each value once. Since m and $j-l$ differ by a multiple of N and $w^N = 1$, we have $w^{mk} = w^{(j-l)k} = w^{jk} w^{-lk}$. Therefore

$$R_k = \sum_l \sum_j x_l \, x_j \, w^{jk} w^{-lk} = \Big(\sum_j x_j w^{jk}\Big)\Big(\sum_l x_l w^{-lk}\Big) = X_k \, \overline{X_k} = |X_k|^2 ,$$

where the second factor is $\overline{X_k}$ because x_l is real and $\overline{w} = w^{-1}$. $\square$

By the inverse DFT (6.33), $r_m = \frac{1}{N} \sum_k |X_k|^2 \, e^{2\pi i m k/N}$: the autocorrelation is completely determined by the periodogram, and vice versa. Setting $m = 0$ gives $r_0 = \sum_j x_j^2 = \frac{1}{N} \sum_k |X_k|^2$, which is Parseval's relation (6.35) for real x. (For complex x, replace $x_l x_{l+m}$ by $\overline{x_l} x_{l+m}$; the same proof goes through.)

Getting the ordinary autocorrelation. Append N zeros to x, making a sequence $\tilde{x}$ of length $2N$. For $0 \le m < N$, a term $\tilde{x}_j \tilde{x}_{(j+m) \bmod 2N}$ of its circular autocorrelation is nonzero only if $j \le N-1$ and the second index is $\le N-1$. Since $j+m \le 2N-2 < 2N$, no wrap-around occurs, so the second index is $j+m$, and we need $j \le N-1-m$. That leaves exactly the terms of c_m in (6.46). So: zero-pad, FFT, take $|\cdot|^2$, inverse FFT, keep lags $0, \dots, N-1$. This is how autocorrelations of long sequences are computed in practice.

Sample Problem 6.10: Wiener–Khinchin by hand

For $x = (1, 1, 0, 0)$ compute the circular autocorrelation r, its DFT R, and $|X_k|^2$, and confirm (6.48).

Step 1: Autocorrelation. Using (6.47) with $N = 4$:

$$\begin{aligned} r_0 &= 1 + 1 + 0 + 0 = 2, \\ r_1 &= x_0 x_1 + x_1 x_2 + x_2 x_3 + x_3 x_0 = 1 + 0 + 0 + 0 = 1, \\ r_2 &= x_0 x_2 + x_1 x_3 + x_2 x_0 + x_3 x_1 = 0, \\ r_3 &= x_0 x_3 + x_1 x_0 + x_2 x_1 + x_3 x_2 = 0 + 1 + 0 + 0 = 1. \end{aligned}$$

So $r = (2, 1, 0, 1)$. This is the sequence of the earlier DFT Sample Problem, so $R = (4, 2, 0, 2)$.

Step 2: DFT of x. With phases $(-i)^{jk}$: $X_0 = 2$, $X_1 = 1 + (-i) = 1 - i$, $X_2 = 1 + (-1) = 0$, $X_3 = 1 + (-i)^3 = 1 + i$.

Step 3: Squared moduli. $|X_0|^2 = 4$, $|1 - i|^2 = 2$, 0, $|1 + i|^2 = 2$: $(4, 2, 0, 2)$. They agree.

$$\boxed{r = (2, 1, 0, 1), \qquad R = |X|^2 = (4, 2, 0, 2).}$$

Taken on trust

Taken on trust: the continuous Wiener–Khinchin theorem. For a signal that fluctuates forever in a statistically steady way (a "stationary random process"), the power spectral density is the Fourier transform of the autocorrelation function. The precise statement needs probability theory beyond this book; see the discussion of correlation functions and energy spectra in [22]. Everything we use in this book is the discrete version (6.48), which we proved.

What the power spectrum does not know. The periodogram fixes the $|X_k|$ but not the phases. Different phases give different sequences with the same periodogram and therefore, by (6.48), the same circular autocorrelation. The shift property is the simplest example: a circularly shifted sequence has X_k multiplied by a phase factor and the same $|X_k|$. But there are many more, as the next section exploits.

Checkpoint 6.5

Two real sequences of the same length have identical periodograms at every DFT frequency. Must they be identical? Must they have the same circular autocorrelation? The same sum of squares?

Answer: They need not be identical (the phases may differ; a circular shift is one example). They must have the same circular autocorrelation, by Eq. (6.48) and the inverse DFT, and hence the same sum of squares r_0.

Key Ideas

- The autocorrelation (ordinary (6.46) or circular (6.47)) measures self-similarity under shifts; the zeros' fluctuations are negatively autocorrelated at lags 1–3.
- Discrete Wiener–Khinchin, proved: the DFT of the circular autocorrelation is $|X_k|^2$, Eq. (6.48). Parseval is its $m = 0$ case. Zero padding gives the ordinary autocorrelation.
- The periodogram and the autocorrelation carry the same information; neither knows the phases.

6.7 Phase randomization and surrogate data

Suppose a model fits your data well. Is it using some specific structure in the data, or would it fit *anything* with the same general look? One way to find out is to build fake data sets, **surrogates**, that share some properties with the real data and lack others, and then run the same analysis on them. If the model does just as well on the surrogates, it is not using what the surrogates lack. Theiler and collaborators [16] introduced a now-standard surrogate: keep the amplitudes of the Fourier coefficients and scramble their phases.

6.7.1 The recipe

Start from a real sequence $x_0, \dots, x_{N-1}$ with DFT X_k.

1. For each k with $1 \le k < N/2$, choose an angle φ_k uniformly at random in $[0, 2\pi)$ and set $X'_k = X_k e^{i\varphi_k}$.
2. Keep $X'_0 = X_0$, and, if N is even, $X'_{N/2} = X_{N/2}$.
3. For $N/2 < k \le N-1$, set $X'_k = \overline{X'_{N-k}}$, to restore the symmetry (6.34).
4. Invert: $x'_j = \frac{1}{N}\sum_k X'_k e^{2\pi ijk/N}$.

The result is real. Conjugate step 4 and use step 3, $\overline{X'_k} = X'_{N-k}$ (which also holds for $k = 0$ and $k = N/2$, because those are real and $N - 0 \equiv 0$):

$$\overline{x'_j} = \frac{1}{N}\sum_k \overline{X'_k} e^{-2\pi ijk/N} = \frac{1}{N}\sum_k X'_{N-k} e^{-2\pi ijk/N}.$$

Let $k' = N - k$ (and $k' = 0$ when $k = 0$). Then $e^{-2\pi ijk/N} = e^{-2\pi ij(N-k')/N} = e^{2\pi ijk'/N}$, so the sum is $\frac{1}{N}\sum_{k'} X'_{k'} e^{2\pi ijk'/N} = x'_j$. A number equal to its own conjugate is real.

What is preserved, exactly.

- Every amplitude: $|X'_k| = |X_k|\,|e^{i\varphi_k}| = |X_k|$. Hence the periodogram, at every DFT frequency.

- The circular autocorrelation, by the Wiener–Khinchin theorem (6.48) and the uniqueness of the inverse DFT; and so, for lags small compared with N, approximately the ordinary one.
- The mean, since $X'_0 = X_0 = \sum x_j$; and the sum of squares, which is r_0.

What is destroyed.

- All phase relations between frequency components, and with them any structure that is not captured by the power spectrum. If the data were, say, a sum of sinusoids all of whose phases are fixed by a law (the explicit formula has every term a pure sine, zero at $t = 0$), that law is gone.
- The *distribution of values* (the "marginal"). The surrogate's histogram generally differs from the original's, as the next Sample Problem shows. As a heuristic, a sum of many oscillations with random phases tends toward a bell-shaped (Gaussian) histogram (Chapter 19).

Sample Problem 6.11: Phase randomization by hand

Phase-randomize $x = (1, 1, 0, 0)$, choosing the random angle to be $\varphi_1 = \pi/4$. As we will see, the new coefficient X'_1 then has phase 0. Check that the result has the same circular autocorrelation but different values.

Step 1: Coefficients. From the previous Sample Problem, $X = (2,\ 1 - i,\ 0,\ 1 + i)$. Only X_1 (and its mirror X_3) can be rotated; $X_0 = 2$ and $X_2 = 0$ stay. Since $X_1 = 1 - i = \sqrt{2}\, e^{-i\pi/4}$, step 1 of the recipe gives $X'_1 = X_1 e^{i\varphi_1} = \sqrt{2}\, e^{-i\pi/4} e^{i\pi/4} = \sqrt{2}$, so $X'_3 = \overline{X'_1} = \sqrt{2}$.

Step 2: Invert. With $N = 4$, $e^{2\pi ijk/4} = e^{i\pi jk/2}$:

$$x'_j = \tfrac{1}{4}\left[2 + \sqrt{2}\, e^{i\pi j/2} + 0 + \sqrt{2}\, e^{3i\pi j/2}\right] = \tfrac{1}{4}\left[2 + \sqrt{2}\,(e^{i\pi j/2} + e^{-i\pi j/2})\right] = \tfrac{1}{2} + \tfrac{\sqrt{2}}{2}\cos\frac{\pi j}{2},$$

using $e^{3i\pi j/2} = e^{-i\pi j/2} e^{2\pi ij} = e^{-i\pi j/2}$.

Step 3: Values. $\cos(\pi j/2) = 1, 0, -1, 0$ for $j = 0, 1, 2, 3$, so $x' = (1.2071,\ 0.5,\ -0.2071,\ 0.5)$.

Step 4: Check. $r'_0 = 1.2071^2 + 0.25 + 0.0429 + 0.25 = 2.000$ and $r'_1 = 1.2071(0.5) + 0.5(-0.2071) + (-0.2071)(0.5) + 0.5(1.2071) = 1.000$, equal to $r_0 = 2$, $r_1 = 1$. (By symmetry $r'_3 = r'_1$, and $r'_2 = 0$; Problem 18 asks you to confirm.) But the original took only the values 0 and 1; the surrogate takes three different values (1.2071, 0.5 twice, and -0.2071), one of them negative.

$$\boxed{x' = (1.2071,\ 0.5,\ -0.2071,\ 0.5):\ \text{same autocorrelation } (2, 1, 0, 1),\ \text{different values.}}$$

(Had we chosen $\varphi_1 = \pi/2$ instead, so that $X'_1 = \sqrt{2}\, e^{i\pi/4}$ has phase $\pi/4$, Step 2 would give $x' = (1, 0, 0, 1)$: just a circular shift of x. A random phase gives, in general, something genuinely new.)

```
import numpy as np

def phase_randomize(x, rng):
    """Real surrogate with the same |DFT| as x and random phases."""
    N = len(x)
    X = np.fft.rfft(x)                  # independent half: X_0 ... X_{N//2}
    phi = rng.uniform(0, 2*np.pi, len(X))
```

```
    phi[0] = 0.0                          # keep X_0 (the mean)
    if N % 2 == 0:
        phi[-1] = 0.0                     # keep the real coefficient X_{N/2}
    return np.fft.irfft(X*np.exp(1j*phi), n=N)   # irfft restores symmetry

def circ_autocorr(x):                     # Wiener--Khinchin in one line
    return np.fft.ifft(np.abs(np.fft.fft(x))**2).real

rng = np.random.default_rng(1)            # a seed makes the draw reproducible
x = np.array([1.0, 1.0, 0.0, 0.0])
xs = phase_randomize(x, rng)
print(xs)
print(circ_autocorr(x), circ_autocorr(xs))           # identical
print(np.abs(np.fft.fft(x)), np.abs(np.fft.fft(xs)))  # identical
```

6.7.2 Two relatives

Random permutation. Shuffle the order of the values. This keeps the marginal distribution exactly (the same numbers, rearranged) and destroys the serial structure, including the autocorrelation. It is the opposite trade to phase randomization.

Amplitude-adjusted surrogates. Schreiber and Schmitz [17] gave an iterative algorithm (IAAFT) that alternates between imposing the original Fourier amplitudes and imposing the original set of values, ending with a sequence that has exactly the original marginal and very nearly the original spectrum. Chapter 26 treats all three and the null hypotheses they test.

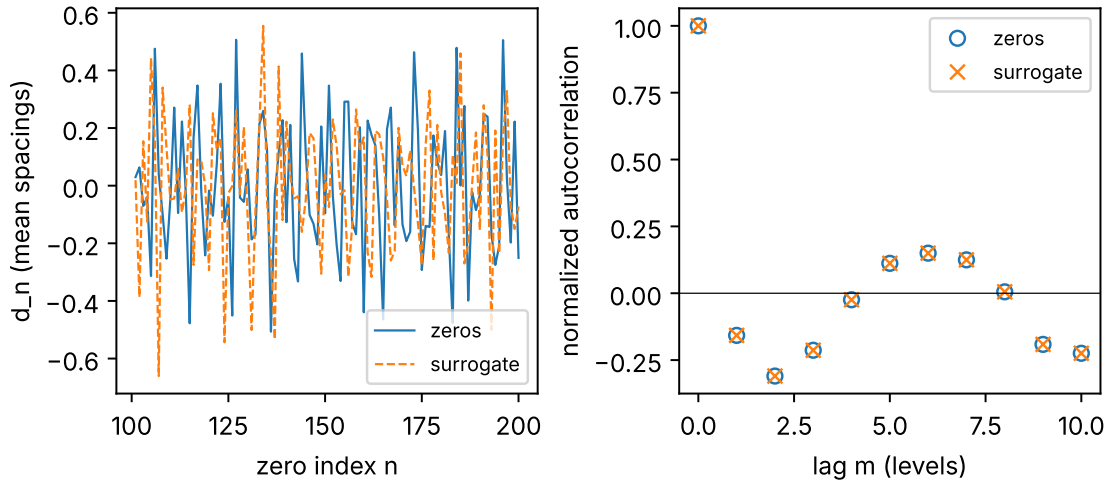


Figure 6.6: Left: the fluctuation $d_n = (t_n - s_n)/\mathrm{gap}_n$ of zeros 101–200 in mean spacings (solid) and the same stretch of one phase-randomized surrogate of the full sequence $n \le 400$ (dashed, seed 1). Right: normalized circular autocorrelation of the mean-subtracted sequences of length 400, real (circles) and surrogate (crosses); they coincide exactly, including the negative values at lags 1–3.

Checkpoint 6.6

Which surrogate keeps the histogram of the data exactly, random permutation or phase randomization? Which keeps the autocorrelation exactly?

Answer: Random permutation keeps the histogram (the same values in a new order) but destroys the

autocorrelation. Phase randomization keeps the circular autocorrelation and periodogram exactly but changes the histogram.

Key Ideas

- Phase randomization multiplies each X_k $(0 < k < N/2)$ by a random $e^{i\varphi_k}$, restores the conjugate symmetry and inverts; the result is real.
- It preserves the periodogram, the circular autocorrelation, the mean and the sum of squares exactly, and destroys phase structure and the marginal distribution.
- Random permutation makes the opposite trade; IAAFT [17] keeps the marginal exactly and the spectrum nearly. Surrogates tell you which properties of the data a method actually uses.

6.8 The spectrum of the zeros: a first look at Landau's formula

So far we have Fourier-analyzed a *function* (y_n) sampled at the zeros. A different and very striking object is the spectral window of the zeros themselves,

$$W(\omega) = \sum_{n:\, t_n \le T} e^{i\omega t_n}, \tag{6.49}$$

the Fourier transform of the set of zeros up to height T. (Careful: here, following the usual statement of Landau's formula, T is the *height* of the top of the range, not the window length. The paper uses T for the training window, for the tensors $T^{(n)}$ and for the Chebyshev polynomials T_k. Since $t_1 = 14.13$, the height and the window length differ by about 14, which is negligible for large T. The sign in the exponent is reversed from (6.28); by (P2) this only conjugates the result.)

If the zeros were scattered at random, (6.49) would be a featureless floor away from $\omega = 0$. It is not.

6.8.1 A heuristic derivation

Let $N(t)$ be the number of zeros with $0 < t_n \le t$ (Chapter 17). Careful: in this section $N(t)$ is the zero-counting function, not the number of samples. Later, x denotes a number greater than 1 (a prime power at the resonances), not a data sequence. Pick a height t_a below t_1, so $N(t_a) = 0$, and let $g(t)$ be any smooth function.

Step A (exact): summation by parts. For each zero $t_n \le T$, $\int_{t_a}^{T} g'(t)\,[t \ge t_n]\,\mathrm{d}t = g(T) - g(t_n)$, where $[t \ge t_n]$ is 1 for $t \ge t_n$ and 0 otherwise. Since $N(t) = \sum_n [t \ge t_n]$, summing over the $N(T)$ zeros gives $\int_{t_a}^{T} g' N\,\mathrm{d}t = N(T)g(T) - \sum_n g(t_n)$, so

$$\sum_{n:\, t_n \le T} g(t_n) = g(T)N(T) - \int_{t_a}^{T} g'(t)\, N(t)\,\mathrm{d}t. \tag{6.50}$$

Step B (exact): split N. Write $N = N_{\mathrm{sm}} + S$, with N_{sm} of paper Eq. (1) and S the fluctuating part. Integrating the N_{sm} piece by parts once more, $\int g' N_{\mathrm{sm}} = [g N_{\mathrm{sm}}]_{t_a}^{T} - \int g N'_{\mathrm{sm}}$, so

$$\sum_{t_n \le T} g(t_n) = g(T)S(T) + g(t_a)N_{\mathrm{sm}}(t_a) + \int_{t_a}^{T} g\, N'_{\mathrm{sm}}\,\mathrm{d}t - \int_{t_a}^{T} g'\, S\,\mathrm{d}t. \tag{6.51}$$

The derivative of Eq. (1) is

$$N'_{\mathrm{sm}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} - \frac{1}{2\pi} = \frac{1}{2\pi}\log\frac{t}{2\pi},$$

the average number of zeros per unit height (this is the density quoted in Section 6.3).

Step C: take $g(t) = e^{i\omega t}$ with $\omega > 0$. The second term, $g(t_a)N_{\mathrm{sm}}(t_a)$, is a constant that does not depend on T. The first term has modulus $|S(T)|$. Numerically $|S|$ is typically below 1 at these heights (Chapter 17). In general it can grow, but only slowly: $S(T) = \mathcal{O}(\log T)$ is a proved bound whose proof is beyond this book (see [6]). Either way this term is small compared with T.

For the integral of $e^{i\omega t}N'_{\mathrm{sm}}$, integrate by parts once more, using $\frac{\mathrm{d}}{\mathrm{d}t}\frac{e^{i\omega t}}{i\omega} = e^{i\omega t}$ and

$$N''_{\mathrm{sm}}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{1}{2\pi}\log\frac{t}{2\pi}\Big) = \frac{1}{2\pi t}:$$

$$\int_{t_a}^{T} e^{i\omega t}N'_{\mathrm{sm}}\,\mathrm{d}t = \Big[\frac{e^{i\omega t}}{i\omega}N'_{\mathrm{sm}}(t)\Big]_{t_a}^{T} - \frac{1}{i\omega}\int_{t_a}^{T}\frac{e^{i\omega t}}{2\pi t}\,\mathrm{d}t.$$

The boundary terms have moduli $\frac{1}{2\pi\omega}\left|\log\frac{T}{2\pi}\right|$ and $\frac{1}{2\pi\omega}\left|\log\frac{t_a}{2\pi}\right|$, using $|e^{i\omega t}| = 1$. The last integral has modulus at most $\frac{1}{2\pi\omega}\int_{t_a}^{T}\frac{\mathrm{d}t}{t} = \frac{1}{2\pi\omega}\log\frac{T}{t_a}$, by the bound $|\int g| \le \int|g|$ of Section 6.1. So the whole integral is of order $\frac{1}{\omega}\log T$, tiny compared with T. The interesting term is $-\int g'S\,\mathrm{d}t = -i\omega\int e^{i\omega t}S(t)\,\mathrm{d}t$.

Step D (heuristic): replace S by the explicit formula. Anticipating Chapter 18, take $S(t) \approx -\frac{1}{\pi}\sum_p\sum_m \frac{\sin(mt\log p)}{mp^{m/2}}$ (paper Eq. (4) with all primes). Suppose $\omega = \log x$ with $x = p^m$ a prime power. The matching ("resonant") term contributes

$$-i\omega\cdot\Big(-\frac{1}{\pi mp^{m/2}}\Big)\int_{t_a}^{T} e^{i\omega t}\sin(\omega t)\,\mathrm{d}t.$$

Since $e^{i\omega t}\sin\omega t = \cos\omega t\sin\omega t + i\sin^2\omega t$ and $\sin^2 = \frac{1}{2}(1-\cos 2\omega t)$, the integral is $iT/2$ plus bounded oscillating pieces (we neglect the small t_a). So the resonant contribution is

$$\frac{i\omega}{\pi mp^{m/2}}\cdot\frac{iT}{2} = -\frac{\omega T}{2\pi mp^{m/2}} = -\frac{T}{2\pi}\cdot\frac{m\log p}{mp^{m/2}} = -\frac{T}{2\pi}\frac{\log p}{\sqrt{x}},$$

using $\omega = \log x = m\log p$ and $p^{m/2} = \sqrt{x}$. Each non-resonant term gives a bounded contribution that does not grow with T. With the **von Mangoldt function** $\Lambda(x) = \log p$ if $x = p^m$ is a prime power and $\Lambda(x) = 0$ otherwise (Chapter 15),

$$\sum_{t_n\le T} e^{it_n\log x} \approx -\frac{T}{2\pi}\frac{\Lambda(x)}{\sqrt{x}} \qquad (x > 1). \tag{6.52}$$

Step D is heuristic: the prime sum does not converge on the critical line, and the infinitely many non-resonant terms are not controlled. Nevertheless a precise version of the conclusion is a theorem (box below). It is stated as a sum of x^ρ over the zeros ρ. It turns into (6.52), with an additional error of order $\log T$ (divided by $\sqrt{x}$), at every height up to which all zeros are known to lie on the critical line. This has been verified numerically for all the zeros used in this book. This error grows far more slowly than the main term, which is proportional to T.

Taken on trust

Taken on trust: Landau's formula. For fixed $x > 1$,

$$\sum_{0<\mathrm{Im}\,\rho\le T} x^\rho = -\frac{T}{2\pi}\Lambda(x) + \mathcal{O}(\log T),$$

where the sum runs over all nontrivial zeros ρ of zeta with imaginary part in $(0, T]$. In this form it is a theorem, and it does *not* assume that the zeros lie on the critical line. Now

suppose every zero up to height T has the form $\rho_n = \frac{1}{2} + it_n$. This is verified numerically for all zeros used in this book; that it holds at every height is the Riemann hypothesis, which is open. Then $x^{\rho_n} = x^{1/2}e^{it_n \log x}$ (Chapter 2), and dividing by $\sqrt{x}$ gives (6.52) plus an error $\mathcal{O}(\log T)/\sqrt{x}$. The proof is in Landau's 1912 paper [18]; Gonek [19] gave a version uniform in x.

So the zeros' spectrum has spikes, growing linearly in T, exactly at $\omega = \log p^m$, and nowhere else (apart from $\omega = 0$). Notice the weight: the periodogram of the *fluctuation* y_n has lines of size $\propto 1/(mp^{m/2})$, decreasing with p, while the zeros' own spectrum has spikes $\propto \log p/p^{m/2}$. The extra factor $\omega = m \log p$ came from g' in Step C: counting zeros is like differentiating the counting function.

Sample Problem 6.12: How tall is the $\log 2$ spike for the first 1000 zeros?

Use (6.52) with $T = t_{1000} = 1419.42$ (paper Sec. II A) to predict $\sum_{n\le 1000} \cos(t_n \log x)$ for $x = 2$ and $x = 3$, and compare with the value 1000 at $\omega = 0$.
Step 1: Prefactor. $T/2\pi = 1419.42/6.2832 = 225.91$.
Step 2: $x = 2$. $\Lambda(2)/\sqrt{2} = 0.69315/1.41421 = 0.49013$, so the prediction is $-225.91 \times 0.49013 = -110.7$.
Step 3: $x = 3$. $\Lambda(3)/\sqrt{3} = 1.09861/1.73205 = 0.63428$, prediction -143.3.
Step 4: Compare. At $\omega = 0$ every term is 1, so the sum is 1000. The $\log 2$ spike is about 11% of that, which is far above what random points would give (of order $\sqrt{1000} \approx 32$). A direct computation with `qmnt` (the listing below, and Figure 6.7) gives -111.1 at $\log 2$, within 1% of the prediction. Agreement this close is partly fortunate, and should not be expected in general. Landau's theorem (box above) has an error term of order $\log T$ with an unspecified constant, and $\log t_{1000} = 7.26$, while the main term is only about 111. Problem 25 repeats the comparison at the much larger height t_{10000}.

$$\boxed{\textstyle\sum_{n\le 1000} \cos(t_n \log 2) \approx -110.7, \qquad \sum_{n\le 1000} \cos(t_n \log 3) \approx -143.3.}$$

```
import numpy as np
import qmnt

t = qmnt.zeros(1000)
T = t[-1]                                   # height of the top zero
for x, Lam in [(2, np.log(2)), (3, np.log(3)), (4, np.log(2)), (5, np.log(5))]:
    w = np.log(x)
    measured = np.sum(np.cos(w*t))
    predicted = -(T/(2*np.pi))*Lam/np.sqrt(x)
    print(x, round(measured, 1), round(predicted, 1))
```

Two consequences for this chapter's tools. First, when you sample a function at the zeros, the spectral window W of (6.28) has these spikes, so a line at ω_0 reappears weakly at $\omega_0 \pm \log q$; for $T = t_{1000}$ the strongest such echoes (at $\log 5$, $\log 7$, $\log 11$, where $\Lambda(q)/\sqrt{q}$ is largest) are about a sixth of the main line. Second, it is the reason the fluctuation of the zeros is *predictable* by an arithmetic formula, the theme of Chapters 18 and 30.

Checkpoint 6.7

Is the spectral window of the zeros flat away from $\omega = 0$, as it would roughly be for random points?

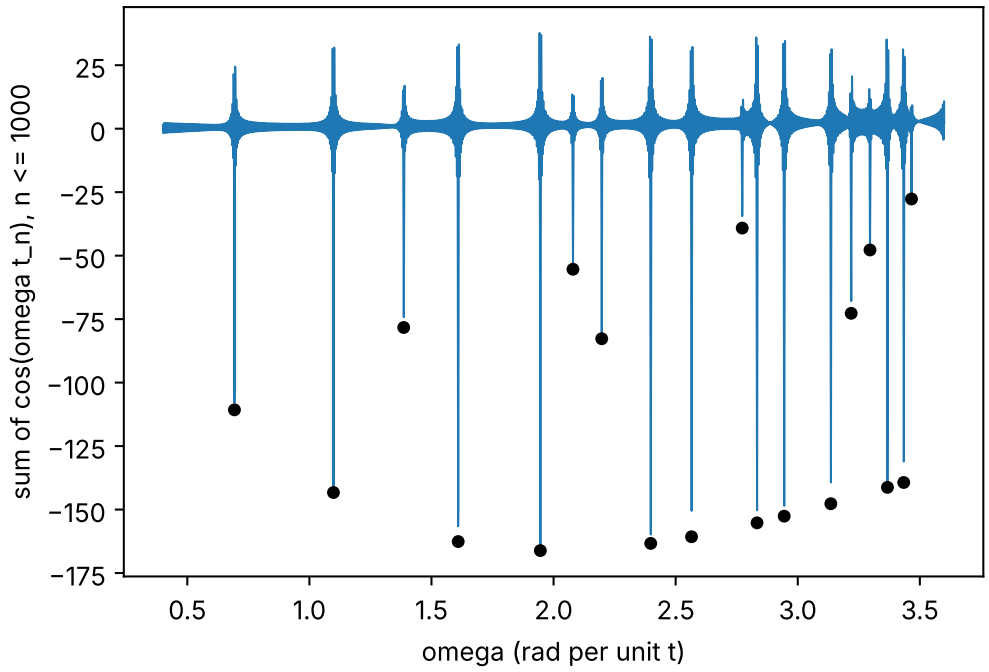


Figure 6.7: The real part of the spectral window of the zeros, $\sum_{n\le 1000}\cos(\omega t_n)$, for $0.4 \le \omega \le 3.6$. Downward spikes appear at $\omega = \log q$ for prime powers q; dots mark Landau's prediction $-(T/2\pi)\Lambda(q)/\sqrt{q}$ of Eq. (6.52) with $T = t_{1000}$. At $\omega = 0$ (off the plot) the sum is 1000.

Answer: No. By Landau's formula it has spikes of depth about $(T/2\pi)\Lambda(q)/\sqrt{q}$ at every $\omega = \log q$ with q a prime power, growing linearly with the height T.

Key Ideas

- Summation by parts, Eq. (6.50), turns sums over zeros into integrals against $N = N_{\mathrm{sm}} + S$; the average density of zeros is $\frac{1}{2\pi}\log\frac{t}{2\pi}$.
- Landau's formula (a theorem; proof beyond this book): for fixed $x > 1$, $\sum_{0<\mathrm{Im}\,\rho\le T} x^{\rho} = -\frac{T}{2\pi}\Lambda(x) + \mathcal{O}(\log T)$. Where all zeros up to height T lie on the critical line (verified numerically for the zeros used here), dividing by $\sqrt{x}$ gives Eq. (6.52) plus an error $\mathcal{O}(\log T)/\sqrt{x}$. The zeros' spectrum is concentrated at $\log p^m$.
- Our derivation of it from the explicit formula is heuristic; the size $\propto \log p/p^{m/2}$ differs from the $1/(mp^{m/2})$ of the fluctuation's lines by the factor ω.

6.9 Paper connection

Here is where each tool of this chapter enters the paper [1]. Section and equation numbers refer to revision R1.

6.9.1 The fitted sinusoidal controls of Sec. II D

Sec. II D (paragraph "Fitted sinusoidal controls") introduces models of the form $N_{\mathrm{sm}}(t) + F(t) = n - \frac{1}{2}$ with $F(t) = \sum_j [a_j \sin(\omega_j t) + b_j \cos(\omega_j t)]$, trained by least squares on exactly the quantity $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ of our Eq. (6.45), at the training zeros only. The fitting is linear least squares, the subject of Chapter 9; the choice of frequencies is Fourier analysis.

C1: a uniform frequency grid. "450 frequencies on a uniform grid (900 coefficients)", with $\omega_j = 8j/450$ (App. A step 20). The grid spacing is $8/450 = 0.01778$. For the fit on zeros 1–400 the Fourier resolution is $2\pi/T \approx 0.0094$ (Sample Problem "Resolving prime-power frequencies"), so the grid spacing is about 1.9 resolution elements. The grid treats every frequency from 0 to 8 alike; it has no preference for $\log p^m$. A line at $\log q$ generally falls between grid points, and by (6.18) the nearest grid frequency, up to 0.0089 away, drifts out of phase with it by up to $0.0089 \times 666 \approx 6$ radians across the window, so the model must build each line from several grid components. Table IV reports C1 at 0.258 with correlation +0.28 on zeros 401–500 (Sec. IV gives the correlation interval $[+0.08, +0.46]$), against a null of 0.263: "capacity without a suitable prior does not generalize" (Sec. IV). Why this happens is a question about extrapolating fitted combinations outside the window, which Chapter 27 analyzes.

C2: a periodogram peak search. "A sparse model whose K frequencies are found one at a time as the largest peak of the periodogram of the current residual on $0.05 < \omega < 8$, with all amplitudes refitted after each addition ($3K$ parameters)" (Sec. II D). Every ingredient is in Section 6.5: the periodogram (6.39) computed by a direct sum at the irregular points t_n; a search range; a coarse grid with "step a quarter of $2\pi/T$" and "refinement on 201 points" (App. A step 20). Our Sample Problem showed that a quarter-bin step catches at least 97.4% of any isolated line's amplitude. For the 400-zero window the coarse grid over $0.05 < \omega < 8$ has a little under 3400 points. The three parameters per frequency are ω_j, a_j and b_j. The procedure of subtracting the fit and searching the residual's periodogram again is a form of matching pursuit, developed in Chapter 27; ridge strength and K are chosen by a forward split inside the training window, also treated there. The paper reports that C2 "selects 24 frequencies (72 parameters) from 400 zeros and predicts zeros 401–500 at 0.034 with correlation +0.99", and from 800 zeros selects 43 frequencies and scores 0.034/+0.99 on 801–1000 (Sec. IV, Table IV). Figure 6.5 shows why such a search can work: the periodogram of y_n is dominated by peaks at $\log p^m$.

Frequencies above the "equivalent Nyquist". For the 800-zero fit, the paper lists the matched prime powers: "those up to 113 except 32, 64, and 81, and six primes between 131 and 211" (Sec. IV). Now $\log 211 = 5.35$, while the equivalent Nyquist frequency of the zeros at the top of that window is $\pi/\text{gap} = \frac{1}{2}\log(s_{800}/2\pi) \approx 2.62$ (Section 6.3). Uniformly spaced samples could not have distinguished these frequencies from aliases below 2.62. That the search recovered them is a numerical observation of the paper. It is compatible with the fact (Section 6.3) that irregular sampling breaks exact aliasing. However, the zeros are not random sample points, and the mechanism is not established here. We claim nothing beyond this compatibility.

6.9.2 "Within half the Fourier resolution" and the alignment check

Sec. IV states: "Every selected frequency lies within half the Fourier resolution of the training window of the logarithm of a prime power." Half the Fourier resolution is π/T; App. A step 23 makes this the matching criterion $|\omega - \log q| < \pi/T$, with $q \le 3000$. The upper limit fits the search range: $e^8 = 2981$, so $q \le 3000$ covers every $\log q$ below the top of the range $\omega < 8$. The criterion is exploratory by the paper's own account: the cutoff-maximized binomial score (24.7 and 50.9 on the real zeros, at most 4.7 on 900 surrogate replicates) was "defined after the data were seen" (App. A step 23), and Sec. IV states that "no significance claim is made here beyond the size of the ensembles." One of the three surrogate families in step 23 is "sorted phase-randomized surrogates". The other two are sparse signals with non-arithmetic frequencies: explicit formulas with 25 displaced prime frequencies, and explicit-formula shapes with 25 random base frequencies. The phase randomization is the construction of Section 6.7. The sorting presumably restores an increasing list that can again be read as heights, but the paper does not spell this out. Why matching within π/T is a sensible tolerance follows from

Section 6.2: for an isolated line the periodogram maximum lies at the line itself. Noise and the sidelobes of neighboring lines displace it by a fraction of the main-lobe half-width $2\pi/T$, so half a resolution element, π/T, is a reasonable tolerance. For the 400-zero window $\pi/T \approx 0.0047$. Chapter 27 discusses how often a random frequency would match some $\log q$ by chance, which is why surrogate replicates were needed at all.

6.9.3 Shuffled and phase-randomized targets in Sec. III

The paragraph "Permutation and surrogate tests" of Sec. III refits the operator protocol (scale 1, $M = 400$, zeros 1–400) to targets in which the fluctuation sequence in mean spacings, $d_n = (\tilde{t}_n - E_n[\text{smooth}])/\text{gap}_n$ (App. A step 18), is replaced by a random permutation ("marginal distribution kept, serial structure destroyed"), a phase-randomized surrogate [16] ("power spectrum kept exactly, no arithmetic content") or an IAAFT surrogate [17] ("marginal kept exactly, spectrum to 1%"). These three phrases are precisely the preserved/destroyed lists of Section 6.7. Step 18 records the construction: "real FFT, uniform phases", seeds 1–10, which is our recipe with `rfft/irfft`. Note that the FFT is legitimate here because d_n is indexed by the integer n and is therefore uniformly sampled in n (pitfall in Section 6.4).

The results: in-sample rms 0.134 ± 0.010, 0.141 ± 0.006 and 0.138 ± 0.005 over ten draws each, against 0.137 for the true zeros and 0.244 for the null (Sec. III). The phase-randomized surrogates keep the autocorrelation of the true d_n, including the negative values at lags 1–3 that Figure 6.6 shows, but destroy the arithmetic phase structure; the permuted ones destroy even the autocorrelation. The fit does equally well on all of them. In the paper's words, "in-sample accuracy measures capacity, not captured structure." Phase randomization keeps the power spectrum of d_n, in frequency per level, because the FFT is taken over the index n. It scrambles the phases, however. So the specific arithmetic pattern in t, sines at $\log p^m$ with phases fixed by the explicit formula, is no longer present; the paper summarizes this as "no arithmetic content". Note that a line at frequency ω in t is not a sharp line in n. Near level n it advances by about $\omega\,\text{gap}_n$ radians per level, and gap_n shrinks across the window, so its frequency per level drifts.

6.9.4 Landau, Gonek and the line spectrum

Sec. IV writes: "That the Fourier transform of the zeros is concentrated at logarithms of prime powers is classical [18, 19]; C2 is a control built on that fact, not a new result." Section 6.8 stated Landau's formula and checked it numerically (Figure 6.7). Keep the paper's hedge: C2 succeeds because "sparse in t-frequency" is "itself a strong and well-chosen prior" (Sec. IV), and the facts of this chapter tell you why the prior is well chosen.

6.9.5 Resolution and the decay of skill

The paper notes that C2's "skill also decays with distance from the training window as the estimated frequencies dephase (0.078/+0.96 on zeros 1001–2000 for the 400-zero fit), which the exact frequencies of the oracle do not" (Sec. IV). Inequality (6.18) is the seed of this: a window of length T fixes a frequency only to a fraction of $2\pi/T$, and an error $\delta\omega$ grows into a phase error $\delta\omega\,\Delta t$ at distance Δt. Chapter 27 derives the consequences. The oracle of paper Eq. (4) uses the exact $m \log p$ and has no such error, which is consistent with its nearly height-independent skill in Table V.

6.9.6 The continuation probe

The "Arithmetic-continuation probe" of Sec. III trains the operator on the prime-2 wave RE2 alone, a signal whose spectrum is three lines (Section 6.2): it represents the window in-sample (rms 0.056, correlation +0.95) and "continues it not at all" (held-out correlation +0.07). From

the Fourier point of view, a model that knew "three lines at $m \log 2$" would continue it exactly; Chapter 5 showed why no finite window's periodic extension can. Figure 1 of the paper shows the same contrast on real data.

6.10 Summary

★ Did you know? *Gauss Did the FFT*

The fast Fourier transform, published by James Cooley and John Tukey in 1965, cut the cost of a discrete Fourier transform of N points from about N^2 to about $N \log N$ operations. Historians later found that Carl Friedrich Gauss had used essentially the same trick around 1805 to interpolate asteroid orbits. His Latin notes appeared only in his posthumous collected works.

- Fourier transform $\hat{f}(\omega) = \int f e^{-i\omega t} \mathrm{d}t$, Eq. (6.6), inverse Eq. (6.7) (proof beyond); properties P1–P5: linearity, $\hat{f}(-\omega) = \overline{\hat{f}(\omega)}$ for real f (6.9), shift (6.10), modulation (6.11), scaling.
- A line seen through a window of length T becomes $T \operatorname{sinc}((\omega - \omega_0)T/2)$, Eq. (6.13); resolution $2\pi/T$, Eq. (6.17); largest sidelobe 0.217; Hann window (6.24) has sidelobes 0.027 but a doubled main lobe.
- Almost periodic sums have line spectra, Eq. (6.20); paper Eq. (4) has lines at $\log p^m$ of size $1/(2\pi m p^{m/2})$.
- Uniform sampling aliases ω with $\omega + 2\pi/\Delta t$; Nyquist frequency $\pi/\Delta t$, Eq. (6.26). Irregular samples: direct sum (6.27), spectral window (6.28).
- DFT (6.30), inverse (6.33) from orthogonality (6.32), symmetry (6.34), Parseval (6.35); FFT cost $N \log_2 N$.
- Periodogram (6.39): line height $NA^2/4$, noise level σ^2, Eq. (6.44); search on a sub-bin grid and refine.
- Discrete Wiener–Khinchin, Eq. (6.48): periodogram and circular autocorrelation carry the same information.
- Phase randomization keeps $|X_k|$, autocorrelation, mean and sum of squares; destroys phases and the marginal.
- Landau's formula (6.52): the zeros' spectrum has spikes at $\log p^m$ of depth $\frac{T}{2\pi}\Lambda(x)/\sqrt{x}$.

6.11 Problems

1. • Using the box transform of the first Sample Problem, find $\hat{f}(\omega)$ at $\omega = \pi/T$, as a multiple of T.

2. • Samples are taken every $\Delta t = 0.25$. Find the Nyquist frequency, and the frequency in $[0, \omega_{\mathrm{N}}]$ at which a component with $\omega = 15$ appears.

3. • What window length T is needed to resolve $\log 3$ from $\log 4$ by the criterion (6.17)?

4. • Compute the DFT of $x = (1, 0, 0, 0)$ and check Parseval's relation.

5. • For $N = 2^{20}$, find the speed-up factor of the FFT over the direct DFT.

6. •• For $f = e^{-a|t|}$ with $a = 2$, find $\hat{f}(0)$ and the FWHM. Verify that the product (FWHM)$\times(1/a)$ is the same as for $a = 0.5$.

7. •• Show that observing $e^{i\omega_0 t}$ on $[0, T]$ instead of $[-T/2, T/2]$ gives a transform with the same modulus as Eq. (6.13), and find the extra phase factor.

8. •• A sinusoid of amplitude $A = 0.05$ is buried in white noise of rms $\sigma = 0.5$, with $N = 1000$ samples. Estimate the ratio of its periodogram peak to the mean noise level.

9. •• Compute the DFT of $x = (1, -1, 1, -1)$. Which frequency is present, and why is it the Nyquist frequency?

10. •• Show that the Hann expression (6.24) tends to $\frac{1}{4}$ as $\delta \to 1$. (Use $\sin(\pi\delta) \approx -\pi(\delta - 1)$ near $\delta = 1$.)

11. •• A coarse grid has step one half of a bin. Find the worst-case fraction of a line's amplitude seen on the grid (rectangular window).

12. •• (Paper.) For the 400-zero window, compare $|\log 128 - \log 127|$ with the matching tolerance π/T of App. A step 23 and with the resolution $2\pi/T$. What does this say about whether the search could distinguish these two prime powers?

13. •• (Paper.) For the fit on zeros 1–800, take $T = t_{800} - t_1$. Using `qmnt`, find $2\pi/T$ and the ratio of the C1 grid spacing $8/450$ to it.

14. •• (Computational.) Using `qmnt` and `numpy`, compute the periodogram of RE2 sampled at zeros 1–400 and list the three largest local maxima in $0.3 < \omega < 2.5$. Compare their positions with $m \log 2$.

15. •• Use Landau's formula (6.52) with $T = t_{1000}$ to predict $\sum_{n \le 1000} \cos(t_n \log 4)$.

16. •• Explain why the mean should be subtracted before computing a periodogram, using the size $N\bar{y}^2$ of the line at $\omega = 0$ and the sidelobes of the spectral window.

17. •• (Computational.) Using `numpy`, phase-randomize $d_n =$`qmnt.fluct`(t_n, n) for $n \le 400$ with seeds 1–10. Verify that the circular autocorrelation is unchanged and compare the largest value of $|d_n|$ before and after.

18. •• Complete the phase-randomization Sample Problem: verify that $r'_2 = 0$ and $r'_3 = 1$ for $x' = (1.2071, 0.5, -0.2071, 0.5)$.

19. •• (Paper.) Sec. III reports in-sample rms 0.134, 0.141, 0.138 on permuted, phase-randomized and IAAFT targets, 0.137 on the true zeros and 0.244 for the null. Compute the fraction of the null removed in each case, and say what this does and does not show.

20. •• For $x = (1, 1, 0, 0)$ compute the ordinary autocorrelation c_m of (6.46) and compare with the circular r_m. Then verify the zero-padding recipe by computing the circular autocorrelation of $(1, 1, 0, 0, 0, 0, 0, 0)$ at lags 0–3.

21. ••• Prove Parseval's relation (6.35) directly from the orthogonality relation (6.32), for complex x.

22. ••• Show from (6.24) that the Hann sidelobes fall like $1/|\delta|^3$ for large $|\delta|$ and find the constant.

23. ••• (Paper.) Compute the equivalent Nyquist frequency $\frac{1}{2}\log(s_{1000}/2\pi)$ at the top of the 1000-zero range used for the operator fits, and compare with $\log 211$.

24. • • • Show that if $t_j = j\Delta t$, the samples of $e^{i\omega t}$ and $e^{i(\omega+2\pi k/\Delta t)t}$ coincide for every integer k, and that for real cosines $\cos(\omega t)$ and $\cos((2\pi/\Delta t - \omega)t)$ coincide.

25. • • • (Computational.) Using `qmnt.zeros(10000)`, compute $\sum_{n\le 10^4} \cos(t_n \log 2)$ and compare with Landau's prediction for $T = t_{10000}$.

6.12 Answers to selected problems

1. $\hat f(\pi/T) = 2\sin(\pi/2)/(\pi/T) = 2T/\pi \approx 0.637\,T$.

2. $\omega_{\mathrm{N}} = \pi/0.25 = 12.57$. The alias period is $2\pi/0.25 = 25.13$; 15 is confused with $25.13 - 15 = 10.13$.

3. $\log(4/3) = 0.2877$, so $T \gtrsim 2\pi/0.2877 = 21.8$.

4. Every phase factor multiplies $x_0 = 1$ only at $j = 0$, so $X_k = 1$ for all k: $X = (1,1,1,1)$. Parseval: $1 = \frac{1}{4}(4)$.

5. $N^2/(N\log_2 N) = N/\log_2 N = 2^{20}/20 = 52428.8$, a speed-up of about 5×10^4.

7. Let g_c be the centered windowed oscillation of Eq. (6.13), and let g be the one observed on $[0,T]$.

For $0 \le t \le T$ we have $g_c(t - T/2) = e^{i\omega_0(t-T/2)}$. Hence $g(t) = e^{i\omega_0 T/2}\, g_c(t - T/2)$ for every t, since both sides vanish outside $[0,T]$.

By linearity (P1) and the shift property (6.10) with $t_0 = T/2$,

$$\hat g(\omega) = e^{i\omega_0 T/2} e^{-i\omega T/2} \hat g_c(\omega) = e^{-i(\omega-\omega_0)T/2}\, \hat g_c(\omega).$$

The extra phase factor $e^{-i(\omega-\omega_0)T/2}$ has modulus 1, so $|\hat g| = |\hat g_c|$.

9. $x_j = (-1)^j = e^{i\pi j}$, so only $k = 2$ survives: $X = (0,0,4,0)$. The frequency $\omega_2 = 2\cdot 2\pi/(4\Delta t) = \pi/\Delta t$, two samples per cycle, is the Nyquist frequency.

11. With a step of half a bin, the worst offset is a quarter bin. Then $s(\frac{1}{4}) = \sin(\pi/4)/(\pi/4) = 0.9003$, that is, 90.0% of the amplitude.

13. With `qmnt`, $T = t_{800} - t_1 \approx 1169$, so $2\pi/T \approx 0.0054$; the C1 grid spacing 0.01778 is about 3.3 resolution elements. A generic grid becomes coarser relative to the resolution as the window grows.

15. $\Lambda(4) = \log 2$, $\sqrt{4} = 2$: prediction $-225.91 \times 0.34657 = -78.3$.

17. The circular autocorrelations agree to rounding error for all seeds (Wiener–Khinchin). The largest $|d_n|$ changes from draw to draw: the marginal distribution is not preserved, which is why IAAFT [17] exists.

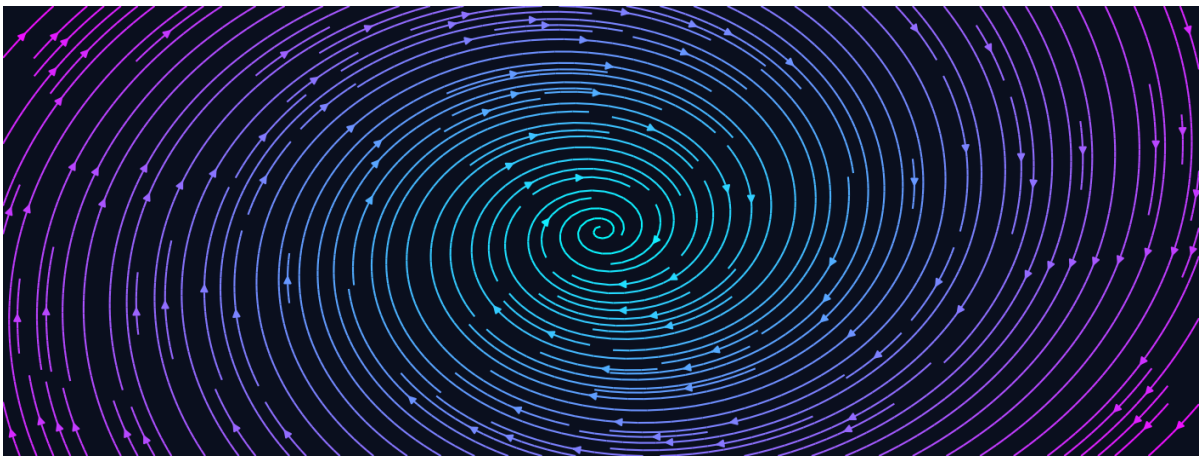
19. Fractions of the null removed: $1 - 0.134/0.244 = 45\%$, 42%, 43%, against 44% for the true zeros. The fit gains the same on targets that lack the arithmetic phase structure (and, for permutation, the autocorrelation). It shows the in-sample gain measures capacity; it does not by itself show anything about held-out skill, which Sec. III tests separately.

20. $c = (2,1,0,0)$ while $r = (2,1,0,1)$: $c_3 = x_0x_3 = 0$, whereas r_3 also contains the wrap-around terms $x_1x_0 + x_2x_1 + x_3x_2$, of which $x_1x_0 = 1$. Padded: $(2,1,0,0)$ at lags 0–3, equal to c.

21. $\sum_k |X_k|^2 = \sum_k \sum_j \sum_l x_j \overline{x_l} e^{-2\pi i k(j-l)/N} = \sum_j \sum_l x_j \overline{x_l}\, N[j = l] = N \sum_j |x_j|^2$, using (6.32) with $m = l - j$.

23. $s_{1000} \approx 1419.4$, so $\frac{1}{2}\log(s_{1000}/2\pi) \approx 2.71$, below $\log 211 = 5.35$ by a factor of about 2.

25. With $t_{10000} \approx 9.88\times 10^3$ from `qmnt.zeros(10000)`, Landau's prediction is $-(t_{10000}/2\pi)\log 2/\sqrt{2} \approx -770.5$. The listing of Section 6.8, with 1000 replaced by 10000, gives a measured sum of about -771.6. The relative discrepancy $|\text{measured} - \text{predicted}|\,/\,|\text{predicted}|$ is 0.14%, against 0.35% at $T = t_{1000}$. The theorem's error term is $\mathcal{O}(\log T)$ while the main term grows like T, so the relative error *bound* shrinks with height. A single height is not guaranteed to show a smaller discrepancy, because the constant in the error term is not specified.



Flow lines of $\dot{\mathbf{x}} = A\mathbf{x}$ for a 2×2 matrix A with complex eigenvalues: every path spirals inward.

Chapter 7

Vectors, Matrices and Linear Systems

↑ Contents

What this chapter is about

In first-year physics a vector is an arrow: a displacement, a velocity, a force, with two or three components. This chapter keeps everything you know about arrows (adding, scaling, the dot product, length, perpendicularity) and drops one restriction, the number of components. A list of 400 numbers turns out to obey the same rules as a list of three, and once you accept that, a great deal of computation becomes geometry. "How far is my prediction from the truth?" becomes the length of a difference vector. "Is my prediction shaped like the truth?" becomes the angle between two vectors. "Can I find coefficients that reproduce these data exactly?" becomes the question of whether a system of linear equations has a solution, and how many.

We also introduce matrices. A matrix is a rectangular table of numbers, but its real meaning is a rule that turns one vector into another in a straight-line (linear) way. We learn to multiply matrices, to undo them (the inverse), to solve the equations they define by elimination, to measure how many independent pieces of information they carry (the rank), and to compute the determinant. We end by treating functions themselves as vectors with very many components, which is exactly what a computer does when it stores a wavefunction on a grid.

Where this shows up in the paper [1]: the fitted correction $\delta V(x) = \sum_k c_k \cos(k\pi x/L)$ of Sec. II C is a basis expansion with a coefficient vector c; the residuals $r_n = \tilde{t}_n - E_n$ form a residual vector; the sensitivity $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\,\mathrm{d}x$ is a matrix with one row per level and one column per basis function; every rms in every table is the length of a vector divided by the square root of its number of components; and statements such as "at 900 coefficients on 400 zeros the least-squares problem is underdetermined" (Sec. IV) are statements about linear systems with more unknowns than equations.

What you need

- Chapter 2: complex numbers $z = a + ib$, the complex conjugate $z^* = a - ib$, the modulus $|z|$ with $|z|^2 = z^* z$, Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$.
- First-year physics: vectors in two and three dimensions, components, the dot product $\boldsymbol{a}\cdot\boldsymbol{b} = a_x b_x + a_y b_y + a_z b_z$, the magnitude $|\boldsymbol{a}| = \sqrt{\boldsymbol{a}\cdot\boldsymbol{a}}$, and the relation $\boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}|\,|\boldsymbol{b}|\cos\theta$.
- Calculus: the integral as a limit of Riemann sums, integration by parts, the identities $\cos^2\theta + \sin^2\theta = 1$, $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ and $\cos A\cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$.
- Appendix A (Python) for the code listings: arrays, `@` for matrix products, `numpy.linalg`.

7.1 From arrows to lists of numbers

A displacement in the laboratory is written $\boldsymbol{r} = (x, y, z)$: three numbers, in a fixed order. Nothing in the rules for adding displacements or stretching them depends on there being exactly three. Add two displacements and you add their components one by one; double a displacement and you double each component. So we make the following definition.

Definition. A *real vector with n components* is an ordered list of n real numbers,

$$\boldsymbol{v} = (v_1, v_2, \ldots, v_n). \tag{7.1}$$

The number v_j is the j-th *component*. The set of all such lists is called $\mathbb{R}^n$ (read "R-n"). When we want to stress the arrangement we write the list as a column,

$$\boldsymbol{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}.$$

In this chapter vectors are printed in bold. The paper prints them in plain type: its coefficient vector is simply c.

Two examples from the running examples of this book:

- The heights of the first ten zeta zeros (running example RE1),

  $$\boldsymbol{t} = (14.134725,\ 21.022040,\ 25.010858,\ 30.424876,\ 32.935062,\ 37.586178,\ 40.918719,\ 43.327073,\ 48.00515$$

  form a vector in $\mathbb{R}^{10}$. You do not need to know yet where these numbers come from (Chapter 16 explains); here they are just ten ordered numbers.

- In the paper's fits, the correction to the potential is described by M numbers $c_0, c_1, \ldots, c_{M-1}$, so the coefficient vector $\boldsymbol{c}$ lies in $\mathbb{R}^M$; for $M = 400$ it has 400 components. (The paper starts this index at 0, not 1; we return to this in a pitfall below.)

Addition and scaling. For $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ and a real number α (a *scalar*) we define

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1,\ u_2 + v_2,\ \ldots,\ u_n + v_n), \tag{7.2}$$

$$\alpha\boldsymbol{v} = (\alpha v_1,\ \alpha v_2,\ \ldots,\ \alpha v_n). \tag{7.3}$$

The *zero vector* $\boldsymbol{0} = (0, \ldots, 0)$ satisfies $\boldsymbol{v} + \boldsymbol{0} = \boldsymbol{v}$, and $-\boldsymbol{v} = (-1)\boldsymbol{v}$ satisfies $\boldsymbol{v} + (-\boldsymbol{v}) = \boldsymbol{0}$. Because each component is an ordinary real number, the familiar rules carry over component by component. For example, the distributive rule $\alpha(\boldsymbol{u} + \boldsymbol{v}) = \alpha\boldsymbol{u} + \alpha\boldsymbol{v}$ holds because its j-th component reads $\alpha(u_j + v_j) = \alpha u_j + \alpha v_j$, which is the distributive law for real numbers. In the same way $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$ and $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$.

Linear combinations. An expression

$$\alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \cdots + \alpha_k \boldsymbol{v}_k \tag{7.4}$$

with scalars α_i and vectors $\boldsymbol{v}_i$ is called a *linear combination* of the $\boldsymbol{v}_i$. Almost everything in this chapter is about linear combinations: which vectors can be built from which, and with what coefficients.

Figure 7.1 (left) shows addition in $\mathbb{R}^2$, where you can still draw arrows: $\boldsymbol{u}+\boldsymbol{v}$ is the diagonal of the parallelogram with sides $\boldsymbol{u}$ and $\boldsymbol{v}$. In $\mathbb{R}^{400}$ you cannot draw the picture, but every algebraic statement we prove holds there just as well, and the two-dimensional picture remains an honest guide to intuition.

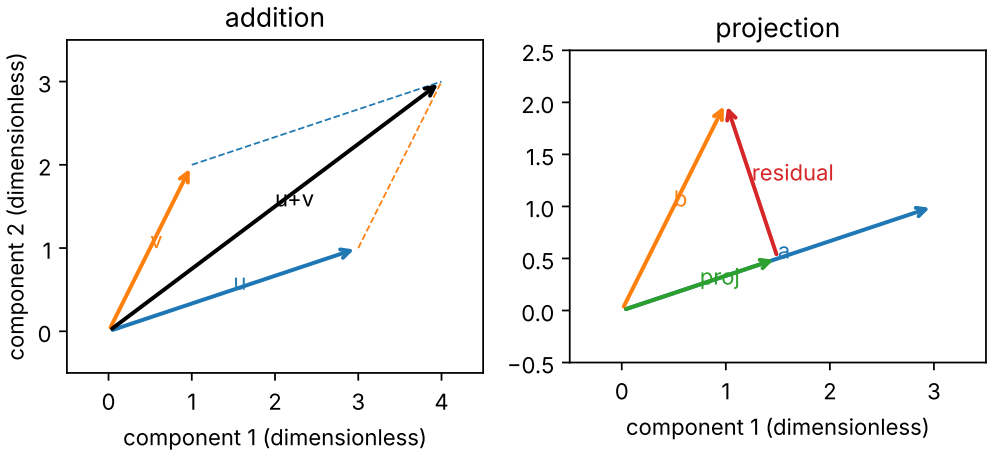


Figure 7.1: Left: addition of $\boldsymbol{u} = (3, 1)$ and $\boldsymbol{v} = (1, 2)$ gives the diagonal $\boldsymbol{u} + \boldsymbol{v} = (4, 3)$ of the parallelogram they span. Right: the projection of $\boldsymbol{b} = (1, 2)$ onto the direction of $\boldsymbol{a} = (3, 1)$ is $(1.5, 0.5)$; the residual $\boldsymbol{b} - (1.5, 0.5) = (-0.5, 1.5)$ is perpendicular to $\boldsymbol{a}$. Components are dimensionless.

Checkpoint 7.1

A model predicts the vector $\boldsymbol{p} = (1.0,\ 2.0,\ 3.0)$ and the truth is $\boldsymbol{d} = (1.5,\ 1.5,\ 3.5)$. What is the error vector $\boldsymbol{p} - \boldsymbol{d}$?

Answer: $\boldsymbol{p}-\boldsymbol{d} = (1.0-1.5,\ 2.0-1.5,\ 3.0-3.5) = (-0.5,\ 0.5,\ -0.5)$, computed component by component from Equations (7.2) and (7.3) with $\boldsymbol{p} - \boldsymbol{d} = \boldsymbol{p} + (-1)\boldsymbol{d}$.

Key Ideas

- A vector in $\mathbb{R}^n$ is an ordered list of n real numbers; addition and scaling act component by component.
- All the algebraic rules of 3D vectors hold in $\mathbb{R}^n$ because they hold for each component.
- A linear combination $\sum_i \alpha_i \boldsymbol{v}_i$ is the basic object of linear algebra.

7.2 Length, angle and the inner product

7.2.1 The inner product and the norm

In three dimensions $\boldsymbol{a} \cdot \boldsymbol{b} = a_x b_x + a_y b_y + a_z b_z$. The obvious generalization is the following.

Definition. The *inner product* (or dot product) of $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ is

$$\boldsymbol{u} \cdot \boldsymbol{v} = \sum_{j=1}^{n} u_j v_j = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n. \tag{7.5}$$

The *norm* (length) of $\boldsymbol{v}$ is

$$\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \tag{7.6}$$

and the *distance* between $\boldsymbol{u}$ and $\boldsymbol{v}$ is $\|\boldsymbol{u} - \boldsymbol{v}\|$.

Three properties follow directly from Equation (7.5):

1. Symmetry: $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$, because $u_j v_j = v_j u_j$ for each j.
2. Linearity: $\boldsymbol{u} \cdot (\alpha \boldsymbol{v} + \beta \boldsymbol{w}) = \sum_j u_j (\alpha v_j + \beta w_j) = \alpha \sum_j u_j v_j + \beta \sum_j u_j w_j = \alpha\, \boldsymbol{u} \cdot \boldsymbol{v} + \beta\, \boldsymbol{u} \cdot \boldsymbol{w}$.
3. Positivity: $\boldsymbol{v} \cdot \boldsymbol{v} = \sum_j v_j^2 \geq 0$, and it equals zero only if every $v_j = 0$, because a sum of squares of real numbers vanishes only when each square does.

A vector with $\|\boldsymbol{v}\| = 1$ is a *unit vector*. Any nonzero vector can be *normalized*: $\hat{\boldsymbol{v}} = \boldsymbol{v} / \|\boldsymbol{v}\|$ has norm $\|\boldsymbol{v}\| / \|\boldsymbol{v}\| = 1$, using $\|\alpha \boldsymbol{v}\| = \sqrt{\sum_j \alpha^2 v_j^2} = |\alpha|\, \|\boldsymbol{v}\|$.

7.2.2 The rms is a norm in disguise

The paper scores every prediction by an "rms", a root mean square error. If a model makes errors $e_1, \ldots, e_n$ (in the paper, each error is measured in units of the local mean spacing between zeros, Chapter 17), then

$$\operatorname{rms}(\boldsymbol{e}) = \sqrt{\frac{1}{n} \sum_{j=1}^{n} e_j^2} = \frac{\|\boldsymbol{e}\|}{\sqrt{n}}. \tag{7.7}$$

The second equality is Equation (7.6) with the factor $1/n$ moved outside the square root as $1/\sqrt{n}$. So the rms is the length of the error vector, rescaled so that it does not grow merely because you have more components. This is why rms values for 100 held-out zeros and for 500 held-out zeros can be compared.

Sample Problem 7.1: Norm and rms of a residual vector

A fit leaves residuals $\boldsymbol{r} = (0.3, -0.1, 0.2, -0.4)$ (in mean spacings) on four levels. Find $\|\boldsymbol{r}\|$ and $\operatorname{rms}(\boldsymbol{r})$.

Step 1: Square and add the components: $\boldsymbol{r} \cdot \boldsymbol{r} = 0.09 + 0.01 + 0.04 + 0.16 = 0.30$.

Step 2: The norm is $\|\boldsymbol{r}\| = \sqrt{0.30} = 0.5477$.

Step 3: By Equation (7.7), $\operatorname{rms}(\boldsymbol{r}) = \|\boldsymbol{r}\| / \sqrt{4} = \sqrt{0.30/4} = \sqrt{0.075}$.

$$\boxed{\|\boldsymbol{r}\| = 0.5477, \qquad \operatorname{rms}(\boldsymbol{r}) = 0.2739.}$$

7.2.3 The Cauchy–Schwarz inequality and angles

In three dimensions $\boldsymbol{a} \cdot \boldsymbol{b} = |\boldsymbol{a}|\, |\boldsymbol{b}| \cos\theta$, so $|\boldsymbol{a} \cdot \boldsymbol{b}| \leq |\boldsymbol{a}|\, |\boldsymbol{b}|$. We want to define an angle in $\mathbb{R}^n$, and for that we first need the same inequality there, proved without pictures.

Theorem (Cauchy–Schwarz). For all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$,

$$|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| . \tag{7.8}$$

Proof. If $\boldsymbol{v} = \boldsymbol{0}$ both sides are zero. Otherwise, for every real number λ the vector $\boldsymbol{u} - \lambda \boldsymbol{v}$ has a non-negative squared norm. Expanding with the linearity and symmetry properties,

$$\begin{aligned} 0 \leq \|\boldsymbol{u} - \lambda \boldsymbol{v}\|^2 &= (\boldsymbol{u} - \lambda \boldsymbol{v}) \cdot (\boldsymbol{u} - \lambda \boldsymbol{v}) \\ &= \boldsymbol{u} \cdot \boldsymbol{u} - \lambda \, \boldsymbol{u} \cdot \boldsymbol{v} - \lambda \, \boldsymbol{v} \cdot \boldsymbol{u} + \lambda^2 \, \boldsymbol{v} \cdot \boldsymbol{v} \\ &= \|\boldsymbol{u}\|^2 - 2\lambda \, \boldsymbol{u} \cdot \boldsymbol{v} + \lambda^2 \, \|\boldsymbol{v}\|^2 . \end{aligned}$$

This holds for every λ; choose $\lambda = (\boldsymbol{u} \cdot \boldsymbol{v}) / \|\boldsymbol{v}\|^2$. Then

$$\begin{aligned} 0 &\leq \|\boldsymbol{u}\|^2 - 2\frac{(\boldsymbol{u} \cdot \boldsymbol{v})^2}{\|\boldsymbol{v}\|^2} + \frac{(\boldsymbol{u} \cdot \boldsymbol{v})^2}{\|\boldsymbol{v}\|^4} \|\boldsymbol{v}\|^2 \\ &= \|\boldsymbol{u}\|^2 - \frac{(\boldsymbol{u} \cdot \boldsymbol{v})^2}{\|\boldsymbol{v}\|^2}. \end{aligned}$$

Multiplying by $\|\boldsymbol{v}\|^2 > 0$ gives $(\boldsymbol{u} \cdot \boldsymbol{v})^2 \leq \|\boldsymbol{u}\|^2 \, \|\boldsymbol{v}\|^2$, and taking the non-negative square root gives Equation (7.8). □

Because of Equation (7.8), for nonzero vectors the ratio $\boldsymbol{u} \cdot \boldsymbol{v} / (\|\boldsymbol{u}\| \, \|\boldsymbol{v}\|)$ lies between -1 and $+1$. We may therefore *define* the angle θ between them by

$$\cos\theta = \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{\|\boldsymbol{u}\| \, \|\boldsymbol{v}\|}, \qquad 0 \leq \theta \leq \pi. \tag{7.9}$$

In $\mathbb{R}^3$ this is the ordinary angle; in $\mathbb{R}^n$ it is the definition. Two vectors are *orthogonal* if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$ (angle $\pi/2$).

Triangle inequality. $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$. Proof: $\|\boldsymbol{u} + \boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + 2\,\boldsymbol{u} \cdot \boldsymbol{v} + \|\boldsymbol{v}\|^2 \leq \|\boldsymbol{u}\|^2 + 2\,\|\boldsymbol{u}\| \, \|\boldsymbol{v}\| + \|\boldsymbol{v}\|^2 = (\|\boldsymbol{u}\| + \|\boldsymbol{v}\|)^2$, where the inequality is Equation (7.8). Take square roots of both (non-negative) sides.

Checkpoint 7.2

Are $(1, 2, 3)$ and $(3, 0, -1)$ orthogonal? Is $(1, 1, 1, 1)$ orthogonal to $(1, -1, 1, -1)$?

Answer: Yes to both: $1 \cdot 3 + 2 \cdot 0 + 3 \cdot (-1) = 0$, and $1 - 1 + 1 - 1 = 0$.

7.2.4 Prediction, truth and the angle between them

The paper reports, for each prediction, both an rms and a correlation. Linear algebra shows that these measure different things and how they are related.

Let $\boldsymbol{d}$ be the vector of true values (say, true fluctuations of the zeros) and $\boldsymbol{p}$ the vector of predicted values. The *null* prediction is $\boldsymbol{p} = \boldsymbol{0}$ ("no fluctuation"); its error vector is $-\boldsymbol{d}$ and its error norm is $\|\boldsymbol{d}\|$. For any prediction, expanding as in the Cauchy–Schwarz proof,

$$\begin{aligned} \|\boldsymbol{p} - \boldsymbol{d}\|^2 &= \|\boldsymbol{p}\|^2 - 2\,\boldsymbol{p} \cdot \boldsymbol{d} + \|\boldsymbol{d}\|^2 \\ &= \|\boldsymbol{p}\|^2 - 2\,\|\boldsymbol{p}\| \, \|\boldsymbol{d}\| \cos\theta + \|\boldsymbol{d}\|^2 , \qquad \text{(by Equation (7.9))} \end{aligned} \tag{7.10}$$

with θ the angle between $\boldsymbol{p}$ and $\boldsymbol{d}$. The prediction beats the null exactly when $\|\boldsymbol{p} - \boldsymbol{d}\|^2 < \|\boldsymbol{d}\|^2$, that is, when $\|\boldsymbol{p}\|^2 < 2\,\|\boldsymbol{p}\| \, \|\boldsymbol{d}\| \cos\theta$. For $\boldsymbol{p} \neq \boldsymbol{0}$ divide by $\|\boldsymbol{p}\|$:

$$\|\boldsymbol{p} - \boldsymbol{d}\| < \|\boldsymbol{d}\| \quad \Longleftrightarrow \quad \|\boldsymbol{p}\| < 2\,\|\boldsymbol{d}\| \cos\theta. \tag{7.11}$$

Two consequences follow at once.

- If $\cos\theta \le 0$, the right side of Equation (7.11) is ≤ 0, which no positive $\|\boldsymbol{p}\|$ can be below. So *every nonzero prediction that is orthogonal or anti-aligned to the truth is worse than predicting nothing.*

- If we keep the direction $\hat{\boldsymbol{p}}$ fixed and choose the signed amplitude α of $\boldsymbol{p} = \alpha\hat{\boldsymbol{p}}$ (a negative α reverses the direction) to minimize the error, Equation (7.10) reads $\alpha^2 - 2\alpha\,\|\boldsymbol{d}\|\cos\theta + \|\boldsymbol{d}\|^2$, a parabola in α with minimum at $\alpha = \|\boldsymbol{d}\|\cos\theta$ (set the derivative $2\alpha - 2\,\|\boldsymbol{d}\|\cos\theta$ to zero). The minimum value is $\|\boldsymbol{d}\|^2 - \|\boldsymbol{d}\|^2\cos^2\theta = \|\boldsymbol{d}\|^2\sin^2\theta$, so the best achievable error in that direction is

$$\min_{\alpha} \|\alpha\hat{\boldsymbol{p}} - \boldsymbol{d}\| = \|\boldsymbol{d}\|\sin\theta. \tag{7.12}$$

The Pearson correlation of statistics (Chapter 19) is $\cos\theta$ computed after subtracting from each vector the mean of its components. If $\bar{p} = \frac{1}{n}\sum_j p_j$ and $\bar{d} = \frac{1}{n}\sum_j d_j$ are the means of the components and $\mathbf{1} = (1, \ldots, 1)$, then, provided neither centered vector is zero (otherwise the denominator vanishes and the correlation is undefined; a constant prediction such as the null has no correlation),

$$\operatorname{corr}(\boldsymbol{p}, \boldsymbol{d}) = \frac{(\boldsymbol{p} - \bar{p}\mathbf{1}) \cdot (\boldsymbol{d} - \bar{d}\mathbf{1})}{\|\boldsymbol{p} - \bar{p}\mathbf{1}\| \, \left\|\boldsymbol{d} - \bar{d}\mathbf{1}\right\|}. \tag{7.13}$$

By Equation (7.8) applied to the centered vectors, $-1 \le \text{corr} \le 1$. When both vectors have means close to zero, subtracting the means changes them very little, so the correlation and the $\cos\theta$ of Equation (7.9) are close, and the geometry above applies approximately to rms and corr. The fluctuations of the zeros are, by construction, deviations from a smooth average, and Chapter 17 discusses why their average over many zeros is small; in this chapter we take that as a stated fact rather than prove it (Problem 25 lets you compute the actual fluctuation vector).

Sample Problem 7.2: Right shape, wrong size; right size, wrong shape

The truth is $\boldsymbol{d} = (1, -1, 2, -2)$. Compare the null, the prediction $\boldsymbol{p} = (0.5, -0.5, 1, -1)$, and the prediction $\boldsymbol{q} = (2, -2, -1, 1)$, by rms error and correlation.
Step 1 (null): error vector $-\boldsymbol{d}$, $\|\boldsymbol{d}\|^2 = 1 + 1 + 4 + 4 = 10$, so rms $= \sqrt{10/4} = 1.5811$.
Step 2 ($\boldsymbol{p}$): $\boldsymbol{p} = \frac{1}{2}\boldsymbol{d}$, so $\boldsymbol{p}\cdot\boldsymbol{d} = \frac{1}{2}\cdot 10 = 5$, $\|\boldsymbol{p}\| = \frac{1}{2}\sqrt{10}$, and $\cos\theta = 5/(\frac{1}{2}\sqrt{10}\cdot\sqrt{10}) = 1$. Both vectors have mean zero, so corr $= 1$. The error is $\boldsymbol{p} - \boldsymbol{d} = (-0.5, 0.5, -1, 1)$ with squared norm 2.5, so rms $= \sqrt{2.5/4} = 0.7906$.
Step 3 ($\boldsymbol{q}$): $\boldsymbol{q}\cdot\boldsymbol{d} = 2+2-2-2 = 0$, mean of $\boldsymbol{q}$ is zero, so corr $= 0$; $\|\boldsymbol{q}\|^2 = 4+4+1+1 = 10 = \|\boldsymbol{d}\|^2$. By Equation (7.10), $\|\boldsymbol{q} - \boldsymbol{d}\|^2 = 10 - 0 + 10 = 20$, so rms $= \sqrt{20/4} = 2.2361$, which is $\sqrt{2}$ times the null.

null 1.5811; $\boldsymbol{p}$: rms $= 0.7906$, corr $= 1$; $\boldsymbol{q}$: rms $= 2.2361$, corr $= 0$.

The perfectly shaped prediction has a large error because its amplitude is wrong; the uncorrelated prediction of the right amplitude is worse than saying nothing.

Key Ideas

- $\boldsymbol{u}\cdot\boldsymbol{v} = \sum_j u_j v_j$, $\|\boldsymbol{v}\| = \sqrt{\boldsymbol{v}\cdot\boldsymbol{v}}$; rms $= \|\boldsymbol{e}\|/\sqrt{n}$ (Equation (7.7)).
- Cauchy–Schwarz (Equation (7.8)) lets us define angles in $\mathbb{R}^n$ (Equation (7.9)).
- Correlation is the cosine of the angle between centered vectors; a prediction with $\cos\theta \le 0$ cannot beat the null (Equation (7.11)), and the best rescaled prediction has error $\|\boldsymbol{d}\|\sin\theta$ (Equation (7.12)).

7.3 Complex vectors

Quantum wavefunctions (Chapter 10) are complex, and in the paper's Sec. II E certain special vectors of a complex matrix (eigenvectors, previewed in Section 7.6 and defined in Chapter 8) have complex components. So we need vectors whose components are complex numbers. Addition and scaling (now by complex scalars) work exactly as before. The inner product needs one change.

Checkpoint 7.3

Using the real formula Equation (7.5) without change, compute $\boldsymbol{u}\cdot\boldsymbol{u}$ for $\boldsymbol{u}=(1,i)$. Would this make a sensible "length squared"?

Answer: $1\cdot 1+i\cdot i=1+i^2=0$. A nonzero vector would have length zero, which is useless as a length. The formula must be modified.

Recall from Chapter 2 that for a complex number z, $z^*z=|z|^2\geq 0$, with equality only when $z=0$. So we put a conjugate on the first factor.

Definition. For $\boldsymbol{u},\boldsymbol{v}\in\mathbb{C}^n$ (lists of n complex numbers) the inner product is

$$\langle \boldsymbol{u},\boldsymbol{v}\rangle=\sum_{j=1}^{n} u_j^* v_j, \tag{7.14}$$

and the norm is $\|\boldsymbol{v}\|=\sqrt{\langle \boldsymbol{v},\boldsymbol{v}\rangle}=\sqrt{\sum_j |v_j|^2}$.

This is the physicists' convention (conjugate on the first slot), the one used with the bra-ket notation $\langle u\,|\,v\rangle$ of Chapter 10. Some mathematics books conjugate the second slot instead; the content is the same. For real vectors $u_j^*=u_j$ and Equation (7.14) reduces to Equation (7.5).

Properties.

1. Conjugate symmetry: $\langle \boldsymbol{u},\boldsymbol{v}\rangle^*=\sum_j (u_j^*v_j)^*=\sum_j u_j v_j^*=\langle \boldsymbol{v},\boldsymbol{u}\rangle$, using $(ab)^*=a^*b^*$ and $(a^*)^*=a$.
2. Linearity in the second slot: $\langle \boldsymbol{u},\alpha\boldsymbol{v}+\beta\boldsymbol{w}\rangle=\alpha\langle \boldsymbol{u},\boldsymbol{v}\rangle+\beta\langle \boldsymbol{u},\boldsymbol{w}\rangle$, exactly as in the real case.
3. Conjugate-linearity in the first slot: $\langle \alpha\boldsymbol{u},\boldsymbol{v}\rangle=\sum_j(\alpha u_j)^*v_j=\alpha^*\langle \boldsymbol{u},\boldsymbol{v}\rangle$.
4. Positivity: $\langle \boldsymbol{v},\boldsymbol{v}\rangle=\sum_j |v_j|^2\geq 0$, zero only for $\boldsymbol{v}=\boldsymbol{0}$.

Property 1 implies that $\langle \boldsymbol{v},\boldsymbol{v}\rangle$ equals its own conjugate, so it is real, as a squared length must be. The Cauchy–Schwarz inequality $|\langle \boldsymbol{u},\boldsymbol{v}\rangle|\leq\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|$ still holds (Problem 21 asks you to prove it). Two complex vectors are orthogonal if $\langle \boldsymbol{u},\boldsymbol{v}\rangle=0$.

Sample Problem 7.3: A complex inner product

Let $\boldsymbol{u}=(1+i,\ 2)$ and $\boldsymbol{v}=(i,\ 1-i)$. Compute $\langle \boldsymbol{u},\boldsymbol{v}\rangle$, $\langle \boldsymbol{v},\boldsymbol{u}\rangle$, $\|\boldsymbol{u}\|$, $\|\boldsymbol{v}\|$, and check Cauchy–Schwarz.
Step 1: Conjugate the components of $\boldsymbol{u}$: $(1-i,\ 2)$.
Step 2: $\langle \boldsymbol{u},\boldsymbol{v}\rangle=(1-i)\,i+2(1-i)=i-i^2+2-2i=i+1+2-2i=3-i$.
Step 3: By conjugate symmetry $\langle \boldsymbol{v},\boldsymbol{u}\rangle=(3-i)^*=3+i$. (Directly: $(-i)(1+i)+(1+i)\cdot 2=-i+1+2+2i=3+i$.)
Step 4: $\|\boldsymbol{u}\|^2=|1+i|^2+|2|^2=2+4=6$; $\|\boldsymbol{v}\|^2=|i|^2+|1-i|^2=1+2=3$.

Step 5: $|3-i|^2 = 9+1 = 10 \le 6\cdot 3 = 18$. The inequality holds.

$$\boxed{\langle \boldsymbol{u},\boldsymbol{v}\rangle = 3-i, \quad \|\boldsymbol{u}\| = \sqrt{6}, \quad \|\boldsymbol{v}\| = \sqrt{3}.}$$

Pitfall

In `numpy`, `np.dot(u, v)` and `u @ v` do *not* conjugate anything. For complex vectors the inner product Equation (7.14) is `np.vdot(u, v)`, which conjugates its first argument, or equivalently `u.conj() @ v`. Using `u @ u` for a complex norm gives nonsense such as the 0 of the checkpoint above.

```
import numpy as np
u = np.array([1+1j, 2])
v = np.array([1j, 1-1j])
print(np.vdot(u, v))            # (3-1j): conjugates u
print(u @ v)                    # no conjugate: a different number
print(np.linalg.norm(u)**2)     # 6.0 (norm conjugates correctly)
r = np.array([0.3, -0.1, 0.2, -0.4])
print(np.linalg.norm(r), np.sqrt(np.mean(r**2)))   # 0.5477, 0.2739
```

Key Ideas

- For complex vectors, $\langle \boldsymbol{u},\boldsymbol{v}\rangle = \sum_j u_j^* v_j$; the conjugate makes $\langle \boldsymbol{v},\boldsymbol{v}\rangle = \sum_j |v_j|^2$ real and non-negative.
- $\langle \boldsymbol{u},\boldsymbol{v}\rangle^* = \langle \boldsymbol{v},\boldsymbol{u}\rangle$; linear in the second slot, conjugate-linear in the first.

7.4 Bases, orthogonality and projection

★ Did you know? *Gram, Schmidt, and Laplace*

The Gram-Schmidt procedure turns any set of independent vectors into an orthonormal basis by subtracting projections one at a time. It is named after Jørgen Gram (1883) and Erhard Schmidt (1907), but Pierre-Simon Laplace had used an equivalent method decades earlier. Schmidt applied it to functions, which is exactly the infinite-dimensional setting of quantum mechanics.

7.4.1 Independence, span and basis

Definitions. Vectors $\boldsymbol{v}_1,\dots,\boldsymbol{v}_k$ are *linearly independent* if the only solution of

$$\alpha_1\boldsymbol{v}_1 + \cdots + \alpha_k\boldsymbol{v}_k = \boldsymbol{0}$$

is $\alpha_1 = \cdots = \alpha_k = 0$. Otherwise they are *dependent*: one of them (any one with a nonzero α_i) can be written as a linear combination of the others. The *span* of the $\boldsymbol{v}_i$ is the set of all their linear combinations. A *basis* of $\mathbb{R}^n$ (or $\mathbb{C}^n$) is a set of independent vectors whose span is the whole space.

The *standard basis* is $\boldsymbol{e}_1 = (1,0,\dots,0)$, $\boldsymbol{e}_2 = (0,1,0,\dots,0)$, …, $\boldsymbol{e}_n = (0,\dots,0,1)$. It spans everything because $\boldsymbol{v} = v_1\boldsymbol{e}_1 + \cdots + v_n\boldsymbol{e}_n$, and it is independent because $\sum \alpha_j\boldsymbol{e}_j = (\alpha_1,\dots,\alpha_n)$ is zero only if every $\alpha_j = 0$.

Taken on trust

Taken on trust: dimension. Every basis of $\mathbb{R}^n$ or $\mathbb{C}^n$ contains exactly n vectors, and any n independent vectors form a basis. The number n is the *dimension*. The proof is not hard but needs a careful counting argument (the "exchange lemma"); see Strang [28], Ch. 3, or Riley, Hobson and Bence [22], Ch. 8. We use the consequence that more than n vectors in $\mathbb{R}^n$ are always dependent.

7.4.2 Orthogonal sets and components

Theorem. Nonzero, mutually orthogonal vectors are linearly independent.

Proof. Suppose $\sum_j \alpha_j \boldsymbol{v}_j = \boldsymbol{0}$ with $\langle \boldsymbol{v}_k, \boldsymbol{v}_j \rangle = 0$ for $j \neq k$. Take the inner product of both sides with $\boldsymbol{v}_k$:

$$0 = \langle \boldsymbol{v}_k, \textstyle\sum_j \alpha_j \boldsymbol{v}_j \rangle = \sum_j \alpha_j \langle \boldsymbol{v}_k, \boldsymbol{v}_j \rangle = \alpha_k \left\| \boldsymbol{v}_k \right\|^2 .$$

Since $\boldsymbol{v}_k \neq \boldsymbol{0}$, $\|\boldsymbol{v}_k\|^2 > 0$, so $\alpha_k = 0$. This holds for every k. $\square$

An *orthonormal* basis is a basis $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ with $\langle \boldsymbol{e}_j, \boldsymbol{e}_k \rangle = \delta_{jk}$, where the Kronecker delta δ_{jk} is 1 if $j = k$ and 0 otherwise. In an orthonormal basis, components are found by inner products. If $\boldsymbol{v} = \sum_j a_j \boldsymbol{e}_j$, then

$$\langle \boldsymbol{e}_k, \boldsymbol{v} \rangle = \sum_j a_j \langle \boldsymbol{e}_k, \boldsymbol{e}_j \rangle = \sum_j a_j \delta_{jk} = a_k . \tag{7.15}$$

And the squared norm is the sum of squared components (a finite form of Parseval's theorem, which Chapter 5 meets for functions):

$$\|\boldsymbol{v}\|^2 = \Big\langle \sum_j a_j \boldsymbol{e}_j, \sum_k a_k \boldsymbol{e}_k \Big\rangle = \sum_j \sum_k a_j^{*} a_k \delta_{jk} = \sum_k |a_k|^2 . \tag{7.16}$$

The middle step uses conjugate-linearity in the first slot and linearity in the second.

Sample Problem 7.4: Components in a rotated orthonormal basis

Let $\boldsymbol{e}_1 = (1,1)/\sqrt{2}$, $\boldsymbol{e}_2 = (1,-1)/\sqrt{2}$. Find the components of $\boldsymbol{v} = (3,1)$ and check Equation (7.16).
Step 1: Check orthonormality: $\boldsymbol{e}_1 \cdot \boldsymbol{e}_1 = (1+1)/2 = 1$, $\boldsymbol{e}_2 \cdot \boldsymbol{e}_2 = 1$, $\boldsymbol{e}_1 \cdot \boldsymbol{e}_2 = (1-1)/2 = 0$.
Step 2: By Equation (7.15), $a_1 = \boldsymbol{e}_1 \cdot \boldsymbol{v} = (3+1)/\sqrt{2} = 2\sqrt{2}$ and $a_2 = \boldsymbol{e}_2 \cdot \boldsymbol{v} = (3-1)/\sqrt{2} = \sqrt{2}$.
Step 3: Rebuild: $2\sqrt{2}\,(1,1)/\sqrt{2} + \sqrt{2}\,(1,-1)/\sqrt{2} = (2,2) + (1,-1) = (3,1)$.
Step 4: $a_1^2 + a_2^2 = 8 + 2 = 10 = 3^2 + 1^2$.

$$\boxed{a_1 = 2\sqrt{2} \approx 2.8284, \qquad a_2 = \sqrt{2} \approx 1.4142.}$$

7.4.3 Projection

Often we want the part of a vector $\boldsymbol{b}$ that points along a given direction $\boldsymbol{a}$. Write $\boldsymbol{b} = \alpha \boldsymbol{a} + \boldsymbol{w}$ and demand that the leftover $\boldsymbol{w}$ be orthogonal to $\boldsymbol{a}$ (for real vectors):

$$0 = \boldsymbol{a} \cdot \boldsymbol{w} = \boldsymbol{a} \cdot (\boldsymbol{b} - \alpha \boldsymbol{a}) = \boldsymbol{a} \cdot \boldsymbol{b} - \alpha\, \boldsymbol{a} \cdot \boldsymbol{a} \quad \Longrightarrow \quad \alpha = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}} .$$

So the *projection* of $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$ is

$$\operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b} = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}}\, \boldsymbol{a} . \tag{7.17}$$

Sample Problem 7.5: Projecting onto a line

Project $\boldsymbol{b} = (1, 2)$ onto $\boldsymbol{a} = (3, 1)$ and check that the residual is perpendicular to $\boldsymbol{a}$.
Step 1: $\boldsymbol{a} \cdot \boldsymbol{b} = 3 + 2 = 5$; $\boldsymbol{a} \cdot \boldsymbol{a} = 9 + 1 = 10$.
Step 2: By Equation (7.17), $\operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b} = \frac{5}{10}(3, 1) = (1.5,\ 0.5)$.
Step 3: Residual $\boldsymbol{w} = \boldsymbol{b} - (1.5, 0.5) = (-0.5,\ 1.5)$; $\boldsymbol{a} \cdot \boldsymbol{w} = -1.5 + 1.5 = 0$.

$$\boxed{\operatorname{proj}_{\boldsymbol{a}} \boldsymbol{b} = (1.5,\ 0.5), \qquad \boldsymbol{w} = (-0.5,\ 1.5) \perp \boldsymbol{a}.}$$

This is the right panel of Figure 7.1.

Projection onto a subspace. A *subspace* S is the span of some set of vectors. Suppose S is spanned by orthonormal vectors $\boldsymbol{e}_1, \dots, \boldsymbol{e}_m$ (with m possibly much smaller than n). Define

$$P\boldsymbol{v} = \sum_{j=1}^{m} \langle \boldsymbol{e}_j, \boldsymbol{v} \rangle \, \boldsymbol{e}_j. \tag{7.18}$$

The residual is orthogonal to S. For each $k \le m$,

$$\langle \boldsymbol{e}_k, \boldsymbol{v} - P\boldsymbol{v} \rangle = \langle \boldsymbol{e}_k, \boldsymbol{v} \rangle - \sum_{j} \langle \boldsymbol{e}_j, \boldsymbol{v} \rangle \langle \boldsymbol{e}_k, \boldsymbol{e}_j \rangle = \langle \boldsymbol{e}_k, \boldsymbol{v} \rangle - \langle \boldsymbol{e}_k, \boldsymbol{v} \rangle = 0,$$

and by linearity $\boldsymbol{v} - P\boldsymbol{v}$ is then orthogonal to every combination of the $\boldsymbol{e}_k$, that is, to all of S.

$P\boldsymbol{v}$ is the point of S closest to $\boldsymbol{v}$. Let $\boldsymbol{w}$ be any vector in S. Then $\boldsymbol{v} - \boldsymbol{w} = (\boldsymbol{v} - P\boldsymbol{v}) + (P\boldsymbol{v} - \boldsymbol{w})$, where the first bracket is orthogonal to S and the second lies in S. For orthogonal $\boldsymbol{x}, \boldsymbol{y}$ we have $\|\boldsymbol{x} + \boldsymbol{y}\|^2 = \|\boldsymbol{x}\|^2 + \langle \boldsymbol{x}, \boldsymbol{y} \rangle + \langle \boldsymbol{y}, \boldsymbol{x} \rangle + \|\boldsymbol{y}\|^2 = \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2$ (Pythagoras), so

$$\|\boldsymbol{v} - \boldsymbol{w}\|^2 = \|\boldsymbol{v} - P\boldsymbol{v}\|^2 + \|P\boldsymbol{v} - \boldsymbol{w}\|^2 \ \ge \ \|\boldsymbol{v} - P\boldsymbol{v}\|^2 , \tag{7.19}$$

with equality only if $\boldsymbol{w} = P\boldsymbol{v}$. This "closest point" property is the heart of least-squares fitting (Chapter 9): fitting a model with a few adjustable coefficients is projecting the data onto the subspace of things the model can produce.

Key Ideas

- A basis is an independent spanning set; in $\mathbb{R}^n$ it has n members.
- Orthogonal nonzero vectors are independent; in an orthonormal basis, components are inner products (Equation (7.15)) and $\|\boldsymbol{v}\|^2 = \sum |a_k|^2$.
- Projection onto a subspace gives the closest point and an orthogonal residual (Equations (7.18) and (7.19)).

7.5 Matrices as linear maps

7.5.1 Linear maps

A *map* (function) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ takes an n-component vector and returns an m-component vector. It is *linear* if, for all vectors $\boldsymbol{u}, \boldsymbol{v}$ and scalars α, β,

$$T(\alpha \boldsymbol{u} + \beta \boldsymbol{v}) = \alpha T(\boldsymbol{u}) + \beta T(\boldsymbol{v}). \tag{7.20}$$

Physical examples: rotating a vector by a fixed angle; stretching the x-component by 2; taking differences of consecutive entries. Differentiation of functions is also linear, $\frac{\mathrm{d}}{\mathrm{d}x}(\alpha f + \beta g) = \alpha f' + \beta g'$, which is why Chapter 8 can turn $-\mathrm{d}^2/\mathrm{d}x^2$ into a matrix.

7.5.2 Matrices and the matrix-vector product

An $m \times n$ *matrix* A is a rectangular array of numbers with m rows and n columns; the entry in row i and column j is A_{ij}. Matrices of the same shape are added entry by entry, $(A+B)_{ij} = A_{ij} + B_{ij}$, and multiplied by a scalar s entry by entry, $(sA)_{ij} = sA_{ij}$. Its product with a vector $\boldsymbol{x} \in \mathbb{R}^n$ is the vector $A\boldsymbol{x} \in \mathbb{R}^m$ with components

$$(A\boldsymbol{x})_i = \sum_{j=1}^{n} A_{ij} x_j, \qquad i = 1, \dots, m. \tag{7.21}$$

There are two ways to read Equation (7.21).

- **Row picture:** $(A\boldsymbol{x})_i$ is the dot product of row i of A with $\boldsymbol{x}$.
- **Column picture:** let $\boldsymbol{a}_j$ be the j-th column of A (the vector with components $A_{1j}, \dots, A_{mj}$). Then
$$A\boldsymbol{x} = x_1 \boldsymbol{a}_1 + x_2 \boldsymbol{a}_2 + \cdots + x_n \boldsymbol{a}_n, \tag{7.22}$$
because the i-th component of the right side is $\sum_j x_j A_{ij}$, which is Equation (7.21).

The column picture says: *$A\boldsymbol{x}$ is a linear combination of the columns of A, with the entries of $\boldsymbol{x}$ as coefficients.* For example,

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

The product is linear. $(A(\alpha \boldsymbol{u} + \beta \boldsymbol{v}))_i = \sum_j A_{ij}(\alpha u_j + \beta v_j) = \alpha (A\boldsymbol{u})_i + \beta (A\boldsymbol{v})_i$.

Every linear map is a matrix. Conversely, let T be linear from $\mathbb{R}^n$ to $\mathbb{R}^m$. Write $\boldsymbol{x} = \sum_j x_j \boldsymbol{e}_j$ in the standard basis. Applying Equation (7.20) repeatedly,

$$T(\boldsymbol{x}) = T\Big(\sum_j x_j \boldsymbol{e}_j\Big) = \sum_j x_j \, T(\boldsymbol{e}_j).$$

Compare with Equation (7.22): $T(\boldsymbol{x}) = A\boldsymbol{x}$ for the matrix A whose j-th column is $T(\boldsymbol{e}_j)$. So a linear map is completely determined by what it does to the basis vectors, and *the columns of its matrix are the images of the basis vectors.* This is a useful way to construct matrices, as the next sample shows.

Sample Problem 7.6: The spacing matrix of the first zeros

The spacings between consecutive zeros of RE1 are $g_1 = t_2 - t_1$, $g_2 = t_3 - t_2$, $g_3 = t_4 - t_3$. Write the map $(t_1, t_2, t_3, t_4) \mapsto (g_1, g_2, g_3)$ as a matrix D and evaluate it on the RE1 heights.

Step 1: The map is linear (differences of linear combinations are linear combinations), from $\mathbb{R}^4$ to $\mathbb{R}^3$, so D is 3×4.

Step 2: Row i produces $g_i = -t_i + t_{i+1}$, so it has -1 in column i, $+1$ in column $i+1$:

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

(Check the column rule: $D\boldsymbol{e}_1 = (-1, 0, 0)$: increasing t_1 alone decreases g_1 and nothing else.)

Step 3: With $\boldsymbol{t} = (14.134725,\ 21.022040,\ 25.010858,\ 30.424876)$,

$$D\boldsymbol{t} = (21.022040 - 14.134725,\ 25.010858 - 21.022040,\ 30.424876 - 25.010858).$$

$$\boxed{D\boldsymbol{t} = (6.887315,\ 3.988818,\ 5.414018).}$$

Extending D to 9×10 for all ten zeros of RE1 gives nine spacings summing to $t_{10} - t_1 = 35.639107$ (each intermediate t_j appears once with $+$ and once with $-$).

Checkpoint 7.4

In the paper's fit on zeros 1–400 with $M = 600$ basis functions, the sensitivity matrix A_{nk} has one row per level and one column per basis function. What is its shape? How many components do $\boldsymbol{c}$ and $A\boldsymbol{c}$ have?

Answer: A is 400×600; $\boldsymbol{c}$ has 600 components; $A\boldsymbol{c}$ has 400 (one predicted shift per level).

Pitfall

Index conventions. The paper numbers levels from $n = 1$ but basis functions from $k = 0$ (the constant $\cos 0 = 1$ is ϕ_0). Python arrays start at 0. So the paper's A_{nk} is stored in `A[n-1, k]`. Off-by-one errors of this kind silently pair the wrong level with the wrong zero; always check one entry by hand.

Key Ideas

- A linear map satisfies $T(\alpha\boldsymbol{u} + \beta\boldsymbol{v}) = \alpha T(\boldsymbol{u}) + \beta T(\boldsymbol{v})$; every linear map $\mathbb{R}^n \to \mathbb{R}^m$ is an $m \times n$ matrix whose columns are the images of the basis vectors.
- $A\boldsymbol{x}$ is a combination of the columns of A weighted by the entries of $\boldsymbol{x}$ (Equation (7.22)).

7.6 Multiplying matrices; transpose and conjugate transpose

7.6.1 The product as composition

If B maps $\mathbb{R}^p \to \mathbb{R}^n$ and A maps $\mathbb{R}^n \to \mathbb{R}^m$, then "first B, then A" maps $\mathbb{R}^p \to \mathbb{R}^m$ and is linear. We define the product AB to be its matrix. To find the entries, apply Equation (7.21) twice:

$$\begin{aligned}(A(B\boldsymbol{x}))_i &= \sum_{j=1}^{n} A_{ij}(B\boldsymbol{x})_j \\ &= \sum_{j=1}^{n} A_{ij} \sum_{k=1}^{p} B_{jk}x_k \\ &= \sum_{k=1}^{p} \Big(\sum_{j=1}^{n} A_{ij}B_{jk} \Big) x_k \qquad \text{(reorder the finite double sum).}\end{aligned}$$

Comparing with Equation (7.21),

$$(AB)_{ik} = \sum_{j=1}^{n} A_{ij}B_{jk}. \tag{7.23}$$

Entry (i, k) of AB is the dot product of row i of A with column k of B. The product exists only if the number of columns of A equals the number of rows of B; an $m \times n$ times an $n \times p$ gives an $m \times p$.

Because AB is defined as a composition, $(AB)C = A(BC)$ (both mean "C, then B, then A"), and $A(B+C) = AB + AC$, because by Equation (7.23), entry by entry, $(A(B+C))_{ik} = \sum_j A_{ij}(B_{jk} + C_{jk}) = \sum_j A_{ij}B_{jk} + \sum_j A_{ij}C_{jk} = (AB)_{ik} + (AC)_{ik}$. But the order matters.

Sample Problem 7.7: Rotate then stretch, or stretch then rotate?

Let $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation by 90° counterclockwise) and $S = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ (stretch x by 2). Compute RS and SR and apply both to $\boldsymbol{e}_1 = (1, 0)$.

Step 1: By Equation (7.23), $RS = \begin{pmatrix} 0\cdot 2 + (-1)\cdot 0 & 0\cdot 0 + (-1)\cdot 1 \\ 1\cdot 2 + 0\cdot 0 & 1\cdot 0 + 0\cdot 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$.

Step 2: $SR = \begin{pmatrix} 2\cdot 0 + 0\cdot 1 & 2\cdot(-1) + 0\cdot 0 \\ 0\cdot 0 + 1\cdot 1 & 0\cdot(-1) + 1\cdot 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$.

Step 3: $RS\,\boldsymbol{e}_1 = (0, 2)$: stretch $(1, 0)$ to $(2, 0)$, then rotate to $(0, 2)$. $SR\,\boldsymbol{e}_1 = (0, 1)$: rotate to $(0, 1)$, whose x-component 0 is unaffected by the stretch.

$$\boxed{RS = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix} \neq SR = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}.}$$

Checkpoint 7.5

Does $AB = BA$ hold for all square matrices of the same size?

Answer: No; the sample above is a counterexample. Some special pairs commute (for example any matrix with the identity, or two diagonal matrices), but in general they do not.

7.6.2 Transpose and conjugate transpose

The *transpose* $A^\top$ of an $m \times n$ matrix is the $n \times m$ matrix with $(A^\top)_{ji} = A_{ij}$: rows become columns. A column vector $\boldsymbol{x}$ is an $n \times 1$ matrix and $\boldsymbol{x}^\top$ is a $1 \times n$ row. With this, the real inner product is a matrix product, $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u}^\top \boldsymbol{v}$ ($1 \times n$ times $n \times 1$ gives 1×1, a number).

Transpose of a product. $(AB)^\top = B^\top A^\top$. Proof, entry (k, i):

$$((AB)^\top)_{ki} = (AB)_{ik} = \sum_j A_{ij}B_{jk} = \sum_j (B^\top)_{kj}(A^\top)_{ji} = (B^\top A^\top)_{ki}.$$

The order reverses, just as taking off shoes and socks reverses putting them on.

For complex matrices the useful operation is the *conjugate transpose* (Hermitian conjugate), written $A^\dagger$ ("A dagger"): $(A^\dagger)_{ji} = A^*_{ij}$. Then $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \boldsymbol{u}^\dagger \boldsymbol{v}$, and by the same proof with conjugates, $(AB)^\dagger = B^\dagger A^\dagger$. A key identity follows:

$$\langle \boldsymbol{u}, A\boldsymbol{v} \rangle = \boldsymbol{u}^\dagger A \boldsymbol{v} = (A^\dagger \boldsymbol{u})^\dagger \boldsymbol{v} = \langle A^\dagger \boldsymbol{u}, \boldsymbol{v} \rangle, \tag{7.24}$$

where the middle step uses $(A^\dagger \boldsymbol{u})^\dagger = \boldsymbol{u}^\dagger (A^\dagger)^\dagger = \boldsymbol{u}^\dagger A$.

A real matrix with $A^\top = A$ is *symmetric*; a complex matrix with $A^\dagger = A$ is *Hermitian*. For such matrices Equation (7.24) says $\langle \boldsymbol{u}, A\boldsymbol{v} \rangle = \langle A\boldsymbol{u}, \boldsymbol{v} \rangle$: the matrix can be moved from one side of the inner product to the other. This property has a consequence that we only preview here. An *eigenvalue* of a square matrix A is a number λ for which some nonzero vector $\boldsymbol{v}$ (an *eigenvector*) satisfies $A\boldsymbol{v} = \lambda \boldsymbol{v}$: the matrix merely rescales that vector. Chapter 8 defines these properly and proves that the "move across the inner product" property forces the eigenvalues of symmetric and Hermitian matrices to be real. That is why the paper's finite-difference

Hamiltonian, a real symmetric matrix (Sec. II B), has real energy levels. "Self-adjoint" is the name for the same property in operators acting on functions rather than on finite lists, which is why the Hilbert–Pólya idea (Chapter 21) asks for a self-adjoint operator.

Quadratic forms. For a square matrix T and a vector $\boldsymbol{c}$, the number $\boldsymbol{c}^{\top} T \boldsymbol{c} = \sum_{k,l} c_k T_{kl} c_l$ is called a *quadratic form*: every term is a product of two components of $\boldsymbol{c}$. Example: $\boldsymbol{c} = (1, 2)$, $T = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ gives $T\boldsymbol{c} = (4, 7)$ and $\boldsymbol{c}^{\top} T \boldsymbol{c} = 1 \cdot 4 + 2 \cdot 7 = 18$. The paper's Eq. (5) writes second-order level shifts as $-c^{\top} T^{(n)} c$, one such form per level n.

Key Ideas

- $(AB)_{ik} = \sum_j A_{ij} B_{jk}$ is the matrix of "first B, then A"; in general $AB \neq BA$.
- $(AB)^{\top} = B^{\top} A^{\top}$, $(AB)^{\dagger} = B^{\dagger} A^{\dagger}$; $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \boldsymbol{u}^{\dagger} \boldsymbol{v}$; $\langle \boldsymbol{u}, A\boldsymbol{v} \rangle = \langle A^{\dagger} \boldsymbol{u}, \boldsymbol{v} \rangle$.
- Symmetric ($A^{\top} = A$) and Hermitian ($A^{\dagger} = A$) matrices can be moved across the inner product.

7.7 The identity and the inverse

The $n \times n$ *identity* I has ones on the diagonal and zeros elsewhere, $I_{ij} = \delta_{ij}$. It does nothing: $(I\boldsymbol{x})_i = \sum_j \delta_{ij} x_j = x_i$, and similarly $IA = AI = A$.

A square matrix A is *invertible* if there is a matrix A^{-1} with

$$A^{-1} A = A A^{-1} = I. \tag{7.25}$$

If $A\boldsymbol{x} = \boldsymbol{b}$ and A is invertible, multiplying on the left by A^{-1} gives $\boldsymbol{x} = A^{-1}\boldsymbol{b}$: the inverse undoes the map.

The inverse is unique. If X and Y both satisfy Equation (7.25), then $X = XI = X(AY) = (XA)Y = IY = Y$.

Inverse of a product. $(AB)^{-1} = B^{-1} A^{-1}$, because $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$, and similarly in the other order.

The 2×2 inverse. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Multiply by the "swapped" matrix:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc)\, I,$$

and the same result appears with the factors in the other order. So if $ad - bc \neq 0$,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \tag{7.26}$$

The number $ad - bc$ is the *determinant* of A (Section 7.10).

When $ad - bc = 0$ there is no inverse. The computation above then reads $A\begin{pmatrix} d \\ -c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ (the first column of the displayed product). If $(d, -c) \neq \mathbf{0}$, A sends a nonzero vector to $\mathbf{0}$. It also sends $\mathbf{0}$ to $\mathbf{0}$. An inverse would have to send $\mathbf{0}$ back to both, which is impossible. If $c = d = 0$, then $A(b, -a) = (ab - ba,\ 0) = \mathbf{0}$, and the same argument applies unless $a = b = 0$ too, in which case A is the zero matrix, which has no inverse, since $X \cdot 0 = 0 \neq I$ for every matrix X. A matrix without an inverse is called *singular*.

Sample Problem 7.8: Inverting a 2×2 matrix and solving

Solve $4x + 7y = 1$, $2x + 6y = 2$ using the inverse.

Step 1: $A = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$, $\boldsymbol{b} = (1, 2)$; $ad - bc = 24 - 14 = 10$.

Step 2: By Equation (7.26), $A^{-1} = \frac{1}{10}\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}$.

Step 3: $\boldsymbol{x} = A^{-1}\boldsymbol{b} = (0.6 - 1.4,\ -0.2 + 0.8) = (-0.8,\ 0.6)$.

Step 4 (check): $4(-0.8) + 7(0.6) = -3.2 + 4.2 = 1$; $2(-0.8) + 6(0.6) = -1.6 + 3.6 = 2$.

$$\boxed{x = -0.8, \qquad y = 0.6.}$$

Pitfall

On a computer, do not form A^{-1} just to solve $A\boldsymbol{x} = \boldsymbol{b}$. Computing the inverse costs more operations and loses more accuracy than solving directly by elimination (Section 7.8). In `numpy`, use `np.linalg.solve(A, b)`, not `np.linalg.inv(A) @ b`. Trefethen and Bau [29] discuss the accuracy question.

Key Ideas

- $I\boldsymbol{x} = \boldsymbol{x}$; A^{-1}, if it exists, is unique and undoes A; $(AB)^{-1} = B^{-1}A^{-1}$.
- A 2×2 matrix is invertible exactly when $ad - bc \neq 0$ (Equation (7.26)); a singular matrix sends some nonzero vector to $\mathbf{0}$.

7.8 Solving linear systems by elimination

A *linear system* of m equations in n unknowns is $A\boldsymbol{x} = \boldsymbol{b}$ with A an $m \times n$ matrix:

$$\begin{aligned} A_{11}x_1 + \cdots + A_{1n}x_n &= b_1 \\ &\vdots \\ A_{m1}x_1 + \cdots + A_{mn}x_n &= b_m. \end{aligned}$$

In the row picture each equation is a "line" (in 2D), "plane" (in 3D), or flat sheet (in higher dimensions), and a solution is a point on all of them (Figure 7.2). In the column picture we ask for the coefficients x_j that combine the columns of A into $\boldsymbol{b}$.

7.8.1 Row operations

We record the system as the *augmented matrix* $[A \,|\, \boldsymbol{b}]$ and apply three *row operations*:

1. swap two equations;
2. multiply an equation by a nonzero number s;

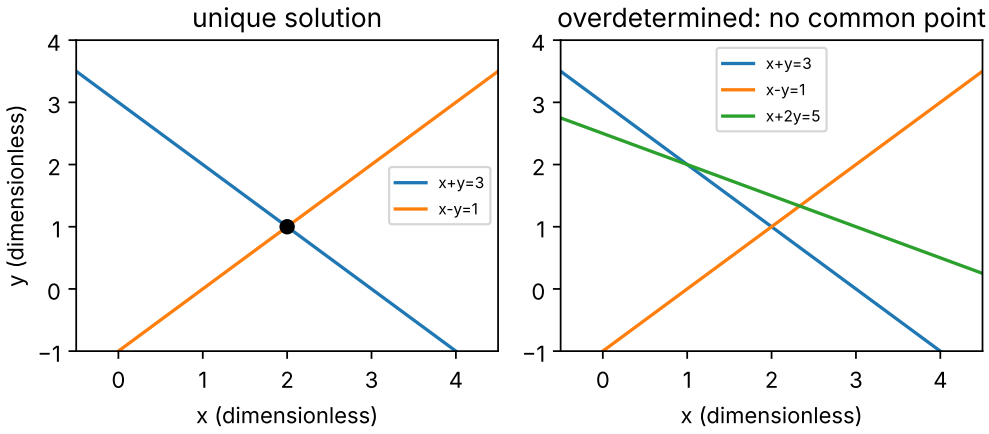


Figure 7.2: Row picture of linear systems in two unknowns. Left: two equations $x + y = 3$ and $x - y = 1$ meet in exactly one point $(2, 1)$. Right: adding a third equation $x + 2y = 5$ makes the system overdetermined; the three lines do not pass through a common point, so no exact solution exists. (Two parallel lines, not shown, would give either no solution or, if they coincide, infinitely many.) Axes dimensionless.

3. add s times one equation to another.

Row operations do not change the set of solutions. For (1) the equations are unchanged, only reordered. For (2), any $\boldsymbol{x}$ satisfying the old equation satisfies the new one (multiply by s), and conversely (divide by $s \neq 0$). For (3), if $\boldsymbol{x}$ satisfies equations i and j it satisfies "equation j plus s times equation i"; conversely, from the new system subtract s times equation i to recover the old equation j. Each operation is reversible, so no solution is gained or lost.

7.8.2 Gaussian elimination and back substitution

Gaussian elimination uses operation (3) to clear the entries below the diagonal, one column at a time. In column 1, pick a row with a nonzero entry (the *pivot*); swap it to the top if needed; subtract multiples of it from the rows below to make their column-1 entries zero. Then repeat with column 2 on the remaining rows, and so on. The result is an *echelon* (staircase) form, upper triangular for a square system, which is solved from the bottom up by *back substitution*.

Sample Problem 7.9: Elimination on a 3×3 system

Solve

$$x + y + z = 6, \qquad 2x + 3y + z = 11, \qquad x + 2y + 3z = 14.$$

Step 1: Augmented matrix $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 11 \\ 1 & 2 & 3 & 14 \end{array}\right]$.

Step 2: Pivot 1 in row 1. $R_2 \leftarrow R_2 - 2R_1$: $(0,\, 1,\, -1 \,|\, -1)$. $R_3 \leftarrow R_3 - R_1$: $(0,\, 1,\, 2 \,|\, 8)$.

Step 3: Pivot 1 in row 2, column 2. $R_3 \leftarrow R_3 - R_2$: $(0,\, 0,\, 3 \,|\, 9)$. The system is now $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 3 & 9 \end{array}\right]$.

Step 4 (back substitution): $3z = 9 \Rightarrow z = 3$. Then $y - z = -1 \Rightarrow y = 2$. Then $x + y + z = 6 \Rightarrow x = 6 - 2 - 3 = 1$.

Step 5 (check): $1+2+3=6$, $2+6+3=11$, $1+4+9=14$.

$$\boxed{(x,y,z)=(1,2,3).}$$

7.8.3 How much work is elimination?

Count multiply-and-add operations for an $n \times n$ system. At step k (clearing column k, $k=1,\dots,n-1$) there are $n-k$ rows below the pivot; each needs its $n-k$ remaining entries and its right-hand side updated, $n-k+1$ multiply-adds per row. The total is

$$\sum_{k=1}^{n-1}(n-k)(n-k+1)=\sum_{j=1}^{n-1} j(j+1) \quad (j=n-k).$$

We need two sums. First, $\sum_{j=1}^{n-1} j = \frac{(n-1)n}{2}$: write the sum forwards and backwards and add term by term; each of the $n-1$ pairs $j+(n-j)$ equals n, so twice the sum is $(n-1)n$. Second, add up the identity $(j+1)^3-j^3=3j^2+3j+1$ for $j=1,\dots,n-1$. The left side telescopes (each intermediate cube appears once with $+$ and once with $-$) to $n^3-1=(n-1)(n^2+n+1)$, so

$$3\sum_{j=1}^{n-1} j^2=(n-1)(n^2+n+1)-3\frac{(n-1)n}{2}-(n-1)=(n-1)\Big[n^2-\frac{n}{2}\Big]=\frac{(n-1)n(2n-1)}{2},$$

that is, $\sum_{j=1}^{n-1} j^2=\frac{(n-1)n(2n-1)}{6}$ (check: $n=3$ gives $1+4=5=\frac{2\cdot3\cdot5}{6}$; $n=4$ gives $1+4+9=14=\frac{3\cdot4\cdot7}{6}$). Using these two results,

$$\sum_{j=1}^{n-1} j(j+1)=\frac{(n-1)n(2n-1)}{6}+\frac{3(n-1)n}{6}=\frac{(n-1)n(2n+2)}{6}=\frac{(n-1)n(n+1)}{3}\approx\frac{n^3}{3}. \tag{7.27}$$

For $n=3$ this gives 8, which you can confirm by counting Step 2 (2 rows $\times$ 3 updates) and Step 3 (1×2) of the sample. Back substitution adds only about $n^2/2$. The n^3 growth matters: a dense 6000×6000 system needs about 7.2×10^{10} multiply-adds. But if most entries are zero, the count collapses. For a *tridiagonal* matrix (nonzero only on the diagonal and its two neighbors) each elimination step touches one row and two entries, so the work grows only like n. The paper's Hamiltonian is tridiagonal (Sec. II B, App. A step 4), and Chapter 8 explains how the same structure makes its eigenvalues cheap.

```
import numpy as np
A = np.array([[1., 1, 1], [2, 3, 1], [1, 2, 3]])
b = np.array([6., 11, 14])
x = np.linalg.solve(A, b)        # Gaussian elimination, done by a standard library
    routine
print(x)                         # [1. 2. 3.]
print(np.abs(A @ x - b).max())   # residual at rounding level
print(np.linalg.matrix_rank(A), np.linalg.det(A))   # 3, 3.0
```

Key Ideas

- Row operations are reversible and preserve the solution set.
- Elimination to echelon form plus back substitution solves $A\boldsymbol{x}=\boldsymbol{b}$; the cost is $\approx n^3/3$ multiply-adds for a dense $n\times n$ system (Equation (7.27)), far less for sparse (e.g. tridiagonal) ones.

7.9 Rank, and systems with too many or too few equations

7.9.1 Pivots, free variables and rank

Run elimination on any $m \times n$ matrix (swapping rows when a column has no usable pivot, and moving on to the next column if a whole column below the current row is zero). The result is an echelon form: each nonzero row starts with a pivot, strictly to the right of the pivot above it, and any all-zero rows are at the bottom.

Definitions. The number r of pivots is the *rank* of A. Unknowns whose columns contain a pivot are *pivot variables*; the other $n - r$ unknowns are *free variables*.

Taken on trust

Taken on trust: rank is well defined. Different sequences of row operations can give different echelon forms, but they always have the same number of pivots. Moreover the rank equals the largest number of linearly independent columns of A, and also the largest number of independent rows ("row rank equals column rank"). Proofs: Strang [28], Ch. 3; Trefethen and Bau [29], Lecture 1.

Since there is at most one pivot per row and one per column, $r \le \min(m, n)$.

The three outcomes. After elimination, consider each all-zero row of the left side. Its equation reads $0 = \beta$, where β is whatever the right-hand side has become.

- If some such $\beta \neq 0$, there is **no solution** (the system is *inconsistent*).
- Otherwise there is at least one solution: assign the free variables any values and solve for the pivot variables by back substitution. If $r = n$ there are no free variables and the solution is **unique**. If $r < n$, each choice of the $n - r$ free variables gives a different solution, so there are **infinitely many**.

7.9.2 The null space and the general solution

The *null space* of A is the set of all $\boldsymbol{x}$ with $A\boldsymbol{x} = \mathbf{0}$. It always contains $\mathbf{0}$, and it is closed under linear combinations: if $A\boldsymbol{x} = \mathbf{0}$ and $A\boldsymbol{y} = \mathbf{0}$ then $A(\alpha\boldsymbol{x} + \beta\boldsymbol{y}) = \alpha A\boldsymbol{x} + \beta A\boldsymbol{y} = \mathbf{0}$. Its size is fixed by the free variables. Take $\boldsymbol{b} = \mathbf{0}$ (always consistent, since $\boldsymbol{x} = \mathbf{0}$ works). For each free variable i ($i = 1, \dots, n - r$), set that free variable to 1 and all other free variables to 0, and solve for the pivot variables by back substitution; call the result $\boldsymbol{x}^{(i)}$, a *special solution*. (i) The special solutions are independent: in $\sum_i \alpha_i \boldsymbol{x}^{(i)}$ the slot of free variable i contains exactly α_i (only $\boldsymbol{x}^{(i)}$ has a nonzero entry there), so the combination is $\mathbf{0}$ only if every $\alpha_i = 0$. (ii) They span the null space: let $A\boldsymbol{x} = \mathbf{0}$ and let f_i be the value of free variable i in $\boldsymbol{x}$. Then $\boldsymbol{y} = \boldsymbol{x} - \sum_i f_i \boldsymbol{x}^{(i)}$ satisfies $A\boldsymbol{y} = \mathbf{0}$ and has every free variable equal to 0; back substitution with zero right side and zero free variables gives every pivot variable 0, so $\boldsymbol{y} = \mathbf{0}$ and $\boldsymbol{x} = \sum_i f_i \boldsymbol{x}^{(i)}$. So the $n - r$ special solutions form a basis of the null space, which is a subspace of dimension $n - r$.

If $\boldsymbol{x}_p$ is any one solution of $A\boldsymbol{x} = \boldsymbol{b}$ (a *particular* solution) and $\boldsymbol{x}$ is another, then $A(\boldsymbol{x} - \boldsymbol{x}_p) = \boldsymbol{b} - \boldsymbol{b} = \mathbf{0}$, so $\boldsymbol{x} - \boldsymbol{x}_p$ is in the null space. Hence

$$\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{x}_h, \qquad A\boldsymbol{x}_h = \mathbf{0}, \tag{7.28}$$

is the general solution: one particular solution plus anything in the null space.

7.9.3 Overdetermined and underdetermined systems

Overdetermined ($m > n$, more equations than unknowns). At most n pivots, so at least $m - n$ rows of the echelon form are zero on the left. Each gives a condition $0 = \beta$ that the right side must satisfy. For a generic right side, especially one containing measurement noise, these conditions fail and there is no exact solution (Figure 7.2, right). The sensible question becomes "which $\boldsymbol{x}$ makes $A\boldsymbol{x}$ *closest* to $\boldsymbol{b}$?", which is least squares (Chapter 9), the projection of Equation (7.19).

Underdetermined ($m < n$, fewer equations than unknowns). At most m pivots, so at least $n - m$ free variables. If the system is consistent at all, it has infinitely many solutions. If moreover $r = m$ (every equation has a pivot), there are no zero rows, so it is consistent for *every* $\boldsymbol{b}$: the columns can combine into any target. An underdetermined model of full rank can therefore reproduce *any* data exactly, and infinitely many coefficient vectors do so.

Checkpoint 7.6

A system has 400 equations and 600 unknowns. Can it have exactly one solution?

Answer: No. The rank is at most 400, so there are at least $600 - 400 = 200$ free variables. Either there is no solution or there are infinitely many.

Sample Problem 7.10: Recovering zeros from spacings

Given only the three spacings $\boldsymbol{g} = (6.887315,\ 3.988818,\ 5.414018)$, find all $\boldsymbol{t} \in \mathbb{R}^4$ with $D\boldsymbol{t} = \boldsymbol{g}$, where D is the spacing matrix of the earlier sample.
Step 1: D is already in echelon form, with pivots -1 in columns 1, 2, 3. So $r = 3$, $n = 4$, and t_4 is free. Three equations, four unknowns: underdetermined, consistent for every $\boldsymbol{g}$.
Step 2: Set $t_4 = s$. Back substitution: $-t_3 + t_4 = g_3 \Rightarrow t_3 = s - g_3$; $-t_2 + t_3 = g_2 \Rightarrow t_2 = s - g_2 - g_3$; $-t_1 + t_2 = g_1 \Rightarrow t_1 = s - g_1 - g_2 - g_3$.
Step 3: In the form Equation (7.28):

$$\boldsymbol{t} = (-g_1 - g_2 - g_3,\ -g_2 - g_3,\ -g_3,\ 0) + s\,(1, 1, 1, 1).$$

The null space is spanned by $(1, 1, 1, 1)$: shifting every zero by the same amount leaves all spacings unchanged.
Step 4: With $g_1 + g_2 + g_3 = 16.290151$ and the choice $s = t_4 = 30.424876$, $t_1 = 30.424876 - 16.290151 = 14.134725$, the true first zero.

$$\boxed{\boldsymbol{t} = \boldsymbol{t}_p + s(1, 1, 1, 1); \quad s = 30.424876 \ \Rightarrow\ t_1 = 14.134725.}$$

Spacings alone cannot tell you where the zeros sit; one extra fact (one zero) is needed.

Key Ideas

- Rank r = number of pivots $\leq \min(m, n)$; there are $n - r$ free variables.
- General solution = particular solution + null space (Equation (7.28)).
- Overdetermined systems usually have no exact solution (least squares instead); underdetermined systems of full row rank have infinitely many exact solutions for *any* right side.

7.10 The determinant

★ Did you know? *Determinants Before Matrices*

Determinants are older than matrices. Seki Takakazu described them in Japan in 1683, and Gottfried Leibniz independently used them in 1693 to solve systems of linear equations. The word *determinant* in its modern sense comes from Augustin-Louis Cauchy in 1812. Matrices as objects in their own right arrived only in the 1850s, more than 150 years after the determinant.

7.10.1 The 2×2 determinant as an area

Let the columns of $A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}$ be $\boldsymbol{a} = (a_1, a_2)$ and $\boldsymbol{b} = (b_1, b_2)$. The parallelogram they span has area $|\boldsymbol{a}|\,|\boldsymbol{b}| \sin\theta$. Square it and use $\sin^2\theta = 1 - \cos^2\theta$:

$$\begin{aligned} \text{area}^2 &= |\boldsymbol{a}|^2 |\boldsymbol{b}|^2 - |\boldsymbol{a}|^2 |\boldsymbol{b}|^2 \cos^2\theta = |\boldsymbol{a}|^2 |\boldsymbol{b}|^2 - (\boldsymbol{a}\cdot\boldsymbol{b})^2 \\ &= (a_1^2 + a_2^2)(b_1^2 + b_2^2) - (a_1 b_1 + a_2 b_2)^2 \\ &= a_1^2 b_1^2 + a_1^2 b_2^2 + a_2^2 b_1^2 + a_2^2 b_2^2 - a_1^2 b_1^2 - 2a_1 b_1 a_2 b_2 - a_2^2 b_2^2 \\ &= a_1^2 b_2^2 - 2a_1 b_2\, a_2 b_1 + a_2^2 b_1^2 \\ &= (a_1 b_2 - a_2 b_1)^2. \end{aligned}$$

So the area is $|a_1 b_2 - a_2 b_1|$, and we define the *determinant*

$$\det \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} = a_1 b_2 - b_1 a_2, \tag{7.29}$$

the same $ad - bc$ that appeared in the inverse. Its sign records orientation: positive if turning from $\boldsymbol{a}$ to $\boldsymbol{b}$ is counterclockwise.

Since $A\boldsymbol{e}_1 = \boldsymbol{a}$ and $A\boldsymbol{e}_2 = \boldsymbol{b}$, the map A sends the unit square (spanned by $\boldsymbol{e}_1, \boldsymbol{e}_2$, area 1) to the parallelogram spanned by $\boldsymbol{a}, \boldsymbol{b}$. *The determinant is the factor by which A scales areas* (Figure 7.3). For a general region, cover it by small squares of side ϵ. By linearity each such square is mapped to a copy of the parallelogram above shrunk by the factor ϵ, of area $|\det A|\,\epsilon^2$, so the total area is multiplied by the same factor $|\det A|$ (in the limit $\epsilon \to 0$). If $\det A = 0$ the parallelogram is flattened onto a line: the columns are parallel (dependent), and some nonzero vector is squashed to $\mathbf{0}$, which is the singular case of Section 7.7.

7.10.2 The 3×3 determinant as a volume

For a 3×3 matrix, expand along the first row:

$$\det A = A_{11}(A_{22}A_{33} - A_{23}A_{32}) - A_{12}(A_{21}A_{33} - A_{23}A_{31}) + A_{13}(A_{21}A_{32} - A_{22}A_{31}). \tag{7.30}$$

Each bracket is the 2×2 determinant left after deleting the row and column of the entry in front, and the signs alternate $+, -, +$. There is a compact way to write Equation (7.30). For the second and third rows $\boldsymbol{r}_2, \boldsymbol{r}_3$ define the vector (called their *cross product*; no prior knowledge of it is needed here) $\boldsymbol{r}_2 \times \boldsymbol{r}_3 = (A_{22}A_{33} - A_{23}A_{32},\ A_{23}A_{31} - A_{21}A_{33},\ A_{21}A_{32} - A_{22}A_{31})$. Dot this with the first row $\boldsymbol{r}_1 = (A_{11}, A_{12}, A_{13})$, component by component:

$$\begin{aligned} \boldsymbol{r}_1 \cdot (\boldsymbol{r}_2 \times \boldsymbol{r}_3) &= A_{11}(A_{22}A_{33} - A_{23}A_{32}) + A_{12}(A_{23}A_{31} - A_{21}A_{33}) + A_{13}(A_{21}A_{32} - A_{22}A_{31}) \\ &= A_{11}(A_{22}A_{33} - A_{23}A_{32}) - A_{12}(A_{21}A_{33} - A_{23}A_{31}) + A_{13}(A_{21}A_{32} - A_{22}A_{31}), \end{aligned}$$

where the second line only rewrites the middle bracket as $A_{23}A_{31} - A_{21}A_{33} = -(A_{21}A_{33} - A_{23}A_{31})$. This is exactly Equation (7.30), so $\det A = \boldsymbol{r}_1 \cdot (\boldsymbol{r}_2 \times \boldsymbol{r}_3)$.

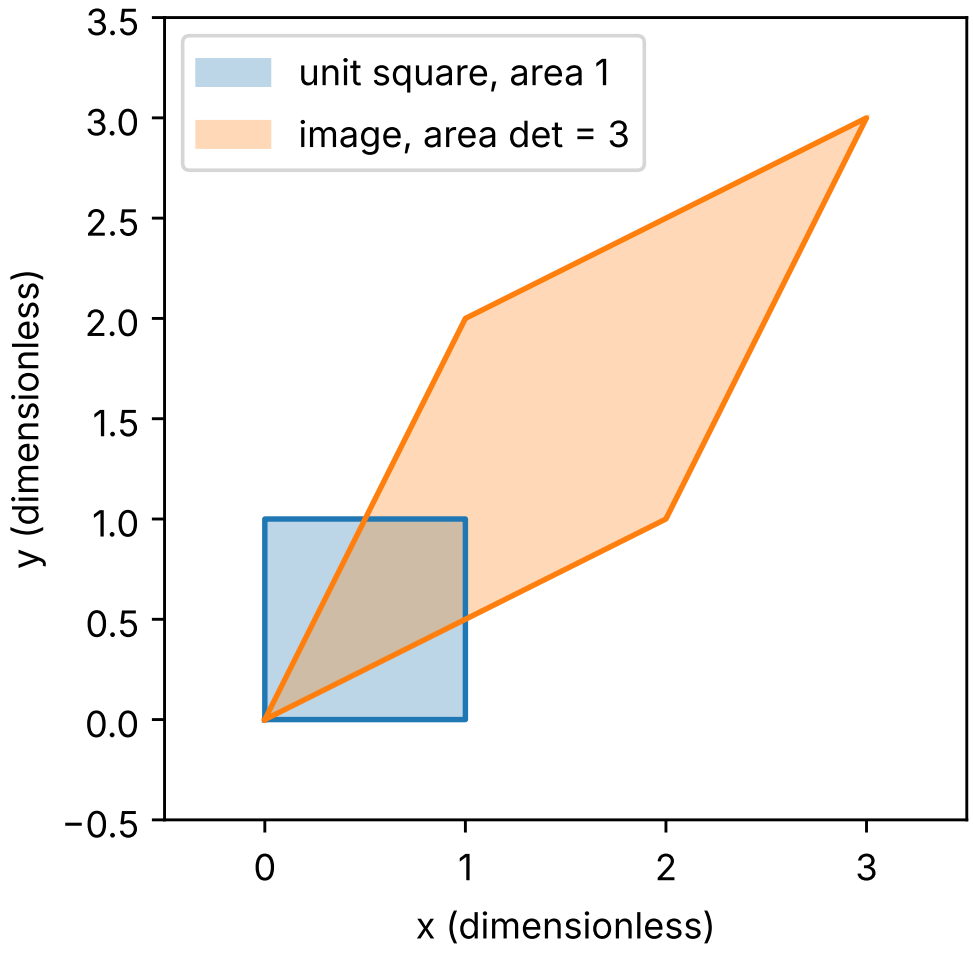


Figure 7.3: The matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ sends the unit square (area 1) to the parallelogram spanned by its columns $(2, 1)$ and $(1, 2)$, whose area is $\det = 2 \cdot 2 - 1 \cdot 1 = 3$. Axes dimensionless.

Taken on trust

Taken on trust: the volume interpretation. The vector $\boldsymbol{r}_2 \times \boldsymbol{r}_3$ is perpendicular to both $\boldsymbol{r}_2$ and $\boldsymbol{r}_3$, and its length is the area of the parallelogram they span. Dotting it with $\boldsymbol{r}_1$ therefore multiplies that base area by the height of $\boldsymbol{r}_1$ above the base (up to sign). Base times height is volume, so $|\det A| = |\boldsymbol{r}_1 \cdot (\boldsymbol{r}_2 \times \boldsymbol{r}_3)|$ is the volume of the parallelepiped spanned by the rows. By property (D7) below, $\det A^{\top} = \det A$, so it is also the volume spanned by the columns, which are the images $A\boldsymbol{e}_1, A\boldsymbol{e}_2, A\boldsymbol{e}_3$ of the edges of the unit cube. Hence $|\det A|$ is the factor by which A scales volumes. Proofs: Riley, Hobson and Bence [22], Chs. 7 and 8.

7.10.3 Properties

Taken on trust

Taken on trust: properties of determinants. The determinant extends to $n \times n$ matrices, and: (D1) $\det I = 1$; (D2) swapping two rows changes the sign; (D3) multiplying one row by s multiplies det by s; (D4) adding a multiple of one row to another leaves det unchanged; (D5) the determinant of a triangular matrix is the product of its diagonal entries; (D6) $\det(AB) = \det A \det B$; (D7) $\det A^{\top} = \det A$; (D8) $\det A \neq 0$ if and only if A is invertible, if and only if its rank is n. Proofs: Strang [28], Ch. 5; Riley, Hobson and Bence [22], Ch. 8. You can check any of them on small cases. For instance (D4) for 2×2:
$\det \begin{pmatrix} a & b \\ c + sa & d + sb \end{pmatrix} = a(d + sb) - b(c + sa) = ad - bc.$

Checkpoint 7.7

Use (D2) to show that a matrix with two equal rows has determinant zero.

Answer: Swapping the two equal rows leaves the matrix unchanged, so its determinant is unchanged; by (D2) it also changes sign. So $\det = -\det$, which forces $\det = 0$.

Properties (D2), (D4), (D5) give a fast way to compute determinants: eliminate to triangular form, keep track of row swaps, and multiply the pivots.

Sample Problem 7.11: A 3×3 determinant two ways

Compute $\det A$ for $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}$, the matrix of the elimination sample.

Step 1 (expansion, Equation (7.30)): $1 \cdot (3 \cdot 3 - 1 \cdot 2) - 1 \cdot (2 \cdot 3 - 1 \cdot 1) + 1 \cdot (2 \cdot 2 - 3 \cdot 1) = 7 - 5 + 1 = 3$.

Step 2 (elimination): The sample used only operations of type (D4) and reached $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{pmatrix}$. By (D4) the determinant is unchanged; by (D5) it is $1 \cdot 1 \cdot 3 = 3$.

$\det A = 3 \neq 0$, so A is invertible and the solution $(1, 2, 3)$ is unique.

For an $n \times n$ matrix, repeated expansion like Equation (7.30) has $n!$ terms ($n!$ grows faster than any power), while elimination needs $\approx n^3/3$ operations. Computers always use elimination.

Key Ideas

- $|\det A|$ is the area (2D) or volume (3D) scaling factor of A; $\det A = 0$ means the columns are dependent and A is singular.
- Compute determinants by elimination: product of pivots, with a sign flip per row swap.

7.11 Functions as vectors

7.11.1 Sampling a function on a grid

A computer cannot store a function $f(x)$; it stores numbers. The paper (Sec. II B, App. A step 3) works on the interval $(-L, L)$ with N equally spaced *interior* points,

$$x_j = -L + jh, \qquad j = 1, \ldots, N, \qquad h = \frac{2L}{N+1}, \tag{7.31}$$

so the walls $x_0 = -L$ and $x_{N+1} = L$ are not stored (the wavefunction is zero there, Chapter 11). The function becomes the vector

$$\boldsymbol{f} = (f(x_1),\, f(x_2),\, \ldots,\, f(x_N)) \in \mathbb{R}^N.$$

(Here N is the grid size; the same letter is used for zero-counting functions elsewhere, and context distinguishes them.) Adding functions or multiplying them by constants corresponds exactly to adding or scaling their sample vectors, so the sampled functions behave like vectors in $\mathbb{R}^N$ (Figure 7.4).

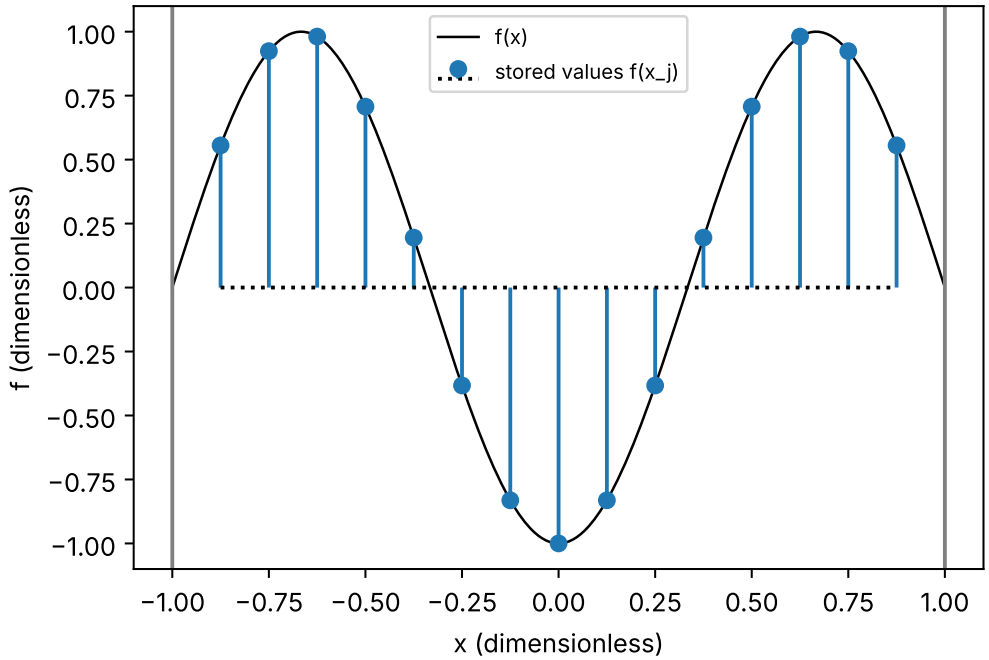


Figure 7.4: A function as a vector. The curve $f(x) = \sin(3\pi(x+1)/2)$ on $(-1, 1)$ is stored as its values (stems) at $N = 15$ interior grid points $x_j = -1 + jh$, $h = 2/16$. The walls $x = \pm 1$ (vertical lines), where $f = 0$, are not stored. x and f dimensionless.

7.11.2 The inner product of functions

For two sampled functions, the inner product of their vectors, multiplied by the spacing h, is

$$h\sum_{j=1}^{N} f(x_j)^* g(x_j),$$

a Riemann sum for $\int f^* g\,\mathrm{d}x$. As $h \to 0$ (more and more points) it tends to the integral, by the definition of the integral as a limit of Riemann sums, for continuous f and g. This motivates the definition

$$\langle f, g\rangle = \int_{-L}^{L} f(x)^*\, g(x)\,\mathrm{d}x, \qquad \|f\| = \sqrt{\langle f, f\rangle} = \sqrt{\int_{-L}^{L} |f(x)|^2\,\mathrm{d}x}. \tag{7.32}$$

It has the same four properties as Equation (7.14) (conjugate symmetry, linearity, conjugate-linearity, positivity), each proved the same way with the sum replaced by an integral. Positivity needs one remark: $\int |f|^2\,\mathrm{d}x = 0$ forces $f = 0$ for a continuous f, since a continuous non-negative function with zero integral must vanish everywhere. A function with $\|f\| = 1$ is *normalized*; the paper normalizes its discrete eigenvectors by $\sum_j \psi_j^2\, h = 1$ (App. A step 9), the grid version of $\int |\psi|^2\,\mathrm{d}x = 1$.

Sample Problem 7.12: A grid sum that is exact

Take $L = 1$, $f(x) = \cos(\pi x/2)$ (which vanishes at the walls). Compute $h\sum_j f(x_j)^2$ for $N = 3$, and show it equals $\int_{-1}^{1} f^2 \mathrm{d}x$ for every N.
Step 1 ($N = 3$): $h = 2/4 = 0.5$, points $-0.5, 0, 0.5$. Values of f^2: $\cos^2(\pi/4) = 0.5$, 1, 0.5. Sum 2, times h: 1.
Step 2 (exact integral): $\int_{-1}^{1} \cos^2(\pi x/2)\mathrm{d}x = \int_{-1}^{1} \frac{1}{2}(1 + \cos \pi x)\mathrm{d}x = 1 + 0 = 1$.
Step 3 (any N): With $x_j = -1 + jh$, $h = 2/(N+1)$: $\pi x_j/2 = -\pi/2 + j\pi/(N+1)$, and $\cos(\theta - \pi/2) = \sin\theta$, so $f(x_j)^2 = \sin^2 \frac{j\pi}{N+1} = \frac{1}{2}(1 - \cos \frac{2\pi j}{N+1})$.
Step 4: Let $z = e^{2\pi i/(N+1)}$ (Chapter 2). The finite geometric sum $S = \sum_{j=0}^{N} z^j$ satisfies $zS - S = z^{N+1} - 1 = e^{2\pi i} - 1 = 0$; since $z \neq 1$ for $N \geq 1$, $S = 0$. Its real part gives $\sum_{j=0}^{N} \cos \frac{2\pi j}{N+1} = 0$, so $\sum_{j=1}^{N} \cos \frac{2\pi j}{N+1} = -1$ (removing the $j = 0$ term, which is 1).
Step 5: $\sum_{j=1}^{N} f(x_j)^2 = \frac{1}{2}(N - (-1)) = \frac{N+1}{2}$, and times $h = \frac{2}{N+1}$ gives 1.

$$\boxed{h\sum_{j=1}^{N} f(x_j)^2 = 1 = \int_{-1}^{1} f^2\,\mathrm{d}x \quad \text{for every } N.}$$

Pitfall

The interior-point sum omits the walls. That is harmless for functions that vanish there (wavefunctions in a box, and products involving them), but not in general. For $f = 1$, $g = x^2$ on $(-1, 1)$, the exact $\langle f, g\rangle = 2/3 \approx 0.6667$, while $h\sum x_j^2 = 0.48$ for $N = 9$ and 0.6468 for $N = 99$; the shortfall is about h times the wall value 1. Adding the two wall points with weight $h/2$ each (the trapezoid rule) removes most of it: the sums become $0.48 + 0.2 = 0.6800$ for $N = 9$ and $0.6468 + 0.02 = 0.6668$ for $N = 99$, errors of 0.0133 and 0.0001 instead of 0.19 and 0.02.

7.11.3 An orthogonal basis of cosines

Consider the functions $\phi_k(x) = \cos(k\pi x/L)$, $k = 0, 1, 2, \ldots$, on $[-L, L]$, the basis the paper uses for δV (Sec. II C). We show they are mutually orthogonal under Equation (7.32).

First, for an integer $q \neq 0$,

$$\int_{-L}^{L} \cos\frac{q\pi x}{L}\mathrm{d}x = \frac{L}{q\pi}\Big[\sin\frac{q\pi x}{L}\Big]_{-L}^{L} = \frac{L}{q\pi}\big(\sin q\pi - \sin(-q\pi)\big) = 0,$$

and for $q = 0$ the integrand is 1, giving $2L$. Now use $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$:

$$\langle\phi_j,\phi_k\rangle = \frac{1}{2}\int_{-L}^{L}\cos\frac{(j-k)\pi x}{L}\mathrm{d}x + \frac{1}{2}\int_{-L}^{L}\cos\frac{(j+k)\pi x}{L}\mathrm{d}x. \tag{7.33}$$

- $j \neq k$ (both ≥ 0): $j-k \neq 0$ and $j+k \neq 0$, so both integrals vanish: $\langle\phi_j,\phi_k\rangle = 0$.
- $j = k \geq 1$: the first integral is $2L$, the second ($q = 2k \neq 0$) vanishes: $\langle\phi_k,\phi_k\rangle = L$.
- $j = k = 0$: $\phi_0 = 1$, and $\langle\phi_0,\phi_0\rangle = 2L$.

Summarizing,

$$\int_{-L}^{L}\cos\frac{j\pi x}{L}\cos\frac{k\pi x}{L}\mathrm{d}x = \begin{cases} 0, & j \neq k, \\ L, & j = k \geq 1, \\ 2L, & j = k = 0. \end{cases} \tag{7.34}$$

They are orthogonal but not normalized; dividing ϕ_k by $\sqrt{L}$ (or ϕ_0 by $\sqrt{2L}$) would make them orthonormal.

7.11.4 Expanding a function in a basis

Suppose an even function can be written as $f(x) = \sum_k a_k\phi_k(x)$. Take the inner product with ϕ_j and use Equation (7.34):

$$\langle\phi_j, f\rangle = \sum_k a_k\langle\phi_j,\phi_k\rangle = a_j\langle\phi_j,\phi_j\rangle,$$

so

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,\mathrm{d}x, \qquad a_k = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{k\pi x}{L}\,\mathrm{d}x \quad (k \geq 1). \tag{7.35}$$

This is Equation (7.15) for functions.

Taken on trust

Taken on trust: completeness. For an infinite sum, moving $\langle\phi_j,\cdot\rangle$ inside the sum, and the claim that every reasonable even function on $[-L,L]$ *can* be written as $\sum_k a_k\cos(k\pi x/L)$, both need convergence theorems for Fourier series. They hold for continuous, piecewise-smooth even functions (convergence in the mean-square sense for much wider classes). See Riley, Hobson and Bence [22], Ch. 12, and Chapter 5. The paper never needs the infinite sum: it uses a finite M, for which no convergence question arises.

Sample Problem 7.13: Cosine coefficients of x^2

Expand $f(x) = x^2$ on $[-1,1]$ ($L = 1$) in $\cos(k\pi x)$. Find a_0, a_1, a_2, a_3 and the partial sums at $x = 0$.
Step 1: $a_0 = \frac{1}{2}\int_{-1}^{1} x^2\mathrm{d}x = \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{3}$.

Step 2: For $k \geq 1$, $a_k = \int_{-1}^{1} x^2 \cos(k\pi x)\mathrm{d}x$. Integrate by parts with $u = x^2$, $\mathrm{d}v = \cos(k\pi x)\mathrm{d}x$, $v = \sin(k\pi x)/(k\pi)$:

$$a_k = \Big[\frac{x^2 \sin k\pi x}{k\pi}\Big]_{-1}^{1} - \frac{2}{k\pi}\int_{-1}^{1} x\sin(k\pi x)\mathrm{d}x = 0 - \frac{2}{k\pi}\int_{-1}^{1} x\sin(k\pi x)\mathrm{d}x,$$

since $\sin(\pm k\pi) = 0$.
Step 3: Again by parts, $u = x$, $\mathrm{d}v = \sin(k\pi x)\mathrm{d}x$, $v = -\cos(k\pi x)/(k\pi)$:

$$\int_{-1}^{1} x\sin(k\pi x)\mathrm{d}x = \Big[-\frac{x\cos k\pi x}{k\pi}\Big]_{-1}^{1} + \frac{1}{k\pi}\int_{-1}^{1}\cos(k\pi x)\mathrm{d}x.$$

The last integral is zero. At $x = 1$ the bracket is $-(-1)^k/(k\pi)$; at $x = -1$ it is $-(-1)\cos(-k\pi)/(k\pi) = (-1)^k/(k\pi)$. Upper minus lower: $-2(-1)^k/(k\pi)$.
Step 4: $a_k = -\frac{2}{k\pi}\cdot\Big(-\frac{2(-1)^k}{k\pi}\Big) = \frac{4(-1)^k}{k^2\pi^2}$. So $a_1 = -4/\pi^2 = -0.40528$, $a_2 = 1/\pi^2 = 0.10132$, $a_3 = -4/(9\pi^2) = -0.04503$.
Step 5: At $x = 0$ every $\cos(0) = 1$. Partial sums: $a_0 + a_1 = -0.0720$; $+a_2$: 0.0294; $+a_3$: -0.0157. They approach the true value 0, overshooting alternately.

$$\boxed{a_0 = \tfrac{1}{3}, \quad a_k = \frac{4(-1)^k}{k^2\pi^2}\ (k \geq 1).}$$

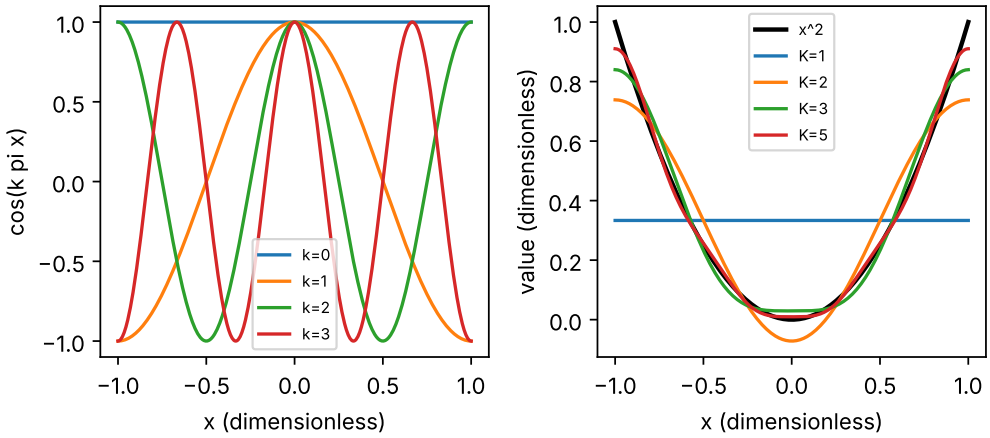


Figure 7.5: Left: the first cosine basis functions $\cos(k\pi x/L)$, $k = 0, 1, 2, 3$, on $[-L, L]$ with $L = 1$; they are mutually orthogonal (Equation (7.34)). Right: x^2 (black) and its partial cosine sums with $K = 1, 2, 3, 5$ terms ($k = 0, \ldots, K-1$), using $a_0 = 1/3$, $a_k = 4(-1)^k/(k^2\pi^2)$. x and function values dimensionless.

7.11.5 Non-orthogonal bases

The paper also uses Chebyshev polynomials $T_k(2|x|/L - 1)$ and piecewise-linear "hat" functions (Sec. II C, "Other bases"; Chapter 24 defines them). Under $\int \mathrm{d}x$ these are not mutually orthogonal (neighboring hats overlap, for instance). The coefficients are then no longer single inner products. If $f = \sum_k a_k\phi_k$ exactly, taking the inner product with each ϕ_j gives

$$\sum_k \underbrace{\langle \phi_j, \phi_k\rangle}_{G_{jk}} a_k = \underbrace{\langle \phi_j, f\rangle}_{b_j} \qquad \Longleftrightarrow \qquad G\boldsymbol{a} = \boldsymbol{b}, \tag{7.36}$$

a linear system with the *Gram matrix* G. For an orthogonal basis G is diagonal and Equation (7.36) collapses to Equation (7.35); otherwise you solve it by elimination. (When f is *not* exactly in the span, require instead that the residual $f - \sum_k a_k\phi_k$ be orthogonal to every ϕ_j. The condition $\langle \phi_j, f - \sum_k a_k\phi_k\rangle = 0$ is exactly Equation (7.36). A residual orthogonal to the span makes $\sum_k a_k\phi_k$ the closest function in the span, by the same Pythagoras argument that gave Equation (7.19); that argument used only the orthogonality of the residual, not the orthonormality of the spanning set. This is the least-squares fit of Chapter 9; Problem 17 is an example.)

Key Ideas

- On the grid $x_j = -L + jh$, $h = 2L/(N+1)$, a function is a vector in $\mathbb{R}^N$; $h\sum f_j^* g_j \to \int f^* g\,\mathrm{d}x$.
- $\cos(k\pi x/L)$ are orthogonal on $[-L, L]$ (Equation (7.34)); expansion coefficients are inner products (Equation (7.35)).
- For non-orthogonal bases, coefficients solve the Gram system $G\boldsymbol{a} = \boldsymbol{b}$ (Equation (7.36)).

7.12 A matrix made of integrals: the sensitivity matrix

In the paper (Sec. II C), the energy levels E_n of a quantum particle in a potential $V(x)$ are adjusted by adding a correction $\delta V(x) = \sum_{k=0}^{M-1} c_k\phi_k(x)$. Chapter 13 derives that, to first order, each level moves by

$$\delta E_n \approx \sum_{k=0}^{M-1} A_{nk}\, c_k, \qquad A_{nk} = \int_{-L}^{L} |\psi_n(x)|^2\, \phi_k(x)\,\mathrm{d}x, \tag{7.37}$$

where ψ_n is the normalized wavefunction of level n ($\int |\psi_n|^2\,\mathrm{d}x = 1$). Here we take Equation (7.37) as given and study it as linear algebra. (The paper also uses the letter A for a non-Hermitian operator in Sec. II E; that is a different object.)

Reading the matrix. Row n belongs to level n, column k to basis function k. In the column picture Equation (7.22), $\delta\boldsymbol{E} = A\boldsymbol{c}$ is a combination of columns: column k is the pattern of level shifts produced by basis function k at unit amplitude. Entry A_{nk} is the inner product $\langle |\psi_n|^2, \phi_k\rangle$ of two real functions: the overlap between where level n "lives" and the shape of basis function k.

The first column is all ones. With $\phi_0 = 1$, $A_{n0} = \int |\psi_n|^2\,\mathrm{d}x = 1$ for every n, by normalization alone, for any potential. So a constant correction c_0 shifts every level by the same c_0, to first order. This is the uniform-shift direction, and it will matter in Chapter 14, where a constant imaginary potential produces no real shift at all.

On a grid. With $W_{nj} = |\psi_n(x_j)|^2$ (an $n_{\text{lev}} \times N$ matrix) and $\Phi_{jk} = \phi_k(x_j)$ (an $N \times M$ matrix), the Riemann sum for Equation (7.37) is

$$A_{nk} \approx h\sum_{j=1}^{N} W_{nj}\Phi_{jk} = h\,(W\Phi)_{nk}, \tag{7.38}$$

a single matrix product, Equation (7.23).

Sample Problem 7.14: The sensitivity matrix of the box

For an infinite square well on $[-L, L]$ (running example RE3), Chapter 11 shows the normalized states are $\psi_n(x) = \frac{1}{\sqrt{L}} \sin \frac{n\pi(x+L)}{2L}$, $n = 1, 2, \ldots$. Compute A_{nk} for the cosine basis.

Step 1 (square): $|\psi_n|^2 = \frac{1}{L} \sin^2 \theta$ with $\theta = \frac{n\pi(x+L)}{2L}$. Using $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ and $2\theta = \frac{n\pi x}{L} + n\pi$,

$$|\psi_n|^2 = \frac{1}{2L}\Big[1 - \cos\Big(\frac{n\pi x}{L} + n\pi\Big)\Big].$$

Step 2: $\cos(\alpha + n\pi) = \cos\alpha \cos n\pi - \sin\alpha \sin n\pi = (-1)^n \cos\alpha$. So

$$|\psi_n|^2 = \frac{1}{2L}\Big[1 - (-1)^n \cos\frac{n\pi x}{L}\Big] = \frac{1}{2L}\big[\phi_0 - (-1)^n \phi_n\big].$$

Step 3: $A_{nk} = \langle |\psi_n|^2, \phi_k \rangle = \frac{1}{2L}\big[\langle \phi_0, \phi_k \rangle - (-1)^n \langle \phi_n, \phi_k \rangle\big]$. Apply Equation (7.34) with $n \geq 1$: for $k = 0$, $\frac{1}{2L}[2L - 0] = 1$; for $k \geq 1$, $\frac{1}{2L}[0 - (-1)^n L\, \delta_{nk}]$.

$$\boxed{A_{nk} = \begin{cases} 1, & k = 0, \\ (-1)^{n+1}/2, & k = n, \\ 0, & \text{otherwise.} \end{cases}}$$

Every row has the 1 in column 0 and at most one more nonzero entry, $+\frac{1}{2}$ for odd n and $-\frac{1}{2}$ for even n. In a truncated basis with columns $k = 0, \ldots, M-1$, the diagonal entry is present only when $n \leq M - 1$; a row with $n \geq M$ contains just the 1 in column 0. Remarkably, the answer does not depend on L.

The box matrix is extremely sparse. For the Wu–Sprung potential that the paper uses (Chapter 22), the wavefunctions are not pure sines, and A fills in (Figure 7.6). The bookkeeping, however, is identical, and the box lets us see the rank questions of Section 7.9 by hand.

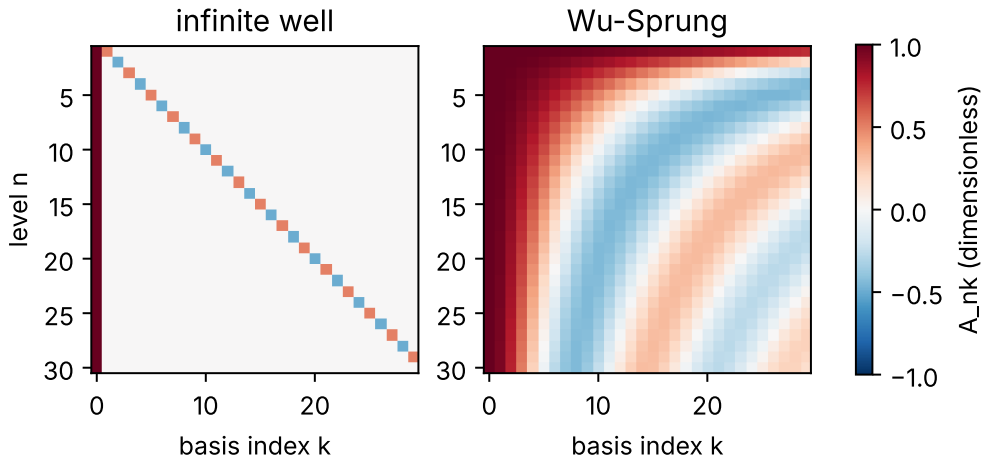


Figure 7.6: Sensitivity matrices $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\, dx$ for the first 30 levels (n, rows) and 30 cosines (k, columns), with $L = 44.883$ (scale 1). Left: infinite square well, from the formula of the box sample (first column 1, diagonal $\pm\frac{1}{2}$). Right: Wu–Sprung potential, computed on a 1200-point finite-difference grid; the first column is again 1, but the rest of the matrix is no longer sparse. Entries are dimensionless.

Sample Problem 7.15: Too many and too few coefficients in the box

Take three box levels with target shifts $\boldsymbol{r} = (0.2, -0.1, 0.3)$. (a) With $M = 4$ coefficients $c_0, \ldots, c_3$, find all exact solutions of $A\boldsymbol{c} = \boldsymbol{r}$. (b) With $M = 2$ (c_0, c_1), show there is no exact solution.

Step 1 (a): From the box formula, rows $n = 1, 2, 3$, columns $k = 0, \ldots, 3$:

$$A = \begin{pmatrix} 1 & \frac{1}{2} & 0 & 0 \\ 1 & 0 & -\frac{1}{2} & 0 \\ 1 & 0 & 0 & \frac{1}{2} \end{pmatrix}.$$

Three equations, four unknowns: underdetermined.

Step 2 (a): Choose $c_0 = s$ as the free variable. (Elimination would make c_3 the free variable, since the pivots fall in the columns of c_0, c_1, c_2. Any one of the four can be used to parametrize the line of solutions, and c_0 is the most convenient here.) Row 1: $s + c_1/2 = 0.2 \Rightarrow c_1 = 2(0.2 - s)$. Row 2: $s - c_2/2 = -0.1 \Rightarrow c_2 = 2(0.1 + s)$. Row 3: $s + c_3/2 = 0.3 \Rightarrow c_3 = 2(0.3 - s)$.

Step 3 (a): In the form Equation (7.28): $\boldsymbol{c} = (0,\ 0.4,\ 0.2,\ 0.6) + s(1, -2, 2, -2)$. Check the null vector: $A(1, -2, 2, -2) = (1 - 1,\ 1 - 1,\ 1 - 1) = \mathbf{0}$. Raising c_0 lifts all three levels, and the other three coefficients exactly cancel it.

Step 4 (b): Now $A = \begin{pmatrix} 1 & \frac{1}{2} \\ 1 & 0 \\ 1 & 0 \end{pmatrix}$. Rows 2 and 3 say $c_0 = -0.1$ and $c_0 = 0.3$. Elimination $(R_3 \leftarrow R_3 - R_2)$ gives the row $0 = 0.4$: inconsistent.

$$\boxed{\text{(a) } \boldsymbol{c} = (0, 0.4, 0.2, 0.6) + s(1, -2, 2, -2),\ s \in \mathbb{R}; \qquad \text{(b) no solution.}}$$

With $M = 2$ levels 2 and 3 are reachable only through c_0, so their different targets cannot both be met; with $M = 4$ every level has its own coefficient plus a spare, and infinitely many coefficient vectors fit perfectly.

Checkpoint 7.8

For the box with n_{lev} levels and $M = n_{\text{lev}} + 1$ coefficients $(k = 0, \ldots, n_{\text{lev}})$, what is the rank of A, and what is the dimension of its null space?

Answer: Columns $1, \ldots, n_{\text{lev}}$ are $\pm\frac{1}{2}$ times the standard basis vectors, which are independent and span $\mathbb{R}^{n_{\text{lev}}}$, so the rank is n_{lev}. The null space has dimension $M - n_{\text{lev}} = 1$, spanned by the vector that combines column 0 with the others to cancel, as in the sample.

```
import numpy as np
L, N = 44.883, 3000                  # RE3 box, scale-1 half-width
h = 2*L/(N + 1)
x = -L + h*np.arange(1, N + 1)       # interior grid points
nlev, M = 6, 6
n = np.arange(1, nlev + 1)           # levels start at 1
k = np.arange(M)                     # cosines start at 0
psi = np.sin(np.outer(x + L, n)*np.pi/(2*L))/np.sqrt(L)    # N x nlev
Phi = np.cos(np.outer(x, k)*np.pi/L)                        # N x M
A = h*(psi**2).T @ Phi               # nlev x M, the grid version of A_nk
print(np.round(A, 6))                # first column 1; A[n-1, n] = +-0.5 for n <= M-1
```

Key Ideas

- The paper's $A_{nk} = \int |\psi_n|^2 \, \phi_k \, \mathrm{d}x$ is a matrix of inner products: rows are levels, columns are basis functions; on a grid it is $h\,W\Phi$.
- Its first column is all ones for any potential: c_0 shifts every level equally.
- For the box, A is sparse and the over/underdetermined cases can be solved by hand.

7.13 Paper connection

★ Did you know? *A Word From Sylvester*

James Joseph Sylvester coined the word *matrix* in 1850, from the Latin for womb, because he saw a rectangular array as the source from which determinants are born. His friend Arthur Cayley published the first systematic algebra of matrices in 1858, including matrix multiplication and inverses. Every numerical calculation later in this book ultimately reduces to operations on matrices.

Here is where the tools of this chapter appear in the paper [1] (revision R1; code and data in [20]).

The correction as a basis expansion (Sec. II C). The fitted correction $\delta V(x) = \sum_{k=0}^{M-1} c_k \cos(k\pi x/L)$ is a finite expansion in the orthogonal cosine basis of Equation (7.34); the vector $c \in \mathbb{R}^M$ is everything the fit can adjust. The "Other bases" paragraph replaces the cosines by Chebyshev polynomials and hat functions, which are not orthogonal under $\int \mathrm{d}x$; the Gram system Equation (7.36) shows that the coefficients of a non-orthogonal basis are coupled to one another. Only the basis changes; the vector bookkeeping is the same.

The sensitivity matrix and the residual vector (Sec. II C, App. A step 6). The paper forms the residual vector r with components $r_n = \tilde{t}_n - E_n$ and the matrix $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\, \mathrm{d}x$, then updates c by a damped least-squares solution of $A\,\delta c \approx r$. The shape of A is (number of training levels)$\times M$. In Table I, the scale-1 fit on zeros 1–400 with $M = 200$ has a 400×200 matrix (overdetermined); the same fit with $M = 600$ has 400×600 (underdetermined, null space of dimension at least 200). The first column is all ones (Section 7.12), so c_0 moves all levels together. The "SVD-damped" step itself is the subject of Chapter 9; the update rule as a whole is explained in Chapter 24.

"More coefficients than levels". Section 7.9 showed that an underdetermined system of full row rank has exact solutions for *every* right-hand side, and infinitely many of them. This is the linear-algebra core of several statements in the paper. Sec. III reports that on the full 500 zeros the fit reaches rms 0.186 at $M = 300$ and reproduces all 500 zeros to rms $< 10^{-4}$ spacings by $M = 900$. Sec. III (Within-height splits) notes that the odd/even splits have 200 training levels at scale 1, "so the rows with $M \geq 200$ are saturated", and the $M = 400$ odd/even rows of Table III indeed have in-sample error $< 10^{-3}$. The square case $M = 200$ still leaves in-sample rms 0.176 and 0.171, so saturation by count does not by itself give an exact fit. Sec. IV says of control B that "at 900 coefficients on 400 zeros the least-squares problem is underdetermined". (Control B's model $F(t) = \sum_j [a_j \sin(\omega_j t) + b_j \cos(\omega_j t)]$ is linear in the amplitudes, so its fit is literally a linear system with one row per training zero and one column per coefficient: 150 primes times three harmonics times two amplitudes gives 900 columns; control C1 has 450 frequencies times two, again 900.) A caution about status: the operator

fit is nonlinear, because ψ_n and E_n change as c changes, so the counting argument applies exactly only to each linearized step. The paper's own statement is empirical and heuristic: the bases "are effectively complete for in-sample level patterns, so they realize essentially arbitrary joint shifts" (Sec. VI). Here "joint shifts" means arbitrary patterns of simultaneous level shifts, not the uniform shift produced by c_0. The surrogate tests of Sec. III, in which permuted, phase-randomized and amplitude-adjusted surrogate targets are fitted as well as the true zeros (in-sample rms 0.134 ± 0.010, 0.141 ± 0.006 and 0.138 ± 0.005 respectively, against 0.137 for the true zeros), are, heuristically, what a nearly full-rank, nearly saturated linear model would lead you to expect. This is an analogy, not a derivation: the paper itself calls its explanation of the failure heuristic ("The failure is intelligible, though heuristically so", Sec. VI).

Shape is not rank. Counting rows and columns gives only an upper bound on what can be fitted: an underdetermined shape guarantees an exact fit only if the matrix has full row rank. In Table I the fits with more coefficients than levels (zeros 1–300 with $M = 400$, 1–400 with $M = 600$, 1–800 with $M = 900$) still have in-sample rms 0.120, 0.064 and 0.103. Sec. III reports that the 500 zeros are reproduced to rms $< 10^{-4}$ only by $M = 900$, not as soon as M exceeds 500. The linearized sensitivity matrix is therefore effectively rank deficient until the cosines can resolve the wavefunctions of the levels being fitted. This is the paper's coverage law $M\pi/L > 2\sqrt{E}$, discussed in Chapter 24, and the notion of "effectively" rank deficient (tiny but nonzero singular values) is made precise in Chapter 9.

rms is a norm; corr is an angle. Every rms in the paper is $\|\boldsymbol{e}\|/\sqrt{n}$ for an error vector measured in local mean spacings (Equation (7.7)). The null rms on the principal holdout, zeros 401–500 (running example RE5), is 0.2631 (App. A, key anchors), so the null error vector over those 100 zeros has norm $0.2631 \times \sqrt{100} = 2.631$. The correlations are the cosines of Equation (7.13). The geometry of Equation (7.11) is consistent with the paper's two statements "the held-out rms never improves on the null" and "no reproducible positive held-out correlation appears" (Sec. III) appearing together. When $\cos\theta$ is zero or negative, no nonzero prediction can beat the null. When $\cos\theta$ is small and positive, only a prediction of small amplitude, $\|\boldsymbol{p}\| < 2\,\|\boldsymbol{d}\|\cos\theta$, can. This is a guide, not an exact explanation, for two reasons. The paper's corr is the centered (Pearson) cosine. The paper also has rows with positive correlation and rms above the null, for example +0.12 with 0.439 in Table I, and Chebyshev $M = 400$ with +0.18 and 0.291 in Table II. In those rows the predictions are too large for their small correlation. Fig. 1 shows the two extremes, the fitted operator with held-out correlation -0.07 and the six-prime formula with +0.96. As a rough consistency check only (it assumes mean-zero vectors and optimal amplitude, and the correlation is rounded to two digits), Equation (7.12) with $\cos\theta = 0.96$ gives a best error of $0.263 \times \sqrt{1 - 0.96^2} = 0.074$ spacings, the same as the six-prime rms 0.074 reported for 401–500 in Table V.

Functions on grids (Sec. II B, App. A steps 3, 4, 9). At scale 1 the paper uses $L = 44.8827$ and a working grid of $N = 6000$ interior points, so every wavefunction is a vector in $\mathbb{R}^{6000}$ with spacing $h = 2L/(N+1) \approx 0.01496$ (Equation (7.31)). Eigenvectors are normalized by $\sum \psi^2\,\mathrm{d}x = 1$, the grid form of $\|\psi\| = 1$ (step 9). The Hamiltonian matrix is real symmetric and tridiagonal (step 4: diagonal $2/\mathrm{d}x^2 + V$, off-diagonal $-1/\mathrm{d}x^2$); Chapters 8 and 23 take it from there.

Vector identities. The no-leakage argument, Eq. (3), $\tilde{t}_n - E_n[\text{fit}] = t_n - (E_n[\text{fit}] - \Delta_n)$, is an identity between vectors: adding the fixed vector Δ to the targets ($\tilde{t} = t + \Delta$, Sec. II B) is the same as subtracting it from the prediction and scoring against the raw zeros. Eq. (5), $\delta E_n = -c^\top T^{(n)} c$, is a quadratic form, one $M \times M$ matrix $T^{(n)}$ per level. The sum rule of

Sec. V A, $\sum_n \delta E_n = 0$, says that the vector of second-order shifts is orthogonal to the all-ones vector $(1, \dots, 1)$.

Complex vectors and overlaps (Sec. II E). In the non-Hermitian study the eigenvectors of $H + iA$ are complex, and they are "matched to the unperturbed basis by overlap": the overlap is the complex inner product Equation (7.14), whose modulus is largest for the unperturbed state the perturbed one most resembles (Chapters 8 and 14). Notice the overloaded symbols: A is the sensitivity matrix in Sec. II C but a non-Hermitian multiplication operator in Sec. II E; N is the grid size in App. A but the zero-counting function in Eq. (1).

7.14 Summary

- Vectors in $\mathbb{R}^n$ and $\mathbb{C}^n$ are ordered lists; addition and scaling act componentwise (Equations (7.2) and (7.3)).
- Real inner product $\boldsymbol{u} \cdot \boldsymbol{v} = \sum u_j v_j$ (Equation (7.5)); complex $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \sum u_j^* v_j$ (Equation (7.14)); norm $\|\boldsymbol{v}\| = \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$; rms $= \|\boldsymbol{e}\| / \sqrt{n}$ (Equation (7.7)).
- Cauchy–Schwarz $|\boldsymbol{u} \cdot \boldsymbol{v}| \le \|\boldsymbol{u}\| \, \|\boldsymbol{v}\|$ (Equation (7.8)) defines angles (Equation (7.9)); correlation is the cosine between centered vectors (Equation (7.13)); a prediction beats the null iff $\|\boldsymbol{p}\| < 2 \, \|\boldsymbol{d}\| \cos\theta$ (Equation (7.11)); best rescaled error $\|\boldsymbol{d}\| \sin\theta$ (Equation (7.12)).
- Orthonormal components are inner products (Equation (7.15)); $\|\boldsymbol{v}\|^2 = \sum |a_k|^2$ (Equation (7.16)); projection gives the closest point (Equations (7.18) and (7.19)).
- Matrices are linear maps; $A\boldsymbol{x}$ is a combination of columns (Equation (7.22)); $(AB)_{ik} = \sum_j A_{ij} B_{jk}$ (Equation (7.23)); $(AB)^\top = B^\top A^\top$; $\langle \boldsymbol{u}, A\boldsymbol{v} \rangle = \langle A^\dagger \boldsymbol{u}, \boldsymbol{v} \rangle$ (Equation (7.24)).
- 2×2 inverse (Equation (7.26)) exists iff $ad - bc \neq 0$; elimination solves $A\boldsymbol{x} = \boldsymbol{b}$ in $\approx n^3/3$ operations (Equation (7.27)).
- Rank = number of pivots; general solution = particular + null space (Equation (7.28)); overdetermined systems usually have no exact solution, underdetermined full-rank systems have infinitely many for every right side.
- Determinants: area/volume factors (Equations (7.29) and (7.30)); $\det A \neq 0$ iff invertible.
- Functions as grid vectors (Equation (7.31)); $\langle f, g \rangle = \int f^* g \, \mathrm{d}x$ (Equation (7.32)); cosine orthogonality (Equation (7.34)); coefficients (Equation (7.35)); Gram system for non-orthogonal bases (Equation (7.36)).
- Sensitivity matrix $A_{nk} = \int |\psi_n|^2 \, \phi_k \, \mathrm{d}x$ (Equation (7.37)), first column all ones; on a grid $A \approx hW\Phi$ (Equation (7.38)).

7.15 Problems

1. • For $\boldsymbol{u} = (2, -1, 3, 0)$ and $\boldsymbol{v} = (1, 4, 0, -2)$, compute $\boldsymbol{u} + \boldsymbol{v}$, $3\boldsymbol{u} - 2\boldsymbol{v}$, $\boldsymbol{u} \cdot \boldsymbol{v}$, $\|\boldsymbol{u}\|$ and $\|\boldsymbol{v}\|$.

2. • Normalize $\boldsymbol{w} = (1, 2, 2)$.

3. • A fit leaves residuals $(0.1, -0.2, 0.2, 0.0, -0.1)$ in mean spacings. What is the rms?

4. • For $\boldsymbol{u} = (i, \, 1 + 2i)$ and $\boldsymbol{w} = (1, \, i)$, compute $\|\boldsymbol{u}\|$, $\|\boldsymbol{w}\|$ and $\langle \boldsymbol{u}, \boldsymbol{w} \rangle$, and verify Cauchy–Schwarz.

5. •• Prove that if $\boldsymbol{u}\cdot\boldsymbol{v}=0$ then $\|\boldsymbol{u}+\boldsymbol{v}\|^2=\|\boldsymbol{u}\|^2+\|\boldsymbol{v}\|^2$, and conversely, for real vectors, that this equality implies $\boldsymbol{u}\cdot\boldsymbol{v}=0$.

6. • Compute $A\boldsymbol{x}$ for $A=\begin{pmatrix}2&1\\0&3\\1&-1\end{pmatrix}$ and $\boldsymbol{x}=(1,2)$, using both the row and the column picture.

7. •• Compute AB and BA for $A=\begin{pmatrix}1&2\\3&4\end{pmatrix}$, $B=\begin{pmatrix}0&1\\1&0\end{pmatrix}$. Describe in words what multiplying by B on the left and on the right does.

8. •• Prove $(AB)^{\dagger}=B^{\dagger}A^{\dagger}$ for complex matrices, using indices.

9. • Find the inverse of $\begin{pmatrix}3&1\\5&2\end{pmatrix}$ and check it.

10. •• Solve by elimination: $2x+y-z=1$, $x+3y+2z=13$, $3x-y+z=4$.

11. •• For the system $x+2y=3$, $2x+ay=b$, find the values of a and b for which there is (i) a unique solution, (ii) no solution, (iii) infinitely many solutions.

12. •• Find the rank of $\begin{pmatrix}1&2&3\\2&4&6\\1&0&1\end{pmatrix}$ and a vector spanning its null space.

13. • Compute $\det\begin{pmatrix}2&0&1\\1&3&2\\1&1&1\end{pmatrix}$. What does your answer say about the columns?

14. • Find the area of the parallelogram spanned by $(3,1)$ and $(1,2)$.

15. •• Compute $\int_{-L}^{L}\cos^2(3\pi x/L)\,\mathrm{d}x$, $\int_{-L}^{L}\cos(\pi x/L)\cos(2\pi x/L)\,\mathrm{d}x$ and $\int_{-L}^{L}\sin(\pi x/L)\cos(\pi x/L)\,\mathrm{d}x$.

16. •• Find the cosine coefficients of $f(x)=|x|$ on $[-1,1]$ ($L=1$). Evaluate the partial sum with $k=0,1$ at $x=0$.

17. ••• The functions 1 and x on $[0,1]$ are not orthogonal. Set up and solve the Gram system Equation (7.36) with $f(x)=x^2$, using $\langle f,g\rangle=\int_0^1 fg\,\mathrm{d}x$. Interpret the resulting function a_0+a_1x.

18. •• (Paper.) App. A step 3 of the paper gives the working grids: $L=44.8827$, $N=6000$ at scale 1, and $L=59.9572$, $N=8000$ at scale 2. Compute the grid spacing h at each scale and state the length of the vector storing one wavefunction.

19. •• (Paper.) Give the shape of the sensitivity matrix for these configurations of Tables I and III, and say whether each linearized system is over- or underdetermined and the minimum dimension of the null space when underdetermined: (a) scale 1, fit 1–300, $M=400$; (b) scale 1, fit 1–400, $M=200$; (c) scale 2, fit 1–800, $M=900$; (d) the scale-1 odd split with $M=400$. Relate (d) to the in-sample entry in Table III.

20. •• (Paper.) Table V reports, on held-out zeros 401–500, a null rms of 0.263 and rms 0.074 (six primes) and 0.043 (25 primes). Convert each into the norm of the error vector over those 100 zeros, and give the ratio of the null norm to the six-prime norm.

21. ••• Prove the Cauchy–Schwarz inequality $|\langle\boldsymbol{u},\boldsymbol{v}\rangle|\le\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|$ for complex vectors. (Hint: expand $\|\boldsymbol{u}-\lambda\boldsymbol{v}\|^2$ with complex $\lambda=\langle\boldsymbol{v},\boldsymbol{u}\rangle/\|\boldsymbol{v}\|^2$.)

22. •• Prove that for any normalized ψ_n, the column $k=0$ of A_{nk} (with $\phi_0=1$) is all ones. What does this imply for the first-order effect of adding a constant to the potential?

23. •• (Computational.) Using numpy, build the box sensitivity matrix on a grid as in the last listing of Section 7.12 (with $L=44.883$, $N=3000$, six levels, six cosines) and compare it with the analytic result of the box sample. How large is the largest deviation?

24. •• (Computational.) Using `numpy.linalg.matrix_rank`, find the ranks of the two box matrices of the sample "Too many and too few coefficients in the box". Then use `numpy.linalg.solve` on a random 5×5 system and print the largest component of $A\boldsymbol{x}-\boldsymbol{b}$.

25. ••• (Computational, paper.) Using qmnt, form the vector of fluctuations $d_n=(t_n-s_n)/\mathrm{gap}_n$ for $n=401,\dots,500$ with `qmnt.zeros` and `qmnt.fluct`, and compute its rms. Compare with the paper's null (App. A, key anchors). Explain why the rms of $\boldsymbol{d}$ is the null's error.

26. •• Compute the Pearson correlation of $\boldsymbol{p}=(1,2,3,4)$ and $\boldsymbol{d}=(2,1,4,3)$ using Equation (7.13).

27. ••• Derive Equation (7.27) for the operation count of elimination, verify it for $n=3$, and evaluate it for $n=6000$. Why is a tridiagonal system so much cheaper?

28. ••• Project $\boldsymbol{v}=(1,2,3)$ onto the plane spanned by the orthonormal vectors $\boldsymbol{e}_1=(1,0,0)$ and $\boldsymbol{e}_2=(0,1,1)/\sqrt{2}$, and check that the residual is orthogonal to the plane.

29. •• Show that the rotation matrix $R(\theta)=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ preserves lengths and has determinant 1.

30. ••• (Paper.) In control B of the paper (Sec. II D and Sec. IV, Table IV), the fluctuation is modeled as $F(t)=\sum_j[a_j\sin(\omega_j t)+b_j\cos(\omega_j t)]$ with $\omega_j=m\log p$ for 150 primes and $m=1,2,3$, fitted to 400 training zeros. Write the fitting problem as a linear system, give the shape of its matrix, the minimum null-space dimension, and explain why ridge regularization (Chapter 9) is needed to pick a single answer.

31. ••• Among all solutions $\boldsymbol{c}=(0,0.4,0.2,0.6)+s(1,-2,2,-2)$ of part (a) of the box sample, find the one with the smallest norm. Show that it is orthogonal to the null vector $(1,-2,2,-2)$, and explain geometrically why it must be.

7.16 Answers to selected problems

Problem 1. Componentwise: $\boldsymbol{u}+\boldsymbol{v}=(3,3,3,-2)$; $3\boldsymbol{u}-2\boldsymbol{v}=(6-2,\ -3-8,\ 9-0,\ 0+4)=(4,-11,9,4)$; $\boldsymbol{u}\cdot\boldsymbol{v}=2-4+0+0=-2$; $\|\boldsymbol{u}\|=\sqrt{4+1+9}=\sqrt{14}=3.7417$; $\|\boldsymbol{v}\|=\sqrt{1+16+4}=\sqrt{21}=4.5826$.

Problem 2. $\|\boldsymbol{w}\|=\sqrt{1+4+4}=3$, so $\hat{\boldsymbol{w}}=(1/3,\ 2/3,\ 2/3)$, whose norm is $\sqrt{(1+4+4)/9}=1$.

Problem 3. Squares $0.01+0.04+0.04+0+0.01=0.10$; divide by 5: 0.02; square root: rms $=0.1414$.

Problem 4. $\|\boldsymbol{u}\|^2=|i|^2+|1+2i|^2=1+5=6$, $\|\boldsymbol{u}\|=\sqrt{6}$; $\|\boldsymbol{w}\|^2=1+1=2$. $\langle\boldsymbol{u},\boldsymbol{w}\rangle=(-i)(1)+(1-2i)(i)=-i+i-2i^2=2$. Cauchy–Schwarz: $|2|^2=4\le 6\cdot 2=12$.

Problem 5. $\|\boldsymbol{u}+\boldsymbol{v}\|^2=(\boldsymbol{u}+\boldsymbol{v})\cdot(\boldsymbol{u}+\boldsymbol{v})=\|\boldsymbol{u}\|^2+2\,\boldsymbol{u}\cdot\boldsymbol{v}+\|\boldsymbol{v}\|^2$ by linearity and symmetry. If $\boldsymbol{u}\cdot\boldsymbol{v}=0$ the middle term vanishes. Conversely, if $\|\boldsymbol{u}+\boldsymbol{v}\|^2=\|\boldsymbol{u}\|^2+\|\boldsymbol{v}\|^2$, the same expansion forces $2\,\boldsymbol{u}\cdot\boldsymbol{v}=0$.

Problem 6. Rows: $(2\cdot1+1\cdot2,\ 0+3\cdot2,\ 1-2)=(4,6,-1)$. Columns: $1\cdot(2,0,1)+2\cdot(1,3,-1)=(4,6,-1)$.

Problem 7. $AB = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$, $BA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$. Multiplying by B on the right swaps the columns of A; on the left it swaps the rows. So $AB \neq BA$.

Problem 9. $\det = 6 - 5 = 1$, so by Equation (7.26) the inverse is $\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$. Check: $\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 6-5 & -3+3 \\ 10-10 & -5+6 \end{pmatrix} = I$.

Problem 11. $\det = a - 4$. (i) If $a \neq 4$ the matrix is invertible: a unique solution for every b. (ii), (iii) If $a = 4$, subtract twice the first equation from the second: $0 = b - 6$. So $a = 4$, $b \neq 6$ gives no solution; $a = 4$, $b = 6$ gives infinitely many ($x = 3 - 2y$, y free).

Problem 13. By Equation (7.30): $2(3\cdot 1 - 2\cdot 1) - 0 + 1(1\cdot 1 - 3\cdot 1) = 2 - 2 = 0$. The matrix is singular and its columns are dependent; indeed column $3 = (1, 2, 1) = \frac{1}{2}(2, 1, 1) + \frac{1}{2}(0, 3, 1)$, half the sum of the first two.

Problem 14. Area $= |3 \cdot 2 - 1 \cdot 1| = 5$ (Equation (7.29)).

Problem 15. By Equation (7.34): L (case $j = k = 3 \geq 1$) and 0 (case $j \neq k$). For the third, $\sin\alpha\cos\alpha = \frac{1}{2}\sin 2\alpha$ and $\int_{-L}^{L} \sin(2\pi x/L)\,\mathrm{d}x = 0$ because the integrand is odd; so 0.

Problem 17. $G = \begin{pmatrix} \int 1 & \int x \\ \int x & \int x^2 \end{pmatrix} = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}$, $\boldsymbol{b} = (\int x^2, \int x^3) = (1/3, 1/4)$. $\det G = 1/3 - 1/4 = 1/12$, so $G^{-1} = 12\begin{pmatrix} 1/3 & -1/2 \\ -1/2 & 1 \end{pmatrix}$ and $\boldsymbol{a} = 12\,(1/9 - 1/8,\ -1/6 + 1/4) = 12\,(-1/72,\ 1/12) = (-1/6,\ 1)$. The function $x - 1/6$ is the straight line closest to x^2 in the norm $\sqrt{\int_0^1 (\cdot)^2 \mathrm{d}x}$, since x^2 is not in the span of $\{1, x\}$ and the equations select its projection.

Problem 19. (a) 300×400: underdetermined, null space of dimension at least 100. (b) 400×200: overdetermined. (c) 800×900: underdetermined, null space at least 100. (d) 200×400: underdetermined, null space at least 200; this is what Table III calls "saturated", and its in-sample entry $< 10^{-3}$ is what one expects when a nearly full-rank model has more coefficients than training levels. (The fit is nonlinear, so this counting is a guide, not a proof.) Note that (a) and (c) are also underdetermined by shape, yet Table I reports in-sample rms 0.120 and 0.103 for them. Their matrices are not effectively of full row rank, so counting columns gives only an upper bound on what can be fitted.

Problem 21. If $\boldsymbol{v} = \boldsymbol{0}$ both sides vanish. Otherwise, for complex λ, $0 \leq \|\boldsymbol{u} - \lambda\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 - \lambda\langle\boldsymbol{u}, \boldsymbol{v}\rangle - \lambda^*\langle\boldsymbol{v}, \boldsymbol{u}\rangle + |\lambda|^2 \|\boldsymbol{v}\|^2$, using conjugate-linearity in the first slot and linearity in the second. Put $\lambda = \langle\boldsymbol{v}, \boldsymbol{u}\rangle / \|\boldsymbol{v}\|^2$, so $\lambda^* = \langle\boldsymbol{u}, \boldsymbol{v}\rangle / \|\boldsymbol{v}\|^2$. Then $\lambda\langle\boldsymbol{u}, \boldsymbol{v}\rangle = \lambda^*\langle\boldsymbol{v}, \boldsymbol{u}\rangle = |\langle\boldsymbol{u}, \boldsymbol{v}\rangle|^2 / \|\boldsymbol{v}\|^2$ and $|\lambda|^2 \|\boldsymbol{v}\|^2 = |\langle\boldsymbol{u}, \boldsymbol{v}\rangle|^2 / \|\boldsymbol{v}\|^2$. The inequality becomes $0 \leq \|\boldsymbol{u}\|^2 - |\langle\boldsymbol{u}, \boldsymbol{v}\rangle|^2 / \|\boldsymbol{v}\|^2$; multiply by $\|\boldsymbol{v}\|^2$ and take square roots.

Problem 23. Running the listing and subtracting the analytic matrix (first column 1, entries $(-1)^{n+1}/2$ at $k = n$, zeros elsewhere) gives a largest deviation below 10^{-9}, at the level of rounding errors. The agreement is this good because $|\psi_n|^2 \cos(k\pi x/L)$ vanishes at the walls and is a sum of a few cosines, for which the equally spaced sum over a full period is exact (the mechanism is the geometric-sum argument of the sample "A grid sum that is exact").

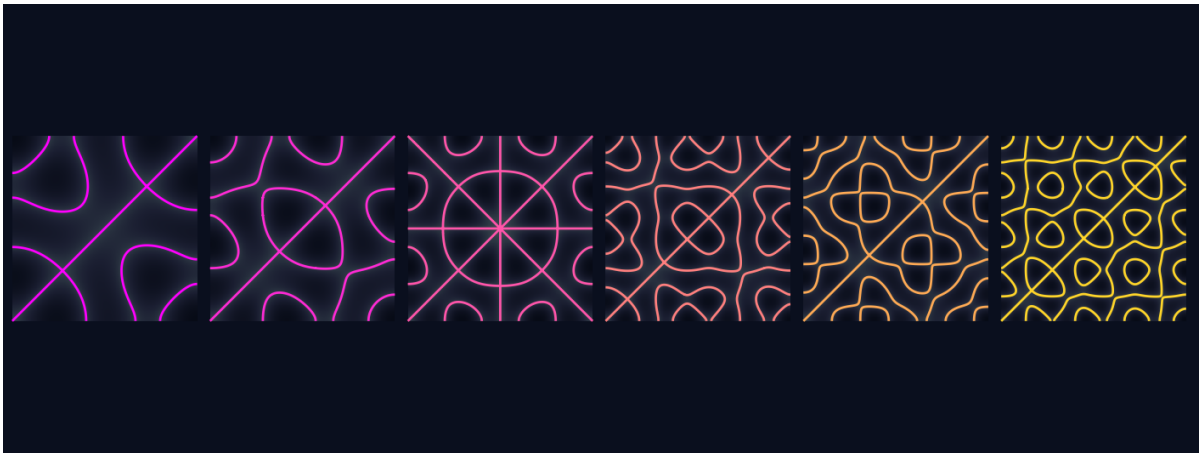
Problem 25. `d = qmnt.fluct(qmnt.zeros(500)[400:], np.arange(401, 501))` and `qmnt.rms(d)` give about 0.263. This is the smooth-zero null, the one the paper uses for the oracle. The paper's operator null (the unperturbed potential's levels scored against bias-corrected targets) is conceptually different; the paper reports that the two coincide numerically on this holdout (both 0.2631, Sec. II C), a numerical coincidence rather than an identity. The smooth-zero null predicts $t_n = s_n$, so its error vector in mean spacings is exactly $-\boldsymbol{d}$, and $\mathrm{rms}(-\boldsymbol{d}) = \mathrm{rms}(\boldsymbol{d})$.

Problem 27. See Equation (7.27): $\sum_{j=1}^{n-1} j(j+1) = (n-1)n(n+1)/3$. For $n = 3$: $2\cdot 3\cdot 4/3 = 8$, matching the count in the elimination sample. For $n = 6000$: $5999 \cdot 6000 \cdot 6001/3 \approx 7.2 \times 10^{10}$. In a tridiagonal matrix each elimination step has a single nonzero below the pivot and only two

entries to update, so the total grows like n, not n^3.

Problem 29. $\|R\boldsymbol{x}\|^2 = (x\cos\theta - y\sin\theta)^2 + (x\sin\theta + y\cos\theta)^2 = x^2(\cos^2\theta + \sin^2\theta) + y^2(\sin^2\theta + \cos^2\theta) + 2xy(-\cos\theta\sin\theta + \sin\theta\cos\theta) = x^2 + y^2$. $\det R = \cos^2\theta + \sin^2\theta = 1$.

Problem 31. $\|\boldsymbol{c}\|^2 = s^2 + (0.4 - 2s)^2 + (0.2 + 2s)^2 + (0.6 - 2s)^2$. Its derivative $2s - 4(0.4 - 2s) + 4(0.2 + 2s) - 4(0.6 - 2s) = 26s - 3.2$ vanishes at $s = 3.2/26 = 8/65 \approx 0.1231$, giving $\boldsymbol{c} \approx (0.1231,\ 0.1538,\ 0.4462,\ 0.3538)$. Its inner product with $(1, -2, 2, -2)$ is $0.1231 - 0.3077 + 0.8923 - 0.7077 = 0$ (to rounding). Geometrically, the solutions form a line parallel to the null vector; by the closest-point argument Equation (7.19), the point of that line nearest the origin is where the line is perpendicular to the vector from the origin, so the minimum-norm solution has no component along the null space. Chapter 9 obtains it from the pseudo-inverse.



Chladni figures: nodal lines of eigenmodes of a vibrating square plate, the 2D cousins of the string's standing waves.

Chapter 8

Eigenvalues, Eigenvectors and Symmetric Matrices

↑ Contents

What this chapter is about

Pluck a guitar string and you hear a note. Touch the string lightly at its midpoint and you hear the octave above. A string fixed at both ends cannot vibrate in any pattern it likes and still keep its shape as it oscillates. It has a list of special shapes, the harmonics, each of which oscillates at its own frequency. Any other motion is a mixture of these. The harmonics and their frequencies are the simplest example of the idea at the center of this chapter: an *eigenvalue problem*. A linear operation (here, the restoring force of the string) acts on most shapes by changing them into different shapes, but on a few special shapes it acts by simple multiplication. Those shapes are eigenvectors, and the multipliers are eigenvalues.

In quantum mechanics this idea is the whole story. The allowed energies of a particle trapped in a potential are the eigenvalues of an operator called the Hamiltonian, and the corresponding states are its eigenvectors. On a computer the Hamiltonian becomes a large matrix, and finding energy levels means finding eigenvalues of that matrix. The matrices that arise from physical energies are *symmetric* (or, with complex entries, *Hermitian*), and for those we will prove two facts that physics depends on: their eigenvalues are real and their eigenvectors are mutually perpendicular. We will also see what goes wrong for matrices that are not symmetric: eigenvalues can become complex and eigenvectors stop being perpendicular.

Where this shows up in the paper [1]: the phrase "spectrum of a self-adjoint operator" in the first sentence of Sec. I is a statement about real eigenvalues; Sec. II B computes energy levels as eigenvalues of a "symmetric tridiagonal matrix" with "eigenpairs from LAPACK"; and Sec. II E (iii) and Sec. V diagonalize non-Hermitian matrices exactly and match their eigenvectors to reference states "by overlap". By the end of this chapter you will understand each of those phrases, and you will be able to compute the levels of a particle in a box on your own computer.

What you need

- Chapter 2: complex numbers, the conjugate z^*, the modulus $|z|$, Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$.
- Chapter 7: vectors in n dimensions (real and complex), the inner product, norm and orthogonality; matrix multiplication, transpose $A^\top$ and conjugate transpose $A^\dagger$; the identity I and the inverse; determinants of 2×2 and 3×3 matrices and their stated properties; the fact that a square matrix B is invertible exactly when $\det B \neq 0$, exactly when $Bv = 0$ has only the solution $v = 0$; linear independence and bases; functions sampled on a grid as vectors.
- First-year physics: standing waves on a string fixed at both ends, harmonics [7]; Taylor expansion of a smooth function.

8.1 Special directions: the eigenvalue equation

Recall from Chapter 7 that an $n\times n$ matrix A is a machine that takes an n-component vector v and returns another n-component vector Av. In two dimensions you can picture this: draw the arrow v, apply A, draw the arrow Av. In general Av points in a different direction from v and has a different length. Some matrices, though, have special directions along which they act very simply: they only stretch or shrink (or flip) the arrow, never turn it.

Take the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{8.1}$$

Apply it to $v = (1,0)^\top$ (we write column vectors as transposed rows to save space):

$$A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\cdot 1 + 1\cdot 0 \\ 1\cdot 1 + 2\cdot 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \tag{8.2}$$

The arrow has been turned: $(1,0)$ points along the horizontal axis, $(2,1)$ points up and to the right. Now apply A to $u = (1,1)^\top$:

$$A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1 \\ 1+2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{8.3}$$

This arrow was not turned at all; it was stretched by a factor of 3. Try $w = (1,-1)^\top$:

$$A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2-1 \\ 1-2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1\cdot\begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{8.4}$$

This arrow is left exactly as it was: stretched by the factor 1. Figure 8.1 shows what A does to every unit vector at once: it turns the unit circle into an ellipse whose long axis lies along $(1,1)$ with half-length 3 and whose short axis lies along $(1,-1)$ with half-length 1.

Definition. Let A be an $n\times n$ matrix, with real or complex entries. A number λ (real or complex) is an *eigenvalue* of A if there is a *nonzero* vector v such that

$$\boxed{Av = \lambda v.} \tag{8.5}$$

Any such nonzero v is an *eigenvector* belonging to λ. The pair (λ, v) is called an *eigenpair*. The set of all eigenvalues of A is its *spectrum*. ("Eigen" is German for "own" or "characteristic": these are the matrix's own special values.)

Two remarks make the definition precise.

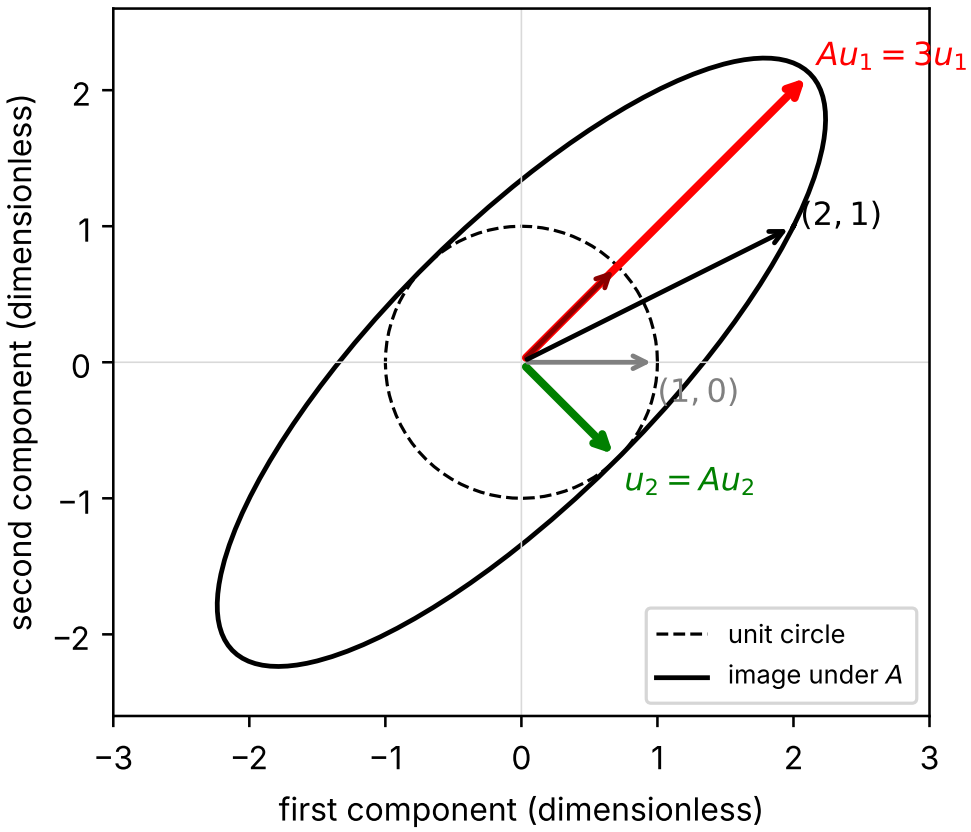


Figure 8.1: The matrix A of Eq. (8.1) applied to every unit vector. The dashed unit circle is mapped to the solid ellipse. The two eigen-directions (red and green) are only stretched, by the eigenvalues 3 and 1; the generic vector $(1, 0)$ (gray) is turned into $(2, 1)$ (black). All quantities are dimensionless.

First, the zero vector is excluded on purpose. For $v = 0$ we have $A0 = 0 = \lambda \cdot 0$ for every number λ, which would make every number an eigenvalue and the idea useless.

Second, an eigenvector is really a *direction*, not a single vector. If $Av = \lambda v$ and a is any nonzero number, then

$$A(av) = a(Av) = a(\lambda v) = \lambda(av), \tag{8.6}$$

where the first step uses linearity of matrix multiplication (Chapter 7). So av is also an eigenvector with the same eigenvalue. It is common to pick the representative of unit length, $\|v\| = 1$; this is called *normalizing* the eigenvector. Even then the sign (for real vectors) or a phase factor $e^{i\alpha}$ (for complex vectors) remains free.

For the matrix (8.1), normalized eigenvectors are

$$u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \ (\lambda_1 = 3), \qquad u_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \ (\lambda_2 = 1). \tag{8.7}$$

Notice something: $u_1 \cdot u_2 = \frac{1}{2}(1 \cdot 1 + 1 \cdot (-1)) = 0$. The two eigenvectors are perpendicular. That is not an accident; it happens because A is symmetric ($A_{12} = A_{21}$), and we will prove in Section 8.4 that it always happens for symmetric matrices.

The physical picture. Think of the matrix as a rule that says how a system responds. For a string, the "vector" is the list of displacements at many points, and the matrix gives the restoring force at each point. A harmonic is a shape for which the restoring force at every point is proportional to the displacement there, with the same constant of proportionality everywhere. Then every point oscillates like a simple harmonic oscillator with the same frequency, and the shape is preserved. The constant of proportionality is the eigenvalue. We will make this precise in Section 8.8.

Checkpoint 8.1

Suppose $Av = 3v$ for a nonzero vector v. What is $A(5v)$? What is $A(Av)$?

Answer: $A(5v) = 5Av = 15v = 3(5v)$, so $5v$ is an eigenvector with the same eigenvalue 3. Also $A(Av) = A(3v) = 3Av = 9v$: applying A twice multiplies by 3^2.

Key Ideas

- An eigenpair (λ, v) of a square matrix A satisfies $Av = \lambda v$ with $v \neq 0$, Eq. (8.5).
- Eigenvectors are directions that A stretches, shrinks or flips without turning; any nonzero multiple of an eigenvector is again an eigenvector with the same eigenvalue.
- Normalizing ($\|v\| = 1$) fixes the length but leaves a sign or phase free.

8.2 The characteristic polynomial

Guessing eigenvectors, as we did above, does not scale. We need a systematic way to find all eigenvalues. Here it is.

Derivation. Write λv as $\lambda I v$, where I is the $n \times n$ identity matrix (ones on the diagonal, zeros elsewhere), so that Eq. (8.5) becomes

$$Av = \lambda I v \tag{8.8}$$

$$Av - \lambda I v = 0 \tag{8.9}$$

$$(A - \lambda I)v = 0. \quad \text{(distributive law for matrices)} \tag{8.10}$$

So λ is an eigenvalue exactly when the matrix $B = A - \lambda I$ sends some *nonzero* vector to zero. Recall from Chapter 7 that a square matrix B has a nonzero solution of $Bv = 0$ if and only if $\det B = 0$. Therefore

$$\boxed{\lambda \text{ is an eigenvalue of } A \iff \det(A - \lambda I) = 0.} \tag{8.11}$$

Equation (8.11) is the *characteristic equation*. When you expand the determinant, each entry on the diagonal of $A - \lambda I$ contains λ once, so $\det(A - \lambda I)$ is a polynomial in λ of degree n (the highest power comes from the product of the n diagonal entries). It is called the *characteristic polynomial* of A, and we write $p_A(\lambda) = \det(A - \lambda I)$.

Once you have an eigenvalue λ, you find its eigenvectors by solving the linear system $(A - \lambda I)v = 0$ by elimination, exactly as in Chapter 7. Because $\det(A - \lambda I) = 0$, elimination will always produce at least one row of zeros, leaving at least one free component that you may choose.

Taken on trust

Taken on trust: the fundamental theorem of algebra. Every polynomial of degree $n \geq 1$ with complex coefficients has exactly n complex roots when each root is counted as many times as it is repeated. Hence every $n \times n$ matrix has exactly n eigenvalues counted with repetition, though some may be complex even when all entries of A are real, and some may coincide. The proof needs complex analysis and is given in many texts, for example [22]. We use only the statement.

The number of times a value λ appears as a root of p_A is its *algebraic multiplicity*. An eigenvalue of multiplicity one is called *simple* or *nondegenerate*; one of multiplicity two or more is *degenerate*.

8.2.1 The 2×2 case in general

For a general 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{8.12}$$

we have

$$A - \lambda I = \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix}, \tag{8.13}$$

and the 2×2 determinant rule "product of diagonal minus product of off-diagonal" gives

$$\begin{aligned} p_A(\lambda) &= (a - \lambda)(d - \lambda) - bc && (8.14)\\ &= ad - a\lambda - d\lambda + \lambda^2 - bc && (8.15)\\ &= \lambda^2 - (a + d)\lambda + (ad - bc). && (8.16) \end{aligned}$$

The coefficient $a + d$ is the sum of the diagonal entries, called the *trace*, $\operatorname{tr} A$. The constant $ad - bc$ is $\det A$. So

$$p_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A. \tag{8.17}$$

If the two roots are λ_1 and λ_2, then $p_A(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$. Comparing coefficients of λ and of λ^0 with Eq. (8.17):

$$\lambda_1 + \lambda_2 = \operatorname{tr} A, \qquad \lambda_1\lambda_2 = \det A. \tag{8.18}$$

These two relations are handy checks on any eigenvalue calculation. (For $n \times n$ matrices the same two facts hold. The determinant part follows in one line. By the fundamental theorem of

algebra (box above), and because the highest power of λ in $\det(A-\lambda I)$ is $(-\lambda)^n$, coming from the product of the diagonal entries, we can write $p_A(\lambda) = (\lambda_1-\lambda)(\lambda_2-\lambda)\cdots(\lambda_n-\lambda)$, with each eigenvalue repeated according to its multiplicity. Setting $\lambda = 0$ gives $\det A = \lambda_1\lambda_2\cdots\lambda_n$. The trace part, $\lambda_1+\cdots+\lambda_n = \operatorname{tr} A$, we state without proof; it follows by comparing the coefficients of λ^{n-1} on the two sides, see [28]. We use the factorization in Section 8.9 and both $n\times n$ facts in the answer to Problem 31.)

The quadratic formula applied to Eq. (8.17) gives

$$\lambda_{\pm} = \frac{\operatorname{tr} A \pm \sqrt{(\operatorname{tr} A)^2 - 4\det A}}{2}. \tag{8.19}$$

For the matrix (8.1): $\operatorname{tr} A = 4$, $\det A = 4-1 = 3$, so $p_A(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda-3)(\lambda-1)$, confirming the eigenvalues 3 and 1.

Sample Problem 8.1: Eigenpairs of a non-symmetric 2×2 matrix

Find all eigenvalues and eigenvectors of $B = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ and check whether the eigenvectors are perpendicular.

Step 1: characteristic polynomial. $\operatorname{tr} B = 4+3 = 7$ and $\det B = 4\cdot 3 - 1\cdot 2 = 10$, so by Eq. (8.17)

$$p_B(\lambda) = \lambda^2 - 7\lambda + 10 = (\lambda-5)(\lambda-2).$$

(Check the factorization: $(\lambda-5)(\lambda-2) = \lambda^2-7\lambda+10$.)

Step 2: eigenvector for $\lambda = 5$. Solve $(B-5I)v = 0$:

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0.$$

The first row says $-v_1 + v_2 = 0$, so $v_2 = v_1$. The second row says $2v_1 - 2v_2 = 0$, the same condition (as it must be, since the determinant is zero). Choosing $v_1 = 1$: $v = (1,1)^{\top}$.

Step 3: eigenvector for $\lambda = 2$. Solve $(B-2I)v = 0$:

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0,$$

so $2v_1 + v_2 = 0$, $v_2 = -2v_1$. Choosing $v_1 = 1$: $v = (1,-2)^{\top}$.

Step 4: check. $B(1,1)^{\top} = (4+1, 2+3)^{\top} = (5,5)^{\top} = 5(1,1)^{\top}$ and $B(1,-2)^{\top} = (4-2, 2-6)^{\top} = (2,-4)^{\top} = 2(1,-2)^{\top}$. Also $\lambda_1+\lambda_2 = 7 = \operatorname{tr} B$ and $\lambda_1\lambda_2 = 10 = \det B$.

Step 5: perpendicular? $(1,1)\cdot(1,-2) = 1-2 = -1 \neq 0$.

$$\boxed{\lambda = 5,\ v = (1,1)^{\top};\quad \lambda = 2,\ v = (1,-2)^{\top};\quad \text{eigenvectors not perpendicular.}}$$

The matrix is not symmetric ($B_{12} = 1 \neq B_{21} = 2$), and its eigenvectors are not orthogonal.

8.2.2 The 3×3 case

For a 3×3 matrix the characteristic polynomial is cubic. You expand $\det(A-\lambda I)$ along the first row (the cofactor expansion of Chapter 7). The following example is important, because the same matrix (in a bigger size) will be our model of the second derivative in Section 8.7.

Sample Problem 8.2: The three-point second-difference matrix

Find the eigenvalues and eigenvectors of

$$D_3 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Step 1: set up. Write $a = 2 - \lambda$ to keep the algebra short. Then

$$D_3 - \lambda I = \begin{pmatrix} a & -1 & 0 \\ -1 & a & -1 \\ 0 & -1 & a \end{pmatrix}.$$

Step 2: expand along the first row. The cofactor expansion gives

$$\det(D_3 - \lambda I) = a \det \begin{pmatrix} a & -1 \\ -1 & a \end{pmatrix} - (-1) \det \begin{pmatrix} -1 & -1 \\ 0 & a \end{pmatrix} + 0.$$

The two 2×2 determinants are $a \cdot a - (-1)(-1) = a^2 - 1$ and $(-1)a - (-1) \cdot 0 = -a$. So

$$\det(D_3 - \lambda I) = a(a^2 - 1) + (-a) = a^3 - a - a = a^3 - 2a = a(a^2 - 2).$$

Step 3: roots. $a(a^2 - 2) = 0$ gives $a = 0$ or $a = \pm\sqrt{2}$. Since $\lambda = 2 - a$:

$$\lambda = 2 - \sqrt{2} \approx 0.585786, \qquad \lambda = 2, \qquad \lambda = 2 + \sqrt{2} \approx 3.414214.$$

Step 4: eigenvectors. For each eigenvalue solve $(D_3 - \lambda I)v = 0$, i.e. the three equations $av_1 - v_2 = 0$, $-v_1 + av_2 - v_3 = 0$, $-v_2 + av_3 = 0$.
From the first, $v_2 = av_1$. From the third, $v_2 = av_3$. If $a \neq 0$, these give $v_3 = v_1$. Then the second equation, $-v_1 + a(av_1) - v_1 = (a^2 - 2)v_1 = 0$, holds automatically because $a^2 = 2$. Choosing $v_1 = 1$:

- $a = \sqrt{2}$ ($\lambda = 2 - \sqrt{2}$): $v = (1, \sqrt{2}, 1)^\top$;
- $a = -\sqrt{2}$ ($\lambda = 2 + \sqrt{2}$): $v = (1, -\sqrt{2}, 1)^\top$.

If $a = 0$ ($\lambda = 2$), the first and third equations give $v_2 = 0$, and the second gives $-v_1 - v_3 = 0$, so $v_3 = -v_1$: $v = (1, 0, -1)^\top$.
Step 5: check orthogonality. $(1, \sqrt{2}, 1) \cdot (1, 0, -1) = 1 - 1 = 0$; $(1, \sqrt{2}, 1) \cdot (1, -\sqrt{2}, 1) = 1 - 2 + 1 = 0$; $(1, 0, -1) \cdot (1, -\sqrt{2}, 1) = 1 - 1 = 0$.

$$\boxed{\lambda = 2 - \sqrt{2},\ 2,\ 2 + \sqrt{2}; \quad v = (1, \sqrt{2}, 1),\ (1, 0, -1),\ (1, -\sqrt{2}, 1).}$$

The matrix is symmetric, and again the eigenvectors are mutually perpendicular. Look at the shapes: the first has no change of sign, the second has one, the third has two. Keep this in mind; it is the discrete version of the harmonics of a string.

8.2.3 Two easy families: diagonal and triangular matrices

If A is *diagonal*, $A = \operatorname{diag}(d_1, \ldots, d_n)$, then $A - \lambda I = \operatorname{diag}(d_1 - \lambda, \ldots, d_n - \lambda)$ and its determinant is the product $(d_1 - \lambda) \cdots (d_n - \lambda)$ (the determinant of a diagonal matrix is the product of its diagonal entries, Chapter 7). So the eigenvalues are the diagonal entries, and the eigenvector for d_k is the k-th standard basis vector e_k (a 1 in position k, zeros elsewhere): $Ae_k = d_k e_k$.

If A is *triangular* (all entries below the diagonal zero, or all above), then $A - \lambda I$ is also triangular, and the determinant of a triangular matrix is again the product of its diagonal entries. So the eigenvalues of a triangular matrix are its diagonal entries too, although the eigenvectors are no longer the standard basis vectors.

In physics, a potential energy $V(x)$ acting on a wavefunction simply multiplies it point by point. Sampled on a grid, that is a diagonal matrix $\operatorname{diag}(V(x_1), \dots, V(x_N))$. We will meet it in Section 8.7.

Pitfall

Do not compute eigenvalues of large matrices by writing down the characteristic polynomial and finding its roots numerically. The coefficients of a high-degree polynomial are extremely sensitive to rounding, and the computed roots can be wildly wrong. The characteristic polynomial is a tool for understanding and for small hand calculations; computers use the methods of Section 8.9.

Checkpoint 8.2

A square matrix A has 0 as one of its eigenvalues. Is A invertible?

Answer: No. If $Av = 0 \cdot v = 0$ for some $v \neq 0$, then A sends a nonzero vector to zero, which (Chapter 7) means $\det A = 0$ and A has no inverse. Equivalently, $p_A(0) = \det A = 0$.

Checkpoint 8.3

A 2×2 matrix has eigenvalues 1 and 4. What are its trace and determinant?

Answer: By Eq. (8.18), $\operatorname{tr} A = 1 + 4 = 5$ and $\det A = 1 \cdot 4 = 4$.

Key Ideas

- λ is an eigenvalue of A exactly when $\det(A - \lambda I) = 0$, Eq. (8.11); this is a polynomial equation of degree n.
- For 2×2 matrices, $p_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A$; the eigenvalues sum to the trace and multiply to the determinant, Eq. (8.18).
- Eigenvectors are found by solving $(A - \lambda I)v = 0$ by elimination.
- Diagonal and triangular matrices have their diagonal entries as eigenvalues.

8.3 When eigenvalues are complex

★ Did you know? *Eigen Means Own*

The prefix *eigen* is German for own or characteristic. David Hilbert used *Eigenwert* and *Eigenfunktion* in his 1904 work on integral equations, and the half-translated hybrid eigenvalue stuck in English. A real two-dimensional rotation matrix has no real eigenvectors unless the angle is 0 or π. Its eigenvalues are $e^{\pm i\theta}$, a first hint that complex eigenvalues signal rotation.

A matrix with real entries can have complex eigenvalues. The simplest example is a rotation.

The matrix

$$R_{90} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{8.20}$$

turns every vector in the plane by 90° counterclockwise: $R_{90}(1,0)^\top = (0,1)^\top$ and $R_{90}(0,1)^\top = (-1,0)^\top$. A rotation by 90° turns *every* real arrow, so there can be no real eigenvector. The algebra agrees: $\operatorname{tr} R_{90} = 0$, $\det R_{90} = 0 - (-1)(1) = 1$, so $p(\lambda) = \lambda^2 + 1$, whose roots are $\lambda = \pm i$. The eigenvalues exist, but they are complex, and so are the eigenvectors. For $\lambda = i$, solve $(R_{90} - iI)v = 0$: the first row gives $-iv_1 - v_2 = 0$, so $v_2 = -iv_1$ and $v = (1, -i)^\top$. Check: $R_{90}(1,-i)^\top = (0 \cdot 1 + (-1)(-i),\, 1 \cdot 1 + 0)^\top = (i, 1)^\top$, and $i(1,-i)^\top = (i, -i^2)^\top = (i,1)^\top$. It works.

Recall from Chapter 2 that multiplying a complex number by $e^{i\theta}$ rotates it by angle θ in the complex plane. The eigenvalue $i = e^{i\pi/2}$ is exactly the rotation by 90°, now expressed as multiplication.

Sample Problem 8.3: A rotation combined with a stretch

Find the eigenvalues and eigenvectors of $C = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$ and interpret their modulus and argument.

Step 1: characteristic polynomial. $\operatorname{tr} C = 2$, $\det C = 1 \cdot 1 - (-2)(2) = 5$, so $p_C(\lambda) = \lambda^2 - 2\lambda + 5$.

Step 2: roots. By Eq. (8.19), $\lambda = \frac{1}{2}\left(2 \pm \sqrt{4 - 20}\right) = \frac{1}{2}(2 \pm \sqrt{-16}) = 1 \pm 2i$.

Step 3: eigenvector for $\lambda = 1 + 2i$. $C - \lambda I = \begin{pmatrix} -2i & -2 \\ 2 & -2i \end{pmatrix}$. The first row gives $-2iv_1 - 2v_2 = 0$, so $v_2 = -iv_1$. Take $v = (1, -i)^\top$.

Step 4: check. $Cv = (1 \cdot 1 + (-2)(-i),\, 2 \cdot 1 + 1 \cdot (-i))^\top = (1 + 2i,\, 2 - i)^\top$. And $\lambda v = ((1+2i),\, (1+2i)(-i))^\top = (1 + 2i,\, -i - 2i^2)^\top = (1 + 2i,\, 2 - i)^\top$. Equal.

Step 5: the other eigenpair. Because C is real, take the complex conjugate of $Cv = \lambda v$: $Cv^* = \lambda^* v^*$ (see the argument just below). So $\lambda = 1 - 2i$ with $v = (1, i)^\top$.

Step 6: interpretation. $|1 \pm 2i| = \sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.236$ and $\arg(1+2i) = \arctan 2 \approx 63.43°$. Indeed $C = \sqrt{5}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ with $\cos\theta = 1/\sqrt{5}$, $\sin\theta = 2/\sqrt{5}$: a rotation by $\theta \approx 63.43°$ followed by a stretch by $\sqrt{5}$.

$$\boxed{\lambda = 1 \pm 2i, \quad v = (1, \mp i)^\top, \quad |\lambda| = \sqrt{5} \approx 2.236, \ \arg\lambda = \pm 63.43°.}$$

Complex eigenvalues of real matrices come in conjugate pairs. Suppose A has real entries and $Av = \lambda v$. Take the complex conjugate of every entry of both sides. Since $A^* = A$ (its entries are real) and the conjugate of a product is the product of conjugates (Chapter 2),

$$(Av)^* = A^* v^* = Av^*, \qquad (\lambda v)^* = \lambda^* v^*, \tag{8.21}$$

so $Av^* = \lambda^* v^*$. Since $v \neq 0$, also $v^* \neq 0$, so λ^* is an eigenvalue with eigenvector v^*. Non-real eigenvalues of a real matrix therefore always come in pairs λ, λ^*.

Non-symmetric matrices: other things that can go wrong. Sample Problem "Eigenpairs of a non-symmetric 2×2 matrix" showed that a real non-symmetric matrix may have real eigenvalues whose eigenvectors are not perpendicular. Worse, a matrix may not have enough eigenvectors at all. Take

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{8.22}$$

It is triangular with both diagonal entries equal to 1, so $\lambda = 1$ is a double root of $p_J(\lambda) = (1-\lambda)^2$. But $(J - I)v = 0$ reads $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v = 0$, i.e. $v_2 = 0$, so every eigenvector is a multiple of $(1, 0)^\top$. There is only one eigen-direction for a double eigenvalue. Such a matrix is called *defective*. You cannot build a basis of the plane out of its eigenvectors.

None of these troubles (complex eigenvalues, non-perpendicular eigenvectors, missing eigenvectors) can happen for the matrices of the next section. That is why physicists love symmetric matrices.

Key Ideas

- Real matrices can have complex eigenvalues; rotations are the basic example. Complex eigenvalues of a real matrix come in conjugate pairs λ, λ^*.
- The modulus of a complex eigenvalue measures stretching and its argument measures rotation.
- Non-symmetric matrices can have non-orthogonal eigenvectors, or too few eigenvectors to form a basis (defective matrices).

8.4 Symmetric and Hermitian matrices

Definitions. A real square matrix is *symmetric* if $A^\top = A$, i.e. $A_{ij} = A_{ji}$ for all i, j: the matrix is its own mirror image across the diagonal. A complex square matrix is *Hermitian* if $A^\dagger = A$, where (recall from Chapter 7) the conjugate transpose $A^\dagger$ has entries $(A^\dagger)_{ij} = A^*_{ji}$. For a Hermitian matrix the diagonal entries must be real ($A_{ii} = A^*_{ii}$) and the entry below the diagonal is the conjugate of the one above. A real symmetric matrix is a special case of a Hermitian matrix (conjugation does nothing to real numbers). So we prove everything for Hermitian matrices and get the symmetric case free.

Inner product. We use the inner product of complex vectors from Chapter 7,

$$u^\dagger v = \sum_{i=1}^{n} u_i^* v_i, \tag{8.23}$$

with the conjugate on the *first* vector. Physicists write the same number as $\langle u \,|\, v \rangle$. Its key property is that $v^\dagger v = \sum_i |v_i|^2 = \|v\|^2$ is real and positive whenever $v \neq 0$. Two vectors are *orthogonal* if $u^\dagger v = 0$.

We need one rule about conjugate transposes of products: $(AB)^\dagger = B^\dagger A^\dagger$. Recall it from Chapter 7, or verify it entry by entry:

$$\begin{aligned}
\big[(AB)^\dagger\big]_{ij} &= \big[(AB)_{ji}\big]^* \quad \text{(definition of } \dagger) && (8.24)\\
&= \Big(\sum_k A_{jk} B_{ki}\Big)^* \quad \text{(matrix product)} && (8.25)\\
&= \sum_k B^*_{ki} A^*_{jk} \quad \text{(conjugate of sum and product)} && (8.26)\\
&= \sum_k (B^\dagger)_{ik} (A^\dagger)_{kj} = \big[B^\dagger A^\dagger\big]_{ij}. && (8.27)
\end{aligned}$$

A column vector is an $n \times 1$ matrix, so in particular $(Au)^\dagger = u^\dagger A^\dagger$.

8.4.1 Theorem 1: Hermitian matrices have real eigenvalues

Theorem (proved). If $A^\dagger = A$ and $Av = \lambda v$ with $v \neq 0$, then λ is real.

Proof. Multiply $Av = \lambda v$ on the left by the row vector $v^\dagger$:

$$v^\dagger A v = \lambda\, v^\dagger v. \tag{8.28}$$

The left side is a single number. Take its complex conjugate. A single number equals its own transpose, so its complex conjugate equals its conjugate transpose:

$$\begin{aligned} (v^\dagger A v)^* &= (v^\dagger A v)^\dagger \quad \text{(a } 1\times 1 \text{ matrix)} && (8.29)\\ &= v^\dagger A^\dagger (v^\dagger)^\dagger \quad \text{(product rule, applied twice)} && (8.30)\\ &= v^\dagger A v. \quad (A^\dagger = A \text{ and } (v^\dagger)^\dagger = v) && (8.31) \end{aligned}$$

A number equal to its own conjugate is real. So $v^\dagger A v$ is real. And $v^\dagger v = \|v\|^2$ is real and strictly positive because $v \neq 0$. Dividing Eq. (8.28) by it,

$$\lambda = \frac{v^\dagger A v}{v^\dagger v} \tag{8.32}$$

is a ratio of two real numbers, hence real. ■

8.4.2 Theorem 2: eigenvectors for different eigenvalues are orthogonal

Theorem (proved). If $A^\dagger = A$, $Au = \lambda u$, $Av = \mu v$ with $u, v \neq 0$ and $\lambda \neq \mu$, then $u^\dagger v = 0$.

Proof. Compute the number $u^\dagger A v$ in two ways. First, using $Av = \mu v$:

$$u^\dagger A v = u^\dagger(\mu v) = \mu\, u^\dagger v. \tag{8.33}$$

Second, using $A = A^\dagger$ and the product rule, $u^\dagger A = u^\dagger A^\dagger = (Au)^\dagger$, so

$$\begin{aligned} u^\dagger A v &= (Au)^\dagger v && (8.34)\\ &= (\lambda u)^\dagger v && (8.35)\\ &= \lambda^* u^\dagger v \quad \text{(conjugate transpose of a number times a vector)} && (8.36)\\ &= \lambda\, u^\dagger v. \quad (\lambda \text{ real, by Theorem 1}) && (8.37) \end{aligned}$$

Subtracting the two expressions, $0 = (\mu - \lambda)\, u^\dagger v$. Since $\mu - \lambda \neq 0$, we must have $u^\dagger v = 0$. ■

Real symmetric matrices have real eigenvectors. If A is real and symmetric, Theorem 1 says λ is real, but the proof did not say v is real. It can always be chosen real. Write $v = a + ib$ with a, b real vectors. Then $Av = \lambda v$ reads $Aa + iAb = \lambda a + i\lambda b$. All of A, λ, a, b are real, so the real and imaginary parts must match separately: $Aa = \lambda a$ and $Ab = \lambda b$. Since $v \neq 0$, at least one of a, b is nonzero, and that one is a real eigenvector. For real symmetric matrices we may therefore work entirely with real numbers, which is what the computer does in the paper's Hermitian fits.

Sample Problem 8.4: A complex Hermitian matrix

Find the eigenvalues and eigenvectors of $H = \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}$, check that H is Hermitian, and verify Theorems 1 and 2.

Step 1: Hermitian? $(H^\dagger)_{12} = H^*_{21} = (-i)^* = i = H_{12}$ and $(H^\dagger)_{21} = H^*_{12} = -i = H_{21}$; the diagonal entries are real. So $H^\dagger = H$. Note that H is *not* symmetric: $H_{12} = i \neq$

$H_{21} = -i$.
Step 2: characteristic polynomial. $\det(H - \lambda I) = (2-\lambda)^2 - (i)(-i) = (2-\lambda)^2 - 1$, because $(i)(-i) = -i^2 = 1$.
Step 3: roots. $(2-\lambda)^2 = 1$ gives $2 - \lambda = \pm 1$, so $\lambda = 1$ or $\lambda = 3$. Both real, as Theorem 1 promises.
Step 4: eigenvectors. For $\lambda = 3$: $H - 3I = \begin{pmatrix} -1 & i \\ -i & -1 \end{pmatrix}$, first row $-v_1 + iv_2 = 0$, so $v_1 = iv_2$; take $u = (i, 1)^\top$. For $\lambda = 1$: $H - I = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$, first row $v_1 + iv_2 = 0$, so $v_1 = -iv_2$; take $w = (-i, 1)^\top$.
Step 5: orthogonality. $u^\dagger w = (i)^*(-i) + (1)^*(1) = (-i)(-i) + 1 = i^2 + 1 = -1 + 1 = 0$.

$$\boxed{\lambda = 3,\ u = (i,1)^\top; \quad \lambda = 1,\ w = (-i,1)^\top; \quad u^\dagger w = 0.}$$

Pitfall

For complex vectors the orthogonality test must use the conjugate, $u^\dagger w = \sum u_i^* w_i$. The plain sum $\sum u_i w_i$ for the vectors above is $i(-i) + 1 = 1 + 1 = 2 \neq 0$, which would wrongly suggest they are not orthogonal. In numpy, `np.vdot(u, w)` conjugates the first argument; `np.dot(u, w)` does not.

Notation used in physics and in the paper. Physicists write a vector as $|v\rangle$, its conjugate transpose as $\langle v|$, and $\langle u \,|\, v\rangle = u^\dagger v$. When the vectors are the eigenvectors of an unperturbed Hamiltonian, labelled by their index, one writes $|n\rangle$ for the n-th eigenvector and

$$\langle n|A|m\rangle = u_n^\dagger A\, u_m \tag{8.38}$$

for the "matrix element" of another matrix A between them. The paper's Eq. (5) uses exactly this notation, $A_{nm} = \langle n|A|m\rangle$.

Checkpoint 8.4

Can a real symmetric 2×2 matrix have complex eigenvalues? Answer without using Theorem 1, by looking at the discriminant in Eq. (8.19).

Answer: No. For $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ the discriminant is $(a+d)^2 - 4(ad - b^2) = a^2 - 2ad + d^2 + 4b^2 = (a-d)^2 + 4b^2 \geq 0$, so the square root is real.

Key Ideas

- Hermitian ($A^\dagger = A$) includes real symmetric ($A^\top = A$).
- Theorem 1: Hermitian matrices have real eigenvalues, $\lambda = v^\dagger A v / v^\dagger v$.
- Theorem 2: eigenvectors of a Hermitian matrix belonging to different eigenvalues are orthogonal.
- For real symmetric matrices the eigenvectors can be chosen real.

8.5 The spectral theorem and diagonalization

Theorems 1 and 2 have a powerful consequence: a Hermitian matrix with distinct eigenvalues provides a complete set of perpendicular axes, and in those axes the matrix becomes as simple as possible, a diagonal matrix.

Lemma (proved): orthogonal nonzero vectors are linearly independent. Let $u_1, \dots, u_k$ be nonzero and mutually orthogonal, and suppose $a_1 u_1 + \cdots + a_k u_k = 0$. Take the inner product with u_j:

$$0 = u_j^\dagger \Big(\sum_{l=1}^{k} a_l u_l \Big) = \sum_{l=1}^{k} a_l \, u_j^\dagger u_l = a_j \, u_j^\dagger u_j = a_j \, \|u_j\|^2 , \tag{8.39}$$

because every term with $l \neq j$ vanishes by orthogonality. Since $\|u_j\|^2 > 0$, $a_j = 0$. This holds for every j, so the only combination giving zero is the trivial one: the vectors are linearly independent. ■

Spectral theorem, distinct-eigenvalue case (proved). Let A be an $n \times n$ Hermitian matrix with n distinct eigenvalues $\lambda_1, \dots, \lambda_n$. Choose a normalized eigenvector u_k for each. Then:

1. the u_k form an *orthonormal basis*: $u_j^\dagger u_k = \delta_{jk}$ (1 if $j = k$, 0 otherwise), and every vector in $\mathbb{C}^n$ is a combination of them;
2. with U the matrix whose k-th column is u_k and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$,

$$\boxed{A = U \Lambda U^\dagger, \qquad U^\dagger U = I.} \tag{8.40}$$

Proof. (1) By Theorem 2 the u_k are mutually orthogonal, and we normalized them, so $u_j^\dagger u_k = \delta_{jk}$. By the Lemma they are linearly independent. Recall from Chapter 7 that n linearly independent vectors in n-dimensional space form a basis. (2) The (j,k) entry of $U^\dagger U$ is (row j of $U^\dagger$) times (column k of U), which is $u_j^\dagger u_k = \delta_{jk}$; so $U^\dagger U = I$. The columns of U are linearly independent, so U is invertible (Chapter 7). Multiplying $U^\dagger U = I$ on the right by U^{-1} gives $U^\dagger = U^{-1}$, and hence $UU^\dagger = UU^{-1} = I$. Next compute AU column by column: column k of AU is $Au_k = \lambda_k u_k$, which is column k of $U\Lambda$ (multiplying U on the right by a diagonal matrix multiplies column k by λ_k). So $AU = U\Lambda$. Multiply on the right by $U^\dagger$: $AUU^\dagger = U\Lambda U^\dagger$, and $UU^\dagger = I$ gives $A = U\Lambda U^\dagger$. ■

A matrix with $U^\dagger U = I$ is called *unitary*; if it is real, $U^\top U = I$ and it is called *orthogonal*. Unitary matrices preserve lengths: $\|Ux\|^2 = (Ux)^\dagger (Ux) = x^\dagger U^\dagger U x = x^\dagger x = \|x\|^2$. So Eq. (8.40) says: a Hermitian matrix acts by (i) rotating into its eigenvector axes ($U^\dagger$), (ii) stretching each axis by its eigenvalue (Λ), (iii) rotating back (U). That is precisely the ellipse of Figure 8.1.

Taken on trust

Taken on trust: the spectral theorem with repeated eigenvalues. Every Hermitian matrix, even one with repeated eigenvalues, has an orthonormal basis of eigenvectors, so Eq. (8.40) always holds. Within a repeated eigenvalue, Theorem 2 gives no orthogonality, but one can always choose orthogonal eigenvectors inside that eigenspace, and there are always as many independent eigenvectors as the multiplicity (Hermitian matrices are never defective). The proof uses an induction on the size of the matrix; see [28], [29] or [22]. The simplest example is the identity matrix I: its only eigenvalue is 1, with multiplicity n, and every nonzero vector is an eigenvector, so any orthonormal basis will do.

Diagonalization in general. The idea extends beyond Hermitian matrices. If an $n \times n$ matrix A (Hermitian or not) has n linearly independent eigenvectors $v_1, \dots, v_n$, place them as the columns of a matrix S. Exactly as in the proof above, $AS = S\Lambda$. A square matrix with independent columns is invertible (Chapter 7), so

$$A = S\Lambda S^{-1}. \tag{8.41}$$

This is called *diagonalizing* A. For non-Hermitian A, S is generally not unitary (Sample Problem "Eigenpairs of a non-symmetric 2×2 matrix" had non-perpendicular columns), and for defective A such as J of Section 8.3 no invertible S exists.

The spectral decomposition. Writing out Eq. (8.40) column by column gives a useful sum. For any vector x, $U^\dagger x$ is the list of numbers $u_k^\dagger x$; multiplying by Λ scales them by λ_k; multiplying by U adds up $\sum_k u_k(\lambda_k u_k^\dagger x)$. Therefore

$$Ax = \sum_{k=1}^{n} \lambda_k \, (u_k^\dagger x) \, u_k, \qquad A = \sum_{k=1}^{n} \lambda_k \, u_k u_k^\dagger. \tag{8.42}$$

Each $u_k u_k^\dagger$ is an $n \times n$ matrix (column times row) that projects onto the direction u_k (Chapter 7). In words: to apply A, split x into its components along the eigenvectors, scale each component by its eigenvalue, and add them back.

Powers and functions of a matrix. From $A = U\Lambda U^\dagger$ and $U^\dagger U = I$,

$$A^2 = U\Lambda U^\dagger U\Lambda U^\dagger = U\Lambda^2 U^\dagger, \tag{8.43}$$

and by repeating the same cancellation, $A^k = U\Lambda^k U^\dagger$ for every positive integer k. Raising a diagonal matrix to a power just raises each diagonal entry to that power.

The Rayleigh quotient (proved). For a Hermitian A and any nonzero x, define $\mathcal{R}(x) = x^\dagger Ax / x^\dagger x$. Expand $x = \sum_k a_k u_k$ in the eigenbasis. When the eigenvalues are distinct this is possible by part 1 of the theorem. When some are repeated it relies on the general spectral theorem taken on trust in the box above, so in that case the result inherits that status. Then, using $u_j^\dagger u_k = \delta_{jk}$,

$$x^\dagger x = \sum_{j,k} a_j^* a_k \, u_j^\dagger u_k = \sum_k |a_k|^2 , \tag{8.44}$$

$$x^\dagger Ax = \sum_{j,k} a_j^* a_k \lambda_k \, u_j^\dagger u_k = \sum_k \lambda_k \, |a_k|^2 . \tag{8.45}$$

So $\mathcal{R}(x) = \sum_k w_k \lambda_k$ with weights $w_k = |a_k|^2 \,/\, \sum_j |a_j|^2$. The denominator $\sum_j |a_j|^2 = x^\dagger x$ is positive because $x \neq 0$, so the weights are well defined. Each $w_k \geq 0$, and $\sum_k w_k = 1$. Hence $\mathcal{R}(x)$ is a weighted average of the eigenvalues. Replacing every λ_k by $\lambda_{\min}$ can only decrease the sum, because each weight is non-negative. Replacing every λ_k by $\lambda_{\max}$ can only increase it. So $\lambda_{\min} \sum_k w_k \leq \sum_k w_k \lambda_k \leq \lambda_{\max} \sum_k w_k$, and with $\sum_k w_k = 1$ this reads

$$\lambda_{\min} \leq \frac{x^\dagger Ax}{x^\dagger x} \leq \lambda_{\max}. \tag{8.46}$$

If x is an eigenvector, $\mathcal{R}(x)$ is its eigenvalue (Eq. (8.32)). In quantum mechanics $\mathcal{R}(x)$ will be the average energy of a state (Chapter 10), and Eq. (8.46) says the average energy can never fall below the lowest level.

Sample Problem 8.5: Diagonalizing a matrix and computing its tenth power

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, write $A = U\Lambda U^{\top}$ explicitly, check the spectral decomposition, and compute A^{10}.

Step 1: eigenpairs. From Section 8.1, $\lambda_1 = 3$, $u_1 = \frac{1}{\sqrt{2}}(1,1)^{\top}$; $\lambda_2 = 1$, $u_2 = \frac{1}{\sqrt{2}}(1,-1)^{\top}$.

Step 2: build U and Λ.

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \Lambda = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Check $U^{\top}U = \frac{1}{2}\begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = I$.

Step 3: spectral decomposition. $u_1 u_1^{\top} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $u_2 u_2^{\top} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Then

$$3u_1u_1^{\top} + 1\,u_2u_2^{\top} = \frac{1}{2}\begin{pmatrix} 3+1 & 3-1 \\ 3-1 & 3+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = A.$$

Step 4: the k-th power. By $A^k = U\Lambda^k U^{\top} = 3^k u_1u_1^{\top} + 1^k u_2u_2^{\top}$,

$$A^k = \frac{1}{2}\begin{pmatrix} 3^k+1 & 3^k-1 \\ 3^k-1 & 3^k+1 \end{pmatrix}.$$

Step 5: $k = 10$. $3^{10} = 59049$, so $(3^{10}+1)/2 = 29525$ and $(3^{10}-1)/2 = 29524$:

$$\boxed{A^{10} = \begin{pmatrix} 29525 & 29524 \\ 29524 & 29525 \end{pmatrix}}.$$

Multiplying A by itself ten times would take nine matrix products; the eigen-route takes one line. Notice also that for large k the $\lambda = 3$ part dominates: repeated application of a matrix pulls any vector toward the eigenvector of the largest eigenvalue.

Key Ideas

- A Hermitian matrix with distinct eigenvalues has an orthonormal eigenbasis and $A = U\Lambda U^{\dagger}$ with U unitary, Eq. (8.40); the same holds with repeated eigenvalues (taken on trust).
- Spectral decomposition: $A = \sum_k \lambda_k u_k u_k^{\dagger}$, Eq. (8.42).
- Powers: $A^k = U\Lambda^k U^{\dagger}$.
- The Rayleigh quotient of a Hermitian matrix lies between its smallest and largest eigenvalue, Eq. (8.46).

8.6 Tridiagonal matrices

A matrix is *tridiagonal* if its only nonzero entries lie on the main diagonal and on the two diagonals immediately above and below it. A symmetric tridiagonal $N \times N$ matrix is described

by two lists: the diagonal $d_1, \dots, d_N$ and the off-diagonal $b_1, \dots, b_{N-1}$:

$$T = \begin{pmatrix} d_1 & b_1 & & & \\ b_1 & d_2 & b_2 & & \\ & b_2 & d_3 & \ddots & \\ & & \ddots & \ddots & b_{N-1} \\ & & & b_{N-1} & d_N \end{pmatrix}. \tag{8.47}$$

(Blank entries are zero.) Instead of N^2 numbers, only $2N-1$ need to be stored. For $N = 6000$, the working grid size used in the paper at scale 1, that is 11999 numbers instead of 3.6×10^7.

Tridiagonal matrices arise whenever each point of a system interacts only with its nearest neighbors: masses on a chain connected by springs, a string sampled at equally spaced points, and (Section 8.7) the quantum Hamiltonian on a grid.

8.6.1 Nondegeneracy

Theorem (proved). If T is symmetric tridiagonal and every off-diagonal entry b_j is nonzero, then each eigenvalue of T has, up to a multiplicative constant, only one eigenvector.

Proof. Fix an eigenvalue λ and write out $(T - \lambda I)v = 0$ row by row:

$$\text{row } 1: \quad (d_1 - \lambda)v_1 + b_1 v_2 = 0, \tag{8.48}$$

$$\text{row } j: \quad b_{j-1}v_{j-1} + (d_j - \lambda)v_j + b_j v_{j+1} = 0, \quad j = 2, \dots, N-1. \tag{8.49}$$

Because $b_1 \neq 0$, row 1 determines $v_2 = -(d_1 - \lambda)v_1/b_1$. Because $b_j \neq 0$, row j determines $v_{j+1} = -[b_{j-1}v_{j-1} + (d_j - \lambda)v_j]/b_j$ from the two previous components. So once v_1 is chosen, every component is fixed, and doubling v_1 doubles every component. If $v_1 = 0$, then $v_2 = 0$, then $v_3 = 0$, and so on: $v = 0$, which is not an eigenvector. Hence every eigenvector is a nonzero multiple of the single vector obtained with $v_1 = 1$. ■

Consequence (relies on a result taken on trust). Combined with the general spectral theorem (taken on trust in the box of Section 8.5), which says a symmetric matrix has as many independent eigenvectors as the multiplicity of each eigenvalue, this shows that a repeated eigenvalue is impossible here: *all N eigenvalues of such a T are distinct.* We can therefore list them unambiguously in increasing order, $\lambda_1 < \lambda_2 < \cdots < \lambda_N$, and speak of "the n-th eigenvalue" and "the n-th eigenvector" without ambiguity. That is what makes "level n" a well-defined object in the paper.

If some $b_j = 0$, the matrix splits into two independent blocks and degeneracy becomes possible (Problem 16).

8.6.2 The second-difference matrix: a complete solution

The most important tridiagonal matrix in this book is

$$D_N = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} \qquad (N \times N), \tag{8.50}$$

with $d_j = 2$ and $b_j = -1$. For $N = 3$ it is the matrix D_3 of Sample Problem "The three-point second-difference matrix". Remarkably, its eigenpairs can be found exactly for every N.

Derivation. Row j of $D_N v = \lambda v$ reads

$$-v_{j-1} + 2v_j - v_{j+1} = \lambda v_j, \tag{8.51}$$

valid for $j = 2, \dots, N-1$. For the first and last rows the term v_0 or v_{N+1} is missing. We can make Eq. (8.51) valid for *all* $j = 1, \dots, N$ by inventing two extra components and insisting that

$$v_0 = 0, \qquad v_{N+1} = 0, \tag{8.52}$$

because then the "missing" terms are zero anyway. These are *boundary conditions*, the discrete version of a string tied down at both ends.

Inspired by the string, try a sampled sine: $v_j = \sin(j\theta)$ for some angle θ to be found. The left side of Eq. (8.51) needs $\sin((j-1)\theta) + \sin((j+1)\theta)$. The trigonometric identity $\sin(\alpha+\beta) + \sin(\alpha-\beta) = 2\sin\alpha\cos\beta$ (add the two angle-sum formulas $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$) with $\alpha = j\theta$, $\beta = \theta$ gives

$$\sin((j+1)\theta) + \sin((j-1)\theta) = 2\sin(j\theta)\cos\theta. \tag{8.53}$$

Therefore

$$\begin{aligned} -v_{j-1} + 2v_j - v_{j+1} &= 2\sin(j\theta) - [\sin((j-1)\theta) + \sin((j+1)\theta)] && (8.54)\\ &= 2\sin(j\theta) - 2\sin(j\theta)\cos\theta && (8.55)\\ &= (2 - 2\cos\theta)\, v_j. && (8.56) \end{aligned}$$

So Eq. (8.51) holds for every j with $\lambda = 2 - 2\cos\theta$. The left boundary condition holds automatically: $v_0 = \sin 0 = 0$. The right one requires $\sin((N+1)\theta) = 0$, i.e. $(N+1)\theta = k\pi$ for an integer k. For $1 \le k \le N$ the first component $\sin(k\pi/(N+1))$ is nonzero because $0 < k\pi/(N+1) < \pi$, so $v \neq 0$ is a genuine eigenvector. The values $k = 1, 2, \dots, N$ give N different eigenvalues (because $\cos\theta$ is strictly decreasing on $0 < \theta < \pi$), and since an $N \times N$ matrix has only N eigenvalues, these are all of them. (The value $k = 0$ gives $v = 0$; $k = N+1$ also gives $v_j = \sin(j\pi) = 0$ for all j; larger k repeat earlier vectors up to sign.) Using $1 - \cos\theta = 2\sin^2(\theta/2)$:

$$\boxed{\lambda_k = 2 - 2\cos\frac{k\pi}{N+1} = 4\sin^2\frac{k\pi}{2(N+1)}, \qquad (v_k)_j = \sin\frac{jk\pi}{N+1}, \qquad k = 1, \dots, N.} \tag{8.57}$$

Check with $N = 3$: $k = 1$ gives $2 - 2\cos(\pi/4) = 2 - \sqrt{2}$ and $v = (\sin\frac{\pi}{4}, \sin\frac{\pi}{2}, \sin\frac{3\pi}{4}) = (\frac{\sqrt{2}}{2}, 1, \frac{\sqrt{2}}{2}) \propto (1, \sqrt{2}, 1)$, exactly as in the Sample Problem.

Figure 8.2 shows several eigenvectors for $N = 20$. They are the harmonics of a string sampled at 20 points. Eigenvector k changes sign $k-1$ times as j runs from 1 to N (counting a passage through a zero component, as in $(1, 0, -1)$, as one change), because $\sin(jk\pi/(N+1))$ passes through $k-1$ zeros strictly between $j = 0$ and $j = N+1$. In Chapter 11 you will meet the same "node counting" for quantum states. All eigenvalues lie strictly between 0 and 4 (Problem 25 proves this without the formula).

Checkpoint 8.5

In Eq. (8.57), why is $k = N+1$ not allowed?

Answer: For $k = N+1$ the components are $\sin(jk\pi/(N+1)) = \sin(j\pi) = 0$ for every j, so the "eigenvector" is the zero vector, which is excluded by definition.

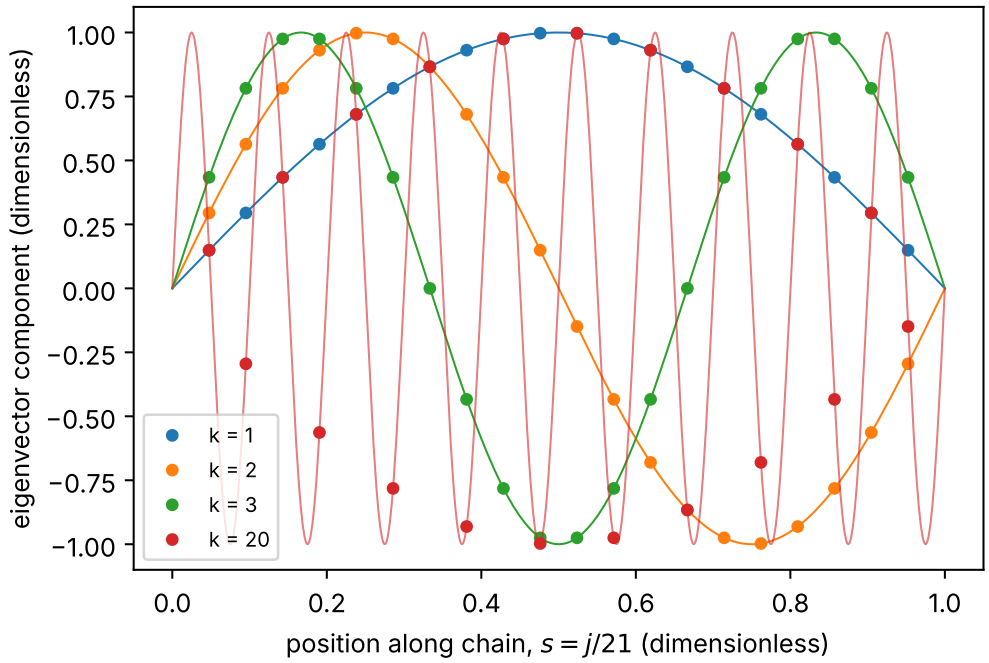


Figure 8.2: Eigenvectors of the 20×20 second-difference matrix D_{20} computed numerically (dots), against the sine curves $\sin(k\pi s)$ (lines), with $s = j/21$ the position along the chain. Modes $k = 1, 2, 3$ look like the first three harmonics of a string; the top mode $k = 20$ alternates in sign from point to point and is a poor imitation of any smooth wave. Components are dimensionless.

Key Ideas

- Symmetric tridiagonal matrices need only $2N - 1$ stored numbers.
- If all off-diagonal entries are nonzero, every eigenvalue is simple, so eigenvalues can be numbered unambiguously in increasing order.
- The second-difference matrix D_N has eigenvalues $4\sin^2(k\pi/2(N+1))$ and sampled-sine eigenvectors, Eq. (8.57); mode k has $k - 1$ sign changes.

8.7 The second derivative as a matrix

★ Did you know? *A String Inside a Matrix*

The matrix with 2 on the diagonal and -1 just above and below is the finite-difference version of $-d^2/dx^2$. For N grid points its eigenvalues are known exactly, $4\sin^2[k\pi/(2N+2)]$, and its eigenvectors are sampled sine waves $\sin(jk\pi/(N+1))$. These are the standing waves of a guitar string, so a simple matrix already contains the physics of vibrating strings.

Recall from Chapter 7 that a function $f(x)$ on an interval can be represented by its values on a grid. Here we use the grid the paper uses. Divide $[-L, L]$ into $N + 1$ equal steps of width

$$h = \frac{2L}{N+1}, \tag{8.58}$$

and keep the N *interior* points

$$x_j = -L + jh, \qquad j = 1, 2, \ldots, N. \tag{8.59}$$

The endpoints $x_0 = -L$ and $x_{N+1} = L$ are the walls. Write $f_j = f(x_j)$.

The second difference. Expand f about x_j in a Taylor series (first-year calculus), assuming f has at least six continuous derivatives. We use big-O notation: $\mathcal{O}(h^5)$, read "order h^5", stands for a quantity whose size is at most a fixed constant times $|h|^5$ for all sufficiently small h. (Similarly, an algorithm whose cost is $\mathcal{O}(n^3)$ needs at most a fixed constant times n^3 operations for large n.) The expansions are:

$$f(x_j + h) = f_j + hf'_j + \tfrac{h^2}{2} f''_j + \tfrac{h^3}{6} f'''_j + \tfrac{h^4}{24} f''''_j + \mathcal{O}(h^5), \tag{8.60}$$

$$f(x_j - h) = f_j - hf'_j + \tfrac{h^2}{2} f''_j - \tfrac{h^3}{6} f'''_j + \tfrac{h^4}{24} f''''_j + \mathcal{O}(h^5). \tag{8.61}$$

Adding the two lines, the odd powers of h cancel:

$$f_{j+1} + f_{j-1} = 2f_j + h^2 f''_j + \tfrac{h^4}{12} f''''_j + \mathcal{O}(h^6), \tag{8.62}$$

where the remainder is $\mathcal{O}(h^6)$ because the h^5 terms are odd in h and cancel as well. Subtract $2f_j$ and divide by h^2:

$$\frac{f_{j+1} - 2f_j + f_{j-1}}{h^2} = f''_j + \frac{h^2}{12} f''''_j + \mathcal{O}(h^4). \tag{8.63}$$

So the "second difference" on the left approximates $f''(x_j)$ with an error of order h^2. Chapter 23 studies this error carefully; here we need only the structure.

The matrix. Multiply Eq. (8.63) by -1 and impose the wall conditions $f_0 = f_{N+1} = 0$ (the function vanishes at the walls, like a string tied down there). Then the list of values $-f''(x_j)$ is approximated by

$$-f''(x_j) \approx \frac{1}{h^2}(-f_{j-1} + 2f_j - f_{j+1}) = \frac{1}{h^2}(D_N f)_j, \tag{8.64}$$

with D_N of Eq. (8.50). In the first row, $f_0 = 0$ removes the term $-f_0$; in the last row, $f_{N+1} = 0$ removes $-f_{N+1}$. So the operator "take minus the second derivative of a function vanishing at $\pm L$" becomes the matrix D_N/h^2.

Adding a potential. In quantum mechanics (Chapter 10) the energy operator, the Hamiltonian, in the units $\hbar = 2m = 1$ that the paper uses, is

$$H = -\frac{d^2}{dx^2} + V(x). \tag{8.65}$$

The potential $V(x)$ multiplies the wavefunction point by point, which on the grid is the diagonal matrix $\mathrm{diag}(V(x_1), \ldots, V(x_N))$. Adding,

$$\boxed{H_{jj} = \frac{2}{h^2} + V(x_j), \qquad H_{j,j\pm1} = -\frac{1}{h^2}, \qquad \text{all other entries } 0.} \tag{8.66}$$

This is a real symmetric tridiagonal matrix with every off-diagonal entry equal to $-1/h^2 \neq 0$. Everything proved so far applies: its eigenvalues are real (Theorem 1), distinct (nondegeneracy theorem of Section 8.6), and its eigenvectors form an orthonormal basis of real vectors (spectral theorem). The eigenvalues approximate the energy levels; the eigenvectors are the wavefunctions sampled on the grid.

The empty box exactly. With $V = 0$ the eigenvalues of D_N/h^2 follow from Eq. (8.57) by dividing by h^2. Since $N + 1 = 2L/h$ by Eq. (8.58), the angle $k\pi/(2(N+1))$ equals $k\pi h/(4L)$, and

$$\lambda_k^{\text{disc}} = \frac{4}{h^2}\sin^2\frac{k\pi h}{4L}, \qquad (v_k)_j = \sin\frac{jk\pi}{N+1} = \sin\frac{k\pi(x_j+L)}{2L}. \tag{8.67}$$

For the second form of the eigenvector we used $x_j + L = jh = 2Lj/(N+1)$, so $jk\pi/(N+1) = k\pi(x_j+L)/(2L)$. We will see in the next section that $\sin(k\pi(x+L)/2L)$ is exactly the k-th eigenfunction of the continuous problem. So the discrete eigenvectors are the exact continuous eigenfunctions sampled at the grid points, and only the eigenvalues carry a discretization error.

Key Ideas

- The second difference $(f_{j+1} - 2f_j + f_{j-1})/h^2$ approximates f'' with error $\mathcal{O}(h^2)$, Eq. (8.63).
- With walls at $\pm L$, $-d^2/dx^2$ becomes D_N/h^2, and $H = -d^2/dx^2 + V$ becomes the symmetric tridiagonal matrix of Eq. (8.66).
- For the empty box the discrete eigenvalues are $(4/h^2)\sin^2(k\pi h/4L)$ and the eigenvectors are sampled sines, Eq. (8.67).

8.8 Eigenvalue problems for differential operators

Now forget the grid and ask the continuous question. Which functions $y(x)$ on $[-L, L]$, vanishing at both ends, satisfy

$$-y''(x) = \lambda\, y(x), \qquad y(-L) = y(L) = 0, \tag{8.68}$$

for some number λ, with y not identically zero? This is an eigenvalue problem in which the "matrix" is the operator $-d^2/dx^2$ and the "vectors" are functions. It is also the equation of a standing wave on a string of length $2L$ fixed at both ends: for a shape $y(x)\cos(\omega t)$ the wave equation reduces to Eq. (8.68) with $\lambda = \omega^2/c_{\text{w}}^2$, where ω is the angular frequency of the standing wave and c_{w} the wave speed [7]. And in Chapter 11 it will be the Schrödinger equation for a particle in a box, with λ the energy.

The eigenvalues are real and non-negative. Before solving, we show that λ must be real. Allow y and λ to be complex for the moment. Multiply both sides of $-y'' = \lambda y$ by y^*, the complex conjugate of y, and integrate over $[-L, L]$:

$$\lambda\int_{-L}^{L}|y|^2\,\mathrm{d}x = -\int_{-L}^{L} y^* y''\,\mathrm{d}x \tag{8.69}$$

$$= \left[-y^* y'\right]_{-L}^{L} + \int_{-L}^{L}(y^*)'\,y'\,\mathrm{d}x \quad (\text{parts: } u = y^*,\ \mathrm{d}v = -y''\mathrm{d}x) \tag{8.70}$$

$$= \int_{-L}^{L}|y'|^2\,\mathrm{d}x. \quad (y(\pm L) = 0 \text{ and } (y^*)' = (y')^*) \tag{8.71}$$

Both integrals are real and non-negative. Also $\int |y|^2\,\mathrm{d}x > 0$, because y is not identically zero. So $\lambda = \int |y'|^2\,\mathrm{d}x / \int |y|^2\,\mathrm{d}x$ is real and $\lambda \geq 0$. This is the continuous analogue of Eq. (8.32).

Solving it completely. Knowing that λ is real, we still consider all three possible signs of λ in turn, which also confirms that $\lambda \leq 0$ gives nothing.

Case $\lambda = 0$. Then $y'' = 0$, so $y = a + bx$. The conditions give $a - bL = 0$ and $a + bL = 0$; adding, $2a = 0$, so $a = 0$; then $bL = 0$, so $b = 0$. Only $y \equiv 0$: no eigenvalue.

Case $\lambda < 0$. Write $\lambda = -\kappa^2$ with $\kappa > 0$. Then $y'' = \kappa^2 y$, with general solution $y = ae^{\kappa x} + be^{-\kappa x}$ (check by differentiating twice). The conditions are $ae^{-\kappa L} + be^{\kappa L} = 0$ and $ae^{\kappa L} + be^{-\kappa L} = 0$. From the first, $a = -be^{2\kappa L}$. Substituting into the second, $-be^{3\kappa L} + be^{-\kappa L} = b(e^{-\kappa L} - e^{3\kappa L}) = 0$. Since $\kappa L > 0$, $e^{3\kappa L} > e^{-\kappa L}$, so $b = 0$ and then $a = 0$. No eigenvalue.

Case $\lambda > 0$. Write $\lambda = q^2$ with $q > 0$. Then $y'' = -q^2 y$, which is the simple-harmonic-motion equation with x in place of time, with general solution $y = a\cos(qx) + b\sin(qx)$. The conditions, using cos even and sin odd, are

$$y(L) = a\cos(qL) + b\sin(qL) = 0, \tag{8.72}$$

$$y(-L) = a\cos(qL) - b\sin(qL) = 0. \tag{8.73}$$

Adding gives $2a\cos(qL) = 0$; subtracting gives $2b\sin(qL) = 0$. For a nonzero solution, either $a \neq 0$ and $\cos(qL) = 0$, i.e. $qL = \pi/2, 3\pi/2, \ldots$ (odd multiples of $\pi/2$), with $b = 0$ because $\sin(qL) = \pm 1 \neq 0$ there; or $b \neq 0$ and $\sin(qL) = 0$, i.e. $qL = \pi, 2\pi, \ldots$ (even multiples of $\pi/2$), with $a = 0$. Both families together are $qL = k\pi/2$ with $k = 1, 2, 3, \ldots$. Hence

$$\boxed{\lambda_k = \left(\frac{k\pi}{2L}\right)^2, \qquad y_k(x) = \sin\frac{k\pi(x+L)}{2L}, \qquad k = 1, 2, 3, \ldots} \tag{8.74}$$

The single formula for y_k covers both families: $\sin(k\pi x/(2L) + k\pi/2)$ equals $\pm\cos(k\pi x/2L)$ for odd k and $\pm\sin(k\pi x/2L)$ for even k. In the language of the string, the wavelength of mode k is $2 \cdot 2L/k$, and the wavenumber $q = 2\pi/\text{wavelength} = k\pi/(2L)$, the familiar harmonics [7].

Unlike a matrix, the operator has infinitely many eigenvalues, growing without bound like k^2. The $N \times N$ matrix D_N/h^2 can only capture the first N of them, and (as we will compute) only the first part of those accurately.

The operator is symmetric. Why are the eigenvalues real and the eigenfunctions orthogonal, exactly as for symmetric matrices? Because $-d^2/dx^2$ with these boundary conditions has the continuous version of the symmetry property. Recall from Chapter 7 that for real functions the inner product is $\int_{-L}^{L} f(x)g(x)\,dx$. Let f, g be twice differentiable with $f(\pm L) = g(\pm L) = 0$. Integrate by parts:

$$\int_{-L}^{L} f\,(-g'')\,dx = [-fg']_{-L}^{L} + \int_{-L}^{L} f'g'\,dx \quad (\text{parts: } u = f,\ dv = -g''dx) \tag{8.75}$$

$$= \int_{-L}^{L} f'g'\,dx \quad (f(\pm L) = 0) \tag{8.76}$$

$$= [f'g]_{-L}^{L} - \int_{-L}^{L} f''g\,dx \quad (\text{parts: } u = g,\ dv = f'dx) \tag{8.77}$$

$$= \int_{-L}^{L} (-f'')\,g\,dx. \quad (g(\pm L) = 0) \tag{8.78}$$

This is the function version of $u^\top Av = (Au)^\top v$. Adding a real potential keeps the property, since $\int f\,(Vg)\,dx = \int (Vf)\,g\,dx$ trivially. Now repeat the proof of Theorem 2 with integrals: if $-y_k'' = \lambda_k y_k$ and $-y_l'' = \lambda_l y_l$, Eq. (8.78) with $f = y_k$, $g = y_l$ gives $\lambda_l \int y_k y_l\,dx = \lambda_k \int y_k y_l\,dx$, so $\int y_k y_l\,dx = 0$ when $\lambda_k \neq \lambda_l$. Readers who have studied Fourier series (Chapter 5) will recognize this as the orthogonality of the sines of Eq. (8.74). Here it has been derived without evaluating any trigonometric integral.

Taken on trust

Taken on trust: "self-adjoint" in infinite dimensions. For operators on infinitely many dimensions, such as $-d^2/dx^2$ acting on functions, the property of Eq. (8.78) (called

"symmetric" or "Hermitian") is not by itself enough to guarantee all the conclusions of the spectral theorem. One needs a stronger condition, *self-adjointness*, which also specifies precisely which functions the operator acts on (its "domain"; here, the boundary conditions). For a Schrödinger operator on a finite interval with walls, the two notions coincide, and the conclusions (real eigenvalues, orthonormal eigenfunctions forming a basis) hold. The precise definitions and proofs belong to functional analysis and are beyond this book; a physicist's account is in [30]. When the paper says "self-adjoint operator", you can read "the infinite-dimensional analogue of a Hermitian matrix, whose eigenvalues are therefore real".

Sample Problem 8.6: The box of RE3: continuum versus grid

Take the box of the running example RE3, $L = 44.883$, which is the half-width the paper uses at scale 1. (a) Find the first three eigenvalues of Eq. (8.68). (b) For the grid the paper uses at scale 1, $N = 6000$, compare the exact λ_{500} with the matrix eigenvalue of D_N/h^2, and express the difference in units of the local spacing between exact eigenvalues.
Step 1: lowest eigenvalues. $\pi/(2L) = \pi/89.766 = 0.0349976$, so $\lambda_1 = (\pi/2L)^2 = 1.2248 \times 10^{-3}$, $\lambda_2 = 4\lambda_1 = 4.8993 \times 10^{-3}$, $\lambda_3 = 9\lambda_1 = 1.10235 \times 10^{-2}$.
Step 2: grid spacing. $h = 2L/(N+1) = 89.766/6001 = 0.014958$.
Step 3: exact λ_{500}. $500\pi/(2L) = 17.4988$, so $\lambda_{500} = (17.4988)^2 = 306.208$.
Step 4: matrix eigenvalue. By Eq. (8.67) with the angle $500\pi h/(4L) = 500\pi/(2 \cdot 6001) = 0.130878$:

$$\lambda_{500}^{\text{disc}} = \frac{4}{h^2}\sin^2(0.130878) = 304.463.$$

A direct numerical eigensolve of the 6000×6000 matrix (Listing in Section 8.9) gives the same number.
Step 5: difference and spacing. $306.208 - 304.463 = 1.744$. The spacing of exact eigenvalues near $k = 500$ is $\lambda_{501} - \lambda_{500} = (\pi/2L)^2(501^2 - 500^2) = (\pi/2L)^2 \cdot 1001 = 1.2261$. The ratio is $1.744/1.2261 = 1.42$.

$\lambda_{1,2,3} = 1.2248 \times 10^{-3},\ 4.8993 \times 10^{-3},\ 1.10235 \times 10^{-2}$; $\lambda_{500} = 306.208$ vs 304.463, off by 1.42 spacings.

Even with 6000 grid points, level 500 of the matrix is more than a full level spacing below the true level. This is a warning that you will hear again: the paper (Sec. II C, paragraph "No leakage through the bias correction") reports that for its potential at levels 401–500 on the working grid the finite-difference error "reaches two mean spacings". Chapter 23 explains how the paper removes it.

Table 8.1 and Figure 8.3 show the pattern. Expanding $\sin y = y - y^3/6 + \dots$ gives $\sin^2 y = y^2 - y^4/3 + \dots$, so with $y = k\pi h/4L$,

$$\lambda_k^{\text{disc}} = \left(\frac{k\pi}{2L}\right)^2 \left[1 - \frac{1}{3}\left(\frac{k\pi h}{4L}\right)^2 + \dots\right]. \tag{8.79}$$

the relative error grows like k^2h^2. This agrees with Eq. (8.63): for $y = \sin(qx + \dots)$, $y'''' = q^4 y$, so the error term $\frac{h^2}{12}y''''$ changes q^2 by $-\frac{h^2}{12}q^4$, a relative change $h^2q^2/12 = k^2\pi^2h^2/(48L^2)$, which is the same as $\frac{1}{3}(k\pi h/4L)^2$.

Table 8.1: Eigenvalues of $-d^2/dx^2$ on $[-L, L]$ with $L = 44.883$: exact $(k\pi/2L)^2$ versus the $N = 6000$ matrix D_N/h^2, Eq. (8.67). Relative error $= 1 - \lambda_k^{\text{disc}}/\lambda_k$.

k	exact	matrix	relative error
1	1.2248×10^{-3}	1.2248×10^{-3}	2.28×10^{-8}
10	0.12248	0.12248	2.28×10^{-6}
100	12.2483	12.2455	2.28×10^{-4}
500	306.208	304.463	5.70×10^{-3}

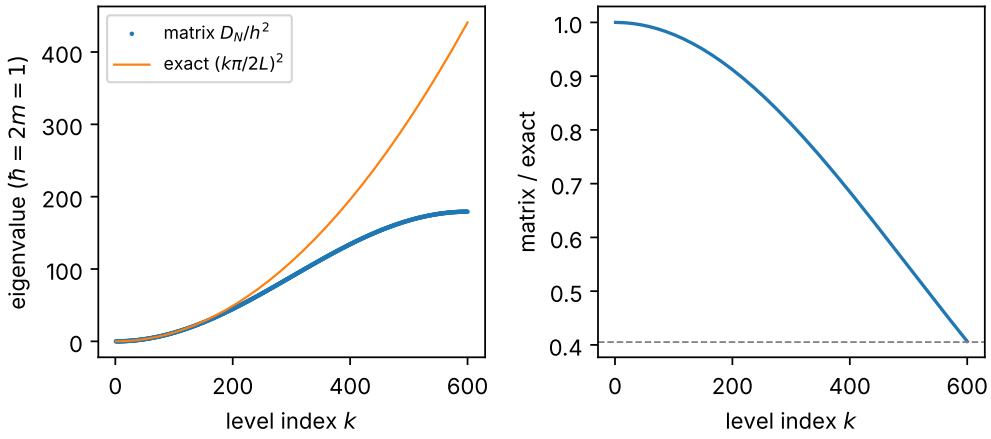


Figure 8.3: Left: all $N = 600$ eigenvalues of the matrix D_N/h^2 for the box $[-L, L]$ with $L = 44.883$ (dots) against the exact continuum eigenvalues $(k\pi/2L)^2$ (line); energies in units $\hbar = 2m = 1$. Right: their ratio, which is 1 for low k and falls to about 0.41 at $k = N$. A coarse grid is used so that the deviation is visible; it is the same phenomenon as in Table 8.1.

Key Ideas

- $-y'' = \lambda y$ with $y(\pm L) = 0$ has eigenvalues $(k\pi/2L)^2$ and eigenfunctions $\sin(k\pi(x + L)/2L)$, Eq. (8.74): the harmonics of a string.
- With these boundary conditions $-d^2/dx^2 + V$ is symmetric, Eq. (8.78), so its eigenvalues are real and its eigenfunctions orthogonal; "self-adjoint" is the precise infinite-dimensional notion.
- The grid matrix reproduces low eigenvalues well, but the relative error grows like k^2h^2; at $N = 6000$, level 500 of the empty box is off by about 1.4 level spacings.

8.9 Numerical eigensolvers

For a 6000×6000 matrix no human will expand a determinant. Computers find eigenvalues by iterative algorithms, packaged in libraries that you can treat as black boxes, provided you know what goes in, what comes out, and roughly how long it takes.

Why iteration? Finding eigenvalues of an $n \times n$ matrix is equivalent to finding the roots of a polynomial of degree n. For degree 5 and higher there is no general formula built from arithmetic operations and roots (see the box below). So every eigenvalue algorithm for large matrices is iterative: it produces a sequence of better and better approximations and stops when the error is at the level of rounding.

Taken on trust

Taken on trust: the Abel–Ruffini theorem. There is no formula that expresses the roots of every polynomial of degree 5 or higher in terms of its coefficients using only addition, subtraction, multiplication, division and k-th roots. The proof belongs to Galois theory and is found in standard texts on abstract algebra. Every polynomial with leading coefficient 1 is, up to sign, the characteristic polynomial of some matrix (a so-called companion matrix), so no such formula can exist for the eigenvalues of general $n \times n$ matrices with $n \geq 5$ either.

What LAPACK does. LAPACK (Linear Algebra PACKage) is the standard, freely available library of linear algebra routines that numpy and scipy call internally. For a dense real symmetric or complex Hermitian matrix, its strategy has three stages [29]:

1. *Reduce to tridiagonal form.* A sequence of carefully chosen unitary matrices Q (Householder reflections) turns A into $Q^\dagger AQ = T$, tridiagonal. A transformation of this form does not change eigenvalues (Problem 29 shows that $S^{-1}AS$ always has the same characteristic polynomial as A, and $Q^\dagger = Q^{-1}$). This stage costs about $\frac{4}{3}n^3$ arithmetic operations [29].

2. *Solve the tridiagonal problem* by an iterative method (the QR iteration, divide-and-conquer, or bisection; see below).

3. *Transform the eigenvectors back* by multiplying by Q.

For a dense non-Hermitian matrix (such as the complex matrix $H + iA$ of the paper's Sec. II E), LAPACK first reduces to a form with zeros below the first sub-diagonal ("Hessenberg form") and then applies the QR iteration; the cost again grows like n^3, with a larger constant. Because of stage 1, the cost of a dense eigensolve is $\mathcal{O}(n^3)$: doubling n multiplies the work by about 8.

If your matrix is *already* tridiagonal, as Eq. (8.66) is, you skip stage 1 entirely and the cost drops to $\mathcal{O}(n^2)$ for all eigenvalues, and less if you want only some of them. To see why, look at bisection.

Counting eigenvalues below a number: the Sturm count. For the symmetric tridiagonal matrix (8.47) and a trial number x, compute the sequence

$$q_1 = d_1 - x, \qquad q_j = d_j - x - \frac{b_{j-1}^2}{q_{j-1}}, \quad j = 2, \dots, N. \tag{8.80}$$

These q_j are the pivots that Gaussian elimination (Chapter 7) produces when it reduces $T - xI$ to triangular form without swapping rows: eliminating the entry b_{j-1} below the pivot q_{j-1} subtracts $(b_{j-1}/q_{j-1})\times$ row $j-1$ from row j, which changes the diagonal entry $d_j - x$ into $d_j - x - b_{j-1}^2/q_{j-1}$. Their product is therefore $\det(T - xI)$, which equals $\prod_k(\lambda_k - x)$ by the factorization of the characteristic polynomial in Section 8.2.

Taken on trust

Taken on trust: Sturm count (Sylvester's law of inertia). If no q_j is exactly zero, the number of negative values among $q_1, \dots, q_N$ equals the number of eigenvalues of T that are less than x. The product formula above shows only that the two counts have the same parity (both even or both odd); the full statement is a consequence of Sylvester's law of inertia, proved in [29] and [27].

Each Sturm count costs $\mathcal{O}(N)$ operations. To pin down eigenvalue number k, start with an interval known to contain it and *bisect* it, that is, keep halving it and retain the half that contains the target: at the midpoint, the count tells you whether λ_k lies to the left or right. Each step halves the interval, so reaching double precision from a reasonable starting interval takes a few dozen steps. All N eigenvalues then cost $N \times$ (a few dozen) $\times\, \mathcal{O}(N) = \mathcal{O}(N^2)$. The eigenvectors can be obtained afterwards by solving a few tridiagonal linear systems per eigenvalue ("inverse iteration"), each $\mathcal{O}(N)$. The same count also answers the question "how many levels lie below E?" in $\mathcal{O}(N)$ operations, which is one standard way (bisection with Sturm counts) for a solver to return "all eigenvalues in a range".

```
import numpy as np

def sturm_count(d, b, x):
    """Number of eigenvalues < x of the symmetric tridiagonal
    matrix with diagonal d (length N) and off-diagonal b (N-1)."""
    count = 0
    q = d[0] - x
    for j in range(len(d)):
        if j > 0:
            q = d[j] - x - b[j-1]**2 / q
        if q == 0.0:          # exact zero pivot: nudge it
            q = -1e-300
        if q < 0:
            count += 1
    return count

d = np.array([2.0, 2.0, 2.0]); b = np.array([-1.0, -1.0])
print(sturm_count(d, b, 2.5), sturm_count(d, b, 0.6))   # 2 1
```

Sample Problem 8.7: Counting levels by hand

Use the Sturm count (8.80) to find how many eigenvalues of D_3 lie below $x = 2.5$ and below $x = 0.6$, and compare with the exact eigenvalues 0.585786, 2, 3.414214.
Step 1: $x = 2.5$. $d_j - x = -0.5$, $b_j^2 = 1$. Then $q_1 = -0.5$; $q_2 = -0.5 - 1/(-0.5) = -0.5 + 2 = 1.5$; $q_3 = -0.5 - 1/1.5 = -1.16667$. Negative pivots: q_1, q_3, two of them.
Step 2: compare. The eigenvalues below 2.5 are 0.585786 and 2: two. Agreement.
Step 3: $x = 0.6$. $d_j - x = 1.4$. Then $q_1 = 1.4$; $q_2 = 1.4 - 1/1.4 = 0.685714$; $q_3 = 1.4 - 1/0.685714 = 1.4 - 1.458333 = -0.058333$. One negative pivot.
Step 4: compare. Only $0.585786 < 0.6$: one. Agreement, and the small size of q_3 reflects how close 0.6 is to an eigenvalue.

$$\boxed{\text{count}(2.5) = 2, \qquad \text{count}(0.6) = 1.}$$

The tools you will actually call. Table 8.2 lists the functions used in this book. All of them call LAPACK.

Table 8.2: Eigensolvers in numpy and scipy.

function	for	returns
`np.linalg.eigh(A)`	dense Hermitian/symmetric	real eigenvalues, ascending;
`np.linalg.eig(A)`	any dense square matrix	complex eigenvalues, *no pa*
`scipy.linalg.eigh_tridiagonal(d, e, ...)`	real symmetric tridiagonal	ascending eigenvalues, optio

For `eigh_tridiagonal`, `d` is the diagonal and `e` the off-diagonal. The option `select='i', select_range=(0, k-1)` returns the lowest k eigenpairs (indices start at 0 in Python); `select='v', select_range=(lo, hi)` returns those with $\text{lo} < \lambda \le \text{hi}$. The eigenvectors are returned as the columns of a matrix.

```
import numpy as np
from scipy.linalg import eigh_tridiagonal

L, N = 44.883, 6000                      # RE3 box, paper's scale-1 grid
h = 2*L/(N + 1)
V = np.zeros(N)                          # empty box; put V(x_j) here
d = 2/h**2 + V                           # diagonal, Eq. (eigen-Hmatrix)
e = -np.ones(N - 1)/h**2                 # off-diagonal
E, psi = eigh_tridiagonal(d, e, select='i', select_range=(0, 499))
psi = psi/np.sqrt(h)                     # now sum(psi**2)*h = 1
k = np.arange(1, 501)
exact = (k*np.pi/(2*L))**2
print(E[499], exact[499])                # 304.463...  306.208...
```

Pitfall

Normalization conventions. Solvers return eigenvectors with $\sum_j v_j^2 = 1$. A wavefunction sampled on a grid should instead satisfy $\sum_j \psi_j^2 h \approx \int \psi^2 \mathrm{d}x = 1$. The two differ by the factor $1/\sqrt{h}$, as in the listing above. Forgetting the rescaling makes every such integral, computed as $\sum_j v_j^2 \phi_k(x_j) h$, too small by a factor h. Rescaling but forgetting the h in the sum makes it too large by a factor $1/h$.

Pitfall

Signs and order. The sign (or complex phase) of each returned eigenvector is arbitrary and may change between runs or library versions; never compare eigenvectors without fixing it or using quantities like $\left|u^\dagger w\right|^2$ that do not depend on it. `np.linalg.eig` returns eigenvalues in no particular order; always sort, or match (Section 8.10). Finally, do not use `eig` on a symmetric matrix: rounding can produce tiny spurious imaginary parts; use `eigh`.

How accurate are computed eigenvalues? The computer works in floating-point arithmetic with machine epsilon $\epsilon \approx 2.2 \times 10^{-16}$ in double precision. This is the relative gap between neighboring representable numbers near 1, so each arithmetic operation can introduce a relative rounding error of about this size.

Taken on trust

Taken on trust: stability of Hermitian eigenvalues. The LAPACK symmetric eigensolvers are *backward stable*: the eigenvalues they return are the exact eigenvalues of a slightly perturbed matrix $A + E$ with the size of E about ϵ times the size of A. For Hermitian matrices, Weyl's inequality guarantees that perturbing A by E moves each eigenvalue by at most the size (largest singular value, Chapter 9) of E. Together: computed eigenvalues of a Hermitian matrix are accurate to about ϵ times the largest $|\lambda|$. Proofs in [29]. Non-Hermitian matrices enjoy no such guarantee; their eigenvalues can be far more sensitive.

For the matrix of the listing, the largest eigenvalue is close to $4/h^2 \approx 1.79 \times 10^4$. Adding the Wu–Sprung potential, which reaches 1000 at scale 1, raises it by at most 1000. So rounding errors in the eigenvalues are a few times 10^{-12}: utterly negligible against the $\mathcal{O}(1)$ discretization errors of Table 8.1. The numerical eigensolver is not the weak link; the grid is.

Figure 8.4 shows what the same code produces when V is the Wu–Sprung potential of the paper's Eq. (2), which you will study in Chapter 22.

Checkpoint 8.6

Roughly how many times more arithmetic does a dense symmetric eigensolve of a 6000×6000 matrix need than computing all eigenvalues of a 6000×6000 tridiagonal matrix, if the costs are n^3 and n^2 up to similar constants?

Answer: The ratio is $n^3/n^2 = n = 6000$. Using the tridiagonal structure saves a factor of thousands.

Key Ideas

- Eigenvalues of large matrices are found iteratively by library routines (LAPACK) that you may treat as black boxes.
- Dense Hermitian or non-Hermitian eigensolves cost $\mathcal{O}(n^3)$; symmetric tridiagonal ones cost $\mathcal{O}(n^2)$ for all eigenvalues.
- The Sturm count, Eq. (8.80), counts eigenvalues below x in $\mathcal{O}(N)$ operations and powers bisection and "eigenvalues in a range".
- Mind conventions: eigenvectors come with $\sum v^2 = 1$ (rescale by $1/\sqrt{h}$ for wavefunctions), arbitrary signs, and (for `eig`) no ordering.

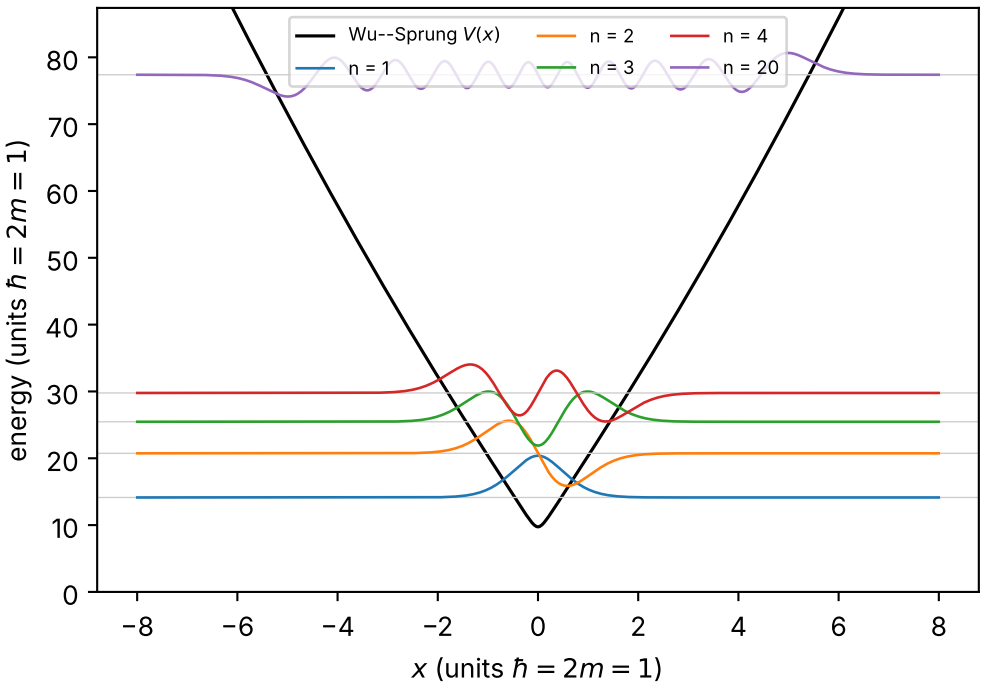


Figure 8.4: Eigenvectors of the tridiagonal matrix (8.66) with V the Wu–Sprung potential (paper Eq. (2)) on $[-L, L]$, $L = 44.883$, $N = 4000$. Each eigenvector (normalized so that $\sum \psi^2 h = 1$, then magnified for visibility) is drawn at the height of its eigenvalue; only $|x| \le 8$ is shown. Levels $n = 1, 2, 3, 4$ and $n = 20$ are shown. Each state oscillates where $E_n > V$ and decays beyond its classical turning points; state n has $n - 1$ sign changes; states alternate between even and odd because V is even. Units $\hbar = 2m = 1$.

- Computed Hermitian eigenvalues are accurate to about ϵ times the largest eigenvalue.

8.10 Ordering eigenvalues and matching eigenvectors

In the paper, "level n" of a fitted potential is compared with "zero n". That requires a labelling of eigenvalues by an index n that means the same thing before and after the potential is changed. For Hermitian matrices there is a natural labelling; for non-Hermitian ones there is not, and one must match eigenvectors instead.

Hermitian case: order by size. Real eigenvalues can be sorted, $\lambda_1 < \lambda_2 < \dots$. For the tridiagonal Hamiltonian (8.66) they are all distinct (Section 8.6) for *every* potential, so the n-th eigenvalue is well defined. Even more is true: as you change V smoothly, the eigenvalues move continuously. This follows from Weyl's inequality (taken on trust in the box of Section 8.9): changing V by at most δ at every grid point adds a diagonal matrix whose largest singular value is at most δ, so each eigenvalue moves by at most δ. Two of them can never meet, because meeting would be a degeneracy. So the n-th level stays the n-th level; levels cannot overtake one another.

But "the n-th level" and "the state that looks like the original state n" need not be the same thing. Consider

$$H(\varepsilon) = \begin{pmatrix} \varepsilon & g \\ g & -\varepsilon \end{pmatrix}, \qquad g > 0 \text{ fixed}, \tag{8.81}$$

as the parameter ε is swept from negative to positive. Here $\operatorname{tr} H = 0$ and $\det H = -\varepsilon^2 - g^2$, so Eq. (8.19) gives $\lambda_\pm = \pm\sqrt{\varepsilon^2 + g^2}$. Without the coupling ($g = 0$) the two levels $\pm\varepsilon$ would cross at $\varepsilon = 0$; with it, they come no closer than $2g$. This is an *avoided crossing* (Figure 8.5, left). Far to the left ($\varepsilon \ll -g$) the lower eigenvector is almost $e_1 = (1, 0)^\top$; far to the right it is almost e_2 (Problem 27 gives the exact formula). By ordering, the "lower level" is followed smoothly; by character, the states have exchanged identities. Which labelling is right depends on what question you ask. The paper's operator fits use ordering, which is natural for one-dimensional Hamiltonians (Chapter 11: the n-th level always has $n - 1$ nodes).

Non-Hermitian case: no natural order. If eigenvalues are complex, "sort by size" has no meaning: complex numbers have no natural order. Sorting by real part is a common guess, but it can mislabel states, for example when two eigenvalues have equal real parts (right panel of Figure 8.5, $g > 1$), or when a large imaginary part accompanies a small real shift. The robust approach is to compare *eigenvectors*.

Matching by overlap. Let $e_1, \dots, e_n$ be an orthonormal reference basis, typically the eigenvectors of the unperturbed Hermitian Hamiltonian, and let $w_1, \dots, w_n$ be the eigenvectors of the new (perturbed) matrix. Define the *overlap matrix*

$$O_{km} = \frac{\left| e_k^\dagger w_m \right|^2}{w_m^\dagger w_m}. \tag{8.82}$$

It does not depend on the arbitrary length or phase of w_m (multiplying w_m by a multiplies the numerator and denominator by $|a|^2$). Since the e_k form an orthonormal basis, expanding $w_m = \sum_k (e_k^\dagger w_m) e_k$ gives $w_m^\dagger w_m = \sum_k \left| e_k^\dagger w_m \right|^2$, so every *column* sums to one:

$$\sum_{k=1}^{n} O_{km} = 1. \tag{8.83}$$

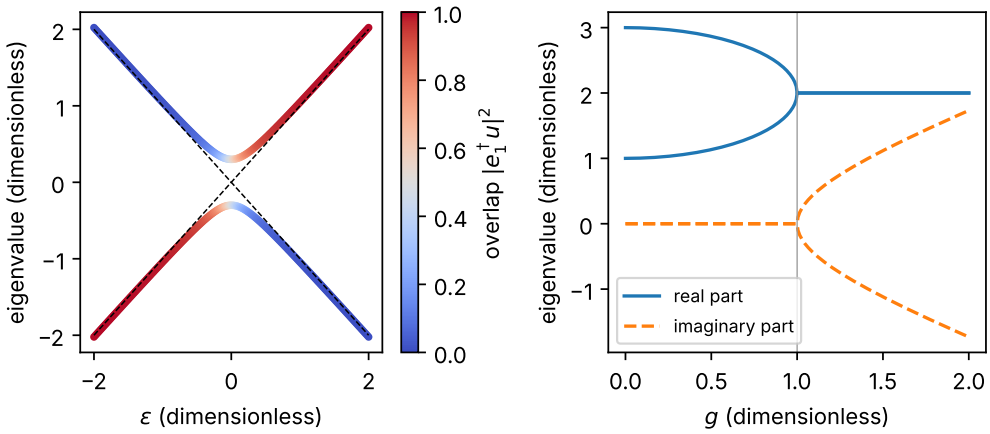


Figure 8.5: Left: eigenvalues of the Hermitian matrix (8.81) with $g = 0.3$ versus ε; color gives the overlap $\left|e_1^\dagger u\right|^2$ of each eigenvector with e_1. The levels avoid each other and exchange character. Dashed: the uncoupled levels $\pm\varepsilon$. Right: the non-Hermitian matrix $\begin{pmatrix} 1 & ig \\ ig & 3 \end{pmatrix}$ versus g: real parts (solid) attract and merge at $g = 1$, after which the eigenvalues become a complex-conjugate pair (imaginary parts, dashed). All quantities dimensionless.

O_{km} is the fraction of w_m that lies along reference state e_k. The matching rule: assign each new eigenvector w_m to the reference state k with the largest O_{km}, and call the matching *one-to-one* if no two new eigenvectors are assigned to the same reference state.

When is the matching guaranteed one-to-one? (proved) If an entry $O_{km} > \frac{1}{2}$, it is the largest in its column, by Eq. (8.83). If moreover the new matrix is Hermitian, its normalized eigenvectors $w_m / \|w_m\|$ are also an orthonormal basis, and the same argument with the roles exchanged shows that every *row* also sums to one: $\sum_m O_{km} = \|e_k\|^2 = 1$. Then two different columns $m \neq m'$ cannot both have an entry above $\frac{1}{2}$ in the same row k, because those two alone would sum to more than 1. So for Hermitian matrices, if every column has an entry above $\frac{1}{2}$, the assignment is automatically one-to-one. For non-Hermitian matrices the eigenvectors are not orthogonal (Section 8.3), rows need not sum to one, and two eigenvectors can both lean mostly on the same reference state. Then the matching fails, which is a sign that the perturbation has mixed the states strongly.

Sample Problem 8.8: Matching a Hermitian and a non-Hermitian perturbation

The reference matrix is $H_0 = \mathrm{diag}(1, 3)$ with eigenvectors e_1, e_2. (a) For the Hermitian $H_a = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 3 \end{pmatrix}$ find the eigenvalues and the overlap of the lower eigenvector with e_1. (b) Do the same for the non-Hermitian $H_b = \begin{pmatrix} 1 & 0.5i \\ 0.5i & 3 \end{pmatrix}$, and compute the normalized inner product of its two eigenvectors.

Step 1: eigenvalues of H_a. $\mathrm{tr} = 4$, $\det = 3 - 0.25 = 2.75$, so $\lambda = 2 \pm \sqrt{4 - 2.75} = 2 \pm \sqrt{1.25} = 2 \pm 1.118034$: 0.881966 and 3.118034.

Step 2: lower eigenvector of H_a. Row 1 of $(H_a - \lambda I)v = 0$: $(1 - \lambda)v_1 + 0.5v_2 = 0$, so $v_2 = -2(1 - \lambda)v_1 = -0.236068\, v_1$ for $\lambda = 0.881966$. With $v_1 = 1$, $v^\dagger v = 1 + 0.0557281 = 1.0557281$ and $O_{11} = 1/1.0557281 = 0.947214$. So $O_{21} = 0.052786$, and the lower state is matched to e_1.

Step 3: eigenvalues of H_b. $\mathrm{tr} = 4$, $\det = 3 - (0.5i)^2 = 3 + 0.25 = 3.25$, so $\lambda =$

$2 \pm \sqrt{4 - 3.25} = 2 \pm \sqrt{0.75} = 2 \pm 0.866025$: 1.133975 and 2.866025. Still real, but now the levels have moved *toward* each other, whereas in (a) they were pushed apart.
Step 4: eigenvectors of H_b. Row 1: $(1-\lambda)v_1 + 0.5iv_2 = 0$, so $v_2 = (\lambda - 1)v_1/(0.5i) = -2i(\lambda - 1)v_1$. For $\lambda = 1.133975$: $u = (1, -0.267949i)^\top$; for $\lambda = 2.866025$: $w = (1, -3.732051i)^\top$. Overlaps: $O_{11} = 1/(1 + 0.267949^2) = 1/1.0717968 = 0.933013$ for u, and $O_{22} = 3.732051^2/(1 + 3.732051^2) = 13.928203/14.928203 = 0.933013$ for w. So u is matched to e_1 and w to e_2, and the matching is one-to-one.
Step 5: are u and w orthogonal? $u^\dagger w = 1 \cdot 1 + (0.267949i)(-3.732051i) = 1 + 0.267949 \times 3.732051 = 1 + 1 = 2$. Normalized: $\left|u^\dagger w\right| / (\|u\|\,\|w\|) = 2/\sqrt{1.0717968 \times 14.928203} = 2/\sqrt{16} = 0.5$.

(a) 0.881966, 3.118034, $O_{11} = 0.947$; (b) 1.133975, 2.866025, $O_{11} = 0.933$, $\frac{|u^\dagger w|}{\|u\|\|w\|} = 0.5$.

H_b is symmetric ($H_b^\top = H_b$) but not Hermitian ($H_b^\dagger$ has $-0.5i$ off the diagonal), so Theorem 2 does not apply, and indeed its eigenvectors are far from orthogonal. Chapter 13 will explain the level shifts: the approximations $1 - 0.25/2 = 0.875$ (Hermitian, levels repel) and $1 + 0.25/2 = 1.125$ (imaginary coupling, levels attract) are second-order perturbation theory, and Chapter 14 derives the sign flip that $i^2 = -1$ produces.

If you increase the imaginary coupling in H_b to ig with general g, the characteristic polynomial becomes $\lambda^2 - 4\lambda + 3 + g^2$, whose roots are $\lambda = 2 \pm \sqrt{1 - g^2}$ (Problem 15). At $g = 1$ the two eigenvalues merge, and for $g > 1$ they are complex, for example $2 \pm 1.118034\,i$ at $g = 1.5$ (Figure 8.5, right). Such a merging point is called an *exceptional point*; Chapter 14 connects it to the breaking of PT symmetry.

```
import numpy as np

def match_by_overlap(E_ref_vecs, W):
    """Columns of E_ref_vecs: orthonormal reference states.
    Columns of W: new eigenvectors. Returns (assignment, O, ok)."""
    O = np.abs(E_ref_vecs.conj().T @ W)**2          # |e_k^dagger w_m|^2
    O = O / np.sum(np.abs(W)**2, axis=0)             # divide by w_m^dagger w_m
    assign = np.argmax(O, axis=0)                    # best reference per column
    ok = len(set(assign)) == W.shape[1]              # one-to-one?
    return assign, O, ok

H0 = np.diag([1.0, 3.0])
Hb = np.array([[1, 0.5j], [0.5j, 3]])
lam, W = np.linalg.eig(Hb)                           # unordered!
assign, O, ok = match_by_overlap(np.eye(2), W)
print(lam, assign, ok)
```

Checkpoint 8.7

You diagonalize a non-Hermitian matrix with `np.linalg.eig` and sort the eigenvalues by real part. Does this guarantee that the n-th sorted eigenvalue belongs to the state that was the n-th unperturbed state?

Answer: No. Complex numbers have no natural order; ties or near-ties in real part (as for $g > 1$ in Figure 8.5) and strong mixing can scramble the labels. Matching eigenvectors by overlap with the unperturbed states is the reliable check, and it can itself fail when the mixing is strong.

Key Ideas

- For Hermitian tridiagonal Hamiltonians, eigenvalues are distinct and keep their order as the potential changes, so "level n" is well defined.
- Avoided crossings exchange the character of states while preserving their order.
- Complex eigenvalues have no natural order; label them by the overlap matrix O_{km}, Eq. (8.82), whose columns sum to one.
- For Hermitian matrices, overlaps above $\frac{1}{2}$ guarantee a one-to-one matching; for non-Hermitian matrices matching can fail.

8.11 Paper connection

★ Did you know? *Jacobi's Rotations*

In 1846 Carl Jacobi published a method for diagonalizing a symmetric matrix through a sequence of plane rotations, each one zeroing an off-diagonal element. He applied it to astronomical problems computed by hand. The method was rediscovered for electronic computers in the 1950s and is still valued for its accuracy, because rotations do not amplify rounding errors.

Here is where each idea of this chapter enters the paper [1]. Section, equation, table and figure numbers refer to revision R1.

"The spectrum of a self-adjoint operator" (Sec. I, first sentence). The Hilbert–Pólya conjecture proposes that the heights t_n of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ are the eigenvalues of a self-adjoint operator. Why would that matter? A zero ρ lies on the critical line $\operatorname{Re}\rho = \frac{1}{2}$ exactly when the number t in $\rho = \frac{1}{2} + it$ is real (Chapters 2 and 16). Theorem 1 of Section 8.4, in its infinite-dimensional form (box in Section 8.8), says that a self-adjoint operator has only real eigenvalues. So if such an operator existed, its eigenvalues would be real, and the zeros would all lie on the critical line: that is the Riemann hypothesis. No such operator is known; the Hilbert–Pólya conjecture and the Riemann hypothesis are both open problems (Chapter 21). The paper does not attempt to settle them; it tests whether *fitted* operators capture transferable information about the zeros.

The forward solver (Sec. II B; App. A steps 3–4). The paper's Hamiltonian is "$H = -d^2/dx^2 + V(x)$ with V even" on $[-L, L]$, with $L = 44.883$ at scale 1 and 59.957 at scale 2. The sentence "Second-order finite differences on N uniform interior points (Dirichlet walls) give a symmetric tridiagonal matrix; eigenpairs from LAPACK" is exactly Sections 8.7 and 8.9: "interior points" are the x_j of Eq. (8.59), "Dirichlet walls" are the conditions $\psi(\pm L) = 0$ that made the first and last rows of D_N lose one neighbor, and the matrix is Eq. (8.66), which App. A step 4 writes as "tridiagonal diag $= 2/dx^2 + V$, off $= -1/dx^2$". Step 4 names the solver, `eigh_tridiagonal`, and gives value ranges: "$[0, 900]$ (Hermitian work, scale 1), $[0, 1000]$ (tensors; 652 states), $[0, 1500]$ (scale 2; 1079 states)". The paper gives these energy windows but not the solver options used to impose them. A window of this kind is naturally requested through a value-range selection such as the `select='v'` option of Section 8.9, and one standard way for a solver to return all eigenvalues in a window is bisection with Sturm counts. The working grids $N = 6000$ (scale 1) and $N = 8000$ (scale 2) are large, but because the matrix is tridiagonal the $\mathcal{O}(N^2)$ cost keeps them practical, and only the levels below the cutoff are computed (652 states below 1000 at scale 1 and 1079 below 1500 at scale 2, as quoted above).

Why "level n" is meaningful. The paper compares the n-th eigenvalue E_n with the n-th zero t_n, and forms residuals $r_n = \tilde{t}_n - E_n$ (Sec. II C). This needs an unambiguous numbering of levels. For the Hermitian fits it is supplied by the nondegeneracy theorem of Section 8.6: all off-diagonal entries of Eq. (8.66) equal $-1/h^2 \neq 0$, so the levels are distinct for every fitted δV, and ascending order labels them.

Grid error in eigenvalues (Sec. II B–C). Sample Problem "The box of RE3" showed that on the scale-1 grid the 500th eigenvalue of the empty box is off by 1.42 level spacings. The paper states that for its potential the finite-difference dispersion "at levels 401–500 on the working grid reaches two mean spacings" (Sec. II C, "No leakage" paragraph), and App. A's key anchors give a "raw-zero null on the uncorrected working grid 2.09 (401–500)". The two kinds of figure use different yardsticks: the empty-box level spacing here, the mean zero spacing gap_n in the paper. They also refer to different potentials, so only the order of magnitude should be compared. This is why the paper computes eigenvalues on two grids and builds a Richardson correction; Chapter 23 treats it in full.

Eigenvectors and normalization (Sec. II C; Eq. (5); App. A step 9). The fit needs eigenvectors, not just eigenvalues: the sensitivity matrix $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\,\mathrm{d}x$ (Sec. II C) and the matrix elements $A_{nm} = \langle n|A|m\rangle$ of Eq. (5) (written in the bra-ket notation of Eq. (8.38)) are integrals of eigenvectors. App. A step 9 states "ψ normalized $\sum \psi^2 dx = 1$", which is the $1/\sqrt{h}$ rescaling of the pitfall in Section 8.9. With this normalization, the $k = 0$ column $A_{n0} = \sum_j \psi_n(x_j)^2 h = 1$ for every n (Problem 18). Note that the paper uses the letter A both for this sensitivity matrix and for the non-Hermitian operator of Sec. II E; context distinguishes them.

Non-Hermitian classes (Sec. II E). There $H \to H + iA$ with A a "real multiplication operator", i.e. on the grid a real diagonal matrix $\mathrm{diag}(A(x_j))$. The matrix $H + iA$ is then complex *symmetric*, $(H + iA)^\top = H + iA$, but *not* Hermitian, $(H + iA)^\dagger = H - iA$, just like H_b in Sample Problem "Matching a Hermitian and a non-Hermitian perturbation". Theorems 1 and 2 do not apply: eigenvalues may be complex and eigenvectors are not orthogonal. The constant mode $\phi_0 = 1$ deserves a remark: $iA = ic_0 I$ shifts every eigenvalue by exactly ic_0 and leaves the eigenvectors untouched (Problems 8 and 20), consistent with Sec. II E (i), which records that this mode has zero second-order real shift; in fact the real shift vanishes exactly, to all orders. An optimizer gains nothing real from c_0, which is why, in Sec. V, a huge value $c_0 \approx 2.2 \times 10^6$ could only be exploiting numerical noise.

Exact diagonalization and matching by overlap (Sec. II E (iii); App. A step 11; Sec. V; Fig. 5). Safeguard (iii) checks final claims "against *exact* dense complex diagonalization of $H + iA$ on a common grid, with eigenstates matched to the unperturbed basis by overlap"; App. A step 11 specifies "dense complex `eig` of $H + iA$ on $N = 2500$ ... eigenstate matching by overlap with the unperturbed basis". Because $H + iA$ is not Hermitian, it cannot use `eigh_tridiagonal`; a dense $\mathcal{O}(N^3)$ solve is needed. This plausibly explains why the exact check uses a smaller grid ($N = 2500$); the paper does not state its reason. Its eigenvalues come back unordered and complex, so they are labelled by the overlap matrix (8.82). Sec. V reports that at the fitted amplitude "3% of eigenstates can no longer be matched one-to-one to unperturbed states": precisely the failure that Section 8.10 showed is impossible for Hermitian matrices with overlaps above $\frac{1}{2}$ but possible for non-Hermitian ones. For the amplitude-controlled $R = 4$ fits Sec. V reports "all states matched". Fig. 5 (left) plots the exact level shifts obtained this way against the second-order surrogate.

Discrete spectrum and Hermiticity as the physical premise. Every operator fit in the paper (Tables I–III) scores eigenvalues of a real symmetric tridiagonal matrix. Its real, ordered, nondegenerate spectrum is what makes the comparison with the real, ordered zeros meaningful at all.

8.12 Summary

- An eigenpair satisfies $Av = \lambda v$, $v \neq 0$ (Eq. (8.5)); eigenvalues are the roots of the characteristic polynomial $\det(A - \lambda I)$ (Eq. (8.11)). For 2×2 matrices, $p_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A$ and $\lambda_1 + \lambda_2 = \operatorname{tr} A$, $\lambda_1 \lambda_2 = \det A$ (Eqs. (8.17), (8.18)).
- Real non-symmetric matrices can have complex eigenvalues (in conjugate pairs), non-orthogonal eigenvectors, or too few eigenvectors.
- Hermitian matrices ($A^\dagger = A$) have real eigenvalues (Theorem 1, Eq. (8.32)) and orthogonal eigenvectors for distinct eigenvalues (Theorem 2); both proved.
- Spectral theorem: $A = U\Lambda U^\dagger$ with U unitary (Eq. (8.40)), proved for distinct eigenvalues; $A = \sum_k \lambda_k u_k u_k^\dagger$ (Eq. (8.42)); the Rayleigh quotient lies between the extreme eigenvalues (Eq. (8.46)).
- Symmetric tridiagonal matrices with nonzero off-diagonals have distinct eigenvalues; D_N has eigenvalues $4\sin^2(k\pi/2(N+1))$ and sine eigenvectors (Eq. (8.57)).
- The second difference approximates f'' to $\mathcal{O}(h^2)$ (Eq. (8.63)); $H = -d^2/dx^2 + V$ with walls becomes the tridiagonal matrix of Eq. (8.66).
- The string/box problem $-y'' = \lambda y$, $y(\pm L) = 0$ has $\lambda_k = (k\pi/2L)^2$ (Eq. (8.74)); the grid reproduces it with relative error $\approx \frac{1}{3}(k\pi h/4L)^2$ (Table 8.1).
- LAPACK eigensolvers are $\mathcal{O}(n^3)$ for dense and $\mathcal{O}(n^2)$ for tridiagonal matrices; the Sturm count (Eq. (8.80)) counts eigenvalues below a number.
- Non-Hermitian eigenvalues have no natural order; match eigenvectors by the overlap matrix (Eq. (8.82)).

8.13 Problems

1. • Show that $(1, 1)^\top$ and $(1, -1)^\top$ are eigenvectors of $\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$ and find their eigenvalues. Check your answer with the trace.

2. • Find the eigenvalues of the triangular matrix $\begin{pmatrix} 1 & 4 & 5 \\ 0 & 3 & 6 \\ 0 & 0 & -2 \end{pmatrix}$ without expanding a cubic.

3. • Find the eigenvalues and eigenvectors of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which swaps the two components of a vector. Interpret the eigenvectors.

4. • A 2×2 matrix has trace 6 and determinant 8. Find its eigenvalues.

5. •• Prove Eq. (8.18) for a general 2×2 matrix and verify it for $B = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

6. •• For the real symmetric matrix $\begin{pmatrix} a & b \\ b & d \end{pmatrix}$, show that the two eigenvalues are equal only if $a = d$ and $b = 0$, i.e. only for a multiple of the identity.

7. •• Show that $\begin{pmatrix} 1 & 1+i \\ 1-i & 2 \end{pmatrix}$ is Hermitian, find its eigenvalues and eigenvectors, and verify that the eigenvectors are orthogonal.

8. •• Let $Av = \lambda v$. (a) Show that $(A + cI)v = (\lambda + c)v$ for any number c. (b) If A is invertible, show $\lambda \neq 0$ and $A^{-1}v = \lambda^{-1}v$. (c) What does (a) say about adding a constant to a potential?

9. •• Find the eigenvalues and eigenvectors of the rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and express them with Euler's formula. Check your result for $\theta = \pi/2$ against Section 8.3.

10. •• Find the eigenvalues and eigenvectors of $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ and compare with D_3.

11. •• Normalize the three eigenvectors of D_3 found in the Sample Problem, assemble them into a matrix U, and verify that $U^{\top}U = I$ and $U^{\top}D_3U = \mathrm{diag}(2-\sqrt{2}, 2, 2+\sqrt{2})$.

12. •• A guitar string has length $0.65\,\mathrm{m}$. Its fundamental frequency is f_1. Using Eq. (8.74) with $2L = 0.65\,\mathrm{m}$, show that the wavenumbers of the harmonics are $q_k = k\pi/(0.65\,\mathrm{m})$ and that the frequencies are $f_k = k f_1$. Which quantity is proportional to k^2?

13. •• For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ compute the Rayleigh quotient $x^{\top}Ax/x^{\top}x$ for $x = (1,0)^{\top}$ and $x = (1,2)^{\top}$, and check Eq. (8.46).

14. •• Show that a unitary matrix preserves inner products, $(Ux)^{\dagger}(Uy) = x^{\dagger}y$, and that its eigenvalues have modulus 1.

15. •• For $H(g) = \begin{pmatrix} 1 & ig \\ ig & 3 \end{pmatrix}$ with real $g \geq 0$, show that the eigenvalues are $2 \pm \sqrt{1-g^2}$. Evaluate them for $g = 0.5$, 1 and 1.5. Is $H(g)$ Hermitian? Symmetric?

16. •• The symmetric tridiagonal 4×4 matrix with diagonal $(2,2,2,2)$ and off-diagonal $(-1,0,-1)$ has a zero off-diagonal entry. Show that it consists of two copies of D_2, find its eigenvalues, and explain why the nondegeneracy theorem of Section 8.6 does not apply.

17. •• Using numpy and scipy, build D_{50} through its diagonal and off-diagonal, compute its eigenvalues with `eigh_tridiagonal`, and compare with $2-2\cos(k\pi/51)$. What is the largest absolute difference?

18. •• (Paper, App. A steps 3 and 9.) At scale 1 the paper uses $L = 44.8827$ and $N = 6000$. (a) Find the grid spacing h and the factor by which eigenvectors returned by `eigh_tridiagonal` must be multiplied to satisfy $\sum \psi^2\,dx = 1$. (b) Show that, with this normalization, the column $k = 0$ of the sensitivity matrix, $A_{n0} = \int |\psi_n|^2 \cdot 1\,dx$, equals 1 for every n, and interpret this using Problem 8(c).

19. •• Using `eigh_tridiagonal`, compute the first five eigenvalues of $H = -d^2/dx^2 + x^2$ on $[-10,10]$ with $N = 2000$ interior points. The exact values in units $\hbar = 2m = 1$ are $1, 3, 5, 7, 9$. How close do you get, and how does the error change with the level index? Explain.

20. •• (Paper, Sec. II E (i) and Sec. V.) Using Problem 8(a), show that adding iA with $A = c_0 \cdot 1$ (the constant mode) to any matrix shifts every eigenvalue by exactly ic_0 and does not change any eigenvector. Explain why an in-sample fit of real level shifts cannot be improved by changing c_0, and why a fitted value $c_0 \approx 2.2 \times 10^6$ (Sec. V) signals a numerical artifact rather than physics.

21. •• Using the Sturm count of Section 8.9 (or the formula (8.67)), find how many eigenvalues of the empty-box matrix D_N/h^2 with $L = 44.883$, $N = 6000$ lie below 1000. Compare with the number of exact eigenvalues $(k\pi/2L)^2$ below 1000, and explain the direction of the difference.

22. •• (Paper, App. A step 4.) Using `qmnt.wu_sprung_V` to sample the Wu–Sprung potential on the grid `qmnt.grid(L, N)` with $L = 44.883$, and `qmnt.fd_levels`, count the eigenvalues below 1000 for $N = 2000$, 4000 and 6000. Compare with the 652 states that the paper reports for its tensor computations, and with your answer to Problem 21. Why are there far fewer states below 1000 for the Wu–Sprung potential than for the empty box of the same width?

23. •• (Paper, Sec. II B and App. A step 11.) Taking the operation counts of a dense eigensolve and of a full tridiagonal eigensolve to be n^3 and n^2 (ignoring constants), compute both for $n = 6000$ and their ratio. Then compute n^3 for the dense complex check at $n = 2500$. Suggest why a smaller grid is a natural choice for the exact non-Hermitian check than for the Hermitian fits. (The paper does not state its reason.)

24. •• A non-Hermitian 3×3 matrix is diagonalized and its eigenvectors give the overlap matrix (rows k = reference states, columns m = new eigenvectors)

$$O = \begin{pmatrix} 0.70 & 0.55 & 0.05 \\ 0.25 & 0.40 & 0.15 \\ 0.05 & 0.05 & 0.80 \end{pmatrix}.$$

Check that the columns sum to one. Apply the matching rule. Is it one-to-one? Could this O come from a Hermitian matrix?

25. ••• (a) Show that for any real vector x, $x^{\top} D_N x = x_1^2 + x_N^2 + \sum_{j=1}^{N-1} (x_j - x_{j+1})^2$. (b) Deduce that every eigenvalue of D_N is positive. (c) Find a similar identity for $x^{\top}(4I - D_N)x$ and deduce that every eigenvalue is less than 4. Check (a) numerically for $x = (1, 2, -1, 3)$.

26. ••• Let A be a real $n \times n$ matrix. Prove that its characteristic polynomial has real coefficients and use this (instead of the argument in Section 8.3) to show that non-real eigenvalues come in conjugate pairs.

27. ••• For the avoided-crossing matrix (8.81), show that the overlap of the lower eigenvector with e_1 is $\tfrac{1}{2}\big(1 - \varepsilon/\sqrt{\varepsilon^2 + g^2}\big)$. Evaluate it for $g = 0.3$ and $\varepsilon = -2, 0, 0.5, 2$, and find the minimum distance between the two eigenvalues.

28. ••• Show that $\int_{-L}^{L} f\,(Hg)\,\mathrm{d}x = \int_{-L}^{L} (Hf)\,g\,\mathrm{d}x$ for $H = -d^2/dx^2 + V(x)$ with real V and real f, g vanishing at $\pm L$. Then prove that the eigenvalues of H are real, following Theorem 1 with integrals in place of sums (use complex f and $\int f^*(Hf)\,\mathrm{d}x$).

29. ••• Prove that A and $S^{-1}AS$ (with S invertible) have the same characteristic polynomial, hence the same eigenvalues. How are their eigenvectors related? (This justifies stage 1 of the LAPACK strategy.)

30. •• (Paper, Sec. II E (iii).) Explain, using the right panel of Figure 8.5, why sorting the eigenvalues of $H + iA$ by real part could mislabel levels, and why the paper matches

eigenstates "by overlap". For $H(g)$ of Problem 15 at $g = 1.5$, what are the real parts of the two eigenvalues?

31. ••• Using numpy, diagonalize $H = \mathrm{diag}(1,2,3) + 0.4i \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ with `np.linalg.eig`. (a) Show by hand that $\lambda = 2$ is an eigenvalue and that the other two are $2 \pm \sqrt{1 - 2g^2}$ with $g = 0.4$. (b) Match the eigenvectors to e_1, e_2, e_3 by overlap and report the largest overlap of each. (c) For which g do the eigenvalues become complex?

32. •• Show that the ratio of the top discrete eigenvalue $\lambda_N^{\mathrm{disc}}$ of D_N/h^2 to the exact $(N\pi/2L)^2$ tends to $4/\pi^2$ as $N \to \infty$. What does this say about the usefulness of the top half of a grid spectrum?

33. ••• Prove that $\sum_{j=1}^{N} \sin^2 \frac{jk\pi}{N+1} = \frac{N+1}{2}$ for $k = 1, \dots, N$. (Hint: use $\sin^2 a = \frac{1}{2}(1 - \cos 2a)$ and sum a geometric series of complex exponentials.) Show that this is consistent with $\int_{-L}^{L} \sin^2 \frac{k\pi(x+L)}{2L}\, \mathrm{d}x = L$ through the factor h.

8.14 Answers to selected problems

1. $\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} (1,1)^{\top} = (5,5)^{\top} = 5(1,1)^{\top}$, so $\lambda = 5$; $(1,-1)^{\top} \mapsto (3-2, 2-3)^{\top} = (1,-1)^{\top}$, so $\lambda = 1$. Trace check: $5 + 1 = 6 = 3 + 3$.

2. $A - \lambda I$ is triangular, so its determinant is the product of its diagonal entries, $(1-\lambda)(3-\lambda)(-2-\lambda)$. The eigenvalues are 1, 3 and -2.

3. $\mathrm{tr} = 0$, $\det = -1$, so $\lambda^2 - 1 = 0$ and $\lambda = \pm 1$. For $\lambda = 1$: $v = (1,1)^{\top}$ (swapping equal components changes nothing). For $\lambda = -1$: $v = (1,-1)^{\top}$ (swapping flips the sign). The eigenvectors are orthogonal, as the matrix is symmetric.

4. By Eq. (8.17), $\lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4) = 0$: $\lambda = 2$ and 4. Check: $2 + 4 = 6$, $2 \cdot 4 = 8$.

5. $p_A(\lambda) = \lambda^2 - (a+d)\lambda + (ad - bc)$ (Eq. (8.16)), and also $p_A(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$. Two polynomials equal for all λ have equal coefficients, giving Eq. (8.18). For B: $5 + 2 = 7 = \mathrm{tr}\, B$ and $5 \cdot 2 = 10 = \det B$.

7. $A_{21} = 1 - i = (1+i)^* = A_{12}^*$ and the diagonal is real, so $A^{\dagger} = A$. $\det(A - \lambda I) = (1-\lambda)(2-\lambda) - (1+i)(1-i) = \lambda^2 - 3\lambda + 2 - 2 = \lambda(\lambda - 3)$: $\lambda = 0, 3$. For $\lambda = 0$: $v_1 + (1+i)v_2 = 0$, $u = (-(1+i), 1)^{\top}$. For $\lambda = 3$: $-2v_1 + (1+i)v_2 = 0$, $w = (1+i, 2)^{\top}$. Then $u^{\dagger}w = (-(1-i))(1+i) + 1 \cdot 2 = -(1-i)(1+i) + 2 = -2 + 2 = 0$.

9. $\det(R - \lambda I) = (\cos\theta - \lambda)^2 + \sin^2\theta = 0$ gives $\cos\theta - \lambda = \pm i \sin\theta$, so $\lambda = \cos\theta \mp i\sin\theta = e^{\mp i\theta}$. For $\lambda = e^{i\theta}$: $R - \lambda I = \begin{pmatrix} -i\sin\theta & -\sin\theta \\ \sin\theta & -i\sin\theta \end{pmatrix}$, so (for $\sin\theta \neq 0$) $-iv_1 - v_2 = 0$ and $v = (1,-i)^{\top}$; for $e^{-i\theta}$, $v = (1,i)^{\top}$. At $\theta = \pi/2$: $\lambda = \pm i$ with $(1, \mp i)^{\top}$, matching R_{90}. Both eigenvalues have modulus 1: a rotation does not stretch.

11. Normalized: $u_1 = \frac{1}{2}(1, \sqrt{2}, 1)^{\top}$, $u_2 = \frac{1}{\sqrt{2}}(1, 0, -1)^{\top}$, $u_3 = \frac{1}{2}(1, -\sqrt{2}, 1)^{\top}$ (lengths: $\frac{1}{4}(1+2+1) = 1$, $\frac{1}{2}(1+1) = 1$). Pairwise dot products vanish (Sample Problem), so $U^{\top}U = I$. Since $D_3 u_k = \lambda_k u_k$, $U^{\top} D_3 U$ has entries $u_j^{\top} \lambda_k u_k = \lambda_k \delta_{jk}$, i.e. $\mathrm{diag}(2 - \sqrt{2}, 2, 2 + \sqrt{2})$. A short numpy computation of $U^{\top}U$ and $U^{\top}D_3U$ confirms both results to rounding error.

13. For $(1,0)$: $x^{\top}Ax = A_{11} = 2$, $x^{\top}x = 1$, quotient 2. For $(1,2)$: $Ax = (4,5)^{\top}$, $x^{\top}Ax = 4 + 10 = 14$, $x^{\top}x = 5$, quotient 2.8. Both lie in $[1,3]$, as Eq. (8.46) requires.

15. $\mathrm{tr} = 4$, $\det = 3 - (ig)^2 = 3 + g^2$, so $\lambda = \frac{1}{2}(4 \pm \sqrt{16 - 4(3+g^2)}) = 2 \pm \sqrt{1 - g^2}$. $g = 0.5$: $2 \pm 0.866025 = 1.133975,\ 2.866025$. $g = 1$: double eigenvalue 2. $g = 1.5$: $2 \pm \sqrt{-1.25} = 2 \pm 1.118034\, i$. $H(g)$ is symmetric ($H^{\top} = H$) but not Hermitian ($H^{\dagger}$ has $-ig$ off the diagonal) for $g \neq 0$.

17. With `d = 2*np.ones(50)`, `e = -np.ones(49)`, `w = eigh_tridiagonal(d, e, eigvals_only=True)` and `k = np.arange(1, 51)`, the largest $|w - (2 - 2\cos(k\pi/51))|$ is below 10^{-13}: rounding level, as the box in Section 8.9 predicts for eigenvalues of size at most 4.

19. Build $d_j = 2/h^2 + x_j^2$, $e_j = -1/h^2$ with $h = 20/2001$ and take the first five eigenvalues. All five agree with $1, 3, 5, 7, 9$ to within 5×10^{-4}, and the discrepancy grows with the level index. Reason: by Eq. (8.63) the error is controlled by $h^2\psi''''$, and higher states oscillate faster (larger local wavenumber), so ψ'''' is larger. The walls at ± 10 are far beyond the turning points ($x^2 = E \le 9$, $|x| \le 3$), so they do not matter.

21. The exact count is the number of k with $k\pi/(2L) < \sqrt{1000}$, i.e. $k < 2L\sqrt{1000}/\pi = 903.6$: 903 levels. For the matrix, $\lambda_k^{\text{disc}} < 1000$ means $\sin(k\pi/12002) < \frac{h}{2}\sqrt{1000} = 0.23651$, i.e. $k < 12002 \arcsin(0.23651)/\pi = 912.2$: 912 levels. The Sturm count gives the same. There are more matrix levels below 1000 because the matrix eigenvalues lie *below* the exact ones (Table 8.1): every level is pulled down, so more of them fit under the cutoff.

23. $6000^3 = 2.16 \times 10^{11}$, $6000^2 = 3.6 \times 10^7$, ratio 6000. $2500^3 = 1.5625 \times 10^{10}$. The Hermitian fits can use the cheap tridiagonal solver on the large grid; the non-Hermitian check $H + iA$ is not Hermitian, so it needs a dense complex solve whose cost grows like n^3, and a smaller grid keeps it affordable. (This is our inference; the paper gives no reason.)

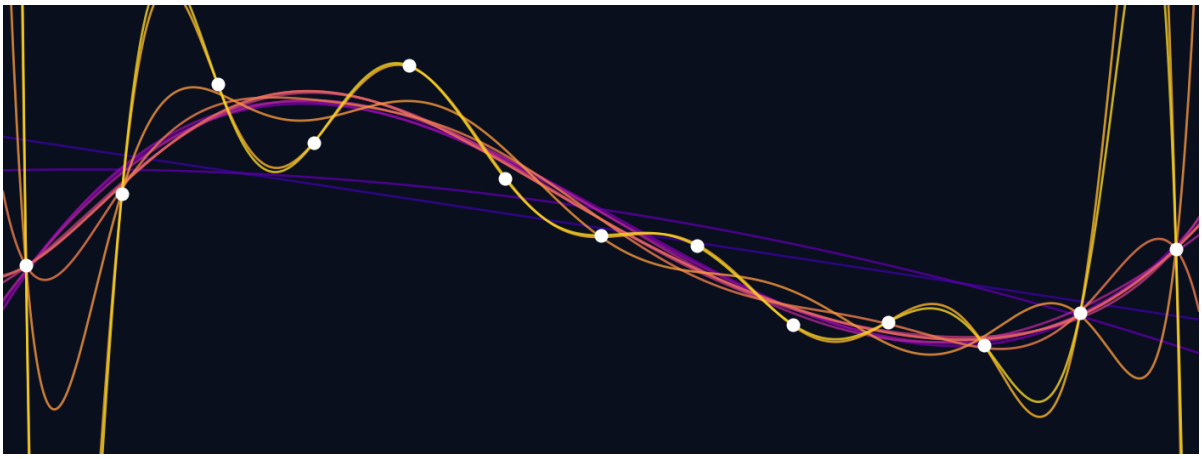
25. (a) $x^\top D_N x = 2\sum_{j=1}^{N} x_j^2 - 2\sum_{j=1}^{N-1} x_j x_{j+1}$. Expanding $\sum_{j=1}^{N-1}(x_j - x_{j+1})^2 = \sum_{j=1}^{N-1} x_j^2 + \sum_{j=2}^{N} x_j^2 - 2\sum_{j=1}^{N-1} x_j x_{j+1}$ and adding $x_1^2 + x_N^2$ gives $2\sum_{j=1}^{N} x_j^2 - 2\sum x_j x_{j+1}$, the same. For $x = (1, 2, -1, 3)$: $D_4 x = (0, 4, -7, 7)^\top$, $x^\top D_4 x = 0 + 8 + 7 + 21 = 36$; the identity gives $1 + 9 + (1 + 9 + 16) = 36$. (b) The right side is a sum of squares, zero only if $x_1 = 0$ and all consecutive differences vanish, i.e. $x = 0$. So for $x \neq 0$ the Rayleigh quotient is positive, and taking x an eigenvector, $\lambda > 0$. (c) $x^\top(4I - D_N)x = 2\sum x_j^2 + 2\sum x_j x_{j+1} = x_1^2 + x_N^2 + \sum_{j=1}^{N-1}(x_j + x_{j+1})^2 > 0$ for $x \neq 0$, so $4 - \lambda > 0$.

27. The lower eigenvalue is $-r$ with $r = \sqrt{\varepsilon^2 + g^2}$. Row 1 of $(H + rI)v = 0$: $(\varepsilon + r)v_1 + g v_2 = 0$, so $v = (g, -(\varepsilon + r))^\top$. Then $\|v\|^2 = g^2 + (\varepsilon + r)^2 = g^2 + \varepsilon^2 + 2\varepsilon r + r^2 = 2r^2 + 2\varepsilon r = 2r(r + \varepsilon)$, and $g^2 = r^2 - \varepsilon^2 = (r - \varepsilon)(r + \varepsilon)$. The overlap is $g^2/\|v\|^2 = (r - \varepsilon)/(2r) = \frac{1}{2}(1 - \varepsilon/r)$. For $g = 0.3$: $\varepsilon = -2$: 0.9945; $\varepsilon = 0$: 0.5; $\varepsilon = 0.5$: 0.0713; $\varepsilon = 2$: 0.0055. Minimum distance $2r$ at $\varepsilon = 0$: $2g = 0.6$.

29. Using $\det(XY) = \det X \det Y$ and $S^{-1}IS = I$: $\det(S^{-1}AS - \lambda I) = \det(S^{-1}(A - \lambda I)S) = \det(S^{-1}) \det(A - \lambda I) \det S = \det(A - \lambda I)$, since $\det(S^{-1}) \det S = \det(S^{-1}S) = 1$. If $Av = \lambda v$, then $(S^{-1}AS)(S^{-1}v) = S^{-1}Av = \lambda S^{-1}v$: the eigenvectors of $S^{-1}AS$ are $S^{-1}v$. You can test this numerically: for random 4×4 matrices A and S, `np.linalg.eigvals` returns the same eigenvalues for A and for $S^{-1}AS$, up to rounding.

31. (a) Write $g = 0.4$. Expanding $\det(H - 2I)$ along the first row gives $-1 \cdot (0 \cdot 1 - (ig)^2) - ig\,(ig \cdot 1 - ig \cdot 0) = -g^2 + g^2 = 0$, so $\lambda = 2$. The trace is 6, so the others are $2 \pm d$; $\det H = 1 \cdot (6 - (ig)^2) - ig \cdot (3ig) = 6 + g^2 + 3g^2 = 6 + 4g^2$ must equal $2(2 + d)(2 - d) = 2(4 - d^2)$, so $d^2 = 1 - 2g^2 = 0.68$ and $\lambda = 1.175379,\ 2,\ 2.824621$. (b) The eigenvector for $\lambda = 2$ is $(ig, 1, -ig)$, overlap with e_2 equal to $1/(1 + 2g^2) = 0.758$; the other two have largest overlaps 0.832 with e_1 and e_3. The matching is one-to-one. (c) Complex when $1 - 2g^2 < 0$, i.e. $g > 1/\sqrt{2} \approx 0.707$.

33. $\sum_{j=1}^{N} \sin^2(ja) = \frac{1}{2}(N - \sum_{j=1}^{N} \cos(2ja))$ with $a = k\pi/(N+1)$. Let $z = e^{2ia} = e^{2\pi ik/(N+1)}$; since $1 \le k \le N$, $z \neq 1$ but $z^{N+1} = 1$. Then $\sum_{j=0}^{N} z^j = (1 - z^{N+1})/(1 - z) = 0$ (finite geometric sum: multiplying $S = \sum_{j=0}^{N} z^j$ by $1 - z$ cancels all middle terms and leaves $1 - z^{N+1}$); its real part gives $\sum_{j=0}^{N} \cos(2ja) = 0$, so $\sum_{j=1}^{N} \cos(2ja) = -1$ (removing the $j = 0$ term, which is 1). Hence the sum is $\frac{1}{2}(N + 1)$. Multiplying by $h = 2L/(N + 1)$ gives L, exactly the integral $\int_{-L}^{L} \sin^2(k\pi(x + L)/2L)\,dx = L$: the grid sum $\sum v_j^2 h$ reproduces the continuum normalization integral.



Polynomials of degree 1 to 12 fitted by least squares to the same 13 noisy points: more freedom, wilder curves.

Chapter 9

Least Squares, the SVD and Regularization

↑ Contents

What this chapter is about

In your first physics laboratory you probably plotted a handful of measured points and drew "the best straight line" through them with a ruler. This chapter turns that ruler into mathematics. We ask what "best" should mean, derive the line that is best in the sense of least squares, and then generalize: from two coefficients (slope and intercept) to hundreds, and from straight lines to any model that is a sum of fixed functions multiplied by unknown coefficients. Along the way we meet the most useful single tool in numerical linear algebra, the singular value decomposition (SVD). It tells you exactly which combinations of coefficients your data can determine, which ones they determine badly, and which ones they cannot see at all.

The SVD also shows why fitting can go wrong. Small singular values amplify noise; problems with more coefficients than data points have infinitely many perfect fits; and a perfect fit to the data you have says nothing, by itself, about data you do not have. The cure for the first two problems is regularization (truncated SVD and ridge regression), which we derive in full. The third problem, memorization, appears here in its simplest form, a polynomial threaded through noisy points, and is taken up properly in Chapter 25.

Where this shows up in the paper: every iteration of the paper's potential fit is a damped least-squares problem solved with the SVD (Sec. II C and Appendix A step 6 of [1]); the damping formula $\sigma/(\sigma^2 + \lambda\sigma_1^2)$ in that step is the ridge filter of Section 9.8; the ridge rows of Tables I and II and the ridge-regularized interpolation fits of Sec. III are ridge regression; and the sinusoidal controls B and C1 of Secs. II D and IV are linear least-squares fits, one of which the paper calls "underdetermined" at 900 coefficients on 400 zeros.

What you need

- Chapter 7: vectors in n dimensions, dot (inner) product, norm, orthogonality, orthonormal bases, matrices and matrix multiplication, transpose, identity and inverse matrices, rank, solving $Ax = b$, sampling functions on a grid.
- Chapter 8: eigenvalues and eigenvectors; real symmetric matrices have real eigenvalues and an orthonormal basis of eigenvectors (the spectral theorem); diagonalization.
- Recalled briefly where used, not required: relative error and machine epsilon (2.2×10^{-16} in double precision, the relative precision of a double-precision floating-point number; Chapter 4); the cosine functions $\cos(k\pi x/L)$ on $[-L, L]$ and the prime-2 wave RE2 (Chapter 5), whose formula is restated in Equation (9.38).
- First-year physics and calculus: derivatives, the condition $f'(x) = 0$ at a smooth minimum, plotting data, the idea of measurement error.

9.1 Fitting a straight line

Suppose a cart rolls along a track and you record its position at four times:

Table 9.1: Positions of a cart (data for the first Sample Problem of this chapter, "The best line for the rolling cart").

time t_i (s)	0	1	2	3
position y_i (m)	1	3	4	4

You believe the cart moves at roughly constant velocity, so you want a model $y \approx c_0 + c_1 t$, where c_0 is the starting position (the intercept) and c_1 the velocity (the slope). No straight line passes through all four points. We need a rule that picks one line.

Residuals. For a trial line with coefficients (c_0, c_1), the *residual* of point i is the measured value minus the model value,

$$r_i = y_i - (c_0 + c_1 t_i), \qquad i = 1, \ldots, m, \tag{9.1}$$

where m is the number of data points ($m = 4$ here). A residual is the vertical distance from the point to the line, positive if the point lies above. A good line has small residuals. But there are m residuals and we want a single number to minimize.

Why squares? The least-squares rule chooses the line that minimizes the *sum of squared residuals*

$$S(c_0, c_1) = \sum_{i=1}^{m} r_i^2 = \sum_{i=1}^{m} (y_i - c_0 - c_1 t_i)^2. \tag{9.2}$$

Four reasons make squares the standard choice. (1) Squares are never negative, so positive and negative residuals cannot cancel. (2) S is a smooth function of the coefficients, so calculus finds its minimum; the sum of absolute values $\sum |r_i|$ has corners where some $r_i = 0$, and derivatives fail there. (3) Squaring penalizes a residual of 2 four times as much as a residual of 1, so the rule works hard to avoid large misses. (4) When measurement errors follow the Gaussian (bell-curve) distribution, least squares picks the coefficients under which the observed data are most probable (see the box below).

Taken on trust

The statistical justification of least squares. Gauss showed that if the errors in y_i are independent and Gaussian with a common width, the coefficients that minimize S are the ones that make the observed data most probable (maximum likelihood), and the Gauss–Markov theorem states that among all unbiased estimators that are linear in the data, least squares has the smallest variance. We do not prove these statements here; they need the probability of Chapter 19. Proofs are in [31]. Nothing in this chapter depends on them.

Functions of two variables and partial derivatives. S depends on two numbers, c_0 and c_1. The *partial derivative* $\partial S/\partial c_0$ is the ordinary derivative of S with respect to c_0 while c_1 is held fixed (treated as a constant); $\partial S/\partial c_1$ is defined the same way with the roles swapped. At a minimum of a smooth function of two variables, both partial derivatives vanish. The reason is a one-variable argument you already know: if $(\hat{c}_0, \hat{c}_1)$ minimizes S, then the one-variable function $g(c_0) = S(c_0, \hat{c}_1)$ has a minimum at $c_0 = \hat{c}_0$, so $g'(\hat{c}_0) = 0$, and $g'(\hat{c}_0)$ is exactly $\partial S/\partial c_0$ at the minimum. The same holds for c_1. (Vanishing partial derivatives are necessary for a minimum; we prove in Section 9.2 that for least squares they are also sufficient.)

Deriving the normal equations. Differentiate Equation (9.2) term by term with the chain rule. For c_0, the derivative of $(y_i - c_0 - c_1 t_i)^2$ is $2(y_i - c_0 - c_1 t_i) \times (-1)$:

$$\frac{\partial S}{\partial c_0} = \sum_{i=1}^{m} 2\,(y_i - c_0 - c_1 t_i)\,(-1) = -2\sum_{i=1}^{m} r_i, \tag{9.3}$$

$$\frac{\partial S}{\partial c_1} = \sum_{i=1}^{m} 2\,(y_i - c_0 - c_1 t_i)\,(-t_i) = -2\sum_{i=1}^{m} t_i r_i. \tag{9.4}$$

Setting both to zero gives two conditions on the residuals of the best line:

$$\sum_{i=1}^{m} r_i = 0, \qquad \sum_{i=1}^{m} t_i r_i = 0. \tag{9.5}$$

Now expand each sum using Equation (9.1). The first condition reads $\sum_i y_i - \sum_i c_0 - \sum_i c_1 t_i = 0$. Since c_0 and c_1 do not depend on i, $\sum_i c_0 = m c_0$ and $\sum_i c_1 t_i = c_1 \sum_i t_i$:

$$m\,c_0 + \Big(\sum_i t_i\Big) c_1 = \sum_i y_i. \tag{9.6}$$

The second condition reads $\sum_i t_i y_i - c_0 \sum_i t_i - c_1 \sum_i t_i^2 = 0$, that is

$$\Big(\sum_i t_i\Big) c_0 + \Big(\sum_i t_i^2\Big) c_1 = \sum_i t_i y_i. \tag{9.7}$$

Equations (9.6) and (9.7) are the *normal equations* for the straight line: two linear equations for two unknowns. To solve them, multiply Equation (9.6) by $\sum_i t_i$ and Equation (9.7) by m:

$$m\Big(\sum t_i\Big) c_0 + \Big(\sum t_i\Big)^2 c_1 = \Big(\sum t_i\Big)\Big(\sum y_i\Big),$$
$$m\Big(\sum t_i\Big) c_0 + m\Big(\sum t_i^2\Big) c_1 = m \sum t_i y_i.$$

Subtract the first line from the second; the c_0 terms cancel:

$$c_1 = \frac{m \sum t_i y_i - (\sum t_i)(\sum y_i)}{m \sum t_i^2 - (\sum t_i)^2}, \qquad c_0 = \frac{\sum y_i - c_1 \sum t_i}{m}. \tag{9.8}$$

The second formula is Equation (9.6) solved for c_0. Splitting its fraction term by term shows $c_0 = \bar{y} - c_1\bar{t}$, where $\bar{t} = \frac{1}{m}\sum t_i$ and $\bar{y} = \frac{1}{m}\sum y_i$ are the averages: the best line always passes through the point of averages $(\bar{t}, \bar{y})$.

When is the denominator safe? The denominator in Equation (9.8) equals $m\sum_i (t_i - \bar{t})^2$. To see this, expand the square:

$$\begin{aligned}\sum_i (t_i - \bar{t})^2 &= \sum_i t_i^2 - 2\bar{t}\sum_i t_i + m\bar{t}^2 \\ &= \sum_i t_i^2 - 2\Big(\frac{1}{m}\sum_i t_i\Big)\sum_i t_i + m\Big(\frac{1}{m}\sum_i t_i\Big)^2 \quad \text{(substitute } \bar{t} = \tfrac{1}{m}\textstyle\sum t_i\text{)} \\ &= \sum_i t_i^2 - 2\frac{(\sum t_i)^2}{m} + \frac{(\sum t_i)^2}{m} \quad \text{(since } m\cdot\tfrac{1}{m^2} = \tfrac{1}{m}\text{)} \\ &= \sum_i t_i^2 - \frac{(\sum t_i)^2}{m}.\end{aligned}$$

Multiplying by m gives the denominator. A sum of squares is zero only if every term is zero, so the denominator vanishes only if all the t_i are equal. That makes sense: if every measurement is taken at the same time, the data say nothing about the slope.

Sample Problem 9.1: The best line for the rolling cart

Find the least-squares line $y = c_0 + c_1 t$ for the data of Table 9.1, its residuals, and the minimum value of S.

Step 1: sums. With $m = 4$: $\sum t_i = 0+1+2+3 = 6$; $\sum t_i^2 = 0+1+4+9 = 14$; $\sum y_i = 1+3+4+4 = 12$; $\sum t_i y_i = 0+3+8+12 = 23$.

Step 2: normal equations. From Equations (9.6) and (9.7): $4c_0 + 6c_1 = 12$ and $6c_0 + 14c_1 = 23$.

Step 3: solve. The first equation gives $c_0 = 3 - 1.5c_1$. Substituting into the second: $6(3 - 1.5c_1) + 14c_1 = 23$, so $18 + 5c_1 = 23$ and $c_1 = 1$. Then $c_0 = 3 - 1.5 = 1.5$.

Step 4: residuals. The line predicts $1.5, 2.5, 3.5, 4.5$ at $t = 0, 1, 2, 3$, so $r = (1-1.5,\ 3-2.5,\ 4-3.5,\ 4-4.5) = (-0.5,\ 0.5,\ 0.5,\ -0.5)$ m.

Step 5: check Equation (9.5). $\sum r_i = 0$ and $\sum t_i r_i = 0 + 0.5 + 1.0 - 1.5 = 0$. Both hold.

Step 6: minimum value. $S = 4 \times 0.25 = 1.0\ \text{m}^2$.

$$\boxed{y = 1.5\ \text{m} + (1.0\ \text{m/s})\,t, \qquad S_{\min} = 1.0\ \text{m}^2}$$

Pitfall

Least squares as set up here measures *vertical* distances, because the model treats t as exact and y as the noisy quantity. If you swap the roles (fit t as a function of y), you get a different line in general. Decide first which variable carries the error.

Checkpoint 9.1

You add 2 m to every measured position in Table 9.1 (you moved the origin). What happens to the best-fit slope and intercept?

Answer: The slope is unchanged and the intercept increases by 2 m. In Equation (9.8), $\sum y_i$ grows by $2m$ and $\sum t_i y_i$ by $2\sum t_i$, so the numerator of c_1 changes by $2m\sum t_i - 2m\sum t_i = 0$; then $c_0 = \bar{y} - c_1\bar{t}$ grows by 2.

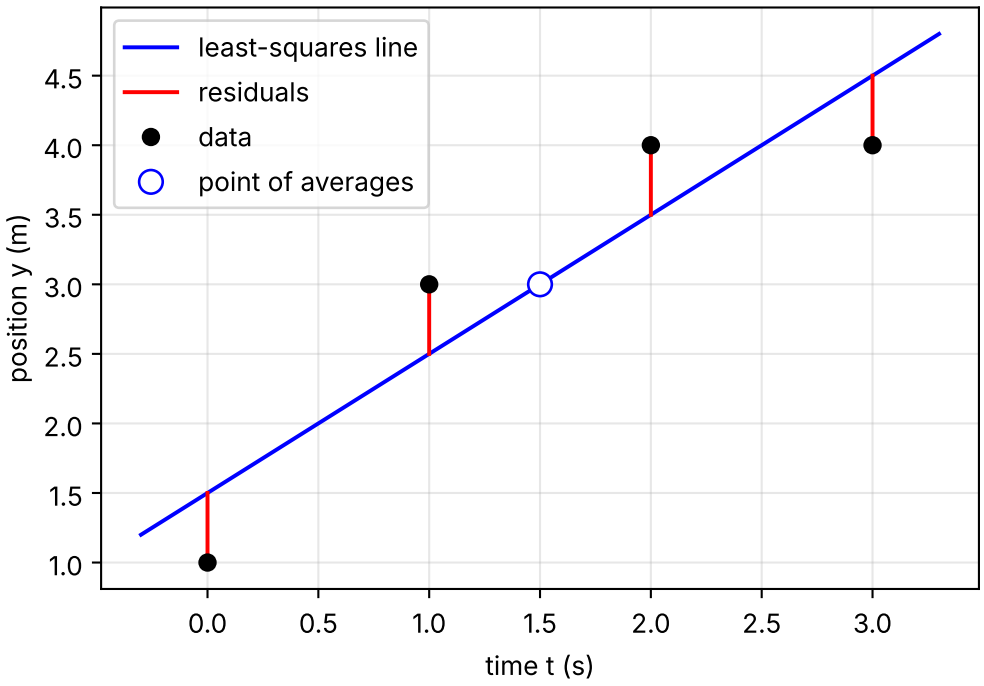


Figure 9.1: Least-squares line through the four cart positions of Table 9.1. The vertical segments are the residuals r_i; least squares minimizes the sum of their squares. The line passes through the point of averages $(\bar{t}, \bar{y}) = (1.5\ \text{s}, 3\ \text{m})$ (open circle).

Key Ideas

- A residual is data minus model; least squares minimizes the sum of squared residuals S.
- Setting the partial derivatives of S to zero gives the normal equations (Equations (9.6) and (9.7)), which say $\sum r_i = 0$ and $\sum t_i r_i = 0$.
- The best line passes through the point of averages; it is unique unless all t_i coincide.

9.2 Linear least squares in general

★ Did you know? *Finding a Lost Planet*

On 1 January 1801 Giuseppe Piazzi discovered Ceres. He tracked it for only about six weeks before it was lost in the Sun's glare. The 24-year-old Carl Friedrich Gauss used those few observations to predict where it would reappear, and astronomers found it close to his prediction around New Year 1802. Adrien-Marie Legendre was the first to publish least squares, in 1805.

Linear models. The straight line is the simplest case of a *linear model*

$$y(t) \approx \sum_{k=1}^{M} c_k\, \phi_k(t), \tag{9.9}$$

where the $\phi_k(t)$ are fixed, known *basis functions* and the c_k are M unknown coefficients. "Linear" refers to the coefficients, not to t: the model is a sum of the c_k each multiplied by something

that does not depend on any c. Examples:

- the line: $M = 2$, $\phi_1 = 1$, $\phi_2 = t$;
- polynomials of degree d: $\phi_k = t^{k-1}$, $M = d + 1$;
- sinusoids with chosen angular frequencies ω_j: $\phi = \sin(\omega_j t)$ and $\cos(\omega_j t)$ (Chapter 5);
- the cosine basis $\cos(k\pi x/L)$ on $[-L, L]$ (Chapter 5), the paper's default basis for its potential correction $\delta V(x)$.

A model such as $y = c_1 \sin(c_2 t)$ is *not* linear in c_2; fitting it is a nonlinear problem (Chapter 24). The paper's potential fit is nonlinear too, but it is solved by a sequence of linear problems of exactly the kind treated here (Section 9.10).

(In this chapter the coefficient index k runs from 1 to M. For polynomials we will sometimes label coefficients from 0 for convenience, and the paper labels its cosine coefficients $c_0, \ldots, c_{M-1}$. Only the counting convention differs.)

The design matrix. We have data (t_i, y_i) for $i = 1, \ldots, m$. Collect the data values into a vector $y = (y_1, \ldots, y_m)$ and the coefficients into $c = (c_1, \ldots, c_M)$. Define the $m \times M$ *design matrix* A with entries

$$A_{ik} = \phi_k(t_i), \tag{9.10}$$

so row i holds all basis functions evaluated at the ith data point and column k holds the kth basis function sampled at all data points ("functions as vectors", Chapter 7). Then the model values at the data points are the components of the matrix-vector product Ac, because $(Ac)_i = \sum_k A_{ik} c_k = \sum_k c_k \phi_k(t_i)$. The *residual vector* is

$$r = y - Ac, \tag{9.11}$$

and the least-squares problem is to minimize

$$S(c) = \|r\|^2 = r^\top r = \sum_{i=1}^{m} \Big(y_i - \sum_{k=1}^{M} A_{ik} c_k \Big)^2. \tag{9.12}$$

Recall from Chapter 7 that $\|r\|$ is the length of the vector r, $r^\top$ is its transpose (the vector written as a row), and $r^\top r$ is the dot product of r with itself.

For the straight line, A has a column of ones and a column of the t_i:

$$A = \begin{pmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{pmatrix}.$$

The general normal equations. Take the partial derivative of Equation (9.12) with respect to one coefficient c_ℓ, holding the others fixed. Only the inner sum depends on c_ℓ, and $\partial(\sum_k A_{ik} c_k)/\partial c_\ell = A_{i\ell}$ (every term with $k \neq \ell$ is a constant). By the chain rule,

$$\begin{aligned} \frac{\partial S}{\partial c_\ell} &= \sum_{i=1}^{m} 2\Big(y_i - \sum_k A_{ik} c_k \Big)\,(-A_{i\ell}) \\ &= -2 \sum_{i=1}^{m} A_{i\ell}\, r_i \\ &= -2\,(A^\top r)_\ell. \quad \text{(definition of } A^\top\text{: } (A^\top)_{\ell i} = A_{i\ell}) \end{aligned} \tag{9.13}$$

Setting all M partial derivatives to zero gives the vector equation $A^\top r = 0$. Substituting $r = y - Ac$ and moving the c term to the left,

$$\boxed{A^\top A\, c = A^\top y} \tag{9.14}$$

These are the *normal equations*: M linear equations for the M coefficients. The matrix $A^\top A$ is $M \times M$ with entries $(A^\top A)_{k\ell} = \sum_i \phi_k(t_i)\phi_\ell(t_i)$. For the line, $A^\top A = \begin{pmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{pmatrix}$ and $A^\top y = (\sum y_i,\ \sum t_i y_i)$, which reproduces Equations (9.6) and (9.7).

The solution really is a minimum. A solution always exists, even when the columns of A are dependent: in Section 9.5 we construct one, c^+, and check that it satisfies the normal equations. Let $\hat{c}$ be any solution of Equation (9.14) and $\hat{r} = y - A\hat{c}$ its residual, so $A^\top \hat{r} = 0$. For any other coefficient vector c,

$$y - Ac = (y - A\hat{c}) + (A\hat{c} - Ac) = \hat{r} + A(\hat{c} - c).$$

Take the squared length. For any vectors a, b, $\|a + b\|^2 = \|a\|^2 + 2a^\top b + \|b\|^2$ (expand $(a + b)^\top (a + b)$). Hence

$$\begin{aligned} S(c) &= \|\hat{r}\|^2 + 2\left(A(\hat{c} - c)\right)^\top \hat{r} + \|A(\hat{c} - c)\|^2 \\ &= \|\hat{r}\|^2 + 2\,(\hat{c} - c)^\top A^\top \hat{r} + \|A(\hat{c} - c)\|^2 \quad (\text{since } (AB)^\top = B^\top A^\top) \\ &= \|\hat{r}\|^2 + \|A(\hat{c} - c)\|^2 \quad (\text{since } A^\top \hat{r} = 0). \end{aligned}$$

So

$$S(c) = S(\hat{c}) + \|A(\hat{c} - c)\|^2 \ \geq\ S(\hat{c}). \tag{9.15}$$

No coefficient vector does better than a solution of the normal equations. Equality holds exactly when $A(\hat{c} - c) = 0$.

Uniqueness. The *null space* of A is the set of vectors z with $Az = 0$. By Equation (9.15), the minimizer is unique precisely when the null space contains only $z = 0$, which is the statement that the columns of A are linearly independent (no nontrivial combination $\sum_k z_k a_k$ of the columns a_k vanishes), that is, $\operatorname{rank}(A) = M$ (Chapter 7). In that case $A^\top A$ is invertible. Proof: suppose $A^\top A z = 0$. Multiply on the left by $z^\top$: $0 = z^\top A^\top A z = (Az)^\top (Az) = \|Az\|^2$, so $Az = 0$, so $z = 0$ by independence. A square matrix whose only null vector is zero is invertible (Chapter 7).

The same calculation shows that for any z, $z^\top A^\top A z = \|Az\|^2 \geq 0$. A symmetric matrix with this property is called *positive semidefinite*. ($A^\top A$ is symmetric because $(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$.)

Counting equations and unknowns. The data give m "equations" $y_i \approx (Ac)_i$ for M unknowns.

- $m > M$, *overdetermined*: the usual laboratory case. In general no c makes $r = 0$; least squares picks the best compromise.
- $m = M$ with independent columns: A is square and invertible, $c = A^{-1} y$ fits every point exactly, $S = 0$.
- $m < M$, *underdetermined*: the rank of A is at most $m < M$, so the columns cannot be independent and the minimizer is not unique. We will see in Section 9.5 that there are then typically infinitely many exact fits.

Checkpoint 9.2

You fit a parabola $c_1 + c_2 t + c_3 t^2$ to 7 points. What are the sizes of A, $A^\top A$ and $A^\top y$?

Answer: A is 7×3; $A^\top A$ is 3×3; $A^\top y$ has 3 components.

Key Ideas

- A linear model $\sum_k c_k \phi_k(t)$ evaluated at the data points is Ac with design matrix $A_{ik} = \phi_k(t_i)$.
- Least squares gives the normal equations $A^\top A c = A^\top y$, equivalently $A^\top r = 0$ (Equation (9.14)).
- Every solution of the normal equations is a global minimum (Equation (9.15)); it is unique if and only if the columns of A are independent.

9.3 The geometry of least squares: projection

The condition $A^\top r = 0$ has a picture. Write the columns of A as vectors $a_1, \dots, a_M$ in m-dimensional space. Then $Ac = \sum_k c_k a_k$ is a combination of the columns. The set of all such combinations, for all possible c, is the *column space* of A: a line, a plane, or a higher-dimensional flat subspace through the origin. The data vector y generally does not lie in it (that is why no model fits exactly).

Component k of $A^\top r$ is the dot product $a_k^\top r$. So $A^\top r = 0$ says: *the residual is perpendicular to every column, and therefore to the whole column space.* The best model vector $\hat{y} = A\hat{c}$ is the foot of the perpendicular dropped from y onto the column space, the *orthogonal projection* of y. Think of a stick held above a table under the noon sun: its shadow is its projection onto the table, and the light rays (the residual) are perpendicular to the table. Equation (9.15) is Pythagoras' theorem: moving the model point away from the foot of the perpendicular, by $A(\hat{c} - c)$ within the plane, can only lengthen the distance to y.

Because $\hat{y}^\top \hat{r} = \hat{c}^\top A^\top \hat{r} = 0$, the data vector splits into two perpendicular pieces, and Pythagoras gives

$$\|y\|^2 = \|\hat{y}\|^2 + \|\hat{r}\|^2 . \tag{9.16}$$

The projection matrix. When the columns are independent, $\hat{c} = (A^\top A)^{-1} A^\top y$, so

$$\hat{y} = Py, \qquad P = A(A^\top A)^{-1} A^\top . \tag{9.17}$$

P is an $m \times m$ matrix that sends any vector to its projection. Projecting twice changes nothing more, and the algebra confirms it: $P^2 = A(A^\top A)^{-1}(A^\top A)(A^\top A)^{-1}A^\top = A(A^\top A)^{-1}A^\top = P$.

Sample Problem 9.2: Projecting a vector onto a plane

Let A have columns $a_1 = (1, 0, 1)$ and $a_2 = (0, 1, 1)$, and let $y = (1, 2, 0)$. Find the least-squares coefficients, the projection $\hat{y}$, the residual, and check Equation (9.16).

Step 1: normal-equation matrix. $A^\top A = \begin{pmatrix} a_1^\top a_1 & a_1^\top a_2 \\ a_2^\top a_1 & a_2^\top a_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Step 2: right-hand side. $A^\top y = (a_1^\top y,\ a_2^\top y) = (1 + 0 + 0,\ 0 + 2 + 0) = (1, 2)$.

Step 3: solve. The determinant of $A^\top A$ is $2 \cdot 2 - 1 \cdot 1 = 3$, so $(A^\top A)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ and $\hat{c} = \frac{1}{3}(2 \cdot 1 - 1 \cdot 2,\ -1 \cdot 1 + 2 \cdot 2) = (0, 1)$.

Step 4: projection and residual. $\hat{y} = 0\,a_1 + 1\,a_2 = (0,1,1)$, and $\hat{r} = y - \hat{y} = (1,1,-1)$.
Step 5: perpendicularity. $a_1^\top \hat{r} = 1 + 0 - 1 = 0$ and $a_2^\top \hat{r} = 0 + 1 - 1 = 0$.
Step 6: Pythagoras. $\|y\|^2 = 1 + 4 + 0 = 5$; $\|\hat{y}\|^2 = 0 + 1 + 1 = 2$; $\|\hat{r}\|^2 = 1 + 1 + 1 = 3$; and $2 + 3 = 5$.

$$\hat{c} = (0,1), \quad \hat{y} = (0,1,1), \quad \hat{r} = (1,1,-1), \quad S_{\min} = 3$$

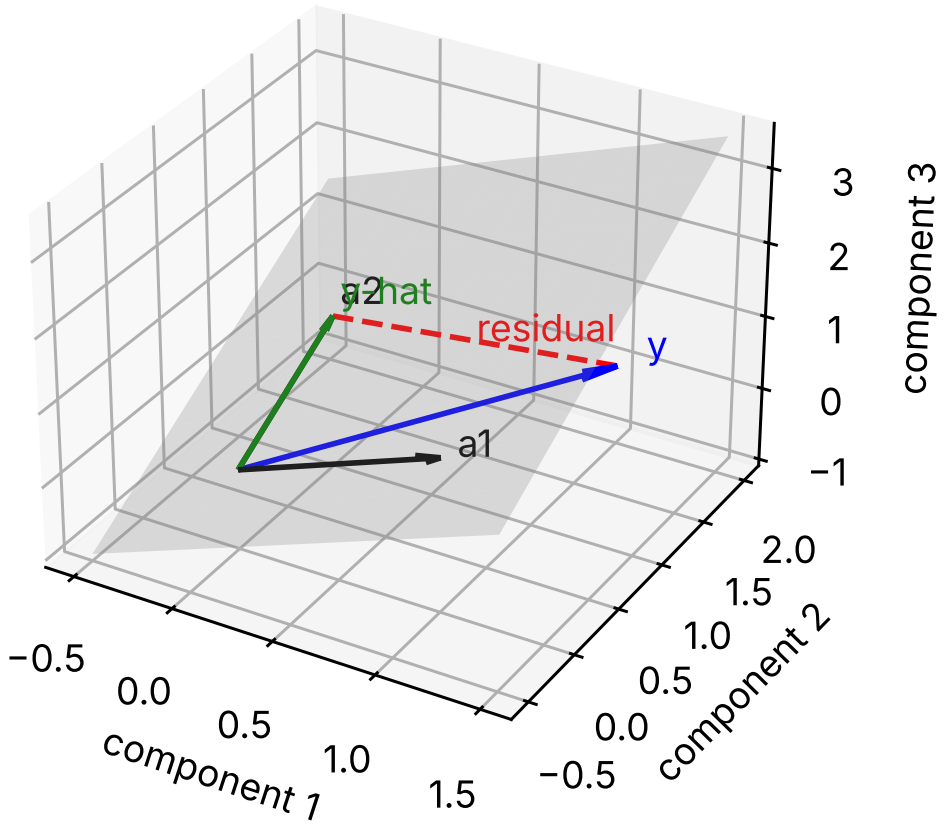


Figure 9.2: The geometry of Sample Problem "Projecting a vector onto a plane". The shaded plane is the column space of A (all combinations of $a_1 = (1,0,1)$ and $a_2 = (0,1,1)$). The least-squares model vector $\hat{y} = (0,1,1)$ is the foot of the perpendicular from the data vector $y = (1,2,0)$; the residual $\hat{r} = (1,1,-1)$ is perpendicular to the plane.

Checkpoint 9.3

In the sample problem, is the residual $\hat{r}$ perpendicular to the data vector y?

Answer: No: $y^\top \hat{r} = 1 + 2 + 0 = 3 \neq 0$. The residual is perpendicular to the column space (to a_1, a_2 and $\hat{y}$), not to y.

Key Ideas

- The model vectors Ac fill the column space of A; least squares picks the point $\hat{y}$ of that space closest to y.
- The residual is perpendicular to the column space; $\|y\|^2 = \|\hat{y}\|^2 + \|\hat{r}\|^2$.
- With independent columns, $\hat{y} = Py$ with the projection matrix $P = A(A^\top A)^{-1}A^\top$, and $P^2 = P$.

9.4 The singular value decomposition

The normal equations solve least squares, but they hide the information you most want: *which* combinations of coefficients the data pin down well and which they pin down badly. The singular value decomposition exposes exactly this.

What a matrix does to a circle. Take a 2×2 matrix A and apply it to every unit vector c (every point on the unit circle). The images Ac trace out an ellipse (Figure 9.3). The ellipse has a longest axis and a shortest axis, perpendicular to each other. Their half-lengths $\sigma_1 \geq \sigma_2$ are the *singular values* of A: the largest and smallest stretch factors the matrix applies. The unit vectors v_1, v_2 that land on the axes are the *right singular vectors*, and the unit vectors u_1, u_2 along the axes are the *left singular vectors*. The whole action of A is then: rotate (so that v_1, v_2 line up with the coordinate axes), stretch each axis by σ_j, rotate again (so the axes line up with u_1, u_2). We now make this precise in any number of dimensions.

Orthogonal matrices. A real square matrix Q is *orthogonal* if $Q^\top Q = I$, where I is the identity matrix. Entry (j, ℓ) of $Q^\top Q$ is the dot product of columns j and ℓ of Q, so $Q^\top Q = I$ says the columns are orthonormal (unit length, mutually perpendicular). Two consequences:

- $Q^\top$ is the inverse of Q, so also $QQ^\top = I$ (a left inverse of a square matrix is also a right inverse, Chapter 7).
- Q preserves lengths: $\|Qx\|^2 = (Qx)^\top(Qx) = x^\top Q^\top Qx = x^\top x = \|x\|^2$.

Rotations and reflections are orthogonal matrices: each preserves lengths and angles, so it carries the standard unit vectors $(1, 0, \dots)$, $(0, 1, \dots)$, $\dots$, which are orthonormal, to an orthonormal set of images, and those images are exactly its columns. Recall from Chapter 8 that the spectral theorem diagonalizes a real symmetric matrix B as $B = QDQ^\top$ with Q orthogonal (columns the orthonormal eigenvectors) and D diagonal (the eigenvalues).

Constructing the SVD. Let A be any real $m \times M$ matrix. We build its SVD in six steps.

Step 1. The $M \times M$ matrix $A^\top A$ is symmetric (Section 9.2). By the spectral theorem it has orthonormal eigenvectors $v_1, \dots, v_M$ with real eigenvalues $\nu_1, \dots, \nu_M$, which we order so that $\nu_1 \geq \nu_2 \geq \cdots \geq \nu_M$. (We call these eigenvalues ν_j rather than λ_j because λ will soon mean a damping parameter, as in the paper.)

Step 2. The eigenvalues are not negative: $\nu_j = \nu_j v_j^\top v_j = v_j^\top (A^\top A v_j) = \|Av_j\|^2 \geq 0$. Define the singular values $\sigma_j = \sqrt{\nu_j}$, so $\|Av_j\| = \sigma_j$. Let ϱ be the number of strictly positive σ_j. (We write the rank as ϱ because the paper uses r for residuals.)

Step 3. For $j \leq \varrho$ define $u_j = Av_j/\sigma_j$. These are orthonormal:

$$u_j^\top u_\ell = \frac{(Av_j)^\top (Av_\ell)}{\sigma_j \sigma_\ell} = \frac{v_j^\top (A^\top A v_\ell)}{\sigma_j \sigma_\ell} = \frac{\nu_\ell\, v_j^\top v_\ell}{\sigma_j \sigma_\ell} = \begin{cases} \nu_j/\sigma_j^2 = 1, & j = \ell, \\ 0, & j \neq \ell. \end{cases}$$

Since $u_1, \dots, u_\varrho$ are ϱ orthonormal vectors in m-dimensional space, and at most m vectors there can be orthonormal (Chapter 7), we have $\varrho \le m$; also $\varrho \le M$, since $A^\top A$ has only M eigenvalues. In particular, when $M > m$ every σ_j with $j > m$ is zero.

Step 4. For $j > \varrho$, $\|Av_j\| = \sigma_j = 0$, so $Av_j = 0$: these v_j lie in the null space.

Step 5. If $\varrho < m$, extend $u_1, \dots, u_\varrho$ by further unit vectors $u_{\varrho+1}, \dots, u_m$ to an orthonormal basis of m-dimensional space (always possible, by the Gram–Schmidt procedure; see the box below).

Step 6. Collect the vectors as columns of the orthogonal matrices $U = (u_1, \dots, u_m)$ $(m \times m)$ and $V = (v_1, \dots, v_M)$ $(M \times M)$, and let Σ be the $m \times M$ matrix with $\Sigma_{jj} = \sigma_j$ for $j \le q = \min(m, M)$ and zeros elsewhere. Compare AV with $U\Sigma$ column by column. Column j of AV is Av_j. Column j of $U\Sigma$ is $\sigma_j u_j$ if $j \le q$ and zero if $j > q$. For $j \le \varrho$ these agree by the definition in Step 3; for $j > \varrho$ both are zero by Step 4 (either $\sigma_j = 0$, or the column of Σ is empty). So $AV = U\Sigma$. Multiply on the right by $V^\top$ and use $VV^\top = I$:

$$\boxed{A = U\,\Sigma\,V^\top} \tag{9.18}$$

This is the *singular value decomposition*. Every real matrix has one.

Taken on trust

What we took on trust. The construction above used two results: the spectral theorem for real symmetric matrices including the case of repeated eigenvalues (stated in Chapter 8, with the general proof beyond this book), and the fact that any orthonormal set in m dimensions can be extended to an orthonormal basis (Gram–Schmidt). Full proofs of the existence of the SVD are in [28] and [29]. For complex matrices the same theorem holds with U and V unitary and $V^\top$ replaced by the conjugate transpose $V^\dagger$. Numerically the SVD is never computed by forming $A^\top A$ (that loses accuracy, Section 9.6); library routines use a reduction to bidiagonal form (Golub–Kahan) at a cost proportional to mM^2 for $m \ge M$ [29, 27].

The outer-product form. Write out $U\Sigma V^\top$. The product $U\Sigma$ has columns $\sigma_1 u_1, \dots, \sigma_q u_q$ followed by zero columns, and multiplying a matrix of columns by $V^\top$ (whose rows are $v_j^\top$) gives the sum of each column times the corresponding row. Dropping the zero terms,

$$A = \sum_{j=1}^{\varrho} \sigma_j\, u_j\, v_j^\top. \tag{9.19}$$

Each term $u_j v_j^\top$ is an $m \times M$ matrix of rank one. Applied to a vector,

$$Ac = \sum_{j=1}^{\varrho} \sigma_j\, (v_j^\top c)\, u_j : \tag{9.20}$$

measure the component of c along each v_j, multiply it by σ_j, and lay it down along u_j. That is the rotate, stretch, rotate picture in words.

What the SVD tells you.

- The rank of A is ϱ, the number of nonzero singular values; the column space is spanned by $u_1, \dots, u_\varrho$ (by Equation (9.20), every Ac is a combination of them, and each $u_j = A(v_j/\sigma_j)$ is in it).
- The null space is spanned by $v_{\varrho+1}, \dots, v_M$. A coefficient change along these directions changes nothing at the data points.

- A coefficient change along v_j with small σ_j changes the model values only a little. The data barely "see" these directions, and, as we shall see, noise in the data pushes the fitted coefficients far along them.
- The vectors u_j are eigenvectors of $AA^\top$ with the same eigenvalues σ_j^2: $AA^\top u_j = AA^\top Av_j/\sigma_j = A(\sigma_j^2 v_j)/\sigma_j = \sigma_j^2 u_j$.

The singular values are unique: they are the square roots of the eigenvalues of $A^\top A$, and those eigenvalues are fixed by A (they are the roots of the characteristic polynomial of $A^\top A$). When the *nonzero* σ_j are distinct, each σ_j^2 is a simple eigenvalue of the symmetric matrix $A^\top A$, so its unit eigenvector v_j is unique up to sign, and $u_j = Av_j/\sigma_j$ is then fixed by v_j (flipping the sign of both u_j and v_j leaves Equation (9.19) unchanged). Singular vectors belonging to a repeated singular value, or to $\sigma_j = 0$ (the null directions), are not unique: any orthonormal basis of the corresponding space will do.

For $m \geq M$ one often keeps only the first M columns of U; this *thin SVD* $A = U_M \Sigma_M V^\top$ with Σ_M square diagonal contains everything in Equation (9.19).

Sample Problem 9.3: SVD of a 2×2 matrix by hand

Find the SVD of $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$, and verify the outer-product form.

Step 1: form $A^\top A$. $A^\top = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}$, so $A^\top A = \begin{pmatrix} 9+16 & 0+20 \\ 0+20 & 0+25 \end{pmatrix} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$.

Step 2: eigenvalues. The characteristic polynomial (Chapter 8) is $(25-\nu)^2 - 400 = 0$, so $25-\nu = \pm 20$, giving $\nu_1 = 45$, $\nu_2 = 5$. The singular values are $\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.708$ and $\sigma_2 = \sqrt{5} \approx 2.236$.

Step 3: right singular vectors. For $\nu = 45$: $(25-45)x + 20y = 0$ gives $x = y$, so $v_1 = (1,1)/\sqrt{2}$. For $\nu = 5$: $20x + 20y = 0$, so $v_2 = (1,-1)/\sqrt{2}$.

Step 4: left singular vectors. $Av_1 = (3,\,9)/\sqrt{2}$, so $u_1 = Av_1/\sigma_1 = (3,9)/(\sqrt{2}\cdot 3\sqrt{5}) = (1,3)/\sqrt{10}$. $Av_2 = (3,\,4-5)/\sqrt{2} = (3,-1)/\sqrt{2}$, so $u_2 = (3,-1)/(\sqrt{2}\sqrt{5}) = (3,-1)/\sqrt{10}$. Check: $u_1^\top u_2 = (3-3)/10 = 0$.

Step 5: outer products. $\sigma_1 u_1 v_1^\top = \frac{3\sqrt{5}}{\sqrt{20}} \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$ (since $\sqrt{20} = 2\sqrt{5}$) and $\sigma_2 u_2 v_2^\top = \frac{\sqrt{5}}{\sqrt{20}} \begin{pmatrix} 3 & -3 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 3 & -3 \\ -1 & 1 \end{pmatrix}$. Their sum is $\begin{pmatrix} 1.5+1.5 & 1.5-1.5 \\ 4.5-0.5 & 4.5+0.5 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} = A$.

Step 6: a check. For a square matrix, $|\det A| = \sigma_1\sigma_2$ (the determinants of U and V are ± 1, Chapter 7). Here $\det A = 15$ and $\sigma_1\sigma_2 = 3\sqrt{5}\cdot\sqrt{5} = 15$.

$$\boxed{\sigma_1 = 3\sqrt{5} \approx 6.708,\ \sigma_2 = \sqrt{5} \approx 2.236;\ v_{1,2} = \tfrac{(1,\pm 1)}{\sqrt{2}},\ u_1 = \tfrac{(1,3)}{\sqrt{10}},\ u_2 = \tfrac{(3,-1)}{\sqrt{10}}}$$

```
import numpy as np
A = np.array([[3.0, 0.0],
              [4.0, 5.0]])
U, s, Vh = np.linalg.svd(A)     # s holds sigma_1 >= sigma_2
print(s)                        # [6.70820393 2.23606798]
V = Vh.T                        # numpy returns V transposed!
print(np.allclose(U @ np.diag(s) @ Vh, A))    # True
print(s[0] * s[1], np.linalg.det(A))        # both 15 (det up to sign)
```

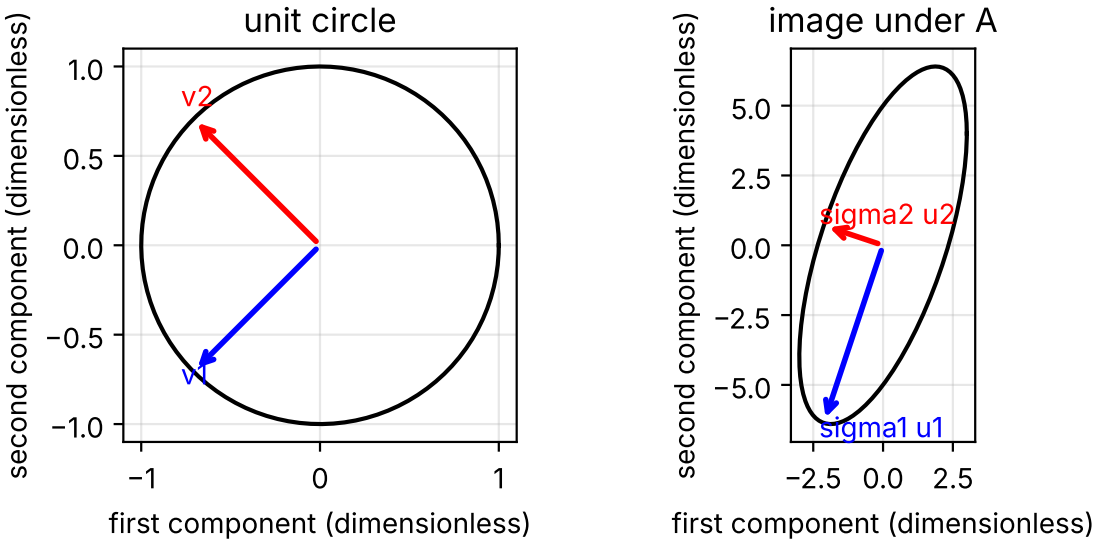


Figure 9.3: The SVD as rotate, stretch, rotate, for $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$. Left: the unit circle and the right singular vectors v_1, v_2. Right: the image of the circle under A is an ellipse whose semi-axes are $\sigma_1 u_1$ and $\sigma_2 u_2$, with $\sigma_1 = 3\sqrt{5}$ and $\sigma_2 = \sqrt{5}$. Both axes are dimensionless.

Pitfall

`numpy.linalg.svd` returns $V^\top$ (often called `Vh`), not V. The right singular vectors are the *rows* of `Vh`, or the columns of `Vh.T`. Mixing these up gives wrong answers that still have the right shapes, so no error message warns you.

Checkpoint 9.4

What are the singular values of (a) a 3×3 orthogonal matrix Q, (b) $\mathrm{diag}(3, -2)$?

Answer: (a) All three equal 1: $Q^\top Q = I$ has every eigenvalue 1. (b) 3 and 2: $A^\top A = \mathrm{diag}(9, 4)$. The minus sign is absorbed into U (take $u_2 = (0, -1)$).

Key Ideas

- Every real matrix factors as $A = U\Sigma V^\top$ (Equation (9.18)): orthogonal U, V and nonnegative singular values $\sigma_1 \geq \sigma_2 \geq \cdots$ on the diagonal of Σ.
- σ_j^2 are the eigenvalues of $A^\top A$; v_j its eigenvectors; $u_j = Av_j/\sigma_j$.
- $A = \sum_j \sigma_j u_j v_j^\top$: the direction v_j in coefficient space produces the pattern u_j in data space with gain σ_j. Zero singular values mark directions the data cannot see.

9.5 Solving least squares with the SVD: the pseudo-inverse

Diagonalizing the problem. Insert $A = U\Sigma V^\top$ into the objective. Because $U^\top$ is orthogonal it preserves lengths, so

$$S(c) = \|y - Ac\|^2 = \left\|U^\top (y - U\Sigma V^\top c)\right\|^2 = \left\|U^\top y - \Sigma V^\top c\right\|^2,$$

using $U^\top U = I$. Introduce new coordinates: $\beta = U^\top y$ (so $\beta_j = u_j^\top y$, the component of the data along pattern u_j) and $\xi = V^\top c$ (so $\xi_j = v_j^\top c$, the component of the coefficients along direction v_j). Since V is orthogonal, $c = V\xi = \sum_j \xi_j v_j$ and $\|c\| = \|\xi\|$. The vector $\Sigma\xi$ has components $\sigma_j\xi_j$ for $j \le q$ and zeros below. Hence, using $\sigma_j = 0$ for $\varrho < j \le q$,

$$S = \sum_{j=1}^{\varrho} (\beta_j - \sigma_j\xi_j)^2 + \sum_{j=\varrho+1}^{m} \beta_j^2. \tag{9.21}$$

In these coordinates the problem has fallen apart into independent one-variable pieces.

- Each term in the first sum is minimized (made zero) by $\xi_j = \beta_j/\sigma_j$, for $j \le \varrho$.
- The second sum does not depend on c at all: it is the part of the data that no model can reach. So $S_{\min} = \sum_{j>\varrho} \beta_j^2$.
- The coefficients ξ_j with $\varrho < j \le M$ do not appear in S. They are completely undetermined by the data.

The minimum-norm solution. Every vector $c = \sum_{j\le\varrho}(\beta_j/\sigma_j)v_j + \sum_{j>\varrho}\xi_j v_j$, with any values of the free ξ_j, minimizes S. Its squared length is $\|c\|^2 = \sum_{j\le\varrho}(\beta_j/\sigma_j)^2 + \sum_{j>\varrho}\xi_j^2$, smallest when every free ξ_j is zero. The shortest least-squares solution is therefore

$$\boxed{c^+ = \sum_{j=1}^{\varrho} \frac{u_j^\top y}{\sigma_j} v_j = A^+ y, \qquad A^+ = V\Sigma^+ U^\top} \tag{9.22}$$

where Σ^+ is the $M \times m$ matrix with $1/\sigma_j$ in position (j,j) for $j \le \varrho$ and zeros elsewhere. A^+ is the (Moore–Penrose) *pseudo-inverse* of A. Read Equation (9.22) as a recipe: project the data onto each pattern u_j, divide by the gain σ_j, and build the coefficients from the directions v_j.

The pseudo-inverse solves the normal equations. Using $A^\top = V\Sigma^\top U^\top$, $A^\top A c^+ = V\Sigma^\top U^\top U\Sigma V^\top V\Sigma^+ U^\top y$. With $U^\top U = I$ and $V^\top V = I$ this is $V(\Sigma^\top\Sigma\Sigma^+)U^\top y$. The $M \times m$ matrix $\Sigma^\top\Sigma\Sigma^+$ has entry $\sigma_j^2 \cdot (1/\sigma_j) = \sigma_j$ in position (j,j) for $j \le \varrho$ and zeros elsewhere. That is exactly $\Sigma^\top$, whose diagonal entries with $j > \varrho$ are zero. Hence $A^\top A c^+ = V\Sigma^\top U^\top y = A^\top y$: the normal equations always have at least one solution, as promised in Section 9.2.

Agreement with the normal equations. If the columns of A are independent ($\varrho = M \le m$), then c^+ equals $(A^\top A)^{-1}A^\top y$. Proof: $A^\top A = V\Sigma^\top U^\top U\Sigma V^\top = V(\Sigma^\top\Sigma)V^\top$, where $\Sigma^\top\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_M^2)$ is invertible. Its inverse is $V(\Sigma^\top\Sigma)^{-1}V^\top$ (multiply out and use $V^\top V = I$). Then

$$(A^\top A)^{-1}A^\top = V(\Sigma^\top\Sigma)^{-1}V^\top\, V\Sigma^\top U^\top = V(\Sigma^\top\Sigma)^{-1}\Sigma^\top U^\top = V\Sigma^+ U^\top,$$

because $(\Sigma^\top\Sigma)^{-1}\Sigma^\top$ has diagonal entries $\sigma_j/\sigma_j^2 = 1/\sigma_j$. If A is square and invertible, $A^+ = A^{-1}$.

Underdetermined problems: infinitely many exact fits. Suppose $m < M$ and the rank is as large as possible, $\varrho = m$ (the rows are independent). Then the second sum in Equation (9.21) is empty, $S_{\min} = 0$, and every point is fitted exactly. The free directions $v_{m+1}, \ldots, v_M$ span a null space of dimension $M - m$. Any vector z in it can be added to c^+ without changing a single model value, so

$$c = c^+ + z, \qquad Az = 0, \tag{9.23}$$

is an exact fit for every z: infinitely many perfect fits. In general (any rank) the null space has dimension $M - \varrho \ge M - m$. Because c^+ is built from $v_1, \ldots, v_\varrho$ and z from $v_{\varrho+1}, \ldots, v_M$, the

two are perpendicular and $\|c\|^2 = \|c^+\|^2 + \|z\|^2$: the minimum-norm solution does nothing at all in the directions the data cannot see.

Sample Problem 9.4: One equation, two unknowns

Find all exact solutions of $c_1 + 2c_2 = 5$ and the minimum-norm one, first directly and then from the SVD.

Step 1: all solutions. Here $A = (1 \;\; 2)$ is 1×2. Setting $c_2 = s$ gives $c_1 = 5 - 2s$, so the solutions are $c = (5 - 2s,\; s)$, a whole line.

Step 2: shortest. $\|c\|^2 = (5-2s)^2 + s^2 = 25 - 20s + 5s^2$. Its derivative $-20 + 10s$ vanishes at $s = 2$, giving $c = (1, 2)$ with $\|c\|^2 = 5$.

Step 3: SVD. $A^\top A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ has eigenvalues 5 and 0; $\sigma_1 = \sqrt{5}$, $v_1 = (1,2)/\sqrt{5}$, $u_1 = Av_1/\sigma_1 = (5/\sqrt{5})/\sqrt{5} = 1$. The null direction is $v_2 = (2,-1)/\sqrt{5}$.

Step 4: pseudo-inverse solution. $c^+ = (u_1 y/\sigma_1)\, v_1 = (5/\sqrt{5})(1,2)/\sqrt{5} = (1,2)$, as in Step 2. Every other exact solution is $(1,2) + s'(2,-1)$, a null vector added; for example $(5,0)$ has $s' = 2$ and norm $5 > \sqrt{5}$.

$$\boxed{c^+ = (1,2), \quad \left\|c^+\right\| = \sqrt{5} \approx 2.236}$$

```
import numpy as np
A = np.array([[1.0, 2.0]])
y = np.array([5.0])
c_lstsq = np.linalg.lstsq(A, y, rcond=None)[0]  # returns the min-norm solution
c_pinv  = np.linalg.pinv(A) @ y                  # same thing via A^+
print(c_lstsq, c_pinv)                           # [1. 2.] [1. 2.]
```

Checkpoint 9.5

A model has $M = 5$ coefficients and you have $m = 3$ data points. At most how many nonzero singular values does A have, and at least how large is its null space?

Answer: At most $\min(3,5) = 3$ nonzero singular values, so the rank is at most 3 and the null space has dimension at least $5 - 3 = 2$.

Key Ideas

- In SVD coordinates, least squares splits into independent pieces: $\xi_j = \beta_j/\sigma_j$ for $\sigma_j > 0$; the rest are free (Equation (9.21)).
- The pseudo-inverse $A^+ = V\Sigma^+ U^\top$ gives the shortest least-squares solution (Equation (9.22)); with independent columns it equals $(A^\top A)^{-1} A^\top$.
- With more coefficients than independent data, there are infinitely many exact fits, differing by null-space vectors (Equation (9.23)).

9.6 Condition number and noise amplification

★ Did you know? *Turing Named the Problem*

Alan Turing introduced the term *condition number* in a 1948 paper on rounding errors in matrix computations. It measures how much a matrix can amplify small errors in the data. A famous troublemaker is the Hilbert matrix, with entries $1/(i+j-1)$. The 10×10 version has a condition number around 10^{13}, so solving with it can wipe out most of a computer's digits.

Gains are bounded by the singular values. For any coefficient vector c with SVD coordinates $\xi = V^\top c$, Equation (9.20) and the orthonormality of the u_j give $\|Ac\|^2 = \sum_{j \le \varrho} \sigma_j^2 \xi_j^2$. Replacing every σ_j by the largest one can only increase the sum, so $\|Ac\|^2 \le \sigma_1^2 \sum_j \xi_j^2 = \sigma_1^2 \|c\|^2$:

$$\|Ac\| \le \sigma_1 \|c\| . \tag{9.24}$$

If A is square ($m = M$) and invertible, all $\sigma_j > 0$ (a zero σ_j would give $Av_j = 0$ with $v_j \neq 0$). Then $A^{-1} = V\Sigma^{-1}U^\top$; check: $(V\Sigma^{-1}U^\top)(U\Sigma V^\top) = V\Sigma^{-1}\Sigma V^\top = VV^\top = I$. This product has the form orthogonal matrix times diagonal matrix with positive entries $1/\sigma_j$ times the transpose of an orthogonal matrix, so after reordering the entries into decreasing order (and the columns of V and U with them) it is an SVD of A^{-1}. Its singular values are the $1/\sigma_j$, the largest being $1/\sigma_M$. Applying Equation (9.24) to A^{-1}:

$$\left\|A^{-1}w\right\| \le \frac{\|w\|}{\sigma_M}. \tag{9.25}$$

The condition number. The *condition number* of a square invertible matrix is the ratio of the largest to the smallest stretch,

$$\kappa(A) = \frac{\sigma_1}{\sigma_M} \ge 1. \tag{9.26}$$

For a rectangular matrix with independent columns we use the same ratio σ_1/σ_M. Now suppose $Ac = y$ and the data change by δy, so the solution changes by δc with $A(c + \delta c) = y + \delta y$. Subtracting, $A\,\delta c = \delta y$, so $\delta c = A^{-1}\delta y$ and by Equation (9.25) $\|\delta c\| \le \|\delta y\| / \sigma_M$. Also, by Equation (9.24), $\|y\| = \|Ac\| \le \sigma_1 \|c\|$, so $1/\|c\| \le \sigma_1 / \|y\|$. Multiplying the two inequalities,

$$\frac{\|\delta c\|}{\|c\|} \le \kappa(A) \frac{\|\delta y\|}{\|y\|}. \tag{9.27}$$

A relative error in the data can be magnified by up to κ in the solution. A matrix with large κ is called *ill-conditioned*.

Sample Problem 9.5: An ill-conditioned pair of equations

Solve $c_1 + c_2 = 2$, $c_1 + 1.001c_2 = 2.001$; then change the second right-hand side to 2.002 and solve again. Compare the relative changes with the bound Equation (9.27).

Step 1: original solution. Subtracting the equations, $0.001c_2 = 0.001$, so $c_2 = 1$ and $c_1 = 1$: $c = (1, 1)$.

Step 2: perturbed solution. Now $0.001c_2 = 0.002$, so $c_2 = 2$ and $c_1 = 0$: $c = (0, 2)$. The data changed in the fourth significant figure; the solution changed completely.

Step 3: relative changes. $\|\delta y\| = 0.001$ and $\|y\| = \sqrt{2^2 + 2.001^2} = \sqrt{8.004001} \approx 2.8291$, so $\|\delta y\| / \|y\| \approx 3.535 \times 10^{-4}$. $\|\delta c\| = \|(-1, 1)\| = \sqrt{2}$ and $\|c\| = \sqrt{2}$, so $\|\delta c\| / \|c\| = 1$. The amplification is $1/(3.535 \times 10^{-4}) \approx 2829$.

Step 4: condition number. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1.001 \end{pmatrix}$ is symmetric with positive eigenvalues, so its singular values equal its eigenvalues. (Reason: if B is symmetric with eigenvalues $\beta_j \geq 0$ and orthonormal eigenvectors q_j, then $B^\top B = B^2$ and $B^2 q_j = \beta_j^2 q_j$, so the eigenvalues of $B^\top B$ are β_j^2 and the singular values are $\sqrt{\beta_j^2} = \beta_j$.) The trace is 2.001 and the determinant 0.001, so the eigenvalues are $\frac{1}{2}\left(2.001 \pm \sqrt{2.001^2 - 0.004}\right)$, i.e. $\sigma_1 \approx 2.0005$ and $\sigma_2 = 0.001/\sigma_1 \approx 4.999 \times 10^{-4}$. Then $\kappa = \sigma_1/\sigma_2 = \sigma_1^2/0.001 \approx 4002$.
Step 5: compare. The observed amplification, about 2829, is below the bound $\kappa \approx 4002$, as Equation (9.27) requires.

$$\kappa \approx 4.0 \times 10^3, \qquad \text{observed amplification} \approx 2.8 \times 10^3$$

Noise amplification in least squares. Real data contain noise: $y = y_{\text{true}} + e$, where e is the vector of measurement errors. Insert this into Equation (9.22):

$$c^+ = \sum_{j=1}^{\varrho} \frac{u_j^\top y_{\text{true}}}{\sigma_j} v_j \; + \; \sum_{j=1}^{\varrho} \frac{u_j^\top e}{\sigma_j} v_j. \tag{9.28}$$

The first sum is what we want. The second is the noise, and each of its components is divided by σ_j. What follows is a heuristic, true in many problems but not in all, and we do not prove it. Measurement noise usually has comparable components along every pattern u_j (it has no reason to prefer one), while a smooth true signal often has components $u_j^\top y_{\text{true}}$ that become small for the patterns with small σ_j, which in many problems are the rapidly wiggling ones. Whether this holds must be checked problem by problem. When it does, for those j the noise term dominates and is magnified by the large factor $1/\sigma_j$. The fitted coefficients then contain large, meaningless excursions along the poorly seen directions v_j. This is the precise sense in which "small singular values amplify noise".

Floating point and the normal equations. Even exact data carry rounding error of relative size about machine epsilon, 2.2×10^{-16} in double precision (Chapter 4). By Equation (9.27), a useful rule of thumb is that you can lose up to about $\log_{10} \kappa$ significant digits. Forming the normal equations makes things worse: the singular values of the symmetric positive semidefinite matrix $A^\top A$ are its eigenvalues σ_j^2 (by the remark in Step 4 of the sample above), so

$$\kappa(A^\top A) = \frac{\sigma_1^2}{\sigma_M^2} = \kappa(A)^2. \tag{9.29}$$

A problem with $\kappa(A) = 10^8$ is solvable with about eight good digits by an SVD-based (or QR-based) method, but $\kappa(A^\top A) = 10^{16}$ wipes out essentially every digit of a double-precision normal-equations solution. Library routines such as `numpy.linalg.lstsq` therefore work with the SVD, and the paper's update (Appendix A step 6) is also written in SVD form, which in addition makes the damping filter explicit.

Pitfall

The normal equations are the right way to *derive* least squares and the wrong way to *compute* it when A is ill-conditioned. Forming $A^\top A$ squares the condition number Equation (9.29). Use `numpy.linalg.lstsq` or an explicit SVD.

Checkpoint 9.6

A least-squares problem has $\kappa(A) = 10^5$. Roughly how many significant digits can you expect from an SVD solution in double precision, and from the normal equations?

Answer: Double precision carries about 16 digits. Losing about 5 leaves roughly 11 with the SVD; the normal equations have $\kappa = 10^{10}$ and leave roughly 6. These are rule-of-thumb estimates, not guarantees.

Key Ideas

- $\kappa = \sigma_1/\sigma_M$ bounds how much relative data errors are magnified (Equation (9.27)).
- In the least-squares solution the noise along pattern u_j is divided by σ_j (Equation (9.28)); small singular values amplify noise.
- $\kappa(A^\top A) = \kappa(A)^2$, so compute with the SVD, not the normal equations.

9.7 Truncated SVD

The simplest defense against noise amplification is to ignore the directions that the data see badly. Keep only the J largest singular values:

$$c^{(J)} = \sum_{j=1}^{J} \frac{u_j^\top y}{\sigma_j} v_j, \qquad J \le \varrho. \tag{9.30}$$

This is the *truncated SVD* solution. It is convenient to write every such solution in one form,

$$c = \sum_{j=1}^{\varrho} f_j \, \frac{u_j^\top y}{\sigma_j} \, v_j, \tag{9.31}$$

where the numbers f_j are *filter factors*. The pseudo-inverse has $f_j = 1$ for every j; the truncated SVD has $f_j = 1$ for $j \le J$ and $f_j = 0$ for $j > J$.

Truncation is a trade. It removes the amplified noise in the discarded directions, but it also discards whatever true signal lay along them: the truncated solution is systematically pulled toward zero in those directions (statisticians call this *bias*, and the noise-driven scatter *variance*; see Chapter 25). A sensible cut keeps the components with σ_j large enough that $\left|u_j^\top y_{\text{true}}\right|$ stands above the noise; finding that cut requires some knowledge of the noise level or a validation test.

Checkpoint 9.7

What truncated-SVD solution do you get with $J = \varrho$? With $J = 0$?

Answer: $J = \varrho$ keeps every nonzero singular value: the pseudo-inverse solution c^+. $J = 0$ keeps nothing: $c = 0$, the model with no fitted correction at all.

Key Ideas

- Regularized solutions have the form (Equation (9.31)) with filter factors $0 \le f_j \le 1$.
- The truncated SVD uses a sharp cutoff: $f_j = 1$ for the J largest σ_j, else 0.
- Truncation trades noise (variance) for systematic shrinkage toward zero (bias).

9.8 Ridge regularization

A sharp cutoff is crude: a singular value just above the cut is kept whole, one just below is thrown away. Ridge regularization (also called Tikhonov regularization) replaces the step with a smooth filter.

The ridge objective. Add to the sum of squared residuals a penalty on the size of the coefficients,

$$S_\alpha(c) = \|y - Ac\|^2 + \alpha \|c\|^2, \qquad \alpha > 0. \tag{9.32}$$

Think of each coefficient as attached to the origin by a spring of stiffness proportional to α: the data pull the coefficients toward a good fit, the springs pull them toward zero, and the minimum of S_α is the compromise. Large excursions along poorly seen directions buy little reduction in the first term but cost a lot in the second, so they are suppressed.

The ridge normal equations. Differentiate with respect to c_ℓ. The first term gives $-2(A^\top r)_\ell$ as in Section 9.2; the second, $\alpha \sum_k c_k^2$, gives $2\alpha c_\ell$. Setting the sum to zero: $-2A^\top(y - Ac) + 2\alpha c = 0$, that is

$$\boxed{(A^\top A + \alpha I)\, c = A^\top y} \tag{9.33}$$

The matrix $A^\top A + \alpha I$ is invertible for every $\alpha > 0$, whatever A is: if $(A^\top A + \alpha I)z = 0$ then $0 = z^\top(A^\top A + \alpha I)z = \|Az\|^2 + \alpha \|z\|^2$, a sum of two nonnegative terms, so $\alpha \|z\|^2 = 0$ and $z = 0$. The ridge solution is therefore always unique, even for underdetermined problems.

Ridge is ordinary least squares in disguise. Stack the matrix A on top of $\sqrt{\alpha} I$ and the data y on top of M zeros:

$$\tilde{A} = \begin{pmatrix} A \\ \sqrt{\alpha}\, I \end{pmatrix}, \qquad \tilde{y} = \begin{pmatrix} y \\ 0 \end{pmatrix}.$$

Then $\tilde{y} - \tilde{A}c$ has top block $y - Ac$ and bottom block $-\sqrt{\alpha}\, c$, so $\left\|\tilde{y} - \tilde{A}c\right\|^2 = \|y - Ac\|^2 + \alpha \|c\|^2 = S_\alpha(c)$. Everything in Section 9.2 applies: the solution of Equation (9.33) is a global minimum by Equation (9.15), since $\tilde{A}^\top \tilde{A} = A^\top A + \alpha I$ and $\tilde{A}^\top \tilde{y} = A^\top y$.

The SVD form and the filter factors. Insert $A = U\Sigma V^\top$. Using $V^\top V = VV^\top = I$,

$$A^\top A + \alpha I = V\Sigma^\top \Sigma V^\top + \alpha V V^\top = V(\Sigma^\top \Sigma + \alpha I)V^\top,$$

where $\Sigma^\top \Sigma + \alpha I$ is the $M \times M$ diagonal matrix with entries $\sigma_j^2 + \alpha$ (with $\sigma_j = 0$ for $j > \varrho$). Its inverse is $V(\Sigma^\top \Sigma + \alpha I)^{-1} V^\top$. Also $A^\top y = V\Sigma^\top U^\top y$. Therefore

$$c_\alpha = V(\Sigma^\top \Sigma + \alpha I)^{-1} V^\top V \Sigma^\top U^\top y = V(\Sigma^\top \Sigma + \alpha I)^{-1} \Sigma^\top U^\top y.$$

The middle product is diagonal with entries $\sigma_j/(\sigma_j^2 + \alpha)$, so

$$\boxed{c_\alpha = \sum_{j=1}^{\varrho} \frac{\sigma_j}{\sigma_j^2 + \alpha} (u_j^\top y)\, v_j} \tag{9.34}$$

(terms with $\sigma_j = 0$ vanish). Comparing with Equation (9.31), the ridge filter factors are

$$f_j = \frac{\sigma_j^2}{\sigma_j^2 + \alpha}. \tag{9.35}$$

For $\sigma_j \gg \sqrt{\alpha}$, $f_j \approx 1$: well-seen directions are fitted almost as by the pseudo-inverse. For $\sigma_j \ll \sqrt{\alpha}$, $f_j \approx \sigma_j^2/\alpha \ll 1$: poorly seen directions are strongly suppressed. The changeover is smooth, centered at $\sigma_j = \sqrt{\alpha}$ where $f_j = \frac{1}{2}$ (Figure 9.4).

The relative form used in the paper. If you multiply A by a constant a (say, measure the data in different units), every σ_j is multiplied by $|a|$, and a fixed α would filter differently. It is therefore natural to measure α relative to the largest singular value, $\alpha = \lambda\sigma_1^2$, with λ a dimensionless number. Then

$$c_\lambda = \sum_j \frac{\sigma_j}{\sigma_j^2 + \lambda\sigma_1^2}(u_j^\top y)\, v_j = V\,\mathrm{diag}\left(\frac{\sigma_j}{\sigma_j^2 + \lambda\sigma_1^2}\right) U^\top y, \tag{9.36}$$

and the filter factors $\sigma_j^2/(\sigma_j^2 + \lambda\sigma_1^2) = (\sigma_j/\sigma_1)^2/((\sigma_j/\sigma_1)^2 + \lambda)$ depend only on the ratios σ_j/σ_1, which do not change when A is rescaled. The matrix form on the right of Equation (9.36) is exactly the form of the update in the paper's Appendix A step 6 (Section 9.10).

The two limits.

- $\alpha \to 0$: for each $\sigma_j > 0$, $\sigma_j/(\sigma_j^2+\alpha) \to 1/\sigma_j$, so $c_\alpha \to c^+$, the minimum-norm least-squares solution. (Ridge never moves along null directions v_j with $\sigma_j = 0$, at any α; that is why its limit is the *minimum-norm* solution.)
- $\alpha \to \infty$: every factor $\sigma_j/(\sigma_j^2 + \alpha) \to 0$, so $c_\alpha \to 0$. Infinitely stiff springs: the fit does nothing. In the paper's language this is "the null", the model with no fitted correction.

Between the limits, $\|c_\alpha\|$ decreases steadily as α grows, because each component $\left|\sigma_j u_j^\top y\right|/(\sigma_j^2+\alpha)$ decreases.

Ridge caps the noise gain. In the pseudo-inverse, data along u_j are multiplied by the gain $1/\sigma_j$, which is unbounded as $\sigma_j \to 0$. The ridge gain is $g(\sigma) = \sigma/(\sigma^2 + \alpha)$. By the quotient rule,

$$g'(\sigma) = \frac{(\sigma^2+\alpha)\cdot 1 - \sigma\cdot 2\sigma}{(\sigma^2+\alpha)^2} = \frac{\alpha - \sigma^2}{(\sigma^2+\alpha)^2},$$

which is positive for $\sigma < \sqrt{\alpha}$ and negative for $\sigma > \sqrt{\alpha}$. So g has its maximum at $\sigma = \sqrt{\alpha}$, where

$$g_{\max} = \frac{\sqrt{\alpha}}{\alpha + \alpha} = \frac{1}{2\sqrt{\alpha}}. \tag{9.37}$$

No component of the data, signal or noise, is ever amplified by more than $1/(2\sqrt{\alpha})$.

Sample Problem 9.6: Ridge versus truncation on a two-coefficient problem

Take $A = \mathrm{diag}(1,\ 0.01)$ and true coefficients $c_{\text{true}} = (1, 1)$, so the true data are $(1,\ 0.01)$. The measurement of the second component carries an error of $+0.01$: $y = (1,\ 0.02)$. Compare the plain solution, the truncated SVD with $J = 1$, and ridge with $\alpha = 10^{-5}, 10^{-4}, 10^{-3}$.

Step 1: SVD. A is diagonal with positive entries, so $U = V = I$, $\sigma_1 = 1$, $\sigma_2 = 0.01$, $\kappa = 100$, and $u_j^\top y = y_j$.

Step 2: plain solution. $c^+ = (y_1/\sigma_1,\ y_2/\sigma_2) = (1,\ 2)$. The noise 0.01 was divided by $\sigma_2 = 0.01$: an error of 1 in c_2. Error norm $\|c - c_{\text{true}}\| = 1$.

Step 3: truncated SVD. Keeping only $j = 1$: $c^{(1)} = (1,\ 0)$. The noise is gone, and so is the true value of c_2. Error norm 1.

Step 4: ridge. By Equation (9.34), $c_1 = 1\cdot 1/(1+\alpha)$ and $c_2 = 0.01\cdot 0.02/(10^{-4}+\alpha) = 2\times 10^{-4}/(10^{-4}+\alpha)$:

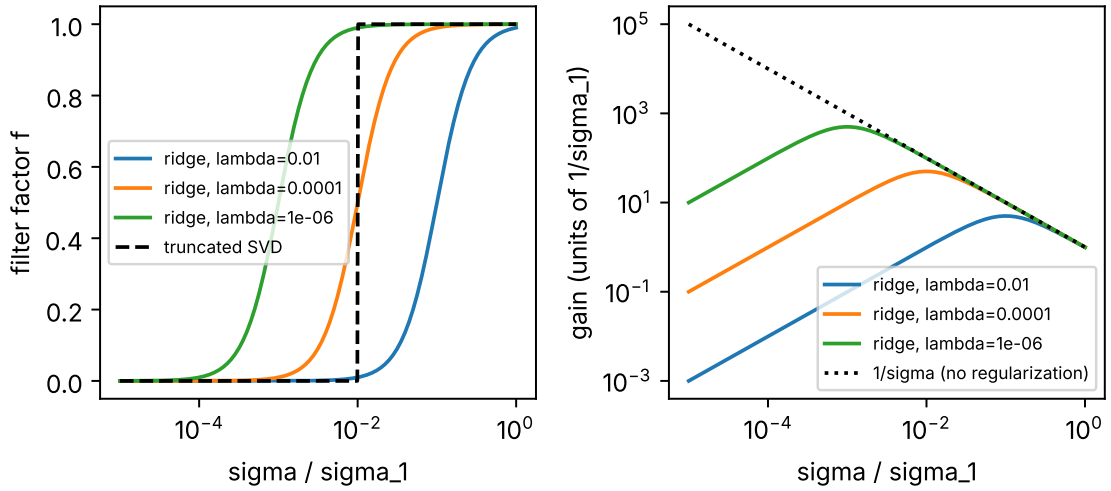


Figure 9.4: Left: filter factors f against σ/σ_1 for ridge regularization, $f = \sigma^2/(\sigma^2 + \lambda\sigma_1^2)$, with $\lambda = 10^{-2}, 10^{-4}, 10^{-6}$ (solid), and for a truncated SVD that keeps $\sigma/\sigma_1 \geq 10^{-2}$ (dashed step). Right: the gain applied to the data component along u_j, in units of $1/\sigma_1$: the unregularized $1/\sigma$ grows without bound, while the ridge gain $\sigma/(\sigma^2 + \lambda\sigma_1^2)$ peaks at $\sigma = \sqrt{\lambda}\,\sigma_1$ with height $1/(2\sqrt{\lambda}\,\sigma_1)$. All quantities dimensionless.

α	c_1	c_2	$\|c_\alpha - c_{\text{true}}\|$
10^{-5}	0.99999	1.8182	0.818
10^{-4}	0.99990	1.0000	1.0×10^{-4}
10^{-3}	0.99900	0.1818	0.818

Step 5: interpret. c_1 barely notices the regularization (its σ is far above $\sqrt{\alpha}$). For c_2, too little regularization leaves the noise in, too much removes the signal. At $\alpha = 10^{-4}$ the filter factor for $j = 2$ is exactly $\frac{1}{2}$, which here happens to cancel the noise exactly, because in this example the noise (0.01) equals the true signal (0.01). Such a perfect result is a coincidence of this example; Problem 25 finds the best α when the noise is known only on average.

plain: $(1, 2)$; TSVD: $(1, 0)$; ridge $\alpha = 10^{-4}$: $(0.9999, 1.0000)$

Choosing the regularization strength. The sample shows that the best α depends on the noise, which in practice you do not know. Using the data you are fitting to choose α fails (the in-sample error is smallest at $\alpha = 0$). The standard remedy is *validation*: fit with several values of α on part of the data and pick the value that best predicts the rest. Chapters 25 and 27 develop this; the paper's fitted controls choose their ridge strength by such a forward split (Section 9.10).

Pitfall

Ridge penalizes $\|c\|^2$, so "small" is measured in your chosen parametrization. Multiply one basis function by 10 and the coefficient needed to produce the same model drops tenfold, which the penalty treats as much cheaper. Ridge (and the minimum-norm solution) therefore depend on how the basis is scaled and which basis is used. Regularization

expresses a preference; it does not by itself discover the right model.

```python
import numpy as np

def ridge_svd(A, y, lam):
    """Ridge solution with alpha = lam * sigma_1**2, Eq. (ridge-rel)."""
    U, s, Vh = np.linalg.svd(A, full_matrices=False)
    gain = s / (s**2 + lam * s[0]**2)      # sigma/(sigma^2 + lam sigma_1^2)
    return Vh.T @ (gain * (U.T @ y))

A = np.diag([1.0, 0.01])
y = np.array([1.0, 0.02])
for lam in [1e-5, 1e-4, 1e-3]:
    print(lam, ridge_svd(A, y, lam))       # reproduces the sample's table
```

Checkpoint 9.8

As α is increased from 0 to very large values, what happens to the in-sample residual $\|y - Ac_\alpha\|$?

Answer: It never decreases. At $\alpha = 0$ the residual is as small as any c can make it; as $\alpha \to \infty$, $c_\alpha \to 0$ and the residual approaches $\|y\|$, the residual of doing nothing. Ridge trades in-sample accuracy for smaller coefficients.

Key Ideas

- Ridge minimizes $\|y - Ac\|^2 + \alpha \|c\|^2$; its solution $(A^\top A + \alpha I)^{-1} A^\top y$ always exists and is unique (Equation (9.33)).
- In SVD form the data along u_j are multiplied by $\sigma_j/(\sigma_j^2 + \alpha)$ (Equation (9.34)); filter factors $\sigma_j^2/(\sigma_j^2 + \alpha)$.
- With $\alpha = \lambda\sigma_1^2$ the filter is $\sigma/(\sigma^2 + \lambda\sigma_1^2)$, the paper's form (Equation (9.36)).
- $\alpha \to 0$ gives the minimum-norm solution; $\alpha \to \infty$ gives $c = 0$ (the null); the gain never exceeds $1/(2\sqrt{\alpha})$.

9.9 Overfitting: the memorizing polynomial

Least squares makes the residuals on the data as small as possible. That is a statement about the data you have. This section shows, with the book's running example RE4, why it is not a statement about the data you do not have.

The data. We sample $\sin t$ at ten equally spaced times $t_j = 0.7j$, $j = 0, \ldots, 9$, and add fixed "measurement errors" ε_j of size 0.10 to 0.15 (Table 9.2). We chose the errors to alternate in sign, which makes the effect below easy to see; Problem 18 asks you to repeat the experiment with random errors.

Polynomial models and their conditioning. We fit polynomials of degree d, which have $M = d + 1$ coefficients. Rather than powers of t we use powers of the rescaled variable $\tau = t/3.15 - 1$, which runs from -1 to 1 over the data. (We write τ rather than u because u_j already denotes the left singular vectors.) A polynomial of degree d in τ is a polynomial of

Table 9.2: The RE4 data: $y_j = \sin t_j + \varepsilon_j$ at $t_j = 0.7j$.

j	0	1	2	3	4	5	6	7	8	9
t_j	0	0.7	1.4	2.1	2.8	3.5	4.2	4.9	5.6	6.3
$\sin t_j$	0.0000	0.6442	0.9854	0.8632	0.3350	−0.3508	−0.8716	−0.9825	−0.6313	0.0168
ε_j	0.15	−0.12	0.10	−0.15	0.12	−0.10	0.14	−0.13	0.11	−0.14
y_j	0.1500	0.5242	1.0855	0.7132	0.4550	−0.4508	−0.7316	−1.1125	−0.5213	−0.1232

degree d in t and vice versa, so this changes only the coordinates in coefficient space, not the set of models or the fitted curve. It does change the conditioning. With powers of t, the column t^9 has entries up to $6.3^9 \approx 1.6 \times 10^7$ while the column 1 has length $\sqrt{10}$, and Problem 27 shows this forces $\kappa > 10^6$; in the variable τ the condition number is far smaller: the listing below prints both condition numbers, and the first exceeds the second by more than a factor 100. Choosing a well-scaled basis is the first line of defense of Section 9.6.

```
import numpy as np
t = 0.7 * np.arange(10)          # the RE4 times
tau = t / 3.15 - 1               # rescaled to [-1, 1]
print(np.linalg.cond(np.vander(t, 10)))   # columns t^9, ..., t, 1
print(np.linalg.cond(np.vander(tau, 10)))   # columns tau^9, ..., tau, 1
```

Three fits. Figure 9.5 shows the least-squares polynomials of degree 1, 3 and 9.

- Degree 1 (a line) is too stiff to follow one full oscillation of $\sin t$: it *underfits*.
- Degree 3 follows the overall shape and smooths through the errors.
- Degree 9 has $M = 10$ coefficients for $m = 10$ points. The design matrix is square, and it is invertible because the times are distinct: if $Ac = 0$, the polynomial with coefficients c vanishes at ten distinct points, but a nonzero polynomial of degree at most 9 has at most 9 roots, so $c = 0$. (The root count follows from the factor theorem of school algebra: if t_0 is a root, the polynomial equals $(t - t_0)$ times a polynomial of one lower degree, so each distinct root uses up one degree.) Hence the degree-9 polynomial passes *exactly* through every data point, errors included. To do so it must wiggle, and between the data points, especially near the ends of the range, it swings far from $\sin t$.

Training error versus test error. The root mean square (rms) of numbers $r_1, \dots, r_m$ is $\sqrt{\frac{1}{m}\sum_i r_i^2} = \|r\|/\sqrt{m}$. Call the rms residual on the ten data points the *training error*, and the rms difference between the fitted polynomial and $\sin t$ over the whole interval $0 \le t \le 6.3$ the *test error* (it measures how well the fit predicts values it was not given). Figure 9.6 plots both against degree.

The training error can never increase with degree. Proof: every polynomial of degree $d-1$ is also a polynomial of degree d (with top coefficient zero), so the minimum of S over degree-d models is taken over a larger set and can only be smaller or equal. At $d = 9$ it reaches zero (up to rounding). The test error behaves differently: it falls from degree 1 to degree 3, and at degree 9 it is more than twice as large as at degree 3. Driving the training error to zero did not make the predictions better; it made them worse, because the extra coefficients were spent fitting the errors ε_j. This is *overfitting*, and a model that reproduces its training data exactly, errors included, has *memorized* them.

If we go further, to degree 15 (16 coefficients, 10 data), the problem is underdetermined: by Section 9.5 there is a null space of dimension at least 6 and infinitely many polynomials

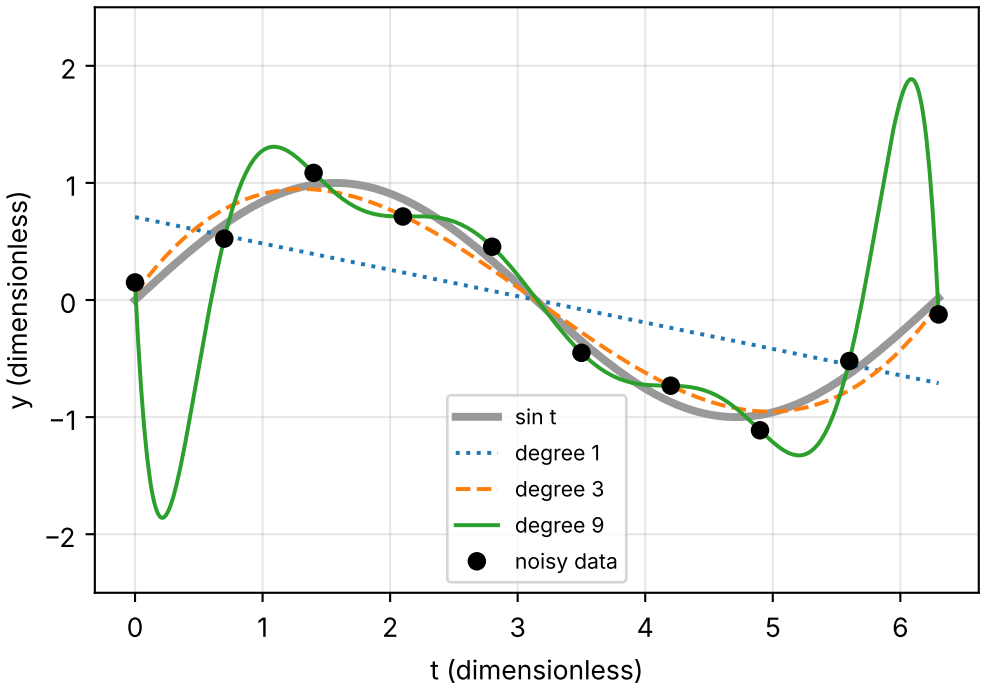


Figure 9.5: The memorizing polynomial (RE4). Points: the ten noisy samples of Table 9.2. Gray: $\sin t$. Least-squares polynomials of degree 1 (underfits), 3 (follows the shape) and 9 (passes through every point, errors included, and oscillates between them). Horizontal axis: t (dimensionless); vertical axis: y (dimensionless).

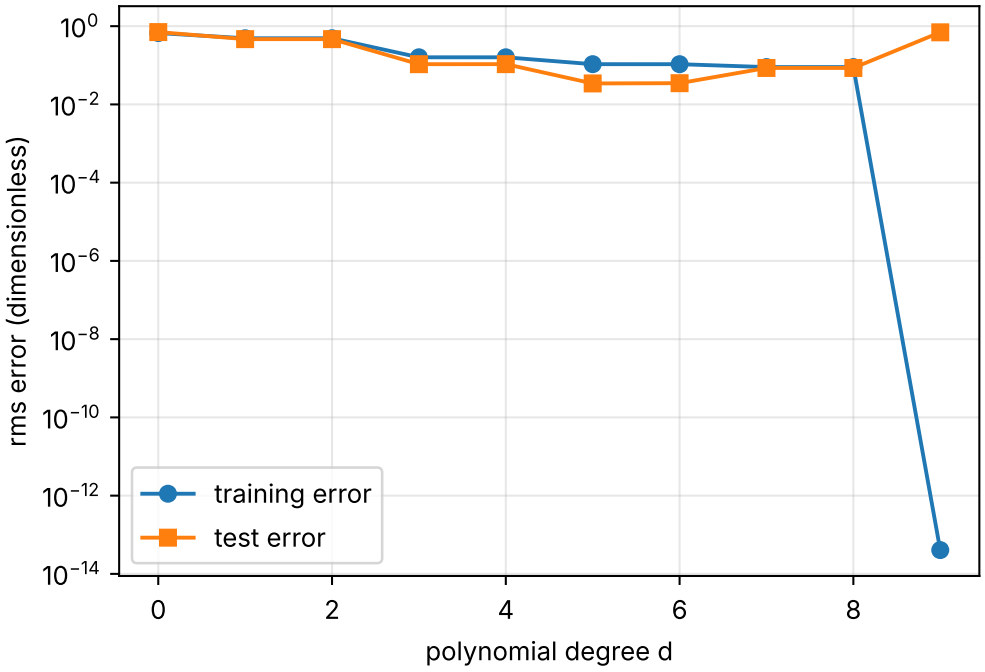


Figure 9.6: Training error (rms residual on the ten RE4 points) and test error (rms deviation from $\sin t$ on $0 \le t \le 6.3$) of least-squares polynomials against degree d. The training error never increases and reaches zero, to rounding, at $d = 9$ (plotted at a floor of 10^{-16}); the test error at $d = 9$ is more than twice that at $d = 3$. Errors dimensionless, log scale.

that pass exactly through all ten points. `lstsq` returns the minimum-norm one. They all have training error zero, so the training data cannot tell them apart at all, yet between and beyond the points they differ arbitrarily (add any null vector). Ridge regularization of the degree-9 or degree-15 fit pulls the coefficients toward zero, at the cost of a nonzero training error; whether that improves the predictions is examined, with the tools of statistics, in Chapter 25.

Sample Problem 9.7: Interpolation versus extrapolation with three points

Three points of $\sin t$ are known exactly: $(0,0)$, $(\pi/2,1)$, $(\pi,0)$. Fit (a) a least-squares line and (b) the parabola through all three, and use each to predict $\sin t$ at $t=\pi/4$ (inside the data range) and at $t=3\pi/2$ (outside it).

Step 1: line sums. $m=3$; $\sum t_i = 3\pi/2$; $\sum t_i^2 = 0+\pi^2/4+\pi^2 = 5\pi^2/4$; $\sum y_i = 1$; $\sum t_i y_i = \pi/2$.

Step 2: line normal equations. $3c_0 + \frac{3\pi}{2}c_1 = 1$ and $\frac{3\pi}{2}c_0 + \frac{5\pi^2}{4}c_1 = \frac{\pi}{2}$. Multiply the first by $\pi/2$: $\frac{3\pi}{2}c_0 + \frac{3\pi^2}{4}c_1 = \frac{\pi}{2}$. Subtract from the second: $\frac{\pi^2}{2}c_1 = 0$, so $c_1 = 0$ and $c_0 = \frac{1}{3}$. The line is $y = \frac{1}{3}$.

Step 3: parabola. Write $y = c_0 + c_1 t + c_2 t^2$. Through $(0,0)$: $c_0 = 0$. Through $(\pi,0)$: $c_1\pi + c_2\pi^2 = 0$, so $c_1 = -c_2\pi$. Through $(\pi/2,1)$: $c_1\frac{\pi}{2} + c_2\frac{\pi^2}{4} = 1$, i.e. $-c_2\frac{\pi^2}{2} + c_2\frac{\pi^2}{4} = 1$, so $c_2 = -4/\pi^2$ and $c_1 = 4/\pi$. Hence $y = \frac{4}{\pi^2}\,t(\pi - t)$.

Step 4: predictions at $t=\pi/4$ (truth $\sin\frac{\pi}{4} \approx 0.7071$). Line: 0.3333, error 0.374. Parabola: $\frac{4}{\pi^2}\cdot\frac{\pi}{4}\cdot\frac{3\pi}{4} = \frac{3}{4} = 0.75$, error 0.043.

Step 5: predictions at $t=3\pi/2$ (truth -1). Line: 0.3333, error 1.333. Parabola: $\frac{4}{\pi^2}\cdot\frac{3\pi}{2}\cdot\left(-\frac{\pi}{2}\right) = -3$, error 2.0.

at $\frac{\pi}{4}$: line 0.333, parabola 0.75;	at $\frac{3\pi}{2}$: line 0.333, parabola -3

Inside the data range the flexible exact fit wins easily; outside it, both models fail and the exact fit fails worse. Neither "knows" that the function oscillates. Predicting outside the range of the data, *extrapolation*, needs a model whose form matches the truth.

The right basis. What happens if the basis does contain the true form? The next sample uses the book's running example RE2, the prime-2 wave of Chapter 5,

$$F_2(t) = -\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 2)}{m\,2^{m/2}}, \tag{9.38}$$

which repeats with period $2\pi/\log 2 \approx 9.065$ in t. (Here m is the harmonic index, as in the paper's Eq. (4), not the number of data points.)

Sample Problem 9.8: Least squares recovers the prime-2 wave

Sample F_2 of Equation (9.38) at the first ten zeta-zero heights of RE1 ($t = 14.134725,\ldots,49.773832$) and fit the six-coefficient model $\sum_{m=1}^{3}[a_m\sin(mt\log 2) + b_m\cos(mt\log 2)]$ by least squares. What coefficients come out, and how well does the fit predict F_2 at other t?

Step 1: design matrix. Ten rows (one per t_i), six columns: $\sin(t_i\log 2)$, $\cos(t_i\log 2)$, $\sin(2t_i\log 2)$, $\cos(2t_i\log 2)$, $\sin(3t_i\log 2)$, $\cos(3t_i\log 2)$. Applying `numpy.linalg.matrix_rank` to this matrix (the listing after this sample) returns 6: its columns are independent, so the least-squares solution is unique.

Step 2: the true coefficients fit exactly. The data are $F_2(t_i)$, which is itself of the model form with $a_m = -1/(\pi m 2^{m/2})$ and $b_m = 0$. With these coefficients every residual is zero, so $S = 0$, the smallest possible value. A minimizer of S is therefore this coefficient

vector, and by uniqueness it is the only one.

Step 3: values. $a_1 = -1/(\pi\sqrt{2}) \approx -0.22508$, $a_2 = -1/(4\pi) \approx -0.07958$, $a_3 = -1/(6\sqrt{2}\,\pi) \approx -0.03751$, $b_1 = b_2 = b_3 = 0$. A numerical `lstsq` reproduces these to rounding error.

Step 4: prediction. With the exact coefficients the model *is* F_2, so it predicts F_2 at every t, far outside the sampled window included.

$$\boxed{a = (-0.22508,\ -0.07958,\ -0.03751), \quad b = 0;\ \text{prediction exact for all } t}$$

```
import numpy as np
t = np.array([14.134725, 21.022040, 25.010858, 30.424876, 32.935062,
              37.586178, 40.918719, 43.327073, 48.005151, 49.773832])
w = np.log(2) * np.array([1, 2, 3])
D = np.hstack([np.sin(np.outer(t, w)), np.cos(np.outer(t, w))])
print(np.linalg.matrix_rank(D))                # 6
F = -sum(np.sin(m*t*np.log(2))/(m*2**(m/2)) for m in (1, 2, 3))/np.pi
print(np.linalg.lstsq(D, F, rcond=None)[0])    # a1, a2, a3, then b's near 0
```

Contrast this with a generic basis. Figure 9.7 fits the same wave, sampled at 80 equally spaced points in $10 \le t \le 60$, with 120 sinusoids whose frequencies lie on a uniform grid that does not contain $\log 2$, $2\log 2$ or $3\log 2$. With 120 coefficients and 80 data the problem is underdetermined, and the minimum-norm solution fits the samples in the window essentially exactly. Beyond the window it drifts away from the true wave, while the three-frequency model continues it perfectly. The moral, developed in Chapters 25 and 27: *a fit generalizes when its basis encodes the right structure, not when it has many coefficients.*

Checkpoint 9.9

A friend fits 50 data points with a 60-coefficient model and reports "zero residual on all 50 points, so the model is correct." What is wrong with the argument?

Answer: With more coefficients than data, an exact fit is expected for almost any data (Section 9.5), and there are infinitely many exact fits differing by null-space vectors. Zero training residual says nothing about predictions at new points; only a test on held-out data can.

Key Ideas

- Training error never increases as you add basis functions; with as many coefficients as data it is zero.
- A model that fits its training data exactly, errors included, has memorized them; its predictions elsewhere can be poor (RE4).
- In these examples extrapolation succeeded when the model family contained the true form (RE2 with the right frequencies) and failed for a large generic basis, which fits the window but not beyond it; more coefficients did not help.

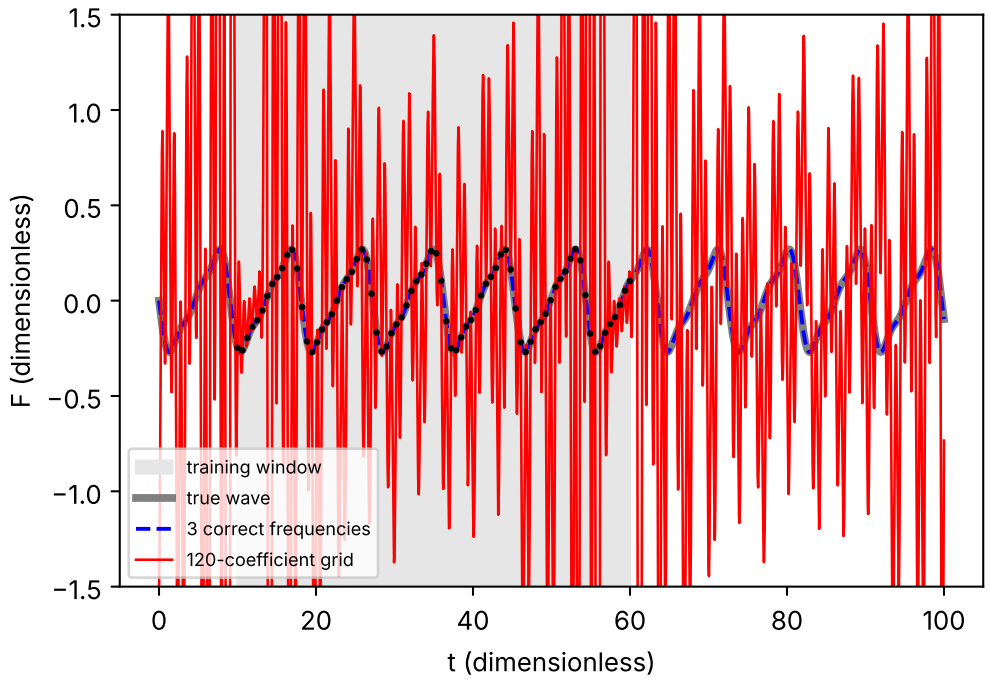


Figure 9.7: Fitting the prime-2 wave RE2 (Equation (9.38)) from 80 samples in the shaded window $10 \le t \le 60$. Gray: the true wave. Dashed: least-squares fit with the three correct frequencies $m \log 2$ (6 coefficients), which continues the wave exactly. Solid: minimum-norm fit with 120 coefficients on a uniform frequency grid $\omega_j = 8j/60$, $j = 1, \dots, 60$, which matches the samples in the window but drifts away from the wave outside it. t and F dimensionless.

9.10 Paper connection

★ Did you know? *Taming Ill-Posed Problems*

Andrey Tikhonov studied unstable inverse problems from the 1940s, and in 1963 he proposed adding a penalty term to stabilize them, now called Tikhonov regularization. Statisticians Arthur Hoerl and Robert Kennard independently popularized the same idea in 1970 as ridge regression. Both trade a little bias for a large reduction in noise amplification, the trade you will meet when fitting potentials to spectra.

This section reads the least-squares content of the paper [1] line by line. Quantities such as energy levels E_n and wavefunctions ψ_n are explained in Part II (Chapters 10 and 13) and in Chapter 23; here we need only that they are numbers and vectors computed from a matrix.

The fit is a sequence of least-squares problems (Sec. II C). The paper adds to the Wu–Sprung potential a correction $\delta V(x) = \sum_{k=0}^{M-1} c_k \cos(k\pi x/L)$, a linear model in the coefficients c_k with the cosine basis of Chapter 5. The energy levels E_n depend on c nonlinearly, so the fit proceeds by "iterated linearization": compute the levels E_n and wavefunctions ψ_n for the current c, form residuals $r_n = \tilde{t}_n - E_n$ (target minus model, exactly as in Equation (9.11); $\tilde{t}_n$ are the bias-corrected zero heights of Chapter 23), and form the first-order sensitivity matrix

$$A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\, \mathrm{d}x.$$

Taken on trust

Taken on trust until Chapter 13. The energy levels E_n and normalized wavefunctions ψ_n are defined in Chapters 10 and 11; here treat them as numbers and functions computed from the current potential. First-order perturbation theory, derived in Chapter 13, states that a small change δc of the coefficients changes level n by approximately $\sum_k A_{nk}\delta c_k$, with an error of second order in δc. We use this fact here without proof.

To first order, then, choosing the change δc that best removes the residuals is the linear least-squares problem

$$A\,\delta c \approx r, \tag{9.39}$$

with A playing the role of the design matrix (rows: training levels n; columns: basis functions k) and r the role of the data. (Do not confuse this matrix A_{nk} with the non-Hermitian operator A of the paper's Sec. II E; the paper uses the letter for both.)

Decoding Appendix A step 6. The paper's record states the update as

$$c \mathrel{+}= V\Big(\Sigma/(\Sigma^2 + \lambda\Sigma_1^2)\Big)\,U^\top r,$$

with a "try-list $(0.2\lambda, \lambda, 5\lambda, 25\lambda)$". In the notation of this chapter, $A = U\Sigma V^\top$ is the SVD of the sensitivity matrix, Σ the diagonal of singular values and $\Sigma_1 = \sigma_1$ the largest one, and the division is entry by entry. Here V is the matrix of right singular vectors, *not* the potential $V(x)$: another overloaded letter. Comparing with Equation (9.36), the step is

$$\delta c = \sum_j \frac{\sigma_j}{\sigma_j^2 + \lambda\sigma_1^2}\,(u_j^\top r)\,v_j, \tag{9.40}$$

which is precisely the ridge solution of Equation (9.39): the δc that minimizes $\|r - A\delta c\|^2 + \lambda\sigma_1^2\,\|\delta c\|^2$. Damping the *step* in this way is the Levenberg–Marquardt idea [32], developed in Chapter 24. In this step λ is the Levenberg–Marquardt damping (the book's notation, Appendix B: λ for damping, μ for ridge strength). Two limits make its role concrete (Problem 17):

- $\lambda \to 0$: $\delta c \to A^+ r$, the full minimum-norm linearized step (Gauss–Newton);
- large λ: $\sigma_j/(\sigma_j^2 + \lambda\sigma_1^2) \approx \sigma_j/(\lambda\sigma_1^2)$, so $\delta c \approx V\Sigma^\top U^\top r/(\lambda\sigma_1^2) = A^\top r/(\lambda\sigma_1^2)$, a short step along $A^\top r$. The *gradient* of S is the vector of its partial derivatives, $(\partial S/\partial c_1, \ldots, \partial S/\partial c_M) = -2A^\top r$ by Equation (9.13), so $A^\top r$ points opposite to the gradient. A small step in this direction decreases S fastest (shown in Chapter 24), which is why it is called the direction of *steepest descent*.

Because each component of Equation (9.40) shrinks as λ grows, larger λ always means a shorter, more cautious step. According to the record, a step is accepted only if it lowers the rms by the factor 0.999 ("accept only improvements (×0.999)"; one row of Table I, scale 2 with $M = 300$, used the looser threshold $1 - 10^{-5}$, and the paper reports that row as a sensitivity result); on success $\lambda \leftarrow \max(0.5\lambda_{\text{acc}}, 10^{-7})$, on failure of every value in the try-list $\lambda \leftarrow 30\lambda$, and the iteration stops when $\lambda > 10^4$. Chapter 24 explains this schedule; the linear algebra inside each step is this chapter's.

Sample Problem 9.9: One damped step in miniature

Apply the step Equation (9.40) with $\lambda = 0.01$ to the matrix $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$ of the earlier SVD sample and residual $r = (1, 1)$, and compare with the undamped step $A^{-1}r$.

Step 1: SVD (from the earlier sample). $\sigma_1 = 3\sqrt{5}$, $\sigma_2 = \sqrt{5}$, $u_1 = (1,3)/\sqrt{10}$, $u_2 = (3,-1)/\sqrt{10}$, $v_1 = (1,1)/\sqrt{2}$, $v_2 = (1,-1)/\sqrt{2}$.
Step 2: data components. $u_1^\top r = 4/\sqrt{10} \approx 1.26491$; $u_2^\top r = 2/\sqrt{10} \approx 0.63246$.
Step 3: damped gains. $\lambda\sigma_1^2 = 0.01 \times 45 = 0.45$. Gain 1: $\sigma_1/(\sigma_1^2 + 0.45) = 6.70820/45.45 \approx 0.147595$. Gain 2: $\sigma_2/(\sigma_2^2 + 0.45) = 2.23607/5.45 \approx 0.410288$.
Step 4: assemble. $\delta c = 0.147595 \times 1.26491\, v_1 + 0.410288 \times 0.63246\, v_2 = 0.186695\, v_1 + 0.259489\, v_2$. With $v_{1,2} = (1, \pm 1)/\sqrt{2}$: $\delta c = ((0.186695 + 0.259489)/\sqrt{2},\ (0.186695 - 0.259489)/\sqrt{2}) \approx (0.3155,\ -0.0515)$.
Step 5: undamped step. Solving $A\delta c = r$: $3\delta c_1 = 1$ gives $\delta c_1 = 1/3$; then $4/3 + 5\delta c_2 = 1$ gives $\delta c_2 = -1/15 \approx -0.0667$.

$$\boxed{\delta c_{\lambda=0.01} \approx (0.3155,\ -0.0515) \quad \text{vs} \quad A^{-1} r = (0.3333,\ -0.0667)}$$

The damped step is shorter; the component along the weaker direction v_2 is shrunk more (gain 0.410 instead of $1/\sigma_2 = 0.447$) than the one along v_1 (gain 0.1476 instead of $1/\sigma_1 = 0.1491$).

What the sensitivity matrix looks like. Figure 9.8 shows the singular values of A_{nk} computed with the companion library `qmnt` for the unperturbed potential (the first iteration of the fit, $c = 0$) in the scale-1 box of RE3 ($L = 44.883$, the paper's working grid $N = 6000$), for the first 400 levels and $M = 400$ or 900 cosines. One column is known exactly: for $k = 0$, $\cos 0 = 1$ and $A_{n0} = \int |\psi_n|^2 \, dx = 1$ for every level, because the wavefunctions are normalized. A constant shift of the potential raises every level by the same amount. A 400×900 matrix has at most 400 nonzero singular values, so at $M = 900$ at least 500 directions of coefficient space are invisible to the 400 training levels, to first order. Read the figure with Section 9.6 in mind. Directions v_j whose normalized singular value σ_j/σ_1 lies far below 1 are poorly constrained by the training levels, and the damping filter of Equation (9.40) shrinks the step along them much more than along the well-constrained directions.

Ridge in the operator fits (Tables I and II, Sec. III). The record says the "ridge variant adds $\mu\Sigma_1^2$ to the filter denominator". In the language of Section 9.8, μ is a ridge strength measured relative to σ_1^2, entering the same filter denominator as λ in Equation (9.36), but playing a different role. The damping λ is adapted during the iteration and driven small after successful steps, so it controls only how long each step is: iterated to convergence, a damped step need not change where the fit ends up. The ridge term $\mu\sigma_1^2$ is fixed and stays in the filter at every step, so the steps never become full Gauss–Newton steps along poorly constrained directions. This chapter does not derive how a fixed μ in the step filter changes the converged fit. The full specification of how the ridge enters the iteration is in the paper's Zenodo record [20], and Chapter 24 discusses it. The effect matches the limits derived in Section 9.8. In Table I (scale 1, fit zeros 1–400, $M = 400$), the unregularized fit has in-sample rms 0.137 and held-out rms 0.369; with $\mu = 0.03$ these become 0.142 and 0.271; with $\mu = 0.3$, 0.206 and 0.265, against a null of 0.263 on zeros 401–500 (RE5). As ridge strength grows, the in-sample error rises (the fit memorizes less) and the held-out error approaches the null from above: the fitted correction stays closer to $c = 0$, "doing nothing". The paper summarizes: "regularization moves them smoothly from 'memorize more' to 'do nothing' and never creates skill" (Sec. III). Table II shows the same for the Chebyshev basis ($\mu = 0.3$: held-out 0.264), and in the interpolation split of Sec. III ridge brings the gap error from 2.64 spacings back to 0.262 ($\mu = 0.03$) and 0.257 ($\mu = 0.3$), close to that split's null 0.255. This behavior is consistent with the picture of Section 9.6: ridge suppresses large excursions along poorly constrained directions (Equation (9.28)), and it cannot add information the parametrization lacks. The paper's own

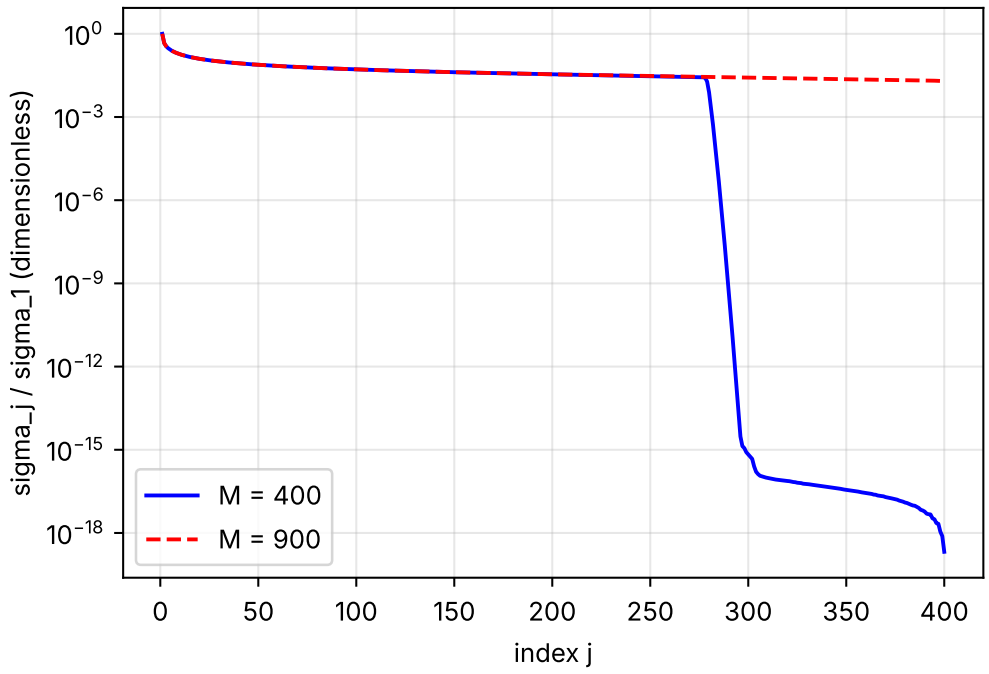


Figure 9.8: Normalized singular values σ_j/σ_1 of the first-order sensitivity matrix $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\,\mathrm{d}x$ of the paper's Sec. II C, for the first 400 levels of the unperturbed Wu–Sprung potential in the scale-1 box ($L = 44.883$, $N = 6000$ interior grid points, units $\hbar = 2m = 1$), with $M = 400$ and $M = 900$ cosine basis functions. The 400×900 matrix has only 400 singular values; the remaining 500 directions of its coefficient space form (at least) its null space. Computed with `qmnt`.

explanation of the gap damage is heuristic and stated in its own terms: the optimizer "places large structure in the part of the potential that only the held-out levels probe strongly, because doing so costs nothing in-sample" (Sec. III). The paper does not attribute the damage to noise amplification, and neither do we.

Saturated fits (Sec. III, Table III). Sec. III reports that on all 500 scale-1 zeros the fit reaches rms 0.186 at $M = 300$ and reproduces all 500 to rms $< 10^{-4}$ spacings by $M = 900$. With 900 coefficients and 500 levels, the linearized problem Equation (9.39) is underdetermined: if the 500×900 matrix has full row rank, exact fits exist and differ by vectors of a null space of dimension at least 400 (Section 9.5). The underlying problem is nonlinear, so this is a first-order heuristic, but it suggests why near-perfect in-sample agreement is unsurprising and, by analogy with RE4, why such agreement need not carry information about held-out levels. (This argument is ours; the paper reports the observation, not this derivation.) Similarly, Sec. III notes that the odd/even splits of Table III have 200 training levels at scale 1, "so the rows with $M \geq 200$ are saturated"; at $M = 400$ the in-sample error is below 10^{-3}.

The fitted sinusoidal controls (Secs. II D and IV, Table IV, Appendix A step 20). The controls fit $F(t) = \sum_j [a_j \sin(\omega_j t) + b_j \cos(\omega_j t)]$ to $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ at the training zeros "by least squares". (Here N_{sm} is the smooth zero-counting function of the paper's Eq. (1), Chapter 17; for the least squares it is just a known function of t.) With fixed frequencies this is linear least squares exactly as in the RE2 sample, with 2 coefficients per frequency. Control B supplies the frequencies $m \log p$ ($m \leq 3$): with 25 primes that is $25 \times 3 \times 2 = 150$ coefficients, which "learned from 400 zeros give 0.028". With 150 primes it is 900 coefficients, and the paper writes: "At 900 coefficients on 400 zeros the least-squares problem is underdetermined and the model degrades to 0.180 with correlation +0.78" (Sec. IV); Section 9.5 explains the word: 400 equations, 900 unknowns, a null space of dimension at least 500, regularized only by the ridge. With 800 training zeros it gives 0.100/+0.94 (Table IV); note that even then $900 > 800$, so the problem is still underdetermined, with a null space of dimension at least 100 (Problem 15). Control C1 uses 450 frequencies on the uniform grid $\omega_j = 8j/450$, again 900 coefficients, and stays within 0.005 of the null (0.258 against 0.263): the generic-basis lesson of Figure 9.7 at full scale.

Appendix A step 20 gives the controls' "ridge filter $\Sigma/(\Sigma^2 + \lambda\Sigma_1^2)$, $\lambda = 10^{-8}, \ldots, 10^3$ in half decades", which is Equation (9.36) with λ now a ridge strength (in step 6 the same letter is the damping). "In half decades" means exponents $-8, -7.5, \ldots, 3$: 23 candidate values. The value is chosen by a forward split inside the training window ("first 75% fit, last 25% validate", Sec. II D), the validation idea of Section 9.8; held-out zeros are not used. Table IV's B row at 900 coefficients is also marked root-ambiguous; that issue concerns the level equation, not the least squares, and is treated in Chapter 27.

The lesson in one sentence. Least squares guarantees the best fit to the data it is given, and the SVD tells you which directions of the model those data constrain. Everything the paper measures beyond that, whether a fit predicts held-out zeros, lies outside what least squares can promise, which is why the paper's central test is out of sample (Chapter 25).

9.11 Summary

- Least squares minimizes $S(c) = \|y - Ac\|^2$, the sum of squared residuals of a linear model $\sum_k c_k \phi_k$ with design matrix $A_{ik} = \phi_k(t_i)$ (Equations (9.10) and (9.12)).
- Minimizers satisfy the normal equations $A^\top A c = A^\top y$ (Equation (9.14)); every solution is a global minimum by Equation (9.15); it is unique iff the columns of A are independent.

For a line, Equation (9.8).

- Geometrically, $A\hat{c}$ is the orthogonal projection of y onto the column space; $\|y\|^2 = \|\hat{y}\|^2 + \|\hat{r}\|^2$ (Equation (9.16)).
- Every matrix has an SVD $A = U\Sigma V^\top = \sum_j \sigma_j u_j v_j^\top$ (Equations (9.18) and (9.19)), with σ_j^2 the eigenvalues of $A^\top A$.
- The minimum-norm least-squares solution is $c^+ = A^+ y = \sum_j (u_j^\top y/\sigma_j) v_j$ (Equation (9.22)); underdetermined problems have infinitely many exact fits $c^+ + z$, $Az = 0$ (Equation (9.23)).
- The condition number $\kappa = \sigma_1/\sigma_M$ bounds error amplification (Equation (9.27)); noise along u_j is divided by σ_j (Equation (9.28)); $\kappa(A^\top A) = \kappa^2$ (Equation (9.29)).
- Truncated SVD and ridge are filters (Equation (9.31)). Ridge solves $(A^\top A + \alpha I)c = A^\top y$ (Equation (9.33)), multiplies the data along u_j by $\sigma_j/(\sigma_j^2 + \alpha)$ (Equation (9.34)), and with $\alpha = \lambda\sigma_1^2$ takes the paper's form (Equation (9.36)); its gain never exceeds $1/(2\sqrt{\alpha})$ (Equation (9.37)).
- Training error never increases with more basis functions; an exact fit of noisy data is memorization (RE4); in the RE2 example, extrapolation worked with the right frequencies and failed with a large generic basis.
- The paper's fit step (App. A step 6) is the ridge-damped linearized step (Equation (9.40)); its ridge rows return toward the null; its controls B and C1 are linear least squares, underdetermined at 900 coefficients on 400 zeros.

9.12 Problems

1. • Find the least-squares line $y = c_0 + c_1 t$ through $(0,0)$, $(1,1)$, $(2,3)$. Give the residuals and $S_{\min}$.

2. • Show that for any least-squares line with an intercept, the residuals sum to zero. Does this remain true for the proportional model $y = c_1 t$ (no intercept)?

3. • For the design matrix $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$ compute $A^\top A$, and for $y = (0,1,3)$ compute $A^\top y$. Compare with Problem 1.

4. • Find the singular values and condition number of $A = \mathrm{diag}(-4, 2)$. Where does the minus sign go in the SVD?

5. • Find the minimum-norm solution of $3c_1 + 4c_2 = 10$ using the SVD, and its length.

6. • Project $y = (1,2,3)$ onto the line spanned by $a = (1,1,1)$. Show that fitting a constant model $y \approx c$ by least squares gives the average of the data.

7. •• Derive the least-squares slope of the proportional model $y = kt$, and evaluate it for the data of Problem 1.

8. •• For A with independent columns and $P = A(A^\top A)^{-1}A^\top$, prove that $P^\top = P$, $P^2 = P$, and $(I - P)A = 0$. Interpret $(I - P)y$.

9. •• Find the SVD of $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, its pseudo-inverse, the minimum-norm least-squares solution for $y = (2,3)$, and the residual.

10. •• Repeat the ridge sample problem with $y = (1,\ 0.015)$. Which α makes the ridge estimate of c_2 exactly equal to the true value 1?

11. •• A least-squares problem has $\sigma_1 = 10$ and is solved by ridge in the relative form with $\lambda = 10^{-4}$. What is the largest gain any data component can receive, and at which singular value?

12. •• Compute the condition number of $\begin{pmatrix} 1 & 2 \\ 2 & 4.001 \end{pmatrix}$. Solve $Ac = (3,\ 6.001)$, then change the right-hand side to $(3,\ 6.002)$ and solve again.

13. •• RE1 meets least squares. Fit a line $t_n \approx c_0 + c_1 n$ to the first ten zeta-zero heights of RE1 ($n = 1, \dots, 10$). Compare the slope with the local mean spacing $2\pi/\log(t/2\pi)$ (Chapter 17) evaluated at the average height of the ten zeros.

14. •• Paper. Using Table I of the paper (scale 1, fit zeros 1–400, $M = 400$), compute the ratio of held-out rms to the null 0.263 for the unregularized fit and for $\mu = 0.03$ and $\mu = 0.3$. Explain, using the limits of Section 9.8, why the ratio approaches 1 and why the in-sample error rises.

15. •• Paper. Count the fitted coefficients of control B with 25 primes and with 150 primes ($m \le 3$, a sine and a cosine amplitude per frequency) and of control C1 (Sec. II D). For each training set (400 or 800 zeros) give the minimum dimension of the null space of the design matrix. Which configurations are underdetermined?

16. •• Paper. Appendix A step 20 scans $\lambda = 10^{-8}, \dots, 10^{3}$ in half decades. How many values is that? For $\sigma = \sigma_1$ and $\lambda = 10^{3}$, and for $\sigma = 0.01\sigma_1$ and $\lambda = 10^{-8}$, evaluate the filter factor $\sigma^2/(\sigma^2 + \lambda\sigma_1^2)$.

17. •• Paper. Show that the step Equation (9.40) tends to $A^{+}r$ as $\lambda \to 0$ and that for large λ it is approximately $A^{\top}r/(\lambda\sigma_1^2)$. Check the large-$\lambda$ formula numerically for the miniature sample ($A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$, $r = (1, 1)$) at $\lambda = 10^{6}$.

18. •• Using numpy, fit polynomials of degree 0 to 9 to the RE4 data of Table 9.2 and reproduce Figure 9.6. Then replace the fixed errors by random ones from `np.random.default_rng(seed).normal(0, 0.12, 10)` for five seeds. Does the qualitative picture survive?

19. •• Using numpy, reproduce the prime-2 sample: sample F_2 at the RE1 heights, fit the six-coefficient model, and print the coefficients. Then reproduce Figure 9.7 and compute the rms error of each model on $60 < t \le 100$.

20. •• Using qmnt, build the sensitivity matrix A_{nk} for the first 100 levels of the Wu–Sprung potential at scale 1 on a reduced grid of $N = 2000$ points, with $M = 150$ cosines (normalize eigenvectors so $\sum \psi^2 h = 1$). Verify that $A_{n0} = 1$ for all n, and state the largest possible rank and the smallest possible null-space dimension.

21. •• Prove $\kappa(A^{\top}A) = \kappa(A)^2$ for A with independent columns. Using numpy, check it numerically for the 30×6 design matrix of a degree-5 polynomial in τ on 30 equally spaced points of $[-1, 1]$.

22. •• Redo the interpolation-versus-extrapolation sample with the middle point measured as 1.2 instead of 1. Find the new line and parabola, and their predictions at $t = \pi/4$ and $t = 3\pi/2$.

23. ••• Prove from Equation (9.34) that $c_\alpha \to c^{+}$ as $\alpha \to 0$, that c_α is perpendicular to the null space of A for every $\alpha > 0$, and that $\|c_\alpha\|$ decreases as α increases.

24. • • • Show that the augmented matrix $\tilde{A} = \begin{pmatrix} A \\ \sqrt{\alpha} I \end{pmatrix}$ has singular values $\sqrt{\sigma_j^2 + \alpha}$ (with $\sigma_j = 0$ for $j > \varrho$), and hence that its condition number is $\sqrt{(\sigma_1^2 + \alpha)/(\sigma_M^2 + \alpha)}$.

25. • • • Best ridge strength. For a single component with singular value σ, true coefficient x and data $\sigma x + e$, the ridge estimate is $\sigma(\sigma x + e)/(\sigma^2 + \alpha)$. Suppose the error e averages to zero over repeated measurements and its square averages to ε^2. Show that the average squared error of the estimate is $(\alpha^2 x^2 + \sigma^2 \varepsilon^2)/(\sigma^2 + \alpha)^2$, that it is minimized at $\alpha^* = \varepsilon^2 / x^2$ (independent of σ), and evaluate the minimum for $\sigma = 0.01$, $\varepsilon = 0.01$, $x = 1$. Compare with $\alpha = 0$ and with truncation.

26. • • • Paper. In Sec. III, all 500 scale-1 levels are reproduced to rms $< 10^{-4}$ spacings by $M = 900$, and in Table III the odd split at $M = 400$ (200 training levels) has in-sample error $< 10^{-3}$. Using the linearized problem Equation (9.39), give the minimum null-space dimension in each case, explain why such in-sample accuracy is expected, and explain why it carries no information about held-out levels. State clearly which part of your argument is heuristic.

27. • • • For the RE4 data, show without computing an SVD that the design matrix with columns $1, t, \ldots, t^9$ has $\kappa > 10^6$. (Hint: $\sigma_1 \geq \|Ae_k\|$ and $\sigma_{\min} \leq \|Ae_k\|$ for every unit vector e_k; prove both from the SVD.)

9.13 Answers to selected problems

1. $m = 3$, $\sum t = 3$, $\sum t^2 = 5$, $\sum y = 4$, $\sum ty = 7$. Normal equations: $3c_0 + 3c_1 = 4$, $3c_0 + 5c_1 = 7$. Subtracting, $2c_1 = 3$, so $c_1 = 1.5$ and $c_0 = (4 - 4.5)/3 = -1/6$. Predictions $-1/6,\ 4/3,\ 17/6$; residuals $(1/6,\ -1/3,\ 1/6)$, which sum to zero; $S_{\min} = 1/36 + 1/9 + 1/36 = 1/6 \approx 0.167$.

2. The condition $\partial S/\partial c_0 = 0$ is $-2\sum r_i = 0$ Equation (9.3). In general the normal equations say $a_k^\top r = 0$ for each column; the intercept's column is all ones, so $\sum_i r_i = 0$. For $y = c_1 t$ there is no column of ones; the only condition is $\sum t_i r_i = 0$, and $\sum r_i$ need not vanish (for Problem 7's fit, residuals are $0,\ -0.4,\ 0.2$, sum -0.2).

3. $A^\top A = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$ (entries: 1+1+1; 0+1+2; 0+1+4) and $A^\top y = (0{+}1{+}3,\ 0{+}1{+}6) = (4, 7)$. These are exactly the sums of Problem 1: the matrix form Equation (9.14) reproduces the line's normal equations.

4. $A^\top A = \mathrm{diag}(16, 4)$, so $\sigma_1 = 4$, $\sigma_2 = 2$, $\kappa = 2$. Take $V = I$; then $u_1 = Av_1/\sigma_1 = (-4, 0)/4 = (-1, 0)$ and $u_2 = (0, 1)$. The minus sign sits in U; singular values are never negative.

5. $A = (3\ \ 4)$, $A^\top A = \begin{pmatrix} 9 & 12 \\ 12 & 16 \end{pmatrix}$ with eigenvalues 25 and 0. So $\sigma_1 = 5$, $v_1 = (3, 4)/5$, $u_1 = Av_1/5 = (25/5)/5 = 1$. Then $c^+ = (u_1 y/\sigma_1)v_1 = (10/5)(3, 4)/5 = (1.2,\ 1.6)$, with $\|c^+\| = \sqrt{1.44 + 2.56} = 2$. Check: $3(1.2) + 4(1.6) = 10$.

6. The design matrix is the single column a; $a^\top a = 3$, $a^\top y = 6$, so $c = 2$ and the projection is $(2, 2, 2)$, with residual $(-1, 0, 1)$ perpendicular to a. For a general constant model the column is all ones, $A^\top A = m$, $A^\top y = \sum y_i$, so $c = \frac{1}{m}\sum y_i = \bar{y}$.

7. $S(k) = \sum (y_i - kt_i)^2$; $\mathrm{d}S/\mathrm{d}k = -2\sum t_i(y_i - kt_i) = 0$ gives $k = \sum t_i y_i / \sum t_i^2$. For Problem 1's data, $k = 7/5 = 1.4$, with residuals $0,\ -0.4,\ 0.2$.

9. $A^\top A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has eigenvalues 2, 0; $\sigma_1 = \sqrt{2}$, $v_1 = (1, 1)/\sqrt{2}$, $u_1 = Av_1/\sqrt{2} = (1, 0)$, null direction $v_2 = (1, -1)/\sqrt{2}$. $A^+ = v_1 u_1^\top / \sigma_1 = \begin{pmatrix} 0.5 & 0 \\ 0.5 & 0 \end{pmatrix}$. Then $c^+ = A^+ y = (1, 1)$,

$Ac^+ = (2, 0)$, residual $(0, 3)$: the second data component lies outside the column space and no model can reach it.

11. $\alpha = \lambda\sigma_1^2 = 10^{-4} \times 100 = 0.01$. By Equation (9.37) the maximum gain is $1/(2\sqrt{0.01}) = 5$, reached at $\sigma = \sqrt{\alpha} = 0.1$ (that is, $\sigma/\sigma_1 = 0.01$).

13. With $\bar{n} = 5.5$, $\sum(n - \bar{n})^2 = 82.5$, $\sum t_n = 343.1385$ (so $\bar{t} = 34.3139$) and $\sum(n - \bar{n})t_n = 318.674$, the slope is $c_1 = 318.674/82.5 \approx 3.863$ and $c_0 = \bar{t} - 5.5c_1 \approx 13.07$. The mean spacing at $t = 34.31$ is $2\pi/\log(34.31/2\pi) \approx 2\pi/1.698 \approx 3.70$. The ratio is $3.863/3.70 \approx 1.044$, so the fitted slope is about 4% larger: the slope of a line through ten zeros is an average spacing over the range, the zeros fluctuate around their smooth positions, and the spacing itself shrinks with height. Ten points are few.

15. B, 25 primes: $25 \times 3 \times 2 = 150$ coefficients; B, 150 primes: 900; C1: $450 \times 2 = 900$. With 400 training zeros: 150 coefficients give a null space of dimension ≥ 0 (overdetermined, it can be trivial); 900 give ≥ 500. With 800: ≥ 0 and ≥ 100. The 900-coefficient models (B with 150 primes, C1) are underdetermined at both training sizes; the ridge makes their solutions unique.

17. For each $\sigma_j > 0$, $\sigma_j/(\sigma_j^2 + \lambda\sigma_1^2) \to 1/\sigma_j$ as $\lambda \to 0$, giving $\sum_j (u_j^\top r/\sigma_j)v_j = A^+ r$. For large λ, $\sigma_j^2 \ll \lambda\sigma_1^2$ for all j, so the gain is $\approx \sigma_j/(\lambda\sigma_1^2)$ and $\delta c \approx \sum_j \sigma_j(u_j^\top r)v_j/(\lambda\sigma_1^2) = A^\top r/(\lambda\sigma_1^2)$ by the transposed outer-product form $A^\top = \sum_j \sigma_j v_j u_j^\top$. Numerically, $A^\top r = (7, 5)$ and $\lambda\sigma_1^2 = 4.5 \times 10^7$, giving $(1.556 \times 10^{-7},\ 1.111 \times 10^{-7})$; the exact damped step agrees with this to a relative difference below 10^{-5}, consistent with corrections of relative order $\sigma_j^2/(\lambda\sigma_1^2) \leq 10^{-6}$.

19. The fitted coefficients agree with $a = (-0.22508, -0.07958, -0.03751)$, $b = 0$ to rounding. On $60 < t \leq 100$ the three-frequency model has rms error below 10^{-10} (rounding level). The 120-coefficient grid model fits the 80 in-window samples to rms below 10^{-6}, yet on $60 < t \leq 100$ its rms error exceeds 0.015. For comparison, the long-time rms of the wave is $\sqrt{(a_1^2 + a_2^2 + a_3^2)/2} \approx 0.171$ (each sinusoid of amplitude a has mean square $a^2/2$, and cross terms between different frequencies average to zero). The grid model's in-window agreement therefore does not carry over to a faithful continuation, while the three-frequency model continues the wave exactly.

21. $A^\top A$ is symmetric positive semidefinite with eigenvalues $\sigma_j^2 \geq 0$; for such a matrix the singular values equal the eigenvalues (its SVD is its spectral decomposition). So $\kappa(A^\top A) = \sigma_1^2/\sigma_M^2 = \kappa(A)^2$. Numerically, for the 30×6 matrix the ratio $\kappa(A^\top A)/\kappa(A)^2$ equals 1 to within 10^{-6}.

23. By Equation (9.34), $c_\alpha = \sum_{j \leq \varrho} g_j(\alpha)(u_j^\top y)v_j$ with $g_j(\alpha) = \sigma_j/(\sigma_j^2 + \alpha)$. (i) For each $\sigma_j > 0$, $g_j(\alpha) \to 1/\sigma_j$ as $\alpha \to 0$; a finite sum of convergent terms converges, so $c_\alpha \to \sum_{j \leq \varrho}(u_j^\top y/\sigma_j)v_j = c^+$. (ii) c_α is a combination of $v_1, \ldots, v_\varrho$; the null space is spanned by $v_{\varrho+1}, \ldots, v_M$, and the v_j are orthonormal, so c_α is perpendicular to every null vector. (iii) By orthonormality of the v_j, $\|c_\alpha\|^2 = \sum_{j \leq \varrho} \sigma_j^2 (u_j^\top y)^2/(\sigma_j^2 + \alpha)^2$. Each term has a fixed nonnegative numerator and a denominator that increases with α, so each term, and therefore the sum, decreases (strictly, unless every $u_j^\top y$ with $j \leq \varrho$ vanishes).

24. $\tilde{A}^\top \tilde{A} = A^\top A + \alpha I = V(\Sigma^\top \Sigma + \alpha I)V^\top$, whose eigenvalues are $\sigma_j^2 + \alpha$ for $j = 1, \ldots, M$ (zero σ_j included). The singular values of $\tilde{A}$ are their square roots, and the ratio of largest to smallest is $\sqrt{(\sigma_1^2 + \alpha)/(\sigma_M^2 + \alpha)}$, finite even when $\sigma_M = 0$.

25. The error of the estimate is $\frac{\sigma(\sigma x + e)}{\sigma^2 + \alpha} - x = \frac{\sigma^2 x + \sigma e - \sigma^2 x - \alpha x}{\sigma^2 + \alpha} = \frac{\sigma e - \alpha x}{\sigma^2 + \alpha}$. Its square is $(\sigma^2 e^2 - 2\sigma\alpha x e + \alpha^2 x^2)/(\sigma^2 + \alpha)^2$. Averaging, with e averaging to 0 and e^2 to ε^2, gives $E(\alpha) = (\alpha^2 x^2 + \sigma^2\varepsilon^2)/(\sigma^2 + \alpha)^2$. By the quotient rule,

$$E'(\alpha) = \frac{2\alpha x^2(\sigma^2 + \alpha)^2 - 2(\sigma^2 + \alpha)(\alpha^2 x^2 + \sigma^2\varepsilon^2)}{(\sigma^2 + \alpha)^4} = \frac{2[\alpha x^2(\sigma^2 + \alpha) - \alpha^2 x^2 - \sigma^2\varepsilon^2]}{(\sigma^2 + \alpha)^3} = \frac{2\sigma^2(\alpha x^2 - \varepsilon^2)}{(\sigma^2 + \alpha)^3}.$$

This is negative for $\alpha < \varepsilon^2/x^2$ and positive above, so the minimum is at $\alpha^* = \varepsilon^2/x^2$, independent of σ. Substituting, $\alpha^{*2}x^2 + \sigma^2\varepsilon^2 = \varepsilon^2\alpha^* + \sigma^2\varepsilon^2 = \varepsilon^2(\alpha^* + \sigma^2)$, so $E(\alpha^*) = \varepsilon^2/(\sigma^2 + \alpha^*)$. For

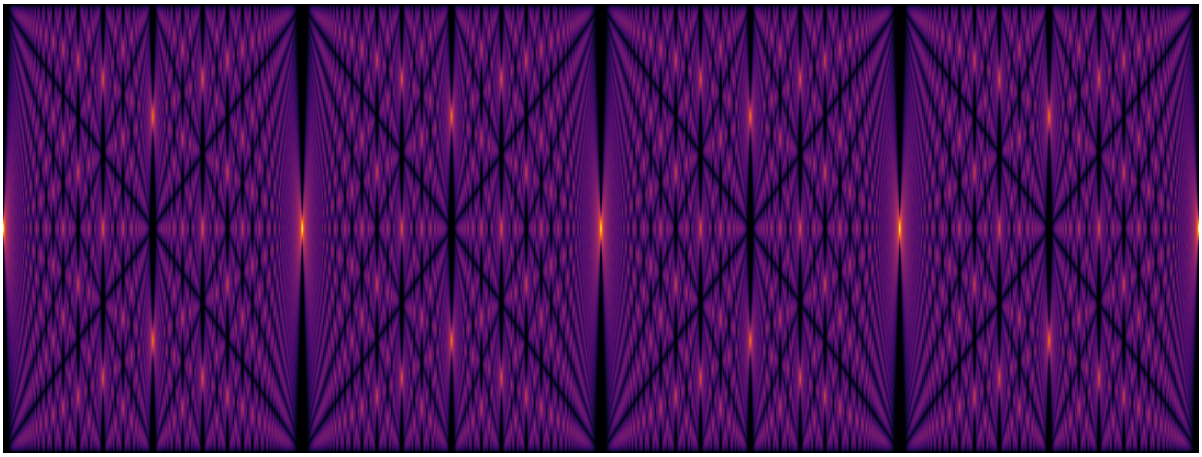
$\sigma = \varepsilon = 0.01$ and $x = 1$: $\alpha^* = 10^{-4}$ and $E = 10^{-4}/(2 \times 10^{-4}) = 0.5$. At $\alpha = 0$, $E = \varepsilon^2/\sigma^2 = 1$; truncation (estimate 0) has squared error $x^2 = 1$. The best ridge halves the average squared error of both.

26. $M = 900$ with 500 levels: null space of dimension ≥ 400; the odd split at $M = 400$ with 200 levels: ≥ 200. If the sensitivity matrix has full row rank, the linearized problem has exact solutions (Section 9.5), so iterating such steps can drive in-sample error to nearly zero for almost any targets, consistent with the surrogate fits of Sec. III. Held-out levels are not rows of the matrix: the fit is free to do anything along directions they alone would constrain, exactly as the degree-15 RE4 polynomials differ arbitrarily away from their data. Heuristic part: the real problem is nonlinear and the sensitivity matrix changes along the iteration; the rank and null-space statements hold only for each linearization.

27. For a unit vector e_k with SVD coordinates $\xi = V^\top e_k$ ($\|\xi\| = 1$), $\|Ae_k\|^2 = \sum_j \sigma_j^2 \xi_j^2$, which lies between σ_M^2 and σ_1^2 because $\sum \xi_j^2 = 1$. Hence $\sigma_1 \geq \|Ae_k\|$ and $\sigma_M \leq \|Ae_k\|$. The t^9 column has norm at least its largest entry $6.3^9 \approx 1.563 \times 10^7$; the column of ones has norm $\sqrt{10} \approx 3.162$. So $\kappa \geq 1.563 \times 10^7/3.162 \approx 4.9 \times 10^6 > 10^6$.

Part II

Quantum Mechanics of a Particle on a Line



A quantum carpet: $|\psi(x,t)|^2$ for a wave packet bouncing in a box (time runs left to right); the pattern is pure interference.

Chapter 10

From Waves to Wavefunctions

↑Contents

What this chapter is about

Pluck a guitar string and you hear a definite note, together with a few overtones at two, three, four times its frequency. You do not hear a smear of every possible pitch, because the string is pinned at both ends and only waves that fit between the pins survive. Heat a gas of hydrogen and it glows in a few sharp colors, not in a continuous rainbow. Early in the twentieth century physicists realized that these two facts are cousins: an electron bound in an atom behaves like a wave confined to a region, and only certain wave patterns, with certain energies, fit. The set of allowed energies is called the *spectrum* of the system. This chapter builds the mathematics of that idea from what you already know about waves on strings. We meet the de Broglie relation that assigns a wavelength to a particle, construct (by motivated guessing, which is honestly all anyone can do) the Schrödinger equation, separate it into a time part and a space part, and arrive at the eigenvalue equation $H\psi = E\psi$. We learn what the wavefunction ψ means (the Born rule), how to compute averages of measured quantities, why the operators of quantum mechanics must be *Hermitian* (self-adjoint; a property defined precisely in Sec. 10.8), and why that guarantees real energies. Finally we adopt the units $\hbar = 2m = 1$ that the paper uses throughout.

Where this shows up in the paper: the very first sentence of the paper speaks of "the spectrum of a self-adjoint operator"; Sec. II B of the paper writes "In units $\hbar = 2m = 1$, $H = -d^2/dx^2 + V(x)$"; Sec. II C works with eigenpairs (E_n, ψ_n) and weights $|\psi_n|^2$; and Secs. II E and V deliberately break Hermiticity with $H \to H + iA$. After this chapter each of those phrases will be a precise statement you can check.

What you need

- First-year physics: traveling waves $y = y_m \sin(kx - \omega t)$, wave speed, the wave equation for a string, standing waves and harmonics on a string fixed at both ends; kinetic energy $K = \frac{1}{2}mv^2 = p^2/2m$; the electron-volt.
- Calculus: derivatives, the chain and product rules, integration by parts, integrals of $\sin^2$ and of exponentials.
- Chapter 2: complex numbers, modulus $|z|$, complex conjugate z^*, Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, $|e^{i\theta}| = 1$.
- Chapter 5: superposition, beats, orthogonality of sines and cosines, expanding a function in a series of sines.
- Chapter 7: inner products $\int f^* g\,dx$ of functions, orthonormal bases.
- Chapter 8: eigenvalues and eigenvectors, symmetric and Hermitian matrices have real eigenvalues and orthogonal eigenvectors, the second-derivative operator as a tridiagonal matrix, the fixed-end string problem $-y'' = \lambda y$ on $[-L, L]$ with eigenvalues $(k\pi/2L)^2$.

10.1 Waves you already know

★ Did you know? *Light Makes Stripes*

In the early 1800s Thomas Young showed that light passing two nearby openings produces bright and dark fringes, just as water waves interfere. It was strong evidence that light is a wave, against Newton's influential particle picture. Much later, the same two-slit pattern was built up from single electrons arriving one dot at a time. It became the classic demonstration of quantum wave behavior.

We start on familiar ground, because every new idea in this chapter is an old idea about strings wearing new clothes.

10.1.1 Traveling waves and the dispersion relation

A sinusoidal wave traveling along a stretched string in the $+x$ direction has transverse displacement

$$y(x,t) = y_m \sin(kx - \omega t), \tag{10.1}$$

where y_m is the amplitude, $k = 2\pi/\lambda$ is the *angular wavenumber* (radians per meter, with λ the wavelength) and $\omega = 2\pi f$ is the *angular frequency* (radians per second, with f the frequency). A crest sits where the phase $kx - \omega t$ is constant, so it moves at the *phase velocity*

$$v_{\text{ph}} = \frac{\omega}{k}. \tag{10.2}$$

Newton's second law applied to a small piece of string (HRW, the chapter on waves [7]) gives the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}, \qquad v = \sqrt{\tau/\mu_s}, \tag{10.3}$$

with τ the tension and μ_s the mass per unit length. (The symbols $\partial/\partial x$ and $\partial/\partial t$ are partial derivatives: differentiate with respect to one variable while holding the other fixed.)

Let us check what Eq. (10.3) demands of the sinusoid (10.1). Differentiating twice in x,

$$\frac{\partial y}{\partial x} = k\, y_m \cos(kx - \omega t), \tag{10.4}$$

$$\frac{\partial^2 y}{\partial x^2} = -k^2\, y_m \sin(kx - \omega t) = -k^2 y. \tag{10.5}$$

Differentiating twice in t (the chain rule brings down $-\omega$ each time),

$$\frac{\partial y}{\partial t} = -\omega\, y_m \cos(kx - \omega t), \tag{10.6}$$

$$\frac{\partial^2 y}{\partial t^2} = -\omega^2\, y_m \sin(kx - \omega t) = -\omega^2 y. \tag{10.7}$$

Substituting into Eq. (10.3) gives $-k^2 y = -(\omega^2/v^2)y$, which holds for all x, t only if

$$\omega^2 = v^2 k^2, \qquad \text{that is} \qquad \omega = vk \quad (\omega, k > 0). \tag{10.8}$$

A relation between ω and k that a wave must satisfy is called a *dispersion relation*. For the string it is linear, so every wavelength travels at the same speed v. Keep this in mind: the dispersion relation is the fingerprint of a wave equation, and in Sec. 10.3 we will run the logic backwards, starting from a dispersion relation and asking which equation produces it.

10.1.2 Standing waves and a discrete set of frequencies

Now pin the string at $x = 0$ and $x = \ell$. Two waves traveling in opposite directions superpose into a standing wave (Chapter 5)

$$y(x, t) = 2y_m \sin(kx)\cos(\omega t). \tag{10.9}$$

The factor $\sin(kx)$ vanishes at $x = 0$ automatically. Requiring $y(\ell, t) = 0$ for all t requires $\sin(k\ell) = 0$, so

$$k\ell = n\pi, \qquad n = 1, 2, 3, \ldots \tag{10.10}$$

and, using $\omega = vk$ from Eq. (10.8) and $f = \omega/2\pi$,

$$f_n = \frac{\omega_n}{2\pi} = \frac{v}{2\pi} \cdot \frac{n\pi}{\ell} = \frac{nv}{2\ell}. \tag{10.11}$$

These are the harmonics of HRW. Two features matter for us. First, the *boundary conditions* (pinned ends) are what turn a continuum of possible frequencies into a discrete list. Second, the list is the set of eigenvalues of a differential operator: in Chapter 8 you met the fixed-end problem $-y'' = \lambda y$, whose eigenvalues are a discrete ladder. We will call such a list a *spectrum*. The left panel of Figure 10.1 shows the first three harmonics.

10.1.3 The complex form

Recall from Chapter 2 that Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ lets us write sinusoids as real parts of complex exponentials:

$$y_m \cos(kx - \omega t) = \mathrm{Re}\left[y_m e^{i(kx - \omega t)}\right]. \tag{10.12}$$

For the string, the complex form is a bookkeeping convenience: the physical displacement is the real part. In quantum mechanics we will find that the complex form is *not* optional. That is the first genuinely new thing in this chapter, and we arrive at it in Sec. 10.3.

Key Ideas

- A sinusoid $\sin(kx - \omega t)$ solves the string wave equation only if $\omega = vk$; a relation $\omega(k)$ forced by a wave equation is its dispersion relation.
- Boundary conditions (pinned ends) select a discrete set of allowed wavenumbers and frequencies: a spectrum.

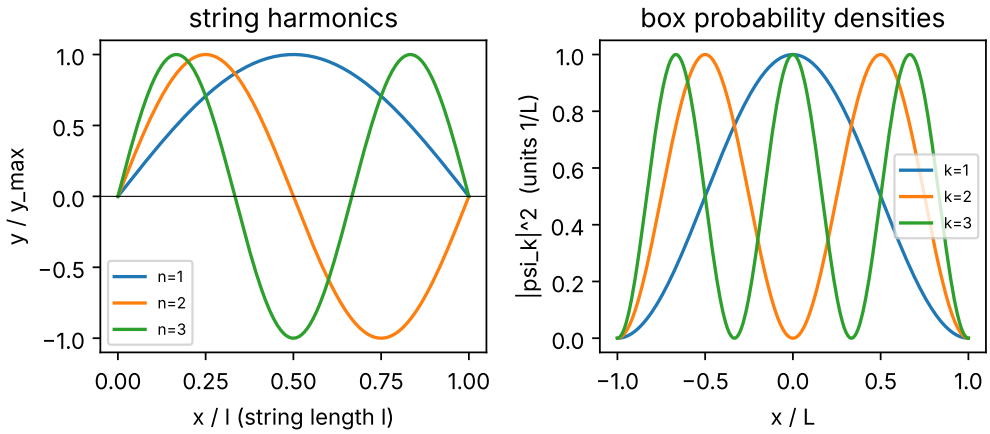


Figure 10.1: Left: the first three standing-wave shapes $\sin(n\pi x/\ell)$ of a string pinned at $x = 0$ and $x = \ell$. Right: the first three probability densities $|\psi_k|^2$ of a quantum particle in a box with impenetrable walls at $x = \pm L$ (derived in Sec. 10.5 and Sec. 10.6). The shapes are the same family of sines; the quantum picture plots the squared modulus, for reasons explained by the Born rule.

- Complex exponentials $e^{i(kx-\omega t)}$ package sines and cosines together; for strings the physics is in the real part.

10.2 Matter waves: the de Broglie relation

10.2.1 Photons first

We need two experimental facts from the early twentieth century. We take them on trust; HRW [7] describes the experiments in its chapters on photons and matter waves. First, Planck (1900) and Einstein (1905) concluded that light of frequency f exchanges energy with matter only in packets, called photons, of energy $E = hf$. Einstein was explaining the photoelectric effect, in which light knocks electrons out of a metal. Here $h = 6.62607015 \times 10^{-34}$ J s is Planck's constant. Second, Compton's 1923 experiments, scattering X-rays from electrons, showed that each photon also carries momentum $p = h/\lambda$, with λ the wavelength of the light. It is convenient to define the *reduced Planck constant*

$$\hbar = \frac{h}{2\pi} = 1.054571817 \times 10^{-34}\ \text{J s} \tag{10.13}$$

(read "h-bar"). Using $\omega = 2\pi f$ and $k = 2\pi/\lambda$,

$$E = hf = \frac{h}{2\pi}\,2\pi f = \hbar\omega, \qquad p = \frac{h}{\lambda} = \frac{h}{2\pi}\cdot\frac{2\pi}{\lambda} = \hbar k. \tag{10.14}$$

Energy is tied to angular frequency, momentum to angular wavenumber, and the exchange rate is $\hbar$ in both cases.

10.2.2 De Broglie's hypothesis

In 1924 Louis de Broglie proposed that the same relations hold for particles with mass: a particle of momentum p has an associated wave of wavelength

$$\lambda = \frac{h}{p}, \qquad \text{equivalently} \qquad p = \hbar k, \quad E = \hbar\omega. \tag{10.15}$$

This was a hypothesis, not a deduction. It was confirmed in 1927 when Davisson and Germer, and independently G. P. Thomson, observed electrons diffracting from crystals exactly as waves of wavelength h/p would. Diffraction is an interference effect, and interference is the signature of superposition: amplitudes add, and the intensity is computed from the sum.

Sample Problem 10.1: De Broglie wavelengths of an electron and a baseball

Problem. (a) Find the de Broglie wavelength of an electron with kinetic energy $K = 100$ eV. (b) Find that of a baseball of mass 0.145 kg moving at 40 m/s. Use $m_e = 9.1093837 \times 10^{-31}$ kg and $1\ \text{eV} = 1.602176634 \times 10^{-19}$ J.

Step 1: momentum of the electron. At 100 eV the electron is far from relativistic (its rest energy is about half a million eV), so $K = p^2/2m_e$ and $p = \sqrt{2m_eK}$. In SI units $K = 100 \times 1.602176634 \times 10^{-19}\ \text{J} = 1.602 \times 10^{-17}\ \text{J}$, so

$$p = \sqrt{2(9.109 \times 10^{-31})(1.602 \times 10^{-17})}\ \text{kg m/s} = 5.403 \times 10^{-24}\ \text{kg m/s}.$$

Step 2: wavelength of the electron.

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}}{5.403 \times 10^{-24}}\ \text{m} = 1.226 \times 10^{-10}\ \text{m}.$$

This is comparable to the spacing of atoms in a crystal (a few tenths of a nanometer), which is why crystals diffract electrons.

Step 3: the baseball. $p = mv = (0.145)(40) = 5.8$ kg m/s, so

$$\lambda = \frac{6.626 \times 10^{-34}}{5.8}\ \text{m} = 1.14 \times 10^{-34}\ \text{m},$$

some twenty orders of magnitude smaller than an atomic nucleus. No slit or crystal can diffract it, which is why you never see a baseball interfere with itself.

Answer.

$$\boxed{\lambda_e = 0.1226\ \text{nm}, \qquad \lambda_{\text{ball}} = 1.14 \times 10^{-34}\ \text{m}}$$

Checkpoint 10.1

A (nonrelativistic) electron's speed is doubled. By what factor does its de Broglie wavelength change? By what factor does its kinetic energy change?

Answer: $p = mv$ doubles, so $\lambda = h/p$ halves. The kinetic energy $\frac{1}{2}mv^2$ grows by a factor 4. Equivalently $\lambda \propto 1/\sqrt{K}$.

Pitfall

Two common slips. (1) Using h where $\hbar$ is meant, or the reverse: $\lambda = h/p$ but $k = p/\hbar$. They differ by 2π. (2) Using the total energy, or $p = E/c$ (which is for light), for a massive slow particle. For a slow particle use $p = \sqrt{2mK}$ with K the kinetic energy.

10.2.3 The free-particle dispersion relation

For a free particle (no forces) the energy is all kinetic, $E = p^2/2m$. Substituting $E = \hbar\omega$ and $p = \hbar k$ from Eq. (10.15),

$$\hbar\omega = \frac{(\hbar k)^2}{2m} \quad \Longrightarrow \quad \omega = \frac{\hbar k^2}{2m}. \tag{10.16}$$

Compare with the string's $\omega = vk$ in Eq. (10.8): here ω grows as the *square* of k. Matter waves are *dispersive*: their phase velocity $\omega/k = \hbar k/2m$ depends on the wavelength. The right panel of Figure 10.2, which appears in Sec. 10.3 and uses the units of Sec. 10.9, contrasts the two.

Key Ideas

- De Broglie: a particle of momentum p carries a wave with $\lambda = h/p$; equivalently $p = \hbar k$ and $E = \hbar\omega$, with $\hbar = h/2\pi$.
- The wavelength is observable (electron diffraction) only when it is comparable to some length in the experiment.
- For a free particle $E = p^2/2m$ turns into the dispersion relation $\omega = \hbar k^2/2m$, quadratic rather than linear in k.

10.3 Building a wave equation

If particles are waves, there should be a wave equation that tells the wave how to evolve, just as Eq. (10.3) tells a string. Can we derive it? Honestly, no. We can only list requirements, guess the simplest equation that meets them, and then test the guess against experiment. That is what Erwin Schrödinger did in 1926, and it is what we do now.

Taken on trust

Taken on trust: the Schrödinger equation itself. The equation we arrive at below is a *postulate* of quantum mechanics. Our argument shows that it is the simplest linear equation consistent with the de Broglie relations and $E = p^2/2m + V$; it does not prove that nature obeys it. Its justification is the agreement of its predictions (atomic spectra, chemistry, semiconductors, and much more) with experiment. Some deeper frameworks (for instance, deriving it from symmetry principles or from relativistic quantum field theory in a low-speed limit) exist but are beyond this book; see Griffiths and Schroeter [30] for the standard presentation and its experimental basis.

10.3.1 The requirements

We want an equation for a wave function $\Psi(x, t)$ (capital psi) such that:

(R1) It is *linear*: if Ψ_1 and Ψ_2 are solutions, so is $a\Psi_1 + b\Psi_2$ for any constants a, b. Electron diffraction shows interference, and interference requires that amplitudes superpose.

(R2) For a free particle, plane waves of wavenumber k and angular frequency ω are solutions exactly when ω and k obey Eq. (10.16).

(R3) The equation's coefficients do not depend on k or ω: one equation must serve every wavelength, so that superpositions of different k also solve it.

10.3.2 The string equation fails

Try Eq. (10.3). We showed it forces $\omega = vk$, linear in k. No constant v can make vk equal $\hbar k^2/2m$ for all k (the ratio $\hbar k^2/2m \div k = \hbar k/2m$ changes with k). So the string equation cannot satisfy (R2) with a single constant v, and (R3) forbids letting v depend on k.

10.3.3 Counting derivatives

Each time derivative of a plane wave brings down one factor of ω; each space derivative brings down one factor of k. The dispersion relation (10.16) has ω to the first power and k to the second. So we should try an equation with *one* time derivative and *two* space derivatives:

$$\frac{\partial \Psi}{\partial t} = \alpha \frac{\partial^2 \Psi}{\partial x^2}, \tag{10.17}$$

with α a constant to be found.

10.3.4 Real sinusoids cannot work

Try a real wave, $\Psi = \cos(kx - \omega t)$. The left side of Eq. (10.17) is

$$\frac{\partial \Psi}{\partial t} = \omega \sin(kx - \omega t),$$

and the right side is

$$\alpha \frac{\partial^2 \Psi}{\partial x^2} = -\alpha k^2 \cos(kx - \omega t).$$

A sine cannot equal a multiple of a cosine for all x and t (at a point where the cosine is zero and the sine is not, the right side vanishes and the left side does not), unless both coefficients vanish: $\omega = 0$ and $\alpha k^2 = 0$. That is no wave at all. The same happens for $\sin(kx - \omega t)$. (Problem 4 at the end of this chapter redoes the cosine case in full, directly against the free Schrödinger equation.) One time derivative turns a cosine into a sine, while two space derivatives turn a cosine back into a cosine, and nothing real can reconcile them.

10.3.5 Complex exponentials do work

Now try the complex plane wave

$$\Psi(x,t) = e^{i(kx-\omega t)} = \cos(kx - \omega t) + i \sin(kx - \omega t). \tag{10.18}$$

We need the derivatives of $e^{i\theta}$. Using Euler's formula and differentiating the real and imaginary parts separately,

$$\frac{\mathrm{d}}{\mathrm{d}\theta} e^{i\theta} = -\sin\theta + i\cos\theta = i(\cos\theta + i\sin\theta) = ie^{i\theta},$$

where in the middle step we used $i \cdot i \sin\theta = -\sin\theta$. With the chain rule, $\theta = kx - \omega t$ gives $\partial\theta/\partial x = k$ and $\partial\theta/\partial t = -\omega$, so

$$\frac{\partial \Psi}{\partial t} = -i\omega\, \Psi, \tag{10.19}$$

$$\frac{\partial \Psi}{\partial x} = ik\, \Psi, \tag{10.20}$$

$$\frac{\partial^2 \Psi}{\partial x^2} = (ik)(ik)\Psi = -k^2 \Psi. \tag{10.21}$$

Unlike the real case, both sides of Eq. (10.17) are now multiples of the *same* function Ψ. Substituting,

$$-i\omega\Psi = -\alpha k^2 \Psi \quad \Longrightarrow \quad \omega = \frac{\alpha k^2}{i} = -i\alpha k^2.$$

We want $\omega = \hbar k^2/2m$, so we need $-i\alpha = \hbar/2m$, that is $\alpha = i\hbar/2m$ (check: $-i \cdot i\hbar/2m = \hbar/2m$, because $-i \cdot i = -i^2 = 1$). With this α, Eq. (10.17) becomes $\partial\Psi/\partial t = (i\hbar/2m)\partial^2\Psi/\partial x^2$. Multiply both sides by $i\hbar$ and use $i\hbar \cdot i\hbar/2m = -\hbar^2/2m$:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2}. \tag{10.22}$$

This is the free Schrödinger equation. The factor i is essential, and with it the wave function is necessarily complex.

There is a nice way to read Eq. (10.22). On a plane wave, Eq. (10.19) says $i\hbar\,\partial\Psi/\partial t = \hbar\omega\Psi = E\Psi$, and Eq. (10.21) says $-(\hbar^2/2m)\partial^2\Psi/\partial x^2 = (\hbar^2k^2/2m)\Psi = (p^2/2m)\Psi$. So the equation is the energy statement $E = p^2/2m$, with the replacements

$$E \;\longrightarrow\; i\hbar\frac{\partial}{\partial t}, \qquad p \;\longrightarrow\; -i\hbar\frac{\partial}{\partial x}. \tag{10.23}$$

(Check the second: $-i\hbar\,\partial\Psi/\partial x = -i\hbar\cdot ik\Psi = \hbar k\Psi = p\Psi$.) From here on we sometimes abbreviate $\partial/\partial t$ as ∂_t, $\partial/\partial x$ as ∂_x and $\partial^2/\partial x^2$ as ∂_x^2, and similarly for other variables. We also write y_{xx} for $\partial^2 y/\partial x^2$, y_{tt} for $\partial^2 y/\partial t^2$ and Ψ_t for $\partial\Psi/\partial t$.

10.3.6 Adding a potential

A particle moving in a potential energy $V(x)$ has $E = p^2/2m + V(x)$. Applying the same replacements, and letting $V(x)$ simply multiply Ψ, gives the *time-dependent Schrödinger equation*

$$\boxed{i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V(x)\,\Psi} \tag{10.24}$$

For a free particle this is the simplest equation meeting (R1)–(R3), given the convention $e^{i(kx-\omega t)}$ for a right-moving wave; it is still a guess. With $V \neq 0$ we have *guessed* further, by analogy; this is the step that experiment, not logic, must validate.

Linearity. Suppose Ψ_1 and Ψ_2 both solve Eq. (10.24) and $\Psi = a\Psi_1 + b\Psi_2$ with constants a, b. Derivatives are linear, so

$$\begin{aligned} i\hbar\frac{\partial\Psi}{\partial t} &= a\,i\hbar\frac{\partial\Psi_1}{\partial t} + b\,i\hbar\frac{\partial\Psi_2}{\partial t} \\ &= a\left(-\frac{\hbar^2}{2m}\Psi_1'' + V\Psi_1\right) + b\left(-\frac{\hbar^2}{2m}\Psi_2'' + V\Psi_2\right) \quad \text{(each solves the equation)} \\ &= -\frac{\hbar^2}{2m}\left(a\Psi_1 + b\Psi_2\right)'' + V\left(a\Psi_1 + b\Psi_2\right) \quad \text{(collect terms)}, \end{aligned}$$

where a prime means $\partial/\partial x$. So Ψ solves it too: requirement (R1) holds.

Checkpoint 10.2

Why can a guitar string get away with a real displacement $y(x,t)$, while the matter wave Ψ must be complex?

Answer: The string equation has two time derivatives, which turn a cosine back into a cosine, matching the two space derivatives. The Schrödinger equation has one time derivative (forced by $\omega \propto k^2$), which turns a cosine into a sine; only the complex combination $e^{i\theta}$, whose derivative is a multiple of itself, can satisfy it.

10.3.7 Phase velocity, group velocity and the particle's speed

Here is a puzzle that tests the equation. A free electron with momentum p moves at $v = p/m$. What does the wave do?

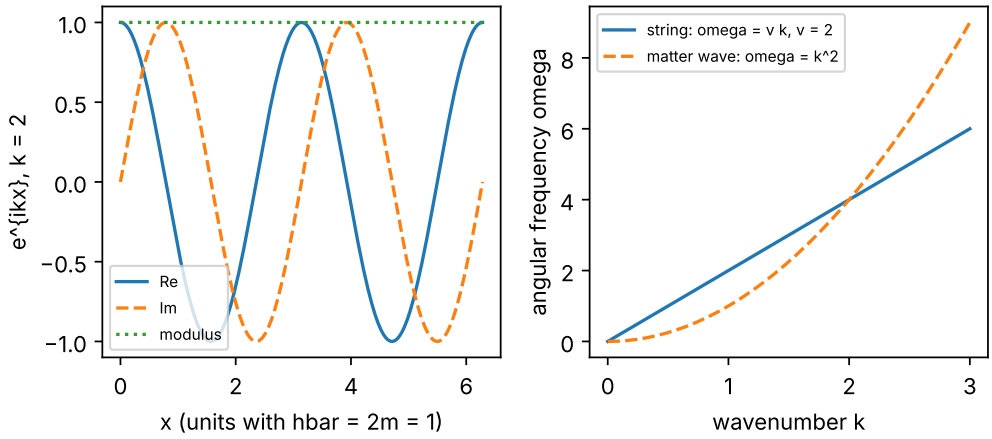


Figure 10.2: Left: real part (solid) and imaginary part (dashed) of the plane wave e^{ikx} at $t = 0$, with $k = 2$ (in units $\hbar = 2m = 1$ of Sec. 10.9); the modulus $\left|e^{ikx}\right| = 1$ (dotted) is constant. Right: dispersion relations. A string has $\omega = vk$ (here $v = 2$); a free matter wave has $\omega = \hbar k^2/2m$, which is $\omega = k^2$ in units $\hbar = 2m = 1$.

Sample Problem 10.2: A free electron wave

Problem. An electron is described by a plane wave with $k = 1.0 \times 10^{10}$ m^{-1}. Find (a) its kinetic energy in eV, (b) ω, (c) the phase velocity, and (d) the classical speed p/m. Use $\hbar = 1.054571817 \times 10^{-34}$ J s and $m_e = 9.1093837 \times 10^{-31}$ kg.

Step 1: energy. $K = p^2/2m = \hbar^2 k^2/2m$:

$$K = \frac{(1.0546 \times 10^{-34})^2 (1.0 \times 10^{20})}{2(9.109 \times 10^{-31})} \text{ J} = 6.104 \times 10^{-19} \text{ J} = 3.81 \text{ eV}.$$

Step 2: angular frequency. $\omega = K/\hbar = \hbar k^2/2m$:

$$\omega = \frac{6.104 \times 10^{-19}}{1.0546 \times 10^{-34}} \text{ rad/s} = 5.79 \times 10^{15} \text{ rad/s}.$$

Step 3: phase velocity. $v_{\text{ph}} = \omega/k = 5.79 \times 10^{15}/10^{10}$ m/s $= 5.79 \times 10^5$ m/s.

Step 4: classical speed. $p/m = \hbar k/m = (1.0546 \times 10^{-24})/(9.109 \times 10^{-31})$ m/s $= 1.16 \times 10^6$ m/s.

Answer.

$$\boxed{K = 3.81 \text{ eV}, \quad \omega = 5.79 \times 10^{15} \text{ rad/s}, \quad v_{\text{ph}} = 5.79 \times 10^5 \text{ m/s}, \quad p/m = 1.16 \times 10^6 \text{ m/s}}$$

The crests move at exactly *half* the particle's speed: $v_{\text{ph}} = \hbar k/2m$ while $p/m = \hbar k/m$.

The resolution of the factor of two uses beats (Chapter 5). A real particle is not an infinite plane wave but a localized bump, a superposition of nearby wavenumbers. Take two complex plane waves $e^{i(k_1 x - \omega_1 t)}$ and $e^{i(k_2 x - \omega_2 t)}$, with $k_1 = k - \frac{1}{2}\Delta k$ and $k_2 = k + \frac{1}{2}\Delta k$. Their frequencies ω_1, ω_2 are given by the dispersion relation, and we write $\Delta\omega = \omega_2 - \omega_1$. By linearity their sum solves the free Schrödinger equation. Look at the real part of the sum, $\cos A + \cos B$; the imaginary part, $\sin A + \sin B$, behaves the same way. The identity $\cos A + \cos B = 2\cos\frac{A-B}{2}\cos\frac{A+B}{2}$ with $A = k_2 x - \omega_2 t$, $B = k_1 x - \omega_1 t$ gives

$$\cos A + \cos B = 2\cos\left(\frac{\Delta k\, x - \Delta\omega\, t}{2}\right)\cos\left(\frac{(k_1 + k_2)x - (\omega_1 + \omega_2)t}{2}\right).$$

The first factor is a slowly varying *envelope*; its shape is unchanged when $\Delta k\,x - \Delta\omega\,t$ is constant, so the envelope moves at $\Delta\omega/\Delta k$. As $\Delta k \to 0$ this becomes the *group velocity*

$$v_{\mathrm{g}} = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{\mathrm{d}}{\mathrm{d}k}\frac{\hbar k^2}{2m} = \frac{\hbar k}{m} = \frac{p}{m}. \tag{10.25}$$

The bump, which is where the particle is likely to be found, travels at exactly the classical speed. The Schrödinger equation passes its first test.

Pitfall

The phase velocity of a matter wave is not the particle's speed; the group velocity is. For the string the two coincide because $\omega = vk$ is linear, which is why first-year physics never needed to distinguish them.

Key Ideas

- The simplest linear equation whose plane-wave solutions obey $\hbar\omega = \hbar^2k^2/2m$ has one time derivative and two space derivatives, and it requires the factor i: Eq. (10.22).
- With a potential, $E = p^2/2m + V$ and the replacements $E \to i\hbar\,\partial_t$, $p \to -i\hbar\,\partial_x$ give the time-dependent Schrödinger equation (10.24), a postulate checked by experiment.
- The equation is linear, so solutions superpose. Wave packets move at the group velocity $\mathrm{d}\omega/\mathrm{d}k = p/m$, the classical speed.

10.4 Separation of variables and stationary states

Equation (10.24) involves both x and t. When V depends on x only (as it always will in this book), a standard trick splits it into two simpler equations.

10.4.1 The separation

Look for solutions that are a product of a function of x and a function of t:

$$\Psi(x,t) = \psi(x)\,\varphi(t). \tag{10.26}$$

Then $\partial\Psi/\partial t = \psi\,\mathrm{d}\varphi/\mathrm{d}t$ and $\partial^2\Psi/\partial x^2 = \varphi\,\mathrm{d}^2\psi/\mathrm{d}x^2$. Substituting into Eq. (10.24),

$$i\hbar\,\psi\frac{\mathrm{d}\varphi}{\mathrm{d}t} = -\frac{\hbar^2}{2m}\varphi\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + V\psi\varphi.$$

Divide both sides by $\psi\varphi$ (at points where it is not zero):

$$i\hbar\,\frac{1}{\varphi}\frac{\mathrm{d}\varphi}{\mathrm{d}t} = \frac{1}{\psi}\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + V\psi\right). \tag{10.27}$$

The left side depends only on t; the right side depends only on x. Fix x and vary t: the right side does not change, so the left side cannot change either. Fix t and vary x: by the same argument the right side is constant. So both sides equal a single constant, which we call E (we will see in a moment that it is the energy). That gives two ordinary differential equations:

$$i\hbar\frac{\mathrm{d}\varphi}{\mathrm{d}t} = E\,\varphi, \tag{10.28}$$

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + V(x)\psi = E\,\psi. \tag{10.29}$$

10.4.2 The time factor

Equation (10.28) says $\mathrm{d}\varphi/\mathrm{d}t = (E/i\hbar)\varphi = -i(E/\hbar)\varphi$, using $1/i = -i$. We claim the solution is

$$\varphi(t) = e^{-iEt/\hbar}. \tag{10.30}$$

Check, with $\omega = E/\hbar$: $e^{-i\omega t} = \cos\omega t - i\sin\omega t$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}e^{-i\omega t} = -\omega\sin\omega t - i\omega\cos\omega t = -i\omega(\cos\omega t - i\sin\omega t) = -i\omega e^{-i\omega t},$$

where we used $-i\omega\cdot(-i\sin\omega t) = i^2\omega\sin\omega t = -\omega\sin\omega t$. So $\varphi' = -i(E/\hbar)\varphi$ as required. (Any constant multiple of φ also works; we absorb that constant into ψ. There are no other solutions. If φ solves the equation, then by the product rule $\frac{\mathrm{d}}{\mathrm{d}t}\left(\varphi e^{iEt/\hbar}\right) = \varphi' e^{iEt/\hbar} + \frac{iE}{\hbar}\varphi e^{iEt/\hbar} = \left(-\frac{iE}{\hbar} + \frac{iE}{\hbar}\right)\varphi e^{iEt/\hbar} = 0$. So $\varphi e^{iEt/\hbar}$ is a constant, and φ is that constant times $e^{-iEt/\hbar}$.) The time factor rotates in the complex plane at angular frequency $\omega = E/\hbar$, exactly the de Broglie relation $E = \hbar\omega$, which justifies calling E the energy.

10.4.3 Why such states are called stationary

For the separated solution $\Psi = \psi(x)e^{-iEt/\hbar}$, with E real (we prove E is real in Sec. 10.8),

$$|\Psi(x,t)|^2 = \Psi^*\Psi = \psi^*(x)e^{+iEt/\hbar}\,\psi(x)e^{-iEt/\hbar} = |\psi(x)|^2. \tag{10.31}$$

The squared modulus does not depend on time. In Sec. 10.6 we will see that $|\Psi|^2$ is what is measured, so nothing observable about such a state changes in time. These solutions are called *stationary states*; each one is a standing wave whose overall phase turns like the hand of a clock.

10.4.4 The time-independent Schrödinger equation

Equation (10.29) is the *time-independent Schrödinger equation*. It contains no time at all: it asks for functions $\psi(x)$ and numbers E such that a certain combination of ψ and its second derivative is a multiple E of ψ itself. That is an eigenvalue problem, the subject of Sec. 10.5.

Sample Problem 10.3: How fast does the phase of a stationary state turn?

Problem. A stationary state has $E = 1.00$ eV. How long does the phase factor $e^{-iEt/\hbar}$ take to make one full turn around the unit circle?
Step 1: One full turn means $Et/\hbar = 2\pi$, so the period is $T_{\mathrm{ph}} = 2\pi\hbar/E = h/E$.
Step 2: With $E = 1.602176634\times10^{-19}$ J,

$$T_{\mathrm{ph}} = \frac{6.62607\times10^{-34}}{1.602177\times10^{-19}}\ \mathrm{s} = 4.14\times10^{-15}\ \mathrm{s}.$$

Answer.

$$\boxed{T_{\mathrm{ph}} = 4.14\times10^{-15}\ \mathrm{s}}$$

A few femtoseconds. But since $|\Psi|^2$ is unchanged, no experiment on this single state can see the rotation; only *differences* of energies between superposed states produce observable time dependence (Sec. 10.7).

10.4.5 General solutions from stationary states

Suppose we have found many solutions (ψ_n, E_n), $n = 1, 2, \ldots$, of Eq. (10.29). Each product $\psi_n(x)e^{-iE_nt/\hbar}$ solves Eq. (10.24) (that is how we built them), and by linearity so does any sum

$$\Psi(x,t) = \sum_n c_n\,\psi_n(x)\,e^{-iE_nt/\hbar}, \tag{10.32}$$

with constants c_n. At $t = 0$ this is $\sum_n c_n \psi_n(x)$. If the ψ_n can build any initial wave function (the expansion question of Sec. 10.7, similar to a Fourier series in Chapter 5), then Eq. (10.32) solves the whole time-dependent problem: find the stationary states once, then attach the clock factors. Given completeness of the stationary states (taken on trust in the `beyond` box of Sec. 10.8), the whole time evolution of the system is encoded in the stationary states and their energies. The paper goes further and never considers time evolution at all: it uses only the eigenvalues E_n and eigenfunctions ψ_n of H.

Checkpoint 10.3

Is a superposition $\Psi = \psi_1 e^{-iE_1 t/\hbar} + \psi_2 e^{-iE_2 t/\hbar}$ with $E_1 \neq E_2$ a stationary state?

Answer: No. It solves the Schrödinger equation, but it is not of the form $\psi(x)\varphi(t)$, and its $|\Psi|^2$ contains a cross term oscillating at $(E_2 - E_1)/\hbar$ (computed in Sec. 10.7), so observable properties change in time.

Key Ideas

- For $V = V(x)$, the product $\psi(x)e^{-iEt/\hbar}$ solves the Schrödinger equation whenever ψ and E solve the time-independent equation (10.29).
- Such stationary states have time-independent $|\Psi|^2$; their phase rotates at $\omega = E/\hbar$.
- Superpositions of stationary states with clock factors attached give the general time evolution, Eq. (10.32).

10.5 The Hamiltonian as an operator

10.5.1 Operators

An *operator* is a rule that takes a function and returns another function. We write operators with hats or capital letters. Three examples:

- the derivative operator D: $Df = \mathrm{d}f/\mathrm{d}x$;
- a multiplication operator: $(\hat{V}f)(x) = V(x)f(x)$, "multiply by the function V";
- the second-derivative operator $D^2 f = \mathrm{d}^2 f/\mathrm{d}x^2$.

An operator Q is *linear* if $Q(af + bg) = aQf + bQg$ for all functions f, g and constants a, b. All three examples are linear (derivatives of sums are sums of derivatives, and $V(af + bg) = aVf + bVg$). Linear operators on functions are the continuous cousins of matrices acting on vectors (Chapter 7); in Sec. 10.8 we will make the analogy concrete by sampling functions on a grid.

10.5.2 The Hamiltonian

Define the *Hamiltonian operator*

$$H = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x), \qquad H\psi = -\frac{\hbar^2}{2m}\psi'' + V\psi. \tag{10.33}$$

It is named after W. R. Hamilton, whose classical "Hamiltonian" is the energy written as a function of position and momentum, $p^2/2m + V(x)$; the replacement $p \to -i\hbar\,\mathrm{d}/\mathrm{d}x$ of Eq. (10.23)

turns p^2 into $(-i\hbar)^2 \mathrm{d}^2/\mathrm{d}x^2 = -\hbar^2 \mathrm{d}^2/\mathrm{d}x^2$, which is how the first term arises. With this notation the time-independent equation (10.29) is simply

$$\boxed{H\psi = E\psi} \tag{10.34}$$

and the time-dependent equation (10.24) is $i\hbar\,\partial\Psi/\partial t = H\Psi$.

Compare with a matrix eigenvalue problem $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ (Chapter 8). The operator H plays the role of $\mathbf{A}$, the function ψ the role of the eigenvector $\mathbf{v}$, and E the role of λ. The solutions ψ are called *eigenfunctions* of H and the numbers E its *eigenvalues*, or *energy levels*. A pair (E_n, ψ_n) is an *eigenpair*.

10.5.3 Boundary conditions belong to the operator

As with the string, the allowed E depend on boundary conditions. In this book (and in the paper) the particle lives in a box $[-L, L]$ with impenetrable walls, which means $V = +\infty$ outside the box and the wave function must vanish at the walls:

$$\psi(-L) = \psi(L) = 0. \tag{10.35}$$

Conditions that fix the value of the function at the boundary are called *Dirichlet* conditions. When we speak of "the operator H on $[-L, L]$ with Dirichlet walls" we mean the formula (10.33) *together with* the rule that it acts only on functions obeying Eq. (10.35). Changing the boundary conditions changes the operator, and therefore its eigenvalues. (Why ψ must vanish at an infinite wall is argued physically in Chapter 11; here we take it as the definition of the box.)

Sample Problem 10.4: Eigenfunctions of the empty box

Problem. Take $V = 0$ inside $[-L, L]$ with Dirichlet walls. Show that for every positive integer k the function

$$\psi_k(x) = \sin\!\left(\frac{k\pi(x + L)}{2L}\right)$$

is an eigenfunction of H, and find its eigenvalue.

Step 1: boundary conditions. At $x = -L$ the argument is 0 and $\sin 0 = 0$. At $x = L$ the argument is $k\pi \cdot 2L/2L = k\pi$, and $\sin(k\pi) = 0$ for integer k. Both walls are satisfied.

Step 2: first derivative. Write $q = k\pi/2L$, so $\psi_k = \sin(q(x + L))$. By the chain rule $\psi_k' = q\cos(q(x + L))$.

Step 3: second derivative. $\psi_k'' = -q^2 \sin(q(x + L)) = -q^2\psi_k$.

Step 4: apply H. With $V = 0$, $H\psi_k = -\frac{\hbar^2}{2m}\psi_k'' = \frac{\hbar^2}{2m}q^2\psi_k$.

Answer.

$$\boxed{E_k = \frac{\hbar^2}{2m}\left(\frac{k\pi}{2L}\right)^2, \qquad k = 1, 2, 3, \ldots}$$

This is the fixed-end string problem of Chapter 8 again, with eigenvalue $(k\pi/2L)^2$ multiplied by $\hbar^2/2m$. Chapter 11 solves the box problem fully, including the question of whether there are any other eigenfunctions. Note $k = 0$ gives $\psi = 0$, which is not allowed (an eigenfunction must be nonzero), and negative k gives $-\psi_{|k|}$, the same state up to sign.

We record the empty-box levels as a numbered equation:

$$E_k = \frac{\hbar^2}{2m}\left(\frac{k\pi}{2L}\right)^2, \qquad k = 1, 2, 3, \ldots \tag{10.36}$$

The integer k that labels the empty-box states should not be confused with the wavenumber k of a plane wave; for the box state ψ_k the wavenumber is $q = k\pi/2L$. The paper labels levels

by n and reserves k for its cosine basis functions $\cos(k\pi x/L)$. In Sec. 10.11 we switch to n for levels.

10.5.4 The momentum operator

Equation (10.23) suggests defining the *momentum operator*

$$\hat{p} = -i\hbar \frac{\mathrm{d}}{\mathrm{d}x}. \tag{10.37}$$

The plane wave e^{ikx} is an eigenfunction: $\hat{p}\, e^{ikx} = -i\hbar \cdot ik\, e^{ikx} = \hbar k\, e^{ikx}$, with eigenvalue $\hbar k = p$. Then $\hat{p}^2 = \hat{p}\hat{p} = (-i\hbar)^2 \mathrm{d}^2/\mathrm{d}x^2 = -\hbar^2 \mathrm{d}^2/\mathrm{d}x^2$ and

$$H = \frac{\hat{p}^2}{2m} + V(x), \tag{10.38}$$

the operator version of "kinetic plus potential energy". A box state ψ_k is *not* an eigenfunction of $\hat{p}$: writing $\sin\theta = (e^{i\theta} - e^{-i\theta})/2i$ (Chapter 2) shows it is a superposition of waves moving left and right, just like the standing wave (10.9).

Checkpoint 10.4

Is the function $\psi(x) = \cos(\pi x/2L)$ an eigenfunction of the empty-box Hamiltonian on $[-L, L]$? If so, which k does it correspond to?

Answer: Yes. $\sin(\pi(x+L)/2L) = \sin(\pi x/2L + \pi/2) = \cos(\pi x/2L)$, so it is ψ_1, with $E_1 = \frac{\hbar^2}{2m}(\pi/2L)^2$. It vanishes at $x = \pm L$ because $\cos(\pm\pi/2) = 0$.

Key Ideas

- An operator maps functions to functions; the Hamiltonian $H = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)$ is the energy operator.
- The time-independent Schrödinger equation is the eigenvalue problem $H\psi = E\psi$: eigenfunctions are stationary states, eigenvalues are energy levels.
- Boundary conditions are part of the operator. In the empty box with Dirichlet walls, $E_k = \frac{\hbar^2}{2m}(k\pi/2L)^2$.
- The momentum operator is $\hat{p} = -i\hbar\, \mathrm{d}/\mathrm{d}x$, and $H = \hat{p}^2/2m + V$.

10.6 What the wavefunction means: the Born rule

★ Did you know? *Physics in a Footnote*

Max Born's 1926 paper on collisions introduced the idea that the wavefunction gives *probabilities*. The crucial point, that the probability is proportional to the *square* of the wavefunction, appeared in a footnote added in proof. Born received the Nobel Prize in Physics for this statistical interpretation in 1954, nearly three decades later.

10.6.1 The Born rule

So far Ψ is a complex wave with no stated meaning. The interpretation used by every working physicist was proposed by Max Born in 1926:

Born rule. If a particle has wave function $\Psi(x,t)$, the probability of finding it between x and $x+\mathrm{d}x$ when its position is measured at time t is $|\Psi(x,t)|^2\,\mathrm{d}x$. The probability of finding it in an interval $[a,b]$ is

$$P_{[a,b]}(t) = \int_a^b |\Psi(x,t)|^2\,\mathrm{d}x. \tag{10.39}$$

So $|\Psi|^2$ is a *probability density*: probability per unit length, with units of 1/length in one dimension. The wave function itself is called a *probability amplitude*. Like the Schrödinger equation, the Born rule is a postulate.

Taken on trust

Taken on trust: the Born rule and measurement. Why squared moduli give probabilities, and what exactly happens during a measurement, are questions about the foundations of quantum mechanics that remain debated. Nothing in this book or in the paper depends on how they are answered; we need only the rule (10.39) and its extension to energy measurements (Sec. 10.7). See Griffiths and Schroeter [30], Chapter 1 and the Afterword, for a careful discussion.

Two consequences follow at once. Because the total probability of finding the particle *somewhere* is one,

$$\int_{\text{all } x} |\Psi(x,t)|^2\,\mathrm{d}x = 1. \tag{10.40}$$

A wave function obeying this is *normalized*. And because $|e^{i\alpha}\Psi|^2 = |e^{i\alpha}|^2\,|\Psi|^2 = |\Psi|^2$ for any real constant α, multiplying a wave function by a constant phase $e^{i\alpha}$ changes no prediction: Ψ and $e^{i\alpha}\Psi$ describe the same physical state. (This is why the overall sign of a single eigenvector returned by a numerical solver does not affect quantities such as $|\psi_n|^2$. Relative signs between different states are another matter, as a checkpoint in Sec. 10.7 shows.)

10.6.2 Normalizing a wave function

The Schrödinger equation is linear, so if ψ is a solution then $A\psi$ is too, for any constant A. Normalization fixes $|A|$: choose it so that Eq. (10.40) holds.

Sample Problem 10.5: Normalizing a cusp

Problem. A wave function on the whole line is $\psi(x) = A\,e^{-|x|/a}$ with $a>0$. (a) Find $A>0$ that normalizes it. (b) Find the probability that the particle is found within $|x|<a$.

Step 1: the normalization integral. $|\psi|^2 = A^2 e^{-2|x|/a}$, an even function, so the integral over the line is twice the integral over $x>0$:

$$\int_{-\infty}^{\infty} A^2 e^{-2|x|/a}\,\mathrm{d}x = 2A^2\int_0^{\infty} e^{-2x/a}\,\mathrm{d}x.$$

Step 2: evaluate. $\int_0^\infty e^{-2x/a}\mathrm{d}x = \left[-\frac{a}{2}e^{-2x/a}\right]_0^\infty = 0-(-\frac{a}{2}) = \frac{a}{2}$. So the integral is $2A^2\cdot a/2 = A^2 a$.

Step 3: set it to 1. $A^2 a = 1$, so $A = 1/\sqrt{a}$.

Step 4: probability within $|x|<a$. Again use evenness:

$$P = 2\cdot\frac{1}{a}\int_0^a e^{-2x/a}\mathrm{d}x = \frac{2}{a}\left[-\frac{a}{2}e^{-2x/a}\right]_0^a = \frac{2}{a}\cdot\frac{a}{2}\left(1-e^{-2}\right) = 1-e^{-2}.$$

Answer.

$$\boxed{A = \frac{1}{\sqrt{a}}, \qquad P(|x| < a) = 1 - e^{-2} = 0.8647}$$

The answer does not depend on a: a only sets the scale. Figure 10.3 shows the density and the shaded probability.

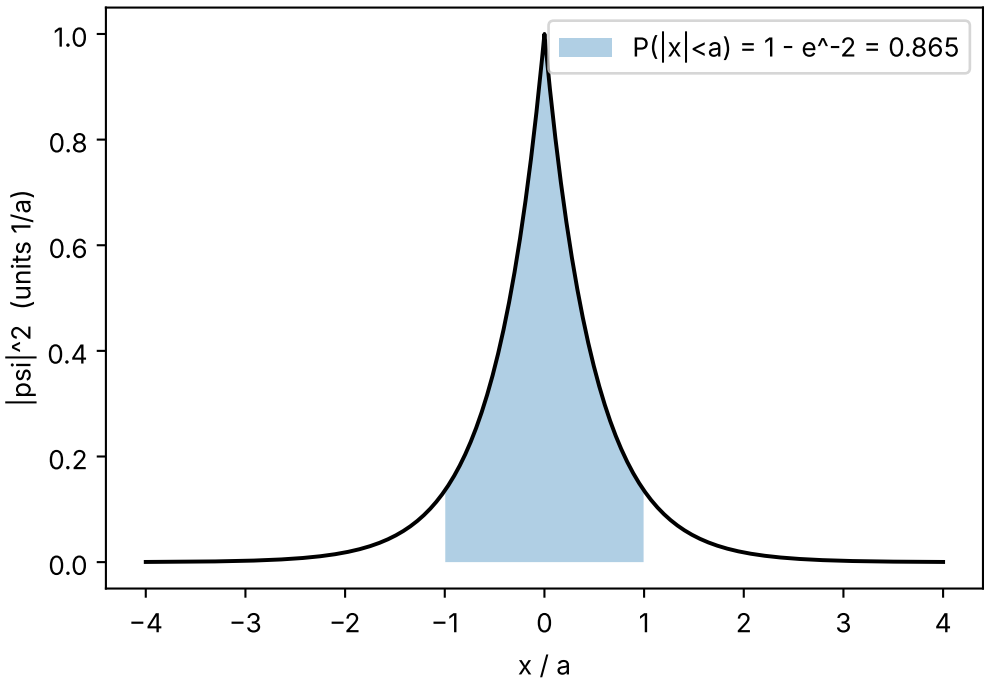


Figure 10.3: The Born rule for the normalized wave function $\psi = a^{-1/2}e^{-|x|/a}$ (here $a = 1$, length in units of a). The curve is the probability density $|\psi|^2 = a^{-1}e^{-2|x|/a}$ (units $1/a$); the total area under it is 1, and the shaded area, the probability of finding the particle in $|x| < a$, is $1 - e^{-2} \approx 0.865$.

Sample Problem 10.6: The box ground state

Problem. The empty-box ground state on $[-L, L]$ is $\psi_1 = A\sin(\pi(x + L)/2L)$. (a) Normalize it. (b) Find the probability that the particle is in the middle half, $-L/2 < x < L/2$.

Step 1: substitution. Let $u = \pi(x + L)/2L$. Then $\mathrm{d}x = (2L/\pi)\mathrm{d}u$, and $x = -L$, L map to $u = 0$, π.

Step 2: normalization integral. Using $\sin^2 u = \frac{1}{2}(1 - \cos 2u)$,

$$\int_{-L}^{L} A^2 \sin^2\!\left(\frac{\pi(x+L)}{2L}\right) \mathrm{d}x = A^2\frac{2L}{\pi}\int_0^{\pi} \sin^2 u\,\mathrm{d}u = A^2\frac{2L}{\pi}\left[\frac{u}{2} - \frac{\sin 2u}{4}\right]_0^{\pi} = A^2\frac{2L}{\pi}\cdot\frac{\pi}{2} = A^2 L.$$

So $A = 1/\sqrt{L}$.

Step 3: limits for the middle half. $x = -L/2$ gives $u = \pi(L/2)/2L = \pi/4$; $x = L/2$ gives $u = \pi(3L/2)/2L = 3\pi/4$.

Step 4: integrate.

$$P = \frac{1}{L}\cdot\frac{2L}{\pi}\left[\frac{u}{2} - \frac{\sin 2u}{4}\right]_{\pi/4}^{3\pi/4} = \frac{2}{\pi}\left[\left(\frac{3\pi}{8} - \frac{\sin(3\pi/2)}{4}\right) - \left(\frac{\pi}{8} - \frac{\sin(\pi/2)}{4}\right)\right].$$

With $\sin(3\pi/2) = -1$ and $\sin(\pi/2) = 1$, the bracket is $\frac{3\pi}{8} + \frac{1}{4} - \frac{\pi}{8} + \frac{1}{4} = \frac{\pi}{4} + \frac{1}{2}$.

Answer.

$$\boxed{A = \frac{1}{\sqrt{L}}, \qquad P = \frac{2}{\pi}\left(\frac{\pi}{4} + \frac{1}{2}\right) = \frac{1}{2} + \frac{1}{\pi} = 0.8183}$$

A classical particle bouncing at constant speed would spend exactly half its time in the middle half. The quantum ground state piles probability toward the center, where $\sin^2$ peaks (Figure 10.1, right).

The same calculation works for every box state. For $\psi_k = A\sin(k\pi(x+L)/2L)$ substitute $u = k\pi(x+L)/2L$, so that $\mathrm{d}x = (2L/k\pi)\mathrm{d}u$ and u runs from 0 to $k\pi$. Since

$$\int_0^{k\pi} \sin^2 u \,\mathrm{d}u = \left[\frac{u}{2} - \frac{\sin 2u}{4}\right]_0^{k\pi} = \frac{k\pi}{2} \quad (\text{because } \sin 2k\pi = 0),$$

the normalization integral is $A^2 \cdot \frac{2L}{k\pi} \cdot \frac{k\pi}{2} = A^2 L$, and again $A = 1/\sqrt{L}$. So the normalized empty-box states are

$$\psi_k(x) = \frac{1}{\sqrt{L}} \sin\left(\frac{k\pi(x+L)}{2L}\right), \qquad k = 1, 2, \ldots \tag{10.41}$$

Checkpoint 10.5

Is $\psi(x) = 2\sin(\pi(x+L)/2L)$ normalized on $[-L, L]$? If not, what is $\int |\psi|^2 \,\mathrm{d}x$?

Answer: No. Doubling ψ multiplies $|\psi|^2$ by 4, so the integral is $4L$ instead of 1 (using Step 2 of the previous Sample Problem with $A = 2$). It describes the same physical state as the normalized ψ_1 but must be rescaled before applying the Born rule.

10.6.3 Normalization is preserved in time

There is a consistency question. Normalization is imposed at one time. If the Schrödinger equation let $\int |\Psi|^2$ drift, total probability would stop being one. Let us prove it does not drift, and see exactly which property of H is responsible.

Start from Eq. (10.24) and divide by $i\hbar$, using $1/i = -i$:

$$\frac{\partial \Psi}{\partial t} = \frac{i\hbar}{2m}\frac{\partial^2 \Psi}{\partial x^2} - \frac{i}{\hbar} V\Psi. \tag{10.42}$$

(Check the first coefficient: $-\frac{\hbar^2}{2m} \cdot \frac{1}{i\hbar} = -\frac{\hbar}{2m} \cdot (-i) = \frac{i\hbar}{2m}$.) Take the complex conjugate of every term, assuming V is *real* so that $V^* = V$:

$$\frac{\partial \Psi^*}{\partial t} = -\frac{i\hbar}{2m}\frac{\partial^2 \Psi^*}{\partial x^2} + \frac{i}{\hbar} V\Psi^*. \tag{10.43}$$

By the product rule, $\partial |\Psi|^2 / \partial t = \partial(\Psi^*\Psi)/\partial t = \Psi^* \,\partial\Psi/\partial t + \Psi \,\partial\Psi^*/\partial t$. Substituting Eqs. (10.42) and (10.43),

$$\begin{aligned} \frac{\partial |\Psi|^2}{\partial t} &= \frac{i\hbar}{2m}\left(\Psi^*\frac{\partial^2 \Psi}{\partial x^2} - \Psi\frac{\partial^2 \Psi^*}{\partial x^2}\right) + \left(-\frac{i}{\hbar}V + \frac{i}{\hbar}V\right)|\Psi|^2 \\ &= \frac{i\hbar}{2m}\left(\Psi^*\frac{\partial^2 \Psi}{\partial x^2} - \Psi\frac{\partial^2 \Psi^*}{\partial x^2}\right) \quad (\text{the } V \text{ terms cancel}). \end{aligned} \tag{10.44}$$

The bracket is a total x-derivative. Indeed, differentiating $\Psi^*\Psi' - \Psi\Psi^{*\prime}$ (primes are $\partial/\partial x$) with the product rule,

$$\frac{\partial}{\partial x}\left(\Psi^*\Psi' - \Psi\Psi^{*\prime}\right) = \Psi^{*\prime}\Psi' + \Psi^*\Psi'' - \Psi'\Psi^{*\prime} - \Psi\Psi^{*\prime\prime} = \Psi^*\Psi'' - \Psi\Psi^{*\prime\prime}.$$

Integrate Eq. (10.44) over the box. On the left, the time derivative may be taken outside the integral (the limits $\pm L$ do not depend on t). On the right, the integral of a derivative is the difference of end values:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{-L}^{L} |\Psi|^2\,\mathrm{d}x = \frac{i\hbar}{2m}\Big[\Psi^*\Psi' - \Psi\Psi^{*\prime}\Big]_{x=-L}^{x=L} = 0,$$

because Ψ and Ψ^* vanish at both walls. (On the whole line, the same holds if $\Psi \to 0$ at $\pm\infty$, as it must for a normalizable function.) So $\int |\Psi|^2\,\mathrm{d}x$ is constant: *normalize once, normalized forever*. The only property of the potential we used is that it is real.

10.6.4 What a complex potential does

To see that the reality of V really matters, replace V by $V - i\Gamma/2$ with Γ a real positive constant. Equation (10.42) gains the extra term $-\frac{i}{\hbar}\left(-\frac{i\Gamma}{2}\right)\Psi = \frac{i^2\Gamma}{2\hbar}\Psi = -\frac{\Gamma}{2\hbar}\Psi$. Its conjugate equation gains the conjugate term $-\frac{\Gamma}{2\hbar}\Psi^*$ (the coefficient is real). In $\partial\,|\Psi|^2/\partial t$ these contribute $\Psi^*\left(-\frac{\Gamma}{2\hbar}\Psi\right) + \Psi\left(-\frac{\Gamma}{2\hbar}\Psi^*\right) = -\frac{\Gamma}{\hbar}\,|\Psi|^2$. Everything else is unchanged, so with $P(t) = \int |\Psi|^2\,\mathrm{d}x$,

$$\frac{\mathrm{d}P}{\mathrm{d}t} = -\frac{\Gamma}{\hbar}P \qquad \Longrightarrow \qquad P(t) = P(0)\,e^{-\Gamma t/\hbar}, \tag{10.45}$$

the law of exponential decay. To check the solution, differentiate: $\frac{\mathrm{d}}{\mathrm{d}t}\left[P(0)e^{-\Gamma t/\hbar}\right] = -\frac{\Gamma}{\hbar}P(0)e^{-\Gamma t/\hbar}$, which is $-\Gamma/\hbar$ times itself. It is also the only solution. If P obeys the equation, then $\frac{\mathrm{d}}{\mathrm{d}t}\left(Pe^{\Gamma t/\hbar}\right) = P'e^{\Gamma t/\hbar} + \frac{\Gamma}{\hbar}Pe^{\Gamma t/\hbar} = 0$. So $Pe^{\Gamma t/\hbar}$ is constant, equal to its value $P(0)$ at $t = 0$. A negative imaginary potential *absorbs* probability: it models a particle that can be lost, for example by decaying. Such potentials will be studied in Chapter 14.

The width may also depend on position: $V(x) - i\Gamma(x)/2$ with $\Gamma(x)$ real. The same steps go through with Γ replaced by $\Gamma(x)$, because $\Gamma(x)^* = \Gamma(x)$. The extra contribution to $\partial\,|\Psi|^2/\partial t$ is then $-\frac{\Gamma(x)}{\hbar}\,|\Psi|^2$ at each point. Integrating over the box gives

$$\frac{\mathrm{d}P}{\mathrm{d}t} = -\frac{1}{\hbar}\int \Gamma(x)\,|\Psi(x,t)|^2\;\mathrm{d}x. \tag{10.46}$$

Probability is drained wherever $\Gamma(x) > 0$ and created wherever $\Gamma(x) < 0$. This position-dependent form is the one the paper uses for its "absorptive-even" class modelling $-i\Gamma/2$ (Sec. II E).

Pitfall

$|\Psi(x)|^2$ is a probability *density*, not a probability. It can exceed 1 (for the cusp with small a, $|\psi|^2 = 1/a$ at $x = 0$), and it has units of inverse length. Only its integral over an interval is a probability. Also, never forget the complex conjugate: $|\Psi|^2 = \Psi^*\Psi$, not Ψ^2, which is complex in general.

Key Ideas

- Born rule: $|\Psi|^2\,\mathrm{d}x$ is the probability of finding the particle in $[x, x + \mathrm{d}x]$; normalized wave functions satisfy $\int |\Psi|^2\,\mathrm{d}x = 1$.

- A constant overall phase $e^{i\alpha}$ is unobservable.
- With a real potential and walls (or decay at infinity), the Schrödinger equation conserves $\int |\Psi|^2 \, \mathrm{d}x$. A potential with negative imaginary part $-i\Gamma/2$ makes the norm decay as $e^{-\Gamma t/\hbar}$.
- Normalized empty-box states: $\psi_k = L^{-1/2} \sin(k\pi(x+L)/2L)$.

10.7 Operators, expectation values and energy measurements

10.7.1 Averages of position

First, a fact about averages. Suppose a measurement has possible outcomes $x_1, x_2, \ldots$. In many repeated trials, outcome x_i occurs a fraction P_i of the time, its probability, so $\sum_i P_i = 1$. In N_{tr} trials, outcome x_i occurs $P_i N_{\mathrm{tr}}$ times. The plain average of all results is therefore

$$\frac{1}{N_{\mathrm{tr}}} \sum_i (P_i N_{\mathrm{tr}}) \, x_i = \sum_i P_i x_i,$$

which is each outcome weighted by its probability. Now apply this to position. Slice the line into thin slices of width $\mathrm{d}x$. By the Born rule the slice at x has probability $|\Psi|^2 \, \mathrm{d}x$, and in the limit of thin slices the sum over slices becomes an integral. So if a measurement of position is repeated on many identically prepared particles, the average result is the probability-weighted average of x (an average weighted by $|\Psi|^2$, like the center of mass of a rod with linear density $|\Psi|^2$):

$$\langle x \rangle = \int x \, |\Psi(x,t)|^2 \, \mathrm{d}x. \tag{10.47}$$

The angle brackets denote this *expectation value*. More generally, for any function $f(x)$,

$$\langle f \rangle = \int f(x) \, |\Psi(x,t)|^2 \, \mathrm{d}x. \tag{10.48}$$

Equation (10.48) is the single most important formula for reading the paper's Sec. II C; hold on to it.

10.7.2 Averages of other operators

Momentum is not a function of x, so Eq. (10.48) cannot give $\langle p \rangle$. The rule of quantum mechanics (another postulate, consistent with everything above) is that the expectation value of the quantity represented by a linear operator Q is

$$\langle Q \rangle = \int \Psi^*(x,t) \, (Q\Psi)(x,t) \, \mathrm{d}x. \tag{10.49}$$

For a multiplication operator $Q = f(x)$, this gives $\int \Psi^* f \Psi \, \mathrm{d}x = \int f \, |\Psi|^2 \, \mathrm{d}x$, which is Eq. (10.48) again: the new rule contains the old one. The order matters: Q acts on Ψ first, and the result is multiplied by Ψ^*.

Bra-ket notation. Physicists abbreviate the inner product of two functions (Chapter 7) as

$$\langle f \,|\, g \rangle = \int f^*(x) \, g(x) \, \mathrm{d}x, \tag{10.50}$$

and write $\langle f|Q|g\rangle = \int f^* \, (Qg) \, \mathrm{d}x$. In this notation $\langle Q \rangle = \langle \Psi|Q|\Psi \rangle$, and the normalization condition is $\langle \Psi \,|\, \Psi \rangle = 1$. When the functions are eigenfunctions ψ_n, ψ_m, one writes simply

$\langle n|Q|m\rangle = \int \psi_n^* Q\,\psi_m\,\mathrm{d}x$ and calls it a *matrix element* of Q. Notice $\langle f\,|\,g\rangle^* = \langle g\,|\,f\rangle$, since $(f^*g)^* = g^*f$. The symbol $|g\rangle$ (a "ket") stands for the function g, and $\langle f|$ (a "bra") for "take the inner product with f".

Sample Problem 10.7: Two expectation values that need no integral tables

Problem. Let ψ be a real, normalized eigenfunction of H with eigenvalue E, in a box with Dirichlet walls. Show that (a) $\langle p\rangle = 0$ and (b) $\langle H\rangle = E$.

Step 1: write out $\langle p\rangle$. By Eq. (10.49) with $\hat{p} = -i\hbar\,\mathrm{d}/\mathrm{d}x$ and $\psi^* = \psi$ (real):

$$\langle p\rangle = \int_{-L}^{L} \psi\,(-i\hbar\psi')\,\mathrm{d}x = -i\hbar\int_{-L}^{L}\psi\psi'\,\mathrm{d}x.$$

Step 2: recognize a derivative. By the chain rule $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}\psi^2\right) = \psi\psi'$, so

$$\int_{-L}^{L}\psi\psi'\,\mathrm{d}x = \left[\tfrac{1}{2}\psi^2\right]_{-L}^{L} = 0 - 0 = 0.$$

Hence $\langle p\rangle = 0$. (Physically: a real wave is an equal mix of left- and right-moving waves.)

Step 3: $\langle H\rangle$. Since $H\psi = E\psi$,

$$\langle H\rangle = \int \psi^*(H\psi)\,\mathrm{d}x = \int \psi^* E\psi\,\mathrm{d}x = E\int|\psi|^2\,\mathrm{d}x = E\cdot 1.$$

Answer.

$$\boxed{\langle p\rangle = 0, \qquad \langle H\rangle = E}$$

10.7.3 Expanding in eigenfunctions, and measuring energy

In Sec. 10.8 we prove that eigenfunctions of H belonging to different eigenvalues are orthogonal, $\langle\psi_m\,|\,\psi_n\rangle = 0$ for $m \neq n$. With normalization this gives an *orthonormal* set:

$$\langle\psi_m\,|\,\psi_n\rangle = \delta_{mn} = \begin{cases} 1 & m = n, \\ 0 & m \neq n, \end{cases} \tag{10.51}$$

where δ_{mn} is called the Kronecker delta. For the empty box this is the orthogonality of sines from Chapter 5.

Suppose a wave function can be written as a combination of eigenfunctions,

$$\Psi = \sum_n c_n\psi_n. \tag{10.52}$$

To find a coefficient, take the inner product with ψ_m and use Eq. (10.51):

$$\langle\psi_m\,|\,\Psi\rangle = \sum_n c_n\langle\psi_m\,|\,\psi_n\rangle = \sum_n c_n\delta_{mn} = c_m. \tag{10.53}$$

(For a finite sum, moving the integral inside the sum is just linearity of the integral. For an infinite sum it needs a convergence argument, which is part of the `beyond` box in Sec. 10.8.) The normalization of Ψ becomes, writing $\Psi^* = \sum_m c_m^*\psi_m^*$,

$$\langle\Psi\,|\,\Psi\rangle = \sum_m\sum_n c_m^* c_n\langle\psi_m\,|\,\psi_n\rangle = \sum_m\sum_n c_m^* c_n\delta_{mn} = \sum_n |c_n|^2\,, \tag{10.54}$$

which is Parseval's theorem of Chapter 5 in new clothes. Similarly, since $H\Psi = \sum_n c_n E_n\psi_n$,

$$\langle H\rangle = \langle\Psi|H|\Psi\rangle = \sum_m\sum_n c_m^* c_n E_n\delta_{mn} = \sum_n |c_n|^2\,E_n. \tag{10.55}$$

Equations (10.54) and (10.55) are exactly what you would get if a measurement of energy returned E_n with probability $|c_n|^2$. To see this, apply the weighted-average rule $\sum_i P_i x_i$ from the start of this section, with outcomes E_n and probabilities $P_n = |c_n|^2$. These probabilities sum to 1 by Eq. (10.54). That is the form of the Born rule for energy, and it is accepted as a postulate:

> **Born rule for energy.** A measurement of the energy of a normalized state $\Psi = \sum c_n \psi_n$ yields one of the eigenvalues E_n, with probability $|c_n|^2$.

Only eigenvalues can ever be measured. This is the precise sense in which the eigenvalues of H are "the energy levels" of the system.

Sample Problem 10.8: Energy measurement in the paper's box (RE3)

Problem. A particle is in the empty box on $[-L, L]$ with $L = 44.883$ (the paper's scale-1 half-width, Sec. II B), in units where $\hbar^2/2m = 1$ (introduced in Sec. 10.9). Its state is $\Psi = \frac{\sqrt{3}}{2}\psi_1 + \frac{1}{2}\psi_2$. Find the possible results of an energy measurement, their probabilities, and $\langle H \rangle$.

Step 1: levels. With $\hbar^2/2m = 1$, $E_k = (k\pi/2L)^2$. Here $\pi/2L = \pi/89.766 = 0.0349976$, so $E_1 = 1.2248 \times 10^{-3}$ and $E_2 = 4E_1 = 4.8993 \times 10^{-3}$.

Step 2: check normalization. $|c_1|^2 + |c_2|^2 = \frac{3}{4} + \frac{1}{4} = 1$.

Step 3: probabilities. E_1 with probability $\frac{3}{4}$; E_2 with probability $\frac{1}{4}$; no other result is possible.

Step 4: average. $\langle H \rangle = \frac{3}{4}E_1 + \frac{1}{4} \cdot 4E_1 = \frac{7}{4}E_1$.

Answer.

$$\boxed{P(E_1) = \tfrac{3}{4},\ P(E_2) = \tfrac{1}{4}, \qquad \langle H \rangle = \tfrac{7}{4}E_1 = 2.143 \times 10^{-3}}$$

Note that $\langle H \rangle$ is *not* itself one of the possible outcomes. It is an average over many repetitions.

10.7.4 Time dependence comes from energy differences

Attach clock factors as in Eq. (10.32). For two real normalized eigenfunctions,

$$\Psi(x,t) = \frac{1}{\sqrt{2}}\left(\psi_1 e^{-iE_1 t/\hbar} + \psi_2 e^{-iE_2 t/\hbar}\right).$$

Then

$$\begin{aligned} |\Psi|^2 &= \frac{1}{2}\left(\psi_1 e^{+iE_1 t/\hbar} + \psi_2 e^{+iE_2 t/\hbar}\right)\left(\psi_1 e^{-iE_1 t/\hbar} + \psi_2 e^{-iE_2 t/\hbar}\right) \\ &= \frac{1}{2}\left[\psi_1^2 + \psi_2^2 + \psi_1\psi_2\left(e^{i(E_2-E_1)t/\hbar} + e^{-i(E_2-E_1)t/\hbar}\right)\right] \\ &= \frac{1}{2}\left[\psi_1^2 + \psi_2^2 + 2\psi_1\psi_2 \cos\!\left(\frac{(E_2 - E_1)t}{\hbar}\right)\right], \end{aligned} \tag{10.56}$$

using $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ (Chapter 2). The overall phases cancel; only the *difference* $E_2 - E_1$ survives, as a beat at angular frequency $(E_2 - E_1)/\hbar$. In the box the cross term $\psi_1\psi_2$ is odd about $x = 0$, so the probability sloshes from one side to the other (Figure 10.4). This is how energy differences become observable: an atom oscillating between two levels radiates light of angular frequency $(E_2 - E_1)/\hbar$.

Checkpoint 10.6

A state is $\Psi = \frac{1}{\sqrt{2}}(\psi_1 + \psi_2)$ at $t = 0$. Someone multiplies ψ_2 by -1 (a legitimate choice of eigenfunction sign). Do the energy probabilities change? Does $|\Psi|^2$ at $t = 0$ change?

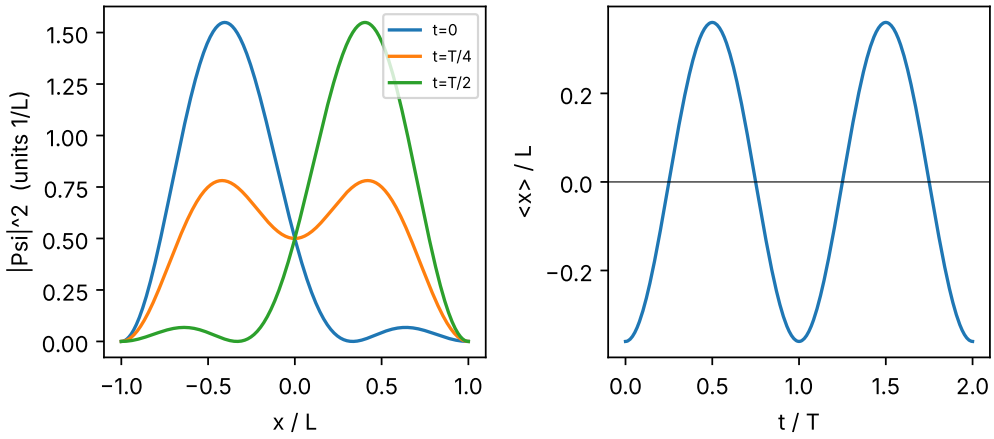


Figure 10.4: A superposition $(\psi_1 e^{-iE_1 t} + \psi_2 e^{-iE_2 t})/\sqrt{2}$ of the two lowest empty-box states on $[-L, L]$, units $\hbar = 2m = 1$ with $L = 1$. Left: $|\Psi|^2$ at $t = 0$, $T/4$ and $T/2$, where $T = 2\pi/(E_2 - E_1)$. Right: the expectation value $\langle x \rangle$ oscillates as $\cos((E_2 - E_1)t)$ with amplitude $32L/(9\pi^2) \approx 0.360L$ (Problem 10 at the end of this chapter).

Answer: The probabilities $|c_n|^2 = \frac{1}{2}$ do not change. But $|\Psi|^2$ does: the cross term $\psi_1\psi_2$ flips sign, so the probability is now piled on the other side. A sign of one eigenfunction is a convention, but a *relative* sign inside a superposition is physical: $\psi_1 + \psi_2$ and $\psi_1 - \psi_2$ are different states.

Key Ideas

- Expectation values: $\langle Q \rangle = \int \Psi^* Q \Psi \, dx = \langle \Psi | Q | \Psi \rangle$; for a function $f(x)$ this is the $|\Psi|^2$-weighted average $\int f \, |\Psi|^2 \, dx$.
- Expanding $\Psi = \sum c_n \psi_n$ in orthonormal eigenfunctions gives $c_n = \langle \psi_n \, | \, \Psi \rangle$, $\sum |c_n|^2 = 1$ and $\langle H \rangle = \sum |c_n|^2 \, E_n$.
- Energy measurements return eigenvalues E_n with probabilities $|c_n|^2$.
- Observable time dependence comes only from energy differences, as beats.

10.8 Hermitian operators and real eigenvalues

10.8.1 Why we need a special class of operators

Measured energies are real numbers. If quantum mechanics is to make sense, the operator H must have only real eigenvalues, and its eigenfunctions should be orthogonal so that the expansion of Sec. 10.7 works. Recall from Chapter 8 that a matrix with complex entries is *Hermitian* if $H_{ij} = H_{ji}^*$ (equal to its conjugate transpose), and that Hermitian matrices have real eigenvalues and orthogonal eigenvectors. We now define the analogous property for operators and prove the same two results.

10.8.2 Definition

A linear operator Q, acting on functions that satisfy given boundary conditions, is *Hermitian* if

$$\langle f | Q | g \rangle = \int f^* \, (Qg) \, dx = \int (Qf)^* \, g \, dx \tag{10.57}$$

for all such functions f and g. In words: Q may be moved from the right-hand function to the left-hand one, provided we conjugate it there. Apply Eq. (10.57) to any two allowed functions ψ_m, ψ_n, for example members of an orthonormal basis. Then

$$\langle m|Q|n\rangle = \int (Q\psi_m)^* \psi_n \,\mathrm{d}x = \left(\int \psi_n^* \, Q\psi_m \,\mathrm{d}x \right)^* = \langle n|Q|m\rangle^*,$$

where the middle step uses $(ab)^* = a^* b^*$ inside the integral. This is the matrix condition $Q_{mn} = Q_{nm}^*$.

10.8.3 The Hamiltonian with Dirichlet walls is Hermitian

Take $H = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)$ on $[-L, L]$ with $f(\pm L) = g(\pm L) = 0$, and V real. Split H into its two terms.

Potential term. $\int f^* V g \,\mathrm{d}x = \int (Vf)^* g \,\mathrm{d}x$ requires $f^* V = (Vf)^* = V^* f^*$, true for all f exactly when $V^* = V$, that is, V real.

Kinetic term. Integrate by parts twice. The first integration by parts, with $u = f^*$ and $\mathrm{d}v = g''\mathrm{d}x$:

$$\int_{-L}^{L} f^* g'' \,\mathrm{d}x = \Big[f^* g' \Big]_{-L}^{L} - \int_{-L}^{L} f^{*\prime} g' \,\mathrm{d}x = - \int_{-L}^{L} f^{*\prime} g' \,\mathrm{d}x,$$

because $f^*(\pm L) = 0$. The second integration by parts, with $u = f^{*\prime}$ and $\mathrm{d}v = g'\mathrm{d}x$ (so $v = g$):

$$- \int_{-L}^{L} f^{*\prime} g' \,\mathrm{d}x = - \Big[f^{*\prime} g \Big]_{-L}^{L} + \int_{-L}^{L} f^{*\prime\prime} g \,\mathrm{d}x = \int_{-L}^{L} f^{*\prime\prime} g \,\mathrm{d}x,$$

because $g(\pm L) = 0$. Combining, $\int f^* g'' \mathrm{d}x = \int f^{*\prime\prime} g \,\mathrm{d}x = \int (f'')^* g \,\mathrm{d}x$; multiplying by the real constant $-\hbar^2/2m$ gives the kinetic part of Eq. (10.57). Adding the two parts:

Result. With real V and Dirichlet walls, $\langle f|H|g\rangle = \int (Hf)^* g \,\mathrm{d}x$: *$H$ is Hermitian.*

Both ingredients matter. The boundary terms vanished because of the walls, and the potential term was symmetric because V is real. Change either and Hermiticity can fail.

10.8.4 Two theorems

Theorem 1 (proved). The eigenvalues of a Hermitian operator are real. *Proof.* Let $Q\psi = \lambda\psi$ with $\psi \neq 0$. Compute $\langle\psi|Q|\psi\rangle$ in two ways:

$$\int \psi^* (Q\psi) \,\mathrm{d}x = \int \psi^* \lambda \psi \,\mathrm{d}x = \lambda \int |\psi|^2 \,\mathrm{d}x,$$
$$\int (Q\psi)^* \psi \,\mathrm{d}x = \int \lambda^* \psi^* \psi \,\mathrm{d}x = \lambda^* \int |\psi|^2 \,\mathrm{d}x.$$

Hermiticity says the two left sides are equal, so $(\lambda - \lambda^*) \int |\psi|^2 \,\mathrm{d}x = 0$. Since $\psi \neq 0$, $\int |\psi|^2 \,\mathrm{d}x > 0$, hence $\lambda = \lambda^*$: λ is real. ■

Theorem 2 (proved). Eigenfunctions of a Hermitian operator belonging to different eigenvalues are orthogonal. *Proof.* Let $Q\psi_m = \lambda_m \psi_m$ and $Q\psi_n = \lambda_n \psi_n$ with $\lambda_m \neq \lambda_n$. Then

$$\int \psi_n^* (Q\psi_m) \,\mathrm{d}x = \lambda_m \langle \psi_n \,|\, \psi_m \rangle, \qquad \int (Q\psi_n)^* \psi_m \,\mathrm{d}x = \lambda_n^* \langle \psi_n \,|\, \psi_m \rangle = \lambda_n \langle \psi_n \,|\, \psi_m \rangle,$$

using Theorem 1 ($\lambda_n^* = \lambda_n$) in the last step. Hermiticity equates the left sides, so $(\lambda_m - \lambda_n)\langle \psi_n \,|\, \psi_m \rangle = 0$, and since $\lambda_m \neq \lambda_n$, $\langle \psi_n \,|\, \psi_m \rangle = 0$. ■

These are word for word the proofs for Hermitian matrices in Chapter 8, with sums replaced by integrals. They justify everything in Sec. 10.7: real measured energies and orthonormal expansions.

Taken on trust

Taken on trust: "self-adjoint" and completeness. For operators on infinite-dimensional function spaces, mathematicians distinguish a *Hermitian* (or *symmetric*) operator, which satisfies Eq. (10.57) on its stated domain, from a *self-adjoint* one, for which the domains of the operator and its adjoint also coincide. Self-adjointness is what guarantees a full set of real eigenvalues (or, more generally, a real spectrum) and the expansion theorem: every square-integrable function can be expanded as in Eq. (10.52), with the infinite sum converging in the mean-square sense. For the operators in this book ($-\mathrm{d}^2/\mathrm{d}x^2 + V$ on a finite interval with Dirichlet walls and a bounded real V) both properties hold; this is part of Sturm–Liouville theory, treated in Riley, Hobson and Bence [22]. Physicists, and the paper, often use "Hermitian" and "self-adjoint" interchangeably; we will too, except when the distinction matters (Chapter 21).

10.8.5 Momentum is Hermitian; the plain derivative is not

Sample Problem 10.9: Why the $-i$ in $\hat{p}$ is needed

Problem. On $[-L, L]$ with Dirichlet walls, show that $\hat{p} = -i\hbar\,\mathrm{d}/\mathrm{d}x$ is Hermitian but $D = \mathrm{d}/\mathrm{d}x$ is not.

Step 1: integrate D by parts. For f, g vanishing at $\pm L$:

$$\int f^* g'\,\mathrm{d}x = \Big[f^* g\Big]_{-L}^{L} - \int f^{*\prime} g\,\mathrm{d}x = -\int (f')^* g\,\mathrm{d}x.$$

So $\langle f|D|g\rangle = -\int (Df)^* g\,\mathrm{d}x$: moving D across costs a minus sign. The boundary term $\Big[f^* g\Big]_{-L}^{L}$ vanishes as soon as *either* function vanishes at the walls; Problem 12 uses this. D is *anti*-Hermitian, not Hermitian.

Step 2: include $-i\hbar$. $\langle f|\hat{p}|g\rangle = -i\hbar\int f^* g'\,\mathrm{d}x = -i\hbar\,(-\int (f')^* g\,\mathrm{d}x) = i\hbar\int (f')^* g\,\mathrm{d}x$.

Step 3: compare with $\int (\hat{p}f)^* g$. $(\hat{p}f)^* = (-i\hbar f')^* = +i\hbar (f')^*$, so $\int (\hat{p}f)^* g\,\mathrm{d}x = i\hbar\int (f')^* g\,\mathrm{d}x$, the same as Step 2.

Answer.

$$\boxed{\hat{p} = -i\hbar\,\tfrac{\mathrm{d}}{\mathrm{d}x} \text{ is Hermitian;} \quad \tfrac{\mathrm{d}}{\mathrm{d}x} \text{ is anti-Hermitian}}$$

The factor i converts the minus sign from integration by parts into the minus sign of complex conjugation. Without it, momentum expectation values $\langle\Psi|\hat{p}|\Psi\rangle$ would not be real. (In the box, $\hat{p}$ has no eigenfunctions obeying the wall conditions, because e^{ikx} never vanishes; its eigenfunctions e^{ikx} belong to the whole line. This is one of the domain subtleties mentioned in the `beyond` box above.)

10.8.6 A non-Hermitian Hamiltonian has complex eigenvalues

Return to the complex potential $V - i\Gamma/2$ with constant real Γ. If $H\psi = E\psi$ with real E, then

$$\left(H - \frac{i\Gamma}{2}\right)\psi = E\psi - \frac{i\Gamma}{2}\psi = \left(E - \frac{i\Gamma}{2}\right)\psi.$$

The eigenvalue is now the complex number $E - i\Gamma/2$. The time factor becomes $e^{-i(E-i\Gamma/2)t/\hbar} = e^{-iEt/\hbar}e^{-\Gamma t/2\hbar}$, and $|\Psi|^2$ decays as $e^{-\Gamma t/\hbar}$, in agreement with Eq. (10.45). The real part of the energy is unchanged; only an imaginary part appears. We have found, by a two-line calculation, a fact the paper relies on in Sec. II E: "a constant iA shifts all eigenvalues by the same imaginary constant". Chapter 14 develops non-Hermitian perturbations in full.

Checkpoint 10.7

Is $H = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x) - \frac{i\Gamma}{2}$, with real V and constant $\Gamma \neq 0$, Hermitian? What property of the eigenvalues tells you so?

Answer: No: the potential term fails the test because $(V - i\Gamma/2)^* \neq V - i\Gamma/2$. Its eigenvalues $E - i\Gamma/2$ are not real, which by Theorem 1 is impossible for a Hermitian operator.

10.8.7 Hermiticity on a computer

Sample a function on N equally spaced interior points $x_j = -L + jh$, $j = 1, \ldots, N$, with spacing $h = 2L/(N+1)$, so the walls sit at $x_0 = -L$ and $x_{N+1} = L$ where $\psi = 0$. Recall from Chapter 8 that the second derivative $\mathrm{d}^2/\mathrm{d}x^2$ becomes the tridiagonal matrix with $-2/h^2$ on the diagonal and $1/h^2$ just above and below. The Hamiltonian contains $-\mathrm{d}^2/\mathrm{d}x^2$, so we multiply every entry by -1: the operator $-\mathrm{d}^2/\mathrm{d}x^2$ becomes the matrix with $+2/h^2$ on the diagonal and $-1/h^2$ just above and below. The potential multiplies each sample $\psi(x_j)$ by $V(x_j)$, which adds $V(x_j)$ to the j-th diagonal entry. So the Hamiltonian (with $\hbar^2/2m = 1$) becomes the matrix

$$H_{jj} = \frac{2}{h^2} + V(x_j), \qquad H_{j,j\pm1} = -\frac{1}{h^2}, \qquad \text{all other entries } 0. \tag{10.58}$$

This matrix is real and symmetric (each off-diagonal entry equals its mirror image), so it is Hermitian, and by Chapter 8 its eigenvalues are real and its eigenvectors orthogonal. The Dirichlet walls are built in by leaving out the boundary points. A real potential is what makes the diagonal real. Chapter 23 studies the accuracy of Eq. (10.58) in detail; for now we just use it.

```
import numpy as np
from scipy.linalg import eigh_tridiagonal

L, N = 44.883, 2000               # half-width of the box, interior points
h = 2*L/(N+1)                     # grid spacing
x = -L + h*np.arange(1, N+1)      # interior points
V = np.zeros(N)                   # empty box
diag = 2/h**2 + V                 # diagonal of H, Eq. (schrodinger-fd)
off = -np.ones(N-1)/h**2          # off-diagonal
E, psi = eigh_tridiagonal(diag, off, select='i', select_range=(0, 4))
exact = (np.arange(1, 6)*np.pi/(2*L))**2
print(E/exact)                    # ratios very close to 1
psi = psi/np.sqrt(np.sum(psi**2, axis=0)*h)    # normalize: sum psi^2 h = 1
print(np.sum(psi[:, 0]*psi[:, 1])*h)           # orthogonality: about 0
```

Key Ideas

- An operator is Hermitian if $\int f^* Q g \,\mathrm{d}x = \int (Qf)^* g \,\mathrm{d}x$ for all allowed f, g; for H this needs a real potential and boundary conditions that kill the boundary terms (Dirichlet walls do).
- Hermitian operators have real eigenvalues and orthogonal eigenfunctions (proved above, exactly as for Hermitian matrices).
- $\hat{p} = -i\hbar\,\mathrm{d}/\mathrm{d}x$ is Hermitian; $\mathrm{d}/\mathrm{d}x$ alone is not.
- A complex potential $-i\Gamma/2$ breaks Hermiticity and makes eigenvalues complex, $E - i\Gamma/2$.
- On a grid, H becomes a real symmetric tridiagonal matrix.

10.9 Units with $\hbar = 2m = 1$

★ Did you know? *Planck's Natural Units*

In 1899, a year before his famous quantum hypothesis, Max Planck proposed units of length, mass, time and temperature built only from fundamental constants such as G, c and his own new constant. He argued they would keep their meaning for any civilization. Setting $\hbar = 2m = 1$ in this book follows the same spirit on a smaller scale: choose units so the equations carry no clutter.

10.9.1 Choosing natural units

The constants $\hbar$ and m clutter every formula. Physicists remove them by choosing units adapted to the problem. Pick any length scale ℓ and define the energy scale

$$\varepsilon = \frac{\hbar^2}{2m\ell^2}. \tag{10.59}$$

Measure positions in units of ℓ by writing $x = \ell\xi$, with ξ ("xi") a pure number. By the chain rule $\frac{\mathrm{d}}{\mathrm{d}x} = \frac{\mathrm{d}\xi}{\mathrm{d}x}\frac{\mathrm{d}}{\mathrm{d}\xi} = \frac{1}{\ell}\frac{\mathrm{d}}{\mathrm{d}\xi}$, and applying it twice, $\frac{\mathrm{d}^2}{\mathrm{d}x^2} = \frac{1}{\ell^2}\frac{\mathrm{d}^2}{\mathrm{d}\xi^2}$. The time-independent equation (10.29) becomes

$$-\frac{\hbar^2}{2m\ell^2}\frac{\mathrm{d}^2\psi}{\mathrm{d}\xi^2} + V\psi = E\psi \quad\Longleftrightarrow\quad -\varepsilon\frac{\mathrm{d}^2\psi}{\mathrm{d}\xi^2} + V\psi = E\psi.$$

Divide by ε and define the pure numbers $\tilde{V} = V/\varepsilon$, $\tilde{E} = E/\varepsilon$:

$$-\frac{\mathrm{d}^2\psi}{\mathrm{d}\xi^2} + \tilde{V}\psi = \tilde{E}\psi.$$

Now drop the tildes and rename ξ back to x:

$$\boxed{H = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x), \qquad H\psi = E\psi} \tag{10.60}$$

This is the form used throughout the paper (Sec. II B). It is called "units $\hbar = 2m = 1$" because setting $\hbar = 1$ and $m = \frac{1}{2}$ in Eq. (10.33) gives the same thing: $\hbar^2/2m = 1/(2\cdot\frac{1}{2}) = 1$. Both descriptions mean: lengths are pure numbers in units of some ℓ, and energies are pure numbers in units of $\hbar^2/2m\ell^2$.

The time-dependent equation simplifies too. Writing $t = (\hbar/\varepsilon)\tau$, we have $\partial/\partial t = (\varepsilon/\hbar)\partial/\partial\tau$, so $i\hbar\,\partial\Psi/\partial t = i\varepsilon\,\partial\Psi/\partial\tau$, and dividing Eq. (10.24) by ε gives $i\,\partial\Psi/\partial\tau = -\partial^2\Psi/\partial\xi^2 + \tilde{V}\Psi$. Time is measured in units of $\hbar/\varepsilon$.

In these units some formulas become very clean:

- free particle: $E = k^2$ (from $E = \hbar^2k^2/2m$), so $k = \sqrt{E}$; and $\omega = E/\hbar = k^2$ (the right panel of Figure 10.2);
- particle speed $v = p/m = \hbar k/m = 2k$ (from $\hbar = 1$, $m = \frac{1}{2}$);
- empty box: $E_k = (k\pi/2L)^2$, as in Chapter 8;
- in a potential, the local wavenumber where $E > V(x)$ is $\sqrt{E - V(x)}$, the starting point of Chapter 12.

Sample Problem 10.10: An electron in a 1 nm box, in and out of natural units

Problem. (a) Find the energy unit ε for an electron with $\ell = 1$ nm, in eV. (b) Find the ground-state energy of an electron in an empty box of total width 1 nm.
Step 1: energy unit. $\varepsilon = \hbar^2/(2m_e\ell^2)$:

$$\varepsilon = \frac{(1.054572 \times 10^{-34})^2}{2(9.109384 \times 10^{-31})(10^{-9})^2}\ \text{J} = 6.104 \times 10^{-21}\ \text{J} = 0.0381\ \text{eV}.$$

Step 2: box in natural units. Total width 1 nm means $2L = 1$, so $L = 0.5$ (in units of ℓ). Then $E_1 = (\pi/2L)^2 = \pi^2 = 9.870$ in units of ε.
Step 3: back to eV. $E_1 = 9.870 \times 0.0381$ eV $= 0.376$ eV.
Answer.

$$\boxed{\varepsilon = 0.0381\ \text{eV}, \qquad E_1 = \pi^2\varepsilon = 0.376\ \text{eV}}$$

This agrees with the familiar SI formula $h^2/(8m_ea^2)$ for a box of width a, since $\frac{\hbar^2}{2m}\left(\frac{\pi}{a}\right)^2 = \frac{h^2}{4\pi^2}\cdot\frac{\pi^2}{2ma^2} = \frac{h^2}{8ma^2}$.

Sample Problem 10.11: The phase of a zeta-height level

Problem. In units $\hbar = 2m = 1$, a stationary state has energy equal to the height of the first zeta zero, $E = t_1 = 14.134725$ (the first of the ten zeros listed in Chapter 1). How much time τ (in the natural time unit) does its phase factor take to rotate by π, that is, to change sign?
Step 1: With $\hbar = 1$ the phase factor is $e^{-iE\tau}$. It equals -1 when $E\tau = \pi$.
Step 2: $\tau = \pi/E = 3.141593/14.134725$.
Answer.

$$\boxed{\tau = 0.2223}$$

In the paper no particle and no physical time appear at all: only the eigenvalues matter, and they are pure numbers compared directly with the pure numbers t_n.

Checkpoint 10.8

In units $\hbar = 2m = 1$, a free particle has $E = 100$. What are its wavenumber and wavelength (in units of ℓ)?

Answer: $k = \sqrt{E} = 10$ and $\lambda = 2\pi/k = 2\pi/10 \approx 0.628$.

Pitfall

After switching to natural units, all numbers are pure numbers, and a formula like $E_1 = (\pi/2L)^2$ only makes sense if L is measured in the same unit ℓ used to define ε. Mixing a length in meters with an energy in units of ε gives nonsense.

Key Ideas

- Measuring lengths in a unit ℓ and energies in $\varepsilon = \hbar^2/2m\ell^2$ turns the Schrödinger equation into $-\psi'' + V\psi = E\psi$; this is what "units $\hbar = 2m = 1$" means.
- In these units a free particle has $k = \sqrt{E}$, $\omega = k^2$, speed $2k$; the empty box has $E_k = (k\pi/2L)^2$.
- The paper's energies are pure numbers on the same scale as the zero heights t_n.

10.10 Energy levels as a spectrum

10.10.1 What a spectrum is

The *spectrum* of an operator is (for the operators in this book) the set of its eigenvalues. For a particle confined by walls or by a potential that rises without bound, the spectrum is a discrete list $E_1 < E_2 < E_3 < \cdots$ of energy levels, ordered by size and labeled by the *level index* $n = 1, 2, \ldots$. For a free particle on the whole line, every $E \geq 0$ is allowed (each k gives a plane wave), and the spectrum is continuous. (The plane waves are not normalizable (Problem 3), so strictly these are not eigenvalues in the sense used above. The precise notion of a continuous spectrum is beyond this book, and we do not need it: every operator in the paper lives in a box.) Confinement creates discreteness, just as pinning a string creates harmonics. Table 10.1 collects the analogy.

Table 10.1: The string and the quantum particle in a box, side by side.

	string pinned at both ends	particle in a box
what waves	displacement $y(x,t)$, real	amplitude $\Psi(x,t)$, complex
equation	$y_{xx} = y_{tt}/v^2$	$i\hbar\Psi_t = H\Psi$
dispersion	$\omega = vk$	$\omega = \hbar k^2/2m$
boundary	$y = 0$ at the ends	$\psi = 0$ at the walls
spectrum	frequencies $f_n = nv/2\ell$	energies $E_k = \frac{\hbar^2}{2m}(k\pi/2L)^2$
observed	sound at f_n	light at $(E_m - E_n)/h$

The historical spectrum is that of hydrogen. Its levels are $E_n = -13.6\ \text{eV}/n^2$. We simply quote this textbook value (HRW [7]); it comes from solving the Schrödinger equation in three dimensions, which this book does not do. A transition from $n = 3$ to $n = 2$ releases a photon of energy $13.6\left(\frac{1}{4} - \frac{1}{9}\right)\ \text{eV} = 13.6 \cdot \frac{5}{36}\ \text{eV} = 1.89\ \text{eV}$, and wavelength $\lambda = hc/\Delta E = (1240\ \text{eV nm})/(1.89\ \text{eV}) \approx 656$ nm, the red Balmer line seen in every hydrogen discharge. Measuring line positions is reading off differences of eigenvalues of H.

10.10.2 Counting levels

When there are many levels, a convenient summary is the *counting function*

$$N(E) = \text{number of eigenvalues } E_n \leq E. \tag{10.61}$$

It is a staircase: it jumps by one at each level and is flat in between. (The letter N is also used for the number of grid points; the paper overloads it the same way, and context decides.)

For the empty box with $\hbar^2/2m = 1$, $E_k = (k\pi/2L)^2 \leq E$ is equivalent to $k \leq 2L\sqrt{E}/\pi$, so

$$N_{\text{box}}(E) = \left\lfloor \frac{2L\sqrt{E}}{\pi} \right\rfloor, \tag{10.62}$$

where $\lfloor y \rfloor$ (the "floor") is the largest integer not exceeding y. Dropping the floor gives a smooth curve $2L\sqrt{E}/\pi$. At $E = E_k$ it equals exactly k, where the staircase jumps up to k, so it touches the top corner of each step. Subtracting $\frac{1}{2}$ gives a curve through the middle of the staircase. Either way the slope is the same. The derivative $\mathrm{d}N/\mathrm{d}E = L/(\pi\sqrt{E})$ is the *level density*, the number of levels per unit energy. The reciprocal of the density, $\pi\sqrt{E}/L$, is the *mean spacing*. Equivalently, since the level index is $k = N$, the spacing is $\mathrm{d}E/\mathrm{d}k$. Differentiating $E_k = (k\pi/2L)^2$ with respect to k and eliminating $k = 2L\sqrt{E}/\pi$,

$$\frac{\mathrm{d}E}{\mathrm{d}k} = 2k\left(\frac{\pi}{2L}\right)^2 = 2 \cdot \frac{2L\sqrt{E}}{\pi} \cdot \frac{\pi^2}{4L^2} = \frac{\pi\sqrt{E}}{L}. \tag{10.63}$$

The empty-box levels spread *apart* as E grows.

Sample Problem 10.12: Counting the paper's box

Problem. For the empty box with $L = 44.883$ (units $\hbar = 2m = 1$), how many levels lie below $E = 1000$, and what is the mean spacing near $E = 1000$?
Step 1: count. $\sqrt{1000} = 31.6228$, so $2L\sqrt{E}/\pi = 89.766 \times 31.6228/3.14159 = 903.57$. The floor is 903.
Step 2: spacing. By Eq. (10.63), $\pi\sqrt{E}/L = 3.14159 \times 31.6228/44.883 = 2.213$.
Answer.

$$\boxed{N_{\text{box}}(1000) = 903, \qquad \text{spacing} \approx 2.21}$$

10.10.3 The zeros as a would-be spectrum

Now compare with the zeta zeros of Chapter 1. Their counting function $N(T)$, the number of zeros with $0 < t_n \le T$, is also a staircase, and on average it follows the smooth function of paper Eq. (1),

$$N_{\text{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{10.64}$$

which Chapter 17 derives (log is the natural logarithm). For instance $N_{\text{sm}}(100) = 29.0$, and there are exactly 29 zeros below 100; $N_{\text{sm}}(1000) = 648.6$, and there are 649 zeros below 1000. Differentiating Eq. (10.64),

$$\frac{\mathrm{d}N_{\text{sm}}}{\mathrm{d}t} = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} - \frac{1}{2\pi} = \frac{1}{2\pi}\log\frac{t}{2\pi},$$

so the mean spacing near height t is $2\pi/\log(t/2\pi)$, which is 1.239 at $t = 1000$ and *shrinks* slowly as t grows. Figure 10.5 shows both staircases.

Here is the first concrete hint of the program the paper tests. If the zeros were energy levels of some $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$, the potential could not be the empty box: box levels spread apart with energy, while the zeros crowd together. A potential would have to be chosen so that its levels become denser at high energy. The Wu–Sprung potential is constructed (Chapter 22) so that its semiclassical (WKB) counting function equals N_{sm}. Its actual levels then follow the smooth zeros closely but not exactly. Whether the zeros are *exactly* the spectrum of some self-adjoint operator is the Hilbert–Pólya question (Chapter 21); it is an open conjecture, not a fact.

Checkpoint 10.9

A particle in an empty box of half-width L and the zeta zeros both have about 650 "levels" below 1000 if L is chosen suitably (what L?). Would the two staircases then look the same near 1000?

Answer: Setting $2L\sqrt{1000}/\pi \approx 649$ gives $L \approx 649\pi/(2\cdot 31.62) \approx 32$. But no: the box's spacing grows like $\sqrt{E}$ while the zeros' spacing shrinks like $1/\log t$, so the shapes differ even if the totals match. Matching a whole counting function requires shaping the potential.

Key Ideas

- The spectrum of H is its set of eigenvalues; confinement makes it discrete, labeled by the level index n.
- The counting function $N(E)$ is a staircase; its smooth trend gives the level density and mean spacing.

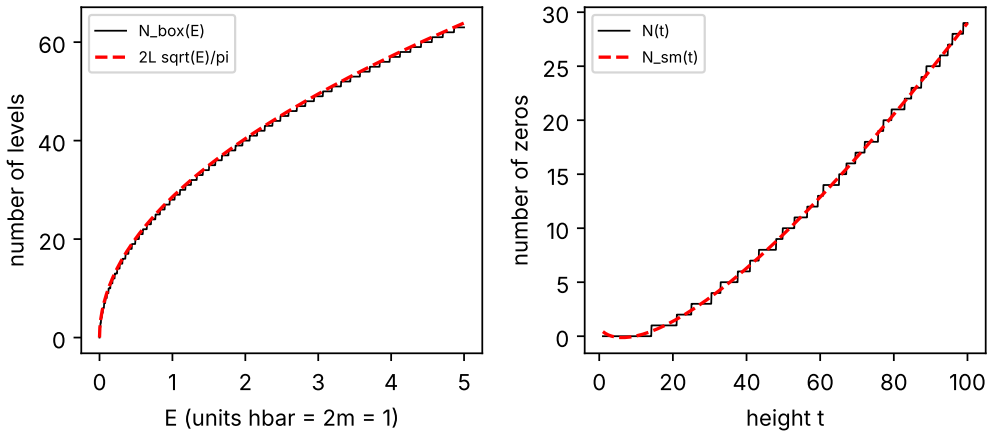


Figure 10.5: Counting functions as staircases. Left: empty-box levels $E_k = (k\pi/2L)^2$ with $L = 44.883$ (units $\hbar = 2m = 1$), the staircase $N_{\text{box}}(E)$ and the smooth curve $2L\sqrt{E}/\pi$, which touches the top corner of each step (a curve through the middle would be $2L\sqrt{E}/\pi - \frac{1}{2}$); the steps widen with E. Right: the zeta zeros, $N(t)$ (staircase) against the smooth counting function $N_{\text{sm}}(t)$ of paper Eq. (1), for $t \leq 100$, which runs through the middle of the steps on average (it equals $n - \frac{1}{2}$ at the smooth zero s_n); the steps narrow with t. There are 29 zeros below $t = 100$.

- Empty-box levels spread apart ($\mathrm{d}E/\mathrm{d}k = \pi\sqrt{E}/L$); the zeta zeros crowd together (mean spacing $2\pi/\log(t/2\pi)$). A potential must be shaped to mimic the zeros.

10.11 Paper connection

We now read specific sentences of the paper [1] with the tools of this chapter. Section, equation, table and figure numbers refer to revision R1.

10.11.1 "The spectrum of a self-adjoint operator" (Sec. I, first sentence)

The paper opens: "The Hilbert–Pólya conjecture proposes that the imaginary parts t_n of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of the Riemann zeta function are the spectrum of a self-adjoint operator." Each phrase now has content:

- *Spectrum*: the set of eigenvalues (Sec. 10.10).
- *Self-adjoint*: Hermitian in the sense of Eq. (10.57), with the domain subtleties of the `beyond` box in Sec. 10.8.
- *Why it would matter*: by Theorem 1, a self-adjoint operator has only real eigenvalues. If every t_n (defined so that the zero is $\rho_n = \frac{1}{2} + it_n$) were such an eigenvalue, every t_n would be real, which means every nontrivial zero would have real part exactly $\frac{1}{2}$. That is the Riemann hypothesis (Chapter 16). The logic runs in one direction only: *if* such an operator existed, RH would follow. No such operator is known; both the conjecture and RH remain open.

The paper's own fitted Hamiltonians are Hermitian by construction (real potential, Dirichlet walls), so their eigenvalues are real automatically. This says nothing about RH: the zeros the paper fits were computed numerically and are already known to lie on the critical line, and a Hermitian operator fitted to real numbers proves nothing about zeros it was not given. The

paper's question (Sec. I) is different and empirical: does such a fit *predict* zeros it was not shown?

10.11.2 "In units $\hbar = 2m = 1$, $H = -d^2/dx^2 + V(x)$ with V even" (Sec. II B)

This is Eq. (10.60). In the paper no electron or physical length is involved: x and V are pure numbers (the interpolation-split paragraph of Sec. III, where Fig. 4 is cited, says explicitly that "x and V are in the units $\hbar = 2m = 1$ of Sec. II B"). The units are chosen so that the eigenvalues E_n of H are numbers that can be compared directly with the zero heights t_n; every residual in the paper, $r_n = \tilde{t}_n - E_n$ (Sec. II C), subtracts an eigenvalue from a target height. Here $\tilde{t}_n$ is the paper's bias-corrected target: the zero t_n plus a small correction for grid error, explained in Chapter 23.

The paper's domain is $[-L, L]$ with $L = 44.883$ at scale 1 and $L = 59.957$ at scale 2, "Dirichlet walls" (Sec. II B), exactly the boundary conditions (10.35) that make H Hermitian. The potential is the Wu–Sprung potential defined by paper Eq. (2), with bottom value $V_0 = 3.10073\pi \approx 9.741$ at $x = 0$ (Chapter 22 derives it). "Even" means $V(-x) = V(x)$; Chapter 11 explains what evenness does to the eigenfunctions. The paper solves $H\psi = E\psi$ by the finite-difference matrix of Eq. (10.58): "Second-order finite differences on N uniform interior points (Dirichlet walls) give a symmetric tridiagonal matrix; eigenpairs from LAPACK" (Sec. II B; App. A step 4 records the diagonal $2/dx^2 + V$ and off-diagonal $-1/dx^2$, which are our $2/h^2 + V_j$ and $-1/h^2$). "Symmetric" is the Hermiticity of Sec. 10.8, and "eigenpairs" are the pairs (E_n, ψ_n) of Sec. 10.5.

Figure 10.6 shows the Wu–Sprung potential with its lowest stationary states, computed with the matrix (10.58). The paper reports (Sec. II B) that, after a grid-error correction, the levels of this potential differ from the smooth zeros s_n (the solutions of $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$) by 0.005 mean spacings on average and 0.063 at worst, "worst at low n". So the unperturbed potential already reproduces the *smooth* trend of the zeros; the paper's fits then try to add the fluctuations.

The paper scores "the first 500 levels" at scale 1 and 1000 at scale 2 (Sec. II B). In App. A step 4 it records that the scale-1 potential has 652 states with $E \le 1000$; compare 903 for the empty box (our Sample Problem in Sec. 10.10) and $N_{\mathrm{sm}}(1000) = 648.6$ for the zeros. The potential is positive inside the box (it never drops below V_0), so at energy E the local wavenumber $\sqrt{E - V(x)}$ (Sec. 10.9) is smaller than the empty-box value $\sqrt{E}$, and fewer half-wavelengths fit below a given E (Problem 17).

10.11.3 Eigenpairs and Born-rule averages (Sec. II C)

The paper's fit loop (Sec. II C) is: "solve for (E_n, ψ_n); form residuals $r_n = \tilde{t}_n - E_n$ and the first-order sensitivity $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\, dx$." Compare with Eq. (10.48): A_{nk} is the *expectation value* of the function $\phi_k(x) = \cos(k\pi x/L)$ in the stationary state ψ_n,

$$A_{nk} = \int |\psi_n|^2 \, \phi_k \, \mathrm{d}x = \langle n|\phi_k|n\rangle = \langle \phi_k \rangle_n, \tag{10.65}$$

the Born-rule average of ϕ_k over where the particle in level n is likely to be. Chapter 13 shows that it is exactly the first-order change of E_n when the potential changes by ϕ_k. For this to be right the eigenfunctions must be normalized; App. A step 9 states "ψ normalized $\sum \psi^2 dx = 1$", the discrete version of Eq. (10.40) on a grid of spacing $dx = h$ (and the ψ_n are real, so $|\psi_n|^2 = \psi_n^2$).

(The paper also uses the letter A for a different object, the non-Hermitian operator of Sec. II E; context decides.)

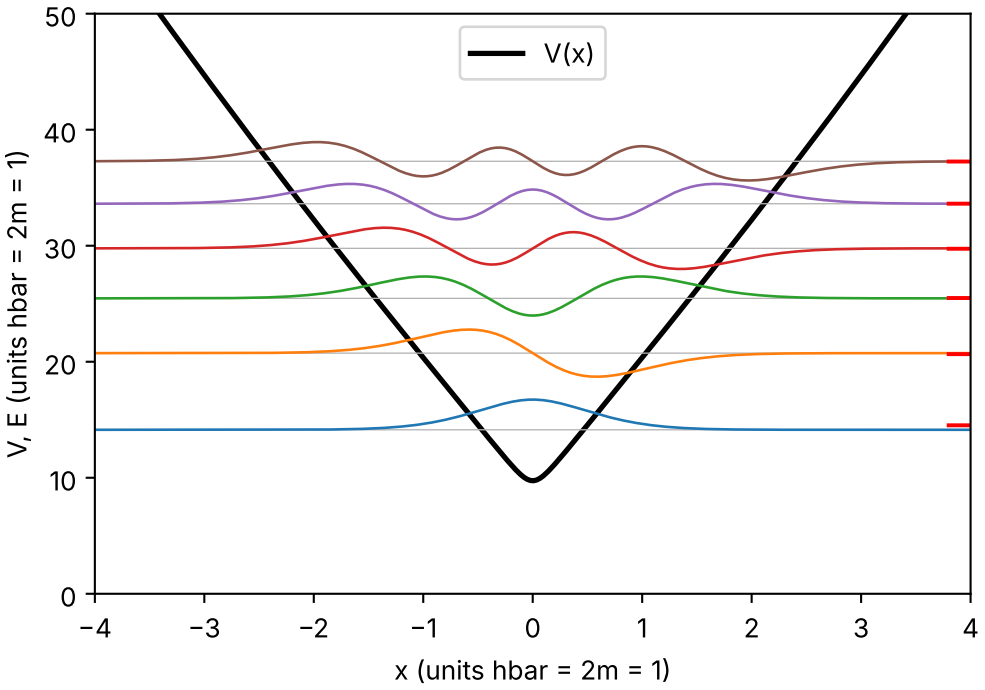


Figure 10.6: The Wu–Sprung potential $V(x)$ of paper Eq. (2) (thick curve; units $\hbar = 2m = 1$; bottom $V_0 = 3.10073\pi$ at $x = 0$) in the paper's scale-1 box $[-L, L]$ with $L = 44.883$, of which only the central region is shown. Thin curves: the six lowest eigenfunctions ψ_n, normalized so that $\sum_j \psi_n(x_j)^2 h = 1$, drawn offset vertically to their energies E_n (horizontal lines) and scaled for visibility. Even and odd states alternate, and the n-th state has $n - 1$ nodes. Short ticks at the right edge mark the smooth zeros s_n, which the levels track closely.

Sample Problem 10.13: The sensitivity matrix of the empty box

Problem. For the empty box on $[-L, L]$, compute $A_{1k} = \int |\psi_1|^2 \cos(k\pi x/L)\,\mathrm{d}x$ for all integers $k \geq 0$, using the normalized ψ_1 of Eq. (10.41).
Step 1: simplify ψ_1. $\sin(\pi(x+L)/2L) = \sin(\pi x/2L + \pi/2) = \cos(\pi x/2L)$, so $|\psi_1|^2 = \frac{1}{L}\cos^2(\pi x/2L)$.
Step 2: double-angle formula. $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ with $\theta = \pi x/2L$ gives $|\psi_1|^2 = \frac{1}{2L}(1+\cos(\pi x/L))$.
Step 3: integrate against $\cos(k\pi x/L)$.

$$A_{1k} = \frac{1}{2L}\int_{-L}^{L} \cos\frac{k\pi x}{L}\,\mathrm{d}x + \frac{1}{2L}\int_{-L}^{L} \cos\frac{\pi x}{L}\cos\frac{k\pi x}{L}\,\mathrm{d}x.$$

By the orthogonality of cosines over $[-L, L]$ (Chapter 5), the first integral is $2L$ if $k = 0$ and 0 otherwise; the second is L if $k = 1$ and 0 otherwise.
Answer.

$$\boxed{A_{10} = 1, \qquad A_{11} = \tfrac{1}{2}, \qquad A_{1k} = 0 \quad (k \geq 2)}$$

$A_{10} = 1$ for *every* level, because $\phi_0 = 1$ and $\int |\psi_n|^2\,\mathrm{d}x = 1$: adding a constant c_0 to the potential shifts every level by c_0. In the empty box, apart from the constant ϕ_0, the cosine with $k = n$ affects only level n (Problem 18). In the Wu–Sprung potential the eigenfunctions are no longer pure sines and each cosine reaches many levels; how that works, and why it lets the fit move levels almost independently, is the subject of Chapters 13 and 24.

10.11.4 Breaking Hermiticity on purpose (Secs. II E and V)

Sec. II E of the paper perturbs $H \to H + iA$ with $A = \sum_k c_k\phi_k$ a real function multiplying ψ. By Sec. 10.8, the potential $V + iA$ is complex, so $H + iA$ is not Hermitian and its eigenvalues are in general complex. The "absorptive-even" class models "$-i\Gamma/2$": by Eq. (10.46), a negative imaginary potential $-i\Gamma(x)/2$ drains probability wherever $\Gamma(x) > 0$, so the physical requirement is $\Gamma \geq 0$, "i.e. $A \leq 0$ pointwise" (Sec. II E). The paper's safeguard (i) uses exactly the two-line calculation of Sec. 10.8: "a constant iA shifts all eigenvalues by the same imaginary constant" (and, as Chapter 14 shows, it has zero second-order real shift). Paper Eq. (5) uses matrix elements $A_{nm} = \langle n|A|m\rangle$, which in the notation of Eq. (10.50) means $\int \psi_n^* A\,\psi_m\,\mathrm{d}x$. Finally, Sec. V insists that "the object of any physical claim is the exact spectrum of $H + iA$", the set of eigenvalues of the actual (non-Hermitian) operator, not an approximation to it.

10.11.5 What this chapter does not yet explain

Why the eigenfunctions in Figure 10.6 have the shapes they do (Chapter 11); why each level "probes" the potential most strongly near its classical turning points, the paper's "position maps to level index through the classical turning point" (Sec. III, interpolation split; Chapter 12); how V is built from N_{sm} (Chapter 22); and how accurate the grid eigenvalues are, including the "raw-zero null 2.09" of App. A (Chapter 23).

10.12 Summary

- String waves obey $\omega = vk$; pinned ends give a discrete spectrum of harmonics, Eq. (10.11).
- De Broglie: $\lambda = h/p$, $p = \hbar k$, $E = \hbar\omega$, Eq. (10.15); a free particle has $\omega = \hbar k^2/2m$, Eq. (10.16).

- The simplest linear equation with these plane-wave solutions is complex, Eq. (10.22); adding V gives the time-dependent Schrödinger equation (10.24), a postulate. Wave packets move at $v_g = p/m$, Eq. (10.25).
- Separation of variables gives stationary states $\psi(x)e^{-iEt/\hbar}$ and the eigenvalue problem $H\psi = E\psi$, Eqs. (10.29) and (10.34), with H of Eq. (10.33); general solutions are superpositions, Eq. (10.32).
- Empty box with Dirichlet walls: $\psi_k = L^{-1/2}\sin(k\pi(x+L)/2L)$, $E_k = \frac{\hbar^2}{2m}(k\pi/2L)^2$, Eqs. (10.41) and (10.36).
- Born rule: $|\Psi|^2$ is a probability density, Eq. (10.39); normalization (10.40) is conserved when V is real, and decays as $e^{-\Gamma t/\hbar}$ for $V - i\Gamma/2$, Eq. (10.45).
- Expectation values $\langle Q\rangle = \int \Psi^* Q \Psi \, \mathrm{d}x$, Eq. (10.49); energy measurements give E_n with probability $|c_n|^2$, and $\langle H\rangle = \sum |c_n|^2 E_n$, Eq. (10.55).
- Hermitian operators, Eq. (10.57), have real eigenvalues and orthogonal eigenfunctions (proved); H is Hermitian with real V and Dirichlet walls; on a grid it is a real symmetric tridiagonal matrix, Eq. (10.58).
- In units $\hbar = 2m = 1$, $H = -\mathrm{d}^2/\mathrm{d}x^2 + V$, Eq. (10.60); energies are pure numbers comparable with the t_n.
- Counting functions are staircases; the empty box has $N(E) = \lfloor 2L\sqrt{E}/\pi \rfloor$ and spreading levels, Eqs. (10.62)–(10.63); the zeros follow N_{sm} and crowd together.
- The paper's sensitivity A_{nk} is a Born-rule expectation value, Eq. (10.65).

10.13 Problems

Constants: $h = 6.62607015 \times 10^{-34}$ J s, $\hbar = h/2\pi$, $m_e = 9.1093837 \times 10^{-31}$ kg, $m_n = 1.6749275 \times 10^{-27}$ kg, 1 eV $= 1.602176634 \times 10^{-19}$ J.

1. • Find the de Broglie wavelength of an electron with kinetic energy 10 eV. How does it compare with the 100 eV result of the first Sample Problem, and why?

2. • Thermal neutrons in a reactor move at about 2200 m/s. Find their de Broglie wavelength in nanometers.

3. • Show that the plane wave $\Psi = e^{i(kx-\omega t)}$ has $|\Psi|^2 = 1$ everywhere. Explain why it cannot be normalized on the whole line, and what that means physically.

4. •• Show in full that $\Psi = \cos(kx - \omega t)$ does not satisfy the free Schrödinger equation (10.22) for any real $k \neq 0$, by evaluating both sides at $x = 0$ for two well-chosen times.

5. • In units $\hbar = 2m = 1$, a free particle has $E = 25$. Find its wavenumber k, wavelength λ, angular frequency ω and speed v.

6. •• Show that if Ψ_1 and Ψ_2 solve the time-independent equation $H\psi = E\psi$ with the *same* E, then $a\Psi_1 + b\Psi_2$ does too. Show by an example (two box states) that this fails if their energies differ.

7. •• In the empty box on $[-L, L]$ (units $\hbar = 2m = 1$), consider the trial function $\psi = A(L^2 - x^2)$. (a) Show it satisfies the Dirichlet conditions. (b) Normalize it. (c) Compute $\langle H\rangle$ and compare with $E_1 = \pi^2/4L^2$. Which is larger, and by what factor?

8. •• For the normalized empty-box ground state, show that $\langle x\rangle = 0$ and $\langle x^2\rangle = L^2\left(\frac{1}{3} - \frac{2}{\pi^2}\right)$. (Hint: after the substitution $u = \pi x/2L$, use $\cos^2 u = \frac{1}{2}(1+\cos 2u)$ and integrate $u^2\cos 2u$ by parts twice.)

9. •• Using the result of the box ground-state Sample Problem and symmetry, find the probability that the particle in the empty-box ground state is found in $L/2 < x < L$.

10. ••• For $\Psi = (\psi_1 e^{-iE_1 t/\hbar} + \psi_2 e^{-iE_2 t/\hbar})/\sqrt{2}$ in the empty box with real normalized ψ_1, ψ_2 from Eq. (10.41): (a) using Eq. (10.56), show $\langle x\rangle(t) = I\cos((E_2 - E_1)t/\hbar)$ with $I = \int x\psi_1\psi_2\,\mathrm{d}x$; (b) show $|I| = 32L/(9\pi^2)$; (c) in units $\hbar = 2m = 1$ with $L = 44.883$, find the period of the oscillation.

11. ••• Repeat the norm-conservation argument of Sec. 10.6 for $V - i\Gamma/2$ with constant $\Gamma > 0$, showing every step, and derive Eq. (10.45). In units with $\hbar = 1$, find the time for the norm to halve when $\Gamma = 0.1$.

12. •• Show that if $V(x)$ is real and $Q = V(x)$ (a multiplication operator), then Q is Hermitian for any boundary conditions. Then show that $\hat{p}^2 = -\hbar^2\mathrm{d}^2/\mathrm{d}x^2$ is Hermitian by applying the momentum Sample Problem twice. (Hint: the boundary term $[f^* g]_{-L}^{L}$ in Step 1 of that Sample Problem vanishes if *either* function vanishes at the walls, so the result may be used with $\hat{p}g$, which need not vanish there, on the right.)

13. •• Verify Theorem 2 directly for the empty box: show $\int_{-L}^{L}\psi_1\psi_2\,\mathrm{d}x = 0$ using a product-to-sum identity.

14. •• An electron is confined to an empty box of total width 0.2 nm. Find its ground-state energy in eV, using natural units with $\ell = 1$ nm as in the Sample Problem of Sec. 10.9.

15. • (Paper) Sec. II B of the paper writes $H = -d^2/dx^2 + V(x)$ without $\hbar$ or m. Explain why this is legitimate, and say what the numbers on the x and V axes of the paper's Fig. 4 mean.

16. •• (Paper) Compute the ground-state energy of the *empty* box at the paper's two scales, $L = 44.883$ and $L = 59.957$ (units $\hbar = 2m = 1$). What is the ratio of the two, and why does it equal $(44.883/59.957)^2$?

17. •• (Paper) Using Eq. (10.62), count the empty-box levels with $E \le 1000$ for $L = 44.883$. The paper (App. A step 4) reports 652 states with $E \le 1000$ for the Wu–Sprung potential in the same box, and $N_{\mathrm{sm}}(1000) = 648.6$. Explain physically why the potential reduces the count, using the local wavenumber $\sqrt{E - V(x)}$.

18. •• (Paper) For the empty box on $[-L, L]$, show that the paper's sensitivity $A_{nk} = \int |\psi_n|^2\cos(k\pi x/L)\,\mathrm{d}x$ equals 1 for $k = 0$, $(-1)^{n+1}/2$ for $k = n$, and 0 otherwise. (Hint: show $|\psi_n|^2 = \frac{1}{2L}\left(1 - (-1)^n\cos(n\pi x/L)\right)$.) What does this say about which basis function moves which level, to first order?

19. •• (Paper) Explain the chain of reasoning "self-adjoint operator ⇒ zeros on the critical line" behind the first sentence of the paper. Then explain why the fact that the paper's fitted H is Hermitian, with real eigenvalues matching many t_n, is *not* evidence for the Riemann hypothesis.

20. •• (Paper) In the paper's absorptive-even class (Sec. II E), the constant basis function gives $iA = ic_0$. Using Sec. 10.8, find the exact eigenvalues of $H + ic_0$ in terms of those of H, say what happens to the norm of a state in time for $c_0 < 0$, and explain why this mode cannot change any real level.

21. ●●● (Computational) Using numpy and `scipy.linalg.eigh_tridiagonal`, build the matrix (10.58) for the empty box with $L = 1$ and $N = 500$. Compute the five lowest eigenvalues and their relative errors against $(k\pi/2)^2$. What is the relative error for $k = 5$? (Chapter 23 derives the exact discrete formula $\frac{4}{h^2}\sin^2\frac{k\pi h}{4L}$; check that it matches your numbers.)

22. ●● (Computational) Using numpy, sample $\psi = e^{-|x|}$ on a uniform grid on $[-20, 20]$ with spacing 0.001, normalize it numerically with $\sum \psi^2 h = 1$, and compute the probability of $|x| < 1$. Compare with the Sample Problem.

23. ●●● (Computational) Using `qmnt.wu_sprung_V` to build the potential on $N = 6000$ interior points of $(-L, L)$ with $L = 44.883$ (`qmnt.grid`), and `qmnt.fd_levels` for the eigenvalues, compute the five lowest levels. Compare them with the smooth zeros `qmnt.smooth_zeros` and express each difference in mean spacings $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$. Compare with the paper's reported residuals (Sec. II B).

24. ●● (Computational) Using `qmnt.zeros` and `qmnt.Nsm`, plot the staircase $N(t)$ of the zeros and $N_{\mathrm{sm}}(t)$ for $t \le 100$, and count the zeros below $t = 100$.

25. ●●● Show that in natural units the time-dependent equation becomes $i\,\partial\Psi/\partial\tau = -\partial^2\Psi/\partial\xi^2 + \tilde{V}\Psi$ with $t = (\hbar/\varepsilon)\tau$, and compute the time unit $\hbar/\varepsilon$ in seconds for an electron with $\ell = 1$ nm.

26. ●●● For the normalized empty-box ground state, compute $\langle p^2\rangle$ directly from $\hat{p}^2 = -\hbar^2\mathrm{d}^2/\mathrm{d}x^2$, and show $\langle H\rangle = \langle p^2\rangle/2m = E_1$. Then explain why $\langle p^2\rangle \neq \langle p\rangle^2$ (recall $\langle p\rangle = 0$), and what this says about the spread of momentum.

27. ●● A box state is $\Psi = \frac{1}{\sqrt{5}}(\psi_1 + 2i\psi_3)$. Check that it is normalized, give the probabilities of measuring E_1 and E_3, and compute $\langle H\rangle$ as a multiple of E_1.

28. ●●● Generalize the beat argument of Sec. 10.3 to a matter wave in units $\hbar = 2m = 1$: superpose $e^{i(k_1x - k_1^2t)}$ and $e^{i(k_2x - k_2^2t)}$, find the exact speed of the envelope of the real part of this two-wave superposition, and show that it equals the classical speed $2k$ with $k = (k_1 + k_2)/2$, with no approximation. Compare with the phase speed.

10.14 Answers to selected problems

★ Did you know? *Father and Son, Particle and Wave*

Louis de Broglie proposed in his 1924 thesis that electrons have wavelength h/p. In 1927 Clinton Davisson and Lester Germer at Bell Labs, and independently George Paget Thomson, observed electron diffraction. Davisson and Thomson shared the 1937 Nobel Prize. The twist is that G. P. Thomson's father, J. J. Thomson, had won the 1906 Nobel Prize for discovering the electron as a particle.

1. $K = 10$ eV $= 1.602 \times 10^{-18}$ J, $p = \sqrt{2m_eK} = 1.708 \times 10^{-24}$ kg m/s, $\lambda = h/p = 3.88 \times 10^{-10}$ m $= 0.388$ nm. Since $\lambda \propto 1/\sqrt{K}$, a tenfold drop in energy multiplies λ by $\sqrt{10}$: $0.1226 \times \sqrt{10} = 0.388$ nm.

2. $p = m_nv = 1.6749 \times 10^{-27} \times 2200 = 3.685 \times 10^{-24}$ kg m/s; $\lambda = h/p = 1.80 \times 10^{-10}$ m $= 0.180$ nm, comparable to atomic spacings, which is why thermal neutrons diffract from crystals.

3. $|\Psi|^2 = \Psi^*\Psi = e^{-i(kx-\omega t)}e^{i(kx-\omega t)} = e^0 = 1$. Then $\int_{-\infty}^{\infty}|\Psi|^2\,\mathrm{d}x = \int_{-\infty}^{\infty}1\,\mathrm{d}x$ diverges, so no constant A can make $\int|A\Psi|^2 = 1$. A plane wave is an idealization of a particle with perfectly sharp momentum spread over all space; real particles are wave packets (superpositions of plane waves) that are normalizable.

5. With $\hbar = 2m = 1$: $E = k^2$ gives $k = 5$; $\lambda = 2\pi/5 = 1.257$; $\omega = E/\hbar = 25$ (equal to k^2); $v = p/m = \hbar k/m = 5/(1/2) = 10$.

7. (a) $L^2 - x^2 = 0$ at $x = \pm L$. (b) $\int_{-L}^{L}(L^2 - x^2)^2 \mathrm{d}x = 2\int_0^L (L^4 - 2L^2x^2 + x^4)\mathrm{d}x = 2\left(L^5 - \frac{2}{3}L^5 + \frac{1}{5}L^5\right) = \frac{16}{15}L^5$, so $A^2 = 15/(16L^5)$. (c) $\psi'' = -2A$, so $H\psi = 2A$ and $\langle H\rangle = \int A(L^2 - x^2)\cdot 2A\,\mathrm{d}x = 2A^2\left(2L^3 - \frac{2}{3}L^3\right) = \frac{8}{3}A^2L^3 = \frac{5}{2L^2}$. Since $E_1 = \pi^2/4L^2 = 2.467/L^2$, $\langle H\rangle/E_1 = 10/\pi^2 = 1.013$: the trial function's average energy is about 1.3% above the true ground state. (It can never be below E_1: by Eq. (10.55), $\langle H\rangle = \sum |c_n|^2 E_n \geq E_1 \sum |c_n|^2 = E_1$.)

9. The ground-state density is even, so the probabilities of $[-L, -L/2]$ and $[L/2, L]$ are equal. Together with the middle half they total 1, so each outer quarter has $\frac{1}{2}\left(1 - \left(\frac{1}{2} + \frac{1}{\pi}\right)\right) = \frac{1}{4} - \frac{1}{2\pi} = 0.0908$.

11. The extra term in $\partial\Psi/\partial t$ is $-\frac{i}{\hbar}(-\frac{i\Gamma}{2})\Psi = -\frac{\Gamma}{2\hbar}\Psi$; in $\partial\Psi^*/\partial t$ it is $-\frac{\Gamma}{2\hbar}\Psi^*$. The kinetic terms integrate to boundary terms that vanish as before, and the real-V terms cancel as before; the extra terms give $\frac{\mathrm{d}P}{\mathrm{d}t} = -\frac{\Gamma}{\hbar}P$, whose solution (check by differentiating) is $P(0)e^{-\Gamma t/\hbar}$. Half-life: $e^{-\Gamma t/\hbar} = \frac{1}{2}$ gives $t = \hbar \ln 2/\Gamma = \ln 2/0.1 = 6.93$ (in units with $\hbar = 1$).

13. $\psi_1\psi_2 = \frac{1}{L}\sin u \sin 2u$ with $u = \pi(x + L)/2L \in [0, \pi]$ and $\mathrm{d}x = (2L/\pi)\mathrm{d}u$. With $\sin u \sin 2u = \frac{1}{2}(\cos u - \cos 3u)$, $\int_0^\pi \frac{1}{2}(\cos u - \cos 3u)\mathrm{d}u = \frac{1}{2}\left[\sin u - \frac{\sin 3u}{3}\right]_0^\pi = 0$.

15. Measuring lengths in a unit ℓ and energies in $\varepsilon = \hbar^2/2m\ell^2$ turns $-\frac{\hbar^2}{2m}\psi'' + V\psi = E\psi$ into $-\psi'' + V\psi = E\psi$ (Sec. 10.9). In the paper there is no physical particle, so ℓ is never specified: x and V on the axes of Fig. 4 are pure numbers in these units, and V (like E_n) is on the same numerical scale as the zero heights t_n; for example the potential reaches 1000 at the scale-1 wall $x = 44.883$.

17. $2L\sqrt{1000}/\pi = 903.57$, so 903 empty-box levels lie below 1000, against 652 for the Wu–Sprung potential (App. A step 4) and $N_{\mathrm{sm}}(1000) = 648.6$ (there are 649 zeros below 1000). Inside the box the potential is positive, rising from $V_0 \approx 9.74$ at the center to 1000 at the walls, so at energy E the local wavenumber $\sqrt{E - V(x)}$ is smaller than the free value $\sqrt{E}$, and near the walls (where $V > E$) there is no oscillation at all. Fewer half-wavelengths fit, hence fewer levels below each E.

19. If t_n were eigenvalues of a self-adjoint operator, Theorem 1 would make each t_n real; since the zero is $\rho_n = \frac{1}{2} + it_n$, a real t_n means $\mathrm{Re}\,\rho_n = \frac{1}{2}$ for all nontrivial zeros, which is RH. The paper's H is Hermitian by construction (real V, Dirichlet walls), so its eigenvalues would be real whatever was fitted; the t_n it fits were computed numerically and already verified to be real (on the line). Matching known real numbers with real eigenvalues adds no information about zeros not fitted, and in any case the paper finds that, for the operator families and fitting protocol it tested, the fits show no reproducible skill at predicting held-out zeros (Sec. III). RH and the Hilbert–Pólya conjecture remain open.

21. The ratios of numerical to exact eigenvalues are slightly below 1, and the relative error grows with k; for $k = 5$ it is 8.19×10^{-5}. The discrete formula gives relative error $1 - \sin^2 a/a^2$ with $a = k\pi h/4L = 5\pi/1002$, which agrees, and is approximately $a^2/3$, growing as k^2. (This growth with level index is the "dispersion" that becomes several mean spacings at levels 401–500 in the paper; Chapter 23.)

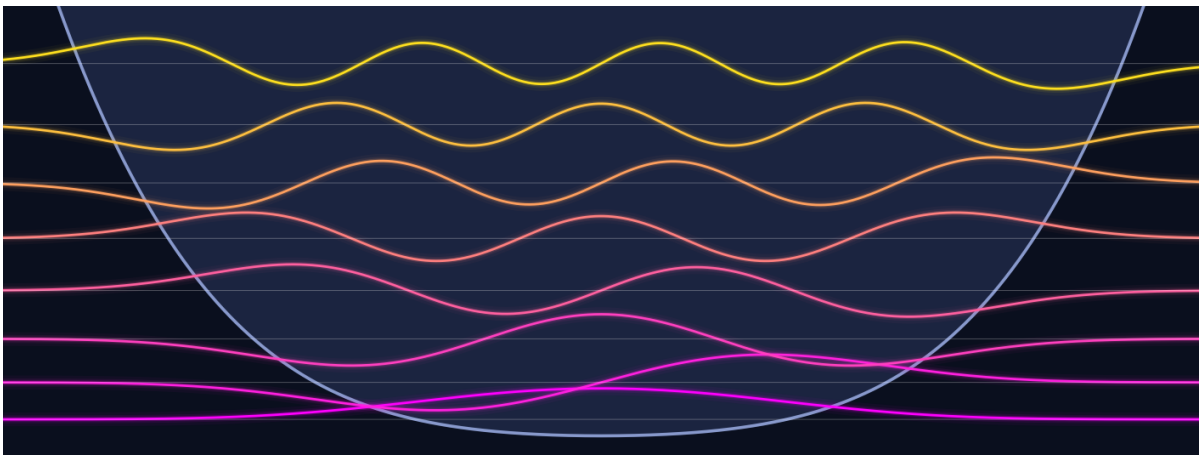
23. All five levels lie within 0.1 mean spacings of $s_1, \ldots, s_5$. This is consistent with the paper's statement (Sec. II B) that the (grid-corrected) levels of the smooth potential differ from the smooth zeros by 0.005 spacings on average and 0.063 at worst, with the worst cases at low n. A sketch of the code:

```
import numpy as np, qmnt
L, N = 44.883, 6000
x = qmnt.grid(L, N)
V = np.array([qmnt.wu_sprung_V(xi) for xi in x])
E = qmnt.fd_levels(V, L, k=5)
```

```
s = np.array([qmnt.smooth_zeros(n) for n in range(1, 6)])
gap = 2*np.pi/np.log(s/(2*np.pi))
print((E - s)/gap)        # each well below 0.1 in magnitude
```

25. With $t = (\hbar/\varepsilon)\tau$, the chain rule gives $\partial/\partial t = (\varepsilon/\hbar)\partial/\partial\tau$, so $i\hbar\,\partial_t\Psi = i\varepsilon\,\partial_\tau\Psi$; with $x = \ell\xi$, $-\frac{\hbar^2}{2m}\partial_x^2 = -\varepsilon\partial_\xi^2$; dividing by ε gives the stated equation. For $\ell = 1$ nm, $\varepsilon = 6.104\times10^{-21}$ J and $\hbar/\varepsilon = 1.0546\times10^{-34}/6.104\times10^{-21}$ s $= 1.73\times10^{-14}$ s.

27. $\left|1/\sqrt{5}\right|^2 + \left|2i/\sqrt{5}\right|^2 = \frac{1}{5} + \frac{4}{5} = 1$ (note $|2i|^2 = 4$). $P(E_1) = \frac{1}{5}$, $P(E_3) = \frac{4}{5}$. With $E_3 = 9E_1$, $\langle H\rangle = \frac{1}{5}E_1 + \frac{4}{5}\cdot 9E_1 = \frac{37}{5}E_1 = 7.4\,E_1$.



The first eight bound states of an anharmonic well, each drawn at its energy.

Chapter 11

Bound States in One Dimension

↑ Contents

What this chapter is about

A guitar string clamped at both ends can vibrate only in a discrete set of patterns: the fundamental, the second harmonic, the third, and so on. You met this in first-year physics, and you met its mathematical skeleton in Chapter 8, where the equation $-y'' = \lambda y$ with fixed ends had eigenvalues $(k\pi/2L)^2$. A quantum particle trapped in a well behaves the same way. Its wavefunction must fit inside the well, and only special energies allow that. These special energies are the *bound-state energies*, and the list of them is the particle's *spectrum*.

In this chapter we solve the two exactly solvable wells that every physicist knows, the infinite square well (the "box") and the harmonic oscillator. We then establish general facts. Bound-state energies in one dimension are never degenerate (proved from a uniqueness theorem we take on trust). Every state of an even well is even or odd (proved). The n-th state has $n-1$ nodes (stated; the proof is beyond this book). Then we ask the question that matters most for this book: how does the *shape* of a well control the *spacing* of its levels? We find a simple and physical answer: the level density equals the classical period divided by 2π, so a well that widens quickly with energy, that is a potential that grows slowly, makes its levels crowd together.

Where this shows up in the paper: the paper's Hamiltonian $H = -d^2/dx^2 + V(x)$ lives in a box $[-L, L]$ with Dirichlet walls (Sec. II B of [1]); its potential, the Wu–Sprung potential of Eq. (2), is even; its levels are labeled from $n = 1$ and the first 500 are scored at scale 1; and the slowly rising level density of the Riemann zeros suggests, at the heuristic semiclassical level of Section 11.7, that such a potential must grow a little more slowly than a harmonic well. The parity facts of this chapter also explain why the paper's "PT-odd" perturbations have vanishing diagonal matrix elements (Sec. II E).

What you need

- Chapter 10: the time-independent Schrödinger equation $H\psi = E\psi$; the Born rule ($|\psi|^2$ is a probability density); normalization; Hermitian operators have real eigenvalues; the units $\hbar = 2m = 1$, in which $H = -d^2/dx^2 + V(x)$.
- Chapter 8: eigenvalues and eigenvectors; eigenvectors of a symmetric (Hermitian) matrix belonging to different eigenvalues are orthogonal; the string with fixed ends $-y'' = \lambda y$; the second-derivative operator as a tridiagonal matrix; `scipy.linalg.eigh_tridiagonal` as a black-box eigensolver.
- Chapter 5: orthogonality of sines and cosines; even and odd functions.
- Chapter 3: power series, term-by-term differentiation, the series of e^x, and ratios of successive coefficients.
- Chapter 2: complex conjugate.
- First-year physics: potential-energy curves, turning points, simple harmonic motion (its period does not depend on amplitude), standing waves on a string.

11.1 What is a bound state?

11.1.1 The classical picture

Think of a marble rolling inside a bowl. If its total energy E is less than the height of the bowl's rim, the marble rolls back and forth forever between two *turning points* x_1 and x_2, where its potential energy $V(x)$ equals E and its kinetic energy is zero. Between the turning points $E > V(x)$; this is the *classically allowed region*. Outside them $E < V(x)$, which would require negative kinetic energy; this is the *classically forbidden region*. A classical particle that stays forever in a bounded region is said to be *bound*.

Classically, any energy above the bottom of the bowl is possible. Quantum mechanically, as we will now see, only special energies are.

11.1.2 The quantum equation and the curvature rule

Recall from Chapter 10 that in units with $\hbar = 2m = 1$ the time-independent Schrödinger equation for a particle on a line is

$$-\psi''(x) + V(x)\,\psi(x) = E\,\psi(x), \tag{11.1}$$

where $\psi(x)$ is the wavefunction, $\psi'' = d^2\psi/dx^2$, $V(x)$ is the potential energy and E is the energy. The Born rule says $|\psi(x)|^2\,\mathrm{d}x$ is the probability of finding the particle between x and $x + \mathrm{d}x$, so a physical state must be *normalizable*:

$$\int |\psi(x)|^2\,\mathrm{d}x = 1, \tag{11.2}$$

the integral running over the whole region available to the particle.

Sample Problem 11.1: Real wavefunctions are enough

Show that if ψ solves Equation (11.1) with real V and real E, then so do its real part and its imaginary part.

Step 1: Take the complex conjugate of Equation (11.1). Since V and E are real (recall from Chapter 10 that the eigenvalues of the Hermitian operator H are real), conjugation acts only on ψ: $-(\psi^*)'' + V\psi^* = E\psi^*$. So ψ^* is a solution.

Step 2: The equation is linear: a sum of solutions with the same E is a solution, and so is a constant multiple of one. Hence $\operatorname{Re}\psi = (\psi + \psi^*)/2$ and $\operatorname{Im}\psi = (\psi - \psi^*)/(2i)$ are solutions.
Step 3: At least one of them is not identically zero (otherwise $\psi = 0$).

For bound states we may always choose ψ real.

From now on all bound-state wavefunctions in this chapter are taken real, so $|\psi|^2 = \psi^2$.

Rewrite Equation (11.1) as

$$\psi''(x) = [V(x) - E]\,\psi(x). \tag{11.3}$$

The second derivative is the *curvature* of the graph of ψ. Equation (11.3) is a rule for the shape of the wavefunction:

- In an allowed region ($E > V$) the bracket is negative, so ψ'' has the sign opposite to ψ. Where $\psi > 0$ the graph bends down, where $\psi < 0$ it bends up: it always curves back toward the axis. The wavefunction *oscillates*. If $E - V = k^2$ is constant, the solutions are $\cos kx$ and $\sin kx$; for example $(\cos kx)'' = -k^2 \cos kx$.
- In a forbidden region ($E < V$) the bracket is positive, so ψ'' has the same sign as ψ: the graph curves away from the axis. If $V - E = \kappa^2$ is constant, the solutions are $e^{\kappa x}$ and $e^{-\kappa x}$; for example $(e^{-\kappa x})'' = \kappa^2 e^{-\kappa x}$. A generic solution is a mixture of a growing and a decaying exponential.

11.1.3 Why only special energies work: shooting

Take the bowl-shaped potential $V(x) = x^2$, which is even, and look for an even solution. An even function has $\psi'(0) = 0$ (its slope at the origin is zero), so start at $x = 0$ with $\psi(0) = 1$, $\psi'(0) = 0$ and integrate Equation (11.3) outward, like firing a projectile and watching where it lands. This is called *shooting*. On a computer you step outward in small steps Δx. From ψ and ψ' at x, the rule $\psi'' = (V - E)\psi$ gives the change in slope, $\psi'(x + \Delta x) \approx \psi'(x) + \psi''(x)\Delta x$. Then $\psi(x + \Delta x) \approx \psi(x) + \psi'(x)\Delta x$, and you repeat. For x beyond the turning point $\sqrt{E}$ the region is forbidden and the solution is some mixture of a growing and a decaying piece. For almost every E the growing piece is present, and ψ runs off to $+\infty$ or $-\infty$; such a ψ violates Equation (11.2). Figure 11.1 shows what happens: for $E = 0.9$ the solution has not curved enough and shoots up; for $E = 1.1$ it has curved too much and shoots down. Between them, at exactly $E = 1$, it decays to zero.

There is a counting argument behind this picture. A second-order equation has two independent solutions. Decay at the right end is one condition, and decay at the left end is another. The overall size of ψ is free (it is fixed afterward by normalization), so only one combination of the two solutions is at our disposal, and it cannot generally meet two conditions. The energy E is the extra knob that must be tuned. This is a heuristic; the rigorous statement is in the box below.

Taken on trust

Taken on trust: discreteness of the spectrum. Call a potential *confining* if $V(x) \to +\infty$ as $|x| \to \infty$, or if the particle is confined to a finite interval by walls. For a confining potential that is continuous (or has finitely many jumps) inside the region, the normalizable solutions of Equation (11.1) exist only for a discrete sequence of real energies $E_1 < E_2 < E_3 < \cdots$ with $E_n \to \infty$, and the corresponding eigenfunctions form

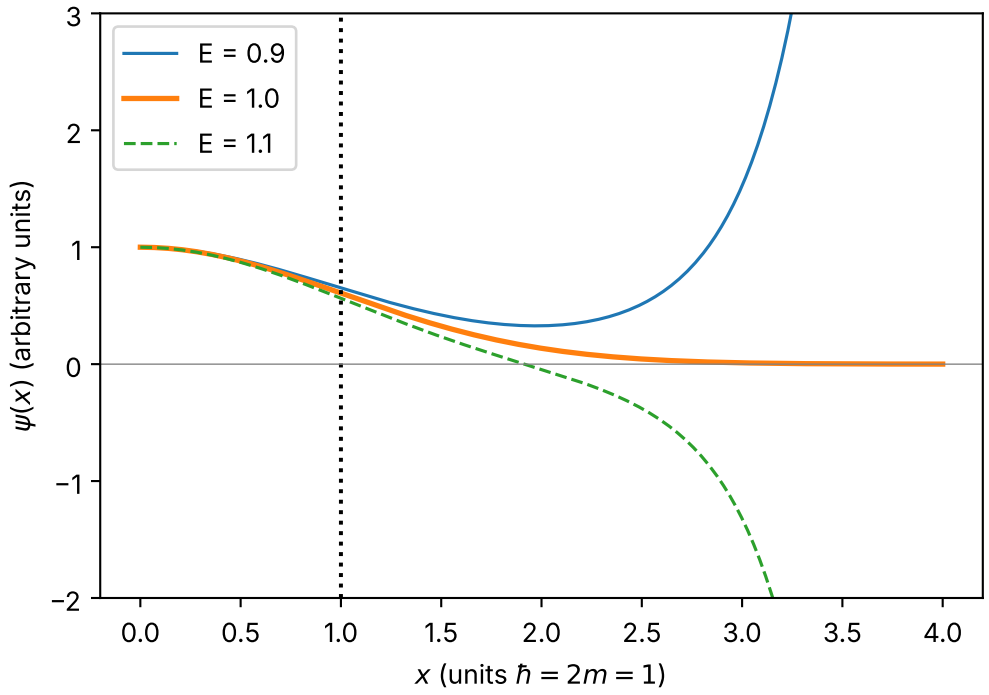


Figure 11.1: Shooting in the potential $V = x^2$ (units $\hbar = 2m = 1$). Each curve solves $\psi'' = (x^2 - E)\psi$ with $\psi(0) = 1$, $\psi'(0) = 0$. For $E = 0.9$ the solution diverges upward, for $E = 1.1$ downward; only at $E = 1$ does it decay, giving a normalizable (bound) state. The dotted vertical line is the classical turning point $x = 1$ of the $E = 1$ state.

a complete orthonormal set. The proof belongs to Sturm–Liouville theory; see [22] for the theory of such eigenvalue problems and [30] for the physics. The finite-difference version of this statement (Chapter 8) is the fact that a symmetric $N \times N$ matrix has N real eigenvalues with orthogonal eigenvectors.

A normalizable solution of Equation (11.1) is called a *bound state*. In this book, as in the paper, we label the bound states in order of increasing energy starting from $n = 1$: E_1 is the *ground state*, E_2 the first excited state, and so on.

11.1.4 The energy is always above the bottom of the well

Sample Problem 11.2: A lower bound on every bound-state energy

Let $V_{\min}$ be the smallest value of V on the region. Show that every bound state has $E > V_{\min}$. Assume, as holds for bound states, that $\psi\psi' \to 0$ at the ends of the region (at walls $\psi = 0$; at infinity both ψ and ψ' tend to zero).

Step 1: Multiply Equation (11.1) by the real, normalized ψ:

$$-\psi\psi'' + V\psi^2 = E\psi^2.$$

Step 2: Integrate over the region $a < x < b$ and use Equation (11.2):

$$-\int_a^b \psi\psi''\,\mathrm{d}x + \int_a^b V\psi^2\,\mathrm{d}x = E.$$

Step 3: Integrate the first term by parts with $u = \psi$, $\mathrm{d}v = \psi''\,\mathrm{d}x$:

$$-\int_a^b \psi\psi''\,\mathrm{d}x = -\big[\psi\psi'\big]_a^b + \int_a^b (\psi')^2\,\mathrm{d}x = \int_a^b (\psi')^2\,\mathrm{d}x \quad \text{(boundary term zero)}.$$

Step 4: Therefore

$$E = \int_a^b (\psi')^2\,\mathrm{d}x + \int_a^b V\psi^2\,\mathrm{d}x. \tag{11.4}$$

The first integral is the expectation value of the kinetic energy; the second is the expectation value $\langle V\rangle$.
Step 5: Since $V \geq V_{\min}$ everywhere and $\psi^2 \geq 0$, $\int V\psi^2\,\mathrm{d}x \geq V_{\min}\int \psi^2\,\mathrm{d}x = V_{\min}$.
Step 6: $\int(\psi')^2\,\mathrm{d}x \geq 0$, with equality only if $\psi' = 0$ everywhere, that is ψ constant. A constant that vanishes at the ends (or at infinity) is zero, which is not normalizable. So the first integral is strictly positive.

$$\boxed{E > V_{\min}}$$

This is the quantum version of "the marble cannot sit still at the bottom": confinement always costs kinetic energy. The lowest energy E_1 minus $V_{\min}$ is called the *zero-point energy*.

Checkpoint 11.1

The paper's Wu–Sprung potential has its minimum $V_0 = 3.10073\pi \approx 9.741$ at $x = 0$ (Sec. II B of [1]). The paper wants the levels of this potential (after fitted corrections) to track the zeta zeros. Is there any inconsistency between the minimum V_0 and the first zero $t_1 = 14.134725$?

Answer: No inconsistency arises. Every bound-state energy must exceed $V_{\min} = 9.741$, and $t_1 = 14.13 > 9.741$; a target below 9.741 could never have been reached by a potential with this minimum. (This says nothing about how accurately the lowest level matches; Section 11.6 shows that semiclassical constructions are weakest at low n.)

Key Ideas

- A bound state is a normalizable solution of $-\psi'' + V\psi = E\psi$; for real V it can be chosen real.
- Curvature rule: $\psi'' = (V - E)\psi$. Allowed regions ($E > V$): oscillation. Forbidden regions ($E < V$): exponential growth or decay.
- Only special energies give decay on both sides; for a confining potential they form a discrete list $E_1 < E_2 < \cdots$ (theorem taken on trust).
- $E = \int(\psi')^2\mathrm{d}x + \int V\psi^2\mathrm{d}x$, so every level lies strictly above the bottom of the well.

11.2 The infinite square well

11.2.1 Setting up the box

The simplest well is a region of zero potential with impenetrable walls:

$$V(x) = \begin{cases} 0, & -L < x < L, \\ +\infty, & |x| \geq L, \end{cases} \tag{11.5}$$

where L is the *half-width* of the box. (We center the box at the origin because the paper does: its domain is $[-L, L]$.) The particle cannot be in the walls, so $\psi = 0$ there. Physically, you can picture the infinite wall as the limit of a very high finite wall: inside a wall of height V_w the wavefunction decays like $e^{-\kappa|x|}$ with $\kappa = \sqrt{V_w - E}$, and as $V_w \to \infty$ the decay becomes instantaneous. We adopt as the definition of the infinite well that ψ is continuous and vanishes at the walls:

$$\psi(-L) = 0, \qquad \psi(L) = 0. \tag{11.6}$$

Conditions that fix the value of ψ at the boundary are called *Dirichlet* boundary conditions, and the walls are called Dirichlet walls. (The derivative ψ' jumps at an infinite wall; only ψ itself is required to be continuous there.)

Inside the box, Equation (11.1) reads

$$-\psi'' = E\psi, \qquad -L < x < L. \tag{11.7}$$

This is exactly the fixed-string equation of Chapter 8 with $\lambda = E$. We now solve it completely, checking every case of the sign of E.

11.2.2 Solving: no solutions for $E \leq 0$

The result $E > V_{\min} = 0$ of the previous section already excludes $E \leq 0$, but it is instructive to see the failure directly.

Case $E = 0$. Then $\psi'' = 0$, so $\psi = a + bx$. The walls give $a - bL = 0$ and $a + bL = 0$. Adding, $2a = 0$; subtracting, $2bL = 0$. So $a = b = 0$: no state.

Case $E < 0$. Write $E = -\kappa^2$ with $\kappa > 0$. Then $\psi'' = \kappa^2\psi$. The functions $\cosh \kappa x = (e^{\kappa x} + e^{-\kappa x})/2$ and $\sinh \kappa x = (e^{\kappa x} - e^{-\kappa x})/2$ are two independent solutions, so $\psi = a \cosh \kappa x + b \sinh \kappa x$. Since cosh is even and sinh is odd, the walls give

$$a \cosh \kappa L + b \sinh \kappa L = 0, \qquad a \cosh \kappa L - b \sinh \kappa L = 0.$$

Adding: $2a \cosh \kappa L = 0$, and $\cosh \kappa L > 0$, so $a = 0$. Subtracting: $2b \sinh \kappa L = 0$, and $\sinh \kappa L > 0$ for $\kappa L > 0$, so $b = 0$. No state.

11.2.3 Solving: the levels for $E > 0$

Write $E = k^2$ with $k > 0$. The general solution of $\psi'' = -k^2\psi$ is

$$\psi(x) = A \cos kx + B \sin kx.$$

Apply Equation (11.6). Because cos is even and sin is odd,

$$\psi(L) = A \cos kL + B \sin kL = 0, \tag{11.8}$$
$$\psi(-L) = A \cos kL - B \sin kL = 0. \tag{11.9}$$

Add the two equations and subtract them:

$$2A \cos kL = 0, \qquad 2B \sin kL = 0.$$

The numbers $\cos kL$ and $\sin kL$ cannot both vanish, since $\cos^2 kL + \sin^2 kL = 1$. So exactly one of two cases occurs.

Case (a): $\cos kL = 0$. Then $\sin kL = \pm 1 \neq 0$, so the second equation forces $B = 0$, and $A \neq 0$ for a nonzero state. The zeros of cosine are $kL = \pi/2, 3\pi/2, 5\pi/2, \ldots$, that is $kL = (2j-1)\pi/2$ with $j = 1, 2, 3, \ldots$, so

$$k = \frac{(2j-1)\pi}{2L}, \qquad \psi \propto \cos kx.$$

Case (b): $\sin kL = 0$. Then $\cos kL = \pm 1$, so $A = 0$ and $B \neq 0$. The zeros of sine with $k > 0$ are $kL = j\pi$, $j = 1, 2, \ldots$, so

$$k = \frac{2j\pi}{2L}, \qquad \psi \propto \sin kx.$$

Case (a) gives k equal to an *odd* multiple of $\pi/2L$, case (b) an *even* multiple. Together they give every positive integer multiple exactly once:

$$k_n = \frac{n\pi}{2L}, \qquad E_n = k_n^2 = \left(\frac{n\pi}{2L}\right)^2, \qquad n = 1, 2, 3, \ldots \tag{11.10}$$

(Negative k adds nothing new: $\cos(-kx) = \cos kx$ and $\sin(-kx) = -\sin kx$ are the same functions up to sign.) This agrees with the fixed-string eigenvalues $(k\pi/2L)^2$ of Chapter 8. The condition behind it is the standing-wave condition from first-year physics: the box length $2L$ must hold a whole number n of half-wavelengths, $2L = n\lambda_n/2$, with $\lambda_n = 2\pi/k_n$.

11.2.4 Normalization

For odd n, $\psi_n = A\cos(n\pi x/2L)$. Using $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$,

$$\int_{-L}^{L} A^2 \cos^2\frac{n\pi x}{2L}\,\mathrm{d}x = \frac{A^2}{2}\int_{-L}^{L}\Big(1 + \cos\frac{n\pi x}{L}\Big)\mathrm{d}x = \frac{A^2}{2}\Big[2L + \frac{L}{n\pi}\big(\sin n\pi - \sin(-n\pi)\big)\Big] = A^2 L,$$

because $\sin(\pm n\pi) = 0$. Setting this to 1 gives $A = 1/\sqrt{L}$. For even n, the same steps with $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ give $B = 1/\sqrt{L}$. So

$$\psi_n(x) = \begin{cases} \dfrac{1}{\sqrt{L}}\cos\dfrac{n\pi x}{2L}, & n \text{ odd}, \\[2ex] \dfrac{1}{\sqrt{L}}\sin\dfrac{n\pi x}{2L}, & n \text{ even}, \end{cases} \qquad -L < x < L. \tag{11.11}$$

A single formula covers both cases:

$$\psi_n(x) = \frac{1}{\sqrt{L}}\sin\left(\frac{n\pi(x + L)}{2L}\right). \tag{11.12}$$

To see this, expand the argument as $\frac{n\pi x}{2L} + \frac{n\pi}{2}$ and use $\sin(\theta + \frac{n\pi}{2})$: for $n = 1$ it is $\cos\theta$, for $n = 2$ it is $-\sin\theta$, for $n = 3$ it is $-\cos\theta$, for $n = 4$ it is $\sin\theta$, and the pattern repeats with period 4. So Equation (11.12) equals Equation (11.11) up to an overall sign, and an overall sign does not change a state.

Pitfall

Many textbooks put the box on $[0, a]$ and write $\psi_n = \sqrt{2/a}\,\sin(n\pi x/a)$. Our box has width $a = 2L$, so $\sqrt{2/a} = 1/\sqrt{L}$, consistent with Equation (11.12). Do not use $\sqrt{2/L}$ on $[-L, L]$: that would give total probability 2.

11.2.5 Orthogonality

Chapter 8 proved that eigenvectors of a symmetric matrix with different eigenvalues are orthogonal. The same argument applies to the Hermitian operator H, with integration by parts playing the role of the matrix transpose. Here we confirm orthogonality directly for the box. Substitute $u = \pi(x + L)/(2L)$ in Equation (11.12): as x runs from $-L$ to L, u runs from 0 to π, and $\mathrm{d}x = (2L/\pi)\mathrm{d}u$. Then

$$\int_{-L}^{L} \psi_m\psi_n\,\mathrm{d}x = \frac{1}{L}\cdot\frac{2L}{\pi}\int_0^{\pi}\sin mu\,\sin nu\,\mathrm{d}u.$$

Using $\sin mu \sin nu = \frac{1}{2}[\cos(m-n)u - \cos(m+n)u]$ (Chapter 5), for $m \neq n$ both cosines integrate to $\sin(j\pi)/j = 0$; for $m = n$ the integral is $\frac{1}{2}\pi$. Hence

$$\int_{-L}^{L} \psi_m(x)\psi_n(x)\,\mathrm{d}x = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \tag{11.13}$$

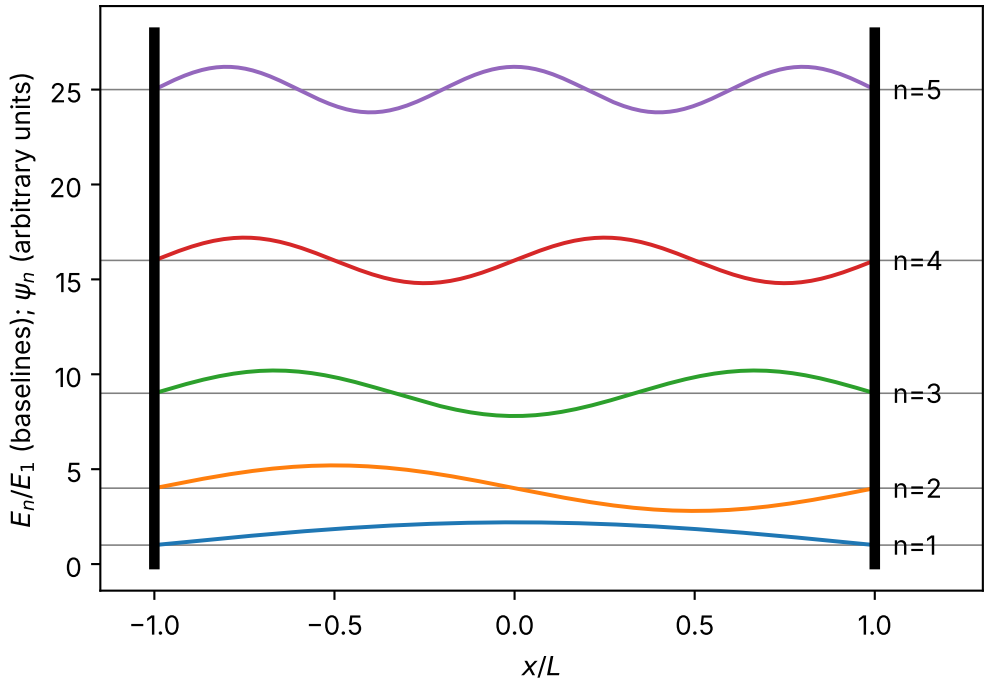


Figure 11.2: The first five states of the infinite square well on $[-L, L]$. Each wavefunction is drawn on a baseline at its energy $E_n/E_1 = n^2$ (gray lines), with $E_1 = (\pi/2L)^2$. The n-th state holds n half-wavelengths and has $n-1$ nodes; odd n gives an even function, even n an odd one. Thick lines: the Dirichlet walls.

Figure 11.2 shows the first five states. Three patterns are visible: the energies are all different; the states are alternately even and odd; and the n-th state crosses zero $n-1$ times inside the box. The next section does three things. It proves that the first pattern holds for every one-dimensional well. It proves that every state of an even well is even or odd. It states the third pattern (the node theorem), with its proof taken on trust. Combined with the node theorem, the alternation of parity then follows for every even well.

11.2.6 Spacing

From Equation (11.10),

$$E_{n+1} - E_n = \frac{\pi^2}{4L^2}\left[(n+1)^2 - n^2\right] = \frac{(2n+1)\pi^2}{4L^2} = (2n+1)E_1. \tag{11.14}$$

The spacing grows linearly with n: in a box, levels *spread out* as you go up. Keep this in mind; the Riemann zeros do the opposite.

Sample Problem 11.3: The empty box of the paper (RE3)

The paper's scale-1 box has half-width $L = 44.883$ (Sec. II B of [1]). With no potential inside (running example RE3), find (a) E_1, (b) E_{500}, (c) how many levels lie at or below $E = 1000$, (d) $E_{501} - E_{500}$.

Step 1: $E_1 = \pi^2/(4L^2)$. Here $L^2 = 44.883^2 = 2014.48$ and $4L^2 = 8057.93$, so $E_1 = 9.8696/8057.94 = 1.2248 \times 10^{-3}$.

Step 2: $E_{500} = 500^2 E_1 = 250000 \times 1.2248 \times 10^{-3} = 306.2$.

Step 3: $E_n \le 1000$ means $n\pi/(2L) \le \sqrt{1000}$, that is $n \le 2L\sqrt{1000}/\pi = 89.766 \times 31.623/3.1416 = 903.6$. The largest integer is 903.

Step 4: By Equation (11.14), $E_{501} - E_{500} = 1001\, E_1 = 1.226$.

$$\boxed{E_1 = 1.2248 \times 10^{-3}, \quad E_{500} = 306.2, \quad 903 \text{ levels } \le 1000, \quad E_{501} - E_{500} = 1.226}$$

In the paper, the Wu–Sprung potential fills this box. We will see in Section 11.8 that the filled box holds far fewer levels below 1000.

Sample Problem 11.4: Where is the particle?

In the ground state of the box, what is the probability of finding the particle in the middle half, $-L/2 < x < L/2$?

Step 1: By the Born rule, $P = \int_{-L/2}^{L/2} \psi_1^2 \, dx = \frac{1}{L} \int_{-L/2}^{L/2} \cos^2 \frac{\pi x}{2L} \, dx$.

Step 2: Use $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$: $P = \frac{1}{2L} \int_{-L/2}^{L/2} (1 + \cos \frac{\pi x}{L}) dx$.

Step 3: The first term gives $\frac{1}{2L} \cdot L = \frac{1}{2}$.

Step 4: The second gives $\frac{1}{2L} \cdot \frac{L}{\pi} \left[\sin \frac{\pi x}{L}\right]_{-L/2}^{L/2} = \frac{1}{2\pi}(\sin \frac{\pi}{2} - \sin(-\frac{\pi}{2})) = \frac{1}{\pi}$.

$$\boxed{P = \tfrac{1}{2} + \tfrac{1}{\pi} \approx 0.818}$$

A classical particle bouncing at constant speed would spend exactly half its time in the middle half. The quantum ground state piles probability in the center. (Problem 31 shows that for high n the answer returns to $\frac{1}{2}$, as the correspondence principle requires.)

Checkpoint 11.2

If you double the half-width L of the box, what happens to E_1 and to every spacing $E_{n+1} - E_n$?

Answer: Both scale as $1/L^2$ by Equations (11.10) and (11.14), so both drop to one quarter. A bigger box means longer wavelengths and lower energies.

Key Ideas

- Box on $[-L, L]$ with Dirichlet walls: $E_n = (n\pi/2L)^2$, $\psi_n = L^{-1/2} \sin(n\pi(x + L)/2L)$, $n = 1, 2, \ldots$
- Odd n: even cosine states; even n: odd sine states. The states are orthonormal.
- Box spacing $(2n + 1)E_1$ grows with n: levels spread out.

11.3 Uniqueness, parity and nodes

★ Did you know? *Counting Nodes*

In the 1830s Charles Sturm and Joseph Liouville proved that for a large class of one-dimensional wave equations, the n-th eigenfunction, counting from $n = 0$, has exactly n zeros inside the interval. This oscillation theorem is why the ground state of any 1D potential well has no nodes, the first excited state has one, and so on. They proved it long before quantum mechanics existed.

11.3.1 One-dimensional bound states are never degenerate

Two different states with the same energy are called *degenerate*. In three dimensions degeneracy is common (think of the hydrogen atom). In one dimension it never happens for bound states. The proof uses one fact about differential equations that we take on trust.

Taken on trust

Taken on trust: uniqueness for the initial-value problem. If V is continuous on an interval and x_0 is a point in it, then for any numbers a, b there is exactly one solution of $\psi'' = (V - E)\psi$ on the interval with $\psi(x_0) = a$ and $\psi'(x_0) = b$. In particular, a solution with $\psi(x_0) = \psi'(x_0) = 0$ is identically zero. This is the existence and uniqueness theorem for linear second-order equations; see [22]. Physically: knowing the position and slope of the wavefunction at one point, the curvature rule Equation (11.3) determines the rest, just as knowing a projectile's position and velocity determines its path.

Theorem (proved, given the box above). If u and w are bound states of the same potential with the same energy E, then $w = c\,u$ for a constant c. (We call the two functions u and w rather than ψ_1 and ψ_2, because elsewhere in this chapter ψ_1, ψ_2 denote the ground and first excited states, which have different energies.)

Proof. Define the *Wronskian* $W(x) = uw' - wu'$.

Step 1: W is constant. By the product rule,

$$W' = u'w' + uw'' - w'u' - wu'' = uw'' - wu''.$$

Insert $u'' = (V - E)u$ and $w'' = (V - E)w$ from Equation (11.3):

$$W' = u(V - E)w - w(V - E)u = 0.$$

Step 2: $W = 0$. At a Dirichlet wall $x = L$ both functions vanish, so $W(L) = 0$. For a potential on the whole line, we use the property of bound states already assumed in Sample Problem "A lower bound on every bound-state energy": both the function and its slope tend to zero at infinity. Hence $W \to 0$. Heuristically, deep in the forbidden region a normalizable solution must be the decaying branch. Its slope is about $-\sqrt{V - E}$ times the function, and the decay of the function wins over the growth of $\sqrt{V - E}$. The oscillator shows this: there $\psi' = -x\psi$ with $\psi \propto e^{-x^2/2}$. A proof belongs to the Sturm–Liouville theory taken on trust in Section 11.1. Either way, the constant is zero.

Step 3: compare at one point. Pick a point x_0 with $u(x_0) \neq 0$ (one exists, since u is not identically zero) and set $c = w(x_0)/u(x_0)$. Define $\phi = w - cu$, which solves the same equation (linearity). Then $\phi(x_0) = w(x_0) - cu(x_0) = 0$.

Step 4: the slope also vanishes. From $W(x_0) = 0$: $u(x_0)w'(x_0) = w(x_0)u'(x_0)$. Divide by $u(x_0)$: $w'(x_0) = c\,u'(x_0)$. Hence $\phi'(x_0) = w'(x_0) - cu'(x_0) = 0$.

Step 5. By uniqueness, $\phi \equiv 0$, so $w = cu$. □

11.3.2 Even potentials have even and odd states

A potential is *even* if $V(-x) = V(x)$: its graph is a mirror image about $x = 0$. The box on $[-L, L]$, the oscillator $V = x^2$, and the paper's Wu–Sprung potential are all even.

Theorem (proved). If V is even (and the region and boundary conditions are symmetric under $x \to -x$), every bound state is either even, $\psi(-x) = \psi(x)$, or odd, $\psi(-x) = -\psi(x)$.

Proof.

Step 1: the mirror image is a solution. Let ψ solve Equation (11.1) and define $\chi(x) = \psi(-x)$. By the chain rule, $\chi'(x) = -\psi'(-x)$ and $\chi''(x) = \psi''(-x)$ (two factors of -1 cancel). So

$$-\chi''(x) + V(x)\chi(x) = -\psi''(-x) + V(-x)\psi(-x) = E\psi(-x) = E\chi(x),$$

where the first equality used $V(x) = V(-x)$ and the second is Equation (11.1) evaluated at the point $-x$. Also χ is normalizable and satisfies the mirrored boundary conditions, which are the same conditions. So χ is a bound state with energy E.

Step 2: nondegeneracy. By the previous theorem, $\chi = c\psi$, that is $\psi(-x) = c\psi(x)$ for all x.

Step 3: $c = \pm 1$. Apply Step 2 twice: $\psi(x) = \psi(-(-x)) = c\,\psi(-x) = c^2\psi(x)$. Since $\psi \neq 0$ somewhere, $c^2 = 1$, so $c = 1$ (even) or $c = -1$ (odd). □

Two consequences are used again and again.

- An odd state has $\psi(0) = 0$, because $\psi(0) = -\psi(0)$. An even state has $\psi'(0) = 0$, because the derivative of an even function is odd.
- For either parity, ψ^2 is even. So the probability density is symmetric, the particle is equally likely to be on either side, and $\langle x \rangle = \int x\psi^2\,\mathrm{d}x = 0$ (an odd integrand over a symmetric interval).

11.3.3 Counting nodes

A *node* is a point strictly inside the region where ψ crosses zero. In Figure 11.2 the n-th box state has $n - 1$ nodes. This is a general law.

Taken on trust

Taken on trust: the oscillation (node) theorem. For a one-dimensional confining potential, the n-th bound state ($n = 1$ the ground state) has exactly $n - 1$ nodes. The proof uses Sturm's comparison method from the theory of Sturm–Liouville problems (see [22] for that theory). Intuitively, raising E makes $\psi'' = (V - E)\psi$ curve more strongly, so the wavefunction turns over more often, and each additional allowed state squeezes in one more node.

Combine the node theorem with parity. In an even potential:

- An odd state has a node at $x = 0$, and its other nodes come in pairs $\pm x_j$. So it has an *odd* number of nodes.
- An even state cannot have a node at $x = 0$: it would have $\psi(0) = 0$ and $\psi'(0) = 0$, hence be identically zero by uniqueness. Its nodes come in pairs, so it has an *even* number of nodes.

State n has $n - 1$ nodes. Therefore state n is even when $n - 1$ is even, that is when n is odd, and odd when n is even. The ground state of an even well is even (no nodes), and the parities then alternate: even, odd, even, odd, ... The box confirms this: odd n gave cosines.

Sample Problem 11.5: Parity and nodes of the third box state

Find the parity and the nodes of ψ_3 in the box on $[-L, L]$, and check both general theorems.
Step 1: $n = 3$ is odd, so by Equation (11.11) $\psi_3 = L^{-1/2}\cos(3\pi x/2L)$, an even function.
Step 2: Nodes: $\cos(3\pi x/2L) = 0$ when $3\pi x/2L = \pm\pi/2$ (the next zeros, $\pm 3\pi/2$, give $x = \pm L$, the walls, which are not interior). So $x = \pm L/3$.
Step 3: Two nodes $= n - 1$, as the node theorem demands; an even number, as an even state must have; none at $x = 0$.

$$\boxed{\psi_3 \text{ is even, with nodes at } x = \pm L/3}$$

Checkpoint 11.3

In an even potential, can the ground state be odd? How many nodes does the seventh state have, and what is its parity?

Answer: No: an odd state has a node at $x = 0$, but the ground state has none. The seventh state has $7 - 1 = 6$ nodes, an even number, so it is even.

Key Ideas

- One-dimensional bound states are nondegenerate (Wronskian proof).
- In an even potential each state is even or odd; odd states vanish at $x = 0$, even states have zero slope there; ψ^2 is always even.
- The n-th state has $n - 1$ nodes (taken on trust); in an even well, odd n means even state and even n means odd state.

11.4 The harmonic oscillator

11.4.1 The equation

The harmonic oscillator is the quantum version of the mass on a spring. In ordinary units its potential is $\frac{1}{2}m\omega^2x^2$. In our units $m = \frac{1}{2}$, so the potential is $\frac{1}{4}\omega^2x^2$. The simplest choice is $\omega = 2$, which gives

$$-\psi'' + x^2\psi = E\psi. \tag{11.15}$$

Any other strength reduces to this one by a change of scale (Sample Problem "Stretching the oscillator" below).

11.4.2 Peeling off the Gaussian

For large $|x|$ the term $x^2\psi$ dominates $E\psi$. Try $e^{-x^2/2}$: its derivative is $-xe^{-x^2/2}$ and its second derivative is $(x^2 - 1)e^{-x^2/2}$, so $-\psi'' + x^2\psi = e^{-x^2/2}$, which is Equation (11.15) with $E = 1$. We have already found a bound state! To find the rest, write

$$\psi(x) = h(x)\, e^{-x^2/2} \tag{11.16}$$

and find the equation for h. By the product rule,

$$\begin{aligned}\psi' &= (h' - xh)e^{-x^2/2},\\ \psi'' &= (h'' - h - xh')e^{-x^2/2} - x(h' - xh)e^{-x^2/2}\\ &= (h'' - 2xh' - h + x^2h)e^{-x^2/2}.\end{aligned}$$

(In the second line the first bracket is the derivative of $h' - xh$; the second term is $h' - xh$ times the derivative $-x$ of the exponent.) So

$$-\psi'' + x^2\psi = (-h'' + 2xh' + h - x^2h + x^2h)e^{-x^2/2} = (-h'' + 2xh' + h)e^{-x^2/2}.$$

Setting this equal to $Eh\,e^{-x^2/2}$ and dividing by $-e^{-x^2/2}$ gives *Hermite's equation*

$$h'' - 2xh' + (E-1)h = 0. \tag{11.17}$$

11.4.3 Power series and the recursion

Look for h as a power series $h(x) = \sum_{j=0}^{\infty} a_j x^j$. Term by term,

$$h' = \sum_j j a_j x^{j-1}, \qquad h'' = \sum_j j(j-1)a_j x^{j-2}.$$

In h'' rename the index $j \to j+2$, so that every sum runs over powers x^j:

$$h'' = \sum_{j=0}^{\infty}(j+2)(j+1)a_{j+2}x^j, \qquad -2xh' = \sum_{j=0}^{\infty}(-2j)a_j x^j.$$

Insert into Equation (11.17):

$$\sum_{j=0}^{\infty}\Big[(j+2)(j+1)a_{j+2} - 2ja_j + (E-1)a_j\Big]x^j = 0.$$

A power series that vanishes for all x has every coefficient zero. To see this, set $x = 0$: every term except the constant vanishes, so the constant coefficient is zero. Differentiate once term by term (Chapter 3) and set $x = 0$ again: the next coefficient is zero. Continue in the same way. Therefore

$$a_{j+2} = \frac{2j + 1 - E}{(j+2)(j+1)}\,a_j. \tag{11.18}$$

Choosing a_0 starts a chain of even powers; choosing a_1 starts a chain of odd powers. Even potential, even and odd solutions: exactly as the parity theorem promised.

11.4.4 Quantization: the series must stop

Suppose the chain never stops. For large j, Equation (11.18) gives $a_{j+2}/a_j \approx 2j/j^2 = 2/j$. Compare $e^{x^2} = \sum_k x^{2k}/k!$: with $j = 2k$, the ratio of successive coefficients is $\frac{1/(k+1)!}{1/k!} = \frac{1}{k+1} = \frac{2}{j+2} \approx \frac{2}{j}$. For the odd chain, compare instead with $x\,e^{x^2} = \sum_k x^{2k+1}/k!$. With $j = 2k+1$ the ratio of successive coefficients is again $\frac{1}{k+1} = \frac{2}{j+1} \approx \frac{2}{j}$. So a nonterminating h grows at large $|x|$ like e^{x^2} (times at most a power of x), and then $\psi = he^{-x^2/2}$ grows like $e^{+x^2/2}$: not normalizable. This is the growing branch that wrecked the shooting curves in Figure 11.1. (This standard comparison of tails is a heuristic as written; a careful version bounds the coefficients from below for all large j. See [30].)

The only escape is for the chain to stop. The numerator of Equation (11.18) vanishes at $j = J$ when $E = 2J + 1$; then $a_{J+2} = a_{J+4} = \cdots = 0$ and h is a polynomial of degree J. (The *other* chain must be switched off by setting its starting coefficient to zero.) With our labeling $n = J + 1$ starting from the ground state $n = 1$:

$$E_n = 2n - 1, \qquad n = 1, 2, 3, \ldots \tag{11.19}$$

The levels $1, 3, 5, 7, \ldots$ are *equally spaced*, with spacing 2. The polynomial h for level n is, up to a constant, the Hermite polynomial H_{n-1}:

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x.$$

For example, $E = 5$ ($J = 2$): with $a_1 = 0$, Equation (11.18) gives $a_2 = \frac{1-5}{2} a_0 = -2a_0$ and $a_4 = 0$, so $h = a_0(1 - 2x^2)$, proportional to H_2.

Taken on trust

Taken on trust: these are all the oscillator states. The argument above shows, at the heuristic level of the tail comparison, that a nonterminating series gives a non-normalizable ψ. That no other energies give bound states, and that the functions $H_{n-1}(x)e^{-x^2/2}$ form a complete orthogonal set, is a theorem whose careful proof can be found in [30] and [22]; we take it on trust.

Pitfall

Most quantum texts label oscillator levels from zero, $E_\nu = \hbar\omega(\nu + \frac{1}{2})$, $\nu = 0, 1, 2, \ldots$ In this book, as in the paper, level indices start at $n = 1$, so the ground state is $n = 1$ and $E_n = 2n - 1$ in our units. With $\hbar = 1$ and $\omega = 2$, $\hbar\omega(\nu + \frac{1}{2}) = 2\nu + 1 = 2n - 1$ for $\nu = n - 1$: the same numbers, a different label.

11.4.5 Normalizing: the Gaussian integral

Taken on trust

Taken on trust: the Gaussian integral. $\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}$. A short proof squares the integral, writes it as a double integral over the plane, and switches to polar coordinates; it needs double integrals, which this book does not assume. You can confirm it numerically: `scipy.integrate.quad(lambda x: np.exp(-x**2), -np.inf, np.inf)` returns $1.7724539\ldots = \sqrt{\pi}$.

From it, integration by parts gives the moments we need. With $u = x$ and $\mathrm{d}v = xe^{-x^2}\mathrm{d}x$ (so $v = -\frac{1}{2}e^{-x^2}$):

$$\int_{-\infty}^{\infty} x^2 e^{-x^2}\mathrm{d}x = \left[-\tfrac{1}{2}xe^{-x^2}\right]_{-\infty}^{\infty} + \tfrac{1}{2}\int_{-\infty}^{\infty} e^{-x^2}\mathrm{d}x = \frac{\sqrt{\pi}}{2}, \tag{11.20}$$

since $xe^{-x^2} \to 0$ at both ends. With $u = x^3$ and the same $\mathrm{d}v$:

$$\int_{-\infty}^{\infty} x^4 e^{-x^2}\mathrm{d}x = \tfrac{3}{2}\int_{-\infty}^{\infty} x^2 e^{-x^2}\mathrm{d}x = \frac{3\sqrt{\pi}}{4}. \tag{11.21}$$

So the ground state is $\psi_1 = \pi^{-1/4}e^{-x^2/2}$, since $\int \psi_1^2 \mathrm{d}x = \pi^{-1/2}\int e^{-x^2}\mathrm{d}x = 1$.

Sample Problem 11.6: Checking an oscillator state

Show that $\psi(x) = (2x^2 - 1)e^{-x^2/2}$ solves Equation (11.15), find its energy and level index, normalize it, and locate its nodes and classical turning points.
Step 1: Here $h = 2x^2 - 1$, $h' = 4x$, $h'' = 4$. Insert into Equation (11.17): $4 - 2x(4x) + (E-1)(2x^2-1) = [2(E-1) - 8]x^2 + [4 - (E-1)]$.
Step 2: Both brackets vanish if and only if $E = 5$. So ψ is a solution with $E = 5$, which is $2n - 1$ with $n = 3$: the third level.
Step 3: $\int (2x^2-1)^2 e^{-x^2}\,dx = \int (4x^4 - 4x^2 + 1)e^{-x^2}\,dx = 4 \cdot \frac{3\sqrt{\pi}}{4} - 4 \cdot \frac{\sqrt{\pi}}{2} + \sqrt{\pi} = 2\sqrt{\pi}$, using Equations (11.20) and (11.21). So $\psi_3 = (2\sqrt{\pi})^{-1/2}(2x^2-1)e^{-x^2/2}$.
Step 4: Nodes where $2x^2 - 1 = 0$: $x = \pm 1/\sqrt{2} = \pm 0.7071$. Two nodes, $n - 1 = 2$. Even function, as $n = 3$ odd requires.
Step 5: Turning points where $x^2 = E = 5$: $x = \pm\sqrt{5} = \pm 2.2361$.

$$E_3 = 5, \quad \psi_3 = \frac{(2x^2-1)e^{-x^2/2}}{(2\sqrt{\pi})^{1/2}}, \quad \text{nodes } \pm 0.7071, \quad \text{turning points } \pm 2.2361$$

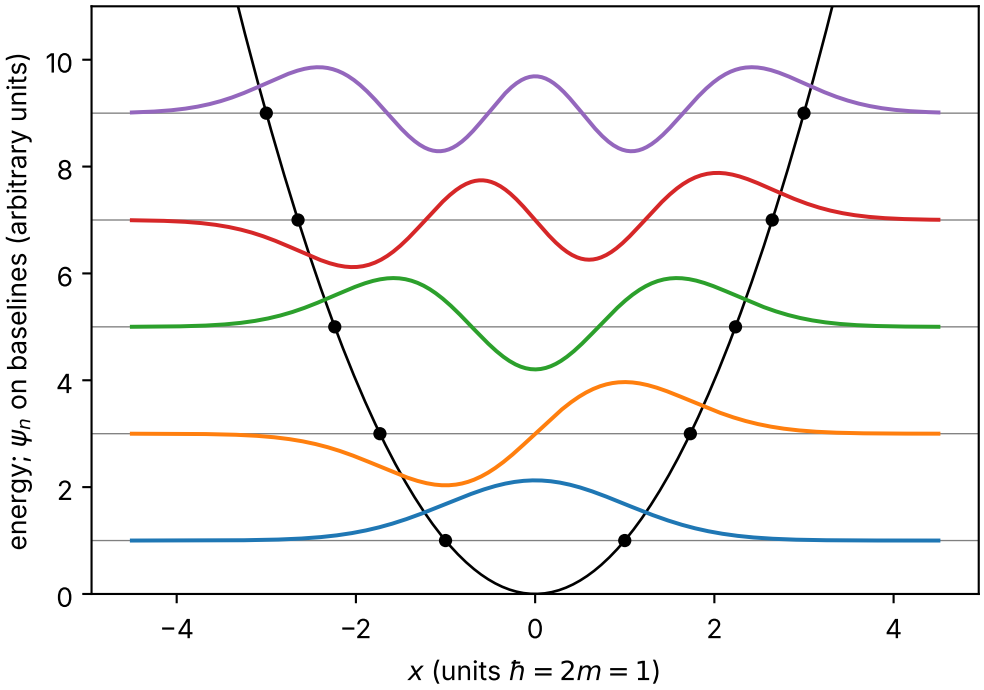


Figure 11.3: The harmonic oscillator $V = x^2$ (units $\hbar = 2m = 1$). The first five normalized states are drawn on baselines at their energies $E_n = 2n - 1$. Dots mark the classical turning points $x = \pm\sqrt{E_n}$. Each state leaks a little past its turning points into the forbidden region, and the levels are equally spaced.

Sample Problem 11.7: Stretching the oscillator

Find the levels of $-\psi'' + ax^2\psi = E\psi$ for any $a > 0$, and evaluate them for $a = \frac{1}{4}$.
Step 1: Change variable $x = \beta y$ with $\beta > 0$ to be chosen. By the chain rule $d/dx =$

$(1/\beta)\,d/dy$, so $d^2\psi/dx^2 = (1/\beta^2)\,d^2\psi/dy^2$. The equation becomes

$$-\frac{1}{\beta^2}\frac{d^2\psi}{dy^2} + a\beta^2 y^2\psi = E\psi.$$

Step 2: Choose β so that the two coefficients are equal: $a\beta^2 = 1/\beta^2$, that is $\beta^4 = 1/a$. Then

$$\frac{1}{\beta^2}\Big(-\frac{d^2\psi}{dy^2} + y^2\psi\Big) = E\psi, \qquad \text{so} \qquad -\frac{d^2\psi}{dy^2} + y^2\psi = \beta^2 E\,\psi.$$

Step 3: This is Equation (11.15) in the variable y, so $\beta^2 E = 2n - 1$, and $1/\beta^2 = \sqrt{a}$:

$$E_n = \sqrt{a}\,(2n - 1).$$

Step 4: For $a = \frac{1}{4}$: $E_n = \frac{1}{2}(2n-1) = 0.5,\ 1.5,\ 2.5, \ldots$ This is $\frac{1}{4}\omega^2 x^2$ with $\omega = 1$, and the result is $\omega(n - \frac{1}{2})$, the familiar $\hbar\omega(\nu + \frac{1}{2})$.

$$\boxed{E_n = \sqrt{a}\,(2n-1); \quad a = \tfrac{1}{4}:\ E_n = 0.5,\ 1.5,\ 2.5, \ldots}$$

You can confirm Equation (11.19) on a computer with the tridiagonal matrix of Chapter 8. Chapter 23 analyzes this method in depth; here we use it as a black box. (In the code, `h` is the grid spacing, not the polynomial $h(x)$ above.)

```
import numpy as np
from scipy.linalg import eigh_tridiagonal

L, N = 8.0, 2000                # box half-width, number of interior points
h = 2*L/(N + 1)                 # grid spacing
x = -L + h*np.arange(1, N + 1)  # interior grid points
V = x**2                        # oscillator potential, units hbar = 2m = 1
diag = 2/h**2 + V               # diagonal: second difference plus V
off = -np.ones(N - 1)/h**2      # off-diagonal of the second difference
E = eigh_tridiagonal(diag, off, eigvals_only=True,
                     select='i', select_range=(0, 9))
# select='i' picks eigenvalues by index; indices start at 0,
# so (0, 9) means the lowest ten levels
exact = 2*np.arange(1, 11) - 1  # 1, 3, 5, ..., 19
print(E)
print(E - exact)                # small, and growing with n
```

The walls at $x = \pm 8$ lie far beyond the turning points of the first ten levels (the tenth has $E = 19$ and turns at $\sqrt{19} \approx 4.4$), so the box barely matters. The small differences that you see, below 0.01 for all ten levels and growing with n, come from the grid, not the walls; Chapter 23 explains them.

Checkpoint 11.4

If the oscillator potential x^2 is replaced by $4x^2$, what happens to the level spacing?

Answer: With $a = 4$, $E_n = 2(2n-1)$, so the spacing becomes 4: it doubles. A stiffer spring vibrates at twice the angular frequency, and the spacing is $\hbar\omega$.

Key Ideas

- $-\psi'' + x^2\psi = E\psi$ has $E_n = 2n - 1$ and $\psi_n \propto H_{n-1}(x)e^{-x^2/2}$; the levels are equally spaced.
- Quantization comes from requiring the power series to terminate; otherwise ψ grows like $e^{x^2/2}$.
- Scaling: $V = ax^2$ gives $E_n = \sqrt{a}(2n - 1)$.

11.5 The shape of a state in a general well

Most wells, including the paper's, cannot be solved in closed form. The curvature rule still tells us what the states look like, and it tells us when a distant wall can be ignored.

11.5.1 Reading a wavefunction

Pick a level E in a smooth, even, confining well. The turning points are $\pm x_E$, where $V(x_E) = E$.

Between the turning points ψ oscillates. Where $E - V$ is large (near the bottom of the well) the curvature is strong and the wiggles are short; near the turning points $E - V$ is small and the wiggles stretch out. The de Broglie relation (Chapter 10) makes this quantitative. Classically, in our units the kinetic energy is $p^2/2m = p^2$ (since $2m = 1$), so $p(x) = \sqrt{E - V(x)}$ and the local wavelength is

$$\lambda(x) = \frac{2\pi\hbar}{p(x)} = \frac{2\pi}{\sqrt{E - V(x)}}. \tag{11.22}$$

This local wavelength is meaningful only where V changes little over one wavelength, $|\lambda\, dV/dx| \ll E - V$. It fails close to the turning points, where $\lambda \to \infty$. This is why the counting rule of Section 11.6 is only approximate and needs the correction γ. The amplitude also varies: it is larger where the particle is slow. A classical particle spends time $\mathrm{d}t = \mathrm{d}x/v$ in each slice $\mathrm{d}x$, so its classical probability density is proportional to $1/v$; the quantum density, averaged over the wiggles, follows this trend for high levels. In Figure 11.3 the high states are largest near their turning points, where a classical oscillator is slowest.

Beyond the turning points ψ decays. The local decay rate is $\kappa(x) = \sqrt{V(x) - E}$; the farther above E the potential rises, the faster the decay. The wavefunction does *not* vanish at the turning point; it leaks into the forbidden region (look at Figure 11.3).

Pitfall

The classical turning point is not a node. The wavefunction is continuous and smooth there; it merely changes from curving toward the axis to curving away. A quantum particle has a nonzero probability of being found in the classically forbidden region (Problem 13 computes it for the oscillator ground state).

11.5.2 How much does a distant wall matter?

The paper puts its Wu–Sprung potential in a box. A wall placed far into the forbidden region should change the levels very little, since ψ is already tiny there. Let us estimate how little, using the oscillator, where everything is known.

Sample Problem 11.8: How much does a distant wall matter?

Put the oscillator Equation (11.15) inside Dirichlet walls at $\pm L$, with L larger than the turning point $x = 1$ of the ground state. Estimate the rise of the ground-state energy above 1.

Step 1 (two equations): Let $\psi = \pi^{-1/4}e^{-x^2/2}$ be the ground state without walls, energy 1, and ϕ the ground state with walls, energy E_b, $\phi(\pm L) = 0$. Then

$$-\psi'' + x^2\psi = \psi, \qquad -\phi'' + x^2\phi = E_b\phi.$$

Step 2 (combine): Multiply the second by ψ, the first by ϕ, and subtract; the x^2 terms cancel:

$$\psi''\phi - \phi''\psi = (E_b - 1)\phi\psi.$$

Step 3 (total derivative): By the product rule, $\frac{d}{dx}(\psi'\phi - \phi'\psi) = \psi''\phi + \psi'\phi' - \phi''\psi - \phi'\psi' = \psi''\phi - \phi''\psi$.

Step 4 (integrate from $-L$ to L): $[\psi'\phi - \phi'\psi]_{-L}^{L} = (E_b - 1)\int_{-L}^{L}\phi\psi\,\mathrm{d}x$. Since $\phi(\pm L) = 0$ the left side is $-\phi'(L)\psi(L) + \phi'(-L)\psi(-L)$. Both states are even, so $\psi(-L) = \psi(L)$ and $\phi'(-L) = -\phi'(L)$, and the left side is $-2\phi'(L)\psi(L)$. This step is exact:

$$E_b - 1 = \frac{-2\phi'(L)\psi(L)}{\int_{-L}^{L}\phi\psi\,\mathrm{d}x}. \tag{11.23}$$

Step 5 (approximate, heuristic): The two states differ only near the walls, where both are tiny, so $\int\phi\psi\,\mathrm{d}x \approx 1$. Near $x = L$, ϕ is ψ minus just enough of the growing solution to make it vanish at the wall. In the forbidden region, the decaying solution has $\psi'/\psi = -x$ exactly. For the growing one, try $g = e^{+x^2/2}$: $g' = xg$ and $g'' = (x^2 + 1)g$, so $-g'' + x^2 g = -g$. Thus g solves the oscillator equation exactly, but with energy -1 instead of $E_b \approx 1$. Where $x^2 \gg E_b$ (deep in the forbidden region), replacing $x^2 - E_b$ by $x^2 + 1$ is a small relative change, so the true growing solution behaves like g, with logarithmic slope $g'/g = +x$ up to corrections of relative size $\mathcal{O}(1/x^2)$. This is the heuristic part of the estimate. So $\phi'(L) \approx \psi'(L) - L\psi(L) = -L\psi(L) - L\psi(L) = -2L\psi(L)$. This approximation requires $L^2 \gg E$, so that the potential dominates near the wall.

Step 6 (result): $E_b - 1 \approx 4L\psi(L)^2 = \dfrac{4L}{\sqrt{\pi}}e^{-L^2}$.

Step 7 (numbers): For $L = 2$: $\frac{8}{\sqrt{\pi}}e^{-4} = 0.083$. A numerical solution (the tridiagonal method above) gives $E_b \approx 1.075$, so the estimate is about 10% high, reasonable since $L^2 = 4$ is only four times E. For $L = 4$: $\frac{16}{\sqrt{\pi}}e^{-16} \approx 1.0 \times 10^{-6}$. Numerically the shift is also about 1.0×10^{-6}, a few percent below the estimate. (This computation uses the same grid spacing with and without walls, so that the grid errors cancel.)

$$\boxed{E_b - 1 \approx \frac{4L}{\sqrt{\pi}}e^{-L^2}: \quad L = 2 \to 0.083 \text{ (numerics } 0.075), \quad L = 4 \to\sim 10^{-6}}$$

The lesson is general: the effect of a wall placed beyond a turning point falls off exponentially with the amount of forbidden region in between, roughly like $\psi(L)^2$. Figure 11.4 compares the estimate with numerics.

Key Ideas

- Inside the allowed region the local wavelength is $2\pi/\sqrt{E - V}$; outside, the decay rate is $\sqrt{V - E}$.

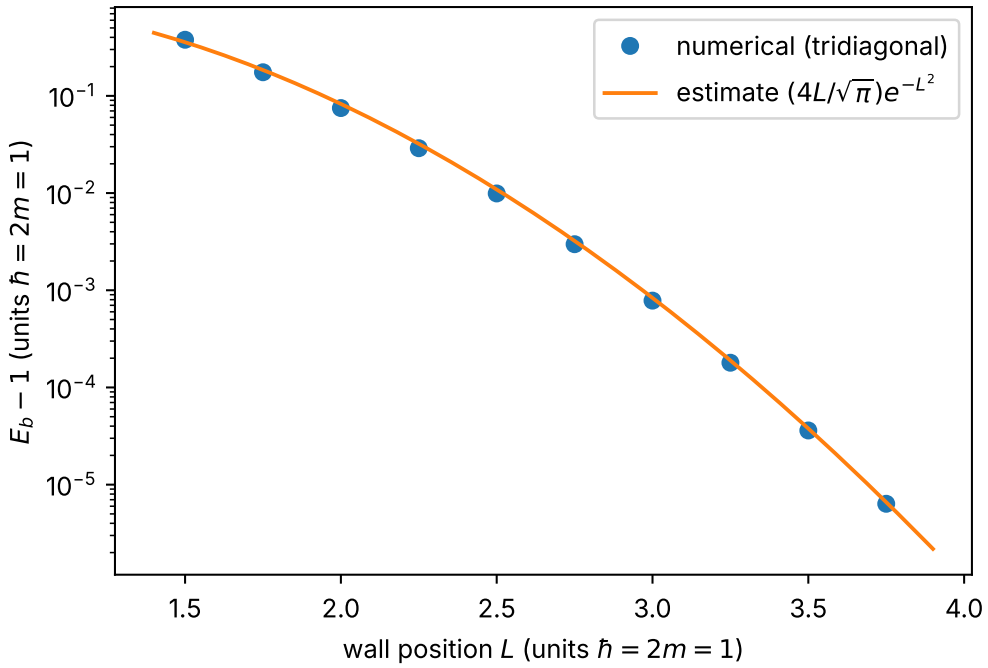


Figure 11.4: Rise of the oscillator ground-state energy, $E_b - 1$, when Dirichlet walls are placed at $\pm L$ (units $\hbar = 2m = 1$; logarithmic vertical axis). Dots: numerical eigenvalues of the tridiagonal matrix. Line: the estimate $(4L/\sqrt{\pi})e^{-L^2}$ of Equation (11.23) and Step 6. The effect of the wall dies off like e^{-L^2}, the square of the wavefunction at the wall.

- Wavefunctions leak past turning points; turning points are not nodes.
- A Dirichlet wall deep in the forbidden region shifts a level by an amount that is exponentially small (for the oscillator ground state, about $(4L/\sqrt{\pi})e^{-L^2}$).

11.6 How the spacing depends on the shape of the well

The box has spacings that grow, $(2n+1)E_1$; the oscillator has constant spacing 2. What decides? We now build a simple counting rule that answers this for any well. It is a *heuristic* here; Chapter 12 derives it properly as the Bohr–Sommerfeld rule.

11.6.1 Counting half-wavelengths

In the box, the n-th state holds exactly n half-wavelengths. In a general well the wavelength Equation (11.22) varies with position, so count half-wavelengths piece by piece: a slice of width $\mathrm{d}x$ contains $\mathrm{d}x/(\lambda/2) = \sqrt{E-V}\,\mathrm{d}x/\pi$ of them. Adding up the slices between the turning points x_1 and x_2:

$$\frac{1}{\pi}\int_{x_1}^{x_2}\sqrt{E_n - V(x)}\,\mathrm{d}x \;\approx\; n - \gamma. \tag{11.24}$$

The constant γ accounts for how the wavefunction meets the boundary. At a hard wall it is pinned to zero; at a soft turning point it leaks into the forbidden region, so the oscillation inside is a little short of a full extra half-wavelength. The precise result (Chapter 12) is that each soft turning point contributes $\frac{1}{4}$ to γ and each hard wall contributes 0. So $\gamma = 0$ for two hard walls, $\gamma = \frac{1}{4}$ for one hard wall and one soft turning point, and $\gamma = \frac{1}{2}$ for two soft turning points. Here we test the rule on wells we already know.

Box (hard walls, $\gamma = 0$): $V = 0$ on $[-L, L]$, so $\frac{1}{\pi} \cdot 2L\sqrt{E_n} = n$, which gives $E_n = (n\pi/2L)^2$. Exactly Equation (11.10).

Oscillator (soft turning points at $\pm\sqrt{E}$, $\gamma = \frac{1}{2}$): substitute $x = \sqrt{E}\sin\theta$, $\mathrm{d}x = \sqrt{E}\cos\theta\,\mathrm{d}\theta$, with θ from $-\pi/2$ to $\pi/2$:

$$\int_{-\sqrt{E}}^{\sqrt{E}} \sqrt{E - x^2}\,\mathrm{d}x = E\int_{-\pi/2}^{\pi/2} \cos^2\theta\,\mathrm{d}\theta = E \cdot \frac{\pi}{2}.$$

So Equation (11.24) reads $E_n/2 = n - \frac{1}{2}$, that is $E_n = 2n - 1$. Exactly Equation (11.19). (For the oscillator this is a lucky exactness; for most wells the rule is only approximate.)

11.6.2 Power-law wells

Now take $V = |x|^\alpha$ with $\alpha > 0$. The turning points are $\pm a$ with $a = E^{1/\alpha}$. Substitute $x = au$:

$$\int_{-a}^{a} \sqrt{E - |x|^\alpha}\,\mathrm{d}x = 2a\int_0^1 \sqrt{E - a^\alpha u^\alpha}\,\mathrm{d}u = 2a\sqrt{E}\int_0^1 \sqrt{1 - u^\alpha}\,\mathrm{d}u = 2E^{\frac{1}{\alpha}+\frac{1}{2}} I_\alpha,$$

where $I_\alpha = \int_0^1 \sqrt{1 - u^\alpha}\,\mathrm{d}u$ is a pure number and we used $a^\alpha = E$ and $a = E^{1/\alpha}$. The counting rule becomes

$$\frac{2I_\alpha}{\pi} E_n^{\frac{1}{2}+\frac{1}{\alpha}} \approx n - \tfrac{1}{2}, \qquad \text{so} \qquad E_n \propto \left(n - \tfrac{1}{2}\right)^{\frac{2\alpha}{\alpha+2}}. \tag{11.25}$$

Checks: $\alpha = 2$ gives $I_2 = \pi/4$ (a quarter of the unit disk) and $E_n \propto n$; as $\alpha \to \infty$ the well becomes a box of half-width 1, $I_\alpha \to 1$, and $E_n \propto n^2$. (Only the power of n is being checked here. In the limit the turning points become hard walls, so γ changes from $\frac{1}{2}$ to 0 and $n - \frac{1}{2}$ becomes n.) In general the spacing behaves as

$$E_{n+1} - E_n \approx \frac{dE_n}{dn} \propto n^{\frac{2\alpha}{\alpha+2}-1} = n^{\frac{\alpha-2}{\alpha+2}} :$$

- $\alpha > 2$ (walls steeper than a parabola): spacing *grows* with n; the box is the extreme case.
- $\alpha = 2$: constant spacing.
- $\alpha < 2$ (walls gentler than a parabola): spacing *shrinks*; the levels crowd together as you climb.

11.6.3 The V-shaped well

The linear well $V = |x|$ ($\alpha = 1$) is exactly solvable with a special function.

Taken on trust

Taken on trust: Airy functions. For $x > 0$, $-\psi'' + x\psi = E\psi$ becomes $\psi'' = y\psi$ with $y = x - E$. This is Airy's equation. Its solution that decays as $y \to +\infty$ is a named special function, $\mathrm{Ai}(y)$, which oscillates for $y < 0$ and has zeros at $y = a_1 > a_2 > \cdots$ (all negative), while its derivative Ai' has zeros at $a_1' > a_2' > \cdots$. These are tabulated and available as `scipy.special.ai_zeros`. Taking $\psi = \mathrm{Ai}(x - E)$ for $x > 0$ and extending by parity: odd states need $\psi(0) = 0$, so $E = -a_k$; even states need $\psi'(0) = 0$, so $E = -a_k'$. The construction is complete except for the properties of Ai, which we take from the tables.

Table 11.1: The first six levels of three even wells (units $\hbar = 2m = 1$). Box: half-width $L = \pi/2$, so $E_n = n^2$. Oscillator: $V = x^2$. V-shaped: $V = |x|$, levels from Airy zeros. The parity column holds for all three wells.

n	parity	nodes	box $L = \pi/2$	$V = x^2$	$V = \|x\|$
1	even	0	1	1	1.0188
2	odd	1	4	3	2.3381
3	even	2	9	5	3.2482
4	odd	3	16	7	4.0879
5	even	4	25	9	4.8201
6	odd	5	36	11	5.5206

Sample Problem 11.9: Counting half-wavelengths in a V-shaped well

Use Equation (11.24) with $\gamma = \frac{1}{2}$ to estimate E_1, E_2, E_9, E_{10} of $V = |x|$, and compare with the Airy values.

Step 1: Here $\alpha = 1$, $I_1 = \int_0^1 \sqrt{1-u}\,du = \left[-\frac{2}{3}(1-u)^{3/2}\right]_0^1 = \frac{2}{3}$. So Equation (11.25) gives $\frac{4}{3\pi}E_n^{3/2} = n - \frac{1}{2}$, that is

$$E_n \approx \left[\tfrac{3\pi}{4}\left(n - \tfrac{1}{2}\right)\right]^{2/3}.$$

Step 2: $n = 1$: $(3\pi/8)^{2/3} = 1.1781^{2/3} = 1.115$. $n = 2$: $3.5343^{2/3} = 2.320$. $n = 9$: $20.028^{2/3} = 7.375$. $n = 10$: $22.384^{2/3} = 7.942$.

Step 3: Airy values: odd n is even parity, so $E_n = -a'_{(n+1)/2}$; even n is odd parity, $E_n = -a_{n/2}$. That gives $E_1 = 1.019$, $E_2 = 2.338$, $E_9 = -a'_5 = 7.372$, $E_{10} = -a_5 = 7.944$.

n	1	2	9	10
counting	1.115	2.320	7.375	7.942
exact	1.019	2.338	7.372	7.944

The rule is off by about 10% for the ground state and by at most about 0.003 at $n = 9, 10$. This is typical of semiclassical counting: weak for the lowest levels, excellent for high ones. The paper meets the same pattern: the Wu–Sprung construction is worst "at low n, the known semiclassical weakness of the construction" (Sec. II B of [1]).

Checkpoint 11.5

For $V = |x|^3$, do the levels crowd or spread as n grows?

Answer: They spread, slowly: $\alpha = 3 > 2$, so $E_n \propto n^{6/5}$ and the spacing grows like $n^{1/5}$.

Key Ideas

- Heuristic counting rule: $\frac{1}{\pi}\int \sqrt{E_n - V}\,dx \approx n - \gamma$ ($\gamma = 0$ hard walls, $\frac{1}{2}$ soft turning points); exact for the box and oscillator, accurate at high n in general, weakest at low n.
- For $V = |x|^\alpha$: $E_n \propto n^{2\alpha/(\alpha+2)}$. Steeper than a parabola: spacing grows. Gentler: spacing shrinks.

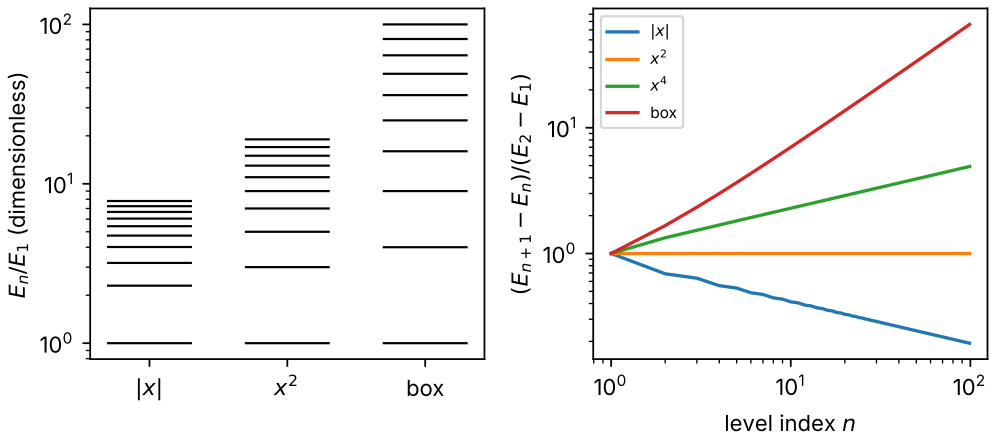


Figure 11.5: Left: the first ten levels of three even wells, scaled by their ground-state energy (logarithmic axis): $V = |x|$ (levels crowd), $V = x^2$ (equally spaced), and the box (levels spread). Right: spacing $E_{n+1} - E_n$, divided by the first spacing, versus n on logarithmic axes, for $V = |x|$, x^2, x^4 and the box. The slopes approach $(\alpha - 2)/(\alpha + 2)$: $-1/3$, 0, $1/3$ and 1. Energies in units $\hbar = 2m = 1$; the scaled quantities are dimensionless.

11.7 Level density, and why slow growth means crowding

★ Did you know? ***Can You Hear a Drum?***

Hermann Weyl proved in 1911 that the number of vibration modes of a drum below a given frequency grows in proportion to its area. In 1966 Mark Kac asked whether the full list of frequencies determines the drum's shape. In 1992 Carolyn Gordon, David Webb and Scott Wolpert answered no by constructing two differently shaped drums with identical spectra. Level counts reveal a lot, but not everything.

11.7.1 Counting function and density

When levels are numerous, it is convenient to describe them collectively. The *counting function* $N(E)$ is the number of levels with $E_n \le E$. It is a staircase that jumps by 1 at each level. The counting rule Equation (11.24) supplies a smooth function $\bar{N}(E)$ that follows the staircase:

$$\bar{N}(E) = \frac{1}{\pi} \int_{x_1(E)}^{x_2(E)} \sqrt{E - V(x)}\, \mathrm{d}x + \gamma,$$

with $\bar{N}(E_n) \approx n$ (the γ is the same constant as before). Strictly, a function with $\bar{N}(E_n) \approx n$ passes near the *top* of each step. The staircase jumps from $n-1$ to n at E_n, so its running average there is $n - \frac{1}{2}$. The average of the staircase therefore lies half a unit lower, at $\bar{N}(E) - \frac{1}{2}$. This is the convention behind the paper's smooth zeros, $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$. A constant shift does not change the derivative, so the level density below is the same either way. The *level density* is the derivative

$$\rho(E) = \frac{d\bar{N}}{dE}, \tag{11.26}$$

the number of levels per unit energy near E. Its inverse $1/\rho$ is the *mean spacing* near E. Do not confuse $\rho(E)$ with the zeros $\rho_n = \frac{1}{2} + it_n$, nor the counting function $N(E)$ with the number of grid points N in the listings; context distinguishes them.

Pitfall

The level density $\rho(E)$ counts levels per unit *energy*; the probability density $|\psi(x)|^2$ is probability per unit *length* for one state. They are unrelated quantities that share a name.

11.7.2 Density equals classical period over 2π

Differentiate $\bar{N}$. The limits $x_{1,2}(E)$ depend on E, and so does the integrand, so we need a rule for differentiating such an integral.

Taken on trust

Taken on trust: Leibniz's rule. Let $f(x, E)$ be continuous, with a continuous derivative $\partial f/\partial E$ (the derivative with respect to E at fixed x), and let the limits $x_1(E)$, $x_2(E)$ be differentiable. Then

$$\frac{d}{dE}\int_{x_1(E)}^{x_2(E)} f(x,E)\,\mathrm{d}x = \int_{x_1}^{x_2}\frac{\partial f}{\partial E}\,\mathrm{d}x + f(x_2,E)\frac{dx_2}{dE} - f(x_1,E)\frac{dx_1}{dE}.$$

The two limit terms come from the fundamental theorem of calculus and the chain rule. The first term is differentiation under the integral sign, whose proof we take from [22]. In our use $f=\sqrt{E-V}$ and $\partial f/\partial E = 1/(2\sqrt{E-V})$. This derivative is infinite at the moving limits, so the hypotheses fail at the endpoints. That the result Equation (11.27) below still holds for potentials smooth near their turning points is also taken on trust from [22]. In the main text we verify it exactly for power-law wells.

Here $f=\sqrt{E-V}$. At a turning point the integrand $\sqrt{E-V}$ is zero, so the limit terms vanish, and

$$\rho(E) = \frac{1}{\pi}\int_{x_1}^{x_2}\frac{\mathrm{d}x}{2\sqrt{E-V(x)}} = \frac{1}{2\pi}\int_{x_1}^{x_2}\frac{\mathrm{d}x}{\sqrt{E-V(x)}}. \tag{11.27}$$

(The integrand blows up at the turning points, but like $1/\sqrt{\text{distance}}$, which integrates to a finite value.) Because the hypotheses of Leibniz's rule fail at the endpoints (see the box), we check the step for $V=|x|^\alpha$. The substitution $x=au$, $a=E^{1/\alpha}$, of Section 11.6 turns Equation (11.27) into $\rho=\frac{1}{\pi}E^{\frac{1}{\alpha}-\frac{1}{2}}J_\alpha$ with $J_\alpha=\int_0^1 \mathrm{d}u/\sqrt{1-u^\alpha}$, while differentiating $\bar{N}=\frac{2I_\alpha}{\pi}E^{\frac{1}{2}+\frac{1}{\alpha}}+\gamma$ directly gives $\frac{2I_\alpha}{\pi}(\frac{1}{2}+\frac{1}{\alpha})E^{\frac{1}{\alpha}-\frac{1}{2}}$. They agree if $J_\alpha=(1+\frac{2}{\alpha})I_\alpha$. For $\alpha=2$: $J_2=\pi/2$ and $2I_2=2\cdot\pi/4=\pi/2$. For $\alpha=4$, numerical integration gives $J_4=1.3110$ and $\frac{3}{2}I_4=1.3110$.

The classical meaning. With $m=\frac{1}{2}$, kinetic energy is $\frac{1}{2}mv^2=v^2/4=E-V$, so the classical speed is $v=2\sqrt{E-V}$. The time to cross from x_1 to x_2 is $\int \mathrm{d}x/v$, and a full round trip takes twice that:

$$T_{\mathrm{cl}}(E) = 2\int_{x_1}^{x_2}\frac{\mathrm{d}x}{2\sqrt{E-V}} = \int_{x_1}^{x_2}\frac{\mathrm{d}x}{\sqrt{E-V}}.$$

Comparing with Equation (11.27):

$$\rho(E) = \frac{T_{\mathrm{cl}}(E)}{2\pi}, \qquad \text{equivalently} \qquad \text{mean spacing} = \frac{2\pi}{T_{\mathrm{cl}}} = \omega_{\mathrm{cl}}. \tag{11.28}$$

In ordinary units this is $\Delta E \approx \hbar\omega_{\mathrm{cl}}$: the level spacing is $\hbar$ times the classical angular frequency of the orbit at that energy. This is a heuristic (semiclassical) result, accurate for high levels.

Check it against first-year physics. Simple harmonic motion has a period that does *not* depend on amplitude, so the oscillator's spacing should be constant: for $V=x^2$, $\omega_{\mathrm{cl}}=2$ (from $\frac{1}{4}\omega^2=1$), and the spacing is indeed 2. In the box a particle of speed $v=2\sqrt{E}$ makes a

round trip of length $4L$ in time $T_{\rm cl} = 4L/(2\sqrt{E}) = 2L/\sqrt{E}$, so $\rho = L/(\pi\sqrt{E})$; differentiating $\bar{N} = 2L\sqrt{E}/\pi$ gives the same. For power laws, $\rho \propto E^{\frac{1}{\alpha}-\frac{1}{2}}$ (differentiate Equation (11.25)).

11.7.3 Why a slowly growing potential crowds its levels

Equation (11.28) gives the physical answer to the question of this section. A potential that grows slowly lets a more energetic particle travel much farther before turning around. If the distance it gains grows faster than its speed, its round trip takes *longer* at higher energy, $T_{\rm cl}$ grows, ρ grows, and the levels crowd together. A parabola is the borderline: the extra distance and the extra speed exactly balance and the period is constant. Steeper walls (the box) shorten the period as E grows and spread the levels.

11.7.4 The density of the Riemann zeros

The smooth counting function of the zeta zeros is the paper's Eq. (1) (derived in Chapter 17):

$$N_{\rm sm}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}.$$

Differentiate, using the product rule on the first term:

$$\frac{dN_{\rm sm}}{dt} = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} - \frac{1}{2\pi} = \frac{1}{2\pi}\log\frac{t}{2\pi}.$$

So if the zeros are to be the levels of a well, that well must have

$$\rho(E) = \frac{1}{2\pi}\log\frac{E}{2\pi}, \tag{11.29}$$

a density that *grows*, but only logarithmically: from 0.129 per unit at $t_1 = 14.13$ to 0.863 at $t_{1000} = 1419.42$. The levels crowd, so the classical period must grow with energy, so the potential must grow more slowly than a parabola. But the growth is gentle, so the potential is only a little slower than a parabola. Figure 11.6 compares the densities.

Sample Problem 11.10: What shape of well do the zeros want?

Treat the zeros' density as if it came locally from a power law $|x|^\alpha$, and find the effective exponent α near the 500th zero, $t_{500} = 811.18$ (value from App. A of [1]). Also find the mean spacing there and compare with the empty box of RE3.

Step 1: For a power law $\rho \propto E^{1/\alpha - 1/2}$, so $\frac{d\log\rho}{d\log E} = \frac{1}{\alpha} - \frac{1}{2}$.

Step 2: For Equation (11.29), $\log\rho = \log\log\frac{t}{2\pi} - \log 2\pi$. Since $\frac{d}{d\log t} = t\frac{d}{dt}$, the chain rule gives

$$\frac{d\log\rho}{d\log t} = t\cdot\frac{1}{\log(t/2\pi)}\cdot\frac{1}{t} = \frac{1}{\log(t/2\pi)}.$$

Step 3: At $t = 811.18$: $t/2\pi = 129.10$, $\log 129.10 = 4.8606$, so $\frac{1}{\log(t/2\pi)} = 0.2057$.

Step 4: $\frac{1}{\alpha} = \frac{1}{2} + 0.2057 = 0.7057$, so $\alpha = 1.417$.

Step 5: Density $\rho = 4.8606/(2\pi) = 0.774$, mean spacing $1/\rho = 1.293$. The empty box at the same energy has $\rho = L/(\pi\sqrt{E}) = 44.883/(\pi\times 28.481) = 0.502$, spacing 1.99.

$$\boxed{\alpha_{\rm eff} \approx 1.42 \text{ at } t \approx 811; \quad \text{mean spacing } 1.293 \text{ (box: } 1.99)}$$

The effective exponent lies between the V-shaped well ($\alpha = 1$) and the parabola ($\alpha = 2$), and since $1/\log(t/2\pi) \to 0$ it creeps toward 2 as t grows, but only logarithmically slowly. This is a heuristic local description, not an exact property of any potential.

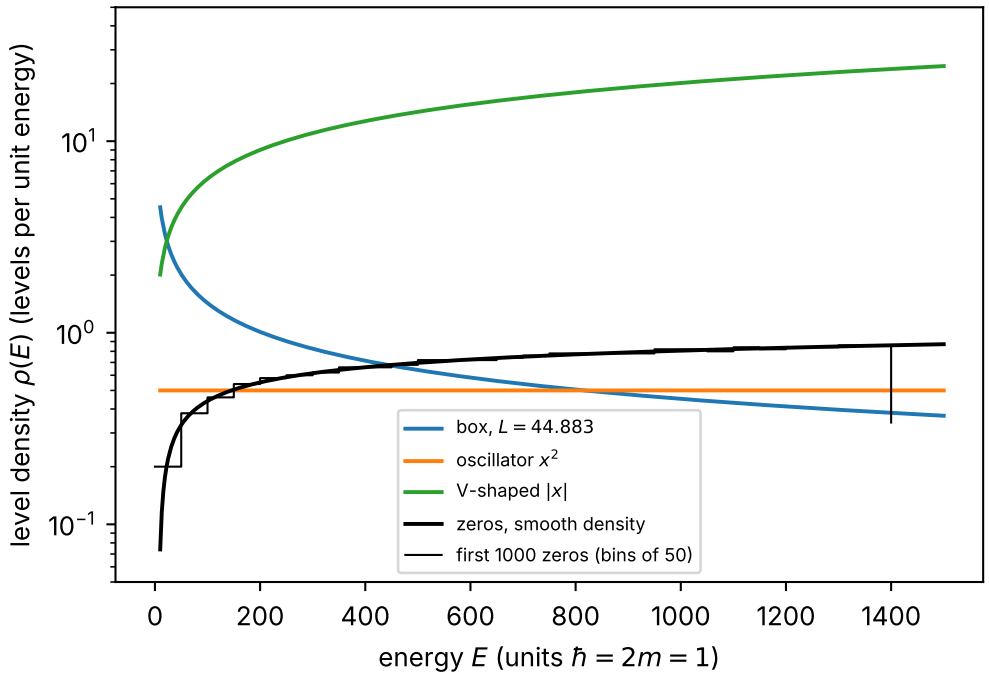


Figure 11.6: Level density $\rho(E)$ (levels per unit energy, logarithmic axis) versus energy (units $\hbar = 2m = 1$). Box with $L = 44.883$: $L/(\pi\sqrt{E})$, falling. Oscillator $V = x^2$: constant $\frac{1}{2}$. V-shaped well $V = |x|$: $(2/\pi)\sqrt{E}$, rising fast. Riemann zeros: the smooth density $\frac{1}{2\pi}\log(t/2\pi)$ (line) and a histogram of the first 1000 zeros in bins of width 50 (steps). The zeros' density rises, but much more slowly than for the V-shaped well.

Sample Problem 11.11: A parabola slowed down by a logarithm

Suppose an even well, increasing for $x > 0$, has for large V the half-width $x(V) = \frac{\sqrt{V}}{\pi}(\log V + b)$, with b a constant. Problem 23 shows that the paper's Wu–Sprung potential has this form with $b = \log\frac{2}{\pi e^2}$. Find the level density at large E from Equation (11.28).

Step 1: By symmetry, $T_{\rm cl} = 2\int_0^{x_E} \mathrm{d}x/\sqrt{E - V(x)}$. On $x > 0$ the potential increases, so we may use V as the variable: $\mathrm{d}x = x'(V)\,\mathrm{d}V$. We use the large-$V$ formula all the way down to $V = 0$. This is heuristic: the region near the bottom of the well contributes a relative error that shrinks as E grows, so only the leading large-E behavior is kept.

Step 2: By the product rule, $x'(V) = \frac{1}{\pi}\left[\frac{\log V + b}{2\sqrt{V}} + \sqrt{V}\cdot\frac{1}{V}\right] = \frac{\log V + b + 2}{2\pi\sqrt{V}}$.

Step 3: Hence $T_{\rm cl} = \frac{1}{\pi}\int_0^E \frac{\log V + b + 2}{\sqrt{V(E-V)}}\,\mathrm{d}V$.

Step 4: Substitute $V = E\sin^2\theta$, so $\mathrm{d}V = 2E\sin\theta\cos\theta\,\mathrm{d}\theta$ and $\sqrt{V(E-V)} = E\sin\theta\cos\theta$, with θ from 0 to $\pi/2$. Since $\log V = \log E + 2\log\sin\theta$,

$$T_{\rm cl} = \frac{2}{\pi}\int_0^{\pi/2}(\log E + b + 2 + 2\log\sin\theta)\mathrm{d}\theta = \log E + b + 2 + \frac{4}{\pi}\int_0^{\pi/2}\log\sin\theta\,\mathrm{d}\theta.$$

Step 5: Use $\int_0^{\pi/2}\log\sin\theta\,\mathrm{d}\theta = -\frac{\pi}{2}\log 2$ (taken on trust; see the box after this Sample Problem). Then $T_{\rm cl} = \log E + b + 2 - 2\log 2$.

Step 6: $\rho = T_{\rm cl}/2\pi = \frac{1}{2\pi}(\log E + b + 2 - 2\log 2)$. With $b = \log 2 - \log\pi - 2$ this becomes $\frac{1}{2\pi}(\log E - \log 2 - \log\pi)$.

$$\boxed{\rho(E) \approx \frac{1}{2\pi} \log \frac{E}{2\pi} \quad \text{(large } E\text{)}}$$

This is exactly Equation (11.29). A parabola slowed down by a logarithm gives a density that grows logarithmically, which is what the zeros require. Like everything in this section, the result is heuristic (semiclassical, leading order in E).

Taken on trust

Taken on trust: a classical definite integral. $\int_0^{\pi/2} \log \sin\theta \, d\theta = -\frac{\pi}{2} \log 2 = -1.088793\ldots$ Its proof uses a symmetry trick with sin and cos that we do not reproduce. You can confirm it numerically with `scipy.integrate.quad(lambda th: np.log(np.sin(th)), 0, np.pi/2)`.

Checkpoint 11.6

The paper scores errors in units of the "local mean spacing" gap_n (Sec. II A of [1]). In the language of this section, what is gap_n?

Answer: It is $1/\rho$ evaluated at the n-th smooth zero, $2\pi/\log(s_n/2\pi)$: the average distance between neighboring levels at that height. Chapter 17 derives it.

Key Ideas

- Counting function $N(E)$, smooth trend $\bar{N}(E)$, level density $\rho = d\bar{N}/dE$, mean spacing $1/\rho$.
- Heuristic: $\rho = T_{\text{cl}}/2\pi$; mean spacing $= \hbar\omega_{\text{cl}}$.
- Slowly growing potential ⇒ longer classical period at higher energy ⇒ crowding levels.
- The zeros have $\rho = \frac{1}{2\pi}\log\frac{t}{2\pi}$: gently increasing, so their potential grows a little more slowly than a parabola (effective exponent about 1.4 near $t \approx 800$).

11.8 Paper connection

11.8.1 The box and its walls

Section II B of [1] states the setting: "In units $\hbar = 2m = 1$, $H = -d^2/dx^2 + V(x)$ with V even." The Wu–Sprung potential of Eq. (2) is defined for $V \geq V_0 = 3.10073\pi \approx 9.741$; it equals V_0 at $x = 0$ (both terms of Eq. (2) vanish when $V = V_0$, because $\sqrt{V - V_0} = 0$ and the logarithm becomes $\log 1$) and rises on both sides. The problem is posed on "the domain $[-L, L]$ with $L = x(V{=}1000) = 44.883$" at scale 1 and $L = x(V{=}1500) = 59.957$ at scale 2, with "Dirichlet walls" in the finite-difference solver. So the paper's operator is exactly the situation of this chapter: a smooth even well, closed off by the walls of Section 11.2. The empty box of RE3 (Sample Problem "The empty box of the paper") is what you get by deleting the potential.

Why are the walls harmless? The walls are placed where the potential has climbed to 1000 (scale 1). The scored levels are "the first 500" (Sec. II B), and the 500th zero is $t_{500} = 811.1843588465$ (App. A, key anchors). So every scored level lies well below the potential at the wall. By the reasoning of Section 11.5 (a heuristic transfer of the oscillator estimate, not a computation for this potential), each scored state decays through a stretch of forbidden region before reaching the wall: for level 500 the turning point, which we computed from Eq. (2) with

`qmnt.wu_sprung_x`, is at $x(811.18) \approx 38.5$, about 6.4 units inside the wall at 44.883, where $V - E$ climbs from 0 to about 190. We therefore expect the wall's influence on such a state to be exponentially small, as Figure 11.4 shows for the oscillator. Levels with energies close to 1000 would feel the wall strongly; the paper's solver computes more levels than it scores (App. A step 4: eigenvalue range $[0, 900]$ for the Hermitian work at scale 1).

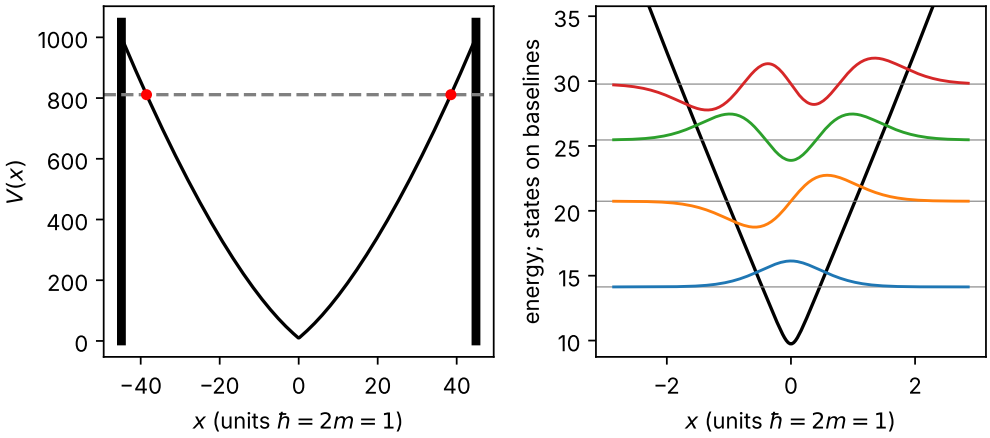


Figure 11.7: The paper's scale-1 setting (units $\hbar = 2m = 1$). Left: the Wu–Sprung potential of Eq. (2) of [1] on $[-L, L]$, $L = 44.883$, with Dirichlet walls (thick lines) where $V = 1000$. The dashed line is the energy $t_{500} = 811.18$ of the last scored level; dots mark its turning points, about 6.4 units inside the walls. Right: close-up near the bottom of the well ($V_0 \approx 9.741$), with the first four eigenfunctions drawn on baselines at their computed energies; they alternate even, odd, even, odd and have $0, 1, 2, 3$ nodes.

11.8.2 Counting levels in the filled box

The empty box holds 903 levels at or below $E = 1000$ (Sample Problem above). The paper reports 652 states with $E \leq 1000$ (Sec. II E and App. A step 4: "$[0, 1000]$ (tensors; 652 states)"); these are the eigenstates of the smooth (unperturbed) Wu–Sprung potential on the finite-difference grid, used as the basis for the second-order state sums of Sec. II E. The Wu–Sprung potential occupies the box and pushes the levels up, so far fewer fit below 1000. How many should fit? The potential was built so that its levels follow the smooth zeros, so the count should be close to $N_{\mathrm{sm}}(1000) = 648.6$. Precisely, $s_n \leq 1000$ exactly when $n - \frac{1}{2} \leq N_{\mathrm{sm}}(1000)$, so the number of smooth zeros $s_n \leq 1000$ is $\lfloor N_{\mathrm{sm}}(1000) + \frac{1}{2} \rfloor = 649$. The agreement with 652 is close but not exact; we do not analyze the small difference here.

11.8.3 Level index and turning points

The paper labels levels from $n = 1$, as we do. Because the n-th state lives mostly between its turning points and is most sensitive to the potential there (Chapter 13 makes this precise), each level index corresponds to a position. Section III (Interpolation split) uses exactly this: "Position maps to level index through the classical turning point, and levels 201–300 turn at $x = 22.5$–28.5." Figure 4 of the paper shades that band. Problem 22 asks you to reproduce these numbers.

11.8.4 Why the potential is even, and what evenness buys

The counting rule Equation (11.24) depends on the potential only through how much room the particle has at each energy, not on how that room is arranged. (The box on $[-L, L]$ and the

box on $[0, 2L]$ have identical levels; to leading semiclassical order, the same is true of any two wells whose allowed intervals have the same lengths at every energy.) So a list of levels cannot by itself fix the potential, and Wu and Sprung [12] fixed the ambiguity by choosing an even potential. Chapter 22 carries out the construction. Evenness has three consequences visible in the paper.

- *Alternating parity.* By Section 11.3, the states alternate even, odd, even, ..., starting from an even ground state (Figure 11.7, right).

- *Even corrections.* The fitted correction $\delta V(x) = \sum_k c_k \cos(k\pi x/L)$ of Sec. II C uses even functions, and the alternative bases of Sec. II C, Chebyshev $T_k(2|x|/L - 1)$ and hat functions of $|x|/L$, are even because they depend on $|x|$. So $V + \delta V$ stays even.

- *Odd perturbations have no diagonal elements.* Since ψ_n^2 is even, $\int \psi_n^2 f \, \mathrm{d}x = 0$ for any odd f (an odd integrand on a symmetric interval). The paper's PT-odd class uses $\phi_k = \sin((k+1)\pi x/L)$, which is odd, and Sec. II E notes "$A_{nn} = 0$": this is exactly that parity fact (Problem 27). Beware that the paper uses A for two different objects. Here $A_{nn} = \langle n|A|n\rangle$ is a matrix element of the operator A of Sec. II E, whereas A_{nk} of Sec. II C is the sensitivity matrix. Conversely, for the Hermitian fit, the first-order sensitivity $A_{nk} = \int |\psi_n|^2 \phi_k \, \mathrm{d}x$ of Sec. II C would vanish for any odd basis function, so restricting δV to even functions loses nothing at first order (Problem 26 uses the first-order formula of Chapter 13).

11.8.5 Shape and spacing

At the heights studied, the zeros' density Equation (11.29) rises slowly, so the potential must grow more slowly than a parabola. Eq. (2) shows how: for large V it gives x growing like $\sqrt{V}\log V$ (Problem 23 derives $x \approx \frac{\sqrt{V}}{\pi}\log\frac{2V}{\pi e^2}$, which gives 44.86 at $V = 1000$ against the exact 44.883). Inverting, V grows roughly like $(x/\log x)^2$: a parabola slowed down by a logarithm, just as Section 11.7 predicted. Section II B also reports that the Richardson-extrapolated levels of the smooth potential match the smooth zeros to 0.005 mean spacings on average and 0.063 at worst, "worst at low n, the known semiclassical weakness of the construction": the same weakness of semiclassical counting at low n that you saw for the V-shaped well.

You can compute the lowest levels of the paper's smooth potential yourself:

```
import numpy as np
import qmnt

L = qmnt.L_of(1000)            # 44.883, scale 1 (paper Sec. II B)
N = 3000                       # interior grid points (paper uses 6000)
x = qmnt.grid(L, N)            # interior points of (-L, L)
V = np.array([qmnt.wu_sprung_V(xi) for xi in x])
E = qmnt.fd_levels(V, L, k=10) # lowest 10 levels, Dirichlet walls
s = qmnt.smooth_zeros(np.arange(1, 11))  # smooth zeros s_1..s_10
t = qmnt.zeros(10)             # true zeros t_1..t_10 (RE1)
print(np.round(E, 3))
print(np.round(s, 3))
print(np.round(t, 3))
```

The levels follow the smooth zeros s_n, not the true zeros t_n: the smooth potential knows only the average density. When you run the listing, every E_n lies within 0.15 mean spacings of s_n. For $n = 9$, by contrast, the true zero lies more than 0.15 mean spacings from E_9. The differences $t_n - s_n$, the fluctuations, are what the paper's fitted corrections try to learn.

11.9 Summary

- Bound states are normalizable solutions of $-\psi'' + V\psi = E\psi$ (Equations (11.1) and (11.2)); they can be taken real; the curvature rule Equation (11.3) gives oscillation where $E > V$ and exponential behavior where $E < V$.
- Every level lies above the bottom of the well: $E = \int(\psi')^2 + \int V\psi^2 > V_{\min}$ (Equation (11.4)).
- Infinite square well on $[-L, L]$: $E_n = (n\pi/2L)^2$, $\psi_n = L^{-1/2}\sin(n\pi(x+L)/2L)$ (Equations (11.10) and (11.12)); spacing $(2n+1)E_1$ (Equation (11.14)).
- One-dimensional bound states are nondegenerate; in an even well they are even or odd, alternating from an even ground state; the n-th has $n-1$ nodes (Section 11.3).
- Harmonic oscillator $V = x^2$: $E_n = 2n-1$, $\psi_n \propto H_{n-1}e^{-x^2/2}$ (Equation (11.19)); $V = ax^2$ gives $\sqrt{a}(2n-1)$.
- A wall deep in the forbidden region shifts levels exponentially little; for the oscillator ground state about $(4L/\sqrt{\pi})e^{-L^2}$ (Equation (11.23)).
- Heuristic counting rule $\frac{1}{\pi}\int\sqrt{E_n - V}\,\mathrm{d}x \approx n - \gamma$ (Equation (11.24)); power laws give $E_n \propto n^{2\alpha/(\alpha+2)}$ (Equation (11.25)).
- Level density $\rho = T_{\mathrm{cl}}/2\pi$ (Equation (11.28)): slowly growing potentials crowd their levels. The zeros have $\rho = \frac{1}{2\pi}\log\frac{t}{2\pi}$ (Equation (11.29)), so their potential grows a little more slowly than a parabola.
- In the paper: Dirichlet walls at $\pm L$ where $V = 1000$ (or 1500), far above the scored levels; even potential and even corrections; $A_{nn} = 0$ for odd perturbations by parity.

11.10 Problems

1. • A box on $[-1, 1]$ ($L = 1$). Compute E_1, E_2, E_3, E_4 and the three spacings between them.

2. • By what factor do all levels of a box change if L is doubled? If L is halved?

3. • Write ψ_5 for the box on $[-L, L]$. Is it even or odd? Find all its nodes and check the node theorem.

4. •• Show by direct integration that ψ_1 and ψ_3 of the box are orthogonal.

5. •• Compute $\langle x^2\rangle = \int x^2\psi_1^2\,\mathrm{d}x$ for the ground state of the box on $[-L, L]$.

6. •• For the second box state, find the probability of the middle half $-L/2 < x < L/2$.

7. •• Let V be even. Prove that if $\psi(x)$ solves Equation (11.1), so does $\psi(-x)$. Deduce that in any bound state of an even well the probability of $x > 0$ is exactly $\frac{1}{2}$.

8. ••• Show that the even states of an even well on $[-L, L]$ are the states of the half-interval $[0, L]$ with $\psi'(0) = 0$, $\psi(L) = 0$, and the odd states those with $\psi(0) = 0$, $\psi(L) = 0$. Use this to re-derive Equation (11.10) by solving only on $[0, L]$.

9. • List the first five energies of $V = x^2$, with the parity and the number of nodes of each state.

10. •• Use the recursion Equation (11.18) to find the polynomial h for $E = 7$ and locate the nodes of that state.

11. •• For the oscillator ground state $\psi_1 = \pi^{-1/4} e^{-x^2/2}$ compute $\langle x^2 \rangle$ and $\langle V \rangle$, and then use Equation (11.4) to find the average kinetic energy.

12. •• Find the first three levels of $-\psi'' + \frac{1}{4}x^2\psi = E\psi$ and of $-\psi'' + 9x^2\psi = E\psi$.

13. •• Using numpy or scipy, compute the probability that the oscillator ground state is found in the classically forbidden region $|x| > 1$.

14. •• Using numpy and the listing in Section 11.4, compute the first ten levels of $V = x^2$ on $[-8, 8]$ with $N = 2000$ and with $N = 4000$. How does the error change? (Chapter 23 explains the pattern.)

15. •• Show by rescaling x that the levels of $V = g|x|$ satisfy $E_n(g) = g^{2/3} E_n(1)$. Find the ground-state energy for $g = 8$ using Table 11.1.

16. •• For $V = |x|^3$, find how E_n, the spacing and the level density depend on n or E at large n.

17. •• Using scipy, evaluate $I_4 = \int_0^1 \sqrt{1 - u^4}\, du$ and use the counting rule with $\gamma = \frac{1}{2}$ to estimate E_{10} of $V = x^4$. Then compute E_{10} numerically with the tridiagonal method and compare.

18. ••• Using scipy, compute the first 12 levels of $V = |x|$ with the tridiagonal method and compare them with $-a_k$ and $-a'_k$ from `scipy.special.ai_zeros`. Which levels does the half-wavelength rule predict worst?

19. •• (Paper.) The paper's scale-1 box has $L = 44.883$ and its Wu–Sprung potential has 652 states with $E \le 1000$ (App. A step 4 of [1]). How many levels does the empty box hold at or below 1000? Compute $N_{\mathrm{sm}}(1000)$ and compare all three numbers.

20. •• (Paper.) Repeat Problem 19 for scale 2: $L = 59.957$, energies up to 1500, against the 1079 states of App. A step 4.

21. •• (Paper.) At the 500th zero, $t_{500} = 811.18$ (App. A of [1]), compute the zeros' smooth density and mean spacing from Equation (11.29), and the level density and spacing of the empty box of RE3 at the same energy. Which spectrum is denser there?

22. •• (Paper.) Using qmnt, compute the classical turning points $x =$`qmnt.wu_sprung_x`(t_n) of levels $n = 201$ and $n = 300$, using `qmnt.zeros`. Compare with the band $x = 22.5$–28.5 quoted in Sec. III of the paper.

23. •• (Paper.) Show that for $V \gg V_0$ the Wu–Sprung relation Eq. (2) becomes $x \approx \frac{\sqrt{V}}{\pi} \log \frac{2V}{\pi e^2}$. Evaluate it at $V = 1000$ and $V = 1500$ and compare with the paper's L values.

24. ••• Repeat the effective-exponent calculation of Sample Problem "What shape of well do the zeros want?" at $t_1 = 14.1347$ and at $t_{1000} = 1419.42$. Why should you distrust the value at t_1?

25. ••• Using Equation (11.23) and the approximation of Step 5, estimate $E_b - 1$ for the oscillator with walls at ± 3, and compare with a numerical computation.

26. •• Assume the first-order shift formula $\delta E_n = \int \psi_n^2\, \delta V\, dx$ of Chapter 13. Show that for an even unperturbed potential, only the even part of δV shifts levels at first order.

27. •• (Paper.) Section II E of [1] states that for the PT-odd class $\phi_k = \sin((k+1)\pi x/L)$ the diagonal elements $A_{nn} = \langle n|A|n\rangle$ vanish. Prove it, and say which property of the unperturbed potential you used.

28. •• Show that adding a constant C to any potential shifts every level by exactly C and leaves every eigenfunction unchanged.

29. ••• Complete the Wronskian proof of nondegeneracy for the box: show directly that if u, w both solve $-\psi'' = E\psi$ with $u(\pm L) = w(\pm L) = 0$, then $w = cu$.

30. •• At what height t does the smooth density of the zeros equal the oscillator density $\frac{1}{2}$?

31. •• For odd n, show that the probability of the middle half of the box is $\frac{1}{2} + \sin(n\pi/2)/(n\pi)$. Evaluate for $n = 1, 3, 5$ and find the limit as $n \to \infty$. Interpret.

11.11 Answers to selected problems

1. $E_n = n^2\pi^2/4$: $E_1 = 2.467$, $E_2 = 9.870$, $E_3 = 22.21$, $E_4 = 39.48$. Spacings $3E_1 = 7.402$, $5E_1 = 12.34$, $7E_1 = 17.27$, growing linearly as in Equation (11.14).

2. $E_n \propto 1/L^2$, so doubling L divides every level by 4 and halving L multiplies every level by 4.

3. $n = 5$ is odd, so $\psi_5 = L^{-1/2}\cos(5\pi x/2L)$, even. Zeros where $5\pi x/2L = \pm\pi/2, \pm 3\pi/2$, that is $x = \pm L/5, \pm 3L/5$ (the next, $\pm 5\pi/2$, is the wall). Four nodes $= n - 1$, an even number, none at 0.

5. $\langle x^2\rangle = \frac{1}{2L}\left[\int_{-L}^{L} x^2 \mathrm{d}x + \int_{-L}^{L} x^2 \cos\frac{\pi x}{L}\mathrm{d}x\right]$ using $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$. The first integral is $2L^3/3$. For the second, integrate by parts twice with $k = \pi/L$: an antiderivative of $x^2\cos kx$ is $\frac{x^2\sin kx}{k} + \frac{2x\cos kx}{k^2} - \frac{2\sin kx}{k^3}$. At $x = \pm L$, $\sin kL = 0$ and $\cos kL = -1$, so only the middle term survives: $\frac{2L(-1)}{k^2} - \frac{2(-L)(-1)}{k^2} = -\frac{4L}{k^2} = -\frac{4L^3}{\pi^2}$. Hence $\langle x^2\rangle = L^2(\frac{1}{3} - \frac{2}{\pi^2}) = 0.1307\,L^2$.

7. The chain-rule argument of Section 11.3, Step 1, shows $\chi(x) = \psi(-x)$ solves the equation. Then $\psi(-x) = \pm\psi(x)$, so ψ^2 is even and $\int_0^L \psi^2\mathrm{d}x = \int_{-L}^{0}\psi^2\mathrm{d}x$ (substitute $x \to -x$). The two halves add to 1, so each is $\frac{1}{2}$. (For a whole-line well replace L by ∞.)

9. $E = 1, 3, 5, 7, 9$. Parities even, odd, even, odd, even. Nodes $0, 1, 2, 3, 4$.

11. $\langle x^2\rangle = \pi^{-1/2}\int x^2 e^{-x^2}\mathrm{d}x = \pi^{-1/2}\cdot\frac{\sqrt{\pi}}{2} = \frac{1}{2}$ by Equation (11.20). Since $V = x^2$, $\langle V\rangle = \frac{1}{2}$. By Equation (11.4), kinetic $= E - \langle V\rangle = 1 - \frac{1}{2} = \frac{1}{2}$: kinetic and potential energy share the energy equally, as for a classical oscillator averaged over a period.

13. $P = 2\pi^{-1/2}\int_1^\infty e^{-x^2}\mathrm{d}x$, which is the complementary error function erfc(1): `scipy.special.erfc(1)` or numerical integration gives $P = 0.1573$. About 16% of the time the particle is where a classical particle could never be.

15. Put $x = \beta y$: $-\beta^{-2}\psi_{yy} + g\beta|y|\psi = E\psi$. Choose $g\beta = \beta^{-2}$, so $\beta = g^{-1/3}$ and both coefficients equal $g^{2/3}$. Dividing by $g^{2/3}$ gives the $g = 1$ problem with energy $E/g^{2/3}$, so $E_n(g) = g^{2/3}E_n(1)$. For $g = 8$, $g^{2/3} = 4$ and $E_1 = 4 \times 1.0188 = 4.075$.

17. $I_4 = 0.8740$. The rule gives $\frac{2I_4}{\pi}E^{3/4} = n - \frac{1}{2}$, so $E_{10} \approx [\pi(9.5)/(2I_4)]^{4/3} = 17.07^{4/3} = 43.96$. The tridiagonal method (for example $L = 6$, $N = 4000$) gives $E_{10} = 43.98$: agreement to about 0.05%.

19. Empty box: $n \le 2L\sqrt{1000}/\pi = 903.6$, so 903 levels. $N_{\mathrm{sm}}(1000) = \frac{1000}{2\pi}\log\frac{1000}{2\pi} - \frac{1000}{2\pi} + \frac{7}{8} = 648.6$. The paper's filled box has 652. The Wu–Sprung potential raises the levels so that about 250 fewer fit below 1000, and its count is close to the smooth count of zeros, as intended. Exactly $\lfloor N_{\mathrm{sm}}(1000) + \frac{1}{2}\rfloor = 649$ smooth zeros lie at or below 1000, so the paper's grid count exceeds the smooth count by 3. These few extra states are not analyzed in this chapter. Chapter 23 studies the finite-difference bias of high levels, which is one effect a reader could examine.

21. $\rho = \log(811.18/2\pi)/2\pi = 4.8606/6.2832 = 0.774$ per unit energy, mean spacing 1.293. Empty box: $\rho = L/(\pi\sqrt{E}) = 44.883/(\pi \times 28.481) = 0.502$, spacing 1.99. At this one energy the zeros (and hence the Wu–Sprung levels, which follow them on average) are about 1.5 times denser than the empty box: $0.774/0.502 = 1.54$. This compares densities (levels per unit energy) at a single energy, not cumulative counts of levels below it.

23. For $V \gg V_0$: $\sqrt{V - V_0} \approx \sqrt{V}$; the denominator of the logarithm is $\sqrt{V} - \sqrt{V - V_0} = \frac{V_0}{\sqrt{V} + \sqrt{V - V_0}} \approx \frac{V_0}{2\sqrt{V}}$ (multiply and divide by $\sqrt{V} + \sqrt{V - V_0}$); the numerator is $\approx 2\sqrt{V}$; so

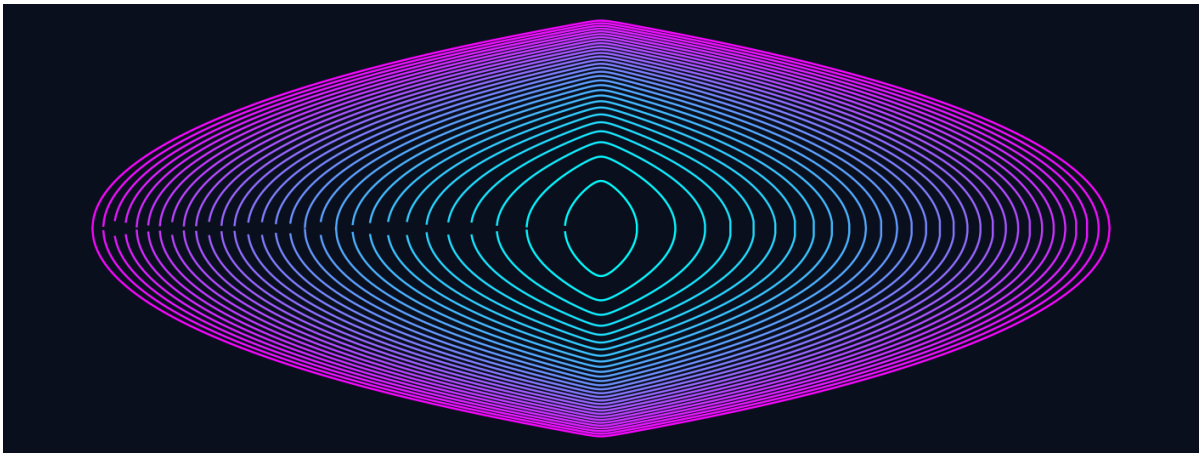
the ratio is $\approx 4V/V_0$. Then $x \approx \frac{\sqrt{V}}{\pi}[\log\frac{V_0}{2\pi e^2} + \log\frac{4V}{V_0}] = \frac{\sqrt{V}}{\pi}\log\frac{4V}{2\pi e^2} = \frac{\sqrt{V}}{\pi}\log\frac{2V}{\pi e^2}$; the constant V_0 cancels. At $V = 1000$: $\frac{31.623}{\pi}(6.4562 - 2) = 44.86$ versus 44.883. At $V = 1500$: $\frac{38.730}{\pi}(6.8616 - 2) = 59.93$ versus 59.957. The neglected terms cost about 0.03.

25. Estimate $\frac{4L}{\sqrt{\pi}}e^{-L^2} = \frac{12}{\sqrt{\pi}}e^{-9} = 8.4 \times 10^{-4}$. The numerical shift (tridiagonal matrix, fine grid) is about 7.9×10^{-4}, some 6% lower; the neglected corrections in Step 5 are of relative size $1/L^2$.

27. $A_{nn} = \int_{-L}^{L} \psi_n^2 \sum_k c_k \sin\frac{(k+1)\pi x}{L}\mathrm{d}x$. Each sin is odd. Because the unperturbed potential (Wu–Sprung) is even, each ψ_n is even or odd (Section 11.3), so ψ_n^2 is even, and even times odd is odd. An odd function integrates to zero over $[-L, L]$ (split at 0 and substitute $x \to -x$ in one half). So $A_{nn} = 0$. The property used is the evenness of the unperturbed potential (together with the symmetric box).

29. Both solutions have the form $A\cos kx + B\sin kx$ with $k = \sqrt{E}$ (for $E \le 0$ only the zero solution exists, Section 11.2). The Wronskian $W = uw' - wu'$ has $W' = uw'' - wu'' = -Euw + Ewu = 0$, and $W(L) = 0$ because both vanish at L. So $W \equiv 0$. At a point x_0 with $u(x_0) \neq 0$, set $c = w(x_0)/u(x_0)$; then $\phi = w - cu$ has $\phi(x_0) = 0$ and, from $W(x_0) = 0$, $\phi'(x_0) = 0$. For $\phi = A\cos kx + B\sin kx$, these two conditions read $A\cos kx_0 + B\sin kx_0 = 0$ and $-A\sin kx_0 + B\cos kx_0 = 0$; multiplying the first by $\cos kx_0$, the second by $-\sin kx_0$ and adding gives $A = 0$, and similarly $B = 0$. So $\phi \equiv 0$ and $w = cu$, without needing the general uniqueness theorem.

31. $\psi_n^2 = \frac{1}{L}\cos^2\frac{n\pi x}{2L} = \frac{1}{2L}(1 + \cos\frac{n\pi x}{L})$. Integrating over $[-L/2, L/2]$: $\frac{1}{2} + \frac{1}{2L}\cdot\frac{L}{n\pi}[\sin\frac{n\pi}{2} - \sin(-\frac{n\pi}{2})] = \frac{1}{2} + \frac{\sin(n\pi/2)}{n\pi}$. For $n = 1$: 0.818; $n = 3$: 0.394; $n = 5$: 0.564. The correction alternates in sign and shrinks like $1/n$, so the probability tends to $\frac{1}{2}$, the classical value for a particle bouncing at constant speed: the correspondence principle.



Quantized orbits in phase space: the curves $p^2 + V(x) = E_n$ for the Wu–Sprung potential, one for each of the first 30 levels.

Chapter 12

The Semiclassical Approximation: WKB and Counting Levels

↑Contents

What this chapter is about

In Chapter 11 you solved the time-independent Schrödinger equation exactly for two potentials: the infinite square well and the harmonic oscillator. They are almost the only one-dimensional potentials with exact formulas for every level. The potential at the center of the paper, the Wu–Sprung potential, is not one of them. It is defined only implicitly, through Eq. (2) of the paper [1], and nobody can write its eigenfunctions in closed form. Yet the paper needs its first 500 to 1000 levels. It also needs to know how each of those levels responds when the potential is changed.

The good news is that high levels are exactly where a powerful approximation works. When a wavefunction oscillates many times across a well, the potential barely changes over one wavelength. Locally the wave then behaves like the wave of a free particle, with a wavelength that depends on position. This idea is called the WKB approximation, after Wentzel, Kramers and Brillouin. It gives us four tools: a formula for the wavefunction; the Bohr–Sommerfeld rule, which predicts the levels from a single integral; a counting function, which says how many levels lie below any energy; and, by running the counting function backwards, a way to build a potential whose levels have a prescribed average density. That last step is called Abel inversion.

Where this shows up in the paper. The Wu–Sprung potential of Eq. (2) is the Abel inversion of the smooth zero-counting function (Sec. II B). The excess error at low n that Sec. II B calls "the known semiclassical weakness of the construction" is the WKB error that you will measure in Section 12.8. The sentence "position maps to level index through the classical turning point" in the interpolation-split paragraph of Sec. III (and Fig. 4) is the turning-point picture of Section 12.9. Finally, the coverage law of Sec. III and the remark in Sec. V that the prime-2 period spans 6–7 levels are both read in this book through WKB ideas: the local wavenumber for the coverage law and the mean level spacing for the prime-2 period.

What you need

- Chapter 10: the time-independent Schrödinger equation $H\psi = E\psi$ with $H = -\frac{d^2}{dx^2} + V(x)$ in the units $\hbar = 2m = 1$; normalization $\int |\psi|^2 \, dx = 1$ and the Born rule (probability density $|\psi|^2$).
- Chapter 11: the infinite well on $[-L, L]$ with levels $(k\pi/2L)^2$; the harmonic oscillator levels; the n-th level has $n-1$ nodes; even and odd states in an even potential; exponential decay in classically forbidden regions.
- Chapter 2: $e^{i\theta} = \cos\theta + i\sin\theta$.
- Chapter 4: the notation $\sim$ and $\mathcal{O}$; inverting a monotone function; bracketed root finding (bisection, Brent).
- First-year physics: energy conservation, potential-energy curves and classical turning points; waves on a string, wavelength λ and wavenumber $2\pi/\lambda$.
- Calculus: substitution (including trigonometric substitution), integration by parts, the fundamental theorem of calculus.

12.1 Why we need an approximation

★ Did you know? *WKB Has More Letters*

The WKB approximation is named after Gregor Wentzel, Hendrik Kramers and Léon Brillouin, who applied it to the new Schrödinger equation in 1926. But Harold Jeffreys had found a general version in 1923, and Joseph Liouville and George Green had used similar ideas in 1837. Some books therefore write JWKB, and mathematicians often call it the Liouville-Green approximation.

Picture a wave travelling along a rope whose mass per unit length changes gradually from one end to the other. You learned in first-year physics that the wave speed on a string is $v = \sqrt{\tau/\mu}$, where τ is the tension and μ the mass per unit length. If μ changes slowly, the wave adapts as it goes. Where the rope is heavy the wave slows down, its wavelength shortens and its amplitude changes, yet hardly any of the wave is reflected. Over any short stretch the rope looks uniform, so locally the wave looks like a plain sinusoid.

A quantum particle in a slowly varying potential behaves the same way. In a region where V is constant, the time-independent Schrödinger equation in our units,

$$-\psi''(x) + V\psi(x) = E\psi(x), \tag{12.1}$$

becomes $\psi'' = -(E-V)\psi$. When $E > V$ its solutions are $\cos(kx)$ and $\sin(kx)$ with $k = \sqrt{E-V}$. (Here a prime means d/dx.) If V changes slowly with x, it is natural to guess that the solution is still a wave, now with a *local* wavenumber $k(x) = \sqrt{E - V(x)}$. This chapter makes that guess precise, finds out when it can be trusted, and extracts its consequences.

Why should the guess work for the levels the paper cares about? Recall from Chapter 11 that the n-th bound state has $n-1$ nodes. Consecutive nodes of a wave are half a wavelength apart, so level $n = 500$ fits roughly 250 wavelengths into the well. Over one of those wavelengths the potential changes by only a tiny fraction of $E - V$. The approximation is therefore at its best for high levels and at its worst for the lowest ones. We will see this quantitatively in Section 12.8.

Units and the meaning of "semiclassical". In ordinary units the Schrödinger equation reads $-\frac{\hbar^2}{2m}\psi'' + V\psi = E\psi$. Dividing by $\hbar^2/2m$ shows that the local wavenumber is

$$k(x) = \frac{\sqrt{2m\,(E - V(x))}}{\hbar}. \tag{12.2}$$

In the paper's units, $\hbar = 1$ and $2m = 1$, so $\hbar^2/2m = 1$ and $k(x) = \sqrt{E - V(x)}$. The wavelength $2\pi/k$ is short when $\hbar$ is small or when $E - V$ is large. The WKB method is an expansion in powers of $\hbar$, and its leading term reproduces classical mechanics. That is why it is called *semiclassical*. In units where $\hbar = 1$ there is no small constant left to adjust; the role of the small parameter is played by the reciprocal of the number of wavelengths in the well.

Key Ideas

- In a slowly varying potential the wavefunction looks locally like a free wave with local wavenumber $k(x) = \sqrt{E - V(x)}$ (units $\hbar = 2m = 1$).
- The approximation improves as the number of wavelengths in the well grows, so it is best for high levels.
- "Semiclassical" means: leading order in an expansion in $\hbar$, which reproduces classical motion.

12.2 Classical motion, turning points and the local wavenumber

Start with the classical particle. In the units $\hbar = 2m = 1$ the mass is $m = \frac{1}{2}$, so the kinetic energy $p^2/2m$ equals p^2. Energy conservation reads

$$E = p^2 + V(x), \tag{12.3}$$

where p is the momentum. Solving for p gives

$$p(x) = \pm\sqrt{E - V(x)}. \tag{12.4}$$

The velocity is $v = p/m = 2p$, because $m = \frac{1}{2}$.

Allowed and forbidden regions. The square root in Equation (12.4) is real only where $E \geq V(x)$. That set is the *classically allowed region*. Where $V(x) > E$ the particle would need negative kinetic energy, so classically it can never be there; this is the *classically forbidden region*. The points where $V(x) = E$ separate the two. At such a point the momentum vanishes, the particle stops and reverses, and so these points are called *turning points*. You met them in first-year physics on potential-energy curves (Figure 12.1).

Local wavenumber. The de Broglie relation (Chapter 10) says that a particle of momentum p has wavelength $\lambda = 2\pi\hbar/p$. With $\hbar = 1$, the wavenumber $2\pi/\lambda$ equals p. This motivates the definition

$$k(x) = \sqrt{E - V(x)} \qquad \text{(allowed region)}, \qquad \lambda(x) = \frac{2\pi}{k(x)}. \tag{12.5}$$

In the forbidden region we define instead the *local decay constant*

$$\kappa(x) = \sqrt{V(x) - E} \qquad \text{(forbidden region)}. \tag{12.6}$$

At a turning point both k and κ vanish, and the local wavelength becomes infinite. That last fact is the warning sign: near a turning point the wavelength is not short compared with anything, and the local-wave picture must fail there.

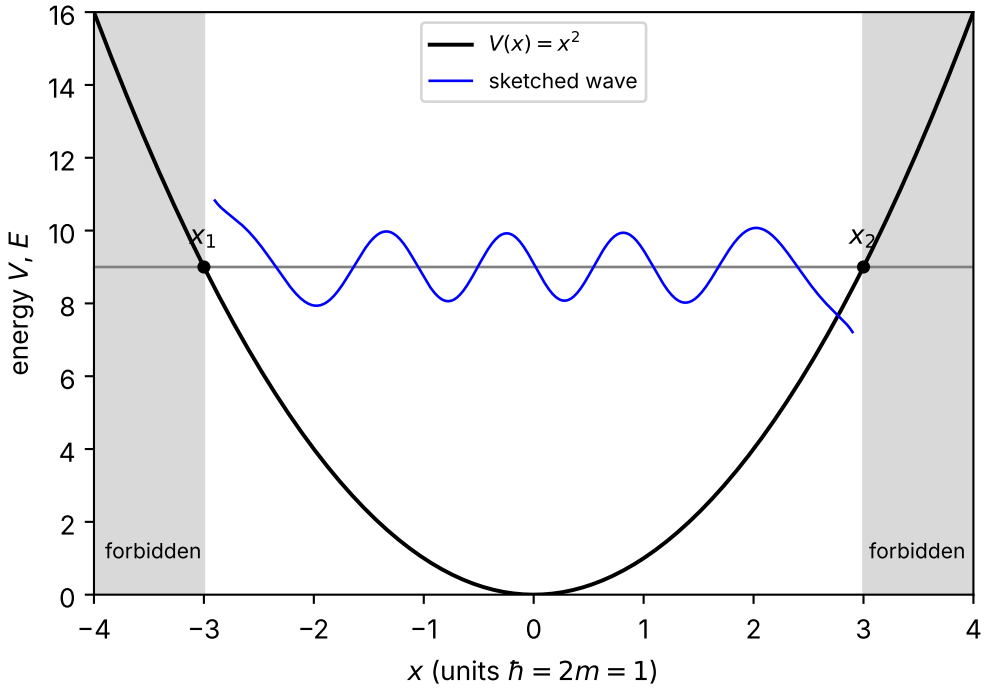


Figure 12.1: A potential-energy curve $V(x)$ (solid) and an energy E (horizontal line). The classically allowed region lies between the turning points x_1 and x_2, where $V = E$; the shaded regions are classically forbidden. The sketched wave, drawn on top of the energy line, has a short wavelength where $E - V$ is large (near the bottom of the well) and a long wavelength and large amplitude near the turning points, where the classical particle moves slowly. Units $\hbar = 2m = 1$.

Sample Problem 12.1: Local wavelength in a linear well

A particle moves in the linear ("V-shaped") well $V(x) = |x|$ with energy $E = 10$, in units $\hbar = 2m = 1$. Find the turning points and the local wavenumber and wavelength at $x = 0$, $x = 5$ and $x = 9.9$.
Step 1: Turning points. Set $V(x) = E$: $|x| = 10$, so $x = \pm 10$.
Step 2: At $x = 0$. $k = \sqrt{10 - 0} = 3.162$ and $\lambda = 2\pi/3.162 = 1.987$.
Step 3: At $x = 5$. $k = \sqrt{10 - 5} = \sqrt{5} = 2.236$ and $\lambda = 2\pi/2.236 = 2.810$.
Step 4: At $x = 9.9$. $k = \sqrt{0.1} = 0.316$ and $\lambda = 2\pi/0.316 = 19.87$.
Step 5: Interpret. At $x = 9.9$ the "local wavelength" is about 200 times the distance 0.1 to the turning point. Calling the potential locally constant over one wavelength is meaningless there.

$$\boxed{x = \pm 10;\ \lambda(0) = 1.987,\ \lambda(5) = 2.810,\ \lambda(9.9) = 19.87}$$

Checkpoint 12.1

A particle in the even well $V(x) = x^2$ has energy $E = 9$. At which point of the allowed region is its local wavelength shortest?

Answer: At $x = 0$, where $E - V$ is largest: $k(0) = 3$ and $\lambda(0) = 2\pi/3$. The wavelength grows without bound as $x \to \pm 3$.

Key Ideas

- In units $\hbar = 2m = 1$: $E = p^2 + V$, velocity $v = 2p$, and the local wavenumber is $k(x) = \sqrt{E - V(x)}$.
- Turning points satisfy $V(x) = E$. There $k = 0$ and the local wavelength is infinite.
- In forbidden regions the analogous quantity is the decay constant $\kappa = \sqrt{V - E}$.

12.3 The WKB wavefunction

We now make the "local wave" guess precise. In the allowed region we write Equation (12.1) as

$$\psi'' + k(x)^2 \psi = 0, \qquad k(x)^2 = E - V(x). \tag{12.7}$$

This is exact: we have only moved $V\psi$ and $E\psi$ to one side.

The amplitude and phase form. We look for solutions of the form of a real amplitude $A(x) > 0$ times a phase factor:

$$\psi(x) = A(x)\, e^{iS(x)}, \tag{12.8}$$

with A and S real functions. Such solutions never vanish, so they describe travelling waves; real standing-wave solutions, which have nodes, will be built below by adding two of them. (In this chapter only, $A(x)$ and $S(x)$ denote the WKB amplitude and phase; they are unrelated to the matrix A_{nk} and to the fluctuation $S(t)$ used elsewhere in the book.) For a free particle, A is constant and $S = kx$. We now compute the derivatives one at a time. By the product rule

and the chain rule,

$$\psi' = A'e^{iS} + iAS'e^{iS}, \tag{12.9}$$

$$\psi'' = A''e^{iS} + iA'S'e^{iS} + iA'S'e^{iS} + iAS''e^{iS} + (iS')(iAS')e^{iS} \tag{12.10}$$

$$= (A'' + 2iA'S' + iAS'' - AS'^2)e^{iS}. \quad \text{(using } i^2 = -1\text{)} \tag{12.11}$$

Substituting into Equation (12.7) gives

$$(A'' + 2iA'S' + iAS'' - AS'^2 + k^2A)e^{iS} = 0. \tag{12.12}$$

The factor e^{iS} has modulus 1 and is never zero, so we may divide by it:

$$A'' - AS'^2 + k^2A + i(2A'S' + AS'') = 0. \tag{12.13}$$

A complex number is zero only if its real and imaginary parts both vanish (Chapter 2). Since A and S are real, this gives two real equations:

$$\text{real part:} \quad A'' - AS'^2 + k^2A = 0, \tag{12.14}$$

$$\text{imaginary part:} \quad 2A'S' + AS'' = 0. \tag{12.15}$$

So far nothing has been approximated.

The amplitude equation is solved exactly. Multiply Equation (12.15) by A:

$$2AA'S' + A^2S'' = 0. \tag{12.16}$$

The left side is exactly the derivative of A^2S', because $(A^2S')' = 2AA'S' + A^2S''$. Hence $(A^2S')' = 0$, so A^2S' is a constant, which we call C^2. Taking $S' > 0$ for now,

$$A(x) = \frac{C}{\sqrt{S'(x)}}. \tag{12.17}$$

This has a physical meaning. The *probability current* $j = (\hbar/m)\,\mathrm{Im}(\psi^*\psi')$ measures the rate at which probability flows past a point. (We take this definition on trust here; it follows from the time-dependent Schrödinger equation.) With $\hbar = 1$ and $m = \frac{1}{2}$ it is $j = 2\,\mathrm{Im}(\psi^*\psi')$. For the form Equation (12.8), $\psi^*\psi' = Ae^{-iS}(A' + iAS')e^{iS} = AA' + iA^2S'$, whose imaginary part is A^2S'. So $j = 2A^2S'$, and Equation (12.17) says that the probability current is the same at every x, as it must be in a steady state.

The one approximation. Divide Equation (12.14) by A and rearrange:

$$S'^2 = k^2 + \frac{A''}{A}. \tag{12.18}$$

This is still exact. The WKB approximation consists of neglecting A''/A compared with k^2:

$$\left|\frac{A''}{A}\right| \ll k^2 \quad \Longrightarrow \quad S' \approx \pm k(x), \qquad S(x) \approx \pm \int^x k(x')\,\mathrm{d}x'. \tag{12.19}$$

Inserting $S' = \pm k$ into Equation (12.17) gives $A = C/\sqrt{k}$; the sign of S' does not affect A, because we take $C^2/\,|S'|$. The two WKB solutions are therefore

$$\psi_\pm(x) \approx \frac{C}{\sqrt{k(x)}} \exp\Bigl(\pm i \int^x k(x')\,\mathrm{d}x'\Bigr). \tag{12.20}$$

Bound-state wavefunctions in one dimension can be chosen real (Chapter 11). Because Equation (12.7) is linear, $\frac{1}{2}e^{i\varphi}\psi_+ + \frac{1}{2}e^{-i\varphi}\psi_-$ is also a solution, for any real φ. Writing $\theta(x) = \int^x k(x')\,\mathrm{d}x'$, it equals

$$\frac{C}{\sqrt{k}}\cdot\frac{1}{2}\Big(e^{i(\theta+\varphi)} + e^{-i(\theta+\varphi)}\Big) = \frac{C}{\sqrt{k}}\cos(\theta+\varphi),$$

by $\cos\alpha = (e^{i\alpha}+e^{-i\alpha})/2$ (Chapter 2). This is the real WKB solution

$$\psi(x) \approx \frac{C}{\sqrt{k(x)}}\cos\Big(\int^x k(x')\,\mathrm{d}x' + \varphi\Big), \tag{12.21}$$

with a real amplitude C and phase φ. Two features matter. The phase advances by $k\,\mathrm{d}x$ over a distance $\mathrm{d}x$, exactly as for a free wave of wavenumber k. And the amplitude is proportional to $k^{-1/2}$, so it is large where the classical particle moves slowly. A slow classical particle spends more time in a region, so finding it there is more probable.

When is the neglect justified? We now turn the condition $|A''/A| \ll k^2$ into a condition on V. With $A = Ck^{-1/2}$,

$$A' = -\tfrac{1}{2}Ck^{-3/2}k', \tag{12.22}$$

$$A'' = \tfrac{3}{4}Ck^{-5/2}k'^2 - \tfrac{1}{2}Ck^{-3/2}k'', \tag{12.23}$$

$$\frac{A''}{A} = \frac{3}{4}\Big(\frac{k'}{k}\Big)^2 - \frac{1}{2}\frac{k''}{k}. \tag{12.24}$$

Dividing by k^2, the condition becomes

$$\frac{3}{4}\Big(\frac{k'}{k^2}\Big)^2 \ll 1 \quad\text{and}\quad \frac{1}{2}\left|\frac{k''}{k^3}\right| \ll 1. \tag{12.25}$$

The first of these says that $|k'|/k^2 \ll 1$. Since $\lambda = 2\pi/k$ gives $\lambda' = -2\pi k'/k^2$, this is the same as $|\lambda'| \ll 2\pi$. Over one wavelength the wavelength changes by about $|\lambda'|\,\lambda$, so, up to the factor 2π, the condition says that this change is small compared with λ itself. To express it through V, differentiate $k = \sqrt{E-V}$, which gives $k' = -V'/(2k)$. Then

$$\frac{|k'|}{k^2} = \frac{|V'(x)|}{2\,(E-V(x))^{3/2}} \ll 1. \tag{12.26}$$

The condition on k'' can also be written in terms of V. Differentiate $k' = -V'/(2k)$ once more with the quotient rule:

$$k'' = -\frac{V''}{2k} + \frac{V'k'}{2k^2} = -\frac{V''}{2k} - \frac{V'^2}{4k^3}, \tag{12.27}$$

where the last step inserts $k' = -V'/(2k)$. Dividing by k^3 and using $|a+b| \le |a| + |b|$,

$$\frac{1}{2}\left|\frac{k''}{k^3}\right| \le \frac{|V''|}{4k^4} + \frac{V'^2}{8k^6} = \frac{|V''|}{4(E-V)^2} + \frac{1}{2}\Big(\frac{|V'|}{2(E-V)^{3/2}}\Big)^2. \tag{12.28}$$

The second term is small whenever Equation (12.26) holds. The first term adds one requirement, $|V''| \ll 4(E-V)^2$: the curvature of V must be small. For smooth wells at high energy it holds wherever Equation (12.26) holds comfortably. At low energy it can fail even where $V' = 0$: for $V = x^2$ at $x = 0$, where $V'' = 2$ and $E - V = E$, it requires $E^2 \gg \frac{1}{2}$. Like Equation (12.26), it fails near turning points. (The kink of $V = |x|$ at $x = 0$, where V'' is not defined, is a separate issue, discussed in the answer to Problem 6.) We treat it as part of the assumption that V is smooth, and use Equation (12.26) as the practical test. It fails near every turning point, because the denominator vanishes there while V' does not.

Forbidden regions. Where $V > E$, Equation (12.1) reads $\psi'' = \kappa^2\psi$. Repeating every step above with $\psi = Ae^{\pm\int\kappa}$ (or simply replacing k by $\mp i\kappa$) gives

$$\psi(x) \approx \frac{D}{\sqrt{\kappa(x)}}\exp\Big(\pm\int^{x}\kappa(x')\,\mathrm{d}x'\Big), \tag{12.29}$$

valid when $|V'|/(2(V-E)^{3/2}) \ll 1$. For a bound state only the solution that decays away from the well is acceptable, as in Chapter 11.

Sample Problem 12.2: Where WKB can be trusted in the linear well

For $V = |x|$ and $E = 10$, evaluate the validity parameter $|V'|/(2(E-V)^{3/2})$ of Equation (12.26) at $x = 0$, 5 and 9.9. Then find the region $x \geq 0$ in which it is below 0.1.

Step 1: The derivative. For $x > 0$, $V' = 1$. (At $x = 0$ we use the one-sided value $|V'| = 1$; the kink at the origin is discussed in Problem 6 and its answer.)

Step 2: Evaluate. At $x = 0$: $1/(2\cdot 10^{3/2}) = 0.0158$. At $x = 5$: $1/(2\cdot 5^{3/2}) = 0.0447$. At $x = 9.9$: $1/(2\cdot 0.1^{3/2}) = 15.8$.

Step 3: Region where the parameter is below 0.1**.** We need $2(10-x)^{3/2} > 10$, that is $(10-x)^{3/2} > 5$. Raising both sides to the power $2/3$ gives $10 - x > 5^{2/3} = 2.924$, so $x < 7.08$.

Step 4: Interpret. The "bad zone" $7.08 < x < 10$ has width $5^{2/3} = 2.924$. This width does not depend on E, while the well half-width equals E. The fraction of the well where WKB is poor therefore shrinks like $1/E$ as the energy rises.

$$\boxed{0.0158,\ 0.0447,\ 15.8;\quad \text{parameter} < 0.1 \text{ for } 0 \leq x < 7.08}$$

Pitfall

The WKB amplitude $C/\sqrt{k}$ diverges at a turning point. That divergence is not physical. The true wavefunction is finite everywhere; the formula has simply been used where its own validity condition Equation (12.26) fails. Never evaluate Equation (12.20) or Equation (12.29) at, or very near, a turning point.

Checkpoint 12.2

In which part of the allowed region is the WKB amplitude largest, and why does this make classical sense?

Answer: Near the turning points, where k is smallest. The classical particle moves slowly there ($v = 2k$), so it spends the most time there, and the probability of finding it is highest.

Key Ideas

- Writing $\psi = Ae^{iS}$ splits the Schrödinger equation exactly into $A^2S' = \text{const}$ (constant current) and $S'^2 = k^2 + A''/A$.
- WKB neglects A''/A, which gives $\psi \approx Ck^{-1/2}\cos(\int k\,\mathrm{d}x + \varphi)$ in allowed regions and $D\kappa^{-1/2}e^{\pm\int\kappa\,\mathrm{d}x}$ in forbidden ones (Equations (12.21) and (12.29)).
- Validity: $|V'|/(2|E-V|^{3/2}) \ll 1$ (Equation (12.26)). This fails near turning points.

12.4 Connection formulas at a turning point

The WKB solutions on the two sides of a turning point are separately fine, but they break down in between. To build a bound state we must know which oscillating solution inside the well joins onto the decaying solution outside. The answer is a *connection formula*. Its one surprising ingredient is a phase of $\pi/4$.

Zooming in on the turning point. Let $x = b$ be a turning point with the forbidden region to its right, so $V(x) > E$ for x just above b. Close to b, a smooth potential is well approximated by its tangent line:

$$V(x) \approx E + F\,(x-b), \qquad F = V'(b) > 0. \tag{12.30}$$

(The neglected term is of order $V''(b)(x-b)^2$; this linearization is the approximation.) The Schrödinger equation $\psi'' = (V-E)\psi$ becomes

$$\psi'' = F\,(x-b)\,\psi. \tag{12.31}$$

Introduce the dimensionless variable $z = F^{1/3}(x-b)$. By the chain rule $d\psi/dx = F^{1/3}\,d\psi/dz$, so $\psi'' = F^{2/3}\,d^2\psi/dz^2$. On the right side, $F(x-b) = F\cdot F^{-1/3}z = F^{2/3}z$. Dividing both sides by $F^{2/3}$ gives the *Airy equation*

$$\frac{d^2\psi}{dz^2} = z\,\psi. \tag{12.32}$$

A second-order equation has two independent solutions. We take on trust (see the box below) that one of them, called $\mathrm{Ai}(z)$, decays as $z \to +\infty$ (deep into the forbidden region), and that every solution independent of it, such as the standard one called $\mathrm{Bi}(z)$, grows there. The bound state must therefore be proportional to Ai near b.

Taken on trust

Taken on trust: the decaying solution Ai and its asymptotic forms. For large positive argument the Airy function behaves as

$$\mathrm{Ai}(z) \approx \frac{1}{2\sqrt{\pi}}\,z^{-1/4}\exp\!\Big(-\tfrac{2}{3}z^{3/2}\Big), \qquad \mathrm{Ai}(-w) \approx \frac{1}{\sqrt{\pi}}\,w^{-1/4}\sin\!\Big(\tfrac{2}{3}w^{3/2}+\tfrac{\pi}{4}\Big), \tag{12.33}$$

for $z \to +\infty$ and $w \to +\infty$. Deriving these needs the integral representation of Ai and the method of steepest descents, which are beyond this book. A full treatment of the connection formulas is given in [30] (the chapter on the WKB approximation) and in [22]. You can test Equation (12.33) numerically with `scipy.special.airy`; Figure 12.2 is such a test.

Matching. Now we compare Equation (12.33) with the WKB forms. To the right of b, the linearized potential gives $\kappa = \sqrt{V-E} = F^{1/2}(x-b)^{1/2}$ and

$$\int_b^x \kappa\,dx' = F^{1/2}\cdot\tfrac{2}{3}(x-b)^{3/2} = \tfrac{2}{3}\big(F^{1/3}(x-b)\big)^{3/2} = \tfrac{2}{3}z^{3/2}. \tag{12.34}$$

Also $z^{-1/4} = F^{-1/12}(x-b)^{-1/4}$ and $\kappa^{-1/2} = F^{-1/4}(x-b)^{-1/4}$, so $z^{-1/4} = F^{1/6}\kappa^{-1/2}$. The first form in Equation (12.33) is therefore

$$\mathrm{Ai}(z) \approx \frac{F^{1/6}}{2\sqrt{\pi}}\,\frac{1}{\sqrt{\kappa}}\exp\!\Big(-\int_b^x \kappa\,dx'\Big), \tag{12.35}$$

which is exactly the decaying WKB solution Equation (12.29). To the left of b, put $w = -z = F^{1/3}(b-x) > 0$. Then $k = \sqrt{E-V} = F^{1/2}(b-x)^{1/2}$, and the same two steps give

$\int_x^b k\,\mathrm{d}x' = \frac{2}{3}w^{3/2}$ and $w^{-1/4} = F^{1/6}k^{-1/2}$. Finally, using $\sin\alpha = \cos(\pi/2-\alpha)$ and the evenness of cosine, $\sin(\theta+\pi/4) = \cos(\pi/2-\theta-\pi/4) = \cos(\pi/4-\theta) = \cos(\theta-\pi/4)$. The second form in Equation (12.33) becomes

$$\mathrm{Ai}(z) \approx \frac{F^{1/6}}{\sqrt{\pi}}\,\frac{1}{\sqrt{k}}\cos\Big(\int_x^b k\,\mathrm{d}x' - \frac{\pi}{4}\Big). \tag{12.36}$$

Both expressions are asymptotic forms of the same exact function $\mathrm{Ai}(z)$, valid at large $|z|$ on its two sides. The matching therefore holds in the asymptotic sense, not as an identity at every x: it tells us which WKB wave on the left belongs to which WKB wave on the right. Comparing the two prefactors, the oscillating side carries twice the amplitude of the decaying side. Writing $D = F^{1/6}/(2\sqrt{\pi})$, and then multiplying both sides by the same arbitrary constant (allowed because the Schrödinger equation is linear, so any multiple of a solution is a solution), we arrive at the connection formula for a turning point with the forbidden region on the right:

$$\boxed{\frac{2D}{\sqrt{k(x)}}\cos\Big(\int_x^b k\,\mathrm{d}x' - \frac{\pi}{4}\Big)\quad (x<b) \quad\longleftrightarrow\quad \frac{D}{\sqrt{\kappa(x)}}\exp\Big(-\int_b^x \kappa\,\mathrm{d}x'\Big)\quad (x>b).} \tag{12.37}$$

For a turning point a with the forbidden region on the left, the mirror-image argument (Problem 25) gives the same formula with $\int_x^b k$ replaced by $\int_a^x k$ and $\int_b^x \kappa$ replaced by $\int_x^a \kappa$.

When is the matching legitimate? The WKB forms hold only some distance from b, while the linearization Equation (12.30) holds only close to b. The argument needs an overlap zone in which both are true. The natural length scale of Equation (12.32) is $F^{-1/3}$ (the "Airy length"), and the asymptotic forms are good once $|z|$ is a few units. So the potential must stay close to its tangent line over a few Airy lengths on each side. For high levels in smooth wells this holds well. For the lowest levels it holds only roughly, which is one source of the WKB errors studied in Section 12.8.

Hard walls. The paper's box has Dirichlet walls at $x = \pm L$, where the wavefunction must vanish. A hard wall is not a smooth turning point, and its rule is simpler. If the wall is at $x = a$, then $\psi \approx Ck^{-1/2}\sin(\int_a^x k\,\mathrm{d}x')$ vanishes there. Since $\sin\theta = \cos(\theta-\pi/2)$, a hard wall behaves like a turning point with phase $\pi/2$ instead of $\pi/4$.

Pitfall

Use Equation (12.37) from right to left: require decay in the forbidden region, then read off the phase inside the well. The reverse direction is unsafe. A tiny error in the phase inside adds a small multiple of the *growing* exponential outside, and that piece eventually dominates.

Key Ideas

- Near a turning point the Schrödinger equation becomes the Airy equation $\mathrm{d}^2\psi/\mathrm{d}z^2 = z\psi$ with $z = F^{1/3}(x-b)$.
- Requiring decay in the forbidden region fixes the inside wave to be $\propto k^{-1/2}\cos(\int_x^b k\,\mathrm{d}x' - \pi/4)$ (Equation (12.37)). A smooth turning point contributes phase $\pi/4$; a hard wall contributes $\pi/2$.
- The asymptotic forms of the Airy function are taken on trust; the matching argument itself is shown in full.

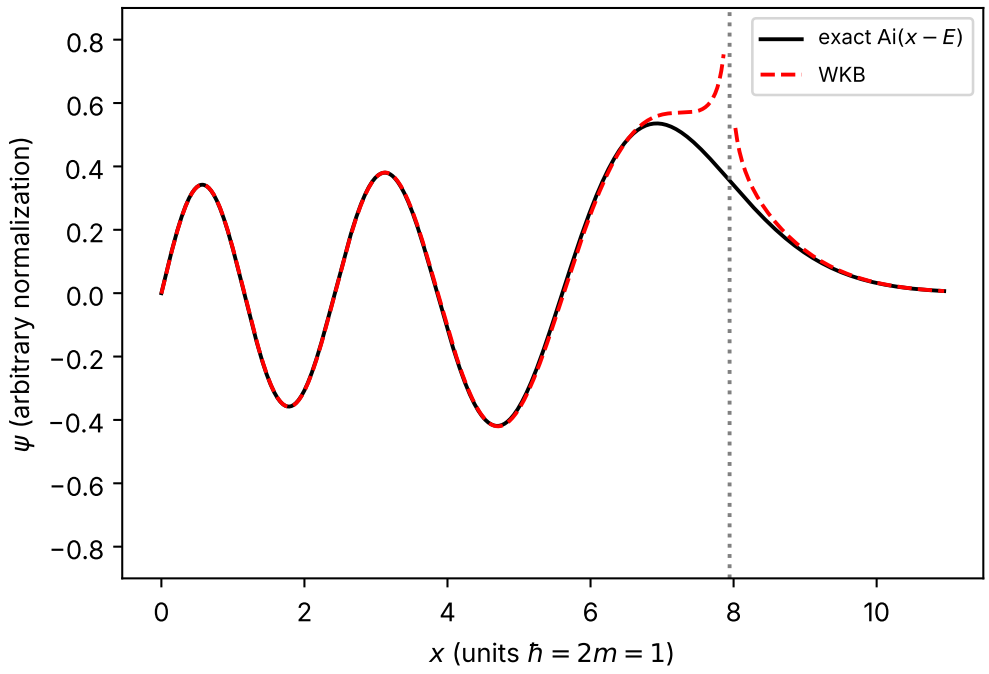


Figure 12.2: Exact eigenfunction (solid) of the linear well $V = |x|$ for the level $n = 10$, which is odd, plotted for $x \geq 0$, against its WKB approximation (dashed) built from the connection formula Equation (12.37). The exact function is $\mathrm{Ai}(x - E)$ with E the fifth zero of Ai (sign reversed). Away from the turning point $x = E$ (dotted line) the two curves are hard to tell apart; at the turning point the WKB amplitude diverges while the exact function stays finite. Units $\hbar = 2m = 1$; arbitrary normalization.

12.5 Bohr–Sommerfeld quantization

Now we put two turning points together and obtain the rule that predicts the energy levels.

Consider a well with turning points $a < b$ at energy E, the regions $x < a$ and $x > b$ being forbidden. Define the *phase integral*

$$\Phi(E) = \int_a^b \sqrt{E - V(x)}\,\mathrm{d}x = \int_a^b k(x)\,\mathrm{d}x, \tag{12.38}$$

which is the total phase that the local wave accumulates in crossing the well. Inside the well, let $\theta(x) = \int_a^x k\,\mathrm{d}x'$; it rises from 0 at $x = a$ to Φ at $x = b$.

Step 1: Two descriptions of one wavefunction. Decay to the right of b, through Equation (12.37), forces

$$\psi(x) \approx \frac{2D}{\sqrt{k}} \cos\Big(\int_x^b k\,\mathrm{d}x' - \frac{\pi}{4}\Big) = \frac{2D}{\sqrt{k}} \cos\Big(\Phi - \theta(x) - \frac{\pi}{4}\Big), \tag{12.39}$$

where we used $\int_x^b = \int_a^b - \int_a^x$. Decay to the left of a, through the mirror-image formula, forces

$$\psi(x) \approx \frac{2D'}{\sqrt{k}} \cos\Big(\theta(x) - \frac{\pi}{4}\Big). \tag{12.40}$$

Both must hold at every x inside the well.

Step 2: Rewrite the first. Cosine is even, so $\cos(\Phi - \theta - \pi/4) = \cos(\theta + \pi/4 - \Phi)$. Write the argument as $\alpha - \beta$ with

$$\alpha = \theta - \frac{\pi}{4}, \qquad \beta = \Phi - \frac{\pi}{2}. \tag{12.41}$$

(Check: $\alpha - \beta = \theta - \pi/4 - \Phi + \pi/2 = \theta + \pi/4 - \Phi$.)

Step 3: Demand proportionality. We need $D\cos(\alpha - \beta) = D'\cos\alpha$ for every value of α in the range covered as x crosses the well. Expand the left side: $D\cos(\alpha - \beta) = D\cos\beta\cos\alpha + D\sin\beta\sin\alpha$. The functions $\cos\alpha$ and $\sin\alpha$ are not proportional to each other, so the coefficient of $\sin\alpha$ must vanish: $\sin\beta = 0$, that is $\beta = j\pi$ for some integer j. (This also needs the well to be many wavelengths wide, so that α really does sweep through a range of values; for the lowest level this holds only roughly.)

Step 4: Which integers? Since $\Phi > 0$, $\beta = \Phi - \pi/2 > -\pi/2$, so $j \geq 0$. Writing $j = n - 1$ with $n = 1, 2, 3, \ldots$ gives $\Phi - \pi/2 = (n-1)\pi$, which is

$$\boxed{\Phi(E_n) = \int_a^b \sqrt{E_n - V(x)}\,\mathrm{d}x = \Big(n - \frac{1}{2}\Big)\pi, \qquad n = 1, 2, 3, \ldots} \tag{12.42}$$

This is the *Bohr–Sommerfeld quantization rule*, with the $\frac{1}{2}$ supplied by the two $\pi/4$ phases. Then $\cos(\alpha - (n-1)\pi) = (-1)^{n-1}\cos\alpha$, so $D' = (-1)^{n-1}D$.

Node count. The WKB wavefunction $\cos(\theta - \pi/4)$ vanishes where $\theta - \pi/4 = \pi/2 + j\pi$, that is $\theta = 3\pi/4 + j\pi$ with $j = 0, 1, 2, \ldots$. These lie inside the well when $3\pi/4 + j\pi < (n - \frac{1}{2})\pi$, that is $j < n - \frac{5}{4}$, so $j = 0, 1, \ldots, n-2$. That makes $n - 1$ nodes, in agreement with Chapter 11. The integer n in Equation (12.42) is therefore the level index, starting at $n = 1$ for the ground state, as in the paper.

General rule with walls. Each smooth turning point contributes $\pi/4$ and each hard wall $\pi/2$. Repeat Steps 1–4 with phase φ_a at a and φ_b at b. The two descriptions of the wavefunction are now proportional to $\cos(\Phi-\theta-\varphi_b)$ and $\cos(\theta-\varphi_a)$. Cosine is even, so the first equals $\cos(\theta+\varphi_b-\Phi)=\cos(\alpha-\beta)$ with $\alpha=\theta-\varphi_a$ and $\beta=\Phi-\varphi_a-\varphi_b$ (check: $\alpha-\beta=\theta+\varphi_b-\Phi$). Step 3 again forces $\sin\beta=0$. Since $\Phi>0$ and $\varphi_a+\varphi_b\le\pi$, we have $\beta>-\pi$, so Step 4 gives $\beta=(n-1)\pi$ with $n\ge 1$, that is $\Phi-\varphi_a-\varphi_b=(n-1)\pi$:

$$\Phi(E_n)=(n-1)\pi+\varphi_a+\varphi_b=\begin{cases}(n-\frac{1}{2})\pi & \text{two smooth turning points,}\\ (n-\frac{1}{4})\pi & \text{one smooth turning point, one hard wall,}\\ n\pi & \text{two hard walls.}\end{cases} \tag{12.43}$$

Restoring $\hbar$. In ordinary units, $\int k\,\mathrm{d}x=\int p\,\mathrm{d}x/\hbar$, and Equation (12.42) becomes $\int_a^b p\,\mathrm{d}x=(n-\frac{1}{2})\pi\hbar$. Over a full back-and-forth orbit the integral doubles. On the return trip from b to a both p and $\mathrm{d}x$ are negative, so $p\,\mathrm{d}x$ is positive and the return contributes the same $\int_a^b p\,\mathrm{d}x$ again: $\oint p\,\mathrm{d}x=2\pi\hbar\,(n-\frac{1}{2})$. This is the form usually found in textbooks.

Sample Problem 12.3: The harmonic oscillator: Bohr–Sommerfeld is exact

For the oscillator $V(x)=\frac{1}{4}\omega^2x^2$ (which is $\frac{1}{2}m\omega^2x^2$ with $m=\frac{1}{2}$), find the Bohr–Sommerfeld levels and compare with the exact levels $(n-\frac{1}{2})\hbar\omega$ of Chapter 11.

Step 1: Turning points. $\frac{1}{4}\omega^2x^2=E$ gives $x=\pm x_E$ with $x_E=2\sqrt{E}/\omega$.

Step 2: Rewrite the integrand. $E-\frac{1}{4}\omega^2x^2=\frac{1}{4}\omega^2(x_E^2-x^2)$, so $\sqrt{E-V}=\frac{\omega}{2}\sqrt{x_E^2-x^2}$.

Step 3: Integrate. Substitute $x=x_E\sin u$, $\mathrm{d}x=x_E\cos u\,\mathrm{d}u$, with u from $-\pi/2$ to $\pi/2$:

$$\int_{-x_E}^{x_E}\sqrt{x_E^2-x^2}\,\mathrm{d}x=x_E^2\int_{-\pi/2}^{\pi/2}\cos^2u\,\mathrm{d}u=x_E^2\cdot\frac{\pi}{2}.$$

(The last integral uses $\cos^2u=\frac{1}{2}(1+\cos 2u)$.) Hence $\Phi(E)=\frac{\omega}{2}\cdot\frac{\pi x_E^2}{2}=\frac{\pi\omega}{4}\cdot\frac{4E}{\omega^2}=\frac{\pi E}{\omega}$.

Step 4: Quantize. $\pi E_n/\omega=(n-\frac{1}{2})\pi$ gives $E_n=(n-\frac{1}{2})\omega$. In our units $\hbar=1$, so this is $(n-\frac{1}{2})\hbar\omega$: identical to the exact result for every n. For $\omega=2$, that is $V=x^2$, the levels are $E_n=2n-1=1,3,5,\ldots$

$$\boxed{E_n=\left(n-\tfrac{1}{2}\right)\omega\quad\text{(exact for every } n)}$$

The oscillator is special: the Bohr–Sommerfeld energies happen to be exact there. (The WKB wavefunctions are still only approximate; they diverge at the turning points.) Do not expect this in general.

Sample Problem 12.4: The linear well: Bohr–Sommerfeld against the exact levels

Find the Bohr–Sommerfeld levels of $V(x)=|x|$ and compare them with the exact levels.

Step 1: Phase integral. The turning points are $\pm E$, and by symmetry

$$\Phi(E)=2\int_0^E\sqrt{E-x}\,\mathrm{d}x=2\Big[-\tfrac{2}{3}(E-x)^{3/2}\Big]_0^E=\tfrac{4}{3}E^{3/2}.$$

Step 2: Quantize. $\frac{4}{3}E_n^{3/2}=(n-\frac{1}{2})\pi$ gives

$$E_n^{\mathrm{WKB}}=\left[\tfrac{3\pi}{4}(n-\tfrac{1}{2})\right]^{2/3}.$$

Step 3: Exact levels. For $x > 0$ the equation is $\psi'' = (x - E)\psi$, so after the shift $z = x - E$ it is the Airy equation Equation (12.32) with $F = 1$. The solution decaying at large x is $\mathrm{Ai}(x - E)$. Even states need $\psi'(0) = 0$, so $\mathrm{Ai}'(-E) = 0$. Odd states need $\psi(0) = 0$, so $\mathrm{Ai}(-E) = 0$. The exact levels are therefore the zeros of Ai' and of Ai with their signs reversed, taken alternately (computed with `scipy.special.ai_zeros`).
Step 4: Compare. The local mean spacing follows from Step 2: $E^{3/2} = \frac{3\pi}{4}(n - \frac{1}{2})$, so $\frac{3}{2}E^{1/2}\,dE/dn = \frac{3\pi}{4}$ and $dE/dn = \pi/(2\sqrt{E})$. Table 12.1 lists the results. Relative to the exact level, the error falls from 9.5% at $n = 1$ (0.0967/1.0188) to below 0.1% by $n = 6$ (0.0034/5.5206 = 0.06%).

$$\boxed{E_n^{\mathrm{WKB}} = \left[\tfrac{3\pi}{4}(n - \tfrac{1}{2})\right]^{2/3};\ \text{error 0.065 spacings at } n = 1,\ 0.005 \text{ at } n = 6}$$

Table 12.1: Bohr–Sommerfeld levels of the linear well $V = |x|$ against the exact levels (zeros of Ai' for odd n, which are even states; zeros of Ai for even n, which are odd states). The last column divides the error by the local mean spacing $\pi/(2\sqrt{E_n^{\mathrm{WKB}}})$.

n	parity	E_n^{WKB}	E_n^{exact}	error	error / spacing
1	even	1.1155	1.0188	+0.0967	+0.065
2	odd	2.3203	2.3381	−0.0179	−0.017
3	even	3.2616	3.2482	+0.0134	+0.015
4	odd	4.0818	4.0879	−0.0061	−0.008
5	even	4.8263	4.8201	+0.0062	+0.009
6	odd	5.5172	5.5206	−0.0034	−0.005

```python
import numpy as np
from scipy.special import ai_zeros

a, ap, _, _ = ai_zeros(50)              # zeros of Ai and of Ai' (negative)
E_exact = np.sort(np.concatenate([-a, -ap]))   # levels of V = |x|
n = np.arange(1, 21)
E_wkb = (0.75*np.pi*(n - 0.5))**(2/3)          # Bohr-Sommerfeld
spacing = np.pi/(2*np.sqrt(E_wkb))             # dE/dn from WKB
err = (E_wkb - E_exact[:20])/spacing           # error in mean spacings
for k in range(6):
    print(n[k], round(E_wkb[k], 4), round(E_exact[k], 4), round(err[k], 3))
```

Sample Problem 12.5: The box: hard walls

Apply the hard-wall rule to the infinite well on $[-L, L]$ (running example RE3) with $L = 44.883$. Find E_1 and the number of levels with $E \le 1000$.
Step 1: Phase integral. Inside the box $V = 0$, so $k = \sqrt{E}$ and $\Phi = 2L\sqrt{E}$.
Step 2: Two hard walls. By Equation (12.43), $2L\sqrt{E_n} = n\pi$, so $E_n = (n\pi/2L)^2$. This is the exact result of Chapter 11. It is exact because k is constant, so $A = Ck^{-1/2}$ is constant, $A'' = 0$, and nothing was neglected in Equation (12.18).
Step 3: Numbers. $E_1 = (\pi/89.766)^2 = 1.225 \times 10^{-3}$. Levels with $E_n \le 1000$ need $n \le 2L\sqrt{1000}/\pi = 903.57$, so there are 903 of them.

$$\boxed{E_1 = 1.225 \times 10^{-3},\quad 903 \text{ levels below } E = 1000}$$

Checkpoint 12.3

A well consists of a hard wall at $x = 0$ and the smooth potential $V = x$ for $x > 0$. Which right-hand side of Equation (12.43) applies?

Answer: $(n - \frac{1}{4})\pi$: one hard wall (phase $\pi/2$) and one smooth turning point (phase $\pi/4$). Problem 7 works it out.

Key Ideas

- Requiring decay on both sides gives Bohr–Sommerfeld: $\int_a^b \sqrt{E_n - V}\,\mathrm{d}x = (n - \frac{1}{2})\pi$, $n = 1, 2, \ldots$ (Equation (12.42)).
- Each smooth turning point contributes $\pi/4$ and each hard wall $\pi/2$ (Equation (12.43)).
- The Bohr–Sommerfeld levels are exact for the box and for the oscillator. For the linear well it is accurate to a few percent of a spacing at low n, improving with n.

12.6 Phase-space area

★ Did you know? *Quantizing an Area*

Before Schrödinger's equation, the Bohr-Sommerfeld rule said allowed orbits enclose a phase-space area $\oint p\,dq = nh$. In 1916 Arnold Sommerfeld applied it to relativistic elliptical orbits and derived a formula for the fine structure of hydrogen's spectral lines. It agrees with the later result of the exact Dirac equation, a striking success for counting quantum states by area.

Classical mechanics offers a picture of Equation (12.42) that is easy to remember. The state of a classical particle on a line is a point (x, p) in the *phase plane*. As the particle moves, the point traces a closed curve, the *orbit*, on which $p^2 + V(x) = E$. The upper half of the orbit is $p = +\sqrt{E - V}$ (moving right) and the lower half is $p = -\sqrt{E - V}$ (moving left).

The area enclosed by the orbit is the area between the two halves:

$$\mathcal{A}(E) = \int_a^b \Big[\sqrt{E - V} - \big(-\sqrt{E - V}\big)\Big]\mathrm{d}x = 2\Phi(E). \tag{12.44}$$

The Bohr–Sommerfeld rule therefore says

$$\mathcal{A}(E_n) = 2\pi\Big(n - \frac{1}{2}\Big). \tag{12.45}$$

Consecutive quantized orbits enclose areas differing by exactly 2π. In ordinary units this is $2\pi\hbar = h$, Planck's constant. Each quantum state "occupies" one cell of area h in the phase plane, and the quantized orbit runs through the middle of its cell (Figure 12.3).

Sample Problem 12.6: Counting cells for the oscillator

For $V = x^2$, find the area enclosed by the orbit at $E = 5$ and the number of Planck cells it contains.

Step 1: The orbit. $p^2 + x^2 = 5$ is a circle of radius $\sqrt{5}$.

Step 2: The area. $\mathcal{A} = \pi(\sqrt{5})^2 = 5\pi = 15.71$.

Step 3: Cells. $\mathcal{A}/2\pi = 2.5 = n - \frac{1}{2}$ with $n = 3$. So $E = 5$ is the third level, in agreement with $E_n = 2n - 1$.

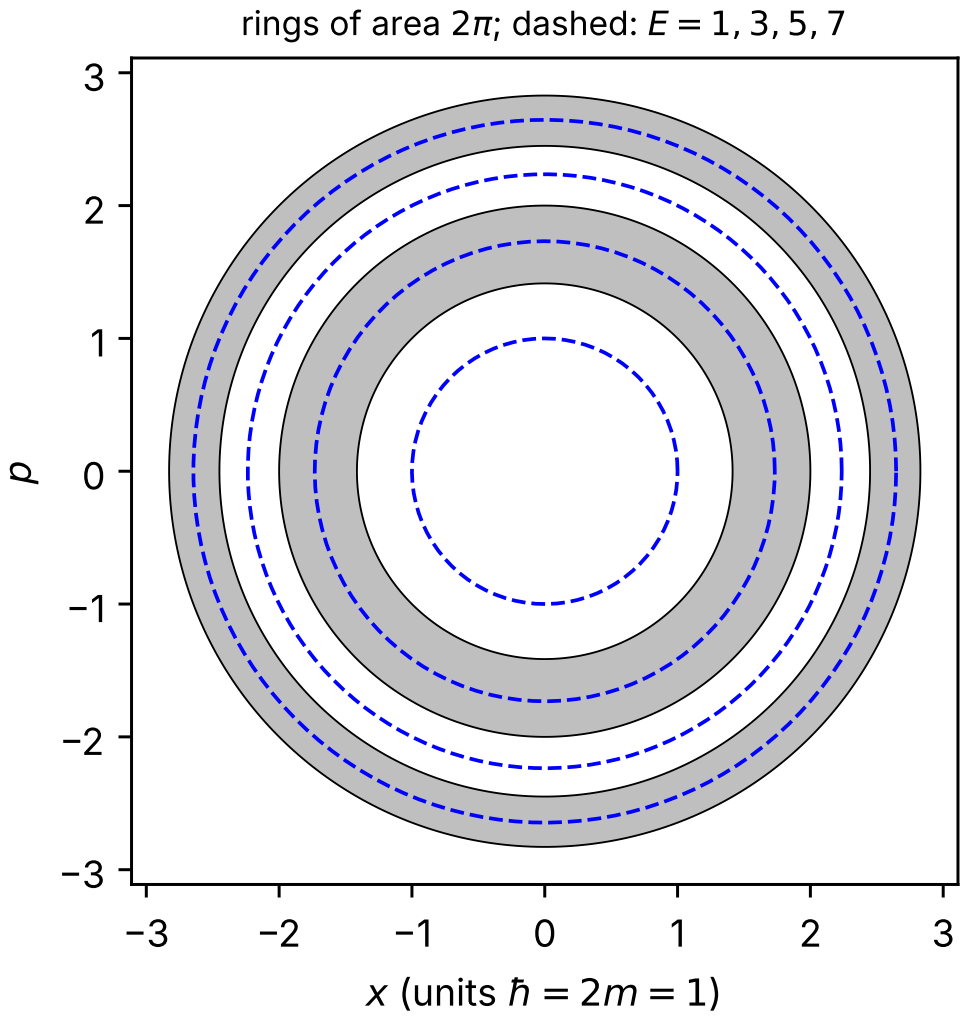


Figure 12.3: Phase plane of the oscillator $V = x^2$ (units $\hbar = 2m = 1$). Orbits $x^2 + p^2 = E$ are circles of radius $\sqrt{E}$. Alternately shaded rings, bounded by the circles $E = 0, 2, 4, 6, 8$, each have area 2π. The Bohr–Sommerfeld orbits (dashed), at $E = 1, 3, 5, 7$, sit in the middle of the rings: the orbit of level n encloses area $2\pi(n - \frac{1}{2})$.

$$\boxed{\mathcal{A} = 5\pi = 15.71, \quad 2.5 \text{ cells } (n = 3)}$$

Key Ideas

- The orbit at energy E encloses phase-space area $\mathcal{A} = 2\Phi(E)$ (Equation (12.44)).
- Bohr–Sommerfeld: $\mathcal{A}(E_n) = 2\pi(n - \frac{1}{2})$. Each state occupies an area $2\pi\hbar = h$.

12.7 Counting levels

The paper never needs a single level in isolation. It needs *how many* levels lie below a given energy, and how that number grows. This section turns Bohr–Sommerfeld into a counting function.

The staircase. Define $\mathcal{N}(E)$ as the number of levels E_n with $E_n \le E$. It is a staircase: 0 below E_1, and jumping by 1 at each level. (We use the script letter so as not to confuse it with the grid size N or with the zero-counting function $N(T)$ of later chapters.) Write $\lfloor y \rfloor$ for the *floor* of y, the largest integer not exceeding y; for example $\lfloor 3.7 \rfloor = 3$.

The function $\Phi(E)$ increases with E: raising E raises $E - V$ at every point and widens the allowed region. So by Equation (12.42), $E_n \le E$ exactly when $(n - \frac{1}{2})\pi \le \Phi(E)$, that is when $n \le \Phi(E)/\pi + \frac{1}{2}$. The number of positive integers n obeying this is the largest such integer:

$$\mathcal{N}_{\mathrm{WKB}}(E) = \Big\lfloor \frac{1}{\pi} \int_a^b \sqrt{E - V(x)}\, \mathrm{d}x + \frac{1}{2} \Big\rfloor. \tag{12.46}$$

This is the WKB counting function: $\frac{1}{\pi}\int \sqrt{E-V}\,\mathrm{d}x + \frac{1}{2}$, rounded down to an integer. (If $\Phi < \pi/2$ the floor is 0, as it should be.)

The smooth count. Write $y = \Phi(E)/\pi$. The staircase $\lfloor y + \frac{1}{2} \rfloor$ jumps by one at $y = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$. Between jumps it differs from y by a sawtooth that runs from $+\frac{1}{2}$ down to $-\frac{1}{2}$, and the sawtooth averages to zero. The smooth curve through the middle of the staircase is therefore

$$\boxed{\bar{\mathcal{N}}(E) = \frac{\Phi(E)}{\pi} = \frac{1}{\pi}\int_a^b \sqrt{E - V(x)}\, \mathrm{d}x, \qquad \bar{\mathcal{N}}(E_n) = n - \frac{1}{2}.} \tag{12.47}$$

At each level the smooth count equals $n - \frac{1}{2}$, the midpoint of the jump from $n-1$ to n (Figure 12.4). This is the same convention as the paper's smooth zeros, defined by $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$. We return to this match in Section 12.11.

Pitfall

Do not confuse the two expressions. $\Phi/\pi + \frac{1}{2}$, once rounded down, counts levels, and at a level it equals n. Φ/π is the smooth average, and at a level it equals $n - \frac{1}{2}$. When a potential is built to reproduce a smooth count, as in the paper, it is Φ/π that is matched.

12.7.1 A tool: differentiating an integral

To find how fast levels accumulate we must differentiate $\Phi(E)$, an integral whose integrand and limits both depend on E. Here is the rule, derived.

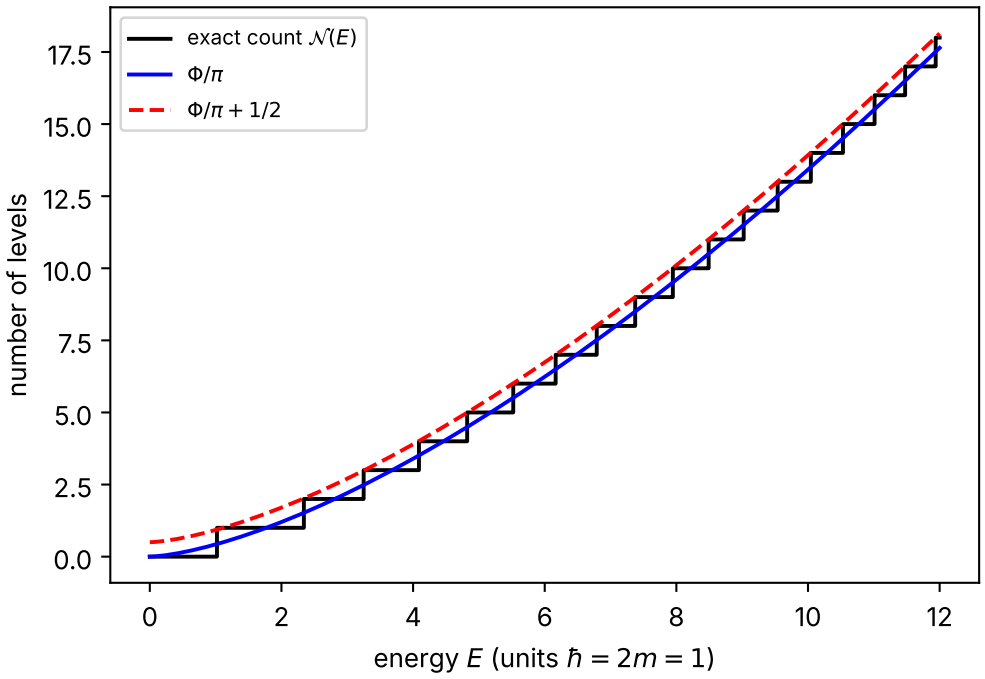


Figure 12.4: Counting levels in the linear well $V = |x|$ (units $\hbar = 2m = 1$). Solid staircase: the exact count $\mathcal{N}(E)$ from Airy zeros. Smooth curve: $\bar{\mathcal{N}}(E) = \Phi(E)/\pi = \frac{4}{3\pi}E^{3/2}$, which passes close to the midpoint of every step. Dashed: $\Phi/\pi + \frac{1}{2}$, whose integer part is the WKB count Equation (12.46).

Let $I(s) = \int_{a(s)}^{b(s)} G(s,u)\,\mathrm{d}u$, where s is a parameter and $a(s)$, $b(s)$ are general limits (below they will be the turning points). Write $\partial G/\partial s$ for the derivative of G with respect to s with u held fixed (a *partial derivative*). Change s by a small ϵ and split the new range of integration into three pieces:

$$I(s+\epsilon) - I(s) = \int_{a(s)}^{b(s)} [G(s+\epsilon,u) - G(s,u)]\,\mathrm{d}u + \int_{b(s)}^{b(s+\epsilon)} G(s+\epsilon,u)\,\mathrm{d}u - \int_{a(s)}^{a(s+\epsilon)} G(s+\epsilon,u)\,\mathrm{d}u. \tag{12.48}$$

In the first integral, $G(s+\epsilon,u) - G(s,u) \approx \epsilon\,\partial G/\partial s$. The second integral runs over a short interval of length $b(s+\epsilon) - b(s) \approx \epsilon\, b'(s)$, on which $G \approx G(s,b)$; similarly for the third. Dividing by ϵ and letting $\epsilon \to 0$ gives *Leibniz's rule*:

$$\frac{dI}{ds} = G(s,b)\,b'(s) - G(s,a)\,a'(s) + \int_a^b \frac{\partial G}{\partial s}\,\mathrm{d}u. \tag{12.49}$$

Taken on trust

What is taken on trust. Turning the "$\approx$" signs into an exact limit requires G and $\partial G/\partial s$ to be continuous on the closed region of integration; the careful proof is in [22]. Below we also apply Equation (12.49) when $\partial G/\partial s$ has an integrable square-root singularity at an endpoint. The result still holds there, but the proof needs a more careful limit argument that we do not give. We check the answers it produces against exactly solvable cases.

12.7.2 Density of states and the classical period

Apply Equation (12.49) to $\Phi(E) = \int_{a(E)}^{b(E)} \sqrt{E - V(x)}\,dx$. The integrand vanishes at both turning points, so the endpoint terms are zero. The derivative of the integrand with respect to E is $1/(2\sqrt{E-V})$. Hence

$$\Phi'(E) = \int_a^b \frac{dx}{2\sqrt{E - V(x)}} = \int_a^b \frac{dx}{2k(x)}. \tag{12.50}$$

Now compare with the classical motion. The velocity is $v = 2k$ (Section 12.2), so the time to cross from a to b is $\int_a^b dx/v = \int_a^b dx/(2k)$. The period T_{cl} of a full back-and-forth oscillation is twice that:

$$T_{\mathrm{cl}}(E) = \int_a^b \frac{dx}{k(x)} = 2\Phi'(E). \tag{12.51}$$

The *density of states*, the average number of levels per unit energy, is

$$\bar{\mathcal{N}}'(E) = \frac{\Phi'(E)}{\pi} = \frac{T_{\mathrm{cl}}(E)}{2\pi}, \tag{12.52}$$

and its reciprocal is the local mean spacing between levels:

$$\Delta E \approx \frac{1}{\bar{\mathcal{N}}'(E)} = \frac{2\pi}{T_{\mathrm{cl}}(E)}. \tag{12.53}$$

In ordinary units this reads $\Delta E = 2\pi\hbar/T_{\mathrm{cl}} = \hbar\omega_{\mathrm{cl}}$, where ω_{cl} is the classical angular frequency. This is Bohr's correspondence principle: high levels are spaced by $\hbar$ times the classical frequency.

Check with the oscillator $V = x^2$: here $\Phi(E) = \pi E/\omega$ with $\omega = 2$, so $\Phi' = \pi/2$ and $T_{\mathrm{cl}} = \pi$. A classical oscillator with angular frequency $\omega = 2$ has period $2\pi/\omega = \pi$, and the level spacing is $2\pi/T_{\mathrm{cl}} = 2$, which matches $E_n = 2n - 1$. *Check with the linear well*: differentiating $\Phi = \frac{4}{3}E^{3/2}$ directly gives $\Phi' = 2\sqrt{E}$. The integral form Equation (12.50) gives the same result:

$$\int_{-E}^{E} \frac{dx}{2\sqrt{E - |x|}} = 2\int_0^E \frac{dx}{2\sqrt{E-x}} = \Big[-2\sqrt{E-x}\Big]_0^E = 2\sqrt{E}.$$

This confirms, in this case, that applying Leibniz's rule despite the square-root singularity of $\partial G/\partial E$ at the endpoints gives the right answer. Then $T_{\mathrm{cl}} = 4\sqrt{E}$, so the spacing is $2\pi/(4\sqrt{E}) = \pi/(2\sqrt{E})$. This is the spacing used in Table 12.1.

Sample Problem 12.7: Density of states in the box

For the box on $[-L, L]$ with $L = 44.883$ (RE3), find the mean level spacing near $E = 1000$ from Equation (12.53), and confirm it from the exact levels.
Step 1: Phase integral. $\Phi = 2L\sqrt{E}$, so $\Phi' = L/\sqrt{E}$.
Step 2: Spacing. $\Delta E = \pi/\Phi' = \pi\sqrt{E}/L = \pi \times 31.623/44.883 = 2.2135$.
Step 3: Exact check. $E_n = (n\pi/2L)^2$ gives $dE_n/dn = n\pi^2/(2L^2)$. Since $\sqrt{E_n} = n\pi/(2L)$, this equals $\pi\sqrt{E_n}/L$, the same expression as in Step 2.

$$\boxed{\Delta E \approx 2.2135 \text{ at } E = 1000}$$

12.7.3 How the shape of the well sets the spacing

For a power-law well $V = |x|^\alpha$, the substitution $x = E^{1/\alpha}y$ gives

$$\Phi(E) = 2\int_0^{E^{1/\alpha}} \sqrt{E - x^\alpha}\,dx = 2E^{1/2}E^{1/\alpha}\int_0^1 \sqrt{1 - y^\alpha}\,dy = c_\alpha E^{\frac{1}{2}+\frac{1}{\alpha}}, \tag{12.54}$$

where $c_\alpha = 2\int_0^1 \sqrt{1-y^\alpha}\,dy$ is a constant. Quantization then gives $E_n \propto (n - \frac{1}{2})^{2\alpha/(\alpha+2)}$, and the spacing behaves as $dE_n/dn \propto n^{(\alpha-2)/(\alpha+2)}$:

- $\alpha = 2$ (oscillator): the spacing is constant;
- $\alpha > 2$ (steeper walls, with the box as $\alpha \to \infty$): the spacing grows with n;
- $\alpha < 2$ (for example $\alpha = 1$): the spacing shrinks with n, so the levels crowd together.

A well that widens quickly with height has a long classical period at high energy, and by Equation (12.53) that means a small spacing. Looking ahead (the shape is derived in Chapter 22): the Wu–Sprung potential grows roughly like $(x/\log x)^2$, a little more slowly than x^2, so the power-law rule suggests slowly shrinking spacings. The precise statement needs no power-law argument. The potential is built so that its smooth WKB count equals $N_{\rm sm}(E)$, so by Equation (12.53) its WKB mean spacing is $1/N'_{\rm sm}(E) = 2\pi/\log(E/2\pi)$, the mean spacing of the Riemann zeros, by construction.

Checkpoint 12.4

A potential grows like $V \propto |x|^{1.5}$. Does the level spacing grow or shrink with n?

Answer: It shrinks: the exponent is $(\alpha - 2)/(\alpha + 2) = -0.5/3.5 < 0$.

Key Ideas

- WKB counting: $\mathcal{N}_{\rm WKB}(E) = \lfloor \Phi(E)/\pi + \frac{1}{2} \rfloor$. The smooth count is $\bar{\mathcal{N}} = \Phi/\pi$, which equals $n - \frac{1}{2}$ at level n (Equations (12.46) and (12.47)).
- Density of states: $\bar{\mathcal{N}}' = T_{\rm cl}/2\pi$. Mean spacing: $\Delta E = 2\pi/T_{\rm cl}$ (Equations (12.52) and (12.53)).
- Wells that widen fast with energy crowd their levels; $V = |x|^\alpha$ gives spacing $\propto n^{(\alpha-2)/(\alpha+2)}$.

12.8 How accurate is WKB?

The derivation used two approximations: neglecting A''/A away from turning points, and linearizing V near them. Both get better as the number of wavelengths in the well grows. The error is therefore largest at the lowest levels and shrinks with n. This is a heuristic statement, not a theorem; the precise error depends on the potential.

Table 12.1 shows this concretely for $V = |x|$. Measured in local mean spacings, the error is 0.065 at $n = 1$, 0.017 at $n = 2$ and 0.005 at $n = 6$. The sign alternates between even and odd states. We record this as a numerical observation rather than derive it, but a heuristic reading is available. The odd states vanish at $x = 0$, so they are exactly the states of the half-well $x > 0$ with a hard wall at the origin (Problem 7); for them the hard-wall phase is exact, and WKB is approximate only at the one smooth turning point $x = E$. The even states instead obey $\psi'(0) = 0$ at the kink. In the WKB form $Ck^{-1/2}\cos\theta$ the derivative contains a term from $k'(0^+) = -V'(0^+)/(2k) \neq 0$, so the condition $\psi'(0) = 0$ is met only approximately. The two families therefore carry different boundary errors, and nothing forces those errors to have the same sign.

Two exceptions are worth remembering: for the box and for the oscillator, the Bohr–Sommerfeld levels are exact (Section 12.5). For the box the WKB wavefunctions are exact too; for the oscillator they are not.

Why measure errors in spacings? Whether an error matters depends on how far apart the levels are. An error is serious when it is a sizable fraction of the distance to the next level,

because then the predicted level could be mistaken for its neighbor. Dividing by the local mean spacing makes errors at different heights comparable. This is also why the paper quotes every error "in units of the local mean spacing". For example, in the box of RE3 an error of 0.01 is negligible at $E = 1000$, where the spacing is 2.2, but amounts to 2.7 spacings near the ground state, where $E_2 - E_1 = 0.0037$. For the zeros the trend is the opposite: the mean spacing $2\pi/\log(t/2\pi)$ shrinks slowly with height, from 7.75 at $t = t_1$ to 1.24 at $t = 1000$, so a fixed error in t matters somewhat more higher up.

Pitfall

"WKB works for high levels" is a statement about levels high *in their own well*. A potential with a sharp feature, such as a narrow spike, can violate Equation (12.26) at any energy if V' is large enough there. The paper's fitted corrections δV have large, rapidly varying structure (Sec. III, Fig. 4), so the smooth WKB picture describes only the background, not the fitted potentials. The paper computes every level it scores by finite-difference diagonalization (Chapter 23); WKB enters only through the construction of the smooth background potential.

Key Ideas

- WKB errors, in units of the mean spacing, are largest at low n and decrease with n (for $V = |x|$: 0.065 at $n = 1$, 0.005 at $n = 6$).
- This is a heuristic expectation. The Bohr–Sommerfeld levels are exact for the box and the oscillator, and the approximation fails for sharply varying potentials.

12.9 Where in the well does a level live?

The WKB probability density. Square Equation (12.21): $|\psi|^2 = \frac{C^2}{k}\cos^2(\theta + \varphi)$. Over many wavelengths, $\cos^2$ averages to $\frac{1}{2}$. The locally averaged density is therefore

$$\overline{|\psi|^2} \approx \frac{C^2}{2k(x)}. \tag{12.55}$$

Normalization requires $\int \overline{|\psi|^2}\,\mathrm{d}x = \frac{C^2}{2}\int_a^b \frac{\mathrm{d}x}{k} = \frac{C^2}{2}T_{\mathrm{cl}} = 1$, using Equation (12.51). Hence $C^2 = 2/T_{\mathrm{cl}}$ and

$$\overline{|\psi(x)|^2} \approx \frac{1}{T_{\mathrm{cl}}\,k(x)} = \frac{2}{T_{\mathrm{cl}}}\cdot\frac{1}{v(x)}. \tag{12.56}$$

The right-hand side is the classical probability density: the fraction of a period spent in $\mathrm{d}x$, counting both directions, is $2\,\mathrm{d}x/(v\,T_{\mathrm{cl}})$. (Replacing $\cos^2$ by its average is a heuristic step. It neglects the region within a few Airy lengths of the turning points, where the true density is finite but the formula diverges, though integrably.)

The turning point labels the level. Since $\overline{|\psi|^2} \propto (E-V)^{-1/2}$, the density piles up near the turning points $\pm x_E$, where $V(x_E) = E$. How does a level respond when the potential changes by a small $\delta V(x)$? Chapter 13 treats this question in full; here we derive directly the one result we need. Take a real normalized bound state, multiply $-\psi_n'' + V\psi_n = E_n\psi_n$ by ψ_n and integrate. Integrating $-\psi_n\psi_n''$ by parts (the boundary term vanishes because $\psi_n \to 0$ at the ends) gives

$$E_n = \int (\psi_n'^2 + V\psi_n^2)\mathrm{d}x. \tag{12.57}$$

The same identity holds for the perturbed level, with $V \to V + \delta V$ and $\psi_n \to \psi_n + \delta\psi$. Subtracting and keeping only terms of first order in the small quantities δV and $\delta\psi$:

$$\begin{aligned}\delta E_n &= \int (2\psi_n'\delta\psi' + 2V\psi_n\delta\psi + \psi_n^2\delta V)\,\mathrm{d}x \\ &= \int (-2\psi_n'' + 2V\psi_n)\delta\psi\,\mathrm{d}x + \int \psi_n^2\delta V\,\mathrm{d}x \quad \text{(by parts; } \delta\psi \to 0 \text{ at the ends)} \\ &= 2E_n \int \psi_n\delta\psi\,\mathrm{d}x + \int \psi_n^2\delta V\,\mathrm{d}x. \quad \text{(Schrödinger equation)} \end{aligned} \tag{12.58}$$

The perturbed state is also normalized, $\int(\psi_n+\delta\psi)^2\mathrm{d}x = 1$, which to first order says $2\int \psi_n\delta\psi\,\mathrm{d}x = 0$. Hence

$$\delta E_n \approx \int |\psi_n|^2\,\delta V\,\mathrm{d}x. \tag{12.59}$$

With Equation (12.56), this becomes the *classical time average of δV along the orbit*:

$$\delta E_n \approx \frac{2}{T_{\mathrm{cl}}}\int_a^b \frac{\delta V(x)}{v(x)}\,\mathrm{d}x. \tag{12.60}$$

The weight $1/v(x)$ is largest near the turning points, so a given δV placed there counts for more than the same δV placed near the center. But the weight is not concentrated there. The level responds to δV everywhere inside $[a, b]$; in the linear well, for example, the outer tenth of the distance carries only about a third of the weight (Sample Problem below). What changes sharply at the turning point is the response to δV *beyond* it: outside $[a, b]$ the wavefunction decays exponentially, so level n barely feels δV there. In an even, monotone well each level index n is thus attached to a position x_{E_n}, the outer edge of the region it probes. Lower levels probe only the center, higher levels reach farther out, and a correction placed at x_* affects only levels with $x_{E_n} \gtrsim x_*$, with the turning-point enhancement for those that turn near x_* (Figure 12.5). This labeling is heuristic, since it rests on the averaged density Equation (12.56); it is the sense in which the paper says that position maps to level index through the turning point.

Sample Problem 12.8: Time spent near the turning point

For $V = |x|$ at any energy E, what fraction of the classical period does the particle spend in the outer tenth of each half of the well, $0.9E \le |x| \le E$?

Step 1: Time in $[x_1, x_2]$ for $x > 0$. The time is $\int \mathrm{d}x/v = \frac{1}{2}\int \mathrm{d}x/\sqrt{E-x}$, and $\int \mathrm{d}x/\sqrt{E-x} = -2\sqrt{E-x}$.

Step 2: Outer tenth. $\frac{1}{2}\cdot 2\sqrt{E-0.9E} = \sqrt{0.1E}$.

Step 3: Whole half-crossing. $\frac{1}{2}\cdot 2\sqrt{E} = \sqrt{E}$.

Step 4: Ratio. By symmetry the same ratio holds for the full period: $\sqrt{0.1} = 0.316$.

31.6% of the time in the outer 10% of the distance

Checkpoint 12.5

You add a narrow bump to an even confining potential at $x = \pm x_*$. Which levels move most?

Answer: Levels whose turning points lie at or just beyond x_*, that is levels with $E_n \gtrsim V(x_*)$. Levels with $E_n < V(x_*)$ barely move, because the bump lies in their forbidden region.

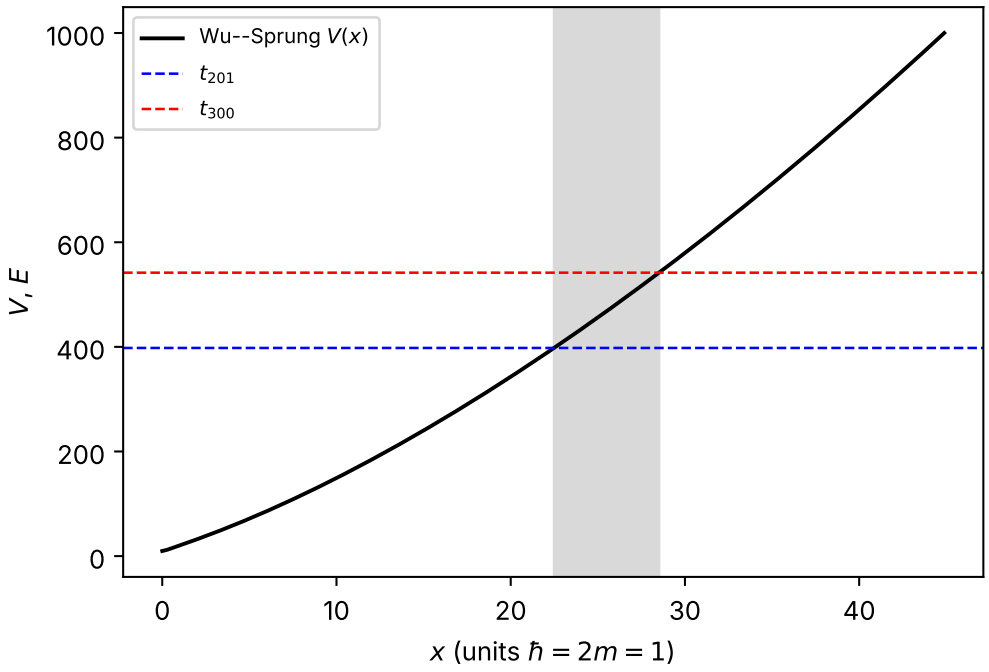


Figure 12.5: The Wu–Sprung potential of the paper (Eq. (2), solid, $x > 0$), with $V_0 = 3.10073\pi$. Horizontal lines mark the energies t_{201} and t_{300} of the first and last zeros of the paper's interpolation holdout. The shaded band is where those levels have their turning points; the WKB density Equation (12.56) of each such level peaks there. Units $\hbar = 2m = 1$.

Key Ideas

- Averaged WKB density: $\overline{|\psi|^2} = 1/(T_{\mathrm{cl}}k)$, which is the classical probability density (Equation (12.56)).
- First-order level shifts are classical time averages of δV, weighted toward the turning points (Equation (12.60)).
- In an even monotone well each level index n maps to the turning-point position x_{E_n}.

12.10 Inversion: from a counting function to a potential

So far we went from V to levels. Now we go backwards. Suppose we are told only the smooth count $\bar{\mathcal{N}}(E)$. Can we find a potential that produces it?

Non-uniqueness. Not uniquely. For a single well with minimum $V_{\min}$, let $x_L(V) < x_R(V)$ be the two turning points at height V and $w(V) = x_R(V) - x_L(V)$ the width. Changing variable from x to V separately on the two sides of the minimum (exactly as in Step 1 below) gives $\Phi(E) = \int_{V_{\min}}^{E} \sqrt{E-V}\,[x_R'(V) - x_L'(V)]\,\mathrm{d}V = \int_{V_{\min}}^{E} \sqrt{E-V}\,w'(V)\,\mathrm{d}V$. So Φ depends only on the width at each height. Sliding each horizontal slice of the well sideways by an amount that depends on height (a shear, which keeps $w(V)$ fixed) changes the shape without changing Φ. Example: $V = |x|$ has $w = 2V$; the lopsided well $V = -2x$ for $x < 0$, $V = \frac{2}{3}x$ for $x > 0$ has $x_L = -V/2$, $x_R = 3V/2$, so also $w = 2V$, and both have $\Phi = \frac{4}{3}E^{3/2}$. To get a definite answer we impose two conditions: V is *even*, and V *increases monotonically* for $x > 0$ from its minimum $V_{\min} = V(0)$. Then V restricted to $x \geq 0$ has an inverse function $x(V)$, the turning-point

position at height V, with $x(V_{\min}) = 0$.

Step 1: Rewrite Φ as an integral over V. By evenness, $\Phi(E) = 2\int_0^{x_E} \sqrt{E - V(x)}\,\mathrm{d}x$. Change the variable of integration from x to V using $\mathrm{d}x = x'(V)\,\mathrm{d}V$; the limits $x = 0$ and $x = x_E$ become $V = V_{\min}$ and $V = E$:

$$\Phi(E) = 2\int_{V_{\min}}^{E} \sqrt{E - V}\, x'(V)\,\mathrm{d}V. \tag{12.61}$$

Step 2: Differentiate. By Leibniz's rule Equation (12.49), the upper-limit term is $2\sqrt{E - E}\,x'(E) = 0$, so

$$\Phi'(E) = \int_{V_{\min}}^{E} \frac{x'(V)}{\sqrt{E - V}}\,\mathrm{d}V. \tag{12.62}$$

This is *Abel's integral equation*. The left side is known (it is $\pi\bar{\mathcal{N}}'(E)$), and the unknown $x'(V)$ sits under the integral.

Step 3: Multiply by $(W - E)^{-1/2}$ and integrate. Pick any $W > V_{\min}$. Multiply both sides of Equation (12.62) by $1/\sqrt{W - E}$ and integrate over E from $V_{\min}$ to W:

$$\int_{V_{\min}}^{W} \frac{\Phi'(E)}{\sqrt{W - E}}\,\mathrm{d}E = \int_{V_{\min}}^{W} \mathrm{d}E \int_{V_{\min}}^{E} \mathrm{d}V\, \frac{x'(V)}{\sqrt{W - E}\,\sqrt{E - V}}. \tag{12.63}$$

Step 4: Swap the order. The double integral covers the triangle $V_{\min} \le V \le E \le W$. Integrating over E first, for fixed V, means E runs from V to W:

$$\mathrm{RHS} = \int_{V_{\min}}^{W} x'(V) \left[\int_{V}^{W} \frac{\mathrm{d}E}{\sqrt{(W - E)(E - V)}} \right] \mathrm{d}V. \tag{12.64}$$

Taken on trust

Swapping the order of an iterated integral whose integrand has integrable square-root singularities on the boundary is justified by a theorem of analysis (Fubini's theorem for absolutely integrable functions); see [22]. We take it on trust. The samples below check the final formula against cases with known answers.

Step 5: The inner integral equals π. Substitute $E = V + (W - V)\sin^2 u$, with u from 0 to $\pi/2$. Then $\mathrm{d}E = 2(W - V)\sin u \cos u\,\mathrm{d}u$, $E - V = (W - V)\sin^2 u$ and $W - E = (W - V)\cos^2 u$, so $\sqrt{(W - E)(E - V)} = (W - V)\sin u \cos u$. The inner integral becomes $\int_0^{\pi/2} 2\,\mathrm{d}u = \pi$, whatever the values of V and W.

Step 6: Finish. The right side is now $\pi\int_{V_{\min}}^{W} x'(V)\,\mathrm{d}V = \pi[x(W) - x(V_{\min})] = \pi x(W)$. Dividing by π and using $\Phi' = \pi\bar{\mathcal{N}}'$:

$$\boxed{x(W) = \frac{1}{\pi}\int_{V_{\min}}^{W} \frac{\Phi'(E)}{\sqrt{W - E}}\,\mathrm{d}E = \int_{V_{\min}}^{W} \frac{\bar{\mathcal{N}}'(E)}{\sqrt{W - E}}\,\mathrm{d}E.} \tag{12.65}$$

This is the *Abel inversion formula*. It gives the turning-point position at every height W, that is the inverse of the potential, from the density of states alone.

There is one consistency requirement. $\Phi(V_{\min}) = 0$, because the allowed region shrinks to a point at the bottom of the well, so a smooth count that comes from a well must vanish at its bottom. A prescribed $\bar{\mathcal{N}}$ that does not vanish there is not exactly the WKB count of any even monotone well. How the paper's construction handles its constant V_0 is the subject of Chapter 22.

Sample Problem 12.9: Recovering the linear well

Given $\bar{\mathcal{N}}(E) = \frac{4}{3\pi}E^{3/2}$ with $V_{\min} = 0$, find $V(x)$.
Step 1: Density. $\bar{\mathcal{N}}'(E) = \frac{2}{\pi}E^{1/2}$.
Step 2: Abel integral. $x(W) = \frac{2}{\pi}\int_0^W \frac{\sqrt{E}}{\sqrt{W-E}}\,\mathrm{d}E$.
Step 3: Substitute $E = W\sin^2 u$**.** Then $\sqrt{E} = \sqrt{W}\sin u$, $\sqrt{W-E} = \sqrt{W}\cos u$ and $\mathrm{d}E = 2W\sin u\cos u\,\mathrm{d}u$. The integral becomes $\frac{2}{\pi}\int_0^{\pi/2} 2W\sin^2 u\,\mathrm{d}u = \frac{2}{\pi}\cdot 2W\cdot\frac{\pi}{4} = W$.
Step 4: Invert. $x = W$, so $V(x) = |x|$, the well we started from in Section 12.5.

$$\boxed{V(x) = |x|}$$

Sample Problem 12.10: Recovering the oscillator

Given $\bar{\mathcal{N}}(E) = E/2$ with $V_{\min} = 0$, find $V(x)$.
Step 1: $\bar{\mathcal{N}}' = \frac{1}{2}$.
Step 2: $x(W) = \frac{1}{2}\int_0^W \frac{\mathrm{d}E}{\sqrt{W-E}} = \frac{1}{2}\left[-2\sqrt{W-E}\right]_0^W = \sqrt{W}$.
Step 3: $V = x^2$. Comparing with Section 12.5: $V = x^2$ has $\Phi = \pi E/2$, so $\bar{\mathcal{N}} = E/2$. Consistent.

$$\boxed{V(x) = x^2}$$

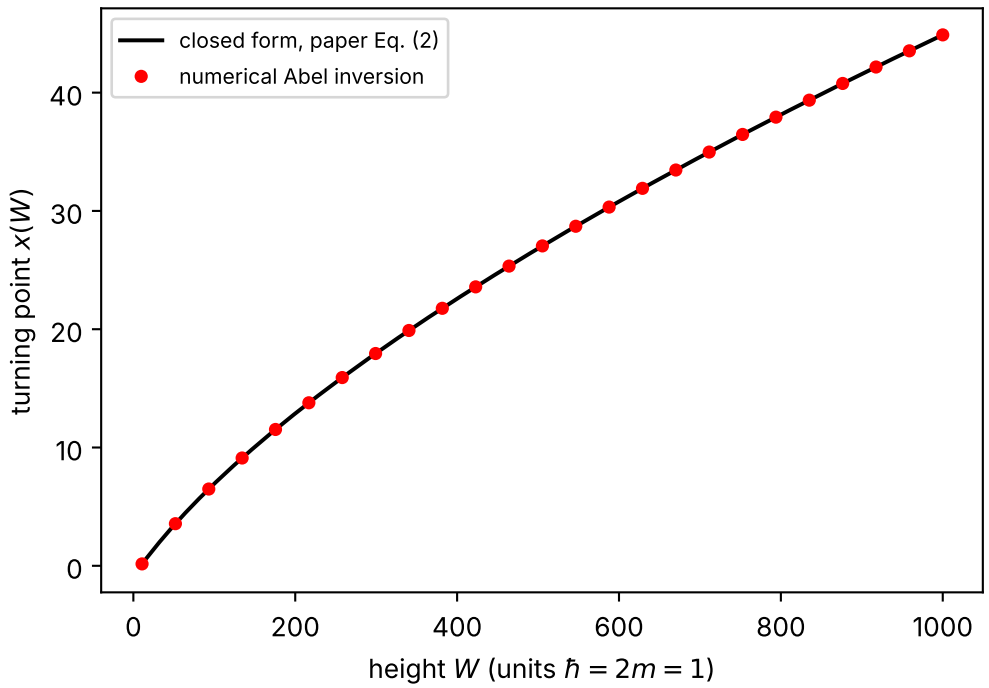


Figure 12.6: Numerical Abel inversion Equation (12.65). Points: $x(W)$ computed by numerical integration from the smooth count N_{sm} of the paper's Eq. (1), with lower limit $V_0 = 3.10073\pi$. Curve: the closed form of the paper's Eq. (2). They agree to the accuracy of the quadrature. Axes: height W and turning point x, units $\hbar = 2m = 1$.

Checkpoint 12.6

If the prescribed density of states $\bar{\mathcal{N}}'(E)$ is doubled at every energy, what happens to $x(W)$?

Answer: It doubles: Equation (12.65) is linear in $\bar{\mathcal{N}}'$. A well twice as wide at every height has twice the density of states.

Key Ideas

- For an even, monotone potential, the smooth count determines the potential through Abel's equation Equation (12.62).
- Inversion: $x(W) = \int_{V_{\min}}^{W} \bar{\mathcal{N}}'(E)(W-E)^{-1/2}\,\mathrm{d}E$ (Equation (12.65)). It recovers $|x|$ and x^2 exactly.
- Without evenness and monotonicity the inverse problem is not unique. Only the *smooth* count is inverted; fluctuations in the level positions are not encoded by this construction.

12.11 Paper connection

★ Did you know? *Exactly Right by Luck?*

For the harmonic oscillator, the WKB rule with the extra $\frac{1}{2}$, $\oint p\,dx = (n+\frac{1}{2})h$, gives $E_n = (n+\frac{1}{2})\hbar\omega$, the *exact* energies for every n, not just large ones. The $\frac{1}{2}$ comes from phase shifts at the two turning points, systematized by Victor Maslov in the 1960s as the Maslov index. The older Bohr-Sommerfeld rule, lacking it, missed the zero-point energy.

Eq. (1) and the $n-\frac{1}{2}$ convention. The paper defines smooth zeros by $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ (Sec. II A). By Equation (12.47) this is exactly the value of a smooth WKB count at the n-th level. So if a potential has $\Phi(E)/\pi = N_{\mathrm{sm}}(E)$, its Bohr–Sommerfeld levels are precisely the smooth zeros s_n.

Eq. (2) is an Abel inversion. Sec. II B defines the Wu–Sprung potential by $x(V)$ in Eq. (2). Inserting $\bar{\mathcal{N}} = N_{\mathrm{sm}}$ of Eq. (1) into Equation (12.65) with lower limit V_0, and doing the integral numerically at $W = 1000$, gives $x = 44.883$. This is the paper's L for scale 1, and Figure 12.6 shows the agreement at all heights. The closed-form evaluation is Problem 16; the role of V_0 and the full construction are Chapter 22.

The semiclassical weakness at low n. Sec. II B reports that the Richardson-extrapolated levels of the smooth potential differ from the smooth zeros by 0.005 spacings on average and 0.063 at most, "worst at low n". That pattern is the one you saw in Table 12.1. The potential was built so that its *WKB* levels are s_n, and WKB is least accurate for the lowest levels.

Turning points and the interpolation split. Sec. III (paragraph "Interpolation split") and Fig. 4 state that "position maps to level index through the classical turning point, and levels 201–300 turn at $x = 22.5$–28.5." This is Equation (12.60) read heuristically. A correction placed in that band is felt with the turning-point enhancement by the levels that turn there, which are exactly the held-out levels. The lower training levels 1–200 turn at smaller x, so the band lies in their forbidden region and they barely feel it. The higher training levels 301–500

do feel it, but only as one part of a broad time average that other coefficients can compensate. This is the heuristic behind the paper's remark that such structure "costs nothing in-sample". Figure 12.5 reproduces the band from Eq. (2) (Problem 13).

The coverage law. Sec. III says the cosine family becomes effectively complete for joint level shifts once $M\pi/L > 2\sqrt{E}$. Here $\sqrt{E}$ is the local WKB wavenumber k near the well center, and $|\psi|^2$ oscillates as $\cos^2(\int k)$, at wavenumber $2k$. A basis must contain wavenumbers up to $2k$ to resolve that structure; Chapter 24 makes this argument. The relevant energy is that of the highest fitted level, $E = t_{500} = 811.18$ (the paper's anchor). At scale 1 the threshold there is $M > 813.8$, which lies below $M = 900$, consistent with the paper's report of rms below 10^{-4} by $M = 900$. At the top of the box, $E = 1000$, the threshold would be 903.6. Notice that $2L\sqrt{E}/\pi$ is also the hard-wall count of box levels below E (Section 12.5): the threshold is reached when the number of cosines matches the number of box states below E.

Levels per prime period. Sec. V notes that the $p = 2$ oscillation, of period $2\pi/\log 2 \approx 9$ in t (RE2), spans "only ∼6–7 levels at these heights". The mean spacings there, from $\text{gap}_n = 2\pi/\log(s_n/2\pi)$, are $\text{gap}_{200} = 1.52$ and $\text{gap}_{400} = 1.34$. Dividing 9.065 by them gives about 6.0 near $n = 200$ and 6.8 near $n = 400$ (Problem 15). The mean spacing is itself the reciprocal of the WKB density of states Equation (12.53) of a potential built to reproduce N_{sm}.

The oracle as a "quantization condition". Eq. (4)'s predicted zeros solve $N_{\text{sm}}(t) + N_{\text{fl}}(t) = n - \frac{1}{2}$. This has the form of Equation (12.47) with a fluctuating correction added to the smooth count. This is only an analogy (Chapter 21); the oracle involves no potential at all.

12.12 Summary

- Local wavenumber $k = \sqrt{E - V}$ (units $\hbar = 2m = 1$); turning points where $V = E$ (Equation (12.5)).
- WKB wavefunction $Ck^{-1/2}\cos(\int k\,\mathrm{d}x + \varphi)$, valid when $|V'|/(2|E - V|^{3/2}) \ll 1$ (Equations (12.21) and (12.26)).
- Connection formula with phase $\pi/4$ (Equation (12.37)); Airy asymptotics taken on trust.
- Bohr–Sommerfeld rule $\int_a^b \sqrt{E_n - V}\,\mathrm{d}x = (n - \frac{1}{2})\pi$, with variants for walls (Equations (12.42) and (12.43)); phase-space area $2\pi(n - \frac{1}{2})$ (Equation (12.45)).
- Counting: $\mathcal{N}_{\text{WKB}} = \lfloor \Phi/\pi + \frac{1}{2} \rfloor$; smooth count $\Phi/\pi = n - \frac{1}{2}$ at levels (Equations (12.46) and (12.47)).
- Density of states $T_{\text{cl}}/2\pi$; spacing $2\pi/T_{\text{cl}}$ (Equations (12.52) and (12.53)).
- Averaged density $1/(T_{\text{cl}}k)$; level shifts are orbit time averages of δV, with weight $1/v$ enhanced (but not concentrated) near the turning points and negligible beyond them (Equations (12.56) and (12.60)).
- Abel inversion $x(W) = \int \bar{\mathcal{N}}'(E)(W - E)^{-1/2}\,\mathrm{d}E$ for even monotone wells (Equation (12.65)); applied to Eq. (1) it gives the paper's Eq. (2).

12.13 Problems

1. • For $V = x^2$ and $E = 16$, find the local wavenumber and wavelength at $x = 0$ and $x = 2$.

2. • How many levels of the box on $[-1, 1]$ lie below $E = 100$?

3. • How many levels of $V = x^2$ lie below $E = 10$? Compare with $\lfloor \Phi/\pi + \tfrac{1}{2} \rfloor$.

4. • For $V = |x|$ at $E = 8$, find the enclosed phase-space area and the smooth count Φ/π.

5. •• Find the classical period of $V = |x|$ at $E = 9$ and the corresponding mean level spacing.

6. •• The potential $|x|$ has a kink at $x = 0$. Discuss whether this invalidates WKB there.

7. •• A hard wall at $x = 0$ is followed by $V = x$ for $x > 0$. Find the WKB levels and compare E_1 with the first odd level of $|x|$ in Table 12.1.

8. •• Show that for $V = |x|^\alpha$, $E_n \propto (n - \tfrac{1}{2})^{2\alpha/(\alpha+2)}$.

9. •• Verify Leibniz's rule Equation (12.49) for $I(s) = \int_0^s su\,\mathrm{d}u$.

10. •• Show that the box spacing $\pi\sqrt{E}/L$ grows like n, and evaluate it at $E = 1000$ for $L = 44.883$.

11. •• Using numpy, find the Bohr–Sommerfeld ground state of $V = x^4$ and compare it with `qmnt.fd_levels` on a grid with $L = 6$.

12. •• Using scipy, regenerate Table 12.1 and extend it to $n = 20$. How does the error scale with n?

13. •• (Paper) Using `qmnt.zeros` and `qmnt.wu_sprung_x`, find the turning points of the levels at heights t_{201} and t_{300}. Compare with Sec. III.

14. •• (Paper) Evaluate the coverage threshold $2L\sqrt{E}/\pi$ at scale 2 for $E = t_{1000} = 1419.42$ and $L = 59.957$.

15. •• (Paper) Using `qmnt.mean_gap`, compute $9.065/\mathrm{gap}_n$ for $n = 200$ and 400. Relate the result to Sec. V.

16. ••• Evaluate the Abel integral Equation (12.65) in closed form for $\bar{\mathcal{N}} = N_{\mathrm{sm}}$ with lower limit V_0, and show that it gives the paper's Eq. (2). (Hint: substitute $u = \sqrt{W - E}$ and integrate by parts.)

17. ••• Derive the normalization $C^2 = 2/T_{\mathrm{cl}}$ and Equation (12.56). State the approximation involved.

18. ••• Invert $\bar{\mathcal{N}} = cE$ with $V_{\min} = 0$. Check your answer with $c = \tfrac{1}{2}$.

19. •• Using `qmnt.fd_levels` on the Wu–Sprung potential (fine grid, scale 1), compare the first 20 levels with `qmnt.smooth_zeros` in units of `mean_gap`. Where is the error largest?

20. • Explain why WKB is least reliable for the ground state.

21. •• (Paper) Compute $N_{\mathrm{sm}}(1000)$ and the WKB count of smooth-potential levels below $E = 1000$. Sec. II E uses 652 states with $E \le 1000$ on its grid. Explain the difference.

22. ••• Show, using Equation (12.51) and Equation (12.65), that within WKB the only even monotone potentials with $V(0) = 0$ whose classical period is independent of energy are $V = cx^2$ with $c > 0$, and that the period π selects $V = x^2$.

23. •• For $V = x^2$ and $E = 101$, evaluate the validity parameter of Equation (12.26) at $x = 5$.

24. ••• Using numpy, invert $\bar{\mathcal{N}} = \frac{4}{3\pi}E^{3/2}$ numerically by quadrature and check that you recover $|x|$.

25. •• Derive the connection formula for a turning point a with the forbidden region on the left.

26. • (Paper) Why does the paper define smooth zeros by $n - \frac{1}{2}$ rather than n?

27. •• (Paper) Explain in what sense the oracle equation $N_{\mathrm{sm}} + N_{\mathrm{fl}} = n - \frac{1}{2}$ resembles Equation (12.47), and why this is only an analogy.

12.14 Answers to selected problems

1. $k(0) = 4$, $\lambda = 1.571$; $k(2) = \sqrt{12} = 3.464$, $\lambda = 1.814$.

2. $n \le 2L\sqrt{E}/\pi = 20/\pi = 6.37$, so 6 levels.

3. $E_n = 2n - 1 \le 10$ gives $n \le 5$, so 5 levels. $\Phi/\pi = E/2 = 5$, and $\lfloor 5.5 \rfloor = 5$. They agree.

4. $\Phi(8) = \frac{4}{3}\cdot 8^{3/2} = 30.17$, so $\mathcal{A} = 2\Phi = 60.34$ and $\Phi/\pi = \frac{4}{3\pi}8^{3/2} = 9.60$. By Equation (12.47), level n has $\Phi/\pi = n - \frac{1}{2}$: level 10 has 9.5 and level 11 has 10.5, so $E = 8$ lies between levels 10 and 11. The WKB count is $\lfloor 10.10 \rfloor = 10$, matching the exact count, since the exact levels are $E_{10} = 7.944$ and $E_{11} = 8.488$.

5. $T_{\mathrm{cl}} = 4\sqrt{9} = 12$; spacing $2\pi/12 = 0.5236$.

6. At $x = 0$ the slope V' jumps from -1 to $+1$, so V'' is not defined there (it behaves like an infinitely tall, narrow spike) and the curvature part of the validity condition fails at that single point. Away from the origin V is linear and the validity parameter is small (second Sample Problem of Section 12.3), so the WKB form is good there. The Bohr–Sommerfeld rule still works because, by symmetry, the states split into odd ones with $\psi(0) = 0$ and even ones with $\psi'(0) = 0$, each a half-well problem on $x > 0$. For the odd states $\psi(0) = 0$ is a hard wall, which WKB treats exactly (Problem 7). For the even states the WKB form meets $\psi'(0) = 0$ only approximately, which is one reason the two families have different errors in Table 12.1 (Section 12.8).

7. $\Phi = \frac{2}{3}E^{3/2} = (n - \frac{1}{4})\pi$, so $E_n = [\frac{3\pi}{2}(n - \frac{1}{4})]^{2/3}$ and $E_1 = 2.3203$. This is identical to the WKB value for $n = 2$ of $|x|$; the exact value is 2.3381. (The odd states of $|x|$ vanish at 0, so they are exactly the states of the half-well with a wall.)

9. $I = s^3/2$, so $I' = 3s^2/2$. With $G(s, u) = su$, Leibniz gives $G(s, s)\cdot 1 + \int_0^s u\,\mathrm{d}u = s^2 + s^2/2$. They agree.

11. $\Phi = 2E^{3/4}\int_0^1 \sqrt{1 - y^4}\,\mathrm{d}y = 1.74804\,E^{3/4}$, so $E_1^{\mathrm{WKB}} = (\pi/(2 \times 1.74804))^{4/3} = 0.8672$. Diagonalization gives 1.0604, an error of about 18% for the ground state.

13. $x(t_{201}) \approx 22.5$ and $x(t_{300}) \approx 28.5$, matching Sec. III.

14. $2 \times 59.957 \times \sqrt{1419.42}/\pi = 1438.06$. The largest scale-2 basis in the paper, $M = 900$, lies below this value at the top of the scored range.

15. $\mathrm{gap}_{200} = 1.52$ and $\mathrm{gap}_{400} = 1.34$, so $9.065/\mathrm{gap}_n$ is about 6.0 at $n = 200$ and 6.8 at $n = 400$: one $p = 2$ period spans 6–7 levels, as Sec. V says.

17. Average $\cos^2$ to $\frac{1}{2}$ over the fast oscillations. Then $\int C^2/(2k)\,\mathrm{d}x = C^2 T_{\mathrm{cl}}/2 = 1$. This neglects the oscillation and the Airy regions near the turning points.

19. By analogy with Table 12.1, expect the largest error among the lowest few levels, falling off as n grows. We do not quote values, because your numbers depend on the grid and on whether you apply Richardson extrapolation. The paper reports this pattern for its Richardson-extrapolated levels: mean residual 0.005 and maximum 0.063 spacings, "worst at low n" (Sec. II B).

20. The well holds less than one wavelength, so A''/A is not small anywhere, and the two Airy regions overlap.

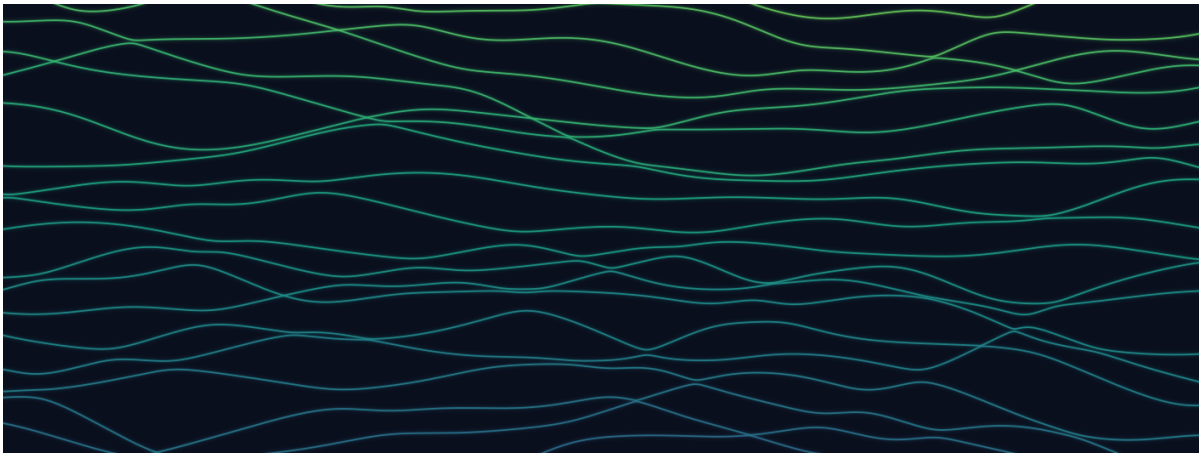
21. $N_{\rm sm}(1000) = 648.61$, so the WKB count is $\lfloor 649.11 \rfloor = 649$. The three-level difference is not computed in this chapter. The main effect is expected to be finite-difference dispersion, which lowers high levels on a grid (Chapter 23) and so lets extra levels fall below 1000. The Dirichlet wall at $L = x(1000)$, which pushes the top levels up, and the WKB error itself are expected to contribute at the level of a state or so.

23. $|V'| = 2x = 10$ and $E - V = 101 - 25 = 76$, so the parameter is $10/(2 \cdot 76^{3/2}) = 5/76^{3/2} = 0.00755$: WKB is excellent there.

25. Near a, $V(x) \approx E + V'(a)(x-a) = E + F(a-x)$ with $F = -V'(a) > 0$. Put $z = F^{1/3}(a-x)$; since $(dz/dx)^2 = F^{2/3}$, the equation $\psi'' = F(a-x)\psi$ becomes $\mathrm{d}^2\psi/\mathrm{d}z^2 = z\psi$, and decay as $x \to -\infty$ (that is $z \to +\infty$) selects $\mathrm{Ai}(z)$. For $x < a$: $\kappa = F^{1/2}(a-x)^{1/2}$, $\int_x^a \kappa \, \mathrm{d}x' = \frac{2}{3} z^{3/2}$ and $z^{-1/4} = F^{1/6}\kappa^{-1/2}$. For $x > a$: put $w = -z = F^{1/3}(x-a)$, so $k = F^{1/2}(x-a)^{1/2}$, $\int_a^x k \, \mathrm{d}x' = \frac{2}{3} w^{3/2}$ and $w^{-1/4} = F^{1/6} k^{-1/2}$. Inserting into Equation (12.33) and using $\sin(\theta + \pi/4) = \cos(\theta - \pi/4)$ gives $\frac{2D}{\sqrt{k}} \cos(\int_a^x k \, \mathrm{d}x' - \frac{\pi}{4}) \leftrightarrow \frac{D}{\sqrt{\kappa}} e^{-\int_x^a \kappa \, \mathrm{d}x'}$, the mirror image of Equation (12.37).

26. Because a smooth count passes through $n - \frac{1}{2}$ at the n-th step (Equation (12.47)).

27. Equation (12.47) says that at a level the smooth count equals $n - \frac{1}{2}$; the oracle says that at a predicted zero the smooth count $N_{\rm sm}$ plus a correction $N_{\rm fl}$ equals $n - \frac{1}{2}$. So the oracle reads like a quantization condition with a fluctuating term added to the smooth count. But in Equation (12.47) the count comes from a potential through $\Phi(E)/\pi$, whereas $N_{\rm fl}$ is a sum over primes (Eq. (4)); no potential, Hamiltonian or wavefunction appears. Nobody has exhibited an operator whose quantization condition is the oracle equation, so the resemblance is formal only (Chapter 21).



Eigenvalues of $\cos\lambda\,A + \sin\lambda\,B$ *for two random symmetric matrices, as* λ *runs from* 0 *to* π*: levels approach, then swerve apart, never crossing.*

Chapter 13

Perturbation Theory

↑ Contents

What this chapter is about

Tune a guitar string by a tiny twist of the peg and every harmonic moves a little; press a small piece of tape onto the string and the pitch drops slightly. In both cases you do not need to re-solve the whole problem of the vibrating string to know roughly what happens. You need only to know the unchanged string and the size and location of the small change. Quantum mechanics has the same situation. We know the energy levels and wavefunctions of a few systems exactly (the box of Chapter 11 is the main one), and we constantly want to know what happens when the potential is changed by a small amount $\delta V(x)$. *Perturbation theory* answers this question as a series of corrections: a first-order shift that is a simple weighted average of δV, a second-order shift built from the way δV couples different states, and so on.

This chapter derives the first- and second-order formulas line by line, tests them against exact numerical answers, finds the condition under which they can be trusted, and then turns them around: if we know how each level responds to each piece of δV, we can adjust δV to move the levels to targets we choose. That is the idea behind fitting a potential to a spectrum.

Where this shows up in the paper: the Hermitian fit of Sec. II C of the paper [1] is driven by the first-order sensitivity matrix $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\,dx$, which is exactly the first-order formula of this chapter; the non-Hermitian models of Sec. II E use the second-order formula, the paper's Eq. (5); and the whole of Sec. V is a case study of what goes wrong when an optimizer drives a perturbative formula outside the regime where it is valid.

What you need

- Chapter 8: eigenvalues and eigenvectors, real symmetric and Hermitian matrices have real eigenvalues and orthogonal eigenvectors, tridiagonal matrices, `scipy.linalg.eigh_tridiagonal`, matching eigenvectors by overlap.
- Chapter 10: the Hamiltonian $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$ in units $\hbar = 2m = 1$, the time-independent equation $H\psi = E\psi$, normalization, expectation values, Hermitian operators.
- Chapter 11: the infinite square well on $[-L, L]$, its levels $(k\pi/2L)^2$ and normalized states, parity of states in an even potential.
- Chapter 7: inner products of functions, expanding a function in a basis, solving a square linear system.
- Chapter 5: orthogonality of sines and cosines over a period.
- Chapter 4: Newton's method for root finding; the expansion $\sqrt{1+y} \approx 1 + y/2$ for small y.
- First-year physics: potential energy curves and turning points; a classical particle moves slowest near its turning points; standing waves on a string.

13.1 A small change to a solved problem

Suppose you have solved the time-independent Schrödinger equation for some potential $V(x)$. Recall from Chapter 10 that in the units $\hbar = 2m = 1$ used throughout this book the Hamiltonian is

$$H_0 = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x), \tag{13.1}$$

an operator that acts on a wavefunction $\psi(x)$ by differentiating it twice, changing the sign, and adding $V(x)\psi(x)$. We write H_0 ("H nought") because it is the *unperturbed* Hamiltonian. Solving the problem means that you know its energies $E_n^{(0)}$ and its normalized wavefunctions $\psi_n^{(0)}(x)$, $n = 1, 2, 3, \ldots$, which satisfy

$$H_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}. \tag{13.2}$$

The superscript (0) means "zeroth order", that is, before any change.

Now change the potential: $V(x) \to V(x) + \delta V(x)$, where δV is a function we call the *perturbation*. The new Hamiltonian is

$$H = H_0 + \delta V. \tag{13.3}$$

Its levels E_n are different from the $E_n^{(0)}$. If δV is "small" (we will make this precise in Section 13.6), the new levels are close to the old ones, and we would like formulas for the differences

$$\delta E_n = E_n - E_n^{(0)}$$

that use only what we already know: the old levels, the old wavefunctions, and δV.

Think of a ball rolling in a valley. If you add a small bump somewhere along the valley floor, the ball's motion changes most if it spends a lot of time where the bump is, and hardly at all if it rarely goes there. We will find the quantum version of exactly this statement: to first order, a level moves by the average of δV, weighted by the probability density $|\psi_n^{(0)}|^2$ of finding the particle at each place.

Why should you care, beyond convenience? Three reasons, all of which matter in the paper.

- Perturbation theory tells you *how sensitive* each level is to each part of the potential. That sensitivity is what an algorithm needs to fit a potential to a list of target levels.

- The second-order formula reveals a qualitative effect, *level repulsion*: two levels coupled by a perturbation are pushed apart.
- A perturbative formula is an *approximation* with a range of validity. Using it outside that range can produce numbers that look meaningful and are not. Sec. V of the paper is built around this danger.

Key Ideas

- The unperturbed problem $H_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$ is assumed solved.
- The perturbation $\delta V(x)$ changes the Hamiltonian to $H = H_0 + \delta V$; we want the level shifts δE_n.
- Perturbation theory expresses δE_n through the old levels, old wavefunctions and δV only.

13.2 Tools: matrix elements, Hermiticity and a bookkeeping parameter

13.2.1 Orthonormality and matrix elements

Recall from Chapter 10 that the eigenfunctions of a Hermitian Hamiltonian belonging to different energies are orthogonal, and that we normalize them. Together:

$$\int_{-L}^{L} \psi_n^{(0)*}(x)\,\psi_m^{(0)}(x)\,\mathrm{d}x = \delta_{nm}, \tag{13.4}$$

where δ_{nm} (the Kronecker delta) equals 1 if $n = m$ and 0 otherwise, and the star denotes the complex conjugate (Chapter 2). We integrate over $[-L, L]$ because all the problems in this chapter live in a box with walls at $x = \pm L$, where every wavefunction vanishes (Dirichlet walls, Chapter 11).

For any function $W(x)$ we define the *matrix element* of W between states n and m:

$$W_{nm} \equiv \langle n|W|m\rangle \equiv \int_{-L}^{L} \psi_n^{(0)*}(x)\,W(x)\,\psi_m^{(0)}(x)\,\mathrm{d}x. \tag{13.5}$$

The notation $\langle n|W|m\rangle$ is only a shorthand for the integral on the right; you never need more than that integral in this book. The numbers W_{nm}, for $n, m = 1, 2, \ldots$, form an (infinite) matrix, the matrix of W in the basis of unperturbed states. The diagonal entry

$$W_{nn} = \int_{-L}^{L} |\psi_n^{(0)}(x)|^2\,W(x)\,\mathrm{d}x$$

is the expectation value of W in state n (Chapter 10).

If $W(x)$ is real, the matrix is Hermitian. To see this, take the complex conjugate of Equation (13.5), using that the conjugate of a product is the product of the conjugates and that $W^* = W$:

$$\begin{aligned} W_{nm}^* &= \int_{-L}^{L} \psi_n^{(0)}(x)\,W(x)\,\psi_m^{(0)*}(x)\,\mathrm{d}x \\ &= \int_{-L}^{L} \psi_m^{(0)*}(x)\,W(x)\,\psi_n^{(0)}(x)\,\mathrm{d}x \quad \text{(reorder the factors)} \\ &= W_{mn}. \end{aligned} \tag{13.6}$$

In the problems of this book the unperturbed wavefunctions are real (one-dimensional bound states can always be chosen real; Chapter 11), so for a real W the matrix is real and symmetric: $W_{nm} = W_{mn}$.

13.2.2 Hermiticity of H_0, reviewed

We will use one property of H_0 repeatedly: for any two functions f and g that vanish at both walls,

$$\int_{-L}^{L} f^*\,(H_0 g)\,\mathrm{d}x = \int_{-L}^{L} (H_0 f)^*\,g\,\mathrm{d}x. \tag{13.7}$$

Recall from Chapter 10 that this is what "H_0 is Hermitian" means. Because the property is so central, here is the proof again. The potential part is immediate, because V is real: $f^* V g = (Vf)^* g$. For the kinetic part, integrate by parts twice:

$$\begin{aligned}
\int_{-L}^{L} f^*\,(-g'')\,\mathrm{d}x &= \Big[-f^* g'\Big]_{-L}^{L} + \int_{-L}^{L} f^{*\prime}\,g'\,\mathrm{d}x \quad \text{(parts, with } u = f^*,\ \mathrm{d}v = -g''\mathrm{d}x\text{)} \\
&= 0 + \Big[f^{*\prime} g\Big]_{-L}^{L} - \int_{-L}^{L} f^{*\prime\prime}\,g\,\mathrm{d}x \quad (f^* = 0 \text{ at walls; parts again}) \\
&= \int_{-L}^{L} (-f'')^*\,g\,\mathrm{d}x \quad (g = 0 \text{ at walls}).
\end{aligned}$$

Adding the potential part gives Equation (13.7).

13.2.3 Expanding in the unperturbed states

We will need to write an unknown function as a combination of the unperturbed wavefunctions,

$$f(x) = \sum_m a_m\,\psi_m^{(0)}(x), \qquad a_m = \int_{-L}^{L} \psi_m^{(0)*} f\,\mathrm{d}x. \tag{13.8}$$

The formula for the coefficients follows by multiplying the sum by $\psi_k^{(0)*}$, integrating, and using Equation (13.4): only the $m = k$ term survives. For the box, the $\psi_m^{(0)}$ are sines (see below), and Equation (13.8) is a Fourier sine series of the kind studied in Chapter 5.

Taken on trust

Completeness. That *every* reasonable function vanishing at the walls can be expanded as in Equation (13.8) (the eigenfunctions are "complete") is a theorem. For the box it is the completeness of the Fourier sine series. For a general confining potential it is part of Sturm–Liouville theory, treated in [22]. We take it on trust. In the computations of this chapter the question disappears, because we work with a finite number of states and check numerically that adding more states does not change the answer.

13.2.4 The box, our test bench

Recall from Chapter 11 that the infinite square well on $[-L, L]$ has

$$E_n^{(0)} = \Big(\frac{n\pi}{2L}\Big)^2, \qquad \psi_n^{(0)}(x) = \frac{1}{\sqrt{L}} \sin\Big(\frac{n\pi(x+L)}{2L}\Big), \qquad n = 1, 2, 3, \ldots \tag{13.9}$$

It is often convenient to use the shifted coordinate $u = x + L$, which runs from 0 to $2L$; then $\psi_n^{(0)} = \sin(n\pi u/2L)/\sqrt{L}$. Writing $\sin(n\pi x/2L + n\pi/2)$ shows that odd n gives $\pm\cos(n\pi x/2L)$ (even in x) and even n gives $\pm\sin(n\pi x/2L)$ (odd in x), the alternation of parity from Chapter 11. The spacing between neighbouring levels is

$$E_{n+1}^{(0)} - E_n^{(0)} = \Big(\frac{\pi}{2L}\Big)^2 [(n+1)^2 - n^2] = \Big(\frac{\pi}{2L}\Big)^2 (2n+1), \tag{13.10}$$

which *grows* with n. Keep this in mind; it will matter for validity.

13.2.5 A bookkeeping parameter

To organize the calculation, write the perturbation as

$$\delta V(x) = \varepsilon\, W(x), \tag{13.11}$$

where W is a fixed function and ε is a number we imagine turning up from zero. The parameter ε is pure bookkeeping: at the end we set $\varepsilon W = \delta V$ again. Its job is to let us sort terms by how many factors of the perturbation they contain.

Taken on trust

The power-series assumption. We will assume that, for small enough ε, each energy and wavefunction can be written as a power series in ε: $E_n = E_n^{(0)} + \varepsilon E_n^{(1)} + \varepsilon^2 E_n^{(2)} + \dots$ and $\psi_n = \psi_n^{(0)} + \varepsilon\psi_n^{(1)} + \varepsilon^2\psi_n^{(2)} + \dots$. For a finite matrix with a nondegenerate eigenvalue (no other eigenvalue equal to it) this is a theorem of analytic perturbation theory; for differential operators the conditions are more delicate. The theory, including when the series converges and when it does not, is in advanced texts; [30] gives the derivation at the level we use. We also use the fact that bound states of a one-dimensional confining potential are nondegenerate (a problem in [30] asks you to prove it), so no two $E_n^{(0)}$ are equal and the denominators $E_n^{(0)} - E_m^{(0)}$ below never vanish.

Key Ideas

- Matrix elements $W_{nm} = \int \psi_n^{(0)*} W \psi_m^{(0)}\, \mathrm{d}x$; for real W and real states the matrix is symmetric.
- H_0 is Hermitian, Equation (13.7), because of two integrations by parts and the walls.
- Any function vanishing at the walls can be expanded in the unperturbed states, Equation (13.8).
- The box, Equation (13.9), is our test bench; its spacing grows with n.

13.3 The first-order energy shift

13.3.1 Derivation

Insert the power series into the eigenvalue equation $(H_0 + \varepsilon W)\psi_n = E_n\psi_n$. We keep terms up to ε^2 because we will need the second order later. The left side is

$$\begin{aligned}(H_0 + \varepsilon W)(\psi_n^{(0)} + \varepsilon\psi_n^{(1)} + \varepsilon^2\psi_n^{(2)}) = {} & H_0\psi_n^{(0)} + \varepsilon(H_0\psi_n^{(1)} + W\psi_n^{(0)}) \\ & + \varepsilon^2(H_0\psi_n^{(2)} + W\psi_n^{(1)}) + \mathcal{O}(\varepsilon^3),\end{aligned}$$

and the right side is

$$\begin{aligned}(E_n^{(0)} + \varepsilon E_n^{(1)} + \varepsilon^2 E_n^{(2)})(\psi_n^{(0)} + \varepsilon\psi_n^{(1)} + \varepsilon^2\psi_n^{(2)}) = {} & E_n^{(0)}\psi_n^{(0)} + \varepsilon(E_n^{(0)}\psi_n^{(1)} + E_n^{(1)}\psi_n^{(0)}) \\ & + \varepsilon^2(E_n^{(0)}\psi_n^{(2)} + E_n^{(1)}\psi_n^{(1)} + E_n^{(2)}\psi_n^{(0)}) + \mathcal{O}(\varepsilon^3).\end{aligned}$$

These two expressions are equal for every small ε. A power series in ε that vanishes for all small ε has every coefficient zero: setting $\varepsilon = 0$ kills the constant term; dividing what remains by ε and setting $\varepsilon \to 0$ kills the next; and so on. So we may equate the coefficients of each power of ε separately.

Order ε^0: $H_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$, which is Equation (13.2), nothing new.

Order ε^1: moving terms so that the unknown $\psi_n^{(1)}$ sits on the left,

$$(H_0 - E_n^{(0)})\psi_n^{(1)} = (E_n^{(1)} - W)\psi_n^{(0)}. \tag{13.12}$$

Order ε^2:

$$(H_0 - E_n^{(0)})\psi_n^{(2)} = (E_n^{(1)} - W)\psi_n^{(1)} + E_n^{(2)}\psi_n^{(0)}. \tag{13.13}$$

Now the key trick: multiply Equation (13.12) by $\psi_k^{(0)*}$ (for some fixed k) and integrate over the box. The left side becomes, using the Hermiticity Equation (13.7) with $f = \psi_k^{(0)}$ and $g = \psi_n^{(1)}$ (both vanish at the walls: $\psi_n(\pm L) = 0$ for every ε, so by the coefficient argument just given every coefficient of its power series, including $\psi_n^{(1)}(\pm L)$, is zero),

$$\begin{aligned}\int \psi_k^{(0)*}(H_0 - E_n^{(0)})\psi_n^{(1)}\,\mathrm{d}x &= \int (H_0\psi_k^{(0)})^*\psi_n^{(1)}\,\mathrm{d}x - E_n^{(0)}\int \psi_k^{(0)*}\psi_n^{(1)}\,\mathrm{d}x \\ &= E_k^{(0)}\int \psi_k^{(0)*}\psi_n^{(1)}\,\mathrm{d}x - E_n^{(0)}\int \psi_k^{(0)*}\psi_n^{(1)}\,\mathrm{d}x \quad (E_k^{(0)} \text{ real}) \\ &= (E_k^{(0)} - E_n^{(0)})\,a_k,\end{aligned}$$

where we named the overlap

$$a_k \equiv \int_{-L}^{L} \psi_k^{(0)*}\,\psi_n^{(1)}\,\mathrm{d}x, \tag{13.14}$$

the coefficient of $\psi_k^{(0)}$ in the expansion of $\psi_n^{(1)}$, Equation (13.8). The right side of Equation (13.12) becomes, by Equations (13.4) and (13.5),

$$E_n^{(1)}\int \psi_k^{(0)*}\psi_n^{(0)}\,\mathrm{d}x - \int \psi_k^{(0)*}W\psi_n^{(0)}\,\mathrm{d}x = E_n^{(1)}\delta_{kn} - W_{kn}.$$

So for every k:

$$(E_k^{(0)} - E_n^{(0)})a_k = E_n^{(1)}\delta_{kn} - W_{kn}. \tag{13.15}$$

Choose $k = n$. The left side is zero (the bracket vanishes) and $\delta_{nn} = 1$, so

$$\boxed{E_n^{(1)} = W_{nn}} \qquad \text{and so} \qquad \delta E_n \approx \varepsilon W_{nn} = \int_{-L}^{L} |\psi_n^{(0)}(x)|^2\,\delta V(x)\,\mathrm{d}x. \tag{13.16}$$

This is the first-order energy shift. Notice that the derivation never used that W is real or Hermitian; only H_0 had to be Hermitian. We will return to this in the paper connection, where W is imaginary.

13.3.2 What it means

Because $|\psi_n^{(0)}|^2 \geq 0$ and integrates to 1, Equation (13.16) says: to first order, *a level moves by the average value of δV, weighted by the probability density of the unperturbed state.* A bump placed where the particle is likely to be moves the level; a bump where the particle never goes does not. Classically, a particle oscillating in a potential well moves slowest near its turning points (first-year physics: kinetic energy is smallest there), so it spends more time per unit length there than anywhere else. For a smoothly varying potential one expects, by analogy with this time-of-stay argument, that in a state of high n the density $|\psi_n^{(0)}|^2$, averaged over its rapid oscillations, is likewise largest near the classical turning points. We do not need to derive this here; it is the subject of Chapter 12. This is a heuristic statement about smooth potentials, not a theorem. It does not apply to the box, whose walls are vertical and whose averaged density $1/(2L)$ is uniform. We will meet this "a level probes the potential near its turning points" idea again when we discuss the paper.

Checkpoint 13.1

Suppose $\delta V(x) \geq 0$ everywhere in the box. Can any level move down at first order?

Answer: No. The first-order shift is $\int |\psi_n^{(0)}|^2 \delta V \, dx$, an integral of a product of two non-negative functions, so it is ≥ 0 for every level.

Sample Problem 13.1: A constant shift of the potential

Add a constant $\delta V(x) = c$ to the box potential. Find the first-order shift of every level, and compare with the exact answer.

Step 1: First order. By Equation (13.16), $\delta E_n^{(1)} = \int |\psi_n^{(0)}|^2 c \, dx = c \int |\psi_n^{(0)}|^2 dx = c$, using normalization.

Step 2: Exact. With $\delta V = c$, the equation $(H_0 + c)\psi = E\psi$ is the same as $H_0\psi = (E - c)\psi$. Its solutions are the old $\psi_n^{(0)}$ with $E - c = E_n^{(0)}$, that is $E_n = E_n^{(0)} + c$.

Step 3: Compare. First order gives the exact answer. All higher-order corrections must vanish; we will see in Section 13.5 that they do, because the off-diagonal matrix elements $c \int \psi_n^{(0)*} \psi_m^{(0)} dx = c\,\delta_{nm}$ are zero.

$$\boxed{\delta E_n = c \text{ for every } n, \text{ exactly.}}$$

The constant function is special: it moves every level by the same amount and changes no wavefunction. In the paper's cosine basis it is the $k = 0$ mode, $\cos(0) = 1$.

13.3.3 Cosine perturbations of the box

The paper's corrections are built from cosines $\cos(k\pi x/L)$, so let us compute the first-order shift of box level n due to

$$\delta V(x) = c \cos\Big(\frac{q\pi x}{L}\Big), \qquad q = 1, 2, 3, \ldots$$

Use $u = x + L$. Then

$$\cos\Big(\frac{q\pi x}{L}\Big) = \cos\Big(\frac{q\pi u}{L} - q\pi\Big) = \cos\Big(\frac{q\pi u}{L}\Big)\cos(q\pi) + \sin\Big(\frac{q\pi u}{L}\Big)\sin(q\pi) = (-1)^q \cos\Big(\frac{q\pi u}{L}\Big),$$

because $\cos(q\pi) = (-1)^q$ and $\sin(q\pi) = 0$. The density, with $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ and $\theta = n\pi u/2L$, is

$$|\psi_n^{(0)}|^2 = \frac{1}{L}\sin^2\Big(\frac{n\pi u}{2L}\Big) = \frac{1}{2L}\Big[1 - \cos\Big(\frac{n\pi u}{L}\Big)\Big].$$

Therefore

$$\int_{-L}^{L} |\psi_n^{(0)}|^2 \cos\Big(\frac{q\pi x}{L}\Big) dx = \frac{(-1)^q}{2L}\left[\int_0^{2L} \cos\Big(\frac{q\pi u}{L}\Big) du - \int_0^{2L} \cos\Big(\frac{n\pi u}{L}\Big)\cos\Big(\frac{q\pi u}{L}\Big) du\right]. \quad (13.17)$$

The first integral is $[\frac{L}{q\pi}\sin(q\pi u/L)]_0^{2L} = 0$, since $\sin(2q\pi) = 0$. For the second, use $\cos a \cos b = \frac{1}{2}[\cos(a-b) + \cos(a+b)]$: the product becomes $\frac{1}{2}[\cos((n-q)\pi u/L) + \cos((n+q)\pi u/L)]$. Each cosine with a nonzero integer multiple of $\pi u/L$ integrates to zero over $[0, 2L]$ for the same reason. Since $n \geq 1$ and $q \geq 1$, the multiple $n + q \geq 2$ is never zero, so the $(n+q)$ term always vanishes, and only the $(n-q)$ term with $n - q = 0$ survives, and it gives $\frac{1}{2} \cdot 2L = L$. The second integral is thus $L\,\delta_{nq}$ (this is the orthogonality of cosines from Chapter 5). Putting it together,

$$\int_{-L}^{L} |\psi_n^{(0)}|^2 \cos\Big(\frac{q\pi x}{L}\Big) dx = \frac{(-1)^q}{2L}(0 - L\,\delta_{nq}) = \frac{(-1)^{q+1}}{2}\,\delta_{nq}. \quad (13.18)$$

This is a striking result. In the box, the cosine with index q shifts *only* level $n = q$ at first order, by $+c/2$ if q is odd and $-c/2$ if q is even. Every other level is untouched.

Why? The density $|\psi_n^{(0)}|^2$ is a constant plus a ripple $\cos(n\pi u/L)$. A cosine perturbation averages to zero against the constant part, and against the ripple unless its wavelength matches. Only the perturbation that "resonates" with the ripple of level n moves level n. Figure 13.1 shows the matching and non-matching cases.

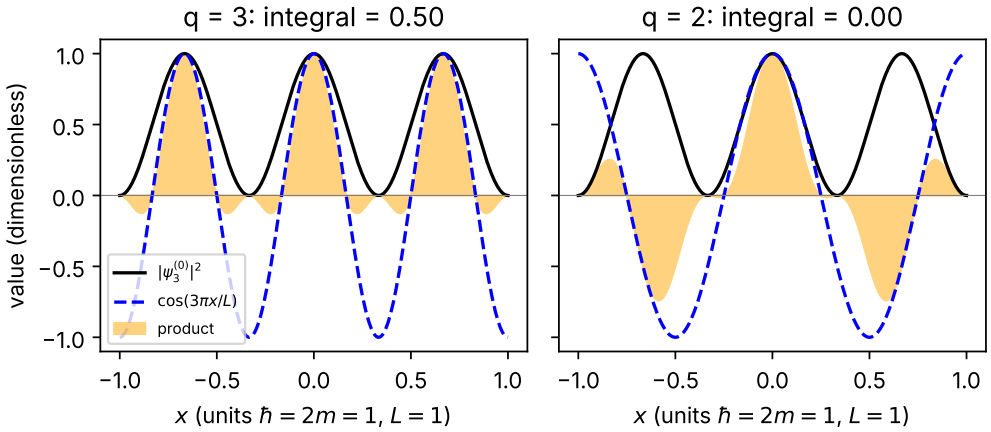


Figure 13.1: The first-order shift is an overlap integral. Both panels show the density $|\psi_3^{(0)}|^2$ of the third box state ($L = 1$), a cosine perturbation, and their product (shaded). Left: $\cos(3\pi x/L)$ has the same ripple as the density, the product is mostly positive, and the integral is $1/2$. Right: $\cos(2\pi x/L)$ does not match, positive and negative areas cancel, and the integral is 0, as Equation (13.18) predicts.

Sample Problem 13.2: One cosine in the running-example box

Take the box of running example RE3, $L = 44.883$, and add $\delta V = c\cos(3\pi x/L)$ with $c = 10^{-3}$. Find the first-order shift of level 3, express it as a fraction of the spacing to level 4, and state what happens to levels 1, 2 and 4.

Step 1: Unperturbed numbers. $\pi/(2L) = \pi/89.766 = 0.0349976$, so $(\pi/2L)^2 = 1.22483 \times 10^{-3}$. Then $E_3^{(0)} = 9(\pi/2L)^2 = 0.0110235$ and, by Equation (13.10), $E_4^{(0)} - E_3^{(0)} = 7(\pi/2L)^2 = 0.0085738$.

Step 2: First-order shift. By Equation (13.18) with $q = n = 3$ (odd), $\delta E_3^{(1)} = +c/2 = 5.0 \times 10^{-4}$.

Step 3: Compare with the spacing. $5.0 \times 10^{-4}/0.0085738 = 0.058$ of a spacing.

Step 4: Other levels. For $n \neq 3$ the first-order shift is zero.

$$\boxed{\delta E_3^{(1)} = 5.0 \times 10^{-4} \approx 0.058 \text{ spacings}; \quad \delta E_{1,2,4}^{(1)} = 0.}$$

In Section 13.5 we will find that the second-order correction to level 3 is -2.83×10^{-6}, less than 1% of the first-order shift; an exact computation (Section 13.6) agrees with first plus second order to better than 10^{-7}.

13.3.4 When parity kills the first order

If the potential V is even, $V(-x) = V(x)$, every state is either even or odd (Chapter 11), so the density $|\psi_n^{(0)}(x)|^2$ is even in x. If δV is odd, $\delta V(-x) = -\delta V(x)$, the integrand of

Equation (13.16) is odd, and the integral over the symmetric interval $[-L, L]$ vanishes: the part from $[-L, 0]$ cancels the part from $[0, L]$.

Checkpoint 13.2

A uniform force field adds $\delta V = Fx$ to the symmetric box (a "tilt"). What are the first-order shifts of all levels?

Answer: All zero. $|\psi_n^{(0)}|^2$ is even and Fx is odd, so $\int |\psi_n^{(0)}|^2 Fx\,\mathrm{d}x = 0$. The levels do move, but only at second order, as Section 13.5 shows.

Key Ideas

- First-order shift: $\delta E_n^{(1)} = \int |\psi_n^{(0)}|^2 \delta V\,\mathrm{d}x$, Equation (13.16), a density-weighted average of δV.
- A constant $\delta V = c$ shifts every level by exactly c.
- In the box, $\cos(q\pi x/L)$ shifts only level q at first order, by $(-1)^{q+1}c/2$, Equation (13.18).
- In an even potential, an odd δV gives zero first-order shift for every level.

13.4 The sensitivity matrix and linearization

★ Did you know? *An Undergraduate's Theorem*

The Hellmann-Feynman theorem says that if a Hamiltonian depends on a parameter λ, then $dE/d\lambda = \langle\psi|\partial H/\partial\lambda|\psi\rangle$. Richard Feynman derived it in his 1939 undergraduate thesis at MIT on forces in molecules, and Hans Hellmann had published it in 1937. It gives exactly the sensitivity of an energy level to a change in the potential.

13.4.1 Many basis functions at once

In the paper the correction is not one cosine but a sum,

$$\delta V(x) = \sum_{k=0}^{M-1} c_k\,\phi_k(x), \tag{13.19}$$

where the ϕ_k are fixed *basis functions* (by default $\phi_k = \cos(k\pi x/L)$), M is the number of them, and the real numbers c_k are the coefficients we may adjust. The first-order shift Equation (13.16) is linear in δV: the integral of a sum is the sum of the integrals. Hence

$$\begin{aligned}\delta E_n^{(1)} &= \int_{-L}^{L} |\psi_n^{(0)}|^2 \sum_{k=0}^{M-1} c_k\phi_k\,\mathrm{d}x = \sum_{k=0}^{M-1}\left(\int_{-L}^{L} |\psi_n^{(0)}|^2\phi_k\,\mathrm{d}x\right) c_k \\ &= \sum_{k=0}^{M-1} A_{nk}\,c_k,\end{aligned} \tag{13.20}$$

with the *sensitivity matrix*

$$A_{nk} = \int_{-L}^{L} |\psi_n(x)|^2\,\phi_k(x)\,\mathrm{d}x. \tag{13.21}$$

Row n of A tells you how level n responds to each basis function; column k tells you how basis function k moves every level. If we collect the shifts into a vector $\delta\boldsymbol{E}$ and the coefficients into a

vector $\boldsymbol{c}$ (Chapter 7), Equation (13.20) is the matrix equation $\delta\boldsymbol{E} \approx A\boldsymbol{c}$. Replacing the true, complicated dependence of the levels on $\boldsymbol{c}$ by this linear one is called *linearization*. Its error is of second order in $\boldsymbol{c}$, the subject of the next section.

Two facts hold for any normalized state:

- The constant basis function $\phi_0 = 1$ has $A_{n0} = \int |\psi_n|^2 \mathrm{d}x = 1$ for every n: column 0 of A is all ones.
- For the box with cosines, Equation (13.18) gives

$$A^{\text{box}}_{nk} = \delta_{k0} + \frac{(-1)^{k+1}}{2}\,\delta_{nk} \qquad (k \geq 0,\ n \geq 1), \tag{13.22}$$

  a column of ones plus a diagonal of $\pm\frac{1}{2}$.

Checkpoint 13.3

In the box with the $M = 10$ cosines $k = 0, \ldots, 9$, which levels can be moved *individually* at first order, and how does level 50 respond?

Answer: Levels 1 to 9: level $n \leq 9$ is the only level that responds to c_n. Level 50 responds only to c_0, which moves all levels together, so at first order $\delta E_{50} = c_0$.

13.4.2 A first look at "memorizing"

Sample Problem 13.3: Tuning three box levels with three cosines

Box with $L = 1$; basis $\phi_0 = 1$, $\phi_1 = \cos(\pi x/L)$, $\phi_2 = \cos(2\pi x/L)$. Find c_0, c_1, c_2 such that, at first order, levels 1, 2, 3 shift by 0.01, -0.02 and 0.005. Then predict, at first order, the shift of level 4.

Step 1: The sensitivity matrix. By Equation (13.22), with rows $n = 1, 2, 3$ and columns $k = 0, 1, 2$:

$$A = \begin{pmatrix} 1 & \frac{1}{2} & 0 \\ 1 & 0 & -\frac{1}{2} \\ 1 & 0 & 0 \end{pmatrix}.$$

Step 2: Write the equations. $A\boldsymbol{c} = \delta\boldsymbol{E}$ reads

$$c_0 + \tfrac{1}{2}c_1 = 0.01, \qquad c_0 - \tfrac{1}{2}c_2 = -0.02, \qquad c_0 = 0.005.$$

Step 3: Solve. The third equation gives $c_0 = 0.005$. Then $c_1 = 2(0.01 - 0.005) = 0.01$ and $c_2 = -2(-0.02 - 0.005) = 0.05$.

Step 4: Predict level 4. Row 4 of A^{box} is $(1, 0, 0)$, so $\delta E^{(1)}_4 = c_0 = 0.005$, and the same holds for every $n \geq 3$.

$$\boxed{c_0 = 0.005,\ c_1 = 0.01,\ c_2 = 0.05; \qquad \delta E^{(1)}_4 = 0.005.}$$

An exact computation (Section 13.6) confirms the shifts of levels 1 to 4 to within 10^{-4}. Notice what the fit has learned about level 4: nothing that came from levels 1 and 2. Whatever pattern you ask levels 1 to 3 to follow, level 4 receives only the common offset c_0. The coefficients reproduce the three targets perfectly and carry no information about the next level. This is the simplest possible picture of a model that memorizes its training data; Chapter 25 develops the idea in general.

13.4.3 The sensitivity matrix is the exact derivative

Equation (13.20) is a statement about small $\boldsymbol{c}$ near $\boldsymbol{c} = 0$. But nothing in the derivation required the unperturbed problem to be the box. Suppose we are at some coefficient vector $\boldsymbol{c}$, with Hamiltonian $H(\boldsymbol{c}) = H_0 + \sum_k c_k \phi_k$, levels $E_n(\boldsymbol{c})$ and normalized states $\psi_n(\boldsymbol{c})$. Increase one coefficient, $c_k \to c_k + \eta$. Relative to $H(\boldsymbol{c})$ this is a perturbation $\eta\phi_k$, and Equation (13.16), applied with $H(\boldsymbol{c})$ as the "unperturbed" Hamiltonian, gives

$$E_n(\boldsymbol{c} + \eta\,\hat{\boldsymbol{e}}_k) = E_n(\boldsymbol{c}) + \eta \int |\psi_n(\boldsymbol{c})|^2 \phi_k \,\mathrm{d}x + \mathcal{O}(\eta^2),$$

where $\hat{\boldsymbol{e}}_k$ is the vector with 1 in place k and zeros elsewhere. Subtract $E_n(\boldsymbol{c})$, divide by η, and let $\eta \to 0$:

$$\left.\frac{\partial E_n}{\partial c_k}\right|_{\boldsymbol{c}} = \int_{-L}^{L} |\psi_n(\boldsymbol{c})|^2 \,\phi_k \,\mathrm{d}x = A_{nk}(\boldsymbol{c}). \tag{13.23}$$

This is the *Hellmann–Feynman theorem*: the derivative of an eigenvalue with respect to a parameter in the Hamiltonian is the expectation value of the derivative of the Hamiltonian. (Problem 9 asks you to derive it directly, without perturbation theory.) The symbol $\partial/\partial c_k$ is a partial derivative: the rate of change with respect to c_k with all other coefficients held fixed. The matrix $A_{nk}(\boldsymbol{c})$ evaluated with the *current* wavefunctions is therefore not an approximation at all: it is the exact Jacobian (matrix of first derivatives) of the levels with respect to the coefficients. This is why it is the natural tool for fitting, Section 13.7.

13.4.4 Computing A on a grid

On a computer, wavefunctions are represented by their values $\psi_n(x_j)$ on grid points x_j with spacing h (Chapter 8; full details in Chapter 23). Integrals become sums, $\int f\,\mathrm{d}x \approx \sum_j f(x_j)h$. The following listing builds the finite-difference Hamiltonian, finds the first 100 eigenpairs with `eigh_tridiagonal`, and forms A as one matrix product.

```
import numpy as np
from scipy.linalg import eigh_tridiagonal

L, N, nlev, M = 44.883, 3000, 100, 100
h = 2 * L / (N + 1)                          # grid spacing
x = -L + h * np.arange(1, N + 1)             # interior grid points
V = np.zeros(N)                              # the box; put any V(x) here
d = 2 / h**2 + V                             # diagonal
e = -np.ones(N - 1) / h**2                   # off-diagonal
E, psi = eigh_tridiagonal(d, e, select='i',
                          select_range=(0, nlev - 1))
psi = psi / np.sqrt(h)                       # now sum(psi**2) * h = 1
k = np.arange(M)
phi = np.cos(np.outer(x, k) * np.pi / L)     # phi[j, k] = cos(k pi x_j / L)
A = (psi**2 * h).T @ phi                     # A[n-1, k]
print(A[0, 0], A[2, 3], A[2, 2])             # about 1, 0.5 and 0
```

Pitfall

`eigh_tridiagonal` returns eigenvectors normalized so that $\sum_j \psi_j^2 = 1$, the normalization of a vector. A wavefunction needs $\sum_j \psi_j^2 h = 1$, the discrete version of $\int |\psi|^2 \mathrm{d}x = 1$. Forgetting the division by $\sqrt{h}$ multiplies every A_{nk} by h, so the constant column is h instead of 1. Every predicted shift is then too small by the factor h, and any step computed from A is too large by the factor $1/h$. The paper states this normalization explicitly (App. A, step 9: "ψ normalized $\sum \psi^2 dx = 1$").

Figure 13.2 compares the sensitivity matrix of the box with that of the Wu–Sprung potential. That potential (defined by the paper's Eq. (2) and derived in Chapter 22) is even, has its minimum $V_0 = 3.10073\pi$ at $x = 0$, and rises to $V = 1000$ at $x = \pm 44.883$. All you need here is that its low levels are confined to a central region much narrower than the box: level 20, near $E \approx 77$, turns at $|x| \approx 5.4$, less than an eighth of L. Over that region $\cos(k\pi x/L) \approx 1$ for the first few k, so those columns behave almost like the constant column for low levels: for example $A_{20,1} > 0.9$. The clean diagonal of the box is gone; level 20 has sizeable sensitivity (above 0.1) to at least three different cosines with $k \geq 1$, and each cosine moves many levels.

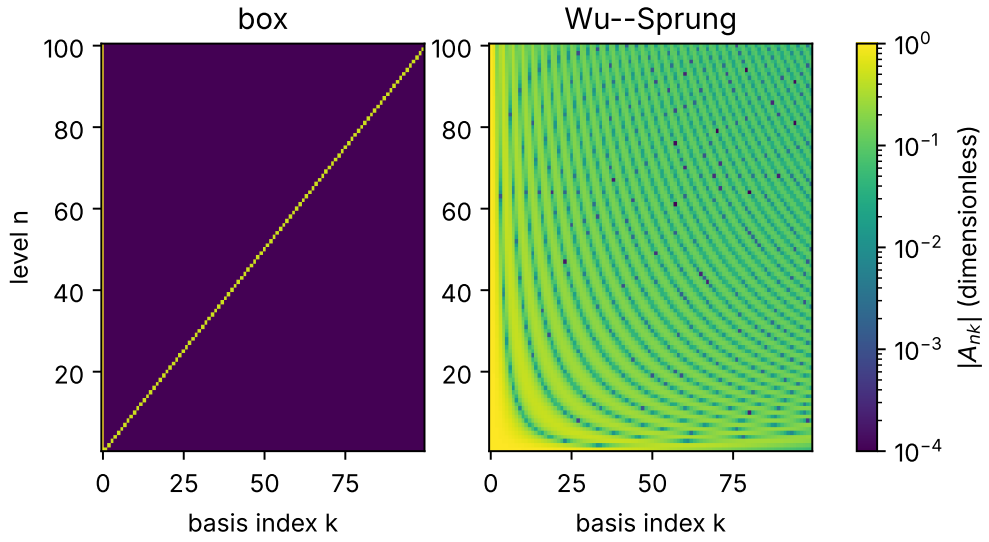


Figure 13.2: Magnitude $|A_{nk}|$ of the first-order sensitivity matrix, Equation (13.21), for cosine basis functions $\cos(k\pi x/L)$, $L = 44.883$, levels $n = 1$–100, computed on a 3000-point finite-difference grid (logarithmic color scale). Left: the box, where A is a column of ones ($k = 0$) plus a diagonal of magnitude $1/2$, Equation (13.22). Right: the Wu–Sprung potential, where each level responds to a broad band of cosines and low cosines act almost like the constant mode for low levels.

Key Ideas

- For $\delta V = \sum_k c_k \phi_k$, the first-order shifts are linear in $\boldsymbol{c}$: $\delta \boldsymbol{E} \approx A\boldsymbol{c}$, with $A_{nk} = \int |\psi_n|^2 \phi_k \, \mathrm{d}x$, Equations (13.20) and (13.21).
- $A_{n0} = 1$ for the constant mode; in the box with cosines, A is otherwise diagonal.
- Evaluated with the current wavefunctions, A_{nk} is the exact derivative $\partial E_n/\partial c_k$ (Hellmann–Feynman, Equation (13.23)).
- On a grid, $A = (\psi^2 h)^\top \phi$ with $\sum_j \psi_j^2 h = 1$.

13.5 Second order and level repulsion

13.5.1 The first-order wavefunction

Return to Equation (13.15) with $k \neq n$. Then $\delta_{kn} = 0$ and

$$a_k = \frac{W_{kn}}{E_n^{(0)} - E_k^{(0)}} \qquad (k \neq n). \tag{13.24}$$

(We divided by $E_k^{(0)} - E_n^{(0)}$ and moved the minus sign into the denominator.) This determines every coefficient of $\psi_n^{(1)}$ except a_n. To fix a_n, use the freedom in normalization: any constant multiple of an eigenfunction is still an eigenfunction, so we may choose the scale of ψ_n by requiring $\int \psi_n^{(0)*}\psi_n \,\mathrm{d}x = 1$ exactly, for all ε. Inserting the series, this says $1 + \varepsilon a_n + \mathcal{O}(\varepsilon^2) = 1$, so $a_n = 0$ (and likewise the overlap of $\psi_n^{(0)}$ with $\psi_n^{(2)}$ vanishes). This choice, called intermediate normalization, does not affect any energy: the energies are fixed by the eigenvalue equation, not by the scale of ψ_n. If a normalized wavefunction is needed, we rescale at the end. Hence, by Equation (13.8),

$$\psi_n^{(1)} = \sum_{k \neq n} \frac{W_{kn}}{E_n^{(0)} - E_k^{(0)}} \psi_k^{(0)}. \tag{13.25}$$

The perturbation mixes a little of every other state into state n; the amount is large when the coupling W_{kn} is large and when the two levels are close.

13.5.2 Derivation of the second-order shift

Multiply the order-ε^2 equation Equation (13.13) by $\psi_n^{(0)*}$ and integrate. The left side is $(E_n^{(0)} - E_n^{(0)}) \int \psi_n^{(0)*}\psi_n^{(2)}\mathrm{d}x = 0$, by exactly the Hermiticity argument used for Equation (13.15) with $k = n$. The right side gives three terms:

$$\begin{aligned} 0 &= E_n^{(1)} \int \psi_n^{(0)*}\psi_n^{(1)}\mathrm{d}x - \int \psi_n^{(0)*} W \psi_n^{(1)}\mathrm{d}x + E_n^{(2)} \int \psi_n^{(0)*}\psi_n^{(0)}\mathrm{d}x \\ &= E_n^{(1)} a_n - \int \psi_n^{(0)*} W \psi_n^{(1)}\mathrm{d}x + E_n^{(2)} \quad \text{(by Equations (13.4) and (13.14))} \\ &= -\int \psi_n^{(0)*} W \psi_n^{(1)}\mathrm{d}x + E_n^{(2)} \quad (a_n = 0). \end{aligned}$$

So $E_n^{(2)} = \int \psi_n^{(0)*} W \psi_n^{(1)}\mathrm{d}x$. Substitute Equation (13.25) and exchange the sum and the integral (each term is a finite integral; for an infinite sum this exchange is part of the trust placed in the power-series assumption):

$$E_n^{(2)} = \sum_{k \neq n} \frac{W_{kn}}{E_n^{(0)} - E_k^{(0)}} \int \psi_n^{(0)*} W \psi_k^{(0)}\mathrm{d}x = \sum_{k \neq n} \frac{W_{nk}\, W_{kn}}{E_n^{(0)} - E_k^{(0)}}. \tag{13.26}$$

This general form needs only H_0 to be Hermitian. If W is also real, Equation (13.6) gives $W_{nk} = W_{kn}^*$, so $W_{nk}W_{kn} = |W_{kn}|^2$. Restoring $\delta V = \varepsilon W$ and writing $\delta V_{nm} = \varepsilon W_{nm} = \langle n|\delta V|m\rangle$ (many books write simply V_{nm}), the energy through second order is

$$\boxed{E_n \approx E_n^{(0)} + \delta V_{nn} + \sum_{m \neq n} \frac{|\delta V_{nm}|^2}{E_n^{(0)} - E_m^{(0)}}}. \tag{13.27}$$

Pitfall

Watch the order in the denominator: it is $E_n^{(0)} - E_m^{(0)}$, the level you are computing minus the other level. Reversing it flips the sign of every term. And the term $m = n$ is excluded; including it would divide by zero.

13.5.3 Level repulsion

Each term of the sum in Equation (13.27) has a non-negative numerator. When the numerator is nonzero, the sign of the term is the sign of the denominator:

- a level m *above* n ($E_m^{(0)} > E_n^{(0)}$) contributes a negative term: it pushes level n down;

- a level m *below* n contributes a positive term: it pushes level n up.

Coupled levels push each other apart, and close levels push hardest. This is *level repulsion*. Its statistical consequences for large spectra, including the Riemann zeros, are the subject of Chapter 20.

Checkpoint 13.4

What is the sign of the second-order shift of the ground state $n = 1$, whatever the (real) perturbation?

Answer: Negative or zero. Every other level lies above the ground state, so every denominator $E_1^{(0)} - E_m^{(0)}$ is negative while every numerator $|\delta V_{1m}|^2$ is non-negative.

The cleanest way to see repulsion is the two-level system, which we can solve exactly. Recall from Chapter 8 the eigenvalues of a 2×2 Hermitian matrix:

$$H = \begin{pmatrix} E_1 & v \\ v^* & E_2 \end{pmatrix}, \qquad E_1 < E_2.$$

The characteristic equation $\det(H - \lambda I) = 0$ is $(E_1 - \lambda)(E_2 - \lambda) - |v|^2 = 0$, that is

$$\lambda^2 - (E_1 + E_2)\lambda + E_1 E_2 - |v|^2 = 0.$$

By the quadratic formula,

$$\lambda_\pm = \frac{E_1 + E_2}{2} \pm \sqrt{\frac{(E_1 + E_2)^2}{4} - E_1 E_2 + |v|^2} = \frac{E_1 + E_2}{2} \pm \sqrt{\Delta^2 + |v|^2}, \tag{13.28}$$

where $\Delta = (E_2 - E_1)/2 > 0$ and we used $\tfrac{1}{4}(E_1 + E_2)^2 - E_1 E_2 = \tfrac{1}{4}(E_2 - E_1)^2$. (This Δ is only a local name for the half-gap; it has nothing to do with the grid bias Δ_n of the paper, Chapter 23.) When $|v| \ll \Delta$, write $\sqrt{\Delta^2 + |v|^2} = \Delta\sqrt{1 + y}$ with $y = |v|^2/\Delta^2 \ll 1$ and use $\sqrt{1+y} \approx 1 + y/2$ (Chapter 4; the neglected term is of order y^2):

$$\sqrt{\Delta^2 + |v|^2} \approx \Delta + \frac{|v|^2}{2\Delta} = \frac{E_2 - E_1}{2} + \frac{|v|^2}{E_2 - E_1}.$$

Then

$$\lambda_- \approx E_1 - \frac{|v|^2}{E_2 - E_1}, \qquad \lambda_+ \approx E_2 + \frac{|v|^2}{E_2 - E_1},$$

which is exactly Equation (13.27) for two levels (the diagonal of the perturbation is zero here; $\delta V_{12} = v$). The exact formula shows something the expansion does not: as you vary the diagonal energies so that the two levels approach each other, the gap $\lambda_+ - \lambda_- = 2\sqrt{\Delta^2 + |v|^2}$ never falls below $2|v|$. The levels *avoid crossing* (Figure 13.3).

Sample Problem 13.4: Two levels, exact and perturbative

Take $E_1 = 1$, $E_2 = 3$, $v = 0.5$. Compute the exact eigenvalues and the second-order estimates.

Step 1: Exact. $\Delta = (3 - 1)/2 = 1$, so by Equation (13.28) $\lambda_\pm = 2 \pm \sqrt{1 + 0.25} = 2 \pm 1.118034$, that is 0.881966 and 3.118034.

Step 2: Second order. $|v|^2/(E_2 - E_1) = 0.25/2 = 0.125$, so $\lambda_- \approx 1 - 0.125 = 0.875$ and $\lambda_+ \approx 3 + 0.125 = 3.125$.

Step 3: Compare. The errors are 0.0070, about 6% of the shift. Here $|v|/\Delta = 0.5$ is not very small, and the neglected term in $\sqrt{1+y}$ (of relative size $y/4 = 1/16$) is visible.

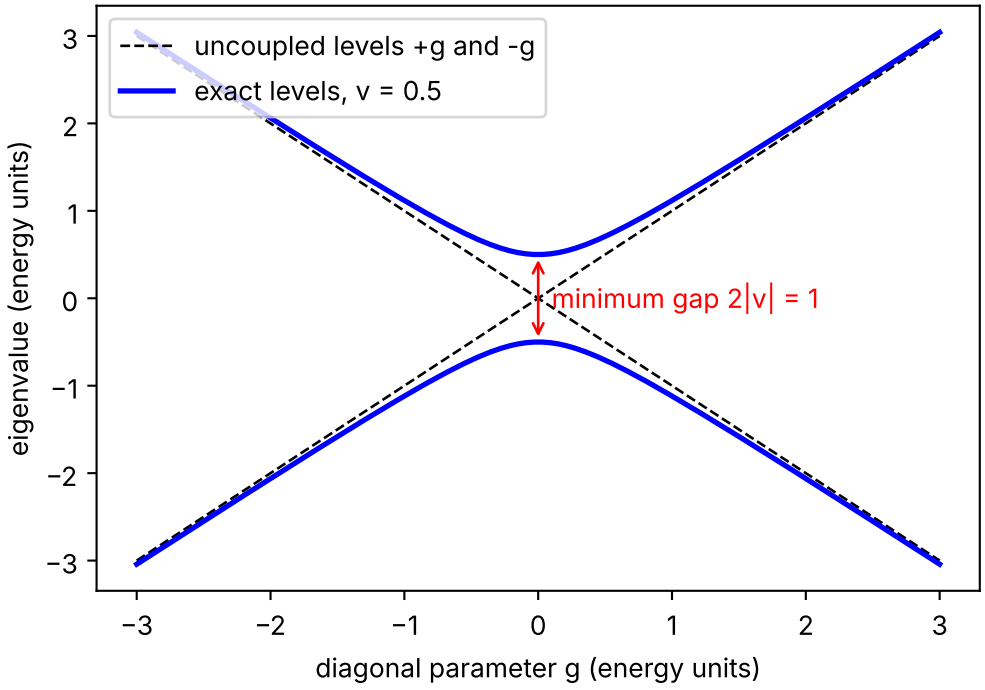


Figure 13.3: Level repulsion in the two-level system whose matrix has diagonal entries g and $-g$ and off-diagonal entries v; the parameter g is varied. Dashed: the uncoupled levels $\pm g$, which cross at $g = 0$. Solid: the exact eigenvalues $\pm\sqrt{g^2 + v^2}$ for $v = 0.5$, which avoid crossing with minimum gap $2|v| = 1$. Far from the crossing the coupled levels approach the uncoupled ones, pushed apart by $v^2/(2|g|)$ each, as second-order perturbation theory predicts.

exact 0.881966, 3.118034;	second order 0.875, 3.125.

13.5.4 The box with a tilt

To do the tilt we need the matrix elements $x_{nm} = \int \psi_n^{(0)} x\, \psi_m^{(0)} \mathrm{d}x$ of the box. With $u = x + L$ and the box width $\ell = 2L$, $\psi_n^{(0)} = \sqrt{2/\ell}\,\sin(n\pi u/\ell)$ and $x = u - L$, so

$$x_{nm} = \frac{2}{\ell}\int_0^\ell u\,\sin\frac{n\pi u}{\ell}\sin\frac{m\pi u}{\ell}\,\mathrm{d}u - L\,\delta_{nm}.$$

We need one integral, for a nonzero integer j; integrating by parts with $U = u$ and $\mathrm{d}w = \cos(j\pi u/\ell)\mathrm{d}u$, so that $w = \frac{\ell}{j\pi}\sin(j\pi u/\ell)$:

$$\begin{aligned}\int_0^\ell u\cos\frac{j\pi u}{\ell}\mathrm{d}u &= \Big[u\,\frac{\ell}{j\pi}\sin\frac{j\pi u}{\ell}\Big]_0^\ell - \frac{\ell}{j\pi}\int_0^\ell \sin\frac{j\pi u}{\ell}\mathrm{d}u \\ &= 0 + \frac{\ell}{j\pi}\Big[\frac{\ell}{j\pi}\cos\frac{j\pi u}{\ell}\Big]_0^\ell = \Big(\frac{\ell}{j\pi}\Big)^2[(-1)^j - 1].\end{aligned} \tag{13.29}$$

Now let $n \neq m$ and use $\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$:

$$x_{nm} = \frac{1}{\ell}\left[\Big(\frac{\ell}{(n-m)\pi}\Big)^2((-1)^{n-m}-1) - \Big(\frac{\ell}{(n+m)\pi}\Big)^2((-1)^{n+m}-1)\right].$$

Since $n-m$ and $n+m$ differ by the even number $2m$, $(-1)^{n-m} = (-1)^{n+m}$. If $n+m$ is even both brackets vanish and $x_{nm} = 0$. If $n+m$ is odd both brackets equal -2, and

$$x_{nm} = -\frac{2\ell}{\pi^2}\left[\frac{1}{(n-m)^2} - \frac{1}{(n+m)^2}\right] = -\frac{2\ell}{\pi^2}\,\frac{4nm}{(n^2-m^2)^2},$$

because $(n+m)^2 - (n-m)^2 = 4nm$ and $(n-m)^2(n+m)^2 = (n^2-m^2)^2$. With $\ell = 2L$:

$$x_{nm} = \begin{cases} -\dfrac{16Lnm}{\pi^2(n^2-m^2)^2}, & n+m \text{ odd}, \\ 0, & n+m \text{ even}. \end{cases} \tag{13.30}$$

The diagonal x_{nn} vanishes by parity (Section 13.3). States of the same parity are not coupled by the odd function x; states of opposite parity are.

Sample Problem 13.5: The quadratic shift of a tilted box

Box with $L = 1$, $\delta V = Fx$ with $F = 1$. Find the second-order shift of the ground state and compare with an exact computation.

Step 1: Which states couple. By Equation (13.30), $n = 1$ couples only to even m: $x_{1m} = -16m/[\pi^2(m^2-1)^2]$. For $m = 2$, $x_{12} = -32/(9\pi^2) = -0.36025$.

Step 2: Denominators. $E_1^{(0)} - E_m^{(0)} = (\pi/2L)^2(1-m^2) = -\frac{\pi^2}{4}(m^2-1)$ for $L = 1$.

Step 3: Assemble. By Equation (13.27),

$$E_1^{(2)} = F^2\sum_{m\ \text{even}}\frac{x_{1m}^2}{E_1^{(0)} - E_m^{(0)}} = -F^2\sum_{m\ \text{even}}\frac{256m^2}{\pi^4(m^2-1)^4}\cdot\frac{4}{\pi^2(m^2-1)} = -F^2\frac{1024}{\pi^6}\,S,$$

with $S = \sum_{m=2,4,6,\ldots} m^2/(m^2-1)^5$. The terms fall off fast: $m = 2$ gives $4/243 = 0.016461$, $m = 4$ gives 2.1×10^{-5}, and the full sum is $S = 0.016483$. (A closed form for S exists; see the box after this Sample Problem. We use only the numerical value.) Since

$1024/\pi^6 = 1.06513$, $E_1^{(2)} = -0.017556\,F^2$.

Step 4: Compare. First order predicts no change. Second order predicts $E_1 \approx 2.467401 - 0.017556 = 2.449845$. Exact diagonalization (the method of Section 13.6) gives 2.44987: the two differ by less than 3×10^{-5}.

$$\boxed{E_1^{(2)} = -0.017556\,F^2 \quad (L=1); \qquad E_1(F{=}1) = 2.44987 \text{ (exact diagonalization).}}$$

The ground state moves down, as every ground state must at second order, and quadratically in F (the shift does not care which way the box is tilted). In atomic physics the analogous effect is the quadratic Stark effect.

Taken on trust

A closed form for S. The sum $S = \sum_{m=2,4,6,\ldots} m^2/(m^2-1)^5$ equals $(15-\pi^2)\pi^2/3072$. Deriving this requires partial fractions together with known values of sums of inverse powers of integers, techniques this book does not develop (see [22]). We take the closed form on trust; Problem 22 checks it numerically, and nothing in this chapter depends on it.

13.5.5 The box with a cosine, at second order

For $\delta V = c\cos(q\pi x/L)$ we also need off-diagonal elements. With $\theta = \pi u/2L$ (which runs from 0 to π as u runs over the box), $\psi_n^{(0)}\psi_m^{(0)} = \frac{1}{L}\sin n\theta \sin m\theta = \frac{1}{2L}[\cos(n-m)\theta - \cos(n+m)\theta]$ and $\cos(q\pi x/L) = (-1)^q \cos 2q\theta$. We need one integral over $[0,\pi]$, for integers $j \geq 0$ and $l \geq 1$:

$$\int_0^\pi \cos j\theta \cos l\theta \,\mathrm{d}\theta = \frac{\pi}{2}\,\delta_{jl}.$$

To see it, product-to-sum turns the integrand into $\frac{1}{2}[\cos(j-l)\theta + \cos(j+l)\theta]$. A cosine $\cos k\theta$ with a nonzero integer k integrates over $[0,\pi]$ to $\sin(k\pi)/k = 0$. Since $j + l \geq 1$, the second term always gives zero; the first gives zero unless $j = l$, in which case it gives $\frac{1}{2}\pi$. If $j = 0$ the result is 0, because $l \geq 1$ means $j \neq l$. Now compute, with $\mathrm{d}u = (2L/\pi)\mathrm{d}\theta$ and $\cos(n-m)\theta = \cos(|n-m|\theta)$:

$$\begin{aligned}
\langle n|\cos\tfrac{q\pi x}{L}|m\rangle &= \int_0^{2L} \frac{1}{2L}\big[\cos(|n-m|\theta) - \cos((n+m)\theta)\big](-1)^q\cos(2q\theta)\,\mathrm{d}u \\
&= \frac{(-1)^q}{2L}\cdot\frac{2L}{\pi}\left[\int_0^\pi \cos(|n-m|\theta)\cos(2q\theta)\,\mathrm{d}\theta - \int_0^\pi \cos((n+m)\theta)\cos(2q\theta)\,\mathrm{d}\theta\right] \\
&= \frac{(-1)^q}{\pi}\left[\frac{\pi}{2}\,\delta_{|n-m|,2q} - \frac{\pi}{2}\,\delta_{n+m,2q}\right].
\end{aligned}$$

In the last line we used the integral above with $l = 2q \geq 2$, and $j = |n-m|$ in the first term (this includes the diagonal case $n = m$, $j = 0$, which gives 0) and $j = n+m$ in the second. Hence

$$\langle n|\cos\tfrac{q\pi x}{L}|m\rangle = \frac{(-1)^q}{2}\Big[\delta_{|n-m|,2q} - \delta_{n+m,2q}\Big] \qquad (q \geq 1). \tag{13.31}$$

So the cosine with index q couples level n to the levels $n \pm 2q$ (with $+\frac{1}{2}(-1)^q$) and to the level $2q - n$ (with $-\frac{1}{2}(-1)^q$). Setting $m = n$ recovers Equation (13.18): only $n = q$ has a diagonal element.

Sample Problem 13.6: Ground state with $\cos(\pi x/L)$

Box with $L = 1$, $\delta V = c\cos(\pi x/L)$, $c = 1$. Compute the ground-state energy at first and second order, and compare with the exact value.

Step 1: Matrix elements. With $q = 1$, Equation (13.31): $\delta V_{11} = -\frac{1}{2}(-1)c = c/2$ (from $n+m=2$), and $\delta V_{13} = \frac{1}{2}(-1)c = -c/2$ (from $|n-m|=2$). No other m couples to $n = 1$.

Step 2: First order. $E_1 \approx \pi^2/4 + c/2 = 2.467401 + 0.5 = 2.967401$.

Step 3: Second order. $E_1^{(0)} - E_3^{(0)} = \frac{\pi^2}{4}(1-9) = -2\pi^2$, so the shift is $(c^2/4)/(-2\pi^2) = -c^2/(8\pi^2) = -0.012665$ (for general L, $-c^2L^2/8\pi^2$). Thus $E_1 \approx 2.954736$.

Step 4: Exact. Exact diagonalization gives $E_1 = 2.954413$.

first order 2.967401,	second order 2.954736,	exact 2.954413.

The first-order error is 1.3×10^{-2}, the second-order error 3.2×10^{-4}. Most of the remainder has a simple origin: the first-order shift $+c/2$ moved level 1 up, closer to level 3, so the true denominator is nearer to $E_1^{(0)} + c/2 - E_3^{(0)}$. Using that denominator gives 2.954407, within 10^{-5} of the exact value. (Systematically, this is part of the third-order correction, which we do not derive.)

Returning to the RE3 box sample: for $q = n = 3$, Equation (13.31) couples level 3 to level 9 with $\delta V_{39} = -c/2$, and $E_3^{(0)} - E_9^{(0)} = (\pi/2L)^2(9-81) = -0.088188$, so the second-order shift is $(c/2)^2/(-0.088188) = -2.83\times10^{-6}$ for $c = 10^{-3}$, as promised.

Key Ideas

- First-order wavefunction: $\psi_n^{(1)} = \sum_{k\neq n} W_{kn}\psi_k^{(0)}/(E_n^{(0)} - E_k^{(0)})$, Equation (13.25).
- Second-order shift: $\sum_{m\neq n} \delta V_{nm}\delta V_{mn}/(E_n^{(0)} - E_m^{(0)})$, which equals $\sum_{m\neq n} |\delta V_{nm}|^2/(E_n^{(0)} - E_m^{(0)})$ for real δV, Equations (13.26) and (13.27).
- Levels above push down, levels below push up: level repulsion; the ground state always moves down at second order.
- Box matrix elements: x_{nm} in Equation (13.30), cosines in Equation (13.31).

13.6 Exact diagonalization and the validity condition

13.6.1 The matrix method

How did we get the "exact" numbers above? Expand the unknown eigenfunction of $H = H_0 + \delta V$ in the unperturbed states, $\psi = \sum_m a_m\psi_m^{(0)}$ (Equation (13.8)). Insert into $H\psi = E\psi$:

$$\sum_m a_m(E_m^{(0)} + \delta V)\psi_m^{(0)} = E\sum_m a_m\psi_m^{(0)}.$$

Multiply by $\psi_k^{(0)*}$ and integrate; by Equations (13.4) and (13.5),

$$\sum_m \Big(E_m^{(0)}\delta_{km} + \delta V_{km}\Big)a_m = E\,a_k \qquad \text{for every } k. \tag{13.32}$$

This is a matrix eigenvalue problem (Chapter 8) for the matrix $H_{km} = E_k^{(0)}\delta_{km} + \delta V_{km}$, the unperturbed energies on the diagonal plus the matrix of the perturbation. It is exact but

infinite. We keep the first N_b states and diagonalize the $N_b \times N_b$ matrix; then we repeat with a larger N_b and accept the answer when it no longer changes. Truncation works well here because a state far above level n couples to it through a large denominator. This approach is often called the Galerkin or Rayleigh–Ritz method. For the box with cosines we know every matrix element in closed form, Equation (13.31), so the method is easy:

```
import numpy as np
L, Nb = 1.0, 60                     # box half-width, number of sine states
n = np.arange(1, Nb + 1)
E0 = (n * np.pi / (2 * L))**2       # unperturbed levels
D = np.abs(n[:, None] - n[None, :]) # |n - m|
S = n[:, None] + n[None, :]         # n + m

def cos_matrix(q):
    """<n|cos(q pi x / L)|m> for q >= 1, Eq. (cosmat)."""
    return 0.5 * (-1)**q * ((D == 2*q) * 1.0 - (S == 2*q) * 1.0)

c = 1.0
H = np.diag(E0) + c * cos_matrix(1)
E = np.linalg.eigvalsh(H)           # exact levels, sorted
first  = E0[0] + c / 2
second = first - c**2 * L**2 / (8 * np.pi**2)
print(E[0], first, second)          # 2.954413  2.967401  2.954736
```

With $N_b = 60$ the ground-state energy agrees with $N_b = 200$ to many digits; the CHECKS script for this chapter also confirms it independently with a finite-difference grid (the method of Chapter 23).

Checkpoint 13.5

Why does the matrix of Equation (13.32) come out real and symmetric for the box with a real δV, and why does that matter for the solver?

Answer: The unperturbed states are real and δV is real, so $\delta V_{km} = \delta V_{mk}$ is real (Equation (13.6)); the diagonal $E_k^{(0)}$ is real. A real symmetric matrix has real eigenvalues and orthogonal eigenvectors (Chapter 8), so we can use the fast, stable symmetric solver `eigvalsh`.

13.6.2 When does perturbation theory work?

Figure 13.4 plots the ground-state energy of the $L = 1$ box with $\delta V = c\cos(\pi x/L)$ as c grows. For small c the three curves agree; second order hugs the exact curve much longer than first order; and eventually both depart.

Where does the expansion break down? Look at the first-order wavefunction Equation (13.25): it mixes state k into state n with amplitude $\delta V_{kn}/(E_n^{(0)} - E_k^{(0)})$. The whole expansion assumes the perturbed state is *mostly* the unperturbed one, so these amplitudes must be small. The same ratio controls the size of each second-order term compared with the coupling itself: $|\delta V_{nm}|^2/|E_n^{(0)} - E_m^{(0)}| = |\delta V_{nm}| \times$ (mixing amplitude), and each higher order brings one more such factor. This gives the practical *validity condition*:

$$|\delta V_{nm}| \ll |E_n^{(0)} - E_m^{(0)}| \qquad \text{for all } m \neq n, \text{ in particular for the nearest coupled levels.} \tag{13.33}$$

Status: this is a heuristic rule of thumb, not a theorem with a sharp threshold; whether and how fast the full series converges is a harder question (see the box in Section 13.2). The Sample Problem "Ground state with $\cos(\pi x/L)$" adds one refinement: diagonal shifts δV_{nn} that are

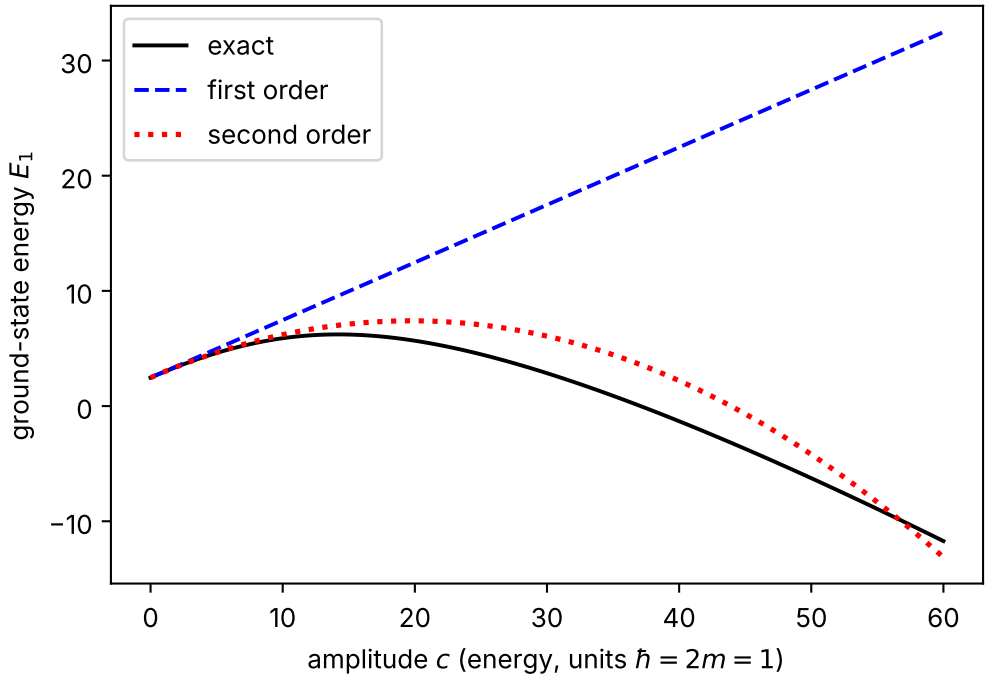


Figure 13.4: Ground-state energy of the box ($L = 1$) with $\delta V = c\cos(\pi x/L)$, as a function of the amplitude c. Solid: exact diagonalization, Equation (13.32). Dashed: first order, $E_1^{(0)} + c/2$. Dotted: second order, $E_1^{(0)} + c/2 - c^2/8\pi^2$. The coupled level is $E_3^{(0)}$, at a distance $2\pi^2 \approx 19.7$; perturbation theory is accurate while the coupling $c/2$ is small compared with that distance and degrades as $c/2$ approaches it.

comparable to spacings change the effective gaps, so they should be small too. In words: *the perturbation must be small compared with the spacing between the levels it couples.*

Sample Problem 13.7: How large may the cosine be?

For the box and $\delta V = c\cos(\pi x/L)$, estimate the largest c for which perturbation theory is reliable for level $n \geq 3$, and for $n = 1$.

Step 1: Couplings. By Equation (13.31) with $q = 1$, level $n \geq 3$ couples to $n-2$ and $n+2$ with $|\delta V| = c/2$ (the partner $2-n$ does not exist).

Step 2: Nearest coupled distance. $E_n^{(0)} - E_{n-2}^{(0)} = (\pi/2L)^2[n^2 - (n-2)^2] = 4(n-1)(\pi/2L)^2$, smaller than $E_{n+2}^{(0)} - E_n^{(0)} = 4(n+1)(\pi/2L)^2$.

Step 3: Apply Equation (13.33). $c/2 \ll 4(n-1)(\pi/2L)^2$, that is $c \ll 8(n-1)(\pi/2L)^2$. For $n = 1$ the only partner is level 3, at distance $8(\pi/2L)^2$, so $c \ll 16(\pi/2L)^2$, which for $L = 1$ is $c \ll 4\pi^2 = 39.5$.

$$\boxed{c \ll 8(n-1)\Big(\frac{\pi}{2L}\Big)^2 \ (n \geq 3), \qquad c \ll 16\Big(\frac{\pi}{2L}\Big)^2 \ (n = 1).}$$

In the box, higher levels are *more* robust, because the spacing grows with n, Equation (13.10). The figure is consistent with this. A condition written with "$\ll$" sets no sharp threshold, so it cannot single out a breakdown point. What exact diagonalization shows is that the first-order error reaches 0.1 near $c = 3$ and the second-order error reaches 0.1 near $c = 7$, both well below 40, as "$\ll$" requires.

Checkpoint 13.6

In the box the spacing grows with n. In the Wu–Sprung potential the spacing *shrinks* slowly with n (its levels follow the Riemann zeros on average, which crowd together as they rise). For a perturbation of fixed size, which levels are most fragile in each case?

Answer: In the box, the low levels (smallest spacings). In the Wu–Sprung potential, the high levels, because there a fixed coupling becomes a larger fraction of the shrinking spacing.

13.6.3 Many levels at once

Figure 13.5 tests first-order perturbation theory on 100 levels at once, as a fit would use it. A random combination of 20 cosines is added to the RE3 box; the left panel uses a small amplitude, the right panel one 100 times larger. We plot the exact shift (from Equation (13.32), levels matched by their order) against the first-order prediction $\sum_k A_{nk}c_k$, both in units of the local level spacing. At small amplitude the points lie on the diagonal. At large amplitude many points, especially for the low levels whose spacing is smallest, scatter far from it: the linear model no longer describes the operator. The paper's Fig. 5 (left) shows the same kind of plot for its second-order surrogate, and the same kind of breakdown.

Pitfall

Truncated state sums. In a computer, the second-order sum $\sum_{m\neq n}$ runs over a finite set of states. Terms from distant states are small, but terms from *nearby* states are the largest of all. If level n sits near the top of the states you kept, its nearest neighbours above are missing, and the truncated sum is badly biased. A drastic example is Problem 14: for box level 2 under a tilt ($L = 1$), the full second-order sum is $+0.0052F^2$ (Problem 8; the terms $m = 1, 3, 5, 7$ already give this to four decimals), but keeping only states 1 and

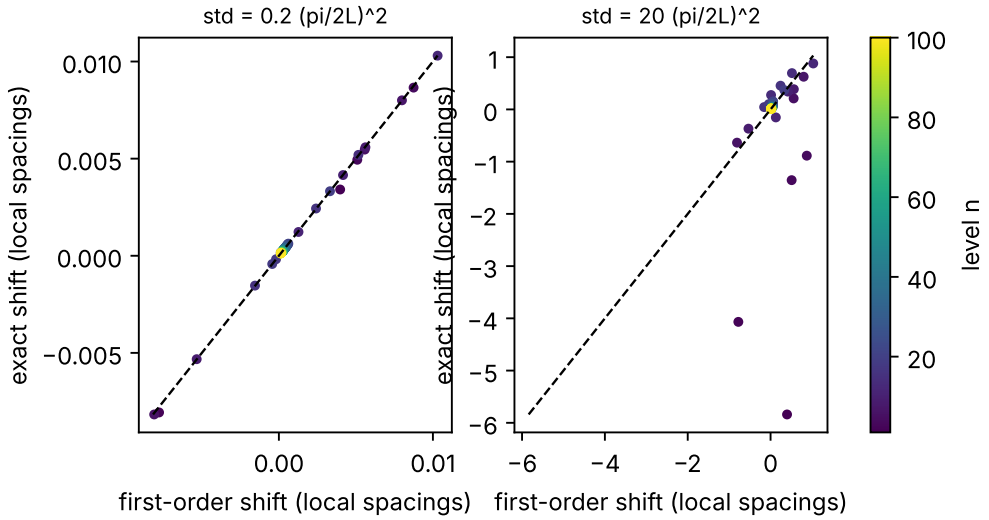


Figure 13.5: Exact level shifts versus first-order predictions for levels $n = 1$–100 of the box $L = 44.883$ perturbed by $\delta V = \sum_{k=0}^{19} c_k \cos(k\pi x/L)$, in units of the local spacing $(\pi/2L)^2(2n+1)$, Equation (13.10). The coefficients are drawn at random (with `numpy.random.default_rng(0)`) from a bell-shaped (Gaussian) distribution; the same draws are used in both panels, scaled to a typical size (standard deviation, Chapter 19) of $0.2(\pi/2L)^2$ (left) and $20(\pi/2L)^2$ (right). Exact shifts come from Equation (13.32) with 300 box states. Color encodes n. Left: the points lie on the diagonal. Right: the first-order model fails, most strongly for the lowest levels, whose spacing is smallest.

2 drops the downward push of level 3 and gives $+0.0175F^2$, more than three times too large. The paper meets exactly this issue, Section 13.8.

Key Ideas

- Exact levels come from diagonalizing $H_{km} = E_k^{(0)}\delta_{km} + \delta V_{km}$ in a truncated basis, Equation (13.32), checking convergence in N_b.
- Perturbation theory is reliable when couplings (and diagonal shifts) are small compared with the spacings of the coupled levels, Equation (13.33); this is a heuristic rule.
- Spacings decide fragility: in the box low levels are fragile, in a potential with shrinking spacings high levels are.
- Truncated state sums bias levels near the top of the kept set.

13.7 Fitting by iterated linearization

★ Did you know? ***Damping the Steps***

Pure Gauss-Newton fitting can overshoot when the linear approximation is poor. In 1944 Kenneth Levenberg proposed blending each step with a cautious gradient step, and in 1963 Donald Marquardt refined the idea. The Levenberg-Marquardt algorithm is still the default nonlinear least-squares solver in many scientific software libraries.

13.7.1 Turning the question around

So far we asked: given δV, how do the levels move? A fit asks the reverse: given target values $\tilde{t}_n$ for the levels, which coefficients $\boldsymbol{c}$ make $E_n(\boldsymbol{c}) = \tilde{t}_n$? The relation between $\boldsymbol{c}$ and the levels is nonlinear, so we cannot solve it in one step. But we know its exact derivative, the sensitivity matrix Equation (13.23). That is precisely what Newton's method needs.

13.7.2 One parameter: Newton's method

Recall from Chapter 4 that Newton's method solves $f(c) = 0$ by replacing f with its tangent line at the current guess c_j and jumping to where the tangent crosses zero: $c_{j+1} = c_j - f(c_j)/f'(c_j)$. To make level n equal a target $\tilde{t}$, take $f(c) = E_n(c) - \tilde{t}$; its derivative is $A_{nk}(c)$ by Equation (13.23). Writing the residual $r = \tilde{t} - E_n(c_j)$,

$$c_{j+1} = c_j + \frac{r}{A_{nk}(c_j)}. \tag{13.34}$$

Sample Problem 13.8: Fitting one level with one cosine

Box with $L = 1$, $\delta V = c\cos(\pi x/L)$. Find c such that the ground state has energy $\tilde{t} = 3$, by Newton's method starting at $c = 0$.

Step 1: First step. At $c = 0$: $E_1 = \pi^2/4 = 2.467401$ and $A_{11} = +\frac{1}{2}$ (Equation (13.18)). The residual is $r = 3 - 2.467401 = 0.532599$, so $c_1 = 0 + 0.532599/0.5 = 1.065198$. This is pure first-order perturbation theory: it would hit the target exactly if the levels were linear in c.

Step 2: Evaluate exactly. Exact diagonalization at c_1 gives $E_1(c_1) = 2.985239$, short of the target by $r = 0.014761$, because the second-order shift (always downward for the ground state) was ignored.

Step 3: Recompute the sensitivity. With the *new* ground-state wavefunction, $A_{11}(c_1) = \int |\psi_1(c_1)|^2 \cos(\pi x/L)\mathrm{d}x = 0.4719$. It is smaller than $\frac{1}{2}$ because the perturbation has mixed some of level 3 into the ground state.

Step 4: Second step. $c_2 = 1.065198 + 0.014761/0.4719 = 1.0965$.

Step 5: Check. At c_2, the exact ground-state energy is 3.0000 to four decimal places.

$$\boxed{c \approx 1.0965 \text{ after two Newton steps; } E_1 = 3.0000.}$$

Figure 13.6 shows the two tangent lines. The first step undershoots in c by about 3%, the second step is already within 10^{-4} in energy: the quadratic convergence of Newton's method from Chapter 4.

```
import numpy as np
L, Nb = 1.0, 60
n = np.arange(1, Nb + 1)
E0 = (n * np.pi / (2 * L))**2
D = np.abs(n[:, None] - n[None, :]); S = n[:, None] + n[None, :]
C1 = -0.5 * ((D == 2) * 1.0 - (S == 2) * 1.0)   # <n|cos(pi x/L)|m>

def ground(c):
    w, v = np.linalg.eigh(np.diag(E0) + c * C1)
    return w[0], v[:, 0] @ C1 @ v[:, 0]          # E_1 and dE_1/dc

c, target = 0.0, 3.0
for step in range(4):
    E1, A11 = ground(c)
    print(step, c, E1, A11)
```

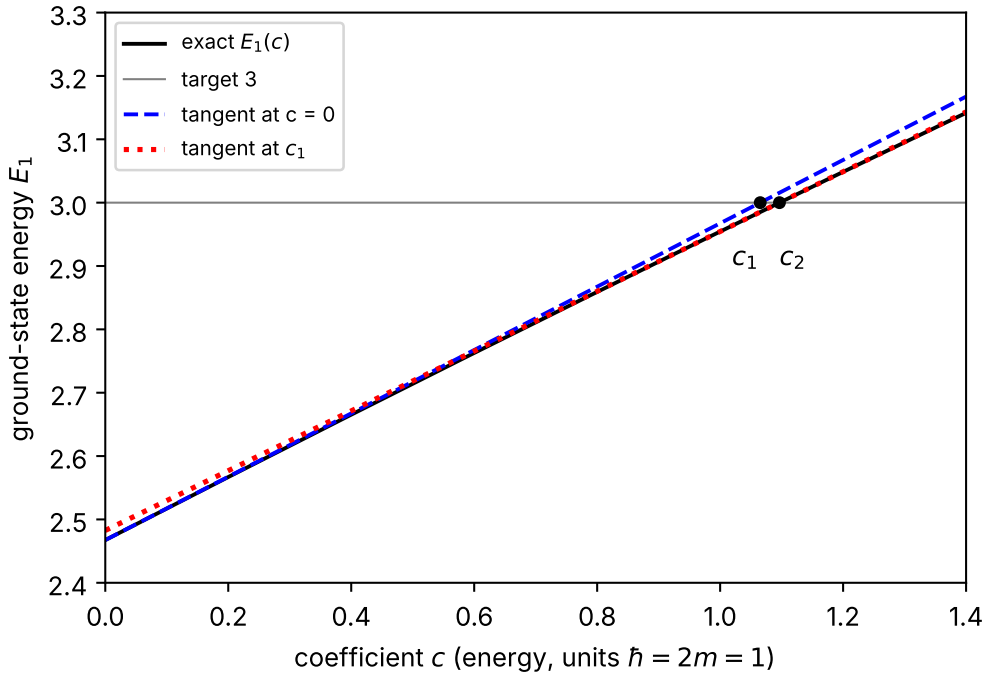


Figure 13.6: Newton's method for fitting one level. Solid: the exact ground-state energy $E_1(c)$ of the box ($L = 1$) with $\delta V = c\cos(\pi x/L)$. Horizontal line: the target 3. Dashed: the tangent at $c = 0$, whose slope is the first-order sensitivity $A_{11} = 1/2$; it crosses the target at $c_1 = 1.0652$. Dotted: the tangent at c_1, slope $A_{11}(c_1) = 0.4719$, crossing the target at $c_2 = 1.0965$, where E_1 already equals 3 to four decimals.

```
    c = c + (target - E1) / A11                    # Newton update
```

The derivative in the listing, $v^\top C_1 v$, is Equation (13.21) written in the basis of box states: if $\psi_1 = \sum_m a_m \psi_m^{(0)}$ then $\int |\psi_1|^2 \cos(\pi x/L)\mathrm{d}x = \sum_{m,l} a_m a_l \langle m|\cos\frac{\pi x}{L}|l\rangle$.

13.7.3 Many parameters: the Gauss–Newton idea

With many levels and many coefficients the same idea becomes a loop:

1. At the current $\boldsymbol{c}$, solve the eigenvalue problem of $H(\boldsymbol{c})$ *exactly* (numerically): levels $E_n(\boldsymbol{c})$ and states $\psi_n(\boldsymbol{c})$.
2. Form the residuals $r_n = \tilde{t}_n - E_n(\boldsymbol{c})$.
3. Form the sensitivity matrix $A_{nk}(\boldsymbol{c}) = \int |\psi_n(\boldsymbol{c})|^2 \phi_k \,\mathrm{d}x$.
4. Find a step $\Delta\boldsymbol{c}$ that makes the *linearized* levels hit the targets: $\sum_k A_{nk}\Delta c_k \approx r_n$.
5. Set $\boldsymbol{c} \leftarrow \boldsymbol{c} + \Delta\boldsymbol{c}$ and return to step 1.

If the number of target levels equals the number of coefficients and A is invertible, step 4 is a square linear system (Chapter 7), and the loop is Newton's method in many dimensions. In practice there are more levels than coefficients, or fewer, and step 4 becomes a least-squares problem (Chapter 9); with that change the loop is called the Gauss–Newton method. When a full step is too bold (the linearization fails over the step's length), one shortens or damps it; Chapter 24 develops this into the Levenberg–Marquardt method that the paper uses.

Checkpoint 13.7

Why recompute A at every iteration instead of using the A of the unperturbed potential throughout?

Answer: Because $A_{nk}(\boldsymbol{c})$ is the exact derivative only at the current $\boldsymbol{c}$. As $\boldsymbol{c}$ grows the wavefunctions change (in the sample, A_{11} dropped from 0.5 to 0.4719), and using stale derivatives makes the steps aim at the wrong place; with the current A the square (Newton) case converges quadratically near the solution, and the least-squares case converges fast when the final residuals are small.

13.7.4 Why this works, and what it does not need

Notice what the loop asks of perturbation theory. It does *not* require the final δV to be small compared with level spacings. Each step only needs the linearization to be good over the length of that *step*, and step 1 evaluates the levels exactly, so an inaccurate step is detected immediately and can be rejected or shortened. Perturbation theory is used as a *step proposer*, never as the *model*. The final coefficients can describe a correction far outside the perturbative regime of Equation (13.33), and the levels reported are still the true eigenvalues of the fitted operator.

The opposite strategy is to write the levels *as* the perturbative formula, for example $E_n \approx E_n^{(0)} + \sum_k A_{nk} c_k +$ (second-order terms), and to optimize that formula directly. This is cheaper (no eigensolve per step), but nothing stops the optimizer from choosing coefficients for which the formula no longer describes the operator being fitted. The formula then reports fits that the actual operator does not achieve. Sec. V of the paper is an example, discussed next.

Key Ideas

- Fitting reverses perturbation theory: choose $\boldsymbol{c}$ to hit target levels.
- The sensitivity matrix is the exact Jacobian, so Newton (one level) and Gauss–Newton (many levels) iterations use it directly.
- Iterated linearization with exact eigensolves needs perturbation theory to hold only for each step; optimizing a perturbative formula as the model needs it to hold everywhere the optimizer goes.

13.8 Paper connection

Throughout this section we use the paper's scoring vocabulary. "rms" is the root-mean-square error, quoted in units of the local mean spacing of the zeros. "corr" is the Pearson correlation (Chapter 19) between two lists of numbers, for example predicted and true shifts: a number between -1 and $+1$ that measures whether the two lists rise and fall together. The "null" is the prediction with no fitted correction. "In-sample" means evaluated on the levels used in the fit; "out-of-sample" or "held-out" means evaluated on levels that were not used. An "unregularized" fit is one without the extra penalty on large coefficients (ridge regularization) introduced in Chapter 9.

The sensitivity matrix of Sec. II C. The paper's Hermitian fit perturbs the Wu–Sprung potential, $V \to V + \delta V$ with $\delta V(x) = \sum_{k=0}^{M-1} c_k \cos(k\pi x/L)$, and fits the coefficients "by iterated linearization: solve for (E_n, ψ_n); form residuals $r_n = \tilde{t}_n - E_n$ and the first-order sensitivity $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\, dx$; update c by SVD-damped least squares with a Levenberg–Marquardt schedule, accepting only steps that improve the rms" (Sec. II C). Every ingredient is in this

chapter: A_{nk} is Equation (13.21); evaluated with the current eigenfunctions it is the exact Jacobian, Equation (13.23); and the loop is the one of Section 13.7, with step 4 solved by damped least squares (App. A, step 6, explained line by line in Chapter 24). The targets $\tilde{t}_n$ are the Riemann zeros corrected for the grid bias of the finite-difference solver, a correction explained in Chapter 23. App. A, step 4, records that the eigenpairs come from `eigh_tridiagonal`, so the normalization pitfall of Section 13.4 applies.

Exact spectrum at every step. Because each iteration begins with an exact eigensolve and accepts only improving steps, the Hermitian fit uses perturbation theory as a step proposer only. This matters, because the fitted corrections are not small. In the interpolation split of Sec. III, the fitted correction has "rms $\delta V = 40$" in the band of turning points of the held-out levels and "a maximum $|\delta V| = 199$ where the smooth potential is about 500" (Sec. III, Fig. 4). For comparison, near levels 201–300 the mean spacing of the zeros is about 1.46 (it is the quantity $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$ of Sec. II A, studied in Chapter 17). A single linearization from $\boldsymbol{c} = 0$ is plausibly inadequate at that size (our rough estimate from Equation (13.33), not a statement of the paper; strictly, the condition concerns the matrix elements δV_{nm}, which can be smaller than the rms of δV, so the comparison is only indicative); the paper's scores are nonetheless computed from actual numerical eigenvalues of the fitted (discretized) operators: each iteration begins with an exact eigensolve, and the held-out score uses "a single plain eigensolve of the fitted potential" (Sec. II C).

A level probes the potential near its turning points. The first-order formula says a level responds to δV where $|\psi_n|^2$ is large, and Section 13.3 noted that for high levels the density is largest near the classical turning points. The paper uses this in Sec. III: "Position maps to level index through the classical turning point, and levels 201–300 turn at $x = 22.5$–28.5," which is exactly where the unregularized fit placed its largest structure. Chapter 12 makes the turning-point picture quantitative.

The coverage law. Sec. III reports that the fit reaches rms 0.186 on all 500 zeros at $M = 300$ and below 10^{-4} spacings by $M = 900$, "consistent with a basis-coverage law (the cosine family becomes effectively complete for joint level shifts once $M\pi/L > 2\sqrt{E}$)". The box gives an exact miniature of this law. In the box $2\sqrt{E_n} = n\pi/L$, so $M\pi/L > 2\sqrt{E_n}$ means $M > n$: by Equation (13.22), level n can be moved independently only if the cosine with $k = n$ is in the basis. Reading the condition as $E < (M\pi/2L)^2$ with the scale-1 half-width $L = 44.883$ (the paper's coverage numbers refer to scale 1) gives $E < 110$ for $M = 300$, $E < 196$ for $M = 400$, $E < 441$ for $M = 600$, and $E < 992$ for $M = 900$; the 500th zero is at $t_{500} = 811.18$ (App. A, key anchors), inside the last range only. This is consistent with the paper's numbers, but remember the paper calls it a law with which the results are "consistent", and Chapter 24 derives it only heuristically. Figure 13.2 also shows why the Wu–Sprung problem is less tidy than the box: low cosines act almost like the constant mode for low levels, so several columns of A are nearly alike, the kind of near-redundancy that makes a linear system ill-conditioned (Chapter 9).

The second-order formula of Eq. (5). Sec. II E perturbs $H \to H + iA$, where $A = \sum_k c_k \phi_k$ is a real function and i is the imaginary unit. Our general results Equations (13.16) and (13.26) needed only H_0 to be Hermitian, so they apply with $W = iA$. First order gives iA_{nn}, purely imaginary, so the leading *real* shift is second order:

$$\sum_{m \neq n} \frac{(iA)_{nm}(iA)_{mn}}{E_n - E_m} = i^2 \sum_{m \neq n} \frac{A_{nm}A_{mn}}{E_n - E_m} = -\sum_{m \neq n} \frac{|A_{nm}|^2}{E_n - E_m},$$

which is the paper's Eq. (5); the sign is opposite to the Hermitian Equation (13.27) because $i^2 = -1$. Since $A_{nm} = \sum_k c_k (\phi_k)_{nm}$ with $(\phi_k)_{nm} = \langle n|\phi_k|m\rangle$, the shift is a quadratic form in $\boldsymbol{c}$: $-\boldsymbol{c}^\top T^{(n)} \boldsymbol{c}$ with $T^{(n)}_{kl} = \sum_{m\neq n} (\phi_k)_{nm}(\phi_l)_{nm}/(E_n - E_m)$. Chapter 14 works all of this out carefully, including decay and PT symmetry. Two details already make sense here. (a) The class the paper calls PT-odd (the name is explained in Chapter 14; here it only matters that its basis functions are odd in x) uses $\phi_k = \sin((k+1)\pi x/L)$, odd functions, and the paper notes "$A_{nn} = 0$": this is the parity argument of Section 13.3, exactly as for our tilt. (b) The state sum runs "over all 652 basis states with $E \leq 1000$ (truncating at 500 states measurably biases the top levels)": this is the truncation pitfall of Section 13.6.

Sec. V: leaving the perturbative regime. In Sec. V the second-order formula is not a step proposer: it is the model that the optimizer minimizes (the "surrogate"). The paper then checks it against exact diagonalization of $H + iA$, as recommended in Section 13.7. At the fitted amplitude, "corr(exact, second-order) drops to +0.14/+0.18 (odd/even), the surrogate overstates shift magnitudes twofold (Fig. 5, left), and 3% of eigenstates can no longer be matched one-to-one to unperturbed states." Scoring the exact spectra in-sample gives rms 0.2523 and 0.2477, worse than the null 0.2438. "At quarter amplitude the surrogate is accurate (corr +0.88/+0.94)", and amplitude-controlled fits with $\|c\| \leq R$, $R = 4$, "pass exact verification (corr +0.93/+0.995, all states matched)". This is Figure 13.5 at full scale: small amplitude, agreement; large amplitude, the formula no longer describes the operator being fitted. Hence the paper's standard (ii) in Sec. VI: "scoring of the actual operator spectrum, not a surrogate."

The constant mode. Sample Problem "A constant shift" showed that $\phi_0 = 1$ shifts every level equally and couples no states. For $H + iA$ a constant therefore shifts all levels by the same imaginary constant and has zero second-order real shift (Sec. II E, safeguard (i)); its tensor rows are exactly zero. Sec. V reports that before this was enforced, the optimizer "had driven the exactly-null constant mode to $c_0 \approx 2.2 \times 10^6$, harvesting in-sample improvement from float32 noise". Chapter 31 dissects the episode.

A warning about the symbol A. The paper uses A for two different things: the sensitivity matrix A_{nk} of Sec. II C and the non-Hermitian operator A of Sec. II E, whose matrix elements are $A_{nm} = \langle n|A|m\rangle$. They are related (both are integrals of products of wavefunctions and basis functions) but not the same object. In this chapter A_{nk} always means the sensitivity matrix.

Memorization, previewed. Sample Problem "Tuning three box levels" fitted three targets exactly while predicting nothing about level 4 beyond a common offset. The paper's central negative result (Sec. III, Table I) is a far richer version of this phenomenon, analysed in Chapters 25 and 29; do not read the toy as an explanation of it, only as a first picture.

13.9 Summary

- Write $H = H_0 + \delta V$, $\delta V = \varepsilon W$, and expand energies and states in powers of ε (assumption justified for nondegenerate levels; Section 13.2). Matrix elements: $W_{nm} = \int \psi_n^{(0)*} W \psi_m^{(0)} \mathrm{d}x$, Equation (13.5).
- First order: $\delta E_n^{(1)} = \delta V_{nn} = \int |\psi_n^{(0)}|^2 \delta V \, \mathrm{d}x$, Equation (13.16): a density-weighted average.
- Constant δV: exact uniform shift. Odd δV in an even potential: zero first-order shift. Box with cosines: $\cos(q\pi x/L)$ moves only level q, by $(-1)^{q+1} c/2$, Equation (13.18).

- For $\delta V = \sum_k c_k \phi_k$: $\delta \boldsymbol{E} \approx A\boldsymbol{c}$, $A_{nk} = \int |\psi_n|^2 \phi_k \mathrm{d}x$, Equations (13.20) and (13.21); with current states $A_{nk} = \partial E_n / \partial c_k$ exactly, Equation (13.23).
- First-order state: Equation (13.25). Second order: $\sum_{m \neq n} \delta V_{nm} \delta V_{mn} / (E_n^{(0)} - E_m^{(0)})$, equal to $\sum_{m \neq n} |\delta V_{nm}|^2 / (E_n^{(0)} - E_m^{(0)})$ for real δV, Equations (13.26) and (13.27).
- Level repulsion: levels above push down, below push up; exact two-level formula Equation (13.28); avoided crossings.
- Box matrix elements: x_{nm}, Equation (13.30); cosines, Equation (13.31).
- Exact levels from the matrix Equation (13.32) in a truncated basis.
- Validity (heuristic): couplings and shifts small compared with spacings of coupled levels, Equation (13.33).
- Fitting: Newton, Equation (13.34), and Gauss–Newton loops using A as the Jacobian with exact eigensolves; optimizing a perturbative formula itself is safe only inside its validity range (paper Sec. V).

13.10 Problems

1. • A box with $L = 2$ has its potential raised by the constant $\delta V = 0.3$. Find the first-order and second-order shifts of level 5, and the exact shift.

2. • In the symmetric box, $\delta V = Fx^3$. Without computing any integral, find the first-order shift of every level.

3. • Show that for any normalized state the sensitivity to the constant basis function $\phi_0 = 1$ is $A_{n0} = 1$. What does the column of ones mean physically?

4. • Box with $L = 1$ and $\delta V = c\cos(2\pi x/L)$. Find the first-order shifts of levels 1, 2 and 3.

5. • A two-level system has $E_1 = 0$, $E_2 = 2$ and coupling $v = 0.1$. Find the second-order estimates of both levels and the exact eigenvalues.

6. •• Prove that the second-order shift of the ground state is never positive. For a finite $N_b \times N_b$ matrix, what can you say about the sign of the second-order shift of the highest level?

7. •• Box with $\delta V = c\cos(\pi x/L)$. Show that level 2 has zero first-order shift and that its second-order shift is $-c^2 L^2/(12\pi^2)$. Evaluate for $L = 1$, $c = 1$.

8. •• For the tilted box ($L = 1$, $\delta V = Fx$), compute the second-order shift of level 2, summing the contributions of $m = 1, 3, 5, 7$. Explain its sign using level repulsion.

9. •• Derive the Hellmann–Feynman theorem directly: if $H(\lambda)\psi_n(\lambda) = E_n(\lambda)\psi_n(\lambda)$ with $\int |\psi_n|^2 \mathrm{d}x = 1$ for all λ, show that $\mathrm{d}E_n/\mathrm{d}\lambda = \int \psi_n^* (\partial H/\partial\lambda)\psi_n \mathrm{d}x$, using Equation (13.7).

10. •• For $\delta V = c\cos(2\pi x/L)$ in the box, use Equation (13.31) to list the levels coupled to level n, and use Equation (13.33) to find the condition on c for level $n \geq 5$ and for level $n = 1$. Are higher levels again more robust?

11. •• (Paper.) In Table I of the paper, the scale-1 fits on zeros 1–400 use $M = 200$, 400 and 600, and a scale-2 fit on zeros 1–800 uses $M = 900$. Give the dimensions of the sensitivity matrix A_{nk} restricted to the training levels in each case, and say which linearized systems have more unknowns than equations.

12. •• (Paper.) In the paper's interpolation split (Sec. III, Fig. 4), the fitted correction has rms 40 in the turning-point band of levels 201–300, while the mean spacing there is about 1.46. Is a single first-order linearization from $\boldsymbol{c} = 0$ adequate to describe this fitted potential? Why are the paper's predicted levels nonetheless trustworthy as eigenvalues of the fitted operator?

13. •• (Paper.) Sec. V of the paper reports that at the fitted amplitude the correlation between exact and second-order shifts is only $+0.14/+0.18$, but at quarter amplitude it is $+0.88/+0.94$. Assuming the perturbation series converges, explain with a scaling argument why reducing all coefficients by a factor 4 should make the second-order formula more accurate.

14. •• (Paper.) Compute the second-order shift of box level 2 under a tilt ($L = 1$) keeping only states 1 and 2 in the sum, and compare with your result of Problem 8. Use the comparison to explain the paper's remark (Sec. II E) that truncating the state sum at 500 states "measurably biases the top levels", and why it sums over all 652 states with $E \le 1000$.

15. •• Using numpy and the matrix method of Section 13.6, compute the error of the second-order formula for the box ground state with $\delta V = c\cos(\pi x/L)$, $L = 1$, at $c = 1$ and $c = 0.5$. Show that the ratio of the errors is close to 8, and explain why.

16. •• Using qmnt (`qmnt.L_of`, `qmnt.grid`, `qmnt.wu_sprung_V`, `qmnt.fd_levels` with `return_vectors=True`) or the listing of Section 13.4, compute A_{nk} for the Wu–Sprung potential for $n \le 50$ and $k \le 50$ on a 2000-point grid. Verify that column $k = 0$ equals 1, and reproduce the right panel of Figure 13.2 qualitatively. Then set the potential to zero and recover Equation (13.22) approximately.

17. •• For the two-level matrix $H(g) = \begin{pmatrix} g & v \\ v & -g \end{pmatrix}$, with g and v real, find the exact eigenvalues, the minimum gap, and the behavior for $|g| \gg |v|$. Show that the latter agrees with second-order perturbation theory.

18. •• For $\delta V = c\cos(q\pi x/L)$ in the box, show that level $n = q$ (with $q \ge 1$) couples only to level $3q$, and that its second-order shift is $-c^2L^2/(8\pi^2q^2)$.

19. ••• *Sum rule.* Let W be real and consider a finite set of N_b nondegenerate levels. Show that $\sum_{n=1}^{N_b} E_n^{(2)} = 0$ (a) by pairing the (n, m) and (m, n) terms, and (b) by using the fact that the trace of a symmetric matrix (the sum of its diagonal entries) equals the sum of its eigenvalues. For a 2×2 matrix this follows from the characteristic polynomial $\lambda^2 - (\text{trace})\lambda + \det$, whose roots sum to the trace; you may take the general case on trust. (This sum rule reappears for $H + iA$ in Sec. V A of the paper.)

20. ••• Using the Hellmann–Feynman theorem and the first-order wavefunction Equation (13.25), show that for $H(\lambda) = H_0 + \lambda W$ one has $\mathrm{d}^2E_n/\mathrm{d}\lambda^2\big|_{\lambda=0} = 2E_n^{(2)}$.

21. ••• For the tilted ground state of the $L = 1$ box, use Hellmann–Feynman to show that the expectation value of position is $\langle x\rangle = \mathrm{d}E_1/\mathrm{d}F$, and evaluate it to first order in F. In which direction does the particle move, and why does that make physical sense?

22. •• Using numpy, sum the series $S = \sum_{m=2,4,6,\ldots} m^2/(m^2-1)^5$ to $m = 1000$ and check that it agrees with $(15 - \pi^2)\pi^2/3072$.

23. •• (Paper.) App. A of the paper uses $L = 44.8827$ and a working grid of $N = 6000$ interior points at scale 1. Compute the grid spacing $h = 2L/(N+1)$. If you forgot to divide the eigenvectors from `eigh_tridiagonal` by $\sqrt{h}$, by what factor would every A_{nk} be wrong, and what would that do to an undamped Gauss–Newton step?

24. ••• (Paper.) Contrast the role of perturbation theory in the paper's Hermitian fit (Sec. II C, App. A step 6) with its role in the non-Hermitian fit (Sec. II E, App. A step 10). Explain why the first is protected against the failure documented in Sec. V and the second is not, and how the safeguards (ii) and (iii) of Sec. II E repair the second.

25. •• Using numpy and the Newton listing of Section 13.7, fit the box ground state ($L = 1$, $\delta V = c\cos(\pi x/L)$) to the target 3.5. Give the first Newton iterate by hand, and count the iterations needed until $|E_1 - 3.5| < 10^{-10}$.

13.11 Answers to selected problems

1. First order: $\int |\psi_5^{(0)}|^2(0.3)\mathrm{d}x = 0.3$. Second order: the off-diagonal elements $0.3\int \psi_5^{(0)}\psi_m^{(0)}\mathrm{d}x = 0.3\,\delta_{5m}$ vanish, so the shift is 0. Exact: $H_0 + 0.3$ has the same eigenfunctions with every energy raised by 0.3 (as in the first Sample Problem), so the exact shift is 0.3.

2. All zero. In the symmetric box every $|\psi_n^{(0)}|^2$ is even and x^3 is odd, so the integrand of Equation (13.16) is odd and integrates to zero over $[-L, L]$.

3. $A_{n0} = \int |\psi_n|^2 \cdot 1\,\mathrm{d}x = 1$ by normalization. Physically, raising the potential by the constant c_0 raises every level by c_0 at first order (exactly, in fact), whatever the state.

4. By Equation (13.18) with $q = 2$: only $n = 2$ shifts, by $(-1)^3 c/2 = -c/2$. Levels 1 and 3: 0.

5. Second order: $\mp|v|^2/(E_2 - E_1) = \mp 0.01/2$, so -0.005 and 2.005. Exact: $\Delta = 1$, $\lambda_\pm = 1 \pm \sqrt{1.01} = 1 \pm 1.0049876$, that is -0.0049876 and 2.0049876. The agreement is good because $|v|/\Delta = 0.1$.

7. With $q = 1$, Equation (13.31) gives a diagonal element only for $n = 1$, so level 2 has none. Level 2 couples to m with $|2 - m| = 2$, i.e. $m = 4$ ($m = 0$ does not exist), and $2 + m = 2$ is impossible for $m \geq 1$. $\delta V_{24} = -c/2$, and $E_2^{(0)} - E_4^{(0)} = (\pi/2L)^2(4 - 16) = -3\pi^2/L^2$. The shift is $(c^2/4)/(-3\pi^2/L^2) = -c^2L^2/(12\pi^2)$, which for $L = c = 1$ is -0.0084434.

9. We assume ψ_n depends smoothly on λ, so that the λ-derivative may be taken inside the integral. Differentiate $E_n = \int \psi_n^* H \psi_n \mathrm{d}x$ with the product rule: $\mathrm{d}E_n/\mathrm{d}\lambda = \int \psi_n^{*\prime} H \psi_n + \int \psi_n^* H' \psi_n + \int \psi_n^* H \psi_n'$, primes denoting $\partial/\partial\lambda$. The first term is $E_n \int \psi_n^{*\prime}\psi_n$. Since $\psi_n(\pm L; \lambda) = 0$ for every λ, its λ-derivative ψ_n' also vanishes at the walls, so Equation (13.7) applies with $f = \psi_n$ and $g = \psi_n'$ ($H(\lambda)$ is H_0 plus a real function, so the identity holds for it too). In the third term it gives $\int \psi_n^* H \psi_n' = \int (H\psi_n)^* \psi_n' = E_n \int \psi_n^* \psi_n'$ (E_n real). Together the first and third give $E_n\,\mathrm{d}/\mathrm{d}\lambda \int |\psi_n|^2 \mathrm{d}x = E_n \cdot 0$, since the normalization is constant. What remains is $\int \psi_n^* H' \psi_n \mathrm{d}x$.

11. Rows are levels, columns are basis functions. Scale 1, fit 1–400: 400×200 (more equations than unknowns), 400×400 (square), 400×600 (more unknowns than equations). Scale 2, fit 1–800, $M = 900$: 800×900 (more unknowns than equations). For the last two the linearized step has infinitely many solutions unless something (damping, regularization, the minimum-norm choice of Chapter 9) selects one.

13. If the series converges, the error of stopping after second order is dominated by the third-order term, which contains three factors of the perturbation and so scales as c^3, while the second-order term scales as c^2. The relative error is roughly the ratio of the two, $c^3/c^2 = c$, so it falls in proportion to the amplitude. Reducing every coefficient by a factor 4 multiplies the second-order shift by 1/16 and the leading neglected term by 1/64, so the neglected part shrinks four times faster than the part kept. In terms of Equation (13.33), the ratios $|\delta V_{nm}|/|E_n - E_m|$ all fall by 4, moving the fit back toward the perturbative regime. That is why the paper's quarter-amplitude check, corr +0.88/+0.94 (Sec. V), succeeds where the full amplitude fails. If the series does not converge (or converges very slowly) at the fitted amplitude, the argument says nothing there. This is part of the reason exact diagonalization is the only safe judge.

15. With $N_b = 60$: at $c = 1$ the error of the second-order value is $2.954736 - 2.954413 = 3.2 \times 10^{-4}$. At $c = 0.5$ the error is 4.0×10^{-5}, and the ratio of the two errors is 8.0. The ratio

is close to $2^3 = 8$ because the leading neglected term is third order (it contains, for example, the product $\delta V_{11}|\delta V_{13}|^2$), which scales as c^3. Halving c divides it by 8.

17. $\det(H - \mu I) = (g - \mu)(-g - \mu) - v^2 = \mu^2 - g^2 - v^2 = 0$, so the eigenvalues are $\mu_\pm = \pm\sqrt{g^2 + v^2}$. The gap $2\sqrt{g^2 + v^2}$ is smallest at $g = 0$, where it equals $2|v|$. For $|g| \gg |v|$ use $\sqrt{1 + y} \approx 1 + y/2$ with $y = v^2/g^2$: $\mu_\pm \approx \pm(|g| + v^2/(2|g|))$. Perturbatively, level g couples to level $-g$ with denominator $g - (-g) = 2g$, so its shift is $v^2/(2g)$. For $g > 0$ this pushes the upper level up by $v^2/2|g|$; the other level is pushed down by the same amount. This matches the expansion of the exact result.

19. (a) The second-order sum over all levels is $\sum_n \sum_{m \neq n} |W_{nm}|^2/(E_n^{(0)} - E_m^{(0)})$. Group the terms (n, m) and (m, n). They have the same numerator ($|W_{nm}| = |W_{mn}|$, by Equation (13.6)) and opposite denominators, so each pair sums to zero. (b) The trace of $H_{km} = E_k^{(0)}\delta_{km} + \varepsilon W_{km}$ is $\sum_k E_k^{(0)} + \varepsilon \sum_k W_{kk}$ exactly, and the trace equals the sum of the eigenvalues (the fact stated in the problem). Therefore $\sum_n E_n = \sum_n E_n^{(0)} + \varepsilon \sum_n W_{nn}$ holds for every ε. Comparing powers of ε, the sum over n of every correction beyond first order vanishes, and in particular $\sum_n E_n^{(2)} = 0$.

21. Here $\partial H/\partial F = x$, so Hellmann–Feynman, Equation (13.23), gives $\mathrm{d}E_1/\mathrm{d}F = \int |\psi_1|^2 x \, \mathrm{d}x = \langle x \rangle$. Through second order, $E_1 \approx E_1^{(0)} - 0.017556F^2$, so $\langle x \rangle \approx -0.035112F$. For $F > 0$ the potential Fx is lower on the left, and the particle shifts toward negative x. This is the direction of the force $-\mathrm{d}(Fx)/\mathrm{d}x = -F$.

23. The spacing is $h = 2(44.8827)/6001 = 0.014958$. Without the division, every A_{nk} is multiplied by h, so it is too small by the factor $1/h = 66.85$. An undamped step solves $A\Delta\boldsymbol{c} = \boldsymbol{r}$, so $\Delta\boldsymbol{c}$ would be 66.85 times too large. The linearization would fail badly over such a step. With the paper's accept-only-improvements rule, the step would be rejected and the damping would increase until the step shrank back, wasting iterations. Without that rule, the fit would diverge.

25. The first iterate is $c_1 = (3.5 - 2.467401)/0.5 = 2.065198$. The Newton loop reaches $|E_1 - 3.5| < 10^{-10}$ after 4 iterations (four updates of c). Each iteration roughly squares the error, the quadratic convergence of Chapter 4.

Real parts of the eigenvalues of a PT-symmetric chain (gain on one half, loss on the other) as the strength grows left to right: pairs of real levels meet and turn complex (brighter = larger imaginary part).

Chapter 14

Beyond Hermitian: Complex Potentials and PT Symmetry

↑Contents

What this chapter is about

Every Hamiltonian you have met so far had a real potential $V(x)$. That was no accident. A real potential makes the Hamiltonian Hermitian, and Hermitian operators have real eigenvalues and orthogonal eigenvectors (Chapter 8). Physically, a real potential conserves probability: a particle in a box stays in the box forever. Some physical systems do not conserve probability. A neutron passing through a nucleus can be absorbed. Light travelling down an optical fiber with a lossy coating fades. A laser medium amplifies. Physicists describe such systems by letting the potential have an *imaginary* part. In this chapter you will learn what happens to energy levels when you do that: they become complex numbers, their imaginary parts describe decay or growth, and the familiar rules of perturbation theory acquire a crucial minus sign.

That minus sign has consequences. Under an imaginary perturbation, energy levels *attract* instead of repelling; the real shifts of all levels add up to exactly zero; a perturbation of a special symmetry (called PT symmetry) keeps the spectrum real until two levels collide; and beyond that collision the spectrum turns complex and perturbation theory fails completely. We will derive each of these facts line by line and check them on examples small enough to do by hand.

Where this shows up in the paper: Sec. II E of the paper defines two non-Hermitian model classes, a “PT-odd” class and an “absorptive-even” class, whose real level shifts are given by the second-order formula of its Eq. (5). Sec. V and Sec. V A dissect what went wrong when those models were fitted to the zeros: an optimizer harvested numerical noise through an “exactly-null constant mode”, and second-order perturbation theory was used far outside its range of validity, as Fig. 5 shows. The formulas and safeguards of those sections are explained here; the fitting machinery itself is in Chapter 24 and the full reading of Sec. V in Chapter 31.

What you need

- Chapter 2: complex numbers, real and imaginary parts, complex conjugate $\bar{z}$ or z^*, modulus $|z|$, $e^{i\theta}$, and $|e^{i\theta}| = 1$.
- Chapter 3: Taylor series and radius of convergence.
- Chapter 4: big-O notation, floating-point numbers, machine epsilon (2.2×10^{-16} in double precision, 1.2×10^{-7} in single precision, float32).
- Chapter 7: matrices, transpose, conjugate transpose, inner products, norms.
- Chapter 8: eigenvalues and eigenvectors, the characteristic polynomial for 2×2 and 3×3 matrices, Hermitian matrices have real eigenvalues, numerical eigensolvers, matching eigenvectors by overlap.
- Chapter 10: the time-dependent and time-independent Schrödinger equations in units $\hbar = 2m = 1$, normalization, Hermitian operators.
- Chapter 11: the infinite square well on $[-L, L]$, parity of states in an even potential, the harmonic oscillator.
- Chapter 13: first- and second-order perturbation theory for a real (Hermitian) perturbation, matrix elements, the condition for validity.
- First-year physics: exponential decay, half-life.

14.1 Complex potentials and decay

★ Did you know? *Escaping Through a Wall*

In 1928 George Gamow, and independently Ronald Gurney and Edward Condon, explained alpha decay as quantum tunneling through the nuclear barrier. Gamow described the decaying state with a complex energy whose imaginary part sets the decay rate. Tunneling explains why alpha-emitter half-lives range from under a microsecond to billions of years, since small changes in energy change the escape probability enormously.

14.1.1 Why a potential might be complex

Recall from Chapter 10 the time-dependent Schrödinger equation. In the units used throughout this book and in the paper, $\hbar = 1$ and $2m = 1$, it reads

$$i\,\frac{\partial \Psi}{\partial t} \;=\; -\frac{\partial^2 \Psi}{\partial x^2} + V(x)\,\Psi. \tag{14.1}$$

Here $\Psi(x,t)$ is the wavefunction, x is position and t is time. (In this chapter t is a time; in the rest of the book t usually denotes the height of a zeta zero. The context always makes clear which is meant.) By the Born rule, $|\Psi(x,t)|^2$ is the probability density for finding the particle at x, so the total probability of finding the particle anywhere is

$$P_{\mathrm{tot}}(t) \;=\; \int_{-L}^{L} |\Psi(x,t)|^2 \,\mathrm{d}x, \tag{14.2}$$

for a particle confined to the box $[-L, L]$ by walls at which $\Psi(\pm L, t) = 0$ (Dirichlet walls, Chapter 11).

In Chapter 10 you saw that P_{tot} stays equal to 1 forever when V is real. Now let us allow the potential to have a real part $V_R(x)$ and an imaginary part $V_I(x)$, both real-valued functions:

$$V(x) = V_R(x) + i\,V_I(x). \tag{14.3}$$

We want to know how P_{tot} changes in time.

14.1.2 The rate of change of the total probability

Differentiate (14.2) with respect to t. The limits do not depend on t, so the derivative goes inside the integral. Since $|\Psi|^2 = \Psi^*\Psi$, the product rule gives

$$\frac{\mathrm{d}P_{\mathrm{tot}}}{\mathrm{d}t} = \int_{-L}^{L} \left(\Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial t} \right) \mathrm{d}x. \tag{14.4}$$

From (14.1), dividing by i (that is, multiplying by $-i$, since $1/i = -i$),

$$\frac{\partial \Psi}{\partial t} = -i \left(-\Psi'' + V_R \Psi + i V_I \Psi \right), \tag{14.5}$$

where a prime denotes $\partial/\partial x$. Taking the complex conjugate of both sides (conjugation turns i into $-i$ and leaves the real functions V_R, V_I alone):

$$\frac{\partial \Psi^*}{\partial t} = +i \left(-\Psi^{*\prime\prime} + V_R \Psi^* - i V_I \Psi^* \right). \tag{14.6}$$

Now multiply the first by Ψ^* and the second by Ψ:

$$\Psi^* \frac{\partial \Psi}{\partial t} = i\Psi^*\Psi'' - iV_R|\Psi|^2 + V_I|\Psi|^2, \quad (\text{using } -i \cdot i = 1) \tag{14.7}$$

$$\Psi \frac{\partial \Psi^*}{\partial t} = -i\Psi\Psi^{*\prime\prime} + iV_R|\Psi|^2 + V_I|\Psi|^2. \quad (\text{using } i \cdot (-i) = 1) \tag{14.8}$$

Add them. The V_R terms cancel, and the V_I terms double:

$$\Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial t} = i \left(\Psi^*\Psi'' - \Psi\Psi^{*\prime\prime} \right) + 2V_I|\Psi|^2. \tag{14.9}$$

The bracket is a total derivative. By the product rule,

$$\frac{\partial}{\partial x} \left(\Psi^*\Psi' - \Psi\Psi^{*\prime} \right) = \Psi^{*\prime}\Psi' + \Psi^*\Psi'' - \Psi'\Psi^{*\prime} - \Psi\Psi^{*\prime\prime} = \Psi^*\Psi'' - \Psi\Psi^{*\prime\prime}. \tag{14.10}$$

So its integral over $[-L, L]$ is $\left[\Psi^*\Psi' - \Psi\Psi^{*\prime}\right]_{-L}^{L}$, which is zero because Ψ and Ψ^* vanish at both walls. Substituting into (14.4):

$$\boxed{\frac{\mathrm{d}P_{\mathrm{tot}}}{\mathrm{d}t} = 2 \int_{-L}^{L} V_I(x) \, |\Psi(x,t)|^2 \, \mathrm{d}x.} \tag{14.11}$$

This is the key result. The real part of the potential never changes the total probability. The imaginary part does: where $V_I < 0$ probability drains away (absorption, or loss), and where $V_I > 0$ it is created (gain). If $V_I = 0$ everywhere we recover conservation of probability.

The commonest special case is a constant negative imaginary part, conventionally written

$$V_I(x) = -\frac{\Gamma}{2}, \qquad \Gamma \text{ a real constant.} \tag{14.12}$$

Then (14.11) becomes $\mathrm{d}P_{\mathrm{tot}}/\mathrm{d}t = -\Gamma P_{\mathrm{tot}}$, whose solution (first-year physics: exponential decay) is

$$P_{\mathrm{tot}}(t) = P_{\mathrm{tot}}(0)\, e^{-\Gamma t}. \tag{14.13}$$

For $\Gamma > 0$ the particle disappears at rate Γ; its mean lifetime is $\tau = 1/\Gamma$ and its half-life is $t_{1/2} = \log 2/\Gamma$ (natural log, as always in this book). The quantity Γ is called the *width*. (The name comes from spectroscopy: a decaying state emits light whose spectral line has a frequency spread proportional to Γ. We use the word only as a label.) The physical requirement that a lossy medium absorbs rather than amplifies is $\Gamma \geq 0$, that is, $V_I \leq 0$.

14.1.3 Stationary states with complex energy

Recall from Chapter 10 that separation of variables, $\Psi(x,t) = \psi(x)e^{-iEt}$, turns (14.1) into the time-independent equation $H\psi = E\psi$ with $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$. Nothing in that step used the reality of V, so it still works; but now nothing forces E to be real. Write

$$E = E_R - i\,\frac{\Gamma}{2}, \tag{14.14}$$

with E_R and Γ real. Then

$$e^{-iEt} = e^{-iE_Rt}\,e^{-i(-i\Gamma/2)t} = e^{-iE_Rt}\,e^{-\Gamma t/2}, \tag{14.15}$$

because $-i\cdot(-i) = i^2 = -1$. Since $|e^{-iE_Rt}| = 1$ (Chapter 2),

$$|\Psi(x,t)|^2 = |\psi(x)|^2\,e^{-\Gamma t}. \tag{14.16}$$

So a stationary state with complex energy oscillates at angular frequency E_R (its "energy level" in the usual sense) while its probability decays at rate $\Gamma = -2\,\mathrm{Im}\,E$. The real part of a complex eigenvalue is the level; minus twice the imaginary part is the width.

14.1.4 An exact formula for the imaginary part

There is a neat exact statement. Suppose $H\psi = E\psi$ with $\psi(\pm L) = 0$. Multiply by ψ^* and integrate:

$$\int_{-L}^{L}\psi^*\,(-\psi'')\,\mathrm{d}x + \int_{-L}^{L}V_R|\psi|^2\mathrm{d}x + i\int_{-L}^{L}V_I|\psi|^2\mathrm{d}x = E\int_{-L}^{L}|\psi|^2\mathrm{d}x. \tag{14.17}$$

Integrate the first term by parts: $\int\psi^*(-\psi'')\mathrm{d}x = -\big[\psi^*\psi'\big]_{-L}^{L} + \int\psi^{*\prime}\psi'\mathrm{d}x = \int|\psi'|^2\mathrm{d}x$, since the boundary term vanishes at the walls. This is real. The second term is real. Only the third is imaginary. Taking the imaginary part of both sides and dividing by $\int|\psi|^2\mathrm{d}x$:

$$\mathrm{Im}\,E = \frac{\int_{-L}^{L}V_I(x)|\psi(x)|^2\,\mathrm{d}x}{\int_{-L}^{L}|\psi(x)|^2\,\mathrm{d}x}. \tag{14.18}$$

The imaginary part of every eigenvalue is a weighted average of V_I, weighted by the probability density of that eigenstate. Two immediate consequences:

- If $V_I(x) \le 0$ everywhere, then $\mathrm{Im}\,E \le 0$ for *every* eigenvalue, so every width is nonnegative. A purely absorbing potential cannot produce a growing state.
- If V_I is a constant $-\Gamma/2$, then $\mathrm{Im}\,E = -\Gamma/2$ for every level, exactly. In fact in this case $H = H_R - i\Gamma/2$ where H_R is the real Hamiltonian, so every eigenfunction of H_R with level E_n is an eigenfunction of H with eigenvalue $E_n - i\Gamma/2$. A constant imaginary potential moves every eigenvalue straight down in the complex plane by the same amount and changes nothing else.

14.1.5 Hermitian or not?

Recall from Chapter 10 that an operator H is Hermitian if $\int f^*(Hg)\,\mathrm{d}x = \int(Hf)^*g\,\mathrm{d}x$ for all allowed f, g. The kinetic part $-\mathrm{d}^2/\mathrm{d}x^2$ passes this test (two integrations by parts), and so does multiplication by a real function. Multiplication by iV_I fails: $\int f^*(iV_Ig)\mathrm{d}x = i\int f^*V_Ig\,\mathrm{d}x$, while $\int(iV_If)^*g\,\mathrm{d}x = -i\int f^*V_Ig\,\mathrm{d}x$. These differ by a sign unless $\int f^*V_Ig\,\mathrm{d}x = 0$. So a Hamiltonian with $V_I \neq 0$ is *non-Hermitian*, and the guarantee of real eigenvalues proved in Chapter 8 no longer applies. Equation (14.18) shows exactly how far from real the eigenvalues can be.

Sample Problem 14.1: A leaky level in the running box

A particle is in the tenth level of the infinite well on $[-L, L]$ with $L = 44.883$ (running example RE3). A uniform absorber $V_I = -\Gamma/2$ with $\Gamma = 0.02$ is switched on. Find the complex energy, the fraction of probability left at $t = 50$, and the half-life.

Step 1: the real level. From Chapter 11, the levels of the box are $E_k = (k\pi/2L)^2$. For $k = 10$: $E_{10} = (10\pi/89.766)^2 = (0.34998)^2 \approx 0.1225$.

Step 2: the complex energy. A constant imaginary part shifts every eigenvalue by exactly $-i\Gamma/2 = -0.01i$, so $E = 0.1225 - 0.01i$.

Step 3: survival. By (14.13), $P_{\text{tot}}(50)/P_{\text{tot}}(0) = e^{-0.02\times 50} = e^{-1} = 0.3679$.

Step 4: half-life. $t_{1/2} = \log 2/\Gamma = 0.6931/0.02 = 34.66$.

$$\boxed{E \approx 0.1225 - 0.01i, \quad P_{\text{tot}}(50) = 0.368\, P_{\text{tot}}(0), \quad t_{1/2} = 34.66}$$

The width does not depend on which level the particle occupies, because the absorber is uniform.

Checkpoint 14.1

A potential has $V_I > 0$ (gain) in the left half of the box and $V_I < 0$ (loss) in the right half. Must the total probability decrease?

Answer: No. By (14.11) the rate is $2\int V_I|\Psi|^2\,\mathrm{d}x$; it is positive if the particle is mostly on the left and negative if it is mostly on the right. The sign depends on where the probability sits.

Key Ideas

- The real part of a potential conserves probability; the imaginary part changes it at the rate $2\int V_I|\Psi|^2\,\mathrm{d}x$, (14.11).
- A stationary state with $E = E_R - i\Gamma/2$ oscillates at E_R and decays as $e^{-\Gamma t}$; $\Gamma = -2\,\mathrm{Im}\,E$ is the width.
- $\mathrm{Im}\,E$ is the probability-weighted average of V_I, (14.18); if $V_I \le 0$ everywhere, all widths are nonnegative.
- A constant imaginary potential shifts all eigenvalues by the same imaginary constant and leaves their real parts and eigenfunctions unchanged.

14.2 Non-Hermitian matrices

14.2.1 The Hamiltonian in the unperturbed basis

The paper works with Hamiltonians of the form

$$H = H_0 + iA, \qquad H_0 = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x), \tag{14.19}$$

where V is real (the Wu–Sprung potential, Chapter 22, or in our examples the box) and $A(x)$ is a *real* function. "A is a real multiplication operator" simply means that A acts on a wavefunction by multiplying it by the real function $A(x)$. Comparing with (14.3), iA is an imaginary potential with $V_I = A$; an absorber has $A = -\Gamma/2 \le 0$.

Pitfall

The letter A is overloaded in the paper. In Sec. II C (and in Chapters 13 and 24) A_{nk} is the *sensitivity matrix* $\int |\psi_n|^2 \phi_k \, dx$. In Sec. II E, Eq. (5), and in this chapter, A is the real multiplication operator in $H + iA$, and $A_{nm} = \langle n|A|m\rangle$ are its matrix elements between two unperturbed states. Read the subscripts: A_{nm} with two level indices is a matrix element of the operator; A_{nk} with a basis index k is a sensitivity.

Let ψ_n, $n = 1, 2, \ldots$, be the normalized eigenfunctions of H_0 with levels E_n. Because V is real, the ψ_n can be chosen real (here is the argument: E is real because H_0 is Hermitian, Chapter 10; if $H_0\psi = E\psi$, conjugating gives $H_0\psi^* = E\psi^*$, because H_0 contains no i; hence $\operatorname{Re}\psi = (\psi + \psi^*)/2$ and $\operatorname{Im}\psi = (\psi - \psi^*)/(2i)$ are real solutions with the same E; they cannot both be zero, since $\psi \neq 0$, and a nonzero one can be used as the eigenfunction). Expand any wavefunction as $\psi = \sum_m v_m \psi_m$. The set of all ψ_m is infinite, so this is an infinite sum. Then $H\psi = E\psi$ becomes, after multiplying by ψ_n and integrating over $[-L, L]$ (the inner product with the real function ψ_n), the matrix equation $\sum_m M_{nm} v_m = E v_n$ with

$$M_{nm} = \int \psi_n \, (H_0 + iA) \, \psi_m \, dx = E_n \delta_{nm} + i A_{nm}, \qquad A_{nm} = \int_{-L}^{L} \psi_n(x) \, A(x) \, \psi_m(x) \, dx. \tag{14.20}$$

Here δ_{nm} is the Kronecker delta: 1 if $n = m$, 0 otherwise. We used $H_0\psi_m = E_m\psi_m$ and $\int \psi_n \psi_m \, dx = \delta_{nm}$ (orthonormality). The exact matrix M_{nm} is infinite. On a computer we keep only the lowest K states, $m = 1, \ldots, K$. This truncation is an approximation: it is accurate for levels well below E_K when their couplings to the discarded states are small, and we will see in Section 14.5 what it does to the levels near the cutoff. In this truncated basis write $D = \operatorname{diag}(E_1, \ldots, E_K)$ and let A be the $K \times K$ matrix of the A_{nm}. Then

$$M = D + i\mathsf{A}. \tag{14.21}$$

Two properties of A follow at once from (14.20). Every factor in the integrand is real, so A_{nm} is **real**. And swapping n and m swaps two real factors that commute, so $A_{mn} = A_{nm}$: A is **symmetric**. Hence $M^\top = D + i\mathsf{A}^\top = M$: the matrix is symmetric. But its conjugate transpose is $M^\dagger = D - i\mathsf{A} \neq M$ (unless $\mathsf{A} = 0$). A *complex symmetric* matrix is not Hermitian, and none of the theorems of Chapter 8 about real eigenvalues apply to it.

14.2.2 Eigenvalues of a 2×2 non-Hermitian matrix

Recall from Chapter 8 that the eigenvalues of a matrix M are the roots of $\det(M - \lambda I) = 0$. For a general 2×2 matrix,

$$\det \begin{pmatrix} M_{11} - \lambda & M_{12} \\ M_{21} & M_{22} - \lambda \end{pmatrix} = \lambda^2 - (M_{11} + M_{22})\lambda + (M_{11}M_{22} - M_{12}M_{21}) = 0, \tag{14.22}$$

so by the quadratic formula

$$\lambda_\pm = \frac{M_{11} + M_{22}}{2} \pm \frac{1}{2}\sqrt{(M_{11} - M_{22})^2 + 4M_{12}M_{21}}. \tag{14.23}$$

(To get the discriminant: $(M_{11} + M_{22})^2 - 4(M_{11}M_{22} - M_{12}M_{21}) = (M_{11} - M_{22})^2 + 4M_{12}M_{21}$.) When the discriminant is a negative real number or a complex number, the eigenvalues are complex; the square root of a complex number is defined as in Chapter 2.

Sample Problem 14.2: Eigenvalues of a complex symmetric matrix

Find the eigenvalues of $M = \begin{pmatrix} 1 & 2i \\ 2i & 3 \end{pmatrix}$ and check that they add up to the sum of the diagonal entries.

Step 1. $M_{11} + M_{22} = 4$ and $(M_{11} - M_{22})^2 = 4$.
Step 2. $4M_{12}M_{21} = 4(2i)(2i) = 16i^2 = -16$.
Step 3. The discriminant is $4 - 16 = -12$, whose square root is $i\sqrt{12} = 2\sqrt{3}\,i$.
Step 4. $\lambda_\pm = 2 \pm \frac{1}{2}(2\sqrt{3}\,i) = 2 \pm \sqrt{3}\,i$.
Step 5. $\lambda_+ + \lambda_- = 4 = 1 + 3$.

$$\boxed{\lambda_\pm = 2 \pm 1.7321\,i}$$

Compare the Hermitian matrix $\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$: its discriminant is $4 + 16 = 20 > 0$ and its eigenvalues $2 \pm \sqrt{5}$ are real. Multiplying the off-diagonal entries by i changed $+16$ into -16. That single sign is the theme of this chapter.

14.2.3 Three general facts about eigenvalues

Fact 1: the trace is the sum of the eigenvalues. The *trace* of a square matrix, $\operatorname{tr} M = \sum_n M_{nn}$, is the sum of its diagonal entries. For 2×2 matrices, adding the two roots in (14.23) cancels the square roots and gives $\lambda_+ + \lambda_- = M_{11} + M_{22}$. The same is true for any size.

Taken on trust

Taken on trust: for any $K \times K$ matrix (real or complex, Hermitian or not), the K eigenvalues, counted with multiplicity, satisfy $\lambda_1 + \cdots + \lambda_K = \operatorname{tr} M$. The proof compares the coefficient of λ^{K-1} in two expressions for $\det(\lambda I - M)$: the factorization $\prod_i (\lambda - \lambda_i)$ (which needs the fundamental theorem of algebra) gives $-\sum_i \lambda_i$, and the permutation expansion of the determinant gives $-\sum_n M_{nn}$, because only the product of diagonal entries contains $K - 1$ factors of λ. See [28] or [22]. You prove the 3×3 case by hand in Problem 27.

For $M = D + i\mathsf{A}$ the trace is $\sum_n E_n + i\sum_n A_{nn}$. Taking real parts:

$$\sum_{\text{all eigenvalues}} \operatorname{Re}\lambda \;=\; \sum_{n=1}^{K} E_n \qquad \text{(exact, for any } \mathsf{A}\text{)}. \tag{14.24}$$

However large the imaginary perturbation, the real parts of the eigenvalues of the truncated matrix always add up to the sum of the unperturbed levels. We will meet this again as the sum rule of Section 14.5.

Fact 2: conjugating the matrix conjugates the eigenvalues. Let $\bar{M}$ be the matrix whose entries are the complex conjugates of those of M. If $Mv = \lambda v$, conjugate every component: $\bar{M}\bar{v} = \bar{\lambda}\bar{v}$. So $\bar{\lambda}$ is an eigenvalue of $\bar{M}$. Now apply this to $M(c) = D + ic\mathsf{A}$ with a real parameter c multiplying the perturbation. Since D and A are real,

$$\overline{M(c)} = D - ic\mathsf{A} = M(-c). \tag{14.25}$$

So the eigenvalues of $M(-c)$ are the conjugates of those of $M(c)$. Conjugation keeps real parts and flips imaginary parts. Therefore:

$$\text{the set of real parts is even in } c, \qquad \text{the set of imaginary parts is odd in } c. \tag{14.26}$$

This is exact, not perturbative. It already tells you that if real shifts can be expanded in powers of c, they contain only even powers: c^2, c^4, and so on. There can be no real shift linear in c.

Fact 3: an absorbing A gives nonnegative widths. The matrix version of (14.18) is short. If $Mv = \lambda v$, multiply on the left by the conjugate transpose $v^\dagger$: $v^\dagger D v + i\, v^\dagger \mathsf{A} v = \lambda\, v^\dagger v$. Both $v^\dagger D v = \sum E_n |v_n|^2$ and $v^\dagger \mathsf{A} v$ are real (the latter because A is real symmetric, hence Hermitian). Divide both sides by $v^\dagger v = \sum_n |v_n|^2$, which is a positive real number because $v \neq 0$:

$$\lambda = \frac{v^\dagger D v}{v^\dagger v} + i\,\frac{v^\dagger \mathsf{A} v}{v^\dagger v}.$$

The first term is real and the second is i times a real number, so taking the imaginary part of both sides gives $\operatorname{Im}\lambda = v^\dagger \mathsf{A} v / v^\dagger v$. Moreover $v^\dagger \mathsf{A} v = \int |\sum_n v_n \psi_n|^2 A(x)\,dx$ by (14.20), which is ≤ 0 if $A(x) \le 0$ everywhere. So even in a truncated basis, a pointwise-absorbing A gives $\operatorname{Im}\lambda \le 0$ for every eigenvalue.

Checkpoint 14.2

For $M(c) = D + ic\mathsf{A}$ with real D and A, a computer reports an eigenvalue $3.0 + 0.2i$ at $c = 0.1$. Name an eigenvalue of $M(-0.1)$ without computing anything.

Answer: $3.0 - 0.2i$, the complex conjugate, by Fact 2.

Key Ideas

- In the real eigenbasis of H_0, the Hamiltonian $H_0 + iA$ becomes $M = D + i\mathsf{A}$ with A real and symmetric: M is complex symmetric, not Hermitian.
- Multiplying off-diagonal couplings by i turns $+4M_{12}M_{21}$ into a negative number in the discriminant, so eigenvalues can become complex.
- The real parts of all eigenvalues add up to $\sum E_n$ exactly, (14.24).
- Real parts are even in the perturbation strength and imaginary parts odd, (14.26).

14.3 Perturbation theory for a non-Hermitian perturbation

14.3.1 Rederiving second order without assuming Hermiticity

In Chapter 13 you derived, for a real perturbation δV with matrix elements $V_{nm} = \langle n|\delta V|m\rangle$,

$$\delta E_n \approx V_{nn} + \sum_{m \neq n} \frac{|V_{nm}|^2}{E_n - E_m}. \tag{14.27}$$

It is tempting to substitute $\delta V \to iA$ and write $|V_{nm}|^2 = |iA_{nm}|^2 = |A_{nm}|^2$. That gives the *wrong sign*. The reason is that the derivation of (14.27) used Hermiticity at one point, to rewrite a product $V_{nm}V_{mn}$ as $|V_{nm}|^2$. For a non-Hermitian perturbation that step is false. So let us redo the derivation from scratch, for an arbitrary perturbation matrix W, in the matrix language of (14.21).

Setup. Let $M = D + \epsilon W$, where $D = \operatorname{diag}(E_1, \dots, E_K)$ has distinct entries, W is any $K \times K$ matrix (complex, not necessarily Hermitian or symmetric), and ϵ is a small bookkeeping parameter that we set to 1 at the end. Fix one level n. At $\epsilon = 0$, the eigenvector belonging to E_n is the unit vector e_n (1 in position n, zeros elsewhere). We assume, as in Chapter 13, that for small ϵ there is an eigenvalue $E(\epsilon)$ close to E_n whose eigenvector $v(\epsilon)$ is close to e_n, both depending smoothly on ϵ.

Taken on trust

Taken on trust: for a *nondegenerate* eigenvalue (one that is a simple root of the characteristic polynomial), the eigenvalue and a suitably normalized eigenvector are smooth (indeed analytic) functions of ϵ near $\epsilon = 0$. This is a theorem of matrix perturbation theory; see [29]. It fails at degeneracies, which is exactly where we will see perturbation theory break down in Section 14.8.

Normalization choice. Since v is close to e_n, its n-th component is not zero, and we can scale v so that $v_n = 1$ exactly. (Eigenvectors can be multiplied by any nonzero number.)

Row n of the eigen-equation. The n-th component of $Mv = Ev$ is $\sum_m M_{nm} v_m = E\, v_n$. With $M_{nm} = E_n \delta_{nm} + \epsilon W_{nm}$ and $v_n = 1$:

$$E_n + \epsilon W_{nn} + \epsilon \sum_{m \neq n} W_{nm} v_m = E. \tag{14.28}$$

Row $m \neq n$. The m-th component is $E_m v_m + \epsilon \sum_l W_{ml} v_l = E v_m$. Split off the term $l = n$ (where $v_n = 1$) and collect the v_m terms:

$$(E - E_m)\, v_m = \epsilon W_{mn} + \epsilon \sum_{l \neq n} W_{ml} v_l. \tag{14.29}$$

Size of v_m. At $\epsilon = 0$ the right side of (14.29) vanishes and $E - E_m = E_n - E_m \neq 0$, so $v_m = 0$; by smoothness, $v_m = \mathcal{O}(\epsilon)$ for every $m \neq n$. Then the sum on the right of (14.29) is $\epsilon \times \mathcal{O}(\epsilon) = \mathcal{O}(\epsilon^2)$. Also, from (14.28), $E - E_n = \mathcal{O}(\epsilon)$, so

$$\frac{1}{E - E_m} = \frac{1}{(E_n - E_m)\left(1 + \frac{E - E_n}{E_n - E_m}\right)} = \frac{1}{E_n - E_m}\big(1 + \mathcal{O}(\epsilon)\big). \tag{14.30}$$

Dividing (14.29) by $E - E_m$ therefore gives

$$v_m = \frac{\epsilon W_{mn}}{E_n - E_m} + \mathcal{O}(\epsilon^2), \qquad m \neq n. \tag{14.31}$$

Back into row n. Substitute (14.31) into (14.28). The term $\epsilon \sum W_{nm} v_m$ becomes $\epsilon^2 \sum W_{nm} W_{mn}/(E_n - E_m) + \mathcal{O}(\epsilon^3)$:

$$\boxed{E = E_n + \epsilon\, W_{nn} + \epsilon^2 \sum_{m \neq n} \frac{W_{nm} W_{mn}}{E_n - E_m} + \mathcal{O}(\epsilon^3).} \tag{14.32}$$

Nowhere did we use Hermiticity. For a Hermitian W, $W_{mn} = W_{nm}^*$, so $W_{nm} W_{mn} = |W_{nm}|^2$ and we recover (14.27) with $\epsilon = 1$. For a non-Hermitian W the numerator is $W_{nm} W_{mn}$, a product of two *different* matrix elements, and it need not be positive.

14.3.2 Specializing to $W = iA$

Now set $W = i\mathsf{A}$, so $W_{nm} = iA_{nm}$, and put $\epsilon = 1$.

First order. $W_{nn} = iA_{nn}$. Since $A_{nn} = \int \psi_n^2 A \, dx$ is real, the first-order shift is *purely imaginary*:

$$\delta E_n^{(1)} = iA_{nn}, \qquad \Gamma_n^{(1)} = -2 \operatorname{Im} \delta E_n^{(1)} = -2A_{nn} = -2 \int \psi_n^2 \, A \, dx. \tag{14.33}$$

The level itself (the real part) does not move at first order; only a width appears. This is (14.18) with the unperturbed ψ_n in place of the exact eigenfunction, and it agrees with Fact 2 of Section 14.2: no real shift linear in A.

Second order. $W_{nm}W_{mn} = (iA_{nm})(iA_{mn}) = i^2 A_{nm}A_{mn} = -A_{nm}A_{mn}$. Because A is real symmetric, $A_{mn} = A_{nm}$, so $A_{nm}A_{mn} = A_{nm}^2 = |A_{nm}|^2$. (More generally, for any Hermitian A, $A_{mn} = A_{nm}^*$ and $A_{nm}A_{mn} = |A_{nm}|^2$; the result below does not even need real eigenfunctions.) Therefore

$$\boxed{\delta E_n = -\sum_{m\neq n} \frac{|A_{nm}|^2}{E_n - E_m}} \qquad \text{(leading real shift)}. \tag{14.34}$$

This is real. Combined with (14.33), the complex eigenvalue to second order is $E_n + iA_{nn} - \sum_{m\neq n} |A_{nm}|^2/(E_n - E_m)$. Equation (14.34) is the paper's Eq. (5).

Pitfall

Never write $|W_{nm}|^2$ in the second-order formula for a non-Hermitian perturbation. The correct numerator is $W_{nm}W_{mn}$. For $W = iA$ that is $-|A_{nm}|^2$, and the overall sign of the second-order shift flips relative to the Hermitian case.

14.3.3 Level attraction

Compare (14.34) with the Hermitian second-order term in (14.27). Consider two levels $E_n < E_m$ coupled by a matrix element of size g.

- Hermitian (δV with $V_{nm} = g$): the lower level gets $g^2/(E_n - E_m) < 0$, the upper gets $g^2/(E_m - E_n) > 0$. They are pushed apart: *level repulsion* (Chapter 13).
- Non-Hermitian (iA with $A_{nm} = g$): the lower level gets $-g^2/(E_n - E_m) > 0$, the upper gets $-g^2/(E_m - E_n) < 0$. They are pulled together: *level attraction*.

Two consequences are worth stating explicitly. The ground state $n = 1$ has every other level above it, so every term $-|A_{1m}|^2/(E_1 - E_m)$ is nonnegative: *any* imaginary perturbation raises the real part of the ground level at second order (for nondegenerate levels, and as long as second order is valid). And in a truncated basis of K states, the top state $n = K$ has every other state below it, so it is always lowered. We will see that the second fact causes a bias at the top of a truncated state sum.

14.3.4 When is it valid?

The derivation assumed that the corrections are small: from (14.31), the admixture v_m is small only if $|A_{nm}| \ll |E_n - E_m|$. As in Chapter 13, the condition is that the coupling matrix elements be small compared with the level spacings. Here the condition is even more important than in the Hermitian case, because, as the next example shows, attracting levels can collide, and at the collision the expansion fails.

Sample Problem 14.3: Two levels: exact against second order

Consider $M = \begin{pmatrix} E_1 & ia \\ ia & E_2 \end{pmatrix}$ with $\Delta = E_2 - E_1 > 0$ and real $a \geq 0$. This is $D + i\mathsf{A}$ with $A_{12} = A_{21} = a$ and $A_{11} = A_{22} = 0$. Find the exact eigenvalues, the second-order prediction for the shift of the lower level, and compare them for $\Delta = 1$ and $a = 0.1$, 0.4 and 1.0.

Step 1: exact eigenvalues. In (14.23), $(M_{11} - M_{22})^2 = \Delta^2$ and $4M_{12}M_{21} = 4(ia)^2 = -4a^2$. With $\bar{E} = (E_1 + E_2)/2$,

$$\lambda_\pm = \bar{E} \pm \tfrac{1}{2}\sqrt{\Delta^2 - 4a^2}. \tag{14.35}$$

Step 2: exact shift of the lower level, while $2a < \Delta$. Factor Δ^2 out of the square root (allowed since $\Delta > 0$): $\sqrt{\Delta^2 - 4a^2} = \Delta\sqrt{1 - 4a^2/\Delta^2} = \Delta\sqrt{1-x}$ with $x = 4a^2/\Delta^2$. So $\lambda_- = \bar{E} - \frac{\Delta}{2}\sqrt{1-x}$. Since $E_1 = \bar{E} - \Delta/2$, the lower eigenvalue moves by

$$\lambda_- - E_1 = \frac{\Delta}{2}(1 - \sqrt{1-x}), \qquad x = \frac{4a^2}{\Delta^2}. \tag{14.36}$$

Step 3: second order. By (14.34), $\delta E_1 = -a^2/(E_1 - E_2) = a^2/\Delta$. The upper level gets $-a^2/\Delta$. (They attract.)

Step 4: numbers with $\Delta = 1$. For $a = 0.1$: $x = 0.04$, exact $= \frac{1}{2}(1 - \sqrt{0.96}) = 0.010102$; second order $= 0.0100$. For $a = 0.4$: $x = 0.64$, exact $= \frac{1}{2}(1 - 0.6) = 0.2000$; second order $= 0.1600$. For $a = 1.0$: $x = 4 > 1$, so the square root in (14.35) is $\sqrt{1-4} = \sqrt{3}\,i$ and $\lambda_\pm = 0.5 \pm 0.8660\,i$. Both eigenvalues have real part 0.5, so the real shift of the lower level is exactly 0.5; second order predicts 1.0.

a	0.1	0.4	1.0
exact	0.0101	0.200	0.5 (and Im $= \pm 0.866$)
2nd order	0.0100	0.160	1.0

At small coupling the formula is excellent. At $a = 0.4$ it is already off by 20%, and at $a = \Delta/2$ (exact 0.5, second order 0.25) by 50%. Beyond $a = \Delta/2$ the levels have merged into a complex pair, the exact real shift is stuck at $\Delta/2$, and the second-order formula, which keeps growing like a^2, overstates the real shift by a factor of two at $a = \Delta$.

Sample Problem 14.4: The i^2 flip in the harmonic oscillator

Indexing note: in this sample, in the box that follows it, and in Problem 28, oscillator levels are labelled from $n = 0$, as is traditional (ground state $E_0 = 1$, then $E_1 = 3$, in general $E_n = 2n+1$). These E_n are oscillator energies, not the box levels $E_n = (n\pi/2L)^2$, $n \geq 1$, used everywhere else in this chapter. In units $\hbar = 2m = 1$, the oscillator $H_0 = -\mathrm{d}^2/\mathrm{d}x^2 + x^2$ has ground state $\psi_0 = \pi^{-1/4}e^{-x^2/2}$ with $E_0 = 1$ and first excited state $\psi_1 = \pi^{-1/4}\sqrt{2}\,x\,e^{-x^2/2}$ with $E_1 = 3$ (Chapter 11). Compare the ground-state energy for the Hermitian perturbation ϵx and the imaginary perturbation $i\epsilon x$, exactly and at second order, with $\epsilon = 0.6$.

Step 1: the matrix elements of x from the ground state. Multiply ψ_0 by x: $x\psi_0 = \pi^{-1/4}xe^{-x^2/2} = \psi_1/\sqrt{2}$. Since the ψ_m are orthonormal, $\langle m|x|0\rangle = \int \psi_m\, x\psi_0\,\mathrm{d}x = \delta_{m1}/\sqrt{2}$. Only $m = 1$ couples, with $x_{01} = 1/\sqrt{2}$.

Step 2: second order. Hermitian ϵx: by (14.27), $\delta E_0 = \epsilon x_{00} + \epsilon^2|x_{01}|^2/(E_0 - E_1) = 0 + \epsilon^2(\frac{1}{2})/(-2) = -\epsilon^2/4$. Imaginary $i\epsilon x$ (that is, $A = \epsilon x$): by (14.34), $\delta E_0 = -\epsilon^2(\frac{1}{2})/(-2) = +\epsilon^2/4$. Also $A_{00} = \epsilon x_{00} = 0$ because $\psi_0^2 x$ is odd.

Step 3: exact, Hermitian case. Complete the square: $x^2 + \epsilon x = (x + \epsilon/2)^2 - \epsilon^2/4$. The potential is the oscillator moved left by $\epsilon/2$ and lowered by $\epsilon^2/4$, so $E_0 = 1 - \epsilon^2/4$ exactly.

Step 4: exact, imaginary case. Try $\chi(x) = e^{-(x+i\epsilon/2)^2/2}$. Then $\chi' = -(x + i\epsilon/2)\chi$ and $\chi'' = -\chi + (x + i\epsilon/2)^2\chi$. Since $(x + i\epsilon/2)^2 = x^2 + i\epsilon x - \epsilon^2/4$,

$$-\chi'' + (x^2 + i\epsilon x)\chi = \chi - (x^2 + i\epsilon x - \tfrac{\epsilon^2}{4})\chi + (x^2 + i\epsilon x)\chi = \Big(1 + \frac{\epsilon^2}{4}\Big)\chi.$$

And $|\chi|^2 = e^{-x^2+\epsilon^2/4}$ is normalizable, so χ is a genuine eigenfunction with $E = 1 + \epsilon^2/4$, a *real* eigenvalue of a non-Hermitian operator.

Step 5: numbers. With $\epsilon = 0.6$, $\epsilon^2/4 = 0.09$.

$$\boxed{E_0 = 0.91 \text{ (Hermitian } \epsilon x), \qquad E_0 = 1.09 \ (i\epsilon x)}$$

Here second order happens to be exact in both cases, and the only difference is the sign, coming from $i^2 = -1$. (The CHECKS script confirms 1.09 by direct numerical diagonalization of the complex matrix.)

Taken on trust

Taken on trust: for every level, $-\mathrm{d}^2/\mathrm{d}x^2 + x^2 + i\epsilon x$ has eigenvalue $2n + 1 + \epsilon^2/4$, with eigenfunctions obtained from the oscillator's by the complex translation $x \to x + i\epsilon/2$. Justifying that such a translation preserves the whole spectrum requires complex analysis beyond this book; Step 4 checks only the ground state directly. Problem 28 asks you to recover $+\epsilon^2/4$ for every level from (14.34), given the oscillator matrix elements $\langle n|x|n+1\rangle = \sqrt{(n+1)/2}$ in these units [30].

14.3.5 First-order widths in the box

To use (14.33) and (14.34) in the box, we need matrix elements of cosines and sines between box states. Recall from Chapter 11 that on $[-L, L]$ with Dirichlet walls,

$$\psi_n(x) = \frac{1}{\sqrt{L}} \sin\frac{n\pi(x+L)}{2L}, \qquad E_n = \left(\frac{n\pi}{2L}\right)^2, \qquad n = 1, 2, \ldots \tag{14.37}$$

Change variable to $u = (x + L)/(2L)$, which runs from 0 to 1; then $x = L(2u - 1)$, $\mathrm{d}x = 2L\,\mathrm{d}u$, and $\psi_n = L^{-1/2}\sin(n\pi u)$.

Cosine elements. For an integer $k \geq 1$, $\cos(k\pi x/L) = \cos(2k\pi u - k\pi) = (-1)^k \cos(2k\pi u)$, using $\cos(\theta - k\pi) = \cos\theta\cos k\pi + \sin\theta \sin k\pi = (-1)^k \cos\theta$. Then

$$(\cos_k)_{nm} \equiv \int_{-L}^{L} \psi_n \psi_m \cos\frac{k\pi x}{L}\,\mathrm{d}x = 2(-1)^k \int_0^1 \sin(n\pi u)\sin(m\pi u)\cos(2k\pi u)\,\mathrm{d}u. \tag{14.38}$$

Use $\sin a \sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$ to write the integrand as $\frac{1}{2}[\cos((n-m)\pi u) - \cos((n+m)\pi u)]\cos(2k\pi u)$. For integers q and $k \geq 1$,

$$\int_0^1 \cos(q\pi u)\cos(2k\pi u)\,\mathrm{d}u = \frac{1}{2}\int_0^1 \big[\cos((q-2k)\pi u) + \cos((q+2k)\pi u)\big]\mathrm{d}u = \frac{1}{2}(\delta_{q,2k} + \delta_{q,-2k}), \tag{14.39}$$

because $\int_0^1 \cos(r\pi u)\,\mathrm{d}u = \sin(r\pi)/(r\pi) = 0$ for a nonzero integer r and equals 1 for $r = 0$. With $q = n - m$ (which can be $\pm 2k$) and $q = n + m$ (positive, so only $+2k$),

$$(\cos_k)_{nm} = \frac{(-1)^k}{2}\Big[\delta_{|n-m|,2k} - \delta_{n+m,2k}\Big], \quad k \geq 1; \qquad (\cos_0)_{nm} = \delta_{nm}. \tag{14.40}$$

The last statement is orthonormality, since $\cos 0 = 1$. On the diagonal, $|n - n| = 0 \neq 2k$ and $n + n = 2k$ only for $n = k$, so $(\cos_k)_{nn} = -\frac{(-1)^k}{2}\delta_{nk}$.

Sample Problem 14.5: Widths from a center-weighted absorber

In the RE3 box ($L = 44.883$), place the absorber $A(x) = -g[1 + \cos(\pi x/L)]$ with $g = 0.01$. It is nonpositive everywhere (since $\cos \geq -1$), strongest at the center and zero at the walls. Find the first-order widths and lifetimes of all levels.

Step 1: diagonal elements. $A_{nn} = -g[(\cos_0)_{nn} + (\cos_1)_{nn}]$. From (14.40), $(\cos_0)_{nn} = 1$ and $(\cos_1)_{nn} = -\frac{(-1)}{2}\delta_{n1} = \frac{1}{2}\delta_{n1}$.
Step 2: widths. $A_{11} = -\frac{3}{2}g$ and $A_{nn} = -g$ for $n \geq 2$. By (14.33), $\Gamma_1 = 3g = 0.03$ and $\Gamma_n = 2g = 0.02$ for $n \geq 2$.
Step 3: lifetimes. $\tau_1 = 1/0.03 = 33.3$, $\tau_n = 1/0.02 = 50$.
Step 4: physical check. The ground state $\psi_1 \propto \cos(\pi x/2L)$ is concentrated at the center, where absorption is strongest, so it should decay fastest. Indeed, from (14.37), $\psi_1 = L^{-1/2}\sin(\pi x/2L + \pi/2) = L^{-1/2}\cos(\pi x/2L)$ (using $\sin(\theta + \pi/2) = \cos\theta$), so $|\psi_1|^2 = \frac{1}{L}\cos^2(\pi x/2L) = \frac{1}{2L}(1 + \cos(\pi x/L))$ (using $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$), which has the same shape as the absorber.

$$\boxed{\Gamma_1 = 0.03,\ \tau_1 = 33.3; \qquad \Gamma_{n\geq 2} = 0.02,\ \tau_n = 50}$$

Real parts are unchanged at first order. They would shift at second order through the off-diagonal elements $A_{n,n\pm 2} = g/2$ that (14.40) provides.

Checkpoint 14.3

At second order, does an imaginary perturbation iA push the ground level up or down? What does a real perturbation δV do to it at second order?

Answer: Up: every term $-|A_{1m}|^2/(E_1 - E_m)$ is ≥ 0 because $E_1 < E_m$. A real perturbation pushes it down at second order, since every term $|V_{1m}|^2/(E_1 - E_m)$ is ≤ 0.

Key Ideas

- For any perturbation W, the second-order shift is $\sum_{m\neq n} W_{nm}W_{mn}/(E_n - E_m)$, (14.32); Hermiticity is what turns the numerator into $|W_{nm}|^2$.
- For $W = iA$: first order is purely imaginary, iA_{nn} (a width); the leading real shift is second order and has a minus sign, (14.34).
- Imaginary couplings make levels attract; the ground level always rises and the top of a truncated basis always falls.
- Validity requires $|A_{nm}| \ll |E_n - E_m|$; in the two-level model the formula fails completely beyond $a = \Delta/2$.

14.4 The quadratic form and the tensors $T^{(n)}$

★ Did you know? *Upside-Down but Stable*

In 1998 Carl Bender and Stefan Boettcher found numerically that Hamiltonians like $p^2 - (ix)^N$ with $N \geq 2$ have entirely real, positive spectra even though they are not Hermitian. The case $N = 4$ is the upside-down quartic $-x^4$, with suitable boundary conditions in the complex plane. Patrick Dorey, Clare Dunning and Roberto Tateo proved that the spectrum is real in 2001.

14.4.1 From a function to coefficients

In the paper, A is not a fixed function; it is a model with adjustable coefficients,

$$A(x) = \sum_{k=0}^{M-1} c_k \, \phi_k(x), \tag{14.41}$$

with real coefficients c_k, collected in a vector c, and fixed basis functions ϕ_k (Chapter 24). Because A_{nm} in (14.20) is linear in A,

$$A_{nm} = \sum_k c_k \, (\phi_k)_{nm}, \qquad (\phi_k)_{nm} = \int \psi_n \phi_k \psi_m \, \mathrm{d}x. \tag{14.42}$$

Square it, writing the square of a sum as a double sum:

$$A_{nm}^2 = \Big(\sum_k c_k (\phi_k)_{nm}\Big)\Big(\sum_l c_l (\phi_l)_{nm}\Big) = \sum_k \sum_l c_k c_l \, (\phi_k)_{nm} (\phi_l)_{nm}. \tag{14.43}$$

Insert into (14.34) and exchange the order of the finite sums:

$$\delta E_n = -\sum_{m \neq n} \frac{1}{E_n - E_m} \sum_{k,l} c_k c_l (\phi_k)_{nm} (\phi_l)_{nm} = -\sum_{k,l} c_k \, T^{(n)}_{kl} \, c_l, \tag{14.44}$$

where

$$\boxed{T^{(n)}_{kl} = \sum_{m \neq n} \frac{(\phi_k)_{nm} (\phi_l)_{nm}}{E_n - E_m}, \qquad \delta E_n = -c^{\top} T^{(n)} c.} \tag{14.45}$$

For each level n there is one $M \times M$ matrix $T^{(n)}$; the collection of all of them is what the paper calls the "tensors". They depend only on the unperturbed problem and the basis, not on c, so they can be computed once and reused. Then every evaluation of the model is just a quadratic form, which is why the paper's Eq. (5) writes $\delta E_n = -c^{\top} T^{(n)} c$.

Pitfall

The letter T is also overloaded. In this chapter $T^{(n)}$ is the second-order tensor; in Section 14.7 we will need the time-reversal operation, which we therefore write as a calligraphic $\mathcal{T}$; and elsewhere in the paper T is the length of a training window and T_k is a Chebyshev polynomial.

14.4.2 Properties of the quadratic form

- **Symmetric.** Swapping k and l in (14.45) swaps two factors in a product, so $T^{(n)}_{lk} = T^{(n)}_{kl}$.
- **Even in c.** Replacing c by $-c$ leaves $c^{\top} T^{(n)} c$ unchanged; doubling c quadruples it. This is (14.26) again. A fit to real level data therefore always has at least two equally good solutions, c and $-c$.
- **Gradient.** For a symmetric matrix T, $\partial (c^{\top} T c)/\partial c_k = 2 (Tc)_k$. (Proof: $c^{\top} T c = \sum_{a,b} c_a T_{ab} c_b$; differentiating with respect to c_k gives $\sum_b T_{kb} c_b + \sum_a c_a T_{ak} = 2 \sum_b T_{kb} c_b$, using symmetry in the second sum.) So $\partial \, \delta E_n / \partial c_k = -2 (T^{(n)} c)_k$. The matrix of these partial derivatives, with one row per level n and one column per coefficient k, is called the *Jacobian* of the model; an optimizer such as Levenberg–Marquardt uses it to decide how to change c at each step (Chapter 24).

- **Indefinite, with a sign pattern.** For $n = 1$ every denominator $E_1 - E_m$ is negative, and $c^{\top}T^{(1)}c = \sum_{m\neq 1} A_{1m}^2/(E_1 - E_m) \le 0$, so $\delta E_1 \ge 0$ for every c. For middle levels the terms from above and below have opposite signs, so $c^{\top}T^{(n)}c$ can take either sign depending on c; with several basis functions $T^{(n)}$ typically has eigenvalues of both signs.

- **Nonconvex fitting.** A fit chooses c to minimize the squared error $f(c) = \sum_n (y_n + c^{\top}T^{(n)}c)^2$ for real targets y_n. Each term is a fourth-degree polynomial in the components of c, and $f(-c) = f(c)$ because the form is even. Suppose a best point $c_\star \neq 0$ improves on doing nothing, $f(c_\star) < f(0) = \sum_n y_n^2$. Then $c_\star$ and $-c_\star$ are equally good, while their midpoint 0 is worse than both. A function with this property is called *nonconvex*: it has separate valleys, so a downhill search can end in different places depending on where it starts. (A convex function, one shaped like a single bowl, always satisfies $f(\text{midpoint}) \le$ the average of the two end values, which fails here.) This is consistent with the paper's use of random restarts together with Levenberg–Marquardt, a damped downhill method explained in Chapter 24.

14.4.3 Sine elements in the box

For the next example we need the matrix elements of an odd function. Let $j \ge 1$ be an integer and $\phi(x) = \sin(j\pi x/L)$. With the substitution of Section 14.3, $\sin(j\pi x/L) = \sin(2j\pi u - j\pi) = (-1)^j \sin(2j\pi u)$, using $\sin(\theta - j\pi) = \sin\theta\cos j\pi - \cos\theta \sin j\pi$. Then, with $\sin a \sin b = \tfrac{1}{2}[\cos(a-b) - \cos(a+b)]$,

$$(\sin_j)_{nm} = 2(-1)^j \int_0^1 \sin(n\pi u)\sin(m\pi u)\sin(2j\pi u)\,\mathrm{d}u = (-1)^j\big[I(n-m) - I(n+m)\big], \tag{14.46}$$

with $I(q) = \int_0^1 \cos(q\pi u)\sin(2j\pi u)\,\mathrm{d}u$. Using $\sin b\cos a = \tfrac{1}{2}[\sin(b+a) + \sin(b-a)]$,

$$I(q) = \frac{1}{2}\int_0^1 \big[\sin((2j+q)\pi u) + \sin((2j-q)\pi u)\big]\mathrm{d}u. \tag{14.47}$$

For a nonzero integer r, $\int_0^1 \sin(r\pi u)\,\mathrm{d}u = \big[-\cos(r\pi u)/(r\pi)\big]_0^1 = (1-(-1)^r)/(r\pi)$, which is $2/(r\pi)$ for odd r and 0 for even r (and the integral is 0 for $r = 0$). If q is even, both $2j \pm q$ are even and $I(q) = 0$. If q is odd,

$$I(q) = \frac{1}{2}\left[\frac{2}{(2j+q)\pi} + \frac{2}{(2j-q)\pi}\right] = \frac{1}{\pi}\,\frac{(2j-q)+(2j+q)}{4j^2 - q^2} = \frac{4j}{\pi(4j^2 - q^2)}. \tag{14.48}$$

Since $n - m$ and $n + m$ have the same parity,

$$(\sin_j)_{nm} = \begin{cases} (-1)^j \dfrac{4j}{\pi}\left[\dfrac{1}{4j^2 - (n-m)^2} - \dfrac{1}{4j^2 - (n+m)^2}\right], & n+m \text{ odd}, \\ 0, & n+m \text{ even}. \end{cases} \tag{14.49}$$

In particular the diagonal elements vanish ($n+n$ is even): an odd function produces no first-order width. We return to the reason in Section 14.7.

Sample Problem 14.6: Four box states and one sine

Take the box with $L = \pi/2$, so that (14.37) gives the convenient levels $E_n = n^2$: $1, 4, 9, 16$. Keep only these four states and let $A(x) = c\sin(\pi x/L)$ ($M = 1$, $j = 1$). Compute $T^{(n)}$ for $n = 1, \dots, 4$ and the second-order shifts, and add them up.

Step 1: matrix elements. From (14.49) with $j = 1$ (prefactor $-4/\pi$), the nonzero pairs have $n + m$ odd:

$$(\sin_1)_{12} = -\tfrac{4}{\pi}\left[\tfrac{1}{3} - \tfrac{1}{4-9}\right] = -\tfrac{4}{\pi}\cdot\tfrac{8}{15} = -0.6791, \quad (\sin_1)_{14} = -\tfrac{4}{\pi}\left[\tfrac{1}{4-9} - \tfrac{1}{4-25}\right] = +0.1940,$$
$$(\sin_1)_{23} = -\tfrac{4}{\pi}\left[\tfrac{1}{3} - \tfrac{1}{4-25}\right] = -0.4850, \qquad (\sin_1)_{34} = -\tfrac{4}{\pi}\left[\tfrac{1}{3} - \tfrac{1}{4-49}\right] = -0.4527.$$

The pairs $(1,3)$ and $(2,4)$ vanish.
Step 2: the 1×1 tensors. With one basis function, $T^{(n)} = \sum_{m\neq n}(\sin_1)^2_{nm}/(E_n - E_m)$:

$$T^{(1)} = \tfrac{0.6791^2}{1-4} + \tfrac{0.1940^2}{1-16} = -0.15371 - 0.00251 = -0.1562,$$
$$T^{(2)} = \tfrac{0.6791^2}{4-1} + \tfrac{0.4850^2}{4-9} = 0.15371 - 0.04705 = 0.1067,$$
$$T^{(3)} = \tfrac{0.4850^2}{9-4} + \tfrac{0.4527^2}{9-16} = 0.04705 - 0.02928 = 0.0178,$$
$$T^{(4)} = \tfrac{0.1940^2}{16-1} + \tfrac{0.4527^2}{16-9} = 0.00251 + 0.02928 = 0.0318.$$

Step 3: shifts. $\delta E_n = -c^2 T^{(n)}$:

$$\boxed{\delta E_1 = +0.1562\,c^2, \quad \delta E_2 = -0.1067\,c^2, \quad \delta E_3 = -0.0178\,c^2, \quad \delta E_4 = -0.0318\,c^2}$$

Step 4: the sum. $0.1562 - 0.1067 - 0.0178 - 0.0318 = -0.0001$, which is zero to the rounding of the four-digit entries (to full precision it is zero, as the CHECKS script confirms). The ground level rose, as it must, and the top level fell, as it must in a truncated basis. Level 4 is pushed down partly because its partners above (states 5, 7, 9, ..., those with $4 + m$ odd) were left out; Problem 17 shows how much it changes when state 5 is added.

Checkpoint 14.4

A coefficient vector c produces second-order shifts δE_n. What shifts do $-c$ and $3c$ produce?

Answer: $-c$ produces the same shifts δE_n (the form is quadratic); $3c$ produces $9\,\delta E_n$.

Key Ideas

- With $A = \sum_k c_k \phi_k$, the real second-order shift is the quadratic form $\delta E_n = -c^\top T^{(n)} c$ with the symmetric tensors (14.45).
- The form is even in c, has gradient $-2T^{(n)}c$, and can take either sign for middle levels; fitting with it is nonconvex (the equally good points c and $-c$ are separated by the worse point 0).
- In the box, odd sines couple only states of opposite parity and have zero diagonal elements, (14.49).

14.5 The sum rule and locality

14.5.1 The sum rule

The four-state example ended with shifts adding to zero. That was not luck.

Theorem (sum rule). Let S be a finite set of nondegenerate levels, and suppose that for every $n \in S$ the state sum in (14.34) runs over $m \in S$, $m \neq n$ (the same closed set). Then

$$\sum_{n\in S} \delta E_n = 0. \tag{14.50}$$

Proof. Write the total as a double sum over ordered pairs:

$$\sum_{n\in S} \delta E_n = -\sum_{n\in S} \sum_{\substack{m\in S \\ m\neq n}} \frac{A_{nm}^2}{E_n - E_m}. \tag{14.51}$$

Each unordered pair $\{n, m\}$ with $n \neq m$ appears exactly twice, once as (n, m) and once as (m, n). Its two contributions are

$$-\frac{A_{nm}^2}{E_n - E_m} - \frac{A_{mn}^2}{E_m - E_n} = -\frac{A_{nm}^2}{E_n - E_m} + \frac{A_{nm}^2}{E_n - E_m} = 0, \tag{14.52}$$

using $A_{mn} = A_{nm}$ and $E_m - E_n = -(E_n - E_m)$. The contribution of each pair is *antisymmetric* under exchange, so every pair cancels, and the total is zero. ■

The same holds for each tensor combination: $\sum_n T^{(n)} = 0$ as a matrix, by the identical pairing argument applied to $(\phi_k)_{nm}(\phi_l)_{nm}/(E_n - E_m)$.

14.5.2 An exact version

The sum rule is not just a property of second-order perturbation theory. Equation (14.24) says that for the exact eigenvalues of the truncated matrix $D + i\mathsf{A}$, the real parts add up to $\sum E_n$ *exactly*, at any amplitude. Write the exact eigenvalue near E_n as $E_n + \Delta E_n^{\text{exact}}$; then $\sum_n \operatorname{Re} \Delta E_n^{\text{exact}} = 0$. Expanding in powers of the amplitude, each order must cancel separately. The first order has no real part, and the second order is (14.50).

Compare the Hermitian case. There the trace rule gives $\sum_n \Delta E_n^{\text{exact}} = \sum_n V_{nn}$: the total shift is the first-order term, and the Hermitian second-order terms, which have the same pairwise antisymmetry, also add to zero. The difference is that a real δV can move the whole spectrum up or down through $\sum V_{nn}$, while an imaginary iA has no real first-order term. Its real shifts can only *redistribute* level positions: whatever it pushes up somewhere, it pushes down somewhere else by the same total amount.

Pitfall

The sum rule needs a closed state set. If you compute δE_n for levels $1, \dots, 400$ but let each state sum run over 652 states, the pairs (n, m) with $m > 400$ contribute once but their partners (m, n) are not in your total, and the sum need not vanish. Conversely, if you stop every state sum at a cutoff, the sum rule holds but the levels just below the cutoff are biased, because they lose all their partners above (Problem 17).

14.5.3 Locality

Which states m dominate the sum for a given level n? Define the weights

$$w_{nm} = \frac{|A_{nm}|^2}{|E_n - E_m|}, \tag{14.53}$$

the magnitudes of the individual terms. Two effects make w_{nm} small when $|n - m|$ is large. The denominator grows. And the matrix element shrinks if A is smooth.

The second effect is a Fourier statement (Chapter 5), which we give as a *heuristic*. A level n has a local wavenumber k_n; in the box $k_n = n\pi/(2L)$. The product $\psi_n\psi_m$ contains oscillations at the difference wavenumber $k_n - k_m$ and the sum $k_n + k_m$. Integrating it against $A(x)$ picks out the Fourier components of A at those wavenumbers. A smooth A has little weight at high wavenumber, so it couples mainly states whose wavenumbers differ by little, that is, nearby levels. The box makes this exact: $\cos(k\pi x/L)$ has wavenumber $k\pi/L = 2k \cdot \pi/(2L)$, and (14.40) shows it couples only $|n - m| = 2k$ (apart from the single pair with $n + m = 2k$). Higher basis functions reach further in the index. For the sine, (14.49) decays like $1/(n - m)^2$ for large $|n - m|$.

For a concrete number, take $A = c\sin(\pi x/L)$ in the box with $E_n = n^2$ and level $n = 20$. Summing (14.53) over all partners $m \leq 400$ (the range used in the CHECKS script), about 89% of that total weight comes from the two immediate neighbors $m = 19$ and $m = 21$. Figure 14.1 shows the weights.

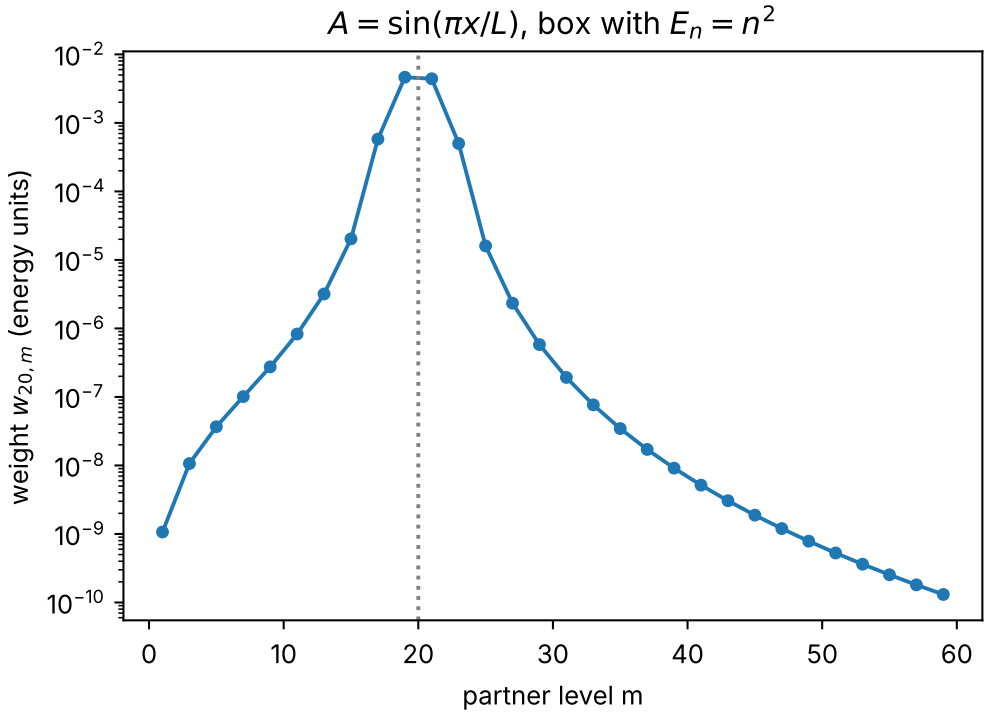


Figure 14.1: Locality of the second-order redistribution. Weights $w_{20,m} = A^2_{20,m}/|E_{20} - E_m|$ for $A = \sin(\pi x/L)$ in the box with $E_n = n^2$ (dimensionless energy units, $\hbar = 2m = 1$, $L = \pi/2$). Only partners with $20 + m$ odd couple. The two neighbors $m = 19, 21$ carry about 89% of the total weight summed over partners $m \leq 400$ (only $m \leq 60$ are plotted); the weights fall by orders of magnitude within a few levels.

Locality plus the sum rule give a picture: the imaginary perturbation moves levels toward each other in local groups, and within each neighborhood the pushes approximately cancel. Heuristically, a net shift of a long block of consecutive levels (say 50 of them) is suppressed: most pair contributions cancel inside the block, and only pairs straddling its edges contribute. This is a tendency, not a theorem; the sum rule itself is global, and levels near the edges of a block can exchange shift with partners outside it. This picture motivated a specific hypothesis in the paper, which we discuss in Section 14.9.

Checkpoint 14.5

Can the second-order real shifts of an imaginary perturbation raise every one of 652 levels (with the state sum closed over those same 652 levels) by 0.1?

Answer: No. The sum rule (14.50) forces the 652 shifts to add to zero, so some must be negative.

Key Ideas

- Pair contributions to second-order real shifts are antisymmetric, so they add to zero over any closed set of states, (14.50).
- The exact real parts of $D + i\mathsf{A}$ also add up to $\sum E_n$, by the trace; imaginary perturbations can only redistribute levels.
- Smooth A couples mainly nearby levels: the redistribution is local, with range set by the highest wavenumber present in A.

14.6 The null mode

14.6.1 A basis function that does nothing

The absorptive-even basis of the paper is $\phi_k = \cos(k\pi x/L)$, $k = 0, 1, \ldots$, and its first member is $\phi_0 = \cos 0 = 1$, a constant. What does a constant imaginary potential ic_0 do?

Exactly: by Section 14.1, $H_0 + ic_0$ has eigenvalues $E_n + ic_0$ and the same eigenfunctions. The real parts do not move at all, at any amplitude.

In the tensors: $(\phi_0)_{nm} = \int \psi_n \psi_m \, dx = \delta_{nm}$. In (14.45) the sum runs over $m \neq n$, where $(\phi_0)_{nm} = 0$. Therefore

$$T^{(n)}_{0l} = T^{(n)}_{l0} = 0 \qquad \text{for every } n \text{ and } l. \tag{14.54}$$

The coefficient c_0 drops out of $c^\top T^{(n)} c$ entirely: it is a *null mode*, or "dead parameter", of the model. Any value of c_0 gives the same predictions.

14.6.2 When zero is not zero

On a computer the integral is replaced by a sum over grid points. Recall from Chapter 8 that on a grid of points x_j with spacing h the second-derivative operator becomes a tridiagonal matrix; its eigenvectors give samples $\psi_n(x_j)$, which we scale so that $\sum_j \psi_n(x_j)^2 h = 1$, the discrete version of normalization (the method is treated in detail later, in Chapter 23). The overlaps are then $(\phi_0)_{nm} = \sum_j \psi_n(x_j)\psi_m(x_j)\, h$, and rounding makes the off-diagonal ones small but not exactly zero. In double precision they come out below 10^{-11} in the example of Figure 14.2. If the eigenvectors are stored in single precision (float32, machine epsilon 1.2×10^{-7}; Chapter 4), the same overlaps are nonzero at the level of float32 rounding, many orders of magnitude larger.

```
import numpy as np
from scipy.linalg import eigh_tridiagonal
L, N = np.pi/2, 400
h = 2*L/(N+1)
d = np.full(N, 2/h**2)              # V = 0 inside the box
e = np.full(N-1, -1/h**2)
E, psi = eigh_tridiagonal(d, e, select="i", select_range=(0, 59))
psi /= np.sqrt(h)                   # now sum(psi**2)*h = 1
O64 = psi.T @ psi * h               # (phi_0)_{nm}: should be the identity
p32 = psi.astype(np.float32)
O32 = (p32.T @ p32) * np.float32(h)
```

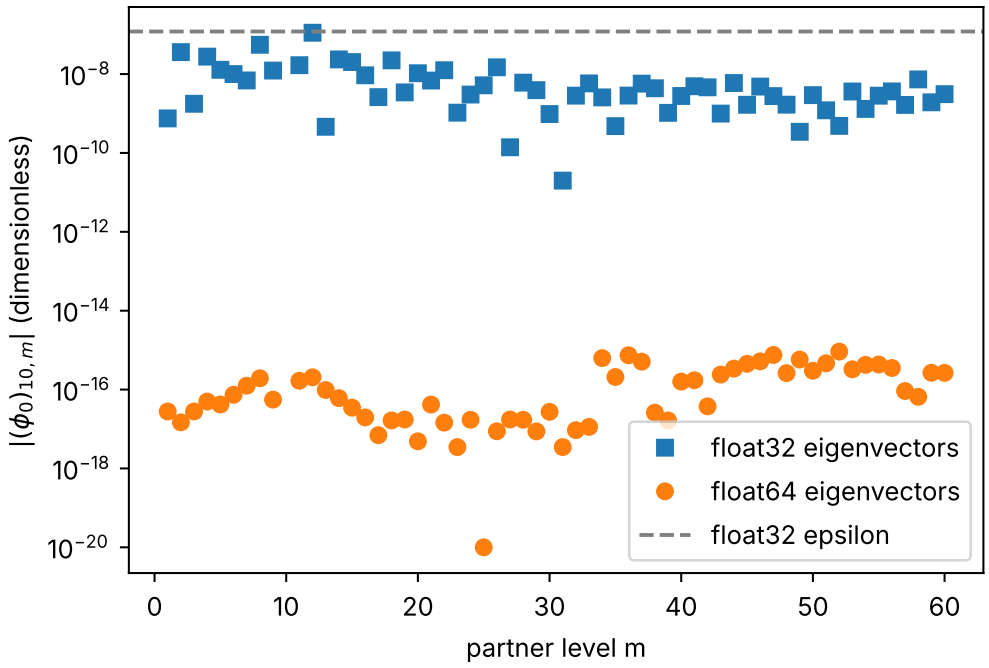


Figure 14.2: Overlaps $|(\phi_0)_{10,m}| = |\sum_j \psi_{10}(x_j)\psi_m(x_j)h|$ for $m \neq 10$, which are exactly zero analytically, computed from finite-difference box eigenvectors ($L = \pi/2$, $N = 400$ grid points, normalization $\sum_j \psi^2 h = 1$) in double precision (float64) and after rounding the eigenvectors to single precision (float32). The float32 values sit near the float32 rounding scale; the float64 values are many orders of magnitude smaller. Values below 10^{-20} are plotted at 10^{-20}.

```
off = ~np.eye(60, dtype=bool)
print("max off-diagonal, float64:", np.abs(O64[off]).max())
print("max off-diagonal, float32:", np.abs(O32[off]).max())
```

Now suppose these noisy overlaps η_{nm} are fed into (14.45). The "zero" rows become $T_{0l}^{(n)} = \sum_{m\neq n} \eta_{nm}(\phi_l)_{nm}/(E_n - E_m)$, tiny but nonzero, and $T_{00}^{(n)}$ becomes of order η^2. An optimizer minimizing an in-sample error does not know these entries are noise. If making c_0 enormous lowers the error slightly, it will make c_0 enormous.

Sample Problem 14.7: How large can a dead parameter grow?

A toy model: one level, one dead coefficient, $\delta E = -c_0^2\, T_{00}$ with a noise entry $T_{00} = 10^{-14}$ (roughly the square of a 10^{-7} overlap divided by a spacing of order one). The fit wants $\delta E = y = -0.01$. What value of c_0 does an unconstrained least-squares fit choose?
Step 1. Set $-c_0^2 T_{00} = y$, so $c_0^2 = -y/T_{00} = 0.01/10^{-14} = 10^{12}$.
Step 2. $c_0 = \pm 10^6$.

$$\boxed{|c_0| = 10^6}$$

A parameter that should be irrelevant becomes the largest number in the model, and the "improvement" it buys is entirely an artifact of rounding. In a real fit the cross terms $2c_0 \sum_l c_l T_{0l}^{(n)}$, linear in c_0, also become order-one numbers when $c_0 \sim 10^6$ multiplies entries of order 10^{-7}, so the noise acts as an extra set of fitting functions. This toy is only an order-of-magnitude illustration; it does not reproduce any specific number in the paper.

Pitfall

When a model contains a parameter that is exactly null by analysis, set its contribution to exactly zero in code (here: zero the rows and columns (14.54), or drop ϕ_0 from the basis). Do not trust the computer to compute zero. A large fitted value for a parameter that should not matter is a warning sign, not a discovery.

Checkpoint 14.6

Why does a constant imaginary potential ic_0 leave the real parts of all levels unchanged, even exactly and at large c_0?

Answer: Because $H_0 + ic_0$ has the same eigenfunctions as H_0, with eigenvalues $E_n + ic_0$: the constant only adds an imaginary number to each eigenvalue.

Key Ideas

- A constant imaginary term shifts all eigenvalues by the same imaginary constant; its tensor rows $T^{(n)}_{0l}$ vanish identically.
- Numerically, analytically-zero entries become rounding noise (∼float32 epsilon in single precision), and an optimizer can drive a dead parameter to huge values to exploit it.
- Zero null contributions exactly in code.

14.7 PT symmetry

14.7.1 Parity and time reversal

We have seen that some non-Hermitian operators, such as $-\mathrm{d}^2/\mathrm{d}x^2 + x^2 + i\epsilon x$, have entirely real spectra. A symmetry explains this. Define two operations on wavefunctions.

Parity $\mathcal{P}$ reflects space: $(\mathcal{P}f)(x) = f(-x)$.

Time reversal $\mathcal{T}$ is complex conjugation: $(\mathcal{T}f)(x) = f(x)^*$. Why does conjugation reverse time? Suppose $\Psi(x,t)$ solves (14.1) with a *real* V, and define $\Phi(x,t) = \Psi(x,-t)^*$. Conjugating (14.1) gives $-i\,\partial_t\Psi^* = -\Psi^{*\prime\prime} + V\Psi^*$. Evaluate it at time $-t$. By the chain rule, $\partial_t\Phi(x,t) = -(\partial_t\Psi^*)(x,-t)$, so the left side becomes $i\,\partial_t\Phi$, and the equation reads $i\,\partial_t\Phi = -\Phi'' + V\Phi$. So Φ is also a solution: the movie run backwards, conjugated, is a valid motion. If V is complex, the same calculation produces V^* in place of V: running a lossy system backwards turns loss into gain.

Combine them: $(\mathcal{PT}f)(x) = f(-x)^*$. Two properties are used below. It is *antilinear*: $\mathcal{PT}(\alpha f) = \alpha^*\,\mathcal{PT}f$ for a complex number α. And applying it twice gives back f: $(\mathcal{PT})^2 f = f$.

14.7.2 PT-symmetric Hamiltonians

$H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$ is *PT-symmetric* if $\mathcal{PT}$ commutes with it: $\mathcal{PT}(Hf) = H(\mathcal{PT}f)$ for all f. Let us find the condition on V. On one side,

$$(\mathcal{PT}(Hf))(x) = \left[-f''(-x) + V(-x)f(-x)\right]^* = -f''(-x)^* + V(-x)^* f(-x)^*. \tag{14.55}$$

On the other, let $\chi(x) = f(-x)^*$. Differentiating twice, each derivative brings a factor -1 from the chain rule, so $\chi''(x) = f''(-x)^*$, and

$$(H\chi)(x) = -f''(-x)^* + V(x)f(-x)^*. \tag{14.56}$$

The two agree for all f if and only if

$$\boxed{V(-x)^* = V(x)} \qquad \Longleftrightarrow \qquad V_R \text{ even and } V_I \text{ odd.} \tag{14.57}$$

(Write $V = V_R + iV_I$: $V(-x)^* = V_R(-x) - iV_I(-x)$, which equals $V_R(x) + iV_I(x)$ exactly when $V_R(-x) = V_R(x)$ and $V_I(-x) = -V_I(x)$.) Examples: $x^2 + i\epsilon x$ is PT-symmetric; so is the famous family $x^2(ix)^\varepsilon$ studied by Bender and Boettcher [33].

14.7.3 Real eigenvalues when the symmetry is unbroken

Theorem. Let H be PT-symmetric.

(a) If E is an eigenvalue, so is E^*. So eigenvalues are real or come in complex-conjugate pairs.

(b) If an eigenfunction ψ is also an eigenfunction of $\mathcal{PT}$, that is, $\mathcal{PT}\psi = \lambda\psi$ for some number λ, then E is real.

Proof of (a). Let $H\psi = E\psi$. Then $H(\mathcal{PT}\psi) = \mathcal{PT}(H\psi) = \mathcal{PT}(E\psi) = E^*\,\mathcal{PT}\psi$, using commutation and then antilinearity. And $\mathcal{PT}\psi \neq 0$, because applying $\mathcal{PT}$ again gives back $\psi \neq 0$. So $\mathcal{PT}\psi$ is an eigenfunction with eigenvalue E^*.

Proof of (b). First, $\lambda \neq 0$: $\psi = (\mathcal{PT})^2\psi = \mathcal{PT}(\lambda\psi) = \lambda^*\mathcal{PT}\psi = \lambda^*\lambda\psi = |\lambda|^2\psi$, so $|\lambda| = 1$. Next compute $\mathcal{PT}(H\psi)$ two ways. Using $H\psi = E\psi$ and antilinearity: $\mathcal{PT}(E\psi) = E^*\lambda\psi$. Using commutation: $H(\mathcal{PT}\psi) = H(\lambda\psi) = \lambda E\psi$. Subtracting, $(E^* - E)\lambda\psi = 0$, and since $\lambda\psi \neq 0$, $E^* = E$. ■

When every eigenfunction is also a $\mathcal{PT}$ eigenfunction, the symmetry is called *unbroken* and the whole spectrum is real. When some eigenfunctions are not, the symmetry is *broken*, and by (a) those eigenvalues appear in conjugate pairs.

Taken on trust

Taken on trust: for the Bender–Boettcher family $-\mathrm{d}^2/\mathrm{d}x^2 + x^2(ix)^\varepsilon$, the spectrum is entirely real and positive for $\varepsilon \geq 0$ (with suitable boundary conditions in the complex plane), and PT symmetry is broken for $\varepsilon < 0$. This was found numerically in [33]; a rigorous proof was given later, in the mathematical literature that followed that paper, and it is far beyond this book.

14.7.4 The PT-odd class

Now take the paper's situation: V is real and even (the Wu–Sprung potential, or the box), and A is real. By (14.57), $H_0 + iA$ is PT-symmetric exactly when A is *odd*. The paper's "PT-odd" class uses the odd basis $\phi_k = \sin((k+1)\pi x/L)$, so every $A = \sum c_k\phi_k$ is odd and every model in the class is PT-symmetric. (The name refers to the imaginary part being odd under parity; the full Hamiltonian is then PT-symmetric.)

For such A the first-order imaginary shift vanishes. In an even potential, each ψ_n is even or odd (Chapter 11), so ψ_n^2 is even, and $\psi_n^2 A$ is odd. The integral of an odd function g over $[-L, L]$ vanishes: $\int_{-L}^{0} g(x)\,\mathrm{d}x = \int_0^L g(-y)\,\mathrm{d}y = -\int_0^L g(y)\,\mathrm{d}y$ (substituting $x = -y$), which cancels $\int_0^L g$. Hence $A_{nn} = 0$, as we found for the box sines in (14.49). The same parity argument shows that an odd A couples only states of opposite parity, and an even A only states of the same parity.

So for the PT-odd class the eigenvalue is $E_n - \sum_{m\neq n} |A_{nm}|^2/(E_n - E_m) + \cdots$ with no imaginary part at first or second order; this is what the paper means by "real spectrum to leading order". By the theorem, while the symmetry is unbroken the levels stay exactly real. How does it break? Because eigenvalues move continuously with the amplitude and complex eigenvalues must come in conjugate pairs, a simple real eigenvalue cannot leave the real axis on its own. Here is why. Suppose an eigenvalue E is real for amplitudes below some value and

complex just above it. By part (a), E^* is then also an eigenvalue just above that value, and as the amplitude decreases back to the departure point both E and E^* approach the same real number. So at the departure point two eigenvalues coincide: a real eigenvalue can leave the axis only by first colliding with another real eigenvalue. This argument uses continuity of the eigenvalues through a coincidence, which the earlier 'beyond' box (smoothness for nondegenerate eigenvalues only) does not supply:

Taken on trust

Taken on trust: the eigenvalues of a matrix, being the roots of its characteristic polynomial, depend continuously on the matrix entries, even at amplitudes where two of them coincide (though not smoothly there). This is the continuity of polynomial roots in the coefficients; see [29].

Level attraction makes such collisions happen.

Sample Problem 14.8: PT symmetry breaking with two levels

Return to $M = \begin{pmatrix} E_1 & ia \\ ia & E_2 \end{pmatrix}$ of Section 14.3, now read as an even state 1 and an odd state 2 coupled by an odd A. Show that M is PT-symmetric, and describe its spectrum as a grows, for $\Delta = E_2 - E_1 = 1$.

Step 1: PT in this basis. Parity multiplies the even component by $+1$ and the odd one by -1: it is the matrix $P = \mathrm{diag}(1, -1)$. Time reversal conjugates components (the basis functions are real). PT symmetry of M means $P\bar{M}P = M$.

Step 2: check. $\bar{M} = \begin{pmatrix} E_1 & -ia \\ -ia & E_2 \end{pmatrix}$; multiplying on both sides by P flips the sign of the off-diagonal entries, giving back $\begin{pmatrix} E_1 & ia \\ ia & E_2 \end{pmatrix} = M$.

Step 3: spectrum. From (14.35), $\lambda_\pm = \bar{E} \pm \frac{1}{2}\sqrt{1 - 4a^2}$. For $a < \frac{1}{2}$: two real eigenvalues approaching each other (unbroken phase). At $a = \frac{1}{2}$: they coincide at $\bar{E}$, and the eigenvectors coincide too (Sample in Section 14.8); this is called an *exceptional point*. For $a > \frac{1}{2}$: a conjugate pair $\bar{E} \pm \frac{i}{2}\sqrt{4a^2 - 1}$ (broken phase).

real for $a < \Delta/2$, exceptional point at $a = \Delta/2$, conjugate pair for $a > \Delta/2$

Contrast the Hermitian coupling a (without the i): $\bar{E} \pm \frac{1}{2}\sqrt{1 + 4a^2}$, always real and always separating. Figure 14.3 plots both parts.

14.7.5 Many levels

In the box, where the spacing $E_{n+1} - E_n = 2n + 1$ grows with n while the couplings stay comparable, the lowest levels break first, because they are closest together relative to the coupling. In a spectrum whose spacing shrinks with height (such as the Wu–Sprung levels, Chapter 22) the opposite ordering is expected for comparable couplings. In the box with $E_n = n^2$ and $A = c\sin(\pi x/L)$, the spacing between levels 1 and 2 is 3, while the coupling is $0.679\,c$ by (14.49); the two-level estimate puts the break where $0.679\,c \approx 1.5$, that is, at $c \approx 2.2$. This is only an estimate: in the full truncation each level also couples to other partners, so the true threshold can differ. The numerics below establish only that it lies between $c = 1$ and $c = 4$; Problem 18 lets you probe $c = 2.5$. Higher pairs have larger spacings and slightly smaller couplings. Figure 14.4 shows the exact eigenvalues of a 60-state truncation for several

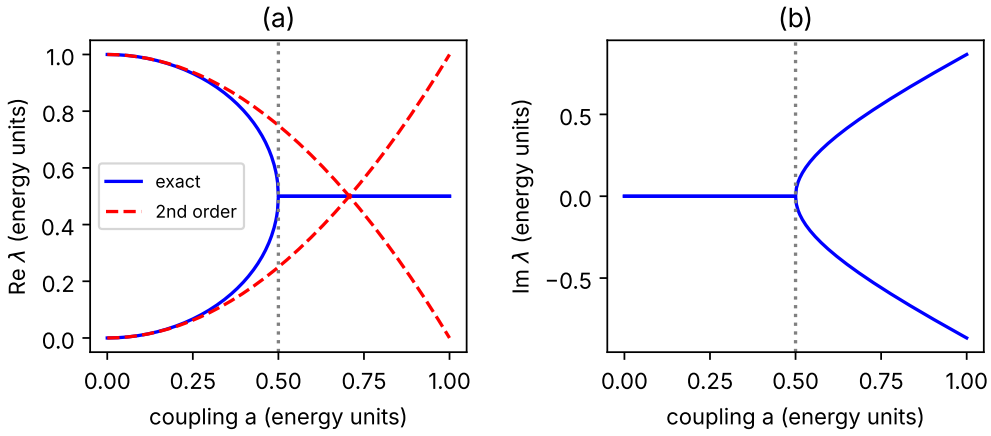


Figure 14.3: The two-level model $\begin{pmatrix} 0 & ia \\ ia & 1 \end{pmatrix}$ (levels $E_1 = 0$, $E_2 = 1$, so $\Delta = 1$; dimensionless units). (a) Real parts of the exact eigenvalues (solid) and the second-order predictions $E_1 + a^2/\Delta$, $E_2 - a^2/\Delta$ (dashed). The levels attract and merge at the exceptional point $a = \Delta/2$ (dotted line); beyond it the exact real parts stay at $1/2$ while the second-order curves keep moving. (b) Imaginary parts: zero below the exceptional point, a conjugate pair above it.

amplitudes, computed as in Section 14.8: at $c = 1$ all 60 eigenvalues are real to rounding error, and at $c = 4$ a complex-conjugate pair has formed near the bottom of the spectrum. The right panel shows the absorptive-even case $A = -g[1 + \cos(\pi x/L)] \leq 0$: every eigenvalue lies on or below the real axis, as Fact 3 of Section 14.2 guarantees.

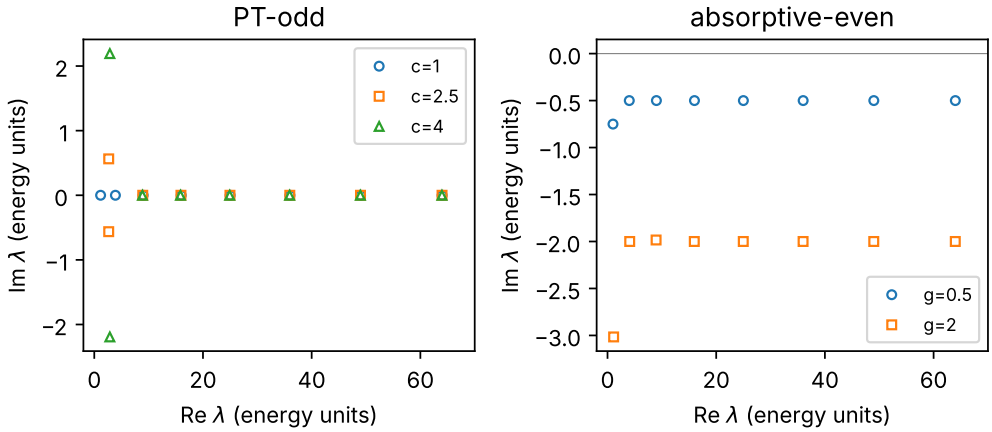


Figure 14.4: Exact eigenvalues of $D + i\mathsf{A}$ in a 60-state truncation of the box with $E_n = n^2$ (dimensionless units, $\hbar = 2m = 1$, $L = \pi/2$); only the lowest part of the spectrum is shown. Left: PT-odd perturbation $A = c\sin(\pi x/L)$ for $c = 1$, 2.5 and 4. For $c = 1$ all eigenvalues are real; for $c = 4$ a complex-conjugate pair has formed near the bottom. Right: absorptive perturbation $A = -g[1 + \cos(\pi x/L)]$ for $g = 0.5$ and $g = 2$; all eigenvalues have $\mathrm{Im}\,\lambda \leq 0$.

Checkpoint 14.7

Which of these potentials are PT-symmetric: (a) $x^2 + ix$; (b) $x^2 + ix^2$; (c) a real even $V(x)$ plus $i\cos(\pi x/L)$?

Answer: Only (a). In (b) and (c) the imaginary part is even, which violates (14.57). Case (c) is the absorptive-even type of the paper.

Key Ideas

- $\mathcal{PT}f(x) = f(-x)^*$; H is PT-symmetric iff $V(-x)^* = V(x)$ (real part even, imaginary part odd).
- Eigenvalues of a PT-symmetric H are real or in conjugate pairs; they are real when the symmetry is unbroken.
- On an even background, odd A gives a PT-symmetric $H_0 + iA$ with $A_{nn} = 0$: the paper's PT-odd class.
- Attraction drives levels together; at an exceptional point they merge and the symmetry breaks into a conjugate pair.

14.8 When second order fails: exact diagonalization and matching

★ Did you know? *PT Symmetry in the Lab*

Non-Hermitian physics is not just theory. In 2010 Christian Rüter and collaborators coupled two optical waveguides, one with gain and one with loss, and observed the predicted PT symmetry breaking. As the gain and loss grew past a threshold, the light's behavior changed abruptly. Related ideas now appear in research on lasers and sensors.

14.8.1 The radius of the perturbation series

The two-level model shows precisely how far second order can be trusted. Expand the exact shift (14.36) using the Taylor series of $\sqrt{1-x}$ (Chapter 3). With $f(x) = (1-x)^{1/2}$: $f(0) = 1$, $f'(x) = -\frac{1}{2}(1-x)^{-1/2}$ so $f'(0) = -\frac{1}{2}$, and $f''(x) = -\frac{1}{4}(1-x)^{-3/2}$ so $f''(0)/2 = -\frac{1}{8}$. Hence $\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} - \cdots$ and

$$\lambda_- - E_1 = \frac{\Delta}{2}\left(\frac{x}{2} + \frac{x^2}{8} + \cdots\right) = \frac{a^2}{\Delta} + \frac{a^4}{\Delta^3} + \cdots, \qquad x = \frac{4a^2}{\Delta^2}. \tag{14.58}$$

The second-order formula is the first term. The next term, a^4/Δ^3, is smaller by the factor $a^2/\Delta^2 = x/4$. So the relative error of second order is about $x/4$ when x is small. The series has radius of convergence 1 in x (Chapter 3), so it converges for $|x| < 1$, that is, for $a < \Delta/2$, and diverges for $|x| > 1$, that is, for $a > \Delta/2$. The radius is set by the exceptional point: at $a = \Delta/2$ the square root vanishes and is not differentiable, and no power series in a can describe the eigenvalue beyond it. This is a general lesson.

- Second order is accurate when every relevant coupling is small compared with the spacing to the coupled level.
- As the amplitude grows, second order first becomes inaccurate (here by $x/4$), and then meaningless: past an exceptional point the true real shift stops growing while the formula keeps growing like the square of the amplitude.

14.8.2 Exact diagonalization

The remedy is to compute the eigenvalues of $D + i\mathsf{A}$, or of the finite-difference matrix of $H_0 + iA$ (Chapter 23), exactly. The matrix is complex and not Hermitian, so the tridiagonal Hermitian solvers of Chapter 8 do not apply. One uses a general dense eigensolver, `scipy.linalg.eig`, which handles any square complex matrix. Its cost grows like $\mathcal{O}(K^3)$ for a $K \times K$ matrix, as for dense Hermitian problems, but with a larger constant, and it returns eigenvalues in no particular order.

```
import numpy as np
from scipy.linalg import eig
K, c = 60, 1.0
n = np.arange(1, K+1)
E = n**2.0                                  # box with L = pi/2
def s1(a, b):                               # (sin_1)_{ab}, Eq. (sinelem)
    if (a+b) % 2 == 0: return 0.0
    return -4/np.pi*(1/(4-(a-b)**2) - 1/(4-(a+b)**2))
A = np.array([[s1(a, b) for b in n] for a in n])
lam, vec = eig(np.diag(E) + 1j*c*A)
match = np.argmax(np.abs(vec)**2, axis=0)  # unperturbed state of each eigvec
order = np.argsort(match)
print(lam[order][:5])
print("one-to-one:", len(set(match)) == K)
```

14.8.3 Matching eigenstates by overlap

The solver returns eigenvalues in an arbitrary order. To compare an exact eigenvalue with the perturbative shift of level n, you must know which exact eigenvalue "belongs" to level n. Sorting by real part fails as soon as two levels cross or merge. The robust method (Chapter 8) is to *match by overlap*: write each exact eigenvector v in the unperturbed basis (for $D + i\mathsf{A}$ that is simply its components v_m), and assign it to the state m with the largest $|v_m|^2$. For small perturbations, (14.31) shows that the eigenvector near E_n is e_n plus small admixtures, so the largest component is the n-th and the match is unambiguous.

The matching must be *one-to-one*: each unperturbed state receives exactly one exact eigenvector. When two eigenvectors claim the same state (and another state is claimed by none), the perturbed states are no longer identifiable with unperturbed ones. That happens near exceptional points, as the next sample shows, and it is a clear signal that perturbation theory has left its domain.

Sample Problem 14.9: The eigenvector at an exceptional point

For $M = \begin{pmatrix} E_1 & ia \\ ia & E_2 \end{pmatrix}$ at $a = \Delta/2$, find the eigenvectors and show that overlap matching fails.

Step 1. The double eigenvalue is $\bar{E}$. Then $M - \bar{E}I = \begin{pmatrix} -\Delta/2 & i\Delta/2 \\ i\Delta/2 & \Delta/2 \end{pmatrix}$, using $E_1 - \bar{E} = -\Delta/2$ and $E_2 - \bar{E} = \Delta/2$.

Step 2. The first row gives $-\frac{\Delta}{2}v_1 + i\frac{\Delta}{2}v_2 = 0$, so $v_1 = iv_2$. The second row gives $i\frac{\Delta}{2}v_1 + \frac{\Delta}{2}v_2 = i\frac{\Delta}{2}(iv_2) + \frac{\Delta}{2}v_2 = -\frac{\Delta}{2}v_2 + \frac{\Delta}{2}v_2 = 0$, consistent. So there is only one independent eigenvector, $v = (i, 1)$.

Step 3. $|v_1|^2 = |v_2|^2 = 1$: the single eigenvector overlaps both unperturbed states equally. There is no second eigenvector for the other state.

Step 4. Note also $v^\top v = i^2 + 1 = 0$ (the transpose, not the conjugate transpose). A

nonzero vector whose unconjugated product with itself vanishes is called "self-orthogonal"; for complex symmetric matrices this happens at exceptional points, as here.

$$\boxed{v = (i, 1), \quad |v_1|^2 = |v_2|^2, \quad \text{only one eigenvector}}$$

A 2×2 matrix with only one eigenvector cannot be diagonalized, which is why exceptional points are where perturbation theory and overlap matching both break down.

14.8.4 Exact against second order in many levels

Figure 14.5 repeats the comparison of the two-level sample for the 60-state box. At small amplitude ($c = 0.5$) the exact real shifts and the second-order predictions agree closely (their correlation over levels 1–50 exceeds 0.99, checked in the CHECKS script). At $c = 4$, beyond the point where the lowest pair has merged, the second-order predictions no longer describe the operator: they overstate the shifts of the low levels, which in reality cannot move past each other. A fit that optimizes the second-order formula at such amplitudes is optimizing a fiction.

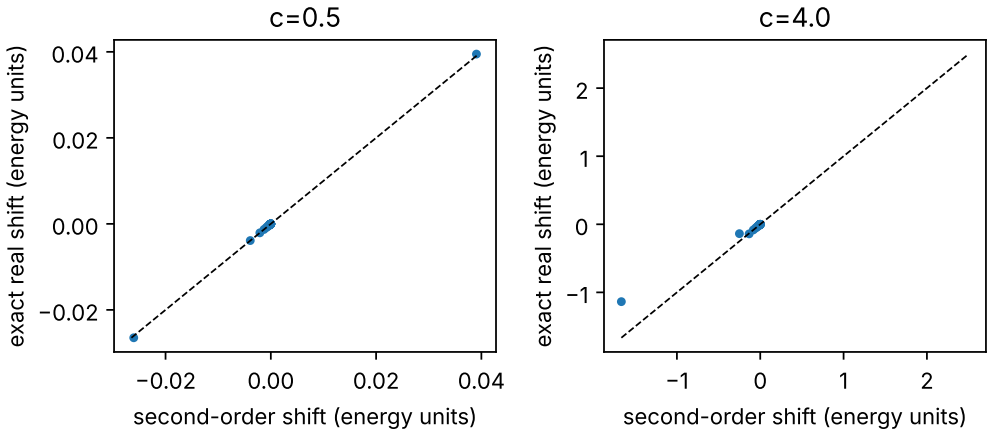


Figure 14.5: Exact real shifts $\mathrm{Re}\,\lambda_n - E_n$ (eigenvectors matched by overlap) against the second-order formula $-c^2 T^{(n)}$, for $A = c\sin(\pi x/L)$ in a 60-state box truncation with $E_n = n^2$, levels 1–50 (dimensionless energy units). Left, $c = 0.5$: points lie on the diagonal. Right, $c = 4$: the lowest levels have merged into a complex pair and the second-order predictions overshoot. Dashed line: exact = second order.

Checkpoint 14.8

In the two-level model with $\Delta = 1$, roughly how large is the relative error of the second-order shift at $a = 0.1$?

Answer: About $x/4 = a^2/\Delta^2 = 0.01$, i.e. 1%, matching 0.0100 versus 0.0101 in the earlier sample.

Key Ideas

- Second order is the first term of a series whose radius is set by the nearest exceptional point; its relative error is about (coupling/spacing)^2.
- Beyond an exceptional point the exact real shift saturates while the second-order formula keeps growing.

- Non-Hermitian matrices need a general eigensolver (`scipy.linalg.eig`); eigenvalues come unordered and are matched to unperturbed states by largest overlap, which must be one-to-one.
- Failure of one-to-one matching signals that perturbation theory has broken down.

14.9 Paper connection

Vocabulary. Secs. V and V A of the paper use a few terms from later chapters. Each model is fitted to the zeros $1, \ldots, 400$. The misfit is measured as an *in-sample rms*: the root mean square, over the fitted zeros, of the difference between model level and target, each difference divided by the local mean spacing gap_n (the average distance between neighboring zeros near zero n; Chapter 17), so errors are quoted "in mean spacings". The *target fluctuation* fluc_n is the amount by which level n must move away from its unperturbed value E_n to land on its target. The *null* is the misfit with no perturbation at all ($c = 0$). The *floor* is the lowest in-sample rms the optimizer reaches. The *surrogate* is the second-order formula $-c^\top T^{(n)} c$, used during fitting in place of the exact spectrum.

The two model classes (Sec. II E). The paper perturbs the Wu–Sprung Hamiltonian H (Chapter 22) as $H \to H + iA$ with $A = \sum_k c_k \phi_k$ a real multiplication operator, exactly our (14.19) and (14.41). The *PT-odd* class uses $\phi_k = \sin((k+1)\pi x/L)$; every such A is odd (a sum of sines); since the Wu–Sprung potential is even, Section 14.7 shows that $H + iA$ is then PT-symmetric and $A_{nn} = 0$, which is the paper's "real spectrum to leading order". The *absorptive-even* class uses $\phi_k = \cos(k\pi x/L)$, "modelling $-i\Gamma/2$": by (14.33) it produces first-order widths $\Gamma_n = -2A_{nn}$, which are nonnegative for every state if $A \le 0$ pointwise (Fact 3 of Section 14.2). That is the paper's "physical constraint $\Gamma \ge 0$ (i.e. $A \le 0$ pointwise)".

Eq. (5). The paper's Eq. (5), $\delta E_n = -\sum_{m \ne n} |A_{nm}|^2/(E_n - E_m) = -c^\top T^{(n)} c$, is our (14.34) and (14.45); the minus sign is the i^2 of Section 14.3. The state sum runs "over all 652 basis states with $E \le 1000$", and the paper notes that "truncating at 500 states measurably biases the top levels". Section 14.3 explains the direction: partners above a level push it up (attraction), so cutting them off removes upward pushes from the levels near the cutoff and biases them downward (the topmost level keeps only downward pushes). App. A step 9 records the tensor computation, with eigenvectors normalized as $\sum \psi^2 dx = 1$. The fit (App. A step 10) minimizes residuals $(\text{fluc}_n + c^\top T^{(n)} c)/\text{gap}_n$ using the Jacobian $2(T^{(n)}c)_k/\text{gap}_n$, the gradient derived in Section 14.4, with six random restarts because the problem is nonconvex.

Safeguard (i) and the null mode (Sec. II E, Sec. V). The paper says the even basis "contains the exactly-null constant mode $\phi_0 = 1$ (a constant iA shifts all eigenvalues by the same imaginary constant and has zero second-order real shift); its analytically-vanishing tensor rows are zeroed exactly rather than left as float32 noise". This is Section 14.6. Sec. V reports what happened before the repair: the absorptive-even floor sat at 0.2322, "spuriously close to the PT-odd value" 0.2321, because "the optimizer had driven the exactly-null constant mode to $c_0 \approx 2.2 \times 10^6$, harvesting in-sample improvement from float32 noise". With the artifact removed, the absorptive-even floor is 0.2358; the null is 0.2438 (all in-sample rms on zeros 1–400, in mean spacings). Our toy sample produced $|c_0| \sim 10^6$ by the same mechanism; Problem 16 estimates why such a value turns noise into order-one shifts.

Safeguard (iii), exact diagonalization (Sec. V, Fig. 5). Under "The optimizer left the perturbative regime", Sec. V reports that at the fitted amplitude "corr(exact, second-order)

drops to +0.14/+0.18 (odd/even), the surrogate overstates shift magnitudes twofold (Fig. 5, left), and 3% of eigenstates can no longer be matched one-to-one to unperturbed states". Each phrase now has a meaning for you: the exact spectrum comes from dense complex `eig` on a grid of $N = 2500$ points (App. A step 11), states are matched by overlap as in Section 14.8, and loss of one-to-one matching signals strong mixing of unperturbed states, of which exceptional points (Section 14.7) are the extreme case; the paper does not identify which occurs. The factor of two recalls our two-level sample, where at $a = \Delta$ second order gave 1.0 against an exact 0.5 (a toy illustration, not an explanation of the paper's specific factor). Scoring the exact spectra in-sample gives 0.2523 and 0.2477, "worse than the null 0.2438"; "the 0.232 floor was a property of the surrogate, not of any operator". At quarter amplitude the surrogate is accurate (corr +0.88/+0.94); at quarter amplitude every coupling A_{nm} is 4 times smaller, so the coupling-to-spacing ratio falls fourfold, the second-order shifts fall 16-fold (they scale as c^2, Section 14.4), and the relative error of second order, of order (coupling/spacing)^2 by Section 14.8, falls about 16-fold. Sec. V also states that "58% of its widths have the wrong sign" for the unconstrained absorptive-even fit, which by Fact 3 means the fitted A was positive (gain) somewhere, and by $\Gamma_n = -2A_{nn}$ that its average weighted by $|\psi_n|^2$ was positive for more than half the states.

Safeguard (ii), amplitude balls. The fair refits restrict $\|c\| \le R$ (App. A step 10: projection after each step, restarts with spread $R/20$, sign penalty $10^4 \int \max(A, 0)^2\, dx$) so that second order stays valid. Their floors are within 0.4–4.8% of the null (Fig. 5, right); the $R = 4$ solutions pass exact verification (corr +0.93/+0.995, all states matched), with exact-spectrum in-sample gains of 1.9% (odd) and 0.6% (even). The paper concludes that these families, "treated fairly, are empirically inert on this problem".

Sum rule and locality (Sec. V, V A). The paper verifies $\sum_n \delta E_n = 0$ to 3.6×10^{-15} (odd) and 1.4×10^{-14} (even) at the fitted coefficients (Sec. V A), our theorem (14.50), and reports locality with exactly our weights (14.53): half the weight within $|n - m| \le 5$ (odd) and ≤ 10 (even), 90% within about 60 levels. The "block-coherence" hypothesis was that, since the redistribution is local and sums to zero, the model can only produce locally mean-free patterns and the floor equals the rms of the locally averaged part of the target. Sec. V A refutes it: the block-coherent rms is $0.108, 0.049, 0.033, 0.035, 0.019$ for windows $w = 3, 5, 7, 9, 13$, "far below the observed floors". The paper's own summary is that "the floor is simply the null minus a small capture".

Status. The paper calls Sec. V "a methodological stress test of a second model class"; its Hermitian conclusions do not depend on it, and Sec. VI states that this analysis "was not rerun for this revision". Its general lesson is the one this chapter's mathematics makes concrete: "A surrogate improvement is not an operator-level result until the exact spectrum of the fitted operator confirms it."

14.10 Summary

- The imaginary part of a potential changes total probability at rate $2 \int V_I |\Psi|^2 \, \mathrm{d}x$, (14.11); a constant $-\Gamma/2$ gives decay $e^{-\Gamma t}$, (14.13).
- $\operatorname{Im} E$ is the state-weighted average of V_I, (14.18); $V_I \le 0$ implies all widths $\Gamma = -2 \operatorname{Im} E \ge 0$.
- In the real eigenbasis, $H_0 + iA$ is the complex symmetric matrix $D + i\mathsf{A}$, (14.21); its eigenvalues' real parts sum to $\sum E_n$, (14.24), and are even in the amplitude, (14.26).

- General second order: $\sum_{m\neq n} W_{nm}W_{mn}/(E_n - E_m)$, (14.32). For $W = iA$: first order iA_{nn}, (14.33); leading real shift $-\sum |A_{nm}|^2/(E_n - E_m)$, (14.34): level attraction.
- Quadratic form $\delta E_n = -c^\top T^{(n)} c$, (14.45); box matrix elements (14.40), (14.49).
- Sum rule $\sum \delta E_n = 0$ over a closed set, (14.50); locality of weights (14.53).
- The constant mode is null: $T^{(n)}_{0l} = 0$, (14.54); zero it in code.
- PT symmetry iff $V(-x)^* = V(x)$, (14.57); spectra real or in conjugate pairs; breaking at exceptional points.
- Second order is a series with radius set by the nearest exceptional point, (14.58); verify with exact diagonalization and one-to-one overlap matching.

14.11 Problems

1. • A uniform absorber with $\Gamma = 0.1$ acts on a particle in a box. Find the half-life and the fraction remaining at $t = 20$.

2. • An eigenvalue is $E = 5 - 0.3i$. What are the level, the width, and the mean lifetime?

3. • Find the eigenvalues of $\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ and of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Comment.

4. • Using (14.57), decide which are PT-symmetric: (a) ix^3; (b) $x^4 + ix^3$; (c) $x^2 + i\sin x$; (d) $x^2 - i$.

5. • Explain in one sentence why, at second order, an imaginary perturbation lowers the highest state of a truncated basis.

6. •• In the RE3 box, the absorber is $A = -g[1 - \cos(\pi x/L)]$ (zero at the center, strongest at the walls). Find the first-order widths of all levels. Which level is longest-lived, and why physically?

7. •• Starting from (14.11), derive (14.13) and show that the decay rate does not depend on the state.

8. •• Show that at the exceptional point the 2×2 matrix of Section 14.8 satisfies $(M - \bar{E}I)^2 = 0$ but $M - \bar{E}I \neq 0$.

9. •• In the two-level model with $\Delta = 1$ and $a = 0.3$, compute the exact shift of the lower level and the second-order prediction.

10. •• Continue Problem 9: add the fourth-order term a^4/Δ^3 of (14.58). How close is the sum to the exact shift?

11. •• For the four-state sample of Section 14.4, show algebraically (without numbers) that $T^{(1)} + T^{(2)} + T^{(3)} + T^{(4)} = 0$.

12. •• Prove that in an even potential an odd A has $A_{nm} = 0$ whenever ψ_n and ψ_m have the same parity.

13. •• Verify numerically the gradient formula $\partial(c^\top Tc)/\partial c_k = 2(Tc)_k$ for $T = \begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix}$ and $c = (0.5, 1)$ by finite differences.

14. •• Find the eigenvalues of $\begin{pmatrix} 1 & i \\ i & 2 \end{pmatrix}$ and check the trace rule (14.24).

15. •• (Paper) Sec. II E says that truncating the state sum of Eq. (5) at 500 states instead of 652 "measurably biases the top levels". Predict the sign of the bias on levels near 400–500.

16. •• (Paper) With $c_0 \approx 2.2 \times 10^6$ (Sec. V) and noisy null-row entries of order 10^{-7}, estimate the size of a cross term $2c_0 T^{(n)}_{0l} c_l$ for $c_l \sim 1$, and compare with a mean spacing (of order one at these heights).

17. ••• Add state 5 ($E_5 = 25$) to the four-state sample. Compute the new $T^{(4)}$ and δE_4, and check that the five shifts still sum to zero.

18. •• Using numpy, rebuild Figure 14.4 and find, for $c = 2.5$, how many eigenvalues of the 60-state truncation have $|\operatorname{Im}\lambda| > 10^{-6}$.

19. •• Using numpy, form the 60-state tensors $T^{(n)}$ for the two-function basis $\sin(\pi x/L)$, $\sin(2\pi x/L)$ in the box with $E_n = n^2$, using (14.49) with $j = 1, 2$. Draw a random $c = (c_0, c_1)$ and verify that $\sum_n c^\top T^{(n)} c = 0$ to rounding.

20. •• Using qmnt or numpy, repeat the null-mode experiment of Figure 14.2 with $N = 2000$. Does the float32 noise grow or shrink?

21. ••• In the two-level model, show that for $a \geq \Delta/2$ the ratio of the second-order real shift to the exact real shift is $2a^2/\Delta^2$.

22. •• (Paper) Sec. V reports that 58% of the unconstrained absorptive-even widths had the wrong sign. Using Fact 3 of Section 14.2, what does this say about the fitted $A(x)$?

23. •• (Paper) Sec. V checks the surrogate at quarter amplitude. By what factor are all second-order shifts reduced?

24. •• (Paper) The fair $R = 4$ odd fit gained 1.9% over the null 0.2438 in exact-spectrum in-sample rms. What rms does this imply, and compare with the exact-verified anchor of App. A.

25. ••• Show that for any A, the sum of first-order widths over a closed set of K states is $-2\sum_n A_{nn}$, and that for $A = -g$ constant it equals $2gK$.

26. •• Write down a 3×3 PT-symmetric matrix (even, odd, even states) with an odd coupling and verify $P\bar{M}P = M$.

27. ••• Prove the trace rule for 3×3 matrices by expanding $\det(\lambda I - M)$ and comparing the λ^2 coefficient with $\prod(\lambda - \lambda_i)$.

28. ••• Use $\langle n|x|n+1\rangle = \sqrt{(n+1)/2}$ and $E_n = 2n + 1$ to show from (14.34) that every oscillator level shifts by $+\epsilon^2/4$ under $i\epsilon x$.

29. •• (Paper) Sec. V A verifies the sum rule to 3.6×10^{-15} with 652 states. Assuming the individual shifts are of order one in the paper's units, compare with 652 times double-precision machine epsilon. Is the verification consistent with rounding?

30. ••• (Paper, open-ended) Explain in your own words why locality plus the sum rule suggested the block-coherence hypothesis, and why the numbers in Sec. V A refute it.

31. • What width Γ makes the probability fall by a factor e in time 100?

14.12 Answers to selected problems

1. $t_{1/2} = \log 2/0.1 = 6.931$; remaining $e^{-2} = 0.1353$.

2. Level 5, width $\Gamma = -2\,\mathrm{Im}\,E = 0.6$, lifetime $1/0.6 = 1.667$.

3. Discriminant $4(i)(i) = -4$, so $\lambda = \pm i$; the real version gives ± 1. The factor i on the couplings turns repulsion into a complex pair.

4. (a) yes (odd imaginary part, zero real part); (b) yes; (c) yes (sin is odd); (d) no (constant imaginary part is even).

5. All its partners lie below it, so each term $-|A_{Km}|^2/(E_K - E_m)$ is negative.

7. With $V_I = -\Gamma/2$, $\mathrm{d}P/\mathrm{d}t = -\Gamma \int |\Psi|^2 = -\Gamma P$; the solution is $P(0)e^{-\Gamma t}$ for any initial state.

9. $x = 0.36$; exact $\frac{1}{2}(1 - 0.8) = 0.1$; second order 0.09.

11. Each pair (n, m) enters $T^{(n)}$ with $1/(E_n - E_m)$ and $T^{(m)}$ with $1/(E_m - E_n)$, same numerator; they cancel.

13. $Tc = (2.5, -2)$, so gradient $(5, -4)$; central differences reproduce it to rounding.

15. Missing partners above would push the top levels up; without them the top levels are biased downward.

17. $(\sin_1)_{45} = -0.4410$, $(\sin_1)_{25} = 0.2264$; $T^{(4)} = 0.0318 - 0.4410^2/9 = 0.0102$, so $\delta E_4 = -0.0102\,c^2$. The five shifts still sum to zero.

19. The total is zero to rounding error: its magnitude is below 10^{-12} times the largest $|\delta E_n|$.

21. Exact real shift $\Delta/2$; second order a^2/Δ; ratio $2a^2/\Delta^2$ ($= 2$ at $a = \Delta$).

23. Shifts scale as c^2: factor $1/16 = 0.0625$.

25. $\Gamma_n = -2A_{nn}$; summing gives $-2\sum A_{nn} = -2\,\mathrm{tr}\,\mathsf{A}$; for $A = -g$, $A_{nn} = -g$ and the sum is $2gK$.

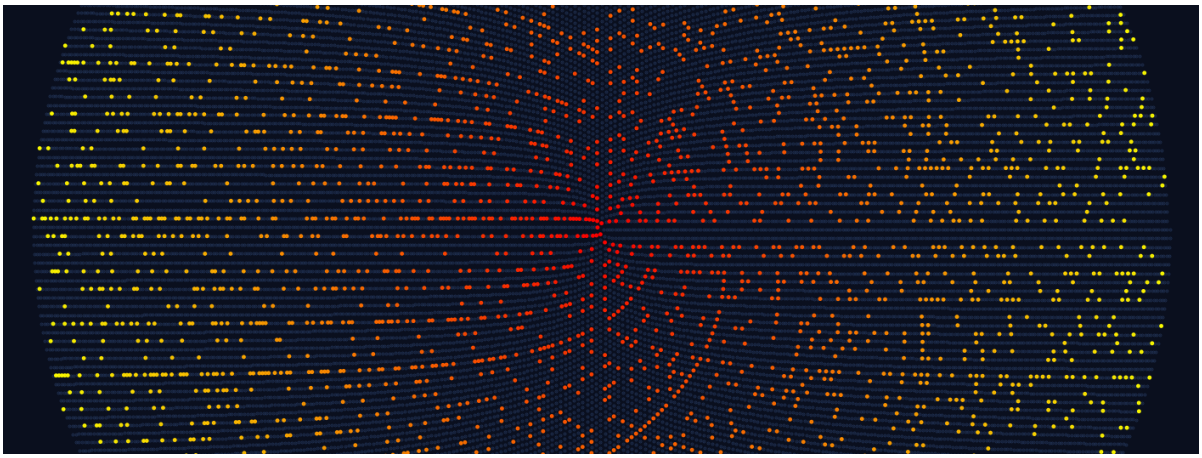
27. Both expressions give $\lambda^3 - (\sum \lambda_i)\lambda^2 + \ldots$ and $\lambda^3 - (M_{11} + M_{22} + M_{33})\lambda^2 + \ldots$; equate coefficients.

29. Assuming shifts of order one, the rounding scale is about $652 \times 2.2 \times 10^{-16} = 1.4 \times 10^{-13}$; the verified 3.6×10^{-15} is below that scale, consistent with pure rounding (under that assumption, since the paper does not report the size of the individual shifts).

31. $\Gamma = 1/100 = 0.01$.

Part III

Primes and the Zeta Function



The Sacks spiral: the integer n sits at radius $\sqrt{n}$ and angle $2\pi\sqrt{n}$; primes are lit. Curves of primes appear.

Chapter 15

The Primes

↑ Contents

What this chapter is about

In chemistry every substance is built from a short list of atoms, and every molecule has a definite recipe: water is two hydrogens and one oxygen, never anything else. Whole numbers have their own atoms. Every whole number larger than 1 is either one of these atoms (a *prime*) or a product of them, and the recipe is unique: 60 is $2 \cdot 2 \cdot 3 \cdot 5$ and nothing else. This chapter is about those atoms. We define them, prove that the recipe always exists and is unique, prove that there is no last prime, learn to list them with a two-thousand-year-old algorithm, and then ask the question that leads directly to the Riemann zeros: *how many* primes are there below a given size?

The answer, the prime number theorem, says that near a large number x roughly one number in $\log x$ is prime. Along the way we meet two weighted ways of counting primes, the von Mangoldt function $\Lambda(k)$ and Chebyshev's function $\psi(x)$, which turn out to be the natural language of the zeta function. We end by explaining why the logarithm $\log p$, rather than p itself, is the natural "size" of a prime.

Where this shows up in the paper. The paper's arithmetic predictor, Eq. (4) of [1], is a sum over a set P of primes and over powers p^m with $m = 1, 2, 3$, of waves $\sin(mt \log p)$ with weights $1/(m\, p^{m/2})$. Its Table V is organized by "prime budgets" ($\{2\}$, 3 primes, 6 primes, 25 primes); Sec. I A b and Sec. II D mention constructions over "primes $p \leq 13$"; and Sec. IV reports that a model with no arithmetic input found frequencies at the logarithms of "those [prime powers] up to 113 except 32, 64, and 81". After this chapter you will be able to list every one of those numbers yourself and understand why their logarithms are the relevant frequencies.

What you need

- Chapter 3: partial sums, telescoping sums, the fact that the logarithm of a product is the sum of the logarithms, the divergence of the harmonic series.
- Chapter 4: the symbols $\sim$, $\mathcal{O}$ and little-o; leading-order behavior; comparing a sum with an integral.
- Chapter 2 (in Section 15.10 and the Paper connection): $a^{-it} = e^{-it\log a}$ for real $a > 0$.
- First-year calculus: natural logarithm and exponential, integration by parts, the binomial expansion $(1+y)^n = \sum_{k=0}^{n} \binom{n}{k} y^k$ for a positive integer n.

Throughout this chapter, as everywhere in this book and in the paper, log means the *natural* logarithm (base e). When we need base 10 we write $\log_{10}$.

15.1 What is a prime?

★ Did you know? *Euclid's Endless Primes*

Around 300 BC, Euclid's *Elements* (Book IX, Proposition 20) proved that the primes outnumber any given finite collection. Multiply all the primes on your list and add 1. The result has a prime factor that is not on your list. The proof remains one of the most admired in mathematics, and it is short enough to check in a minute.

15.1.1 Divisibility

We work with the *integers* $\ldots, -2, -1, 0, 1, 2, \ldots$ and especially with the *positive integers* $1, 2, 3, \ldots$. If a and b are integers with $a \neq 0$, we say that a *divides* b, written $a \mid b$, when $b = ac$ for some integer c. In that case a is a *divisor* (or *factor*) of b and b is a *multiple* of a. For example $3 \mid 12$ because $12 = 3 \cdot 4$, while $5 \nmid 12$ ("5 does not divide 12") because $12/5 = 2.4$ is not an integer.

Two facts about divisibility follow straight from the definition and are used repeatedly below.

- If $a \mid b$ and $a \mid c$, then $a \mid (b \pm c)$. (Proof: $b = ab'$ and $c = ac'$, so $b \pm c = a(b' \pm c')$, and $b' \pm c'$ is an integer.)
- If $a \mid b$ with a, b positive, then $a \leq b$. (Proof: $b = ac$ with c a positive integer, so $c \geq 1$ and $b = ac \geq a$.)

15.1.2 The definition

Every positive integer n is divisible by 1 and by itself. Most numbers have other divisors too; the primes are the ones that do not.

> **Definition.** A *prime* is an integer $p \geq 2$ whose only positive divisors are 1 and p. An integer $n \geq 2$ that is not prime is *composite*; equivalently, n is composite when $n = ab$ for integers a, b with $1 < a < n$ (and then automatically $1 < b < n$).

The number 1 is neither prime nor composite. This is a convention, but a carefully chosen one; we will see in Section 15.2 that calling 1 a prime would destroy the uniqueness of factorization. The primes below 30 are

$$2,\ 3,\ 5,\ 7,\ 11,\ 13,\ 17,\ 19,\ 23,\ 29.$$

The number 2 is the only even prime: any even $n > 2$ has the divisor 2 with $1 < 2 < n$.

15.1.3 Two useful lemmas

To decide whether a number is prime we need a test. The following two lemmas supply one.

Lemma 1 (proved). *The smallest divisor larger than 1 of any integer $n \geq 2$ is a prime.*

Proof. The number n has at least one divisor larger than 1, namely n itself, so it has a smallest one; call it d. Suppose d were composite: $d = ab$ with $1 < a < d$. Then $a \mid d$ and $d \mid n$, so $n = d\,c = a\,(bc)$ and $a \mid n$. But $1 < a < d$, so a is a divisor of n larger than 1 and smaller than d, contradicting the choice of d. Hence d is prime. ■

Lemma 2 (proved). *If $n \geq 2$ is composite, then n has a prime divisor $p \leq \sqrt{n}$. Equivalently: if no prime $p \leq \sqrt{n}$ divides n, then n is prime.*

Proof. Write $n = ab$ with $1 < a < n$, and label the two factors so that $a \leq b$. Then

$$a^2 = a \cdot a \leq a \cdot b = n, \qquad \text{so} \qquad a \leq \sqrt{n}. \tag{15.1}$$

By Lemma 1 the smallest divisor of a larger than 1 is a prime p, and $p \leq a \leq \sqrt{n}$. Since $p \mid a$ and $a \mid n$, we have $p \mid n$. ■

Lemma 2 is the basis of *trial division*: to test n, divide it by every prime up to $\sqrt{n}$. If none divides it, n is prime.

Sample Problem 15.1: Testing 221 and 1009 for primality

Decide whether 221 and 1009 are prime.

Step 1: bound the search for 221. $\sqrt{221} \approx 14.87$, so by Lemma 2 we need only try the primes $2, 3, 5, 7, 11, 13$.

Step 2: try them. 221 is odd, so $2 \nmid 221$. Its digit sum is 5, not a multiple of 3, so $3 \nmid 221$. It does not end in 0 or 5, so $5 \nmid 221$. $221 = 7 \cdot 31 + 4$, so $7 \nmid 221$. $221 = 11 \cdot 20 + 1$, so $11 \nmid 221$. Finally $13 \cdot 17 = 221$. So 221 is composite.

Step 3: bound the search for 1009. $\sqrt{1009} \approx 31.76$, so the candidates are the primes up to 31: $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31$.

Step 4: try them. The remainders of 1009 on division by these primes are $1, 1, 4, 1, 8, 8, 6, 2, 20, 23, 17$ respectively (for instance $1009 = 31 \cdot 32 + 17$). None is zero.

Step 5: conclude. By Lemma 2,

$$\boxed{221 = 13 \cdot 17 \text{ is composite;} \qquad 1009 \text{ is prime.}}$$

Checkpoint 15.1

To test whether $n = 10403$ is prime by trial division, what is the largest prime you would have to try? (Hint: $\sqrt{10403} \approx 101.99$.)

Answer: The largest prime not exceeding 101.99, namely 101. (In fact $10403 = 101 \cdot 103$, so you would find a factor only at the very last trial.)

Pitfall

Do not stop early. You must try every prime up to and including the largest prime $\leq \sqrt{n}$. For 10403 the factor appears only at 101. And do not include 1 among the primes: 1 divides everything, so "$1 \mid n$" tells you nothing.

Key Ideas

- $a \mid b$ means $b = ac$ for an integer c. A prime is an integer $p \geq 2$ with no positive divisors other than 1 and p; 1 is not prime.
- The smallest divisor larger than 1 of any $n \geq 2$ is prime (Lemma 1).
- A composite n has a prime factor $\leq \sqrt{n}$, so trial division up to $\sqrt{n}$ decides primality (Lemma 2, Equation (15.1)).

15.2 The fundamental theorem of arithmetic

Fundamental theorem of arithmetic. Every integer $n \geq 2$ can be written as a product of primes,

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}, \qquad p_1 < p_2 < \cdots < p_k \text{ primes}, \quad a_i \geq 1 \text{ integers}, \tag{15.2}$$

and this representation is unique: any two ways of writing n as a product of primes contain the same primes the same number of times, and differ at most in the order of the factors.

Equation (15.2) is called the *canonical factorization* of n, and a_i is the *exponent* of p_i in n. For example $360 = 2^3 \cdot 3^2 \cdot 5$.

The theorem has two halves, existence and uniqueness. We prove existence completely. For uniqueness we prove everything except one lemma, which we state in a box and take on trust.

15.2.1 Existence

The proof uses the *well-ordering principle*: every nonempty set of positive integers has a smallest element. (Picture walking up $1, 2, 3, \ldots$; the first member of the set you meet is the smallest.)

Proof of existence (proved). Suppose, for contradiction, that some integers $n \geq 2$ are *not* products of primes. By well-ordering there is a smallest such integer; call it n_*. Then:

1. n_* is not prime, because a prime is a product of primes (a product with a single factor).
2. So n_* is composite: $n_* = ab$ with $1 < a < n_*$ and $1 < b < n_*$.
3. Both a and b are at least 2 and smaller than n_*, so by the choice of n_* each *is* a product of primes: $a = q_1 \cdots q_r$, $b = q'_1 \cdots q'_s$.
4. Then $n_* = ab = q_1 \cdots q_r \, q'_1 \cdots q'_s$ is a product of primes, contradicting the choice of n_* as an integer that is not.

So no such n_* exists, and every $n \geq 2$ is a product of primes. ■

15.2.2 Uniqueness

Uniqueness rests on a property of primes that is not obvious from the definition.

Taken on trust

Euclid's lemma. If p is prime and $p \mid ab$ for integers a, b, then $p \mid a$ or $p \mid b$.
What we take on trust. The standard proof uses *Bézout's identity*: if $p \nmid a$, there exist integers x, y with $ax + py = 1$. Granting this, multiply by b: $b = abx + pby$. Both terms on the right are multiples of p (the first because $p \mid ab$), so $p \mid b$. Bézout's identity itself is

proved with division with remainder (the Euclidean algorithm); see [24], Chapter 2, or [23], Chapter 1. Problem 21 outlines a different proof of uniqueness that avoids Euclid's lemma altogether.

Applying Euclid's lemma repeatedly gives the version we need: *if a prime p divides a product $b_1 b_2 \cdots b_k$, then p divides at least one b_i.* (Write the product as $b_1 \cdot (b_2 \cdots b_k)$; either $p \mid b_1$, or $p \mid b_2 \cdots b_k$ and we repeat with one fewer factor.)

Proof of uniqueness (given Euclid's lemma). Suppose

$$p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s \tag{15.3}$$

with all p_i and q_j prime, repetitions allowed. We show by induction on r that the two lists are the same up to order. (*Induction* works in two stages. First prove the statement for $r = 1$. Then show that its truth for $r - 1$ forces its truth for r. It then holds for $r = 2, 3, 4, \ldots$ in turn, hence for every r.)

- *Case $r = 1$.* The left side is the prime p_1. If $s \geq 2$, then q_1 would be a divisor of p_1 with $1 < q_1 < p_1$ (since $q_1 \geq 2$ and the remaining factors multiply to at least 2), which is impossible for a prime. So $s = 1$ and $q_1 = p_1$.

- *Step.* Assume the statement for products of $r - 1$ primes. In Equation (15.3), p_1 divides the left side, hence the right side. By the repeated Euclid's lemma, $p_1 \mid q_j$ for some j. The only positive divisors of the prime q_j are 1 and q_j, and $p_1 \neq 1$, so $p_1 = q_j$. Suppose $s = 1$. Then $q_1 = p_1 p_2 \cdots p_r$ with $r \geq 2$, so p_1 would be a divisor of the prime q_1 with $1 < p_1 < q_1$ (the remaining factors multiply to at least 2). That is impossible, so $s \geq 2$, and after cancelling both sides are still nonempty products of primes. Cancel this common factor from both sides of Equation (15.3). The left side becomes a product of $r - 1$ primes, the right side a product of $s - 1$ primes, and by the induction hypothesis these two lists agree up to order. Restoring $p_1 = q_j$ to both lists, the original lists agree. ■

15.2.3 Uniqueness is not automatic

Why fuss over uniqueness? Because it can fail in number systems that look perfectly reasonable. Consider only the *even* positive integers $2, 4, 6, 8, \ldots$. Call an even number an "E-prime" if it is not the product of two even numbers. The product of two even numbers is divisible by 4, so the E-primes are exactly the even numbers not divisible by 4: $2, 6, 10, 14, 18, 22, 26, 30, \ldots$ Now

$$60 = 2 \cdot 30 = 6 \cdot 10,$$

and $2, 30, 6, 10$ are all E-primes. The number 60 has two genuinely different E-prime factorizations. The existence proof above works word for word in this system; what fails is Euclid's lemma (6 "divides" $2 \cdot 30 = 6 \cdot 10$ within the even numbers, but 6 divides neither 2 nor 30 there). The ordinary integers are special.

This is also why 1 is not a prime. If it were, $6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3$ would have infinitely many factorizations into "primes".

Sample Problem 15.2: Factorizing 5040

Find the canonical factorization of 5040 and use it to write $\log 5040$ as a sum.
Step 1: divide out 2 as often as possible. $5040 \to 2520 \to 1260 \to 630 \to 315$. Four divisions, so 2^4 is a factor and 315 is odd.
Step 2: divide out 3. $315 \to 105 \to 35$; 35 is not a multiple of 3. So 3^2.
Step 3: divide out 5. $35 \to 7$. So 5^1.

Step 4: what remains. 7 is prime. So $5040 = 2^4 \cdot 3^2 \cdot 5 \cdot 7$. Check: $16 \cdot 9 \cdot 5 \cdot 7 = 144 \cdot 35 = 5040$.
Step 5: take the logarithm. Since the log of a product is the sum of the logs (Chapter 3), and $\log(p^a) = a \log p$,

$$\log 5040 = 4\log 2 + 2\log 3 + \log 5 + \log 7 = 2.7726 + 2.1972 + 1.6094 + 1.9459 = 8.5252.$$

(The four rounded terms add to 8.5251. The value 8.5252 is $\log 5040$ correctly rounded; the last digit differs only because of rounding.)

$$\boxed{5040 = 2^4 \cdot 3^2 \cdot 5 \cdot 7, \qquad \log 5040 = 4\log 2 + 2\log 3 + \log 5 + \log 7 \approx 8.525.}$$

15.2.4 A consequence: logarithms of primes are independent

Taking logarithms of Equation (15.2) gives

$$\log n = a_1 \log p_1 + a_2 \log p_2 + \cdots + a_k \log p_k. \tag{15.4}$$

Multiplication of integers turns into addition of these sums. Uniqueness then has a striking consequence for the numbers $\log 2, \log 3, \log 5, \ldots$, which will be the frequencies of the paper's Eq. (4).

Theorem (proved). *Let $p_1, \ldots, p_k$ be distinct primes and $a_1, \ldots, a_k$ integers, not all zero. Then $a_1 \log p_1 + \cdots + a_k \log p_k \neq 0$.*

Proof. Suppose the sum were zero. Move the terms with negative coefficients to the right-hand side:

$$\sum_{a_i > 0} a_i \log p_i = \sum_{a_j < 0} |a_j| \log p_j.$$

Exponentiate both sides:

$$\prod_{a_i > 0} p_i^{a_i} = \prod_{a_j < 0} p_j^{|a_j|}.$$

Both sides are positive integers, and the primes appearing on the left are different from those on the right. If one side is an empty product (equal to 1) the other side is a nonempty product of primes, hence at least 2, a contradiction. Otherwise we have one integer with two prime factorizations that share no prime, contradicting uniqueness. ■

Corollary. $\log 3 / \log 2$ *is irrational.* If $\log 3 / \log 2 = a/b$ with positive integers a, b, then $b \log 3 - a \log 2 = 0$, which the theorem forbids. (More concretely: it would mean $3^b = 2^a$, an odd number equal to an even one.)

Here is why a physicist should care. The waves $\sin(t \log 2)$ and $\sin(t \log 3)$ have periods $2\pi/\log 2$ and $2\pi/\log 3$. A common period P would need $P \log 2 = 2\pi k$ and $P \log 3 = 2\pi l$ for positive integers k, l, which would give $\log 3/\log 2 = l/k$, a rational number. So the two waves never fall back into step. Sums of such incommensurate waves are the "almost periodic" functions of Chapter 5, and the paper's Eq. (4) is one of them.

Checkpoint 15.2

Can $2^a = 5^b$ for positive integers a, b? Can $\log 10/\log 2$ be rational?

Answer: No: $2^a = 5^b$ would give one integer two different prime factorizations. And $\log 10/\log 2 = (\log 2 + \log 5)/\log 2 = 1 + \log 5/\log 2$; if it were rational, so would be $\log 5/\log 2$, which the theorem rules out. So $\log 10/\log 2$ is irrational.

Key Ideas

- Every $n \geq 2$ is a product of primes (proved by the smallest-counterexample argument), uniquely up to order (proved from Euclid's lemma, which rests on Bézout's identity, taken on trust).
- Uniqueness can fail in other number systems (the even numbers), and it is the reason 1 is not called prime.
- Taking logs, $\log n = \sum a_i \log p_i$ (Equation (15.4)); uniqueness implies that $\log 2, \log 3, \log 5, \ldots$ obey no integer relation, so waves at these frequencies never share a period.

15.3 There are infinitely many primes

Theorem (Euclid, proved). *There are infinitely many primes.*

Proof. We show that any finite list of primes $p_1, p_2, \ldots, p_k$ misses at least one prime. Form

$$N = p_1 p_2 \cdots p_k + 1. \tag{15.5}$$

Then $N \geq 3$, so by Lemma 1 it has a prime divisor q. Could q be one of the p_i? If $q = p_i$, then q divides the product $p_1 \cdots p_k$ and also divides N, so it divides their difference, $N - p_1 \cdots p_k = 1$. No prime divides 1. Hence q is a prime not on the list. Since no finite list contains all primes, there are infinitely many. ■

Notice what the proof does *not* say: it does not say N itself is prime.

Sample Problem 15.3: Euclid's numbers

Compute $N_k = 2 \cdot 3 \cdot 5 \cdots p_k + 1$ (the product of the first k primes, plus one) for $k = 1, \ldots, 6$ and identify which are prime.

Step 1: compute. $N_1 = 2+1 = 3$; $N_2 = 6+1 = 7$; $N_3 = 30+1 = 31$; $N_4 = 210+1 = 211$; $N_5 = 2310 + 1 = 2311$; $N_6 = 30030 + 1 = 30031$.

Step 2: test the first five. $3, 7, 31$ are prime. $\sqrt{211} < 15$ and none of $2, 3, 5, 7, 11, 13$ divides 211, so it is prime. $\sqrt{2311} < 49$ and none of the primes up to 47 divides 2311, so it is prime. (By construction none of N_k is divisible by the first k primes, so only primes larger than p_k need be tried.)

Step 3: test N_6. Trying primes from 17 upward, we find $30031 = 59 \cdot 509$.

Step 4: interpret. N_6 is composite, yet its prime factors 59 and 509 are both missing from the list $2, 3, 5, 7, 11, 13$, exactly as the proof promises.

$$\boxed{N_1, \ldots, N_5 = 3, 7, 31, 211, 2311 \text{ prime}; \quad N_6 = 30031 = 59 \cdot 509.}$$

15.3.1 Other routes, and what is still open

Euclid's argument is not the only one. In Chapter 16 you will see Euler's argument: the harmonic series $\sum 1/k$ diverges (Chapter 3), while a finite set of primes would make it equal to a finite product; so the primes cannot be finite in number. That argument is the seed of the whole zeta-function story.

Two further facts help calibrate what is known.

Gaps between primes can be as long as you like (proved). For any integer $n \geq 2$, the $n - 1$ consecutive numbers $n! + 2,\ n! + 3,\ \ldots,\ n! + n$ are all composite: for $2 \leq j \leq n$, j divides $n!$ (it is one of the factors $1 \cdot 2 \cdots n$) and divides j, hence divides $n! + j$, and $1 < j < n! + j$. For example $n = 5$ gives the run $122, 123, 124, 125$, all composite.

Taken on trust

Primes in arithmetic progressions (proved; proof beyond this book). Dirichlet proved in 1837 that if a and b have no common factor larger than 1, there are infinitely many primes of the form $ak + b$. Some special cases yield to Euclid-type arguments (Problem 8 does $4k + 3$), but the general proof needs the L-functions mentioned at the end of this book; see [25] or [23], Chapter 7.

Twin primes (open). Pairs of primes differing by 2, such as $(11, 13)$ or $(29, 31)$, are called twin primes. Whether there are infinitely many is an open problem, the twin prime conjecture.

Checkpoint 15.3

Someone claims: "Euclid's proof shows that $p_1 \cdots p_k + 1$ is always prime." What is wrong?

Answer: The proof only shows that every prime factor of N is new. N itself may be composite, as $30031 = 59 \cdot 509$ shows.

Key Ideas

- There are infinitely many primes (Euclid, Equation (15.5)); the number N built in the proof need not be prime.
- Prime gaps can be arbitrarily long $(n! + 2, \ldots, n! + n)$. Primes in progressions: proved by Dirichlet. Infinitely many twin primes: open.

15.4 Finding primes: the sieve of Eratosthenes

Trial division tests one number at a time. To list *all* primes up to some x there is a far better method, attributed to Eratosthenes of Cyrene (third century BCE).

The sieve.

1. Write down the integers $2, 3, \ldots, x$. All are "unmarked".
2. Let p be the smallest unmarked number not yet used. It is prime.
3. Mark (cross out) the multiples p^2, $p^2 + p$, $p^2 + 2p, \ldots$ up to x.
4. If $p^2 \le x$ for the next unused unmarked p, return to step 2. Otherwise stop: the numbers still unmarked are exactly the primes up to x.

Why it works. A number $n \le x$ is marked only when it is a multiple of some smaller number p found in step 2, so marked numbers are composite. Conversely, let $n \le x$ be composite and let p be its *smallest* prime divisor. By Lemma 2, n has some prime divisor $\le \sqrt{n}$, so the smallest one satisfies $p \le \sqrt{n} \le \sqrt{x}$. Write $n = p \cdot c$. Then $c = n/p \ge n/\sqrt{n} = \sqrt{n} \ge p$. Hence $n = pc \ge p^2$ and n lies on the list $p^2, p^2 + p, \ldots$, so it is marked when p is processed. Two consequences: we may start crossing out at p^2 (smaller multiples kp with $k < p$ have a smaller prime factor and were already marked), and we may stop once $p^2 > x$.

Sample Problem 15.4: Sieving up to 50

List the primes up to $x = 50$ with the sieve and count them.

Step 1: decide when to stop. We need passes only for primes p with $p^2 \le 50$, namely $2, 3, 5, 7$ (since $7^2 = 49 \le 50 < 121 = 11^2$).

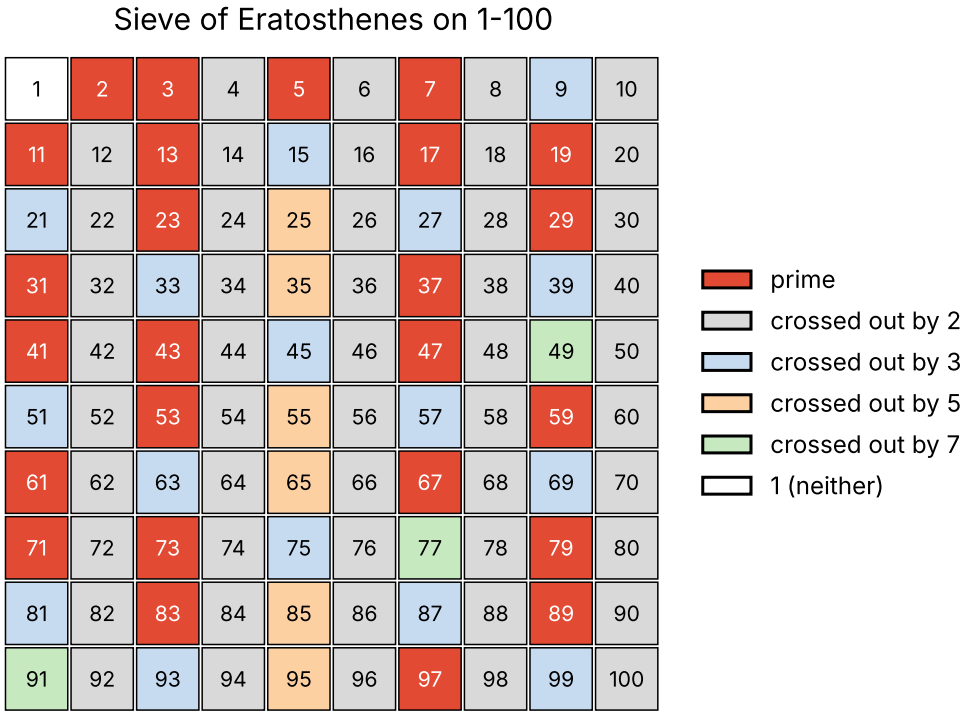


Figure 15.1: The sieve of Eratosthenes on 1–100. Primes are shown in red. Every composite number is colored by its smallest prime factor, which is the prime whose pass of the sieve first crosses it out. Because $\sqrt{100} = 10$, the passes for $2, 3, 5, 7$ already remove every composite, in agreement with Lemma 2.

Step 2: pass for 2. Cross out $4, 6, 8, \ldots, 50$.
Step 3: pass for 3. Cross out $9, 12, 15, \ldots, 48$ (many already gone).
Step 4: pass for 5. Cross out $25, 30, 35, 40, 45, 50$.
Step 5: pass for 7. Cross out 49 (the only multiple of 7 that is at least 49 and at most 50).
Step 6: read off the survivors. $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47$.

15 primes up to 50.

```
import numpy as np

def sieve(x):
    """Return a numpy array of all primes <= x."""
    is_prime = np.ones(x + 1, dtype=bool)   # start: everything unmarked
    is_prime[:2] = False                    # 0 and 1 are not prime
    p = 2
    while p * p <= x:
        if is_prime[p]:                     # p survived, so p is prime
            is_prime[p*p::p] = False        # cross out p^2, p^2+p, ...
        p += 1
    return np.nonzero(is_prime)[0]          # indices still marked True

primes = sieve(100)
print(len(primes), primes[-1])              # prints: 25 97
```

The slice `is_prime[p*p::p]` means "every p-th entry, starting at index p^2", which is exactly step 3. The same job is done by `qmnt.primes_upto(x)`.

How much work is the sieve? The pass for p crosses out about x/p numbers, so the total is about $x \sum_{p \le \sqrt{x}} 1/p$. A theorem of Mertens, which we do not prove (see [24], Chapter 22), says $\sum_{p \le y} 1/p$ grows like $\log \log y$, extraordinarily slowly. So in practice the sieve costs only a little more than writing the list down. Listing the first 10^6 integers' primes takes a fraction of a second.

Checkpoint 15.4

When sieving up to 200, which is the last prime whose multiples you must cross out?

Answer: 13: since $13^2 = 169 \le 200$ but the next prime has $17^2 = 289 > 200$.

Key Ideas

- The sieve lists all primes up to x by crossing out multiples of each surviving p, starting at p^2, for $p \le \sqrt{x}$.
- It is correct because every composite $n \le x$ has a smallest prime factor p with $p^2 \le n$.
- Its cost grows only slightly faster than x.

15.5 Counting primes: the function $\pi(x)$

Definition. For a real number x, the *prime counting function* $\pi(x)$ is the number of primes p with $p \le x$.

So $\pi(10) = 4$ (the primes $2, 3, 5, 7$), $\pi(30) = 10$, $\pi(2.5) = 1$ and $\pi(1) = 0$. As x increases, $\pi(x)$ stays constant until x reaches a prime, where it jumps up by exactly 1. It is a *step function* (Figure 15.2).

Pitfall

In $\pi(x)$ the letter π is the name of a function, not the number $3.14159\ldots$ Context always decides; in this chapter, $\pi(x)$ with an argument means the prime counting function, while the number π appears in expressions like $2\pi/\log 2$. A second trap: $\pi(n)$ is *how many* primes are at most n, while p_n denotes the *n-th prime* ($p_1 = 2$, $p_2 = 3$, $p_6 = 13$, $p_{25} = 97$). The two are inverse to each other in the sense that $\pi(p_n) = n$.

Counting with the sieve gives Table 15.1. Look at the last column, the *average spacing* $x/\pi(x)$ between primes up to x. Each time x grows by a factor 10, the average spacing grows by about the same amount: the increments are 1.50, 1.95, 2.18, 2.29, 2.31, creeping toward $\log 10 = 2.3026$. That is exactly how a function of the form $\log x$ behaves, since $\log(10x) = \log x + \log 10$. This pattern, that near x primes are spaced by about $\log x$, is what the young Gauss noticed in his own tables around 1792.

Table 15.1: The prime counting function and its approximations. $\mathrm{Li}(x) = \int_2^x \mathrm{d}t/\log t$ is defined in Equation (15.7). All entries computed by the sieve and by numerical integration (verified in this chapter's check script).

x	$\pi(x)$	$x/\log x$	$\mathrm{Li}(x)$	$\dfrac{\pi(x)}{x/\log x}$	$\pi(x) - \mathrm{Li}(x)$	$x/\pi(x)$
10	4	4.343	5.120	0.921	−1.12	2.500
10^2	25	21.715	29.081	1.151	−4.08	4.000
10^3	168	144.765	176.564	1.161	−8.56	5.952
10^4	1229	1085.736	1245.092	1.132	−16.09	8.137
10^5	9592	8685.890	9628.764	1.104	−36.76	10.425
10^6	78498	72382.414	78626.504	1.084	−128.50	12.739

```
import numpy as np, mpmath, qmnt
primes = qmnt.primes_upto(10**6)
for e in range(1, 7):
    x = 10**e
    pi_x = np.searchsorted(primes, x, side="right")   # number of primes <= x
    Li = float(mpmath.li(x, offset=True))              # integral from 2 to x
    print(x, pi_x, round(x/np.log(x), 3), round(Li, 3), round(x/pi_x, 3))
```

Here `np.searchsorted(primes, x, side="right")` returns the position at which x would be inserted into the sorted array of primes, which is the number of primes $\le x$. The option `offset=True` makes `mpmath.li` integrate from 2 rather than from 0.

Checkpoint 15.5

At which values of x in $[10, 20]$ does $\pi(x)$ jump, and what is $\pi(20)$?

Answer: It jumps by 1 at $x = 11, 13, 17, 19$. Since $\pi(10) = 4$, we get $\pi(20) = 8$.

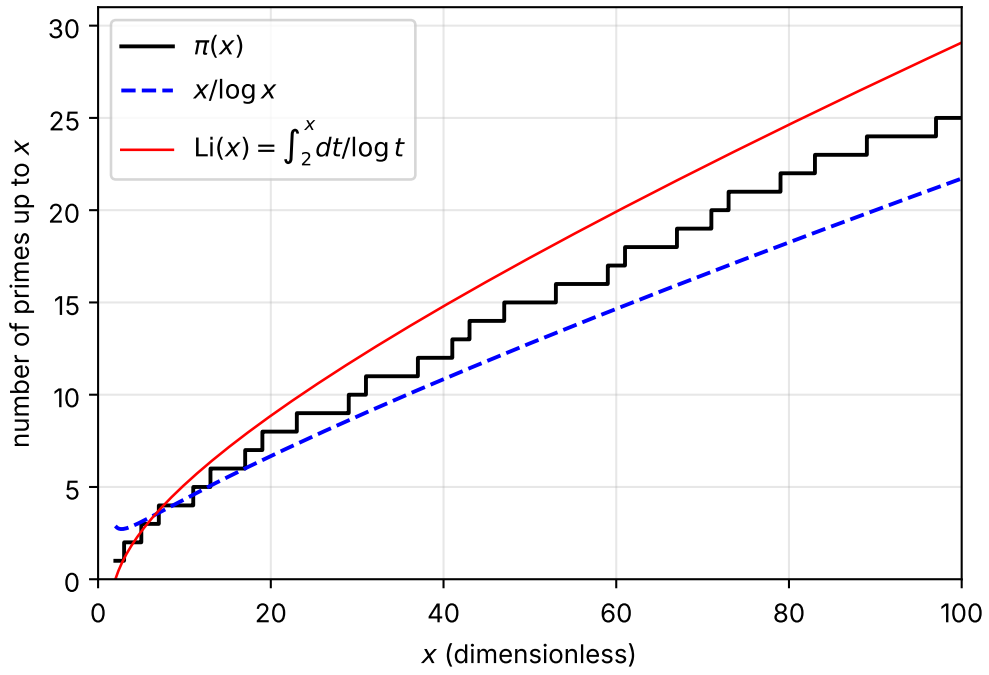


Figure 15.2: The prime counting function $\pi(x)$ (staircase) for $2 \le x \le 100$, with the two smooth approximations of the prime number theorem: $x/\log x$ (below the staircase over most of this range) and $\mathrm{Li}(x) = \int_2^x \mathrm{d}t/\log t$ (above the staircase over most of this range). At such small x, Li is not yet the better approximation; its advantage shows only at larger x (see Figure 15.3). At $x = 100$ the three values are 25, 21.7 and 29.1.

Key Ideas

- $\pi(x)$ counts primes up to x; it is a step function with unit jumps at the primes.
- Numerically, the average spacing $x/\pi(x)$ grows by about $\log 10$ per decade (Table 15.1): primes near x are spaced by roughly $\log x$.

15.6 The prime number theorem

15.6.1 Statement

Recall from Chapter 4 that $f(x) \sim g(x)$ ("f is asymptotic to g") means $f(x)/g(x) \to 1$ as $x \to \infty$.

Prime number theorem (PNT).

$$\pi(x) \sim \frac{x}{\log x} \qquad (x \to \infty). \tag{15.6}$$

Taken on trust

Status: proved, proof beyond this book. The PNT was proved in 1896, independently, by Jacques Hadamard and Charles de la Vallée Poussin. Both proofs use the zeta function of Chapter 16 as a function of a complex variable, and the key input is that $\zeta(s)$ has no zeros on the line $\mathrm{Re}\, s = 1$. An "elementary" proof (no complex analysis, but not simpler) was found by Erdős and Selberg in 1949. Readable accounts: [23], Chapter 13;

[25], Chapters 13 and 18; [5], Chapter 5.

15.6.2 History in brief

The theorem was conjectured long before it was proved. Gauss, as a teenager around 1792, counted primes in blocks of 1000 and concluded that their density near x is about $1/\log x$; he described this in a letter of 1849. Legendre published in 1798 (refined 1808) the empirical formula $x/(\log x - 1.08366)$. Around 1850 Chebyshev proved that the ratio $\pi(x)/(x/\log x)$ is trapped, for all sufficiently large x (not for every x), between two constants close to 1 (about 0.921 and 1.106; the first happens to be close to the value of the ratio at $x = 10$ in Table 15.1, a coincidence), and that *if* the ratio has a limit, the limit is 1; Section 15.7 reproduces the spirit of his argument. In 1859 Riemann's eight-page paper [2] showed that the fine structure of $\pi(x)$ is governed by the zeros of the zeta function. That paper is where the Riemann hypothesis was first stated.

15.6.3 The logarithmic integral

Gauss's density idea suggests a better approximation than $x/\log x$. If a number near t is prime "with probability" about $1/\log t$, then the expected number of primes in a short interval $[t, t+\mathrm{d}t]$ is $\mathrm{d}t/\log t$, and adding up these contributions gives the *logarithmic integral*

$$\mathrm{Li}(x) = \int_2^x \frac{\mathrm{d}t}{\log t}. \tag{15.7}$$

(The "probability" language is a heuristic: primes are not random. It is a useful bookkeeping device, not a theorem.)

Pitfall

Many tables use instead $\mathrm{li}(x) = \int_0^x \mathrm{d}t/\log t$, where the singularity at $t = 1$ is handled by a symmetric limit. The two differ by a constant: $\mathrm{li}(x) = \mathrm{Li}(x) + \mathrm{li}(2)$ with $\mathrm{li}(2) \approx 1.045$. For large x the constant is irrelevant; for small x make sure you know which one a table means. This book uses Li.

$\mathrm{Li}(x) \sim x/\log x$ **(proved).** We show that the two approximations agree to leading order, so Equation (15.6) may equally be written $\pi(x) \sim \mathrm{Li}(x)$.

Step 1: integrate by parts. In Equation (15.7) take $u = 1/\log t$ and $\mathrm{d}v = \mathrm{d}t$, so that $\mathrm{d}u = -\mathrm{d}t/(t\log^2 t)$ and $v = t$:

$$\begin{aligned} \mathrm{Li}(x) &= \Big[\frac{t}{\log t}\Big]_2^x - \int_2^x t \cdot \Big(-\frac{1}{t\log^2 t}\Big)\mathrm{d}t \\ &= \frac{x}{\log x} - \frac{2}{\log 2} + \int_2^x \frac{\mathrm{d}t}{\log^2 t}. \end{aligned} \tag{15.8}$$

Step 2: bound the leftover integral. Split it at $\sqrt{x}$ (assume $x > 4$ so that $\sqrt{x} > 2$). On $[2, \sqrt{x}]$ the integrand is at most $1/\log^2 2$ and the interval has length less than $\sqrt{x}$. On $[\sqrt{x}, x]$ the integrand is at most $1/\log^2\sqrt{x} = 4/\log^2 x$ and the interval has length less than x. So

$$0 < \int_2^x \frac{\mathrm{d}t}{\log^2 t} < \frac{\sqrt{x}}{\log^2 2} + \frac{4x}{\log^2 x}.$$

Step 3: compare with $x/\log x$. Divide the bound by $x/\log x$:

$$\frac{\sqrt{x}}{\log^2 2}\cdot\frac{\log x}{x} + \frac{4x}{\log^2 x}\cdot\frac{\log x}{x} = \frac{\log x}{\sqrt{x}\,\log^2 2} + \frac{4}{\log x} \longrightarrow 0.$$

(The first term tends to zero because $\log x$ grows more slowly than any positive power of x, Chapter 4.) Also $(2/\log 2)/(x/\log x) \to 0$. Dividing Equation (15.8) by $x/\log x$ therefore gives $\mathrm{Li}(x)/(x/\log x) \to 1$. ■

Step 3 also tells us something about *how fast* the ratio approaches 1. It bounds the relative correction by $4/\log x$ plus a term that is even smaller. The asymptotic series below shows that the correction is in fact about $1/\log x$, which shrinks painfully slowly. Since $\pi(x) \approx \mathrm{Li}(x)$, $\pi(x)$ exceeds $x/\log x$ by about the same fraction. At $x = 10^6$, $1/\log x \approx 0.072$, against the observed excess of about 8% (ratio 1.084 in Table 15.1). Repeating the integration by parts (Problem 25) gives the *asymptotic series*

$$\mathrm{Li}(x) = \frac{x}{\log x} + \frac{x}{\log^2 x} + \frac{2x}{\log^3 x} + \dots,$$

whose first terms improve the approximation, although (like Stirling's series in Chapter 4) the full series does not converge.

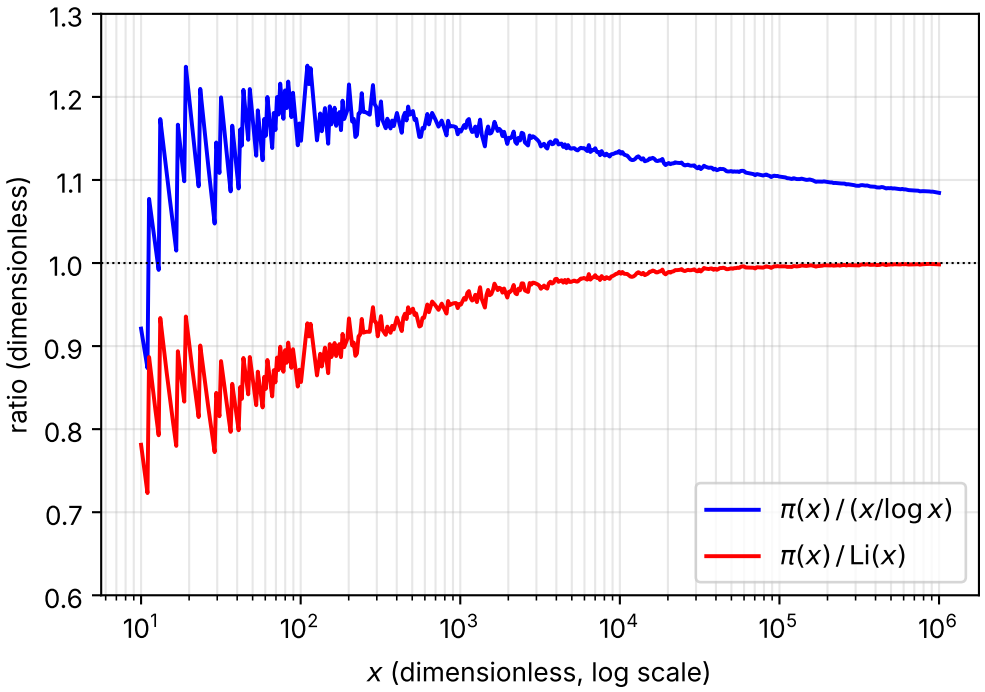


Figure 15.3: Ratios $\pi(x)/(x/\log x)$ and $\pi(x)/\mathrm{Li}(x)$ for $10 \le x \le 10^6$ (logarithmic x-axis). Both tend to 1, as the prime number theorem requires, but $\pi(x)/\mathrm{Li}(x)$ is within 2% of 1 from $x = 10^4$ on and within 0.2% at $x = 10^6$, while $\pi(x)/(x/\log x)$ is still about 1.08 at $x = 10^6$.

Sample Problem 15.5: How good are the two approximations?

Using Table 15.1, compute the relative errors of $x/\log x$ and $\mathrm{Li}(x)$ as approximations to $\pi(x)$ at $x = 10^3$ and $x = 10^6$. Then test the density picture on the interval $[10^4, 10^5]$.

Step 1: $x = 10^3$**.** Relative error of $x/\log x$: $(144.765 - 168)/168 = -0.138$. Relative error of Li: $(176.564 - 168)/168 = +0.051$.

Step 2: $x = 10^6$**.** $(72382.414 - 78498)/78498 = -0.078$ and $(78626.504 - 78498)/78498 = +0.0016$.

Step 3: an interval. The density picture predicts that the number of primes in $(10^4, 10^5]$ is about $\int_{10^4}^{10^5} dt/\log t = \mathrm{Li}(10^5) - \mathrm{Li}(10^4) = 9628.764 - 1245.092 = 8383.67$. The true

count is $\pi(10^5) - \pi(10^4) = 9592 - 1229 = 8363$.

$x/\log x$: -13.8% (10^3), -7.8% (10^6); Li : $+5.1\%$ (10^3), $+0.16\%$ (10^6) primes in $(10^4, 10^5]$: 8363 actual vs 8383.7 predicted

Both approximations are asymptotically correct. From $x = 10^3$ upward in Table 15.1, Li is far better. At $x = 10$ and $x = 100$ it is not: its errors there, $+1.12$ and $+4.08$, are larger in size than those of $x/\log x$, which are $+0.34$ and -3.29.

15.6.4 Mean spacing of primes

If primes near x have density $1/\log x$, their *local mean spacing* is the reciprocal, $\log x$. At $x = 10^6$ that is $\log 10^6 = 13.82$. The average spacing over the *whole* range up to 10^6 is smaller, $10^6/78498 = 12.74$, because it includes the crowded small primes. Compare Chapter 17, where the Riemann zeros get a local mean spacing gap_n in exactly the same way, as the reciprocal of a smooth density.

15.6.5 The error term and the Riemann hypothesis

How large is $\pi(x) - \text{Li}(x)$? In Table 15.1 it is negative and grows in size, but slowly compared with $\pi(x)$ itself.

Taken on trust

What is known. De la Vallée Poussin proved $\pi(x) = \text{Li}(x) + \mathcal{O}(x\, e^{-c\sqrt{\log x}})$ for some constant $c > 0$, an error smaller than $x/\log^k x$ for every k (proof: [25], Chapter 18).

The Riemann hypothesis connection. It is a theorem (von Koch, 1901) that the Riemann hypothesis of Chapter 16 is *equivalent* to the statement $\pi(x) = \text{Li}(x) + \mathcal{O}(\sqrt{x}\log x)$. The Riemann hypothesis itself is **open**: it has been neither proved nor disproved. So the best possible error term for counting primes is an open problem, and it is controlled by the zeros ([5], Chapter 5).

A warning about numerical evidence. Classical tables compare $\pi(x)$ with $\text{li}(x) = \text{Li}(x) + \text{li}(2)$ (see the pitfall above). In every case computed directly, $\pi(x) < \text{li}(x)$ for $x \geq 2$; this chapter's check script confirms it for all $x \leq 10^6$. Our Li is smaller than li by the constant 1.045. For Li the inequality fails at a few small x (for example $\text{Li}(7) \approx 3.71 < \pi(7) = 4$), but it holds for every x from 10 to 10^6, including all of Table 15.1. Yet Littlewood proved in 1914 that $\pi(x) - \text{li}(x)$ changes sign infinitely often. The oscillations he found grow without bound, so $\pi(x) - \text{Li}(x)$, which differs from $\pi(x) - \text{li}(x)$ only by a constant, also changes sign infinitely often. No sign change of $\pi(x) - \text{li}(x)$ has been found by direct computation; the literature proves that the first one lies below an explicit but astronomically large bound, far beyond any table. Tables, however large, can mislead.

Checkpoint 15.6

"Since $\pi(x) \sim x/\log x$, the difference $\pi(x) - x/\log x$ tends to zero." True or false?

Answer: False. $\sim$ says the *ratio* tends to 1. The difference grows without bound: it is 6115.6 at $x = 10^6$ (from Table 15.1) and is of order $x/\log^2 x$.

Key Ideas

- PNT: $\pi(x) \sim x/\log x \sim \mathrm{Li}(x)$ (Equations (15.6) and (15.7)); proved in 1896, proof beyond this book.
- $\mathrm{Li}(x) \sim x/\log x$ is proved here by integration by parts (Equation (15.8)); the relative difference is of order $1/\log x$.
- Near x the primes have density $1/\log x$ and mean spacing $\log x$ (heuristic picture, consistent with the PNT).
- The size of $\pi(x) - \mathrm{Li}(x)$ is governed by the zeta zeros; the Riemann hypothesis, which would make it $\mathcal{O}(\sqrt{x}\log x)$, is open.

15.7 A proven bound: Chebyshev's argument

★ Did you know? *Always a Prime Between*

In 1845 Joseph Bertrand conjectured that for every $n > 1$ there is a prime between n and $2n$, and he checked it for n up to about three million. Pafnuty Chebyshev proved it around 1850 using clever bounds on binomial coefficients. In 1932 the 19-year-old Paul Erdős published a famously simple proof.

The PNT is beyond us, but a weaker statement of the right shape is not. It introduces a function that appears throughout the rest of the chapter.

Definition. Chebyshev's *theta function* is

$$\theta(x) = \sum_{p \le x} \log p, \tag{15.9}$$

the sum of the logarithms of all primes up to x. Equivalently $\theta(x) = \log\left(\prod_{p \le x} p\right)$.

So $\theta(10) = \log(2 \cdot 3 \cdot 5 \cdot 7) = \log 210$. Where $\pi(x)$ counts each prime with weight 1, $\theta(x)$ counts it with weight $\log p$, its "size". This θ is unrelated to the Riemann–Siegel theta function $\theta(T)$ of Chapter 17; context distinguishes them.

Theorem (proved, given Euclid's lemma, which was taken on trust in Section 15.2). *For every integer $n \ge 1$,*

$$\theta(2n) - \theta(n) < n \log 4, \tag{15.10}$$

and for every real $x \ge 1$, $\theta(x) < (4 \log 2)\, x \approx 2.77\, x$.

Proof of Equation (15.10). Recall the binomial coefficient $\binom{2n}{n} = \frac{(2n)!}{n!\,n!}$, the number of ways of choosing n objects from $2n$, which is a positive integer.

1. *Every prime p with $n < p \le 2n$ divides $\binom{2n}{n}$.* Such a p is one of the factors of $(2n)! = 1 \cdot 2 \cdots (2n)$, so $p \mid (2n)! = \binom{2n}{n}(n!)^2$. By Euclid's lemma p divides $\binom{2n}{n}$ or $(n!)^2$. It cannot divide $(n!)^2 = (1 \cdot 2 \cdots n)^2$: by the repeated Euclid's lemma it would then divide one of the factors $1, \ldots, n$, all of which are positive and smaller than p. So $p \mid \binom{2n}{n}$.

2. *Their product divides $\binom{2n}{n}$.* Write the canonical factorization $\binom{2n}{n} = r_1^{b_1} \cdots r_j^{b_j}$ (Equation (15.2)). If a prime p divides $\binom{2n}{n}$, then by the repeated Euclid's lemma p divides some r_i, so $p = r_i$: every such p is one of the r_i, with exponent $b_i \ge 1$. The primes in $(n, 2n]$ are distinct, so they are distinct members of the list $r_1, \ldots, r_j$. Hence $\binom{2n}{n} = \left(\prod_{n < p \le 2n} p\right) \cdot c$,

where c is the product of all remaining prime factors (each r_i taken $b_i - 1$ more times if it is one of our primes, b_i times otherwise), a positive integer. So $\prod_{n<p\le 2n} p$ divides $\binom{2n}{n}$, and therefore $\prod_{n<p\le 2n} p \le \binom{2n}{n}$.

3. $\binom{2n}{n} < 4^n$. By the binomial expansion, $4^n = (1+1)^{2n} = \sum_{k=0}^{2n}\binom{2n}{k}$, a sum of positive terms of which $\binom{2n}{n}$ is one, and for $n \ge 1$ there are other terms (e.g. $\binom{2n}{0} = 1$).

4. *Take logs.* $\theta(2n) - \theta(n) = \sum_{n<p\le 2n} \log p = \log \prod_{n<p\le 2n} p \le \log \binom{2n}{n} < n \log 4$.

Proof of $\theta(x) < (4\log 2)x$. Given $x \ge 1$, choose the integer $k \ge 1$ with $2^{k-1} \le x < 2^k$. Since θ never decreases and $\theta(1) = 0$, a telescoping sum (Chapter 3) gives

$$\begin{aligned}
\theta(x) \le \theta(2^k) &= \sum_{j=1}^{k} [\theta(2^j) - \theta(2^{j-1})] \\
&< \sum_{j=1}^{k} 2^{j-1} \log 4 \quad \text{(Equation (15.10) with } n = 2^{j-1}\text{)} \\
&= (2^k - 1)\log 4 \quad \text{(geometric series)} \\
&< 2^k \log 4 \le 2x \log 4 = (4\log 2)\, x \quad \text{(since } 2^{k-1} \le x\text{)}. \qquad \blacksquare
\end{aligned}$$

Corollary (proved): $\pi(x) = \mathcal{O}(x/\log x)$. For $x > 1$, the primes in $(\sqrt{x}, x]$ each contribute more than $\log\sqrt{x} = \frac{1}{2}\log x$ to $\theta(x)$, so

$$\theta(x) \;\ge \sum_{\sqrt{x}<p\le x} \log p \;\ge\; \left[\pi(x) - \pi(\sqrt{x})\right] \tfrac{1}{2}\log x.$$

Rearranging and using $\pi(\sqrt{x}) \le \sqrt{x}$ and $\theta(x) < (4\log 2)x$,

$$\pi(x) \;\le\; \pi(\sqrt{x}) + \frac{2\,\theta(x)}{\log x} \;<\; \sqrt{x} + (8\log 2)\frac{x}{\log x}. \tag{15.11}$$

We now show that the right side has the form $\mathcal{O}(x/\log x)$. The function $\log x/\sqrt{x}$ has derivative $(1 - \frac{1}{2}\log x)/x^{3/2}$, which vanishes only at $x = e^2$, where the function takes its maximum value $2/e < 1$. So $\log x \le \sqrt{x}$ for all $x > 0$. Then for $x > 1$ we have $\sqrt{x} = x/\sqrt{x} \le x/\log x$. Hence the right side of Equation (15.11) is at most $(1 + 8\log 2)\, x/\log x$, which is the $\mathcal{O}$-statement of the corollary.

The constant $8\log 2 \approx 5.55$ is generous (at $x = 10^6$ the bound is about 402,000 against the true 78,498), but the *shape* $x/\log x$ is exactly the one in the PNT. The argument used nothing but unique factorization and Euclid's lemma (taken on trust in Section 15.2). Chebyshev's sharper version of this reasoning gave the constants 0.921 and 1.106 quoted above; Problem 23 proves a lower bound of the same shape.

Sample Problem 15.6: Chebyshev's divisibility for $n = 10$

Check each step of the proof of Equation (15.10) for $n = 10$.
Step 1: the primes in $(10, 20]$**.** They are $11, 13, 17, 19$, with product $11 \cdot 13 \cdot 17 \cdot 19 = 46189$.
Step 2: the binomial coefficient. $\binom{20}{10} = 184756$.
Step 3: divisibility. $184756/46189 = 4$ exactly, so $184756 = 2^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19$; the product of the four primes divides it, as claimed.
Step 4: the bound. $\theta(20) - \theta(10) = \log 46189 = 10.7405$, while $10\log 4 = 13.8629$.

$$\boxed{\theta(20) - \theta(10) = \log 46189 \approx 10.74 < 10\log 4 \approx 13.86.}$$

Key Ideas

- $\theta(x) = \sum_{p \le x} \log p$ weights each prime by its logarithm (Equation (15.9)).
- Using only unique factorization and the binomial coefficient $\binom{2n}{n} < 4^n$, one proves $\theta(x) < (4 \log 2)x$ and $\pi(x) < \sqrt{x} + (8 \log 2)x/\log x$ (Equations (15.10) and (15.11)): primes thin out at the rate the PNT predicts, up to a constant.

15.8 Prime powers and the von Mangoldt function

15.8.1 Prime powers

A *prime power* is a number of the form $q = p^m$ with p prime and $m \ge 1$ an integer. Primes themselves are prime powers with $m = 1$. The prime powers up to 30 are

$$2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29,$$

where $4 = 2^2$, $8 = 2^3$, $9 = 3^2$, $16 = 2^4$, $25 = 5^2$ and $27 = 3^3$ are the ones that are not prime. By unique factorization, a prime power determines its p and m: $q = p^m$ has exactly one prime factor.

Why single out prime powers? The paper's Eq. (4) contains the waves $\sin(mt \log p)$, and $m \log p = \log(p^m)$, so its frequencies are precisely the logarithms of prime powers. The next definition gives each prime power its natural weight.

15.8.2 The von Mangoldt function

Definition. The *von Mangoldt function* is defined for positive integers k by

$$\Lambda(k) = \begin{cases} \log p, & \text{if } k = p^m \text{ for a prime } p \text{ and an integer } m \ge 1, \\ 0, & \text{otherwise.} \end{cases} \tag{15.12}$$

So Λ is zero except at prime powers, and at a power of p it equals $\log p$, whatever the exponent. Table 15.2 lists the first values and Figure 15.4 plots them.

Table 15.2: The von Mangoldt function for $k = 1, \dots, 12$.

k	1	2	3	4	5	6	7	8	9	10	11	12
$\Lambda(k)$	0	$\log 2$	$\log 3$	$\log 2$	$\log 5$	0	$\log 7$	$\log 2$	$\log 3$	0	$\log 11$	0
value	0	0.693	1.099	0.693	1.609	0	1.946	0.693	1.099	0	2.398	0

Pitfall

$\Lambda(1) = 0$: the number $1 = p^0$ is *not* counted as a prime power, because the definition requires $m \ge 1$. And $\Lambda(8) = \log 2$, not $\log 8$: the weight is the log of the underlying prime.

15.8.3 The key identity

The weights in Equation (15.12) look odd until you see what they add up to.

Theorem (proved). *For every positive integer n,*

$$\sum_{d \mid n} \Lambda(d) = \log n, \tag{15.13}$$

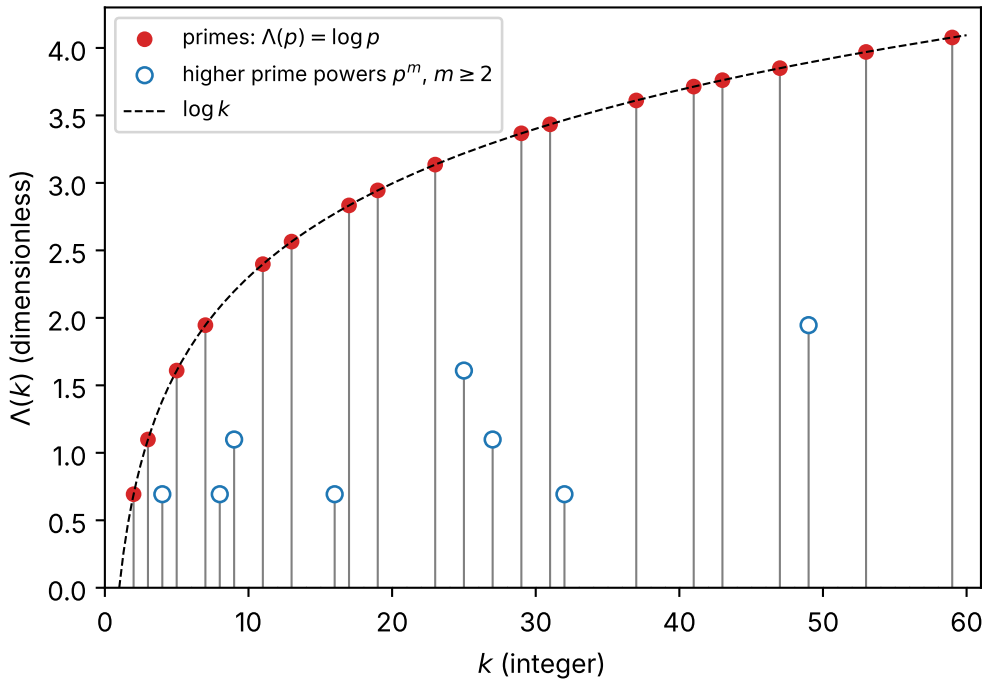


Figure 15.4: The von Mangoldt function $\Lambda(k)$ for $1 \le k \le 60$. Primes (filled markers) sit on the curve $\log k$ (dashed); higher prime powers such as $4, 8, 16, 32$ (open markers) sit lower, at the height $\log p$ of their prime. At all other integers $\Lambda(k) = 0$.

where the sum runs over all positive divisors d of n.

Proof. For $n = 1$ the only divisor is 1 and both sides are 0. For $n \ge 2$ write the canonical factorization $n = p_1^{a_1} \cdots p_k^{a_k}$. Only divisors that are prime powers contribute to the sum. A prime power p^j divides n exactly when p is one of the p_i and $1 \le j \le a_i$ (by uniqueness of factorization, the prime powers dividing n are of this form, and each of these does divide n). For each i there are a_i such divisors $p_i^1, \dots, p_i^{a_i}$, each contributing $\log p_i$. Therefore

$$\sum_{d \mid n} \Lambda(d) = \sum_{i=1}^{k} a_i \log p_i = \log n,$$

where the last step is Equation (15.4). ■

In words: Λ is the weighting that makes the prime-power divisors of n add up exactly to the size $\log n$ of n. It is "the logarithm, broken into its prime pieces". This is why Λ appears in the logarithmic derivative of the zeta function (Chapter 18).

Sample Problem 15.7: Checking the identity for $n = 12$

Verify Equation (15.13) for $n = 12$.
Step 1: list the divisors. $1, 2, 3, 4, 6, 12$.
Step 2: their Λ values (from Table 15.2): $0,\ \log 2,\ \log 3,\ \log 2,\ 0,\ 0$.
Step 3: add. $\log 2 + \log 3 + \log 2 = \log(2 \cdot 3 \cdot 2) = \log 12$.
Step 4: numerical value.

$$\boxed{\sum_{d \mid 12} \Lambda(d) = 2\log 2 + \log 3 = \log 12 \approx 2.4849.}$$

Key Ideas

- A prime power is $q = p^m$, $m \geq 1$; the frequencies $m \log p$ of the paper are the numbers $\log q$.
- $\Lambda(q) = \log p$ at $q = p^m$ and 0 elsewhere (Equation (15.12)); $\Lambda(1) = 0$.
- $\sum_{d|n} \Lambda(d) = \log n$ (Equation (15.13)): Λ splits $\log n$ into prime-power pieces.

15.9 Chebyshev's function $\psi(x)$

Definition. Chebyshev's *psi function* is

$$\psi(x) = \sum_{k \leq x} \Lambda(k) = \sum_{p^m \leq x} \log p, \tag{15.14}$$

the sum over all prime powers $p^m \leq x$ (with $m \geq 1$) of $\log p$.

Like $\pi(x)$, $\psi(x)$ is a step function, but its steps sit at all prime powers and have height $\log p$ (Figure 15.5a).

15.9.1 Three properties

(a) ψ is built from θ. Group the terms of Equation (15.14) by the exponent m. The prime powers $p^m \leq x$ with a fixed m are exactly those with $p \leq x^{1/m}$, and they contribute $\sum_{p \leq x^{1/m}} \log p = \theta(x^{1/m})$. So

$$\psi(x) = \theta(x) + \theta(x^{1/2}) + \theta(x^{1/3}) + \dots \tag{15.15}$$

The sum is finite: once $m > \log x / \log 2$ we have $x^{1/m} < 2$ and $\theta(x^{1/m}) = 0$. The $m = 1$ term dominates; the others add up to something much smaller (Problem 22 shows the extra amount is $\mathcal{O}(\sqrt{x} \log^2 x)$).

(b) ψ is the logarithm of a least common multiple. For a positive integer N, let $\operatorname{lcm}(1, \dots, N)$ be the smallest positive integer divisible by each of $1, 2, \dots, N$. Write $\lfloor y \rfloor$ (the *floor* of y) for the largest integer not exceeding y. A prime p appears in $\operatorname{lcm}(1, \dots, N)$ with exponent equal to the largest m with $p^m \leq N$ (p^m is itself one of the numbers $1, \dots, N$, so the lcm must be divisible by it; and no number $\leq N$ is divisible by $p^{m+1} > N$, so no higher power is needed), namely $m = \lfloor \log N / \log p \rfloor$. Taking logs,

$$\log \operatorname{lcm}(1, \dots, N) = \sum_{p \leq N} \left\lfloor \frac{\log N}{\log p} \right\rfloor \log p.$$

On the other hand, in Equation (15.14) each prime $p \leq N$ contributes $\log p$ once for each $m \geq 1$ with $p^m \leq N$, that is $\lfloor \log N / \log p \rfloor$ times. The two expressions agree:

$$\psi(N) = \log \operatorname{lcm}(1, 2, \dots, N). \tag{15.16}$$

(c) $\psi(x) \leq \pi(x) \log x$. From the counting in (b), each prime $p \leq x$ contributes $\lfloor \log x / \log p \rfloor \log p \leq (\log x / \log p) \log p = \log x$. There are $\pi(x)$ such primes, so

$$\psi(x) \leq \pi(x) \log x. \tag{15.17}$$

15.9.2 The PNT in Chebyshev's form

Taken on trust

Proved, proof beyond this book. The statements $\pi(x) \sim x/\log x$, $\theta(x) \sim x$ and $\psi(x) \sim x$ are equivalent: any one implies the others. One direction is suggested by Equation (15.17); the full argument uses a summation-by-parts technique ([23], Theorem 4.4). So the PNT can be stated as

$$\psi(x) \sim x. \tag{15.18}$$

This is the most natural form of the PNT: the prime powers, each weighted by $\log p$, accumulate at an average rate of exactly 1 per unit of x.

Table 15.3 shows how quickly $\psi(x)/x$ approaches 1, much faster than $\pi(x)/(x/\log x)$ did in Table 15.1. Panel (b) of Figure 15.5 shows that, for $x \le 10^5$, $\psi(x) - x$ wanders on the scale of $\sqrt{x}$. This is a numerical observation, not a theorem. A bound $\psi(x) - x = \mathcal{O}(\sqrt{x}\log^2 x)$ valid for all large x is equivalent to the Riemann hypothesis (von Koch, 1901), which is open. Without that hypothesis, only much weaker bounds are proved, of the de la Vallée Poussin type quoted for $\pi(x)$ in Section 15.6.

Table 15.3: Chebyshev's function $\psi(x)$ compared with x (computed in the check script).

x	$\psi(x)$	$\psi(x)/x$	$\psi(x) - x$
10	7.832	0.783	−2.168
10^2	94.045	0.940	−5.955
10^3	996.681	0.997	−3.319
10^4	10013.4	1.001	+13.4

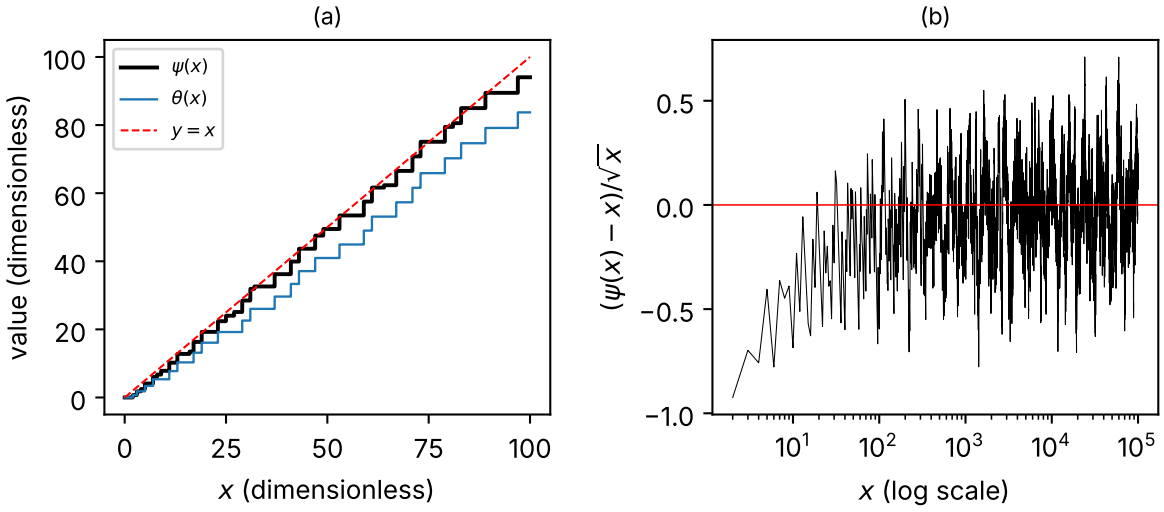


Figure 15.5: (a) Chebyshev's functions $\psi(x)$ (steps at every prime power) and $\theta(x)$ (steps at primes only) for $x \le 100$, with the line $y = x$ that the PNT says ψ follows. (b) The scaled difference $(\psi(x) - x)/\sqrt{x}$ at integers x up to 10^5: in this range the deviation of $\psi(x)$ from x fluctuates on the scale $\sqrt{x}$ (a numerical observation). This is the scale tied to the critical line $\mathrm{Re}\, s = \frac{1}{2}$ of the zeta function (defined in Chapter 16; read this as a preview).

Sample Problem 15.8: $\psi(10)$ and $\psi(30)$ as least common multiples

Compute $\psi(10)$ and $\psi(30)$ using Equation (15.16).
Step 1: $\operatorname{lcm}(1,\dots,10)$**.** The largest powers of each prime not exceeding 10 are $2^3 = 8$, $3^2 = 9$, 5, 7. So lcm $= 8 \cdot 9 \cdot 5 \cdot 7 = 2520$ and $\psi(10) = \log 2520 = 7.8320$. Check with Equation (15.14): the prime powers up to 10 are $2, 3, 4, 5, 7, 8, 9$, contributing $\log 2 + \log 3 + \log 2 + \log 5 + \log 7 + \log 2 + \log 3 = \log(2^3 \cdot 3^2 \cdot 5 \cdot 7)$, the same.
Step 2: $\operatorname{lcm}(1,\dots,30)$**.** Largest prime powers up to 30: $2^4 = 16$, $3^3 = 27$, $5^2 = 25$, and the primes $7, 11, 13, 17, 19, 23, 29$.
Step 3: add logarithms.

$$\psi(30) = 4\log 2 + 3\log 3 + 2\log 5 + \log 7 + \log 11 + \log 13 + \log 17 + \log 19 + \log 23 + \log 29.$$

Numerically: $2.7726 + 3.2958 + 3.2189 + 1.9459 + 2.3979 + 2.5649 + 2.8332 + 2.9444 + 3.1355 + 3.3673 = 28.4764$. Carried at full precision, the sum is 28.4765; the last digit differs because of rounding.
Step 4: compare with x.

$$\boxed{\psi(10) = \log 2520 \approx 7.832, \qquad \psi(30) \approx 28.48, \qquad \psi(30)/30 \approx 0.949.}$$

```python
import numpy as np, qmnt
x = 10000
Lam = np.zeros(x + 1)                  # Lam[k] will hold Lambda(k)
for p in qmnt.primes_upto(x):
    q = int(p)
    while q <= x:                      # q runs over p, p^2, p^3, ...
        Lam[q] = np.log(p)
        q = q * int(p)
psi = np.cumsum(Lam)                   # psi[k] = Lambda(1) + ... + Lambda(k)
print(psi[10], np.log(2520))           # both 7.832...
print(psi[1000], psi[10000] / 10000)   # 996.68..., 1.0013...
```

Taken on trust

A preview: $\psi(x)$ and the zeros. Riemann's ideas, made rigorous by von Mangoldt in 1895, give an exact formula: for $x > 1$ not a prime power,

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \tfrac{1}{2}\log(1 - x^{-2}),$$

where ρ runs over the nontrivial zeros of the zeta function, summed in pairs $\rho, \bar{\rho}$ in order of increasing $|\operatorname{Im}\rho|$ (proof: [25], Chapter 17; [5], Chapter 3). Consider a zero $\rho_n = \frac{1}{2} + it_n$ on the critical line. All zeros used in this book lie on it; whether all zeros do is the Riemann hypothesis (Chapter 16). Such a zero contributes $-x^{\rho_n}/\rho_n = -x^{1/2}e^{it_n \log x}/\rho_n$, a wave in the variable $\log x$ of size $\sqrt{x}/|\rho_n|$. In this sense the zeros are frequencies of the primes. Chapter 18 turns this duality around, making the prime powers the frequencies of the zeros, which is the content of the paper's Eq. (4).

Checkpoint 15.7

What are $\psi(16) - \psi(15)$ and $\psi(18) - \psi(17)$?

Answer: $\psi(16) - \psi(15) = \Lambda(16) = \log 2 \approx 0.693$ since $16 = 2^4$; $\psi(18) - \psi(17) = \Lambda(18) = 0$ since $18 = 2 \cdot 3^2$

is not a prime power.

Key Ideas

- $\psi(x) = \sum_{k \le x} \Lambda(k)$ (Equation (15.14)) $= \theta(x) + \theta(x^{1/2}) + \dots$ (Equation (15.15)) $= \log \operatorname{lcm}(1, \dots, \lfloor x \rfloor)$ (Equation (15.16)).
- $\psi(x) \le \pi(x) \log x$ (proved). PNT in Chebyshev's form: $\psi(x) \sim x$ (equivalence proved, beyond this book).
- Numerically ($x \le 10^5$), $\psi(x) - x$ fluctuates on the scale $\sqrt{x}$. An $\mathcal{O}(\sqrt{x} \log^2 x)$ bound for all x is equivalent to the (open) Riemann hypothesis. The von Mangoldt explicit formula writes $\psi(x) - x$ as a sum over the zeta zeros.

15.10 $\log p$ as the natural size of a prime

We have now met $\log p$ in five roles. It is worth collecting them, because together they explain why the paper's frequencies are $\log p$ and not p.

1. *Logs turn multiplication into addition.* By Equation (15.4), every integer is a point $\log n = \sum a_i \log p_i$ built from the "directions" $\log 2, \log 3, \log 5, \dots$, and these directions obey no integer relation (Section 15.2).
2. *Density.* Near x, primes have density $1/\log x$ (Section 15.6).
3. *Natural weight.* In θ, ψ and Λ, each prime counts with weight $\log p$, and with these weights the PNT becomes the clean statement $\psi(x) \sim x$, and Λ splits $\log n$ exactly (Equation (15.13)).
4. *Digits.* $\log_{10} p = \log p / \log 10$ is, up to rounding, the number of decimal digits of p: the information needed to write p down.
5. *Frequency.* For real t, $p^{-it} = e^{-it \log p}$ (Chapter 2). As t increases, this complex number rotates around the unit circle at angular frequency $\log p$. More generally $(p^m)^{-it} = e^{-imt \log p}$ rotates at $m \log p = \log(p^m)$. So in any expression containing powers n^{-s} with $s = \sigma + it$, each prime power q becomes a wave in t with angular frequency $\log q$ and period

$$P_q = \frac{2\pi}{\log q}. \tag{15.19}$$

Sample Problem 15.9: Periods of the first prime-power waves; the prime-2 wave (RE2)

Compute the angular frequency $\log q$ and period $P_q = 2\pi/\log q$ for $q = 2, 3, 4, 5, 7, 8, 9$. Then find the period of the running example RE2, $-\frac{1}{\pi} \sum_{m=1}^{3} \frac{\sin(mt \log 2)}{m\, 2^{m/2}}$.
Step 1: logarithms. $\log 2 = 0.693147$, $\log 3 = 1.098612$, $\log 4 = 1.386294$, $\log 5 = 1.609438$, $\log 7 = 1.945910$, $\log 8 = 2.079442$, $\log 9 = 2.197225$.
Step 2: periods ($2\pi = 6.283185$). $P_2 = 9.0647$, $P_3 = 5.7192$, $P_4 = 4.5324$, $P_5 = 3.9040$, $P_7 = 3.2289$, $P_8 = 3.0216$, $P_9 = 2.8596$.
Step 3: the three terms of RE2. They are the $q = 2, 4, 8$ waves, with periods P_2, $P_4 = P_2/2$ and $P_8 = P_2/3$. Shifting t by P_2 changes the argument $mt \log 2$ by $mP_2 \log 2 = 2\pi m$, a whole number of turns for each m, so every term repeats after P_2.
Step 4: what we have and have not shown. Step 3 proves that P_2 is *a* period of RE2. That no shorter shift leaves the sum unchanged is also true, but proving it requires

separating the $m = 1$ component from the others, which is done with the orthogonality of sines in Chapter 5; we do not use minimality here.

$$\boxed{P_2 = 2\pi/\log 2 \approx 9.065 \text{ (a period of RE2)}, \quad P_3 \approx 5.719,\ P_5 \approx 3.904,\ P_7 \approx 3.229.}$$

The $q = 2$ wave is the slowest and, as we will see next, the strongest. When more primes are included, the waves at $\log 3, \log 5, \ldots$ are incommensurate with it (Section 15.2), and the sum never repeats.

15.10.1 A preview of the zeta connection

Here is a piece of pure bookkeeping that prepares the paper connection. Consider the double sum over primes p and exponents $m \geq 1$ of p^{-ms}/m. Each pair (p, m) corresponds to exactly one prime power $n = p^m$, and for that n,

$$\frac{\Lambda(n)}{\log n} = \frac{\log p}{\log(p^m)} = \frac{\log p}{m \log p} = \frac{1}{m}. \tag{15.20}$$

So for any s for which the sums converge *absolutely*, so that their terms may be regrouped (Chapter 3), we have the identity below. The condition $\operatorname{Re} s > 1$ suffices: $|p^{-ms}/m| \leq p^{-m \operatorname{Re} s}$, and by unique factorization these are distinct terms of the convergent series $\sum_k k^{-\operatorname{Re} s}$. The identity reads

$$\sum_p \sum_{m=1}^{\infty} \frac{p^{-ms}}{m} = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{\log n}\, n^{-s}, \tag{15.21}$$

where on the right the terms with $\Lambda(n) = 0$ simply vanish. In Chapter 18 you will prove that the left side is $\log \zeta(s)$; here we only note that the right side is a sum over prime powers n with weights $\Lambda(n)/\log n$. On the line $s = \frac{1}{2} + it$ the term $n^{-s} = n^{-1/2} e^{-it \log n}$ is a wave of frequency $\log n$ and amplitude $n^{-1/2}\Lambda(n)/\log n$. Each term makes sense on this line, but the sum no longer converges absolutely there; see Section 15.11 and Chapter 18.

Checkpoint 15.8

Which is the larger weight in Figure 15.6: that of $q = 4 = 2^2$ or that of $q = 13$?

Answer: $q = 4$: $1/(2 \cdot 2^1) = 0.25$; $q = 13$: $1/\sqrt{13} \approx 0.277$. So 13 has the (slightly) larger weight, even though 4 is the smaller number. The weight depends on both p and m.

Key Ideas

- $\log p$ is the natural size of a prime: additive coordinate, inverse density, weight in θ, ψ, Λ, digit count, and frequency.
- $p^{-it} = e^{-it \log p}$ rotates at angular frequency $\log p$; prime power q gives period $2\pi/\log q$ (Equation (15.19)); $2\pi/\log 2 \approx 9.065$.
- $\Lambda(p^m)/\log(p^m) = 1/m$ (Equation (15.20)), so double sums over (p, m) are sums over prime powers (Equation (15.21)).

15.11 Paper connection

Everything in the paper's arithmetic side is built from the objects of this chapter. Here is where each appears.

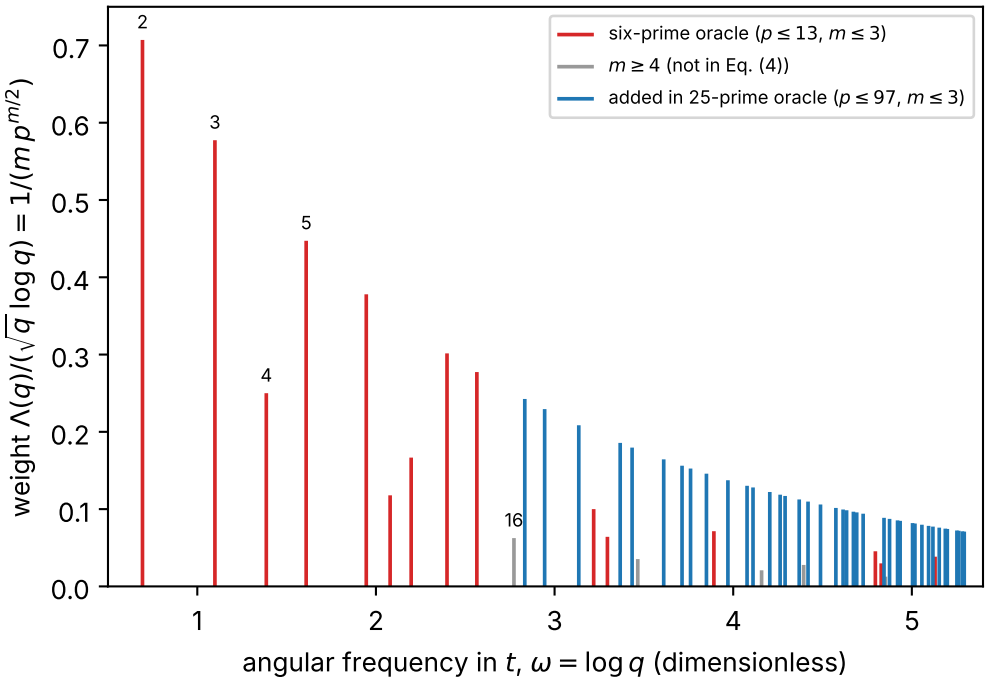


Figure 15.6: The "prime-power spectrum": a vertical line at angular frequency $\omega = \log q$ for every prime power $q \le 200$, of height $\Lambda(q)/(\sqrt{q}\log q) = 1/(m\,p^{m/2})$ for $q = p^m$. Red: the 15 terms of the six-prime explicit formula ($p \le 13$, $m \le 3$) with $q \le 200$. Its other three terms, $7^3 = 343$, $11^3 = 1331$ and $13^3 = 2197$, lie off the plot. Blue: further terms with $m \le 3$, $p \le 97$ and $q \le 200$, included in the 25-prime formula, which also has terms beyond $q = 200$. Gray: powers with $m \ge 4$ (such as 16, 32, 64, 81), which Eq. (4) of the paper omits. The weights decay roughly like $q^{-1/2}$ while the lines crowd together at large ω.

Eq. (4): a sum over prime powers. The paper's oracle (Sec. II D) is

$$N_{\text{fl}}(t) = -\frac{1}{\pi} \sum_{p \in P} \sum_{m=1}^{3} \frac{\sin(mt \log p)}{m\, p^{m/2}},$$

where P is a finite set of primes. By Equation (15.20), $1/m = \Lambda(p^m)/\log(p^m)$, so with $q = p^m$,

$$N_{\text{fl}}(t) = -\frac{1}{\pi} \sum_{\substack{q = p^m,\ p \in P \\ 1 \le m \le 3}} \frac{\Lambda(q)}{\sqrt{q}\, \log q} \sin(t \log q).$$

Every term is one line of Figure 15.6: a sinusoid in the zero height t with angular frequency $\log q$ and amplitude $\Lambda(q)/(\pi\sqrt{q}\log q)$. The sine (rather than a cosine) and the factor $-1/\pi$ come from taking an imaginary part of Equation (15.21) on the critical line; Chapter 18 derives this, and is careful about its formal status. Two things are already visible here. First, the largest amplitude belongs to $q = 2$, $m = 1$: $1/\sqrt{2} = 0.707$ before the factor $1/\pi$. Second, the amplitudes shrink only like $q^{-1/2}$, while by the PNT the primes up to x number about $x/\log x$; summing the $m = 1$ amplitudes with density $1/\log t$ gives roughly $\int_2^x \mathrm{d}t/(\sqrt{t}\log t)$, which grows without bound (heuristic; Problem 26). So the full sum cannot converge absolutely on the critical line, and truncating to a finite P is a genuine approximation, as anticipated in Chapter 3.

The prime budgets of Table V and Fig. 1. Table V's columns are the sets $P = \{2\}$; "3 primes" $= \{2, 3, 5\}$; "6 primes" $= \{2, 3, 5, 7, 11, 13\}$; and "25 primes" = all primes up to $p_{25} = 97$. We take these to be the first k primes. The paper does not list them, but this choice reproduces its numbers: this chapter's check script recomputes the 401–500 row with qmnt for all four budgets. With $m \le 3$ these contain 3, 9, 18 and 75 terms. On the principal holdout RE5 (zeros 401–500, null 0.263), they give rms 0.198, 0.116, 0.074 and 0.043 mean spacings, with correlations +0.66, +0.90, +0.96, +0.99 (Table V, first row). The $\{2\}$ column is RE2 used as a predictor: the single slowest prime-power family already beats the null. Fig. 1 shows the six-prime formula, held-out correlation +0.96. "Zero fitted parameters" (Sec. II D) means exactly this: the set P is chosen by size, and its members are fixed facts that the sieve of Section 15.4 produces without error.

"Primes $p \le 13$". Sec. I A b and Sec. II D describe Connes' construction [21], which extremizes a truncated Weil quadratic form over primes $p \le 13$. That is the same prime set as the six-prime oracle: $\{2, 3, 5, 7, 11, 13\}$, $\pi(13) = 6$. The paper places this construction "on the arithmetic side" because its input is this list of primes, not zeros.

Budgets in Table IV and Sec. II D. Control A uses "up to 300 primes, i.e. 900 terms": 300 primes times 3 values of m. The 300th prime is $p_{300} = 1987$, so the fastest wave in that oracle has angular frequency $3\log 1987 \approx 22.78$. Control B fits two amplitudes (a_j for the sine, b_j for the cosine) per frequency $m \log p$: 25 primes give 75 frequencies and 150 fitted parameters, and 150 primes (up to $p_{150} = 863$) give 450 frequencies and 900 fitted parameters, the "150" and "900" of Table IV's "fitted" column. Control C1's 450 uniform frequencies likewise carry 900 coefficients, and C2's $3K$ parameters count a frequency and two amplitudes per selected wave ($K = 24$ gives 72).

The prime powers found by C2 (Sec. IV). The sparse frequency search C2, given no arithmetic input, selected 43 frequencies from 800 zeros, each lying within half the Fourier resolution of some $\log q$: "those up to 113 except 32, 64, and 81, and six primes between 131 and 211". You can now check the count (Sample Problem below). Two further observations follow from this chapter:

- The list contains $16 = 2^4$, whose frequency $\log 16 = 4 \log 2$ is *absent* from Eq. (4), which stops at $m = 3$. C2's list omits $32 = 2^5$, $64 = 2^6$ and $81 = 3^4$, whose weights $1/(m\,p^{m/2})$ are 0.035, 0.021 and 0.028, smaller than those of the six primes between 131 and 211 (for example $1/\sqrt{211} = 0.069$), while 16 has weight $1/16 = 0.0625$. This weight ordering is consistent with part of what C2 found, but it does not explain the whole list: only six of the sixteen primes in $[131, 211]$ were selected, although all sixteen have weights $1/\sqrt{p} \geq 1/\sqrt{211} = 0.069$, larger than those of 32, 64 and 81. The paper does not offer this interpretation; treat it as our heuristic observation.

- App. A step 23 matches a frequency ω to a prime power $q \leq 3000$ when $|\omega - \log q| < \pi/T$. Here T is the length in t of the training window. For zeros 1–800 the window runs from t_1 to t_{800}, so $T \approx 1170$. Then $\pi/T \approx 0.0027$ ("half the Fourier resolution"), and the full matching window has width $2\pi/T \approx 0.0054$. Chapter 6 derives the resolution $2\pi/T$. (This T is a window length, not the height T used in other chapters.) There are 466 prime powers up to 3000 ($\pi(3000) = 430$ primes plus 36 higher powers). A plausible reason for the cutoff, ours rather than the paper's (the paper does not give one), is that C2 searches $0.05 < \omega < 8$ (Sec. II D) and $e^8 \approx 2981$. But the numbers $\log q$ crowd together: near q, consecutive prime powers are about $\log q$ apart (the PNT spacing), so consecutive values of $\log q$ are about $\log q/q$ apart, roughly 0.007 near $q = 1000$ (heuristic). Once that spacing is smaller than the matching window $2\pi/T$, almost any ω matches some q. The paper gives its own reason for checking the alignment on surrogates: "Because the frequencies are the output of a search" (Sec. IV). It reruns the search on 900 surrogate replicates, uses "a statistic that does not depend on a chosen frequency cutoff", and labels the check exploratory. The crowding of the $\log q$ is our own observation; it is a further reason why counting matches alone would be weak evidence. Chapters 27 and 30 return to this.

"Logarithms of prime powers" and Landau–Gonek. Sec. IV notes that "the Fourier transform of the zeros is concentrated at logarithms of prime powers" and cites [18, 19]. Those peaks sit at the frequencies $\log q$ of Figure 15.6, with strengths governed by $\Lambda(q)$; Chapter 18 states Landau's formula and Chapter 27 recovers the peaks numerically.

Table V's caption. The caption attributes the mild growth of the error at fixed prime budget partly to "the increasing number of arithmetically relevant primes at height". A heuristic reading in this chapter's terms: at larger t the mean spacing of the zeros, $2\pi/\log(t/2\pi)$ (Chapter 17), shrinks. A wave $\sin(t \log q)$ therefore completes only $\log q/\log(t/2\pi)$ cycles per spacing. More prime powers (roughly those with q below $t/2\pi$) then vary slowly on the scale of one spacing and shift individual zeros coherently. A fixed prime set therefore leaves out more of what matters. The paper states this qualitatively, and so do we.

Why logarithms, not primes. Finally, Sec. III's arithmetic-continuation probe uses the $p = 2$ part of Eq. (4) (RE2), whose period $2\pi/\log 2 \approx 9.065$ in t appears again in Sec. V. The reason the target is "almost periodic" rather than periodic once more primes are added is the independence of $\log 2, \log 3, \ldots$ proved in Section 15.2.

Sample Problem 15.10: Counting the 43 prime powers of Sec. IV

Verify that "prime powers up to 113 except 32, 64, and 81, and six primes between 131 and 211" are 43 in number, and list the term counts of Eq. (4) for the Table V budgets.
Step 1: primes up to 113. Sieving (or extending the list of primes below 100, which has $\pi(100) = 25$ members, by $101, 103, 107, 109, 113$) gives $\pi(113) = 30$.

Step 2: higher prime powers up to 113. Powers of 2: $4, 8, 16, 32, 64$ ($128 > 113$). Powers of 3: $9, 27, 81$. Powers of 5: 25 ($125 > 113$). Powers of 7: 49. Powers of 11: none ($121 > 113$). Total $5 + 3 + 1 + 1 = 10$.
Step 3: assemble. $30 + 10 = 40$ prime powers up to 113; remove $32, 64, 81$: 37; add six primes between 131 and 211: 43.
Step 4: term counts of Eq. (4). Three terms ($m = 1, 2, 3$) per prime: 3×1, 3×3, 3×6, 3×25.

$30 + 10 - 3 + 6 = 43$; Table V budgets: 3, 9, 18, 75 terms.

15.12 Summary

★ Did you know? *A 41-Million-Digit Prime*

In October 2024 the record for the largest known prime went to the Mersenne prime $2^{136279841} - 1$, with 41,024,320 decimal digits. Luke Durant discovered it in the Great Internet Mersenne Prime Search, using graphics processors rented in the cloud. Mersenne numbers $2^p - 1$ dominate the record list because the Lucas-Lehmer test checks them far faster than primality tests for general numbers.

- A prime is an integer $p \geq 2$ divisible only by 1 and p. Trial division up to $\sqrt{n}$ decides primality (Equation (15.1)).
- Fundamental theorem of arithmetic: $n = p_1^{a_1} \cdots p_k^{a_k}$ uniquely (Equation (15.2)); existence proved, uniqueness proved from Euclid's lemma (Bézout taken on trust). Consequently $\log n = \sum a_i \log p_i$ (Equation (15.4)) and the logs of distinct primes satisfy no integer relation; $\log 3/\log 2$ is irrational.
- Infinitely many primes (Euclid, Equation (15.5)). Twin primes: open.
- The sieve of Eratosthenes lists all primes up to x, crossing out from p^2 for $p \leq \sqrt{x}$.
- $\pi(x)$ counts primes up to x. PNT: $\pi(x) \sim x/\log x \sim \mathrm{Li}(x)$ (Equations (15.6) and (15.7)); proved 1896, proof beyond this book. $\mathrm{Li}(x) \sim x/\log x$ proved by parts (Equation (15.8)). Local density $1/\log x$, mean spacing $\log x$.
- Chebyshev (proved here): $\theta(x) = \sum_{p \leq x} \log p < (4 \log 2)x$ and $\pi(x) < \sqrt{x} + (8 \log 2)x/\log x$ (Equations (15.10) and (15.11)).
- Prime powers $q = p^m$; von Mangoldt $\Lambda(q) = \log p$, zero elsewhere (Equation (15.12)); $\sum_{d|n} \Lambda(d) = \log n$ (Equation (15.13)).
- $\psi(x) = \sum_{k \leq x} \Lambda(k) = \sum_m \theta(x^{1/m}) = \log \mathrm{lcm}(1, \ldots, \lfloor x \rfloor)$ (Equations (15.14) to (15.16)); $\psi(x) \leq \pi(x) \log x$ (Equation (15.17)); PNT $\Leftrightarrow \psi(x) \sim x$ (Equation (15.18)).
- The error terms of prime counting are governed by the zeta zeros; the Riemann hypothesis, equivalent to $\pi(x) = \mathrm{Li}(x) + \mathcal{O}(\sqrt{x} \log x)$, is open.
- $\log p$ is the natural size of a prime; p^{-it} rotates at angular frequency $\log p$, period $2\pi/\log p$ (Equation (15.19)). $\Lambda(p^m)/\log p^m = 1/m$ (Equation (15.20)), so the paper's Eq. (4) is a sum over prime powers q of $\Lambda(q) \sin(t \log q)/(\sqrt{q} \log q)$.

15.13 Problems

1. • Which of 91, 97, 119, 127, 133 are prime? For each composite one give the factorization, and for each prime say which trial divisors were needed.

2. • Find the canonical factorizations of 1001, 1024, 2310 and 9999.

3. • Explain, with the example $n = 6$, why the fundamental theorem of arithmetic would be false if 1 were counted as a prime.

4. • Give $\Lambda(1)$, $\Lambda(27)$, $\Lambda(28)$, $\Lambda(29)$, $\Lambda(32)$ and $\Lambda(121)$, exactly and to four decimal places.

5. • Which of the functions π, θ, ψ jump at $x = 16$, at $x = 17$ and at $x = 18$, and by how much?

6. •• Show that every prime $p > 3$ leaves remainder 1 or 5 on division by 6, i.e. is of the form $6k \pm 1$. (Consider the six possible remainders.)

7. •• For $k = 1, \dots, 6$ compute $N_k = p_1 p_2 \cdots p_k + 1$ and determine which N_k are prime. For the composite one, show that its prime factors are not among $p_1, \dots, p_k$.

8. •• Prove that there are infinitely many primes of the form $4k + 3$. (Given primes $p_1, \dots, p_r$ of this form, consider $N = 4p_1 \cdots p_r - 1$, and show that N must have a prime factor of the form $4k + 3$, using the fact that a product of numbers of the form $4k + 1$ is again of that form.)

9. •• Let p_n denote the n-th prime. Assuming the PNT, show that $p_n \sim n \log n$. (Set $x = p_n$ in Equation (15.6), take logarithms, and use $\log\log x = o(\log x)$.) Compare with $p_{1000} = 7919$, and also evaluate the refined estimate $n(\log n + \log\log n - 1)$ at $n = 1000$.

10. •• List the divisors of 360 that are prime powers and verify Equation (15.13) for $n = 360$.

11. •• Prove Equation (15.16) for $N = 12$ directly: compute $\operatorname{lcm}(1, \dots, 12)$ and $\psi(12)$ separately and compare. Then explain in one sentence why $\psi(N)$ jumps exactly when N is a prime power.

12. •• List the twin prime pairs $(p, p + 2)$ with $p + 2 \le 100$. What is the status of the twin prime conjecture?

13. •• (Paper.) Using Eq. (4) of the paper and Sec. II D, compute: the number of sinusoidal terms in the oracle with 6, 25 and 300 primes; the number of fitted real parameters of control B with 25 and with 150 primes; and explain the "72" of control C2 in Table IV, given that C2 selected 24 frequencies.

14. • (Paper.) Which primes make up the "3 primes", "6 primes" and "25 primes" budgets of the paper's Table V? Which of these coincides with the prime set of Connes' construction mentioned in Sec. I A b?

15. •• (Paper.) Reproduce the count "43 distinct prime powers" of Sec. IV from the description "those up to 113 except 32, 64, and 81, and six primes between 131 and 211". How many primes are there between 131 and 211 inclusive?

16. •• (Paper.) Using Equation (15.20), rewrite the paper's Eq. (4) as a single sum over prime powers q, and show that for $P = \{2\}$ the three amplitudes (before the factor $-1/\pi$) are $1/\sqrt{2}$, $1/4$ and $1/(6\sqrt{2})$.

17. •• (Paper.) Compute the periods $2\pi/\log q$ for $q = 11, 13, 16, 97$. What is the longest period among the terms of the 25-prime oracle, and the shortest?

18. •• Using qmnt or numpy, compute $\pi(10^7)$, $10^7/\log 10^7$ and $\mathrm{Li}(10^7)$, and extend Table 15.1 by one row. Does the column $x/\pi(x)$ still increase by about $\log 10$?

19. •• Using numpy, compute $\psi(x)/x$ at $x = 100, 1000, 10^4$, and check the inequality $\psi(x) \le \pi(x)\log x$ (Equation (15.17)) at those three points by evaluating both sides.

20. •• Using qmnt or numpy, compute $\max_{2\le x\le 10^5} |\psi(x)-x|/\sqrt{x}$ over integers x, and plot $(\psi(x)-x)/\sqrt{x}$ against $\log x$. Does the scale of the fluctuations change with x?

21. ••• (A proof of uniqueness without Euclid's lemma.) Suppose some integer has two different prime factorizations and let n be the smallest such integer, $n = p_1\cdots p_r = q_1\cdots q_s$ with $p_1 \le \cdots \le p_r$ and $q_1 \le \cdots \le q_s$. (a) Show that no p_i equals any q_j. (b) Assume $p_1 < q_1$ and consider $m = n - p_1 q_2 \cdots q_s$. Show that $0 < m < n$, that $m = (q_1 - p_1)q_2\cdots q_s$ and that $p_1 \mid m$. (c) Derive a contradiction.

22. ••• Use Equation (15.15) to prove $0 \le \psi(x) - \theta(x) \le \dfrac{\log x}{\log 2}\cdot\dfrac{\sqrt{x}\log x}{2}$ for $x \ge 2$, so that $\psi(x)-\theta(x) = \mathcal{O}(\sqrt{x}\log^2 x)$.

23. ••• (Chebyshev's lower bound.) (a) Show that $\binom{2n}{n} \ge 4^n/(2n+1)$. (b) You may use *Legendre's formula*: the exponent of a prime p in $\binom{2n}{n}$ is $\sum_{j\ge 1}(\lfloor 2n/p^j\rfloor - 2\lfloor n/p^j\rfloor)$, where each term is 0 or 1. Deduce that every prime power p^a in the canonical factorization of $\binom{2n}{n}$ satisfies $p^a \le 2n$. (c) Conclude that $\binom{2n}{n} \le (2n)^{\pi(2n)}$ and hence $\pi(2n) \ge (n\log 4 - \log(2n+1))/\log(2n)$. (d) Evaluate the bound at $n = 50$ and compare with $\pi(100)$.

24. •• Prove that $\log 5/\log 3$ is irrational, and more generally that $\log a/\log b$ is irrational whenever $a, b \ge 2$ are integers with no common prime factor.

25. •• Integrate Equation (15.8) by parts once more to show $\mathrm{Li}(x) = \dfrac{x}{\log x} + \dfrac{x}{\log^2 x} + 2\displaystyle\int_2^x \frac{\mathrm{d}t}{\log^3 t} + C$ for a constant C, and guess the next term. Evaluate $\dfrac{x}{\log x}$, $\dfrac{x}{\log x} + \dfrac{x}{\log^2 x}$ and $\dfrac{x}{\log x} + \dfrac{x}{\log^2 x} + \dfrac{2x}{\log^3 x}$ at $x = 10^6$ and compare with $\pi(10^6)$ and $\mathrm{Li}(10^6)$ from Table 15.1.

26. •• Using qmnt or numpy, compute $\sum_{p\le x} p^{-1/2}$ for $x = 13$, 97, 1987 and 10^6, and compare with the heuristic $2\sqrt{x}/\log x$ obtained from $\int_2^x \mathrm{d}t/(\sqrt{t}\log t)$. What does this say about truncating the paper's Eq. (4)?

27. •• Using qmnt (`qmnt.Nfl(t, [2], mmax=3)`), evaluate the prime-2 wave RE2 at the first ten zeros $t_1, \dots, t_{10}$ of RE1 and at $t_n + 2\pi/\log 2$. What do you find, and why?

28. ••• Using qmnt (`qmnt.oracle_zeros` and `qmnt.score`), reproduce the row "401–500" of the paper's Table V for the budgets $\{2\}$, 3 primes and 6 primes, and compute the same quantities for 12 primes, which the paper does not report. Plot rms against the number of primes on a log axis.

29. •• In the even numbers $2, 4, 6, \dots$ (Section 15.2), verify that 2, 6, 10 and 30 are E-primes and that $60 = 2\cdot 30 = 6\cdot 10$. Which step of the uniqueness proof fails in this system?

30. •• For $n = 15$, find the product of the primes in $(15, 30]$ and verify that it divides $\binom{30}{15}$. What is the quotient?

31. •• Estimate the local mean spacing of primes near 10^6 from the PNT and compare it with the average spacing $10^6/\pi(10^6)$. Why is the second number smaller?

32. ••• (Paper, open-ended.) The sparse search C2 of the paper found the frequency $\log 16$ but not $\log 32$, $\log 64$ or $\log 81$ (Sec. IV). Compute the weights $\Lambda(q)/(\sqrt{q}\log q)$ for $q = 16, 32, 64, 81$ and for the primes 131 and 211, and discuss whether ordering the prime powers by weight explains the list. Why must any such explanation be labeled heuristic? (Hint: consider the frequency resolution π/T and how close $\log q$ values are near $q = 200$.)

15.14 Answers to selected problems

1. $91 = 7 \cdot 13$; 97 is prime ($\sqrt{97} < 10$; not divisible by $2, 3, 5, 7$); $119 = 7 \cdot 17$; 127 is prime ($\sqrt{127} < 11.3$; not divisible by $2, 3, 5, 7, 11$); $133 = 7 \cdot 19$.

2. $1001 = 7 \cdot 11 \cdot 13$; $1024 = 2^{10}$; $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$; $9999 = 3^2 \cdot 11 \cdot 101$.

3. If 1 were prime, then $6 = 2 \cdot 3 = 1 \cdot 2 \cdot 3 = 1 \cdot 1 \cdot 2 \cdot 3 = \dots$ would have infinitely many prime factorizations differing in more than order, so uniqueness would fail. Excluding 1 keeps the theorem true.

4. $\Lambda(1) = 0$; $\Lambda(27) = \log 3 = 1.0986$ ($27 = 3^3$); $\Lambda(28) = 0$ ($28 = 2^2 \cdot 7$); $\Lambda(29) = \log 29 = 3.3673$; $\Lambda(32) = \log 2 = 0.6931$; $\Lambda(121) = \log 11 = 2.3979$.

5. At $16 = 2^4$: only ψ jumps, by $\log 2$. At 17 (prime): π jumps by 1, θ and ψ by $\log 17 \approx 2.833$. At 18: none, since 18 is not a prime power.

7. $N_k = 3, 7, 31, 211, 2311, 30031$. The first five are prime (trial division up to $\sqrt{N_k}$). $30031 = 59 \cdot 509$; both factors exceed 13, so neither is among $2, 3, 5, 7, 11, 13$, as Euclid's argument requires.

9. From $\pi(x) \sim x/\log x$ with $x = p_n$: $n \sim p_n/\log p_n$. Taking logs, $\log n = \log p_n - \log\log p_n + o(1)$, and since $\log\log p_n = o(\log p_n)$, $\log n \sim \log p_n$. Hence $p_n \sim n \log p_n \sim n \log n$. At $n = 1000$: $1000 \log 1000 = 6907.76$, ratio $7919/6907.76 = 1.146$; the refined estimate gives $1000(6.9078 + 1.9326 - 1) = 7840.4$, within 1%.

11. $\operatorname{lcm}(1, \dots, 12) = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 = 27720$. The prime powers up to 12 are $2, 3, 4, 5, 7, 8, 9, 11$, so $\psi(12) = 3\log 2 + 2\log 3 + \log 5 + \log 7 + \log 11 = \log 27720 \approx 10.23$. The lcm gains a factor (namely p) exactly when N reaches a new power of p, i.e. when N is a prime power.

13. Three terms per prime: 18, 75 and 900 terms (the last is control A's "900 terms"). Control B: two amplitudes per frequency, $3 \cdot 25 \cdot 2 = 150$ and $3 \cdot 150 \cdot 2 = 900$ fitted parameters, as in Table IV. Control C2 fits a frequency and two amplitudes per selected wave: $3 \cdot 24 = 72$.

14. 3 primes: $\{2, 3, 5\}$; 6 primes: $\{2, 3, 5, 7, 11, 13\}$; 25 primes: all primes up to 97. The six-prime set is exactly Connes' set $p \le 13$.

15. $\pi(113) = 30$ primes plus 10 higher prime powers $(4, 8, 16, 32, 64, 9, 27, 81, 25, 49)$ give 40; removing $32, 64, 81$ leaves 37; adding six primes gives 43. There are 16 primes in $[131, 211]$, so C2 found six of these sixteen.

17. $2\pi/\log 11 = 2.620$, $2\pi/\log 13 = 2.450$, $2\pi/\log 16 = 2.266$, $2\pi/\log 97 = 1.373$. In the 25-prime oracle the longest period is the $q = 2$ term, 9.065; the shortest is the $q = 97^3$ term, $2\pi/(3\log 97) = 0.458$.

19. $\psi(x)/x = 0.9405,\ 0.9967,\ 1.0013$. The inequality: $94.05 \le 25\log 100 = 115.13$; $996.68 \le 168\log 1000 = 1160.50$; $10013.4 \le 1229\log 10^4 = 11319.51$. All hold.

21. (a) If some $p_i = q_j$, cancel it to get a smaller integer with two different factorizations, contradicting minimality. (b) $p_1 < q_1$ gives $p_1 q_2 \cdots q_s < q_1 q_2 \cdots q_s = n$, so $m > 0$, and $m < n$ because $p_1 q_2 \cdots q_s > 0$. Factoring, $m = (q_1 - p_1) q_2 \cdots q_s$; also $m = p_1(p_2 \cdots p_r - q_2 \cdots q_s)$, so $p_1 \mid m$. (c) We have $m < n$, and $m \ge 2$ because $p_1 \mid m$ and $m > 0$, so m has a unique factorization. From the second form, p_1 appears in it. Consider the first form. If $q_1 - p_1 = 1$, then $m = q_2 \cdots q_s$ is a factorization of m that does not contain p_1 (no q_j equals p_1, by (a)). That is already a contradiction. Otherwise $q_1 - p_1 \ge 2$. Factor it into primes; then m is the product of those primes and $q_2, \dots, q_s$. By uniqueness p_1 is in this list, and it is not a q_j, so it is one of the prime factors of $q_1 - p_1$, i.e. $p_1 \mid q_1 - p_1$. Since also $p_1 \mid p_1$, this gives $p_1 \mid q_1$,

impossible for distinct primes. (If $s = 1$, $n = q_1$ is prime and has only one factorization.) So no counterexample exists.

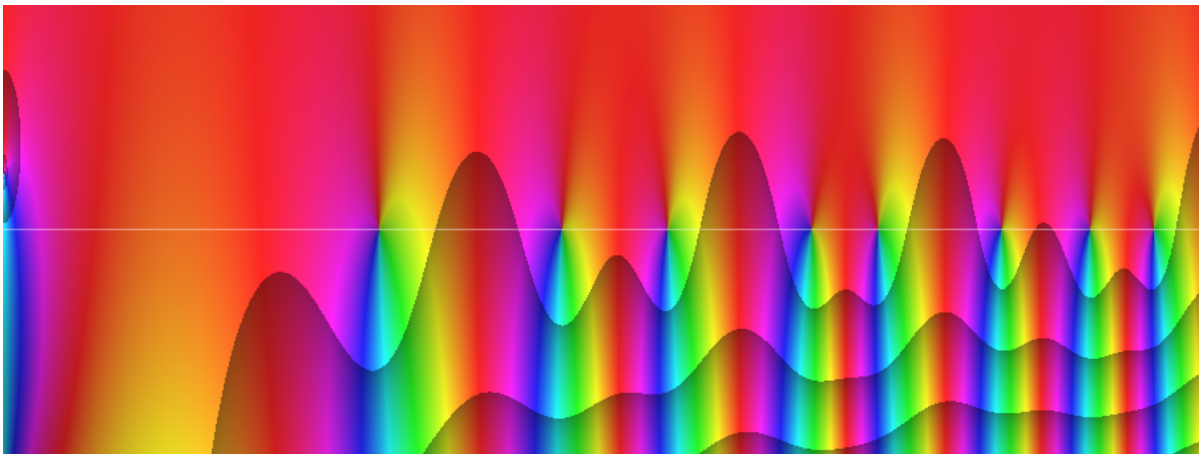
23. (a) $4^n = \sum_{k=0}^{2n} \binom{2n}{k}$ has $2n+1$ terms, of which $\binom{2n}{n}$ is the largest. To see this, note that $\binom{2n}{k+1}/\binom{2n}{k} = (2n-k)/(k+1)$ is ≥ 1 exactly when $k < n$, so the terms rise up to $k = n$ and fall afterwards. Hence $4^n \leq (2n+1)\binom{2n}{n}$. (b) Each term of Legendre's sum vanishes once $p^j > 2n$, so the exponent a is at most the number of j with $p^j \leq 2n$, i.e. $p^a \leq 2n$. (c) $\binom{2n}{n}$ is a product over the primes $p \leq 2n$ of factors $p^a \leq 2n$, so it is at most $(2n)^{\pi(2n)}$; combining with (a) and taking logs, $n \log 4 - \log(2n+1) \leq \pi(2n) \log(2n)$. (d) $n = 50$: $(69.315 - 4.615)/4.605 = 14.05 \leq \pi(100) = 25$.

25. With $u = 1/\log^2 t$, $\mathrm{d}v = \mathrm{d}t$: $\int_2^x \mathrm{d}t/\log^2 t = x/\log^2 x - 2/\log^2 2 + 2\int_2^x \mathrm{d}t/\log^3 t$, which gives the stated form with $C = -2/\log 2 - 2/\log^2 2$. Repeating, the next term is $6x/\log^4 x$ (the pattern is $(k-1)!\,x/\log^k x$). At $x = 10^6$: 72382.4, then 77621.7, then 78380.1, against $\pi = 78498$ and $\mathrm{Li} = 78626.5$. Each term helps here, but the series is asymptotic, not convergent.

27. The values at t_n and $t_n + 2\pi/\log 2$ agree to rounding error (about 10^{-15}), because each term $\sin(mt \log 2)$ changes its argument by $2\pi m$ under the shift (Sample Problem in Section 15.10).

29. An even number is an E-prime when it is not a product of two even numbers, i.e. when it is not divisible by 4; $2, 6, 10, 30$ qualify. $2 \cdot 30 = 60 = 6 \cdot 10$. The step that fails is Euclid's lemma: 6 E-divides $60 = 2 \cdot 30$ but E-divides neither 2 nor 30 (the quotients $1/3$ and 5 are not even), so "p_1 divides some q_j" breaks down.

31. Local spacing $\approx \log 10^6 = 13.8$; average spacing over $[2, 10^6]$ is $10^6/78498 = 12.74$. The average includes the small primes, which are packed more closely (spacing $\log t$ with $t < 10^6$), so it is smaller than the local value at the top of the range.



Domain colouring of $\zeta(s)$ for $-1.5 \le \sigma \le 2.5$, $0 \le t \le 45$ (turned on its side): every point where all colours meet is a zero.

Chapter 16

The Riemann Zeta Function

↑ Contents

What this chapter is about

In Chapter 1 we met a list of numbers,

$$t_1 = 14.134725, \quad t_2 = 21.022040, \quad t_3 = 25.010858, \quad \dots,$$

and were told that they are the "heights" of the zeros of a function built from the primes. This chapter builds that function. It is the Riemann zeta function $\zeta(s)$, defined by a sum over all positive integers,

$$\zeta(s) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \cdots .$$

Euler discovered that this sum over *integers* equals a product over *primes*. That single identity is the bridge between the primes of Chapter 15 and everything else in this book, including the prime "frequencies" $\log p$ that the paper's arithmetic oracle uses.

There is a twist. The zeros we care about sit at points $s = \frac{1}{2} + it$ where the defining sum does *not* converge. To give a meaning to the statement "$\zeta(\frac{1}{2} + 14.134725\,i) = 0$" we need *analytic continuation*: a way of extending a function beyond the region where its first formula works. We will carry this out completely for $\operatorname{Re} s > 0$, using an alternating version of the series, and then take Riemann's functional equation on trust to reach the rest of the complex plane. With the function in hand we will find where its zeros can and cannot lie, state the Riemann hypothesis precisely (it is an open problem), and compute the zeros on a computer.

Where this shows up in the paper: the first sentence of Sec. I of the paper [1] refers to "the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of the Riemann zeta function"; Sec. II A and Appendix A step (1) describe computing $t_1, \dots, t_{10^4}$ with `mpmath`. Every table and figure of the paper is built on these numbers.

What you need

- Chapter 2: complex numbers $s = \sigma + it$, modulus, conjugate, Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$, the definition $a^s = e^{s\log a}$ for real $a > 0$, and $|p^{-s}| = p^{-\sigma}$.
- Chapter 3: convergence of sequences and series, the geometric series (real and complex), divergence of the harmonic series, the integral test and the fact that $\sum k^{-s}$ converges exactly when $s > 1$, absolute convergence implies convergence, the alternating series test with its error bound, conditional convergence.
- Nothing from Chapter 4 is required. The Gamma function and bisection are recalled where they are used; that chapter treats both more fully.
- Chapter 15: primes, Euclid's theorem, the fundamental theorem of arithmetic (every integer $k \geq 2$ is a product of primes in exactly one way, up to order).
- First-year physics and calculus: derivatives, the chain and product rules, improper integrals, small-angle approximations, rotating arrows (phasors).

16.1 A sum over all the positive integers

Imagine a long row of springs, the k-th of which contributes an amount $1/k^s$ to some total. If s is large the contributions die off quickly and the total is finite; if s is small they die off so slowly that the total runs away. The zeta function is this total, viewed as a function of the decay exponent s.

Definition. For a real number $s > 1$, the *Riemann zeta function* is

$$\zeta(s) \;=\; \sum_{k=1}^{\infty} \frac{1}{k^s} \;=\; 1 + \frac{1}{2^s} + \frac{1}{3^s} + \cdots . \tag{16.1}$$

Here k runs over the positive integers and s is the variable. Recall from Chapter 3 (integral test) that this series converges exactly when $s > 1$; at $s = 1$ it is the harmonic series, which diverges.

How big is $\zeta(s)$? We can trap $\zeta(s)$ between two simple numbers. Let $f(x) = x^{-s}$ with $s > 1$. Its derivative is $f'(x) = -s\,x^{-s-1} < 0$, so f is positive and decreasing for $x \geq 1$. Take an integer $k \geq 2$. For every x in the interval $[k-1, k]$ we have

$$f(k) \;\leq\; f(x) \;\leq\; f(k-1) \qquad \text{(because } f \text{ decreases)}.$$

Integrate over this interval, which has length 1:

$$f(k) \;\leq\; \int_{k-1}^{k} f(x)\,\mathrm{d}x \;\leq\; f(k-1).$$

Now add these inequalities for $k = 2, 3, \ldots, K$. The integrals join into one integral from 1 to K:

$$\sum_{k=2}^{K} f(k) \;\leq\; \int_{1}^{K} f(x)\,\mathrm{d}x \;\leq\; \sum_{k=1}^{K-1} f(k).$$

The integral is

$$\int_1^K x^{-s}\,\mathrm{d}x = \left[\frac{x^{1-s}}{1-s}\right]_1^K = \frac{K^{1-s}-1}{1-s} = \frac{1-K^{1-s}}{s-1}.$$

Because $1 - s < 0$, $K^{1-s} \to 0$ as $K \to \infty$, so the integral tends to $1/(s-1)$. Letting $K \to \infty$ in both inequalities (the left sum tends to $\zeta(s) - 1$, the right sum to $\zeta(s)$) gives

$$\zeta(s) - 1 \;\leq\; \frac{1}{s-1} \;\leq\; \zeta(s),$$

which we rearrange as

$$\frac{1}{s-1} \;\le\; \zeta(s) \;\le\; 1 + \frac{1}{s-1} \qquad (s>1). \tag{16.2}$$

Two consequences are immediate. First, as $s \to 1^+$, $\zeta(s)$ grows without bound, like $1/(s-1)$. Second, the first term of Equation (16.1) is 1 and the rest are positive, so $1 \le \zeta(s) \le 1 + 1/(s-1)$. The right side tends to 1 as $s \to \infty$, so $\zeta(s) \to 1$.

The same argument applied to the *tail* $\sum_{k>N} k^{-s}$ (sum the inequalities $\int_k^{k+1} f \le f(k) \le \int_{k-1}^{k} f$ over $k \ge N+1$) gives

$$\frac{(N+1)^{1-s}}{s-1} \;\le\; \sum_{k=N+1}^{\infty} \frac{1}{k^s} \;\le\; \frac{N^{1-s}}{s-1}. \tag{16.3}$$

This tells you how many terms you need for a given accuracy.

Table 16.1: Values of $\zeta(s)$ for real s together with the bounds of Equation (16.2). The values for $s = 2, 4$ are exact formulas (see the box below); all others are computed numerically with `mpmath.zeta`.

s	lower bound $1/(s-1)$	$\zeta(s)$	upper bound $1 + 1/(s-1)$
1.5	2.000000	2.612375	3.000000
2	1.000000	$\pi^2/6 = 1.644934$	2.000000
3	0.500000	1.202057	1.500000
4	0.333333	$\pi^4/90 = 1.082323$	1.333333

Taken on trust

Exact values at even integers. Euler showed in 1735 that $\zeta(2) = \pi^2/6$ (the "Basel problem"), and more generally that $\zeta(2n)$ is a rational multiple of π^{2n}, for example $\zeta(4) = \pi^4/90$. We use these values but do not prove them here. If you have read Chapter 5, here is the route for $\zeta(2)$: the function $f(x) = x$ on $(-\pi, \pi)$ has the Fourier series $\sum_{k\ge1} b_k \sin kx$ with $b_k = 2(-1)^{k+1}/k$, and Parseval's theorem $\frac{1}{\pi}\int_{-\pi}^{\pi} f^2\,dx = \sum b_k^2$ reads $\frac{2\pi^2}{3} = \sum_k \frac{4}{k^2}$, which is $\zeta(2) = \pi^2/6$. For odd integers no such formula is known; Apéry proved in 1978 that $\zeta(3) = 1.202057\ldots$ is irrational. Proofs of the even-integer formulas are in [23]; Apéry's theorem is beyond this book and is not in the cited texts.

Sample Problem 16.1: Partial sums of $\zeta(2)$

Add the first ten terms of $\zeta(2)$, then use Equation (16.3) to trap $\zeta(2)$ in an interval. Is $\pi^2/6$ inside?

Step 1: The partial sum is

$$S_{10} = 1 + \tfrac{1}{4} + \tfrac{1}{9} + \tfrac{1}{16} + \tfrac{1}{25} + \tfrac{1}{36} + \tfrac{1}{49} + \tfrac{1}{64} + \tfrac{1}{81} + \tfrac{1}{100} = 1.549768.$$

Step 2: With $s = 2$, $N = 10$, Equation (16.3) gives $\frac{1}{11} \le \sum_{k\ge11} k^{-2} \le \frac{1}{10}$.

Step 3: Adding the tail bounds to S_{10}:

$$1.549768 + 0.090909 \le \zeta(2) \le 1.549768 + 0.1.$$

Step 4: So $\boxed{1.640677 \le \zeta(2) \le 1.649768}$, and $\pi^2/6 = 1.644934$ lies inside.

Notice how slowly the raw partial sums converge: the tail after N terms is at least $1/(N+1)$, so to get within 10^{-6} by brute force you need at least 10^6 terms. Bounding the tail with an integral is far more efficient than adding more terms.

Checkpoint 16.1

Without a computer, decide whether $\zeta(1.001)$ is larger or smaller than 1000.

Answer: Larger (or equal at worst): Equation (16.2) with $s-1=0.001$ gives $1000 \le \zeta(1.001) \le 1001$.

Key Ideas

- $\zeta(s)=\sum_{k\ge 1} k^{-s}$ converges for real $s>1$ and diverges at $s=1$.
- Comparing the sum with $\int x^{-s}\,dx$ traps it: $1/(s-1) \le \zeta(s) \le 1+1/(s-1)$.
- $\zeta(s)\to\infty$ as $s\to 1^+$ and $\zeta(s)\to 1$ as $s\to\infty$; $\zeta(2)=\pi^2/6$.

16.2 Complex s: a sum of spinning arrows

The zeros of the paper live at complex values of s, so we must let s be complex. Write

$$s=\sigma+it,$$

where $\sigma=\operatorname{Re} s$ (real part) and $t=\operatorname{Im} s$ (imaginary part) are real numbers. Recall from Chapter 2 that for a real number $a>0$ we define $a^s=e^{s\log a}$, where log is the natural logarithm. Apply this to k^{-s}:

$$\begin{aligned} k^{-s} &= e^{-s\log k} \\ &= e^{-(\sigma+it)\log k} \\ &= e^{-\sigma\log k}\,e^{-it\log k} && \text{(since } e^{a+b}=e^a e^b\text{)} \\ &= k^{-\sigma}\left[\cos(t\log k)-i\sin(t\log k)\right] && \text{(Euler's formula).} \end{aligned} \tag{16.4}$$

The picture. Each term k^{-s} is an arrow in the complex plane of length

$$|k^{-s}|=k^{-\sigma}$$

pointing at angle $-t\log k$. Hold σ fixed and increase t: every arrow rotates clockwise, the k-th at angular frequency $\log k$ (radians per unit of t). The $k=1$ arrow never moves (since $\log 1=0$); the $k=2$ arrow makes one full turn every $2\pi/\log 2=9.065$ units of t; the $k=3$ arrow every $2\pi/\log 3$; and so on. The value $\zeta(s)$ is the tip-to-tail sum of infinitely many shrinking arrows spinning at different rates $\log k$. (These rates are not all incommensurate, since $\log 4=2\log 2$ and $\log 6=\log 2+\log 3$. It is the rates $\log p$ of distinct primes that are mutually incommensurate, a fact used in Chapter 5 and proved for $p=2,3$ in Problem 26.) Where the sum converges ($\sigma>1$) the arrows never cancel completely; we prove this in Section 16.3. The zeros we want lie where this sum diverges, and making sense of them is the job of analytic continuation (Section 16.4). The period 9.065 will come back: it is the period of the "prime-2 wave" (running example RE2) that the paper uses in its continuation probe.

Where the series converges. The sum of the arrow lengths is

$$\sum_{k=1}^{\infty}|k^{-s}|=\sum_{k=1}^{\infty}k^{-\sigma}=\zeta(\sigma),$$

which is finite exactly when $\sigma>1$. By Chapter 3, an absolutely convergent complex series converges. So Equation (16.1) defines $\zeta(s)$ for every complex s with $\operatorname{Re} s>1$, a *half-plane*. The triangle inequality, applied to partial sums and then passed to the limit, gives the bound

$$|\zeta(s)| \;\le\; \sum_{k=1}^{\infty}|k^{-s}| \;=\; \zeta(\sigma) \qquad (\sigma>1). \tag{16.5}$$

Multiplying powers. For positive integers k and m,

$$(km)^{-s} = e^{-s\log(km)} = e^{-s(\log k+\log m)} = e^{-s\log k}\,e^{-s\log m} = k^{-s}m^{-s}.$$

We will use this repeatedly.

Pitfall

The rule $(ab)^s = a^s b^s$ is safe when a and b are *positive real* numbers, because then $\log(ab) = \log a + \log b$ exactly. For complex a, b the logarithm has branches (Chapter 2) and the rule can fail. In this chapter the bases are always positive integers, so we are safe.

Mirror symmetry. Recall that the conjugate of $e^{x+iy} = e^x(\cos y + i\sin y)$ is $e^x(\cos y - i\sin y) = e^{x-iy}$, that is, $\overline{e^w} = e^{\bar w}$. Since $\log k$ is real,

$$\overline{k^{-s}} = \overline{e^{-s\log k}} = e^{-\bar s\log k} = k^{-\bar s}.$$

The conjugate of a partial sum is the sum of the conjugates, and conjugation does not change distances ($|\bar a - \bar b| = |a-b|$), so conjugates of convergent sequences converge to the conjugate limit. Therefore

$$\zeta(\bar s) = \overline{\zeta(s)} \qquad (\sigma > 1). \tag{16.6}$$

Reflecting s in the real axis reflects $\zeta(s)$ in the real axis. In particular $\zeta(s)$ is real when s is real, as it should be.

Analytic functions. A complex function f is *analytic* on a region (an open, connected set of the complex plane) if at every point s of the region the limit

$$f'(s) = \lim_{h\to 0}\frac{f(s+h)-f(s)}{h}$$

exists, with the *complex* number h allowed to approach 0 from any direction. This is a strong demand. For $f(z) = z^2$, $\frac{(z+h)^2-z^2}{h} = 2z + h \to 2z$, fine. For $f(z) = \bar z$, $\frac{\bar h}{h}$ equals 1 when h is real and -1 when h is imaginary, so no limit exists and $\bar z$ is not analytic, although it is perfectly smooth as a map of the plane.

Taken on trust

ζ is analytic for $\operatorname{Re} s > 1$. Each term $k^{-s} = e^{-s\log k}$ is analytic with derivative $-\log k\, k^{-s}$. Because the series converges uniformly (the error after N terms is below a bound that depends on N but not on s) on every half-plane $\operatorname{Re} s \ge \sigma_0 > 1$ (its tail is bounded by the tail of $\zeta(\sigma_0)$, independently of s), a theorem of Weierstrass says the sum is analytic and may be differentiated term by term: $\zeta'(s) = -\sum_k \log k\, k^{-s}$. Proof: [6], Chapter 1, or the complex-variable chapters of [22].

Sample Problem 16.2: Two arrows at $s = 2 + 3i$

Find 2^{-s} and 3^{-s} for $s = 2+3i$ in the form $a + ib$, and bound $|\zeta(2+3i)|$ from above.
Step 1: By Equation (16.4), $|2^{-s}| = 2^{-2} = 0.25$ and $|3^{-s}| = 3^{-2} = 1/9 = 0.1111$.
Step 2: The angles are $-3\log 2 = -2.0794$ rad and $-3\log 3 = -3.2958$ rad.
Step 3: $2^{-s} = 0.25[\cos(2.0794) - i\sin(2.0794)] = 0.25[-0.4870 - 0.8734\,i]$, so $2^{-s} = -0.1217 - 0.2184\,i$.
Step 4: $3^{-s} = \frac{1}{9}[\cos(3.2958) - i\sin(3.2958)] = \frac{1}{9}[-0.9881 + 0.1536\,i]$, so $3^{-s} = -0.1098 + 0.0171\,i$.

Step 5: By Equation (16.5), $|\zeta(2+3i)| \le \zeta(2)$:

$$\boxed{2^{-s} = -0.1217 - 0.2184\,i, \quad 3^{-s} = -0.1098 + 0.0171\,i, \quad |\zeta(2+3i)| \le 1.644934.}$$

Checkpoint 16.2

Does the series Equation (16.1) converge at $s = 3 + 100i$? Give an upper bound for $|\zeta(3+100i)|$.

Answer: Yes, because $\operatorname{Re} s = 3 > 1$ (the size of t is irrelevant). The bound is $|\zeta(3+100i)| \le \zeta(3) = 1.202057$. A lower bound follows in Section 16.3.

Key Ideas

- $k^{-s} = k^{-\sigma}e^{-it\log k}$: an arrow of length $k^{-\sigma}$ rotating at angular frequency $\log k$ in t.
- The series converges absolutely in the half-plane $\operatorname{Re} s > 1$, where $|\zeta(s)| \le \zeta(\sigma)$.
- $\zeta(\bar{s}) = \overline{\zeta(s)}$; ζ is analytic for $\operatorname{Re} s > 1$.

16.3 The Euler product: integers from primes

Every positive integer is built from primes in exactly one way (Chapter 15). Euler saw in 1737 that this fact can be written as an identity between a sum and a product. Here is the idea with only the primes 2 and 3 and $s = 2$:

$$\left(1 + \tfrac{1}{2^2} + \tfrac{1}{4^2} + \tfrac{1}{8^2} + \cdots\right)\left(1 + \tfrac{1}{3^2} + \tfrac{1}{9^2} + \cdots\right) = \sum_{a,b\ge 0} \frac{1}{(2^a 3^b)^2}.$$

Multiplying out produces exactly one term $1/k^2$ for every integer k of the form $2^a 3^b$ (1, 2, 3, 4, 6, 8, 9, 12, ...), each exactly once because factorizations are unique. Including *all* primes should produce every integer exactly once, that is, $\zeta(2)$.

Statement. For $\operatorname{Re} s > 1$,

$$\zeta(s) = \prod_{p} \frac{1}{1-p^{-s}} \;=\; \lim_{P\to\infty} E_P(s), \qquad E_P(s) = \prod_{p\le P} \frac{1}{1-p^{-s}}, \tag{16.7}$$

where the product runs over primes p and E_P is the finite product over primes not exceeding P.

Proof. Fix s with $\sigma = \operatorname{Re} s > 1$. We use two sets of integers: $\mathcal{I}_P$, the positive integers all of whose prime factors are at most P (this includes $k = 1$, which has no prime factors), and $\mathcal{I}_{P,J}$, the members of $\mathcal{I}_P$ in whose factorization every exponent is at most J.

(a) Each factor is a geometric series. Since $|p^{-s}| = p^{-\sigma} \le 2^{-\sigma} < \tfrac{1}{2} < 1$, the complex geometric series (Chapter 3) gives

$$\frac{1}{1-p^{-s}} = \sum_{j=0}^{\infty} p^{-js} = \lim_{J\to\infty} \sum_{j=0}^{J} p^{-js}.$$

(b) A finite version of the identity. Consider the finite product of finite sums

$$\prod_{p\le P} \sum_{j=0}^{J} p^{-js}.$$

Expanding it (finitely many terms, so ordinary algebra applies) gives one term for each choice of an exponent $j_p \in \{0, \dots, J\}$ for every prime $p \le P$. That term is $\prod_{p\le P} p^{-j_p s} = \left(\prod_{p\le P} p^{j_p}\right)^{-s}$, using the multiplication rule of Section 16.2. The integer $k = \prod p^{j_p}$ belongs to $\mathcal{I}_{P,J}$; conversely each $k \in \mathcal{I}_{P,J}$ arises from its own factorization, and by uniqueness of factorization from no other choice. Hence

$$\prod_{p\le P} \sum_{j=0}^{J} p^{-js} = \sum_{k\in\mathcal{I}_{P,J}} k^{-s}.$$

For example, with $P = 3$, $J = 2$:

$$(1 + 2^{-s} + 4^{-s})(1 + 3^{-s} + 9^{-s}) = 1 + 2^{-s} + 3^{-s} + 4^{-s} + 6^{-s} + 9^{-s} + 12^{-s} + 18^{-s} + 36^{-s}.$$

(c) Let $J \to \infty$. On the left, each of the finitely many factors tends to $(1 - p^{-s})^{-1}$ by (a), so the left side tends to $E_P(s)$. On the right, the series $\sum_{k\in\mathcal{I}_P} k^{-s}$ converges absolutely (its terms are a subset of those of ζ). An element of $\mathcal{I}_P$ missing from $\mathcal{I}_{P,J}$ has some exponent larger than J, hence is at least 2^{J+1}. So

$$\Big|\sum_{k\in\mathcal{I}_P} k^{-s} - \sum_{k\in\mathcal{I}_{P,J}} k^{-s}\Big| \le \sum_{k\ge 2^{J+1}} k^{-\sigma} \xrightarrow[J\to\infty]{} 0,$$

because this is a tail of the convergent series $\zeta(\sigma)$. Therefore $E_P(s) = \sum_{k\in\mathcal{I}_P} k^{-s}$.

(d) Let $P \to \infty$. Every integer $k \le P$ belongs to $\mathcal{I}_P$ (its prime factors are at most $k \le P$). So the integers *not* in $\mathcal{I}_P$ are all larger than P, and

$$|\zeta(s) - E_P(s)| = \Big|\sum_{k\notin\mathcal{I}_P} k^{-s}\Big| \le \sum_{k>P} k^{-\sigma} \le \frac{P^{1-\sigma}}{\sigma - 1},$$

using Equation (16.3) at the last step (with $N = P$). The right side tends to 0 as $P \to \infty$, which proves Equation (16.7). ■

Sample Problem 16.3: Building $\zeta(2)$ prime by prime

Compute $E_P(2)$ for $P = 2, 3, 5, 7, 11, 13$ and compare the error with the bound of step (d).

Step 1: The factor for prime p at $s = 2$ is $1/(1 - p^{-2}) = p^2/(p^2 - 1)$: these are $\frac{4}{3}, \frac{9}{8}, \frac{25}{24}, \frac{49}{48}, \frac{121}{120}, \frac{169}{168}$.

Step 2: Multiply them in turn (Table 16.2).

Step 3: The proof's bound with $\sigma = 2$ is $P^{1-2}/(2 - 1) = 1/P$. Every error in the table respects it.

Step 4: With six primes,

$$\boxed{E_{13}(2) = 1.617918, \quad \zeta(2) - E_{13}(2) = 0.027016 \le \tfrac{1}{13} = 0.076923.}$$

Consequence 1: no zeros in $\operatorname{Re} s > 1$. For each prime, $|1 - p^{-s}| \le 1 + |p^{-s}| = 1 + p^{-\sigma}$ (triangle inequality). Hence

$$\frac{1}{|E_P(s)|} = \prod_{p\le P} |1 - p^{-s}| \le \prod_{p\le P} (1 + p^{-\sigma}).$$

Expanding the right side as in step (b), with exponents restricted to 0 or 1, gives a sum of $k^{-\sigma}$ over some distinct positive integers k (the square-free members of $\mathcal{I}_P$), which is at most

Table 16.2: Truncated Euler products at $s = 2$, against $\zeta(2) = \pi^2/6 = 1.644934$.

P	factor	$E_P(2)$	$\zeta(2) - E_P(2)$	bound $1/P$
2	4/3	1.333333	0.311601	0.500000
3	9/8	1.500000	0.144934	0.333333
5	25/24	1.562500	0.082434	0.200000
7	49/48	1.595052	0.049882	0.142857
11	121/120	1.608344	0.036590	0.090909
13	169/168	1.617918	0.027016	0.076923

$\zeta(\sigma)$. So $|E_P(s)| \geq 1/\zeta(\sigma)$ for every P. Since $E_P(s) \to \zeta(s)$ and the modulus is continuous, $|\zeta(s)| \geq 1/\zeta(\sigma)$. Combining with Equation (16.5):

$$\frac{1}{\zeta(\sigma)} \;\leq\; |\zeta(s)| \;\leq\; \zeta(\sigma) \qquad (\sigma = \operatorname{Re} s > 1). \tag{16.8}$$

Theorem (proved): $\zeta(s) \neq 0$ whenever $\operatorname{Re} s > 1$. So every zero of ζ, whatever it is, lies in $\operatorname{Re} s \leq 1$, exactly where our series fails. This is why we will need analytic continuation.

Consequence 2: infinitely many primes. At $s = 1$ the sum diverges; the product must then also run away, and that is impossible with finitely many primes. Here is a version with no limits in s. Fix $x \geq 2$ and choose an integer J with $2^J \geq x$. Repeating step (b) with real $s = 1$,

$$\prod_{p \leq x} \frac{1}{1 - 1/p} \;\geq\; \prod_{p \leq x} \sum_{j=0}^{J} \frac{1}{p^j} \;=\; \sum_{k \in \mathcal{I}_{x,J}} \frac{1}{k} \;\geq\; \sum_{k=1}^{\lfloor x \rfloor} \frac{1}{k}.$$

The first inequality holds because each geometric series of positive terms exceeds its truncation. The last holds because every integer $k \leq x$ is in $\mathcal{I}_{x,J}$: its primes are at most x, and an exponent j satisfies $2^j \leq p^j \leq k \leq x \leq 2^J$, so $j \leq J$. Thus

$$\prod_{p \leq x} \Bigl(1 - \frac{1}{p}\Bigr)^{-1} \;\geq\; \sum_{k \leq x} \frac{1}{k}. \tag{16.9}$$

The right side grows without bound (harmonic series, Chapter 3). If there were only finitely many primes, the left side would stop changing once x passed the largest prime, and so would be bounded. Contradiction: there are infinitely many primes. This is a second proof of Euclid's theorem, and the first hint that analytic properties of ζ encode facts about primes.

Sample Problem 16.4: The product beats the harmonic sum

Check Equation (16.9) at $x = 10$.
Step 1: The primes up to 10 are 2, 3, 5, 7, with factors $\frac{1}{1-1/p} = \frac{p}{p-1}$: 2, $\frac{3}{2}$, $\frac{5}{4}$, $\frac{7}{6}$.
Step 2: Product: $\frac{2 \cdot 3 \cdot 5 \cdot 7}{1 \cdot 2 \cdot 4 \cdot 6} = \frac{210}{48} = 4.375$.
Step 3: Harmonic sum: $1 + \frac{1}{2} + \cdots + \frac{1}{10} = 2.928968$.
Step 4: $\boxed{4.375 \geq 2.928968}$, as required.

Pitfall

The Euler product is proved only for $\operatorname{Re} s > 1$. On the critical line $\operatorname{Re} s = \frac{1}{2}$, where the zeros live, it does *not* converge, and you cannot locate zeros by asking when a factor vanishes (no factor $(1 - p^{-s})^{-1}$ is ever zero). Chapter 18 uses the logarithm of the Euler

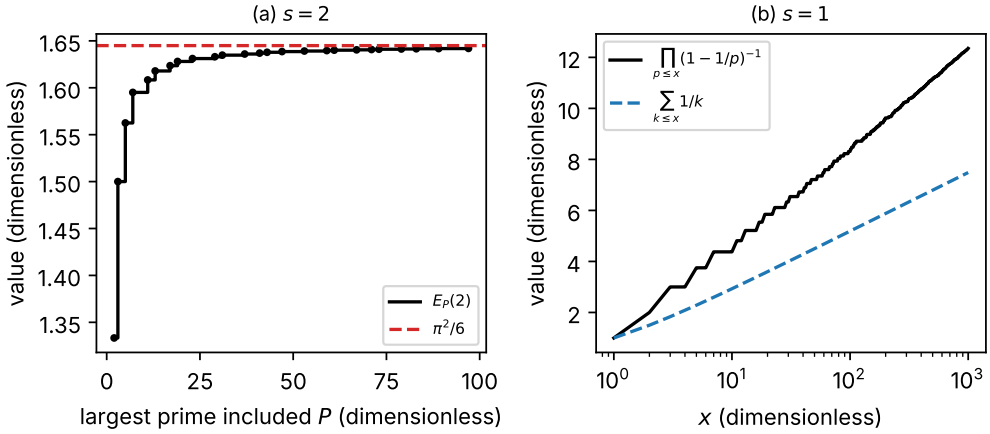


Figure 16.1: The Euler product. (a) Truncated products $E_P(2)$ over primes $p \le P$ approach $\zeta(2) = \pi^2/6$ (dashed) from below as more primes are included. (b) At $s = 1$, the product over primes $p \le x$ (solid) stays above the harmonic sum $\sum_{k \le x} 1/k$ (dashed), and both grow without bound, so there must be infinitely many primes. All quantities are dimensionless.

product *formally* on the critical line to obtain the paper's Eq. (4); that step is heuristic and is labelled as such there.

Checkpoint 16.3

Using Equation (16.8), give a lower bound for $|\zeta(3 + 100i)|$.

Answer: $|\zeta(3 + 100i)| \ge 1/\zeta(3) = 1/1.202057 = 0.831907$. Together with the earlier checkpoint, $0.831907 \le |\zeta(3 + 100i)| \le 1.202057$.

Key Ideas

- For $\operatorname{Re} s > 1$, $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$; the proof is unique factorization plus geometric series.
- The product shows $\zeta(s) \neq 0$ for $\operatorname{Re} s > 1$, with $1/\zeta(\sigma) \le |\zeta(s)| \le \zeta(\sigma)$.
- Divergence at $s = 1$ forces infinitely many primes.

16.4 Analytic continuation

★ Did you know? *Euler's Strange Sums*

Long before analytic continuation was understood, Leonhard Euler manipulated divergent series in work presented in 1749. In effect he obtained the value $-\frac{1}{12}$ for $1 + 2 + 3 + \cdots$, and he guessed a relation between values at s and $1 - s$. That relation is the functional equation Riemann proved in 1859. Today we write $\zeta(-1) = -\frac{1}{12}$ in the continued sense, which does not mean the series literally adds up to it.

The paper's zeros are at $s = \frac{1}{2} + it_n$. There, $|k^{-s}| = k^{-1/2}$ and $\sum k^{-1/2}$ diverges; our defining series makes no sense. What can "$\zeta(\frac{1}{2} + 14.134725\,i) = 0$" possibly mean? The answer is analytic continuation. We first see it in the simplest possible case.

16.4.1 A model: the geometric series

For complex z with $|z| < 1$ the geometric series converges (Chapter 3):

$$G(z) = \sum_{k=0}^{\infty} z^k = \frac{1}{1-z} \qquad (|z| < 1). \tag{16.10}$$

The series defines G only inside the unit disc. The formula $1/(1-z)$, however, makes sense for every $z \neq 1$, and it is analytic there: its derivative is $1/(1-z)^2$. Inside the disc the two agree. We say $1/(1-z)$ is the *analytic continuation* of the series to the whole plane minus the point $z = 1$. The point $z = 1$, where $1/(1-z)$ blows up, is a *pole*.

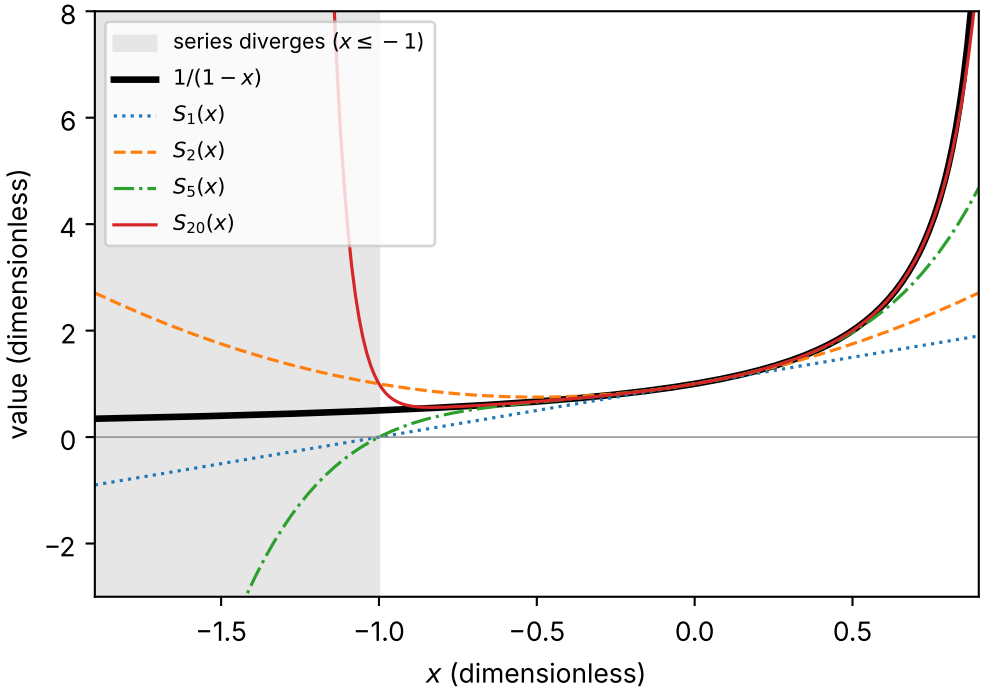


Figure 16.2: Partial sums $S_N(x) = \sum_{k=0}^{N} x^k$ of the geometric series for real x (thin lines, $N = 1, 2, 5, 20$) and the function $1/(1-x)$ (thick). For $-1 < x < 1$ the partial sums approach $1/(1-x)$; for $x < -1$ (shaded) they oscillate with growing amplitude (at $x = -1$ they hop between 1 and 0) and the series diverges, yet $1/(1-x)$ is perfectly finite there; the series also diverges for $x \geq 1$. All quantities are dimensionless.

Sample Problem 16.5: The geometric series at $z = -2$

Compare the partial sums of $\sum z^k$ at $z = -2$ with $1/(1-z)$.

Step 1: The partial sum of a geometric series is $S_N(z) = \sum_{k=0}^{N} z^k = \frac{1-z^{N+1}}{1-z}$ (Chapter 3). At $z = -2$, $S_N(-2) = \frac{1-(-2)^{N+1}}{3}$.

Step 2: The first few are $S_0 = 1$, $S_1 = -1$, $S_2 = 3$, $S_3 = -5$, $S_4 = 11$: they oscillate with growing size and the series diverges.

Step 3: The continuation is $1/(1-(-2)) = 1/3$. Note that $S_N = \frac{1}{3} - \frac{(-2)^{N+1}}{3}$: the continuation is the "center" about which the partial sums swing.

Step 4: series diverges; continued value $G(-2) = 1/3$.

16.4.2 Why continuation is unique

Definition. Let F be analytic on a region D. An *analytic continuation* of F to a larger region $D' \supset D$ is a function analytic on D' that equals F on D.

You might worry that there are many ways to extend a function, giving many "values" at $\frac{1}{2} + 14.134725\,i$. For complex-differentiable functions this cannot happen.

Taken on trust

Identity theorem (uniqueness of analytic continuation). If two functions are analytic on a connected region and agree on a small disc (or even on a short line segment) inside it, they agree everywhere in the region. Consequently, an analytic continuation, if it exists, is unique: every method of extending ζ analytically gives the same values. Proof: [6], or the complex-variable chapters of [22]. Nothing like this holds for merely smooth real functions: the function equal to 0 for $x \le 0$ and to $e^{-1/x}$ for $x > 0$ has derivatives of all orders everywhere, yet agrees with the zero function on $x \le 0$ without being zero. Complex differentiability is what rules this out.

So the program is: find *some* formula, analytic on a bigger region, that agrees with $\sum k^{-s}$ where the series converges. Uniqueness then makes it *the* zeta function there.

16.4.3 The alternating zeta function

Flip the sign of every second term:

$$\eta(s) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^s} = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \cdots . \tag{16.11}$$

This is the *Dirichlet eta function*, or alternating zeta function. Write $S_N(s) = \sum_{k=1}^{N} (-1)^{k+1} k^{-s}$ for its partial sums. The cancellation between neighboring terms makes it converge on a larger region.

Real $s > 0$. The terms k^{-s} are positive, decrease, and tend to 0. By the alternating series test (Chapter 3) the series converges, the partial sums alternate around the limit ($S_2 \le S_4 \le \cdots \le \eta(s) \le \cdots \le S_3 \le S_1$), and

$$|\eta(s) - S_N(s)| \le (N+1)^{-s}.$$

Complex s with $\sigma = \operatorname{Re} s > 0$. The alternating series test needs real, decreasing terms, so we need a different argument. We first prove a lemma.

Lemma. For real $0 < a < b$ and complex s,

$$|a^{-s} - b^{-s}| \;\le\; |s| \int_a^b x^{-\sigma-1}\,\mathrm{d}x. \tag{16.12}$$

Proof. Write $x^{-s} = e^{-su}$ with $u = \log x$. The u-derivative of $e^{-su} = e^{-\sigma u}[\cos(tu) - i\sin(tu)]$ is, by the product rule,

$$-\sigma e^{-\sigma u}[\cos tu - i\sin tu] + e^{-\sigma u}[-t\sin tu - it\cos tu] = -(\sigma + it)\,e^{-\sigma u}[\cos tu - i\sin tu] = -s\,e^{-su},$$

where in the middle step we used $-t\sin tu - it\cos tu = -it(\cos tu - i\sin tu)$. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}x} x^{-s} = -s\,e^{-su}\cdot\frac{1}{x} = -s\,x^{-s-1}$. The fundamental theorem of calculus, applied separately to real and imaginary parts, gives

$$a^{-s} - b^{-s} = -\int_a^b \frac{\mathrm{d}}{\mathrm{d}x} x^{-s}\,\mathrm{d}x = s\int_a^b x^{-s-1}\,\mathrm{d}x.$$

The modulus of an integral of a complex function is at most the integral of its modulus (each Riemann sum obeys the triangle inequality, and the bound survives the limit), and $|x^{-s-1}| = x^{-\sigma-1}$. This gives Equation (16.12). ■

Theorem (proved). The series Equation (16.11) converges for every complex s with $\operatorname{Re} s > 0$.

Proof. Group the terms in pairs, $D_j = (2j-1)^{-s} - (2j)^{-s}$ for $j = 1, 2, \ldots$, so that $S_{2J} = \sum_{j=1}^{J} D_j$. By the lemma with $a = 2j-1$, $b = 2j$,

$$\sum_{j=1}^{J} |D_j| \le |s| \sum_{j=1}^{J} \int_{2j-1}^{2j} x^{-\sigma-1}\,\mathrm{d}x \le |s| \int_{1}^{\infty} x^{-\sigma-1}\,\mathrm{d}x = \frac{|s|}{\sigma}.$$

The second inequality holds because the intervals $[2j-1, 2j]$ are disjoint pieces of $[1, \infty)$ and the integrand is positive; the integral is $[-x^{-\sigma}/\sigma]_1^{\infty} = 1/\sigma$ since $\sigma > 0$. So $\sum_j D_j$ converges absolutely, and S_{2J} tends to a limit, which we call $\eta(s)$. The odd partial sums are $S_{2J+1} = S_{2J} + (2J+1)^{-s}$, and $|(2J+1)^{-s}| = (2J+1)^{-\sigma} \to 0$. So $S_N(s) \to \eta(s)$ along all N. ■

Pitfall

For $0 < \sigma \le 1$ the convergence of η is only *conditional*: $\sum |(-1)^{k+1} k^{-s}| = \sum k^{-\sigma}$ diverges. As Chapter 3 warned, the terms of a conditionally convergent series must be added in their given order; rearranging them can change the sum. Our pairing kept the order, which is why it is legitimate.

Taken on trust

η is analytic on $\operatorname{Re} s > 0$. The pairing argument gives a bound $|s|/\sigma$ that is uniform on any region where $|s|$ is bounded and σ is bounded away from 0 (that is, $\sigma \ge \sigma_0$ for some fixed $\sigma_0 > 0$); by the same Weierstrass theorem as before, η is analytic on the half-plane $\operatorname{Re} s > 0$, and in particular continuous there. Proof: [6], Chapter 2.

16.4.4 From η back to ζ

For $\operatorname{Re} s > 1$ both series converge absolutely, and we may subtract them term by term:

$$\begin{aligned}
\zeta(s) - \eta(s) &= \sum_{k=1}^{\infty} \left[1 - (-1)^{k+1}\right] k^{-s} \\
&= \sum_{k \text{ even}} 2\,k^{-s} && \text{(the bracket is 0 for odd } k\text{, 2 for even } k\text{)} \\
&= 2\sum_{j=1}^{\infty} (2j)^{-s} && \text{(write } k = 2j\text{)} \\
&= 2 \cdot 2^{-s} \sum_{j=1}^{\infty} j^{-s} && \text{(multiplication rule)} \\
&= 2^{1-s}\zeta(s).
\end{aligned}$$

Rearranging,

$$\eta(s) = \left(1 - 2^{1-s}\right)\zeta(s) \qquad (\operatorname{Re} s > 1). \tag{16.13}$$

Now turn this around. The right side of

$$\zeta(s) = \frac{\eta(s)}{1 - 2^{1-s}} \tag{16.14}$$

is defined and analytic for all $\operatorname{Re} s > 0$ at which the denominator is not zero, and it agrees with $\sum k^{-s}$ for $\operatorname{Re} s > 1$. By uniqueness, Equation (16.14) *is* the zeta function on this larger region. In particular, it is what gives meaning to $\zeta(\frac{1}{2} + 14.134725\,i)$.

Where the denominator vanishes. $1-2^{1-s}=0$ means $e^{(1-s)\log 2}=1$. Recall from Chapter 2 that $e^{x+iy}=e^x(\cos y+i\sin y)$ equals 1 exactly when $e^x=1$ and y is a multiple of 2π, that is, when $x+iy=2\pi in$ for an integer n. So $(1-s)\log 2=2\pi in$, or

$$s=1+\frac{2\pi i\,m}{\log 2},\qquad m=0,\pm1,\pm2,\dots$$

(we renamed $m=-n$). These points lie on the line $\operatorname{Re}s=1$, spaced by $2\pi/\log 2=9.0647$ in t, the same number as the period of the 2^{-it} arrow. At $m=0$ we get $s=1$, discussed next. At $m\neq 0$ the denominator vanishes, so Equation (16.14) does not define ζ there; whether ζ is finite depends on whether η also vanishes (it does; see the box).

Taken on trust

ζ is finite at $s=1+2\pi im/\log 2$, $m\neq 0$. In this section we take on trust that ζ has no pole at these points (equivalently, that η vanishes there, so the $0/0$ has a finite limit). The book does not leave this unproved: Problems 25 and 26 construct a second continuation $\zeta=\eta_3/(1-3^{1-s})$ whose denominator vanishes at different points, which shows that ζ is finite at every point of $\operatorname{Re}s>0$ except $s=1$. Once that is known, Equation (16.13), which holds on all of $\operatorname{Re}s>0$ by uniqueness of continuation, gives $\eta=(1-2^{1-s})\zeta=0$ at these points. A direct treatment is in [6], Chapter 2.

The pole at $s=1$. Near $s=1$ write $u=(s-1)\log 2$, a small complex number. Then $2^{1-s}=e^{-u}=1-u+\frac{u^2}{2}-\cdots$ (Taylor series of the exponential, Chapter 2), so

$$1-2^{1-s}=u-\frac{u^2}{2}+\cdots=(s-1)\log 2\Big[1-\frac{(s-1)\log 2}{2}+\cdots\Big].$$

Meanwhile η is continuous at 1, and $\eta(1)=1-\frac12+\frac13-\cdots=\log 2$. (To see the value: $S_{2n}(1)=\sum_{k=1}^{2n}\frac1k-2\sum_{j=1}^{n}\frac{1}{2j}=\sum_{k=n+1}^{2n}\frac1k$, and comparing with integrals as in Section 16.1, $\int_{n+1}^{2n+1}\frac{dx}{x}\le\sum_{k=n+1}^{2n}\frac1k\le\int_n^{2n}\frac{dx}{x}$; both bounds tend to $\log 2$.) Hence

$$(s-1)\,\zeta(s)=\frac{(s-1)\,\eta(s)}{1-2^{1-s}}\longrightarrow\frac{\log 2}{\log 2}=1\qquad(s\to 1).\tag{16.15}$$

So near $s=1$, $\zeta(s)\approx 1/(s-1)$. A point s_0 where a function behaves like $c/(s-s_0)$ with $c\neq 0$ is called a *simple pole* with *residue* c. The zeta function has a simple pole at $s=1$ with residue 1, matching the real-variable bounds Equation (16.2), which give $1\le(s-1)\zeta(s)\le s$.

Sample Problem 16.6: Close to the pole

Estimate $\zeta(1.01)$.
Step 1: Equation (16.2) with $s-1=0.01$: $100\le\zeta(1.01)\le 101$.
Step 2: Equation (16.15) predicts $\zeta(1.01)\approx 1/0.01=100$, with a correction of order 1 (the bounds say the correction lies between 0 and 1).
Step 3: A computer (`mpmath.zeta(1.01)`) gives $\zeta(1.01)=100.578$, so $(s-1)\zeta(s)=1.00578$, close to the limit 1.
Step 4: The linearization behind Equation (16.15) can be checked directly: $1-2^{-0.01}=0.0069075$ while $(s-1)\log 2=0.0069315$; they differ by about 0.35%, the size of the neglected term $(s-1)\log 2/2$.
$\boxed{100\le\zeta(1.01)\le 101;\ \zeta(1.01)=100.578.}$

Sample Problem 16.7: The value $\zeta(\frac{1}{2})$

Use Equation (16.14) to find $\zeta(\frac{1}{2})$, a point where $\sum k^{-s}$ diverges.
Step 1: At $s = \frac{1}{2}$, $2^{1-s} = 2^{1/2} = 1.414214$, so $1 - 2^{1-s} = -0.414214$.
Step 2: $\eta(\frac{1}{2}) = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \cdots$ converges, slowly: the alternating series bound after $N = 1000$ terms is $(1001)^{-1/2} = 0.0316$.
Step 3: Because the partial sums alternate around the limit, η lies between S_N and S_{N+1}, so their average is within half the bound, 0.0158. In practice the average is far better: averaging S_{1000} and S_{1001} reproduces $\eta(\frac{1}{2}) = 0.604899$ to within 10^{-5} (see the listing that follows this sample problem).
Step 4: $\zeta(\frac{1}{2}) = \eta(\frac{1}{2})/(1 - \sqrt{2}) = 0.604899/(-0.414214)$:

$$\boxed{\zeta(\tfrac{1}{2}) = -1.460355.}$$

The value is *negative*, although every term of $\sum k^{-1/2}$ is positive. There is no contradiction: that series diverges, and -1.460355 is the value of the continued function, not a sum.

```
import numpy as np

def eta_avg(s, N):
    """Average of the partial sums S_N and S_{N+1} of eta(s)."""
    k = np.arange(1, N + 2)                  # k = 1, ..., N+1
    terms = (-1.0)**(k + 1) * np.exp(-s * np.log(k))
    S = np.cumsum(terms)                     # S[j] = S_{j+1}
    return 0.5 * (S[N - 1] + S[N])

s = 0.5
eta = eta_avg(s, 1000)
print(eta)                                   # about 0.604899
print(eta / (1 - 2**(1 - s)))                # zeta(1/2), about -1.460355
```

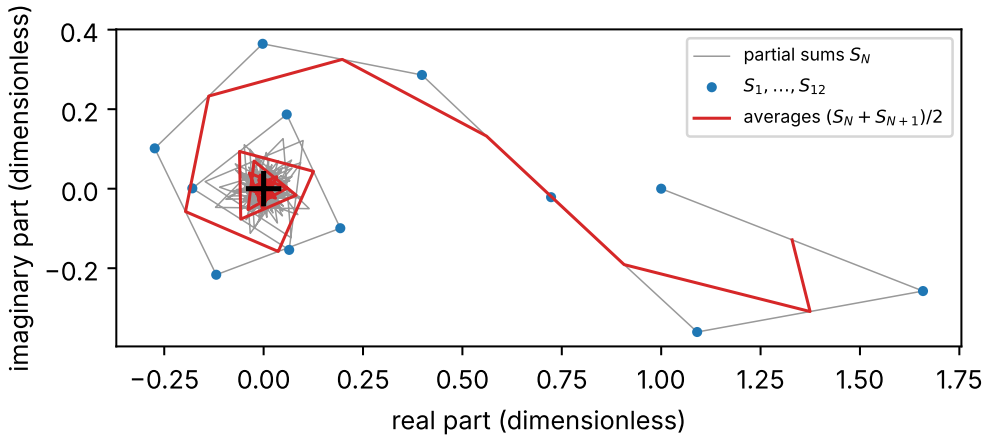


Figure 16.3: Partial sums S_N of the alternating series $\eta(s)$ at $s = \frac{1}{2} + i\,t_1$, $t_1 = 14.134725$, drawn in the complex plane for $N = 1$ to 300 (gray path; the first twelve partial sums are dots). The averages $(S_N + S_{N+1})/2$ (red) converge much faster. Both head to the origin (cross), because $\eta(\rho_1) = (1 - 2^{1-\rho_1})\zeta(\rho_1) = 0$. Axes are the real and imaginary parts (dimensionless).

Pitfall

You may have seen the claim "$1+2+3+4+\cdots=-\frac{1}{12}$". The series diverges; it has no sum. What is true is that the analytically continued zeta function satisfies $\zeta(-1)=-\frac{1}{12}$ (we compute it in Section 16.5). At $s=-1$ the formula $\sum k^{-s}$ would read $1+2+3+\cdots$, but that formula is not valid there. Never substitute s into $\sum k^{-s}$ unless $\operatorname{Re} s>1$.

Checkpoint 16.4

At $s=1+9.0647\,i$ the denominator $1-2^{1-s}$ in Equation (16.14) is zero. Is ζ infinite there?

Answer: No. The numerator $\eta(s)$ also vanishes at these points, and ζ is finite there; the only pole of ζ is at $s=1$. Problems 25 and 26 prove this with a second formula whose denominator vanishes at different points.

Key Ideas

- Analytic continuation extends a function beyond the domain of its first formula; it is unique (identity theorem).
- $\eta(s)=\sum(-1)^{k+1}k^{-s}$ converges for $\operatorname{Re} s>0$ and equals $(1-2^{1-s})\zeta(s)$; so $\zeta=\eta/(1-2^{1-s})$ extends ζ to $\operatorname{Re} s>0$.
- ζ has a simple pole at $s=1$ with residue 1: $(s-1)\zeta(s)\to 1$.
- Values like $\zeta(\frac{1}{2})=-1.460355$ are values of the continuation, not sums of divergent series.

16.5 The whole plane: the functional equation

To reach $\operatorname{Re} s\le 0$ we need Riemann's great discovery of 1859 [2]: a symmetry relating $\zeta(s)$ to $\zeta(1-s)$. It involves the Gamma function, which we now recall (Chapter 4 treats it more fully). For $\operatorname{Re} z>0$ it is defined by

$$\Gamma(z)=\int_0^\infty x^{z-1}e^{-x}\,dx, \qquad x^{z-1}=e^{(z-1)\log x}.$$

Integrate by parts with $u=x^z$ and $dv=e^{-x}\,dx$. The derivative formula proved in the lemma of Section 16.4 gives $\frac{d}{dx}x^z=z\,x^{z-1}$, so

$$\Gamma(z+1)=\Big[-x^ze^{-x}\Big]_0^\infty+z\int_0^\infty x^{z-1}e^{-x}\,dx=z\,\Gamma(z).$$

The boundary term vanishes because $|x^z|=x^{\operatorname{Re} z}\to 0$ as $x\to 0$, and because e^{-x} decays faster than any power grows as $x\to\infty$. Since $\Gamma(1)=\int_0^\infty e^{-x}\,dx=1$, repeated use of $\Gamma(z+1)=z\,\Gamma(z)$ gives $\Gamma(n+1)=n!$ for $n=0,1,2,\dots$. In this sense Γ generalizes the factorial.

Taken on trust

Facts about Γ used here. The Gamma function extends to all complex z as an analytic function except for simple poles at $z=0,-1,-2,\dots$, and it is never zero. (Proofs in [6] and [22].)

The functional equation (Riemann, 1859). The zeta function extends to an analytic function on the whole complex plane except for the simple pole at $s=1$, and for all s

other than $0, 1, 2, 3, \ldots$ (at those points the right side contains a pole of $\Gamma(1-s)$ or of $\zeta(1-s)$, and it is understood as a limit),

$$\zeta(s) = 2^s\,\pi^{s-1}\,\sin\!\Big(\frac{\pi s}{2}\Big)\,\Gamma(1-s)\,\zeta(1-s). \tag{16.16}$$

An equivalent, more symmetric form uses $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, which is analytic everywhere and satisfies

$$\xi(s) = \xi(1-s). \tag{16.17}$$

Powers such as 2^s and π^{s-1} mean $e^{s\log 2}$ and $e^{(s-1)\log\pi}$, and $\sin w = (e^{iw} - e^{-iw})/(2i)$ for complex w. Proofs: [2] (original), [5] Chapter 1, [6] Chapter 2. You can verify both numerically with `mpmath` at any sample point, for example $s = 0.3 + 7i$.

How it extends ζ. If $\operatorname{Re} s < 0$, then $\operatorname{Re}(1-s) = 1 - \operatorname{Re} s > 1$, so the right side of Equation (16.16) uses $\zeta(1-s)$ where the original series converges. The equation thus *defines* ζ on the left half-plane in terms of known values on the right. The strip $0 < \operatorname{Re} s < 1$ is already covered by Equation (16.14). In fact Equation (16.14) covers all of $\operatorname{Re} s > 0$, including the line $\operatorname{Re} s = 1$ apart from $s = 1$. For $\operatorname{Re} s \le 0$, $s \neq 0$, the functional equation uses $\zeta(1-s)$ with $\operatorname{Re}(1-s) \ge 1$, which is already known. The theorem in the box says these pieces join into one analytic function.

Sample Problem 16.8: $\zeta(-1)$ from the functional equation

Compute $\zeta(-1)$ using Equation (16.16) and $\zeta(2) = \pi^2/6$.
Step 1: Put $s = -1$: $2^s = \frac{1}{2}$, $\pi^{s-1} = \pi^{-2}$, $\sin(\pi s/2) = \sin(-\pi/2) = -1$.
Step 2: $\Gamma(1-s) = \Gamma(2) = 1! = 1$, and $\zeta(1-s) = \zeta(2) = \pi^2/6$.
Step 3: Multiply:

$$\zeta(-1) = \frac{1}{2}\cdot\frac{1}{\pi^2}\cdot(-1)\cdot 1\cdot\frac{\pi^2}{6}.$$

Step 4: The π^2 cancels: $\boxed{\zeta(-1) = -\frac{1}{12} = -0.083333.}$

Sample Problem 16.9: $\zeta(0)$ from the pole

Compute $\zeta(0)$. At $s = 0$ the factor $\sin(\pi s/2)$ vanishes while $\zeta(1-s) = \zeta(1)$ is infinite, so we must take a limit.
Step 1: Let $s \to 0$. Then $2^s \to 1$, $\pi^{s-1} \to 1/\pi$, $\Gamma(1-s) \to \Gamma(1) = 1$.
Step 2: For small s, $\sin(\pi s/2) \approx \pi s/2$ (small-angle approximation; the neglected term is of order s^3).
Step 3: Put $u = 1 - s \to 1$. By Equation (16.15), $(u-1)\zeta(u) \to 1$, that is, $(-s)\zeta(1-s) \to 1$, so $\zeta(1-s) \approx -1/s$.
Step 4: Multiply: $\zeta(s) \approx \frac{1}{\pi}\cdot\frac{\pi s}{2}\cdot 1\cdot\left(-\frac{1}{s}\right) = -\frac{1}{2}$. So $\boxed{\zeta(0) = -\frac{1}{2}.}$
The series for η converges only for $\operatorname{Re} s > 0$, so $\eta(0)$ needs a definition. For $\operatorname{Re} s \le 0$ we define the continuation of η by $\eta(s) = (1 - 2^{1-s})\zeta(s)$. This function is analytic: at $s = 1$ the zero of $1 - 2^{1-s}$ cancels the pole of ζ. It agrees with the series for $\operatorname{Re} s > 0$, because there Equation (16.14) says exactly this. By the identity theorem it is therefore the continuation of η. With this definition $\eta(0) = (1-2)\zeta(0) = \frac{1}{2}$: the continued value of $1 - 1 + 1 - 1 + \cdots$ is $\frac{1}{2}$, the same as the geometric continuation $1/(1-z)$ at $z = -1$.

The trivial zeros. When is $\sin w = 0$ for complex w? Since $\sin w = (e^{iw} - e^{-iw})/(2i)$, it vanishes exactly when $e^{iw} = e^{-iw}$, that is, $e^{2iw} = 1$, that is, $2iw = 2\pi i n$, so $w = n\pi$ with n an integer. There are no complex zeros of the sine beyond the familiar real ones. In Equation (16.16), $\sin(\pi s/2) = 0$ exactly when s is an even integer.

Take $s = -2, -4, -6, \ldots$, that is $s = -2j$ with $j \geq 1$. Then $\sin(-j\pi) = 0$, while $2^s \pi^{s-1}$ is finite, $\Gamma(1 + 2j) = (2j)!$ is finite, and $\zeta(1 + 2j)$ is a convergent series. Hence

$$\zeta(-2) = \zeta(-4) = \zeta(-6) = \cdots = 0.$$

These are the *trivial zeros*, "trivial" because they come from the sine factor and are completely understood. (At $s = 0, 2, 4, \ldots$ the vanishing sine is compensated by a pole: of $\zeta(1-s)$ at $s = 0$, as in the sample problem, and of $\Gamma(1 - s)$ at $s = 2, 4, \ldots$; so these are not zeros.)

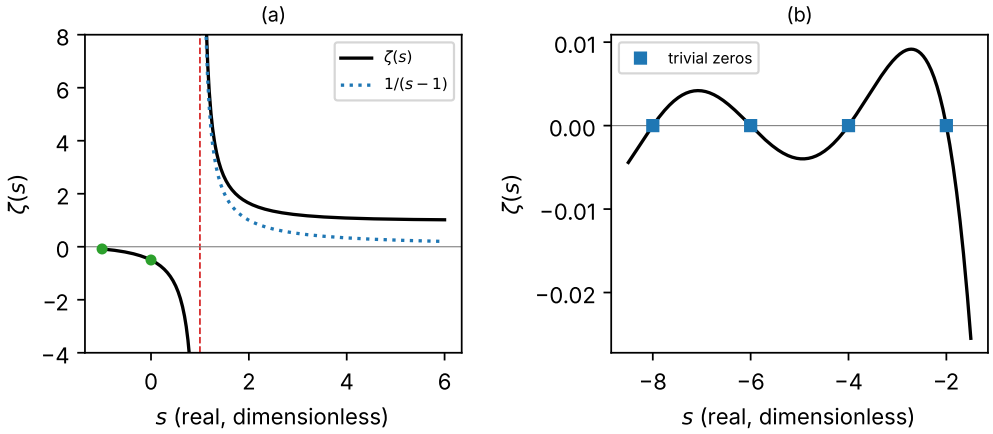


Figure 16.4: The continued zeta function on the real axis. (a) $-1 \leq s \leq 6$: the simple pole at $s = 1$, with $1/(s-1)$ dotted for comparison; $\zeta(0) = -\frac{1}{2}$, $\zeta(-1) = -\frac{1}{12}$, and $\zeta(s) \to 1$ for large s. (b) Zoom on $-8.5 \leq s \leq -1.5$: the trivial zeros at $s = -2, -4, -6, -8$ (squares). Between them the values are small, for example $\zeta(-3) = 1/120$. All quantities are dimensionless.

Checkpoint 16.5

Is $s = -3$ a trivial zero?

Answer: No. $\sin(-3\pi/2) = +1 \neq 0$, and the other factors in Equation (16.16) are finite and nonzero, so $\zeta(-3) \neq 0$. In fact $\zeta(-3) = \frac{1}{120}$ (Problem 13). Only negative *even* integers are trivial zeros.

Key Ideas

- The functional equation $\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1 - s) \zeta(1 - s)$ (taken on trust) extends ζ to the whole plane, analytic except for the pole at $s = 1$.
- $\zeta(0) = -\frac{1}{2}$, $\zeta(-1) = -\frac{1}{12}$.
- The trivial zeros are $s = -2, -4, -6, \ldots$, from the sine factor.

16.6 Where the zeros can be

We now collect what is known about the location of the zeros of ζ, labelling each statement's status.

1. **No zeros in** $\operatorname{Re} s > 1$**.** Theorem, proved in Section 16.3 from the Euler product.

2. **In** $\operatorname{Re} s < 0$ **the only zeros are** $-2, -4, -6, \ldots$**.** Theorem; deduced here from Equation (16.16) and the Γ facts in the box, which are themselves taken on trust (proofs beyond this book). For $\operatorname{Re} s < 0$ each factor on the right of Equation (16.16) is examined: $2^s\pi^{s-1}$ is an exponential, never zero; $\Gamma(1-s)$ is never zero; $\zeta(1-s) \neq 0$ by item 1, since $\operatorname{Re}(1-s) > 1$; and $\sin(\pi s/2) = 0$ only at even integers. So the zeros in $\operatorname{Re} s < 0$ are exactly the negative even integers.

3. **No zeros on the line** $\operatorname{Re} s = 1$**.** Theorem, proved by Hadamard and de la Vallée Poussin in 1896; it is the key step in their proof of the prime number theorem (Chapter 15). The proof is beyond this book; see [25] or [6].

4. **No zeros on the line** $\operatorname{Re} s = 0$**.** Theorem; deduced here from item 3 and Equation (16.16), both taken on trust (proofs beyond this book): at $s = it$ with $t \neq 0$, Equation (16.16) gives $\zeta(it)$ as a product of nonzero factors, because $\sin(\pi it/2) \neq 0$ (the argument $\pi it/2$ is not a real multiple of π) and $\zeta(1-it) \neq 0$ by item 3. At $s = 0$, $\zeta(0) = -\frac{1}{2} \neq 0$.

Therefore every zero other than the trivial ones lies in the open **critical strip**

$$0 < \operatorname{Re} s < 1.$$

These are the **nontrivial zeros**. The vertical line $\operatorname{Re} s = \frac{1}{2}$ through the middle of the strip is the **critical line**.

Taken on trust

There are infinitely many nontrivial zeros; many are on the critical line. Hardy proved in 1914 that infinitely many zeros lie on the critical line, and Conrey proved in 1989 that more than two fifths of all nontrivial zeros do. Chapter 17 counts the zeros up to height T: to leading order, roughly $\frac{T}{2\pi}\log\frac{T}{2\pi}$ of them. Proofs of Hardy's theorem and of the count: [6], Chapters 9–10. Conrey's result postdates that text; we quote it without a reference from the book's bibliography.

Symmetries of the nontrivial zeros. Two symmetries organize them.

Mirror (conjugate) symmetry. For $\operatorname{Re} s > 0$, Equation (16.14) gives ζ as $\eta/(1-2^{1-s})$. Exactly the argument that proved Equation (16.6) shows $\eta(\bar{s}) = \overline{\eta(s)}$ (term by term, $\overline{k^{-s}} = k^{-\bar{s}}$), and $\overline{1-2^{1-s}} = 1-2^{1-\bar{s}}$. Dividing, $\zeta(\bar{s}) = \overline{\zeta(s)}$ throughout $\operatorname{Re} s > 0$, except at the points where the denominator vanishes. At $s = 1 + 2\pi i m/\log 2$ with $m \neq 0$ the identity follows by continuity of ζ, which is finite there (box in Section 16.4). The remaining point $s = 1$ is the pole, which is real. So if $\zeta(\rho) = 0$ then $\zeta(\bar{\rho}) = \bar{0} = 0$.

Reflection symmetry. For s in the open strip, every factor in Equation (16.16) other than $\zeta(1-s)$ is finite and nonzero: s is not an even integer, so the sine is nonzero; $\operatorname{Re}(1-s)$ lies in $(0,1)$, so $1-s$ is not a pole of Γ, and Γ is never zero; exponentials are never zero. Hence $\zeta(s) = 0$ exactly when $\zeta(1-s) = 0$.

Combining: if $\rho = \beta + i\gamma$ is a nontrivial zero ($0 < \beta < 1$), so are

$$\rho = \beta + i\gamma, \qquad \bar{\rho} = \beta - i\gamma, \qquad 1-\rho = (1-\beta) - i\gamma, \qquad 1-\bar{\rho} = (1-\beta) + i\gamma. \tag{16.18}$$

Off the critical line ($\beta \neq \frac{1}{2}$) these are four distinct points, symmetric about both the real axis and the critical line. On the critical line ($\beta = \frac{1}{2}$) we have $1-\bar{\rho} = \rho$ and $1-\rho = \bar{\rho}$, so the four collapse to a conjugate pair $\frac{1}{2} \pm i\gamma$. Put differently, an off-line zero always comes with a mirror partner at the same height on the other side of the critical line.

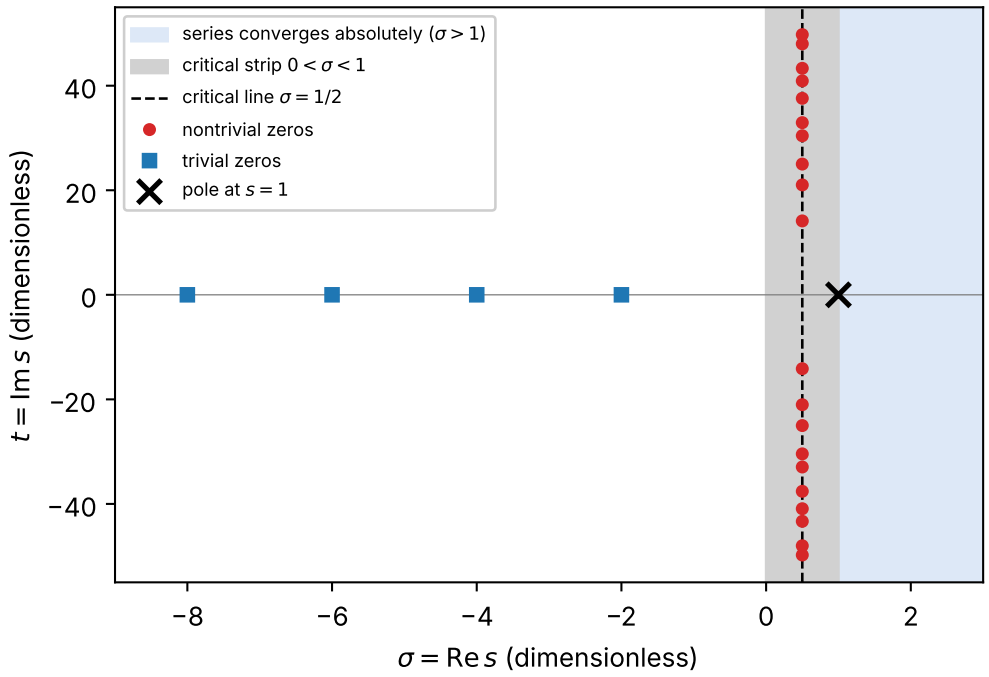


Figure 16.5: Map of the zeta function in the complex s plane, $s = \sigma + it$. The original series converges absolutely for $\sigma > 1$ (blue); the critical strip $0 < \sigma < 1$ (gray) contains all nontrivial zeros; the dashed line is the critical line $\sigma = \frac{1}{2}$. Red dots: the first ten nontrivial zeros $\frac{1}{2} \pm it_n$, $n = 1, \ldots, 10$ (conjugate pairs). Squares: trivial zeros at $-2, -4, -6, -8$. Cross: the pole at $s = 1$. Both axes are dimensionless.

Sample Problem 16.10: Partners of a zero

(a) List the zeros forced by the first zero $\rho_1 = \frac{1}{2} + 14.134725\, i$. (b) Suppose, hypothetically, that $\zeta(0.6 + 20i) = 0$. Which other zeros would follow?

Step 1: By Equation (16.18), (a) gives $\bar{\rho}_1 = \frac{1}{2} - 14.134725\, i$, $1 - \rho_1 = \frac{1}{2} - 14.134725\, i$, $1 - \bar{\rho}_1 = \frac{1}{2} + 14.134725\, i$.

Step 2: Only two of these are distinct: $\frac{1}{2} \pm 14.134725\, i$.

Step 3: For (b), $\beta = 0.6$, $\gamma = 20$: the partners are $0.6 - 20i$, $0.4 - 20i$, $0.4 + 20i$.

Step 4:

$$\boxed{\text{(a) } \tfrac{1}{2} \pm 14.134725\, i; \qquad \text{(b) } 0.6 \pm 20i,\ 0.4 \pm 20i.}$$

(No zero at $0.6 + 20i$ exists; the first two zeros above height 20 are at $t = 21.022040$ and 25.010858 on the critical line.)

Checkpoint 16.6

Why does the paper (and this book) only ever list zeros with $t_n > 0$?

Answer: Because of conjugate symmetry: zeros with negative imaginary part are exactly the mirror images $\bar{\rho}_n$ of those with positive imaginary part, so they carry no new information. Also there are no zeros on the real segment $0 < s < 1$ (Problem 22 asks you to explain why).

Key Ideas

- Zero-free: $\operatorname{Re} s > 1$ (Euler product), $\operatorname{Re} s = 1$ and $\operatorname{Re} s = 0$ (deep theorem), $\operatorname{Re} s < 0$ except $-2, -4, \ldots$ (functional equation).
- Nontrivial zeros lie in the open strip $0 < \operatorname{Re} s < 1$, in symmetric quadruples $\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}$, which on the critical line are pairs $\frac{1}{2} \pm i\gamma$.

16.7 The Riemann hypothesis

★ Did you know? *A Million-Dollar Question*

In 1900 David Hilbert included the Riemann hypothesis in Problem 8 of his famous list of 23 problems for the new century. In 2000 the Clay Mathematics Institute made it one of seven Millennium Prize Problems, each worth one million US dollars. As of this writing, only one of the seven, the Poincaré conjecture, has been solved, by Grigori Perelman, who declined the prize.

Statement. The **Riemann hypothesis** (RH) is the conjecture:

$$\textit{Every nontrivial zero } \rho \textit{ of } \zeta \textit{ has real part exactly } \tfrac{1}{2}.$$

Equivalently: every zero in the critical strip lies on the critical line.

Status: open. Riemann wrote in 1859 that it is "very probable" that this is so, and that he had put aside the search for a proof [2, 5]. It has been neither proved nor disproved since. It is one of the Clay Mathematics Institute's Millennium Prize Problems. Nothing in this book, and nothing in the paper, proves or disproves it.

What is known.

- *Proved:* all nontrivial zeros lie in $0 < \operatorname{Re} s < 1$ (Section 16.6); infinitely many lie on the critical line (Hardy, 1914); more than two fifths lie on it (Conrey, 1989).
- *Reported computational fact:* more than 10^{13} zeros, starting from the lowest, have been checked to lie on the critical line and to be simple, that is, of multiplicity one: near each such zero ρ, $\zeta(s) \approx c\,(s-\rho)$ with $c \neq 0$ (see Section 16.8). Large computations were reported by X. Gourdon in 2004 and by D. Platt and T. Trudgian in 2021. This is a finite check, not a proof: it says nothing about zeros higher up.
- *Numerical observation:* every zero ever computed is simple (not a double zero). That all zeros are simple is a separate open conjecture.

Why it matters for primes. RH is equivalent to the statement that the prime counting function $\pi(x)$ differs from $\operatorname{li}(x)$ (Chapter 15) by at most a constant times $\sqrt{x}\log x$ (a theorem of von Koch, 1901; see [25]). The zeros control the fluctuations of the primes, and, as Chapter 18 shows, the primes control the fluctuations of the zeros. This two-way street is the explicit formula behind the paper's oracle.

Notation used from now on. We list the nontrivial zeros in the upper half-plane by increasing height, $0 < t_1 < t_2 < \cdots$, and write

$$\rho_n = \tfrac{1}{2} + i\,t_n.$$

Writing the real part as $\frac{1}{2}$ is a statement about each *specific* zero we use: each of them has been verified, by the method of Section 16.8, to lie on the critical line. It is not an assumption that RH is true in general. The heights t_n are the "mysterious numbers" of Chapter 1.

Key Ideas

- RH: all nontrivial zeros have $\operatorname{Re}\rho = \frac{1}{2}$. It is open.
- More than 10^{13} zeros have been verified on the line (a reported finite computation).
- We write $\rho_n = \frac{1}{2} + it_n$ with $t_1 < t_2 < \cdots$ for zeros individually verified on the line.

16.8 Computing the zeros

16.8.1 Values and zeros with `mpmath`

The Python library `mpmath` evaluates the continued zeta function at any complex point to any requested precision, using formulas far more efficient than Equation (16.14). Its function `mpmath.zetazero(n)` returns the n-th zero ρ_n as a complex number.

```
import mpmath
mpmath.mp.dps = 20                  # work with 20 significant digits

for n in range(1, 11):
    rho = mpmath.zetazero(n)        # complex number 1/2 + i t_n
    print(n, rho.real, mpmath.nstr(rho.imag, 10))

rho1 = mpmath.zetazero(1)
print(abs(mpmath.zeta(rho1)))       # tiny: rho1 really is a zero
```

Table 16.3: The first ten nontrivial zeros (running example RE1) and the gaps between neighbors. Real parts are all $\frac{1}{2}$; the spacings are irregular.

n	t_n	$t_{n+1} - t_n$
1	14.134725	6.887315
2	21.022040	3.988818
3	25.010858	5.414018
4	30.424876	2.510186
5	32.935062	4.651116
6	37.586178	3.332541
7	40.918719	2.408354
8	43.327073	4.678078
9	48.005151	1.768681
10	49.773832	

Pitfall

`mpmath.zetazero(n)` returns the complex number ρ_n, not the height. Use `.imag` to get t_n. Indices start at $n = 1$, as in the paper; the book's library `qmnt.zeros(nmax)` returns the array $t_1, \dots, t_{n_{\max}}$, whose Python index 0 holds t_1.

16.8.2 The Riemann–Siegel Z function

How can a computer prove that a zero lies *exactly* on the critical line, when every computed number is rounded? The trick is to find a *real* function whose zeros are the zeros of ζ on the line. Then the intermediate value theorem does the work: a continuous real function that is negative at one point and positive at another must vanish in between.

Define the Riemann–Siegel theta function and Z function for real t:

$$\theta(t) = \operatorname{Im} \log \Gamma\Big(\frac{1}{4} + \frac{it}{2}\Big) - \frac{t}{2} \log \pi, \tag{16.19}$$

$$Z(t) = e^{i\theta(t)}\, \zeta\Big(\tfrac{1}{2} + it\Big). \tag{16.20}$$

In Equation (16.19) the imaginary part of $\log \Gamma$ is the phase (argument) of $\Gamma(\frac{1}{4} + \frac{it}{2})$, followed continuously from $\theta(0) = 0$. Chapter 17 derives from Stirling's formula the approximation $\theta(t) \approx \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8}$, which is the origin of the paper's Eq. (1).

Since $|e^{i\theta}| = 1$,

$$|Z(t)| = |\zeta(\tfrac{1}{2} + it)|,$$

so $Z(t) = 0$ exactly when $\zeta(\frac{1}{2} + it) = 0$. The factor $e^{i\theta(t)}$ rotates $\zeta(\frac{1}{2} + it)$ back onto the real axis.

Taken on trust

$Z(t)$ is real and continuous for real t. This follows from the functional equation Equation (16.17) together with the mirror symmetry $\zeta(\bar{s}) = \overline{\zeta(s)}$: on the critical line, $1 - s = \bar{s}$, and the phase of $\zeta(\frac{1}{2} + it)$ is exactly $-\theta(t)$ modulo π. Proof: [5], Chapter 6; [6], Chapter 4. A quick `mpmath` evaluation of $e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ at, say, $t = 20$ confirms that its imaginary part vanishes to working precision. In `mpmath` the functions are `siegeltheta(t)` and `siegelz(t)`.

At $t = 0$, $\theta(0) = 0$ and $Z(0) = \zeta(\frac{1}{2}) = -1.460355 < 0$. By definition t_1 is the lowest positive height of a nontrivial zero, and $\zeta(\frac{1}{2}) \neq 0$, so Z has no zero on $[0, t_1)$. Being continuous, it therefore stays negative there: a sign change would force a zero, by the intermediate value theorem.

A zero t_0 of Z is called *simple* (multiplicity one) if $Z'(t_0) \neq 0$; Z is differentiable because it is built from analytic functions. Near a simple zero $Z(t) \approx Z'(t_0)(t - t_0)$, which changes sign at t_0. More generally, a zero of *multiplicity* m is one near which $Z(t) \approx c\,(t - t_0)^m$ with $c \neq 0$. Such a zero changes sign only for odd m. The same words apply to zeros of ζ in the complex plane, with $(s - \rho)^m$ in place of $(t - t_0)^m$.

Sample Problem 16.11: Trapping the first zero by bisection

Start from $Z(14) < 0$ and $Z(15) > 0$, and use `mpmath.siegelz` to evaluate the sign of Z at each midpoint. Locate t_1 by eight steps of bisection: halve the bracket at each step and keep the half on which Z changes sign (see also Chapter 4). Bound the error.

Step 1: Start with $[a, b] = [14, 15]$. At each step evaluate Z at the midpoint m and keep the half on which Z changes sign.

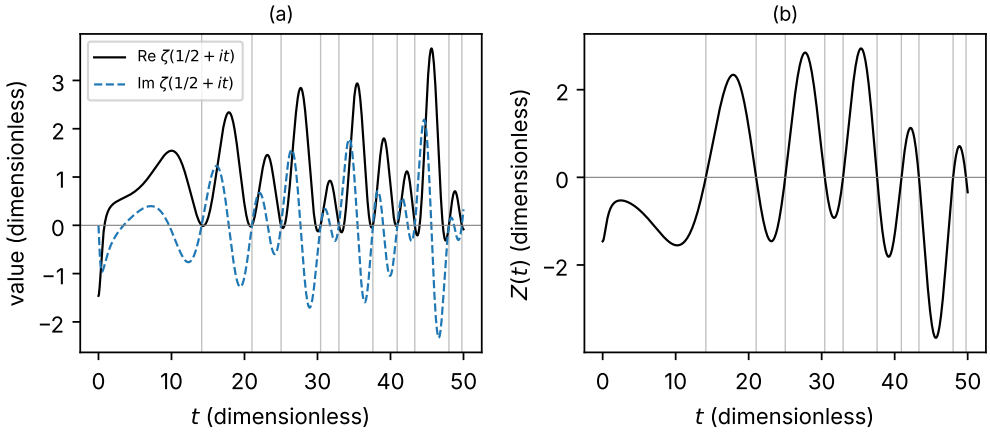


Figure 16.6: The zeta function on the critical line, $0 \le t \le 50$. (a) Real part (solid) and imaginary part (dashed) of $\zeta(\frac{1}{2}+it)$; at a zero both vanish together (gray lines mark $t_1, \ldots, t_{10}$). (b) The Riemann–Siegel function $Z(t)$, a real function with $|Z(t)| = |\zeta(\frac{1}{2}+it)|$, which changes sign at every zero. The variable t and the function values are dimensionless.

Step 2: The eight steps are listed in Table 16.4. (The signs are computed by the listing below. As a consistency check only, they agree with what the known value $t_1 = 14.134725$ implies: $Z < 0$ below it and $Z > 0$ above it, up to $t_2 = 21.02$.)
Step 3: After eight halvings the final bracket is $[14.1328125,\ 14.13671875]$, of width $1/2^8 = 0.00390625$. Our estimate is the midpoint of this final bracket, 14.134765625. (This is not the step-8 midpoint 14.13671875 listed in the table; that point is now the right endpoint of the bracket.) The zero lies in the bracket, so the estimate is within half the width, 0.001953, of the zero.
Step 4:

$$\boxed{t_1 \approx 14.134766,\ \text{true error } 4.0 \times 10^{-5} \le 0.001953.}$$

Each step gains one binary digit; about 12 more steps (20 in total) would shrink the bracket to $2^{-20} \approx 9.5 \times 10^{-7}$ of the starting width.
What, exactly, is proved? A computed sign can be trusted only when the computed $|Z|$ is larger than a guaranteed bound on the rounding and truncation error of the evaluation. The proof therefore uses two points where $|Z|$ is safely nonzero with opposite signs, for example $t = 14$ and $t = 15$. The intermediate value theorem then gives a zero of Z between them, and hence a zero of ζ exactly on the critical line. Bisection refines the location, not the proof. Midpoints close to the zero have tiny $|Z|$, and their computed signs are precisely the ones that need the error bound.

```
import mpmath

a, b = 14.0, 15.0
Za = mpmath.siegelz(a)                  # negative
for step in range(1, 9):
    m = 0.5 * (a + b)
    Zm = mpmath.siegelz(m)
    if Zm * Za < 0:                     # sign change in [a, m]
        b = m
    else:                               # sign change in [m, b]
        a, Za = m, Zm
    print(step, m, "+" if Zm > 0 else "-", (a, b))
```

Table 16.4: Bisection for the first zero on $[14, 15]$.

step	midpoint m	sign of $Z(m)$	new bracket
1	14.5	+	[14, 14.5]
2	14.25	+	[14, 14.25]
3	14.125	−	[14.125, 14.25]
4	14.1875	+	[14.125, 14.1875]
5	14.15625	+	[14.125, 14.15625]
6	14.140625	+	[14.125, 14.140625]
7	14.1328125	−	[14.1328125, 14.140625]
8	14.13671875	+	[14.1328125, 14.13671875]

```
print("estimate", 0.5 * (a + b))
```

How RH is checked numerically. Sign changes of Z on $[0, T]$ give a *lower bound* on the number of zeros on the critical line up to height T. They can miss zeros only in pairs (two close zeros between sample points) or zeros of even multiplicity. Separately, the argument principle (Chapter 17) gives the *total* number $N(T)$ of zeros in the whole strip up to height T, wherever they are. If the two numbers agree, every zero up to height T is on the critical line and simple. This is the logic of the large verifications, and `mpmath.zetazero` locates zeros with the same ingredients: sign changes of Z, organized with the help of the counting function, followed by refinement.

Taken on trust

Making the count rigorous. Turning the comparison above into a proof requires an exact, certified value of $N(T)$ at the chosen heights. A method due to A. M. Turing (1953) provides a checkable way to confirm the count using values of Z alone. We take this method on trust; it is described in [5].

Checkpoint 16.7

Your program finds $Z(30) > 0$ and $Z(31) < 0$. What can you conclude, and what can you not conclude?

Answer: You can conclude that there is at least one zero of ζ exactly on the critical line with $30 < t < 31$ (here it is $t_4 = 30.424876$). You cannot conclude that there is only one: an odd number of zeros, counted with multiplicity, lies in the interval. Nor do sign changes say anything about zeros off the line; that needs the count $N(T)$ of Chapter 17.

Key Ideas

- `mpmath.zeta` evaluates the continued function; `mpmath.zetazero(n)` returns ρ_n.
- $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real, with $|Z| = |\zeta|$ on the line; its sign changes prove zeros exactly on the critical line.
- Sign changes plus the zero count $N(T)$ verify RH up to height T, a finite check.

16.9 Paper connection

This chapter supplies the objects that the paper [1] takes as its data. Here is where each idea appears.

Sec. I, first sentence. "The Hilbert–Pólya conjecture proposes that the imaginary parts t_n of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of the Riemann zeta function are the spectrum of a self-adjoint operator." You can now read every word. "Nontrivial" excludes the zeros $-2, -4, \ldots$ of Section 16.5; these play no role in the paper. "$\rho_n = \frac{1}{2} + it_n$" records that each zero used lies on the critical line, which is verified for every zero in the paper's range (at most 10^4 zeros, far inside the more than 10^{13} checked; Section 16.7). The Hilbert–Pólya idea, developed in Chapter 21, is a proposed route to RH: a self-adjoint operator has real eigenvalues (Chapter 10), so if the t_n were its eigenvalues, every ρ_n would have real part exactly $\frac{1}{2}$. RH and Hilbert–Pólya are both open, and the paper does not claim otherwise; Sec. VI states that "Nothing here touches the analytic Hilbert–Pólya program."

Sec. II A and Appendix A step (1): the data. The paper computes "zeros $t_1, \ldots, t_{1000} \ldots$ with `mpmath` at 20 significant digits ($t_1 = 14.134725141734695$, $t_{1000} = 1419.42248094599$) and zeros 10^3–10^4 at 15 digits", via `mpmath.zetazero(n)` (Appendix A, step 1). The "key anchors" in Appendix A add $t_{500} = 811.1843588465$. These are exactly the computations of Section 16.8, and running `mpmath.zetazero` at 20 digits reproduces t_{500} and t_{1000} to the digits printed, and t_1 to all but the last printed digit (see the reading lesson that follows). A small reading lesson: to 20 digits, $t_1 = 14.13472514173469379\ldots$; the paper's printed 14.134725141734695 agrees with it to about 10^{-15}, the resolution of a double-precision number near 14 (a double carries about 16 significant digits; see also Chapter 4). The paper notes that "both precisions exceed every use below by many orders": errors in the paper are quoted in units of the local mean spacing, which near t_{1000} is about 1.159 (Chapter 17; Problem 21), and the smallest error the paper quotes is an rms below 10^{-4} spacings (Sec. III).

Only positive heights. By the mirror symmetry of Section 16.6, the zeros below the real axis are the conjugates $\bar{\rho}_n$, so the paper's list $t_1 < t_2 < \cdots$ of positive heights is the complete information. Every figure of the paper (Fig. 1: fluctuations of zeros 370–460; Fig. 2: spacings of the first 10^4 zeros) and every table (Tables I–V, with holdouts up to zero 10^4 in Table V) uses this list.

The Euler product behind the oracle. The paper's arithmetic oracle, Eq. (4), is a sum of terms $\sin(mt \log p)/(m\, p^{m/2})$ over primes p and powers m. You have now seen where $\log p$ and $p^{-1/2}$ come from: by Equation (16.4), the prime factor p^{-s} on the critical line is an arrow of length $p^{-1/2}$ rotating at angular frequency $\log p$ in t. Chapter 18 turns the Euler product Equation (16.7) into Eq. (4) by a formal (heuristic) step; the paper itself says the formula "has no fitted parameters and no novelty" (Sec. II D). The arrow for $p = 2$ turns once every $2\pi/\log 2 = 9.0647$ in t, which is the "period $2\pi/\log 2 \approx 9$ in t" of Sec. V and the target of the continuation probe in Sec. III.

Prime-only constructions. Sec. I A(b) and Sec. II D cite Connes [21]: "extremizing the truncated Weil quadratic form over primes $p \le 13$ yields approximations to the first 50 zeros with accuracies from 10^{-3} down to 2.6×10^{-55}, provably located on the critical line." In Connes's construction the approximating values lie on the critical line by a proof internal to that construction, which is beyond this book. They are approximations to the first 50 zeros, so this does not prove RH.

What the paper does not use. The paper never needs values of ζ off the critical line, the trivial zeros, or the functional equation directly. It needs the heights t_n (this chapter), their counting function (Chapter 17, paper Eq. (1)), and the prime sum (Chapter 18, paper Eq. (4)).

16.10 Summary

- $\zeta(s) = \sum_{k\geq 1} k^{-s}$ converges for real $s > 1$ (Equation (16.1)), with $1/(s-1) \leq \zeta(s) \leq 1 + 1/(s-1)$ (Equation (16.2)) and tail bounds Equation (16.3).
- For complex $s = \sigma + it$, $k^{-s} = k^{-\sigma}e^{-it\log k}$ (Equation (16.4)): an arrow of length $k^{-\sigma}$ rotating at angular frequency $\log k$. The series converges absolutely for $\sigma > 1$, with $|\zeta(s)| \leq \zeta(\sigma)$ (Equation (16.5)) and $\zeta(\bar{s}) = \overline{\zeta(s)}$ (Equation (16.6)).
- Euler product: $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ for $\sigma > 1$ (Equation (16.7)), proved from unique factorization. Consequences: $1/\zeta(\sigma) \leq |\zeta(s)| \leq \zeta(\sigma)$, so no zeros in $\sigma > 1$ (Equation (16.8)); infinitely many primes (Equation (16.9)).
- Analytic continuation is unique (identity theorem). The model case is $\sum z^k = 1/(1-z)$ (Equation (16.10)).
- $\eta(s) = \sum(-1)^{k+1}k^{-s}$ converges for $\sigma > 0$ (proof via Equation (16.12)), and $\eta = (1-2^{1-s})\zeta$ (Equation (16.13)), so $\zeta = \eta/(1-2^{1-s})$ continues ζ to $\sigma > 0$ (Equation (16.14)). Simple pole at $s = 1$ with residue 1 (Equation (16.15)).
- Functional equation (taken on trust): Equations (16.16) and (16.17). It gives $\zeta(0) = -\frac{1}{2}$, $\zeta(-1) = -\frac{1}{12}$, and the trivial zeros $-2, -4, -6, \ldots$
- Nontrivial zeros lie in $0 < \operatorname{Re} s < 1$ and come in quadruples Equation (16.18), which collapse to conjugate pairs on the critical line.
- RH (open): all nontrivial zeros have real part $\frac{1}{2}$. More than 10^{13} zeros have been verified on the line.
- $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$ (Equations (16.19) and (16.20)) is real; its sign changes prove zeros exactly on the line. `mpmath.zetazero(n)` returns $\rho_n = \frac{1}{2} + it_n$; the first ten heights are in Table 16.3.

16.11 Problems

1. • Compute the partial sum $S_5 = \sum_{k=1}^{5} k^{-2}$ and use Equation (16.3) to give an interval containing $\zeta(2)$. Does it contain $\pi^2/6$?

2. • Find $|4^{-s}|$ at $s = 1.5 + 40i$. Does the value of $t = 40$ affect your answer?

3. • On the critical line $s = \frac{1}{2} + it$, write 5^{-s} in polar form. What are its modulus and its angular frequency in t?

4. • Which of the following are zeros of ζ, and of which kind (trivial or nontrivial)? $s = -2$, -3, -10, 0, 1, $\frac{1}{2} + 14.134725\,i$.

5. • At which of these points does the series $\sum k^{-s}$ converge absolutely: (a) $s = 0.9$; (b) $s = 1.1$; (c) $s = 2 + 50i$; (d) $s = \frac{1}{2} + 14i$; (e) $s = 1 + i$?

6. • Using Table 16.3, how many nontrivial zeros ρ have $0 < \operatorname{Im}\rho \leq 30$? How many have $-30 \leq \operatorname{Im}\rho \leq 30$?

7. •• Prove that $\zeta(s)$ is strictly decreasing for real $s > 1$, and that $\zeta(s) \to 1$ as $s \to \infty$.

8. •• Compute the truncated Euler product $E_5(4) = \prod_{p \le 5} (1 - p^{-4})^{-1}$ and compare with $\zeta(4) = \pi^4/90$. Is the error within the bound $P^{1-\sigma}/(\sigma - 1)$ of the proof?

9. •• (a) Explain each inequality in Equation (16.9). (b) Evaluate both sides at $x = 20$.

10. •• Use Equation (16.2) to bracket $\zeta(1.1)$, and compare with the computed value $\zeta(1.1) = 10.584$. What does Equation (16.15) predict?

11. •• Use Equation (16.13) to find $\eta(2) = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \cdots$ exactly and numerically.

12. •• The series $1 - 1 + 1 - 1 + \cdots$ diverges. (a) Show that the geometric continuation $1/(1 - z)$ at $z = -1$ gives $\frac{1}{2}$. (b) Use the continuation $\eta(s) = (1 - 2^{1-s})\zeta(s)$ defined in the sample problem on $\zeta(0)$, together with $\zeta(0) = -\frac{1}{2}$, to show $\eta(0) = \frac{1}{2}$, agreeing with (a). (c) Explain why $(1 - 2^{1-s})\zeta(s)$ is the unique analytic continuation of the series for η, even though Equation (16.13) was first derived for $\operatorname{Re} s > 1$.

13. •• Use the functional equation and $\zeta(4) = \pi^4/90$ to compute $\zeta(-3)$.

14. •• Give a complete proof that $\zeta(\bar{s}) = \overline{\zeta(s)}$ for $0 < \operatorname{Re} s \le 1$, $s \ne 1$, using Equation (16.14). (Hint: at the points $1 + 2\pi i m/\log 2$, $m \ne 0$, where the denominator vanishes, use continuity of ζ.)

15. •• Suppose a nontrivial zero were found at $\rho = 0.8 + 30i$. List all the zeros that would be forced by symmetry, and say what this would mean for the Riemann hypothesis. Why are there four and not two?

16. •• Show that if w is complex and $\sin w = 0$, then w is a real multiple of π. Where in this chapter was this used?

17. •• Using `mpmath`, compute $\rho_1, \ldots, \rho_{10}$. Check that every real part is 0.5, that the heights match Table 16.3, and compute the ten values $|\zeta(\rho_n)|$.

18. •• Using `qmnt.primes_upto(10000)` and numpy, compute the Euler product at $s = 2$ over all primes up to 10^4, its error against $\pi^2/6$, and the bound of step (d) of the proof with $P = 10^4$. Compare, and explain why the bound is pessimistic.

19. •• Using `mpmath.siegelz`, evaluate $Z(t)$ on a grid $t = 0, 0.05, 0.10, \ldots, 50$ and count the sign changes. Compare with Table 16.3. Why does a step of 0.05 suffice here?

20. •• Call the function `eta_avg` of Section 16.4 with the complex argument $s = \frac{1}{2} + i\,t_1$ and $N = 1000$. The function works unchanged; explain why. Show that the result has modulus below 10^{-3}, and explain why the exact value is 0.

21. •• (Paper, Sec. II A.) The paper states $t_{1000} = 1419.42248094599$ and says its precisions "exceed every use below by many orders." (a) Evaluate the mean-spacing formula $2\pi/\log(t/2\pi)$ of Chapter 17 at $t = t_{1000}$. (b) Roughly what absolute error in t corresponds to 20 significant digits at this height, and to 15 significant digits at heights below 10^4? (c) Compare these, in units of the mean spacing, with the smallest error the paper reports (Sec. III).

22. •• (Paper, Sec. I.) The paper writes $\rho_n = \frac{1}{2} + i t_n$. (a) What does writing the real part as $\frac{1}{2}$ assert about the zeros used, and how is it justified without assuming RH? (b) Why are the conjugate zeros $\bar{\rho}_n$ not treated as separate data? (c) Why can no nontrivial zero lie on the real segment $0 < s < 1$? (Hint: use η on real s and the alternating series bounds.)

23. •• (Paper, Eq. (4) and Sec. V.) (a) For $s = \frac{1}{2} + it$, show $\operatorname{Im}(2^{-s}) = -2^{-1/2} \sin(t \log 2)$ and give the amplitude and period in t. (b) Compare the period with the paper's "period

$2\pi/\log 2 \approx 9$ in t" in Sec. V and with the running example RE2. (c) Show that the points where $1 - 2^{1-s} = 0$ are spaced by the same amount in t.

24. • • • (Paper, Sec. I A(b) and II D.) Connes's prime-only construction gives approximations to the first 50 zeros "provably located on the critical line" [21]. (a) Leave Connes's method aside (it is beyond this book). Explain in your own words how a sign change of the real function $Z(t)$, computed at two points where $|Z|$ exceeds a guaranteed error bound, certifies a zero of ζ *exactly* on the line, even though all computations are rounded. (b) Explain why no finite computation of this kind can prove RH. Explain also why Connes's result, which concerns approximations to the first 50 zeros, does not prove it either. (c) Why does the paper classify such constructions as being "on the arithmetic side" of its comparison?

25. • • • Let $c_k = 1$ if $3 \nmid k$ and $c_k = -2$ if $3 \mid k$, and $\eta_3(s) = \sum_{k\geq 1} c_k k^{-s}$. (a) Show that $\eta_3(s) = (1 - 3^{1-s})\zeta(s)$ for $\operatorname{Re} s > 1$. (b) Prove that the series converges for $\operatorname{Re} s > 0$, by grouping terms in threes and using Equation (16.12).

26. • • • (a) Prove that $\log 2/\log 3$ is irrational, using unique factorization. (b) Deduce that the zeros of $1 - 2^{1-s}$ and of $1 - 3^{1-s}$ have only $s = 1$ in common. (c) Combine with Problem 25 to show that ζ extends analytically to $\operatorname{Re} s > 0$ with the single exception $s = 1$, without assuming that η vanishes at $1 + 2\pi i m/\log 2$.

27. • • • Prove that $0 \leq \zeta(s) - \frac{1}{s-1} \leq 1$ for all real $s > 1$, and deduce directly that $(s-1)\zeta(s) \to 1$ as $s \to 1^+$. Check the inequality at $s = 1.5$ using Table 16.1.

28. • • • Sign changes of Z give a lower bound on the number of zeros on the critical line. (a) Give two ways a sign-change count on a grid can undercount. (b) Explain why an exact count $N(T)$ of all zeros in the strip up to height T (Chapter 17) closes the gap. (c) Explain why a zero off the critical line could never be detected as a sign change of Z.

16.12 Answers to selected problems

1. $S_5 = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} = 1.463611$. With $N = 5$, $s = 2$, the tail lies between $1/6$ and $1/5$, so $1.630278 \leq \zeta(2) \leq 1.663611$. Yes, $\pi^2/6 = 1.644934$ is inside.

2. $|4^{-s}| = 4^{-1.5} = 1/8 = 0.125$. The imaginary part only rotates the arrow (Equation (16.4)); it does not change its length.

3. $5^{-s} = 5^{-1/2} e^{-it\log 5}$. Modulus $1/\sqrt{5} = 0.447214$; it rotates clockwise at angular frequency $\log 5 = 1.609438$ radians per unit t.

4. -2 and -10 are trivial zeros (negative even integers). -3 is not a zero ($\zeta(-3) = 1/120$). 0 is not ($\zeta(0) = -\frac{1}{2}$). 1 is the pole, not a zero. $\frac{1}{2} + 14.134725\,i$ is (to the digits given) the first nontrivial zero ρ_1.

5. Absolute convergence holds exactly when $\operatorname{Re} s > 1$: (b) and (c) only. At (e), $\operatorname{Re} s = 1$ and $\sum |k^{-s}| = \sum 1/k$ diverges.

6. Three ($t = 14.13, 21.02, 25.01$; the next is $30.42 > 30$). By conjugate symmetry, six with $-30 \leq \operatorname{Im}\rho \leq 30$.

7. If $1 < s_1 < s_2$, then for every $k \geq 2$, $k^{-s_2} = e^{-s_2 \log k} < e^{-s_1 \log k} = k^{-s_1}$ because $\log k > 0$. Summing (the $k = 1$ terms are equal, the $k = 2$ terms differ strictly) gives $\zeta(s_2) < \zeta(s_1)$. For the limit, Equation (16.2) gives $1 \leq \zeta(s) \leq 1 + \frac{1}{s-1}$ (the lower bound 1 because the first term is 1 and the rest are positive), and $\frac{1}{s-1} \to 0$.

9. (a) First inequality: each factor $(1 - 1/p)^{-1} = \sum_{j\geq 0} p^{-j}$ exceeds its truncation at $j = J$, and all quantities are positive. Middle equality: expanding the finite product gives each integer in $\mathcal{I}_{x,J}$ exactly once (unique factorization). Last inequality: every $k \leq x$ is in $\mathcal{I}_{x,J}$ because its exponents satisfy $2^j \leq k \leq x \leq 2^J$, and the extra members of $\mathcal{I}_{x,J}$ only add positive terms. (b)

Product over $p \in \{2,3,5,7,11,13,17,19\}$: $2 \cdot \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{7}{6} \cdot \frac{11}{10} \cdot \frac{13}{12} \cdot \frac{17}{16} \cdot \frac{19}{18} = 5.847132$; $H_{20} = 3.597740$. Indeed $5.847132 \geq 3.597740$.

11. $\eta(2) = (1 - 2^{1-2})\zeta(2) = \frac{1}{2} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12} = 0.822467$.

13. At $s = -3$: $2^{-3} = \frac{1}{8}$, π^{-4}, $\sin(-3\pi/2) = +1$, $\Gamma(4) = 3! = 6$, $\zeta(4) = \pi^4/90$. So $\zeta(-3) = \frac{1}{8} \cdot \frac{1}{\pi^4} \cdot 1 \cdot 6 \cdot \frac{\pi^4}{90} = \frac{6}{720} = \frac{1}{120} = 0.008333$.

15. With $\beta = 0.8$, $\gamma = 30$: $0.8 - 30i$, $0.2 - 30i$, $0.2 + 30i$, plus $0.8 + 30i$ itself. RH would be false. There are four because $\beta \neq \frac{1}{2}$: reflection $\rho \to 1 - \bar{\rho}$ moves the zero to $0.2 + 30i$, a different point; only on the critical line does reflection map a zero to itself.

17. Every real part prints as 0.5; the heights agree with Table 16.3 to all six decimals; the values $|\zeta(\rho_n)|$ are tiny (below 10^{-17} with 20 working digits, at the level of the working precision), confirming they are zeros.

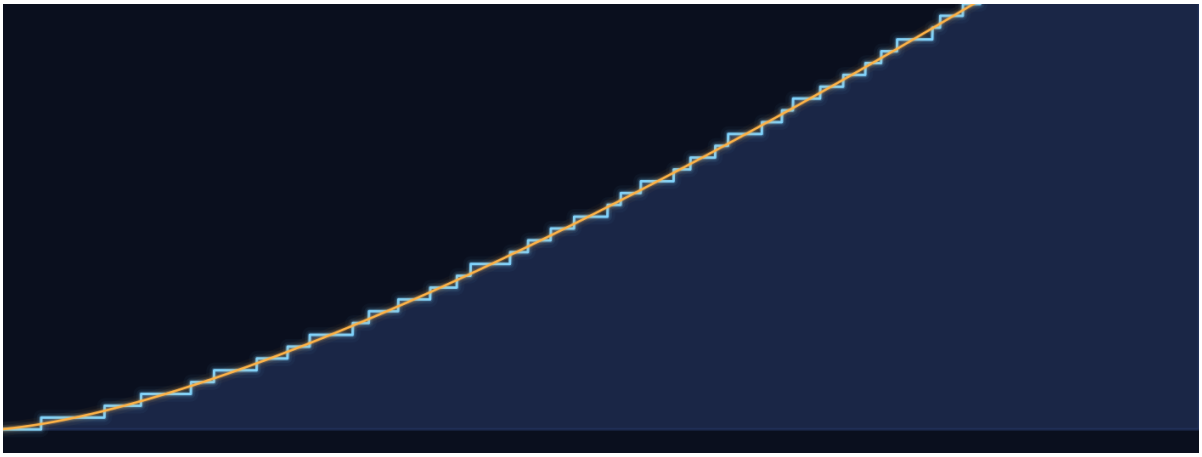
19. Ten sign changes, at $t_1, \ldots, t_{10}$ (all below 50; $t_{11} > 50$, see Table 16.3 for $t_{10} = 49.773832$). A step of 0.05 suffices because the smallest gap among these zeros is 1.768681 (between t_9 and t_{10}), so no grid interval can contain two zeros.

21. (a) $2\pi/\log(1419.42/2\pi) = 2\pi/\log(225.91) = 6.2832/5.4201 = 1.159$. (b) The number 1419.42… has four digits before the decimal point, so 20 significant digits reach the 16th decimal place; the rounding error is at most 0.5×10^{-16}, that is, of order 10^{-17} to 10^{-16}. The zeros 10^3–10^4 have heights between $t_{1000} = 1419.42$ and $t_{10^4} < 10^4$, all with four digits before the decimal point. So 15 significant digits give 11 decimal places, and a rounding error of at most 0.5×10^{-11}. (c) The mean spacing $2\pi/\log(t/2\pi)$ decreases with height. It is 1.159 at t_{1000} by part (a), and still 0.852 at $t = 10^4$. So in spacings the errors are of order 10^{-16} for the first 1000 zeros and below 10^{-11} beyond. The smallest error reported in the paper is the coverage rms "$< 10^{-4}$ spacings" at $M = 900$ (Sec. III). That is at least seven orders of magnitude larger, so the precision of the data never matters.

23. (a) $2^{-s} = 2^{-1/2} e^{-it\log 2} = 2^{-1/2}[\cos(t\log 2) - i\sin(t\log 2)]$, so $\mathrm{Im}(2^{-s}) = -2^{-1/2}\sin(t\log 2)$, amplitude 0.707107, angular frequency $\log 2 = 0.693147$, period $2\pi/\log 2 = 9.064720$. (b) This is the "≈ 9" of Sec. V and the 9.065 of RE2: the $m = 1$ term of the $p = 2$ part of the paper's Eq. (4) is $-\frac{1}{\pi} \cdot 2^{-1/2}\sin(t\log 2)$, which is $\frac{1}{\pi}\,\mathrm{Im}(2^{-s})$. (c) $1 - 2^{1-s} = 0$ at $s = 1 + 2\pi i m/\log 2$; consecutive values of m differ by $2\pi/\log 2 = 9.064720$ in t.

25. (a) For $\mathrm{Re}\, s > 1$, $\zeta(s) - 3 \cdot 3^{-s}\zeta(s) = \sum_k k^{-s} - 3\sum_j (3j)^{-s}$ (using $(3j)^{-s} = 3^{-s} j^{-s}$). The subtracted sum removes $3k^{-s}$ from each multiple of 3, turning its coefficient 1 into -2. So $(1 - 3^{1-s})\zeta(s) = \sum_k c_k k^{-s}$. (b) Group $B_j = (3j-2)^{-s} + (3j-1)^{-s} - 2(3j)^{-s} = [(3j-2)^{-s} - (3j)^{-s}] + [(3j-1)^{-s} - (3j)^{-s}]$. By Equation (16.12), $|B_j| \leq |s| \int_{3j-2}^{3j} x^{-\sigma-1}\mathrm{d}x + |s| \int_{3j-1}^{3j} x^{-\sigma-1}\mathrm{d}x \leq 2|s| \int_{3j-2}^{3j} x^{-\sigma-1}\mathrm{d}x$. Summing over j (disjoint intervals in $[1,\infty)$) gives $\sum |B_j| \leq 2|s|/\sigma$, so the grouped series converges absolutely. A partial sum $\sum_{k \leq N} c_k k^{-s}$ with $3J < N < 3J + 3$ differs from the complete-group sum $\sum_{j \leq J} B_j$ by the term $(3J+1)^{-s}$ and possibly $(3J+2)^{-s}$, whose total modulus is at most $2(3J+1)^{-\sigma}$; this tends to 0 as $J \to \infty$ because $\sigma > 0$. So all partial sums converge to the same limit.

27. The integral comparison of Section 16.1 gives $\frac{1}{s-1} \leq \zeta(s) \leq 1 + \frac{1}{s-1}$, which is $0 \leq \zeta(s) - \frac{1}{s-1} \leq 1$. Multiply by $s - 1 > 0$: $0 \leq (s-1)\zeta(s) - 1 \leq s - 1 \to 0$. At $s = 1.5$: $\zeta(1.5) - 2 = 0.612375$, which lies in $[0, 1]$.



The staircase $N(t)$ counting zeros up to height 140, and the smooth curve $N_{\mathrm{sm}}(t)$ it hugs.

Chapter 17

Counting the Zeros

↑ Contents

What this chapter is about

In Chapter 16 you met the nontrivial zeros of the Riemann zeta function, the points $\rho_n = \frac{1}{2} + it_n$ on the critical line. Their heights t_n look irregular: 14.13, 21.02, 25.01, 30.42, ..., with no evident rule taking you from one to the next. Suppose someone hands you these numbers one at a time. Could you say in advance, roughly, where the next one will be? Surprisingly, yes. The individual heights are hard to predict, but their *average density* is known exactly. A smooth, explicit function of t tells you how many zeros lie below any height, with an error that stays small: at the heights in this book it is a fraction of a zero. This chapter derives that function. We count zeros with a tool from complex analysis, the argument principle, which turns "how many zeros are inside this region?" into "how many times does the phase of the function wind around?". Applied to the zeta function, it gives the Riemann–von Mangoldt formula $N(T) = \theta(T)/\pi + 1 + S(T)$. Stirling's formula (Chapter 4) then turns $\theta(T)$ into an elementary expression, and out drops the smooth counting function with its curious constant $\frac{7}{8}$. From it we build the "smooth zeros" s_n, the local mean spacing gap_n, and the idea of measuring every error in units of that spacing.

Where this shows up in the paper. The smooth counting function is the paper's Eq. (1); the smooth zeros and mean spacing are defined in its Sec. II A; and every rms number in every table of the paper (Tables I–V) is in units of gap_n. The principal "null" error 0.263, against which the models in the paper are judged on its principal holdout, is simply the rms size of the fluctuation $(t_n - s_n)/\mathrm{gap}_n$ over zeros 401–500, a quantity you will compute yourself at the end of this chapter.

What you need

- Chapter 2: complex numbers in polar form, modulus and argument, the argument of a product is the sum of the arguments, $a^s = e^{s\log a}$, the logarithm of a complex number and its branches.
- Chapter 3: the series $\log(1+y) = y - \frac{y^2}{2} + \dots$ for $|y| < 1$, the geometric series, alternating series and their sums.
- Chapter 4: big-$\mathcal{O}$ notation; Stirling's formula for $\log\Gamma$, including the complex form stated there; inverting a monotone function; bisection, Newton and Brent root finding; the Lambert W function.
- Chapter 16: $\zeta(s)$, the Euler product and the fact that $\zeta(s) \neq 0$ for $\operatorname{Re} s > 1$; the alternating zeta $\eta(s) = (1 - 2^{1-s})\zeta(s)$; the pole at $s = 1$; trivial zeros; the critical strip and line; the functional equation (stated there); the Riemann–Siegel Z function; computing zeros with `mpmath.zetazero`.
- First-year physics and calculus: derivatives, the chain rule, Taylor expansion to second order, and reading a graph.

17.1 The staircase $N(T)$

★ Did you know? *Fifteen Zeros by Hand*

In 1903 the Danish mathematician Jørgen Gram computed the first fifteen nontrivial zeros of zeta and confirmed that they lie on the critical line. The first sits at height $t \approx 14.1347$. Gram points, where a related phase passes a multiple of π, are still used to count zeros efficiently. They usually, but not always, alternate with the zeros.

17.1.1 Definition

Recall from Chapter 16 that a nontrivial zero of ζ is written $\rho_n = \frac{1}{2} + it_n$ with $t_n > 0$ its *height* (strictly, the imaginary part of the zero; writing the real part as $\frac{1}{2}$ assumes the zero is on the critical line, which has been verified numerically for every zero used in this book). The zeros with negative imaginary part are mirror images of these, so we only ever count the upper half.

> **Definition.** For $T > 0$, the *zero-counting function* $N(T)$ is the number of nontrivial zeros ρ of ζ with $0 < \operatorname{Im}\rho \le T$, each counted as many times as its multiplicity.

"Multiplicity" means this: if ζ vanished like $(s-\rho)^2$ near ρ we would count ρ twice. Every zero ever computed has turned out to be simple (multiplicity one). That is a *numerical observation*; nobody has proved that all zeros are simple.

$N(T)$ is a staircase. It is zero for $T < t_1$, jumps to 1 at t_1, stays flat until t_2, jumps to 2, and so on. Our running example RE1 lists the first ten heights:

$$t_1, \dots, t_{10} = 14.134725,\ 21.022040,\ 25.010858,\ 30.424876,\ 32.935062,\ 37.586178,\ 40.918719,\ 43.327073,\ 48.005151,\ 49.773832. \tag{17.1}$$

Sample Problem 17.1: Reading the staircase

Using RE1, find $N(30)$, $N(40)$ and $N(50)$, and find the average spacing of the first ten zeros.

Step 1: $N(30)$ counts heights ≤ 30. From the list, t_1, t_2, t_3 are below 30 and $t_4 = 30.42 > 30$. So $N(30) = 3$.

Step 2: For $T = 40$: $t_6 = 37.59 < 40 < t_7 = 40.92$, so $N(40) = 6$.

Step 3: For $T = 50$: $t_{10} = 49.77 < 50$, and the next zero (not in the list) is above 50, as you can confirm with `mpmath.zetazero(11)`, whose imaginary part is 52.97. So $N(50) = 10$.
Step 4: The average spacing of the first ten zeros is the total distance divided by the number of gaps:

$$\frac{t_{10} - t_1}{9} = \frac{49.773832 - 14.134725}{9} = \frac{35.639107}{9} = 3.9599.$$

$$\boxed{N(30) = 3, \quad N(40) = 6, \quad N(50) = 10, \quad \text{mean spacing of } t_1, \ldots, t_{10} \approx 3.96.}$$

Ten zeros between heights 14 and 50 give spacings of about 4. If you look higher up, at zeros near number 1000, the spacing is closer to 1. The zeros crowd together as you go up. We want a formula that captures this crowding exactly.

17.1.2 The smooth curve through the staircase

Figure 17.1 shows $N(T)$ up to $T = 100$ together with the smooth function

$$N_{\text{sm}}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{17.2}$$

which is the paper's Eq. (1). Throughout, log is the natural logarithm. The smooth curve threads through the staircase, passing close to the middle of each vertical riser. The whole purpose of Secs. 17.2–17.4 is to derive Eq. (17.2) and to say exactly what the difference $N(T) - N_{\text{sm}}(T)$ is.

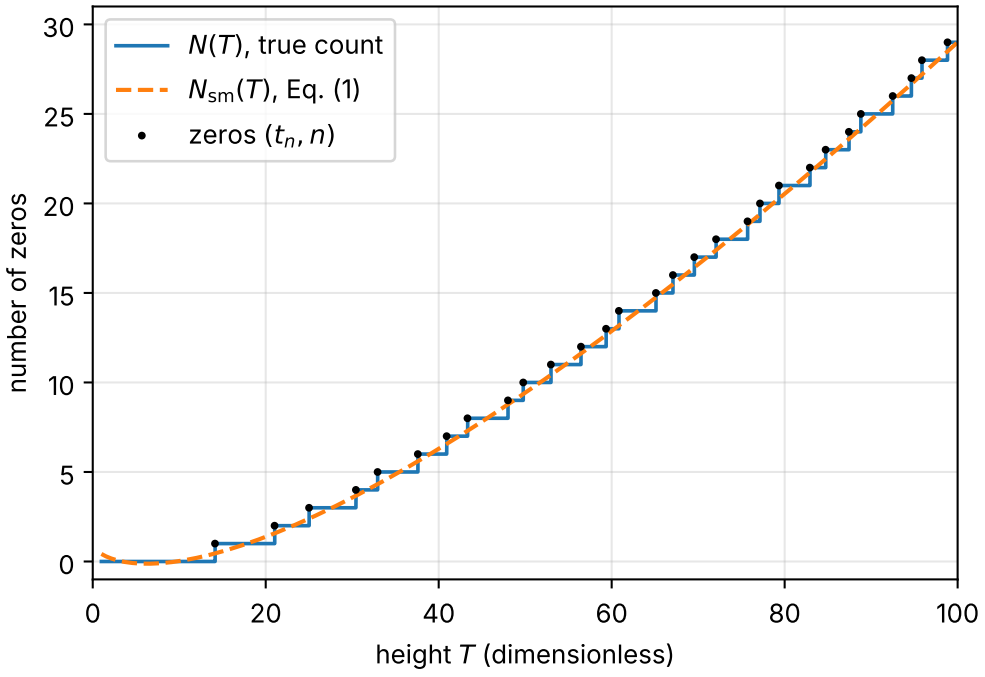


Figure 17.1: The zero-counting function $N(T)$ (staircase) for $0 < T \le 100$ and the smooth counting function $N_{\text{sm}}(T)$ of Eq. (17.2) (dashed). Dots mark the zeros t_n at height n. The smooth curve passes near the midpoint of each riser; at $T = 100$ it gives 29.002 while $N(100) = 29$.

Sample Problem 17.2: Evaluating the smooth count

Evaluate $N_{\rm sm}(50)$ and compare with $N(50) = 10$.
Step 1: Compute $t/2\pi$. With $2\pi = 6.283185$, $50/2\pi = 7.957747$.
Step 2: Its logarithm is easiest as a difference: $\log(50/2\pi) = \log 50 - \log(2\pi) = 3.912023 - 1.837877 = 2.074146$.
Step 3: First term: $7.957747 \times 2.074146 = 16.505529$.
Step 4: Subtract $t/2\pi$ and add $7/8$:

$$N_{\rm sm}(50) = 16.505529 - 7.957747 + 0.875 = 9.422782.$$

Step 5: Compare: $N(50) - N_{\rm sm}(50) = 10 - 9.4228 = 0.5772$.

$$\boxed{N_{\rm sm}(50) \approx 9.423, \qquad N(50) - N_{\rm sm}(50) \approx 0.577.}$$

The smooth formula is off by about half a zero at this height, even though nothing in it "knows" where the individual zeros are.

The same arithmetic gives $N_{\rm sm}(30) = 3.565$ against $N(30) = 3$, $N_{\rm sm}(40) = 6.293$ against $N(40) = 6$, and $N_{\rm sm}(100) = 29.002$ against $N(100) = 29$. The difference wanders up and down but stays well below 1. Those differences are the subject of Sec. 17.5.

```
import numpy as np, qmnt
t = qmnt.zeros(100)                  # heights t_1 ... t_100
def N(T):                            # number of zeros with 0 < t_n <= T
    return np.searchsorted(t, T, side="right")
def Nsm(T):                          # paper Eq. (1)
    u = T/(2*np.pi)
    return u*np.log(u) - u + 7/8
for T in [30, 40, 50, 100]:
    print(T, N(T), round(Nsm(T), 4), round(N(T) - Nsm(T), 4))
```

17.1.3 Why a smooth curve should cross $n - \frac{1}{2}$ at the n-th zero

A staircase that jumps from $n-1$ to n at t_n has, at that point, a riser running from height $n-1$ to height n. A smooth curve that "averages" the staircase should cut each riser near its middle, at height $n - \frac{1}{2}$. This suggests a natural smooth stand-in for the n-th zero: the height s_n at which the smooth curve reaches $n - \frac{1}{2}$. We return to this definition in Sec. 17.6. For the same reason we introduce, for a simple zero, the *midpoint value* of the staircase at t_n:

$$N^{\rm mid}(t_n) = \tfrac{1}{2}\big[N(t_n^-) + N(t_n^+)\big] = \tfrac{1}{2}\big[(n-1) + n\big] = n - \tfrac{1}{2},$$

the average of the value just below the zero $(n-1)$ and the value just above it (n). This is a separate symbol: by the definition of $N(T)$, which includes T itself, $N(t_n) = n$.

Checkpoint 17.1

Is $N(T)$ a continuous function of T? By how much does it jump at a zero of multiplicity 2?
Answer: No. It is constant between zeros and jumps at each zero. At a zero of multiplicity 2 it jumps by 2, because each zero is counted with its multiplicity. (No such zero is known.)

Key Ideas

- $N(T)$ counts the zeros with $0 < t_n \le T$; it is a staircase.
- The zeros crowd together with height: spacings near 4 at the bottom, near 1 around zero 1000.
- The smooth function $N_{\text{sm}}(t)$ of Eq. (17.2) threads the staircase, crossing each riser near its midpoint $n - \frac{1}{2}$.

17.2 Counting by winding: the argument principle

How could one count zeros of a function without finding them? For real functions of a real variable you count sign changes. For complex functions there is a much more powerful idea: walk around a closed loop and watch the *phase* of the function.

17.2.1 Following the argument continuously

Recall from Chapter 2 that any nonzero complex number w can be written $w = |w|\, e^{i \arg w}$, where the argument $\arg w$ is defined only up to adding multiples of 2π. Now let z move along a path C in the complex plane, and suppose $f(z) \neq 0$ everywhere on C. As z moves, the point $f(z)$ moves, and we can choose values of $\arg f(z)$ that change *continuously*, never jumping by 2π. (Picture following a runner on a circular track and counting laps rather than reporting only the compass direction.) The *net change of the argument along* C is

$$\Delta_C \arg f = (\text{final value of } \arg f) - (\text{initial value of } \arg f), \tag{17.3}$$

computed with this continuous choice. It does not depend on which value we start with, because changing the start by $2\pi k$ changes the finish by the same $2\pi k$.

17.2.2 One factor at a time

Take the simplest function with a zero, $f(z) = z - a$, and let C be a circle traversed once counterclockwise.

Case 1: a inside the circle. First suppose the circle is centered at a: $z = a + re^{i\phi}$, $0 \le \phi \le 2\pi$. Then $f(z) = re^{i\phi}$, whose argument is ϕ itself, so $\Delta_C \arg f = 2\pi$. If the circle is not centered at a but still encloses it, a geometric argument settles the matter (look at the left panel of Fig. 17.2). A disk is convex, so every ray leaving the interior point a crosses the circle exactly once. Hence, as z goes once round the circle, the direction of $z - a$ takes every compass direction exactly once. It can never turn back, because turning back would make it take some direction twice. So it turns steadily the same way through one full turn, and the net change is again 2π.

Case 2: a outside the circle. Draw the two tangent lines from a to the circle. The whole circle lies inside the wedge between them, and the opening angle of that wedge is less than π. The direction of $z - a$ therefore always stays inside a range of angles narrower than π. A continuously followed argument that never leaves such a range cannot gain a full turn, so it returns to its starting value: $\Delta_C \arg f = 0$. (Both cases are geometric arguments for circles; the general theorem is stated below.)

Now take a product $f(z) = (z - a_1)(z - a_2)\cdots(z - a_k)$. From Chapter 2, the argument of a product is the sum of the arguments of the factors. Therefore the net change of $\arg f$ around C is the sum of the net changes of the factors: 2π for every a_j inside C and 0 for every a_j outside. Dividing by 2π:

$$\frac{1}{2\pi}\Delta_C \arg f = \text{number of zeros of } f \text{ inside } C. \tag{17.4}$$

A zero repeated twice, $(z-a)^2$, contributes $2\times 2\pi$, so the count includes multiplicity automatically.

Sample Problem 17.3: Winding of a cubic

Let $f(z)=(z-1)(z-2)(z+3)$ and let C be the circle $|z|=2.5$ traversed counterclockwise. Predict $\Delta_C \arg f/2\pi$ and confirm it numerically.

Step 1: Locate the zeros: $z=1$, $z=2$, $z=-3$. Their distances from the origin are 1, 2 and 3.

Step 2: The circle has radius 2.5, so 1 and 2 are inside and -3 is outside.

Step 3: By Eq. (17.4), the factors $z-1$ and $z-2$ each contribute 2π, and $z+3$ contributes 0. So $\Delta_C \arg f = 4\pi$.

Step 4 (numerical check): Sample $z=2.5e^{i\phi}$ at many values of ϕ, compute the argument of f at each, remove the artificial 2π jumps (`np.unwrap`), and take last minus first. The listing below prints 2.0000 for the ratio.

$$\boxed{\frac{1}{2\pi}\Delta_C \arg f = 2.}$$

```
import numpy as np
phi = np.linspace(0, 2*np.pi, 20001)      # one counterclockwise turn
z = 2.5*np.exp(1j*phi)
f = (z - 1)*(z - 2)*(z + 3)
ph = np.unwrap(np.angle(f))               # continuous argument
print((ph[-1] - ph[0])/(2*np.pi))         # number of enclosed zeros
```

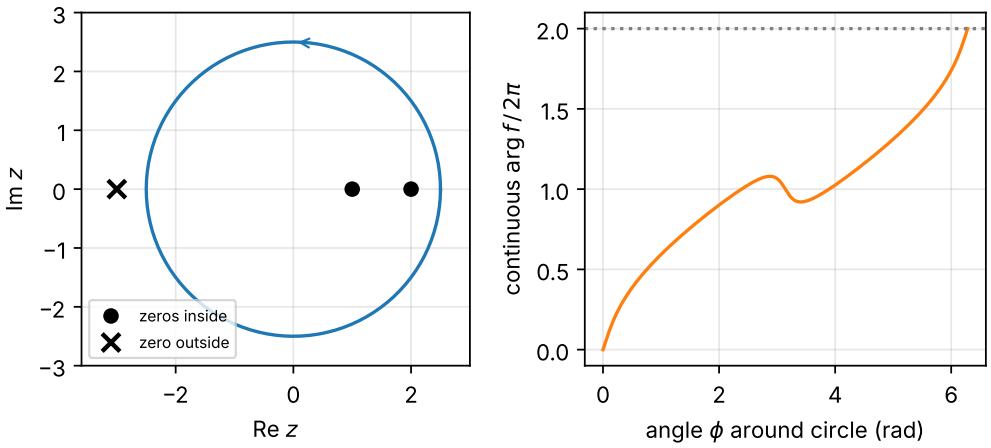


Figure 17.2: The argument principle for $f(z)=(z-1)(z-2)(z+3)$. Left: the circle $|z|=2.5$ (traversed counterclockwise) and the three zeros; two lie inside. Right: the continuously followed argument of f along the circle, divided by 2π, against the angle ϕ around the circle. It rises by exactly 2 over one turn, the number of enclosed zeros.

Pitfall

`np.angle` returns the *principal* argument, in $(-\pi,\pi]$. Each time the true argument passes π the principal value jumps down by 2π. If you subtract first from last principal values you always get something in $(-2\pi,2\pi)$, which hides the winding. Always apply `np.unwrap` (which adds back the 2π jumps) and sample the path finely enough that the true argument changes by much less than π between samples.

17.2.3 The general theorem

Most functions are not polynomials, so we cannot factor them into finitely many $(z - a_j)$. The intuition still holds. An *analytic* function (one that has a complex derivative at every point of a region) behaves near each zero like a constant times $(z - a)^k$, with k a positive integer; this local form is itself part of what we take on trust below. Away from its zeros its argument can be followed continuously inside the region without net winding. Making this precise is a theorem of complex analysis that we take on trust.

Taken on trust

The argument principle (proved; proof beyond this book). Let f be analytic (complex-differentiable) at every point inside and on a simple closed curve C, traversed counterclockwise, and suppose f has no zeros on C. Then the number of zeros of f inside C, counted with multiplicity, equals $\frac{1}{2\pi}\Delta_C \arg f$. The proof uses contour integration of f'/f (the residue theorem). See the chapter on complex variables in [22].

Two requirements of the theorem matter below: f must be analytic everywhere inside and on C (no poles), and f must not vanish on C itself (otherwise the argument is undefined there).

Checkpoint 17.2

You walk around a circle that encloses no zero of an analytic function f. What is $\Delta_C \arg f$? What goes wrong if the circle passes exactly through a zero?

Answer: The net change is 0. If the circle passes through a zero, $f = 0$ there and its argument is undefined, so the count is not defined; the theorem excludes this case.

Key Ideas

- Following the phase of f continuously around a closed loop and dividing the net change by 2π counts the zeros inside, with multiplicity.
- For products, phases add, so each enclosed zero contributes one full turn.
- The general argument principle requires f analytic inside and on the loop and nonzero on the loop.

17.3 The Riemann–von Mangoldt formula

We now apply the argument principle to count the zeros of ζ in a tall rectangle. There is one obstacle: ζ has a pole at $s = 1$ and trivial zeros at $s = -2, -4, \dots$, and the theorem needs a function analytic everywhere inside the loop whose zeros are exactly the ones we want. Riemann's remedy is to multiply ζ by factors that cancel the pole and the trivial zeros.

17.3.1 The completed zeta function

Define

$$\xi(s) = \tfrac{1}{2}\, s(s-1)\, \pi^{-s/2}\, \Gamma\!\left(\tfrac{s}{2}\right) \zeta(s), \tag{17.5}$$

where Γ is the gamma function (the generalized factorial of Chapter 4) and $\pi^{-s/2} = e^{-(s/2)\log \pi}$ as in Chapter 2. The properties we need are collected here.

Taken on trust

Properties of ξ (proved; proofs beyond this book; see [5], Chs. 1–2, or [6], Ch. 2).

(a) ξ extends to a function analytic at every point of the complex plane. The pole of ζ at $s = 1$ is canceled by the factor $s - 1$; the poles of $\Gamma(s/2)$ at $s = 0, -2, -4, \dots$ are canceled by the factor s (at $s = 0$) and by the trivial zeros of ζ (at $s = -2, -4, \dots$).

(b) Functional equation in symmetric form: $\xi(s) = \xi(1 - s)$. (This is the functional equation of Chapter 16 rewritten.)

(c) Reflection: $\xi(\bar{s}) = \overline{\xi(s)}$, because ξ is real on the real axis.

(d) $\Gamma(z)$ is never zero, and $\Gamma(x) > 0$ for real $x > 0$.

(e) $\zeta(s) \neq 0$ for $\operatorname{Re} s \geq 1$. (For $\operatorname{Re} s > 1$ this follows from the Euler product, Chapter 16; on the line $\operatorname{Re} s = 1$ it is a deeper theorem, closely tied to the prime number theorem.)

Consequences: the zeros of ξ are exactly the nontrivial zeros of ζ, all in the strip $0 < \operatorname{Re} s < 1$. Indeed, by (a), (d) and (e), none of the factors in Eq. (17.5) vanishes for $\operatorname{Re} s \geq 1$ except $s - 1$ at $s = 1$, where it cancels the pole of ζ and leaves $\xi(1) = \frac{1}{2} \neq 0$ (shown below); by (b), ξ has no zeros with $\operatorname{Re} s \leq 0$ either; inside the strip the extra factors are nonzero, so ξ vanishes exactly where ζ does, with the same multiplicity.

One more property we can prove ourselves, using only Chapter 16.
Claim: $\xi(\sigma) > 0$ for every real σ with $-1 \leq \sigma \leq 2$.
Proof. Split into ranges.

1. $1 < \sigma \leq 2$. Here the constant $\frac{1}{2}$ is positive, $\zeta(\sigma) = \sum_k k^{-\sigma} > 0$ (a sum of positive terms), $\sigma(\sigma - 1) > 0$, $\pi^{-\sigma/2} > 0$ and $\Gamma(\sigma/2) > 0$ by (d). The product of these five positive numbers is positive.

2. $0 < \sigma < 1$. Recall $\eta(\sigma) = 1 - 2^{-\sigma} + 3^{-\sigma} - 4^{-\sigma} + \cdots$, an alternating series whose terms decrease in size (because $\sigma > 0$, $k^{-\sigma}$ decreases as k increases) and tend to zero. From Chapter 3, its sum lies between any two consecutive partial sums, in particular between $1 - 2^{-\sigma}$ and 1. Since $2^{-\sigma} < 1$, the sum satisfies $\eta(\sigma) \geq 1 - 2^{-\sigma} > 0$. Also $1 - 2^{1-\sigma} < 0$ because $1 - \sigma > 0$ makes $2^{1-\sigma} > 1$. Hence $\zeta(\sigma) = \eta(\sigma)/(1 - 2^{1-\sigma}) < 0$. Meanwhile $\sigma(\sigma - 1) < 0$ (positive times negative). In Eq. (17.5) the two negative factors multiply to a positive number, and the remaining factors are positive, so $\xi(\sigma) > 0$.

3. $\sigma = 1$ and $\sigma = 0$. Near $s = 1$, $(s - 1)\zeta(s) \to 1$ (the pole has residue 1, Chapter 16), and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, so $\xi(1) = \frac{1}{2} \cdot 1 \cdot \pi^{-1/2} \cdot \sqrt{\pi} \cdot 1 = \frac{1}{2} > 0$. By (b), $\xi(0) = \xi(1) = \frac{1}{2}$.

4. $-1 \leq \sigma < 0$. By (b), $\xi(\sigma) = \xi(1 - \sigma)$, and $1 - \sigma$ lies in $(1, 2]$, where we showed $\xi > 0$. □

17.3.2 The contour

Fix a height $T > 0$ that is *not* the height of any zero. Let R be the rectangle with corners -1, 2, $2 + iT$, $-1 + iT$ (Fig. 17.3), traversed counterclockwise. Check the conditions of the argument principle for $f = \xi$:

- ξ is analytic everywhere, by (a).
- No zeros on the bottom edge (the real segment $[-1, 2]$): we just proved $\xi > 0$ there.

- No zeros on the right edge $\operatorname{Re} s = 2$ or the left edge $\operatorname{Re} s = -1$: all zeros of ξ have $0 < \operatorname{Re} s < 1$.
- No zeros on the top edge $\operatorname{Im} s = T$: we chose T not to be the height of a zero.

Inside R the zeros of ξ are the nontrivial zeros with $0 < \operatorname{Im} \rho < T$; there are $N(T)$ of them. So

$$N(T) = \frac{1}{2\pi} \Delta_{\partial R} \arg \xi, \tag{17.6}$$

where ∂R denotes the boundary of R traversed counterclockwise.

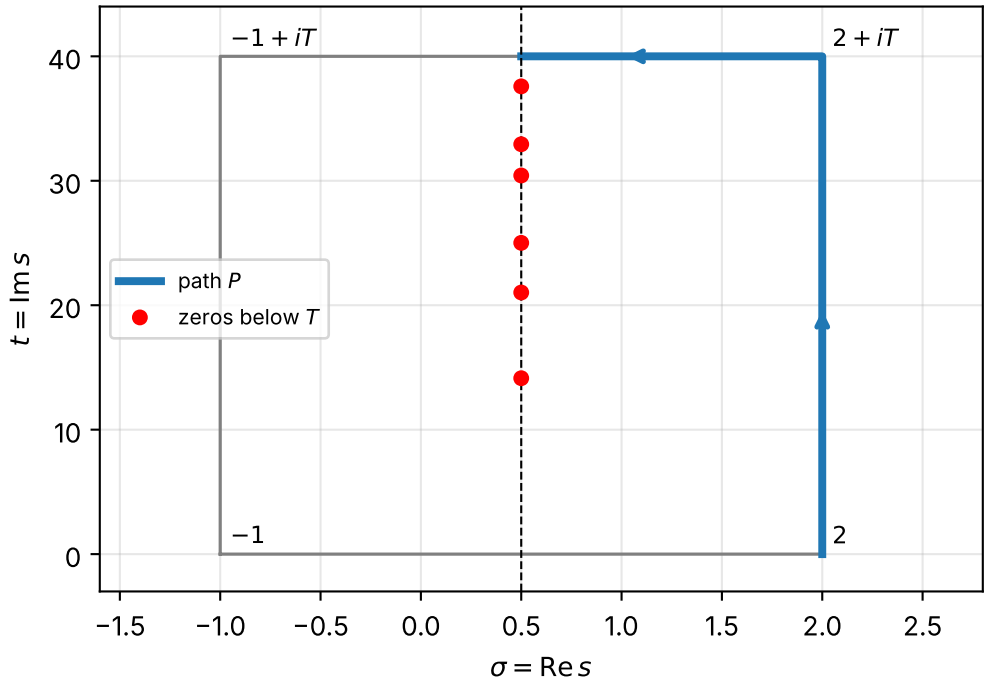


Figure 17.3: The contour used to count zeros of ξ up to height $T = 40$. The rectangle has corners -1, 2, $2 + iT$ and $-1 + iT$ in the complex s-plane ($\sigma = \operatorname{Re} s$, $t = \operatorname{Im} s$). Dots: the six zeros $\frac{1}{2} + it_n$ below height 40. The thick path P runs from 2 up to $2 + iT$ and left to $\frac{1}{2} + iT$; by symmetry, the change of $\arg \xi$ around the whole rectangle is twice its change along P. The dashed line is the critical line $\sigma = \frac{1}{2}$.

17.3.3 Halving the contour by symmetry

Split ∂R into three pieces:

1. the bottom edge, from -1 to 2;
2. the right half P: from 2 up to $2 + iT$, then left to $\frac{1}{2} + iT$;
3. the left half P': from $\frac{1}{2} + iT$ left to $-1 + iT$, then down to -1.

(The top edge is split at $\frac{1}{2} + iT$ between P and P'.)

Bottom edge. There ξ is real and positive, so its argument can be taken to be 0 all along. The net change is 0.

Left half. Consider the map $s \mapsto s^* = 1 - \bar{s}$, which reflects a point across the critical line $\operatorname{Re} s = \frac{1}{2}$ (it sends $\sigma + it$ to $(1 - \sigma) + it$). Combining properties (b) and (c):

$$\xi(1 - \bar{s}) = \xi(\bar{s}) = \overline{\xi(s)}. \tag{17.7}$$

The first equality is (b) applied at the point $\bar{s}$; the second is (c). Now the reflection sends $2 \mapsto -1$, $2+iT \mapsto -1+iT$ and $\frac{1}{2}+iT \mapsto \frac{1}{2}+iT$. So P', which runs $\frac{1}{2}+iT \to -1+iT \to -1$, is the mirror image of P run backwards ($\frac{1}{2}+iT \to 2+iT \to 2$). Two things happen to the argument:

- Running P backwards reverses the sign of the net change: $-\Delta_P \arg \xi$.
- By Eq. (17.7), the values of ξ on P' are the complex conjugates of the values at the mirror points, and conjugation negates the argument. This reverses the sign again.

The two sign reversals cancel: $\Delta_{P'} \arg \xi = +\Delta_P \arg \xi$. Adding the three pieces,

$$\Delta_{\partial R} \arg \xi = 0 + \Delta_P \arg \xi + \Delta_{P'} \arg \xi = 2\,\Delta_P \arg \xi, \tag{17.8}$$

and Eq. (17.6) becomes

$$N(T) = \frac{1}{\pi} \Delta_P \arg \xi. \tag{17.9}$$

17.3.4 Factor by factor

Since ξ is a product, $\Delta_P \arg \xi$ is the sum of the changes of the arguments of the factors in Eq. (17.5) (Chapter 2). We take them one at a time along P, which starts at $s = 2$ and ends at $s = \frac{1}{2}+iT$.

The constant $\frac{1}{2}$. Its argument is always 0: change 0.

The factor $s(s-1)$. At the start, $s = 2$ and $s-1 = 1$ are both positive reals, with argument 0. Along the vertical part of P, both s and $s-1$ have positive real part, so their arguments stay in $(-\pi/2, \pi/2)$ without jumps. Along the horizontal part, the imaginary part is fixed at $T > 0$, so the arguments of s and $s-1$ stay in $(0, \pi)$, again without jumps. At the end:

$$\begin{aligned} \arg s &= \arg\Big(\tfrac{1}{2}+iT\Big) = \arctan(2T), \\ \arg(s-1) &= \arg\Big(-\tfrac{1}{2}+iT\Big) = \pi - \arctan(2T), \end{aligned}$$

where the second line uses that $-\frac{1}{2}+iT$ is the mirror image of $\frac{1}{2}+iT$ across the imaginary axis. Adding, the change of $\arg[s(s-1)]$ is

$$\arctan(2T) + \pi - \arctan(2T) = \pi. \tag{17.10}$$

The factor $\pi^{-s/2}$. With $s = \sigma + it$, $\pi^{-s/2} = e^{-(\sigma/2)\log\pi}\, e^{-i(t/2)\log\pi}$. The first factor is a positive real number; the second has argument $-(t/2)\log\pi$. This argument is 0 at $s = 2$ and $-(T/2)\log\pi$ at the end. Change: $-\frac{T}{2}\log\pi$.

The factor $\Gamma(s/2)$. As s runs along P, $s/2$ runs from 1 to $1+iT/2$ to $\frac{1}{4}+iT/2$, always in the right half-plane $\operatorname{Re} z > 0$, where Γ is analytic and never zero. At the start $\Gamma(1) = 1$, argument 0. The change is therefore $\arg\Gamma(\frac{1}{4}+\frac{iT}{2})$, where the argument is followed continuously from $\Gamma(1)$. Equivalently, it is $\operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{iT}{2})$, where $\log\Gamma$ is the branch of the logarithm that is real on the positive real axis and continuous in the right half-plane. (Because Γ has no zeros in the right half-plane, any path from 1 to $\frac{1}{4}+iT/2$ inside it gives the same continuous argument; this is the branch used by Stirling's formula in Sec. 17.4.)

The factor $\zeta(s)$. We give its change a name. Along P, ζ does not vanish: on $\operatorname{Re} s = 2$ by the Euler product, and on the top edge because T is not the height of a zero. At the start $\zeta(2) = \pi^2/6 > 0$, argument 0.

Definitions. The *Riemann–Siegel theta function* is

$$\theta(T) = \operatorname{Im}\log\Gamma\Big(\tfrac{1}{4}+\tfrac{iT}{2}\Big) - \frac{T}{2}\log\pi, \tag{17.11}$$

with the continuous branch just described (so $\theta(0) = 0$). The function $S(T)$ is

$$S(T) = \frac{1}{\pi}\Delta_P \arg\zeta = \frac{1}{\pi}\arg\zeta\Big(\tfrac{1}{2} + iT\Big), \tag{17.12}$$

where $\arg\zeta(\frac{1}{2} + iT)$ means the value reached by continuous variation along P, starting from $\arg\zeta(2) = 0$.

The factors $\frac{1}{2}$, $s(s-1)$, $\pi^{-s/2}$ and $\Gamma(s/2)$ together contribute $0 + \pi - \frac{T}{2}\log\pi + \operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{iT}{2}) = \pi + \theta(T)$, and ζ contributes $\pi S(T)$. Insert into Eq. (17.9):

$$N(T) = \frac{1}{\pi}\big[\pi + \theta(T) + \pi S(T)\big] \quad\Longrightarrow\quad \boxed{N(T) = \frac{\theta(T)}{\pi} + 1 + S(T)} \tag{17.13}$$

for every $T > 0$ that is not the height of a zero. This is the **Riemann–von Mangoldt formula**. Riemann stated the leading behavior in 1859 [2]; von Mangoldt gave the first complete proof decades later [5, 6]. Its status in this book: a theorem. The proof above is complete modulo the results taken on trust in the boxes: the argument principle, and properties (a)–(e) of ξ and Γ (of which (e) on the line $\operatorname{Re} s = 1$ is a deep theorem).

Notice what we have achieved. The left side counts zeros, which are hard to find. On the right, $\theta(T)$ is a property of the gamma function, which we understand very well, and all the difficulty is pushed into $S(T)$, the argument of ζ on the critical line.

Checkpoint 17.3

Why does the formula require that T not be the height of a zero? Name the step of the derivation that fails.

Answer: If $T = t_n$, then ξ (and ζ) vanish at a point of the top edge of the rectangle, so $\arg\xi$ is undefined there and the argument principle does not apply. Equivalently, $S(T)$ in Eq. (17.12) is undefined at $T = t_n$.

Key Ideas

- $\xi(s)$ removes the pole and trivial zeros of ζ and is symmetric under $s \to 1 - s$; its zeros are exactly the nontrivial zeros.
- Counting zeros of ξ in a rectangle with the argument principle, and halving by symmetry, gives $N(T) = \theta(T)/\pi + 1 + S(T)$, Eq. (17.13).
- The $+1$ comes from the factor $s(s-1)$, whose argument changes by π; θ comes from $\pi^{-s/2}\Gamma(s/2)$; S comes from ζ itself.

17.4 The theta function and the constant $\frac{7}{8}$

17.4.1 Stirling's formula for complex argument

Recall from Chapter 4 Stirling's formula, stated there for complex argument:

Taken on trust

Stirling's formula (proved; proof beyond this book; see [22], or any text on the gamma function). As $|z| \to \infty$ with $|\arg z| \le \pi - \delta$ for some fixed $\delta > 0$,

$$\log\Gamma(z) = \Big(z - \tfrac{1}{2}\Big)\log z - z + \tfrac{1}{2}\log(2\pi) + \frac{1}{12z} + \mathcal{O}\Big(|z|^{-3}\Big), \tag{17.14}$$

where $\log z$ is the principal logarithm and $\log\Gamma$ is the branch real on the positive real axis.

We apply it with $z = \frac{1}{4} + \frac{it}{2}$, $t > 0$ large. Then $|z| \approx t/2$ is large and $\arg z$ is close to $\pi/2$, safely inside the allowed range.

17.4.2 Derivation of the asymptotic formula for θ

We need $\operatorname{Im}\log\Gamma(z)$, term by term. Throughout, $t > 0$.

Step A: modulus of z.

$$|z|^2 = \frac{1}{16} + \frac{t^2}{4} = \frac{t^2}{4}\left(1 + \frac{1}{4t^2}\right). \tag{17.15}$$

Taking logarithms and halving,

$$\log|z| = \log\frac{t}{2} + \frac{1}{2}\log\left(1 + \frac{1}{4t^2}\right). \tag{17.16}$$

Using $\log(1+y) = y - \frac{y^2}{2} + \cdots$ with $y = 1/(4t^2)$ (Chapter 3),

$$\log|z| = \log\frac{t}{2} + \frac{1}{8t^2} + \mathcal{O}(t^{-4}). \tag{17.17}$$

Step B: argument of z. Since $\operatorname{Re} z = \frac{1}{4} > 0$, $\arg z = \arctan(\operatorname{Im} z/\operatorname{Re} z) = \arctan(2t)$. For $a > 0$, if $\varphi = \arctan a \in (0, \pi/2)$ then $\tan(\pi/2 - \varphi) = \cot\varphi = 1/a$, so $\arctan a = \pi/2 - \arctan(1/a)$. Hence

$$\arg z = \frac{\pi}{2} - \arctan\frac{1}{2t}. \tag{17.18}$$

For small y, $\arctan y = y - \frac{y^3}{3} + \cdots$: this follows by integrating $\frac{\mathrm{d}}{\mathrm{d}y}\arctan y = \frac{1}{1+y^2} = 1 - y^2 + y^4 - \cdots$ (a geometric series in $-y^2$, Chapter 3) term by term from 0. With $y = 1/(2t)$,

$$\arg z = \frac{\pi}{2} - \frac{1}{2t} + \mathcal{O}(t^{-3}). \tag{17.19}$$

Step C: the term $(z - \frac{1}{2})\log z$. Write $z - \frac{1}{2} = -\frac{1}{4} + \frac{it}{2}$ and $\log z = \log|z| + i\arg z$. For two complex numbers $(a+ib)(c+id)$ the imaginary part is $ad + bc$. Here $a = -\frac{1}{4}$, $b = \frac{t}{2}$, $c = \log|z|$, $d = \arg z$:

$$\operatorname{Im}\left[\left(z - \tfrac{1}{2}\right)\log z\right] = -\frac{1}{4}\arg z + \frac{t}{2}\log|z|. \tag{17.20}$$

Substitute Eqs. (17.17) and (17.19):

$$\frac{t}{2}\log|z| = \frac{t}{2}\log\frac{t}{2} + \frac{1}{16t} + \mathcal{O}(t^{-3}), \tag{17.21}$$

$$-\frac{1}{4}\arg z = -\frac{\pi}{8} + \frac{1}{8t} + \mathcal{O}(t^{-3}). \tag{17.22}$$

Step D: the term $-z$. $\operatorname{Im}(-z) = -\dfrac{t}{2}$.

Step E: the constant. $\frac{1}{2}\log(2\pi)$ is real: imaginary part 0.

Step F: the term $1/(12z)$. Multiply numerator and denominator by $\bar z$: $\dfrac{1}{z} = \dfrac{\bar z}{|z|^2} = \dfrac{\frac{1}{4} - \frac{it}{2}}{|z|^2}$.

So

$$\operatorname{Im}\frac{1}{12z} = -\frac{t/2}{12|z|^2} = -\frac{t}{24|z|^2} = -\frac{t}{6t^2\left(1 + \frac{1}{4t^2}\right)} = -\frac{1}{6t} + \mathcal{O}(t^{-3}), \tag{17.23}$$

where we used $24\,|z|^2 = 24 \cdot \frac{t^2}{4}(1 + \frac{1}{4t^2}) = 6t^2(1 + \frac{1}{4t^2})$ and $1/(1+y) = 1 - y + \cdots$.

Step G: the remainder. $\mathcal{O}(|z|^{-3}) = \mathcal{O}(t^{-3})$.

Step H: add everything. Collecting Steps C to G,

$$\operatorname{Im} \log \Gamma\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) = \frac{t}{2}\log\frac{t}{2} - \frac{t}{2} - \frac{\pi}{8} + \left(\frac{1}{16} + \frac{1}{8} - \frac{1}{6}\right)\frac{1}{t} + \mathcal{O}(t^{-3}). \tag{17.24}$$

The bracket is $\frac{3}{48} + \frac{6}{48} - \frac{8}{48} = \frac{1}{48}$.

Step I: subtract $\frac{t}{2}\log\pi$ (Eq. (17.11)) and combine logarithms, $\frac{t}{2}\log\frac{t}{2} - \frac{t}{2}\log\pi = \frac{t}{2}\log\frac{t}{2\pi}$:

$$\boxed{\theta(t) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \mathcal{O}(t^{-3}).} \tag{17.25}$$

Status: a proved asymptotic expansion (given Stirling). Carrying Stirling one term further shows that the $\mathcal{O}(t^{-3})$ term is $\frac{7}{5760\,t^3}$; you can confirm this coefficient numerically in Problem 23.

17.4.3 Where $\frac{7}{8}$ comes from

Divide Eq. (17.25) by π:

$$\frac{\theta(T)}{\pi} = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} - \frac{1}{8} + \frac{1}{48\pi T} + \mathcal{O}(T^{-3}). \tag{17.26}$$

Add 1 and use Eq. (17.13):

$$N(T) = \underbrace{\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8}}_{N_{\mathrm{sm}}(T)} + S(T) + \frac{1}{48\pi T} + \mathcal{O}(T^{-3}). \tag{17.27}$$

The constant is $-\frac{1}{8} + 1 = \frac{7}{8}$: the $-\frac{1}{8}$ is the $-\pi/8$ of Stirling divided by π, and the $+1$ is the half turn of $s(s-1)$. This is exactly the paper's Eq. (1). The leftover $1/(48\pi T)$ is tiny: at $T = t_1 = 14.13$ it is 4.7×10^{-4}, and it shrinks further with height. So, up to an $\mathcal{O}(1/T)$ term,

$$N(T) \approx N_{\mathrm{sm}}(T) + S(T). \tag{17.28}$$

Sample Problem 17.4: The theta function at $T = 50$

Evaluate $\theta(50)$ from Eq. (17.25), compare with the exact value from `mpmath.siegeltheta`, and deduce $S(50)$ from Eq. (17.13).

Step 1: From the earlier Sample Problem, $\log(50/2\pi) = 2.0741459$. So $\frac{50}{2}\log\frac{50}{2\pi} = 25 \times 2.0741459 = 51.853648$.

Step 2: Subtract $\frac{t}{2} = 25$: 26.853648.

Step 3: Subtract $\pi/8 = 0.392699$: 26.460949.

Step 4: Add $\frac{1}{48\times 50} = \frac{1}{2400} = 0.000417$: $\theta(50) \approx 26.461366$. The exact value, `mpmath.siegeltheta(50)`, agrees to all digits shown; the next correction $7/(5760 \times 50^3)$ is about 10^{-8}.

Step 5: $\theta(50)/\pi = 26.461366/3.141593 = 8.422914$.

Step 6: From Eq. (17.13) with $N(50) = 10$: $S(50) = 10 - 1 - 8.422914 = 0.577086$.

$$\boxed{\theta(50) \approx 26.4614, \qquad S(50) \approx +0.577.}$$

This agrees with `mpmath.backlunds(50)`, which computes S directly from the argument of ζ, and with $N(50) - N_{\mathrm{sm}}(50) = 0.577$ found earlier (the two differ only by $1/(48\pi \cdot 50) = 1.3 \times 10^{-4}$).

```
import numpy as np, mpmath as mp
def theta_asym(t):                          # Eq. (counting-zeros-thetaasym)
    return t/2*np.log(t/(2*np.pi)) - t/2 - np.pi/8 + 1/(48*t)
for t in [10.0, 30.0, 50.0]:
    exact = float(mp.siegeltheta(t))
    print(t, exact, theta_asym(t), exact - theta_asym(t))
# Riemann--von Mangoldt check: S(T) = N(T) - 1 - theta(T)/pi
print(10 - 1 - float(mp.siegeltheta(50))/np.pi, float(mp.backlunds(50)))
```

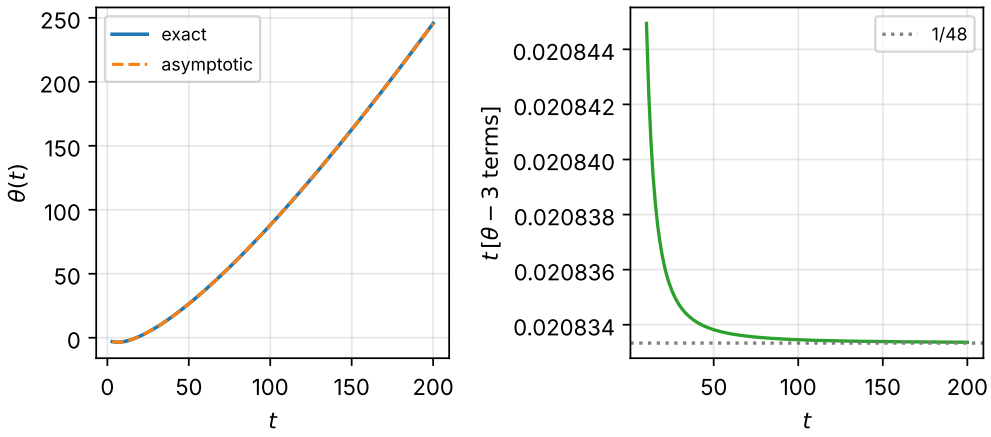


Figure 17.4: Left: the Riemann–Siegel theta function $\theta(t)$ computed exactly with `mpmath.siegeltheta` (solid) and the asymptotic formula Eq. (17.25) without its $\mathcal{O}(t^{-3})$ remainder (dashed); the two are indistinguishable except at small t. Right: t times the difference between the exact $\theta(t)$ and the first three terms $\frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8}$, against t; it approaches $1/48$ (horizontal line), confirming the $1/(48t)$ term derived in Steps A to I. Both axes are dimensionless.

Pitfall

Do not confuse $\theta(T)/\pi + 1$ with $N_{\mathrm{sm}}(T)$. They differ by $1/(48\pi T) + \mathcal{O}(T^{-3})$. This book puts that tiny term into the "fluctuating part"; the paper simply uses Eq. (1) as its smooth part and never mentions the term. Some texts instead define the fluctuating part exactly as $S(T)$ and keep the $1/(48\pi T)$ in the smooth part. At the heights of this book the difference is below 5×10^{-4} of a zero, far below every effect we study.

Checkpoint 17.4

Where does each part of the constant $\frac{7}{8}$ come from?
Answer: $-\frac{1}{8}$ is the $-\pi/8$ in Stirling's expansion of $\operatorname{Im}\log\Gamma(\frac{1}{4} + \frac{iT}{2})$, divided by π; it comes from $-\frac{1}{4}\arg z$ with $\arg z \to \pi/2$. The $+1$ is the change π in the argument of the factor $s(s-1)$ along the path, divided by π.

Key Ideas

- Stirling's formula gives $\theta(t) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \mathcal{O}(t^{-3})$, derived term by term in Steps A to I.
- Hence $N(T) = N_{\mathrm{sm}}(T) + S(T) + \frac{1}{48\pi T} + \mathcal{O}(T^{-3})$, with N_{sm} the paper's Eq. (1).

- The constant $\frac{7}{8} = -\frac{1}{8} + 1$ has a precise origin in the gamma factor and the factor $s(s-1)$.

17.5 The fluctuating part $S(T)$

★ Did you know? *The Most Slowly Growing Noise*

In 1946 Atle Selberg proved that the fluctuating part $S(T)$ of the zero count behaves statistically like a Gaussian random variable with variance about $\frac{1}{2\pi^2}\log\log T$. That growth is absurdly slow: even at $T = 10^{20}$, $\log\log T$ is only about 3.8. This is why $S(T)$ stays small in every computation, although it is known to be unbounded.

17.5.1 Shape of $S(T)$

Rearranging Eq. (17.13),

$$S(T) = N(T) - 1 - \frac{\theta(T)}{\pi}. \tag{17.29}$$

Two properties follow at once.

At a zero, S jumps up. θ is continuous, but N jumps by 1 at a simple zero, so S jumps by $+1$ there (by k at a zero of multiplicity k).

Between zeros, S decreases steadily. Between consecutive zeros N is constant, so

$$\frac{\mathrm{d}S}{\mathrm{d}T} = -\frac{\theta'(T)}{\pi}. \tag{17.30}$$

Differentiating an asymptotic expansion term by term is not automatically legitimate: a small remainder can still have a large derivative. Here it is legitimate, because Stirling's expansion of $\log\Gamma(z)$ may itself be differentiated term by term in the same sector, giving the corresponding expansion of $\Gamma'(z)/\Gamma(z)$ with remainder $\mathcal{O}(|z|^{-4})$; this is a proved property of the gamma function that we take on trust (it is proved in texts on the gamma function). With that understood, differentiating Eq. (17.25) term by term (product rule on the first term: $\frac{\mathrm{d}}{\mathrm{d}T}\left[\frac{T}{2}\log\frac{T}{2\pi}\right] = \frac{1}{2}\log\frac{T}{2\pi} + \frac{T}{2}\cdot\frac{1}{T} = \frac{1}{2}\log\frac{T}{2\pi} + \frac{1}{2}$),

$$\theta'(T) = \frac{1}{2}\log\frac{T}{2\pi} + \frac{1}{2} - \frac{1}{2} - \frac{1}{48T^2} + \mathcal{O}(T^{-4}) = \frac{1}{2}\log\frac{T}{2\pi} - \frac{1}{48T^2} + \mathcal{O}(T^{-4}), \tag{17.31}$$

so

$$\frac{\mathrm{d}S}{\mathrm{d}T} = -\frac{1}{2\pi}\log\frac{T}{2\pi} + \frac{1}{48\pi T^2} + \mathcal{O}(T^{-4}). \tag{17.32}$$

For $T > 2\pi$ the slope is negative. So $S(T)$ is a *sawtooth*: it drifts down between zeros and jumps up by 1 at each zero (Fig. 17.5). If the zeros were perfectly evenly spaced at the local mean spacing, the downward drift between two zeros would exactly cancel each jump, and S would oscillate between fixed limits. The actual irregular placement of the zeros is what makes S wander.

Sample Problem 17.5: Values of S at several heights

Using N_{sm} and the counts $N(30) = 3$, $N(40) = 6$, $N(100) = 29$, find $S(30)$, $S(40)$ and $S(100)$.

Step 1: From Eq. (17.27), $S(T) = N(T) - N_{\mathrm{sm}}(T) - \frac{1}{48\pi T}$ to excellent accuracy (error $\mathcal{O}(T^{-3})$, below 10^{-7} here).

Step 2: $T = 30$: $\log(30/2\pi) = \log 30 - \log 2\pi = 3.401197 - 1.837877 = 1.563320$. Then

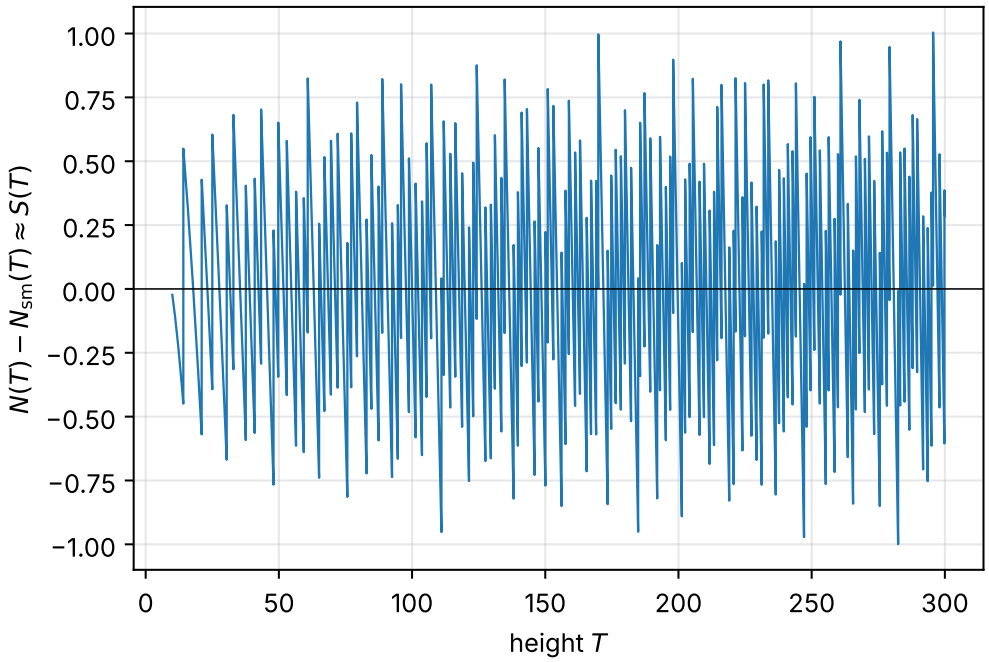


Figure 17.5: The fluctuating part $N(T) - N_{\mathrm{sm}}(T)$, which equals $S(T)$ up to the term $1/(48\pi T)$, for $10 \le T \le 300$ (dimensionless count against height T). It jumps by $+1$ at every zero and decreases between zeros with slope $-\frac{1}{2\pi}\log\frac{T}{2\pi}$. Despite the jumps, it stays within a band of width about 2 around zero over the whole range shown.

$N_{\mathrm{sm}}(30) = 4.774648\times 1.563320 - 4.774648 + 0.875 = 7.464305 - 4.774648 + 0.875 = 3.564656$. And $\frac{1}{48\pi\times 30} = 0.000221$. So $S(30) = 3 - 3.564656 - 0.000221 = -0.564877$.

Step 3: $T = 40$: $\log(40/2\pi) = 3.688879 - 1.837877 = 1.851002$; $N_{\mathrm{sm}}(40) = 6.366198 \times 0.851002 + 0.875 = 5.417650 + 0.875 = 6.292650$. (Here we used $u\log u - u = u(\log u - 1)$ with $u = T/2\pi$.) Then $S(40) = 6 - 6.292650 - 0.000166 = -0.292816$.

Step 4: $T = 100$: $\log(100/2\pi) = 4.605170 - 1.837877 = 2.767293$; $N_{\mathrm{sm}}(100) = 15.915494 \times 1.767293 + 0.875 = 28.127344 + 0.875 = 29.002344$. Then $S(100) = 29 - 29.002344 - 0.000066 = -0.002410$.

$$\boxed{S(30) \approx -0.565, \qquad S(40) \approx -0.293, \qquad S(100) \approx -0.002.}$$

Each value agrees with `mpmath.backlunds` (see the CHECKS script). All three, and $S(50) = +0.577$ from the previous Sample Problem, are well below 1 in magnitude.

Checkpoint 17.5

Just after a zero, is $S(T)$ larger or smaller than just before it? Between two zeros, does it increase or decrease?

Answer: Larger: it jumps up by 1 at a simple zero. Between zeros it decreases, with slope $-\frac{1}{2\pi}\log\frac{T}{2\pi}$ (for $T > 2\pi$).

17.5.2 How big is $S(T)$?

The numbers above suggest that S is a quantity of order a few tenths. What is known for certain is weaker but still striking.

Taken on trust

Size of $S(T)$ (proved; proofs beyond this book; see [6], Ch. 9). (i) $S(T) = \mathcal{O}(\log T)$: it grows at most logarithmically. (ii) $S(T)$ is not bounded: it takes arbitrarily large positive and negative values as $T \to \infty$, but extremely slowly. (iii) Selberg proved that its mean square grows like $\frac{1}{2\pi^2}\log\log T$, so its typical size grows like $\sqrt{\log\log T}$. The function $\log\log T$ is famously sluggish: it is about 1.9 at $T = 10^3$ and about 2.2 at $T = 10^4$.

So at the heights used in the paper (up to about $t \approx 9900$, zero 10^4) we expect S to be of order a few tenths, growing only very gently with height. The paper's own numbers confirm this: the rms size of the fluctuation at the zeros (defined precisely in Sec. 17.7) grows from 0.263 on zeros 401–500 to 0.296 on zeros 5001–10000 (paper Table V, "null" column).

Historically, Backlund in 1914 used exactly this machinery to show that there are precisely 79 zeros below height 200 [5]; you can confirm with `qmnt.zeros` that $t_{79} < 200 < t_{80}$.

17.5.3 Counting zeros on the line: how RH is checked numerically

Chapter 16 introduced the Riemann–Siegel Z function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$, which is real for real t; its θ is the same theta function as Eq. (17.11). Because $|Z(t)| = \left|\zeta(\frac{1}{2}+it)\right|$, every sign change of Z between two heights marks a zero *on the critical line*. Now combine two facts:

- Counting sign changes of Z in $(0, T)$ gives a *lower bound* on the number of zeros on the line below T (each sign change needs at least one zero of odd multiplicity).
- The Riemann–von Mangoldt formula gives the *total* number of zeros below T, on or off the line, with multiplicity.

If the two numbers agree, then every zero below T lies on the line and is simple: there is no room left for a zero off the line or a double zero. The catch is that Eq. (17.13) needs the exact integer $N(T)$, and $S(T)$ is not given by any closed formula.

Taken on trust

Turing's method (proved; details beyond this book; see [5], Ch. 8). Turing showed how to determine the integer $N(T)$ rigorously from computed values of Z, using a bound on the average of $S(t)$ over an interval. This is the method by which RH has been verified numerically for very many zeros (Chapter 16 reports the scale). It verifies RH up to a height; it does not prove RH, which remains open.

Key Ideas

- $S(T) = N(T) - 1 - \theta(T)/\pi$ is a sawtooth: jumps of $+1$ at zeros, downward drift of slope $-\frac{1}{2\pi}\log\frac{T}{2\pi}$ between them.
- $S(T) = \mathcal{O}(\log T)$ is proved; its typical size grows like $\sqrt{\log\log T}$, and at the heights in this book it is a few tenths.
- Comparing $N(T)$ with sign changes of $Z(t)$ is how zeros are verified to lie on the critical line.

17.6 Smooth zeros and the mean spacing

17.6.1 Definition and existence

Following the "middle of the riser" idea of Sec. 17.1:

Definition. The n-th *smooth zero* s_n is the solution of

$$N_{\mathrm{sm}}(s_n) = n - \tfrac{1}{2}, \qquad s_n > 2\pi. \tag{17.33}$$

Why is there exactly one such solution? Differentiate Eq. (17.2). With the product rule on the first term,

$$N'_{\mathrm{sm}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} - \frac{1}{2\pi} = \frac{1}{2\pi}\log\frac{t}{2\pi}. \tag{17.34}$$

This is negative for $t < 2\pi$, zero at $t = 2\pi$ and positive for $t > 2\pi$. So N_{sm} has its minimum at $t = 2\pi$, where $N_{\mathrm{sm}}(2\pi) = 1\cdot\log 1 - 1 + \frac{7}{8} = -\frac{1}{8}$, and it increases for $t > 2\pi$ without bound (since $N'_{\mathrm{sm}}(t) \to \infty$). An increasing continuous function that starts at $-\frac{1}{8}$ and grows without bound takes every value above $-\frac{1}{8}$ exactly once; in particular it takes the value $n - \frac{1}{2} \geq \frac{1}{2}$ exactly once on $(2\pi, \infty)$. That is why we add the condition $s_n > 2\pi$: on the decreasing branch $t < 2\pi$ the function also reaches small positive values, but that branch has nothing to do with counting zeros (there are no zeros below $t_1 = 14.13$).

17.6.2 Computing s_n

There are two routes: numerical root finding on a monotone function (Chapter 4), and an exact inversion using the Lambert W function, defined briefly in that chapter. We derive the second route here and use both.

Root finding. Bisection or Brent's method needs a bracket $[a, b]$ with $N_{\mathrm{sm}}(a) < n - \frac{1}{2} < N_{\mathrm{sm}}(b)$. Take $a = 2\pi$, where $N_{\mathrm{sm}} = -\frac{1}{8}$. For b take $2\pi(n + 10)$: with $u = n + 10 \geq 11 > e^2$ we have $\log u - 1 > 1$, so $N_{\mathrm{sm}}(b) = u(\log u - 1) + \frac{7}{8} > u > n - \frac{1}{2}$. This guarantees a valid bracket for every $n \geq 1$.

Lambert W. Recall that $W(x)$ is the solution w of $we^w = x$; for $x \geq -1/e$ the principal branch gives the solution with $w \geq -1$. Equation (17.33) reads

$$\frac{s}{2\pi}\left(\log\frac{s}{2\pi} - 1\right) = n - \frac{1}{2} - \frac{7}{8} = n - \frac{11}{8}. \tag{17.35}$$

Let $v = \log\frac{s}{2\pi} - 1$, so $\frac{s}{2\pi} = e^{1+v}$. Substituting, $e^{1+v}v = n - \frac{11}{8}$, that is

$$ve^v = \frac{n - \frac{11}{8}}{e} \equiv x_n. \tag{17.36}$$

Since $s > 2\pi$ means $v > -1$, v is the principal branch: $v = W(x_n)$. Then $\frac{s}{2\pi} = e\cdot e^v$, and from $ve^v = x_n$ we have $e^v = x_n/v$, so $\frac{s}{2\pi} = \frac{e\,x_n}{v} = \frac{n-\frac{11}{8}}{W(x_n)}$. Hence

$$\boxed{s_n = \frac{2\pi\left(n - \frac{11}{8}\right)}{W\!\left(\frac{n-11/8}{e}\right)}.} \tag{17.37}$$

This is an exact solution of Eq. (17.33), not an approximation. For $n = 1$, $n - \frac{11}{8} = -\frac{3}{8}$ and $x_1 = -0.138$, which is above $-1/e = -0.368$, so the principal branch exists and gives $W(x_1) < 0$; numerator and denominator are both negative and $s_1 > 0$.

For large x, take logarithms of $we^w = x$: $w + \log w = \log x$. For large x (so large w), $\log w$ is much smaller than w, so to leading order $w \approx \log x$, that is $W(x) \approx \log x$. For large n, $n - \frac{11}{8} \approx n$ and $\log x_n \approx \log(n/e) = \log n - 1 \approx \log n$, which gives the leading behavior $s_n \approx 2\pi n/\log n$: the n-th zero sits at a height growing slightly slower than linearly in n.

Sample Problem 17.6: The first smooth zero by Newton's method

Find s_1 by Newton's method, starting from $t_1 = 14.134725$, and compare with t_1.
Step 1: Evaluate $N_{\rm sm}(t_1)$. With $u = t_1/2\pi = 2.249611$, $\log u = 0.810757$, so $N_{\rm sm}(t_1) = 2.249611 \times (0.810757 - 1) + 0.875 = -0.425722 + 0.875 = 0.449278$.
Step 2: The Newton update for solving $g(t) = N_{\rm sm}(t) - \frac{1}{2} = 0$ is $t \to t - g(t)/g'(t)$ with $g' = N'_{\rm sm}$ from Eq. (17.34). At t_1: $N'_{\rm sm}(t_1) = 0.810757/6.283185 = 0.129036$, so the step is $+0.050722/0.129036 = 0.393085$, giving 14.527810.
Step 3: Repeat at 14.527810: now $u = 14.527810/2\pi = 2.312173$ and $\log u = 0.838188$, so $N_{\rm sm} = 2.312173 \times (0.838188 - 1) + 0.875 = -0.374138 + 0.875 = 0.500862$, and $N'_{\rm sm} = 0.838188/6.283185 = 0.133402$. The step is $-0.000862/0.133402 = -0.006462$, giving 14.521348.
Step 4: One more iteration changes the value only in the sixth decimal place, to 14.521346; Newton's method has converged.
Step 5: Compare: $t_1 - s_1 = 14.134725 - 14.521346 = -0.386621$. The true first zero lies below its smooth position.

$$\boxed{s_1 \approx 14.5213, \qquad t_1 - s_1 \approx -0.387.}$$

Equation (17.37) gives the same s_1 with $W(-0.13795) = -0.16226$.

```
import numpy as np
from scipy.optimize import brentq
from scipy.special import lambertw
def Nsm(t):
    u = t/(2*np.pi); return u*np.log(u) - u + 7/8
def s_brent(n):          # bracket [2 pi, 2 pi (n+10)] always works
    return brentq(lambda t: Nsm(t) - (n - 0.5), 2*np.pi, 2*np.pi*(n + 10),
                  xtol=1e-13)
def s_lambert(n):        # exact inversion, Eq. (counting-zeros-lambert)
    w = lambertw((n - 11/8)/np.e).real
    return 2*np.pi*(n - 11/8)/w
for n in [1, 10, 100, 1000]:
    print(n, s_brent(n), s_lambert(n))
```

17.6.3 The local mean spacing

How far apart are consecutive smooth zeros? Treat n as a continuous variable and differentiate both sides of $N_{\rm sm}(s_n) = n - \frac{1}{2}$ with respect to n. By the chain rule,

$$N'_{\rm sm}(s_n)\,\frac{{\rm d}s_n}{{\rm d}n} = 1. \tag{17.38}$$

Using Eq. (17.34),

$$\boxed{\text{gap}_n \equiv \frac{{\rm d}s_n}{{\rm d}n} = \frac{1}{N'_{\rm sm}(s_n)} = \frac{2\pi}{\log(s_n/2\pi)}.} \tag{17.39}$$

This is the *local mean spacing*: the average distance between consecutive zeros near the n-th one. Its reciprocal $N'_{\rm sm}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$ is the *mean density* of zeros, the expected number per unit height. Because the density grows logarithmically, the spacing shrinks logarithmically: slowly but without limit.

Since $\log(s_n/2\pi) = 1 + v = 1 + W(x_n)$ in the Lambert notation, we also have $\text{gap}_n = 2\pi/(1 + W(x_n))$.

The derivative $\mathrm{d}s_n/\mathrm{d}n$ and the actual difference $s_{n+1} - s_n$ are not identical, because gap changes slightly from one n to the next. The difference is tiny except at the very lowest n; we return to this in the Paper connection, because the paper computes gap_n by finite differences.

Sample Problem 17.7: The tenth smooth zero and its spacing

Use Eq. (17.37) to find s_{10} and gap_{10}, and compute $(t_{10} - s_{10})/\mathrm{gap}_{10}$.
Step 1: $n - \frac{11}{8} = 8.625$ and $x_{10} = 8.625/e = 3.172960$.
Step 2: Solve $we^w = 3.172960$. Try $w = 1.0788$: $e^{1.0788} = 2.941150$ and $1.0788 \times 2.941150 = 3.172913$, slightly low. One Newton step ($\frac{\mathrm{d}}{\mathrm{d}w} we^w = e^w(1+w) = 6.114$) adds $0.0000473/6.114 = 0.0000077$: $W(x_{10}) = 1.078808$.
Step 3: $s_{10} = 2\pi \times 8.625/1.078808 = 54.192473/1.078808 = 50.2337$.
Step 4: $\mathrm{gap}_{10} = 2\pi/(1 + 1.078808) = 6.283185/2.078808 = 3.0225$.
Step 5: $t_{10} - s_{10} = 49.773832 - 50.2337 = -0.4598$, so $(t_{10} - s_{10})/\mathrm{gap}_{10} = -0.4598/3.0225 = -0.152$.

$$\boxed{s_{10} \approx 50.234, \qquad \mathrm{gap}_{10} \approx 3.022, \qquad \frac{t_{10} - s_{10}}{\mathrm{gap}_{10}} \approx -0.152.}$$

There is no contradiction with the average spacing 3.96 of the first ten zeros (first Sample Problem). That average is taken over the whole range $n = 1, \ldots, 10$, where the local mean spacing falls from about 7.5 at $n = 1$ to 3.02 at $n = 10$; an average of values running from 7.5 down to 3.0 naturally lies above the last of them.

Table 17.1 collects the smooth count and the local mean spacing $2\pi/\log(t/2\pi)$ at heights that appear in this book and in the paper.

Table 17.1: The smooth count $N_{\mathrm{sm}}(t)$ of Eq. (17.2) and the local mean spacing $2\pi/\log(t/2\pi)$ at selected heights. The heights 811.184 and 1419.422 are t_{500} and t_{1000} as quoted in the paper (App. A key anchors and Sec. II A). Spacing in units of t.

height t	$N_{\mathrm{sm}}(t)$	mean spacing $2\pi/\log(t/2\pi)$
50	9.423	3.029
100	29.002	2.271
811.184 (t_{500})	499.296	1.293
1000	648.616	1.239
1419.422 (t_{1000})	999.418	1.159
10000	10142.965	0.852

Sample Problem 17.8: How many zeros below height 10^4?

Estimate $N(10^4)$ and the mean spacing there.
Step 1: $u = 10^4/2\pi = 1591.549$, $\log u = \log 10^4 - \log 2\pi = 9.210340 - 1.837877 = 7.372463$.
Step 2: $N_{\mathrm{sm}}(10^4) = u(\log u - 1) + \frac{7}{8} = 1591.549 \times 6.372463 + 0.875 = 10142.090 + 0.875 = 10142.965$.
Step 3: By Eq. (17.28), $N(10^4) = N_{\mathrm{sm}}(10^4) + S(10^4)$ up to $\mathcal{O}(10^{-5})$, and S is of order a few tenths. So $N(10^4)$ is the integer within roughly one of 10143.
Step 4: Mean spacing $2\pi/7.372463 = 0.852$.

$$\boxed{N(10^4) \approx 10143, \qquad \text{mean spacing at } t = 10^4 \approx 0.852.}$$

This is consistent with the paper's use of "zeros 10^3–10^4" (Sec. II A): the ten-thousandth zero lies just below height 10^4.

Checkpoint 17.6

The mean spacing at $t = 2\pi e^2 \approx 46.43$ is a famous number. Which?
Answer: $\log(t/2\pi) = \log e^2 = 2$, so the spacing is $2\pi/2 = \pi$.

Key Ideas

- $N_{\rm sm}$ is increasing for $t > 2\pi$, so $N_{\rm sm}(s_n) = n - \frac{1}{2}$ has exactly one solution $s_n > 2\pi$, the n-th smooth zero.
- s_n can be found by Brent's method with bracket $[2\pi, 2\pi(n+10)]$ or exactly by $s_n = 2\pi(n - \frac{11}{8})/W((n - \frac{11}{8})/e)$.
- The local mean spacing is $\mathrm{gap}_n = \mathrm{d}s_n/\mathrm{d}n = 2\pi/\log(s_n/2\pi)$, shrinking from about 7.5 at $n = 1$ to about 0.85 at height 10^4.

17.7 Unfolding and fluctuations in mean spacings

17.7.1 Fluctuations

The smooth zeros tell us where the zeros "should" be on average. The interesting part, everything the paper tries to predict, is the deviation from that average. To compare deviations at different heights we must remove the shrinking spacing. Define the *fluctuation* of zero n, in units of the local mean spacing:

$$\boxed{\delta_n \equiv \frac{t_n - s_n}{\mathrm{gap}_n}.} \tag{17.40}$$

This is what `qmnt.fluct(t, n)` computes. A value $\delta_n = +0.3$ means zero n sits 0.3 of a local spacing above its smooth position. (The symbol δ_n is local to this chapter; the paper writes "fluctuations in units of the local mean spacing" in words, and in App. A step 18 writes d_n for a closely related quantity.)

Sample Problem 17.9: Fluctuations of the first and tenth zeros

Find δ_1 and δ_{10}.
Step 1: From the Newton Sample Problem, $t_1 - s_1 = -0.386621$ and $s_1 = 14.521346$. The spacing is $\mathrm{gap}_1 = 2\pi/\log(s_1/2\pi) = 6.283185/0.837743 = 7.5001$.
Step 2: $\delta_1 = -0.386621/7.5001 = -0.0515$.
Step 3: From the Lambert Sample Problem, $\delta_{10} = -0.152$.

$$\boxed{\delta_1 \approx -0.052, \qquad \delta_{10} \approx -0.152.}$$

Both zeros sit a little below their smooth positions, by a small fraction of a spacing.

17.7.2 Three faces of the same quantity

There are three ways to talk about the same fluctuation, and the paper uses all three.

(1) In terms of the smooth count at the zero. Expand $N_{\rm sm}$ around s_n to second order (Taylor, with $N''_{\rm sm}(t) = \frac{1}{2\pi t}$ from differentiating Eq. (17.34)):

$$N_{\rm sm}(t_n) = N_{\rm sm}(s_n) + N'_{\rm sm}(s_n)(t_n - s_n) + \tfrac{1}{2}N''_{\rm sm}(s_n)(t_n - s_n)^2 + \cdots . \tag{17.41}$$

Use $N_{\rm sm}(s_n) = n - \frac{1}{2}$ and $N'_{\rm sm}(s_n) = 1/\mathrm{gap}_n$:

$$N_{\rm sm}(t_n) - \left(n - \tfrac{1}{2}\right) = \delta_n + \frac{(t_n - s_n)^2}{4\pi s_n} + \cdots . \tag{17.42}$$

The correction term is tiny: for $n = 1$ it is $0.386621^2/(4\pi \times 14.521) = 8.2 \times 10^{-4}$, and it shrinks with height. Indeed $N_{\rm sm}(t_1) - \frac{1}{2} = 0.449278 - 0.5 = -0.050722$, against $\delta_1 = -0.051548$.

(2) In terms of S. Equation (17.27) holds just below and just above t_n. Average the two versions. On the left, the average of N is the midpoint value $N^{\rm mid}(t_n) = n - \frac{1}{2}$ of Sec. 17.1; on the right, $N_{\rm sm}$ and the $1/(48\pi T)$ term are continuous, and the average of S from below and above is what we write as $S(t_n)$. This gives

$$S(t_n) = n - \tfrac{1}{2} - N_{\rm sm}(t_n) - \frac{1}{48\pi t_n} + \mathcal{O}(t_n^{-3}) \approx -\delta_n, \tag{17.43}$$

using Eq. (17.42) in the last step. (Strictly, S is undefined at a zero; here $S(t_n)$ denotes the average of its limits from just below and just above.) So *the fluctuation of a zero, in mean spacings, is minus the value of S at that zero.* For $n = 1$: $S(t_1) = \frac{1}{2} - 1 - \theta(t_1)/\pi = 0.0503$, against $-\delta_1 = 0.0515$.

(3) Unfolding. The map $t \mapsto N_{\rm sm}(t)$ stretches the t axis so that the zeros acquire unit mean spacing. The *unfolded zeros* are $x_n = N_{\rm sm}(t_n)$, and by Eq. (17.42), $x_n \approx n - \frac{1}{2} + \delta_n$. The *unfolded spacings* are

$$x_{n+1} - x_n \approx 1 + \delta_{n+1} - \delta_n, \tag{17.44}$$

whose average over many n is 1, because the δ_n stay bounded in practice while the number of spacings grows. This is the unfolding behind the spacing histogram of the paper's Fig. 2, which you will study in Chapter 20.

Sample Problem 17.10: Unfolding the first ten zeros

Compute the unfolded zeros x_1 and x_{10} and the mean unfolded spacing of the first ten zeros.

Step 1: $x_1 = N_{\rm sm}(t_1) = 0.449278$ (Newton Sample Problem, Step 1).

Step 2: $x_{10} = N_{\rm sm}(49.773832)$: $u = 7.921751$, $\log u = 2.069612$, so $x_{10} = 7.921751 \times 1.069612 + 0.875 = 8.473203 + 0.875 = 9.348203$.

Step 3: The nine unfolded spacings add up to $x_{10} - x_1$ (the sum telescopes), so their mean is $(9.348203 - 0.449278)/9 = 8.898925/9 = 0.98877$.

$$\boxed{\text{mean unfolded spacing of } t_1, \ldots, t_{10} \approx 0.989.}$$

Compare: the raw spacings of the same zeros average 3.96, but the local mean spacing varies from 7.5 to 3.0 over this range. After unfolding the average is close to 1, as it should be; it misses 1 by $(\delta_{10} - \delta_1)/9 \approx -0.011$.

17.7.3 Why errors are quoted in mean spacings

Suppose a method predicts zero n at $\hat{t}_n$. Its error in units of t is $\hat{t}_n - t_n$. Why does the paper always divide by gap_n?

1. **Comparability across heights.** The spacing shrinks from 7.5 to 0.85 between $n = 1$ and $t = 10^4$ (Table 17.1). An error of 0.3 in t is a small error at $n = 1$ (4% of a spacing) and a large one at $t = 10^4$ (35% of a spacing). Dividing by gap_n makes errors at different heights comparable.

2. **A natural scale for "getting it right".** An error of half a spacing means you can no longer tell which zero you have predicted. Errors much smaller than 1 in these units mean the method resolves individual zeros.

3. **The trivial prediction has a fixed size in these units.** The simplest prediction, "zero n is at s_n", has error δ_n in mean spacings. Its rms over a block of zeros is a slowly growing number of order 0.25–0.3 (Sec. 17.5), almost the same at every height. In units of t it would shrink with height and mislead you.

4. **Dimensionless.** The ratio has no units, so it is the same whether heights are thought of as energies, frequencies or just numbers.

17.7.4 A numerical illustration

Figure 17.6 shows gap_n and δ_n for the first 1000 zeros. The fluctuations look like noise of roughly constant size, with no visible trend: all the regular growth has been absorbed into s_n and gap_n. The listing below computes the rms of δ_n over the block 401–500, the paper's principal holdout (running example RE5).

```
import numpy as np, qmnt
t = qmnt.zeros(2000)
def null_rms(n1, n2):                    # rms of (t_n - s_n)/gap_n, n1..n2
    n = np.arange(n1, n2 + 1)
    d = qmnt.fluct(t[n - 1], n)          # fluctuation in mean spacings
    return qmnt.rms(d)
for blk in [(401, 500), (501, 1000), (1001, 2000)]:
    print(blk, round(null_rms(*blk), 3))
```

The three values printed are 0.263, 0.270 and 0.279, the first three entries of the "null" column of the paper's Table V. The CHECKS script for this chapter confirms all three to three decimal places.

Pitfall

Mind the sign convention. In this chapter $\delta_n = (t_n - s_n)/\mathrm{gap}_n$ is "truth minus smooth". The quantity $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ used as a fitting target in the paper's Sec. II D has the *opposite* sign to first order ($y_n \approx -\delta_n \approx S(t_n)$). The rms is the same either way, but a correlation of $+0.9$ with one becomes -0.9 with the other. Always check which convention a formula uses.

Checkpoint 17.7

A prediction misses zero n by 0.2 in units of t. If you move to a height where the mean spacing is half as large, and the miss in t stays 0.2, what happens to the error in mean spacings?

Answer: It doubles, from 0.2/gap to 0.2/(gap/2) = 0.4/gap. The same absolute error is a larger fraction of a spacing where zeros are denser.

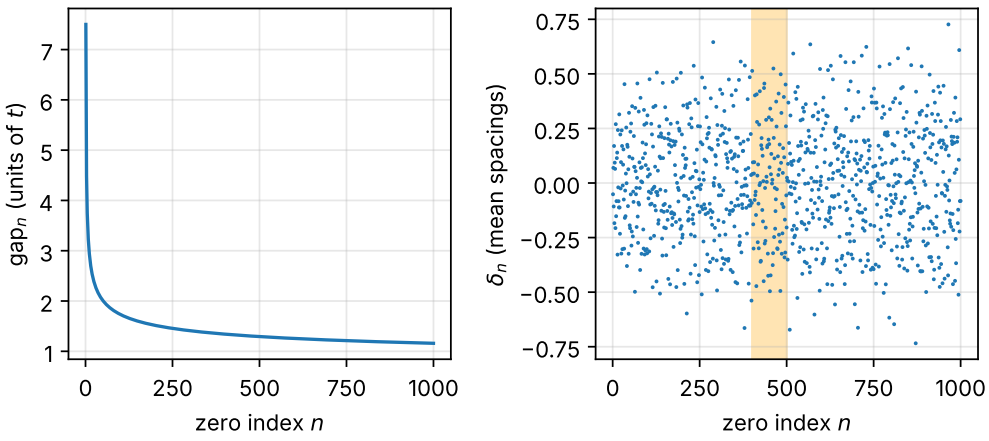


Figure 17.6: Left: the local mean spacing $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$ against zero index n for $n \le 1000$, in units of t; it falls from 7.5 at $n = 1$ to about 1.16 at $n = 1000$. Right: the fluctuations $\delta_n = (t_n - s_n)/\mathrm{gap}_n$ (dimensionless, in mean spacings) for $n \le 1000$. The shaded band is the paper's principal holdout, zeros 401–500, over which the rms of δ_n is 0.263: the "null" error of the paper.

Key Ideas

- The fluctuation $\delta_n = (t_n - s_n)/\mathrm{gap}_n$ measures each zero's displacement from its smooth position in local mean spacings.
- To excellent accuracy, $\delta_n \approx N_{\mathrm{sm}}(t_n) - (n - \frac{1}{2}) \approx -S(t_n)$, and the unfolded zeros $N_{\mathrm{sm}}(t_n)$ have unit mean spacing.
- Errors in mean spacings are comparable across heights; the rms of δ_n over zeros 401–500 is 0.263, the paper's null.

17.8 Paper connection

This chapter supplies the yardstick for the entire paper [1]. Here is where each idea appears.

Eq. (1) and its constant. The paper's Eq. (1), $N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$, is introduced in Sec. II A as "the Riemann–von Mangoldt counting function". You have now derived it: Eq. (17.27) shows it is $\theta(T)/\pi + 1$ with the $1/(48\pi T)$ term dropped, and the $\frac{7}{8}$ is $-\frac{1}{8} + 1$. The paper does not name the difference $N(t) - N_{\mathrm{sm}}(t)$; in this book's notation it is $S(t)$, by Eq. (17.28), up to the $\mathcal{O}(1/t)$ term. The paper only approximates it, by the prime sum N_{fl} of its Eq. (4).

Smooth zeros and the mean spacing (Sec. II A, App. A step 2). The paper defines smooth zeros by "$N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$" and the local mean spacing by "$\mathrm{gap}_n = ds_n/dn$", exactly Eqs. (17.33) and (17.39). App. A step 2 says the smooth zeros were found by Brent's method on $N_{\mathrm{sm}}(t) = n - \frac{1}{2}$ (our first route) and that gap was computed as `np.gradient` of the array of smooth zeros, a central finite difference $(s_{n+1} - s_{n-1})/2$ for interior n. That agrees with the analytic derivative $2\pi/\log(s_n/2\pi)$ to better than one part in 10^4 at interior indices such as $n = 450$ (checked in the CHECKS script); only at the ends of the array, where `np.gradient` switches to a one-sided difference, do the two definitions differ appreciably. The companion library `qmnt.mean_gap` uses the analytic form.

The Lambert-W asymptotic (Sec. I A, paragraph b). The paper notes that França and LeClair [3] derived an equation for the n-th zero "whose smooth part is the Lambert-W asymptotic of the counting function". That smooth part is our Eq. (17.37): setting $S = 0$ in the counting relation and solving for the height gives s_n in closed form.

Every rms in the paper is in mean spacings. Sec. II A states: "All errors below are quoted in units of the local mean spacing gap_n", and App. A repeats: "All rms values in units of local mean spacing from the smooth zeros of Eq. (1)". So every rms in Tables I–V, and every rms interval quoted in Secs. III–V (the correlation intervals are pure numbers), is an error measured in units of gap_n as in Eq. (17.40). To translate into units of t, multiply by the local spacing: near zero 500 (paper's $t_{500} = 811.184$, App. A key anchors) the spacing is 1.293 (Table 17.1), so the null 0.263 corresponds to about 0.34 in t.

The null 0.263 (Sec. II C, Tables I and V, App. A). The paper's baseline prediction is "no fluctuation": predict $t_n \approx s_n$. Its error in mean spacings is exactly δ_n, so its rms over zeros 401–500 is the rms of δ_n there, 0.263 (App. A key anchor 0.2631; Table V first row). Sec. II C explains that the operator null (the unperturbed potential's levels) coincides numerically with this smooth-zero null on 401–500, "both 0.2631", which is why one baseline serves the whole paper. You reproduced this number in Sec. 17.7.

The slow growth of the null (Table V, Sec. II C "Within-height splits"). The null column of Table V grows from 0.263 (401–500) to 0.270, 0.279, 0.288 and 0.296 (5001–10000). This is the slow growth of the typical size of S described in Sec. 17.5. The paper calls it a "covariate shift" in its within-height-splits paragraph: zeros held out above the training range have slightly larger fluctuations. The nulls of Table III, 0.23–0.27, are the analogous baseline (the unperturbed potential's error) on other subsets of zeros; the paper reports that this operator null coincides numerically with the rms of δ_n on the principal holdout 401–500.

Unfolding and Fig. 2. Sec. II A says "Unfolded spacings of the first 10^4 zeros show textbook GUE level repulsion (Fig. 2)". The unfolding is the map $t \mapsto N_{\mathrm{sm}}(t)$ of Sec. 17.7, which makes the mean spacing 1. The level statistics themselves are Chapter 20.

The oracle and the fitted controls (Sec. II D, Eq. (4)). The paper's arithmetic oracle predicts zero n by solving $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$. Compare with the Riemann–von Mangoldt formula at the midpoint of the n-th riser, $N_{\mathrm{sm}}(t_n) + S(t_n) \approx n - \frac{1}{2}$ (Eq. (17.43)): the oracle is this relation (exact up to the $1/(48\pi t_n)$ term) with the unknown S replaced by an explicit approximation N_{fl} built from primes (Chapter 18). The fitted sinusoidal controls of Sec. II D train on "$y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$", which by Eq. (17.43) is $S(t_n)$ up to $1/(48\pi t_n)$. So everything in the paper's Sec. IV is, in the language of this chapter, an attempt to predict $S(t)$ at unseen heights.

Discretization errors in mean spacings (Sec. II C, App. A). The paper's leakage paragraph notes that finite-difference dispersion "at levels 401–500 on the working grid reaches two mean spacings", and App. A records a "raw-zero null on the uncorrected working grid 2.09". Both are in units of gap_n; near zero 450 two mean spacings are about 2.6 in t. Chapter 23 explains where these errors come from.

The data and the critical line (Sec. II A). The paper computes $t_1, \ldots, t_{1000}$ with `mpmath` at 20 digits, quoting $t_1 = 14.134725141734695$ and $t_{1000} = 1419.42248094599$. From Table 17.1, $N_{\mathrm{sm}}(t_{1000}) = 999.418$, so $\delta_{1000} \approx 999.418 - 999.5 = -0.082$: zero 1000 sits 0.08 spacings below

its smooth position. Writing every zero as $\frac{1}{2} + it_n$ relies on the numerical verification of RH described in Sec. 17.5.

Overloaded symbols. In this chapter N is the counting function and T a height. In the paper, N also denotes the number of grid points (Sec. II B: "$N = 6000$ at scale 1"), and T also denotes the length of a training window in t (App. A steps 20 and 23, "$2\pi/T$", "π/T"); T further labels the second-order tensors $T^{(n)}$ of the paper's Eq. (5) and the Chebyshev polynomials T_k of its Sec. II C. Context always tells you which is meant.

17.9 Summary

★ Did you know? *Riemann's Hidden Notes*

Riemann published no numerical work on the zeros, so for decades many assumed his hypothesis was a hunch. In 1932 Carl Ludwig Siegel studied Riemann's unpublished papers in the Göttingen library and found a powerful formula for computing zeta on the critical line. Now called the Riemann-Siegel formula, it showed that Riemann had computed zeros himself, and it remains at the core of large-scale zero computations.

- $N(T)$ counts zeros with $0 < t_n \le T$; it is a staircase rising by 1 at each (simple) zero (Sec. 17.1).
- Argument principle: the net change of $\arg f$ around a closed curve, divided by 2π, counts the zeros inside, Eq. (17.4) (general theorem taken on trust).
- Riemann–von Mangoldt formula, Eq. (17.13): $N(T) = \theta(T)/\pi + 1 + S(T)$, with θ of Eq. (17.11) and $S(T) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + iT)$, Eq. (17.12).
- Stirling gives $\theta(t) = \frac{t}{2}\log\frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \mathcal{O}(t^{-3})$, Eq. (17.25).
- Hence $N(T) = N_{\mathrm{sm}}(T) + S(T) + \frac{1}{48\pi T} + \mathcal{O}(T^{-3})$ with N_{sm} the paper's Eq. (1), Eqs. (17.2) and (17.27); $\frac{7}{8} = -\frac{1}{8} + 1$.
- $S(T)$ jumps by $+1$ at zeros and drifts down with slope $-\frac{1}{2\pi}\log\frac{T}{2\pi}$ between them, Eq. (17.32); $S = \mathcal{O}(\log T)$; typical size of a few tenths at the paper's heights.
- Smooth zeros: $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$, Eq. (17.33), exactly $s_n = 2\pi(n - \frac{11}{8})/W((n - \frac{11}{8})/e)$, Eq. (17.37).
- Mean spacing: $\mathrm{gap}_n = \mathrm{d}s_n/\mathrm{d}n = 2\pi/\log(s_n/2\pi)$, Eq. (17.39).
- Fluctuation in mean spacings: $\delta_n = (t_n - s_n)/\mathrm{gap}_n \approx N_{\mathrm{sm}}(t_n) - (n - \frac{1}{2}) \approx -S(t_n)$, Eqs. (17.40)–(17.43); its rms over zeros 401–500 is the paper's null 0.263.

17.10 Problems

1. • Using the ten heights of RE1, Eq. (17.1), find $N(20)$, $N(35)$ and $N(48.1)$.

2. • Evaluate $N_{\mathrm{sm}}(2\pi)$ and $N_{\mathrm{sm}}(2\pi e)$ exactly. Compare each with the true count N at the same height ($2\pi e \approx 17.08$). What does this tell you about using N_{sm} at very low heights?

3. • Show from Eq. (17.2) that $N'_{\mathrm{sm}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$, showing the product-rule step. Find the mean spacing at $t = 2\pi e^2 \approx 46.43$.

4. • Compute the local mean spacing $2\pi/\log(t/2\pi)$ at $t = 1000$.

5. • A prediction misses a zero near height $t = 1000$ by 0.1 in units of t. Express the error in local mean spacings.

6. • Use the argument principle to find $\frac{1}{2\pi}\Delta_C \arg(z^3 - 1)$ when C is (a) the circle $|z| = 2$ and (b) the circle $|z - 1| = \frac{1}{2}$, both traversed counterclockwise.

7. • Suppose (contrary to every computation) that ζ had a double zero at height t^*. By how much would $N(T)$ and $S(T)$ jump at $T = t^*$? Would the number of sign changes of $Z(t)$ near t^* change?

8. •• Verify the Riemann–von Mangoldt formula at $T = 30$: compute $\theta(30)$ from Eq. (17.25), then $S(30)$ from Eq. (17.29) using $N(30) = 3$. Compare with `mpmath.backlunds(30)`.

9. •• Use Eq. (17.37) to compute s_{100} and gap_{100}. Given $t_{100} = 236.524230$, find the fluctuation δ_{100}.

10. •• Show that $\mathrm{gap}_n = 2\pi/[1 + W((n - \frac{11}{8})/e)]$.

11. •• (Paper.) The paper's Sec. II A quotes $t_{1000} = 1419.42248094599$. Compute $N_{\mathrm{sm}}(t_{1000})$, the approximate fluctuation δ_{1000} using Eq. (17.42), and the local mean spacing there.

12. •• (Paper.) The null error on zeros 401–500 is 0.263 mean spacings (paper Table V). Using the paper's anchor $t_{500} = 811.1843588465$ (App. A), estimate this error in units of t.

13. •• Using Eq. (17.32), find the slope $\mathrm{d}S/\mathrm{d}T$ at $T = 30$. Over a gap of one local mean spacing, how much does S fall? Comment.

14. •• Show that N_{sm} is decreasing on $(0, 2\pi)$ and increasing on $(2\pi, \infty)$, and that its minimum value is $-\frac{1}{8}$. Explain why this makes the condition $s_n > 2\pi$ in Eq. (17.33) necessary for uniqueness.

15. •• (Paper, computational.) Using `qmnt`, compute the rms of δ_n over zeros 401–500, 501–1000 and 1001–2000. Compare with the null column of the paper's Table V.

16. •• (Computational.) Using `qmnt.zeros(1000)` and `qmnt.fluct`, find the largest $|\delta_n|$ for $n \le 1000$ and the index at which it occurs. Is it consistent with the statement that S is "a few tenths" at these heights? Explain the difference between a typical value and a maximum.

17. •• How many zeros do you expect, on average, between heights 1000 and 1001?

18. •• For $n = 1$, evaluate the second-order term $(t_1 - s_1)^2/(4\pi s_1)$ in Eq. (17.42) and compare it with the difference between $N_{\mathrm{sm}}(t_1) - \frac{1}{2}$ and δ_1.

19. •• A *Gram point* g_k is defined by $\theta(g_k) = k\pi$. Using Eq. (17.25) and Newton's method, find g_0 (the solution with $g_0 > 2\pi$). Check with `mpmath.grampoint(0)`.

20. •• (Paper.) The paper's Sec. II C says that finite-difference dispersion at levels 401–500 on the working grid "reaches two mean spacings". Using Table 17.1, estimate this in units of energy (which equal units of t in the paper's conventions).

21. •• (Computational.) Show that the mean of the unfolded spacings $x_{n+1} - x_n$, $n = 1, \dots, N-1$, equals $(x_N - x_1)/(N-1)$. Compute it for $N = 10$ from RE1, and then with `qmnt` for $N = 1000$.

22. •• (Paper.) The fitted controls of the paper's Sec. II D train on $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$. Show that $y_n \approx S(t_n) \approx -\delta_n$, compute y_1, and explain why a model that predicts y_n well automatically predicts the zeros well.

23. ••• (Computational.) Using `mpmath` with at least 30 digits, compute $t^3\big[\theta(t) - \frac{t}{2}\log\frac{t}{2\pi} + \frac{t}{2} + \frac{\pi}{8} - \frac{1}{48t}\big]$ at $t = 50$ and $t = 200$, and compare with $7/5760$.

24. ••• Fill in the symmetry argument of Sec. 17.3 in detail: parametrize P and P' explicitly, and show that $\arg\xi$ at the point of P' corresponding to parameter value τ equals $-\arg\xi$ at the mirror point of P, and hence that $\Delta_{P'}\arg\xi = \Delta_P \arg\xi$.

25. ••• Explain, step by step, why the following procedure proves that all zeros with $0 < t < T$ lie on the critical line and are simple: (i) find k sign changes of $Z(t)$ in $(0, T)$; (ii) establish that $N(T) = k$ exactly. Which step needs $S(T)$, and why can't one simply evaluate $N_{\rm sm}(T)$ and round?

26. •• Compute $\xi(2)$ exactly from Eq. (17.5) using $\zeta(2) = \pi^2/6$ and $\Gamma(1) = 1$. Then, using `mpmath` (`mp.zeta`, `mp.gamma`), evaluate the right side of Eq. (17.5) at $s = -1$ and confirm that $\xi(-1) = \xi(2)$, as the functional equation requires. Which two factors are negative at $s = -1$?

27. ••• (Paper, computational.) The paper cites the Lambert-W asymptotic of França and LeClair (Sec. I A, paragraph b). Compute s_n for $n = 1, \dots, 1000$ both from Eq. (17.37) and with `qmnt.smooth_zeros`, and find the largest absolute difference. Explain why the two must agree to within root-finding tolerance.

28. ••• Selberg's theorem says the mean square of S grows like $\frac{1}{2\pi^2}\log\log T$. Compute $\sqrt{\log\log T/(2\pi^2)}$ at $T = 10^3$ and $T = 10^4$ and the ratio of the two. Compare with the ratio of the paper's nulls 0.296/0.263 (Table V). Discuss why you should not expect close agreement (consider lower-order terms, and the difference between S at all heights and S at the zeros).

17.11 Answers to selected problems

1. $t_1 = 14.13 < 20 < t_2 = 21.02$, so $N(20) = 1$. $t_5 = 32.94 < 35 < t_6 = 37.59$, so $N(35) = 5$. $t_9 = 48.005 < 48.1 < t_{10} = 49.77$, so $N(48.1) = 9$.

2. $N_{\rm sm}(2\pi) = 1\cdot\log 1 - 1 + \frac{7}{8} = -\frac{1}{8}$. $N_{\rm sm}(2\pi e) = e\log e - e + \frac{7}{8} = \frac{7}{8}$. True counts: $N(2\pi) = 0$ and $N(2\pi e) = N(17.08) = 1$. At these low heights $N_{\rm sm}$ is off by $\frac{1}{8}$ of a zero and is even negative at 2π; it is an asymptotic formula, and its relative accuracy improves with height. (Its absolute error, S, does not shrink.)

3. $\frac{\rm d}{{\rm d}t}\left[\frac{t}{2\pi}\log\frac{t}{2\pi}\right] = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t}$ by the product rule (the derivative of $\log(t/2\pi)$ is $1/t$). Subtracting the derivative $\frac{1}{2\pi}$ of $\frac{t}{2\pi}$ leaves $\frac{1}{2\pi}\log\frac{t}{2\pi}$. At $t = 2\pi e^2$, $\log(t/2\pi) = 2$, so the mean spacing is $2\pi/2 = \pi \approx 3.14$.

4. $\log(1000/2\pi) = 6.907755 - 1.837877 = 5.069878$; spacing $6.283185/5.069878 = 1.239$.

5. $0.1/1.239 = 0.0807$ mean spacings.

6. The zeros are 1, $e^{2\pi i/3}$ and $e^{-2\pi i/3}$, all of modulus 1. (a) All three lie inside $|z| = 2$: answer 3. (b) The distance from 1 to $e^{\pm 2\pi i/3} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ is $\sqrt{(3/2)^2 + 3/4} = \sqrt{3} > \frac{1}{2}$, so only $z = 1$ is inside: answer 1.

7. N would jump by 2 (zeros are counted with multiplicity), and since θ is continuous, S would also jump by 2. Near a double zero $Z(t)$ behaves like a constant times $(t - t^*)^2$ and does not change sign; a sign-change count would see no zero there at all, while $N(T)$ would count two. This mismatch is exactly what the procedure of Problem 25 detects.

9. $x_{100} = 98.625/e = 36.282110$; solving $we^w = x_{100}$ gives $W = 2.625901$. Then $s_{100} = 2\pi \times 98.625/2.625901 = 619.679151/2.625901 = 235.987$; ${\rm gap}_{100} = 2\pi/3.625901 = 1.733$; $\delta_{100} = (236.524230 - 235.987266)/1.733 = 0.536964/1.733 = +0.310$. Zero 100 sits about a third of a spacing above its smooth position.

11. $u = 1419.422481/2\pi = 225.908136$, $\log u = 5.420128$; $N_{\rm sm}(t_{1000}) = 225.908136 \times 4.420128 + 0.875 = 998.542976 + 0.875 = 999.418$. By Eq. (17.42), $\delta_{1000} \approx 999.418 - 999.5 = -0.082$ (the second-order term is about 2×10^{-6}). Spacing: $2\pi/5.420128 = 1.159$.

13. $\mathrm{d}S/\mathrm{d}T = -\frac{1}{2\pi}\log\frac{30}{2\pi} + \frac{1}{48\pi\cdot 900} = -\frac{1.563320}{6.283185} + 0.000007 = -0.249$. Over one local mean spacing gap $= 1/N'_{\mathrm{sm}}(30)$, S falls by $N'_{\mathrm{sm}}(30) \times \mathrm{gap} = 1$ (to the accuracy of the tiny $1/(48\pi T^2)$ term). This is why perfectly evenly spaced zeros would leave S oscillating between fixed limits: each jump of $+1$ is exactly undone by the drift across one mean spacing.

15. The rms values are 0.263, 0.270, 0.279, matching the paper's Table V nulls for 401–500, 501–1000 and 1001–2000. The paper computes gap_n by `np.gradient`, `qmnt` analytically; the difference is invisible at three decimals.

17. The expected number is $N_{\mathrm{sm}}(1001) - N_{\mathrm{sm}}(1000) = \int_{1000}^{1001} N'_{\mathrm{sm}}(t)\,\mathrm{d}t \approx N'_{\mathrm{sm}}(1000.5) = \frac{1}{2\pi}\log\frac{1000.5}{2\pi} = \frac{5.070378}{6.283185} = 0.807$. (The midpoint rule is accurate to about 10^{-8} here because N''_{sm} is tiny.)

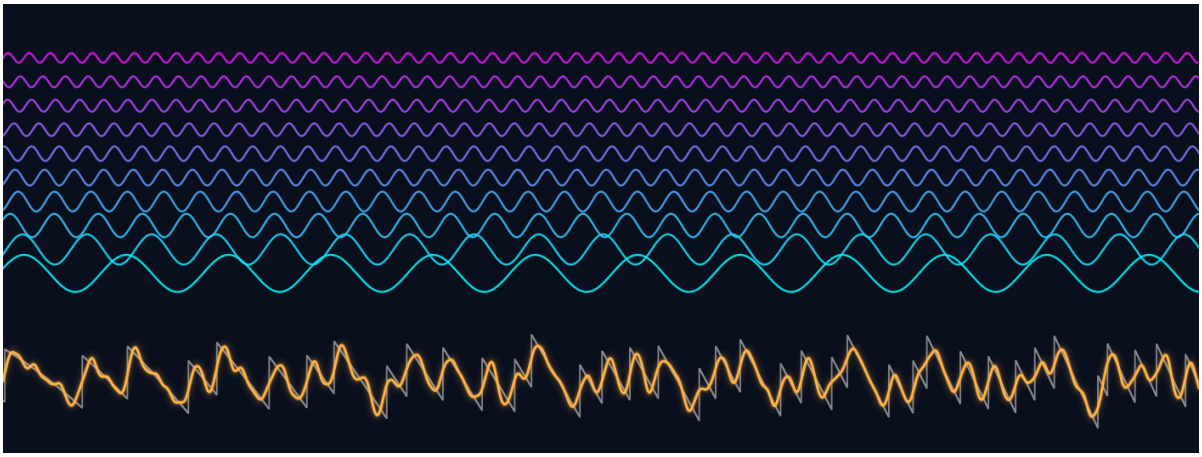
19. Solve $f(g) = \frac{g}{2}\log\frac{g}{2\pi} - \frac{g}{2} - \frac{\pi}{8} + \frac{1}{48g} = 0$ with $f'(g) \approx \frac{1}{2}\log\frac{g}{2\pi}$. Starting from $g = 17.8$, $f(17.8) \approx -0.0238$ and $f'(17.8) \approx 0.5207$, so the first step gives $17.8 + 0.0238/0.5207 \approx 17.8457$. A second step changes this only in the fifth decimal, giving $g_0 = 17.8456$. Because f is increasing and convex there ($f'' \approx 1/(2g) > 0$), a Newton step started to the left of the root lands slightly to its right, by about 6×10^{-5} here. `mpmath.grampoint(0)` gives 17.8456 as well.

21. The sum $\sum_{n=1}^{N-1}(x_{n+1} - x_n)$ telescopes to $x_N - x_1$, so the mean is $(x_N - x_1)/(N-1)$. For $N = 10$: $(9.348203 - 0.449278)/9 = 0.989$ (Sample Problem in Sec. 17.7). For $N = 1000$ the same formula with $x_{1000} = N_{\mathrm{sm}}(t_{1000}) = 999.418$ gives $(999.418 - 0.449)/999 = 0.99997$; the departure from 1 is $(\delta_{1000} - \delta_1)/999$, which shrinks as N grows because the δ_n do not grow appreciably.

23. At $t = 50$ the bracket times t^3 is 0.001215, and at $t = 200$ it is closer still to $7/5760 = 0.0012153$. The next term in the expansion is of order t^{-5}, so $t^3\times$(remainder) approaches $7/5760$ with a correction of order t^{-2}, about 10^{-7} at $t = 50$.

25. Each sign change of the real function $Z(t)$ in $(0, T)$ requires a zero of $\zeta(\frac{1}{2} + it)$ of odd multiplicity on the critical line, so there are at least k zeros on the line, counted with multiplicity. If $N(T) = k$ exactly, the total number of zeros in the strip below T (on or off the line, with multiplicity) is also k. So there is no zero left over to lie off the line, and no zero can be multiple (a multiple zero would contribute more than 1 to N while producing at most one sign change). Step (ii) needs $S(T)$, because $N(T) = \theta(T)/\pi + 1 + S(T)$. Rounding $N_{\mathrm{sm}}(T)$ fails because $S(T)$ is of order a few tenths and can exceed $\frac{1}{2}$ in magnitude (for example $S(50) = 0.577$), so the nearest integer to $N_{\mathrm{sm}}(T)$ can be wrong. Turing's method supplies a rigorous way to pin down the integer.

27. The largest difference is at the level of the root finder's tolerance (about 10^{-9} or smaller). Both methods solve the same equation $N_{\mathrm{sm}}(s) = n - \frac{1}{2}$ on the branch $s > 2\pi$, where the solution is unique (Problem 14); Eq. (17.37) is the exact solution, so any difference is numerical error in the root finder or in the evaluation of W.



One prime wave each for $p = 2, 3, 5, \ldots, 29$, and below them their sum, which tracks the true fluctuation of the zero count.

Chapter 18

The Explicit Formula: Primes as Frequencies

↑Contents

What this chapter is about

In Chapter 17 we split the zero-counting function into a smooth part $N_{\text{sm}}(t)$, which we derived, and a small wiggly part $S(t)$, which we could only measure. This chapter explains the wiggles. The explanation, which is exact in a smoothed form and heuristic in the simple form we will use, is this: $S(t)$ is, to a good approximation, a superposition of sine waves. There is one wave for every prime power p^m. Its angular frequency in t is $m \log p$ and its amplitude is $1/(\pi m p^{m/2})$. The primes, which you met as indivisible integers, reappear here as *frequencies*.

We derive this from the Euler product of Chapter 16, using nothing beyond the logarithm series of Chapter 3 and the complex exponential of Chapter 2. One step of the derivation is *formal*: it moves a formula into a region where the series no longer converges. We say exactly which step that is, what goes wrong, and why the truncated formula still works as an approximation. Then we use the truncated formula to predict individual zeros, measure how the accuracy improves as more primes are included, and see the deeper statement behind it: an exact duality in which the zeros are the frequencies of the primes and the primes are the frequencies of the zeros.

Where this shows up in the paper: the truncated formula is the paper's Eq. (4), its "arithmetic oracle" (Sec. II D). Its predictions are the explicit-formula curve of Fig. 1 and every entry of Table V; enlarged and fitted versions of it are the controls A and B of Table IV; its single $p = 2$ term is the target of the arithmetic-continuation probe of Sec. III; and the fact that the Fourier transform of the zeros peaks at $\log p^m$ (Landau's formula, Sec. IV) is what the sparse control C2 rediscovers.

What you need

- Complex numbers, Euler's formula, $a^s = e^{s\log a}$, so that $p^{-s} = p^{-\sigma}e^{-it\log p}$ for $s = \sigma + it$; arguments and logarithms of complex numbers (Chapter 2).
- Geometric series, absolute convergence, comparison and integral tests, the series of $\log(1-x)$, infinite products (Chapter 3).
- Big-O notation, Stirling's formula, Lambert's W, root finding by bisection and Brent's method with brackets (Chapter 4).
- Fourier series, orthogonality of sines over a period, convergence of a Fourier series to the midpoint of a jump, almost-periodic sums (Chapter 5).
- The Fourier transform, the finite-window resolution $2\pi/T$, peaks in a spectrum (Chapter 6).
- Primes, prime powers, the von Mangoldt function $\Lambda(k)$ and Chebyshev's $\psi(x)$ (Chapter 15).
- The zeta function, the Euler product, the pole at $s = 1$, the critical line, the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ (Chapter 16).
- $N(T)$, the Riemann–von Mangoldt formula, $S(T) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + iT)$, smooth zeros s_n, mean spacing gap_n, fluctuations in mean spacings (Chapter 17).
- First-year physics: superposition of waves, beats, standing waves.

18.1 From a product over primes to a sum over primes

You know from Chapter 16 that for every complex $s = \sigma + it$ with $\sigma > 1$,

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s} = \prod_{p} \frac{1}{1 - p^{-s}}, \tag{18.1}$$

where the product runs over all primes $p = 2, 3, 5, 7, \ldots$ in increasing order. A product is awkward to analyze; a sum is much friendlier, because we can split it into pieces and study each piece separately. A physicist's reflex, when facing a product, is to take a logarithm: the logarithm of a product is the sum of the logarithms. That single move turns the Euler product into a sum over primes, and in the next section each term of that sum will turn out to be a wave.

For real positive numbers the move is safe. For complex numbers we must be careful, because a nonzero complex number has infinitely many logarithms, differing by integer multiples of $2\pi i$ (Chapter 2). We will therefore not simply "take log" of Equation (18.1). Instead we will *construct* a specific series and prove that its exponential equals $\zeta(s)$. That series is then, by definition, our $\log\zeta(s)$.

18.1.1 The logarithm series for a complex argument

Recall from Chapter 3 that for real x with $|x| < 1$,

$$-\log(1-x) = \sum_{m=1}^{\infty} \frac{x^m}{m} = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots. \tag{18.2}$$

We need the same statement with x replaced by a complex number w.

Sample Problem 18.1: The logarithm series for complex w

Let w be a complex number with $|w| < 1$ and define

$$\ell(w) = \sum_{m=1}^{\infty} \frac{w^m}{m}.$$

Show that the series converges absolutely and that $e^{\ell(w)} = \dfrac{1}{1-w}$.

Step 1: absolute convergence. The absolute value of the m-th term is $|w|^m/m \le |w|^m$. The geometric series $\sum_m |w|^m$ converges because $|w| < 1$. By the comparison test (Chapter 3), $\sum_m |w|^m/m$ converges, so $\ell(w)$ converges absolutely.

Step 2: a helper function. For real r in $[0,1]$ define

$$f(r) = \sum_{m=1}^{\infty} \frac{(rw)^m}{m}, \qquad g(r) = (1-rw)\,e^{f(r)}.$$

Then $f(1) = \ell(w)$, and $f(0) = 0$, so $g(0) = 1 \cdot e^0 = 1$.

Step 3: differentiate f. Here f is a power series in the real variable r with complex coefficients w^m/m. Its radius of convergence is $1/|w| > 1$ (infinite if $w = 0$; in that case $f \equiv 0$ and everything below is trivially true). Write $w^m = a_m + ib_m$ with a_m, b_m real. Then $f(r) = \sum_m a_m r^m/m + i\sum_m b_m r^m/m$ is a pair of real power series, and each converges for $|r| < 1/|w|$ by comparison, because $|a_m| \le |w|^m$ and $|b_m| \le |w|^m$. A real power series may be differentiated term by term inside its radius of convergence (a theorem of calculus; see [22]). Applying this to the real and imaginary parts separately and recombining them,

$$f'(r) = \sum_{m=1}^{\infty} \frac{m\,w^m r^{m-1}}{m} = w\sum_{m=1}^{\infty} (rw)^{m-1} = \frac{w}{1-rw},$$

where the last step is the geometric series, valid because $|rw| < 1$.

Step 4: the chain rule for e^f with complex f. Write $f = u + iv$ with u, v real functions of r. By Euler's formula $e^f = e^u(\cos v + i\sin v)$. Differentiating with the product and chain rules,

$$\frac{d}{dr}e^f = e^u u'(\cos v + i\sin v) + e^u(-\sin v + i\cos v)v' = (u' + iv')e^u(\cos v + i\sin v) = f'\,e^f.$$

Step 5: differentiate g. By the product rule and Step 4,

$$g'(r) = -w\,e^{f(r)} + (1-rw)f'(r)e^{f(r)} = -w\,e^{f(r)} + (1-rw)\frac{w}{1-rw}e^{f(r)} = 0.$$

Step 6: conclude. A function of a real variable whose derivative vanishes on an interval is constant (apply this to its real and imaginary parts separately). So $g(1) = g(0) = 1$, that is $(1-w)e^{\ell(w)} = 1$:

$$\boxed{e^{\ell(w)} = \frac{1}{1-w} \quad \text{for } |w| < 1.}$$

So $\ell(w)$ is *a* logarithm of $1/(1-w)$, namely the one that equals 0 at $w = 0$ and varies continuously with w. For real w it coincides with Equation (18.2).

18.1.2 The logarithm of the zeta function

Fix $s = \sigma + it$ with $\sigma > 1$. For each prime p put $w = p^{-s}$. By Chapter 2, $|p^{-s}| = p^{-\sigma} \le 2^{-\sigma} < \frac{1}{2}$, so the sample problem applies:

$$\frac{1}{1-p^{-s}} = \exp\Big(\sum_{m=1}^{\infty} \frac{p^{-ms}}{m}\Big). \tag{18.3}$$

Multiply Equation (18.3) over all primes $p \le P$, for some cutoff P. The exponential of a sum is the product of exponentials, so

$$\prod_{p \le P} \frac{1}{1-p^{-s}} = \exp\Big(\sum_{p \le P} \sum_{m=1}^{\infty} \frac{p^{-ms}}{m}\Big). \tag{18.4}$$

Now let $P \to \infty$. The left side tends to $\zeta(s)$ by the Euler product. For the right side we need the double sum to converge. Bound the absolute values:

$$\sum_{m=1}^{\infty} \frac{p^{-m\sigma}}{m} \le \sum_{m=1}^{\infty} p^{-m\sigma} = \frac{p^{-\sigma}}{1-p^{-\sigma}} \le 2p^{-\sigma} \quad \text{(geometric series; } p^{-\sigma} \le \tfrac{1}{2}\text{)},$$

and $\sum_p 2p^{-\sigma} \le 2\sum_{k \ge 2} k^{-\sigma}$, which converges for $\sigma > 1$ (Chapter 3). So the double series converges absolutely. Absolute convergence also means the order of summation does not matter (Chapter 3). Because the exponential function is continuous, the right side of Equation (18.4) tends to the exponential of the full double sum. We have proved:

$$\boxed{\log \zeta(s) = \sum_{p} \sum_{m=1}^{\infty} \frac{p^{-ms}}{m}, \qquad \operatorname{Re} s > 1,} \tag{18.5}$$

in the precise sense that the right side converges absolutely and its exponential is $\zeta(s)$. This is a **theorem (proved)**. For real $s > 1$ every term is positive, so this $\log \zeta(s)$ is the ordinary real logarithm.

18.1.3 The von Mangoldt form

Each pair (p, m) corresponds to exactly one prime power $k = p^m$, and by unique factorization (Chapter 15) each prime power arises from exactly one pair. Recall the von Mangoldt function: $\Lambda(k) = \log p$ if $k = p^m$ for a prime p and some $m \ge 1$, and $\Lambda(k) = 0$ otherwise. For $k = p^m$,

$$\frac{\Lambda(k)}{\log k} = \frac{\log p}{\log p^m} = \frac{\log p}{m \log p} = \frac{1}{m}.$$

So Equation (18.5) can be rewritten as a single sum over integers,

$$\log \zeta(s) = \sum_{k=2}^{\infty} \frac{\Lambda(k)}{\log k} k^{-s}, \qquad \operatorname{Re} s > 1. \tag{18.6}$$

Compare this with $\zeta(s) = \sum_k k^{-s}$: taking the logarithm keeps only the prime powers and weights them by $\Lambda(k)/\log k$.

Sample Problem 18.2: How fast do the primes build $\log \zeta(2)$?

Using $\zeta(2) = \pi^2/6$ (a standard result), compare $\log \zeta(2)$ with the contribution of the primes $2, 3, 5, 7$ to Equation (18.5).

Step 1: the target. $\pi^2/6 = 1.6449341$, so $\log \zeta(2) = 0.4977003$.

Step 2: sum over m for one prime. At $s = 2$, $\sum_m p^{-2m}/m = -\log(1-p^{-2}) = \log \frac{p^2}{p^2-1}$ by Equation (18.2) with $x = p^{-2}$.

Step 3: evaluate.

$$p = 2:\ \log(4/3) = 0.2876821, \qquad p = 3:\ \log(9/8) = 0.1177830,$$
$$p = 5:\ \log(25/24) = 0.0408220, \qquad p = 7:\ \log(49/48) = 0.0206193.$$

Step 4: add. The four primes give 0.4669064, which is 93.8% of the target; the infinitely many remaining primes supply the missing 0.0307943.

$$\boxed{\textstyle\sum_{p\le 7}\sum_m \frac{p^{-2m}}{m} = 0.46691 \quad \text{vs} \quad \log\zeta(2) = 0.49770.}$$

At $\sigma = 2$ the large primes matter little because their terms are of size p^{-2}. On the critical line, $\sigma = \frac{1}{2}$, the terms will only shrink like $p^{-1/2}$, and the story changes completely (Section 18.3).

Checkpoint 18.1

In Equation (18.6), which of the integers $k = 6, 8, 9, 10, 16$ have a nonzero coefficient, and what is it?

Answer: Only prime powers survive: $k = 8 = 2^3$ has coefficient $1/3$, $k = 9 = 3^2$ has $1/2$, $k = 16 = 2^4$ has $1/4$. The integers 6 and 10 are not prime powers, so $\Lambda = 0$ and the coefficient is 0.

Key Ideas

- For $|w| < 1$ the series $\ell(w) = \sum_m w^m/m$ converges absolutely and satisfies $e^{\ell(w)} = 1/(1-w)$.
- Applying this to each Euler factor gives $\log\zeta(s) = \sum_p\sum_m p^{-ms}/m$ for $\operatorname{Re} s > 1$, an absolutely convergent series whose exponential is $\zeta(s)$ (Equation (18.5)).
- Equivalently $\log\zeta(s) = \sum_k \frac{\Lambda(k)}{\log k} k^{-s}$: only prime powers contribute, with weight $1/m$ for $k = p^m$.

18.2 Each prime power is a wave in t

Now we look at a single term of Equation (18.5) as we move up a vertical line in the complex plane, that is, as t changes with σ fixed. Using $a^s = e^{s\log a}$ (Chapter 2),

$$\begin{aligned} p^{-ms} &= e^{-ms\log p} = e^{-m\sigma\log p}\, e^{-imt\log p} \\ &= p^{-m\sigma}\left[\cos(mt\log p) - i\sin(mt\log p)\right]. \end{aligned} \tag{18.7}$$

As a function of t, this is a phasor (an arrow in the complex plane representing the number $re^{i\phi}$, with length r and angle ϕ; Chapter 2) of fixed length $p^{-m\sigma}$ turning clockwise at **angular frequency** $\omega = m\log p = \log p^m$ (radians per unit of t). Its imaginary part is a sine wave:

$$\operatorname{Im}\frac{p^{-ms}}{m} = -\frac{\sin(mt\log p)}{m\,p^{m\sigma}}. \tag{18.8}$$

So the imaginary part of $\log\zeta(\sigma+it)$ is a superposition of sine waves, one per prime power, with frequencies $\log 2, \log 3, \log 4, \log 5, \log 7, \log 8, \log 9, \ldots$ and amplitudes $1/(m\,p^{m\sigma})$ that decrease overall as the prime power $k = p^m$ grows. The decrease is not monotonic, because higher powers m carry the extra factor $1/m$: on the critical line, for example, $k = 4$ has amplitude 0.25 while $k = 5$ has 0.4472 (Table 18.1). In the language of Chapter 5, it is an *almost-periodic* sum: the frequencies $\log p$ for different primes are incommensurate (Problem 19 asks you to prove it), so the sum never repeats exactly.

Table 18.1 lists the first few waves on the critical line $\sigma = \frac{1}{2}$, where the amplitude of the $k = p^m$ wave is $1/(m\sqrt{k})$.

Table 18.1: The first prime-power waves in $\operatorname{Im}\log\zeta(\frac{1}{2}+it)$ as they would appear if Equation (18.8) were used at $\sigma=\frac{1}{2}$. Angular frequency $\omega=\log k$, period $2\pi/\omega$ in t, amplitude $1/(m\sqrt{k})$. Dividing the amplitude by π gives the amplitude in $S(t)$ (Section 18.3).

$k=p^m$	(p,m)	$\omega=\log k$	period $2\pi/\omega$	amplitude $1/(m\sqrt{k})$
2	(2,1)	0.6931	9.0647	0.7071
3	(3,1)	1.0986	5.7192	0.5774
4	(2,2)	1.3863	4.5324	0.2500
5	(5,1)	1.6094	3.9040	0.4472
7	(7,1)	1.9459	3.2289	0.3780
8	(2,3)	2.0794	3.0216	0.1179
9	(3,2)	2.1972	2.8596	0.1667
11	(11,1)	2.3979	2.6203	0.3015
13	(13,1)	2.5649	2.4496	0.2774

Sample Problem 18.3: One term on the critical line

Evaluate 3^{-s} at $s=\frac{1}{2}+10i$, and the corresponding contribution $-\frac{1}{\pi}\frac{\sin(t\log 3)}{\sqrt{3}}$ at $t=10$.

Step 1: modulus. $|3^{-s}|=3^{-1/2}=0.577350$.

Step 2: phase. $t\log 3=10\times 1.0986123=10.986123$. Subtract $2\pi+\frac{3\pi}{2}=10.995574$: the angle is $\frac{3\pi}{2}+2\pi+\delta$ with $\delta=-0.009451$. Using $\cos(\frac{3\pi}{2}+\delta)=\sin\delta$ and $\sin(\frac{3\pi}{2}+\delta)=-\cos\delta$,

$$\cos(10.986123)=\sin(-0.009451)=-0.009451,\qquad \sin(10.986123)=-\cos(0.009451)=-0.999955.$$

Step 3: assemble Equation (18.7).

$$3^{-s}=0.577350\,(\cos -i\sin)=0.577350\,(-0.009451+0.999955\,i)=-0.005457+0.577324\,i.$$

Step 4: the wave. $-\frac{1}{\pi}\frac{\sin(10\log 3)}{\sqrt{3}}=-\frac{1}{\pi}(-0.999955)(0.577350)$:

$$\boxed{3^{-s}=-0.005457+0.577324\,i,\qquad -\tfrac{1}{\pi}\sin(10\log 3)/\sqrt{3}=+0.18377.}$$

At $t=10$ the $p=3$ wave is almost exactly at a crest.

18.2.1 Running example RE2: the prime-2 wave

Keep only $p=2$ and $m=1,2,3$, and divide by π as we will do for $S(t)$. This is the running example

$$N_{\text{fl}}^{p=2}(t)=-\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 2)}{m\,2^{m/2}}=-\frac{1}{\pi}\Big[\frac{\sin(t\log 2)}{\sqrt{2}}+\frac{\sin(2t\log 2)}{4}+\frac{\sin(3t\log 2)}{6\sqrt{2}}\Big]. \tag{18.9}$$

The three terms have periods $2\pi/\log 2=9.0647$, $2\pi/(2\log 2)=4.5324$ and $2\pi/(3\log 2)=3.0216$. The second and third are harmonics of the first (their frequencies are integer multiples of $\log 2$), so the sum is exactly periodic with period 9.0647, like a plucked string's fundamental plus overtones (Figure 18.1).

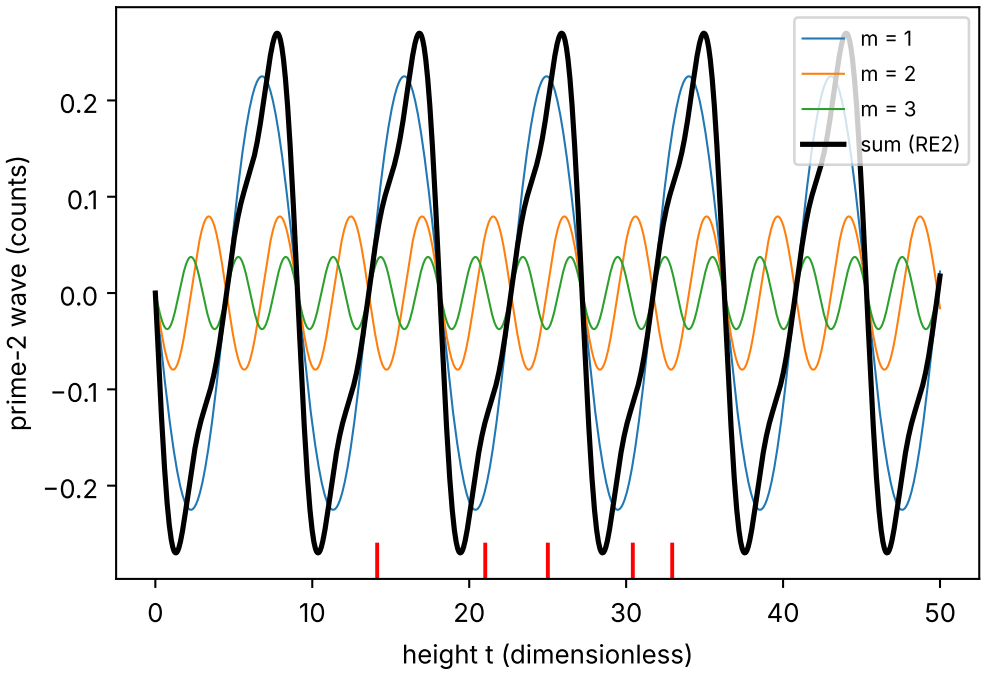


Figure 18.1: Running example RE2. The three harmonics $m = 1, 2, 3$ of the prime $p = 2$ (thin lines) and their sum $N_{\mathrm{fl}}^{p=2}(t)$ of Equation (18.9) (thick line), plotted against height t (dimensionless). The sum repeats with period $2\pi/\log 2 \approx 9.065$. Vertical ticks mark the first five zeta zeros.

Sample Problem 18.4: The rms size of the prime-2 wave

Find the root mean square of $N_{\mathrm{fl}}^{p=2}(t)$ over one period.

Step 1: mean squares of products. Let $\omega_0 = \log 2$ and $T_0 = 2\pi/\omega_0$. By the orthogonality of sines over a period (Chapter 5),

$$\frac{1}{T_0}\int_0^{T_0} \sin(a\omega_0 t)\sin(b\omega_0 t)\,\mathrm{d}t = \begin{cases} \tfrac{1}{2}, & a = b, \\ 0, & a \neq b, \end{cases} \qquad a, b \in \{1, 2, 3\}.$$

Step 2: square the sum. Write $N_{\mathrm{fl}}^{p=2} = \sum_m c_m \sin(m\omega_0 t)$ with $c_m = -1/(\pi m 2^{m/2})$. Expanding the square and averaging term by term, all cross terms vanish by Step 1, leaving

$$\overline{\left(N_{\mathrm{fl}}^{p=2}\right)^2} = \tfrac{1}{2}\sum_{m=1}^{3} c_m^2 = \frac{1}{2\pi^2}\left(\frac{1}{2} + \frac{1}{16} + \frac{1}{72}\right) = \frac{0.5763889}{2\pi^2}.$$

Step 3: take the square root. $\sqrt{0.5763889/2} = 0.536837$, divided by π:

$$\boxed{\mathrm{rms}\left(N_{\mathrm{fl}}^{p=2}\right) = 0.1709.}$$

Keep this number in mind. In the paper's continuation probe (Sec. III) the target fluctuation is exactly this wave, and the paper's App. A lists the "probe-null" (the error of predicting no fluctuation on levels 401–500) as 0.1708. Evaluating $N_{\mathrm{fl}}^{p=2}$ at the smooth zeros $s_{401}, \ldots, s_{500}$ and taking the rms gives the same value to within 0.003 (verified in this chapter's check script). The reason is that these 100 points span about 14 periods of the wave, and their spacing, about 1.3 to 1.4 in t, is less than half the shortest period 3.0216 of the three harmonics; so the samples

resolve all three harmonics, and their mean square approximates the time average.

Checkpoint 18.2

At which angular frequency does the prime power $25 = 5^2$ appear, what is its amplitude $1/(m\sqrt{k})$, and is this wave included in the paper's Eq. (4) when $5 \in P$?

Answer: $\omega = \log 25 = 2\log 5 = 3.2189$; amplitude $1/(2 \cdot 5) = 0.1$; yes, Eq. (4) keeps $m = 1, 2, 3$ for every prime in P, so $m = 2$ for $p = 5$ is included.

Key Ideas

- On a vertical line $\operatorname{Re} s = \sigma$, the term p^{-ms}/m is a phasor of length $p^{-m\sigma}/m$ rotating at angular frequency $m \log p$ in t; its imaginary part is $-\sin(mt \log p)/(mp^{m\sigma})$.
- $\operatorname{Im} \log \zeta(\sigma + it)$ is therefore a superposition of waves at the frequencies $\log p^m$, an almost-periodic function of t.
- The prime-2 part with $m \le 3$ (RE2) is periodic with period $2\pi/\log 2 \approx 9.065$ and has rms 0.1709 after division by π.

18.3 The formal step to the critical line

★ Did you know? *Proving Riemann's Formula*

Riemann's 1859 paper sketched an explicit formula expressing prime counting as a sum over zeta zeros, but did not fully prove it. Hans von Mangoldt gave a rigorous proof in 1895, in a cleaner form for the weighted count $\psi(x) = x - \sum_\rho x^\rho/\rho - \log 2\pi - \frac{1}{2}\log(1 - x^{-2})$. Each zero contributes one oscillation, so the primes become a kind of music played by the zeros.

18.3.1 What we want to explain

Recall from Chapter 17 the Riemann–von Mangoldt formula,

$$N(T) = \frac{\theta(T)}{\pi} + 1 + S(T), \qquad S(T) = \frac{1}{\pi}\arg \zeta\left(\tfrac{1}{2} + iT\right), \tag{18.10}$$

where $N(T)$ is the number of zeros with $0 < t \le T$, θ is the Riemann–Siegel theta function, and $\theta(T)/\pi + 1 = N_{\mathrm{sm}}(T) + \mathcal{O}(1/T)$ with N_{sm} the smooth counting function of the paper's Eq. (1). The argument in $S(T)$ is defined by *continuous variation*: start at $s = 2$, where $\zeta(2) > 0$ and we take $\arg \zeta(2) = 0$; move straight up to $2 + iT$; then move straight left to $\frac{1}{2} + iT$; follow the angle continuously along the way. (This works when T is not the height of a zero.) So $S(T) \approx N(T) - N_{\mathrm{sm}}(T)$: the actual count minus the smooth count.

18.3.2 The argument where the series converges

For $\sigma > 1$, take the imaginary part of Equation (18.5) term by term using Equation (18.8):

$$\Phi_\sigma(t) \equiv \operatorname{Im} \log \zeta(\sigma + it) = -\sum_p \sum_{m=1}^{\infty} \frac{\sin(mt \log p)}{m\, p^{m\sigma}}. \tag{18.11}$$

(Taking the imaginary part term by term is allowed because the series converges absolutely: the imaginary parts of the partial sums are the partial sums of the imaginary parts.)

Is $\Phi_\sigma(t)$ the same as the continuously varied argument used to define S? Three facts settle it.

1. $\Phi_\sigma(t)$ is *an* argument of $\zeta(\sigma+it)$. Indeed, writing $\log\zeta = R + i\Phi$ with R, Φ real, $\zeta = e^{\log\zeta} = e^R e^{i\Phi}$, which is polar form with angle Φ.

2. $\Phi_\sigma(t)$ is continuous in σ and t for $\sigma \geq \sigma_0 > 1$. Each term is bounded in absolute value by $p^{-m\sigma_0}/m$, a fixed summable number independent of σ and t, so the convergence is uniform; a uniformly convergent series of continuous functions has a continuous sum (see the box below).

3. At $s = 2$ every sine vanishes, so $\Phi_2(0) = 0$, the agreed starting value.

For $\sigma > 1$ the function ζ never vanishes: by Equation (18.5), $\zeta(s) = e^{\log\zeta(s)}$, and an exponential is never zero because $|e^z| = e^{\operatorname{Re} z} > 0$. Two continuous arguments of the same nonvanishing function along a path differ by $2\pi\times$(an integer), and that integer, being a continuous integer-valued function, is constant. Starting from the same value 0, they agree everywhere. So on the vertical segment from 2 to $2+iT$, and on the horizontal segment from $2+iT$ to $\sigma+iT$ for any $\sigma > 1$, the continuous argument *is* Φ. This much is a **theorem (proved)**.

Taken on trust

Taken on trust: uniform limits are continuous. Suppose continuous functions f_K converge to f uniformly on a set D. Here D may be an interval, or a region of the plane such as $\sigma \geq \sigma_0$; the statement and its proof are the same for functions of two variables. Uniform convergence means that the largest value (strictly, the least upper bound) of $|f_K - f|$ over D tends to 0. Then f is continuous on D. The "M-test" used above (if $|\text{term}_j| \leq M_j$ with $\sum M_j < \infty$, convergence is uniform) is its standard companion. Both are proved in any real-analysis text and in [22].

18.3.3 The formal step

Here is the step that is **not** justified. The path defining $S(T)$ continues from $\sigma+iT$ (with $\sigma > 1$) to $\frac{1}{2}+iT$. If we *pretend* that Equation (18.11) still holds at $\sigma = \frac{1}{2}$ and divide by π, we obtain

$$S(t) \text{ “=” } -\frac{1}{\pi}\sum_p \sum_{m=1}^{\infty} \frac{\sin(mt\log p)}{m\,p^{m/2}} = -\frac{1}{\pi}\sum_{k=2}^{\infty} \frac{\Lambda(k)}{\sqrt{k}\,\log k}\sin(t\log k). \qquad \text{(formal)} \qquad (18.12)$$

The second form uses $\Lambda(k)/\log k = 1/m$ and $p^{m/2} = \sqrt{k}$ for $k = p^m$. The quotation marks are deliberate. Equation (18.12) is a **heuristic**: it is the right *shape*, and truncations of it approximate S well (Section 18.4), but as written it is not a theorem. Three separate obstacles stand in the way.

Obstacle 1: no absolute convergence. At $\sigma = \frac{1}{2}$ the $m = 1$ amplitudes are $p^{-1/2}$. We now prove that $\sum_p p^{-1/2}$ diverges, by proving the stronger fact that $\sum_p 1/p$ diverges.

Sample Problem 18.5: The sum of the reciprocals of the primes diverges

Show that $\sum_p 1/p$ diverges, and deduce that $\sum_p p^{-1/2}$ diverges.
Step 1: split $\log\zeta(\sigma)$ for real $\sigma > 1$. By Equation (18.5),

$$\log\zeta(\sigma) = \sum_p p^{-\sigma} + \sum_p \sum_{m\geq 2} \frac{p^{-m\sigma}}{m}.$$

Step 2: the $m \geq 2$ part is bounded. Since $p^{-m\sigma} \leq p^{-m}$ and $1/m \leq 1$,

$$\sum_{m\geq 2} \frac{p^{-m\sigma}}{m} \leq \sum_{m\geq 2} p^{-m} = \frac{p^{-2}}{1-p^{-1}} = \frac{1}{p(p-1)}.$$

Summing over primes and then enlarging to all integers $k \geq 2$,

$$\sum_{p} \frac{1}{p(p-1)} \leq \sum_{k=2}^{\infty} \frac{1}{k(k-1)} = \sum_{k=2}^{\infty} \Big(\frac{1}{k-1} - \frac{1}{k}\Big) = 1 \quad \text{(telescoping, Chapter 3)}.$$

Step 3: $\zeta(\sigma) \to \infty$ as $\sigma \to 1^+$. Each term $k^{-\sigma}$ is at least $\int_k^{k+1} x^{-\sigma}\mathrm{d}x$ because $x^{-\sigma}$ decreases, so $\zeta(\sigma) \geq \int_1^{\infty} x^{-\sigma}\mathrm{d}x = \frac{1}{\sigma-1}$.
Step 4: combine. From Steps 1–3, $\sum_p p^{-\sigma} \geq \log\zeta(\sigma) - 1 \geq \log\frac{1}{\sigma-1} - 1$, which grows without bound as $\sigma \to 1^+$.
Step 5: conclude. For every $\sigma > 1$ and every prime, $1/p \geq p^{-\sigma}$, so $\sum_p 1/p \geq \sum_p p^{-\sigma}$. The right side can be made as large as we like, hence

$$\boxed{\sum_{p} \frac{1}{p} = \infty, \qquad \text{and therefore} \qquad \sum_{p} \frac{1}{\sqrt{p}} = \infty \quad (\text{since } p^{-1/2} \geq p^{-1}).}$$

Even the $m = 2$ amplitudes, $1/(2p)$, have a divergent sum by the same result. Only the $m \geq 3$ terms, of size at most $p^{-3/2}/3$, converge absolutely (Problem 6). Figure 18.2 shows the contrast.

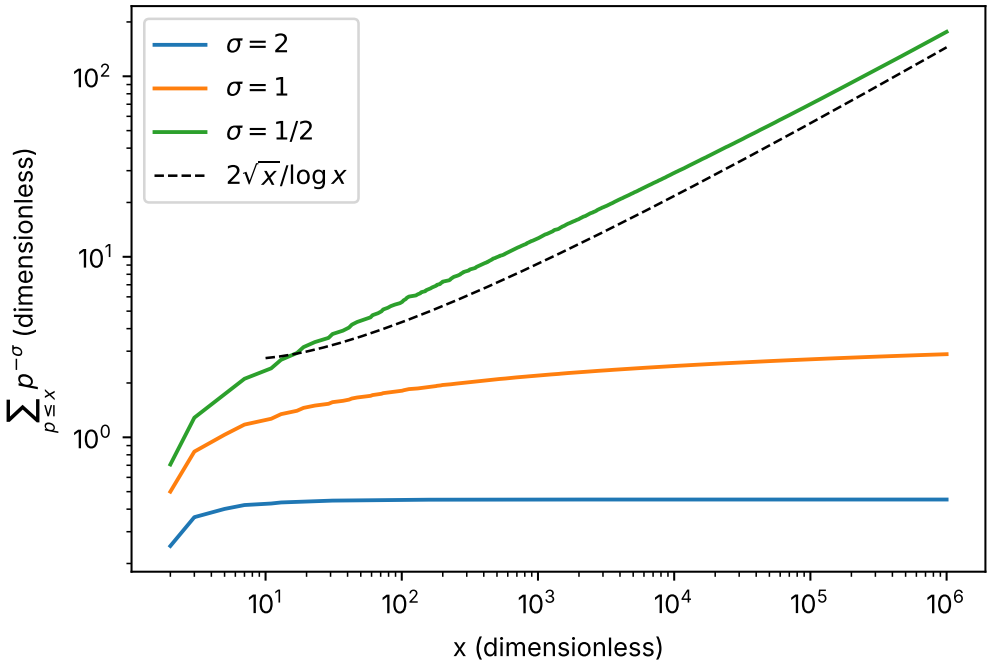


Figure 18.2: Partial sums $\sum_{p\leq x} p^{-\sigma}$ over primes up to x (both axes logarithmic, dimensionless). For $\sigma = 2$ the sum levels off; for $\sigma = 1$ it creeps upward without bound (like $\log\log x$, by Mertens' theorem, which is proved but whose proof is beyond this book; we proved only the divergence); for $\sigma = \frac{1}{2}$, the critical line, it grows like $2\sqrt{x}/\log x$ for large x (dashed curve; a consequence of the prime number theorem, Chapter 15). The amplitudes of the waves in Equation (18.12) are not summable.

Sample Problem 18.6: How big do the critical-line amplitudes get?

Compute $\sum_{p\le 100} p^{-1/2}$, the sum of the $m = 1$ amplitudes for the 25 primes below 100 (the prime set of the paper's "25 primes" oracle).
Step 1: list. The primes are $2, 3, 5, \dots, 97$, with $1/\sqrt{p}$ from 0.707107 down to 0.101535.
Step 2: add (term by term, or with `numpy`):

$$\sum_{p\le 100} \frac{1}{\sqrt{p}} = 5.5365.$$

If all 25 waves happened to peak together, the $m = 1$ part alone of Equation (18.12) would reach $5.5365/\pi \approx 1.76$ (about 1.76 mean spacings), far more than the typical size $|S| < 1$ found in Chapter 17. The waves must mostly cancel.

Obstacle 2: the strip $\frac{1}{2} \le \sigma \le 1$. The series Equation (18.5) was established only for $\sigma > 1$. At $\sigma = 1$ the function ζ has its pole (Chapter 16); our path at height T avoids $s = 1$, but nothing we have proved says that the series represents $\log\zeta$ anywhere with $\sigma \le 1$. The value of $\arg\zeta$ on the critical line is determined by the analytic continuation of ζ (Chapter 16), not by the series.

Obstacle 3: jumps. As T passes a zero t_n (assumed simple, which is verified numerically for all zeros used here), $N(T)$ jumps by 1 while $\theta(T)$ is smooth. By Equation (18.10), $S(T)$ jumps by $+1$ at every zero. Between zeros it decreases steadily, with slope $-N'_{\mathrm{sm}}(T)$ up to $\mathcal{O}(1/T^2)$. Every truncation of Equation (18.12) is a finite sum of sines, hence continuous. By the box above, a sequence of continuous functions cannot converge *uniformly* to a function with a jump. So no sequence of truncations can converge to S uniformly on any interval containing a zero.

Taken on trust

Taken on trust: rigorous versions. (i) Everything in Section 18.1 and in Equation (18.11) for $\sigma > 1$ is proved above. (ii) Whether Equation (18.12) converges at a given point t of the critical line is not established by anything in this book, and the paper does not rely on it. (iii) What *is* proved, by A. Selberg, is that $S(t)$ is well approximated on average over t by short, smoothly weighted sums over primes $p \le x$ of the form $-\frac{1}{\pi}\sum_p \sin(t\log p)/\sqrt{p}$, with explicit control of the error; a consequence is that $S(t)$ has mean square growing like $\frac{1}{2\pi^2}\log\log t$ and an approximately Gaussian distribution. These theorems are presented in [6]. (iv) The exact cure for the jumps is *smoothing*: replace S by an average of $N(t)$ against a smooth weight. Then the sum over prime powers becomes an exact identity with a convergent, smoothly cut-off prime sum: the Guinand–Weil explicit formula of Section 18.7.

Pitfall

Do not write Equation (18.12) with an equals sign and treat it as an identity. It is a formal continuation of a series that converges only for $\mathrm{Re}\, s > 1$. Its truncations, used with care, are good *approximations*; the paper itself presents its Eq. (4) as a truncation used as a baseline, with "no novelty" and with the predictions checked against actual zeros.

Checkpoint 18.3

The $m = 3$ terms of Equation (18.12) have amplitudes $1/(3p^{3/2})$. Does $\sum_p 1/(3p^{3/2})$ converge? What about the $m = 2$ amplitudes?

Answer: The $m = 3$ sum converges: it is bounded by $\frac{1}{3}\sum_{k\geq 2} k^{-3/2}$, a convergent p-series with exponent $3/2 > 1$. The $m = 2$ sum $\sum_p \frac{1}{2p}$ diverges, because $\sum_p 1/p$ diverges.

Key Ideas

- For $\sigma > 1$, $\operatorname{Im} \log \zeta(\sigma + it) = -\sum_p \sum_m \sin(mt \log p)/(mp^{m\sigma})$ exactly, and it equals the continuously varied argument.
- Setting $\sigma = \frac{1}{2}$ formally gives $S(t)$ "=" $-\frac{1}{\pi}\sum_p \sum_m \sin(mt \log p)/(mp^{m/2})$ (Equation (18.12)): a heuristic, not a theorem.
- Obstacles: the amplitudes are not summable ($\sum_p 1/p = \infty$, proved), the series is not established for $\sigma \leq 1$, and S jumps by 1 at each zero while finite sine sums are continuous.
- Rigorous replacements exist: Selberg's smoothed prime sums for $S(t)$, and the exact Guinand–Weil formula with smooth test functions.

18.4 Why a truncated sum still works

After the warnings of the previous section, Figure 18.3 may surprise you. It plots the measured $S(t) \approx N(t) - N_{\text{sm}}(t)$, a sawtooth staircase, together with truncations of Equation (18.12) keeping one prime, six primes and 25 primes (always with $m \leq 3$). The truncated sums are smooth, of course, but they follow the *middle* of the sawtooth: wherever the jumps sit above average, the smooth curve rises; where they sit below, it falls.

18.4.1 What a zero "sees"

At a zero, $N(t)$ jumps from $n - 1$ to n. Adopt the convention (as in Chapter 17) that $N(t_n) = n - \frac{1}{2}$, the average of its left and right limits; correspondingly $S(t_n)$ means the midpoint of its jump. Then, up to the neglected $\mathcal{O}(1/t)$ term of θ/π, every zero satisfies

$$N_{\text{sm}}(t_n) + S(t_n) = n - \tfrac{1}{2}. \tag{18.13}$$

To predict t_n we therefore need only the *midpoint value* of S at the zero, not the whole sawtooth.

18.4.2 The midpoint heuristic

We need a fact about Fourier series: Dirichlet's theorem, which is **proved, but its proof is beyond this book** (see [22]). For a piecewise smooth periodic function with a jump, the partial sums of its Fourier series converge at the jump to the *midpoint* of the two one-sided values. There is Gibbs overshoot nearby (Chapter 5), but not at the jump itself. Example: the square wave equal to -1 on $(-\pi, 0)$ and $+1$ on $(0, \pi)$ has the sine series $\frac{4}{\pi}\sum_{j\geq 0} \frac{\sin((2j+1)x)}{2j+1}$. At the jump $x = 0$ every term vanishes, so every partial sum equals 0, the midpoint of -1 and $+1$. The prime sum Equation (18.12) plays the role of a Fourier expansion of $S(t)$, with the prime powers as the frequencies. If the analogy holds, its truncations should approximate S at a zero by the midpoint of the jump, which is exactly the quantity in Equation (18.13).

A complementary picture: decompose S into a slowly varying part and a sawtooth that jumps by $+1$ at each zero and ramps down in between. If the zeros were evenly spaced with

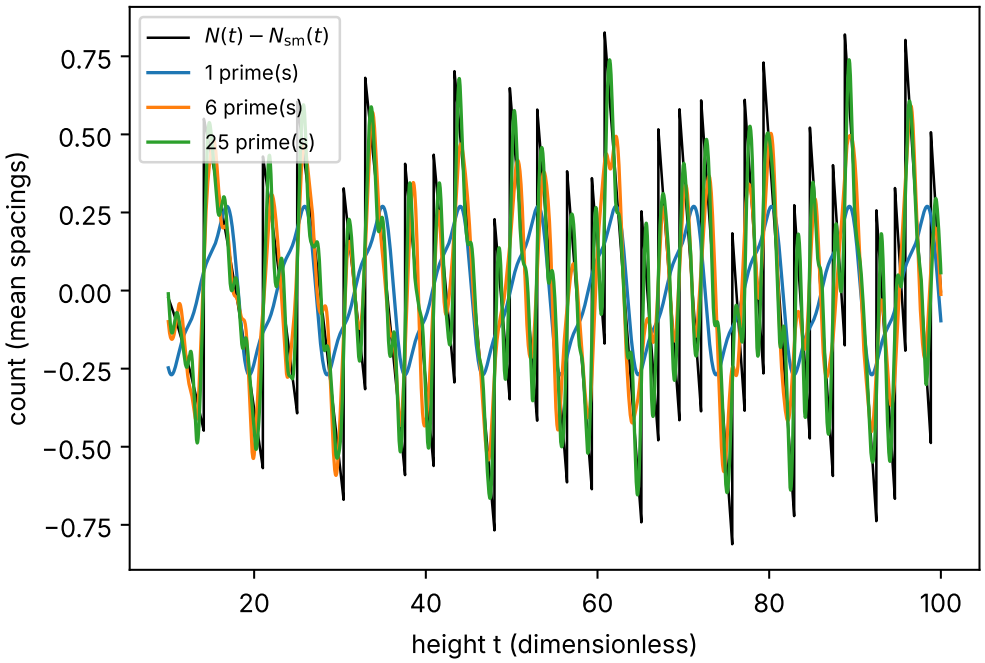


Figure 18.3: The measured fluctuating count $S(t) \approx N(t) - N_{\text{sm}}(t)$ (black step-and-ramp curve, jumping by $+1$ at each zero) and truncations of the explicit formula with $m \leq 3$: primes $\{2\}$, the first 6 primes, and the first 25 primes. Horizontal axis: height t (dimensionless); vertical axis: count (equivalently, mean spacings). The smooth curves follow the midpoints of the jumps.

spacing gap, the sawtooth would be periodic with zero average and with its lowest frequency at $2\pi/\text{gap}$; its midpoint value at each zero would be 0. A sum of waves with frequencies well below $2\pi/\text{gap}$ cannot represent the sawtooth at all; it represents only the slow part, and at a zero the slow part equals the midpoint value. With gap $= 2\pi/\log(t/2\pi)$, the condition "well below $2\pi/\text{gap}$" reads $\log p^m \ll \log(t/2\pi)$. Because the real zeros are not evenly spaced, this picture is only qualitative, and including more primes than this keeps helping in practice (Section 18.6).

Status: this whole subsection is **heuristic**. The rigorous statements are those of the boxes in Section 18.3 and Section 18.7; the practical evidence is the numerical accuracy of the resulting predictions.

Sample Problem 18.7: The truncated formula at the first zero

Evaluate the one-prime truncation $N_{\text{fl}}^{p=2}(t)$ of Equation (18.9) at $t = t_1 = 14.134725$ and compare it with $S(t_1)$ as defined by Equation (18.13).

Step 1: phases. $t_1 \log 2 = 9.797445$, $2t_1 \log 2 = 19.594890$, $3t_1 \log 2 = 29.392335$. Reducing by 3π, 6π and 9π: $9.797445 = 3\pi + 0.372667$, $19.594890 = 6\pi + 0.745334$, $29.392335 = 9\pi + 1.118001$.

Step 2: sines. Using $\sin(x + 3\pi) = -\sin x$, $\sin(x + 6\pi) = \sin x$, $\sin(x + 9\pi) = -\sin x$:

$$\sin(t_1 \log 2) = -0.364101, \quad \sin(2t_1 \log 2) = 0.678217, \quad \sin(3t_1 \log 2) = -0.899228.$$

Step 3: weights. Divide by $m2^{m/2} = 1.414214,\ 4,\ 8.485281$: the three terms are -0.257458, 0.169554, -0.105975, summing to -0.193879.

Step 4: multiply by $-1/\pi$. $N_{\text{fl}}^{p=2}(t_1) = 0.06171$.

Step 5: the measured midpoint. With $t_1/2\pi = 2.2496113$ and $\log(2.2496113) =$

0.8107574,

$$N_{\mathrm{sm}}(t_1) = 2.2496113 \times 0.8107574 - 2.2496113 + 0.875 = 0.44928.$$

So $S(t_1) = \frac{1}{2} - 0.44928 = 0.05072$ (neglecting corrections of order $1/t$, here $1/(48\pi t_1) \approx 0.0005$).

$$\boxed{N_{\mathrm{fl}}^{p=2}(t_1) = 0.0617 \quad \text{vs} \quad S(t_1) = 0.0507.}$$

A single prime gets the sign and rough size right.

Checkpoint 18.4

Just below a zero, $S = -0.2$; just above it, $S = +0.8$. What value of S enters Equation (18.13), and is the zero earlier or later than the smooth zero s_n?

Answer: The midpoint, 0.3. Then $N_{\mathrm{sm}}(t_n) = n - \frac{1}{2} - 0.3 < N_{\mathrm{sm}}(s_n)$, and since N_{sm} increases, $t_n < s_n$: the zero is early, by about 0.3 mean spacings.

Key Ideas

- Up to the neglected $\mathcal{O}(1/t)$ term, a zero satisfies $N_{\mathrm{sm}}(t_n) + S(t_n) = n - \frac{1}{2}$ with $S(t_n)$ the midpoint of its jump; prediction needs only that midpoint.
- Heuristically, truncated prime sums behave like truncated Fourier series, which converge to midpoints at jumps; low frequencies carry the slow part of S, which is what moves zeros early or late.
- This explanation is heuristic; its support is rigorous smoothed versions plus numerical accuracy.

18.5 Predicting the n-th zero

18.5.1 The level equation

Replace the exact but unknown $S(t_n)$ in Equation (18.13) by a finite truncation. Following the paper's Eq. (4), for a finite set of primes P define

$$N_{\mathrm{fl}}(t) = -\frac{1}{\pi} \sum_{p \in P} \sum_{m=1}^{3} \frac{\sin(mt \log p)}{m\, p^{m/2}}, \tag{18.14}$$

and define the predicted n-th zero as the solution t of

$$\boxed{N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \tfrac{1}{2}.} \tag{18.15}$$

Two truncations are made: primes outside P are dropped, and powers $m \geq 4$ are dropped (their total amplitude for $p = 2$ is at most 0.0487, Problem 8). Nothing is fitted: P is chosen in advance and every coefficient is fixed by Equation (18.12).

18.5.2 A first-order solution

Recall from Chapter 17 that the smooth zero s_n solves $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$, that $N'_{\mathrm{sm}}(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}$, and that the mean spacing is $\mathrm{gap}_n = ds_n/dn = 1/N'_{\mathrm{sm}}(s_n) = 2\pi/\log(s_n/2\pi)$. Write the

predicted zero as $t = s_n + \Delta$. Expanding to first order,

$$N_{\mathrm{sm}}(s_n + \Delta) \approx N_{\mathrm{sm}}(s_n) + N'_{\mathrm{sm}}(s_n)\Delta = n - \tfrac{1}{2} + \frac{\Delta}{\mathrm{gap}_n}, \tag{18.16}$$

$$N_{\mathrm{fl}}(s_n + \Delta) \approx N_{\mathrm{fl}}(s_n) + N'_{\mathrm{fl}}(s_n)\Delta. \tag{18.17}$$

Substituting both into Equation (18.15) and cancelling $n - \frac{1}{2}$,

$$\frac{\Delta}{\mathrm{gap}_n} + N_{\mathrm{fl}}(s_n) + N'_{\mathrm{fl}}(s_n)\Delta = 0 \quad \Longrightarrow \quad \Delta = -\frac{N_{\mathrm{fl}}(s_n)}{1/\mathrm{gap}_n + N'_{\mathrm{fl}}(s_n)}. \tag{18.18}$$

If moreover $|N'_{\mathrm{fl}}| \ll 1/\mathrm{gap}_n$ (the smooth count rises much faster than the fluctuation changes), the predicted fluctuation in mean spacings is simply

$$\frac{t - s_n}{\mathrm{gap}_n} \approx -N_{\mathrm{fl}}(s_n). \tag{18.19}$$

A positive N_{fl} means an early zero, as in the checkpoint above. Neglected: second-order terms in Δ (the curvature of N_{sm} and of N_{fl}).

Sample Problem 18.8: Predicting t_1 from the prime 2

Predict the first zero using $P = \{2\}$, and compare with $t_1 = 14.134725$ and with the null prediction s_1.

Step 1: the smooth zero. Solve $N_{\mathrm{sm}}(s) = \frac{1}{2}$ by Newton's method (Chapter 4), $s \mapsto s + (\frac{1}{2} - N_{\mathrm{sm}}(s))/N'_{\mathrm{sm}}(s)$, starting from $s = t_1$. There $N_{\mathrm{sm}}(t_1) = 0.44928$ (previous sample) and $N'_{\mathrm{sm}}(t_1) = \frac{1}{2\pi}\log\frac{t_1}{2\pi} = 0.8107574/2\pi = 0.12904$, so the first step is $0.05072/0.12904 = 0.3931$ and the first iterate is 14.5278. At the first iterate, $N_{\mathrm{sm}}(14.5278) - \frac{1}{2} = 8.5 \times 10^{-4}$ and $N'_{\mathrm{sm}}(14.5278) = 0.1334$, so the second step is $-8.5 \times 10^{-4}/0.1334 = -0.0064$, giving $s_1 = 14.5214$; a further step changes this by less than 10^{-5}. Then $\mathrm{gap}_1 = 2\pi/\log(s_1/2\pi) = 2\pi/0.837746 = 7.500$.

Step 2: N_{fl} and its slope at s_1. As in the previous sample, reduce the phases: $s_1 \log 2 = 10.0654 = 3\pi + 0.6407$, $2s_1 \log 2 = 6\pi + 1.2813$, $3s_1 \log 2 = 9\pi + 1.9220$. Shifting by 3π or 9π reverses the sign of both sine and cosine, and shifting by 6π changes neither. So the sines are -0.5977, 0.9584, -0.9390 and the cosines are -0.8017, 0.2855, 0.3440 (for $m = 1, 2, 3$). Dividing the sines by $m2^{m/2} = 1.414214,\ 4,\ 8.485281$ gives -0.4227, 0.2396, -0.1107, which sum to -0.2937; multiplying by $-1/\pi$ gives $N_{\mathrm{fl}}^{p=2}(s_1) = 0.09350$. Differentiating Equation (18.9) gives $N'_{\mathrm{fl}}(t) = -\frac{\log 2}{\pi}\sum_{m=1}^{3}\frac{\cos(mt\log 2)}{2^{m/2}}$. Dividing the cosines by $2^{m/2} = 1.414214,\ 2,\ 2.828427$ gives -0.5669, 0.1427, 0.1216, which sum to -0.3025; multiplying by $-\log 2/\pi = -0.22064$ gives $N'_{\mathrm{fl}}(s_1) = 0.0667$.

Step 3: crude first order, Equation (18.19). $t \approx s_1 - \mathrm{gap}_1 N_{\mathrm{fl}}(s_1) = 14.5214 - 0.7012 = 13.820$. This overshoots badly.

Step 4: why, and the better first order. At $t \approx 14$, $1/\mathrm{gap}_1 = 0.1333$ while $N'_{\mathrm{fl}}(s_1) = 0.0667$: the condition $|N'_{\mathrm{fl}}| \ll 1/\mathrm{gap}$ fails. Using Equation (18.18), $\Delta = -0.09350/(0.13333 + 0.06675) = -0.4673$, so $t \approx 14.054$.

Step 5: solve Equation (18.15) exactly. Let $f(t) = N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) - \frac{1}{2}$ and evaluate it at the two estimates already in hand. At $t = 14.1347$ (essentially t_1) the previous sample gives $N_{\mathrm{sm}} = 0.4493$ and $N_{\mathrm{fl}} = 0.0617$, so $f(14.1347) = 0.4493 + 0.0617 - 0.5 = +0.0110$. At $t = 13.8202$: $13.8202/2\pi = 2.199553$, so $N_{\mathrm{sm}}(13.8202) = 2.199553\log(2.199553) - 2.199553 + 0.875 = 0.4093$. The three RE2 phases reduce to $3\pi + 0.15465$, $6\pi + 0.30931$ and $9\pi + 0.46396$, so (with the sign rules of the previous sample) the sines are -0.1540, 0.3044, -0.4475; dividing by 1.414214, 4, 8.485281, adding and multiplying by $-1/\pi$ gives $N_{\mathrm{fl}}(13.8202) = 0.0272$. Hence $f(13.8202) = 0.4093 + 0.0272 - 0.5 = -0.0635$. This is a sign

change, so a root lies between (bisection logic, Chapter 4). Linear interpolation between these two points gives $13.8202 + 0.3145 \times 0.0635/(0.0635 + 0.0110) = 14.0883$, where $f = +0.00036$. Repeat the linear interpolation, now between 13.8202 and 14.0883. The slope is $(0.00036 + 0.0635)/0.2681 = 0.238$, so the correction is $-0.00036/0.238 = -0.0015$, giving 14.0868, that is,

$$\boxed{t_1^{\text{pred}} = 14.087.}$$

Step 6: compare in mean spacings. True fluctuation $(t_1 - s_1)/\text{gap}_1 = -0.0516$; predicted $(14.087 - 14.5214)/7.500 = -0.0580$; null 0. The prediction error is 0.0064 spacings against the null's 0.0516.

18.5.3 How many roots?

Equation (18.15) asks where the increasing function N_{sm} plus a wiggle crosses a horizontal line. If the wiggle's slope can be more negative than the rise of N_{sm}, the sum can turn down and cross the same line two or three times (Figure 18.4). A safe condition for a single crossing is that $N'_{\text{sm}}(t) + N'_{\text{fl}}(t) > 0$ everywhere near the crossing.

Sample Problem 18.9: When is the prime-2 root unique?

Find a height above which the level equation with $P = \{2\}$ has at most one root.
Step 1: bound the wiggle's slope. Differentiating Equation (18.9) and using $|\cos| \le 1$,

$$|N'_{\text{fl}}(t)| \le \frac{\log 2}{\pi}\Big(\frac{1}{\sqrt{2}} + \frac{1}{2} + \frac{1}{2\sqrt{2}}\Big) = \frac{0.693147 \times 1.560660}{\pi} = 0.34434.$$

Step 2: compare with the smooth slope. $N'_{\text{sm}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$ exceeds 0.34434 when $\log\frac{t}{2\pi} > 2\pi \times 0.34434 = 2.16353$, i.e. $\frac{t}{2\pi} > e^{2.16353} = 8.7018$.
Step 3: conclude. For $t > 2\pi \times 8.7018$, the function $N_{\text{sm}} + N_{\text{fl}}$ is strictly increasing, so it crosses each level $n - \frac{1}{2}$ at most once:

$$\boxed{t > 54.7 \;\Rightarrow\; \text{unique root for } P = \{2\}.}$$

Below 54.7 this bound does not guarantee uniqueness. On the interval $[13.82, 14.14]$ used in the previous sample, however, evaluating $N'_{\text{sm}} + N'_{\text{fl}}$ on a fine grid shows that it stays positive throughout (this chapter's check script), so the root near 14 found there is unique on that interval. With more primes the bound on $|N'_{\text{fl}}|$ grows, so the guaranteed height rises; for $P = \{2, 3\}$ it is about 617 (Problem 9). The bound is a worst case: all waves would have to reach their steepest descent simultaneously.

Pitfall

When N_{fl} has large amplitude or contains steep high-frequency terms, Equation (18.15) can have several roots near s_n. A root finder then returns *one* of them, determined by the bracket, and the "prediction" depends on that arbitrary choice. Always check for multiple roots. The paper flags exactly this for one of its fitted controls (Sec. IV): the level equation "has several roots for about a third of the held-out levels".

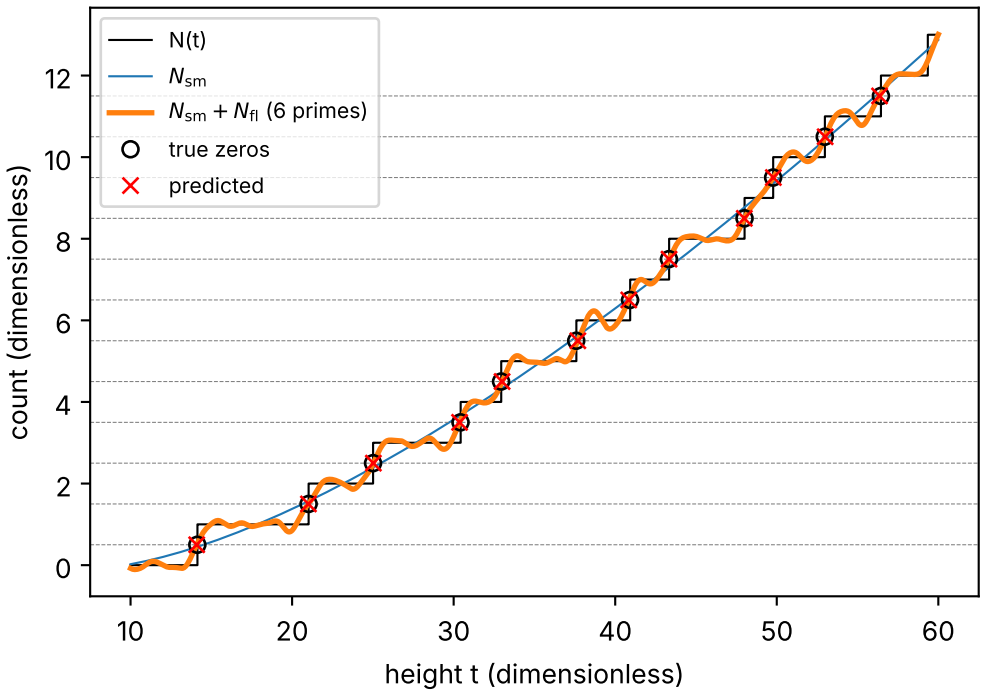


Figure 18.4: The level equation. Staircase: the true count $N(t)$. Thin smooth line: $N_{\mathrm{sm}}(t)$. Thick line: $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t)$ with the first six primes. Dashed horizontal lines: the levels $n - \frac{1}{2}$. Open circles: true zeros; crosses: predicted zeros (roots of Equation (18.15)). Horizontal axis height t, vertical axis count (dimensionless).

18.5.4 The paper's procedure

The paper's App. A, step (8), solves Equation (18.15) with Brent's method (Chapter 4) bracketed by ± 3 around s_n (we take this, as `qmnt` does, to mean three mean spacings), with the bracket expanded if the function does not change sign at its ends. Here is the same procedure in a few lines.

```
import numpy as np, qmnt
from scipy.optimize import brentq

P = qmnt.first_primes(6)                    # 2, 3, 5, 7, 11, 13

def level_fn(t, n):
    """N_sm(t) + N_fl(t) - (n - 1/2): its root is the predicted zero."""
    return qmnt.Nsm(t) + qmnt.Nfl(t, P, mmax=3) - (n - 0.5)

n = 450
s = brentq(lambda t: qmnt.Nsm(t) - (n - 0.5), 10.0, 5000.0)   # smooth zero
g = 2*np.pi/np.log(s/(2*np.pi))                               # mean gap
a, b = s - 3*g, s + 3*g                     # bracket +-3 gaps (App. A step 8)
while level_fn(a, n)*level_fn(b, n) > 0:     # expand on sign failure
    a, b = a - g, b + g
t_pred = brentq(level_fn, a, b, args=(n,))
t_true = qmnt.zeros(n)[-1]
print("predicted fluctuation", (t_pred - s)/g)
print("true fluctuation     ", (t_true - s)/g)
```

18.5.5 The França–LeClair form and Lambert's W

França and LeClair [3, 4] wrote the exact condition Equation (18.13) as a transcendental equation for the n-th zero. Since $N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}$, it reads

$$\frac{t_n}{2\pi}\log\frac{t_n}{2\pi e} + S(t_n) = n - \frac{11}{8}, \tag{18.20}$$

where $S(t_n)$ is understood as the appropriate limit at the zero (they take a limit from slightly off the critical line). Drop S and solve the smooth part exactly. Put $A = n - \frac{11}{8}$ and $y = t/(2\pi e)$, so that $\frac{t}{2\pi} = ey$ and the equation becomes $e\,y\log y = A$. Put $y = e^u$:

$$u\,e^u = \frac{A}{e} \quad\Longrightarrow\quad u = W\!\left(\frac{A}{e}\right),$$

where W is Lambert's function, defined by $W(z)e^{W(z)} = z$ (Chapter 4). Then $y = e^u = \frac{A/e}{W(A/e)}$ (because $e^{W(z)} = z/W(z)$), and

$$s_n = 2\pi e\,y = \frac{2\pi\,(n - \frac{11}{8})}{W\!\left((n - \frac{11}{8})/e\right)}. \tag{18.21}$$

For $n = 1$, $A = -\frac{3}{8}$ and $W(-0.13795) = -0.16226$ on the principal branch, giving $s_1 = 14.521$, in agreement with the Newton value above. This is the "Lambert-W asymptotic of the counting function" mentioned in the paper's Sec. I A. The prime corrections of França and LeClair are truncations of the classical explicit formula, the same idea as Equation (18.15).

Checkpoint 18.5

Why is s_n the natural center for the bracket in the paper's root finding, and why is ± 3 spacings wide enough?

Answer: $|N_{\mathrm{fl}}|$ is typically well below one count, so the predicted zero lies within a fraction of a spacing of s_n (Equation (18.19)); that makes s_n the natural center. N_{sm} rises by about one count per mean spacing. So the level function $N_{\mathrm{sm}} + N_{\mathrm{fl}} - (n - \frac{1}{2})$ is about $-3 + N_{\mathrm{fl}}$ at $s_n - 3\mathrm{gap}_n$ and about $+3 + N_{\mathrm{fl}}$ at $s_n + 3\mathrm{gap}_n$. For the prime sets of Table V the sum of all amplitudes in N_{fl} is below 3, so the function changes sign across the bracket, as Brent's method requires, without the bracket being needlessly wide. The real danger is a different one: extra crossings of the *same* level, the root ambiguity of the pitfall above.

Key Ideas

- The predicted n-th zero solves $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ (Equation (18.15)), with zero fitted parameters.
- To first order, the predicted fluctuation in mean spacings is $-N_{\mathrm{fl}}(s_n)$, valid when $|N'_{\mathrm{fl}}| \ll 1/\mathrm{gap}_n$; otherwise use Equation (18.18) or solve exactly.
- The root is unique where $N'_{\mathrm{sm}} + N'_{\mathrm{fl}} > 0$; for $P = \{2\}$ this is guaranteed above $t \approx 54.7$. Large or steep corrections create root ambiguity.
- Dropping S gives the smooth zeros in closed form, $s_n = 2\pi(n - \frac{11}{8})/W((n - \frac{11}{8})/e)$.

18.6 How many primes?

★ Did you know? *A Teenager's Guess*

In an 1849 letter to Johann Encke, Carl Friedrich Gauss recalled that around 1792 or 1793, as a teenager, he noticed that primes near x occur with density about $1/\log x$. He kept checking this against prime tables for decades. The prime number theorem, $\pi(x) \sim x/\log x$, was finally proved in 1896, independently, by Jacques Hadamard and Charles de la Vallée Poussin, using the zeta function.

18.6.1 Scoring a prediction

Recall the paper's scoring (Chapter 17): for each held-out index n, convert both the true and the predicted zero into fluctuations in mean spacings, $(t_n - s_n)/\mathrm{gap}_n$ and $(t_n^{\mathrm{pred}} - s_n)/\mathrm{gap}_n$. Report the **rms** of their difference and the Pearson **correlation** ("corr") between them. For two sequences a_n and b_n (here the true and the predicted fluctuations) with averages $\bar{a}$ and $\bar{b}$ over the holdout,

$$\mathrm{corr}(a,b) = \frac{\sum_n (a_n - \bar{a})(b_n - \bar{b})}{\sqrt{\sum_n (a_n - \bar{a})^2 \, \sum_n (b_n - \bar{b})^2}}. \tag{18.22}$$

It is a number between -1 and $+1$; the proof, via the Cauchy–Schwarz inequality, comes in Chapter 19. The value $+1$ means that the predicted fluctuations are an exact increasing linear function of the true ones, and 0 means no linear relation at all. So corr measures agreement of *shape*, not of size. The **null** predicts $t_n^{\mathrm{pred}} = s_n$, so its rms is the rms of the true fluctuations; on zeros 401–500 (running example RE5) it is 0.263.

18.6.2 Accuracy versus the prime budget

Table 18.2 reproduces the first two rows of the paper's Table V. Every entry was recomputed with `qmnt.oracle_zeros` and `qmnt.score` and agrees with the paper to the digits shown (this chapter's check script asserts this).

Table 18.2: Accuracy of the truncated explicit formula (rms in mean spacings / corr), zero fitted parameters, $m \le 3$. Quoted from the paper's Table V and reproduced with `qmnt`. "6 primes" means $P = \{2,3,5,7,11,13\}$; "25 primes" means all primes below 100.

holdout	null	{2}	3 primes	6 primes	25 primes
401–500	0.263	0.198 / +0.66	0.116 / +0.90	0.074 / +0.96	0.043 / +0.99
501–1000	0.270	0.208 / +0.64	0.127 / +0.88	0.083 / +0.95	0.050 / +0.99

```
import numpy as np, qmnt

n = np.arange(401, 501)                          # RE5, the principal holdout
z = qmnt.zeros(500)
print("null rms", qmnt.rms(qmnt.fluct(z[n-1], n)))   # about 0.263

for k in [1, 3, 6, 25]:
    P = qmnt.first_primes(k)
    t_pred = qmnt.oracle_zeros(n, P, mmax=3)   # roots of N_sm + N_fl = n - 1/2
    r, c = qmnt.score(t_pred, n)               # rms and corr in mean spacings
    print(f"{k:2d} primes: rms {r:.3f}  corr {c:+.2f}")
```

Figure 18.5 shows the six-prime predictions against the truth on zeros 401–500 and the rms error as a function of the number of primes. Three features stand out.

1. **One prime already helps.** With $P = \{2\}$ the error drops from 0.263 to 0.198 and the correlation is +0.66: the prime-2 wave alone carries a real share of the fluctuation.
2. **Slow, steady improvement.** The amplitudes decay only like $1/\sqrt{p}$ (Section 18.3), so each additional prime contributes a little; the error falls gradually rather than collapsing.
3. **Weak height dependence.** Going from zeros 401–500 to 501–1000, both the null and the oracle errors grow slightly, and the ratio of oracle error to null changes little (six primes: from $0.074/0.263 = 0.28$ to $0.083/0.270 = 0.31$). Through zero 10^4 (the paper's Table V) the ratio drifts to $0.119/0.296 = 0.40$ at a fixed six-prime budget, while the correlation stays at +0.92 or above. The paper describes relative skill as nearly constant and attributes the mild growth to the growth of the null and to the increasing number of arithmetically relevant primes at height.

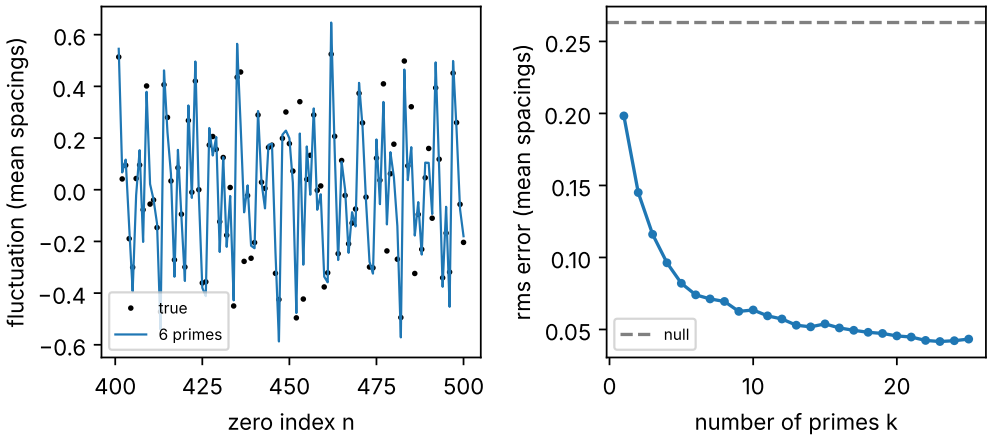


Figure 18.5: Left: true fluctuations $(t_n - s_n)/\text{gap}_n$ of zeros 401–500 (dots) and the six-prime prediction (line), both in mean spacings, against the zero index n. Right: rms error on zeros 401–500 against the number of primes in P (the first k primes, $m \le 3$); dashed line: the null.

Checkpoint 18.6

Two predictors on the same holdout have corr +0.66 and +0.96. Which can you say is more accurate in rms without further information, and why might a predictor with high corr still have a large rms?

Answer: Neither statement follows from correlation alone: correlation measures shape, not amplitude (Equation (18.22); more in Chapter 19). A predictor with the right shape but twice the true amplitude, or a constant offset, has high corr and large rms. In Table V both metrics improve together, so there the six-prime row is better on both counts.

Key Ideas

- Scores are rms and corr of predicted vs true fluctuations in mean spacings, relative to the null (0.263 on zeros 401–500).
- With zero fitted parameters, the truncated formula gives 0.198/ + 0.66 (one prime), 0.074/ + 0.96 (six primes), 0.043/ + 0.99 (25 primes) on zeros 401–500; qmnt reproduces

these.

- Improvement with more primes is gradual, reflecting the slow $1/\sqrt{p}$ decay of the amplitudes, and only weakly dependent on height.

18.7 The explicit formula as a duality

Equation (18.12) says that the prime powers are the *frequencies* of the zeros' fluctuations. Riemann's 1859 memoir [2] (see [5] for a guided reading) already contained the mirror statement: the zeros are the frequencies of the primes' fluctuations. Both are faces of one exact identity.

18.7.1 The primes, built from the zeros

Recall Chebyshev's function $\psi(x) = \sum_{k \le x} \Lambda(k)$ from Chapter 15: a staircase that jumps by $\log p$ at each prime power p^m, and grows like x by the prime number theorem. Let $\psi_0(x)$ be the same staircase with the value at each jump replaced by the midpoint.

Taken on trust

The Riemann–von Mangoldt explicit formula (proved; proof beyond this book). For $x > 1$,

$$\psi_0(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \tfrac{1}{2}\log(1 - x^{-2}), \tag{18.23}$$

where the sum runs over all nontrivial zeros ρ, taken in symmetric pairs $\rho, \bar{\rho}$ in order of increasing $|\operatorname{Im}\rho|$. The sum converges (conditionally) for each x. A proof is given in [25]; see also [5].

Pair each zero $\rho_n = \frac{1}{2} + it_n$ with its mirror image $\bar{\rho}_n$. Since $x^{\bar{\rho}}/\bar{\rho}$ is the complex conjugate of x^{ρ}/ρ, the pair contributes $2\operatorname{Re}(x^{\rho_n}/\rho_n)$. Using $x^{\rho_n} = x^{1/2}e^{it_n \log x}$,

$$-2\operatorname{Re}\frac{x^{\rho_n}}{\rho_n} = -\frac{2\sqrt{x}}{|\rho_n|}\cos(t_n \log x - \arg\rho_n). \tag{18.24}$$

As a function of $\log x$, each zero on the critical line contributes a wave of angular frequency t_n and amplitude $2\sqrt{x}/|\rho_n|$. All zeros computed so far lie on the critical line. That every nontrivial zero does is the Riemann hypothesis, which is open, so writing every term of Equation (18.23) this way assumes it. Compare: in Equation (18.12), each prime power contributes a wave in t of angular frequency $\log p^m$. The roles of "position" and "frequency" are swapped. Figure 18.6 shows how adding more zero-waves sharpens the prime staircase.

Sample Problem 18.10: The size of the first zero's wave in $\psi(x)$

Find the amplitude $2\sqrt{x}/|\rho_1|$ of the first zero pair's wave at $x = 10$ and at $x = 100$.
Step 1: $|\rho_1|$**.** $|\rho_1|^2 = \frac{1}{4} + t_1^2 = 0.25 + 199.790451 = 200.040451$, so $|\rho_1| = 14.143566$.
Step 2: $x = 10$**.** $2\sqrt{10} = 6.324555$, divided by 14.143566: 0.4472.
Step 3: $x = 100$**.** $2 \times 10/14.143566 = 1.4141$.

$$\boxed{0.4472 \text{ at } x = 10, \qquad 1.4141 \text{ at } x = 100.}$$

The waves grow like $\sqrt{x}$, while $\psi(x)$ itself grows like x; relative to ψ, the zero-waves shrink like $1/\sqrt{x}$. This $\sqrt{x}$ is the $\operatorname{Re}\rho = \frac{1}{2}$ of the critical line made visible: a zero off the line, with $\operatorname{Re}\rho = \beta > \frac{1}{2}$, would produce a wave growing like x^{β}.

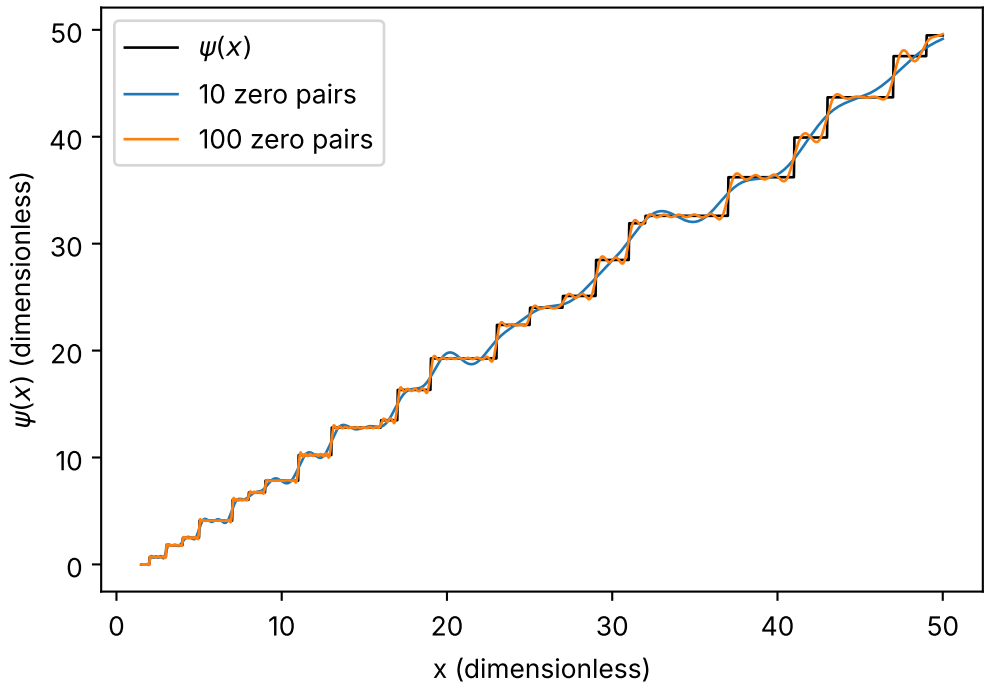


Figure 18.6: The primes from the zeros. Staircase: $\psi(x)$, jumping by $\log p$ at each prime power. Smooth curves: the right side of Equation (18.23) truncated to the first 10 and first 100 pairs of zeros. Horizontal axis x, vertical axis ψ (both dimensionless).

18.7.2 The Guinand–Weil form

The formula Equation (18.23) sums a smooth function of the zeros (the waves) to produce a jumpy function of x. Its mirror version sums a function of the zeros to produce prime terms, and it becomes an exact, absolutely convergent identity once both sides are smoothed.

Taken on trust

The Guinand–Weil explicit formula (proved; proof beyond this book). Let $h(r)$ be an even function that extends analytically to the strip $|\operatorname{Im} r| \le \frac{1}{2} + \epsilon$ and decays fast enough there, and let $g(u) = \frac{1}{2\pi}\int_{-\infty}^{\infty} h(r)e^{-iru}\mathrm{d}r$ be its Fourier transform. Write each nontrivial zero as $\rho = \frac{1}{2} + i\gamma$ (with $\gamma = \pm t_n$ for zeros on the critical line). Then, in one common normalization,

$$\sum_{\rho} h(\gamma) = h\left(\tfrac{i}{2}\right) + h\left(-\tfrac{i}{2}\right) - g(0)\log\pi + \frac{1}{2\pi}\int_{-\infty}^{\infty} h(r)\,\operatorname{Re}\frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{ir}{2}\right)\mathrm{d}r - 2\sum_{k=2}^{\infty}\frac{\Lambda(k)}{\sqrt{k}}\,g(\log k). \tag{18.25}$$

The left side is a sum over zeros; the last term is a sum over prime powers. Proofs appear, in varying normalizations, in advanced texts on analytic number theory; the special case Equation (18.23) is proved in [25].

Read Equation (18.25) as a duality: probe the zeros with a function h, and the primes respond through its Fourier transform g, evaluated at the frequencies $\log k$. A smooth h has a rapidly decaying g, so the prime sum effectively stops at some finite k: this is the exact version of "truncating the prime sum".

A formal check: recovering $N(T)$. Choose h to be the box: $h(r) = 1$ for $|r| \le T$, 0 otherwise. This h is not smooth, so the following is **formal**, but it shows where Equation (18.12) comes from.

1. *Left side.* Zeros come in pairs $\pm t_n$, so $\sum_\rho h(\gamma) = 2N(T)$.
2. *Transform.* $g(u) = \frac{1}{2\pi}\int_{-T}^{T} e^{-iru}\mathrm{d}r = \frac{1}{2\pi}\cdot\frac{2\sin(Tu)}{u} = \frac{\sin(Tu)}{\pi u}$, and $g(0) = T/\pi$ (the limit $u \to 0$).
3. *Gamma term.* Here $\frac{\Gamma'}{\Gamma}(z) = \frac{d}{dz}\log\Gamma(z)$ is the logarithmic derivative of the Gamma function. Stirling's formula for complex arguments (Chapter 4) says that $\log\Gamma(z) \approx (z-\frac{1}{2})\log z - z + \frac{1}{2}\log 2\pi$ for large $|z|$ away from the negative real axis. We differentiate it term by term; that this approximation may be differentiated is taken on trust, like Stirling's formula itself. This gives $\frac{\Gamma'}{\Gamma}(z) \approx \log z + \frac{z-\frac{1}{2}}{z} - 1 = \log z - \frac{1}{2z} \approx \log z$ for large $|z|$. Since $\log z = \log|z| + i\arg z$ (Chapter 2), $\operatorname{Re}\log z = \log|z|$. So $\operatorname{Re}\frac{\Gamma'}{\Gamma}(\frac{1}{4}+\frac{ir}{2}) \approx \log\left|\frac{1}{4}+\frac{ir}{2}\right| \approx \log\frac{|r|}{2}$. Hence the integral is approximately $\frac{1}{2\pi}\cdot 2\int_0^T \log\frac{r}{2}\,\mathrm{d}r = \frac{1}{\pi}\left[T\log\frac{T}{2} - T\right]$, using $\int \log\frac{r}{2}\,\mathrm{d}r = r\log\frac{r}{2} - r$.
4. *Constant terms.* Combining $-g(0)\log\pi = -\frac{T}{\pi}\log\pi$ with step 3 gives $\frac{T}{\pi}(\log\frac{T}{2\pi} - 1)$. The terms $h(\pm\frac{i}{2})$ and the corrections to Stirling contribute amounts that do not grow with T (in a careful treatment they produce the $\frac{7}{8}$ of Chapter 17); we do not track them here.
5. *Prime term.* $-2\sum_k \frac{\Lambda(k)}{\sqrt{k}}\frac{\sin(T\log k)}{\pi\log k}$.
6. *Divide by 2.*

$$N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + (\text{const}) - \frac{1}{\pi}\sum_k \frac{\Lambda(k)}{\sqrt{k}\log k}\sin(T\log k).$$

The first two terms are $N_{\text{sm}}(T)$ without its constant, and the last is exactly the formal series Equation (18.12). The non-smooth box is precisely what makes the prime sum fail to converge absolutely; smoothing h repairs it.

18.7.3 Landau's formula: the spectrum of the zeros

Now probe the zeros with a pure wave: compute $\sum_n x^{\rho_n}$ over the zeros up to height T. If the zeros were randomly placed, this would be a random walk of size about $\sqrt{N(T)}$. Landau proved that it is not.

Taken on trust

Landau's formula (proved; proof beyond this book). For fixed $x > 1$, as $T \to \infty$,

$$\sum_{0<t_n\le T} x^{\rho_n} = -\frac{T}{2\pi}\Lambda(x) + \mathcal{O}(\log T), \tag{18.26}$$

where the sum runs over the zeros with $0 < t_n \le T$, each counted once (the conjugates $\bar\rho_n$ are not included); T here is the upper height of the zeros included, not a training window (one of the other meanings of T in the paper); and $\Lambda(x) = 0$ unless x is an integer prime power [18]. A version uniform in x is due to Gonek [19].

Write $x = e^\omega$ with $\omega > 0$. Then $x^{\rho_n} = x^{1/2}x^{it_n} = e^{\omega/2}e^{i\omega t_n}$, so the real part of the left side of Equation (18.26) is $e^{\omega/2}\sum_{0<t_n\le T}\cos(\omega t_n)$, while the main term on the right is real. Dividing

both sides by the positive factor $e^{\omega/2}$ (which also divides the $\mathcal{O}(\log T)$ error by $e^{\omega/2}$) gives

$$\sum_{0<t_n\le T} \cos(\omega t_n) \approx -\frac{T}{2\pi}\frac{\Lambda(e^{\omega})}{e^{\omega/2}}. \tag{18.27}$$

The left side is the (cosine) Fourier transform of the set of zeros, viewed as a comb of spikes at the heights t_n (Chapter 6). The right side is zero except at $\omega = \log p^m$, where it is a large negative spike growing linearly with T. So *the spectrum of the zeros consists of lines at the logarithms of prime powers.* With a finite window the lines have width about $2\pi/T$, the resolution of Chapter 6.

Sample Problem 18.11: Predicted spectral lines for the first 1000 zeros

Using Equation (18.27) with $T = t_{1000} = 1419.42248$, predict $\sum_{n\le 1000}\cos(\omega t_n)$ at $\omega = \log 2$, $\log 3$ and $\log 4$.
Step 1: the common factor. $T/2\pi = 225.908$.
Step 2: the weights $\Lambda(k)/\sqrt{k}$. $k = 2$: $0.693147/1.414214 = 0.490129$. $k = 3$: $1.098612/1.732051 = 0.634285$. $k = 4$: $\Lambda(4) = \log 2$, so $0.693147/2 = 0.346574$.
Step 3: multiply.

$$\boxed{\sum \cos(\omega t_n) \approx -110.7\ (\omega = \log 2), \quad -143.3\ (\log 3), \quad -78.3\ (\log 4).}$$

These are predictions; the computed sum, plotted in Figure 18.7, shows dips of comparable size at these frequencies. The error term in Equation (18.26) is $\mathcal{O}(\log T)$ with a constant that depends on x, so the formula does not tell us the exact size of the discrepancy. At $\omega = \log 2$, the check script confirms that the computed sum agrees with -110.7 to within 30%. Notice that the line at $\log 3$ is deeper than the line at $\log 2$: the weight $\log p/\sqrt{p}$ is largest near $p = 7$ (Problem 11).

```
import numpy as np, qmnt

t = qmnt.zeros(1000)                    # heights t_1 ... t_1000
T = t[-1]
omega = np.linspace(0.3, 3.5, 6401)     # step 5e-4, finer than 2*pi/T
spec = -np.array([np.cos(w*t).sum() for w in omega])
for k, lab in [(2, "log 2"), (3, "log 3"), (4, "log 4")]:
    i = np.argmin(abs(omega - np.log(k)))
    lam = np.log(2) if k in (2, 4) else np.log(3)
    print(lab, spec[i], "Landau:", T/(2*np.pi)*lam/np.sqrt(k))
```

This is the fact that the paper's sparse control C2 uses without being told: a model that only assumes "a few sinusoids in t" and looks for the largest periodogram peaks of the zeros' fluctuations finds frequencies at $\log p^m$ (Chapter 27).

18.7.4 How far can prime-only constructions go?

Our truncation Equation (18.15) is deliberately simple. Much more refined prime-only constructions exist. As the paper reports (Sec. I A, paragraph b), Connes [21] extremizes a truncated form of Weil's quadratic form built from the primes $p \le 13$ and obtains approximations to the first 50 zeros with accuracies from 10^{-3} down to 2.6×10^{-55}, provably on the critical line. We mention this only to locate our baseline: the paper uses the simple truncation as an "information-existence baseline", not as the best possible arithmetic predictor.

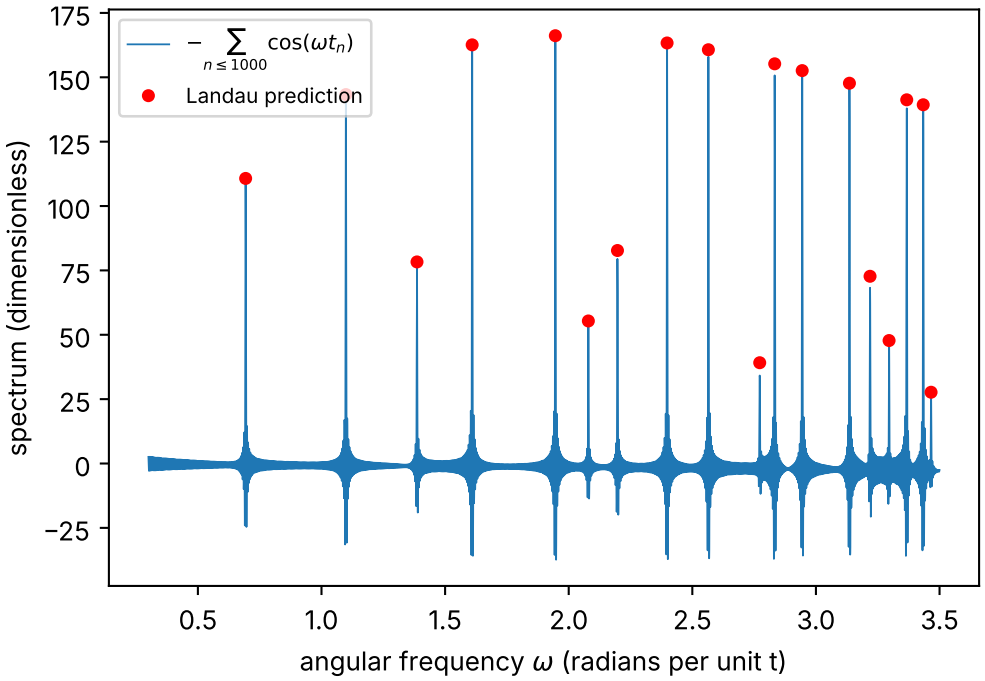


Figure 18.7: The spectrum of the zeros. Line: $-\sum_{n\le 1000}\cos(\omega t_n)$ against angular frequency ω (radians per unit t). Dots: Landau's prediction $\frac{T}{2\pi}\Lambda(e^{\omega})e^{-\omega/2}$ at each prime power $e^{\omega}=p^m$, with $T=t_{1000}$. The peaks sit at $\log 2, \log 3, \log 4, \log 5, \log 7, \ldots$.

Checkpoint 18.7

In Equation (18.24) the angular frequency (in $\log x$) of the n-th wave is t_n; in Equation (18.12) the angular frequency (in t) of the p^m wave is $\log p^m$. What plays the role of the amplitude in each?

Answer: In $\psi(x)$: $2\sqrt{x}/\,|\rho_n|$, decreasing like $1/t_n$ for high zeros. In $S(t)$: $1/(\pi m p^{m/2})$, decreasing like $1/\sqrt{p^m}$ for large prime powers.

Key Ideas

- The explicit formula is an exact duality: $\psi_0(x)=x-\sum_\rho x^\rho/\rho-\log 2\pi-\frac{1}{2}\log(1-x^{-2})$ builds the primes from waves with frequencies t_n (in $\log x$).
- The Guinand–Weil formula, Equation (18.25), relates sums over zeros of a smooth h to sums over prime powers of its transform $g(\log k)$; a sharp cutoff formally reproduces $N(T)\approx N_{\mathrm{sm}}(T)+$ the formal series.
- Landau's formula: $\sum_{t_n\le T} x^{\rho_n}\approx -\frac{T}{2\pi}\Lambda(x)$, so the Fourier spectrum of the zeros has lines at $\log p^m$ of depth $\frac{T}{2\pi}\Lambda(p^m)/p^{m/2}$.
- Far more accurate prime-only constructions exist (Connes, first 50 zeros); the simple truncation is a baseline.

18.8 Paper connection

Eq. (4) is Equation (18.14). The paper's arithmetic oracle (Sec. II D) is the truncation of the formal series Equation (18.12) to a prime set P with $m \leq 3$, and its predicted zeros solve $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$, our Equation (18.15). The paper is explicit that the formula "has no fitted parameters and no novelty" and cites França and LeClair [3, 4]; Sec. I A (b) adds that these are "truncations of the classical Riemann–von Mangoldt/Guinand–Weil explicit formula" (our Section 18.7) and that "the Lambert-W asymptotic of the counting function" is the smooth part (our Equation (18.21)). The paper also stresses that the oracle "is not a fair ML competitor (it encodes the known functional form)": its role is to show that predictive information exists.

App. A step (8): how the roots are found. "Eq. (4), $m \leq 3$, Brent bracketed ± 3 around the smooth zero (bracket expanded on sign failure)." That is the procedure of our listing in Section 18.5; the uniqueness discussion there explains why a bracket must be chosen with care.

Table V. Each column is Equation (18.15) with a different P: $\{2\}$, the first 3, 6, and 25 primes. The rows 401–500 and 501–1000 are reproduced in our Table 18.2. The paper's caption notes that "the mild growth at fixed prime budget mirrors the growth of the null", which grows from 0.263 to 0.296 over the table, and that "relative skill is nearly constant"; the six-prime correlation intervals are $[+0.95, +0.97]$ on 401–500 and $[+0.91, +0.92]$ on 5001–10000.

Fig. 1. The paper's Fig. 1 is a single panel showing zeros 370–460. It overlays the predictions of the fitted operator and of the six-prime explicit formula, and shades the held-out region past zero 400. Its explicit-formula curve is the same prediction as the left panel of our Figure 18.5, which shows the full holdout 401–500; its held-out correlation +0.96 matches Table V. The fitted operator's held-out correlation in the same figure is −0.07 (Chapters 29 and 25).

Sec. IV and Table IV, controls A and B. Control A is Equation (18.15) with up to 300 primes (900 terms), still with nothing fitted; it reaches 0.018 on zeros 401–500, continuing the slow improvement of our Figure 18.5. Control B keeps the frequencies $m \log p$ of Equation (18.14) but fits the amplitudes; with 25 primes (150 coefficients) it gives 0.028. With 150 primes (900 coefficients on 400 zeros) it degrades to 0.180 and is flagged root-ambiguous: the level equation "has several roots for about a third of the held-out levels" and "the score varies by at most 0.012 with the choice of root". This is the multiple-root situation of our Figure 18.4 and pitfall: fitted amplitudes can be large enough that $N'_{\mathrm{sm}} + F'$ changes sign (Problem 14). The paper's Sec. IV also states that "one prime beats every fitted operator's holdout performance by a factor $\gtrsim 1.3$" (Problem 13).

Sec. IV, control C2 and Landau–Gonek. The sparse control selects frequencies that all lie "within half the Fourier resolution of the training window of the logarithm of a prime power" (for the 800-zero fit: prime powers up to 113 except 32, 64, 81, and six primes between 131 and 211). The paper calls this classical, citing [18, 19]: it is our Equation (18.27) and Figure 18.7. Its skill decays with distance (0.078/ + 0.96 on zeros 1001–2000) "as the estimated frequencies dephase ..., which the exact frequencies of the oracle do not" (Chapter 27).

Sec. III, arithmetic-continuation probe. The synthetic target is "solely of the p=2 term of Eq. (4)", our RE2, Equation (18.9). App. A step (15) defines the targets as $E_n[\mathrm{smooth}] - N_{\mathrm{fl}}^{p=2}(s_n)\mathrm{gap}_n$, so the target fluctuation in mean spacings is defined as $-N_{\mathrm{fl}}^{p=2}(s_n)$. This has the form of the first-order relation Equation (18.19). It coincides with what the one-prime oracle would predict only to first order (Problem 23). The probe-null 0.1708 (App. A key anchors) is the rms of RE2 at the smooth zeros, which our sample problem predicted as 0.1709 from the

time average. The fitted potential reproduces this wave in-sample (rms 0.0555, corr +0.95) but not out of sample (rms 0.1958, worse than the probe-null, corr +0.07). By contrast, any model that contains the frequency $\log 2$ continues the wave by construction.

Sec. V. The paper notes that "the dominant p=2 oscillation, period $2\pi/\log 2 \approx 9$ in t, is only ∼6–7 levels at these heights". The period is our 9.0647; the number of levels per period is this period divided by the local mean spacing.

Sec. I A (b) and Sec. II D: Connes. Both passages cite Connes' truncated Weil quadratic form over $p \le 13$ (first 50 zeros, accuracies from 10^{-3} to 2.6×10^{-55}, provably on the critical line), which we placed at the end of Section 18.7. The paper's point is that this construction "consumes only prime data" and is therefore on the arithmetic side of its comparison.

Overloaded symbols. In this chapter T is a height on the critical line, as in $N(T)$. The paper also uses T for the length of a training window (Sec. II D, App. A step 20), for the tensors $T^{(n)}$ (Sec. II E), and for Chebyshev polynomials T_k; and it uses N both for the counting function and for the grid size.

18.9 Summary

- For $|w| < 1$, $\ell(w) = \sum_m w^m/m$ satisfies $e^{\ell(w)} = 1/(1-w)$. Applied to the Euler product, this proves $\log\zeta(s) = \sum_p \sum_m p^{-ms}/m = \sum_k \frac{\Lambda(k)}{\log k} k^{-s}$ for $\operatorname{Re} s > 1$ (Equations (18.5) and (18.6)).
- Each term p^{-ms} is a phasor rotating at angular frequency $\log p^m$ in t, with imaginary part $-\sin(mt\log p)/p^{m\sigma}$ (Equations (18.7) and (18.8)).
- For $\sigma > 1$ the imaginary part of the series is the continuously varied argument of ζ (Equation (18.11)). Setting $\sigma = \frac{1}{2}$ formally gives $S(t)$ "=" $-\frac{1}{\pi}\sum_p \sum_m \frac{\sin(mt\log p)}{mp^{m/2}}$ (Equation (18.12)), a heuristic: the amplitudes are not summable ($\sum_p 1/p = \infty$), the series is not established for $\sigma \le 1$, and S jumps at zeros.
- Truncations still approximate the midpoint value of S at a zero (heuristic: Fourier sums converge to midpoints); rigorous versions use smoothing (Selberg; Guinand–Weil).
- Prediction: solve $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ (Equation (18.15)); to first order the predicted fluctuation is $-N_{\mathrm{fl}}(s_n)$ (Equation (18.19)), with the slope-corrected form Equation (18.18). Roots are unique where $N'_{\mathrm{sm}} + N'_{\mathrm{fl}} > 0$.
- Smooth zeros in closed form: $s_n = 2\pi(n - \frac{11}{8})/W((n - \frac{11}{8})/e)$ (Equation (18.21)).
- Accuracy on zeros 401–500 with zero fitted parameters: null 0.263; {2}: 0.198/ + 0.66; 6 primes: 0.074/ + 0.96; 25 primes: 0.043/ + 0.99 (Table 18.2).
- Duality: $\psi_0(x) = x - \sum_\rho x^\rho/\rho - \log 2\pi - \frac{1}{2}\log(1 - x^{-2})$ (Equation (18.23)); Guinand–Weil (Equation (18.25)); Landau: $\sum_{t_n \le T} x^{\rho_n} \approx -\frac{T}{2\pi}\Lambda(x)$, so the spectrum of the zeros has lines at $\log p^m$ (Equations (18.26) and (18.27)).

18.10 Problems

1. • Give the coefficient $\Lambda(k)/\log k$ of k^{-s} in Equation (18.6) for $k = 9$, $k = 12$ and $k = 16$.

2. • Show that $|p^{-s}| = p^{-\sigma}$ for $s = \sigma + it$, and evaluate it for $p = 13$ at $\sigma = \frac{1}{2}$ and at $\sigma = 2$.

3. • Find the period in t of the wave $\sin(mt\log p)$ for $(p,m)=(3,1)$, $(2,2)$ and $(13,1)$.

4. • Using the paper's Table V, by what factor is the six-prime oracle's rms smaller than the null on zeros 401–500, and on zeros 5001–10000?

5. • Explain in two sentences why no finite sum of sine waves can equal $S(t)$ on an interval containing a zero.

6. •• Show that the part of Equation (18.12) with $m\geq 3$ converges absolutely for every t, while the parts with $m=1$ and $m=2$ do not.

7. •• Following the RE2 sample problem, compute the time-averaged rms of the $p=3$ part of Equation (18.14), $-\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 3)}{m3^{m/2}}$.

8. •• The paper's Eq. (4) stops at $m=3$. Using the series $-\log(1-x)=\sum_m x^m/m$, find the exact value of $\frac{1}{\pi}\sum_{m\geq 4}\frac{1}{m2^{m/2}}$, an upper bound on the size of the neglected $p=2$ terms.

9. •• Repeat the uniqueness sample problem for $P=\{2,3\}$: find the height above which $N_{\mathrm{sm}}+N_{\mathrm{fl}}$ is guaranteed to be strictly increasing.

10. •• Verify Equation (18.21) for $n=1$: compute $W(-3/(8e))$ on the principal branch to four digits by trial (solve $we^{w}=-0.13795$) and deduce s_1.

11. •• Using Landau's formula with $T=t_{1000}=1419.42248$, predict the depth of the spectral lines at $\omega=\log 5$ and $\omega=\log 7$. Among the prime powers $k\leq 20$, which has the largest weight $\Lambda(k)/\sqrt{k}$?

12. •• Show that the coefficient $\Lambda(k)/(\sqrt{k}\log k)$ in the prime sum of the Guinand–Weil formula, for $k=p^m$, equals the amplitude $1/(mp^{m/2})$ of Equation (18.12).

13. •• The paper (Sec. IV) says one prime beats every fitted operator's holdout performance by a factor $\gtrsim 1.3$. Using Table I (best operator OOS rms on zeros 401–500) and Table V, compute the factor for the 401–500 holdout.

14. •• Compute $N'_{\mathrm{sm}}(t)$ at $t=1000$. Explain, using it, why a fitted correction $F(t)=\sum_j[a_j\sin\omega_j t+b_j\cos\omega_j t]$ with 450 frequencies $m\log p$ reaching $3\log 863\approx 20$ (the paper's 150-prime control B) can make the level equation root-ambiguous, while the 25-prime oracle mostly is not.

15. •• Using `qmnt.oracle_zeros` and `qmnt.score`, reproduce the 3-prime entry of Table V for zeros 401–500 and report the rms and corr you obtain.

16. •• Using `qmnt.zeros` and `qmnt.fluct`, compute the null rms on zeros 401–500 and on 501–1000, and compare with Table V.

17. •• In the paper's Fig. 1 the six-prime formula has held-out correlation $+0.96$ and the fitted operator -0.07. Express the six-prime rms on 401–500 (Table V) as a fraction of the null and explain in words what a correlation of -0.07 says about the operator's predictions.

18. ••• Show that for $\mathrm{Re}\,s=\sigma>1$, $|\log\zeta(s)|\leq\log\zeta(\sigma)$, with $\log\zeta$ defined by Equation (18.5). Deduce that $|\zeta(s)|\geq 1/\zeta(\sigma)$, so ζ has no zeros with $\mathrm{Re}\,s>1$.

19. ••• Prove that $\log 2/\log 3$ is irrational. Deduce that $\sin(t\log 2)+\sin(t\log 3)$ is not periodic in t (the frequencies are incommensurate), so the six-prime N_{fl} never repeats exactly.

20. ••• Differentiate Equation (18.6) term by term (allowed for $\mathrm{Re}\,s>1$; take this on trust) to show $-\zeta'(s)/\zeta(s)=\sum_{k\geq 2}\Lambda(k)k^{-s}$.

21. • • • Using `qmnt.zeros(1000)` and `numpy`, compute $\sum_{n\le 1000}\cos(t_n \log 2)$ and compare it with Landau's prediction -110.7. Then scan ω from 0.3 to 3.5 in steps of 5×10^{-4} and list the five deepest dips.

22. • • • Using `qmnt`, compute the rms error on zeros 401–500 for P equal to the first k primes, $k = 1, \dots, 25$. What is the smallest k for which the rms falls below 0.1? Is the decrease monotone in k?

23. • • • In the paper's continuation probe (Sec. III, App. A step 15) the target fluctuation is $-N_{\text{fl}}^{p=2}(s_n)$. Would the one-prime oracle of Equation (18.15) with $P = \{2\}$ predict this synthetic target exactly? Explain why or why not, and contrast with the fitted operator's probe scores (OOS 0.1958, corr +0.07, probe-null 0.1708).

24. • • • (a) Compute the amplitude $2\sqrt{x}/|\rho_2|$ of the wave contributed to $\psi_0(x)$ by the second zero pair ($t_2 = 21.022040$) at $x = 1000$. (b) Show that a hypothetical zero $\rho = \beta + i\gamma$ off the critical line would contribute, together with its conjugate, a wave of amplitude $2x^{\beta}/|\rho|$, so that relative to $\psi(x) \sim x$ it decays like $x^{\beta-1}$. (c) For $\beta = 0.6$ at the height $\gamma = t_2$, by what factor would its amplitude exceed that of the on-line zero ρ_2 at $x = 10^6$?

25. • • • The formal box-function computation in Section 18.7 dropped $h(\pm i/2)$. Explain why these are undefined for a box but well defined for a Gaussian $h(r) = e^{-r^2/(2a^2)}$, and compute $g(u)$ for the Gaussian. Describe how fast the prime sum $\sum_k \Lambda(k)k^{-1/2}g(\log k)$ is cut off.

18.11 Answers to selected problems

★ Did you know? *A Sign Change Far Away*

For every x ever checked, $\pi(x) < \mathrm{li}(x)$, so many believed this always holds. In 1914 J. E. Littlewood proved that the difference changes sign infinitely often. In 1933 Stanley Skewes showed, assuming the Riemann hypothesis, that a first crossing occurs below about $10^{10^{10^{34}}}$. Later work lowered the bound to about 1.4×10^{316}, still far beyond any direct computation.

1. $9 = 3^2$: $\Lambda(9)/\log 9 = \log 3/(2\log 3) = \frac{1}{2}$. $12 = 2^2\cdot 3$ is not a prime power: 0. $16 = 2^4$: $\frac{1}{4}$.

2. $|p^{-s}| = |p^{-\sigma}|\,\left|e^{-it\log p}\right| = p^{-\sigma}\cdot 1$ because $\left|e^{i\phi}\right| = 1$ for real ϕ. For $p = 13$: $13^{-1/2} = 0.27735$ at $\sigma = \frac{1}{2}$; $13^{-2} = 1/169 = 0.005917$ at $\sigma = 2$.

3. Period $= 2\pi/(m\log p)$: $(3,1)$: $2\pi/1.098612 = 5.719$; $(2,2)$: $2\pi/1.386294 = 4.532$; $(13,1)$: $2\pi/2.564949 = 2.450$.

4. $0.263/0.074 = 3.55$ on 401–500; $0.296/0.119 = 2.49$ on 5001–10000. The relative advantage shrinks somewhat with height at a fixed budget, as the caption of Table V explains (more primes become relevant at greater height).

5. $S(t)$ jumps by $+1$ at each zero, because $N(t)$ does and N_{sm} is continuous. A finite sum of sines is continuous, so it cannot have a jump; the best it can do is approximate the midpoint at the jump.

7. By orthogonality, the mean square is $\frac{1}{2\pi^2}\sum_{m=1}^{3}\frac{1}{m^2 3^m} = \frac{1}{2\pi^2}(\frac{1}{3}+\frac{1}{36}+\frac{1}{243}) = \frac{0.3652263}{2\pi^2}$. The rms is $\sqrt{0.1826132}/\pi = 0.4273326/\pi = 0.1360$.

8. With $x = 1/\sqrt{2}$, $\sum_{m\ge 1} x^m/m = -\log(1-0.7071068) = 1.2279471$. Subtracting $m = 1,2,3$: $0.7071068 + 0.25 + 0.1178511 = 1.0749579$, leaving 0.1529892. Dividing by π: 0.0487. So the neglected $p = 2$ terms can shift the count by at most 0.0487, and typically much less since their phases vary.

9. $|N_{\text{fl}}'| \le \frac{1}{\pi}\left[\log 2(\frac{1}{\sqrt{2}}+\frac{1}{2}+\frac{1}{2\sqrt{2}}) + \log 3(\frac{1}{\sqrt{3}}+\frac{1}{3}+\frac{1}{3\sqrt{3}})\right] = \frac{1.081766+1.211915}{\pi} = 0.73010$. Requiring $\frac{1}{2\pi}\log\frac{t}{2\pi} > 0.73010$ gives $\log\frac{t}{2\pi} > 4.58736$, $\frac{t}{2\pi} > 98.235$, $t > 617.2$.

10. Trial values: $w=-0.1623$ gives $we^{w}=-0.13798$; $w=-0.16226$ gives -0.13795. So $W\approx -0.16226$ and $s_1=2\pi(-0.375)/(-0.16226)=14.521$.

11. $\log 5/\sqrt{5}=0.719760$ gives depth $225.908\times 0.719760=162.6$; $\log 7/\sqrt{7}=0.735485$ gives 166.2. Among $k\le 20$ the weights are 0.490 (2), 0.634 (3), 0.347 (4), 0.720 (5), 0.735 (7), 0.245 (8), 0.366 (9), 0.723 (11), 0.711 (13), 0.173 (16), 0.687 (17), 0.676 (19): the largest is at $k=7$. (The function $\log x/\sqrt{x}$ peaks at $x=e^2\approx 7.4$.)

12. $\frac{\Lambda(p^m)}{\sqrt{p^m}\log p^m}=\frac{\log p}{p^{m/2}\,m\log p}=\frac{1}{mp^{m/2}}$.

13. The best operator OOS rms on 401–500 in Table I is 0.265 (ridge $\mu=0.3$); the one-prime oracle gives 0.198. Factor $0.265/0.198=1.34$. Every other operator row is worse, so the factor is at least 1.34 for this holdout.

14. $N'_{\rm sm}(1000)=\frac{1}{2\pi}\log\frac{1000}{2\pi}=\frac{5.0699}{2\pi}=0.8069$. The slope of a fitted term $a\sin\omega t$ is up to $|a|\,\omega$. With frequencies up to $3\log 863=20.28$ (the 150th prime is 863), an amplitude of only 0.04 already gives a slope of 0.81, comparable to $N'_{\rm sm}$, and many such terms can add. Least-squares amplitudes in an underdetermined fit (900 coefficients, 400 zeros) are not small, so $N'_{\rm sm}+F'$ changes sign and the level equation has several roots. The oracle's amplitudes are fixed at $1/(\pi m p^{m/2})$ and its frequencies are at most $3\log 97\approx 13.7$ only for the weakest terms; its total slope is usually below $N'_{\rm sm}$ at these heights, though the worst-case bound is not.

15. You should obtain rms 0.116 and corr +0.90, matching Table V (this chapter's check script asserts it).

17. $0.074/0.263=0.281$: the six-prime formula's error is 28% of the null's. A correlation of -0.07 means the operator's predicted fluctuations are essentially unrelated to the true ones (if anything very slightly opposite in sign); whatever variation it predicts carries no usable information about which zeros are early or late.

19. Suppose $\log 2/\log 3=a/b$ with positive integers a,b. Then $b\log 2=a\log 3$, so $2^b=3^a$. The left side is even and the right side is odd, a contradiction (equivalently, unique factorization). Now let $a=\log 2$ and $b=\log 3$, and suppose $f(t)=\sin(at)+\sin(bt)$ had a period $\tau>0$, so $f(t+\tau)=f(t)$ for all t. Differentiating this identity twice shows that f'' also has period τ, and so does $f''+b^2f$. But $f''=-a^2\sin(at)-b^2\sin(bt)$, so $f''+b^2f=(b^2-a^2)\sin(at)$. Since $b^2\neq a^2$, $\sin(at)$ has period τ. Setting $t=0$ in $\sin(a(t+\tau))=\sin(at)$ gives $\sin(a\tau)=0$. Differentiating the same identity and setting $t=0$ gives $\cos(a\tau)=1$. Together these give $a\tau=2\pi j$ for an integer j, positive because $a,\tau>0$. The same argument with $f''+a^2f=(a^2-b^2)\sin(bt)$ gives $b\tau=2\pi l$. Dividing, $\log 2/\log 3=j/l$, which is impossible.

Now the six-prime $N_{\rm fl}$. Its frequencies $m\log p$ ($p\le 13$, $m\le 3$) are 18 distinct positive numbers: $m\log p=m'\log p'$ would mean $p^m=p'^{m'}$, which by unique factorization forces $(p,m)=(p',m')$. Apply the operation $\frac{d^2}{dt^2}+\omega^2$ once for each frequency $\omega\neq\log 2$. Since $(\frac{d^2}{dt^2}+\omega^2)\sin(\nu t)=(\omega^2-\nu^2)\sin(\nu t)$, this removes every term except the $\sin(t\log 2)$ term. That term is multiplied by the nonzero number $\prod(\omega^2-(\log 2)^2)$, and its coefficient $-1/(\pi\sqrt{2})$ is nonzero. So a period τ of $N_{\rm fl}$ would again force $\tau\log 2\in 2\pi\mathbb{Z}$, and likewise $\tau\log 3\in 2\pi\mathbb{Z}$: the same contradiction.

21. The computed sum should be close to -110.7; how close depends on the $\mathcal{O}(\log T)$ error term, whose constant depends on x, and on leakage from neighboring lines. This chapter's check script confirms agreement within 30%. For the scan, Landau's formula predicts a dip of depth $\frac{T}{2\pi}\Lambda(k)/\sqrt{k}=225.908\,\Lambda(k)/\sqrt{k}$ at each prime power $k\le e^{3.5}\approx 33$. Ranking the weights (those of answer 11, plus 0.654 for $k=23$ and smaller values for $29,31,25,27,32$), the five deepest predicted lines are at $\log 7$, $\log 11$, $\log 5$, $\log 13$, $\log 17$. The predicted depths of the first four differ from one another by less than 6, small compared with the discrepancies the $\mathcal{O}(\log T)$ term can produce, so in your computed spectrum these four may appear in a different order. In the computed spectrum itself (check script, peaks of $-\sum\cos(\omega t_n)$ located with a minimum separation of 0.01), each of the five deepest dips lies within 0.005 of the logarithm of a prime power $k\le 33$, as Landau's formula requires. The line at $\log 3$ (weight 0.634) is not among the

predicted five, because $\log 17$, $\log 19$ and $\log 23$ all have larger weights.

23. Almost but not exactly. The oracle with $P = \{2\}$ contains exactly the right function, so it predicts the fluctuation $(t - s_n)/\text{gap}_n$ that solves $N_{\text{sm}}(t) + N_{\text{fl}}^{p=2}(t) = n - \frac{1}{2}$, which by Equation (18.18) equals $-N_{\text{fl}}^{p=2}(s_n)$ only when $|N_{\text{fl}}'| \ll 1/\text{gap}_n$. On zeros 401–500, $\text{gap}_n \le 1.341$ and $|N_{\text{fl}}'| \le 0.34434$, so the slope correction factor $\text{gap}_n\,|N_{\text{fl}}'|$ can reach about 0.46; the correction is not small at every level. Averaged over the holdout the effect is moderate: the rms difference between the oracle's predicted fluctuations and $-N_{\text{fl}}^{p=2}(s_n)$ is below 0.08 spacings (check script), against the wave's own rms of about 0.17. Crucially, its accuracy does not degrade beyond any window, because it has no window: it contains the frequency $\log 2$ itself. The fitted operator, by contrast, scored 0.1958 out of sample, worse than predicting no fluctuation (0.1708), with corr +0.07: it painted the wave onto its training levels but did not continue it.

24. (a) $|\rho_2| = \sqrt{0.25 + t_2^2} = 21.0280$ and $2\sqrt{1000} = 63.2456$, so the amplitude is 3.008. (b) For $\rho = \beta + i\gamma$, $|x^\rho| = x^\beta \left|e^{i\gamma \log x}\right| = x^\beta$. So $-2\,\text{Re}(x^\rho/\rho)$ (the zero together with its conjugate, exactly as in Equation (18.24)) is a cosine wave of amplitude $2x^\beta/|\rho|$. Dividing by $\psi(x) \sim x$ gives a relative size proportional to $x^{\beta-1}$, which decays like $1/\sqrt{x}$ on the critical line and more slowly for $\beta > \frac{1}{2}$. (c) The ratio is $\frac{2x^{0.6}/|0.6+it_2|}{2x^{0.5}/|0.5+it_2|} = x^{0.1}\frac{|0.5+it_2|}{|0.6+it_2|}$. At $x = 10^6$, $x^{0.1} = 10^{0.6} = 3.981$. The ratio of moduli differs from 1 by about 10^{-4}, so the factor is 3.981.

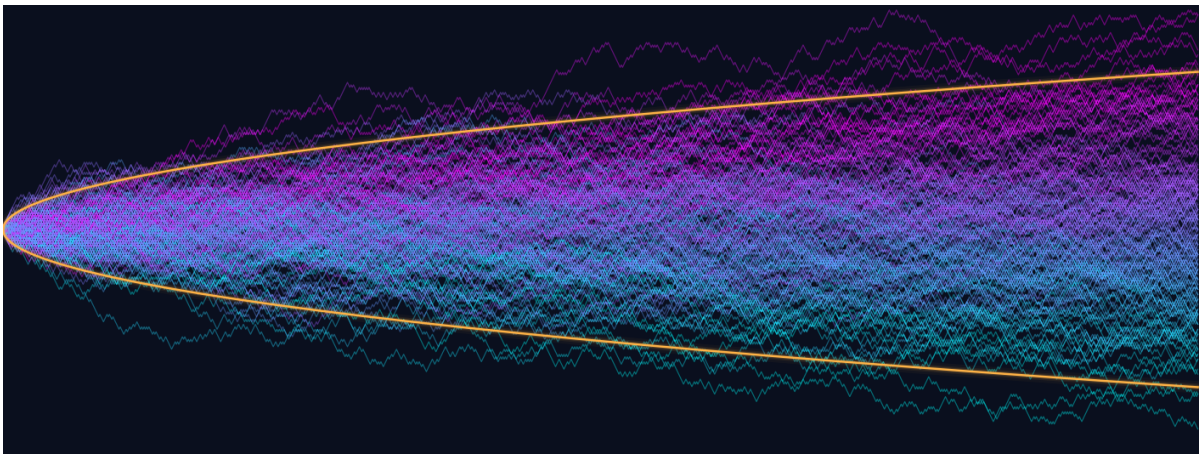
25. A box h is defined only for real r: it is 1 on $[-T, T]$ and 0 outside. It is not the restriction of any function analytic in a strip, because an analytic function that vanishes on an interval of the real axis vanishes identically (proved; proof beyond this book). So $h(\pm i/2)$ has no meaning. The Gaussian is given by the same formula for complex r: $h(\pm i/2) = e^{-(\pm i/2)^2/(2a^2)} = e^{1/(8a^2)}$. For the transform, the sine part of $e^{-iru} = \cos(ru) - i\sin(ru)$ integrates to zero because h is even. So $g(u) = I(u)/(2\pi)$ with $I(u) = \int_{-\infty}^{\infty} e^{-r^2/(2a^2)}\cos(ru)\,\mathrm{d}r$. Differentiating under the integral gives $I'(u) = -\int re^{-r^2/(2a^2)}\sin(ru)\,\mathrm{d}r$. Integrate by parts with $v = -a^2 e^{-r^2/(2a^2)}$, so that $\mathrm{d}v = re^{-r^2/(2a^2)}\mathrm{d}r$; the boundary terms vanish, and we get $\int re^{-r^2/(2a^2)}\sin(ru)\,\mathrm{d}r = a^2 u\,I(u)$. Hence $I'(u) = -a^2 u\,I(u)$ with $I(0) = a\sqrt{2\pi}$ (the Gaussian integral), so $I(u) = a\sqrt{2\pi}\,e^{-a^2u^2/2}$ and

$$g(u) = \frac{a}{\sqrt{2\pi}}\,e^{-a^2u^2/2}.$$

The prime sum therefore carries the factor $k^{-1/2}\Lambda(k)\,e^{-a^2(\log k)^2/2}$, a Gaussian in $\log k$, which cuts it off effectively at $\log k$ of a few times $1/a$. For $a = 1$ the factor $e^{-(\log k)^2/2}$ is 0.786 at $k = 2$ but 2.5×10^{-5} at $k = 100$.

Part IV

Spectra, Randomness and Operators



Two hundred random walks of 1000 steps fanning out from the same start: their spread grows like the square root of time.

Chapter 19

Probability and Statistics Essentials

↑Contents

What this chapter is about

Every number that the paper reports about prediction comes in a pair. Table I of the paper, for instance, says that one fitted potential predicts the held-out zeros 401–500 with an "rms" of 0.369 and a "corr" of −0.07, and that the do-nothing prediction (the "null") has an rms of 0.263. To read such a pair you need to know what a root mean square error is, what a correlation coefficient is, why the two can disagree, and how much either number would wobble if the experiment were repeated on different data. Those are questions of probability and statistics, and this chapter answers them from scratch.

We start with random variables and their distributions, define the mean, the variance and the root mean square, and meet two distributions you must know: the Gaussian, and the exponential, which is the Poisson law plotted in the paper's Fig. 2. Then we turn to pairs of quantities: covariance, the Pearson correlation coefficient (and a proof that it lies between −1 and +1), and a single identity that ties correlation to rms error. That identity explains at a glance why a prediction that is uncorrelated with the truth must do worse than the null, and why the paper's zero-parameter prime formula sits almost exactly where the identity says the best possible predictor with its correlation must sit (an observation we make from the paper's rounded numbers, not a claim the paper makes). We finish with autocorrelation of a sequence, estimators, standard errors, the law of large numbers, and the random-number seeds that make computations repeatable.

Where this shows up in the paper: every "rms / corr" pair in Tables I–V, the bootstrap intervals of Sec. III, the ranges over seeds in Table III, the negative autocorrelation invoked in Sec. III ("Within-height splits"), the exponential law e^{-s} in Fig. 2, and the arithmetic-continuation probe, whose "probe-null" 0.1708 we will predict by hand.

What you need

- Chapter 3: limits of sequences, partial sums, the geometric series, the Taylor series of $\log(1-x)$.
- Chapter 4: big-O notation and the idea of a leading-order approximation.
- First-year physics: integrals, substitution, integration by parts, exponentials and logarithms, trigonometric identities (in particular $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$), the dot product of vectors.
- Nothing about the zeta function is needed for the teaching sections. The paper connection uses the fluctuations of the zeros; we define them there in a few lines.

19.1 Random variables and their distributions

19.1.1 Outcomes, events and probabilities

Throw a die. Before it lands you cannot say which face will come up, but you can say a great deal: each of the six faces is equally likely, an even number comes up about half the time, and over many throws the average face value settles near 3.5. Probability theory is the bookkeeping that makes such statements precise.

The possible results of an experiment are called *outcomes*. For the die they are $1, 2, 3, 4, 5, 6$. A collection of outcomes is an *event*; "the face is even" is the event $\{2, 4, 6\}$. A *probability* P assigns to each event a number with three properties:

1. $0 \le P(\text{event}) \le 1$;
2. the probability that *something* happens is 1: $P(\text{all outcomes}) = 1$;
3. if two events cannot both happen (they are *mutually exclusive*), the probability that one or the other happens is the sum of their probabilities.

For a fair die each outcome has probability $1/6$, and property 3 gives $P(\text{even}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$.

What does a probability *mean* physically? The working interpretation used throughout science is the long-run frequency: if you repeat the experiment many times under the same conditions, the fraction of trials in which the event occurs approaches its probability. We will make that statement precise and prove a version of it (the law of large numbers) in Section 19.7.

19.1.2 Discrete random variables

A *random variable* is a number attached to each outcome. The face value X of a die is a random variable taking the values $1, \dots, 6$. A random variable that takes a list of separate values $x_1, x_2, \dots$ is called *discrete*, and its *distribution* is the list of probabilities

$$p_j = P(X = x_j), \qquad p_j \ge 0, \qquad \sum_j p_j = 1. \tag{19.1}$$

The second condition is property 2 above: the values x_j exhaust all outcomes, and they are mutually exclusive.

The *expectation* or *mean* of X is the probability-weighted average of its values,

$$\mu = E[X] = \sum_j x_j \, p_j. \tag{19.2}$$

More generally, for any function g, the expectation of the random variable $g(X)$ is

$$E[g(X)] = \sum_j g(x_j) \, p_j. \tag{19.3}$$

This is the same weighted average, applied to the values $g(x_j)$ instead of x_j.

Physically, μ is the balance point. If you put a mass p_j at position x_j on a weightless rod, the center of mass is at $\sum_j x_j p_j / \sum_j p_j = \mu$, because the total mass $\sum_j p_j$ is 1.

19.1.3 Continuous random variables and densities

Some quantities vary continuously: the time until a radioactive nucleus decays, the exact landing point of a dart, the gap between two consecutive zeta zeros. For such a quantity the probability of any single exact value is zero (the probability of an interval shrinks to zero with its width, Equation (19.4)), and what has meaning is the probability of landing in an interval. We describe it by a *probability density* $f(x)$:

$$P(a < X < b) = \int_a^b f(x)\,\mathrm{d}x, \qquad f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1. \tag{19.4}$$

The density is probability per unit x, exactly as mass density is mass per unit length. For a small interval of width $\mathrm{d}x$ around x, $P(x < X < x + \mathrm{d}x) \approx f(x)\,\mathrm{d}x$, with an error that becomes negligible compared with $\mathrm{d}x$ as $\mathrm{d}x \to 0$ when f is continuous.

Expectations become integrals:

$$\mu = E[X] = \int_{-\infty}^{\infty} x\, f(x)\,\mathrm{d}x, \qquad E[g(X)] = \int_{-\infty}^{\infty} g(x)\, f(x)\,\mathrm{d}x. \tag{19.5}$$

Each formula is the continuous analogue of Equations (19.2) and (19.3): the sum over values becomes an integral, and p_j becomes $f(x)\,\mathrm{d}x$.

The *cumulative distribution function* (CDF) is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u. \tag{19.6}$$

By the fundamental theorem of calculus, $F'(x) = f(x)$ wherever f is continuous. The CDF rises from 0 at $x = -\infty$ to 1 at $x = +\infty$, never decreasing.

The uniform distribution. The simplest density is constant on an interval. A random variable *uniform on* $[0, 1]$ has $f(x) = 1$ for $0 \leq x \leq 1$ and $f(x) = 0$ elsewhere. Its mean is

$$\mu = \int_0^1 x \cdot 1\,\mathrm{d}x = \left[\tfrac{x^2}{2}\right]_0^1 = \tfrac{1}{2}.$$

A random *phase* θ uniform on $[0, 2\pi)$ has $f(\theta) = 1/(2\pi)$ there, since the density must integrate to 1 over an interval of length 2π. The paper's phase-randomized surrogates (Sec. III and App. A step 18) draw exactly such uniform phases.

Checkpoint 19.1

Can a probability density be larger than 1? Can a probability?

Answer: A density can: a variable uniform on $[0, 0.1]$ has $f = 10$ there, and still $\int f\,\mathrm{d}x = 10 \times 0.1 = 1$. A probability cannot exceed 1. A density is probability per unit x, so only its integral over an interval is a probability.

Key Ideas

- A random variable attaches a number to each outcome; its distribution is either a list of probabilities p_j (discrete) or a density $f(x)$ (continuous), normalized to total

probability 1.

- The mean $E[X]$ is the probability-weighted average, Equations (19.2) and (19.5); it is the balance point of the distribution.
- For a continuous variable only intervals have nonzero probability, Equation (19.4); the CDF F satisfies $F' = f$.

19.2 Mean, variance, standard deviation and rms

19.2.1 Linearity of expectation

Two properties of E are used constantly. For constants a and b,

$$E[aX + b] = aE[X] + b. \tag{19.7}$$

Proof for a discrete variable, using Equation (19.3) with $g(x) = ax + b$:

$$\begin{aligned} E[aX + b] &= \sum_j (ax_j + b)\, p_j \\ &= a \sum_j x_j p_j + b \sum_j p_j \quad \text{(split the sum)} \\ &= a\mu + b \cdot 1 \quad \text{(definition of } \mu\text{; normalization Equation (19.1)).} \end{aligned}$$

The continuous proof is identical with integrals. The second property, that the expectation of a sum of two random variables is the sum of their expectations, needs the idea of a joint distribution; we prove it in Section 19.4.

19.2.2 Variance and standard deviation

The mean tells you where a distribution sits. The *variance* tells you how widely it spreads around that point:

$$\mathrm{Var}(X) = \sigma^2 = E\big[(X - \mu)^2\big]. \tag{19.8}$$

It is the expected squared distance from the mean. Its square root σ is the *standard deviation*, which has the same units as X. In the mechanical picture, σ^2 is the moment of inertia about the center of mass (per unit total mass).

A shortcut formula is often easier to evaluate. Expand the square and use Equation (19.7):

$$\begin{aligned} \sigma^2 &= E\big[X^2 - 2\mu X + \mu^2\big] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \quad \text{(expectation is linear; } \mu \text{ is a constant)} \\ &= E[X^2] - 2\mu^2 + \mu^2 \quad (E[X] = \mu) \\ &= E[X^2] - \mu^2. \end{aligned} \tag{19.9}$$

This is the probabilistic version of the parallel-axis theorem: the moment of inertia about the origin, $E[X^2]$, equals that about the center of mass, σ^2, plus μ^2.

Scaling and shifting a variable changes the variance in a simple way. With $Y = aX + b$, the mean of Y is $a\mu + b$ by Equation (19.7), so

$$\mathrm{Var}(aX + b) = E\big[(aX + b - a\mu - b)^2\big] = E\big[a^2(X - \mu)^2\big] = a^2\sigma^2. \tag{19.10}$$

A shift does nothing to the spread; a factor a multiplies the standard deviation by $|a|$.

Uniform example. For X uniform on $[0, 1]$, $E[X^2] = \int_0^1 x^2\,dx = \frac{1}{3}$, so by Equation (19.9) $\sigma^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$ and $\sigma = 1/\sqrt{12} \approx 0.2887$.

Sample Problem 19.1: Mean and spread of a fair die

Find the mean, variance and standard deviation of the face value X of a fair six-sided die.
Step 1: Mean. Each $p_j = 1/6$. By Equation (19.2),

$$\mu = \tfrac{1}{6}(1+2+3+4+5+6) = \tfrac{21}{6} = 3.5.$$

Step 2: Second moment. By Equation (19.3) with $g(x) = x^2$,

$$E[X^2] = \tfrac{1}{6}(1+4+9+16+25+36) = \tfrac{91}{6}.$$

Step 3: Variance. By Equation (19.9),

$$\sigma^2 = \tfrac{91}{6} - \left(\tfrac{7}{2}\right)^2 = \tfrac{182}{12} - \tfrac{147}{12} = \tfrac{35}{12} \approx 2.9167.$$

Step 4: Standard deviation. $\sigma = \sqrt{35/12}$.

$$\boxed{\mu = 3.5, \quad \sigma^2 = 35/12 \approx 2.917, \quad \sigma \approx 1.708}$$

Notice that the mean 3.5 is not a possible face value. The mean is a balance point, not a "typical outcome".

19.2.3 Statistics of a list of numbers

In practice you rarely know a distribution; you have data, a list $x_1, \dots, x_n$. The *sample mean* (or simply "mean") of the list is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{19.11}$$

For the spread we will use two versions, and it pays to name them separately. The *spread of the list* is

$$\sigma_x = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}, \tag{19.12}$$

and the *sample standard deviation* (the estimator of an underlying σ, Section 19.7) is

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{19.13}$$

The two differ only by the factor $\sqrt{n/(n-1)}$, which tends to 1 for large n. In `numpy` they are `x.std(ddof=0)` and `x.std(ddof=1)`. Why anyone would divide by $n-1$ is explained, with proof, in Section 19.7.

There is a useful way to see that list statistics are a special case of the expectations above. Imagine picking one entry of the list at random, each with probability $1/n$. This defines a discrete random variable, its *empirical distribution*. Its mean, by Equation (19.2), is $\sum_i x_i \cdot \frac{1}{n} = \bar{x}$, and its variance is σ_x^2. So every identity proved for E and Var holds for $\bar{x}$ and σ_x^2 too. In particular, the shortcut Equation (19.9) becomes

$$\sigma_x^2 = \overline{x^2} - \bar{x}^2, \qquad \overline{x^2} = \frac{1}{n}\sum_{i=1}^{n} x_i^2. \tag{19.14}$$

Sample Problem 19.2: Spacings of the first ten zeros (RE1)

The first ten heights of the zeta zeros (Running Example RE1) are 14.134725, 21.022040, 25.010858, 30.424876, 32.935062, 37.586178, 40.918719, 43.327073, 48.005151, 49.773832. Find the mean of the nine consecutive spacings $t_{n+1} - t_n$, and both σ_x and $\hat{\sigma}$ for these spacings.

Step 1: Spacings. Subtracting neighbors gives 6.887315, 3.988818, 5.414018, 2.510186, 4.651116, 3.332541, 2.408354, 4.678078, 1.768681.

Step 2: Mean. The sum of the spacings telescopes (Chapter 3): $\sum_{n=1}^{9}(t_{n+1} - t_n) = t_{10} - t_1 = 35.639107$. So

$$\bar{x} = 35.639107/9 = 3.959901.$$

Step 3: Deviations and their squares. The deviations $x_i - \bar{x}$ are 2.927414, 0.028917, 1.454117, -1.449715, 0.691215, -0.627360, -1.551547, 0.718177, -2.191220. Their squares sum to 21.3826.

Step 4: The two spreads.

$$\sigma_x = \sqrt{21.3826/9} = \sqrt{2.37584} \approx 1.5414, \qquad \hat{\sigma} = \sqrt{21.3826/8} = \sqrt{2.67282} \approx 1.6349.$$

$$\boxed{\bar{x} \approx 3.960, \quad \sigma_x \approx 1.541, \quad \hat{\sigma} \approx 1.635}$$

The spacings scatter by about 40% of their mean. These low zeros are far apart; Chapter 17 shows that the mean spacing shrinks slowly with height.

19.2.4 The root mean square

The paper scores every prediction by a *root mean square* (rms). For a list $a_1, \ldots, a_n$,

$$\mathrm{rms}(a) = \sqrt{\frac{1}{n}\sum_{i=1}^{n} a_i^2} = \sqrt{\overline{a^2}}. \tag{19.15}$$

The recipe is in the name, read from right to left: square, take the mean, take the root. Squaring makes negative and positive entries count alike, and the root restores the original units. You met rms values in first-year physics: the rms current of an alternating supply is the peak current divided by $\sqrt{2}$, which is Equation (19.15) applied to a sinusoid (we derive the $\sqrt{2}$ in Section 19.6).

When the list is a list of *errors* $e_i = (\text{prediction})_i - (\text{truth})_i$, the rms is the typical size of an error. How is it related to the mean and spread of the errors? Rearranging Equation (19.14) with $x \to e$,

$$\mathrm{rms}(e)^2 = \bar{e}^{\,2} + \sigma_e^2. \tag{19.16}$$

The mean error $\bar{e}$ is the *bias* (a systematic offset); σ_e is the *scatter* about that offset. The rms counts both. The standard deviation of the errors counts only the scatter and is blind to a systematic offset. That is why prediction errors are scored by rms rather than by standard deviation: a forecaster who is always wrong by exactly the same amount has $\sigma_e = 0$, but is still wrong.

Sample Problem 19.3: Bias, scatter and rms

A model predicts three quantities with errors $e = (0.1, 0.3, 0.2)$. Find $\bar{e}$, σ_e and $\mathrm{rms}(e)$, and verify Equation (19.16).

Step 1: Mean. $\bar{e} = (0.1 + 0.3 + 0.2)/3 = 0.2$.

Step 2: rms. $\overline{e^2} = (0.01 + 0.09 + 0.04)/3 = 0.14/3 = 0.046667$, so $\mathrm{rms}(e) = \sqrt{0.046667} =$

0.21602.

Step 3: Spread. The deviations are -0.1, 0.1, 0, with squares 0.01, 0.01, 0; so $\sigma_e^2 = 0.02/3 = 0.006667$ and $\sigma_e = 0.08165$.

Step 4: Check. $\bar{e}^{\,2} + \sigma_e^2 = 0.04 + 0.006667 = 0.046667 = \mathrm{rms}(e)^2$. ✓

$$\boxed{\bar{e} = 0.2, \quad \sigma_e \approx 0.0816, \quad \mathrm{rms}(e) \approx 0.216}$$

Most of this model's error is bias. Subtracting 0.2 from every prediction would cut the rms to 0.0816.

Checkpoint 19.2

When is the rms of a list equal to its spread σ_x?

Answer: Exactly when the mean of the list is zero, by Equation (19.16) (or Equation (19.14)). For lists whose mean is small compared with the spread, the two are nearly equal.

```
import numpy as np

t = np.array([14.134725, 21.022040, 25.010858, 30.424876, 32.935062,
              37.586178, 40.918719, 43.327073, 48.005151, 49.773832])
gaps = np.diff(t)                  # the 9 consecutive spacings
print(gaps.mean())                 # sample mean, Eq. (sample mean)
print(gaps.std(ddof=0))            # spread of the list, divide by n
print(gaps.std(ddof=1))            # estimator, divide by n-1
print(np.sqrt(np.mean(gaps**2)))   # rms of the list

e = np.array([0.1, 0.3, 0.2])      # errors of a biased model
print(np.sqrt(np.mean(e**2)), e.mean(), e.std())
```

19.2.5 Histograms

A list of many numbers is best seen as a *histogram*. Divide the axis into bins of width Δ; count the number c_j of entries in bin j; draw a bar of height c_j. To compare a histogram with a density $f(x)$ you must put both on the same footing. The density estimate in bin j is

$$\hat{f}_j = \frac{c_j}{n\,\Delta}, \tag{19.17}$$

the fraction of entries in the bin, c_j/n, divided by the width. With this normalization the bar areas $\hat{f}_j\Delta$ add up to $\sum_j c_j/n = 1$, just as $\int f\,\mathrm{d}x = 1$. In `matplotlib` this is the option `density=True`.

Choosing Δ is a compromise. Very narrow bins contain few counts each, so the bar heights jump around randomly; very wide bins smear out real features of the shape. A common starting point is a few dozen bins for a few thousand entries.

Figure 19.1 is a histogram of the kind the paper deals with: the fluctuations of the first 1000 zeta zeros about their smooth positions, in units of the local mean spacing (defined in Section 19.9). They pile up near zero and rarely exceed half a spacing, which is why errors of a few tenths of a spacing are the natural scale of the paper's tables.

Key Ideas

- Variance is the expected squared deviation from the mean; $\sigma^2 = E[X^2] - \mu^2$, Equation (19.9); $\mathrm{Var}(aX + b) = a^2\sigma^2$, Equation (19.10).

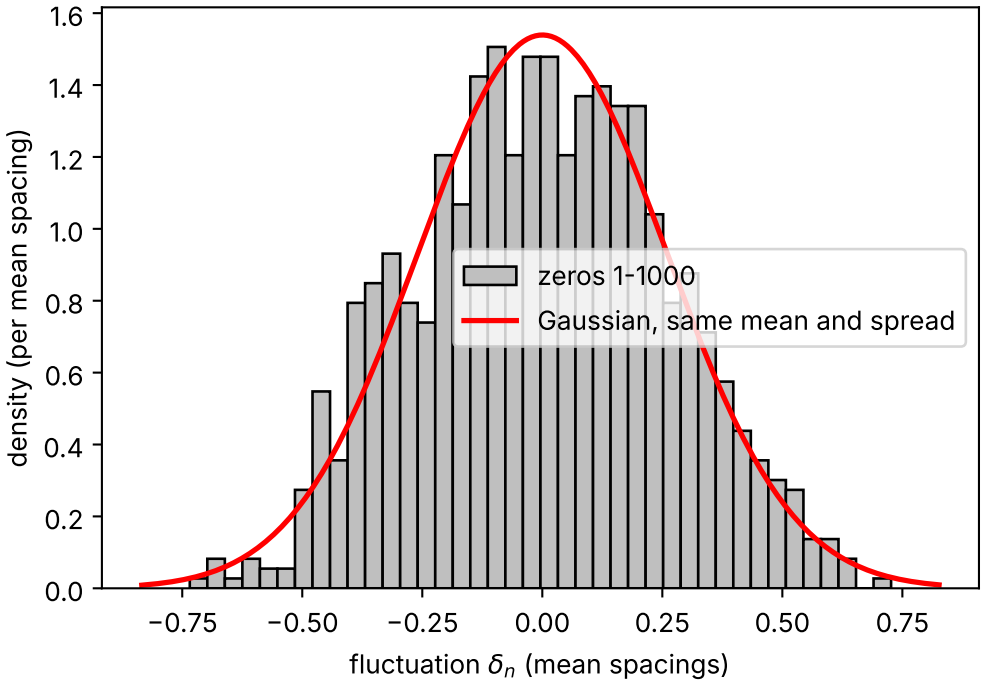


Figure 19.1: Histogram, normalized to a density by Equation (19.17), of the fluctuations $\delta_n = (t_n - s_n)/\mathrm{gap}_n$ of the zeta zeros $n = 1$–1000 about their smooth positions s_n, measured in local mean spacings gap_n. The red curve is a Gaussian with the same mean and spread σ_x. The fluctuations are confined to a few tenths of a mean spacing.

- For a list, σ_x divides by n and $\hat{\sigma}$ by $n-1$; both measure spread.
- The rms of errors combines bias and scatter: $\mathrm{rms}(e)^2 = \bar{e}^{\,2} + \sigma_e^2$, Equation (19.16). It is the right score for prediction error.
- A histogram becomes comparable with a density after dividing counts by $n\Delta$, Equation (19.17).

19.3 Two distributions you must know

★ Did you know? ***Death by Horse Kick***

In 1898 Ladislaus Bortkiewicz showed that the yearly number of Prussian cavalrymen killed by horse kicks, recorded across army corps over 20 years, closely followed a Poisson distribution. Rare, independent events spread over many opportunities produce Poisson counts, whatever their cause. His book, *The Law of Small Numbers*, made this grim dataset one of the most famous examples in statistics.

19.3.1 The Gaussian

The *Gaussian* (or *normal*) density with mean μ and standard deviation σ is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\!\Big[-\frac{(x-\mu)^2}{2\sigma^2}\Big]. \tag{19.18}$$

It is the bell curve: symmetric about μ, falling off faster than any power of x. With the substitution $u = (x-\mu)/\sigma$ (so $\mathrm{d}x = \sigma\,\mathrm{d}u$) every Gaussian becomes the *standard normal* density

$$\varphi(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}, \tag{19.19}$$

because $f(x)\,\mathrm{d}x = \frac{1}{\sigma\sqrt{2\pi}} e^{-u^2/2}\,\sigma\,\mathrm{d}u = \varphi(u)\,\mathrm{d}u$. The variable u is called the *z-score*: it measures distance from the mean in units of σ.

Taken on trust

Taken on trust: the normalization $\int_{-\infty}^{\infty} e^{-u^2/2}\,\mathrm{d}u = \sqrt{2\pi}$. There is no elementary antiderivative of $e^{-u^2/2}$, so this needs a trick that uses integrals over a plane, which this book does not develop. The idea: call the integral I. Then I^2 is the integral of $e^{-(u^2+v^2)/2}$ over the whole (u, v) plane. The integrand depends only on the distance $r = \sqrt{u^2+v^2}$ from the origin, so slice the plane into thin rings of radius r, width $\mathrm{d}r$ and area $2\pi r\,\mathrm{d}r$. Then $I^2 = \int_0^{\infty} e^{-r^2/2}\,2\pi r\,\mathrm{d}r = 2\pi\left[-e^{-r^2/2}\right]_0^{\infty} = 2\pi$. The careful version of this argument is in [22].

Given the normalization, the mean and variance follow by calculus you know.

Mean. Using $u = (x-\mu)/\sigma$, so $x = \mu + \sigma u$,

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} (\mu + \sigma u)\,\varphi(u)\,\mathrm{d}u \\ &= \mu \int_{-\infty}^{\infty} \varphi(u)\,\mathrm{d}u + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u\, e^{-u^2/2}\,\mathrm{d}u \\ &= \mu \cdot 1 + \frac{\sigma}{\sqrt{2\pi}} \left[-e^{-u^2/2}\right]_{-\infty}^{\infty} \quad \text{(antiderivative of } ue^{-u^2/2}\text{)} \\ &= \mu + 0 = \mu. \end{aligned}$$

Variance. $\mathrm{Var}(X) = E[(X-\mu)^2] = \sigma^2 \int u^2 \varphi(u)\,\mathrm{d}u$. Integrate by parts, writing $u^2 e^{-u^2/2} = u \cdot (ue^{-u^2/2})$ and using $\int ue^{-u^2/2}\,\mathrm{d}u = -e^{-u^2/2}$:

$$\begin{aligned} \int_{-\infty}^{\infty} u^2 e^{-u^2/2}\,\mathrm{d}u &= \left[u \cdot (-e^{-u^2/2})\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} 1 \cdot (-e^{-u^2/2})\,\mathrm{d}u \\ &= 0 + \sqrt{2\pi} \quad (ue^{-u^2/2} \to 0 \text{ at } \pm\infty\text{; normalization}). \end{aligned}$$

Dividing by $\sqrt{2\pi}$ gives $\int u^2 \varphi\,\mathrm{d}u = 1$, so $\mathrm{Var}(X) = \sigma^2$. The parameters μ and σ in Equation (19.18) really are the mean and standard deviation.

Tail probabilities. The probability of landing within k standard deviations of the mean, $P(|X-\mu| < k\sigma) = \int_{-k}^{k} \varphi(u)\,\mathrm{d}u$, has no elementary closed form; it is tabulated, or computed in Python as `scipy.special.erf(k/np.sqrt(2))`. The standard values are

$$P(|u|<1) = 0.6827, \qquad P(|u|<2) = 0.9545, \qquad P(|u|<3) = 0.9973, \qquad P(|u|<1.960) = 0.9500. \tag{19.20}$$

The last entry is why "$\pm 1.96\,\sigma$" appears in 95% intervals (Section 19.7). Figure 19.2 shows the first two.

Why does the Gaussian appear so often? The reason is a theorem about sums of many independent random contributions, which we state in Section 19.7 (the central limit theorem). Measurement errors that are the sum of many small independent disturbances tend to be Gaussian; that is a tendency, not a law, and Figure 19.1 should be read as a comparison, not as a claim that the zeros' fluctuations are exactly Gaussian.

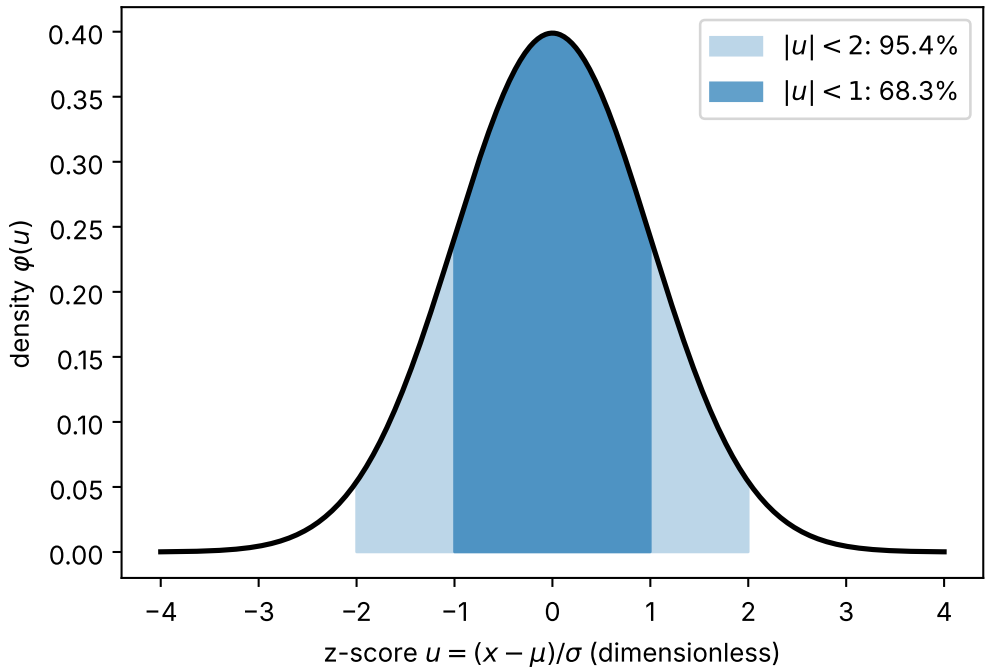


Figure 19.2: The standard normal density $\varphi(u)$ of Equation (19.19). The dark band $|u| < 1$ holds 68.3% of the probability; together with the light bands, $|u| < 2$ holds 95.4%. A Gaussian with any μ and σ looks the same with $u = (x - \mu)/\sigma$ on the horizontal axis.

Pitfall

The rule "95% within $1.96\,\sigma$" is a property of the Gaussian. For other distributions the fraction can be very different. The only guarantee valid for *every* distribution with finite variance is the much weaker Chebyshev bound, proved later in this chapter (Section 19.7): at least 75% within 2σ.

19.3.2 The exponential distribution and the Poisson process

Now a distribution that you can derive from a physical picture. Imagine points scattered along a line "completely at random": clicks of a Geiger counter in time, raindrops landing along a gutter. Make three assumptions, which together define a *Poisson process* of unit rate:

1. the probability of a point in any tiny interval of length $\mathrm{d}s$ is $\mathrm{d}s$ (so the mean density of points is 1 per unit length, and the mean gap is 1);

2. the probability of two or more points in a tiny interval is negligible compared with $\mathrm{d}s$;

3. what happens in non-overlapping intervals is independent: knowing where the points are in one interval tells you nothing about any other.

Assumption 3 is the heart of it: the points know nothing about one another. "Independent" will be defined precisely in Section 19.4; for now it means that probabilities of events in different intervals multiply.

Probability of an empty interval. What is the probability $P_0(s)$ that an interval of length s contains no point? Split it into n equal pieces of length s/n. For large n each piece contains a

point with probability s/n (assumption 1) and is empty with probability $1-s/n$. By assumption 3 the pieces are independent, so the probability that all n are empty is the product

$$P_0(s) \approx \Big(1 - \frac{s}{n}\Big)^n.$$

The approximation comes from assumptions 1 and 2 holding only for tiny pieces. Precisely, each piece is empty with probability $1 - s/n + \epsilon_n$, where assumptions 1 and 2 say only that ϵ_n is negligible compared with $1/n$, that is, $n\epsilon_n \to 0$. We show below that $n\log(1-x) \to -s$ for $x = s/n$; the argument uses only that $nx \to s$ and $x \to 0$, and $x = s/n - \epsilon_n$ satisfies both. So the small errors ϵ_n do not accumulate over the n pieces, and the approximation becomes exact as $n \to \infty$. To find the limit take the logarithm and use the Taylor series $\log(1-x) = -\sum_{j\ge1} x^j/j$ from Chapter 3, valid for $0 \le x < 1$, with $x = s/n$:

$$\begin{aligned} n\log\Big(1 - \frac{s}{n}\Big) &= -n\sum_{j=1}^{\infty} \frac{(s/n)^j}{j} \\ &= -s - n\sum_{j=2}^{\infty} \frac{(s/n)^j}{j} \quad \text{(separate the } j=1 \text{ term)}. \end{aligned}$$

The remainder is small. Each term obeys $(s/n)^j/j \le (s/n)^j$, so by the geometric series (Chapter 3), for $n > s$,

$$0 \le n\sum_{j=2}^{\infty} \frac{(s/n)^j}{j} \le n\,\frac{(s/n)^2}{1 - s/n} = \frac{s^2}{n-s} \;\longrightarrow\; 0 \quad (n \to \infty).$$

Hence $n\log(1 - s/n) \to -s$, and since the exponential function is continuous,

$$P_0(s) = \lim_{n\to\infty} \Big(1 - \frac{s}{n}\Big)^n = e^{-s}. \tag{19.21}$$

Table 19.1 shows how quickly the limit is approached for $s = 1$.

Table 19.1: Approach of $(1 - 1/n)^n$ to e^{-1}, the probability that an interval of one mean gap in a Poisson process is empty.

n	10	100	1000	∞
$(1-1/n)^n$	0.348678	0.366032	0.367695	$e^{-1} = 0.367879$

The gap distribution. Stand on one point of the process and look forward. The next point lies farther than s exactly when the interval of length s ahead of you is empty. By assumption 3 the interval ahead does not care that a point sits where you stand, so $P(\text{gap} > s) = e^{-s}$, and the CDF of the gap is $F(s) = P(\text{gap} \le s) = 1 - e^{-s}$ for $s \ge 0$. Differentiating, the gap density is

$$p(s) = F'(s) = e^{-s} \qquad (s \ge 0). \tag{19.22}$$

This is the *exponential distribution*, and it is the Poisson law plotted in the paper's Fig. 2.

Mean and variance. Integration by parts gives

$$E[s] = \int_0^\infty s\,e^{-s}\,\mathrm{d}s = \big[-se^{-s}\big]_0^\infty + \int_0^\infty e^{-s}\,\mathrm{d}s = 0 + 1 = 1,$$

as it must, since the rate was 1. Once more by parts, $E[s^2] = \int_0^\infty s^2 e^{-s}\,\mathrm{d}s = \big[-s^2e^{-s}\big]_0^\infty + 2\int_0^\infty se^{-s}\,\mathrm{d}s = 2$, so $\mathrm{Var}(s) = 2 - 1^2 = 1$.

Look at the shape of Equation (19.22): it is largest at $s = 0$. In a Poisson process the most likely gap is no gap at all; points that ignore each other often land almost on top of each other. The zeta zeros do the opposite. Figure 19.3 compares the gaps of a simulated Poisson process with the *unfolded* spacings of the first 10^4 zeros (spacings rescaled to have mean 1; Chapter 17). The zeros almost never come close together: they repel. Chapter 20 derives the curve the zeros do follow.

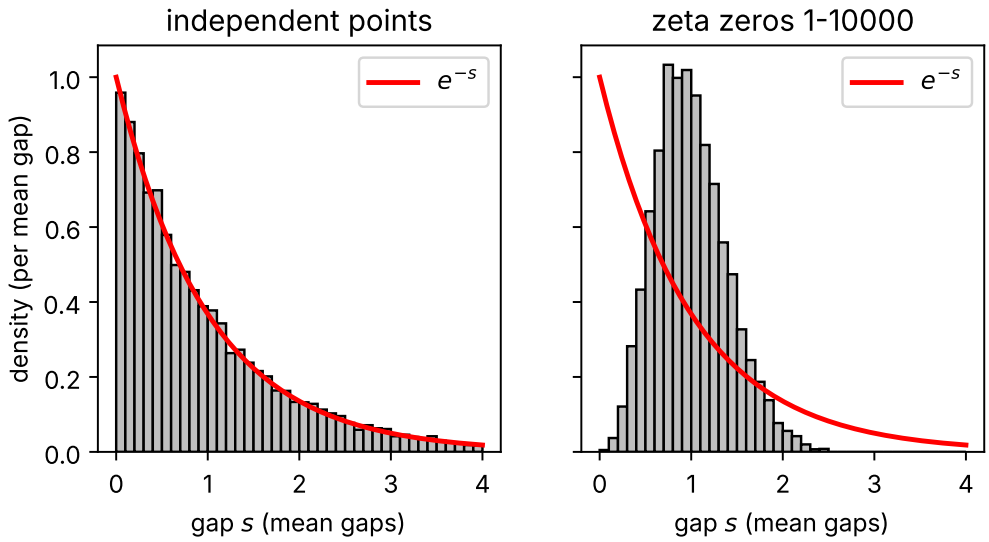


Figure 19.3: Left: gaps between 10^4 points dropped independently and uniformly on an interval of length 10^4 (seed 1; for 10^4 points this is indistinguishable in practice from a unit-rate Poisson process, although strictly a Poisson process has a random number of points), against the exponential law e^{-s} of Equation (19.22). Right: unfolded spacings of the first 10^4 zeta zeros against the same law. Small gaps are the most common for independent points but are strongly suppressed for the zeros.

Sample Problem 19.4: Gap probabilities in a Poisson process

For a unit-rate Poisson process, find (a) the probability that a gap is shorter than 0.1 mean gaps, (b) the probability that it is longer than 2 mean gaps, (c) the median gap.
Step 1: Small gaps. $P(\text{gap} < 0.1) = F(0.1) = 1 - e^{-0.1} = 1 - 0.904837 = 0.095163$.
Step 2: Long gaps. $P(\text{gap} > 2) = e^{-2} = 0.135335$.
Step 3: Median. The median m splits the probability in half: $F(m) = \frac{1}{2}$, so $e^{-m} = \frac{1}{2}$ and $m = \log 2 = 0.693147$.

$$\boxed{(a)\ 0.0952, \quad (b)\ 0.1353, \quad (c)\ \log 2 \approx 0.693}$$

Nearly one gap in ten is shorter than a tenth of the mean. The median is below the mean 1 because the long tail pulls the mean upward.

Checkpoint 19.3

For an exponential gap distribution, where is the density largest? What does that say about the zeros, whose spacing histogram in Figure 19.3 is near zero at $s = 0$?

Answer: At $s = 0$, since e^{-s} decreases. Uncorrelated points cluster. A histogram that vanishes at $s = 0$ shows that the zeros are not independent: they avoid each other (level repulsion, Chapter 20).

Key Ideas

- The Gaussian Equation (19.18) has mean μ and variance σ^2; 68.3% of probability lies within 1σ, 95% within 1.96σ, Equation (19.20).
- Independent points of unit density leave empty intervals of length s with probability e^{-s}, Equation (19.21); their gaps are exponentially distributed with mean 1 and variance 1, Equation (19.22).
- The zeta zeros are not like independent points: their small spacings are suppressed.

19.4 Covariance and correlation

19.4.1 Two random variables at once

Often two quantities are measured together: height and weight, the true and the predicted position of a zero. Their *joint distribution* for discrete variables is the list $p(x_j, y_k) = P(X = x_j \text{ and } Y = y_k)$. Summing over all values of Y recovers the distribution of X alone, $p_X(x_j) = \sum_k p(x_j, y_k)$, because the events "$Y = y_k$" are mutually exclusive and exhaust all possibilities; similarly for p_Y.

Expectation of a sum. For any function $g(x, y)$, $E[g(X, Y)] = \sum_{j,k} g(x_j, y_k)\, p(x_j, y_k)$. With $g = x + y$:

$$\begin{aligned} E[X+Y] &= \sum_{j,k} (x_j + y_k)\, p(x_j, y_k) \\ &= \sum_j x_j \sum_k p(x_j, y_k) + \sum_k y_k \sum_j p(x_j, y_k) \quad \text{(split, reorder)} \\ &= \sum_j x_j\, p_X(x_j) + \sum_k y_k\, p_Y(y_k) = E[X] + E[Y]. \end{aligned}$$

This holds for *any* two random variables, related or not.

Independence. X and Y are *independent* if $p(x_j, y_k) = p_X(x_j)\, p_Y(y_k)$ for all j, k: knowing one tells you nothing about the other. Then

$$E[XY] = \sum_{j,k} x_j y_k\, p_X(x_j) p_Y(y_k) = \Big(\sum_j x_j p_X(x_j)\Big)\Big(\sum_k y_k p_Y(y_k)\Big) = E[X]\, E[Y]. \tag{19.23}$$

Taken on trust

Taken on trust: continuous pairs. For two continuous variables the joint distribution is a *joint density* $f(x, y)$, and probabilities of rectangles are iterated integrals, $P(a < X < b \text{ and } c < Y < d) = \int_a^b \Big[\int_c^d f(x, y)\, \mathrm{d}y\Big] \mathrm{d}x$: first integrate over y with x held fixed, then integrate the result over x. Sums over j and k become such integrals, the density of X alone is $f_X(x) = \int f(x, y)\, \mathrm{d}y$, and independence means $f(x, y) = f_X(x) f_Y(y)$. With these replacements every result of this section holds by the same steps; the justification that iterated integrals may be split and reordered like sums is in [22]. We use the continuous results only in the same form as the discrete ones.

19.4.2 Covariance

The *covariance* measures whether two variables tend to be above their means together:

$$\operatorname{Cov}(X,Y) = E\big[(X-\mu_X)(Y-\mu_Y)\big] = E[XY] - \mu_X\mu_Y. \tag{19.24}$$

The second form follows by expanding the product and using linearity, exactly as in Equation (19.9). Note $\operatorname{Cov}(X,X) = \operatorname{Var}(X)$. By Equation (19.23), independent variables have zero covariance. (The converse is false: zero covariance does not imply independence; Problem 28.)

The variance of a sum follows. Write $X' = X - \mu_X$, $Y' = Y - \mu_Y$; the sum $X+Y$ has mean $\mu_X + \mu_Y$, so

$$\begin{aligned}\operatorname{Var}(X+Y) &= E\big[(X'+Y')^2\big] = E[X'^2] + 2E[X'Y'] + E[Y'^2] \\ &= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).\end{aligned} \tag{19.25}$$

For independent variables the covariance term vanishes and variances simply add. This one fact drives the whole of Section 19.7.

Correlation coefficient of two random variables. For random variables with $\sigma_X > 0$ and $\sigma_Y > 0$, the *correlation coefficient* is

$$\rho = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y}. \tag{19.26}$$

For the empirical distribution of two paired lists (each pair (x_i, y_i) picked with probability $1/n$), Cov becomes the list covariance and ρ becomes the Pearson coefficient r defined below. The proof of $|r| \le 1$ given below also proves $|\rho| \le 1$: replace $\sum u_i v_i$ by $E[X'Y']$, $\sum u_i^2$ by $E[X'^2]$ and $\sum v_i^2$ by $E[Y'^2]$; the proof uses only that the expectation of a square is nonnegative and that expectation is linear.

19.4.3 The Pearson correlation coefficient

Covariance has awkward units (units of X times units of Y) and changes if you rescale either variable. Dividing by the two standard deviations removes both defects. For two lists $x_1, \dots, x_n$ and $y_1, \dots, y_n$, the *Pearson correlation coefficient* is

$$\operatorname{corr}(x,y) = r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}. \tag{19.27}$$

If you divide numerator and denominator by n, the numerator is the list covariance $c_{xy} = \frac{1}{n}\sum(x_i-\bar{x})(y_i-\bar{y})$ and the denominator is $\sigma_x\sigma_y$, so $r = c_{xy}/(\sigma_x\sigma_y)$. The factors of n (or $n-1$) cancel, so it does not matter which convention you use for the spreads, provided you use the same one top and bottom.

A geometric reading. Define the *centered* lists $u_i = x_i - \bar{x}$ and $v_i = y_i - \bar{y}$, and think of them as vectors with n components. The numerator of Equation (19.27) is their dot product $\mathbf{u}\cdot\mathbf{v} = \sum u_i v_i$; the denominator is the product of their lengths. In three dimensions you know that $\mathbf{u}\cdot\mathbf{v} = |\mathbf{u}|\,|\mathbf{v}|\cos\theta$, so r is the cosine of the angle between the centered data vectors. That already suggests $-1 \le r \le 1$. In n dimensions we need a proof.

19.4.4 Why $-1 \le r \le 1$: the Cauchy–Schwarz inequality

Theorem (Cauchy–Schwarz, proved). For any real lists $u_1, \dots, u_n$ and $v_1, \dots, v_n$,

$$\Big(\sum_{i=1}^{n} u_i v_i\Big)^2 \le \Big(\sum_{i=1}^{n} u_i^2\Big)\Big(\sum_{i=1}^{n} v_i^2\Big), \tag{19.28}$$

with equality if and only if one list is a constant multiple of the other.

Proof. If every $v_i = 0$ both sides are zero and there is nothing to prove; so suppose $\sum v_i^2 > 0$. For any real number λ, a sum of squares is never negative:

$$\begin{aligned} 0 \le g(\lambda) &= \sum_{i=1}^{n} (u_i - \lambda v_i)^2 \\ &= \sum u_i^2 - 2\lambda \sum u_i v_i + \lambda^2 \sum v_i^2 \quad \text{(expand the square)}. \end{aligned}$$

The inequality $g(\lambda) \ge 0$ holds for every λ, so we may pick the most useful one, the λ at which g is smallest. Setting the derivative to zero, $g'(\lambda) = -2\sum u_i v_i + 2\lambda \sum v_i^2 = 0$, gives $\lambda^* = \sum u_i v_i / \sum v_i^2$. (The proof uses only $g(\lambda^*) \ge 0$; that λ^* minimizes g merely explains why it is the best choice.) Substituting,

$$\begin{aligned} 0 \le g(\lambda^*) &= \sum u_i^2 - 2\frac{(\sum u_i v_i)^2}{\sum v_i^2} + \frac{(\sum u_i v_i)^2}{(\sum v_i^2)^2} \sum v_i^2 \\ &= \sum u_i^2 - \frac{(\sum u_i v_i)^2}{\sum v_i^2}. \end{aligned}$$

Multiplying by the positive number $\sum v_i^2$ gives Equation (19.28). Equality holds exactly when $g(\lambda^*) = 0$, that is, when every term $(u_i - \lambda^* v_i)^2$ vanishes, i.e. $u_i = \lambda^* v_i$ for all i. □

Applying Equation (19.28) to the centered lists and taking square roots,

$$\Big|\sum (x_i - \bar{x})(y_i - \bar{y})\Big| \le \sqrt{\sum (x_i - \bar{x})^2}\sqrt{\sum (y_i - \bar{y})^2},$$

which is precisely $|r| \le 1$. So:

- $r = +1$ exactly when $y_i - \bar{y} = \lambda(x_i - \bar{x})$ with $\lambda > 0$: the points lie on a rising straight line;
- $r = -1$ exactly when they lie on a falling straight line;
- $r = 0$ when there is no *linear* tendency for one to rise with the other.

Note the word "linear". Correlation is blind to the slope's size, and blind to nonlinear relations.

Sample Problem 19.5: Correlation by hand

Find r for $x = (1, 2, 3, 4, 5)$ and $y = (2, 1, 4, 3, 5)$.
Step 1: Means. $\bar{x} = 15/5 = 3$ and $\bar{y} = 15/5 = 3$.
Step 2: Centered lists. $u = (-2, -1, 0, 1, 2)$ and $v = (-1, -2, 1, 0, 2)$.
Step 3: Sums. $\sum u_i v_i = 2 + 2 + 0 + 0 + 4 = 8$; $\sum u_i^2 = 4 + 1 + 0 + 1 + 4 = 10$; $\sum v_i^2 = 1 + 4 + 1 + 0 + 4 = 10$.
Step 4: Coefficient. $r = 8/\sqrt{10 \cdot 10} = 0.8$.

$$\boxed{r = 0.8}$$

The two lists rise together but not along a perfect line.

Checkpoint 19.4

You multiply every entry of a prediction p by 2. What happens to $\mathrm{corr}(p, y)$? What if you multiply by -2?

Answer: Multiplying by 2 multiplies the centered list by 2, so the numerator of Equation (19.27) and the factor $\sqrt{\sum(p_i - \bar{p})^2}$ both double: r is unchanged. Multiplying by -2 flips the sign of the numerator but not of the square root: $r \to -r$.

Key Ideas

- Expectations of sums add for any variables; for independent variables $E[XY] = E[X]E[Y]$ and variances add, Equation (19.25).
- The Pearson coefficient Equation (19.27) is the cosine of the angle between centered data vectors.
- Cauchy–Schwarz Equation (19.28) proves $-1 \le r \le 1$, with ± 1 only for an exact linear relation.
- Correlation ignores the size of the slope: it is unchanged by rescaling either list by a positive factor.

19.5 Correlation versus rms error

19.5.1 One identity that ties them together

The paper reports both an rms error and a correlation for each prediction because they answer different questions. The rms asks: *how far off is the prediction?* The correlation asks: *does the prediction go up when the truth goes up?* A single identity tells us exactly how they are related.

Let $y_1, \dots, y_n$ be the true values and $p_1, \dots, p_n$ the predictions. Split each list into its mean and a centered part, $p_i = \bar{p} + p'_i$ and $y_i = \bar{y} + y'_i$, where $\sum p'_i = \sum y'_i = 0$. Then

$$p_i - y_i = (\bar{p} - \bar{y}) + (p'_i - y'_i),$$
$$(p_i - y_i)^2 = (\bar{p} - \bar{y})^2 + 2(\bar{p} - \bar{y})(p'_i - y'_i) + (p'_i - y'_i)^2.$$

Average over i. The middle term averages to zero, because $\frac{1}{n}\sum(p'_i - y'_i) = 0$. The last term is $\overline{p'^2} + \overline{y'^2} - 2\,\overline{p'y'} = \sigma_p^2 + \sigma_y^2 - 2c_{py}$, where c_{py} is the list covariance. Finally $c_{py} = r\,\sigma_p\sigma_y$ with $r = \mathrm{corr}(p, y)$. So

$$\mathrm{rms}(p - y)^2 = (\bar{p} - \bar{y})^2 + \sigma_p^2 + \sigma_y^2 - 2r\,\sigma_p\sigma_y. \tag{19.29}$$

This is exact: no approximation, no assumption about distributions. It is the law of cosines in disguise: if the centered vectors $\mathbf{p}'$ and $\mathbf{y}'$ make an angle θ with $\cos\theta = r$, the squared length of their difference is $|\mathbf{p}'|^2 + |\mathbf{y}'|^2 - 2\,|\mathbf{p}'|\,|\mathbf{y}'|\cos\theta$.

19.5.2 Three consequences

Suppose, as for the fluctuations scored in the paper, that the truth has mean close to zero, $\bar{y} \approx 0$, so that $\sigma_y \approx \mathrm{rms}(y)$ by Equation (19.16). The *null* prediction $p = 0$ has error $\mathrm{rms}(0 - y) = \mathrm{rms}(y)$.

1. Right shape, wrong amplitude. Let $p = ay$ with $a > 0$. Then $r = +1$ (a rising straight line), yet the error is $\mathrm{rms}(ay - y) = |a - 1|\,\mathrm{rms}(y)$. A prediction can be perfectly correlated with the truth and still have a large rms. For $a = 2$ the error equals the null; for $a > 2$ it is *worse* than predicting nothing.

2. Uncorrelated predictions are worse than the null. Put $r = 0$ and $\bar{p} = \bar{y} = 0$ in Equation (19.29):

$$\mathrm{rms}(p - y)^2 = \sigma_y^2 + \sigma_p^2 \;\geq\; \sigma_y^2 = (\mathrm{null})^2. \tag{19.30}$$

A prediction that wiggles independently of the truth adds its own variance to the error. The bigger its wiggles, the worse it does. Predicting nothing is better than predicting noise.

3. The best possible rms at a given correlation. Fix r and σ_y and ask which prediction minimizes Equation (19.29). The first term is minimized (to zero) by $\bar{p} = \bar{y}$. The σ_p terms are a quadratic, which we complete to a square:

$$\sigma_p^2 - 2r\sigma_y\sigma_p = (\sigma_p - r\sigma_y)^2 - r^2\sigma_y^2.$$

If $r > 0$, the square vanishes at $\sigma_p = r\sigma_y$, and then

$$\mathrm{rms}(p - y)_{\min} = \sigma_y\sqrt{1 - r^2} \qquad (r > 0). \tag{19.31}$$

If $r \leq 0$, the expression $\sigma_p^2 - 2r\sigma_y\sigma_p$ is positive for every $\sigma_p > 0$ (both terms are nonnegative and the first is positive), so the rms exceeds σ_y and only approaches σ_y as $\sigma_p \to 0$. (The value $\sigma_p = 0$ itself is excluded, because there the correlation is undefined; see the pitfall below.) No prediction with nonpositive correlation beats the null. Two lessons follow. The optimal amplitude is r times the true amplitude, smaller than the truth whenever the correlation is imperfect: a cautious predictor should shrink its fluctuations. And a correlation of 0.96, which sounds nearly perfect, can at best cut the error to $\sqrt{1 - 0.96^2} = 0.28$ of the null.

Sample Problem 19.6: Correct shape, wrong amplitude

The true fluctuations on five levels are $y = (0.3, -0.2, 0.1, -0.4, 0.2)$ (in mean spacings). Compare the null with the predictions $p = 0.5y$, $p = 2y$ and $p = 3y$.
Step 1: Null. $\bar{y} = 0$. $\overline{y^2} = (0.09 + 0.04 + 0.01 + 0.16 + 0.04)/5 = 0.34/5 = 0.068$, so the null rms is $\mathrm{rms}(y) = \sqrt{0.068} = 0.2608$.
Step 2: Correlations. Each p is a positive multiple of y, so $r = +1$ in all three cases.
Step 3: Errors. $p - y = (a - 1)y$, so $\mathrm{rms}(p - y) = |a - 1| \times 0.2608$: for $a = 0.5$, 0.1304; for $a = 2$, 0.2608; for $a = 3$, 0.5215.

null 0.261; $a = 0.5:$ 0.130; $a = 2:$ 0.261; $a = 3:$ 0.522; $r = +1$ throughout

The $a = 3$ prediction is perfectly correlated with the truth and twice as bad as the null. Correlation and rms are not interchangeable scores.

Checkpoint 19.5

A prediction has correlation +1 with the truth. Must it beat the null?

Answer: No. With $p = ay$, the rms is $|a - 1|\,\mathrm{rms}(y)$, which exceeds the null whenever $a > 2$. Correlation measures shape, rms measures size; you need both.

Sample Problem 19.7: How good can a prediction with correlation r be?

On the held-out zeros 401–500 the paper's null rms is 0.263 (Table V). The zero-parameter prime formula with the prime 2 only has correlation +0.66; with 3 primes, +0.90; with 6 primes, +0.96 (Table V). Use Equation (19.31) to find the smallest rms any prediction with these correlations could have, and compare with the rms values in Table V: 0.198, 0.116, 0.074.

Step 1: Assumption. Take $\sigma_y \approx 0.263$. This holds if the true fluctuations on the holdout have mean close to zero; if their mean were not small, σ_y would be below 0.263 and the bound below would be lower.
Step 2: The prime 2. $\sqrt{1-0.66^2} = \sqrt{0.5644} = 0.7513$; bound $0.263 \times 0.7513 = 0.1976$.
Step 3: Three primes. $\sqrt{1-0.81} = \sqrt{0.19} = 0.4359$; bound 0.1146.
Step 4: Six primes. $\sqrt{1-0.9216} = \sqrt{0.0784} = 0.28$; bound 0.0736.

bounds 0.198, 0.115, 0.074 vs. Table V 0.198, 0.116, 0.074

The formula, which has no fitted parameters at all, sits essentially on the bound: its fluctuations have nearly the optimal amplitude for its correlation. One caution: the correlations are rounded to two digits, and near $r = 1$ the bound is very sensitive to rounding. For the 25-prime column ($r = +0.99$, rms 0.043) the bound with $r = 0.99$ is 0.037, but with $r = 0.985$, which also rounds to 0.99, it is 0.045. So no conclusion should be drawn from the last column's comparison.

Figure 19.4 shows the two extremes graphically, using the prime-2 wave of Running Example RE2 as the "truth".

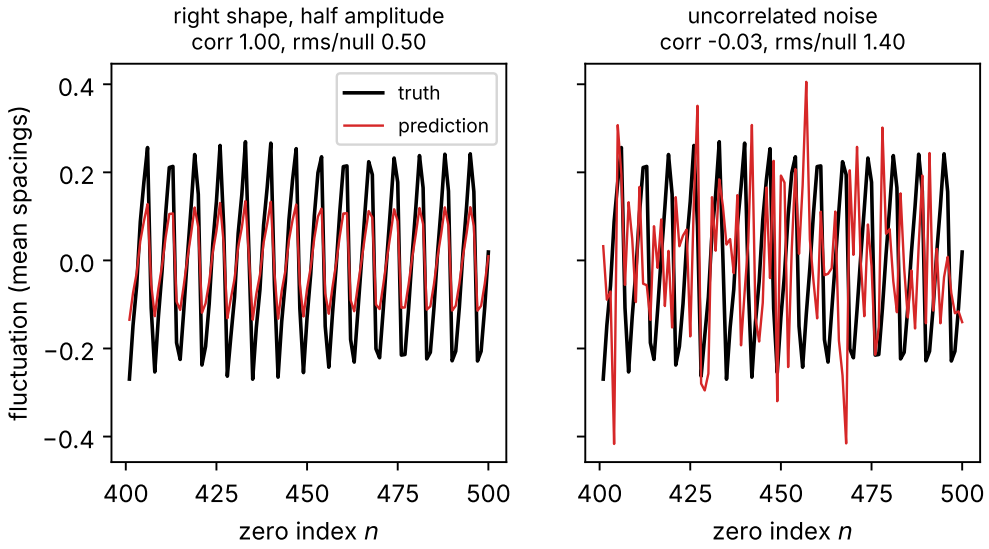


Figure 19.4: Two ways to miss. Truth (black): the prime-2 wave RE2, $-\frac{1}{\pi}\sum_{m=1}^{3}\sin(mt\log 2)/(m2^{m/2})$, evaluated at the smooth zeros s_n, $n = 401$–500. Left: a prediction with the right shape and half the amplitude; correlation $+1$, rms error exactly half the null. Right: Gaussian noise with the same spread as the truth (seed 2); correlation near zero, rms error close to $\sqrt{2}$ times the null, as Equation (19.30) predicts.

Pitfall

The correlation of the null prediction is undefined. If $p_i = 0$ for every i, the factor $\sqrt{\sum(p_i - \bar{p})^2}$ in Equation (19.27) is zero and r is 0/0. Consistently, the paper's tables give a null rms but no null correlation. `numpy.corrcoef` returns `nan` (with a warning) in this case.

Key Ideas

- The exact identity Equation (19.29) expresses mean-square error through means, spreads and correlation.
- Correct shape with wrong amplitude gives $r = +1$ but rms $|a - 1|\,\mathrm{rms}(y)$.
- For zero-mean truth, an uncorrelated prediction is worse than the null by its own variance, Equation (19.30).
- At correlation $r > 0$ the smallest possible rms is $\sigma_y\sqrt{1 - r^2}$, reached by shrinking the prediction to amplitude $r\sigma_y$, Equation (19.31).

19.6 Autocorrelation of a sequence

★ Did you know? *Sunspots and Memory*

In 1927 the statistician George Udny Yule modeled yearly sunspot numbers by letting each year's value depend on the previous two years plus random shocks. This was the first autoregressive model, and it showed how a correlated random sequence can mimic periodic behavior. Autocorrelation measures how strongly a sequence remembers its own past, and it is the tool for telling real rhythms from noise.

19.6.1 Definition

So far the order of a list did not matter: shuffling $x_1, \ldots, x_n$ leaves $\bar{x}$, σ_x and $\mathrm{rms}(x)$ unchanged. But the zeros come in order, and their fluctuations are not shuffled at random: neighbors are related. *Autocorrelation* measures how a sequence relates to a shifted copy of itself. The sample autocorrelation at lag k is

$$r_k = \frac{\sum_{i=1}^{n-k}(x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad k = 0, 1, 2, \ldots \tag{19.32}$$

It is the Pearson correlation of the sequence with itself shifted by k places, with the convention that the overall mean and the full sum of squares are used for normalization. Then $r_0 = 1$, and $|r_k| \le 1$ follows from Cauchy–Schwarz (Problem 29). A positive r_1 says "if this entry is above average, the next tends to be above average too"; a negative r_1 says "highs tend to be followed by lows".

Autocorrelation is most useful when the sequence is (at least roughly) *stationary*: its statistical behavior does not depend on where in the sequence you look, so the relation between x_i and x_{i+k} depends on k but not on i. For the zeros' fluctuations this is an approximation (their amplitude grows slowly with height, Sec. II C of the paper); over a window of a few hundred zeros it is reasonable.

Sample Problem 19.8: An alternating sequence

Compute r_1 and r_2 for $x = (1, -1, 1, -1, 1, -1)$.
Step 1: Mean and denominator. $\bar{x} = 0$; $\sum x_i^2 = 6$.
Step 2: Lag 1. The products $x_i x_{i+1}$ for $i = 1, \ldots, 5$ are all -1; their sum is -5. So $r_1 = -5/6$.
Step 3: Lag 2. The products $x_i x_{i+2}$ for $i = 1, \ldots, 4$ are all $+1$; sum 4. So $r_2 = 4/6$.

$$\boxed{r_1 = -0.833, \quad r_2 = +0.667}$$

The magnitudes are below 1 only because fewer pairs $(n - k)$ enter the numerator than

terms (n) enter the denominator; for a long alternating sequence $r_1 \to -1$ and $r_2 \to +1$.

19.6.2 The autocorrelation of a sinusoid

Take $y(t) = a\sin(\omega t + \phi)$. To see what Equation (19.32) measures for such a signal, consider first the idealized case of samples at many closely and evenly spaced times t_i over a window of many periods. Then each sum in Equation (19.32), divided by its number of terms, is a Riemann sum for a time average, and a lag of k samples is a fixed time shift τ. This is an approximation: it neglects the discreteness of the sampling and the incomplete period at the ends of the window, both of which become negligible as the sampling gets finer and the window longer. Because an average over many whole periods equals the average over one period $P = 2\pi/\omega$, we compute averages over one period. First, the mean is zero, because $\int_0^P \sin(\omega t + \phi)\,\mathrm{d}t = 0$. Next, by the identity $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$ with $A = \omega t + \phi$ and $B = \omega t + \omega\tau + \phi$,

$$y(t)\,y(t+\tau) = \frac{a^2}{2}\left[\cos(\omega\tau) - \cos(2\omega t + \omega\tau + 2\phi)\right].$$

The second cosine completes two full cycles in one period, so it averages to zero; the first does not depend on t. Therefore

$$\langle y(t)\,y(t+\tau)\rangle = \frac{a^2}{2}\cos(\omega\tau), \tag{19.33}$$

where $\langle\cdot\rangle$ denotes the average over a period. Setting $\tau = 0$ gives the mean square $a^2/2$, so $\mathrm{rms}(y) = a/\sqrt{2}$: this is the $\sqrt{2}$ of AC circuits. Dividing Equation (19.33) by its value at $\tau = 0$, the autocorrelation of a sinusoid at lag τ is $\cos(\omega\tau)$: it oscillates with the same period as the signal and never dies out. When the signal is sampled instead at the smooth zeros s_n, as in Figure 19.5, a lag of k levels corresponds to a time shift $\tau \approx k$ times the local mean spacing (defined in Section 19.9), which changes slowly with height. Over a window where that spacing is nearly constant, the lag-in-levels autocorrelation still oscillates like $\cos(\omega\tau)$, but only approximately.

The same tools give the mean square of a sum of sinusoids whose frequencies are $\omega, 2\omega, 3\omega, \ldots$. For positive integers $m \neq m'$, the identity above turns $\sin(m\omega t)\sin(m'\omega t)$ into $\frac{1}{2}[\cos((m - m')\omega t) - \cos((m + m')\omega t)]$; both cosines complete a whole number of cycles in the period P and average to zero. Cross terms vanish, and

$$\left\langle \Big(\sum_m a_m \sin(m\omega t)\Big)^2 \right\rangle = \sum_m \frac{a_m^2}{2}. \tag{19.34}$$

(When the frequencies are not integer multiples of one common ω there is no common period, but the cross terms still average to zero over a long window; this is the long-window orthogonality of sinusoids with different frequencies treated in Chapter 5. We do not need it here, because the frequencies of RE2 below are harmonics.)

Sample Problem 19.9: The rms of the prime-2 wave (RE2)

Running Example RE2 is $y(t) = -\frac{1}{\pi}\sum_{m=1}^{3} \frac{\sin(mt\log 2)}{m\,2^{m/2}}$. Find its long-run rms.

Step 1: Amplitudes. $a_m = 1/(\pi m 2^{m/2})$ (the overall minus sign does not affect squares): $a_1 = 1/(\pi\sqrt{2}) = 0.225079$, $a_2 = 1/(4\pi) = 0.079577$, $a_3 = 1/(6\sqrt{2}\,\pi) = 0.037513$.

Step 2: Mean square. The frequencies are $\omega, 2\omega, 3\omega$ with $\omega = \log 2$, so Equation (19.34) applies:

$$\overline{y^2} = \tfrac{1}{2}(0.050661 + 0.006333 + 0.001407) = \tfrac{1}{2}(0.058400) = 0.029200.$$

Step 3: Root. $\mathrm{rms}(y) = \sqrt{0.029200} = 0.17088$.

$$\boxed{\mathrm{rms(RE2)} \approx 0.1709}$$

The paper's arithmetic-continuation probe uses this wave (with the opposite sign, since a level shift is minus the counting fluctuation; the rms is unaffected) as a synthetic target, sampled at the smooth zeros 401–500, and records the rms of predicting no fluctuation as the "probe-null" 0.1708 (App. A, "Key anchors"). The window covers about 15 periods of the wave, and the long-run value 0.1709 agrees closely with the paper's 0.1708. Evaluating the wave directly at the 100 smooth zeros 401–500 with qmnt, `qmnt.rms(qmnt.Nfl(qmnt.smooth_zeros(n), [2]))`, reproduces the paper's 0.1708 to within 0.001. A small difference from the long-run value is to be expected, because on the holdout the wave is sampled at 100 discrete, unevenly spaced points rather than averaged continuously.

19.6.3 Predicting a value from its neighbors

Autocorrelation decides whether you can predict a missing entry from its neighbors. Suppose a stationary sequence has mean zero, variance σ^2, lag-1 autocorrelation ρ_1 and lag-2 autocorrelation ρ_2, meaning $E[x_i x_{i\pm1}] = \rho_1\sigma^2$ and $E[x_{i-1}x_{i+1}] = \rho_2\sigma^2$. Predict x_i by the average of its two neighbors, $\hat{x}_i = \frac{1}{2}(x_{i-1} + x_{i+1})$. What is the correlation between prediction and truth?

Covariance with the truth. By linearity, $E[x_i\hat{x}_i] = \frac{1}{2}(E[x_i x_{i-1}] + E[x_i x_{i+1}]) = \frac{1}{2}(2\rho_1\sigma^2) = \rho_1\sigma^2$.

Variance of the prediction. By Equation (19.25) and Equation (19.10),

$$\mathrm{Var}(\hat{x}_i) = \tfrac{1}{4}[\sigma^2 + \sigma^2 + 2\rho_2\sigma^2] = \frac{\sigma^2(1+\rho_2)}{2}.$$

Correlation. Dividing the covariance by the two standard deviations, as in the correlation coefficient ρ of two random variables, Equation (19.26),

$$\mathrm{corr}(\hat{x}_i, x_i) = \frac{\rho_1\sigma^2}{\sigma\cdot\sigma\sqrt{(1+\rho_2)/2}} = \frac{\rho_1}{\sqrt{(1+\rho_2)/2}}. \tag{19.35}$$

The sign is the sign of ρ_1. If neighbors are negatively correlated, a smooth interpolation between them is anticorrelated with the truth. For example, $\rho_1 = -0.3$ and $\rho_2 = -0.1$ give $\mathrm{corr} = -0.3/\sqrt{0.45} = -0.447$. This little calculation is the mechanism the paper invokes in Sec. III for negative held-out correlations; Chapter 25 develops the general best linear predictor.

Figure 19.5 shows autocorrelations of two sequences: the RE2 wave sampled at the smooth zeros, which oscillates because its $m = 1$ term dominates (it carries 87% of the mean square in Step 2 of the sample problem above; the $m = 2, 3$ harmonics distort the shape slightly away from a pure cosine), and independent noise, which is near zero at every nonzero lag.

```
import numpy as np

def autocorr(x, kmax):
    """Sample autocorrelation r_0..r_kmax, Eq. (acf)."""
    x = np.asarray(x, float) - np.mean(x)
    denom = np.sum(x**2)
    return np.array([np.sum(x[:len(x)-k] * x[k:]) / denom
                     for k in range(kmax + 1)])

print(autocorr([1, -1, 1, -1, 1, -1], 2))  # 1, -5/6, 4/6
```

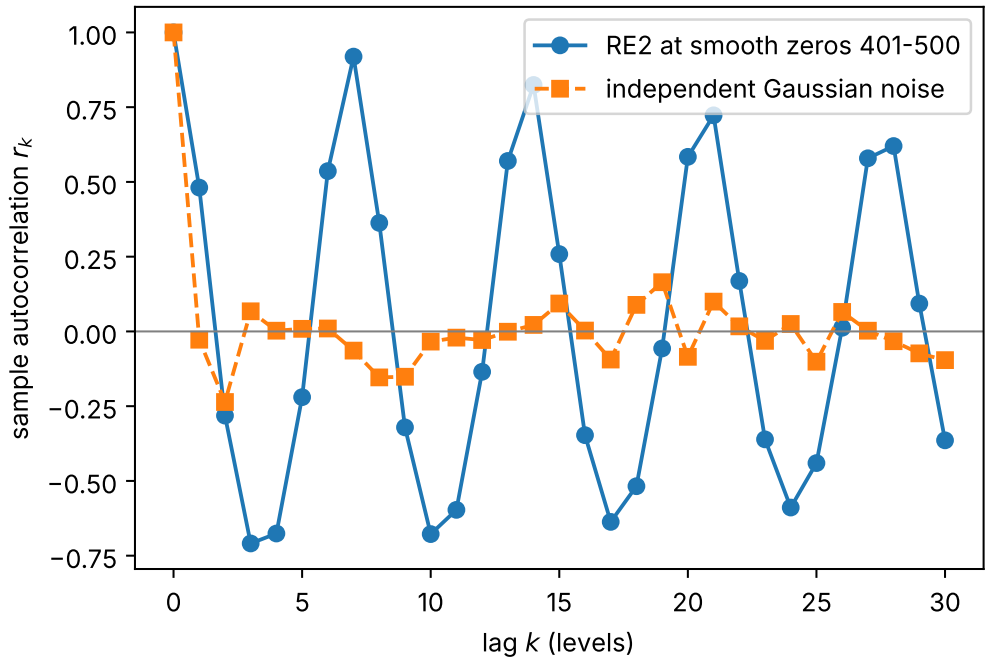


Figure 19.5: Sample autocorrelation Equation (19.32) against lag in levels. Circles: the prime-2 wave RE2 sampled at the smooth zeros s_n, $n = 401$–500; it oscillates, dominated by its $m = 1$ term (the $m = 2, 3$ harmonics modify the shape), with a period of roughly 6–7 levels, the level-count of one period $2\pi/\log 2 \approx 9.07$ in t at these heights (compare paper Sec. V). Squares: 100 independent Gaussian numbers (seed 1), whose autocorrelations scatter near zero.

Key Ideas

- The lag-k autocorrelation Equation (19.32) correlates a sequence with its own shifted copy; it captures the order information that $\bar{x}$ and σ_x ignore.
- A sinusoid has autocorrelation $\cos(\omega\tau)$ and rms $a/\sqrt{2}$; harmonically related sinusoids add in mean square, Equation (19.34).
- The long-run rms of RE2 is 0.1709, close to the paper's probe-null 0.1708.
- Averaging neighbors predicts with correlation $\rho_1/\sqrt{(1+\rho_2)/2}$, Equation (19.35): negative if neighbors are negatively correlated.

19.7 Sampling, estimators and the standard error

19.7.1 Estimators

In practice we want properties of a distribution (its mean μ, its variance σ^2, the correlation between two quantities) but we see only a finite *sample* $x_1, \ldots, x_n$ drawn from it. An *estimator* is a rule that turns the sample into a guess; $\bar{x}$ is an estimator of μ. Because the sample is random, the estimator is itself a random variable: draw another sample and you get another value. Two questions matter. Is it right on average (is it *unbiased*, E[estimator] = true value)? And how much does it scatter from sample to sample?

Throughout this section the draws are *independent and identically distributed*: each x_i has the same distribution, with mean μ and variance σ^2, and different draws are independent.

We need the two-variable results of Section 19.4 for n variables. Applying them repeatedly

(first to $x_1 + x_2$, then to $(x_1 + x_2) + x_3$, and so on; for independent draws each partial sum is independent of the next draw) gives $E[\sum_i x_i] = \sum_i E[x_i]$ for any draws, and $\mathrm{Var}(\sum_i x_i) = \sum_i \mathrm{Var}(x_i)$ for independent ones.

The sample mean is unbiased. By linearity, $E[\bar{x}] = \frac{1}{n}\sum_i E[x_i] = \frac{1}{n} \cdot n\mu = \mu$.

Its variance shrinks like $1/n$. By Equation (19.10) with $a = 1/n$, and then Equation (19.25) with all covariances zero (independence),

$$\mathrm{Var}(\bar{x}) = \frac{1}{n^2}\mathrm{Var}\Big(\sum_i x_i\Big) = \frac{1}{n^2}\sum_i \mathrm{Var}(x_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. \tag{19.36}$$

The standard deviation of the estimator is called its *standard error*:

$$\mathrm{SE}(\bar{x}) = \frac{\sigma}{\sqrt{n}} \approx \frac{\hat{\sigma}}{\sqrt{n}}. \tag{19.37}$$

The second form replaces the unknown σ by its estimate. To halve the standard error you need four times as many data.

Why divide by $n-1$. Let us compute the expectation of $Q = \sum_i (x_i - \bar{x})^2$. First an algebraic rearrangement:

$$\begin{aligned} Q &= \sum_i x_i^2 - 2\bar{x}\sum_i x_i + n\bar{x}^2 \\ &= \sum_i x_i^2 - 2\bar{x}\,(n\bar{x}) + n\bar{x}^2 \quad \Big(\sum_i x_i = n\bar{x}\Big) \\ &= \sum_i x_i^2 - n\bar{x}^2. \end{aligned}$$

Now take expectations. By Equation (19.9) applied to each x_i, $E[x_i^2] = \sigma^2 + \mu^2$. By the same identity applied to $\bar{x}$, together with $E[\bar{x}] = \mu$ and Equation (19.36), $E[\bar{x}^2] = \sigma^2/n + \mu^2$. Hence

$$E[Q] = n(\sigma^2 + \mu^2) - n\Big(\frac{\sigma^2}{n} + \mu^2\Big) = (n-1)\sigma^2.$$

So $E[Q/(n-1)] = \sigma^2$: the $n-1$ version $\hat{\sigma}^2$ is unbiased, while $\sigma_x^2 = Q/n$ is too small on average by the factor $(n-1)/n$. The reason is that $\bar{x}$ is computed from the same data and sits closer to them than the true μ does. (This is *Bessel's correction*.) For example, for $n = 2$ throws of a fair die, averaging over all 36 equally likely pairs, $Q/1$ averages to exactly $35/12$, the true variance found earlier, while $Q/2$ averages to $35/24 \approx 1.458$ (Problem 30).

Pitfall

Formula Equation (19.37) assumes *independent* draws. If neighboring values are correlated, as the zeros' fluctuations are (Section 19.6), the covariance terms in Equation (19.25) no longer vanish and $\sigma/\sqrt{n}$ can be badly wrong. Chapter 26 treats this with the block bootstrap; the paper does the same in Sec. III ("Other bases").

19.7.2 The law of large numbers

How sure can we be that $\bar{x}$ is close to μ? A beautiful one-line inequality answers this for every distribution with finite variance.

Chebyshev's inequality (proved). For any random variable with mean μ, variance σ^2, and any $a > 0$,

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}. \tag{19.38}$$

Proof (discrete case; the continuous case replaces sums by integrals). Start from the definition of variance and keep only the values far from the mean:

$$\begin{aligned}\sigma^2 = \sum_j p_j (x_j - \mu)^2 &\geq \sum_{j:\,|x_j-\mu|\geq a} p_j (x_j - \mu)^2 \quad \text{(dropped terms are } \geq 0) \\ &\geq \sum_{j:\,|x_j-\mu|\geq a} p_j\, a^2 \quad \text{(each kept } (x_j-\mu)^2 \geq a^2) \\ &= a^2\, P(|X - \mu| \geq a).\end{aligned}$$

Divide by a^2. □

With $a = 2\sigma$, at most $1/4$ of the probability lies 2σ or more from the mean, for any distribution. (For a Gaussian the true fraction is 0.0455; the bound is crude but universal.)

Weak law of large numbers (proved, for finite variance). Apply Equation (19.38) to $\bar{x}$, whose mean is μ and variance σ^2/n:

$$P(|\bar{x} - \mu| \geq a) \leq \frac{\sigma^2}{n\,a^2} \;\longrightarrow\; 0 \quad (n \to \infty), \tag{19.39}$$

for every fixed $a > 0$, however small. With enough data the sample mean is, with probability as close to 1 as you like, within any chosen distance of the true mean. As an instance: for exponential gaps ($\sigma^2 = 1$), $n = 1000$ and $a = 0.1$, the bound says the sample mean of 1000 gaps misses 1 by 0.1 or more with probability at most $1/(1000 \times 0.01) = 0.1$. This theorem is what justifies the long-run-frequency reading of probability: the fraction of trials in which an event happens is the sample mean of a variable that is 1 when it happens and 0 otherwise, whose mean is the event's probability. A stronger version also holds.

Taken on trust

Taken on trust: the strong law of large numbers. For independent, identically distributed draws with finite mean, $\bar{x} \to \mu$ with probability 1, not merely with probability approaching 1. Its proof needs measure-theoretic probability and is beyond this book; we never use it.

Figure 19.6 shows running means of exponential samples for three seeds. Each wanders early, and each settles near 1, with deviations of the size $1/\sqrt{n}$ set by Equation (19.37).

Checkpoint 19.6

After 10^4 exponential draws, is the running mean exactly 1?

Answer: Almost surely not. The law of large numbers says it is *close* to 1 with high probability; its typical distance from 1 is one standard error, $1/\sqrt{10^4} = 0.01$.

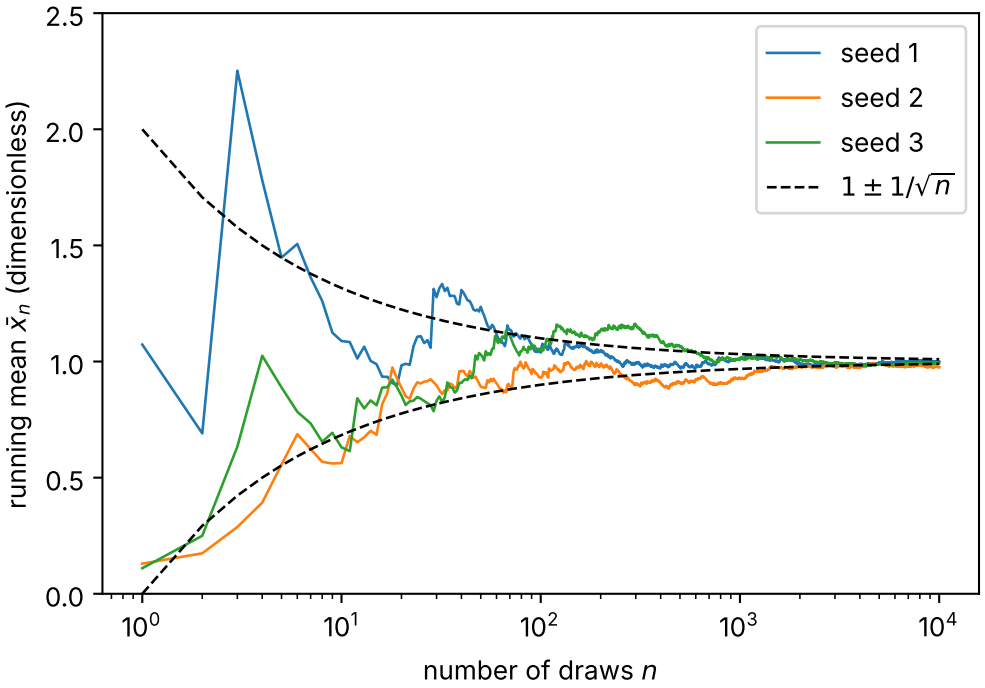


Figure 19.6: Law of large numbers. Running sample mean $\bar{x}_n$ of n exponential draws of mean 1, for seeds 1, 2, 3 of `numpy.random.default_rng`. The dashed lines are $1 \pm 1/\sqrt{n}$, one standard error Equation (19.37) either side of the true mean.

19.7.3 Confidence intervals

Taken on trust

Taken on trust: the central limit theorem. If $x_1, \ldots, x_n$ are independent and identically distributed with mean μ and finite variance σ^2, then for large n the distribution of $(\bar{x} - \mu)/(\sigma/\sqrt{n})$ approaches the standard normal φ of Equation (19.19), whatever the distribution of the individual x_i. How large n must be depends on the distribution (more for strongly skewed ones). A proof is in [22].

Granting the theorem, $\bar{x}$ lies within $1.96\,\mathrm{SE}$ of μ with probability about 0.95, by Equation (19.20). Turn the statement around: the interval

$$[\bar{x} - 1.96\,\mathrm{SE},\ \bar{x} + 1.96\,\mathrm{SE}] \tag{19.40}$$

is an approximate *95% confidence interval* for μ. Its meaning is a statement about the procedure: if you repeated the whole experiment many times and built the interval each time, about 95% of the intervals would contain the true μ. It is not a statement that μ "has a 95% chance" of being in your particular interval; μ is a fixed number, and it either is or is not. For small samples with $\hat{\sigma}$ in place of σ, statisticians replace 1.96 by a slightly larger number (from Student's t distribution: 2.06 for $n = 25$); we ignore that refinement here.

The paper builds its intervals differently, by the bootstrap (Chapter 26), which does not rely on the central limit theorem but shares the same interpretation.

Sample Problem 19.10: A confidence interval for g

A lab group measures the free-fall acceleration 25 times, getting $\bar{x} = 9.79$ m/s^2 and $\hat{\sigma} = 0.15$ m/s^2. Assuming independent measurements, give the standard error and an approximate 95% confidence interval.
Step 1: Standard error. By Equation (19.37), SE $= 0.15/\sqrt{25} = 0.03$ m/s^2.
Step 2: Half-width. $1.96 \times 0.03 = 0.0588$ m/s^2.
Step 3: Interval. By Equation (19.40), $[9.79 - 0.0588,\ 9.79 + 0.0588]$.

$$\boxed{\text{SE} = 0.03 \text{ m/s}^2, \qquad [9.731,\ 9.849] \text{ m/s}^2}$$

The standard value 9.81 m/s^2 lies inside. Notice the distinction: the measurements scatter by 0.15, but the mean is uncertain by only 0.03.

Taken on trust

Taken on trust: the scatter of a sample correlation. For n independent pairs from a joint Gaussian distribution with true correlation ρ, the sample correlation r has, for large n, a standard deviation of approximately $(1 - \rho^2)/\sqrt{n}$. This is a standard normal-theory result, usually derived through Fisher's z-transformation; its derivation is beyond this book. A comparison of this normal-theory standard error with bootstrap estimates on a real data set is given in [34]. Two lessons: correlations near zero are uncertain by about $\pm 1/\sqrt{n}$, and correlations near ± 1 are pinned down much more tightly.

Key Ideas

- An estimator is a random variable. $\bar{x}$ is unbiased with standard error $\sigma/\sqrt{n}$, Equation (19.37), for independent draws.
- Dividing by $n - 1$ makes the variance estimator unbiased (Bessel's correction).
- Chebyshev's inequality Equation (19.38) implies the weak law of large numbers Equation (19.39).
- With the central limit theorem, $\bar{x} \pm 1.96\,\text{SE}$ is an approximate 95% confidence interval, Equation (19.40); it describes the long-run behavior of the procedure.

19.8 Random numbers and seeds

Computers are deterministic machines, so where do the random numbers in Figures 19.3 and 19.6 come from? From a *pseudo-random number generator*: an algorithm that, starting from an internal state, produces a long sequence of numbers that pass statistical tests for randomness (uniform distribution, no detectable correlation between successive values) but is entirely determined by its starting state. The integer that sets the starting state is the *seed*.

Two consequences matter for science. First, *reproducibility*: with the same seed, the same code produces exactly the same "random" numbers on every run, so a result that involved randomness can be regenerated digit for digit. Second, *exploring variability*: running with several different seeds shows how much a result depends on the particular random draws.

```
import numpy as np

rng = np.random.default_rng(1)        # seed 1
a = rng.exponential(1.0, size=5)
rng = np.random.default_rng(1)        # same seed again
```

```
b = rng.exponential(1.0, size=5)
print(np.array_equal(a, b))            # True: identical draws

rng2 = np.random.default_rng(2)        # a different seed
print(rng2.exponential(1.0, size=5))   # different numbers

phases = np.random.default_rng(7).uniform(0, 2*np.pi, size=10**5)
print(np.mean(np.cos(phases)), np.mean(np.cos(phases)**2))   # near 0, 0.5
```

The last lines check by simulation two expectations for a uniform random phase θ on $[0, 2\pi)$ with density $1/(2\pi)$:

$$E[\cos\theta] = \frac{1}{2\pi}\int_0^{2\pi} \cos\theta \, \mathrm{d}\theta = \frac{1}{2\pi}[\sin\theta]_0^{2\pi} = 0, \qquad E[\cos^2\theta] = \frac{1}{2\pi}\int_0^{2\pi} \frac{1+\cos 2\theta}{2} \, \mathrm{d}\theta = \frac{1}{2}.$$

These are the facts behind phase randomization: scrambling phases keeps each sinusoid's mean square ($a^2/2$) but destroys its timing.

Pitfall

Results from different seeds are not independent new data if they all use the *same* underlying data set. Varying the seed of an optimizer, or of a random train/test split drawn from the same zeros, tells you how robust a result is to those random choices. It does not tell you how the result would change on fresh zeros. A range over seeds is therefore not a standard error.

Checkpoint 19.7

A study fits a model ten times with seeds 1–10, always to zeros 1–400, and reports the spread of the ten held-out scores. Does that spread estimate how the score would vary on a new set of zeros?

Answer: No. All ten runs share the same zeros, so they probe sensitivity to the random choices (initial values, which zeros are held out), not sampling variability of the data. This is exactly the qualification in the caption of the paper's Table III.

Key Ideas

- Pseudo-random generators are deterministic; the seed fixes the whole sequence, which makes random computations reproducible.
- Different seeds explore the sensitivity of a result to random choices; with fixed data they are not independent samples.

19.9 Paper connection

We can now read every "rms / corr" entry in the paper. First, the quantities being scored.

What is scored. Recall from Chapter 17 that each zero height t_n has a *smooth position* s_n, defined by $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ with N_{sm} the smooth counting function of paper Eq. (1), and a *local mean spacing* $\mathrm{gap}_n = ds_n/dn$. The fluctuation of zero n, in mean spacings, is

$$\delta_n = \frac{t_n - s_n}{\mathrm{gap}_n}. \tag{19.41}$$

(Here the letter s means smooth zeros, not the spacing variable of Section 19.3; the paper uses both, and context tells them apart.) A model that predicts height $\hat{t}_n$ has predicted fluctuation $\hat{\delta}_n = (\hat{t}_n - s_n)/\mathrm{gap}_n$ and error, in spacings,

$$e_n = \frac{\hat{t}_n - t_n}{\mathrm{gap}_n} = \hat{\delta}_n - \delta_n.$$

The paper's "rms" is rms(e) of Equation (19.15) over the scored zeros, and its "corr" is the Pearson coefficient Equation (19.27) between $\hat{\delta}$ and δ (Sec. II C: "reporting rms error in local spacings and the correlation between predicted and true fluctuations"). The *null* predicts $\hat{t}_n = s_n$, so $\hat{\delta}_n = 0$ and its error is rms(δ): the null rms is simply the rms of the true fluctuations. On the principal holdout, zeros 401–500 (Running Example RE5), it is 0.2631 (Sec. II C and App. A, "Key anchors"), and it grows slowly with height, from 0.263 to 0.296 over the holdouts of Table V. For the fitted potentials the fluctuations are measured relative to the unperturbed potential's levels instead of s_n; the paper notes that this null coincides numerically with the smooth-zero null on 401–500, which is what allows one baseline for both comparisons (Sec. II C). As the pitfall in Section 19.5 explains, the null has no correlation, and indeed none appears in the tables.

Why "rms worse than the null, corr near zero" is one fact, not two. Table I reports, for the fitted potentials, held-out rms values from 0.265 to 0.439 against nulls of 0.261–0.274, with correlations between −0.17 and +0.12. Sec. III states that "the fitted models emit fluctuations of 0.16–0.62× the true amplitude that are uncorrelated with truth". Equation (19.30) says that a zero-mean prediction uncorrelated with zero-mean truth has rms $\sqrt{\sigma_y^2 + \sigma_p^2}$, always above the null. With $\sigma_y = 0.263$ and σ_p between 0.16 and 0.62 times that, the identity gives 0.266–0.309 (Problem 19): above the null, as every unregularized row of Table I is. This is a toy reading; actual predictions also have mean offsets, nonzero sample correlations and amplitudes that vary along the holdout, so the table entries are not computed from this formula. It does explain the pattern: a prediction with no correlation cannot beat the null, and the more it fluctuates the worse it does. It also explains the paper's remark that ridge regularization "moves them smoothly from 'memorize more' to 'do nothing' and never creates skill" (Sec. III): shrinking an uncorrelated prediction lowers σ_p toward zero and the rms toward the null, never below it.

The prime formula sits on the optimal-amplitude bound. The sample problem in Section 19.5 showed that for the zero-parameter formula of paper Eq. (4), the rms entries of Table V on 401–500 (0.198, 0.116, 0.074 for 1, 3, 6 primes) match the best rms Equation (19.31) possible at their correlations (0.198, 0.115, 0.074), under the stated assumption that the true fluctuations have small mean. This is our observation from the paper's numbers, not a claim the paper makes, and the rounding caveat applies near $r = 1$.

The arithmetic-continuation probe. In Sec. III the paper fits a potential to a synthetic target, the prime-2 wave RE2 (with opposite sign, App. A step 15) sampled on levels 1–400, and asks it to continue on 401–500. In-sample: rms 0.056, correlation +0.95. Held out: correlation +0.07, "rms worse than predicting zero fluctuation"; App. A gives the held-out rms 0.1958 against the probe-null 0.1708. That probe-null is the rms of RE2 sampled on the holdout, whose long-run value we derived by hand as 0.1709 in Section 19.6. With the identity Equation (19.29) you can even estimate the amplitude the fitted potential emitted on the holdout (Problem 20).

Seeds and ranges. The paper uses recorded seeds throughout: random restarts of the non-Hermitian fit (App. A step 10, seed 0), permutations (step 12, seeds 1, 2), random 80/20 splits (step 17, seeds 1–10 and 1, 2), surrogates (step 18, seeds 1–10). The revision computations

(steps 16–23) and the permutations of step 12 draw from `numpy.random.default_rng(seed)`; for step 10 the paper records the seed 0 without naming the generator. Sec. II F stresses that every number is regenerated "deterministically, or from recorded random seeds". The caption of Table III states the lesson of Section 19.8 exactly: "The seeds reuse the same zeros, so the ranges indicate robustness and are not standard errors." The caption of Table I reports optimizer variability the same way: four random-initialization repeats of one configuration gave held-out rms 0.365–0.379.

Intervals. Sec. III gives bootstrap 95% intervals, for example $[0.33, 0.43]$ for the held-out rms of the scale-1, $M = 400$ random-restart replicate against the null 0.263, with correlation interval $[-0.21, +0.08]$, and $[+0.95, +0.97]$ for the six-prime formula's correlation on the same holdout. These are confidence intervals in the sense of Section 19.7, built by the bootstrap of Chapter 26. Their widths agree in order of magnitude with the Gaussian rule of thumb $(1-\rho^2)/\sqrt{n}$ of the `beyond` box on the scatter of a sample correlation in Section 19.7: about ± 0.2 (± 1.96 standard deviations) near $r = 0$, against a half-width of about 0.15 for the paper's interval $[-0.21, +0.08]$, and about ± 0.015 near $r = 0.96$ for $n = 100$ (Problem 23). Sec. III ("Other bases") warns that the intervals in Table II "resample held-out levels independently although the fluctuations are serially correlated", the pitfall of Section 19.7; the paper's block-bootstrap recomputation widens them slightly without changing any conclusion (Sec. III, "Other bases").

Autocorrelation and the within-height splits. Sec. III ("Within-height splits") explains the negative held-out correlations of Table III for $M \le 200$: "the fluctuation sequence has negative autocorrelation at lags 1–3, so any smooth interpolation between neighbouring training levels is anticorrelated with the truth (the mean of the two nearest training fluctuations gives -0.26 to -0.41)." Equation (19.35) is the simplest version of that mechanism: the neighbor average has the sign of ρ_1. Chapter 25 returns to this with the best linear predictor ("kriging"), which the paper reports reaching 0.117 with correlation $+0.87$ within one height range.

Fig. 2. The paper's Fig. 2 overlays the Poisson law e^{-s}, which we derived in Equation (19.22), on the unfolded spacings of the first 10^4 zeros, together with a second curve, the GUE Wigner surmise, which Chapter 20 derives. Our Figure 19.3 reproduces the comparison with the Poisson law.

19.10 Summary

★ Did you know? *Random Matrices Began in Statistics*

Random matrices first appeared in statistics, not physics. In 1928 John Wishart derived the distribution of sample covariance matrices built from Gaussian data, now called the Wishart distribution. In the 1950s Eugene Wigner used random matrices to model nuclear energy levels, and in the 1970s they turned up in the Riemann zeros. The same mathematics serves surveys, nuclei and primes.

- A random variable has a discrete distribution p_j or a density $f(x)$; expectations are weighted averages, Equations (19.2) and (19.5).
- Variance $\sigma^2 = E[(X-\mu)^2] = E[X^2] - \mu^2$, Equations (19.8) and (19.9); $\mathrm{Var}(aX+b) = a^2\sigma^2$, Equation (19.10).
- For a list: mean Equation (19.11), spreads σ_x and $\hat{\sigma}$, Equations (19.12) and (19.13); rms Equation (19.15); $\mathrm{rms}(e)^2 = \bar{e}^{\,2} + \sigma_e^2$, Equation (19.16).

- Gaussian Equation (19.18): mean μ, variance σ^2; probabilities Equation (19.20).
- Poisson process: $P(\text{empty interval } s) = e^{-s}$, Equation (19.21); gap density e^{-s} with mean and variance 1, Equation (19.22).
- Covariance Equation (19.24); variance of a sum Equation (19.25); Pearson r Equation (19.27); Cauchy–Schwarz Equation (19.28) gives $|r| \le 1$.
- Exact link between errors and correlation: Equation (19.29). Uncorrelated predictions lose to the null, Equation (19.30); best rms at correlation r is $\sigma_y\sqrt{1-r^2}$, Equation (19.31).
- Autocorrelation Equation (19.32); sinusoids: Equations (19.33) and (19.34); rms of RE2 = 0.1709; neighbor-average correlation Equation (19.35).
- $\bar{x}$ is unbiased with SE $= \sigma/\sqrt{n}$ for independent draws, Equations (19.36) and (19.37); Chebyshev Equation (19.38) and the weak law Equation (19.39); 95% intervals Equation (19.40).
- Seeds make random computations reproducible; ranges over seeds on fixed data measure robustness, not sampling error.

19.11 Problems

1. • A fair die is thrown once. What is the probability that the face shown is a prime number?

2. • For the list $(2, 4, 4, 4, 5, 5, 7, 9)$ find the mean, σ_x, $\hat{\sigma}$ and the rms. Check that $\text{rms}^2 = \bar{x}^2 + \sigma_x^2$.

3. • Two models make errors $(0.2, -0.2, 0.2, -0.2)$ and $(0.2, 0.2, 0.2, 0.2)$. Compute the mean, spread σ_e and rms of each. What does this tell you about using the rms alone?

4. •• Prove $\text{Var}(aX + b) = a^2\text{Var}(X)$ starting from the shortcut Equation (19.9) rather than from the definition.

5. • For a unit-rate Poisson process, what is the probability that a gap is shorter than half the mean gap?

6. •• A measured quantity is rounded to the nearest integer. Model the rounding error as uniform on $[-\frac{1}{2}, \frac{1}{2}]$. Find its mean and its rms.

7. • Using Equation (19.20), what fraction of a Gaussian distribution lies more than 2σ from its mean?

8. •• Compute the Pearson correlation of $x = (1, 2, 3, 4)$ and $y = (2, 1, 4, 3)$.

9. •• Show that $\text{corr}(ax + b, y) = \text{corr}(x, y)$ for $a > 0$ and $= -\text{corr}(x, y)$ for $a < 0$, for any constant b.

10. •• Compare the Chebyshev bound for the probability of landing 3σ or more from the mean with the exact Gaussian value from Equation (19.20).

11. •• Suppose the fluctuations δ_n of the zeros on some holdout were Gaussian with mean 0 and standard deviation 0.263. What fraction would exceed half a mean spacing in magnitude? (Use `scipy.stats.norm` or a table.)

12. •• Compute r_1 and r_2 of Equation (19.32) for $x = (1, 1, -1, -1, 1, 1, -1, -1)$.

13. •• Four hundred independent measurements have $\hat{\sigma} = 0.25$. Find the standard error of their mean and the half-width of an approximate 95% confidence interval.

14. •• In a histogram of 1000 numbers with bin width 0.1, one bin holds 38 entries. What density does Equation (19.17) assign to that bin?

15. • Why does rerunning a numpy program with `default_rng(5)` reproduce exactly the same "random" numbers?

16. •• Verify by integration by parts that the exponential density Equation (19.22) has $E[s] = 1$ and $E[s^2] = 2$.

17. ••• Prove the equality case of Cauchy–Schwarz in detail: if $(\sum u_i v_i)^2 = \sum u_i^2 \sum v_i^2$ and $\sum v_i^2 > 0$, then $u_i = \lambda v_i$ for one constant λ and all i.

18. •• (Paper.) The caption of Table III says that the ranges over seeds "indicate robustness and are not standard errors." Explain in your own words why, and say what kind of data would be needed to estimate a genuine standard error of the held-out rms.

19. •• (Paper.) Sec. III states that the fitted potentials "emit fluctuations of 0.16–0.62× the true amplitude that are uncorrelated with truth." Model truth and prediction as zero-mean with $\sigma_y = 0.263$ and $r = 0$. Use Equation (19.30) to find the held-out rms at the two ends of this range, and compare with the null.

20. ••• (Paper.) In the arithmetic-continuation probe (Sec. III and App. A), the held-out rms is 0.1958, the probe-null 0.1708 and the held-out correlation +0.07. Assuming both truth and prediction have zero mean on the holdout, solve Equation (19.29) for the spread σ_p of the fitted operator's held-out fluctuations, and express it as a fraction of the target's spread.

21. •• (Paper.) Using Equation (19.31) with $\sigma_y = 0.263$, find the smallest rms possible at correlations +0.66, +0.90, +0.96, and compare with the rms values of the corresponding columns of Table V on 401–500. State the assumption you need.

22. ••• (Paper.) (a) Using Equation (19.35), find the correlation of the neighbor-average predictor when $\rho_1 = \rho_2 = -0.2$. (b) With $\rho_2 = 0$, what ρ_1 would give -0.41, the most negative value quoted in Sec. III ("Within-height splits")? (c) Why should the formula be applied to the paper's numbers only as an illustration?

23. •• (Paper.) Use the rule of thumb $(1-\rho^2)/\sqrt{n}$ of the `beyond` box on the scatter of a sample correlation in Section 19.7 to estimate the standard deviation of the sample correlation of the six-prime formula ($\rho = 0.96$) on the $n = 100$ held-out zeros 401–500. Form a ± 1.96 interval and compare with the paper's bootstrap interval $[+0.95, +0.97]$ (Sec. III).

24. •• Using numpy, draw 10^4 exponential numbers of mean 1 with `default_rng(0)`. Compute their mean and variance. Compare the mean with the exact value 1 in units of its standard error; for the variance, just report the relative deviation from the exact value 1.

25. •• Using qmnt, compute the null rms $\mathrm{rms}(\delta)$ of Equation (19.41) on zeros 401–500 and on 501–1000, and compare with the null column of Table V.

26. ••• Using qmnt, compute the held-out rms and correlation of the one-prime formula, `qmnt.oracle_zeros(n, [2])`, on zeros 401–500. Compare with Table V and with the bound Equation (19.31) evaluated at your correlation.

27. ••• Using numpy, evaluate RE2 on a fine grid of t covering 1000 periods $2\pi/\log 2$ and compute its rms. Compare with the value derived from Equation (19.34).

28. •• Let θ be uniform on $[0, 2\pi)$, $X = \cos\theta$ and $Y = \sin\theta$. Show that $\mathrm{Cov}(X, Y) = 0$ although X and Y are not independent (hint: $X^2 + Y^2 = 1$).

29. ••• Apply Cauchy–Schwarz to the lists $u_i = x_i - \bar{x}$ for $i = 1, \dots, n-k$ and $v_i = x_{i+k} - \bar{x}$ to prove that $|r_k| \le 1$ for the autocorrelation Equation (19.32).

30. •• For $n = 2$ independent throws of a fair die, list or reason over all 36 equally likely pairs to show that $Q = \sum_i (x_i - \bar{x})^2$ averages to $35/12$, confirming Bessel's correction.

19.12 Answers to selected problems

1. The prime faces are 2, 3, 5. By the addition rule $P = 3 \times \frac{1}{6} = \frac{1}{2}$.

2. Sum 40, so $\bar{x} = 5$. Deviations $-3, -1, -1, -1, 0, 0, 2, 4$; squares sum to 32. $\sigma_x = \sqrt{32/8} = 2$; $\hat{\sigma} = \sqrt{32/7} = 2.138$. Mean square $= (4 + 16 + 16 + 16 + 25 + 25 + 49 + 81)/8 = 232/8 = 29$, so rms $= \sqrt{29} = 5.385$. Check: $\bar{x}^2 + \sigma_x^2 = 25 + 4 = 29$. ✓

3. First model: mean 0, $\sigma_e = 0.2$, rms 0.2. Second: mean 0.2, $\sigma_e = 0$, rms 0.2. Both have the same rms, one entirely from scatter and one entirely from bias, Equation (19.16). The rms tells you the size of the errors but not their nature; to diagnose a model you also look at $\bar{e}$ (or at the correlation).

5. $F(0.5) = 1 - e^{-0.5} = 1 - 0.606531 = 0.3935$.

7. $1 - 0.9545 = 0.0455$, about 1 in 22.

9. Let $x'_i = ax_i + b$. Then $\overline{x'} = a\bar{x} + b$, so $x'_i - \overline{x'} = a(x_i - \bar{x})$. The numerator of Equation (19.27) is multiplied by a; the factor $\sqrt{\sum (x'_i - \overline{x'})^2} = \sqrt{a^2 \sum (x_i - \bar{x})^2} = |a| \sqrt{\sum (x_i - \bar{x})^2}$ is multiplied by $|a|$. The ratio is multiplied by $a/|a| = \pm 1$.

11. The z-score is $u = 0.5/0.263 = 1.901$. The two-sided tail probability is $2[1 - \Phi(1.901)] = 0.0573$, where Φ is the standard normal CDF; about 6% of the fluctuations. (Whether the real fluctuations are Gaussian is a separate question; Figure 19.1 lets you judge.)

13. SE $= 0.25/\sqrt{400} = 0.0125$; half-width $1.96 \times 0.0125 = 0.0245$.

15. The generator is a deterministic algorithm. The seed sets its initial internal state, and from a given state every subsequent number is fixed. Same seed, same state, same sequence.

17. Write $S_{uv} = \sum u_i v_i$, $S_{uu} = \sum u_i^2$, $S_{vv} = \sum v_i^2 > 0$. From the proof of Equation (19.28), with $\lambda^* = S_{uv}/S_{vv}$, $g(\lambda^*) = S_{uu} - S_{uv}^2/S_{vv}$. If $S_{uv}^2 = S_{uu}S_{vv}$, this is zero. But $g(\lambda^*) = \sum (u_i - \lambda^* v_i)^2$ is a sum of nonnegative terms; it can vanish only if every term vanishes, so $u_i = \lambda^* v_i$ for all i. Conversely, if $u_i = \lambda v_i$, both sides of Equation (19.28) equal $\lambda^2 S_{vv}^2$.

19. rms $= 0.263\sqrt{1 + (\sigma_p/\sigma_y)^2}$. For 0.16: $0.263\sqrt{1.0256} = 0.2663$. For 0.62: $0.263\sqrt{1.3844} = 0.3094$. Both exceed the null 0.263; uncorrelated output can only add error. The toy model ignores mean offsets and sample correlations, so it is not expected to reproduce Table I row by row.

21. Assuming the true fluctuations on 401–500 have mean near zero, so $\sigma_y \approx 0.263$: bounds $0.263\sqrt{1 - r^2} = 0.1976$, 0.1146, 0.0736, against Table V's 0.198, 0.116, 0.074. The formula's rms is at the bound to the precision shown, meaning its amplitude is close to the optimal $r\sigma_y$.

23. $(1 - 0.96^2)/\sqrt{100} = 0.0784/10 = 0.0078$. Then $1.96 \times 0.0078 = 0.0154$, giving roughly $[0.945, 0.975]$, comparable with the bootstrap $[+0.95, +0.97]$. The rule assumes independent Gaussian pairs, which the serially correlated fluctuations are not, so only the order of magnitude should be compared.

25. About 0.263 on 401–500 and 0.270 on 501–1000, matching the null column of Table V. Code: `n = np.arange(401, 501); d = qmnt.fluct(qmnt.zeros(1000)[n-1], n); print(qmnt.rms(d))`, and similarly for 501–1000.

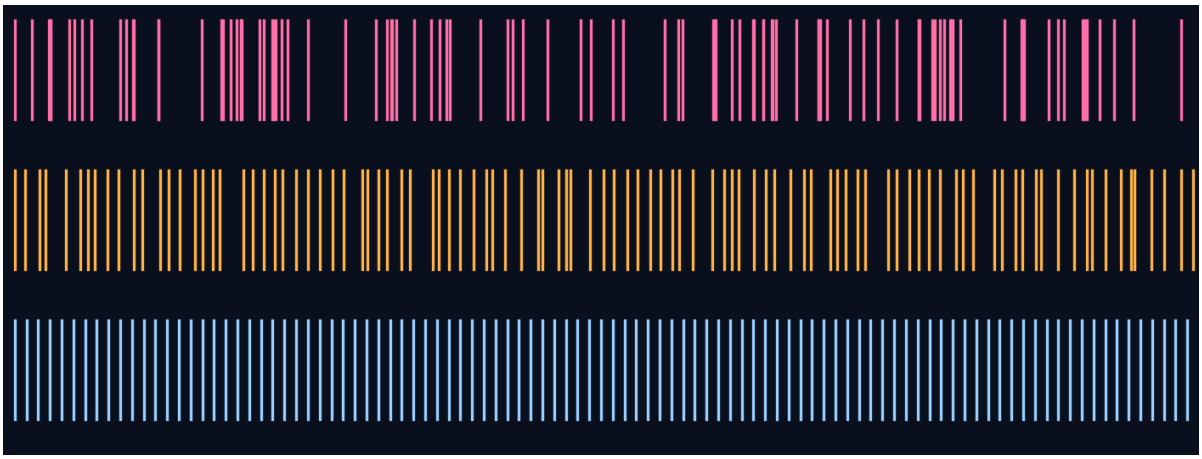
27. The rms is 0.1709, in agreement with $\sqrt{(a_1^2 + a_2^2 + a_3^2)/2} = 0.17088$ from Equation (19.34). Over 1000 whole periods the end corrections are negligible.

29. Let $D = \sum_{i=1}^{n} (x_i - \bar{x})^2 > 0$. Cauchy–Schwarz Equation (19.28) applied to $u_i = x_i - \bar{x}$

and $v_i = x_{i+k} - \bar{x}$, $i = 1, \ldots, n-k$, gives

$$\Big|\sum_{i=1}^{n-k} u_i v_i\Big| \le \sqrt{\sum_{i=1}^{n-k} (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^{n-k} (x_{i+k} - \bar{x})^2}.$$

Each sum under a root contains only some of the nonnegative terms of D, so each is at most D. Hence the right side is at most $\sqrt{D}\sqrt{D} = D$, and dividing by D gives $|r_k| \le 1$.



Three barcodes with the same mean spacing: independent random levels (top), Riemann zeros 1000 to 1100 (middle), a perfect lattice (bottom).

Chapter 20

Level Statistics and Random Matrices

↑ Contents

What this chapter is about

In the 1950s, nuclear physicists measuring how heavy nuclei absorb slow neutrons found long lists of sharp resonances. Each resonance is an energy level of a nucleus built from more than two hundred interacting protons and neutrons. Nobody could compute those levels one by one, and nobody can today. Eugene Wigner proposed a change of question. Instead of asking *where* a particular level lies, ask *how the levels are arranged*: how often two neighbors are unusually close together, how often they are unusually far apart. To answer that, he replaced the unknown Hamiltonian by a matrix with random entries and worked out the statistics of its eigenvalues. Those statistics matched experiment, and they turned out to depend on almost nothing except the symmetry of the problem [35].

About two decades later the same statistics appeared somewhere nobody expected. In 1972 the number theorist Hugh Montgomery had found a formula for how pairs of Riemann zeros are correlated, and Freeman Dyson recognized it as the formula for eigenvalues of random complex Hermitian matrices [9]. Andrew Odlyzko's computations then showed the agreement in great detail [8]. This is a large part of the reason physicists take the Hilbert–Pólya idea (that the zeros might be the energy levels of some quantum system) seriously.

In this chapter you will learn how to measure the statistics of any spectrum: how to remove its changing density (unfolding), how to define the distribution of spacings between neighbors, and why uncorrelated levels give the law e^{-s} while random matrices give laws that vanish at $s = 0$ (level repulsion). You will derive the Wigner surmises completely, including the formula $\frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ printed in the paper's Fig. 2, meet Montgomery's pair correlation, reproduce the paper's Fig. 2 with a few lines of Python, and finally see why consecutive fluctuations of the zeros are negatively correlated. **Where this shows up in the paper:** the evidence sentence of Sec. I (Refs. 1–2 of the paper), the dataset check of Sec. II A and Fig. 2, the unit of every rms in the paper, and the explanation in Sec. III (paragraph "Within-height splits") of why smooth interpolation is anticorrelated with the truth.

What you need

- Chapter 8: eigenvalues of a 2×2 matrix from its characteristic polynomial; real symmetric and Hermitian matrices have real eigenvalues; the string (box) eigenvalues $(k\pi/2L)^2$.
- Chapter 17: the counting function $N(T)$, the smooth part $N_{\text{sm}}(t)$ of paper Eq. (1), the fluctuating part $S(t)$, smooth zeros s_n, the mean spacing gap_n, fluctuations measured in mean spacings.
- Chapter 19: probability densities, mean, variance, standard deviation, rms, independence, the Gaussian and exponential distributions, histograms, Pearson correlation, autocorrelation of a sequence, random seeds.
- Chapter 3 (Taylor series of $\log(1-x)$, the geometric series) and Chapter 2 (modulus of a complex number).
- First-year physics and calculus: integration by parts, substitution, the circumference $2\pi r$ of a circle and the surface area $4\pi r^2$ of a sphere.

20.1 Spectra as patterns of points

A spectrum is a list of numbers on a line: the energy levels $E_1 < E_2 < E_3 < \cdots$ of a quantum system, or the heights $t_1 < t_2 < t_3 < \cdots$ of the zeta zeros. You can ask two very different kinds of questions about such a list.

Position questions ask about individual members: where is level 500? Given levels 1–400, where are levels 401–500? This is the paper's central question. Its Fig. 1 is about positions.

Statistical questions ask about the pattern formed by many members, without caring which member is which: what fraction of neighboring pairs are closer than half the average distance? Is a large gap more likely to be followed by a small one? The paper's Fig. 2 is about statistics.

An analogy helps. Imagine three fields seen from above.

- In an orchard, trees stand in exact rows. Every gap between neighbors is the same. Physicists call such a spectrum a **picket fence**.
- In a meadow, wildflower seeds have been scattered by the wind, each one landing independently of all the others. Some land almost on top of each other; some leave wide bare patches. This is a **Poisson** pattern (you met the Poisson process in Chapter 19).
- In a colony of nesting birds, each pair defends a little territory. Nests are never very close together, yet they do not form a perfect grid. This pattern shows **repulsion**.

Figure 20.1 shows three such patterns on a line. The middle one is a stretch of the Riemann zeros, after the rescaling ("unfolding") that Sec. 20.2 explains. Even by eye you can see that the zeros behave like the birds: no pair is very close, and the gaps vary much less than in the Poisson row.

Statistical questions have one great advantage: they can have universal answers. Two completely different systems can share the same statistics while having no individual level in common. That advantage is also a limitation, and it is central to the paper: matching the statistics of a spectrum says nothing about predicting its individual members. We return to this in Sec. 20.8.

Key Ideas

- A spectrum can be studied through *positions* (where is each level?) or through *statistics* (what pattern do many levels form?).
- Three reference patterns: picket fence (equal gaps), Poisson (independent points,

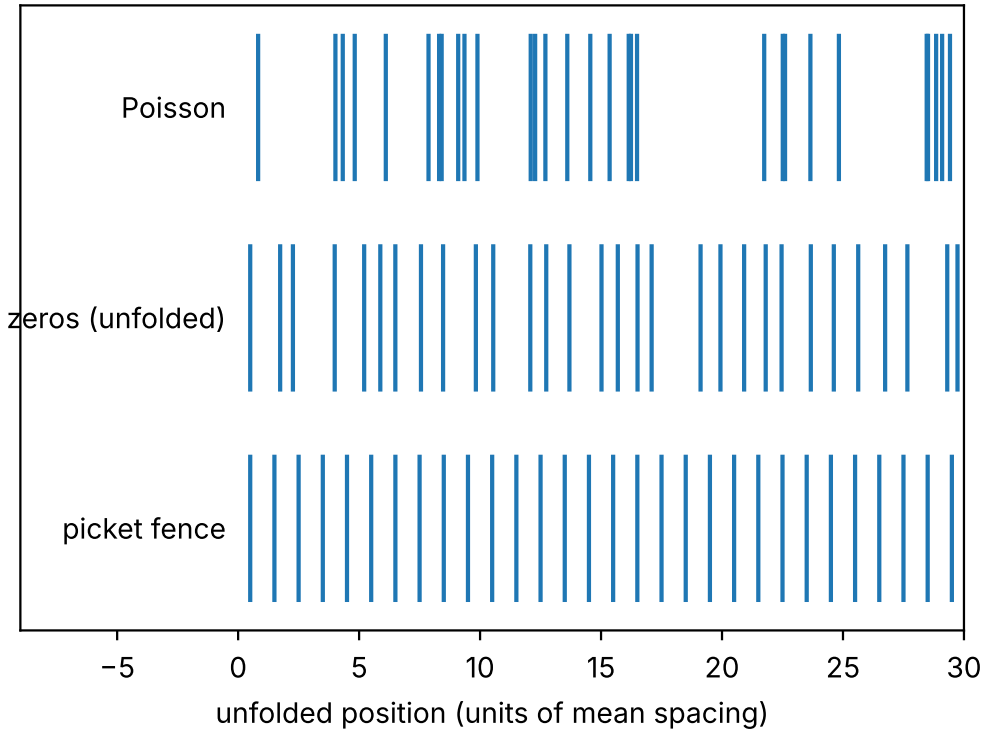


Figure 20.1: Three point patterns with the same average density (one point per unit length). Top: uniformly random independent points (Poisson); note the near-collisions and the wide empty stretches. Middle: the Riemann zeros starting at zero number 1001, unfolded so that their mean spacing is one. Bottom: a picket fence. The zeros avoid close pairs (level repulsion) without being regularly spaced.

frequent near-collisions), and repulsive patterns (close pairs suppressed).

- The Riemann zeros, suitably rescaled, look like a repulsive pattern.

20.2 Unfolding: making the mean spacing equal to one

20.2.1 The problem of a changing density

Before comparing patterns we must deal with an annoying fact: the average distance between neighbors usually changes along the spectrum. Recall from Chapter 17 that the smooth counting function of the zeros is (paper Eq. (1))

$$N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{20.1}$$

and that its derivative gives the local density of zeros. Differentiating with the product rule,

$$\begin{aligned}\frac{\mathrm{d}N_{\mathrm{sm}}}{\mathrm{d}t} &= \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} - \frac{1}{2\pi} \quad \text{(derivative of } \log(t/2\pi) \text{ is } 1/t\text{)} \\ &= \frac{1}{2\pi}\log\frac{t}{2\pi}.\end{aligned} \tag{20.2}$$

The mean spacing is the reciprocal of the density:

$$\mathrm{gap}(t) = \frac{2\pi}{\log(t/2\pi)}. \tag{20.3}$$

(The paper writes gap_n, meaning this quantity evaluated at the smooth zero s_n; recall $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$.) At the first zero, $t_1 = 14.134725$, Eq. (20.3) gives gap ≈ 7.750; at zero number 1000, $t_{1000} = 1419.42248$ (paper Sec. II A), it gives gap ≈ 1.159. The zeros are more than six times denser there.

If you made a histogram of the raw differences $t_{n+1} - t_n$ of the first thousand zeros, you would be mixing spacings of typical size 7 with spacings of typical size 1. The histogram would show the change of density, not the local pattern. We need to rescale.

20.2.2 The smooth counting function and the definition of unfolding

Let $E_1 < E_2 < \cdots$ be any spectrum and let $N(E)$ be its counting function: for an energy E that is not itself a level, $N(E)$ is the number of levels below E. At a level N jumps by one. There we adopt the **midpoint convention** $N(E_n) = n - \frac{1}{2}$, halfway between the value $n-1$ just below the jump and the value n just above it.

A **smooth counting function** $\bar{N}(E)$ is a smooth increasing function that follows the staircase $N(E)$ on average, so that the difference $N(E) - \bar{N}(E)$ stays small and oscillates about zero. For the zeros, $\bar{N} = N_{\mathrm{sm}}$ and the difference is (up to an $\mathcal{O}(1/t)$ term) the function $S(t)$ of Chapter 17.

Unfolding replaces each level E_n by the value of the smooth counting function at that level:

$$\boxed{u_n = \bar{N}(E_n)} \tag{20.4}$$

The numbers u_n are the **unfolded levels**, and $\Delta u_n = u_{n+1} - u_n$ is the n-th **unfolded spacing**. In words: where the levels are dense, $\bar{N}$ rises steeply and the levels are pulled apart; where they are sparse, $\bar{N}$ rises slowly and they are pushed together. Figure 20.2 shows this for the zeros. In the unfolded variable the steps of the staircase all have mean width one.

Pitfall

In this book s_n (with a subscript) always means the n-th smooth zero of Chapter 17. To avoid a clash we write the n-th unfolded spacing as Δu_n. The bare letter s, with no subscript, will be the *variable* of a spacing distribution $p(s)$, exactly as in the caption of the paper's Fig. 2.

20.2.3 Why the unfolded spacings have mean one

Define the fluctuation of level n by

$$\delta_n = u_n - \left(n - \tfrac{1}{2}\right) = \bar{N}(E_n) - N(E_n), \tag{20.5}$$

where the second form uses the midpoint convention $N(E_n) = n - \frac{1}{2}$. Now add up k consecutive unfolded spacings. The sum telescopes (Chapter 3):

$$\begin{aligned} \sum_{j=n}^{n+k-1} \Delta u_j &= u_{n+k} - u_n \\ &= \left(n + k - \tfrac{1}{2} + \delta_{n+k}\right) - \left(n - \tfrac{1}{2} + \delta_n\right) \quad \text{(Eq. (20.5))} \\ &= k + \delta_{n+k} - \delta_n. \end{aligned} \tag{20.6}$$

Dividing by k, the average of k consecutive spacings is

$$\frac{1}{k} \sum_{j=n}^{n+k-1} \Delta u_j = 1 + \frac{\delta_{n+k} - \delta_n}{k}. \tag{20.7}$$

If the fluctuations stay bounded, say $|\delta_n| \le C$ for all n in the range of interest, then the correction term is at most $2C/k$ in size, and it goes to zero as k grows. So *the mean unfolded spacing is one*, whatever the original density was. This is the whole point of unfolding. For the zeros, $|\delta_n|$ is typically below one (Chapter 17). This is an observed typical size, not a proved bound. Taking $C \approx 1$ as a rough size, over ten thousand zeros the mean spacing differs from one by at most about $2C/k = 2 \times 10^{-4}$, that is, by something of order 10^{-4}.

20.2.4 The local-gap version, and the link to the paper's fluctuations

Often one uses a simpler recipe: divide each raw spacing by the local mean spacing,

$$\Delta u_n \approx \frac{E_{n+1} - E_n}{\mathrm{gap}(\bar{E})}, \qquad \bar{E} = \tfrac{1}{2}(E_n + E_{n+1}). \tag{20.8}$$

This is the first-order Taylor approximation of Eq. (20.4): $\bar{N}(E_{n+1}) - \bar{N}(E_n) \approx \bar{N}'(\bar{E})\,(E_{n+1} - E_n)$, and $\bar{N}' = 1/\mathrm{gap}$ by Eq. (20.3).

How large is the neglected part? Write $\Delta = E_{n+1} - E_n$, so that $E_{n+1} = \bar{E} + \Delta/2$ and $E_n = \bar{E} - \Delta/2$. Taylor expansion about $\bar{E}$ gives

$$\bar{N}\left(\bar{E} \pm \tfrac{\Delta}{2}\right) = \bar{N}(\bar{E}) \pm \bar{N}'\frac{\Delta}{2} + \bar{N}''\frac{\Delta^2}{8} \pm \bar{N}'''\frac{\Delta^3}{48} + \cdots, \tag{20.9}$$

with all derivatives evaluated at $\bar{E}$. Subtracting the lower sign from the upper sign, the even-order terms cancel:

$$\bar{N}(E_{n+1}) - \bar{N}(E_n) = \bar{N}'\Delta + \bar{N}'''\frac{\Delta^3}{24} + \cdots. \tag{20.10}$$

Because we expand about the midpoint, the curvature term $\bar{N}''$ drops out, and the relative error of the shortcut is approximately $\bar{N}'''\Delta^2/(24\,\bar{N}')$. For the zeros, $N'_{\rm sm}(t) = \log(t/2\pi)/(2\pi)$ (Eq. (20.2)), $N''_{\rm sm}(t) = 1/(2\pi t)$ and $N'''_{\rm sm}(t) = -1/(2\pi t^2)$, so

$$\text{relative error} \approx -\frac{\Delta^2}{24\,\bar{t}^{\,2}\log(\bar{t}/2\pi)}. \tag{20.11}$$

The shortcut is therefore accurate whenever the spacing Δ is small compared with the height $\bar{t}$, which is true for every zero and becomes better and better higher up.

The same Taylor step links Eq. (20.5) to the fluctuation used throughout the paper, $(t_n - s_n)/\text{gap}_n$ (Chapter 17). Since $N_{\rm sm}(s_n) = n - \frac{1}{2}$,

$$\begin{aligned}\delta_n &= N_{\rm sm}(t_n) - N_{\rm sm}(s_n)\\ &\approx N'_{\rm sm}(s_n)\,(t_n - s_n) \quad \text{(first-order Taylor about } s_n)\\ &= \frac{t_n - s_n}{\text{gap}_n}.\end{aligned} \tag{20.12}$$

So δ_n is the paper's "fluctuation in mean spacings" to first order. When the paper reports an rms of 0.263 on zeros 401–500 for the null prediction $t_n \approx s_n$ (paper Sec. II C and Table V), it is reporting the rms of δ_n over those zeros.

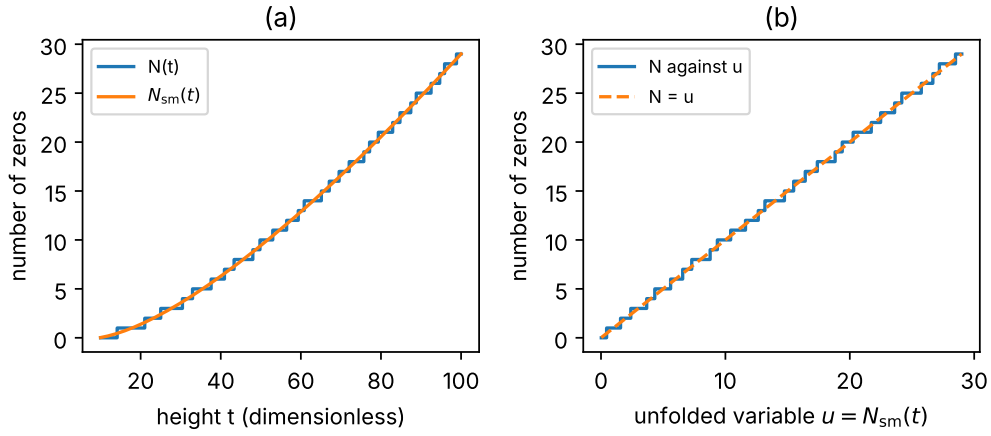


Figure 20.2: Unfolding the zeros. (a) The staircase $N(t)$ (number of zeros up to height t) and the smooth counting function $N_{\rm sm}(t)$ of paper Eq. (1), for $10 \le t \le 100$; the steps get narrower as t grows. (b) The same staircase drawn against the unfolded variable $u = N_{\rm sm}(t)$: now the steps have mean width one everywhere, and the straight line $N = u$ passes through the middle of the steps (because $N_{\rm sm}(s_n) = n - \frac{1}{2}$ and $N(t_n) = n - \frac{1}{2}$ by the midpoint convention).

Sample Problem 20.1: Unfolding the box: a perfect picket fence

The levels of a particle in a box $[-L, L]$ with Dirichlet walls are the eigenvalues $E_k = (k\pi/2L)^2$, $k = 1, 2, \ldots$ of $-y'' = \lambda y$ (Chapter 8). Take $L = 44.883$ (the paper's scale-1 box, running example RE3). Find E_1, E_{500}, the raw spacings at the bottom and near level 500, and the unfolded spacings.

Step 1: raw levels. $\pi/(2L) = \pi/89.766 = 0.0349976$, so $E_1 = (0.0349976)^2 = 1.2248 \times 10^{-3}$ and $E_{500} = 500^2 E_1 = 306.2$.

Step 2: raw spacings. $E_{k+1} - E_k = [(k+1)^2 - k^2]\,E_1 = (2k+1)E_1$. At the bottom, $E_2 - E_1 = 3E_1 = 3.674 \times 10^{-3}$; near level 500, $E_{501} - E_{500} = 1001\,E_1 = 1.226$. The raw spacings grow linearly with k.

Step 3: smooth counting function. Level k lies below E when $k\pi/2L < \sqrt{E}$, that is $k < 2L\sqrt{E}/\pi$. So away from the jumps the staircase is $N(E) = \lfloor 2L\sqrt{E}/\pi \rfloor$ (the floor function rounds down). At a level itself the midpoint convention gives $k - \frac{1}{2}$ instead. Between levels k and $k+1$ the quantity $2L\sqrt{E}/\pi$ runs from k to $k+1$ while $N = k$, so the staircase lies on average half a step below it. The smooth function about which N oscillates is therefore $\bar{N}(E) = 2L\sqrt{E}/\pi - \frac{1}{2}$, and $N - \bar{N}$ stays between $-\frac{1}{2}$ and $+\frac{1}{2}$.

Step 4: unfold. $u_k = \bar{N}(E_k) = \dfrac{2L}{\pi} \cdot \dfrac{k\pi}{2L} - \dfrac{1}{2} = k - \dfrac{1}{2}$, so $\delta_k = u_k - (k - \frac{1}{2}) = 0$ for every k.

Answer:

$$\boxed{\Delta u_k = 1 \text{ for every } k}$$

Although the raw spacings differ by a factor 333.67 between the bottom and level 500, the unfolded spectrum is an exact picket fence: zero variance, no fluctuation.

Sample Problem 20.2: Unfolding the first ten zeros

Using paper Eq. (1), unfold the first ten zeros (running example RE1) and list the unfolded spacings.

Step 1: one value in full. For $t_1 = 14.134725$: $t_1/2\pi = 2.249611$, $\log 2.249611 = 0.810757$, so $N_{\text{sm}}(t_1) = 2.249611 \times 0.810757 - 2.249611 + 0.875 = 1.823889 - 2.249611 + 0.875 = 0.449$.

Step 2: all ten. Repeating Step 1 (or running the listing below) gives Table 20.1.

Step 3: spacings and fluctuations. Subtract consecutive u_n for Δu_n, and subtract $n - \frac{1}{2}$ for δ_n.

Step 4: mean spacing. The nine spacings sum to $u_{10} - u_1 = 9.348 - 0.449 = 8.899$, so their mean is 0.989, close to one as Eq. (20.7) promised.

Answer:

$$\boxed{\Delta u_n = 1.121,\ 0.823,\ 1.278,\ 0.646,\ 1.276,\ 0.972,\ 0.729,\ 1.476,\ 0.578}$$

Notice two things. No spacing is small (the smallest is 0.578), and long and short spacings tend to alternate. Both are features of the pattern you will learn to quantify.

Table 20.1: Unfolding the first ten zeros (RE1) with paper Eq. (1). $u_n = N_{\text{sm}}(t_n)$, $\delta_n = u_n - (n - \frac{1}{2})$, $\Delta u_n = u_{n+1} - u_n$.

n	t_n	u_n	δ_n	Δu_n
1	14.134725	0.449	−0.051	1.121
2	21.022040	1.570	+0.070	0.823
3	25.010858	2.393	−0.107	1.278
4	30.424876	3.671	+0.171	0.646
5	32.935062	4.317	−0.183	1.276
6	37.586178	5.593	+0.093	0.972
7	40.918719	6.565	+0.065	0.729
8	43.327073	7.294	−0.206	1.476
9	48.005151	8.771	+0.271	0.578
10	49.773832	9.348	−0.152	

```
import numpy as np, qmnt
```

```
t = qmnt.zeros(10)                # first ten zeros t_1..t_10
u = qmnt.Nsm(t)                   # unfolded levels u_n = Nsm(t_n)
n = np.arange(1, 11)
print(np.round(u, 3))
print(np.round(u - (n - 0.5), 3))    # fluctuations delta_n
print(np.round(np.diff(u), 3))       # unfolded spacings
```

How good is the local-gap shortcut of Eq. (20.8) here? The raw spacing is $t_2 - t_1 = 6.887$; at the midpoint $\bar{t} = 17.578$ the mean gap is 6.107, so the shortcut gives 1.128 instead of the exact 1.121. The relative difference, $(1.121-1.128)/1.128 = -0.0062$, is what Eq. (20.11) predicts: with $\Delta^2 = 47.43$, $\bar{t}^{\,2} = 309.0$ and $\log(\bar{t}/2\pi) = 1.0288$, it gives $-47.43/(24 \times 309.0 \times 1.0288) = -0.0062$. Higher up the error shrinks rapidly, because Δ shrinks and $\bar{t}$ grows.

Pitfall

Unfolding with the wrong smooth function does not merely shift numbers: it changes the shape of every statistic. If $\bar{N}$ grows too fast in some region, spacings there are inflated and the histogram acquires a spurious tail. Always check that your unfolded spacings have mean one over each part of the spectrum you study, not just overall.

Checkpoint 20.1

A spectrum has mean spacing 0.5 in one region and 2 in another. After correct unfolding, what is the mean spacing in each region?

Answer: One in both regions. Unfolding divides out the local density, so every region has unit mean spacing; only the pattern of fluctuations about that mean remains.

Key Ideas

- Unfolding $u_n = \bar{N}(E_n)$ with a smooth counting function $\bar{N}$ removes the changing density; unfolded spacings have mean one (Eq. (20.7)).
- For the zeros, $\bar{N} = N_{\mathrm{sm}}$ of paper Eq. (1), and the mean gap is $2\pi/\log(t/2\pi)$ (Eq. (20.3)).
- The fluctuation $\delta_n = u_n - (n - \frac{1}{2})$ equals the paper's $(t_n - s_n)/\mathrm{gap}_n$ to first order (Eq. (20.12)). "Units of mean spacing" means unfolded units.

20.3 The nearest-neighbor spacing distribution

★ Did you know? *Wigner's Inspired Guess*

In 1956 Eugene Wigner proposed that spacings s between neighboring energy levels of heavy nuclei follow $p(s) = \frac{\pi s}{2} e^{-\pi s^2/4}$, a formula derived from mere 2×2 random matrices. It vanishes at $s = 0$, so levels repel each other. A companion formula from 2×2 complex Hermitian matrices, with s^2 in place of s, describes Riemann zero spacings remarkably well.

Once a spectrum is unfolded, the simplest statistic is the distribution of the gaps between neighbors.

Definition. The **nearest-neighbor spacing distribution** is the probability density $p(s)$ such that, for a spacing chosen at random from a long unfolded spectrum,

$$\Pr\left(s < \Delta u \le s + \mathrm{d}s\right) = p(s)\,\mathrm{d}s. \tag{20.13}$$

Because it is a probability density on $s \geq 0$ and because unfolding forces the mean spacing to be one, every spacing distribution in this chapter obeys two conditions:

$$\int_0^\infty p(s)\,\mathrm{d}s = 1, \qquad \int_0^\infty s\,p(s)\,\mathrm{d}s = 1. \tag{20.14}$$

The first is normalization (Chapter 19); the second is the unit mean. These two conditions will fix all the constants in the Wigner surmises. Keep in mind which laws are exact. The Poisson law will be derived exactly for independent levels, and the Wigner surmises exactly for 2×2 random matrices. For large random matrices, and for the zeros, the surmises are only close approximations to the true spacing distributions (see the box on large matrices in Sec. 20.6).

Estimating $p(s)$ from data. Given K unfolded spacings, divide the s axis into bins of width b. If c_j spacings fall in bin j, the histogram estimate of p at the center of that bin is

$$\hat{p}_j = \frac{c_j}{K\,b}, \tag{20.15}$$

the fraction of spacings in the bin divided by the bin width, so that $\sum_j \hat{p}_j\,b = 1$, the discrete version of the first condition. This is what `density=True` does in a Matplotlib histogram, and it is how the bars of paper Fig. 2 are scaled.

Besides the mean, two summaries of $p(s)$ are useful: its variance $\int (s-1)^2 p(s)\,\mathrm{d}s = \int s^2 p\,\mathrm{d}s - 1$, which measures how irregular the spacings are, and its behavior near $s = 0$, which measures how often levels nearly collide. A picket fence has every spacing equal to one, so its variance is zero.

Checkpoint 20.2

Ten thousand unfolded spacings are binned with width $b = 0.1$, and 1100 of them fall in the bin $0.9 < s \leq 1.0$. What is the histogram estimate of p in that bin?

Answer: $\hat{p} = 1100/(10000 \times 0.1) = 1.1$. Note that a density can exceed one; only its integral must equal one.

Key Ideas

- $p(s)\,\mathrm{d}s$ is the probability that a random unfolded spacing lies in $[s, s + \mathrm{d}s]$.
- Every unfolded spacing density satisfies $\int p = 1$ and $\int sp = 1$ (Eq. (20.14)).
- Histograms estimate p by count/($K \times$bin width).

20.4 Uncorrelated levels: the Poisson law

What would the spacings look like if levels knew nothing about each other? Model this by throwing K points independently and uniformly onto an interval of length K (we write K because N already denotes the counting function), so the mean density is one point per unit length (unfolded from the start).

20.4.1 Derivation from the finite model

Pick one of the points; call its position y, and suppose y is not within distance s of the right end. This costs nothing in the limit. A reference point chosen at random lies within s of the right end with probability s/K (the length of that strip divided by the total length), and for fixed s this tends to zero as $K \to \infty$. Because all K points were thrown independently, knowing where the reference point landed tells us nothing about the other $K - 1$: each of them is still

uniform on the whole interval. The nearest neighbor to the right is farther than s exactly when none of the other $K-1$ points lands in the interval $(y, y+s]$.

One other point. Because it is uniform on a length K, it lands in an interval of length s with probability s/K, and misses it with probability $1-s/K$.

All other points. They are independent, so the probabilities multiply (Chapter 19):

$$Q_K(s) \equiv \Pr(\text{spacing} > s) = \left(1 - \frac{s}{K}\right)^{K-1}. \tag{20.16}$$

The limit $K \to \infty$. Take logarithms and use the Taylor series $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$, valid for $|x| < 1$ (Chapter 3), with $x = s/K$:

$$\begin{aligned} \log Q_K(s) &= (K-1)\log\left(1 - \frac{s}{K}\right) \\ &= (K-1)\left(-\frac{s}{K} - \frac{s^2}{2K^2} - \cdots\right) \\ &= -s + \frac{s}{K} - \frac{(K-1)s^2}{2K^2} - \cdots . \end{aligned} \tag{20.17}$$

Each term after $-s$ carries at least one factor $1/K$. There are infinitely many of them, so we bound their sum rather than treating them one at a time. For $0 \le x \le \frac{1}{2}$,

$$|\log(1-x) + x| = \frac{x^2}{2} + \frac{x^3}{3} + \cdots \le \frac{x^2}{2}\left(1 + x + x^2 + \cdots\right) = \frac{x^2}{2(1-x)} \le x^2. \tag{20.18}$$

The first inequality replaces each coefficient $1/j$ ($j \ge 3$) by the larger $1/2$. The next step sums the geometric series (Chapter 3), and the last uses $1 - x \ge \frac{1}{2}$. Now write $\log Q_K(s) = (K-1)\,[\log(1-x) + x] - (K-1)x$ and note that $(K-1)x = s - s/K$. Then, for $K \ge 2s$ (so that $x \le \frac{1}{2}$),

$$|\log Q_K(s) + s| \le (K-1)x^2 + \frac{s}{K} \le \frac{s^2}{K} + \frac{s}{K} \to 0 \qquad (K \to \infty). \tag{20.19}$$

Hence $\log Q_K(s) \to -s$, and

$$Q(s) = e^{-s}. \tag{20.20}$$

From survival to density. The probability that the spacing lies between s and $s + \mathrm{d}s$ is the probability that it exceeds s minus the probability that it exceeds $s + \mathrm{d}s$:

$$p(s)\,\mathrm{d}s = Q(s) - Q(s+\mathrm{d}s) = -\frac{\mathrm{d}Q}{\mathrm{d}s}\,\mathrm{d}s \quad \Longrightarrow \quad \boxed{p_{\mathrm{P}}(s) = e^{-s}} \tag{20.21}$$

This is the exponential distribution that Chapter 19 introduced for gaps of a Poisson process, and the curve labeled "Poisson" in paper Fig. 2.

20.4.2 Check of the two conditions, and the variance

$\int_0^\infty e^{-s}\mathrm{d}s = 1$. By parts, $\int_0^\infty s e^{-s}\mathrm{d}s = [-se^{-s}]_0^\infty + \int_0^\infty e^{-s}\mathrm{d}s = 0 + 1 = 1$. Again by parts, $\int_0^\infty s^2 e^{-s}\mathrm{d}s = [-s^2 e^{-s}]_0^\infty + 2\int_0^\infty s e^{-s}\mathrm{d}s = 2$. So the variance is $2 - 1 = 1$: Poisson spacings are very irregular, with standard deviation equal to the mean.

20.4.3 The most probable spacing is zero

The density e^{-s} is largest at $s = 0$. In an uncorrelated spectrum, near-coincidences are the single most common kind of spacing. This is **clumping**: independence does not spread points out evenly, it produces clusters and voids. (Raindrops on a dry pavement show the same thing: some spots get hit twice before others are hit at all.)

Pitfall

"Random" does not mean "evenly spread." Many people, asked to draw random dots, space them too evenly. A truly uncorrelated pattern has more close pairs than any other kind of spacing. An evenly spread pattern is evidence of correlation (in this case repulsion), not of randomness.

Sample Problem 20.3: How rare are small and large gaps?

For an unfolded spectrum, compare the probability that a spacing is below 0.1, and the probability that it exceeds 2, under the Poisson law and under the GUE surmise $p_{\mathrm{GUE}}(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ (derived in Sec. 20.6).

Step 1: Poisson, small gaps. $\Pr(s < 0.1) = 1 - Q(0.1) = 1 - e^{-0.1} = 0.0952$.

Step 2: GUE, small gaps. With $B = 4/\pi$, expand $e^{-Bs^2} = 1 - Bs^2 + \cdots$ and integrate term by term: $\int_0^{0.1} s^2(1 - Bs^2)\,\mathrm{d}s = \frac{(0.1)^3}{3} - B\frac{(0.1)^5}{5} = 3.3333 \times 10^{-4} - 2.546 \times 10^{-6}$. The next term is about 10^{-8}. Multiplying by $32/\pi^2 = 3.2423$ gives $\Pr(s < 0.1) = 1.073 \times 10^{-3}$.

Step 3: ratio. $0.0952/0.00107 \approx 89$: small gaps are about 89 times rarer under GUE.

Step 4: large gaps, Poisson. $\Pr(s > 2) = Q(2) = e^{-2} = 0.135$.

Step 5: large gaps, GUE. Write $A = 32/\pi^2$ and $B = 4/\pi$, and note that $A = 2B^2$ (because $2 \cdot 16/\pi^2 = 32/\pi^2$). Then $\Pr(s > 2) = A\int_2^\infty s \cdot s e^{-Bs^2}\,\mathrm{d}s$. Integrate by parts with $f = s$ and $\mathrm{d}g = s e^{-Bs^2}\mathrm{d}s$, so $g = -e^{-Bs^2}/(2B)$:

$$\int_2^\infty s^2 e^{-Bs^2}\mathrm{d}s = \left[-\frac{s e^{-Bs^2}}{2B}\right]_2^\infty + \frac{1}{2B}\int_2^\infty e^{-Bs^2}\mathrm{d}s = \frac{e^{-4B}}{B} + \frac{1}{2B}\int_2^\infty e^{-Bs^2}\mathrm{d}s.$$

In the last integral substitute $w = \sqrt{B}\,s$: it becomes $\frac{1}{\sqrt{B}}\int_{2\sqrt{B}}^\infty e^{-w^2}\mathrm{d}w = \frac{\sqrt{\pi}}{2\sqrt{B}}\operatorname{erfc}(2\sqrt{B})$, where $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-w^2}\mathrm{d}w$ is the complementary error function, a tabulated function (`scipy.special.erfc`). Multiplying by $A = 2B^2$:

$$\Pr(s > 2) = 2Be^{-4B} + \frac{\sqrt{\pi B}}{2}\operatorname{erfc}(2\sqrt{B}) = \frac{8}{\pi}e^{-16/\pi} + \operatorname{erfc}\left(\frac{4}{\sqrt{\pi}}\right),$$

using $\sqrt{\pi B} = 2$. Numerically this is $0.0156 + 0.0014 = 0.017$, about 8 times rarer than Poisson.

Answer:

$$\boxed{\Pr(s < 0.1):\ 0.0952\ \text{(Poisson) vs}\ 1.07 \times 10^{-3}\ \text{(GUE)};\quad \Pr(s > 2):\ 0.135\ \text{vs}\ 0.017}$$

A repulsive spectrum suppresses both extremes: it is more regular than a Poisson one at both ends.

Key Ideas

- Independent levels give $p_{\mathrm{P}}(s) = e^{-s}$ (Eq. (20.21)), derived from $(1 - s/N)^{N-1} \to e^{-s}$.
- Poisson spacings have mean 1, variance 1, and most probable value 0: uncorrelated levels clump.

20.5 Level repulsion and avoided crossings

Why would eigenvalues of a matrix avoid each other? The answer is already visible in 2×2 matrices.

20.5.1 Eigenvalues of a real symmetric 2×2 matrix

Take

$$H = \begin{pmatrix} a & b \\ b & d \end{pmatrix}, \qquad a, b, d \text{ real.} \tag{20.22}$$

Its characteristic polynomial (Chapter 8) is

$$\begin{aligned} \det(H - \lambda I) &= (a - \lambda)(d - \lambda) - b^2 \\ &= \lambda^2 - (a + d)\lambda + ad - b^2. \end{aligned} \tag{20.23}$$

Setting it to zero and using the quadratic formula,

$$\begin{aligned} \lambda_{\pm} &= \frac{(a + d) \pm \sqrt{(a + d)^2 - 4(ad - b^2)}}{2} \\ &= \frac{a + d}{2} \pm \frac{1}{2}\sqrt{a^2 + 2ad + d^2 - 4ad + 4b^2} \\ &= \frac{a + d}{2} \pm \frac{1}{2}\sqrt{(a - d)^2 + 4b^2}. \end{aligned} \tag{20.24}$$

The spacing is therefore

$$\lambda_+ - \lambda_- = \sqrt{(a - d)^2 + 4b^2}. \tag{20.25}$$

The two eigenvalues coincide only if $(a - d)^2 + 4b^2 = 0$, and since both terms are squares of real numbers this requires *two* conditions at once: $a = d$ and $b = 0$.

20.5.2 Avoided crossings

Let the matrix depend on a control parameter τ (think of a field strength you are turning up; we avoid λ, which denotes the eigenvalues), with a fixed coupling g:

$$H(\tau) = \begin{pmatrix} \tau & g \\ g & -\tau \end{pmatrix}. \tag{20.26}$$

By Eq. (20.24) with $a = \tau$, $d = -\tau$, $b = g$, the levels are $\pm\sqrt{\tau^2 + g^2}$. Without coupling ($g = 0$) the levels $\pm\tau$ cross at $\tau = 0$. With any nonzero coupling they never meet: the smallest separation is $2|g|$, reached at $\tau = 0$ (Fig. 20.3). This is an **avoided crossing**. Mechanically, it is like two pendulums joined by a weak spring: when their natural frequencies are tuned past each other, the frequencies of the coupled system approach and then turn away.

Sample Problem 20.4: Eigenvalues, spacing, and an avoided crossing

(a) Find the eigenvalues and spacing of $H = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1.2 \end{pmatrix}$. (b) For $H(\tau)$ above with $g = 0.05$, find the spacing at $\tau = 0$ and $\tau = 0.2$, and compare with the uncoupled spacing $2|\tau|$.

Step 1: In (a), $a = 1$, $d = 1.2$, $b = 0.3$. The center is $(a + d)/2 = 1.1$.

Step 2: Spacing from Eq. (20.25): $\sqrt{(1 - 1.2)^2 + 4(0.3)^2} = \sqrt{0.04 + 0.36} = \sqrt{0.40} = 0.6325$.

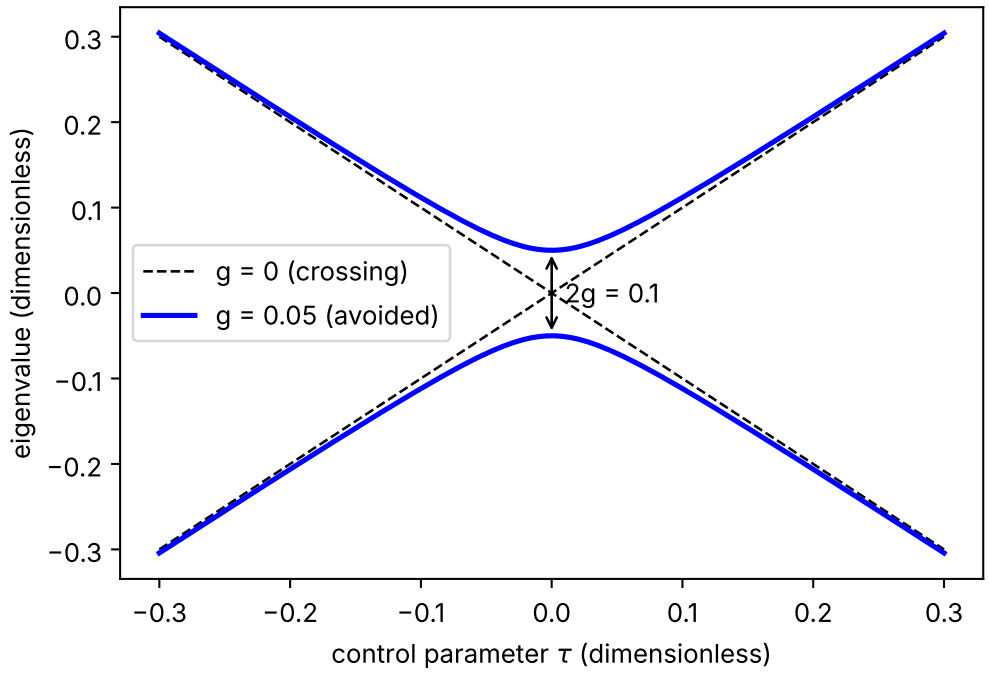


Figure 20.3: Eigenvalues of $H(\tau) = \begin{pmatrix} \tau & g \\ g & -\tau \end{pmatrix}$ as functions of the dimensionless control parameter τ. Dashed: $g = 0$, the levels $\pm\tau$ cross. Solid: $g = 0.05$, the levels $\pm\sqrt{\tau^2 + g^2}$ never meet; their minimum separation is $2g = 0.1$ at $\tau = 0$.

Step 3: Eigenvalues 1.1 ± 0.3162, i.e. 0.7838 and 1.4162. Check: their sum 2.2 equals the trace $a + d$, and their product $1.1^2 - 0.3162^2 = 1.11$ equals $\det H = 1.2 - 0.09 = 1.11$.
Step 4: In (b), the spacing is $2\sqrt{\tau^2 + g^2}$. At $\tau = 0$: $2g = 0.1$. At $\tau = 0.2$: $2\sqrt{0.04 + 0.0025} = 0.4123$, compared with 0.4 without coupling.
Answer:

$\lambda_\pm = 0.7838,\ 1.4162;$ spacing $0.6325;$ avoided crossing: $0.1\ (\tau = 0),\ 0.4123\ (\tau = 0.2)$

Far from the crossing the coupling hardly matters; near it, it dominates.

20.5.3 Counting conditions

Here is the key idea behind level repulsion. A real symmetric 2×2 matrix has three free numbers (a, b, d), but a degeneracy needs two of them to be tuned (the combinations $a - d$ and b). If the matrix is "generic" (its entries are random and smoothly distributed), hitting two conditions by accident is very unlikely, much less likely than hitting one. For a complex Hermitian matrix the off-diagonal entry b is complex, with a real and an imaginary part, so a degeneracy needs *three* conditions. In the next section we turn this counting into exact formulas.

By contrast, consider a *diagonal* random matrix ($b = 0$ always). Then the spacing is $|a - d|$, a single condition, and nothing prevents near-coincidences: the eigenvalues are just two independent numbers. Repulsion is caused by the off-diagonal mixing.

Checkpoint 20.3

Two levels belong to two completely separate systems (for example, two boxes far apart), so their Hamiltonian is block diagonal and no matrix element connects them. Do these two levels repel?

Answer: No. Without a coupling matrix element there is no avoided crossing; their spacing behaves like the difference of two independent numbers. When the spectra of unrelated systems are superposed, close pairs are therefore no longer suppressed. With many superposed spectra the statistics approach Poisson; this is a known result that we state without proof (see [35]).

Key Ideas

- For a real symmetric 2×2 matrix, the spacing is $\sqrt{(a-d)^2 + 4b^2}$ (Eq. (20.25)); a degeneracy needs two conditions.
- Coupling turns crossings into avoided crossings with minimum gap $2|g|$.
- Repulsion comes from off-diagonal mixing; levels of uncoupled systems do not repel.

20.6 Random 2×2 matrices and the Wigner surmises

20.6.1 A toolkit of Gaussian integrals

We will need the integrals

$$I_k(B) = \int_0^\infty s^k e^{-Bs^2}\,\mathrm{d}s, \qquad B > 0, \quad k = 0, 1, 2, \ldots \tag{20.27}$$

I_0. Recall from Chapter 19 that the Gaussian density integrates to one; equivalently $G \equiv \int_{-\infty}^{\infty} e^{-Bx^2}\mathrm{d}x = \sqrt{\pi/B}$. For completeness, here is why. Write G^2 as a product of two copies, one in x and one in y. As Riemann sums, $\left(\sum_i e^{-Bx_i^2}\Delta x\right)\left(\sum_j e^{-By_j^2}\Delta y\right) = \sum_{i,j} e^{-B(x_i^2+y_j^2)}\Delta x\,\Delta y$, a sum over little squares tiling the plane. The integrand depends only on the distance $r = \sqrt{x^2+y^2}$ from the origin, so group the squares into thin rings of radius r and width $\mathrm{d}r$, each of area $2\pi r\,\mathrm{d}r$:

$$G^2 = \int_0^\infty e^{-Br^2}\,2\pi r\,\mathrm{d}r = 2\pi\left[-\frac{e^{-Br^2}}{2B}\right]_0^\infty = \frac{\pi}{B}. \tag{20.28}$$

So $G = \sqrt{\pi/B}$, and because the integrand is even, $I_0 = \frac{1}{2}\sqrt{\pi/B}$.

I_1. Substituting directly, $I_1 = \left[-\dfrac{e^{-Bs^2}}{2B}\right]_0^\infty = \dfrac{1}{2B}$.

Recursion. For $k \geq 0$, integrate $I_{k+2} = \int_0^\infty s^{k+1}\cdot s e^{-Bs^2}\mathrm{d}s$ by parts with $f = s^{k+1}$ and $\mathrm{d}g = s e^{-Bs^2}\mathrm{d}s$, so $g = -e^{-Bs^2}/(2B)$:

$$I_{k+2} = \left[-\frac{s^{k+1}e^{-Bs^2}}{2B}\right]_0^\infty + \frac{k+1}{2B}\int_0^\infty s^k e^{-Bs^2}\mathrm{d}s = \frac{k+1}{2B}I_k. \tag{20.29}$$

The boundary term vanishes at $s = 0$ because $s^{k+1} = 0$ there, and at infinity because the exponential decays faster than any power grows. Hence

$$I_2 = \frac{\sqrt{\pi}}{4B^{3/2}}, \qquad I_3 = \frac{1}{2B^2}, \qquad I_4 = \frac{3\sqrt{\pi}}{8B^{5/2}}. \tag{20.30}$$

20.6.2 The real symmetric case: a spacing density proportional to s

Any real symmetric 2×2 matrix can be written as

$$H = \begin{pmatrix} c + \frac{x}{2} & \frac{y}{2} \\ \frac{y}{2} & c - \frac{x}{2} \end{pmatrix}, \qquad c = \frac{a+d}{2},\ x = a - d,\ y = 2b. \tag{20.31}$$

By Eq. (20.25), its spacing is

$$s = \sqrt{x^2 + y^2}, \tag{20.32}$$

the distance of the point (x, y) from the origin of a plane. The center c shifts both levels together and does not affect s.

Now make the matrix random: let x and y be independent Gaussian random variables with mean zero and the same variance σ^2. Independence means the probability of landing in a small rectangle $\mathrm{d}x\,\mathrm{d}y$ is the product of the two one-dimensional probabilities:

$$\frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/2\sigma^2} \mathrm{d}x \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-y^2/2\sigma^2} \mathrm{d}y = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}\, \mathrm{d}x\, \mathrm{d}y. \tag{20.33}$$

This joint density depends only on the distance s from the origin. So the probability that the spacing lies between s and $s + \mathrm{d}s$ is the density on the ring of radius s times the area $2\pi s\,\mathrm{d}s$ of that ring:

$$p(s)\,\mathrm{d}s = \frac{1}{2\pi\sigma^2} e^{-s^2/2\sigma^2} \cdot 2\pi s\,\mathrm{d}s \quad \Longrightarrow \quad p(s) = \frac{s}{\sigma^2}\, e^{-s^2/2\sigma^2}. \tag{20.34}$$

The factor s is the circumference of the ring. That is level repulsion in its purest form: small spacings need (x, y) to fall in a small disk around the origin, and a small disk has tiny area.

Normalization check. With $B = 1/(2\sigma^2)$, $\int p\,\mathrm{d}s = \frac{1}{\sigma^2} I_1 = \frac{1}{\sigma^2} \cdot \frac{1}{2B} = \frac{2\sigma^2}{2\sigma^2} = 1$.

Mean spacing. $\langle s \rangle = \frac{1}{\sigma^2} I_2 = \frac{1}{\sigma^2} \cdot \frac{\sqrt{\pi}\,(2\sigma^2)^{3/2}}{4} = \frac{\sqrt{\pi}\, 2\sqrt{2}\,\sigma^3}{4\sigma^2} = \sigma\sqrt{\frac{\pi}{2}}$.

20.6.3 Unfolding the 2×2 ensemble: rescaling a density

A 2×2 matrix has only one spacing, so "unfolding" means dividing by the ensemble mean: $s' = s/\langle s \rangle$. How does a density change under a rescaling $s' = s/\ell$? The event "s' in $[s', s' + \mathrm{d}s']$" is the same event as "s in $[\ell s', \ell s' + \ell\,\mathrm{d}s']$", so $p'(s')\,\mathrm{d}s' = p(\ell s')\,\ell\,\mathrm{d}s'$, i.e.

$$p'(s') = \ell\, p(\ell s'). \tag{20.35}$$

With $\ell = \langle s \rangle = \sigma\sqrt{\pi/2}$ and Eq. (20.34):

$$\begin{aligned} p'(s') &= \ell \cdot \frac{\ell s'}{\sigma^2} e^{-\ell^2 s'^2/2\sigma^2} \\ &= \frac{\ell^2}{\sigma^2}\, s'\, e^{-(\ell^2/2\sigma^2)s'^2} \\ &= \frac{\pi}{2}\, s'\, e^{-\pi s'^2/4} \qquad (\text{using } \ell^2/\sigma^2 = \pi/2). \end{aligned} \tag{20.36}$$

The variance σ^2 has dropped out completely. Dropping the prime:

$$\boxed{p_{\mathrm{GOE}}(s) = \frac{\pi}{2}\, s\, e^{-\pi s^2/4}} \tag{20.37}$$

This is the **Wigner surmise** for real symmetric matrices.

Cross-check by the two conditions. Suppose we only knew that $p(s) = A s\, e^{-Bs^2}$. Then Eq. (20.14) gives $AI_1 = A/(2B) = 1$, so $A = 2B$, and $AI_2 = 2B\sqrt{\pi}/(4B^{3/2}) = \sqrt{\pi}/(2\sqrt{B}) = 1$, so $\sqrt{B} = \sqrt{\pi}/2$, $B = \pi/4$, $A = \pi/2$, the same result. The two conditions determine A and B uniquely.

20.6.4 The complex Hermitian case: a spacing density proportional to s^2

A complex Hermitian 2×2 matrix (Chapter 8) has real diagonal entries and complex conjugate off-diagonal entries. Write it as

$$H=\begin{pmatrix} c+\frac{x}{2} & \frac{y-\mathrm{i}z}{2} \\ \frac{y+\mathrm{i}z}{2} & c-\frac{x}{2}\end{pmatrix}, \qquad x,y,z,c \text{ real}. \tag{20.38}$$

Its characteristic polynomial is

$$\begin{aligned}\det(H-\lambda I) &= \left(c-\lambda+\tfrac{x}{2}\right)\left(c-\lambda-\tfrac{x}{2}\right)-\frac{(y-\mathrm{i}z)(y+\mathrm{i}z)}{4}\\ &= (c-\lambda)^2-\frac{x^2}{4}-\frac{y^2+z^2}{4},\end{aligned} \tag{20.39}$$

using $(y-\mathrm{i}z)(y+\mathrm{i}z)=y^2+z^2$ (Chapter 2). Setting it to zero, $\lambda=c\pm\frac{1}{2}\sqrt{x^2+y^2+z^2}$, so the spacing is

$$s=\sqrt{x^2+y^2+z^2}, \tag{20.40}$$

the length of the vector (x,y,z) in ordinary three-dimensional space. A degeneracy needs three conditions, $x=y=z=0$.

Let x,y,z be independent Gaussians with mean zero and variance σ^2. The joint density is $(2\pi\sigma^2)^{-3/2}e^{-(x^2+y^2+z^2)/2\sigma^2}$, constant on spheres. A thin spherical shell of radius s and thickness $\mathrm{d}s$ has volume $4\pi s^2\,\mathrm{d}s$, so

$$p(s)=\frac{4\pi s^2}{(2\pi\sigma^2)^{3/2}}e^{-s^2/2\sigma^2}=\sqrt{\frac{2}{\pi}}\,\frac{s^2}{\sigma^3}\,e^{-s^2/2\sigma^2}, \tag{20.41}$$

where $4\pi/(2\pi)^{3/2}=2/\sqrt{2\pi}=\sqrt{2/\pi}$. (Physicists know this shape as the Maxwell speed distribution, for the same geometric reason.)

Rescaling to unit mean. By Eq. (20.35), rescaling a density of the form $s^2e^{-Bs^2}$ produces another density of the same form, $As^2e^{-B's^2}$. So the unfolded density is $As^2e^{-Bs^2}$ with A and B fixed by the two conditions:

$$\int_0^\infty As^2e^{-Bs^2}\mathrm{d}s = AI_2=\frac{A\sqrt{\pi}}{4B^{3/2}}=1, \tag{20.42}$$

$$\int_0^\infty As^3e^{-Bs^2}\mathrm{d}s = AI_3=\frac{A}{2B^2}=1. \tag{20.43}$$

From Eq. (20.43), $A=2B^2$. Substituting into Eq. (20.42):

$$\frac{2B^2\sqrt{\pi}}{4B^{3/2}}=\frac{\sqrt{\pi}}{2}B^{1/2}=1 \quad\Longrightarrow\quad B^{1/2}=\frac{2}{\sqrt{\pi}} \quad\Longrightarrow\quad B=\frac{4}{\pi}. \tag{20.44}$$

Then $A=2B^2=2\cdot\dfrac{16}{\pi^2}=\dfrac{32}{\pi^2}$. Therefore

$$\boxed{\,p_{\mathrm{GUE}}(s)=\frac{32}{\pi^2}\,s^2\,e^{-4s^2/\pi}\,} \tag{20.45}$$

This is exactly the curve in the caption of paper Fig. 2. Figure 20.4 compares it with the GOE surmise and the Poisson law.

20.6.5 Why these ensembles, and their names

You may object that we *chose* x and y (or x, y, z) to have equal variances. Here is the natural choice that produces it. Let the entries a, d of a real symmetric matrix be independent Gaussians of variance σ_0^2, and let b have variance $\sigma_0^2/2$. Then the joint density is proportional to

$$e^{-a^2/2\sigma_0^2}\, e^{-d^2/2\sigma_0^2}\, e^{-b^2/\sigma_0^2} = e^{-(a^2+d^2+2b^2)/2\sigma_0^2} = e^{-\mathrm{tr}(H^2)/2\sigma_0^2}, \tag{20.46}$$

because the **trace** (sum of diagonal entries) of H^2 is $a^2+b^2+b^2+d^2$. (Recall the matrix product from Chapter 7: $(XY)_{ik} = \sum_j X_{ij}Y_{jk}$; for H^2 the diagonal entries are a^2+b^2 and b^2+d^2. A rotation of the plane is a real matrix O with $O^\top O = I$, and $OHO^\top$ is the same linear map written in the rotated basis, Chapter 7.) The trace has the property $\mathrm{tr}(XY) = \sum_i \sum_j X_{ij}Y_{ji} = \sum_j \sum_i Y_{ji}X_{ij} = \mathrm{tr}(YX)$, so for any rotation O (with $O^\top O = I$), $\mathrm{tr}\big((OHO^\top)^2\big) = \mathrm{tr}(OH^2O^\top) = \mathrm{tr}(H^2O^\top O) = \mathrm{tr}(H^2)$. Probabilities are densities times volumes, so one more fact is needed: the volume element $\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}d$ must also be unchanged. It is. The map $H \to OHO^\top$ changes the triple $(a, \sqrt{2}\,b, d)$ linearly (each new entry is a fixed combination of the old ones), and it preserves $a^2 + 2b^2 + d^2 = \mathrm{tr}(H^2)$, the squared length of that triple in a three-dimensional space. A linear map that preserves all lengths is a rotation or a reflection, and these preserve volumes. The ensemble therefore has no preferred basis: rotating every matrix leaves the probabilities unchanged. It is called the **Gaussian Orthogonal Ensemble (GOE)**.

Now convert to the variables of Eq. (20.31). Since $a = c + x/2$ and $d = c - x/2$,

$$a^2 + d^2 = \left(c + \tfrac{x}{2}\right)^2 + \left(c - \tfrac{x}{2}\right)^2 = 2c^2 + \tfrac{x^2}{2}, \tag{20.47}$$

so the density in (c, x) is proportional to $e^{-c^2/\sigma_0^2}\, e^{-x^2/4\sigma_0^2}$. (The change of variables is the linear map $a = c + \frac{1}{2}x$, $d = c - \frac{1}{2}x$, with matrix $J = \begin{pmatrix} 1 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}$. A linear map with matrix J sends a small rectangle of area $\mathrm{d}c\,\mathrm{d}x$ to a parallelogram of area $|\det J|\,\mathrm{d}c\,\mathrm{d}x$; this is the meaning of the 2×2 determinant as an area factor (Chapter 7). A probability is a density times an area, so a density $f(a, d)$ becomes $f\,|\det J|$ when expressed in (c, x). Here $\det J = 1 \cdot (-\frac{1}{2}) - \frac{1}{2} \cdot 1 = -1$, so $|\det J| = 1$, and the density only needs re-expressing in the new variables.) The density factorizes, so x is a Gaussian of variance $2\sigma_0^2$, independent of c. And $y = 2b$ has variance $4 \cdot \sigma_0^2/2 = 2\sigma_0^2$ (scaling rule $\mathrm{Var}(kX) = k^2\mathrm{Var}(X)$, Chapter 19). So x and y are independent with equal variance $\sigma^2 = 2\sigma_0^2$, exactly our assumption.

The complex case. Now let b denote the lower-left entry $(y + \mathrm{i}z)/2$ of Eq. (20.38). It is complex, $b = b_r + \mathrm{i}b_i$, with b_r, b_i independent Gaussians each of variance $\sigma_0^2/2$, and the upper-right entry is its conjugate $b_r - \mathrm{i}b_i$. Comparing with Eq. (20.38), $y = 2b_r$ and $z = 2b_i$, so each has variance $4 \cdot \sigma_0^2/2 = 2\sigma_0^2$. The variable $x = a - d$ has variance $2\sigma_0^2$ and is independent of c, exactly as above. Finally x, y, z are independent of each other because they are built separately from the independent numbers (a, d), b_r and b_i. So again all three components have the common variance $\sigma^2 = 2\sigma_0^2$. The joint density is proportional to

$$e^{-(a^2+d^2)/2\sigma_0^2}\, e^{-(b_r^2+b_i^2)/\sigma_0^2} = e^{-(a^2+d^2+2|b|^2)/2\sigma_0^2} = e^{-\mathrm{tr}(H^2)/2\sigma_0^2},$$

because the diagonal entries of H^2 are now $a^2+|b|^2$ and $|b|^2+d^2$. This is the **Gaussian Unitary Ensemble (GUE)**. Its density is unchanged under unitary changes of basis $H \to UHU^\dagger$ with $U^\dagger U = I$, by the same trace argument with $O^\top$ replaced by $U^\dagger$ (Problem 5 asks you to write out the trace step; the corresponding statement about volumes for complex entries we take on trust from [35]).

Some physics background, which we state rather than derive. A Hamiltonian is the Hermitian matrix or operator whose eigenvalues are the energy levels of a quantum system (Chapter 10 develops this). Time reversal means running the motion backward; for a wavefunction it

amounts to complex conjugation. Real symmetric Hamiltonians arise for systems that are invariant under time reversal (no magnetic field, for instance), and complex Hermitian ones for systems where time-reversal symmetry is broken [35]. The exponent of s near zero is called the **Dyson index** β: $\beta = 1$ for GOE, $\beta = 2$ for GUE, and $\beta = 4$ for a third ("symplectic") ensemble treated in Problem 11. Poisson corresponds to $\beta = 0$. The rule behind all of them is the counting of Sec. 20.5: if a degeneracy needs D independent conditions, then $p(s) \propto s^{D-1}$ near zero, with $D = 2$ (GOE), 3 (GUE), 5 (symplectic). The first two follow from the counting above: one condition from the diagonal ($a = d$) plus one (real b) or two (complex b) from the off-diagonal entry. The symplectic value we quote rather than derive: in that ensemble the off-diagonal element of the 2×2 (quaternion) matrix has four real components, so a degeneracy needs $1 + 4 = 5$ conditions [35].

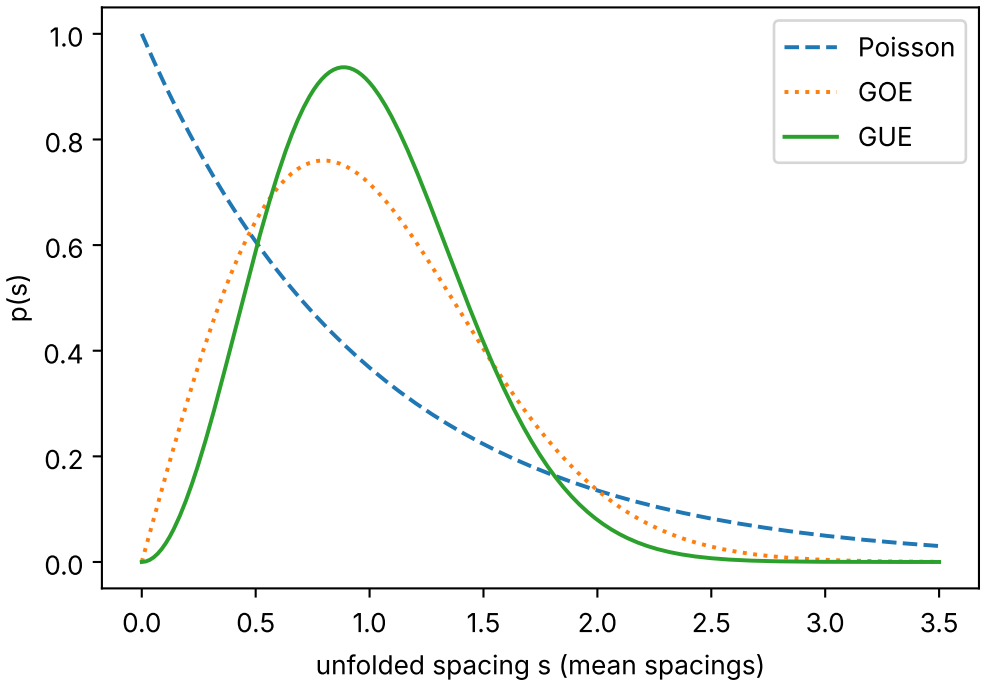


Figure 20.4: The three reference spacing distributions for unfolded spectra (unit mean spacing): Poisson e^{-s}, GOE Wigner surmise $\frac{\pi}{2} s e^{-\pi s^2/4}$, and GUE Wigner surmise $\frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$. The random-matrix laws vanish at $s = 0$ (linearly for GOE, quadratically for GUE) and are much narrower than Poisson.

Sample Problem 20.5: Shape of the three distributions

Find the most probable spacing and the variance for the Poisson, GOE and GUE distributions.

Step 1: Poisson. e^{-s} is decreasing, so the most probable spacing is 0; the variance is 1 (Sec. 20.4).

Step 2: GOE mode. $\frac{\mathrm{d}}{\mathrm{d}s}\left(s e^{-\pi s^2/4}\right) = e^{-\pi s^2/4}\left(1 - \frac{\pi s^2}{2}\right) = 0$ gives $s = \sqrt{2/\pi} = 0.798$.

Step 3: GUE mode. $\frac{\mathrm{d}}{\mathrm{d}s}\left(s^2 e^{-4s^2/\pi}\right) = e^{-4s^2/\pi}\left(2s - \frac{8s^3}{\pi}\right) = 0$ gives $s^2 = \pi/4$, $s = 0.886$.

Step 4: GOE variance. $\langle s^2 \rangle = \frac{\pi}{2} I_3(\pi/4) = \frac{\pi}{2} \cdot \frac{1}{2(\pi/4)^2} = \frac{\pi}{2} \cdot \frac{8}{\pi^2} = \frac{4}{\pi}$, so the variance is $\frac{4}{\pi} - 1 = 0.273$.

Step 5: GUE variance. $\langle s^2 \rangle = \frac{32}{\pi^2} I_4(4/\pi) = \frac{32}{\pi^2} \cdot \frac{3\sqrt{\pi}}{8(4/\pi)^{5/2}}$. Since $(4/\pi)^{5/2} = 32/\pi^{5/2}$,

this is $\frac{32}{\pi^2}\cdot\frac{3\sqrt{\pi}\,\pi^{5/2}}{256}=\frac{3\pi^3}{8\pi^2}=\frac{3\pi}{8}$, so the variance is $\frac{3\pi}{8}-1=0.178$ (standard deviation 0.422).

Answer:

mode: 0, 0.798, 0.886; variance: 1, 0.273, 0.178 (Poisson, GOE, GUE)

Stronger repulsion means a more regular spectrum: GUE spacings are the least spread out of the three.

Sample Problem 20.6: The small-spacing laws

Find $\Pr(s<\varepsilon)$ for small ε under each law, and evaluate at $\varepsilon=0.1$.

Step 1: Poisson. $1-e^{-\varepsilon}\approx\varepsilon$ for small ε (first-order Taylor). At $\varepsilon=0.1$: exact 0.0952.

Step 2: GOE, exact. $\int_0^\varepsilon \frac{\pi}{2}se^{-\pi s^2/4}\mathrm{d}s=\left[-e^{-\pi s^2/4}\right]_0^\varepsilon=1-e^{-\pi\varepsilon^2/4}\approx\frac{\pi}{4}\varepsilon^2$. At $\varepsilon=0.1$: exact 0.00782, approximation 0.00785.

Step 3: GUE. For small s, $e^{-4s^2/\pi}\approx1$, so $\int_0^\varepsilon\frac{32}{\pi^2}s^2\mathrm{d}s=\frac{32}{3\pi^2}\varepsilon^3=1.081\varepsilon^3$. At $\varepsilon=0.1$: 1.081×10^{-3}, against the exact 1.073×10^{-3} found earlier.

Answer:

$$\boxed{\Pr(s<\varepsilon)\approx\varepsilon\ \text{(Poisson)},\quad \tfrac{\pi}{4}\varepsilon^2\ \text{(GOE)},\quad \tfrac{32}{3\pi^2}\varepsilon^3\ \text{(GUE)}}$$

Halving ε halves the Poisson probability, quarters the GOE one, and divides the GUE one by eight. This is the most sensitive fingerprint of the symmetry class.

```
import numpy as np
rng = np.random.default_rng(0)          # fixed seed: reproducible
n = 200000                              # number of random 2x2 matrices
x, y, z = rng.normal(size=(3, n))       # independent N(0,1) numbers
s_real = np.hypot(x, y)                 # GOE-type: s = sqrt(x^2 + y^2)
s_cplx = np.sqrt(x**2 + y**2 + z**2)    # GUE-type: s = |(x, y, z)|
for s in (s_real, s_cplx):
    s = s / s.mean()                    # unfold: unit mean spacing
    print(np.mean(s < 0.1), np.mean(s**2))
```

With this seed the printed small-gap fractions agree with the exact values 0.00782 (GOE) and 1.07×10^{-3} (GUE) of the sample problems to within three binomial standard errors. Here the standard error is $\sqrt{p(1-p)/n}$ with $n=2\times10^5$, about 2×10^{-4} for GOE and 7×10^{-5} for GUE (Chapter 19). The values of $\langle s^2\rangle$ agree with $4/\pi$ and $3\pi/8$ to within 0.01. Figure 20.5 shows the histograms.

Taken on trust

Large matrices. The surmises are exact for 2×2 Gaussian matrices (proved above). For $N\times N$ GOE or GUE matrices with $N\to\infty$, the exact spacing distribution in the middle of the spectrum (after unfolding) is a more complicated function, found by Gaudin and Mehta and expressed through objects beyond this book. Numerically it is very close to the corresponding surmise, which is why the surmise is the standard benchmark, as in paper Fig. 2. Derivations are in [35]. We take the closeness on trust.

Checkpoint 20.4

Why does the GUE density start like s^2 while the GOE density starts like s?

Answer: A GUE degeneracy needs three real conditions ($x=y=z=0$), a GOE one only two. The

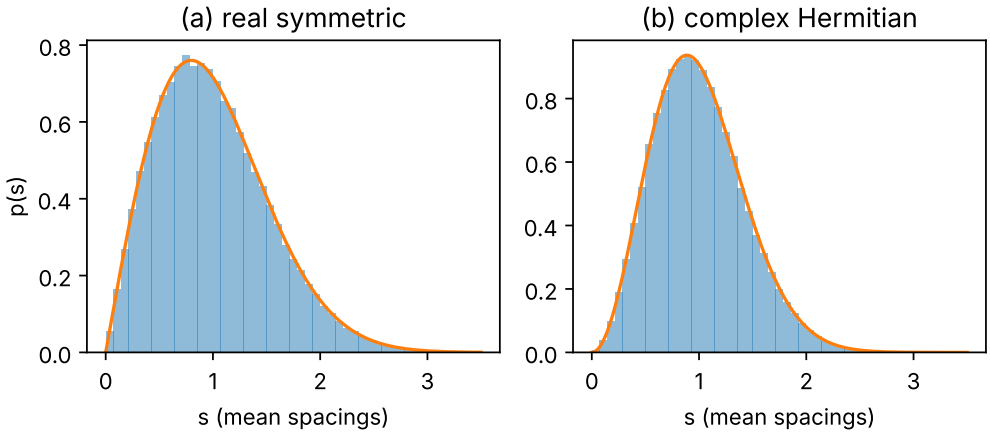


Figure 20.5: Spacings of 2×10^5 random 2×2 matrices, each rescaled by the sample mean spacing. (a) Real symmetric matrices of the form of Eq. (20.31) with independent Gaussian x, y: the histogram follows the GOE surmise. (b) Complex Hermitian matrices of the form of Eq. (20.38): the histogram follows the GUE surmise. Spacings are dimensionless (units of mean spacing).

probability of a spacing below s is the probability of landing in a ball of radius s around the origin: area $\propto s^2$ in 2D, volume $\propto s^3$ in 3D. Differentiating gives densities $\propto s$ and $\propto s^2$.

Key Ideas

- Spacing of a 2×2 matrix = length of a Gaussian vector: 2 components (real symmetric) or 3 (complex Hermitian).
- Geometry (ring circumference $2\pi s$, shell area $4\pi s^2$) gives $p \propto s\,e^{-Bs^2}$ and $p \propto s^2 e^{-Bs^2}$; the two conditions of Eq. (20.14) give $p_{\mathrm{GOE}} = \frac{\pi}{2} s e^{-\pi s^2/4}$ and $p_{\mathrm{GUE}} = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$.
- The Dyson index β (0, 1, 2, 4) controls small-spacing behavior $\Pr(s < \varepsilon) \propto \varepsilon^{\beta+1}$.

20.7 Beyond nearest neighbors: pair correlation and Montgomery's discovery

★ Did you know? *Zeros at Dizzying Heights*

Starting in the 1980s, Andrew Odlyzko at Bell Labs computed huge batches of zeta zeros at enormous heights, first near the 10^{12}-th zero and later near the 10^{20}-th and beyond. Histograms of their spacings and pair correlations match random-matrix predictions with stunning precision. His results turned Montgomery's conjecture into one of the best-tested patterns in mathematics.

20.7.1 The pair correlation function

The spacing distribution looks only at neighbors. A broader statistic asks about *all* pairs. For an unfolded spectrum, the **pair correlation function** $R_2(v)$ is defined so that

$$R_2(v)\,\mathrm{d}v = \text{expected number of } \textit{other} \text{ levels in } [u+v,\, u+v+\mathrm{d}v], \text{ given a level at } u, \quad (20.48)$$

for $v > 0$, averaged over the reference level u. Because the mean density is one, $R_2(v) \to 1$ when v is large: far away, knowing there is a level at u tells you nothing.

Poisson. Other levels are independent of the reference level, so the expected number in $[u+v, u+v+\mathrm{d}v]$ is just the density times the length: $R_2(v) = 1$ for all v.

Relation to $p(s)$. Let $p_k(v)$ be the density of the distance from a level to its k-th neighbor on the right ($p_1 = p$). Every other level is some k-th neighbor, so $R_2(v) = p_1(v)+p_2(v)+p_3(v)+\cdots$. For small v, reaching the second neighbor within distance v requires two levels in a tiny interval, which is much rarer, so $R_2(v) \approx p(v)$ as $v \to 0$. Check with Poisson: $R_2(0) = 1$ and $p_{\mathrm{P}}(0) = e^0 = 1$.

20.7.2 The GUE pair correlation

For large GUE matrices, unfolded, the pair correlation is

$$\boxed{R_2(v) = 1 - \left(\frac{\sin \pi v}{\pi v}\right)^2} \tag{20.49}$$

For small v, use $\sin x = x - x^3/6 + \cdots$ with $x = \pi v$: $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \cdots$, so $\left(\frac{\sin x}{x}\right)^2 = 1 - \frac{x^2}{3} + \cdots$ and $R_2(v) \approx \frac{\pi^2 v^2}{3}$. So R_2 vanishes like v^2, the same power as the GUE spacing density. The coefficients are close too: $\pi^2/3 = 3.290$ for R_2 against $32/\pi^2 = 3.242$ for the surmise.

Taken on trust

Status of Eq. (20.49) for random matrices. That the large-N GUE pair correlation is exactly $1 - (\sin \pi v/\pi v)^2$ is a theorem; its proof uses orthogonal (Hermite) polynomials and is given in [35]. We take it on trust.

20.7.3 Montgomery, Dyson and Odlyzko

In the early 1970s Montgomery studied the differences $t_m - t_n$ between zeta zeros, normalized by the mean spacing (equivalently, differences of unfolded zeros $u_m - u_n$). His result, as the story is usually told, came up at afternoon tea at the Institute for Advanced Study in Princeton in 1972. When Montgomery described the function he had found, Dyson recognized it at once as the pair correlation of eigenvalues of random Hermitian matrices, Eq. (20.49). Montgomery published the result in 1973 [9].

You must be careful about what was proved.

Taken on trust

Montgomery's theorem and the pair correlation conjecture. Assuming the Riemann hypothesis (still open, Chapter 16), Montgomery proved that the pair correlation of the normalized zeros agrees with $1 - (\sin \pi v/\pi v)^2$ when it is tested against a restricted class of smooth weighting functions (those whose Fourier transform, Chapter 6, vanishes outside one specific fixed interval, $(-1, 1)$ in Montgomery's normalization). The statement that the agreement holds without this restriction is the **pair correlation conjecture**, which is open. See [9], and the review [11].

Odlyzko then tested these ideas numerically at great heights. His 1987 paper [8] computed large samples of zeros, including samples very high up the critical line. The spacing distribution and pair correlation of the high zeros matched the GUE predictions with striking accuracy. This match is a **numerical observation**, not a theorem. Figure 20.6 shows the same test for the first 10^4 zeros, computed with qmnt.

Sample Problem 20.7: Reading the pair correlation

Evaluate $R_2(v)$ of Eq. (20.49) at $v = 0.1,\ 0.5,\ 1$; compare $v = 0.1$ with the small-v approximation; and find the expected number of other levels within distance 0.5 on one side of a given level, comparing with Poisson.

Step 1: $v = 0.1$**.** $\pi v = 0.314159$, $\sin(0.314159) = 0.309017$, ratio 0.983632, squared 0.967532, so $R_2 = 0.0325$. The approximation $\pi^2 v^2/3$ gives 0.0329.

Step 2: $v = 0.5$**.** $\sin(\pi/2) = 1$, ratio $1/(\pi/2) = 0.636620$, squared 0.405285, so $R_2 = 0.5947$.

Step 3: $v = 1$**.** $\sin\pi = 0$, so $R_2 = 1$ exactly. (Integer separations are where the correction term vanishes.)

Step 4: expected number. By Eq. (20.48), the expected number of levels in $(u, u+0.5]$ is $\int_0^{0.5} R_2(v)\,dv = 0.5 - \int_0^{0.5} \left(\frac{\sin\pi v}{\pi v}\right)^2 dv$. The last integral has no elementary antiderivative; numerical integration (`scipy.integrate.quad`) gives 0.38685, so the result is 0.113. For Poisson it is $\int_0^{0.5} 1\,dv = 0.5$.

Answer:

$$R_2(0.1) = 0.0325,\ R_2(0.5) = 0.5947,\ R_2(1) = 1;\quad \int_0^{0.5} R_2 = 0.113 \text{ vs } 0.5 \text{ (Poisson)}$$

A level of a GUE spectrum has, on average, less than a quarter as many companions within half a spacing as a Poisson level does.

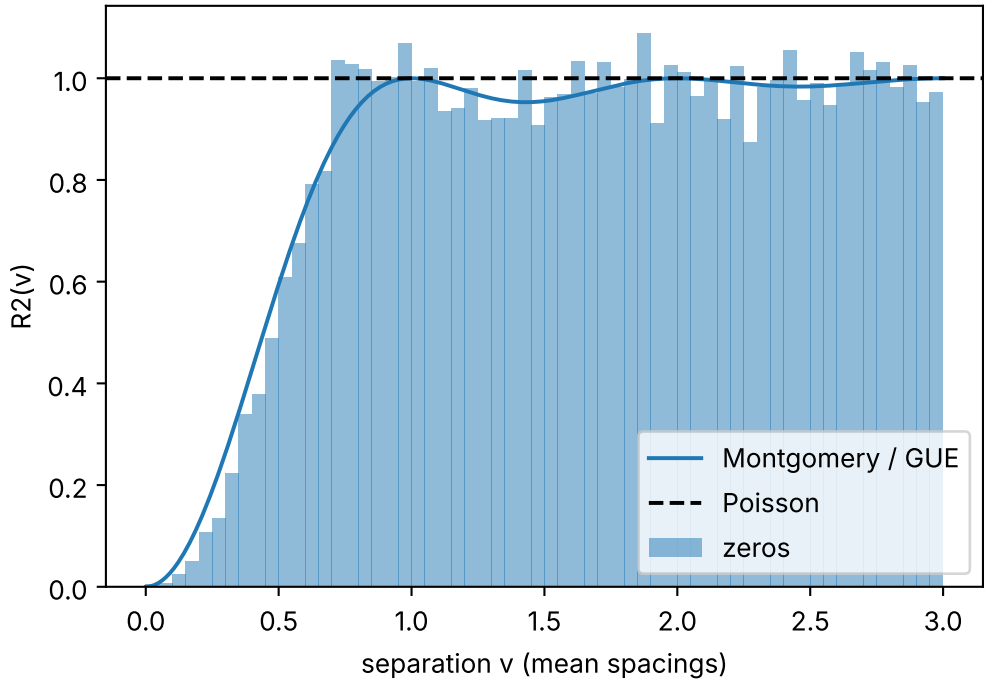


Figure 20.6: Pair correlation of the first 10^4 unfolded zeros: histogram of all differences $u_m - u_n$ with $0 < u_m - u_n < 3$ (bin width 0.05), divided by the number of reference zeros and the bin width, against Montgomery's form $1 - (\sin\pi v/\pi v)^2$ and the Poisson value 1. The separation v is in units of the mean spacing.

Key Ideas

- $R_2(v)$ counts all other levels at distance v; Poisson gives $R_2 = 1$; for small v, $R_2 \approx p$.
- Large-N GUE: $R_2(v) = 1 - (\sin \pi v/\pi v)^2$ (theorem, proof in [35]).
- For the zeros: Montgomery's theorem (assuming RH, restricted test functions), the pair correlation conjecture (open), Odlyzko's numerical agreement [8].

20.8 The Riemann zeros as a GUE-like spectrum

20.8.1 Reproducing paper Fig. 2

You now have every tool needed to reproduce the paper's Fig. 2.

```
import numpy as np, qmnt
import matplotlib.pyplot as plt
t = qmnt.zeros(10000)            # first 10^4 zeros
u = qmnt.Nsm(t)                  # unfold with paper Eq. (1)
ds = np.diff(u)                  # 9999 unfolded spacings
print(ds.mean(), ds.var(), np.mean(ds < 0.1))
s = np.linspace(0, 3.5, 400)
plt.hist(ds, bins=np.linspace(0, 3.5, 36), density=True)
plt.plot(s, 32/np.pi**2 * s**2 * np.exp(-4*s**2/np.pi), label="GUE")
plt.plot(s, np.pi/2 * s * np.exp(-np.pi*s**2/4), ":", label="GOE")
plt.plot(s, np.exp(-s), "--", label="Poisson")
plt.legend(); plt.xlabel("unfolded spacing s"); plt.ylabel("p(s)")
plt.savefig("gue.pdf")
```

Three lines do all the physics: `qmnt.zeros` supplies the data, `qmnt.Nsm` unfolds them with paper Eq. (1), and `np.diff` forms the spacings. Everything else is plotting. When you run it you should find three things. First, a mean spacing within 2×10^{-4} of one. Eq. (20.7) explains this: the mean of all 9999 spacings is exactly $1 + (\delta_{10000} - \delta_1)/9999$, and both end fluctuations are observed to be below one in size. Second, a variance closer to the GUE value 0.178 than to the GOE value 0.273, and far from the Poisson value 1. Third, a fraction of spacings below 0.1 that is below 0.01, far smaller than the Poisson fraction 0.0952. Figure 20.7 shows the result. It is the paper's Fig. 2 recomputed, with the GOE surmise added for comparison (the paper shows only the GUE and Poisson curves).

Why does the paper include this figure at all? Not as evidence about operators, but as a *dataset check*. If the zeros had been computed wrongly (a zero missed, a zero duplicated, a precision problem), the histogram would show it at once. A missed zero produces a spacing near 2, which GUE makes very rare; a duplicated zero produces a spacing 0, which GUE forbids. If the unfolding function had the wrong density (for example a wrong factor in the logarithmic term), the mean spacing would drift away from one and the histogram would be distorted. A wrong additive constant, such as an error in the $\frac{7}{8}$, would however cancel in every spacing and go unnoticed. A clean match to the surmise therefore confirms "the dataset and unfolding" (paper Sec. II A).

20.8.2 What GUE statistics do and do not imply

It is tempting to read Fig. 20.7 as saying "the zeros are the eigenvalues of some Hermitian operator without time-reversal symmetry". Be careful. Here is the status of each statement.

- **Numerical observation:** the statistics of the unfolded zeros agree closely with GUE statistics [8].

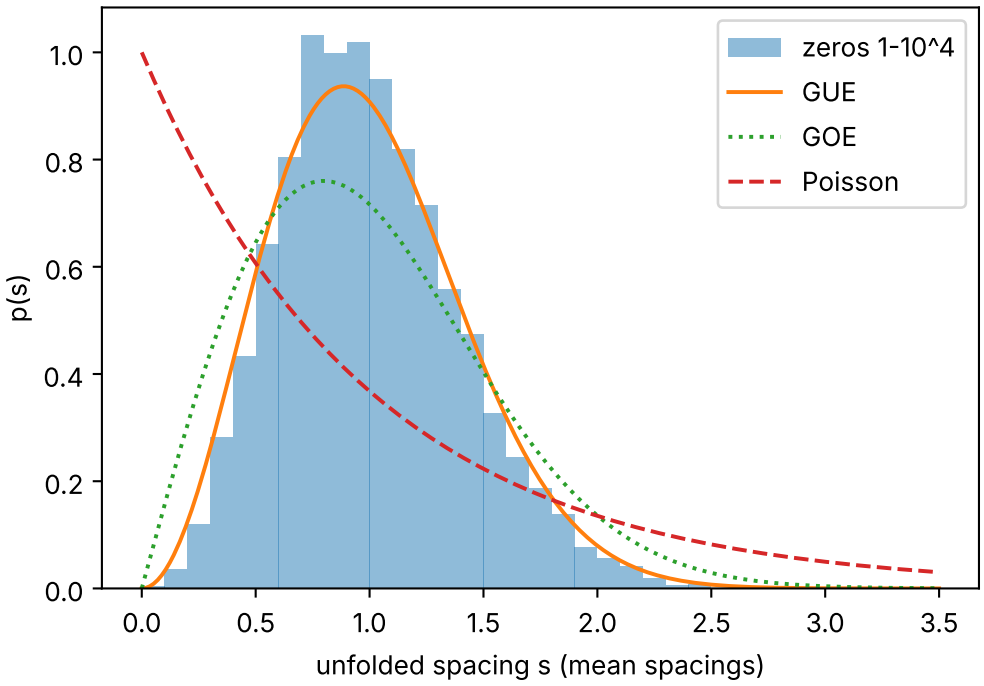


Figure 20.7: Reproduction of paper Fig. 2: histogram of the 9999 unfolded nearest-neighbor spacings of the first 10^4 zeros (unfolded with paper Eq. (1)), against the GUE Wigner surmise $\frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ (solid), the GOE surmise (dotted) and the Poisson law e^{-s} (dashed). Spacings are in units of the local mean spacing.

- **Theorem (conditional):** Montgomery's restricted pair correlation result, assuming RH [9].
- **Heuristic/hope:** that this agreement points toward a self-adjoint operator whose eigenvalues are the t_n (Hilbert–Pólya idea, Chapter 21). This is open.
- **Not implied:** any *specific* operator. A GUE matrix of the right size, unfolded, has, to the accuracy of the comparisons above, the same local statistics as the zeros, yet its individual eigenvalues have nothing to do with the individual zeros.

The last point is precisely the gap between statistics and positions that the paper exploits. Statistics are invariant under a vast set of changes to the spectrum: reflect it, shift it by a fraction of a spacing, replace it by another random draw, and the histogram does not change. A model can therefore reproduce the statistics, even reproduce the training zeros themselves, and still predict nothing about the next zero. Predicting position is a much more demanding test, and it is the test the paper applies (Chapter 25).

Pitfall

"Has GUE statistics" is a property shared by every generic complex Hermitian random matrix. It cannot single out one operator. Do not quote a matching spacing histogram as evidence that a particular fitted Hamiltonian "is" the zeta operator.

20.8.3 Negative correlations of consecutive fluctuations

Look again at the column δ_n of Table 20.1: $-0.051, +0.070, -0.107, +0.171, -0.183, \ldots$ In this short sample of ten values the signs mostly alternate: a zero pushed above its smooth

position tends to be followed by one pushed below. Ten values are only suggestive. Fig. 20.8 below measures the effect over 400 zeros, and the paper reports the same thing (negative autocorrelation at lags 1–3, paper Sec. III). This is a **numerical observation** about the low zeros, not something we derive. In particular it is *not* a consequence of level repulsion alone. If zero n moves up and repels zero $n+1$, then zero $n+1$ is pushed up as well, which by itself would favor a *positive* correlation of consecutive fluctuations. What we can do exactly is relate the sign of the correlation to two measurable quantities, through the identity below.

We can quantify it. Recall from Chapter 19 that the autocovariance at lag k of a sequence with mean zero is $C_k = \langle \delta_n \delta_{n+k} \rangle$, and the autocorrelation is $r_k = C_k/C_0$. The sequence is called **stationary** when averages such as $\langle \delta_n \rangle$ and $\langle \delta_n \delta_{n+k} \rangle$ do not depend on n, so that C_k depends only on the lag k. The unfolded spacing is $\Delta u_n = 1 + \delta_{n+1} - \delta_n$ (from Eq. (20.5)), so

$$\begin{aligned} \mathrm{Var}(\Delta u) &= \left\langle (\delta_{n+1} - \delta_n)^2 \right\rangle \\ &= \langle \delta_{n+1}^2 \rangle - 2\langle \delta_{n+1}\delta_n \rangle + \langle \delta_n^2 \rangle \\ &= 2\,(C_0 - C_1)\,. \end{aligned} \tag{20.50}$$

This is an identity, valid for any stationary sequence. Solved for C_1 it reads

$$C_1 = C_0 - \tfrac{1}{2}\mathrm{Var}(\Delta u), \tag{20.51}$$

so the lag-1 covariance is negative exactly when the fluctuation variance C_0 is smaller than half the spacing variance. For the first thousand zeros, computing both sides with qmnt (unfold as in the listing of Sec. 20.2), this is what happens: $C_0 < \frac{1}{2}\mathrm{Var}(\Delta u)$, so $C_1 < 0$, and the spacings vary more than $2C_0$ alone would give. Nothing forces this in general.

Taken on trust

For large GUE matrices the variance of the fluctuations δ_n grows slowly (logarithmically) with the matrix size, while the variance of the unfolded spacings stays near 0.178. By Eq. (20.51), C_1 then becomes positive. We quote this growth without derivation; see [35].

So the negative lag-1 correlation is a property of the zeros at the heights studied here, not a universal consequence of repulsion.

Consecutive spacings. The alternation of long and short spacings noticed in Table 20.1 is measured by the covariance of consecutive spacings. The mean spacing is one, so $\mathrm{Cov}(\Delta u_n, \Delta u_{n+1}) = \langle (\Delta u_n - 1)(\Delta u_{n+1} - 1) \rangle$, and $\Delta u_n - 1 = \delta_{n+1} - \delta_n$. Hence

$$\begin{aligned} \mathrm{Cov}(\Delta u_n, \Delta u_{n+1}) &= \langle (\delta_{n+1} - \delta_n)(\delta_{n+2} - \delta_{n+1}) \rangle \\ &= \langle \delta_{n+1}\delta_{n+2} \rangle - \langle \delta_{n+1}^2 \rangle - \langle \delta_n \delta_{n+2} \rangle + \langle \delta_n \delta_{n+1} \rangle \\ &= C_1 - C_0 - C_2 + C_1 \quad \text{(stationarity)} \\ &= -C_0 + 2C_1 - C_2. \end{aligned} \tag{20.52}$$

The term $-C_0$ is always negative, and $2C_1$ is negative whenever $C_1 < 0$. For the first thousand zeros, computed with qmnt, the correlation of consecutive unfolded spacings is negative: a long spacing tends to be followed by a short one. The sample value of the left side of Eq. (20.52) agrees with $-C_0 + 2C_1 - C_2$ computed from the same zeros to better than 0.01. This is the spacing counterpart of the negative fluctuation autocorrelation that the paper reports at lags 1–3.

A simple model that shows the consequence. Let ε_n be independent random numbers with mean zero and variance σ^2, and let

$$\delta_n = \varepsilon_n - \alpha\,\varepsilon_{n-1}, \qquad 0 < \alpha \le 1. \tag{20.53}$$

Because different ε's are uncorrelated, only matching indices contribute to averages of products:

$$C_0 = \langle(\varepsilon_n - \alpha\varepsilon_{n-1})^2\rangle = \sigma^2 + \alpha^2\sigma^2 = (1+\alpha^2)\sigma^2, \tag{20.54}$$

$$C_1 = \langle(\varepsilon_n - \alpha\varepsilon_{n-1})(\varepsilon_{n+1} - \alpha\varepsilon_n)\rangle = -\alpha\langle\varepsilon_n^2\rangle = -\alpha\sigma^2, \tag{20.55}$$

$$C_2 = \langle(\varepsilon_n - \alpha\varepsilon_{n-1})(\varepsilon_{n+2} - \alpha\varepsilon_{n+1})\rangle = 0. \tag{20.56}$$

So $r_1 = -\alpha/(1+\alpha^2) < 0$ and $r_k = 0$ for $k \geq 2$.

Now suppose you know δ_{n-1} and δ_{n+1} and "interpolate smoothly" by predicting $\hat{\delta}_n = \tfrac{1}{2}(\delta_{n-1} + \delta_{n+1})$. How well does $\hat{\delta}_n$ correlate with the truth δ_n? Using the definitions of Chapter 19:

$$\mathrm{Cov}(\delta_n, \hat{\delta}_n) = \tfrac{1}{2}\left(\langle\delta_n\delta_{n-1}\rangle + \langle\delta_n\delta_{n+1}\rangle\right) = \tfrac{1}{2}(C_1 + C_1) = C_1, \tag{20.57}$$

$$\mathrm{Var}(\hat{\delta}_n) = \tfrac{1}{4}\left(C_0 + 2C_2 + C_0\right) = \tfrac{1}{2}C_0 \quad (C_2 = 0), \tag{20.58}$$

$$\mathrm{corr}(\delta_n, \hat{\delta}_n) = \frac{C_1}{\sqrt{C_0 \cdot \tfrac{1}{2}C_0}} = \sqrt{2}\, r_1 = -\frac{\sqrt{2}\,\alpha}{1+\alpha^2}. \tag{20.59}$$

For $\alpha = 1$ this is -0.707. The prediction made from the nearest neighbors points in the *wrong direction*. Smooth interpolation, which assumes that nearby values are similar, is anticorrelated with a sequence whose nearby values are dissimilar.

Sample Problem 20.8: The neighbor-average predictor

In the model of Eq. (20.53) with $\alpha = 0.6$, find r_1 and the correlation of the neighbor average with the truth.

Step 1: $r_1 = -0.6/(1+0.36) = -0.6/1.36 = -0.441$.

Step 2: By Eq. (20.59), $\mathrm{corr} = \sqrt{2} \times (-0.441) = -0.624$.

Answer:

$$\boxed{r_1 = -0.441, \quad \mathrm{corr}(\delta_n, \hat{\delta}_n) = -0.624}$$

The real zeros are more complicated (their autocorrelation is negative at several lags and contains the slow oscillations of the prime sum, Chapter 18), but the mechanism is the same. The paper reports exactly this: "the fluctuation sequence has negative autocorrelation at lags 1–3, so any smooth interpolation between neighbouring training levels is anticorrelated with the truth (the mean of the two nearest training fluctuations gives -0.26 to -0.41)" (paper Sec. III, "Within-height splits"). Figure 20.8 shows the autocorrelation of δ_n for zeros 1–400.

Checkpoint 20.5

If consecutive fluctuations were *positively* correlated ($r_1 > 0$, $r_2 = 0$), would the neighbor average be correlated or anticorrelated with the truth?

Answer: Correlated: Eq. (20.59) gives $\mathrm{corr} = \sqrt{2}\, r_1 > 0$. The sign of the interpolation's skill follows the sign of the lag-1 correlation.

Key Ideas

- Paper Fig. 2 is reproduced in a few lines; it validates the data and the unfolding.
- GUE statistics are a numerical observation; they do not identify an operator and say nothing about individual positions.
- For the low zeros, consecutive fluctuations are observed to be negatively correlated (a numerical observation, not derived from repulsion). Whenever $r_1 < 0$ and $r_2 \approx 0$,

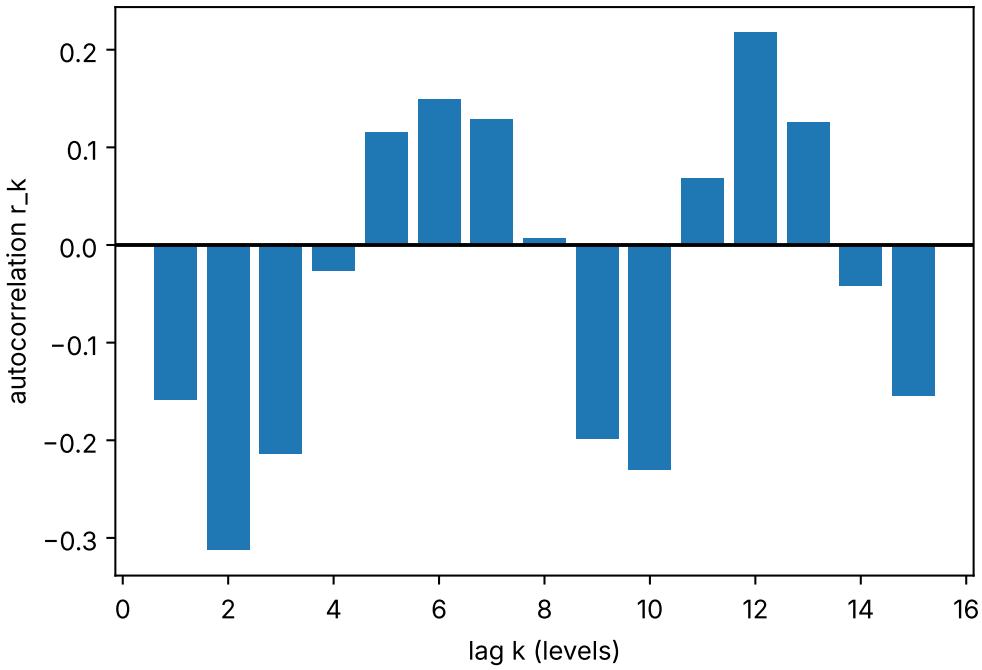


Figure 20.8: Autocorrelation r_k of the fluctuations $\delta_n = (t_n - s_n)/\mathrm{gap}_n$ (in mean spacings) of zeros 1–400, for lags $k = 1, \ldots, 15$. The first lags are negative: consecutive zeros tend to be displaced in opposite directions from their smooth positions.

neighbor interpolation is anticorrelated with the truth (corr $= \sqrt{2}\, r_1$ in the MA(1) model, Eq. (20.59)).

20.9 Paper connection

Sec. I, first paragraph. "Montgomery's pair-correlation theorem and Odlyzko's numerics match Gaussian unitary ensemble (GUE) eigenvalue statistics [9, 8]." You can now unpack this: the theorem is Montgomery's conditional, restricted result on R_2 (Sec. 20.7); the numerics are Odlyzko's spacing and pair-correlation histograms; GUE is the ensemble of Sec. 20.6. The paper calls this "circumstantial evidence", which is the honest status.

Sec. II A and Fig. 2. "Unfolded spacings of the first 10^4 zeros show textbook GUE level repulsion (Fig. 2), confirming the dataset and unfolding." The unfolding is Eq. (20.4) with paper Eq. (1); the curves in the caption are Eqs. (20.45) and (20.21); you reproduced the figure in Sec. 20.8. Its role is validation, not argument.

The unit of every rms. "All errors below are quoted in units of the local mean spacing $\mathrm{gap}_n = ds_n/dn$" (Sec. II A). Measuring in mean spacings is unfolding (Eq. (20.12)). That is why errors at zero 400 and zero 10^4 can be compared in Table V, and why the null grows only mildly, from 0.263 to 0.296 (Table V; Sec. II C, "Within-height splits").

Sec. III, "Within-height splits". The paper observes negative autocorrelation at lags 1–3; this is a numerical observation, which Fig. 20.8 reproduces for zeros 1–400. Given that observation, Sec. 20.8 shows why neighbor interpolation is anticorrelated with the truth (Eq. (20.59) in the simple model), which accounts for the paper's neighbor-mean correlation of -0.26 to -0.41. This chapter does not derive the negative autocorrelation itself from repulsion. The paper uses this to call the negative correlations of Table III at $M \leq 200$ "mechanical".

Sec. V, "The proposed mechanism fails quantitatively". The dominant $p = 2$

oscillation has period $2\pi/\log 2 \approx 9$ in t, "only ∼6–7 levels at these heights". Dividing a period in t by the mean spacing (Eq. (20.3)) is again unfolding (Problem 17).

What this chapter does *not* support. Nothing in Fig. 2 bears on whether a fitted potential predicts. Statistics are shared by all GUE-like spectra; the paper's test (held-out positions, Figs. 1, 3, Tables I–V) is of a different and stronger kind.

20.10 Summary

- Unfolding $u_n = \bar{N}(E_n)$ gives unit mean spacing (Eqs. (20.4), (20.7)); for the zeros the mean gap is $2\pi/\log(t/2\pi)$ (Eq. (20.3)), and $\delta_n \approx (t_n - s_n)/\mathrm{gap}_n$ (Eq. (20.12)).
- Spacing densities obey $\int p = \int sp = 1$ (Eq. (20.14)).
- Independent levels: $p_{\mathrm{P}} = e^{-s}$ (Eq. (20.21)); mean 1, variance 1.
- 2×2 spacing $\sqrt{(a-d)^2 + 4b^2}$ (Eq. (20.25)); avoided crossings.
- Wigner surmises: $p_{\mathrm{GOE}} = \frac{\pi}{2} s e^{-\pi s^2/4}$, $p_{\mathrm{GUE}} = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ (Eqs. (20.37), (20.45)); variances 0.273, 0.178.
- GUE pair correlation $1 - (\sin \pi v/\pi v)^2$ (Eq. (20.49)); Montgomery (conditional, restricted), conjecture open, Odlyzko numerics.
- $\mathrm{Var}(\Delta u) = 2(C_0 - C_1)$ (Eq. (20.50)) and $\mathrm{Cov}(\Delta u_n, \Delta u_{n+1}) = -C_0 + 2C_1 - C_2$ (Eq. (20.52)); for the low zeros $C_1 < 0$ is observed, not derived from repulsion; neighbor interpolation has corr $= \sqrt{2} r_1$ in the MA(1) model (Eq. (20.59)).

20.11 Problems

1. • Unfold the box levels $E_k = (k\pi/2L)^2$ for the scale-2 box $L = 59.957$. What are the unfolded spacings?

2. • For Poisson levels, what is the probability that a spacing exceeds 1?

3. • Find the mean spacing of the zeros at $t = 100$ from Eq. (20.3).

4. •• For the GOE surmise, find $\Pr(s > 1)$ exactly.

5. •• Let U be a complex 2×2 matrix with $U^\dagger U = I$. Show that $\mathrm{tr}((UHU^\dagger)^2) = \mathrm{tr}(H^2)$ for every Hermitian H, so that the exponent $-\mathrm{tr}(H^2)/2\sigma_0^2$ of the GUE density does not depend on the basis.

6. •• Using the small-ε law, estimate $\Pr(s < 0.2)$ under GUE.

7. •• For $H(\tau)$ with $g = 0.2$, find the minimum separation of the levels and the separation at $\tau = 0.5$.

8. •• Compare the small-v coefficients of R_2 for GUE and of p_{GUE}. By what percentage do they differ?

9. •• In the model of Eq. (20.53) with $\alpha = 0.5$, compute r_1 and the neighbor-average correlation.

10. •• Show that if spacings were exactly 1 (picket fence), δ_n would be constant and $\mathrm{Var}(\Delta u) = 0$; check against Eq. (20.50).

11. ••• A symplectic degeneracy needs five conditions, so $p(s) = As^4 e^{-Bs^2}$. Using Eq. (20.14) find A and B.

12. •• Using qmnt, compute the variance of the unfolded spacings of zeros 1–1000 and of zeros 9001–10000. Does it drift toward the GUE value?

13. •• Using numpy, throw 2×10^4 uniform points on $[0, 2\times10^4]$, sort, and compute the fraction of spacings below 0.1. Compare with theory.

14. ••• Using numpy, build a 400×400 GUE matrix ($H=(G+G^\dagger)/2$ with G complex Gaussian), take the central 100 eigenvalues, unfold by dividing spacings by their mean, and compare the variance with 0.178.

15. • The paper's null rms on zeros 401–500 is 0.263 (Sec. II C). In the language of this chapter, what quantity's rms is this?

16. •• Paper Fig. 2 compares with GUE, not GOE. Which feature of the histogram near $s=0$ would distinguish the two?

17. •• Using qmnt, find the mean gap at t_{400} and the number of levels in one period $2\pi/\log 2$ of the $p=2$ wave (RE2). Compare with the paper's "6–7 levels" (Sec. V).

18. ••• Using qmnt, compute $\mathrm{corr}(\delta_n, \frac12(\delta_{n-1}+\delta_{n+1}))$ for $n=2,\dots,399$ and compare with the paper's range -0.26 to -0.41 (Sec. III).

19. •• A spectrum is obtained by reflecting the unfolded zeros ($u\to-u$). Does its spacing histogram change? Can it predict the zeros? What does this say about Fig. 2?

20. • Estimate $\Pr(s<0.05)$ for GUE using the small-ε law.

21. ••• In the model of Eq. (20.53), use Eqs. (20.50) and (20.52) to show that the correlation of consecutive unfolded spacings is $-(1+\alpha)^2/[2(1+\alpha+\alpha^2)]$, and evaluate it for $\alpha=1$.

22. •• Verify $I_2(B)=\sqrt{\pi}/(4B^{3/2})$ numerically for $B=4/\pi$.

20.12 Answers to selected problems

1. $\bar N(E)=2L\sqrt{E}/\pi-\frac12$ gives $u_k=k-\frac12$; every $\Delta u_k=1$ and every $\delta_k=0$, independent of L.

2. $Q(1)=e^{-1}=0.368$.

3. $\log(100/2\pi)=\log 15.9155=2.7674$; gap $=2\pi/2.7674=2.270$.

5. $(UHU^\dagger)^2=UHU^\dagger UHU^\dagger=UH(U^\dagger U)HU^\dagger=UH^2U^\dagger$. With $\mathrm{tr}(XY)=\mathrm{tr}(YX)$ for $X=U$, $Y=H^2U^\dagger$: $\mathrm{tr}(UH^2U^\dagger)=\mathrm{tr}(H^2U^\dagger U)=\mathrm{tr}(H^2)$.

7. $\pm\sqrt{\tau^2+g^2}$: minimum $2g=0.4$; at $\tau=0.5$, $2\sqrt{0.29}=1.077$.

9. $r_1=-0.5/1.25=-0.4$; corr $=\sqrt{2}(-0.4)=-0.566$.

11. $I_5=\frac{4}{2B}I_3=\frac{1}{B^3}$, so $A=B^3$; then $AI_4=\frac{3\sqrt{\pi}}{8}B^{1/2}=1$, $B=\frac{64}{9\pi}=2.264$, $A=\left(\frac{64}{9\pi}\right)^3=\frac{2^{18}}{3^6\pi^3}=11.60$.

13. The fraction should be near $1-e^{-0.1}=0.0952$, within about 0.004 (binomial standard error $\sqrt{0.095\times0.905/(2\times10^4)}\approx0.002$).

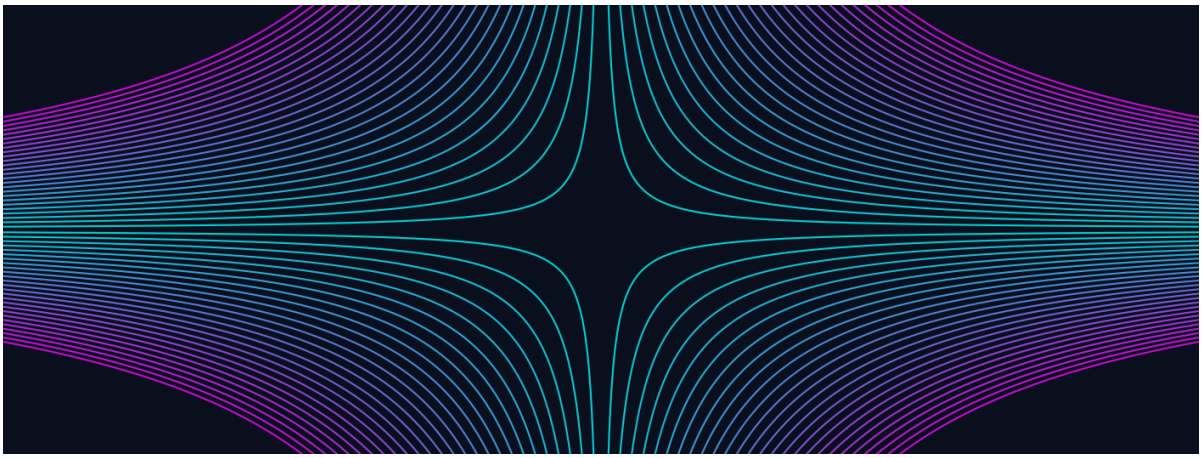
15. The rms over zeros 401–500 of the fluctuation $\delta_n=(t_n-s_n)/\mathrm{gap}_n$, i.e. of unfolded deviations from the smooth positions.

17. `qmnt.zeros(400)[-1]` gives $t_{400}\approx680$. Eq. (20.3) then gives gap ≈1.341 (it changes by only about 0.0004 per unit of t here), and one period of RE2 spans $9.065/1.341\approx6.8$ levels, inside "6–7".

19. Reflection reverses the order of spacings but keeps the multiset of spacings, so the histogram is unchanged; the reflected positions are useless for predicting the zeros. Fig. 2 tests the statistics, not positions.

20. $\frac{32}{3\pi^2}(0.05)^3=1.351\times10^{-4}$.

21. From the text, $C_0 = (1+\alpha^2)\sigma^2$, $C_1 = -\alpha\sigma^2$, $C_2 = 0$. Then $\mathrm{Cov} = -C_0 + 2C_1 - C_2 = -(1+\alpha^2+2\alpha)\sigma^2 = -(1+\alpha)^2\sigma^2$ and $\mathrm{Var}(\Delta u) = 2(C_0 - C_1) = 2(1+\alpha+\alpha^2)\sigma^2$. The ratio is $-(1+\alpha)^2/[2(1+\alpha+\alpha^2)]$; for $\alpha = 1$ it is $-4/6 = -0.667$.



The flow of the Berry–Keating Hamiltonian $H = xp$ in phase space: hyperbolas $xp = E$, the classical orbits behind a conjectured zeta operator.

Chapter 21

The Hilbert–Pólya Dream

↑ Contents

What this chapter is about

Suppose someone hands you the list 14.134725, 21.022040, 25.010858, 30.424876, ... and tells you two things. First, these numbers are the heights t_n of the nontrivial zeros of the Riemann zeta function, a function built entirely out of the prime numbers. Second, the gaps between them are observed numerically to follow, very closely, the statistics of the energy levels of a chaotic quantum system without time-reversal symmetry (GUE statistics, Chapter 20). As a physicist, you would ask at once: *which* quantum system? What Hamiltonian has these numbers as its energy levels? The hope that such a Hamiltonian exists, and that it is self-adjoint, is called the Hilbert–Pólya conjecture. If someone found it and proved that its spectrum is exactly the list of zeros, the Riemann hypothesis would follow in two lines. Nobody has found it. The conjecture is open, and so is the Riemann hypothesis.

This chapter explains what the conjecture says and why it would settle the Riemann hypothesis; why physicists find it plausible (the resemblance between the explicit formula of Chapter 18 and the trace formulas of quantum chaos, in which periodic orbits play the role of primes); the simplest concrete attempt, the Berry–Keating Hamiltonian $H = xp$, whose semiclassical level count reproduces the smooth counting function N_{sm} to leading order; and the more recent operator proposals. We end by separating two very different research programs: the *analytic* one, which specifies an operator first and then predicts zeros, and the *empirical* one, which fits an operator to zeros that are already known.

Where this shows up in the paper: the first sentence of Sec. I of the paper [1] states the conjecture; the rest of that paragraph lists the circumstantial evidence (random-matrix statistics, the trace-formula resemblance, operator candidates); Sec. I A d ("Operator proposals") and the first paragraph of Sec. VI say explicitly that the paper's results constrain only the empirical program. This chapter gives you the background needed to read those passages with understanding and to see exactly where the paper's claims stop.

What you need

- Chapter 10: the Hamiltonian $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$ in units $\hbar = 2m = 1$; Hermitian (self-adjoint) operators have real eigenvalues; de Broglie relation $p = \hbar k$.
- Chapter 16: the zeta function, its nontrivial zeros ρ_n, the critical strip and critical line, the symmetry of the zeros, the precise statement of the Riemann hypothesis (RH) and its open status.
- Chapter 17: the counting function $N(t)$, the smooth part $N_{\mathrm{sm}}(t)$ of paper Eq. (1), the mean spacing $2\pi/\log(t/2\pi)$.
- Chapter 18: the truncated explicit formula $N_{\mathrm{fl}}(t)$ of paper Eq. (4), its heuristic status, and the idea that prime powers act as frequencies $m \log p$ in t.
- Chapter 20: nearest-neighbor spacings, GUE statistics, and the link between real symmetric matrices (GOE) and complex Hermitian matrices (GUE).
- Through those chapters: complex numbers and $a^s = e^{s \log a}$ (Chapter 2), Fourier series (Chapter 5), eigenvalues of 2×2 matrices (Chapter 8).
- Chapters 11 and 12 are referred to for context only. The few results quoted from them (harmonic-oscillator levels, the WKB phase at a turning point) are stated where they are used, and nothing here depends on their derivations.
- First-year physics: energy conservation, potential-energy curves and turning points, the period of an oscillation, standing waves on a string.

21.1 What the conjecture says

★ Did you know? *A Conjecture Without a Paper*

Neither Hilbert nor Pólya ever published the idea that the zeta zeros are eigenvalues of a self-adjoint operator. Our main evidence is a 1982 letter from George Pólya to Andrew Odlyzko. Pólya recalled that Edmund Landau had asked him in Göttingen, around 1912 to 1914, for a physical reason the hypothesis should hold. Hilbert's role rests on oral tradition.

21.1.1 From zeros to heights

Recall from Chapter 16 that the nontrivial zeros of $\zeta(s)$ lie in the critical strip $0 < \mathrm{Re}\, s < 1$, and that the Riemann hypothesis asserts that every one of them lies on the critical line $\mathrm{Re}\, s = \frac{1}{2}$. Write a nontrivial zero as $\rho = \sigma + i\tau$, with σ and τ real. We now define, for *every* nontrivial zero, a number t by

$$\rho = \tfrac{1}{2} + it, \qquad \text{equivalently} \qquad t = \frac{\rho - \frac{1}{2}}{i} = -i\left(\rho - \tfrac{1}{2}\right). \tag{21.1}$$

Nothing in this definition assumes RH: t is simply a relabeling of ρ, and for a general zero it is a complex number. Let us compute it. Substituting $\rho = \sigma + i\tau$,

$$\begin{aligned} t &= -i\left(\sigma - \tfrac{1}{2} + i\tau\right) \\ &= -i\left(\sigma - \tfrac{1}{2}\right) - i \cdot i\,\tau \\ &= \tau - i\left(\sigma - \tfrac{1}{2}\right) \quad (\text{since } -i \cdot i = -i^2 = 1). \end{aligned} \tag{21.2}$$

So the real part of t is the height τ of the zero, and the imaginary part of t is $\frac{1}{2} - \sigma$, the (signed) distance of the zero from the critical line. This gives the key equivalence:

$$t \text{ is real} \iff \sigma = \tfrac{1}{2}. \tag{21.3}$$

In this language RH says: *all the numbers t defined by Eq. (21.1) are real.* For the zeros on the line, t is the familiar height t_n of the paper's notation, $\rho_n = \frac{1}{2} + it_n$. Figure 21.1 shows the map.

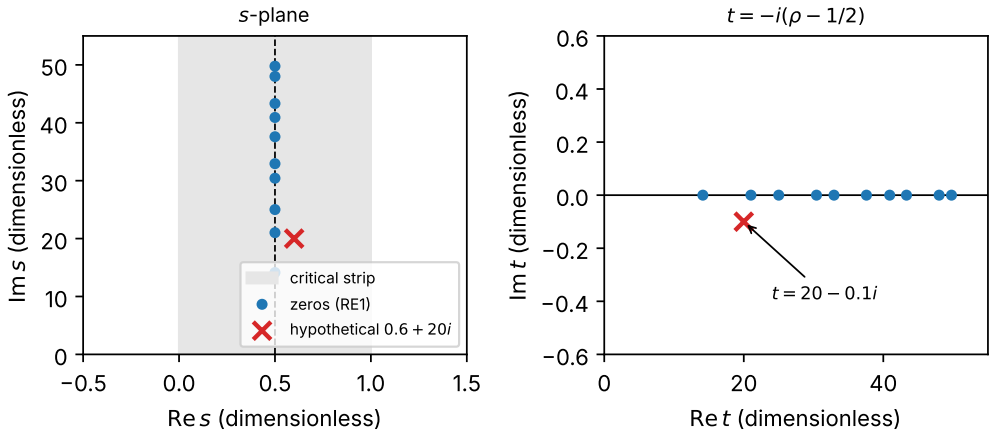


Figure 21.1: Left: the complex s-plane near the critical strip $0 < \operatorname{Re} s < 1$ (shaded), with the first ten nontrivial zeros on the critical line $\operatorname{Re} s = \frac{1}{2}$ (dashed) and a hypothetical zero at $0.6 + 20i$ off the line (cross; no such zero exists at this height). Right: the same points after the relabeling $t = -i(\rho - \frac{1}{2})$ of Eq. (21.1). Zeros on the line land on the real t axis; the hypothetical off-line zero lands at $t = 20 - 0.1i$, below the axis. Both axes are dimensionless.

Sample Problem 21.1: Relabeling a zero

(a) Find t for the first zero $\rho_1 = \frac{1}{2} + 14.134725\,i$. (b) Find t for a hypothetical zero at $\rho = 0.6 + 20i$, and say how far it would sit from the real t axis.

Step 1: For (a), $\rho_1 - \frac{1}{2} = 14.134725\,i$. Multiply by $-i$: $t = -i \cdot 14.134725\,i = 14.134725\,(-i^2) = 14.134725$. The value is real, as Eq. (21.3) predicts for a zero with $\sigma = \frac{1}{2}$.

Step 2: For (b), $\sigma = 0.6$ and $\tau = 20$. Eq. (21.2) gives $t = 20 - i(0.6 - 0.5) = 20 - 0.1i$.

Step 3: The distance from the real axis is $|\operatorname{Im} t| = \left|\frac{1}{2} - \sigma\right| = 0.1$.

$$\boxed{t_1 = 14.134725, \qquad t = 20 - 0.1i \ \text{ (distance 0.1 from the real axis)}}$$

The symmetry of the t values. Chapter 16 showed that nontrivial zeros come in symmetric sets: if ρ is a zero, so are $1 - \rho$ and the complex conjugate $\bar{\rho}$. Let us see what this does to t. For $1 - \rho$:

$$t(1-\rho) = -i\left(1 - \rho - \tfrac{1}{2}\right) = -i\left(\tfrac{1}{2} - \rho\right) = +i\left(\rho - \tfrac{1}{2}\right) = -t(\rho). \tag{21.4}$$

For $\bar{\rho} = \sigma - i\tau$, Eq. (21.2) with $\tau \to -\tau$ gives $t(\bar{\rho}) = -\tau - i(\sigma - \frac{1}{2})$, which is $-\overline{t(\rho)}$, because $\overline{t(\rho)} = \tau + i(\sigma - \frac{1}{2})$. So the full set of t values is symmetric under $t \to -t$ and under $t \to \bar{t}$. On the critical line this just says that the zeros come in pairs $\pm t_n$. An operator that wants to have "the zeros" as its spectrum must therefore either have the symmetric spectrum $\{\pm t_n\}$ or be designed for the positive heights only; both choices appear in the literature.

21.1.2 The conjecture

Conjecture (Hilbert–Pólya; open). There exists a self-adjoint operator H, acting on some space of functions, whose eigenvalues are exactly the numbers t of Eq. (21.1), one for each nontrivial zero of ζ, counted with multiplicity.

Status: conjecture. No such operator is known. Neither Hilbert nor Pólya published the idea; it is known through later recollections, and its history and the many attempts to realize it are reviewed in [11]. The statement is also deliberately vague about *which* space of functions and *which* operator: making it precise is part of the problem.

Why would such an operator prove RH? The argument has two links.

Link 1: self-adjoint operators have real eigenvalues. You met this in Chapter 8 for matrices and in Chapter 10 for operators. Here is the matrix version again, line by line. Let H be a Hermitian matrix, meaning $H^\dagger = H$, where $\dagger$ is the conjugate transpose. Let $v \neq 0$ be an eigenvector, $Hv = \lambda v$. Multiply on the left by $v^\dagger$:

$$v^\dagger H v = \lambda\, v^\dagger v. \tag{21.5}$$

The number $v^\dagger v = \sum_j |v_j|^2$ is real and positive because $v \neq 0$. The number $q = v^\dagger H v$ is a 1×1 matrix, that is, a single number. Transposing a 1×1 matrix does nothing, so its conjugate transpose $q^\dagger$ is just its complex conjugate $\bar{q}$. Therefore:

$$\bar{q} = q^\dagger = (v^\dagger H v)^\dagger = v^\dagger H^\dagger (v^\dagger)^\dagger = v^\dagger H^\dagger v = v^\dagger H v = q, \tag{21.6}$$

using $(ABC)^\dagger = C^\dagger B^\dagger A^\dagger$ and $H^\dagger = H$. So q is real, and $\lambda = q/(v^\dagger v)$ is a real number divided by a positive real number: real.

Link 2: real t means the zero is on the line. This is Eq. (21.3).

Chaining the links: if every t is an eigenvalue of a self-adjoint H, every t is real, so every nontrivial zero has $\sigma = \frac{1}{2}$, which is RH. Notice what the argument needs and what it does not. It needs the operator to be self-adjoint *and* needs a proof that its eigenvalues are *all* the t values. It does not need any particular physical interpretation; and strictly, the conclusion only uses the reality of the eigenvalues, so any operator with a proven real spectrum equal to the set of t's would do. Self-adjointness is simply the most familiar guarantee of reality.

Sample Problem 21.2: Real eigenvalues from Hermiticity, and their absence without it

(a) Find the eigenvalues of the Hermitian matrix $H = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$. (b) Find the eigenvalues of the non-Hermitian matrix $B = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$.

Step 1: Check Hermiticity of H. The conjugate transpose swaps the off-diagonal entries and conjugates them: the $(1,2)$ entry of $H^\dagger$ is $\overline{1+i} = 1-i$, equal to H_{12}. The diagonal entries are real. So $H^\dagger = H$.

Step 2: The characteristic polynomial of a 2×2 matrix is $\lambda^2 - (\text{trace})\lambda + \det = 0$ (Chapter 8). Here the trace is $2+3=5$ and $\det H = 2\cdot 3 - (1-i)(1+i) = 6 - (1-i^2) = 6-2 = 4$.

Step 3: Solve $\lambda^2 - 5\lambda + 4 = 0$: $\lambda = \frac{1}{2}\left(5 \pm \sqrt{25-16}\right) = \frac{1}{2}(5 \pm 3)$, so $\lambda = 1$ or 4. Both real.

Step 4: For B, the trace is 2 and $\det B = 1\cdot 1 - 2\cdot(-1) = 3$. Solve $\lambda^2 - 2\lambda + 3 = 0$: $\lambda = 1 \pm \sqrt{1-3} = 1 \pm i\sqrt{2}$. Complex. B is real but not symmetric ($B_{12} = 2 \neq B_{21} = -1$), so Link 1 does not apply.

$$\boxed{H:\ \lambda = 1,\ 4 \qquad B:\ \lambda = 1 \pm i\sqrt{2}}$$

Checkpoint 21.1

Does a matrix need to be Hermitian in order to have only real eigenvalues? Think of the upper-triangular matrix $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

Answer: No. That matrix is not Hermitian (its $(1,2)$ and $(2,1)$ entries are 1 and 0), yet its characteristic polynomial is $(1-\lambda)(2-\lambda)-1\cdot 0=(1-\lambda)(2-\lambda)$, with roots 1 and 2, both real. Hermiticity is sufficient for real eigenvalues, not necessary. This is why some operator proposals (Section 21.5) try non-Hermitian operators with other symmetries.

21.1.3 The infinite-dimensional subtlety

For matrices, Link 1 is airtight. For differential operators acting on functions, the analogue of "$H^\dagger = H$" involves integration by parts, and integration by parts produces boundary terms. Whether the boundary terms vanish depends on which functions you allow. Here is the cleanest example of what can go wrong.

Sample Problem 21.3: A Hermitian-looking operator with complex eigenvalues

Consider the momentum operator $\hat{p} = -i\,\mathrm{d}/\mathrm{d}x$ (units $\hbar = 1$) acting on functions on the half-line $x \geq 0$. (a) Show that $f(x) = e^{ikx}$ with $k = 2 + i$ satisfies $\hat{p}f = kf$ and is normalizable. (b) Show where the usual argument for real eigenvalues breaks.

Step 1: Differentiate: $\mathrm{d}f/\mathrm{d}x = ik\,e^{ikx}$, so $\hat{p}f = -i\cdot ik\,e^{ikx} = k\,e^{ikx} = kf$. The eigenvalue is $k = 2 + i$, which is complex.

Step 2: Normalizability: $\left|e^{ikx}\right|^2 = \left|e^{i(2+i)x}\right|^2 = |e^{2ix}|^2\,|e^{-x}|^2 = 1\cdot e^{-2x}$. So $\int_0^\infty |f|^2\,\mathrm{d}x = \int_0^\infty e^{-2x}\mathrm{d}x = \frac{1}{2}$, finite.

Step 3: The "Hermiticity" of $\hat{p}$ means $\int_0^\infty \bar{g}\,(\hat{p}f)\,\mathrm{d}x = \int_0^\infty \overline{(\hat{p}g)}\,f\,\mathrm{d}x$. Integrate the left side by parts:

$$\begin{aligned}\int_0^\infty \bar{g}\,(-if')\,\mathrm{d}x &= -i\,[\bar{g}f]_0^\infty + i\int_0^\infty \bar{g}'\,f\,\mathrm{d}x \\ &= i\,\bar{g}(0)f(0) + \int_0^\infty \overline{(-ig')}\,f\,\mathrm{d}x,\end{aligned}$$

where we used that $f, g \to 0$ at infinity and $\overline{-ig'} = i\bar{g}'$. The extra term $i\bar{g}(0)f(0)$ vanishes only if $f(0) = 0$ or $g(0) = 0$.

Step 4: Our eigenfunction has $f(0) = 1 \neq 0$, so for it the boundary term does not vanish, the operator is not symmetric on the space containing f, and Link 1 fails.

$$\boxed{\text{eigenvalue } 2+i, \quad \int_0^\infty |f|^2\,\mathrm{d}x = \tfrac{1}{2}, \text{ broken by the boundary term } i\bar{g}(0)f(0)}$$

Taken on trust

Symmetric versus self-adjoint. In infinite dimensions mathematicians distinguish a *symmetric* operator (the integration-by-parts identity holds for a chosen class of nice functions) from a *self-adjoint* one (it is symmetric *and* its domain is exactly as large as the domain of its adjoint, so no eigenfunctions can escape through boundary terms). Only self-adjoint operators come with the full guarantee that their spectrum is real. The precise definitions belong to functional analysis and are beyond this book. What matters here: any operator proposed for the Hilbert–Pólya program must come with a proof of

self-adjointness (or some other proof of reality) on a specified domain, and proposals are often most delicate exactly at this point. Half-line operators such as $\hat{p}$ above are relevant because the xp models of Section 21.4 live naturally on $x > 0$; see [36] and [11] for discussions of these domain questions.

21.1.4 What else the operator must do

Suppose you were designing the operator. Besides self-adjointness, the facts collected in earlier chapters impose requirements. These are *heuristic* design targets, discussed in [10] and reviewed in [11], not theorems.

1. **The right average density.** The number of eigenvalues below E must grow like $N_{\text{sm}}(E) = \frac{E}{2\pi}\log\frac{E}{2\pi} - \frac{E}{2\pi} + \frac{7}{8}$ (paper Eq. (1), Chapter 17). This is unusual: a particle in a box has $N(E) \propto \sqrt{E}$, an oscillator has $N(E) \propto E$; the zeros grow like $E \log E$.
2. **The right fluctuations.** The deviation of the true count from N_{sm} must reproduce the prime-power oscillations of the explicit formula (Chapter 18). Section 21.3 develops what this suggests.
3. **The right statistics.** Unfolded spacings are observed numerically to follow GUE statistics closely (a numerical observation, supported by Montgomery's partial theoretical result; Chapter 20; paper Fig. 2). In random-matrix theory, complex Hermitian matrices (GUE) model systems *without* time-reversal symmetry, while real symmetric matrices (GOE) model systems *with* it. If the zeros are the spectrum of a chaotic quantum system, the heuristic reading is that its Hamiltonian is not time-reversal invariant.

Time reversal, concretely. Run a film of a classical motion backward: each position is unchanged, each velocity (hence momentum) is reversed. So time reversal acts as $x \to x$, $p \to -p$. A classical Hamiltonian $H(x,p)$ is time-reversal invariant if $H(x,-p) = H(x,p)$. For $H = p^2 + V(x)$ (our units, $2m = 1$) this holds, because $(-p)^2 = p^2$. In quantum mechanics, for a particle without spin, time reversal acts on wavefunctions by complex conjugation, $\psi \to \bar{\psi}$. Here is why conjugation is the right operation. The plane wave e^{ikx} has momentum $p = k$. Its conjugate e^{-ikx} has momentum $-k$, so conjugation reverses the momentum. More generally, $\hat{p} = -i\,\mathrm{d}/\mathrm{d}x$ contains an explicit i. Therefore $\overline{\hat{p}\psi} = \overline{-i\psi'} = i\bar{\psi}' = -\hat{p}\bar{\psi}$, while multiplication by x is unchanged by conjugation. Conjugation thus keeps x and flips p, exactly as the backward film does. If V is real, then $H = -\mathrm{d}^2/\mathrm{d}x^2 + V$ has real coefficients, and taking the complex conjugate of $H\psi = E\psi$ (with E real) gives $H\bar{\psi} = E\bar{\psi}$: the conjugate of a solution is again a solution. In matrix form, a real potential on a grid gives a real symmetric matrix (Chapter 8), the GOE symmetry class.

Pitfall

Do not conclude from item 3 that "the zeros are not the spectrum of any real potential." The random-matrix heuristic is about *generic chaotic* systems. A one-dimensional system $p^2 + V(x)$ is not chaotic at all (Section 21.3), so the heuristic simply does not apply to it, and, as the next Sample Problem shows, any finite list of real numbers can be the spectrum of a real symmetric matrix. Statistics constrain natural operators, not fitted ones.

21.1.5 The trap of finite lists

There is a trivial way to "realize" any finite list of zeros as a spectrum, and understanding why it is trivial is the key to the last section of this chapter and to the whole paper.

Sample Problem 21.4: An operator for two zeros

Construct a 2×2 real symmetric matrix, not diagonal, whose eigenvalues are $t_1 = 14.134725$ and $t_2 = 21.022040$ (RE1). Use a rotation by $\theta = 30°$.

Step 1: Start from the diagonal matrix $D = \operatorname{diag}(t_1, t_2)$; its eigenvalues are t_1, t_2 with eigenvectors along the axes.

Step 2: Let $R = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}$ with $c = \cos 30° = \sqrt{3}/2$, $s = \sin 30° = \frac{1}{2}$. R is orthogonal, $R^\top R = I$. Set $H = RDR^\top$. If $De = \lambda e$ then $H(Re) = RDR^\top Re = RDe = \lambda Re$, so H has the same eigenvalues. Also $H^\top = (R^\top)^\top D^\top R^\top = RDR^\top = H$, so H is symmetric.

Step 3: Multiply out. $(RDR^\top)_{ij} = \sum_k R_{ik} d_k R_{jk}$, so

$$H_{11} = c^2 t_1 + s^2 t_2 = 0.75(14.134725) + 0.25(21.022040) = 15.856554,$$
$$H_{22} = s^2 t_1 + c^2 t_2 = 0.25(14.134725) + 0.75(21.022040) = 19.300211,$$
$$H_{12} = cs\, t_1 + (-s)c\, t_2 = cs(t_1 - t_2) = 0.4330127 \times (-6.887315) = -2.982295.$$

Step 4: Check: trace $= 35.156765 = t_1 + t_2$, as it must be.

$$\boxed{H = \begin{pmatrix} 15.856554 & -2.982295 \\ -2.982295 & 19.300211 \end{pmatrix}}$$

The same recipe works for 10^4 zeros: a diagonal $10^4 \times 10^4$ matrix, rotated by any orthogonal matrix you like. Such a matrix is "a self-adjoint operator whose spectrum is the first 10^4 zeros." It proves nothing, predicts nothing about zero $10^4 + 1$, and contains no information that was not typed into it. The Hilbert–Pólya conjecture has content only because it asks for an operator that is *specified independently of the zeros* (by geometry, by the primes, by a physical model) and whose spectrum can then be *proved* to be all of them.

Checkpoint 21.2

A colleague announces a real symmetric 1000×1000 matrix whose eigenvalues agree with $t_1, \dots, t_{1000}$ to twelve digits. Is this evidence for RH, or for the existence of a Hilbert–Pólya operator?

Answer: Neither, on its own. The diagonal matrix $\operatorname{diag}(t_1, \dots, t_{1000})$ already does this, and every rotation of it does too. Agreement with data that were used to build the matrix shows only that the construction can represent that data. What would count is a prediction: an operator fixed without looking at the zeros, or at least one that predicts zeros it was not built from.

Key Ideas

- Relabel each nontrivial zero by $t = -i(\rho - \frac{1}{2})$; then $\operatorname{Im} t = \frac{1}{2} - \operatorname{Re}\rho$, so RH is the statement that every t is real (Eq. (21.3)).
- Hilbert–Pólya conjecture (open): the t's are the eigenvalues of a self-adjoint operator. With a proof that the spectrum is exactly the t's, real eigenvalues would give RH.
- In infinite dimensions, "looks Hermitian" is not enough; boundary terms can produce complex eigenvalues. Self-adjointness on a specified domain is required.
- Heuristic design targets: counting function N_{sm}, prime-power fluctuations, GUE statistics (suggesting no time-reversal symmetry).

- Any finite list of reals is the spectrum of some symmetric matrix; reproducing finitely many zeros is not evidence by itself.

21.2 Counting quantum states with classical phase space

To compare operators with the zeros we need a quick way to estimate how many eigenvalues an operator has below an energy E. Physics offers a beautiful one: count area in classical phase space. Chapter 12 develops a refined version (Bohr–Sommerfeld quantization with its $\frac{1}{2}$); here we need only the leading rule, which we derive from the particle in a box.

21.2.1 Phase space and Hamilton's equations

The *phase space* of a particle on a line is the plane with coordinates (x, p): position horizontally, momentum vertically. A state of classical motion is a point; as time runs, the point moves along a curve. For a conserved energy the curve is a level curve $H(x, p) = E$ of the Hamiltonian, the energy written as a function of x and p.

The motion along the curve is given by *Hamilton's equations*. We need a *partial derivative*: $\partial H/\partial p$ means "differentiate H with respect to p while holding x fixed," and similarly for $\partial H/\partial x$. Hamilton's equations are

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \frac{\partial H}{\partial p}, \qquad \frac{\mathrm{d}p}{\mathrm{d}\tau} = -\frac{\partial H}{\partial x}. \tag{21.7}$$

(Here τ is time. We write τ rather than t because in this book t is reserved for the height of a zero; the $\tau = \operatorname{Im}\rho$ of Section 21.1 does not appear again, so from here on τ always means time.) Check that these reproduce Newton for $H = p^2/(2m) + V(x)$: the first gives $\mathrm{d}x/\mathrm{d}\tau = p/m$, the definition of momentum; the second gives $\mathrm{d}p/\mathrm{d}\tau = -\mathrm{d}V/\mathrm{d}x$, which is Newton's second law with force $-\mathrm{d}V/\mathrm{d}x$. In the paper's units $2m = 1$, $H = p^2 + V$ and $\mathrm{d}x/\mathrm{d}\tau = 2p$.

21.2.2 One state per area $2\pi\hbar$

Take a box of length ℓ with hard walls. Standing waves (Chapter 10, and standing waves on a string from first-year physics) fit an integer number of half-wavelengths: $\ell = k\lambda/2$, $k = 1, 2, 3, \ldots$ By de Broglie, $p = h/\lambda = 2\pi\hbar/\lambda$ (recall $\hbar = h/2\pi$), so the allowed momenta are

$$p_k = \frac{2\pi\hbar}{\lambda_k} = \frac{2\pi\hbar\, k}{2\ell} = \frac{\pi\hbar k}{\ell}. \tag{21.8}$$

The number of states with momentum magnitude at most P is the number of positive integers k with $\pi\hbar k/\ell \le P$, i.e. the integer part of $P\ell/(\pi\hbar)$.

Now compute the phase-space area of the classical region with the same energy bound: $0 \le x \le \ell$ and $-P \le p \le P$ (a classical particle bounces, so both signs of p occur). It is a rectangle of area $\ell \cdot 2P$. Divide by $2\pi\hbar$:

$$\frac{2P\ell}{2\pi\hbar} = \frac{P\ell}{\pi\hbar}, \tag{21.9}$$

exactly the count, up to rounding down. The box is a motivating example only; extending the rule to other Hamiltonians is a heuristic generalization, whose status is stated in the box below. The semiclassical counting rule reads:

$$N(E) \approx \frac{\text{area of } \{(x,p) : H(x,p) \le E\}}{2\pi\hbar}. \tag{21.10}$$

Taken on trust

Status of Eq. (21.10). We derived it for the box; for a general smooth one-dimensional potential it is the leading term of the WKB counting function of Chapter 12, valid for large N with an error of order one level (the $\frac{1}{2}$ in Bohr–Sommerfeld is part of that error). In higher dimensions the analogous statement, with $(2\pi\hbar)^d$, is Weyl's law; its proof is beyond this book. For our purposes the rule is a *heuristic* that is accurate to leading order at high energy.

From now on $\hbar = 1$, so each state occupies area 2π. (The area rule needs only $\hbar = 1$, which the paper's units $\hbar = 2m = 1$ include. The paper's second convention, $2m = 1$, is a separate matter: it makes $H = p^2 + V$, which changes the shape of the region $H \le E$ but not the rule "area divided by 2π.")

Sample Problem 21.5: Counting states in the paper's box

Use RE3: an infinite well on $[-L, L]$ with $L = 44.883$, in units $\hbar = 2m = 1$, so $E = p^2$. Estimate the number of states with $E \le 1000$ by Eq. (21.10), and compare with the exact count from $E_k = (k\pi/2L)^2$ (the squares of the allowed momenta $p_k = \pi k/\ell$ derived above, with $\ell = 2L$ and $\hbar = 1$).

Step 1: With $E = p^2$, the bound $E \le 1000$ means $|p| \le P = \sqrt{1000} = 31.6228$.

Step 2: The region is $-L \le x \le L$, $|p| \le P$: area $= 2L \cdot 2P = 4LP$. Divide by 2π: $N \approx 2LP/\pi = 89.766 \times 31.6228/\pi = 903.57$.

Step 3: Exact count: $E_k \le 1000$ means $k\pi/(2L) \le \sqrt{1000}$, i.e. $k \le 2L\sqrt{1000}/\pi = 903.57$. So $k = 1, \ldots, 903$.

$$\boxed{N_{\text{rule}} \approx 903.57, \qquad N_{\text{exact}} = 903}$$

For comparison, when the Wu–Sprung potential is placed in this same box, it pushes the levels up and only about 650 remain below 1000: the paper reports 652 basis states with $E \le 1000$ for its discretized smooth potential (Sec. II E). That potential was built so that its semiclassical count follows $N_{\text{sm}}(E)$ (Chapter 22); the value $N_{\text{sm}}(1000) \approx 648.62$ is computed in a Sample Problem of Section 21.4. The difference of about three levels between 652 and $N_{\text{sm}}(1000)$ is plausibly mainly a discretization effect. The 652 states are eigenvalues of a finite-difference matrix, and finite-difference levels lie below the continuum ones by an amount that grows with level index (Chapter 23). The paper reports that on its working grid this dispersion already reaches about two mean spacings at levels 401–500 (Sec. II C). The paper's Richardson-extrapolated levels, by contrast, match the smooth zeros to within 0.063 spacings (Sec. II B).

Checkpoint 21.3

A classical region of phase space has area 20π (units $\hbar = 1$). About how many quantum states does it hold?

Answer: About $20\pi/(2\pi) = 10$.

21.2.3 Area, period and density of levels

Two facts about closed orbits will be used repeatedly. Let $A(E)$ be the phase-space area enclosed by the closed orbit $H = E$, and $T_\gamma(E)$ its period (the subscript γ labels the orbit; do not confuse it with the paper's training window T).

Fact 1: $\mathrm{d}A/\mathrm{d}E = T_\gamma$**.** For $H = p^2 + V(x)$ with turning points $x_1 < x_2$ (where $V = E$, as

on any potential-energy curve), the orbit is the curve $p = \pm\sqrt{E - V(x)}$, so

$$A(E) = 2\int_{x_1(E)}^{x_2(E)} \sqrt{E - V(x)}\,\mathrm{d}x. \tag{21.11}$$

Differentiate with respect to E. Both the limits and the integrand depend on E, so we need the rule for differentiating such an integral (the Leibniz rule):

$$\frac{\mathrm{d}}{\mathrm{d}E}\int_{a(E)}^{b(E)} g(x,E)\,\mathrm{d}x = g(b,E)\,b'(E) - g(a,E)\,a'(E) + \int_a^b \frac{\partial g}{\partial E}\,\mathrm{d}x. \tag{21.12}$$

Here is where it comes from. Write $F(a,b,E) = \int_a^b g(x,E)\,\mathrm{d}x$.

- By the fundamental theorem of calculus, $\partial F/\partial b = g(b,E)$ and $\partial F/\partial a = -g(a,E)$.
- Differentiating under the integral sign gives $\partial F/\partial E = \int_a^b (\partial g/\partial E)\,\mathrm{d}x$. We take this step on trust; it holds for the integrands met here and is stated in [22].
- The chain rule for $F(a(E), b(E), E)$ adds the three contributions, which gives Eq. (21.12).

Apply the rule with $g = \sqrt{E - V(x)}$, $a = x_1(E)$ and $b = x_2(E)$. The integrand vanishes at both limits ($V = E$ there), so the two boundary terms are zero and only the derivative of the integrand survives:

$$\frac{\mathrm{d}A}{\mathrm{d}E} = 2\int_{x_1}^{x_2} \frac{1}{2\sqrt{E - V(x)}}\,\mathrm{d}x = \int_{x_1}^{x_2} \frac{\mathrm{d}x}{\sqrt{E - V(x)}}. \tag{21.13}$$

(The integrand blows up at the turning points like $1/\sqrt{\text{distance}}$, which is integrable.) The speed is $\mathrm{d}x/\mathrm{d}\tau = 2p = 2\sqrt{E - V}$ from Eq. (21.7), so the time to go from x_1 to x_2 is $\int \mathrm{d}x/(2\sqrt{E - V})$, and the period (there and back) is twice that, $\int \mathrm{d}x/\sqrt{E - V}$. Hence $\mathrm{d}A/\mathrm{d}E = T_\gamma(E)$.

Fact 2: the mean level density is $T_\gamma/2\pi$. Differentiate Eq. (21.10): $\mathrm{d}N/\mathrm{d}E \approx (\mathrm{d}A/\mathrm{d}E)/(2\pi) = T_\gamma(E)/2\pi$. The mean spacing between levels is the reciprocal, $2\pi/T_\gamma$. In words: slow classical orbits mean densely packed quantum levels.

The *action* of a closed orbit is $S_\gamma(E) = \oint p\,\mathrm{d}x$, which for a one-dimensional orbit is exactly the enclosed area $A(E)$. Fact 1 therefore also says $\mathrm{d}S_\gamma/\mathrm{d}E = T_\gamma$. A consequence we will need: *if the period is independent of energy, then $S_\gamma(E) = ET_\gamma$ up to a constant.*

Key Ideas

- Phase space is the (x, p) plane; motion follows Hamilton's equations (21.7).
- Semiclassical rule (heuristic, leading order): one quantum state per phase-space area $2\pi\hbar$, Eq. (21.10).
- For a closed orbit, $\mathrm{d}A/\mathrm{d}E = \mathrm{d}S/\mathrm{d}E = T_\gamma$, so the mean level density is $T_\gamma/2\pi$.

21.3 Trace formulas: periodic orbits and primes

The most compelling reason physicists take the Hilbert–Pólya idea seriously is a resemblance of *formulas*. In quantum chaos, the fluctuating part of the level count is a sum over classical periodic orbits. In number theory, the fluctuating part of the zero count is a sum over primes. Let us build up to this comparison with an example you can check completely.

21.3.1 A trace formula you can verify: equally spaced levels

Take the equally spaced spectrum $E_k = k - \frac{1}{2}$, $k = 1, 2, 3, \ldots$ This is the harmonic-oscillator spectrum $(k-\frac{1}{2})\hbar\omega$ in units where $\hbar\omega = 1$. We quote these levels; they are derived in Chapter 11. Nothing below depends on that derivation, since the trace formula is worked out for the list $E_k = k - \frac{1}{2}$ itself. Its counting function $N(E)$, the number of levels with $E_k \le E$, is a staircase that jumps by 1 at $E = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$ For $E \ge 0$, $N(E) = \lfloor E + \frac{1}{2} \rfloor$, where $\lfloor y \rfloor$ is the integer part ("floor") of y.

We now write this staircase as "smooth part plus oscillations," exactly as $N(t) = N_{\rm sm}(t) + S(t)$ in Chapter 17. Write $\{y\} = y - \lfloor y \rfloor$ for the fractional part. Then $\lfloor y \rfloor = y - \{y\} = y - \frac{1}{2} - \left(\{y\} - \frac{1}{2}\right)$. The function $f(y) = \{y\} - \frac{1}{2}$ is the sawtooth, periodic with period 1. Its Fourier series (Chapter 5) on $0 < y < 1$ has coefficients

$$a_r = 2\int_0^1 \left(y - \tfrac{1}{2}\right)\cos(2\pi r y)\,\mathrm{d}y, \qquad b_r = 2\int_0^1 \left(y - \tfrac{1}{2}\right)\sin(2\pi r y)\,\mathrm{d}y, \tag{21.14}$$

for $r \ge 1$ (the Fourier coefficients for a function of period 1, Chapter 5), and $a_0 = 0$ because $\int_0^1 (y - \frac{1}{2})\mathrm{d}y = 0$. For b_r, the constant $\frac{1}{2}$ integrates to zero against a full period of the sine, and integration by parts gives

$$\int_0^1 y\sin(2\pi r y)\,\mathrm{d}y = \left[-\frac{y\cos(2\pi r y)}{2\pi r}\right]_0^1 + \int_0^1 \frac{\cos(2\pi r y)}{2\pi r}\,\mathrm{d}y = -\frac{1}{2\pi r} + 0, \tag{21.15}$$

so $b_r = -1/(\pi r)$. For a_r, integration by parts gives $\left[(y - \frac{1}{2})\sin(2\pi r y)/(2\pi r)\right]_0^1 - \int_0^1 \sin(2\pi r y)\mathrm{d}y/(2\pi r) = 0 - 0$, so $a_r = 0$ for all $r \ge 1$.

Taken on trust

The sawtooth jumps at every integer. We take on trust Dirichlet's theorem (stated in [22]; the proof is found in standard texts on Fourier analysis): the Fourier series of a piecewise-smooth periodic function converges to the function at every point where the function is continuous, and to the midpoint of the jump at a discontinuity, with the Gibbs overshoot nearby (Chapter 5).

Hence, at every non-integer y,

$$\{y\} - \tfrac{1}{2} = -\frac{1}{\pi}\sum_{r=1}^{\infty}\frac{\sin(2\pi r y)}{r}. \tag{21.16}$$

Put $y = E + \frac{1}{2}$. Using $\sin(2\pi r E + \pi r) = (-1)^r \sin(2\pi r E)$,

$$\begin{aligned} N(E) = \lfloor E + \tfrac{1}{2} \rfloor &= E + \tfrac{1}{2} - \tfrac{1}{2} + \frac{1}{\pi}\sum_{r=1}^{\infty}\frac{\sin\left(2\pi r(E + \frac{1}{2})\right)}{r} \\ &= \underbrace{E}_{\text{smooth}} + \underbrace{\frac{1}{\pi}\sum_{r=1}^{\infty}\frac{(-1)^r}{r}\sin(2\pi r E)}_{\text{oscillating}}. \end{aligned} \tag{21.17}$$

This is an exact *trace formula* for this spectrum, valid at every E that is not a level. Now interpret it classically. In units $\hbar = 2m = 1$, an oscillator with $\hbar\omega = 1$ has Hamiltonian $H = p^2 + \omega^2 x^2/4$ with $\omega = 1$, that is $H = p^2 + x^2/4$. Its orbit at energy E is an ellipse of area $A = 2\pi E/\omega = 2\pi E$ (Problem 19 derives this for general ω). So:

- the smooth part $E = A/2\pi$ is the phase-space count of Eq. (21.10);

- each oscillating term has phase $r \cdot 2\pi E = rS_\gamma(E)$, where $S_\gamma = A$ is the action of the periodic orbit and $r = 1, 2, 3, \ldots$ counts how many times the orbit is traversed (its *repetitions*);
- the amplitude is $1/r$, and the factor $(-1)^r$ is a phase shift of π per traversal. We derived it from the shift $y = E + \frac{1}{2}$ inside the floor function; that it agrees with a phase loss of $\pi/2$ at each of the two turning points (Chapter 12) is a consistency remark, not a separate derivation.

The discrete levels are encoded in a single periodic orbit and its repetitions. Figure 21.2 shows the truncated sums building up the staircase.

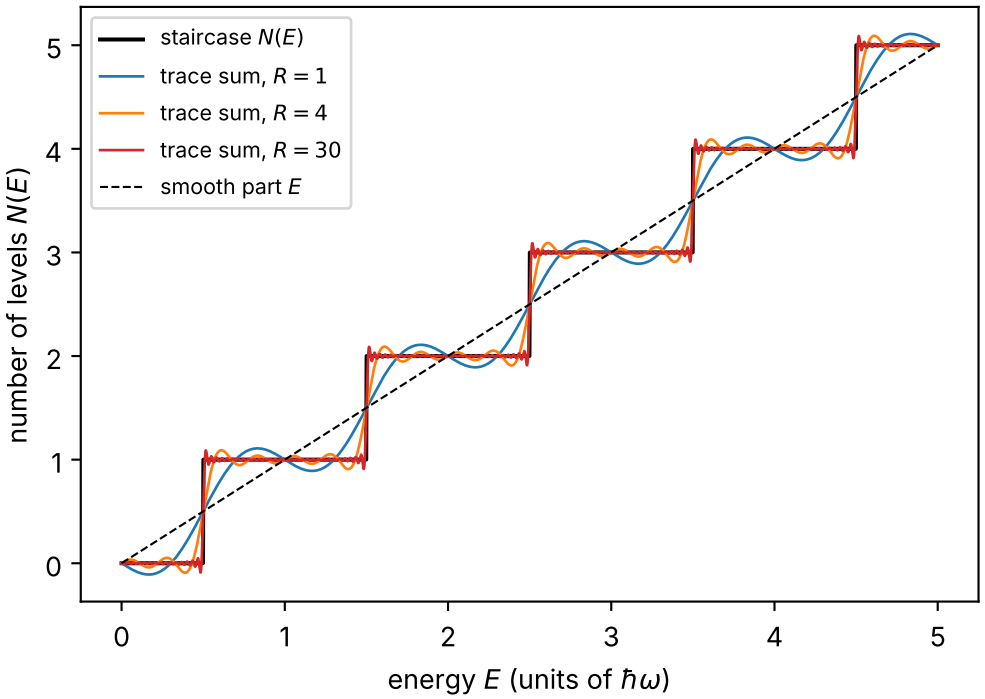


Figure 21.2: The level staircase $N(E) = \lfloor E + \frac{1}{2} \rfloor$ of the equally spaced spectrum $E_k = k - \frac{1}{2}$ (black), and the trace formula (21.17) truncated after $R = 1$, 4 and 30 repetitions of the single periodic orbit. With one term the sum is a smooth wave; with more terms it sharpens into steps at the levels. Energy in units of $\hbar\omega$.

Sample Problem 21.6: Summing the orbit series at $E = 9/4$

Evaluate the oscillating sum of Eq. (21.17) at $E = \frac{9}{4}$, show that it gives $N = 2$ exactly, and give the partial sums through $r = 1, 3, 5$.

Step 1: The levels below $\frac{9}{4}$ are $\frac{1}{2}$ and $\frac{3}{2}$, so the answer should be $N = 2$.

Step 2: The phase is $2\pi r \cdot \frac{9}{4} = \frac{9\pi r}{2}$. For even r this is a multiple of π (since $9r/2$ is then an integer), so $\sin = 0$.

Step 3: For odd $r = 2j + 1$ ($j = 0, 1, 2, \ldots$): $\frac{9\pi(2j+1)}{2} = 9\pi j + \frac{9\pi}{2} = 9\pi j + 4\pi + \frac{\pi}{2}$. Removing 4π, $\sin(9\pi j + \frac{\pi}{2}) = \cos(9\pi j) = (-1)^{9j} = (-1)^j$. With $(-1)^r = -1$ for odd r, the term is $-(-1)^j/(2j+1)$.

Step 4: The sum is $-\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$. The bracket is the Leibniz series. It converges, because it is alternating with terms decreasing to zero (Chapter 3). To find its value, start from the geometric series (Chapter 3) with ratio $-u^2$: for $|u| < 1$,

$\frac{1}{1+u^2} = 1 - u^2 + u^4 - u^6 + \cdots$. Integrate both sides from 0 to x, with $|x| < 1$. The left side gives $\arctan x$, because $\frac{\mathrm{d}}{\mathrm{d}x} \arctan x = \frac{1}{1+x^2}$ and $\arctan 0 = 0$; the right side, integrated term by term, gives

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \qquad (|x| < 1).$$

At $x = 1$ the series becomes the Leibniz bracket, and $\arctan 1 = \pi/4$.

Taken on trust

Two steps are taken on trust here: integrating a power series term by term inside its interval of convergence, and passing to the endpoint $x = 1$, where the series still converges (Abel's theorem: if a power series converges at an endpoint of its interval, its sum there is the limit of the function). Abel's theorem and term-by-term integration are stated in [22]; proofs are in standard real-analysis texts. Numerically, two million terms of the Leibniz series agree with $\pi/4 = 0.785398$ to six decimal places.

So

$$N\left(\tfrac{9}{4}\right) = \tfrac{9}{4} + \frac{1}{\pi}\left(-\frac{\pi}{4}\right) = \tfrac{9}{4} - \tfrac{1}{4} = 2.$$

Step 5: Partial sums: through $r = 1$, $\frac{9}{4} - \frac{1}{\pi} = 1.931690$; through $r = 3$, add $\frac{1}{3\pi} = 0.106103$ to get 2.037793; through $r = 5$, subtract $\frac{1}{5\pi} = 0.063662$ to get 1.974131.

$$\boxed{N(9/4) = 2; \quad \text{partial sums } 1.931690,\ 2.037793,\ 1.974131}$$

The partial sums oscillate around 2 and approach it slowly. Orbit sums of this kind converge only conditionally, if at all; this echoes the convergence issues you met in Chapter 18.

21.3.2 Gutzwiller's trace formula

For a chaotic system there is no single orbit family; at each energy there are infinitely many isolated periodic orbits, each unstable. Gutzwiller found the analogous formula for chaotic systems. It has the same structure (a smooth term plus a sum over orbits and their repetitions), but it is asymptotic rather than exact. Each orbit also carries a stability factor that has no counterpart in Eq. (21.17).

Taken on trust

Gutzwiller's trace formula (an asymptotic approximation for small $\hbar$, derived by the stationary-phase method; derivation beyond this book, see [37]). For a quantum system with two degrees of freedom (for example a particle moving in a plane, so that phase space is four-dimensional) whose classical limit is chaotic with isolated unstable periodic orbits, the level count is approximately

$$N(E) \approx \bar{N}(E) + \frac{1}{\pi} \sum_{\gamma} \sum_{r=1}^{\infty} \frac{1}{r} \frac{\sin\!\left(r S_\gamma(E)/\hbar - r\mu_\gamma \pi/2\right)}{\left|\det\left(M_\gamma^r - I\right)\right|^{1/2}}. \tag{21.18}$$

Here $\bar{N}$ is the smooth phase-space count; γ runs over *primitive* periodic orbits (orbits traversed once), r over repetitions; S_γ is the action; μ_γ is an integer (the Maslov index)

counting turning-point-like phase losses; and M_γ is the *stability matrix*, the 2×2 matrix (for a two-dimensional system) that maps a small displacement away from the orbit, measured across the orbit, to its value one period later. The formula is usually written for the level density $\mathrm{d}N/\mathrm{d}E$, whose oscillating terms are $\frac{1}{\pi\hbar}\,T_\gamma\cos(rS_\gamma/\hbar - r\mu_\gamma\pi/2)/\left|\det(M_\gamma^r - I)\right|^{1/2}$. To pass to the count, integrate in E. By the chain rule and $\mathrm{d}S_\gamma/\mathrm{d}E = T_\gamma$ (we derived this relation only for one-dimensional motion; it holds for periodic orbits in general, which we take on trust, see [37]), $\frac{\mathrm{d}}{\mathrm{d}E}\sin(rS_\gamma/\hbar - r\mu_\gamma\pi/2) = \frac{rT_\gamma}{\hbar}\cos(rS_\gamma/\hbar - r\mu_\gamma\pi/2)$. Treating T_γ and the determinant as constant over one oscillation (this is the approximation: their slow energy dependence is neglected), $\int T_\gamma\cos(\cdots)\,\mathrm{d}E \approx \frac{\hbar}{r}\sin(\cdots)$, so the prefactor becomes $\frac{1}{\pi\hbar}\cdot\frac{\hbar}{r} = \frac{1}{\pi r}$, which is the form above. For chaotic systems the orbit sum does not converge absolutely; it is meaningful as an asymptotic statement and after smoothing.

The stability factor of an unstable orbit. For an unstable (hyperbolic) orbit, a small displacement grows by a factor $\Lambda = e^{\lambda T_\gamma} > 1$ per period along one direction and shrinks by $1/\Lambda$ along another; λ is the Lyapunov exponent (the growth rate). This λ is neither the eigenvalue λ of Section 21.1 nor the wavelength λ of the box count in Section 21.2. The stability matrix then has eigenvalues Λ and $1/\Lambda$. (This Λ is a growth factor; it has nothing to do with the von Mangoldt function $\Lambda(k)$ of Chapter 15.)

Taken on trust

That the two eigenvalues multiply to 1, i.e. $\det M_\gamma = 1$, follows from the fact that Hamiltonian motion preserves phase-space area (Liouville's theorem). We take this on trust; see [37].

After r traversals the eigenvalues of M_γ^r are Λ^r and Λ^{-r}. For any 2×2 matrix B with eigenvalues β_1, β_2, the characteristic polynomial is $\det(B - zI) = (\beta_1 - z)(\beta_2 - z)$. Both sides are quadratics in z with leading coefficient 1 and roots β_1, β_2, so they are equal. Putting $z = 1$,

$$\begin{aligned}\det\left(M_\gamma^r - I\right) &= (\Lambda^r - 1)(\Lambda^{-r} - 1)\\ &= \Lambda^r\Lambda^{-r} - \Lambda^r - \Lambda^{-r} + 1\\ &= 2 - \Lambda^r - \Lambda^{-r} = -\left(\Lambda^{r/2} - \Lambda^{-r/2}\right)^2,\end{aligned} \tag{21.19}$$

where the last step expands $(\Lambda^{r/2} - \Lambda^{-r/2})^2 = \Lambda^r - 2 + \Lambda^{-r}$. Therefore

$$\left|\det\left(M_\gamma^r - I\right)\right|^{1/2} = \Lambda^{r/2} - \Lambda^{-r/2} = e^{r\lambda T_\gamma/2} - e^{-r\lambda T_\gamma/2} \approx e^{r\lambda T_\gamma/2} \quad (\text{for } r\lambda T_\gamma \gg 1). \tag{21.20}$$

Unstable orbits are exponentially suppressed in the sum: the more unstable, the weaker their contribution.

21.3.3 The explicit formula as a trace formula

Now put the zeros' version next to it. From Chapter 18, the fluctuating part of $N(t)$ is approximated (heuristically; the series does not converge on the critical line, and the truncation is an approximation whose accuracy is measured numerically) by paper Eq. (4):

$$N(t) \approx N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t), \qquad N_{\mathrm{fl}}(t) = -\frac{1}{\pi}\sum_{p\in P}\sum_{m=1}^{3}\frac{\sin(mt\log p)}{m\,p^{m/2}}, \tag{21.21}$$

with P a finite set of primes. The comparison with Gutzwiller's formula below concerns the untruncated formal sum $-\frac{1}{\pi}\sum_p\sum_{m=1}^{\infty}\sin(mt\log p)/(m\,p^{m/2})$, taken over all primes and all m.

Eq. (21.21) is a truncation of that sum. Term by term the two are the same, so the dictionary can be read off from either. Read Eqs. (21.18) and (21.21) side by side, with energy $E \leftrightarrow$ height t and $\hbar = 1$. (The comparison is with Gutzwiller's formula for a hypothetical chaotic system with two degrees of freedom, whose stability matrices are 2×2; each prime is treated as a primitive orbit of such a system. It is not a comparison with a one-dimensional potential, which is not chaotic.)

- Both have the form "smooth part $+\frac{1}{\pi}\sum_{\text{primitive}}\sum_{\text{repetitions}} \frac{1}{\text{rep.}} \times \frac{\text{oscillation}}{\text{stability}}$."
- The phase $mt\log p$ matches $rS_\gamma(E)$ if each prime p is a primitive orbit with $S_p(E) = E\log p$, and m is the repetition number. By Fact 1 of Section 21.2 ($T = \mathrm{d}S/\mathrm{d}E$), $S_p(E) = E\log p$ gives $T_p = \log p$, a period independent of energy.
- The amplitude $p^{-m/2}$ matches $e^{-r\lambda T_\gamma/2}$ (the large-$r\lambda T$ form of Eq. (21.20)) if $T_\gamma = \log p$ and $\lambda = 1$: then $e^{m\log p/2} = p^{m/2}$. Every "prime orbit" would have the same Lyapunov exponent, 1.
- The smooth part N_{sm} would be the phase-space count $\bar{N}$.

Table 21.1 collects this dictionary.

Table 21.1: The trace-formula dictionary between a chaotic quantum system (Gutzwiller, Eq. (21.18)) and the Riemann zeros (explicit formula, Eq. (21.21)). This is an *analogy*: no dynamical system with these properties is known. The last two rows are where the analogy is imperfect.

chaotic system	Riemann zeros
energy E	height t
smooth count $\bar{N}(E)$	$N_{\text{sm}}(t)$, paper Eq. (1)
primitive periodic orbit γ	prime p
period T_γ	$\log p$
repetition number r	power m (prime power p^m)
action $rS_\gamma(E) = rET_\gamma$	$mt\log p$
stability $\lvert\det(M^r - I)\rvert^{1/2}$	$p^{m/2}$ (matches only for large p^m, with $\lambda = 1$)
phase $-r\mu_\gamma\pi/2$	overall minus sign (not of Maslov form)

Sample Problem 21.7: The $p = 2$ orbit

In the dictionary, find the period, the frequency in t, the period in t, and the counting-function amplitude of the $p = 2$, $m = 1$ term (RE2). Then compare its stability factor with the exact hyperbolic value of Eq. (21.20).

Step 1: Period $T_2 = \log 2 = 0.693147$. The term $\sin(t\log 2)$ oscillates in t with angular frequency $\log 2$, hence with period in t equal to $2\pi/\log 2 = 9.064720$, the RE2 value.

Step 2: Amplitude in N_{fl}: $\frac{1}{\pi}\cdot\frac{1}{1\cdot 2^{1/2}} = \frac{1}{\pi\sqrt{2}} = 0.225079$. So this single term moves the count up and down by about 0.23 of a zero.

Step 3: The explicit formula's stability factor is $2^{1/2} = 1.414214$. The exact hyperbolic factor with $\Lambda = e^{\lambda T} = e^{\log 2} = 2$ is $\Lambda^{1/2} - \Lambda^{-1/2} = 1.414214 - 0.707107 = 0.707107$.

Step 4: Ratio: $1.414214/0.707107 = 2$. In general the ratio is $p^{m/2}/(p^{m/2} - p^{-m/2}) = 1/(1 - p^{-m})$, which is 2 for $p^m = 2$ and tends to 1 for large p^m.

$$\boxed{T_2 = 0.693147, \quad 2\pi/\log 2 = 9.064720, \quad \text{amplitude } 0.225079, \quad \text{stability ratio } 2}$$

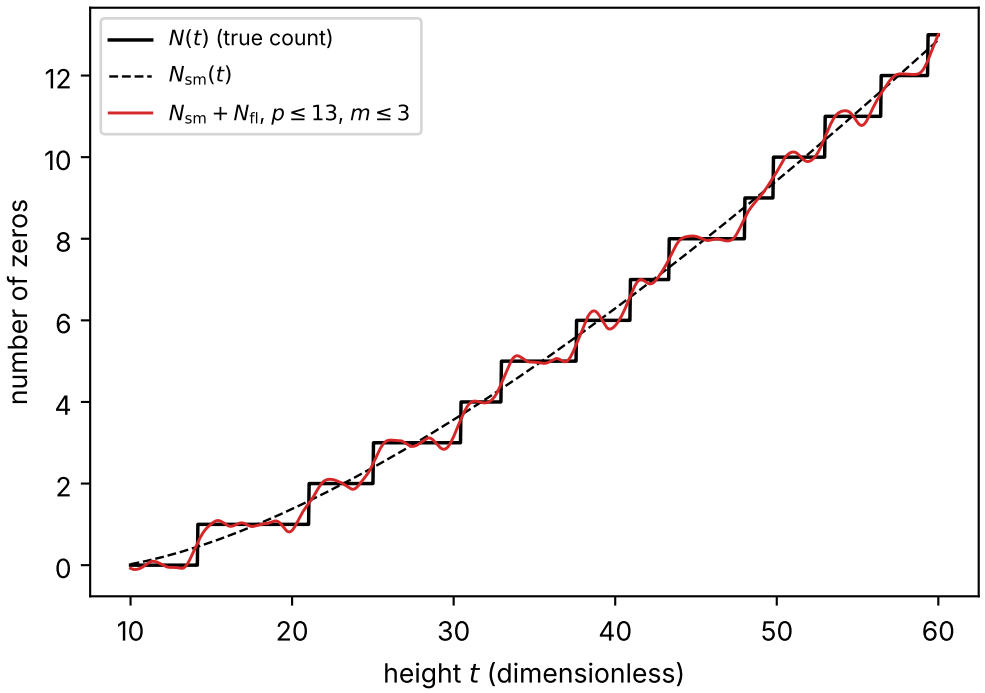


Figure 21.3: The zero-counting staircase $N(t)$ for $10 \le t \le 60$ (black; it jumps by 1 at each zero t_n of RE1 and beyond), the smooth count $N_{\mathrm{sm}}(t)$ of paper Eq. (1) (dashed), and $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t)$ with the six primes $p \le 13$ and $m \le 3$ (paper Eq. (4), solid color). The "prime-orbit" sum bends the smooth curve toward the steps. Height t is dimensionless.

21.3.4 Where the analogy is imperfect

An honest dictionary records its failures. Three stand out.

The sign. Eq. (21.21) has an overall minus sign for *every* p and m. In Eq. (21.18) the corresponding quantity is the phase shift $-r\mu_\gamma\pi/2$ inside the sine. Since $\sin(\theta - \phi) = \sin\theta\cos\phi - \cos\theta\sin\phi$ equals $-\sin\theta$ for all θ exactly when $\cos\phi = -1$ and $\sin\phi = 0$, that is, when $\phi \equiv \pi \pmod{2\pi}$, a constant sign -1 needs $r\mu\pi/2 \equiv \pi \pmod{2\pi}$ for every r. For $r = 1$ this means $\mu\pi/2 \equiv \pi$, so $\mu \equiv 2 \pmod 4$. For $r = 2$ it means $\mu\pi \equiv \pi$, so μ is odd. These two conditions contradict each other (Problem 10). This mismatch is discussed in the literature; Connes interprets the zeros as an *absorption* spectrum (lines missing from a continuum) rather than as emitted eigenvalues [38].

The stability factor. As the Sample Problem showed, $p^{m/2}$ is only the large-p^m limit of the hyperbolic factor; for $p = 2$, $m = 1$ they differ by a factor 2. The naive dictionary works asymptotically, not exactly.

No dynamics. Nobody has exhibited a classical system whose primitive periodic orbits have periods $\log 2, \log 3, \log 5, \ldots$, all with Lyapunov exponent 1. The dictionary is a description of what such a system would have to look like.

Checkpoint 21.4

In the dictionary, which "orbit" does the term $\sin(2t\log 3)/(2\cdot 3)$ of N_{fl} correspond to, and what is its total traversal time?

Answer: The primitive orbit $p = 3$ traversed $m = 2$ times (the prime power 9). Its primitive period is $\log 3$, so two traversals take $2\log 3 = \log 9 \approx 2.197$. The amplitude $1/(2\cdot 3)$ is $1/(mp^{m/2})$ with $p^{m/2} = 3$.

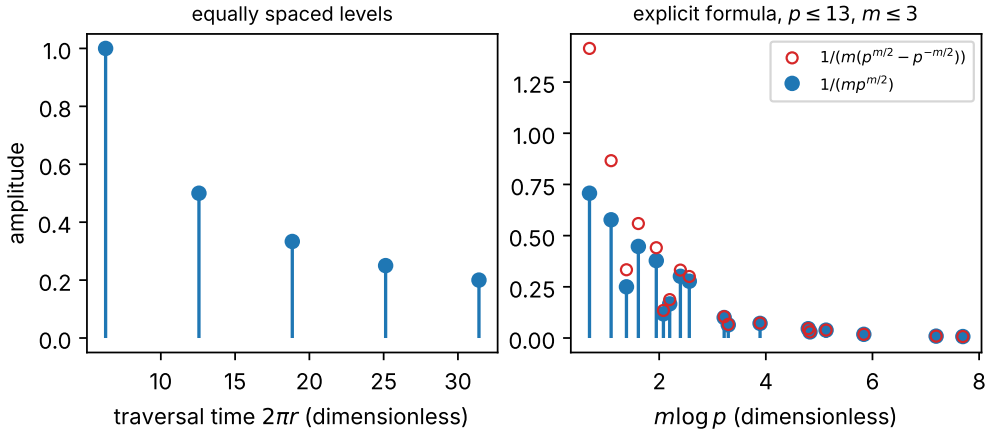


Figure 21.4: "Orbit spectra": the amplitude of each oscillating term in a trace formula, plotted at its action per unit energy (that is, at its total traversal time, the angular frequency of the term in E or t). Left: the equally spaced spectrum of Eq. (21.17), one orbit of period 2π and its repetitions, amplitudes $1/r$. Right: the explicit formula, Eq. (21.21), with $p \le 13$, $m \le 3$: stems at $m \log p$ with heights $1/(mp^{m/2})$; open circles show the heights $1/(m(p^{m/2} - p^{-m/2}))$ that an exactly hyperbolic orbit with $\lambda = 1$ would give. The overall factor $1/\pi$ and the signs are omitted. Horizontal axes are dimensionless.

21.3.5 A heuristic consequence for one-dimensional potentials

Here is an observation, *heuristic* and clearly labeled, that becomes important in the paper. A one-dimensional potential $p^2 + V(x)$ has, at each energy, exactly one periodic orbit: the back-and-forth motion between the two turning points. Its trace formula is therefore of the oscillator type, Eq. (21.17), with $2\pi E$ replaced by the action $S(E) = A(E)$: a single orbit and its repetitions. Those terms produce the steps of the staircase; they do not produce independent oscillations at frequencies $\log 2, \log 3, \ldots$ in E. So a one-dimensional potential can only reproduce the zeros' fluctuations by building them into the *shape* of V, so that the phase-space area $A(E)/2\pi$ itself contains $N_{\rm sm}(E) + N_{\rm fl}(E)$. This is the spirit of what Wu and Sprung did (Chapter 22): a smooth potential from $N_{\rm sm}$, plus fitted fluctuating corrections that make the levels reproduce the low zeros. In such a construction the fluctuations sit in the shape of the potential as fine wiggles; they are not generated dynamically. The fluctuation at energy E is then controlled by the potential near the turning points of that energy, that is, locally in energy. We will meet this picture again in the paper's Sec. VI.

Key Ideas

- A trace formula writes the level count as a smooth phase-space term plus a sum over periodic orbits and their repetitions; Eq. (21.17) is an exact, checkable example.
- Gutzwiller's formula (21.18) (asymptotic; proof beyond this book) weights unstable orbits by $1/\left|\det(M^r - I)\right|^{1/2} \approx e^{-r\lambda T/2}$.
- The explicit formula (21.21) has the same shape with primes as primitive orbits, periods $\log p$, repetitions m, and stability $p^{m/2}$ (Table 21.1). This is an analogy, not a theorem.
- The analogy is imperfect: an overall minus sign not of Maslov form, $p^{m/2}$ matching the hyperbolic factor only for large p^m, and no known dynamics.
- Heuristically, a 1D potential has one orbit per energy and can carry prime oscillations

only in its shape.

21.4 The Berry–Keating Hamiltonian $H = xp$

★ Did you know? *Hyperbolas and Primes*

In 1999 Michael Berry and Jonathan Keating proposed the classical Hamiltonian $H = xp$ as a candidate. Its orbits are hyperbolas, and with suitable cutoffs in phase space, counting states gives $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi}$ plus a constant, the smooth part of Riemann's zero count. In the same year Alain Connes proposed a related construction. Neither approach produces the zeros themselves.

The dictionary asks for unstable orbits, all with Lyapunov exponent 1, and no time-reversal symmetry. Berry and Keating [10] noticed that the simplest conceivable Hamiltonian has all three features:

$$H(x,p) = xp. \tag{21.22}$$

21.4.1 Classical motion

Hamilton's equations (21.7) with $H = xp$: $\partial H/\partial p = x$ and $\partial H/\partial x = p$, so

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = x, \qquad \frac{\mathrm{d}p}{\mathrm{d}\tau} = -p. \tag{21.23}$$

These are the growth and decay equations of first-year physics, with solutions $x(\tau) = x_0 e^{\tau}$ and $p(\tau) = p_0 e^{-\tau}$ (τ is time, as in Section 21.2). Their product $x(\tau)p(\tau) = x_0 p_0$ is constant: energy is conserved and the orbits are hyperbolas $xp = E$ (Figure 21.5). Every orbit stretches x by e^{τ} and squeezes p by $e^{-\tau}$ in time τ: a small displacement grows at rate $\lambda = 1$, the same for every orbit. That is the exponent the dictionary wants.

Time reversal sends $p \to -p$: $H(x,-p) = -xp \neq H(x,p)$. So xp is *not* time-reversal invariant, consistent with the GUE heuristic of Section 21.1.

The trouble is equally visible: the hyperbolas are not closed. A particle runs off to $x \to \infty$ and never returns. There are no periodic orbits at all, and the phase-space region $xp \le E$ has infinite area.

21.4.2 Regularizing and counting

Berry and Keating cut phase space off by hand: keep only $x \ge \ell_x$ and $p \ge \ell_p$ for two positive constants, and count the region of the quadrant $x > 0$, $p > 0$ with $xp \le E$ (Figure 21.5). For the region to be non-empty we need $E > \ell_x \ell_p$. At a given x, the allowed p run from ℓ_p up to E/x; the largest x is where these meet, $x = E/\ell_p$. So

$$\begin{aligned} A(E) &= \int_{\ell_x}^{E/\ell_p} \left(\frac{E}{x} - \ell_p\right) \mathrm{d}x \\ &= E\big[\log x\big]_{\ell_x}^{E/\ell_p} - \ell_p\left(\frac{E}{\ell_p} - \ell_x\right) \\ &= E\log\frac{E}{\ell_x\ell_p} - E + \ell_x\ell_p. \end{aligned} \tag{21.24}$$

The cutoffs enter only through their product $a = \ell_x\ell_p$, a phase-space area. Berry and Keating chose $a = 2\pi\hbar = 2\pi$, one Planck cell. The motivation is the counting rule itself: a quantum state occupies an area $2\pi\hbar$, so phase-space structure on a smaller area cannot be resolved, and

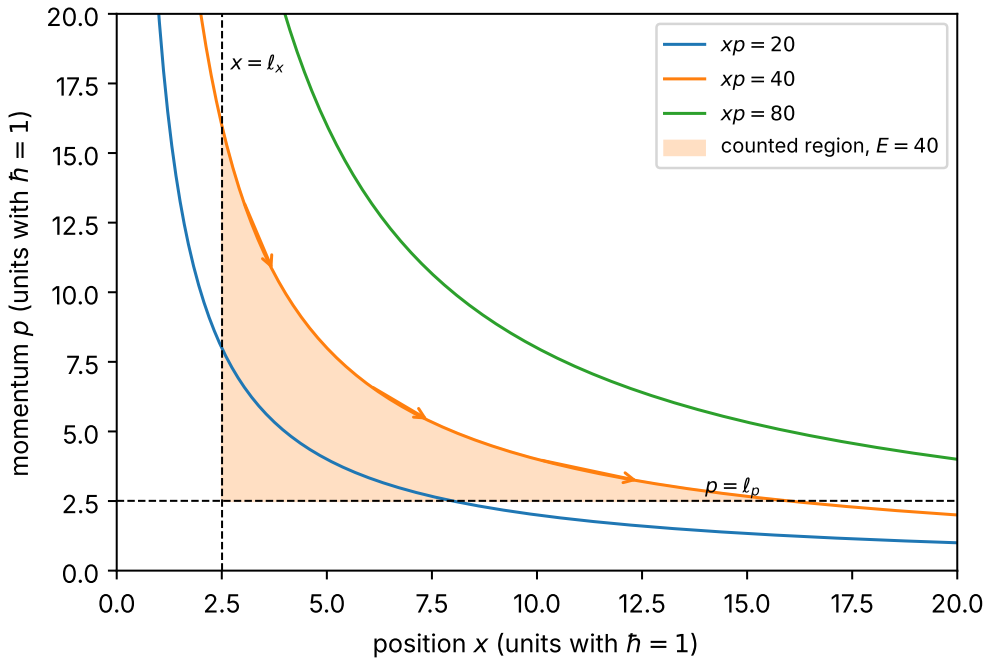


Figure 21.5: Phase space of $H = xp$ in the quadrant $x > 0$, $p > 0$ (units $\hbar = 1$). Orbits are hyperbolas $xp = E$ (curves for $E = 20, 40, 80$), traversed in the direction of the arrows (x grows, p shrinks). The Berry–Keating regularization keeps only $x \geq \ell_x$ and $p \geq \ell_p$ with $\ell_x \ell_p = 2\pi$ (dashed lines, drawn with $\ell_x = \ell_p = \sqrt{2\pi}$); the shaded region is the part of $xp \leq 40$ that is counted. Its area divided by 2π gives Eq. (21.25).

a cutoff of one cell is the natural minimal choice. (Problem 7 shows that any other choice adds a term growing like E and spoils the agreement below). With $a = 2\pi$ and the counting rule (21.10):

$$N_{\mathrm{BK}}(E) = \frac{A(E)}{2\pi} = \frac{E}{2\pi}\log\frac{E}{2\pi} - \frac{E}{2\pi} + 1. \tag{21.25}$$

Compare with paper Eq. (1): $N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$. The two growing terms agree exactly; the constants differ:

$$N_{\mathrm{BK}}(E) - N_{\mathrm{sm}}(E) = 1 - \tfrac{7}{8} = \tfrac{1}{8}. \tag{21.26}$$

A crude classical count with one Planck-cell cutoff reproduces the leading asymptotics of the zero count, the $E\log E$ growth that no box or oscillator produces. Be careful about what is independent here. The coefficient $\frac{1}{2\pi}$ of $E\log E$ does not depend on the cutoff area a (Problem 7), so it is a genuine output of the xp dynamics. The linear term $-\frac{E}{2\pi}$ matches only because $a = 2\pi$ was chosen, so that agreement is a consistency check of the choice, not independent evidence. This is why xp is regarded as a suggestive starting point, not as a solution. The difference of $\frac{1}{8}$ of a level is well within the order-one error of the counting rule itself, so the computation says nothing about the constant.

Sample Problem 21.8: The Berry–Keating count at $E = 1000$

Evaluate $N_{\mathrm{sm}}(1000)$ and $N_{\mathrm{BK}}(1000)$, and compare with the true number of zeros up to height 1000.

Step 1: $1000/(2\pi) = 159.15494$ and $\log 159.15494 = 5.069878$.

Step 2: The growing part: $159.15494 \times (5.069878 - 1) = 159.15494 \times 4.069878 = 647.7412$.

Step 3: $N_{\mathrm{sm}}(1000) = 647.7412 + 0.875 = 648.6162$ and $N_{\mathrm{BK}}(1000) = 647.7412 + 1 = 648.7412$.
Step 4: Counting the zeros (with `qmnt.zeros`; see the listing that follows this Sample Problem) gives $N(1000) = 649$: the 649th zero lies below 1000 and the 650th above. The fluctuating part at this height is $N(1000) - N_{\mathrm{sm}}(1000) \approx 0.38$.

$$\boxed{N_{\mathrm{sm}}(1000) \approx 648.62, \quad N_{\mathrm{BK}}(1000) \approx 648.74, \quad N(1000) = 649}$$

```
import numpy as np, qmnt
t = qmnt.zeros(700)                    # t_1 ... t_700 (t_700 > 1000)
def Nsm(E):  return E/(2*np.pi)*np.log(E/(2*np.pi)) - E/(2*np.pi) + 7/8
def NBK(E):  return Nsm(E) + 1/8       # Eq. (hilbert-polya-nbk)
for E in [100.0, 500.0, 1000.0]:
    N = np.sum(t <= E)                  # true count N(E)
    print(E, N, round(Nsm(E), 3), round(NBK(E), 3))
# last line printed: 1000.0 649 648.616 648.741
```

21.4.3 The orbit time and the level density

Fact 2 of Section 21.2 says the mean level density is (orbit time)$/2\pi$. Does it hold for the regularized xp? The orbit enters the counted region at $x = \ell_x$ and leaves at $x = E/\ell_p$. Since $x = x_0 e^{\tau}$, the time spent inside is $T(E)$ (a transit time; again not the paper's training window T) with $e^{T} = (E/\ell_p)/\ell_x$:

$$T(E) = \log \frac{E}{\ell_x \ell_p} = \log \frac{E}{2\pi}. \tag{21.27}$$

Meanwhile, differentiating Eq. (21.24) with $\ell_x \ell_p = 2\pi$,

$$\frac{\mathrm{d}A}{\mathrm{d}E} = \log \frac{E}{2\pi} + E \cdot \frac{1}{E} - 1 = \log \frac{E}{2\pi} = T(E), \tag{21.28}$$

in agreement with Fact 1. Note that here $T(E)$ is a *transit time* between the two cutoffs, not the period of a closed orbit (the hyperbolas never close); the relation $\mathrm{d}A/\mathrm{d}E = T$ is verified directly by the two computations just made. The density $\mathrm{d}N_{\mathrm{BK}}/\mathrm{d}E = \frac{1}{2\pi}\log\frac{E}{2\pi}$ is the mean density of the zeros, and its reciprocal $2\pi/\log(E/2\pi)$ is the mean spacing of Chapter 17, the paper's gap_n evaluated at height E. The zeros crowd together because the orbit time grows logarithmically.

Sample Problem 21.9: Orbit time and spacing at $E = 1000$

Find $T(1000)$ and the mean spacing of the zeros at height 1000.
Step 1: $T(1000) = \log(1000/2\pi) = \log 159.15494 = 5.069878$.
Step 2: Mean spacing $= 2\pi/T = 6.283185/5.069878 = 1.239317$.

$$\boxed{T(1000) \approx 5.0699, \qquad \text{spacing} \approx 1.2393}$$

Near height 1000, consecutive zeros are about 1.24 apart on average, compared with $t_2 - t_1 \approx 6.9$ at the bottom.

21.4.4 The quantum xp operator

To make xp a quantum operator we replace p by $\hat{p} = -i\,\mathrm{d}/\mathrm{d}x$ (with $\hbar = 1$). The classical product xp does not fix the order of the quantum operators, and the order matters. From

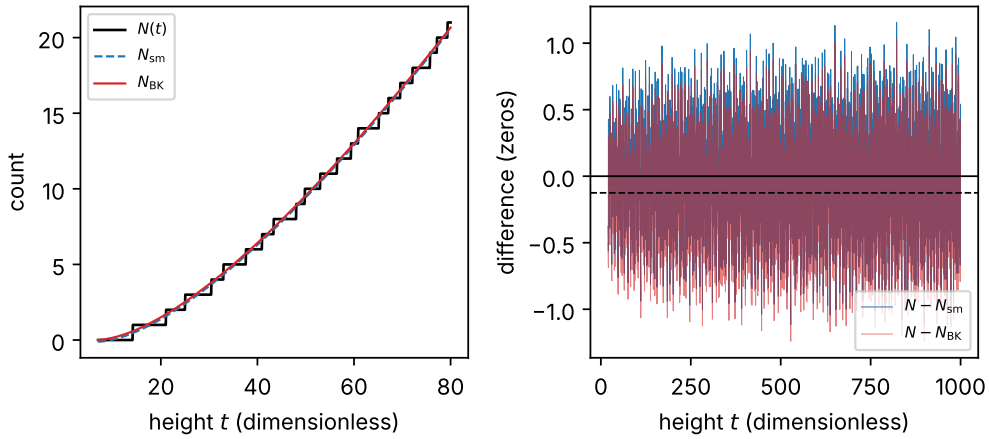


Figure 21.6: Left: the true count $N(t)$ (staircase), $N_{\rm sm}(t)$ of paper Eq. (1) and the Berry–Keating count $N_{\rm BK}(t)$ of Eq. (21.25) for $0 < t \le 80$; the two smooth curves differ by the constant $\frac{1}{8}$. Right: $N(t) - N_{\rm sm}(t)$ and $N(t) - N_{\rm BK}(t)$ for $20 \le t \le 1000$. Both fluctuate (the fluctuation is $S(t)$ of Chapter 17), around 0 and $-\frac{1}{8}$ respectively; neither smooth curve drifts away from the data. All quantities dimensionless.

the lines below, $\hat{p}\hat{x}\psi - \hat{x}\hat{p}\psi = -i(\psi + x\psi') + ix\psi' = -i\psi$. Moreover, neither ordering alone is Hermitian. Formally, $(\hat{x}\hat{p})^\dagger = \hat{p}^\dagger\hat{x}^\dagger = \hat{p}\hat{x}$, using $(AB)^\dagger = B^\dagger A^\dagger$ and the fact that $\hat{x}$ and $\hat{p}$ are each formally Hermitian; this differs from $\hat{x}\hat{p}$. Their average is formally Hermitian, because taking $\dagger$ just swaps its two terms. So one symmetrizes:

$$\hat{H} = \tfrac{1}{2}\left(\hat{x}\hat{p} + \hat{p}\hat{x}\right). \tag{21.29}$$

Acting on a function $\psi(x)$:

$$\begin{aligned} \hat{x}\hat{p}\,\psi &= -ix\psi', \\ \hat{p}\hat{x}\,\psi &= -i(x\psi)' = -i(\psi + x\psi'), \\ \hat{H}\psi &= \tfrac{1}{2}\left(-ix\psi' - i\psi - ix\psi'\right) = -i\left(x\psi' + \tfrac{1}{2}\psi\right). \end{aligned} \tag{21.30}$$

Sample Problem 21.10: Eigenfunctions of the quantum xp

Show that $\psi_E(x) = x^{-1/2+iE}$ $(x > 0)$ satisfies $\hat{H}\psi_E = E\psi_E$ for every real E, and show that it is not normalizable.

Step 1: For complex powers use $x^a = e^{a\log x}$ (Chapter 2); then $\frac{\mathrm{d}}{\mathrm{d}x}x^a = a\,x^{a-1}$ as for real a, since $\frac{\mathrm{d}}{\mathrm{d}x}e^{a\log x} = \frac{a}{x}e^{a\log x}$.

Step 2: With $a = -\frac{1}{2} + iE$: $x\psi_E' = a\,x^a = \left(-\frac{1}{2} + iE\right)\psi_E$.

Step 3: So $x\psi_E' + \frac{1}{2}\psi_E = iE\,\psi_E$, and Eq. (21.30) gives $\hat{H}\psi_E = -i\cdot iE\,\psi_E = E\psi_E$.

Step 4: Modulus: $\left|x^{iE}\right| = \left|e^{iE\log x}\right| = 1$, so $|\psi_E|^2 = x^{-1}$. Then $\int_1^X x^{-1}\mathrm{d}x = \log X \to \infty$ as $X \to \infty$, and similarly $\int_\epsilon^1 x^{-1}\mathrm{d}x = \log(1/\epsilon) \to \infty$ as $\epsilon \to 0$.

$$\boxed{\hat{H}\,x^{-1/2+iE} = E\,x^{-1/2+iE} \text{ for all real } E; \quad \int_0^\infty |\psi_E|^2\,\mathrm{d}x = \infty}$$

Every real number is a (generalized, non-normalizable) eigenvalue: the spectrum of the plain quantum xp is the whole real line, continuous, like a free particle's. (In the variable $u = \log x$, $\psi_E = e^{-u/2}e^{iEu}$ is a plane wave in u, which explains the analogy.) The regularization that

produced discrete counting in Eq. (21.25) was a classical cutoff drawn by hand, not a property of an operator. Turning those cutoffs into genuine boundary conditions or interactions that make the spectrum discrete, with the right density, and then showing that the eigenvalues are the zeros, is the unsolved part. Sierra [36] reviews attempts along this line, including modified xp Hamiltonians designed to make the orbits bounded and the spectrum discrete.

Checkpoint 21.5

Is $H = xp$ time-reversal invariant? Is $H = p^2 + x^2$?

Answer: Under $p \to -p$, $xp \to -xp$, so xp is not invariant. $p^2 + x^2 \to (-p)^2 + x^2 = p^2 + x^2$, so the oscillator is invariant.

Key Ideas

- $H = xp$ has hyperbolic orbits $x \propto e^{\tau}$, $p \propto e^{-\tau}$ (τ = time), Lyapunov exponent 1, and broken time-reversal symmetry: the features the dictionary asks for.
- With phase-space cutoffs $\ell_x \ell_p = 2\pi$, the semiclassical count is $N_{\mathrm{BK}} = \frac{E}{2\pi}\log\frac{E}{2\pi} - \frac{E}{2\pi} + 1$, matching N_{sm} except for the constant (1 versus $\frac{7}{8}$), Eqs. (21.25)–(21.26).
- The orbit time $\log(E/2\pi)$ gives the mean density of zeros.
- The quantum $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$ has continuous spectrum (eigenfunctions $x^{-1/2+iE}$); the cutoffs are not an operator. The xp idea reproduces averages, not the zeros.

21.5 Other routes: Connes, Bender–Brody–Müller, Sierra and recent proposals

The analytic program has produced many candidate constructions. We describe the ones the paper cites, at the level of what they attempt and what their authors report, without verdicts beyond what is published. Throughout, keep in mind the one fact not in dispute: *the Riemann hypothesis remains open*, so no construction has been accepted as a proof.

Connes: trace formulas in noncommutative geometry. Connes [38] works not with a Schrödinger operator but with an action of the multiplicative group on a space built from all the primes at once (the "adèle class space," an object from number theory whose definition is beyond this book). In that setting he obtains a spectral interpretation of the critical zeros, as an *absorption* spectrum (zeros appear as missing lines), and shows that RH is equivalent to the validity of a certain trace formula. That trace formula is closely related to the Weil form of the explicit formula (the duality between zeros and prime powers stated in Chapter 18); Weil had shown that RH is equivalent to the positivity of a quadratic form built from that explicit formula, and proving such a positivity is the goal of this line of work. The paper also cites a 2026 contribution [21]: as the paper reports (Sec. I A b and Sec. II D), extremizing the truncated Weil quadratic form over primes $p \le 13$ yields approximations to the first 50 zeros with accuracies from 10^{-3} down to 2.6×10^{-55}, provably located on the critical line. Note the logic: this uses only prime data, never a zero.

Zeta spectral triples. Connes, Consani and Moscovici [39] construct operators from Euler products over primes, with numerically observed spectral convergence (the paper's description, Sec. I A d). Again the input is arithmetic, not zero data.

Bender, Brody and Müller. Bender, Brody and Müller [40] proposed an operator built from the symmetrized $\hat{x}\hat{p} + \hat{p}\hat{x}$ of Section 21.4 combined with the shift operator $e^{-i\hat{p}}$ (which shifts a function's argument by one unit). Its formal eigenfunctions are related to the Hurwitz zeta function, a relative of ζ, and, as the authors describe, a boundary condition at $x = 0$ selects eigenvalues that correspond to the zeros of ζ (see [40] for the construction). The operator is not Hermitian in the ordinary sense; the authors argued that it has a symmetry of the kind studied by Bender and Boettcher for non-Hermitian Hamiltonians with real spectra [33] (PT symmetry; Chapter 14), and that if a suitable inner product can be constructed, the eigenvalues are real. Recall the Checkpoint of Section 21.1: reality does not require ordinary Hermiticity, which is why this route is conceivable. Bellissard published a comment [41] disputing aspects of this argument; we do not summarize its technical content here. Whatever one's view of the exchange, RH remains open.

Sierra. Sierra [36] reviews the program of realizing the zeros as a spectrum, centered on xp-type Hamiltonians and their modifications, aimed at replacing the hand-drawn cutoffs of Eq. (21.24) by genuine quantum boundary conditions or interactions, and discusses what such constructions would and would not imply for RH.

Recent proposals. Yakaboylu [42] proposes a Hamiltonian intended to realize the Hilbert–Pólya conjecture; Tamburini [43] proposes a relativistic approach based on a Majorana particle in a (1+1)-dimensional Rindler spacetime. The paper cites these, with the others above, as evidence that "explicit operator candidates continue to be proposed" (Sec. I). We do not assess them.

Table 21.2: Analytic operator proposals cited in the paper, by their input and their published status. "Input" records whether zero data are used in building the operator. None of these constructions is accepted as a proof of RH, which remains open.

proposal	input	what is reported
Berry–Keating xp [10]	classical model	leading count N_{sm}; quantum spectrum continuous
Connes [38]	primes (adèles)	RH equivalent to a trace formula
Connes 2026 [21]	primes $p \leq 13$	first 50 zeros, accuracies to 2.6×10^{-55}
Connes–Consani–Moscovici [39]	Euler products	numerically observed spectral convergence
Bender–Brody–Müller [40]	xp and shift	eigenvalues tied to zeros; reality argued; disputed in [41]
Sierra [36]	modified xp	review and modified models
Yakaboylu [42]; Tamburini [43]	models	recent proposals

Checkpoint 21.6

Which of the constructions in Table 21.2 use zero data as input?

Answer: None. They are all specified from other principles (classical dynamics, primes, Euler products, symmetry), which is what makes their output a prediction rather than a fit. This is the defining feature of the analytic program.

Key Ideas

- Connes gives a spectral (absorption) interpretation in which RH is equivalent to the validity of a trace formula. By Weil's earlier criterion, RH is also equivalent to the positivity of a quadratic form built from the explicit formula. A 2026 construction

from primes $p \le 13$ reproduces the first 50 zeros to very high accuracy (as reported in paper Sec. I A b).

- Bender–Brody–Müller propose a non-Hermitian, PT-type operator linked to the Hurwitz zeta function; Bellissard published a comment disputing aspects of the argument.
- Sierra and others refine xp; new proposals continue to appear.
- No operator is known whose spectrum is proved to be the zeros and that is proved self-adjoint (or proved to have real spectrum). RH and the Hilbert–Pólya conjecture remain open.

21.6 Two programs: analytic and empirical

There is a second, quite different way to pursue the dream. Instead of specifying an operator and computing its spectrum, you *fit* an operator to the zeros you already know. Wu and Sprung [12] built a smooth one-dimensional potential whose semiclassical count is N_{sm} (Chapter 22) and then added fitted fluctuating corrections so that its levels reproduce the low zeros; van Zyl and Hutchinson [13], Slater [14] and Jaksch [15] refined such reconstructions. Call this the *empirical* program.

The two programs differ in the direction of the arrow (Figure 21.7):

- **Analytic:** principles → operator → spectrum → compare with zeros. Every agreement is a prediction, because the zeros were not used. A disagreement refutes the proposal.
- **Empirical:** zeros → fit operator parameters → spectrum → compare with the same zeros. Agreement is guaranteed to the extent the family is flexible enough; it measures capacity.

The implicit hope of the empirical program is that a potential reproducing many zeros accurately has captured something about the putative zeta operator. The Sample Problem "An operator for two zeros" shows why agreement on the training data cannot by itself test that hope: a rotated diagonal matrix reproduces any finite list exactly. What *can* test it is the standard question of every data-driven science: does the fitted operator predict zeros it was not shown? If it has captured structure, it should; if it has only stored the data, it should not. This is the question of memorization versus generalization (Chapter 25), and it is the question the paper asks.

The trace-formula picture of Section 21.3 also suggests, heuristically, what to expect. The structure that would transfer to unseen zeros is the prime-orbit sum: oscillations at fixed frequencies $m \log p$ in t that continue forever. A one-dimensional potential addresses each level through its turning points, locally in energy, and has only one orbit per energy. A fit can bend the potential to reproduce a window of oscillations, but nothing in the parametrization says "keep oscillating at frequency $\log 2$ above the window." Whether this heuristic is right is an empirical question, and Part VI shows how the paper tests it.

Checkpoint 21.7

A fitted potential reproduces zeros 1–400 to rms 0.14 mean spacings, while predicting no fluctuation at all gives 0.24. Does this show the potential has learned the zeros' structure?

Answer: No. Those are in-sample numbers, and a flexible enough family can reproduce any sequence of that size. What matters is how it does on zeros it was not fitted to, compared with the null on those same zeros. (These particular numbers are the paper's: in-sample rms 0.137 at $M = 400$ and in-sample null 0.2438, App. A key anchors; Section 21.7 tells you what happened out of sample.)

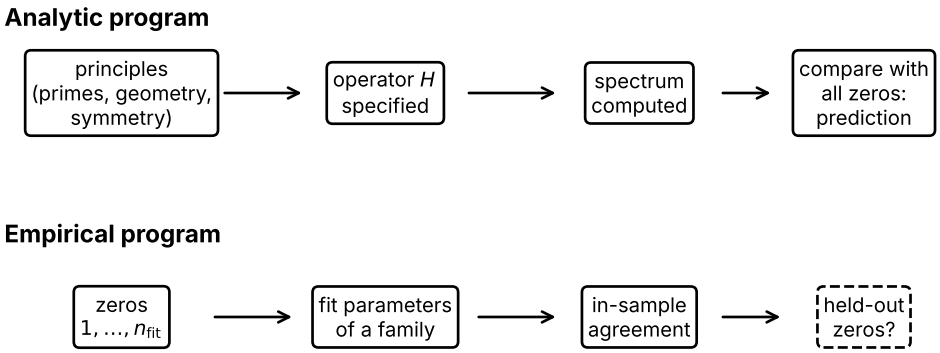


Figure 21.7: The two programs. Top (analytic): an operator is specified from principles that do not involve the zeros, its spectrum is computed and compared with all zeros, so every agreement is a prediction. Bottom (empirical): operator parameters are fitted to the known zeros $1, \ldots, n_{\text{fit}}$, and agreement on those zeros is the usual criterion of success. The dashed box is the test this program had lacked and that the paper applies: prediction of held-out zeros.

Key Ideas

- Analytic program: specify, then predict; agreement is evidence. Empirical program: fit, then compare; in-sample agreement measures capacity.
- The decisive test for a fitted operator is held-out prediction relative to its own null.
- Heuristically, the transferable structure is a sum of fixed-frequency oscillations in t, which a locally acting 1D potential has no built-in way to continue.

21.7 Paper connection

Sec. I, first paragraph. The paper opens with the conjecture in exactly the form of Section 21.1: the t_n of $\rho_n = \frac{1}{2} + it_n$ "are the spectrum of a self-adjoint operator." It then lists the circumstantial evidence in three items, each of which this chapter (with its prerequisites) unpacks:

1. "Montgomery's pair-correlation theorem and Odlyzko's numerics match Gaussian unitary ensemble (GUE) eigenvalue statistics" [9, 8]. This is Chapter 20, and it is the statistical design target of Section 21.1, with its heuristic reading as broken time-reversal symmetry. The paper's Fig. 2 shows the GUE spacing histogram for the first 10^4 zeros and uses it only to validate the dataset and unfolding (Sec. II A), not as evidence for any operator.

2. "the zero-counting function resembles a semiclassical trace formula in which the primes play the role of periodic orbits" [10, 11]. This is Section 21.3 and Table 21.1. The word "resembles" is exactly right: it is an analogy with known imperfections.

3. "explicit operator candidates continue to be proposed" [38, 40, 41, 36, 42, 43, 39]. These are the seven references of Section 21.5 and Table 21.2. Note that the paper includes Bellissard's comment in the same citation list without taking sides, and so do we.

Sec. I, second paragraph and the scope statement. "Alongside this analytic program runs an empirical one," citing [12, 13, 14, 15]: this is the distinction of Section 21.6. The paper states the implicit logic ("a potential reproducing many zeros accurately has captured something about the putative zeta operator") and then the scope: "We do not test, and make no claim against, analytic Hilbert–Pólya constructions in which an operator is specified independently of the zeros and its spectrum is then predicted." After this chapter you can see why the scope is natural: an analytic construction already makes predictions about every zero, so the held-out question is built into it; the fitted constructions are the ones for which in-sample agreement was the criterion. The scope is also narrower than "all empirical constructions": only fluctuation potentials added to the Wu–Sprung background, in three bases (cosine, Chebyshev, hat; paper Sec. II C), were tested.

Sec. I A d, "Operator proposals." The paper says the analytic program "seeks exact operators and is not our target," singles out the zeta spectral triples [39] as "prime-driven construction, not a fit to zero data," and concludes: "Our results constrain only the empirical shortcut: the hope that fitting flexible operator families to finitely many zeros captures transferable structure." The "Input" column of Table 21.2 is the criterion being used.

Sec. I A b and Sec. II D: arithmetic constructions. The paper places Connes' 2026 construction [21] "on the arithmetic side of our comparison," because it "consumes only prime data." Its role in the paper is to show that "the predictive information is arithmetic and abundantly accessible without fitting." The paper's own arithmetic baseline is the much simpler truncated orbit sum, paper Eq. (4), our Eq. (21.21) with $m \leq 3$ and a finite prime set, which the paper calls an "information-existence baseline" and for which it explicitly claims no novelty. In trace-formula language, the oracle is "the prime-orbit sum, truncated to the shortest orbits."

Fig. 1 and Table V. Fig. 1 of the paper shows the contrast this chapter prepares you for: on zeros past the training boundary, a fitted operator (fit on zeros 1–400, scale 1, $M = 400$) has held-out correlation -0.07, while the six-prime explicit formula, with zero fitted parameters, has $+0.96$. The six-prime row of Table V gives rms 0.074 and correlation $+0.96$ on zeros 401–500, against the null 0.263 (RE5). You can reproduce that row in a few seconds:

```
import numpy as np, qmnt
n    = np.arange(401, 501)                          # RE5 holdout
true = qmnt.zeros(500)[n - 1]
null = qmnt.rms(qmnt.fluct(true, n))                # predict t_n = s_n
pred = qmnt.oracle_zeros(n, qmnt.first_primes(6), mmax=3)  # p <= 13
rms, corr = qmnt.score(pred, n)
print(round(null, 4), round(rms, 3), round(corr, 2))   # 0.2631 0.074 0.96
```

The "prime orbits" $2, 3, 5, 7, 11, 13$, the shortest periods in Figure 21.4, already carry most of the predictable fluctuation.

Sec. VI, first and second paragraphs. "Nothing here touches the analytic Hilbert–Pólya program [10, 38, 40, 39, 21]. What is constrained is a shortcut." Read with this chapter, the sentence is precise: the analytic program's hard problems (self-adjointness on a domain, a proof that the spectrum is all the zeros) are untouched by any numerical fit. The second paragraph offers a *heuristic* reading of why the fits fail: the fitted quantity "is, by the explicit formula, an almost-periodic function of t that is sparse in frequency, whereas a potential addresses the levels through their turning points, locally in energy." That is the picture of the end of Section 21.3 and of Section 21.6: one orbit per energy in 1D, versus fixed prime-orbit frequencies in t. The paper is careful to call this a heuristic, to say it is "evidence about the tested parametrizations

and optimizer together, not a statement about what a potential can represent," and to list as open "whether the memorization result can be derived rather than measured." Keep those hedges when you retell the result.

The falsifiability standard. Finally, the paper's proposal (Sec. VI, items (i)–(iv)) applies to "any future claim that a constructed or learned operator 'captures' the Riemann zeros." In our reading (the paper does not say this), an analytically specified operator meets the spirit of item (i) (held-out rms and correlation against its own smooth part) once its spectrum is actually computed and scored, since no zero was used to build it. The paper's list makes no exemption. For a fitted operator, item (i) is the whole question. Chapters 28–32 read these items in detail.

21.8 Summary

- Relabel each nontrivial zero by $t = -i(\rho - \frac{1}{2})$; then $\operatorname{Im} t = \frac{1}{2} - \operatorname{Re}\rho$ (Eq. (21.2)), so RH $\iff$ all t real (Eq. (21.3)). The t values are symmetric under $t \to -t$ and $t \to \bar{t}$.
- Hilbert–Pólya conjecture (open): the t values are the eigenvalues of a self-adjoint operator. Hermitian $\Rightarrow$ real eigenvalues (Eqs. (21.5)–(21.6)), so a proved realization would give RH. In infinite dimensions self-adjointness on a domain is essential; boundary terms can produce complex eigenvalues.
- Any finite list of reals is the spectrum of a symmetric matrix; finite agreement is not evidence.
- Semiclassical counting: one state per phase-space area $2\pi\hbar$ (Eq. (21.10)); $\mathrm{d}A/\mathrm{d}E = \mathrm{d}S/\mathrm{d}E = T_\gamma$; mean density $T_\gamma/2\pi$.
- Trace formulas: exact for equally spaced levels (Eq. (21.17)); Gutzwiller for chaotic systems (Eq. (21.18), asymptotic), with hyperbolic stability factor $\Lambda^{r/2} - \Lambda^{-r/2}$ (Eq. (21.20)).
- The explicit formula (Eq. (21.21), paper Eq. (4)) resembles a trace formula with primes as orbits of period $\log p$, repetitions m, stability $p^{m/2}$ (Table 21.1); the analogy fails in the sign and at small p^m, and no dynamics is known.
- Berry–Keating xp (Eq. (21.22)): with cutoffs $\ell_x\ell_p = 2\pi$, $N_{\mathrm{BK}} = \frac{E}{2\pi}\log\frac{E}{2\pi} - \frac{E}{2\pi} + 1 = N_{\mathrm{sm}} + \frac{1}{8}$ (Eqs. (21.25)–(21.26)); orbit time $\log(E/2\pi)$; the quantum operator has continuous spectrum (Eq. (21.30)).
- Connes, Bender–Brody–Müller (with Bellissard's comment), Sierra, Yakaboylu, Tamburini, Connes–Consani–Moscovici: analytic proposals built without zero data; none accepted as a proof.
- Analytic program (specify, predict) versus empirical program (fit, compare): the paper tests only the latter, by held-out prediction.

21.9 Problems

1. • Find $t = -i(\rho - \frac{1}{2})$ for (a) $\rho = \frac{1}{2} + 21.022040\,i$ (the second zero of RE1), and (b) the hypothetical point $\rho = 0.55 + 30i$. For (b), give $\operatorname{Im} t$ and state what its sign tells you.

2. • Show that $H = \begin{pmatrix} 3 & 2i \\ -2i & 3 \end{pmatrix}$ is Hermitian and find its eigenvalues.

3. • Which of the following classical Hamiltonians are time-reversal invariant ($p \to -p$)? (a) $p^2 + x^4$; (b) xp; (c) x^2p^2; (d) p^3.

4. • A box of length $\ell = 10$ in units $\hbar = 2m = 1$ has levels $E_k = (k\pi/\ell)^2$. Estimate the number of states with $E \le 100$ by the area rule (21.10) and compare with the exact count.

5. •• In the level density $\mathrm{d}N_{\mathrm{fl}}/\mathrm{d}t$, the term from prime p with $m = 1$ has amplitude $\frac{1}{\pi}\log p/\sqrt{p}$. (a) Derive this from Eq. (21.21). (b) Compute $\log p/\sqrt{p}$ for $p = 2, 3, 5, 7, 11, 13$. (c) Show that the function $\log y/\sqrt{y}$ has its maximum at $y = e^2$, and say which prime gives the largest density amplitude.

6. •• Repeat the Sample Problem "Summing the orbit series" at $E = \frac{7}{4}$: show that the oscillating sum of Eq. (21.17) reduces to $+\frac{1}{\pi}(1 - \frac{1}{3} + \frac{1}{5} - \cdots)$ and that $N(\frac{7}{4}) = 2$.

7. •• Redo the Berry–Keating count with a general cutoff area $a = \ell_x\ell_p$. Show that $N(E) = \frac{E}{2\pi}\log\frac{E}{2\pi} - \frac{E}{2\pi} + \frac{E}{2\pi}\log\frac{2\pi}{a} + \frac{a}{2\pi}$, and explain why only $a = 2\pi$ is compatible with N_{sm} at large E.

8. •• Using $\mathrm{d}N_{\mathrm{BK}}/\mathrm{d}E$, compute the mean spacing of the zeros at height $t_{500} = 811.1843588$ (paper App. A key anchors). Compare with the spacing at height 1000 found in the text.

9. •• Compute the ratio $1/(1 - p^{-m})$ between the explicit formula's stability factor $p^{m/2}$ and the exact hyperbolic factor $p^{m/2} - p^{-m/2}$ for (a) $p = 2$, $m = 3$ and (b) $p = 13$, $m = 1$. Which terms of the explicit formula are best described by the hyperbolic-orbit dictionary?

10. •• Show that no integer Maslov index μ makes the phase shift $r\mu\pi/2$ congruent to π modulo 2π (equivalently $e^{-ir\mu\pi/2} = -1$) for both $r = 1$ and $r = 2$. Conclude that the explicit formula's overall sign cannot be produced by the phase term of Eq. (21.18).

11. •• (a) Verify by direct differentiation that $\psi_E(x) = x^{-1/2+iE}$ satisfies $-i(x\psi' + \psi/2) = E\psi$. (b) Compute $\int_1^X |\psi_E|^2\,\mathrm{d}x$ and its limit as $X \to \infty$. (c) In the variable $u = \log x$, what kind of function is $\sqrt{x}\,\psi_E$?

12. •• (a) Give a 2×2 real matrix that is not symmetric but has real eigenvalues. (b) Explain why the Hilbert–Pólya argument needs only the reality of the eigenvalues, and why self-adjointness is nevertheless the usual requirement.

13. •• (Paper.) Read the scope statement in Sec. I of the paper. (a) List what is tested and what is explicitly not tested. (b) Classify each of the following as analytic or empirical in the sense of Section 21.6: Wu–Sprung [12], van Zyl–Hutchinson [13], Slater [14], Jaksch [15], Berry–Keating [10], Connes [38], Bender–Brody–Müller [40], Connes–Consani–Moscovici [39].

14. •• (Paper.) Sec. I A b of the paper reports accuracies from 10^{-3} down to 2.6×10^{-55} for Connes' 2026 construction. (a) How many orders of magnitude does this range span? (b) Why does the paper say this construction "strengthens the paper's conclusion" rather than competing with it?

15. •• (Paper.) Using Table V of the paper, compute the ratio (oracle rms)/(null rms) for six primes and for 25 primes on the holdouts 401–500 and 5001–10000. How do the corresponding correlations change? What reason does the table caption give for the mild growth of rms at a fixed prime budget?

16. •• (Computational, paper.) Using `qmnt`, reproduce the 3-prime and 25-prime entries of Table V of the paper on the holdout 401–500, and check the null 0.263.

17. ••• (Computational.) Using `qmnt.zeros`, compute $D_1(T) = N(T) - N_{\mathrm{sm}}(T)$ and $D_2(T) = N(T) - N_{\mathrm{BK}}(T)$ on a fine grid of T in $[100, 1000]$, and their averages over the grid. Explain your results using Eq. (21.26).

18. ••• (Computational.) Using numpy, compute the partial sums of Eq. (21.17) at $E = \frac{9}{4}$ for $R = 1, \dots, 200$ repetitions. Plot the error $|N_R - 2|$ against R and show that it stays below $1/(\pi R)$ for odd R. What does the slow decay say about truncating orbit sums?

19. ••• For $H = p^2 + \omega^2 x^2/4$ (an oscillator in units $2m = 1$): (a) show the orbit $H = E$ is an ellipse and compute its area $A(E)$; (b) verify $\mathrm{d}A/\mathrm{d}E = T_\gamma$ with $T_\gamma = 2\pi/\omega$; (c) deduce from the area rule that the level spacing is ω (that is, $\hbar\omega$).

20. ••• (a) For the stable orbit of a rotation, $M = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$, compute $\det(M^r - I)$. (b) Contrast with the hyperbolic result (21.20): which kind of orbit gives amplitudes that decay exponentially with r?

21. •• Construct a 3×3 real symmetric, non-diagonal matrix whose eigenvalues are t_1, t_2, t_3 of RE1, and explain in two sentences why such a matrix is no evidence for any Hilbert–Pólya operator.

22. •• (Paper.) Explain which of the paper's criteria (i)–(iv) in Sec. VI an analytically specified operator, such as one built from primes alone, satisfies "automatically," and which it still needs to address.

23. •• (Paper.) The paper's Hermitian fitted potentials are real, so their Hamiltonians are time-reversal invariant, whereas the zeros show GUE statistics (paper Fig. 2). Explain why this does not prevent the fits from matching the training zeros closely in-sample (paper Sec. III), and why GUE statistics alone do not single out any particular operator.

24. ••• (Paper.) Using the heuristic of Section 21.3 (one periodic orbit per energy for a 1D potential), explain the paper's Sec. VI reading that "a potential addresses the levels through their turning points, locally in energy," and relate it to the arithmetic-continuation probe of paper Sec. III, in which a fit to a window of the $p = 2$ oscillation (RE2) did not continue it (held-out correlation $+0.07$). State clearly what is heuristic in your argument.

25. • Label each statement as theorem (proved), numerical observation, analogy/heuristic, or open conjecture: (a) Hermitian matrices have real eigenvalues; (b) RH; (c) the Hilbert–Pólya conjecture; (d) the unfolded spacings of the first 10^4 zeros follow the GUE Wigner surmise closely; (e) primes are the periodic orbits of a chaotic system; (f) with cutoffs $\ell_x \ell_p = 2\pi$, the classical area of $xp \le E$ divided by 2π is $N_{\mathrm{sm}}(E) + \frac{1}{8}$.

26. •• (Computational.) Using `qmnt`, evaluate $N(t)$, $N_{\mathrm{sm}}(t)$ and $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t)$ (six primes, $m \le 3$) at $t = 100, 500, 1000$ and comment on how much of the fluctuation $N - N_{\mathrm{sm}}$ the six "prime orbits" capture at each height.

21.10 Answers to selected problems

★ Did you know? ***Geodesics Play Primes***

In 1956 Atle Selberg found a trace formula for surfaces of constant negative curvature. In it, the lengths of closed geodesics play the role that primes play in the explicit formula, and Laplacian eigenvalues play the role of zeta zeros. For the resulting Selberg zeta functions, the analogue of the Riemann hypothesis essentially holds. This analogy is a strong reason

to hope for a spectral explanation.

Problem 1. (a) $\rho - \frac{1}{2} = 21.022040\,i$, so $t = -i \cdot 21.022040\,i = 21.022040$, real. (b) Eq. (21.2): $t = 30 - i(0.55 - 0.5) = 30 - 0.05i$, so $\operatorname{Im} t = -0.05$. The negative sign means $\sigma > \frac{1}{2}$: the point is to the right of the critical line.

Problem 2. $H^\dagger$: the $(1,2)$ entry is $\overline{-2i} = 2i = H_{12}$, diagonal real, so Hermitian. Trace 6, determinant $9 - (2i)(-2i) = 9 - 4 = 5$. $\lambda^2 - 6\lambda + 5 = 0$ gives $\lambda = 1, 5$.

Problem 3. (a) $(-p)^2 + x^4$: invariant. (b) $-xp$: not invariant. (c) $x^2(-p)^2 = x^2p^2$: invariant. (d) $(-p)^3 = -p^3$: not invariant.

Problem 4. $E = p^2 \le 100$ means $|p| \le 10$. Area $\ell \cdot 2 \cdot 10 = 200$; divided by 2π: 31.83. Exact: $k\pi/10 \le 10$, so $k \le 100/\pi = 31.83$, i.e. 31 states.

Problem 5. (a) $\frac{\mathrm{d}}{\mathrm{d}t} \sin(t \log p) = \log p \cos(t \log p)$, so the $m = 1$ term of $\mathrm{d}N_{\mathrm{fl}}/\mathrm{d}t$ is $-\frac{1}{\pi} \frac{\log p}{\sqrt{p}} \cos(t \log p)$. (b) 0.490, 0.634, 0.720, 0.735, 0.723, 0.711 for $p = 2, 3, 5, 7, 11, 13$. (c) $\frac{\mathrm{d}}{\mathrm{d}y} \frac{\log y}{\sqrt{y}} = \frac{1}{y\sqrt{y}} - \frac{\log y}{2y\sqrt{y}} = \frac{2 - \log y}{2y^{3/2}}$, zero at $\log y = 2$, positive below and negative above: a maximum at $y = e^2 \approx 7.39$. Among primes, $p = 7$ gives the largest amplitude. (In the counting function itself the amplitude $1/\sqrt{p}$ decreases with p; differentiation multiplies by the frequency $\log p$.)

Problem 7. Eq. (21.24) with general a: $A = E \log(E/a) - E + a$. Write $\log(E/a) = \log(E/2\pi) + \log(2\pi/a)$ and divide by 2π to get the stated form. Compared with N_{sm}, there is an extra term $\frac{E}{2\pi} \log \frac{2\pi}{a}$, which grows linearly in E and would eventually exceed any constant error unless $\log(2\pi/a) = 0$, i.e. $a = 2\pi$.

Problem 9. (a) $1/(1 - 2^{-3}) = 8/7 \approx 1.143$. (b) $1/(1 - 13^{-1}) = 13/12 \approx 1.083$. The dictionary is best for large prime powers p^m, where $p^{-m/2}$ is negligible next to $p^{m/2}$; it is worst for $p^m = 2$ (ratio 2).

Problem 11. (a) $\psi' = (-\frac{1}{2} + iE)x^{-3/2+iE}$, so $x\psi' + \frac{1}{2}\psi = iE\psi$ and $-i(iE\psi) = E\psi$. (b) $|\psi_E|^2 = 1/x$, so the integral is $\log X$, which tends to infinity. (c) $\sqrt{x}\,\psi_E = x^{iE} = e^{iEu}$: a plane wave in u with wavenumber E, which is why every real E is allowed.

Problem 13. (a) Tested: empirical inverse-spectral constructions with parameters fitted to finite sets of zeros, specifically fluctuation potentials added to the Wu–Sprung background in cosine, Chebyshev and hat bases. Not tested: other backgrounds, other parametrizations (Hermite functions, wavelets, neural-network potentials), and analytic Hilbert–Pólya constructions. (b) Empirical: Wu–Sprung, van Zyl–Hutchinson, Slater, Jaksch. Analytic: Berry–Keating, Connes, Bender–Brody–Müller, Connes–Consani–Moscovici.

Problem 15. Six primes: $0.074/0.263 = 0.281$ (401–500) and $0.119/0.296 = 0.402$ (5001–10000), with correlations $+0.96$ and $+0.92$. Twenty-five primes: $0.043/0.263 = 0.163$ and $0.067/0.296 = 0.226$, correlations $+0.99$ and $+0.98$. The correlation, which does not depend on the overall size of the fluctuations, falls only slightly. The caption attributes the mild growth at fixed budget to the growth of the null and to "the increasing number of arithmetically relevant primes at height": a fixed truncation omits more of the relevant prime orbits as t grows.

Problem 17. The average of D_1 over the grid is close to zero (magnitude below 0.05), because $N - N_{\mathrm{sm}}$ is the fluctuating part $S(t)$ (up to a tiny $\mathcal{O}(1/t)$ term), which oscillates about zero. The average of D_2 is close to $-\frac{1}{8}$, because $D_2 = D_1 - \frac{1}{8}$ exactly by Eq. (21.26). The data cannot distinguish the two smooth counts by eye (Figure 21.6), but averaging over many zeros does.

Problem 19. (a) $p^2 + \omega^2 x^2/4 = E$ is an ellipse with semi-axes $\sqrt{E}$ in p and $2\sqrt{E}/\omega$ in x; the area of an ellipse is $\pi \times$(product of semi-axes), so $A = \pi\sqrt{E} \cdot 2\sqrt{E}/\omega = 2\pi E/\omega$. (b) $\mathrm{d}A/\mathrm{d}E = 2\pi/\omega$. Hamilton's equations give $\mathrm{d}x/\mathrm{d}\tau = 2p$ and $\mathrm{d}p/\mathrm{d}\tau = -\omega^2 x/2$. Differentiating the first and substituting the second, $\mathrm{d}^2x/\mathrm{d}\tau^2 = 2\,\mathrm{d}p/\mathrm{d}\tau = -\omega^2 x$: simple harmonic motion with period $2\pi/\omega$, equal to $\mathrm{d}A/\mathrm{d}E$. (c) $N(E) \approx A/2\pi = E/\omega$, so consecutive levels differ by ω (restoring $\hbar$: $\hbar\omega$).

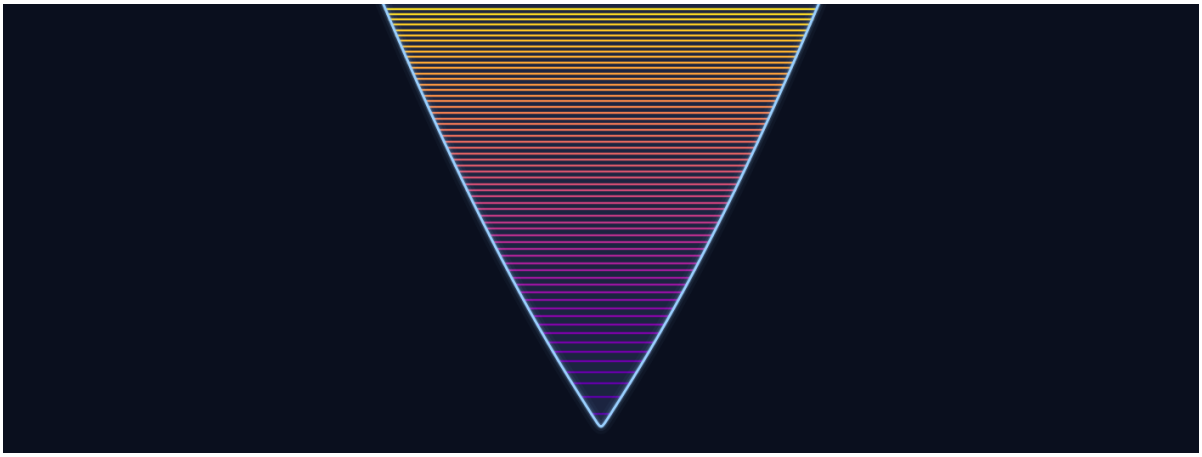
Problem 21. Take the 2×2 block of the Sample Problem "An operator for two zeros" and add t_3 on the diagonal:

$$H = \begin{pmatrix} 15.856554 & -2.982295 & 0 \\ -2.982295 & 19.300211 & 0 \\ 0 & 0 & 25.010858 \end{pmatrix}.$$

It is $QDQ^\top$ with $D = \mathrm{diag}(t_1, t_2, t_3)$ and Q a rotation in the 1–2 plane, so its eigenvalues are t_1, t_2, t_3. It is no evidence because it was built from the three numbers it reproduces; the same recipe works for any list, true zeros or random numbers, and it says nothing about t_4.

Problem 23. In-sample fitting adjusts many parameters until the eigenvalues match the given numbers; as Problem 21 shows, a symmetric (time-reversal-invariant) operator can match any finite list, so symmetry class places no obstacle on fitting. GUE statistics describe the *distribution* of spacings; infinitely many operators (indeed random matrices themselves) share them, so the statistics constrain the symmetry class of a natural chaotic operator, heuristically, but do not identify any operator.

Problem 25. (a) Theorem. (b) Open conjecture. (c) Open conjecture. (d) Numerical observation (paper Fig. 2). (e) Analogy/heuristic: no such system is known. (f) Theorem in the sense of an exact calculation of an area (Eqs. (21.24)–(21.25)); its relevance to any quantum operator is heuristic.



The Wu–Sprung potential, built so that its levels follow the smooth zeros; the first 60 levels are drawn across the well.

Chapter 22

Building a Potential from the Zeros: the Wu–Sprung Construction

↑ Contents

What this chapter is about

Until now the arrow has pointed one way. You were handed a potential $V(x)$ and asked for its energy levels: solve $H\psi = E\psi$, find the list $E_1, E_2, \ldots$. In this chapter we reverse the arrow. Someone hands you a list of numbers and asks: *what potential well would have these numbers as its energy levels?* This is the *inverse spectral problem*, and it is the heart of the empirical side of the Hilbert–Pólya program you met in Chapter 21.

In 1993 Wu and Sprung [12] asked this question for the Riemann zeros. They did not try to reproduce every zero at once. They first asked for a smooth, even potential whose levels have exactly the *average* spacing of the zeros, the spacing described by the smooth counting function N_{sm} of Chapter 17. Using the semiclassical counting of Chapter 12, this question turns into an integral equation first solved by Abel, and its solution is a closed formula for the potential. We derive that formula line by line, find out where its single constant V_0 comes from, study the shape of the resulting well, and check numerically that its levels really do sit on the smooth zeros.

Where this shows up in the paper [1]: the formula we derive is the paper's Eq. (2) in Sec. II B; the box half-widths $L = 44.883$ and 59.957 come from it; it is the "smooth potential" and the null model of every operator fit in Secs. II C, III and V; the turning-point map we build explains the paper's statement (Sec. III, interpolation split) that "position maps to level index through the classical turning point"; and the closing section explains why the whole empirical program, of which the paper is a test, needs held-out data.

What you need

- Chapter 11: levels of the infinite well on $[-L, L]$, $E_k = (k\pi/2L)^2$; the harmonic oscillator levels $1, 3, 5, \ldots$ for $V = x^2$ in units $\hbar = 2m = 1$; even potentials; slowly growing potentials have crowded levels.
- Chapter 12: the local wavenumber $\sqrt{E - V(x)}$, classical turning points, Bohr–Sommerfeld quantization with the $(n - \frac{1}{2})\pi$ rule, the semiclassical counting function, the idea of inverting a counting function by an Abel integral equation.
- Chapter 17: the smooth counting function $N_{\mathrm{sm}}(t)$, the smooth zeros s_n, the mean spacing gap_n, and fluctuations measured in mean spacings.
- Chapter 21: the Hilbert–Pólya conjecture (open), and the distinction between analytic operator proposals and empirical fits.
- Chapter 4: big-$\mathcal{O}$ notation, root finding (Brent), floating-point round-off and machine epsilon; an exponential grows faster than any power.
- Chapter 10: the Hamiltonian $H = -d^2/dx^2 + V$ in units $\hbar = 2m = 1$ and the meaning of an energy level.
- Chapter 3: the Taylor series of $\log(1 \pm r)$.
- Chapter 8: tridiagonal matrices and numerical eigensolvers used as black boxes.
- Chapter 18: the explicit formula, as background for the fluctuations discussed in Section 22.7.
- Chapter 23 (a later chapter): its results are used once, in Section 22.6, and are stated there explicitly as taken on trust.
- First-year physics: potential-energy curves, turning points, energy conservation $\frac{1}{2}mv^2 + V = E$.
- Calculus: substitution, integration by parts, the antiderivative $z \log z - z$ of $\log z$, the fundamental theorem of calculus.

22.1 The inverse spectral problem

22.1.1 Forward and inverse

Recall from Chapter 10 that in units $\hbar = 2m = 1$ the Hamiltonian of a particle on a line is

$$H = -\frac{d^2}{dx^2} + V(x), \tag{22.1}$$

where $V(x)$ is the potential energy, and that an energy level E is a number for which $H\psi = E\psi$ has a normalizable solution ψ. In these units an energy and a "height" t on the critical line are measured in the same way, which is why we can compare levels E_n directly with zeros t_n.

The **forward problem** is: given V, find the levels. The **inverse spectral problem** is: given the levels, find V. You have a physical feel for the inverse problem already. A trained ear can tell a large bell from a small one, or a steel drum from a timpani, from the frequencies alone. In 1966 the mathematician Mark Kac popularized the question "Can one hear the shape of a drum?", asking whether the frequencies of a drumhead determine its outline.

For us the "drum" is a one-dimensional well and the "frequencies" are the levels. The first lesson of the inverse problem is that the answer is usually *not unique*, and we must decide what extra assumptions pin it down.

22.1.2 Three ways to get the same spectrum

Translation. On the whole line, shift the potential sideways: $\tilde{V}(x) = V(x - x_0)$ for a constant x_0. If ψ solves $-\psi''(x) + V(x)\psi(x) = E\psi(x)$ for all x, define $\varphi(x) = \psi(x - x_0)$. By the chain rule $\varphi''(x) = \psi''(x - x_0)$, so evaluating the original equation at the point $x - x_0$ gives

$$-\varphi''(x) + V(x - x_0)\varphi(x) = E\varphi(x).$$

So $\tilde{V}$ has the same level E, with the same normalization (shifting does not change $\int |\varphi|^2 \mathrm{d}x$). Every level of V is a level of $\tilde{V}$ and, running the argument backwards with $-x_0$, vice versa.

Reflection. Let $\tilde{V}(x) = V(-x)$ and $\varphi(x) = \psi(-x)$. Then $\varphi'(x) = -\psi'(-x)$ and $\varphi''(x) = \psi''(-x)$, so again $-\varphi'' + \tilde{V}\varphi = E\varphi$. A well and its mirror image have the same spectrum.

Shearing (semiclassical). This one is less obvious and is the important one. We show next that, within the semiclassical approximation of Chapter 12, the levels of a single well depend on the potential only through how *wide* the well is at each height, not on how that width is placed left and right.

22.1.3 Slicing the well horizontally

Consider a single well: V has one minimum value V_0 and rises on both sides. For each height $V \geq V_0$ let $x_-(V) \leq x_+(V)$ be the two points where the potential equals V, and define the **width function**

$$w(V) = x_+(V) - x_-(V), \tag{22.2}$$

the length of the horizontal chord of the well at height V. At energy E the classical turning points are $x_\pm(E)$.

Recall from Chapter 12 the phase integral, which we call

$$\Phi(E) = \int_{x_-(E)}^{x_+(E)} \sqrt{E - V(x)}\, \mathrm{d}x, \tag{22.3}$$

and the Bohr–Sommerfeld rule: the n-th level E_n ($n = 1, 2, \dots$) satisfies

$$\Phi(E_n) = \left(n - \tfrac{1}{2}\right)\pi. \tag{22.4}$$

This is an approximation (Chapter 12): accurate for high levels, weaker for the lowest few.

Now the trick. For a fixed x inside the well, write the square root as an integral over heights:

$$\sqrt{E - V(x)} = \int_{V(x)}^{E} \frac{\mathrm{d}V'}{2\sqrt{E - V'}}. \tag{22.5}$$

Check it with the fundamental theorem of calculus: an antiderivative of $1/(2\sqrt{E - V'})$ with respect to V' is $-\sqrt{E - V'}$; evaluated between $V(x)$ and E it gives $0 - \left(-\sqrt{E - V(x)}\right) = \sqrt{E - V(x)}$. The integrand blows up at $V' = E$, but only like an inverse square root, and the integral is finite (it equals a square root), so nothing is wrong.

Put (22.5) into (22.3):

$$\Phi(E) = \int_{x_-(E)}^{x_+(E)} \left[\int_{V(x)}^{E} \frac{\mathrm{d}V'}{2\sqrt{E - V'}} \right] \mathrm{d}x. \tag{22.6}$$

Read this as a sum over tiny rectangles of size $\mathrm{d}x \times \mathrm{d}V'$ in the (x, V') plane, each carrying the weight $1/(2\sqrt{E - V'})$. Which rectangles? Those with x between the turning points and V' between the floor $V(x)$ and the ceiling E: that is, exactly the region *inside the well and below the energy* E (the shaded region you would draw on a potential-energy diagram from first-year physics). We have added up this region in vertical strips: first up a column at fixed x, then across all columns.

We can add the same rectangles in horizontal strips instead: first along a row at fixed height V', then up all rows from V_0 to E. At fixed height V' the row runs over the x where $V(x) \leq V'$, which for a single well is the interval $[x_-(V'), x_+(V')]$ of length $w(V')$. The weight does not depend on x, so the row contributes $w(V')/(2\sqrt{E - V'})\,\mathrm{d}V'$, and

$$\boxed{\Phi(E) = \int_{V_0}^{E} \frac{w(V)}{2\sqrt{E - V}}\, \mathrm{d}V} \tag{22.7}$$

(we renamed V' to V).

Taken on trust

Swapping the order of integration. We used the fact that adding up a positive weight over a region gives the same answer whether you sum columns first or rows first. For positive integrands this is a theorem of measure theory (Tonelli's theorem), valid even when the weight has an integrable singularity on the boundary as ours does at $V = E$. Double integrals and changes of order are treated in Riley, Hobson and Bence [22]. We take the theorem on trust; nothing in this chapter needs more than "rows-first equals columns-first for a positive weight".

Equation (22.7) says something remarkable: *the phase integral, and hence every Bohr–Sommerfeld level, depends on the well only through its width function* $w(V)$. Two wells with the same chord length at every height have the same semiclassical spectrum, however the chords are slid left or right. Figure 22.1 shows an example.

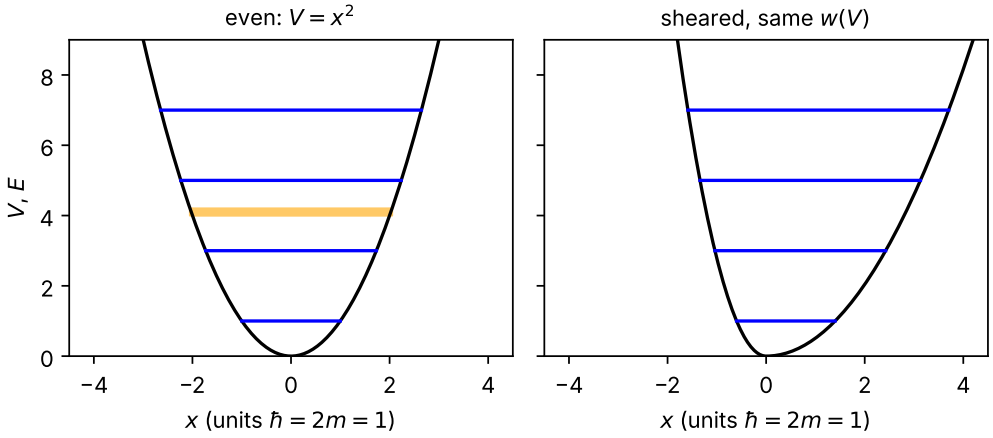


Figure 22.1: Two single wells with the same width function $w(V) = 2\sqrt{V}$. Left: the even harmonic well $V = x^2$. Right: a "sheared" well with $x_+(V) = 1.4\sqrt{V}$ and $x_-(V) = -0.6\sqrt{V}$, so it is steep on the left and gentle on the right. Horizontal segments are the Bohr–Sommerfeld levels $E = 1, 3, 5, 7$ drawn between their turning points; they are identical in both panels because each chord has the same length. The shaded strip in the left panel is one horizontal slice of height $\mathrm{d}V$ used in Eq. (22.7). Units $\hbar = 2m = 1$.

Sample Problem 22.1: A lopsided oscillator with an ordinary spectrum

Take $0 < \beta < 1$ and the potential $V(x) = x^2/(1+\beta)^2$ for $x \geq 0$ and $V(x) = x^2/(1-\beta)^2$ for $x < 0$. Find its Bohr–Sommerfeld levels.

Step 1: turning points. For $x \geq 0$, $V = x^2/(1+\beta)^2$ gives $x_+(V) = (1+\beta)\sqrt{V}$. For $x < 0$, $V = x^2/(1-\beta)^2$ gives $x_-(V) = -(1-\beta)\sqrt{V}$.

Step 2: width. $w(V) = x_+ - x_- = (1+\beta)\sqrt{V} + (1-\beta)\sqrt{V} = 2\sqrt{V}$, independent of β.

Step 3: compare. For $\beta = 0$ the potential is $V = x^2$, whose width is also $2\sqrt{V}$. By Eq. (22.7) the two wells have the same $\Phi(E)$ for every E, and therefore the same Bohr–Sommerfeld levels. For $V = x^2$ these are $E_n = 2n - 1$ (we verify this in the next section's Sample Problem).

$$\boxed{E_n = 2n - 1 \quad \text{for every } 0 \leq \beta < 1 \text{ (semiclassically)}}$$

The exact quantum levels of the lopsided well need not equal $2n - 1$; the statement is about the semiclassical spectrum.

22.1.4 Fixing the ambiguity: even and monotone

Since the semiclassical spectrum determines only $w(V)$, we must add assumptions to get a single answer. The natural ones are:

1. **Evenness**, $V(-x) = V(x)$. Then $x_-(V) = -x_+(V)$. Writing $x(V) \equiv x_+(V)$ we get $w(V) = 2x(V)$, so knowing w means knowing $x(V)$. Evenness also removes translation (the well is centered at 0) and reflection (the mirror image is the same well).

2. **Monotonicity**: V increases for $x > 0$. Then the well is a single well, each height V is reached at exactly one positive x, and $x(V)$ is a genuine function, the inverse of $V(x)$ on $x \geq 0$.

With both assumptions, Eq. (22.7) becomes

$$\boxed{\Phi(E) = \int_{V_0}^{E} \frac{x(V)}{\sqrt{E - V}}\, \mathrm{d}V} \tag{22.8}$$

This is the equation we must solve: $\Phi(E)$ is prescribed by the levels we want, and $x(V)$ is unknown.

Taken on trust

The exact (non-semiclassical) inverse problem. Even when the spectrum is computed exactly rather than by WKB, a single spectrum does not in general determine a potential: there exist distinct "isospectral" potentials. A classical theorem of G. Borg (1946) states that for a potential on a finite interval with Dirichlet ends that is symmetric about the midpoint, the spectrum does determine the potential uniquely. We do not use this theorem; the Wu–Sprung construction is semiclassical from the start. It is mentioned only so that you know evenness is the right kind of extra assumption in the exact theory as well.

Checkpoint 22.1

A friend claims to have measured the first 200 levels of a single-well potential to great accuracy, and to have deduced from them that the well is steeper on the left than on the right. Within the semiclassical approximation, can the levels support that claim?

Answer: No. Semiclassically the levels depend only on the width $w(V)$ (Eq. (22.7)); any redistribution of each chord between left and right leaves them unchanged. The asymmetry claim needs information beyond the levels, for example an assumption or a measurement of the wavefunctions.

Key Ideas

- The inverse spectral problem (levels $\to$ potential) has non-unique answers: translation, reflection, and, semiclassically, any shear that preserves chord lengths.
- Slicing the well horizontally gives $\Phi(E) = \int_{V_0}^{E} w(V)\, \mathrm{d}V/(2\sqrt{E - V})$, Eq. (22.7): WKB levels see only the width function.
- Assuming an even, monotone single well fixes $w = 2x(V)$ and gives the integral equation (22.8) for the unknown $x(V)$.

22.2 The target: a level density that grows like a logarithm

22.2.1 What the zeros ask for, on average

Recall from Chapter 17 the smooth counting function of the Riemann zeros, the paper's Eq. (1):

$$N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{22.9}$$

where log is the natural logarithm. It describes the *average* number of nontrivial zeros $\rho_n = \frac{1}{2}+it_n$ with height $0 < t_n \le t$. Recall also the **smooth zeros** s_n, defined by

$$N_{\mathrm{sm}}(s_n) = n - \tfrac{1}{2}, \tag{22.10}$$

which are where the zeros would sit if there were no fluctuations, and the **local mean spacing** $\mathrm{gap}_n = \mathrm{d}s_n/\mathrm{d}n$.

Differentiate (22.9) with the product rule. The first term gives $\frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t}$, the second gives $-\frac{1}{2\pi}$, the constant gives 0:

$$N'_{\mathrm{sm}}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{1}{2\pi} - \frac{1}{2\pi} = \frac{1}{2\pi}\log\frac{t}{2\pi}. \tag{22.11}$$

$N'_{\mathrm{sm}}(t)$ is the **level density**: the average number of zeros per unit height. Differentiating (22.10) with respect to n gives $N'_{\mathrm{sm}}(s_n)\,\mathrm{d}s_n/\mathrm{d}n = 1$, so

$$\mathrm{gap}_n = \frac{1}{N'_{\mathrm{sm}}(s_n)} = \frac{2\pi}{\log(s_n/2\pi)}. \tag{22.12}$$

Sample Problem 22.2: Density and spacing at height 1000

Find the level density and the mean spacing of the zeros at $t = 1000$, and the smooth number of zeros below 1000.
Step 1: $1000/2\pi = 159.155$ and $\log 159.155 = 5.0699$.
Step 2: By (22.11), $N'_{\mathrm{sm}}(1000) = 5.0699/(2\pi) = 0.8069$ zeros per unit height.
Step 3: By (22.12), the spacing is $1/0.8069 = 2\pi/5.0699 = 1.239$.
Step 4: By (22.9), $N_{\mathrm{sm}}(1000) = 159.155\times 5.0699 - 159.155 + 0.875 = 648.6$.

$$\boxed{N'_{\mathrm{sm}}(1000)\approx 0.807, \qquad \mathrm{gap}\approx 1.239, \qquad N_{\mathrm{sm}}(1000)\approx 648.6}$$

The density *increases* slowly with height (like a logarithm), so the zeros crowd together slowly as you go up.

22.2.2 Which wells have growing level density?

Recall from Chapter 11 that the spacing of levels depends on how fast the potential rises. Let us make that quantitative for the power-law family $V(x) = |x|^\alpha$ with $\alpha > 0$.

Sample Problem 22.3: Phase integrals of power-law wells

Compute $\Phi(E)$ for $V = |x|^\alpha$, then specialize to the harmonic ($\alpha = 2$) and linear ($\alpha = 1$) wells and find their Bohr–Sommerfeld levels.
Step 1: turning points. $|x|^\alpha = E$ gives $x_\pm = \pm E^{1/\alpha}$.
Step 2: use evenness. $\Phi(E) = 2\int_0^{E^{1/\alpha}} \sqrt{E - x^\alpha}\,\mathrm{d}x$.

Step 3: scale out E. Substitute $x = E^{1/\alpha}u$, so $\mathrm{d}x = E^{1/\alpha}\mathrm{d}u$ and $E - x^\alpha = E(1-u^\alpha)$:

$$\Phi(E) = 2\int_0^1 \sqrt{E}\sqrt{1-u^\alpha}\; E^{1/\alpha}\mathrm{d}u = 2E^{\frac{1}{2}+\frac{1}{\alpha}} I_\alpha, \qquad I_\alpha = \int_0^1 \sqrt{1-u^\alpha}\,\mathrm{d}u.$$

Step 4: harmonic. $I_2 = \int_0^1 \sqrt{1-u^2}\,\mathrm{d}u$ is the area of a quarter of the unit disk, $\pi/4$. So $\Phi = 2E\cdot\pi/4 = \pi E/2$. Bohr–Sommerfeld (22.4): $\pi E_n/2 = (n-\frac{1}{2})\pi$, so $E_n = 2n-1$. These are exactly the oscillator levels $1, 3, 5, \ldots$ of Chapter 11.

Step 5: linear. $I_1 = \int_0^1 (1-u)^{1/2}\mathrm{d}u = \left[-\frac{2}{3}(1-u)^{3/2}\right]_0^1 = \frac{2}{3}$. So $\Phi = \frac{4}{3}E^{3/2}$ and $E_n = \left[\frac{3\pi}{4}(n-\frac{1}{2})\right]^{2/3}$.

$$\boxed{\Phi(E) = 2I_\alpha E^{\frac{1}{2}+\frac{1}{\alpha}}; \quad \alpha = 2:\ E_n = 2n-1; \quad \alpha = 1:\ E_n = \left[\tfrac{3\pi}{4}\left(n-\tfrac{1}{2}\right)\right]^{2/3}}$$

By Bohr–Sommerfeld the smooth number of levels below E is $\Phi(E)/\pi$ up to a constant, so for $|x|^\alpha$ it grows like $E^{1/2+1/\alpha}$, and the level density $\mathrm{d}(\Phi/\pi)/\mathrm{d}E$ grows like $E^{1/\alpha-1/2}$:

- $\alpha > 2$ (steeper than harmonic; the box is $\alpha \to \infty$): density falls, levels spread apart;
- $\alpha = 2$: constant density, equally spaced levels;
- $\alpha < 2$ (softer than harmonic): density grows, levels crowd together.

The zeros have a density that grows, but only logarithmically, Eq. (22.11). So we should expect a potential that is softer than harmonic, though not by a fixed power. We will see in Section 22.5 what the answer actually looks like.

Pitfall

Do not confuse "a potential that grows slowly" with "levels that are far apart". It is the opposite: a slowly growing potential is a wide well at high energy, and wide wells have closely spaced levels (think of the box: doubling L quarters the levels). Soft wells crowd their levels.

22.2.3 The matching condition

Here is the precise demand. Put side by side the two equations that define the two lists of numbers:

$$\text{Bohr–Sommerfeld:}\quad \frac{\Phi(E_n)}{\pi} = n - \tfrac{1}{2}, \qquad \text{smooth zeros:}\quad N_{\mathrm{sm}}(s_n) = n - \tfrac{1}{2}.$$

Both have the form "a smooth increasing function equals $n - \frac{1}{2}$". If the two functions are equal,

$$\boxed{\Phi(E) = \pi\, N_{\mathrm{sm}}(E) \quad \text{for all } E \geq V_0,} \tag{22.13}$$

then $E_n = s_n$ for every n. Strictly, we only need agreement at the discrete points s_n; demanding it for all E is our choice of the smoothest interpolation, with no wiggles between levels.

Pitfall

Two half-integer conventions. Chapter 12 wrote the semiclassical counting function as $N(E) = \Phi(E)/\pi + \frac{1}{2}$, a curve that passes through the integer n at the level E_n. The paper's N_{sm} passes through $n - \frac{1}{2}$ at s_n, the midpoint of the jump of the true staircase (Chapter 17). The two conventions differ by exactly $\frac{1}{2}$. If you set $N(E) = N_{\mathrm{sm}}(E)$ instead

of (22.13) you would shift every level by half a spacing. Matching the two quantization conditions directly, as above, avoids the trap.

Notice what Eq. (22.13) asks for and what it does not. It asks for levels at the *smooth* zeros s_n, not at the true zeros t_n. The potential we are about to build knows the average spacing of the zeros and nothing about their individual fluctuations.

Checkpoint 22.2

Below $t = 2\pi$, Eq. (22.11) gives a negative level density. Should this worry us?

Answer: Not much. $N_{\rm sm}$ is an asymptotic description of the zeros for large t (Chapter 17); the first zero is at $t_1 = 14.13$, well above $2\pi \approx 6.28$. But it does tell us the well must start above 2π; we make this precise in Section 22.4.

Key Ideas

- The zeros' level density is $N'_{\rm sm}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$, Eq. (22.11), and their mean spacing is $\text{gap} = 2\pi/\log(t/2\pi)$, Eq. (22.12).
- Power-law wells $|x|^\alpha$ have level density $\propto E^{1/\alpha - 1/2}$; a growing density needs a potential softer than harmonic.
- The Wu–Sprung requirement is $\Phi(E) = \pi N_{\rm sm}(E)$, Eq. (22.13), which puts the Bohr–Sommerfeld levels on the smooth zeros s_n.

22.3 Solving the Abel equation

22.3.1 The equation and its inversion

Combining (22.8) and (22.13) we must find $x(V)$ from

$$\int_{V_0}^{E} \frac{x(V)}{\sqrt{E-V}}\,\mathrm{d}V = \Phi(E) \qquad (E \geq V_0), \tag{22.14}$$

with Φ given. Because the unknown sits inside an integral, this is an *integral equation*. Equations with the kernel $1/\sqrt{E-V}$ are called **Abel equations**, after N. H. Abel, who met one in 1823 while finding the shape of a slide down which a bead takes the same time from any starting height. You met the inversion in Chapter 12; we redo it here in full, in exactly the form we need.

Step A: multiply and integrate. Pick a height $W > V_0$. Multiply (22.14) by $1/\sqrt{W-E}$ and integrate over E from V_0 to W. Call the result $J(W)$:

$$J(W) \equiv \int_{V_0}^{W} \frac{\Phi(E)}{\sqrt{W-E}}\,\mathrm{d}E = \int_{V_0}^{W} \left[\int_{V_0}^{E} \frac{x(V)\,\mathrm{d}V}{\sqrt{E-V}}\right] \frac{\mathrm{d}E}{\sqrt{W-E}}. \tag{22.15}$$

Step B: swap the order. The double integral runs over the triangle $V_0 \leq V \leq E \leq W$ in the (V, E) plane, with the positive weight $x(V)/\sqrt{(E-V)(W-E)}$ (positive because $x(V) \geq 0$). Summing rows first instead of columns first (the swap justified in Section 22.1): for fixed V, E runs from V to W. So

$$J(W) = \int_{V_0}^{W} x(V)\,K(V,W)\,\mathrm{d}V, \qquad K(V,W) = \int_{V}^{W} \frac{\mathrm{d}E}{\sqrt{(W-E)(E-V)}}. \tag{22.16}$$

Step C: the kernel integral is π. Substitute $E = V + (W - V)\sin^2\varphi$ with $0 \le \varphi \le \pi/2$. Then

$$\begin{aligned} E - V &= (W - V)\sin^2\varphi, \\ W - E &= (W - V)(1 - \sin^2\varphi) = (W - V)\cos^2\varphi, \\ \mathrm{d}E &= 2(W - V)\sin\varphi\cos\varphi\,\mathrm{d}\varphi, \\ \sqrt{(W - E)(E - V)} &= (W - V)\sin\varphi\cos\varphi \quad (\text{both factors are } \ge 0 \text{ on } [0, \pi/2]). \end{aligned}$$

The integrand becomes $2\,\mathrm{d}\varphi$, and φ runs from 0 (at $E = V$) to $\pi/2$ (at $E = W$):

$$K(V, W) = \int_0^{\pi/2} 2\,\mathrm{d}\varphi = \pi, \tag{22.17}$$

independent of V and W. You can check this numerically with the short listing below. It evaluates $K(2, 5)$ by adaptive quadrature, and the printed value agrees with $\pi = 3.14159265\ldots$ to better than 10^{-9}.

```
import numpy as np
from scipy.integrate import quad
V, W = 2.0, 5.0
K = quad(lambda E: 1/np.sqrt((W-E)*(E-V)), V, W)[0]
print(K, np.pi)      # the two numbers agree
```

Step D: undo the integral. With $K = \pi$, Eq. (22.16) reads $J(W) = \pi \int_{V_0}^{W} x(V)\,\mathrm{d}V$. By the fundamental theorem of calculus,

$$x(W) = \frac{1}{\pi}\,\frac{\mathrm{d}J}{\mathrm{d}W}. \tag{22.18}$$

Step E: compute $\mathrm{d}J/\mathrm{d}W$. The integrand of J blows up at $E = W$, which makes direct differentiation awkward. Remove the singularity first: in (22.15) substitute $E = W - u^2$ with $u \ge 0$. Then $\sqrt{W - E} = u$, $\mathrm{d}E = -2u\,\mathrm{d}u$, $E = V_0$ corresponds to $u = \sqrt{W - V_0}$ and $E = W$ to $u = 0$:

$$J(W) = \int_{u=\sqrt{W-V_0}}^{0} \frac{\Phi(W - u^2)}{u}(-2u)\,\mathrm{d}u = \int_0^{\sqrt{W-V_0}} 2\,\Phi(W - u^2)\,\mathrm{d}u.$$

Now the integrand is smooth. We need one piece of notation from multivariable calculus. For a function $f(u, W)$ of two variables, the *partial derivative* $\partial f/\partial W$ is the ordinary derivative with respect to W while u is held fixed. For example, if $f(u, W) = 2\Phi(W - u^2)$, then holding u fixed and using the chain rule gives $\partial f/\partial W = 2\Phi'(W - u^2)$. Leibniz's rule for a variable upper limit $g(W)$ says: if f and $\partial f/\partial W$ are continuous, then

$$\frac{\mathrm{d}}{\mathrm{d}W}\int_0^{g(W)} f(u, W)\,\mathrm{d}u = f(g(W), W)\,g'(W) + \int_0^{g(W)} \frac{\partial f}{\partial W}\,\mathrm{d}u.$$

The first term is the fundamental theorem of calculus combined with the chain rule: it accounts for moving the upper limit. The second term accounts for the change of the integrand with the limits held fixed.

Taken on trust

Differentiation under the integral sign. That the change of the integrand at fixed limits contributes $\int \partial f/\partial W\,\mathrm{d}u$ (that is, that one may differentiate inside the integral) is a theorem of analysis that needs the continuity hypotheses just stated. A proof is in

Riley, Hobson and Bence [22]. We take it on trust.

Are the hypotheses met? For the harmonic test below, Φ' is a constant. For the Riemann target, $\Phi'(E) = \frac{1}{2}\log(E/2\pi)$ (Eq. (22.21) in the next subsection) is continuous for every $E > 0$, hence on the whole interval $[V_0, W]$ because $V_0 > 0$; and Φ itself is continuous there. So the rule applies. Here $f = 2\Phi(W - u^2)$, $g(W) = \sqrt{W - V_0}$, $g'(W) = 1/(2\sqrt{W - V_0})$, and $f(g(W), W) = 2\Phi(W - (W - V_0)) = 2\Phi(V_0)$. Hence

$$\frac{\mathrm{d}J}{\mathrm{d}W} = \frac{\Phi(V_0)}{\sqrt{W - V_0}} + \int_0^{\sqrt{W-V_0}} 2\,\Phi'(W - u^2)\,\mathrm{d}u. \tag{22.19}$$

Step F: the boundary term. For any actual single well, $\Phi(V_0) = 0$: at the bottom energy the allowed region has zero length. Then the first term of (22.19) vanishes. (Hold on to this remark: it is where the constant V_0 will be decided, in Section 22.4.)

Step G: change back to E. In the remaining integral substitute back $u = \sqrt{W - E}$, which reverses Step E: $\int_0^{\sqrt{W-V_0}} 2\Phi'(W - u^2)\mathrm{d}u = \int_{V_0}^{W} \Phi'(E)\,\mathrm{d}E/\sqrt{W - E}$. Putting this into (22.18) and renaming $W \to V$:

$$\boxed{x(V) = \frac{1}{\pi}\int_{V_0}^{V} \frac{\Phi'(E)}{\sqrt{V - E}}\,\mathrm{d}E \qquad \text{provided } \Phi(V_0) = 0.} \tag{22.20}$$

Sample Problem 22.4: Testing the inversion on the harmonic oscillator

In the previous section the well $V = x^2$ gave $\Phi(E) = \pi E/2$ with $V_0 = 0$. Feed this Φ into (22.20) and check that the formula returns $V = x^2$.
Step 1: $\Phi(0) = 0$, so the proviso holds; $\Phi'(E) = \pi/2$.
Step 2: $x(V) = \frac{1}{\pi}\int_0^V \frac{\pi/2}{\sqrt{V-E}}\mathrm{d}E = \frac{1}{2}\int_0^V \frac{\mathrm{d}E}{\sqrt{V-E}}$.
Step 3: $\int_0^V (V - E)^{-1/2}\mathrm{d}E = \left[-2\sqrt{V - E}\right]_0^V = 2\sqrt{V}$.
Step 4: $x(V) = \frac{1}{2}\cdot 2\sqrt{V} = \sqrt{V}$, that is $V = x^2$ for $x \geq 0$, extended evenly.

$$\boxed{x(V) = \sqrt{V} \iff V(x) = x^2}$$

The inversion recovers the well we started from.

22.3.2 Inverting the zeta counting function

Now insert the Riemann target. If $\Phi(E) = \pi N_{\mathrm{sm}}(E)$, then by (22.11)

$$\Phi'(E) = \pi N'_{\mathrm{sm}}(E) = \frac{1}{2}\log\frac{E}{2\pi}, \tag{22.21}$$

and (22.20) gives

$$x(V) = \frac{1}{2\pi}\int_{V_0}^{V} \frac{\log(E/2\pi)}{\sqrt{V - E}}\,\mathrm{d}E. \tag{22.22}$$

(We postpone the proviso $\Phi(V_0) = 0$ to the next section.) Now we evaluate the integral exactly. Introduce the shorthand

$$a = \sqrt{V - V_0}, \qquad b = \sqrt{V}, \qquad \text{so that} \qquad b^2 - a^2 = V_0, \quad 0 \leq a < b. \tag{22.23}$$

Step 1: remove the singularity. Substitute $E = V - u^2$: $\sqrt{V - E} = u$, $\mathrm{d}E = -2u\,\mathrm{d}u$, $E = V_0 \leftrightarrow u = a$, $E = V \leftrightarrow u = 0$:

$$x(V) = \frac{1}{2\pi}\int_a^0 \log\frac{V - u^2}{2\pi}\,\frac{-2u\,\mathrm{d}u}{u} = \frac{1}{\pi}\int_0^a \log\frac{V - u^2}{2\pi}\,\mathrm{d}u. \tag{22.24}$$

Step 2: factor. Since $V-u^2=b^2-u^2=(b-u)(b+u)$ and both factors are positive for $0\le u\le a<b$,

$$\log\frac{V-u^2}{2\pi}=\log(b+u)+\log(b-u)-\log 2\pi.$$

So $x(V)=\frac{1}{\pi}\,(Q_1+Q_2-Q_3)$ with

$$Q_1=\int_0^a\log(b+u)\,\mathrm{d}u,\qquad Q_2=\int_0^a\log(b-u)\,\mathrm{d}u,\qquad Q_3=\int_0^a\log 2\pi\,\mathrm{d}u.$$

Step 3: Q_1. With $z=b+u$ and the antiderivative $z\log z-z$ of $\log z$ (check: its derivative is $\log z+1-1$):

$$Q_1=\big[(b+u)\log(b+u)-(b+u)\big]_0^a=(b+a)\log(b+a)-(b+a)-b\log b+b=(b+a)\log(b+a)-b\log b-a.$$

Step 4: Q_2. With $z=b-u$, $\mathrm{d}z=-\mathrm{d}u$, the limits $u=0$ and $u=a$ become $z=b$ and $z=b-a$:

$$Q_2=\int_b^{b-a}\log z\,(-\mathrm{d}z)=\int_{b-a}^{b}\log z\,\mathrm{d}z=\big[z\log z-z\big]_{b-a}^{b}=b\log b-(b-a)\log(b-a)-a.$$

(The two linear terms combine as $-b+(b-a)=-a$.)

Step 5: Q_3. The integrand is constant, so $Q_3=a\log 2\pi$.

Step 6: add. In Q_1+Q_2 the terms $\pm b\log b$ cancel:

$$Q_1+Q_2-Q_3=(b+a)\log(b+a)-(b-a)\log(b-a)-2a-a\log 2\pi.$$

Step 7: regroup by b and by a. Expand the two products:

$$\begin{aligned}(b+a)\log(b+a)-(b-a)\log(b-a)&=b\big[\log(b+a)-\log(b-a)\big]+a\big[\log(b+a)+\log(b-a)\big]\\&=b\log\frac{b+a}{b-a}+a\log(b^2-a^2)\\&=b\log\frac{b+a}{b-a}+a\log V_0\quad\text{(by Eq. (22.23)).}\end{aligned}$$

Step 8: collect the a terms. $a\log V_0-2a-a\log 2\pi=a\big[\log V_0-\log e^2-\log 2\pi\big]=a\log\dfrac{V_0}{2\pi e^2}$.

Step 9: the result. Dividing by π and restoring $a=\sqrt{V-V_0}$, $b=\sqrt{V}$:

$$\boxed{x(V)=\frac{1}{\pi}\left[\sqrt{V-V_0}\,\log\frac{V_0}{2\pi e^2}+\sqrt{V}\,\log\frac{\sqrt{V}+\sqrt{V-V_0}}{\sqrt{V}-\sqrt{V-V_0}}\right],\qquad V\ge V_0.}\tag{22.25}$$

This is exactly the paper's Eq. (2). The potential $V(x)$ itself is the inverse function: for each $x\ge 0$ solve $x(V)=x$ for V (a one-dimensional root-finding problem, Chapter 4), and set $V(-x)=V(x)$. The paper calls this "inverted pointwise by bracketed root finding".

Pitfall

In Step 2 we split $\log\big[(b+u)(b-u)\big]$ into two logs. That is legal only because both factors are positive. They are, because $u\le a<b$. Had we allowed $V<V_0$ (so a imaginary) the formula would stop making sense; Eq. (22.25) is valid only for $V\ge V_0$.

Checkpoint 22.3

Set $V=V_0$ in Eq. (22.25). What is $x(V_0)$, and why must it be that value?

Answer: $a=0$, so the first term vanishes, and the logarithm becomes $\log(b/b)=0$. So $x(V_0)=0$: the bottom of the well is at the origin, as it must be for an even single well whose minimum value is V_0.

Key Ideas

- Abel's equation $\int_{V_0}^{E} x(V)\,\mathrm{d}V/\sqrt{E-V} = \Phi(E)$ is inverted by $x(V) = \frac{1}{\pi}\int_{V_0}^{V} \Phi'(E)\,\mathrm{d}E/\sqrt{V-E}$, Eq. (22.20), provided $\Phi(V_0) = 0$. The key fact is the kernel integral $K = \pi$.
- Inserting $\Phi = \pi N_{\mathrm{sm}}$ and evaluating the integral exactly gives the Wu–Sprung formula (22.25), the paper's Eq. (2).

22.4 Where V_0 comes from

★ Did you know? *Abel's Sliding Bead*

In the 1820s Niels Henrik Abel solved a generalized tautochrone problem. Given how long a bead takes to slide down a curve from each starting height, find the shape of the curve. His integral equation and its inversion formula are exactly what turn a smooth level-counting function into the smooth Wu–Sprung potential. A 19th-century mechanics puzzle builds a quantum well for the Riemann zeros.

22.4.1 The consistency condition

The inversion formula (22.20) was derived under the proviso $\Phi(V_0) = 0$ (Step F). With $\Phi = \pi N_{\mathrm{sm}}$ the proviso reads

$$N_{\mathrm{sm}}(V_0) = 0. \tag{22.26}$$

Physically: the number of levels below the bottom of the well must be zero. If $N_{\mathrm{sm}}(V_0)$ were not zero, the target would demand a nonzero phase integral at an energy where the well has no width, which no well can provide. Equation (22.26) is the condition that makes the construction consistent.

Write $y = V_0/2\pi$. Then (22.9) turns (22.26) into

$$f(y) \equiv y\log y - y = -\tfrac{7}{8}. \tag{22.27}$$

The function f has $f'(y) = \log y$, so it decreases for $y < 1$, increases for $y > 1$, and has its minimum $f(1) = -1$. Two limits complete the picture. As $y \to 0^+$, put $y = e^{-u}$, so that $u \to \infty$; then $y\log y = -ue^{-u} \to 0$, because an exponential grows faster than any power (Chapter 4), and also $y \to 0$, so $f(y) \to 0$. As $y \to \infty$, $f(y) = y(\log y - 1)$ is a product of two factors that both grow without bound, so $f \to \infty$. Now count roots. On $(0, 1]$ the continuous function f decreases strictly from values arbitrarily close to 0 down to $f(1) = -1$; since $-1 < -\frac{7}{8} < 0$, it takes the value $-\frac{7}{8}$ exactly once there. On $[1, \infty)$ it increases strictly from -1 without bound, so it takes the value $-\frac{7}{8}$ exactly once more. There are exactly two roots, one on each side of $y = 1$ (Figure 22.2).

22.4.2 Which root?

The well must have $x(V) > 0$ just above the bottom; otherwise it is not a single well with its minimum at $x = 0$. Expand (22.25) for V slightly above V_0, where $a \ll b$. Put $r = a/b$ and use the series $\log\frac{1+r}{1-r} = 2r + \frac{2r^3}{3} + \cdots$ (from $\log(1 \pm r) = \pm r - \frac{r^2}{2} \pm \frac{r^3}{3} - \cdots$, Chapter 3):

$$b\log\frac{b+a}{b-a} = b\log\frac{1+r}{1-r} = b\Big(\frac{2a}{b} + \frac{2a^3}{3b^3} + \cdots\Big) = 2a + \mathcal{O}(a^3).$$

So

$$x(V) = \frac{1}{\pi}\Big[a\Big(\log\frac{V_0}{2\pi} - 2\Big) + 2a\Big] + \mathcal{O}(a^3) = \frac{a}{\pi}\log\frac{V_0}{2\pi} + \mathcal{O}(a^3). \tag{22.28}$$

This is positive only if $\log(V_0/2\pi) > 0$, that is $V_0 > 2\pi$, or $y > 1$. The small root gives a well that bends the wrong way at the bottom, so we take the root with $y > 1$.

Sample Problem 22.5: Solving for V_0

Find the root of (22.27) with $y > 1$ and compare with the paper's $V_0 = 3.10073\pi$.
Step 1: bracket. $f(1.5) = 1.5(\log 1.5 - 1) = 1.5(0.40547 - 1) = -0.8918 < -0.875$; $f(1.6) = 1.6(0.47000 - 1) = -0.8480 > -0.875$. The root lies in $[1.5, 1.6]$.
Step 2: refine (Brent's method, Chapter 4, or Newton with $f' = \log y$): $y = 1.54013$.
Step 3: convert. $V_0 = 2\pi y = 3.0803\pi$.
Step 4: compare. The paper's value corresponds to $y = 3.10073/2 = 1.550365$, and there $N_{\mathrm{sm}}(V_0) = \frac{7}{8} + f(1.550365) = 0.0045$, not exactly zero.

$$\boxed{V_0^{\mathrm{root}} \approx 3.0803\pi, \qquad N_{\mathrm{sm}}(3.10073\pi) \approx 0.0045}$$

How should we read the small difference? The paper states $V_0 = 3.10073\pi$ without derivation, and we do not know how that value was obtained. It differs slightly from the root of the consistency condition. So be clear about what this section does and does not do. It derives the condition that fixes V_0 approximately. It does not reproduce the last digits of the paper's value, and we will not guess at their origin. What we can say precisely is what any V_0 near the root does. Formula (22.25) came from (22.20), which drops the boundary term. What, then, is the actual phase integral of the well defined by (22.25)? Call it $\Phi_{\mathrm{WS}}(E)$. Steps A–G showed only one direction: *if* a well has a phase integral Φ with $\Phi(V_0) = 0$, *then* its $x(V)$ is given by (22.20). We now need the converse for our particular $x(V)$.

Because this well is even and monotone (Problem 5), Eq. (22.8) applies: $\Phi_{\mathrm{WS}}(E) = \int_{V_0}^{E} x(V)\,\mathrm{d}V/\sqrt{E - V}$. At $E = V_0$ this is an integral over an interval of zero length, so $\Phi_{\mathrm{WS}}(V_0) = 0$ and the proviso of Step F holds automatically. We take Φ'_{WS} to be continuous, as Step E requires; this is plausible because $x(V)$ is smooth for $V > V_0$, but we do not prove it here. Steps A–G applied to Φ_{WS} then give

$$\text{(i)} \quad x(W) = \frac{1}{\pi}\int_{V_0}^{W} \frac{\Phi'_{\mathrm{WS}}(E)}{\sqrt{W - E}}\,\mathrm{d}E.$$

On the other hand, our $x(V)$ was constructed from Eqs. (22.22) and (22.21), which say

$$\text{(ii)} \quad x(W) = \frac{1}{\pi}\int_{V_0}^{W} \frac{\pi N'_{\mathrm{sm}}(E)}{\sqrt{W - E}}\,\mathrm{d}E.$$

Subtract (ii) from (i), multiply by π, and write $D(E) = \Phi'_{\mathrm{WS}}(E) - \pi N'_{\mathrm{sm}}(E)$:

$$\int_{V_0}^{W} \frac{D(E)}{\sqrt{W - E}}\,\mathrm{d}E = 0 \qquad \text{for every } W > V_0.$$

This is an Abel equation of the form (22.14), with the unknown D in place of x and zero on the right. Apply Steps A–D to it: multiply by $1/\sqrt{Y - W}$ and integrate over W from V_0 to Y, swap the order of integration, and use the kernel integral $K = \pi$. (The swap in Step B was justified for a positive weight; for a weight of either sign, the same theorem in the form known as Fubini's theorem requires only that the integral of the absolute value be finite. That holds here because D is continuous and the only singularities are inverse square roots, which are integrable.) The result is

$$\pi\int_{V_0}^{Y} D(E)\,\mathrm{d}E = 0 \qquad \text{for every } Y > V_0.$$

Differentiating with respect to Y (fundamental theorem of calculus) gives $D(Y) = 0$. So $\Phi'_{\mathrm{WS}} = \pi N'_{\mathrm{sm}}$ everywhere above V_0. Integrating from V_0 and using $\Phi_{\mathrm{WS}}(V_0) = 0$ gives

$$\Phi_{\mathrm{WS}}(E) = \int_{V_0}^{E} \pi N'_{\mathrm{sm}}(E')\,\mathrm{d}E' = \pi\big[N_{\mathrm{sm}}(E) - N_{\mathrm{sm}}(V_0)\big]. \tag{22.29}$$

Its Bohr–Sommerfeld levels therefore satisfy $N_{\mathrm{sm}}(E_n) = n - \frac{1}{2} + N_{\mathrm{sm}}(V_0)$: every level is pushed up by $N_{\mathrm{sm}}(V_0)$ mean spacings (to first order, since N_{sm} increases by 1 per spacing). With the paper's V_0 this is a uniform shift of about 0.0045 spacings. Compare the paper's own statement (Sec. II B) that its extrapolated levels miss the smooth zeros by 0.005 spacings on average: a shift of this size is at the level at which the paper itself says the construction reproduces s_n. In short, the requirement that the well have no levels below its floor fixes V_0 up to such tiny shifts; it does not by itself single out the last digits of the paper's value.

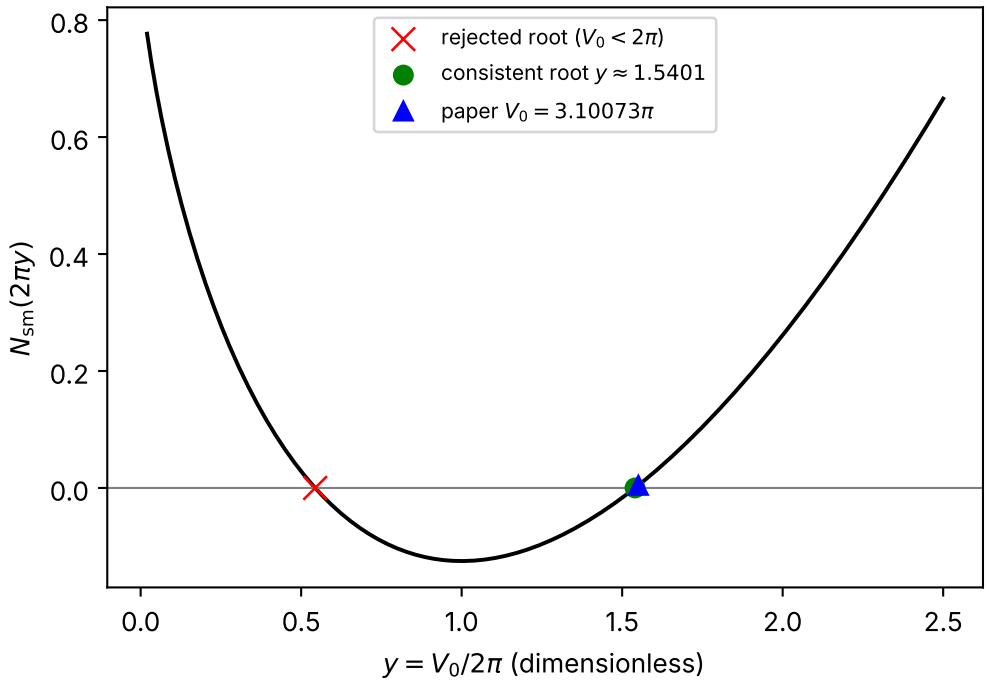


Figure 22.2: The function $N_{\mathrm{sm}}(2\pi y) = y\log y - y + \frac{7}{8}$ against $y = V_0/2\pi$. It has two roots. The left one ($y < 1$, i.e. $V_0 < 2\pi$) gives a well with $x(V) < 0$ just above its bottom and is rejected (Eq. (22.28)). The right root, $y \approx 1.5401$ ($V_0 \approx 3.0803\pi$), is the consistent bottom; the paper's $V_0 = 3.10073\pi$ (marked) lies just beside it, where $N_{\mathrm{sm}} \approx 0.0045$.

Sample Problem 22.6: The harmonic bottom of the well

Find the shape of the Wu–Sprung potential near $x = 0$, with $V_0 = 3.10073\pi$.

Step 1: From (22.28), $x \approx \frac{\sqrt{V - V_0}}{\pi}\log\frac{V_0}{2\pi}$.

Step 2: Solve for V: $V - V_0 \approx \left(\dfrac{\pi x}{\log(V_0/2\pi)}\right)^2$.

Step 3: Numbers: $V_0/2\pi = 1.550365$, $\log 1.550365 = 0.43849$, $\pi/0.43849 = 7.1645$, squared 51.33.

$$\boxed{V(x) \approx V_0 + 51.3\,x^2 \quad \text{near } x = 0}$$

The bottom is a harmonic well. This harmonic bottom is a by-product of the semiclassical

construction. Only the lowest few levels are sensitive to it, and those are the levels for which Bohr–Sommerfeld, and hence the construction, is least accurate (the paper's maximum residual, 0.063 spacings, occurs there).

Checkpoint 22.4

Suppose the constant $\frac{7}{8}$ in $N_{\rm sm}$ were replaced by 1. What would happen to the consistency condition?

Answer: It would become $f(y) = -1$, whose only solution is $y = 1$, i.e. $V_0 = 2\pi$, the minimum of f. Then $\log(V_0/2\pi) = 0$, the linear term of (22.28) vanishes and the bottom is no longer harmonic (Problem 7). The constant $\frac{7}{8}$ genuinely shapes the well's floor.

Key Ideas

- The natural consistency condition for V_0 is $N_{\rm sm}(V_0) = 0$: no levels below the floor. Of its two roots, only the one with $V_0 > 2\pi$ gives a well with its minimum at $x = 0$.
- The root is $V_0 \approx 3.0803\pi$. The paper's 3.10073π is close to this root but not equal to it, and its exact origin is not derived here; it gives $N_{\rm sm}(V_0) \approx 0.0045$, a uniform level shift of 0.0045 spacings, Eq. (22.29).
- Near the origin the well is harmonic, $V \approx V_0 + 51.3\,x^2$.

22.5 The shape of the Wu–Sprung well

22.5.1 Growth at large V

For large V, $a = \sqrt{V - V_0}$ and $b = \sqrt{V}$ are nearly equal. Rationalize the ratio in the second logarithm by multiplying top and bottom by $b + a$:

$$\frac{b+a}{b-a} = \frac{(b+a)^2}{b^2 - a^2} = \frac{(b+a)^2}{V_0}.$$

For $V \gg V_0$, $b + a \approx 2\sqrt{V}$, so $\frac{b+a}{b-a} \approx \frac{4V}{V_0}$. Also $a \approx b = \sqrt{V}$ in the first term. Then

$$x(V) \approx \frac{\sqrt{V}}{\pi}\left[\log\frac{V_0}{2\pi e^2} + \log\frac{4V}{V_0}\right] = \frac{\sqrt{V}}{\pi}\log\frac{2V}{\pi e^2} = \frac{\sqrt{V}}{\pi}\Big(\log\frac{2V}{\pi} - 2\Big). \tag{22.30}$$

Note that V_0 has dropped out: far up the well, only the leading logarithmic growth of $N_{\rm sm}$ matters. The neglected terms come from $a = \sqrt{V}(1 - V_0/2V + \cdots)$ and from the correction to $b + a$; they are smaller by a factor of order V_0/V.

So $x \sim \sqrt{V}\log V$: the chord grows a little faster than for the harmonic well ($x = \sqrt{V}$). Inverting roughly, $\sqrt{V} \approx \pi x/\log V$, and since $\log V \approx 2\log x$ to leading order,

$$V(x) \approx \left(\frac{\pi x}{2\log x}\right)^2 \qquad (x \text{ large, leading order}),$$

so V grows roughly like $(x/\log x)^2$: slower than harmonic by a factor $(\log x)^2$ (equivalently, the chord $x(V)$ is wider than the harmonic chord $\sqrt{V}$ by one logarithm). That matches the expectation of Section 22.2: a density growing like $\log E$ needs a well that is softer than harmonic, but only logarithmically so.

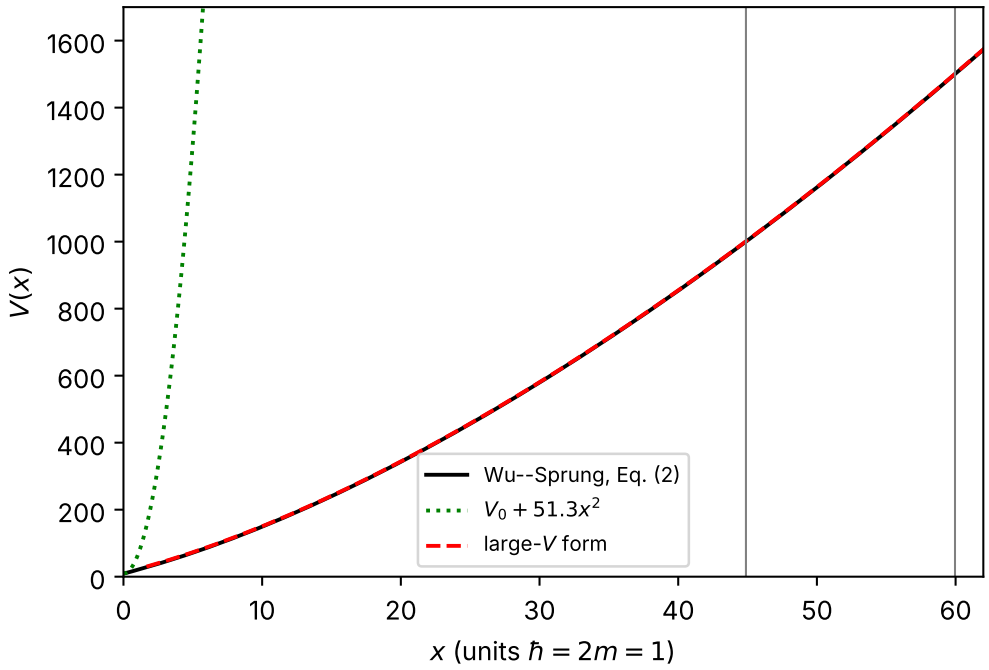


Figure 22.3: The Wu–Sprung potential (solid), Eq. (22.25) inverted, for $0 \le x \le 62$, with its harmonic bottom $V_0 + 51.3x^2$ (dotted) and the large-V form from Eq. (22.30) (dashed). Vertical lines: the box half-widths $L = x(1000) = 44.883$ (scale 1) and $L = x(1500) = 59.957$ (scale 2) of the paper. Units $\hbar = 2m = 1$.

22.5.2 The box: $L = x(V_{\max})$

A computer cannot handle the whole infinite line. The paper puts the well inside a box $[-L, L]$ with hard walls (Dirichlet conditions, Chapter 11). The walls must be far enough out that the levels of interest never feel them; the natural choice is to place the wall where the potential reaches a chosen ceiling $V_{\max}$ well above those levels: $L = x(V_{\max})$.

Sample Problem 22.7: The paper's two box sizes (RE3)

Compute $L = x(1000)$ and $L = x(1500)$ from Eq. (22.25) with $V_0 = 3.10073\pi$.
Step 1: constants. $V_0 = 9.7412$, $\log \frac{V_0}{2\pi e^2} = 0.43849 - 2 = -1.56151$.
Step 2: $V = 1000$. $a = \sqrt{990.259} = 31.468$ and $b = 31.623$. Their difference suffers from cancellation (Chapter 4), so compute it as $b - a = (b^2 - a^2)/(b + a) = V_0/(b + a) = 9.7412/63.091 = 0.1544$. Then $\frac{b+a}{b-a} = \frac{63.091}{0.1544}$, whose log is 6.0128. So $\pi x = 31.468(-1.5615) + 31.623(6.0128) = -49.14 + 190.14 = 141.00$, and $x = 141.00/\pi \approx 44.88$.
Step 3: precise evaluation (carrying more digits, or with `qmnt.L_of`): $x(1000) = 44.883$, and in the same way $x(1500) = 59.957$.
Step 4: compare with the asymptotic form (22.30): $\frac{\sqrt{1000}}{\pi}(\log \frac{2000}{\pi} - 2) = 44.855$, about 0.06% low.

$$\boxed{L_{\text{scale 1}} = 44.883, \qquad L_{\text{scale 2}} = 59.957}$$

These are the paper's values (Sec. II B).

Is $V_{\max} = 1000$ high enough at scale 1? The paper scores the first 500 levels there, and $s_{500} \approx 811$ (the true $t_{500} = 811.18$, App. A key anchors). A level at $E \approx 811$ has its turning point inside the box, and its wavefunction decays exponentially (Chapter 12) in the forbidden

region between the turning point and the wall, where V climbs to 1000. The wall affects such levels only through that exponentially small tail.

22.5.3 Turning points: each level probes a place

The turning point of level n is the point where $V(x) = E_n$; since $E_n \approx s_n$ for this potential, it is

$$x_n = x(s_n). \tag{22.31}$$

In WKB the wavefunction amplitude is proportional to $(E - V)^{-1/4}$ (Chapter 12): large where the particle moves slowly, which is near the turning points. So $|\psi_n|^2$ is largest near $\pm x_n$, and a small change δV near x_n shifts level n more than the same change elsewhere. In this sense *each level probes the potential most strongly near its turning points*, and the map $n \mapsto x_n$ converts level indices into positions. This does not mean that level n ignores the rest of the well: $|\psi_n|^2$ is spread over the whole allowed region $|x| < x_n$, so level n feels a change anywhere inside its turning points, only less strongly. What level n hardly feels is a change beyond x_n, where its wavefunction is exponentially small.

Sample Problem 22.8: Where levels 201–300 turn

Find the turning points of levels 201 and 300 of the Wu–Sprung well.
Step 1: smooth zeros. Solve $N_{\rm sm}(s) = n - \frac{1}{2}$ by Brent's method: $s_{201} \approx 398.3$ and $s_{300} \approx 542.5$. Here is a quick check of the first value. We have $N_{\rm sm}(400) = 63.66 \log 63.66 - 63.66 + 0.875 \approx 201.6$, which exceeds the target 200.5 by about 1.1. The level density there is $N'_{\rm sm}(400) = \log(63.66)/2\pi \approx 0.661$, so s_{201} lies about $1.1/0.661 \approx 1.7$ below 400, consistent with 398.3.
Step 2: evaluate Eq. (22.25). $x(s_{201}) \approx 22.5$ and $x(s_{300}) \approx 28.5$.

$$\boxed{\text{levels 201–300 turn at } x \approx 22.5\text{–}28.5}$$

This agrees with the band $x = 22.5$–28.5 that the paper's interpolation split (Sec. III) quotes for the same levels.

Checkpoint 22.5

A fitting algorithm changes V only in the region $40 < x < 44$. Which levels of the scale-1 problem does this affect most?

Answer: Those whose turning points x_n lie in $40 < x < 44$. Since x_n increases with n and x_{500} is about 38.5 (Problem 10), these are levels above 500, which scale 1 does not score at all. A change there costs almost nothing on levels 1–500 (only through the exponentially small tails of their wavefunctions).

Key Ideas

- For large V, $x(V) \approx \frac{\sqrt{V}}{\pi}(\log \frac{2V}{\pi} - 2)$, Eq. (22.30), so V grows roughly like $(x/\log x)^2$: softer than harmonic by a factor $(\log x)^2$.
- The box half-width is $L = x(V_{\rm max})$: 44.883 for $V_{\rm max} = 1000$, 59.957 for 1500.
- Level n turns at $x_n = x(s_n)$, where its wavefunction is largest; the monotone map $n \mapsto x_n$ links level index and position.

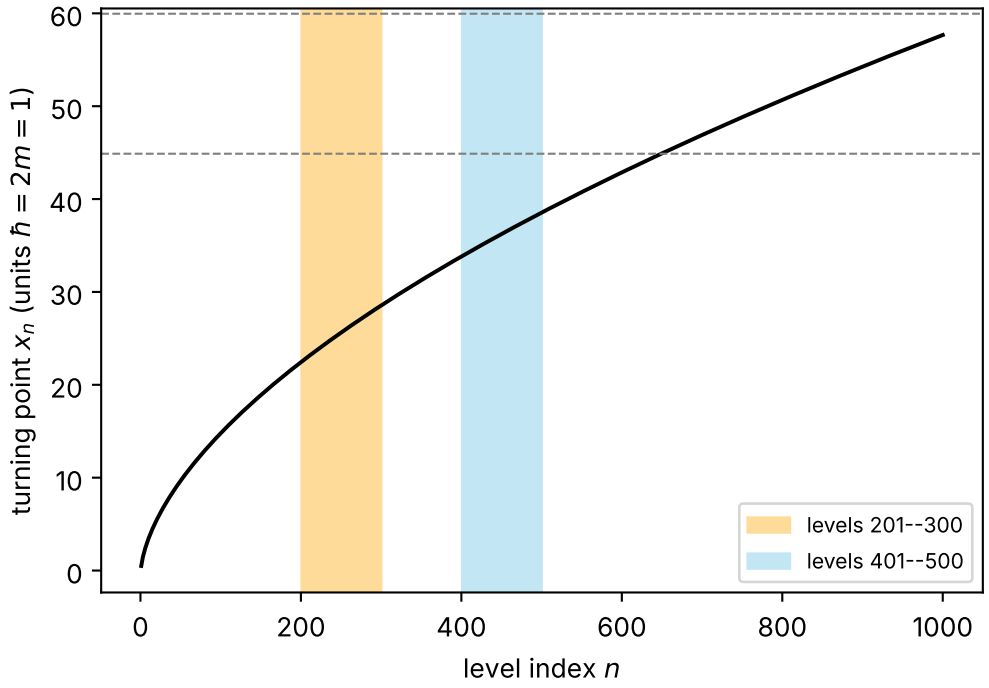


Figure 22.4: The turning-point map $n \mapsto x_n = x(s_n)$ of the Wu–Sprung potential for $n = 1, \ldots, 1000$. Shaded bands: levels 201–300 (the held-out gap of the paper's interpolation split) and levels 401–500 (the principal holdout RE5). Dashed lines: the box walls $L = 44.883$ and 59.957. Because the map is monotone, a band of levels corresponds to a band of positions.

22.6 Do the levels really sit on the smooth zeros?

We have made two approximations: Bohr–Sommerfeld, and the choice of V_0. Let us check them separately.

Check 1: the semiclassical levels. Using only the formula (22.25), compute the phase integral numerically, $\Phi(E) = 2\int_0^{\sqrt{E-V_0}} x(E-u^2)\,\mathrm{d}u$ (Eq. (22.8) after the substitution of Step E), and evaluate $\Phi(s_n)/\pi - (n - \frac{1}{2})$. Equation (22.29) predicts the constant $-N_{\mathrm{sm}}(V_0) \approx -0.0045$ for every n. A numerical evaluation with adaptive quadrature at $n = 1, 2, 10, 100, 250, 500$ confirms this to better than 10^{-6} (Problem 13 asks you to do all 500): by construction, the WKB levels of the Wu–Sprung well are the smooth zeros, shifted by 0.0045 spacings.

Check 2: the true quantum levels. Bohr–Sommerfeld is itself an approximation. To test it we need the exact eigenvalues of H in the box. We compute them with a method that is developed only later, in Chapter 23; here we use it as a black box and state exactly what we take on trust.

Taken on trust

Finite differences, taken on trust (forward reference to Chapter 23). We assume three facts proved there. (1) Sampling V at N equally spaced interior points x_j of $(-L, L)$, with spacing $h = 2L/(N+1)$ and $\psi = 0$ at the walls, turns H into a symmetric tridiagonal matrix with diagonal entries $2/h^2 + V(x_j)$ and off-diagonal entries $-1/h^2$. Its lowest eigenvalues, found with `scipy.linalg.eigh_tridiagonal` (Chapter 8), approximate the lowest levels of H in the box. (2) For a fixed level, the error of the corresponding eigenvalue is of order h^2. (3) Hence the Richardson combination $E^{\mathrm{Rich}} = (4E_{h/2} - E_h)/3$ of results on grids of spacing h and $h/2$ cancels the h^2 term. Note that this grid size N is

not the counting function $N(E)$ of Chapter 12; the paper uses the same letter for both.

```
import numpy as np, qmnt
from scipy.linalg import eigh_tridiagonal
V0 = qmnt.V0
def xws(V):                                   # Eq. (2)
    a, b = np.sqrt(V - V0), np.sqrt(V)
    return (a*np.log(V0/(2*np.pi*np.e**2)) + b*np.log((b+a)/(b-a)))/np.pi
aa = np.linspace(0, np.sqrt(1200 - V0), 400001)
VV = V0 + aa**2                               # fine table of heights V
XX = np.concatenate([[0.0], xws(VV[1:])])     # and their x(V)
L = qmnt.L_of(1000)                   # 44.883
def levels(N, k=500):
    x = qmnt.grid(L, N); h = 2*L/(N+1)
    V = np.interp(np.abs(x), XX, VV)   # table lookup of V(x)
    d = 2/h**2 + V; e = -np.ones(N-1)/h**2
    return eigh_tridiagonal(d, e, select='i',
                            select_range=(0, k-1), eigvals_only=True)
Eh, Eh2 = levels(11999), levels(23999)  # h and h/2
ER = (4*Eh2 - Eh)/3                     # Richardson
n = np.arange(1, 501)
s = qmnt.smooth_zeros(n); g = qmnt.mean_gap(n)
dev = (ER - s)/g
print(np.mean(dev), np.max(np.abs(dev)))
```

Figure 22.5 shows the result. The exact levels track s_n to a small fraction of a spacing over all 500 levels; the largest deviations are at the lowest n. That pattern is expected: Bohr–Sommerfeld is in general only asymptotically accurate, improving as n grows (Chapter 12); the harmonic well is a special case where it is exact for every n. The paper reports the same comparison at scale 1 (Sec. II B): mean residual 0.005 spacings, maximum 0.063, "worst at low n, the known semiclassical weakness of the construction". We ran exactly the listing above, in which the potential is taken from a fine table of Eq. (22.25) and interpolated linearly onto the grid. Our run gives a signed mean deviation (the average of $(E_n - s_n)/\mathrm{gap}_n$, keeping its sign) of 0.004 spacings and a maximum of $|E_n - s_n|/\mathrm{gap}_n$ of 0.05 spacings, attained at the lowest levels. These are close to the paper's 0.005 and 0.063. The paper does not say whether its mean is signed or a mean of absolute values, and the last digits depend on details such as the interpolation between table points, so we compare only to the digits shown. Our signed mean is numerically close to the WKB offset 0.0045 of Check 1.

The paper's mean of 0.005 is not the same quantity as the WKB offset of 0.0045 found in Check 1. The quantum deviation contains that uniform offset, but it also contains the genuine departures of the exact levels from Bohr–Sommerfeld, which are largest at low n. The two numbers are of similar size, but we do not claim that one explains the other.

Pitfall

"The Wu–Sprung levels sit on the zeros" is false. They sit on the *smooth* zeros s_n. The true zeros t_n differ from s_n by fluctuations of typical size a quarter of a spacing (the null rms 0.263 on zeros 401–500, RE5). The smooth well reproduces exactly the part of the zeros that a single formula, N_{sm}, already knew.

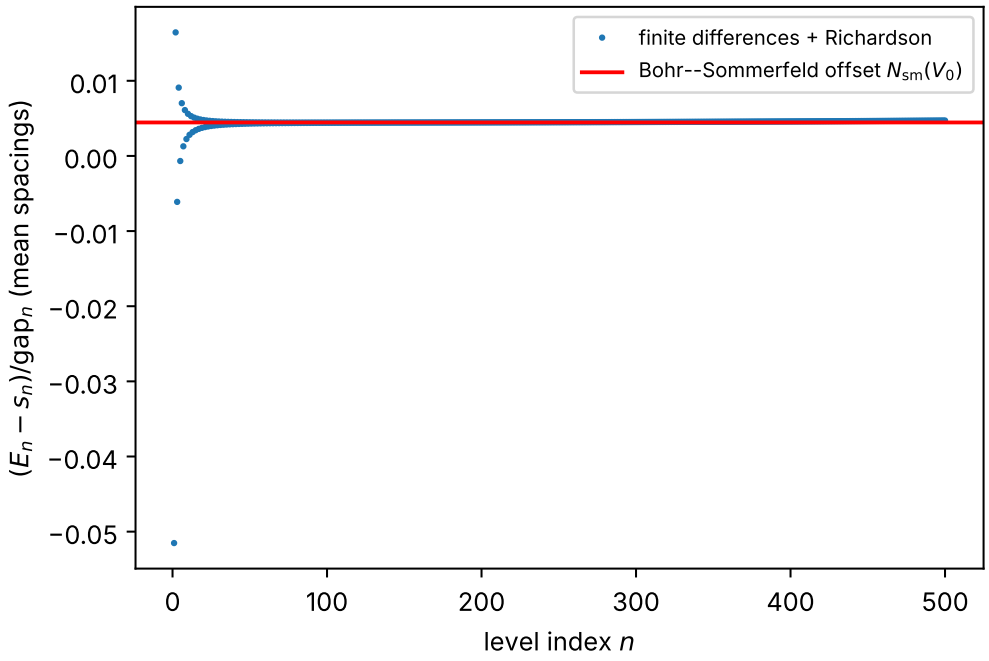


Figure 22.5: Deviation of the levels of the Wu–Sprung well from the smooth zeros, $(E_n - s_n)/\mathrm{gap}_n$, for $n = 1, \ldots, 500$ in the box $L = 44.883$. Dots: Richardson-extrapolated finite-difference eigenvalues (grids of 11999 and 23999 interior points). Line: Bohr–Sommerfeld levels of the same potential, a constant offset $N_{\mathrm{sm}}(V_0) \approx 0.0045$. The quantum levels differ from WKB mainly at low n.

Key Ideas

- The WKB levels of the Wu–Sprung well are s_n up to a uniform 0.0045 spacings, an exact consequence of Eq. (22.29).
- The exact quantum levels deviate by small fractions of a spacing, mostly at low n; the paper reports mean 0.005, max 0.063 spacings.
- The well reproduces the smooth zeros, not the zeros: the fluctuations are not in it.

22.7 Fluctuations, fractals and the empirical program

22.7.1 What Wu and Sprung did next

Having built the smooth well, Wu and Sprung [12] asked for more: a potential whose levels are the actual zeros t_n. They added a correction to the smooth potential and adjusted it until the low zeros were reproduced. They described the fitted potential as *fractal*, hence the title of their paper "Riemann zeros and a fractal potential". That a potential reproducing the zeros must be irregular is plausible on general grounds: by the explicit formula (Chapter 18) the fluctuations $t_n - s_n$ contain oscillations at every frequency $m \log p$ with slowly decaying amplitudes, and a potential that reproduces all of them must carry structure at many scales.

Later work continued the program. Van Zyl and Hutchinson [13] and Slater [14] extended the fractal fits; and Jaksch [15] generated successive zeros from a Wu–Sprung-type potential and traced the fractal corrections to the error of the Riemann–von Mangoldt approximation. Schumayer and Hutchinson [11] review this tradition. We describe these works only to the extent the paper does (its Sec. I A a).

22.7.2 The implicit logic and why it needs a test

The paper states the implicit logic of this empirical program (Sec. I): "a potential reproducing many zeros accurately has captured something about the putative zeta operator." Recall from Chapter 21 that the Hilbert–Pólya conjecture (open) asks for a self-adjoint operator whose spectrum is the zeros. A fitted potential is a self-adjoint operator. If it reproduces 500 zeros, is it a candidate?

Count the ingredients. The smooth part is fixed by $N_{\rm sm}$ alone; we built it in this chapter from one formula and one constant. The fluctuating correction is adjusted to the data. If the correction has M adjustable numbers and you fit n levels, then when M is large enough you can make the agreement as good as you like whether or not the potential "knows" anything. The paper finds in-sample rms 0.137 spacings with $M = 400$ on 400 zeros (Table I) and below 10^{-4} spacings with $M = 900$ on all 500 zeros (Sec. III); the threshold is set by the coverage law of Chapter 24, not simply by $M = n$. A useful analogy, but only an analogy because fitting a potential is a nonlinear problem, is the linear result of Chapter 9: an underdetermined linear fit has infinitely many exact solutions. Chapter 25 develops the memorizing polynomial (RE4) in full. Agreement on the zeros used to build a model is therefore not evidence of structure. The test that can separate structure from capacity is to predict zeros the model was *not* shown. This is exactly the question the paper puts to this program.

Two lessons of this chapter prepare that test.

1. **The null.** Before any fit, the Wu–Sprung well already predicts every level to within the fluctuations: its levels are s_n. Any fitted correction must beat "predict s_n", and the size of that target is the typical fluctuation (about 0.26 spacings on zeros 401–500).

2. **Locality.** A correction δV near position x affects all levels whose turning points lie at or beyond x, most strongly those turning near x, and hardly at all the levels that turn well inside x, whose wavefunctions are exponentially small there (Section 22.5). Levels 1–400 all turn inside $|x| < x_{400}$, so they say almost nothing about δV in the region $x_{400} < |x| < x_{500}$, which only higher levels reach. Conversely, a correction placed in a band of positions also shifts every level turning beyond that band, not only the levels turning inside it; this is why the paper's interpolation-split result (Sec. III) is not as simple as "each level controls its own patch".

Checkpoint 22.6

Why is it not a contradiction that the smooth Wu–Sprung potential has no fitted parameters at all (only V_0, fixed approximately by consistency), yet predicts the zeros with an rms error of about a quarter spacing?

Answer: Because the information is in the target $N_{\rm sm}$, which was derived from the zeta function (Chapter 17), not from fitting. The potential merely re-encodes $N_{\rm sm}$ as a well; it contains exactly the average behavior of the zeros, and the remaining quarter spacing is the fluctuation that $N_{\rm sm}$ does not describe.

Key Ideas

- Wu and Sprung added fitted corrections to the smooth well and found fractal-looking structure; van Zyl–Hutchinson, Slater and Jaksch continued the program.
- Fitting many zeros with many parameters shows capacity, not structure; the test is out-of-sample prediction against the null "levels at s_n".
- The turning-point map tells you which part of the potential each level constrains: level n feels changes anywhere inside its turning points, most strongly near them, and

hardly any change beyond them.

22.8 Paper connection

Eq. (2), Sec. II B. Our Eq. (22.25) is the paper's Eq. (2), term for term, including the constant $V_0 = 3.10073\pi$ and the condition $V \geq V_0$. The paper's phrase "defined implicitly" means that Eq. (2) gives x as a function of V, and "inverted pointwise by bracketed root finding" (App. A step 3: "potential by Brent inversion of Eq. (2) on a dense reference grid, cubic-spline interpolated") is the inversion we discussed. The paper does not say how V_0 was obtained. Our Section 22.4 gives the natural consistency condition $N_{\rm sm}(V_0) = 0$, whose root is 3.0803π; the paper's 3.10073π is close to but not equal to this root, and its exact origin is not derived here.

The box, Sec. II B and App. A step 3. "Scale 1 uses the domain $[-L, L]$ with $L = x(V{=}1000) = 44.883$ and the first 500 levels scored; scale 2 extends to $L = x(V{=}1500) = 59.957$ with 1000 levels." Sample Problem "The paper's two box sizes" reproduces both L values from Eq. (2). App. A step 3 quotes them to more digits as 44.8827 and 59.9572.

Accuracy of the construction, Sec. II B and App. A key anchors. The paper's statement that the Richardson levels match the smooth zeros to "0.005 spacings mean, 0.063 max (worst at low n, the known semiclassical weakness of the construction)" is the comparison of Figure 22.5. The anchors give the same 0.005/0.063 at scale 2. Our analysis adds one detail. With the paper's V_0, the WKB levels of the potential are offset uniformly by $N_{\rm sm}(V_0) \approx 0.0045$ spacings. The published numbers do not let us separate how this offset combines with the non-WKB deviations in the paper's quoted mean.

The null model, Sec. II C. "The null model is the unperturbed potential itself; numerically its holdout error coincides with that of the smooth zeros used as the oracle's null (both 0.2631 on zeros 401–500)." This chapter is why: the unperturbed Wu–Sprung levels are the smooth zeros to within a few thousandths of a spacing on average (the maximum deviation, 0.063 in the paper and 0.05 in our run, occurs at the lowest levels, far from the holdout 401–500), so "predict the Wu–Sprung levels" and "predict s_n" have the same error to the digits quoted. The paper correctly stresses that the two nulls are conceptually distinct (an eigenvalue spectrum versus the solution of the mean counting function); their numerical coincidence is a consequence of the construction derived here.

Evenness of δV, Sec. II C "Other bases". The paper fits only even corrections (cosine $\cos(k\pi x/L)$, Chebyshev $T_k(2|x|/L - 1)$, hat functions in $|x|/L$). The paper does not state why it restricts to even corrections. A natural reading, from Section 22.1, is that the semiclassical spectrum cannot distinguish left from right, so odd parts of δV are not constrained by the levels. Evenness does not by itself make the fitted problem unique: the corrected potential need not be monotone, and the paper's point is precisely that many different δV fit equally well.

Turning points, Sec. III interpolation split and Fig. 4. "Position maps to level index through the classical turning point, and levels 201–300 turn at $x = 22.5$–28.5." This is our map (22.31). The Sample Problem "Where levels 201–300 turn" gives 22.5–28.5, in agreement with the paper. Fig. 4(a),(b) of the paper shades exactly this band and shows that the fitted correction is largest there (rms $\delta V = 40$ in the band, maximum $|\delta V| = 199$ "where the smooth potential is about 500"). You can check the last remark with our formula. Across the band, $V(x)$ runs from about 398 to about 543, because the band lies between the turning points of $s_{201} \approx 398.3$ and $s_{300} \approx 542.5$.

Sec. V ($p = 2$ period remark). Sec. V (paragraph "The proposed mechanism fails quantitatively") states that the $p = 2$ oscillation, period $2\pi/\log 2 \approx 9$ in t, is "only ∼6–7 levels at these heights"; that conversion uses the mean spacing (22.12) (Problem 14).

Sec. I and Sec. I A a. The paper's description of the empirical program ("Wu and Sprung showed that a smooth one-dimensional potential, fixed by the mean counting function, reproduces

the low zeros after fitted fluctuating corrections; successors refined the reconstructions") and its literature strand (a) are the subject of Section 22.7. The paper's scope statement matters here: it tests only fluctuation potentials *on the Wu–Sprung background* in three bases, and "other backgrounds and other parametrizations . . . are not tested". Its title says "on the Wu–Sprung background" for this reason.

22.9 Summary

★ Did you know? *A Fractal Well*

In 1993 Hua Wu and Donald Sprung published, in *Physical Review E*, a one-dimensional potential whose lowest levels reproduce Riemann zeros. As they fitted more zeros, the correction to the smooth potential became rough and jagged, and they reported a fractal structure with dimension close to 1.5. The fluctuations of the zeros had become fluctuations in the shape of the well.

- Semiclassically, a single well's levels depend only on its width function, $\Phi(E) = \int_{V_0}^{E} w(V)\,\mathrm{d}V/(2\sqrt{E-V})$, Eq. (22.7); evenness and monotonicity make the inverse problem unique, Eq. (22.8).
- The target $\Phi(E) = \pi N_{\mathrm{sm}}(E)$, Eq. (22.13), puts Bohr–Sommerfeld levels on the smooth zeros.
- Abel inversion: $x(V) = \frac{1}{\pi}\int_{V_0}^{V} \Phi'(E)\,\mathrm{d}E/\sqrt{V-E}$ if $\Phi(V_0) = 0$, Eq. (22.20); the kernel integral is π, Eq. (22.17).
- The Wu–Sprung potential, Eq. (22.25) (paper Eq. (2)), follows by exact integration.
- The natural consistency condition for V_0 is $N_{\mathrm{sm}}(V_0) = 0$ with $V_0 > 2\pi$: root $\approx 3.0803\pi$. The paper's 3.10073π is close to but not equal to this root (its exact origin is not derived here) and shifts levels uniformly by ≈ 0.0045 spacings, Eq. (22.29).
- Shape: harmonic bottom $V \approx V_0 + 51.3x^2$; $x \approx \frac{\sqrt{V}}{\pi}(\log\frac{2V}{\pi} - 2)$ at large V, Eq. (22.30); $L = 44.883$ and 59.957.
- Turning-point map $x_n = x(s_n)$, Eq. (22.31): levels 201–300 turn at $x \approx 22.5$–28.5.
- The well reproduces s_n, not t_n; fitted fluctuation corrections must be judged by held-out prediction against this null.

22.10 Problems

1. • For the linear well $V = |x|$ find the width function $w(V)$ and use Eq. (22.7) to compute $\Phi(E)$. Compare with the Sample Problem on power-law wells.

2. • Compute the mean spacing of the zeros at height $t = 100$ from Eq. (22.12).

3. • Show directly from Eq. (22.25) that $x(V_0) = 0$ for any $V_0 > 0$.

4. •• An infinite well $[-L, L]$ with $V = 0$ inside has levels $(k\pi/2L)^2$ and $\Phi(E) = 2L\sqrt{E}$. Apply the Abel inversion (22.20) with $V_0 = 0$ and interpret the result $x(V) = L$.

5. •• Differentiate Eq. (22.24) with Leibniz's rule to show $\dfrac{\mathrm{d}x}{\mathrm{d}V} = \dfrac{1}{2\pi a}\log\dfrac{V_0}{2\pi} + \dfrac{1}{\pi}\displaystyle\int_0^a \frac{\mathrm{d}u}{V-u^2}$, and conclude that the Wu–Sprung potential is monotone for $x > 0$ when $V_0 > 2\pi$.

6. • Which of the two assumptions, evenness or monotonicity, removes the reflection ambiguity? Which removes multiple wells?

7. •• Replace $\frac{7}{8}$ by 1 in N_{sm}. Show that the consistency condition forces $V_0 = 2\pi$ and find the leading behavior of $x(V)$ near the bottom. Is the bottom harmonic?

8. •• Using numpy and scipy, code Eq. (22.25) and compute $L = x(1000)$ and $x(1500)$. Compare with the paper's Sec. II B and with `qmnt.L_of`.

9. •• (Paper, Sec. III.) Compute the turning points $x(s_{201})$ and $x(s_{300})$ and compare with the paper's band $x = 22.5$–28.5 for the held-out levels of the interpolation split.

10. •• Using qmnt, compute the turning points of levels 401 and 500 (the principal holdout, RE5). Which part of the scale-1 box do these levels probe?

11. •• (Paper, Sec. II C.) Explain why the paper can say that the null of the operator fits and the null of the arithmetic oracle coincide numerically (0.2631 on zeros 401–500), and why it still calls them conceptually distinct.

12. ••• Derive Eq. (22.30) including the next correction: show that $x(V) = \frac{\sqrt{V}}{\pi}(\log\frac{2V}{\pi} - 2) + \mathcal{O}(V^{-1/2}\log V)$.

13. ••• Using scipy's `quad`, compute $\Phi(s_n) = 2\int_0^{\sqrt{s_n - V_0}} x(s_n - u^2)\,\mathrm{d}u$ for $n = 1, \dots, 500$ and show that $\Phi(s_n)/\pi - (n - \frac{1}{2})$ is the same constant for all n. Identify the constant.

14. •• (Paper, Sec. V.) Show that one period $2\pi/\log 2$ of the $p = 2$ wave (RE2) corresponds to about 6–7 levels near $t = 500$, as stated in the paper. (The paper's "these heights" refers to the training zeros 1–400, which span t up to about 680; check that the count stays between 6 and 7 there too.)

15. •• Construct a non-even potential with exactly the same Bohr–Sommerfeld levels as the Wu–Sprung well.

16. ••• Show that for $\Phi(E) = cE^{\gamma}$ ($\gamma > 0$, $V_0 = 0$) the Abel inversion gives $x(V) \propto V^{\gamma - 1/2}$, and find the proportionality constant in terms of $\int_0^{\pi/2} \sin^{2\gamma-1}\varphi\,\mathrm{d}\varphi$.

17. ••• Use the Abel inversion to recover the linear well from $\Phi(E) = \frac{4}{3}E^{3/2}$.

18. •• (Paper, Sec. I.) In a paragraph, explain why reproducing the first 500 zeros with a 900-coefficient correction to the Wu–Sprung well is not, by itself, evidence for the Hilbert–Pólya conjecture.

19. • Compute $N_{\mathrm{sm}}(t_1)$ for $t_1 = 14.134725$ (RE1). Is it close to $\frac{1}{2}$, as Eq. (22.10) would suggest for $n = 1$?

20. ••• Using numpy and scipy, reproduce Figure 22.5 at the smaller grid pair $N = 5999$ and 11999. Does Richardson extrapolation still bring the levels within a small fraction of a spacing of s_n?

22.11 Answers to selected problems

1. $x_{\pm} = \pm V$, so $w = 2V$. Then $\Phi = \int_0^E \frac{2V\,\mathrm{d}V}{2\sqrt{E-V}}$; with $V = E - u^2$ this is $2\int_0^{\sqrt{E}}(E - u^2)\mathrm{d}u = 2(E^{3/2} - E^{3/2}/3) = \frac{4}{3}E^{3/2}$, matching $2I_1E^{3/2}$ with $I_1 = \frac{2}{3}$.

2. $100/2\pi = 15.9155$ and $\log 15.9155 = 2.7673$, so gap $= 2\pi/2.7673 = 2.2705$.

3. At $V = V_0$, $a = 0$: the first term is 0 and the second is $\sqrt{V_0}\log(\sqrt{V_0}/\sqrt{V_0}) = 0$.

5. From (22.24), $x = \frac{1}{\pi}\int_0^a \log\frac{V-u^2}{2\pi}\mathrm{d}u$ with $a = \sqrt{V - V_0}$, $\mathrm{d}a/\mathrm{d}V = 1/(2a)$. Leibniz: the boundary term is $\frac{1}{\pi}\log\frac{V-a^2}{2\pi}\cdot\frac{1}{2a} = \frac{1}{2\pi a}\log\frac{V_0}{2\pi}$, and $\partial_V \log(V-u^2) = 1/(V-u^2)$. For $V_0 > 2\pi$ both terms are positive (the integrand $1/(V-u^2) > 0$ since $u \le a < \sqrt{V}$), so x increases with V and the inverse $V(x)$ increases with x.

6. Evenness removes reflection (and translation). Monotonicity on $x > 0$ ensures a single well, so each height has one chord.

7. $f(y) = y\log y - y = -1$ has the unique solution $y = 1$ (the minimum of f), so $V_0 = 2\pi$. Then $\log\frac{V_0}{2\pi e^2} = -2$ and, with $\log\frac{1+r}{1-r} = 2r + \frac{2}{3}r^3 + \cdots$, $\pi x = -2a + 2a + \frac{2a^3}{3V_0} + \cdots$, so $x \approx \frac{2a^3}{3\pi V_0}$. Hence $V - V_0 = a^2 \propto x^{2/3}$: a cusp at the bottom, not harmonic.

9. $s_{201} \approx 398.3$ and $s_{300} \approx 542.5$ give $x(s_{201}) \approx 22.5$ and $x(s_{300}) \approx 28.5$. This agrees with the paper's 22.5–28.5.

11. The unperturbed Wu–Sprung levels equal s_n to within a few thousandths of a spacing on average (Sec. 22.6; the largest deviations, a few hundredths, occur at the lowest levels, not in the holdout), so the error of predicting them for t_n equals the error of predicting s_n, to the digits shown. Conceptually one is an eigenvalue list of a particular operator, the other the solution of $N_{\mathrm{sm}}(s) = n - \frac{1}{2}$; they coincide only because the Wu–Sprung construction was designed to make them coincide, and the coincidence is approximate.

13. The constant is $-N_{\mathrm{sm}}(V_0) \approx -0.0045$ for $V_0 = 3.10073\pi$, by Eq. (22.29); numerically it is the same for all n to quadrature accuracy.

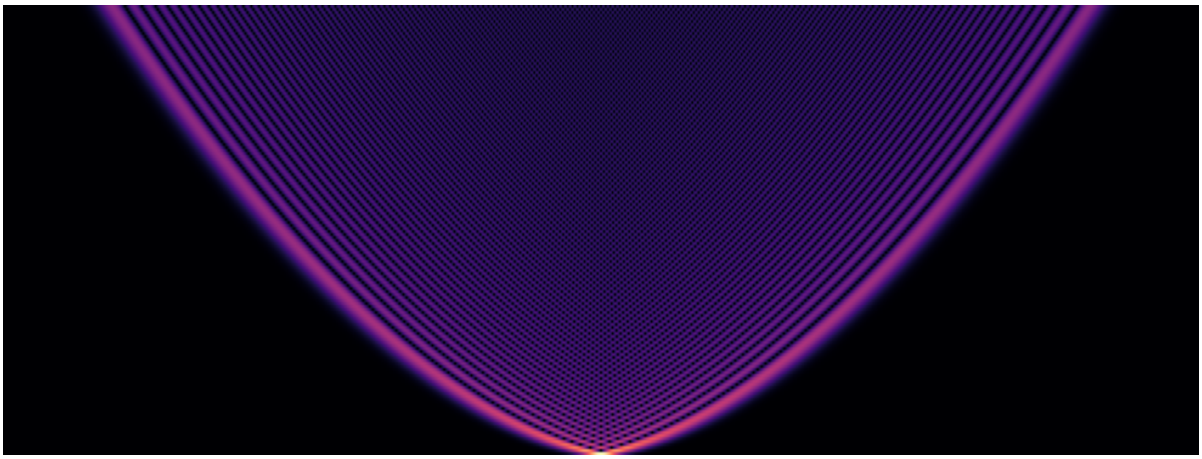
15. Keep the width $2x(V)$ but share it unequally: for $0 < \beta < 1$ set $x_+(V) = (1+\beta)x(V)$ and $x_-(V) = -(1-\beta)x(V)$, i.e. $\tilde{V}(x) = V_{\mathrm{WS}}(x/(1+\beta))$ for $x \ge 0$ and $V_{\mathrm{WS}}(x/(1-\beta))$ for $x < 0$. Chords have length $2x(V)$, so by Eq. (22.7) the Bohr–Sommerfeld levels are unchanged.

17. $\Phi' = 2\sqrt{E}$, so $x = \frac{2}{\pi}\int_0^V \sqrt{E}\,\mathrm{d}E/\sqrt{V-E}$. With $E = V\sin^2\varphi$, $\mathrm{d}E = 2V\sin\varphi\cos\varphi\,\mathrm{d}\varphi$, the integral is $\int_0^{\pi/2} 2V\sin^2\varphi\,\mathrm{d}\varphi = \pi V/2$. Hence $x = V$, i.e. $V = |x|$.

19. $t_1/2\pi = 2.24961$ and $\log 2.24961 = 0.81076$, so $N_{\mathrm{sm}} = 2.24961 \times 0.81076 - 2.24961 + 0.875 = 0.449$. This is close to $\frac{1}{2}$. Because N_{sm} is increasing and $N_{\mathrm{sm}}(t_1) < N_{\mathrm{sm}}(s_1) = \frac{1}{2}$, we have $t_1 < s_1$. The difference $\frac{1}{2} - 0.449 \approx 0.05$ means that the first zero lies about 0.05 of a mean spacing below s_1.

Part V

Computation, Fitting and Generalization



Probability densities $|\psi_n(x)|^2$ of the first 150 levels of the discretized Wu–Sprung Hamiltonian, one row per level.

Chapter 23

Solving the Schrödinger Equation on a Computer

↑Contents

What this chapter is about

In Chapter 11 you solved the infinite square well with pencil and paper: you wrote down sines and cosines, imposed the walls, and read off the energies $E_k = (k\pi/2L)^2$. That worked because the potential inside the box was zero. The potential this book cares about, the Wu–Sprung potential, has no such luck. Nobody can write its eigenfunctions in closed form. To get its energy levels we have to hand the problem to a computer.

A computer cannot take a derivative of a function it only knows at a finite list of points. What it can do is subtract neighboring numbers. In this chapter you will learn to replace $-\mathrm{d}^2\psi/\mathrm{d}x^2$ by a difference of neighboring values, and you will watch the Schrödinger equation turn into a matrix eigenvalue problem of exactly the kind Chapter 8 taught you to solve. That replacement has a price. The computed levels are slightly wrong, and the error grows quickly with the level number. For the levels that matter in the paper the error is not small at all: at levels 401–500 it is several times larger than the tiny fluctuations the paper is trying to predict. So we will learn how big the error is, why it has the size it has, how to remove it by Richardson extrapolation, and how to keep track of normalization when integrals become sums.

Where this shows up in the paper. Section II B of the paper [1] describes the "forward solver": second-order finite differences on N interior points with Dirichlet walls, a symmetric tridiagonal matrix, eigenpairs from LAPACK, and a Richardson extrapolation used to correct the *targets* rather than the operator. Section II C proves, in Eq. (3), that this correction leaks no information from the held-out zeros. Appendix A steps 3–5 record the grids, the solver and the corrected targets, and step 9 the normalization of the eigenvectors; the appendix's list of key anchors contains a striking number: without the correction the null model's error on zeros 401–500 is 2.09 mean spacings instead of 0.263. By the end of this chapter you will be able to explain every one of those statements and reproduce the underlying computations at small scale.

What you need

- Chapter 8: eigenvalues and eigenvectors, the characteristic polynomial, real symmetric matrices have real eigenvalues and orthonormal eigenvectors, tridiagonal matrices, `scipy.linalg.eigh_tridiagonal` as a black-box solver.
- Chapter 11: the infinite square well on $[-L, L]$ with levels $E_k = (k\pi/2L)^2$ in units $\hbar = 2m = 1$, normalized eigenfunctions, parity of states in even potentials; the harmonic oscillator levels.
- Chapter 10: the time-independent equation $H\psi = E\psi$ with $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$, normalization $\int |\psi|^2 \, \mathrm{d}x = 1$, expectation values.
- Chapter 7: vectors in N dimensions, inner products, a function sampled on a grid as a vector.
- Chapter 4: big-O notation, machine epsilon (2.2×10^{-16} in double precision, 1.2×10^{-7} in float32).
- Chapter 2: $e^{i\theta} = \cos\theta + i\sin\theta$. Chapter 3: the geometric series.
- First-year calculus: Taylor series of a smooth function, the trigonometric identities $\sin(a+b) + \sin(a-b) = 2\sin a\cos b$ and $1 - \cos\theta = 2\sin^2(\theta/2)$.

23.1 From derivatives to differences

A computer stores a function as a list of numbers: the values $f(x_1), f(x_2), \ldots$ at a set of points. It has no idea what happens between the points. If we want it to compute a derivative, we must express the derivative in terms of those stored values. The tool for this is the Taylor series you met in first-year calculus.

23.1.1 Taylor series on both sides of a point

Let f be a smooth function (one with as many continuous derivatives as we need), and let $h > 0$ be a small step. Write f', f'', f''', f'''' for the first four derivatives at x, and $f^{(5)}$, $f^{(6)}$ for the fifth and sixth. Taylor's theorem gives the value one step to the right,

$$f(x+h) = f + hf' + \frac{h^2}{2}f'' + \frac{h^3}{6}f''' + \frac{h^4}{24}f'''' + \frac{h^5}{120}f^{(5)} + \frac{h^6}{720}f^{(6)} + \cdots, \tag{23.1}$$

where every f on the right is evaluated at x. Replacing h by $-h$ flips the sign of every odd power:

$$f(x-h) = f - hf' + \frac{h^2}{2}f'' - \frac{h^3}{6}f''' + \frac{h^4}{24}f'''' - \frac{h^5}{120}f^{(5)} + \frac{h^6}{720}f^{(6)} - \cdots. \tag{23.2}$$

23.1.2 The forward difference: first order

The definition of the derivative suggests the simplest approximation, the *forward difference*. Subtract f from both sides of (23.1):

$$f(x+h) - f(x) = hf' + \frac{h^2}{2}f'' + \frac{h^3}{6}f''' + \cdots.$$

Divide by h:

$$\frac{f(x+h) - f(x)}{h} = f' + \frac{h}{2}f'' + \frac{h^2}{6}f''' + \cdots. \tag{23.3}$$

The left side is computable from stored values. It equals f' plus an error whose leading term is $\frac{h}{2}f''$. In the language of Chapter 4, the error is $\mathcal{O}(h)$: halve h and the error roughly halves. We call such a formula *first-order accurate*.

23.1.3 The central second difference: second order

The Schrödinger equation needs the second derivative. Add (23.1) and (23.2). The odd powers cancel in pairs ($+hf'$ with $-hf'$, $+\frac{h^3}{6}f'''$ with $-\frac{h^3}{6}f'''$, and so on), and the even powers double:

$$f(x+h)+f(x-h)=2f+2\cdot\frac{h^2}{2}f''+2\cdot\frac{h^4}{24}f''''+2\cdot\frac{h^6}{720}f^{(6)}+\cdots .$$

Simplify the coefficients ($2/2=1$, $2/24=1/12$, $2/720=1/360$):

$$f(x+h)+f(x-h)=2f+h^2f''+\frac{h^4}{12}f''''+\frac{h^6}{360}f^{(6)}+\cdots .$$

Subtract $2f$ from both sides:

$$f(x+h)-2f(x)+f(x-h)=h^2f''+\frac{h^4}{12}f''''+\frac{h^6}{360}f^{(6)}+\cdots .$$

Divide by h^2:

$$\boxed{D_h^2 f(x)\equiv\frac{f(x+h)-2f(x)+f(x-h)}{h^2}=f''(x)+\frac{h^2}{12}f''''(x)+\frac{h^4}{360}f^{(6)}(x)+\cdots} \tag{23.4}$$

The symbol D_h^2 is our name for the *central second difference* with step h. Its error starts at $\frac{h^2}{12}f''''$, so it is $\mathcal{O}(h^2)$: halve the step and the error falls by a factor of about four. This is called *second-order accuracy*, and it is the "second-order finite differences" of the paper's Sec. II B.

Two features of (23.4) will matter all chapter. First, only even powers of h appear in the error, because the formula is symmetric under $h\to -h$. Second, the error is controlled by the *fourth* derivative of f. A function that wiggles rapidly has a large fourth derivative, so rapid wiggles are where the formula struggles.

A word on status. The series in (23.4) need not converge for every function, but Taylor's theorem with remainder (first-year calculus) guarantees that if f has a continuous sixth derivative, the terms we kept are exact and the error in stopping after the h^2 term is bounded by a constant times h^4. That is all we use.

Sample Problem 23.1: The second difference of $\sin x$

Approximate $f''(1)$ for $f(x)=\sin x$ using D_h^2 with $h=0.1$ and with $h=0.05$. Compare the errors with the prediction $\frac{h^2}{12}f''''(1)$, and find the ratio of the two errors.

Step 1: Simplify the numerator. With $f=\sin$, the identity $\sin(a+b)+\sin(a-b)=2\sin a\cos b$ with $a=1$, $b=h$ gives $\sin(1+h)+\sin(1-h)=2\sin 1\cos h$. So

$$D_h^2\sin(1)=\frac{2\sin 1\cos h-2\sin 1}{h^2}=-\sin 1\,\frac{2(1-\cos h)}{h^2}.$$

The exact value is $f''(1)=-\sin 1=-0.8414710$.

Step 2: $h=0.1$. From the cosine series, $1-\cos 0.1=0.005-4.1667\times10^{-6}+\cdots=0.0049958$, so $2(1-\cos 0.1)/0.01=0.9991669$ and $D_{0.1}^2\sin(1)=-0.8407700$. The error $D_h^2 f-f''$ is $-0.8407700+0.8414710=7.010\times10^{-4}$.

Step 3: $h=0.05$. Similarly $2(1-\cos 0.05)/0.0025=0.9997917$, so $D_{0.05}^2\sin(1)=-0.8412957$ and the error is 1.753×10^{-4}.

Step 4: Compare with the prediction. Since $f''''=\sin$, the leading error is $\frac{h^2}{12}\sin 1$: for $h=0.1$ this is $\frac{0.01}{12}(0.8414710)=7.012\times10^{-4}$, matching Step 2 to three digits. The

ratio of the two errors is

$$\boxed{\frac{7.010 \times 10^{-4}}{1.753 \times 10^{-4}} = 4.00}$$

the factor $2^2 = 4$ that second-order accuracy predicts, to three digits (the small h^4 terms keep it from being exactly 4).

23.1.4 When smaller is not better: roundoff

Figure 23.1 plots the error of $D_h^2 \sin(1)$ against h on logarithmic axes. For moderate h the points follow a straight line of slope 2, the signature of $\mathcal{O}(h^2)$. For very small h, however, the error turns around and grows. Chapter 4 explained why: each stored value $f(x \pm h)$ carries a relative rounding error of order machine epsilon, $\epsilon \approx 2.2 \times 10^{-16}$. The numerator $f(x+h) - 2f(x) + f(x-h)$ is a tiny difference of three numbers of size about 1, so it carries an absolute error of a few times ϵ, and dividing by h^2 magnifies that to roughly ϵ/h^2. The total error behaves like

$$\text{error} \approx \frac{h^2}{12}\,|f''''| + (\text{a few}) \times \frac{\epsilon\,|f|}{h^2}. \tag{23.5}$$

The first term falls as h shrinks, the second rises. For the grids in the paper ($h \approx 0.015$, as we will compute) the roundoff term is utterly negligible, but it is worth knowing that "make h tiny" is not a free lunch.

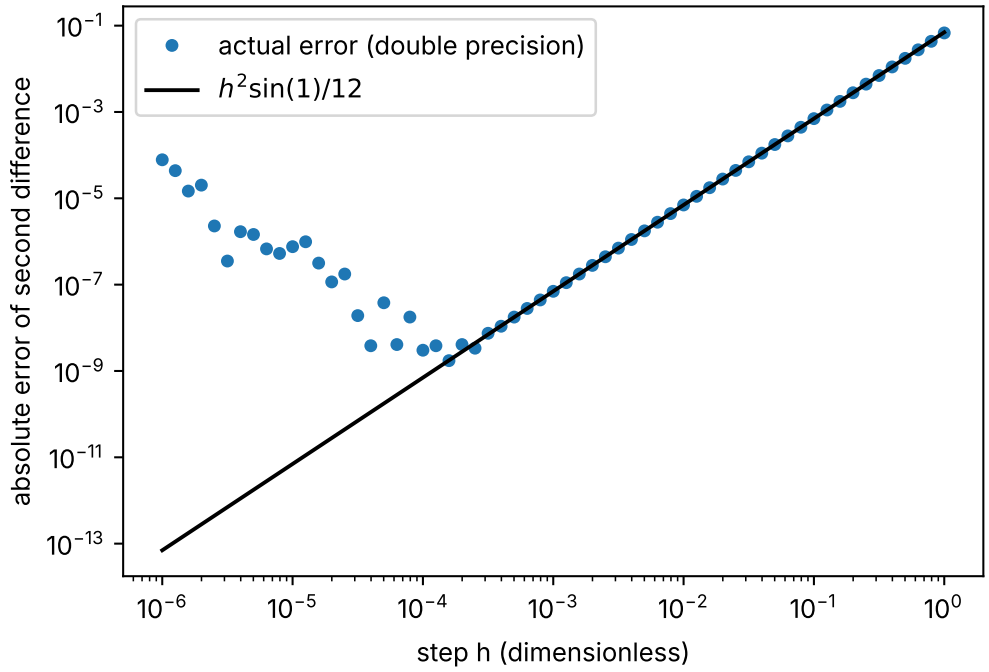


Figure 23.1: Error of the central second difference $D_h^2 \sin(1)$ as a function of the step h (dimensionless). Dots: actual error computed in double precision. Line: the leading truncation error $h^2 \sin(1)/12$ from Eq. (23.4). For h above about 10^{-3} the dots follow the line of slope 2; for very small h rounding error, of order ϵ/h^2, takes over and the error grows again.

Checkpoint 23.1

You compute a second derivative with D_h^2 and get an error of 8×10^{-6}. You repeat with h divided by 10, still far from the roundoff regime. What error do you expect?

Answer: About 8×10^{-8}: the error scales as h^2, so dividing h by 10 divides the error by 100.

Pitfall

Do not confuse "second-order" (error $\propto h^2$) with "second derivative". The forward difference (23.3) approximates a first derivative and is first-order; the central second difference approximates a second derivative and is second-order. The central *first* difference $[f(x+h) - f(x-h)]/(2h)$ is also second-order (Problem 21).

Key Ideas

- The central second difference $D_h^2 f = [f(x+h) - 2f(x) + f(x-h)]/h^2$ equals $f'' + \frac{h^2}{12} f'''' + \mathcal{O}(h^4)$, Eq. (23.4).
- Its error contains only even powers of h and is controlled by the fourth derivative: rapidly wiggling functions are hardest.
- Rounding error grows like ϵ/h^2, so very small steps eventually make things worse.

23.2 The grid, the walls and the matrix

23.2.1 Placing the points

We want the levels of $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$ in a box $[-L, L]$ with infinitely high walls at $x = \pm L$ (units $\hbar = 2m = 1$, as in Chapter 10). Recall from Chapter 11 that infinite walls force the wavefunction to vanish there: $\psi(-L) = \psi(L) = 0$. A condition that fixes the value of the unknown function on the boundary is called a *Dirichlet* boundary condition, so these are "Dirichlet walls".

Divide $[-L, L]$ into $N+1$ equal intervals. That creates $N+2$ points,

$$x_j = -L + jh, \qquad j = 0, 1, \ldots, N+1, \qquad h = \frac{2L}{N+1}. \tag{23.6}$$

The spacing formula follows because $N+1$ intervals of width h must fill the length $2L$. The two end points $x_0 = -L$ and $x_{N+1} = L$ sit on the walls. The other N points, $x_1, \ldots, x_N$, are the *interior points*; for grids, the paper and `qmnt` use "N" for the number of interior points. Do not confuse it with the counting functions $N(T)$, N_{sm} and N_{fl}, which the paper writes with the same letter. Write ψ_j for our approximation to $\psi(x_j)$ and $V_j = V(x_j)$. The walls give

$$\psi_0 = 0, \qquad \psi_{N+1} = 0, \tag{23.7}$$

so only the N interior values are unknown.

23.2.2 One equation per interior point

At each interior point x_j, replace $\psi''(x_j)$ in $-\psi'' + V\psi = E\psi$ by the second difference (23.4) with step h, dropping the error terms:

$$-\frac{\psi_{j+1} - 2\psi_j + \psi_{j-1}}{h^2} + V_j \psi_j = E\psi_j, \qquad j = 1, \ldots, N. \tag{23.8}$$

Expand the left side to collect the coefficient of each unknown:

$$-\frac{1}{h^2}\psi_{j-1} + \left(\frac{2}{h^2} + V_j\right)\psi_j - \frac{1}{h^2}\psi_{j+1} = E\psi_j. \tag{23.9}$$

Now look at the two ends. For $j = 1$ the equation involves ψ_0, which is zero by (23.7), so that term disappears:

$$\left(\frac{2}{h^2} + V_1\right)\psi_1 - \frac{1}{h^2}\psi_2 = E\psi_1.$$

For $j = N$ the term in ψ_{N+1} disappears in the same way:

$$-\frac{1}{h^2}\psi_{N-1} + \left(\frac{2}{h^2} + V_N\right)\psi_N = E\psi_N.$$

23.2.3 The matrix form

Collect the unknowns into a column vector $\boldsymbol{\psi} = (\psi_1, \ldots, \psi_N)^\top$ (Chapter 7). The N equations (23.9), with the two end versions, say exactly $\mathsf{H}\boldsymbol{\psi} = E\boldsymbol{\psi}$, where H is the $N \times N$ matrix

$$\mathsf{H} = \begin{pmatrix} \frac{2}{h^2} + V_1 & -\frac{1}{h^2} & 0 & \cdots & 0 \\ -\frac{1}{h^2} & \frac{2}{h^2} + V_2 & -\frac{1}{h^2} & \cdots & 0 \\ 0 & -\frac{1}{h^2} & \frac{2}{h^2} + V_3 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -\frac{1}{h^2} \\ 0 & 0 & \cdots & -\frac{1}{h^2} & \frac{2}{h^2} + V_N \end{pmatrix}. \tag{23.10}$$

Row j of H holds the three coefficients of (23.9): $-1/h^2$ in column $j-1$, $2/h^2 + V_j$ on the diagonal, and $-1/h^2$ in column $j+1$. The first and last rows have only two entries because the would-be third neighbor is a wall value that vanished.

Three properties follow at once from the pattern of entries.

1. H is *tridiagonal*: nonzero only on the main diagonal and the two adjacent diagonals (Chapter 8). It has N diagonal entries and $N-1$ entries on each of the two off-diagonals, so $N + 2(N-1) = 3N - 2$ nonzero numbers.

2. H is *real symmetric*: the entry in row j, column $j+1$ is $-1/h^2$, and so is the entry in row $j+1$, column j. By the theorem of Chapter 8, its eigenvalues are real and it has N orthonormal eigenvectors. This is the discrete counterpart of H being Hermitian (Chapter 10).

3. The continuous problem has infinitely many levels; the matrix has exactly N. Only the lowest ones, whose wavefunctions vary slowly on the scale of h, approximate physical levels. We will see in Section 23.3 how fast the approximation deteriorates.

Sample Problem 23.2: A three-point box

Take the box $[-1, 1]$ ($L = 1$) with $V = 0$ and $N = 3$ interior points. Build H, find its eigenvalues from the characteristic polynomial, and compare them with the exact levels $E_k = (k\pi/2L)^2$ of Chapter 11.

Step 1: Spacing and points. By (23.6), $h = 2L/(N+1) = 2/4 = 0.5$, and the interior points are $x_1 = -0.5$, $x_2 = 0$, $x_3 = 0.5$.

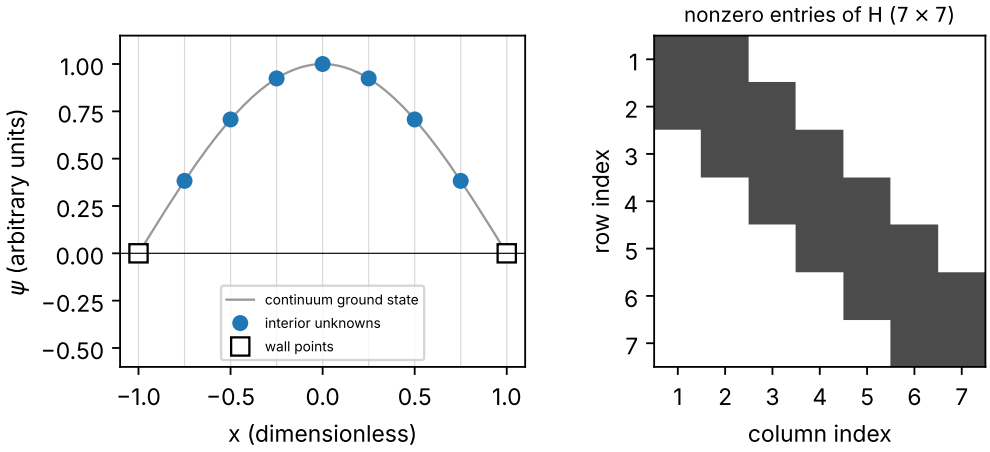


Figure 23.2: Left: the grid of Eq. (23.6) for $N = 7$ interior points on $[-1, 1]$ (units $\hbar = 2m = 1$, x dimensionless). Hollow squares: wall points where $\psi = 0$; filled dots: the N unknowns, here the sampled ground state of the box. Right: the pattern of the 7×7 matrix H of Eq. (23.10) with $V = 0$; only three diagonals are nonzero.

Step 2: The matrix. $2/h^2 = 2/0.25 = 8$ and $1/h^2 = 4$, so

$$\mathsf{H} = \begin{pmatrix} 8 & -4 & 0 \\ -4 & 8 & -4 \\ 0 & -4 & 8 \end{pmatrix}.$$

Step 3: Characteristic polynomial. Write $a = 8 - \lambda$. Expanding $\det(\mathsf{H} - \lambda I)$ along the first row (Chapter 7),

$$\begin{aligned} \det \begin{pmatrix} a & -4 & 0 \\ -4 & a & -4 \\ 0 & -4 & a \end{pmatrix} &= a\,(a^2 - 16) - (-4)\big((-4)a - (-4)(0)\big) + 0 \\ &= a^3 - 16a - 16a \\ &= a\,(a^2 - 32). \end{aligned}$$

Step 4: Roots. $a = 0$ gives $\lambda = 8$; $a^2 = 32$ gives $\lambda = 8 \pm \sqrt{32} = 8 \pm 5.65685$. In increasing order the eigenvalues are 2.34315, 8, 13.65685.

Step 5: Compare. The exact levels are $(\pi/2)^2 = 2.46740$, $\pi^2 = 9.86960$ and $(3\pi/2)^2 = 22.20661$. The relative errors are

$$\boxed{\frac{2.46740 - 2.34315}{2.46740} = 5.04\%, \quad \frac{9.86960 - 8}{9.86960} = 18.9\%, \quad \frac{22.20661 - 13.65685}{22.20661} = 38.5\%.}$$

Every computed level is too low, and the error grows rapidly with the level index. Three points per box is far too few; the next sections explain the pattern and how to beat it.

Checkpoint 23.2

Why does the first row of H contain only two nonzero entries, while a middle row contains three?

Answer: Row j couples ψ_j to its neighbors ψ_{j-1} and ψ_{j+1}. For $j = 1$ the left neighbor is the wall value

ψ_0, which is zero by the Dirichlet condition, so its coefficient never enters the matrix.

23.2.4 Code

The matrix is never stored as an $N \times N$ array; we hand the solver its two diagonals. Chapter 8 introduced `scipy.linalg.eigh_tridiagonal`, which calls the LAPACK routines for symmetric tridiagonal matrices and can return only the eigenvalues with indices in a requested range.

```
import numpy as np
from scipy.linalg import eigh_tridiagonal

def fd_levels(V, L, k):
    """Lowest k levels of -d2/dx2 + V on (-L, L) with Dirichlet walls.
    V holds the potential at the N interior points."""
    N = len(V)
    h = 2 * L / (N + 1)                    # Eq. (grid)
    diag = 2.0 / h**2 + V                  # main diagonal
    off = -np.ones(N - 1) / h**2           # the two off-diagonals
    return eigh_tridiagonal(diag, off, eigvals_only=True,
                            select='i', select_range=(0, k - 1))

print(fd_levels(np.zeros(3), 1.0, 3))     # 2.34315, 8, 13.65685
```

The library `qmnt` provides the same thing as `qmnt.fd_levels(Vvals, L, k)`, together with `qmnt.grid(L, N)` for the interior points. In what follows we use whichever is more convenient.

Key Ideas

- With $h = 2L/(N+1)$ and $\psi_0 = \psi_{N+1} = 0$, the equation $-\psi'' + V\psi = E\psi$ becomes the $N \times N$ eigenproblem $\mathsf{H}\boldsymbol{\psi} = E\boldsymbol{\psi}$.
- H is real symmetric tridiagonal: diagonal $2/h^2 + V_j$, off-diagonals $-1/h^2$, Eq. (23.10).
- The matrix has only N eigenvalues; only the low ones approximate physical levels.

23.3 A test we can solve exactly: the box on a grid

★ Did you know? *Weather by Hand*

Lewis Fry Richardson's 1922 book *Weather Prediction by Numerical Process* replaced the equations of the atmosphere with finite differences on a grid. His hand-computed six-hour forecast came out badly wrong, largely because of problems with the initial data. He imagined a forecast factory of 64,000 human computers. Electronic computers later made his grid approach the basis of modern forecasting. He also invented Richardson extrapolation, which you meet in this chapter.

Before trusting a numerical method on a hard problem, test it on a problem whose answer you know. The infinite well with $V = 0$ is perfect, because, remarkably, the *discrete* problem can also be solved exactly. That will tell us precisely how the error depends on h and on the level index.

23.3.1 Exact eigenvectors of the discrete box

With $V = 0$, (23.8) reads

$$-\frac{\psi_{j+1} - 2\psi_j + \psi_{j-1}}{h^2} = E\psi_j, \qquad \psi_0 = \psi_{N+1} = 0. \tag{23.11}$$

The continuum box has sine solutions, so try $\psi_j = \sin(j\theta)$ for some number θ to be found.

The interior equation. Use $\sin(a+b)+\sin(a-b) = 2\sin a\cos b$ with $a = j\theta$, $b = \theta$:

$$\psi_{j+1} + \psi_{j-1} = \sin((j+1)\theta) + \sin((j-1)\theta) = 2\sin(j\theta)\cos\theta.$$

Subtract $2\psi_j = 2\sin(j\theta)$:

$$\psi_{j+1} - 2\psi_j + \psi_{j-1} = 2\sin(j\theta)\cos\theta - 2\sin(j\theta) = -2(1-\cos\theta)\sin(j\theta).$$

Divide by $-h^2$:

$$-\frac{\psi_{j+1} - 2\psi_j + \psi_{j-1}}{h^2} = \frac{2(1-\cos\theta)}{h^2}\,\psi_j.$$

So $\sin(j\theta)$ satisfies (23.11) at every interior point with $E = 2(1-\cos\theta)/h^2$, *whatever* θ is.

The walls. At $j = 0$, $\psi_0 = \sin 0 = 0$ automatically. At $j = N+1$ we need $\sin((N+1)\theta) = 0$, so $(N+1)\theta$ must be a whole multiple of π:

$$\theta = \frac{k\pi}{N+1}, \qquad k = 1, 2, \ldots, N.$$

The eigenvalues. Using $1-\cos\theta = 2\sin^2(\theta/2)$,

$$E = \frac{2}{h^2}\cdot 2\sin^2\frac{\theta}{2} = \frac{4}{h^2}\sin^2\frac{k\pi}{2(N+1)}.$$

Finally, $N+1 = 2L/h$ from (23.6), so $\frac{k\pi}{2(N+1)} = \frac{k\pi h}{4L}$ and

$$\boxed{E_k^{(h)} = \frac{4}{h^2}\sin^2\!\left(\frac{k\pi h}{4L}\right), \qquad k = 1, \ldots, N.} \tag{23.12}$$

Have we found them all? For $k = 1, \ldots, N$, the angle $k\pi/(2(N+1))$ lies strictly between 0 and $\pi/2$, where $\sin^2$ is strictly increasing, so the N values in (23.12) are distinct. An $N \times N$ matrix has exactly N eigenvalues counted with multiplicity (Chapter 8), so these are all of them. (Problem 6 shows that $k = N+1$ gives the zero vector and that larger k only repeat the same vectors.) Check with Sample Problem "A three-point box": $h = 0.5$, $L = 1$, $k = 1$ gives $16\sin^2(\pi/8) = 2.34315$, as found there.

The eigenvectors are exact samples. Since $j = (x_j + L)/h$ and $(N+1)h = 2L$,

$$j\theta = \frac{jk\pi}{N+1} = \frac{k\pi(x_j+L)}{2L}.$$

So the discrete eigenvector is $\psi_j = \sin\big(k\pi(x_j+L)/(2L)\big)$: precisely the continuum eigenfunction of Chapter 11, sampled at the grid points. For the box, the grid gets the *shape* of every state exactly right and only the *energy* wrong. That is special to $V = 0$, but it is a very clean laboratory.

23.3.2 How wrong is the energy?

Compare (23.12) with the exact level $E_k = (k\pi/2L)^2$. Introduce $y = k\pi h/(4L)$. Then

$$\frac{4}{h^2}y^2 = \frac{4}{h^2}\frac{k^2\pi^2h^2}{16L^2} = \left(\frac{k\pi}{2L}\right)^2 = E_k,$$

so $E_k = (4/h^2)y^2$ and $E_k^{(h)} = (4/h^2)\sin^2 y$. Expand $\sin^2 y$ using $\sin^2 y = (1-\cos 2y)/2$ and the cosine series $\cos u = 1 - u^2/2 + u^4/24 - u^6/720 + \cdots$ with $u = 2y$:

$$\begin{aligned}\sin^2 y &= \frac{1}{2}\left[\frac{(2y)^2}{2} - \frac{(2y)^4}{24} + \frac{(2y)^6}{720} - \cdots\right] \\ &= \frac{4y^2}{4} - \frac{16y^4}{48} + \frac{64y^6}{1440} - \cdots \\ &= y^2 - \frac{y^4}{3} + \frac{2y^6}{45} - \cdots .\end{aligned}$$

Multiply by $4/h^2$:

$$E_k^{(h)} = \frac{4}{h^2}y^2 - \frac{4}{3h^2}y^4 + \frac{8}{45h^2}y^6 - \cdots .$$

Express each power of y through E_k, using $y^2 = h^2 E_k/4$, so $y^4 = h^4 E_k^2/16$ and $y^6 = h^6 E_k^3/64$:

$$\begin{aligned}\frac{4}{3h^2}y^4 &= \frac{4}{3h^2}\cdot\frac{h^4E_k^2}{16} = \frac{h^2E_k^2}{12}, \\ \frac{8}{45h^2}y^6 &= \frac{8}{45h^2}\cdot\frac{h^6E_k^3}{64} = \frac{h^4E_k^3}{360}.\end{aligned}$$

Therefore

$$\boxed{E_k^{(h)} = E_k - \frac{h^2}{12}E_k^2 + \frac{h^4}{360}E_k^3 - \cdots} \tag{23.13}$$

Three lessons are packed into this formula.

- The grid levels are *below* the true levels for every k: exactly so, because $\sin^2 y < y^2$ for every $y > 0$ in Eq. (23.12); the series gives the leading size of the deficit.
- The error is $\mathcal{O}(h^2)$, as (23.4) promised, with only even powers of h.
- At fixed h the error grows like E_k^2, that is like k^4. The relative error, $h^2E_k/12$, grows like k^2. High levels are far worse than low ones.

The expansion is useful only when $y = k\pi h/(4L)$ is small, that is when the level index k is small compared with the number of points N. When k approaches N the exact formula (23.12) must be used.

23.3.3 The same thing in the language of waves: numerical dispersion

There is a more physical way to see (23.12). Feed the complex wave e^{iqx} (Chapter 2) into the second difference:

$$\begin{aligned}D_h^2 e^{iqx} &= \frac{e^{iq(x+h)} - 2e^{iqx} + e^{iq(x-h)}}{h^2} \\ &= e^{iqx}\,\frac{e^{iqh} + e^{-iqh} - 2}{h^2} \\ &= e^{iqx}\,\frac{2\cos(qh) - 2}{h^2} \qquad (\text{since } e^{i\theta} + e^{-i\theta} = 2\cos\theta) \\ &= -\frac{4}{h^2}\sin^2\!\left(\frac{qh}{2}\right)e^{iqx} \qquad (\text{since } 1-\cos\theta = 2\sin^2(\theta/2)).\end{aligned}$$

The exact operator $-\mathrm{d}^2/\mathrm{d}x^2$ multiplies e^{iqx} by q^2: a free particle of wavenumber q has kinetic energy q^2 in our units. The grid operator multiplies it instead by

$$K_h(q) = \frac{4}{h^2}\sin^2\!\left(\frac{qh}{2}\right). \tag{23.14}$$

A box state is a standing wave with $q_k = k\pi/(2L)$ (Chapter 11), and indeed $K_h(q_k)$ is exactly (23.12). A rule relating energy (or frequency) to wavenumber is called a *dispersion relation*, so the error is often called *numerical dispersion*: the grid gets the kinetic energy of long waves right and systematically underestimates it for short waves.

The natural measure of "short" is the number of grid points per wavelength, $n_{\mathrm{pw}} = \lambda/h = 2\pi/(qh)$. Then $qh/2 = \pi/n_{\mathrm{pw}}$ and

$$\frac{K_h(q)}{q^2} = \left[\frac{\sin(\pi/n_{\mathrm{pw}})}{\pi/n_{\mathrm{pw}}}\right]^2 . \tag{23.15}$$

Table 23.1 evaluates the relative error $1 - K_h/q^2$. The shortest wave a grid can carry alternates sign from point to point, $n_{\mathrm{pw}} = 2$, $qh = \pi$; there $K_h = 4/h^2$, the largest value (23.14) can take.

Table 23.1: Relative underestimate $1 - K_h(q)/q^2$ of the kinetic energy of a wave sampled with n_{pw} grid points per wavelength, Eq. (23.15).

n_{pw}	4	6	8	12	24	50
$1 - K_h/q^2$	18.9%	8.81%	5.04%	2.26%	0.570%	0.13%

The $n_{\mathrm{pw}} = 4$ entry reproduces the middle level of the three-point box: there $k = 2$, $q = \pi$, $h = 0.5$, so $qh = \pi/2$ and four points per wavelength, error 18.9%.

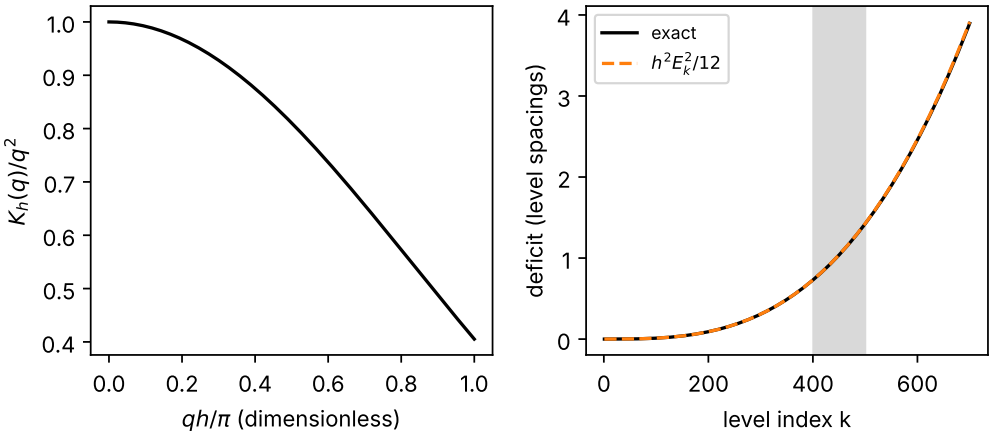


Figure 23.3: Left: the grid dispersion relation $K_h(q)/q^2$ of Eq. (23.14) against qh/π (both dimensionless); the right end, $qh = \pi$, is the zigzag wave with two points per wavelength. Right: the box of running example RE3, $L = 44.883$, $N = 6000$ (units $\hbar = 2m = 1$): the deficit $E_k - E_k^{(h)}$ in units of the local box level spacing $E_{k+1} - E_k$, exact (solid) and leading-order estimate $h^2E_k^2/12$ (dashed). The shaded band marks $k = 401$–500.

Sample Problem 23.3: The paper's grid applied to the empty box (RE3)

Running example RE3 is the box $[-L, L]$ with $L = 44.883$, the half-width the paper uses at scale 1. Put $N = 6000$ interior points in it, the paper's working grid at scale 1, with $V = 0$. Find the error in level $k = 500$ in energy units and in units of the local level spacing of the box, and the number of grid points per wavelength.

Step 1: Spacing. $h = 2L/(N+1) = 89.766/6001 = 0.0149585$, so $h^2 = 2.23757 \times 10^{-4}$ and $4/h^2 = 17876.5$.

Step 2: Exact level. $E_{500} = (500\pi/89.766)^2 = 306.21$.

Step 3: Grid level. The argument in (23.12) is $k\pi h/(4L) = 500\pi/12002 = 0.130878$, and $\sin^2(0.130878) = 0.0170314$, so $E^{(h)}_{500} = 17876.5 \times 0.0170314 = 304.46$.
Step 4: Error. $E_{500} - E^{(h)}_{500} = 1.744$. The leading term of (23.13), $h^2E^2_{500}/12 = 1.748$, agrees to better than half a percent; the relative error is 0.570%.
Step 5: In units of the level spacing. Box levels are spaced by $E_{k+1} - E_k = (2k+1)(\pi/2L)^2$, which for $k = 500$ is $1001 \times 1.22483 \times 10^{-3} = 1.226$. So

$$\boxed{E_{500} - E^{(h)}_{500} = 1.74 \text{ energy units} = 1.42 \text{ level spacings.}}$$

Step 6: Points per wavelength. With $q = 500\pi/(2L)$, $n_{\text{pw}} = 2\pi/(qh) = 4L/(500h) = 2(N+1)/500 = 24.0$. Table 23.1 says 24 points per wavelength costs 0.570%, matching Step 4.
By contrast the ground state, $E_1 = 1.22483 \times 10^{-3}$, has error $h^2E_1^2/12 \approx 3 \times 10^{-11}$, completely negligible. A grid that is superb for low levels can misplace level 500 by more than a level spacing.

Checkpoint 23.3

On a uniform grid, is the computed ground-state energy of the infinite well above or below the exact value? Does the answer change for level 500?

Answer: Below, for every level: $\sin^2 y < y^2$ for $0 < y \le \pi/2$, so $E^{(h)}_k < E_k$ by Eq. (23.12). The deficit is just much larger for level 500.

Key Ideas

- The discrete box is exactly solvable: eigenvectors are sampled sines and $E^{(h)}_k = (4/h^2)\sin^2(k\pi h/4L)$, Eq. (23.12).
- $E^{(h)}_k = E_k - h^2E_k^2/12 + h^4E_k^3/360 - \cdots$: always too low, error growing like E_k^2.
- In wave language, the grid replaces q^2 by $(4/h^2)\sin^2(qh/2)$ (numerical dispersion); accuracy is set by points per wavelength.
- On the paper's scale-1 grid, box level 500 is off by about 1.4 level spacings.

23.4 General potentials: estimating the grid bias

For the box we had an exact formula. For a general potential we need an estimate. The idea is simple: the grid operator is the true operator plus a small extra piece, and a small extra piece in a matrix shifts each eigenvalue by its "expectation value". Chapter 13 develops this idea, perturbation theory, at length; here we need only its first step, and we prove it directly for matrices.

23.4.1 How an eigenvalue moves when a symmetric matrix changes

Let $\mathsf{H}(\varepsilon) = \mathsf{H}_0 + \varepsilon\mathsf{W}$, where H_0 and W are real symmetric $N \times N$ matrices and ε is a real number. Suppose $E(\varepsilon)$ is an eigenvalue that stays distinct from all the others as ε varies, with a unit eigenvector $\mathbf{u}(\varepsilon)$, so that

$$\mathsf{H}(\varepsilon)\mathbf{u}(\varepsilon) = E(\varepsilon)\mathbf{u}(\varepsilon), \qquad \mathbf{u}^{\top}\mathbf{u} = 1.$$

Taken on trust

That a non-repeated eigenvalue and its eigenvector depend smoothly (differentiably) on ε is a standard theorem of matrix perturbation theory, which we take on trust. Its proof applies the implicit function theorem to the characteristic equation $\det(\mathsf{H}(\varepsilon) - E\,I) = 0$ at a simple root (where the derivative of the determinant with respect to E is nonzero), and then to the linear equations for the eigenvector; it is found in standard texts on matrix perturbation theory. Everything below is then ordinary calculus.

Differentiate the eigen-equation with respect to ε, using the product rule and writing a prime for $\mathrm{d}/\mathrm{d}\varepsilon$:

$$\mathsf{W}\mathbf{u} + \mathsf{H}\mathbf{u}' = E'\mathbf{u} + E\mathbf{u}'.$$

Multiply on the left by the row vector $\mathbf{u}^{\top}$:

$$\mathbf{u}^{\top}\mathsf{W}\mathbf{u} + \mathbf{u}^{\top}\mathsf{H}\mathbf{u}' = E'\,\mathbf{u}^{\top}\mathbf{u} + E\,\mathbf{u}^{\top}\mathbf{u}'.$$

Because H is symmetric, $\mathbf{u}^{\top}\mathsf{H} = (\mathsf{H}\mathbf{u})^{\top} = E\mathbf{u}^{\top}$, so the second term on the left is $E\,\mathbf{u}^{\top}\mathbf{u}'$, which cancels the last term on the right. With $\mathbf{u}^{\top}\mathbf{u} = 1$ what remains is

$$\boxed{\frac{\mathrm{d}E}{\mathrm{d}\varepsilon} = \mathbf{u}^{\top}\mathsf{W}\mathbf{u} = \sum_{i,j} u_i W_{ij} u_j.} \tag{23.16}$$

For a small change $\varepsilon\mathsf{W}$, the eigenvalue moves by approximately $\varepsilon\,\mathbf{u}^{\top}\mathsf{W}\mathbf{u}$. The continuum version, which Chapter 13 derives, reads $\delta E \approx \int \psi\,\delta H\,\psi\,\mathrm{d}x$. Keep (23.16) in mind: we will use it twice more in this chapter.

Sample Problem 23.4: An eigenvalue nudged by hand

Take the three-point box of Sample Problem "A three-point box" ($L = 1$, $h = 0.5$, $V = 0$) and add ε to the middle diagonal entry only, so $\mathsf{W} = \mathrm{diag}(0, 1, 0)$. Predict $\mathrm{d}E/\mathrm{d}\varepsilon$ for the lowest eigenvalue from Eq. (23.16) and test the prediction at $\varepsilon = 0.01$.

Step 1: The unit eigenvector. By Section 23.3, the lowest eigenvector is proportional to $(\sin\frac{\pi}{4}, \sin\frac{\pi}{2}, \sin\frac{3\pi}{4}) = (0.7071, 1, 0.7071)$. Its squares sum to $0.5 + 1 + 0.5 = 2$, so dividing by $\sqrt{2}$ gives the unit vector $\mathbf{u} = (0.5, 0.7071, 0.5)$.

Step 2: Predict. $\mathbf{u}^{\top}\mathsf{W}\mathbf{u} = u_2^2 = 0.5$, so the lowest eigenvalue should rise by about $0.5\varepsilon = 5.000 \times 10^{-3}$, from 2.343146 to 2.348146.

Step 3: Exact. The lowest eigenvalue of the matrix with rows $(8, -4, 0)$, $(-4, 8.01, -4)$, $(0, -4, 8)$, computed with `np.linalg.eigvalsh`, is 2.3481435, a rise of 4.998×10^{-3}.

$$\boxed{\frac{\mathrm{d}E}{\mathrm{d}\varepsilon} = 0.5; \quad \text{predicted shift } 5.000 \times 10^{-3},\ \text{exact } 4.998 \times 10^{-3}.}$$

The tiny difference is the next term of the expansion in ε, of order ε^2.

23.4.2 The grid as a perturbed Hamiltonian

By (23.4), applied to a smooth wavefunction, $-D_h^2\psi = -\psi'' - \frac{h^2}{12}\psi'''' - \cdots$. So, acting on smooth functions, the grid Hamiltonian looks like the true Hamiltonian plus an extra operator,

$$H_{\text{grid}} \approx H + W, \qquad W = -\frac{h^2}{12}\frac{\mathrm{d}^4}{\mathrm{d}x^4}.$$

This step is heuristic, for three reasons: Eq. (23.16) was proved for matrices, not for differential operators; the replacement of the grid operator by $H + W$ holds only when acting on smooth

functions; and W is not a small operator in general (it is large on rapidly varying functions). We nevertheless treat W as a small perturbation and carry Eq. (23.16) over to functions. If ψ is normalized, $\int \psi^2 \,\mathrm{d}x = 1$, then the grid vector with entries $u_j = \sqrt{h}\,\psi(x_j)$ is approximately a unit vector, because $\sum_j u_j^2 = \sum_j \psi(x_j)^2 h$ is a Riemann sum for $\int \psi^2 \,\mathrm{d}x$. Likewise $\mathbf{u}^\top \mathsf{W}\mathbf{u} = \sum_j \psi(x_j)\,(W\psi)(x_j)\,h$ is a Riemann sum for $\int \psi\, W\psi \,\mathrm{d}x$. So the continuum form of Eq. (23.16) is $\delta E \approx \int \psi\, \delta H\, \psi \,\mathrm{d}x$ (Chapter 13 reaches the same formula directly in the continuum). Then the grid level should sit at

$$E_n^{\mathrm{grid}} - E_n \approx \int_{-L}^{L} \psi_n\, W\psi_n \,\mathrm{d}x = -\frac{h^2}{12}\int_{-L}^{L} \psi_n\, \psi_n'''' \,\mathrm{d}x,$$

where ψ_n is the normalized true eigenfunction (real, as bound states in one dimension can be chosen, Chapter 11). Integrate by parts twice:

$$\begin{aligned}\int_{-L}^{L} \psi_n \psi_n'''' \,\mathrm{d}x &= \Big[\psi_n \psi_n'''\Big]_{-L}^{L} - \int_{-L}^{L} \psi_n' \psi_n''' \,\mathrm{d}x \\ &= \Big[\psi_n \psi_n'''\Big]_{-L}^{L} - \Big[\psi_n' \psi_n''\Big]_{-L}^{L} + \int_{-L}^{L} (\psi_n'')^2 \,\mathrm{d}x.\end{aligned}$$

At the walls $\psi_n = 0$, so the first bracket vanishes. For the second, use the Schrödinger equation, $\psi_n'' = (V - E_n)\psi_n$, which also vanishes at the walls, provided V is finite and smooth up to the walls, so that ψ_n'' is continuous up to the wall (true for ours: the Wu–Sprung potential is smooth and only about 1000 at $x = \pm L$). So both brackets are zero and

$$\int \psi_n \psi_n'''' \,\mathrm{d}x = \int (\psi_n'')^2 \,\mathrm{d}x = \int (E_n - V)^2 \psi_n^2 \,\mathrm{d}x.$$

Therefore

$$\boxed{E_n^{\mathrm{grid}} - E_n \approx -\frac{h^2}{12}\int_{-L}^{L} \big(E_n - V(x)\big)^2 \psi_n(x)^2 \,\mathrm{d}x \equiv -\frac{h^2}{12}\big\langle (E_n - V)^2 \big\rangle_n.} \tag{23.17}$$

Pitfall

Equation (23.17) is a **heuristic, leading-order estimate**, not a theorem. The operator $\mathrm{d}^4/\mathrm{d}x^4$ is not small for rapidly varying functions, so the estimate is trustworthy only when the wavelength of the state is long compared with h (many points per wavelength everywhere the state lives). It neglects the h^4 terms and the change of the wavefunction caused by the grid. For the box it reproduces the exact leading term of (23.13): with $V = 0$, $\langle (E_n - 0)^2 \rangle = E_n^2$.

23.4.3 What the estimate says

The quantity $E_n - V(x)$ is the local kinetic energy, and in our units it equals the square of the local wavenumber, $q(x)^2 = E_n - V(x)$, inside the classically allowed region. So (23.17) says: the grid bias of a level is $h^2/12$ times the average of $q(x)^4$ over the state. Where the particle moves fast (deep in the well, high above the potential) the wave is short and the grid underestimates it most. Near a classical turning point $q \to 0$ and the grid does well. This is the general version of the box statement "error grows like E^2".

Sample Problem 23.5: Grid bias for the harmonic oscillator

Recall from Chapter 11 that in units $\hbar = 2m = 1$ the Hamiltonian $H = -\mathrm{d}^2/\mathrm{d}x^2 + x^2$ has ground state $\psi_0 \propto e^{-x^2/2}$ with $E_0 = 1$. (Oscillator states are traditionally numbered

from $n = 0$; do not confuse this ψ_0 with the wall value $\psi_0 = 0$ of Eq. (23.7).) Estimate the grid bias of E_0 for $h = 0.1$ and predict the computed value.

Step 1: The integrand. $E_0 - V = 1 - x^2$, so $(E_0 - V)^2 = 1 - 2x^2 + x^4$ and

$$\langle (E_0 - V)^2 \rangle = 1 - 2\langle x^2 \rangle + \langle x^4 \rangle.$$

Step 2: The moments. With $\psi_0^2 = \pi^{-1/2} e^{-x^2}$ and the Gaussian integrals $\int e^{-x^2} \mathrm{d}x = \sqrt{\pi}$, $\int x^2 e^{-x^2} \mathrm{d}x = \sqrt{\pi}/2$, $\int x^4 e^{-x^2} \mathrm{d}x = 3\sqrt{\pi}/4$ (Problem 10 asks you to derive them by integration by parts), $\langle x^2 \rangle = 1/2$ and $\langle x^4 \rangle = 3/4$.

Step 3: Combine. $\langle (E_0 - V)^2 \rangle = 1 - 1 + \frac{3}{4} = \frac{3}{4}$, so by (23.17)

$$E_0^{\text{grid}} - E_0 \approx -\frac{h^2}{12} \cdot \frac{3}{4} = -\frac{h^2}{16} = -6.25 \times 10^{-4}.$$

Step 4: Predict.

$$\boxed{E_0^{\text{grid}} \approx 0.999375.}$$

Solving the tridiagonal problem on $[-10, 10]$ with $N = 199$ (so $h = 20/200 = 0.1$; the walls are so far out that ψ_0 is negligible there) gives this value to within a few times 10^{-6}, as Problem 12 lets you confirm. Notice that the box formula $h^2 E_0^2/12 = 8.3 \times 10^{-4}$ would overestimate the bias: the potential slows the particle down, and the state spends time near its turning points.

23.4.4 Why levels 401–500 are hard

Now we can see the size of the problem the paper faced. At scale 1 the paper uses $L = 44.883$ and $N = 6000$ interior points, the same $h = 0.0149585$ as our box example. Its levels 401–500 lie at heights roughly between t_{401} and $t_{500} = 811.18$ (the latter from the paper's Appendix A key anchors). Two things conspire:

- The bias (23.17) grows roughly like the square of the local kinetic energy, and these are high levels. At the bottom of the Wu–Sprung well, where $V = V_0 = 3.10073\pi$, a level at $E = 811.18$ has $q = \sqrt{E - V_0} = 28.31$ and therefore only $2\pi/(qh) = 14.8$ points per wavelength. This is the *shortest* local wavelength of the state, reached at the bottom of the well where the kinetic energy is largest; elsewhere the wave is longer and better resolved, so this is a worst case, not a typical value.
- The quantity we care about is not the energy error but the error in units of the *mean spacing of the zeros*, which near height t is $2\pi/\log(t/2\pi)$ (Chapter 17 derives this) and is only about 1.3 near $t = 811$. The spacing shrinks while the bias grows.

The paper reports the result: the finite-difference dispersion "at levels 401–500 on the working grid reaches two mean spacings" (Sec. II C). The fluctuations of the zeros that the paper studies have an rms of only about 0.26 mean spacings on this range (the null of 0.263). A bias about eight times larger than the signal ($2/0.263 \approx 7.6$) cannot be left in.

Checkpoint 23.4

Two levels of the same potential are computed on the same grid: level 10 and level 400. Which is more accurate in energy units, and why?

Answer: Level 10. The bias is $h^2/12$ times the average of $(E - V)^2$, the fourth power of the local wavenumber; level 400 has far more kinetic energy, hence far shorter waves and far fewer points per wavelength.

Key Ideas

- For a symmetric matrix, a small change $\varepsilon \mathsf{W}$ moves a non-repeated eigenvalue by $\varepsilon\, \mathbf{u}^{\top} \mathsf{W} \mathbf{u}$, Eq. (23.16).
- Heuristically, the grid bias of level n is $-\frac{h^2}{12}\langle (E_n - V)^2 \rangle$, Eq. (23.17): $h^2/12$ times the mean fourth power of the local wavenumber.
- On the paper's working grid this bias reaches about two mean spacings at levels 401–500 (paper Sec. II C), far larger than the fluctuations being studied.

23.5 Richardson extrapolation

We could shrink the bias by brute force: more points. Because the error is $\mathcal{O}(h^2)$, cutting it by a factor of 100 costs ten times as many points. There is a much cheaper trick, named after the meteorologist Lewis Fry Richardson. It exploits the fact that we know *how* the error depends on h, even though we do not know its size.

23.5.1 The derivation

For a given level, write its computed value on a grid of spacing h as $E(h)$. For the box we proved, in Eq. (23.13), that the error is a series in even powers of h. For a general potential, Eq. (23.4) makes this plausible, since the local error of D_h^2 contains only even powers of h; but that equation describes the difference operator acting on a fixed function, not the eigenvalues of the grid matrix, so for a general smooth potential the eigenvalue expansion is an assumption here (see the box below). We assume

$$E(h) = E + Ch^2 + Dh^4 + \mathcal{O}(h^6), \tag{23.18}$$

where E is the exact level and C, D are numbers that depend on the level and the potential but not on h.

Taken on trust

For a smooth potential on a finite interval with Dirichlet walls, the eigenvalues of the second-order finite-difference matrix do have, for each fixed level, an expansion $E(h) = E + Ch^2 + Dh^4 + \cdots$ in even powers of h. We take this on trust. Its proof belongs to the numerical analysis of finite-difference eigenvalue problems and is found in advanced texts on that subject; in this book it is proved only for the box, Eq. (23.13). You need not trust it blindly: condition 2 below gives an empirical test (successive differences should shrink by a factor close to 4), and Problem 25 applies it.

Now compute the same level on a grid with half the spacing:

$$E(h/2) = E + C\frac{h^2}{4} + D\frac{h^4}{16} + \mathcal{O}(h^6).$$

Multiply this by 4:

$$4E(h/2) = 4E + Ch^2 + D\frac{h^4}{4} + \mathcal{O}(h^6).$$

Subtract (23.18); the Ch^2 terms cancel:

$$4E(h/2) - E(h) = 3E + D\left(\frac{1}{4} - 1\right) h^4 + \mathcal{O}(h^6) = 3E - \frac{3}{4} Dh^4 + \mathcal{O}(h^6).$$

Divide by 3:

$$\boxed{E^{\text{Rich}} \equiv \frac{4E(h/2) - E(h)}{3} = E - \frac{D}{4}h^4 + \mathcal{O}(h^6).} \tag{23.19}$$

The unknown h^2 error has been eliminated without ever knowing C. The combination (23.19) is what `qmnt.richardson(E_h, E_h2)` returns. For the box, $D = E^3/360$ by (23.13), so the Richardson error is

$$E^{\text{Rich}} - E \approx -\frac{h^4 E^3}{1440} \qquad \text{(box, leading order)}. \tag{23.20}$$

Unequal refinement. If the fine spacing is h/r rather than $h/2$, repeat the argument: $E(h/r) = E + Ch^2/r^2 + \cdots$, multiply by r^2 and subtract $E(h)$ to get $r^2E(h/r) - E(h) = (r^2 - 1)E + \mathcal{O}(h^4)$, so

$$E^{\text{Rich}} = \frac{r^2E(h/r) - E(h)}{r^2 - 1} + \mathcal{O}(h^4). \tag{23.21}$$

With $r = 2$ this is (23.19).

23.5.2 Conditions for Richardson to work

Every step above relied on something; here is the list.

1. **The same level on both grids.** $E(h)$ and $E(h/2)$ must be the *same* physical level. The grid matrix has no repeated eigenvalues. Proof: suppose $\mathsf{H}\mathbf{v} = \lambda\mathbf{v}$. The first row of H gives $(2/h^2 + V_1 - \lambda)v_1 - v_2/h^2 = 0$, which fixes v_2 once v_1 is known; row j, Eq. (23.9), then fixes v_{j+1} from v_j and v_{j-1}, because the coefficient $-1/h^2$ of v_{j+1} is not zero. So $v_1 = 0$ forces $\mathbf{v} = 0$, and every eigenvector for λ is a multiple of the one with $v_1 = 1$: each eigenspace is one-dimensional. For a symmetric matrix the number of times an eigenvalue is repeated equals the dimension of its eigenspace (spectral theorem, Chapter 8), so all N eigenvalues are distinct. (The continuum counterpart is the non-degeneracy of one-dimensional bound states, Chapter 11.) The solver returns them sorted, so "the n-th smallest eigenvalue on each grid" identifies the level, as long as n is far below N.

2. **The asymptotic regime.** The expansion (23.18) must be dominated by its first term, which requires many points per wavelength. A practical test: the ratio $[E(h) - E(h/2)]/[E(h/2) - E(h/4)]$ should be close to 4.

3. **Smoothness.** The derivation of (23.4) needs a smooth wavefunction, which needs a smooth potential. A kink in V (for example $|x|$ at $x = 0$) makes ψ'' jump in slope there, so the Taylor argument fails at the kink, the clean even-power expansion is no longer guaranteed, and neither is Richardson's cancellation; the ratio test of condition 2 is the practical way to detect such trouble.

4. **The right weights.** The *finer* grid gets the weight 4. Swapping them makes things worse, not better (Problem 31).

5. **Exact halving.** Halving h means doubling $N + 1$, not N. Doubling N gives a ratio r slightly below 2; (23.21) shows how small the resulting mistake is (Problem 11).

Sample Problem 23.6: Richardson on the three-point box

Using $L = 1$, $V = 0$, combine the ground-state energies on grids with $N = 3$ ($h = 0.5$) and $N = 7$ ($h = 0.25$) into a Richardson estimate, and compare all three errors with the exact $E_1 = \pi^2/4 = 2.467401$.

Step 1: Coarse grid. From the three-point box Sample Problem, $E(0.5) = 16\sin^2(\pi/8) = 2.343146$.

Step 2: Fine grid. $N = 7$ gives $h = 2/8 = 0.25$, exactly half (because $N+1$ doubled from 4 to 8). Then $4/h^2 = 64$ and $k\pi h/(4L) = \pi/16$, so $E(0.25) = 64\sin^2(\pi/16) = 2.435855$.
Step 3: Combine.

$$E^{\text{Rich}} = \frac{4(2.435855) - 2.343146}{3} = \frac{9.743420 - 2.343146}{3} = 2.466758.$$

Step 4: Errors. $E_1 - E(0.5) = 0.1243$; $E_1 - E(0.25) = 0.0315$; $E_1 - E^{\text{Rich}} = 6.4 \times 10^{-4}$. The prediction (23.20) gives $h^4 E_1^3/1440 = (0.0625)(15.02)/1440 = 6.5 \times 10^{-4}$ for $h = 0.5$. So

$E^{\text{Rich}} = 2.46676$, error 6.4×10^{-4}, about 50 times smaller than on the finer grid alone.

```
import numpy as np, qmnt
L = 1.0
E_h  = qmnt.fd_levels(np.zeros(3), L, k=1)     # h = 0.5
E_h2 = qmnt.fd_levels(np.zeros(7), L, k=1)     # h = 0.25, exactly half
E_R  = qmnt.richardson(E_h, E_h2)              # (4*E_h2 - E_h)/3
print(E_h, E_h2, E_R, np.pi**2 / 4)
```

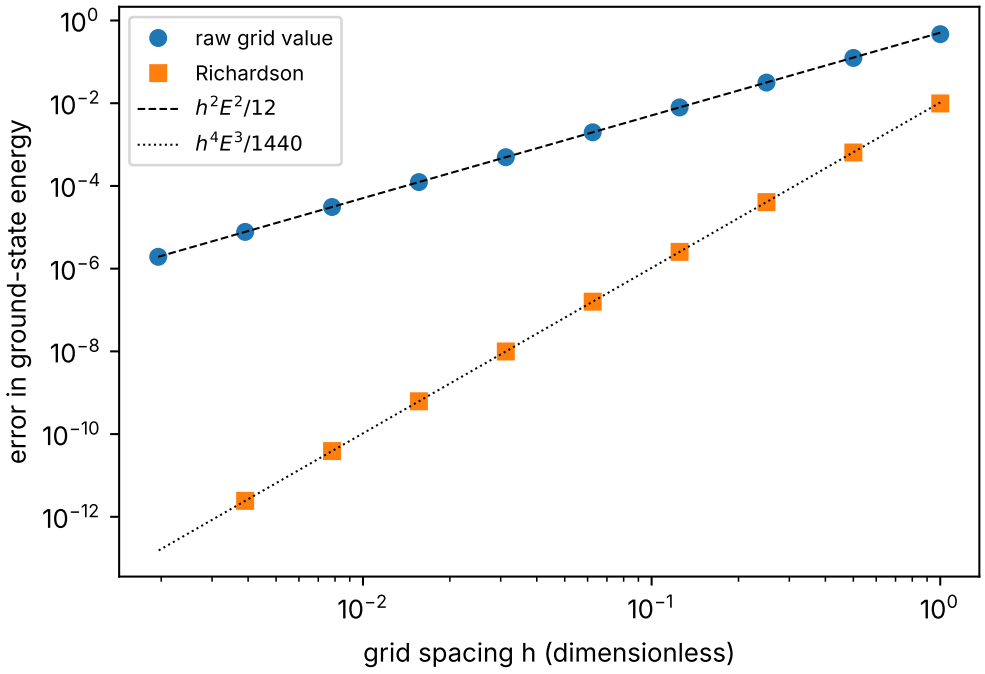


Figure 23.4: Error in the ground-state energy of the box $[-1, 1]$ (units $\hbar = 2m = 1$) against grid spacing h, with $N + 1 = 2, 4, 8, \ldots$ so that h halves exactly. Circles: raw grid value, Eq. (23.12), slope 2. Squares: Richardson combination of each grid with the next finer one, Eq. (23.19), slope 4. Dashed lines: the leading-order predictions $h^2E^2/12$ and $h^4E^3/1440$.

Sample Problem 23.7: Brute force versus Richardson at level 500 (RE3)

For the empty box with $L = 44.883$, how many interior points would a single grid need to bring the error of level 500 below 0.01 level spacings? What error does Richardson achieve with $N = 6000$ and its exact halving $N = 12001$?

Step 1: Target. From the RE3 Sample Problem, the level spacing at $k = 500$ is 1.226, so we want an error below 0.01226.
Step 2: Brute force. Using the leading term $h^2E^2/12$ with $E = 306.21$: $h^2 < 12{\times}0.01226/306.21^2 = 1.569{\times}10^{-6}$, so $h < 1.253{\times}10^{-3}$ and $N{+}1 > 89.766/(1.253{\times}10^{-3})$, that is $N \approx 71{,}700$ interior points.
Step 3: Richardson. With $h = 0.0149585$, (23.20) gives $h^4E^3/1440 = (5.007 \times 10^{-8})(2.871 \times 10^7)/1440 = 1.0 \times 10^{-3}$, which is 8×10^{-4} level spacings. Computing both grids with the exact formula (23.12) and combining them confirms this.

Brute force: $N \approx 71{,}700$. Richardson with $N = 6000, 12001$: error $\approx 8 \times 10^{-4}$ spacings.

Two modest grids beat one huge grid by more than a factor of ten in accuracy.

Checkpoint 23.5

In $E^{\text{Rich}} = (4E_a - E_b)/3$, which grid is a: the coarse one or the fine one?

Answer: The fine one. Its h^2 error is a quarter of the coarse grid's, so four times the fine value minus the coarse value cancels the h^2 term.

Key Ideas

- If $E(h) = E + Ch^2 + \mathcal{O}(h^4)$, then $(4E(h/2) - E(h))/3 = E + \mathcal{O}(h^4)$, Eq. (23.19); for refinement ratio r use Eq. (23.21).
- It requires: the same level on both grids, the asymptotic regime, a smooth potential, the weight 4 on the finer grid.
- For the box the residual error is $\approx -h^4E^3/1440$; at level 500 on the paper's grid this is below 10^{-3} spacings.

23.6 Normalization and integrals on a grid

★ Did you know? *Kepler and the Wine Barrels*

In 1615 Johannes Kepler published a book on measuring the volume of wine barrels, prompted by buying wine for his 1613 wedding and doubting the merchants' measuring method. One of his formulas is equivalent to what we now call Simpson's rule, which Thomas Simpson published in 1743. Grid integration is old: weight a few sample values cleverly and you get an excellent estimate of an area.

23.6.1 From integrals to sums

The continuum wavefunction is normalized by $\int_{-L}^{L} \psi^2 \, \mathrm{d}x = 1$ (Chapter 10; we take ψ real). On the grid the natural replacement is the trapezoid rule. Replace the area under g over each strip $[x_j, x_{j+1}]$ by the area $h(g_j + g_{j+1})/2$ of the trapezoid through the two end values, and add the $N + 1$ strips. Each interior value belongs to two strips and so appears with total weight h; each end value belongs to one strip and appears with weight $h/2$. (For smooth g the error of the rule is $\mathcal{O}(h^2)$, a standard result of numerical integration [27] that we do not need below.) The rule approximates the integral of a function g by

$$\int_{-L}^{L} g \, \mathrm{d}x \approx h \left[\frac{g_0}{2} + g_1 + g_2 + \cdots + g_N + \frac{g_{N+1}}{2} \right].$$

For $g = \psi^2$ the two end values are zero by the Dirichlet condition, so the half-weights disappear and

$$\int_{-L}^{L} \psi^2 \, dx \approx \sum_{j=1}^{N} \psi_j^2 \, h. \tag{23.22}$$

So the grid normalization is

$$\boxed{\sum_{j=1}^{N} \psi_j^2 \, h = 1.} \tag{23.23}$$

23.6.2 What the eigensolver returns

An eigensolver knows nothing about h. It returns *unit vectors* $\mathbf{u}$ in the sense of Chapter 7: $\sum_j u_j^2 = 1$. To convert, set

$$\psi_j = \frac{u_j}{\sqrt{h}}, \tag{23.24}$$

and check: $\sum_j \psi_j^2 h = \sum_j (u_j^2/h)h = \sum_j u_j^2 = 1$. Without this rescaling a plotted wavefunction is too small by a factor $\sqrt{h}$, which on the paper's grid is about 0.12.

Two more facts carry over from Chapter 8. Eigenvectors of the symmetric matrix H for different eigenvalues are orthogonal, $\sum_j u_{n,j} u_{m,j} = 0$, hence $\sum_j \psi_{n,j}\psi_{m,j} h = 0$: the grid version of $\int \psi_n \psi_m \, dx = 0$. And each eigenvector is determined only up to an overall sign, because $-\mathbf{u}$ is a unit eigenvector whenever $\mathbf{u}$ is.

23.6.3 Expectation values and the sensitivity matrix

For a multiplication operator $f(x)$ the expectation value becomes

$$\int_{-L}^{L} \psi_n^2 f \, dx \approx \sum_{j=1}^{N} \psi_{n,j}^2 f(x_j) \, h = \sum_{j=1}^{N} u_{n,j}^2 f(x_j). \tag{23.25}$$

The factors of h cancel when you work with the unit vectors directly.

This has a neat consequence for fitting (Chapter 24). Suppose the grid potential is changed by $\delta V(x) = \sum_k c_k \phi_k(x)$, so the matrix becomes $\mathsf{H}_0 + \sum_k c_k \, \mathrm{diag}(\phi_k(x_1), \ldots, \phi_k(x_N))$. By (23.16), with $\varepsilon = c_k$ and $\mathsf{W} = \mathrm{diag}(\phi_k(x_j))$, the derivative of the grid eigenvalue is

$$\frac{\partial E_n}{\partial c_k} = \mathbf{u}_n^{\top} \, \mathrm{diag}(\phi_k) \, \mathbf{u}_n = \sum_j u_{n,j}^2 \phi_k(x_j) = \sum_j \psi_{n,j}^2 \phi_k(x_j) \, h. \tag{23.26}$$

The right side is the trapezoid version of the paper's first-order sensitivity $A_{nk} = \int |\psi_n|^2 \, \phi_k \, dx$ (Sec. II C). So when A_{nk} is computed this way from the grid eigenvectors it is not merely an approximation of a continuum integral: it is the *exact* derivative of the grid eigenvalue that the fit actually uses. Because only $u_{n,j}^2$ appears, the sign ambiguity of eigenvectors is harmless here.

Sample Problem 23.8: Normalizing the three-point ground state

For the three-point box ($L = 1$, $N = 3$, $h = 0.5$), find the unit eigenvector of the lowest eigenvalue, rescale it by (23.24), compare with the continuum ground state $\psi_1(x) = \cos(\pi x/2)$ (normalized on $[-1, 1]$), and compute A_{11} for $\phi_1 = \cos(\pi x/L)$ both ways.

Step 1: Unnormalized eigenvector. From Section 23.3, $\psi_j \propto \sin(j\pi/4)$ for $k = 1$, $N + 1 = 4$: the values $\sin(\pi/4)$, $\sin(\pi/2)$, $\sin(3\pi/4)$, that is $(0.7071, 1, 0.7071)$.

Step 2: Unit vector. The sum of squares is $0.5 + 1 + 0.5 = 2$, so $\mathbf{u} = (0.5, 0.7071, 0.5)$.

Step 3: Rescale. $1/\sqrt{h} = \sqrt{2}$, so $\boldsymbol{\psi} = (0.7071, 1, 0.7071)$. The continuum ground state at $x = -0.5, 0, 0.5$ is $\cos(\pi/4)$, $\cos 0$, $\cos(\pi/4)$, the same numbers. (Its normalization:

$\int_{-1}^{1}\cos^2(\pi x/2)\,dx = \int_{-1}^{1}\frac{1}{2}(1+\cos\pi x)\,dx = 1.$)

Step 4: The sensitivity. On the grid, (23.25) with $\cos(\pi x_j) = (0, 1, 0)$ gives $A_{11} = 0.25(0) + 0.5(1) + 0.25(0) = 0.5$. In the continuum,

$$\int_{-1}^{1}\cos^2\frac{\pi x}{2}\cos(\pi x)\,dx = \int_{-1}^{1}\frac{1+\cos\pi x}{2}\cos\pi x\,dx = \frac{1}{2}\int_{-1}^{1}\cos\pi x\,dx + \frac{1}{2}\int_{-1}^{1}\cos^2\pi x\,dx = 0 + \frac{1}{2}.$$

$$\boxed{\psi = (0.7071,\ 1,\ 0.7071), \qquad A_{11} = 0.5 \text{ on the grid and in the continuum.}}$$

(For the box this agreement is not an accident: the grid vectors are exact samples of the continuum states, and the equal-weight grid sum reproduces exactly the integral of a product of a few low-frequency sines and cosines, as Problem 9 illustrates for $\sin^2$. For a general potential the grid sum and the continuum integral differ by $\mathcal{O}(h^2)$.)

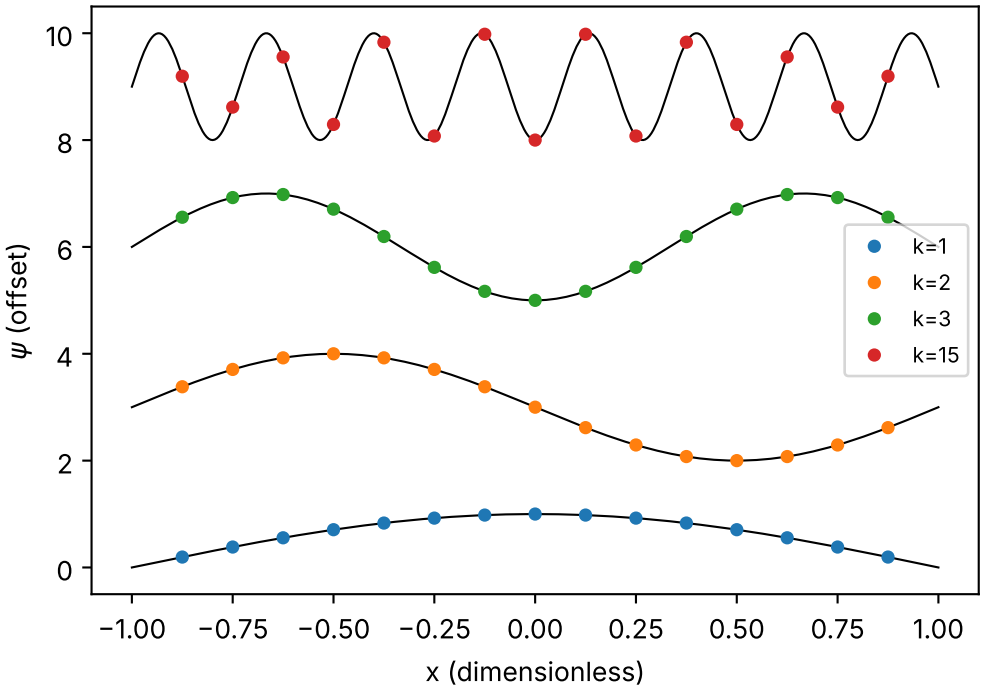


Figure 23.5: Grid eigenvectors of the box $[-1, 1]$ with $N = 15$, rescaled so that $\sum_j \psi_j^2 h = 1$ (dots), against the continuum eigenfunctions $\sin(k\pi(x + L)/2L)/\sqrt{L}$ (curves), for $k = 1, 2, 3$ and $k = N = 15$; curves are offset vertically for clarity (x dimensionless). For the box the grid vectors are exact samples. The $k = 15$ vector is the nearly zigzag mode with about two points per wavelength, whose energy the grid underestimates most.

Checkpoint 23.6

`eigh_tridiagonal` returns an eigenvector with $\sum_j u_j^2 = 1$ on a grid with $h = 0.01$. By what factor must you multiply it to obtain a wavefunction normalized as $\int \psi^2\,dx = 1$?

Answer: By $1/\sqrt{h} = 10$, Eq. (23.24).

Key Ideas

- With Dirichlet walls, the trapezoid rule makes the grid normalization $\sum_j \psi_j^2 h = 1$, Eq. (23.23).
- Solvers return unit vectors; divide by $\sqrt{h}$ to get wavefunctions. Signs are arbitrary.
- Expectation values of $f(x)$ are $\sum_j u_j^2 f(x_j)$; computed this way, A_{nk} is the exact derivative of the grid eigenvalue, Eq. (23.26).

23.7 Correcting the targets instead of the operator

23.7.1 Three ways to deal with grid bias

Suppose you want to fit a potential so that its levels match a list of target numbers t_n (in the paper, the zeros). A fit needs many eigensolves: every iteration solves for the levels and eigenvectors of the current potential, and an iteration may try several candidate steps (Chapter 24). You have three options for the grid bias.

1. **A very fine grid every time.** Accurate but expensive, as the brute-force Sample Problem showed.
2. **Richardson every time.** Two fine eigensolves per trial step instead of one working-grid solve.
3. **Correct the targets once.** Compute the bias for the *unperturbed* potential once, before fitting, and move the targets by that amount. Then fit on a cheap working grid.

The paper uses option 3. Let us define it precisely.

23.7.2 Definitions

Let V be the smooth (unperturbed) potential. Compute its levels three times: on the working grid, giving $E_n^g[\text{smooth}]$, and on two finer grids related by (nearly) exact halving (see Section 23.8 and Problem 11 for the paper's pairs), combined by (23.19) into $E_n^{\text{Rich}}[\text{smooth}]$, an estimate of the true continuum levels with only $\mathcal{O}(h^4)$ error. Define the *grid bias*

$$\Delta_n = E_n^g[\text{smooth}] - E_n^{\text{Rich}}[\text{smooth}] \tag{23.27}$$

and the *bias-corrected targets*

$$\tilde{t}_n = t_n + \Delta_n. \tag{23.28}$$

Since grid levels sit below the true ones (exactly for the box, Section 23.3, and to leading order by the heuristic estimate of Section 23.4 for smooth potentials), Δ_n is negative in practice and the corrected targets are lowered to where the working grid would place a level that is truly at t_n. The fit then compares working-grid eigenvalues $E_n[\text{fit}]$ with $\tilde{t}_n$.

23.7.3 The approximation being made

Option 3 assumes that the grid bias of the *fitted* potential equals that of the smooth potential, $E_n^g[\text{fit}] - E_n^{\text{true}}[\text{fit}] \approx \Delta_n$. This is an approximation, and we should say when it holds.

- **Exactly true for a constant shift.** If $\delta V = c$ everywhere, the grid matrix changes by c times the identity, so every grid eigenvalue shifts by exactly c; the continuum levels and the Richardson estimate also shift by exactly c. The difference Δ_n is unchanged.

- **Approximately true for a small varying correction.** By (23.17) the bias depends on $\langle (E_n - V)^2 \rangle$. Adding δV changes $E_n - V$ to $E_n + \delta E_n - V - \delta V$. Problem 29 shows the bias then changes by approximately $-\frac{h^2}{6}\langle (E_n - V)(\delta E_n - \delta V) \rangle$, which is a small fraction of Δ_n as long as $\delta V - \delta E_n$ is small compared with the local kinetic energy $E_n - V$ over the region the level occupies.

The neglected piece is therefore the change of a correction that is itself of order two mean spacings, multiplied by a ratio like $2\,|\delta V - \delta E_n|\,/(E_n - V)$ (Problem 29). As an order-of-magnitude estimate: if $\delta V - \delta E_n$ is 3% of the local kinetic energy, the change is about $2 \times 0.03 \times 2 = 0.12$ spacings, roughly a tenth of a spacing; it is smaller where δV oscillates and its effect averages out over the state. Keep in mind that this is an estimate built on the heuristic (23.17).

23.7.4 Why the correction leaks nothing: the identity of the paper's Eq. (3)

Here is a worry a careful referee raises. In an out-of-sample test, the fit sees only training zeros, and its predictions are scored on held-out zeros. But the corrected targets $\tilde{t}_n$ exist for *every* index n, including the held-out ones. Could the correction smuggle information about the held-out zeros into the score?

Write the residual of the fitted level against the corrected target and substitute (23.28):

$$\begin{aligned} \tilde{t}_n - E_n[\text{fit}] &= (t_n + \Delta_n) - E_n[\text{fit}] \qquad \text{(definition of } \tilde{t}_n\text{)} \\ &= t_n - E_n[\text{fit}] + \Delta_n \qquad \text{(reorder)} \\ &= t_n - (E_n[\text{fit}] - \Delta_n). \qquad \text{(factor the minus sign)} \end{aligned} \tag{23.29}$$

This is the paper's Eq. (3). Read the two ends of the chain as two different procedures:

- *Left side:* compare the fitted level with the corrected target.
- *Right side:* first correct the *prediction*, replacing $E_n[\text{fit}]$ by $E_n[\text{fit}] - \Delta_n$, then compare with the raw zero t_n.

They are the same number for every n, so every rms and every correlation computed from them is identical. Now ask what Δ_n is made of. By (23.27) it is built only from eigenvalues of the *unperturbed* potential on three grids. No zero, training or held-out, enters it. So the correction is a fixed, data-independent adjustment of the model's output, of the same kind as choosing a more accurate solver. Nothing about the held-out t_n reaches the fit or the prediction except through the final comparison, where it must enter.

There is a second illuminating rewriting. The paper measures fluctuations relative to the unperturbed levels. The "true fluctuation" of level n is then

$$\begin{aligned} \tilde{t}_n - E_n^g[\text{smooth}] &= t_n + (E_n^g[\text{smooth}] - E_n^{\text{Rich}}[\text{smooth}]) - E_n^g[\text{smooth}] \\ &= t_n - E_n^{\text{Rich}}[\text{smooth}]. \end{aligned}$$

The working-grid levels cancel, and what remains is the raw zero minus the continuum-accurate level of the smooth potential. That is exactly the quantity a physicist would want, freed of any grid artifact.

Sample Problem 23.9: The leakage identity with numbers

Take the paper's $t_{500} = 811.1843588$ (Appendix A, key anchors). Suppose, purely for illustration (these are not the paper's values), that $\Delta_{500} = -2.50$ and a fitted potential has $E_{500}[\text{fit}] = 808.50$. Compute the residual both ways.

Step 1: Corrected target. $\tilde{t}_{500} = 811.1843588 - 2.50 = 808.6843588$.

Step 2: Left side of (23.29). $\tilde{t}_{500} - E_{500}[\text{fit}] = 808.6843588 - 808.50 = 0.1843588$.
Step 3: Right side. Corrected prediction $E_{500}[\text{fit}] - \Delta_{500} = 808.50 + 2.50 = 811.00$; residual $811.1843588 - 811.00 = 0.1843588$.

Both procedures give the residual 0.1843588.

Checkpoint 23.7

Does the grid bias Δ_n depend on the value of any Riemann zero?

Answer: No. By Eq. (23.27) it is a difference of eigenvalues of the unperturbed potential computed on different grids; no zero enters.

Pitfall

"No leakage" is a statement about *information*, not about accuracy. The identity (23.29) guarantees that the correction cannot help a model cheat. It does not guarantee that the correction is accurate for the fitted potential; that is the separate approximation discussed above, which affects all models and the null in the same way.

Key Ideas

- The paper computes $\Delta_n = E_n^g[\text{smooth}] - E_n^{\text{Rich}}[\text{smooth}]$ once and fits against $\tilde{t}_n = t_n + \Delta_n$ on a cheap working grid.
- This assumes the fitted potential has nearly the same grid bias as the smooth one: exact for constant shifts, approximate otherwise.
- $\tilde{t}_n - E_n[\text{fit}] = t_n - (E_n[\text{fit}] - \Delta_n)$, Eq. (23.29): scoring against corrected targets equals correcting the prediction by a zero-free quantity. No held-out information leaks.

23.8 Paper connection

Everything in this chapter was built to read the paper's "forward solver". Here it is, statement by statement.

The solver (Sec. II B, App. A steps 3–4). The paper works in units $\hbar = 2m = 1$ with $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$, V even. It uses "second-order finite differences on N uniform interior points (Dirichlet walls)", which "give a symmetric tridiagonal matrix; eigenpairs from LAPACK". That is Section 23.2 exactly: Appendix A step 4 records the matrix as "tridiagonal diag $= 2/dx^2 + V$, off $= -1/dx^2$" (the paper's dx is our h) and the solver as `eigh_tridiagonal`, the routine used in our listings. Step 3 records the grids: interior-point uniform on $(-L, L)$, scale 1 with $L = 44.8827$ and working grid $N = 6000$, scale 2 with $L = 59.9572$ and $N = 8000$. With $h = 2L/(N+1)$ both scales have nearly the same spacing, 0.0149585 and 0.0149874 (Problem 1). The paper evaluates only the eigenvalues it needs: step 4 lists ranges $[0, 900]$ for the scale-1 Hermitian work, $[0, 1000]$ for the tensors of Sec. II E (652 states) and $[0, 1500]$ at scale 2 (1079 states).

The potential on the grid (Sec. II B, App. A step 3). The Wu–Sprung potential is given implicitly by $x(V)$, Eq. (2), which Chapter 22 derives. The paper inverts it "by Brent inversion of Eq. (2) on a dense reference grid, cubic-spline interpolated". Richardson needs a

smooth potential (condition 3 of Section 23.5), so it is reassuring to check the bottom of the well. Put $V = V_0 + \eta^2$ with $\eta = \sqrt{V - V_0} \geq 0$ (a new symbol, unrelated to machine epsilon ϵ and to the perturbation parameter ε of Eq. (23.16)). First rationalize the argument of the second logarithm by multiplying numerator and denominator by $\sqrt{V} + \eta$:

$$\frac{\sqrt{V} + \eta}{\sqrt{V} - \eta} = \frac{(\sqrt{V} + \eta)^2}{V - \eta^2} = \frac{(\sqrt{V} + \eta)^2}{V_0},$$

so Eq. (2) reads

$$x = \frac{1}{\pi}\left[\eta \log \frac{V_0}{2\pi e^2} + 2\sqrt{V_0 + \eta^2}\, \log \frac{\sqrt{V_0 + \eta^2} + \eta}{\sqrt{V_0}}\right].$$

Now let η be small. Then $\sqrt{V_0 + \eta^2} \approx \sqrt{V_0}$, and with $\log(1 + u) \approx u$ for small u, $\log\left[(\sqrt{V_0} + \eta)/\sqrt{V_0}\right] = \log(1 + \eta/\sqrt{V_0}) \approx \eta/\sqrt{V_0}$, so

$$x \approx \frac{1}{\pi}\left[\eta \log \frac{V_0}{2\pi e^2} + 2\eta\right] = \frac{\eta}{\pi} \log \frac{V_0}{2\pi} \qquad (\text{since } \log e^2 = 2).$$

Solving for $\eta^2 = V - V_0$, $V \approx V_0 + \left(\pi/\log(V_0/2\pi)\right)^2 x^2 \approx V_0 + 51.3\, x^2$. This leading-order parabola does not by itself exclude higher terms such as $|x|^3$, which would be a hidden kink. To exclude them, read the formula for x as a function of η allowed to take either sign. The factor $\sqrt{V_0 + \eta^2}$ is even in η. The logarithm is odd in η, because $(\sqrt{V_0 + \eta^2} - \eta)(\sqrt{V_0 + \eta^2} + \eta) = V_0$ gives $\log\left[(\sqrt{V_0 + \eta^2} - \eta)/\sqrt{V_0}\right] = -\log\left[(\sqrt{V_0 + \eta^2} + \eta)/\sqrt{V_0}\right]$. The first term is odd as well. So $x(\eta)$ is an odd smooth function with nonzero slope $\log(V_0/2\pi)/\pi$ at $\eta = 0$ (here $V_0/2\pi = 1.550$, whose logarithm is positive). A smooth function with nonzero slope has a smooth inverse (first-year calculus), so η is a smooth odd function of x, and $V = V_0 + \eta^2$ is a smooth even function of x: no kink at $x = 0$. The code below reproduces the paper's construction with a spline, and evaluating `x_of_V(1000.0)` returns 44.883, the paper's scale-1 half-width L.

```
import numpy as np
from scipy.interpolate import CubicSpline
V0 = 3.10073 * np.pi

def x_of_V(V):                          # paper Eq. (2); note (r+s)/(r-s) = (r+s)^2/V0
    s, r = np.sqrt(V - V0), np.sqrt(V)
    return (s * np.log(V0 / (2*np.pi*np.e**2))
            + r * np.log((r + s)**2 / V0)) / np.pi

u = np.linspace(0, np.sqrt(1100 - V0), 40001)    # V = V0 + u^2, dense near bottom
spline = CubicSpline(x_of_V(V0 + u**2), u)
V_of_x = lambda x: V0 + spline(np.abs(x))**2     # even in x
L = x_of_V(1000.0)                               # about 44.883
```

Correcting the targets (Sec. II B, App. A step 5). The paper states that "discretization bias is removed from the *targets* rather than the operator": all fitting runs on the working grid against $\tilde{t}_n = t_n + (E_n^{\text{grid}}[\text{smooth}] - E_n^{\text{Rich}}[\text{smooth}])$. That is (23.27)–(23.28). The Richardson pairs are 12000/24000 (scale 1) and 16000/32000 (scale 2). Because $h = 2L/(N + 1)$, doubling N from 12000 to 24000 halves h only approximately ($r = 24001/12001$); Problem 11 shows the leftover h^2 error is below 3×10^{-5} of the coarse grid's, utterly negligible.

Pitfall

The paper writes the extrapolation compactly as "$E^{\text{Rich}} = (4E_{2h} - E_h)/3$ ... over grid doubling". The derivation of Section 23.5 shows that the weight 4 must go to the *finer*

grid. We therefore read the paper's subscripts (this is our interpretation; the paper does not spell it out) as labels for the grid with twice as many points (the second member of each listed pair, 24000 or 32000), not as a spacing of $2h$. In terms of spacings the formula is $(4E_{h/2} - E_h)/3$, Eq. (23.19). If you reconstruct the pipeline, give the weight 4 to the finer grid; Problem 31 shows that the opposite assignment is worse than no extrapolation at all.

How good is the smooth potential after correction (Sec. II B). The paper reports that "the residual of E^{Rich} against the smooth zeros is 0.005 spacings mean, 0.063 max (worst at low n, the known semiclassical weakness of the construction)". Read this carefully: once the grid bias is removed, what remains is not a grid error but a property of the Wu–Sprung potential itself. That potential is built from a semiclassical approximation (Chapter 12) that is least accurate for the lowest levels (Chapter 22). Richardson cannot and should not remove it. Figure 23.6 shows both quantities computed with the spline construction above.

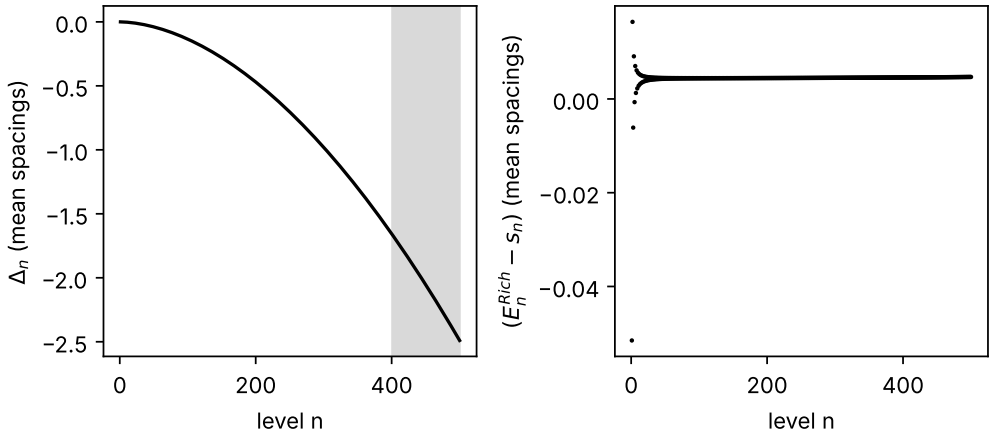


Figure 23.6: The Wu–Sprung potential at scale 1 ($L = x(1000)$, units $\hbar = 2m = 1$), computed with the spline construction of this section. Left: grid bias $\Delta_n = E_n^g - E_n^{\mathrm{Rich}}$ on the working grid $N = 6000$, with E^{Rich} from $N = 12000$ and 24000, in units of the local mean spacing $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$; the paper reports that this bias reaches about two mean spacings at levels 401–500 (Sec. II C). Right: $(E_n^{\mathrm{Rich}} - s_n)/\mathrm{gap}_n$, the residual of the extrapolated levels against the smooth zeros; the paper reports mean 0.005 and maximum 0.063 spacings, largest at low n (Sec. II B).

No leakage (Sec. II C, Eq. (3)). The paragraph "No leakage through the bias correction" is the identity (23.29): "scoring the fit against $\tilde{t}_n$ is algebraically identical to correcting the *prediction* by the fit-independent quantity Δ_n and scoring against the raw zeros"; Δ_n "is built solely from eigenvalues of the *unperturbed* potential on two grids; no zero, in-sample or held-out, enters it". (The paper's "two grids" refers to the Richardson pair. The full Δ_n of Eq. (23.27) uses three grids: the working grid supplies E_n^g and the pair supplies E_n^{Rich}. None of the three involves a zero.) The paper also states why the correction cannot be skipped: finite-difference dispersion "at levels 401–500 on the working grid reaches two mean spacings and would otherwise dominate both the null and every model identically". Appendix A (step 20, revision computations) adds an empirical counterpart for the sinusoidal controls: "Overwriting held-out data with noise before fitting leaves all fitted coefficients bit-identical". The sentence follows step 20, which describes the sinusoidal controls, and we read it as referring to them; the paper does not say explicitly whether the operator fits were tested in the same way. For the operator fits the

guarantee we can verify ourselves is the algebraic identity (23.29) together with the zero-free construction of Δ_n.

The raw-zero null 2.09 (App. A, key anchors). The null model of Sec. II C is the unperturbed potential itself. Scored against corrected targets, its error on zeros 401–500 is 0.2631 mean spacings (key anchors; 0.263 in the tables). By the rewriting in Section 23.7, this is the rms of $t_n - E_n^{\mathrm{Rich}}[\mathrm{smooth}]$, the genuine fluctuation of the zeros about the smooth levels. Scored instead against the *raw* zeros on the uncorrected working grid, the same null gives 2.09. That number is dominated by the grid bias of about two spacings reported by the paper (Sec. II C) and plotted in Figure 23.6; the real fluctuations of size 0.26 are buried underneath it (Problem 14). This is also why the paper's two nulls can coincide numerically (Sec. II C): after correction, $E_n^{\mathrm{Rich}}[\mathrm{smooth}]$ agrees with the smooth zeros s_n to 0.005 spacings on average, so "unperturbed levels" and "smooth zeros" are, as predictions, nearly the same.

Normalization and the sensitivity matrix (Sec. II C, App. A steps 6, 9). The fit uses $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\,\mathrm{d}x$, and Appendix A step 9 records "ψ normalized $\sum \psi^2 dx = 1$", Eq. (23.23). By (23.26), the grid sum is the exact derivative of the grid eigenvalue with respect to c_k, which is what an iterated linearization needs (Chapter 24).

Exact checks on a coarser grid (App. A step 11). The non-Hermitian checks of Sec. V diagonalize the complex matrix of $H + iA$ densely on $N = 2500$, a coarser common grid; dense complex eigensolves cost far more than tridiagonal ones (Chapters 8 and 14). Those checks compare two calculations on the same grid, so the grid bias is common to both.

The interpolation split (Sec. III, Fig. 4). When the paper says "x and V are in the units $\hbar = 2m = 1$ of Sec. II B", and reports a fitted correction with maximum $|\delta V| = 199$ where the smooth potential is about 500, it is describing a potential on exactly the grid of this chapter. A correction of that size is a large fraction of the local kinetic energy, the regime where the target-correction approximation of Section 23.7 is least accurate. The paper does not quantify how much Δ_n changes there. By the estimate of Problem 29 the change could be a sizeable fraction of Δ_n itself, which at levels 201–300 is presumably below its size of about two spacings at 401–500 (the bias grows with energy). That would still be smaller than the 2.64-spacing gap error the paper reports, but we have not verified it numerically: read this as a plausibility argument, not a result.

23.9 Summary

- The central second difference satisfies $D_h^2 f = f'' + \frac{h^2}{12} f'''' + \frac{h^4}{360} f^{(6)} + \cdots$, Eq. (23.4): second-order accurate, even powers of h only. Roundoff adds an error of order ϵ/h^2, Eq. (23.5).

- On N interior points with $h = 2L/(N+1)$ and Dirichlet walls, $-\psi'' + V\psi = E\psi$ becomes $\mathsf{H}\boldsymbol{\psi} = E\boldsymbol{\psi}$ with the symmetric tridiagonal H of Eq. (23.10): diagonal $2/h^2 + V_j$, off-diagonals $-1/h^2$.

- The discrete box is exact: $E_k^{(h)} = (4/h^2)\sin^2(k\pi h/4L)$, Eq. (23.12), eigenvectors are sampled continuum sines, and $E_k^{(h)} = E_k - h^2E_k^2/12 + h^4E_k^3/360 - \cdots$, Eq. (23.13).

- Numerical dispersion: the grid gives a wave of wavenumber q the kinetic energy $(4/h^2)\sin^2(qh/2)$, Eq. (23.14); accuracy is governed by points per wavelength, Table 23.1.

- For a symmetric matrix, $\mathrm{d}E/\mathrm{d}\varepsilon = \mathbf{u}^\top \mathsf{W}\mathbf{u}$, Eq. (23.16); heuristically the grid bias of level n is $-\frac{h^2}{12}\langle (E_n - V)^2\rangle$, Eq. (23.17).
- Richardson: $(4E(h/2) - E(h))/3 = E + \mathcal{O}(h^4)$, Eq. (23.19); general ratio, Eq. (23.21); box residual $\approx -h^4E^3/1440$, Eq. (23.20).
- Grid normalization $\sum_j \psi_j^2 h = 1$, Eq. (23.23); solver unit vectors are divided by $\sqrt{h}$, Eq. (23.24); A_{nk} computed by grid sums is the exact grid derivative, Eq. (23.26).
- The paper corrects targets: $\tilde{t}_n = t_n + \Delta_n$ with $\Delta_n = E_n^g - E_n^{\mathrm{Rich}}$ of the smooth potential, Eqs. (23.27)–(23.28); the leakage identity $\tilde{t}_n - E_n[\mathrm{fit}] = t_n - (E_n[\mathrm{fit}] - \Delta_n)$, Eq. (23.29), is the paper's Eq. (3).
- Without the correction, the grid bias (about two mean spacings at levels 401–500) swamps the fluctuations: the paper's raw-zero null is 2.09 against 0.263 corrected.

23.10 Problems

1. • Compute the grid spacing $h = 2L/(N+1)$ for the paper's two working grids: scale 1 with $L = 44.883$, $N = 6000$, and scale 2 with $L = 59.957$, $N = 8000$. What do you notice?

2. • How many nonzero entries does the $N \times N$ matrix (23.10) have? For $N = 6000$, what fraction of all N^2 entries is that?

3. • Use the forward difference (23.3) with $h = 0.01$ to estimate the derivative of e^x at $x = 0$. Compare the error with the prediction $\frac{h}{2}f''$.

4. • By what factor does the error of a computed box level fall when h is halved (a) for the raw grid value, (b) for the Richardson combination? Assume the asymptotic regime.

5. •• Apply Richardson's idea to the second difference itself: show that $\frac{1}{3}[4D_h^2 f - D_{2h}^2 f]$ equals $\big[-f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h)\big]/(12h^2)$, and explain why this five-point formula has an error of order h^4.

6. •• For the discrete box, show that $k = N+1$ in $\psi_j = \sin(jk\pi/(N+1))$ gives the zero vector, and that k and $2(N+1) - k$ give the same vector up to sign. Conclude that there are only N distinct eigenvectors.

7. •• Take the box $[-1, 1]$ with only $N = 1$ interior point. Find the single grid eigenvalue and its error. Then combine it with the $N = 3$ ground state (2.343146) by Richardson. What is the error now?

8. •• For RE3 ($L = 44.883$, $N = 6000$), use the leading term of (23.13) to estimate the error of box level $k = 250$, in energy units and in units of the box level spacing $(2k+1)(\pi/2L)^2$.

9. •• Prove that $\sum_{j=1}^{N} \sin^2\big(jk\pi/(N+1)\big) = (N+1)/2$ for $1 \le k \le N$. (Hint: write $\sin^2 = \frac{1}{2}(1 - \cos)$ and sum the cosines as the real part of a geometric series of $z = e^{2\pi ik/(N+1)}$.) Deduce that the normalized discrete box eigenvector is $\psi_j = \sin(jk\pi/(N+1))/\sqrt{L}$, matching the continuum normalization.

10. •• Derive the Gaussian moments used in the harmonic-oscillator Sample Problem by integration by parts, and then estimate the grid bias of the first excited state $\psi_1 \propto xe^{-x^2/2}$, $E_1 = 3$, for $h = 0.1$.

11. •• The paper's Richardson pair at scale 1 is $N = 12000$ and $N = 24000$ with L fixed. Find the exact refinement ratio r of the spacings, and use (23.21) to show that applying the $r = 2$ formula leaves a fraction $(4/r^2 - 1)/3$ of the coarse grid's h^2 error uncancelled. Evaluate it.

12. ••• Using numpy and `eigh_tridiagonal`, compute the first 21 levels of $H = -\mathrm{d}^2/\mathrm{d}x^2 + x^2$ on $[-10, 10]$ with $h = 0.1$. You may use the known result $\langle x^4\rangle_n = \frac{3}{4}(2n^2 + 2n + 1)$ and $\langle x^2\rangle_n = n + \frac{1}{2}$ for this oscillator. Show that (23.17) predicts a bias $-\frac{h^2}{32}(E_n^2 + 1)$ with $E_n = 2n + 1$, and compare with your numbers. Why is it smaller than the box formula $-h^2E^2/12$ at the same energy?

13. •• (Paper.) A student argues: "The corrected targets $\tilde{t}_n$ are defined for held-out n, so the out-of-sample score in the paper is contaminated." Using Eq. (3) of the paper (our Eq. (23.29)) and the definition of Δ_n, explain precisely why the student is wrong. Would the conclusion change if Δ_n were computed from the fitted potential rather than the smooth one?

14. •• (Paper.) The paper's raw-zero null on 401–500 is 2.09 spacings, the corrected null 0.263. Model the uncorrected residual as a constant bias b plus a fluctuation with rms 0.263 and zero mean. What value of b reproduces 2.09? Compare with the paper's statement that the dispersion "reaches two mean spacings". What does the model leave out?

15. •• Using `qmnt.fd_levels` with $V = 0$, $L = 44.883$, $N = 6000$, compute level 500 and compare with (23.12) and with the continuum value.

16. •• (Paper.) Using the spline construction of Section 23.8 or `qmnt`, compute the Wu–Sprung levels on $N = 6000$, 12000, 24000 at scale 1, form Δ_n, and express it in units of gap_n (`qmnt.mean_gap`). Compare the size on 401–500 with the paper's "two mean spacings".

17. ••• Show that every eigenvalue of the discrete box lies strictly between 0 and $4/h^2$. Find the largest eigenvalue for $L = 44.883$, $N = 6000$. What does its eigenvector look like, and is it a good approximation of any physical level?

18. •• How many continuum box levels with $L = 44.883$ lie below $E = 1000$? How many grid levels with $N = 6000$? Explain the direction of the difference. (Compare with the paper's count of 652 Wu–Sprung states below $E = 1000$, App. A step 4.)

19. • Check the leakage identity numerically with $t_n = 50.000$, $\Delta_n = -0.30$, $E_n[\mathrm{fit}] = 49.65$.

20. •• Show that if V is even and the grid symmetric, the matrix H commutes with the reversal $j \to N + 1 - j$, and hence that every eigenvector is either symmetric or antisymmetric under reversal. Relate this to the parity of states in Chapter 11.

21. •• Show that the central first difference $[f(x+h) - f(x-h)]/(2h)$ equals $f' + \frac{h^2}{6}f''' + \mathcal{O}(h^4)$, and test it on $\sin x$ at $x = 1$ with $h = 0.1$.

22. ••• Starting from $\sin^2 y = y^2 - y^4/3 + 2y^6/45 - y^8/315 + \cdots$, derive the h^4 and h^6 terms of $E_k^{(h)}$ for the box, and show that the Richardson combination leaves $-h^4E^3/1440 + E^4h^6/64512 + \cdots$.

23. •• (Paper.) Apply the box estimate $h^2E^2/12$ with the paper's scale-1 spacing to $E = t_{500} = 811.18$, and express it in units of the mean spacing $2\pi/\log(t/2\pi)$ at that height. Compare with the paper's "two mean spacings". Using (23.17), explain the difference.

24. ••• Write your own solver (no `qmnt.fit_potential`) for the Wu–Sprung levels at scale 1 and compute $(E_n^{\mathrm{Rich}} - s_n)/\mathrm{gap}_n$ for $n = 1, \dots, 500$. Compare the mean and maximum with the paper's 0.005 and 0.063. Where is the maximum, and why there?

25. •• For the box $[-1, 1]$, compute the ground-state error on grids with $N + 1 = 4, 8, 16, 32$ and the ratios of successive errors. What value should the ratios approach, and how does this test the "asymptotic regime" condition of Richardson extrapolation?

26. •• (Paper.) Appendix A step 4 asks `eigh_tridiagonal` only for eigenvalues in $[0, 900]$ at scale 1, out of $N = 6000$. Give two reasons not to compute all of them.

27. • Why does the paper fit on a working grid of $N = 6000$ rather than directly on the $N = 24000$ grid used for Richardson?

28. •• Show that adding a constant c to the potential leaves Δ_n exactly unchanged, by examining the grid matrix and the Richardson combination.

29. ••• Using (23.17), and neglecting the change of ψ_n, show that a correction δV that shifts level n by δE_n changes the grid bias by approximately $-\frac{h^2}{6}\langle (E_n - V)(\delta E_n - \delta V)\rangle$. State a condition under which the paper's target correction remains accurate for a fitted potential.

30. •• Using numpy only (`np.diag`), build the 5×5 matrix for the box $[-1, 1]$, compute its eigenvalues with `np.linalg.eigvalsh`, and check them against (23.12).

31. •• (Paper.) Suppose a reconstruction used the weights the wrong way round, $(4E_{\text{coarse}} - E_{\text{fine}})/3$. Evaluate this for the box $[-1, 1]$ with the $N = 3$ and $N = 7$ ground states, and compare with the correct Richardson value and with the raw values.

32. •• For the box $[-L, L]$, compute $A_{2,1} = \int \psi_2^2 \cos(\pi x/L)\,\mathrm{d}x$ and $A_{2,2} = \int \psi_2^2 \cos(2\pi x/L)\,\mathrm{d}x$, where $\psi_2 = \sin(\pi x/L)/\sqrt{L}$. Verify them with a grid sum (23.25) for $L = 1$, $N = 99$.

23.11 Answers to selected problems

★ Did you know? *Surprise in the Simulation*

In 1955 Enrico Fermi, John Pasta, Stanislaw Ulam and Mary Tsingou, who programmed the computer, simulated a chain of masses joined by slightly nonlinear springs on the MANIAC at Los Alamos. They expected the energy to spread evenly among all modes. Instead it sloshed among a few modes and almost returned to its starting state. It was among the first discoveries made by numerical experiment.

1. Scale 1: $h = 89.766/6001 = 0.0149585$. Scale 2: $h = 119.914/8001 = 0.0149874$. The two spacings agree to about 0.2%: the paper increased N in proportion to L, so the grid resolution is essentially the same at both scales. (The scale-2 levels go higher, so their bias is still larger.)

2. N diagonal entries plus $N - 1$ on each off-diagonal: $3N - 2$. For $N = 6000$: 17998 entries out of 3.6×10^7, a fraction 5.0×10^{-4}. This is why the matrix is passed to the solver as two diagonals.

3. $(e^{0.01} - 1)/0.01 = 1.0050167$. The exact derivative is 1, so the error is 0.0050167; the prediction $\frac{h}{2}f''(0) = 0.005$ accounts for it, the remainder being the $h^2 f'''/6$ term.

4. (a) By 4, since the error is $\mathcal{O}(h^2)$. (b) By 16, since the Richardson error is $\mathcal{O}(h^4)$, Eq. (23.19).

5. $D_{2h}^2 f = [f(x + 2h) - 2f(x) + f(x - 2h)]/(4h^2)$. Then

$$\begin{aligned} 4D_h^2 f - D_{2h}^2 f &= \frac{16f(x+h) - 32f(x) + 16f(x-h)}{4h^2} - \frac{f(x+2h) - 2f(x) + f(x-2h)}{4h^2} \\ &= \frac{-f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h)}{4h^2}, \end{aligned}$$

and dividing by 3 gives the stated formula. By (23.4), $D_h^2 f = f'' + \frac{h^2}{12}f'''' + \mathcal{O}(h^4)$ and $D_{2h}^2 f = f'' + \frac{4h^2}{12}f'''' + \mathcal{O}(h^4)$, so $4D_h^2 f - D_{2h}^2 f = 3f'' + \mathcal{O}(h^4)$: the h^2 terms cancel exactly as in Section 23.5.

7. $N = 1$ gives $h = 1$ and the 1×1 matrix $2/h^2 = 2$. The exact value is $\pi^2/4 = 2.467401$, so the error is 0.467 (18.9%; one interior point means four points per wavelength counting the walls, as in Table 23.1). Richardson with $h = 1$ and $h = 0.5$: $(4 \times 2.343146 - 2)/3 = 2.457528$, error 0.00987, almost fifty times smaller than the raw coarse error.

9. With $\sin^2 a = \frac{1}{2}(1 - \cos 2a)$,

$$\sum_{j=1}^{N} \sin^2 \frac{jk\pi}{N+1} = \frac{N}{2} - \frac{1}{2} \sum_{j=1}^{N} \cos \frac{2jk\pi}{N+1}.$$

Let $z = e^{2\pi ik/(N+1)}$. For $1 \le k \le N$, $z \ne 1$ but $z^{N+1} = e^{2\pi ik} = 1$, so the geometric series (Chapter 3) gives $\sum_{j=0}^{N} z^j = (1 - z^{N+1})/(1 - z) = 0$. Removing the $j = 0$ term, $\sum_{j=1}^{N} z^j = -1$; its real part is $\sum_{j=1}^{N} \cos(2jk\pi/(N+1)) = -1$. Hence the sum is $N/2 + 1/2 = (N+1)/2$. (Check: $N = 3$, $k = 1$: $0.5 + 1 + 0.5 = 2$.) For $\psi_j = A \sin(\cdots)$, (23.23) requires $A^2(N+1)h/2 = 1$, and $(N+1)h = 2L$, so $A^2 = 1/L$: exactly the continuum normalization constant of the box.

11. $h_c/h_f = (24001)/(12001) = r = 1.99992$. With $E_c = E + Ch_c^2$ and $E_f = E + Ch_c^2/r^2$, the $r = 2$ formula gives $(4E_f - E_c)/3 = E + Ch_c^2(4/r^2 - 1)/3$. Numerically $4/r^2 = 1.0000833$, so the uncancelled fraction is 2.78×10^{-5} of the coarse ($N = 12000$) grid's h^2 error. Since that error is of order a fraction of a mean spacing at these levels, the leftover is of order 10^{-5} spacings: negligible.

13. By Eq. (23.29), scoring against $\tilde{t}_n$ is identical, number by number, to scoring the adjusted prediction $E_n[\text{fit}] - \Delta_n$ against the raw zeros. The adjustment Δ_n is computed from eigenvalues of the smooth potential on three grids, before any fit, without any zero. So the held-out zeros enter only in the final comparison, which is what out-of-sample scoring means. If Δ_n were computed from the fitted potential instead, it would still contain no zero (the fitted potential depends only on training zeros), so there would still be no leakage; it would merely be a slightly more accurate, fit-dependent correction, at the cost of extra fine-grid eigensolves.

15. `qmnt.fd_levels(np.zeros(6000), 44.883, k=500)[-1]` gives 304.46, matching (23.12) (agreement to better than 10^{-8}); the continuum value is 306.21, so the grid is 1.74 too low, 1.42 box level spacings (RE3 Sample Problem).

17. The eigenvalues are $(4/h^2)\sin^2(k\pi/(2(N+1)))$ with $0 < k\pi/(2(N+1)) < \pi/2$ for $1 \le k \le N$, so $0 < \sin^2 < 1$ and $0 < E_k^{(h)} < 4/h^2$. The largest is $k = N$: $(4/h^2)\cos^2(\pi/(2(N+1)))$, which for $N = 6000$ is 17876.5 (the cosine factor differs from 1 by less than 10^{-7}). Its eigenvector $\sin(jN\pi/(N+1)) = (-1)^{j+1}\sin(j\pi/(N+1))$ alternates in sign from point to point, the zigzag mode of Figure 23.5. Its continuum partner would have energy $(N\pi/2L)^2 \approx 4/h^2 \times (\pi/2)^2$, about 2.5 times larger; it is not a usable approximation of any physical level. Only levels with $k \ll N$ are physical.

19. $\tilde{t}_n = 50.000 - 0.30 = 49.70$, and $\tilde{t}_n - E_n[\text{fit}] = 0.05$. Corrected prediction $49.65 + 0.30 = 49.95$, and $t_n - 49.95 = 0.05$. Same residual.

21. Subtract (23.2) from (23.1): even powers cancel, odd powers double, $f(x+h) - f(x-h) = 2hf' + \frac{h^3}{3}f''' + \mathcal{O}(h^5)$; divide by $2h$ to get $f' + \frac{h^2}{6}f''' + \mathcal{O}(h^4)$. For sin at 1 with $h = 0.1$: $[\sin 1.1 - \sin 0.9]/0.2 = \cos 1 \, \sin(0.1)/0.1 = 0.5394022$, against $\cos 1 = 0.5403023$: error -9.0005×10^{-4}, prediction $\frac{h^2}{6}(-\cos 1) = -9.0050 \times 10^{-4}$.

23. $h^2 = 2.23757 \times 10^{-4}$, $E^2 = 811.18^2 = 6.580 \times 10^5$, so $h^2E^2/12 = 12.3$ energy units. The mean spacing near $t = 811$ is $2\pi/\log(811.18/2\pi) = 1.293$, so the box estimate is 9.5 spacings, several times the paper's "two mean spacings". The box formula assumes all of E is kinetic energy everywhere. In the Wu–Sprung well, Eq. (23.17) averages $(E - V)^2$. In the Wu–Sprung well V rises from $V_0 \approx 9.7$ to E across the allowed region, so $E - V$ is well below E over most of it and tends to zero at the turning points, where the state spends much of its time. Both effects make $\langle (E-V)^2 \rangle$ much smaller than E^2.

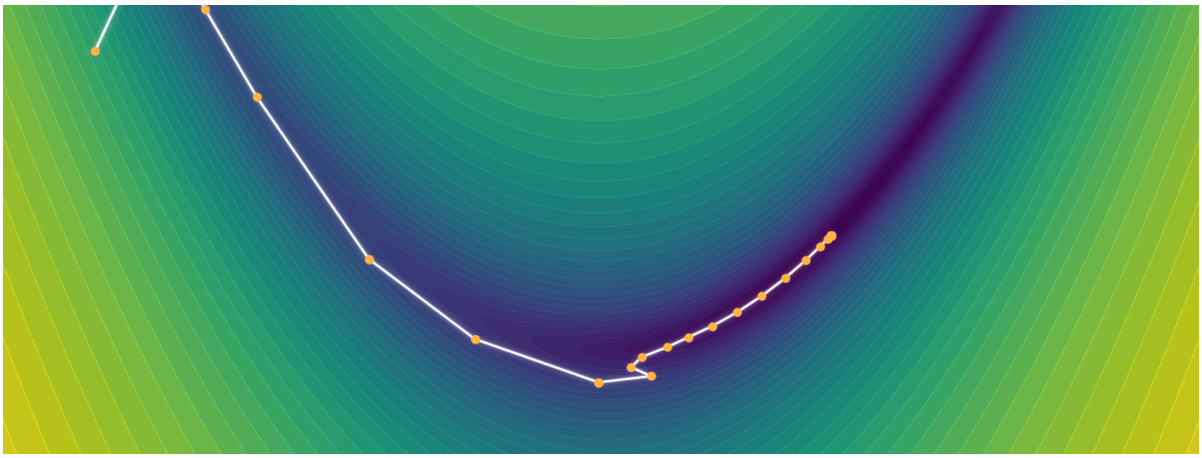
25. With $E_1 = \pi^2/4 = 2.4674011$: errors 0.12426 ($N + 1 = 4$), 0.031546 (8), 0.0079170 (16), 0.0019811 (32). Ratios 3.94, 3.98, 4.00, approaching $4 = 2^2$. Ratios close to 4 show the h^2 term

dominates, so Richardson's cancellation is valid; a ratio far from 4 would warn that the grid is too coarse (or the potential not smooth).

27. Each fit iteration requires at least one eigensolve with eigenvectors for all fitted levels, and several trial steps (Chapter 24). The cost and memory of such a solve grow at least in proportion to N, so the $N = 24000$ grid would make every one of these solves several times more expensive. The target correction delivers essentially the accuracy of the Richardson grids at the price of three eigensolves of the smooth potential, done once, while the fit itself runs on the cheap grid.

29. To first order in small quantities, $(E_n+\delta E_n-V-\delta V)^2 = (E_n-V)^2+2(E_n-V)(\delta E_n-\delta V)+\cdots$. Inserting into (23.17) with ψ_n held fixed, the bias changes by $-\frac{h^2}{12}\cdot 2\langle(E_n-V)(\delta E_n-\delta V)\rangle = -\frac{h^2}{6}\langle(E_n-V)(\delta E_n-\delta V)\rangle$. Compared with the bias itself, $-\frac{h^2}{12}\langle(E_n-V)^2\rangle$, this is a relative change of order $2\,|\delta E_n-\delta V|\,/(E_n-V)$. So the correction stays accurate when δV departs from a constant by much less than the local kinetic energy E_n-V over the region the level occupies. For a constant δV, $\delta E_n = \delta V$ and the change vanishes, consistent with Problem 28.

31. $(4\times 2.343146 - 2.435855)/3 = 2.312243$, error 0.155, worse than either raw value (errors 0.124 and 0.032), and far worse than the correct Richardson value 2.466758 (error 6.4×10^{-4}). Putting the weight on the coarse grid amplifies the h^2 error instead of cancelling it.



Levenberg–Marquardt descending a curved valley: the path bends between careful gradient steps and bold Gauss–Newton jumps.

Chapter 24

Fitting a Potential to a Spectrum

↑ Contents

What this chapter is about

In Chapter 23 you learned to run quantum mechanics forward: you are given a potential $V(x)$, you build a matrix, and a computer returns the energy levels. This chapter runs it backward. You are handed a list of target levels, and you must adjust a potential until its levels match the list. This is a *fitting* problem, of the same family as drawing the best straight line through measured points, except that the "line" is a whole function $\delta V(x)$ and each "point" is an eigenvalue that depends on that function in a nonlinear way.

You already own every ingredient. First-order perturbation theory (Chapter 13) tells you how much each level moves when you nudge the potential. Least squares and the singular value decomposition (Chapter 9) tell you the best nudge to make when there are many levels and many knobs. The finite-difference solver (Chapter 23) tells you where the levels really are after the nudge. Put together and repeated, with a safety mechanism called damping, these become the Levenberg–Marquardt algorithm. We build it from scratch, then take apart, clause by clause, the exact recipe the paper uses. We also meet three ways of writing the correction δV (cosines, Chebyshev polynomials and hat functions) and derive a rule of thumb, the *coverage law*, that tells you how many basis functions are needed before a fit can reproduce essentially any list of levels.

Where this shows up in the paper: Sec. II C ("Hermitian fluctuation fit and out-of-sample protocol", including the paragraph "Other bases"), Appendix A step 6 (the damping schedule), the first paragraph of Sec. III (the coverage law and the in-sample numbers 0.186 at $M = 300$ and $< 10^{-4}$ at $M = 900$), and the in-sample columns of Tables I, II and III. This chapter explains how the paper's fits are *made*. Whether they *predict* anything is a different question, taken up in Chapter 25 and Chapter 29.

What you need

- Chapter 9: linear least squares, normal equations, the singular value decomposition (SVD), the pseudo-inverse and minimum-norm solutions, ridge regularization and filter factors.
- Chapter 13: the first-order energy shift $\delta E_n = \int |\psi_n|^2 \, \delta V \, \mathrm{d}x$, the sensitivity matrix A_{nk}, the second-order shift, and the condition that matrix elements be small compared with level spacings.
- Chapter 23: the tridiagonal finite-difference Hamiltonian with Dirichlet walls, the discrete normalization $\sum_j \psi_j^2 h = 1$, Richardson extrapolation and bias-corrected targets $\tilde{t}_n$.
- Chapter 8: symmetric matrices, real eigenvalues, positive definite matrices.
- Chapter 5: the cosine basis $\cos(k\pi x/L)$ on $[-L, L]$ for even functions; beats.
- Chapter 4: Newton's method for a single equation; Taylor's theorem with remainder.
- Chapter 11: eigenfunctions and levels of the infinite square well.
- Chapter 12: the local wavenumber $\sqrt{E - V(x)}$, turning points and the WKB wavefunction (restated where used).
- Chapter 17: the smooth counting function N_{sm}, the smooth zeros s_n and the mean spacing gap_n.
- First-year physics: potential energy curves and classical turning points; standing waves on a string.

24.1 From levels to a potential: the inverse problem as least squares

24.1.1 The forward map

Throughout this chapter we use the units of the paper, $\hbar = 2m = 1$, so the Hamiltonian of a particle on a line is

$$H(c) \;=\; -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x) + \delta V(x; c), \tag{24.1}$$

where $V(x)$ is a fixed *background* potential (in the paper, the Wu–Sprung potential of paper Eq. (2), derived in Chapter 22) and δV is an adjustable *correction*. We write the correction as a finite sum of chosen *basis functions* $\phi_k(x)$:

$$\delta V(x; c) \;=\; \sum_{k=0}^{M-1} c_k \, \phi_k(x). \tag{24.2}$$

Here M is the number of basis functions (the *basis size*), and the real numbers $c_0, c_1, \ldots, c_{M-1}$ are the *coefficients*, which we collect into a column vector c with M entries. The default basis in the paper is $\phi_k(x) = \cos(k\pi x/L)$, where L is the half-width of the box $[-L, L]$ whose walls are at $x = \pm L$.

For each choice of c, the finite-difference solver of Chapter 23 returns the energy levels $E_1(c) < E_2(c) < E_3(c) < \cdots$ of $H(c)$. The rule "c in, list of levels out" is called the *forward map*. We only ever need the lowest n_{fit} levels, where n_{fit} is the number of levels we try to match (the paper's n_{fit}, for example 400).

24.1.2 Targets, residuals and the objective

The targets are numbers $\tilde{t}_1, \ldots, \tilde{t}_{n_{\mathrm{fit}}}$ that we want the levels to equal. In the paper they are the bias-corrected zeros of Chapter 23: recall that $\tilde{t}_n = t_n + (E_n^{\mathrm{grid}}[\mathrm{smooth}] - E_n^{\mathrm{Rich}}[\mathrm{smooth}])$, where t_n is the height of the n-th Riemann zero, $E_n^{\mathrm{grid}}[\mathrm{smooth}]$ is the n-th level of the uncorrected

background on the working grid, and $E_n^{\mathrm{Rich}}[\mathrm{smooth}]$ is its Richardson extrapolant. For this chapter all that matters is that $\tilde{t}_n$ is a fixed list of numbers.

The *residual* of level n is the mismatch

$$r_n(c) \;=\; \tilde{t}_n - E_n(c), \qquad n = 1, \dots, n_{\mathrm{fit}}. \tag{24.3}$$

A positive residual means the level is too low and must be pushed up.

The paper quotes every error in units of the *local mean spacing* gap_n, the average distance between neighboring zeros near zero n. Recall (Chapter 17 derives it) that $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$, where s_n is the n-th "smooth zero", the solution of $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ with N_{sm} the smooth counting function of paper Eq. (1). Dividing by gap_n makes errors at different heights comparable: an error of 0.1 in energy is large where zeros are 1.3 apart and small where they are 7.5 apart. So we define the *scaled residual*

$$\hat{r}_n(c) \;=\; \frac{r_n(c)}{\mathrm{gap}_n}, \tag{24.4}$$

and the *root mean square* (rms) error in mean spacings,

$$\mathrm{rms}(c) \;=\; \sqrt{\frac{1}{n_{\mathrm{fit}}} \sum_{n=1}^{n_{\mathrm{fit}}} \hat{r}_n(c)^2}. \tag{24.5}$$

The quantity we minimize is the *objective*

$$\Phi(c) \;=\; \frac{1}{2} \sum_{n=1}^{n_{\mathrm{fit}}} \hat{r}_n(c)^2 \;=\; \frac{n_{\mathrm{fit}}}{2}\, \mathrm{rms}(c)^2. \tag{24.6}$$

The factor $\frac{1}{2}$ is a convenience that removes a factor 2 from derivatives. Because $\mathrm{rms} = \sqrt{2\Phi/n_{\mathrm{fit}}}$ and the square root is an increasing function, any c that lowers Φ lowers the rms, and the two have the same minimizers.

Pitfall

The paper's Appendix A writes its update step in terms of "r", the residual vector, without saying whether each row has been divided by gap_n. Dividing (*weighting*) makes the least-squares step minimize exactly the quantity that is reported, the rms in mean spacings, so that is what we do in this chapter. The two choices need not give identical fits: gap_n falls from about 7.5 at $n = 1$ to about 1.3 at $n = 500$, so weighting changes how much the low levels count. The Zenodo scripts [20] settle what the paper's code does; our conclusions about the algorithm do not depend on it.

Sample Problem 24.1: Residuals and rms for three levels

A trial potential gives the three lowest levels $E = (14.30, 20.90, 25.10)$. The targets are the first three zeros of RE1, $t = (14.134725, 21.022040, 25.010858)$, and the local mean spacings there are $\mathrm{gap}_1 = 7.500$, $\mathrm{gap}_2 = 5.279$, $\mathrm{gap}_3 = 4.486$ (computed from $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$). Find the residuals, the scaled residuals and the rms.
Step 1: residuals. $r_n = t_n - E_n$: $r_1 = 14.134725 - 14.30 = -0.165275$, $r_2 = 21.022040 - 20.90 = 0.122040$, $r_3 = 25.010858 - 25.10 = -0.089142$.
Step 2: scaled residuals. $\hat{r}_1 = -0.165275/7.500 = -0.02204$, $\hat{r}_2 = 0.122040/5.279 = 0.02312$, $\hat{r}_3 = -0.089142/4.486 = -0.01987$.
Step 3: squares and mean. $\hat{r}_1^2 = 4.856 \times 10^{-4}$, $\hat{r}_2^2 = 5.344 \times 10^{-4}$, $\hat{r}_3^2 = 3.948 \times 10^{-4}$. Their mean is $(4.856 + 5.344 + 3.948) \times 10^{-4}/3 = 4.716 \times 10^{-4}$.

Step 4: square root.

$$\boxed{\text{rms} \approx 0.0217 \text{ mean spacings}}$$

The levels are off by about two hundredths of a spacing, a very good fit in the paper's units.

24.1.3 Why the problem is nonlinear

If each level moved in exact proportion to each coefficient, the residuals would be linear functions of c and Chapter 9 would finish the job in one line. They are not. Recall from Chapter 13 that a small perturbation δV shifts level n by

$$\delta E_n \;=\; \int |\psi_n|^2 \, \delta V \, \mathrm{d}x \;+\; \sum_{m \neq n} \frac{\left|\langle m|\delta V|n\rangle\right|^2}{E_n - E_m} \;+\; \cdots , \tag{24.7}$$

where ψ_n is the normalized eigenfunction of level n and $\langle m|\delta V|n\rangle = \int \psi_m \, \delta V \, \psi_n \, \mathrm{d}x$. The first term is linear in c; the second is quadratic in c, because each matrix element is linear in c and it appears squared. Doubling c doubles the first term and quadruples the second. So $E_n(c)$ is a nonlinear function of c, and the inverse problem is a *nonlinear least-squares problem*: minimize a sum of squares of functions that are not linear in the unknowns.

Checkpoint 24.1

You double every coefficient c_k of a correction that already shifts level 7 by $+0.04$. Is the new shift exactly $+0.08$? Under what condition is it approximately $+0.08$?

Answer: Not exactly: the second-order term of Eq. (24.7) quadruples while the first-order term doubles. The new shift is close to $+0.08$ only if the second-order part is small, which by Chapter 13 requires the matrix elements $\langle m|\delta V|7\rangle$ to be small compared with the spacings $|E_7 - E_m|$.

24.1.4 Which level is "level n"?

To compare E_n with $\tilde{t}_n$ we match by index: the lowest level with the first target, and so on. This is unambiguous only if no two levels coincide, so that sorting by size is well defined and each level moves continuously as c changes. For the matrices we use this is a theorem with a short proof.

The finite-difference Hamiltonian is a real symmetric tridiagonal $N \times N$ matrix with diagonal entries $d_j = 2/h^2 + V_j$ and off-diagonal entries $e = -1/h^2 \neq 0$, where h is the grid spacing. Suppose $v = (v_1, \ldots, v_N)$ is an eigenvector with eigenvalue E. Row 1 of the eigen-equation reads $d_1 v_1 + e\, v_2 = E v_1$, so $v_2 = (E - d_1) v_1 / e$: v_2 is fixed by v_1. Row j reads $e\, v_{j-1} + d_j v_j + e\, v_{j+1} = E v_j$, so $v_{j+1} = [(E - d_j) v_j - e\, v_{j-1}]/e$: each component is fixed by the two before it. Hence every eigenvector with eigenvalue E is v_1 times one fixed vector, namely the vector that the recursion generates when it is started from $v_1 = 1$. The value $v_1 = 0$ is not available to a genuine eigenvector: if $v_1 = 0$, row 1 gives $v_2 = 0$, then row 2 gives $v_3 = 0$, and so on, so $v = 0$, which is not an eigenvector. A nonzero eigenvector therefore has $v_1 \neq 0$ and is a nonzero multiple of that one fixed vector. (The last row, $e\, v_{N-1} + d_N v_N = E v_N$, has no v_{N+1} to solve for; it is a condition that only certain values of E satisfy, and it adds no freedom.) So the eigenspace is one-dimensional. A real symmetric matrix has a full set of orthogonal eigenvectors (Chapter 8), so an eigenvalue of multiplicity two would need a two-dimensional eigenspace. Therefore all N eigenvalues are distinct. (A similar argument for the differential equation itself, which we do not give here, shows that bound states of the continuum problem in one dimension are nondegenerate.)

Key Ideas

- The inverse problem: choose coefficients c of $\delta V = \sum_k c_k \phi_k$ so that the levels $E_n(c)$ match targets $\tilde{t}_n$.
- Residuals $r_n = \tilde{t}_n - E_n$, scaled by gap_n; the objective $\Phi = \frac{1}{2}\sum \hat{r}_n^2$ has the same minimizers as the rms in mean spacings.
- $E_n(c)$ is nonlinear in c (second-order perturbation theory), so this is a nonlinear least-squares problem.
- Levels of the tridiagonal Hamiltonian are never degenerate, so matching level n to target n by sorting is well defined.

24.2 Linearization: the sensitivity matrix

24.2.1 Derivative of a level with respect to a coefficient

Every method in this chapter rests on one question: if we change c by a small amount δc, how do the levels move? Suppose the current coefficients are c, and let E_n and ψ_n be the eigenvalues and normalized eigenfunctions of $H(c)$. Changing c to $c + \delta c$ changes the Hamiltonian by

$$H(c+\delta c) - H(c) = \sum_k (c_k + \delta c_k)\phi_k - \sum_k c_k \phi_k = \sum_{k=0}^{M-1} \delta c_k \, \phi_k(x).$$

This is a perturbing potential added to $H(c)$. First-order perturbation theory (Chapter 13), applied with $H(c)$ as the unperturbed operator, gives

$$\begin{aligned} E_n(c+\delta c) &= E_n(c) + \int |\psi_n(x)|^2 \sum_k \delta c_k \, \phi_k(x)\, \mathrm{d}x + \mathcal{O}(\|\delta c\|^2) \\ &= E_n(c) + \sum_k \delta c_k \int |\psi_n(x)|^2 \phi_k(x)\, \mathrm{d}x + \mathcal{O}(\|\delta c\|^2) \quad \text{(integral of a sum)} \\ &= E_n(c) + \sum_k A_{nk}(c)\, \delta c_k + \mathcal{O}(\|\delta c\|^2), \end{aligned} \tag{24.8}$$

with the *sensitivity matrix*

$$A_{nk}(c) \;=\; \int_{-L}^{L} |\psi_n(x;c)|^2 \, \phi_k(x)\, \mathrm{d}x. \tag{24.9}$$

Here $\|\delta c\|$ is the length of the vector δc and $\mathcal{O}(\|\delta c\|^2)$ collects terms at least quadratic in δc (the second-order term of Eq. (24.7) and beyond). Comparing Eq. (24.8) with the definition of a partial derivative,

$$\frac{\partial E_n}{\partial c_k} \;=\; A_{nk}(c).$$

In words: the rate at which level n moves when you turn knob k is the overlap of the knob's shape ϕ_k with the probability density $|\psi_n|^2$ of the current state. This is often called the Hellmann–Feynman relation. In general form: if a Hamiltonian depends on a parameter α, a nondegenerate level obeys $\mathrm{d}E_n/\mathrm{d}\alpha = \langle \psi_n | \partial H/\partial \alpha | \psi_n \rangle$. This is nothing but first-order perturbation theory (Chapter 13) applied to the small change $(\partial H/\partial \alpha)\,\delta\alpha$ of the Hamiltonian, divided by $\delta\alpha$. Here $\alpha = c_k$, and by Eqs. (24.1) and (24.2), $\partial H/\partial c_k = \phi_k(x)$. Note that A depends on c, because the eigenfunctions ψ_n do.

The $n_{\text{fit}} \times M$ matrix A is the *Jacobian* of the forward map (the matrix of all first partial derivatives). From Eq. (24.3), residuals move the opposite way to levels:

$$r(c+\delta c) \;\approx\; r(c) - A(c)\,\delta c, \qquad \hat{r}(c+\delta c) \;\approx\; \hat{r}(c) - \hat{A}(c)\,\delta c, \tag{24.10}$$

where $\hat{A}_{nk} = A_{nk}/\text{gap}_n$ is the sensitivity matrix with row n divided by the same spacing as residual n.

Taken on trust

We take on trust that each nondegenerate eigenvalue $E_n(c)$ is a differentiable (indeed smooth) function of c, so that the remainder in Eq. (24.8) really is of order $\|\delta c\|^2$. For finite matrices with distinct eigenvalues, which we proved in Section 24.1, this follows from the implicit function theorem applied to the characteristic polynomial; texts on the perturbation theory of linear operators give the full proof. Chapter 13 already gave the first and second terms of the expansion explicitly.

24.2.2 The constant mode

If one basis function is the constant $\phi_0(x) = 1$ (true for the cosine basis with $k = 0$), then

$$A_{n0} = \int |\psi_n|^2 \cdot 1 \,\mathrm{d}x = 1 \qquad \text{for every } n,$$

because each ψ_n is normalized. A constant added to the potential shifts every level by the same amount. In fact, $H + c_0$ has exactly the same eigenvectors as H with every eigenvalue raised by exactly c_0, so this one knob acts linearly to all orders.

24.2.3 The discrete version and a normalization trap

On a grid of N interior points $x_j = -L + jh$, $h = 2L/(N+1)$, the integral in Eq. (24.9) becomes a sum, $A_{nk} = \sum_j |\psi_n(x_j)|^2 \phi_k(x_j)\, h$, with the grid normalization $\sum_j |\psi_n(x_j)|^2 h = 1$ of Chapter 23 (paper Appendix A step 9). A numerical eigensolver, however, returns eigenvectors w_n of unit *Euclidean* length, $\sum_j w_{jn}^2 = 1$. (We write w, not v, for these eigenvectors, because v_i will denote singular vectors in Section 24.5.) The two are related by $\psi_n(x_j) = w_{jn}/\sqrt{h}$, so

$$A_{nk} = \sum_j \frac{w_{jn}^2}{h}\, \phi_k(x_j)\, h = \sum_j w_{jn}^2\, \phi_k(x_j). \tag{24.11}$$

The factors of h cancel.

Pitfall

If you rescale the solver's vectors to $\psi = w/\sqrt{h}$ and then forget the factor h in the sum, every entry of A comes out too large by $1/h$, which on the paper's working grid ($N = 6000$, $L = 44.883$, so $1/h \approx 67$) is a factor of about 67. Gauss–Newton steps would then be 67 times too small. A one-line test catches this: with $\phi_0 = 1$, every A_{n0} must equal 1.

Sample Problem 24.2: The sensitivity matrix of the box

Take RE3, the infinite square well on $[-L, L]$ ($V = 0$, Dirichlet walls), with the cosine basis $\phi_k = \cos(k\pi x/L)$. Compute A_{nk} exactly.

Step 1: eigenfunctions. From Chapter 11, $\psi_n(x) = \frac{1}{\sqrt{L}} \sin\!\left(\frac{n\pi(x+L)}{2L}\right)$, $n = 1, 2, \ldots$, with $E_n = (n\pi/2L)^2$.
Step 2: the density. Using $\sin^2\alpha = \frac{1}{2}(1 - \cos 2\alpha)$,

$$|\psi_n|^2 = \frac{1}{2L}\left[1 - \cos\!\left(\frac{n\pi(x+L)}{L}\right)\right].$$

Since $\cos(\theta + n\pi) = (-1)^n \cos\theta$, the bracket is $1 - (-1)^n \cos(n\pi x/L)$, so

$$|\psi_n|^2 = \frac{1}{2L}\left[1 - (-1)^n \cos\frac{n\pi x}{L}\right].$$

Step 3: overlaps. For $k = 0$: $A_{n0} = \frac{1}{2L}\left[2L - (-1)^n \int_{-L}^{L} \cos\frac{n\pi x}{L}\mathrm{d}x\right] = \frac{1}{2L}[2L - 0] = 1$. For $k \geq 1$: $\int_{-L}^{L} \cos\frac{k\pi x}{L}\mathrm{d}x = 0$, and by the orthogonality of Chapter 5, $\int_{-L}^{L} \cos\frac{n\pi x}{L}\cos\frac{k\pi x}{L}\mathrm{d}x = L\,\delta_{nk}$ for $n, k \geq 1$, where δ_{nk} is 1 if $n = k$ and 0 otherwise. So $A_{nk} = \frac{1}{2L}\left[0 - (-1)^n L\,\delta_{nk}\right]$.
Step 4: result.

$$\boxed{A_{n0} = 1, \qquad A_{nk} = \frac{(-1)^{n+1}}{2}\,\delta_{nk} \quad (k \geq 1)}$$

To first order, cosine k moves only level $n = k$ (up by $c_k/2$ for odd n, down for even n), and c_0 moves all levels together. For the finite-difference box with N interior grid points the same values hold exactly, provided $n + k < N + 1$ (Problem 8); on fine grids this covers every n and k of interest.

The physical reason for the result is worth remembering. The density $|\psi_n|^2$ is a constant plus a ripple $\cos(n\pi x/L)$, a standing wave with n humps across the box. A cosine perturbation can push on level n only if it has the same ripple, like pushing a child on a swing in time with the swing. In the box, the wavenumber of the ripple of level n is $n\pi/L = 2\sqrt{E_n}$. Keep that factor of 2 in mind: it returns in Section 24.8.

Sample Problem 24.3: One linearized step in the box

In the box of RE3 with $M = 4$ cosines ($k = 0, 1, 2, 3$), we want first-order shifts $d = (0.01, -0.02, 0.03)$ of levels 1, 2, 3. Find the coefficient vector of smallest length that produces them.
Step 1: equations. By the Sample Problem "The sensitivity matrix of the box", $\delta E_n = c_0 + \frac{1}{2}(-1)^{n+1} c_n$, so

$$c_0 + \tfrac{1}{2}c_1 = 0.01, \qquad c_0 - \tfrac{1}{2}c_2 = -0.02, \qquad c_0 + \tfrac{1}{2}c_3 = 0.03.$$

Three equations, four unknowns: the system is underdetermined (Chapter 9) and has infinitely many solutions.
Step 2: express in terms of c_0. Writing $\epsilon_n = (-1)^{n+1}$, the equations give $c_n = 2\epsilon_n(d_n - c_0)$, so $c_n^2 = 4(d_n - c_0)^2$.
Step 3: minimize the length. $\|c\|^2 = c_0^2 + 4\sum_{n=1}^{3}(d_n - c_0)^2$. Setting the derivative with respect to c_0 to zero:

$$2c_0 - 8\sum_n (d_n - c_0) = 0 \;\Rightarrow\; 2c_0 - 8\sum_n d_n + 24c_0 = 0 \;\Rightarrow\; c_0 = \frac{4}{13}\sum_n d_n = \frac{4 \times 0.02}{13}.$$

Step 4: numbers. $c_0 = 0.0061538$, $c_1 = 2(0.01 - 0.0061538) = 0.0076923$, $c_2 = -2(-0.02 - 0.0061538) = 0.0523077$, $c_3 = 2(0.03 - 0.0061538) = 0.0476923$.

$$\boxed{c \approx (0.00615,\ 0.00769,\ 0.05231,\ 0.04769)}$$

This is exactly what the pseudo-inverse of Chapter 9 returns (you can confirm it with `numpy.linalg.pinv`).

Pitfall

Do not expect the step of the Sample Problem "One linearized step in the box" to produce the shifts it was designed for. In the box with $L = 44.883$, the lowest levels are extremely close together: $E_2 - E_1 = 3\pi^2/(4L^2) \approx 0.00367$ and $E_3 - E_1 = 2\pi^2/L^2 \approx 0.00980$. The off-diagonal matrix elements are as large as the diagonal ones. For example, with $\theta = \pi(x + L)/(2L)$, $\psi_1\psi_3 = \frac{1}{2L}(\cos 2\theta - \cos 4\theta)$ and $\cos(2\pi x/L) = \cos 4\theta$, so

$$\langle 1|\cos\tfrac{2\pi x}{L}|3\rangle = \frac{1}{2L}\cdot\frac{2L}{\pi}\int_0^\pi (\cos 2\theta - \cos 4\theta)\cos 4\theta\,\mathrm{d}\theta = \frac{1}{\pi}\left(0 - \frac{\pi}{2}\right) = -\frac{1}{2}.$$

The second-order term this coupling contributes to level 1 is of size $(c_2/2)^2/(E_3 - E_1) \approx (0.026)^2/0.0098 \approx 0.07$, seven times the first-order shift of 0.01 we aimed for. The validity condition of Chapter 13 fails badly. The cure is to take a linearized step, recompute the true levels, and repeat, with steps short enough that each one is trustworthy. That is the subject of the next three sections.

24.2.4 The sensitivity matrix of the Wu–Sprung well

Figure 24.1 compares $|A_{nk}|$ for the box (left) and for the Wu–Sprung potential in the same box (right). In the box the only nonzero entries with $k \geq 1$ lie on the diagonal $k = n$. In the Wu–Sprung well each level responds to a whole range of cosines, and the range widens as n grows. Even for the lowest level the range is not small: the cutoff introduced below sits at $k = 2L\sqrt{E_1 - V_0}/\pi \approx 60$ if we take $E_1 \approx t_1 = 14.13$, because the lowest level lies about 4.4 above the bottom of the well. The dashed curves mark $k = 2L\sqrt{E_n - V_{\min}}/\pi$, with $V_{\min}$ the bottom of the well (0 for the box, $V_0 = 3.10073\pi \approx 9.7412$ for Wu–Sprung). Above the curve the entries fall off sharply. Section 24.8 explains why.

Checkpoint 24.2

For the Wu–Sprung potential with the cosine basis, what is $A_{37,0}$? What would it be for a hat basis if you summed $A_{37,j}$ over all hats j?

Answer: $A_{37,0} = \int |\psi_{37}|^2\mathrm{d}x = 1$ by normalization. For hats, $\sum_j \phi_j(x) = 1$ at every x (Section 24.7), so $\sum_j A_{37,j} = \int |\psi_{37}|^2 \sum_j \phi_j\,\mathrm{d}x = 1$ as well.

Key Ideas

- $\partial E_n/\partial c_k = A_{nk} = \int |\psi_n|^2\phi_k\,\mathrm{d}x$, evaluated with the *current* eigenfunctions; A is the Jacobian of the forward map.
- Residuals linearize as $r(c + \delta c) \approx r(c) - A\,\delta c$ (Eq. (24.10)).
- With solver eigenvectors w_n of unit Euclidean norm, $A_{nk} = \sum_j w_{jn}^2\phi_k(x_j)$; test with $A_{n0} = 1$.
- In the box, cosine k moves only level k to first order; in the Wu–Sprung well each level responds to many cosines.
- Linearized steps can be badly wrong when level spacings are small, so steps must be iterated and controlled.

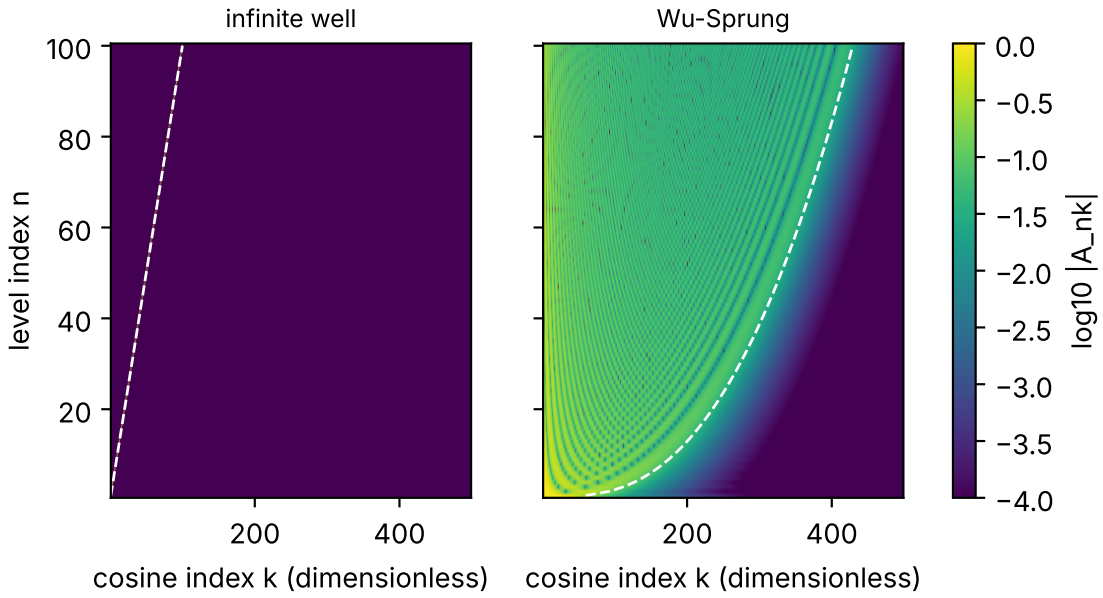


Figure 24.1: Magnitude of the first-order sensitivity matrix $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\,\mathrm{d}x$ (color, $\log_{10}$ scale, entries below 10^{-4} shown dark) for the lowest 100 levels and $k = 1, \ldots, 499$, in a box of half-width $L = 44.883$ (units $\hbar = 2m = 1$). Left: infinite well, where only $k = n$ responds. Right: the Wu–Sprung potential, where level n responds to all k up to roughly the dashed curve $k = 2L\sqrt{E_n - V_0}/\pi$ and very weakly beyond it.

24.3 Gauss–Newton: iterate the linear fit

★ Did you know? *One Spectrum Is Not Enough*

In 1946 Göran Borg proved that one list of energy levels does not in general determine a one-dimensional potential, but two spectra with different boundary conditions do. There is an important exception: if the potential is known to be symmetric, one spectrum suffices. Israel Gel'fand and Boris Levitan gave a constructive inversion method in 1951. Fitting a potential to levels is an old inverse problem.

24.3.1 The idea

At the current c, the linearization Eq. (24.10) turns the nonlinear problem into a linear one: choose δc to minimize

$$\frac{1}{2}\left\|\hat{r} - \hat{A}\,\delta c\right\|^2 .$$

This is exactly the linear least-squares problem of Chapter 9, with $\hat{A}$ as the design matrix and $\hat{r}$ as the data. Its solution satisfies the normal equations

$$\hat{A}^\top \hat{A}\,\delta c \;=\; \hat{A}^\top \hat{r}. \tag{24.12}$$

Taking that step, recomputing the true levels at $c + \delta c$, and repeating, is the *Gauss–Newton method*:

1. Start from $c^{(0)}$ (the paper starts from $c = 0$).
2. At $c^{(i)}$: solve the eigenproblem for E_n and ψ_n; form $\hat{r}$ and $\hat{A}$.
3. Solve Eq. (24.12) for δc.

4. Set $c^{(i+1)} = c^{(i)} + \delta c$ and return to step 2.

With one level and one coefficient, Eq. (24.12) reads $J^2\delta c = Jr$ with $J = \partial E/\partial c$, so $\delta c = r/J$. That is Newton's method (Chapter 4) for solving $E(c) = \tilde{t}$. Gauss–Newton is the many-equation, least-squares generalization of Newton's method.

If the levels happen to depend linearly on c (for instance, a constant shift c_0 only), the linearization is exact and one Gauss–Newton step lands on the minimum.

24.3.2 A two-level model you can do by hand

We need a nonlinear example small enough to iterate on paper. Take the 2×2 real symmetric Hamiltonian

$$H(c) = \begin{pmatrix} 1 & c \\ c & 3 \end{pmatrix}.$$

Its characteristic polynomial (Chapter 8) is $(1-E)(3-E) - c^2 = E^2 - 4E + 3 - c^2 = 0$, so

$$E_\pm(c) = 2 \pm \sqrt{4 - 3 + c^2} = 2 \pm \sqrt{1 + c^2}.$$

We fit the single coefficient c so that the lower level equals the target 0.5. Exact answer: $2 - \sqrt{1+c^2} = 0.5 \Rightarrow \sqrt{1+c^2} = 1.5 \Rightarrow c^2 = 1.25$, so $c_* = \pm 1.118034$.

The derivative is $J(c) = \mathrm{d}E_-/\mathrm{d}c = -c/\sqrt{1+c^2}$. Hellmann–Feynman gives the same thing: $\partial H/\partial c = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so $J = \langle\psi|\partial H/\partial c|\psi\rangle = 2\psi_1\psi_2$ for the normalized lower eigenvector (ψ_1, ψ_2).

Sample Problem 24.4: Gauss–Newton on the two-level model

Starting from $c^{(0)} = 0.5$, take three Gauss–Newton steps toward $E_-(c) = 0.5$.
Step 1: at $c^{(0)} = 0.5$. $\sqrt{1.25} = 1.118034$, so $E_- = 0.881966$, $r = 0.5 - 0.881966 = -0.381966$, $J = -0.5/1.118034 = -0.447214$. As a check, the lower eigenvector satisfies $(1-E)\psi_1 + c\psi_2 = 0$, so $\psi_2/\psi_1 = (E-1)/c = -0.236068$; normalizing gives $(\psi_1, \psi_2) = (0.973249, -0.229753)$ and $2\psi_1\psi_2 = -0.447214 = J$.
Step 2: first step. $\delta c = r/J = 0.854102$, so $c^{(1)} = 1.354102$. There $\sqrt{1+c^2} = 1.683328$, $E_- = 0.316672$: we overshot.
Step 3: second step. $r = 0.183328$, $J = -1.354102/1.683328 = -0.804419$, $\delta c = -0.227901$, so $c^{(2)} = 1.126201$.
Step 4: third step. At $c^{(2)}$: $E_- = 0.493903$, $r = 0.006097$, $J = -0.747761$, $\delta c = -0.008153$, so $c^{(3)} = 1.118047$.

$$\boxed{c^{(1)} = 1.3541, \quad c^{(2)} = 1.1262, \quad c^{(3)} = 1.11805 \ \ (c_* = 1.118034)}$$

The errors $c^{(i)} - c_*$ are -0.618, 0.236, 0.0082, 0.000013: once close, each error is roughly a constant times the square of the previous one. The error analysis of Newton's method predicts the constant; here is the argument. Write $f(c) = E_-(c) - 0.5$. Since $r = -f$ and $J = f'$, the Gauss–Newton step is $c_{i+1} = c_i - f(c_i)/f'(c_i)$. Taylor's theorem about c_i gives, for some ξ between c_i and c_*,

$$0 = f(c_*) = f(c_i) + f'(c_i)(c_* - c_i) + \tfrac{1}{2}f''(\xi)(c_* - c_i)^2.$$

Dividing by $f'(c_i)$ and moving terms,

$$c_i - \frac{f(c_i)}{f'(c_i)} - c_* = \frac{f''(\xi)}{2f'(c_i)}(c_i - c_*)^2.$$

The left side is $c_{i+1} - c_* = e_{i+1}$, with $e_i = c_i - c_*$. Close to the solution $f'(c_i) \approx f'(c_*)$ and $f''(\xi) \approx f''(c_*)$; also $f' = E'_-$ and $f'' = E''_-$. Hence $e_{i+1} \approx [E''_-(c_*)/2E'_-(c_*)]\, e_i^2$, with constant of size $|E''_-(c_*)/2E'_-(c_*)|$. Here $E'_-(c) = -c(1+c^2)^{-1/2}$, and differentiating once more,

$$E''_-(c) = -(1+c^2)^{-1/2} + c^2(1+c^2)^{-3/2} = -(1+c^2)^{-3/2}.$$

At c_*, $1 + c_*^2 = 2.25$, so $E'_-(c_*) = -1.118034/1.5 = -0.7454$ and $E''_-(c_*) = -1/1.5^3 = -0.2963$, giving the constant $0.2963/(2 \times 0.7454) = 0.199$. Indeed $0.199 \times 0.0082^2 = 1.3 \times 10^{-5}$, the last error. The first two errors are too large for this asymptotic rule to apply. Quadratic convergence of this kind is typical of Newton-type methods when the fit can be made exact. But the first step overshot by a lot.

Figure 24.2 shows the landscape. The absolute residual $|r(c)|$ (equivalently the objective $\Phi = \frac{1}{2}r^2$) has two equally good minima at $c = \pm 1.118$ and a local *maximum* at $c = 0$.

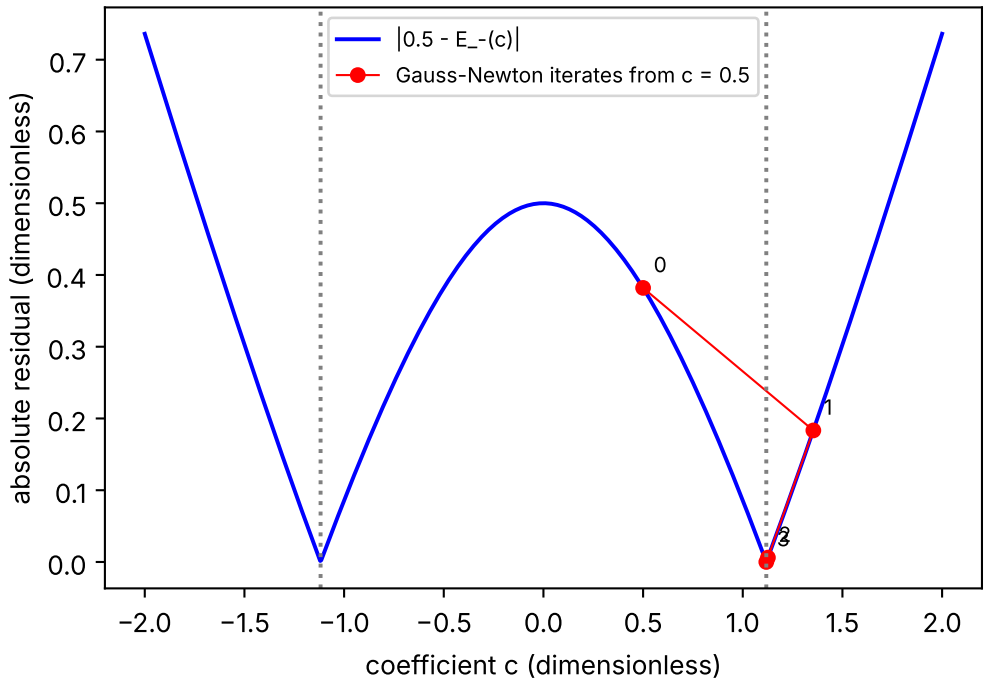


Figure 24.2: The two-level model $H = \begin{pmatrix} 1 & c \\ c & 3 \end{pmatrix}$ fitted so that its lower eigenvalue equals 0.5. Curve: absolute residual $|0.5 - E_-(c)|$ (dimensionless). Dots: Gauss–Newton iterates from $c = 0.5$ (Sample Problem "Gauss–Newton on the two-level model"), which overshoot once and then converge rapidly. The two minima at $c = \pm\sqrt{1.25}$ fit equally well; at $c = 0$ the slope vanishes and a Gauss–Newton step is undefined.

24.3.3 How Gauss–Newton fails

The two-level model already displays the three failure modes that matter for the paper's problem.

Zero sensitivity. At $c = 0$, $J = 0$ and $\delta c = r/J$ is undefined. In the many-parameter case the analogue is a matrix $\hat{A}^\top \hat{A}$ with zero or tiny eigenvalues, so Eq. (24.12) has no unique solution or an enormous one.

Overshoot. Far from the solution the linear model is poor and a full step can land somewhere worse, or (in harder problems) diverge. The pitfall at the end of Section 24.2 showed that the

Wu–Sprung problem, with its many closely spaced levels, is prone to this.

Too many knobs. When $M > n_{\text{fit}}$ the $M \times M$ matrix $\hat{A}^{\top}\hat{A}$ has rank at most $n_{\text{fit}} < M$ (Chapter 9), so it is singular and Eq. (24.12) has infinitely many solutions. The paper's fits include such cases, for instance $M = 900$ on 500 levels (Sec. III), $M = 400$ on 200 odd-indexed levels (Table III), and $M = 400$ on zeros 1–300 (Table I).

All three are cured by one modification, which we now derive.

Checkpoint 24.3

Your fit has a single knob, a constant shift c_0 with $\phi_0 = 1$. How many Gauss–Newton steps does it need, and what is the answer?

Answer: One step: the levels depend linearly on c_0 with $A_{n0} = 1$, so the linearization is exact. Unweighted, the normal equation gives $n_{\text{fit}}\, c_0 = \sum_n r_n$, the mean residual; with weights it is the weighted mean $\sum_n r_n/\text{gap}_n^2 / \sum_n 1/\text{gap}_n^2$.

Key Ideas

- Gauss–Newton: at each iterate solve the linear least-squares problem $\min \left\| \hat{r} - \hat{A}\delta c \right\|$, via $\hat{A}^{\top}\hat{A}\,\delta c = \hat{A}^{\top}\hat{r}$, then recompute.
- It is the least-squares version of Newton's method; fast near a solution.
- It fails with zero or tiny sensitivities, overshoots far from the solution, and its step is not unique when $M > n_{\text{fit}}$.

24.4 Gradient descent and the Levenberg–Marquardt compromise

24.4.1 The downhill direction

A completely different way to lower Φ is to walk downhill. The gradient $\nabla\Phi$ is the vector of partial derivatives $\partial\Phi/\partial c_k$. From Eq. (24.6) and the chain rule,

$$\begin{aligned} \frac{\partial \Phi}{\partial c_k} &= \sum_n \hat{r}_n \frac{\partial \hat{r}_n}{\partial c_k} \\ &= \sum_n \hat{r}_n \left(-\frac{1}{\text{gap}_n} \frac{\partial E_n}{\partial c_k} \right) \quad \text{(Eq. (24.4), } \tilde{t}_n \text{ fixed)} \\ &= -\sum_n \hat{r}_n \hat{A}_{nk}, \end{aligned}$$

that is,

$$\nabla\Phi = -\hat{A}^{\top}\hat{r}. \tag{24.13}$$

Steepest descent takes $\delta c = \eta\, \hat{A}^{\top}\hat{r}$ with a small positive step size η. To first order the change in Φ is $\nabla\Phi \cdot \delta c = -\eta \left\| \hat{A}^{\top}\hat{r} \right\|^2 < 0$, so a small enough step always goes downhill (unless the gradient is already zero). The price is speed: in a long narrow valley the gradient points mostly across the valley, and the walker zigzags.

24.4.2 Damped least squares

Gauss–Newton is fast but reckless; steepest descent is safe but slow. In 1963 Marquardt [32], building on earlier work of Levenberg, proposed a single formula that slides between them.

Keep the linear model, but add a penalty for long steps:

$$Q(\delta c) = \frac{1}{2}\left\|\hat{r} - \hat{A}\,\delta c\right\|^2 + \frac{1}{2}\lambda' \left\|\delta c\right\|^2, \qquad \lambda' > 0. \tag{24.14}$$

The *damping parameter* λ' sets how strongly long steps are penalized. To minimize Q, write it out in components, $Q = \frac{1}{2}\sum_n (\hat{r}_n - \sum_l \hat{A}_{nl}\delta c_l)^2 + \frac{1}{2}\lambda' \sum_l \delta c_l^2$, and set each partial derivative to zero:

$$\begin{aligned} \frac{\partial Q}{\partial \delta c_k} &= \sum_n \Big(\hat{r}_n - \sum_l \hat{A}_{nl}\delta c_l\Big)(-\hat{A}_{nk}) + \lambda'\delta c_k = 0 \\ &\Rightarrow \sum_l \Big(\sum_n \hat{A}_{nk}\hat{A}_{nl}\Big)\delta c_l + \lambda'\delta c_k = \sum_n \hat{A}_{nk}\hat{r}_n. \end{aligned}$$

In matrix form,

$$(\hat{A}^\top \hat{A} + \lambda' I)\,\delta c = \hat{A}^\top \hat{r}, \tag{24.15}$$

with I the $M \times M$ identity. This is the *Levenberg–Marquardt* (LM) step. Notice its family resemblance to ridge regression (Chapter 9): the same $+\lambda' I$, but penalizing the *step* δc, not the coefficients c.

The matrix $B = \hat{A}^\top \hat{A} + \lambda' I$ is always invertible for $\lambda' > 0$. For any nonzero vector z, $z^\top B z = z^\top \hat{A}^\top \hat{A} z + \lambda' z^\top z = \left\|\hat{A}z\right\|^2 + \lambda' \left\|z\right\|^2 > 0$, so B is symmetric and positive definite, its eigenvalues are all positive (Chapter 8), and it has an inverse. So the LM step exists even when $M > n_{\text{fit}}$ or when sensitivities vanish.

24.4.3 The two limits

Small damping. As $\lambda' \to 0$, Eq. (24.15) becomes the Gauss–Newton equation (24.12).

Large damping. Factor out λ': $B = \lambda'(I + \hat{A}^\top \hat{A}/\lambda')$. When λ' is much larger than every eigenvalue of $\hat{A}^\top \hat{A}$, the second term is small and $B^{-1} \approx I/\lambda'$, so

$$\delta c \approx \frac{1}{\lambda'} \hat{A}^\top \hat{r},$$

which by Eq. (24.13) is steepest descent with step size $\eta = 1/\lambda'$. What is neglected is the term $\hat{A}^\top \hat{A}/\lambda'$, of relative size σ_1^2/λ' where σ_1^2 is the largest eigenvalue of $\hat{A}^\top \hat{A}$.

24.4.4 Every LM step points downhill

At first order the change in Φ along the LM step is

$$\nabla\Phi \cdot \delta c = -(\hat{A}^\top \hat{r})^\top B^{-1} (\hat{A}^\top \hat{r}).$$

B^{-1} is positive definite (its eigenvalues are the reciprocals of those of B), so this is negative whenever $\hat{A}^\top \hat{r} \neq 0$. Moreover, as λ' grows the step shrinks like $1/\lambda'$, its first-order decrease $\approx \left\|\hat{A}^\top \hat{r}\right\|^2 / \lambda'$ shrinks like $1/\lambda'$, and the neglected second-order terms shrink like $1/\lambda'^2$. So for λ' large enough the first-order decrease wins and Φ really goes down. This is the guarantee that makes the algorithm safe: *if a step fails, increase the damping and try again; eventually a step succeeds*, unless you are already at a point where the gradient vanishes.

Taken on trust

Convergence theorems for Levenberg–Marquardt (that the iterates approach a point where $\nabla\Phi = 0$ under suitable conditions, and converge quickly when the residuals there are small) are beyond this book. See Marquardt's original paper [32] and the chapter

on nonlinear models in [27]. Nothing guarantees that the point reached is the global minimum.

Sample Problem 24.5: Damped steps on the two-level model

At $c = 0.5$ in the two-level model, compute the LM step for $\lambda' = 0,\ 0.04,\ 0.2,\ 1,\ 5$, and the new residual for each.

Step 1: ingredients. From the Sample Problem "Gauss–Newton on the two-level model", $J = -0.447214$, $r = -0.381966$, so $J^2 = 0.2$ and $Jr = 0.170820$.

Step 2: the formula. With one parameter, Eq. (24.15) reads $(J^2 + \lambda')\delta c = Jr$, so $\delta c = 0.170820/(0.2 + \lambda')$.

Step 3: the steps. $\lambda' = 0$: 0.85410 (Gauss–Newton). $\lambda' = 0.04$: 0.71175. $\lambda' = 0.2$: 0.42705. $\lambda' = 1$: 0.14235. $\lambda' = 5$: 0.03285. The steepest-descent estimate $Jr/\lambda' = 0.03416$ is already close for $\lambda' = 5$, which is 25 times J^2.

Step 4: new residuals. Evaluating $|0.5 - E_-(0.5 + \delta c)|$: $\lambda' = 0.04$ gives $c = 1.21175$ and $|r| = 0.0711$; $\lambda' = 0.2$ gives 0.1364; $\lambda' = 1$ gives 0.3115; $\lambda' = 5$ gives 0.3669.

all damped steps improve on $\lvert r\rvert = 0.381966$; $\lambda' = 0.04$ improves most

More damping means shorter, safer, but less productive steps.

Figure 24.3(b) shows the same trade-off in two dimensions. As λ' runs from very large to zero, the LM step traces a curve that starts out along the steepest-descent direction and ends at the Gauss–Newton point at the bottom of the quadratic model's valley.

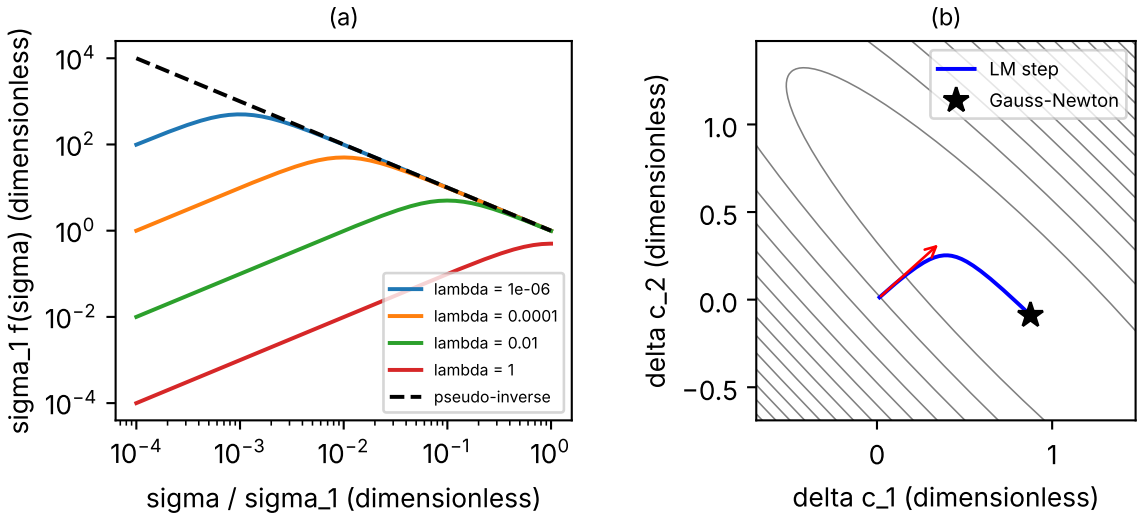


Figure 24.3: (a) Filter factors $\sigma_1 f(\sigma) = \sigma_1\sigma/(\sigma^2 + \lambda\sigma_1^2)$ of the damped SVD step against σ/σ_1 (both dimensionless) for $\lambda = 10^{-6}, 10^{-4}, 10^{-2}, 1$; the undamped pseudo-inverse σ_1/σ is dashed. Damping caps the amplification of small singular directions. (b) Contours of a quadratic model $\frac{1}{2}\|r - A\delta c\|^2$ in the plane of a two-component step $(\delta c_1, \delta c_2)$, with an elongated valley. The arrow is the steepest-descent direction $A^\top r$ at the origin; the solid curve is the Levenberg–Marquardt step as the damping λ' decreases from 10^2 to 10^{-4}, ending at the Gauss–Newton step (star).

Checkpoint 24.4

What happens to the LM step as $\lambda' \to \infty$: its length, and its direction?

Answer: Its length tends to zero like $\left\|\hat{A}^\top \hat{r}\right\| / \lambda'$, and its direction tends to the steepest-descent direction $\hat{A}^\top \hat{r} = -\nabla\Phi$.

Key Ideas

- The gradient of Φ is $-\hat{A}^\top \hat{r}$; steepest descent is safe but slow.
- The Levenberg–Marquardt step solves $(\hat{A}^\top \hat{A} + \lambda' I)\delta c = \hat{A}^\top \hat{r}$: Gauss–Newton for $\lambda' \to 0$, short steepest descent for $\lambda' \to \infty$.
- The LM matrix is positive definite for every $\lambda' > 0$, so the step always exists and always points downhill; enough damping always produces a decrease unless the gradient vanishes.

24.5 The damped step through the SVD

24.5.1 Derivation

Recall the singular value decomposition from Chapter 9. Any real $n_{\text{fit}} \times M$ matrix $\hat{A}$ of rank p can be written

$$\hat{A} = \boldsymbol{U}\Sigma\boldsymbol{V}^\top, \tag{24.16}$$

where $\boldsymbol{U}$ is $n_{\text{fit}} \times p$ with orthonormal columns $u_1, \dots, u_p$ ($\boldsymbol{U}^\top\boldsymbol{U} = I_p$), $\boldsymbol{V}$ is $M \times p$ with orthonormal columns $v_1, \dots, v_p$ ($\boldsymbol{V}^\top\boldsymbol{V} = I_p$), and $\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_p)$ holds the positive singular values in decreasing order, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p > 0$.

Pitfall

The paper's Appendix A writes this decomposition with a plain V for the matrix of right singular vectors. That V has nothing to do with the potential $V(x)$. In this chapter we write the matrix in bold, $\boldsymbol{V}$, to keep them apart. (The paper overloads other letters too: A is both this sensitivity matrix and, in Sec. II E, a non-Hermitian operator; T is a training window, a tensor $T^{(n)}$, and a Chebyshev polynomial T_k.)

Claim. The LM step is

$$\delta c = \boldsymbol{V} \operatorname{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda'}\right) \boldsymbol{U}^\top \hat{r}. \tag{24.17}$$

Proof. Call $D = \operatorname{diag}(\sigma_i/(\sigma_i^2 + \lambda'))$ and $z = \boldsymbol{V}D\boldsymbol{U}^\top\hat{r}$. First compute the two sides of Eq. (24.15) in SVD form:

$$\begin{aligned} \hat{A}^\top\hat{A} &= \boldsymbol{V}\Sigma\boldsymbol{U}^\top\boldsymbol{U}\Sigma\boldsymbol{V}^\top = \boldsymbol{V}\Sigma^2\boldsymbol{V}^\top \quad (\boldsymbol{U}^\top\boldsymbol{U} = I_p), \\ \hat{A}^\top\hat{r} &= \boldsymbol{V}\Sigma\boldsymbol{U}^\top\hat{r}. \end{aligned}$$

Now apply the LM matrix to z:

$$\begin{aligned} (\hat{A}^\top\hat{A} + \lambda' I)z &= \boldsymbol{V}\Sigma^2\boldsymbol{V}^\top\boldsymbol{V}D\boldsymbol{U}^\top\hat{r} + \lambda'\boldsymbol{V}D\boldsymbol{U}^\top\hat{r} \\ &= \boldsymbol{V}\Sigma^2 D\boldsymbol{U}^\top\hat{r} + \boldsymbol{V}\lambda' D\boldsymbol{U}^\top\hat{r} \quad (\boldsymbol{V}^\top\boldsymbol{V} = I_p) \\ &= \boldsymbol{V}(\Sigma^2 + \lambda' I_p)D\,\boldsymbol{U}^\top\hat{r}. \end{aligned}$$

The diagonal matrix $(\Sigma^2 + \lambda' I_p)D$ has entries $(\sigma_i^2 + \lambda') \cdot \sigma_i/(\sigma_i^2 + \lambda') = \sigma_i$, so it equals Σ, and the right side is $\boldsymbol{V}\Sigma\boldsymbol{U}^\top \hat{r} = \hat{A}^\top \hat{r}$. Hence z solves Eq. (24.15); since the LM matrix is invertible (Section 24.4), the solution is unique and $\delta c = z$. ■

Read Eq. (24.17) right to left, like a recipe: $\boldsymbol{U}^\top \hat{r}$ resolves the residual into its components $u_i^\top \hat{r}$ along the "level patterns" u_i; each component is multiplied by a *filter factor* $f_i = \sigma_i/(\sigma_i^2 + \lambda')$; and $\boldsymbol{V}$ converts the result into a change of coefficients along the "coefficient patterns" v_i. Without damping, $f_i = 1/\sigma_i$, the pseudo-inverse of Chapter 9.

24.5.2 The paper's scaling: $\lambda' = \lambda\sigma_1^2$

The paper measures damping relative to the largest singular value. It sets $\lambda' = \lambda\sigma_1^2$ with a dimensionless λ, so its step (Appendix A step 6) is

$$c \;+\!=\; \boldsymbol{V}\,\mathrm{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda\sigma_1^2}\right)\boldsymbol{U}^\top r. \tag{24.18}$$

(The symbol $+\!=$ is programmers' shorthand for "add the right side to c".) Here we copy the paper's symbols; in our weighted convention read $r \to \hat{r}$ and take σ_i, $\boldsymbol{U}$ and $\boldsymbol{V}$ from the SVD of $\hat{A}$. Why the relative scaling? If every basis function is multiplied by a constant α, then $\hat{A} \to \alpha\hat{A}$, every $\sigma_i \to \alpha\sigma_i$, and with $\lambda' = \lambda\sigma_1^2$ each filter factor becomes $\alpha\sigma_i/(\alpha^2\sigma_i^2 + \lambda\alpha^2\sigma_1^2) = f_i/\alpha$. The step in coefficients shrinks by exactly $1/\alpha$, so the change $\delta V = \sum_k \delta c_k(\alpha\phi_k)$ is unchanged. A given value of λ therefore means the same amount of damping whatever the units or normalization of the basis.

24.5.3 What the filter does

Four properties follow from $f(\sigma) = \sigma/(\sigma^2 + \lambda\sigma_1^2)$.

Large singular values pass almost untouched. For $\sigma^2 \gg \lambda\sigma_1^2$, $f \approx 1/\sigma$.

Small singular values are suppressed. For $\sigma^2 \ll \lambda\sigma_1^2$, $f \approx \sigma/(\lambda\sigma_1^2) \to 0$, instead of the huge $1/\sigma$.

Amplification is capped. $\mathrm{d}f/\mathrm{d}\sigma = [(\sigma^2 + \lambda\sigma_1^2) - 2\sigma^2]/(\sigma^2 + \lambda\sigma_1^2)^2 = (\lambda\sigma_1^2 - \sigma^2)/(\sigma^2 + \lambda\sigma_1^2)^2$, which vanishes at $\sigma = \sqrt{\lambda}\,\sigma_1$. There f takes its largest value $\sqrt{\lambda}\sigma_1/(2\lambda\sigma_1^2) = 1/(2\sqrt{\lambda}\,\sigma_1)$. No component of the residual is ever amplified by more than this.

The step shrinks as damping grows. Since the v_i are orthonormal, $\|\delta c\|^2 = \sum_i f_i^2 (u_i^\top \hat{r})^2$, and every f_i decreases as λ increases (its denominator grows). So $\|\delta c\|$ decreases steadily with λ.

Sample Problem 24.6: Filter factors

A sensitivity matrix has singular values $\sigma = (10,\ 1,\ 0.01)$ and the damping is $\lambda = 0.01$. Compute the filter factors and compare with the pseudo-inverse.

Step 1: the damping term. $\lambda\sigma_1^2 = 0.01 \times 100 = 1$.

Step 2: the factors. $f_1 = 10/(100+1) = 0.0990$, $f_2 = 1/(1+1) = 0.5$, $f_3 = 0.01/(0.0001+1) = 0.0099990$.

Step 3: compare. The pseudo-inverse would use $1/\sigma = (0.1,\ 1,\ 100)$.

Step 4: the cap. The largest possible factor is $1/(2\sqrt{\lambda}\sigma_1) = 1/(2\times 0.1\times 10) = 0.5$, reached at $\sigma = \sqrt{\lambda}\sigma_1 = 1$.

$$\boxed{f = (0.0990,\ 0.5,\ 0.0100) \text{ instead of } (0.1,\ 1,\ 100)}$$

The weakest direction, which the pseudo-inverse would amplify 100-fold, is now multiplied by one hundredth.

24.5.4 Too many coefficients

When $M > n_{\text{fit}}$ the rank p is at most n_{fit}, and $\boldsymbol{V}$ has only p columns. Every step Eq. (24.17) is a combination of $v_1, \ldots, v_p$, the directions in coefficient space that actually move some level. The remaining $M - p$ directions (the *null space* of $\hat{A}$, Chapter 9) change no level to first order, and the step never moves along them. As $\lambda \to 0$ the step becomes the minimum-norm step of Chapter 9, as in the Sample Problem "One linearized step in the box". Because $\hat{A}$ changes from iterate to iterate, the final c after many steps is not exactly the minimum-norm solution of anything; it depends on the path.

24.5.5 The ridge variant

The paper also runs fits with a "ridge variant" that "adds $\mu\Sigma_1^2$ to the filter denominator" (Appendix A step 6), with ridge strength $\mu = 0.03$ or 0.3 (Table I). Taken literally, the step becomes

$$c \;+\!=\; \boldsymbol{V}\,\mathrm{diag}\left(\frac{\sigma_i}{\sigma_i^2 + (\lambda+\mu)\sigma_1^2}\right)\boldsymbol{U}^\top r, \tag{24.19}$$

which is an LM step whose damping can never fall below μ. Compare the Tikhonov (ridge) regularization of Chapter 9, which adds a penalty $\frac{1}{2}\mu\sigma_1^2\,\|c\|^2$ on the coefficients themselves and so changes *where* the minimum is. The two are not the same. Under this literal reading of the record, the step of Eq. (24.19) vanishes exactly where the unregularized gradient vanishes, so the stationary points are unchanged; the variant acts by shortening steps, which, combined with the accept rule and iteration cap of Section 24.6, stops the iteration closer to its starting point $c = 0$. This is our reading of the one-line description in the record; Problem 23 works out the difference, and the scripts in [20] are the authority on the exact implementation.

Key Ideas

- The LM step in SVD form: $\delta c = \boldsymbol{V}\,\mathrm{diag}(\sigma_i/(\sigma_i^2 + \lambda'))\boldsymbol{U}^\top \hat{r}$, proved by substitution.
- The paper uses $\lambda' = \lambda\sigma_1^2$, which makes λ independent of how the basis is normalized.
- Filter factors pass strong directions, suppress weak ones, and never exceed $1/(2\sqrt{\lambda}\sigma_1)$; the step length falls as λ rises.
- With $M > n_{\text{fit}}$ the step only moves within the span of $v_1, \ldots, v_p$.
- Read literally, the paper's ridge variant raises the floor of the damping, which would differ from a Tikhonov penalty on c; the archived scripts decide.

24.6 The paper's damping schedule, line by line

An LM method needs a rule for choosing λ at each iteration. Here is the paper's rule, quoted from Appendix A step 6 (with the revision settings stated just after step 15, "start $c = 0$, at most 60 iterations, acceptance 0.999"):

> iterate {eigensolve with vectors; $A_{nk} = \int |\psi_n|^2 \phi_k$; SVD-damped step $c \mathrel{+}= V(\Sigma/(\Sigma^2 + \lambda\Sigma_1^2))U^\top r$ with try-list $(0.2\lambda, \lambda, 5\lambda, 25\lambda)$; on success $\lambda \leftarrow \max(0.5\lambda_{\text{acc}}, 10^{-7})$, on total failure $\lambda \leftarrow 30\lambda$, stop at $\lambda > 10^4$; accept only improvements ($\times 0.999$; the scale-2 M=300 row used $1 - 10^{-5}$)}

Table 24.1 translates each clause.

Two details are not spelled out in the quoted text: the initial value of λ, and whether the try-list is scanned in order with the first acceptable trial taken, or all four are evaluated and

Table 24.1: The clauses of the paper's fitting loop (Appendix A step 6) and what each one does.

clause	meaning
eigensolve with vectors	compute E_n and eigenvectors w_n of $H(c)$ for the fitted levels (Chapter 23)
$A_{nk} = \int \lvert\psi_n\rvert^2 \phi_k$	build the sensitivity matrix, Eq. (24.11)
SVD-damped step	Eq. (24.18): one SVD per iteration, then each trial step costs only a few matrix products plus one eigenvalue-only solve to score it
try-list $(0.2\lambda, \lambda, 5\lambda, 25\lambda)$	candidate damping values, from bolder to more cautious
accept only improvements ($\times 0.999$)	a trial is accepted only if its rms is below 0.999 times the current rms
on success $\lambda \leftarrow \max(0.5\lambda_{\text{acc}}, 10^{-7})$	λ_{acc} is the damping of the accepted trial; next time start bolder, but never below 10^{-7}
on total failure $\lambda \leftarrow 30\lambda$	no trial accepted: keep c, damp much harder
stop at $\lambda > 10^4$	give up when even very short steps cannot improve by 0.1%
at most 60 iterations	a cap on the number of passes through the loop (stated for the revision computations, App. A after step 15; not stated for the original fits)

the best taken. In this book's reading, and in our code, the list is scanned in order and the first trial that passes is accepted, and we start from $\lambda = 1$; both are our choices, and the archived scripts [20] record the paper's. The logic of the schedule is the same either way.

24.6.1 Why the schedule works

The accept rule makes the in-sample rms a non-increasing sequence: c changes only when the rms drops by at least 0.1%. The try-list explores a factor of 125 in damping around the current λ at the cost of at most four eigenvalue solves (each trial needs only eigenvalues, not eigenvectors). After a success the next iteration starts at half the accepted damping, so the algorithm grows bolder while things go well and approaches Gauss–Newton speed. After a total failure, λ jumps by 30, and by Section 24.4 enough damping always produces a decrease unless the gradient vanishes.

The threshold 0.999 also provides the stopping mechanism. Near a minimum, any improvement available is tiny; the trials fail the 0.1% test, λ is multiplied by 30 again and again, and once it exceeds 10^4 the loop stops. A stricter threshold stops earlier, a looser one later. The paper reports one case where this matters: the scale-2 $M = 300$ row of Table I "required relaxing the step-acceptance threshold; under the strict protocol the optimizer accepts no step at all" (Table I caption), and it used $1 - 10^{-5}$ instead.

```
import numpy as np

def lm_fit(c, resid, jac, lam=1.0, maxiter=60, accept=0.999):
    # resid(c): r_n/gap_n;  jac(c): A_nk/gap_n  (paper App. A, step 6)
    r = resid(c)
    cur = np.sqrt(np.mean(r**2))                # current rms
    for it in range(maxiter):
        U, s, Vt = np.linalg.svd(jac(c), full_matrices=False)
        Utr = U.T @ r
        accepted = False
        for lam_try in (0.2*lam, lam, 5*lam, 25*lam):
            f = s / (s**2 + lam_try * s[0]**2)    # filter factors
            c_try = c + Vt.T @ (f * Utr)          # Vt.T is the matrix V
            r_try = resid(c_try); rms_try = np.sqrt(np.mean(r_try**2))
            if rms_try < accept * cur:            # accept only improvements
```

```
                c, r, cur = c_try, r_try, rms_try
                lam = max(0.5 * lam_try, 1e-7)
                accepted = True
                break
        if not accepted:                          # total failure
            lam = 30 * lam
            if lam > 1e4:
                break                             # stop rule
    return c, cur

# example: two-level model, target 0.5, start c = 0.5
Em = lambda c: 2 - np.sqrt(1 + c[0]**2)
res = lambda c: np.array([0.5 - Em(c)])
jac = lambda c: np.array([[-c[0] / np.sqrt(1 + c[0]**2)]])
print(lm_fit(np.array([0.5]), res, jac))  # c close to 1.118034, rms near 0
```

The example call fits the two-level model of Section 24.3. It returns a coefficient within 10^{-6} of the exact answer $c_* = 1.118034$ and an rms that is zero to within rounding; with `maxiter=1` it returns $c = 1.21175$, the single iteration worked out by hand in the Sample Problem "One pass of the schedule on the two-level model".

Sample Problem 24.7: One pass of the schedule on the two-level model

Run the first iteration of the schedule (code above) on the two-level model of Section 24.3, starting from $c = 0.5$ with $\lambda = 1$. There is one parameter, so the "SVD" of the 1×1 matrix J has $\sigma_1 = |J|$.

Step 1: current state. From the Sample Problem "Gauss–Newton on the two-level model", $r = -0.381966$, $J = -0.447214$, $\sigma_1^2 = J^2 = 0.2$. The current rms is $|r| = 0.381966$, and a trial passes only if its rms is below $0.999 \times 0.381966 = 0.381584$.

Step 2: first trial. $\lambda_{\text{try}} = 0.2\lambda = 0.2$, so the effective damping is $\lambda' = \lambda_{\text{try}}\sigma_1^2 = 0.04$. By the Sample Problem "Damped steps on the two-level model" this gives $\delta c = 0.71175$, $c = 1.21175$, and $|r| = 0.071095$.

Step 3: the test. $0.071095 < 0.381584$, so the trial is accepted and the remaining trials are not tried.

Step 4: update the damping. $\lambda \leftarrow \max(0.5 \times 0.2,\ 10^{-7}) = 0.1$.

$$\boxed{c = 1.21175, \quad \text{rms} = 0.0711, \quad \lambda = 0.1 \text{ after one iteration}}$$

The next iteration starts with trial damping 0.02, close to Gauss–Newton, which is appropriate because the iterate is now near the solution.

24.6.2 Random restarts

The paper's protocol "is deterministic from $c = 0$; optimizer variability is probed with randomized initial c" (Sec. II C). Starting from a random $c^{(0)}$ instead of 0 tests whether the answer depends on where the descent began. The two-level model shows why it might: Figure 24.2 has two minima, and the one you reach depends on the sign of the starting c. In a problem with hundreds of coefficients the landscape can have many valleys, and an LM method only ever finds a point where the gradient is (nearly) zero near where it started. The paper reports that four random-init repeats of the scale-1, $M = 400$, zeros 1–400 configuration "give identical in-sample rms 0.1370 and OOS 0.365–0.379" (Table I caption): the in-sample error did not depend on the start, but the held-out behavior varied a little. That pattern (many different coefficient vectors that fit equally well) is what you expect when there are far more ways to fit the data than data, a theme of Chapter 25.

Checkpoint 24.5

Why does the accept rule guarantee that the final in-sample rms is never worse than the starting one?

Answer: c is changed only when a trial's rms is below 0.999 times the current rms. So the sequence of accepted rms values strictly decreases, and if nothing is ever accepted the final c is the starting c.

Key Ideas

- Each iteration: one eigensolve with vectors, one SVD, then up to four trial steps needing only eigenvalues.
- Accept a trial only if it lowers the rms by at least 0.1%; after success relax damping to half the accepted value, after total failure multiply it by 30; stop when $\lambda > 10^4$ (and, in the revision computations, after at most 60 iterations).
- Random starting points probe whether the result depends on the optimizer's path.

24.7 Three bases for δV

★ **Did you know?** ***Runge's Wiggles***

In 1901 Carl Runge showed that fitting a polynomial through more and more *equally spaced* points can make things worse. For $1/(1+25x^2)$ on $[-1,1]$, the interpolating polynomials oscillate wildly near the ends, and the error grows without bound. Clustering the points toward the ends, as Chebyshev nodes do, cures the problem. The choice of basis and sampling can make or break a fit.

The choice of basis functions ϕ_k decides which shapes δV can take. The Wu–Sprung background is even, $V(-x) = V(x)$, and the paper keeps the correction even too, so that the fitted potential stays even and its states keep definite parity (Chapter 11). All three bases below are even functions of x. Figure 24.4 plots the first few members of each.

Cosines. $\phi_k(x) = \cos(k\pi x/L)$, $k = 0, 1, \ldots, M-1$. By Chapter 5 these are orthogonal on $[-L, L]$ and form the natural Fourier basis for even functions. Each is spread over the whole box (a *global* basis), with uniform resolution: member k has wavelength $2L/k$ everywhere.

Chebyshev polynomials. The Chebyshev polynomials $T_k(y)$ on $-1 \le y \le 1$ are defined by

$$T_k(y) = \cos(k \arccos y), \tag{24.20}$$

that is, if $y = \cos\theta$ with $0 \le \theta \le \pi$, then $T_k(y) = \cos k\theta$. They are polynomials: $T_0 = 1$, $T_1 = y$, and from the identity $\cos((k+1)\theta)+\cos((k-1)\theta) = 2\cos\theta\cos k\theta$ we get the recurrence $T_{k+1}(y) = 2yT_k(y) - T_{k-1}(y)$, giving $T_2 = 2y^2 - 1$, $T_3 = 4y^3 - 3y$. The paper uses $\phi_k(x) = T_k(2|x|/L - 1)$, $k = 0, \ldots, M-1$: the variable $y = 2|x|/L - 1$ runs from -1 at $x = 0$ to $+1$ at $|x| = L$. Since $y = \cos\theta$ changes slowly near $\theta = 0$ and $\theta = \pi$, equal steps in θ crowd together near $y = \pm 1$; the oscillations of T_k are therefore densest near $x = 0$ and $|x| = L$. This is the paper's "resolution concentrated near $x = 0$ and $x = L$" (Sec. II C, "Other bases"). The absolute value makes ϕ_k even, at the price of a kink at $x = 0$ for every $k \ge 1$. To see this, differentiate $T_k(\cos\theta) = \cos k\theta$ with respect to θ: $T_k'(y)\,(-\sin\theta) = -k\sin k\theta$, so $T_k'(y) = k\sin k\theta/\sin\theta$. As $y \to -1$, $\theta \to \pi$ and both sines vanish; l'Hôpital's rule gives $T_k'(-1) = k \cdot k\cos k\pi/\cos\pi = (-1)^{k+1}k^2$. Just right of $x = 0$, $\mathrm{d}\phi_k/\mathrm{d}x = (2/L)T_k'(y) \to (2/L)(-1)^{k+1}k^2$; by evenness the slope just left of $x = 0$ is the negative of this, so the slope jumps by $4k^2/L \neq 0$ (visible in Figure 24.4).

Hat functions. Put M equally spaced nodes $u_j = j/(M-1)$, $j = 0, \dots, M-1$, on the interval $0 \le u \le 1$, where $u = |x|/L$. (The paper says only "M uniform nodes of $|x|/L$" and does not specify their placement; we take nodes that include both ends, and the archived scripts [20] record the paper's choice. The variable u and the nodes u_j have nothing to do with the left singular vectors u_i of Section 24.5.) The hat function centered at node j is

$$\phi_j(x) = \max\bigl(0,\ 1 - (M-1)|u - u_j|\bigr), \qquad u = \frac{|x|}{L}. \tag{24.21}$$

It equals 1 at its own node, 0 at every other node, and is a straight line in between neighbors. A combination $\sum_j c_j \phi_j$ is the piecewise-linear function that passes through the values c_j at the nodes. Hats are *local*: each is nonzero only within one node spacing of its center, so a coefficient changes the potential only in one small region. The paper calls this "the simplest stand-in for a wavelet-type basis". Hats sum to one: if u lies between nodes u_j and u_{j+1}, only ϕ_j and ϕ_{j+1} are nonzero there, equal to $1-(M-1)(u-u_j)$ and $1-(M-1)(u_{j+1}-u)$; their sum is $2-(M-1)(u_{j+1}-u_j) = 2-1 = 1$.

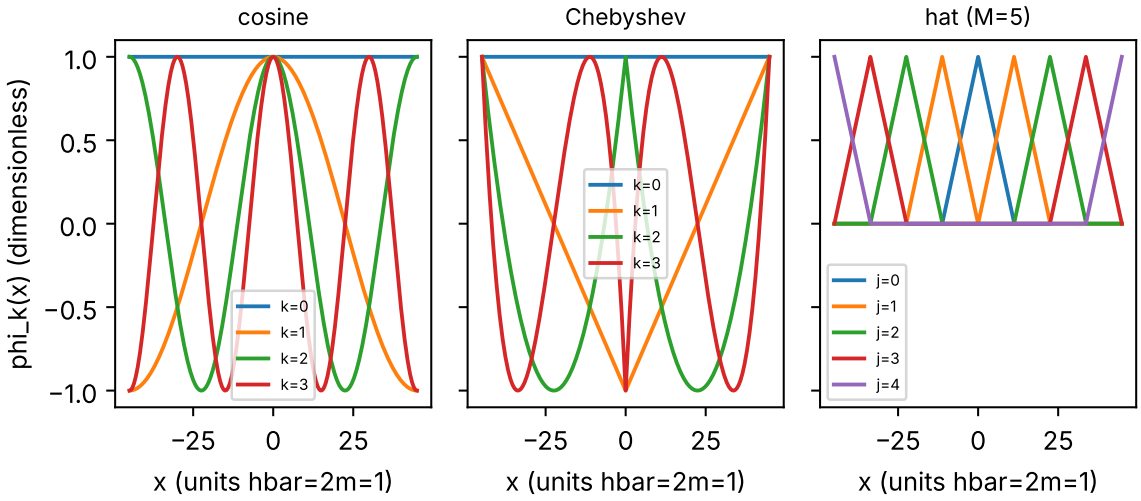


Figure 24.4: The three even bases used for δV in the paper, on $[-L, L]$ with $L = 44.883$ (units $\hbar = 2m = 1$; ϕ is dimensionless). Left: cosines $\cos(k\pi x/L)$, $k = 0, \dots, 3$. Middle: Chebyshev polynomials $T_k(2|x|/L - 1)$, $k = 0, \dots, 3$, with kinks at $x = 0$ for every $k \ge 1$. Right: the five hat functions for $M = 5$ nodes in $|x|/L$.

Sample Problem 24.8: Evaluating Chebyshev and hat functions

(a) Evaluate $\phi_3(x) = T_3(2|x|/L - 1)$ at $x = L/4$ two ways. (b) For $M = 5$ hats, find all nonzero ϕ_j at $|x|/L = 0.6$.

Step 1: the variable. At $x = L/4$, $y = 2 \cdot \frac{1}{4} - 1 = -\frac{1}{2}$.

Step 2: polynomial. $T_3(-\frac{1}{2}) = 4(-\frac{1}{8}) - 3(-\frac{1}{2}) = -\frac{1}{2} + \frac{3}{2} = 1$.

Step 3: cosine form. $\arccos(-\frac{1}{2}) = 2\pi/3$, so $T_3 = \cos(3 \cdot 2\pi/3) = \cos 2\pi = 1$. The two agree.

Step 4: hats. Nodes are $0, 0.25, 0.5, 0.75, 1$, and $M - 1 = 4$. Only the neighbors $u_2 = 0.5$ and $u_3 = 0.75$ are within 0.25 of 0.6: $\phi_2 = 1 - 4(0.1) = 0.6$, $\phi_3 = 1 - 4(0.15) = 0.4$.

$$\boxed{T_3(-\tfrac{1}{2}) = 1; \qquad \phi_2 = 0.6,\ \phi_3 = 0.4,\ \text{others } 0 \text{ (sum } 1)}$$

Key Ideas

- Cosines: global, orthogonal, uniform resolution.
- Chebyshev $T_k(2|x|/L-1)$: global, resolution concentrated near $x=0$ and $|x|=L$.
- Hats: local piecewise-linear functions on M nodes of $|x|/L$; they sum to one.
- All three are even, so the fitted potential stays even; only the basis changes between the paper's three families.

24.8 The coverage law: how many basis functions are enough?

24.8.1 Which cosines can push which levels

The Wu–Sprung panel of Figure 24.1 shows that level n responds to cosines up to a cutoff. We now derive that cutoff heuristically. The argument uses the WKB picture of Chapter 12 and is approximate, not a theorem.

Step 1: the WKB density. A warning on notation first: in this section $k(x)$ is the local wavenumber, a function of position, while the basis index of $\cos(k\pi x/L)$ is a plain integer k; to keep them apart we write the cosine's wavenumber as $q=k\pi/L$. Recall from Chapter 12 that between its turning points (the points where $E=V(x)$, at which a classical particle would stop and turn back) a level of energy E has local wavenumber $k(x)=\sqrt{E-V(x)}$ and WKB wavefunction $\psi \approx C\,k(x)^{-1/2}\cos(\theta(x))$, where $\theta(x)=\int^x k(x')\,\mathrm{d}x'-\pi/4$ and C is a constant. (The wave has the local wavelength $2\pi/k(x)$ set by the classical momentum; the amplitude factor $k^{-1/2}$ makes $|\psi|^2$ proportional to the time a classical particle spends near x; the phase $-\pi/4$ comes from the connection formulas of Chapter 12.) Squaring and using $\cos^2\theta=\frac{1}{2}(1+\cos 2\theta)$,

$$|\psi|^2 \approx \frac{C^2}{2k(x)}\Big[1+\cos 2\theta(x)\Big].$$

The density is a slowly varying part plus a ripple whose phase 2θ advances at the local rate $\mathrm{d}(2\theta)/\mathrm{d}x=2k(x)$. So the ripple has local wavenumber $2k(x)$.

Step 2: overlap with one cosine. The sensitivity $A_{nk}=\int|\psi_n|^2\cos(qx)\,\mathrm{d}x$ with $q=k\pi/L$ has two pieces. The slow part of $|\psi|^2$ overlaps only with cosines of low q. The ripple contributes $\int\frac{C^2}{2k}\cos 2\theta\cos qx\,\mathrm{d}x=\frac{C^2}{4}\int\frac{1}{k}[\cos(2\theta-qx)+\cos(2\theta+qx)]\mathrm{d}x$. Each integrand oscillates rapidly, and its positive and negative lobes cancel, except near a point where its phase is stationary, $\mathrm{d}(2\theta-qx)/\mathrm{d}x=2k(x)-q=0$. This is called *stationary-phase* reasoning. A small example shows why the stationary point dominates. With a phase that never stands still, $\int_{-X}^{X}\cos(ax)\,\mathrm{d}x=\big[\sin(ax)/a\big]_{-X}^{X}=2\sin(aX)/a$, at most $2/a$ in size: the lobes cancel and only an endpoint remnant of order $1/a$ survives. With the phase ax^2, which is stationary at $x=0$, the integrand $\cos(ax^2)$ stays positive throughout $ax^2<\pi/2$, a central interval of width $2\sqrt{\pi/2a}$ that shrinks only like $1/\sqrt{a}$; this uncancelled region contributes an amount of order $1/\sqrt{a}$, far more than $1/a$ when a is large. For $a=100$ and $X=3$ the first integral is at most 0.02 in size, while numerically the second is about 0.13.

Taken on trust

The infinite-range value $\int_{-\infty}^{\infty}\cos(ax^2)\,\mathrm{d}x=\sqrt{\pi/2a}$ (about 0.125 for $a=100$) is a Fresnel integral, and the general statement that an oscillatory integral is dominated by its points of stationary phase is the method of stationary phase. We take both on trust; they are treated with the method of steepest descents in [22].

Step 3: the cutoff. A matching point $2k(x) = q$ exists only if q is no larger than the maximum of $2k(x)$. For a well with its minimum V_0 at $x = 0$, that maximum is $2\sqrt{E - V_0}$. Hence level n responds strongly only to cosines with

$$\frac{k\pi}{L} \lesssim 2\sqrt{E_n - V_0}, \tag{24.22}$$

which is the dashed curve of Figure 24.1. Above the cutoff the condition $2k(x) = q$ has no real solution anywhere in the well, so the positive and negative lobes of the integrand cancel along the whole range of integration and the overlap is small, though not exactly zero. That is why the figure shows a sharp fall, not a hard edge.

Taken on trust

How small is the overlap above the cutoff? A refinement of the stationary-phase method, the method of steepest descents, treats the condition $2k(x) = q$ as an equation for a *complex* x and shows that above the cutoff the overlap decreases exponentially as q grows. We take this on trust; the method is developed in the chapter on asymptotic expansions of [22]. In this chapter we use only the qualitative statement that the overlap is small above the cutoff, which is what Figure 24.1 shows.

Check against the box. With $V_0 = 0$ and $E_n = (n\pi/2L)^2$, Eq. (24.22) reads $k\pi/L \lesssim n\pi/L$, i.e. $k \lesssim n$. The Sample Problem "The sensitivity matrix of the box" found the exact answer: in the box only $k = n$ (and $k = 0$) respond. The box is special because its ripple has a single wavenumber everywhere, so only the matching cosine survives; in the Wu–Sprung well $2k(x)$ ranges from 0 at the turning points up to $2\sqrt{E - V_0}$, and every cosine in that range finds a matching point.

24.8.2 The law

To move each of the levels up to $E_{\max}$ independently of its neighbors, the basis must contain cosines that reach the finest ripple of the top level. The largest wavenumber in a basis of M cosines is $(M - 1)\pi/L \approx M\pi/L$. So the basis is effectively complete for joint level shifts up to $E_{\max}$ when

$$\frac{M\pi}{L} > 2\sqrt{E_{\max}}, \tag{24.23}$$

which is the paper's coverage law (Sec. III, first paragraph). We have used $\sqrt{E}$ rather than $\sqrt{E - V_0}$, as the paper does; for small u the first-order Taylor expansion $\sqrt{1 - u} \approx 1 - u/2$ gives $\sqrt{E - V_0} = \sqrt{E}\,\sqrt{1 - V_0/E} \approx \sqrt{E}\,(1 - V_0/2E)$, so the relative difference is about $V_0/(2E)$. At the top scored level, $E \approx 811$, that is 0.6%, so the paper's form is harmless where the law matters. At the bottom of the spectrum the difference is large (at $E = t_1 = 14.13$, $\sqrt{E - V_0} \approx 0.56\sqrt{E}$), but the coverage law is set by the highest level, not the lowest. Two further conditions are implicit: you also need at least as many coefficients as levels, $M \geq n_{\text{fit}}$, to shift n_{fit} levels independently, and the optimizer must actually find the fit. Equation (24.23) is a heuristic rule of thumb, derived from WKB and stationary phase, not a proven statement about the minimum of Φ. In particular it is not a necessary condition for a small in-sample error. It concerns shifting *all consecutive* levels up to $E_{\max}$ independently. A level above the reach of the basis still responds to all M cosines (its own cutoff lies above M), so if only some levels are fitted (every other one, say), or if n_{fit} is well below M, the sensitivity matrix can still have full row rank and the fit can reach very small residuals even though $M\pi/L < 2\sqrt{E_{\max}}$. The paper's Table III shows this: at scale 1, $M = 400$ fits the 200 odd-indexed (or even-indexed) zeros among zeros 1–400 to rms $< 10^{-3}$, whereas covering all consecutive levels up to $t_{400} \approx 679.7$ would, by Eq. (24.23), need $M \gtrsim 745$.

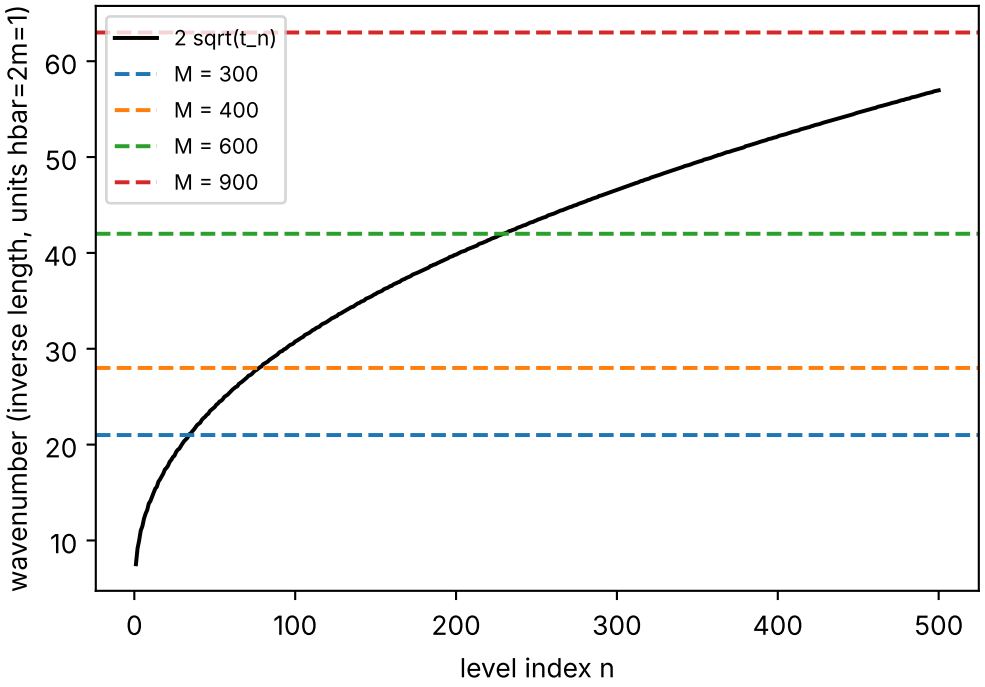


Figure 24.5: Coverage law at scale 1 ($L = 44.883$, units $\hbar = 2m = 1$). Solid curve: the finest ripple wavenumber $2\sqrt{t_n}$ of level n (in units of inverse length), using the zero heights t_n as the level energies. Horizontal lines: the highest cosine wavenumber $M\pi/L$ for $M = 300, 400, 600, 900$. A basis covers all levels below the crossing point; only $M = 900$ clears all 500 scored levels.

Sample Problem 24.9: How many cosines for 500 zeros?

At scale 1 ($L = 44.883$) the top scored level is $t_{500} = 811.1843588465$ (paper Appendix A, "Key anchors"). (a) How many cosines does Eq. (24.23) require? (b) Up to which energy, and roughly which level index, does $M = 400$ reach?

Step 1: required wavenumber. $2\sqrt{811.18} = 2 \times 28.481 = 56.96$.

Step 2: required M. $M > 56.96\, L/\pi = 56.96 \times 44.883/3.14159 = 813.8$.

Step 3: reach of $M = 400$. Solving $400\pi/L = 2\sqrt{E}$ gives $E = (400\pi/2L)^2 = (13.999)^2 = 196.0$.

Step 4: level index. The number of zeros below 196 is approximately $N_{\rm sm}(196) = \frac{196}{2\pi}\log\frac{196}{2\pi} - \frac{196}{2\pi} + \frac{7}{8} \approx 77.0$.

$M \gtrsim 814$ for all 500 levels; $M = 400$ gives full coverage only up to $E \approx 196$, about level 77

"Full coverage" means that every consecutive level up to that energy can be shifted independently. Levels above $E \approx 196$ still respond to all 400 cosines, so the law does not say they cannot be fitted; it says only that $M = 400$ does not guarantee independent shifts of every consecutive level there. This is consistent with the paper: $M = 900$ reproduces all 500 zeros to rms $< 10^{-4}$, while $M = 300$ leaves 0.186 (Sec. III).

Checkpoint 24.6

An $M = 400$ cosine fit on zeros 1–400 reaches in-sample rms 0.137 (Table I). By the coverage law, which levels would you expect to be fitted well and which poorly?

Answer: The law alone does not tell you. It guarantees independent shifts only for the consecutive levels up to about level 77. Levels above that still respond to all 400 cosines, and with as many coefficients as training levels the sensitivity matrix can have full row rank, so they may well be fitted too. The law is heuristic and is not a necessary condition for a small error: Table III shows $M = 400$ fitting 200 odd-indexed levels as high as $n = 399$ to rms $< 10^{-3}$. Where the residual of the 1–400 fit is concentrated can only be found by computing it, which this chapter does not do.

24.8.3 In-sample error can only fall with M

The paper reports that "in-sample error falls steadily with basis size M" (Fig. 3 caption) in every basis (Table II). Here is the reason, and its limit. The bases are *nested*: the cosine family with M members is contained in the family with $M + 1$ members (set $c_M = 0$). Every potential available with M coefficients is available with $M + 1$, so the smallest achievable Φ cannot increase with M. For hats, the families with different M are not exactly nested (node positions move), so the argument holds only approximately. And the argument is about the *best* fit; an optimizer that stops early could, in principle, report a larger error at larger M. Figure 24.6 shows the steady fall in a small toy problem you can run in seconds.

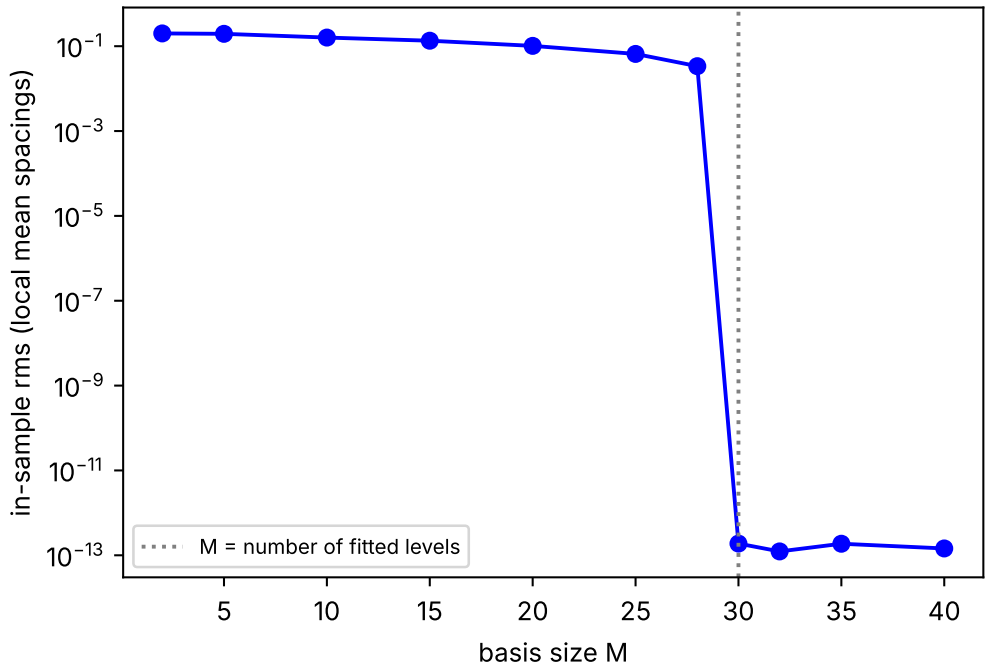


Figure 24.6: Toy version of the paper's fit: a box of half-width $L = 10$ with $V = 0$ on $N = 300$ grid points (units $\hbar = 2m = 1$), cosine basis, and targets equal to the lowest 30 levels displaced by random amounts of rms 0.2 local spacings (seed 0). The paper-style damped schedule of Section 24.6 is run for each basis size M. In-sample rms (in local spacings, log scale) falls with M and collapses to essentially zero once M reaches the number of fitted levels, $M = 30$, where the first-order equations first become exactly solvable (Problem 24).

Key Ideas

- By WKB and stationary phase, level n responds to cosines with $k\pi/L \lesssim 2\sqrt{E_n - V_0}$ (heuristic).
- Coverage law: $M\pi/L > 2\sqrt{E_{\max}}$ makes the cosine family effectively complete for joint

shifts of all levels up to $E_{\max}$; at scale 1 this needs $M \gtrsim 814$.

- For nested bases the best achievable in-sample error cannot rise with M.
- Coverage is about what can be fitted, not about what will be predicted.

24.9 Paper connection

Sec. II C, first paragraph. The paper writes $\delta V(x) = \sum_{k=0}^{M-1} c_k \cos(k\pi x/L)$ (our Eq. (24.2)), forms residuals $r_n = \tilde{t}_n - E_n$ (Eq. (24.3)) and "the first-order sensitivity $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\, dx$" (Eq. (24.9), Section 24.2), and updates "c by SVD-damped least squares with a Levenberg–Marquardt schedule, accepting only steps that improve the rms". You can now read each phrase: the SVD step is Eq. (24.18), its derivation from damped least squares is Sections 24.4–24.5, and the schedule is Table 24.1. "Iterated linearization" is Gauss–Newton (Section 24.3) made safe by damping.

Appendix A step 6. Every symbol of the step $c \mathrel{+}= V(\Sigma/(\Sigma^2 + \lambda\Sigma_1^2))U^\top r$ has been defined: the plain V there is our $\boldsymbol{V}$ of right singular vectors, Σ_1 is σ_1, and the ridge variant "adds $\mu\Sigma_1^2$ to the filter denominator" (literal reading: Eq. (24.19)). Appendix A step 9 states the normalization $\sum \psi^2 dx = 1$ used in our Eq. (24.11). Step 7, "plain eigensolve of the fitted potential, scored on held-out zeros", is the out-of-sample evaluation; it uses no part of the fitting machinery except the final c.

Sec. II C, "Other bases"; Table II. The three bases of Section 24.7 are exactly the paper's: cosines, $T_k(2|x|/L - 1)$, and hats "on M uniform nodes of $|x|/L$". "Only the basis changes." Table II's in-sample column falls with M in each basis (Chebyshev $0.230, 0.162, 0.111$; hat $0.221, 0.145, 0.066$ at $M = 200, 400, 600$), as Section 24.8 predicts for the best achievable error in the nested cosine and Chebyshev families, and approximately for hats.

Sec. III, first paragraph. "On the full 500 zeros the fit reaches rms 0.186 at $M = 300$ and reproduces all 500 zeros to rms $< 10^{-4}$ spacings by $M = 900$, consistent with a basis-coverage law (the cosine family becomes effectively complete for joint level shifts once $M\pi/L > 2\sqrt{E}$)." The Sample Problem "How many cosines for 500 zeros?" shows $M \gtrsim 814$ is needed at scale 1, so 900 clears the bar and 300 does not. The paper calls this "consistent with", and we have derived the law only heuristically; keep both qualifications.

Table I and Fig. 3. The in-sample column of Table I is the final rms of the schedule: for zeros 1–400 at scale 1 it falls 0.226, 0.137, 0.064 for $M = 200$, 400, 600. The two ridge rows ($\mu = 0.03$, 0.3) end at higher in-sample rms, 0.142 and 0.206, which is consistent with either reading of the ridge variant (a damping floor that, with the accept rule and iteration cap, stops the iteration early, or a Tikhonov penalty that moves the minimum); see Section 24.5. The Table I caption reports that the scale-2 $M = 300$ row "required relaxing the step-acceptance threshold"; in Table 24.1 terms, every trial in every iteration failed the 0.999 test, so (in our reading of the schedule) λ was multiplied by 30 until it exceeded 10^4, with c still zero. With $1 - 10^{-5}$ the fit moved.

Table III, saturated rows. The odd/even splits at scale 1 have 200 training levels, so rows with $M \geq 200$ have at least as many coefficients as levels. At $M = 400$ the in-sample error is "$< 10^{-3}$": with twice as many knobs as levels, the underdetermined damped steps (Section 24.5) drive the residuals to essentially zero. This does not contradict the coverage law: the law

concerns independent shifts of all consecutive levels up to $E_{\max}$, while here only every other level is fitted, and a 200×400 sensitivity matrix can have full row rank (Section 24.8). The paper calls these rows "stress tests of memorization".

What this chapter does not decide. Every number above is in-sample. The paper's central result is that the out-of-sample columns of Tables I–III show no reproducible skill against the null; the tools for that comparison are in Chapter 25 and the reading in Chapter 29. A low in-sample error, by itself, tells you that the coverage law is satisfied and the optimizer works, nothing more.

24.10 Summary

- The inverse problem: choose c in $\delta V = \sum_k c_k \phi_k$ (Eq. (24.2)) to minimize $\Phi = \frac{1}{2}\sum_n (r_n/\mathrm{gap}_n)^2$ (Eqs. (24.3)–(24.6)). It is nonlinear because of second-order perturbation terms (Eq. (24.7)).
- The Jacobian is the sensitivity matrix $A_{nk} = \partial E_n/\partial c_k = \int |\psi_n|^2 \phi_k \, \mathrm{d}x$ (Eqs. (24.8), (24.9)); discretely $A_{nk} = \sum_j w_{jn}^2 \phi_k(x_j)$ (Eq. (24.11)).
- Gauss–Newton solves $\hat{A}^\top \hat{A}\, \delta c = \hat{A}^\top \hat{r}$ (Eq. (24.12)) and iterates; the gradient is $-\hat{A}^\top \hat{r}$ (Eq. (24.13)).
- Levenberg–Marquardt: $(\hat{A}^\top \hat{A} + \lambda' I)\delta c = \hat{A}^\top \hat{r}$ (Eq. (24.15)), interpolating between Gauss–Newton and steepest descent, always defined and downhill.
- SVD form: $\delta c = \boldsymbol{V}\, \mathrm{diag}(\sigma_i/(\sigma_i^2 + \lambda\sigma_1^2))\boldsymbol{U}^\top \hat{r}$ (Eqs. (24.17), (24.18)); filter factors capped at $1/(2\sqrt{\lambda}\sigma_1)$; ridge variant Eq. (24.19).
- The paper's schedule (Table 24.1, our reading): try-list $(0.2\lambda, \lambda, 5\lambda, 25\lambda)$, accept only 0.1% improvements, relax or multiply by 30, stop at $\lambda > 10^4$.
- Bases: cosine, Chebyshev $T_k(2|x|/L - 1)$ (Eq. (24.20)), hats (Eq. (24.21)).
- Coverage law (heuristic): $M\pi/L > 2\sqrt{E_{\max}}$ (Eq. (24.23)); for nested bases the best in-sample error cannot rise with M.

24.11 Problems

1. • A fitted level has residual $r_{12} = -0.3$. Is E_{12} above or below its target? To first order, which sign of c_0 (with $\phi_0 = 1$) would reduce this residual?

2. • Three scaled residuals are $(0.1, -0.2, 0.2)$. Compute the rms and Φ.

3. • Explain why $A_{n0} = 1$ for every level when $\phi_0 = 1$, and show that the best (unweighted) constant shift is the mean residual.

4. •• In the box of RE3, take a single basis function $\phi(x) = x$ (an odd tilt). Show that $A_n = \int |\psi_n|^2 x \, \mathrm{d}x = 0$ for every n. What does this imply about fitting with odd basis functions to first order?

5. • Singular values $\sigma = (4, 2, 1)$, damping $\lambda = 0.25$. Compute the filter factors and the cap $1/(2\sqrt{\lambda}\sigma_1)$.

6. •• Prove that the length of the damped step Eq. (24.18) decreases as λ increases, for fixed $\hat{A}$ and $\hat{r}$.

7. •• A single level depends on one coefficient as $E(c) = c^2$, and the target is 2. Starting from $c = 1$, take two Gauss–Newton steps. Compare with $\sqrt{2}$.

8. • • • For the discrete box (Dirichlet walls, N interior points, $V = 0$), the normalized eigenvectors are $w_{jn} = \sqrt{2/(N+1)}\sin(n\pi j/(N+1))$. Using $\sum_{j=1}^{N}\cos(2\pi mj/(N+1)) = -1$ for m not a multiple of $N+1$, prove that Eq. (24.11) gives exactly $A_{n0} = 1$ and $A_{nk} = \frac{1}{2}(-1)^{n+1}\delta_{nk}$ for $1 \le n, k \le N$ with $n + k < N + 1$.

9. •• In the two-level model, the target for the lower level is 1.5. Show that no c reaches it, find the best-fitting c, and describe what the paper's schedule does if started at $c = 0.5$.

10. •• Table I gives in-sample rms 0.226, 0.137, 0.064 for $M = 200$, 400, 600 (scale 1, zeros 1–400). Explain why the *best possible* in-sample rms cannot increase with M for the cosine basis, and why the reported optimizer values still might, in principle.

11. •• Apply the coverage law at scale 2 ($L = 59.957$) to the fits of Table I, which are trained on zeros 1–800. Take the top training level to be near the smooth zero s_{800}, the solution of $N_{\mathrm{sm}}(s_{800}) = 799.5$. Find s_{800} numerically, then the number of cosines the law requires, and the energy and approximate level index that $M = 900$ reaches. Compare with the in-sample rms 0.103 at $M = 900$ (Table I).

12. •• Evaluate $T_4(y)$ from the recurrence and check $T_4(\frac{1}{2}) = \cos(4\pi/3)$.

13. • With $M = 3$ hat nodes $0, 0.5, 1$ in $u = |x|/L$, evaluate all ϕ_j at $u = 0.3$.

14. •• Show that if every basis function is replaced by $\alpha\phi_k$, the paper's step Eq. (24.18) gives the same $\delta V(x)$.

15. •• The Table I caption says that for the scale-2 $M = 300$ row, under the strict protocol "the optimizer accepts no step at all". Trace the schedule of Table 24.1 to explain how the fit ends, and what final c and in-sample rms it reports.

16. • • • Using the argument of Section 24.4, show that if $\nabla\Phi \neq 0$ at the current c, the schedule cannot stop by the rule $\lambda > 10^4$ unless the decrease available at $\lambda = 10^4$ is smaller than 0.1%. Discuss what this says about the meaning of "converged".

17. •• Using numpy, implement the `lm_fit` listing and run it on the two-level model with target 0.5, starting from $c = 0.5$ and $\lambda = 1$. Report c after the first iteration and at the end.

18. •• Using qmnt, build the Wu–Sprung potential on `qmnt.grid(44.883, 3000)`, compute the lowest 50 eigenvectors and A_{nk} for $k < 500$. For $n = 50$, find the largest k with $|A_{nk}| > 10^{-3}$ and compare with $2L\sqrt{E_{50} - V_0}/\pi$.

19. •• Using numpy, evaluate $f(\sigma) = \sigma/(\sigma^2 + \lambda\sigma_1^2)$ on a fine grid for $\sigma_1 = 1$, $\lambda = 0.01$, and confirm its maximum.

20. •• In Table III the scale-1 odd and even splits with $M = 400$ reach in-sample rms $< 10^{-3}$, while with $M = 100$ they reach about 0.22. Explain both using Section 24.5.

21. • • • The paper says Chebyshev polynomials $T_k(2|x|/L - 1)$ have "resolution concentrated near $x = 0$ and $x = L$". Find the zeros of $T_M(y)$ and the corresponding $|x|$, and show their spacing near $|x| = 0$ and $|x| = L$ is smaller than near $|x| = L/2$ by a factor of order $1/M$.

22. •• Why does the paper repeat fits from random initial c? Using Figure 24.2, describe a case where two starts give the same in-sample error but different coefficients.

23. • • • Compare the literal reading of the ridge variant, Eq. (24.19), with a Tikhonov penalty, i.e. minimizing $\Phi(c) + \frac{1}{2}\mu\sigma_1^2 \|c\|^2$. Show that the stationary points of the variant are those of Φ, while Tikhonov's satisfy $\hat{A}^\top \hat{r} = \mu\sigma_1^2 c$.

24. • • • Using numpy and scipy, build the box of Figure 24.6 and show numerically that with M cosines ($k = 0, \dots, M-1$) the first-order model can produce any shifts of levels $1, \dots, M$ exactly, but not arbitrary shifts of levels $1, \dots, M+1$ (hint: compute the rank of the sensitivity matrix). Explain why levels M and $M+1$ are reached only through c_0 at first order, and connect the result with Figure 24.6.

24.12 Answers to selected problems

1. $r = \tilde{t} - E < 0$, so E_{12} is above its target by 0.3. Since $\partial E/\partial c_0 = 1$, a negative c_0 lowers the level and reduces $|r|$.

2. rms $= \sqrt{(0.01 + 0.04 + 0.04)/3} = \sqrt{0.03} = 0.1732$; $\Phi = \frac{1}{2}(0.09) = 0.045$.

3. $A_{n0} = \int |\psi_n|^2 \mathrm{d}x = 1$ by normalization. With one knob the normal equation is $(\sum_n 1)c_0 = \sum_n r_n$, so $c_0 = \frac{1}{n_{\mathrm{fit}}} \sum_n r_n$.

5. $\lambda\sigma_1^2 = 4$: $f = (4/20,\ 2/8,\ 1/5) = (0.2,\ 0.25,\ 0.2)$. Cap $1/(2 \cdot 0.5 \cdot 4) = 0.25$, reached at $\sigma = \sqrt{\lambda}\sigma_1 = 2$.

7. $J = 2c$. From $c = 1$: $\delta c = (2-1)/2 = 0.5$, $c = 1.5$. Then $\delta c = (2 - 2.25)/3 = -0.08333$, $c = 1.41667$, against $\sqrt{2} = 1.41421$.

9. $E_- = 2 - \sqrt{1 + c^2} \le 1$, with equality only at $c = 0$, so 1.5 is out of reach and the best fit is $c = 0$ with $|r| = 0.5$. From $c = 0.5$ every trial moving toward 0 lowers $|r|$, so the schedule walks toward $c = 0$; there $J = 0$ and the gradient vanishes, no trial passes the 0.1% test, λ grows by 30 per iteration until it exceeds 10^4, and the fit stops near $c = 0$.

11. Solving $N_{\mathrm{sm}}(t) = 799.5$ by bisection or Brent's method (Chapter 4) gives $s_{800} \approx 1184$. Then $M > 2\sqrt{1184} \times 59.957/\pi = 68.8 \times 19.085 \approx 1313$. The reach of $M = 900$ is $E = (900\pi/2L)^2 = (23.58)^2 \approx 556$, and $N_{\mathrm{sm}}(556) \approx 309$, so only about the lowest 309 of the 800 training levels are covered. Although $900 > 800$ coefficients, the coverage condition fails, which is consistent with the in-sample rms remaining at 0.103 rather than near zero.

13. $M - 1 = 2$: $\phi_0 = 1 - 2(0.3) = 0.4$, $\phi_1 = 1 - 2(0.2) = 0.6$, $\phi_2 = 0$; sum 1.

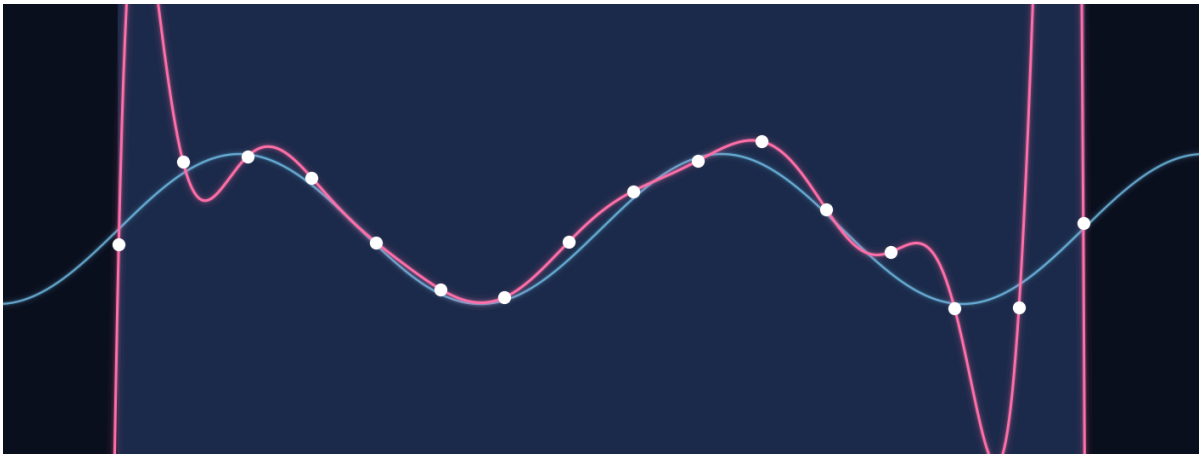
15. In our reading of the schedule (the starting λ and the trial order are our choices), in every iteration all four trials fail the 0.999 test, so c stays 0 and $\lambda \to 30\lambda$; this repeats until $\lambda > 10^4$ and the loop stops. The final $c = 0$ is the unperturbed potential and the reported in-sample rms is the null's. (The paper notes this configuration "remains at the null" under the strict protocol.)

17. After one iteration $c = 1.21175$ (Sample Problem "One pass of the schedule on the two-level model"). The loop then continues with Gauss–Newton-like steps and ends near $c_* = 1.118034$, where further improvements of 0.1% become impossible because the residual is essentially zero.

19. The maximum is at $\sigma = \sqrt{0.01} = 0.1$, where $f = 0.1/(0.01 + 0.01) = 5 = 1/(2\sqrt{\lambda}\sigma_1)$.

21. $T_M(\cos\theta) = \cos M\theta = 0$ at $\theta_j = (j - \frac{1}{2})\pi/M$, so $y_j = \cos\theta_j$ and $|x|_j = \frac{L}{2}(1 + y_j)$. Spacing $\Delta y \approx \sin\theta_j\, \pi/M$: about π/M at $\theta = \pi/2$ ($|x| = L/2$), but about $\theta_1\pi/M \approx \pi^2/(2M^2)$ near $\theta = 0$ or π ($|x| = L$ or 0). The ratio is of order $\pi/(2M)$.

23. Under the literal reading, the variant's step is zero if and only if every $f_i\, u_i^\top \hat{r} = 0$, i.e. $\boldsymbol{U}^\top \hat{r} = 0$, i.e. $\hat{A}^\top \hat{r} = \boldsymbol{V}\boldsymbol{\Sigma}\boldsymbol{U}^\top \hat{r} = 0$: the stationary points of Φ. For Tikhonov, $\nabla(\Phi + \frac{1}{2}\mu\sigma_1^2 \|c\|^2) = -\hat{A}^\top \hat{r} + \mu\sigma_1^2 c = 0$, which moves the stationary point toward $c = 0$. The variant regularizes only through shorter steps, the accept rule and the iteration cap.



Memorization: a degree-15 polynomial passes through every training point (shaded window) and goes wild the moment it leaves them.

Chapter 25

Memorization and Generalization

↑Contents

What this chapter is about

Picture two students preparing for a physics final. The first collects every past exam paper and memorizes every answer, word for word. The second works through the same papers but tries to understand why each answer is what it is. On the past papers both score 100%. You cannot tell them apart by testing them on questions they have already seen. You can tell them apart instantly by giving them a question neither has seen.

This chapter turns that observation into a working method. You will learn what it means for a fitted model to "learn" something, how to split data into a part you fit and a part you hold back, why every claim of skill must be measured against a reference prediction that knows nothing (the *null*), why a model with enough adjustable parameters can reproduce any data set and therefore proves nothing by doing so, and how the arrangement of training and test points (above, below, or interleaved) changes what a test measures. We will also derive two results that are less obvious: a smooth interpolation between neighbors can be *anticorrelated* with the truth when consecutive values are negatively correlated, and the best linear predictor built from the correlations (simple kriging) flips that sign.

Where this shows up in the paper: the paper's central negative result is precisely the distinction between memorizing and predicting. Its out-of-sample protocol (Sec. II C), the within-height splits (Sec. II C and Table III), the twelve Hermitian configurations of Table I, the memorization curves of Fig. 3, the interpolation-gap damage of Fig. 4, and the heuristic explanation and falsifiability standard of Sec. VI all rest on the ideas of this chapter [1].

What you need

- Chapter 7: vectors, matrices, rank, solving $Ax = b$, overdetermined and underdetermined systems.
- Chapter 9: linear least squares, the normal equations, the singular value decomposition (SVD), minimum-norm solutions of underdetermined systems, ridge regularization with filter factors $\sigma/(\sigma^2 + \lambda\sigma_1^2)$, and the memorizing polynomial (running example RE4).
- Chapter 19: mean, variance, standard deviation, covariance, Pearson correlation and its range $[-1, 1]$, rms, autocorrelation of a sequence, random seeds.
- Chapter 24: the fit of a potential correction $\delta V(x) = \sum_k c_k \phi_k(x)$ to a list of levels, the sensitivity matrix A_{nk}, the damped least-squares step, and the coverage law $M\pi/L > 2\sqrt{E}$.
- Used with short reminders at the point of use: the smooth zeros s_n and the local mean spacing gap_n (Chapter 17); the Wigner surmise for unfolded spacings (Chapter 20); classical turning points and where a level feels the potential (Chapter 12); first-order perturbation theory (Chapter 13); the bias-corrected targets (Chapter 23); the truncated explicit formula (Chapter 18); almost-periodic sums and phase randomization (Chapters 5 and 6); bootstrap and surrogate controls (Chapter 26, forward references only).
- First-year physics: graphs, averages, and the idea of a measurement uncertainty.

25.1 What does a model learn?

★ Did you know? *The Horse Who Read Faces*

In the early 1900s a Berlin horse called Clever Hans seemed to do arithmetic, tapping out answers with his hoof. In 1907 the psychologist Oskar Pfungst showed that Hans was reading tiny, unconscious cues in the questioner's posture and face, and that he failed when the questioner did not know the answer. A model can score perfectly while learning something other than what you intended.

25.1.1 Data, models and parameters

Every problem in this chapter has the same ingredients.

- **Data.** A list of pairs (x_i, y_i), $i = 1, \dots, n$. The x_i are *inputs* (a time, a position, an index) and the y_i are *targets* (the measured values we want to reproduce).
- **A model.** A family of functions $f(x; \boldsymbol{c})$, one function for each choice of the *parameter vector* $\boldsymbol{c} = (c_0, c_1, \dots, c_{M-1})$. The integer M is the number of parameters.
- **Fitting** (also called *training*). A rule that chooses $\boldsymbol{c}$ from the data, usually by making the errors $f(x_i; \boldsymbol{c}) - y_i$ small in the least-squares sense of Chapter 9.

Here are four models we will meet again.

1. *A straight line*: $f(x; \boldsymbol{c}) = c_0 + c_1 x$, with $M = 2$.
2. *A polynomial of degree* $M - 1$: $f(x; \boldsymbol{c}) = c_0 + c_1 x + \cdots + c_{M-1} x^{M-1}$.
3. *A fitted potential* (Chapter 24). The input is a level index n. The parameters are the coefficients c_k of a correction $\delta V(x) = \sum_{k=0}^{M-1} c_k \cos(k\pi x/L)$ added to the Wu–Sprung potential $V(x)$ in a box $[-L, L]$. The output $f(n; \boldsymbol{c}) = E_n(\boldsymbol{c})$ is the n-th eigenvalue of the Hamiltonian $H = -d^2/dx^2 + V(x) + \delta V(x)$ (units $\hbar = 2m = 1$). The targets are the heights t_n of the zeta zeros (in practice the bias-corrected targets $\tilde{t}_n$ of Chapter 23).
4. *The truncated explicit formula* (Chapter 18). The output is the n-th root of $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$, built from a list of primes. It has *no* fitted parameters at all: $M = 0$.

25.1.2 Two meanings of "learning"

When we say a fitted model has "learned" from data, we might mean either of two very different things.

Storing. The model's parameters hold a copy of the data. A lookup table that lists the first 400 zeros reproduces them perfectly, yet nobody would say it has learned anything about where zero 401 is.

Capturing a rule. The model has picked up the mechanism (or a good stand-in for it) that generates the data, so it can tell you about data it was never shown.

Fitting quality alone cannot separate these two. Both the lookup table and a correct physical law reproduce the training data. What separates them is the question the second student can answer and the first cannot: *what happens at a point you did not fit?* This leads to the operational definition used throughout this book and in the paper.

> **Operational definition.** A fitted model has *captured structure* (rather than stored data) to the extent that it predicts targets it was not shown better than a reference prediction that uses no information about them.

The paper states it this way: a model fitted to zeros $1, \ldots, n_{\text{fit}}$ "has captured structure, rather than data, only to the extent that it predicts zeros it was not shown," and "capture" is "predictive transfer to held-out zeros, measured by rms error and correlation relative to the model's own smooth baseline" (Sec. I) [1]. The words "rather than data" are the memorizing student; the words "relative to the model's own smooth baseline" are the reference prediction, which we will call the *null*.

Checkpoint 25.1

A friend fits a model with 500 parameters to 500 measured numbers and reports that every number is reproduced to six decimal places. Has the friend shown that the model has captured structure?

Answer: No. With as many parameters as data points, an exact reproduction can be a disguised lookup table. Only a prediction of numbers that were not used in the fit, compared with a null, can show captured structure.

Key Ideas

- A model is a family $f(x; \boldsymbol{c})$; fitting chooses $\boldsymbol{c}$ from data.
- Reproducing the fitted data cannot distinguish storing the data from capturing the rule that generated it.
- Operationally, a model has captured structure only to the extent that it predicts unseen data better than a null.

25.2 Training sets, test sets and two kinds of error

25.2.1 The split

Divide the indices $\{1, \ldots, n\}$ into two disjoint sets: a *training set* I_{tr} and a *test set* (also called the *held-out set* or *holdout*) I_{te}. Fit the model using only the pairs with $i \in I_{\text{tr}}$. Then, once and only once, evaluate the fitted model on I_{te}.

- The **in-sample error** is the rms error on the training set,

$$E_{\text{in}} = \left[\frac{1}{|I_{\text{tr}}|} \sum_{i \in I_{\text{tr}}} \left(f(x_i; \boldsymbol{c}) - y_i \right)^2 \right]^{1/2}, \tag{25.1}$$

  where $|I_{\text{tr}}|$ is the number of indices in the training set.

- The **out-of-sample (OOS) error** is the rms error on the test set,

$$E_{\text{out}} = \left[\frac{1}{|I_{\text{te}}|} \sum_{i \in I_{\text{te}}} \left(f(x_i; \boldsymbol{c}) - y_i \right)^2 \right]^{1/2}. \tag{25.2}$$

- The **generalization gap** is $E_{\text{out}} - E_{\text{in}}$.

Recall from Chapter 19 that "rms" means *root mean square*: square each error, average the squares, take the square root. An rms error has the same units as the targets.

The one rule that makes the test meaningful is this: **nothing about the test targets may influence any choice made before scoring.** That includes the coefficients $\boldsymbol{c}$, the number of parameters M, any regularization strength, the number of optimizer iterations, and the decision of which runs to report. If you look at the test error, change M, and look again, the test set has quietly become part of the training set.

Sample Problem 25.1: Train, test and null on four points

A quantity is measured at $t = 1, 2, 3, 4$ (arbitrary units) with results $y = 1.0$, 1.4, 1.2, 1.3. Fit on the first three points only and predict the fourth. Compare three models: a constant $f = c_0$, a line $f = c_0 + c_1 t$, and a quadratic $f = c_0 + c_1 t + c_2 t^2$. Treat the constant model as the null ("predict the typical value").

Step 1: Constant model. Least squares for a constant gives the mean (Chapter 9): $c_0 = (1.0 + 1.4 + 1.2)/3 = 1.2$. Residuals are -0.2, $+0.2$, 0, so

$$E_{\text{in}} = \sqrt{\frac{0.04 + 0.04 + 0}{3}} = \sqrt{0.02667} = 0.1633.$$

Prediction at $t = 4$: 1.2; error $|1.3 - 1.2| = 0.1$.

Step 2: Line. With $\bar{t} = 2$ and $\bar{y} = 1.2$, the least-squares slope is

$$c_1 = \frac{\sum (t_i - \bar{t})(y_i - \bar{y})}{\sum (t_i - \bar{t})^2} = \frac{(-1)(-0.2) + (0)(0.2) + (1)(0)}{1 + 0 + 1} = \frac{0.2}{2} = 0.1,$$

so $f(t) = 1.2 + 0.1(t - 2)$. Fitted values 1.1, 1.2, 1.3; residuals -0.1, $+0.2$, -0.1;

$$E_{\text{in}} = \sqrt{\frac{0.01 + 0.04 + 0.01}{3}} = \sqrt{0.02} = 0.1414.$$

Prediction at $t = 4$: $1.2 + 0.1(2) = 1.4$; error $|1.3 - 1.4| = 0.1$.

Step 3: Quadratic. Three parameters, three points: the quadratic passes through all of them. Write $f(t) = 1.4 + b(t - 2) + a(t - 2)^2$ (so that $f(2) = 1.4$). Then $f(1) = 1.4 - b + a = 1.0$ and $f(3) = 1.4 + b + a = 1.2$. Adding the two equations: $2.8 + 2a = 2.2$, so $a = -0.3$. Subtracting: $2b = 0.2$, so $b = 0.1$. In-sample error: $E_{\text{in}} = 0$. Prediction at $t = 4$:

$$f(4) = 1.4 + 0.1(2) - 0.3(2)^2 = 1.4 + 0.2 - 1.2 = 0.4,$$

error $|1.3 - 0.4| = 0.9$.

Step 4: Compare (Table 25.1). The quadratic has the best in-sample error (zero) and by far the worst test error.

$$E_{\text{in}}: \ 0.163,\ 0.141,\ 0; \qquad E_{\text{out}}: \ 0.1,\ 0.1,\ 0.9$$

Table 25.1: The three models of the Sample Problem "Train, test and null on four points". More parameters always lower (or keep) the in-sample error; they need not lower the out-of-sample error.

model	parameters M	E_{in}	E_{out} (point $t = 4$)
constant (null)	1	0.163	0.1
line	2	0.141	0.1
quadratic	3	0	0.9

Why does in-sample error never increase as we go from constant to line to quadratic? Because the models are *nested*: every constant is a line with $c_1 = 0$, and every line is a quadratic with $c_2 = 0$. The least-squares quadratic minimizes the error over a set that contains every line, so its minimum cannot be larger than the best line's. This argument holds for any nested family, including cosine expansions δV with growing M.

Pitfall

Reporting only the in-sample error of a flexible model. A falling in-sample error as M grows is guaranteed by nesting, whether or not the model is learning anything. It is a statement about the size of the family, not about the data.

Pitfall

Tuning on the test set. If you try $M = 200, 400, 600$ and report whichever gives the best test error, the reported number is biased downward, because the test set was used to make a choice. The cure is a third split: choose settings on a *validation* part of the training data, then score the test set once. The paper does exactly this for its fitted sinusoidal controls: "Ridge strength and K are chosen by a forward split inside the training window (first 75% fit, last 25% validate)" (Sec. II D).

Checkpoint 25.2

In the four-point Sample Problem, suppose we had fitted the quadratic to all four points with least squares (four points, three parameters). Would the resulting error at $t = 4$ be a legitimate out-of-sample error?

Answer: No. The point $t = 4$ would then be part of the training set; its residual would be an in-sample error.

Key Ideas

- Split indices into disjoint training and test sets; fit on the first, score once on the second.
- E_{in} can only fall (or stay the same) as a nested family grows; E_{out} can rise.
- No test target may influence any choice made before scoring.

25.3 The null model: every claim is relative

25.3.1 Why a reference is needed

Suppose someone tells you their model predicts held-out zeros "to 0.3 mean spacings." Is that good? You cannot say until you know how well a prediction that *knows nothing about the fluctuations* would do. If that trivial prediction already achieves 0.26, then 0.3 is worse than knowing nothing.

Definition. The *null model* for a prediction task is the best prediction available without using the information the model claims to capture. A model shows *skill* only if it beats the null on held-out data.

25.3.2 The null for the zeta zeros

Recall from Chapter 17 the smooth counting function $N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$ (paper Eq. (1)), the *smooth zeros* s_n defined by $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$, and the *local mean spacing* $\mathrm{gap}_n = ds_n/dn = 2\pi/\log(s_n/2\pi)$. The smooth zeros are where the zeros would sit if they were perfectly evenly spread according to their average density. The true zero t_n deviates from s_n; we measure that deviation in units of the local spacing,

$$d_n = \frac{t_n - s_n}{\mathrm{gap}_n}, \tag{25.3}$$

called the *fluctuation* of zero n (`qmnt.fluct`).

A predictor produces predicted heights $\hat{t}_n$, hence predicted fluctuations $\hat{d}_n = (\hat{t}_n - s_n)/\mathrm{gap}_n$. Subtracting,

$$\hat{d}_n - d_n = \frac{\hat{t}_n - s_n}{\mathrm{gap}_n} - \frac{t_n - s_n}{\mathrm{gap}_n} = \frac{\hat{t}_n - t_n}{\mathrm{gap}_n}, \tag{25.4}$$

so the error in the fluctuation equals the error in the height, in mean spacings. Over a held-out set H the paper reports

$$\mathrm{rms} = \left[\frac{1}{|H|}\sum_{n\in H}\left(\frac{\hat{t}_n - t_n}{\mathrm{gap}_n}\right)^2\right]^{1/2}, \qquad \mathrm{corr} = \text{Pearson correlation of } \{\hat{d}_n\} \text{ and } \{d_n\},\ n \in H. \tag{25.5}$$

The null predicts "no fluctuation": $\hat{t}_n = s_n$, so $\hat{d}_n = 0$ and, by Eq. (25.4), the null rms is simply the rms of the true fluctuations:

$$\mathrm{rms}_{\mathrm{null}} = \left[\frac{1}{|H|}\sum_{n\in H} d_n^2\right]^{1/2}. \tag{25.6}$$

The correlation of the null is undefined (a constant has no variance), which is natural: the null makes no claim about the pattern.

For the fitted potential, the paper's null is the unperturbed Wu–Sprung potential itself, and fluctuations are measured relative to its levels. On the principal holdout (zeros 401–500) the two nulls give the same number, 0.2631, which is why the operator and the explicit-formula comparisons can share one baseline (Sec. II C).

```
import numpy as np, qmnt
t = qmnt.zeros(500)              # t_1 ... t_500
n = np.arange(401, 501)          # the principal holdout (RE5)
d = qmnt.fluct(t[400:500], n)    # true fluctuations in mean spacings
print("null rms on 401-500:", qmnt.rms(d))
```

25.3.3 A skill ratio

A convenient summary is the ratio of a model's held-out rms to the null rms,

$$\text{ratio} = \frac{\text{rms}_{\text{model}}}{\text{rms}_{\text{null}}}, \qquad \text{skill} = 1 - \text{ratio}. \tag{25.7}$$

A ratio below 1 (positive skill) means the model beats the null; a ratio above 1 (negative skill) means it does harm.

Sample Problem 25.2: The null on RE5 and two predictions measured against it

(a) Compute the null rms on zeros 401–500 with `qmnt`. (b) The paper's Table I reports held-out rms 0.369 for the scale-1 cosine fit with $M = 400$ trained on zeros 1–400, and Table V reports 0.074 for the six-prime explicit formula on the same holdout. Compute the ratio and skill of each, using the paper's rounded null 0.263.
Step 1: Run the listing above. It evaluates Eq. (25.6) with d_n from Eq. (25.3) and prints 0.2631, matching the paper's value (Sec. II C and the key anchors of App. A).
Step 2: Fitted operator: ratio $= 0.369/0.263 = 1.403$, skill $= 1 - 1.403 = -0.403$.
Step 3: Six-prime formula: ratio $= 0.074/0.263 = 0.281$, skill $= 1 - 0.281 = 0.719$.

$\text{rms}_{\text{null}} = 0.2631$; operator: ratio 1.40 (skill -0.40); formula: ratio 0.281 (skill $+0.72$)

The fitted operator is about 40% *worse* than predicting no fluctuation at all; the formula removes most of the null error. (You can reproduce the formula's numbers yourself with `qmnt.oracle_zeros` and `qmnt.score`; Problem 30 does this for one prime.)

25.3.4 rms and correlation measure different things

Why does the paper always report *two* numbers? Because a prediction can have the right shape and the wrong size, or the right size and no shape. Let us make that precise.

Suppose that over the held-out set the true fluctuations d have mean zero and standard deviation σ, and the predicted fluctuations $\hat{d}$ have mean zero, standard deviation $a\sigma$ (so a is the *amplitude ratio*) and correlation ρ with d. Recall from Chapter 19 that $\text{Cov}(\hat{d}, d) = \rho\,(a\sigma)(\sigma)$. Because both means are zero, the average of a square is a variance and the average of a product is a covariance. The mean squared error is

$$\begin{aligned} \text{MSE} &= \overline{(\hat{d} - d)^2} \\ &= \overline{\hat{d}^{\,2}} - 2\,\overline{\hat{d}\,d} + \overline{d^2} && \text{(expand the square)} \\ &= a^2\sigma^2 - 2\rho a\sigma^2 + \sigma^2 && \text{(zero means)} \\ &= \sigma^2(1 + a^2 - 2\rho a). \end{aligned} \tag{25.8}$$

The null has $\text{MSE} = \sigma^2$, so

$$\frac{\text{rms}_{\text{model}}}{\text{rms}_{\text{null}}} = \sqrt{1 + a^2 - 2\rho a}. \tag{25.9}$$

Three consequences follow directly.

1. If $\rho \le 0$, every $a > 0$ makes the ratio exceed 1: an uncorrelated (or anticorrelated) prediction with any nonzero amplitude is worse than the null.

2. For fixed ρ, the ratio is smallest when $\frac{d}{da}(1 + a^2 - 2\rho a) = 2a - 2\rho = 0$, that is $a = \rho$. The minimum value is $\sqrt{1 + \rho^2 - 2\rho^2} = \sqrt{1 - \rho^2}$. No rescaling of a prediction can beat $\sqrt{1 - \rho^2}$.

3. A prediction with positive correlation can still be worse than the null if its amplitude is too large: the ratio exceeds 1 when $a^2 - 2\rho a > 0$, i.e. when $a > 2\rho$.

The assumption of zero means is an idealization; a constant offset between $\hat{d}$ and d adds its square to the MSE and makes things worse still.

Sample Problem 25.3: Uncorrelated output always adds error

The paper reports that the fitted Hermitian models "emit fluctuations of 0.16–0.62× the true amplitude that are uncorrelated with truth" (Sec. III). Using Eq. (25.9) with null 0.263, find the held-out rms of a prediction with (a) $a = 0.62$, $\rho = 0$; (b) $a = 0.62$, $\rho = -0.07$; (c) $a = 0.16$, $\rho = 0$.
Step 1: (a) $\sqrt{1 + 0.62^2} = \sqrt{1.3844} = 1.1766$; rms $= 1.1766 \times 0.263 = 0.309$.
Step 2: (b) $\sqrt{1 + 0.3844 - 2(-0.07)(0.62)} = \sqrt{1.3844 + 0.0868} = \sqrt{1.4712} = 1.2129$; rms $= 1.2129 \times 0.263 = 0.319$.
Step 3: (c) $\sqrt{1 + 0.0256} = 1.0127$; rms $= 0.266$.

$$\boxed{\text{(a) } 0.309, \quad \text{(b) } 0.319, \quad \text{(c) } 0.266}$$

All three are worse than the null 0.263. This is an idealized calculation, not a reconstruction of any table row: real held-out errors (for example 0.369 in Table I) also include offsets and amplitude variations that Eq. (25.9) ignores. The lesson is general: output that is uncorrelated with the truth can only add error.

Checkpoint 25.3

A prediction has correlation $\rho = +0.5$ with the truth but twice the true amplitude ($a = 2$). Is it better or worse than the null?

Answer: Worse: the ratio is $\sqrt{1 + 4 - 2(0.5)(2)} = \sqrt{3} = 1.73$. Since $a = 2 > 2\rho = 1$, positive correlation does not save it.

Key Ideas

- The null is the best prediction that uses none of the information a model claims to capture; skill means beating it on held-out data.
- For the zeros, the null is $\hat{t}_n = s_n$ (or the unperturbed potential's levels), with rms equal to the rms of the true fluctuations, 0.2631 on zeros 401–500.
- $\text{rms}_{\text{model}}/\text{rms}_{\text{null}} = \sqrt{1 + a^2 - 2\rho a}$: correlation ρ measures shape, amplitude ratio a measures size; uncorrelated output always hurts, and the best possible ratio at correlation ρ is $\sqrt{1 - \rho^2}$.

25.4 Overfitting and capacity: the memorizing polynomial

25.4.1 RE4 revisited

Recall running example RE4 from Chapter 9: a few noisy samples of $\sin t$ fitted by a polynomial. Figure 25.1 shows ten such samples on $0 \le t \le 2\pi$ fitted by polynomials of degree 1, 3 and 9. The degree-9 polynomial has ten coefficients and passes exactly through all ten points. Inside the training range it wiggles to chase the noise; just beyond it, it shoots off.

Definition. A fit *overfits* when it lowers its in-sample error by adapting to features of the training data that do not recur in new data (noise, or accidents of the particular sample), so

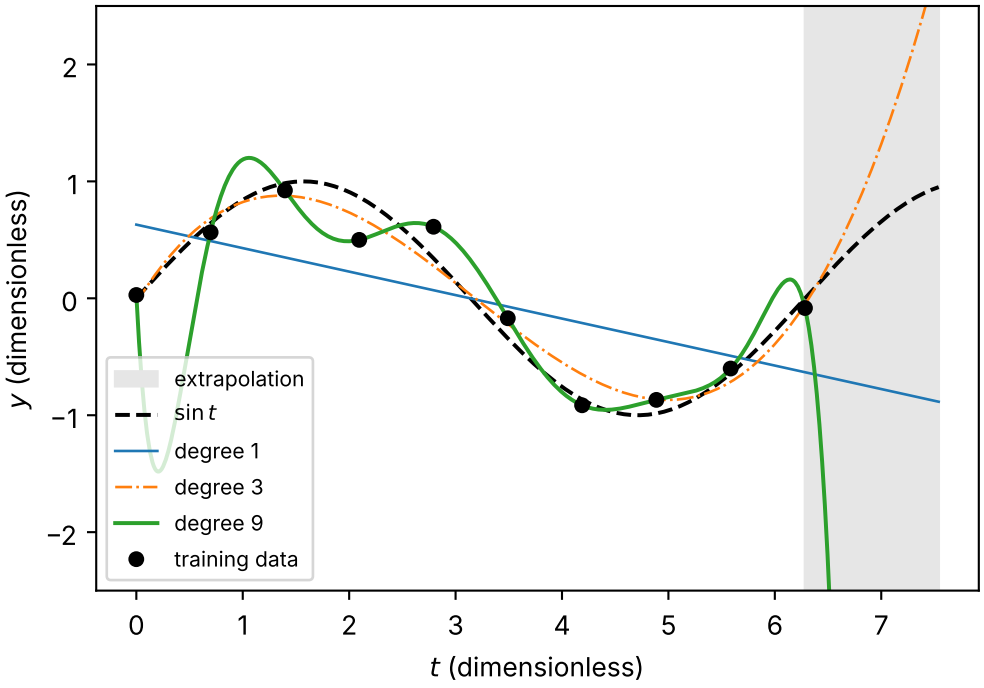


Figure 25.1: Running example RE4. Ten noisy samples (dots) of $\sin t$ (dashed) on $0 \le t \le 2\pi$, fitted by least-squares polynomials of degree 1, 3 and 9. The degree-9 polynomial has as many coefficients as points and passes through every one of them; in the shaded extrapolation region $t > 2\pi$ it departs rapidly from the truth. Horizontal axis: t (dimensionless); vertical axis: y (dimensionless).

that its out-of-sample error rises.

Figure 25.2 shows the error curves for a larger version of RE4 (15 training points). The in-sample error falls with degree, as nesting guarantees. The out-of-sample error behaves differently for test points inside the training range and for test points beyond it. Inside the range (squares), the test error is large at degrees 0 to 2, which cannot follow a full period of the sine. It reaches its minimum at a moderate degree (between 3 and 12 for the data of the listing below) and is larger again at degree 14, where the fit chases the noise. Beyond the range (triangles), the error grows rapidly with degree: at degree 14 it is more than ten times its value at degree 3.

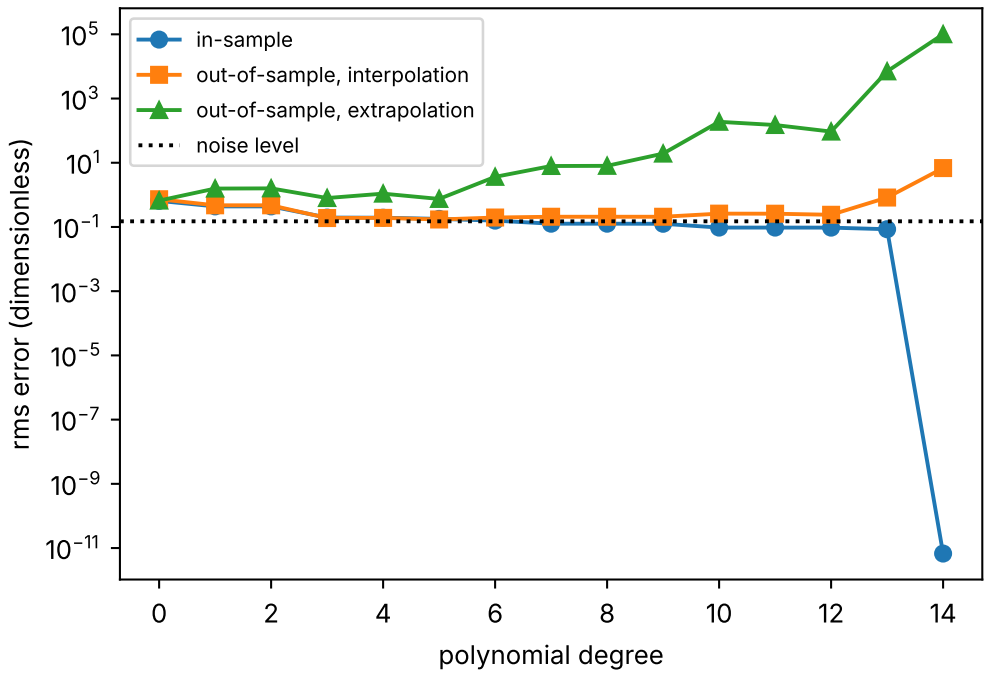


Figure 25.2: Error versus polynomial degree for RE4 with 15 noisy training points on $0 \le t \le 2\pi$ (noise standard deviation 0.15). In-sample rms (circles) never increases with degree because the models are nested, and falls to a negligible value (below 10^{-6}) at degree 14, where the fit interpolates. Out-of-sample rms is shown for fresh noisy points inside the training range (squares, interpolation) and for points with $2\pi < t \le 2.4\pi$ (triangles, extrapolation). Dotted line: the noise level. Vertical axis logarithmic; all quantities dimensionless.

```
import numpy as np
rng = np.random.default_rng(3)
t_tr = np.linspace(0, 2*np.pi, 15)
y_tr = np.sin(t_tr) + 0.15*rng.standard_normal(15)
t_in = rng.uniform(0, 2*np.pi, 200)            # interpolation test
t_out = np.linspace(2*np.pi, 2.4*np.pi, 50)    # extrapolation test
y_in = np.sin(t_in) + 0.15*rng.standard_normal(200)
y_out = np.sin(t_out) + 0.15*rng.standard_normal(50)
rms = lambda e: np.sqrt(np.mean(e**2))
for deg in range(15):
    p = np.polynomial.Polynomial.fit(t_tr, y_tr, deg)
    print(deg, rms(p(t_tr)-y_tr), rms(p(t_in)-y_in), rms(p(t_out)-y_out))
```

Sample Problem 25.4: Interpolation works, extrapolation fails

Take three noisy samples of $\sin t$: $(0,\ 0.1)$, $(\pi/2,\ 0.9)$, $(\pi,\ 0.1)$. (The true values are $0, 1, 0$; the noise is $+0.1, -0.1, +0.1$.) Fit the unique quadratic through them. Predict at $t = \pi/4$ (inside the range) and at $t = 3\pi/2$ (outside). Compare with the truth and with the null "predict zero".

Step 1: The data are symmetric about $t = \pi/2$, so try $q(t) = 0.9 - a(t - \pi/2)^2$, which already passes through the middle point. At $t = 0$: $0.9 - a\pi^2/4 = 0.1$, so $a = 0.8 \times 4/\pi^2 = 3.2/\pi^2 = 0.324$. By symmetry $q(\pi) = 0.1$ too, so the quadratic fits all three points: $E_{\text{in}} = 0$.

Step 2: Interpolation, $t = \pi/4$: $(t - \pi/2)^2 = \pi^2/16$, so $q = 0.9 - \frac{3.2}{\pi^2} \cdot \frac{\pi^2}{16} = 0.9 - 0.2 = 0.7$. True value $\sin(\pi/4) = 0.7071$. Error 0.0071.

Step 3: Extrapolation, $t = 3\pi/2$: $(t - \pi/2)^2 = \pi^2$, so $q = 0.9 - 3.2 = -2.3$. True value $\sin(3\pi/2) = -1$. Error $|-1 - (-2.3)| = 1.3$. The null "predict zero" has error $|-1 - 0| = 1$.

Step 4: For comparison, the least-squares line through the three points has slope 0 (the data are symmetric) and intercept equal to the mean $1.1/3 = 0.3667$; its error at $3\pi/2$ is 1.3667.

at $\pi/4$: error 0.0071;	at $3\pi/2$: error 1.3 (null 1)

The same fitted curve is excellent between its training points and worse than the null beyond them. Whether a fit generalizes depends on *where* you ask.

25.4.2 Capacity and saturation

How much a model family can fit is called its *capacity*. For a model that is linear in its parameters, $f(x; \boldsymbol{c}) = \sum_k c_k \phi_k(x)$, capacity has a precise meaning. Arrange the training inputs into the *design matrix* X_{tr} with entries $(X_{\text{tr}})_{ik} = \phi_k(x_i)$, $i \in I_{\text{tr}}$; then the vector of predictions on the training set is $X_{\text{tr}}\boldsymbol{c}$. Recall from Chapter 7 that the *rank* of a matrix is the number of linearly independent columns (equivalently, rows). The rank of X_{tr} is the number of independent patterns of training values the model can produce.

Definition. The model is *saturated* on a training set when the rank of X_{tr} equals the number of training points n_{tr}. A saturated model can fit *any* training targets exactly: rank n_{tr} means the columns of X_{tr} span all of $\mathbb{R}^{n_{\text{tr}}}$, so every target vector $\boldsymbol{y}_{\text{tr}}$ is a combination $X_{\text{tr}}\boldsymbol{c}$ of them. A necessary condition is $M \geq n_{\text{tr}}$.

For the fitted potential, the model is not linear in $\boldsymbol{c}$: the levels $E_n(\boldsymbol{c})$ depend on $\boldsymbol{c}$ through an eigenvalue problem. Recall from Chapters 13 and 24 that for small changes it is linear to first order, $\delta E_n \approx \sum_k A_{nk} c_k$ with $A_{nk} = \int |\psi_n|^2 \phi_k \, dx$. The sensitivity matrix A plays the role of X, and "$M \geq$ number of training levels" is the warning sign of saturation. (The paper uses the letter A also for a non-Hermitian operator in Sec. II E; here it always means the sensitivity matrix.)

25.4.3 A saturated model's predictions are not constrained by its data

Here is the central mathematical fact of this chapter.

Result (proved). Let a linear model have training design matrix X_{tr} $(n_{\text{tr}} \times M)$ and test design matrix X_{te} $(n_{\text{te}} \times M)$. Suppose the stacked matrix $X = \begin{pmatrix} X_{\text{tr}} \\ X_{\text{te}} \end{pmatrix}$, of size $(n_{\text{tr}} + n_{\text{te}}) \times M$, has rank $n_{\text{tr}} + n_{\text{te}}$. Then for *any* training targets $\boldsymbol{y}_{\text{tr}}$ and *any* vector $\boldsymbol{z}$ of desired test predictions, there is a parameter vector $\boldsymbol{c}$ with $X_{\text{tr}}\boldsymbol{c} = \boldsymbol{y}_{\text{tr}}$ and $X_{\text{te}}\boldsymbol{c} = \boldsymbol{z}$.

Proof. The rank of X equals its number of rows, so its columns span $\mathbb{R}^{n_{\text{tr}}+n_{\text{te}}}$ (Chapter 7). Hence the stacked vector $\begin{pmatrix} \boldsymbol{y}_{\text{tr}} \\ \boldsymbol{z} \end{pmatrix}$ is a combination of the columns: there is $\boldsymbol{c}$ with $X\boldsymbol{c} = \begin{pmatrix} \boldsymbol{y}_{\text{tr}} \\ \boldsymbol{z} \end{pmatrix}$. Reading off the top n_{tr} rows gives $X_{\text{tr}}\boldsymbol{c} = \boldsymbol{y}_{\text{tr}}$; the bottom n_{te} rows give $X_{\text{te}}\boldsymbol{c} = \boldsymbol{z}$. □

In words: among the parameter vectors that fit the training data *perfectly*, there is one that predicts any held-out values you like. The training data cannot select among them. Whatever the model ends up predicting on the test set is decided by something other than the data: the optimizer's starting point, its step rule, a regularizer, the choice of basis. That "something other" is called the model's *inductive bias*, and we return to it in Sec. 25.9.

Sample Problem 25.5: Infinitely many perfect fits, one chosen by the minimum norm

Fit $f(t) = c_0 + c_1 t + c_2 t^2$ to two points, $f(0) = 1$ and $f(1) = 2$, and predict at $t = 2$. (a) Describe all exact fits and their predictions. (b) Find the minimum-norm exact fit (Chapter 9) and its prediction.

Step 1: The conditions are $c_0 = 1$ and $c_0 + c_1 + c_2 = 2$, so $c_1 + c_2 = 1$, i.e. $c_1 = 1 - c_2$ with c_2 free.

Step 2: Prediction at $t = 2$: $f(2) = c_0 + 2c_1 + 4c_2 = 1 + 2(1 - c_2) + 4c_2 = 3 + 2c_2$. As c_2 ranges over all real numbers, so does the prediction. The two data points say nothing about $f(2)$.

Step 3: Minimum norm. With $c_0 = 1$ fixed, minimize $c_0^2 + c_1^2 + c_2^2 = 1 + (1 - c_2)^2 + c_2^2$. The derivative with respect to c_2 is $-2(1 - c_2) + 2c_2 = 4c_2 - 2$, which vanishes at $c_2 = \frac{1}{2}$; then $c_1 = \frac{1}{2}$. So $\boldsymbol{c} = (1, \frac{1}{2}, \frac{1}{2})$.

Step 4: Check with the formula of Chapter 9, $\boldsymbol{c} = X_{\text{tr}}^\top (X_{\text{tr}} X_{\text{tr}}^\top)^{-1} \boldsymbol{y}_{\text{tr}}$. Here $X_{\text{tr}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$, $X_{\text{tr}} X_{\text{tr}}^\top = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$, whose inverse is $\frac{1}{2}\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$. Then $(X_{\text{tr}} X_{\text{tr}}^\top)^{-1} \boldsymbol{y}_{\text{tr}} = \frac{1}{2}\begin{pmatrix} 3 - 2 \\ -1 + 2 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$ and $X_{\text{tr}}^\top \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 1 \\ 0.5 \\ 0.5 \end{pmatrix}$, as found.

Step 5: Prediction: $f(2) = 1 + 2(0.5) + 4(0.5) = 4$.

$$\boxed{f(2) = 3 + 2c_2 \text{ for any } c_2; \qquad \text{minimum norm: } \boldsymbol{c} = (1, \tfrac{1}{2}, \tfrac{1}{2}),\ f(2) = 4}$$

A straight line through the two points would predict 3. The minimum-norm rule predicts 4, not because the data demand it, but because "small coefficients" is the rule we chose.

Checkpoint 25.4

A cosine model with $M = 400$ coefficients is fitted to 200 training levels. Is it necessarily saturated? What extra condition is needed?

Answer: Not necessarily: $M \geq n_{\text{tr}}$ is necessary but not sufficient. Saturation requires the (linearized) training sensitivity matrix to have rank 200, i.e. 200 independent level patterns among its columns.

Key Ideas

- Overfitting: lowering in-sample error by fitting features that do not recur, at the cost of out-of-sample error.
- Capacity of a linear model is the rank of its design matrix; it is saturated when that rank equals the number of training points, and then fits any targets exactly.

- If training and test rows together have full row rank, exact training fits exist with *any* test predictions; the data cannot choose, the inductive bias does.

25.5 Interpolation, extrapolation and covariate shift

★ Did you know? *Two Million Wrong Answers*

In 1936 the *Literary Digest* polled about 2.4 million people and predicted that Alf Landon would defeat Franklin Roosevelt. Roosevelt won in a landslide. The mailing lists, drawn largely from telephone directories and car registrations, did not represent all voters. George Gallup, with a far smaller sample, called the winner. A model built on one population can fail badly on another.

25.5.1 Where the test points lie

The RE4 Sample Problem showed that the same fit can be good inside its training range and bad outside. We name the two situations.

- **Interpolation**: test inputs lie among the training inputs (between training points).
- **Extrapolation**: test inputs lie outside the range of the training inputs.

A related idea concerns the *statistics* of the test data rather than its location.

Definition. *Covariate shift* occurs when the test inputs come from a different region (or distribution) than the training inputs, so that the targets there may behave differently (different typical size, different noise level) even if the underlying rule is the same.

For the zeros, a split by index "train on $1, \ldots, 400$, test on $401, \ldots, 500$" is extrapolation in height, and it involves a mild covariate shift: the typical size of the fluctuations d_n grows slowly with height. The paper notes "the null grows from 0.263 to 0.296 over Table V" (Sec. II C). You can watch this happen:

```
import numpy as np, qmnt
t = qmnt.zeros(10000)
for a, b in [(401, 500), (501, 1000), (1001, 2000), (2001, 5000), (5001, 10000)]:
    n = np.arange(a, b + 1)
    print(a, b, round(qmnt.rms(qmnt.fluct(t[a-1:b], n)), 3))
```

This prints the null column of the paper's Table V: 0.263, 0.270, 0.279, 0.288, 0.296. A model that fails a forward test could, in principle, be failing because the test region is "harder" (larger fluctuations), not because it captured nothing. To rule this out you need a test that has no covariate shift.

25.5.2 Four ways to split

Figure 25.3 and Table 25.2 show the split types used in the paper.

Sample Problem 25.6: Counting levels and checking saturation in the within-height splits

At scale 1 the paper's within-height splits are drawn inside zeros 1–400 (Sec. II C). (a) How many levels are trained and held out in a random 80/20 split? (b) In an odd/even split? (c) For which of the basis sizes $M = 100, 200, 400$ is the odd/even fit at or above saturation?

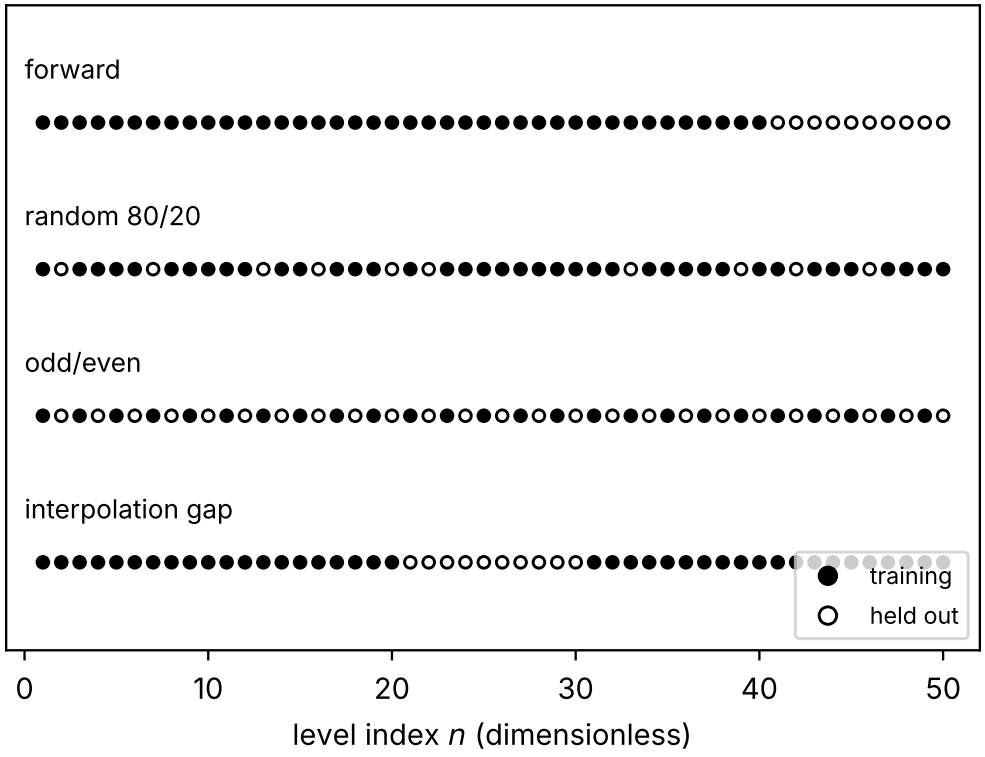


Figure 25.3: Schematic of four train/test splits on 50 level indices. Filled circles: training levels; open circles: held-out levels. Forward split: hold out the highest indices (extrapolation in height, with covariate shift). Random 80/20: hold out a random fifth (interpolation within one height range). Odd/even: train on odd indices, score on even (interpolation; every held-out level has training neighbors on both sides). Interpolation gap: hold out a contiguous interior block. The horizontal axis is the level index n (dimensionless).

Table 25.2: Split types and what each tests. "Paper" gives where the split is used in the paper [1].

split	held-out levels	tests	paper
forward	above training range	extrapolation in height	Tables I, II, IV, V
random 80/20	random fifth, same range	interpolation, no height shift	Table III, Sec. II C
odd/even	alternate levels	interpolation from both neighbors	Table III
interpolation gap	interior block	interpolation over a gap	Sec. III, Fig. 4

Step 1: Random 80/20 of 400: held out $400/5 = 80$ (the paper draws them with `rng.choice(n, n/5, replace=False)`, App. A step 17); trained $400 - 80 = 320$.
Step 2: Odd/even: the odd indices $1, 3, \ldots, 399$ are 200 levels; the even indices $2, 4, \ldots, 400$ are another 200. Each direction trains on 200 and scores 200.
Step 3: Saturation requires at least as many coefficients as training levels: $M \geq 200$. So $M = 200$ and $M = 400$ are at or above the threshold; $M = 100$ is below it.

(a) 320/80; (b) 200/200; (c) $M = 200, 400$

This is why the paper calls the odd/even rows with $M \geq 200$ "saturated" and "stress tests of memorization," and adds an $M = 100$ row (Table III).

Pitfall

Each split has its own null. Because the held-out levels differ, the null rms differs too: in Table III it is 0.238 for the even-indexed held-out levels at scale 1 ("odd" rows, trained on odd), 0.250 for the odd-indexed ones, and 0.23–0.26 across random seeds. Always compare a model's held-out error with the null *on the same held-out set*.

Checkpoint 25.5

A model fails a forward split (trained on 1–400, scored on 401–500) but succeeds on a random 80/20 split inside 1–400. Which explanation does this pattern support: covariate shift, or failure to capture structure?

Answer: It points toward the forward failure being (at least partly) a problem of extrapolation or covariate shift, since the model does capture something usable within one height range. The reverse pattern (failure within height as well) rules covariate shift out as the explanation.

Key Ideas

- Interpolation (test among training points) and extrapolation (test beyond them) can give opposite verdicts for the same fit.
- Forward splits of the zeros involve a mild covariate shift: the null grows from 0.263 (401–500) to 0.296 (5001–10000).
- Within-height splits (random, odd/even) remove the height shift; each split has its own null.

25.6 Serial correlation: when smooth interpolation is anticorrelated with truth

25.6.1 Stationary sequences and their autocorrelation

A natural way to predict a held-out value in a sequence is to look at its neighbors. Whether that works depends on how neighbors are correlated.

Consider a sequence $d_1, d_2, \ldots$ (for us, the fluctuations of Eq. (25.3)). Treat it as random, and suppose it is *stationary*: its statistics do not depend on the position n. Concretely we assume

$$\langle d_n \rangle = 0, \qquad \langle d_n^2 \rangle = \sigma^2, \qquad \langle d_n d_{n+k} \rangle = \sigma^2 \rho_k, \tag{25.10}$$

where $\langle\cdot\rangle$ denotes an average (expectation) and ρ_k is the *autocorrelation at lag k* (Chapter 19); $\rho_0 = 1$ and $\rho_{-k} = \rho_k$. For the zeros, stationarity is an approximation that holds within a limited height range: the null rms drifts slowly (Sec. 25.5), and we neglect that drift.

25.6.2 The neighbor-mean predictor

The simplest smooth interpolator predicts a held-out value as the average of its two neighbors:

$$\hat{d}_n = \tfrac{1}{2}(d_{n-1} + d_{n+1}). \tag{25.11}$$

We compute its correlation with the truth and its mean squared error, using only Eq. (25.10).

Covariance with the truth:

$$\begin{aligned} \langle \hat{d}_n d_n \rangle &= \tfrac{1}{2}\langle d_{n-1} d_n \rangle + \tfrac{1}{2}\langle d_{n+1} d_n \rangle && \text{(linearity of the average)} \\ &= \tfrac{1}{2}\sigma^2\rho_1 + \tfrac{1}{2}\sigma^2\rho_1 && \text{(both pairs are at lag 1)} \\ &= \sigma^2\rho_1. \end{aligned} \tag{25.12}$$

Variance of the prediction:

$$\begin{aligned} \langle \hat{d}_n^2 \rangle &= \tfrac{1}{4}\langle (d_{n-1} + d_{n+1})^2 \rangle \\ &= \tfrac{1}{4}\left(\langle d_{n-1}^2 \rangle + 2\langle d_{n-1} d_{n+1} \rangle + \langle d_{n+1}^2 \rangle\right) && \text{(expand the square)} \\ &= \tfrac{1}{4}(\sigma^2 + 2\sigma^2\rho_2 + \sigma^2) && (n-1 \text{ and } n+1 \text{ are at lag 2}) \\ &= \tfrac{1}{2}\sigma^2(1 + \rho_2). \end{aligned} \tag{25.13}$$

Correlation (covariance divided by the product of standard deviations):

$$\operatorname{corr}(\hat{d}, d) = \frac{\sigma^2\rho_1}{\sigma \cdot \sigma\sqrt{(1+\rho_2)/2}} = \rho_1\sqrt{\frac{2}{1+\rho_2}}. \tag{25.14}$$

Mean squared error:

$$\begin{aligned} \langle (\hat{d}_n - d_n)^2 \rangle &= \langle \hat{d}_n^2 \rangle - 2\langle \hat{d}_n d_n \rangle + \langle d_n^2 \rangle \\ &= \sigma^2\left[\frac{1+\rho_2}{2} - 2\rho_1 + 1\right]. \end{aligned} \tag{25.15}$$

The conclusion is immediate from Eq. (25.14): since $\sqrt{2/(1+\rho_2)} > 0$, **the neighbor mean has the sign of** ρ_1. If consecutive values are negatively correlated, interpolating between neighbors is anticorrelated with the truth, and by Eq. (25.15) its error exceeds the null's (σ^2) whenever $(1+\rho_2)/2 - 2\rho_1 > 0$, which certainly holds when $\rho_1 < 0$ (because $1 + \rho_2 \geq 0$ for any valid autocorrelation).

The same holds for any interpolator that averages nearby neighbors with positive weights, $\hat{d}_n = \sum_{k\geq 1} w_k(d_{n-k} + d_{n+k})$ with $w_k \geq 0$: by the same steps, $\langle \hat{d}_n d_n \rangle = 2\sigma^2 \sum_k w_k\rho_k$, which is negative when the weights sit mostly on lags where $\rho_k < 0$.

Sample Problem 25.7: Anticorrelation by construction

A stationary sequence has $\rho_1 = -0.3$ and $\rho_2 = -0.1$. Find the correlation of the neighbor-mean predictor with the truth and the ratio of its rms error to the null.

Step 1: Correlation from Eq. (25.14): $-0.3\sqrt{2/0.9} = -0.3 \times 1.4907 = -0.447$.

Step 2: MSE from Eq. (25.15): $\sigma^2[0.9/2 + 0.6 + 1] = 2.05\,\sigma^2$.

Step 3: rms ratio: $\sqrt{2.05} = 1.432$.

$$\boxed{\text{corr} = -0.447, \qquad \text{rms/rms}_{\text{null}} = 1.43}$$

25.6.3 Why the zeros have negative lag-1 correlation

Is $\rho_1 < 0$ plausible for the zeros? There is a simple approximate relation behind it. Define the unfolded spacing $u_n = (t_{n+1} - t_n)/\text{gap}_n$. Using $t_n = s_n + \text{gap}_n d_n$ and the approximation $s_{n+1} - s_n \approx \text{gap}_n$, $\text{gap}_{n+1} \approx \text{gap}_n$ (the mean spacing changes slowly from one level to the next; we neglect its change),

$$u_n \approx \frac{s_{n+1} - s_n}{\text{gap}_n} + d_{n+1} - d_n \approx 1 + d_{n+1} - d_n. \tag{25.16}$$

Then, using Eq. (25.10),

$$\begin{aligned} \text{Var}(u) &= \langle (d_{n+1} - d_n)^2 \rangle \\ &= \langle d_{n+1}^2 \rangle - 2\langle d_{n+1} d_n \rangle + \langle d_n^2 \rangle \\ &= 2\sigma^2(1 - \rho_1), \end{aligned} \tag{25.17}$$

and solving for ρ_1,

$$\rho_1 = 1 - \frac{\text{Var}(u)}{2\sigma^2}. \tag{25.18}$$

For a perfectly regular lattice with independent jitter of each level, d_n independent gives $\rho_1 = 0$ and $\text{Var}(u) = 2\sigma^2$, consistent with Eq. (25.18). Eq. (25.18) says when ρ_1 is negative: $\rho_1 < 0$ if and only if $\text{Var}(u) > 2\sigma^2$. The sign is therefore decided by comparing two numbers, the variance of the unfolded spacings and twice the variance of the fluctuations. For the zeros the next Sample Problem gives $\text{Var}(u) \approx 0.178$ (from the Wigner surmise, Chapter 20) against $2\sigma^2 \approx 0.138$ (from the measured null rms 0.263), so the inequality holds and $\rho_1 < 0$. In words: the positions stay close to their smooth values (small σ, often called *rigidity*), while consecutive spacings vary by more than independent jitter of that size would produce. The two facts are compatible only if neighboring zeros tend to be displaced in opposite directions (when one sits above its smooth position the next tends to sit below), which is what a negative ρ_1 means.

Sample Problem 25.8: Estimating ρ_1 for the zeros from the Wigner surmise

Estimate ρ_1 for the zero fluctuations using Eq. (25.18), taking $\text{Var}(u)$ from the GUE Wigner surmise $p(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ (paper Fig. 2; Chapter 20) and $\sigma = 0.263$, the null rms on zeros 401–500.

Step 1: Recall from Chapter 20 that the Wigner surmise $p(s)$ is an approximate probability density for the unfolded spacings: $\int_0^\infty p(s)\, ds = 1$, and its mean is 1. We check the mean. Write $b = 4/\pi$. Then $\langle s \rangle = \frac{32}{\pi^2} \int_0^\infty s^3 e^{-bs^2}\, ds$. Substitute $u = s^2$, $du = 2s\, ds$: $\int_0^\infty s^3 e^{-bs^2}\, ds = \frac{1}{2} \int_0^\infty u\, e^{-bu}\, du = \frac{1}{2} \cdot \frac{1}{b^2}$ (integration by parts). With $b^2 = 16/\pi^2$ this is $\pi^2/32$, so $\langle s \rangle = \frac{32}{\pi^2} \cdot \frac{\pi^2}{32} = 1$. Taking the spacings u to follow the surmise, $\text{Var}(u) = \langle s^2 \rangle - \langle s \rangle^2 = \langle s^2 \rangle - 1$, with $\langle s^2 \rangle = \frac{32}{\pi^2} \int_0^\infty s^4 e^{-bs^2}\, ds$.

Step 2: Start from the Gaussian integral $\int_0^\infty e^{-bs^2} ds = \frac{1}{2}\sqrt{\pi}\, b^{-1/2}$ (Chapter 19). Differentiating with respect to b under the integral sign (a step we take on trust here; it is justified in [22]) brings down a factor $-s^2$ each time. Once:

$$\int_0^\infty s^2 e^{-bs^2} ds = -\frac{d}{db}\left(\tfrac{1}{2}\sqrt{\pi}\, b^{-1/2}\right) = \tfrac{1}{4}\sqrt{\pi}\, b^{-3/2}.$$

Twice:

$$\int_0^\infty s^4 e^{-bs^2}\,ds = -\frac{d}{db}\left(\tfrac{1}{4}\sqrt{\pi}\,b^{-3/2}\right) = \tfrac{3}{8}\sqrt{\pi}\,b^{-5/2}.$$

With $b^{-5/2} = (\pi/4)^{5/2} = \pi^{5/2}/32$:

$$\langle s^2\rangle = \frac{32}{\pi^2}\cdot\frac{3}{8}\sqrt{\pi}\cdot\frac{\pi^{5/2}}{32} = \frac{3\pi^3}{8\pi^2} = \frac{3\pi}{8} = 1.1781.$$

So $\mathrm{Var}(u) = 0.1781$.
Step 3: $2\sigma^2 = 2(0.263)^2 = 0.1383$.
Step 4: $\rho_1 \approx 1 - 0.1781/0.1383 = 1 - 1.287 = -0.287$.

$$\boxed{\rho_1 \approx -0.29}$$

This is an estimate, not a measurement: the surmise is an approximation to the true spacing law, Eq. (25.16) neglects the slow change of gap_n, and stationarity is approximate. It does show that negative ρ_1 is the expected sign. How large is the measured value? The paper's measured correlations of the mean of the two nearest training fluctuations with the truth are -0.26 to -0.41 (Sec. III). Inverting Eq. (25.14) with $\rho_2 = 0$ gives $\rho_1 = \mathrm{corr}/\sqrt{2}$, between about -0.18 and -0.29; a negative ρ_2, which the paper also reports, would make $|\rho_1|$ somewhat smaller. (This inversion is approximate: in the random splits the nearest training neighbors are not always at lag 1.) So the estimate -0.29 has the right sign and the right size, at the large end of the range. You can measure ρ_1 directly with the listing below (Problem 8). The paper reports that "the fluctuation sequence has negative autocorrelation at lags 1–3" (Sec. III).

Figure 25.4 shows the real fluctuations of zeros 1–400 computed with `qmnt`, the neighbor-mean prediction for the even-indexed levels from their odd neighbors, and the sample autocorrelation. When you run the listing below, you will find that the measured lag-1 autocorrelation of zeros 1–400 is negative, that the neighbor-mean prediction for the even-indexed levels is anticorrelated with the truth, and that its rms error is above the null of those levels, as Eqs. (25.14) and (25.15) predict for $\rho_1 < 0$.

```
import numpy as np, qmnt
t = qmnt.zeros(400); n = np.arange(1, 401)
d = qmnt.fluct(t, n)
dc = d - d.mean()
for k in range(1, 6):
    print("lag", k, np.sum(dc[:-k] * dc[k:]) / np.sum(dc**2))
ev = np.arange(1, 399, 2)               # positions of n = 2, 4, ..., 398
pred = 0.5 * (d[ev - 1] + d[ev + 1])    # mean of the two odd neighbors
print("neighbor mean: rms", qmnt.rms(pred - d[ev]), "corr", qmnt.corr(pred, d[ev]))
print("null rms", qmnt.rms(d[ev]))
```

Checkpoint 25.6

If a sequence had $\rho_1 = +0.8$, would the neighbor-mean predictor be correlated or anticorrelated with the truth?

Answer: Positively correlated: by Eq. (25.14) the correlation has the sign of ρ_1.

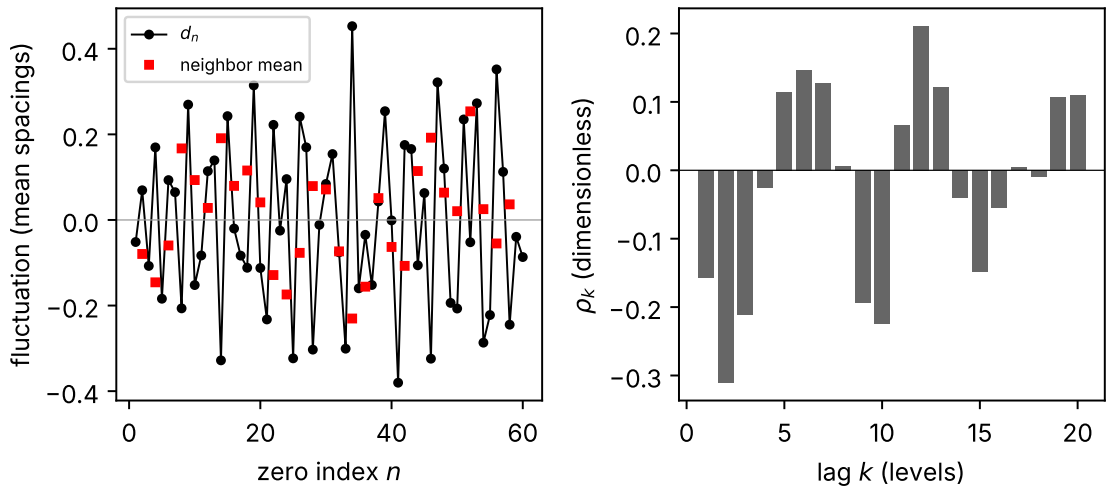


Figure 25.4: Left: fluctuations $d_n = (t_n - s_n)/\text{gap}_n$ of zeros 1–60 (black, in mean spacings) and the neighbor-mean prediction $\frac{1}{2}(d_{n-1} + d_{n+1})$ for even n (red squares). Where d_n is high its neighbors tend to be low, so the prediction tends to have the wrong sign. Right: sample autocorrelation ρ_k of d_n for zeros 1–400 against lag k (in levels); the lag-1 value is negative (the paper reports negative values at lags 1–3, Sec. III).

Key Ideas

- For a stationary sequence, the neighbor mean has correlation $\rho_1\sqrt{2/(1+\rho_2)}$ with the truth: its sign is the sign of ρ_1.
- The identity $\rho_1 = 1 - \text{Var}(u)/(2\sigma^2)$ links lag-1 correlation to spacing variance and position rigidity; for the zeros it predicts $\rho_1 < 0$ (estimate ≈ -0.29).
- Hence any interpolator that averages nearby neighboring zeros with positive weights tends to be anticorrelated with the truth; a correlation-aware predictor (Sec. 25.7) need not be.

25.7 The best linear predictor: simple kriging

25.7.1 Deriving the normal equations

If neighbor averaging has the wrong sign, what is the *right* way to use neighbors? Ask for the linear combination of training values that minimizes the expected squared error. This is called the *best linear predictor*, or *simple kriging* (a name from geostatistics, where it was developed to interpolate ore grades between boreholes; "simple" means the mean is known, here zero).

Let the value to predict be d_0 and the available training values be $d_1, \ldots, d_K$ (any positions). Write $C_{ij} = \langle d_i d_j \rangle$ for their covariances, collected in the $K \times K$ matrix C, and $\boldsymbol{c}_0$ for the vector with entries $C_{i0} = \langle d_i d_0 \rangle$. (In this section only, $\boldsymbol{c}_0$ is this covariance vector, not a coefficient of

δV.) The predictor is $\hat{d}_0 = \sum_{i=1}^{K} w_i d_i = \boldsymbol{w}^\top \boldsymbol{d}$. Its mean squared error is

$$\begin{aligned}\mathcal{E}(\boldsymbol{w}) &= \Big\langle \Big(d_0 - \sum_i w_i d_i\Big)^2 \Big\rangle \\ &= \langle d_0^2 \rangle - 2\sum_i w_i \langle d_i d_0 \rangle + \sum_i \sum_j w_i w_j \langle d_i d_j \rangle \qquad \text{(expand the square)} \\ &= C_{00} - 2\sum_i w_i C_{i0} + \sum_i \sum_j w_i w_j C_{ij}. \end{aligned} \tag{25.19}$$

Differentiate with respect to one weight w_k, holding all the other weights fixed (a partial derivative: ordinary differentiation in which the other weights are treated as constants). The middle term gives $-2C_{k0}$. In the double sum, w_k appears when $i = k$ (giving $\sum_j w_j C_{kj}$) and when $j = k$ (giving $\sum_i w_i C_{ik}$); since $C_{ik} = C_{ki}$ these are equal, totalling $2\sum_j C_{kj} w_j$. Setting the derivative to zero:

$$-2C_{k0} + 2\sum_j C_{kj} w_j = 0 \quad \Longrightarrow \quad \sum_{j=1}^{K} C_{kj} w_j = C_{k0}, \quad k = 1, \ldots, K, \tag{25.20}$$

or in matrix form

$$C\boldsymbol{w} = \boldsymbol{c}_0. \tag{25.21}$$

These are the *kriging normal equations*. They have the same structure as the least-squares normal equations of Chapter 9, with covariances in place of sums of products. To see that the solution is a minimum and not some other stationary point, change the weights from $\boldsymbol{w}$ to $\boldsymbol{w} + \boldsymbol{v}$ for an arbitrary vector $\boldsymbol{v}$. From Eq. (25.19) in matrix form, $\mathcal{E}(\boldsymbol{w}) = C_{00} - 2\boldsymbol{w}^\top \boldsymbol{c}_0 + \boldsymbol{w}^\top C \boldsymbol{w}$, so

$$\begin{aligned}\mathcal{E}(\boldsymbol{w} + \boldsymbol{v}) &= C_{00} - 2(\boldsymbol{w} + \boldsymbol{v})^\top \boldsymbol{c}_0 + (\boldsymbol{w} + \boldsymbol{v})^\top C (\boldsymbol{w} + \boldsymbol{v}) \\ &= \mathcal{E}(\boldsymbol{w}) - 2\boldsymbol{v}^\top \boldsymbol{c}_0 + 2\boldsymbol{v}^\top C \boldsymbol{w} + \boldsymbol{v}^\top C \boldsymbol{v} \qquad (C \text{ symmetric, so } \boldsymbol{w}^\top C \boldsymbol{v} = \boldsymbol{v}^\top C \boldsymbol{w}) \\ &= \mathcal{E}(\boldsymbol{w}) + 2\boldsymbol{v}^\top (C\boldsymbol{w} - \boldsymbol{c}_0) + \boldsymbol{v}^\top C \boldsymbol{v}. \end{aligned}$$

If $\boldsymbol{w}$ solves $C\boldsymbol{w} = \boldsymbol{c}_0$, the middle term vanishes, and $\boldsymbol{v}^\top C \boldsymbol{v} = \sum_i \sum_j v_i v_j \langle d_i d_j \rangle = \langle (\boldsymbol{v}^\top \boldsymbol{d})^2 \rangle \geq 0$. Hence $\mathcal{E}(\boldsymbol{w} + \boldsymbol{v}) \geq \mathcal{E}(\boldsymbol{w})$ for every $\boldsymbol{v}$: the solution is a minimum.

Minimum error. At the solution, $\boldsymbol{w}^\top C \boldsymbol{w} = \boldsymbol{w}^\top \boldsymbol{c}_0$ (multiply Eq. (25.21) by $\boldsymbol{w}^\top$), so Eq. (25.19) becomes

$$\mathcal{E}_{\min} = C_{00} - 2\boldsymbol{w}^\top \boldsymbol{c}_0 + \boldsymbol{w}^\top \boldsymbol{c}_0 = C_{00} - \boldsymbol{w}^\top \boldsymbol{c}_0. \tag{25.22}$$

Orthogonality. The normal equations say $\langle (d_0 - \hat{d}_0) d_k \rangle = C_{k0} - \sum_j w_j C_{kj} = 0$ for every k: the error of the best linear predictor is uncorrelated with every piece of data it used. If it were not, some multiple of that piece could be added to reduce the error further.

25.7.2 Two neighbors: the sign flips

Predict d_n from d_{n-1} and d_{n+1}. With Eq. (25.10), $C = \sigma^2 \begin{pmatrix} 1 & \rho_2 \\ \rho_2 & 1 \end{pmatrix}$ and $\boldsymbol{c}_0 = \sigma^2 \begin{pmatrix} \rho_1 \\ \rho_1 \end{pmatrix}$. The equations are $w_1 + \rho_2 w_2 = \rho_1$ and $\rho_2 w_1 + w_2 = \rho_1$. Subtracting the second from the first gives $(1 - \rho_2)(w_1 - w_2) = 0$, so $w_1 = w_2 \equiv w$ (provided $\rho_2 \neq 1$). Then $w(1 + \rho_2) = \rho_1$:

$$w = \frac{\rho_1}{1 + \rho_2}, \qquad \hat{d}_n = w\,(d_{n-1} + d_{n+1}). \tag{25.23}$$

From Eq. (25.22), $\mathcal{E}_{\min} = \sigma^2 - 2w\sigma^2\rho_1 = \sigma^2\Big(1 - \dfrac{2\rho_1^2}{1 + \rho_2}\Big)$, which is never larger than the null σ^2.

Now compare with the neighbor mean. The kriging predictor is $\hat{d}_n = 2w \times \frac{1}{2}(d_{n-1} + d_{n+1})$, a constant multiple of the neighbor mean. Multiplying a variable by a constant $\alpha \neq 0$ multiplies its correlation with anything by the sign of α. To see this, recall from Chapter 19 that $\mathrm{Cov}(\alpha X, Y) = \alpha \,\mathrm{Cov}(X, Y)$ and that the standard deviation of αX is $|\alpha|$ times that of X. Then

$$\mathrm{corr}(\alpha X, Y) = \frac{\alpha \,\mathrm{Cov}(X, Y)}{|\alpha| \,\mathrm{sd}(X)\,\mathrm{sd}(Y)} = \frac{\alpha}{|\alpha|}\,\mathrm{corr}(X, Y) = \mathrm{sign}(\alpha)\,\mathrm{corr}(X, Y).$$

The magnitude of the correlation is unchanged; only its sign can change. Here $\alpha = 2w$. When $\rho_1 < 0$ we have $w < 0$, so

Result (proved). When $w < 0$, the two-neighbor kriging prediction and the neighbor mean have correlations with the truth that are equal in magnitude and opposite in sign. With $\rho_1 < 0$ the neighbor mean is anticorrelated and kriging is positively correlated, $\mathrm{corr} = |\rho_1| \sqrt{2/(1+\rho_2)}$.

The best linear use of negatively correlated neighbors is to predict the *opposite* of their average, shrunk toward zero.

Sample Problem 25.9: Kriging versus the neighbor mean

Use $\rho_1 = -0.3$, $\rho_2 = -0.1$ as in the earlier Sample Problem "Anticorrelation by construction". Find the kriging weight, the error ratio and the correlation, and compare.
Step 1: $w = -0.3/(1 - 0.1) = -0.3/0.9 = -0.3333$.
Step 2: $\mathcal{E}_{\min}/\sigma^2 = 1 - 2(0.09)/0.9 = 1 - 0.2 = 0.8$; rms ratio $\sqrt{0.8} = 0.894$.
Step 3: Correlation. $\mathrm{Var}(\hat{d}) = w^2 \cdot 2\sigma^2(1+\rho_2) = \frac{1}{9} \cdot 2(0.9)\sigma^2 = 0.2\sigma^2$ and $\langle \hat{d} d \rangle = 2w\sigma^2\rho_1 = 2(-\frac{1}{3})(-0.3)\sigma^2 = 0.2\sigma^2$. So $\mathrm{corr} = 0.2\sigma^2/(\sigma\sqrt{0.2}\,\sigma) = \sqrt{0.2} = 0.447$.

$$\boxed{w = -\tfrac{1}{3}, \quad \mathrm{rms}/\mathrm{rms}_{\mathrm{null}} = 0.894, \quad \mathrm{corr} = +0.447 \text{ (neighbor mean: } -0.447,\ 1.43)}$$

Note that $\mathrm{corr}^2 = 0.2 = 1 - \mathcal{E}_{\min}/\sigma^2$, which agrees with the optimum $\sqrt{1-\rho^2}$ of Eq. (25.9): kriging automatically chooses the best amplitude.

25.7.3 The nugget

In practice the covariances are estimated from data, and the estimated C may be nearly singular or even slightly inconsistent. Geostatisticians add a *nugget*: a constant τ^2 on the diagonal,

$$(C + \tau^2 I)\,\boldsymbol{w} = \boldsymbol{c}_0. \tag{25.24}$$

There are two ways to read it. (i) If each observed training value carries independent measurement noise of variance τ^2, the covariances among the *observations* are $C_{ij} + \tau^2\delta_{ij}$ (where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise), while their covariances with the noiseless target are unchanged; Eq. (25.24) is then the exact best predictor. (ii) As a regularizer, it is the kriging cousin of ridge regression (Chapter 9): a larger nugget shrinks $\boldsymbol{w}$ toward zero and the prediction toward the null. In the two-neighbor case the weight becomes $w = \rho_1/(1 + \rho_2 + \tau^2/\sigma^2)$.

Sample Problem 25.10: The price of a small nugget

With $\rho_1 = -0.3$, $\rho_2 = -0.1$, add a nugget $\tau^2/\sigma^2 = 0.05$ as a pure regularizer (the data themselves are noiseless). Find w and the true MSE.
Step 1: $w = -0.3/(0.9 + 0.05) = -0.3158$.
Step 2: For noiseless data, Eq. (25.19) with two equal weights gives $\mathcal{E}/\sigma^2 = 1 - 4w\rho_1 + 2w^2(1+\rho_2)$.

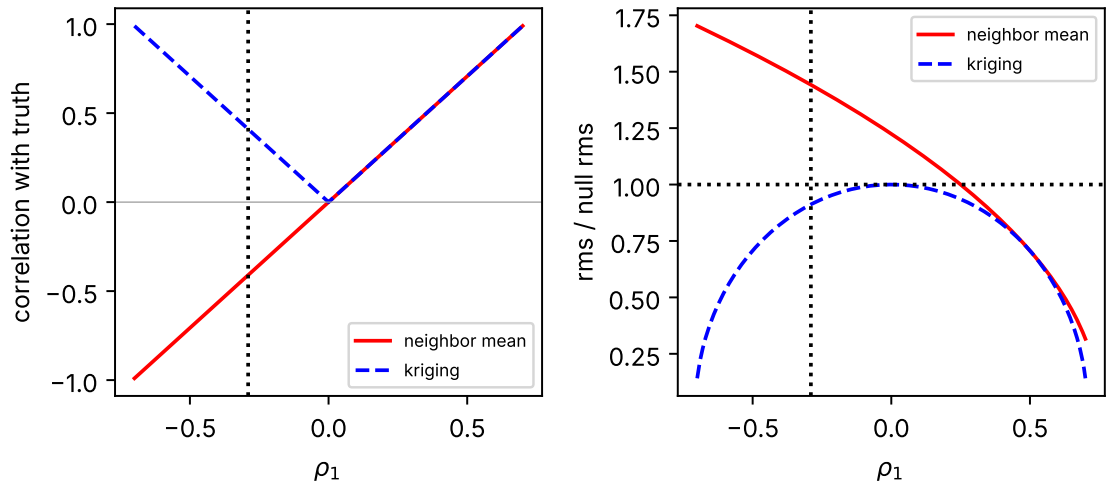


Figure 25.5: Neighbor mean versus two-neighbor kriging for a stationary sequence with lag-2 correlation $\rho_2 = 0$, as functions of the lag-1 correlation ρ_1 (dimensionless). Left: correlation with the truth, Eq. (25.14) and its kriging counterpart. Right: rms error relative to the null (dotted line at 1). The vertical dotted line marks the estimate $\rho_1 \approx -0.29$ for the zeros from the Sample Problem on the Wigner surmise. For $\rho_1 < 0$ the neighbor mean is anticorrelated and worse than the null; kriging is positively correlated and better than the null. The plotted range is $|\rho_1| \le 1/\sqrt{2}$: with $\rho_2 = 0$ the neighbor-mean correlation is $\sqrt{2}\,\rho_1$, which must lie in $[-1, 1]$; equivalently, the correlation matrix of (d_{n-1}, d_n, d_{n+1}), whose determinant is $1 - 2\rho_1^2$, is a valid covariance only if $1 - 2\rho_1^2 \ge 0$.

Step 3: $-4w\rho_1 = -4(-0.3158)(-0.3) = -0.3789$ and $2w^2(0.9) = 2(0.09972)(0.9) = 0.1795$, so $\mathcal{E}/\sigma^2 = 1 - 0.3789 + 0.1795 = 0.8006$.

$$\boxed{w = -0.316, \qquad \mathcal{E}/\sigma^2 = 0.8006 \ \ \text{(optimum 0.8)}}$$

A small nugget costs almost nothing in accuracy and buys numerical stability. The paper's kriging used a nugget of 0.05, added once "after a first version without them was numerically unstable, and not tuned" (App. A step 19).

Now try two-neighbor kriging on the real zeros. The listing below makes a random 80/20 split of zeros 1–400 (seed 1), estimates ρ_1 and ρ_2 from pairs of training levels only, and predicts every held-out level whose two neighbors are both training levels, first by the neighbor mean and then by Eq. (25.23). It subtracts the training mean first, because the sample mean of the fluctuations is not exactly zero. (In the listing the variable `mu` is that mean, not a ridge strength.)

```
import numpy as np, qmnt
t = qmnt.zeros(400); n = np.arange(1, 401); d = qmnt.fluct(t, n)
rng = np.random.default_rng(1)
test = np.zeros(400, bool); test[rng.choice(400, 80, replace=False)] = True
train = ~test
mu = d[train].mean()
def acf(k):                          # autocorrelation from training pairs only
    i = np.arange(400 - k)
    ok = train[i] & train[i + k]
    a, b = d[i[ok]] - mu, d[i[ok] + k] - mu
    return np.mean(a * b) / np.var(d[train])
```

```
r1, r2 = acf(1), acf(2)
w = r1 / (1 + r2)
# held-out levels whose two neighbors are both training levels
j = np.array([i for i in range(1, 399) if test[i] and train[i-1] and train[i+1]])
s = (d[j-1] - mu) + (d[j+1] - mu)
for name, pred in [("neighbor mean", mu + 0.5*s), ("kriging", mu + w*s)]:
    print(name, qmnt.rms(pred - d[j]), qmnt.corr(pred, d[j]))
print("null", qmnt.rms(d[j]))
```

When you run it, the estimated weight w comes out negative, the neighbor mean is anticorrelated with the held-out truth, and the kriging prediction is positively correlated with it, with exactly the same magnitude of correlation, as the Result above requires. Compare the two printed rms values with the printed null yourself.

Pitfall

Estimating correlations with the test data. In the listing above the autocorrelations come only from pairs of *training* levels. In an odd/even split this matters sharply: no two training levels are adjacent, so ρ_1 cannot be estimated from the training set at all. Using all levels to estimate ρ_1 would leak held-out information into the predictor.

Taken on trust

For a sequence whose joint distribution is Gaussian (a "Gaussian process"), the best linear predictor is also the best predictor of *any* form, linear or not: the conditional mean of d_0 given the data is exactly $\boldsymbol{w}^\top \boldsymbol{d}$ with $\boldsymbol{w}$ from Eq. (25.21). We do not use this; it is proved in standard texts on multivariate statistics and Gaussian processes. Everything in this section is proved for linear predictors only.

Key Ideas

- The best linear predictor solves the kriging normal equations $C\boldsymbol{w} = \boldsymbol{c}_0$; its error is uncorrelated with the data used.
- With two neighbors, $w = \rho_1/(1+\rho_2)$; for $\rho_1 < 0$ the prediction is a negative, shrunken multiple of the neighbor sum, and its correlation equals that of the neighbor mean with the sign flipped.
- A nugget $(C + \tau^2 I)\boldsymbol{w} = \boldsymbol{c}_0$ stabilizes the solve and shrinks toward the null, like ridge.

25.8 Regularization: the road back to the null

★ Did you know? *Fitting an Elephant*

Freeman Dyson recalled that in 1953 Enrico Fermi dismissed a complicated model by quoting John von Neumann: *with four parameters I can fit an elephant, and with five I can make him wiggle his trunk.* In 2010 three physicists actually drew an elephant using four complex parameters. Regularization is the disciplined way to stop extra parameters from fitting elephants.

25.8.1 Ridge in one parameter

Recall ridge (Tikhonov) regularization from Chapter 9. We write it in the paper's convention: minimize $\|X\boldsymbol{c} - \boldsymbol{y}\|^2 + \mu\sigma_1^2\|\boldsymbol{c}\|^2$, where σ_1 is the largest singular value of the design matrix

X. The penalty punishes large coefficients; scaling it by σ_1^2 makes μ a pure number that measures the penalty relative to the most strongly determined direction of the data. In the simplest case, a single parameter c and the model $f(x) = cx$, the design matrix is the single column $(x_1, \ldots, x_n)^\top$, and its only singular value is the length of that column, $\sigma_1 = (\sum_i x_i^2)^{1/2}$ (Chapter 9). The penalized error is therefore $S(c) = \sum_i (cx_i - y_i)^2 + \mu(\sum_j x_j^2)c^2$. Setting $dS/dc = 2\sum_i x_i(cx_i - y_i) + 2\mu(\sum_j x_j^2)c = 0$ gives $c\,(1+\mu)\sum_i x_i^2 = \sum_i x_i y_i$, that is

$$c(\mu) = \frac{\sum_i x_i y_i}{(1+\mu)\sum_i x_i^2}. \tag{25.25}$$

At $\mu = 0$ this is ordinary least squares; as $\mu \to \infty$, $c(\mu) \to 0$ and every prediction $c(\mu)x$ tends to zero. If "zero" is the null (no fluctuation), then *strong regularization returns any model to the null.*

Sample Problem 25.11: A ridge path

Data: $(x, y) = (1, 2.1), (2, 3.9), (3, 6.2)$. Fit $f = cx$ with ridge strengths $\mu = 0$, 1 and 10 (in the convention of Eq. (25.25)); give c, the in-sample rms for $\mu = 0$ and $\mu = 1$, and the prediction at $x = 4$.
Step 1: $\sum x_i y_i = 2.1 + 7.8 + 18.6 = 28.5$; $\sum x_i^2 = 1 + 4 + 9 = 14$.
Step 2: $c(0) = 28.5/14 = 2.0357$; $c(1) = 28.5/(2 \times 14) = 28.5/28 = 1.0179$; $c(10) = 28.5/(11 \times 14) = 28.5/154 = 0.1851$.
Step 3: In-sample residuals at $\mu = 0$: $2.1 - 2.0357 = 0.0643$, $3.9 - 4.0714 = -0.1714$, $6.2 - 6.1071 = 0.0929$; rms $= \sqrt{(0.00413 + 0.02939 + 0.00862)/3} = 0.119$. At $\mu = 1$: residuals 1.0821, 1.8643, 3.1464; rms $= \sqrt{(1.1710 + 3.4756 + 9.9000)/3} = 2.20$.
Step 4: Predictions at $x = 4$: 8.143, 4.071, 0.740.

$$\boxed{c = 2.036,\ 1.018,\ 0.185; \quad E_{\text{in}}(0) = 0.119,\ E_{\text{in}}(1) = 2.20; \quad f(4) = 8.14,\ 4.07,\ 0.74}$$

In-sample error rises with μ, and the prediction slides toward zero.

25.8.2 Ridge with many parameters

Recall the SVD solution with filter factors from Chapter 9: writing $X = U\Sigma V^\top$ with singular values $\sigma_1 \geq \sigma_2 \geq \ldots$, the one-shot ridge solution in the convention above has the form

$$\boldsymbol{c} = V\,\mathrm{diag}\left(\frac{\sigma_j}{\sigma_j^2 + \mu\sigma_1^2}\right) U^\top \boldsymbol{y}. \tag{25.26}$$

This is the one-shot, linear analogue of what the paper does. The paper's fit of the potential is iterative (App. A step 6): at each iteration it recomputes the sensitivity matrix, takes its SVD $U\Sigma V^\top$, and updates

$$\boldsymbol{c} \;\leftarrow\; \boldsymbol{c} + V\,\mathrm{diag}\left(\frac{\sigma_j}{\sigma_j^2 + (\lambda+\mu)\sigma_1^2}\right) U^\top \boldsymbol{r},$$

where $\boldsymbol{r}$ is the current residual vector and λ the Levenberg–Marquardt damping (Chapter 24). The paper's ridge variant "adds $\mu\Sigma_1^2$ to the filter denominator" alongside $\lambda\Sigma_1^2$; it does not replace λ. Its μ is in the same σ_1^2-scaled convention as ours. Each filter factor satisfies $0 \leq \sigma_j/(\sigma_j^2 + \mu\sigma_1^2) \leq \sigma_j/(\mu\sigma_1^2)$, which tends to zero as $\mu \to \infty$. In the one-shot form every component of $\boldsymbol{c}$ therefore tends to zero. In the iterated form the same bound (with $\lambda + \mu$ in place of μ) makes every step tend to zero, so $\boldsymbol{c}$ stays at its starting value $\boldsymbol{c} = \boldsymbol{0}$. Either way $\delta V \to 0$, and the predicted levels tend to the unperturbed (null) levels. For moderate μ, the

factor is suppressed most for small σ_j, i.e. for the parameter directions that the training data constrain least. Those are exactly the directions that, by the Result of Sec. 25.4, can produce arbitrary held-out behavior.

Figure 25.6 shows this for RE4.

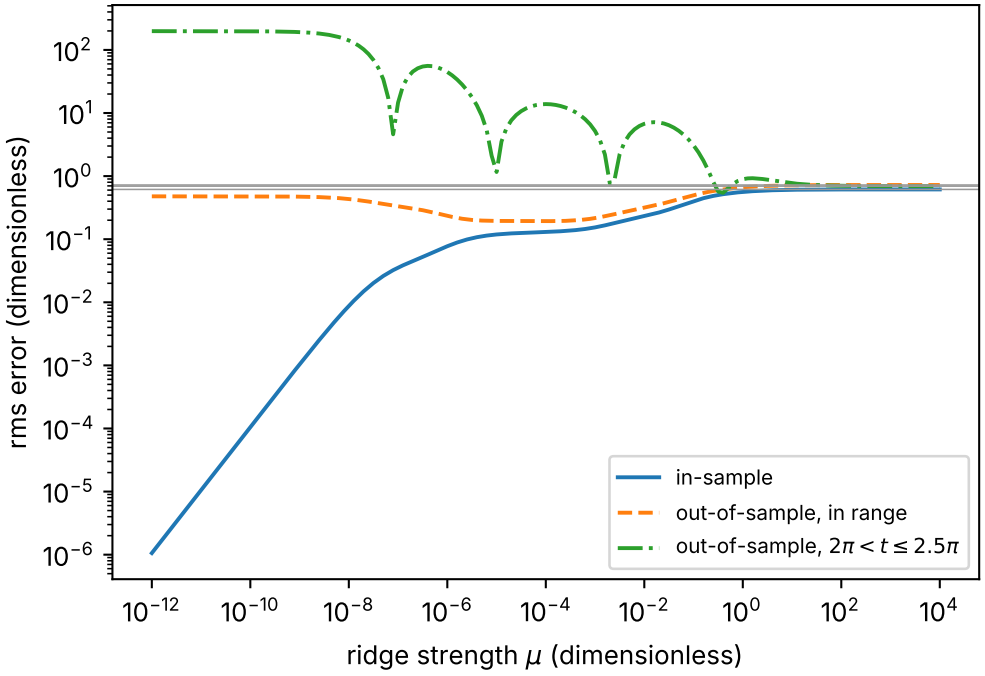


Figure 25.6: Ridge path for RE4: a degree-9 polynomial (10 coefficients) fitted to 10 noisy samples of $\sin t$ on $0 \le t \le 2\pi$ with the filter of Eq. (25.26), as a function of the ridge strength μ (dimensionless, logarithmic). Solid: in-sample rms; dashed: out-of-sample rms inside the training range; dash-dotted: out-of-sample rms on $2\pi < t \le 2.5\pi$. Thin horizontal lines: the corresponding nulls (rms of the test targets, i.e. the error of predicting zero). As μ grows every curve approaches its null.

Pitfall

Believing "regularization creates skill." Regularization can only *select* among the behaviors the model already offers. If the family has the right structure (a good inductive bias), ridge helps by suppressing noise: the paper's control B, with arithmetic frequencies given and amplitudes fitted with ridge, predicts well (Table IV; Chapter 27). If the family has no useful structure, ridge can only move it from "memorize more" toward "do nothing." For the fitted potentials that is what the paper observed (Sec. III); it is an empirical finding about those families, not a theorem about ridge.

Checkpoint 25.7

As $\mu \to \infty$ in Eq. (25.26), what held-out rms does a ridge-fitted potential approach?

Answer: The null rms, the error of the unperturbed potential's levels (for example 0.263 on zeros 401–500), because $\boldsymbol{c} \to 0$.

Key Ideas

- Ridge shrinks coefficients; as $\mu \to \infty$ the fit tends to the null.
- The filter $\sigma_j/(\sigma_j^2 + \mu\sigma_1^2)$ suppresses poorly constrained directions first.
- Regularization helps a model with a good inductive bias; it cannot supply a bias the model lacks.

25.9 Inductive bias: a model that can fit anything

25.9.1 Definition

Definition. The *inductive bias* of a fitting procedure is everything, apart from the training data, that determines its predictions on new inputs: the choice of model family, the basis, the starting point, the optimizer, the regularizer, the stopping rule.

Some bias is unavoidable. Two training points do not determine a value at a third point (Sample Problem "Infinitely many perfect fits"); some rule must break the tie. "Fit a straight line" is a bias toward linear behavior; "minimize the norm" is a bias toward small coefficients; "a few sinusoids" is a bias toward sparsity in frequency. A procedure generalizes when its bias matches the process that generated the data.

25.9.2 Why fitting anything means predicting nothing

Combine the Result of Sec. 25.4 with this definition. If a model family can fit *any* training targets, and its training and test rows are jointly of full rank, then the set of perfect fits contains every possible held-out prediction. The data then carry no information about which prediction to make; the choice is made entirely by the bias. Unless that bias happens to match the truth, the predictions are, from the data's point of view, arbitrary. A family that can fit anything encodes no preference for the truth over any of its other perfect fits.

A sharp test follows. If a model fits *scrambled* versions of the data (randomly permuted, or with randomized Fourier phases, Chapter 6) as well as it fits the real data, then its in-sample accuracy measures its capacity, not anything specific to the real data. Chapter 26 develops these surrogate tests.

Taken on trust

In machine learning there are general "no free lunch" theorems, stating roughly that averaged over all possible target functions, no learning procedure outperforms any other on unseen inputs; any advantage comes from matching assumptions to the problem. We do not need or prove the general versions; the elementary linear Result of Sec. 25.4 is the precise form we use. A textbook discussion of the role of model assumptions in prediction is [31].

25.9.3 The fitted potential through this lens

For the fitted potential, recall the first-order relation $\delta E_n \approx \sum_k A_{nk} c_k$ (Chapters 13 and 24). The levels feel the correction δV through $|\psi_n|^2$, and recall from Chapter 12 that a level of energy E has *classical turning points* at the positions where $V(x) = E$, where a classical particle of that energy would stop and reverse. The particle moves slowest there, and correspondingly $|\psi_n|^2$ is largest near the turning points, so a high level is most sensitive to δV near them. A cosine correction with many terms can therefore shift the levels in almost any joint pattern, provided

its basis is rich enough to resolve the local wavelength at the relevant energies. Chapter 24 expressed this as the (heuristic) coverage law $M\pi/L > 2\sqrt{E}$.

Sample Problem 25.12: When is the cosine family complete for the scored levels?

At scale 1 the box half-width is $L = 44.883$ and the highest scored level sits near $E \approx t_{500} = 811.184$ (paper App. A key anchors). Use the coverage law to estimate the basis size M beyond which the cosine family can realize essentially arbitrary shifts of all 500 levels. Compare $M = 300$ and $M = 900$.
Step 1: Solve $M\pi/L > 2\sqrt{E}$ for M: $M > 2\sqrt{E}\,L/\pi$.
Step 2: $\sqrt{811.184} = 28.481$, so $2\sqrt{E} = 56.963$.
Step 3: $M > 56.963 \times 44.883/\pi = 2556.7/\pi = 813.8$.
Step 4: For $M = 300$: $M\pi/L = 300\pi/44.883 = 21.0 < 56.96$, not covered. For $M = 900$: $900\pi/44.883 = 63.0 > 56.96$, covered.

$$\boxed{M \gtrsim 814}$$

The paper reports in-sample rms 0.186 on all 500 zeros at $M = 300$ and below 10^{-4} at $M = 900$, "consistent with a basis-coverage law" (Sec. III). The law is a heuristic estimate, not a sharp threshold.

How far does the Result of Sec. 25.4 carry over? Its hypothesis is that the stacked training-plus-test matrix has full row rank. That requires $M \geq n_{\mathrm{tr}} + n_{\mathrm{te}}$; saturation on the training levels alone is not enough. Some of the paper's configurations meet the counting condition: $M = 600$ fitted on zeros 1–400 and scored on 401–500 (500 rows, 600 columns), and the $M = 400$ odd/even splits (400 rows). For these, provided the joint (linearized) sensitivity matrix has full row rank, exact training fits exist with any held-out shifts, and the data cannot choose among them. For the principal configuration, $M = 400$ fitted on 400 zeros and scored on 100, the stacked matrix has 500 rows but rank at most 400, so the Result does not apply. There the held-out shifts are not completely free. They are only partly constrained, because parameter directions with small singular values of the training sensitivity matrix are weakly determined by the training levels, and such directions can move the held-out levels at little in-sample cost. In both cases, what the fitted potential predicts for the held-out levels is largely set by the optimizer: starting from $\boldsymbol{c} = \mathbf{0}$, damped least-squares steps of the kind in Eq. (25.26) build $\boldsymbol{c}$ from combinations of the rows of the training sensitivity matrix, and the held-out shifts are whatever those combinations happen to do to the held-out levels. This is a first-order, heuristic picture of a nonlinear fit; it explains the observed behavior without proving it.

25.9.4 The right parametrization

The explicit formula of Chapter 18 tells us what the fluctuations look like as functions of height t: a sum of sinusoids $\sin(mt \log p)$, an almost-periodic function (Chapter 5) that is *sparse in frequency*. A bias toward "a few sinusoids in t" matches that structure, and a model with only that bias, and no knowledge of primes, can learn the frequencies from the zeros (Chapter 27). A cosine expansion of $\delta V(x)$ in position addresses each level locally through its turning points. Nothing in it says "the pattern you saw at levels 1–400 continues at 401–500." The paper offers exactly this as a heuristic reading of its results (Sec. VI); in Sec. 25.10 we see the experiment that isolates it.

Checkpoint 25.8

Two models fit the same 400 zeros to the same in-sample rms. Model P is a 400-coefficient potential; model S is a sum of 24 sinusoids in t with fitted frequencies and amplitudes (72 parameters). Without looking at held-out data, can you say which will predict better?

Answer: Not from the in-sample numbers alone; equal in-sample error says nothing about generalization. You can argue which inductive bias better matches the explicit formula (S), but the claim must be settled by held-out scores against the null.

Key Ideas

- Inductive bias is everything besides the data that determines predictions on new inputs.
- A family that can fit anything (including scrambled data) has its held-out predictions set largely by its bias, not by its data; fully so when training and test rows are jointly of full row rank ($M \geq n_{\text{tr}} + n_{\text{te}}$).
- The coverage law estimates when the cosine family becomes effectively complete: $M \gtrsim 814$ for all 500 scale-1 levels.
- Generalization requires a bias that matches the data-generating structure; for the zeros, sparsity in t-frequency is such a bias.

25.10 Paper connection

We now read the parts of the paper [1] that rest on this chapter. Numbers are quoted exactly from R1.

25.10.1 The out-of-sample protocol (Sec. II C)

The paper's protocol is the split of Sec. 25.2 applied to the fitted potential: "fit on zeros $1, \ldots, n_{\text{fit}}$ only; then a single plain eigensolve of the fitted potential is scored on the held-out zeros, reporting rms error in local spacings and the correlation between predicted and true fluctuations (both measured relative to the unperturbed smooth-potential levels)." Three features deserve attention.

1. *One eigensolve, no further adjustment.* After fitting, nothing is tuned on the held-out zeros. That is the "no test target influences any choice" rule.

2. *Two numbers.* rms and correlation, Eq. (25.5), because (Sec. 25.3) a prediction can fail by having no shape (correlation near zero) or by having the wrong size.

3. *A null.* "The null model is the unperturbed potential itself," with holdout error 0.2631 on zeros 401–500, numerically the same as the smooth-zero null used for the explicit formula. The paper stresses that the two nulls are conceptually distinct and that only their numerical coincidence on the principal holdout permits a single baseline.

The bias-corrected targets $\tilde{t}_n$ touch held-out indices, which might look like a violation of the split. The paper's Eq. (3) shows it is not: scoring against $\tilde{t}_n$ is the same as correcting the prediction by a quantity Δ_n computed only from the unperturbed potential, with no zero entering (Chapter 23). This is the leakage check Sec. 25.2 asks you to make.

25.10.2 Table I and Fig. 3: in-sample falls, held-out does not

Table I lists twelve Hermitian configurations: two scales, five train/test splits, M from 200 to 900, two ridge strengths. Reading its columns with this chapter's tools:

- *In-sample column.* At scale 1, fit 1–400, the in-sample rms goes 0.226 ($M = 200$), 0.137 ($M = 400$), 0.064 ($M = 600$): the nesting argument of Sec. 25.2. Figure 3 of the paper plots this fall at both scales.
- *OOS column.* The same rows give 0.274, 0.369, 0.313 against the null 0.263. In the paper's words, "the held-out rms never improves on the null and no reproducible positive held-out correlation appears." Figure 3 shows the held-out curves at or above the dashed null.
- *OOS corr column.* Values from -0.17 to $+0.12$. The bootstrap interval for the scale-1, $M = 400$ random-restart replicate is $[-0.21, +0.08]$ (Sec. III); bootstrap methods are the subject of Chapter 26.
- *Ridge rows.* $\mu = 0.03$ gives OOS 0.271, $\mu = 0.3$ gives 0.265, approaching the null 0.263 while the in-sample error rises from 0.137 to 0.142 and 0.206. This is Sec. 25.8 in action: "regularization moves them smoothly from 'memorize more' to 'do nothing' and never creates skill" (Sec. III).

The paper describes the fitted models' output as fluctuations of 0.16–0.62× the true amplitude, uncorrelated with truth. Eq. (25.9) explains why such output cannot beat the null: with $\rho \approx 0$ the ratio is $\sqrt{1 + a^2} > 1$. The same formula gives a consistency check on the explicit formula: under the idealized zero-mean model of Eq. (25.9), the best achievable ratio at $\rho = 0.96$ is $\sqrt{1 - 0.96^2} = 0.280$, and Table V reports $0.074/0.263 = 0.281$ for six primes, which is consistent with a near-optimal amplitude under that idealized model (Problem 5). This is not evidence of optimality, and the paper makes no such claim.

At scale 1 the coverage estimate $M \gtrsim 814$ (Sec. 25.9) says the family is effectively complete for all 500 levels by $M = 900$, and the paper reports in-sample rms $< 10^{-4}$ there. The family has the capacity to memorize; Table I shows it does not transfer.

25.10.3 Covariate shift and the within-height splits (Sec. II C, Table III)

Every split in Table I is forward in height, so the held-out zeros have slightly larger fluctuations (the null grows from 0.263 to 0.296 across Table V). Sec. 25.5 explains why a forward failure alone could be blamed on this shift. The paper therefore adds within-height splits, "random 80/20 splits (ten seeds) and odd/even splits in both directions, inside zeros 1–400 at scale 1 and inside zeros 1–800 at scale 2."

Table III reads as follows.

- *Random splits* are "the principal evidence": every held-out error is above the null, by a factor 1.5–3.0 at scale 1 and 1.8–1.9 at scale 2, with no reliably positive correlation.
- *Odd/even splits* have 200 training levels at scale 1, so rows with $M \geq 200$ are saturated (Sample Problem "Counting levels"). The $M = 400$ odd/even rows have in-sample error $< 10^{-3}$, a lookup table in disguise, and held-out errors 0.351 and 0.368. With $M = 100$, fewer coefficients than levels, the held-out error is still 0.312 and 0.311 against nulls 0.238 and 0.250.
- *Sign of the correlations.* At $M \leq 200$ the correlations are all negative: -0.26 to -0.39 for the odd/even rows, and -0.40 to -0.14 across the random splits at $M = 200$. The paper calls this "mechanical": the fluctuation sequence has negative autocorrelation at lags 1–3,

> "so any smooth interpolation between neighbouring training levels is anticorrelated with the truth (the mean of the two nearest training fluctuations gives −0.26 to −0.41)." That is Eq. (25.14) with $\rho_1 < 0$, and Eq. (25.18) explains why $\rho_1 < 0$ is expected for a rigid spectrum.

The paper concludes that the failure is "not a covariate shift in height" and is "in fact worse within height than in forward extrapolation." On the same held-out sets the six-prime explicit formula scores 0.062–0.063 with correlation +0.97.

25.10.4 Kriging: the serial structure is there to be used (Sec. III)

Sec. 25.7 predicts that a correlation-aware linear predictor can turn negative lag correlations into positive skill. The paper reports exactly this, within one height range: "a linear interpolator (simple kriging from the training fluctuations) predicts held-out true zeros at 0.117 with correlation +0.87 and permuted targets at the null." Its version uses autocovariances from training pairs up to lag 40, a $\cos^2$ taper, a positive-semidefinite projection and a nugget 0.05 (App. A step 19), so it is richer than our two-neighbor example, and its skill is far larger. Two neighbors with $\rho_1 = -0.3$, $\rho_2 = -0.1$ give only corr ≈ 0.45 and an rms ratio of about 0.89 (Sample Problem "Kriging versus the neighbor mean"), whereas the paper reports +0.87. The difference comes from the richer predictor: it uses training neighbors out to lag 40, whose joint correlations carry information that lags 1 and 2 alone do not. Note the role of the permuted targets: permutation destroys serial correlation, so kriging falls to the null, as it should. The fitted operator, by contrast, "fails alike on true zeros and on surrogates (0.57–0.70; one split, illustrative)." The paper labels this single-split comparison as illustrative, and so should we.

25.10.5 The interpolation gap and Fig. 4: damage, not just ignorance

Fitting zeros 1–200 and 301–500 and predicting the gap 201–300 gives in-sample rms 0.136 and a gap error of 2.64 spacings, "ten times the null 0.255," with correlation −0.02 (Sec. III). This is worse than the "no information" outcome; the fit does harm. Sec. 25.4 explains how this is possible: parameter directions that matter mainly for the held-out levels are weakly constrained by the training levels, so the optimizer can move along them at little in-sample cost. The paper's Fig. 4 shows where: levels 201–300 turn at $x = 22.5$–28.5, and the fitted correction is largest exactly there (rms $\delta V = 40$ in that band, maximum $|\delta V| = 199$ where $V \approx 500$), displacing the gap levels by −6 to +2 spacings. Ridge "removes the damage and nothing else": the gap error returns to 0.262 ($\mu = 0.03$) and 0.257 ($\mu = 0.3$). Keep the paper's hedge: "this does not show that the family could not interpolate under another fitting procedure."

25.10.6 The arithmetic-continuation probe

The cleanest illustration of inductive bias in the paper replaces the targets by the $p = 2$ term of Eq. (4) alone (running example RE2), fits levels 1–400, and asks for the continuation on 401–500. In-sample: rms 0.056, correlation +0.95. Held out: correlation +0.07, rms 0.1958 against the probe null 0.1708 (App. A key anchors), i.e. worse than predicting zero fluctuation. The answer is known exactly here, yet the family does not continue it: "the arithmetic frequency is not in its vocabulary" (Sec. III). In this chapter's language, the bias of a cosine-in-x potential fitted by this optimizer does not include "keep oscillating in t."

25.10.7 Sec. VI: the heuristic explanation and the standard

Sec. VI summarizes the failure in terms that are now familiar: the bases "realize essentially arbitrary joint shifts, which is why they fit permuted and phase-randomized targets as well as the true ones. A family that can fit anything encodes no inductive bias toward the truth,

and its held-out behaviour is set by the optimizer rather than by the data: unregularized, it places large structure where only held-out levels probe the potential and does harm . . . ; regularized, it returns to the null and not below it." Each clause matches a section here: capacity and saturation (Sec. 25.4), inductive bias (Sec. 25.9), the interpolation damage (Fig. 4), and regularization (Sec. 25.8). The paper calls this reading heuristic and says "Whether the memorization result can be derived rather than measured is an open problem." It also limits its scope: one background potential, three bases, operator fits on at most the first 1000 zeros.

Finally, the first item of the paper's proposed falsifiability standard is this chapter in one sentence: any claim that an operator captures the zeros should come with "(i) a train/test split with held-out rms and correlation against the null of its own smooth part, including a split within one height range."

25.11 Summary

- A model has captured structure only to the extent that it predicts held-out data better than a null (Sec. 25.1).
- In-sample and out-of-sample errors, Eqs. (25.1) and (25.2); in-sample error cannot increase along a nested family.
- For the zeros, fluctuations $d_n = (t_n - s_n)/\mathrm{gap}_n$ (Eq. (25.3)); the error in fluctuation equals the height error in mean spacings (Eq. (25.4)); the null rms is the rms of d_n (Eq. (25.6)), 0.2631 on zeros 401–500.
- rms ratio from amplitude and correlation: $\sqrt{1 + a^2 - 2\rho a}$, minimum $\sqrt{1 - \rho^2}$ at $a = \rho$ (Eq. (25.9)).
- A saturated linear model fits any targets; if training and test rows are jointly full rank, exact fits exist with any held-out prediction, so the inductive bias decides.
- Forward splits test extrapolation with covariate shift; random and odd/even splits test interpolation within height (Table 25.2).
- Neighbor-mean correlation with truth: $\rho_1\sqrt{2/(1+\rho_2)}$ (Eq. (25.14)); its MSE, Eq. (25.15).
- Lag-1 identity $\rho_1 = 1 - \mathrm{Var}(u)/(2\sigma^2)$ (Eq. (25.18)); estimate $\rho_1 \approx -0.29$ for the zeros.
- Kriging normal equations $C\boldsymbol{w} = \boldsymbol{c}_0$ (Eq. (25.21)), minimum error Eq. (25.22); two-neighbor weight $w = \rho_1/(1+\rho_2)$ (Eq. (25.23)) flips the sign of the neighbor mean's correlation; nugget Eq. (25.24).
- Ridge (paper's $\mu\sigma_1^2$ convention): $c(\mu) = \sum xy/((1+\mu)\sum x^2)$ in one parameter (Eq. (25.25)); SVD filter Eq. (25.26); $\mu \to \infty$ returns to the null.
- Coverage law estimate $M \gtrsim 2\sqrt{E}\,L/\pi \approx 814$ for scale 1.

25.12 Problems

1. • In your own words, define in-sample error, out-of-sample error and the null model. Which of the two errors can a saturated model always drive to zero?

2. • Training data $y = 3, 5, 4$ at $t = 1, 2, 3$; test point $y = 6$ at $t = 4$. Fit a constant model. Find its in-sample rms and its error at $t = 4$.

3. • Among the scale-1 rows of the paper's Table I, find the smallest held-out rms. Is it below the null 0.263? Compute the ratio of Eq. (25.7).

4. • A forecaster predicts $\hat{d}_n = -0.5\, d_n$ for every held-out level. Find the correlation with the truth and the rms ratio to the null.

5. •• (a) Starting from Eq. (25.9), show that at fixed correlation ρ the smallest rms ratio is $\sqrt{1-\rho^2}$. (b) Evaluate it for $\rho = 0.96$ and compare with the six-prime explicit formula of the paper's Table V on zeros 401–500 (0.074 against null 0.263). What does the comparison suggest about the formula's amplitude?

6. •• Using numpy, reproduce the error curves of Fig. 25.2 with the listing of Sec. 25.4. Then change the extrapolation window to $2\pi < t \le 3\pi$ and describe how the extrapolation curve changes.

7. •• A sequence is built as $d_n = e_n - \theta e_{n-1}$ from independent e_n of unit variance. (a) Show that $\rho_1 = -\theta/(1+\theta^2)$ and $\rho_2 = 0$. (b) For $\theta = 0.5$, find the neighbor-mean correlation with the truth and its rms ratio to the null.

8. •• Using qmnt, compute the sample autocorrelation of d_n for zeros 1–400 at lags 1 to 10. Repeat for zeros 601–1000. Which lags are negative in each window?

9. •• Use the nugget in its measurement-noise reading: each observed neighbor is $d_i + \eta_i$ with independent noise of variance $\tau^2 = 0.05\sigma^2$. With $\rho_1 = -0.3$, $\rho_2 = -0.1$, find the two-neighbor weight and the rms ratio to the null.

10. •• At scale 2 the paper's within-height splits use zeros 1–800. How many levels train in an odd/even split and in a random 80/20 split? Which of these is saturated at $M = 600$?

11. •• Fit $f(t) = c_0 + c_1 t + c_2 t^2$ to $f(-1) = 0$ and $f(1) = 2$. (a) Find all exact fits and their prediction at $t = 2$. (b) Find the minimum-norm fit and its prediction.

12. •• Data $(x, y) = (1, 1.2), (2, 1.8), (4, 4.1)$. Fit $f = cx$ with ridge strength μ. Find $c(\mu)$ and the prediction at $x = 5$ for $\mu = 0, 1, 10$ (convention of Eq. (25.25)), and describe the trend.

13. •• Use the coverage law $M\pi/L > 2\sqrt{E}$ to estimate the basis size needed to cover all 1000 scale-2 levels, with $L = 59.957$ and $E \approx t_{1000} = 1419.422$ (paper Sec. II A). Is $M = 900$ above or below this estimate?

14. ••• Prove the Result of Sec. 25.4 in the form: if X_{tr} has full row rank n_{tr} and the stacked matrix X has full row rank, then the set of exact training fits is an affine subspace (one particular solution plus any vector of a subspace) on which the map $\boldsymbol{c} \mapsto X_{\mathrm{te}}\boldsymbol{c}$ is onto $\mathbb{R}^{n_{\mathrm{te}}}$ (every vector of $\mathbb{R}^{n_{\mathrm{te}}}$ is reached). Show also that when X does *not* have full row rank, the training data can constrain the predictions; give a 2-parameter example.

15. •• From the paper's Table III, compute the ratio of held-out error to null for the scale-1, $M = 100$ odd and even rows. Why are these rows more informative than the $M = 400$ odd/even rows?

16. •• Using qmnt, run the odd/even neighbor-mean listing of Sec. 25.6 with the roles reversed (predict odd n from even neighbors). Compare its correlation with the paper's range -0.26 to -0.41 (Sec. III).

17. ••• (a) Derive Eq. (25.18) step by step, stating every approximation. (b) If $\mathrm{Var}(u) = 0.178$ and $\sigma = 0.25$, find ρ_1. (c) What would ρ_1 be for a sequence of independent d_n?

18. •• Using qmnt, extend the kriging listing of Sec. 25.7 to use four neighbors ($n \pm 1$, $n \pm 2$), solving the 4×4 normal equations with autocorrelations at lags 1 to 4 estimated from training pairs. Report rms and correlation for seeds 1, 2, 3, and compare with the two-neighbor version and the null.

19. •• The paper's App. A gives a "raw-zero null on the uncorrected working grid" of 2.09 on zeros 401–500, against the corrected null 0.2631. Compute the ratio. Explain why a comparison of a model with the wrong null could make a useless model look skillful, or a skillful one look useless.

20. •• In the paper's interpolation split, the gap error is 2.6434 against a gap null of 0.255. Compute the ratio. Explain in terms of Sec. 25.4 why an error larger than the null is possible at all.

21. ••• Prove the orthogonality property of the best linear predictor: if $\boldsymbol{w}$ solves $C\boldsymbol{w} = \boldsymbol{c}_0$, then $\langle (d_0 - \boldsymbol{w}^\top \boldsymbol{d})\, \boldsymbol{v}^\top \boldsymbol{d} \rangle = 0$ for every vector $\boldsymbol{v}$. Use it to show that no other linear predictor has smaller mean squared error.

22. •• Using numpy, reproduce the ridge path of Fig. 25.6. Find (roughly) the value of μ that minimizes the in-range out-of-sample error, and check that at very large μ every error equals its null to three digits.

23. •• For the paper's arithmetic-continuation probe, compute the ratio of the held-out rms 0.1958 to the probe null 0.1708 (App. A key anchors). Why is this probe a test of inductive bias rather than of noise?

24. ••• Polynomials of degree $n - 1$ interpolate n points uniquely. Show that allowing degree n (one more coefficient) produces a one-parameter family of interpolants whose value at any point t^* not among the data points can be any real number. (Hint: add a multiple of $\prod_i (t - t_i)$.)

25. • Why is "held-out correlation near zero together with held-out rms above the null" the signature of memorization rather than of a weak but genuine signal?

26. •• For each of the four split types of Table 25.2, say whether it involves interpolation or extrapolation and whether it involves covariate shift in height. Which would you choose to test a claim that a model "learns the local structure" of the zeros?

27. •• The paper's Table II reports Chebyshev $M = 400$: OOS rms 0.291 with interval $[0.25, 0.33]$ and correlation $+0.18$ with interval $[-0.02, +0.37]$, against null 0.263. Compute the rms ratio and explain why the paper calls this row "marginal" rather than evidence of prediction. (Note that these bootstrap intervals resample held-out levels independently, although the fluctuations are serially correlated; the paper reports that a moving-block bootstrap widens them slightly and changes no conclusion.)

28. ••• Design a test, using only the ideas of this chapter and surrogates (Chapter 26), that distinguishes a model whose in-sample accuracy reflects captured structure from one whose accuracy reflects capacity. State the null hypothesis, the surrogates, and the outcome that would favor each interpretation.

29. •• Compute the skill $1 - \mathrm{rms}/\mathrm{rms}_{\mathrm{null}}$ on zeros 401–500 for: the 25-prime oracle (0.043, Table V), control C1 (0.258, Table IV) and the cosine operator at $M = 400$ (0.369, Table I), all with null 0.263.

30. ••• Using qmnt, score the one-prime explicit formula ($P = \{2\}$) on zeros 401–500 with `qmnt.oracle_zeros` and `qmnt.score`, and compare with Table V ($0.198/+0.66$). Then

fit, by least squares on zeros 1–400 only, a single amplitude α multiplying the $P = \{2\}$ fluctuation, and score the rescaled prediction on 401–500. Discuss the result using Eq. (25.9).

25.13 Answers to selected problems

★ Did you know? *A Million-Dollar Holdout*

From 2006 to 2009 Netflix offered one million US dollars to anyone who could beat its movie-rating predictions by 10%. To stop teams from tuning to the scoring data, the public leaderboard used one hidden set, while the prize was judged on a different hidden set. The team BellKor's Pragmatic Chaos won in 2009, and holdout testing became a widely known idea.

1. In-sample error: rms error of the fitted model on the data used to fit it. Out-of-sample error: rms error on data not used in any fitting decision. Null model: the best prediction that uses none of the information the model claims to capture (for the zeros, "no fluctuation"). A saturated model can always drive the *in-sample* error to zero; its out-of-sample error is not controlled by the data (Sec. 25.4).

2. The constant is the mean, $c_0 = 4$. Residuals $-1, +1, 0$; $E_{\text{in}} = \sqrt{2/3} = 0.816$. Error at $t = 4$: $|6 - 4| = 2$.

3. The smallest scale-1 OOS value is 0.265 (fit 1–400, $M = 400$, ridge $\mu = 0.3$). It is above the null 0.263; ratio $0.265/0.263 = 1.008$. Even the best row, the most heavily regularized, only approaches the null.

4. $\hat{d} = -0.5d$ is a negative multiple of the truth, so corr $= -1$. The error is $\hat{d} - d = -1.5d$, so the rms ratio is 1.5.

5. (a) The quantity $1 + a^2 - 2\rho a$ has derivative $2a - 2\rho$ in a, zero at $a = \rho$; the second derivative is $2 > 0$, so this is a minimum, with value $1 - \rho^2$. (b) $\sqrt{1 - 0.9216} = \sqrt{0.0784} = 0.280$; the oracle achieves $0.074/0.263 = 0.281$. Within rounding the oracle is at the best ratio its correlation allows, suggesting its amplitude is close to optimal. (This is a consistency check, not a proof; nonzero mean offsets would also raise the ratio.)

7. (a) $\langle d_n^2 \rangle = 1 + \theta^2$; $\langle d_n d_{n+1} \rangle = \langle (e_n - \theta e_{n-1})(e_{n+1} - \theta e_n) \rangle = -\theta$ (only the e_n product survives); $\langle d_n d_{n+2} \rangle = 0$ (no shared e). So $\rho_1 = -\theta/(1+\theta^2)$, $\rho_2 = 0$. (b) $\rho_1 = -0.5/1.25 = -0.4$. Correlation $-0.4\sqrt{2} = -0.566$. MSE ratio $1 + \frac{1}{2} - 2(-0.4) = 2.3$; rms ratio $\sqrt{2.3} = 1.517$.

9. With noisy observations the normal equations are $(C + \tau^2 I)\boldsymbol{w} = \boldsymbol{c}_0$, giving $w = -0.3/(0.9 + 0.05) = -0.3158$. The error (now against the noiseless target, with noisy inputs) is $\mathcal{E}/\sigma^2 = 1 - 2\rho_1^2/(1 + \rho_2 + \tau^2/\sigma^2) = 1 - 0.18/0.95 = 0.8105$; rms ratio $\sqrt{0.8105} = 0.900$.

11. (a) $c_0 - c_1 + c_2 = 0$ and $c_0 + c_1 + c_2 = 2$ give $c_1 = 1$ and $c_0 + c_2 = 1$. Prediction $f(2) = c_0 + 2 + 4c_2 = 3 + 3c_2$, any real number. (b) Minimize $c_0^2 + 1 + c_2^2$ with $c_0 = 1 - c_2$: derivative $-2(1 - c_2) + 2c_2 = 0$ gives $c_2 = \frac{1}{2}$, so $\boldsymbol{c} = (\frac{1}{2}, 1, \frac{1}{2})$ and $f(2) = 4.5$.

13. $M > 2\sqrt{1419.422} \times 59.957/\pi = 2(37.675)(59.957)/\pi = 1438$. $M = 900$ is below this estimate, so by this heuristic the cosine family at scale 2 is not complete for the top levels; this is in line with (though it does not prove) the scale-2 in-sample errors in Table I not reaching zero.

15. Odd: $0.312/0.238 = 1.311$. Even: $0.311/0.250 = 1.244$. With $M = 100$ there are fewer coefficients than the 200 training levels, so the model is not saturated and cannot be a lookup table; failing to beat the null there shows the failure is not just an artifact of saturation.

17. (a) See Eqs. (25.16)–(25.18); approximations: $s_{n+1} - s_n \approx \text{gap}_n$ and $\text{gap}_{n+1} \approx \text{gap}_n$ (slowly varying mean spacing), and stationarity of d_n over the window. (b) $\rho_1 = 1 - 0.178/(2 \times 0.0625) = 1 - 1.424 = -0.424$. (c) Independent d_n have $\rho_1 = 0$, and then $\text{Var}(u) = 2\sigma^2$.

19. $2.09/0.2631 = 7.94$. The raw null is dominated by finite-difference dispersion (about two mean spacings at levels 401–500 on the working grid, paper Sec. II C), not by the zeros'

fluctuations. Against the inflated null, any model that merely corrected the grid bias would look highly "skillful"; conversely a real signal could be hidden under the bias. Skill is meaningful only against the null of the model's own smooth part.

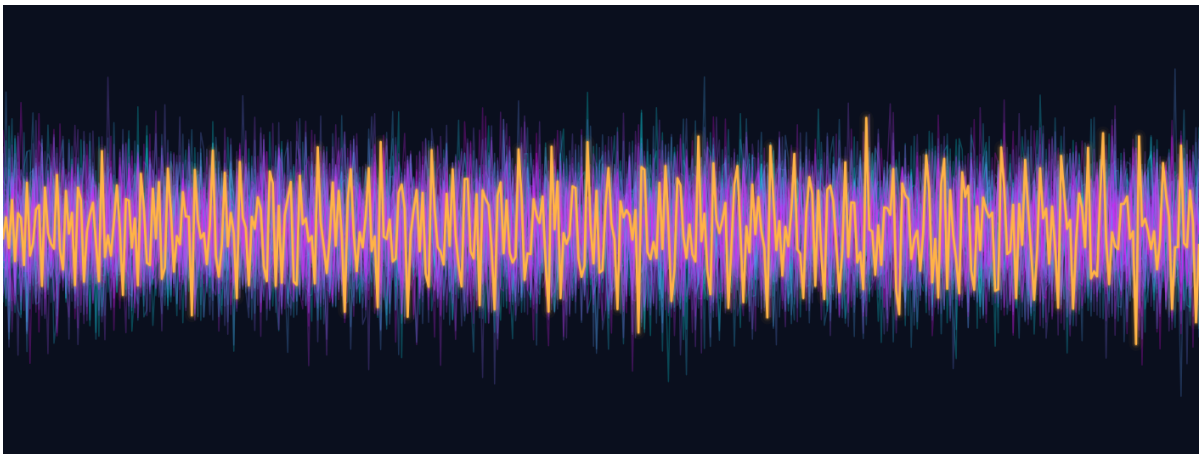
21. Let $e = d_0 - \boldsymbol{w}^\top \boldsymbol{d}$. Then $\langle e\, \boldsymbol{v}^\top \boldsymbol{d}\rangle = \sum_k v_k \langle e\, d_k\rangle = \sum_k v_k (C_{k0} - \sum_j C_{kj} w_j) = 0$ by Eq. (25.21). For any other weights $\tilde{\boldsymbol{w}} = \boldsymbol{w} + \boldsymbol{v}$: $\langle (d_0 - \tilde{\boldsymbol{w}}^\top \boldsymbol{d})^2\rangle = \langle (e - \boldsymbol{v}^\top \boldsymbol{d})^2\rangle = \langle e^2\rangle - 2\langle e\, \boldsymbol{v}^\top \boldsymbol{d}\rangle + \langle (\boldsymbol{v}^\top \boldsymbol{d})^2\rangle = \langle e^2\rangle + \langle (\boldsymbol{v}^\top \boldsymbol{d})^2\rangle \geq \langle e^2\rangle$.

23. $0.1958/0.1708 = 1.146$: about 15% worse than predicting zero. The synthetic target is a single known sinusoid with no noise, so there is nothing to "overfit" in the noise sense; the only question is whether the family's bias continues the pattern. It does not.

25. A weak genuine signal would show up as a small but positive correlation, and, at a sensible amplitude, an rms slightly below the null. Correlation near zero means the predicted pattern carries no information about the truth, and by Eq. (25.9) any nonzero amplitude of uncorrelated output pushes the rms above the null. That combination is what output driven by fitted training data, rather than by a transferable rule, looks like.

27. $0.291/0.263 = 1.106$. The rms interval $[0.25, 0.33]$ contains the null, so the rms is not distinguishable from the null; the correlation interval $[-0.02, +0.37]$ has its lower end at (about) zero. Its neighbours $M = 200$ and 600 show no trend, and it is one row among eight. The paper therefore reports it as marginal: "not evidence of prediction, but a small positive correlation for this row is not excluded" (Sec. III).

29. 25-prime oracle: $1 - 0.043/0.263 = 0.837$. C1: $1 - 0.258/0.263 = 0.019$. Operator $M = 400$: $1 - 0.369/0.263 = -0.403$.



Forty phase-randomized surrogates (faint) of the real zero fluctuations (bright): same power spectrum, no arithmetic.

Chapter 26

Uncertainty and Controls

↑ Contents

What this chapter is about

Suppose you time ten swings of a pendulum and get 14.2 s, and a friend timing the same pendulum gets 14.5 s. Did the pendulum change, or did the two of you press the stopwatch at slightly different moments? You cannot answer until each number carries an error bar. First-year laboratories teach this lesson for measurements made with rulers and clocks (see [7]). This chapter teaches it for the numbers produced by a computational study: an rms error on a set of held-out zeros, a correlation between a prediction and the truth, an in-sample fit quality.

We address three questions. First, *how uncertain is a score computed from a finite set of data?* The tools are the standard error, the bootstrap, and its block version for data whose neighbors are correlated. Second, *when a model fits its training data well, did it find structure or can it fit anything?* The tools are permutation tests, surrogate data (shuffled, phase-randomized, amplitude-adjusted) and the general idea of positive and negative controls. Third, *which analyses were planned before the data were seen?* Here we distinguish confirmatory from exploratory analysis and explain why a statistic invented after looking at the data cannot carry a significance claim.

Where this shows up in the paper [1]: every bracketed interval in Secs. III and IV and in Tables II, IV and V is a bootstrap percentile interval; the block-bootstrap check in Sec. III uses block lengths 3 to 20; the "Permutation and surrogate tests" paragraph of Sec. III refits the operator to shuffled and phase-randomized targets; Appendix A steps 12, 13, 18, 22 and 23 record exactly how; and Sec. IV qualifies its frequency-alignment check with the words "the statistic was defined after the data were seen, so the check is exploratory". After this chapter you will know what each of those sentences means and why it is there.

What you need

- Chapter 19: random variables, mean, variance, standard deviation, rms, covariance, Pearson correlation, autocorrelation of a sequence, standard error, the idea of a confidence interval, the law of large numbers, random number generators and seeds, the exponential distribution of gaps in a Poisson process.
- Chapter 6: the discrete Fourier transform (DFT) and the FFT, power spectrum, the Wiener–Khinchin connection between autocorrelation and power spectrum, and the basic idea of phase randomization.
- Chapter 25: training and held-out sets, in-sample and out-of-sample error, the null model, saturated models, serial correlation of the zero fluctuations, simple kriging.
- Chapter 3: the geometric series and its sum.
- Chapter 9: linear least squares (used in one toy demonstration).
- Self-contained reminders, restated where used: the smooth counting function N_{sm}, smooth zeros s_n and mean spacing gap_n (Chapter 17); the truncated explicit formula, Eq. (4) of the paper (Chapter 18); the density of primes (Chapter 15). You do not need those chapters to follow this one.
- First-year physics: measurement uncertainty as an idea, averaging repeated measurements.

26.1 Sampling variability: every score is one draw

Imagine scoring a model on zeros 401–500 and getting rms error 0.369 (in units of the local mean spacing; recall from Chapter 25 that the rms error is $\sqrt{\frac{1}{n}\sum_j(\mathrm{pred}_j - \mathrm{true}_j)^2}$). Had you scored on a different hundred zeros from the same height range, you would have gotten a somewhat different number. The score depends on *which* levels happened to be in the test set. This dependence is called **sampling variability**, and the first task of this chapter is to measure it.

26.1.1 Estimators and their sampling distributions

We model the data $x_1, \dots, x_n$ as the observed values of random variables $X_1, \dots, X_n$. In the simplest model the X_j are **independent and identically distributed** (iid): each is drawn from the same distribution, with mean μ and variance σ^2, and knowing one tells you nothing about another.

A **statistic** (or **estimator**) is any number computed from the data, written $\hat{\theta} = \hat{\theta}(x_1, \dots, x_n)$. The sample mean $\bar{x} = \frac{1}{n}\sum_j x_j$ is one; so are the rms error and the correlation of a prediction with the truth. Because the data are random, so is $\hat{\theta}$. Its probability distribution over repeated experiments is the **sampling distribution**, and the standard deviation of the sampling distribution is the **standard error** (SE) of $\hat{\theta}$.

26.1.2 The standard error of a mean

For the mean of iid data we can compute the standard error exactly. Write $\bar{X} = \frac{1}{n}\sum_{j=1}^{n} X_j$. Its expected value is

$$E[\bar{X}] = \frac{1}{n}\sum_{j=1}^{n} E[X_j] = \frac{1}{n}\cdot n\mu = \mu. \tag{26.1}$$

For the variance, start from the definition $\mathrm{Var}(\bar{X}) = E[(\bar{X}-\mu)^2]$ and write $\bar{X}-\mu = \frac{1}{n}\sum_j (X_j-\mu)$:

$$\begin{aligned}
\mathrm{Var}(\bar{X}) &= E\Big[\Big(\frac{1}{n}\sum_{i=1}^{n}(X_i-\mu)\Big)\Big(\frac{1}{n}\sum_{j=1}^{n}(X_j-\mu)\Big)\Big] \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} E[(X_i-\mu)(X_j-\mu)] \quad \text{(expand, expectation is linear)} \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(X_i,X_j) \quad \text{(definition of covariance, Ch. 19)} \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 \quad \text{(independence: Cov} = 0 \text{ for } i\neq j;\ \mathrm{Cov}(X_i,X_i)=\sigma^2) \\
&= \frac{\sigma^2}{n}. \qquad (26.2)
\end{aligned}$$

So the standard error of the mean is $\sigma/\sqrt{n}$. Quadrupling the data halves the error bar; this is the familiar rule from averaging repeated measurements.

We do not know σ, so we estimate it from the data. Two estimates are in common use:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{j=1}^{n}(x_j-\bar{x})^2 \quad \text{(plug-in)}, \qquad s^2 = \frac{1}{n-1}\sum_{j=1}^{n}(x_j-\bar{x})^2 \quad \text{(sample variance)}. \qquad (26.3)$$

Chapter 19 explained that s^2 is unbiased ($E[s^2]=\sigma^2$) while $\hat{\sigma}^2 = \frac{n-1}{n}s^2$ is slightly too small on average. The estimated standard error of the mean is

$$\widehat{\mathrm{SE}}(\bar{x}) = \frac{s}{\sqrt{n}}. \qquad (26.4)$$

Sample Problem 26.1: Standard error of a small sample

Five measurements are $x = (3, 7, 4, 9, 2)$. Find $\bar{x}$, s, $\widehat{\mathrm{SE}}(\bar{x})$ and the plug-in variance $\hat{\sigma}^2$.
Step 1: Mean. $\bar{x} = (3+7+4+9+2)/5 = 25/5 = 5$.
Step 2: Deviations $x_j-\bar{x}$: -2, 2, -1, 4, -3. Squares: $4, 4, 1, 16, 9$, sum 34.
Step 3: $s^2 = 34/4 = 8.5$, so $s = \sqrt{8.5} = 2.9155$.
Step 4: $\widehat{\mathrm{SE}} = s/\sqrt{5} = 2.9155/2.2361 = 1.3038$.
Step 5: $\hat{\sigma}^2 = 34/5 = 6.8$.

$$\boxed{\bar{x} = 5, \quad s = 2.916, \quad \widehat{\mathrm{SE}}(\bar{x}) = 1.304, \quad \hat{\sigma}^2 = 6.8}$$

26.1.3 Why we need something more general

Equation (26.4) is special to the mean. The scores in the paper are an rms error, $\sqrt{\frac{1}{n}\sum_j r_j^2}$, and a Pearson correlation, $\mathrm{corr}(a,b) = \sum_j (a_j-\bar{a})(b_j-\bar{b}) / \sqrt{\sum_j (a_j-\bar{a})^2 \sum_j (b_j-\bar{b})^2}$. Neither has a simple exact standard-error formula, and the approximate formulas found in statistics books assume Gaussian data. We want a method that works for *any* statistic you can compute. That method is the bootstrap.

Checkpoint 26.1

A held-out set is enlarged from 100 to 400 levels drawn from the same height range. If the levels were independent, by roughly what factor would the standard error of the held-out mean fluctuation shrink?

Answer: By $\sqrt{400/100} = 2$, from Eq. (26.2): the SE scales as $1/\sqrt{n}$.

Key Ideas

- Any number computed from data is a random draw from its sampling distribution; its standard deviation is the standard error.
- For iid data, $\mathrm{Var}(\bar{X}) = \sigma^2/n$ (Eq. (26.2)); estimate σ by s.
- rms errors and correlations need a more general method.

26.2 The bootstrap

26.2.1 The plug-in idea

If we knew the distribution F from which the data were drawn, we could estimate the standard error of any statistic by simulation: draw a fresh data set of size n from F, compute $\hat{\theta}$, repeat thousands of times, and take the standard deviation. We do not know F. The **bootstrap** (Efron, 1979; see [34]) replaces F by its best data-based estimate, the **empirical distribution** $\hat{F}$, which puts probability $1/n$ on each observed value $x_1, \ldots, x_n$.

Drawing n values from $\hat{F}$ means: choose one of the n data points uniformly at random, write it down, put it back, and repeat n times. This is **resampling with replacement**. The result $x^* = (x_1^*, \ldots, x_n^*)$ is a **bootstrap resample**; some original points appear twice or more, some not at all. Computing the statistic on it gives a **bootstrap replicate** $\hat{\theta}^*$.

The bootstrap principle is the assumption that

$$\text{the distribution of } \hat{\theta}^* \text{ around } \hat{\theta} \quad \approx \quad \text{the distribution of } \hat{\theta} \text{ around the true } \theta. \tag{26.5}$$

A physical analogy: you cannot rerun the universe to get another set of zeros of the same height, so you treat the hundred zeros you have as a miniature population and rerun the sampling experiment inside it.

26.2.2 The ideal bootstrap for the mean

There are n^n ordered resamples, all equally likely. The **ideal bootstrap** averages over all of them exactly. For the mean we can do this in closed form.

A single resampled value X^* equals x_j with probability $1/n$ for each j. Hence

$$E^*[X^*] = \sum_{j=1}^{n} \frac{1}{n} x_j = \bar{x}, \tag{26.6}$$

$$\mathrm{Var}^*(X^*) = \sum_{j=1}^{n} \frac{1}{n} (x_j - \bar{x})^2 = \hat{\sigma}^2, \tag{26.7}$$

where the star means "in the bootstrap world", with the data held fixed. The n resampled values are drawn independently of each other, so Eq. (26.2) applies inside the bootstrap world with σ^2 replaced by $\hat{\sigma}^2$:

$$\mathrm{Var}^*(\bar{x}^*) = \frac{\hat{\sigma}^2}{n} = \frac{n-1}{n} \cdot \frac{s^2}{n} \quad (\text{using } \hat{\sigma}^2 = \tfrac{n-1}{n} s^2, \text{ Eq. (26.3)}). \tag{26.8}$$

The bootstrap standard error of the mean is therefore $\sqrt{(n-1)/n}$ times the usual one. For $n = 100$ the factor is 0.995; for $n = 5$ it is 0.894. The bootstrap is slightly optimistic for tiny samples, a first hint that it is a large-sample method.

Sample Problem 26.2: The ideal bootstrap by enumeration

Data $x = (1, 2, 6)$. List the bootstrap distribution of the mean exactly, and check its variance against Eq. (26.8).

Step 1: A resample is fixed by how many times each value appears: counts (k_1, k_2, k_6) with $k_1 + k_2 + k_6 = 3$. The number of ordered resamples with those counts is the multinomial coefficient $3!/(k_1!\,k_2!\,k_6!)$; the total over all count patterns is $3^3 = 27$.

Step 2: The sum of a resample is $k_1 + 2k_2 + 6k_6$, and the mean is the sum divided by 3. Tabulating all ten count patterns gives Table 26.1.

Step 3: Expected mean. Multiply each resample sum by its weight and add: $3 \cdot 1 + 4 \cdot 3 + 5 \cdot 3 + 6 \cdot 1 + 8 \cdot 3 + 9 \cdot 6 + 10 \cdot 3 + 13 \cdot 3 + 14 \cdot 3 + 18 \cdot 1 = 243$. Divide by the 27 resamples, and by 3 to turn sums into means: $243/(27 \times 3) = 3$. This equals $\bar{x} = 3$, as Eq. (26.6) requires.

Step 4: Expected squared mean: $\frac{1}{27}\sum(\text{weight} \times \text{mean}^2) = 285/27 = 10.556$. So $\text{Var}^* = 10.556 - 3^2 = 1.556$.

Step 5: Formula: deviations from $\bar{x} = 3$ are $-2, -1, 3$, squares sum to 14, so $\hat{\sigma}^2 = 14/3$ and $\hat{\sigma}^2/n = 14/9 = 1.556$. They agree.

$$\boxed{\text{Var}^*(\bar{x}^*) = 14/9 = 1.556, \qquad \text{SE}_{\text{boot}} = 1.247}$$

Table 26.1: The complete bootstrap distribution of the mean for the data $(1, 2, 6)$. "Weight" is the number of the 27 equally likely ordered resamples with the given counts; the last column is the cumulative probability F^*.

counts (k_1, k_2, k_6)	example	sum	mean	weight	F^*(mean)
$(3, 0, 0)$	$1, 1, 1$	3	1	1	1/27
$(2, 1, 0)$	$1, 1, 2$	4	4/3	3	4/27
$(1, 2, 0)$	$1, 2, 2$	5	5/3	3	7/27
$(0, 3, 0)$	$2, 2, 2$	6	2	1	8/27
$(2, 0, 1)$	$1, 1, 6$	8	8/3	3	11/27
$(1, 1, 1)$	$1, 2, 6$	9	3	6	17/27
$(0, 2, 1)$	$2, 2, 6$	10	10/3	3	20/27
$(1, 0, 2)$	$1, 6, 6$	13	13/3	3	23/27
$(0, 1, 2)$	$2, 6, 6$	14	14/3	3	26/27
$(0, 0, 3)$	$6, 6, 6$	18	6	1	27/27

26.2.3 How many resamples are there?

The table has ten rows because ten count patterns exist. In general, a count pattern is a list of n non-negative integers summing to n. Picture n stars (the draws) separated by $n - 1$ bars into n bins (the data points). Every arrangement of $2n - 1$ symbols, n of them stars, gives one pattern and vice versa, so the number of distinct resamples is

$$\binom{2n-1}{n}. \tag{26.9}$$

For $n = 3$ this is $\binom{5}{3} = 10$, matching the table. For $n = 100$ held-out zeros it is $\binom{199}{100} \approx 4.5 \times 10^{58}$. Enumeration is hopeless, so in practice we use the Monte Carlo bootstrap.

26.2.4 The Monte Carlo bootstrap

Draw B resamples at random (with a seeded generator, so the result is reproducible; Chapter 19), compute $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$, and use their standard deviation as the standard error. The Monte Carlo step adds its own small error. How large is that error? We need one standard statistical result, which we take on trust.

Taken on trust

For B independent Gaussian draws with standard deviation σ, the sample variance s_B^2 has variance $2\sigma^4/(B-1)$. This follows from the fact that $(B-1)s_B^2/\sigma^2$ has a chi-squared distribution with $B-1$ degrees of freedom; the proof is found in statistics texts and is beyond this book. We apply it to bootstrap replicates, which is accurate only when their distribution is close to Gaussian.

From this result, the relative standard deviation of s_B^2 is therefore $\sqrt{2/(B-1)}$. The standard deviation is $s_B = \sqrt{s_B^2}$, and a small relative change ϵ in s_B^2 produces a relative change of about $\epsilon/2$ in s_B, because $\sqrt{1+\epsilon} \approx 1 + \epsilon/2$ to first order in Taylor's expansion. So the relative error of s_B is about $\frac{1}{2}\sqrt{2/(B-1)} \approx 1/\sqrt{2B}$. This is a heuristic: it holds when the bootstrap distribution is close to Gaussian. For $B = 2 \times 10^4$ it gives 0.5%, which is negligible next to the sampling uncertainty itself.

```
import numpy as np
rng = np.random.default_rng(0)          # seeded: reproducible
x = np.array([3., 7., 4., 9., 2.])
B = 20000
idx = rng.integers(0, len(x), size=(B, len(x)))  # B resamples of positions
means = x[idx].mean(axis=1)             # one bootstrap mean per row
print(means.std())                      # close to the exact 1.166
print(np.percentile(means, [2.5, 97.5]))  # percentile interval (see below)
```

For the five data of the first Sample Problem, Eq. (26.8) gives the exact ideal-bootstrap standard error $\sqrt{6.8/5} = \sqrt{1.36} = 1.166$, and the listing reproduces it to within about one percent.

26.2.5 Percentile intervals

A standard error summarizes spread by one number. Often we want an interval. The **percentile interval** at level $1-\alpha$ (for example 95%, $\alpha = 0.05$) takes the $\alpha/2$ and $1-\alpha/2$ quantiles of the bootstrap distribution. With the cumulative bootstrap distribution $F^*(v) = P^*(\hat{\theta}^* \le v)$, the q-quantile is the smallest value v with $F^*(v) \ge q$. With B sorted Monte Carlo replicates, it is (up to small interpolation conventions, such as the one `np.percentile` uses) the replicate at position $\lceil qB \rceil$; for $B = 2 \times 10^4$ the 2.5% point is the 500th sorted value and the 97.5% point the 19 500th.

Sample Problem 26.3: Percentile intervals from the enumeration

Using Table 26.1, find the 80% and 95% percentile intervals for the mean of $(1, 2, 6)$.
Step 1: 80% interval: quantiles 0.10 and 0.90. The smallest mean with $F^* \ge 0.10$: $F^*(1) = 1/27 = 0.037$, $F^*(4/3) = 4/27 = 0.148$. So the lower end is $4/3$.
Step 2: Upper end: $F^*(13/3) = 23/27 = 0.852 < 0.90$, $F^*(14/3) = 26/27 = 0.963 \ge 0.90$. So the upper end is $14/3$.
Step 3: 95% interval: quantiles 0.025 and 0.975. $F^*(1) = 0.037 \ge 0.025$ gives lower end 1; $F^*(14/3) = 0.963 < 0.975$ and $F^*(6) = 1$ give upper end 6.

$$\boxed{80\%: \ [4/3,\ 14/3], \qquad 95\%: \ [1,\ 6]}$$

The 95% interval is the whole range of possible means: with $n = 3$ the bootstrap has nothing to say at that level. Figure 26.1 shows the distribution.

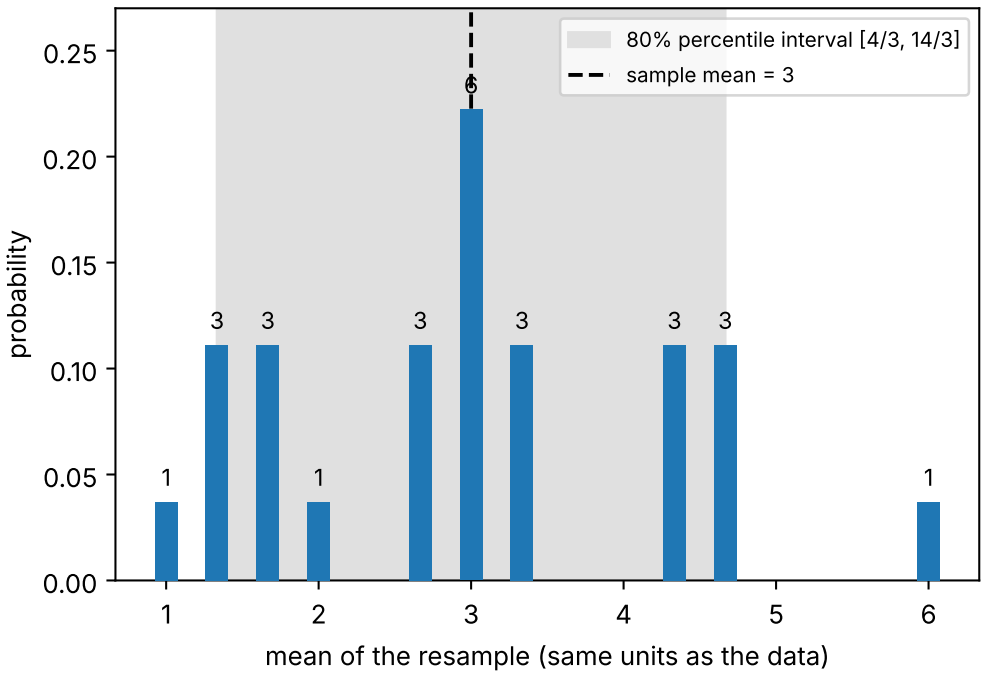


Figure 26.1: The exact bootstrap distribution of the mean of the three data $(1, 2, 6)$, from Table 26.1. Bar labels give the number of the 27 ordered resamples producing each mean. The dashed line is the sample mean 3; the shaded band is the 80% percentile interval $[4/3, 14/3]$.

Taken on trust

When does the bootstrap work? For iid data and statistics that are smooth functions of averages (the mean, the variance, the rms, and the Pearson correlation, which is a smooth function of five averages), it is a theorem (under the condition that the relevant moments of the data are finite; for the rms and the correlation, finite fourth moments) that the bootstrap distribution of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ approaches the true sampling distribution of $\sqrt{n}(\hat{\theta} - \theta)$ as $n \to \infty$, and the coverage of percentile intervals approaches the nominal level. The proofs are beyond this book; Efron and Tibshirani [34] give the ideas and references, and [31] discusses the bootstrap in a learning context. For finite n, coverage is approximate, and improved intervals (bias-corrected and accelerated, or "BCa", and others) exist [34]. The bootstrap fails, or needs modification, for very small samples (as the Sample Problem shows), for statistics like the sample maximum, and for dependent data (Sec. 26.3).

26.2.6 A bootstrap interval from the paper, rerun

Table V of the paper reports that the six-prime explicit formula (Eq. (4) of the paper with $P = \{2, 3, 5, 7, 11, 13\}$) predicts zeros 401–500 with rms 0.074 and correlation +0.96, and its caption gives the bootstrap 95% correlation interval $[+0.95, +0.97]$. We can rerun this. We restate the

ingredients so this section is self-contained (they are developed in Chapters 17 and 18). t_n are the true zeros. $N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$ is the smooth counting function (Eq. (1) of the paper), and s_n the smooth zeros solving $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$. $\mathrm{gap}_n = \mathrm{d}s_n/\mathrm{d}n$ is the local mean spacing. The truncated explicit formula (Eq. (4) of the paper) is $N_{\mathrm{fl}}(t) = -\frac{1}{\pi}\sum_{p\in P}\sum_{m=1}^{3}\frac{\sin(mt\log p)}{m\,p^{m/2}}$, and its predicted zeros solve $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$. The fluctuation of a zero (true or predicted) is $(t - s_n)/\mathrm{gap}_n$.

Sample Problem 26.4: Bootstrapping the oracle's held-out correlation

Using `qmnt`, compute the six-prime oracle's held-out rms and correlation on zeros 401–500, and a 95% percentile interval for the correlation from $B = 2 \times 10^4$ resamples of the 100 held-out levels.

Step 1: Predictions. Solve $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ for each n with the six smallest primes (`qmnt.oracle_zeros`).

Step 2: Fluctuations. Convert true and predicted zeros to fluctuations with `qmnt.fluct`.

Step 3: Score. `qmnt.score` returns rms 0.074 and correlation 0.96 (to the digits of Table V).

Step 4: Resample. Draw 100 positions with replacement, compute the correlation of the resampled pairs, repeat B times. Pairs are resampled together: truth and prediction for the same level stay attached.

Step 5: Percentiles. The 2.5% and 97.5% points of the replicates agree with the paper's $[+0.95, +0.97]$ to within 0.01 (verified in this chapter's check script; the exact third digit depends on the seed).

$$\boxed{\text{rms } 0.074,\ \text{corr } 0.96,\ \text{interval} \approx [+0.95, +0.97]}$$

```
import numpy as np, qmnt
n = np.arange(401, 501)                      # held-out indices
t = qmnt.zeros(500)[n - 1]                   # true zeros t_401..t_500
tp = qmnt.oracle_zeros(n, qmnt.first_primes(6))
ft, fp = qmnt.fluct(t, n), qmnt.fluct(tp, n)
print(qmnt.score(tp, n))                     # (rms, corr), Table V
rng = np.random.default_rng(1)
B = 20000
idx = rng.integers(0, 100, size=(B, 100))    # resampled positions
a = ft[idx] - ft[idx].mean(axis=1, keepdims=True)
b = fp[idx] - fp[idx].mean(axis=1, keepdims=True)
r = (a * b).sum(1) / np.sqrt((a * a).sum(1) * (b * b).sum(1))
print(np.percentile(r, [2.5, 97.5]))
```

26.2.7 What a held-out bootstrap measures, and what it does not

Be precise about the question the bootstrap answers in the paper. The model has already been fitted; its predictions for levels 401–500 are fixed numbers. Resampling the held-out levels asks: *if the score had been computed on a different set of 100 levels behaving like these, how much would it move?* It treats the held-out levels as a sample from a conceptual population of "levels at this height". The zeros themselves are not random; they are fixed numbers determined by the zeta function. The randomness is a modeling device that lets us express how representative a finite window is.

The held-out bootstrap does *not* include: variability from a different training set; variability from the optimizer (different random starts); or variability from a different choice of basis size. Those are probed by other means (Sec. 26.9).

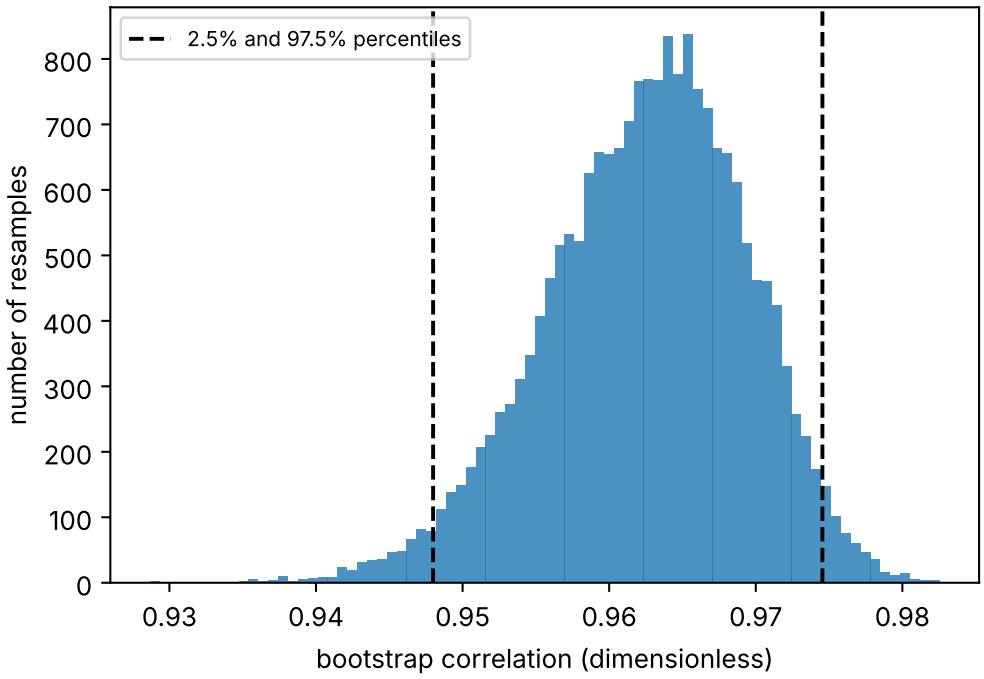


Figure 26.2: Bootstrap distribution of the held-out correlation between the six-prime explicit-formula prediction and the true fluctuations of zeros 401–500 (2×10^4 resamples of the 100 levels, seed 1). Vertical lines: the 2.5% and 97.5% percentiles, which agree with the paper's interval $[+0.95, +0.97]$ (Table V caption) to within 0.01.

Pitfall

Resample *pairs*, not the two columns separately. If you resample the true fluctuations and the predictions independently, you destroy their association, and the bootstrap correlation scatters around zero whatever the model's skill. Step 4 of the last Sample Problem keeps each prediction attached to its own level.

Checkpoint 26.2

A model's held-out rms is 0.369 with bootstrap interval $[0.32, 0.42]$, and the null's rms on the same levels is 0.263. Does the interval say the model is worse than the null?

Answer: It says the model's rms is unlikely to be as low as 0.263 on comparable levels: the whole interval lies above the null value. Strictly, a paired comparison (bootstrapping the difference of the two rms values on the same resamples) is cleaner, because the null's rms also varies from resample to resample; Problem 24 explores this.

Key Ideas

- The bootstrap replaces the unknown distribution by the empirical one and reruns the sampling experiment by resampling with replacement.
- For the mean, the ideal bootstrap variance is exactly $\hat{\sigma}^2/n$ (Eq. (26.8)).
- Percentile intervals take quantiles of the bootstrap replicates; they are approximate and need n not too small.

- A held-out bootstrap measures uncertainty due to which levels were scored, nothing else.

26.3 When neighbors are correlated

★ Did you know? *Galton Names Correlation*

In 1888 Francis Galton introduced a numerical measure of what he called co-relation, studying how the forearm length and height of people vary together. Karl Pearson soon gave the formula its modern form. When neighboring data points are correlated, they carry less independent information than their number suggests, so naive error bars come out too small.

The bootstrap of Sec. 26.2 assumes independence. The fluctuations of consecutive zeros are *not* independent: Chapter 25 showed (and the paper's Sec. III states) that they have negative autocorrelation at lags 1–3. How much does dependence change the standard error?

26.3.1 Variance of the mean of a correlated sequence

Model the sequence as **stationary**: every X_j has the same mean μ and variance γ_0, and the covariance between two terms depends only on their separation,

$$\gamma_k = \mathrm{Cov}(X_j, X_{j+k}), \qquad \rho_k = \gamma_k/\gamma_0. \tag{26.10}$$

Here γ_k is the **autocovariance** at lag k and ρ_k the **autocorrelation** (Chapter 19). Repeat the first three lines of Eq. (26.2), but now keep the off-diagonal terms:

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\gamma_{|i-j|}. \tag{26.11}$$

Count how many index pairs (i, j) have $|i - j| = k$. For $k = 0$ there are n pairs ($i = j$). For $1 \le k \le n-1$, the pairs with $j = i + k$ need $1 \le i \le n - k$, giving $n - k$ pairs, and the pairs with $i = j + k$ give another $n - k$. So

$$\begin{aligned}
\mathrm{Var}(\bar{X}) &= \frac{1}{n^2}\Big[n\gamma_0 + 2\sum_{k=1}^{n-1}(n-k)\gamma_k\Big] \\
&= \frac{\gamma_0}{n}\Big[1 + 2\sum_{k=1}^{n-1}\Big(1 - \frac{k}{n}\Big)\rho_k\Big] \quad \text{(factor out } n\gamma_0\text{)} \\
&\equiv \frac{\gamma_0}{n}\, f_n.
\end{aligned} \tag{26.12}$$

The number f_n is the **variance inflation factor**. Independent data have all $\rho_k = 0$ and $f_n = 1$, recovering Eq. (26.2). If the weighted sum $\sum_k (1 - k/n)\rho_k$ is positive, $f_n > 1$ (the data are partly redundant); if it is negative (for example when negative short-lag correlations dominate), $f_n < 1$ (errors partly cancel between neighbors). The **effective sample size** is

$$n_{\mathrm{eff}} = n/f_n, \tag{26.13}$$

the number of independent observations that would give the same variance of the mean.

26.3.2 A worked model: the AR(1) sequence

The simplest correlated sequence is the **autoregressive process of order one**, AR(1):

$$X_j = \varphi X_{j-1} + e_j, \qquad |\varphi| < 1, \tag{26.14}$$

where the e_j are independent, with mean zero and variance σ_e^2, and e_j is independent of all earlier X. Each term remembers a fraction φ of the previous one and adds fresh noise.

Variance. Take the variance of both sides of Eq. (26.14). The two terms on the right are independent, so their variances add: $\gamma_0 = \varphi^2\gamma_0 + \sigma_e^2$ (stationarity: $\mathrm{Var}(X_{j-1}) = \mathrm{Var}(X_j) = \gamma_0$). Solving,

$$\gamma_0 = \frac{\sigma_e^2}{1-\varphi^2}. \tag{26.15}$$

Autocovariance. Take $\mu = 0$ (it does not affect covariances). Multiply Eq. (26.14) by X_{j-k} for $k \geq 1$ and take expectations:

$$E[X_j X_{j-k}] = \varphi\, E[X_{j-1}X_{j-k}] + E[e_j X_{j-k}]. \tag{26.16}$$

The last term vanishes because e_j is independent of the earlier X_{j-k} and has mean zero. The left side is γ_k and the first term on the right is $\varphi\gamma_{k-1}$. So $\gamma_k = \varphi\gamma_{k-1}$, and applying this k times,

$$\gamma_k = \varphi^k \gamma_0, \qquad \rho_k = \varphi^k. \tag{26.17}$$

Inflation factor for large n. When n is large, the weights $1 - k/n$ are close to 1 for all k where φ^k is not negligible, so

$$f_\infty = 1 + 2\sum_{k=1}^{\infty} \varphi^k = 1 + \frac{2\varphi}{1-\varphi} = \frac{1+\varphi}{1-\varphi}, \tag{26.18}$$

using the geometric series $\sum_{k\geq 1}\varphi^k = \varphi/(1-\varphi)$ (Chapter 3).

Inflation factor for finite n. We need also $\sum_{k\geq 1} k\varphi^k$. Call it S. Then $\varphi S = \sum_{k\geq 1} k\varphi^{k+1} = \sum_{k\geq 2}(k-1)\varphi^k$. Subtracting term by term,

$$S - \varphi S = \sum_{k\geq 1}[k-(k-1)]\varphi^k = \sum_{k\geq 1}\varphi^k = \frac{\varphi}{1-\varphi}, \qquad \text{so} \qquad S = \frac{\varphi}{(1-\varphi)^2}. \tag{26.19}$$

Next, replace the finite sums $\sum_{k=1}^{n-1}$ by infinite ones. This adds the tail $\sum_{k\geq n}(1-k/n)\varphi^k$. Write $k = n + j$, so that $|1 - k/n| = j/n$. The magnitude of the tail is then at most $\sum_{j\geq 0}(j/n)\,|\varphi|^{n+j} = \frac{|\varphi|^{n+1}}{n(1-|\varphi|)^2}$, using Eq. (26.19) with $|\varphi|$ in place of φ. This is negligible when $|\varphi|^n \ll 1$. Making the replacement,

$$f_n \approx 1 + 2\Big[\frac{\varphi}{1-\varphi} - \frac{1}{n}\,\frac{\varphi}{(1-\varphi)^2}\Big]. \tag{26.20}$$

Sample Problem 26.5: How much does correlation change the error bar?

An AR(1) sequence has $n = 200$ terms. Find f_n, the factor by which the true SE of the mean differs from the naive $\sqrt{\gamma_0/n}$, and n_{eff}, for (a) $\varphi = 0.5$ and (b) $\varphi = -0.5$.
Step 1 (a): $\varphi/(1-\varphi) = 0.5/0.5 = 1$ and $\varphi/(1-\varphi)^2 = 0.5/0.25 = 2$.
Step 2 (a): Eq. (26.20): $f_{200} = 1 + 2[1 - 2/200] = 1 + 2(0.99) = 2.98$. SE ratio $\sqrt{2.98} = 1.726$. $n_{\text{eff}} = 200/2.98 = 67.1$.
Step 3 (b): $\varphi/(1-\varphi) = -0.5/1.5 = -1/3$ and $\varphi/(1-\varphi)^2 = -0.5/2.25 = -0.2222$.
Step 4 (b): $f_{200} = 1 + 2[-0.33333 + 0.2222/200] = 1 + 2(-0.33222) = 0.3356$. SE ratio

$\sqrt{0.3356} = 0.579$. $n_{\text{eff}} = 200/0.3356 = 596$.

(a) $f = 2.98$, $n_{\text{eff}} = 67$; (b) $f = 0.336$, $n_{\text{eff}} = 596$

The large-n limits from Eq. (26.18) are 3 and 1/3. Positive correlation makes 200 points worth 67; negative correlation makes them worth nearly 600, *for the mean*.

26.3.3 What the independent bootstrap gets wrong

The independent bootstrap resamples points as if they were iid. Inside the bootstrap world the resampled points really are independent, so it reports $\sqrt{\hat{\sigma}^2/n} \approx \sqrt{\gamma_0/n}$ and misses the factor $\sqrt{f_n}$ entirely. With positive correlation the bootstrap interval is too narrow (overconfident); with negative correlation it is too wide (conservative). Figure 26.3 shows sample paths and the size of the effect.

For the paper, two cautions apply. First, the zero fluctuations are negatively correlated at short lags, which for the *mean* would make the independent bootstrap conservative. Second, the paper's scores are rms errors and correlations of prediction errors, not means of the fluctuations, and Eq. (26.12) does not apply to them directly. The paper did not guess: it reran the intervals with a block bootstrap and reports that they "widen slightly and no conclusion changes" (Sec. III). The next section explains the method.

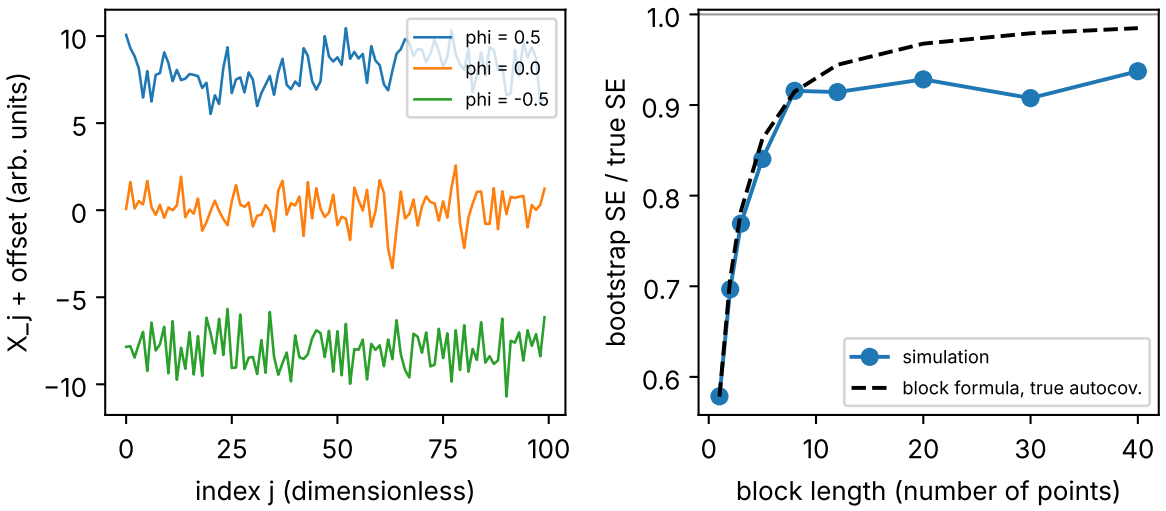


Figure 26.3: Left: sample paths of AR(1) sequences (Eq. (26.14), unit noise variance) with $\varphi = 0.5$, 0 and -0.5, offset vertically. Positive φ gives slow wandering; negative φ gives zig-zag. Right: for $\varphi = 0.5$, $n = 400$, the average circular-block-bootstrap standard error of the mean divided by the true one (Eq. (26.12)), against block length ℓ (40 simulated series, 300 resamples each). $\ell = 1$ is the independent bootstrap, short by about the factor $1/\sqrt{f_n}$. The dashed curve is the exact block formula Eq. (26.22) evaluated with the true autocovariances.

Checkpoint 26.3

The fluctuation sequence of the zeros has negative autocorrelation at lags 1–3 (and, suppose, negligible correlation beyond). Would the independent bootstrap overstate or understate the uncertainty of the *mean* fluctuation of a block of zeros?

Answer: Overstate. Negative short-lag autocorrelation gives $f_n < 1$ in Eq. (26.12), so the true variance of the mean is smaller than the independent formula.

Key Ideas

- For a stationary sequence, $\mathrm{Var}(\bar{X}) = (\gamma_0/n) f_n$ with $f_n = 1 + 2\sum_k (1 - k/n)\rho_k$ (Eq. (26.12)).
- AR(1): $\rho_k = \varphi^k$ and $f_\infty = (1+\varphi)/(1-\varphi)$.
- The independent bootstrap ignores f_n: overconfident for positive, conservative for negative correlation (for the mean).

26.4 The block bootstrap

26.4.1 Resampling blocks instead of points

If short-range dependence is the problem, keep short stretches of the data intact. Choose a **block length** ℓ, cut the sequence into overlapping blocks of ℓ consecutive observations, and resample whole blocks.

Moving block bootstrap. The blocks are $\mathcal{B}_j = (x_j, x_{j+1}, \ldots, x_{j+\ell-1})$ for $j = 1, \ldots, n-\ell+1$. Draw $b = \lceil n/\ell \rceil$ blocks uniformly with replacement, concatenate them, and keep the first n values.

Circular block bootstrap. Wrap the data around a circle, so that after x_n comes x_1 again. Now a block can start at any $j = 1, \ldots, n$, and its entries are $x_j, x_{j+1}, \ldots$ with indices taken modulo n. Resampling proceeds as before.

Figure 26.4 draws both for $n = 12$, $\ell = 3$. Inside each block the original neighbors stay neighbors, so the block carries the short-range correlations; only at the joins between blocks is the order broken. These methods were introduced and analyzed in the 1980s and 1990s (the moving block version by H. Künsch, the circular version by D. Politis and J. Romano).

Sample Problem 26.6: Counting blocks

A held-out set has $n = 100$ levels and the block length is $\ell = 5$. (a) How many blocks are available in each method, and how many are drawn per resample? (b) In the moving block version, how many blocks contain observation 1, observation 3, and observation 50? (c) What is the expected number of copies of observation 1 and of observation 50 in one resample, in each method?

Step 1 (a): Moving: $n - \ell + 1 = 96$ blocks. Circular: $n = 100$. Drawn per resample: $\lceil 100/5 \rceil = 20$.

Step 2 (b): Observation i lies in block $\mathcal{B}_j$ when $j \le i \le j + 4$, i.e. $i - 4 \le j \le i$, and also $1 \le j \le 96$. The count is $\min(i, 96) - \max(1, i-4) + 1$. For $i = 1$: $1 - 1 + 1 = 1$. For $i = 3$: $3 - 1 + 1 = 3$. For $i = 50$: $50 - 46 + 1 = 5$.

Step 3 (c), moving: Each drawn block contains observation i with probability (number of blocks containing i)/96, and contributes one copy when it does. Twenty independent blocks give expected copies $20 \times 1/96 = 0.208$ for $i = 1$ and $20 \times 5/96 = 1.042$ for $i = 50$.

Step 4 (c), circular: Every observation is in exactly 5 of the 100 blocks: expected copies $20 \times 5/100 = 1$ for every i.

moving: 0.208 vs 1.042 copies; circular: 1 copy for every observation

The moving version under-represents the ends; the circular version treats every observation equally, which is why it is often preferred.

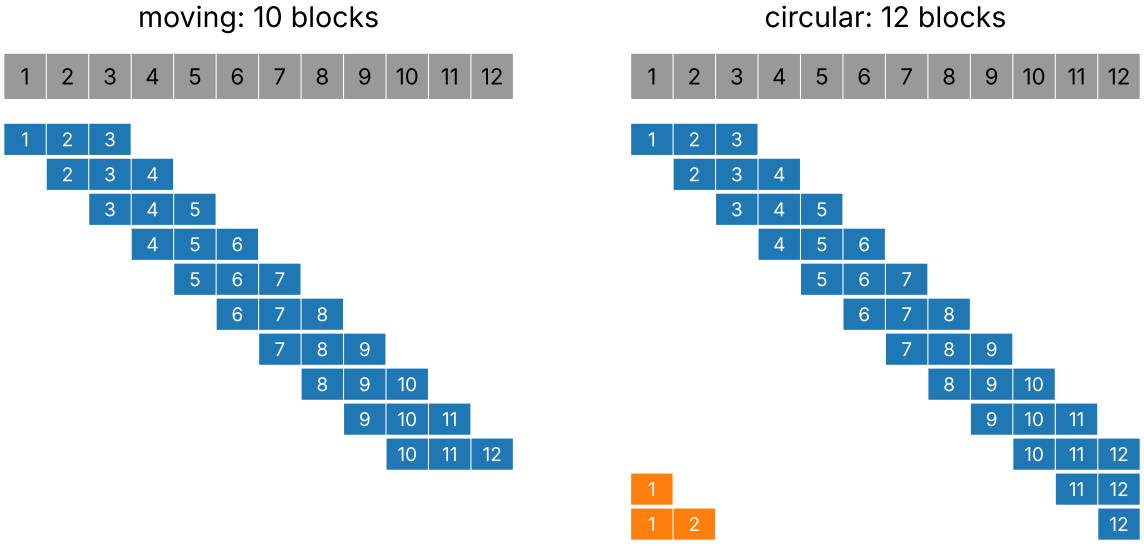


Figure 26.4: Blocks of length $\ell = 3$ for $n = 12$ observations (gray boxes, numbered). Each row below is one block available for resampling. Left: the moving block bootstrap has $n - \ell + 1 = 10$ blocks, so observations near the ends belong to fewer blocks. Right: the circular block bootstrap wraps around (orange cells), giving $n = 12$ blocks and putting every observation in exactly ℓ of them.

26.4.2 What the block bootstrap does to the variance of a mean

For the mean we can compute the circular block bootstrap variance exactly, which shows what the method captures. Assume $n = b\ell$ exactly, so b blocks fill the resample with no truncation.

Step (i). The bootstrap mean is the average of the b block means: $\bar{x}^* = \frac{1}{b}\sum_{i=1}^{b} \bar{\mathcal{B}}_i^*$, where $\bar{\mathcal{B}}_i^*$ is the mean of the i-th drawn block.

Step (ii). The drawn blocks are independent, so $\mathrm{Var}^*(\bar{x}^*) = \frac{1}{b}\mathrm{Var}^*(\bar{\mathcal{B}}^*)$ (Eq. (26.2) with b terms).

Step (iii). A drawn block mean equals $\bar{\mathcal{B}}_j$ with probability $1/n$ for each start j. The average of the $\bar{\mathcal{B}}_j$ over j is $\bar{x}$, because every observation sits in exactly ℓ blocks with weight $1/\ell$ in each. Hence $\mathrm{Var}^*(\bar{\mathcal{B}}^*) = \frac{1}{n}\sum_{j=1}^{n}(\bar{\mathcal{B}}_j - \bar{x})^2$.

Step (iv). Write $y_j = x_j - \bar{x}$ (indices modulo n). Then $\bar{\mathcal{B}}_j - \bar{x} = \frac{1}{\ell}\sum_{a=0}^{\ell-1} y_{j+a}$ and

$$\frac{1}{n}\sum_{j=1}^{n}(\bar{\mathcal{B}}_j - \bar{x})^2 = \frac{1}{\ell^2}\sum_{a=0}^{\ell-1}\sum_{c=0}^{\ell-1}\Big[\frac{1}{n}\sum_{j=1}^{n} y_{j+a}y_{j+c}\Big] = \frac{1}{\ell^2}\sum_{a,c}\tilde{\gamma}_{|c-a|}, \tag{26.21}$$

where $\tilde{\gamma}_k = \frac{1}{n}\sum_{j=1}^{n} y_j y_{j+k}$ is the **circular sample autocovariance**. (Shifting the summation index j around the circle does not change the inner sum, so it depends only on $c - a$; and $\tilde{\gamma}_k = \tilde{\gamma}_{-k}$.)

Step (v). Counting pairs (a, c) with $|c - a| = k$ exactly as in Eq. (26.12), but with ℓ in place of n: $\frac{1}{\ell^2}\sum_{a,c}\tilde{\gamma}_{|c-a|} = \frac{1}{\ell}\big[\tilde{\gamma}_0 + 2\sum_{k=1}^{\ell-1}(1 - k/\ell)\tilde{\gamma}_k\big]$.

Step (vi). Combine with (ii) and $b = n/\ell$:

$$\mathrm{Var}^*_{\mathrm{CBB}}(\bar{x}^*) = \frac{1}{n}\Big[\tilde{\gamma}_0 + 2\sum_{k=1}^{\ell-1}\Big(1 - \frac{k}{\ell}\Big)\tilde{\gamma}_k\Big]. \tag{26.22}$$

Compare with Eq. (26.12). The block bootstrap uses the data's own autocovariances up to

lag $\ell - 1$, with weights $1 - k/\ell$ instead of $1 - k/n$, and drops all lags $\geq \ell$. For $\ell = 1$ it reduces to $\tilde{\gamma}_0/n = \hat{\sigma}^2/n$, the independent bootstrap. Two consequences follow:

- ℓ must be longer than the range over which correlations matter, or they are cut off (bias).
- ℓ must be much shorter than n, so that there are many blocks to resample (otherwise the estimate is noisy, and the weights $1 - k/\ell$ still discount real correlations).

For the AR(1) with $\varphi = 0.5$, $n = 400$ and $\ell = 20$, inserting the true autocovariances into Eq. (26.22) gives a factor $1 + 2(1 - 2/20) = 2.8$ instead of the true $f_{400} = 1 + 2(1 - 2/400) = 2.99$ (Eq. (26.20)), so the block standard error is about 3% low, while the independent bootstrap is about 42% low (factor $1/\sqrt{2.99} = 0.578$). The right panel of Fig. 26.3 compares these predictions with a simulation. The simulated ratios sit close to the dashed curve, slightly below it, because the bootstrap uses autocovariances measured about the sample mean.

In practice nobody knows the right ℓ, so the honest procedure is to report a range: compute intervals for several block lengths and check whether conclusions change. That is exactly what the paper does with "block lengths 3 to 20" (Sec. III; Appendix A step 22).

```python
import numpy as np

def cbb_indices(n, ell, rng):
    """Positions for one circular-block-bootstrap resample."""
    nb = -(-n // ell)                          # ceil(n / ell) blocks
    starts = rng.integers(0, n, nb)            # any start; wrap around
    idx = (starts[:, None] + np.arange(ell)[None, :]) % n
    return idx.ravel()[:n]                     # truncate to length n

rng = np.random.default_rng(2)
x = rng.normal(size=100)                       # stand-in data
reps = [x[cbb_indices(100, 5, rng)].mean() for _ in range(5000)]
print(np.std(reps))
```

Taken on trust

For stationary sequences whose correlations die out fast enough, the moving and circular block bootstraps are consistent provided $\ell \to \infty$ and $\ell/n \to 0$ as $n \to \infty$. The proofs, and rules for choosing ℓ (it should grow roughly like $n^{1/3}$ for variance estimation), are in the specialized literature and beyond this book. For $n = 100$ held-out levels and correlations lasting a few lags, block lengths 3 to 20 span the plausible range around the $n^{1/3} \approx 5$ rule of thumb ($100^{1/3} = 4.64$).

Pitfall

A block bootstrap on $n = 100$ points with $\ell = 20$ draws only five blocks per resample. Its replicates are then made from a handful of pieces and the interval can be erratic. Report a range of ℓ, not one hand-picked value.

Checkpoint 26.4

What does the circular block bootstrap with $\ell = 1$ reduce to?

Answer: The ordinary independent bootstrap: blocks of one point resampled with replacement. Eq. (26.22) then gives $\tilde{\gamma}_0/n = \hat{\sigma}^2/n$, as in Eq. (26.8).

Key Ideas

- Block bootstraps resample stretches of ℓ consecutive points, preserving dependence within blocks.
- The circular version wraps around so every point appears in exactly ℓ blocks.
- For the mean, the circular block variance is Eq. (26.22): autocovariances up to lag $\ell - 1$, weighted by $1 - k/\ell$ (triangular, or Bartlett, weights).
- Choose ℓ longer than the correlation range, much shorter than n; report sensitivity over a range.

26.5 Permutation tests

Uncertainty intervals tell you how precisely a score is known. A different kind of question is: *could this result have arisen if there were no effect at all?* Statistics answers it with a test.

26.5.1 The vocabulary of a test

- The **null hypothesis** H_0 is a precise statement of "no effect", precise enough that you can compute (or simulate) what data would look like under it.
- The **test statistic** T is a number computed from the data that tends to be large when there is an effect.
- The **p-value** is the probability, computed assuming H_0, of a statistic at least as extreme as the one observed: $p = P_{H_0}(T \geq T_{\text{obs}})$.

A small p means the data would be surprising under H_0. It is *not* the probability that H_0 is true; it is a statement about data given a hypothesis.

A note on symbols. Several letters do double duty in this chapter, as in the paper. T is a test statistic here, but the length of a training window in t in Sec. 26.8. N in Eq. (26.23) is a number of surrogates, not a grid size or the counting function. p is a p-value, except in p^m and $\log p$, where it is a prime. q is a DFT index, a quantile level, a prime power or an exceedance probability, depending on context. μ is a mean (not the ridge strength), ρ_k an autocorrelation (not a zeta zero), and ϕ_q a random phase (not a basis function). The surrounding text always says which meaning is intended.

26.5.2 Exchangeability and the permutation idea

A **permutation test** (introduced by R. A. Fisher in the 1930s) builds the null distribution from the data themselves. Suppose H_0 says that some labels attached to the data are irrelevant: for example, "treatment" and "control" labels in an experiment, or the pairing between a prediction and a true value. Then under H_0 every rearrangement of the labels was equally likely to have produced the data we see. This property is called **exchangeability**. Computing T for every rearrangement gives the exact distribution of T under H_0, and the p-value is the fraction of rearrangements with $T \geq T_{\text{obs}}$.

Sample Problem 26.7: An exact permutation test

Group A has values $(3, 6, 8)$ and group B has $(1, 2, 4)$. Test H_0: "the group labels are irrelevant" with statistic $D = \bar{x}_A - \bar{x}_B$.
Step 1: Observed: $\bar{x}_A = 17/3$, $\bar{x}_B = 7/3$, $D_{\text{obs}} = 10/3 = 3.333$.

Step 2: Pool the six values $\{1, 2, 3, 4, 6, 8\}$, total 24. Under H_0 any 3 of the 6 could have been group A: $\binom{6}{3} = 20$ equally likely relabelings.
Step 3: If group A has sum S_A, then $D = \frac{S_A - (24 - S_A)}{3} = \frac{2S_A - 24}{3}$. So $D \geq 10/3$ exactly when $S_A \geq 17$.
Step 4: Three-element subsets with sum ≥ 17: $\{4, 6, 8\}$ (18) and $\{3, 6, 8\}$ (17). The next largest, $\{2, 6, 8\}$, has 16. Two subsets.
Step 5: One-sided $p = 2/20 = 0.1$. For a two-sided test count also $D \leq -10/3$, i.e. $S_A \leq 7$: $\{1, 2, 3\}$ (6) and $\{1, 2, 4\}$ (7). Two-sided $p = 4/20 = 0.2$.

$$\boxed{p_{\text{one-sided}} = 0.10, \qquad p_{\text{two-sided}} = 0.20}$$

With three values per group, even a clear-looking separation is not surprising: one relabeling in ten does as well.

26.5.3 Testing a correlation

To test whether a prediction a_j is associated with the truth b_j, take H_0: "the pairing is irrelevant". Randomly permute the b_j among the levels, recompute the correlation, and repeat. The fraction of permuted correlations at least as large as the observed one is the p-value.

Be careful about what this H_0 includes. Permuting the b_j assumes that, absent association, *every* ordering of the b_j was equally likely. If the true fluctuations are serially correlated (they are) and the predictions are also serially correlated (for example, a smooth interpolation), then unpermuted pairs can share slow trends that random permutations destroy. A permutation test can then report "significant" association produced by shared serial structure rather than by skill. The cure is the same as for the bootstrap: respect the dependence (for example, by circularly shifting one sequence relative to the other instead of shuffling it), or use surrogates that preserve the serial structure (Sec. 26.6).

26.5.4 Monte Carlo tests and the size of an ensemble

Usually there are too many rearrangements to enumerate, so we draw N random ones (or N surrogate data sets). Let r be the number of them whose statistic is at least T_{obs}. The Monte Carlo p-value is

$$p = \frac{1 + r}{N + 1}. \tag{26.23}$$

Why $1 + r$ *over* $N + 1$*?* Under H_0, the real data and the N simulated ones are generated by the same mechanism, so the real statistic is equally likely to have any of the $N + 1$ ranks in the combined list (ignoring ties). The probability that it ranks among the top $r + 1$ is therefore $(r + 1)/(N + 1)$. This is the probability of a result at least as extreme as the one observed, which is the definition of p.

An immediate consequence: **the smallest p-value an ensemble of size N can ever produce is** $1/(N + 1)$. Even if no simulated statistic comes close, you can claim no more.

Sample Problem 26.8: What an ensemble of 200 can show

A statistic is computed on real data and on 200 surrogate data sets; none of the 200 reaches the real value. What is the Monte Carlo p-value? If the surrogates had numbered 100?
Step 1: $r = 0$, $N = 200$: $p = 1/201 = 0.004975$.
Step 2: $N = 100$: $p = 1/101 = 0.0099$.
Step 3: Interpretation: the result is limited by the ensemble size, not by how far the

real value is from the surrogates. To claim $p < 10^{-4}$ you need at least 10^4 surrogates.

$$\boxed{p = 1/201 \approx 0.005 \quad (N = 200), \qquad p = 1/101 \approx 0.010 \quad (N = 100)}$$

Pitfall

"$p = 0.01$" does not mean "the null hypothesis has a 1% chance of being true", nor "the effect is large". It means that data this extreme would arise in about 1% of experiments *if* H_0 held. And it is only meaningful if the test was chosen before seeing the data (Sec. 26.8).

Checkpoint 26.5

In a permutation test of a correlation, which hypothesis is being tested: "the model has no skill" or "the truth values are exchangeable with respect to the predictions"?

Answer: The second. "No skill" is translated into exchangeability. If exchangeability fails for another reason (serial correlation), the test can reject even when the model has no skill.

Key Ideas

- A test needs a precise null hypothesis, a statistic, and a p-value $P_{H_0}(T \geq T_{\text{obs}})$.
- Permutation tests generate the null distribution by rearranging labels; they test exchangeability.
- Monte Carlo p-value: $(1+r)/(N+1)$; an ensemble of size N cannot show $p < 1/(N+1)$.

26.6 Surrogate data

26.6.1 The idea

A permutation test destroys *all* order in the data. Sometimes that is too much: you want to destroy one specific property while keeping others. **Surrogate data** are synthetic data sets constructed to share chosen properties $\mathcal{P}$ with the real data while lacking a property $\mathcal{Q}$ under investigation [16]. The null hypothesis becomes: "the real data are a typical member of the class of sequences with properties $\mathcal{P}$". If an analysis cannot tell the real data from their surrogates, then whatever it measures is explained by $\mathcal{P}$ alone and is not evidence for $\mathcal{Q}$.

We describe three families, each preserving different things. Throughout, $d_1, \ldots, d_n$ is a real sequence, for instance the fluctuations of zeros in mean spacings.

26.6.2 Random permutation

Reorder $d_1, \ldots, d_n$ randomly. The **marginal distribution** (the histogram of values, ignoring order) is preserved exactly, because the surrogate contains the very same numbers. All serial structure is destroyed: on average the surrogate's autocorrelation at any nonzero lag is close to zero (precisely, $-1/(n-1)$ for a mean-zero sequence; Problem 8).

26.6.3 Phase randomization

Recall from Chapter 6 the DFT of a length-n sequence,

$$D_q = \sum_{j=0}^{n-1} d_j \, e^{-2\pi i j q/n}, \qquad d_j = \frac{1}{n} \sum_{q=0}^{n-1} D_q \, e^{2\pi i j q/n}, \tag{26.24}$$

(indices from 0 here, as in `numpy`). For real d, replacing q by $n-q$ gives $e^{-2\pi i j(n-q)/n} = e^{-2\pi i j} e^{2\pi i j q/n} = e^{2\pi i j q/n}$, hence $D_{n-q} = \sum_j d_j e^{2\pi i j q/n}$. On the other hand, conjugating D_q term by term gives $D_q^* = \sum_j d_j^* e^{2\pi i j q/n} = \sum_j d_j e^{2\pi i j q/n}$, because d_j is real. The two are equal:

$$D_{n-q} = D_q^*, \tag{26.25}$$

where the star is the complex conjugate. In particular $D_0 = \sum_j d_j$ is real, and for even n, $D_{n/2} = \sum_j (-1)^j d_j$ is real.

The **phase-randomized surrogate** (Theiler and co-workers [16]) multiplies each D_q by a random phase factor $e^{i\phi_q}$, ϕ_q uniform in $[0, 2\pi)$, with two constraints that keep the result real: $\phi_{n-q} = -\phi_q$ (so Eq. (26.25) still holds), and $\phi_0 = 0$ (and $\phi_{n/2} = 0$ for even n), so the real coefficients stay real. Then invert the DFT. Every magnitude $|D_q|$ is unchanged.

What is preserved. The mean (because D_0 is untouched), the power spectrum $|D_q|^2$ by construction, and hence the circular autocorrelation. To see the last claim, define the circular lag-product sum (an unnormalized, uncentered autocorrelation) $c_k = \sum_{j=0}^{n-1} d_j d_{(j+k) \bmod n}$ and compute its DFT:

$$C_q = \sum_{k=0}^{n-1} c_k \, e^{-2\pi i k q/n} = \sum_{j=0}^{n-1} d_j \sum_{k=0}^{n-1} d_{(j+k) \bmod n} \, e^{-2\pi i k q/n}. \tag{26.26}$$

Put $l = (j+k) \bmod n$. As k runs over $0, \ldots, n-1$ with j fixed, l runs over the same values once each, and $e^{-2\pi i k q/n} = e^{-2\pi i (l-j) q/n}$ (adding a multiple of n to the exponent's numerator changes nothing, since $e^{-2\pi i q} = 1$). So

$$C_q = \Big(\sum_j d_j e^{2\pi i j q/n} \Big) \Big(\sum_l d_l e^{-2\pi i l q/n} \Big) = D_q^* D_q = |D_q|^2. \tag{26.27}$$

This is the discrete Wiener–Khinchin theorem of Chapter 6. Since c_k is recovered from C_q by the inverse DFT, two sequences with the same $|D_q|$ have the same c_k at every lag. By Parseval's theorem, $\sum_j d_j^2 = c_0 = \frac{1}{n} \sum_q |D_q|^2$, so the variance is preserved too. Finally, with $\bar{d}$ the mean, $\sum_j (d_j - \bar{d})(d_{(j+k) \bmod n} - \bar{d}) = c_k - n\bar{d}^2$, so the centered, normalized circular autocorrelation $(c_k - n\bar{d}^2)/(c_0 - n\bar{d}^2)$ (the circular version of ρ_k) is built from preserved quantities and is preserved as well.

What is destroyed. The specific phases: *where* each oscillation is at each index, and how oscillations at different frequencies line up. A sawtooth, for example, owes its shape to harmonics locked in phase; randomizing the phases turns it into a smooth, irregular wiggle with the same spectrum. The marginal distribution generally changes; a sum of many components with random phases tends to look Gaussian (see the box).

Taken on trust

The tendency of a sum of many independent random contributions to have a nearly Gaussian distribution is the central limit theorem. We use it here only qualitatively and take it on trust; [22] states and discusses it, and full proofs are in probability texts.

Sample Problem 26.9: Phase randomization by hand

Let $d = (1, 2, 0, -1)$, $n = 4$. Compute its DFT, rotate the phase of D_1 by $\phi = \pi/4$ (and D_3 by $-\pi/4$), invert, and check that the circular autocorrelation is unchanged.
Step 1: With $n = 4$, $e^{-2\pi i jq/4} = (-i)^{jq}$. $D_0 = 1 + 2 + 0 - 1 = 2$. $D_2 = 1 - 2 + 0 + 1 = 0$. $D_1 = 1 + 2(-i) + 0(-1) + (-1)(i) = 1 - 3i$. $D_3 = D_1^* = 1 + 3i$.
Step 2: For real d with these symmetries the inverse DFT simplifies to $d_j = \frac{1}{4}[D_0 + 2\,\mathrm{Re}(D_1 i^j) + D_2(-1)^j]$. Check: $j = 1$: $D_1 i = i + 3$, real part 3, $d_1 = \frac{1}{4}(2 + 6) = 2$. Correct.
Step 3: New coefficient $D_1' = (1-3i)e^{i\pi/4} = (1-3i)\frac{1+i}{\sqrt{2}} = \frac{1+i-3i+3}{\sqrt{2}} = \frac{4-2i}{\sqrt{2}} = 2\sqrt{2} - \sqrt{2}\,i$. Its magnitude is $\sqrt{8+2} = \sqrt{10} = |D_1|$.
Step 4: Write $D_1' = a + ib$ with $a = 2\sqrt{2}$, $b = -\sqrt{2}$. Then $\mathrm{Re}(D_1' i^j)$ is a, $-b$, $-a$, b for $j = 0, 1, 2, 3$, and $d_j' = \frac{1}{4}(2 + 2\,\mathrm{Re}(D_1' i^j))$:

$$d' = (\tfrac{1}{2} + \sqrt{2},\ \tfrac{1}{2} + \tfrac{\sqrt{2}}{2},\ \tfrac{1}{2} - \sqrt{2},\ \tfrac{1}{2} - \tfrac{\sqrt{2}}{2}) = (1.914,\ 1.207,\ -0.914,\ -0.207).$$

Step 5: Autocorrelation of d: $c_0 = 1 + 4 + 0 + 1 = 6$; $c_1 = d_0d_1 + d_1d_2 + d_2d_3 + d_3d_0 = 2 + 0 + 0 - 1 = 1$; $c_2 = d_0d_2 + d_1d_3 + d_2d_0 + d_3d_1 = 0 - 2 + 0 - 2 = -4$; $c_3 = c_1 = 1$.
Step 6: Autocorrelation of d': $c_0' = (\frac{1}{2} + \sqrt{2})^2 + (\frac{1}{2} + \frac{\sqrt{2}}{2})^2 + (\frac{1}{2} - \sqrt{2})^2 + (\frac{1}{2} - \frac{\sqrt{2}}{2})^2 = 3.664 + 1.457 + 0.836 + 0.043 = 6.000$. The check script confirms $c_1' = 1$ and $c_2' = -4$ as well.

$$\boxed{d' = (1.914,\ 1.207,\ -0.914,\ -0.207), \quad (c_0, c_1, c_2) = (6, 1, -4) \text{ for both}}$$

The values changed (the marginal is different: $\{1, 2, 0, -1\}$ became four irrational numbers), the mean ($\frac{1}{2}$) and autocorrelation did not.

26.6.4 Amplitude-adjusted surrogates: IAAFT

Sometimes we want a surrogate that keeps *both* the marginal and the spectrum. In general no sequence other than a rearrangement of the original has exactly the original values, and almost no rearrangement has exactly the original spectrum, so both cannot usually hold at once. Schreiber and Schmitz [17] proposed an iteration that keeps the marginal exactly and the spectrum approximately: the **iterative amplitude-adjusted Fourier transform** (IAAFT).

1. Store the original Fourier magnitudes $|D_q|$ and the sorted original values.
2. Start from a random permutation y of d.
3. *Spectrum step:* take the DFT of y, keep its phases, replace its magnitudes by $|D_q|$, invert.
4. *Rank step:* replace the smallest value of the result by the smallest original value, the second smallest by the second smallest, and so on ("rank-order remapping"). Now the values are exactly the original ones.
5. Repeat steps 3 and 4 until the spectrum stops improving (the paper uses 200 iterations, Appendix A step 18).

Ending on a rank step makes the marginal exact; the spectrum is then close but not exact (the paper reports agreement "to 1%", Sec. III).

```
import numpy as np

def phase_randomize(d, rng):
    D = np.fft.rfft(d)                          # q = 0 .. n//2
```

```
    ph = rng.uniform(0, 2*np.pi, len(D))
    ph[0] = 0.0                              # keep the mean
    if len(d) % 2 == 0:
        ph[-1] = 0.0                         # keep Nyquist term real
    return np.fft.irfft(D * np.exp(1j*ph), n=len(d))

def iaaft(d, rng, iters=200):
    amp = np.abs(np.fft.rfft(d))
    srt = np.sort(d)
    y = rng.permutation(d)
    for _ in range(iters):
        Y = np.fft.rfft(y)
        y = np.fft.irfft(amp * np.exp(1j*np.angle(Y)), n=len(d))
        y = srt[np.argsort(np.argsort(y))]   # rank-order remap
    return y
```

(`np.fft.rfft` returns only $q = 0, \ldots, \lfloor n/2 \rfloor$; `irfft` fills in the rest using Eq. (26.25), so the constraint $\phi_{n-q} = -\phi_q$ is automatic.)

Table 26.2 summarizes, and Fig. 26.5 shows all three applied to running example RE2, the prime-2 wave. For a zero sequence, the prime-2 term alone moves the n-th zero to first order by δ with $N'_{\mathrm{sm}}(s_n)\delta + N_{\mathrm{fl}}^{p=2}(s_n) = 0$; since $N'_{\mathrm{sm}}(s_n) = 1/\mathrm{gap}_n$, the fluctuation in mean spacings is $\delta/\mathrm{gap}_n = -N_{\mathrm{fl}}^{p=2}(s_n)$, where $N_{\mathrm{fl}}^{p=2}(t) = -\frac{1}{\pi}\sum_{m=1}^{3} \frac{\sin(mt\log 2)}{m2^{m/2}}$.

Table 26.2: What each surrogate keeps (✓), keeps approximately (≈), or destroys (×).

property	permutation	phase-randomized	IAAFT
marginal distribution (set of values)	✓	×	✓
mean and variance	✓	✓	✓
power spectrum / circular autocorrelation	×	✓	≈
specific phases (where each oscillation is)	×	×	×

Pitfall

The DFT treats a sequence as one period of a periodic signal. If d_0 and d_{n-1} are far apart, the periodic extension has a jump, the jump contributes spurious high-frequency power, and the phase-randomized surrogates inherit that power. For zero *positions* (as opposed to fluctuations) there is a further constraint: a zero sequence must be increasing, and the paper's step 23 uses "sorted phase-randomized surrogates", presumably for this reason (the construction is documented in the reconstruction record, not in the paper).

Checkpoint 26.6

You want to test whether an analysis responds to the *phases* of the zero fluctuations, not merely to their spectrum. Which surrogate do you use?

Answer: A phase-randomized (or IAAFT) surrogate: it keeps the spectrum and scrambles the phases, so any difference between real and surrogate results must come from the phases.

Key Ideas

- Surrogates keep chosen properties $\mathcal{P}$ and destroy a property $\mathcal{Q}$; the null is "real data are typical of class $\mathcal{P}$".

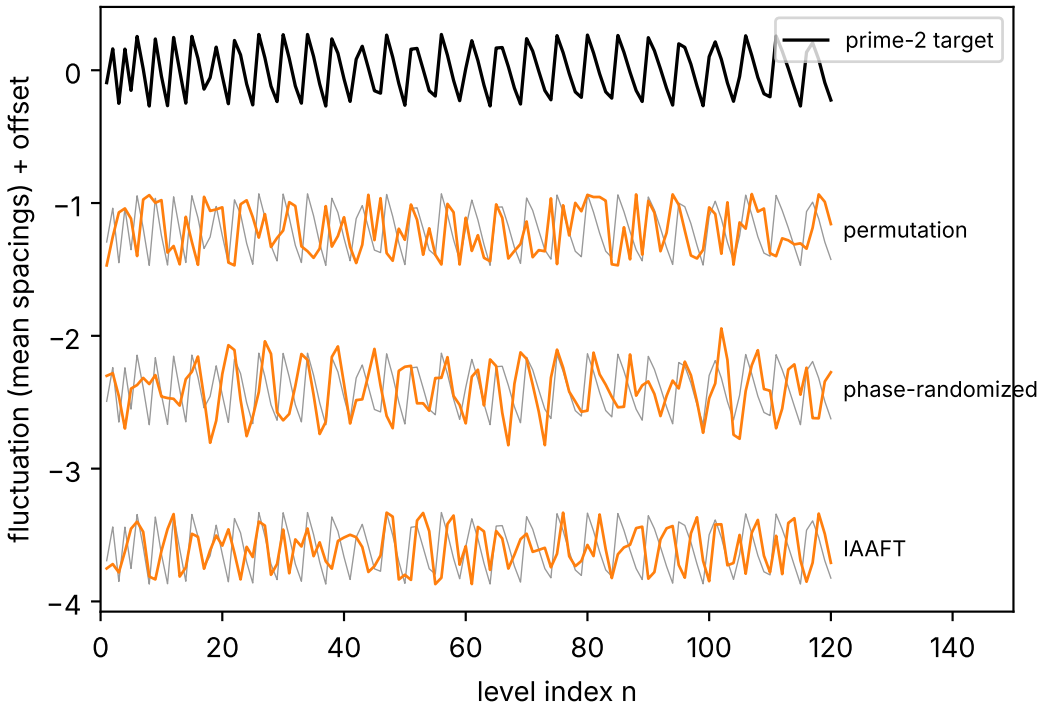


Figure 26.5: Surrogates of the prime-2 target (running example RE2): the fluctuation $-N_{\mathrm{fl}}^{p=2}(s_n)$ in mean spacings for levels $n = 1$–400 (first 120 shown, black). Orange, top to bottom: a random permutation (same values, no order), a phase-randomized surrogate (same power spectrum, new waveform), and an IAAFT surrogate (same values and nearly the same spectrum). None of the surrogates continues the black curve: the information about *where* the oscillation is has been removed.

- Permutation keeps the marginal; phase randomization keeps the spectrum and autocorrelation (Eq. (26.27)); IAAFT keeps the marginal exactly and the spectrum approximately.

26.7 Controls: structure or capacity?

★ Did you know? *Twelve Sailors and Citrus*

In 1747 the naval surgeon James Lind took twelve sailors with scurvy aboard HMS *Salisbury* and gave pairs of them different treatments, including cider, vinegar, seawater, and oranges with lemons. The pair eating citrus recovered dramatically. By comparing treatments side by side under similar conditions, Lind ran one of the earliest controlled trials. A control tells you whether a result means anything.

26.7.1 Positive and negative controls

In a biology laboratory, a new test for a virus is run alongside two extra samples: one known to contain the virus (a **positive control**) and one known not to (a **negative control**). If the negative control lights up, the test responds to something other than the virus; if the positive control does not light up, the test is broken. Either way you learn how to read the real result.

Computational studies need the same thing. For a fitted model:

- A **negative control** is a data set known to lack the structure the model is claimed to learn (a surrogate). If the model fits it as well as the real data, the fit quality is not evidence of that structure.
- A **positive control** is a data set where the right answer is known exactly (a synthetic target with known continuation). If the model fails there, it would also fail on the real problem.
- A **leakage control** checks that held-out data cannot influence the fit, for example by overwriting the held-out data with noise and confirming that the fitted parameters do not change.

26.7.2 The logic of a surrogate refit

Suppose a family of models reaches in-sample error e_{real} on the real targets. Refit the same family, with the same protocol, to surrogate targets and record e_{surr}. Then:

- If $e_{\text{surr}} \approx e_{\text{real}}$, the family fits sequences with properties $\mathcal{P}$ equally well whether or not they have $\mathcal{Q}$. The in-sample error measures **capacity** (Chapter 25), not captured structure.
- If $e_{\text{surr}} \gg e_{\text{real}}$, the family responds to $\mathcal{Q}$. This is necessary for having learned $\mathcal{Q}$, but not sufficient for predicting held-out data; that still requires an out-of-sample test.

26.7.3 A toy demonstration

Take the RE2 target for levels $n = 101$–200, a sequence of 100 numbers. Fit it with two linear models by least squares (Chapter 9):

- **Generic model:** the first M functions $\cos(\pi k(j + \frac{1}{2})/100)$, $k = 0, \ldots, M-1$, of the position $j = 0, \ldots, 99$ in the sequence. It knows nothing about primes; it is a flexible basis, like the cosine basis for δV.

- **Structured model:** the six functions $\sin(ms_n \log 2)$, $\cos(ms_n \log 2)$, $m = 1, 2, 3$, of the smooth height s_n. It knows the prime-2 frequencies.

Fit each to the real target, to 20 phase-randomized surrogates and to 20 permutations. Figure 26.6 shows the result.

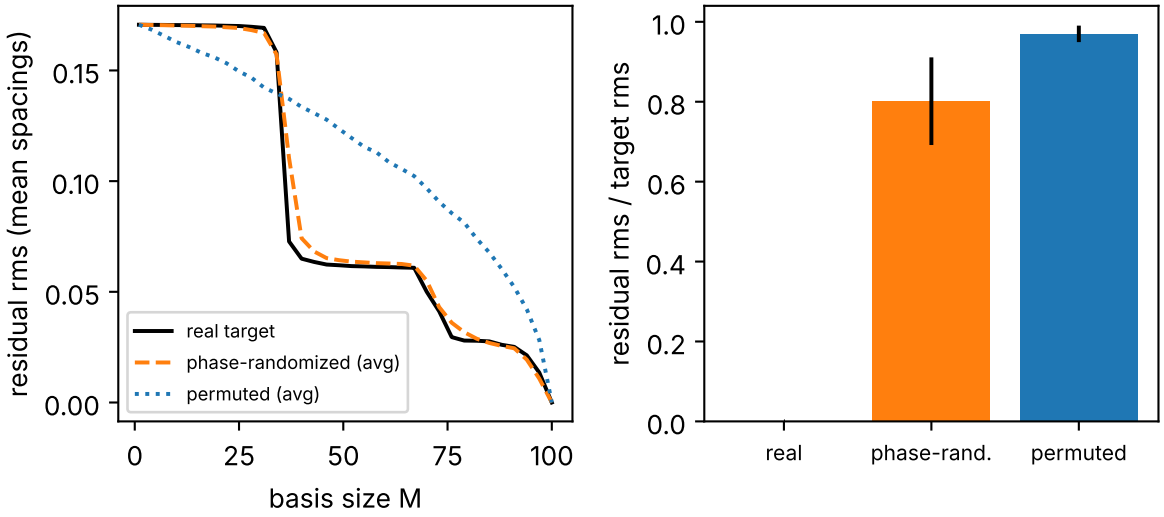


Figure 26.6: In-sample fits of the prime-2 target (levels 101–200, fluctuation in mean spacings) and its surrogates. Left: residual rms of the generic cosine model against its size M; curves for surrogates are averages over 20 draws. All reach zero at $M = 100$ (saturated), and the phase-randomized surrogates track the real target closely, because a generic basis responds mainly to the power spectrum. Right: residual rms of the six-parameter structured model, divided by the target's rms. It fits the real target exactly (the target lies in its span) and leaves a large fraction of each surrogate unexplained (on average more than half of its rms).

The generic model cannot tell the real target from its phase-randomized surrogates (in this chapter's check script, its residual curves for the real target and for the average phase-randomized surrogate differ by less than 0.2 of the target's rms at $M = 0, 10, \ldots, 100$), and at $M = 100$ it fits everything exactly. Why does it treat the phase-randomized surrogates almost the same? A cosine basis in the sequence index is close to a Fourier basis, and the fraction of a target that the first M basis functions capture depends on how the target's power is spread over frequency, which phase randomization preserves (this is a heuristic; the cosine and DFT bases are not identical, so the agreement is approximate). The structured model, by contrast, fits the real target to round-off precision and leaves, on average over the 20 draws of each family, more than half of each surrogate's rms unexplained. Only a model with the right inductive bias makes fit quality informative.

Sample Problem 26.10: Reading the paper's surrogate refits

The paper (Sec. III, "Permutation and surrogate tests") refits the scale-1, $M = 400$ operator on zeros 1–400 to ten draws each of three surrogate families. Reported in-sample rms: permutation 0.134 ± 0.010, phase-randomized 0.141 ± 0.006, IAAFT 0.138 ± 0.005, against 0.137 for the true zeros and 0.244 for the null. (a) How many "$\pm$" units from the true-zero value is each surrogate mean? (b) What fraction of the null's error does the fit remove on the true zeros, and on each surrogate family?

Step 1 (a): Permutation: $(0.137 - 0.134)/0.010 = 0.3$. Phase-randomized: $(0.137 - 0.141)/0.006 = -0.67$. IAAFT: $(0.137 - 0.138)/0.005 = -0.2$.

Step 2 (b): True zeros: $1 - 0.137/0.244 = 0.439$. Permutation: $1 - 0.134/0.244 = 0.451$. Phase-randomized: $1 - 0.141/0.244 = 0.422$. IAAFT: $1 - 0.138/0.244 = 0.434$.

all within 0.7 spreads; the fit removes 42–45% of the null error on every target

The fit is equally good on sequences with no arithmetic content. The paper concludes: "in-sample accuracy measures capacity, not captured structure."

Checkpoint 26.7

Suppose a fitted potential had reached in-sample rms 0.05 on the true zeros but only 0.20 on phase-randomized surrogates. Would that prove it predicts held-out zeros?

Answer: No. It would show the family responds to the phases of the real fluctuations (necessary for having learned them), but prediction can only be established by scoring held-out zeros against the null.

Key Ideas

- Negative controls (surrogates) test whether a result needs the structure claimed; positive controls (known answers) test whether the method can work at all; leakage controls test that held-out data cannot influence the fit.
- If a family fits surrogates as well as real data, its in-sample quality measures capacity.
- A generic basis responds mainly to the power spectrum; only the right inductive bias makes fit quality discriminating.

26.8 Exploratory and confirmatory analysis

26.8.1 Two kinds of analysis

A **confirmatory** analysis fixes the hypothesis, the statistic, the data and the decision rule *before* looking at the outcome; its p-values mean what Sec. 26.5 says they mean. An **exploratory** analysis looks at the data, notices patterns, and invents ways to quantify them. Exploration is essential (it is how most discoveries start), but its numbers cannot be read as significance levels, because the choices that produced them were steered by the data.

26.8.2 Many looks

The simplest way the steering inflates evidence: each look at the data is a chance to be fooled. If you perform k independent tests, each with false-positive probability α when its null is true, the probability that at least one fires is

$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^k, \tag{26.28}$$

because each test stays silent with probability $1 - \alpha$ and independence multiplies these. The **Bonferroni correction** runs each test at α/k. By the same formula, the family-wise probability is then $1 - (1 - \alpha/k)^k$, and this is at most α.

To see why, use **Bernoulli's inequality**: $(1 + u)^k \geq 1 + ku$ for $u \geq -1$ and every integer $k \geq 1$. We prove it by induction.

- For $k = 1$ it holds with equality.

- Suppose it holds for some k. Then $(1+u)^{k+1} = (1+u)^k(1+u) \geq (1+ku)(1+u) = 1 + (k+1)u + ku^2 \geq 1 + (k+1)u$. The first inequality multiplies both sides by $1+u \geq 0$; the second drops the term $ku^2 \geq 0$. So it holds for $k+1$.

With $u = -\alpha/k$, Bernoulli's inequality gives $(1-\alpha/k)^k \geq 1-\alpha$. Hence $1-(1-\alpha/k)^k \leq \alpha$.

Sample Problem 26.11: Twenty cutoffs

A researcher computes a statistic for 20 different cutoffs and reports the one that crosses $p < 0.05$. If none of the effects is real and the tests are independent, how often will something cross? What per-test threshold restores a 5% family-wise rate?
Step 1: One test: 0.05. Five tests: $1 - 0.95^5 = 1 - 0.7738 = 0.2262$.
Step 2: Twenty tests: $0.95^{20} = 0.3585$, so $1 - 0.3585 = 0.6415$.
Step 3: Bonferroni: $0.05/20 = 0.0025$ per test, giving $1 - (0.9975)^{20} = 0.0488 \leq 0.05$.

$$\boxed{P = 0.64 \text{ for 20 looks;} \quad \text{Bonferroni threshold } 0.0025}$$

Choosing the best of 20 cutoffs after seeing the data turns a 5% false-alarm rate into roughly 64%. (Real cutoffs are usually correlated, which lowers this number, but in an unknown way.)

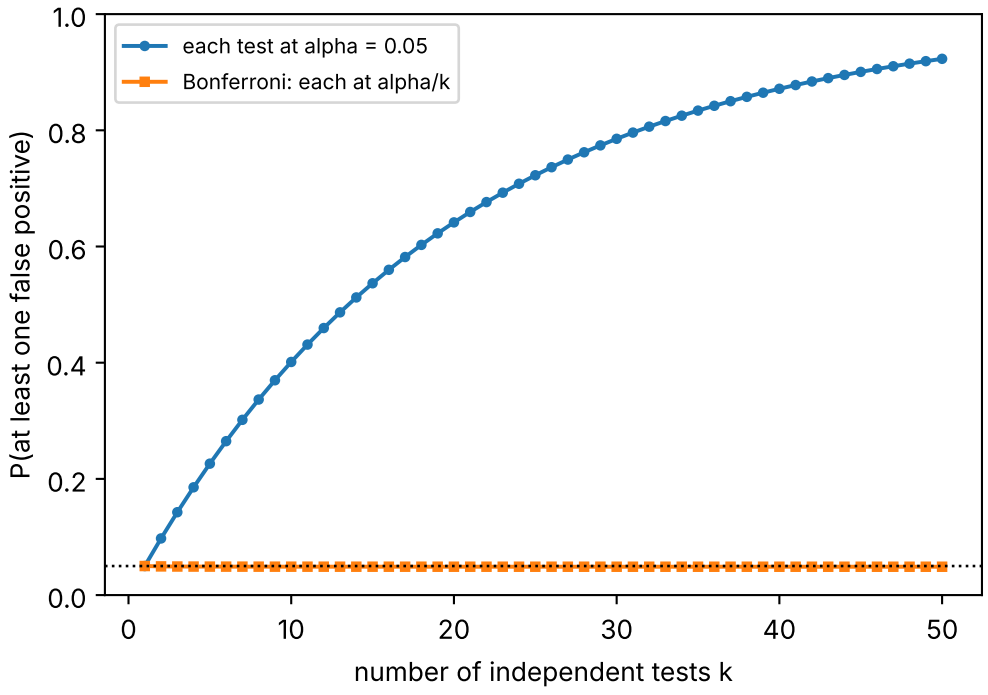


Figure 26.7: Probability of at least one false positive among k independent tests at level $\alpha = 0.05$ (Eq. (26.28)), and with the Bonferroni threshold α/k. The dotted line marks 0.05.

26.8.3 Statistics defined after the data were seen

A subtler version: the statistic itself is invented after looking. Suppose you notice that the frequencies picked out by a sparse fit of the zeros all sit near logarithms of prime powers. (The sparse fit is a procedure, described later in Chapter 27, that fits the fluctuation $n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ by a sum of a few sinusoids in t and returns their angular frequencies ω; here you need only

that it outputs a list of frequencies.) Suppose you then design a score for "how prime-like" a set of frequencies is. Your design choices (which matching window, which prime powers, which frequency cutoff, how to combine matches into one number) were all made knowing the result for the real data. Even if every choice was reasonable, there were many reasonable alternatives you did not take, and nobody can compute the probability of the score over the space of analyses you might have done. This is sometimes called the "garden of forking paths".

Generating surrogates and running *exactly the same* procedure on each (including any maximization over cutoffs) does correct for the steps that are applied identically to real and surrogate data. It does not correct for the choice of the statistic, made after the real outcome was known. That is why such a check is exploratory: it can show that the real data are far from every surrogate under *this* statistic, but it does not license a p-value.

Taken on trust

A density we quote. The next Sample Problem uses the prime number theorem in the form "near x, the primes have density about $1/\log x$ per unit length of x" (Chapter 15). Its proof is beyond this book. Prime powers p^m with $m \geq 2$ are neglected in the estimate. This is justified for large q, where they are far rarer than primes. Near $q \approx 13$ the prime powers 8, 9, 16 are as common as the primes, and the density $1/\log x$ is itself crude at such small x, so result (a) below is only an order-of-magnitude guide, good to a factor of about two; result (b), at $q \approx 3000$, is reliable.

Sample Problem 26.12: Why matching windows need a cutoff

A selected frequency ω is said to "match" if $|\omega - \log q| < \pi/T$ for some prime power q, where T is the length of the training window in t. Take $T = t_{400} - t_1 \approx 666$. Consider a frequency ω picked without reference to the primes (for instance, one returned by a search that knows nothing about them), lying (a) near $\omega \approx \log 13 \approx 2.56$, (b) near $\omega \approx 8$. Estimate the expected number of prime powers q whose $\log q$ falls inside its matching window, and the probability of a chance match in each case. (For a window centered exactly on $\log 13$ a match would be certain; what we estimate is the chance-match rate for a generic frequency in that region.)

Step 1: Window width in $\log q$: $2\pi/T = 2\pi/666 = 0.00943$.

Step 2: Density of primes per unit of $\log q$. Near q there are about $\mathrm{d}q/\log q$ primes in $\mathrm{d}q$. Since $\mathrm{d}(\log q) = \mathrm{d}q/q$, we have $\mathrm{d}q = q\,\mathrm{d}(\log q)$, so the count per unit of $\log q$ is $q/\log q = e^{\omega}/\omega$ at $\omega = \log q$.

Step 3 (a): $e^{\omega}/\omega = 13/2.565 = 5.07$, expected count $5.07 \times 0.00943 = 0.048$.

Step 4 (b): $e^{8}/8 = 2981/8 = 372.6$, expected count $372.6 \times 0.00943 = 3.5$.

Step 5: Treating the $\log q$ as scattered like a Poisson process at this density (a heuristic; Chapter 19 gives the probability of no point in an interval with expected count λ as $e^{-\lambda}$), the chance-match probability is $1 - e^{-\lambda}$: (a) $1 - e^{-0.048} = 0.047$; (b) $1 - e^{-3.52} = 0.970$.

(a) ≈ 0.05 expected, $P \approx 0.05$;	(b) ≈ 3.5 expected, $P \approx 0.97$

Near $\omega = 8$ almost any frequency matches some prime power by accident, so an alignment statistic needs a frequency cutoff, and *where* the cutoff goes strongly affects the score. A cutoff chosen to maximize the score after seeing the data is a post hoc choice of exactly the kind this section warns about.

Pitfall

Do not report an exploratory statistic with a p-value or the word "significant". Report what was done, say that the statistic was defined after the data were seen, and if possible confirm on fresh data (here: a new range of zeros never looked at) with the statistic frozen in advance.

Checkpoint 26.8

If you apply the same cutoff maximization to every surrogate as to the real data, which part of the post hoc problem have you fixed, and which remains?

Answer: Fixed: the inflation from choosing the best cutoff, since surrogates get the same advantage. Remaining: the choice of the statistic itself (matching rule, prime-power range, scoring formula), which was made after seeing the real result.

Key Ideas

- Confirmatory: everything fixed in advance; p-values meaningful. Exploratory: data-steered; descriptive only.
- k independent looks at level α give false-alarm probability $1-(1-\alpha)^k$; Bonferroni uses α/k.
- A statistic designed after seeing the data cannot support a significance claim, even with surrogates.

26.9 Ensembles, seeds, ranges and standard errors

26.9.1 Sources of variability

A computational result can change because of: (1) the data (which zeros were used); (2) the train/test split (which zeros were held out, when chosen at random); (3) the optimizer (random starting coefficients); (4) the surrogate draw. Each source is probed by repeating the computation with different **seeds** of the random number generator, producing an **ensemble of replicates**.

Three different summaries of an ensemble of R replicate values $v_1, \ldots, v_R$ are common, and they answer different questions:

- the **range** $[\min v, \max v]$: "how different can the result look?";
- the **standard deviation** s_v: "how spread out are individual replicates?";
- the **standard error of the ensemble mean** $s_v/\sqrt{R}$: "how precisely is the average known?"; valid only if the replicates are independent draws from the variability one cares about.

26.9.2 When replicates are not independent samples

If ten seeds randomly split the *same* 400 zeros into training and test sets, the ten results share the same underlying data. Their spread shows how sensitive the result is to the split, but it says nothing about how the result would change with different zeros. Dividing their standard deviation by $\sqrt{10}$ would give a misleadingly small number. That is why the paper's Table III caption says: "The seeds reuse the same zeros, so the ranges indicate robustness and are not standard errors."

26.9.3 From a range to a spread

For R independent draws from a Gaussian distribution with standard deviation σ, the expected range is $d_R\sigma$, where d_R is a tabulated constant from quality-control practice: $d_2 = 1.128$, $d_4 = 2.059$, $d_{10} = 3.078$. It can be written as $d_R = \int_{-\infty}^{\infty} [1 - \Phi(x)^R - (1 - \Phi(x))^R] \mathrm{d}x$, with Φ the standard Gaussian cumulative distribution. The box below explains where this formula comes from, and the check script evaluates it numerically. So a range over ten seeds is roughly three standard deviations wide, and a range over four repeats about two.

Taken on trust

Where the formula for d_R comes from. Let $Z_1, \ldots, Z_R$ be independent standard Gaussians.

- The maximum is $\le x$ exactly when every $Z_i \le x$, so by independence $P(\max \le x) = \Phi(x)^R$.
- The minimum is $> x$ exactly when every $Z_i > x$, so $P(\min > x) = (1 - \Phi(x))^R$.

We use the identity $E[Y] = \int_0^{\infty} P(Y > y)\,\mathrm{d}y - \int_{-\infty}^{0} P(Y \le y)\,\mathrm{d}y$ for a random variable Y with finite mean, density g and cumulative distribution G. Proof: integrate by parts with $u = 1 - G(y)$, $\mathrm{d}v = \mathrm{d}y$, so $\mathrm{d}u = -g\,\mathrm{d}y$:

$$\int_0^{\infty} (1 - G)\,\mathrm{d}y = [y(1 - G(y))]_0^{\infty} + \int_0^{\infty} y\,g(y)\,\mathrm{d}y = \int_0^{\infty} y\,g(y)\,\mathrm{d}y.$$

The boundary term vanishes at 0 trivially and at ∞ because $0 \le y(1 - G(y)) \le \int_y^{\infty} u\,g(u)\,\mathrm{d}u \to 0$ when the mean is finite. Likewise, with $u = G(y)$,

$$\int_{-\infty}^{0} G\,\mathrm{d}y = [yG(y)]_{-\infty}^{0} - \int_{-\infty}^{0} y\,g(y)\,\mathrm{d}y = -\int_{-\infty}^{0} y\,g(y)\,\mathrm{d}y,$$

the boundary term at $-\infty$ vanishing by the same argument. Subtracting the second from the first gives $\int_{-\infty}^{\infty} y\,g(y)\,\mathrm{d}y = E[Y]$. Apply it to the maximum and to the minimum, and subtract.

- On $(0, \infty)$ the integrands combine to $[1 - \Phi^R] - (1 - \Phi)^R$.
- On $(-\infty, 0)$ they combine to $-\Phi^R + [1 - (1 - \Phi)^R]$, which is the same expression.

So $E[\max - \min] = \int_{-\infty}^{\infty} [1 - \Phi(x)^R - (1 - \Phi(x))^R]\,\mathrm{d}x = d_R$. For a Gaussian with standard deviation σ every value scales by σ, so the expected range is $d_R\sigma$.

Sample Problem 26.13: Optimizer spread versus sampling uncertainty

Table I's caption reports that four random-initialization repeats of the scale-1, $M = 400$, zeros 1–400 fit give held-out rms from 0.365 to 0.379. Sec. III gives the bootstrap 95% interval $[0.33, 0.43]$ for the held-out rms of a random-restart replicate of this configuration. (The deterministic run from $c = 0$, rms 0.369, has the similar interval $[0.32, 0.42]$ in Table II.) Both intervals are 0.10 wide. Compare the two sources of variability.
Step 1: Range of the four repeats: $0.379 - 0.365 = 0.014$. Estimated standard deviation: $0.014/d_4 = 0.014/2.059 = 0.0068$.
Step 2: A 95% interval spans about ± 1.96 standard errors if the bootstrap distribution is near-Gaussian, so the held-out sampling SE is about $0.10/3.92 = 0.0255$.
Step 3: Ratio: $0.0255/0.0068 = 3.75$.

$$\boxed{\text{sampling SE} \approx 0.026,\ \text{optimizer spread} \approx 0.007,\ \text{ratio} \approx 3.8}$$

Which held-out levels were scored matters several times more than where the optimizer started. Both are small next to the gap between 0.369 and the null 0.263. (These are rough conversions, valid for near-Gaussian spreads; they are meant only for comparing orders of magnitude.)

26.9.4 Good reporting practice

- State what was varied (seed for split, for initialization, for surrogate), how many replicates, and how they were summarized.
- Use "mean ± SD over R draws" or a range when replicates share data; reserve standard errors for independent samples.
- Report seeds, so anyone can regenerate the ensemble exactly.
- When a result depends on an analysis choice (block length, ridge strength, a threshold), report a range of choices and say whether conclusions change.
- Record changes made after a first attempt (for instance, "added once, after a first version was unstable, and not tuned").

Checkpoint 26.9

Ten surrogate draws give in-sample rms with mean 0.134 and spread 0.010. Is 0.010 an uncertainty on the mean 0.134?

Answer: Not directly; it describes how much individual surrogate fits differ. If the ten draws are independent and 0.010 is their standard deviation, the uncertainty of the mean is about $0.010/\sqrt{10} = 0.003$.

Key Ideas

- Range, SD and SE of an ensemble answer different questions; replicates sharing the same data give robustness ranges, not standard errors.
- For R Gaussian draws the expected range is $d_R\sigma$ ($d_4 = 2.059$, $d_{10} = 3.078$).
- Report seeds and the sensitivity of conclusions to analysis choices.

26.10 Paper connection

The paper [1] uses every tool of this chapter. Table 26.3 maps them; we then go through each use.

Bootstrap intervals for operator fits (Sec. III, Table II). Appendix A step 13 records the recipe: "2×10^4 resamples of held-out indices; percentile 95% intervals for rms and correlation", exactly the Monte Carlo percentile bootstrap of Sec. 26.2, with prediction–truth pairs resampled together. Sec. III reports, for the scale-1, $M = 400$ random-restart replicate, OOS rms $[0.33, 0.43]$ against the null 0.263, and correlation $[-0.21, +0.08]$; for scale 2, $M = 600$, rms $[0.26, 0.30]$ against null 0.274 and correlation $[-0.18, +0.08]$. The paper describes the first as "significantly worse than the null" and the second as "consistent with its null". In the language of this chapter:

Table 26.3: Where the tools of this chapter appear in the paper (R1 numbering).

tool	paper location
independent bootstrap, percentile 95%	Sec. III; Tables II, IV, V captions; App. A steps 13, 22
circular / moving block bootstrap	Sec. III ("Other bases"); App. A step 22
permutation surrogate	Sec. III; App. A steps 12, 18
phase-randomized and IAAFT surrogates	Sec. III; App. A step 18; refs. [16, 17]
surrogate ensembles for an alignment check	Sec. IV; App. A step 23
positive controls	Sec. III (continuation probe); Sec. II D (oracle)
leakage controls	Sec. II C (Eq. (3)); App. A step 20
seeds, ranges vs. standard errors	Tables I and III captions; App. A step 17
exploratory qualification	Sec. IV; App. A step 23
proposed standard	Sec. VI, items (i)–(iv)

the first interval lies entirely above the null value; the second contains it. Both correlation intervals contain zero. On the same 401–500 holdout, the six-prime oracle's correlation interval is $[+0.95, +0.97]$, which you reproduced in the fourth Sample Problem.

Table II attaches intervals to every alternative-basis row. Note the Chebyshev $M = 400$ row, 0.291 $[0.25, 0.33]$ and $+0.18$ $[-0.02, +0.37]$: the rms interval contains the null 0.263, and the correlation interval's lower end is at (slightly below) zero. The paper reports this honestly as "marginal": "one such row among eight is not evidence of prediction, but a small positive correlation for this row is not excluded." With eight rows examined, one interval whose lower end touches zero is unsurprising (Sec. 26.8); the paper accordingly declines to read it as prediction while not excluding a small positive correlation.

Serial correlation and the block bootstrap (Sec. III, App. A step 22). The Table II caption says the brackets come "from independent resampling of the held-out levels", and Sec. III warns that the intervals "resample held-out levels independently although the fluctuations are serially correlated" and continues: "recomputed with a moving-block bootstrap (block lengths 3 to 20) they widen slightly and no conclusion changes (for the cosine row, at most $[0.30, 0.44]$)". Appendix A step 22 specifies a circular block bootstrap on zeros 401–500. (The main text says "moving-block" and the reconstruction record "circular"; the two differ only in the treatment of the ends, as in Fig. 26.4, and the record is the more specific description.) This is the sensitivity-over-ℓ procedure of Sec. 26.4. The Discussion (Sec. VI) restates the limitation: "Bootstrap intervals resample held-out levels independently except where stated."

Intervals for the arithmetic models (Sec. IV, Tables IV and V). The Table IV caption gives intervals on 401–500: A (900 terms) $[0.015, 0.020]$, B (900) $[0.158, 0.201]$, C1 $[0.231, 0.285]$, C2 $[0.028, 0.041]$, against null 0.263. Sec. IV adds C1's correlation interval $[+0.08, +0.46]$. Reading them: the A, B and C2 intervals lie far below the null; the C1 rms interval contains the null, so "capacity without a suitable prior does not generalize", even though its correlation interval sits above zero. The Table V caption gives the six-prime oracle's correlation intervals $[+0.95, +0.97]$ (401–500) and $[+0.91, +0.92]$ (5001–10000); the second is narrower because it uses 5000 held-out levels instead of 100, as the $1/\sqrt{n}$ law of Sec. 26.1 leads you to expect.

Surrogate refits: structure or capacity (Sec. III; App. A steps 12, 18). Step 18 defines the surrogates on the fluctuation sequence in mean spacings, $d_n = (\tilde{t}_n - E_n[\text{smooth}])/\text{gap}_n$ for $n \le 400$ (here $\tilde{t}_n$ are the bias-corrected targets and $E_n[\text{smooth}]$ the levels of the unperturbed Wu–Sprung potential): "permutation; phase randomization (real FFT, uniform phases); IAAFT (200 iterations); seeds 1–10; targets $E_n[\text{smooth}] + d'_n \text{gap}_n$, refit with $M = 400$". Your

`phase_randomize` and `iaaft` functions implement the same recipes. The results (the Sample Problem "Reading the paper's surrogate refits" in Sec. 26.7) show the operator family fits all three surrogates as well as the true zeros. Sec. III also explains a change from the original submission: permuting fluctuations in energy units "also moves amplitude across heights" and gave 0.148, 0.149 (Appendix A step 12 describes the procedure; its "Key anchors" list gives 0.1483/0.1493 for seeds 1 and 2); permuting in spacing units keeps the amplitude at each height, which is "the cleaner control". Choosing which properties a surrogate keeps is the core design decision of Sec. 26.6.

Held-out surrogates and the kriging positive control (Sec. III; App. A steps 18, 19). The paper does not stop at in-sample refits. A linear interpolator (simple kriging, Chapter 25) "predicts held-out true zeros at 0.117 with correlation +0.87 and permuted targets at the null", which is a positive control: it shows the serial structure of the zeros is real and usable, and the permuted negative control confirms the interpolator is not exploiting anything else. The fitted operator, held out the same way, "fails alike on true zeros and on surrogates (0.57–0.70; one split, illustrative)". Note the honest qualifiers: one split, illustrative. Step 19 also records a forking path transparently: "projection and nugget were added once, after a first version without them was numerically unstable, and not tuned."

Positive controls with known answers (Secs. II D, III). The arithmetic-continuation probe of Sec. III replaces the targets by the pure prime-2 oscillation (running example RE2; Appendix A step 15). The right continuation is known exactly; the family "represents it almost perfectly in-sample (rms 0.056, correlation +0.95) and continues it not at all (held-out correlation +0.07, rms worse than predicting zero fluctuation)". The oracle (Sec. II D) plays the complementary role: it is "an information-existence baseline" showing that predictable structure exists. A negative result ("the operator does not predict") is only informative when such positive controls show that prediction is possible.

Leakage controls (Sec. II C, App. A step 20). Two controls guard against held-out data influencing the fit. One is algebraic: Eq. (3) shows that scoring against bias-corrected targets equals correcting the prediction by a fit-independent quantity Δ_n built from unperturbed eigenvalues only. The other is empirical and is recorded for the sinusoidal controls of Sec. II D (step 20): "Overwriting held-out data with noise before fitting leaves all fitted coefficients bit-identical." This is the leakage control of Sec. 26.7 in its cleanest form. (The paper does not report such an overwrite test for the operator fits; for those, the algebraic argument of Eq. (3) is the safeguard.)

The surrogate alignment check, and why it is exploratory (Sec. IV; App. A step 23). The sparse sinusoidal control C2 selects frequencies from the zeros, and every one lies near some $\log p^m$. Because "the frequencies are the output of a search", the paper reruns "the complete procedure on surrogate zero sets: phase-randomized surrogates and two families of sparse signals with non-arithmetic frequencies, 900 replicates in all". Step 23 gives the composition: 200 (fit 1–400) and 100 (fit 1–800) replicates of each of three families, $3 \times (200 + 100) = 900$, with match criterion $|\omega - \log q| < \pi/T$, $q \leq 3000$, and a "cutoff-maximized binomial score 24.7 and 50.9 on the real zeros, at most 4.7 on surrogates (score defined after the data were seen)". The paper's own words are measured: "the qualifications that apply (the statistic was defined after the data were seen, so the check is exploratory) are documented in the reconstruction record [20]; no significance claim is made here beyond the size of the ensembles." After Sec. 26.5 you can unpack the last phrase: with no surrogate reaching the real score, an ensemble of 200 could support no p-value smaller than $1/201 \approx 0.005$, and one of 100 none smaller than $1/101 \approx 0.01$; and because the score was designed post hoc (Sec. 26.8), even those bounds are not offered as

p-values. The paper refers to the reconstruction record for the construction of the surrogates and of the score; this book does not reproduce them. Note also, from the chance-match Sample Problem, why a frequency cutoff is needed at all: near $\omega = 8$ a window of width $2\pi/T$ catches a prime power by chance with probability close to 1. Finally, the paper stresses that the underlying fact, that the Fourier transform of the zeros concentrates at logarithms of prime powers, is classical [18, 19]; the check concerns only whether the search procedure recovers it.

Seeds and ensembles (Tables I and III captions; App. A step 17). Table I's caption reports "four random-init repeats ... give identical in-sample rms 0.1370 and OOS 0.365–0.379" and "three repeats of the (scale 2, $M = 600$) row give OOS 0.279–0.283": ranges, summarizing optimizer variability, which the last Sample Problem showed to be several times smaller than held-out sampling uncertainty. The within-height random splits use "`rng.choice(n, n/5, replace=False)`, seeds 1–10" (step 17), and Table III reports ranges over seeds with the explicit caveat that they "are not standard errors".

The proposed standard (Sec. VI). The paper proposes that any claim that an operator "captures" the zeros come with: (i) a train/test split, including one within a height range; (ii) scoring of the actual operator spectrum; (iii) "evidence that performance survives regularization and restarts" (an ensemble over seeds and settings, Sec. 26.9); and (iv) "a refit to permuted and phase-randomized targets" (negative controls, Sec. 26.7). Two of the four items are exactly this chapter's subject.

Rerunning at small scale. With `qmnt`, the null rms on zeros 401–500 (smooth zeros as prediction) is 0.2631, matching the paper's anchor (Appendix A, "null OOS (401–500) 0.2631"), and the six-prime oracle and its bootstrap interval are reproduced by the listing in Sec. 26.2. The surrogate and operator refits require `qmnt.fit_potential` and take minutes per fit; Problems 22 and 32 ask for related small-scale computations that do not need `fit_potential`.

26.11 Summary

★ Did you know? *Where 0.05 Came From*

The habit of calling a result significant when $p < 0.05$ traces largely to Ronald Fisher's 1925 book *Statistical Methods for Research Workers*, where he described one in twenty as a convenient threshold. Fisher meant it as a rough guide, not a law. Today many statisticians urge reporting effect sizes and uncertainties instead of leaning on a single cutoff.

- Every score is a draw from a sampling distribution; for iid data the SE of a mean is $\sigma/\sqrt{n}$ (Eq. (26.2)), estimated by $s/\sqrt{n}$ (Eq. (26.4)).
- The bootstrap resamples the data with replacement; for the mean its exact variance is $\hat{\sigma}^2/n$ (Eq. (26.8)); there are $\binom{2n-1}{n}$ distinct resamples (Eq. (26.9)); percentile intervals take quantiles of the replicates.
- For a stationary sequence, $\mathrm{Var}(\bar{X}) = (\gamma_0/n) f_n$ with $f_n = 1 + 2\sum_k (1 - k/n)\rho_k$ (Eq. (26.12)); AR(1) gives $\rho_k = \varphi^k$ and $f_\infty = (1+\varphi)/(1-\varphi)$ (Eq. (26.18)).
- The circular block bootstrap keeps dependence within blocks of length ℓ; for the mean its variance is Eq. (26.22). Report results over a range of ℓ.
- Permutation tests test exchangeability; Monte Carlo p-values are $(1+r)/(N+1)$ (Eq. (26.23)), bounded below by $1/(N+1)$.

- Surrogates: permutation keeps the marginal; phase randomization keeps spectrum and circular autocorrelation (Eq. (26.27)); IAAFT keeps the marginal exactly and the spectrum approximately (Table 26.2).
- Negative, positive and leakage controls decide whether a fit reflects structure or capacity; a family that fits surrogates as well as real data shows capacity only.
- k looks at level α give $1-(1-\alpha)^k$ false-alarm probability (Eq. (26.28)); statistics defined after seeing the data are exploratory.
- Ranges over seeds that reuse the same data indicate robustness, not standard errors; for R Gaussian draws the expected range is $d_R\sigma$.

26.12 Problems

1. • Data $(2, 5, 8)$. Compute $\bar{x}$, s, $\widehat{\mathrm{SE}}(\bar{x}) = s/\sqrt{n}$ and the ideal bootstrap SE $\sqrt{\hat{\sigma}^2/n}$.

2. • How many distinct bootstrap resamples (as multisets) exist for $n = 4$ data points? How many ordered resamples?

3. • Which of the three surrogate families of Table 26.2 would you use to keep (a) exactly the same set of values, (b) exactly the same power spectrum, (c) both, as nearly as possible?

4. • In a Monte Carlo test, 3 of 99 surrogates have a statistic at least as large as the real one. What is the p-value?

5. •• Enumerate the bootstrap distribution of the mean for the data $(0, 1)$ $(n = 2)$. Compute its variance and check Eq. (26.8).

6. •• For an AR(1) sequence with $\varphi = 0.8$, find the large-n inflation factor f_∞ and the effective sample size of $n = 1000$ observations. Interpret.

7. •• Evaluate f_{100} for $\varphi = -0.3$ using Eq. (26.20), and the corresponding effective sample size. Compare with the large-n limit f_∞.

8. •• A mean-zero sequence $x_1, \ldots, x_n$ is randomly permuted. Show that the expected value of the lag-1 product $x_{\pi(i)}x_{\pi(i+1)}$ is $[(\sum_j x_j)^2 - \sum_j x_j^2]/[n(n-1)] = -\sum_j x_j^2/[n(n-1)]$, so the expected normalized lag-1 autocorrelation is about $-1/(n-1)$. Evaluate it for $n = 400$.

9. •• Show that phase randomization preserves the mean and the sum of squares $\sum_j d_j^2$ of a real sequence. (Use D_0 and Parseval's theorem for the DFT, $\sum_j d_j^2 = \frac{1}{n}\sum_q |D_q|^2$.)

10. •• For $n = 20$ and $\ell = 4$, how many moving blocks are there? How many blocks contain observations 1, 2 and 10? How many circular blocks are there, and how many contain each observation?

11. •• Using Table II of the paper, list the rows whose bootstrap rms interval excludes the null 0.263, and those whose correlation interval excludes zero. What do the latter say?

12. •• Using the Table IV caption intervals of the paper, decide for each of A (900 terms), B (900), C1 and C2 whether the rms interval lies below, contains, or lies above the null 0.263.

13. •• Appendix A of the paper anchors the original-submission permutation test (energy units) at in-sample rms 0.1483 and 0.1493 (seeds 1 and 2), against 0.1370 on true targets. By what percentages are these worse? Explain, using the fact that the mean spacing gap_n decreases with n, why permuting in energy units changes the amplitude profile in spacing units.

14. • With $k = 10$ independent looks at $\alpha = 0.05$, what is the probability of at least one false positive?

15. •• None of $N = 200$ surrogates reaches the real statistic. (a) Give the Monte Carlo p-value. (b) Find the largest exceedance probability q consistent, at 95% confidence, with seeing zero exceedances, by solving $(1 - q)^{200} = 0.05$, and compare with the "rule of three" $3/N$.

16. •• Table III of the paper reports random-split in-sample rms 0.20–0.22 over ten seeds (scale 1, $M = 200$). Estimate the seed-to-seed standard deviation using d_{10}, and explain why you would not divide it by $\sqrt{10}$.

17. •• Sec. IV of the paper says "no significance claim is made here beyond the size of the ensembles". For each fit range in Appendix A step 23, what is the smallest Monte Carlo p-value that a single surrogate family could support if no replicate reached the real score?

18. ••• Derive Eq. (26.12) for a stationary sequence, then derive Eq. (26.20) for AR(1) without neglecting the tail, i.e. find $\sum_{k=1}^{n-1}(1 - k/n)\varphi^k$ exactly.

19. ••• Using numpy, simulate AR(1) sequences with $\varphi = 0.5$, $n = 400$ and unit noise variance. For block lengths $\ell = 1, 5, 20$ compute the average circular-block-bootstrap SE of the mean over many series and compare with the true SE from Eq. (26.12) and the prediction of Eq. (26.22) with the true autocovariances.

20. •• Using numpy, phase-randomize $d = (1, 2, 0, -1)$ with $\phi = \pi/2$ and with $\phi = \pi$ applied to D_1. Show that both surrogates are circular shifts of d, and explain why a sequence with a single nonzero non-real Fourier pair can only produce shifted (or interpolated-shift) surrogates.

21. •• Using qmnt, compute the null rms (smooth zeros as prediction) on zeros 401–500, and the six-prime oracle's rms and correlation there. Compare with Table V of the paper.

22. ••• Using qmnt and numpy, compute circular-block-bootstrap 95% intervals for the six-prime oracle's correlation on zeros 401–500, for $\ell = 3, 5, 10, 20$, and compare with the independent bootstrap. Does any conclusion change?

23. ••• Using numpy, generate an IAAFT surrogate of the RE2 target (Fig. 26.5) with 200 iterations. Verify that its sorted values equal the original's exactly and compute the relative rms difference between its Fourier magnitudes and the original's. How does that difference evolve with the number of iterations?

24. •• Explain how a *paired* bootstrap would compare a model's held-out rms with the null's: what is resampled, what statistic is computed on each resample, and why this is preferable to comparing the model's interval with a fixed null value.

25. •• The paper's alignment check (Sec. IV; App. A step 23) runs the full sparse-frequency search on 900 surrogates and computes a "cutoff-maximized" score on each. Which part of the post hoc problem does this design handle, and which remains? Why does the paper call the check exploratory?

26. • Ten independent surrogate draws have in-sample rms standard deviation 0.010. What is the standard error of their mean?

27. ••• Prove that for two groups of equal size m, the exact permutation distribution of the difference of means $D = \bar{x}_A - \bar{x}_B$ is symmetric about zero, whatever the data.

28. •• In the paper's within-height analysis (Sec. III), the six-prime oracle scores 0.062–0.063 on the same held-out sets where operator fits score above the null. Which role (positive or

negative control, or neither) does the oracle play there, and why is it needed to interpret the operator's failure?

29. ••• Repeat the chance-match estimate of Sec. 26.8 for a frequency picked without reference to the primes, near $\omega \approx \log 101 \approx 4.62$, with $T = 666$. What does the result suggest about the frequency range in which an alignment with prime-power logarithms is informative?

30. •• The kriging interpolator of the paper predicts held-out true zeros with correlation $+0.87$ and permuted targets at the null (Sec. III). Explain which part of this is a positive control and which a negative control, and what each establishes.

31. •• With $B = 2 \times 10^4$ Monte Carlo bootstrap replicates, which sorted positions give the 2.5% and 97.5% percentile endpoints? Roughly how accurate is the bootstrap SE as an estimate of the ideal bootstrap SE?

32. ••• Using numpy, reproduce Fig. 26.6. Then replace the six-function structured model by one with the wrong frequencies (say $m \log 3$, $m = 1, 2, 3$) and show that it fits neither the real target nor the surrogates well. What does this say about using surrogate refits with a misspecified model?

26.13 Answers to selected problems

1. $\bar{x} = 5$; deviations $-3, 0, 3$, squares sum 18; $s^2 = 18/2 = 9$, $s = 3$; $\widehat{\mathrm{SE}} = 3/\sqrt{3} = 1.732$. Plug-in $\hat{\sigma}^2 = 18/3 = 6$, ideal bootstrap variance $6/3 = 2$, SE $= \sqrt{2} = 1.414$ (ratio $\sqrt{2/3}$, as Eq. (26.8) predicts).

2. Eq. (26.9): $\binom{7}{4} = 35$ multisets; ordered resamples $4^4 = 256$.

3. (a) Permutation (or IAAFT). (b) Phase randomization. (c) IAAFT: exact marginal, approximate spectrum. No standard method gives both exactly.

4. $r = 3$, $N = 99$: $p = (1 + 3)/(99 + 1) = 0.04$.

5. Ordered resamples $(0,0), (0,1), (1,0), (1,1)$, means $0, \frac{1}{2}, \frac{1}{2}, 1$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Mean $\frac{1}{2}$; $E[\bar{x}^{*2}] = \frac{1}{4}\cdot 0 + \frac{1}{2}\cdot\frac{1}{4} + \frac{1}{4}\cdot 1 = \frac{3}{8}$; variance $\frac{3}{8} - \frac{1}{4} = \frac{1}{8}$. Formula: $\hat{\sigma}^2 = \frac{1}{2}[(\frac{1}{2})^2 + (\frac{1}{2})^2] = \frac{1}{4}$, divided by $n = 2$: $\frac{1}{8}$. Agreement.

7. $\varphi/(1-\varphi) = -0.3/1.3 = -0.23077$, $\varphi/(1-\varphi)^2 = -0.3/1.69 = -0.17751$; $f_{100} = 1 + 2[-0.23077 + 0.17751/100] = 1 + 2(-0.22899) = 0.5420$; $n_{\mathrm{eff}} = 100/0.5420 = 184.5$. The large-$n$ limit is $f_\infty = 0.7/1.3 = 0.5385$, so the finite-$n$ correction is under 1%.

9. The mean is D_0/n and $\phi_0 = 0$, so D_0 is unchanged. Each $|D_q|$ is unchanged, so by Parseval $\sum_j d_j'^2 = \frac{1}{n}\sum_q |D_q'|^2 = \frac{1}{n}\sum_q |D_q|^2 = \sum_j d_j^2$. Mean and sum of squares fixed imply the variance is fixed.

11. rms interval excludes 0.263 (all lie above it): cosine 400 $[0.32, 0.42]$; Chebyshev 200 $[0.32, 0.42]$; Chebyshev 600 $[0.64, 0.82]$; hat 200 $[0.30, 0.38]$; hat 400 $[0.29, 0.39]$; hat 600 $[0.29, 0.37]$. Six rows. The Chebyshev 400 $[0.25, 0.33]$ and Chebyshev $\mu = 0.3$ $[0.24, 0.29]$ intervals contain the null. Only hat 400 has a correlation interval excluding zero, $[-0.37, -0.01]$, and it is *negative*, so it is not evidence of skill; with eight intervals examined, one excluding zero is also unremarkable (Sec. 26.8).

13. $0.1483/0.1370 = 1.082$ and $0.1493/0.1370 = 1.090$: about 8% and 9% worse. In energy units a fluctuation is $d_n \mathrm{gap}_n$. Low-n levels have large gap_n, so their energy-unit fluctuations are large; permuting moves some of them to high n, where dividing by the smaller gap_n

makes them large in spacing units, and moves small ones to low n. The surrogate's amplitude in spacings then varies with height, unlike the real one, so the control changes more than the serial order.

14. $1 - 0.95^{10} = 1 - 0.5987 = 0.4013$.

15. (a) $p = 1/201 = 0.004975$. (b) $\ln(1-q) = \ln(0.05)/200 = -0.014979$, so $q = 1 - e^{-0.014979} = 0.01487$; the rule of three gives $3/200 = 0.015$, nearly the same because $-\ln 0.05 = 2.996 \approx 3$ and $\ln(1-q) \approx -q$ for small q.

17. Each family has 200 replicates for the 1–400 fit and 100 for the 1–800 fit (Appendix A step 23). If no replicate reaches the real score, Eq. (26.23) gives $p = 1/201 \approx 0.005$ and $p = 1/101 \approx 0.010$, respectively; no smaller p-value can be supported. And because the score was defined after the data were seen, the paper does not offer even these as p-values.

19. True SE: $\gamma_0 = 1/(1-0.25) = 4/3$, $f_{400} = 1+2(1-2/400) = 2.99$, SE $= \sqrt{(4/3)(2.99)/400} = 0.0998$. Independent bootstrap ($\ell = 1$): about $\sqrt{(4/3)/400} = 0.0577$. Eq. (26.22) with true autocovariances: $\ell = 5$: $1 + 2(0.8 \cdot 0.5 + 0.6 \cdot 0.25 + 0.4 \cdot 0.125 + 0.2 \cdot 0.0625) = 2.225$, SE 0.0861; $\ell = 20$: factor 2.8, SE 0.0966. Your simulated averages should sit close to these, slightly lower because the sample mean is estimated (Fig. 26.3, right).

21. Null rms 0.2631, matching the paper's anchor and the 0.263 entry of Table V; six-prime oracle rms 0.074 and correlation +0.96, matching Table V's "6 primes" column, row 401–500.

23. The sorted values agree exactly because the last step of each iteration is the rank-order remap, which places the original values in the new order. The relative magnitude mismatch falls quickly during the first iterations and then levels off at a small value (the paper reports spectra matched "to 1%" for its surrogates, Sec. III); it cannot reach zero in general, because an exact marginal and an exact spectrum are not simultaneously attainable for generic data.

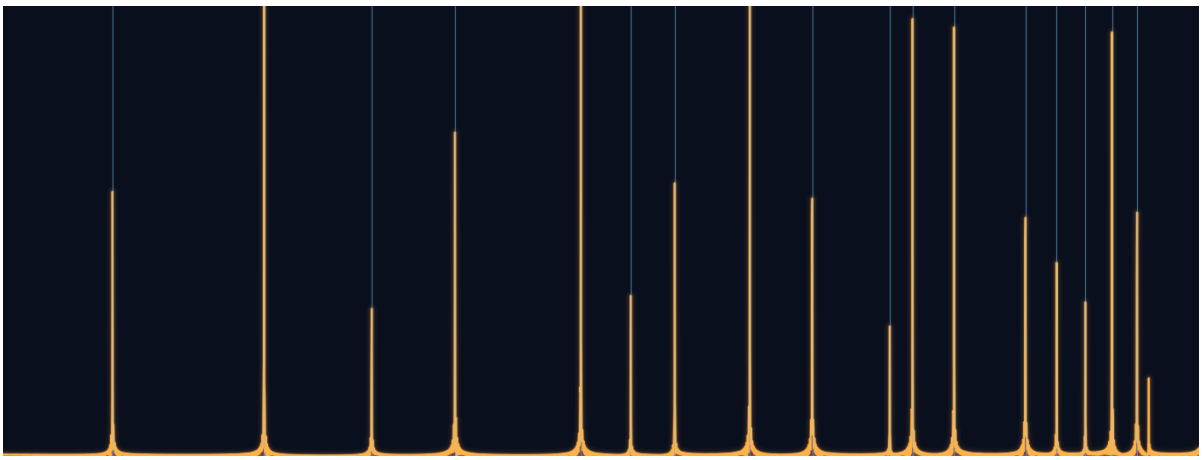
25. Running the same search and the same cutoff maximization on every surrogate gives the surrogates the same opportunity to look aligned, which accounts for the search over frequencies and the choice of cutoff. What remains is the choice of the statistic itself (matching criterion, prime-power range, binomial scoring), designed after the real result was known; its distribution over the analyses one might have chosen is unknown. Hence "exploratory", and "no significance claim ... beyond the size of the ensembles".

26. If the ten draws are independent and 0.010 is their standard deviation, SE $= 0.010/\sqrt{10} = 0.0032$.

27. Every relabeling chooses a set A of m positions; its complement B also has m positions. The map "swap A and B" is a one-to-one correspondence of the set of $\binom{2m}{m}$ relabelings onto itself, and it sends $D = \bar{x}_A - \bar{x}_B$ to $-D$. So for every value D the number of relabelings giving D equals the number giving $-D$, and the distribution is symmetric about zero.

29. $e^{\omega}/\omega = 101/4.615 = 21.9$; times $2\pi/666 = 0.00943$ gives about 0.21 expected prime powers in the window, so a chance match has probability roughly $1 - e^{-0.21} \approx 0.19$. Already at $q \approx 100$ one frequency in five would match by accident, against about one in twenty near $q = 13$ and almost all near $q = 3000$: alignment carries most of its information at low frequencies, and the statistic must weigh matches by how unlikely they are by chance.

31. Positions $\lceil 0.025 \times 20000 \rceil = 500$ and $\lceil 0.975 \times 20000 \rceil = 19500$. For a near-Gaussian bootstrap distribution the relative error of the SE estimate is about $1/\sqrt{2B} = 1/200 = 0.5\%$.



The spectrum of the first 10 000 zeros, $|\sum_n e^{i\omega t_n}|$: sharp peaks stand at $\omega = \log 2, \log 3, \log 4, \log 5, \ldots$

Chapter 27

Learning Frequencies from Data

↑ Contents

What this chapter is about

Strike two tuning forks at once and record the sound. If someone hands you the recording and tells you only that it was made by "a few tuning forks", you can recover their pitches: you look for the strongest periodic component, take it out, look again, and stop when nothing periodic is left. If instead you were told nothing, and you tried to describe the recording with a huge, flexible set of wiggles, you could reproduce the recording perfectly and still have no idea what the next second will sound like.

This chapter is about that difference. In Chapter 18 you learned that the fluctuating part of the zero-counting function of the Riemann zeta function is, approximately, a sum of sinusoids in the height t, with angular frequencies $m \log p$ fixed by the primes. Here we ask a learning question: if you did not know that formula, could you discover those frequencies from the zeros alone? We build three kinds of fitted sinusoidal model: one told the frequencies, one given a dense uniform grid of frequencies, and one told only that a *few* frequencies suffice. We explain, with exact structural facts where available and labelled heuristics elsewhere, why the first and third are observed to generalize to unseen zeros and the second is not, how precisely a frequency can be learned from a finite window, why skill decays far from that window, how to decide whether a learned frequency "is" the logarithm of a prime power, and why turning a fitted sum of sinusoids back into predicted zeros can be ambiguous.

Where this shows up in the paper: these are the "fitted sinusoidal controls" of Sec. II D of the paper [1], labelled B, C1 and C2, whose results fill Table IV and the paragraph "Comparable budgets and fitted controls" of Sec. IV; their recipe is step (20) of Appendix A, and the surrogate check of their frequencies is step (23). They are what lets the paper say that "the obstacle is the parametrization, not the parameter count".

What you need

- Chapter 6: the Fourier transform of a finite-length sinusoid (a sinc), the resolution $\Delta\omega \approx 2\pi/T$ of a window of length T, the periodogram, leakage, phase randomization.
- Chapter 9: linear least squares, normal equations, the singular value decomposition (SVD), the condition number, ridge regularization with filter factors $\sigma/(\sigma^2 + \lambda\sigma_1^2)$, underdetermined systems and the minimum-norm solution.
- Chapter 18: the truncated explicit formula $N_{\text{fl}}(t)$ (paper Eq. (4)), predicting zeros by solving $N_{\text{sm}}(t) + N_{\text{fl}}(t) = n - \frac{1}{2}$, and Landau's formula that puts peaks at $\log p^m$.
- Chapter 25: training and held-out data, the null model, saturated models, interpolation versus extrapolation, inductive bias.
- Chapter 26: bootstrap intervals, surrogate data, exploratory versus confirmatory analysis.
- First-year physics: sinusoids, superposition, beats.

27.1 What it means to learn a frequency

27.1.1 The data: one number per zero

Recall the setting. The n-th nontrivial zero of the Riemann zeta function is $\rho_n = \frac{1}{2} + \mathrm{i}t_n$, and $t_n > 0$ is its *height*; the first is $t_1 = 14.134725$ (running example RE1). The counting function $N(t)$ is the number of zeros with $0 < t_n \le t$. Recall from Chapter 17 that it splits as

$$N(t) = N_{\text{sm}}(t) + S(t), \qquad N_{\text{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{27.1}$$

up to a term of order $1/t$, where N_{sm} (paper Eq. (1)) is the smooth part and $S(t)$ is the small fluctuating part. Throughout, log is the natural logarithm. The *smooth zeros* s_n solve $N_{\text{sm}}(s_n) = n - \frac{1}{2}$, and the local mean spacing is $\text{gap}_n = \mathrm{d}s_n/\mathrm{d}n$.

At $t = t_n$ the step function $N(t)$ jumps from $n-1$ to n. The midpoint of the jump is $n - \frac{1}{2}$, so if we evaluate the decomposition (27.1) at the midpoint of the jump we get $n - \frac{1}{2} = N_{\text{sm}}(t_n) + S(t_n)$, where $S(t_n)$ means the midpoint value of S at its jump. This motivates the definition of the **training target**

$$y_n \;=\; n - \tfrac{1}{2} - N_{\text{sm}}(t_n) \;\approx\; S(t_n). \tag{27.2}$$

Each zero supplies one number y_n, attached to one position t_n. The positions are irregular: they are the zeros themselves.

How does y_n relate to the fluctuation measured in mean spacings, $(t_n - s_n)/\text{gap}_n$, which is how the paper scores everything? Expand N_{sm} about s_n by Taylor's theorem (Chapter 4):

$$\begin{aligned} N_{\text{sm}}(t_n) &= N_{\text{sm}}(s_n) + N'_{\text{sm}}(s_n)(t_n - s_n) + \tfrac{1}{2}N''_{\text{sm}}(\xi)(t_n - s_n)^2 \quad (\text{some } \xi \text{ between } s_n \text{ and } t_n) \\ &= n - \tfrac{1}{2} + N'_{\text{sm}}(s_n)(t_n - s_n) + \tfrac{1}{2}N''_{\text{sm}}(\xi)(t_n - s_n)^2. \end{aligned} \tag{27.3}$$

To find $N'_{\text{sm}}(s_n)$, treat n as continuous and differentiate the identity $N_{\text{sm}}(s_n) = n - \frac{1}{2}$ with respect to n using the chain rule:

$$N'_{\text{sm}}(s_n)\,\frac{\mathrm{d}s_n}{\mathrm{d}n} = 1 \quad\Longrightarrow\quad N'_{\text{sm}}(s_n) = \frac{1}{\text{gap}_n}. \tag{27.4}$$

Substituting and rearranging,

$$y_n = n - \tfrac{1}{2} - N_{\text{sm}}(t_n) \approx -\frac{t_n - s_n}{\text{gap}_n}. \tag{27.5}$$

The neglected term is $\frac{1}{2}N''_{\text{sm}}(t_n - s_n)^2$. Differentiating (27.1) twice gives $N''_{\text{sm}}(t) = 1/(2\pi t)$, so for a displacement of one full spacing at zero 400 ($t \approx 680$, gap ≈ 1.34) the neglected term is $1.34^2/(4\pi \cdot 680) \approx 2 \times 10^{-4}$, utterly negligible. So: *y_n is minus the fluctuation in mean spacings*, and an error of 0.01 in y is an error of about 0.01 mean spacings in the zero.

27.1.2 What the explicit formula says the data look like

Chapter 18 gave the (heuristic, truncated) approximation

$$S(t) \approx N_{\mathrm{fl}}(t) = -\frac{1}{\pi} \sum_{p \in P} \sum_{m=1}^{3} \frac{\sin(mt \log p)}{m\, p^{m/2}}, \tag{27.6}$$

the paper's Eq. (4), where P is a set of primes p. Each term is a sinusoid in t with angular frequency $\omega = m \log p$, that is, $\omega = \log q$ for the prime power $q = p^m$. (Angular frequency "in t" means the term repeats when t grows by $2\pi/\omega$; for $p = 2$, $m = 1$ that period is $2\pi/\log 2 \approx 9.065$, running example RE2.) So the data y_n are samples, at irregular points t_n, of a function that is (approximately) a sum of sinusoids with frequencies we know.

The question of this chapter is: *suppose we pretend not to know those frequencies.* How much would a learner have to be told in order to predict zeros it has not seen?

27.1.3 Three levels of supplied information

The paper (Sec. IV) organizes its models by what they are told. Table 27.1 lists them. The truncated explicit formula, which the paper calls the oracle, is told everything and fits nothing. At the other end, a fitted potential (Chapters 24 and 25) or a dense grid of frequencies is told nothing structural at all. In between sits a model told only "the answer is a sum of a few sinusoids in t".

Table 27.1: The three levels of supplied information in the paper's Sec. IV, with the model labels of its Sec. II D and Table IV.

level	what the model is told	models
(i)	the frequencies $m \log p$	A (oracle, nothing fitted), B (amplitudes fitted)
(ii)	only "few frequencies" (sparsity)	C2 (greedy sparse search)
(iii)	nothing structural (generic basis)	C1 (dense uniform grid), fitted potentials

All the fitted controls share one shape. A function

$$F(t) = \sum_{j=1}^{J} [a_j \sin(\omega_j t) + b_j \cos(\omega_j t)] \tag{27.7}$$

is fitted so that $F(t_n) \approx y_n$ on the training zeros, and then the n-th zero is predicted by solving the **level equation**

$$N_{\mathrm{sm}}(t) + F(t) = n - \tfrac{1}{2} \tag{27.8}$$

for t, exactly as the oracle solves $N_{\mathrm{sm}} + N_{\mathrm{fl}} = n - \frac{1}{2}$. What differs between B, C1 and C2 is only how the frequencies ω_j are chosen.

Why is fitting $F(t_n) \approx y_n$ the right thing to do if we care about zeros? Let $\hat{t}_n$ solve (27.8). Subtract the definition $n - \frac{1}{2} = N_{\mathrm{sm}}(t_n) + y_n$ from (27.8):

$$N_{\mathrm{sm}}(\hat{t}_n) - N_{\mathrm{sm}}(t_n) = y_n - F(\hat{t}_n). \tag{27.9}$$

To first order in $\hat{t}_n - t_n$, the left side is $N'_{\mathrm{sm}}(t_n)(\hat{t}_n - t_n)$. The slope $N'_{\mathrm{sm}}(t_n)$ differs from $N'_{\mathrm{sm}}(s_n) = 1/\mathrm{gap}_n$ of (27.4) by about $N''_{\mathrm{sm}} \cdot (t_n - s_n)$, and since $N''_{\mathrm{sm}} = 1/(2\pi t)$ is tiny this is a small relative change: for a displacement of one spacing at zero 400 it is $1.34/(2\pi \cdot 680 \cdot 0.745) \approx 4 \times 10^{-4}$, using $N'_{\mathrm{sm}}(680) \approx 0.745$ (computed in Section 27.9). So the left side is $(\hat{t}_n - t_n)/\mathrm{gap}_n$, and $F(\hat{t}_n) \approx F(t_n) + F'(t_n)(\hat{t}_n - t_n)$. Collecting the terms in $\hat{t}_n - t_n$,

$$\frac{\hat{t}_n - t_n}{\mathrm{gap}_n} \approx \frac{y_n - F(t_n)}{1 + F'(t_n)\,\mathrm{gap}_n}. \tag{27.10}$$

In general the denominator $1 + F'(t_n)\mathrm{gap}_n$ must be kept. Only when $|F'|\,\mathrm{gap}_n \ll 1$ is the error of the predicted zero, in mean spacings, simply the fitting residual $y_n - F(t_n)$, and for an accurate model this condition fails. The function being approximated, $S(t)$, is a staircase. Between zeros it falls with slope $-N'_{\mathrm{sm}}(t) \approx -1/\mathrm{gap}$, and at each zero it jumps up by 1. A smooth approximation to it therefore falls between zeros and rises steeply near each zero, so at a zero $F'(t_n)\mathrm{gap}_n$ is typically positive and of order one or larger. The zero error is then smaller than the residual. The residual is thus only a proxy for the paper's score, which is computed from the solved zeros $\hat{t}_n$. We will compare F' with N'_{sm} again in Section 27.9, where the comparison decides whether (27.8) has one root or several.

Key Ideas

- The data are one number per zero, $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n) \approx S(t_n)$, sampled at the irregular heights t_n; to excellent accuracy y_n is minus the fluctuation in mean spacings, (27.5).
- The explicit formula says y_n is approximately a sum of sinusoids with frequencies $\log q$, q a prime power.
- All fitted controls fit $F(t)$ of form (27.7) to y_n and predict zeros from $N_{\mathrm{sm}}(t) + F(t) = n - \frac{1}{2}$; they differ in how much they are told about the frequencies.

27.2 Sinusoidal regression with known frequencies

27.2.1 Linear in amplitudes, nonlinear in frequency

Look at (27.7) with the frequencies ω_j fixed. Then $F(t)$ depends on the $2J$ numbers a_j, b_j *linearly*: doubling every a_j and b_j doubles F. Fitting them to data is therefore linear least squares (Chapter 9). Stack the n_{fit} targets into a vector $\boldsymbol{y}$ and the unknowns into

$$\boldsymbol{c} = (a_1, \dots, a_J,\ b_1, \dots, b_J)^{\top}. \tag{27.11}$$

The **design matrix** X has one row per training zero and one column per unknown:

$$X_{n,j} = \sin(\omega_j t_n), \qquad X_{n,J+j} = \cos(\omega_j t_n), \qquad n = 1, \dots, n_{\mathrm{fit}},\ j = 1, \dots, J. \tag{27.12}$$

Then the vector of model values at the training zeros is $X\boldsymbol{c}$, and least squares minimizes $\|\boldsymbol{y} - X\boldsymbol{c}\|^2$. Recall from Chapter 9 that the minimizer satisfies the normal equations

$$X^{\top}X\,\boldsymbol{c} = X^{\top}\boldsymbol{y}. \tag{27.13}$$

If instead the frequencies are unknown, F depends on each ω_j through $\sin(\omega_j t)$, which is not linear in ω_j; frequency estimation is a nonlinear problem. The strategy of this whole chapter is to *choose* frequencies by some rule and then solve the easy linear problem for the amplitudes.

Checkpoint 27.1

Is the model $a\sin(\omega t) + b\cos(\omega t)$ linear in ω? Is it linear in (a, b)?

Answer: Not in ω: replacing ω by 2ω does not double the function. Yes in (a, b): the function is a times one fixed function plus b times another.

27.2.2 Amplitude and phase

The pair $a\sin\omega t + b\cos\omega t$ is a single sinusoid in disguise. Write $a = R\cos\varphi$ and $b = R\sin\varphi$ with

$$R = \sqrt{a^2 + b^2}, \qquad \varphi = \operatorname{atan2}(b, a), \tag{27.14}$$

where $\operatorname{atan2}(b, a)$ is the angle of the point (a, b) in the plane, measured from the positive horizontal axis. Then, by the addition formula $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$,

$$R\sin(\omega t + \varphi) = R\cos\varphi\,\sin\omega t + R\sin\varphi\,\cos\omega t = a\sin\omega t + b\cos\omega t. \tag{27.15}$$

Fitting (a, b) is therefore the same as fitting an amplitude R and a phase φ, but in a form where the unknowns enter linearly.

Sample Problem 27.1: A four-point fit with a known frequency

Fit $a\sin t + b\cos t$ (so $\omega = 1$) to the data $y = 1.1,\ 2.0,\ -0.9,\ -2.2$ at $t = 0,\ \pi/2,\ \pi,\ 3\pi/2$. Find a, b, the amplitude, the phase and the rms residual.
Step 1: design matrix. Row n is $(\sin t_n, \cos t_n)$: $(0,1)$, $(1,0)$, $(0,-1)$, $(-1,0)$.
Step 2: $X^\top X$. The $(1,1)$ entry is $\sum \sin^2 t_n = 0+1+0+1 = 2$; the $(2,2)$ entry is $\sum\cos^2 t_n = 1+0+1+0 = 2$; the off-diagonal entry is $\sum \sin t_n \cos t_n = 0$, since every row has a zero. So $X^\top X = \operatorname{diag}(2, 2)$.
Step 3: $X^\top \boldsymbol{y}$. Sine column: $0(1.1) + 1(2.0) + 0(-0.9) + (-1)(-2.2) = 4.2$. Cosine column: $1(1.1) + 0(2.0) + (-1)(-0.9) + 0(-2.2) = 2.0$.
Step 4: solve (27.13). $a = 4.2/2 = 2.1$, $b = 2.0/2 = 1.0$.
Step 5: amplitude and phase. $R = \sqrt{2.1^2 + 1.0^2} = \sqrt{5.41} = 2.326$, $\varphi = \operatorname{atan2}(1.0, 2.1) = 0.444$ rad.
Step 6: residuals. The fitted values are $b,\ a,\ -b,\ -a = 1.0,\ 2.1,\ -1.0,\ -2.1$, so the residuals $y - \hat{y}$ are $0.1,\ -0.1,\ 0.1,\ -0.1$, rms 0.1.

$$\boxed{a = 2.1, \quad b = 1.0, \quad R \approx 2.326, \quad \varphi \approx 0.444 \text{ rad}, \quad \text{rms residual} = 0.1}$$

27.2.3 When are the columns nearly orthogonal?

In the sample, $X^\top X$ came out diagonal, so each amplitude was found independently of the other. With many frequencies and many irregular samples this is not exact, but it is nearly true when the frequencies are well separated. Here is why.

A typical entry of $X^\top X$, divided by the number of samples, is an average such as $\frac{1}{n_{\text{fit}}}\sum_n \sin(\omega_i t_n)\sin(\omega_j t_n)$. Use $\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$:

$$\frac{1}{n_{\text{fit}}}\sum_n \sin(\omega_i t_n)\sin(\omega_j t_n) = \frac{1}{2}\Big[\frac{1}{n_{\text{fit}}}\sum_n \cos(\delta_- t_n) - \frac{1}{n_{\text{fit}}}\sum_n \cos(\delta_+ t_n)\Big], \quad \delta_\mp = \omega_i \mp \omega_j. \tag{27.16}$$

Approximation: if the samples are spread fairly evenly over a window of length T centred at t_c, and many samples fall within each period of the cosines, the average over samples is close to the average over the window. (For the zeros the sample density is not uniform: it grows with height. That changes the weights but not the conclusion below.) The window average is

$$\begin{aligned}\frac{1}{T}\int_{t_c - T/2}^{t_c + T/2} \cos(\delta t)\,\mathrm{d}t &= \frac{1}{\delta T}\Big[\sin\big(\delta(t_c + \tfrac{T}{2})\big) - \sin\big(\delta(t_c - \tfrac{T}{2})\big)\Big] \\ &= \frac{1}{\delta T}\,2\cos(\delta t_c)\sin\big(\tfrac{\delta T}{2}\big) \quad (\sin A - \sin B = 2\cos\tfrac{A+B}{2}\sin\tfrac{A-B}{2}) \\ &= \cos(\delta t_c)\,\frac{\sin(\delta T/2)}{\delta T/2}. \end{aligned} \tag{27.17}$$

The factor $\sin(x)/x$ with $x = \delta T/2$ is the sinc function of Chapter 6: it equals 1 at $x = 0$ and is at most $1/|x| = 2/(|\delta|\,T)$ in magnitude for $x \neq 0$. So:

- For $i = j$, $\delta_- = 0$ and the diagonal entry is $\frac{1}{2}[1 - (\text{small})] \approx \frac{1}{2}$.
- For $i \neq j$ with $|\omega_i - \omega_j| \gg 2\pi/T$, both terms are small, and the columns are nearly orthogonal.
- For $|\omega_i - \omega_j| \lesssim 2\pi/T$, the first term is close to $\frac{1}{2}\cos(\delta_- t_c)$; the two columns are nearly parallel, $X^\top X$ is nearly singular, and (Chapter 9) the condition number is large and noise is amplified.

This is the **resolution limit** of Chapter 6, now in least-squares language: two frequencies closer than about $2\pi/T$ cannot be told apart by a window of length T.

27.2.4 Control B: the frequencies $m \log p$ given

The paper's control B takes the frequencies of the oracle, $\omega = m \log p$ for $m = 1, 2, 3$ and a set of primes, and fits the amplitudes to the training zeros with ridge regularization. Recall the ridge solution from Chapter 9: with the SVD $X = U\Sigma V^\top$ and singular values $\sigma_1 \geq \sigma_2 \geq \ldots$,

$$\boldsymbol{c} = V \operatorname{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda\sigma_1^2}\right) U^\top \boldsymbol{y}, \tag{27.18}$$

which is exactly the filter of the paper's Appendix A step (20). Directions with $\sigma_i^2 \gg \lambda\sigma_1^2$ are fitted almost as plain least squares; directions with $\sigma_i^2 \ll \lambda\sigma_1^2$ are suppressed.

Sample Problem 27.2: Budgets of the known-frequency controls

The paper's Table IV lists control B with 25 primes (150 fitted parameters) and with 150 primes (900 fitted parameters), and the oracle A with 300 primes (900 terms). (a) Confirm these counts. (b) Which primes are the 25th, 150th and 300th? (c) What is the largest frequency in the 25-prime set? (d) For 900 coefficients and 400 training zeros, is the least-squares problem over- or underdetermined?

Step 1: count. Each prime contributes $m = 1, 2, 3$: three frequencies. Each frequency carries two coefficients (a, b). With 25 primes: 75 frequencies, 150 coefficients. With 150 primes: 450 frequencies, 900 coefficients. The oracle with 300 primes has $300 \times 3 = 900$ terms, none fitted.

Step 2: primes. Listing primes (Chapter 15), the 25th is 97, the 150th is 863 and the 300th is 1987.

Step 3: largest frequency. $3 \log 97 = 3(4.5747) = 13.724$.

Step 4: determinacy. Unknowns per equation: $900/400 = 2.25 > 1$. There are more unknowns than equations, so (Chapter 9) there are infinitely many exact fits and the data alone cannot pin down the coefficients.

150 and 900 coefficients; $p_{25} = 97$, $p_{150} = 863$, $p_{300} = 1987$; $\omega_{\max} = 13.72$; underdetermined (ratio 2.25)

The paper reports (Sec. IV and Table IV) that B with 25 primes gives held-out rms 0.028 and correlation +0.99 on zeros 401–500 from 400 training zeros, *better* than the unfitted 25-prime oracle (0.043/+0.99). A plausible reading (ours, not the paper's, and not tested here) is that fitting the amplitudes helps since the truncated formula leaves out everything beyond $p \leq 97$ and $m \leq 3$, and fitted amplitudes can partly absorb leftover structure at nearby frequencies. With 900 coefficients on 400 zeros, B degrades to 0.180/+0.78, yet it stays below the null 0.263

and positively correlated. Why does an underdetermined fit still predict something? The paper reports the numbers but does not offer an explanation; what follows is our heuristic reading. Every candidate F in this family, however badly its amplitudes are pinned, is built only from the correct frequencies, so whatever pattern it fits in the window is continued at the right frequencies outside it. This is not a guarantee: large, partly cancelling amplitudes that cancel only inside the window can extrapolate badly, and the degradation of this row from 0.028 to 0.180 shows that some such damage does occur. With 800 training zeros the same 900-coefficient model scores 0.100/+0.94 on the different holdout 801–1000 (null 0.274), plausibly because more data constrain the amplitudes.

```
import numpy as np, qmnt
def Nsm(t):
    u = np.asarray(t)/(2*np.pi); return u*np.log(u) - u + 7/8
nfit = 400
t = qmnt.zeros(500)
n = np.arange(1, 501)
y = n - 0.5 - Nsm(t)                        # targets y_n, approx S(t_n)
P = qmnt.first_primes(25)
om = np.array([m*np.log(p) for p in P for m in (1, 2, 3)])
def design(tt, om):
    tt = np.atleast_1d(tt)
    return np.hstack([np.sin(np.outer(tt, om)), np.cos(np.outer(tt, om))])
X = design(t[:nfit], om)                    # 400 x 150 design matrix
U, s, Vt = np.linalg.svd(X, full_matrices=False)
lam = 1e-6                                  # ridge strength (times sigma_1^2)
c = Vt.T @ (s/(s**2 + lam*s[0]**2) * (U.T @ y[:nfit]))
F = lambda tt: design(tt, om) @ c           # the fitted F(t)
res = y[:nfit] - F(t[:nfit])
print("in-sample rms (counting units):", np.sqrt(np.mean(res**2)))
```

To turn the fitted F into predicted zeros, solve the level equation (27.8) for each held-out n by bracketed root finding (Brent's method, Chapter 4), with the paper's bracket of ± 3 mean spacings around s_n (Appendix A step (8)):

```
from scipy.optimize import brentq
def predict_zero(k, F):
    s = brentq(lambda x: Nsm(x) - (k - 0.5), 2*np.pi, 1e6)  # smooth zero s_k
    g = 2*np.pi/np.log(s/(2*np.pi))                          # gap_k
    G = lambda x: Nsm(x) + F(x)[0] - (k - 0.5)               # level equation
    a, b = s - 3*g, s + 3*g
    while G(a)*G(b) > 0:                                     # expand on sign failure
        a, b = a - g, b + g
    return brentq(G, a, b)
hold = np.arange(401, 501)
tp = np.array([predict_zero(k, F) for k in hold])
print(qmnt.score(tp, hold))        # (rms, corr) in mean spacings
```

Your numbers will not match the paper's 0.028 exactly, because the paper chooses the ridge strength by forward validation (Section 27.5) while this listing fixes it; Problem 18 asks you to add the validation.

Key Ideas

- With frequencies fixed, a sum of sinusoids is linear in its amplitudes, and fitting it is ordinary (or ridge) least squares, (27.13), (27.18).
- Columns at frequencies separated by much more than $2\pi/T$ are nearly orthogonal,

(27.17); closer frequencies make the problem ill-conditioned. This is the resolution limit.

- Control B is told the frequencies $m \log p$. Even underdetermined, it keeps positive skill in the paper's Table IV; a heuristic reason is that every member of its family oscillates at the right frequencies.

27.3 A dense grid of frequencies: control C1

★ Did you know? *Periodograms for Messy Data*

Astronomers rarely get evenly spaced data, because clouds, daylight and seasons interrupt observing. Nicholas Lomb in 1976 and Jeffrey Scargle in 1982 developed a periodogram that fits sinusoids directly to irregularly timed points. The Lomb-Scargle periodogram is now used to find periods of variable stars and exoplanet signals by scanning a dense grid of trial frequencies.

27.3.1 The model

Now take away the arithmetic. Control C1 of the paper uses 450 frequencies on a uniform grid,

$$\omega_j = j\,\Delta\omega, \qquad \Delta\omega = \frac{8}{450}, \qquad j = 1, \ldots, 450, \tag{27.19}$$

(Appendix A step (20)), so the grid runs up to $\omega = 8$ and carries 900 coefficients, the same nominal budget as the largest fitted potential. Its amplitudes are fitted by ridge least squares on 400 zeros. The paper reports (Table IV) held-out rms 0.258 with correlation +0.28 on zeros 401–500, against a null of 0.263, so it "stays within 0.005 of the null" (Sec. IV). With 800 training zeros it scores 0.267/+0.22 on 801–1000, which lies 0.007 below its null 0.274. Why does a model with this many sinusoids, which includes frequencies close to every $\log q$ below 8, learn so little? This section derives two exact facts that explain it.

27.3.2 Fact 1: a uniform grid of frequencies is periodic

Every frequency in (27.19) is an integer multiple of $\Delta\omega$. Shift t by $T_{\text{rep}} = 2\pi/\Delta\omega$. (We write T_{rep}, for "repeat", to keep P for the prime set and T for the window length.)

$$\sin\big(\omega_j(t + T_{\text{rep}})\big) = \sin\big(\omega_j t + j\Delta\omega \cdot \tfrac{2\pi}{\Delta\omega}\big) = \sin(\omega_j t + 2\pi j) = \sin(\omega_j t), \tag{27.20}$$

and likewise for cosine. Every term repeats after T_{rep}, so the whole model does too:

$$F(t + T_{\text{rep}}) = F(t), \qquad T_{\text{rep}} = \frac{2\pi}{\Delta\omega} = 2\pi \cdot \frac{450}{8} \approx 353.4. \tag{27.21}$$

Whatever amplitudes are fitted, the C1 model is a periodic function with period about 353.4 in t. The true fluctuation is not periodic. It is a sum of sinusoids whose base frequencies $\log p$, for distinct primes p, are mutually incommensurate, together with their harmonics $m \log p$. The base frequencies are incommensurate because $\log p / \log p' = a/b$ with integers a, b would give $p^b = p'^a$, which is impossible for distinct primes by unique factorization (Chapter 15). Such a sum is almost periodic and never repeats (Chapter 5). A grid frequency close to $\log q$ can mimic it over a stretch of t, but the mismatch accumulates.

27.3.3 Fact 2: the ridge fit is a kernel interpolator

The second fact shows what the fitted C1 model actually *does* with its training data. Write $\boldsymbol{x}(t)$ for the row vector of all basis functions at height t, so $F(t) = \boldsymbol{x}(t)\boldsymbol{c}$, and the rows of X are $\boldsymbol{x}(t_n)$. Write $\kappa = \lambda\sigma_1^2$. The ridge solution with filter factors $\sigma_i/(\sigma_i^2 + \kappa)$, which are those of (27.18), is $\boldsymbol{c} = (X^\top X + \kappa I)^{-1} X^\top \boldsymbol{y}$. (We avoid the symbol μ here, because the paper uses μ for the dimensionless ridge strength that multiplies σ_1^2.) We now rewrite this solution.

First, both $X^\top X + \kappa I$ and $XX^\top + \kappa I$ are invertible for $\kappa > 0$. To see this for the first, take any nonzero vector $\boldsymbol{v}$. Then $\boldsymbol{v}^\top(X^\top X + \kappa I)\boldsymbol{v} = \|X\boldsymbol{v}\|^2 + \kappa\,\|\boldsymbol{v}\|^2 > 0$, so $(X^\top X + \kappa I)\boldsymbol{v} \neq \boldsymbol{0}$. No nonzero vector is sent to zero, so zero is not an eigenvalue, and the square matrix is invertible (Chapter 8). The same argument with $X^\top$ in place of X covers $XX^\top + \kappa I$.

Next, start from the identity

$$(X^\top X + \kappa I)X^\top = X^\top XX^\top + \kappa X^\top = X^\top(XX^\top + \kappa I). \tag{27.22}$$

Multiply on the left by $(X^\top X + \kappa I)^{-1}$ and on the right by $(XX^\top + \kappa I)^{-1}$:

$$X^\top(XX^\top + \kappa I)^{-1} = (X^\top X + \kappa I)^{-1} X^\top. \tag{27.23}$$

Therefore

$$\boldsymbol{c} = X^\top \boldsymbol{\alpha}, \qquad \boldsymbol{\alpha} = (XX^\top + \kappa I)^{-1}\boldsymbol{y}, \tag{27.24}$$

where $\boldsymbol{\alpha}$ has one entry per training zero. The prediction at any height is

$$F(t) = \boldsymbol{x}(t)X^\top\boldsymbol{\alpha} = \sum_{n=1}^{n_{\text{fit}}} [\boldsymbol{x}(t)\cdot\boldsymbol{x}(t_n)]\,\alpha_n. \tag{27.25}$$

Compute the dot product using $\cos(A-B) = \cos A\cos B + \sin A\sin B$:

$$\boldsymbol{x}(t)\cdot\boldsymbol{x}(t_n) = \sum_{j=1}^{J} [\sin\omega_j t\,\sin\omega_j t_n + \cos\omega_j t\,\cos\omega_j t_n] = \sum_{j=1}^{J}\cos(\omega_j(t-t_n)) \equiv D(t-t_n). \tag{27.26}$$

So

$$F(t) = \sum_{n=1}^{n_{\text{fit}}} \alpha_n\, D(t-t_n), \qquad D(\tau) = \sum_{j=1}^{J}\cos(j\Delta\omega\,\tau). \tag{27.27}$$

The fitted model is a weighted sum of copies of one fixed shape D, called the **kernel** (we write D rather than K, which denotes the number of selected frequencies), centred on the training zeros.

Now sum the kernel in closed form. Using Euler's formula (Chapter 2), $1 + 2\sum_{j=1}^{J}\cos(jx) = \sum_{j=-J}^{J} e^{\mathrm{i}jx}$. This is a geometric series (Chapter 3) with first term $e^{-\mathrm{i}Jx}$, ratio $e^{\mathrm{i}x}$ and $2J+1$ terms. For x not a multiple of 2π the ratio is not 1, and the sum is as follows. (At multiples of 2π every term is 1 and the sum is $2J+1$.)

$$\begin{aligned}
\sum_{j=-J}^{J} e^{\mathrm{i}jx} &= e^{-\mathrm{i}Jx}\,\frac{1-e^{\mathrm{i}(2J+1)x}}{1-e^{\mathrm{i}x}} \\
&= \frac{e^{-\mathrm{i}Jx} - e^{\mathrm{i}(J+1)x}}{1-e^{\mathrm{i}x}} \\
&= \frac{e^{-\mathrm{i}(J+\frac{1}{2})x} - e^{\mathrm{i}(J+\frac{1}{2})x}}{e^{-\mathrm{i}x/2} - e^{\mathrm{i}x/2}} \quad \text{(numerator and denominator times } e^{-\mathrm{i}x/2}\text{)} \\
&= \frac{-2\mathrm{i}\sin\big((J+\frac{1}{2})x\big)}{-2\mathrm{i}\sin(x/2)} = \frac{\sin\big((J+\frac{1}{2})x\big)}{\sin(x/2)}.
\end{aligned} \tag{27.28}$$

Hence

$$D(\tau) = \frac{1}{2}\left[\frac{\sin((J+\frac{1}{2})\Delta\omega\,\tau)}{\sin(\Delta\omega\,\tau/2)} - 1\right]. \tag{27.29}$$

This is the Dirichlet kernel. Its properties explain the structure of C1:

- At $\tau = 0$, both sines vanish; their ratio tends to $(J+\frac{1}{2})\Delta\omega\tau/(\Delta\omega\tau/2) = 2J+1$, so $D(0) = J = 450$.
- The central peak is narrow. The numerator first vanishes at $\tau = \pi/((J+\frac{1}{2})\Delta\omega) = \pi/8.009 \approx 0.39$, where $D = -\frac{1}{2}$, tiny compared with the peak.
- Away from the peak, for $\Delta\omega\,|\tau| \ll 1$, $\sin(\Delta\omega\tau/2) \approx \Delta\omega\tau/2$, so D oscillates with an envelope falling like $1/(\Delta\omega\,|\tau|)$: slowly, but steadily.
- D repeats with period $T_{\text{rep}} \approx 353.4$, as it must by Fact 1.

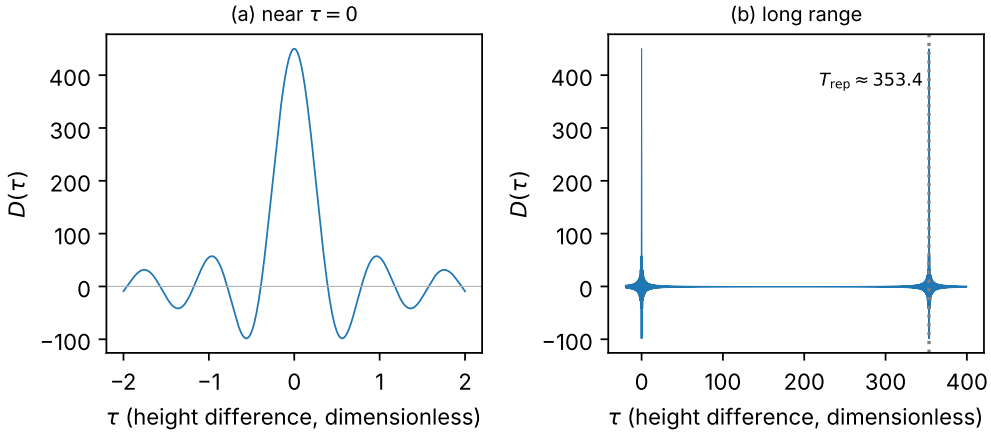


Figure 27.1: The kernel $D(\tau) = \sum_{j=1}^{450}\cos(j\Delta\omega\,\tau)$ of the dense uniform-grid model C1, $\Delta\omega = 8/450$. (a) Near $\tau = 0$: a narrow central peak of height 450 whose first zero lies at $\tau \approx 0.39$ in t. (b) Over a long range: the peak recurs at $\tau = T_{\text{rep}} \approx 353.4$, because every grid frequency is a multiple of $\Delta\omega$. By (27.27), the fitted C1 model at height t is built from training values near t and near its periodic images $t \pm T_{\text{rep}},\ t \pm 2T_{\text{rep}}, \ldots$

Put the two facts together. By (27.27), the C1 prediction at a held-out height t is dominated by the training values y_n at heights t_n within a fraction of a unit of t, or of one of its images $t - T_{\text{rep}}$, $t - 2T_{\text{rep}}$, and so on. Just beyond the training window there are no training zeros within a fraction of a unit; the nearest information is at the images, roughly 353 or 707 units earlier. Our interpretation of the kernel form (it is not an analysis given in the paper) is that C1 acts, in effect, as a *local interpolator plus a periodic copying rule*; how strongly each part acts depends on the weights α_n, which the ridge strength shrinks. It encodes no notion that a frequency found in the window should persist. That is the sense in which it "behaves like a generic basis": like the fitted potentials of Chapter 25, with 900 coefficients on 400 zeros it can fit anything in-sample, and outside it falls back on whatever its structure dictates, which here is local smoothing and repetition. Ridge regularization (chosen by forward validation) shrinks the α_n, pulling F toward zero, that is, toward the null.

Sample Problem 27.3: The C1 grid compared with the window

For the 400-zero training window, take its length as $T = t_{400} - t_1 \approx 666$. (a) Compare the C1 grid spacing with the resolution $2\pi/T$. (b) Which grid frequency lies nearest $\log 2$, and by how much does it miss? (c) How many periods of the $p = 2$ wave fit into one period T_{rep} of the C1 model?

Step 1: spacing and resolution. $\Delta\omega = 8/450 = 0.01778$. $2\pi/T = 2\pi/666 = 0.0094$. The grid spacing is about 1.9 resolution cells: the grid is *coarser* than the window's resolution.

Step 2: nearest to $\log 2$. $\log 2/\Delta\omega = 0.693147 \times 450/8 = 38.99$, so $j = 39$ and $\omega_{39} = 39 \times 8/450 = 0.693333$, which misses $\log 2 = 0.693147$ by 1.86×10^{-4}.

Step 3: periods. $T_{\text{rep}}/(2\pi/\log 2) = T_{\text{rep}} \log 2/(2\pi) = (450/8)\log 2 = 38.99$.

$$\Delta\omega \approx 1.9 \times \tfrac{2\pi}{T}; \quad \omega_{39} = 0.693333, \text{ miss } 1.86 \times 10^{-4}; \quad T_{\text{rep}} \text{ holds } 38.99 \text{ periods of } \log 2$$

The near-integer 38.99 is a coincidence of the paper's choice of grid: it means the $p = 2$ wave very nearly repeats after one C1 period. This is our own observation, not an analysis in the paper; whether it contributes to C1's small positive held-out correlation (+0.28, interval $[+0.08, +0.46]$ in Table IV) has not been checked, and Problem 6 invites you to explore it.

Checkpoint 27.2

A model uses only frequencies that are integer multiples of $\Delta\omega$. Can it represent a sinusoid of frequency $\log 2$ accurately over an arbitrarily long stretch of t?

Answer: No. The model is exactly periodic with period $2\pi/\Delta\omega$, while $\sin(t \log 2)$ is periodic with period $2\pi/\log 2$; unless $\log 2$ were an integer multiple of $\Delta\omega$ the two drift apart, so any match on one stretch degrades on later stretches.

Pitfall

"The dense grid contains frequencies near every $\log q$, so it contains the answer." Containing approximately the right frequencies is not enough: the fit must also *prefer* them. With more coefficients than data, infinitely many coefficient vectors fit the window exactly (Chapter 9). As the ridge strength tends to zero, the ridge solution tends to the exact fit of smallest norm. For positive strength it trades a little misfit for an even smaller norm. Either way the weight is spread over all 450 frequencies rather than concentrated on the few that are right.

Key Ideas

- A sum of sinusoids on a uniform frequency grid of spacing $\Delta\omega$ is exactly periodic with period $2\pi/\Delta\omega$, (27.21); for C1 that period is about 353.4.
- The ridge fit of any linear model can be rewritten as a weighted sum of kernels centred at the training points, (27.27); for C1 the kernel is the Dirichlet kernel (27.29), narrow and periodic.
- So C1 predicts by local interpolation and periodic copying: a generic basis with no preference for persistent frequencies. Its near-null score in Table IV shows that capacity without a suitable prior does not generalize.

27.4 Greedy sparse frequency search: control C2

27.4.1 The periodogram of irregular samples

Control C2 is told only that a few sinusoids suffice. To find them it uses the tool of Chapter 6: the periodogram. For values r_n at irregular heights t_n, $n = 1, \dots, n_{\text{fit}}$, define

$$\mathcal{P}(\omega) = \frac{1}{n_{\text{fit}}} \Big| \sum_n r_n \, e^{-\mathrm{i}\omega t_n} \Big|^2 = \frac{1}{n_{\text{fit}}} \Big[\Big(\sum_n r_n \cos \omega t_n \Big)^2 + \Big(\sum_n r_n \sin \omega t_n \Big)^2 \Big]. \tag{27.30}$$

The second form follows from $e^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$ and $|u - \mathrm{i}v|^2 = u^2 + v^2$. (Other normalizations exist, for example one that divides by the squared norm of each sine and cosine column; the exact convention of the paper is in its archived code [20]. The peak positions, which are all we use, are nearly the same for well-spread samples.)

Sample Problem 27.4: Height and width of a periodogram peak

Let $r_n = A\cos(\omega_0 t_n + \varphi)$ with $n_{\text{fit}} = 400$ samples spread over a window of length T, many periods long. Find $\mathcal{P}(\omega_0)$ for $A = 0.225$ (the amplitude of the $p = 2$ term, Table 27.2), and the offset δ at which $\mathcal{P}(\omega_0 + \delta)$ first vanishes.

Step 1: expand the sum at $\omega = \omega_0$. Using $\cos\theta = \tfrac{1}{2}(e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta})$,

$$r_n e^{-\mathrm{i}\omega_0 t_n} = \tfrac{A}{2}\big[e^{\mathrm{i}\varphi} + e^{-\mathrm{i}\varphi} e^{-2\mathrm{i}\omega_0 t_n} \big].$$

Step 2: sum over n. The first term adds to $\tfrac{A}{2} n_{\text{fit}} e^{\mathrm{i}\varphi}$. The second is a sum of unit-size phasors rotating at frequency $2\omega_0$. To see that it stays small, assume, as an idealization (the zeros are in fact denser at greater height), that the samples are equally spaced, $t_n = t_1 + (n-1)\Delta$. Then, apart from the constant factor $e^{-2\mathrm{i}\omega_0 t_1}$ of modulus 1, the second sum is a geometric series with ratio $z = e^{-2\mathrm{i}\omega_0 \Delta}$, and

$$\Big| \sum_{n=1}^{n_{\text{fit}}} z^{n-1} \Big| = \frac{|1 - z^{n_{\text{fit}}}|}{|1 - z|} \le \frac{2}{2\,|\sin(\omega_0 \Delta)|} = \frac{1}{|\sin(\omega_0 \Delta)|},$$

using $|1 - z^{n_{\text{fit}}}| \le 2$ and $\big|1 - e^{\mathrm{i}\theta}\big| = 2\,|\sin(\theta/2)|$ with $\theta = -2\omega_0\Delta$. The bound does not grow with n_{fit} (unless $\omega_0\Delta$ is close to a multiple of π), while the first term grows like n_{fit}. For the irregular zeros we carry the same conclusion over as a heuristic. Neglecting the second term, $\sum_n r_n e^{-\mathrm{i}\omega_0 t_n} \approx \tfrac{A}{2} n_{\text{fit}} e^{\mathrm{i}\varphi}$.

Step 3: square. $\mathcal{P}(\omega_0) \approx \frac{1}{n_{\text{fit}}} \big(\tfrac{A}{2} n_{\text{fit}}\big)^2 = \frac{n_{\text{fit}} A^2}{4} = \frac{400(0.225)^2}{4} = 5.06$.

Step 4: off the peak. At $\omega = \omega_0 + \delta$ the surviving term is $\tfrac{A}{2} e^{\mathrm{i}\varphi} \sum_n e^{-\mathrm{i}\delta t_n}$. Replace the sum by (n_{fit}/T) times the integral over the window, as in (27.17). With $u = t - t_c$,

$$\frac{1}{T} \int_{t_c - T/2}^{t_c + T/2} e^{-\mathrm{i}\delta t} \, \mathrm{d}t = \frac{e^{-\mathrm{i}\delta t_c}}{T} \int_{-T/2}^{T/2} e^{-\mathrm{i}\delta u} \, \mathrm{d}u = \frac{e^{-\mathrm{i}\delta t_c}}{T} \, \frac{e^{-\mathrm{i}\delta T/2} - e^{\mathrm{i}\delta T/2}}{-\mathrm{i}\delta} = e^{-\mathrm{i}\delta t_c} \, \frac{\sin(\delta T/2)}{\delta T/2},$$

where the last step uses $e^{-\mathrm{i}a} - e^{\mathrm{i}a} = -2\mathrm{i}\sin a$ with $a = \delta T/2$. Since $\big|e^{-\mathrm{i}\delta t_c}\big| = 1$, the magnitude of the surviving term is $\tfrac{A}{2} n_{\text{fit}} \, |\sin(\delta T/2)/(\delta T/2)|$, so

$$\mathcal{P}(\omega_0 + \delta) \approx \frac{n_{\text{fit}} A^2}{4} \left[\frac{\sin(\delta T/2)}{\delta T/2} \right]^2, \tag{27.31}$$

which first vanishes when $\delta T/2 = \pi$.

$$\boxed{\mathcal{P}(\omega_0) \approx n_{\text{fit}} A^2/4 \approx 5.06, \qquad \text{first zero at } \delta = 2\pi/T}$$

So a sinusoid of amplitude A produces a peak of height proportional to A^2, centred at its frequency, with width set by the window. The largest peak belongs to the largest-amplitude sinusoid, provided the peaks are separated by more than their widths.

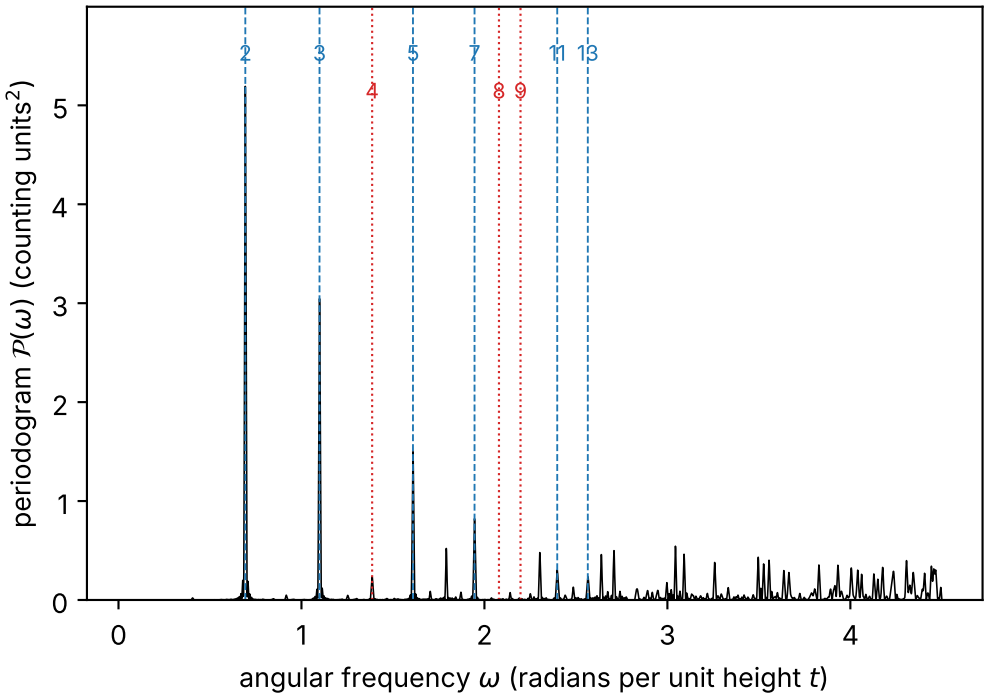


Figure 27.2: Periodogram (27.30) of the targets $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ for the first 400 zeros, for $0.05 < \omega < 4.5$. Dashed lines mark $\log p$ for primes $p \le 13$ (labelled); dotted lines mark the squares and cubes $\log 4, \log 8, \log 9$. The zeros were never told these frequencies; compare the positions of the tallest peaks with the marked lines. The peak widths are of order $2\pi/T \approx 0.0094$.

27.4.2 The greedy algorithm

Control C2 builds its model one frequency at a time (paper Sec. II D; Appendix A step (20)):

1. Start with the residual equal to the data, $\boldsymbol{r}^{(0)} = \boldsymbol{y}$, and an empty frequency list.
2. **Search.** Compute the periodogram of the current residual on a coarse grid covering $0.05 < \omega < 8$, with step a quarter of the resolution, $\frac{1}{4} \cdot 2\pi/T$. Take the grid point with the largest value.
3. **Refine.** Recompute the periodogram on a fine grid of 201 points around that coarse peak and take the maximum. (The record gives the count, 201; in our listings we take the fine grid to span one coarse step on each side, a natural reading; the exact span is in the archived code.)
4. **Refit jointly.** Append the refined frequency to the list and refit *all* amplitudes a_j, b_j of the chosen frequencies together by ridge least squares, (27.18).
5. **Update.** Set the new residual $\boldsymbol{r} = \boldsymbol{y} - X\boldsymbol{c}$ and return to step 2, until K frequencies have been chosen.

Each chosen frequency contributes three fitted numbers: ω_j (chosen by the search) and a_j, b_j. A model with K frequencies has $3K$ parameters, which is why Table IV lists C2 with "72, 129" parameters for $K = 24$ and $K = 43$.

Why does the search never pick the same frequency twice? After a plain least-squares refit, the residual is orthogonal to every column of X (Chapter 9: $X^\top(\boldsymbol{y} - X\boldsymbol{c}) = \mathbf{0}$ is the normal equation (27.13) rearranged). The columns include $\cos\omega_j t_n$ and $\sin\omega_j t_n$ for each chosen ω_j, so both sums in (27.30) vanish and $\mathcal{P}(\omega_j) = 0$ there. With a small ridge the orthogonality is approximate, and the periodogram is nearly zero at chosen frequencies.

Sample Problem 27.5: How much peak can the coarse grid miss?

The coarse grid step is $\frac{1}{4} \cdot 2\pi/T$. In the worst case the true peak lies halfway between grid points. By what factor is the periodogram there reduced relative to the true peak height, and why is the refinement step worth having?

Step 1: worst offset. Half a coarse step: $\delta = \frac{1}{8} \cdot \frac{2\pi}{T} = \frac{\pi}{4T}$.

Step 2: sinc argument. $\delta T/2 = \pi/8 = 0.3927$.

Step 3: apply (27.31). $\sin(\pi/8)/(\pi/8) = 0.38268/0.39270 = 0.9745$; squared, 0.9496.

Step 4: interpret. The coarse grid sees at least about 95% of every peak, so it rarely misses a strong one, but two peaks whose heights differ by less than about 5% could be ranked in the wrong order, and the coarse frequency can be off by up to $\pi/(4T)$. The refinement pins the frequency down much more finely than the coarse step before it is frozen into the model, which matters because frequency errors grow into phase errors outside the window (Section 27.6).

$$\boxed{\text{worst-case peak height} \approx 0.9745^2 \approx 0.950 \text{ of the true peak}}$$

```
def periodogram(tt, r, w):
    C = np.cos(np.outer(w, tt)); S = np.sin(np.outer(w, tt))
    return ((C @ r)**2 + (S @ r)**2) / len(tt)

def greedy(tt, yy, K, lam=1e-8):
    T = tt[-1] - tt[0]
    step = 0.25*2*np.pi/T                          # quarter of the resolution
    coarse = np.arange(0.05, 8.0, step)
    om, r = [], yy.copy()
    for k in range(K):
        w0 = coarse[np.argmax(periodogram(tt, r, coarse))]
        fine = np.linspace(w0 - step, w0 + step, 201)
        om.append(fine[np.argmax(periodogram(tt, r, fine))])
        X = design(tt, np.array(om))               # joint refit of all amplitudes
        U, s, Vt = np.linalg.svd(X, full_matrices=False)
        c = Vt.T @ (s/(s**2 + lam*s[0]**2) * (U.T @ yy))
        r = yy - X @ c
    return np.array(om), c

om24, c24 = greedy(t[:400], y[:400], 24)
print(np.round(om24[:5], 4), np.log([2, 3, 5]))
```

If you run this listing, the first frequency chosen lies within π/T of $\log 2 = 0.6931$: the largest peak of the zeros' fluctuation is the dominant term of running example RE2.

27.4.3 Matching pursuit: the general idea

The greedy search is an instance of a general method called **matching pursuit**. You have a *dictionary* of candidate shapes (here: sinusoids of every frequency in the band), and you build an approximation of the data by repeatedly picking the dictionary element that best matches the current residual. Two variants:

- *Plain matching pursuit*: pick the element most aligned with the residual, add its projection, keep all earlier coefficients frozen.
- *Orthogonal matching pursuit*: pick the element the same way, then refit *all* chosen coefficients jointly by least squares. This is what C2 does.

Each element of the dictionary is called an **atom**. The two samples below work in tiny dimensions: the first shows the orthonormal case, where plain and orthogonal pursuit agree; the second shows, with overlapping atoms, why the joint refit matters.

Sample Problem 27.6: Matching pursuit with an orthonormal dictionary

In four dimensions take the orthonormal dictionary $\boldsymbol{d}_1 = \frac{1}{2}(1,1,1,1)$, $\boldsymbol{d}_2 = \frac{1}{2}(1,-1,1,-1)$, $\boldsymbol{d}_3 = \frac{1}{\sqrt{2}}(1,0,-1,0)$, $\boldsymbol{d}_4 = \frac{1}{\sqrt{2}}(0,1,0,-1)$, and data $\boldsymbol{y} = (3,1,-1,1)$. Run matching pursuit.

Step 1: inner products with the data. $\boldsymbol{y}\cdot\boldsymbol{d}_1 = \frac{1}{2}(3+1-1+1) = 2$; $\boldsymbol{y}\cdot\boldsymbol{d}_2 = \frac{1}{2}(3-1-1-1) = 0$; $\boldsymbol{y}\cdot\boldsymbol{d}_3 = \frac{1}{\sqrt{2}}(3+1) = 2.828$; $\boldsymbol{y}\cdot\boldsymbol{d}_4 = \frac{1}{\sqrt{2}}(1-1) = 0$.

Step 2: first pick. The largest is $\boldsymbol{d}_3$, coefficient 2.828. Residual: $\boldsymbol{y} - 2.828\,\boldsymbol{d}_3 = (3,1,-1,1) - (2,0,-2,0) = (1,1,1,1)$.

Step 3: second pick. $\boldsymbol{r}\cdot\boldsymbol{d}_1 = 2$, all others zero. Add $2\boldsymbol{d}_1 = (1,1,1,1)$; the residual is now $\boldsymbol{0}$.

Step 4: compare with joint refit. Because the dictionary is orthonormal, the coefficient of each element does not change when others are added, so plain and orthogonal pursuit agree.

$$\boxed{\boldsymbol{y} = 2.828\,\boldsymbol{d}_3 + 2\,\boldsymbol{d}_1 \text{ (two atoms, exact)}}$$

Sample Problem 27.7: Why the joint refit matters

In the plane take unit vectors $\boldsymbol{a} = (1,0)$ and $\boldsymbol{b} = (\cos 60^\circ, \sin 60^\circ) = (0.5, 0.866)$, and data $\boldsymbol{y} = \boldsymbol{a} + 0.8\,\boldsymbol{b} = (1.4, 0.6928)$. Run two steps of plain matching pursuit, then orthogonal matching pursuit.

Step 1: first pick. $\boldsymbol{y}\cdot\boldsymbol{a} = 1.4$, $\boldsymbol{y}\cdot\boldsymbol{b} = 1.4(0.5) + 0.6928(0.866) = 0.7 + 0.6 = 1.3$. Pick $\boldsymbol{a}$ with coefficient 1.4 (not the true 1). Residual $\boldsymbol{r} = (0, 0.6928)$.

Step 2 (plain): second pick. $\boldsymbol{r}\cdot\boldsymbol{b} = 0.6928(0.866) = 0.6$. Add $0.6\,\boldsymbol{b}$, keeping 1.4 on $\boldsymbol{a}$. Approximation $1.4\,\boldsymbol{a} + 0.6\,\boldsymbol{b} = (1.7, 0.5196)$; residual $(-0.3, 0.1732)$, length $\sqrt{0.09 + 0.03} = 0.346$.

Step 2 (orthogonal): refit both. Solve $\alpha\boldsymbol{a} + \beta\boldsymbol{b} = \boldsymbol{y}$: the second component gives $0.866\beta = 0.6928$, so $\beta = 0.8$; the first gives $\alpha + 0.5(0.8) = 1.4$, so $\alpha = 1$. Residual $\boldsymbol{0}$.

Step 3: interpret. When dictionary elements overlap, the first coefficient is contaminated by the part of the data that belongs to elements not yet chosen. Freezing it leaves a permanent error; the joint refit removes it. Nearby frequencies overlap in exactly this way when they are within a few resolution cells, (27.17).

$$\boxed{\text{plain: } (1.4, 0.6),\ \text{residual } 0.346; \quad \text{orthogonal: } (1, 0.8),\ \text{residual } 0}$$

Taken on trust

When does greedy selection provably find the right few elements? For dictionaries whose elements are nearly orthogonal and signals that are truly sparse, there are theorems (from the field called sparse approximation or compressed sensing) guaranteeing that orthogonal

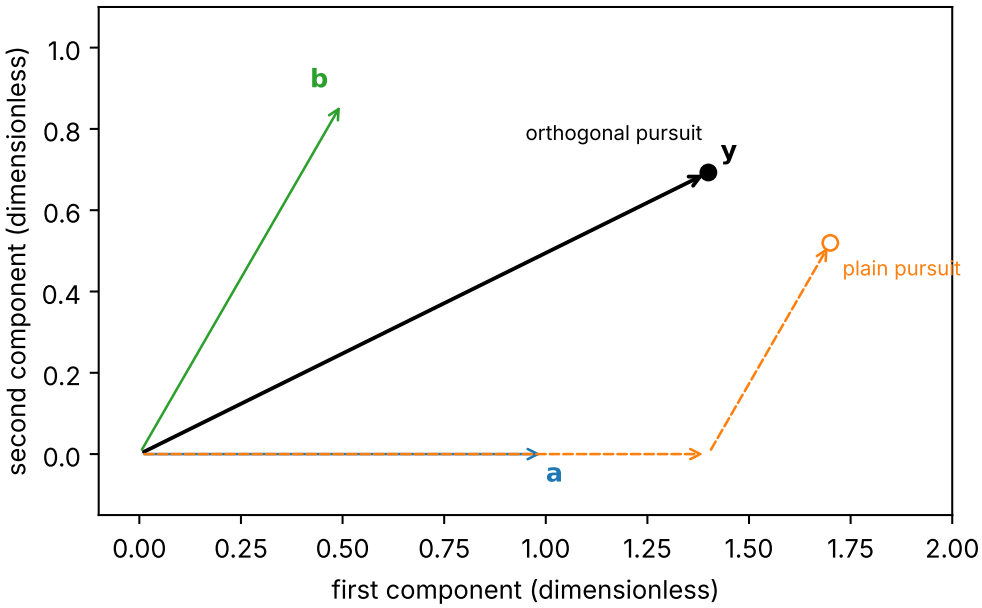


Figure 27.3: Plain versus orthogonal matching pursuit for the vectors of the sample above. The data $\boldsymbol{y} = \boldsymbol{a} + 0.8\boldsymbol{b}$ (black). Plain pursuit takes the projection $1.4\,\boldsymbol{a}$ onto $\boldsymbol{a}$, then adds the projection of the residual onto $\boldsymbol{b}$, ending at $1.4\,\boldsymbol{a} + 0.6\,\boldsymbol{b}$ (open circle) with a leftover error. Orthogonal pursuit refits both coefficients jointly and lands exactly on $\boldsymbol{y}$.

matching pursuit recovers the correct elements; they require bounds on the overlaps between elements and on the noise. Those theorems are beyond this book, and we do not claim that their hypotheses hold for the zeros. For C2 the evidence that the right frequencies were found is empirical: they land at $\log q$ (Section 27.7) and they predict held-out zeros.

Pitfall

Aliasing. Take samples equally spaced by Δ, $t_n = n\Delta$. Two frequencies that differ by $2\pi/\Delta$ give identical samples: $e^{\mathrm{i}(\omega+2\pi/\Delta)t_n} = e^{\mathrm{i}\omega t_n}e^{2\pi \mathrm{i}n} = e^{\mathrm{i}\omega t_n}$, because n is an integer. Also, $-\omega$ gives the same cosine samples as ω and sine samples of opposite sign, and the free sign of a fitted amplitude absorbs that. So every frequency has an alias in $[0, \pi/\Delta]$, and frequencies above π/Δ cannot be told apart from lower ones. The mean spacing of the zeros at the top of the 400-zero window is about 1.34, giving $\pi/1.34 \approx 2.34$, yet the paper's band reaches $\omega = 8$ and the 800-zero search recovered frequencies above $\log 131 \approx 4.88$. This is not a contradiction: the zeros are irregular samples whose mean spacing shrinks severalfold across the window (Chapter 17), so an alias that matches the samples at one height fails at another. Do not carry rules for uniform sampling over to irregular sampling without thought; equally, do not assume irregular sampling removes all ambiguity.

Checkpoint 27.3

Before any frequency is chosen, which frequency do you expect the greedy search to pick first for the zeros, and why?

Answer: $\log 2$. The periodogram peak height grows like the square of the amplitude, and among the

terms of (27.6) the $p = 2$, $m = 1$ term has the largest amplitude, $1/(\pi\sqrt{2}) = 0.225$ (Table 27.2).

Key Ideas

- A sinusoid of amplitude A gives a periodogram peak of height about $n_{\text{fit}}A^2/4$ and width $2\pi/T$, (27.31).
- Control C2: repeatedly take the largest periodogram peak of the residual (coarse grid of step $\frac{1}{4} \cdot 2\pi/T$, then refinement), and refit all amplitudes jointly; K frequencies cost $3K$ parameters.
- This is orthogonal matching pursuit. Joint refitting matters when dictionary elements overlap, as nearby frequencies do.

27.5 Choosing K and the ridge: forward validation

The greedy search can go on adding frequencies until it fits the training data perfectly. Training error can only fall as K grows: each model contains the previous one (set the new amplitudes to zero), so the least-squares minimum cannot rise. Chapter 25 told you that this proves nothing about prediction. We need a rule for stopping that uses only training data.

The paper's rule (Sec. II D): split the training window *forward*. Fit on the first 75% of the training zeros, score on the last 25%, choose the ridge strength λ and the number of frequencies K that score best on that last quarter, and finally refit the chosen configuration on the whole training window. Held-out zeros are never touched before scoring.

Why forward and not a random quarter? Because the real task is forward extrapolation: predicting zeros above the window. A validation split should imitate the task. A random split would place validation zeros between training zeros, where a model can score well by interpolating neighbours, which says little about extrapolation; and, as Chapter 25 showed, the serial correlation of the fluctuations makes interpolation behave quite differently from extrapolation.

The record (Appendix A step (20)) adds that the procedure was checked for leakage directly: overwriting the held-out data with noise before fitting leaves all fitted coefficients bit-identical. This is a cheap and decisive test you can apply to any pipeline: if the fit changes when held-out data are corrupted, held-out data are leaking into the fit.

Sample Problem 27.8: The bookkeeping of the validation search

For training windows of 400 and 800 zeros, find (a) the sizes of the inner fit and validation sets, (b) the number of ridge strengths in the paper's grid $\lambda = 10^{-8}, \ldots, 10^{3}$ in half decades, (c) the cap on K from the rule $K \le \min(300, n_{\text{fit}}/3)$, and (d) the approximate number of coarse search frequencies on $0.05 < \omega < 8$ for each window, taking $T \approx 666$ and $T \approx 1170$.

Step 1: split. $0.75 \times 400 = 300$ fit, 100 validate; $0.75 \times 800 = 600$ fit, 200 validate.

Step 2: ridge grid. The exponents run $-8, -7.5, \ldots, 3$, a span of 11 decades in steps of 0.5: $11/0.5 + 1 = 23$ values.

Step 3: cap on K. $n_{\text{fit}}/3 = 133.3$ for 400 zeros, so $K \le 133$; $800/3 = 266.7$, so $K \le 266$; both are below 300. The cap keeps $3K \le n_{\text{fit}}$: never more parameters than data.

Step 4: coarse grid size. Step $\frac{1}{4} \cdot 2\pi/T$. Number of points $\approx 7.95/(\frac{1}{4} \cdot 2\pi/T) = 7.95 \times 4T/(2\pi)$: for $T \approx 666$ about 3,400; for $T \approx 1170$ about 5,900.

300/100 and 600/200; 23 ridge values; $K \le 133$ and $K \le 266$; $\approx$ 3,400 and $\approx$ 5,900 coarse frequencies

How should you search over K efficiently? One run of the greedy algorithm on the inner 75% up to the cap produces the whole nested sequence of models with $K = 1, 2, 3, \ldots$; you can score each on the validation quarter as you go. (This is our suggestion for an implementation, not a statement about the paper's code.) The paper's validation selected $K = 24$ for the 400-zero window and $K = 43$ for the 800-zero window (Sec. IV; Appendix A, revision anchors).

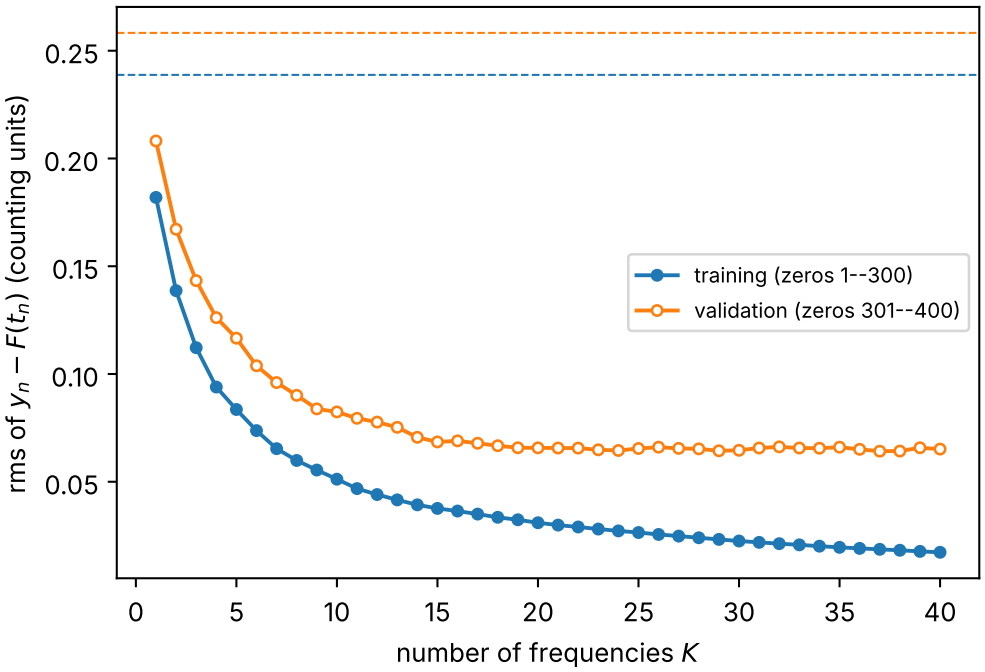


Figure 27.4: Greedy sparse search inside the 400-zero training window, split forward: inner fit on zeros 1–300, validation on 301–400. Training rms (filled) and validation rms (open) of $y_n - F(t_n)$, in counting units, against the number of frequencies K; dashed lines show the null (predicting $F = 0$) on each set. The training error can only fall, because the models are nested; the validation error is what selects K. Here the validation error is measured on y_n at the true zeros, a simplification of the paper's scoring through root solving. By (27.10) the zero error is the residual divided by $1 + F'(t_n)\text{gap}_n$, so the residual is only a proxy for that score.

Checkpoint 27.4

A colleague selects K by scoring on a random 25% of the training zeros instead of the last 25%. Why might the chosen K be poorly suited to predicting zeros above the window?

Answer: Random validation zeros sit between training zeros, so the score rewards models that interpolate well locally; that need not be the same K that extrapolates well. A forward split tests the model on the kind of task it will face: values beyond the data it was fitted on.

Key Ideas

- Training error never rises with K; the choice of K and ridge must come from data the inner fit did not see.
- The paper uses a forward 75/25 split inside the training window, a 23-value ridge grid, and $K \le \min(300, n_{\text{fit}}/3)$, then refits on the whole window. Held-out zeros are untouched, as checked by overwriting them with noise.

27.6 How well can a frequency be learned? Resolution, precision and dephasing

★ Did you know? *The Most Precise Measurement*

Frequency is the quantity physicists measure best. Modern optical atomic clocks reach fractional uncertainties around 10^{-18}, which corresponds to drifting less than a second over the age of the universe. Yet even they obey the basic rule of this section: to resolve two frequencies separated by Δf, you must observe for a time of order $1/\Delta f$.

27.6.1 Resolution is not precision

Recall from Chapter 6 that a window of length T has resolution $2\pi/T$: two frequencies closer than that blur into one peak. Resolution is about *separating* two frequencies. Precision is about *locating* one isolated frequency, and it can be much better than the resolution, because the centre of a well-measured peak can be found far more accurately than its width. Here is a linearized least-squares estimate of how well.

27.6.2 Deriving the precision

Model and assumptions. Measure time from the window centre, $u_n = t_n - t_c$, and suppose

$$y_n = A\sin(\omega u_n + \varphi) + e_n, \tag{27.32}$$

where the e_n are independent errors of mean zero and variance σ^2 (for the zeros, "error" is everything other than the one sinusoid we are estimating, which is not truly independent noise: so what follows is a heuristic scale, not a rigorous bound). Suppose the samples are spread roughly uniformly over the window and A and φ are known well. We want the uncertainty in ω.

Step 1: linearize. Near the true frequency ω_0, write $\omega = \omega_0 + \delta\omega$ and expand to first order (Chapter 4):

$$A\sin\big((\omega_0+\delta\omega)u_n+\varphi\big) \approx A\sin(\omega_0 u_n+\varphi) + g_n\,\delta\omega, \qquad g_n = A\,u_n\cos(\omega_0 u_n+\varphi). \tag{27.33}$$

Step 2: one-parameter least squares. Fitting $\delta\omega$ to the data $d_n = y_n - A\sin(\omega_0 u_n+\varphi)$ means minimizing $\sum_n (d_n - g_n\delta\omega)^2$. Setting the derivative with respect to $\delta\omega$ to zero, $-2\sum_n g_n(d_n - g_n\delta\omega) = 0$, so

$$\widehat{\delta\omega} = \frac{\sum_n g_n d_n}{\sum_n g_n^2}. \tag{27.34}$$

Step 3: its variance. At the true frequency, $d_n = e_n$. A sum $\sum_n w_n e_n$ of independent errors with fixed weights has variance $\sigma^2\sum_n w_n^2$ (Chapter 19). With $w_n = g_n/\sum_m g_m^2$,

$$\operatorname{Var}(\widehat{\delta\omega}) = \sigma^2\frac{\sum_n g_n^2}{\big(\sum_n g_n^2\big)^2} = \frac{\sigma^2}{\sum_n g_n^2}. \tag{27.35}$$

Step 4: evaluate $\sum g_n^2$. Using $\cos^2\theta = \tfrac{1}{2}(1+\cos 2\theta)$,

$$\sum_n g_n^2 = A^2\sum_n u_n^2\cos^2(\omega_0 u_n+\varphi) = \frac{A^2}{2}\sum_n u_n^2 + \frac{A^2}{2}\sum_n u_n^2\cos\big(2(\omega_0 u_n+\varphi)\big). \tag{27.36}$$

The second sum oscillates and is small compared with the first when the window holds many periods. For samples spread uniformly over $[-T/2, T/2]$, the average of u^2 is $\frac{1}{T}\int_{-T/2}^{T/2} u^2\,\mathrm{d}u = \frac{T^2}{12}$, so $\sum_n u_n^2 \approx n_{\mathrm{fit}}\,T^2/12$, with n_{fit} the number of samples. Hence $\sum_n g_n^2 \approx A^2 n_{\mathrm{fit}} T^2/24$.

Step 5: why the phase does not interfere. If φ is fitted too, linearize in both parameters: the model changes by $g_n\,\delta\omega + h_n\,\delta\varphi$ with $h_n = A\cos(\omega_0 u_n + \varphi)$. Least squares for $(\delta\omega, \delta\varphi)$ then gives the 2×2 normal equations

$$\begin{pmatrix} \sum_n g_n^2 & \sum_n g_n h_n \\ \sum_n g_n h_n & \sum_n h_n^2 \end{pmatrix} \begin{pmatrix} \delta\omega \\ \delta\varphi \end{pmatrix} = \begin{pmatrix} \sum_n g_n d_n \\ \sum_n h_n d_n \end{pmatrix}. \tag{27.37}$$

The off-diagonal entry is $\sum_n g_n h_n = A^2 \sum_n u_n \cos^2(\omega_0 u_n+\varphi) = \frac{A^2}{2}\sum_n u_n + \frac{A^2}{2}\sum_n u_n \cos\big(2(\omega_0 u_n + \varphi)\big)$. The first sum is close to zero because u is measured from the window centre, so positive and negative u_n cancel; the second oscillates and is small compared with the diagonal entries. Neglecting the off-diagonal entry, the matrix is diagonal, and its first row reads $(\sum_n g_n^2)\delta\omega = \sum_n g_n d_n$, exactly the one-parameter equation of Step 2. The variance of Step 3 is therefore unchanged. The two parameters decouple, which is why we centred the time variable. The same holds for the amplitude. The derivative of the model with respect to A is $k_n = \sin(\omega_0 u_n + \varphi)$, and its cross term with g_n is $\sum_n g_n k_n = A\sum_n u_n \sin(\omega_0 u_n + \varphi)\cos(\omega_0 u_n + \varphi) = \frac{A}{2}\sum_n u_n \sin\big(2(\omega_0 u_n + \varphi)\big)$. This is an oscillating sum, small for the same reason. Its cross term with h_n, $\frac{A}{2}\sum_n \sin\big(2(\omega_0 u_n + \varphi)\big)$, is small likewise. So A decouples too, and fitting it does not change the variance of $\widehat{\delta\omega}$ to this order.

Result.

$$\operatorname{std}(\widehat{\omega}) \approx \frac{\sqrt{24}\,\sigma}{A\sqrt{n_{\text{fit}}}\,T}. \tag{27.38}$$

Two lessons. The precision improves like $1/T$ with the window length and like $1/\sqrt{n_{\text{fit}}}$ with the number of samples; and it is worse for weak components (small A), which is why the greedy search places weak frequencies less accurately than strong ones.

Taken on trust

Equation (27.38) has the form of the Cramér–Rao lower bound for estimating the frequency of a sinusoid in independent Gaussian noise, a theorem from statistical estimation theory that bounds the variance of any unbiased estimator. Its proof and the precise conditions are beyond this book; for least-squares estimation more generally see [27] or [31]. For the zeros the "noise" is the deterministic sum of all other sinusoids, so treat (27.38) only as an order-of-magnitude scale.

Sample Problem 27.9: Precision of the learned $\log 2$

Estimate the precision of the $p = 2$ frequency learned from 400 zeros, taking $A = 0.225$ (Table 27.2), $\sigma = 0.18$ (an illustrative rms for everything else in y; the full fluctuation of zeros 1–400 has rms 0.2438 according to the paper's Appendix A anchors), $n_{\text{fit}} = 400$ and $T = 666$. Compare with the resolution $2\pi/T$ and the matching tolerance π/T.
Step 1: numerator. $\sqrt{24}\,\sigma = 4.899 \times 0.18 = 0.8818$.
Step 2: denominator. $A\sqrt{n_{\text{fit}}}\,T = 0.225 \times 20 \times 666 = 2997$.
Step 3: divide. $\operatorname{std}(\widehat{\omega}) \approx 0.8818/2997 = 2.94 \times 10^{-4}$.
Step 4: compare. $2\pi/T = 0.00943$, so the precision is about 3% of a resolution cell. $\pi/T = 0.00472$, about 16 times larger than the estimated error.

$$\boxed{\operatorname{std}(\widehat{\omega}) \approx 2.9 \times 10^{-4},\ \text{about } 0.031 \text{ resolution cells, } 1/16 \text{ of } \pi/T}$$

Checkpoint 27.5

If the number of training zeros doubles from 400 to 800, the window grows from $T \approx 666$ to $T \approx 1170$. If A and σ stayed the same, by what factor would (27.38) improve?

Answer: By $\sqrt{2} \times (1170/666) \approx 2.5$: a factor $\sqrt{2}$ from the number of samples and a factor 1170/666 from the window length.

27.6.3 Dephasing: why skill decays far from the window

A learned frequency is never exactly right. Inside the window the fit compensates: the fitted phase is correct near the window centre, and the error of the phase stays small across the window. Outside, there is nothing to compensate.

Setup. Let the true component be $A\sin\theta$ with $\theta = \omega_0(t - t_c) + \varphi$, and the fitted one $A\sin(\theta + \epsilon)$ with the same amplitude and a frequency error $\delta\omega$. The phase error is then

$$\epsilon(t) = \delta\omega\,(t - t_c), \tag{27.39}$$

growing linearly with the distance $\Delta t = t - t_c$ from the window centre.

Prediction error. Over a stretch of t short enough that ϵ is nearly constant but long enough to contain many periods, θ sweeps uniformly through all angles. The difference between prediction and truth is

$$A\sin(\theta + \epsilon) - A\sin\theta = 2A\cos(\theta + \tfrac{\epsilon}{2})\sin\tfrac{\epsilon}{2} \quad (\sin\alpha - \sin\beta = 2\cos\tfrac{\alpha+\beta}{2}\sin\tfrac{\alpha-\beta}{2}). \tag{27.40}$$

The mean of $\cos^2$ over all angles is $\frac{1}{2}$, so the mean square error is $4A^2\sin^2(\epsilon/2)\cdot\frac{1}{2}$, and

$$\text{rms error} = \sqrt{2}\,A\,|\sin(\epsilon/2)|\,. \tag{27.41}$$

The null prediction (zero) has rms error equal to the rms of $A\sin\theta$, which is $A/\sqrt{2}$. The ratio is

$$\frac{\text{rms error}}{\text{null rms}} = 2\,|\sin(\epsilon/2)|\,. \tag{27.42}$$

Correlation. The mean of $\sin\theta\sin(\theta+\epsilon) = \frac{1}{2}[\cos\epsilon - \cos(2\theta+\epsilon)]$ over all angles is $\frac{1}{2}\cos\epsilon$; each factor alone has mean square $\frac{1}{2}$ and mean zero. Dividing the covariance by the product of the standard deviations (Chapter 19),

$$\text{corr} = \frac{\frac{1}{2}A^2\cos\epsilon}{\sqrt{\frac{1}{2}A^2}\sqrt{\frac{1}{2}A^2}} = \cos\epsilon. \tag{27.43}$$

Consequences. At $\epsilon = 0$ the prediction is perfect. As ϵ grows, the correlation falls as $\cos\epsilon$ and the error grows. The prediction is worse than the null once $2\,|\sin(\epsilon/2)| > 1$, that is, once $\epsilon > \pi/3$. At $\epsilon = \pi/2$ the correlation is zero, and at $\epsilon = \pi$ the prediction is anticorrelated and twice as bad as the null. A model with many learned frequencies, each with its own small error, loses skill gradually, the weakest (least precise) components first.

Sample Problem 27.10: How far does a learned frequency carry?

A component is learned with frequency error $\delta\omega = 3 \times 10^{-4}$, about the size found in the previous sample. (a) At a distance $\Delta t = 2000$ from the window centre, find the phase error, the rms ratio (27.42) and the correlation. (b) At what distance does this component's prediction become worse than the null?

Step 1: phase. $\epsilon = 3 \times 10^{-4} \times 2000 = 0.6$ rad.

Step 2: rms ratio. $2\sin(0.3) = 2(0.2955) = 0.591$.

Step 3: correlation. $\cos 0.6 = 0.825$.

Step 4: break-even. $\epsilon = \pi/3$ requires $\Delta t = (\pi/3)/(3 \times 10^{-4}) = 3491$.

$\epsilon = 0.6$, rms ratio ≈ 0.59, corr ≈ 0.83; worse than null beyond $\Delta t \approx 3490$

For the 400-zero window the centre is near $t_c \approx (14 + 680)/2 = 347$, and zero 2000 sits near $t \approx 2515$ (values computed with `qmnt.zeros`), about 2170 units away. So by (27.39), frequency errors of a few times 10^{-4} already cost a noticeable fraction of the skill on zeros 1001–2000. This is consistent with what the paper reports for C2 (Sec. IV): from the 400-zero fit, 0.034/+0.99 on zeros 401–500 but 0.078/+0.96 on zeros 1001–2000, "as the estimated frequencies dephase, which the exact frequencies of the oracle do not". The oracle's exact frequencies have $\delta\omega = 0$, so its relative skill stays nearly constant through zero 10^4 (Table V). Our single-sinusoid model is a picture of the mechanism, not a fit to the paper's numbers.

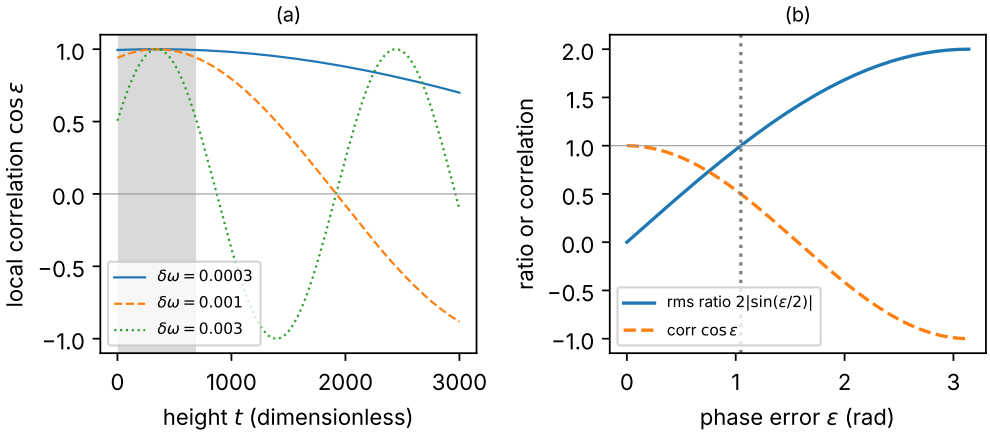


Figure 27.5: Dephasing. (a) Local correlation $\cos\epsilon(t)$ of a learned component with the truth, (27.39) and (27.43), for frequency errors $\delta\omega = 3 \times 10^{-4}$, 10^{-3} and 3×10^{-3}, measured from the window centre $t_c = 347$. The 400-zero training window $14 \le t \le 680$ is shaded, and t is the height (dimensionless). (b) Rms error relative to the null, $2\,|\sin(\epsilon/2)|$, and correlation $\cos\epsilon$, against the phase error ϵ. The prediction is worse than the null beyond $\epsilon = \pi/3$ (dotted).

Key Ideas

- Resolution $(2\pi/T)$ limits separating frequencies; precision of a single strong frequency is much finer, roughly $\sqrt{24}\,\sigma/(A\sqrt{n_{\text{fit}}}\,T)$, (27.38), a heuristic scale for the zeros.
- A frequency error $\delta\omega$ becomes a phase error $\delta\omega\,\Delta t$ that grows with distance from the window, (27.39); correlation falls as $\cos\epsilon$, and the prediction is worse than the null beyond $\epsilon = \pi/3$.
- Learned frequencies therefore lose skill far from the window; exact arithmetic frequencies do not.

27.7 Recovering $\log p^m$ from the zeros

27.7.1 What the zeros contain

The truncated formula (27.6) stops at $m = 3$, but the formal series of Chapter 18 runs over all $m \ge 1$. A term with prime power $q = p^m$ has frequency $\log q$ and amplitude

$$B_q = \frac{1}{\pi\, m\, p^{m/2}} = \frac{1}{\pi\, m\sqrt{q}} = \frac{\Lambda(q)}{\pi \log q\, \sqrt{q}}, \tag{27.44}$$

where the last form uses the von Mangoldt function of Chapter 15, $\Lambda(p^m) = \log p$, so that $\Lambda(q)/\log q = \log p/(m \log p) = 1/m$. Table 27.2 lists the strongest terms.

Table 27.2: Frequencies and amplitudes (27.44) of the first prime powers $q = p^m$ in the fluctuation of the counting function (counting units). Terms with $m \geq 4$ are absent from the truncation (27.6) but present in the zeros.

q	(p, m)	$\log q$	B_q
2	(2,1)	0.6931	0.2251
3	(3,1)	1.0986	0.1838
4	(2,2)	1.3863	0.0796
5	(5,1)	1.6094	0.1424
7	(7,1)	1.9459	0.1203
8	(2,3)	2.0794	0.0375
9	(3,2)	2.1972	0.0531
11	(11,1)	2.3979	0.0960
13	(13,1)	2.5649	0.0883
16	(2,4)	2.7726	0.0199

27.7.2 The matching criterion

Given a frequency ω found by the search, when do we say it "is" $\log q$? The paper's rule (Sec. IV and Appendix A step (23)) is

$$|\omega - \log q| < \frac{\pi}{T} \tag{27.45}$$

for some prime power q (with $q \leq 3000$ in the surrogate check): within *half* the Fourier resolution of the training window. This tolerance is a convention of the paper, not something derived from first principles. Its meaning is as follows. Frequencies closer than about one resolution cell, $2\pi/T$, cannot be separated by the window (Section 27.2), so a learned frequency is only meaningful to within such a cell. Draw a cell of width $2\pi/T$ centred on ω: it extends π/T on each side, so (27.45) says precisely that $\log q$ lies inside the cell centred on ω. A wider tolerance would admit more chance matches (see below); a narrower one would reject frequencies that are correct to within what the window can resolve.

Sample Problem 27.11: The 43 prime powers found from 800 zeros

The paper reports (Sec. IV) that the sparse search on 800 zeros selected 43 frequencies, matching "43 distinct prime powers: those up to 113 except 32, 64, and 81, and six primes between 131 and 211". (a) Verify the count. (b) Compute the amplitudes of the three excluded prime powers and of the weakest included one. (c) What does the list contain that the truncated oracle (27.6) does not?

Step 1: primes up to 113. They are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113$: 30 primes.

Step 2: higher prime powers up to 113. Powers of 2: $4, 8, 16, 32, 64$ ($128 > 113$). Powers of 3: $9, 27, 81$ ($243 > 113$). Of 5: 25 ($125 > 113$). Of 7: 49 ($343 > 113$). Of 11: none ($121 > 113$). That is 10.

Step 3: total. $30 + 10 = 40$; removing $32, 64, 81$ leaves 37; adding six primes from 131–211 gives 43. (There are 16 primes in that range, from 131 to 211 inclusive, so the search took six of them.)

Step 4: amplitudes. From (27.44): $B_{64} = 1/(6{\cdot}8\,\pi) = 0.0066$; $B_{81} = 1/(4{\cdot}9\,\pi) = 0.0088$; $B_{32} = 1/(5\sqrt{32}\,\pi) = 0.0113$. The weakest included term is $q = 16$, $B_{16} = 1/(4 \cdot 4\,\pi) = 0.0199$. The three excluded powers are exactly the three *weakest* prime powers up to 113.
Step 5: beyond the oracle. $16 = 2^4$ has $m = 4$, which the $m \le 3$ truncation omits. The data-driven model found a term the truncated formula leaves out.

$30 + 10 - 3 + 6 = 43$; excluded $B_{64}, B_{81}, B_{32} \approx 0.0066, 0.0088, 0.0113 < B_{16} \approx 0.0199$; includes 2^4

Amplitude is not the whole story: the primes 131–211 have amplitudes between $1/(\pi\sqrt{211}) = 0.0219$ and $1/(\pi\sqrt{131}) = 0.0278$, all larger than B_{16}, yet only six of the sixteen were chosen while 16 was. Finite resolution, overlap with neighbouring frequencies and the "noise" of all other terms affect which peaks win; the greedy search is not an exact ranking by amplitude.

27.7.3 Matching by chance, and why the check is exploratory

How surprising is it that every selected frequency lies within π/T of some $\log q$? That depends on how crowded the numbers $\log q$ are. By the prime number theorem (Chapter 15), the number of primes near x per unit x is about $1/\log x$; with $x = e^\omega$, $\mathrm{d}x = e^\omega \mathrm{d}\omega$, so the number of values $\log p$ per unit of ω near ω is about

$$\frac{e^\omega}{\omega} \tag{27.46}$$

(a heuristic density, ignoring the sparser higher powers). The expected number of $\log q$ values inside one resolution cell of width $2\pi/T \approx 0.0094$ is then about $0.0094\, e^\omega/\omega$, which reaches 1 near $\omega \approx 6.5$. Suppose the $\log q$ were scattered at random with this density (a Poisson-like model). Then a frequency near 6.5 would already match some $\log q$ with probability about $1 - e^{-1} \approx 63\%$, more often than not. That chance rises quickly toward 1 at higher ω, where the expected count per cell grows like e^ω/ω, so matches there carry little information. Below 6.5, matches are increasingly informative; at low ω, where the $\log q$ are far apart, a match is strong evidence.

This is a plausible motivation for the paper's use of "a statistic that does not depend on a chosen frequency cutoff" (Sec. IV). Our reading of its name, "cutoff-maximized binomial score" (Appendix A step (23)), is a score measuring how improbable the observed number of matches would be if each frequency matched only by chance, maximized over the cutoff. The definition itself is in the record. The paper reports scores of 24.7 and 50.9 for the 400- and 800-zero fits and at most 4.7 over 900 surrogate replicates: 200 (fit 1–400) and 100 (fit 1–800) replicates of each of three families, namely phase-randomized surrogates, explicit formulas with 25 displaced prime frequencies, and explicit-formula shapes with 25 random base frequencies (Appendix A step (23); the precise definition of the score is in the record [20]). The paper is explicit about the status: "the statistic was defined after the data were seen, so the check is exploratory", and "no significance claim is made here beyond the size of the ensembles". Recall from Chapter 26 why: a statistic chosen after looking at the data can be tuned, consciously or not, to the feature you already saw.

Taken on trust

That the zeros, viewed as a set of frequencies, have a Fourier transform concentrated at the logarithms of prime powers is classical: it is Landau's formula [18], made uniform by Gonek [19]; see Chapter 18 for its statement. The proofs are beyond this book. C2 is a learning procedure that rediscovers this fact from 400 or 800 zeros; the paper says so explicitly: "C2 is a control built on that fact, not a new result".

Pitfall

"Every selected frequency matches a prime power, so the matches prove arithmetic structure." Not by itself. Near $\omega \approx 6.5$ (for $T \approx 666$), a random frequency already matches some $\log q$ more often than not. A little higher up it almost always does, so matches there are weak evidence. Evidence comes from comparing with surrogates processed through the same pipeline, as the paper does, and even then a post hoc statistic supports only an exploratory conclusion.

Key Ideas

- The zeros contain all prime powers $q = p^m$, with amplitude $1/(\pi m\sqrt{q})$, (27.44); the truncated oracle keeps only $m \leq 3$.
- A learned frequency is matched to $\log q$ if it lies within π/T, (27.45).
- The $\log q$ crowd together at high ω (density about e^{ω}/ω; for $T \approx 666$ a cell of width $2\pi/T$ expects at least one $\log p$ above $\omega \approx 6.5$), so chance matching must be controlled; the paper's surrogate check is cutoff-free but exploratory.

27.8 Why "sparse in frequency" is a strong prior

Compare the three fitted families once more, now as statements about the world.

- C1 says: "the fluctuation is some function on this window". With 900 coefficients on 400 zeros, every sequence of training values is fitted exactly by some member (Chapter 9), so the family excludes nothing in-sample, and its behaviour outside is set by its structure (local kernel, periodic repetition), not by the data.
- C2 says: "the fluctuation is a sum of a few sinusoids, each of which persists for all t". That excludes almost every sequence. With $K = 24$ frequencies, the model has 72 parameters for 400 data values, more than five data values per parameter. If the statement is true, the data pin down the parameters, and the persistence of each sinusoid carries the information beyond the window.
- B says even more: "and the frequencies are these".

In the language of Chapter 25, C2 has a strong **inductive bias**, and here it is the *right* one, because the explicit formula says, heuristically, that the fluctuation is approximately this kind of function (recall from Chapter 18 that the prime sum does not converge on the critical line, and that its truncations are approximations whose accuracy is established numerically, as in Table V). The paper's own words are worth keeping: "sparse in t-frequency" is "itself a strong and well-chosen prior". It is well chosen *because we know the answer*; on data that were not almost periodic it could fail badly. Its success is evidence that the prior matches the data, not evidence that sparse priors are generally superior.

What about a fitted potential $V(x)$? Recall from Chapter 13 that a change δV in the potential shifts the level E_n by $\int |\psi_n|^2 \, \delta V \, \mathrm{d}x$ to first order, and from Chapter 12 that a level is most sensitive to the potential near its classical turning points. So a potential addresses levels *locally*: a bump at position x moves the levels whose turning points lie near x. Nothing in that parametrization says "the pattern found in levels 1–400 continues, with the same frequency in t, in levels 401–500". The paper's continuation probe (Sec. III) tests exactly this: trained on a pure $p = 2$ oscillation (running example RE2), the fitted potential represents it in-sample (rms 0.056, correlation +0.95) and does not continue it (held-out correlation +0.07). This local-versus-global

reading is the paper's heuristic explanation; it states that whether the memorization result "can be derived rather than measured is an open problem".

Checkpoint 27.6

Both C1 and the 900-coefficient B are underdetermined on 400 zeros, yet Table IV gives B 0.180/+0.78 and C1 0.258/+0.28. What does B have that C1 lacks?

Answer: The right frequencies. A heuristic reason for B's positive skill is that every member of its family, whatever its amplitudes, oscillates forever at the frequencies $m \log p$, so the in-window fit is carried forward at the correct frequencies. C1's members are periodic with period about 353 and, through its kernel, predict by local interpolation and copying.

Key Ideas

- A prior is a statement that excludes possibilities. "Few persistent sinusoids" excludes almost everything, and when it is true it lets a small model generalize.
- The parameter count is not the issue: C1 and the fitted potentials have many parameters and do not generalize; B (with 150) and C2 (with 72) have fewer and do. What matters is the parametrization.
- A potential addresses levels locally through their turning points, a heuristic reason it does not express "keep oscillating at $\log 2$".

27.9 From a fitted $F(t)$ back to zeros: root ambiguity

27.9.1 When does the level equation have one root?

To predict zero n we solve $G_n(t) = 0$ with

$$G_n(t) = N_{\mathrm{sm}}(t) + F(t) - (n - \tfrac{1}{2}). \tag{27.47}$$

Its derivative is $G_n'(t) = N_{\mathrm{sm}}'(t) + F'(t)$. Differentiate (27.1) with $u = t/2\pi$: $\frac{\mathrm{d}}{\mathrm{d}t}[u \log u - u] = \frac{1}{2\pi}[\log u + 1 - 1]$, so

$$N_{\mathrm{sm}}'(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}. \tag{27.48}$$

For F of form (27.7), write each pair as $R_j \sin(\omega_j t + \varphi_j)$ by (27.14). Its derivative is $R_j \omega_j \cos(\omega_j t + \varphi_j)$, at most $R_j \omega_j$ in magnitude, so

$$|F'(t)| \le \sum_j R_j\, \omega_j. \tag{27.49}$$

Sufficient condition for a unique root. If

$$\sum_j R_j\, \omega_j < \frac{1}{2\pi} \log \frac{t}{2\pi} \quad \text{throughout the bracket,} \tag{27.50}$$

then $G_n'(t) \ge N_{\mathrm{sm}}'(t) - |F'(t)| > 0$ there, so G_n is strictly increasing. A strictly increasing function crosses zero at most once; if it changes sign across the bracket it crosses exactly once (Chapter 4, bisection). Note the link to (27.10): the same comparison of $|F'|$ with $N_{\mathrm{sm}}' = 1/\mathrm{gap}$ decides whether residuals translate cleanly into zero errors.

The condition is sufficient, not necessary: it assumes every sinusoid has its steepest slope at the same t. When it fails, G_n may decrease somewhere, and then for some n the level equation can have three (or more) roots inside the bracket. A smooth counting model that decreases is also unphysical, since a counting function never decreases.

Sample Problem 27.12: Is the prime-2 wave safe?

For running example RE2, $F(t) = -\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 2)}{m2^{m/2}}$, find the maximum of $|F'|$, the height t^* above which (27.50) holds, and the values of $N'_{\rm sm}$ at t_1 and near zero 400.

Step 1: differentiate. $\frac{\mathrm{d}}{\mathrm{d}t}\frac{\sin(mt\log 2)}{m} = \log 2\,\cos(mt\log 2)$; the m cancels. So $F'(t) = -\frac{\log 2}{\pi}\sum_{m=1}^{3}2^{-m/2}\cos(mt\log 2)$.

Step 2: maximum. The bound is $\frac{\log 2}{\pi}(2^{-1/2}+2^{-1}+2^{-3/2}) = 0.2206\times 1.5607 = 0.3443$. It is attained: at $t = 2\pi k/\log 2$ every cosine equals 1.

Step 3: threshold. Solve $\frac{1}{2\pi}\log\frac{t^*}{2\pi} = 0.3443$: $\log(t^*/2\pi) = 2.1636$, $t^*/2\pi = 8.70$, $t^* = 54.7$.

Step 4: compare. At $t_1 = 14.1347$, $N'_{\rm sm} = \frac{1}{2\pi}\log(2.2496) = 0.129$, well below 0.3443; near zero 400, $t \approx 680$, $N'_{\rm sm} = \frac{1}{2\pi}\log(108.2) = 0.745$, comfortably above.

$\max|F'| = 0.3443$; unique roots guaranteed for $t > t^* \approx 54.7$; $N'_{\rm sm}(t_1) \approx 0.129$, $N'_{\rm sm}(680) \approx 0.745$

For the 25-prime oracle the bound (27.49) exceeds 4, far above 0.745; the sufficient condition fails badly. Failure of a sufficient condition does not by itself mean that roots are ambiguous, since the bound is reached only if all the slopes line up. We have not examined how often the oracle's level equation is non-monotone. The paper flags root ambiguity only for the 900-coefficient B row (below). Fitted amplitudes, however, can be much larger than the true ones: an underdetermined least-squares fit that matches the data exactly may do so with large, partly cancelling amplitudes, and then F' can genuinely exceed $N'_{\rm sm}$.

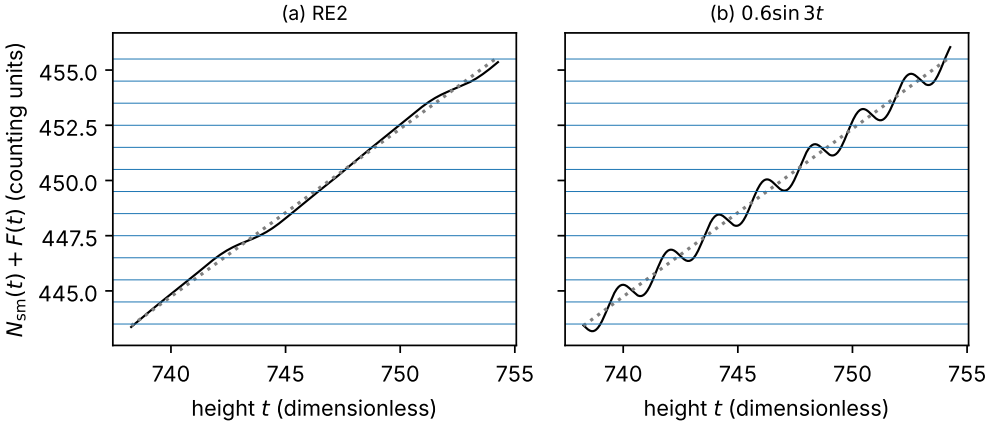


Figure 27.6: Root ambiguity of the level equation near zero 450 (t dimensionless). Solid: $N_{\rm sm}(t) + F(t)$; horizontal lines: the levels $n - \frac{1}{2}$; dotted: $N_{\rm sm}(t)$ alone. (a) F = the prime-2 wave RE2: the curve is increasing, and each level is crossed once. (b) $F = 0.6\sin(3t)$, whose slope reaches 1.8, above $N'_{\rm sm} \approx 0.77$: the curve wiggles back, and some levels are crossed three times, so "the" predicted zero is not defined.

27.9.2 What the paper does about it

The paper's root finder brackets ± 3 mean spacings around s_n and expands the bracket if the sign does not change (Appendix A step (8)); Brent's method then returns *one* root in the bracket, which is a choice, not a uniquely defined answer, when several exist. The paper flags its 900-coefficient B row as root-ambiguous: "the level equation has several roots for about a third of the held-out levels", and "the score varies by at most 0.012 with the choice of root"

(Sec. IV; Table IV, dagger). Reporting the sensitivity to the arbitrary choice, as the paper does, is the honest response: the row is "not a clean predictive score", but its conclusion (below the null, positively correlated) does not hinge on the choice.

Checkpoint 27.7

If $G_n'(t) > 0$ everywhere in a bracket $[a, b]$ and $G_n(a) < 0 < G_n(b)$, how many roots does G_n have in the bracket?

Answer: Exactly one. A sign change guarantees at least one crossing, and a strictly increasing function cannot return to zero after crossing it.

Key Ideas

- The level equation $N_{\mathrm{sm}}(t) + F(t) = n - \frac{1}{2}$ has a unique root whenever $\sum_j R_j\omega_j < N_{\mathrm{sm}}'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$, (27.50); this is sufficient, not necessary.
- Large fitted amplitudes, typical of underdetermined fits, can make the model counting function non-monotone and the predicted zero ambiguous.
- The paper flags such a row and reports how much the score depends on the choice of root.

27.10 Paper connection

This chapter supplies the tools behind one paragraph of methods and one paragraph of results in the paper [1], together with Table IV and two steps of its reconstruction record.

Sec. II D, "Fitted sinusoidal controls". The paper explains why it adds these controls: the oracle of Eq. (4) "is given the functional form", so it is "not a fair ML competitor". All controls have "the shape $N_{\mathrm{sm}}(t) + F(t) = n - \frac{1}{2}$ with $F(t) = \sum_j [a_j \sin(\omega_j t) + b_j \cos(\omega_j t)]$", are "trained by least squares on $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ at the training zeros only", and are "scored exactly as the oracle". These are our (27.7), (27.2) and (27.8). Section 27.1 showed how residuals in y are related to zero errors in mean spacings, through the factor $1 + F'(t_n)\mathrm{gap}_n$ of (27.10). Control A is the oracle with up to 300 primes (900 terms, nothing fitted); B fits amplitudes at the given frequencies $m \log p$ (Section 27.2); C1 uses 450 frequencies on a uniform grid (Section 27.3); C2 finds K frequencies by greedy periodogram search on $0.05 < \omega < 8$ with joint refits, $3K$ parameters (Section 27.4). Ridge strength and K are chosen by the forward 75/25 split of Section 27.5.

Appendix A step (20). This step records the details we used: the ridge filter $\Sigma/(\Sigma^2 + \lambda\Sigma_1^2)$, i.e. (27.18), with $\lambda = 10^{-8}, \ldots, 10^3$ in half decades (23 values); the uniform grid $\omega_j = 8j/450$; the sparse search with step a quarter of $2\pi/T$, refinement on 201 points, joint refit after each added frequency, and $K \le \min(300, n_{\mathrm{fit}}/3)$; scoring as in step (8) (Brent, bracket ± 3 around the smooth zero, expanded on sign failure); and the leakage test (held-out data overwritten with noise, coefficients bit-identical).

Sec. IV and Table IV. The three levels of supplied information are our Table 27.1. Reading the rows of Table IV with this chapter's tools:

- A, 300 primes: 0.018/+1.00 on 401–500 and 0.020/+1.00 on 801–1000, interval $[0.015, 0.020]$; the oracle "improves monotonically" with budget. Exact frequencies, no dephasing (Section 27.6).

- B, 25 primes: 0.028/+0.99 and 0.040/+0.99 with 150 fitted amplitudes. B, 150 primes: 0.180/+0.78 and 0.100/+0.94 with 900, underdetermined on 400 zeros (Section 27.2) and root-ambiguous (Section 27.9), interval $[0.158, 0.201]$.
- C1: 0.258/+0.28 (interval on the rms $[0.231, 0.285]$, on the correlation $[+0.08, +0.46]$) and 0.267/+0.22. Within 0.005 of the null 0.263: "capacity without a suitable prior does not generalize, and in this respect the operator family is not special". Section 27.3 derived the kernel form (27.27) and the periodicity (27.21) that make C1 a generic basis.
- C2: 24 frequencies (72 parameters) from 400 zeros, 0.034/+0.99, interval $[0.028, 0.041]$, "the level of the 25-prime oracle" (0.043/+0.99); 43 frequencies (129 parameters) from 800 zeros, 0.034/+0.99 on 801–1000.

The operator rows (0.369/−0.07 at $M = 400$; 0.313/−0.16 and 0.279/−0.03 at $M = 600$; 0.302/+0.08 at $M = 900$) come from the paper's Table I.

The prime-power alignment and Appendix A step (23). The paper reports that "every selected frequency lies within half the Fourier resolution of the training window of the logarithm of a prime power", our criterion (27.45), and lists the 43 prime powers we verified in Section 27.7. Because the frequencies are the output of a search, the alignment was rerun on 900 surrogate replicates; the cutoff-maximized binomial score is 24.7 and 50.9 on the real zeros and at most 4.7 on surrogates, with the explicit qualification that the score was "defined after the data were seen". Our density estimate e^{ω}/ω suggests why a cutoff matters and why a cutoff-free statistic is useful.

Dephasing. The paper observes that C2's skill "decays with distance from the training window as the estimated frequencies dephase (0.078/+0.96 on zeros 1001–2000 for the 400-zero fit), which the exact frequencies of the oracle do not". Section 27.6 derived the mechanism, (27.39)–(27.43). The paper's Sec. VI limits note that "the fitted sinusoidal controls were scored only below zero 2000".

"The obstacle is the parametrization, not the parameter count." This sentence of the abstract, and the paragraph closing Sec. IV ("what fails is a potential $V(x)$ as the parametrization"), rest on the combination: C1 (900 coefficients, generic) fails like the potentials; C2 (72 parameters, sparse) succeeds without being told any arithmetic. Section 27.8 spelled out the inductive-bias reading, together with the paper's own hedges: sparsity in frequency is "a strong and well-chosen prior", C2 is "a control built on" the classical Landau–Gonek fact [18, 19], "not a new result", and the explanation of the potentials' failure is heuristic.

Fig. 1 and Table V. Fig. 1 of the paper shows zeros 370–460 (fit on 1–400) with the six-prime oracle and a fitted operator. The oracle's held-out correlation is +0.96, the 401–500 value of Table V. Table V shows the oracle's skill by holdout and prime budget. These are the level-(i) baselines against which C2's level-(ii) result is measured.

27.11 Summary

★ Did you know? *Prony's 1795 Fit*

In 1795 Gaspard de Prony published a method for fitting data with a sum of exponentials by turning the problem into polynomial root finding. With complex exponents those exponentials become oscillations, so Prony's method extracts frequencies and damping rates from short data records. Variants are still used in power systems and signal processing

more than two centuries later.

- The training target is one number per zero, $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n) \approx S(t_n)$, (27.2); it equals minus the fluctuation in mean spacings to high accuracy, (27.5), and fitting residuals translate into zero errors by (27.10).
- With frequencies fixed, a sum of sinusoids is linear in its amplitudes, fitted by the normal equations (27.13) or the ridge filter (27.18); amplitude and phase follow from (27.14).
- Columns at frequencies separated by much more than $2\pi/T$ are nearly orthogonal, (27.17): the resolution limit in least-squares form.
- A uniform frequency grid is exactly periodic, (27.21); its ridge fit is a kernel interpolator with the Dirichlet kernel, (27.27)–(27.29). This is why control C1 behaves like a generic basis.
- A sinusoid of amplitude A gives a periodogram peak $\approx n_{\mathrm{fit}} A^2/4$ of width $2\pi/T$, (27.30)–(27.31); greedy search with joint refits (orthogonal matching pursuit) is control C2.
- K and the ridge are chosen by a forward split inside the training window; training error alone never selects them.
- Frequency precision scales as $\sqrt{24}\,\sigma/(A\sqrt{n_{\mathrm{fit}}}\,T)$, (27.38); a frequency error dephases as $\epsilon = \delta\omega\,\Delta t$, (27.39), giving correlation $\cos\epsilon$ and rms ratio $2\,|\sin(\epsilon/2)|$, (27.42)–(27.43).
- Learned frequencies are matched to prime powers by $|\omega - \log q| < \pi/T$, (27.45); amplitudes are $1/(\pi m\sqrt{q})$, (27.44); chance matching grows like e^{ω}/ω.
- The level equation has a unique root when $\sum_j R_j\omega_j < \frac{1}{2\pi}\log\frac{t}{2\pi}$, (27.50); large fitted amplitudes can create root ambiguity.

27.12 Problems

1. • Control B uses frequencies $m\log p$ with $m = 1, 2, 3$. How many frequencies and how many fitted coefficients would it have with 50 primes? With 100 primes? Which of these configurations is underdetermined on 400 training zeros, and which on 800?

2. • A fitted pair has $a = 0.3$, $b = -0.4$. Write $a\sin\omega t + b\cos\omega t$ as $R\sin(\omega t + \varphi)$: find R and φ.

3. • For the 800-zero window take $T = 1170$. Compute the resolution $2\pi/T$ and the matching tolerance π/T. Does a learned frequency $\omega = 1.0996$ match $\log 3$?

4. • A learned component has frequency error $\delta\omega = 0.002$. Using (27.42), at what distance from the window centre does its prediction become worse than the null?

5. • A ridge grid runs from 10^{-6} to 10^{2} in half decades. How many values does it contain? Check your method against the paper's grid of 23 values.

6. •• (a) Prove that $F(t) = \sum_{j=1}^{450}[a_j\sin(j\Delta\omega t) + b_j\cos(j\Delta\omega t)]$ with $\Delta\omega = 8/450$ satisfies $F(t + T_{\mathrm{rep}}) = F(t)$ for $T_{\mathrm{rep}} = 2\pi/\Delta\omega$. (b) For held-out heights $680 < t \le 811$, find the two intervals of images $t - T_{\mathrm{rep}}$ and $t - 2T_{\mathrm{rep}}$, and say which of them overlap the training window $14 \le t \le 680$. (c) Exploration: the $p = 2$ wave nearly repeats after T_{rep} (Section 27.3). Compute the phase shift after T_{rep} of the $p = 3$ and $p = 5$ waves. Explain why this observation alone cannot tell you whether the periodic images account for C1's small positive correlation, and what computation would.

7. •• A residual contains two well-separated sinusoids with amplitudes $A_1 = 0.225$ and $A_2 = 0.184$, sampled at 400 points over many periods. Estimate the two periodogram peak heights and their ratio. Which is picked first?

8. •• Derive (27.43): show that the correlation of $\sin\theta$ and $\sin(\theta+\epsilon)$, averaged over uniformly distributed θ, is $\cos\epsilon$.

9. •• Repeat the second matching-pursuit sample with atoms $\boldsymbol{a} = (1, 0)$, $\boldsymbol{b} = (0.6, 0.8)$ and data $\boldsymbol{y} = 0.5\,\boldsymbol{a} + \boldsymbol{b}$. Which atom is picked first? Give the plain matching-pursuit coefficients after two steps, the remaining residual length, and the orthogonal matching-pursuit coefficients.

10. •• For $F(t) = B\sin(\omega t)$ with $B = 0.1$, $\omega = 4$, find the height above which (27.50) guarantees a unique root of the level equation.

11. •• (Paper.) From Table IV, compute the number of training zeros per fitted parameter for C2 on 400 zeros (72 parameters) and on 800 zeros (129 parameters), and for B with 900 coefficients on 400 and on 800 zeros. Relate the B numbers to the improvement from 0.180/+0.78 to 0.100/+0.94.

12. •• (Paper.) C1 on zeros 401–500 has rms interval $[0.231, 0.285]$ and correlation interval $[+0.08, +0.46]$ (Table IV caption and Sec. IV). The null is 0.263. Does C1 predict better than the null? Is its correlation distinguishable from zero? Explain how both statements can be true at once (recall Chapter 19 on rms versus correlation).

13. •• (Paper, toy model.) Suppose all of C2's held-out error came from the dephasing of a single sinusoid, so that (27.42) applies with "null rms" the paper's null. Using C2's 0.034 on 401–500 (null 0.263) and 0.078 on 1001–2000 (null 0.279, Table V), find the implied phase errors. Give two reasons the toy model cannot be taken literally.

14. ••• Derive (27.38) in full when both ω and φ are fitted, writing the 2×2 normal-equation matrix for $(\delta\omega, \delta\varphi)$ and showing that its off-diagonal entry is small when time is measured from the window centre. What happens to the variance of $\widehat{\delta\omega}$ if time is measured from the start of the window instead and the cross-term is not neglected?

15. •• (Paper.) List all prime powers $q = p^m \le 200$ with $m \ge 2$ and compute their amplitudes (27.44). Which five are the weakest? Compare them with the three prime powers up to 113 (32, 64, 81) that the paper's 800-zero sparse search left out, and with the smallest amplitude of any prime up to 200.

16. ••• Using the heuristic density e^{ω}/ω of values $\log p$, find the frequency above which the expected number of $\log p$ values in a matching cell of width $2\pi/T$ exceeds 1, for $T = 666$ and for $T = 1170$. How do the squares and cubes of primes change the estimate?

17. •• Using qmnt and numpy, compute y_n for $n \le 400$ and its periodogram (27.30) on the coarse grid of the paper. Where is the largest peak? Does it satisfy (27.45) for some prime power?

18. •• Using qmnt and numpy, extend the control-B listing of Section 27.2: choose the ridge strength from the paper's 23-value grid by the forward 75/25 split (scoring the validation zeros through the level equation), refit on zeros 1–400, and score zeros 401–500. Compare with the paper's 0.028/+0.99.

19. •• Using `qmnt.oracle_zeros` and `qmnt.score`, reproduce the six-prime oracle entry of Table V on zeros 401–500. Then add your result for the sparse model of Problem 20 to a small table alongside it.

20. ••• Using the greedy listing, implement control C2 with K chosen by forward validation (cap $K \le 133$), score zeros 401–500 and 1001–2000, and compare with the paper's 0.034/+0.99 and 0.078/+0.96. Report how many of your selected frequencies satisfy (27.45).

21. •• For the $p = 3$ wave $F(t) = -\frac{1}{\pi}\sum_{m=1}^{3} \frac{\sin(mt\log 3)}{m\,3^{m/2}}$, find $\max|F'|$ and the threshold height t^* of (27.50). Then find the bound (27.49) and the corresponding t^* for the sum of the $p = 2$ and $p = 3$ waves. In each case, is a unique root guaranteed on the paper's holdout 401–500?

22. •• For a 600-zero training window, give the inner fit and validation sizes and the cap on K. Then compute $T = t_{600} - t_1$ with qmnt and the approximate number of coarse search frequencies on $0.05 < \omega < 8$.

23. ••• Using (27.38), estimate the precision of the $p = 5$ frequency learned from 800 zeros, with $A = 0.142$, $\sigma = 0.2$, $n_{\text{fit}} = 800$, $T = 1170$. Express it as a fraction of π/T. Would you expect weak components such as $q = 16$ to satisfy the matching criterion as comfortably?

24. •• The coarse grid for the 800-zero window has step $\frac{1}{4}\cdot 2\pi/T$. Show that the worst-case loss of peak height from the coarse grid alone is the same for every window length, and explain why the refinement step is needed nonetheless.

25. ••• Write a one-page argument, using the first-order level shift $\delta E_n = \int |\psi_n|^2\,\delta V\,\mathrm{d}x$ and the localization of sensitivity near turning points (Chapters 13 and 12), for why a potential fitted to levels 1–400 has no built-in reason to continue a $\sin(t\log 2)$ pattern to levels 401–500. Then state precisely what the paper does and does not claim about this (Secs. III and VI).

26. •• The prime powers $125 = 5^3$ and 127 give frequencies $\log 125$ and $\log 127$. How many resolution cells apart are they for $T = 666$? Could the 400-zero window separate them? Which of the two has the larger amplitude?

27. • With $T = 666$, which of the frequencies 1.3870, 1.7000, 2.3979 satisfy the matching criterion (27.45), and with which prime powers?

27.13 Answers to selected problems

1. Three frequencies per prime, two coefficients per frequency. 50 primes: 150 frequencies, 300 coefficients. 100 primes: 300 frequencies, 600 coefficients. On 400 training zeros the 300-coefficient model is overdetermined ($300 < 400$) and the 600-coefficient model underdetermined ($600 > 400$). On 800 zeros both are overdetermined ($600 < 800$).

2. $R = \sqrt{0.09 + 0.16} = 0.5$; $\varphi = \operatorname{atan2}(-0.4, 0.3) = -0.927$ rad. Check: $0.5\cos(-0.927) = 0.3$, $0.5\sin(-0.927) = -0.4$.

3. $2\pi/1170 = 0.00537$; $\pi/1170 = 0.002685$. $|1.0996 - 1.098612| = 0.000988 < 0.002685$: yes, it matches $\log 3$.

4. Worse than null when $\epsilon > \pi/3$, i.e. $\Delta t > (\pi/3)/0.002 = 524$ (more precisely 523.6).

5. Exponents $-6, -5.5, \dots, 2$: $(2-(-6))/0.5+1 = 17$ values. The same count for $-8, \dots, 3$ gives $11/0.5 + 1 = 23$, as in the paper.

7. By (27.31) at the peaks, $\mathcal{P}_1 \approx 400(0.225)^2/4 = 5.06$ and $\mathcal{P}_2 \approx 400(0.184)^2/4 = 3.39$, ratio 1.50. The first (A_1) is picked first.

9. $\boldsymbol{y} = (1.1, 0.8)$. $\boldsymbol{y}\cdot\boldsymbol{a} = 1.1$, $\boldsymbol{y}\cdot\boldsymbol{b} = 0.66 + 0.64 = 1.3$: $\boldsymbol{b}$ is picked first, coefficient 1.3. Residual $\boldsymbol{y} - 1.3\boldsymbol{b} = (0.32, -0.24)$; $\boldsymbol{r}\cdot\boldsymbol{a} = 0.32$, so plain pursuit gives coefficients 0.32 on $\boldsymbol{a}$ and 1.3 on $\boldsymbol{b}$, leaving residual $(0, -0.24)$ of length 0.24. Orthogonal pursuit solves $\alpha + 0.6\beta = 1.1$, $0.8\beta = 0.8$: $\beta = 1$, $\alpha = 0.5$, residual zero.

11. C2: $400/72 = 5.56$ and $800/129 = 6.20$ zeros per parameter. B(900): $400/900 = 0.444$ and $800/900 = 0.889$. Both B configurations are underdetermined, but with 800 zeros the data

pin down twice as many directions in coefficient space, so the amplitudes are better determined and the held-out score improves.

13. $0.034/0.263 = 0.129 = 2\sin(\epsilon/2)$ gives $\epsilon \approx 0.129$ rad; $0.078/0.279 = 0.280$ gives $\epsilon \approx 0.280$ rad. Reasons the toy fails: the model has 24 components with different frequency errors and amplitudes, not one; and part of the error is not dephasing at all (omitted terms, amplitude errors, the finite sampling of the window), which is why the error is nonzero even just above the window.

15. The prime powers with $m \geq 2$ up to 200 are $4, 8, 16, 32, 64, 128$; $9, 27, 81$; $25, 125$; 49; 121; 169: 14 in all. With amplitudes $1/(\pi m\sqrt{q})$, the five weakest are $B_{128} \approx 0.0040$, $B_{64} \approx 0.0066$, $B_{81} \approx 0.0088$, $B_{125} \approx 0.0095$ and $B_{32} \approx 0.0113$. Every prime $p \leq 200$ has $B_p = 1/(\pi\sqrt{p}) \geq 1/(\pi\sqrt{199}) \approx 0.0226$, which is larger. So the three prime powers up to 113 that the 800-zero search left out (32, 64, 81) are exactly the three weakest prime powers up to 113, and they are among the five weakest up to 200. The other two (125, 128) lie above 113.

17. The largest peak lies within π/T of $\log 2 = 0.6931$ (you should find it within a few thousandths), so it matches the prime power $q = 2$. This is the dominant term of RE2, as the amplitude ordering of Table 27.2 predicts.

19. You should reproduce Table V's six-prime entry on 401–500: rms 0.074 and correlation +0.96, the paper's values (Table V; also Fig. 1). Small differences in the third decimal can come from your bracket or tolerance choices; anything larger points to an error in the prime list or the $m \leq 3$ truncation.

21. As in the RE2 sample, the factor m cancels on differentiation, so $F'(t) = -\frac{\log 3}{\pi}\sum_{m=1}^{3} 3^{-m/2}\cos(mt\log 3)$. Its maximum magnitude is $\frac{\log 3}{\pi}(3^{-1/2} + 3^{-1} + 3^{-3/2}) = 0.3497 \times 1.1031 = 0.3858$, attained at $t = 2\pi k/\log 3$. Then $t^* = 2\pi e^{2\pi\times 0.3858} \approx 70.9$. For the sum of the $p = 2$ and $p = 3$ waves the bound (27.49) is $0.3443 + 0.3858 = 0.7301$, giving $t^* = 2\pi e^{2\pi\times 0.7301} \approx 617$. The holdout 401–500 lies at $t \geq 680$, where $N'_{\mathrm{sm}} \geq 0.745$. That exceeds both bounds, so a unique root is guaranteed in both cases, though only narrowly for the two-prime sum.

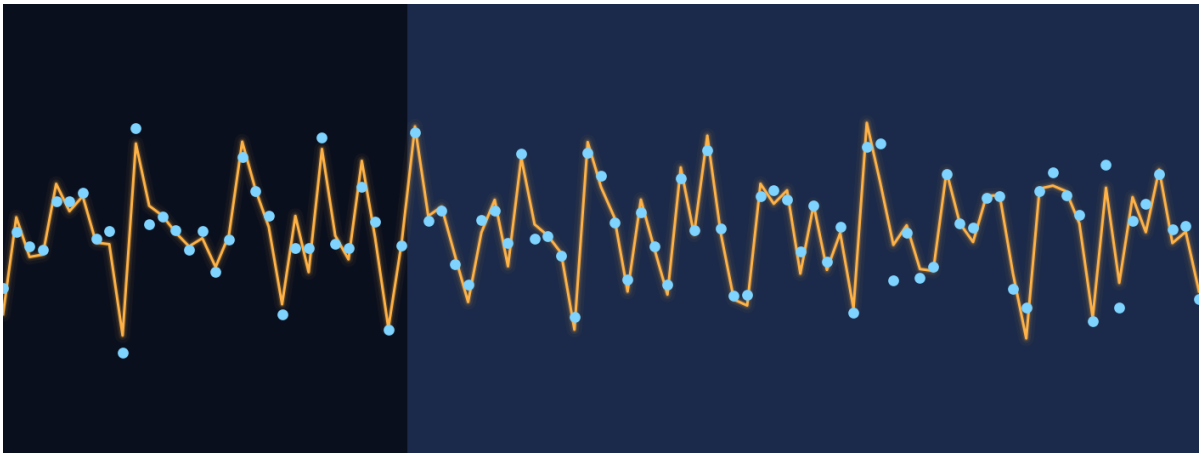
23. $\sqrt{24}\,(0.2) = 0.980$; $0.142 \times \sqrt{800} \times 1170 = 4699$; so $\mathrm{std}(\widehat{\omega}) \approx 2.09 \times 10^{-4}$, which is 0.078 of $\pi/T = 0.002685$. For $q = 16$, $A \approx 0.020$, the same formula gives about 1.5×10^{-3}, roughly 0.55 of π/T. The tolerance is then only about 1.8 standard deviations, so under a Gaussian error model a correct component would miss the criterion about 7% of the time. For $p = 5$ the tolerance is about 13 standard deviations, so a miss essentially never happens.

25. Outline. By first-order perturbation theory, $\delta E_n = \int |\psi_n|^2\,\delta V\,\mathrm{d}x$. For high levels, $|\psi_n|^2$ is largest (classically, the particle is slowest) near the turning points $\pm x_{E_n}$, so δE_n is dominated by δV near x_{E_n}. Levels 1–400 have turning points inside a band $|x| \leq x_{E_{400}}$; levels 401–500 turn further out. The training residuals constrain δV mainly inside the band; outside it, δV is set by whatever the basis functions fitted inside happen to do there, which is fixed by the basis and optimizer, not by a rule that says "continue the oscillation in t". Nothing in $\int |\psi_n|^2\,\delta V$ ties a sinusoid in E (the level heights) to a structure in x that extends itself. The paper claims only what it measured: the probe fit (Sec. III) represents the $p = 2$ wave in-sample (rms 0.056, corr +0.95) and does not continue it (corr +0.07); in Sec. VI it calls the local-versus-global reading heuristic, says the evidence concerns "the tested parametrizations and optimizer together, not a statement about what a potential can represent", and that deriving the memorization result is open.

27. $\pi/T = \pi/666 = 0.004717$. $|1.3870 - \log 4| = |1.3870 - 1.386294| = 0.00071$: matches $q = 4$. 1.7000: the nearest prime powers are 5 ($\log 5 = 1.6094$) and 7 ($\log 7 = 1.9459$), both far: no match. $|2.3979 - \log 11| = 0.000005$: matches $q = 11$.

Part VI

The Paper, Section by Section



Zeros 370 to 460: true fluctuations (dots) and the six-prime explicit formula (line), which knows nothing of the zeros it predicts.

Chapter 28

Reading the Paper I: The Question, the Data and the Protocol

↑ Contents

What this chapter is about

You now have every tool the paper uses: complex numbers and series, Fourier analysis, linear algebra and eigenvalues, quantum mechanics on a line, the zeta function and its explicit formula, level statistics, finite differences, fitting, and the statistics of generalization. This chapter begins Part VI, where we stop learning new mathematics and read the paper [1] itself. We read it the way a careful referee would. For each claim we ask what it says in plain words, which earlier chapters supply the tools, what computation stands behind it (and we rerun that computation at small scale with `qmnt` wherever we can), and how strong it is and where it stops.

This first reading covers the front half of the paper: the abstract, the introduction with its scope statement and its account of prior work (Sec. I and I A), the data and the unfolding (Sec. II A), the background potential and the solver (Sec. II B), the fitting and out-of-sample protocol (Sec. II C), the arithmetic oracle and its fitted controls (Sec. II D), the disclosure of artificial-intelligence use (Sec. II F), Figs. 1 and 2, and steps 1–8 of the reconstruction record in Appendix A. No result tables are interpreted here in depth; that is the job of Chapters 29, 30 and 31. What this chapter gives you is the protocol: exactly what is computed, against what baseline, and why the baseline is fair.

Where this shows up in the paper: everywhere. Every number in Tables I–V is an rms or a correlation computed by the protocol of Secs. II A–D, and every one of them is judged against the null model defined there.

What you need

- Chapter 1: memorizing versus predicting, training versus held-out data.
- Chapters 2–4: complex exponentials, Taylor series, Lambert W, Brent's method, floating-point precision.
- Chapters 5 and 6: almost-periodic sums, Fourier resolution, phase-randomized surrogates.
- Chapters 7–9: vectors and norms, eigenvalues, least squares and ridge.
- Chapters 10–13: $H = -d^2/dx^2 + V$ in units $\hbar = 2m = 1$, bound states in a box, WKB turning points, first-order sensitivity.
- Chapters 15–18: primes, the zeta function, $N(T)$, N_{sm}, smooth zeros, the truncated explicit formula.
- Chapters 19–22: rms, Pearson correlation, the Cauchy–Schwarz inequality, GUE statistics, the Hilbert–Pólya conjecture, the Wu–Sprung potential.
- Chapters 23–27: finite-difference dispersion, Richardson extrapolation, Levenberg–Marquardt fitting, null models, splits, surrogates, sparse frequency search.
- First-year physics: potential energy curves and turning points, standing waves.

28.1 How to read a research paper

A textbook proves things and then moves on. A research paper does something different: it makes *claims* and offers *evidence*, and the evidence is usually a computation that you were not present for. Reading such a paper well means keeping the claim and the evidence apart in your head, and checking that the second actually supports the first.

In this part of the book every claim gets the same four questions.

(a) **What does it say?** Restate the claim in plain words, without the technical vocabulary.

(b) **Which tools?** Name the chapters of this book that supply the mathematics.

(c) **What computation?** Identify the calculation behind the claim and, where it is cheap, rerun a small version yourself.

(d) **How strong, and where does it stop?** Is the statement a theorem, a numerical observation, a heuristic explanation, or an open question? Over what range of models and data was it tested?

Question (d) uses the status vocabulary you have met throughout the book: *theorem (proved)*, *proved but beyond this book*, *heuristic or approximate*, *conjecture or open*, and *numerical observation*. Almost everything in the paper is a numerical observation about a specific, scoped family of models. Two background statements are open conjectures: the Riemann hypothesis (RH) and the Hilbert–Pólya conjecture. The paper never asserts either, and neither will we.

One more piece of reading discipline: the paper reuses some letters. Keep these in mind.

- N is the number of interior grid points of the finite-difference solver (Sec. II B) *and* the counting function $N(T)$ of the zeros (Chapter 17).
- A is the first-order sensitivity matrix A_{nk} (Sec. II C) *and* the non-Hermitian operator in $H + iA$ (Sec. II E).
- T is the length of a training window in t (Sec. II D), *and* the second-order tensors $T^{(n)}$ (Sec. II E), *and* the Chebyshev polynomials T_k (Sec. II C, "Other bases").

Context always tells you which is meant, but only if you are watching for it.

Key Ideas

- Separate claim from evidence; for each claim ask (a) what, (b) which tools, (c) which computation, (d) how strong and how far.
- Nearly every result in the paper is a scoped numerical observation; RH and Hilbert–Pólya are open conjectures.
- The symbols N, A and T each carry more than one meaning in the paper.

28.2 The abstract, sentence by sentence

An abstract compresses a whole paper into a paragraph, so every word is load-bearing. The paper's abstract has eight sentences; we take them one at a time. Table 28.1 collects where each is substantiated.

S1. "Empirical Hilbert–Pólya constructions fit quantum Hamiltonians to the nontrivial Riemann zeros and judge success by the quality of the fit."

(a) There is a line of work that builds a quantum system whose energy levels are supposed to match the heights t_n of the zeta zeros, and those works have measured success by how well the levels match the zeros used to build the system. (b) Chapters 21 and 22. (c) No computation; this is a description of the literature, supported in Sec. I A and by the audit in Appendix A. (d) It is a characterization of prior work. Its strength rests on the literature audit, whose limits we examine in Section 28.4.

S2. "We apply the test this program has lacked: out-of-sample prediction."

(a) The paper hides some zeros from the fitting procedure and asks whether the fitted system predicts them. (b) Chapter 25. (c) The protocol of Sec. II C. (d) "Has lacked" is again a claim about the literature, hedged in the body as "to our knowledge".

S3. "Fluctuation potentials on the Wu–Sprung background with up to 900 fitted coefficients, in cosine, Chebyshev, and local piecewise-linear bases, fit their training zeros closely yet show no reproducible held-out predictive skill relative to the unfitted potential, whether the held-out zeros lie above the training range or are interleaved with it."

(a) Take the smooth Wu–Sprung potential, add a correction built from up to 900 adjustable numbers, tune them so the levels match the training zeros. The match is good, but the tuned system predicts hidden zeros no better than the untuned one, whether the hidden zeros are higher up or mixed in among the training ones. (b) Chapters 22, 23, 24, 25. (c) Tables I–III (paper Sec. III), analysed in Chapter 29. (d) Note the qualifiers. "Wu–Sprung background": one background potential. "Cosine, Chebyshev, and local piecewise-linear": three bases. "Reproducible": one Chebyshev row is marginal (Table II), and the paper says so. "Relative to the unfitted potential": the comparison is with a specific null, which we define in Section 28.8.

S4. "They fit shuffled and phase-randomized surrogate targets as well as the real zeros and, trained on a pure prime-2 explicit-formula oscillation, fail to continue it beyond the training window."

(a) Two diagnostic experiments. First, scramble the target fluctuations (by shuffling, or by keeping the power spectrum and randomizing phases) and refit: the fit is just as good, so fit quality does not detect real structure. Second, replace the target by a single known periodic-looking signal, the $p = 2$ term of the explicit formula (our running example RE2),

and ask whether the fitted potential keeps oscillating past the training window: it does not. (b) Chapters 6, 26, 18. (c) Paper Sec. III, paragraphs "Permutation and surrogate tests" and "Arithmetic-continuation probe". (d) Numerical observations on the scale-1, $M = 400$ configuration; Chapter 29 gives the numbers.

S5. "By contrast, a truncation of the Riemann–von Mangoldt explicit formula with zero fitted parameters predicts held-out zeros with correlation up to $+0.99$ through zero 10^4."

(a) A formula built from a handful of primes, with nothing tuned to the zeros at all, predicts the hidden zeros well, and keeps doing so up to the ten-thousandth zero. (b) Chapter 18. (c) Table V; we reproduce two of its entries in Section 28.9. (d) Numerical observation. The formula is classical and the paper claims no novelty for it (Sec. I A b).

S6. "Controls with fitted sinusoidal models show that the obstacle is the parametrization, not the parameter count: a dense 900-coefficient Fourier model also stays near the null, while a fitted control given only a sparsity prior in frequency, and no arithmetic input, recovers frequencies at $\log p^m$ from 400 zeros and predicts held-out zeros with correlation $+0.99$."

(a) Two more fitted models, neither told about primes. One has 900 coefficients spread over a uniform grid of frequencies and stays within 0.005 of the null. The other is told only "the signal is a sum of a few sinusoids", finds the frequencies by search, and those frequencies turn out to sit at logarithms of prime powers; it predicts as well as the 25-prime formula. So the number of parameters is not what matters; their form is. (b) Chapters 6, 9, 27. (c) Table IV, Sec. IV; Chapter 30. (d) Numerical observation. The paper itself calls "sparse in t-frequency" "a strong and well-chosen prior" and notes the concentration of the zeros' spectrum at $\log p^m$ is classical [18, 19].

S7. "A subsidiary dissection of second-order non-Hermitian perturbations shows how surrogate optimization outside its validity range can manufacture an apparently meaningful in-sample floor; exact diagonalization dissolves it."

(a) A side study: a different model family, fitted through a perturbative approximation, appeared to hit a common error floor; checking the exact spectrum showed the floor was an artifact. (b) Chapters 13 and 14. (c) Sec. V, Fig. 5; Chapter 31. (d) Labelled "subsidiary"; Sec. VI states it was not rerun for the revision.

S8. "We propose held-out prediction as a minimal falsifiability standard for empirical operator claims and provide the protocol, baselines, and a complete reconstruction record."

(a) A recommendation for future work, plus the materials to check this work. (b) Chapters 25 and 26. (c) Sec. VI items (i)–(iv); Appendix A; the Zenodo record [20]. (d) A proposal, not a result; Chapter 32 discusses it.

Checkpoint 28.1

Which sentence of the abstract would become false if the sparse-frequency control C2 had selected frequencies unrelated to prime powers, and which sentences would be unaffected?

Answer: S6 would lose its second half ("recovers frequencies at $\log p^m$"). S1–S5, S7 and S8 do not mention C2 and would be unaffected; in particular S3 and S5 concern the fitted potentials and the oracle, which do not involve C2.

Table 28.1: Where each sentence of the abstract is substantiated in the paper (R1 numbering) and which chapter of this book reads it.

sentence	content	paper location	read in
S1	prior work judged by fit quality	Sec. I, I A, App. A audit	this chapter
S2	the out-of-sample test	Sec. II C	this chapter
S3	fits memorize, do not predict	Sec. III, Tables I–III, Fig. 3	Ch. 29
S4	surrogates and continuation probe	Sec. III	Ch. 29
S5	explicit formula predicts	Sec. IV, Table V, Fig. 1	this chapter, Ch. 30
S6	fitted sinusoidal controls	Sec. II D, IV, Table IV	Ch. 30
S7	spurious non-Hermitian floor	Sec. II E, V, Fig. 5	Ch. 31
S8	falsifiability standard, record	Sec. VI, App. A	Ch. 32

Key Ideas

- The abstract makes three kinds of statement: about prior work (S1, S2), about numerical results (S3–S7), and a methodological proposal (S8).
- Every result sentence carries scope qualifiers (one background, three bases, "reproducible", "relative to the unfitted potential"); keep them when you summarize.

28.3 The question and its scope

28.3.1 The Hilbert–Pólya motivation

The paper's first sentence states the Hilbert–Pólya conjecture: the heights t_n of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of $\zeta(s)$ are the spectrum of a self-adjoint operator. Recall from Chapter 8 that a self-adjoint (Hermitian) operator has real eigenvalues; if the t_n were such eigenvalues they would be real, which is the statement that every nontrivial zero lies on the critical line. That is why the conjecture, if realized by a concrete operator, would prove RH (Chapter 21). The conjecture is open. The notation $\rho_n = \frac{1}{2} + it_n$ presumes the zeros used are on the critical line, which is numerically verified for every zero the paper uses; it presumes nothing about zeros in general.

The paper lists three pieces of "circumstantial evidence" with their sources.

1. Montgomery's pair-correlation theorem and Odlyzko's numerics match GUE eigenvalue statistics [9, 8]. Status: Montgomery's result is a theorem proved under the assumption of RH, and only for test functions whose Fourier transform vanishes outside the interval $(-1, 1)$ (this is the restriction on the test functions; compare the statement of Montgomery's pair correlation in Chapter 20); the full pair-correlation statement, without that restriction, is a conjecture; Odlyzko's agreement is a numerical observation (Chapter 20).

2. The zero-counting function resembles a semiclassical trace formula in which primes play the role of periodic orbits [10, 11]. Status: an analogy (Chapter 21).

3. Explicit operator candidates continue to be proposed [38, 40, 41, 36, 42, 43, 39]. Status: no proposal is accepted as an operator whose spectrum is the zeros (Chapter 21).

None of this is evidence that any particular operator exists, and the paper does not say it is.

28.3.2 The empirical program and its implicit logic

The second paragraph of Sec. I introduces the target. Wu and Sprung [12] built a smooth one-dimensional potential from the average density of zeros (Chapter 22) and added fitted corrections to reproduce the low zeros; successors refined this [13, 14, 15]. The paper names the reasoning that such work implicitly relies on:

> "The implicit logic is that a potential reproducing many zeros accurately has captured something about the putative zeta operator."

The paper tests this logic by the standard you met in Chapter 25: a model fitted to zeros $1, \ldots, n_{\text{fit}}$ "has captured structure, rather than data, only to the extent that it predicts zeros it was not shown." Remember the student who memorizes last year's exam answers (Chapter 1). Reproducing those answers perfectly says nothing about this year's exam. The fit quality on training zeros is the analogue of reproducing last year's answers.

28.3.3 The scope statement

Immediately after naming the test, the paper limits its own claim. Read this paragraph slowly, because every later claim inherits it. What is tested:

- only empirical inverse-spectral constructions whose parameters are fitted to finite sets of zeros;
- among those, only fluctuation potentials added to the Wu–Sprung background;
- in three bases (cosine, Chebyshev, hat; Sec. II C).

What is explicitly not tested:

- other backgrounds;
- other parametrizations: Hermite functions, wavelets, neural-network potentials;
- analytic Hilbert–Pólya constructions, "in which an operator is specified independently of the zeros and its spectrum is then predicted". The paper makes "no claim against" these.

Two definitions close the paragraph. First, "capture" is operational: "predictive transfer to held-out zeros, measured by rms error and correlation relative to the model's own smooth baseline." Second, the novelty claimed is "not the explicit formula or any representation of the zeros, but the controlled out-of-sample comparison and the associated falsifiability protocol."

The operational definition of "capture" is worth dwelling on. It replaces a vague word (has the model "captured" the zeta operator?) with two measurable numbers (rms error and correlation on hidden zeros) compared with a third (the same numbers for the model's own smooth part). We define all three precisely in Sections 28.7 and 28.8.

Pitfall

Do not read the paper as saying "no potential can encode the Riemann zeros" or "the Hilbert–Pólya conjecture is false". Its claim concerns fitted fluctuation potentials on one background, in three bases, under one fitting protocol. Sec. VI is explicit: "This is evidence about the tested parametrizations and optimizer together, not a statement about what a potential can represent."

28.3.4 The headline result and Fig. 1

The third paragraph of Sec. I summarizes the outcome. Smooth potentials with a few hundred basis coefficients reproduce any prefix of the zeros to machine-level accuracy in-sample, yet their out-of-sample predictions show no reproducible correlation with the truth across all the scales, splits, bases, budgets, regularization settings and restarts examined. A formula with zero fitted parameters, the explicit formula truncated to a handful of primes, predicts held-out zeros with high and nearly height-independent skill through zero 10^4. The paragraph ends with a hedge you should keep: "within the operator families tested, the fits do not recover it, although a model with a non-arithmetic sparsity prior does."

Paper Fig. 1 is the whole story in one picture. Zeros 370–460 are shown; the fit used zeros 1–400 at scale 1 with $M = 400$ cosine coefficients. Inside the training region both curves track the true fluctuation. Past index 400 the fitted operator's prediction has held-out correlation -0.07, while the six-prime explicit formula has $+0.96$ (both numbers from the Fig. 1 caption). Neither curve uses any zero past the boundary. We reproduce the oracle half of this figure in Section 28.9 (Figure 28.5); the operator half requires a full potential fit, which Chapter 29 discusses.

Checkpoint 28.2

The scope statement says the paper makes no claim against analytic Hilbert–Pólya constructions. In one sentence, why would an out-of-sample test not even apply to such a construction?

Answer: An analytic construction is specified without fitting to any zeros, so every zero is already "held out"; there is no training set whose memorization could masquerade as prediction.

Key Ideas

- The question: does a potential fitted to zeros $1, \ldots, n_{\text{fit}}$ predict zeros it was not shown?
- The scope: Wu–Sprung background, three bases, fitted parameters; analytic constructions are not targeted.
- "Capture" means held-out rms and correlation compared with the model's own smooth baseline.
- The paper claims novelty only for the controlled comparison and protocol.

28.4 Four strands of prior work and the literature audit

★ Did you know? *Stigler's Law*

In 1980 the statistician Stephen Stigler proposed that no scientific discovery is named after its original discoverer, and he jokingly credited this law to the sociologist Robert Merton. This book has examples: the Gaussian curve appeared in de Moivre's work, and Gauss used the FFT long before Cooley and Tukey. A good literature audit checks who really did what first.

Sec. I A places the work among four strands of literature. Table 28.2 summarizes them by three yes/no questions.

(a) Inverse-spectral fits to the zeros. Wu and Sprung [12] inverted the smooth counting function into a semiclassical potential and added fitted corrections; van Zyl and Hutchinson [13]

and Slater [14] extended the fractal fits; Jaksch [15] generated successive zeros from a Wu–Sprung-type potential and traced the fractal corrections to the Riemann–von Mangoldt approximation error; Schumayer and Hutchinson [11] review the tradition. The paper's claim: these models "have been evaluated primarily by agreement with the zeros used in their construction; to our knowledge, no systematic held-out test of predictive performance has been reported in this literature".

(b) Explicit formulae for the n-th zero. França and LeClair [3, 4] derived a transcendental equation for the n-th zero whose smooth part is the Lambert-W asymptotic of the counting function (we derive this smooth part in Section 28.5), and developed it with prime corrections. These are truncations of the classical Riemann–von Mangoldt/Guinand–Weil explicit formula (Chapter 18). The paper uses such a truncation as an "information-existence baseline" and claims no novelty for it. It also cites Connes [21]: extremizing a truncated Weil quadratic form over primes $p \leq 13$ yields approximations to the first 50 zeros with accuracies from 10^{-3} down to 2.6×10^{-55}, provably on the critical line. The paper places this on the arithmetic side of its comparison (it uses prime data only and fits nothing to zeros) and says its existence strengthens the conclusion that the predictive information is arithmetic.

(c) Machine learning on zeta zeros and arithmetic data. Direct regression of zero positions by generic machine-learning models has been explored [44] "in an interpolation setting without operator structure". He, Lee, Oliver and collaborators have machine-learned number-theoretic structures [45, 46], and the murmurations of elliptic-curve L-functions were discovered through machine-learning-driven exploration [47]. The paper's contrast: that literature asks what is learnable from arithmetic data; this paper asks what a physically structured model family fails to learn from spectral data.

(d) Operator proposals. The analytic program [10, 38, 40, 36, 42, 43, 39] seeks exact operators and "is not our target". The zeta spectral triples of Connes, Consani and Moscovici [39] build operators from Euler products over primes, with numerically observed spectral convergence; like [21] this is prime-driven, not a fit to zero data.

Table 28.2: The four strands of prior work in paper Sec. I A, classified by the paper's own description. "Fitted to zeros" means parameters are adjusted to match computed zeros.

strand	operator?	fitted to zeros?	held-out test reported?
(a) inverse-spectral fits	yes	yes	none found
(b) explicit formulae	no	no	not applicable
(c) machine learning	no	yes for zero regression [44]; others fit arithmetic data	yes (train/test splits)
(d) operator proposals	yes	no	not applicable

The sentence that sums up the positioning: "ML papers that predict zero sequences use train/test splits but involve no operator; inverse-spectral papers have operators but no held-out evaluation; we found the intersection empty." The paper's own work is the combination missing from Table 28.2: operator yes, fitted to zeros yes, held-out test yes.

28.4.1 The literature audit: basis and limits

How does the paper support "no systematic held-out test ... has been reported"? Appendix A records a citation-graph audit: among all works citing Wu–Sprung [12] and van Zyl–Hutchinson [13] indexed by Semantic Scholar (accessed August 2026; 28 and 29 works respectively), none

mentions prediction, generalization, held-out evaluation, or machine learning; Slater [14] and Jaksch [15] had no indexed citations.

Apply question (d). The audit is reproducible, which is a real strength: anyone can rerun the same query. But it is bounded in four ways.

1. **Coverage of the graph.** Only works that cite [12] or [13] are examined. A paper that fitted a potential to zeros and tested it on held-out zeros without citing either would be missed.

2. **Coverage of the index.** Only works indexed by one database at one date are examined.

3. **Keyword method.** The test is whether the listed words are mentioned. A held-out test described in other words (for example "extrapolation to higher zeros") could be missed; a mention of "prediction" in an unrelated sense would count as a mention.

4. **Absence versus nonexistence.** The audit shows that nothing was found inside a bounded search, not that nothing exists.

The paper's wording respects these limits: "to our knowledge" and "we found the intersection empty". When you summarize this part of the paper, keep both phrases. Chapter 32 lists the audit among the points a referee could press.

Checkpoint 28.3

Suppose a 2015 paper fitted a potential to zeros 1–100 and reported its error on zeros 101–120, but cited only Berry and Keating. Would the paper's audit find it?

Answer: No. The audit examines only works citing [12] or [13], so a work citing neither is outside the searched set, however relevant it is.

Key Ideas

- Four strands: inverse-spectral fits (operator, fitted, no held-out test found), explicit formulae (arithmetic, nothing fitted), machine learning (fitted, held-out, no operator), operator proposals (operator, nothing fitted).
- The novelty claim rests on a reproducible but bounded citation-graph keyword audit; the paper hedges it with "to our knowledge".
- Prime-only constructions such as [21] sit on the arithmetic side of the comparison.

28.5 The data: zeros, smooth zeros and fluctuations

28.5.1 The zeros and their precision

Sec. II A and Appendix A step 1: zeros $t_1, \dots, t_{1000}$ were computed with `mpmath.zetazero(n)` at 20 significant digits, with $t_1 = 14.134725141734695$ and $t_{1000} = 1419.42248094599$; zeros 10^3–10^4 at 15 digits. The paper asserts that "both precisions exceed every use below by many orders". You quantify this in Problem 25: the smallest held-out error the paper reports is about 0.02 mean spacings (Table IV), and its tightest in-sample statement is rms $< 10^{-4}$ spacings (Sec. III), while 15 significant digits at height 10^4 fix a zero to roughly 10^{-11} in t, at least seven orders of magnitude below either.

28.5.2 The smooth counting function and the mean spacing

Recall from Chapter 17 that $N(T)$ counts zeros with $0 < t \le T$, and that the Riemann–von Mangoldt formula splits it into a smooth part plus a small oscillating remainder $S(T)$. The paper's Eq. (1) is the smooth part,

$$N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{28.1}$$

with log the natural logarithm. Its derivative will give the mean spacing, so let us compute it line by line.

$$\begin{aligned} \frac{dN_{\mathrm{sm}}}{dt} &= \frac{d}{dt}\Big[\frac{t}{2\pi}\log\frac{t}{2\pi}\Big] - \frac{1}{2\pi} && \text{(the constant } \tfrac{7}{8} \text{ has zero derivative)} \\ &= \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} - \frac{1}{2\pi} && \text{(product rule; } \tfrac{d}{dt}\log\tfrac{t}{2\pi} = \tfrac{1}{t}\text{)} \\ &= \frac{1}{2\pi}\log\frac{t}{2\pi}. \end{aligned} \tag{28.2}$$

Differentiating once more,

$$\frac{d^2N_{\mathrm{sm}}}{dt^2} = \frac{1}{2\pi t}. \tag{28.3}$$

For $t > 2\pi$ the derivative (28.2) is positive, so N_{sm} is increasing there and can be inverted.

The *smooth zeros* s_n are defined (Sec. II A) by

$$N_{\mathrm{sm}}(s_n) = n - \tfrac{1}{2}. \tag{28.4}$$

The half reflects the convention (Chapter 17) that $N(t)$ jumps from $n-1$ to n at t_n, so its midpoint value there is $n - \frac{1}{2}$. Treat n as a continuous variable and differentiate (28.4) with respect to n using the chain rule:

$$N'_{\mathrm{sm}}(s_n)\,\frac{ds_n}{dn} = 1 \quad\Longrightarrow\quad \mathrm{gap}_n \equiv \frac{ds_n}{dn} = \frac{1}{N'_{\mathrm{sm}}(s_n)} = \frac{2\pi}{\log(s_n/2\pi)}, \tag{28.5}$$

where the last step substitutes (28.2). This is the local mean spacing, the paper's gap_n. Appendix A step 2 computes it numerically as `np.gradient` of the array of smooth zeros, a centred difference that agrees closely with (28.5) except at the two ends of the array (see the pitfall below).

The *fluctuation* of zero n, in units of the local mean spacing, is

$$f_n = \frac{t_n - s_n}{\mathrm{gap}_n}. \tag{28.6}$$

This is what every model in the paper tries to predict. "All errors below are quoted in units of the local mean spacing" (Sec. II A), and Appendix A adds that this spacing is computed from the smooth zeros of Eq. (1). Dividing by gap_n makes errors comparable across heights: near zero 1 the mean spacing is several units of t, near zero 1000 it is just over one (see the first and third Sample Problems of this section, below).

Sample Problem 28.1: Unfolding the first zero

Find s_1, gap_1 and the fluctuation f_1 of the first zero $t_1 = 14.134725$ (RE1).
Step 1: set up the equation. By (28.4), s_1 solves $N_{\mathrm{sm}}(s) = \frac{1}{2}$.
Step 2: bracket. At $t = 14$: $t/2\pi = 2.22817$, $\log(2.22817) = 0.80117$, so $N_{\mathrm{sm}}(14) = 2.22817 \times 0.80117 - 2.22817 + 0.875 = 0.432$, which is below $\frac{1}{2}$. At $t = 15$, $t/2\pi = 2.38732$, $\log = 0.87018$, $N_{\mathrm{sm}}(15) = 2.07738 - 2.38732 + 0.875 = 0.565$, above $\frac{1}{2}$. The root is in

$(14, 15)$.
Step 3: Newton steps. With (28.2), $N'_{\rm sm}(14) = 0.80117/2\pi = 0.1275$. Carried to four decimals, $N_{\rm sm}(14) = 0.4320$, so the first iterate is $s^{(1)} = 14 + (0.5 - 0.4320)/0.1275 = 14 + 0.533 = 14.533$. At this point $N_{\rm sm}(14.533) = 0.5016$ and $N'_{\rm sm}(14.533) = 0.1335$, so the second iterate is $s^{(2)} = 14.533 - (0.5016 - 0.5)/0.1335 = 14.533 - 0.0120 = 14.521$. A third Newton step changes s by less than 10^{-5}, so $s_1 \approx 14.521$.
Step 4: spacing. $\text{gap}_1 = 2\pi/\log(14.521/2\pi) = 6.28319/0.83775 = 7.500$.
Step 5: fluctuation. $f_1 = (14.134725 - 14.521)/7.500 = -0.0515$.

$$\boxed{s_1 \approx 14.521, \quad \text{gap}_1 \approx 7.500, \quad f_1 \approx -0.052}$$

The first zero sits about one twentieth of a mean spacing below its smooth position.

28.5.3 The Lambert-W form of the smooth zeros

Sec. I A b says the smooth part of the França–LeClair equation is "the Lambert-W asymptotic of the counting function". Recall from Chapter 4 that the Lambert function $W(z)$ is defined by $W(z)\,e^{W(z)} = z$; for $-1/e \le z < 0$ there are two real values, and the principal branch W_0 is the one with $W_0 \ge -1$.

Sample Problem 28.2: Solving $N_{\rm sm}(s_n) = n - \frac{1}{2}$ in closed form

Derive a closed form for s_n and evaluate it at $n = 1$.
Step 1: collect terms. From (28.1) and (28.4),

$$\frac{s}{2\pi}\log\frac{s}{2\pi} - \frac{s}{2\pi} = n - \tfrac{1}{2} - \tfrac{7}{8} = n - \tfrac{11}{8}, \qquad \text{so} \qquad \frac{s}{2\pi}\Big(\log\frac{s}{2\pi} - 1\Big) = n - \tfrac{11}{8}.$$

Step 2: substitute. Let $u = s/(2\pi e)$. Then $s/2\pi = eu$ and $\log(s/2\pi) - 1 = \log(eu) - 1 = \log u$. The equation becomes $eu\log u = n - \frac{11}{8}$, that is

$$u\log u = z, \qquad z \equiv \frac{n - 11/8}{e}.$$

Step 3: recognize W. Put $w = \log u$, so $u = e^w$: then $we^w = z$, hence $w = W(z)$.
Step 4: back-substitute. From $We^W = z$ we get $e^W = z/W$, so $u = e^{W(z)} = z/W(z)$ and

$$s_n = 2\pi e\,u = \frac{2\pi e\,z}{W(z)} = \frac{2\pi\,(n - 11/8)}{W\big((n - 11/8)/e\big)}. \tag{28.7}$$

Step 5: choose the branch. For $n \ge 2$, $z > 0$ and W has one real value. For $n = 1$, $z = -0.375/e = -0.13795$, which lies in $(-1/e, 0)$, so there are two real branches. The branch $W_{-1} < -1$ gives $u < 1/e$, i.e. $s < 2\pi$, where $N_{\rm sm}$ is decreasing and not a counting function. We need W_0.
Step 6: evaluate. $W_0(-0.13795) = -0.16225$ (check: $-0.16225\,e^{-0.16225} = -0.16225 \times 0.85023 = -0.13795$). Then $s_1 = 2\pi(-0.375)/(-0.16225) = 14.521$.

$$\boxed{s_n = \frac{2\pi(n - 11/8)}{W_0\big((n - 11/8)/e\big)}, \qquad s_1 \approx 14.521}$$

This agrees with the Sample Problem "Unfolding the first zero". The paper itself solves (28.4) by Brent's method (Appendix A step 2); both routes give the same s_n.

Sample Problem 28.3: What "0.263 mean spacings" means in height

The paper's null error on zeros 401–500 is 0.2631 spacings (Sec. II C; Appendix A "Key anchors"). Convert this to units of t near zero 500, and find the mean spacing at zero 1000 too.

Step 1: spacing near zero 500. Use $t_{500} = 811.1843588465$ (Appendix A anchors) in (28.5) as an estimate of s_{500}: $811.18/2\pi = 129.10$, $\log(129.10) = 4.8606$, so the spacing is $2\pi/4.8606 = 1.293$.

Step 2: convert. $0.263 \times 1.293 = 0.340$ units of t.

Step 3: spacing at zero 1000. $1419.42/2\pi = 225.91$, $\log(225.91) = 5.4201$, spacing $= 2\pi/5.4201 = 1.159$.

gap ≈ 1.293 near $n = 500$, 0.263 spacings ≈ 0.340 in t, gap ≈ 1.159 near $n = 1000$

28.5.4 Fluctuations and the counting residual

The sinusoidal models of Sec. II D are trained not on f_n but on

$$y_n = n - \tfrac{1}{2} - N_{\mathrm{sm}}(t_n), \tag{28.8}$$

the amount by which the smooth count misses the midpoint count at the true zero. These two quantities are nearly negatives of each other. To see it, let $\delta_n = t_n - s_n$ and expand N_{sm} about s_n using Taylor's theorem with remainder, which we state here.

Taken on trust

Taylor's theorem with the Lagrange remainder: if g has a continuous second derivative on an interval containing s and $s+\delta$, then $g(s+\delta) = g(s) + g'(s)\,\delta + \frac{1}{2}g''(\xi)\,\delta^2$ for some ξ between s and $s+\delta$. We take it on trust; a proof (by repeated use of the mean value theorem) is in [22].

With $g = N_{\mathrm{sm}}$, $s = s_n$ and $\delta = \delta_n$:

$$\begin{aligned} N_{\mathrm{sm}}(t_n) &= N_{\mathrm{sm}}(s_n) + N'_{\mathrm{sm}}(s_n)\,\delta_n + \tfrac{1}{2}N''_{\mathrm{sm}}(\xi_n)\,\delta_n^2 && (\xi_n \text{ between } s_n \text{ and } t_n) \\ &= n - \tfrac{1}{2} + \frac{\delta_n}{\mathrm{gap}_n} + \frac{\delta_n^2}{4\pi\xi_n} && (\text{by (28.4), (28.5), (28.3)}). \end{aligned}$$

Substituting into (28.8) and using $f_n = \delta_n/\mathrm{gap}_n$,

$$y_n = -f_n - \frac{\delta_n^2}{4\pi\xi_n}. \tag{28.9}$$

The correction is tiny: at $n = 500$, $|y_n + f_n| \approx 7 \times 10^{-6}$; even at $n = 1$ it is about 8×10^{-4} (Problem 15). So, to excellent accuracy, the counting residual is minus the fluctuation in spacings. A zero that lies *above* its smooth position ($f_n > 0$) is one where the smooth count has *overshot* the midpoint count ($y_n < 0$).

Pitfall

Mind the sign in (28.9). The explicit formula approximates $S(t)$, the excess of the true count over the smooth count (Chapter 17); at the zeros, S is approximately y_n, hence approximately $-f_n$. If you plot $N_{\mathrm{fl}}(t_n)$ against f_n and see a strong *negative* correlation, you have not made an error.

Pitfall

`np.gradient` uses centred differences in the interior of an array and one-sided differences at its ends. At $n = 1$ it returns $s_2 - s_1 \approx 6.13$ instead of the analytic $\mathrm{gap}_1 \approx 7.50$ of (28.5). This affects only the end points; recomputing the null rms over the window of zeros 1–400 with the spacings returned by `np.gradient` (one-sided at both ends of that window) changes it by less than 10^{-4}, far below the precision of any table. (This was checked for that window only.) The lesson is general: know what your library does at the boundary.

```
import numpy as np, qmnt
from scipy.special import lambertw

def Nsm(t):                          # paper Eq. (1)
    u = t / (2*np.pi)
    return u*np.log(u) - u + 7/8

def smooth_zero(n):                  # Lambert-W form, principal branch
    z = (n - 11/8) / np.e
    return 2*np.pi*(n - 11/8) / np.real(lambertw(z, 0))

def gap(s):                          # local mean spacing
    return 2*np.pi / np.log(s / (2*np.pi))

t = qmnt.zeros(1000)                 # t_1 ... t_1000
n = np.arange(1, 1001)
s = smooth_zero(n)
f = (t - s) / gap(s)                 # fluctuations in mean spacings

for lo, hi in [(1, 400), (401, 500), (801, 1000)]:
    sel = (n >= lo) & (n <= hi)
    print(lo, hi, np.sqrt(np.mean(f[sel]**2)))   # null rms
```

Running this listing prints null values within 0.002 of the paper's anchor 0.2631 for zeros 401–500, and within 0.004 of its anchors 0.2438 (zeros 1–400) and 0.2739 (zeros 801–1000) (Appendix A). (The paper's anchors are measured against the unperturbed potential's levels; Section 28.8 bounds the difference from the smooth-zero values by 0.063 and discusses why it is in practice much smaller. That the values agree as closely as they do is a numerical observation.)

28.5.5 Validating the dataset: the GUE histogram

The last sentence of Sec. II A: "Unfolded spacings of the first 10^4 zeros show textbook GUE level repulsion (Fig. 2), confirming the dataset and unfolding." Recall from Chapter 20 that the unfolded spacing between consecutive zeros is $N_{\mathrm{sm}}(t_{n+1}) - N_{\mathrm{sm}}(t_n)$, which has mean one by construction, and that the GUE Wigner surmise is

$$p(s) = \frac{32}{\pi^2}\, s^2 e^{-4s^2/\pi}, \tag{28.10}$$

against the Poisson law e^{-s} for uncorrelated levels. Figure 28.2 reproduces paper Fig. 2 from the zeros `qmnt` supplies; the mean unfolded spacing of the first 10^4 zeros is 1 to better than one part in a thousand. (The sum of the 9999 spacings telescopes to $N_{\mathrm{sm}}(t_{10000}) - N_{\mathrm{sm}}(t_1)$; writing $N_{\mathrm{sm}}(t_n) = n - \frac{1}{2} - y_n$ from (28.8), the mean is exactly $1 + (y_1 - y_{10000})/9999$, and the computed deviation from 1 is below 10^{-3}.)

What does this figure prove? Read the paper's verb: "confirming the dataset and unfolding". It is a sanity check. Gross errors, such as missed zeros or a wrong density in the unfolding,

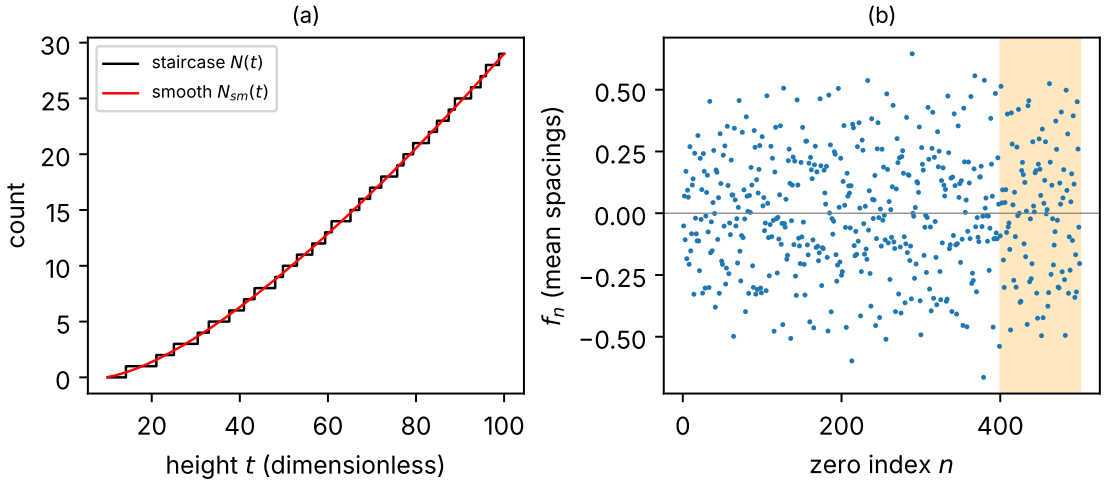


Figure 28.1: Unfolding. (a) The staircase $N(t)$ of the true zeros and the smooth counting function $N_{\rm sm}(t)$ of paper Eq. (1), for $10 \le t \le 100$; the staircase crosses the smooth curve repeatedly. (b) The fluctuations $f_n = (t_n - s_n)/\mathrm{gap}_n$ of the first 500 zeros, in units of the local mean spacing. The shaded band marks zeros 401–500, the paper's principal holdout (RE5).

would typically show up as a non-unit mean or as weight near $s = 0$; the check cannot detect every error (a slightly wrong constant in $N_{\rm sm}$, or a few slightly perturbed zeros, would leave both features intact). It is not evidence that any particular operator generates the zeros: GUE statistics are shared by vast classes of systems (Chapter 20).

Key Ideas

- Eq. (1) gives $N_{\rm sm}$; smooth zeros solve $N_{\rm sm}(s_n) = n - \frac{1}{2}$; the mean spacing is $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$, eq. (28.5).
- Smooth zeros have the closed Lambert-W form (28.7) (principal branch).
- Fluctuations $f_n = (t_n - s_n)/\mathrm{gap}_n$ are what every model predicts; all errors are in mean spacings.
- The counting residual y_n equals $-f_n$ up to a correction of order 10^{-5} to 10^{-3}, eq. (28.9).
- Fig. 2 validates the data and unfolding; it says nothing about which operator, if any, lies behind the zeros.

28.6 The background potential and the forward solver

28.6.1 The Wu–Sprung potential, Eq. (2)

Sec. II B works in units $\hbar = 2m = 1$, so the Hamiltonian is $H = -d^2/dx^2 + V(x)$ and an energy level is directly comparable to a zero height t (Chapter 10). The potential is even, $V(-x) = V(x)$, and is defined implicitly for $V \ge V_0$, $V_0 = 3.10073\,\pi$, by the paper's Eq. (2):

$$x(V) = \frac{1}{\pi}\Big[\sqrt{V - V_0}\,\log\frac{V_0}{2\pi e^2} + \sqrt{V}\,\log\frac{\sqrt{V} + \sqrt{V - V_0}}{\sqrt{V} - \sqrt{V - V_0}}\Big]. \tag{28.11}$$

Chapter 22 derived this from the WKB counting function (Chapter 12) by an Abel inversion: it is the even, monotone potential whose semiclassical level count equals $N_{\rm sm}$. Here we only need to evaluate it.

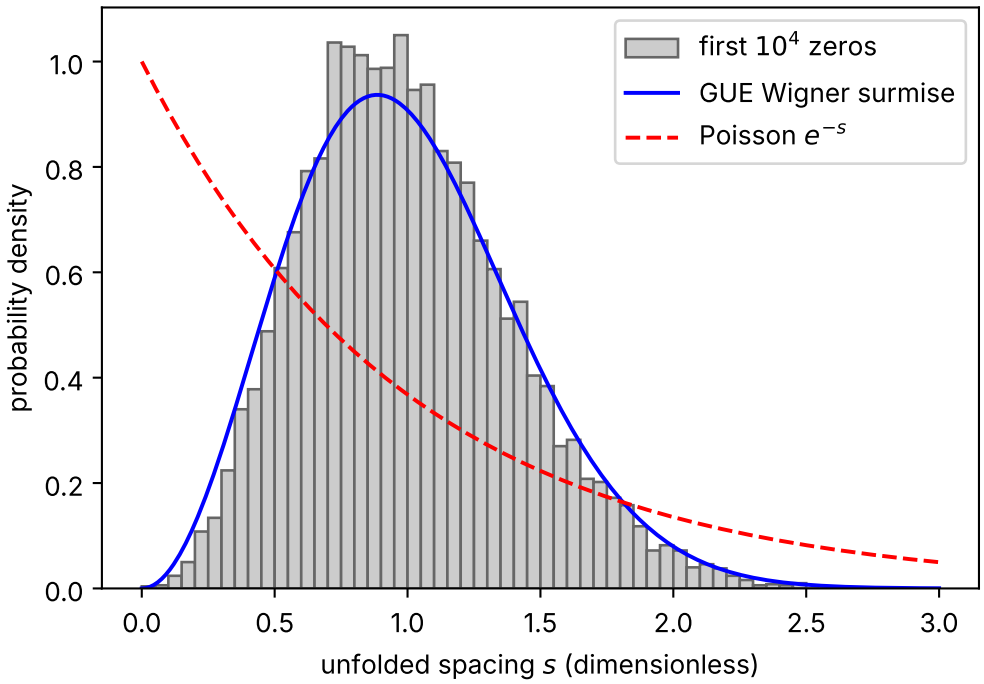


Figure 28.2: Reproduction of paper Fig. 2: histogram of the unfolded nearest-neighbour spacings $N_{\mathrm{sm}}(t_{n+1}) - N_{\mathrm{sm}}(t_n)$ of the first 10^4 zeros, compared with the GUE Wigner surmise $\frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ and the Poisson law e^{-s}. The spacing s is dimensionless (unit mean).

At $V = V_0$ both terms vanish: $\sqrt{V - V_0} = 0$ kills the first, and the second contains $\log(\sqrt{V_0}/\sqrt{V_0}) = \log 1 = 0$. So $x(V_0) = 0$: the bottom of the well is at the origin. The paper inverts (28.11) "pointwise by bracketed root finding" (Appendix A step 3: Brent's method on a dense reference grid, then cubic-spline interpolation onto the working grid; a cubic spline is a smooth piecewise-cubic curve through the reference points, used to evaluate V between them). Bracketing works because $x(V)$ increases strictly with V (Figure 28.3); we prove this after (28.12) below.

The second logarithm in (28.11) suffers from cancellation when you evaluate it in floating point (Chapter 4): for large V the denominator $\sqrt{V} - \sqrt{V - V_0}$ is a small difference of nearly equal numbers. Multiplying top and bottom by $\sqrt{V} + \sqrt{V - V_0}$ removes it:

$$\begin{aligned} \frac{\sqrt{V} + \sqrt{V - V_0}}{\sqrt{V} - \sqrt{V - V_0}} &= \frac{(\sqrt{V} + \sqrt{V - V_0})^2}{(\sqrt{V} - \sqrt{V - V_0})(\sqrt{V} + \sqrt{V - V_0})} \\ &= \frac{(\sqrt{V} + \sqrt{V - V_0})^2}{V - (V - V_0)} \qquad \text{(difference of squares)} \\ &= \frac{(\sqrt{V} + \sqrt{V - V_0})^2}{V_0}, \end{aligned}$$

so

$$\log \frac{\sqrt{V} + \sqrt{V - V_0}}{\sqrt{V} - \sqrt{V - V_0}} = 2 \log \frac{\sqrt{V} + \sqrt{V - V_0}}{\sqrt{V_0}}. \tag{28.12}$$

Now the monotonicity. Write $a = \sqrt{V - V_0}$ and $b = \sqrt{V}$, so that $da/dV = 1/(2a)$ and $db/dV = 1/(2b)$, and let $c = \log(V_0/(2\pi e^2))$. By (28.12), $\pi x = ac + 2b \log((b + a)/\sqrt{V_0})$.

Differentiate term by term for $V > V_0$:

$$\begin{aligned}
\frac{d(ac)}{dV} &= \frac{c}{2a}, \\
\frac{d}{dV}\Big[2b\log\frac{b+a}{\sqrt{V_0}}\Big] &= \frac{1}{b}\log\frac{b+a}{\sqrt{V_0}} + 2b\,\frac{1/(2b)+1/(2a)}{b+a} && \text{(product and chain rules)} \\
&= \frac{1}{b}\log\frac{b+a}{\sqrt{V_0}} + \frac{(a+b)/a}{a+b} && (2b(\tfrac{1}{2b}+\tfrac{1}{2a}) = 1+\tfrac{b}{a}) \\
&= \frac{1}{b}\log\frac{b+a}{\sqrt{V_0}} + \frac{1}{a}.
\end{aligned}$$

Adding the two pieces, and using $1+\frac{c}{2} = 1+\frac{1}{2}\log V_0 - \frac{1}{2}\log(2\pi) - 1 = \frac{1}{2}\log\frac{V_0}{2\pi}$,

$$\pi\frac{dx}{dV} = \frac{1}{\sqrt{V}}\log\frac{\sqrt{V}+\sqrt{V-V_0}}{\sqrt{V_0}} + \frac{\log(V_0/2\pi)}{2\sqrt{V-V_0}}. \tag{28.13}$$

Both terms are positive for $V > V_0$: the first because $\sqrt{V}+\sqrt{V-V_0} > \sqrt{V_0}$, the second because $V_0/2\pi = 3.10073/2 = 1.5504 > 1$. So $x(V)$ is strictly increasing, and its inverse $V(x)$ is well defined for $x \geq 0$.

Sample Problem 28.4: Recovering $L = 44.883$ from Eq. (2)

Scale 1 uses the box $[-L, L]$ with $L = x(V{=}1000) = 44.883$ (Sec. II B). Verify this number.

Step 1: constants. $V_0 = 3.10073\pi = 9.7412$; $\sqrt{V_0} = 3.1211$; $\log(V_0/(2\pi e^2)) = \log(9.7412/46.4268) = \log(0.20982) = -1.5615$.

Step 2: square roots. $\sqrt{1000} = 31.6228$; $\sqrt{1000-9.7412} = \sqrt{990.2588} = 31.4684$.

Step 3: first term. $31.4684 \times (-1.5615) = -49.138$.

Step 4: second term, stable form. By (28.12), the logarithm is $2\log((31.6228 + 31.4684)/3.1211) = 2\log(20.2145) = 2 \times 3.00640 = 6.01279$. Times $\sqrt{1000}$: $31.6228 \times 6.01279 = 190.141$.

Step 5: combine. $x = (190.141 - 49.138)/\pi = 141.003/\pi = 44.883$.

$$\boxed{L = x(1000) \approx 44.883}$$

The same calculation at $V = 1500$ gives the scale-2 half-width 59.957; Appendix A step 3 lists the more precise values 44.8827 and 59.9572.

28.6.2 Why 500 and 1000 levels

Sec. II B says scale 1 scores "the first 500 levels" and scale 2 "1000 levels". The paper does not spell out the reason, so here is our reading, a heuristic based on the turning-point picture of Chapter 12. A level of energy E is classically confined between turning points $\pm x_E$ with $V(x_E) = E$. If $E < V(L)$ the turning points are inside the box, and the Dirichlet walls at $\pm L$ only touch the exponentially small tail of the wavefunction. At scale 1, $V(L) = 1000$, and the 500th zero is $t_{500} = 811.18 < 1000$; at scale 2, $V(L) = 1500$, and $t_{1000} = 1419.42 < 1500$. In both cases every scored level has its turning points inside the box. The margin follows from Eq. (2): at scale 1 the top scored level, $E \approx t_{500}$, turns at $x(811.18) \approx 38.53$, about 6.35 inside the wall at 44.883; at scale 2 the top level turns at $x(1419.42) \approx 57.66$, about 2.29 inside the wall at 59.957. The scale-2 margin is smaller, so the argument that the walls touch only a small tail of the wavefunction is strongest at scale 1; we have not quantified the size of that tail. By contrast, $N_{\rm sm}(1000) \approx 648.6$, so about 649 smooth zeros lie below 1000; levels near the top of

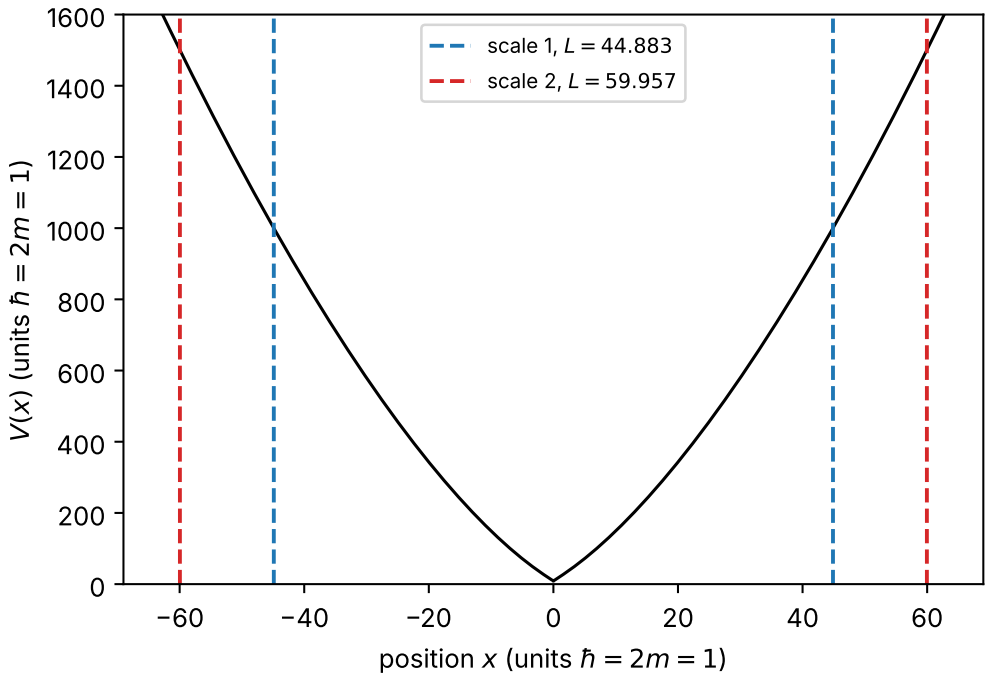


Figure 28.3: The Wu–Sprung potential of paper Eq. (2), plotted from its implicit form $x(V)$ and mirrored to $x < 0$ because V is even. The bottom is $V_0 = 3.10073\pi$ at $x = 0$. Dashed vertical lines: the box walls at ± 44.883 (scale 1, where $V = 1000$) and ± 59.957 (scale 2, where $V = 1500$). Units $\hbar = 2m = 1$; x and V are dimensionless in these units.

that range would feel the walls. Sec. II E reports 652 finite-difference states with $E \leq 1000$, close to this smooth count.

28.6.3 The finite-difference solver

Appendix A steps 3–4 and Sec. II B: the interval $(-L, L)$ carries N uniform interior points with spacing $h = 2L/(N+1)$ and Dirichlet walls (Chapter 23). The second-order difference gives a symmetric tridiagonal matrix with diagonal $2/h^2 + V_j$ and off-diagonal $-1/h^2$; its eigenpairs come from LAPACK through `eigh_tridiagonal` (Chapter 8), restricted to an energy range: $[0, 900]$ for the Hermitian work at scale 1, $[0, 1000]$ for the non-Hermitian tensors (652 states), $[0, 1500]$ at scale 2 (1079 states). The working grid has $N = 6000$ at scale 1 and $N = 8000$ at scale 2; both give $h \approx 0.0150$.

28.6.4 Removing the discretization bias

Chapter 23 showed that the finite-difference eigenvalues of $-d^2/dx^2$ lie below the continuum ones by an amount that grows with energy (dispersion). For the levels the paper scores this matters a great deal.

Sample Problem 28.5: How large is the dispersion error?

Consider the empty box ($V = 0$, RE3) on $[-44.883, 44.883]$ with the scale-1 working grid $N = 6000$. Find the level k with continuum energy near 700 and compare its discrete eigenvalue with the continuum value. Express the shortfall in mean spacings of the zeros at height 700.

Step 1: grid spacing. $h = 2L/(N+1) = 89.766/6001 = 0.0149585$.

Step 2: choose k. Continuum levels are $E_k = (k\pi/2L)^2$ (Chapter 11). Setting $E_k = 700$ gives $k = 2L\sqrt{700}/\pi = 755.98$; take $k = 756$, for which $E_{756} = (756\pi/89.766)^2 = (26.4582)^2 = 700.03$.
Step 3: discrete eigenvalue. From Chapter 23, $E_k^{\text{disc}} = \frac{4}{h^2}\sin^2\frac{k\pi h}{4L}$. Since $h/(4L) = 1/(2(N+1))$, the angle is $\theta = k\pi/(2(N+1)) = 2375.04/12002 = 0.197887$. Then $\sin\theta = 0.196598$, $\sin^2\theta = 0.0386509$, $4/h^2 = 17876.5$, and $E_{756}^{\text{disc}} = 690.94$.
Step 4: shortfall. $700.03 - 690.94 = 9.09$. The mean spacing of the zeros at $t = 700$ is $2\pi/\log(700/2\pi) = 1.333$, so the shortfall is $9.09/1.333 = 6.8$ spacings.

$$\boxed{E_{756}^{\text{disc}} \approx 690.94 \text{ vs } E_{756} \approx 700.03 : \text{ short by} \approx 9.09 \approx 6.8 \text{ spacings}}$$

In the Wu–Sprung potential we expect the bias at a given energy to be smaller. Heuristically (we do not derive this here), the leading error is proportional to h^2 times the average of $k(x)^4 = (E - V(x))^2$ over the state, where $k(x)$ is the local WKB wavenumber of Chapter 12, and in a well this average is smaller than E^2. The paper reports that at levels 401–500 on the working grid the dispersion "reaches two mean spacings" (Sec. II C, "No leakage"). Either way, it is far larger than the 0.263-spacing fluctuations the models are trying to predict.

The paper's remedy is to correct the *targets*, not the operator. Compute the smooth potential's eigenvalues $E_n^{\text{g}}[\text{smooth}]$ on the working grid, and also a Richardson extrapolation $E_n^{\text{Rich}}[\text{smooth}]$ from a finer pair of grids (Appendix A step 3: $N = 12000/24000$ at scale 1, 16000/32000 at scale 2). The grid bias of level n is

$$\Delta_n = E_n^{\text{g}}[\text{smooth}] - E_n^{\text{Rich}}[\text{smooth}], \tag{28.14}$$

and the bias-corrected targets are (Sec. II B; Appendix A step 5)

$$\tilde{t}_n = t_n + \Delta_n. \tag{28.15}$$

The idea: the working-grid eigenvalue of *any* nearby potential is low by roughly Δ_n, so the fit should aim at t_n shifted down by the same amount ($\Delta_n < 0$), not at t_n itself.

The paper reports the quality of the extrapolant: the residual of E^{Rich} against the smooth zeros is 0.005 spacings on average and 0.063 at most, "worst at low n, the known semiclassical weakness of the construction" (Sec. II B; Appendix A gives the same values at scale 2). Recall from Chapter 12 that WKB counting is least accurate for the lowest levels; since Eq. (2) is built to make the *WKB* count equal N_{sm}, the true low levels deviate slightly from s_n.

Sample Problem 28.6: Reading the Richardson formula correctly

Sec. II B writes $E^{\text{Rich}} = (4E_{2h} - E_h)/3$. Test the two possible readings of the subscript on a model with exactly quadratic error, $E_h = E + ah^2$, with $E = 100$, $a = -50$, $h = 0.02$.
Step 1: the standard form. Chapter 23 derived $(4E_{h/2} - E_h)/3$, where $h/2$ is the *finer* grid. Here $E_h = 100 - 50(0.0004) = 99.98$ and $E_{h/2} = 100 - 50(0.0001) = 99.995$, so $(4 \times 99.995 - 99.98)/3 = (399.98 - 99.98)/3 = 100.000$, exact.
Step 2: literal reading with spacing $2h$. $E_{2h} = 100 - 50(0.0016) = 99.92$, and $(4\times 99.92 - 99.98)/3 = (399.68 - 99.98)/3 = 99.900$. In general, $(4E_{2h} - E_h)/3 = E + 5ah^2$: the error is amplified fivefold.

$$\boxed{\text{finer-grid reading: } 100.000 \text{ (exact)}; \qquad \text{spacing-}2h \text{ reading: } 99.900}$$

So the weight 4 must multiply the eigenvalue on the finer grid: only then does the h^2 error cancel. The paper's subscript "$2h$" must therefore be read as "the grid with

twice the points". Appendix A step 3 lists the Richardson pairs 12000/24000 and 16000/32000, consistent with that reading. With $h = 2L/(N+1)$, doubling N halves h only approximately: $h_{24000}/h_{12000} = 12001/24001 = \frac{1}{2}(1+\eta)$ with $\eta = 1/24001 \approx 4.2 \times 10^{-5}$ (smaller still for the scale-2 pair). Repeating Step 2 with $h/2$ replaced by $h(1+\eta)/2$ gives $E + \frac{2\eta+\eta^2}{3}\, ah^2 \approx E + 2.8 \times 10^{-5}\, ah^2$, a negligible residual compared with the coarse-grid error ah^2. The extrapolation is from the finer pair, and the working grid (6000 or 8000) is a third, coarser grid.

Pitfall

Three different grids appear at each scale: the working grid on which all fitting runs, and a finer pair used only for Richardson extrapolation. Δ_n compares the working grid with the extrapolant from the finer pair. Do not confuse it with the difference between the two finer grids.

Key Ideas

- Eq. (2) defines the even Wu–Sprung potential implicitly; $x(V_0) = 0$, $L = x(1000) = 44.883$, $x(1500) = 59.957$; use the stable form (28.12) numerically.
- Every scored level has its turning points inside the box (our reading of why 500 and 1000 levels are scored).
- Finite-difference dispersion at the scored levels is several times larger than the fluctuations; the paper removes it from the targets, $\tilde{t}_n = t_n + \Delta_n$.
- Richardson extrapolation uses the finer grid of a doubling pair; read the paper's subscript accordingly.

28.7 The Hermitian fit and the out-of-sample protocol

★ Did you know? *Leave Some Data Out*

Cross-validation judges a model on data it never saw while fitting. It was formalized in the mid-1970s by the statisticians Mervyn Stone and Seymour Geisser, building on older practice of holding out samples. The idea is simple and powerful. A fit can always be made to match its own training data, so only performance on fresh data measures real predictive skill.

28.7.1 The model and the fit

Sec. II C perturbs the background, $V \to V + \delta V$, with

$$\delta V(x) = \sum_{k=0}^{M-1} c_k \cos\frac{k\pi x}{L}, \tag{28.16}$$

an even function (Chapter 5) with M coefficients c_k. The fit is iterated linearization, which Chapter 24 developed in full. In outline (Appendix A step 6):

1. Solve for eigenpairs (E_n, ψ_n) of the current potential on the working grid.
2. Form residuals $r_n = \tilde{t}_n - E_n$ on the training indices and the sensitivity matrix $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\, dx$ (the first-order energy shift per unit c_k; Chapter 13).

3. Propose $c \mathrel{+}= V\,\mathrm{diag}(\sigma/(\sigma^2+\lambda\sigma_1^2))U^\top r$, an SVD-damped least-squares step (Chapter 9). Here U, σ, V come from the SVD of A (this V is the matrix of right singular vectors, not the potential), σ_1 is the largest singular value and λ the Levenberg–Marquardt damping.

4. Try λ from the list $(0.2\lambda, \lambda, 5\lambda, 25\lambda)$; accept a step only if the new rms is below 0.999 times the old one (the scale-2, $M = 300$ row of Table I used $1 - 10^{-5}$ instead, as its caption notes); on success set $\lambda \leftarrow \max(0.5\lambda_{\mathrm{acc}}, 10^{-7})$, on total failure $\lambda \leftarrow 30\lambda$; stop at $\lambda > 10^4$. The ridge variant adds $\mu\sigma_1^2$ to the denominator.

The protocol is deterministic from $c = 0$; optimizer variability is probed with random initial c.

Other bases. The same protocol runs with two alternative even bases: Chebyshev polynomials $T_k(2|x|/L - 1)$, $k = 0, \ldots, M-1$, where $T_k(u) = \cos(k \arccos u)$ for $-1 \le u \le 1$ (so $u = -1$ at $x = 0$ and $u = 1$ at $|x| = L$, with resolution concentrated near both ends), and piecewise-linear "hat" functions on M uniform nodes of $|x|/L$, each nonzero only between its two neighbouring nodes. "Only the basis changes." Chapter 24 plotted all three.

28.7.2 Out-of-sample scoring

The crucial step is Appendix A step 7: fit on zeros $1, \ldots, n_{\mathrm{fit}}$ only, then perform "a single plain eigensolve of the fitted potential", and score it on the held-out zeros. No further adjustment happens after the fit.

Two numbers are reported. Let $\mathcal{T}$ be the set of held-out indices, with n_{test} members. Let $\hat{t}_n$ be a model's predicted height for zero n, t_n the true height, and R_n a reference (the model's smooth baseline). Define the predicted and true fluctuations relative to the reference,

$$\hat{f}_n = \frac{\hat{t}_n - R_n}{\mathrm{gap}_n}, \qquad f_n^{\mathrm{ref}} = \frac{t_n - R_n}{\mathrm{gap}_n}. \tag{28.17}$$

Then

$$\mathrm{rms} = \Big[\frac{1}{n_{\mathrm{test}}}\sum_{n\in\mathcal{T}}\Big(\frac{\hat{t}_n - t_n}{\mathrm{gap}_n}\Big)^2\Big]^{1/2} = \Big[\frac{1}{n_{\mathrm{test}}}\sum_{n\in\mathcal{T}}(\hat{f}_n - f_n^{\mathrm{ref}})^2\Big]^{1/2}, \qquad \mathrm{corr} = \mathrm{Pearson}(\hat{f}, f^{\mathrm{ref}}), \tag{28.18}$$

where the second form of the rms follows because $\hat{f}_n - f_n^{\mathrm{ref}} = (\hat{t}_n - t_n)/\mathrm{gap}_n$, the reference cancelling. (We write $\mathcal{T}$, n_{test} and hats to avoid clashes with the Hamiltonian H, the number K of frequencies in control C2, the prime set P and the prime p.) For the oracle, $\hat{t}_n$ solves the level equation of Section 28.9 and $R_n = s_n$. For the operator, the fit is scored against $\tilde{t}_n$, which plays the role of the true height, with the reference $E_n^{\mathrm{g}}[\mathrm{smooth}]$ ("both measured relative to the unperturbed smooth-potential levels", Sec. II C). Writing E_n^{g} and E_n^{Rich} for the smooth-potential eigenvalues on the working grid and from Richardson extrapolation, and $\Delta_n = E_n^{\mathrm{g}} - E_n^{\mathrm{Rich}}$ for the grid bias (28.14), this is the same as using the corrected prediction $\hat{t}_n = E_n[\mathrm{fit}] - \Delta_n$ and the reference $R_n = E_n^{\mathrm{Rich}}$ against the raw zeros, because both fluctuations are unchanged:

$$\begin{aligned}(E_n[\mathrm{fit}] - \Delta_n) - E_n^{\mathrm{Rich}} &= E_n[\mathrm{fit}] - E_n^{\mathrm{g}} + E_n^{\mathrm{Rich}} - E_n^{\mathrm{Rich}} = E_n[\mathrm{fit}] - E_n^{\mathrm{g}}, && (\text{insert } \Delta_n)\\ t_n - E_n^{\mathrm{Rich}} &= t_n + \Delta_n - E_n^{\mathrm{g}} = \tilde{t}_n - E_n^{\mathrm{g}} && (\text{add and subtract } E_n^{\mathrm{g}};\ \tilde{t}_n = t_n + \Delta_n).\end{aligned}$$

Dividing by gap_n shows that the predicted and true fluctuations, and hence both scores, are the same in the two descriptions. Recall from Chapter 19 why both numbers are needed: rms measures size of error, correlation measures shape agreement. A prediction with the right shape but the wrong amplitude can have high correlation and a poor rms.

28.7.3 The splits

The paper uses four kinds of train/test split (Figure 28.4).

1. **Forward** (the main protocol): train on zeros $1, \ldots, n_{\text{fit}}$, test on the next window, e.g. 1–400 → 401–500 at scale 1 and 1–800 → 801–1000 at scale 2 (Table I).
2. **Within-height random** (Sec. II C, "Within-height splits"): inside zeros 1–400 (scale 1) or 1–800 (scale 2), hold out a random 20%; ten seeds at scale 1, two at scale 2 (Appendix A step 17: `rng.choice(n, n/5, replace=False)`, residuals restricted to training indices).
3. **Odd/even**: train on odd-indexed zeros and score the even ones, and vice versa.
4. **Interpolation** (Sec. III; Appendix A step 14): train on 1–200 ∪ 301–500, predict the gap 201–300.

Why within-height splits at all? Because forward splits put the test zeros higher up, where the fluctuation amplitude is slightly larger; the paper notes that the null grows from 0.263 to 0.296 over Table V. A model could fail forward because the test data come from a different region (covariate shift, Chapter 25) rather than because it learned nothing. Within-height splits remove that excuse.

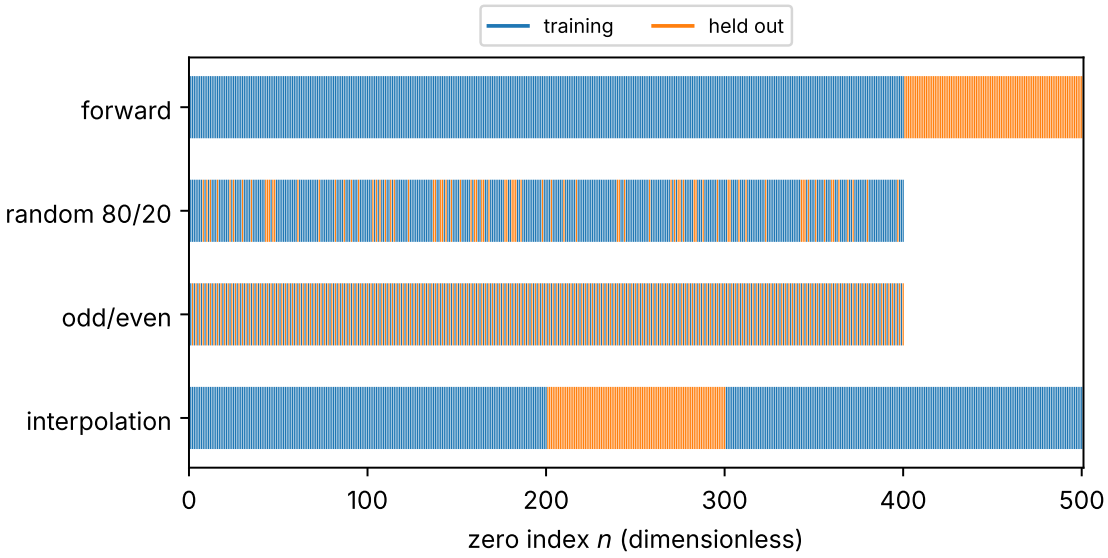


Figure 28.4: The four train/test designs of the paper, drawn along the zero index n at scale 1. Blue: training zeros; orange: held-out zeros. Forward split (Table I), within-height random 80/20 split (one seed shown; Table III), odd/even split (Table III), and the interpolation split of Sec. III.

Checkpoint 28.4

In an odd/even split at scale 1, the model trains on 200 levels. The caption of Table III describes the odd/even rows with M at or above the number of training levels (here $M \geq 200$) as "saturated". Using Chapter 9, say briefly why.

Answer: With M at or above the number of training levels (200 here), the model has at least as many adjustable numbers as data, so counting alone no longer prevents an exact fit (Chapter 9: a square or underdetermined linear system can generically be solved exactly). In practice exactness also requires the basis to cover the levels and the nonlinear iteration to converge (Chapter 24): Table III shows in-sample

rms 0.176 and 0.171 at $M = 200$ but $< 10^{-3}$ at $M = 400$. Either way, a small in-sample error in these rows says nothing about predictive skill.

Key Ideas

- The fitted correction $\delta V = \sum c_k \cos(k\pi x/L)$ is fit by SVD-damped Levenberg–Marquardt steps that are accepted only if they improve the rms.
- Out-of-sample scoring is a single plain eigensolve of the fitted potential, scored on zeros never used in fitting.
- rms and correlation of predicted versus true fluctuations, eq. (28.18); the rms does not depend on the reference, the correlation does (Problem 19).
- Forward, within-height random, odd/even and interpolation splits separate "learned nothing" from "tested elsewhere".

28.8 Two nulls and the leakage argument

28.8.1 Why a null model

An rms of 0.3 spacings is meaningless on its own. Is that good? It depends on how large the thing you are predicting is. The *null* model predicts no fluctuation at all, $\hat{t}_n = R_n$, so $\hat{f}_n = 0$ and, by (28.18),

$$\mathrm{rms}_{\mathrm{null}} = \Big[\frac{1}{n_{\mathrm{test}}} \sum_{n \in \mathcal{T}} \big(f_n^{\mathrm{ref}}\big)^2\Big]^{1/2}, \tag{28.19}$$

the rms size of the true fluctuations themselves. A model has predictive skill only if its rms is below this and its correlation is positive. Every claim in Tables I–V is relative to the null.

Pitfall

The null has no correlation: its predicted fluctuations $\hat{f}_n$ are all zero, a sequence with zero variance, so the Pearson coefficient (covariance divided by the product of standard deviations, Chapter 19) has zero in the denominator. Do not look for a "null corr" column in the tables; there is none.

28.8.2 The two nulls

The paper has two families of models, and each has its own smooth baseline.

- **Operator null:** the unperturbed Wu–Sprung potential, $\delta V = 0$. Its prediction for zero n is its own eigenvalue.
- **Oracle null:** the smooth zeros s_n, the solutions of the mean counting equation (28.4).

These are "conceptually distinct (an eigenvalue spectrum versus the solution of the mean counting function)" (Sec. II C). Yet "numerically its holdout error coincides with that of the smooth zeros used as the oracle's null (both 0.2631 on zeros 401–500), so the operator and oracle comparisons share one effective baseline." Let us see exactly why.

Score the unperturbed potential by the operator protocol: prediction $E_n^{\mathrm{g}}[\mathrm{smooth}]$ against

target $\tilde{t}_n$. The error in height units is

$$\begin{aligned}\tilde{t}_n - E_n^{\mathrm{g}}[\mathrm{smooth}] &= t_n + \Delta_n - E_n^{\mathrm{g}}[\mathrm{smooth}] && \text{(by (28.15))}\\ &= t_n + E_n^{\mathrm{g}}[\mathrm{smooth}] - E_n^{\mathrm{Rich}}[\mathrm{smooth}] - E_n^{\mathrm{g}}[\mathrm{smooth}] && \text{(by (28.14))}\\ &= t_n - E_n^{\mathrm{Rich}}[\mathrm{smooth}]. && (28.20)\end{aligned}$$

The working-grid eigenvalue cancels completely. The operator null is the error of the Richardson-extrapolated (essentially continuum) spectrum of the smooth potential against the raw zeros. Now write $\varepsilon_n = (E_n^{\mathrm{Rich}}[\mathrm{smooth}] - s_n)/\mathrm{gap}_n$ for the residual the paper reports (mean 0.005, max 0.063 spacings). Dividing (28.20) by gap_n,

$$\frac{t_n - E_n^{\mathrm{Rich}}}{\mathrm{gap}_n} = \frac{t_n - s_n}{\mathrm{gap}_n} - \frac{E_n^{\mathrm{Rich}} - s_n}{\mathrm{gap}_n} = f_n - \varepsilon_n. \tag{28.21}$$

The oracle null error is f_n; the operator null error is $f_n - \varepsilon_n$.

How different can their rms values be? Write $\mathrm{rms}(a) = \|a\| / \sqrt{n_{\mathrm{test}}}$ with $\|\cdot\|$ the Euclidean norm of an n_{test}-component vector (Chapter 7). We need the triangle inequality $\|a+b\| \le \|a\| + \|b\|$. It follows from the Cauchy–Schwarz inequality $a \cdot b \le \|a\|\,\|b\|$ (Chapter 19):

$$\begin{aligned}\|a+b\|^2 &= \|a\|^2 + 2\,a\cdot b + \|b\|^2 && \text{(expand the dot product)}\\ &\le \|a\|^2 + 2\,\|a\|\,\|b\| + \|b\|^2 && \text{(Cauchy–Schwarz)}\\ &= (\|a\| + \|b\|)^2,\end{aligned}$$

and taking square roots of the two nonnegative sides gives the inequality. Applied twice, it gives

$$\begin{aligned}\|f\| = \|(f-\varepsilon)+\varepsilon\| &\le \|f-\varepsilon\| + \|\varepsilon\|\,,\\ \|f-\varepsilon\| = \|f+(-\varepsilon)\| &\le \|f\| + \|\varepsilon\|\,.\end{aligned}$$

The first line says $\|f\| - \|f-\varepsilon\| \le \|\varepsilon\|$, the second says $\|f-\varepsilon\| - \|f\| \le \|\varepsilon\|$; together,

$$\bigl|\mathrm{rms}(f-\varepsilon) - \mathrm{rms}(f)\bigr| \le \mathrm{rms}(\varepsilon) \le \max_n |\varepsilon_n|. \tag{28.22}$$

The last inequality holds because a mean of squares cannot exceed the largest square. With the paper's $\max|\varepsilon_n| = 0.063$, the two nulls can differ by at most 0.063 on any holdout. On zeros 401–500 they agree to four digits. This is consistent with ε_n being much smaller there than its worst value (the paper says the worst residuals occur at low n), although agreement of the rms values alone does not prove it: expanding the squares, $\mathrm{rms}(f-\varepsilon)^2 - \mathrm{rms}(f)^2 = \overline{\varepsilon^2} - 2\,\overline{f\varepsilon}$ (bars denote averages over the holdout), so an ε that is uncorrelated with f changes the rms only through the second-order term $\overline{\varepsilon^2}$.

Sample Problem 28.7: The rms bound on a small example

Let $f = (0.30, -0.20, 0.10, -0.40)$ and $\varepsilon = (0.010, -0.020, 0.005, 0.015)$ (illustrative numbers, not the paper's). Check (28.22).

Step 1: $\mathrm{rms}(f)$**.** $(0.09 + 0.04 + 0.01 + 0.16)/4 = 0.075$, so $\mathrm{rms}(f) = 0.2739$.

Step 2: $\mathrm{rms}(f-\varepsilon)$**.** $f - \varepsilon = (0.290, -0.180, 0.095, -0.415)$; squares sum to $0.0841 + 0.0324 + 0.009025 + 0.172225 = 0.29775$; divided by 4 gives 0.074438, so rms $= 0.2728$.

Step 3: $\mathrm{rms}(\varepsilon)$**.** Squares sum to 0.00075; divided by 4 gives 0.0001875; $\mathrm{rms}(\varepsilon) = 0.0137$.

Step 4: compare. $|0.2728 - 0.2739| = 0.0010 \le 0.0137 \le 0.020 = \max|\varepsilon_n|$.

$$\boxed{|\Delta\mathrm{rms}| \approx 0.0010 \;\le\; \mathrm{rms}(\varepsilon) \approx 0.0137 \;\le\; \max|\varepsilon_n| = 0.020}$$

The difference of the rms values is much smaller than the bound, because ε is not aligned

with f.

28.8.3 No leakage through the bias correction

Here is a subtle point a referee would raise. The corrected targets $\tilde{t}_n = t_n + \Delta_n$ are formed for *all* n, including held-out ones. Does the held-out information leak into the fit through them? The paper's answer is its Eq. (3):

$$\tilde{t}_n - E_n[\text{fit}] = t_n - (E_n[\text{fit}] - \Delta_n). \tag{28.23}$$

Derivation, one line at a time:

$$\begin{aligned} \tilde{t}_n - E_n[\text{fit}] &= (t_n + \Delta_n) - E_n[\text{fit}] && \text{(definition (28.15))} \\ &= t_n - E_n[\text{fit}] + \Delta_n && \text{(rearrange)} \\ &= t_n - (E_n[\text{fit}] - \Delta_n) && \text{(factor the minus sign).} \end{aligned}$$

So scoring the fit against $\tilde{t}_n$ is algebraically identical to correcting the *prediction* by Δ_n and scoring against the raw zeros.

The identity by itself proves nothing about leakage; it is true for any Δ_n whatsoever. What carries the argument is the second half of the paragraph: Δ_n "is built solely from eigenvalues of the *unperturbed* potential on two grids; no zero, in-sample or held-out, enters it." Look at (28.14): both terms are eigenvalues of the smooth potential, which is fixed by N_{sm} alone. So Δ_n is a known, data-free correction to the solver, exactly like a calibration constant of an instrument. The identity shows that applying it to the targets is the same as applying it to the predictions; the construction shows it contains no information about any zero. Both halves are needed (Problem 7 asks you to see what would go wrong if the second half failed).

Why bother with the correction at all? Because without it, finite-difference dispersion "at levels 401–500 on the working grid reaches two mean spacings and would otherwise dominate both the null and every model identically". Appendix A's anchors quantify this: the raw-zero null on the uncorrected working grid is 2.09 on zeros 401–500, against 0.2631 after correction. A null of 2.09 would make every model look equally terrible and hide all differences between them.

Sample Problem 28.8: The leakage identity with numbers

Take illustrative values (not from the paper) for one held-out level: $t_n = 700.250$, $E_n^{\text{g}}[\text{smooth}] = 698.300$, $E_n^{\text{Rich}}[\text{smooth}] = 700.100$, and a fitted prediction $E_n[\text{fit}] = 698.600$. Compute the error both ways, and the null error.

Step 1: bias. $\Delta_n = 698.300 - 700.100 = -1.800$.

Step 2: corrected target. $\tilde{t}_n = 700.250 - 1.800 = 698.450$.

Step 3: score against the corrected target. $\tilde{t}_n - E_n[\text{fit}] = 698.450 - 698.600 = -0.150$.

Step 4: score the corrected prediction against the raw zero. $E_n[\text{fit}] - \Delta_n = 698.600 + 1.800 = 700.400$; $t_n - 700.400 = -0.150$. Same, as (28.23) demands.

Step 5: null error. By (28.20), $t_n - E_n^{\text{Rich}} = 700.250 - 700.100 = 0.150$.

$$\boxed{\tilde{t}_n - E_n[\text{fit}] = t_n - (E_n[\text{fit}] - \Delta_n) = -0.150; \qquad \text{null error} = +0.150}$$

In this illustration the fit predicts a fluctuation of $E_n[\text{fit}] - E_n^{\text{g}}[\text{smooth}] = +0.300$ where the true one is $\tilde{t}_n - E_n^{\text{g}}[\text{smooth}] = +0.150$: right sign, twice the size, so its error (0.150 in magnitude) is no smaller than the null's.

A second leakage safeguard appears in Appendix A step 20 for the sinusoidal controls: "Overwriting held-out data with noise before fitting leaves all fitted coefficients bit-identical."

This is the strongest kind of leakage test: physically destroy the held-out data and confirm that nothing in the fit changes.

Checkpoint 28.5

Does the operator null on zeros 401–500 depend on the working grid size $N = 6000$? Use (28.20).

Answer: No. The working-grid eigenvalue cancels, leaving $t_n - E_n^{\mathrm{Rich}}$, which depends only on the finer Richardson pair. Changing the working grid changes Δ_n and the fitted eigenvalues together.

Key Ideas

- Every skill claim is relative to a null; the null rms is the rms of the true fluctuations, and the null has no correlation.
- The operator null equals $t_n - E_n^{\mathrm{Rich}}[\mathrm{smooth}]$, eq. (28.20); it differs from the smooth-zero null by at most $\max|\varepsilon_n| = 0.063$, eq. (28.22), and coincides with it numerically (0.2631, a coincidence of measured values, not an identity) on zeros 401–500.
- No leakage: Eq. (3) shows correcting targets equals correcting predictions, and Δ_n contains no zero.
- Without the correction the null on 401–500 would be 2.09 instead of 0.263.

28.9 The arithmetic oracle and the fitted sinusoidal controls

28.9.1 The oracle, Eq. (4)

Sec. II D's prediction baseline is the paper's Eq. (4),

$$N_{\mathrm{fl}}(t) = -\frac{1}{\pi}\sum_{p\in P}\sum_{m=1}^{3}\frac{\sin(mt\log p)}{m\,p^{m/2}}, \tag{28.24}$$

for a finite prime set P, with predicted zeros solving

$$N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \tfrac{1}{2}. \tag{28.25}$$

Chapter 18 derived (28.24) from the Euler product: $\log\zeta(s) = \sum_p\sum_m p^{-ms}/m$ for $\mathrm{Re}\,s > 1$, and the imaginary part formally continued to $s = \frac{1}{2}+it$ gives $S(t) \approx -\frac{1}{\pi}\sum_p\sum_m \sin(mt\log p)/(mp^{m/2})$, using $p^{-ms} = p^{-m\sigma}e^{-imt\log p}$ (Chapter 2). Each prime contributes oscillations of angular frequency $m\log p$ in t; for $p = 2$ the period is $2\pi/\log 2 \approx 9.065$ (RE2).

Taken on trust

The full series over all primes does not converge on the critical line, so (28.24) is a heuristic truncation, not a convergent approximation with a proved error bound. Rigorous explicit formulae (Riemann–von Mangoldt, Guinand–Weil) relate smoothed sums over zeros to smoothed sums over prime powers; see [5, 6, 25]. That truncations such as (28.24) predict individual zeros well at these heights is a numerical observation, which Table V documents and which França and LeClair [3, 4] studied.

Three features of the oracle matter for reading the paper.

1. **Zero fitted parameters.** Nothing in (28.24) or (28.25) is adjusted to any zero. Every zero is held out.

2. **Not a fair competitor.** The paper says so: "It is not a fair ML competitor (it encodes the known functional form); its role is informational, to prove that transferable structure about unseen zeros exists and to quantify how much of it simple arithmetic captures."

3. **Root finding.** Appendix A step 8 solves (28.25) by Brent's method (Chapter 4) "bracketed ±3 around the smooth zero (bracket expanded on sign failure)". The step does not state the unit of the ±3; `qmnt.oracle_zeros` uses three local mean spacings.

Sample Problem 28.9: Reproducing two entries of Table V

Compute the oracle's rms and correlation on zeros 401–500 with $P = \{2\}$ and with the first six primes, and compare with Table V.
Step 1: data. Indices $n = 401, \dots, 500$; true heights from `qmnt.zeros(500)`.
Step 2: predictions. `qmnt.oracle_zeros(n, primes)` solves (28.25) for each n with $m \le 3$.
Step 3: score. `qmnt.score(t_pred, n)` returns the rms of $(\hat{t}_n - t_n)/\mathrm{gap}_n$ and the Pearson correlation of $\hat{f}_n = (\hat{t}_n - s_n)/\mathrm{gap}_n$ with $f_n^{\mathrm{ref}} = f_n$, as in (28.18) with $R_n = s_n$.
Step 4: results. One prime gives rms 0.198, corr +0.66; six primes give 0.074, +0.96. Both match Table V's 401–500 row.

$\{2\}$: 0.198/+0.66;	$\{2, 3, 5, 7, 11, 13\}$: 0.074/+0.96	(null 0.263)

A single prime, with nothing fitted, already beats the null by a quarter.

```
import numpy as np, qmnt

n = np.arange(401, 501)                    # RE5: principal holdout
t_true = qmnt.zeros(500)[n - 1]
null = qmnt.rms(qmnt.fluct(t_true, n))     # about 0.263

for k in [1, 3, 6, 25]:
    P = qmnt.first_primes(k)
    t_pred = qmnt.oracle_zeros(n, P, mmax=3)
    r, c = qmnt.score(t_pred, n)
    print(k, "primes: rms", round(r, 3), "corr", round(c, 2),
          "null", round(null, 3))
```

Figure 28.5 shows the oracle half of paper Fig. 1. The true fluctuation and the six-prime prediction are plotted for zeros 370–460; nothing changes at the boundary $n = 400$ because the oracle never saw any zero.

28.9.2 The fitted sinusoidal controls

Because the oracle is handed the answer's form, Sec. II D adds three fitted controls that differ in how much of that form they are given. All have the shape $N_{\mathrm{sm}}(t) + F(t) = n - \frac{1}{2}$ with

$$F(t) = \sum_j [a_j \sin(\omega_j t) + b_j \cos(\omega_j t)], \tag{28.26}$$

trained by least squares on y_n of (28.8) at the training zeros only, and scored exactly like the oracle. By (28.9), training on y_n is training on minus the fluctuations.

(A) The oracle with a larger budget: up to 300 primes, i.e. 900 terms ($m \le 3$), still nothing fitted.

(B) Frequencies $m \log p$ given; amplitudes a_j, b_j fitted with ridge regularization (Chapter 9).

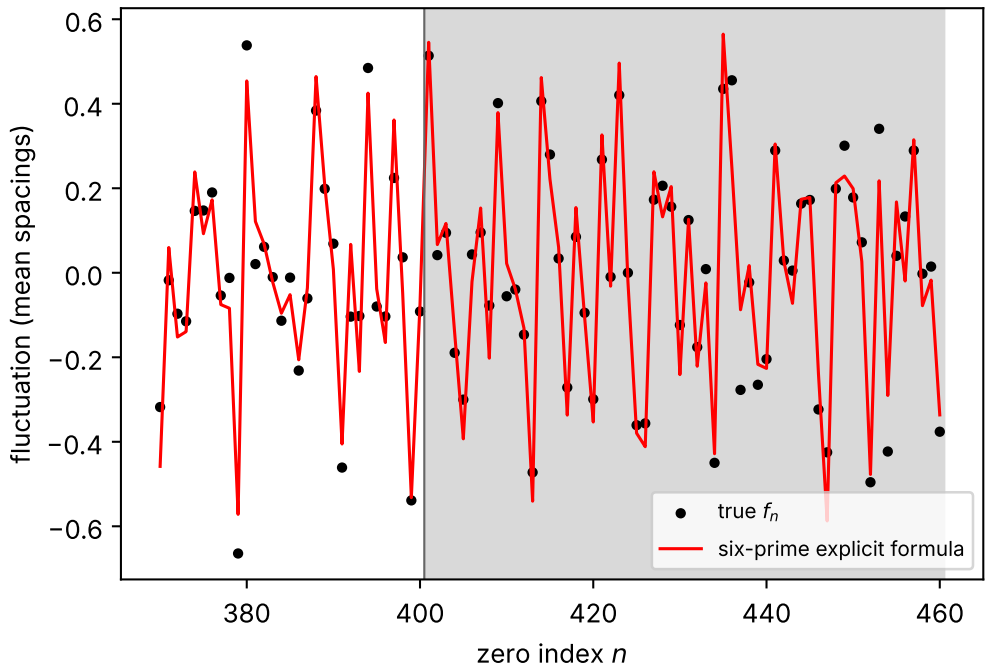


Figure 28.5: The oracle half of paper Fig. 1: true fluctuations $f_n = (t_n - s_n)/\mathrm{gap}_n$ of zeros 370–460 (dots) and the six-prime explicit-formula prediction of paper Eq. (4) (line), both in units of the local mean spacing. The shaded region $n > 400$ is the held-out region of Fig. 1. The oracle uses no zeros at all, so its quality is the same on both sides of the boundary; the paper quotes held-out correlation +0.96. The fitted-operator curve of Fig. 1 is discussed in Chapter 29.

(C1) No arithmetic input: 450 frequencies on a uniform grid, $\omega_j = 8j/450$ (900 coefficients), ridge.

(C2) No arithmetic input: a sparse model whose K frequencies are found one at a time as the largest peak of the residual periodogram on $0.05 < \omega < 8$, all amplitudes refitted after each addition ($3K$ parameters: frequency, sine and cosine amplitude for each).

Ridge strength and K are chosen by a forward split inside the training window (first 75% fit, last 25% validate), then the model is refitted on the whole window; "held-out zeros are not used before scoring." The tools are in Chapters 6 and 27; the results, the root-ambiguity footnote of row B, and the surrogate alignment check are read in Chapter 30.

Checkpoint 28.6

Control C1 has 900 coefficients, the same as the largest operator fit. Which level of "supplied information" (Sec. IV: (i) arithmetic frequencies, (ii) sparsity only, (iii) generic basis) does it belong to, and why is it a useful comparison for the fitted potentials?

Answer: Level (iii), a generic basis with no prior favouring the true frequencies, the same level as the fitted potentials. If C1 also fails, the failure of the potentials cannot be blamed on something peculiar to operators; it is the absence of a suitable prior.

Key Ideas

- The oracle, Eq. (4), is a heuristic truncation of the explicit formula with no fitted parameters; predicted zeros solve $N_{\mathrm{sm}} + N_{\mathrm{fl}} = n - \frac{1}{2}$ by bracketed Brent root finding.
- It is an information-existence baseline, not a fair competitor; one prime already gives 0.198 against the null 0.263 on zeros 401–500.
- Controls A, B, C1, C2 are fitted on y_n and scored like the oracle; their hyperparameters are chosen inside the training window.

28.10 The AI-use disclosure and the reconstruction record

28.10.1 What Sec. II F discloses

Sec. II F states, in accordance with the journal's policy, that "the computations, data analysis, and drafting of this article were carried out with substantial assistance from the large language model Claude Fable 5 (Anthropic . . .), used in agentic coding sessions, and, for the revised version, Claude Fable 5.1." The tool was used (i) to write and debug the Python pipeline of Appendix A, (ii) to analyze the data, including Tables I–V, and (iii) to draft and revise the text. The stated reason: "the study required rapid, iterative construction and cross-checking of a substantial numerical pipeline by a single author."

The disclosure then makes three verifiable commitments.

1. "No reported result depends on stochastic model output": every number, table entry and figure is regenerated by the archived scripted pipeline [20], deterministically or from recorded random seeds.

2. The author "re-ran [the pipeline] to verify every reported value".

3. All AI-assisted content, including all references, "was critically reviewed by the author, who takes full responsibility".

28.10.2 How to weigh it

Apply question (d) neutrally. The disclosure is neither an endorsement nor a discount of the results. What determines whether a computational result is trustworthy is whether it can be regenerated and checked, and the paper's claims about that are concrete and testable:

- Data Availability: all scripts, computed zeros, fitted coefficients and the full parameter record are on Zenodo (DOI 10.5281/zenodo.24035819 for R1; the original submission is 10.5281/zenodo.21842988).
- The analysis of the original submission "was reconstructed from scratch, from that record alone, in a fresh compute environment".
- For the revision, "the complete pipeline (`run_all.sh`) was re-run end to end from a clean copy of the code, and the scripts were re-run by the author on a separate system; both reproduced the stored values" (Appendix A).
- Software versions for the revision computations are recorded (Python 3, `numpy` 2.5.3, `scipy` 1.18.1), as are seeds (`numpy.random.default_rng(seed)`).
- "Key anchors" (Appendix A) give target numbers for checking a reconstruction: $t_1 = 14.1347251417$, $t_{500} = 811.1843588465$, null in-sample (1–400) 0.2438, null OOS (401–500) 0.2631, and so on.

Why do records like this matter, whoever or whatever wrote the code? Because in a computational paper the computation *is* the evidence. Chapter 31 will show how a single dead parameter coupling to float32 noise produced a convincing but spurious result; the artifact was removed by zeroing the analytically vanishing tensor entries, and the remaining floor was shown to be a property of the surrogate only when the exact spectrum was computed. Both repairs depended on being able to rerun and recheck the computation. A reconstruction record turns "trust me" into "check me", and that is the only footing on which a numerical claim can stand.

28.10.3 Appendix A steps 1–8 in one place

The protocol you have now read corresponds to the first eight pipeline steps:

1. zeros by `mpmath.zetazero`, 20 digits for $n \le 10^3$, 15 beyond (Section 28.5);
2. smooth zeros by Brent on $N_{\mathrm{sm}}(t) = n - \frac{1}{2}$; gap by `np.gradient` (Section 28.5);
3. grids and the potential: $L = 44.8827$ / 59.9572; working $N = 6000$ / 8000; Richardson pairs 12000/24000 / 16000/32000; Brent inversion of Eq. (2), cubic spline (Section 28.6);
4. tridiagonal solver, `eigh_tridiagonal`, energy ranges (Section 28.6);
5. bias-corrected targets $\tilde{t} = t + (E^{\mathrm{g}} - E^{\mathrm{Rich}})$ (Section 28.6);
6. the Hermitian fit with its damping schedule (Section 28.7);
7. OOS: plain eigensolve of the fitted potential, scored on held-out zeros (Section 28.7);
8. the oracle, $m \le 3$, Brent bracketed around the smooth zero (Section 28.9).

Figure 28.6 draws the flow.

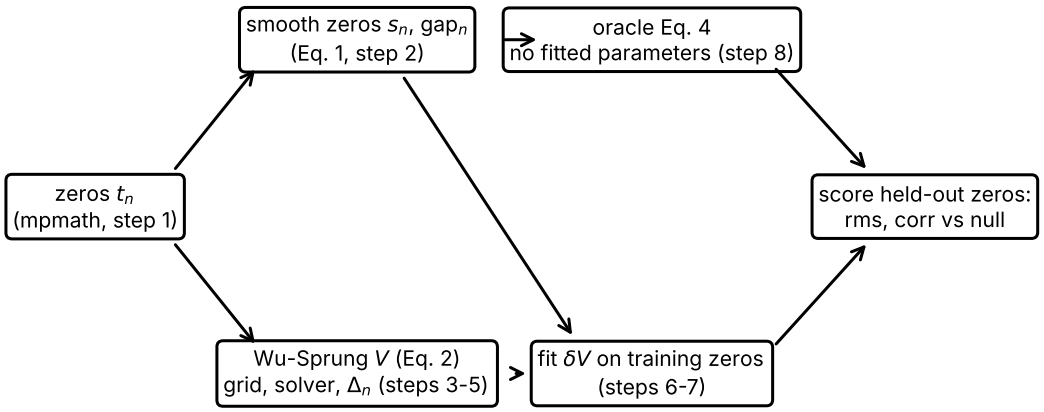


Figure 28.6: Schematic of the paper's protocol (Sec. II A–D, Appendix A steps 1–8). Zeros and the smooth counting function feed two families of predictors: the fitted potential on the Wu–Sprung background (with bias-corrected targets) and the arithmetic oracle of Eq. (4). Both are scored only on held-out zeros, by rms and correlation in mean spacings, against the null of their own smooth part.

Pitfall

Do not treat "produced with AI assistance" as evidence for or against any number in the paper. The relevant questions are the ones you would ask of any computational work: is the pipeline archived, are seeds and versions recorded, has it been rerun independently, and do the anchors reproduce?

Key Ideas

- Sec. II F discloses substantial use of Claude Fable 5 and 5.1 for code, analysis and drafting; the author reviewed all content and takes responsibility.
- The paper commits to deterministic or seeded regeneration of every number from the archived pipeline, rerun from scratch and on a separate system.
- Reproducibility records, not authorship tools, determine whether a numerical claim can be checked.

28.11 Paper connection

★ Did you know? *Trillions of Zeros Checked*

In 2021 David Platt and Timothy Trudgian published a rigorous verification that all zeta zeros with imaginary part up to 3×10^{12} lie on the critical line, more than 10^{13} zeros in all. They used interval arithmetic to control every rounding error. Verified computations like this, together with Odlyzko's high-altitude tables, provide the kind of data such studies rely on.

This whole chapter is a reading of the paper, so this section serves as a map. Table 28.3 lists each part of the paper covered here, where we read it, and which earlier chapters supply

its tools. Use it when you reread the paper with this chapter beside you.

Table 28.3: Map from the front half of the paper (R1 numbering) to the sections of this chapter and the tool chapters.

paper element	read in	tools from
Abstract	Section 28.2	all of Parts I–V
Sec. I, scope statement, Fig. 1	Section 28.3	Chs. 1, 25, 21
Sec. I A (a)–(d); App. A audit	Section 28.4	Chs. 18, 21, 22
Sec. II A, Eq. (1), Fig. 2; App. A steps 1–2	Section 28.5	Chs. 4, 17, 20
Sec. II B, Eq. (2); App. A steps 3–5	Section 28.6	Chs. 12, 22, 23
Sec. II C (fit, other bases, OOS, within-height); App. A steps 6–7	Section 28.7	Chs. 9, 13, 24, 25
Sec. II C "No leakage", Eq. (3); two nulls	Section 28.8	Chs. 7, 19, 23
Sec. II D, Eq. (4), controls A–C2; App. A step 8; Table V entries	Section 28.9	Chs. 6, 18, 27
Sec. II F; Data Availability; App. A record	Section 28.10	Chs. 26, 32

What this chapter has *not* done is interpret the results: Tables I–III and Figs. 3–4 are read in Chapter 29; Tables IV–V and Sec. IV in Chapter 30; Sec. II E, Sec. V and Fig. 5 in Chapter 31; Sec. VI in Chapter 32. The numbers we did reproduce here (the nulls of Appendix A's anchors, 0.2438, 0.2631 and 0.2739, to within a few thousandths, the scale lengths 44.883 and 59.957 of Sec. II B, and the Table V entries 0.198/+0.66 and 0.074/+0.96) confirm that you and the paper are measuring the same things in the same units.

28.12 Summary

- The paper tests whether potentials fitted to zeros $1, \ldots, n_{\text{fit}}$ predict held-out zeros; its scope is fluctuation potentials on the Wu–Sprung background in three bases, and it makes no claim against analytic constructions (Section 28.3).
- "Capture" is operational: held-out rms and correlation relative to the model's own smooth baseline; novelty is claimed only for the controlled comparison and protocol.
- Prior work falls into four strands; the claimed gap (no held-out test of inverse-spectral fits) rests on a reproducible but bounded citation-graph keyword audit (Table 28.2).
- Smooth part: N_{sm} of eq. (28.1), smooth zeros (28.4) with closed form (28.7), mean spacing (28.5), fluctuations (28.6); the counting residual $y_n \approx -f_n$, eq. (28.9).
- The Wu–Sprung potential (28.11), stably evaluated with (28.12), gives $L = 44.883$ and 59.957.
- Bias-corrected targets $\tilde{t}_n = t_n + \Delta_n$, eqs. (28.14)–(28.15), remove finite-difference dispersion; Richardson uses the finer grid of a doubling pair.
- Scoring: rms and correlation of predicted versus true fluctuations, eq. (28.18); the null rms is the rms of the true fluctuations, eq. (28.19).
- The operator null is $t_n - E_n^{\text{Rich}}$, eq. (28.20), within 0.063 of the smooth-zero null by (28.22); both are 0.2631 on zeros 401–500.

- No leakage: Eq. (3) (our (28.23)) plus the fact that Δ_n contains no zero.
- The oracle, Eq. (4) (our (28.24)), has zero fitted parameters; on zeros 401–500 it gives 0.198/+0.66 with one prime and 0.074/+0.96 with six.
- The controls A, B, C1, C2 are fitted on y_n with hyperparameters chosen inside the training window.
- AI assistance is disclosed; reproducibility rests on the archived, rerun pipeline and its anchors.

28.13 Problems

1. • List the four strands of prior work in paper Sec. I A. For each, state whether its models (i) involve an operator, (ii) have parameters fitted to computed zeros, and (iii) according to the paper, have been evaluated on held-out zeros.

2. • Evaluate $N_{\rm sm}(100)$ from Eq. (1). How many zeros actually lie in $0 < t \le 100$? (Use `qmnt.zeros(40)` or the fact that $t_{29} < 100 < t_{30}$.)

3. • Compute the local mean spacing at height $t = 100$ from (28.5).

4. • Show directly from Eq. (2) that $x(V_0) = 0$, and explain why Eq. (2) defines V only for $x \ge 0$ even though the paper's potential lives on $[-L, L]$.

5. •• For $V \gg V_0$, use $\sqrt{V - V_0} \approx \sqrt{V}$ in Eq. (2) and the stable form (28.12) to show $x(V) \approx \frac{\sqrt{V}}{\pi} \log \frac{2V}{\pi e^2}$. Evaluate this at $V = 1000$ and compare with $L = 44.883$.

6. •• Using `scipy.special.lambertw` and (28.7), compute s_{500} and the fluctuation f_{500} of $t_{500} = 811.1843588465$. Compare s_{500} with `qmnt.smooth_zeros`.

7. •• Suppose that, instead of (28.14), the bias correction had been defined as $\Delta'_n = E_n^{\rm g}[\text{smooth}] - t_n$ for all n. Does the identity Eq. (3) still hold with Δ'_n? Is there now leakage? What does this tell you about the logical structure of the paper's no-leakage argument?

8. •• Using the exact discrete eigenvalues of the empty box, $E_k^{\rm disc} = \frac{4}{h^2} \sin^2 \frac{k\pi}{2(N+1)}$, with $L = 44.883$, compute the error of level $k = 100$ for $N = 1000$, and the error of the Richardson extrapolant built from $N = 1000$ and $N = 2001$ (which halves h exactly). Optionally confirm with `qmnt.fd_levels`.

9. • Why does Table V have a "null" column with rms values but no correlations for the null?

10. •• Using qmnt, compute the smooth-zero null rms on the holdouts 401–500, 501–1000 and 1001–2000, and compare with the null column of Table V.

11. •• Using `qmnt.oracle_zeros` and `qmnt.score`, reproduce the {2} and 3-prime entries of Table V for the holdout 501–1000.

12. •• With `rng = np.random.default_rng(1)` and `test = rng.choice(400, 80, replace=False) + 1`, form a within-height random test set inside zeros 1–400. Compute the smooth-zero null on it and compare with the null range in Table III. How many training levels does the split have?

13. •• Compute the smooth-zero null rms over the even-indexed zeros $2, 4, \ldots, 400$ and over the odd-indexed zeros $1, 3, \ldots, 399$. Match them to the "odd" and "even" rows of Table III, recalling that "odd" means trained on odd-indexed zeros and scored on even-indexed ones.

14. • In the interpolation split of Sec. III, how many zeros are in the training set and how many are held out? Is it a forward, within-height, or neither kind of split in the sense of Section 28.7?

15. ••• From (28.9), $y_n + f_n = -\delta_n^2/(4\pi\xi_n)$. (a) Explain why $y_n + f_n \le 0$ always. (b) Estimate $y_{500} + f_{500}$ using $\delta_{500} = t_{500} - s_{500}$ from Problem 6. (c) Estimate $y_1 + f_1$ using the Sample Problem "Unfolding the first zero". Verify both numerically.

16. •• Design an improved literature audit for the claim in Sec. I A a. Name three changes that would address the limits listed in Section 28.4, and one limit that no audit can fully remove.

17. • Match each of Appendix A steps 1–8 to the section of the paper (II A, II B, II C or II D) where it is described.

18. •• Quote the paper's operational definition of "capture" (Sec. I). Turn it into an explicit pass/fail criterion that uses the numbers 0.263 and 0 for the holdout 401–500, and apply it to the scale-1, $M = 400$ operator row of Table I (OOS 0.369, corr -0.07) and the six-prime oracle of Table V.

19. ••• Let a model's predicted and true fluctuations be measured relative to a reference R_n, as in (28.17). Show that the rms error is unchanged if R_n is replaced by any other reference R'_n, but that the correlation can change. Illustrate with $f^{\mathrm{ref}} = (-1, 1, 0)$, $\hat{f} = (0.5, -0.5, 0)$ and a reference change that adds $d = (0, 0, 3)$ to both.

20. •• Show that the GUE surmise (28.10) is normalized and has mean 1, using $\int_0^\infty s^2 e^{-as^2}\, ds = \frac{\sqrt{\pi}}{4a^{3/2}}$ and $\int_0^\infty s^3 e^{-as^2}\, ds = \frac{1}{2a^2}$.

21. •• Using `qmnt.zeros(10000)`, compute the fraction of unfolded spacings below 0.2. Compare with the GUE value $\int_0^{0.2} p(s)\, ds$ and the Poisson value $1 - e^{-0.2}$.

22. •• List what a reader can check, without access to the author, to test the commitments made in Sec. II F. Which of them can you check with `qmnt` in a few minutes?

23. ••• Implement the oracle yourself: write N_{fl} from Eq. (4), solve (28.25) with `scipy.optimize.brentq` on the bracket $s_n \pm 3\mathrm{gap}_n$ (widening it if the signs agree), and score zeros 401–500 with $P = \{2, 3, 5\}$. Compare with Table V.

24. •• Explain, using turning points, why scale 1 scores only the first 500 levels although its box reaches $V = 1000$. Compute $N_{\mathrm{sm}}(1000)$ and the number of smooth zeros below 1000, and compare with the 652 states the paper reports in Sec. II E.

25. •• Quantify the paper's statement that the zero precisions "exceed every use below by many orders". Take t_{10000} from qmnt, estimate the absolute uncertainty of a 15-significant-digit value at that height, convert it to mean spacings, and compare with the smallest rms in Table IV (0.018) and with the in-sample bound rms $< 10^{-4}$ quoted in Sec. III.

28.14 Answers to selected problems

1. (a) Inverse-spectral fits [12, 13, 14, 15]: operator yes, fitted to zeros yes, held-out test none found "to our knowledge". (b) Explicit formulae [3, 4, 21]: no operator in the inverse-spectral sense, nothing fitted to zeros, so the held-out question does not arise. (c) Machine learning [44, 45, 46, 47]: no operator; models are fitted to data and use train/test splits. (d) Operator proposals: operator yes, nothing fitted to zeros, so not applicable. The paper's own work combines operator, fitting and held-out testing.

2. $100/2\pi = 15.9155$, $\log = 2.76729$, so $N_{\rm sm}(100) = 15.9155 \times 2.76729 - 15.9155 + 0.875 = 44.0427 - 15.9155 + 0.875 = 29.00$. Exactly 29 zeros lie below 100, so the smooth count is accurate to about a hundredth here.

3. gap $= 2\pi/\log(15.9155) = 6.28319/2.76729 = 2.271$.

4. At $V = V_0$, $\sqrt{V - V_0} = 0$, so the first term vanishes, and the argument of the second logarithm is $\sqrt{V_0}/\sqrt{V_0} = 1$, whose log is 0. Hence $x(V_0) = 0$. As V increases from V_0, $x(V)$ increases from 0, so Eq. (2) produces only $x \geq 0$; the potential on $x < 0$ is fixed by evenness, $V(-x) = V(x)$.

5. With $\sqrt{V - V_0} \approx \sqrt{V}$, the stable form gives the second log as $2\log(2\sqrt{V}/\sqrt{V_0}) = \log(4V/V_0)$. Then $x \approx \frac{\sqrt{V}}{\pi}[\log\frac{V_0}{2\pi e^2} + \log\frac{4V}{V_0}] = \frac{\sqrt{V}}{\pi}\log\frac{4V}{2\pi e^2} = \frac{\sqrt{V}}{\pi}\log\frac{2V}{\pi e^2}$. At $V = 1000$: $\sqrt{1000}/\pi = 10.0658$, $\log(2000/(\pi e^2)) = \log(86.157) = 4.4562$, so $x \approx 44.86$, about 0.03 below the exact 44.883. The neglected terms are of relative order $V_0/V \approx 0.01$.

6. $z = (500 - 11/8)/e = 183.43$; $s_{500} = 2\pi(498.625)/W(183.43) \approx 811.448$; $\text{gap}_{500} = 2\pi/\log(811.448/2\pi) = 1.2926$; $f_{500} = (811.184 - 811.448)/1.2926 \approx -0.204$.

7. Eq. (3) is pure algebra, true for any Δ, so it still holds. But now Δ'_n contains t_n for held-out n, and the corrected target $\tilde{t}'_n = t_n + \Delta'_n = E_n^{\rm g}[\text{smooth}]$ no longer contains the zero at all, while the "corrected prediction" $E_n[\text{fit}] - \Delta'_n$ contains t_n: held-out truth has been inserted into the prediction. The lesson is that the no-leakage argument has two parts: the identity (correcting targets equals correcting predictions) and the construction of Δ_n from unperturbed eigenvalues only. The identity alone proves nothing.

8. For $N = 1000$, $h = 0.089676$, $E_{100} = 12.2483$, $E_{100}^{\rm disc} = 12.1481$, error ≈ -0.100. With $N = 2001$ the error is about a quarter of that, and $(4E_{h/2} - E_h)/3$ has error $\approx -8 \times 10^{-5}$, matching the next-order estimate $-E^3h^4/1440$. To see where this comes from, note that the angle $k\pi/(2(N+1))$ equals $x = \sqrt{E}\,h/2$ with $E = (k\pi/2L)^2$. Expanding $\sin^2 x = x^2 - x^4/3 + 2x^6/45 - \ldots$ gives $E^{\rm disc} = \frac{4}{h^2}\sin^2 x = E - E^2h^2/12 + E^3h^4/360 - \ldots$. In the Richardson combination the h^2 term cancels and the h^4 term is multiplied by $(4 \cdot \frac{1}{16} - 1)/3 = -\frac{1}{4}$, leaving $-E^3h^4/1440$.

9. The null predicts zero fluctuation for every held-out zero; a constant sequence has zero variance, so the Pearson correlation (covariance over the product of standard deviations) is undefined. The null rms is simply the rms of the true fluctuations, eq. (28.19).

11. You should obtain rms 0.208, corr +0.64 for $\{2\}$ and 0.127, +0.88 for $\{2, 3, 5\}$, the 501–1000 row of Table V.

13. Within 0.006 of 0.238 on the even-indexed zeros and within 0.006 of 0.250 on the odd-indexed zeros (the paper's values are measured against the unperturbed potential's levels, which by (28.22) can differ slightly). The "odd" row of Table III scores the even-indexed zeros, so its null is 0.238; the "even" row scores the odd-indexed zeros, null 0.250, as printed.

14. Training: zeros 1–200 and 301–500, i.e. 400 zeros; held out: 201–300, 100 zeros. It is neither forward (the held-out zeros are below some training zeros) nor random within height; it is a contiguous interior gap, which probes a specific band of turning points.

15. (a) $\delta_n^2 \geq 0$ and $\xi_n > 0$, so $-\delta_n^2/(4\pi\xi_n) \leq 0$. (b) $\delta_{500} = -0.2635$, $\delta^2 = 0.06944$, $4\pi\xi \approx 4\pi(811.3) = 10195$: $y_{500} + f_{500} \approx -6.8 \times 10^{-6}$. (c) $\delta_1 = -0.3866$, $\delta^2 = 0.1495$, $4\pi\xi \approx 4\pi(14.33) = 180.1$: $y_1 + f_1 \approx -8.3 \times 10^{-4}$. The numerical values agree with these estimates.

17. Steps 1–2 (zeros, smooth zeros, gap): Sec. II A. Steps 3–5 (grids and potential, solver, bias-corrected targets): Sec. II B. Steps 6–7 (Hermitian fit, OOS scoring): Sec. II C. Step 8 (oracle): Sec. II D.

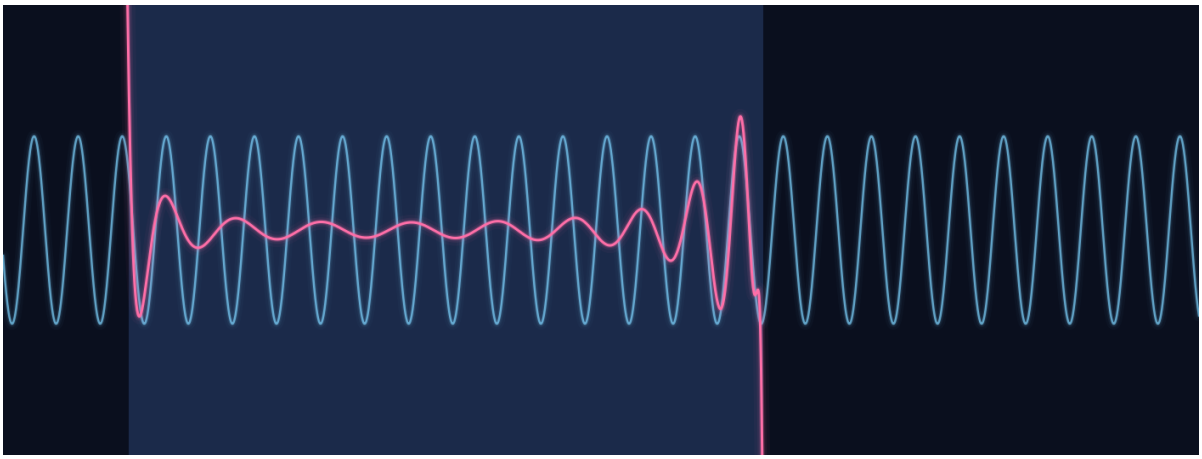
19. With $\hat{f}_n = (\hat{t}_n - R_n)/\text{gap}_n$ and $f_n^{\rm ref} = (t_n - R_n)/\text{gap}_n$, the difference $\hat{f}_n - f_n^{\rm ref} = (\hat{t}_n - t_n)/\text{gap}_n$ does not contain R_n, so the rms is unchanged. Replacing R_n by R'_n adds the same sequence $d_n = (R_n - R'_n)/\text{gap}_n$ to both $\hat{f}$ and $f^{\rm ref}$, which changes their covariance and variances, so the correlation can change. Example: $\hat{f} = (0.5, -0.5, 0)$, $f^{\rm ref} = (-1, 1, 0)$ has correlation -1; after adding $d = (0, 0, 3)$, $\hat{f}' = (0.5, -0.5, 3)$, $f^{\rm ref\prime} = (-1, 1, 3)$, both with mean

1; deviations $(-0.5, -1.5, 2)$ and $(-2, 0, 2)$; covariance sum $1 + 0 + 4 = 5$; variance sums 6.5 and 8; correlation $5/\sqrt{52} = 0.693$. The rms error stays $\sqrt{(1.5^2 + 1.5^2 + 0)/3} = 1.225$. This is why the paper specifies its reference: correlations are meaningful only relative to a stated smooth baseline.

21. The GUE value is $\int_0^{0.2} p(s)\, ds \approx 0.0084$ and the Poisson value is $1 - e^{-0.2} = 0.181$. The fraction from the first 10^4 zeros is below 0.05 (the check script confirms this bound), far closer to GUE than to Poisson: small spacings are strongly suppressed (level repulsion).

23. With the bracket $s_n \pm 3\mathrm{gap}_n$ every root is found without widening at these heights, and the score is rms 0.116, corr +0.90, matching the 3-prime entry of Table V on 401–500. Your N_{fl} needs 9 terms: $p \in \{2, 3, 5\}$, $m = 1, 2, 3$.

25. `qmnt` gives $t_{10000} \approx 9877.78$, where the mean spacing is $2\pi/\log(9877.78/2\pi) \approx 0.854$. Fifteen significant digits with four before the decimal point leave eleven after it, so the rounding error is at most 5×10^{-12} in t, i.e. about 6×10^{-12} spacings. The smallest rms in Table IV is 0.018 spacings, about 3×10^9 times larger. Against the in-sample bound 10^{-4} of Sec. III the ratio is about 1.7×10^7. "Many orders" therefore means at least seven orders of magnitude, and about nine for the held-out errors.



The continuation probe in miniature: a flexible fit paints one window of $\sin(t \log 2)$ *perfectly, then cannot continue it.*

Chapter 29

Reading the Paper II: Fitting Is Easy, Transfer Is Nil

↑ Contents

What this chapter is about

Section III of the paper [1] carries its central negative result. Its title, "Fitting is easy, transfer is nil", makes two claims at once. The first is about *capacity*: a smooth potential with a few hundred adjustable coefficients, added to the Wu–Sprung background, can be tuned until its energy levels sit almost exactly on any prefix of the Riemann zeros. The second is about *transfer*: the same tuned potential predicts the zeros it was not shown no better than doing nothing at all. A model that reproduces its training data but carries no information about new data has memorized, not learned.

In this chapter we read Section III paragraph by paragraph. For each claim we say what it means in plain words, which earlier chapters supply the tools, what computation lies behind it, how much of that computation you can rerun yourself with `qmnt` at small scale, and how strong the evidence is. We give special attention to the paper's "arithmetic-continuation probe", the cleanest single experiment in the paper: the fitted family is shown a pure prime-2 oscillation, whose correct continuation is known exactly, and it fails to continue it.

Where this shows up in the paper: Section III (all five paragraphs: Other bases, Within-height splits, Permutation and surrogate tests, Interpolation split, Arithmetic-continuation probe); Tables I, II and III; Figs. 1, 3 and 4; and Appendix A steps 12–18 and 21–22, with the numerical anchors listed at the end of Appendix A.

What you need

- Smooth zeros s_n, mean spacing gap_n, fluctuations in mean spacings, and the null 0.263 (Chapter 17).
- The truncated explicit formula, paper Eq. (4), and the prime-2 wave RE2 (Chapter 18).
- Rms, Pearson correlation, mean, variance and autocorrelation (Chapter 19); the Wigner surmise for GUE spacings (Chapter 20).
- Classical turning points and the map from level index to position (Chapter 12); the Wu–Sprung potential, paper Eq. (2) (Chapter 22).
- First-order sensitivity A_{nk} (Chapter 13); finite-difference Hamiltonians and the bias correction of paper Eq. (3) (Chapter 23).
- The SVD-damped Levenberg–Marquardt fit, the three bases and the coverage law (Chapter 24); ridge filters and underdetermined problems (Chapter 9).
- Train/test splits, the null model, covariate shift, saturation, serial correlation and simple kriging (Chapter 25).
- Bootstrap intervals, block bootstrap, permutation and surrogate data (Chapter 26); power spectra and phase randomization (Chapter 6).
- Sinusoidal regression with known frequencies and dephasing (Chapter 27).
- First-year physics: standing waves, turning points on a potential energy curve.

29.1 The claim in one paragraph, and the protocol behind it

Before we read any number, we need to be precise about what is being measured. Chapter 28 walks through the paper's protocol in detail; here is the minimum you need for Section III, so that this chapter stands on its own.

The data. Recall from Chapter 17 that t_n is the height of the n-th nontrivial zero $\rho_n = \frac{1}{2} + it_n$ of the zeta function. The smooth counting function of paper Eq. (1),

$$N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \tag{29.1}$$

defines the *smooth zeros* s_n through $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$, and the local mean spacing

$$\mathrm{gap}_n = \frac{\mathrm{d}s_n}{\mathrm{d}n} = \frac{2\pi}{\log(s_n/2\pi)}. \tag{29.2}$$

The *fluctuation* of zero n, measured in mean spacings, is

$$d_n = \frac{t_n - s_n}{\mathrm{gap}_n}. \tag{29.3}$$

A value $d_n = 0.3$ means that zero n sits three tenths of a typical spacing above where the smooth law puts it.

The model. In units $\hbar = 2m = 1$, the Hamiltonian is $H = -\mathrm{d}^2/\mathrm{d}x^2 + V(x)$ on the box $[-L, L]$ with walls at $\pm L$. The background V is the Wu–Sprung potential of paper Eq. (2), whose levels follow the smooth zeros. The fitted model adds a correction

$$\delta V(x) = \sum_{k=0}^{M-1} c_k\,\phi_k(x), \tag{29.4}$$

where the ϕ_k are basis functions ($\cos(k\pi x/L)$ by default), the c_k are M real coefficients, and M is the basis size. The coefficients are chosen to make the eigenvalues E_n of $H + \delta V$ match

bias-corrected targets $\tilde{t}_n$ for the training indices, by iterated linearization with the first-order sensitivity $A_{nk} = \int |\psi_n|^2 \phi_k \, \mathrm{d}x$ and an SVD-damped Levenberg–Marquardt step that accepts only improvements (Chapter 24; paper Sec. II C and Appendix A step 6).

The score. After fitting on zeros $1, \ldots, n_{\mathrm{fit}}$, the paper performs one plain eigensolve of the fitted potential and compares its levels with the *held-out* zeros, the ones the fit never saw. Write p_n for the predicted fluctuation of level n and d_n for the true one, both in mean spacings and both measured relative to the unperturbed potential's levels. Strictly, the operator scores use $d_n = (\tilde{t}_n - E_n[\mathrm{smooth}])/\mathrm{gap}_n$ (Appendix A step 18). This differs from Eq. (29.3) by the Richardson residual $E_n^{\mathrm{Rich}} - s_n$, which has mean 0.005 spacings and is at most 0.063 spacings (paper Sec. II B). That small difference is why the two nulls coincide, and we use the same symbol d_n for both. For a held-out index set $\mathcal{H}$ containing $|\mathcal{H}|$ indices, the two reported numbers are

$$\mathrm{rms} = \sqrt{\frac{1}{|\mathcal{H}|} \sum_{n \in \mathcal{H}} (p_n - d_n)^2}, \qquad \mathrm{corr} = \text{Pearson correlation of } \{p_n\} \text{ and } \{d_n\},\ n \in \mathcal{H}. \tag{29.5}$$

OOS (out-of-sample) means "computed on $\mathcal{H}$"; in-sample means "computed on the training indices".

The null. The *null* prediction is "no fluctuation": $p_n = 0$. Its rms is

$$\mathrm{rms}_{\mathrm{null}} = \sqrt{\frac{1}{|\mathcal{H}|} \sum_{n \in \mathcal{H}} d_n^2}. \tag{29.6}$$

On RE5, the zeros 401–500, this is 0.263 (paper Sec. II C; Appendix A gives 0.2631). The paper notes that the unperturbed potential and the smooth zeros give the same null on this holdout, so the operator and the explicit-formula oracle share one baseline. The correlation of the null is undefined (a constant has no variance), so only its rms is quoted.

The bias correction. Finite-difference levels drift away from the continuum levels as the level index grows (Chapter 23). The paper subtracts that drift from the targets, using only eigenvalues of the *unperturbed* potential on two grids. Paper Eq. (3) shows this is algebraically the same as correcting the prediction by a fit-independent quantity and scoring against raw zeros, so no held-out information leaks into the fit. Without the correction, the raw-zero null on the working grid would be 2.09 on zeros 401–500 (Appendix A, key anchors), dominated by grid error.

The claim. With these definitions, the opening of Section III says, in plain words:

1. **Capacity.** The fitted family can drive the in-sample rms almost to zero.

2. **No transfer.** Across twelve configurations, the OOS rms never beats the null and no reproducible positive OOS correlation appears.

3. **Diagnosis.** Two robustness checks (other bases, within-height splits) and the three probes that, in the paper's words, "complete the diagnosis" (permutation and surrogate tests, the interpolation split, and the arithmetic-continuation probe, the last being the cleanest) show why: the family fits anything, so its in-sample success measures capacity, and it does not express the long-range oscillation that would let it continue.

Checkpoint 29.1

A model predicts $p_n = d_n + 0.263$ for every held-out n (a perfect shape shifted by a constant). What are its OOS rms and correlation on zeros 401–500?

Answer: The error is the constant 0.263, so the rms is exactly 0.263, equal to the null; the correlation is +1, because a constant shift does not change the Pearson correlation. This is why the paper always reports both numbers: they measure different things (Chapter 19).

Key Ideas

- Fluctuations $d_n = (t_n - s_n)/\text{gap}_n$ are measured in local mean spacings; every rms in Section III uses this unit.
- The null predicts $p_n = 0$; on zeros 401–500 its rms is 0.263.
- OOS scores use one plain eigensolve of the fitted potential on zeros never shown to the fit; the bias correction leaks nothing (paper Eq. (3)).
- Section III's claim has three parts: capacity, no transfer, diagnosis.

29.2 Capacity: in-sample fitting is easy

The section opens: "In-sample capacity is not the issue: on the full 500 zeros the fit reaches rms 0.186 at $M = 300$ and reproduces all 500 zeros to rms $< 10^{-4}$ spacings by $M = 900$, consistent with a basis-coverage law (the cosine family becomes effectively complete for joint level shifts once $M\pi/L > 2\sqrt{E}$)."

Plain words. With 900 cosine coefficients, the first 500 levels sit on their zeros with an rms error below one ten-thousandth of a spacing. That is not a prediction; it is a demonstration that the family is flexible enough to realize essentially any pattern of shifts among these levels.

Tools. The coverage law is the heuristic of Chapter 24. Recall the reasoning. Near the center of the well a level of energy E has local wavenumber $\sqrt{E - V} \le \sqrt{E}$, so $|\psi_n|^2$ oscillates with wavenumber up to $2\sqrt{E}$. The shift $\delta E_n = \int |\psi_n|^2 \delta V \, dx$ can respond differently to neighboring levels only if δV contains structure on that scale. The highest cosine in the basis has wavenumber $(M-1)\pi/L \approx M\pi/L$. The heuristic condition for the basis to address levels up to energy E independently is therefore

$$\frac{M\pi}{L} > 2\sqrt{E} \quad \Longleftrightarrow \quad M > \frac{2L\sqrt{E}}{\pi} \quad \Longleftrightarrow \quad E < \left(\frac{M\pi}{2L}\right)^2. \tag{29.7}$$

The three forms are the same inequality: the second multiplies both sides by L/π, and the third divides the first by 2 and squares (both sides are positive). Its status is *heuristic*: it is a wavelength-matching argument, not a theorem, and the word "effectively" in the paper keeps that status.

Sample Problem 29.1: The coverage law at scale 1

At scale 1, $L = 44.883$ and the highest scored level is near $t_{500} = 811.1843588$ (Appendix A key anchors). (a) How large must M be for Eq. (29.7) to hold for all 500 levels? (b) Up to what energy does $M = 900$ reach? (c) Up to what energy does $M = 300$ reach, and how many zeros lie below it? (d) For training on zeros 1–400, with $t_{400} \approx 680$, what M

does the law require?
Step 1: For (a), use the second form: $M > 2L\sqrt{E}/\pi$ with $E = 811.1843588$. We have $\sqrt{811.1843588} = 28.4813$, so $2\sqrt{E} = 56.9626$.
Step 2: $56.9626 \times 44.883 = 2556.6$, and dividing by π gives $M > 813.8$.
Step 3: For (b), use the third form with $M = 900$: $M\pi/(2L) = 2827.43/89.766 = 31.498$, and squaring gives $E < 992.1$. This exceeds 811, so $M = 900$ covers all 500 levels.
Step 4: For (c), with $M = 300$: $M\pi/(2L) = 942.48/89.766 = 10.499$, and squaring gives $E < 110.2$. The zeros below 110.2 are $t_1, \ldots, t_{33}$ (since $t_{33} = 107.17$ and $t_{34} = 111.03$), so 33 zeros.
Step 5: For (d), $\sqrt{680} = 26.08$, so $M > 2 \times 44.883 \times 26.08/\pi \approx 745$.

$$\boxed{\begin{array}{ll} M \gtrsim 814 \text{ for 500 levels;} & M \gtrsim 745 \text{ for 400 levels;} \\ M = 900 \Rightarrow E < 992; & M = 300 \Rightarrow E < 110 \text{ (33 zeros)} \end{array}}$$

Interpretation. The paper's two anchors bracket the law: at $M = 900$ the law is satisfied for every level and the in-sample rms is below 10^{-4}; at $M = 300$ the law covers only the lowest few dozen levels, yet the rms has already fallen from the null to 0.186. Partial coverage still buys partial in-sample improvement. The law is a rough guide to where the fit becomes complete, not a sharp threshold.

Figure 3 of the paper. Paper Fig. 3 plots, at both scales, in-sample and held-out rms against M. In-sample error falls steadily as M grows; held-out error stays at or above the dashed null line. That figure is the visual summary of the whole section: two curves that separate because one measures capacity and the other measures information.

Pitfall

A falling in-sample error is often read as "the model is getting closer to the truth". It means only that the model is getting closer to the *training data*. Suppose the basis is rich enough to address every training level independently. Here that means satisfying the coverage law of Eq. (29.7), which requires $M \gtrsim 745$ for 400 levels; having M merely equal to the number of levels is not enough. Once that holds, the family can fit essentially any targets (compare Chapter 9: an underdetermined linear system has infinitely many exact solutions). At $M = 400$ the paper shows that the in-sample error is the same for shuffled and phase-randomized targets as for the true zeros (Section 29.7). It is plausible, but not tested, that the whole in-sample curve of paper Fig. 3 would look the same for such targets.

Key Ideas

- The cosine family reaches in-sample rms 0.186 at $M = 300$ and below 10^{-4} at $M = 900$ on all 500 scale-1 levels (paper Sec. III, first paragraph).
- The heuristic coverage law $M\pi/L > 2\sqrt{E}$ predicts completeness for the 500 levels at $M \gtrsim 814$, consistent with the $M = 900$ anchor.
- In-sample success is a statement about capacity, not about captured structure.

29.3 The central negative result: twelve configurations

★ Did you know? *The Most Famous Null Result*

In 1887 Albert Michelson and Edward Morley searched for Earth's motion through the supposed luminiferous ether and found essentially nothing. This negative result became a cornerstone of the evidence for special relativity. A carefully designed experiment that finds no effect, with clear error bars, can be one of the most informative results in science.

"Transfer is the central negative result (Fig. 3 and Table I). Across the two scales, five train/test splits, basis budgets from 200 to 900, two ridge strengths, and repeated optimizer restarts examined (twelve configurations), the held-out rms never improves on the null and no reproducible positive held-out correlation appears."

Reading Table I. Table 29.1 reproduces paper Table I. Each row is one configuration: a scale (1: $L = 44.883$, 500 scored levels; 2: $L = 59.957$, 1000 levels), a training set (always a prefix $1, \ldots, n_{\text{fit}}$), a basis size M and possibly a ridge strength μ. The five training prefixes are 1–300, 1–350, 1–400, 1–450 (scale 1) and 1–800 (scale 2). The table caption gives the null references: 0.261–0.270 at scale 1 depending on the holdout (0.263 for 401–500), and 0.274 at scale 2 (801–1000).

Table 29.1: Paper Table I (Hermitian cosine fits), quoted exactly. rms in local mean spacings. Nulls: 0.263 for holdout 401–500 at scale 1 (other scale-1 holdouts 0.261–0.270), and 0.274 for 801–1000 at scale 2.

scale	fit zeros	M / variant	in-sample	OOS	OOS corr
1	1–300	200	0.212	0.317	−0.01
1	1–300	400	0.120	0.320	−0.17
1	1–350	400	0.128	0.273	+0.04
1	1–400	200	0.226	0.274	−0.09
1	1–400	400	0.137	0.369	−0.07
1	1–400	400, ridge $\mu = 0.03$	0.142	0.271	−0.16
1	1–400	400, ridge $\mu = 0.3$	0.206	0.265	−0.14
1	1–400	600	0.064	0.313	−0.16
1	1–450	400	0.147	0.439	+0.12
2	1–800	300	0.242	0.370	−0.01
2	1–800	600	0.170	0.279	−0.03
2	1–800	900	0.103	0.302	+0.08

Three features stand out in Figure 29.1. First, no point lies below the null band. Second, the horizontal position (in-sample quality) does not predict the vertical one: the best in-sample row (0.064) has OOS 0.313, worse than the null. Third, the correlations scatter around zero, from −0.17 to +0.12, nowhere near the oracle's +0.96.

A compact way to put every row on one scale is a *relative skill*

$$S = 1 - \frac{\text{rms}_{\text{model}}}{\text{rms}_{\text{null}}}. \tag{29.8}$$

This is our convenience, not a quantity the paper reports. (Notation warning: this S is unrelated to the fluctuating part $S(t)$ of the counting function in Chapter 17. Likewise, below, ρ and ρ_ℓ denote correlations and are unrelated to the zeros $\rho_n = \frac{1}{2} + it_n$.) $S > 0$ means better than doing nothing, $S = 0$ means no better, $S < 0$ means harm, and $S = 1$ means perfect prediction.

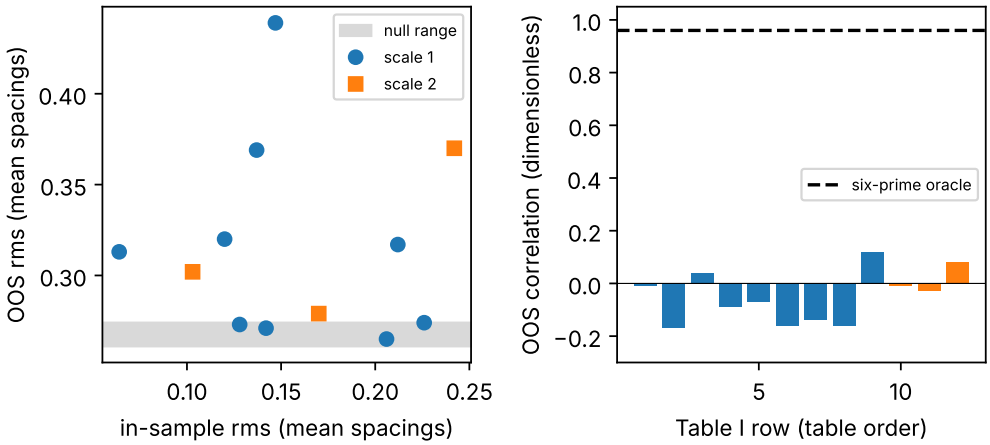


Figure 29.1: Paper Table I drawn as a picture. Left: OOS rms against in-sample rms for the twelve configurations (circles: scale 1; squares: scale 2); the shaded band is the range of null references 0.261–0.274 quoted in the table caption. Every point lies in or above the band, wherever it sits horizontally. Right: OOS correlation for each row (in table order); the dashed line is the six-prime explicit-formula oracle's correlation +0.96 on zeros 401–500 (paper Table V).

Sample Problem 29.2: Relative skill of the operator rows and the oracle

Compute S for the four 1–400 rows of Table I that use $M = 400$ (with $\mu = 0$, 0.03 and 0.3) or $M = 600$ (no ridge), and for the six-prime and one-prime oracles on the same holdout (paper Table V: rms 0.074 and 0.198). Use the null 0.263.

Step 1: $M = 400$: $S = 1 - 0.369/0.263 = 1 - 1.403 = -0.403$.
Step 2: $M = 400$, $\mu = 0.03$: $S = 1 - 0.271/0.263 = 1 - 1.0304 = -0.030$.
Step 3: $M = 400$, $\mu = 0.3$: $S = 1 - 0.265/0.263 = 1 - 1.0076 = -0.0076$.
Step 4: $M = 600$: $S = 1 - 0.313/0.263 = 1 - 1.190 = -0.190$.
Step 5: Six-prime oracle: $S = 1 - 0.074/0.263 = 1 - 0.281 = +0.719$.
Step 6: One-prime oracle: $S = 1 - 0.198/0.263 = 1 - 0.753 = +0.247$.

$$S_{M=400} = -0.40, \quad S_{\mu=0.03} = -0.03, \quad S_{\mu=0.3} = -0.008, \quad S_{M=600} = -0.19;$$
$$S_{6\,\text{primes}} = +0.72, \quad S_{\{2\}} = +0.25$$

The best operator row is essentially at zero skill; the explicit formula with a single prime and no fitted parameters already has positive skill. This is the comparison behind paper Fig. 1 and the statement in Sec. IV that one prime beats every fitted operator's holdout performance by a factor of at least about 1.3 (0.265/0.198 = 1.34).

Fluctuations that are there but wrong. The paper says: "The fitted models emit fluctuations of 0.16–0.62× the true amplitude that are uncorrelated with truth." This sentence explains why the OOS rms sits above the null rather than at it. The next sample problem derives the relation.

Sample Problem 29.3: Why an uncorrelated prediction is worse than the null

Let the held-out truth have values d_n and the prediction have values p_n, with means $\bar{d}$, $\bar{p}$, standard deviations σ_d, σ_p and correlation ρ. (a) Express the mean square error in these

quantities. (b) If $\bar{p} = 0$ and $\rho = 0$, and the prediction amplitude is a times the null rms, how does the OOS rms compare with the null? Evaluate for $a = 0.16$ and $a = 0.62$ with null 0.263. (c) What correlation is needed to beat the null when $\bar{p} = \bar{d} = 0$?
Step 1: Write the error $e_n = p_n - d_n$, and write each variable as mean plus deviation: $p_n = \bar{p} + p'_n$, $d_n = \bar{d} + d'_n$, where the deviations average to zero. Denote averages over $\mathcal{H}$ by $\langle\cdot\rangle$.
Step 2: Expand the square:

$$\langle e^2 \rangle = \langle p^2 \rangle - 2\langle pd \rangle + \langle d^2 \rangle.$$

Step 3: Each term, using $\langle p' \rangle = \langle d' \rangle = 0$:

$$\langle p^2 \rangle = \bar{p}^2 + \sigma_p^2, \qquad \langle d^2 \rangle = \bar{d}^2 + \sigma_d^2, \qquad \langle pd \rangle = \bar{p}\bar{d} + \langle p'd' \rangle = \bar{p}\bar{d} + \rho\sigma_p\sigma_d,$$

where the last step is the definition of the Pearson correlation, $\rho = \langle p'd' \rangle / (\sigma_p \sigma_d)$ (Chapter 19).
Step 4: Substitute and collect $\bar{p}^2 - 2\bar{p}\bar{d} + \bar{d}^2 = (\bar{p} - \bar{d})^2$:

$$\langle e^2 \rangle = (\bar{p} - \bar{d})^2 + \sigma_p^2 + \sigma_d^2 - 2\rho\,\sigma_p\sigma_d. \tag{29.9}$$

Step 5: For (b), set $\bar{p} = 0$ and $\rho = 0$: $\langle e^2 \rangle = \bar{d}^2 + \sigma_d^2 + \sigma_p^2 = \mathrm{rms}_{\mathrm{null}}^2 + \sigma_p^2$, using Eq. (29.6). With $\sigma_p = a\,\mathrm{rms}_{\mathrm{null}}$,

$$\mathrm{rms} = \mathrm{rms}_{\mathrm{null}}\sqrt{1 + a^2}.$$

Step 6: Numbers: $a = 0.16$ gives $0.263\sqrt{1.0256} = 0.263 \times 1.0127 = 0.2663$; $a = 0.62$ gives $0.263\sqrt{1.3844} = 0.263 \times 1.1766 = 0.3094$.
Step 7: For (c), with zero means, $\langle e^2 \rangle < \sigma_d^2$ requires $\sigma_p^2 - 2\rho\sigma_p\sigma_d < 0$, that is (dividing by $\sigma_p\sigma_d > 0$) $\rho > \sigma_p/(2\sigma_d) = a/2$.

$\mathrm{rms} = \mathrm{rms}_{\mathrm{null}}\sqrt{1 + a^2}$: 0.266 ($a = 0.16$) to 0.309 ($a = 0.62$); beating the null needs $\rho > a/2$

Comment. The paper's amplitude range accounts for OOS values from 0.266 to 0.309. Some rows of Table I are higher (0.369, 0.439). By Eq. (29.9), a larger rms at small ρ needs a larger prediction amplitude in those rows or a mean offset $\bar{p} - \bar{d}$. The paper does not report the decomposition row by row, so we cannot say which; Problem 5 explores the arithmetic.

Regularization: from "memorize more" to "do nothing". The paper says regularization "moves them smoothly from 'memorize more' to 'do nothing' and never creates skill". Look at the three 1–400, $M = 400$ rows. As the ridge strength goes $\mu = 0 \to 0.03 \to 0.3$, the in-sample rms rises $0.137 \to 0.142 \to 0.206$ and the OOS rms falls $0.369 \to 0.271 \to 0.265$, approaching the null 0.263 from above. Recall from Chapter 9 that the ridge filter shrinks every coefficient toward zero; as $\mu \to \infty$, $c \to 0$ and the model *is* the null. Ridge removes the harm done by large uncontrolled coefficients, but it can only interpolate between the unregularized fit and the null; it cannot add information that the parametrization does not carry.

Optimizer variability. The table caption reports that four random-initialization repeats of the (scale 1, $M = 400$, 1–400) row give identical in-sample rms 0.1370 and OOS 0.365–0.379 with correlation -0.06 to -0.08, and three repeats of the (scale 2, $M = 600$) row give OOS 0.279–0.283. The scatter across restarts is small compared with the gap to the oracle. The caption also flags that the scale-2 $M = 300$ row needed a relaxed step-acceptance threshold

($1 - 10^{-5}$ instead of 0.999, Appendix A step 6); under the strict rule the optimizer accepts no step and the configuration stays at the null. The paper keeps that row as a sensitivity result, and the conclusion is the same either way.

Uniformity across the holdout. One might worry that the failure is concentrated at the top of the holdout, where levels approach the edge of the basis coverage or the box. The paper checks: "Hermitian OOS error is uniform across the holdout (e.g. $M = 600$, scale 1: rms 0.321/0.304 on its first/second half), ruling out edge-of-coverage effects."

Sample Problem 29.4: Do the two halves add up to the table entry?

The scale-1, $M = 600$, fit 1–400 row has OOS rms 0.313 on zeros 401–500. The first half (401–450) has rms 0.321 and the second half (451–500) has 0.304. Check consistency.
Step 1: The mean square over the union of two sets of equal size is the average of the two mean squares, because $\frac{1}{100}\sum_{\text{all}} e^2 = \frac{1}{2}\left(\frac{1}{50}\sum_{\text{first}} e^2 + \frac{1}{50}\sum_{\text{second}} e^2\right)$.
Step 2: $0.321^2 = 0.103041$ and $0.304^2 = 0.092416$; their average is 0.0977285.
Step 3: $\sqrt{0.0977285} = 0.3126$.

$$\boxed{\text{rms}_{401\text{–}500} = 0.3126 \approx 0.313}$$

The halves reproduce the table entry, and neither half is close to the null: the error is spread evenly, not piled up at the far edge.

Paper Fig. 1. The paper's first figure shows zeros 370–460 for the scale-1, $M = 400$ fit on zeros 1–400. In the training region both the operator and the six-prime formula track the fluctuations; past the boundary the operator's held-out correlation is -0.07 (the 1–400, $M = 400$ row of Table I) while the formula's is $+0.96$ (the 401–500 row of Table V). Neither curve uses any zero past the boundary.

Checkpoint 29.2

In Table I, the row with the lowest in-sample error (0.064) does not have the lowest OOS error. Which row has the lowest OOS error, and why is that not evidence of skill?

Answer: The ridge row $\mu = 0.3$ (OOS 0.265). It is the row closest to the null model, having been shrunk most toward $c = 0$; its OOS rms is still slightly above the null 0.263 and its correlation is -0.14. Being near the null is the absence of harm, not the presence of skill.

Key Ideas

- In all twelve rows of paper Table I the OOS rms is at or above the null, and correlations scatter around zero (-0.17 to $+0.12$).
- By Eq. (29.9), an uncorrelated zero-mean prediction with nonzero amplitude is always worse than the null; to beat the null a zero-mean prediction needs $\rho > a/2$.
- Ridge regularization moves the fit toward the null and stops there; restarts change little; errors are uniform across the holdout.

29.4 How sure are we? Bootstrap intervals

A table entry computed on 100 held-out zeros is a sample statistic; a different set of 100 zeros would give a somewhat different number. Recall from Chapter 26 the *percentile bootstrap*: draw

100 indices from the held-out set with replacement, recompute rms and correlation, repeat many times, and report the 2.5th and 97.5th percentiles of the resampled values as a 95% interval. The paper uses 2×10^4 resamples (Appendix A step 13).

The paper's statements (Sec. III, second paragraph):

- scale 1, $M = 400$, random-restart replicate: OOS rms $[0.33, 0.43]$, correlation $[-0.21, +0.08]$;
- scale 2, $M = 600$: OOS rms $[0.26, 0.30]$, correlation $[-0.18, +0.08]$;
- six-prime oracle on the same 401–500 holdout: correlation $[+0.95, +0.97]$.

How to read them. For scale 1, the whole rms interval lies above 0.263, so the paper says the fit is "significantly worse than the null". For scale 2, the interval $[0.26, 0.30]$ contains 0.274: the fit is "consistent with its null", meaning indistinguishable from doing nothing. Both correlation intervals contain zero and lie far below the oracle's interval. Nothing in these intervals points toward skill.

Taken on trust

The percentile bootstrap gives approximately correct coverage when the resampled units are independent and identically distributed and the sample is not too small; the theory is in Efron and Tibshirani [34]. We take that result on trust. The fluctuations d_n are *not* independent: consecutive values are correlated (Section 29.6). The block bootstrap, which resamples runs of consecutive indices, is the standard repair; its theory is also beyond this book.

The paper addresses the dependence directly in the "Other bases" paragraph: "The intervals in Table II resample held-out levels independently although the fluctuations are serially correlated; recomputed with a moving-block bootstrap (block lengths 3 to 20) they widen slightly and no conclusion changes (for the cosine row, at most $[0.30, 0.44]$)." Appendix A step 22 records a circular block bootstrap with block lengths 3–20 on zeros 401–500. The main text says moving-block and the appendix says circular block. The two schemes differ only in how blocks wrap around at the ends of the series, and the paper does not say which one produced the quoted interval. The Discussion repeats the general caveat: "Bootstrap intervals resample held-out levels independently except where stated."

Checkpoint 29.3

The block-bootstrap interval for the cosine row is at most $[0.30, 0.44]$, slightly wider than the independent interval. Does serial correlation always widen bootstrap intervals?

Answer: No. The effect depends on the sign of the dependence and on the statistic. For a sample mean, positive serial correlation enlarges the true sampling spread and negative correlation shrinks it, and the fluctuations here are negatively correlated at short lags. For the rms, the relevant dependence is that of the squared errors e_n^2: if errors of similar size cluster in runs, independent resampling understates the variability, even when the signed fluctuations alternate. The paper observes only a slight widening. The block bootstrap is the safer choice because it makes no independence assumption (Chapter 26).

Key Ideas

- The principal scale-1 rms interval $[0.33, 0.43]$ lies entirely above the null 0.263: significantly worse than doing nothing. The scale-2 interval contains its null.
- Correlation intervals of the fits contain zero; the oracle's is $[+0.95, +0.97]$.

- Block bootstrap (blocks 3–20) widens intervals slightly and changes no conclusion.

29.5 Other bases

A natural objection is that cosines are a poor basis: they are global, so a change of one coefficient moves δV everywhere. The paper reruns the protocol with two other bases for the even correction (paper Sec. II C, "Other bases"; Appendix A step 16), defined in Chapter 24:

- **Chebyshev** polynomials $T_k(2|x|/L - 1)$, $k = 0, \ldots, M - 1$: global, with resolution concentrated near $x = 0$ and $x = L$. (Notation warning: this T_k is a polynomial, unrelated to the training-window length T or the tensors $T^{(n)}$ of Sec. V.)
- **Hat** functions: piecewise-linear tents on M uniform nodes of $|x|/L$, with local support, "the simplest stand-in for a wavelet-type basis".

Only the basis changes. Paper Table II gives the results for fit 1–400, holdout 401–500, null 0.263. Table 29.2 reproduces it, and Figure 29.2 draws its intervals.

Table 29.2: Paper Table II (other bases, scale 1, fit 1–400, holdout 401–500, null 0.263), quoted exactly. Brackets: bootstrap 95% intervals from independent resampling of the held-out levels. The cosine row is the Table I configuration rerun through the code used for the other bases.

basis	M	in-sample	OOS rms	OOS corr
cosine	400	0.137	0.369 [0.32, 0.42]	−0.07 [−0.22, +0.07]
Chebyshev	200	0.230	0.368 [0.32, 0.42]	−0.05 [−0.25, +0.14]
Chebyshev	400	0.162	0.291 [0.25, 0.33]	+0.18 [−0.02, +0.37]
Chebyshev	600	0.111	0.733 [0.64, 0.82]	+0.01 [−0.20, +0.23]
Chebyshev, $\mu = 0.3$	400	0.211	0.264 [0.24, 0.29]	+0.03 [−0.21, +0.28]
hat	200	0.221	0.342 [0.30, 0.38]	−0.05 [−0.24, +0.14]
hat	400	0.145	0.340 [0.29, 0.39]	−0.19 [−0.37, −0.01]
hat	600	0.066	0.330 [0.29, 0.37]	+0.09 [−0.13, +0.30]

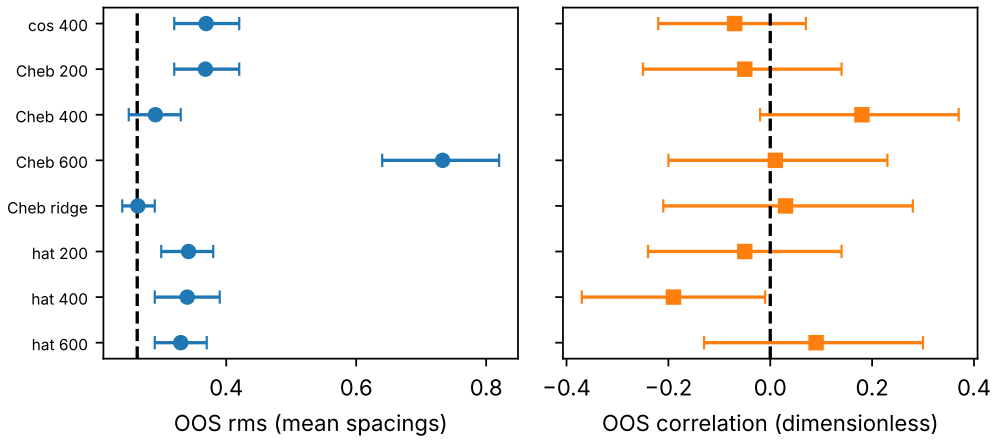


Figure 29.2: Paper Table II as intervals. Left: OOS rms with bootstrap 95% intervals for each basis and M (fit 1–400, holdout 401–500); the dashed line is the null 0.263. Right: OOS correlation with its interval; the dashed line is zero. "Cheb ridge" is Chebyshev $M = 400$ with $\mu = 0.3$. All values quoted from the paper.

What the table shows. In each basis the in-sample error falls with M (Chebyshev 0.230 → 0.162 → 0.111; hat 0.221 → 0.145 → 0.066). Every unregularized OOS rms is above 0.263. Ridge with $\mu = 0.3$ returns the Chebyshev fit to the null (0.264), as for cosines. The cosine row, 0.369 / −0.07, is the Table I configuration rerun through the alternative-basis code (Appendix A anchor: 0.1370 / 0.3685 / −0.0709), a consistency check between two code paths.

The marginal row. The paper is careful here, and we should be too. Chebyshev $M = 400$ gives OOS rms 0.291 with correlation +0.18. The paper says: "its rms interval contains the null and its correlation interval has its lower end at zero." The rms interval is $[0.25, 0.33]$ and the correlation interval is $[-0.02, +0.37]$. The paper's verdict has two halves, and both should be kept: "one such row among eight is not evidence of prediction, but a small positive correlation for this row is not excluded." The supporting observation is that the neighbors $M = 200$ and $M = 600$ show no trend (0.368 and 0.733, correlations −0.05 and +0.01). If +0.18 reflected real structure that Chebyshev polynomials capture, we would expect it not to vanish when the basis is enlarged.

One interval below zero. The hat $M = 400$ row has correlation −0.19 with interval $[-0.37, -0.01]$, which lies entirely below zero. The paper does not single this row out. Read at face value, it indicates anti-skill: a prediction that tends to go the wrong way. The neighbor-interpolation mechanism of Section 29.6 concerns interleaved splits and does not directly apply to this forward holdout. We have no explanation for this row, and one interval among eight excluding zero is not surprising by chance alone. Whatever its cause, a negative correlation is not predictive skill.

Checkpoint 29.4

Of the eight rows in Table II, how many have an OOS rms interval lying entirely above the null 0.263?

Answer: Six: cosine 400 $[0.32, 0.42]$, Chebyshev 200 $[0.32, 0.42]$, Chebyshev 600 $[0.64, 0.82]$, hat 200 $[0.30, 0.38]$, hat 400 $[0.29, 0.39]$, hat 600 $[0.29, 0.37]$. The other two (Chebyshev 400 $[0.25, 0.33]$ and Chebyshev ridge $[0.24, 0.29]$) contain the null. None lies below it.

Key Ideas

- Chebyshev and hat bases reproduce the cosine pattern: in-sample error falls with M, OOS error stays above the null, ridge returns to the null.
- Chebyshev $M = 400$ (0.291, +0.18) is marginal: not evidence of prediction, but a small positive correlation is not excluded. Keep both halves of the hedge.

29.6 Within-height splits

The worry: covariate shift. Every split in Table I trains on low zeros and tests on higher ones. Recall from Chapter 25 that when test data come from a different region than training data (covariate shift), failure can reflect the shift rather than a failure to learn. Here the fluctuation amplitude grows slowly with height: the null grows from 0.263 to 0.296 across paper Table V. To rule this out, the paper also trains and tests *within one height range* (paper Sec. II C, "Within-height splits"; Appendix A step 17):

- random 80/20 splits: 20% of the indices chosen at random are held out (ten seeds at scale 1 inside zeros 1–400; two seeds at scale 2 inside zeros 1–800);

- odd/even splits: train on odd-indexed zeros and score on even-indexed ones, or the reverse.

Results (paper Table III). "The random 80/20 splits are the principal evidence: every held-out error is above the null, by a factor 1.5–3.0 at scale 1 and 1.8–1.9 at scale 2, and no correlation is reliably positive." On the same held-out sets the six-prime oracle scores 0.062–0.063 with correlation +0.97. The table caption stresses that the seeds reuse the same zeros, so the ranges indicate robustness and are not standard errors. Table 29.3 reproduces paper Table III.

Table 29.3: Paper Table III (within-height splits), quoted exactly. Training and held-out zeros come from the same range (zeros 1–400 at scale 1, 1–800 at scale 2). Random splits are 80/20; ranges are over ten seeds (scale 1) and two seeds (scale 2), and for random splits the corr column gives the minimum and maximum. "odd" means trained on odd-indexed zeros and scored on even-indexed ones. These splits have 200 training levels at scale 1 and 400 at scale 2, so rows with M at or above that number are saturated.

scale, M	split	in-sample	held-out	corr	null
1, 100	odd	0.224	0.312	−0.32	0.238
1, 100	even	0.212	0.311	−0.39	0.250
1, 200	random	0.20–0.22	0.37–0.43	−0.40, −0.14	0.23–0.26
1, 200	odd	0.176	0.391	−0.26	0.238
1, 200	even	0.171	0.351	−0.31	0.250
1, 400	random	0.10–0.11	0.50–0.71	−0.11, +0.09	0.23–0.26
1, 400	odd	$< 10^{-3}$	0.351	−0.06	0.238
1, 400	even	$< 10^{-3}$	0.368	−0.05	0.250
2, 600	random	0.14–0.15	0.47–0.49	−0.07, −0.02	0.26–0.27
2, 600	odd	0.078	0.377	+0.03	0.245
2, 600	even	0.069	0.366	−0.06	0.266

Saturation. An odd/even split at scale 1 has 200 training levels. A model with $M \geq 200$ coefficients can fit 200 numbers exactly (Chapter 9), so those rows are *saturated*: the in-sample rms of the $M = 400$ odd and even rows is below 10^{-3}. The paper calls these rows "stress tests of memorization". Saturation is a statement about the count of unknowns, not a guarantee that the optimizer finds an exact fit. The $M = 200$ odd and even rows are nominally saturated ($200 \geq 200$), yet they reach in-sample rms only 0.176 and 0.171. A likely reason (ours, not the paper's) comes from the coverage law, Eq. (29.7). Two hundred cosines satisfy it only for $E < (200\pi/2L)^2 \approx 49$, while the training levels reach heights near 680. The sensitivity matrix is therefore effectively rank-deficient, and the count of unknowns overstates what the basis can address independently. The damped, accept-only-improvements optimizer may also stop early. The informative odd/even rows are those with $M = 100$, fewer coefficients than levels: held-out errors 0.312 and 0.311 against nulls 0.238 and 0.250, still above the null.

Worse within height. The paper notes: "The failure is in fact worse within height than in forward extrapolation." Compare the random-split held-out errors at scale 1, $M = 400$ (0.50–0.71) with the forward 1–400 row of Table I (0.369). Section 29.8 gives the mechanism: the fit does damage in the parts of the potential probed only by the held-out levels.

Why the correlations are negative. The odd/even rows at $M \leq 200$ have correlations -0.26 to -0.39. The paper explains: "The negative correlations at $M \leq 200$ are mechanical: the fluctuation sequence has negative autocorrelation at lags 1–3, so any smooth interpolation between neighbouring training levels is anticorrelated with the truth (the mean of the two nearest training fluctuations gives -0.26 to -0.41)." The next sample problem makes this mechanism quantitative.

Sample Problem 29.5: Mechanical anticorrelation of neighbor averages

Treat the fluctuations d_n as a stationary sequence with mean zero, variance σ^2 and autocorrelations ρ_1, ρ_2 at lags 1 and 2. Stationary means that the variance σ^2 and the correlation ρ_ℓ between values ℓ steps apart do not depend on n, so $\mathrm{Cov}(d_n, d_{n+\ell}) = \rho_\ell \sigma^2$. We also use $\mathrm{Var}(X+Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y + 2\,\mathrm{Cov}(X,Y)$ (Chapter 19). (a) Show that the prediction $q_n = \frac{1}{2}(d_{n-1} + d_{n+1})$ has correlation $\rho_1/\sqrt{(1+\rho_2)/2}$ with d_n. (b) Estimate ρ_1 from the GUE spacing variance and the null $\sigma = 0.263$. (c) Evaluate the correlation of (a) for the illustrative values $\rho_1 = -0.29$, $\rho_2 = -0.10$.

Step 1: Covariance of prediction and truth:

$$\mathrm{Cov}(q_n, d_n) = \tfrac{1}{2}\left[\mathrm{Cov}(d_{n-1}, d_n) + \mathrm{Cov}(d_{n+1}, d_n)\right] = \tfrac{1}{2}[\rho_1\sigma^2 + \rho_1\sigma^2] = \rho_1\sigma^2.$$

Step 2: Variance of the prediction:

$$\mathrm{Var}(q_n) = \tfrac{1}{4}\left[\mathrm{Var}(d_{n-1}) + \mathrm{Var}(d_{n+1}) + 2\,\mathrm{Cov}(d_{n-1}, d_{n+1})\right] = \tfrac{1}{4}[2\sigma^2 + 2\rho_2\sigma^2] = \tfrac{1+\rho_2}{2}\sigma^2,$$

since d_{n-1} and d_{n+1} are two steps apart.

Step 3: Correlation is covariance over the product of standard deviations:

$$\mathrm{corr}(q, d) = \frac{\rho_1\sigma^2}{\sigma\sqrt{(1+\rho_2)/2}\,\sigma} = \frac{\rho_1}{\sqrt{(1+\rho_2)/2}}. \tag{29.10}$$

Its sign is the sign of ρ_1: if neighbors are negatively correlated, averaging them predicts the wrong direction.

Step 4: For (b), consider the difference of consecutive fluctuations. Since $s_{n+1} - s_n \approx \mathrm{gap}_n$ and gap varies slowly,

$$d_{n+1} - d_n \approx \frac{t_{n+1} - t_n}{\mathrm{gap}_n} - \frac{s_{n+1} - s_n}{\mathrm{gap}_n} \approx u_n - 1,$$

where u_n is the unfolded spacing, of mean 1 (Chapter 20).

Step 5: Take variances. On the left, by stationarity, $\mathrm{Var}(d_{n+1} - d_n) = \sigma^2 + \sigma^2 - 2\rho_1\sigma^2 = 2\sigma^2(1-\rho_1)$. On the right, $\mathrm{Var}(u_n - 1) = \mathrm{Var}(u)$. Solving,

$$\rho_1 \approx 1 - \frac{\mathrm{Var}(u)}{2\sigma^2}. \tag{29.11}$$

Step 6: For the GUE Wigner surmise $p(u) = \frac{32}{\pi^2}u^2 e^{-4u^2/\pi}$, the second moment is $\langle u^2\rangle = \frac{32}{\pi^2}\int_0^\infty u^4 e^{-4u^2/\pi}\mathrm{d}u$. (The surmise is normalized to unit total probability and unit mean, Chapter 20.) The needed Gaussian moment follows from $I(\alpha) = \int_0^\infty e^{-\alpha u^2}\mathrm{d}u = \frac{1}{2}\sqrt{\pi}\,\alpha^{-1/2}$ by differentiating twice with respect to α. Each derivative brings down a factor $-u^2$ under the integral, so

$$\int_0^\infty u^4 e^{-\alpha u^2}\mathrm{d}u = \frac{\mathrm{d}^2 I}{\mathrm{d}\alpha^2} = \tfrac{1}{2}\sqrt{\pi}\cdot\left(-\tfrac{1}{2}\right)\left(-\tfrac{3}{2}\right)\alpha^{-5/2} = \frac{3}{8}\sqrt{\pi}\,\alpha^{-5/2}.$$

With $\alpha = 4/\pi$, we get $\alpha^{-5/2} = \pi^{5/2}/32$ and

$$\langle u^2\rangle = \frac{32}{\pi^2}\cdot\frac{3}{8}\sqrt{\pi}\cdot\frac{\pi^{5/2}}{32} = \frac{3\pi^3}{8\pi^2} = \frac{3\pi}{8} = 1.1781.$$

Since the mean of u is 1, $\mathrm{Var}(u) = \langle u^2\rangle - 1^2 = 1.1781 - 1 = 0.1781$.

Step 7: With $\sigma = 0.263$, $2\sigma^2 = 0.13834$, and Eq. (29.11) gives $\rho_1 \approx 1 - 0.1781/0.13834 = 1 - 1.287 = -0.287$.

Step 8: For (c), $\sqrt{(1-0.10)/2} = \sqrt{0.45} = 0.6708$, and $-0.29/0.6708 = -0.432$.

$$\boxed{\mathrm{corr}(q,d) = \frac{\rho_1}{\sqrt{(1+\rho_2)/2}}, \qquad \rho_1 \approx -0.29, \qquad \text{corr} \approx -0.43 \text{ for } \rho_2 = -0.10}$$

Status. Steps 4–8 are a heuristic estimate resting on five approximations:

- the Wigner surmise replaces the exact spacing distribution of the zeros;
- d_n is treated as stationary, although its variance grows slowly with n;
- gap is treated as constant between neighbors;
- σ is borrowed from the holdout 401–500 (with the in-range value 0.2438 for zeros 1–400, the same formula gives $\rho_1 \approx -0.50$, so the magnitude is sensitive to this choice);
- $\rho_2 = -0.10$ is chosen by hand.

The robust point is the sign: the spacing variance 0.178 exceeds $2\sigma^2$ for either choice of σ, so the lag-1 autocorrelation must be negative. The illustrative -0.43 is of the same order as the paper's measured -0.26 to -0.41. This is not a test, because the paper's values are measured on held-out subsets with training neighbors. Problem 15 asks you to measure the actual autocorrelations; the paper states only that they are negative at lags 1–3.

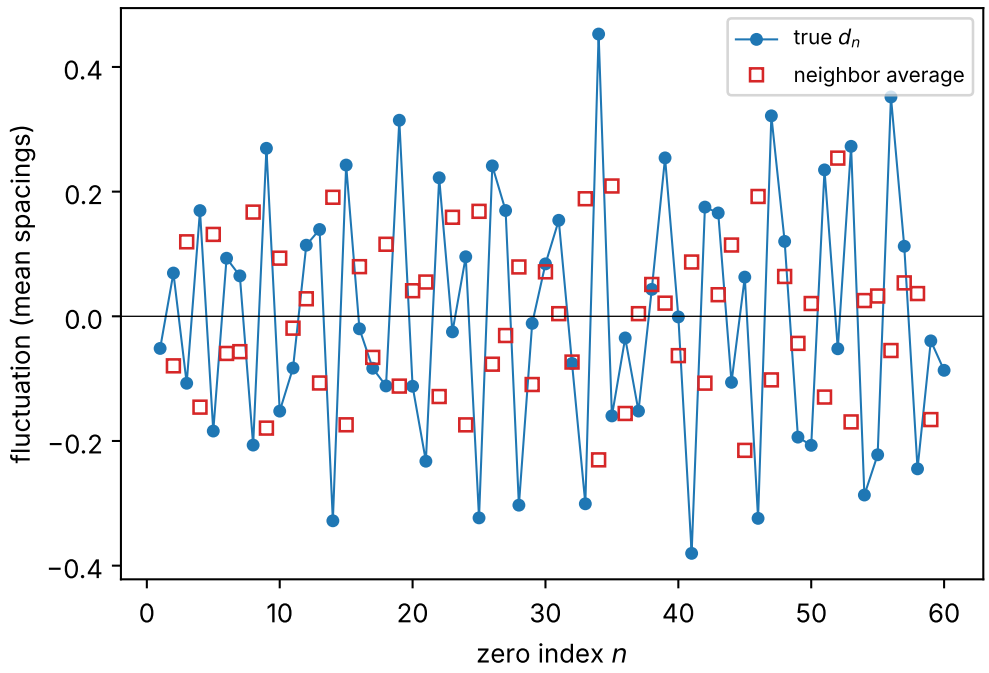


Figure 29.3: The first 60 fluctuations $d_n = (t_n - s_n)/\mathrm{gap}_n$ (filled circles, in mean spacings, computed from the true zeros) and the neighbor-average prediction $\frac{1}{2}(d_{n-1} + d_{n+1})$ (open squares). Where a zero sits high, its neighbors tend to sit low, so their average tends to point the wrong way. This is the "mechanical" anticorrelation of paper Sec. III.

The structure is usable, by the right predictor. A negative ρ_1 is information, and a predictor that learns its sign can use it. Recall *simple kriging* from Chapter 25: the best linear predictor of a held-out value from training values, built from the estimated autocovariance. For two neighbors, the best weight on their average has the sign of ρ_1, so kriging turns anticorrelation into positive correlation (Problem 13). Two neighbors alone give a correlation of at most $|\rho_1|/\sqrt{(1+\rho_2)/2}$, about 0.43 for the illustrative values above. The paper's kriging uses all training values, with an autocovariance estimated to lag 40, so its extra skill must come from that larger set of predictors and not from the sign flip alone. The paper reports that within one height range, kriging from the training fluctuations predicts held-out true zeros at 0.117 with correlation +0.87 (Sec. III, surrogate paragraph). Its construction is recorded honestly in Appendix A step 19: autocovariance to lag 40, a $\cos^2$ taper, a positive-semidefinite projection and a nugget 0.05, with the projection and nugget "added once, after a first version without them was numerically unstable, and not tuned". The fitted potential, held out in the same way, does not achieve this.

Checkpoint 29.5

Why are the odd/even rows with $M = 400$ at scale 1 not counted as evidence against the operator family, even though their held-out errors (0.351, 0.368) are well above the nulls?

Answer: With 200 training levels and 400 coefficients the fit is saturated: the system is underdetermined and has infinitely many exact solutions, so the held-out behavior is set by which solution the optimizer happens to pick. A saturated fit can match its training levels whatever they are, so its held-out behavior tests the optimizer more than the family. The paper uses these rows as stress tests; the evidence comes from the random 80/20 splits and the $M = 100$ odd/even rows.

Key Ideas

- Within-height random 80/20 splits remove covariate shift; held-out errors remain 1.5–3.0 (scale 1) and 1.8–1.9 (scale 2) times the null, while the oracle scores 0.062–0.063 with correlation +0.97.
- Odd/even rows with M at or above the number of training levels are saturated; the $M = 100$ rows still lose to the null.
- Consecutive fluctuations are negatively correlated, so smooth interpolation is anti-correlated with truth, Eq. (29.10); kriging, which learns the sign, reaches 0.117 / +0.87.

29.7 Permutation and surrogate tests

★ Did you know? *The Lady Tasting Tea*

In his 1935 book *The Design of Experiments*, Ronald Fisher imagined testing a lady who claimed she could tell whether milk or tea went into the cup first. With eight cups, four of each kind, there are $\binom{8}{4} = 70$ ways to choose which four had milk first. Guessing all correctly by chance has probability 1/70. Counting rearrangements this way is the heart of a permutation test.

If a model's in-sample accuracy reflects the special structure of the zeros, then destroying that structure should make the fit worse. If the accuracy reflects only capacity, the fit should be just as good on any sequence of the same size. Surrogate data turn this idea into a test (Chapter 26).

The three surrogates. The paper takes the training fluctuations d_n of zeros 1–400 (in mean spacings, Appendix A step 18) and replaces them by:

1. a **random permutation**: the same values in shuffled order. It keeps the marginal distribution (the histogram of values) and destroys all serial structure;
2. a **phase-randomized surrogate** [16]: Fourier transform, replace each phase by a uniform random phase, transform back. It keeps the power spectrum exactly, hence (by the Wiener–Khinchin connection of Chapter 6) the circular autocorrelation, and has no arithmetic content;
3. an **amplitude-adjusted (IAAFT) surrogate** [17]: an iteration that keeps the marginal exactly and the spectrum to about 1%.

New targets are formed as $E_n[\text{smooth}] + d'_n \text{gap}_n$, with d'_n the surrogate, and the scale-1, $M = 400$ protocol is refit.

```
import numpy as np, qmnt
n = np.arange(1, 401)
d = qmnt.fluct(qmnt.zeros(400), n)        # true fluctuations, mean spacings
rng = np.random.default_rng(1)

perm = rng.permutation(d)                 # surrogate 1: same values, shuffled

D = np.fft.rfft(d)                        # surrogate 2: phase randomization
ph = rng.uniform(0, 2*np.pi, len(D))
ph[0] = 0.0; ph[-1] = 0.0                 # keep mean and Nyquist terms as they are
d_phase = np.fft.irfft(D*np.exp(1j*ph), n=len(d))

for name, a in [("true", d), ("perm", perm), ("phase", d_phase)]:
    print(name, qmnt.rms(a))              # all three rms values agree
```

The listing shows the property that makes these surrogates fair: all three sequences have the same rms (the permutation trivially; the phase surrogate by Parseval's theorem of Chapter 5, since the moduli $|D_k|$ are unchanged). The fit faces targets of identical amplitude; only the structure differs.

Results. "Over ten draws each, the in-sample rms is 0.134 ± 0.010, 0.141 ± 0.006, and 0.138 ± 0.005, against 0.137 for the true zeros and 0.244 for the null."

Sample Problem 29.6: Is the true-zero fit special?

Compare the in-sample rms on the true zeros, 0.137, with each surrogate ensemble. Read "±" as the spread over the ten draws (the paper does not say whether it is a standard deviation or a standard error; Problem 23 shows the conclusion does not depend on this). Also compute the fraction of the null removed in-sample for the true zeros and for the permutation.

Step 1: Permutation: $|0.137 - 0.134|/0.010 = 0.30$ spreads.
Step 2: Phase-randomized: $|0.137 - 0.141|/0.006 = 0.67$ spreads.
Step 3: IAAFT: $|0.137 - 0.138|/0.005 = 0.20$ spreads.
Step 4: Fraction of the null removed, true zeros: $(0.244 - 0.137)/0.244 = 0.107/0.244 = 0.439$.
Step 5: Permutation: $(0.244 - 0.134)/0.244 = 0.110/0.244 = 0.451$.

true zeros within 0.2–0.7 spreads of every surrogate; fit removes 44% (true) vs 45% (shuffled)

The fit removes as large a fraction of a shuffled sequence as of the real one. Its in-sample accuracy therefore cannot be attributed to the structure of the zeros. In the paper's words: "in-sample accuracy measures capacity, not captured structure."

A revision note on units. The paper is open about a change: "The original submission permuted fluctuations in energy units, which also moves amplitude across heights and gave 0.148, 0.149; the spacing-unit convention used here is the cleaner control." The point is that a fluctuation of one energy unit is a different number of spacings at different heights, because gap_n shrinks with n. Shuffling energy-unit values therefore mixes amplitude between low and high levels. Shuffling spacing-unit values preserves amplitude everywhere. The older values (Appendix A anchors 0.1483 and 0.1493 for seeds 1 and 2) are only slightly worse than the true-zero fit, so the conclusion survives either convention.

The held-out variant. The paper adds a held-out version (one split seed, five draws): kriging predicts held-out true zeros at 0.117 / +0.87 and permuted targets at the null, as a structure-learning method should, "whereas the fitted operator, held out in the same way, fails alike on true zeros and on surrogates (0.57–0.70; one split, illustrative)". Note the hedge "one split, illustrative"; this is supporting evidence, not a systematic study.

Checkpoint 29.6

Kriging on permuted targets scores at the null. Why is that the *desired* behavior for a method that learns structure?

Answer: A permutation destroys serial correlation, so the training values carry no information about a held-out value at a given position. A method that predicts by exploiting serial correlation should find nothing to exploit and fall back to the null. Failing on permuted data while succeeding on true data is the signature of learning real structure.

Key Ideas

- The fit reaches the same in-sample rms on shuffled, phase-randomized and IAAFT surrogates (0.134, 0.141, 0.138) as on the true zeros (0.137).
- Surrogates are amplitude-matched, so the comparison isolates structure from capacity.
- Kriging distinguishes true from permuted targets; the operator does not.

29.8 The interpolation split: damage at the turning points

The experiment. Train on zeros 1–200 together with 301–500, and predict the interior gap 201–300 (Appendix A step 14). The held-out levels are surrounded by training levels on both sides, the most favorable setting for any model that has captured smooth structure.

Results. In-sample rms 0.136, but a gap error of 2.64 spacings, "ten times the null 0.255", with correlation −0.02. Two fits from random starts give gap errors 2.78 and 2.43 with the largest correction at the same position. Ridge refits bring the gap error back to the null: 0.262 at $\mu = 0.03$ and 0.257 at $\mu = 0.3$, with correlations −0.02 and −0.06 (Appendix A step 21).

The ratio is $2.64/0.255 = 10.4$: the unregularized fit does not merely fail to help; it actively harms the held-out levels.

The mechanism: position maps to level index. Recall from Chapter 12 that a semiclassical level of energy E is confined between turning points $\pm x_E$ where $V(x_E) = E$, and that it is most sensitive to the potential near those turning points (where the particle moves slowest and $|\psi|^2$ is largest). For the Wu–Sprung potential the turning point of level n is $x(t_n)$, with $x(V)$ from paper Eq. (2):

$$x(V) = \frac{1}{\pi}\left[\sqrt{V - V_0}\,\log\frac{V_0}{2\pi e^2} + \sqrt{V}\,\log\frac{\sqrt{V} + \sqrt{V - V_0}}{\sqrt{V} - \sqrt{V - V_0}}\right], \qquad V_0 = 3.10073\pi. \tag{29.12}$$

Sample Problem 29.7: Turning points on the Wu–Sprung potential

(a) Evaluate Eq. (29.12) at $V = 1000$ and confirm the scale-1 half-width $L = 44.883$. (b) Evaluate it at $V = 500$, the height of the smooth potential where the paper's largest $|\delta V|$ occurs. (c) Compare with the turning band 22.5–28.5 the paper gives for levels 201–300.
Step 1: Constants: $V_0 = 3.10073 \times 3.14159 = 9.74123$. Also $2\pi e^2 = 6.28319 \times 7.38906 = 46.4268$, so $V_0/(2\pi e^2) = 0.209819$ and $\log 0.209819 = -1.56151$.
Step 2: A useful identity for the second logarithm: multiply numerator and denominator by $\sqrt{V} + \sqrt{V - V_0}$ and use $(\sqrt{V})^2 - (\sqrt{V - V_0})^2 = V_0$:

$$\frac{\sqrt{V} + \sqrt{V - V_0}}{\sqrt{V} - \sqrt{V - V_0}} = \frac{\left(\sqrt{V} + \sqrt{V - V_0}\right)^2}{V_0}.$$

This avoids subtracting two nearly equal square roots.
Step 3: $V = 1000$: $\sqrt{990.259} = 31.4684$, $\sqrt{1000} = 31.6228$, sum 63.0912, squared 3980.50, divided by V_0: 408.62, logarithm 6.01278.
Step 4: Bracket: $31.4684 \times (-1.56151) + 31.6228 \times 6.01278 = -49.138 + 190.141 = 141.003$. Divide by π: $x(1000) = 44.883$, the paper's L.
Step 5: $V = 500$: $\sqrt{490.259} = 22.1418$, $\sqrt{500} = 22.3607$, sum 44.5025, squared 1980.47, divided by V_0: 203.31, logarithm 5.31473.
Step 6: Bracket: $22.1418 \times (-1.56151) + 22.3607 \times 5.31473 = -34.575 + 118.841 = 84.266$. Divide by π: $x(500) = 26.82$.

$$\boxed{x(1000) = 44.883, \qquad x(500) = 26.82 \in [22.5,\ 28.5]}$$

The position of the largest fitted correction, where $V \approx 500$, lies inside the band of turning points of the held-out levels 201–300. (Problem 11 repeats this for t_{100}. The `qmnt` listing in Section 29.11 evaluates $x(t_{201}) \approx 22.5$ and $x(t_{300}) \approx 28.5$ from the actual zeros.)

What the fit did. Paper Fig. 4 shows the single deterministic fit from $c = 0$: (a) fitted and smooth potentials, (b) the correction δV, (c) true and predicted fluctuations. The paper's wording is: "rms $\delta V = 40$ in that band against 15 inside it and 13 outside it, with a maximum $|\delta V| = 199$ where the smooth potential is about 500"; the predicted gap levels are displaced by -6 to $+2$ spacings. The phrase "inside it" is ambiguous as written, since the band itself is where the rms is 40. The most natural reading, which we adopt only tentatively, is that 15 refers to the region on the inner side of the band ($x < 22.5$) and 13 to the region outside it ($x > 28.5$); the paper does not define the regions further. On any reading, the correction in the band is about $40/15 = 2.7$ and $40/13 = 3.1$ times its rms in the two comparison regions, and its peak is about $199/500 = 40\%$ of the local potential.

Why it costs nothing in-sample. The paper states the conclusion but not a detailed mechanism. What follows in this paragraph, up to the quotation, is our own heuristic reading,

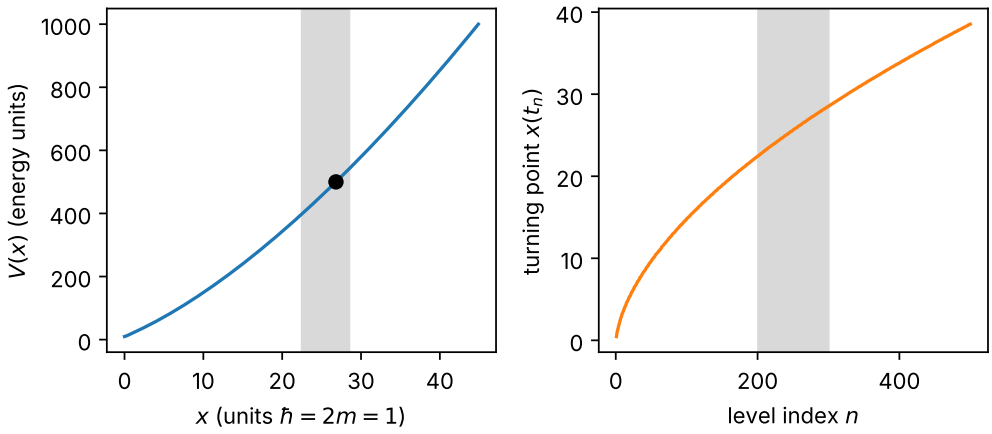


Figure 29.4: Left: the Wu–Sprung potential $V(x)$ for $0 \le x \le L = 44.883$ (units $\hbar = 2m = 1$); the shaded band $22.5 \le x \le 28.5$ contains the turning points of levels 201–300, the held-out gap of the interpolation split; the dot marks $V = 500$ at $x = 26.82$. Right: turning point $x(t_n)$ against level index n for the first 500 zeros, with the held-out gap shaded. Each level "owns" a position on the potential.

not the paper's, and we have not tested it. Training levels with turning points below 22.5 are confined inside the band's inner edge and feel the band only through the exponentially small tails of their wavefunctions in the forbidden region. Training levels 301–500 turn beyond 28.5, so the band lies inside their classically allowed region, where $|\psi|^2$ oscillates rapidly. Their shifts $\int |\psi_n|^2 \delta V \, \mathrm{d}x$ still feel a correction there; it would leave them unchanged only if its contributions largely averaged out against the oscillating densities, or were offset by structure elsewhere that the optimizer adjusts at the same time. Whether the fitted δV is of that kind is not reported in the paper (its Fig. 4 shows only the shape), so this part of the argument is speculation. The optimizer, which sees only in-sample residuals, has no reason to keep δV small there. In the paper's words: "The optimizer places large structure in the part of the potential that only the held-out levels probe strongly, because doing so costs nothing in-sample." This explanation is heuristic, as is the paper's.

The paper's hedge. The paragraph ends: "The unregularized fit is harmful between its training blocks and the regularized fits tried are uninformative there; this does not show that the family could not interpolate under another fitting procedure." Keep it. The experiment tests the family *together with* its optimizer.

The forward split, viewed the same way. The same geometry applies to the forward splits of Table I, as our own reading (the paper states the mechanism for the interpolation split only). Using Eq. (29.12), $x(680) = 33.82$ and $x(811.18) = 38.53$: the training levels 1–400 turn inside $|x| \lesssim 33.8$, while the held-out levels 401–500 turn in $33.8 \lesssim |x| \lesssim 38.5$, a region the training levels reach only through forbidden-region tails. The fit's behavior there is again unconstrained by the data.

Checkpoint 29.7

Ridge regularization brings the gap error from 2.64 back to about 0.26. Does the ridge-regularized fit therefore predict the gap?

Answer: No. 0.262 and 0.257 are at the null 0.255 (slightly above it), and the correlations are −0.02 and

−0.06. Ridge removed the damage by suppressing the large correction, leaving nearly the unperturbed potential there, which is the null. "Removes the damage and nothing else," as the paper says.

Key Ideas

- The interpolation split (train 1–200 and 301–500, predict 201–300) gives gap error 2.64, ten times the null 0.255: the fit does harm.
- Level n probes the potential most strongly near its turning point $x(t_n)$; levels 201–300 turn at $x = 22.5$–28.5, where the fitted δV is largest.
- Ridge returns the gap to the null; the paper does not claim that no other procedure could interpolate.

29.9 The arithmetic-continuation probe

Every experiment so far uses the real zeros, whose fluctuation is a complicated superposition of many prime frequencies plus whatever the truncation misses. A skeptic could say: perhaps the true fluctuation is simply too complicated for a few hundred coefficients to extrapolate. The continuation probe removes that objection. It is the cleanest single experiment in the paper.

The target. Replace the true targets by a synthetic fluctuation made solely of the $p = 2$ term of paper Eq. (4), our running example RE2:

$$N_{\mathrm{fl}}^{p=2}(t) = -\frac{1}{\pi}\sum_{m=1}^{3}\frac{\sin(mt\log 2)}{m\,2^{m/2}}. \tag{29.13}$$

Appendix A step 15 defines the synthetic targets $T_n = E_n[\text{smooth}] - N_{\mathrm{fl}}^{p=2}(s_n)\,\mathrm{gap}_n$. (The letter T_n is a target here, not a Chebyshev polynomial or a tensor.) The minus sign makes the target fluctuation equal $-N_{\mathrm{fl}}^{p=2}(s_n)$ spacings, the same sign convention as real zeros: solving $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ moves the root by $-N_{\mathrm{fl}}$ divided by the slope $N'_{\mathrm{sm}} = 1/\mathrm{gap}$. Fit the same family on levels 1–400 and ask whether the fitted operator *continues* the oscillation on levels 401–500.

Why is this a fair test? Because the right answer is known exactly: the oscillation does not stop at level 400, and a function with three known frequencies $\log 2$, $2\log 2$, $3\log 2$ is as simple a target as arithmetic offers.

Sample Problem 29.8: How big is the prime-2 wave?

Find the long-run rms of RE2, and compare with the probe-null 0.1708 of Appendix A (the rms of predicting zero fluctuation on levels 401–500 for the synthetic target).
Step 1: The amplitudes are $a_m = 1/(m\,2^{m/2})$: $a_1 = 1/\sqrt{2} = 0.707107$, $a_2 = 1/4 = 0.25$, $a_3 = 1/(3\times 2\sqrt{2}) = 0.117851$.
Step 2: The mean square over a window of length T_w in t involves averages of $\sin^2(\omega t)$ and of cross products $\sin(\omega t)\sin(\omega' t)$ with $\omega \neq \omega'$. Using $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ we write

$$\sin(\omega t)\sin(\omega' t) = \tfrac{1}{2}\left[\cos((\omega-\omega')t) - \cos((\omega+\omega')t)\right].$$

For any $\kappa \neq 0$, an antiderivative of $\cos(\kappa t)$ is $\sin(\kappa t)/\kappa$, which vanishes at $t = 0$, so $\int_0^{T_w}\cos(\kappa t)\,\mathrm{d}t = \sin(\kappa T_w)/\kappa$. Applying this with $\kappa = \omega - \omega'$ and with $\kappa = \omega + \omega'$, and

dividing by T_w,

$$\frac{1}{T_w}\int_0^{T_w}\sin(\omega t)\sin(\omega' t)\,\mathrm{d}t = \frac{1}{2T_w}\left[\frac{\sin((\omega-\omega')T_w)}{\omega-\omega'} - \frac{\sin((\omega+\omega')T_w)}{\omega+\omega'}\right],$$

which, because $|\sin| \le 1$, is at most $\frac{1}{2T_w}\left(\frac{1}{|\omega-\omega'|}+\frac{1}{\omega+\omega'}\right)$ in magnitude and tends to 0 as $T_w \to \infty$. With $\omega' = \omega$ the identity reads $\sin^2(\omega t) = \frac{1}{2}[1-\cos(2\omega t)]$; using the cosine integral with $\kappa = 2\omega$, its average over the window is $\frac{1}{2} - \frac{\sin(2\omega T_w)}{4\omega T_w} \to \frac{1}{2}$.

Step 3: So in the long run each term contributes $a_m^2/2$ and the cross terms vanish:

$$\mathrm{rms}\left(N_{\mathrm{fl}}^{p=2}\right) \to \frac{1}{\pi}\sqrt{\frac{1}{2}\sum_{m=1}^{3} a_m^2}. \tag{29.14}$$

Step 4: $\sum a_m^2 = 0.5 + 0.0625 + 0.0138889 = 0.5763889$; half is 0.2881944; square root 0.5368374; divide by π: 0.17088.

rms(RE2) ≈ 0.1709 vs. probe-null 0.1708

Status. Eq. (29.14) is the long-window limit for continuous t. The probe samples RE2 at the 100 irregular points $s_{401}, \ldots, s_{500}$, which span about 14.5 periods (next sample problem), so it is an approximation; the agreement to 10^{-4} shows it is a good one here.

Sample Problem 29.9: How many levels per period?

(a) Show that the number of consecutive levels per period of $\sin(t\log 2)$ is $\log(t/2\pi)/\log 2$. Evaluate at $t = 680$ (near zero 400) and $t = 811.18$ (zero 500). (b) Estimate how many periods lie in the training window 1–400 and in the holdout 401–500, using $t_1 = 14.13$, $t_{400} \approx 680$ and $t_{500} = 811.18$.

Step 1: The period in t is $2\pi/\log 2 = 6.28319/0.693147 = 9.0647$.

Step 2: The spacing at height t is gap $= 2\pi/\log(t/2\pi)$, Eq. (29.2). Levels per period:

$$\frac{2\pi/\log 2}{2\pi/\log(t/2\pi)} = \frac{\log(t/2\pi)}{\log 2} = \log_2\frac{t}{2\pi}.$$

Step 3: At $t = 680$: $\log(680/6.28319) = \log 108.225 = 4.68421$; divided by 0.693147: 6.758.

Step 4: At $t = 811.18$: $\log 129.104 = 4.86062$; divided by 0.693147: 7.012.

Step 5: Periods in training: $(680-14.13)/9.0647 = 665.87/9.0647 = 73.5$. In the holdout: $(811.18-680)/9.0647 = 131.2/9.0647 = 14.5$.

about 6.8–7.0 levels per period; about 73.5 periods in training, 14.5 in the holdout

The training window contains more than seventy full periods of the oscillation. Lack of examples is not the problem.

Results. "It represents it almost perfectly in-sample (rms 0.056, correlation +0.95) and continues it not at all (held-out correlation +0.07, rms worse than predicting zero fluctuation)." Appendix A gives the anchors: in-sample 0.0555, correlation +0.95; OOS 0.1958, correlation +0.07; probe-null 0.1708. The in-sample rms is about a third of the wave's own rms

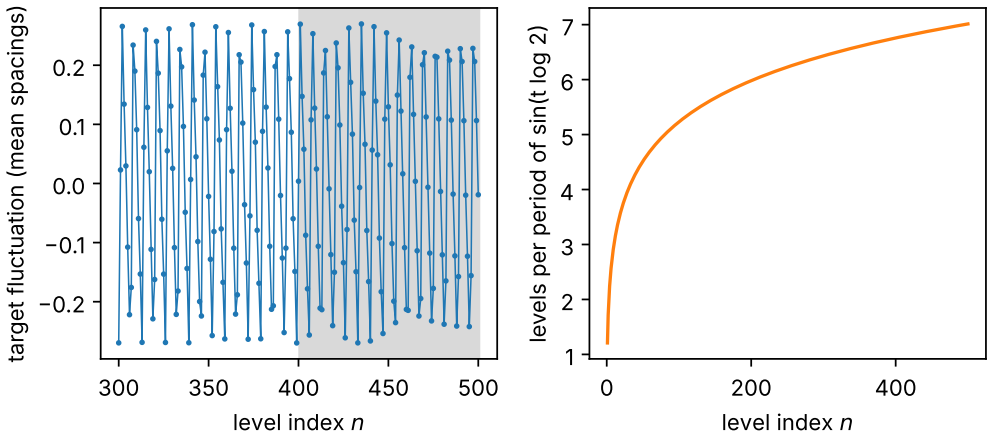


Figure 29.5: The continuation-probe target. Left: the synthetic fluctuation $-N_{\mathrm{fl}}^{p=2}(s_n)$ of Eq. (29.13) in mean spacings, levels 300–500; the shaded region 401–500 is held out. The correct continuation is the same oscillation. Right: levels per period of $\sin(t\log 2)$, $\log_2(t/2\pi)$ evaluated at $t = s_n$, for levels 1–500.

$(0.0555/0.1709 = 0.32)$, while the OOS rms exceeds the probe-null by $0.1958/0.1708 = 1.146$, about 15%.

The contrast: a model that knows the frequency. The following is our own illustration, not an experiment in the paper. Consider instead the six-function model

$$F(t) = \sum_{m=1}^{3} \left[\beta_m \sin(mt\log 2) + \gamma_m \cos(mt\log 2)\right],$$

fitted by least squares to the same 400 target values (Chapter 27, control B). Evaluate the six functions at the 400 sample points $s_1, \ldots, s_{400}$ and collect them into the columns of a 400×6 matrix F. Collect the six coefficients into the vector $\mathbf{b} = (\beta_1, \beta_2, \beta_3, \gamma_1, \gamma_2, \gamma_3)^\top$, and write the target vector as $\mathbf{g}$, with entries $-N_{\mathrm{fl}}^{p=2}(s_n)$. The target is *exactly* in the span: $\mathbf{g} = \mathsf{F}\mathbf{b}_{\mathrm{true}}$ with $\beta_m = +1/(\pi m 2^{m/2})$ and $\gamma_m = 0$. (The minus sign in Eq. (29.13) cancels the minus sign of the target.) We need the six columns to be linearly independent, so that $\mathsf{F}^\top\mathsf{F}$ is invertible. We do not prove this; we checked it numerically. At these 400 points the 6×6 matrix $\mathsf{F}^\top\mathsf{F}$ has condition number below 10^3, so it is safely invertible. The least-squares solution (Chapter 9) is then

$$\hat{\mathbf{b}} = (\mathsf{F}^\top\mathsf{F})^{-1}\mathsf{F}^\top\mathbf{g} = (\mathsf{F}^\top\mathsf{F})^{-1}(\mathsf{F}^\top\mathsf{F})\mathbf{b}_{\mathrm{true}} = \mathbf{b}_{\mathrm{true}}.$$

The model recovers the exact coefficients, and its prediction at *any* height, in or out of the training window, is the exact wave. Six parameters with the right functional form continue perfectly; a few hundred parameters in a potential do not continue at all.

The paper's interpretation, and its hedges. Section III puts it strongly: "The family can paint one window of $\sin(t\log 2)$ onto its levels but cannot express 'continue oscillating': the arithmetic frequency is not in its vocabulary. This is the memorization mechanism exhibited in isolation, on a target where the right continuation is known exactly." And: "the failure on the real zeros is not explained by the complicated mixture of primes in the true fluctuation: a single known arithmetic frequency, learned from a finite window, already defeats the family's extrapolation."

The Discussion (Sec. VI) states the scope more carefully: "Under the fitting protocol used, the parametrizations tested did not recover the long-range continuation of a frequency such

as $\log 2$, as the continuation probe shows on a target where the answer is known. This is evidence about the tested parametrizations and optimizer together, not a statement about what a potential can represent." A careful reader keeps the second version as the claim. "Not in its vocabulary" is an interpretation of an experiment with one background, one basis family at a time and one optimizer; it is not a theorem about potentials. The Discussion also says plainly that "Whether the memorization result can be derived rather than measured is an open problem."

Why might a potential not "speak" in sinusoids of t? Heuristically (the paper's own word): a potential addresses levels through their turning points, locally in energy (Section 29.8), whereas the target is an almost-periodic function of t, sparse in frequency (Chapter 5). The correction a fit places at the turning points of levels 1–400 says nothing, by itself, about the region $33.8 \lesssim x \lesssim 38.5$ where levels 401–500 turn.

A small-scale rerun: one linearized step. The full fit takes minutes per configuration (`qmnt.fit_potential`). You can explore in seconds whether a linear toy shows the same behavior: compute the eigenvectors ψ_n of the Wu–Sprung Hamiltonian once, form $A_{nk} = \int |\psi_n|^2 \cos(k\pi x/L)\mathrm{d}x$ for $n \le 500$ and $k < 400$, and take a single SVD-damped least-squares step from $c = 0$ toward the probe target on levels 1–400. This is the first Gauss–Newton iteration of the paper's fit (Chapter 24) and nothing more: no iteration, no bias correction, a coarse grid. Figure 29.6 shows its output. We do not quote scores for the toy, and you should not expect it to reproduce the paper's numbers. A single linear step without iteration or bias correction is a different procedure from the paper's iterated nonlinear fit, and it may behave quite differently. Problem 18 asks you to compute its scores and judge for yourself whether they resemble the paper's held-out correlation +0.07. Here is the complete toy; the script for Figure 29.6 is the same code plus plotting.

```
import numpy as np, qmnt
L, N, M = 44.883, 4000, 400
x = qmnt.grid(L, N); h = 2*L/(N+1)
V = np.array([qmnt.wu_sprung_V(abs(xi)) for xi in x])
E, psi = qmnt.fd_levels(V, L, k=500, return_vectors=True)
if psi.shape[0] != N: psi = psi.T            # columns = eigenvectors
dens = psi**2/(np.sum(psi**2, axis=0)*h)     # sum psi^2 h = 1
phi = np.cos(np.outer(x, np.arange(M))*np.pi/L)
A = h*dens.T @ phi                           # A[n,k] = int |psi_n|^2 phi_k dx
n = np.arange(1, 501)
s = qmnt.smooth_zeros(n); g = qmnt.mean_gap(n)
f = -qmnt.Nfl(s, [2])                        # target fluctuation, spacings
U, sig, Wt = np.linalg.svd(A[:400], full_matrices=False)
filt = sig/(sig**2 + 1e-6*sig[0]**2)         # SVD-damped filter
c = Wt.T @ (filt*(U.T @ (f[:400]*g[:400])))  # one step from c = 0
p = (A @ c)/g                                # predicted fluctuation
print('in  ', qmnt.rms(p[:400]-f[:400]), qmnt.corr(p[:400], f[:400]))
print('out ', qmnt.rms(p[400:]-f[400:]), qmnt.corr(p[400:], f[400:]))
print('null', qmnt.rms(f[400:]))
```

Checkpoint 29.8

The probe's in-sample correlation is +0.95 and its OOS rms is 0.1958. Why is "rms worse than predicting zero fluctuation" a stronger statement than "correlation +0.07"?

Answer: A correlation near zero says the predicted held-out fluctuation has no linear relation to the truth. The rms comparison adds that the prediction has nonzero amplitude, so by Eq. (29.9) it adds error: the fitted operator does worse than the trivial model that predicts nothing.

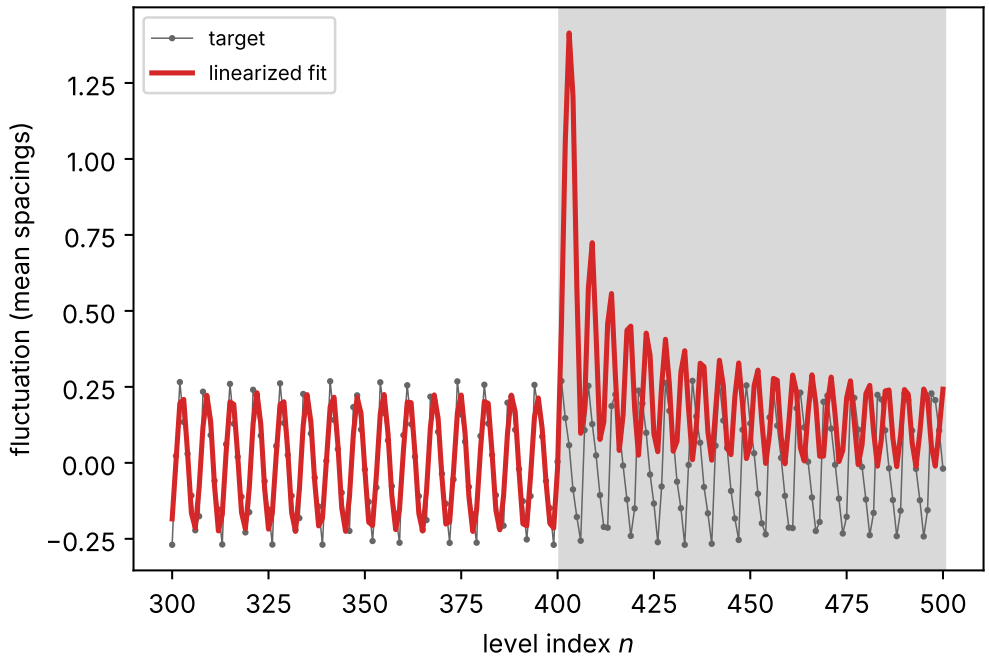


Figure 29.6: A linearized toy of the continuation probe (not the paper's computation). Thin line with dots: the target fluctuation $-N_{\mathrm{fl}}^{p=2}(s_n)$ in mean spacings. Thick line: the first-order prediction $\sum_k A_{nk}c_k/\mathrm{gap}_n$ after one SVD-damped least-squares step ($M = 400$ cosines, damping $10^{-6}\sigma_1^2$) fitted on levels 1–400 only, using eigenvectors of the Wu–Sprung Hamiltonian on 4000 grid points with $L = 44.883$. The shaded region 401–500 was not used. Compare how the prediction behaves on the two sides of the boundary; the paper's full nonlinear fit gives held-out correlation +0.07 (Sec. III).

Key Ideas

- The probe target is RE2 alone, with long-run rms 0.1709, matching the probe-null 0.1708; the training window holds about 73 of its periods.
- The fitted family reproduces it in-sample (0.0555, +0.95) and does not continue it (0.1958 vs. null 0.1708, correlation +0.07).
- A six-function model with the right frequencies recovers the wave exactly from the same data; the obstacle is the form of the model, not the amount of data.
- The careful statement (paper Sec. VI) is about the tested parametrizations and optimizer together; deriving the result is open.

29.10 What Section III establishes, and what it does not

★ Did you know? ***Evidence Can Mislead***

In 1897 Franz Mertens conjectured that $|M(x)| < \sqrt{x}$, where $M(x)$ sums the Möbius function, and computations supported it for every value checked. In 1985 Andrew Odlyzko and Herman te Riele proved it false, yet no explicit counterexample is known, because the first one lies at an astronomically large x. Numerical agreement over a huge range does not prove what happens beyond it.

We now collect the strengths and limits of the section, claim by claim. Chapter 32 returns to these from a referee's point of view.

Established within the tested scope.

- The cosine family reaches in-sample rms below 10^{-4} at $M = 900$ on 500 levels, and in-sample error falls with M in all three bases. The cosine $M = 400$ fit reaches the same in-sample accuracy on surrogate targets as on the true zeros. In-sample agreement alone is therefore not evidence of captured structure. This is the methodological core, and it is robust: it rests on direct refits, ten draws per surrogate type, and agreement across three surrogate constructions.
- Across the twelve configurations of Table I, the eight rows of Table II, and the within-height splits of Table III, no row shows reproducible held-out skill relative to the null, while a zero-parameter arithmetic formula shows strong skill on the same holdouts.
- On a target whose continuation is known exactly (RE2), the tested family under the tested protocol does not continue it.

Limits stated by the paper.

- One background potential (Wu–Sprung); three bases; Hermite, wavelet and neural-network parametrizations not tested (Secs. I and VI).
- Operator fits use at most the first 1000 zeros (Sec. VI).
- One Chebyshev row is marginal (Table II); a small positive correlation for it is not excluded.
- Bootstrap intervals resample independently except where stated; the block bootstrap was run for Table II.

- The interpolation result does not show that the family could not interpolate under another fitting procedure.
- The explanation of the failure is heuristic; whether it can be derived is open.

Things to watch as a reader.

- Several results are single-split or few-seed (the surrogate held-out variant: "one split, illustrative"; scale-2 random splits: two seeds). Ranges over seeds that reuse the same zeros are robustness indications, not standard errors (Table III caption).
- The optimizer and the family are tested together. Different stopping rules, priors or regularizers define different procedures, and the paper's conclusions are about the ones it ran.

Key Ideas

- The robust lesson of Section III is methodological: in-sample spectral agreement demonstrates capacity; held-out tests and surrogate refits are ways to show more, and they are the ones the paper uses.
- Every specific negative result is scoped to the background, bases and optimizer tested.

29.11 Paper connection

Table 29.4 maps each part of this chapter to the paper (R1 numbering) and to the chapters that supply the tools.

Table 29.4: Where each piece of paper Sec. III lives, and which chapters explain it.

This chapter	Paper location	Tools
Protocol recap	Sec. II A–C; Eqs. (1), (3); App. A steps 1–7	Chs. 17, 23, 24
Capacity, coverage law	Sec. III para. 1; Fig. 3; App. A anchors $(300, 0.186)$, $(900, < 10^{-4})$	Ch. 24
Twelve configurations	Sec. III para. 2; Table I; Fig. 1; Fig. 3	Chs. 25, 19
Bootstrap intervals	Sec. III para. 2 and "Other bases"; App. A steps 13, 22	Ch. 26
Other bases	Sec. II C "Other bases"; Sec. III "Other bases"; Table II; App. A step 16	Chs. 24, 26
Within-height splits	Sec. II C "Within-height splits"; Sec. III; Table III; App. A steps 17, 19	Chs. 25, 20
Surrogates	Sec. III "Permutation and surrogate tests"; App. A steps 12, 18	Chs. 26, 6
Interpolation split	Sec. III "Interpolation split"; Fig. 4; Eq. (2); App. A steps 14, 21	Chs. 12, 22
Continuation probe	Sec. III "Arithmetic-continuation probe"; Eq. (4); App. A step 15; Sec. VI	Chs. 18, 27

Numbers you can reproduce cheaply. With `qmnt` (Appendix A of this book teaches the Python) you can reproduce, in seconds, the null rms 0.263 on zeros 401–500 (paper Sec. II C), the six-prime oracle score 0.074 / +0.96 (paper Table V, used in Fig. 1 and throughout Sec. III), the probe-null 0.1708 (paper App. A), the turning band 22.5–28.5 (paper Sec. III), the half-widths $L = 44.883$ and 59.957 (paper Sec. II B), and the sign of the fluctuation autocorrelations (paper Sec. III). The full fits behind Tables I–III require `qmnt.fit_potential` or the Zenodo pipeline [20].

```
import numpy as np, qmnt
n = np.arange(401, 501)                          # RE5: the principal holdout
s = qmnt.smooth_zeros(n)
d = qmnt.fluct(qmnt.zeros(500)[n-1], n)
print("null rms", qmnt.rms(d))                   # paper: 0.2631

P6 = qmnt.first_primes(6)                        # 2, 3, 5, 7, 11, 13
print("oracle", qmnt.score(qmnt.oracle_zeros(n, P6), n))   # Table V: 0.074 / +0.96

print("probe-null", qmnt.rms(qmnt.Nfl(s, [2])))            # App. A: 0.1708
print("turning band", qmnt.wu_sprung_x(qmnt.zeros(300)[[200, 299]]))  # ~22.5, 28.5
```

Reproducibility record. Every number in Section III is regenerated by the scripted pipeline archived on Zenodo [20], which the author reports rerunning end to end (Appendix A). The revision computations (steps 16–23) are included in the updated record. For the sinusoidal controls of paper Sec. IV, Appendix A step 20 records that overwriting held-out data with noise before fitting leaves all fitted coefficients bit-identical. For the operator fits of Section III the no-leakage argument is different. Paper Eq. (3) shows that the bias correction carries no zero information, and the residuals are restricted to the training indices (stated explicitly for the within-height splits in step 17).

29.12 Summary

- Fluctuations are measured as $d_n = (t_n - s_n)/\mathrm{gap}_n$, Eq. (29.3); scores are the held-out rms and correlation of Eq. (29.5); the null predicts $p_n = 0$, Eq. (29.6), with rms 0.263 on zeros 401–500.
- Capacity: the cosine family reaches in-sample rms 0.186 at $M = 300$ and below 10^{-4} at $M = 900$ on 500 levels; the heuristic coverage law Eq. (29.7) predicts completeness at $M \gtrsim 814$.
- Transfer: in all twelve configurations of paper Table I, OOS rms is at or above the null and correlations scatter around zero; relative skill Eq. (29.8) is at most about zero, against +0.72 for the six-prime oracle.
- An uncorrelated prediction of relative amplitude a has rms $\mathrm{rms}_{\mathrm{null}}\sqrt{1+a^2}$; beating the null needs $\rho > a/2$, Eq. (29.9).
- Bootstrap intervals: principal scale-1 rms $[0.33, 0.43]$ (significantly worse than null), correlation $[-0.21, +0.08]$; oracle $[+0.95, +0.97]$; block bootstrap changes no conclusion.
- Chebyshev and hat bases behave like cosines; Chebyshev $M = 400$ is marginal and reported as such.
- Within-height splits fail too (1.5–3.0 times the null at scale 1); negative neighbor correlations, Eqs. (29.10)–(29.11), make smooth interpolation anticorrelated, while kriging reaches 0.117 / +0.87.

- Surrogate targets are fitted as well as the true zeros (0.134, 0.141, 0.138 vs. 0.137): in-sample accuracy measures capacity.
- The interpolation split does harm (gap error 2.64, ten times the null) by placing large δV at the turning points $x = 22.5$–28.5 of the held-out levels, Eq. (29.12).
- The continuation probe: RE2, of long-run rms 0.1709, Eq. (29.14), is fitted in-sample (0.0555, +0.95) and not continued (0.1958 vs. 0.1708, +0.07); a model with the right frequencies would continue it exactly.
- Scope: one background, three bases, one optimizer, at most 1000 zeros; the explanation is heuristic and its derivation open.

29.13 Problems

1. • In paper Table I, the row (scale 1, fit 1–400, $M = 600$) has in-sample rms 0.064 and OOS rms 0.313. Compute the ratio of OOS to in-sample error, and the ratio of OOS error to the null 0.263. What does each ratio measure?

2. • Count the rows of paper Table I at each scale, and list the distinct basis sizes M that appear. Confirm the paper's statement of "twelve configurations" and "five train/test splits".

3. • In paper Table II the hat $M = 400$ row has correlation -0.19 with interval $[-0.37, -0.01]$. Is this evidence of predictive skill? What does an interval entirely below zero indicate, and what caveat about the interval applies?

4. • In paper Table III, odd/even splits at scale 2 have 400 training levels. Which scale-2 odd/even rows are saturated? Explain what "saturated" means in terms of the least-squares problem.

5. •• Use Eq. (29.9) with zero means. (a) If the (scale 1, fit 1–400, $M = 600$) row had correlation exactly zero, what amplitude ratio $a = \sigma_p/\sigma_d$ would reproduce OOS rms 0.313 with null 0.263? (b) Repeat with the reported correlation -0.16. Compare with the paper's range 0.16–0.62.

6. •• Apply the coverage law Eq. (29.7) at scale 2 ($L = 59.957$) to all 1000 scored levels, using $t_{1000} = 1419.42248$ (paper Sec. II A). Is $M = 900$ above or below the requirement? The in-sample rms 0.103 of the scale-2 $M = 900$ row of Table I refers to the training zeros 1–800, so repeat the estimate with t_{800} (from `qmnt.zeros`). Is the in-sample rms consistent with your result?

7. •• Using Eq. (29.14), find the long-run rms of the $m = 1$ term of RE2 alone, and the fraction of RE2's mean square carried by that term.

8. •• Show that the number of levels per period of $\sin(t \log p)$ is $\log(t/2\pi)/\log p$. Evaluate it at $t = 680$ for $p = 3$ and $p = 5$. Relate your answer to the paper's statement in Sec. V that the $p = 2$ oscillation "is only ∼6–7 levels at these heights, itself a local pattern".

9. •• In the interpolation split the in-sample rms is 0.136 on 400 levels and the gap rms is 2.64 on 100 levels. Compute the rms over all 500 levels together. What does this teach about reporting a single pooled error?

10. •• For the $M = 100$ odd/even rows of paper Table III, compute the ratio of held-out error to null. For the six-prime oracle at 0.062 on the odd split (null 0.238), compute the same ratio.

11. •• Evaluate the Wu–Sprung turning point $x(t_{100})$ from Eq. (29.12), with $t_{100} = 236.524230$. Where does level 100 sit relative to the band of Fig. 29.4?

12. •• Within-height kriging scores 0.117 on held-out true zeros, and the operator scores 0.57–0.70 in the same held-out setting (paper Sec. III). Taking 0.245 as a representative within-height null (the middle of the 0.23–0.26 range of Table III), compute the relative skill Eq. (29.8) of kriging and the factor by which the operator exceeds the null.

13. ••• Best two-neighbor predictor (an extension of the sample problem; the paper does not compute it). With the stationary model of the sample problem on neighbor averages, consider $q_n = w\,(d_{n-1} + d_{n+1})$. (a) Find the w that minimizes $\langle (q_n - d_n)^2 \rangle$. (b) Show that the resulting correlation with d_n is $|\rho_1|/\sqrt{(1+\rho_2)/2}$, always nonnegative. (c) Evaluate w and the correlation for $\rho_1 = -0.29$, $\rho_2 = -0.10$. Explain why the best two-neighbor predictor is positively correlated with d_n although the plain neighbor average is anticorrelated, and why two neighbors alone cannot account for the kriging value $+0.87$.

14. ••• Prove that phase randomization with the DC and Nyquist terms unchanged preserves the circular autocorrelation $C_\ell = \frac{1}{N}\sum_j d_j d_{(j+\ell) \bmod N}$ of a real sequence of even length N, for every lag ℓ. (Use the discrete Wiener–Khinchin relation of Chapter 6.) Does it preserve the ordinary (non-circular) autocorrelation exactly?

15. •• Using `qmnt` or `numpy`, compute the lag-1, lag-2 and lag-3 sample autocorrelations of d_n for $n = 1, \ldots, 400$. Compare the lag-1 value with the heuristic Eq. (29.11), and use your ρ_1, ρ_2 in Eq. (29.10). Compare with the paper's -0.26 to -0.41.

16. •• Using `qmnt`, compute the rms of $N_{\mathrm{fl}}^{p=2}(s_n)$ over $n = 1, \ldots, 400$ and over $n = 401, \ldots, 500$. Compare both with the long-run value 0.1709 and with the probe-null 0.1708 of paper Appendix A.

17. •• Using `qmnt.oracle_zeros` and `qmnt.score`, reproduce the six-prime oracle score on zeros 401–500 (paper Table V). Then repeat with only the prime 2. Which row of Table V do you reproduce in each case?

18. •• Using the toy listing of Section 29.9 (the same code generates Fig. 29.6), compute the in-sample and held-out rms and correlation of the linearized toy, and the held-out rms of predicting zero. Vary the damping (10^{-8}, 10^{-6}, 10^{-3}, 10^{-1} times σ_1^2) and M (200, 400, 600). Do your results resemble the paper's probe (0.0555/+0.95 in-sample, 0.1958/+0.07 held-out)? List two differences between the toy and the paper's computation that could matter. (Optionally, run `qmnt.fit_potential` once for a Table I row and compare.)

19. •• Follow the ridge path of the three (scale 1, fit 1–400, $M = 400$) rows of Table I. The null in-sample rms on 1–400 is 0.2438 (paper App. A). (a) What fraction of the unregularized in-sample improvement over the null is retained at $\mu = 0.3$? (b) How far is the $\mu = 0.3$ OOS rms from the null? (c) What happens as $\mu \to \infty$, and why?

20. •• The null grows from 0.263 (401–500) to 0.296 (5001–10000) across paper Table V. By what factor? Explain why the paper calls this a covariate shift and how within-height splits address it.

21. ••• Dephasing. Suppose a sparse model estimates $\log 2$ with error $\delta\omega = \pi/T$ (half the Fourier resolution, the matching tolerance of paper App. A step 23), where $T = t_{400} - t_1$ is the length of the training window. (a) Show that the phase error at distance Δt past the window is $\delta\omega\,\Delta t$. (b) Evaluate it at the end of the holdout 401–500, using $t_1 = 14.13$, $t_{400} \approx 680$, $t_{500} = 811.18$. (c) Why does the exact frequency of the oracle not dephase?

22. • In plain words, explain why the unregularized interpolation fit can put a correction of rms 40 in the band $22.5 < x < 28.5$ "at no cost in-sample". Which levels would have revealed it?

23. •• The surrogate in-sample results are quoted as 0.134 ± 0.010 over ten draws. If 0.010 is the standard deviation of the ten draws, what is the standard error of their mean? Express the difference between the true-zero value 0.137 and 0.134 in units of that standard error. Does the conclusion change?

24. • • • Design an experiment, within the paper's protocol, to decide whether the marginal Chebyshev $M = 400$ row (0.291, correlation +0.18) reflects real predictive skill. Specify the extra holdouts, restarts, bootstrap type and the decision rule you would fix *before* looking at the results, and explain why fixing the rule in advance matters (Chapter 26).

25. • What is the "probe-null" of the continuation probe, and by what factor does the fitted operator's held-out rms (0.1958) exceed it? Why does this justify the paper's phrase "rms worse than predicting zero fluctuation"?

26. • • • Using `numpy`, implement a within-height odd/even test on the true fluctuations d_n, $n \le 400$: estimate ρ_1 and ρ_2 from the odd-indexed values only (use lag-2 pairs among the odd-indexed values to get ρ_2; since no two odd indices are one step apart, take ρ_1 from the heuristic Eq. (29.11) with σ estimated from the odd-indexed values, and discuss what this assumption costs), predict each even-indexed d_n with the best two-neighbor predictor of Problem 13, and compute rms and correlation. Compare with the operator's odd-split rows in Table III.

29.14 Answers to selected problems

Problem 1. $0.313/0.064 = 4.89$: the held-out error is almost five times the training error, a measure of how much the fit's in-sample accuracy fails to carry over. $0.313/0.263 = 1.19$: the fit is 19% worse than doing nothing, a measure of skill relative to the null (here negative).

Problem 2. Scale 1: 9 rows; scale 2: 3 rows; total 12. Distinct M: 200, 300, 400, 600, 900 (five values), with two ridge variants of $M = 400$. Training prefixes: 1–300, 1–350, 1–400, 1–450 at scale 1 and 1–800 at scale 2, five splits.

Problem 3. No. An interval entirely below zero indicates that the predictions tend to move opposite to the truth: anti-skill. A useful predictor needs positive correlation and, by Eq. (29.9), $\rho > a/2$. Caveat: the interval resamples held-out levels independently although they are serially correlated; the block bootstrap would widen it (the paper reports this widening for the cosine row).

Problem 4. The scale-2 $M = 600$ odd and even rows, since $600 \ge 400$. Saturated means the linearized least-squares problem has at least as many unknowns as equations, so (generically) the training residuals can be driven to zero and there are infinitely many exact solutions; the in-sample error says nothing, and held-out behavior is set by which solution the optimizer reaches. In fact these rows reach only 0.078 and 0.069 in-sample (paper Table III). Saturation counts unknowns; it does not guarantee that the nonlinear, damped fit reaches zero residual.

Problem 5. (a) With $\rho = 0$ and zero means, $\mathrm{rms}^2 = \sigma_d^2(1+a^2)$, so $a = \sqrt{(0.313/0.263)^2 - 1} = \sqrt{1.4164 - 1} = 0.645$. (b) With $\rho = -0.16$: $(0.313/0.263)^2 = 1 + a^2 + 0.32a$, so $a^2 + 0.32a - 0.4164 = 0$ and $a = \frac{1}{2}\left(-0.32 + \sqrt{0.1024 + 1.6655}\right) = 0.505$. The second value lies inside the

paper's 0.16–0.62; the first is just above it. A negative correlation lets a smaller amplitude produce the same error.

Problem 6. For all 1000 scored levels, $M > 2L\sqrt{E}/\pi = 2 \times 59.957 \times 37.675/\pi = 1438$. For the in-sample comparison the relevant energy is $t_{800} \approx 1184$, with $\sqrt{1184} = 34.41$, giving $M > 2 \times 59.957 \times 34.41/\pi \approx 1313$. $M = 900$ is below both requirements, so the basis is not complete even for the training levels. That is consistent with an in-sample rms of 0.103 rather than near zero.

Problem 7. The $m = 1$ term alone has rms $\frac{1}{\pi}\sqrt{a_1^2/2} = \frac{1}{\pi} \cdot \frac{1}{2} = 1/(2\pi) = 0.159$. Its share of the mean square is $a_1^2/\sum a_m^2 = 0.5/0.57639 = 0.867$, about 87%.

Problem 9. $\sqrt{(400 \times 0.136^2 + 100 \times 2.64^2)/500} = \sqrt{(7.398 + 696.96)/500} = \sqrt{1.4087} = 1.187$. A pooled number mixes training and held-out levels and can hide both an excellent in-sample fit and a disastrous held-out one; always report the held-out score separately.

Problem 11. $V - V_0 = 226.783$, $\sqrt{226.783} = 15.0593$, $\sqrt{236.524} = 15.3794$; the ratio in the logarithm is $(30.4387)^2/9.74123 = 95.113$ with logarithm 4.55506. Then $x = \frac{1}{\pi}[15.0593 \times (-1.56151) + 15.3794 \times 4.55506] = \frac{1}{\pi}(-23.515 + 70.054) = 14.81$. Level 100 turns at $x \approx 14.8$, well inside the band's inner edge 22.5; it barely feels the band.

Problem 12. Kriging: $S = 1 - 0.117/0.245 = +0.52$. Operator: $0.57/0.245 = 2.33$ to $0.70/0.245 = 2.86$ times the null. (These depend on taking 0.245 as the null, an assumption stated in the problem.)

Problem 13. (a) $\langle (q-d)^2 \rangle = w^2 \cdot 2\sigma^2(1+\rho_2) - 2w \cdot 2\rho_1\sigma^2 + \sigma^2$, using $\mathrm{Var}(d_{n-1}+d_{n+1}) = 2\sigma^2(1+\rho_2)$ and $\mathrm{Cov}(d_{n-1}+d_{n+1}, d_n) = 2\rho_1\sigma^2$. Setting the derivative in w to zero: $4w\sigma^2(1+\rho_2) - 4\rho_1\sigma^2 = 0$, so $w = \rho_1/(1+\rho_2)$. (b) Correlation of $w(d_{n-1} + d_{n+1})$ with d_n is $\mathrm{sign}(w)\,\rho_1/\sqrt{(1+\rho_2)/2}$, by Eq. (29.10) (multiplying a variable by w multiplies its correlation by the sign of w). Since w has the sign of ρ_1 (as $1 + \rho_2 > 0$), this equals $|\rho_1|/\sqrt{(1+\rho_2)/2} \geq 0$. (c) $w = -0.29/0.90 = -0.322$; correlation $0.29/0.6708 = 0.432$. The best predictor *flips the sign* of the neighbor average. Smooth interpolation assumes neighbors resemble each other; kriging learns from the autocovariance that they do not, and uses that. For these illustrative values, two neighbors alone give at most this 0.43. The paper's kriging reaches +0.87 by using all training values, with autocovariance to lag 40 (Appendix A step 19).

Problem 15. Compute d_n with `qmnt.fluct`, subtract the mean, and form $\rho_\ell = \sum_n d_n d_{n+\ell} / \sum_n d_n^2$. Do this for $\ell = 1, 2, 3$. All three come out negative, as the paper states (Sec. III). The heuristic Eq. (29.11) gives $\rho_1 \approx -0.29$ with the holdout $\sigma = 0.263$, and $\rho_1 \approx -0.50$ with the in-range value $\sigma = 0.2438$ appropriate to zeros 1–400. Your measured ρ_1 is the arbiter; its difference from these estimates reflects the Wigner-surmise and stationarity approximations. Because the measured ρ_1 is negative and $1 + \rho_2 > 0$, Eq. (29.10) gives a negative neighbor-average correlation to compare with -0.26 to -0.41 (the paper's values are measured on held-out subsets with training neighbors, so they need not coincide exactly).

Problem 17. `qmnt.score(qmnt.oracle_zeros(n, qmnt.first_primes(6)), n)` with $n = 401, \ldots, 500$ reproduces the six-prime entry of the 401–500 row of Table V, 0.074 / +0.96; with primes [2] you reproduce the {2} entry, 0.198 / +0.66.

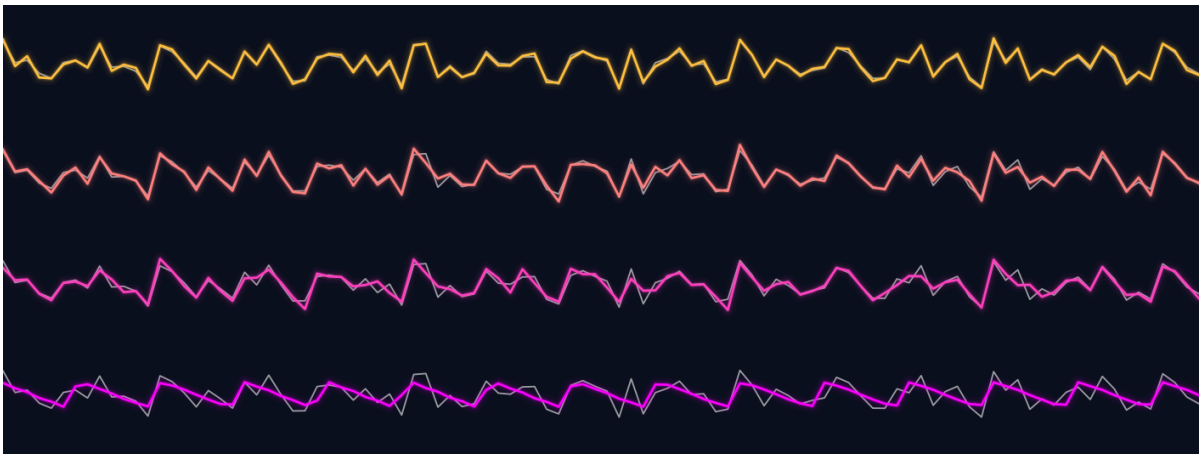
Problem 19. (a) Unregularized improvement $0.2438 - 0.137 = 0.1068$; at $\mu = 0.3$ it is $0.2438 - 0.206 = 0.0378$; retained fraction $0.0378/0.1068 = 0.354$. (b) $0.265 - 0.263 = 0.002$ above the null. (c) As $\mu \to \infty$ the ridge filter $\sigma/(\sigma^2 + \mu\sigma_1^2) \to 0$ for every singular value, so each step vanishes and c stays at 0: the model becomes the unperturbed potential, in-sample rms $\to 0.2438$ and OOS rms $\to 0.263$. Ridge interpolates between memorizing and doing nothing.

Problem 21. (a) A frequency error $\delta\omega$ shifts the phase ωt by $\delta\omega\, t$; if the fit makes the phase correct on average over the window, the error grows as $\delta\omega$ times the distance Δt from the window, $\delta\phi = \delta\omega\, \Delta t$. (b) $T = 680 - 14.13 = 665.9$; $\delta\omega = \pi/665.9 = 0.00472$; $\Delta t = 811.18 - 680 = 131.2$; $\delta\phi = 0.00472 \times 131.2 = 0.62$ rad. (c) The oracle's frequencies $m \log p$ are exact, $\delta\omega = 0$, so no phase error accumulates; this is why the oracle's skill is nearly height-independent in Table V, while the sparse model decays (paper Sec. IV: 0.078 / +0.96 on 1001–2000).

Problem 22. Training levels 1–200 turn below $x = 22.5$ and do not reach the band classically; training levels 301–500 cross it deep inside their allowed region. On our heuristic reading (not stated in the paper), a correction there can be offset, against their rapidly oscillating densities, by structure elsewhere, so their residuals need not penalize it; the paper's own statement is only that the structure "costs nothing in-sample". Levels 201–300, which turn inside the band and are most sensitive there, would have revealed it; they were held out.

Problem 23. Standard error $0.010/\sqrt{10} = 0.0032$. Difference 0.003 is $0.003/0.0032 = 0.95$ standard errors. Even on this stricter yardstick the true-zero fit is not distinguishable from the permuted fits; the conclusion is unchanged.

Problem 25. The probe-null is the held-out rms of predicting zero fluctuation for the synthetic RE2 target on levels 401–500, 0.1708 (paper App. A). The operator's 0.1958 exceeds it by a factor $0.1958/0.1708 = 1.146$. Predicting nothing is better than using the fitted operator, which is exactly "rms worse than predicting zero fluctuation".



Predicted fluctuations of zeros 401 to 500 from 1, 3, 6 and 25 primes (bottom to top), converging on the truth (white).

Chapter 30

Reading the Paper III: The Arithmetic Information Generalizes

↑ Contents

What this chapter is about

Chapter 29 ended on a hard negative result. Fitted potentials on the Wu–Sprung background reproduce their training zeros closely, yet their predictions for zeros they were not shown are no better than the "no fluctuation" null. A skeptic could now say: perhaps nothing predicts the zeros. Perhaps zero number 401 is, for practical purposes, unpredictable from zeros 1 to 400, and the operators fail simply because the task is impossible.

Section IV of the paper answers that skeptic. It has three jobs. First, it shows that the information needed to predict unseen zeros *exists* and is arithmetic: a truncation of the explicit formula with zero fitted parameters predicts held-out zeros with correlation up to $+0.99$ all the way to zero 10^4 (Table V). Second, it separates two explanations for the operators' failure that are easy to confuse: "too few parameters" and "the wrong kind of parameters". Four controls (A, B, C1, C2 of Table IV) separate the two explanations for the model families tested. Third, it checks honestly whether a fitted model with no arithmetic input that nevertheless "finds the primes" is fooling itself; this is the surrogate alignment check, which the paper labels exploratory.

This chapter teaches no new mathematics. It reads paper Sec. IV line by line, together with the parts of Sec. II D, Sec. VI and Appendix A (steps 8, 20 and 23) that it relies on. For every claim we say what it means in plain words, which earlier chapters supply the tools, how the computation works (rerunning it at small scale with `qmnt` where feasible), and how strong it is. Where this shows up in the paper: Sec. IV, Tables IV and V, Eq. (4), Fig. 1, Sec. II D, Sec. VI, and Appendix A steps 8, 20 and 23.

What you need

- Root finding by bracketing and Brent's method; inverting a monotone function (Chapter 4).
- Sinusoids, phase, almost-periodic sums of incommensurate frequencies (Chapter 5).
- Fourier resolution $\Delta\omega \approx 2\pi/T$, the periodogram, irregular sampling, phase randomization (Chapter 6).
- Linear least squares, the SVD, underdetermined systems, ridge regularization with filter factors $\sigma/(\sigma^2 + \lambda\sigma_1^2)$ (Chapter 9).
- Primes, prime powers, the prime number theorem, the von Mangoldt function (Chapter 15).
- The smooth counting function N_{sm}, smooth zeros s_n, mean spacing gap_n, fluctuations in mean spacings (Chapter 17).
- The explicit formula, its truncation Eq. (4), Landau's formula and its peaks at $\log p^m$ (Chapter 18).
- rms, Pearson correlation and the difference between them (Chapter 19).
- Training and test sets, the null model, covariate shift, inductive bias (Chapter 25).
- Bootstrap intervals, surrogate data, exploratory versus confirmatory analysis (Chapter 26).
- Sinusoidal regression, the dense-grid and sparse-search controls, dephasing (Chapter 27).
- The paper's protocol and its Sec. III results (Chapters 28 and 29).
- First-year physics: the phase of an oscillation, and beats between two nearby frequencies.

30.1 What Section IV sets out to show

Recall the situation at the end of paper Sec. III. The fitted operators have rms errors on held-out zeros between about 0.27 and 0.44 mean spacings (Table I), against a null of 0.263 on the principal holdout, zeros 401–500 (running example RE5). Their held-out correlations with the truth hover around zero. Two very different stories could explain this:

1. **The "no information" story.** The fluctuations of zeros beyond 400 are not determined, in any usable way, by anything we could know. Then every model must fail, and the operators are blameless.

2. **The "wrong vocabulary" story.** The information is there, but a potential $V(x)$ expanded in a generic basis cannot express it in a form that continues past the training window.

Section IV is built to decide between these. Its title states the verdict: "The arithmetic information generalizes." Table 30.1 lists the claims we will read, in the order of this chapter.

For each claim we will give (a) what it says in plain words, (b) the chapters that supply the tools, (c) the computation behind it, rerun at small scale where feasible, and (d) its strength and its limits.

Pitfall

The letters A and T are badly overloaded in the paper, and Sec. IV adds to the load. In this chapter, "control A" means the oracle run with a larger prime budget. It has nothing to do with the sensitivity matrix $A_{nk} = \int |\psi_n|^2 \phi_k \, \mathrm{d}x$ of Sec. II C, nor with the non-Hermitian operator iA of Sec. II E. Likewise T here is always the length, in t, of the training window; it is not the second-order tensor $T^{(n)}$ of Eq. (5), nor a Chebyshev polynomial T_k. When you read Sec. IV, mentally replace "A" by "control A" and T by "window length".

Table 30.1: The claims of paper Sec. IV (with supporting text from Secs. II D and VI) read in this chapter. "Where" gives the R1 location.

	Claim (paraphrase)	Where
IV.1	The truncated explicit formula predicts unseen zeros essentially as well at $n = 10^4$ as at $n = 400$.	Sec. IV, Table V
IV.2	One prime beats every fitted operator's holdout performance by a factor $\gtrsim 1.3$, with zero fitted parameters.	Sec. IV
IV.3	Twenty-five primes reach rms 0.04–0.07 with correlation +0.98 to +0.99.	Sec. IV, Table V
IV.4	The oracle is not a fair machine-learning competitor; it shows only that transferable structure exists and is arithmetic.	Secs. II D, IV
IV.5	Control A (the oracle at a larger prime budget): going from 25 primes (0.043) to 300 primes, i.e. 900 terms, still with zero fitted parameters, the oracle improves monotonically, to 0.018.	Sec. IV, Table IV
IV.6	Control B: with frequencies given and amplitudes fitted, 150 coefficients give 0.028; at 900 coefficients on 400 zeros the problem is underdetermined and the score degrades to 0.180/+0.78 (root-ambiguous).	Sec. IV, Table IV
IV.7	Control C1: a dense 900-coefficient Fourier model stays within 0.005 of the null.	Sec. IV, Table IV
IV.8	Control C2: a sparse search with no arithmetic input selects frequencies at $\log p^m$ and predicts at 0.034/+0.99; an exploratory surrogate check supports the alignment.	Sec. IV, App. A step 23
IV.9	C2's skill decays with distance from the training window as its frequencies dephase.	Sec. IV
IV.10	The obstacle is the parametrization, not the parameter count.	Abstract, Secs. IV, VI

Key Ideas

- Section IV distinguishes "the information does not exist" from "the operators cannot express it".
- It does so with one zero-parameter predictor (the oracle, Table V) and four controls that differ in how much of the answer's form they are given (A and B with the arithmetic frequencies, then C2 with only a sparsity prior, then C1 with a generic grid; Table IV).
- Each claim is measured against the null of its own holdout, the same null the operators face: 0.263 on zeros 401–500, 0.274 on 801–1000, and 0.270–0.296 on the other holdouts of Table V.

30.2 The oracle and how it is scored

★ Did you know? *Scoring the Weather*

In 1950 Glenn Brier proposed scoring probability forecasts by the mean squared difference between the forecast probabilities and what actually happened. The Brier score rewards forecasters for being both accurate and honest about uncertainty. Weather services still use it. The same principle applies to any oracle: decide how predictions will be scored *before* looking at the results.

30.2.1 The formula

Recall from Chapter 18 that the fluctuating part of the zero-counting function is approximated by a sum over prime powers. The paper's Eq. (4) truncates it to a prime set P and to $m \le 3$:

$$N_{\mathrm{fl}}(t) = -\frac{1}{\pi} \sum_{p \in P} \sum_{m=1}^{3} \frac{\sin(mt \log p)}{m\, p^{m/2}}. \tag{30.1}$$

Here t is height on the critical line, p runs over the primes in P, m is the power, and log is the natural logarithm. Each term is a sinusoid in t with angular frequency $m \log p$ and amplitude $1/(\pi m p^{m/2})$. We will often call $w(p^m) = 1/(m p^{m/2})$ the *weight* of the prime power $q = p^m$; the amplitude is w/π. Recall also, from Chapter 18, that the untruncated sum does not converge on the critical line, so Eq. (30.1) is a heuristic approximation whose quality must be measured, not assumed. Measuring it is exactly what Table V does.

30.2.2 From formula to predicted zeros

The smooth counting function of paper Eq. (1) is

$$N_{\mathrm{sm}}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}. \tag{30.2}$$

The true counting function $N(t)$ is a staircase that jumps by 1 at each zero. Just below t_n it equals $n-1$, just above it equals n. The midpoint of the jump is $n - \frac{1}{2}$. The oracle predicts the n-th zero as the height t where the smooth-plus-fluctuating count reaches that midpoint:

$$N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \tfrac{1}{2}. \tag{30.3}$$

We call Eq. (30.3) the *level equation*. Appendix A step 8 records how it is solved: Brent's method (Chapter 4) on a bracket of ± 3 mean spacings around the smooth zero s_n, with the bracket

expanded if the function has the same sign at both ends. Recall that s_n solves $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ and that the local mean spacing is

$$\mathrm{gap}_n = \frac{\mathrm{d}s_n}{\mathrm{d}n} = \frac{2\pi}{\log(s_n/2\pi)}. \tag{30.4}$$

Checkpoint 30.1

The oracle's prediction of zero n uses the index n, the smooth zero s_n (to place the bracket) and the prime set P. Does it use any computed zeta zero at all?

Answer: No. s_n comes from Eq. (30.2) alone, and N_{fl} from the primes alone. No zero, in-sample or held-out, enters the prediction. That is why the paper can say the oracle has "zero fitted parameters" and why it needs no training window.

30.2.3 How predictions are scored

The score is the one used everywhere in the paper (Chapter 28). For each held-out index n, convert a zero t into a fluctuation in mean spacings,

$$f_n(t) = \frac{t - s_n}{\mathrm{gap}_n}. \tag{30.5}$$

The true fluctuation is $f_n(t_n)$ and the predicted one is $f_n(t_n^{\mathrm{pred}})$. The rms of their difference over the holdout is the "rms"; the Pearson correlation between the two sequences is the "corr". The null predicts $t_n = s_n$, so its fluctuation is 0 and its rms is the rms of the true fluctuations themselves: 0.263 on zeros 401–500.

30.2.4 What the oracle's prediction is, in mean spacings

It helps to see what Eq. (30.3) does to the fluctuation. Write the predicted zero as $t = s_n + \delta$. We use Taylor's theorem with the Lagrange form of the remainder: if f is twice differentiable, then $f(s+\delta) = f(s) + \delta f'(s) + \frac{1}{2}\delta^2 f''(\xi)$ for some ξ between s and $s+\delta$. (This sharpens the Taylor approximations of Chapter 4 by saying exactly what the neglected term is; its proof, by Rolle's theorem applied to a suitable auxiliary function, is in [22].) Applied to N_{sm} it gives

$$\begin{aligned} N_{\mathrm{sm}}(s_n + \delta) &= N_{\mathrm{sm}}(s_n) + \delta\, N'_{\mathrm{sm}}(s_n) + \tfrac{1}{2}\delta^2\, N''_{\mathrm{sm}}(\xi) \quad (\text{some } \xi \text{ between } s_n \text{ and } s_n + \delta) \\ &= n - \tfrac{1}{2} + \frac{\delta}{\mathrm{gap}_n} + \tfrac{1}{2}\delta^2\, N''_{\mathrm{sm}}(\xi). \end{aligned} \tag{30.6}$$

The second line used two facts. First, $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$ by definition of s_n. Second, differentiating Eq. (30.2),

$$\begin{aligned} N'_{\mathrm{sm}}(t) &= \frac{1}{2\pi}\log\frac{t}{2\pi} + \frac{t}{2\pi}\cdot\frac{1}{t} - \frac{1}{2\pi} \quad (\text{product rule on the first term}) \\ &= \frac{1}{2\pi}\log\frac{t}{2\pi}, \end{aligned} \tag{30.7}$$

which is exactly $1/\mathrm{gap}_n$ at $t = s_n$ by Eq. (30.4). Differentiating once more gives $N''_{\mathrm{sm}}(t) = 1/(2\pi t)$.

Now substitute Eq. (30.6) into the level equation (30.3):

$$\begin{aligned} n - \tfrac{1}{2} + \frac{\delta}{\mathrm{gap}_n} + \tfrac{1}{2}\delta^2 N''_{\mathrm{sm}}(\xi) + N_{\mathrm{fl}}(s_n + \delta) &= n - \tfrac{1}{2} \\ \frac{\delta}{\mathrm{gap}_n} &= -N_{\mathrm{fl}}(s_n + \delta) - \tfrac{1}{2}\delta^2 N''_{\mathrm{sm}}(\xi). \end{aligned} \tag{30.8}$$

The left side is the predicted fluctuation f_n of Eq. (30.5). The curvature term is tiny: with $\delta \approx \text{gap}_n \approx 1.34$ and $t \approx 680$ (the top of the training window), it is $\frac{1}{2}(1.3414)^2/(2\pi \cdot 680) \approx 2 \times 10^{-4}$ mean spacings. So, to that accuracy,

$$f_n^{\text{pred}} \approx -N_{\text{fl}}(t_n^{\text{pred}}). \tag{30.9}$$

In mean spacings, the oracle's predicted fluctuation is minus the prime sum evaluated at the predicted zero. The same argument applied to a true zero shows that the true fluctuation $f_n(t_n)$ is, to the same accuracy, $-y_n$ with

$$y_n = n - \tfrac{1}{2} - N_{\text{sm}}(t_n), \tag{30.10}$$

the amount by which the smooth count misses the midpoint at the true zero. This y_n will reappear as the training target of every fitted control.

Sample Problem 30.1: One oracle step by hand

Use only the prime 2 (the running example RE2, $m = 1, 2, 3$). Pretend that $s = 680$ is a smooth zero, so that the target count is $N_{\text{sm}}(680)$. (The index would not be an integer; the algebra does not care.) Find the oracle's predicted shift δ (a) by neglecting the variation of N_{fl} over the shift, (b) by one Newton step on the full level equation, and (c) exactly.

Step 1: evaluate N_{fl} at s. The phases are $m \cdot 680 \log 2$ reduced modulo 2π. With $\log 2 = 0.693147$, $680 \log 2 = 471.34008$, and $75 \cdot 2\pi = 471.23890$, so the $m = 1$ phase is 0.10118. Doubling and tripling (and removing $150 \cdot 2\pi$ and $225 \cdot 2\pi$) give 0.20237 and 0.30355. Then

$$N_{\text{fl}}(680) = -\frac{1}{\pi}\Big[\frac{\sin 0.10118}{\sqrt{2}} + \frac{\sin 0.20237}{2 \cdot 2} + \frac{\sin 0.30355}{3 \cdot 2^{3/2}}\Big] = -\frac{0.15690}{\pi} = -0.0499.$$

Step 2: the mean spacing and slope of N_{sm}. By Eq. (30.7), $N'_{\text{sm}}(680) = \log(108.225)/2\pi = 0.7455$, so gap $= 1/0.7455 = 1.3414$.

Step 3: approximation (a). Neglect the change of N_{fl} and the curvature in Eq. (30.8): $\delta_a = -\text{gap} \cdot N_{\text{fl}}(680) = 1.3414 \times 0.0499 = 0.0670$.

Step 4: approximation (b). The full level function is $G(t) = N_{\text{sm}}(t) - N_{\text{sm}}(680) + N_{\text{fl}}(t)$ with $G(680) = -0.0499$. Its slope is $G'(680) = N'_{\text{sm}}(680) + N'_{\text{fl}}(680)$, where

$$N'_{\text{fl}}(t) = -\frac{1}{\pi}\sum_{m=1}^{3} \frac{\log 2 \, \cos(mt \log 2)}{2^{m/2}}, \qquad N'_{\text{fl}}(680) = -0.3377.$$

One Newton step gives $\delta_b = -G(680)/G'(680) = 0.0499/(0.7455 - 0.3377) = 0.122$.

Step 5: exact root. Brent's method on $[680 - 3\text{gap}, 680 + 3\text{gap}]$ (one call to `scipy.optimize.brentq`) gives the root at $\delta_c = 0.120$, that is, a predicted fluctuation of $0.120/1.3414 = 0.090$ mean spacings. As Eq. (30.9) promises, this is $-N_{\text{fl}}$ at the root, not at s.

$$\boxed{\delta_a = 0.0670, \quad \delta_b = 0.122, \quad \delta_c = 0.120 \ \ (0.090 \text{ mean spacings})}$$

The lesson: the slope of the prime sum, -0.34, is not small next to the slope of the smooth count, 0.75. Neglecting it nearly halves the shift. This shows why solving the level equation, as the paper does, matters: a single linearized step from s_n would be badly off.

30.2.5 Is the root unique?

Brent's method returns one root of the level equation in its bracket. If $G(t) = N_{\rm sm}(t) + N_{\rm fl}(t) - (n - \frac{1}{2})$ is strictly increasing, there is exactly one. A sufficient condition is that $|N'_{\rm fl}(t)| < N'_{\rm sm}(t)$ everywhere near the bracket. Differentiating Eq. (30.1) term by term,

$$N'_{\rm fl}(t) = -\frac{1}{\pi}\sum_{p\in P}\sum_{m=1}^{3}\frac{m\log p\,\cos(mt\log p)}{mp^{m/2}} = -\frac{1}{\pi}\sum_{p\in P}\sum_{m=1}^{3}\frac{\log p\,\cos(mt\log p)}{p^{m/2}}, \tag{30.11}$$

and since $|\cos| \le 1$,

$$|N'_{\rm fl}(t)| \le B_P \equiv \frac{1}{\pi}\sum_{p\in P}\log p\,(p^{-1/2} + p^{-1} + p^{-3/2}). \tag{30.12}$$

This bound is the worst case, reached only if all cosines equal 1 at once. It is useful because it is simple.

Sample Problem 30.2: When is the oracle's root guaranteed unique?

Compute B_P for $P = \{2\}$ and for the six primes $P = \{2, 3, 5, 7, 11, 13\}$, and find the heights above which $B_P < N'_{\rm sm}(t)$.

Step 1: $P = \{2\}$. $\log 2\,(0.70711 + 0.5 + 0.35355) = 0.69315 \times 1.56066 = 1.08178$, so $B_{\{2\}} = 1.08178/\pi = 0.3443$.

Step 2: threshold for $\{2\}$. By Eq. (30.7) we need $\log(t/2\pi)/2\pi > 0.3443$, that is $\log(t/2\pi) > 2.1636$, so $t/2\pi > e^{2.1636} = 8.702$ and $t > 54.7$.

Step 3: six primes. The six contributions $\log p\,(p^{-1/2} + p^{-1} + p^{-3/2})$ are 1.0818, 1.2119, 1.1856, 1.1185, 1.0067, 0.9634; their sum is 6.568, so $B_6 = 6.568/\pi = 2.09$.

Step 4: threshold for six primes. We need $\log(t/2\pi) > 2\pi \times 2.09 = 13.14$, that is $t > 2\pi e^{13.14} \approx 3.2 \times 10^6$.

$B_{\{2\}} = 0.344 \;\Rightarrow$ unique root for $t > 54.7$; $\qquad B_6 = 2.09 \;\Rightarrow$ guaranteed only for $t \gtrsim 3.2 \times 10^6$

The worst-case bound is far from typical. Because the frequencies are incommensurate, the cosines in Eq. (30.11) do not line up, and the rms of $N'_{\rm fl}$ over t is $\frac{1}{\pi}\big[\sum_p (\log p)^2 \sum_m \frac{1}{2p^m}\big]^{1/2} = 0.415$ for six primes (Problem 28 asks you to derive this formula). That is below $N'_{\rm sm} \approx 0.75$ at the heights of the paper, but only by a factor of about 1.8: for six primes, stretches where G decreases are uncommon, not negligible. For larger budgets they are common. The same formula gives an rms slope of about 0.7 for the 25 primes of Table V, comparable with $N'_{\rm sm}$, and about 1.2 for the 300 primes of control A (Section 30.5), larger than $N'_{\rm sm}$.

A decreasing stretch of G does not by itself produce several roots in the bracket. Several roots require a decreasing excursion that straddles the level $n - \frac{1}{2}$. A heuristic reason why this is much rarer than non-monotonicity: $N_{\rm sm} + N_{\rm fl}$ approximates the true staircase $N(t)$, which rises steeply through $n - \frac{1}{2}$ near the zero and is flat near the integer values $n - 1$ and n on either side, so the wiggles of a good approximation tend to sit near integer values, away from the half-integer levels. The paper reports root ambiguity only for one fitted row of Table IV (control B with 900 coefficients), which we meet in Section 30.6.

30.2.6 Rerunning Table V at small scale

The listing below recomputes the first row of paper Table V. It uses `qmnt.first_primes(k)` for the first k primes, `qmnt.oracle_zeros` for the bracketed Brent solution of Eq. (30.3), and `qmnt.score` for the paper's scoring.

```
import numpy as np, qmnt
n = np.arange(401, 501)                    # held-out indices (RE5)
t_true = qmnt.zeros(500)[400:500]          # t_401 ... t_500
null = qmnt.rms(qmnt.fluct(t_true, n))     # the "no fluctuation" prediction
print("null", round(null, 3))
for k in (1, 3, 6, 25):                    # prime budgets of Table V
    P = qmnt.first_primes(k)
    t_pred = qmnt.oracle_zeros(n, P, mmax=3)
    r, c = qmnt.score(t_pred, n)           # rms (mean spacings), Pearson corr
    print(k, "primes:", round(r, 3), round(c, 2))
```

Running it reproduces the paper's row to the printed precision: null 0.263; one prime 0.198/ + 0.66; three primes 0.116/ + 0.90; six primes 0.074/ + 0.96; twenty-five primes 0.043/ + 0.99. Setting `n = np.arange(1001, 2001)` and `t_true = qmnt.zeros(2000)[1000:2000]` reproduces the third row, for instance 0.091/ + 0.95 for six primes. Figure 30.1 shows the six-prime prediction against the truth around the train/test boundary used in paper Fig. 1.

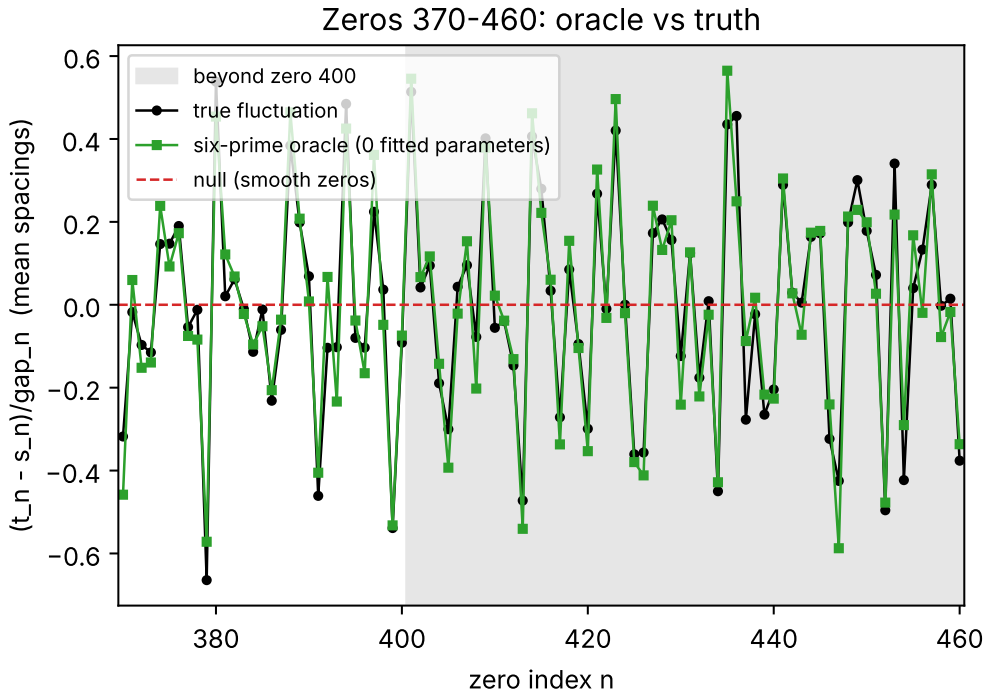


Figure 30.1: A small-scale rerun of the oracle half of paper Fig. 1. Black circles: true fluctuations $(t_n - s_n)/\mathrm{gap}_n$ of zeros 370–460, in mean spacings. Green squares: the six-prime oracle of Eq. (30.1) ($p \le 13$, $m \le 3$), obtained by solving the level equation (30.3) with zero fitted parameters. Dashed red: the null (no fluctuation). The shaded region, beyond zero 400, is where the paper's fitted operator is held out; the oracle has no training window, so nothing changes at the boundary. The paper reports held-out correlations +0.96 (oracle) and −0.07 (operator) for this figure.

Key Ideas

- The oracle solves $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ with no fitted parameters and no zero data, Eq. (30.3).
- In mean spacings its predicted fluctuation is $-N_{\mathrm{fl}}$ at the predicted zero, Eq. (30.9);

the true fluctuation is $-y_n$, Eq. (30.10).

- The slope of N_{fl} is comparable with that of N_{sm}, so the level equation must be solved, not linearized; uniqueness of the root is guaranteed by the simple bound (30.12) for one prime but not for six.
- Table V's oracle rows are reproducible in a few seconds with `qmnt`.

30.3 Table V: prediction through zero ten thousand

30.3.1 What the table says

Table 30.2 copies paper Table V. Every entry has zero fitted parameters. Each cell is "rms / corr" on the stated holdout.

Table 30.2: Paper Table V, quoted exactly: the oracle of Eq. (30.1), rms / correlation on held-out zeros, for prime budgets {2}, the first 3, 6 and 25 primes.

holdout	null	{2}	3 primes	6 primes	25 primes
401–500	0.263	0.198 / +0.66	0.116 / +0.90	0.074 / +0.96	0.043 / +0.99
501–1000	0.270	0.208 / +0.64	0.127 / +0.88	0.083 / +0.95	0.050 / +0.99
1001–2000	0.279	0.219 / +0.62	0.139 / +0.87	0.091 / +0.95	0.054 / +0.98
2001–5000	0.288	0.232 / +0.59	0.156 / +0.84	0.107 / +0.93	0.060 / +0.98
5001–10000	0.296	0.242 / +0.58	0.169 / +0.82	0.119 / +0.92	0.067 / +0.98

Read across a row: more primes, better prediction, with no exception. Read down a column: the rms grows slowly with height. Claim IV.3 is the last column: rms 0.043 to 0.067, which the paper rounds to "0.04–0.07", with correlation +0.98 to +0.99.

Claim IV.1 says the formula predicts "essentially as well at $n = 10^4$ as at $n = 400$". To weigh the word "essentially", divide each rms by the null of its row. The result is Table 30.3: the fraction of the null error that remains.

Table 30.3: Ratio rms/null computed from Table 30.2 (our arithmetic). Smaller is better; 1 would mean no skill.

holdout	{2}	3 primes	6 primes	25 primes
401–500	0.75	0.44	0.28	0.16
501–1000	0.77	0.47	0.31	0.19
1001–2000	0.78	0.50	0.33	0.19
2001–5000	0.81	0.54	0.37	0.21
5001–10000	0.82	0.57	0.40	0.23

Two honest readings are possible. By correlation, skill is nearly constant: +0.99 to +0.98 at 25 primes. By the rms ratio, it drifts upward, from 0.16 to 0.23 at 25 primes. The paper's caption to Table V gives the reason for the drift: "The mild growth at fixed prime budget mirrors the growth of the null and the increasing number of arithmetically relevant primes at height; relative skill is nearly constant." Dividing by the null removes the first effect. What remains in Table 30.3 is the second: at greater heights, a fixed set of primes captures a smaller share of the fluctuation because more primes contribute appreciably. "Nearly constant" is the paper's wording, and it is a fair summary of a ratio that changes by a factor of 1.1 to 1.4, depending on the prime budget, over a factor of 25 in zero index (about 15 in height t, from $t_{400} \approx 680$ to $t_{10000} \approx 9878.7$). Figure 30.2 shows both views.

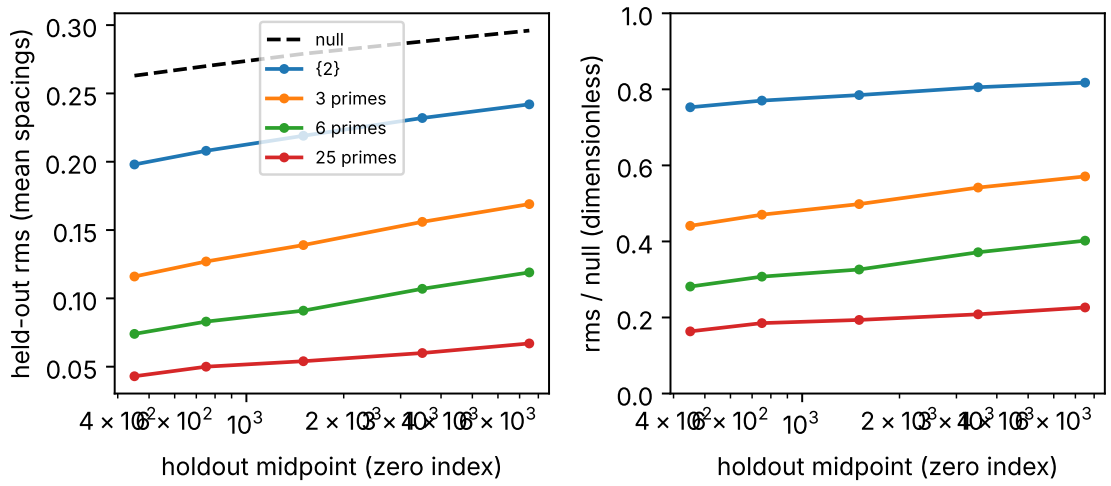


Figure 30.2: Paper Table V plotted against the midpoint of each holdout (log scale). Left: rms in mean spacings for each prime budget, with the null (dashed). Right: rms divided by the null, Table 30.3. Every curve stays far below the null through zero 10^4; the slow upward drift at fixed budget reflects the growing number of relevant primes at height.

Taken on trust

Why the null grows, and why more primes matter higher up. A theorem of Selberg (his central limit theorem for $S(t)$; see the discussion in [6]) states that, as $t \to \infty$, $S(t)$ divided by $\sqrt{\frac{1}{2\pi^2}\log\log t}$ becomes distributed like a standard Gaussian. We take this on trust; its proof is beyond this book. A note on units: $S(t)$ is measured in counts, while the paper's fluctuations are measured in mean spacings. To leading order these are the same: by Eqs. (30.9) and (30.10), a count offset y_n at a zero corresponds to a height offset of about $y_n \mathrm{gap}_n$, which is y_n mean spacings, because the local density of zeros is $1/\mathrm{gap}_n$. So the variance of S applies directly to the fluctuations in mean spacings, up to the small curvature and slope corrections neglected in those equations. Heuristically, the variance $\frac{1}{2\pi^2}\log\log t$ matches the variance $\sum_{p\le x}\frac{1}{2\pi^2 p} \approx \frac{1}{2\pi^2}\log\log x$ of the $m = 1$ prime terms (each sinusoid of amplitude $1/(\pi\sqrt{p})$ has mean square $1/(2\pi^2 p)$), with an effective prime cutoff x that grows with t. This is a heuristic, not a derivation, but it explains both trends in Table V: the null grows very slowly, like $\sqrt{\log\log t}$, and a fixed prime budget covers a shrinking part of the variance. Problem 20 compares the predicted growth with the table.

30.3.2 "One prime beats every fitted operator"

Claim IV.2 is a comparison between Table V and Table I.

Sample Problem 30.3: The factor 1.3

The fitted operators' OOS rms values in paper Table I are 0.317, 0.320, 0.273, 0.274, 0.369, 0.271, 0.265, 0.313, 0.439, 0.370, 0.279, 0.302. The one-prime oracle scores 0.198 on zeros 401–500 and 0.208 on 501–1000 (Table V). Check the paper's "factor $\gtrsim 1.3$".

Step 1: the best operator. The smallest value in the list is 0.265 (scale 1, fit 1–400, $M = 400$, ridge $\mu = 0.3$). Its holdout is 401–500.

Step 2: compare on the same holdout. $0.265/0.198 = 1.34$.
Step 3: the best scale-2 operator. The smallest scale-2 value is 0.279 (holdout 801–1000). The closest Table V row is 501–1000, where one prime gives 0.208; $0.279/0.208 = 1.34$.

$$\boxed{\text{smallest ratio} \approx 1.34 \quad (\gtrsim 1.3)}$$

Two caveats keep this honest. The scale-2 comparison uses neighboring rather than identical holdouts, so it is approximate. And the operators' holdouts for fits on zeros 1–300, 1–350 and 1–450 differ from 401–500; but each of those operators scores ≥ 0.273, and every Table V entry for one prime is ≤ 0.242, so one prime still beats each of them whichever row is used (worst case $0.273/0.242 = 1.13$). The factor of about 1.3 applies to matched or neighboring holdouts, for example $0.273/0.198 = 1.38$ for the 1–350 fit against the 401–500 row.

30.3.3 rms and correlation tell different stories

Throughout Table IV you will want to read rms and correlation together. Chapter 19 showed that they measure different things. Here is the quantitative link we will use repeatedly.

Let Y_n be the true fluctuations on a holdout and P_n the predicted ones, with standard deviations σ_Y, σ_P and correlation ρ. The mean squared error is

$$\begin{aligned}\overline{(P-Y)^2} &= \operatorname{var}(P-Y) + \left(\overline{P-Y}\right)^2 \quad \text{(variance plus squared mean)} \\ &\geq \operatorname{var}(P-Y) = \sigma_P^2 + \sigma_Y^2 - 2\rho\,\sigma_P\sigma_Y. \end{aligned} \tag{30.13}$$

For $\rho > 0$, the right side is a quadratic in σ_P with minimum at $\sigma_P = \rho\sigma_Y$ (set its derivative $2\sigma_P - 2\rho\sigma_Y$ to zero). Substituting,

$$\overline{(P-Y)^2} \;\geq\; \rho^2\sigma_Y^2 + \sigma_Y^2 - 2\rho^2\sigma_Y^2 = \sigma_Y^2(1-\rho^2), \qquad \text{so} \qquad \text{rms} \geq \sigma_Y\sqrt{1-\rho^2}. \tag{30.14}$$

The null's rms is $\sqrt{\overline{Y^2}}$, which equals σ_Y when the true fluctuations average to zero. They nearly do: on zeros 401–500 the mean of the true fluctuations (computed with `qmnt.fluct`) is below 0.05 in magnitude, so its square is under 4% of the mean square 0.263^2, and we use the null as an approximation to σ_Y. (What is neglected: the squared mean of Y over the holdout, which is small compared with σ_Y^2.)

Sample Problem 30.4: Reading a correlation off an rms, and vice versa

(a) Control C1 has correlation +0.28 on zeros 401–500 (Table IV). What is the best rms any prediction with that correlation could have? (b) Control B with 25 primes has rms 0.028 on the same holdout. What correlation does that imply?
Step 1: (a). With $\sigma_Y \approx 0.263$ and $\rho = 0.28$, Eq. (30.14) gives $0.263\sqrt{1-0.0784} = 0.263 \times 0.960 = 0.252$.
Step 2: compare. C1 scores 0.258, close to this best case. This is consistent with predictions of small amplitude (our inference from the two reported numbers; the paper does not report the amplitude). Indeed, if the mean error is negligible, Eq. (30.13) holds with equality, and setting $0.258^2 = \sigma_P^2 + 0.263^2 - 2(0.28)(0.263)\sigma_P$ gives a quadratic in σ_P with roots $\sigma_P \approx 0.02$ and 0.13, both at most about half of σ_Y.
Step 3: (b). Invert Eq. (30.14): $\rho \geq \sqrt{1-(\text{rms}/\sigma_Y)^2} = \sqrt{1-(0.028/0.263)^2} = \sqrt{1-0.0113} = 0.994$.

$$\boxed{\text{(a) best rms} \approx 0.252; \qquad \text{(b) } \rho \gtrsim 0.994}$$

Part (b) explains why Table IV prints "+0.99" next to 0.028 and "+1.00" next to 0.018: at such small errors the correlation is forced toward 1, and two-decimal rounding cannot show the difference.

Checkpoint 30.2

A model has held-out correlation +0.28 and rms almost equal to the null. Is it useless?

Answer: Not entirely. A correlation of +0.28 means its predictions point weakly in the right direction; Eq. (30.14) shows that even the best scaling of such a prediction leaves an rms of about 0.96 times the null. So it carries a small amount of information but removes almost none of the error. This is the situation of control C1.

Key Ideas

- Table V: more primes always help, and skill persists through zero 10^4; correlation is nearly constant, the rms ratio drifts mildly upward (Table 30.3).
- One prime beats the best fitted operator by a factor of about 1.34 (Claim IV.2).
- rms $\geq \sigma_Y\sqrt{1-\rho^2}$, Eq. (30.14): a weak correlation cannot produce a small rms, and a small rms forces a correlation near 1.

30.4 Three levels of supplied information

30.4.1 Why the oracle alone is not enough

The paper is explicit that the oracle "is not a fair ML competitor (it encodes the known functional form); its role is informational" (Sec. II D), and again in Sec. IV: "it is given the functional form, so this comparison shows only that transferable structure exists and is arithmetic." The operators were given a background potential and a basis; the oracle was given the answer's shape. A critic could therefore say: of course a model told the answer's form wins. Claim IV.4 concedes this and the controls are designed to answer the critic.

The paper organizes all models "by the information supplied, at three levels":

(i) the arithmetic frequencies $m \log p$ supplied (the oracle, control A, and control B with fitted amplitudes);

(ii) no arithmetic frequencies, but sparsity in t-frequency supplied (C2);

(iii) a generic basis with no such prior (the dense Fourier model C1, and the fitted potentials).

Figure 30.3 summarizes where each model lands.

30.4.2 The common shape of the fitted controls

All three fitted controls (B, C1, C2) are described in Sec. II D ("Fitted sinusoidal controls") as having the shape

$$N_{\mathrm{sm}}(t) + F(t) = n - \tfrac{1}{2}, \qquad F(t) = \sum_j \left[a_j \sin(\omega_j t) + b_j \cos(\omega_j t)\right], \tag{30.15}$$

"trained by least squares on $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ at the training zeros only, and scored exactly as the oracle." Line by line:

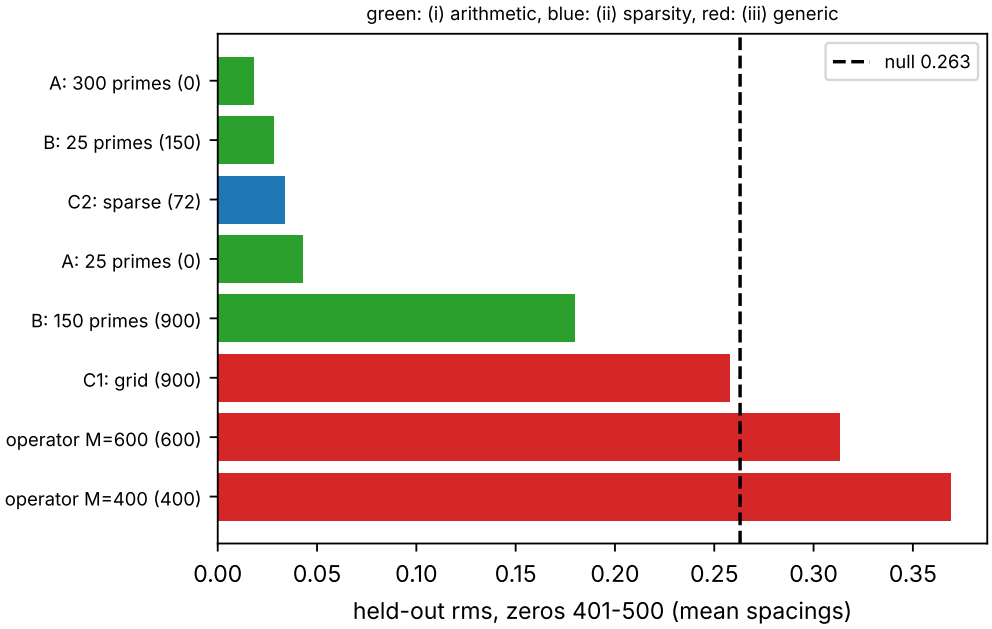


Figure 30.3: The three levels of supplied information of paper Sec. IV, with held-out rms on zeros 401–500 from paper Table IV (null 0.263). Level (i) is given the arithmetic frequencies; level (ii) is told only that the fluctuation is a sum of few sinusoids; level (iii) gets a generic basis. Skill tracks the prior, not the number of fitted parameters.

- *Shape.* Eq. (30.15) is the level equation (30.3) with N_{fl} replaced by a general sum of sinusoids F. The oracle is the special case $\omega_j = m \log p$, $a_j = -1/(\pi m p^{m/2})$, $b_j = 0$.

- *Target.* By Eq. (30.10), y_n is what $F(t_n)$ would have to be for the level equation to hold exactly at the true zero t_n. Fitting $F(t_n) \approx y_n$ at training zeros is the natural regression.

- *Linear in the amplitudes.* Once the frequencies ω_j are fixed, $F(t_n)$ is a linear combination of the known numbers $\sin(\omega_j t_n)$ and $\cos(\omega_j t_n)$. Stack them into a design matrix X with rows indexed by training zeros and columns by $(j, \sin)$ and $(j, \cos)$. The fit is then the linear least-squares problem $X\theta \approx y$ of Chapter 9, with $\theta = (a_1, \dots, b_1, \dots)$.

- *Ridge.* Appendix A step 20 gives the ridge filter $\Sigma/(\Sigma^2 + \lambda\Sigma_1^2)$: if $X = U\Sigma V^\top$ is the SVD with largest singular value σ_1, the solution is $\theta = V\,\mathrm{diag}(\sigma_i/(\sigma_i^2 + \lambda\sigma_1^2))U^\top y$. The grid of strengths is "$\lambda = 10^{-8}, \dots, 10^3$ in half decades": exponents $-8, -7.5, \dots, 3$, which is 23 values.

- *Selection.* "Ridge strength and K are chosen by a forward split inside the training window (first 75% fit, last 25% validate), after which the model is refitted on the whole training window; held-out zeros are not used before scoring." For 400 training zeros: fit 1–300, validate 301–400, pick the best λ (and, for C2, K), then refit on 1–400. The forward split imitates, inside the training window, the forward extrapolation that the real test will demand (Chapter 25).

- *Leakage test.* Appendix A step 20 adds: "Overwriting held-out data with noise before fitting leaves all fitted coefficients bit-identical." This is a direct, mechanical proof that no held-out zero influences any fitted number.

Sample Problem 30.5: Counting parameters and equations

For each model of Table IV, count fitted real parameters and compare with the number of training zeros.

Step 1: control A. Three powers m for each of 300 primes give 900 terms, all with fixed amplitudes: 0 fitted. The 300th prime is 1987, so the highest frequency is $3\log 1987 = 22.78$.

Step 2: control B, 25 primes. The 25th prime is 97. There are $25 \times 3 = 75$ frequencies, each with a sine and a cosine amplitude: 150 parameters, against 400 training zeros (or 800).

Step 3: control B, 150 primes. The 150th prime is 863: $150 \times 3 \times 2 = 900$ parameters against 400 zeros. The system $X\theta = y$ has more unknowns than equations, so (Chapter 9) its exact solutions form a family of dimension at least $900 - 400 = 500$. With 800 zeros it is still underdetermined, by at least 100.

Step 4: C1. 450 frequencies, two amplitudes each: 900, underdetermined in the same way.

Step 5: C2. K frequencies, each with a frequency, a sine and a cosine amplitude: $3K$. With $K = 24, 72$; with $K = 43, 129$. These are the "72, 129" of Table IV.

$A : 0;$	$B_{25} : 150;$	$B_{150} : 900;$	$C1 : 900;$	$C2 : 72$ or 129

Key Ideas

- The fitted controls replace N_{fl} by a sum of sinusoids F fitted to $y_n = n - \frac{1}{2} - N_{\text{sm}}(t_n)$, Eq. (30.15).
- With frequencies fixed, the fit is linear least squares with ridge; strength (and K) are chosen by a 75/25 forward split inside the training window.
- The controls span parameter counts from 0 to 900 and priors from "exact arithmetic" to "generic". That design lets the paper vary count and prior separately.

30.5 Control A: the oracle with a bigger budget

★ **Did you know?** *Paying for Parameters*

In the early 1970s Hirotugu Akaike proposed an information criterion, AIC, that rewards a model for fitting the data but charges a penalty proportional to its number of parameters. Gideon Schwarz's 1978 criterion, BIC, uses a stiffer penalty that grows with sample size. Both encode one lesson: when a bigger parameter budget improves a fit, ask whether it improves it more than chance would.

(a) Plain words. Table V stops at 25 primes. Control A asks what happens at the same *nominal* budget as the largest operator fit, 900 terms, with still nothing fitted. Answer (Sec. IV): "the oracle improves monotonically, to 0.018 on zeros 401–500." Table IV adds $0.020/+1.00$ on 801–1000 and the bootstrap interval $[0.015, 0.020]$ on 401–500.

(b) Tools. The explicit formula and its truncation (Chapter 18); bootstrap intervals (Chapter 26).

(c) Computation. The same loop as in Section 30.2 with `qmnt.first_primes(300)`. The cost is 900 sines per evaluation of N_{fl}, still trivial.

(d) Strength and limits. Control A addresses one specific objection: that the operators lose only because they have fewer "terms". At these heights the oracle with 900 terms is better

than with 75 terms (25 primes), so a large term count is not, in itself, harmful when the terms have the right form. Notice too that A involves no fitting, so it cannot overfit in the statistical sense: no parameter is tuned to the training zeros. That does not make more terms automatically better. On the critical line the full series does not converge (Chapter 18), so whether adding terms helps is an empirical question, answered affirmatively at these heights by Tables IV and V; it is not a theorem. The word "monotonically" refers to intermediate budgets that the paper does not tabulate; we take it as reported.

Pitfall

Do not read control A's 0.018 as "900 parameters generalize". Control A has 900 *terms* and 0 fitted parameters. The informative comparison for parameter count is between the fitted models (B, C1, C2, and the operators).

30.6 Control B: arithmetic frequencies, fitted amplitudes

30.6.1 The 25-prime version

(a) Plain words. Control B is given the frequencies $m \log p$ but not the amplitudes. It learns the amplitudes from the training zeros. With 25 primes (150 coefficients) learned from 400 zeros, it predicts zeros 401–500 at $0.028/+0.99$, "better than the unfitted 25-prime oracle" (0.043). From 800 zeros it scores $0.040/+0.99$ on 801–1000, against the oracle's 0.053 there (Table IV).

(b) Tools. Linear least squares and ridge (Chapter 9); sinusoidal regression with known frequencies (Chapter 27).

(c) Computation. The listing reruns control B with 25 primes. It validates λ by the rms residual of y in the validation block. The paper's text does not specify the validation metric (the reconstruction record [20] does), so this choice is ours.

```
import numpy as np, qmnt
from scipy.optimize import brentq
t = qmnt.zeros(500); n = np.arange(1, 501)
y = n - 0.5 - qmnt.Nsm(t)                              # targets y_n
P = qmnt.first_primes(25)
w = np.array([m*np.log(p) for p in P for m in (1, 2, 3)])   # 75 freqs
X = lambda tt: np.hstack([np.sin(np.outer(tt, w)), np.cos(np.outer(tt, w))])
def ridge(Xm, yv, lam):                                # SVD filter of App. A step 20
    U, s, Vt = np.linalg.svd(Xm, full_matrices=False)
    return Vt.T @ ((s/(s**2 + lam*s[0]**2)) * (U.T @ yv))
lams = 10.0**np.arange(-8, 3.01, 0.5)                  # 23 half decades
val = [np.sqrt(np.mean((X(t[300:400]) @ ridge(X(t[:300]), y[:300], L)
       - y[300:400])**2)) for L in lams]               # forward split 300/100
theta = ridge(X(t[:400]), y[:400], lams[np.argmin(val)])   # refit on 1-400
F = lambda tt: float(np.ravel(X(np.atleast_1d(tt)) @ theta)[0])
nh = np.arange(401, 501)
s_all, g_all = qmnt.smooth_zeros(nh), qmnt.mean_gap(nh)
def predict(k, s, g):
    G = lambda tt: qmnt.Nsm(tt) + F(tt) - (k - 0.5)
    a, b = s - 3*g, s + 3*g
    while G(a)*G(b) > 0:                               # expand bracket on failure
        a, b = a - g, b + g
    return brentq(G, a, b)
t_pred = np.array([predict(k, s, g) for k, s, g in zip(nh, s_all, g_all)])
print(qmnt.score(t_pred, nh))                          # (rms, corr) on 401-500
```

Running the listing gives a held-out rms below the six-prime oracle's 0.074 and a correlation above 0.95. It need not reproduce 0.028 to three decimals, because our validation metric may

differ from the paper's. It also returns a fitted sine amplitude for $\log 2$ within 0.05 of the oracle value $-1/(\pi\sqrt{2}) = -0.225$, and a cosine amplitude within 0.05 of zero.

(d) Strength and limits. Control B shows that, once the frequencies are given, the amplitudes are easy to learn from 400 zeros: 150 coefficients transfer forward with almost no loss. Why does fitting beat the exact amplitudes? The paper does not say. One plausible reading (ours, heuristic) is that a sharp truncation at 25 primes is not the best way to stop a non-convergent sum; rigorous smoothed explicit formulas taper the weights of large primes (Chapter 18), and fitted amplitudes can learn such a taper, and partially absorb the effect of omitted terms, inside the window.

30.6.2 The 150-prime version: underdetermined and root-ambiguous

(a) Plain words. With 150 primes, the fit has 900 coefficients but only 400 equations. The paper: "the least-squares problem is underdetermined and the model degrades to 0.180 with correlation +0.78, still below the null and positively correlated, unlike any operator row; with 800 training zeros the same model gives 0.100/+0.94." The bootstrap interval on 401–500 is $[0.158, 0.201]$.

Why does it degrade? An underdetermined system has infinitely many exact fits (Sample Problem in Section 30.4, Step 3). Ridge picks the one with smallest filtered norm, which spreads amplitude across all 900 columns instead of concentrating it on the correct values; many columns have frequencies that a window of length T cannot separate cleanly from their neighbors (Section 30.7). The fit therefore learns a mixture that agrees with the data in the window but continues imperfectly. Why does it still beat the null by a wide margin? The paper does not say. Our heuristic reading: every column has a correct arithmetic frequency, so whatever mixture the fit chooses is built from sinusoids that would continue correctly if their weights were right. The weights are partly wrong, and wrong weights spread over many columns can still continue poorly, which is presumably part of why the score is 0.180 rather than 0.028; but no column introduces a frequency foreign to the target. This is a plausible explanation, not a demonstrated one. With 800 zeros the problem is less underdetermined and the score improves to 0.100.

The root-ambiguity footnote. The dagger on this row in Table IV reads: "Root-ambiguous (see text); not a clean predictive score." The text explains: "the level equation has several roots for about a third of the held-out levels; the score varies by at most 0.012 with the choice of root." Here is what is going on. The level function $G(t) = N_{\mathrm{sm}}(t) + F(t) - (n - \frac{1}{2})$ is increasing wherever $F'(t) > -N'_{\mathrm{sm}}(t)$. A fitted F with large amplitudes a_j, b_j at many frequencies can have $|F'|$ larger than $N'_{\mathrm{sm}} \approx 0.75$, just as the worst-case oracle bound (30.12) warned. Then G wiggles, and the bracket $s_n \pm 3\mathrm{gap}_n$ can contain several roots. Brent returns one of them, but the "prediction of zero n" is no longer well defined. Figure 30.4 shows a synthetic example.

Sample Problem 30.6: How big must a single wiggle be to create extra roots?

Let $F(t) = A\sin(\omega t)$ with $\omega = 5$, near $t = 680$. Find the smallest A for which $G(t) = N_{\mathrm{sm}}(t) + F(t) - c$ can fail to be monotone.

Step 1: the slope. $G'(t) = N'_{\mathrm{sm}}(t) + A\omega\cos(\omega t)$. Its smallest value over a cycle is $N'_{\mathrm{sm}}(t) - A\omega$.

Step 2: the condition. G is monotone for all phases iff $A\omega < N'_{\mathrm{sm}}(t)$. At $t = 680$, $N'_{\mathrm{sm}} = 0.7455$ (Sample Problem "One oracle step by hand", Step 2).

Step 3: solve. $A > 0.7455/5 = 0.149$.

$$\boxed{A_{\min} \approx 0.149}$$

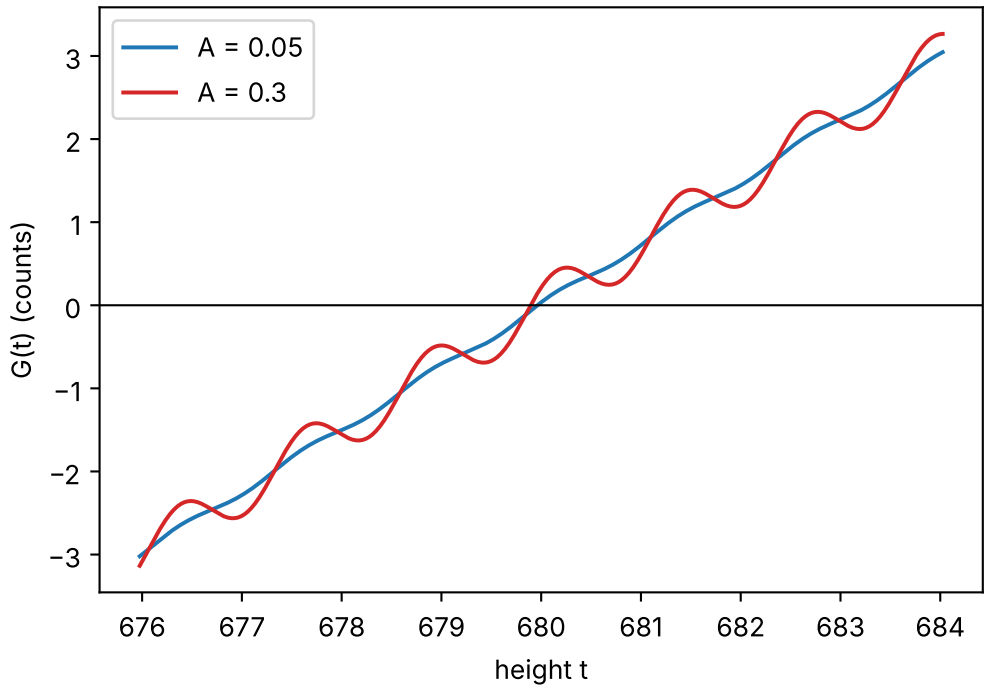


Figure 30.4: Synthetic illustration of root ambiguity (not paper data). The level function $G(t) = N_{\mathrm{sm}}(t) - N_{\mathrm{sm}}(680) + A\sin(5t)$ is plotted over 680 ± 3 mean spacings for two amplitudes. For $A = 0.05$, $|G' - N'_{\mathrm{sm}}| \le 0.25 < N'_{\mathrm{sm}} \approx 0.75$ and the root is unique. For $A = 0.3$, $A\omega = 1.5$ exceeds N'_{sm}, G is not monotone, and the level equation has several roots in the bracket. A true counting function is a nondecreasing staircase, so several roots signal that F is not behaving like a count.

An amplitude of 0.15 is smaller than the amplitude of the $p = 2$, $m = 1$ oracle term (0.225). High frequencies make wiggles cheap: the slope grows with ω. An underdetermined fit with many frequencies up to $3 \log 863 \approx 20.3$ does not need large amplitudes to break monotonicity.

(d) Strength and limits. The paper reports the row with its caveat attached, and the caveat is quantified: at most 0.012. Reading "varies by at most 0.012" in the most pessimistic way, as ± 0.012 around the reported value, the worst choice of root would give at most 0.192, still far below the null 0.263. The qualitative point of the row, that a badly underdetermined model built from the right frequencies still predicts with positive correlation, survives the ambiguity. The row is not a clean predictive score, and the paper says so.

Checkpoint 30.3

Control B with 900 coefficients on 400 zeros is underdetermined, and so is control C1. Why does B still reach 0.180 while C1 stays at 0.258?

Answer: Both pick, among infinitely many exact fits, one regularized combination of their columns. A plausible reading (ours, not the paper's, which gives no mechanism for this row): B's columns all carry frequencies present in the target, so a regularized combination has a chance of continuing beyond the window, although wrong weights can still continue poorly; C1's grid columns do not contain those frequencies exactly, so combinations tuned to the window have no reason to continue correctly. Same count, same underdetermination, different vocabulary.

Key Ideas

- Given the frequencies, amplitudes are easy to learn: 150 coefficients from 400 zeros give 0.028, better than the unfitted 25-prime oracle.
- Underdetermination (900 unknowns, 400 equations) degrades B to $0.180/+0.78$, still well below the null.
- Root ambiguity can arise only where $F' < -N'_{\mathrm{sm}}$, so that G decreases; several roots then occur if such a decreasing excursion straddles the level $n - \frac{1}{2}$. For this row it changes the score by at most 0.012.

30.7 Control C1: a dense uniform grid of frequencies

(a) Plain words. C1 has no arithmetic input: 450 frequencies on a uniform grid, $\omega_j = 8j/450$ for $j = 1, \ldots, 450$ (Appendix A step 20), with sine and cosine amplitudes fitted by ridge: 900 coefficients. The paper: it "stays within 0.005 of the null (0.258, correlation +0.28 with interval $[+0.08, +0.46]$): capacity without a suitable prior does not generalize, and in this respect the operator family is not special." On 801–1000 it scores $0.267/+0.22$ against a null of 0.274. Its rms interval on 401–500 is $[0.231, 0.285]$, which contains the null.

(b) Tools. Fourier resolution (Chapter 6); ridge filter factors (Chapter 9); the dense-grid control (Chapter 27).

(c) Computation and analysis. Recall from Chapter 6 that a window of length T resolves frequencies separated by about $2\pi/T$. For the training window of zeros 1–400, $T = t_{400} - t_1 \approx 666$, so $2\pi/T \approx 0.0094$. The grid spacing is $8/450 = 0.0178$, about 1.9 resolution widths. A grid of this spacing does not contain the arithmetic frequencies; it contains nearby ones.

Sample Problem 30.7: How far is each arithmetic frequency from the C1 grid?

For $\log 2$, $\log 3$ and $\log 5$, find the nearest grid frequency $\omega_j = 8j/450$, the offset $\delta\omega$, and the phase drift $\delta\omega\, T$ accumulated across the training window.
Step 1: $\log 2 = 0.693147$**.** $0.693147 \times 450/8 = 38.99$, so $j = 39$ and $\omega_{39} = 0.693333$; $\delta\omega = 0.000186$; drift $0.000186 \times 666 = 0.12$ rad.
Step 2: $\log 3 = 1.098612$**.** $1.098612 \times 56.25 = 61.80$, so $j = 62$, $\omega_{62} = 1.102222$; $\delta\omega = 0.00361$; drift 2.4 rad.
Step 3: $\log 5 = 1.609438$**.** $1.609438 \times 56.25 = 90.53$, so $j = 91$, $\omega_{91} = 1.617778$; $\delta\omega = 0.00834$; drift 5.6 rad, almost a full cycle.

drifts across the window: 0.12, 2.4, 5.6 rad for $\log 2$, $\log 3$, $\log 5$

A single grid column cannot follow the $p = 5$ term across the window, because it slips nearly a whole cycle. The fit can still mimic the $p = 5$ term inside the window by combining several neighboring columns, the way two nearby tones produce beats (first-year physics). But that combination is tuned to the window, and its continuation outside is a different beat pattern, not the $p = 5$ term.

This analysis is our reading of one mechanism, not the paper's stated explanation. The paper's own statement is more general: "capacity without a suitable prior does not generalize." Two further mechanisms point the same way. First, 900 coefficients on 400 zeros is underdetermined, so the fit chooses among infinitely many exact interpolants. Second, the forward split then selects a ridge strength. Recall the ridge filter: the coefficient along the i-th singular direction is multiplied by $\sigma_i/(\sigma_i^2 + \lambda\sigma_1^2)$ instead of $1/\sigma_i$. For $\lambda\sigma_1^2 \gg \sigma_i^2$ the factor is about $\sigma_i/(\lambda\sigma_1^2)$, tiny. If the validation block shows that the unregularized interpolant extrapolates badly, the selected λ is large and the predictions shrink toward zero, that is, toward the null. Sample Problem "Reading a correlation off an rms" showed that C1's 0.258 is close to the best rms (0.252) compatible with its correlation of $+0.28$, which is consistent with a weak prediction of small amplitude (our inference; the paper does not report the amplitude or the selected λ).

(d) Strength and limits. C1 is the cleanest answer to the parameter-count question: same count as the largest operator (900), same underdetermination, a basis that is even "Fourier" in name, and no skill to speak of. In Sec. VI the paper summarizes: "a dense 900-coefficient Fourier model gains almost nothing over the null." Note the word "almost". C1's correlation interval, $[+0.08, +0.46]$, excludes zero, which no operator row in Table I achieves reproducibly; C1 does extract a weak positive signal. The paper's statement that "in this respect the operator family is not special" refers to generalization of rms, where C1, like the operators, stays at the null.

Checkpoint 30.4

Suppose a reader says: "C1 failed because 450 frequencies are too few to cover $[0, 8]$." Using control C2's result, how would you reply?

Answer: C2 uses only 24 frequencies on the same range and predicts at 0.034. The issue is not how many frequencies, but whether they are the right ones, placed precisely. A dense grid spends its budget everywhere; a sparse search spends it where the data's spectrum actually has power.

30.8 Control C2: a sparse frequency search with no arithmetic input

30.8.1 The algorithm, line by line

Sec. II D describes C2 as "a sparse model whose K frequencies are found one at a time as the largest peak of the periodogram of the current residual on $0.05 < \omega < 8$, with all amplitudes refitted after each addition ($3K$ parameters)." Appendix A step 20 adds the details: "sparse search with step a quarter of $2\pi/T$, refinement on 201 points, joint refit after each added frequency, $K \le \min(300, n_{\text{fit}}/3)$." Unpacked:

1. *Start.* Residual $r_n = y_n$ at the training zeros, Eq. (30.10); no frequencies selected.
2. *Periodogram.* Recall from Chapter 6 the periodogram for irregular samples, $\mathcal{P}(\omega) = \left|\sum_n r_n e^{-i\omega t_n}\right|^2$. It is large at ω when r_n contains a sinusoid of that frequency, because the phases $e^{-i\omega t_n}$ then cancel the oscillation of r_n and the terms add coherently.
3. *Coarse search.* Evaluate $\mathcal{P}$ on a grid over $0.05 < \omega < 8$ with step $\frac{1}{4} \cdot 2\pi/T = \pi/(2T)$. For $T \approx 666$ the step is 0.00236, so the grid has about 3370 points. With four grid points per resolution width, every peak (width about $2\pi/T$) is sampled several times, so the coarse maximum is unlikely to miss a peak; the refinement step then locates it precisely. (This makes a miss unlikely; it does not strictly guarantee against one, for example when two peaks of nearly equal height compete.)
4. *Refinement.* Around the coarse maximum, search 201 finer points and take the maximum there. (The exact refinement interval is in the reconstruction record [20].)
5. *Joint refit.* Add the refined frequency to the list and refit *all* amplitudes a_j, b_j of all selected frequencies together by least squares. Update r_n to the new residual. The joint refit matters: two nearby sinusoids leak into each other's periodogram peaks, and refitting them together removes the leakage that a one-at-a-time subtraction would leave.
6. *Repeat* up to $K_{\max} = \min(300, n_{\text{fit}}/3)$: 133 for 400 training zeros, 266 for 800. The cap $n_{\text{fit}}/3$ ensures $3K \le n_{\text{fit}}$, so the final fit is never underdetermined.
7. *Select K and ridge* by the forward split, then refit on the whole window.

Step 5 is the connection to matching pursuit (Chapter 27): pick the best single atom, refit, repeat.

```
import numpy as np, qmnt
t = qmnt.zeros(400); n = np.arange(1, 401)
y = n - 0.5 - qmnt.Nsm(t)                          # targets y_n
T = t[-1] - t[0]                                    # training window length
d = 0.25*2*np.pi/T                                  # coarse step
coarse = np.arange(0.05, 8.0, d)
def design(w):
    return np.hstack([np.sin(np.outer(t, w)), np.cos(np.outer(t, w))])
def peak(r, grid):                                  # largest periodogram peak
    Pw = np.abs(np.exp(-1j*np.outer(grid, t)) @ r)**2
    return grid[np.argmax(Pw)]
freqs, r = [], y.copy()
for step in range(10):
    w0 = peak(r, coarse)
    w1 = peak(r, np.linspace(w0 - d, w0 + d, 201))   # refinement
    freqs.append(w1)
```

```
    X = design(np.array(freqs))
    coef, *_ = np.linalg.lstsq(X, y, rcond=None)      # joint refit
    r = y - X @ coef
print(np.round(freqs, 4))
print(np.round(np.exp(freqs), 2))                 # compare with prime powers
```

The listing omits ridge and the choice of K, which only matter once many frequencies have been selected. Running it, the first frequency selected is within π/T of $\log 2$, and each of the first five lies within π/T of the logarithm of a prime power. The last line prints e^{ω}, which for a matched frequency is close to the prime power q itself. Figure 30.5 shows the periodogram that the very first step sees.

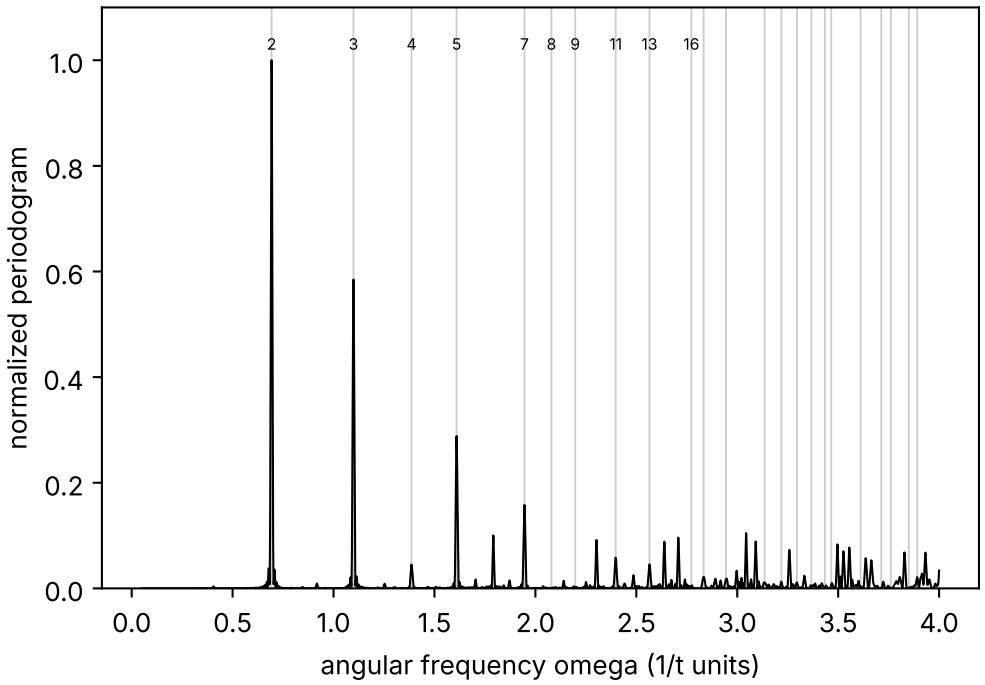


Figure 30.5: Periodogram $\mathcal{P}(\omega) = |\sum_n y_n e^{-i\omega t_n}|^2$ of the targets $y_n = n - \frac{1}{2} - N_{\mathrm{sm}}(t_n)$ at the first 400 zeros, normalized to its maximum, for $0.05 < \omega < 4$ (angular frequency in units of inverse height t). Vertical lines mark $\log q$ for prime powers $q \le 50$, labeled for $q \le 16$. The tallest peak sits at $\log 2$, and the next tallest are at logarithms of other small prime powers: the first step of control C2 sees the primes without being told about them.

30.8.2 What C2 found

The paper (Sec. IV): C2 "selects 24 frequencies (72 parameters) from 400 zeros and predicts zeros 401–500 at 0.034 with correlation +0.99, the level of the 25-prime oracle; from 800 zeros it selects 43 frequencies and scores 0.034/+0.99 on 801–1000." The bootstrap interval on 401–500 is $[0.028, 0.041]$ (Table IV caption). Then: "Every selected frequency lies within half the Fourier resolution of the training window of the logarithm of a prime power (for the 800-zero fit, 43 distinct prime powers: those up to 113 except 32, 64, and 81, and six primes between 131 and 211)."

"Half the Fourier resolution" is $\frac{1}{2} \cdot 2\pi/T = \pi/T$, the matching criterion $|\omega - \log q| < \pi/T$ of Appendix A step 23. For the 800-zero window, $T = t_{800} - t_1 \approx 1170$, so $\pi/T \approx 0.0027$.

Sample Problem 30.8: Checking the list of 43 prime powers

(a) Verify that "prime powers up to 113 except 32, 64, 81, plus six primes" makes 43. (b) Compute the weights $w(q) = 1/(mp^{m/2})$ of the excluded and of some included prime powers, and see whether weight explains which were found.
Step 1: primes up to 113. There are 30 of them $(2, 3, \ldots, 109, 113)$.
Step 2: higher prime powers up to 113. $4, 8, 16, 32, 64$; $9, 27, 81$; 25; 49. ($121 = 11^2$ exceeds 113.) That is 10.
Step 3: count. $30 + 10 - 3 + 6 = 43$. ✓
Step 4: weights of the excluded ones. $w(32) = 1/(5 \cdot 2^{5/2}) = 0.0354$; $w(64) = 1/(6 \cdot 8) = 0.0208$; $w(81) = 1/(4 \cdot 9) = 0.0278$.
Step 5: weights of the faintest included ones. $w(16) = 1/(4 \cdot 4) = 0.0625$; $w(27) = 1/(3 \cdot 27^{1/2}) = 0.0642$; $w(211) = 1/\sqrt{211} = 0.0688$; $w(49) = 1/(2 \cdot 7) = 0.0714$.

$$\boxed{30 + 10 - 3 + 6 = 43; \quad \text{excluded: } w \le 0.0354; \quad \text{included: } w \ge 0.0625}$$

Up to 113 the separation is clean: every prime power of weight at least 0.0625 was found, and the three of weight at most 0.0354 were not. Weight is not the whole story, however. All 17 primes from 127 to 211 have $w \ge 0.0688 > w(16)$, yet only six were found. At higher frequency the logarithms of neighboring primes crowd together: $\log 139 - \log 137 = 0.0145$, only 2.7 resolution widths $2\pi/T \approx 0.0054$, and $\log 128 - \log 127 = 0.0078$, only 1.5. A finite window struggles to separate them. Note also that $16 = 2^4$ has $m = 4$, beyond the $m \le 3$ of Eq. (30.1): C2 found a term the oracle omits, as the untruncated explicit formula says it should. The paper does not say which six primes were selected, so we cannot test this reading further.

Checkpoint 30.5

C2 at 0.034 beats the 25-prime oracle (0.043) on zeros 401–500, although it was never told about primes. Give two reasons this is possible.

Answer: First, it fits amplitudes, which (as control B showed) helps relative to the sharply truncated exact amplitudes. Second, it is not limited to the 75 frequencies $m \log p$ with $p \le 97$, $m \le 3$: it can pick, for example, $\log 16$ ($m = 4$) if the data demand it. Its 24 frequencies are chosen where the residual has the most power.

Key Ideas

- C2: greedy periodogram peaks on $0.05 < \omega < 8$, quarter-resolution coarse grid, 201-point refinement, joint refit, $K \le \min(300, n_{\text{fit}}/3)$; K and ridge by forward split.
- With 72 parameters and no arithmetic input it predicts at $0.034/+0.99$, the level of the 25-prime oracle.
- Its frequencies match $\log p^m$; which prime powers it finds is consistent with (our heuristic) their weights $1/(mp^{m/2})$ and the window's resolution.

30.9 Is the alignment real? The surrogate check

30.9.1 Why a check is needed

"Every selected frequency lies near $\log p^m$" sounds decisive. But the frequencies are the *output of a search*, and the logarithms of prime powers are not sparse at large ω. A frequency chosen

at random could land near some $\log q$ by chance. How likely is that?

Sample Problem 30.9: The chance that a random frequency matches some prime power (heuristic)

Estimate the expected number $\nu(\omega)$ of prime powers q with $|\omega - \log q| < \pi/T$, for the 400-zero window ($2\pi/T \approx 0.0094$), at $\omega = 2$, $\omega = \log 113$ and $\omega = 8$, and the probability of at least one match.

Step 1: density of $\log q$. By the prime number theorem (Chapter 15), the number of primes in $[x, x + \mathrm{d}x]$ is about $\mathrm{d}x/\log x$. With $x = e^\omega$, $\mathrm{d}x = e^\omega \mathrm{d}\omega$, so the number of $\log p$ in $[\omega, \omega + \mathrm{d}\omega]$ is about $e^\omega \mathrm{d}\omega/\omega$. Higher prime powers are much rarer, and we neglect them.

Step 2: expected count. The matching window has width $2\pi/T$, so $\nu(\omega) \approx (2\pi/T)\, e^\omega/\omega$. (We write ν rather than λ, which in this chapter is the ridge strength.)

Step 3: evaluate. $\omega = 2$: $0.0094 \times 7.389/2 = 0.035$. $\omega = \log 113 = 4.727$: $0.0094 \times 113/4.727 = 0.23$. $\omega = 8$: $0.0094 \times 2981/8 = 3.5$.

Step 4: probability of at least one match. Treat the $\log q$ as points scattered independently at random with mean count ν per matching window (a heuristic). For such a Poisson process (Chapter 19) the probability that a window is empty is $e^{-\nu}$, so the probability of at least one match is $\Pr \approx 1 - e^{-\nu}$: 0.034, 0.20, 0.97.

$$\boxed{\Pr(\text{chance match}) \approx 0.034\ (\omega = 2), \quad 0.20\ (\omega = \log 113), \quad 0.97\ (\omega = 8)}$$

Near $\omega = 8$ almost any frequency "matches a prime power" ($e^8 \approx 2981$, which is presumably why the matching uses $q \le 3000$: the search range ends at $\omega = 8$). At low ω a chance match is rare. So a list of matches is impressive mainly for its low-frequency members, and how impressive it is overall depends on details that only a careful null can settle. This is a back-of-envelope estimate, not the paper's statistic.

There is a second, subtler worry. The *procedure* itself, periodogram peaks of a residual sampled at the zeros, might prefer certain frequencies for reasons unrelated to the signal. Recall from Chapter 18 Landau's formula: sums like $\sum_n e^{-i\omega t_n}$ over the zeros themselves peak at $\omega = \log p^m$ [18, 19]. Since the zeros are the sample points, one must rule out that the alignment is produced by the sampling rather than by the target.

30.9.2 What the paper did

The paper: "Because the frequencies are the output of a search, this alignment was checked by rerunning the complete procedure on surrogate zero sets: phase-randomized surrogates and two families of sparse signals with non-arithmetic frequencies, 900 replicates in all. None approaches the alignment of the real zeros." Appendix A step 23 gives the composition: 200 replicates of each family for the 400-zero fit and 100 of each for the 800-zero fit, so $3 \times 200 + 3 \times 100 = 900$. The three families are:

- *Sorted phase-randomized surrogates* (Chapter 26, [16]): phase-randomized copies of the real fluctuation sequence, which preserve its power spectrum before sorting and carry no arithmetic content in their phases. They are then sorted, presumably so that the surrogate zeros form an increasing sequence; sorting changes the sequence, so the spectrum is no longer preserved exactly. The paper's text gives no further detail (see the reconstruction record [20]).
- *Explicit formulas with 25 displaced prime frequencies*: signals with the explicit formula's sparse structure but with each $\log p$ moved off its arithmetic value.

- *Explicit-formula shapes with 25 random base frequencies*: the same kind of sparse, harmonic structure built on random base frequencies.

The two sparse families are the sharp controls: they share the sparse, harmonic structure of the explicit formula but not its arithmetic frequencies (details in the reconstruction record). If C2 still returned frequencies near $\log q$ on them, the alignment would be an artifact of the search or of the sampling.

The result is reported as a "cutoff-maximized binomial score 24.7 and 50.9 on the real zeros, at most 4.7 on surrogates (score defined after the data were seen)." The construction of the score, which does not depend on a chosen frequency cutoff, is documented in the reconstruction record [20]; the paper's text does not give the formula, and we do not guess it.

30.9.3 The status of the check

The paper attaches two explicit qualifications, and you should keep both.

1. "The statistic was defined after the data were seen, so the check is exploratory." Recall from Chapter 26: once a statistic has been chosen while looking at the result, its value on the real data cannot be turned into a significance level, because the choice itself may have favored the data.

2. "No significance claim is made here beyond the size of the ensembles." With N surrogates of one family and none reaching the real value, the most a permutation-style argument could ever give is an empirical tail probability of about $1/(N+1)$; for $N = 200$ that is about 0.005. The paper does not even claim this, because of qualification 1.

Pitfall

Do not describe C2 as "discovering the primes in the zeros". The paper is explicit: "That the Fourier transform of the zeros is concentrated at logarithms of prime powers is classical [18, 19]; C2 is a control built on that fact, not a new result, and 'sparse in t-frequency' is itself a strong and well-chosen prior." C2's role is to show that a fitting procedure *can* learn the transferable structure from 400 zeros when its prior is right; it is not a claim about number theory.

Checkpoint 30.6

Suppose, on the "displaced prime frequency" surrogates, C2 had returned frequencies near the true $\log p$ rather than near the displaced ones. What would that have meant?

Answer: That the procedure is drawn to $\log p$ regardless of the target, for example through the sampling at the zeros (Landau's peaks) or through the matching tolerance. The alignment on the real zeros would then be an artifact, not evidence that C2 learned the target's frequencies. The paper reports that no surrogate approached the real alignment.

30.10 Dephasing: why C2's skill fades with distance

★ Did you know? *Rounding Off the Weather*

Around 1961 Edward Lorenz restarted a weather simulation from numbers he had typed in rounded to three decimal places. Before long the new run diverged completely from the old one. His 1963 paper showed that deterministic systems can lose predictability exponentially fast. Forecast skill that fades with distance is a general phenomenon, from

the weather to sums of many oscillations.

(a) Plain words. The paper: "Its skill also decays with distance from the training window as the estimated frequencies dephase (0.078/+0.96 on zeros 1001–2000 for the 400-zero fit), which the exact frequencies of the oracle do not." Compare Table V on the same holdout: six primes 0.091, twenty-five primes 0.054. On 401–500 C2 beat the 25-prime oracle (0.034 against 0.043); on 1001–2000 it has fallen behind it (0.078 against 0.054).

(b) Tools. Frequency estimation from a finite window (Chapters 6 and 27).

(c) The mechanism, derived. Suppose the true term is $a\sin(\omega t)$ and the model has learned $a\sin(\hat{\omega}t+\hat{\phi})$ with $\hat{\omega}=\omega+\delta\omega$. The learned phase then differs from the true one by $\delta\omega\, t+\hat{\phi}$. Where does least squares put $\hat{\phi}$? A phase error φ costs a mean square error $2a^2\sin^2(\varphi/2)\approx a^2\varphi^2/2$ for small φ (we derive this in Eq. (30.18) below; that derivation does not use the choice of $\hat{\phi}$). So, with the frequency fixed, the fit makes the average of $(\hat{\phi}+\delta\omega\, t_n)^2$ over the training zeros as small as possible. The derivative of that average with respect to $\hat{\phi}$ is $2(\hat{\phi}+\delta\omega\,\bar{t})$, where $\bar{t}$ is the mean height of the training zeros; setting it to zero gives $\hat{\phi}=-\delta\omega\,\bar{t}$, so the phase error vanishes at $\bar{t}$. Call this point t_c. Because zeros are slightly denser higher up, t_c lies somewhat above the midpoint of the window; for round numbers below we use the midpoint, which does not change the order of magnitude. (We also assume the amplitude is learned exactly; an amplitude error adds a separate error that does not grow with distance.) So write the learned term as $a\sin\big(\omega t+\varphi(t)\big)$ with

$$\varphi(t)=\delta\omega\,(t-t_c). \tag{30.16}$$

The error from this one term is, by the identity $\sin X-\sin Y=2\cos\frac{X+Y}{2}\sin\frac{X-Y}{2}$,

$$e(t)=a\big[\sin(\omega t+\varphi)-\sin(\omega t)\big]=2a\cos\Big(\omega t+\tfrac{\varphi}{2}\Big)\sin\tfrac{\varphi}{2}. \tag{30.17}$$

Over one cycle, the mean square of cos is $\frac{1}{2}$, and φ hardly changes during a cycle, so the rms error is

$$e_{\mathrm{rms}}(t)=\sqrt{2}\,a\left|\sin\frac{\varphi(t)}{2}\right|\approx\frac{a}{\sqrt{2}}\,|\delta\omega|\,|t-t_c|\quad(\varphi\ll 1). \tag{30.18}$$

The error grows linearly with distance from the window center until φ approaches π, where the learned term is fully out of phase. The oracle's frequencies are exact, $\delta\omega=0$, so it suffers no dephasing at all; its slow decline in Table V has the different cause discussed in Section 30.3. Figure 30.6 shows the effect with an exaggerated frequency error.

Sample Problem 30.10: How fast does one dephased term lose skill?

Take the dominant term, $p=2$, $m=1$, with amplitude $a=1/(\pi\sqrt{2})=0.225$ mean spacings (by Eq. (30.9), amplitudes of N_{fl} are amplitudes of the predicted fluctuation). Assume a hypothetical frequency error $\delta\omega=10^{-4}$. Compute the rms error at distances $|t-t_c|=400$ and 1600.

Step 1: phases. $\varphi=10^{-4}\times 400=0.04$ rad and $10^{-4}\times 1600=0.16$ rad.

Step 2: prefactor. $\sqrt{2}\,a=\sqrt{2}/(\pi\sqrt{2})=1/\pi=0.3183$.

Step 3: errors. $0.3183\sin(0.02)=0.0064$ and $0.3183\sin(0.08)=0.025$.

$$\boxed{e_{\mathrm{rms}}\approx 0.0064 \text{ at } 400,\qquad e_{\mathrm{rms}}\approx 0.025 \text{ at } 1600}$$

A fourfold increase in distance gives a fourfold increase in error, as Eq. (30.18) predicts for small φ. The distances are representative: the training window 1–400 spans $t\approx 14$ to 680, so $t_c\approx 347$; zeros 401–500 lie a few hundred units above it, and zeros 1001–2000 run from just above $t_{1000}=1419.42$ (paper Sec. II A) to about $t=2515$. The value of $\delta\omega$ is

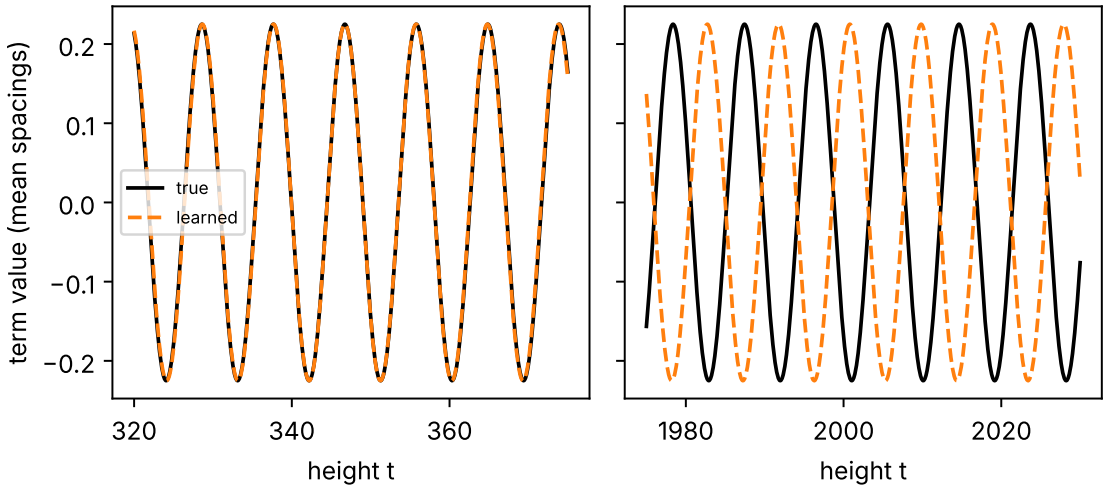


Figure 30.6: Dephasing of a learned frequency (synthetic; the frequency error is exaggerated for visibility). Black: the true $p = 2$ term $a\sin(t\log 2)$ with $a = 1/(\pi\sqrt{2})$, in mean spacings. Orange dashed: a learned copy with frequency error $\delta\omega = 2\times10^{-3}$, phase pinned at the training-window center $t_c = 347$. Left: near the window center the two agree. Right: at $t \approx 2000$, about 1650 units of t away, the phase error $\varphi = \delta\omega(t - t_c)$ is about 3.3 rad and the learned term is nearly inverted.

invented for illustration; the paper does not report C2's frequency errors.

Checkpoint 30.7

The matching criterion allows $|\delta\omega|$ up to $\pi/T \approx 0.0047$ for the 400-zero window. If C2's dominant frequency were off by that much, what would happen at $|t - t_c| = 1600$? What does the observed correlation +0.96 on 1001–2000 then tell you?

Answer: $\varphi \approx 0.0047 \times 1600 \approx 7.5$ rad, more than a full cycle: the dominant term would be completely dephased and the prediction would have lost much of its skill. Since C2 still correlates +0.96 there, its dominant frequencies must be estimated far more precisely than the matching tolerance. The tolerance is a criterion for naming a frequency, not the accuracy actually achieved.

(d) Strength and limits. The paper reports one far holdout (1001–2000) for C2; Sec. VI notes that "the fitted sinusoidal controls were scored only below zero 2000." How C2 behaves at $n = 10^4$, where the oracle is still at +0.98, is not measured. Dephasing is also what you should expect from any model that *estimates* frequencies from a finite window; it is a quantitative cost of not knowing the arithmetic, and it shows that knowing it exactly (the oracle) is worth something that 400 zeros cannot fully supply.

Key Ideas

- A frequency error $\delta\omega$ produces a phase error $\delta\omega(t - t_c)$ that grows with distance from the training window, Eq. (30.16).
- The resulting rms error grows linearly at first, Eq. (30.18); exact frequencies do not dephase.
- C2 falls from 0.034 to 0.078 between 401–500 and 1001–2000, while the oracle does not dephase.

30.11 The obstacle is the parametrization, not the parameter count

30.11.1 Assembling the argument

Claim IV.10 appears in the abstract ("the obstacle is the parametrization, not the parameter count"), in Sec. IV, and in Sec. VI. Here is the argument, piece by piece, with its evidence.

1. *Many terms are not harmful if they have the right form.* Control A improves to 0.018 at 900 terms.
2. *Many fitted parameters are not harmful if the prior is right.* Control B with 150 fitted coefficients reaches 0.028; even underdetermined at 900, it stays far below the null (0.180, positively correlated).
3. *Many fitted parameters with a generic prior did not generalize in any of the generic parametrizations tested* (cosine, Chebyshev and hat bases for the potential; the uniform Fourier grid C1). C1 with 900 coefficients: 0.258, within 0.005 of the null. The operators with 400–900 coefficients: 0.279 to 0.369 in Table IV, never below the null.
4. *Few fitted parameters with a strong but non-arithmetic prior suffice.* C2 with 72: 0.034.
5. *The arithmetic frequencies need not be supplied.* C2 finds them.

Put together: at a fixed count of 900, scores range from 0.018 (A) through 0.180 (B) and 0.258 (C1) to 0.302 (the $M = 900$ operator, on 801–1000). And the best fitted model has only 72 parameters. Count does not organize the results; prior does. Figure 30.7 shows the 401–500 column of Table IV.

Table 30.4: Paper Table IV, quoted exactly: rms / correlation on the first held-out window after the training zeros. "Fitted" counts fitted real parameters. Nulls: 0.263 (401–500) and 0.274 (801–1000). †Root-ambiguous.

model	fitted	fit 1–400, 401–500	fit 1–800, 801–1000
operator, cosine, $M = 400$	400	0.369 / −0.07	
operator, cosine, $M = 600$	600	0.313 / −0.16	0.279 / −0.03
operator, cosine, $M = 900$	900		0.302 / +0.08
A: oracle, 25 primes	0	0.043 / +0.99	0.053 / +0.98
A: oracle, 300 primes (900 terms)	0	0.018 / +1.00	0.020 / +1.00
B: 25 primes, amplitudes fitted	150	0.028 / +0.99	0.040 / +0.99
B: 150 primes, amplitudes fitted†	900	0.180 / +0.78	0.100 / +0.94
C1: uniform frequency grid	900	0.258 / +0.28	0.267 / +0.22
C2: sparse frequency search	72, 129	0.034 / +0.99	0.034 / +0.99

30.11.2 "What fails is a potential $V(x)$ as the parametrization"

The paper closes Sec. IV: "The form is learnable from 400 zeros by a model told only that the fluctuation is a sum of few sinusoids in t; what fails is a potential $V(x)$ as the parametrization, as the continuation probe of Sec. III shows from the other side." Recall the probe from Chapter 29: trained on a pure $p = 2$ oscillation (RE2) on levels 1–400, the operator family fits it in-sample (rms 0.056, correlation +0.95) and continues it not at all on 401–500 (correlation +0.07). C2 and the probe are mirror images. C2: right vocabulary, the frequency is found from the data and continues. Probe: wrong vocabulary, the frequency is painted onto one window and does not continue.

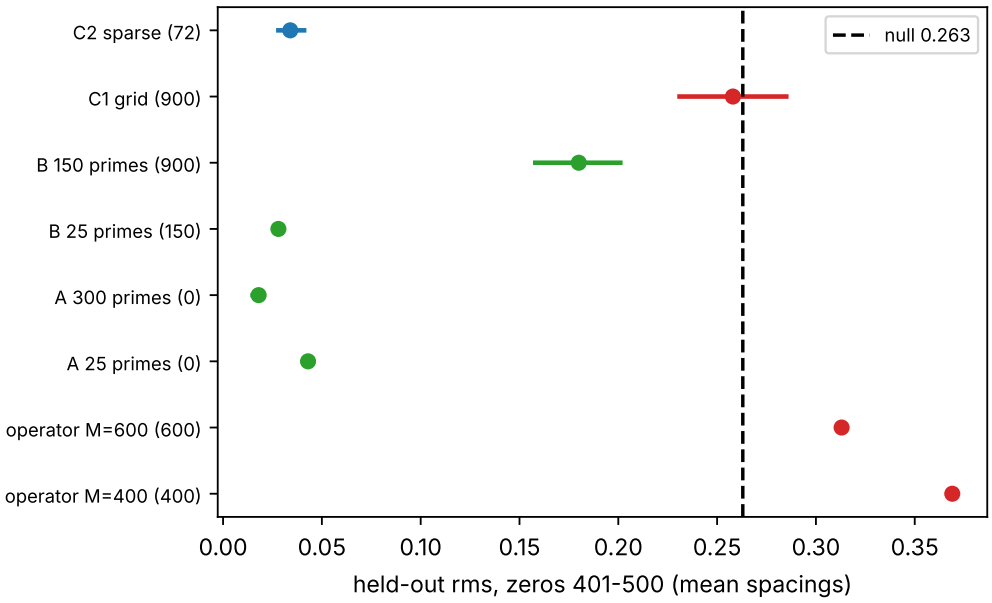


Figure 30.7: Held-out rms on zeros 401–500 (fit on zeros 1–400) for every model of paper Table IV with an entry in that column, colored by level of supplied information (green: arithmetic frequencies; blue: sparsity only; red: generic). Bars show the bootstrap 95% intervals reported in the Table IV caption where available. The dashed line is the null 0.263. The number in each label is the count of fitted parameters; skill is unrelated to it.

Sec. VI gives the heuristic reading, and labels it as heuristic: "the quantity being fitted is, by the explicit formula, an almost-periodic function of t that is sparse in frequency, whereas a potential addresses the levels through their turning points, locally in energy." In the language of Chapters 12 and 5: a bump in δV near the turning point x_E shifts the levels whose energies turn near x_E, a local effect in level index. An almost-periodic function like $\sin(t \log 2) + \sin(t \log 3)$ never repeats (Chapter 5); continuing it requires identifying its frequencies, which is exactly what C2 does from a finite window. A family that addresses levels locally through their turning points has no mechanism, under the protocol tested, that encodes those frequencies, so it can match a window without extending it (a heuristic reading, like the paper's).

30.11.3 Strengths and limits

Strengths.

- The controls vary prior and count separately, on the same targets, splits and scoring as the operators, with the same null.
- Held-out data are provably unused before scoring (the bit-identical test of Appendix A step 20).
- Bootstrap intervals are given for the decisive rows (A, B, C1, C2 on 401–500).
- The one surprising positive result, C2's alignment with $\log p^m$, was checked against 900 surrogates, and its post hoc status is stated.

Limits (most stated by the paper itself, Sec. VI).

- The fitted controls were scored only below zero 2000; the operator fits use at most the first 1000 zeros.

- The alignment statistic was defined after the data were seen; the check is exploratory and carries no significance claim.
- The 900-coefficient B row is root-ambiguous; its score is not clean (though bounded within 0.012).
- "The obstacle is the parametrization" is supported for the parametrizations tested (one background, three bases, one optimizer protocol); Sec. VI is explicit that "this is evidence about the tested parametrizations and optimizer together, not a statement about what a potential can represent," and that whether the memorization result can be derived "is an open problem."
- C2's prior, sparsity in t-frequency, is itself informed by the classical fact it recovers; the paper calls it "strong and well-chosen".
- The bootstrap intervals resample held-out levels independently, although fluctuations are serially correlated (Sec. VI: "except where stated").

Checkpoint 30.8

A reader argues: "C2 only works because sparsity in frequency is secretly arithmetic knowledge, so the comparison with the operators is unfair." Is this objection fatal to Claim IV.10?

Answer: No. The paper concedes that the prior is strong and well chosen, and builds on a classical fact. Claim IV.10 does not say the operators lack a fair chance; it says that their failure is not explained by parameter count, because C1 shows count without the right prior fails and C2 shows the right prior with few parameters succeeds. The objection sharpens the conclusion: the right prior matters, and a potential in a generic basis does not supply it.

Key Ideas

- At a fixed 900 parameters or terms, scores range from 0.018 to 0.302; the best fitted model has 72 parameters.
- Prior, not count, organizes Table IV.
- C2 and the continuation probe of Sec. III show the same point from opposite sides; the explanation in terms of locality in energy versus sparsity in frequency is heuristic and labeled so.

30.12 Paper connection

This chapter is itself a reading of the paper; here we collect the exact correspondences so you can find every item in R1.

- **Abstract.** The sentences "By contrast, a truncation of the Riemann–von Mangoldt explicit formula with zero fitted parameters predicts held-out zeros with correlation up to +0.99 through zero 10^4" and "Controls with fitted sinusoidal models show that the obstacle is the parametrization, not the parameter count" are Claims IV.1, IV.3 and IV.10 (Sections 30.3 and 30.11).
- **Sec. I.** "The information needed to predict unseen zeros is available, but it is arithmetic; within the operator families tested, the fits do not recover it, although a model with a non-arithmetic sparsity prior does (Sec. IV)." Fig. 1 (operator −0.07, six-prime oracle +0.96) is rerun in Figure 30.1.

- **Sec. I A, paragraph "Explicit formulæ for the n-th zero".** The oracle is an "information-existence baseline" with no claimed novelty [3, 4]; Connes' prime-only construction [21] is "on the arithmetic side of our comparison". We kept this framing in Section 30.4.

- **Sec. II D and Eq. (4).** The oracle Eq. (30.1) and level equation Eq. (30.3); paragraph "Fitted sinusoidal controls" defines A, B, C1, C2, the target y_n, and the 75/25 forward split (Section 30.4).

- **Sec. II A and Eq. (1).** N_{sm}, smooth zeros, and "all errors below are quoted in units of the local mean spacing", the scoring of Section 30.2.

- **Sec. IV, first paragraph.** Claims IV.1–IV.4 and the three levels of information (Sections 30.3 and 30.4).

- **Sec. IV, paragraph "Comparable budgets and fitted controls".** Controls A (Section 30.5), B and root ambiguity (Section 30.6), C1 (Section 30.7), C2 (Section 30.8), the surrogate alignment check (Section 30.9), dephasing (Section 30.10), and the closing "what fails is a potential $V(x)$ as the parametrization" (Section 30.11).

- **Table IV.** Quoted as Table 30.4, plotted in Figure 30.7; its caption's bootstrap intervals A $[0.015, 0.020]$, B $[0.158, 0.201]$, C1 $[0.231, 0.285]$, C2 $[0.028, 0.041]$, and its dagger.

- **Table V.** Quoted as Table 30.2, rerun with `qmnt`, plotted in Figure 30.2; caption's six-prime intervals $[+0.95, +0.97]$ (401–500) and $[+0.91, +0.92]$ (5001–10000).

- **Sec. II C, "Within-height splits".** "the null grows from 0.263 to 0.296 over Table V", the first column of Table 30.2.

- **Sec. III, "Arithmetic-continuation probe".** The mirror image of C2 (Section 30.11).

- **Sec. VI.** "It is not the parameter count ... It is not that the arithmetic frequencies must be supplied"; the heuristic reading in terms of almost-periodic, frequency-sparse targets versus turning-point locality; the limits (controls scored only below zero 2000; bootstrap independence).

- **Appendix A step 8.** Oracle root finding: Brent bracketed ± 3 gaps around the smooth zero, bracket expanded on sign failure.

- **Appendix A step 20.** Ridge filter and λ grid; C1 grid $\omega_j = 8j/450$; C2 step, refinement, joint refit, $K \le \min(300, n_{\mathrm{fit}}/3)$; the bit-identical leakage test.

- **Appendix A step 23.** Surrogate families and replicate counts (200 and 100 each), matching criterion $|\omega - \log q| < \pi/T$, $q \le 3000$, scores 24.7 and 50.9 against at most 4.7, "score defined after the data were seen".

- **Appendix A, Key anchors.** Sparse model $K = 24$ and $K = 43$ (also stated in Sec. IV).

- **References used in Sec. IV.** Sec. IV cites [18, 19], for the classical concentration of the zeros' Fourier transform at $\log p^m$, and [20], for the reconstruction record. The surrogate methods [16, 17] are cited in Sec. III (permutation and surrogate tests) and underlie step 23.

30.13 Summary

- The oracle solves $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$, Eqs. (30.1)–(30.3), with no fitted parameters and no zero data; its predicted fluctuation in mean spacings is $-N_{\mathrm{fl}}$ at the predicted zero, Eq. (30.9).
- The slope of N_{fl} is comparable with $N'_{\mathrm{sm}} = \frac{1}{2\pi}\log\frac{t}{2\pi}$, Eq. (30.7), so the level equation must be solved; uniqueness is guaranteed when $B_P < N'_{\mathrm{sm}}$, Eq. (30.12).
- Table V: more primes always help; skill persists through zero 10^4, with correlation nearly constant and the rms ratio drifting mildly (Table 30.3). One prime beats the best operator by about 1.34.
- rms $\geq \sigma_Y\sqrt{1-\rho^2}$, Eq. (30.14), links the two scores.
- The fitted controls fit F of Eq. (30.15) to y_n of Eq. (30.10) by ridge least squares, with strength and K chosen by a forward split.
- A (0 fitted, 900 terms): 0.018. B (150 fitted): 0.028; B (900, underdetermined, root-ambiguous): 0.180. C1 (900, generic grid): 0.258, near the null. C2 (72, sparse search): 0.034, frequencies at $\log p^m$, checked against 900 surrogates in an exploratory analysis.
- Estimated frequencies dephase, Eqs. (30.16)–(30.18); C2 falls to 0.078 on 1001–2000 while the oracle does not dephase.
- The obstacle is the parametrization, not the parameter count, for the families and optimizer tested; the explanation is heuristic.

30.14 Problems

1. • Control B with the first 10 primes and $m = 1, 2, 3$: how many frequencies and how many fitted real parameters does it have? Is it over- or underdetermined with 400 training zeros?

2. • For the 800-zero training window, $T = t_{800} - t_1 \approx 1170$. Compute the Fourier resolution $2\pi/T$ and the matching tolerance π/T. Compare with the 400-zero values 0.0094 and 0.0047.

3. • Using the {2} column of Table 30.2, compute for each holdout the fraction of the null error removed, $1 - \mathrm{rms}/\mathrm{null}$.

4. • Assign each of the following to level (i), (ii) or (iii) of Section 30.4: the six-prime oracle; control B with 25 primes; C1; C2; the scale-1 cosine operator with $M = 600$; the Chebyshev operator of Table II.

5. •• Starting from the level equation (30.3), derive the Newton iteration for the predicted zero, starting from s_n. Show that the first iterate is $s_n - N_{\mathrm{fl}}(s_n)/[N'_{\mathrm{sm}}(s_n) + N'_{\mathrm{fl}}(s_n)]$, and state when it reduces to $s_n - \mathrm{gap}_n N_{\mathrm{fl}}(s_n)$.

6. •• Compute the bound B_P of Eq. (30.12) for $P = \{2, 3\}$ and the height above which the two-prime oracle's root is guaranteed unique.

7. •• Control B with 900 coefficients has correlation +0.78 on zeros 401–500. Using Eq. (30.14) with $\sigma_Y \approx 0.263$, find the best rms consistent with that correlation and compare with the reported 0.180.

8. •• Compute the weights $w(q) = 1/(mp^{m/2})$ for $q = 121, 125, 127, 128, 169$. Which of these would an amplitude-only argument predict C2 to find from 800 zeros? The paper reports

that the six primes found above 113 lie between 131 and 211, so 127 was not among them. Suggest a reason, using $\log 128 - \log 127$ and the resolution $2\pi/T \approx 0.0054$.

9. •• Use the heuristic of the Sample Problem in Section 30.9 to estimate the expected number of prime-power logarithms within π/T of $\omega = \log 211$, for the 800-zero window ($2\pi/T \approx 0.0054$).

10. •• Repeat the dephasing Sample Problem with $\delta\omega = 5 \times 10^{-5}$ at distance $|t - t_c| = 1000$.

11. •• Using `qmnt`, reproduce the 401–500 row of Table V (null and all four prime budgets). Then compute the six-prime oracle on zeros 101–400, inside the operators' training window. Does a zero-parameter predictor have an "in-sample" score? What does your number mean?

12. •• Using `qmnt`, reproduce the 1001–2000 row of Table V for $\{2\}$ and six primes, and compute their rms/null ratios. Compare with Table 30.3.

13. ••• Using numpy and `qmnt`, implement control B with 25 primes as in the listing of Section 30.6. Report your held-out rms and correlation on zeros 401–500, and compare your fitted sine and cosine amplitudes for $\log 2$ with the oracle's $-1/(\pi\sqrt{2})$ and 0.

14. ••• Using numpy and `qmnt`, run the C2 listing for 15 steps on zeros 1–400. For each selected ω, find the nearest $\log q$ over prime powers $q \le 3000$ and record whether $|\omega - \log q| < \pi/T$. Then extend your code to choose K by the 300/100 forward split.

15. •• Using numpy and `qmnt`, compute the periodogram of y_n for the first 400 zeros on $0.05 < \omega < 8$. Where is its tallest peak? Rerun with y_n replaced by a random permutation of itself (seed 1) and describe the change.

16. •• For the C1 grid $\omega_j = 8j/450$, find the nearest grid frequency to $\log 7$, the offset, and the phase drift across a window of length $T \approx 666$.

17. •• Appendix A step 20 states: "Overwriting held-out data with noise before fitting leaves all fitted coefficients bit-identical." Explain exactly what this test rules out, and give one kind of information flow from the test set that it does *not* rule out.

18. •• Appendix A step 23 uses 200 replicates per surrogate family for the 400-zero fit. What is the smallest empirical tail probability that 200 replicates could support, if none reached the real score? Why does the paper decline to quote such a number?

19. • The 900-coefficient B row scores 0.180 and "the score varies by at most 0.012 with the choice of root." Give the range of possible scores and compare with the null.

20. •• Using Selberg's variance $\frac{1}{2\pi^2}\log\log t$ from the `beyond` box in Section 30.3, predict the ratio of the null rms at $t = 8000$ to that at $t = 700$. Compare with the ratio 0.296/0.263 of Table V's first and last nulls, and say what the asymptotic formula neglects.

21. ••• For $F(t) = A\sin(\omega t)$ near $t = 680$ with $\omega = 5$, find the smallest A for which the level function can be non-monotone. For $A = 0.3$, how many times can G change direction within one period $2\pi/\omega$? Relate your answer to Figure 30.4.

22. •• C2 beats the 25-prime oracle on zeros 401–500 (0.034 vs. 0.043) but loses to it on 1001–2000 (0.078 vs. 0.054). Explain both facts with the material of Sections 30.8 and 30.10.

23. ••• Show from the ridge filter factor $\sigma_i/(\sigma_i^2 + \lambda\sigma_1^2)$ that, as $\lambda \to \infty$, every ridge prediction tends to $F \equiv 0$, i.e. to the null. Show also that for $\lambda \to 0$ and an underdetermined system the solution tends to the minimum-norm exact fit. Use these limits to explain why C1's validation-selected ridge leaves it near the null.

24. •• Compute the ratio of C2's rms on 1001–2000 to its rms on 401–500, and the same ratio for the 25-prime oracle, from the numbers quoted in Section 30.10. Which ratio reflects dephasing?

25. • C1's held-out rms interval is $[0.231, 0.285]$ and its correlation interval is $[+0.08, +0.46]$. In plain words, what does each interval say about C1 relative to the null?

26. ••• (Research flavor.) Suppose C2's search range were restricted to $0.05 < \omega < 2$. Which prime powers could it find? Predict, without running, roughly where its held-out rms on 401–500 would fall relative to Table V, and explain. Then, if you like, test your prediction with your code from Problem 14.

27. •• Suppose a revised paper reported that C1, unchanged, scored 0.05 on zeros 401–500. Which sentence of Sec. IV or Sec. VI would this contradict, and would Claim IV.10 survive?

28. •• Starting from Eq. (30.11), show that the mean square of $N'_{\mathrm{fl}}(t)$ over a long range of t is $\frac{1}{\pi^2}\sum_{p\in P}(\log p)^2\sum_{m=1}^{3}\frac{1}{2p^m}$. (Hint: the frequencies $m\log p$ are all distinct, so the cross terms average to zero, and each $\cos^2$ averages to $\frac{1}{2}$.) Evaluate the rms for $P=\{2\}$ and for the six primes $p\le 13$, and compare both with $N'_{\mathrm{sm}}(680)$.

30.15 Answers to selected problems

Problem 1. $10\times 3 = 30$ frequencies, each with a sine and a cosine amplitude: 60 fitted parameters. With 400 equations it is overdetermined (more equations than unknowns), so least squares has a unique best fit if the columns are independent.

Problem 2. $2\pi/1170 \approx 0.0054$ and $\pi/1170 \approx 0.0027$. Both are about 0.57 times the 400-zero values, because the window is about $1170/666 \approx 1.76$ times longer; resolution improves in proportion to window length.

Problem 3. $1-0.198/0.263 = 0.247$; $1-0.208/0.270 = 0.230$; $1-0.219/0.279 = 0.215$; $1-0.232/0.288 = 0.194$; $1-0.242/0.296 = 0.182$. A single prime removes about a quarter to a fifth of the null error at every height in the table.

Problem 4. Six-prime oracle: (i). B with 25 primes: (i). C1: (iii). C2: (ii). Cosine operator $M = 600$: (iii). Chebyshev operator: (iii) (it is another generic basis for δV).

Problem 5. Let $G(t) = N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) - (n-\frac{1}{2})$. Newton's method is $t_{k+1} = t_k - G(t_k)/G'(t_k)$. At $t_0 = s_n$, $G(s_n) = N_{\mathrm{fl}}(s_n)$ because $N_{\mathrm{sm}}(s_n) = n-\frac{1}{2}$, and $G'(s_n) = N'_{\mathrm{sm}}(s_n) + N'_{\mathrm{fl}}(s_n)$. Hence $t_1 = s_n - N_{\mathrm{fl}}(s_n)/[N'_{\mathrm{sm}}(s_n) + N'_{\mathrm{fl}}(s_n)]$. It reduces to $s_n - \mathrm{gap}_n N_{\mathrm{fl}}(s_n)$ when $|N'_{\mathrm{fl}}(s_n)| \ll N'_{\mathrm{sm}}(s_n) = 1/\mathrm{gap}_n$. Sample Problem "One oracle step by hand" shows that this condition fails badly for one prime at $t = 680$ (0.34 vs. 0.75).

Problem 7. $0.263\sqrt{1-0.78^2} = 0.263\times 0.626 = 0.165$. The reported 0.180 is above this bound, as it must be (to the extent that the null approximates σ_Y). The excess is consistent with an amplitude that is not optimally scaled, a nonzero mean error, or both; the paper does not say which, and the bound alone cannot tell them apart.

Problem 9. $\nu \approx 0.0054\times 211/\log 211 = 0.0054\times 39.4 \approx 0.21$. So even the highest frequency in the 800-zero list had only about a one-in-five chance of a match by accident under this heuristic; many such matches together are hard to obtain by chance, which is the intuition the surrogate check tests properly.

Problem 11. You should get null 0.263, and $0.198/+0.66$, $0.116/+0.90$, $0.074/+0.96$, $0.043/+0.99$ for 1, 3, 6, 25 primes, matching Table V. The oracle is not fitted, so a score on zeros 101–400 is not "in-sample" in the sense of training; it is simply another holdout. You should find an rms below 0.1 and a correlation above 0.9: the truncated formula works about as well at these lower heights as on 401–500.

Problem 13. Your rms should be well below the six-prime oracle's 0.074, of the same order as Table IV's 0.028, and your correlation above 0.95; exact values depend on validation details. The fitted sine amplitude for $\log 2$ should lie within about 0.05 of -0.225, and the cosine amplitude within about 0.05 of zero: the data recover the explicit formula's amplitude and phase for the dominant term.

Problem 15. The tallest peak lies within π/T of $\log 2 = 0.693$, with the next tallest at logarithms of other small prime powers (see Figure 30.5). For the permuted sequence (`np.random.default_rng(1).permutation(y)`) the peaks at $\log q$ disappear: permutation keeps the values but destroys their ordering in t, and with it every coherent oscillation. A convenient number to compare is the ratio of the tallest peak to the median of the periodogram on the coarse grid: it drops by more than a factor of 3 after permutation, and what remains is an irregular noise floor with no preferred frequencies.

Problem 17. It rules out any dependence of the fitted coefficients on the values of held-out zeros: if those values could influence the fit, replacing them by noise would change at least one coefficient. It does not rule out influence through *design choices* made by the researchers after seeing held-out results (for example, choosing the frequency range $0.05 < \omega < 8$ or the step size because they worked well on the test set). That kind of leakage lives in the protocol's history, not in the code, and is addressed by fixing the protocol in advance and reporting it.

Problem 19. Reading "varies by at most 0.012" in the most pessimistic way, as ± 0.012 around the reported value, the score lies in $[0.168, 0.192]$, entirely below the null 0.263. (If the phrase means a total spread of 0.012, the range is narrower still.) The qualitative conclusion (positive skill despite underdetermination) does not depend on the root choice.

Problem 21. $G'(t) = N'_{\mathrm{sm}}(t) + A\omega\cos(\omega t)$ can become negative only if $A\omega > N'_{\mathrm{sm}}(680) = 0.7455$, so $A > 0.149$. For $A = 0.3$, $A\omega = 1.5 > 0.7455$, so $G'(t) = 0.7455 + 1.5\cos(5t)$ is negative exactly when $\cos(5t) < -0.7455/1.5 = -0.497$, which happens on one interval of each period, and positive on the rest. Hence G' changes sign exactly twice per period: G has one local maximum $G_{\max}$ (where G' goes from positive to negative) and one local minimum $G_{\min} < G_{\max}$ (where it goes back). So G changes direction twice per period. Extra roots: around one such excursion, $G = 0$ has three roots (one on the rising stretch before the maximum, one on the falling stretch, one on the rising stretch after the minimum) if and only if $G_{\min} < 0 < G_{\max}$; otherwise it has at most one. Each excursion whose range straddles zero therefore adds exactly two extra roots, which is what the $A = 0.3$ curve of Figure 30.4 shows.

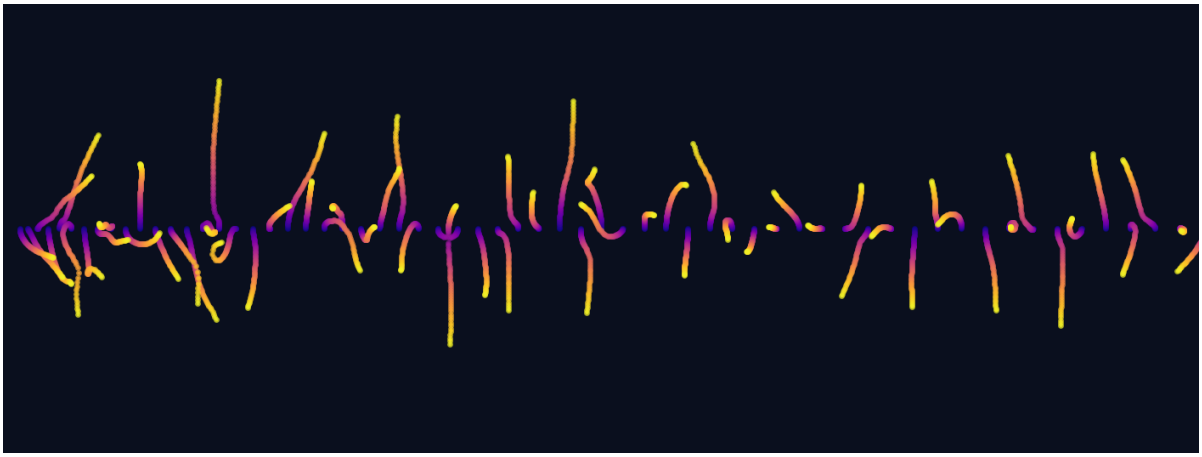
Problem 23. The ridge solution is $\theta = \sum_i \frac{\sigma_i}{\sigma_i^2 + \lambda\sigma_1^2}(u_i^\top y)\, v_i$. As $\lambda \to \infty$ every factor tends to 0, so $\theta \to 0$ and $F \equiv 0$: the prediction is the smooth zero, the null. As $\lambda \to 0$ each factor tends to $1/\sigma_i$ for $\sigma_i > 0$, and the sum runs only over nonzero singular values, which is the pseudo-inverse solution: the exact fit of minimum norm (Chapter 9). For C1, if (as is plausible) the minimum-norm exact fit extrapolates poorly, the forward split would detect this in the validation block and select a large λ, pushing the held-out predictions toward the $\lambda \to \infty$ limit, the null. The paper does not report the selected value, so this is a consistent explanation, not a demonstrated one.

Problem 25. The rms interval contains the null 0.263: C1's rms cannot be distinguished from predicting no fluctuation. The correlation interval excludes zero: its predictions are weakly but genuinely aligned with the truth. Together: a little information, almost no error reduction, as Eq. (30.14) predicts for $\rho \approx 0.28$.

Problem 26. The prime powers with $\log q < 2$, i.e. $q < e^2 \approx 7.39$, are $2, 3, 4, 5, 7$. A C2 restricted to this range can capture the terms $\log 2$, $\log 4 = 2\log 2$, $\log 3$, $\log 5$, $\log 7$, i.e. a subset of the four-prime oracle, with fitted amplitudes. We would expect a held-out rms roughly between the 3-prime (0.116) and 6-prime (0.074) oracle entries of Table V, since it captures more than three primes' main terms but misses $p = 11, 13$ and all higher terms. This is a heuristic expectation, not a result; the fitted amplitudes could do a little better.

Problem 27. It would contradict "a dense 900-coefficient Fourier model also stays near

the null" (abstract) and "a dense 900-coefficient Fourier model gains almost nothing over the null" (Sec. VI). Claim IV.10 would weaken: count without an arithmetic or sparsity prior would then have generalized, and the operators' failure could no longer be blamed on a generic parametrization in general, only on the potential specifically.



Complex eigenvalues of $H + iA$ for a random real A of growing strength: the spectrum lifts off the real axis and scatters.

Chapter 31

Reading the Paper IV: Anatomy of a Spurious Floor

↑ Contents

What this chapter is about

Suppose two lab groups time two quite different pendulums, one with a brass bob on a steel wire and one with a wooden bob on a string. One group reports a period of 2.321 s, the other 2.322 s. Two different systems agreeing to four digits is exciting: it suggests a hidden common cause, maybe a law. Then someone notices that both groups used the same stopwatch, and that the stopwatch had a fault that pushed every reading toward the same value. The "law" was a property of the instrument.

Section V of the paper [1] tells a story with exactly this shape. Two different non-Hermitian model classes, fitted to the Riemann zeros, stalled at in-sample error floors of 0.2321 and 0.2322 mean spacings. The near-identity invited a physical interpretation. The paper then takes the result apart, piece by piece: a dead parameter that fed on single-precision rounding noise, a plausible mechanism that fails by a factor of 2 to 7, an approximation used far outside its range of validity, and finally a fair refit that captures almost nothing. Almost nothing in this chapter is new mathematics. The few short extra arguments (evenness of real parts in the amplitude, the exceptional point of a 2×2 matrix, the moving-average bound) are derived in full where they are used. Every other tool you already have (second-order perturbation theory with an imaginary perturbation, the sum rule, PT symmetry and the null mode from Chapter 14, floating-point precision from Chapter 4, least squares from Chapter 9, Levenberg–Marquardt from Chapter 24). What is new is watching these tools used as a forensic kit.

Where this shows up in the paper: the method is in Sec. II E and Eq. (5); the dissection is Sec. V and its subsection V A, with Fig. 5; the recipe is Appendix A, steps 9–11, plus several "key anchors"; the abstract summarizes it in one sentence; and Sec. VI records that this part was not rerun for the revision.

What you need

- Chapter 14: the operator $H + iA$, imaginary first-order shifts, the second-order real shift, the quadratic form $-c^\top T^{(n)} c$, the sum rule $\sum_n \delta E_n = 0$, PT symmetry, the null mode, exact diagonalization of complex matrices and matching by overlap.
- Chapter 13: Rayleigh–Schrödinger perturbation theory and its validity condition (matrix elements small compared with level spacings).
- Chapter 4: machine epsilon (2.2×10^{-16} in double precision, 1.2×10^{-7} in single precision, "float32") and how analytic zeros become rounding noise.
- Chapters 9 and 24: least squares as projection, the Jacobian, Levenberg–Marquardt damping, random restarts.
- Chapters 25 and 26: in-sample versus out-of-sample (OOS) error, the null model, exploratory versus confirmatory statements.
- Chapters 23, 8 and 11: the finite-difference Hamiltonian, eigenvectors normalized by $\sum \psi^2 h = 1$, parity of states in an even potential, the infinite well.
- Chapter 19: rms, Pearson correlation, covariance.
- Chapter 20: the negative short-range correlation of consecutive zero fluctuations.
- Chapters 28 and 29: the data, the target fluctuations, the in-sample null 0.2438 on zeros 1–400.
- First-year physics: decay laws, the idea of a damped oscillation.

31.1 What Section V is, and what it is not

★ Did you know? *Faster Than Light? No.*

In 2011 the OPERA experiment reported neutrinos apparently arriving about 60 nanoseconds sooner than light would. The team openly asked others to help find an error. In 2012 the anomaly was traced largely to a badly connected fiber-optic cable in the timing system. The right response to a surprising signal is careful dissection, not excitement.

Before reading any claim, read its scope. The paper is explicit. Section V opens with:

> "A subsidiary dissection reinforces the need for exact verification. This section is a methodological stress test of a second model class; the Hermitian conclusions of Secs. III and IV do not depend on it."

and the Discussion (Sec. VI) adds that "the non-Hermitian analysis of Sec. V is inherited from the original computation and was not rerun for this revision." So three facts frame everything below.

1. The main results of the paper (fitted Hermitian potentials memorize, the explicit formula predicts) stand or fall independently of this section. If every number in Sec. V were wrong, Tables I–V would be untouched.

2. Section V is about a *method failure*, not about the zeros. Its lesson is how an optimization can produce an attractive but meaningless number, and how to catch it.

3. Its numbers come from the original submission's computation. The revision did not regenerate them. You should read them as carefully documented results of one pipeline, and note this when you weigh them (Chapter 32 returns to this).

31.1.1 The story in five sentences

(1) The paper fits a non-Hermitian correction iA to the Wu–Sprung Hamiltonian, using the second-order formula Eq. (5) for the real level shifts as a cheap stand-in (a *surrogate*) for the exact spectrum. (2) Unconstrained fits on zeros 1–400 stall at in-sample rms floors 0.2321 (PT-odd class) and 0.2322 (absorptive-even class), against a null of 0.2438. (3) The second number turns out to be an artifact: a coefficient that should have no effect at all grew to $c_0 \approx 2.2 \times 10^6$ and harvested a small improvement from float32 rounding noise; after repair the even floor is 0.2358 and the coincidence disappears. (4) A natural mechanism for "why a floor at 0.232" fails quantitatively, and exact diagonalization of the fitted operators shows that the surrogate no longer describes them: their exact in-sample scores, 0.2523 and 0.2477, are worse than the null. (5) Fits redone inside amplitude limits where the surrogate is verified accurate capture only 1.9% and 0.6% over the null in exact spectra, so these model classes, treated fairly, are “empirically inert on this problem.”

Table 31.1 maps each claim to the section of this chapter that dissects it and to the earlier chapters that supply the tools.

Table 31.1: Claims of the paper’s Sec. V and where this chapter treats them.

Claim (paraphrased)	Paper location	This chapter	Tools from
Second-order shift is a quadratic form in c	Sec. II E, Eq. (5)	Section 31.3	Ch. 13, 14
Unconstrained floors 0.2321, 0.2358; null 0.2438	Sec. V, para. 1	Section 31.4	Ch. 24, 25
Null-mode artifact, $c_0 \approx 2.2 \times 10^6$	Sec. V, para. 1; App. A step 9	Section 31.5	Ch. 4, 9
Sum rule, locality, block coherence fails	Sec. V, V A	Section 31.6	Ch. 14, 19
Surrogate invalid at fitted amplitude	Sec. V; Fig. 5 left; step 11	Section 31.7	Ch. 8, 14
Fair version is inert	Sec. V; Fig. 5 right; step 10	Section 31.8	Ch. 7, 24
Generic lessons	Sec. V, last para.	Section 31.9	Ch. 26

31.1.2 Two words you need: “floor” and “null”

Throughout Sec. V the score is an *in-sample* rms, measured on the training zeros 1–400, in units of the local mean spacing gap_n (recall from Chapter 17 that $\mathrm{gap}_n = ds_n/dn$, with s_n the smooth zeros). The quantity to be explained is the target fluctuation

$$\mathrm{fluc}_n = \tilde{t}_n - E_n[\mathrm{smooth}], \tag{31.1}$$

where $\tilde{t}_n$ is the bias-corrected target (Chapter 23) and $E_n[\mathrm{smooth}]$ is the n-th eigenvalue of the unperturbed Wu–Sprung Hamiltonian on the working grid (App. A, step 12, uses exactly this definition). The **null** is the score of predicting no correction at all, that is, the rms of $\mathrm{fluc}_n/\mathrm{gap}_n$ over $n = 1, \ldots, 400$. The paper’s key anchors give it as 0.2438.

A **floor** is the lowest in-sample rms an optimizer reaches before it stops improving, taken over restarts. A floor is a property of three things together: the model class, the data, and the optimizer. That sentence is the whole chapter in miniature.

Pitfall

Do not confuse this in-sample null (0.2438, zeros 1–400) with the out-of-sample null (0.2631 on zeros 401–500) used in Secs. III and IV of the paper. Both are rms values of the same kind of fluctuation, but over different sets of zeros. Section V's floors are all compared with 0.2438.

Key Ideas

- Section V is a subsidiary methodological stress test; the Hermitian conclusions do not depend on it, and it was not rerun in revision.
- A floor is the in-sample rms at which an optimizer stalls; it reflects model, data and optimizer together.
- All Sec. V scores are in-sample on zeros 1–400 and are compared with the null 0.2438.

31.2 The two non-Hermitian model classes

31.2.1 What is being added to the Hamiltonian

Recall the setting (Chapters 10 and 22): in units $\hbar = 2m = 1$ the unperturbed Hamiltonian is $H = -d^2/dx^2 + V(x)$, with V the even Wu–Sprung potential on the box $[-L, L]$ with Dirichlet walls, $L = 44.883$ at scale 1. Its eigenvalues E_n are real and its normalized eigenfunctions $\psi_n(x)$ can be chosen real, because H is a real symmetric operator (Chapter 8).

Sec. II E of the paper perturbs

$$H \;\to\; H + iA, \qquad A(x) = \sum_{k=0}^{M-1} c_k\, \phi_k(x), \tag{31.2}$$

where $A(x)$ is a *real* function of position, the c_k are real coefficients to be fitted and the ϕ_k are basis functions. "Multiplication operator" means that A acts on a wavefunction by multiplying it pointwise: $(A\psi)(x) = A(x)\psi(x)$. Because of the factor i, the operator $H + iA$ is not Hermitian, and its eigenvalues are in general complex (Chapter 14).

The matrix elements of A in the unperturbed basis are

$$A_{nm} = \langle n|A|m\rangle = \int_{-L}^{L} \psi_n(x)\, A(x)\, \psi_m(x)\, \mathrm{d}x. \tag{31.3}$$

Since ψ_n, ψ_m and A are all real, A_{nm} is real, and swapping n and m does not change the integrand, so $A_{nm} = A_{mn}$. These two facts (real and symmetric) will be used again and again.

Pitfall

The paper uses the letter A twice. In Sec. II C, $A_{nk} = \int |\psi_n|^2 \phi_k \, \mathrm{d}x$ is the first-order *sensitivity matrix* of the Hermitian fit: rows are levels, columns are basis functions. In Sec. II E, A is the *operator* of Equation (31.2), and A_{nm} is a matrix between two *levels*. They are related (the diagonal A_{nn} of the operator equals $\sum_k c_k \times$ the sensitivity entry A_{nk}), but they are different objects. Similarly $T^{(n)}$ below is a tensor, not the training window T of Sec. II D and not a Chebyshev polynomial T_k.

31.2.2 The PT-odd class

The first class uses

$$\phi_k(x) = \sin\!\Big(\frac{(k+1)\pi x}{L}\Big), \qquad k = 0, 1, \ldots, M-1. \tag{31.4}$$

Every ϕ_k is an odd function: $\phi_k(-x) = -\phi_k(x)$. Notice the $k+1$. With k instead, the $k=0$ function would be $\sin 0 = 0$, a basis function that is identically zero. The designers avoided that dead function here; keep this in mind, because the even class has a subtler dead function that was not avoided.

Recall from Chapter 11 that in an even potential every eigenfunction has definite parity, even or odd, alternating with n. Then $|\psi_n|^2$ is even, and for an odd A,

$$A_{nn} = \int_{-L}^{L} \underbrace{\psi_n(x)^2}_{\text{even}} \underbrace{A(x)}_{\text{odd}} \, \mathrm{d}x = 0, \tag{31.5}$$

because the integral of an odd function over a symmetric interval vanishes. This is the paper's "$A_{nn} = 0$". More generally $A_{nm} \neq 0$ only if $\psi_n \psi_m A$ is even, which for odd A requires ψ_n and ψ_m of *opposite* parity.

Why "PT"? Recall from Chapter 14: parity P sends $x \to -x$ and time reversal T sends $i \to -i$ (complex conjugation). Applying P to $iA(x)$ gives $iA(-x) = -iA(x)$; applying T then gives $+iA(x)$. So $H + iA$ is unchanged by the combined operation PT when A is odd. Its first-order shifts iA_{nn} vanish, so its spectrum is real to leading order (the paper's words). We will see in Section 31.7 that the spectrum stays exactly real for small enough amplitude and can turn complex beyond a threshold.

31.2.3 The absorptive-even class

The second class uses the even cosine basis familiar from the Hermitian fits,

$$\phi_k(x) = \cos\!\Big(\frac{k\pi x}{L}\Big), \qquad k = 0, 1, \ldots, M-1, \tag{31.6}$$

and the paper says it models $-i\Gamma/2$. Here is what that means. A complex potential $V(x) - i\Gamma(x)/2$ with $\Gamma(x) \geq 0$ describes absorption: probability leaks out. To see this, use the time-dependent Schrödinger equation $i\,\partial\Psi/\partial\tau = \mathcal{H}\Psi$ (we write the time as τ to keep t for the heights of zeros) with $\mathcal{H} = H - i\Gamma/2$:

$$\begin{aligned}
\frac{d}{d\tau}\int |\Psi|^2 \mathrm{d}x &= \int \Big(\Psi^* \frac{\partial \Psi}{\partial \tau} + \Psi \frac{\partial \Psi^*}{\partial \tau}\Big)\mathrm{d}x \quad \text{(product rule)} \\
&= \int \Big(\Psi^*(-iH\Psi - \tfrac{\Gamma}{2}\Psi) + \Psi(iH\Psi^* - \tfrac{\Gamma}{2}\Psi^*)\Big)\mathrm{d}x \\
&\qquad \text{(insert } \partial_\tau \Psi = -i\mathcal{H}\Psi \text{ and its complex conjugate)} \\
&= -i\int (\Psi^* H\Psi - \Psi H\Psi^*)\mathrm{d}x - \int \Gamma|\Psi|^2 \mathrm{d}x \quad \text{(collect terms)} \\
&= -\int \Gamma(x)|\Psi|^2 \mathrm{d}x \;\leq\; 0.
\end{aligned} \tag{31.7}$$

The first integral vanishes because H is real and Hermitian, so $\int \Psi^* H \Psi \, \mathrm{d}x = \int \Psi H \Psi^* \mathrm{d}x$ (Chapter 10). Comparing $V - i\Gamma/2$ with $V + iA$ gives

$$A(x) = -\tfrac{1}{2}\Gamma(x), \qquad \Gamma \geq 0 \iff A \leq 0 \text{ pointwise.} \tag{31.8}$$

That is the "physical constraint $\Gamma \geq 0$ (i.e. $A \leq 0$ pointwise)" of Sec. II E.

For a single level, the eigenvalue $E_n - i\Gamma_n/2$ gives $\Psi \propto e^{-iE_n\tau} e^{-\Gamma_n \tau/2}$, so the *width* $\Gamma_n = -2\,\mathrm{Im}(\text{eigenvalue})$ is the decay rate of $|\Psi|^2$. To first order the eigenvalue is $E_n + iA_{nn}$, so

$$\Gamma_n \approx -2A_{nn} = \int |\psi_n|^2 \, \Gamma(x) \, \mathrm{d}x. \tag{31.9}$$

If $\Gamma(x) \geq 0$ everywhere, every first-order width is non-negative. A negative width means growth, not decay: the "absorber" would be pumping probability in.

Sample Problem 31.1: Parity kills the diagonal, and picks the partners

Use the infinite well on $[-L, L]$ (RE3, Chapter 11) with $L = \pi/2$, chosen so that the levels are $E_k = (k\pi/2L)^2 = k^2$. The states are $\psi_k(x) = L^{-1/2}\sin\big(k\pi(x+L)/(2L)\big)$. Take one PT-odd mode, $A(x) = c\sin(\pi x/L)$. (a) Show that $A_{11} = A_{22} = 0$. (b) Compute A_{12}. (c) Keeping only levels 1 and 2, estimate the second-order real shift of level 1 for $c = 1$ using Eq. (5) of the paper.

Step 1: parity. Substituting $u = (x+L)/(2L) \in [0,1]$, $\psi_k = L^{-1/2}\sin(k\pi u)$. Reflection $x \to -x$ is $u \to 1-u$, and $\sin(k\pi(1-u)) = \sin(k\pi)\cos(k\pi u) - \cos(k\pi)\sin(k\pi u) = -(-1)^k\sin(k\pi u)$. So ψ_1 is even and ψ_2 is odd. Since $\sin(\pi x/L)$ is odd, $\psi_1^2 A$ and $\psi_2^2 A$ are odd and integrate to zero: $A_{11} = A_{22} = 0$.

Step 2: rewrite the mode in u. $\pi x/L = \pi(2u-1) = 2\pi u - \pi$, so $\sin(\pi x/L) = -\sin(2\pi u)$. Also $\mathrm{d}x = 2L\,\mathrm{d}u$.

Step 3: the integral.

$$A_{12} = c\int_0^1 \frac{\sin(\pi u)\sin(2\pi u)}{L}\big(-\sin 2\pi u\big)\,2L\,\mathrm{d}u = -2c\int_0^1 \sin(\pi u)\sin^2(2\pi u)\,\mathrm{d}u.$$

Use $\sin^2(2\pi u) = \frac{1}{2}(1-\cos 4\pi u)$ and $\sin a\cos b = \frac{1}{2}[\sin(a+b)+\sin(a-b)]$:

$$\int_0^1 \sin(\pi u)\,\mathrm{d}u = \frac{2}{\pi}, \qquad \int_0^1 \sin(\pi u)\cos(4\pi u)\,\mathrm{d}u = \tfrac{1}{2}\Big[\frac{2}{5\pi} - \frac{2}{3\pi}\Big] = -\frac{2}{15\pi}.$$

Hence $\int_0^1 \sin(\pi u)\sin^2(2\pi u)\,\mathrm{d}u = \frac{1}{2}\big(\frac{2}{\pi} + \frac{2}{15\pi}\big) = \frac{16}{15\pi}$, and

$$A_{12} = -\frac{32}{15\pi}\,c \approx -0.6791\,c.$$

Step 4: two-level second-order shift. Eq. (5) of the paper with one partner: $\delta E_1 = -A_{12}^2/(E_1 - E_2) = A_{12}^2/3$. For $c = 1$:

$$\boxed{\delta E_1 \approx +0.1537, \qquad \delta E_2 \approx -0.1537 \text{ (two-level estimate)}}$$

Level 1 is pulled *up* toward level 2 and level 2 pulled down: the opposite of Hermitian level repulsion. (Level 1 also couples to levels 4, 6, ..., all of opposite parity; the two-level number is an estimate.)

Checkpoint 31.1

In the PT-odd class, can level 1 and level 3 of an even potential couple through A?

Answer: No. Levels 1 and 3 have the same parity (both even), so $\psi_1\psi_3$ is even and $\psi_1\psi_3 A$ is odd for odd A; the integral vanishes. The odd class couples only opposite-parity pairs; the even class only same-parity pairs.

Key Ideas

- The perturbation is iA with $A(x) = \sum_k c_k\phi_k(x)$ real; its matrix A_{nm} is real and symmetric.
- PT-odd class: $\phi_k = \sin((k+1)\pi x/L)$, odd, so $A_{nn} = 0$ and only opposite-parity levels couple.
- Absorptive-even class: $\phi_k = \cos(k\pi x/L)$, modeling $-i\Gamma/2$; physics requires $\Gamma \geq 0$, i.e.

$A \leq 0$ pointwise, and first-order widths $\Gamma_n \approx -2A_{nn}$.

31.3 The second-order surrogate and how it was fitted

31.3.1 Re-deriving Eq. (5)

Chapter 14 derived Eq. (5). Because everything in this chapter hinges on what that formula neglects, we re-derive it here in a form that works for any perturbation W, Hermitian or not.

We write H_0 for the unperturbed Hamiltonian, called H in the paper and so far in this chapter, to distinguish it from the perturbed operators below. Let H_0 have real eigenvalues E_n (nondegenerate) and real orthonormal eigenvectors $|n\rangle$. Seek an eigenvalue E of $H_0 + W$ near E_n with eigenvector written as $|\psi\rangle = |n\rangle + \sum_{m\neq n} a_m|m\rangle$ (we fix the coefficient of $|n\rangle$ to be 1; this is allowed because an eigenvector can be rescaled). Write $W_{km} = \langle k|W|m\rangle$.

Project onto $\langle n|$. Using $\langle n|H_0 = E_n\langle n|$ and $\langle n|m\rangle = \delta_{nm}$,

$$E_n + W_{nn} + \sum_{m\neq n} W_{nm}a_m = E. \tag{31.10}$$

Project onto $\langle k|$, $k \neq n$.

$$E_k a_k + W_{kn} + \sum_{m\neq n} W_{km}a_m = Ea_k \quad\Longrightarrow\quad (E - E_k)\, a_k = W_{kn} + \sum_{m\neq n} W_{km}a_m. \tag{31.11}$$

Lowest order. The a_m are at least first order in W, so $\sum W_{km}a_m$ is second order; and $E - E_n$ is first order. Keeping only first-order terms,

$$a_k = \frac{W_{kn}}{E_n - E_k} + \mathcal{O}(W^2). \tag{31.12}$$

Insert into Equation (31.10).

$$E = E_n + W_{nn} + \sum_{m\neq n} \frac{W_{nm}W_{mn}}{E_n - E_m} + \mathcal{O}(W^3). \tag{31.13}$$

Nothing here used $W^\dagger = W$; only the orthonormal basis of H_0 was used. Now set $W = iA$ with A_{nm} real and symmetric:

$$W_{nn} = iA_{nn} \quad \text{(purely imaginary)}, \tag{31.14}$$

$$W_{nm}W_{mn} = (iA_{nm})(iA_{mn}) = i^2A_{nm}^2 = -A_{nm}^2 \quad (\text{using } A_{mn} = A_{nm}). \tag{31.15}$$

Therefore the real part of the eigenvalue, to second order, is $E_n + \delta E_n$ with

$$\boxed{\delta E_n = -\sum_{m\neq n} \frac{A_{nm}^2}{E_n - E_m}} \tag{31.16}$$

which is Eq. (5) of the paper (there written $|A_{nm}|^2$; for real A_{nm} the two coincide). The sign is the reverse of Hermitian second-order theory: a partner above ($E_m > E_n$) pulls level n *up*, a partner below pulls it down. For an imaginary coupling, levels attract.

31.3.2 What the surrogate neglects: only even orders

How large is the first neglected real term? Here is a short argument that it is fourth order, not third. Write $H(a) = H_0 + iaA$, where a scales the whole perturbation. (This a is a new symbol, unrelated to the eigenvector coefficients a_m above.) In the unperturbed basis H_0 and A are real matrices, so taking the complex conjugate of every entry gives

$$H(a)^* = H_0 - iaA = H(-a). \tag{31.17}$$

If $H(a)v = Ev$, conjugating both sides gives $H(-a)v^* = E^*v^*$: the conjugate E^* is an eigenvalue of $H(-a)$. Follow the eigenvalue $E_n(a)$ that starts at the real, nondegenerate E_n when $a = 0$. Its conjugate $E_n(a)^*$ also starts at E_n (as $a \to 0$, $E_n(a)^* \to E_n^* = E_n$) and is an eigenvalue of $H(-a)$. Because E_n is nondegenerate, for small enough $|a|$ the eigenvalues of $H(-a)$ stay close to those of H_0 (continuity, taken on trust in the box below), so exactly one of them lies near E_n, and that one is by definition $E_n(-a)$. Hence $E_n(a)^*$ is the branch $E_n(-a)$:

$$E_n(a)^* = E_n(-a) \quad \Longrightarrow \quad \operatorname{Re} E_n(a) = \frac{E_n(a) + E_n(-a)}{2}, \qquad \operatorname{Im} E_n(a) = \frac{E_n(a) - E_n(-a)}{2i}. \tag{31.18}$$

The real part is an even function of a; the imaginary part is odd. A power series of an even function has only even powers. So

$$\operatorname{Re} E_n(a) = E_n + a^2\,\delta E_n^{(2)} + a^4\,\delta E_n^{(4)} + \dots, \tag{31.19}$$

and the relative error of the surrogate is of order a^2 times (coupling/spacing)^2. Halving the amplitude cuts the relative error by about four; quartering it, by about sixteen. This scaling holds provided the amplitude is small enough that the a^4 term dominates the error. Once some levels have passed an exceptional point (Section 31.7), the series no longer converges for them and the rule is only a rough guide. Remember this when you read "at quarter amplitude the surrogate is accurate."

Taken on trust

We used that a nondegenerate eigenvalue $E_n(a)$ of $H_0 + iaA$ is a smooth (indeed analytic, i.e. given by a convergent power series) function of a near $a = 0$, so that "follow the branch" is meaningful and the series Equation (31.19) exists. This is a theorem of analytic perturbation theory, due to Rellich and Kato; its proof needs complex analysis beyond this book. We take it on trust. The series converges only up to the nearest point (possibly complex) where two eigenvalues collide; Section 31.7 shows such a collision explicitly.

For the PT-odd class there is more. Parity P maps $x \to -x$; since H_0 is even and A is odd, $PH(-a)P = H_0 - iaA(-x) = H_0 + iaA(x) = H(a)$. Since $P^2 = 1$, $P^{-1} = P$, so this says $H(a) = PH(-a)P^{-1}$. Two matrices related this way are called *similar*, and they have the same eigenvalues: if $H(-a)v = Ev$, then $H(a)(Pv) = PH(-a)P^{-1}Pv = P(Ev) = E\,(Pv)$. So $E_n(-a)$ is an eigenvalue of $H(a)$ lying near E_n; by the same uniqueness argument as above, the only such eigenvalue for small $|a|$ is $E_n(a)$, hence $E_n(-a) = E_n(a)$. Combined with Equation (31.18), $E_n(a)^* = E_n(a)$: for small enough $|a|$ the eigenvalues of the PT-odd class are *exactly real*.

The reality persists to larger amplitudes, up to the first collision, by a continuation argument. Take any real a. If E is an eigenvalue of $H(a)$, then E^* is an eigenvalue of $H(-a)$ (conjugation, Equation (31.17)). Also $H(-a)$ has the same eigenvalues as $H(a)$ (similarity, just shown). So the spectrum of $H(a)$ is closed under complex conjugation: non-real eigenvalues come in conjugate pairs E, E^*.

Now follow a simple real eigenvalue as a grows; it moves continuously. Suppose it first leaves the real axis at an amplitude a_*, at the real point E_*. Just beyond a_* it is a non-real E close to

E_*, and its partner E^* is a distinct eigenvalue, also close to E_*. As $a \to a_*$ both tend to E_*, so at a_* two eigenvalues coincide: a collision. A real eigenvalue can therefore become complex only at a collision. Hence the PT-odd spectrum stays exactly real for every amplitude below the first collision on the real a-axis. Beyond it, eigenvalues may pair off into complex conjugates (PT symmetry breaking, Chapter 14).

31.3.3 The quadratic form and the tensors $T^{(n)}$

Insert $A_{nm} = \sum_k c_k (\phi_k)_{nm}$, with $(\phi_k)_{nm} = \int \psi_n \phi_k \psi_m \, \mathrm{d}x$, into Equation (31.16):

$$A_{nm}^2 = \Big(\sum_k c_k (\phi_k)_{nm}\Big)\Big(\sum_l c_l (\phi_l)_{nm}\Big) = \sum_k \sum_l c_k c_l (\phi_k)_{nm} (\phi_l)_{nm},$$

$$\delta E_n = -\sum_k \sum_l c_k \, T_{kl}^{(n)} \, c_l = -c^\top T^{(n)} c, \qquad T_{kl}^{(n)} = \sum_{m \neq n} \frac{(\phi_k)_{nm} (\phi_l)_{nm}}{E_n - E_m}. \tag{31.20}$$

Each level n has its own $M \times M$ symmetric matrix $T^{(n)}$ (swapping k and l leaves the formula unchanged). The tensors depend only on the unperturbed potential, so they are computed once; after that, evaluating all 400 shifts for a trial c costs only matrix–vector products, with no eigensolve. That is the attraction of a surrogate.

The paper adds a practical detail: the state sum m runs "over all 652 basis states with $E \leq 1000$ (truncating at 500 states measurably biases the top levels)." The solvable example below shows why truncation biases the top.

31.3.4 A solvable laboratory: one cosine in the box

The infinite well gives exact matrix elements for the cosine modes. With $u = (x + L)/(2L)$ as in the sample problem above, $\cos(j\pi x/L) = \cos(2j\pi u - j\pi) = (-1)^j \cos(2j\pi u)$, and

$$\begin{aligned} \langle k | \cos \tfrac{j\pi x}{L} | l \rangle &= 2(-1)^j \int_0^1 \sin(k\pi u) \sin(l\pi u) \cos(2j\pi u) \, \mathrm{d}u \\ &= (-1)^j \int_0^1 \big[\cos((k-l)\pi u) - \cos((k+l)\pi u) \big] \cos(2j\pi u) \, \mathrm{d}u. \end{aligned}$$

For integers a and $b \geq 1$, $\int_0^1 \cos(a\pi u) \cos(b\pi u) \, \mathrm{d}u = \frac{1}{2}$ if $|a| = b$ and 0 otherwise (write the product as $\frac{1}{2}[\cos((a-b)\pi u) + \cos((a+b)\pi u)]$ and integrate). Hence, for $j \geq 1$,

$$\langle k | \cos \tfrac{j\pi x}{L} | l \rangle = \frac{(-1)^j}{2} \Big[\delta_{|k-l|,\,2j} - \delta_{k+l,\,2j} \Big]. \tag{31.21}$$

So one cosine couples level k to $k + 2j$, to $k - 2j$ when that is ≥ 1, and to $2j - k$ when that is ≥ 1 (with a minus sign); when $k = j$ the second term sits on the diagonal. Every coupling has magnitude $\frac{1}{2}$.

Now take $A = c \cos(j\pi x/L)$ and levels $E_k = \kappa k^2$, $\kappa = (\pi/2L)^2$. For $n \neq j$ and $n \neq 2j$, level n has exactly two partners, $n + 2j$ and $|n - 2j|$, each with $A_{nm}^2 = c^2/4$. The energy differences are

$$E_n - E_{n+2j} = -4\kappa j (n + j), \qquad E_n - E_{|n-2j|} = \kappa\big(n^2 - (n-2j)^2\big) = 4\kappa j (n - j).$$

Then Equation (31.16) gives

$$\begin{aligned} \delta E_n &= -\frac{c^2}{4}\Big[\frac{1}{-4\kappa j(n+j)} + \frac{1}{4\kappa j(n-j)}\Big] = -\frac{c^2}{16\kappa j}\Big[\frac{1}{n-j} - \frac{1}{n+j}\Big] \\ &= -\frac{c^2}{16\kappa j} \cdot \frac{2j}{n^2 - j^2} = -\frac{c^2}{8\kappa\,(n^2 - j^2)}. \end{aligned} \tag{31.22}$$

The two special levels are $n = j$ (only partner $3j$; also a first-order imaginary diagonal) and $n = 2j$ (only partner $4j$, because $|2j - 2j| = 0$ is not a level). For them we redo the sum. Level $n = j$: the first Kronecker delta in Equation (31.21) needs $|j - l| = 2j$, i.e. $l = 3j$ (since $l = -j$ is not a level); the second needs $j + l = 2j$, i.e. $l = j$, the diagonal, which gives the first-order imaginary shift and is excluded from the $m \neq n$ sum. With $E_j - E_{3j} = \kappa(j^2 - 9j^2) = -8\kappa j^2$,

$$\delta E_j = -\frac{c^2/4}{-8\kappa j^2} = \frac{c^2}{32\kappa j^2}.$$

Level $n = 2j$: $|2j - l| = 2j$ gives $l = 4j$ ($l = 0$ is not a level) and $2j + l = 2j$ gives $l = 0$, again not a level. With $E_{2j} - E_{4j} = \kappa(4j^2 - 16j^2) = -12\kappa j^2$,

$$\delta E_{2j} = -\frac{c^2/4}{-12\kappa j^2} = \frac{c^2}{48\kappa j^2}.$$

Sample Problem 31.2: The shift of level 5 from one cosine

In the box with $L = \pi/2$ (so $\kappa = 1$, $E_k = k^2$), take $A = \cos(\pi x/L)$, i.e. $j = 1$, $c = 1$. Find the second-order real shift of level 5 and check that perturbation theory is valid.
Step 1: partners. By Equation (31.21) with $j = 1$, level 5 couples to levels 3 and 7, each with $A_{nm} = -\frac{1}{2}$, so $A_{nm}^2 = \frac{1}{4}$.
Step 2: sum. $E_5 - E_3 = 25 - 9 = 16$, $E_5 - E_7 = 25 - 49 = -24$:

$$\delta E_5 = -\tfrac{1}{4}\Big[\frac{1}{16} + \frac{1}{-24}\Big] = -\tfrac{1}{4}\cdot\frac{3-2}{48} = -\frac{1}{192}.$$

This agrees with Equation (31.22): $-1/(8(25-1)) = -1/192$.
Step 3: validity. The coupling $\frac{1}{2}$ is small compared with the nearest spacing 16 (ratio $1/32$), so the fourth-order correction is of relative size about $(1/32)^2 \approx 10^{-3}$.

$$\boxed{\delta E_5 = -1/192 \approx -0.00521}$$

Sample Problem 31.3: The sum rule by hand, and the bias at the top

Truncate the same problem ($j = 1$, c arbitrary, $E_k = k^2$) to the five states $k = 1, \ldots, 5$. Compute all five second-order shifts in the truncated basis, check $\sum_n \delta E_n = 0$, and compare levels 4 and 5 with their full-basis values.
Step 1: pairs. With $j = 1$ the couplings are $(1,3)$, $(2,4)$, $(3,5)$ inside the truncated set (and $(4,6)$, $(5,7)$ lost), each with square $c^2/4$.
Step 2: shifts.

$$\delta E_1 = -\tfrac{c^2}{4}\cdot\tfrac{1}{1-9} = \tfrac{c^2}{32}, \qquad \delta E_2 = -\tfrac{c^2}{4}\cdot\tfrac{1}{4-16} = \tfrac{c^2}{48},$$
$$\delta E_3 = -\tfrac{c^2}{4}\Big[\tfrac{1}{9-1} + \tfrac{1}{9-25}\Big] = -\tfrac{c^2}{64}, \qquad \delta E_4 = -\tfrac{c^2}{4}\cdot\tfrac{1}{16-4} = -\tfrac{c^2}{48},$$
$$\delta E_5 = -\tfrac{c^2}{4}\cdot\tfrac{1}{25-9} = -\tfrac{c^2}{64}.$$

Step 3: sum. $\frac{1}{32} + \frac{1}{48} - \frac{1}{64} - \frac{1}{48} - \frac{1}{64} = \frac{1}{32} - \frac{2}{64} = 0$. The sum rule holds exactly.
Step 4: full-basis values. With all levels present, level 4 couples to 2 and 6 and level 5 to 3 and 7, so by Equation (31.22) $\delta E_4 = -c^2/120$ and $\delta E_5 = -c^2/192$. The truncated values $-c^2/48$ and $-c^2/64$ are 2.5 and 3 times too negative: the missing upper partners would have pulled these levels up.

$$\boxed{\text{truncated: } c^2\big(\tfrac{1}{32}, \tfrac{1}{48}, -\tfrac{1}{64}, -\tfrac{1}{48}, -\tfrac{1}{64}\big),\ \text{sum } 0; \quad \text{top two biased low}}$$

Figure 31.1 shows the same effect with 30 states: the bulk agrees with the full basis, while the top two levels, which have lost their upper partners, are biased strongly downward (level 30 couples only to level 28 in the truncated basis, so $\delta E_{30} = -\frac{1}{4}/(900 - 784) = -1/464$, against the full-basis value $-1/(8 \cdot 899) = -1/7192$ from Equation (31.22): a factor of 15.5). This is the mechanism behind the paper's remark that truncating the state sum at 500 states "measurably biases the top levels"; it is why the sum ran over 652 states.

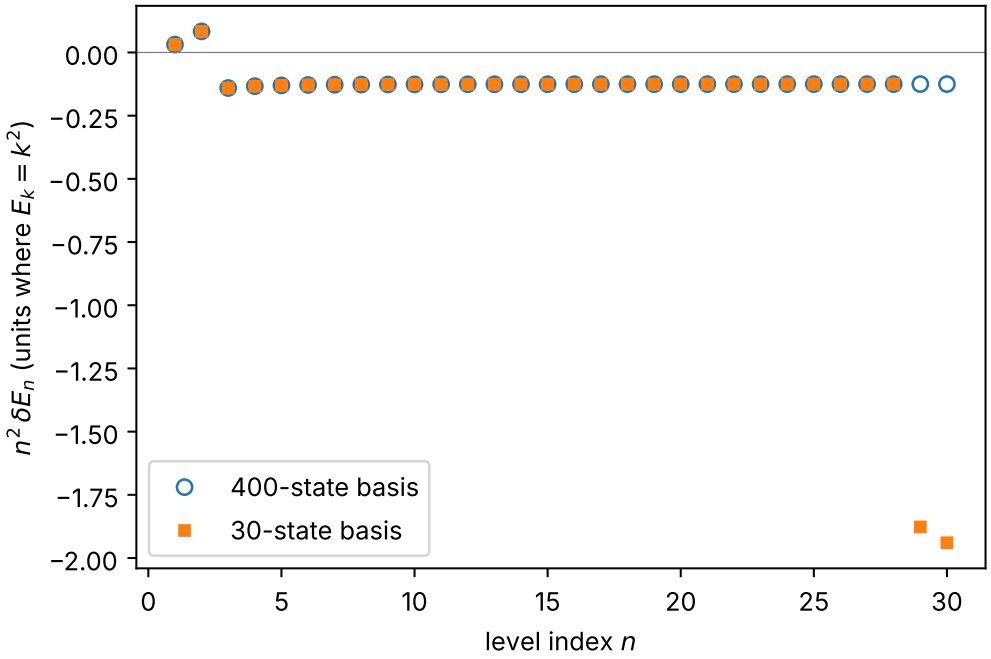


Figure 31.1: Second-order real shifts δE_n (multiplied by n^2 to make them visible) for a single cosine mode $A = \cos(\pi x/L)$ in the infinite well with $L = \pi/2$, computed in a 400-state basis (circles) and a 30-state basis (squares). Energies in units where $E_k = k^2$. The two curves agree except for the top two levels of the 30-state basis, which lack their upper coupling partners. Both sets of shifts satisfy $\sum_n \delta E_n = 0$ over their own basis.

31.3.5 How the surrogate was fitted (Appendix A, step 10)

The fit asks: which c makes the surrogate shifts reproduce the target fluctuations? The residual of App. A step 10 is

$$r_n(c) = \frac{\mathrm{fluc}_n + c^\top T^{(n)} c}{\mathrm{gap}_n}, \qquad n = 1, \ldots, 400. \tag{31.23}$$

Read it as $(\mathrm{fluc}_n - \delta E_n)/\mathrm{gap}_n$ with $\delta E_n = -c^\top T^{(n)} c$: target minus model, in mean spacings. The Jacobian (Chapter 24) needs the derivative of the quadratic form. Writing it out with indices,

$$\frac{\partial}{\partial c_k} \sum_{i,l} c_i T_{il} c_l = \sum_l T_{kl} c_l + \sum_i c_i T_{ik} = 2\,(Tc)_k \quad (\text{using } T_{ik} = T_{ki}), \tag{31.24}$$

so $J_{nk} = \partial r_n/\partial c_k = 2(T^{(n)}c)_k/\mathrm{gap}_n$, as the paper states. The update is Levenberg–Marquardt with "diag scaling" (Marquardt's form, Chapter 24): solve $(J^\top J + \lambda\,\mathrm{diag}(J^\top J))\,\delta c = -J^\top r$, trying damping values $(0.3\lambda, \lambda, 10\lambda)$. Plain damping adds λ times the identity matrix. Replacing the identity by $\mathrm{diag}(J^\top J)$ damps each coefficient in proportion to its own sensitivity, so the step does not change if one coefficient is rescaled (say, measured in different units). Because the model is quadratic in c, the least-squares problem is non-convex (for instance, c and $-c$

give identical shifts) and can have several local minima. The paper uses 6 random restarts with $c \sim \mathcal{N}(0, 0.3^2)$ in each component, seed 0, at most 200 iterations.

Sample Problem 31.4: One residual and its Jacobian

Suppose for one level $\text{fluc}_n = -0.6$, $\text{gap}_n = 1.5$, $M = 2$, $T^{(n)} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and $c = (1, -1)$. Find r_n and the Jacobian row.

Step 1: Tc. $Tc = (2 \cdot 1 + 1 \cdot (-1),\ 1 \cdot 1 + 3 \cdot (-1)) = (1, -2)$.

Step 2: quadratic form. $c^\top Tc = 1 \cdot 1 + (-1)(-2) = 3$, so the surrogate shift is $\delta E_n = -3$.

Step 3: residual. $r_n = (-0.6 + 3)/1.5 = 1.6$.

Step 4: Jacobian. $J_{n\cdot} = 2(1, -2)/1.5 = (1.333, -2.667)$. (A finite-difference check, $[r_n(c + he_k) - r_n(c - he_k)]/2h$, gives the same.)

$$\boxed{r_n = 1.6, \qquad J_{n\cdot} = (1.333,\ -2.667)}$$

```python
import numpy as np
# Box with L = pi/2, so E_k = k^2; one cosine mode j with coefficient c
K, j, c = 60, 1, 1.0
k = np.arange(1, K + 1)
E = k**2.0
kk, ll = np.meshgrid(k, k, indexing="ij")
A = c * (-1)**j / 2 * ((abs(kk - ll) == 2*j).astype(float)
                       - (kk + ll == 2*j).astype(float))
D = E[:, None] - E[None, :]
np.fill_diagonal(D, np.inf)          # exclude m = n
dE2 = -(A**2 / D).sum(axis=1)        # paper Eq. (5)
print("sum rule:", dE2.sum())        # zero to rounding
w, v = np.linalg.eig(np.diag(E) + 1j * A)
# columns of v are eigenvectors; row n holds the weight of unperturbed
# state n in each of them, so argmax along row n picks its eigenvector
idx = np.argmax(abs(v), axis=1)
dE_exact = w[idx].real - E
print(dE2[:5])
print(dE_exact[:5])
```

Key Ideas

- Eq. (5) follows from second-order perturbation theory with $W = iA$; $i^2 = -1$ reverses the Hermitian sign, so coupled levels attract.
- The real part of each eigenvalue is even in the amplitude, so the surrogate's first neglected term is fourth order and its relative error scales as amplitude squared.
- PT-odd eigenvalues are exactly real until two levels collide.
- $\delta E_n = -c^\top T^{(n)} c$ with symmetric tensors computed once; residual Equation (31.23), Jacobian $2T^{(n)}c/\text{gap}_n$.
- Truncating the state sum biases the top levels; the paper summed over 652 states.

31.4 The unconstrained floors

★ Did you know? *Rays That Weren't There*

In 1903 the respected French physicist Prosper-René Blondlot announced N-rays, and many other researchers soon reported seeing them. In 1904 the American physicist Robert Wood visited Blondlot's lab and secretly removed an essential prism, yet Blondlot still reported the effect. Effects that appear only under unconstrained, subjective measurement deserve the hardest scrutiny.

31.4.1 What the paper reports

Sec. V, first paragraph:

> "Unconstrained second-order non-Hermitian fits (Sec. II E) on zeros 1–400 stall at in-sample rms 0.2321 (PT-odd) and, after repairing the null-mode artifact, 0.2358 (absorptive-even), against a null of 0.2438; their OOS predictions are uncorrelated and can be wildly wrong (rms up to 1.4; up to 167 when fitting to 450 zeros, where extrapolated quadratic shifts blow up)."

In plain words. The optimizer improved on doing nothing by only a small amount in-sample, and its predictions for zeros it did not see were useless, sometimes catastrophically so.

Tools. In-sample vs OOS and the null (Chapter 25); LM fitting (Chapter 24); the quadratic surrogate (Section 31.3).

Computation behind it. App. A step 10, as described above, with the tensors of step 9.

Sample Problem 31.5: How much did the floors capture?

Using the paper's numbers, express each floor as a percentage improvement over the null 0.2438: (a) PT-odd 0.2321; (b) absorptive-even after repair 0.2358; (c) absorptive-even before repair 0.2322. (d) By what percentage did the artifact lower the even floor?
Step 1: definition. Capture = (null − floor)/null.
Step 2: compute. (a) $(0.2438 - 0.2321)/0.2438 = 0.0117/0.2438 = 4.80\%$. (b) $0.0080/0.2438 = 3.28\%$. (c) $0.0116/0.2438 = 4.76\%$.
Step 3: artifact. $(0.2358 - 0.2322)/0.2358 = 0.0036/0.2358 = 1.53\%$.

$$\boxed{4.80\%,\ 3.28\%,\ 4.76\%; \quad \text{artifact gain } 1.53\%}$$

These sit inside the paper's later statement that the floor is "the null minus a small capture (3–7% across splits)." For contrast, the Hermitian fit with $M = 400$ on the same zeros reaches in-sample 0.137 (Table I of the paper), a capture of more than 40%.

Why OOS can blow up. In-sample, the data constrain the 400 numbers $c^\top T^{(n)} c$ for $n \le 400$, not the vector c itself. Directions of c that barely affect these 400 quadratic forms can grow without penalty. A held-out level has its own tensor $T^{(n)}$, which can weight those directions strongly; since the shift is quadratic, a coefficient ten times too large gives a shift a hundred times too large. With 450 training zeros the paper reports held-out rms up to 167 mean spacings: far beyond any physical fluctuation, which is typically a fraction of a spacing. This is a heuristic explanation, consistent with the paper's phrase "extrapolated quadratic shifts blow up"; the paper does not give a detailed decomposition.

Strength and limits. The floor values are reported to four digits and are anchored in App. A, which makes them checkable against the Zenodo record [20]. The paper does not list

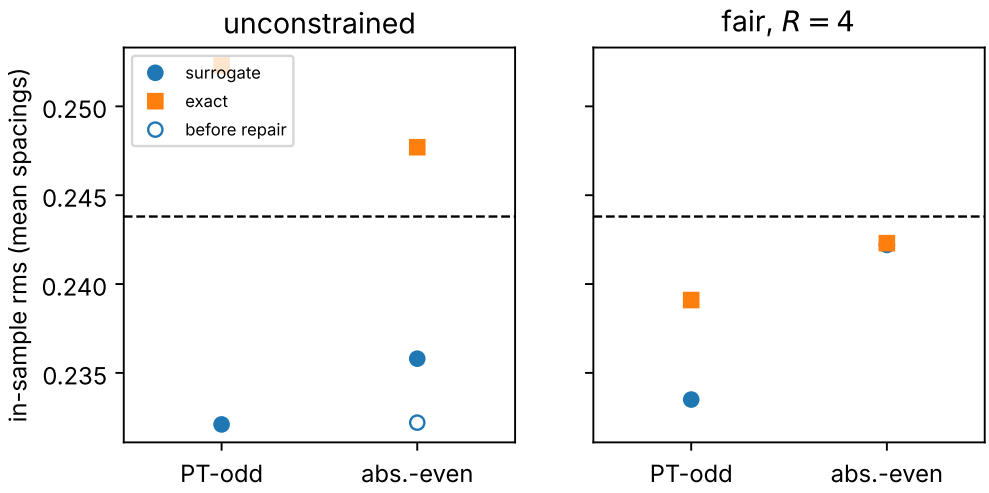


Figure 31.2: The paper's in-sample scores (rms in mean spacings, zeros 1–400) for the two non-Hermitian classes, redrawn from the numbers in Sec. V and the key anchors of Appendix A. "Surrogate" means scored with the second-order formula Eq. (5); "exact" means scored with the exact eigenvalues of the fitted operator. The open marker is the even-class floor before the null-mode repair. Dashed line: null 0.2438. Lower is better.

M for these fits in the main text (App. A gives only the restart distribution $\mathcal{N}(0, 0.3^2)^M$), so do not assume a particular value when reasoning about these fits; and the computation was not rerun in revision. The OOS figures (1.4, 167) are given as maxima ("up to"), not as a table.

Key Ideas

- Unconstrained floors: 0.2321 (odd), 0.2358 (even, repaired) against null 0.2438, captures of 4.8% and 3.3%.
- OOS predictions are uncorrelated with truth and can explode because the shifts are quadratic in coefficients that the training data do not pin down.

31.5 The null-mode artifact

31.5.1 What the paper reports

> "Before repair, the absorptive-even floor sat at 0.2322, spuriously close to the PT-odd value: the optimizer had driven the exactly-null constant mode to $c_0 \approx 2.2 \times 10^6$, harvesting in-sample improvement from float32 noise in analytically-zero tensor entries. The apparent 'identical floors across parity' dissolves."

31.5.2 Why the constant mode should do nothing

The even basis Equation (31.6) starts with $\phi_0 = \cos 0 = 1$, the constant function. It is not identically zero (unlike $\sin 0$), so it looks like a harmless basis function. But consider its effect.

Exactly, with no perturbation theory: if $H\psi_n = E_n\psi_n$, then

$$(H + ic_0)\psi_n = E_n\psi_n + ic_0\psi_n = (E_n + ic_0)\psi_n. \tag{31.25}$$

A constant imaginary potential shifts every eigenvalue by the same imaginary amount ic_0 and changes no eigenfunction. Its effect on every real part is zero to all orders.

In the surrogate, the matrix of ϕ_0 is

$$(\phi_0)_{nm} = \int \psi_n \cdot 1 \cdot \psi_m \, \mathrm{d}x = \delta_{nm} \quad \text{(orthonormality)}, \tag{31.26}$$

so every off-diagonal element is zero and, from Equation (31.20),

$$T_{0l}^{(n)} = \sum_{m \neq n} \frac{(\phi_0)_{nm}(\phi_l)_{nm}}{E_n - E_m} = \sum_{m \neq n} \frac{0 \cdot (\phi_l)_{nm}}{E_n - E_m} = 0 \quad \text{for all } l, \tag{31.27}$$

and by symmetry $T_{l0}^{(n)} = 0$. The coefficient c_0 appears nowhere in $c^\top T^{(n)} c$. It is a *dead parameter*: a null mode of the model.

31.5.3 Why it was not dead in practice

On a computer, $(\phi_0)_{nm}$ is computed as a sum $\sum_j \psi_n(x_j)\psi_m(x_j)h$ over grid points. In exact arithmetic, for exactly orthonormal eigenvectors, this is δ_{nm}. In floating point it is $\delta_{nm} + \eta_{nm}$, where η_{nm} is rounding noise. Recall from Chapter 4 that the unit roundoff of single precision (float32) is about 1.2×10^{-7}, against 2.2×10^{-16} in double precision. The paper's wording implies that the tensors were held in float32 (Sec. II E: "zeroed exactly rather than left as float32 noise"). The paper does not state why; a common reason for such a choice is memory, since float32 halves storage.

Figure 31.3 shows the effect on a small example: the overlaps of 60 finite-difference eigenvectors of the infinite well, computed in float32 and in float64. The analytically-zero off-diagonal overlaps come out many orders of magnitude larger in float32 than in float64. In this example their median magnitude lies between 10^{-10} and 10^{-6} in float32 and below 10^{-12} in float64, with a ratio of more than a thousand.

```
import numpy as np
from scipy.linalg import eigh_tridiagonal
L, N = np.pi/2, 800
h = 2*L/(N + 1)
d = 2/h**2 * np.ones(N)              # V = 0: infinite well
e = -1/h**2 * np.ones(N - 1)
E, psi = eigh_tridiagonal(d, e, select="i", select_range=(0, 59))
psi /= np.sqrt((psi**2).sum(axis=0) * h)   # sum psi^2 h = 1
G64 = psi.T @ psi * h
p32 = psi.astype(np.float32)
G32 = (p32.T @ p32) * np.float32(h)
off = ~np.eye(60, dtype=bool)
print("float64 median |offdiag|:", np.median(abs(G64[off])))
print("float32 median |offdiag|:", np.median(abs(G32[off])))
```

What follows is one possible route by which such noise can act, and the reconstruction is ours. The paper says only that the noise sat in the stored float32 tensor entries $T_{0l}^{(n)}$. That noise may have arisen from rounding of the overlaps, of the state sums, or of both. Suppose, for illustration, that the overlaps carry noise, $(\phi_0)_{nm} = \eta_{nm} \neq 0$ off the diagonal. Then the operator matrix becomes $A_{nm} = c_0\eta_{nm} + \sum_{l \geq 1} c_l(\phi_l)_{nm}$, and the constant mode contributes a *pseudo-random matrix* $c_0\eta_{nm}$. Take entries η of size 10^{-9} to 10^{-7} (an assumed range covering float32 rounding levels). A coefficient $c_0 = 2.2 \times 10^6$ then makes $c_0\eta$ of order 2×10^{-3} to 0.2 in energy units, up to a sizeable fraction of a level spacing (spacings here are of order 1).

Whatever the route, the effect on the fit is the same. The tensor rows $T_{0l}^{(n)}$ stop being zero, and c_0 multiplies a set of meaningless, pseudo-random numbers. Such numbers form a new, arbitrary set of features, and an unconstrained least-squares fit will use any feature that reduces the residual, however meaningless.

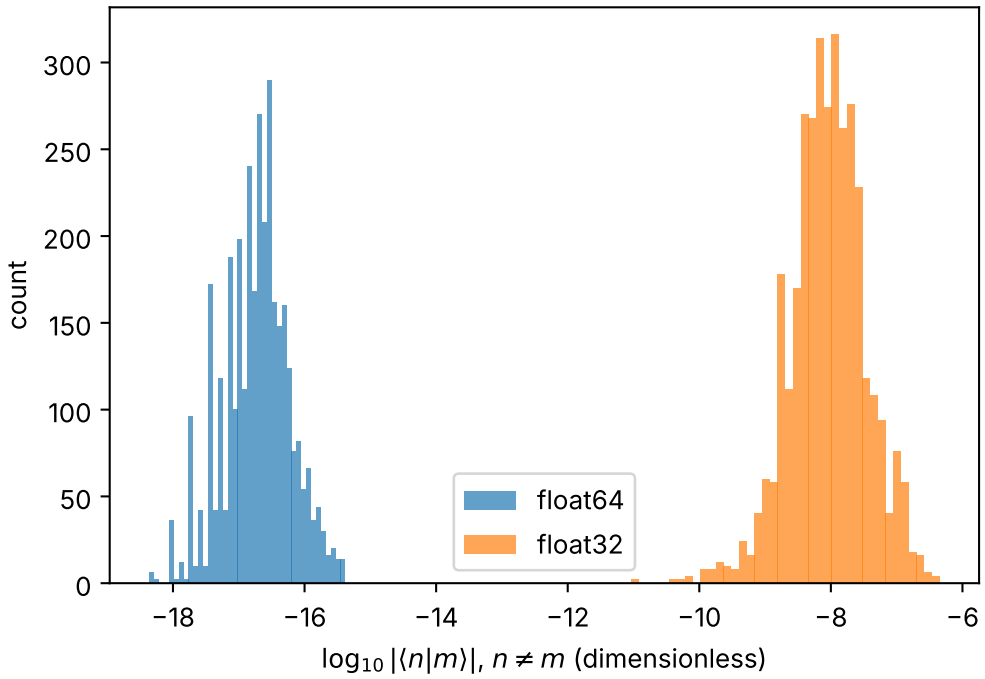


Figure 31.3: Histogram of $\log_{10}|\langle n|m\rangle|$ for $n \neq m$, the analytically zero overlaps of the first 60 finite-difference eigenvectors of the infinite well ($L = \pi/2$, $N = 800$ interior points), each normalized so that $\sum \psi^2 h = 1$. Overlaps computed in float32 (right group) lie many orders of magnitude above the same overlaps computed in float64 (left group), which are at the rounding level of double precision. A dead parameter multiplying such numbers is no longer dead.

31.5.4 Harvesting noise: a least-squares picture

The essence can be seen in plain linear least squares (Chapter 9).

Sample Problem 31.6: Harvesting a noise column

Fit $y = (0.3, -0.1, 0.2, -0.4)$ with two columns: a genuine feature $g = (1, 1, 1, 1)$ and a "noise" column $z_\varepsilon = \varepsilon\,(1, -1, -1, 1)$ with $\varepsilon = 10^{-7}$. Find the least-squares coefficients and the rms residual with and without the noise column.

Step 1: orthogonality. $g \cdot z_\varepsilon = \varepsilon(1 - 1 - 1 + 1) = 0$, so the two coefficients decouple (projection onto orthogonal directions, Chapter 9).

Step 2: genuine coefficient. $\beta_g = g \cdot y / g \cdot g = (0.3 - 0.1 + 0.2 - 0.4)/4 = 0$.

Step 3: noise coefficient. $z_\varepsilon \cdot y = \varepsilon(0.3 + 0.1 - 0.2 - 0.4) = -0.2\varepsilon$ and $z_\varepsilon \cdot z_\varepsilon = 4\varepsilon^2$, so

$$\beta_z = \frac{-0.2\varepsilon}{4\varepsilon^2} = -\frac{0.05}{\varepsilon} = -5 \times 10^5.$$

Step 4: residuals. Without the noise column the residual is y itself: rms $= \sqrt{(0.09 + 0.01 + 0.04 + 0.16)/4} = \sqrt{0.075} = 0.2739$. The noise column removes $(z_\varepsilon \cdot y)^2/(z_\varepsilon \cdot z_\varepsilon) = 0.04\varepsilon^2/4\varepsilon^2 = 0.01$ from the sum of squares, leaving 0.29: rms $= \sqrt{0.0725} = 0.2693$.

$$\boxed{\beta_z = -5 \times 10^5, \quad \text{rms } 0.2739 \to 0.2693}$$

The improvement does not depend on ε at all; only the coefficient does, as $1/\varepsilon$. Smaller noise means a larger coefficient, not a smaller harvest. That is exactly the signature in the paper: an absurdly large c_0 together with a small, real-looking in-sample gain.

Figure 31.4 repeats this at the paper's scale of data: 400 random targets of rms about 0.24, fitted with 20 genuine smooth columns plus a growing number of noise columns of size 10^{-7}. Every added noise column lowers the in-sample rms a little, and every noise coefficient is of order 10^5.

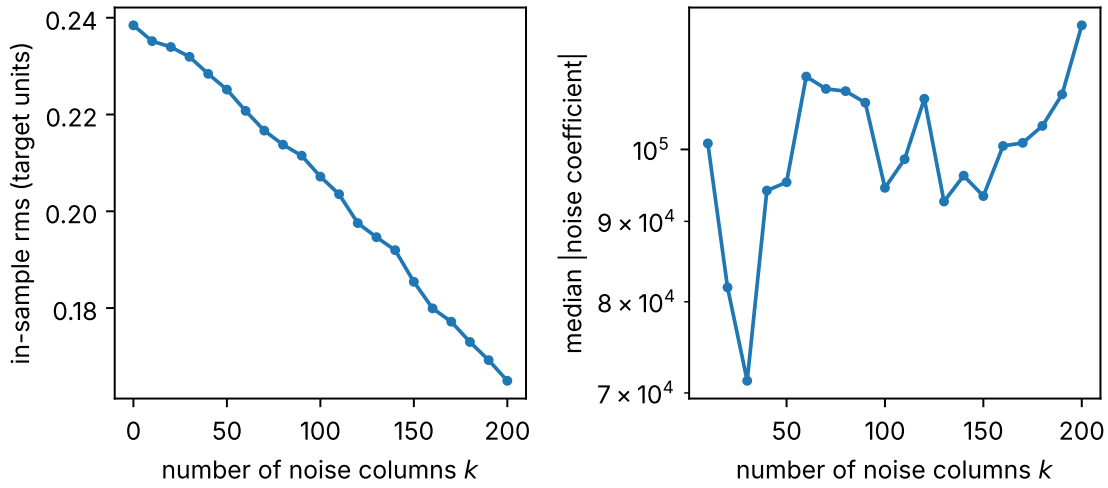


Figure 31.4: Harvesting rounding-level noise by least squares. Left: in-sample rms of a fit to 400 random targets (rms about 0.24) with 20 smooth columns plus k columns of Gaussian noise scaled by 10^{-7}; the rms falls with every added column. Right: median magnitude of the fitted noise coefficients, of order 10^5, i.e. inversely proportional to the noise scale. Data and noise drawn with `numpy.random.default_rng(2)`.

31.5.5 The repair and what it revealed

App. A step 9: "even-class rows $T_{0l}^{(n)}$ zeroed analytically." Setting those entries to their exact value, zero, makes c_0 truly dead again. After repair the even floor rose from 0.2322 to 0.2358, while the odd floor stayed at 0.2321 (the odd basis has no constant mode). The near coincidence 0.2321 versus 0.2322, which had suggested a parity-independent "floor" with physical meaning, was an artifact: it vanished when one input bug was fixed.

Checkpoint 31.2

Why does the constant mode have zero second-order real shift but a nonzero effect on the imaginary parts?

Answer: Its matrix is the identity, $(\phi_0)_{nm} = \delta_{nm}$. The off-diagonal elements, which drive the second-order real shift, are zero; the diagonal elements, which give the first-order imaginary shift ic_0, are 1. Exactly, $H + ic_0$ has eigenvalues $E_n + ic_0$: the shift is purely imaginary, and the real parts do not change.

Checkpoint 31.3

Two variants of a model report floors 0.2321 and 0.2322. List two checks you would make before attributing the agreement to physics.

Answer: (i) Inspect the fitted coefficients for absurd magnitudes (a coefficient near 10^6 on a mode that should be inert is a red flag). (ii) Recompute in higher precision, or zero analytically-null quantities, and see whether the numbers move. Also: score the exact operator rather than the surrogate.

Strength and limits. The artifact claim is unusually strong for a diagnostic: the paper names the dead parameter, its value, the noise source, and the repair, and reports the score before and after. What the paper does not do is show that float32 noise alone produces exactly a 1.5% gain; our least-squares illustration shows that such gains are generic, not that this one is quantitatively explained.

Key Ideas

- The constant mode $\phi_0 = 1$ is a null mode: exactly, $H + ic_0$ only shifts eigenvalues by ic_0; in the surrogate, $T_{0l}^{(n)} = 0$.
- In float32 the analytic zeros become rounding noise, many orders of magnitude above the double-precision level. Multiplied by $c_0 \approx 2.2 \times 10^6$, noise in the range 10^{-9}–10^{-7} would give pseudo-random entries of up to a sizeable fraction of a spacing.
- Least squares harvests a fixed improvement from any noise column, with a coefficient scaling as 1/(noise size).
- After repair the even floor is 0.2358; the "identical floors across parity" dissolve.

31.6 The block-coherence hypothesis and why it fails

31.6.1 The sum rule and locality (Sec. V A)

Recall the sum rule from Chapter 14. If n and m both run over the same truncated set S of states,

$$\begin{aligned}\sum_{n\in S}\delta E_n &= -\sum_{n\in S}\sum_{\substack{m\in S\\ m\neq n}}\frac{A_{nm}^2}{E_n - E_m}\\ &= -\sum_{\substack{n,m\in S\\ n<m}} A_{nm}^2\Big[\frac{1}{E_n - E_m} + \frac{1}{E_m - E_n}\Big] \quad \text{(group each pair once; } A_{mn}^2 = A_{nm}^2\text{)}\\ &= 0. \end{aligned} \tag{31.28}$$

Each pair contributes equal and opposite amounts to its two members. The paper verifies this numerically at the fitted coefficients: 3.6×10^{-15} (odd) and 1.4×10^{-14} (even) in Sec. V A, summarized as "verified to 4×10^{-15}" in the main text. Numbers at this level are rounding of double-precision sums; they confirm both the identity and the code.

Locality: with weights $w_{nm} = |A_{nm}|^2/|E_n - E_m|$, the paper finds that half the total weight lies within $|n - m| \leq 5$ (odd class) and ≤ 10 (even), and 90% within about 60 levels. So the surrogate mostly moves levels by trading shifts between near neighbors.

31.6.2 The hypothesis

Put the two facts together. If shifts sum to zero and are traded only locally, then the model output averaged over a block of w consecutive levels should be close to zero. Such a model cannot produce a slowly varying, block-averaged pattern. So the part of the target that is block-averaged (the "block-coherent" part) should be unreachable, and perhaps the floor *is* that part. This is the hypothesis Sec. V tests: that the floor equals the rms of the locally averaged part of the target fluctuation.

To make it precise, write $d_n = \mathrm{fluc}_n/\mathrm{gap}_n$ and, for an odd window $w = 2q + 1$, define the

centred moving average and its complement:

$$B_w(d)_n = \frac{1}{w}\sum_{i=-q}^{q} d_{n+i}, \qquad F_w(d)_n = d_n - B_w(d)_n. \tag{31.29}$$

B_w is the block-coherent part, F_w the locally mean-free part.

31.6.3 The block part is a lower bound, not a prediction

Here is what the hypothesis can and cannot claim. Suppose an ideal local model produced outputs m_n with $B_w(m) = 0$ exactly. The residual $d-m$ then has $B_w(d-m) = B_w(d) - B_w(m) = B_w(d)$ (moving averages are linear). And a moving average cannot increase rms. Indeed, for each window, by the Cauchy–Schwarz inequality (Chapter 19: the square of an average is at most the average of the squares),

$$\Big(\frac{1}{w}\sum_{i=-q}^{q} x_{n+i}\Big)^2 \le \frac{1}{w}\sum_{i=-q}^{q} x_{n+i}^2, \tag{31.30}$$

and we sum over all $n_{\text{fit}} - w + 1$ full windows. Here $n_{\text{fit}} = 400$ is the number of scored levels; we avoid the letter N, which in this chapter is a grid size. Each x_j^2 appears in at most w windows, each time with weight $1/w$, so $\sum_n B_w(x)_n^2 \le \sum_j x_j^2$. To form mean squares, divide the left side by the number of full windows and the right side by n_{fit}:

$$\operatorname{rms}(x) \;\ge\; \sqrt{\frac{n_{\text{fit}} - w + 1}{n_{\text{fit}}}}\,\operatorname{rms}\big(B_w(x)\big). \tag{31.31}$$

Apply this to $x = d - m$, using $B_w(d-m) = B_w(d)$. The rms of the residual is at least the rms of $B_w(d)$ times the edge factor $\sqrt{(n_{\text{fit}} - w + 1)/n_{\text{fit}}}$, which exceeds 0.98 for $w \le 13$ and $n_{\text{fit}} = 400$. So the block-coherent rms is a *lower bound* on what a perfectly local, sum-preserving model could achieve. The hypothesis was that the observed floor sits at this bound.

31.6.4 The test

Sec. V A decomposes the in-sample target fluctuation (rms 0.2438) by centred w-level moving averages and finds block-coherent rms

$$0.108,\ 0.049,\ 0.033,\ 0.035,\ 0.019 \qquad \text{for } w = 3, 5, 7, 9, 13.$$

All are far below 0.232. The main text summarizes: "that component has rms only 0.03–0.11 spacings for averaging windows of 3–13 levels, never near 0.232," failing "by a factor 2–7." The factor follows from $0.232/0.108 = 2.1$ and $0.232/0.033 = 7.0$. (A careful reader notices that the $w = 13$ value 0.019 lies below the quoted range 0.03–0.11 and would give a factor of about 12; the qualitative conclusion only gets stronger.)

The lower bound is nowhere near binding. The floor is not explained by what the model cannot do; it must be explained by what the model, as fitted, does not do.

Figure 31.5 repeats the decomposition at small scale with qmnt, using $d_n = (t_n - s_n)/\text{gap}_n$ for zeros 1–400. This is the smooth-zero version of the target, not the paper's Richardson-corrected one, so it is close to but not identical with the paper's d_n. Its block-coherent rms stays below 0.15 for every window, so it lies at least a factor 1.5 below the floor 0.232. The individual values are plotted, and Problem 11 asks you to compute them. We quote only this bound in the text because the smooth-zero version is a stand-in for the paper's targets, not a reproduction of them.

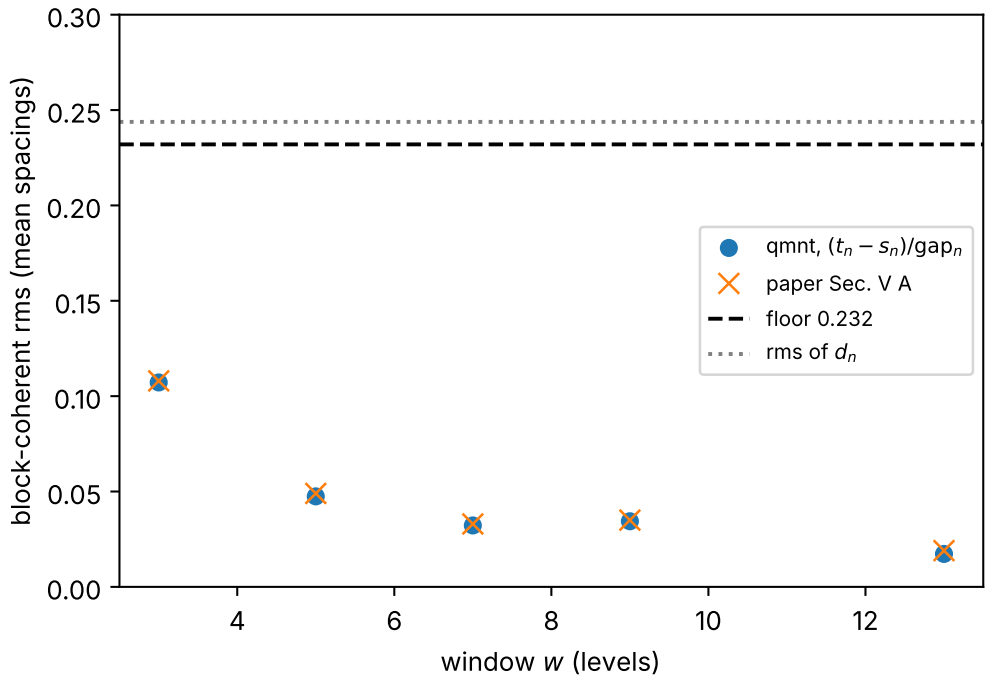


Figure 31.5: Block-coherent rms (rms of the centred w-level moving average) of the zero fluctuations $d_n = (t_n - s_n)/\mathrm{gap}_n$, zeros 1–400, computed with qmnt (circles), and the values reported in the paper's Sec. V A for its corrected targets (crosses). Dashed: the unconstrained floor 0.232. Dotted: rms of d_n itself. Moving averages use only full windows. The paper's values lie a factor 2 to 12 below the floor; the qmnt values all lie below 0.15.

Sample Problem 31.7: Why negatively correlated sequences have small block averages

Let $d_n = (-1)^n d_0$ with a constant amplitude $d_0 > 0$, the extreme of negative lag-1 correlation. Compute $B_w(d)$ and its rms for odd $w = 2q + 1$.

Step 1: one window. $B_w(d)_n = \frac{d_0}{w}\sum_{i=-q}^{q}(-1)^{n+i} = \frac{(-1)^n d_0}{w}\sum_{i=-q}^{q}(-1)^i$.

Step 2: the alternating sum. The terms for i and $-i$ are equal, so $\sum_{i=-q}^{q}(-1)^i = 1 + 2\sum_{i=1}^{q}(-1)^i$. For q odd the inner sum is -1, giving -1; for q even it is 0, giving $+1$. So the sum is $(-1)^q$.

Step 3: rms. $B_w(d)_n = (-1)^{n+q} d_0/w$, so $|B_w| = d_0/w$ for every n:

$$\boxed{\mathrm{rms}\, B_w(d) = \frac{d_0}{w}}$$

For $w = 3$ the block part is one third of the signal. Recall from Chapter 20 and the paper's Sec. III that the zero fluctuations have negative autocorrelation at lags 1–3; that is why their block averages are small.

31.6.5 The $p = 2$ oscillation is itself local

The paper adds: "The dominant $p = 2$ oscillation, period $2\pi/\log 2 \approx 9$ in t, is only ∼6–7 levels at these heights, itself a local pattern." Check it. The period in t is $2\pi/\log 2 = 9.065$ (RE2). The number of levels in one period is the period divided by the mean spacing $\mathrm{gap}_n = 2\pi/\log(s_n/2\pi)$. With qmnt, $\mathrm{gap}_{200} \approx 1.516$ and $\mathrm{gap}_{400} \approx 1.341$, so one period spans about 6.0 levels near

$n = 200$ and about 6.8 levels near $n = 400$. A pattern that repeats every six or seven levels averages out over windows of comparable size, so it contributes little to the block-coherent part; and it is a pattern a local redistribution could in principle produce. Locality is not what stops the model.

31.6.6 Capture, not mechanism

The paper's conclusion: "The floor is simply the null minus a small capture (3–7% across splits), not a physically meaningful constant." And from Sec. V A: "The fitted models' output correlates $\approx +0.3$ with the locally mean-free component but with rms only 0.06–0.075: capture, not mechanism, limits the floor." The next sample shows that these numbers are self-consistent.

Sample Problem 31.8: Is a weak capture enough to make the floor?

Model the target as $d = B + F$ with B and F uncorrelated, $\mathrm{rms}(d) = 0.2438$, $\mathrm{rms}(B) = 0.033$ (the $w = 7$ value). Let the model output m have rms σ_m, correlation $\rho = 0.3$ with F, and zero correlation with B. (These simplifying assumptions are ours, not the paper's.) Find the residual rms for $\sigma_m = 0.07$ and 0.06, and the σ_m that minimizes it. Treat all sequences as mean-zero.

Step 1: rms of F. Uncorrelated parts add in square: $\sigma_F^2 = 0.2438^2 - 0.033^2 = 0.059438 - 0.001089 = 0.058349$, so $\sigma_F = 0.2416$.

Step 2: residual variance. $\sigma_r^2 = \mathrm{var}(d - m) = \sigma_d^2 - 2\,\mathrm{cov}(d, m) + \sigma_m^2$, and $\mathrm{cov}(d, m) = \mathrm{cov}(F, m) + \mathrm{cov}(B, m) = \rho\sigma_F\sigma_m + 0$.

Step 3: numbers. For $\sigma_m = 0.07$: $\sigma_r^2 = 0.059438 - 2(0.3)(0.2416)(0.07) + 0.0049 = 0.054191$, $\sigma_r = 0.2328$. For $\sigma_m = 0.06$: $\sigma_r = 0.2331$.

Step 4: optimum. $d\sigma_r^2/d\sigma_m = -2\rho\sigma_F + 2\sigma_m = 0$ gives $\sigma_m = \rho\sigma_F = 0.0725$ and $\sigma_r^2 = \sigma_d^2 - \rho^2\sigma_F^2$, $\sigma_r = 0.2328$.

$$\boxed{\sigma_r \approx 0.233, \text{ optimum at } \sigma_m \approx 0.072}$$

A model output of the reported size and correlation lands right at the floors 0.232–0.236, and its rms sits close to the least-squares optimum for its correlation. Nothing beyond "small capture" is needed to explain the floor. This back-of-envelope check is an illustration with stated assumptions, not a computation from the paper.

Checkpoint 31.4

Does the sum rule $\sum_n \delta E_n = 0$ restrict the in-sample target the model can fit?

Answer: Yes, in the surrogate: the model's 400 shifts (if the scored levels formed the whole basis) would have mean zero, so any nonzero mean of the target is unreachable. In the paper the sum runs over 652 states while 400 are scored, so the constraint on the scored window is only approximate; together with locality, the constraint is the one behind the block-coherence hypothesis.

Strength and limits. The refutation is decisive in direction: a lower bound two to twelve times below the observed floor cannot be the floor. The positive explanation ("small capture") is descriptive; the paper does not explain *why* the capture is small for these classes, and we should not either.

Key Ideas

- Sum rule and locality imply that a perfect surrogate model can only redistribute shifts locally; the block-coherent rms is then a lower bound on its residual.

- That bound is far below the floor 0.232. The paper's main text quotes 0.03–0.11 for windows 3–13 and a failure by a factor 2–7; its own Sec. V A list gives 0.019 at $w = 13$ (factor about 12), a small inconsistency in the paper that only strengthens the refutation.
- The $p = 2$ oscillation spans only about 6–7 levels at these heights, a local pattern.
- The floor is the null minus a small capture; a weak output (correlation 0.3, rms 0.06–0.075) reproduces it.

31.7 Leaving the perturbative regime

★ Did you know? ***When Series Diverge***

In 1952 Freeman Dyson argued that the perturbation series of quantum electrodynamics cannot converge. If the square of the electric charge were negative, the vacuum would be unstable, so the series must have zero radius of convergence. Yet its first few terms give spectacularly accurate predictions. Perturbation theory can be excellent up to a point and then fail.

31.7.1 The claim

> "The second-order surrogate of Eq. (5) is an optimization device; the object of any physical claim is the exact spectrum of $H + iA$. At the fitted amplitude, exact complex diagonalization disagrees with the second-order surrogate: corr(exact, second-order) drops to +0.14/+0.18 (odd/even), the surrogate overstates shift magnitudes twofold (Fig. 5, left), and 3% of eigenstates can no longer be matched one-to-one to unperturbed states. Scoring the *exact* spectra of the fitted operators in-sample gives rms 0.2523 and 0.2477, *worse than the null* 0.2438."

In plain words. The optimizer pushed the coefficients so far that the formula it was minimizing no longer described the operator. The real operator, solved exactly, does worse than doing nothing.

31.7.2 A two-level model that shows everything

Two of the three symptoms (wrong magnitude, failed matching) appear already in the simplest possible case. The loss of correlation across levels needs many levels and is shown in the box example below. Take two levels with spacing $\Delta > 0$ and an imaginary coupling ia with zero diagonal (a PT-odd pair):

$$H(a) = \begin{pmatrix} E_1 & ia \\ ia & E_1 + \Delta \end{pmatrix}. \tag{31.32}$$

The characteristic equation (Chapter 8) is $(E_1 - \lambda)(E_1 + \Delta - \lambda) - (ia)^2 = 0$, i.e. $(\lambda - E_1)^2 - \Delta(\lambda - E_1) + a^2 = 0$, so

$$\lambda_\pm = E_1 + \frac{\Delta}{2} \pm \sqrt{\frac{\Delta^2}{4} - a^2}. \tag{31.33}$$

Write $x = a/\Delta$. In this subsection x is a dimensionless coupling, not a position, and a is the coupling strength. For $x < \frac{1}{2}$ both eigenvalues are real. The exact shift of the lower level is

$$\delta E_1^{\text{exact}} = \frac{\Delta}{2}\left(1 - \sqrt{1 - 4x^2}\right), \qquad \delta E_1^{(2)} = -\frac{a^2}{E_1 - (E_1 + \Delta)} = x^2\Delta. \tag{31.34}$$

Expanding the square root, $\sqrt{1-4x^2} = 1 - 2x^2 - 2x^4 - \dots$, gives $\delta E_1^{\text{exact}} = \Delta(x^2 + x^4 + \dots)$: second order agrees, and the first correction is fourth order, as Equation (31.19) promised.

At $x = \frac{1}{2}$ the two eigenvalues collide at $E_1 + \Delta/2$: an *exceptional point*. For $x > \frac{1}{2}$ the square root becomes imaginary, $\lambda_\pm = E_1 + \Delta/2 \pm i\sqrt{a^2 - \Delta^2/4}$: PT symmetry is broken, and both real parts are stuck at $E_1 + \Delta/2$. The exact real shift saturates at $\Delta/2$, while the surrogate keeps growing as $x^2\Delta$. Beyond the exceptional point the ratio is surrogate/exact $= x^2\Delta/(\Delta/2) = 2x^2$. So the surrogate still understates for $\frac{1}{2} < x < 1/\sqrt{2} \approx 0.707$, agrees at $x = 1/\sqrt{2}$, and overstates for larger x (twofold at $x = 1$). Below the exceptional point it always understates, because every term $x^4, x^6, \dots$ in the expansion of the exact shift has a positive coefficient.

The eigenvectors tell the matching story. Write an eigenvector with eigenvalue λ as $v = (v_1, v_2)$, its components along unperturbed states 1 and 2. The first row of $(H - \lambda)v = 0$ gives $(E_1 - \lambda)v_1 + iav_2 = 0$, so $|v_2/v_1| = |\lambda - E_1|/a$. Below the exceptional point the lower eigenvector has $|v_2/v_1| < 1$: it is mostly state 1 and is matched to it. At and beyond the exceptional point, $|\lambda - E_1|^2 = \Delta^2/4 + (a^2 - \Delta^2/4) = a^2$, so $|v_2/v_1| = 1$: *both* eigenvectors overlap equally with both unperturbed states. Matching by overlap (Chapter 14) has no answer. Figure 31.6 plots both effects.

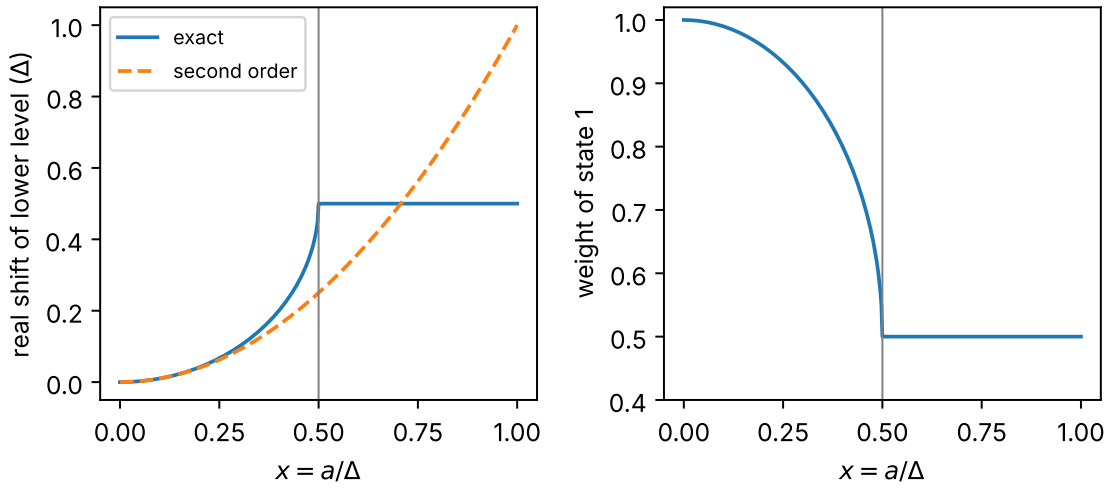


Figure 31.6: The two-level model Equation (31.32). Left: real shift of the lower level, exact (solid) and second order (dashed), in units of the spacing Δ, against coupling $x = a/\Delta$. The exceptional point at $x = 1/2$ is marked; beyond it the exact real shift saturates at $\Delta/2$ while the surrogate keeps growing, overstating the shift twofold at $x = 1$. Here x is the dimensionless coupling, not a position. Right: weight of unperturbed state 1 in the lower-branch eigenvector (v_1, v_2), $|v_1|^2/(|v_1|^2 + |v_2|^2)$; it falls to 1/2 at the exceptional point and stays there, so the eigenstate can no longer be assigned to a single unperturbed level.

Sample Problem 31.9: Exact versus surrogate in the two-level model

For Equation (31.32), compute the exact and second-order real shifts of the lower level, in units of Δ, at $x = \frac{1}{4}, \frac{1}{2}, 1$. Also give $|v_2/v_1|$ for the lower state at $x = \frac{1}{4}$.

Step 1: $x = \frac{1}{4}$. $\sqrt{1 - 4/16} = \sqrt{0.75} = 0.8660$, so $\delta E^{\text{exact}} = \frac{1}{2}(1 - 0.8660) = 0.0670$; surrogate $x^2 = 0.0625$. The surrogate is low by 7%.

Step 2: $x = \frac{1}{2}$. Exact $\frac{1}{2}(1 - 0) = 0.5$; surrogate 0.25. At the exceptional point the surrogate understates by a factor two.

Step 3: $x = 1$. Exact real part stuck at 0.5; surrogate 1. The surrogate now overstates by a factor two.

Step 4: overlap. At $x = \frac{1}{4}$, $|v_2/v_1| = |\lambda - E_1|/a = (\delta E^{\text{exact}}/\Delta)/x = 0.0670/0.25 = 0.268$, so the lower state is clearly state 1.

$$\boxed{(0.0670,\, 0.0625),\ (0.5,\, 0.25),\ (0.5,\, 1);\quad |v_2/v_1| = 0.268}$$

The model does not prove the paper's "twofold" overstatement. It shows only that overstatements of order one (for example, twofold at $x = 1$) arise naturally once a coupling exceeds the spacing; just past the exceptional point the surrogate can even understate.

Sample Problem 31.10: Why quarter amplitude restores agreement

In the two-level model at $x = 0.4$ (below the exceptional point), compute the relative error of the surrogate; repeat at quarter amplitude, $x = 0.1$, and compare the ratio of relative errors with the factor 16 suggested by Equation (31.19).
Step 1: $x = 0.4$. $\sqrt{1 - 0.64} = 0.6$, exact $= 0.2$, surrogate $= 0.16$, relative error $(0.16 - 0.2)/0.2 = -0.200$.
Step 2: $x = 0.1$. $\sqrt{0.96} = 0.97980$, exact $= \frac{1}{2}(0.02020) = 0.010102$, surrogate $= 0.01$, relative error -0.0101.
Step 3: ratio. $0.200/0.0101 = 19.8$.

$$\boxed{\text{relative error } -0.200 \to -0.0101,\ \text{ratio} \approx 19.8}$$

The ratio is near 16 (the leading x^2 scaling), somewhat larger because at $x = 0.4$ the higher terms of the series also matter. Quartering the amplitude is a powerful test: if the code were wrong, agreement would not improve like this.

This is exactly the logic of the paper's next sentence: "At quarter amplitude the surrogate is accurate (corr +0.88/+0.94), confirming the machinery and isolating amplitude as the culprit." The same tensors and the same code were used, and presumably the fitted coefficients scaled by $\frac{1}{4}$, that is, the same coefficient *direction* (the paper does not spell out the construction). Only the size changed, and agreement returned. This supports the correctness of the machinery and points to amplitude as the main cause of the disagreement at full amplitude, although by itself it does not exclude every other cause. Note also that the clean sixteenfold rule of Equation (31.19) applies only when both amplitudes are inside the radius of convergence. At the paper's full amplitude some pairs had already left it (3% of states unmatched), so the improvement need not follow that rule.

31.7.3 The same thing in a many-level box

Figure 31.7 is a small-scale analogue of the paper's Fig. 5 (left): six random cosine modes in the infinite well, with all 80 basis states, scored on the lowest 40 levels. At small amplitude the exact real shifts (obtained by full complex diagonalization and matched by overlap) lie on the diagonal: their relative rms disagreement with the surrogate is below 5%. At large amplitude the points leave the diagonal, with relative disagreement above 10%. This is consistent with the strongest couplings of the low levels approaching the two-level exceptional-point threshold (a coupling of half the corresponding level difference).

31.7.4 Scores worse than the null

How can a fitted operator score worse than doing nothing? Because its exact shifts are not the surrogate shifts the optimizer tuned. With correlation only +0.14 between exact and surrogate,

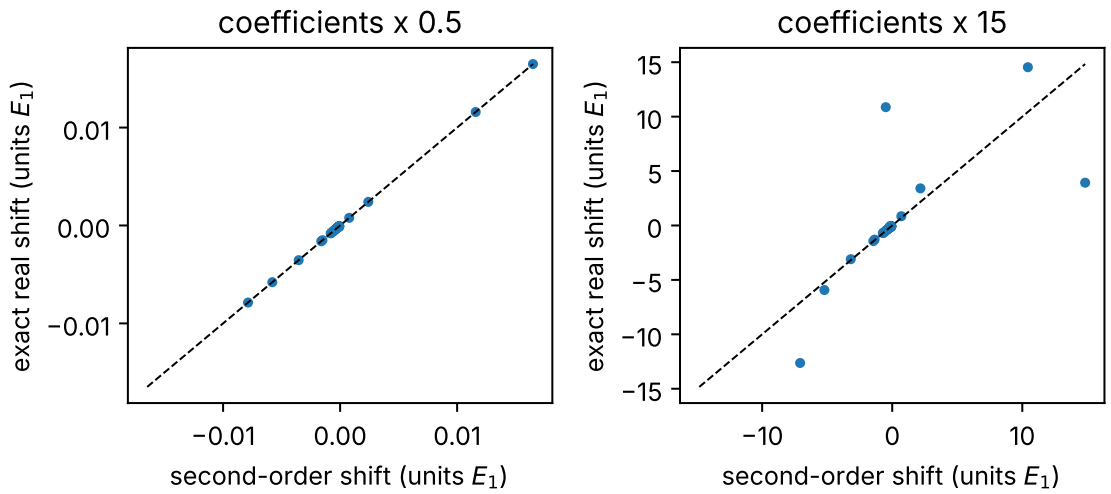


Figure 31.7: Exact versus second-order real shifts for $H_0 + iA$ in the infinite well ($L = \pi/2$, levels $E_k = k^2$, 80 states), with $A = \sum_{j=1}^{6} c_j \cos(j\pi x/L)$ and random c_j (`default_rng(1)`). Exact shifts from dense complex diagonalization, each unperturbed level matched to the eigenvector with largest weight on it; the lowest 40 levels shown. Left: coefficients $0.5c_j$; points on the diagonal. Right: coefficients $15c_j$; the points leave the diagonal (relative rms disagreement above 10%). Energies in units where $E_k = k^2$.

the exact shifts are, for scoring purposes, mostly noise added to the residual.

Sample Problem 31.11: How much noise turns 0.2438 into 0.2523?

If the exact shifts (in mean spacings) were uncorrelated with the target fluctuations and had rms σ, what σ would turn the null 0.2438 into the paper's exact in-sample scores 0.2523 (odd) and 0.2477 (even)?

Step 1: variance addition. For uncorrelated, mean-zero d and m, $\mathrm{rms}(d-m)^2 = \mathrm{rms}(d)^2 + \sigma^2$.

Step 2: solve. $\sigma = \sqrt{0.2523^2 - 0.2438^2} = \sqrt{0.063655 - 0.059438} = \sqrt{0.004217} = 0.0649$; and $\sqrt{0.2477^2 - 0.2438^2} = \sqrt{0.001917} = 0.0438$.

$$\boxed{\sigma \approx 0.065 \text{ (odd)}, \quad \sigma \approx 0.044 \text{ (even)}}$$

Shifts of only about a twentieth of a spacing, uncorrelated with the target, are enough. (The assumption of zero correlation is an idealization for illustration.)

31.7.5 Wrong-sign widths

"The fitted absorptive-even solution also violates its physical premise: 58% of its widths have the wrong sign." The even class was meant to model absorption, $\Gamma(x) \geq 0$. By Equation (31.9), if $\Gamma \geq 0$ everywhere then every first-order width is non-negative; so a negative width proves that $\Gamma(x) < 0$ somewhere, i.e. the fitted "absorber" amplifies. The paper does not specify whether "widths" means the level widths Γ_n or the values of $\Gamma(x)$ on the grid; either reading shows that the unconstrained optimizer ignored the physics the class was designed to represent. That is why the fair version (next section) enforces $\Gamma \geq 0$.

31.7.6 The cost of exactness

App. A step 11: "dense complex `eig` of $H + iA$ on $N = 2500$, second-order sums over the full grid spectrum, eigenstate matching by overlap with the unperturbed basis." A dense eigensolve costs of order N^3 operations (Chapter 8): $2500^3 \approx 1.6 \times 10^{10}$, seconds to minutes on a laptop. Note that this check uses a coarser grid than the fits and sums the surrogate over all 2500 grid states, so that exact and surrogate are compared on one common grid, as the paper states in Sec. II E (iii).

Checkpoint 31.5

The paper scored the exact spectra in-sample, not out-of-sample. Why is an in-sample score the right test here?

Answer: The question is whether the floor is a property of an operator at all. The surrogate claimed an in-sample rms of 0.232; if the operator's own spectrum does not reproduce that on the very zeros it was fitted to, the floor belonged to the surrogate. An in-sample failure (0.2523, 0.2477 against 0.2438) settles this without any appeal to generalization.

Checkpoint 31.6

In the one-cosine laboratory above ($A = c\cos(j\pi x/L)$ in the box, every coupling of magnitude $c/2$), which levels leave the perturbative regime first as c grows?

Answer: The validity condition compares each coupling with the energy difference to its partner. All couplings have magnitude $c/2$, but the differences are $4\kappa j(n + j)$ and $4\kappa j|n - j|$ for a generic level, $8\kappa j^2$ for $n = j$ and $12\kappa j^2$ for $n = 2j$. They are smallest near $n = j$ (for $j = 1$ the smallest, 8κ, joins levels 1 and 3; for $j \geq 2$ it is $4\kappa j$, at $n = j \pm 1$) and grow with n beyond that, so in this example the lowest levels fail first. In another potential, with different spacings and matrix elements, nothing guarantees the same ordering: the condition has to be checked level by level.

Strength and limits. This is the strongest part of the dissection: a direct, exact computation contradicts the surrogate, and a quarter-amplitude control supports the correctness of the machinery. Limits: the exact check used $N = 2500$ rather than the working grid, and, like the rest of Sec. V, it was not rerun in revision.

Key Ideas

- The two-level model shows two of the three symptoms. First, the surrogate is wrong in magnitude: it understates up to and somewhat past the exceptional point (beyond it the ratio surrogate/exact is $2x^2$) and overstates for $x > 1/\sqrt{2}$ (twofold at $a = \Delta$). Second, the eigenvectors become unmatchable. The loss of correlation needs many levels (box figure).
- Because real parts are even in amplitude, quartering a small amplitude cuts the relative error about sixteenfold. Agreement returning at quarter amplitude supports the correctness of the machinery and points to amplitude.
- Exact in-sample scores 0.2523 and 0.2477 are worse than the null; under the idealized assumption of zero correlation with the target (an illustration, not a finding of the paper), shifts of rms 0.065 and 0.044 spacings would do this.
- 58% wrong-sign widths show the absorptive premise was violated.

31.8 The fair version: amplitude balls and sign constraints

31.8.1 The claim

> "Refitting inside amplitude balls where perturbation theory is verified valid, and with $\Gamma \geq 0$ enforced, gives floors within 0.4–4.8% of the null (Fig. 5, right); the $R = 4$ solutions pass exact verification (corr +0.93/+0.995, all states matched) and their exact-spectrum in-sample gains over the null are 1.9% (odd) and 0.6% (even). We conclude that these second-order non-Hermitian families, treated fairly, are empirically inert on this problem."

31.8.2 The machinery (App. A step 10, "fair variant")

Amplitude ball. After every LM step, the coefficient vector is projected onto the ball $\|c\| \leq R$:

$$c \;\to\; c \cdot \min\Big(1, \frac{R}{\|c\|}\Big). \tag{31.35}$$

This is the closest point of the ball to c (Chapter 7: for a point outside a sphere, the nearest point of the sphere lies on the line to the center). Restarts are drawn from $\mathcal{N}(0, (R/20)^2)$ in each of the M components. The squared norm is a sum of M squares, each of mean $(R/20)^2$, so the typical starting norm is about $(R/20)\sqrt{M}$. That lies inside the ball provided $M < 400$. The paper does not state M, so we can only say that the spread of the restarts is scaled with R. R is "calibrated so that second-order perturbation theory is verified accurate", that is, chosen by checking the surrogate against exact diagonalization.

Sign penalty for the absorptive class: the objective gets an extra term $10^4 \int \max(A, 0)^2 \, \mathrm{d}x$, which is zero when $A \leq 0$ everywhere (i.e. $\Gamma \geq 0$) and grows quadratically with any positive excursion.

Sample Problem 31.12: Projection and penalty

(a) Project $c = (3, 4, 12)$ onto the ball $\|c\| \leq 4$. (b) For $A(x) = -1 + 2\cos(\pi x/L)$ on $[-L, L]$ with $L = 44.883$ (RE3), compute the penalty $10^4 \int \max(A, 0)^2 \, \mathrm{d}x$.

Step 1: norm. $\|c\| = \sqrt{9 + 16 + 144} = 13$.

Step 2: scale. $c \to (4/13)(3, 4, 12) = (0.923,\ 1.231,\ 3.692)$.

Step 3: where $A > 0$. $\cos(\pi x/L) > \frac{1}{2}$ means $|\pi x/L| < \pi/3$. Substitute $\theta = \pi x/L$, $\mathrm{d}x = (L/\pi)\mathrm{d}\theta$.

Step 4: integrate. $(2\cos\theta - 1)^2 = 4\cos^2\theta - 4\cos\theta + 1 = 3 + 2\cos 2\theta - 4\cos\theta$, so

$$\int_{-\pi/3}^{\pi/3} (2\cos\theta - 1)^2 \mathrm{d}\theta = \Big[3\theta + \sin 2\theta - 4\sin\theta\Big]_{-\pi/3}^{\pi/3} = 2\Big(\pi + \frac{\sqrt{3}}{2} - 2\sqrt{3}\Big) = 2\pi - 3\sqrt{3} = 1.0870.$$

Step 5: penalty. $\int \max(A, 0)^2 \mathrm{d}x = (L/\pi)(2\pi - 3\sqrt{3}) = 14.287 \times 1.0870 = 15.53$, so the penalty is 1.553×10^5.

$$\boxed{c \to (0.923, 1.231, 3.692); \qquad \text{penalty} \approx 1.55 \times 10^5}$$

A penalty this large dwarfs the least-squares objective (a mean-square residual of about $0.2438^2 \approx 0.059$, or about 24 when summed over 400 levels), so the optimizer is pushed hard toward $A \leq 0$.

31.8.3 The results

From the key anchors of App. A: fair $R = 4$ surrogate floors 0.2335 (odd) and 0.2422 (even); exact-verified in-sample scores 0.2391 and 0.2423. As percentages of the null 0.2438 (an exercise in the Problems section), the surrogate floors capture 4.2% and 0.66% and the exact spectra 1.9% and 0.6%. These sit inside the paper's quoted range "within 0.4–4.8% of the null" for the surrogate floors over the ball sizes tried. At $R = 4$ the surrogate tracks the exact shifts level by level (correlation +0.93/+0.995, every eigenstate matched). Yet for the odd class it still overstates the in-sample gain: 4.2% from the surrogate against 1.9% exact. This is why the exact-spectrum gains, not the surrogate floors, are the final numbers to quote.

In plain words. When the model is kept in the regime where its own formula is correct, and when the absorber is forced to absorb, it improves on doing nothing by about one or two percent. That is what "empirically inert" means.

Strength and limits. The fair refit is the right control: it gives the model class every chance consistent with its own validity. But it tests two classes, one background, one optimizer and one set of ball sizes; the word "inert" is the paper's summary for these, "on this problem," and should not be stretched to non-Hermitian operators in general. The paper itself scopes it: "these second-order non-Hermitian families, treated fairly."

Checkpoint 31.7

Why restart from $\mathcal{N}(0, (R/20)^2)$ rather than from $\mathcal{N}(0, 0.3^2)$ as in the unconstrained fits?

Answer: To start well inside the region where the surrogate is valid. A spread of 0.3 per component could put starts far outside a small ball, where each would immediately be projected onto the surface, and the restarts would no longer sample the interior. Scaling the spread with R keeps the restarts comparable for every ball size. Whether every start actually lies inside the ball depends on M: the typical starting norm is $(R/20)\sqrt{M}$, inside the ball if $M < 400$, and the paper does not state M.

Key Ideas

- The fair version projects c onto $\|c\| \leq R$ after every step, penalizes $A > 0$ with weight 10^4, and restarts inside the ball.
- At $R = 4$ the surrogate is verified (corr +0.93/+0.995, all states matched); exact in-sample gains are 1.9% (odd) and 0.6% (even).
- Conclusion, scoped: these families, treated fairly, are empirically inert on this problem.

31.9 General lessons for computational science

The paper closes Sec. V with a general statement:

> "Each ingredient of the spurious floor is generic in computational and machine-learned science: surrogate optimization outside its validity region, dead parameters coupling to numerical noise, and shared constants across model variants mistaken for physics. A surrogate improvement is not an operator-level result until the exact spectrum of the fitted operator confirms it."

Each ingredient has a cheap detector, and you have now seen all of them work.

1. **Surrogate outside its validity region.** Detector: score the exact model at the fitted parameters, and repeat at reduced amplitude. If agreement returns at reduced amplitude, that supports the correctness of the machinery and points to the amplitude. (Section 31.7.)

2. **Dead parameters coupling to noise.** Detector: list every parameter that should have no effect, check analytically that its derivatives vanish, set them to exact zeros, and look at fitted coefficient magnitudes; a coefficient of 2.2×10^6 is a symptom, not a result. Recompute in higher precision. (Section 31.5.)

3. **Shared constants mistaken for physics.** Detector: when two variants agree suspiciously well, ask what they share (code, optimizer, data type, defaults) and break the shared part. (Section 31.5.)

4. **A plausible mechanism.** Detector: make the mechanism quantitative before believing it. The block-coherence idea was natural and wrong by a factor of 2 to 7. (Section 31.6.)

5. **Physical constraints.** Detector: check that fitted parameters respect the premise of the model (58% wrong-sign widths). (Section 31.7.)

Item 1 reappears as point (ii) of the paper's falsifiability standard in Sec. VI: "scoring of the actual operator spectrum, not a surrogate." Section V is the paper's own evidence for why that point belongs on the list.

Pitfall

"Exact" here means exact for the discretized operator: dense diagonalization of the finite-difference matrix. It does not remove discretization error. That is a separate issue, handled in the Hermitian part of the paper by bias-corrected targets (Chapter 23).

Key Ideas

- The three generic ingredients: surrogate outside validity, dead parameters on noise, shared constants read as physics.
- Each has a cheap detector: exact scoring plus reduced amplitude; analytic zeros and coefficient inspection; breaking shared components.
- Section V motivates point (ii) of the paper's proposed standard.

31.10 Paper connection

This section lists, in paper order, every place the material of this chapter appears in R1 [1].

Abstract. One sentence: "A subsidiary dissection of second-order non-Hermitian perturbations shows how surrogate optimization outside its validity range can manufacture an apparently meaningful in-sample floor; exact diagonalization dissolves it." Every word is now unpacked: "subsidiary" (Section 31.1), "surrogate" (Section 31.3), "outside its validity range" (Section 31.7), "apparently meaningful floor" (Sections 31.4 and 31.5).

Sec. II E, "Second-order non-Hermitian classes, amplitude control, and exact diagonalization." Defines $H \to H + iA$, the PT-odd class $\phi_k = \sin((k+1)\pi x/L)$ with $A_{nn} = 0$, and the absorptive-even class $\phi_k = \cos(k\pi x/L)$ modeling $-i\Gamma/2$ (our Equations (31.4), (31.6) and (31.8)). Eq. (5) is our Equations (31.16) and (31.20); the state sum over 652 states and the 500-state bias are Figure 31.1. Safeguards (i) null mode (Section 31.5), (ii) amplitude balls and $\Gamma \geq 0$ (Section 31.8), (iii) exact diagonalization with overlap matching (Section 31.7). Notation overloads: A (operator) versus A_{nk} of Sec. II C; $T^{(n)}$ versus the window T of Sec. II D and T_k of the Chebyshev basis.

Sec. V, first paragraph. Floors 0.2321, 0.2358 (repaired), 0.2322 (artifact), null 0.2438, OOS rms up to 1.4 and 167, $c_0 \approx 2.2 \times 10^6$: Sections 31.4 and 31.5, Figure 31.2.

Sec. V, "The proposed mechanism fails quantitatively." Sum rule to 4×10^{-15}, locality, block-coherent rms 0.03–0.11 for windows 3–13, factor 2–7, $p = 2$ period ≈ 9 in t and 6–7 levels, capture 3–7%: Section 31.6, Figure 31.5.

Sec. V, "The optimizer left the perturbative regime." Correlations +0.14/+0.18, twofold overstatement, 3% unmatched, exact scores 0.2523/0.2477, quarter-amplitude correlations +0.88/+0.94, 58% wrong-sign widths: Section 31.7, Figures 31.6 and 31.7.

Sec. V, "The fair version captures almost nothing." Floors within 0.4–4.8% of the null, $R = 4$ correlations +0.93/+0.995, exact gains 1.9%/0.6%: Section 31.8.

Sec. V, closing paragraph. The three generic ingredients: Section 31.9.

Sec. V A, "Sum rule and locality details." Sum-rule values 3.6×10^{-15} and 1.4×10^{-14}; weights $w_{nm} = |A_{nm}|^2/|E_n - E_m|$ with half within $|n - m| \leq 5$ (odd) and ≤ 10 (even), 90% within about 60 levels; block-coherent rms 0.108, 0.049, 0.033, 0.035, 0.019 for $w = 3, 5, 7, 9, 13$; output correlation $\approx +0.3$ with rms 0.06–0.075: Section 31.6.

Fig. 5. Left panel (exact versus second-order shifts, unconstrained PT-odd fit) is reproduced in spirit by our Figure 31.7; right panel (in-sample floors under amplitude control, "$R = 4$ exact") corresponds to the right half of our Figure 31.2.

Appendix A, step 9. Tensors with ψ normalized by $\sum \psi^2 \mathrm{d}x = 1$; $T^{(n)}$ over 652 states; even-class rows $T^{(n)}_{0l}$ zeroed analytically. **Step 10.** LM on $r_n = (\mathrm{fluc}_n + c^\top T^{(n)} c)/\mathrm{gap}_n$, Jacobian $2(T^{(n)}c)_k/\mathrm{gap}_n$, try-list $(0.3\lambda, \lambda, 10\lambda)$ with diag scaling, 6 restarts $c \sim \mathcal{N}(0, 0.3^2)^M$, seed 0, at most 200 iterations; fair variant with projection onto $\|c\| \leq R$, restarts $\mathcal{N}(0, (R/20)^2)$, sign penalty $10^4 \int \max(A, 0)^2 \mathrm{d}x$. **Step 11.** Dense complex `eig` on $N = 2500$, matching by overlap. **Step 4** supplies the 652 states (eigenvalue range $[0, 1000]$ for tensors).

Key anchors. Unconstrained floors 0.2321/0.2358 (0.2322 with artifact); exact-spectrum in-sample 0.2523/0.2477; fair $R = 4$ floors 0.2335/0.2422; exact-verified 0.2391/0.2423; in-sample null 0.2438.

Sec. VI. "The non-Hermitian analysis of Sec. V is inherited from the original computation and was not rerun for this revision." Also point (ii) of the proposed standard. Chapter 32 weighs these.

Sec. II F. The pipeline, including this part, was written and analyzed with substantial assistance from a large language model, under the author's direction and review; reproducibility from the archived record [20] is what makes the numbers above checkable.

31.11 Summary

- Section V is a subsidiary method stress test; the Hermitian conclusions do not depend on it; it was not rerun in revision.

- The surrogate Eq. (5), $\delta E_n = -\sum_{m \neq n} A^2_{nm}/(E_n - E_m) = -c^\top T^{(n)} c$, follows from second-order perturbation theory with $W = iA$ (Equations (31.13), (31.16) and (31.20)); coupled levels attract.

- Real parts of eigenvalues are even in the amplitude (Equation (31.18)); the surrogate's relative error grows as amplitude squared; PT-odd spectra stay exactly real until a collision.

- One cosine in the box gives $\delta E_n = -c^2/(8\kappa(n^2 - j^2))$ (Equation (31.22)); truncating the state sum biases the top levels.

- Unconstrained floors 0.2321 (odd) and 0.2358 (even) against null 0.2438 are captures of 4.8% and 3.3%; OOS predictions are useless and can explode.

- The constant mode is exactly null (Equations (31.25) and (31.27)); in float32 it coupled to noise with $c_0 \approx 2.2 \times 10^6$ and produced the spurious 0.2322; least squares harvests a fixed gain from any noise column.
- The sum rule (Equation (31.28)) and locality make the block-coherent rms a lower bound on a local model's residual; it is far below the floor, so the block-coherence mechanism fails; the floor is a small capture.
- The two-level model (Equations (31.33) and (31.34)) shows the exceptional point, twofold overstatement past it, and loss of overlap matching.
- Exact spectra of the unconstrained fits score 0.2523/0.2477, worse than the null; at quarter amplitude the surrogate is accurate.
- The fair refit (Equation (31.35), sign penalty) gains 1.9%/0.6% in exact spectra: these families are empirically inert on this problem.
- Generic lesson: a surrogate improvement is not an operator-level result until the exact spectrum confirms it.

31.12 Problems

1. • Using the paper's numbers (Sec. V), compute by what percentage the exact-spectrum in-sample scores of the unconstrained fits, 0.2523 (odd) and 0.2477 (even), exceed the null 0.2438.

2. • In two or three sentences, explain why the paper calls Eq. (5) a "surrogate" and an "optimization device", and what object a physical claim must be about.

3. • Starting from $V(x) - i\Gamma(x)/2 = V(x) + iA(x)$, show that the physical requirement $\Gamma \geq 0$ is equivalent to $A \leq 0$ pointwise.

4. •• In the box with $L = \pi/2$ ($E_k = k^2$), take a single cosine mode with $j = 2$ and $c = 1$. Using Equation (31.21), list the coupling partners of levels $n = 1, \ldots, 6$ and compute δE_n for each, treating $n = j = 2$ and $n = 2j = 4$ separately.

5. •• Show that $T^{(n)}_{kl}$ in Equation (31.20) is symmetric in k and l, and derive $\partial(c^\top T c)/\partial c_k = 2(Tc)_k$. Then evaluate r_n and the Jacobian row for $T^{(n)} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix}$, $c = (2, 1)$, $\mathrm{fluc}_n = 0.3$, $\mathrm{gap}_n = 1.2$.

6. •• Suppose the computed overlaps $(\phi_0)_{nm}$ carry noise of typical size η. Estimate the size of c_0 needed for the effective matrix $c_0\eta_{nm}$ to reach 0.1 in energy units, for $\eta = 10^{-8}$ and for $\eta = 10^{-16}$. What does the second answer tell you about float64?

7. •• Using qmnt, compute gap_n at $n = 100$, 300 and 400 and the number of levels spanned by one period $2\pi/\log 2$ of the $p = 2$ oscillation at each height. Is the paper's "6–7 levels at these heights" consistent with your numbers for zeros 1–400?

8. •• For the two-level model Equation (31.32) at $x = a/\Delta = 0.3$, compute the exact and second-order real shifts of the lower level in units of Δ, and the relative error of the surrogate.

9. •• Three levels $E = (0, 1, 3)$ are coupled by a real symmetric A with $A_{12} = 0.1$, $A_{23} = 0.2$, $A_{13} = 0.05$ and zero diagonal. Compute the three second-order real shifts under $H \to H + iA$ and verify the sum rule.

10. •• For $d_n = (-1)^n d_0$ with $d_0 = 0.24$, compute the block-coherent rms for $w = 3$ and $w = 5$ using the result of the sample problem on negatively correlated sequences. Compare with the paper's $w = 3$ value 0.108 and comment.

11. •• Using qmnt, compute $d_n = (t_n - s_n)/\mathrm{gap}_n$ for $n = 1, \ldots, 400$ and the rms of its centred w-level moving average (full windows only) for $w = 3, 5, 7, 9, 13$. Compare with the paper's Sec. V A values and with the floor 0.232.

12. •• Using numpy, repeat the "harvesting a noise column" sample with $\varepsilon = 10^{-3}$ and $\varepsilon = 10^{-9}$. Report the coefficient β_z and the rms residual in each case, and explain why the rms does not change.

13. • Using the key anchors of App. A, express the fair $R = 4$ surrogate floors (0.2335 odd, 0.2422 even) and the exact-verified scores (0.2391, 0.2423) as percentage improvements over the null 0.2438.

14. •• For a general 2×2 matrix $\begin{pmatrix} E_1 & ia \\ ia & E_2 \end{pmatrix}$ with $E_2 > E_1$ and real a, find the value of a at which the exceptional point occurs, and the common eigenvalue there.

15. ••• Expand the exact two-level shift Equation (31.34) to sixth order in x. Confirm that no odd powers appear and give the coefficients of x^4 and x^6.

16. •• Using numpy, build $H_0 + iA$ for the box ($E_k = k^2$, 80 states) with $A = c\cos(\pi x/L)$, and compare exact and second-order real shifts of the lowest 10 levels at $c = 4$ and at $c = 1$ (quarter amplitude). (At $c = 4$ the strongest coupling, between levels 1 and 3, is a quarter of their spacing, safely below the two-level exceptional-point threshold of one half.) By roughly what factor does the relative error drop?

17. ••• Generalize the argument of Equations (31.17) and (31.18): show that for any real symmetric H_0 and real symmetric A, the characteristic polynomial $\det(H_0 + iaA - \lambda)$ has real coefficients when a is replaced by ia' with a' real, and use this to explain once more why only even powers of a appear in the real parts.

18. •• Redo the weak-capture sample problem with correlation $\rho = 0.2$ instead of 0.3. What is the optimal output rms and the resulting residual rms? Does this still land near the paper's floors?

19. •• From the paper's Sec. V A block-coherent values (0.108, 0.049, 0.033, 0.035, 0.019 for $w = 3, 5, 7, 9, 13$), compute the ratio $0.232/\mathrm{rms}(B_w)$ for each window. Which windows give the paper's quoted range "2–7"?

20. ••• Design, in words, a test that would have caught the null-mode artifact even if nobody had noticed $c_0 \approx 2.2 \times 10^6$. Your test must use only the fitted model and the data, and should be cheap.

21. ••• The paper reports that 3% of eigenstates could not be matched one-to-one at the fitted amplitude. Using the two-level model, explain why an unmatched state signals that at least one pair of levels has passed (or is near) an exceptional point, and why such levels make the surrogate shifts meaningless for that pair.

31.13 Answers to selected problems

★ Did you know? *Don't Fool Yourself*

In his 1974 Caltech commencement address, later printed as *Cargo Cult Science*, Richard Feynman warned that the first principle is that you must not fool yourself, and that you are the easiest person to fool. He urged scientists to report everything that might make their results invalid. A spurious floor is exactly the kind of trap that this advice protects against.

1. $0.2523/0.2438 = 1.0349$, so 3.49% worse; $0.2477/0.2438 = 1.0160$, so 1.60% worse. Both exact spectra are worse than doing nothing.

2. Eq. (5) is a cheap approximate formula for the real level shifts, used in place of the exact eigenvalues during optimization. It is only valid while couplings are small compared with spacings. A physical claim must be about the exact spectrum of the fitted operator $H + iA$.

3. Equate imaginary parts: $-\Gamma/2 = A$, so $A = -\Gamma/2$. Since $-\frac{1}{2} < 0$, $\Gamma(x) \geq 0$ holds at a point exactly when $A(x) \leq 0$ there.

5. T_{kl} contains the product $(\phi_k)_{nm}(\phi_l)_{nm}$, symmetric under $k \leftrightarrow l$. The derivative is Equation (31.24). Numbers: $Tc = (2.5, 3)$, $c^\top Tc = 8$, $r_n = (0.3 + 8)/1.2 = 6.917$, $J = 2(2.5, 3)/1.2 = (4.167, 5.000)$.

7. qmnt gives $\text{gap}_{100} \approx 1.73$, $\text{gap}_{300} \approx 1.41$, $\text{gap}_{400} \approx 1.34$. Dividing $2\pi/\log 2 = 9.065$ by these gives about 5.2, 6.4 and 6.8 levels per period. The paper's "6–7 levels" describes the upper part of the training range well (from about $n = 200$ up); near $n = 100$ a period spans only about 5 levels, which is even more local.

8. $\sqrt{1 - 0.36} = 0.8$, exact $= 0.1\Delta$; surrogate $= 0.09\Delta$; relative error -10%.

9. $\delta E_1 = -[A_{12}^2/(0-1) + A_{13}^2/(0-3)] = 0.01 + 0.000833 = 0.010833$; $\delta E_2 = -[A_{12}^2/(1-0) + A_{23}^2/(1-3)] = -0.01 + 0.02 = 0.01$; $\delta E_3 = -[A_{13}^2/(3-0) + A_{23}^2/(3-1)] = -0.000833 - 0.02 = -0.020833$. Sum $= 0$, as the pairwise antisymmetry requires.

10. $w = 3$: $q = 1$, rms $d_0/3 = 0.080$; $w = 5$: $d_0/5 = 0.048$. The paper's 0.108 at $w = 3$ is somewhat larger than for a perfectly alternating sequence of the same amplitude: the zeros are negatively correlated but not perfectly alternating, so some slow component survives. Both are far below 0.232.

11. Compute `d = qmnt.fluct(qmnt.zeros(400), np.arange(1, 401))`. Then, for each w, form the full-window moving average `np.convolve(d, np.ones(w)/w, 'valid')` and take its rms. All five values come out below 0.15, hence at least a factor $0.232/0.15 \approx 1.5$ below the floor. They are therefore far below 0.232, like the paper's values (0.108 down to 0.019), and the block-coherence explanation fails for this version of the targets too. Plot them against the paper's values as in the block-coherence figure. Differences from the paper's numbers come from using s_n rather than the paper's Richardson-corrected targets.

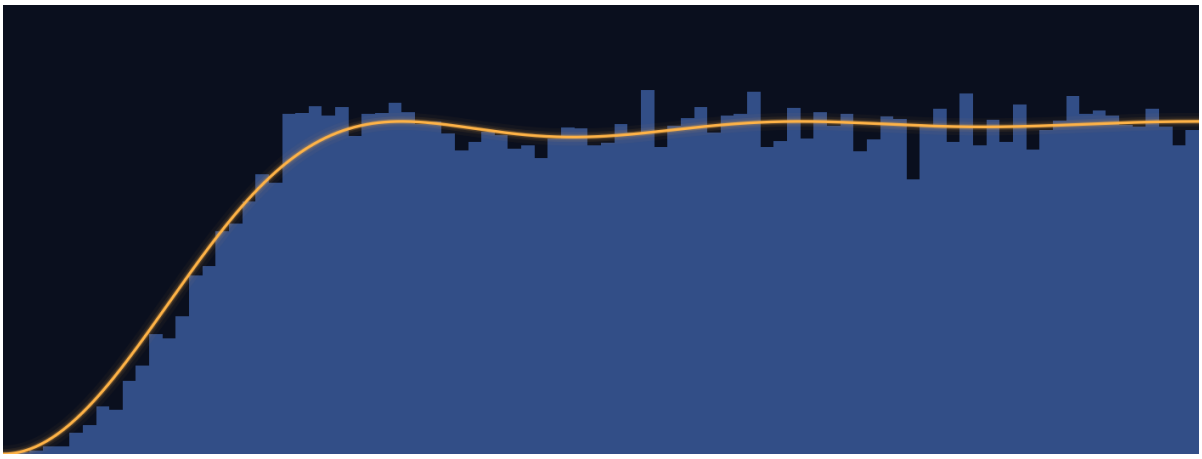
13. Surrogate: $(0.2438 - 0.2335)/0.2438 = 4.22\%$, $(0.2438 - 0.2422)/0.2438 = 0.66\%$. Exact: 1.93% and 0.62%, matching the paper's 1.9% and 0.6%.

15. With $u = 4x^2$, $\sqrt{1-u} = 1 - \frac{u}{2} - \frac{u^2}{8} - \frac{u^3}{16} - \cdots = 1 - 2x^2 - 2x^4 - 4x^6 - \ldots$, so $\delta E_1^{\text{exact}} = \frac{\Delta}{2}(2x^2 + 2x^4 + 4x^6) = \Delta(x^2 + x^4 + 2x^6) + \ldots$. Only even powers; coefficients 1 and 2.

17. Sketch: in the real basis, $(H_0 + iaA)^* = H_0 - iaA$, so the polynomial $p_a(\lambda) = \det(H_0 + iaA - \lambda)$ satisfies $\overline{p_a(\bar{\lambda})} = p_{-a}(\lambda)$. Setting $a = ia'$ makes the matrix $H_0 - a'A$ real, so its coefficients are real. The roots continuing E_n are therefore real-analytic in a'; in terms of a, $a' = -ia$, so terms odd in a carry a factor i and contribute only to imaginary parts, even terms only to real parts. Same conclusion as Equation (31.19).

19. $0.232/0.108 = 2.15$, $/0.049 = 4.73$, $/0.033 = 7.03$, $/0.035 = 6.63$, $/0.019 = 12.21$. Windows 3–9 give 2.1–7.0, the quoted "2–7"; $w = 13$ gives about 12, outside the quoted range but strengthening the conclusion.

21. In the two-level model the lower eigenvector (v_1, v_2) has weight $|v_1|^2/(|v_1|^2 + |v_2|^2) > \frac{1}{2}$ on state 1 below the exceptional point, so overlap matching assigns it uniquely. At and beyond the exceptional point $|v_2/v_1| = 1$: both eigenvectors have weight exactly $\frac{1}{2}$ on both states and no one-to-one assignment exists. So an eigenstate that cannot be matched is one that is (nearly) an equal mixture of two unperturbed states, the signature of a pair at or near an exceptional point. For such a pair the exact real parts have saturated at the midpoint (shift $\Delta/2$) while the surrogate keeps growing as a^2/Δ; the series behind Eq. (5) has passed its radius of convergence, so the surrogate shift for that pair has no meaning.



Pair correlation of the first 10 000 zeros (bars) against Montgomery's $1-(\sin \pi u/\pi u)^2$*: a law no fitted potential was asked to learn.*

Chapter 32

Reading the Paper V: Discussion, Limits, and How to Go Further

↑ Contents

What this chapter is about

You have now followed the paper [1] from its question and protocol (Chapter 28), through its negative result on fitted potentials (Chapter 29) and its positive result on the explicit formula (Chapter 30), to its dissection of a spurious in-sample floor (Chapter 31). This last chapter does three things that a careful reader of any research paper should do once the results are understood.

First, it reads the Discussion (Sec. VI of the paper) sentence by sentence. For each statement it asks what kind of statement it is: a numerical observation, a heuristic explanation, a proposal, or an open problem. Second, it plays referee. It lists the limits the paper states about itself, and then the weak points a skeptical expert could press, and for each it does a small calculation to see how much weight the point carries. Third, it shows you how to rerun the paper's anchor numbers yourself with the `qmnt` library and the Zenodo record [20], and it proposes graded research projects that start where the paper stops.

No new mathematics is introduced. Every tool you need was built in Parts I–V; we only combine those tools in small, honest calculations (probabilities of chance coincidences, worst-case phase errors, coverage thresholds) that help you judge the evidence.

Where this shows up in the paper: Sec. VI (Discussion), Appendix A (the reconstruction record, steps 1–23, the literature audit and the key anchors), the Data Availability statement, Sec. II F (use of artificial intelligence), and the hedges written into the captions of Tables I–V.

What you need

- All of Parts I–V. In particular: the counting function N_{sm}, smooth zeros and mean spacings (Chapter 17); the truncated explicit formula (Chapter 18); the Wu–Sprung potential and its turning points (Chapters 12 and 22); second-order non-Hermitian shifts (Chapter 14); the finite-difference solver and bias correction (Chapter 23); fitting and the coverage law (Chapter 24); held-out testing and nulls (Chapter 25); bootstrap and surrogates (Chapter 26); sparse frequency learning and dephasing (Chapter 27); correlation, independence and the Poisson law (Chapters 19 and 20); the prime number theorem (Chapter 15).
- Chapters 28–31, which read Secs. I–V of the paper.
- First-year physics: the habit of checking a claim by an order-of-magnitude estimate.

32.1 What the Discussion section claims

Before we judge anything we recall the vocabulary. The nontrivial zeros of the zeta function are $\rho_n = \frac{1}{2} + it_n$, and $t_n > 0$ is the height of the n-th zero. The smooth zeros s_n solve $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$, where $N_{\mathrm{sm}}(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$ is the smooth counting function of paper Eq. (1) (Chapter 17). The local mean spacing is $\mathrm{gap}_n = ds_n/dn$. The *fluctuation* of a zero is $(t_n - s_n)/\mathrm{gap}_n$, its displacement from the smooth zero measured in mean spacings. A *prediction* t_n^{pred} has its own fluctuation $(t_n^{\mathrm{pred}} - s_n)/\mathrm{gap}_n$. The paper scores a prediction on a set of held-out (out-of-sample, OOS) zeros by two numbers: the rms, the root mean square of the error in units of mean spacing, and corr, the Pearson correlation of predicted with true fluctuations (Chapter 19). The *null* is the prediction "no fluctuation", $t_n^{\mathrm{pred}} = s_n$ (or, for the operator, the levels of the unperturbed Wu–Sprung potential). Since the null error of zero n is the true fluctuation itself, the null rms is the rms of the true fluctuations; on the principal holdout, zeros 401–500 (running example RE5), it is 0.263 (paper Sec. II C; anchor 0.2631 in App. A).

Sec. VI of the paper has five paragraphs. We take them in order.

32.1.1 Paragraph 1: the scope

The Discussion opens: "Nothing here touches the analytic Hilbert–Pólya program". What is constrained is "a shortcut: the idea that flexible operator families fitted to finitely many zeros capture transferable spectral structure."

In plain words. The Hilbert–Pólya conjecture (Chapter 21) hopes for a self-adjoint operator whose eigenvalues are the t_n. It is open. The paper does not test any operator that was written down *before* looking at the zeros (such as the Berry–Keating, Connes, or Bender–Brody–Müller proposals [10, 38, 40]). It tests only a shortcut: take a flexible family of potentials, tune its parameters until its levels match the first few hundred zeros, and hope the tuned potential "knows" something about the zeros you did not show it.

Tools. The distinction between analytic and empirical programs is Chapter 21; the meaning of "transferable" is the held-out test of Chapter 25.

Computation behind it. The scoreboard of Figure 32.1 collects the held-out rms on zeros 401–500 from paper Tables I, IV and V. The fitted operators sit at or above the null line; every model that uses the arithmetic structure, given or learned, sits below it, and all but the underdetermined, root-ambiguous B row (0.180, Table IV) sit far below it.

It helps to compress each rms and its null into one number. The paper does not define such a number; it reports rms and null side by side. We define, as a convenience for this chapter, the *relative skill*

$$\mathcal{S} \;=\; 1 - \frac{\text{rms of the model}}{\text{rms of the null}}. \tag{32.1}$$

A model that predicts perfectly has $\mathcal{S} = 1$; a model that does exactly as well as the null has $\mathcal{S} = 0$; a model that does worse than predicting nothing has $\mathcal{S} < 0$.

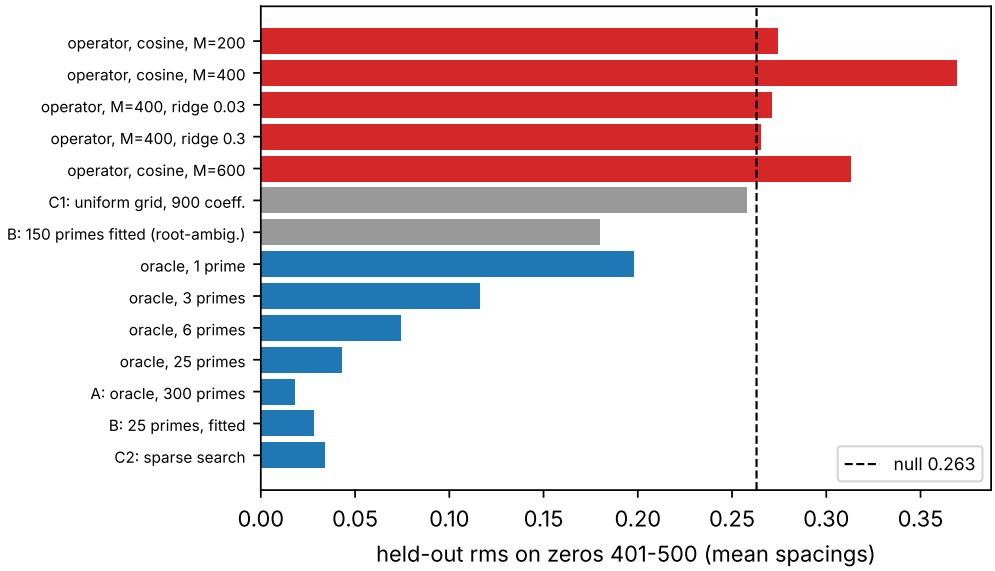


Figure 32.1: Scoreboard of held-out rms errors on zeros 401–500, fit on zeros 1–400, in units of local mean spacing, as reported in paper Tables I, IV and V. Red: fitted potentials (cosine basis). Gray: the dense Fourier control C1 (no arithmetic input, no sparsity prior) and the root-ambiguous B row. Blue: models that use arithmetic frequencies (supplied, as in the oracle and control B, or learned, as in control C2). The dashed line is the null 0.263. Bars to the right of the dashed line are worse than predicting no fluctuation at all.

Sample Problem 32.1: Relative skill of the operator and of the oracle

On zeros 401–500 the fitted cosine potential with $M = 400$ coefficients scores OOS rms 0.369 (paper Table I), and the six-prime oracle scores 0.074 (Table V). The null is 0.263. Compute $\mathcal{S}$ for both.
Step 1: For the operator, the ratio is $0.369/0.263 = 1.403$.
Step 2: Hence $\mathcal{S}_{\text{op}} = 1 - 1.403 = -0.403$. The fitted potential is about 40% worse than predicting no fluctuation.
Step 3: For the oracle, the ratio is $0.074/0.263 = 0.281$.
Step 4: Hence $\mathcal{S}_{\text{or}} = 1 - 0.281 = 0.719$.
$\boxed{\mathcal{S}_{\text{op}} \approx -0.40, \qquad \mathcal{S}_{\text{or}} \approx +0.72}$

Strength and limits. The scope sentence is the most important hedge in the paper, and it is faithful to the experiments: only Wu–Sprung backgrounds, three bases, one optimizer family, fits to at most 1000 zeros. You should carry this scope into every sentence you write about the paper.

32.1.2 Paragraph 2: why the fits fail (a heuristic)

The second paragraph begins "The failure is intelligible, though heuristically so." Notice the word *heuristically*: what follows is an explanation that fits the observations, not a theorem. The argument has five links.

1. *The bases are effectively complete for in-sample level patterns.* With enough coefficients the fitted potential can shift each training level by an essentially arbitrary amount. Evidence: in-sample rms falls below 10^{-4} spacings on all 500 zeros at $M = 900$ (Sec. III), and the fits reach the same in-sample accuracy on permuted and phase-randomized targets as on the

true zeros (Sec. III, "Permutation and surrogate tests"). The coverage law of Chapter 24 predicts when completeness sets in.

2. *A family that can fit anything encodes no inductive bias toward the truth* (Chapter 25). If every target is fitted equally well, the quality of the fit says nothing about which target is real.

3. *Held-out behavior is then set by the optimizer, not the data.* Unregularized, the fit puts large structure where only held-out levels probe the potential and does harm (paper Fig. 4, Table III); regularized, it returns to the null and not below it.

4. *The controls locate the obstacle.* It is not the parameter count: the oracle keeps improving at 900 terms, and a dense 900-coefficient Fourier model (C1) gains almost nothing. It is not that the frequencies must be supplied: the sparse search C2 finds them from 400 zeros (Table IV).

5. *A heuristic reading.* The target is, by the explicit formula, an almost-periodic function of t that is sparse in frequency (Chapters 5 and 18). A potential addresses each level through its classical turning point, that is, locally in energy (Chapter 12). On this heuristic reading, and for the parametrizations and optimizer tested, a local-in-energy vocabulary did not produce "keep oscillating at frequency $\log 2$ forever".

The paragraph closes with two careful sentences: "This is evidence about the tested parametrizations and optimizer together, not a statement about what a potential can represent. Whether the memorization result can be derived rather than measured is an open problem."

Link 1 can be checked by a quick estimate.

Sample Problem 32.2: Coverage threshold at the two scales

The coverage law (Chapter 24; paper Sec. III) says that the cosine family $\cos(k\pi x/L)$, $k = 0, \dots, M-1$, becomes effectively complete for joint level shifts up to energy E once $M\pi/L > 2\sqrt{E}$. (a) At scale 1, $L = 44.883$ and the top scored level has $E \approx t_{500} = 811.18$. Find the smallest M satisfying the law. (b) At scale 2, $L = 59.957$ and the top training level of the 1–800 fits has $E \approx s_{800} \approx 1184$. Repeat.

Step 1: Rearrange the inequality. Multiply both sides of $M\pi/L > 2\sqrt{E}$ by L/π, which is positive:

$$M > \frac{2\sqrt{E}\,L}{\pi}.$$

Step 2: Scale 1. $\sqrt{811.18} = 28.481$, so $2\sqrt{E} = 56.963$. Multiply by L: $56.963 \times 44.883 = 2556.7$. Divide by π: 813.8.

Step 3: Scale 2. $\sqrt{1184} = 34.41$, so $2\sqrt{E} = 68.81$. Multiply by $L = 59.957$: 4126. Divide by π: 1313.

$$\boxed{M_{\text{scale 1}} \gtrsim 814, \qquad M_{\text{scale 2}} \gtrsim 1313}$$

Interpretation: at scale 1 the paper's statement that all 500 zeros are reproduced to rms $< 10^{-4}$ "by $M = 900$" sits just above the threshold, as the law predicts. At scale 2 the largest basis used, $M = 900$, is below the threshold for the top training levels. Turning the law around, $M = 900$ at scale 2 covers energies only up to

$$E = \left(\frac{M\pi}{2L}\right)^2 = \left(\frac{900\pi}{119.914}\right)^2 = (23.58)^2 \approx 556,$$

which is roughly the height of level 310 (because $N_{\text{sm}}(556) + \frac{1}{2} \approx 310$). Levels above about 310 are not covered. This fits the observation that the in-sample rms there is still 0.103 (Table I). The coverage law is itself heuristic (Chapter 24), so these thresholds are estimates, not sharp lines.

Link 3 involves a second fact worth making quantitative: a prediction that is uncorrelated with the truth cannot beat the null, however you rescale it.

Sample Problem 32.3: What a correlation is worth

Let f_j be the true fluctuations and g_j the predicted fluctuations on n_{out} held-out levels, $j = 1, \dots, n_{\text{out}}$, and suppose both have mean close to zero (true to good approximation for the zeros). Find the smallest rms you could reach by multiplying the prediction by the best constant a, and evaluate it for the six-prime oracle (corr $+0.96$) and the $M = 400$ operator (corr -0.07), with null 0.263.

Step 1: The mean squared error of the rescaled prediction is

$$Q(a) = \frac{1}{n_{\text{out}}} \sum_j (f_j - a g_j)^2.$$

Step 2: Differentiate with respect to a and set to zero:

$$\frac{dQ}{da} = -\frac{2}{n_{\text{out}}} \sum_j g_j (f_j - a g_j) = 0 \quad \Longrightarrow \quad a = \frac{\sum_j f_j g_j}{\sum_j g_j^2}.$$

Step 3: Expand Q and substitute this a:

$$Q_{\min} = \frac{1}{n_{\text{out}}} \Big[\sum_j f_j^2 - 2a \sum_j f_j g_j + a^2 \sum_j g_j^2 \Big] = \frac{1}{n_{\text{out}}} \Big[\sum_j f_j^2 - \frac{(\sum_j f_j g_j)^2}{\sum_j g_j^2} \Big].$$

Step 4: Factor out $\sum_j f_j^2 / n_{\text{out}}$, which is the square of the null rms:

$$Q_{\min} = (\text{null rms})^2 \Big[1 - \frac{(\sum_j f_j g_j)^2}{\sum_j f_j^2 \sum_j g_j^2} \Big] = (\text{null rms})^2 \, (1 - r^2),$$

where, for mean-zero sequences, r is the Pearson correlation (Chapter 19).

Step 5: Take the square root: smallest rms $=$ null $\times \sqrt{1 - r^2}$.

Step 6: Oracle: $\sqrt{1 - 0.96^2} = \sqrt{0.0784} = 0.280$, so $0.263 \times 0.280 = 0.0736$. Operator: $\sqrt{1 - 0.07^2} = 0.9975$, so $0.263 \times 0.9975 = 0.2624$.

oracle: 0.074 at best rescaling, operator: 0.262

The oracle's actual rms, 0.074 (Table V), is consistent with its best-rescaled value within the rounding of r. As r runs from 0.965 down to 0.955, the best-rescaled rms $0.263\sqrt{1 - r^2}$ runs from 0.069 up to 0.078, so with two-digit inputs we can say only that its amplitudes are close to right. The mean-zero assumption of the sample is reasonable here: the mean of the true fluctuations on 401–500 is less than a fifth of their rms (check it with `qmnt.fluct`). The operator, even with the best possible rescaling, could only tie the null; its actual rms 0.369 is worse because it emits uncorrelated structure of nonzero size.

The statements in paragraph 2 and elsewhere in Sec. VI have different status. Table 32.1 sorts them, using the status labels of this book.

32.1.3 Paragraphs 3 to 5

Paragraph 3 states the limits of the evidence; we read it in Section 32.3. Paragraph 4 states the broader lesson: "Spectral agreement obtained by fitting an operator to a finite sequence establishes representational capacity, not that the fitted operator contains the mechanism generating the sequence." It quotes the decisive pair of intervals: principal $M = 400$ held-out correlation $[-0.21, +0.08]$ against the oracle's $[+0.95, +0.97]$ on the same holdout (bootstrap

Table 32.1: Status of the main statements of the paper's Sec. VI.

Statement	Status	Evidence in paper
In-sample success coexisted with no reproducible held-out content, in the families examined	numerical observation	Tables I–III, Figs. 1, 3, 4
A zero-parameter arithmetic formula carries abundant transferable content	numerical observation	Table V, Fig. 1
A sparse sinusoidal fit learns that content from the same zeros	numerical observation; frequency alignment check exploratory	Table IV, Sec. IV
The bases are effectively complete in-sample	numerical observation plus heuristic coverage law	Sec. III
A family that fits anything has no inductive bias	general principle (heuristic)	Sec. VI
Almost-periodic target versus local-in-energy parametrization	heuristic explanation	Sec. VI, continuation probe
The memorization result can be derived	open problem	Sec. VI
Riemann hypothesis; Hilbert–Pólya conjecture	open; not addressed	Sec. I

intervals, Chapter 26). Paragraph 5 proposes the falsifiability standard (read in Section 32.2) and closes by concluding that "at the scales accessible to direct numerics, and within the operator families tested here, the predictable structure in the Riemann zeros is arithmetic: fully accessible, learnable from the zeros by a frequency-sparse model, and entirely missed by the inverse-spectral fits tested." Keep both qualifiers (scales, families) whenever you repeat this sentence.

Checkpoint 32.1

Does the paper show that no Hermitian operator can have the Riemann zeros as its spectrum?

Answer: No. It shows that, in three bases on one background with one fitting protocol, potentials fitted to at most 1000 zeros showed no reproducible held-out skill. It says explicitly that this is "not a statement about what a potential can represent" and that it does not touch analytic constructions. The Hilbert–Pólya conjecture remains open.

Key Ideas

- Sec. VI makes scoped numerical observations (fits memorize; arithmetic predicts; a sparse model learns), a heuristic explanation, a methodological proposal, and one open problem.
- Relative skill (32.1) compresses rms and null into one number; the fitted operators have $\mathcal{S} < 0$ on zeros 401–500.
- A prediction uncorrelated with the truth cannot beat the null even after optimal rescaling: the best rms is null$\times\sqrt{1-r^2}$.
- The coverage threshold $M > 2\sqrt{E}L/\pi$ is about 814 at scale 1 and about 1313 at scale 2.

32.2 The proposed falsifiability standard

A claim is *falsifiable* when some possible outcome of an experiment would show it to be wrong. "This fitted potential reproduces the first 400 zeros" is true but almost unfalsifiable once the basis is large: Sec. III shows the same fit reproduces shuffled zeros just as well, so no in-sample outcome could have told against it. The paper therefore proposes (Sec. VI, paragraph 5) that any future claim that a constructed or learned operator "captures" the zeros be accompanied by four pieces of evidence. Figure 32.2 shows them as a checklist.

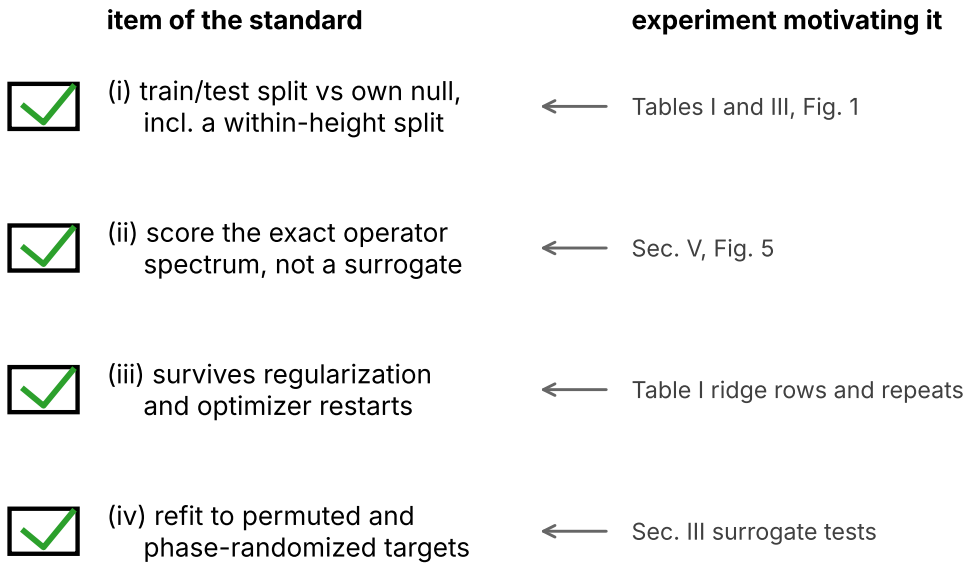


Figure 32.2: The four-item falsifiability standard proposed in Sec. VI of the paper, drawn as a checklist. Left: the item. Right: the experiment in the paper that shows why the item is needed. Passing all four is a minimal (necessary) condition for a "captures the zeros" claim, not a sufficient one.

32.2.1 Item (i): a train/test split, against the model's own null, including a within-height split

What it says. Fit on some zeros, predict others, report held-out rms and corr, and compare with the null of the model's own smooth part (for a fitted potential, the unperturbed background). Include at least one split in which training and held-out zeros come from the same height range.

Why "its own null". Any model built on a smooth background already predicts the mean positions of the zeros. Only the fluctuations are informative. A model must be compared with the version of itself that predicts no fluctuation.

Why a within-height split. A forward split (train low, test high) mixes two effects: failure to capture structure, and *covariate shift* (Chapter 25), because the fluctuation amplitude grows with height (the null rises from 0.263 on 401–500 to 0.296 on 5001–10000, Table V). Random 80/20 and odd/even splits inside one height range (Table III) remove the shift. The paper's finding that the failure is even worse within height (for the random 80/20 splits, held-out errors 1.5–3.0 times the null at scale 1 and 1.8–1.9 times at scale 2) is what rules out covariate shift as the explanation.

32.2.2 Item (ii): score the actual operator spectrum

What it says. If the fit used an approximation of the spectrum (a surrogate such as second-order perturbation theory), the final score must use the exact eigenvalues of the fitted operator.

Recall from Chapter 14 that Sec. V of the paper perturbs $H \to H + iA$, with $A = \sum_k c_k \phi_k$ a real multiplication operator, in two classes. The *PT-odd* class uses odd basis functions $\phi_k = \sin((k+1)\pi x/L)$. The *absorptive-even* class uses even basis functions $\phi_k = \cos(k\pi x/L)$ and models a decay width Γ through $-i\Gamma/2$. In the even class the constant function $\phi_0 = 1$, with coefficient c_0, is the *null mode*: a constant iA shifts every level by the same imaginary constant and has zero second-order real shift. Before it was repaired, the optimizer drove c_0 to about 2.2×10^6 by exploiting float32 noise in tensor entries that are analytically zero (Chapter 31).

Sample Problem 32.4: Why item (ii) matters: the spurious floor in numbers

In Sec. V the second-order surrogate of the PT-odd fit reaches in-sample rms 0.2321, and the absorptive-even fit (null mode repaired) reaches 0.2358, against a null of 0.2438. Exact diagonalization of the same fitted operators gives in-sample rms 0.2523 and 0.2477. Express each as a percentage improvement over the null, defined as (null − rms)/null, so that a positive value means better than the null and a negative value means worse.
Step 1: PT-odd, surrogate: $(0.2438 - 0.2321)/0.2438 = 0.0117/0.2438 = 0.0480$, an apparent improvement of 4.80%.
Step 2: PT-odd, exact: $(0.2438 - 0.2523)/0.2438 = -0.0085/0.2438 = -0.0349$, an improvement of −3.49%, i.e. 3.49% *worse* than the null.
Step 3: Absorptive-even, surrogate: $(0.2438 - 0.2358)/0.2438 = 0.0080/0.2438 = 0.0328$, an apparent improvement of 3.28%.
Step 4: Absorptive-even, exact: $(0.2438 - 0.2477)/0.2438 = -0.0039/0.2438 = -0.0160$, an improvement of −1.60%, i.e. 1.60% worse.
improvement over null: surrogate $+4.80\%, +3.28\%$; exact $-3.49\%, -1.60\%$
The sign flips. A paper that stopped at the surrogate would have reported an improvement that no operator possesses (Chapter 31).

32.2.3 Item (iii): survival under regularization and restarts

What it says. A claimed skill should persist when the fit is mildly regularized (ridge, Chapter 9) and when the optimizer is started from different random initial coefficients.

Why. A skill that appears for one starting point only, or that vanishes under a mild ridge, is a property of the optimizer's path, not of the data. In the paper, the ridge rows of Table I ($\mu = 0.03, 0.3$) and the random-init repeats in the Table I caption (OOS 0.365–0.379 over four repeats) show that the negative result is stable. For the interpolation split, ridge "removes the damage and nothing else" (Sec. III).

Pitfall

Item (iii) is phrased for *positive* claims. When a ridge returns a fit to the null (as in Tables I and II), that does not make the fit predictive; it makes it inert. "Survives regularization" means that the skill survives, not that the error stops being worse than the null.

32.2.4 Item (iv): refit to permuted and phase-randomized targets

What it says. Repeat the whole fit on surrogate targets: the true fluctuation sequence randomly permuted (marginal distribution kept, order destroyed) and phase-randomized [16] (power

spectrum kept, phases scrambled), as in Chapter 26.

Why. If the model fits surrogates as well as the zeros, in-sample accuracy measures capacity. Sec. III reports in-sample rms 0.134 ± 0.010 (permuted), 0.141 ± 0.006 (phase-randomized) and 0.138 ± 0.005 (amplitude-adjusted [17]), against 0.137 on the true zeros. The zeros are not special to the fit.

32.2.5 What the standard does and does not guarantee

The paper calls held-out prediction a "minimal" standard. Two consequences follow.

It is necessary, not sufficient. Passing all four items does not show that a model is "the" zeta operator. The simple kriging interpolator of Sec. III predicts held-out zeros within one height range at rms 0.117, correlation +0.87, and the sparse sinusoidal model C2 predicts at +0.99; neither is an operator at all. Passing the standard shows that a model has captured *some* transferable structure.

It does not apply to analytic constructions in the same way. If an operator is specified without looking at any zero (for example, constructions built only from primes [21, 39]), every zero is held out. The paper notes that for such constructions "no held-out evaluation question arises"; they are judged by direct comparison with all zeros.

Checkpoint 32.2

A hypothetical paper fits a 500-coefficient potential to zeros 1–400 and reports in-sample rms 10^{-5}, held-out rms 0.262 on 401–500 (null 0.263), and held-out corr +0.02. Which items of the standard has it addressed, and does it support a "captures" claim?

Answer: It has addressed half of item (i): a forward split with rms and corr against the null, but no within-height split. Items (ii)–(iv) are not reported. In any case the numbers show no skill: $\mathcal{S} = 1 - 0.262/0.263 \approx 0.004$ and a correlation of +0.02, which is indistinguishable from zero for 100 levels (for 100 independent pairs, the standard error of a correlation near zero is about $1/\sqrt{100} = 0.1$). The claim is not supported.

Checkpoint 32.3

Why must item (ii) be checked separately from item (i)? Could a model pass (i) using surrogate spectra and fail (ii)?

Answer: Yes. Item (i) can be computed with whatever spectrum the fit used. If that spectrum is an approximation (such as the second-order formula, paper Eq. (5)) and the fitted amplitudes are outside its range of validity, the approximate held-out scores need not describe the actual operator. Sec. V shows exactly this for in-sample scores: surrogate 0.2321, exact 0.2523.

Key Ideas

- The standard: (i) held-out rms and corr against the model's own null, including a within-height split; (ii) exact spectra; (iii) robustness to ridge and restarts; (iv) surrogate refits.
- Each item is motivated by a specific experiment in the paper (Tables I and III; Sec. V; Table I ridge rows and repeats; Sec. III surrogates).
- The standard is a minimal necessary condition; non-operator models such as kriging and C2 also pass item (i).

32.3 The limits the paper states

★ Did you know? *Replication Is Hard*

In 2015 the Open Science Collaboration reported repeating 100 published psychology studies. Although 97 of the original studies reported significant results, only about 36% of the replications did. The project helped spark a broad push for preregistration, shared data and openly stated limitations, the habits that make a paper's honest list of limits valuable.

Paragraph 3 of Sec. VI states five limits. A good habit is to ask of each: *what would change if this limit were removed, and in which direction?*

32.3.1 Limit 1: one background, three bases

"One background potential (Wu–Sprung) and three bases were tested; Hermite, wavelet, and neural-network parametrizations were not". The hat basis is described as "the simplest stand-in for a wavelet-type basis" (Sec. II C), not as a wavelet basis. The direction of the uncertainty is one-sided: an untested basis could in principle show skill; nothing tested so far gives reproducible evidence that it would (one Chebyshev row is marginal; see Limit 2). We return to this in Section 32.4 (W2).

32.3.2 Limit 2: one marginal Chebyshev row

Table II has eight rows. One, Chebyshev $M = 400$, has OOS rms 0.291 (interval $[0.25, 0.33]$, which contains the null 0.263) and corr $+0.18$ with interval $[-0.02, +0.37]$, whose lower end is at zero. The paper says "one such row among eight is not evidence of prediction, but a small positive correlation for this row is not excluded." How surprised should we be by one such row?

Sample Problem 32.5: One marginal row among eight

Suppose (heuristically) that every row in Table II had true correlation zero, and that for each row independently the chance that its 95% percentile interval lies entirely at or above zero is $\alpha = 0.025$ (one tail of a 95% interval). What is the probability that at least one of the $k = 8$ rows looks like the Chebyshev row or better? What is the expected number of such rows?

Step 1: The probability that a given row does *not* show this is $1 - \alpha = 0.975$.

Step 2: By independence (Chapter 19), the probability that none of the k rows shows it is the product of the individual probabilities:

$$P(\text{none}) = (1 - \alpha)^k. \tag{32.2}$$

Step 3: "At least one" is the complement:

$$P(\text{at least one}) = 1 - (1 - \alpha)^k. \tag{32.3}$$

Step 4: Numerically, $0.975^8 = 0.8167$, so $P = 1 - 0.8167 = 0.183$.

Step 5: The expected number is $k\alpha = 8 \times 0.025 = 0.2$.

$$\boxed{P(\text{at least one}) \approx 0.18, \qquad \text{expected count} = 0.2}$$

So roughly one time in five a table of eight null rows would contain one row this good. The assumptions are rough: rows share the same training and held-out zeros, so they are not independent; bootstrap intervals are approximate; and the Chebyshev interval's lower end is "at zero", not above it. The calculation supports the paper's wording, neither stronger nor weaker: not evidence of prediction, not excluded.

Table II also contains one row whose correlation interval lies entirely *below* zero: hat, $M = 400$, $[-0.37, -0.01]$. Count excursions of either sign. A 95% interval of a null row then excludes zero with probability $\alpha = 0.05$, and (32.3) gives $1 - 0.95^8 \approx 0.34$ for at least one such row among eight. Excursions of either sign are what a table of null rows produces, and one positive and one negative excursion among eight rows fit that picture.

32.3.3 Limit 3: heights

"The operator fits use at most the first 1000 zeros, and the fitted sinusoidal controls were scored only below zero 2000." The oracle, which has no fitted parameters, was scored through zero 10^4 (Table V).

Why does height matter for the fitted controls? Because estimated frequencies dephase. The paper reports that C2's skill falls from 0.034/+0.99 on 401–500 to 0.078/+0.96 on zeros 1001–2000 (Sec. IV). Chapter 27 explained the mechanism; here is the worst-case size.

Sample Problem 32.6: Worst-case dephasing beyond the training window

Control C2 fitted on zeros 1–400 scores 0.034/+0.99 on 401–500 and 0.078/+0.96 on 1001–2000 (Sec. IV). To see why skill must eventually decay, suppose a model learns the $p = 2$ frequency $\omega = \log 2$ from the window of zeros 1–400, of length $T = t_{400} - t_1$ in t, with an error $\delta\omega$ no larger than half a Fourier resolution cell, $|\delta\omega| \le \pi/T$. This bound is the tolerance the paper uses when matching selected frequencies to prime powers (App. A step 23). It is an upper bound, not a measured error, and Step 6 shows that a good fit cannot come close to it. Assume the learned and true waves agree in phase at the end of the window, $t_* = t_{400}$. Find the phase error of the fundamental $m = 1$ at t_{1001} and at t_{2000}, the two ends of the paper's 1001–2000 holdout.

Step 1: The true phase of the m-th harmonic is $m\omega t+\varphi$, the learned phase is $m(\omega+\delta\omega)t+\hat{\varphi}$. Agreement at t_* fixes $\hat{\varphi} = \varphi - m\,\delta\omega\,t_*$.

Step 2: Subtract the two phases at a general t:

$$\delta\phi_m(t) = m(\omega + \delta\omega)t + \varphi - m\,\delta\omega\,t_* - (m\omega t + \varphi) = m\,\delta\omega\,(t - t_*). \tag{32.4}$$

Step 3: Window: $T = t_{400} - t_1 \approx 666$ (computed with `qmnt.zeros`), so the largest allowed frequency error is $\pi/T \approx 0.00472$.

Step 4: Distances, from `qmnt.zeros(2000)`: $t_{1001} - t_{400} \approx 741$ and $t_{2000} - t_{400} \approx 1836$.

Step 5: By (32.4) with $m = 1$: $\delta\phi_1 \approx 0.00472 \times 741 \approx 3.5$ rad at t_{1001}, and $\delta\phi_1 \approx 0.00472 \times 1836 \approx 8.7$ rad at t_{2000}. The third harmonic, $m = 3$, has three times these phase errors.

$\boxed{\delta\phi_1 \approx 3.5 \text{ rad at } t_{1001}, \qquad \delta\phi_1 \approx 8.7 \text{ rad at } t_{2000}}$

Step 6: Why the bound is far from a typical error. Apply (32.4) inside the window. Between t_1 and t_{400} an error $\delta\omega = \pi/T$ produces a phase drift $\delta\omega\,T = \pi$, half a cycle. A learned wave that slides half a cycle out of step across its own training data cannot fit those data well. So a model that does fit them must have $|\delta\omega|$ well below π/T. Chapter 27 showed that the least-squares frequency error scales like $1/T$ times a factor that shrinks as the number of samples and the signal-to-noise ratio grow. If, for illustration, $|\delta\omega|$ were a tenth of the bound, the fundamental's phase error at t_{2000} would be about 0.87 rad. (We matched the phase at the end of the window for simplicity. A least-squares fit tends to get the phase right nearer the middle of the window. That would add $\delta\omega\,T/2$ to every phase error beyond the window, so matching at the end understates the error rather than exaggerating it.)

What the sample shows is the shape, not the size. The phase error grows linearly with distance from the window, so any learned frequency model has a finite range of validity

in height (C2's rms rises from 0.034 on 401–500 to 0.078 on 1001–2000). The exact frequencies $m \log p$ of the oracle never dephase (Table V).

Figure 32.3 draws this worst case for the prime-2 wave of running example RE2.

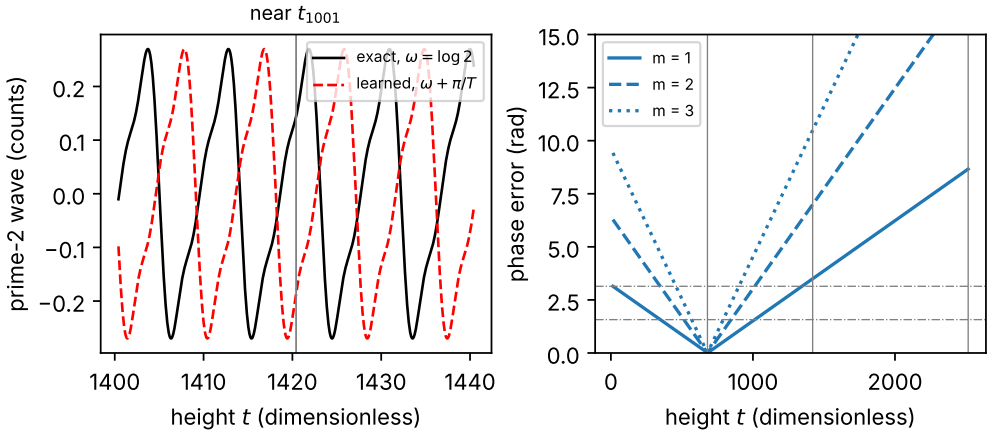


Figure 32.3: Worst-case dephasing of the prime-2 wave (RE2), $-\frac{1}{\pi}\sum_{m=1}^{3}\sin(mt\omega)/(m2^{m/2})$. The frequency is learned from zeros 1–400. Left: near t_{1001}, the exact wave ($\omega = \log 2$, solid) and a learned wave whose frequency is off by the matching tolerance $\delta\omega = \pi/T$, with $T = t_{400} - t_1$ (dashed). The two waves are matched in phase at t_{400}, and the vertical line marks t_{1001}. Right: magnitude of the phase error, $m\,\delta\omega\ |t - t_{400}|$, in radians for $m = 1, 2, 3$, with $\pi/2$ and π marked and vertical lines at t_{400}, t_{1001} and t_{2000}. The horizontal axes are the height t. The tolerance is an upper bound, much larger than the error of a good fit (see text).

32.3.4 Limit 4: Sec. V was not rerun

"The non-Hermitian analysis of Sec. V is inherited from the original computation and was not rerun for this revision." Sec. V's numbers (the floors, the null-mode artifact with $c_0 \approx 2.2 \times 10^6$, the exact-versus-surrogate comparison) therefore rest on the original computation. The paper also states that the Hermitian conclusions of Secs. III and IV "do not depend on" Sec. V. We give a cheap partial check in Section 32.4 (W4).

32.3.5 Limit 5: bootstrap independence

"Bootstrap intervals resample held-out levels independently except where stated." Independent resampling can misestimate uncertainty when neighboring values are correlated (Chapter 26). Consider the mean of n values with standard deviation σ. Its variance is σ^2/n plus $2/n^2$ times the sum of the covariances of all pairs. That variance equals σ^2/n only when those covariances vanish. Positive correlations between neighbors increase it and negative correlations decrease it. Independent resampling ignores these pair terms, so it understates the uncertainty in the first case and can overstate it in the second. The zeros' fluctuations have negative autocorrelation at lags 1–3 (Sec. III), and for statistics such as rms and corr the direction of the effect is not obvious in advance. The paper reran the Table II intervals with a moving-block bootstrap (block lengths 3 to 20) and reports that "they widen slightly and no conclusion changes (for the cosine row, at most $[0.30, 0.44]$)." You can see the effect yourself on the six-prime oracle in Figure 32.4.

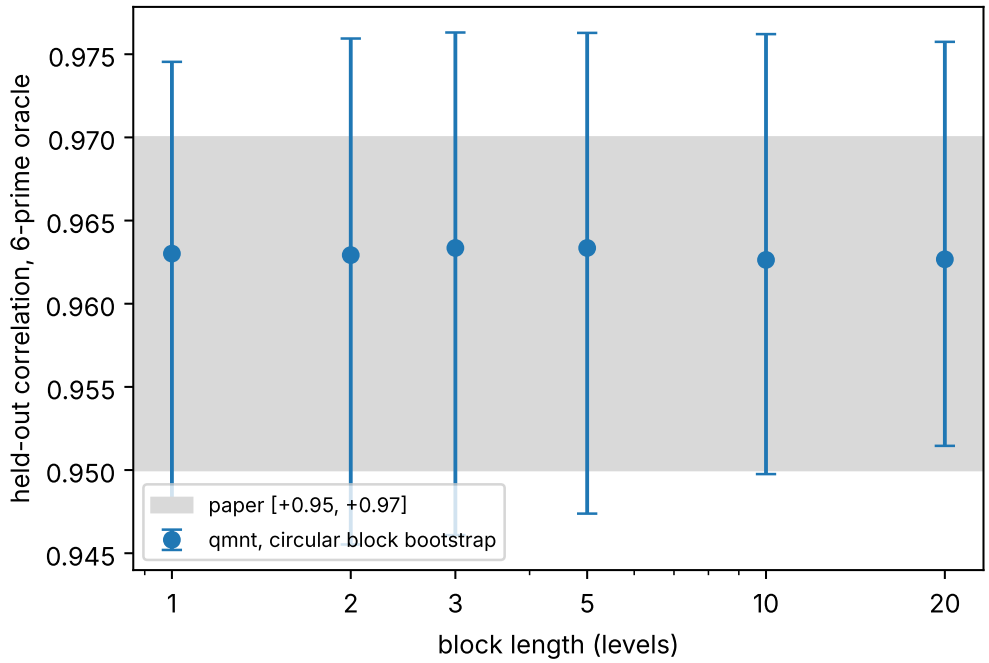


Figure 32.4: Percentile 95% bootstrap intervals for the held-out correlation of the six-prime oracle on zeros 401–500, computed with `qmnt` by circular block bootstrap, as a function of block length (block length 1 is independent resampling, as in most of the paper's intervals). The gray band is the paper's reported interval $[+0.95, +0.97]$ (Sec. III and Table V caption). Compare the widths at different block lengths to see how serial correlation affects the interval.

Checkpoint 32.4

Limit 5 concerns intervals, not point estimates. Could switching to a block bootstrap ever turn the operator's OOS rms 0.369 into a value below the null?

Answer: No. The bootstrap changes the interval around an estimate, not the estimate. The point value 0.369 is computed once from the held-out levels. A wider interval could at most make the comparison with the null less decisive; the paper reports that even with blocks the cosine row's interval is at most $[0.30, 0.44]$, still above 0.263.

Key Ideas

- The paper's stated limits: one background and three bases; one marginal Chebyshev row; operator fits to at most zero 1000 and fitted controls scored below zero 2000; Sec. V not rerun; mostly independent bootstrap.
- One marginal row among eight is expected about one time in five under rough independence, (32.3).
- A learned frequency dephases linearly with distance, (32.4); with the matching tolerance as the worst case, a frequency learned from zeros 1–400 could be about 3.5 rad out of phase at t_{1001}. A good fit does much better, but the linear growth with distance is unavoidable.

32.4 Weak points a referee could press

A referee's job is to find the places where a paper's conclusions are least secure. Below are the points a well-informed skeptic could raise. Several overlap with the stated limits; others are not listed in Sec. VI. For each we say what the paper itself says, how much weight the point carries, and what would resolve it. Raising a point is not the same as refuting a claim.

32.4.1 W1: the literature-gap claim rests on a keyword audit

Sec. I A states: "to our knowledge, no systematic held-out test of predictive performance has been reported in this literature". The basis (App. A, "Literature audit") is: among all works citing Wu–Sprung [12] and van Zyl–Hutchinson [13] indexed by Semantic Scholar (accessed August 2026; 28 and 29 works respectively; Slater [14] and Jaksch [15] had no indexed citations), none mentions prediction, generalization, held-out evaluation, or machine learning.

A referee could press three points. (a) Coverage: a database does not index everything, and relevant work need not cite these two papers. (b) Keywords: a held-out test might be described as "extrapolation" or "testing on higher zeros". (c) Sample size: a few dozen works is a small sample. The paper hedges with "to our knowledge" and records the audit so that it can be checked. How strong is "none of n"?

Sample Problem 32.7: What "none out of n" can support

Treat each citing work, heuristically, as an independent draw that reports a held-out test with unknown probability f. If none of n works does, find the value f_{95} above which observing "none" would have probability below 5%. Evaluate for $n = 29$ (if the two citing sets overlap completely, the distinct works number at least 29) and $n = 57 = 28 + 29$ (no overlap).

Step 1: For one work, the probability of no held-out test is $1 - f$. For n independent works, $P(\text{none}) = (1 - f)^n$, the same reasoning as (32.2). The independence assumption is rough: the 28 and 29 citing works overlap, may cite one another, and are not random draws from the literature, so what follows is an order-of-magnitude guide, not a precise bound.

Step 2: Set $(1 - f_{95})^n = 0.05$ and take the n-th root: $1 - f_{95} = 0.05^{1/n}$, so

$$f_{95} = 1 - 0.05^{1/n}. \tag{32.5}$$

Step 3: Approximate for large n. Write $0.05^{1/n} = e^{(\log 0.05)/n}$ and use $\log 0.05 = -2.996 \approx -3$ and $e^{-y} \approx 1 - y$ for small y (Chapter 4): $f_{95} \approx 3/n$, the "rule of three".

Step 4: $n = 29$: $(\log 0.05)/29 = -0.10330$, $e^{-0.10330} = 0.90186$, so $f_{95} = 0.0981$ (rule of three: $3/29 = 0.103$).

Step 5: $n = 57$: $(\log 0.05)/57 = -0.05256$, $e^{-0.05256} = 0.94880$, so $f_{95} = 0.0512$ (rule of three: $3/57 = 0.053$).

$$\boxed{f_{95} \approx 0.098 \ (n = 29), \qquad f_{95} \approx 0.051 \ (n = 57)}$$

Interpretation: the audit supports "held-out tests are rare in this citation neighborhood (probably under one work in ten, and under about one in twenty if the two citing lists do not overlap)" more firmly than "there are none anywhere". Keyword misses (point b) would make f_{95} larger still (Problem 25). The paper's own wording, "to our knowledge", is the appropriate strength.

32.4.2 W2: one background potential

Every fitted potential is a correction δV added to the Wu–Sprung potential (Chapter 22). The background fixes which part of space each level probes, through the classical turning point x_E with $V(x_E) = E$ (Chapter 12). A referee could ask whether a different background (one with fluctuation structure already built in, such as Wu and Sprung's own fractal corrections, or a different confining shape) would change the conclusion. The paper does not test this and says so.

The turning-point map is easy to compute and makes the question concrete.

Sample Problem 32.8: Turning points of the interpolation gap

The paper states that "levels 201–300 turn at $x = 22.5$–28.5" (Sec. III, interpolation split). Check this with `qmnt`, using the smooth zeros s_{201} and s_{300} as the level energies and the Wu–Sprung relation $x(V)$ of paper Eq. (2).
Step 1: For an even, monotone potential, a level of energy E turns where $V(x) = E$, so the turning point is $x(E)$, the Wu–Sprung function evaluated at $V = E$.
Step 2: Compute s_{201} by solving $N_{\rm sm}(s) = 200.5$ and s_{300} by solving $N_{\rm sm}(s) = 299.5$, with `qmnt.smooth_zeros` (the listing after this sample). This gives $s_{201} \approx 398$ and $s_{300} \approx 543$.
Step 3: Evaluate Eq. (2) at these energies (with `qmnt.wu_sprung_x`, in the same listing): $x(s_{201}) \approx 22.5$ and $x(s_{300}) \approx 28.5$.
$\boxed{x \approx 22.5 \text{ to } 28.5}$
This agrees with the paper. With a different background V, the same levels would turn at different places, and the region where an unregularized fit can "do damage at no in-sample cost" (paper Fig. 4) would move.

```
import numpy as np, qmnt
s = qmnt.smooth_zeros(np.array([201, 300]))    # energies of levels 201 and 300
x = qmnt.wu_sprung_x(s)                        # turning points, paper Eq. (2)
print(s, x)                                    # about [398, 543] and [22.5, 28.5]
```

32.4.3 W3: height range of the operator fits

Operator fits stop at zero 1000. A referee could ask two questions: would larger fits behave differently, and how saturated are the held-out fits? Apply the coverage law of Section 32.1 to the training ranges.

At scale 1 the principal fits train on zeros 1–400. With $E \approx s_{400} \approx 680$, the threshold is $2\sqrt{680} \times 44.883/\pi \approx 745$. The bases used, $M = 200$ to 600, give $M/745 \approx 0.27$ to 0.81.

At scale 2 the fits train on zeros 1–800. The threshold is about 1313, and $M = 300$ to 900 give 0.23 to 0.69.

So the held-out fits at *both* scales are below the heuristic coverage threshold for their top training levels, by comparable factors. Only the in-sample coverage run on all 500 zeros at scale 1, with $M = 900$ above its threshold of about 814, is fully saturated. The negative result held across this whole range of M relative to the threshold: the scale-2 rows sit at or above their null 0.274, as the scale-1 rows sit at or above theirs. Fits at higher zeros are a research project (Section 32.6).

32.4.4 W4: Sec. V inherited, not rerun

The paper states this limit itself. A reader can do one cheap structural check of Sec. V A without rerunning anything: the sum rule $\sum_n \delta E_n = 0$ is the second-order piece of an exact identity.

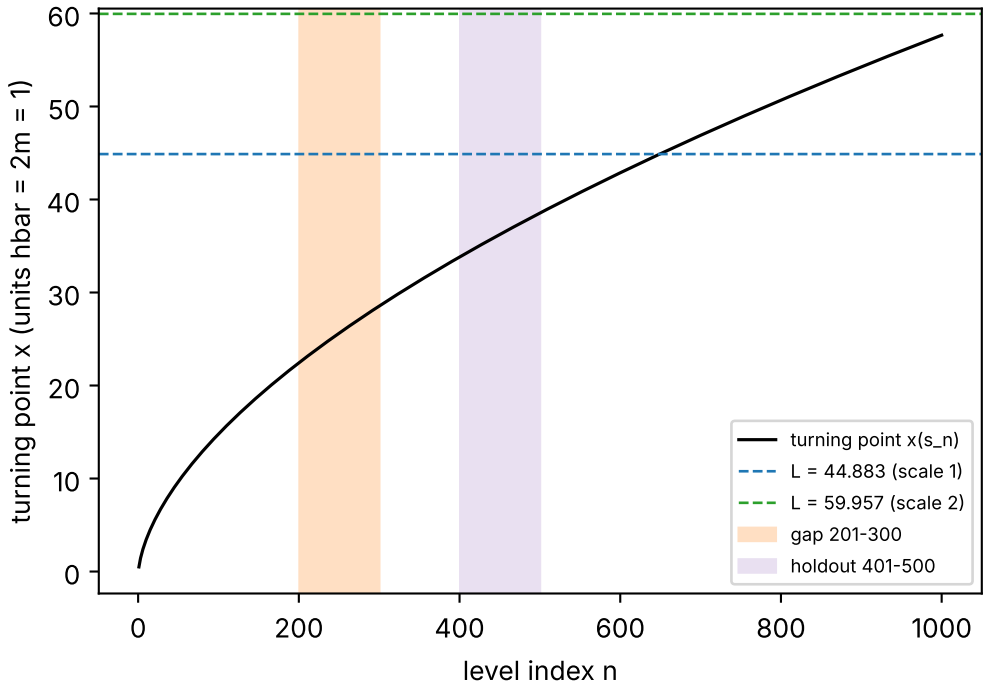


Figure 32.5: Classical turning point $x(s_n)$ of level n in the Wu–Sprung potential (paper Eq. (2)), for $n = 1$ to 1000, in units $\hbar = 2m = 1$. Horizontal lines: the box half-widths $L = 44.883$ (scale 1) and $L = 59.957$ (scale 2). Shaded: the interpolation gap 201–300 and the principal holdout 401–500. Each held-out band of levels probes its own band of positions most strongly; the background potential fixes this map.

Taken on trust

Taken on trust: trace equals the sum of eigenvalues. For any $n \times n$ complex matrix, the sum of its eigenvalues, counted with multiplicity, equals its trace (the sum of its diagonal entries). The proof compares the coefficient of λ^{n-1} in the characteristic polynomial $\det(\lambda I - B)$ written two ways: as $\prod_k (\lambda - \lambda_k)$ and by expanding the determinant. Recall the 2×2 case from Chapter 8, where the characteristic polynomial is $\lambda^2 - (\text{trace})\lambda + \det$, so the two eigenvalues add up to the trace. For the general case see the chapter on eigenvalues and eigenvectors in [28].

Apply it to the truncated matrix of $H + iA$ in the unperturbed eigenbasis: its diagonal entries are $E_n + iA_{nn}$, with $A_{nm} = \langle n|A|m\rangle$ real for a real multiplication operator A. The sum of exact eigenvalues λ_n is therefore

$$\sum_n \lambda_n = \sum_n E_n + i \sum_n A_{nn} \quad \Longrightarrow \quad \sum_n (\operatorname{Re} \lambda_n - E_n) = 0. \tag{32.6}$$

This is a statement about a finite matrix: the λ_n here are the eigenvalues of the truncated matrix of $H + iA$ in the chosen basis of unperturbed states, not of the full operator on the whole (infinite-dimensional) space. For that finite matrix the exact real shifts sum to zero. The second-order sum rule is one piece of this identity, as the following argument shows.

Replace A by ϵA, with ϵ a real bookkeeping parameter. Suppose the unperturbed levels E_n are distinct, so that each eigenvalue can be expanded as a power series in ϵ (Chapter 13):

$$\lambda_n(\epsilon) = E_n + \epsilon a_n + \epsilon^2 b_n + \mathcal{O}(\epsilon^3).$$

Chapter 14 found $a_n = iA_{nn}$. It also found, from second-order perturbation theory with $V \to iA$ (the product of two matrix elements carries $i^2 = -1$), that

$$b_n = -\sum_{m \neq n} \frac{A_{nm}^2}{E_n - E_m},$$

which is the shift δE_n of paper Eq. (5). The trace identity holds for every ϵ:

$$\sum_n \lambda_n(\epsilon) = \sum_n E_n + i\epsilon \sum_n A_{nn}.$$

The right side has no ϵ^2 term. Two power series in ϵ that agree for all small ϵ have equal coefficients. Comparing the coefficients of ϵ^2 therefore gives $\sum_n b_n = \sum_n \delta E_n = 0$. The same comparison at every higher power shows that each order sums to zero separately. Our reading, which is this chapter's interpretation and not a statement in the paper, is therefore that the second-order sum rule verified in Sec. V A (to 3.6×10^{-15} and 1.4×10^{-14}) checks that the code is internally consistent rather than telling us anything about the zeros. The paper itself concludes, on other grounds, that the floor is "not a physically meaningful constant".

A three-level example makes this concrete. Take $E = (1, 2, 4)$ and real symmetric A with zero diagonal, $A_{12} = 0.05$, $A_{13} = 0.1$, $A_{23} = 0.15$. Paper Eq. (5) gives

$$\begin{aligned}
\delta E_1 &= -\Big[\frac{0.0025}{1-2} + \frac{0.01}{1-4}\Big] = -[-0.0025 - 0.003333] = 0.005833, \\
\delta E_2 &= -\Big[\frac{0.0025}{2-1} + \frac{0.0225}{2-4}\Big] = -[0.0025 - 0.01125] = 0.00875, \\
\delta E_3 &= -\Big[\frac{0.01}{4-1} + \frac{0.0225}{4-2}\Big] = -[0.003333 + 0.01125] = -0.014583,
\end{aligned}$$

and the sum is $0.005833 + 0.00875 - 0.014583 = 0$. An exact diagonalization of the 3×3 matrix $\operatorname{diag}(1, 2, 4) + iA$ (for example with `numpy.linalg.eigvals`) gives real parts that agree with $E_n + \delta E_n$ to better than 10^{-3}, and whose sum is exactly the trace, 7.

32.4.5 W5: the frequency-alignment statistic was defined after seeing the data

Sec. IV reports that every frequency selected by C2 lies within half a Fourier resolution cell of some $\log q$, q a prime power, and that a surrogate ensemble (900 replicates) never approaches the real zeros' alignment (App. A step 23: cutoff-maximized binomial scores 24.7 and 50.9 on the real zeros, at most 4.7 on surrogates). The paper is explicit that "the statistic was defined after the data were seen, so the check is exploratory" and that "no significance claim is made here beyond the size of the ensembles."

The paper describes the score only briefly, as a "cutoff-maximized binomial score" and as "a statistic that does not depend on a chosen frequency cutoff". Our reading, in outline, has three steps. First, count how many selected frequencies lie within π/T of $\log q$ for some prime power q up to a cutoff. Second, compare that count with a binomial distribution, which gives the number of chance matches if each frequency matched independently with a chance probability of the kind estimated in the sample below. The less probable the observed count under that model, the larger the score. Third, maximize the score over the cutoff, so that no single cutoff has to be chosen by hand. The exact definition is in the Zenodo record [20], and Chapter 30 discusses it. What matters here is the comparison: 24.7 and 50.9 on the real zeros against at most 4.7 on any surrogate.

Why is a surrogate check needed at all? Because chance alignment is common at high frequency, where prime powers crowd together.

Sample Problem 32.9: How often does a random frequency land near some $\log q$?

Use the prime number theorem (Chapter 15): near x, primes occur with density about $1/\log x$ per unit of x. Estimate the expected number $\bar{m}(\omega)$ of primes q with $|\omega - \log q| < \pi/T$, and the probability of at least one, at $\omega = 2$, 5 and 8, for the 400-zero window $T = t_{400} - t_1 \approx 666$.

Step 1: The number of primes in $[x, x + \mathrm{d}x]$ is about $\mathrm{d}x/\log x$.

Step 2: Change variable to $\omega = \log x$: $x = e^{\omega}$, $\mathrm{d}x = e^{\omega}\,\mathrm{d}\omega$, $\log x = \omega$. The number of primes with $\log q$ in $[\omega, \omega + \mathrm{d}\omega]$ is about $(e^{\omega}/\omega)\,\mathrm{d}\omega$.

Step 3: The matching window has width $2\pi/T$ in ω. If it is narrow compared with the scale on which e^{ω}/ω changes,

$$\bar{m}(\omega) \approx \frac{2\pi}{T}\,\frac{e^{\omega}}{\omega}. \tag{32.7}$$

Prime powers p^m with $m \geq 2$ add only a few more, so we neglect them.

Step 4: Make a heuristic assumption: treat the $\log q$ near ω as if they were scattered at random and independently of one another, that is, as a Poisson process with the density of Step 3 (Chapter 20). This is a model, not a theorem. Real primes have correlations (for example, every prime above 2 is odd, so two primes above 2 are never adjacent integers), and these correlations are neglected. Under the assumption, the probability of no point in a window with expected count $\bar{m}$ is $e^{-\bar{m}}$, so $P(\geq 1) = 1 - e^{-\bar{m}}$.

Step 5: $2\pi/666 = 0.009434$. At $\omega = 2$: $e^2/2 = 3.695$, $\bar{m} = 0.0349$, $P = 0.034$. At $\omega = 5$: $e^5/5 = 29.68$, $\bar{m} = 0.280$, $P = 0.244$. At $\omega = 8$: $e^8/8 = 372.6$, $\bar{m} = 3.52$, $P = 1 - e^{-3.52} = 0.970$.

$$\boxed{P(\text{chance match}) \approx 0.03\ (\omega = 2),\ 0.24\ (\omega = 5),\ 0.97\ (\omega = 8)}$$

So near the top of C2's search band ($\omega < 8$, and $\log 3000 = 8.006$) almost any frequency "matches" some prime power. The evidence of alignment rests on the low-frequency selections and on the collective comparison with surrogates, which is consistent with the paper's decision to check the alignment against a surrogate ensemble (its stated reason:

"the frequencies are the output of a search") and to use a statistic that does not depend on a chosen frequency cutoff. These are heuristic estimates (density from the prime number theorem, Poisson approximation, prime powers neglected).

You can count the actual prime powers in each window with the following listing and compare with (32.7).

```
import numpy as np, qmnt
t = qmnt.zeros(400)
T = t[-1] - t[0]                  # length of the 400-zero window in t
q = []                            # all prime powers up to 3000
for p in qmnt.primes_upto(3000):
    pk = int(p)
    while pk <= 3000:
        q.append(pk)
        pk = pk * int(p)
logq = np.log(np.array(q, dtype=float))
for w in (2.0, 5.0, 8.0):
    count = np.sum(np.abs(logq - w) < np.pi / T)
    estimate = (2 * np.pi / T) * np.exp(w) / w
    print(w, count, round(estimate, 3))
```

32.4.6 W6: the explanation is heuristic, and the optimizer is part of the result

The paper says the explanation is heuristic and that the evidence is "about the tested parametrizations and optimizer together". A referee could press: the Hermitian fits use a specific Levenberg–Marquardt schedule (App. A step 6) with at most 60 iterations in the revision computations; a different optimizer, or a prior on δV that favors long-range oscillation in energy, might behave differently. The paper's within-height and interpolation results show that, under this protocol, the fit actively places structure where only held-out levels look; it does not show that no protocol could interpolate. The paper says this explicitly for the interpolation split.

32.4.7 W7: the memorization result is measured, not derived

"Whether the memorization result can be derived rather than measured is an open problem." A derivation would explain, for example, why held-out correlations cluster near zero rather than at some small positive value. A starting point is suggested in Section 32.6.

32.4.8 W8: smaller points

- *Seed ranges are not standard errors.* Table III's caption says so: the seeds reuse the same zeros.
- *One relaxed row.* The scale-2 $M = 300$ row of Table I used a relaxed acceptance threshold; under the strict protocol "the optimizer accepts no step at all". The paper labels it a sensitivity result.
- *The factor* $\gtrsim 1.3$. Sec. IV says one prime beats every fitted operator's holdout performance by a factor $\gtrsim 1.3$. From the tables we can check this only for operator rows scored on the same holdout as the tabulated one-prime oracle, zeros 401–500, where the oracle scores 0.198 (Table V). Among those rows the lowest Table I OOS rms is 0.265 (ridge $\mu = 0.3$), giving $0.265/0.198 = 1.34$; the lowest Table II row, Chebyshev with ridge $\mu = 0.3$ at 0.264, gives $0.264/0.198 = 1.33$. Both clear 1.3, but only just. Table I rows fitted on 1–300,

1–350 or 1–450, and the scale-2 rows, are scored on other holdouts, for which Table V gives no matching one-prime entry, so this check does not cover them. Problem 6 asks you to redo the computation.

- *Two nulls.* The operator null and the smooth-zero null are conceptually distinct; the paper notes that their numerical coincidence on 401–500 (both 0.2631) is what permits a single baseline (Sec. II C).

- *The oracle is not a fair competitor.* The paper says so itself; its role is to show that transferable information exists, while the fitted controls B, C1, C2 provide the fair comparisons.

32.4.9 W9: substantial AI assistance

Sec. II F discloses that the computations, data analysis and drafting were carried out with substantial assistance from a large language model, used in agentic coding sessions, and that the author reviewed all content and takes responsibility for it. A referee should treat this neutrally: a result is judged by whether it can be reproduced and checked, not by who or what typed the code. The paper's mitigations are the relevant facts: every number is regenerated by an archived scripted pipeline (deterministic or with recorded seeds), the pipeline was rerun end to end from a clean copy, and the scripts were rerun by the author on a separate system (App. A). The next section shows you how to exercise that record yourself.

32.4.10 What a referee should not press

Fairness requires listing the strengths too: explicit nulls; a leakage argument for the operator fits (paper Eq. (3)) and, for the sinusoidal controls, a leakage test ("overwriting held-out data with noise before fitting leaves all fitted coefficients bit-identical", App. A step 20); controls that separate parameter count from functional form (Table IV); surrogate refits; exact verification of the non-Hermitian claims; a clean continuation probe on a target whose answer is known; scoped language throughout; and an archived reconstruction record.

Checkpoint 32.5

Which of W1–W9 could change the paper's *numbers*, and which could only change how far the conclusions extend?

Answer: W4 is the only one that concerns whether reported numbers are correct as computed (Sec. V was not rerun), and the paper states the Hermitian conclusions do not depend on it. W1, W2, W3, W6 and W7 concern scope or explanation: the reported numbers stand, but the conclusions might not extend to other bases, backgrounds, heights, optimizers, or be explained as the paper suggests. W5 concerns the strength of one exploratory statistic. W8 collects small points, most of which the paper states itself in captions or text; the factor-1.3 check is ours and concerns how widely one sentence applies. W9 is addressed by reproducibility.

Checkpoint 32.6

At $\omega = 8$ a random frequency matches some prime with probability about 0.97. Does this mean C2's alignment result is meaningless?

Answer: No. It means the alignment of *high* frequencies alone is weak evidence. The paper does not rest on single matches: it compares the alignment of the whole set of selected frequencies with that of 900 surrogate runs, and the real zeros score far above all surrogates (24.7 and 50.9 versus at most 4.7).

The paper still labels this check exploratory because the statistic was chosen after the data were seen. Independently of alignment, C2's predictive score (0.034, corr +0.99) is a held-out result.

Key Ideas

- W1: "none of 29 to 57 works" supports "rare" (upper bound roughly 5% to 10% per work) more firmly than "none", by the rule of three (32.5).
- W2: the background fixes which positions each level probes; levels 201–300 turn at $x \approx 22.5$–28.5.
- W4: the second-order sum rule is part of the exact identity (32.6); it checks code, not physics.
- W5: chance alignment with some $\log q$ is rare at low ω but nearly certain near $\omega = 8$, (32.7); hence the surrogate ensemble and the exploratory label.
- Strengths should be listed with weaknesses.

32.5 Reproducing the paper

★ Did you know? *Turing Hunts Zeros*

In 1950 Alan Turing used the Manchester Mark 1, one of the first stored-program computers, to check zeta zeros, and he published the results in 1953. He also devised a clever method to certify that no zeros had been missed in a given range. Versions of Turing's method are still used to verify zero counts, so anyone reproducing a zero computation today follows in his footsteps.

32.5.1 What the record contains

According to the Data Availability statement, the Zenodo record [20] (version R1, DOI 10.5281/zenodo.24035819) contains "all scripts, computed zeros, fitted coefficients, and the full parameter record of Appendix A needed for independent reconstruction". The version as first submitted has DOI 10.5281/zenodo.21842988. The analysis of the original submission "was reconstructed from scratch, from that record alone, in a fresh compute environment", and the scripts, coefficients, logs and result files of the revision computations (App. A steps 16–23) are in the updated record, with an audit file that maps scripts, commands, seeds and outputs to every item. The complete pipeline is driven by `run_all.sh`. The stated environment is Python 3 with `numpy`, `scipy` and `mpmath`; the revision computations name `numpy` 2.5.3 and `scipy` 1.18.1.

Using only what the paper states, a reconstruction goes as follows.

1. Download version R1 of the record from DOI 10.5281/zenodo.24035819.
2. Create a Python 3 environment with `numpy`, `scipy` and `mpmath`, preferably at the versions named above.
3. Run `run_all.sh`, which drives the complete pipeline.
4. Compare the outputs with the key anchors of App. A and with the stored result files, using the audit file to find which script produced which number.

The paper names no file other than `run_all.sh` and the audit file, and it gives no run times. Even the reduced-cost operator fit `qmnt.fit_potential` takes minutes, so expect the full-size operator fits to dominate the run time.

Table 32.2 maps each pipeline step to the chapter of this book that explains it.

Table 32.2: App. A pipeline steps and where this book explains them.

Step	Content	Chapters
1	zeros by `mpmath.zetazero`	16
2	smooth zeros by Brent; gap by `np.gradient`	4, 17
3–4	grids, Wu–Sprung inversion, tridiagonal solver	8, 22, 23
5	bias-corrected targets (Richardson)	23
6–7	Hermitian fit (SVD-damped LM), OOS scoring	9, 24, 25
8	oracle, Eq. (4), Brent with bracket expansion	4, 18
9–11	tensors $T^{(n)}$, NH fit, exact check	13, 14
12, 18	permutation and surrogate targets	6, 26
13, 22	bootstrap; circular block bootstrap	26
14, 21	interpolation split, rescoring, ridge refits	25
15	arithmetic-continuation probe	5, 25
16	Chebyshev and hat bases	24
17, 19	within-height splits; kriging	25
20, 23	sinusoidal controls; alignment surrogate check	27, 26

32.5.2 Anchors: checking a reconstruction

App. A lists "key anchors", numbers that any correct reconstruction must reproduce. Some are cheap enough to check here.

Sample Problem 32.10: Reproducing the cheap anchors

Use `qmnt` to reproduce: (a) $t_1 = 14.1347251417$ and $t_{500} = 811.1843588465$ (App. A key anchors) and $t_{1000} = 1419.42248094599$ (Sec. II A); (b) the null OOS rms on 401–500, 0.2631; (c) $L = 44.883$ and 59.957.

Step 1: Zeros. `t = qmnt.zeros(1000)`; then `t[0]`, `t[499]`, `t[999]` are t_1, t_{500}, t_{1000} (Python arrays start at index 0, the paper's indices at 1). They agree with these values to at least eight decimal places (absolute differences below 10^{-8}).

Step 2: Null. The null predicts $t_n^{\text{pred}} = s_n$, so its error in mean spacings is the true fluctuation $(t_n - s_n)/\text{gap}_n$. Compute `f = qmnt.fluct(t[400:500], np.arange(401, 501))` and `qmnt.rms(f)`.

Step 3: Box widths: `qmnt.L_of(1000)` and `qmnt.L_of(1500)`, which evaluate paper Eq. (2) at $V = 1000$ and 1500.

t_1, t_{500}, t_{1000} as anchored; null $= 0.263$; $L = 44.883,\ 59.957$

Checkpoint 32.7

You build your own finite-difference version of the operator null on the scale-1 working grid ($N = 6000$), skip the bias correction, and score the unperturbed levels against the raw zeros 401–500. You get about 2 spacings instead of 0.263. What happened?

Answer: You measured finite-difference dispersion, not the zeros. On the working grid the discrete levels near 401–500 are displaced by about two mean spacings (Chapter 23); App. A records the "raw-zero null on the uncorrected working grid" as 2.09. The bias correction of step 5, which by paper Eq. (3) is the same as correcting the prediction by the fit-independent Δ_n, removes this.

32.5.3 Reproducing a Table V row

```
import numpy as np, qmnt
n = np.arange(401, 501)                # held-out indices (RE5)
t = qmnt.zeros(500)[n - 1]             # true heights
print("null rms", qmnt.rms(qmnt.fluct(t, n)))     # paper: 0.2631
for k in (1, 3, 6, 25):
    P = qmnt.first_primes(k)           # primes 2, 3, 5, ...
    tp = qmnt.oracle_zeros(n, P)       # solve Nsm + Nfl = n - 1/2
    r, c = qmnt.score(tp, n)           # rms and corr in mean spacings
    print(k, "primes: rms", round(r, 3), "corr", round(c, 2))
```

Sample Problem 32.11: Reproducing the six-prime oracle on zeros 401–500

Run the listing above with $k = 6$ (primes 2, 3, 5, 7, 11, 13) and compare with Table V.
Step 1: `qmnt.oracle_zeros` solves $N_{\mathrm{sm}}(t) + N_{\mathrm{fl}}(t) = n - \frac{1}{2}$ for each n, where N_{fl} is paper Eq. (4) with $m \le 3$, by Brent's method in a bracket of ± 3 gaps around s_n (App. A step 8).
Step 2: `qmnt.score` converts predictions and truths to fluctuations in mean spacings and returns the rms error and Pearson correlation.
Step 3: Running the listing gives rms 0.074 and corr +0.96, the Table V entry.
$\boxed{0.074 \ / \ +0.96}$

Figure 32.6 shows the same computation for the three holdouts up to zero 2000, against the paper's full Table V.

32.5.4 A leakage test you can run

App. A step 20 reports, for the sinusoidal controls, a simple but powerful check: overwrite the held-out data with noise, refit, and confirm that every fitted coefficient is bit-identical. If anything changes, held-out information leaked into the fit. Here is the test for a small version of control B (frequencies $m \log p$ given, amplitudes fitted by least squares, Chapter 27).

```
import numpy as np, qmnt
t = qmnt.zeros(500).copy()
n = np.arange(1, 501)
train = n <= 400                           # training indices 1..400
P = qmnt.first_primes(3)
omegas = np.array([m * np.log(p) for p in P for m in (1, 2, 3)])

def fit(heights):
    y = n - 0.5 - qmnt.Nsm(heights)        # targets y_n of Sec. II D
    tt, yy = heights[train], y[train]      # training rows only
    X = np.hstack([np.sin(np.outer(tt, omegas)),
                   np.cos(np.outer(tt, omegas))])
    return np.linalg.lstsq(X, yy, rcond=None)[0]

c1 = fit(t)
rng = np.random.default_rng(0)
t_noisy = t.copy()
t_noisy[~train] = rng.uniform(0, 5000, size=100)   # destroy the holdout
c2 = fit(t_noisy)
print(np.array_equal(c1, c2))              # True: no leakage
```

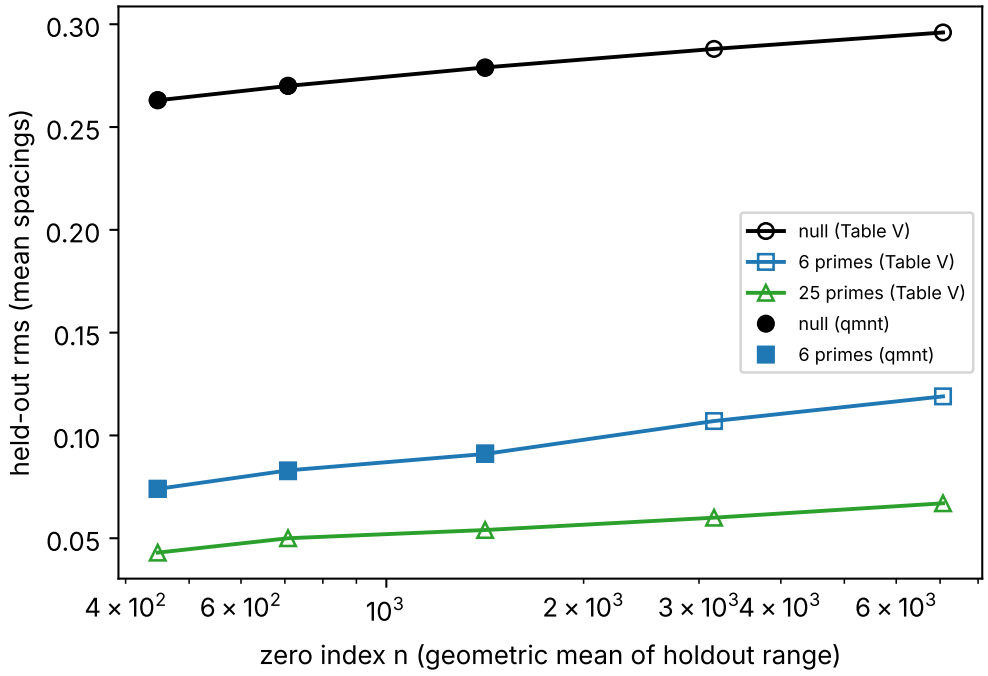


Figure 32.6: Null rms and oracle rms (6 and 25 primes) as functions of height, in units of local mean spacing. Lines with open markers: paper Table V for the five holdouts 401–500 through 5001–10000, plotted at the geometric mean of each index range. Filled markers: recomputed with `qmnt` for the holdouts up to zero 2000 (null and 6 primes). The null grows slowly with height. The oracle's error grows somewhat faster than the null: for six primes, $\mathcal{S}$ of (32.1) falls from 0.72 to 0.60 between the first and last holdout, and for 25 primes from 0.84 to 0.77. The Table V caption describes this as nearly constant relative skill.

The test is only as strong as what it overwrites: here the held-out heights. A pipeline that also reads held-out information from a second source (for example, a precomputed table of fluctuations) would need that source overwritten too.

Key Ideas

- The Zenodo record holds scripts, zeros, coefficients, logs and the parameter record; `run_all.sh` reruns the pipeline.
- Anchors (t_1, t_{500}, null 0.2631, L values, Table V rows) are cheap first checks of any reconstruction.
- Overwriting held-out data and checking bit-identical coefficients is a direct leakage test.

32.6 How to go further: research projects

Each project below starts from a limit or open question of the paper. The rating is $\star$ (a weekend with `qmnt`), $\star\star$ (a few weeks; requires writing your own fitting loop from Chapter 24) or $\star\star\star$ (open-ended; could become a research paper). For each we state what success would look like. Any project that fits an operator should report the paper's four-item standard.

Project 1 ($\star$): extend Table V. Recompute the oracle rows with `qmnt` for 401–500, 501–1000 and 1001–2000, then for 2001–5000 and 5001–10000 (`qmnt.zeros` goes to 10^4). Add 50 and 100 primes. Success: you reproduce the paper's rows and quantify how relative skill (32.1) depends on the prime budget.

Project 2 ($\star$): within-height kriging. Reimplement the simple kriging interpolator of Sec. III (App. A step 19: autocovariance to lag 40, $\cos^2$ taper, positive-semidefinite projection, nugget 0.05) on random 80/20 splits of zeros 1–400. Success: held-out correlation near the paper's +0.87, and the null on permuted targets.

Project 3 ($\star\star$): control C2 from scratch. Implement the greedy sparse frequency search (Chapter 27) on 400 zeros, choosing K by the forward split. List the selected frequencies, and for each compute $\bar{m}(\omega)$ from (32.7), so you can see which matches are individually informative.

Project 4 ($\star\star$): a small operator memorization study. Using `qmnt.fit_potential` (one call per run, slow), reproduce the qualitative shape of paper Fig. 3 at small M: in-sample error falling with M, held-out error at or above the null. Then refit to a permuted target (item (iv)).

Project 5 ($\star\star$): a Hermite basis. Replace $\cos(k\pi x/L)$ by even Hermite functions $\phi_k(x) = H_{2k}(x/\ell)\, e^{-x^2/(2\ell^2)}$, the harmonic-oscillator eigenfunctions of Chapter 11 (up to normalization) with a length scale ℓ you choose. Here H_j is the Hermite polynomial of degree j. It is defined by $H_0(u) = 1$, $H_1(u) = 2u$ and the recurrence $H_{j+1}(u) = 2uH_j(u) - 2jH_{j-1}(u)$; for example, $H_2(u) = 4u^2 - 2$. `scipy.special.eval_hermite` evaluates these polynomials. Write the Gauss–Newton loop of Chapter 24 with this basis. Success: an honest table in the format of paper Table II, with bootstrap intervals. This tests one of the parametrizations the paper names as untested.

Project 6 (⋆⋆⋆): a neural-network potential. Represent $\delta V(x)$ by a small neural network (a composition of linear maps and a fixed nonlinear function) and fit its weights. The paper names this as untested. The difficulty is not the code but the protocol: report restarts, regularization, and surrogates exactly as for the other bases.

Project 7 (⋆⋆): another background. Replace the Wu–Sprung background by a different even confining potential (for example one whose WKB levels track s_n only approximately, to test sensitivity to the background), or add Wu and Sprung's fractal corrections [12] to the background. Recompute the turning-point map (Figure 32.5) first, then rerun the held-out protocol.

Project 8 (⋆⋆): rerun Sec. V at small scale. On a small grid, compute the second-order tensors of paper Eq. (5) for a PT-odd basis, fit, and compare with exact complex diagonalization at full and quarter amplitude. Check the exact sum rule (32.6). This addresses Limit 4.

Project 9 (⋆⋆⋆): higher zeros. Operator fits stopped at zero 1000 and fitted controls were scored below zero 2000. Extend the controls to zeros near 10^4 with `qmnt`, and think about how the cost of operator fits grows with the number of levels (the grid must resolve the local wavelength $\pi/\sqrt{E}$, Chapter 24). Zeros much higher than 10^4 can be computed with `mpmath` slowly, or taken from the tables of zeros that Odlyzko has made publicly available. These tables are not the cited paper; his study of spacings is [8].

Project 10 (⋆⋆⋆, open): derive the memorization result. In the linearized regime of Chapter 13, a fit with training sensitivity matrix A_{in} (rows: training levels) and held-out sensitivity A_{out} predicts held-out shifts $A_{\text{out}}\, A_{\text{in}}^{+}\, r_{\text{in}}$, where A_{in}^{+} is the (regularized) pseudo-inverse of Chapter 9 and r_{in} the training residuals. Skill requires this "transfer matrix" $A_{\text{out}} A_{\text{in}}^{+}$ to map true training fluctuations to true held-out ones. Compute it for the cosine basis on the Wu–Sprung background and study its structure. The paper calls the derivation an open problem; a convincing first-order analysis would be a contribution.

Project 11 (⋆⋆⋆): Dirichlet L-functions. The zeta function is one member of a family. For the character χ_4 modulo 4 ($\chi_4(k) = 0$ for even k, 1 for $k \equiv 1$, -1 for $k \equiv 3 \pmod 4$), the function $L(s, \chi_4) = \sum_k \chi_4(k) k^{-s}$ has its own zeros and its own explicit formula, in which each prime term is weighted by $\chi_4(p)^m$. `mpmath` evaluates such functions (`mpmath.dirichlet`); you must locate zeros yourself by sign changes of a suitable real function and Brent's method, and look up the smooth counting function in a number theory text such as [25]. Then repeat the oracle-versus-fit comparison. Success: the paper's protocol transferred to a second spectrum.

Key Ideas

- Projects 1–4 reproduce and vary the paper's results; 5–8 test what the paper names as untested; 9–11 extend its scope; Project 10 attacks its stated open problem.
- Any operator fit you report should satisfy the four-item standard of Section 32.2.

32.7 Paper connection

This chapter is itself a reading of the paper, so the connection is a map from our sections to R1's numbering.

Three points deserve emphasis. First, the paper's own hedges are its strongest feature: "heuristically", "exploratory", "not excluded", "within the operator families tested". When

Table 32.3: Where each part of this chapter meets the paper (R1 numbering).

This chapter	Paper
Section 32.1, scope	Sec. VI para. 1; Sec. I scope paragraph; Sec. I A d
Section 32.1, heuristic	Sec. VI para. 2; Sec. III (coverage, surrogates, interpolation split, Fig. 4, continuation probe); Table III; Table IV
Relative skill, best rescaling	Tables I, IV, V; Fig. 1 (corr -0.07 vs $+0.96$)
Coverage threshold	Sec. III first paragraph; Table I scale-2 rows; App. A key anchors
Section 32.2	Sec. VI para. 5; items motivated by Fig. 1, Tables I, III; Sec. V and Fig. 5; Table I ridge rows and caption; Sec. III surrogates
Section 32.3	Sec. VI para. 3; Table II (Chebyshev $M = 400$); Sec. IV dephasing ($0.078/+0.96$); Sec. III block bootstrap; App. A steps 13, 22
W1	Sec. I A a; App. A literature audit
W2, W3	Sec. II B, Eq. (2), L values; Sec. III interpolation split; Fig. 4
W4	Sec. V, V A, Eq. (5), Fig. 5; App. A steps 9–11
W5	Sec. IV (alignment, surrogate check); App. A steps 20, 23; refs. [18, 19]
W8, W9	Captions of Tables I, III, IV; Sec. II C two nulls; Sec. II F
Section 32.5	Data Availability; App. A steps 1–23 and key anchors; Eqs. (1), (3), (4); Table V
Section 32.6	Sec. VI limits and open problem; untested bases named in Sec. I and VI

you summarize the paper, keep them. Second, the paper's novelty claim is the controlled out-of-sample comparison and the protocol, not the explicit formula, which it attributes to the classical Riemann–von Mangoldt/Guinand–Weil theory and to França and LeClair [3, 4]. Third, the falsifiability standard (Sec. VI) is the part of the paper most likely to outlast its specific numbers: it is a checklist anyone can apply to the next claim of a fitted "zeta operator".

32.8 Summary

- Sec. VI of the paper constrains only an empirical shortcut, not the analytic Hilbert–Pólya program; its explanation of the failure is heuristic, and whether memorization can be derived is open (Table 32.1).
- Relative skill $\mathcal{S} = 1 - \text{rms/null}$, (32.1): -0.40 for the $M = 400$ operator and $+0.72$ for the six-prime oracle on zeros 401–500.
- The best rms obtainable by rescaling a prediction with correlation r is null$\times\sqrt{1-r^2}$; uncorrelated predictions cannot beat the null.
- Coverage threshold $M > 2\sqrt{E}L/\pi$: about 814 at scale 1 and about 1313 for the top training levels at scale 2.
- The proposed standard: (i) held-out rms and corr against the model's own null, including a within-height split; (ii) exact spectra; (iii) robustness to regularization and restarts; (iv) surrogate refits. It is necessary, not sufficient.
- Multiple comparisons, (32.3): one marginal row in eight is expected about one time in five under rough independence.

- Dephasing, (32.4): phase error $m\,\delta\omega\,(t-t_*)$ grows linearly with distance from the window.
- Rule of three, (32.5): "none of n" bounds a per-item rate by about $3/n$.
- Exact sum rule, (32.6): real shifts of $H+iA$ sum to zero at all orders over a finite basis.
- Chance alignment, (32.7): expected number of primes within π/T of ω is about $(2\pi/T)e^{\omega}/\omega$; near $\omega = 8$ chance matches are almost certain.
- Anchors and a Table V row are reproducible with `qmnt`; a leakage test compares coefficients after overwriting held-out data.

32.9 Problems

1. • List the four items of the falsifiability standard proposed in Sec. VI of the paper. For each, name one experiment in the paper (section, table or figure) that shows why the item is needed.

2. • Using (32.1) with null 0.263, compute the relative skill on zeros 401–500 of: the $M = 600$ cosine operator (OOS 0.313, Table I), control C1 (0.258, Table IV), and control B with 25 primes (0.028, Table IV).

3. • The interpolation split (Sec. III) has gap error 2.64 spacings against a null of 0.255. Compute the ratio and compare with the paper's phrase "ten times the null".

4. • Classify each statement as (A) a limit the paper states in Sec. VI, (B) a weak point a referee could raise that is not among the Sec. VI limits, or (C) a claim the paper does not make: (a) Hermite bases were not tested. (b) The literature-gap claim rests on a keyword search of two citation lists. (c) The truncated explicit formula is a new result. (d) Sec. V was not rerun for the revision.

5. • From the Data Availability statement, list what the Zenodo record contains, and give the DOI of the version as first submitted.

6. • Check the paper's claim (Sec. IV) that one prime beats every fitted operator's holdout performance by a factor $\gtrsim 1.3$. Use the two lowest-rms operator rows scored on 401–500, Table I ridge $\mu = 0.3$ (OOS 0.265) and Table II Chebyshev ridge $\mu = 0.3$ (OOS 0.264), and compare them with the one-prime oracle (0.198, Table V). Why does this check not cover every operator row in Table I?

7. •• In the arithmetic-continuation probe (Sec. III; App. A anchors), the OOS rms is 0.1958 against a probe-null of 0.1708. By what percentage is the fit worse than predicting zero fluctuation? Why is the probe-null smaller than the real-zero null 0.263?

8. •• For the amplitude-controlled $R = 4$ fits of Sec. V, the exact-verified in-sample rms values are 0.2391 (odd) and 0.2423 (even), against the null 0.2438. Compute the percentage gains and compare with the paper's 1.9% and 0.6%.

9. •• Table I has twelve configurations. Using (32.3) with $\alpha = 0.025$, find the probability that at least one would show a spuriously "significant" positive correlation if all had true correlation zero and were independent. Which assumption is most clearly violated?

10. •• Using `qmnt`, compute the null rms on zeros 401–500 and 501–1000 and compare with Table V.

11. •• Using `qmnt`, reproduce the {2} and 3-prime oracle entries of Table V on zeros 401–500. Then compute the relative skill of each.

12. •• Using `qmnt.smooth_zeros` and `qmnt.wu_sprung_x`, find the band of turning points of levels 401–500 (the principal holdout) in the Wu–Sprung potential. Express its ends as fractions of $L = 44.883$.

13. •• In the dephasing sample, at what distance $t - t_*$ beyond the window does the worst-case phase error of the fundamental reach $\pi/2$? Express your answer in terms of T and evaluate it for $T = 666$, the window of zeros 1–400.

14. •• Using (32.7) with $T = 666$, compute the expected number of primes within π/T of $\omega = 3$ and $\omega = 6$, and the probability of at least one.

15. •• Choose your own $E = (E_1, E_2, E_3)$ with distinct values and a real symmetric 3×3 matrix A with nonzero diagonal. Compute the second-order real shifts from paper Eq. (5) and verify that they sum to zero. Then diagonalize $\mathrm{diag}(E) + iA$ with `numpy.linalg.eigvals` and verify (32.6).

16. •• Explain in your own words why item (i) of the standard requires a within-height split, using the growth of the null in Table V and the results of Table III.

17. •• Design a leakage test, in the style of the listing in Section 32.5, for control C2. What exactly must be overwritten, and which outputs must be identical?

18. •• Sec. III reports surrogate in-sample rms 0.134 ± 0.010 (permuted), 0.141 ± 0.006 (phase-randomized), 0.138 ± 0.005 (amplitude-adjusted), against 0.137 on the true zeros. Assuming each $\pm$ is the spread over ten draws, by how many spreads does the true-zero value differ from each surrogate mean? What do you conclude?

19. •• The scale-2 $M = 300$ row of Table I required a relaxed step-acceptance threshold. Explain why the paper calls it a "sensitivity result", and why including it does not change the qualitative conclusion.

20. ••• Using `qmnt`, form the odd/even split of zeros 1–400 and predict each held-out fluctuation by the mean of its two nearest training fluctuations. Compute the correlation with the truth and compare with the paper's range -0.26 to -0.41 (Sec. III, within-height splits). Explain the sign using the negative autocorrelation at lags 1–3.

21. ••• The six primes between 131 and 211 selected by the 800-zero C2 fit lie at $\omega \approx 4.9$–5.4. For the 800-zero window, $T = t_{800} - t_1 \approx 1170$. Compute $\bar{m}(\log 131)$ from (32.7) and the probability of a chance match. Write two sentences, as a referee, on what this does and does not imply for the paper's alignment claim.

22. ••• Using `qmnt` and the circular block bootstrap of Chapter 26, compute 95% intervals for the six-prime oracle's held-out correlation on 401–500 with block lengths 1, 5 and 20. Compare with the paper's $[+0.95, +0.97]$.

23. ••• Connes [21] obtains approximations to the first 50 zeros from primes $p \le 13$ alone. Explain why the paper says that for this construction "no held-out evaluation question arises", and describe what test is appropriate instead.

24. ••• (Research.) For a small grid and the cosine basis with $M = 50$, compute the first-order sensitivity matrices A_{in} (levels 1–40) and A_{out} (levels 41–50) of the unperturbed Wu–Sprung potential using `qmnt.fd_levels` with `return_vectors=True`. Form the transfer matrix $A_{\mathrm{out}} A_{\mathrm{in}}^{+}$ with a small ridge. Apply it to the true training fluctuations and compute the correlation of its output with the true held-out fluctuations. Discuss.

25. • • • Extend the rule-of-three sample: suppose the keyword search detects a relevant work with probability $d = 0.8$ (it misses 20%). With $n = 29$ works and none detected, find the 95% upper bound on the per-work rate f of held-out tests.

32.10 Answers to selected problems

Problem 1. (i) Train/test split with held-out rms and corr against the model's own null, including a within-height split: Fig. 1 and Table I (forward splits), Table III (within-height splits, needed because the null grows with height, Table V). (ii) Score the exact operator spectrum: Sec. V and Fig. 5, where the second-order surrogate floor 0.232 became 0.2523 (worse than the null) under exact diagonalization. (iii) Survival under regularization and restarts: Table I ridge rows and random-init repeats (caption), and the interpolation split under ridge (Sec. III). (iv) Refit to permuted and phase-randomized targets: Sec. III, where surrogates are fitted as well as the true zeros.

Problem 2. $\mathcal{S} = 1 - 0.313/0.263 = 1 - 1.190 = -0.190$ (operator worse than the null); $\mathcal{S} = 1 - 0.258/0.263 = 1 - 0.981 = +0.019$ (C1 essentially at the null); $\mathcal{S} = 1 - 0.028/0.263 = 1 - 0.106 = +0.894$ (B with given frequencies predicts well).

Problem 3. $2.64/0.255 = 10.35$. "Ten times" is accurate. (With the anchor 2.6434 the ratio is 10.37.)

Problem 4. (a) A: stated in Sec. VI para. 3. (b) B: the audit is recorded in App. A but its limits are not listed in Sec. VI. (c) C: the paper explicitly claims no novelty for the formula. (d) A: stated in Sec. VI para. 3.

Problem 5. All scripts, computed zeros, fitted coefficients, and the full parameter record of App. A; for the revision, the scripts, coefficients, logs and result files of steps 16–23, with an audit file mapping them. The first-submission version has DOI 10.5281/zenodo.21842988.

Problem 6. $0.265/0.198 = 1.34$ and $0.264/0.198 = 1.33$. Both are consistent with "$\gtrsim 1.3$", narrowly. The other Table I rows are scored on different holdouts (and the scale-2 rows with a different null). Checking them would need the one-prime oracle on exactly those holdouts, which Table V does not list, so the check covers only the rows scored on 401–500. You could compute the missing one-prime values yourself with `qmnt.oracle_zeros`.

Problem 7. $(0.1958 - 0.1708)/0.1708 = 0.0250/0.1708 = 0.146$, so the fit is about 14.6% worse than predicting zero fluctuation. The probe-null is the rms of the synthetic target itself, which contains only the $p = 2$ terms ($m \le 3$). The real fluctuations also contain the contributions of all other primes, so their rms is larger.

Problem 8. Odd: $(0.2438 - 0.2391)/0.2438 = 0.0047/0.2438 = 0.0193$, i.e. 1.93%. Even: $(0.2438 - 0.2423)/0.2438 = 0.0015/0.2438 = 0.0062$, i.e. 0.62%. Both round to the paper's 1.9% and 0.6%.

Problem 9. By (32.3), $1 - 0.975^{12} = 1 - 0.738 = 0.262$. The most clearly violated assumption is independence. The rows share training and held-out zeros, and several differ only in M or in the ridge strength, so their errors are strongly related.

Problem 10. Compute `qmnt.rms(qmnt.fluct(t[n-1], n))` for `n = np.arange(401, 501)` and `np.arange(501, 1001)`. You obtain 0.263 and 0.270, the Table V nulls.

Problem 11. Run the Table V listing of Section 32.5 with $k = 1$ and $k = 3$. The results are rms 0.198, corr +0.66 and rms 0.116, corr +0.90, the Table V entries. Relative skill: $1 - 0.198/0.263 = 0.247$ and $1 - 0.116/0.263 = 0.559$.

Problem 12. $s_{401} \approx 681$ and $s_{500} \approx 811$, giving $x(s_{401}) \approx 33.9$ and $x(s_{500}) \approx 38.5$. As fractions of L: $33.9/44.883 = 0.755$ and $38.5/44.883 = 0.858$. The principal holdout probes a band in the outer quarter of the box, beyond the interpolation-gap band 22.5–28.5.

Problem 13. The phase error of the fundamental is $\delta\omega\,(t - t_*)$ with $\delta\omega = \pi/T$. Setting it equal to $\pi/2$ gives $(\pi/T)(t - t_*) = \pi/2$, so $t - t_* = T/2$. For $T = 666$ this is 333 in t.

Problem 14. $2\pi/666 = 0.009434$. At $\omega = 3$: $e^3/3 = 6.695$, $\bar{m} = 0.0632$, $P = 1 - e^{-0.0632} = 0.061$. At $\omega = 6$: $e^6/6 = 67.24$, $\bar{m} = 0.634$, $P = 1 - e^{-0.634} = 0.470$.

Problem 15. One example: $E = (1, 2, 4)$, and A with diagonal $(0.3, -0.4, 0.1)$ and off-diagonal entries $A_{12} = 0.2$, $A_{13} = -0.1$, $A_{23} = 0.25$. The diagonal does not enter Eq. (5), because the sum excludes $m = n$; it gives only the imaginary first-order shifts iA_{nn}. Eq. (5) gives

$$\begin{aligned}\delta E_1 &= -[0.04/(-1) + 0.01/(-3)] = 0.04333,\\ \delta E_2 &= -[0.04/1 + 0.0625/(-2)] = -0.00875,\\ \delta E_3 &= -[0.01/3 + 0.0625/2] = -0.03458.\end{aligned}$$

These sum to zero, because each pair (n, m) contributes $-A_{nm}^2/(E_n - E_m)$ to level n and the opposite amount to level m. The real parts of the exact eigenvalues sum to $7 = 1 + 2 + 4$, as (32.6) requires: the trace is $7 + i(0.3 - 0.4 + 0.1) = 7$.

Problem 16. In a forward split the held-out zeros are higher, where the fluctuation amplitude is larger (null 0.263 on 401–500 rising to 0.296 on 5001–10000, Table V). A model could fail because the test data differ in distribution (covariate shift) rather than because it captured no structure. A split within one height range removes the shift. Table III shows the fits fail even then (for the random 80/20 splits, held-out errors 1.5–3.0 times the null at scale 1), so covariate shift is not the explanation, while kriging and the oracle succeed on the same held-out sets, showing that predictable structure is present.

Problem 17. Overwrite every held-out height (all zeros beyond the training window) with noise. Also overwrite any other source of held-out information, such as a precomputed table of fluctuations. Then rerun the *complete* C2 procedure: the periodogram search, the forward split inside the training window that chooses K and the ridge strength, and the final refit on the whole window. The number K, the selected frequencies, the chosen ridge strength and all fitted amplitudes must be bit-identical to those from the clean run. Comparing only the final amplitudes would miss leakage into the model-selection step.

Problem 18. Permuted: $(0.137 - 0.134)/0.010 = 0.3$. Phase-randomized: $(0.141 - 0.137)/0.006 = 0.67$. Amplitude-adjusted: $(0.138 - 0.137)/0.005 = 0.2$. All differences are well under one spread: the fit does no better on the real zeros than on targets with no arithmetic content, so in-sample accuracy measures capacity.

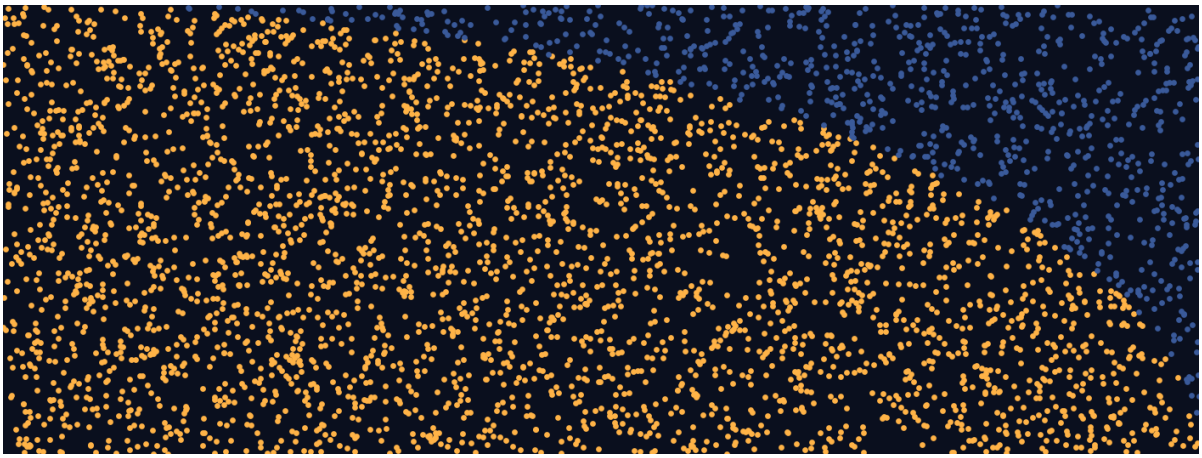
Problem 19. Under the strict acceptance rule (an improvement factor of 0.999, App. A step 6), the optimizer accepts no step for this configuration. The fitted potential therefore stays at $c = 0$, which is the null. To obtain a fitted row at all, the threshold was relaxed to $1 - 10^{-5}$ for this row only. A result that exists only after changing one setting of the protocol measures how sensitive the outcome is to that setting; it is not a default-protocol result, hence "sensitivity result". Including it does not change the conclusion. The relaxed fit scores OOS 0.370 with corr -0.01, worse than the null 0.274, and the strict fit is at the null. Neither shows skill.

Problem 21. $\log 131 = 4.875$, $e^{4.875}/4.875 = 131/4.875 = 26.87$, $2\pi/1170 = 0.005370$, so $\bar{m} = 0.144$ and $P = 1 - e^{-0.144} = 0.134$. Referee: a single match near $\omega \approx 4.9$ has roughly a one-in-seven chance of occurring by accident, so no single high-frequency match is strong evidence. The claim rests instead on the collective alignment compared with a surrogate ensemble, which the paper reports and labels exploratory; and C2's held-out predictive skill stands independently of the alignment statistic.

Problem 23. The construction takes only prime data as input; no zero is used to fit anything. There is no training set, so every zero is effectively held out and the question of transfer from training to test does not arise. The appropriate test is direct: compare the constructed values with the true zeros (the paper reports accuracies from 10^{-3} down to 2.6×10^{-55}) and check the mathematical guarantees (location on the critical line).

Problem 25. A relevant work is detected with probability fd. Then $(1 - fd)^n = 0.05$ gives

$fd = 1 - 0.05^{1/29} = 0.0981$, so $f = 0.0981/0.8 = 0.123$. Keyword misses loosen the bound from about 10% to about 12% per work.



Four thousand random points in the unit square; the fraction inside the quarter circle estimates $\pi/4$. Five lines of Python.

Appendix A

Python for This Book

↑Contents

What this chapter is about

Every chapter contains short Python listings, and every figure and every worked number in this book is produced by a Python script that you can run. This appendix teaches exactly the Python you need to read those listings and to modify them. If you have never programmed, read it once before Chapter 2 and come back to it whenever a listing puzzles you.

A.1 Getting started

Install Python 3 (version 3.10 or later) from `python.org` or through the Anaconda distribution. Then install the four libraries the book uses, from a terminal:

```
pip install numpy scipy matplotlib mpmath
```

Download the book's source repository; the companion library `qmnt` lives in its folder `code/`. To use it, either run your scripts from inside `code/`, or tell Python where it is:

```
import sys
sys.path.insert(0, "path/to/repository/code")
import qmnt
```

You can type Python line by line in an interactive session (run `python` in a terminal, or use a Jupyter notebook), or save lines in a file `myscript.py` and run `python myscript.py`.

A.2 Numbers, variables and arithmetic

A *variable* is a name attached to a value. The equals sign assigns; it does not state an equation.

```
L = 44.883          # a floating-point number (a "float")
n = 400             # an integer
h = 2 * L / (n + 1) # arithmetic: + - * /
E = h ** 2          # ** is "to the power"
print(h, E)         # prints 0.2238553615960100 0.05011122...
```

Everything after `#` on a line is a comment, ignored by Python. Integer division is `//` (so `7 // 2` is `3`) and the remainder is `%` (so `7 % 2` is `1`). Complex numbers are written with `j` for i: `z = 3 + 4j`, and `abs(z)` gives 5.0.

Floats carry about 16 significant digits. Chapter 4 explains why `0.1 + 0.2 == 0.3` is `False` in Python, and why you should compare floats with a tolerance: `abs(a - b) < 1e-12`. (`1e-12` means 10^{-12}.)

A.3 Lists, loops and functions

A *list* holds several values in order; positions are counted from 0.

```
t = [14.134725, 21.022040, 25.010858]   # the first three zeta zeros
print(t[0])         # 14.134725 (first element: index 0)
print(t[-1])        # 25.010858 (last element)
print(len(t))       # 3
for x in t:         # a loop: the indented block runs once per element
    print(x / 2)
```

Indentation (four spaces) is how Python knows which lines belong to the loop. `range(1, 6)` produces $1, 2, 3, 4, 5$ (it stops *before* the second number).

A *function* packages a calculation so that it can be reused:

```
import math
def Nsm(t):
    """Smooth zero-counting function, Eq. (1) of the paper."""
    u = t / (2 * math.pi)
    return u * math.log(u) - u + 7 / 8

print(Nsm(14.134725))   # about 0.449: the first zero comes slightly before s_1 =
    14.52
```

`def` names the function and its inputs; `return` gives back the result. The text in triple quotes is a description for human readers. `import math` makes the standard mathematical functions available as `math.log`, `math.sin`, `math.pi`, and so on. Note that `log` is the natural logarithm, as everywhere in this book.

`if` chooses between alternatives:

```
def is_prime(k):
    if k < 2:
        return False
    for d in range(2, int(k ** 0.5) + 1):
        if k % d == 0:      # d divides k exactly
            return False
    return True
print([k for k in range(30) if is_prime(k)])   # [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]
```

The last line is a *list comprehension*: it builds the list of all k in the range for which the condition holds.

A.4 NumPy: arrays and vectorized arithmetic

NumPy provides *arrays*: lists of numbers on which arithmetic acts element by element, fast.

```
import numpy as np
n = np.arange(1, 11)          # array([1, 2, ..., 10])
x = np.linspace(0, 1, 5)      # 5 equally spaced points: 0, 0.25, 0.5, 0.75, 1
y = np.sin(2 * np.pi * x)     # sin applied to every element
```

```
print(y.round(3))              # [ 0.  1.  0. -1. -0.]
print(n.sum(), n.mean(), n.max())   # 55 5.5 10
print(n[2:5])                  # elements at positions 2, 3, 4: [3 4 5]
print(n[n % 2 == 0])           # the even ones: [ 2  4  6  8 10]
```

The *slice* `a[i:j]` takes positions i up to but not including j. A two-dimensional array is a matrix: `A = np.array([[1, 2], [3, 4]])`; `A @ v` multiplies matrix by vector, `A.T` is the transpose, `A[0, 1]` is the entry in row 0, column 1. The root mean square and the correlation used throughout the book are one line each:

```
err = np.array([0.1, -0.3, 0.2])
rms = np.sqrt(np.mean(err ** 2))          # 0.2160
a, b = np.array([1., 2, 3, 4]), np.array([2., 4, 5, 9])
r = np.corrcoef(a, b)[0, 1]               # Pearson correlation, 0.9648
```

A.5 SciPy and mpmath

SciPy supplies standard numerical algorithms. The two used most in this book are root finding and eigenvalues.

```
from scipy.optimize import brentq
from scipy.linalg import eigh_tridiagonal
root = brentq(lambda t: Nsm(t) - 0.5, 10, 20)   # solve Nsm(t) = 1/2 between 10 and
    20
print(root)                                      # 14.5213...: the first smooth zero
    s_1

N, L = 1000, 1.0
h = 2 * L / (N + 1)
d = np.full(N, 2 / h**2)                         # diagonal of -d2/dx2 on a grid
e = np.full(N - 1, -1 / h**2)                    # off-diagonal
E = eigh_tridiagonal(d, e, eigvals_only=True)[:3]
print(E, (np.arange(1, 4) * np.pi / (2 * L))**2)  # numerical vs exact box levels
```

`lambda t: ...` defines a small unnamed function on the spot. `brentq(f, a, b)` finds a root of f between a and b, provided $f(a)$ and $f(b)$ have opposite signs (Chapter 4).

mpmath does arithmetic with as many digits as you ask for, and knows the zeta function:

```
import mpmath
mpmath.mp.dps = 20                  # work with 20 significant digits
print(mpmath.zeta(2))               # 1.6449340668482264365 = pi^2/6
print(mpmath.zetazero(1))           # (0.5 + 14.134725141734693790j)
```

A.6 Plotting with Matplotlib

```
import matplotlib.pyplot as plt
t = np.linspace(0, 40, 2000)
plt.figure(figsize=(6, 4))
plt.plot(t, np.sin(t * np.log(2)), label="sin(t log 2)")
plt.xlabel("t"); plt.ylabel("amplitude"); plt.legend()
plt.savefig("wave.pdf", bbox_inches="tight")   # or plt.show() on screen
```

A.7 The companion library qmnt

qmnt collects, in one short and readable file (code/qmnt/__init__.py), every computation the book needs. Its functions are listed below; each is explained in the chapter shown. Indices of zeros start at 1, as in the paper, and all energies are in the units $\hbar = 2m = 1$.

Function	What it returns	Chapter
zeros(nmax)	$t_1, \ldots, t_{n_{\max}}$, up to 10^4	16
Nsm(t), smooth_zeros(n), mean_gap(n), fluct(t, n)	smooth counting function, s_n, gap_n, fluctuations	17
primes_upto(x), first_primes(k)	primes	15
Nfl(t, primes), oracle_zeros(n, primes)	explicit-formula prediction, paper Eq. (4)	18
rms(a), corr(a, b), score(t_pred, n)	the paper's scores	19
V0, wu_sprung_x(V), wu_sprung_V(x), L_of(Vmax)	Wu–Sprung potential, paper Eq. (2)	22
grid(L, N), fd_levels(V, L, k), richardson	finite-difference solver	23
basis(x, L, M, kind), fit_potential(nfit, M, ...)	the paper's Hermitian fit	24

A complete small session, which reproduces the first entry of the paper's Table V (the one-prime oracle on zeros 401–500):

```
import numpy as np, qmnt
n = np.arange(401, 501)
null = qmnt.score(qmnt.smooth_zeros(n), n)        # predict t_n = s_n
one = qmnt.score(qmnt.oracle_zeros(n, [2]), n)    # one prime
print(null, one)    # rms 0.263 (corr undefined for the null), then rms 0.198, corr
    +0.66
```

A.8 Reproducibility habits

Three habits make computational results trustworthy, and the paper follows all three. (1) Fix random seeds: rng = np.random.default_rng(1) then rng.normal(size=10) gives the same numbers on every run. (2) Record versions: np.__version__, scipy.__version__. (3) Keep the script that made every number, and rerun it from a clean start before trusting the number.

The symbols of this book.

Appendix B

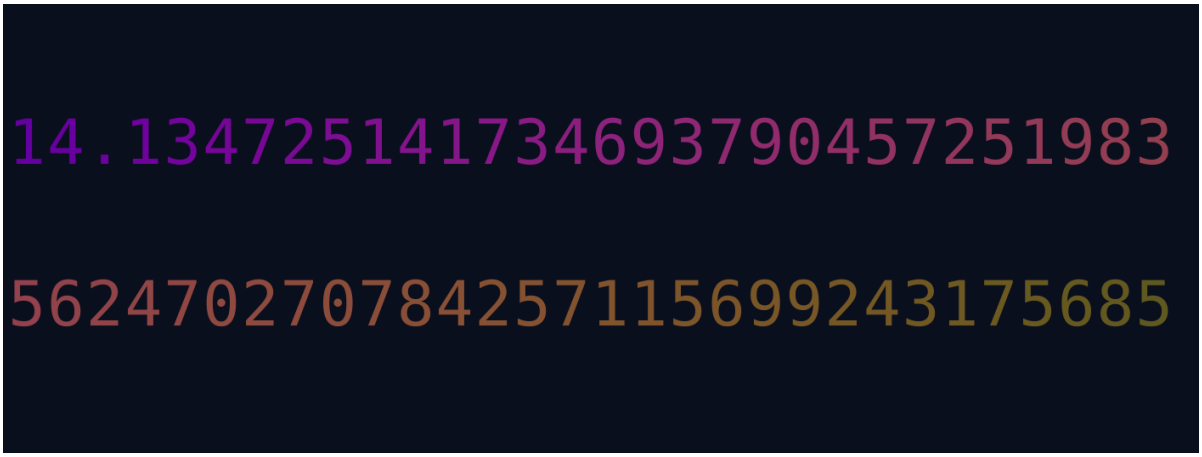
Notation

↑ Contents

Symbols are listed with their meaning and the chapter where they are introduced. The notation follows the paper [1].

Symbol	Meaning	Ch.
i	imaginary unit, $i^2 = -1$	2
$s = \sigma + it$	complex variable of the zeta function	16
$\zeta(s)$	Riemann zeta function	16
$\rho_n = \frac{1}{2} + it_n$	n-th nontrivial zero; $t_n > 0$ its height	16
p, p^m, $\Lambda(k)$	prime, prime power, von Mangoldt function	15
log	natural logarithm	2
$N(T)$	number of zeros with $0 < t \le T$	17
$N_{\mathrm{sm}}(t)$	smooth counting function, paper Eq. (1)	17
$S(t)$	fluctuating part of the count	17
s_n	smooth zeros, $N_{\mathrm{sm}}(s_n) = n - \frac{1}{2}$	17
gap_n	local mean spacing ds_n/dn	17
$N_{\mathrm{fl}}(t)$	truncated explicit formula, paper Eq. (4)	18
P	a finite set of primes	18
$H = -d^2/dx^2 + V(x)$	Hamiltonian in units $\hbar = 2m = 1$	10
E_n, ψ_n	energy levels and normalized eigenfunctions	11
$V_0 = 3.10073\pi$	bottom of the Wu–Sprung potential	22
$x(V)$	Wu–Sprung potential in implicit form, paper Eq. (2)	22
L	half-width of the box $[-L, L]$	11
N (grid), h	number of interior grid points, grid spacing $2L/(N+1)$	23
E^{Rich}	Richardson extrapolant $(4E_{h/2} - E_h)/3$	23
$\tilde{t}_n$, Δ_n	bias-corrected targets, grid bias	23
δV, c_k, ϕ_k, M	fitted correction, its coefficients, basis functions, basis size	24
A_{nk}	sensitivity $\int \lvert\psi_n\rvert^2 \phi_k \, dx$	13
r_n	residual $\tilde{t}_n - E_n$	24
λ, μ	Levenberg–Marquardt damping, ridge strength	9, 24
σ_k	singular values (σ_1 the largest)	9
iA, A_{nm}	non-Hermitian perturbation, its matrix elements	14
$T^{(n)}$	second-order tensor, $\delta E_n = -c^\top T^{(n)} c$	14
Γ, R	decay widths; amplitude-ball radius	14
ω, T, K	angular frequency in t, training-window length, number of selected frequencies	27
rms, corr	root-mean-square error in mean spacings; Pearson correlation	19
null	the prediction "no fluctuation"	25
OOS	out-of-sample (held-out)	25

Overloaded symbols in the paper. The paper uses A both for the sensitivity matrix A_{nk} and for the non-Hermitian operator A; T for the training window, the tensors $T^{(n)}$ and the Chebyshev polynomials T_k; and N both for the number of grid points and for the counting function $N(T)$. The context always decides.



The first nontrivial zero, $\frac{1}{2} + 14.134725141734693790\ldots\, i$.

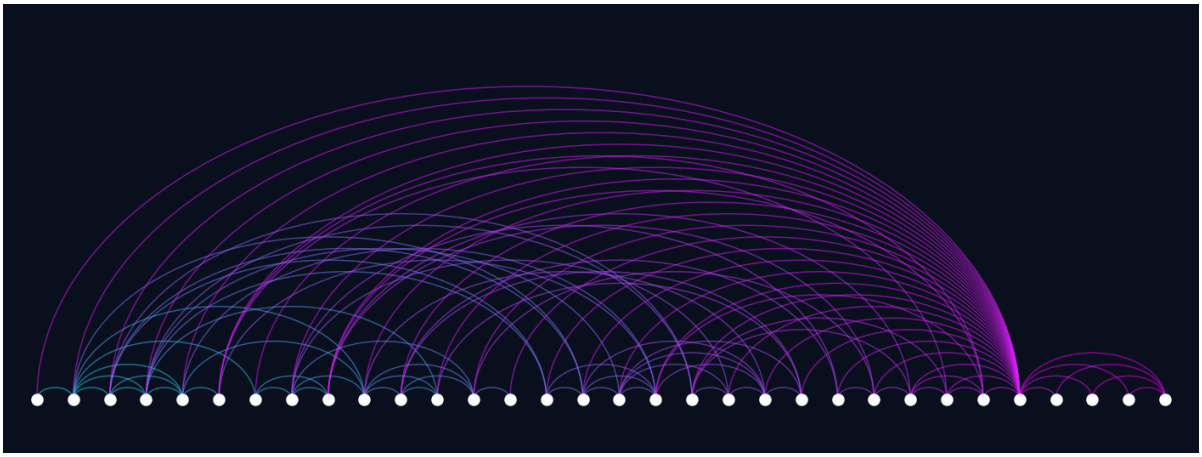
Appendix C

Key Numbers of the Paper

↑ Contents

These are the anchors that the paper's Appendix A lists for checking a reconstruction, with the chapters that explain them. All rms values are in units of the local mean spacing.

Quantity	Value	Ch.
t_1, t_{500}, t_{1000}	14.1347251417, 811.1843588465, 1419.42248094599	16
Box half-widths $L = x(1000)$, $x(1500)$	44.883, 59.957	22
Richardson vs smooth residual	mean 0.005, max 0.063	23
Null, in-sample 1–400 / OOS 401–500	0.2438 / 0.2631	25
Null, scale 2: 1–800 / 801–1000	0.2554 / 0.2739	25
Raw-zero null on uncorrected grid, 401–500	2.09	23
Hermitian coverage on 500 zeros, $M = 300$ / 900	0.186 / $< 10^{-4}$	24
Principal operator fit ($M = 400$, fit 1–400): in / OOS / corr	0.137 / 0.369 / −0.07	29
Six-prime oracle on 401–500, corr interval	$[+0.95, +0.97]$	30
One-prime oracle on 401–500	0.198 / +0.66	18
25-prime oracle on 401–500	0.043 / +0.99	18
Continuation probe: in-sample / OOS / probe-null	0.0555 (+0.95) / 0.1958 (+0.07) / 0.1708	29
Interpolation split: in-sample / gap rms / corr	0.1361 / 2.6434 / −0.02	29
Sparse model (C2): selected frequencies	$K = 24$ (fit 1–400), $K = 43$ (fit 1–800)	30
Non-Hermitian unconstrained floors (odd / even)	0.2321 / 0.2358 (0.2322 with artifact)	31
Exact-spectrum in-sample rms of those fits	0.2523 / 0.2477	31
Fair $R = 4$ floors, exact-verified	0.2391 / 0.2423	31



How the 32 chapters lead into one another: an edge joins each chapter to the chapters it assumes.

Appendix D

Map from the Paper to This Book

↑ Contents

For each part of the paper [1] (revision R1 numbering), the chapters that supply the tools and the chapter of Part VI that reads it.

Paper element	Tools	Read in
Abstract, Sec. I, Fig. 1	1, 25, 18	28
Sec. I A a: inverse-spectral fits	22, 21	28
Sec. I A b: explicit formulae, Connes	18	28
Sec. I A c: machine learning on arithmetic data	25	28
Sec. I A d: operator proposals	21, 14	28
Sec. II A, Eq. (1), Fig. 2: zeros, unfolding	16, 17, 20	28
Sec. II B, Eq. (2): Wu–Sprung potential, solver	12, 22, 23	28
Sec. II C: Hermitian fit, OOS protocol, bases, within-height splits	13, 9, 24, 25	28
Eq. (3): no leakage	23	28
Sec. II D, Eq. (4): oracle and fitted sinusoidal controls	18, 6, 27	30
Sec. II E, Eq. (5): non-Hermitian classes	13, 14	31
Sec. II F: use of AI	none	28, 32
Sec. III, Table I, Fig. 3: memorization	24, 25, 26	29
Table II: other bases	24, 26	29
Table III: within-height splits, kriging	25, 20	29
Permutation and surrogate tests	6, 26	29
Interpolation split, Fig. 4	12, 24, 25	29
Arithmetic-continuation probe	5, 18, 25	29
Sec. IV, Tables IV–V	18, 27, 26	30
Sec. V, V A, Fig. 5: spurious floor	14, 4 (float32)	31
Sec. VI: discussion and proposed standard	all	32
Appendix A: reconstruction record	A, 23, 24	32

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